Mathematics

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Notation List

Collision Index and Scope Rules

bare $A, B, \ldots$; $\mathcal{A}, \mathcal{B}, \ldots$	unadorned Latin and calligraphic letters denote local objects unless a decorated form is listed below; local declarations in a theorem, proof, or subsection take precedence
B, $B_r(x)$, $\overline{B}_r(x)$, $\mathrm{B}(x, y)$, $\mathcal{B}(\mathbb{R}^n)$	matrix or set variable / open ball / closed ball / beta function / Borel σ-algebra; font, arguments, and ambient subject determine the referent
$I_{\mathcal{V}}$, I_n, I^k, I_R	identity operator, identity matrix, k-cell, and power-series interval
P, $\mathcal{P}(X)$, $\mathscr{P}$, $P_{\mathcal{U}}$, P_j	locally declared projection or map / power set / umbral polynomial algebra / orthogonal projection / spectral projection
$\mathcal{F}$, $\mathscr{F}$, $\mathcal{F}(K)$, $\mathcal{F}(\mathcal{V})$	local family of functions or operators / umbral formal-series algebra / finite subsets of K / finite-rank operators
$\mathcal{L}(\mathcal{V}, \mathcal{W})$, $\mathcal{L}^n$, $\mathscr{L}^1$, $\mathrm{LS}(A, \mathbf{y})$, $\mathcal{L}(\theta, \alpha)$	linear maps / Lebesgue measurable sets / integrable functions / least-squares solution set / local Lagrangian or loss notation
$R(f, \Pi, \{c_i\})$, R, $R_k(A)$, $A = QR$	Riemann sum, locally declared radius or scalar, row-sum quantity, and QR factorization; arity and argument type disambiguate the notation
Δx_i, ΔA, $\Delta(A)$, $\Delta(\mathcal{V})$, Δ_{k-1}	increment, matrix perturbation, defect from normality, metric discriminant, and coefficient simplex
Ω, Ω_T, Ω_n, $\mathrm{vol}(\Omega)$	local domain or set / finite spectrum of T / primitive roots of unity or DFT diagonal, as specified by context / volume domain
Q^k, Q_n, Φ_n	standard k-simplex / normalized Fourier matrix / cyclotomic polynomial; cyclotomic polynomials use Φ_n
f^{-1}, $f^{-1}(B)$	inverse function when f is bijective / preimage of B under any function f; bijectivity, argument type, and local declaration determine the meaning
$\wedge$, $\vee$, $x \wedge y$, $x \vee y$, $dx_I \wedge dx_J$	logical conjunction and disjunction / lattice meet and join / exterior product; subject and operand types determine the meaning
$\mathrm{supp}(f)$	nonzero set $\{x : f(x) \neq 0\}$ when the codomain has a distinguished zero / its closure when the domain is topological; the local definition determines the meaning
$\overline{z}$, $\overline{E}$, $\overline{\mathcal{A}}$, A^*, $T^*\omega$, f', $\mathcal{V}'$	conjugation, closure, uniform closure, adjoint or conjugate transpose, pullback, derivative, and dual space; the decorated object determines the operation

Global Conventions and Basic Notation

Convention. Notation is scoped by subject, argument type, font, decoration, and any local definition in force. Unadorned letters denote local objects by default; persistent meanings are carried by subscripts, superscripts, operators, fonts, or declared argument patterns. When a glyph is reused, the distinguishing feature is recorded below.

Set, Function, and Logical Notation

$x \in E$, $x \notin E$	membership / non-membership
$E \subseteq F$, $\emptyset$	subset inclusion, not necessarily proper / empty set
$X \times Y$, $f : X \to Y$	Cartesian product / function with domain X and codomain Y
$f(A)$, $f^{-1}(B)$, f^{-1}	image of A / preimage of B / inverse function when f is bijective; argument type and local context distinguish preimages from inverse-function values
$\mathrm{range}(f)$, $f\|_A$	range of f / restriction of f to A
$g \circ f$, id_X	composition of functions with compatible domain and codomain / identity function on X
$\mathbf{1}_E$	indicator of E, equal to 1 on E and 0 outside E
$[x]_\sim$, $X/\sim$	equivalence class of x / quotient set for the relation $\sim$
$\{A_i\}_{i \in I}$, $\bigcup_{i \in I} A_i$, $\bigcap_{i \in I} A_i$	indexed family of sets / set of points lying in some A_i / set of points lying in every A_i
$\sup E$, $\inf E$	least upper bound / greatest lower bound in the ambient ordered set; extended values may be used when explicitly allowed
$x \wedge y$, $x \vee y$, $\bot$, $\top$	meet / join / bottom / top in a lattice
$\neg$, $\wedge$, $\vee$	logical negation / conjunction / inclusive disjunction
$\Rightarrow$, $\Longrightarrow$, $\Longleftrightarrow$	logical implication (either arrow length) / logical equivalence
$\forall$, $\exists$, $\exists!$	universal / existential / unique-existence quantifier
$:=$	definition by equality

Number, Size, and Scalar Notation

$\mathbb{N}$, $\mathbb{N}_0$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	positive integers / nonnegative integers / integers / rationals / reals / complexes
$n!$, $(n)_r$, $\binom{n}{r}$	factorial / falling factorial / binomial coefficient
$\mathfrak{S}_n$, $\mathrm{sign}(\sigma)$	permutations of $\{1, \ldots, n\}$ / sign of a permutation
$\mathbb{F}$, $\mathbb{F}_0$	ambient scalar field unless fixed as $\mathbb{R}$ or $\mathbb{C}$ / prime subfield of $\mathbb{F}$; probabilities use explicit Pr or locally declared probability spaces
$\|x\|$, $\|E\|$, $\|\|x\|\|$	absolute value or complex modulus for scalars; interval length or cardinality according to argument type; norm of a vector, function, or operator
a, b, c	locally declared scalars or interval endpoints

Linear and Operator Structures Notation

Vector Spaces and Lists

$\mathbb{F},\ \mathbb{F}_0$	ambient scalar field, typically $\mathbb{R}$ or $\mathbb{C}$ / prime subfield of $\mathbb{F}$
$V(n,q),\ \mathbb{F}_q^n$	n-dimensional coordinate space over the finite field with q elements
$\mathbb{F}^{(\mathbb{N})}$	vector space of finitely supported sequences over $\mathbb{F}$
$\mathbf{u},\mathbf{v},\mathbf{w},\mathbf{0}$	vectors / zero vector
$\mathcal{U},\mathcal{V},\mathcal{W}$	vector spaces or subspaces; other calligraphic letters such as $\mathcal{A},\mathcal{B},\mathcal{F},\mathcal{M}$ are local unless listed below; umbral algebras use script $\mathscr{P},\mathscr{F}$
$I_{\mathcal{V}},\ I_n,\ 0$	identity operator on $\mathcal{V}$ / identity matrix / zero map or zero matrix; distinct from the cell I^k and interval I_R
$\beta=(\mathbf{v}_1,\dots,\mathbf{v}_m)$	ordered vector list; a basis iff it is linearly independent and spans
$(),\ \beta_{\widehat{j}}$	empty list / list β with the jth entry omitted
$(\beta,\mathbf{v}),\ \beta_{j\leftarrow\mathbf{v}}$	list with $\mathbf{v}$ appended / list with the jth entry replaced by $\mathbf{v}$
$\mathcal{B}$	locally declared basis set, such as a Hamel, Hilbert, module, or affine basis; $\mathcal{B}(\mathbb{R}^n)$ denotes the Borel σ-algebra in analysis
$\operatorname{span}(E),\ \dim(\mathcal{V})$	linear span of E / dimension of $\mathcal{V}$
$\operatorname{codim}_{\mathcal{V}}(\mathcal{U})$	codimension of $\mathcal{U}$ in $\mathcal{V}$; equal to $\dim(\mathcal{V}/\mathcal{U})$ in finite dimension
$\bigwedge_{i\in I}\mathcal{U}_i,\ \bigvee_{i\in I}\mathcal{U}_i$	meet / join of a family of subspaces
$\sum_{i\in I}\mathcal{U}_i$	sum of an indexed family of subspaces; equivalently the join
$\mathcal{U}+\mathcal{W},\ \mathcal{U}\oplus\mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$\mathcal{V}_1\times\cdots\times\mathcal{V}_m$	product vector space with componentwise operations
$\prod_{i\in I}\mathcal{V}_i,\ \bigoplus_{i\in I}^{\text{ext}}\mathcal{V}_i$	external direct product / external direct sum of finite-support families
$\mathcal{V}\cong\mathcal{W},\ \mathbf{e}_{j,k}$	isomorphic vector spaces / basis vector supported in one product component
$\mathbf{v}+\mathcal{U},\ \pi(\mathbf{v})$	coset modulo $\mathcal{U}$ / locally declared quotient map $\mathcal{V}\to\mathcal{V}/\mathcal{U}$

Linear Maps and Duality

$T:\mathcal{V}\to\mathcal{W},\ \mathcal{L}(\mathcal{V},\mathcal{W})$	linear map / vector space of linear maps; distinct from measure-theoretic $\mathcal{L}^n$, least-squares $\mathrm{LS}(A,\mathbf{y})$, and local loss notation $\mathcal{L}(\theta)$
$T_A:\mathbb{F}^n\to\mathbb{F}^m$	linear map $\mathbf{x}\mapsto A\mathbf{x}$ induced by $A\in\mathrm{Mat}(m,n,\mathbb{F})$
$ST,\ T\vert_{\mathcal{U}},\ T/\mathcal{U}$	composition $S\circ T$ / restriction to $\mathcal{U}$ / induced quotient operator
$T^{-1},\ T^m,\ p(T)$	inverse of an invertible map or operator / operator power / polynomial functional calculus
$\ker(T),\ \operatorname{range}(T)$	kernel and range of T
$\operatorname{rank}(T),\ \operatorname{nullity}(T)$	dimensions of the range and kernel of a finite-domain linear map T
$\rho_{\mathcal{U},\mathcal{W}},\ \rho\perp\sigma$	projection onto $\mathcal{U}$ along $\mathcal{W}$ / algebraically orthogonal idempotents satisfying $\rho\sigma=\sigma\rho=0$; geometric orthogonality is listed separately below
$\mathcal{V}_{\mathbb{C}},\ \mathrm{cpx},\ T_{\mathbb{C}}$	complexification of a real vector space / canonical embedding $\mathbf{v}\mapsto\mathbf{v}+i\mathbf{0}$ / complexification of a real operator
$\operatorname{graph}(T),\ [S,T]$	graph of a map / commutator $ST-TS$
$\mathrm{Comm}(\mathcal{F}),\ \mathrm{Comm}(A)$	commutant of a family / commutant of a single matrix or operator
$A\sim_{\text{row}}B,\ S\sim T$	row-equivalent matrices / operator similarity when declared; distinct from cardinal, a.e., and coefficient equivalence
$\mathcal{V}',\ \beta',\ T'$	dual space, dual basis, and dual map; prime denotes duality in this context
$\varphi,\psi,\ \mathcal{U}^0$	linear functionals / annihilator
$\mathcal{V}'',\ \Lambda_{\mathcal{V}}$	double dual / natural evaluation map $\mathcal{V}\to\mathcal{V}''$
$[B]_{\beta,\gamma},\ \Phi_B,\ \Psi_B$	matrix of a bilinear pairing in ordered bases / induced maps $\mathcal{V}\to\mathcal{W}'$ and $\mathcal{W}\to\mathcal{V}'$ in a pairing context; module coordinate maps redeclare Φ_B locally
$[\mathbf{v}]_\beta,\ [T]_{\gamma\leftarrow\beta}$	coordinate vector in basis β / matrix of T from β-coordinates to γ-coordinates
$C_{\gamma\leftarrow\beta}$	change-of-basis matrix sending β-coordinate columns to γ-coordinate columns

Matrices and Entrywise Structure

$A,B,C,\ \mathrm{Mat}(m,n,\mathbb{F})$	matrix variables / space of $m\times n$ matrices over $\mathbb{F}$; decorated forms such as C_f and C_{ij} have separate meanings
$A_{ij},\ A_{j,\cdot},\ A_{\cdot,k}$	entry in row i and column j / jth row / kth column
$[a_{ij}],\ E_{j,k},\ A\mathbf{x}$	matrix with entries a_{ij} / matrix unit / matrix-vector product
$A^{-1},\ \mathrm{Mat}(n,n,\mathbb{F})^\times$	inverse matrix when it exists / set of invertible $n\times n$ matrices
$A^t,\ A^T,\ A^*$	transpose / alternate transpose notation / conjugate transpose or adjoint
$\operatorname{diag}(\lambda_1,\dots,\lambda_n)$	diagonal matrix with diagonal entries $\lambda_1,\dots,\lambda_n$
$\overline{z},\ \operatorname{Re}z,\ \operatorname{Im}z$	complex conjugate / real part / imaginary part; for sets, overline denotes closure
$\operatorname{tr}(A),\det(A)$	trace and determinant
$\operatorname{adj}(A)$	adjugate matrix
$M_{ij},\ C_{ij},\ A_j(\mathbf{b})$	minor matrix / cofactor / matrix obtained from A by replacing column j with $\mathbf{b}$
$\operatorname{col}(A),\ \operatorname{row}(A)$	column space and row space of A
$\operatorname{rank}(A),\ \operatorname{nullity}(A)$	rank and nullity of A
$A\oplus B,\ A\otimes B$	matrix direct sum / Kronecker product; vector-space direct sums use subspace notation
$A\otimes I_q+I_n\otimes B$	Kronecker sum of square matrices A and B in compatible dimensions
$A\circ B,\ \operatorname{vec}(A)$	Hadamard product / column-stacking vectorisation
$M/A,\ M/D$	Schur complement of the block A or D in a compatible block matrix M; distinct from quotient-space notation
$B,\ L=B^tB$	directed incidence matrix / associated incidence Laplacian in a graph context

Inner Product Geometry

$\|\mathbf{v}\|$, $\|T\|_{\mathrm{op}}$	vector norm / operator norm
$\|\cdot\|_1$, $\|\cdot\|_2$, $\|\cdot\|_\infty$	coordinate one-norm / Euclidean or spectral two-norm according to argument type / sup norm
$\|\cdot\|_F$	Frobenius norm
$\langle \mathbf{u}, \mathbf{v}\rangle$, $\langle A, B\rangle_F$	inner product / Frobenius inner product
$\mathbf{u} \perp \mathbf{v}$, $\mathcal{U}^\perp$, δ_{ij}	orthogonal vectors, orthogonal complement, Kronecker delta
$\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$, $P_{\mathcal{U}}$	orthogonal projection of $\mathbf{u}$ onto $\mathrm{span}(\mathbf{v})$ / orthogonal projection onto $\mathcal{U}$
$\mathrm{LS}(A, \mathbf{y})$	least-squares solution set $\arg\min_{\mathbf{x}\in\mathbb{F}^n} \|A\mathbf{x} - \mathbf{y}\|_2$ for $A \in \mathrm{Mat}(m, n, \mathbb{F})$; roman notation avoids collision with $\mathcal{L}$ and $\mathcal{L}^n$
$d(\mathbf{v}, \mathcal{U})$	distance $\inf_{\mathbf{u}\in\mathcal{U}} \|\mathbf{v} - \mathbf{u}\|$ from $\mathbf{v}$ to $\mathcal{U}$
$\mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram matrix of $\mathbf{u}_1, \ldots, \mathbf{u}_n$
$\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram determinant
$P_{\mathbf{u}}$, $U_{\mathbf{w}}$	rank-one projection matrix / Householder matrix, with defining vector indicated by the subscript
$\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$	ordered basis; orthonormal when declared in an inner-product context
$\mathbf{e} = (1, \ldots, 1)^t$	all-ones vector
$\mathrm{Ray}_A(\mathbf{x})$, $\mathbf{x}^* A\mathbf{x}$	Rayleigh quotient notation; for unit $\mathbf{x}$, $\mathrm{Ray}_A(\mathbf{x}) = \mathbf{x}^* A\mathbf{x}$ / Hermitian quadratic form
$\mathrm{Sym}(B)$, $\mathrm{Skew}(B)$, $\mathrm{Alt}(B)$	predicates that the bilinear form B is symmetric, skew-symmetric, or alternating
$\mathrm{rad}_L(B)$, $\mathrm{rad}_R(B)$	left / right radical of a declared bilinear pairing B
$S^{\perp_R}$, ${}^{\perp_L}T$	right / left orthogonal of a subset under a declared bilinear pairing
M_β, $\Delta(\mathcal{V})$, $\mathrm{rad}(\mathcal{V})$	matrix of a square bilinear form in basis β / discriminant set or square class / radical of a metric vector space
$\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, $\mathcal{H}$, $w(\mathcal{V})$	orthogonal direct sum / hyperbolic space or plane in metric-vector-space contexts / Witt index; affine hyperplanes use $\mathcal{H}(\mathbf{N}, b)$
$\mathcal{O}(\mathcal{V})$, $\mathrm{Sp}(\mathcal{V})$, $\tau_{\mathbf{v},a}$, $\sigma_{\mathbf{u}}$	orthogonal group / symplectic group / symplectic transvection / orthogonal symmetry

Affine, Projective, and Convex Geometry

$\mathcal{A}(\mathcal{V})$, $\mathcal{A}_0(\mathcal{V})$, $\mathrm{Sub}(\mathcal{V})$	flats in $\mathcal{V}$ / $\mathcal{A}(\mathcal{V}) \cup \{\emptyset\}$ / subspaces of $\mathcal{V}$; $\mathcal{A}$ without arguments remains local
$\mathrm{affspan}(E)$, $\mathrm{affhull}(E)$	affine span / least flat containing E
$X \parallel Y$, $X \vee Y$	parallel flats / join of flats or projective subspaces; meet is intersection in the relevant lattice
$\mathbb{R}^n_+$, $\mathbb{R}^n_{++}$, $\mathbf{x} \geq \mathbf{0}$, $\mathbf{x} > \mathbf{0}$, $\mathbf{x} \gg \mathbf{0}$	nonzero nonnegative orthant / coordinatewise positive orthant / coordinatewise nonnegative, nonzero nonnegative, and coordinatewise positive inequalities
$\mathrm{conv}(S)$, Δ_{k-1}, C_A	convex hull / coefficient simplex $\{(t_i) : t_i \geq 0, \sum_i t_i = 1\}$ / nonnegative cone generated by the columns of A; distinct from cofactor and companion notation
$\mathcal{H}(\mathbf{N}, b)$, $\mathcal{H}_\pm(\mathbf{N}, b)$, $\langle \mathbf{N}, X\rangle < b$	affine hyperplane / closed half-spaces / setwise inequality, meaning $\langle \mathbf{N}, \mathbf{x}\rangle < b$ for all $\mathbf{x} \in X$

Spectral and Polynomial Structure

$E_\lambda(T)$, $\mathrm{spec}(T)$	eigenspace $\ker(T - \lambda I_{\mathcal{V}})$ / spectrum
$\mathrm{Inv}(T)$	lattice or set of T-invariant subspaces
$\lambda_i(A)$, $m_A(\lambda)$	eigenvalues in the locally declared order / algebraic multiplicity of λ as a root of p_A; distinct from measure notation $m(E)$
$\mathcal{G}_\lambda(T)$, $J_k(\lambda)$	generalized eigenspace for λ / Jordan block of size k and eigenvalue λ
p_A, q_T, χ_T	matrix characteristic polynomial / operator characteristic polynomial / characteristic polynomial determined by elementary divisors
μ_A, μ_T, $w_p(\lambda, A)$	minimal polynomial of a matrix or operator / Weyr count of λ-Jordan blocks of size at least p; distinct from measure notation $\mu(E)$
P_j, $T = \sum_j \mu_j P_j$	spectral projections / spectral decomposition with distinct eigenvalues μ_j
$\sqrt{T}$, $A^{1/2}$, $\rho(A)$	positive square root of a positive operator or matrix / spectral radius of A; distinct from the interval parameter in K_ρ
$\mathbb{F}[z]$	polynomials in z over $\mathbb{F}$
$\mathscr{P} = \mathbb{C}[x]$, $\mathscr{P}'$, $\mathscr{F} = \mathbb{C}[[t]]$	umbral polynomial algebra / algebraic dual / formal power-series algebra; script notation distinguishes these from $\mathcal{P}(X)$ and local families $\mathcal{F}$
$\mathcal{V}_T$, $\mathcal{C}_T(\mathbf{v})$, $\mathrm{ann}_T(\mathbf{v})$	$\mathbb{F}[z]$-module induced by T / T-cyclic subspace generated by $\mathbf{v}$ / annihilator ideal of $\mathbf{v}$
$m_{\mathbf{v}}$, C_f, $\mathrm{InvFact}(T)$, $\mathrm{ElemDiv}(T)$	order polynomial of $\mathbf{v}$ under T / companion matrix of f / invariant factors / elementary divisors
e^{tA}	matrix exponential $\sum_{k\geq 0} (tA)^k/k!$

Matrix Analysis and Numerical Structure

$A \succeq 0,\ A \succ 0$	Hermitian positive semidefinite / Hermitian positive definite
$\mathrm{in}(A),\ n_+(A), n_-(A), n_0(A)$	inertia of a Hermitian matrix / positive, negative, and zero eigenvalue counts
$A = QR,\ A = LL^*,\ A = LDL^t$	QR factorization / Cholesky factorization / symmetric or Hermitian LDL^t factorization
$A = V\Sigma W^*,\ A = X\Sigma_r Y^*$	singular value decomposition / compact SVD with rank r
$\sigma_j(A),\ \sigma_{\max}(A),\ \sigma_{\min}(A)$	singular values and extreme singular values
$\lvert T\rvert,\ \lvert A\rvert,\ \mathrm{vol}(\Omega)$	operator or matrix modulus / volume of a set; scalar absolute value remains $\lvert x\rvert$
$A^\dagger,\ A_k$	Moore–Penrose pseudoinverse / rank-k truncated SVD approximation in an SVD context
$\kappa_2(A),\ \Delta A,\ \Delta\mathbf{x}$	spectral condition number of an invertible matrix / matrix and solution perturbations
$\Delta(A)$	defect from normality; distinct from perturbation notation ΔA
$R_k(A),\ R_k^{\mathrm{off}}(A),\ R_{\max}(A)$	absolute row sum $\sum_j \lvert a_{kj}\rvert$ / deleted row sum $\sum_{j\neq k} \lvert a_{kj}\rvert$ / maximum absolute row sum
$\mathrm{Ger}_k(A),\ \mathrm{Ger}(A)$	kth Gersgorin disk / union of all Gersgorin disks
$\mathrm{tridiag}(a, b, c)$	tridiagonal matrix with subdiagonal a, diagonal b, and superdiagonal c
$w,\ F_n,\ Q_n,\ \omega_n$	half-bandwidth / unnormalised Fourier matrix / normalised Fourier matrix / primitive nth root used for the DFT
$\mathcal{K}_m(A, \mathbf{b})$	Krylov subspace $\mathrm{span}(\mathbf{b}, A\mathbf{b}, \ldots, A^{m-1}\mathbf{b})$
$Q_m,\ H_m$	Krylov basis matrix / projected Hessenberg or tridiagonal matrix
$A = S - T,\ B = S^{-1}T,\ \mathbf{e}_k$	matrix splitting / stationary iteration matrix / kth error vector
$A = D + L + U,\ B_J,\ B_{\mathrm{GS}}$	diagonal, lower, and upper splitting / Jacobi and Gauss–Seidel iteration matrices
$\omega,\ S_\omega,\ T_\omega,\ B_\omega$	SOR parameter / SOR splitting matrices / SOR iteration matrix; distinct from the differential form ω
$M,\ M^{-1}A,\ AM^{-1}$	preconditioner / left and right preconditioned matrices

Algebraic Structures Notation

Algebraic Language and Categories

$\mathcal{C},\ A \xrightarrow{f} B,\ \mathrm{id}_A$	ambient category / morphism $f : A \to B$ / identity morphism of A; $\mathcal{C}$ remains local unless declared
$\mathrm{Hom}_{\mathcal{C}}(A, B),\ g \circ f$	set of morphisms $A \to B$ in $\mathcal{C}$ / composition for compatible morphisms
$A \simeq B,\ F : \mathcal{C} \to \mathcal{D}$	isomorphic objects / functor with domain $\mathcal{C}$ and codomain $\mathcal{D}$

Groups, Actions, Rings, and Ideals

$G, H, K,\ e_G,\ \langle S\rangle$	groups or subgroups / identity element of G / substructure generated by S in the ambient algebraic category
$H \leq G,\ H \trianglelefteq G,\ G/H,\ [G : H]$	subgroup / normal subgroup / quotient group / subgroup index
$\mathrm{ord}(g),\ \mathrm{Aut}(G)$	order of an element / automorphism group
$R, S,\ 0_R,\ 1_R,\ R^\times$	rings / zero and identity elements of R / unit group of R
$Z(R),\ \mathrm{char}(R)$	center of R / characteristic of R
$I, J,\ \mathfrak{a}, \mathfrak{p},\ R/I$	ideals; fraktur letters denote distinguished ideals, such as prime ideals, when declared / quotient ring
$R[x],\ \mathbb{F}[z],\ \deg f$	polynomial rings in the displayed indeterminate / degree of a polynomial
$U_n,\ \Omega_n,\ \Phi_n(x),\ \varphi(n)$	complex nth roots of unity / primitive nth roots / nth cyclotomic polynomial / Euler totient; Ω_n is algebraic root notation, distinct from domain notation

Modules and Associative Algebras

$M, N,\ S \leq M,\ M/N$	modules / submodule S of M / quotient module
$\mathrm{Hom}_R(M, N),\ \mathrm{End}_R(M)$	set of R-module homomorphisms $M \to N$ / endomorphism ring of M
$\ker\tau,\ \mathrm{im}\,\tau,\ \overline{\tau}$	kernel / image / quotient-induced map $M/\ker\tau \to \mathrm{im}\,\tau$ for a homomorphism $\tau : M \to N$
$\bigoplus_i M_i,\ M = N \oplus H$	external direct sum of modules / internal direct-sum decomposition
$\mathcal{C}_S(N),\ \mathcal{L}_\sigma,\ \mathcal{R}_\sigma$	set $\{H \leq N : N = S \oplus H\}$ of complements / left inverses of σ / right inverses of σ; subscript distinguishes these from $\mathcal{L}$ and $\mathcal{L}^n$
$\mathcal{F}(\mathcal{V})$	finite-rank operators $\{T \in \mathcal{L}(\mathcal{V}) : \dim\mathrm{im}\,T < \infty\}$ in associative-algebra contexts; distinct from local function-family notation

Algebraic Geometry and Commutative Algebra Notation

Status. This part of the notes is still mostly scaffolded. The entries below reserve the notation promised by the section headings and keep it distinct from already-active algebra and analysis notation.

Commutative Algebra

$\mathrm{Spec}\,R,\ \mathfrak{p} \in \mathrm{Spec}\,R$	set of prime ideals of a commutative ring R / prime ideal $\mathfrak{p} \subseteq R$
$D(f),\ V(I)$	standard open set $\{\mathfrak{p} \in \mathrm{Spec}\,R : f \notin \mathfrak{p}\}$ / Zariski closed set $\{\mathfrak{p} : I \subseteq \mathfrak{p}\}$
$S^{-1}R,\ R_{\mathfrak{p}},\ k(\mathfrak{p})$	localization at a multiplicative set S / localization $(R\setminus\mathfrak{p})^{-1}R$ / residue field $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$
$\mathrm{rad}(I),\ \sqrt{I}$	radical ideal $\{r \in R : r^n \in I \text{ for some } n \geq 1\}$; use $\mathrm{rad}(\mathcal{V})$ for metric-vector-space radicals

Affine and Projective Geometry

$\mathbb{A}^n_k,\ \mathbb{P}^n_k$	affine n-space over a field k / projective n-space over k; geometric notation distinct from probability notation
$I(X),\ Z(I),\ k[X]$	vanishing ideal $\{f \in k[x_1, \ldots, x_n] : f\vert_X = 0\}$ / zero locus of an ideal / coordinate ring, typically $k[x_1, \ldots, x_n]/I(X)$
$f^\#,\ \varphi^*$	coordinate-ring pullback, e.g. $f^\# : k[Y] \to k[X]$ for $f : X \to Y$; distinct from adjoints and differential-form pullbacks by context

Analysis and Continuous Structure Notation

Sets and Functions in Analysis

$\mathcal{P}(X)$	power set $\{A : A \subseteq X\}$; projection symbols use $P_{\mathcal{U}}$ or P_j
$J_n,\ \#E,\ A \sim_{\#} B$	finite initial segment $J_0 = \emptyset$, $J_n = \{1, \dots, n\}$ / cardinality of a finite set / existence of a bijection $A \to B$

Order and Completeness

$\lvert x\rvert,\ \lvert a - b\rvert$	real absolute value or complex modulus / usual real-line distance $d(a, b) = \lvert a - b\rvert$
$\overline{\mathbb{R}}$	extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$ ordered by $-\infty < x < +\infty$ for $x \in \mathbb{R}$; the bar is conventional, not topological closure
$x < y,\ x \le y$	strict order / non-strict order in the ambient ordered set

Sequences and Series

$(a_n),\ a_n \to a$	sequence indexed by $n \in \mathbb{N}$ / convergence of the sequence to a
$(a_{n_k}),\ n_1 < n_2 < \cdots$	subsequence indexed by a strictly increasing sequence (n_k) of positive integers
$s_n = \sum_{k=1}^{n} a_k$	partial-sum sequence associated to $\sum a_k$
$\limsup a_n = s^*,\ \liminf a_n = s_*$	upper and lower limits in $\overline{\mathbb{R}}$, equivalently $\inf_N \sup_{n \ge N} a_n$ and $\sup_N \inf_{n \ge N} a_n$ for real sequences
$\sum_{n=1}^{\infty} a_n,\ \prod_{n=1}^{\infty} a_n$	limit of partial sums / limit of finite partial products when defined

Metric and Topological Notation

$d(x, y),\ B_r(x)$	metric distance / open ball $\{y : d(x, y) < r\}$
$B_\varepsilon(a) = V_\varepsilon(a)$	real-line open ball $(a - \varepsilon, a + \varepsilon)$; $V_\varepsilon(a)$ is Abbott's notation
$\overline{B}_r(x),\ \{G_\alpha\}_{\alpha \in A}$	closed ball $\{y : d(x, y) \le r\}$ / indexed open cover
$d_E,\ d_{\max},\ d_{\text{sum}}$	Euclidean, maximum, and sum product metrics on a product space
$E^\circ,\ \overline{E},\ E^{\text{acc}}$	interior / topological closure / set of limit points of E
$\operatorname{bd}(E),\ E^c$	boundary $\overline{E} \setminus E^\circ$ / complement in the ambient space
$\operatorname{diam} E$	diameter $\sup\{d(p, q) : p, q \in E\}$ of a nonempty metric set
$X \simeq Y$	homeomorphic metric spaces, i.e. related by a bijection with continuous inverse
$A \cap \overline{B} = \emptyset,\ \overline{A} \cap B = \emptyset$	definition of separated subsets A and B
$F_\sigma,\ G_\delta,$ meager/thin/first category	countable union of closed sets / countable intersection of open sets / subset of a countable union of nowhere-dense sets
residual/thick, generic	set containing a countable intersection of dense open sets / property whose truth set is residual

Functions, Differentiation, and Integration

$\lim_{x \to p} f(x),\ f(x+),\ f(x-)$	limit of f at p / right and left limits at x
$C(X),\ C([a, b]),\ C_b(X),\ \lVert f\rVert_\infty$	continuous functions on X and $[a, b]$ / bounded continuous functions / supremum norm
$f_n \to f,\ f_n \rightrightarrows f$	pointwise convergence / uniform convergence on the ambient domain
$f'(x),\ Df(a),\ D_j f_i,\ \partial_i f,\ J_f(x)$	one-variable derivative, total derivative at a, component partial $D_j f_i$, scalar partial, and Jacobian determinant; J_f is distinct from J_n and $J_k(\lambda)$
$\nabla f,\ D_{\mathbf{u}} f,\ C^r,\ C^\infty,\ C^\omega$	gradient, directional derivative, and classes C^r, C^∞, C^ω
$\Pi = \{x_0, \dots, x_n\},\ c_i,\ \Delta x_i,\ \Delta\alpha_i$	partition of $[a, b]$, tags $c_i \in [x_{i-1}, x_i]$, and increments $\Delta x_i = x_i - x_{i-1}$, $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$
$R(f, \Pi, \{c_i\}),\ \delta,\ \delta(x)$	$R(f, \Pi, \{c_i\}) = \sum_{i=1}^{n} f(c_i)\Delta x_i$ / constant mesh bound / gauge; uppercase $R(\cdots)$ is local to Riemann sums
$U(\Pi, f, \alpha),\ L(\Pi, f, \alpha)$	Riemann–Stieltjes upper and lower sums with integrator α
$f \in \mathcal{R},\ f \in \mathcal{R}(\alpha)$	ordinary Riemann integrability / Riemann–Stieltjes integrability with respect to α
$\int_a^b f,\ \int_a^b f\, d\alpha,\ \int_E f\, d\mu$	ordinary Riemann integral / Riemann–Stieltjes integral / measure integral
$\int_a^\infty f,\ \int_{-\infty}^{\infty} f$	one-sided and two-sided improper integrals as limits over finite truncations
$\ell(\Pi, \gamma),\ \ell(\gamma)$	polygonal length of γ along Π / curve length
$I^k,\ Q^k,\ \operatorname{supp}(f)$	k-cell, standard k-simplex, and topological support $\overline{\{x : f(x) \neq 0\}}$; I^k is not an identity operator
$dx_I,\ dx_I \wedge dx_J$	basic form indexed by I / wedge product of basic forms
$\omega,\ d\omega,\ T^*\omega,\ \partial_{\text{ch}}\Psi$	differential form, exterior derivative, pullback by T, and chain boundary
$\int_\Psi \omega,\ \partial_{\text{ch}}[\mathbf{p}_0, \dots, \mathbf{p}_k]$	integral of ω over the chain Ψ / oriented boundary $\sum_{j=0}^{k} (-1)^j [\mathbf{p}_0, \dots, \widehat{\mathbf{p}_j}, \dots, \mathbf{p}_k]$
$dV,\ dA,\ ds$	volume, area, and arc-length elements

Power Series, Fourier, and Special Functions

$\sum_{n \ge 0} c_n (x - a)^n,\ R$	power series centered at a / radius of convergence in this context, distinct from $R(\cdots)$ and the R in QR
$I_R,\ K_\rho$	centered interval $I_R = (-R, R)$ / compact subinterval $K_\rho = [-\rho, \rho] \subset I_R$ for $0 < \rho < R$
$\widehat{f}(m),\ s_N(f; x),\ s_N(f)$	$\widehat{f}(m) = (2\pi)^{-1} \int_{-\pi}^{\pi} f(x) e^{-imx}\, dx$ / $s_N(f; x) = \sum_{\lvert m\rvert \le N} \widehat{f}(m) e^{imx}$, viewed pointwise or as a function
$f \sim_{\text{coef}} \sum \widehat{f}(n) e^{inx},\ f \sim_{\text{coef}} \sum c_n \phi_n$	formal Fourier or orthonormal coefficient expansion; no convergence assertion is implied
$D_N(t)$	Dirichlet kernel $D_N(t) = \sum_{m=-N}^{N} e^{imt}$; uppercase D_N is not derivative notation
$\mathsf{E},\ \mathsf{L},\ \mathsf{C},\ \mathsf{S}$	power-series exponential, inverse logarithm, cosine, and sine
$\Gamma(x),\ \mathrm{B}(x, y)$	gamma function / beta function in roman type, distinct from balls $B_r(x)$ and variables B

Measure and Function Spaces

$E \in \mathcal{M},\ E \in \mathcal{M}(\mu)$	E measurable in the ambient σ-algebra / Rudin measurable set generated by μ
$\mathcal{B}(\mathbb{R}^n),\ \mathcal{L}^n$	Borel σ-algebra on $\mathbb{R}^n$ / Lebesgue measurable subsets of $\mathbb{R}^n$; not a basis $\mathcal{B}$, a linear-map space $\mathcal{L}$, or a least-squares set LS
$\mathcal{E},\ \mathcal{M}_F(\mu),\ A \triangle B,\ d_\mu(A,B)$	elementary sets, finitely μ-measurable sets, symmetric difference, and $d_\mu(A,B) = \mu^*(A \triangle B)$; $\mathcal{M}$ is measure-theoretic here
$\mu^*(E),\ m^*(E)$	μ-outer measure / Lebesgue outer measure
$\mathrm{vol}_n(B),\ m_n^*(\Omega)$	n-dimensional box volume / box-cover outer measure $\inf\{\sum_{j\in J} \mathrm{vol}_n(B_j) : (B_j)_{j\in J} \text{ covers } \Omega,\ J \text{ at most countable}\}$
$m(E),\ \mu(E)$	Lebesgue or elementary volume of E / measure of E; not algebraic multiplicity $m_A(\lambda)$ or minimal polynomial μ_A
$\lambda^+,\ \lambda^-,\ \lvert\lambda\rvert,\ \lambda \ll \mu,\ \lambda \perp \mu,\ \frac{d\lambda}{d\mu}$	positive variation, negative variation, total variation, absolute continuity, mutual singularity, and Radon–Nikodym derivative
$\bowtie,\ \{f \bowtie a\}$	relation in $\{>, \geq, <, \leq\}$ / level set $\{x : f(x) \bowtie a\}$
$f^+,\ f^-,\ \mathcal{S}^+(f),\ \mathcal{I}_E(s)$	$f^+ = \max(f,0),\ f^- = -\min(f,0)$ / simple minorants of f / simple-function integral on E
$f \sim_{\text{a.e.}} g$ on E, a.e.	$\mu(\{x \in E : f(x) \neq g(x)\}) = 0$ / outside a null set
$E_x,\ E^y,\ f_x,\ f^y$	product slices $E_x = \{y : (x,y) \in E\}$, $E^y = \{x : (x,y) \in E\}$, and $f_x(y) = f^y(x) = f(x,y)$
$\mathcal{U}f,\ F_a$	undergraph $\{(x,t) : 0 < t < f(x)\}$ for $f \geq 0$ / superlevel set $F_a = \{x : f(x) > a\}$; $\mathcal{U}f$ is not a vector subspace
$[f : A],\ [B : Q],\ \mathrm{dp}(E)$	$[f : A] = m(A)^{-1} \int_A f$ / $[B : Q] = m(B \cap Q)/m(Q)$ / density-point set of E
$\mathscr{L}^1(\mu),\ \mathscr{L}^1_{\text{loc}}(\mathbb{R}^n)$	measurable f with $\int \lvert f\rvert\, d\mu < +\infty$ / locally integrable functions; script $\mathscr{L}$ is distinct from $\mathcal{L}^n$ and $\mathcal{L}$
$L^p(\mu),\ L^p(E),\ \lVert f\rVert_p$	Lebesgue space modulo a.e. equality with respect to μ or the ambient measure on E / $\lVert f\rVert_p = (\int \lvert f\rvert^p\, d\mu)^{1/p}$

Probability and Stochastic Structure Notation

Probability Spaces and Distributions

$(\Omega, \mathcal{F}, \mathbb{P}),\ \Pr(A)$	probability space / event probability $\Pr(A) = \mathbb{P}(A)$ for $A \in \mathcal{F}$; explicit probability notation keeps $\mathbb{P}_k^n$ and $\mathbb{F}_0$ unambiguous
$X, Y,\ X_*\mathbb{P},\ \mathcal{L}(X)$	random variables / pushforward law $X_*\mathbb{P}(B) = \mathbb{P}(X^{-1}(B))$ / law of X when locally declared; not Lebesgue $\mathcal{L}^n$
$X \sim F,\ X \overset{\text{dist}}{=} Y,\ X_1, \ldots, X_n$ iid	X has distribution F / X and Y have the same distribution / independent and identically distributed sample
$F(t),\ g(t),\ \Phi(t)$	distribution function $F(t) = \Pr(X \leq t)$ / density g / standard normal distribution function in probabilistic contexts
$\mathbb{E}[X],\ \mathrm{Var}(X),\ \mathrm{Cov}(X,Y)$	expectation / $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$ / $\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, when finite
$\mathrm{Corr}(X,Y),\ \rho,\ \sigma_{XY}$	correlation coefficient / local correlation parameter / covariance shorthand $\sigma_{XY} = \mathrm{Cov}(X,Y)$ when declared
$\psi_X(t)$	moment-generating function $\psi_X(t) = \mathbb{E}e^{tX}$ on its domain of finiteness
$X \perp\!\!\!\perp Y$	independence of random variables, equivalently independence of $\sigma(X)$ and $\sigma(Y)$ when defined
$X \perp\!\!\!\perp Y \mid Z$	conditional independence of X and Y given Z
$X_n \xrightarrow{\mathbb{P}} X,\ X_n \rightsquigarrow X,\ X_n \xrightarrow{L^p} X,\ X_n \xrightarrow{\text{a.s.}} X,\ X_n \xrightarrow{qm} X$	convergence in probability / convergence in distribution / convergence in L^p / almost-sure convergence / convergence in quadratic mean

Stochastic Processes and Markov Structure

$(X_t)_{t\in T},\ \mathcal{F}_t$	stochastic process indexed by T / filtration $(\mathcal{F}_t)$ when locally declared; this $\mathcal{F}_t$ is not the umbral algebra $\mathscr{F}$
$\mathbf{x},\ \mathbf{e},\ A\mathbf{e} = \mathbf{e}$	probability vector $\mathbf{x} \geq 0$, $\sum_i x_i = 1$ / all-ones vector $\mathbf{e} = (1, \ldots, 1)^t$ / row-stochastic Markov convention $A = [a_{ij}] \geq 0,\ A\mathbf{e} = \mathbf{e}$
$\pi,\ \pi A = \pi,\ A^p$	stationary distribution written as a row probability vector / stationarity equation / p-step Markov matrix
$P_t,\ p_t(x,y)$	transition semigroup / transition probability, kernel, or density from x to y, when introduced

Numerical and Computational Mathematics Notation

Status. Apart from numerical linear algebra, already covered under Linear and Operator Structures, the numerical and computational chapters remain placeholders. The entries below reserve notation only; they do not introduce PDE or machine-learning conventions.

Floating Point, Approximation, and Error

$\mathrm{fl}(x),\ u,\ \varepsilon_{\text{mach}}$	floating-point rounded value, unit roundoff, and machine epsilon
$x_h,\ e_h,\ \mathcal{O}(h^p)$	approximation at discretization scale h, corresponding error, and Landau order as $h \to 0$
$\widehat{x},\ \Delta x,\ r = b - A\widehat{x}$	computed approximation, perturbation or error in x, and residual for $Ax = b$

Monte Carlo and Stochastic Computation

$X_1, \ldots, X_N,\ \widehat{\mu}_N$	samples / sample-mean estimator $\widehat{\mu}_N = N^{-1}\sum_{i=1}^N X_i$ when estimating $\mu = \mathbb{E}X$
$\mathrm{Var}(\widehat{\mu}_N),\ \mathrm{SE}(\widehat{\mu}_N)$	estimator variance / standard error $\mathrm{SE}(\widehat{\mu}_N) = \sqrt{\mathrm{Var}(\widehat{\mu}_N)}$
$w_i,\ \widehat{I}_N$	importance weights / Monte Carlo estimate of an integral or expectation, when locally introduced

Part I
Set Theory, Logic, and Foundations

Foundational prerequisites collect the logical, set-theoretic, functional, order-theoretic, cardinal, and research-practice language used throughout the notes.

1 Sets

Sets provide the membership language used throughout the notes: elements, subsets, products, power sets, operations, and indexed families.

1.1 Introduction to Sets

Definition 1.1.1. *Sets and Membership*
Sets and the membership relation are taken as primitive notions.

(i) *Membership*: Given an object x and a set E, the assertion $x \in E$ means that x is an element of E; its negation is written $x \notin E$.
(ii) *Empty set*: The symbol $\emptyset$ denotes a set satisfying $x \notin \emptyset$ for every object x.

Definition 1.1.2. *Set-Builder Notation*
For a set X and a condition $P(x)$ on its elements, $\{x \in X : P(x)\}$ denotes the set of all $x \in X$ satisfying $P(x)$.

1.2 Cartesian Products

Definition 1.2.1. *Ordered Pairs and Cartesian Products*
Coordinate order is part of both constructions.

(i) *Ordered pair*: For objects a, b, (a, b) has first coordinate a and second coordinate b.
(ii) *Equality*: For objects a, b, a', b', $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$.
(iii) *Cartesian product*: For sets X and Y, $X \times Y := \{(x, y) : x \in X,\ y \in Y\}$.

For intervals A and B, the Cartesian product $A \times B$ has a rectangular geometric representation.

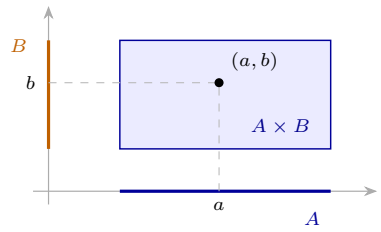

Figure 1: The product region $A \times B$ for horizontal interval A and vertical interval B.

Figure 1 illustrates Definition 1.2.1: $(a, b) \in A \times B$ precisely when $a \in A$ and $b \in B$, in that order.

1.3 Subsets and Power Sets

Definition 1.3.1. *Basic Subset Terminology*
Let A, B be sets.

(i) *Subset*: $A \subseteq B$ if and only if $x \in A$ implies $x \in B$ for every object x.
(ii) *Proper subset*: $A \subsetneq B$ iff $A \subseteq B$ and $A \neq B$.
(iii) *Singleton*: $A = \{x\}$ for some object x.
(iv) *Nonempty*: $A \neq \emptyset$.

Assumption 1.3.2. *Set Extensionality*
Throughout these notes, set equality is governed by extensionality.
For sets A and B, $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof Method 1.3.3. *Proof by Two Containments*
To prove equality of sets A and B, discharge the following obligations.

(i) Fix an arbitrary $x \in A$ and prove $x \in B$; this establishes $A \subseteq B$.
(ii) Fix an arbitrary $x \in B$ and prove $x \in A$; this establishes $B \subseteq A$.
(iii) Invoke Assumption 1.3.2 to conclude $A = B$.

If both sets have explicit membership criteria, the two inclusions may be proved together.
Fix an arbitrary object x and prove $x \in A \iff x \in B$.

Definition 1.3.4. *Power Set*
For a set X, its *power set* is $\mathcal{P}(X) := \{S : S \subseteq X\}$. The two smallest cases are as follows.

(i) The empty set has only itself as a subset, so $\mathcal{P}(\emptyset) = \{\emptyset\}$.
(ii) If $X = \{a, b\}$ with $a \neq b$, then $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, X\}$.

1.4 Union, Intersection, Difference, and Complement

Definition 1.4.1. *Elementary Set Operations*
Let X be a set and let $A, B \subseteq X$.

(i) *Union*: $A \cup B := \{x \in X : x \in A \text{ or } x \in B\}$.
(ii) *Intersection*: $A \cap B := \{x \in X : x \in A \text{ and } x \in B\}$.
(iii) *Difference*: $A \setminus B := \{x \in X : x \in A \text{ and } x \notin B\}$.
(iv) *Complement*: $A^c := X \setminus A$, relative to the ambient set X.

Definition 1.4.2. *Symmetric Difference*
Let X be a set and let $A, B \subseteq X$. The *symmetric difference* is $A \triangle B := (A \setminus B) \cup (B \setminus A)$.
It consists exactly of the elements belonging to one of A and B, but not both.

Proposition 1.4.3. *Basic Set Algebra*
Let X be a set and let $A, B, C \subseteq X$. Then

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
(iii) $(A \cup B)^c = A^c \cap B^c$;
(iv) $(A \cap B)^c = A^c \cup B^c$;
(v) $A \triangle B = (A \cup B) \setminus (A \cap B)$.

Proof. Fix an arbitrary $x \in X$. By set extensionality, each identity follows once its two membership conditions are equivalent.

(i) Membership in $A \cap (B \cup C)$ means that $x \in A$ and that either $x \in B$ or $x \in C$. This holds exactly when either $x \in A \cap B$ or $x \in A \cap C$. Hence it is equivalent to membership in $(A \cap B) \cup (A \cap C)$.
(ii) Membership in $A \cup (B \cap C)$ means that $x \in A$, or that both $x \in B$ and $x \in C$. This holds exactly when two conditions are satisfied. First, $x \in A$ or $x \in B$; second, $x \in A$ or $x \in C$. Hence it is equivalent to membership in $(A \cup B) \cap (A \cup C)$.
(iii) The condition $x \in (A \cup B)^c$ is equivalent to $x \notin A \cup B$. This means that $x \notin A$ and $x \notin B$, or equivalently $x \in A^c \cap B^c$.
(iv) The condition $x \in (A \cap B)^c$ is equivalent to $x \notin A \cap B$. This means that $x \notin A$ or $x \notin B$, or equivalently $x \in A^c \cup B^c$.
(v) The condition $x \in A \triangle B$ is equivalent to $x \in A \setminus B$ or $x \in B \setminus A$. Thus x belongs to one of A and B, but not both. Equivalently, $x \in A \cup B$ and $x \notin A \cap B$. By the definition of difference, this is equivalent to $x \in (A \cup B) \setminus (A \cap B)$.

Structural Commentary: Each identity reduces to an equivalence of membership conditions.
Key Insight: Membership turns set operations into logical connectives. □

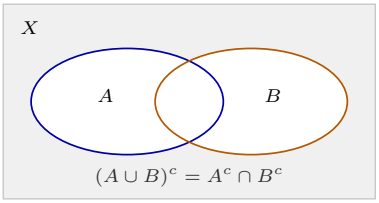

Figure 2: The shaded region represents $(A \cup B)^c = A^c \cap B^c$ inside the ambient set X.

Figure 2 illustrates part (iii). The shaded region lies outside both A and B, so it is outside $A \cup B$.

1.5 Indexed Families and Products

Definition 1.5.1. *Families, Indexed Families, Unions, and Intersections*
Let X be a set and let I be an index set.

(i) *Family*: A set whose members are sets, also called a *collection* of sets.
(ii) *Indexed family*: An assignment $i \mapsto A_i$ over I, written $(A_i)_{i \in I}$ or $\{A_i\}_{i \in I}$. It is an indexed family in X iff $(\forall i \in I)\, A_i \subseteq X$.
(iii) *Indexed union*: For an indexed family $\{A_i\}_{i \in I}$ in X, $\bigcup_{i \in I} A_i := \{x \in X : (\exists i \in I)\, x \in A_i\}$.
(iv) *Indexed intersection*: For an indexed family $\{A_i\}_{i \in I}$ in X, $\bigcap_{i \in I} A_i := \{x \in X : (\forall i \in I)\, x \in A_i\}$.

Theorem 1.5.2. *Arbitrary De Morgan Laws*
Let X and I be sets, and let $\{E_i\}_{i \in I}$ be an indexed family in X. Take all complements relative to X.

(i) $\left(\bigcup_{i \in I} E_i\right)^c = \bigcap_{i \in I} E_i^c$.
(ii) $\left(\bigcap_{i \in I} E_i\right)^c = \bigcup_{i \in I} E_i^c$.

Proof. Fix an arbitrary $x \in X$. Extensionality reduces both identities to pointwise membership equivalences.

(i) $x \in \left(\bigcup_{i \in I} E_i\right)^c \iff (\forall i \in I)\, x \notin E_i \iff (\forall i \in I)\, x \in E_i^c \iff x \in \bigcap_{i \in I} E_i^c$.
(ii) $x \in \left(\bigcap_{i \in I} E_i\right)^c \iff (\exists i \in I)\, x \notin E_i \iff (\exists i \in I)\, x \in E_i^c \iff x \in \bigcup_{i \in I} E_i^c$.

Structural Commentary: The proof checks both identities pointwise and applies quantifier negation.
Key Insight: Complement exchanges existential and universal membership conditions. □

Definition 1.5.3. *Disjoint Families and Parametrizations*
Let I be an index set and let $\mathcal{F}$ be a family of sets. Let $\{A_i\}_{i\in I}$ be an indexed family of sets.

(i) *Pairwise disjoint*: $(\forall E, G \in \mathcal{F})\,(E \neq G \implies E \cap G = \emptyset)$.
(ii) *Parametrization*: $\{A_i\}_{i\in I}$ parametrizes $\mathcal{F}$ iff $\mathcal{F} = \{A_i : i \in I\}$.

Definition 1.5.4. *Cartesian Product of an Indexed Family*
Let I be an index set and let $\{E_i\}_{i\in I}$ be an indexed family of sets.

(i) *Product*: $\prod_{i\in I} E_i := \{(x_i)_{i\in I} : (\forall i \in I)\, x_i \in E_i\}$.
(ii) *Binary case*: If $I = \{1, 2\}$, identify $(x_i)_{i\in I}$ with the ordered pair (x_1, x_2). Under this identification, $\prod_{i\in I} E_i = E_1 \times E_2$.
(iii) *Constant family*: If E is a set and $(\forall i \in I)\, E_i = E$, write $E^I := \prod_{i\in I} E_i$. Thus E^I is the set of I-indexed families in E.
(iv) *Dependence*: Different parametrizations of the same family can yield different products.

1.6 Russell's Paradox and Axioms

Theorem 1.6.1. *Russell's Paradox*
For a condition $P(x)$ on sets, unrestricted comprehension asserts $(\exists R)(\forall x)\,[x \in R \iff P(x)]$.

(i) No set R satisfies $(\forall x)\,[x \in R \iff x \notin x]$.
(ii) The unrestricted comprehension schema is inconsistent.

Proof.

(i) Suppose that such a set R exists. Setting $x = R$ gives $R \in R \iff R \notin R$. If $R \in R$, then $R \notin R$; if $R \notin R$, then $R \in R$. Both cases are contradictory, so no such set exists.
(ii) Take $P(x) := x \notin x$. The corresponding instance asserts $(\exists R)(\forall x)\,[x \in R \iff x \notin x]$. This contradicts (i), proving (ii).

Structural Commentary: Specialize at $x = R$, then apply the matching comprehension instance.
Key Insight: Self-membership makes the defining condition assert its own negation. □

Perspective 1.6.2. *From Comprehension to Axioms*
The obstruction in Theorem 1.6.1 explains the need for controlled set formation.
Separation forms $\{x \in B : P(x)\}$ only after a bounding set B is available. The standard ZFC axioms have the following informal roles.

(i) *Extensionality* makes membership determine set equality.
(ii) *Separation* extracts a condition-defined subset from an existing bounding set.
(iii) *Pairing* supplies $\{a, b\}$ from given sets a and b, including $\{a\}$ when $a = b$.
(iv) *Union* collects into one set the elements of the members of a given set.
(v) *Power set* supplies $\mathcal{P}(A)$, the set of all subsets of a given set A.
(vi) *Infinity* supplies an inductive set from which the natural numbers are constructed.
(vii) *Replacement* forms the image of a set under a definable functional relation.
(viii) *Foundation* requires nonempty sets to contain an $\in$-minimal member.
(ix) *Choice* supplies a choice function on a set-indexed family of nonempty sets.

This perspective is a functional map, not a formal axiom list or a source of proof authority.
Formal statements and dependencies belong to the advanced set-theory section.

2 Logic and Proofs

Logic makes explicit the statement grammar already used informally in the elementary set development and fixes it for the definitions, theorems, and proofs that follow.

2.1 Statements and Well-Formed Language

Definition 2.1.1. *Mathematical Statements and Truth Values*
The *truth values* used here are *true* and *false*.
A *mathematical statement* is a declarative sentence having exactly one truth value.

Principle 2.1.2. *Well-Formed Statements*
A mathematical expression may serve as a statement only when all the following conditions hold.

(i) *Symbols*: All symbols have declared syntactic roles.
(ii) *Meaning*: All constituent expressions have defined meanings.
(iii) *Scope*: All variable occurrences have unambiguous scope.

Failure of any condition makes the expression ill formed rather than false; no truth value is assigned.

2.2 Connectives and Negation

Definition 2.2.1. *Truth-Functional Connectives*
A connective is *truth functional* when its output depends only on input truth values. Let a and b be mathematical statements.

(i) *Negation*: $\neg a$ is true iff a is false.
(ii) *Conjunction*: $a \wedge b$ is true iff both a and b are true.
(iii) *Disjunction*: $a \vee b$ is true iff at least one of a and b is true; both may be true.
(iv) *Implication*: $a \Rightarrow b$ is false iff a is true and b is false. It is true in the remaining cases.
(v) *Equivalence*: $a \iff b$ is true iff a and b have the same truth value.

Proposition 2.2.2. *Negating Truth-Functional Connectives*
For mathematical statements a and b, the following equivalences hold.

(i) $\neg(a \wedge b) \iff (\neg a \vee \neg b)$.

(ii) $\neg(a \vee b) \iff (\neg a \wedge \neg b)$.
(iii) $\neg(a \Rightarrow b) \iff (a \wedge \neg b)$.

Proof. By Definition 2.2.1, the three equivalences reduce to the following truth conditions.

(i) The statement $\neg(a \wedge b)$ is true iff $a \wedge b$ is false. This occurs iff at least one of a and b is false, precisely when $\neg a \vee \neg b$ is true.
(ii) The statement $\neg(a \vee b)$ is true iff $a \vee b$ is false. This occurs iff both a and b are false, precisely when $\neg a \wedge \neg b$ is true.
(iii) The statement $\neg(a \Rightarrow b)$ is true iff $a \Rightarrow b$ is false. This occurs iff a is true and b is false, precisely when $a \wedge \neg b$ is true.

Thus each pair has the same truth value, proving all three equivalences.
Structural Commentary: Each item reduces both sides to one constituent truth-value condition.
Key Insight: Negation selects the cases excluded by the original connective. □

2.3 Quantifiers and Bound Variables

Definition 2.3.1. *Quantifiers, Bound Variables, and Scope*
Let X be a set and let $P(x)$ be a condition on its elements.

(i) *Universal quantifier*: $(\forall x \in X)\, P(x)$ means $\{x \in X : \neg P(x)\} = \emptyset$.
(ii) *Existential quantifier*: $(\exists x \in X)\, P(x)$ means $\{x \in X : P(x)\} \neq \emptyset$.
(iii) *Binding and scope*: A binding declaration governs a region, called the variable's *scope*.
(iv) *Bound occurrence*: An occurrence governed by a binding declaration is *bound*.
(v) *Free occurrence*: An occurrence outside every binding declaration is *free*.
(vi) *Standard binders*: Quantifiers, summation signs, integrals, and set-builder notation bind variables.
(vii) *Renaming*: A bound variable may be renamed consistently within its scope to a fresh symbol.
The statement remains unchanged provided no free occurrence is captured.

Proposition 2.3.2. *Quantifier Order Can Change Truth Value*
There exist sets X, Y and a condition $P(x, y)$ on $X \times Y$ such that

(i) $(\forall x \in X)(\exists y \in Y)\, P(x, y)$ *is true;*
(ii) $(\exists y \in Y)(\forall x \in X)\, P(x, y)$ *is false.*

Consequently, the two quantifier orders are not logically equivalent.

Proof. Take $X = Y = \{0, 1\}$ and let $P(x, y)$ be the condition $x = y$.

(i) The choice $y = x$ verifies $(\forall x \in X)(\exists y \in Y)\, P(x, y)$.
(ii) If $(\exists y \in Y)(\forall x \in X)\, P(x, y)$ were true, choose its witness $y \in Y$. Then $P(0, y) \wedge P(1, y)$, so $y = 0$ and $y = 1$, contradicting $0 \neq 1$.

Thus (i) and (ii) hold, so the quantifier orders are not logically equivalent.
Structural Commentary: The proof separates the orders using equality on a two-point set.
Key Insight: Pointwise witnesses exist, but no single witness works uniformly. □

Remark 2.3.3. *Index Variables*
Let X and I be sets, and let $\{A_i\}_{i \in I}$ be an indexed family in X. For $x \in X$, $x \in \bigcup_{i \in I} A_i \iff (\exists i \in I)\, x \in A_i$.
Likewise, $x \in \bigcap_{i \in I} A_i \iff (\forall i \in I)\, x \in A_i$. The indexed operator binds i in both formulas.
The index may be renamed consistently to any fresh symbol without changing either formula.

2.4 Proof Grammar

Proof Method 2.4.1. *Proof Grammar*
Given a mathematical claim, expose its proof obligations before selecting a proof rule.

(i) *Objects*: Identify the ambient sets or structures and declare the role of each symbol.
(ii) *Logical form*: Separate the hypotheses from the exact conclusion.
(iii) *Quantifiers*: Record the quantifiers in order. Note the permitted dependence of witnesses.
By Proposition 2.3.2, quantifier order can change a statement's truth value.
(iv) *Well-formedness*: Verify that every expression has a defined meaning.
Check that every variable occurrence has unambiguous scope. Principle 2.1.2 records this requirement.

A completed parse specifies the data, hypotheses, quantifier dependencies, and exact conclusion.
An undefined expression is ill formed rather than false and must be repaired before a proof begins.

2.5 Proof by Contradiction

Definition 2.5.1. *Contradiction*
A *contradiction* is a mathematical statement whose truth conditions cannot be satisfied.
For a mathematical statement c, the expression $c \wedge \neg c$ is the canonical contradiction form.
By Definition 2.2.1, its two conjuncts cannot be true simultaneously.

Proof Method 2.5.2. *Proof by Contradiction*
Let a and b be mathematical statements. To prove $a \Rightarrow b$ by classical contradiction, discharge the following obligations.

(i) *Temporary assumption*: Assume $\neg(a \Rightarrow b)$. By Proposition 2.2.2, this is equivalent to assuming $a \wedge \neg b$.
(ii) *Contradiction*: Identify a mathematical statement c. Derive both c and $\neg c$ from the hypotheses and temporary assumption.
(iii) *Conclusion*: Since the temporary assumption yields a contradiction, reject it and conclude $a \Rightarrow b$.

Justify each inference by the hypotheses, definitions, earlier results, or temporary assumption.

2.6 Mathematical Induction and Well-Ordering

Assumption 2.6.1. *Well-Ordering of* $\mathbb{N}$
Throughout these notes, $(\forall S \subseteq \mathbb{N})\,[S \neq \emptyset \Rightarrow (\exists m \in S)(\forall n \in S)\, m \leq n]$.

Theorem 2.6.2. *Mathematical Induction*
Let $(P_n)_{n\in\mathbb{N}}$ *be mathematical statements. Suppose the following conditions hold.*

(i) Basis: P_1 is true.
(ii) Induction step: $(\forall n \in \mathbb{N})\,(P_n \Rightarrow P_{n+1})$.

Then $(\forall n \in \mathbb{N})\, P_n$.

Proof. Let $F := \{n \in \mathbb{N} : \neg P_n\}$ and suppose $F \neq \emptyset$. By the standing well-ordering assumption, choose the least $m \in F$.
By (i), $1 \notin F$, so $m \neq 1$. Hence $m-1 \in \mathbb{N}$ and $m-1 < m$. Minimality gives $m-1 \notin F$, so P_{m-1} is true.
Applying (ii) at $n = m-1$ gives P_m, contradicting $m \in F$. Thus $F = \emptyset$, so $(\forall n \in \mathbb{N})\, P_n$.
Structural Commentary: Well-ordering supports a least-counterexample contradiction.
Key Insight: A least failure has a smaller successful predecessor. □

3 Functions

Functions assign unique outputs; their images, preimages, restrictions, compositions, and inverses organize nearly every later construction.

3.1 Functions, Images, and Preimages

Definition 3.1.1. *Functions, Images, and Preimages*
Let X, Y be sets. A *function* $f : X \to Y$ assigns a unique value $f(x) \in Y$ to each $x \in X$.

(i) $X = \mathrm{dom}(f)$ is the *domain*, and Y is the *codomain*.
(ii) $\mathrm{range}(f) := f(X) := \{f(x) : x \in X\}$ is the *range* of f.
(iii) If $A \subseteq X$, then $f(A) := \{f(x) : x \in A\}$ is the *image* of A.
(iv) If $B \subseteq Y$, then $f^{-1}(B) := \{x \in X : f(x) \in B\}$ is the *preimage* of B; no inverse function is required.

Assumption 3.1.2. *Function Extensionality*
Let X, Y be sets and let $f, g : X \to Y$. If $(\forall x \in X)\, f(x) = g(x)$, then $f = g$.

Proposition 3.1.3. *Image and Preimage Calculus*
Let X, Y, I be sets with $I \neq \emptyset$, and let $f : X \to Y$. Let $\{A_i\}_{i\in I}$ and $\{B_i\}_{i\in I}$ be indexed families in X and Y, respectively.

(i) $f\left(\bigcup_{i\in I} A_i\right) = \bigcup_{i\in I} f(A_i)$.
(ii) $f\left(\bigcap_{i\in I} A_i\right) \subseteq \bigcap_{i\in I} f(A_i)$.
(iii) $f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i)$.
(iv) $f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i)$.
(v) $(\forall B \subseteq Y)\, f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$.
(vi) $(\forall B, C \subseteq Y)\, f^{-1}(B \setminus C) = f^{-1}(B) \setminus f^{-1}(C)$.
(vii) $(\forall B \subseteq Y)\, f(f^{-1}(B)) = B \cap \mathrm{range}(f)$.
(viii) $(\forall A \subseteq X)(\forall B \subseteq Y)\, [f(A) \subseteq B \iff A \subseteq f^{-1}(B)]$.

Proof.

(i) Fix $y \in Y$. Then $y \in f\left(\bigcup_{i\in I} A_i\right) \iff (\exists i \in I)(\exists x \in A_i)\, y = f(x)$. This is equivalent to $y \in \bigcup_{i\in I} f(A_i)$.

(ii) If $y \in f\left(\bigcap_{i\in I} A_i\right)$, choose $x \in \bigcap_{i\in I} A_i$ with $y = f(x)$. Then $(\forall i \in I)\, x \in A_i$, so $(\forall i \in I)\, y \in f(A_i)$ and $y \in \bigcap_{i\in I} f(A_i)$.

(iii) Fix $x \in X$. Then $x \in f^{-1}\left(\bigcup_{i\in I} B_i\right) \iff (\exists i \in I)\, f(x) \in B_i$. This is equivalent to $(\exists i \in I)\, x \in f^{-1}(B_i)$, hence to $x \in \bigcup_{i\in I} f^{-1}(B_i)$.

(iv) Fix $x \in X$. Then $x \in f^{-1}\left(\bigcap_{i\in I} B_i\right) \iff (\forall i \in I)\, f(x) \in B_i$. This is equivalent to $(\forall i \in I)\, x \in f^{-1}(B_i)$, hence to $x \in \bigcap_{i\in I} f^{-1}(B_i)$.

(v) Fix $B \subseteq Y$ and $x \in X$. Then $x \in f^{-1}(Y \setminus B) \iff f(x) \notin B \iff x \notin f^{-1}(B) \iff x \in X \setminus f^{-1}(B)$.

(vi) Fix $B, C \subseteq Y$ and $x \in X$. Then $x \in f^{-1}(B \setminus C) \iff [f(x) \in B \wedge f(x) \notin C]$. This is equivalent to $x \in f^{-1}(B) \setminus f^{-1}(C)$.

(vii) Fix $B \subseteq Y$ and $y \in Y$. Then $y \in f(f^{-1}(B)) \iff (\exists x \in X)\, [f(x) \in B \wedge y = f(x)]$. This is equivalent to $y \in B \cap \mathrm{range}(f)$.

(viii) ($\Rightarrow$) Fix $A \subseteq X$ and $B \subseteq Y$. Suppose $f(A) \subseteq B$ and fix $x \in A$. Then $f(x) \in B$, so $x \in f^{-1}(B)$; hence $A \subseteq f^{-1}(B)$.

(viii) ($\Leftarrow$) Fix $A \subseteq X$ and $B \subseteq Y$. Suppose $A \subseteq f^{-1}(B)$ and fix $y \in f(A)$. Choose $x \in A$ with $y = f(x)$.
Then $x \in f^{-1}(B)$, so $y = f(x) \in B$ and $f(A) \subseteq B$.

Structural Commentary: Membership chains prove the identities; direct containments prove the adjunction.
Key Insight: Preimages translate codomain membership into domain membership. □

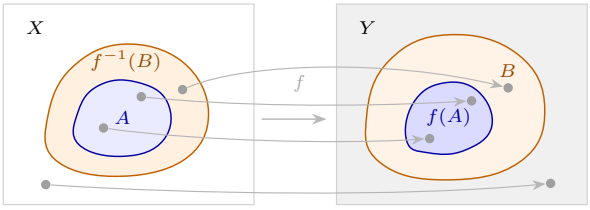

Figure 3: Image–preimage adjunction: $f(A) \subseteq B \iff A \subseteq f^{-1}(B)$.

Figure 3 illustrates Proposition 3.1.3(viii). The labeled regions show that $f(A) \subseteq B$ exactly when $A \subseteq f^{-1}(B)$.

Definition 3.1.4. *Algebraic Support of a Function*
Let $f : X \to Y$ and let $0 \in Y$ be distinguished. The *algebraic support* of f is $\operatorname{supp}(f) := \{x \in X : f(x) \neq 0\}$.

3.2 Injective, Surjective, and Bijective Maps

Definition 3.2.1. *Injective, Surjective, and Bijective Maps*
Let X, Y be sets and let $f : X \to Y$.

(i) *Injective*: $(\forall x_1, x_2 \in X)\,[f(x_1) = f(x_2) \implies x_1 = x_2]$.
(ii) *Surjective*: $\operatorname{range}(f) = Y$.
(iii) *Bijective*: f is both injective and surjective.

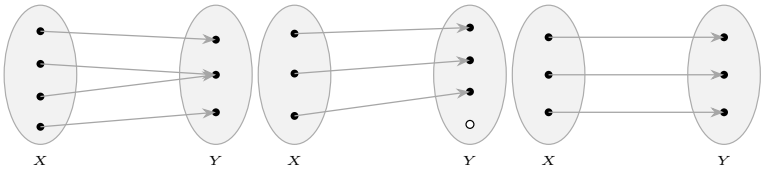

Figure 4: Left: surjective with a collision; center: injective but not surjective; right: bijective.

Figure 4 contrasts the three conditions in Definition 3.2.1. The left map is surjective but not injective, whereas the center map is injective but not surjective. The right map is bijective.

Example 3.2.2. *Image and Preimage for the Square Map*
Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := x^2$. If $x \in [-2, 1]$, then $0 \leq f(x) \leq 4$. Conversely, if $y \in [0, 4]$, then $-\sqrt{y} \in [-2, 1]$ and $f(-\sqrt{y}) = y$.
Hence $f([-2, 1]) = [0, 4]$. For $x \in \mathbb{R}$, $x \in f^{-1}([1, 4]) \iff 1 \leq x^2 \leq 4 \iff 1 \leq |x| \leq 2 \iff x \in [-2, -1] \cup [1, 2]$.
Since $f(-1) = f(1)$ and $-1 \neq 1$, the map is not injective. Since $(\forall x \in \mathbb{R})\, f(x) \geq 0$, one has $-1 \notin \operatorname{range}(f)$, so the map is not surjective.
Key Insight: Repeated values obstruct injectivity; omitted codomain values obstruct surjectivity.

3.3 Composition, Restriction, and Inverse Functions

Definition 3.3.1. *Composition, Restriction, Extension, and Identity*
Let X, Y be sets and let $f : X \to Y$.

(i) *Restriction*: If $A \subseteq X$, then $f|_A : A \to Y$ satisfies $(\forall x \in A)\,(f|_A)(x) = f(x)$.
(ii) *Extension*: If U is a set with $X \subseteq U$, an extension of f is a map $\widetilde{f} : U \to Y$ satisfying $\widetilde{f}|_X = f$.
(iii) *Composition*: If C, Z are sets, $g : C \to Z$, and $f(X) \subseteq C$, then $g \circ f : X \to Z$ satisfies $(\forall x \in X)\,(g \circ f)(x) = g(f(x))$.
(iv) *Identity*: The map $\mathrm{id}_X : X \to X$ satisfies $(\forall x \in X)\,\mathrm{id}_X(x) = x$.

Theorem 3.3.2. *Bijective Means Invertible*
Let X, Y be sets and let $f : X \to Y$.

(i) The map f is bijective iff $(\exists g : Y \to X)\,[g \circ f = \mathrm{id}_X \wedge f \circ g = \mathrm{id}_Y]$.
(ii) If $g, h : Y \to X$ both satisfy the two identities in (i), then $g = h$. When it exists, this map is denoted $f^{-1} : Y \to X$.

Proof.

(i) ($\Rightarrow$) Suppose f is bijective. For $y \in Y$, choose $x_y \in X$ with $f(x_y) = y$ by surjectivity; it is unique by injectivity.
Define $g : Y \to X$ by $g(y) := x_y$. For $x \in X$, $g(f(x)) = x$, and for $y \in Y$, $f(g(y)) = y$.
Thus $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$ by function extensionality.
(i) ($\Leftarrow$) Suppose $g : Y \to X$ satisfies the two identities. If $x_1, x_2 \in X$ and $f(x_1) = f(x_2)$, then $x_1 = g(f(x_1)) = g(f(x_2)) = x_2$.
Hence f is injective. For $y \in Y$, $y = f(g(y)) \in \operatorname{range}(f)$, so f is surjective and hence bijective.
(ii) Suppose $g, h : Y \to X$ satisfy the two identities. For $y \in Y$, $g(y) = g(f(h(y))) = h(y)$. Thus $g = h$ by function extensionality.

Structural Commentary: Construction proves existence; cancellation recovers bijectivity and proves uniqueness.
Key Insight: A bijection pairs each output with exactly one input. □

Proposition 3.3.3. *Composition and Restriction Calculus*
Let X, Y, Z, W be sets, and let $f : X \to Y$, $g : Y \to Z$, and $h : Z \to W$.

(i) $h \circ (g \circ f) = (h \circ g) \circ f$.
(ii) $\mathrm{id}_Y \circ f = f$ and $f \circ \mathrm{id}_X = f$.
(iii) $(\forall A \subseteq X)\,(g \circ f)(A) = g(f(A))$.
(iv) $(\forall C \subseteq Z)\,(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$.
(v) $(\forall A \subseteq X)\,(g \circ f)|_A = g \circ (f|_A)$.
(vi) If f and g are bijective, then $g \circ f$ is bijective.

(vii) If f and g are bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof.

(i) For $x \in X$, both sides evaluate to $h(g(f(x)))$. Hence the two maps are equal by function extensionality.
(ii) For $x \in X$, $(\mathrm{id}_Y \circ f)(x) = f(x) = (f \circ \mathrm{id}_X)(x)$. Hence both identities hold by function extensionality.
(iii) Fix $A \subseteq X$ and $z \in Z$. Then $z \in (g \circ f)(A) \iff (\exists x \in A)\, z = g(f(x)) \iff (\exists y \in f(A))\, z = g(y) \iff z \in g(f(A))$.
(iv) Fix $C \subseteq Z$ and $x \in X$. Then $x \in (g \circ f)^{-1}(C) \iff g(f(x)) \in C \iff f(x) \in g^{-1}(C) \iff x \in f^{-1}(g^{-1}(C))$.
(v) Fix $A \subseteq X$ and $x \in A$. Then $((g \circ f)|_A)(x) = g(f(x)) = (g \circ (f|_A))(x)$, so the restrictions are equal by function extensionality.
(vi) Suppose f and g are bijective. Fix $x_1, x_2 \in X$ and suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $f(x_1) = f(x_2)$ by the injectivity of g. Hence $x_1 = x_2$ by the injectivity of f, so $g \circ f$ is injective.
For $z \in Z$, choose $y \in Y$ with $g(y) = z$ by the surjectivity of g, then choose $x \in X$ with $f(x) = y$ by the surjectivity of f. Thus $(g \circ f)(x) = z$, so $g \circ f$ is surjective and therefore bijective.
(vii) Suppose f and g are bijective. The inverse functions $f^{-1} : Y \to X$ and $g^{-1} : Z \to Y$ exist by Theorem 3.3.2(i).
For $x \in X$, $(f^{-1} \circ g^{-1})((g \circ f)(x)) = f^{-1}(g^{-1}(g(f(x)))) = f^{-1}(f(x)) = x$.
For $z \in Z$, $(g \circ f)((f^{-1} \circ g^{-1})(z)) = g(f(f^{-1}(g^{-1}(z)))) = g(g^{-1}(z)) = z$. Thus $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ by Theorem 3.3.2(ii).

Structural Commentary: Pointwise evaluation proves the calculus; injectivity and surjectivity pass through the factors.
Key Insight: Inverse maps cancel compositions in reverse order. □

4 Counting and Finite Sums

Counting records finite constructions before cardinality and infinite sets are treated abstractly.

4.1 Finite Sums

Definition 4.1.1. *Finite Sums Indexed by Integers*
Let $\mathbb{F}$ be a field and let $(a_i)_{i\in\mathbb{Z}}$ be a family in $\mathbb{F}$. Fix $m, n \in \mathbb{Z}$. If $m \le n$, define $\sum_{i=m}^{n} a_i := a_m + a_{m+1} + \cdots + a_n$.

If $m > n$, set $\sum_{i=m}^{n} a_i := 0$; this is the empty sum.

Proposition 4.1.2. *Basic Laws for Finite Sums*
Let $\mathbb{F}$ be a field. Let $(a_i)_{i\in\mathbb{Z}}$ and $(b_i)_{i\in\mathbb{Z}}$ be families in $\mathbb{F}$. Fix $m, n \in \mathbb{Z}$ with $m \le n$.

(i) Splitting: Fix $r \in \mathbb{Z}$ with $m \le r < n$. Then $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) Additivity: $\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$.

(iii) Homogeneity: For $c \in \mathbb{F}$, $\sum_{i=m}^{n} ca_i = c \sum_{i=m}^{n} a_i$.

(iv) Triangle inequality: If $\mathbb{F} = \mathbb{C}$, then $\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n} |a_i|$.

(v) Monotonicity: Suppose $\mathbb{F} = \mathbb{R}$ and $(\forall i \in \mathbb{Z})\,[m \le i \le n \Rightarrow a_i \le b_i]$. Then $\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i$.

Proof. For (ii)–(v), fix m and set $\ell := n - m + 1$; by Theorem 2.6.2 with induction parameter ℓ, it suffices in each item to verify the case $n = m$ and the step from n to $n + 1$.

(i) By Definition 4.1.1 and associativity of addition in $\mathbb{F}$, $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) If $n = m$, both sides equal $a_m + b_m$. Assume additivity through n. By (i), $\sum_{i=m}^{n+1} (a_i + b_i) = \sum_{i=m}^{n} (a_i + b_i) + a_{n+1} + b_{n+1}$.
Substitute the induction hypothesis and regroup to obtain $\sum_{i=m}^{n+1} a_i + \sum_{i=m}^{n+1} b_i$.

(iii) If $n = m$, both sides equal ca_m. Assume homogeneity through n. By (i), $\sum_{i=m}^{n+1} ca_i = \sum_{i=m}^{n} ca_i + ca_{n+1}$.
The induction hypothesis and distributivity turn the right side into $c(\sum_{i=m}^{n} a_i + a_{n+1})$. By (i), this equals $c \sum_{i=m}^{n+1} a_i$.

(iv) If $n = m$, equality holds. Assume the inequality through n. By (i) and the triangle inequality, $\left|\sum_{i=m}^{n+1} a_i\right| \le \left|\sum_{i=m}^{n} a_i\right| + |a_{n+1}|$.
The induction hypothesis and (i) bound the right side by $\sum_{i=m}^{n+1} |a_i|$.

(v) If $n = m$, the conclusion is the hypothesis $a_m \le b_m$. Assume monotonicity through n. By (i), the sums through $n + 1$ split at n.
The induction hypothesis, order compatibility, and $a_{n+1} \le b_{n+1}$ give $\sum_{i=m}^{n+1} a_i \le \sum_{i=m}^{n+1} b_i$.

Structural Commentary: Splitting reduces each law to one new summand and the scalar axioms.
Key Insight: Finite-sum laws propagate one term at a time. □

4.2 Lists and Counting Principles

Counting represents finite constructions by explicit lists and converts disjoint alternatives or successive choices into cardinality identities.

Definition 4.2.1. *Finite Sets and Standard Index Sets*
Set $J_0 := \emptyset$ and $J_n := \{1, \ldots, n\}$ for $n \in \mathbb{N}$. A set A is *finite* if it admits a bijection $e : J_n \to A$ with $n \in \mathbb{N}_0$.

Proposition 4.2.2. *Finite Cardinality Is Well Defined*
Let A be a set and let $m, n \in \mathbb{N}_0$. If $e : J_m \to A$ and $g : J_n \to A$ are bijections, then $m = n$.
Thus each finite set A admits a unique $n \in \mathbb{N}_0$ such that a bijection $J_n \to A$ exists. Define the finite cardinality of A by $\#A := n$.

Proof. The composite $g^{-1} \circ e : J_m \to J_n$ is bijective.
It therefore suffices to prove that a bijection between two standard index sets forces their indices to agree.
For $r \in \mathbb{N}$, let P_r assert that a bijection $J_{r-1} \to J_k$ with $k \in \mathbb{N}_0$ forces $k = r - 1$. It suffices to prove P_1 and $(\forall r \in \mathbb{N})\,(P_r \Rightarrow P_{r+1})$.
A bijection $J_0 \to J_k$ forces $J_k = \emptyset$, hence $k = 0$; thus P_1 holds. Assume P_r and let $\varphi : J_r \to J_k$ be bijective. Since $J_r \neq \emptyset$, $k \in \mathbb{N}$.
Put $q := \varphi(r)$. The restriction $\varphi|_{J_{r-1}}$ is a bijection $J_{r-1} \to J_k \setminus \{q\}$.
Define $\rho : J_{k-1} \to J_k \setminus \{q\}$ by $\rho(i) := i$ when $i < q$ and $\rho(i) := i + 1$ when $i \geq q$.
Its branch images J_{q-1} and $J_k \setminus J_q$ are disjoint and cover $J_k \setminus \{q\}$. Both branches are injective, so ρ is bijective.
The composite $\rho^{-1} \circ \varphi|_{J_{r-1}} : J_{r-1} \to J_{k-1}$ is therefore bijective. Thus $k - 1 = r - 1$ by P_r. Hence $k = r$, and P_{r+1} holds.
Consequently, $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2. Finally, $n = m$ by P_{m+1} with the bijection $g^{-1} \circ e$.
Structural Commentary: Induction deletes one matched point and canonically relists the remaining index set.
Key Insight: Deletion reduces both finite indices while preserving bijectivity. □

Definition 4.2.3. *Lists, Words, and Product Sets*
Let $p \in \mathbb{N}$ and let $S_1, \ldots, S_p$ be sets.

(i) A *list* of length p in $S_1 \times \cdots \times S_p$ is a tuple $s = (s_1, \ldots, s_p)$ in that product.
(ii) If $S_1 = \cdots = S_p = S$, a list in S^p is a *word* of length p over the alphabet S.
(iii) A counting model is *ordered* when its outcomes are lists and coordinate positions are retained.
Thus lists $s = (s_1, \ldots, s_p)$ and $t = (t_1, \ldots, t_p)$ are distinct iff $(\exists i \in J_p)\, s_i \neq t_i$.

Proposition 4.2.4. *Addition, Subtraction, and Division Principles*

(i) Addition: Let $r \in \mathbb{N}$ and let $A_1, \ldots, A_r$ be pairwise disjoint finite sets. Then $\bigcup_{i=1}^{r} A_i$ is finite and $\#\big(\bigcup_{i=1}^{r} A_i\big) = \sum_{i=1}^{r} \#A_i$.
(ii) Subtraction: If A is finite and $B \subseteq A$, then B and $A \setminus B$ are finite and $\#(A \setminus B) = \#A - \#B$.
(iii) Division: Let $f : Y \to X$ be a surjection between finite sets. Its fibers are finite.
If $d \in \mathbb{N}$ satisfies $(\forall x \in X)\, \#f^{-1}(\{x\}) = d$, then $\#Y = d\#X$ and $\#X = \frac{\#Y}{d}$.

Proof.

(i) For $k \in \mathbb{N}$, let P_k denote claim (i) for k sets. If $k = 1$, then $\bigcup_{i=1}^{1} A_i = A_1$ and $\sum_{i=1}^{1} \#A_i = \#A_1$, so P_1 holds.
Assume P_k and let $A_1, \ldots, A_{k+1}$ be pairwise disjoint finite sets. Put $U := \bigcup_{i=1}^{k} A_i$. By P_k, U is finite and $\#U = \sum_{i=1}^{k} \#A_i$.
Put $q := \#U$ and $n := \#A_{k+1}$, and choose bijections $e : J_q \to U$ and $g : J_n \to A_{k+1}$.
Define $h : J_{q+n} \to U \cup A_{k+1}$ as follows. Set $h(t) := e(t)$ when $t \in J_q$ and $h(t) := g(t - q)$ when $t \in J_{q+n} \setminus J_q$.
The two domain blocks partition J_{q+n}, and their images are the disjoint sets U and A_{k+1}.
Both restrictions of h are bijective, so h is bijective.
Therefore $U \cup A_{k+1}$ is finite with cardinality $q + n = \sum_{i=1}^{k+1} \#A_i$, so P_{k+1} holds. Hence $(\forall k \in \mathbb{N})\, P_k$ by Theorem 2.6.2.

(ii) For $r \in \mathbb{N}$, let P_r assert claim (ii) whenever $\#A = r - 1$ and $B \subseteq A$. If $r = 1$, then $A = B = \emptyset$, so P_1 holds.
Assume P_r, let $\#A = r$, and choose a bijection $e : J_r \to A$. Put $a := e(r)$ and $A' := A \setminus \{a\}$.
The restriction $e|_{J_{r-1}} : J_{r-1} \to A'$ is bijective, so $\#A' = r - 1$. If $a \in B$, put $B' := B \setminus \{a\}$.
By P_r, the sets B' and $A' \setminus B'$ are finite and $\#(A' \setminus B') = \#A' - \#B'$.
By part (i) for $B = B' \cup \{a\}$, B is finite with $\#B = \#B' + 1$.
Since $A \setminus B = A' \setminus B'$, the set $A \setminus B$ is finite and $\#(A \setminus B) = (r - 1) - (\#B - 1) = \#A - \#B$.
If $a \notin B$, then B and $A' \setminus B$ are finite with $\#(A' \setminus B) = \#A' - \#B$ by P_r.
By part (i) for $A \setminus B = (A' \setminus B) \cup \{a\}$, $\#(A \setminus B) = ((r - 1) - \#B) + 1 = \#A - \#B$.
Thus P_{r+1} holds. Consequently, $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2.
An arbitrary finite A and subset $B \subseteq A$ satisfy $P_{\#A+1}$, proving claim (ii).

(iii) Put $m := \#X$. If $m = 0$, then $X = \emptyset$ and hence $Y = \emptyset$ because $f : Y \to X$; both identities follow.
Suppose $m \in \mathbb{N}$, choose a bijection $e : J_m \to X$, and set $F_i := f^{-1}(\{e(i)\})$ for $i \in J_m$.
Each F_i is finite by part (ii). The sets $F_1, \ldots, F_m$ are pairwise disjoint with union Y.
The equality $(\forall i \in J_m)\, \#F_i = d$ holds by the uniform-fiber hypothesis.
The identity $\#Y = \sum_{i=1}^{m} \#F_i$ holds by part (i). Hence $\#Y = \sum_{i=1}^{m} d = md = d\#X$. Since $d \neq 0$, $\#X = \frac{\#Y}{d}$.

Structural Commentary: Induction constructs union and complement listings; fiber decomposition supplies division.
Key Insight: Disjoint blocks convert set decompositions into arithmetic. □

Theorem 4.2.5. *Multiplication Principle*
Let $p \in \mathbb{N}$ and let $n_1, \ldots, n_p \in \mathbb{N}_0$.

(i) Successive choices: Set $H_0 := \{\emptyset\}$.
(ii) At stage $i \in J_p$, let $(C_i(h))_{h \in H_{i-1}}$ be a family of finite sets satisfying $(\forall h \in H_{i-1})\, \#C_i(h) = n_i$.
Set $H_i := \{(h, c) : h \in H_{i-1},\ c \in C_i(h)\}$. Then H_p is finite and $\#H_p = \prod_{i=1}^{p} n_i$.
(iii) Finite products: Let $S_1, \ldots, S_p$ be finite sets satisfying $(\forall i \in J_p)\, \#S_i = n_i$.
Then $S_1 \times \cdots \times S_p$ is finite and $\#(S_1 \times \cdots \times S_p) = \prod_{i=1}^{p} n_i$.

Proof.

(i) For $r \in \mathbb{N}$, let P_r denote claim (i) with $p = r$. The map $(\emptyset, c) \mapsto c$ is a bijection $H_1 \to C_1(\emptyset)$.
Thus H_1 is finite with $\#H_1 = n_1$, so P_1 holds. Assume P_r and consider an $(r + 1)$-stage model.

By P_r for its first r stages, H_r is finite with $\#H_r = \prod_{i=1}^{r} n_i$.

If $H_r = \emptyset$, then $H_{r+1} = \emptyset$ and $\#H_{r+1} = 0 = n_{r+1}\#H_r = \prod_{i=1}^{r+1} n_i$, so P_{r+1} holds.

Suppose $H_r \neq \emptyset$. Put $m := \#H_r$; then $m \in \mathbb{N}$. Choose a bijection $e : J_m \to H_r$.
Define the family $(A_j)_{j \in J_m}$ by $A_j := \{e(j)\} \times C_{r+1}(e(j))$. The sets $A_1, \dots, A_m$ are pairwise disjoint and have union H_{r+1}.
The map $c \mapsto (e(j), c)$ is a bijection $C_{r+1}(e(j)) \to A_j$. Thus $A_1, \dots, A_m$ are finite and $(\forall j \in J_m)\, \#A_j = n_{r+1}$.

Therefore $\#H_{r+1} = \sum_{j=1}^{m} \#A_j = mn_{r+1} = \prod_{i=1}^{r+1} n_i$ by Proposition 4.2.4(i). Thus P_{r+1} holds.

Consequently, $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2; in particular, P_p holds.

(ii) In (i), take the constant families $C_i(h) := S_i$. Then H_p is finite with $\#H_p = \prod_{i=1}^{p} n_i$.

Given $s = (s_1, \dots, s_p) \in S_1 \times \cdots \times S_p$, set $h_0 := \emptyset$. At stage $i \in J_p$, set $h_i := (h_{i-1}, s_i)$, and define $\Psi(s) := h_p$.
The map $\Psi : S_1 \times \cdots \times S_p \to H_p$ is bijective. Indeed, successive first and second coordinates recover the unique tuple s.

Hence $S_1 \times \cdots \times S_p$ is finite. Its cardinality is $\#(S_1 \times \cdots \times S_p) = \#H_p = \prod_{i=1}^{p} n_i$.

Structural Commentary: Induction counts recursive histories; a coordinate bijection transfers the count to products.
Key Insight: Uniform extension fibers multiply the number of histories. □

Example 4.2.6. *Finite Words by the Multiplication Principle*
The following slot counts multiply by Theorem 4.2.5.

(i) A card in a standard deck is a pair consisting of one of 4 suits and one of 13 ranks. Thus the deck has $4 \cdot 13 = 52$ cards.
(ii) For $B := \{0, 1\}$, $\#(B^8) = 2^8 = 256$, so the binary alphabet has 256 words of length 8.
(iii) For $D := \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$, $(\forall N \in \mathbb{N})\, \#(D^N) = 4^N$. At $N = 3$, the DNA alphabet therefore has 64 codons.

Key Insight: Fixed-size coordinate choices multiply, so constant alphabets produce exponential word counts.

4.3 Permutations, Subsets, and Binomial Coefficients

Definition 4.3.1. *Permutations and Sign*
Let $n \in \mathbb{N}_0$. A *permutation* of J_n is a bijection $\sigma : J_n \to J_n$, and $\mathfrak{S}_n$ denotes the set of all such permutations.
Given $\sigma \in \mathfrak{S}_n$, set $\mathrm{Inv}(\sigma) := \{(j, k) \in J_n^2 : j < k,\ \sigma(j) > \sigma(k)\}$ and define its *sign* by $\mathrm{sign}(\sigma) := (-1)^{\#\,\mathrm{Inv}(\sigma)}$.

Definition 4.3.2. *Factorials and Binomial Coefficients*

(i) *Factorial*: Set $0! := 1$ and $n! := \prod_{j=1}^{n} j$ for $n \in \mathbb{N}$.

(ii) *Falling factorial*: Set $(n)_0 := 1$ for $n \in \mathbb{N}_0$. If $n, r \in \mathbb{N}$ and $r \leq n$, set $(n)_r := \prod_{j=0}^{r-1} (n - j)$.

(iii) *Binomial coefficient*: If $n, r \in \mathbb{N}_0$ and $r \leq n$, set $\binom{n}{r} := \frac{n!}{r!(n-r)!}$; this is read *n choose r*.

Proposition 4.3.3. *Ordered Sampling With and Without Replacement*
Let $n, r \in \mathbb{N}_0$ and let S be finite with $\#S = n$. Set $S^0 := \{()\}$ and use $x^0 := 1$ for $x \in \mathbb{N}_0$.
Define $D_r(S) := \{s = (s_1, \dots, s_r) \in S^r : (\forall i, j \in J_r)\, [i < j \Rightarrow s_i \neq s_j]\}$.

(i) With replacement: The ordered samples form S^r, and $\#(S^r) = n^r$.
(ii) Without replacement: If $r \leq n$, the ordered samples with distinct entries form $D_r(S)$, and $\#D_r(S) = (n)_r$.
(iii) Full orderings: $\#D_n(S) = n!$ and $\#\mathfrak{S}_n = n!$.

Proof.

(i) If $r = 0$, then $\#S^0 = 1 = n^0$. If $r \in \mathbb{N}$, then $\#S^r = \prod_{k=1}^{r} n = n^r$ by Theorem 4.2.5(ii).

(ii) Suppose $r \leq n$. If $r = 0$, then $D_0(S) = S^0$ and $\#D_0(S) = 1 = (n)_0$.
Suppose $r \in \mathbb{N}$. At stage $k \in J_r$, an injective prefix $(s_1, \dots, s_{k-1})$ uses a set of cardinality $k - 1$.
The available set $S \setminus \{s_1, \dots, s_{k-1}\}$ is finite with cardinality $n - k + 1$ by Proposition 4.2.4(ii).

Successive-choice histories flatten bijectively to $D_r(S)$, so $\#D_r(S) = \prod_{k=1}^{r} (n - k + 1) = (n)_r$ by Theorem 4.2.5(i).

(iii) The identity $\#D_n(S) = n!$ holds by (ii). Define $\Phi : \mathfrak{S}_n \to D_n(J_n)$ by $\Phi(\sigma) := (\sigma(i))_{i \in J_n}$.
Given $s \in D_n(J_n)$, the map $i \mapsto s_i$ is injective and its image has cardinality n. Its complement in J_n has cardinality 0 by Proposition 4.2.4(ii), so the map is surjective.
Thus $s \mapsto (i \mapsto s_i)$ is the inverse of Φ. Hence Φ is bijective and $\#\mathfrak{S}_n = \#D_n(J_n) = n!$.

Structural Commentary: Product counting handles all lists; available-set histories handle injective lists.
Key Insight: Removing used entries decreases each successive choice count by one. □

Proposition 4.3.4. *Unordered Samples Without Replacement*
Let $n, r \in \mathbb{N}_0$ with $r \leq n$, let S be finite with $\#S = n$, and set $\mathcal{P}_r(S) := \{A \subseteq S : \#A = r\}$.
Then $\mathcal{P}_r(S)$ is finite and $\#\mathcal{P}_r(S) = \binom{n}{r} = \frac{(n)_r}{r!} = \frac{n!}{(n-r)!\, r!}$.

Proof. Choose a bijection $e : J_n \to S$. Define $\chi : \mathcal{P}_r(S) \to \{0, 1\}^n$ by $\chi(A)_i := 1$ when $e(i) \in A$ and $\chi(A)_i := 0$ otherwise.
If $A \neq B$, choose $x \in (A \setminus B) \cup (B \setminus A)$ and then choose $i \in J_n$ with $e(i) = x$. Thus $\chi(A)_i \neq \chi(B)_i$, so χ is injective.
The set $\{0, 1\}^n$ is finite by Proposition 4.3.3(i). Its subset $\chi(\mathcal{P}_r(S))$ is finite by Proposition 4.2.4(ii).
Hence $\mathcal{P}_r(S)$ is finite because χ is a bijection onto its image. Define $\pi : D_r(S) \to \mathcal{P}_r(S)$ by $\pi(s_1, \dots, s_r) := \{s_1, \dots, s_r\}$.
Given $A \in \mathcal{P}_r(S)$, choose a bijection $f : J_r \to A$. Then $(f(i))_{i \in J_r} \in \pi^{-1}(\{A\})$, so π is surjective.
The fiber $\pi^{-1}(\{A\})$ is bijective with $D_r(A)$, so $\#\pi^{-1}(\{A\}) = r!$ by Proposition 4.3.3(iii).
The identity $\#D_r(S) = (n)_r$ holds by Proposition 4.3.3(ii). The equality $\#D_r(S) = r!\,\#\mathcal{P}_r(S)$ holds by Proposition 4.2.4(iii).
Therefore $\#\mathcal{P}_r(S) = \frac{(n)_r}{r!} = \frac{n!}{(n-r)!\, r!} = \binom{n}{r}$.

Structural Commentary: Binary encoding proves finiteness; uniform ordering fibers convert ordered samples to subsets.
Key Insight: Each subset has exactly $r!$ ordered listings. □

Example 4.3.5. *Committees*
Let C be a class of 20 people. The 3-person committees form $\mathcal{P}_3(C)$. Thus $\#\mathcal{P}_3(C) = \binom{20}{3} = \frac{20\cdot 19\cdot 18}{3\cdot 2\cdot 1} = 1140$ by Proposition 4.3.4.
Key Insight: A committee is an unordered subset, so permutations of its members are not recounted.

Proposition 4.3.6. *Edge Values and Symmetry*
Let $n, k \in \mathbb{N}_0$ with $k \le n$.

(i) Edge values: $\binom{n}{0} = \binom{n}{n} = 1$.
(ii) Symmetry: $\binom{n}{k} = \binom{n}{n-k}$.

Proof.

(i) Since $0! := 1$, $\binom{n}{0} = \frac{n!}{0!\,n!} = 1$ and $\binom{n}{n} = \frac{n!}{n!\,0!} = 1$.
(ii) Since $0 \le n-k \le n$, $\binom{n}{n-k} = \frac{n!}{(n-k)!\,k!} = \frac{n!}{k!\,(n-k)!} = \binom{n}{k}$.

Structural Commentary: Direct factorial simplification proves the edge values and symmetry.
Key Insight: The denominator is invariant under $k \leftrightarrow n-k$. □

Proposition 4.3.7. *Binomial Theorem and Subset Count*
Let $n \in \mathbb{N}$ and let $a, b \in \mathbb{C}$.

(i) Binomial theorem: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.

(ii) Subset count: If S is finite with $\#S = n$, then $\mathcal{P}(S)$ is finite and $\#\mathcal{P}(S) = 2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof.

(i) Write $(a+b)^n = \prod_{j=1}^{n} (a+b)$. A subset $I \subseteq J_n$ records the factors contributing a; the corresponding term is $a^{\#I} b^{n-\#I}$.
For $k \in \{0, \dots, n\}$, $\#\mathcal{P}_k(J_n) = \binom{n}{k}$ by Proposition 4.3.4. Grouping terms by $k = \#I$ proves the identity.
(ii) Fix a finite set S with $\#S = n$ and choose a bijection $e : J_n \to S$.
Define $\chi : \mathcal{P}(S) \to \{0,1\}^n$ by $\chi(A)_i := 1$ when $e(i) \in A$ and $\chi(A)_i := 0$ otherwise.
The inverse sends $v \in \{0,1\}^n$ to $\{e(i) : i \in J_n,\ v_i = 1\}$, so χ is bijective.
Hence $\mathcal{P}(S)$ is finite with $\#\mathcal{P}(S) = 2^n$ by Proposition 4.3.3(i).
The sets $\mathcal{P}_0(S), \dots, \mathcal{P}_n(S)$ are pairwise disjoint with union $\mathcal{P}(S)$. Hence $\#\mathcal{P}(S) = \sum_{k=0}^{n} \#\mathcal{P}_k(S)$ by Proposition 4.2.4(i).
For $k \in \{0, \dots, n\}$, $\#\mathcal{P}_k(S) = \binom{n}{k}$ by Proposition 4.3.4. Therefore $\#\mathcal{P}(S) = \sum_{k=0}^{n} \binom{n}{k}$.

Structural Commentary: Expansion and subset partitioning classify the same binary choices by cardinality.
Key Insight: Choosing the a-positions is equivalent to choosing a subset of J_n. □

Definition 4.3.8. *Multisets*
Let S be a set. A *multiset on S* is a pair (S_0, m) such that

(i) $S_0 \subseteq S$ is the *underlying set*;
(ii) $m : S_0 \to \mathbb{N}$ is the *multiplicity function*, and $m(s)$ is the *multiplicity* of $s \in S_0$;
(iii) if S_0 is finite, set $E_m := \{(s, j) : s \in S_0,\ j \in J_{m(s)}\}$ and define the *size* by $|m| := \#E_m$.

The graph $\{(s, m(s)) : s \in S_0\}$ is an equivalent record of (S_0, m).

Example 4.3.9. *Multiset Bookkeeping*
The list (a, a, b, b, b, c) determines the multiset (S_0, m) with $S_0 = \{a, b, c\}$, $m(a) = 2$, $m(b) = 3$, and $m(c) = 1$.
The set E_m has $2 + 3 + 1 = 6$ elements, so $|m| = 6$. The same bookkeeping records eigenvalues counted with algebraic multiplicity.
Key Insight: A multiset forgets order while retaining the number of occurrences of each element.

Definition 4.3.10. *Multinomial Coefficients*
Let $r \in \mathbb{N}$ and let $\mathbf{n} = (n_1, \dots, n_r) \in \mathbb{N}_0^r$. Set $n := \sum_{i=1}^{r} n_i$ and define $\binom{n}{n_1, \dots, n_r} := \frac{n!}{n_1! \cdots n_r!}$.

Proposition 4.3.11. *Multinomial Count and Expansion*
Let $r \in \mathbb{N}$, let $\mathbf{n} = (n_1, \dots, n_r) \in \mathbb{N}_0^r$, and let $x_1, \dots, x_r \in \mathbb{C}$. Set $n := \sum_{i=1}^{r} n_i$.

Let $\mathcal{M}_{\mathbf{n}}$ be the set of tuples $(A_1, \dots, A_r)$ of pairwise disjoint subsets of J_n satisfying $\bigcup_{i=1}^{r} A_i = J_n$ and $(\forall i \in J_r)\, \#A_i = n_i$.

Set $K_{r,n} := \{\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{N}_0^r : \sum_{i=1}^{r} m_i = n\}$.

(i) Multinomial count: $\#\mathcal{M}_{\mathbf{n}} = \binom{n}{n_1, \dots, n_r} = \frac{n!}{n_1! \cdots n_r!}$.

(ii) Multinomial expansion: $(x_1 + \cdots + x_r)^n = \sum_{\mathbf{m} \in K_{r,n}} \binom{n}{m_1, \dots, m_r} \prod_{i=1}^{r} x_i^{m_i}$.

Proof.

(i) Set $N_0 := n$ and $N_j := n - \sum_{i=1}^{j} n_i$ for $j \in J_r$; thus $N_r = 0$. After $A_1, \dots, A_{j-1}$ are chosen, put $R_j := J_n \setminus \bigcup_{i=1}^{j-1} A_i$.
The equality $\#R_1 = N_0$ holds directly. If $j \ge 2$, then $\#R_j = N_{j-1}$ by Proposition 4.2.4(i)–(ii).
Since $N_{j-1} = \sum_{i=j}^{r} n_i \ge n_j$, the number of choices for $A_j \in \mathcal{P}_{n_j}(R_j)$ is $\binom{N_{j-1}}{n_j}$ by Proposition 4.3.4.

Since $N_r = 0$, the successive-choice histories are exactly the elements of $\mathcal{M}_{\mathbf{n}}$. Hence $\#\mathcal{M}_{\mathbf{n}} = \prod_{j=1}^{r} \binom{N_{j-1}}{n_j}$ by Theorem 4.2.5(i).

The factorial factors telescope, so $\#\mathcal{M}_{\mathbf{n}} = \prod_{j=1}^{r} \frac{N_{j-1}!}{n_j!\,N_j!} = \frac{n!}{n_1!\cdots n_r!} = \binom{n}{n_1,\ldots,n_r}$.

(ii) Expand $(x_1 + \cdots + x_r)^n = \prod_{k=1}^{n} (x_1 + \cdots + x_r)$. Each term is determined by a function $\phi : J_n \to J_r$ and equals $\prod_{k=1}^{n} x_{\phi(k)}$.
Set $m_i := \#\phi^{-1}(\{i\})$ for $i \in J_r$. Then $\mathbf{m} = (m_1, \ldots, m_r) \in K_{r,n}$. The fibers $(\phi^{-1}(\{1\}), \ldots, \phi^{-1}(\{r\}))$ form an element of $\mathcal{M}_{\mathbf{m}}$. Conversely, each element of $\mathcal{M}_{\mathbf{m}}$ determines a unique function with those fibers.
Thus the number of terms equal to $\prod_{i=1}^{r} x_i^{m_i}$ is $\binom{n}{m_1,\ldots,m_r}$ by (i). Grouping the terms over $\mathbf{m} \in K_{r,n}$ proves the expansion.

Structural Commentary: Successive subset choices count labeled partitions; fiber profiles classify product terms.
Key Insight: Multinomial coefficients count functions with prescribed fiber sizes. □

Example 4.3.12. *Repeated Letters, Committees, and Pairings*

(i) The position sets of the letters in $(\mathrm{A}, \mathrm{A}, \mathrm{G}, \mathrm{G}, \mathrm{G}, \mathrm{C}, \mathrm{C}, \mathrm{C})$ form a labeled partition of sizes $2, 4, 3$.
Thus the number of arrangements is $\binom{9}{2,4,3} = \frac{9!}{2!\,4!\,3!} = 1260$ by Proposition 4.3.11(i).

(ii) Splitting seven people into labeled subcommittees of sizes $3, 2, 2$ gives $\binom{7}{3,2,2} = \frac{7!}{3!\,2!\,2!} = 210$ assignments by Proposition 4.3.11(i).

(iii) Let $m \in \mathbb{N}$ and let S be finite with $\#S = 2m$.
Let $\mathcal{Q}_m(S)$ be the set of pairwise disjoint m-element families $\mathcal{A} \subseteq \mathcal{P}_2(S)$ satisfying $\bigcup \mathcal{A} = S$; its elements are the pairings of S.
The sets $\mathcal{P}(S)$ and $\mathcal{P}(\mathcal{P}(S))$ are finite by two applications of Proposition 4.3.7(ii). Hence $\mathcal{Q}_m(S)$ is finite by Proposition 4.2.4(ii).
Define $q : D_{2m}(S) \to \mathcal{Q}_m(S)$ by $q(s) := \{\{s_{2j-1}, s_{2j}\} : j \in J_m\}$. Each $s \in D_{2m}(S)$ lists S exactly once, so $q(s) \in \mathcal{Q}_m(S)$.
Given $\mathcal{A} \in \mathcal{Q}_m(S)$, choose a bijection $p : J_m \to \mathcal{A}$. Choose labels $((a_j, b_j))_{j \in J_m}$ satisfying $(\forall j \in J_m)\, p(j) = \{a_j, b_j\}$.
Then $(a_1, b_1, \ldots, a_m, b_m) \in q^{-1}(\{\mathcal{A}\})$, so q is surjective.
Each pairing has $m!\,2^m$ preimages: the m pairs may be ordered arbitrarily, and the two members of each pair may be interchanged.
The identity $\#D_{2m}(S) = (2m)!$ holds by Proposition 4.3.3(iii). Therefore the number of pairings is $\frac{(2m)!}{m!\,2^m}$ by Proposition 4.2.4(iii).

Key Insight: Multinomial counts labeled blocks; quotienting symmetries handles unlabeled blocks.

4.4 Inclusion-Exclusion and Combinatorial Proof

Proposition 4.4.1. *Finite Inclusion-Exclusion for Two and Three Sets*
Let A, B, C be finite sets.

(i) Two sets: The set $A \cup B$ is finite. Its cardinality is $\#(A \cup B) = \#A + \#B - \#(A \cap B)$.

(ii) Three sets: The set $A \cup B \cup C$ is finite.
Its cardinality is $\#(A \cup B \cup C) = \#A + \#B + \#C - \#(A \cap B) - \#(A \cap C) - \#(B \cap C) + \#(A \cap B \cap C)$.

Proof.

(i) Set $D := A \cap B$ and $E := B \setminus D$. The sets D and E are finite with $\#E = \#B - \#D$ by Proposition 4.2.4(ii).
The sets A and E are disjoint with union $A \cup B$. Hence $A \cup B$ is finite and $\#(A \cup B) = \#A + \#E$ by Proposition 4.2.4(i).
Substituting $D = A \cap B$ and $\#E = \#B - \#D$ proves (i).

(ii) The set $A \cup B$ is finite by (i). Set $F := (A \cup B) \cap C$.
The set $A \cup B \cup C$ is finite and $\#(A \cup B \cup C) = \#(A \cup B) + \#C - \#F$ by (i) with $A \cup B$ and C.
The sets $A \cap C$ and $B \cap C$ are finite by two applications of Proposition 4.2.4(ii).
Since $F = (A \cap C) \cup (B \cap C)$, the identity $\#F = \#(A \cap C) + \#(B \cap C) - \#(A \cap B \cap C)$ holds by (i) with $A \cap C$ and $B \cap C$.
The identity $\#(A \cup B) = \#A + \#B - \#(A \cap B)$ holds by (i). Substitution of the three identities proves (ii).

Structural Commentary: The two-set decomposition supplies the three-set formula through distributivity.
Key Insight: Correct each overlap after adding the individual set sizes. □

Method 4.4.2. *Combinatorial Proof by Classification*
Use When: Both sides of a finite identity can be expressed by weighting the same classified outcomes.
Fix target counts $N_L, N_R \in \mathbb{N}_0$.

(i) Choose finite sets Ω and P, a pattern map $p : \Omega \to P$, and weights $w_L, w_R : P \to \mathbb{Z}$.

(ii) Establish the exact fiber decompositions $(\forall \nu \in \{L, R\})\, N_\nu = \sum_{\alpha \in P} w_\nu(\alpha) \#p^{-1}(\{\alpha\})$.

(iii) Prove $(\forall \alpha \in P)\, w_L(\alpha) = w_R(\alpha)$, then substitute into the decompositions to obtain $N_L = N_R$.

Endpoint Check: Confirm that the fibers cover Ω disjointly and that both weight sums record every intended contribution.
Key Idea: Common classification reduces global equality to equality on each pattern.

5 Relations, Orders, and Lattices

Relations package comparisons, identifications, order structures, and quotient constructions.

5.1 Relations and Equivalence

Definition 5.1.1. *Relations*
Let X, Y be sets.

(i) *From X to Y*: A subset $R \subseteq X \times Y$ is a relation from X to Y.

(ii) *Relation notation*: Given a relation R from X to Y, write $xRy \iff (x, y) \in R$ for $(x, y) \in X \times Y$.

(iii) *On X*: A subset $R \subseteq X^2$ is a relation on X.

Definition 5.1.2. *Equivalence Relations*
Let X be a set and let $\sim$ be a relation on X.

(i) *Reflexive*: $\sim$ is reflexive iff $(\forall x \in X)\, x \sim x$.

(ii) *Symmetric*: $\sim$ is symmetric iff $(\forall x, y \in X)\, [x \sim y \Rightarrow y \sim x]$.

(iii) *Transitive*: $\sim$ is transitive iff $(\forall x, y, z \in X)\,[(x \sim y \wedge y \sim z) \Rightarrow x \sim z]$.
(iv) *Equivalence relation*: $\sim$ is an equivalence relation iff conditions (i)–(iii) hold.

Definition 5.1.3. *Equivalence Classes, Invariants, and Canonical Forms*
Let X, Y be sets and let $\sim$ be an equivalence relation on X.

(i) *Equivalence class*: For $x \in X$, set $[x]_\sim := \{y \in X : y \sim x\}$.
(ii) *Invariant*: A map $f : X \to Y$ is an invariant of $\sim$ iff $(\forall x, y \in X)\,[x \sim y \Rightarrow f(x) = f(y)]$.
(iii) *Complete invariant*: A map $f : X \to Y$ is a complete invariant of $\sim$ iff $(\forall x, y \in X)\,[x \sim y \iff f(x) = f(y)]$.
(iv) *Canonical forms*: A subset $\mathcal{C} \subseteq X$ is a set of canonical forms for $\sim$ iff $(\forall x \in X)(\exists! c \in \mathcal{C})\, c \sim x$.

5.2 Partitions and Quotient Sets

Definition 5.2.1. *Partition of a Set*
Let X be a set. A *partition* of X is a set $\mathcal{F} \subseteq \mathcal{P}(X) \setminus \{\emptyset\}$ satisfying $(\forall x \in X)(\exists! B \in \mathcal{F})\, x \in B$.
The members of $\mathcal{F}$ are its *blocks*.

Proposition 5.2.2. *Equivalence Classes Partition the Set*
Let X be a set, let $\sim$ be an equivalence relation on X, and set $\mathcal{F}_\sim := \{[x]_\sim : x \in X\}$. Then $\mathcal{F}_\sim$ is a partition of X.

Proof. For $x \in X$, reflexivity gives $x \in [x]_\sim$; hence the classes are nonempty and cover X.
If $u \in [x]_\sim \cap [y]_\sim$, then $u \sim x$ and $u \sim y$, whence $x \sim y$ and $y \sim x$ by symmetry and transitivity.
If $v \in [x]_\sim$, then $v \sim x \sim y$, so $v \in [y]_\sim$. If $v \in [y]_\sim$, then $v \sim y \sim x$, so $v \in [x]_\sim$.
Thus intersecting classes are equal, proving unique class membership; hence $\mathcal{F}_\sim$ partitions X.
Structural Commentary: Reflexivity supplies classes; symmetry and transitivity collapse intersections.
Key Insight: Two classes sharing an element have the same members. □

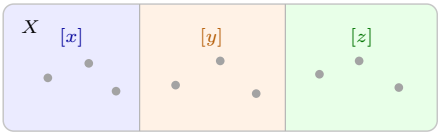

Figure 5: Equivalence classes form disjoint blocks that cover the underlying set.

Figure 5 depicts the partition in Proposition 5.2.2. Every element of X lies in one equivalence class, and distinct classes are disjoint.

Proposition 5.2.3. *Partitions Determine Equivalence Relations*
Let X be a set and let $\mathcal{F}$ be a partition of X. For $x \in X$, let $B_x \in \mathcal{F}$ be the unique block containing x.
For $(x, y) \in X^2$, define $x \sim_{\mathcal{F}} y \iff B_x = B_y$.

(i) Equivalence relation: $\sim_{\mathcal{F}}$ is an equivalence relation on X.
(ii) Classes: $(\forall x \in X)\,[x]_{\sim_{\mathcal{F}}} = B_x$, and $\{[x]_{\sim_{\mathcal{F}}} : x \in X\} = \mathcal{F}$.

Proof.

(i) For $x, y, z \in X$, $B_x = B_x$, so $x \sim_{\mathcal{F}} x$. If $x \sim_{\mathcal{F}} y$, then $B_x = B_y$, hence $B_y = B_x$ and $y \sim_{\mathcal{F}} x$.
Moreover, $(x \sim_{\mathcal{F}} y \wedge y \sim_{\mathcal{F}} z) \Rightarrow B_x = B_y = B_z \Rightarrow x \sim_{\mathcal{F}} z$. Thus $\sim_{\mathcal{F}}$ is reflexive, symmetric, and transitive.
(ii) Fix $x \in X$. If $y \in [x]_{\sim_{\mathcal{F}}}$, then $B_y = B_x$, so $y \in B_y = B_x$. If $y \in B_x$, then B_x is a block containing y.
Uniqueness gives $B_y = B_x$, so $y \sim_{\mathcal{F}} x$ and $y \in [x]_{\sim_{\mathcal{F}}}$. Hence $[x]_{\sim_{\mathcal{F}}} = B_x$. Therefore $\{[x]_{\sim_{\mathcal{F}}} : x \in X\} \subseteq \mathcal{F}$.
If $B \in \mathcal{F}$, then $B \neq \emptyset$; choose $x \in B$. The uniqueness of B_x gives $B = B_x = [x]_{\sim_{\mathcal{F}}}$, so the reverse containment holds.

Structural Commentary: Block equality proves the axioms; uniqueness identifies classes with blocks.
Key Insight: Equivalent elements are assigned to the same block. □

Example 5.2.4. *Rational Equivalence on $\mathbb{R}$*
For $(x, y) \in \mathbb{R}^2$, define $x \sim_{\mathbb{Q}} y \iff x - y \in \mathbb{Q}$. Fix $x, y, z \in \mathbb{R}$.

(i) *Reflexive*: $x - x = 0 \in \mathbb{Q}$.
(ii) *Symmetric*: $x - y \in \mathbb{Q} \Rightarrow y - x = -(x - y) \in \mathbb{Q}$.
(iii) *Transitive*: $(x - y \in \mathbb{Q} \wedge y - z \in \mathbb{Q}) \Rightarrow x - z = (x - y) + (y - z) \in \mathbb{Q}$.

Thus $\sim_{\mathbb{Q}}$ is an equivalence relation, and $[x]_{\sim_{\mathbb{Q}}} = x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}$.
A subset $\mathcal{R} \subseteq \mathbb{R}$ is a representative set iff $(\forall x \in \mathbb{R})(\exists! r \in \mathcal{R})\, r \in x + \mathbb{Q}$.
Key Insight: Rational translation orbits are exactly the equivalence classes.

Definition 5.2.5. *Quotient Set*
Let X be a set and let $\sim$ be an equivalence relation on X. The *quotient set* of X by $\sim$ is $X/\!\sim := \{[x]_\sim : x \in X\}$.

Proposition 5.2.6. *Maps Defined on Equivalence Classes*
Let X, Y be sets, let $\sim$ be an equivalence relation on X, and let $f : X \to Y$.
The assignment $g([x]_\sim) := f(x)$ defines $g : X/\!\sim \to Y$ iff $(\forall x, y \in X)\,[x \sim y \Rightarrow f(x) = f(y)]$.

Proof.

($\Rightarrow$) Suppose g is well-defined, $x \sim y$. Then $x \in [x]_\sim \cap [y]_\sim$, so $[x]_\sim = [y]_\sim$ by Proposition 5.2.2. Thus $f(x) = g([x]_\sim) = g([y]_\sim) = f(y)$.
($\Leftarrow$) Suppose $(\forall x, y \in X)\,[x \sim y \Rightarrow f(x) = f(y)]$. If $[x]_\sim = [y]_\sim$, then $x \in [y]_\sim$, whence $x \sim y$ and $f(x) = f(y)$.
Thus the assignment is independent of the representative. Hence it defines a map $g : X/\!\sim \to Y$.

Structural Commentary: Equal classes prove necessity; representative independence proves sufficiency.
Key Insight: A class function cannot distinguish equivalent representatives. □

5.3 Ordered Sets, Well-Orders, and Lattices

Definition 5.3.1. *Ordered Set Vocabulary*
Let S be a set, let $E, C \subseteq S$, and let $\leq_S \subseteq S^2$. Set $x <_S y \iff (x \leq_S y \wedge x \neq y)$.
The relation $\leq_S$ is *antisymmetric* iff $(\forall x, y \in S)\,[(x \leq_S y \wedge y \leq_S x) \Rightarrow x = y]$. Elements $x, y \in S$ are *comparable* iff $x \leq_S y \vee y \leq_S x$.

(i) *Partial order*: The relation $\leq_S$ is a partial order iff it is reflexive, antisymmetric, and transitive. Then $(S, \leq_S)$ is a *poset*.
(ii) *Total order*: A partial order $\leq_S$ is total iff $(\forall x, y \in S)\,(x \leq_S y \vee y \leq_S x)$. Then $(S, \leq_S)$ is an *ordered set*.
(iii) *Chain*: If $\leq_S$ is a partial order, then C is a chain iff $(\forall x, y \in C)\,(x \leq_S y \vee y \leq_S x)$.
(iv) *Well-order*: The order $\leq_S$ well-orders S iff it is total and the following condition holds. $(\forall Y \in \mathcal{P}(S)\setminus\{\emptyset\})(\exists \ell \in Y)(\forall y \in Y)\,\ell \leq_S y$.
(v) *Bounds*: An element $\beta \in S$ is an upper bound of E iff $(\forall x \in E)\,x \leq_S \beta$. It is a lower bound iff $(\forall x \in E)\,\beta \leq_S x$.
(vi) *Boundedness*: The set E is bounded above iff $(\exists \beta \in S)(\forall x \in E)\,x \leq_S \beta$. It is bounded below iff $(\exists \beta \in S)(\forall x \in E)\,\beta \leq_S x$.
(vii) *Supremum*: For $\alpha \in S$, $\alpha = \sup E$ iff the following two conditions hold.
  1. *Upper bound*: $(\forall x \in E)\,x \leq_S \alpha$.
  2. *Upper bound*: $(\forall x \in E)\,x \leq_S \alpha$.
  3. *Least*: $(\forall \beta \in S)\,[((\forall x \in E)\,x \leq_S \beta) \Rightarrow \alpha \leq_S \beta]$.

(viii) *Infimum*: For $\gamma \in S$, $\gamma = \inf E$ iff the following two conditions hold.
  1. *Lower bound*: $(\forall x \in E)\,\gamma \leq_S x$.
  2. *Greatest*: $(\forall \beta \in S)\,[((\forall x \in E)\,\beta \leq_S x) \Rightarrow \beta \leq_S \gamma]$.

(ix) *Maximal and minimal elements*: An element $m \in E$ is maximal iff $(\nexists x \in E)\,m <_S x$. It is minimal iff $(\nexists x \in E)\,x <_S m$.
(x) *Least-upper-bound property*: This property is characterized by the following condition.
$(\forall A \in \mathcal{P}(S) \setminus \{\emptyset\})\,[((\exists \beta \in S)(\forall x \in A)\,x \leq_S \beta) \Rightarrow (\exists \alpha \in S)\,\alpha = \sup A]$.

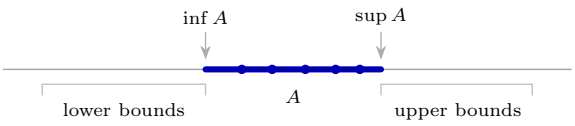

Figure 6: The supremum is the least upper bound, while the infimum is the greatest lower bound.

Figure 6 illustrates the bounds in Definition 5.3.1 for a set A.
It marks $\inf A$ and $\sup A$ as extremal bounds; Example 5.3.2 shows that neither need belong to the set.

Example 5.3.2. *Suprema Need Not Be Elements*
In $\mathbb{Q}$, set $E_1 := \{r \in \mathbb{Q} : r < 0\}$ and $E_2 := \{r \in \mathbb{Q} : r \leq 0\}$. Both sets have supremum 0, but $0 \notin E_1$ and $0 \in E_2$.
For $E := \{\frac{1}{n} : n \in \mathbb{N}\}$, $\sup E = 1 \in E$ and $\inf E = 0 \notin E$.
Key Insight: An extremal bound need not be an element of the set.

Theorem 5.3.3. *Least Upper Bounds Give Greatest Lower Bounds*
Let $(S, \leq)$ be an ordered set with the least-upper-bound property. Let $B \subseteq S$ be nonempty and bounded below. Then $\inf B$ exists in S.

Proof. Set $L := \{\ell \in S : (\forall b \in B)\,\ell \leq b\}$. $L \neq \emptyset$ by the boundedness of B below. Choose $b_0 \in B$.
Then $(\forall \ell \in L)\,\ell \leq b_0$, so L is bounded above. Set $\alpha := \sup L$; it exists in S by the least-upper-bound property.
For $b \in B$, the element b is an upper bound of L, so $\alpha \leq b$. Hence $\alpha \in L$. Moreover, $(\forall \ell \in L)\,\ell \leq \alpha$ by $\alpha = \sup L$.
Thus α is the greatest lower bound of B, and $\alpha = \inf B$.
Structural Commentary: The proof takes the supremum of the lower-bound set.
Key Insight: A nonempty set bounds its lower bounds above. □

Definition 5.3.4. *Lattice Vocabulary*
Let $(L, \leq)$ be a partially ordered set.

(i) *Meet*: For $x, y \in L$, the meet is $x \wedge y := \inf\{x, y\}$ when this infimum exists.
(ii) *Join*: For $x, y \in L$, the join is $x \vee y := \sup\{x, y\}$ when this supremum exists.
(iii) *Lattice*: The partially ordered set L is a lattice iff the following condition holds. $(\forall x, y \in L)\,[x \wedge y \in L \wedge x \vee y \in L]$.
(iv) *Complete lattice*: A lattice L is complete iff $(\forall E \subseteq L)\,[\inf E \in L \wedge \sup E \in L]$.
(v) *Bottom*: An element $\bot \in L$ is a bottom element iff $(\forall x \in L)\,\bot \leq x$.
(vi) *Top*: An element $\top \in L$ is a top element iff $(\forall x \in L)\,x \leq \top$.
(vii) *Modular lattice*: A lattice L is modular iff the following law holds. $(\forall x, y, z \in L)\,[x \leq z \Rightarrow z \wedge (y \vee x) = (z \wedge y) \vee x]$.

5.4 Relations Between Sets and Closures

Definition 5.4.1. *Inverse and Composition of Relations*
Let X, Y, Z be sets, let $R \subseteq X \times Y$, and let $S \subseteq Y \times Z$.

(i) *Inverse*: $R^{-1} := \{(y, x) \in Y \times X : (x, y) \in R\}$.
(ii) *Composition*: $S \circ R := \{(x, z) \in X \times Z : (\exists y \in Y)\,[(x, y) \in R \wedge (y, z) \in S]\}$.
(iii) *Diagonal*: $\Delta_X := \{(x, x) : x \in X\}$.

Proposition 5.4.2. *Inverse and Composition Laws*
Let W, X, Y, Z be sets. Let $R \subseteq X \times Y$, let $S \subseteq Y \times Z$, and let $T \subseteq Z \times W$.

(i) Identity: $\Delta_Y \circ R = R = R \circ \Delta_X$.
(ii) Associativity: $T \circ (S \circ R) = (T \circ S) \circ R$.
(iii) Involution: $(R^{-1})^{-1} = R$.
(iv) Reversal: $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

Proof.

(*i*) For $(x,y) \in X \times Y$, $(x,y) \in \Delta_Y \circ R \iff (\exists u \in Y)\,[(x,u) \in R \wedge u = y] \iff (x,y) \in R$.
For the right identity, $(x,y) \in R \circ \Delta_X \iff (\exists v \in X)\,[v = x \wedge (v,y) \in R] \iff (x,y) \in R$.
(*ii*) ($\Rightarrow$) Suppose $(x,w) \in T \circ (S \circ R)$. Choose $z \in Z$ with $(x,z) \in S \circ R$ and $(z,w) \in T$.
Choose $y \in Y$ with $(x,y) \in R$ and $(y,z) \in S$. Thus $(y,w) \in T \circ S$ and $(x,w) \in (T \circ S) \circ R$.
(*ii*) ($\Leftarrow$) Suppose $(x,w) \in (T \circ S) \circ R$. Choose $y \in Y$ with $(x,y) \in R$ and $(y,w) \in T \circ S$.
Choose $z \in Z$ with $(y,z) \in S$ and $(z,w) \in T$. Thus $(x,z) \in S \circ R$ and $(x,w) \in T \circ (S \circ R)$.
(*iii*) For $(x,y) \in X \times Y$, $(x,y) \in (R^{-1})^{-1} \iff (y,x) \in R^{-1} \iff (x,y) \in R$.
(*iv*) ($\Rightarrow$) Suppose $(z,x) \in (S \circ R)^{-1}$. Then $(x,z) \in S \circ R$.
Choose $y \in Y$ with $(x,y) \in R$ and $(y,z) \in S$. Hence $(z,y) \in S^{-1}$ and $(y,x) \in R^{-1}$, so $(z,x) \in R^{-1} \circ S^{-1}$.
(*iv*) ($\Leftarrow$) Suppose $(z,x) \in R^{-1} \circ S^{-1}$. Choose $y \in Y$ with $(z,y) \in S^{-1}$ and $(y,x) \in R^{-1}$.
Then $(y,z) \in S$ and $(x,y) \in R$, so $(x,z) \in S \circ R$ and $(z,x) \in (S \circ R)^{-1}$.

Structural Commentary: Membership expansion and witness comparison prove the four laws.
Key Insight: Composition records witnesses; inversion reverses their order. □

Definition 5.4.3. *Relation Powers and Closures*
Let X be a set and let $R \subseteq X^2$. Set $R^0 := \Delta_X$ and $R^{n+1} := R \circ R^n$ for $n \in \mathbb{N}_0$.

(i) *Reflexive closure*: $\mathrm{Ref}(R) := R \cup \Delta_X$.
(ii) *Transitive closure*: $R^+ := \bigcup_{n \in \mathbb{N}} R^n$.
(iii) *Reflexive-transitive closure*: $R^* := \bigcup_{n \in \mathbb{N}_0} R^n = \Delta_X \cup R^+$.

Lemma 5.4.4. *Addition Law for Relation Powers*
Let X be a set and let $R \subseteq X^2$. Then $(\forall m, n \in \mathbb{N}_0)\, R^n \circ R^m = R^{m+n}$.

Proof. Fix $m \in \mathbb{N}_0$. For $r \in \mathbb{N}$, let P_r assert $R^{r-1} \circ R^m = R^{m+r-1}$.
The identity $R^0 \circ R^m = \Delta_X \circ R^m = R^m$ holds by Proposition 5.4.2(i), so P_1 holds.
Assume P_r. Then $R^r \circ R^m = R \circ (R^{r-1} \circ R^m)$ by Proposition 5.4.2(ii) and the recursion.
Hence $R^r \circ R^m = R \circ R^{m+r-1} = R^{m+r}$ by P_r and the recursion, so P_{r+1} holds.
Consequently, $(\forall r \in \mathbb{N})\, P_r$ by induction. For $n \in \mathbb{N}_0$, take $r = n+1$ to obtain $R^n \circ R^m = R^{m+n}$.
Structural Commentary: Reindexed induction combines the recursive power law with associativity.
Key Insight: Relation composition adds power indices. □

Theorem 5.4.5. *Minimality of Reflexive and Transitive Closures*
Let X be a set and let $R \subseteq X^2$.

(i) Reflexive: $\mathrm{Ref}(R)$ is the smallest reflexive relation on X containing R.
(ii) Transitive: R^+ is the smallest transitive relation on X containing R.
(iii) Reflexive and transitive: R^* is the smallest reflexive and transitive relation on X containing R.

Proof.

(i) The relation $\mathrm{Ref}(R)$ contains R and Δ_X, so it is reflexive.
If $Q \subseteq X^2$ is reflexive and $R \subseteq Q$, then $\Delta_X \subseteq Q$, whence $\mathrm{Ref}(R) = R \cup \Delta_X \subseteq Q$.
(ii) Since $R = R^1$, $R \subseteq R^+$. Suppose $(x,y) \in R^+$ and $(y,z) \in R^+$. Choose $m, n \in \mathbb{N}$ with $(x,y) \in R^m$ and $(y,z) \in R^n$.
Then $(x,z) \in R^{m+n} \subseteq R^+$ by Lemma 5.4.4, so R^+ is transitive. Let $Q \subseteq X^2$ be transitive with $R \subseteq Q$.
For $n \in \mathbb{N}$, let P_n assert $R^n \subseteq Q$. P_1 holds by $R^1 = R \subseteq Q$. Assume P_n. Then $R^{n+1} = R \circ R^n \subseteq Q \circ Q$ by $R \subseteq Q$ and P_n.
Also $Q \circ Q \subseteq Q$ by the transitivity of Q, so P_{n+1} holds. Thus $(\forall n \in \mathbb{N})\, P_n$ by induction, and $R^+ \subseteq Q$.
(iii) The relation R^* contains R and is reflexive by $R^* = \Delta_X \cup R^+$. Suppose $(x,y) \in R^*$ and $(y,z) \in R^*$.
Choose $m, n \in \mathbb{N}_0$ with $(x,y) \in R^m$ and $(y,z) \in R^n$. Then $(x,z) \in R^{m+n} \subseteq R^*$ by Lemma 5.4.4, so R^* is transitive.
If $Q \subseteq X^2$ is reflexive and transitive with $R \subseteq Q$, then $\Delta_X \subseteq Q$ and $R^+ \subseteq Q$ by (ii). Thus $R^* = \Delta_X \cup R^+ \subseteq Q$.

Structural Commentary: Containment, closure, and minimality are verified in parallel.
Key Insight: Closure is generated by diagonal pairs together with finite composites. □

Example 5.4.6. *Closures of the Successor Relation*
Define $R \subseteq \mathbb{Z}^2$ by $m\,R\,n \iff n = m+1$. For $k \in \mathbb{N}$, let P_k assert $(\forall m, n \in \mathbb{Z})\,[m\,R^k\,n \iff n = m+k]$. P_1 holds by the definition of R.
Assume P_k. For $(m,n) \in \mathbb{Z}^2$, $m\,R^{k+1}\,n \iff (\exists u \in \mathbb{Z})\,[u = m+k \wedge n = u+1] \iff n = m+k+1$.
Thus P_{k+1} holds, and $(\forall k \in \mathbb{N})\, P_k$ by induction.

(i) $\mathrm{Ref}(R) = \{(m,n) \in \mathbb{Z}^2 : n = m \vee n = m+1\}$.
(ii) $R^+ = \{(m,n) \in \mathbb{Z}^2 : m < n\}$.
(iii) $R^* = \{(m,n) \in \mathbb{Z}^2 : m \le n\}$.

Key Insight: Finite successor compositions generate the strict and nonstrict integer orders.

Definition 5.4.7. *Order Isomorphism*
Let $(S, \le_S)$ and $(T, \le_T)$ be partially ordered sets.
A bijection $f : S \to T$ is an *order isomorphism* when the following condition holds. $(\forall x, y \in S)\,[x \le_S y \iff f(x) \le_T f(y)]$.
The two partially ordered sets are *order-isomorphic* when such a map exists.

Example 5.4.8. *Basic Order Isomorphisms*

(i) Define $f : \mathbb{Z} \to \{2n : n \in \mathbb{Z}\}$ by $f(n) := 2n$. Its inverse is $2n \mapsto n$, so f is bijective.
Moreover, $m \le n \iff 2m \le 2n$; hence f is an order isomorphism for the usual orders.
(ii) Let X be a set and define $c : \mathcal{P}(X) \to \mathcal{P}(X)$ by $c(A) := X \setminus A$. Since $c(c(A)) = A$, the map is bijective.
Moreover, $A \subseteq B \iff c(A) \supseteq c(B)$. Hence c is an order isomorphism from $(\mathcal{P}(X), \subseteq)$ to $(\mathcal{P}(X), \supseteq)$.
(iii) Let $(S, \le)$ be a poset and set $\downarrow s := \{x \in S : x \le s\}$. Define $\eta : S \to \{\downarrow s : s \in S\}$ by $\eta(s) :=\downarrow s$.
If $s \le t$, then $\downarrow s \subseteq \downarrow t$ by transitivity. If $\downarrow s \subseteq \downarrow t$, then $s \in \downarrow s \subseteq \downarrow t$, so $s \le t$. Thus $s \le t \iff \eta(s) \subseteq \eta(t)$.
If $\eta(s) = \eta(t)$, then $s \le t$ and $t \le s$, so $s = t$ by antisymmetry. Thus η is injective.
It is surjective by the definition of its codomain, so it is an order isomorphism.

Key Insight: Order isomorphisms preserve comparisons while changing their representation.

6 Cardinality, Choice, and Infinite Sets

Choice and maximality principles come first as foundational tools; cardinality then compares sets by functions rather than by informal size.

6.1 Choice and Zorn's Lemma

Definition 6.1.1. *Choice Function*
Let $\mathcal{F}$ be a family of nonempty sets, let I be a set, and let $(X_i)_{i\in I}$ be an indexed family of nonempty sets.

(i) A *choice function* on $\mathcal{F}$ is a map $c : \mathcal{F} \to \bigcup_{F\in\mathcal{F}} F$ satisfying $(\forall F \in \mathcal{F})\, c(F) \in F$.

(ii) An *indexed choice function* for $(X_i)_{i\in I}$ is a map $c : I \to \bigcup_{i\in I} X_i$ satisfying $(\forall i \in I)\, c(i) \in X_i$.

Principle 6.1.2. *Choice Principle*
Let I be a set and let $(X_i)_{i\in I}$ be an indexed family of nonempty sets. The axiom of choice asserts $(\exists c : I \to \bigcup_{i\in I} X_i)(\forall i \in I)\, c(i) \in X_i$. Equivalently, $\prod_{i\in I} X_i \neq \varnothing$. An explicit c proves existence directly; the principle supplies simultaneous choices without an explicit rule.

Principle 6.1.3. *Zorn's Lemma*
Let $(P, \leq)$ be a poset with $P \neq \varnothing$. Assume the following chain-upper-bound condition.
$(\forall C \subseteq P)\,[((\forall x, y \in C)\,(x \leq y \vee y \leq x)) \Rightarrow (\exists u \in P)(\forall c \in C)\, c \leq u]$. Then $(\exists m \in P)(\forall x \in P)\,(m \leq x \Rightarrow x = m)$.

Definition 6.1.4. *Inclusion and Containment Orders*
Let $\mathcal{F}$ be a family of sets and let $\mathcal{A} \subseteq \mathcal{F}$.

(i) *Inclusion*: for $A, B \in \mathcal{F}$, define $A \leq B$ iff $A \subseteq B$. An upper bound of $\mathcal{A}$ is $U \in \mathcal{F}$ satisfying $(\forall A \in \mathcal{A})\, A \subseteq U$.
An element $M \in \mathcal{F}$ is maximal iff $(\forall B \in \mathcal{F})\,(M \subseteq B \Rightarrow B = M)$.

(ii) *Reverse inclusion*: for $A, B \in \mathcal{F}$, define $A \leq B$ iff $B \subseteq A$. An upper bound of $\mathcal{A}$ is $U \in \mathcal{F}$ satisfying $(\forall A \in \mathcal{A})\, U \subseteq A$.
An element $M \in \mathcal{F}$ is maximal iff $(\forall B \in \mathcal{F})\,(B \subseteq M \Rightarrow B = M)$.

Proposition 6.1.5. *Inclusion Orders Are Partial Orders*
Let $\mathcal{F}$ be a family of sets. Inclusion and reverse inclusion are partial orders on $\mathcal{F}$.

Proof. Fix $A, B, C \in \mathcal{F}$. Since $A \subseteq A$, inclusion is reflexive. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$, so inclusion is transitive. If $A \subseteq B$ and $B \subseteq A$, then $A = B$ by Assumption 1.3.2, so inclusion is antisymmetric. Hence inclusion is a partial order.
For reverse inclusion, $A \supseteq A$ gives reflexivity, $A \supseteq B \supseteq C$ gives $A \supseteq C$, and $A \supseteq B \supseteq A$ gives $A = B$ by Assumption 1.3.2. Thus reverse inclusion is also a partial order.
Structural Commentary: Each order law follows from the corresponding containment law and set extensionality.
Key Insight: Reversing containment preserves reflexivity, transitivity, and antisymmetry. □

Proof Method 6.1.6. *Zorn Maximality Template*
Use this method when the target follows from a candidate admitting no proper admissible extension.
Let P be the set of candidates, and let $\leq$ record admissible extension.

(i) Verify that $P \neq \varnothing$ and that $\leq$ is a partial order on P.
(ii) Fix a chain $C \subseteq P$, construct $u \in P$, and prove $(\forall p \in C)\, p \leq u$.
(iii) Obtain a maximal $m \in P$ by Principle 6.1.3, and prove that maximality forces the target property.

For $C \neq \varnothing$, common candidates are $\bigcup_{A\in C} A$ under inclusion and $\bigcap_{A\in C} A$ under reverse inclusion.
Under reverse inclusion, another object contained in every member of C may be necessary.
If $C = \varnothing$, any element of P is an upper bound. In every case, prove $u \in P$; this is the indispensable admissibility check.
Key Idea: A chain bound must preserve every admissibility condition defining the candidate set.

6.2 Same Cardinality, Finite, Countable, and Uncountable

Definition 6.2.1. *Same Cardinality*
For sets A, B, write $A \sim_{\#} B$ iff a bijection $f : A \to B$ exists. Then A and B have *the same cardinality*, or are *equivalent*.

Proposition 6.2.2. *Same Cardinality Is an Equivalence Relation*
Let A, B, C be sets.

(i) $A \sim_{\#} A$.
(ii) $A \sim_{\#} B \Rightarrow B \sim_{\#} A$.
(iii) $(A \sim_{\#} B \wedge B \sim_{\#} C) \Rightarrow A \sim_{\#} C$.

Consequently, $\sim_{\#}$ is an equivalence relation among sets.

Proof.

(i) The identity $\mathrm{id}_A : A \to A$ is bijective, so $A \sim_{\#} A$.
(ii) Suppose $A \sim_{\#} B$ and choose a bijection $f : A \to B$. The inverse $f^{-1} : B \to A$ exists, and f is its two-sided inverse. Hence f^{-1} is bijective by Theorem 3.3.2(i), so $B \sim_{\#} A$.
(iii) Suppose $A \sim_{\#} B$ and $B \sim_{\#} C$, and choose bijections $f : A \to B$ and $g : B \to C$. The composite $g \circ f : A \to C$ is bijective by Proposition 3.3.3(vi), so $A \sim_{\#} C$.

Structural Commentary: Identity, inversion, and composition verify the three equivalence-relation laws.
Key Insight: Bijections are closed under reversal and composition. □

Theorem 6.2.3. *Schroeder–Bernstein Theorem*
Let A, B be sets, and let $f : A \to B$ and $g : B \to A$ be injections. Then $A \sim_{\#} B$.

Proof. Set $A_0 := A \setminus g(B)$ and recursively set $B_n := f(A_n)$ and $A_{n+1} := g(B_n)$ for $n \in \mathbb{N}_0$.
Put $U_A := \bigcup_{n\in\mathbb{N}_0} A_n$, $U_B := \bigcup_{n\in\mathbb{N}_0} B_n$, $A_* := A \setminus U_A$, and $B_* := B \setminus U_B$.
The restriction $f_* := f|_{U_A} : U_A \to U_B$ is bijective because f is injective and $U_B = f(U_A)$. For $b \in B_*$, $g(b) \notin A_0$.
Suppose $g(b) \in A_{n+1} = g(B_n)$ with $n \in \mathbb{N}_0$. Then injectivity of g gives $b \in B_n \subseteq U_B$, a contradiction. Thus $g(b) \in A_*$.
Conversely, if $a \in A_*$, then $a \notin A_0$, so $a \in g(B)$. Choose $b \in B$ satisfying $a = g(b)$. Suppose $b \in B_n$ with $n \in \mathbb{N}_0$.
Then $a \in A_{n+1} \subseteq U_A$, a contradiction. Hence $b \in B_*$. Therefore $g_* := g|_{B_*} : B_* \to A_*$ is bijective.
Define $h : A \to B$ by $h(x) := f_*(x)$ for $x \in U_A$ and $h(x) := g_*^{-1}(x)$ for $x \in A_*$. The domain parts U_A, A_* and image parts U_B, B_* are disjoint partitions, and both branches are bijective. Hence h is bijective, so $A \sim_{\#} B$.
Structural Commentary: Forward layers are matched by f; the complementary remainders are matched by g^{-1}.
Key Insight: Alternate the injections until the unmatched remainder can be reversed. □

Definition 6.2.4. *Finite, Countable, and Uncountable*
Let A be a set.

(i) A is *finite* iff $(\exists n \in \mathbb{N}_0)\, A \sim_{\#} J_n$.
(ii) A is *infinite* iff it is not finite.
(iii) A is *countable* iff $A \sim_{\#} \mathbb{N}$.
(iv) A is *uncountable* iff it is neither finite nor countable.
(v) A is *at most countable* iff it is finite or countable.

Countable sets are also called denumerable. Here *enumerable* more generally describes the image of a sequence.

Proposition 6.2.5. *The Natural Numbers Are Infinite*

(i) $(\forall m \in \mathbb{N}_0)$, *no injection* $\mathbb{N} \to J_m$ *exists.*
(ii) The set $\mathbb{N}$ *is infinite.*
(iii) If A *is countable, then* A *is infinite.*
(iv) If F *is finite and* $A \subseteq F$, *then* A *is not countable.*

Proof.

(i) For $r \in \mathbb{N}$, let Q_r assert that no injection $\mathbb{N} \to J_{r-1}$ exists. Since $J_0 = \emptyset$, no map $\mathbb{N} \to J_0$ exists, so Q_1 holds.
Assume Q_r and suppose $u : \mathbb{N} \to J_r$ is injective. If $r \notin u(\mathbb{N})$, then $u : \mathbb{N} \to J_{r-1}$ contradicts Q_r.
Otherwise let $s \in \mathbb{N}$ be the unique index satisfying $u(s) = r$.
Define $\tau : \mathbb{N} \to \mathbb{N} \setminus \{s\}$ by $\tau(k) := k$ for $k < s$ and $\tau(k) := k + 1$ for $k \geq s$.
The map τ is injective and avoids s, so $u \circ \tau : \mathbb{N} \to J_{r-1}$ is injective, again contradicting Q_r. Thus Q_{r+1} holds.
Hence $(\forall r \in \mathbb{N})\, Q_r$ by Theorem 2.6.2, proving (i).
(ii) If $\mathbb{N}$ were finite, a bijection $J_m \to \mathbb{N}$ would have an inverse injection $\mathbb{N} \to J_m$, contradicting (i).
(iii) Let A be countable. If A were finite, composing a bijection $\mathbb{N} \to A$ with a bijection $A \to J_m$ would contradict (i).
(iv) If a countable A were contained in a finite F, then $\mathbb{N} \to A \hookrightarrow F \to J_m$ would be injective, contradicting (i).

Structural Commentary: Induction deletes the unique preimage of the final point of a finite target.
Key Insight: Removing one domain point leaves $\mathbb{N}$ injectively equivalent to itself. □

Proposition 6.2.6. *Finite Sums Are Invariant under Reindexing*
Let $n \in \mathbb{N}_0$, *let* $c : J_n \to \mathbb{C}$, *and let* $\sigma : J_n \to J_n$ *be a bijection. Then* $\sum_{i=1}^{n} c(\sigma(i)) = \sum_{i=1}^{n} c(i)$.

Proof. For $r \in \mathbb{N}$, let P_r be the assertion of the proposition with $n = r - 1$. Then P_1 is the empty-sum identity.
Assume P_r, put $m := r - 1$, and fix $c : J_{m+1} \to \mathbb{C}$ and a permutation σ of J_{m+1}.
Put $j := \sigma^{-1}(m + 1)$ and define $\theta : J_m \to J_{m+1} \setminus \{j\}$ by $\theta(i) := i$ for $i < j$ and $\theta(i) := i + 1$ for $i \geq j$.
Then θ and $\sigma \circ \theta : J_m \to J_m$ are bijective. Since $\sigma(j) = m + 1$, commutativity gives $\sum_{i=1}^{m+1} c(\sigma(i)) = c(m + 1) + \sum_{i=1}^{m} c(\sigma(\theta(i)))$.
Latter expression equals $c(m + 1) + \sum_{i=1}^{m} c(i) = \sum_{i=1}^{m+1} c(i)$ by P_r. Thus P_{r+1} holds. Hence $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2, proving P_{n+1}.
Structural Commentary: Induction removes the position mapped to the final index and reindexes the remainder.
Key Insight: A permutation changes positions without changing the multiset of summands. □

Definition 6.2.7. *Summation over a Finite Set*
Let X be finite, put $n := \#X$, let $f : X \to \mathbb{C}$, and choose a bijection $g : J_n \to X$. Define $\sum_{x\in X} f(x) := \sum_{i=1}^{n} f(g(i))$.
If $X = \emptyset$, then $n = 0$ and this value is the empty sum 0. If $k : J_n \to X$ is another bijection, put $\sigma := g^{-1} \circ k$.
Then σ is a permutation of J_n, $k = g \circ \sigma$, and the two listings give equal sums by Proposition 6.2.6.
Key Insight: Reindexing changes the listing, not the sum.

Proposition 6.2.8. *Finite Set Summation Laws*
Let X, Y *be finite sets.*

(i) Well-definedness: If $f : X \to \mathbb{C}$, *then* $\sum_{x\in X} f(x)$ *is independent of the bijection* $J_{\#X} \to X$.
(ii) Substitution: If $h : Y \to X$ *is a bijection and* $f : X \to \mathbb{C}$, *then* $\sum_{x\in X} f(x) = \sum_{y\in Y} f(h(y))$.
(iii) Linearity: If $f, g : X \to \mathbb{C}$, *then* $\sum_{x\in X} (f(x) + g(x)) = \sum_{x\in X} f(x) + \sum_{x\in X} g(x)$. *If* $c \in \mathbb{C}$, *then* $\sum_{x\in X} cf(x) = c \sum_{x\in X} f(x)$.
(iv) Disjoint unions: If $X \cap Y = \emptyset$ *and* $f : X \cup Y \to \mathbb{C}$, *then* $X \cup Y$ *is finite. The identity* $\sum_{z\in X\cup Y} f(z) = \sum_{x\in X} f(x) + \sum_{y\in Y} f(y)$ *holds.*
(v) Finite Fubini: The set $X \times Y$ *is finite. If* $f : X \times Y \to \mathbb{C}$, *then the following identities hold.*
$\sum_{x\in X} \sum_{y\in Y} f(x, y) = \sum_{(x,y)\in X\times Y} f(x, y)$, *and* $\sum_{y\in Y} \sum_{x\in X} f(x, y) = \sum_{(x,y)\in X\times Y} f(x, y)$.

Proof.

(i) Let $g, k : J_{\#X} \to X$ be bijections and put $\sigma := g^{-1} \circ k$. Then $k = g \circ \sigma$, so the two sums agree by Proposition 6.2.6.

(ii) Choose a bijection $g : J_{\#Y} \to Y$. The composite $h \circ g : J_{\#Y} \to X$ is bijective, so $\#Y = \#X$ by Proposition 4.2.2.
After expanding the finite-set sums, $\sum_{x \in X} f(x) = \sum_{i=1}^{\#Y} f(h(g(i))) = \sum_{y \in Y} f(h(y))$ by part (i).

(iii) Put $m := \#X$, choose a bijection $\gamma : J_m \to X$, and set $a_i := f(\gamma(i))$ and $b_i := g(\gamma(i))$ for $i \in J_m$, with $a_i = b_i = 0$ otherwise.
After expanding the three finite-set sums, the first identity holds by Proposition 4.1.2(ii).
After the analogous expansions, the scalar-multiple identity holds by Proposition 4.1.2(iii).

(iv) Put $m := \#X$ and $n := \#Y$. If $m = 0$ or $n = 0$, then one set is empty and both conclusions follow directly.
Suppose $m, n \in \mathbb{N}$, and choose bijections $\alpha : J_m \to X$ and $\beta : J_n \to Y$.
Define $\delta : J_{m+n} \to X \cup Y$ by $\delta(i) := \alpha(i)$ when $i \leq m$ and $\delta(i) := \beta(i - m)$ when $i > m$.
The two branches have disjoint images and make δ bijective because $X \cap Y = \emptyset$. Thus $X \cup Y$ is finite.
The identity in (iv) holds by Proposition 4.1.2(i) after expansion along δ.

(v) Put $m := \#X$ and $n := \#Y$. The set $X \times Y$ is finite with $\#(X \times Y) = mn$ by Theorem 4.2.5(iii).
If $m = 0$ or $n = 0$, all three sums equal 0. Otherwise choose bijections $\alpha : J_m \to X$ and $\beta : J_n \to Y$.
Define bijections $\ell, r : J_{mn} \to X \times Y$ by $\ell((i-1)n + j) := (\alpha(i), \beta(j))$ and $r((j-1)m + i) := (\alpha(i), \beta(j))$ for $i \in J_m$ and $j \in J_n$.
The first identity in (v) holds by Proposition 4.1.2(i) and part (i) after expansion along ℓ.
The second identity in (v) holds by Proposition 4.1.2(i) and part (i) after expansion along r.

Structural Commentary: Reindexing, concatenation, and lexicographic listings transfer ordinary finite-sum laws to sets.
Key Insight: A finite-set sum depends on values, not listing order. □

Definition 6.2.9. *Sequences as Enumerations*
Let A be a set.

(i) A *sequence* in A is a map $x : \mathbb{N} \to A$, written $(x_n)_{n \in \mathbb{N}}$ with $x_n := x(n)$.
(ii) A sequence $x : \mathbb{N} \to A$ *enumerates* A iff $x(\mathbb{N}) = A$.
(iii) A sequence $x : \mathbb{N} \to A$ is a *distinct enumeration* of A iff $x : \mathbb{N} \to A$ is bijective.

Thus A is countable iff it has a distinct enumeration.

Proposition 6.2.10. *The Integers Are Countable*
The set $\mathbb{Z}$ is countable.

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ by $f(2k) := k$ and $f(2k-1) := 1 - k$ for $k \in \mathbb{N}$.
Define $e : \mathbb{Z} \to \mathbb{N}$ by $e(z) := 2z$ for $z \geq 1$ and $e(z) := 1 - 2z$ for $z \leq 0$.
The even and odd branches partition $\mathbb{N}$, while the positive and nonpositive branches partition $\mathbb{Z}$.
Direct substitution gives $e \circ f = \mathrm{id}_{\mathbb{N}}$ and $f \circ e = \mathrm{id}_{\mathbb{Z}}$. Thus f is bijective by Theorem 3.3.2(i), so $\mathbb{Z}$ is countable.
Structural Commentary: Parity separates the positive and nonpositive integer branches.
Key Insight: Interleave the two one-sided integer lists. □

Theorem 6.2.11. *Infinite Subsets of Countable Sets*
Let A be countable and let $E \subseteq A$ be infinite. Then E is countable.

Proof. Choose a bijection $x : \mathbb{N} \to A$, write $x_n := x(n)$, and set $I_E := \{n \in \mathbb{N} : x_n \in E\}$. The restriction $x|_{I_E} : I_E \to E$ is bijective.
Hence I_E is infinite, since otherwise this bijection would make E finite. Each intersection $I_E \cap J_n$ is finite by Proposition 4.2.4(ii).
For $k \in \mathbb{N}$, set $M_k := \{n \in I_E : \#(I_E \cap J_n) = k\}$. We prove $(\forall k \in \mathbb{N})\, M_k \neq \emptyset$ by induction.
Let $m := \min I_E$ by well-ordering. Then $I_E \cap J_m = \{m\}$, so $m \in M_1$. Suppose $M_k \neq \emptyset$ and choose $m \in M_k$.
The set $I_E \setminus J_m$ is nonempty; otherwise $I_E \subseteq J_m$ would be finite by Proposition 4.2.4(ii). Let $r := \min(I_E \setminus J_m)$ by well-ordering.
Then $I_E \cap J_r = (I_E \cap J_m) \dot\cup \{r\}$, so $\#(I_E \cap J_r) = k + 1$ by Proposition 4.2.4(i). Thus $r \in M_{k+1}$; $(\forall k \in \mathbb{N})\, M_k \neq \emptyset$ by Theorem 2.6.2.
If $n < m$ belong to I_E, then $I_E \cap J_m$ is the disjoint union of $I_E \cap J_n$ and the nonempty set $I_E \cap (J_m \setminus J_n)$.
Hence $\#(I_E \cap J_n) < \#(I_E \cap J_m)$ by Proposition 4.2.4(i), so each M_k is a singleton. Let n_k be its unique element.
If $n \in I_E$ and $k := \#(I_E \cap J_n)$, then $n \in M_k$, so $n = n_k$. Thus $k \mapsto n_k$ is a bijection $\mathbb{N} \to I_E$.
The composite $k \mapsto x_{n_k}$ is a bijection $\mathbb{N} \to E$, so E is countable.
Structural Commentary: Finite initial-segment ranks canonically enumerate the indices that hit the subset.
Key Insight: Rank each selected index by the number of earlier selected indices. □

Proposition 6.2.12. *Subsets and Images of At Most Countable Sets*
Let A be at most countable.

(i) If $B \subseteq A$, then B is at most countable.
(ii) If $f : A \to Y$ is a function, then $f(A)$ is at most countable.

Proof.

(i) Suppose $B \subseteq A$. If A is finite, then B is finite by Proposition 4.2.4(ii).
If A is countable, then B is finite or, when infinite, countable by Theorem 6.2.11. Thus B is at most countable.

(ii) Suppose $f : A \to Y$ and first assume A is finite. Put $m := \#A$, choose a bijection $e : J_m \to A$.
Set $I := \{i \in J_m : (\forall j \in J_m)\, (j < i \Rightarrow f(e(j)) \neq f(e(i)))\}$. The set I is finite by Proposition 4.2.4(ii).
The map $i \mapsto f(e(i))$ on I is injective by the definition of I. Fix $y \in f(A)$. The nonempty set $\{i \in J_m : f(e(i)) = y\}$ has a least element by well-ordering, and that element belongs to I. Thus the map is also surjective onto $f(A)$, so $f(A)$ is finite.
Now suppose A is countable. Choose a bijection $e : \mathbb{N} \to A$ and set $I := \{i \in \mathbb{N} : (\forall j \in \mathbb{N})\, (j < i \Rightarrow f(e(j)) \neq f(e(i)))\}$.
The set I is at most countable by (i). The map $i \mapsto f(e(i))$ on I is injective by the definition of I.
Fix $y \in f(A)$. The nonempty set $\{i \in \mathbb{N} : f(e(i)) = y\}$ has a least element by well-ordering, and that element belongs to I.
Thus the map is also surjective onto $f(A)$, so $f(A)$ is at most countable.

Structural Commentary: Subsets restrict a listing; images retain the first index of each attained value.
Key Insight: Deleting entries or repetitions cannot increase cardinality. □

Proposition 6.2.13. *At Most Countable Sets Are Exactly Enumerable Images*
For a set B, the following are equivalent:

(i) B is at most countable.
(ii) $B = \emptyset$ or a surjection $f : \mathbb{N} \to B$ exists.

Thus a nonempty set is at most countable iff it is the image of a sequence.

Proof.

(*i*) $\Rightarrow$ (*ii*) If $B = \emptyset$, then (ii) holds. If B is countable, any bijection $\mathbb{N} \to B$ is surjective.
If B is finite and nonempty, put $m := \#B$, choose a bijection $e : J_m \to B$.
Define $f : \mathbb{N} \to B$ by $f(n) := e(n)$ for $n \le m$ and $f(n) := e(m)$ for $n > m$. Then f is surjective.

(*ii*) $\Rightarrow$ (*i*) If $B = \emptyset$, then B is finite. If $f : \mathbb{N} \to B$ is surjective, then $B = f(\mathbb{N})$ is at most countable by Proposition 6.2.12(ii).

Structural Commentary: A finite list can repeat its final term; a countable list already supplies a surjection.
Key Insight: At most countable sets are sequence images or the empty set. □

Proposition 6.2.14. *Pairs of Positive Integers Are Countable*
The set $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. For $r \in \mathbb{N}_0$, put $t_r := \frac{r(r+1)}{2}$. Define $p : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $p(i,j) := t_{i+j-2} + i$.
Fix $(i,j) \in \mathbb{N} \times \mathbb{N}$ and put $r := i + j - 1$; then $t_{r-1} < p(i,j) \le t_r$. The intervals $(t_{r-1}, t_r]$ are disjoint for $r \in \mathbb{N}$.
Hence equality of two values of p forces equality of the coordinate sums and then equality of the first, and therefore also the second, coordinates. Thus p is injective. Fix $q \in \mathbb{N}$. The set $R_q := \{r \in \mathbb{N} : q \le t_r\}$ is nonempty because $t_q \ge q$.
Let $r := \min R_q$ by well-ordering. Minimality gives $t_{r-1} < q \le t_r$. Set $i := q - t_{r-1}$ and $j := r + 1 - i$.
Then $i, j \in \mathbb{N}$ and $p(i,j) = q$, so p is surjective. Hence p is a bijection and $\mathbb{N} \times \mathbb{N}$ is countable.
Structural Commentary: Triangular intervals separate the diagonals and recover both coordinates from each index.
Key Insight: Coordinate-sum diagonals occupy disjoint finite intervals. □

Theorem 6.2.15. *Countable Union of Countable Sets*
Let $(E_n)_{n\in\mathbb{N}}$ be countable sets and set $S := \bigcup_{n\in\mathbb{N}} E_n$. Then S is countable.

Proof. For $n \in \mathbb{N}$, let $\mathcal{B}_n$ be the set of bijections $\mathbb{N} \to E_n$. Then $(\forall n \in \mathbb{N})\, \mathcal{B}_n \neq \emptyset$.
Choose $(x_n)_{n\in\mathbb{N}}$ satisfying $(\forall n \in \mathbb{N})\, x_n \in \mathcal{B}_n$ by the axiom of choice, and write $x_{n,k} := x_n(k)$.
Choose a bijection $d : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by Proposition 6.2.14. For $t \in \mathbb{N}$, write $d(t) = (n_t, k_t)$ and define $a : \mathbb{N} \to S$ by $a(t) := x_{n_t,k_t}$.
Then $a(\mathbb{N}) = S$, so S is at most countable by Proposition 6.2.13.
The countable set E_1 is contained in S, so S is not finite by Proposition 6.2.5(iv). Hence S is countable.
Structural Commentary: Choice selects row enumerations, and a pairing bijection flattens their two indices.
Key Insight: A countable array becomes one sequence after pairing its indices. □

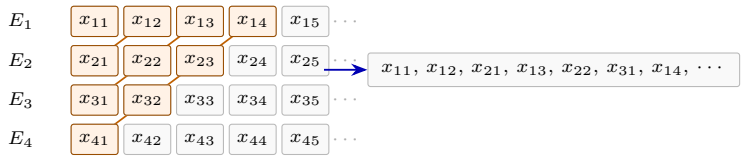

Figure 7: Diagonal scan of countable rows into one surjective sequence.

Figure 7 illustrates the scan used in Theorem 6.2.15.
Every indexed entry occurs on a finite diagonal; repeated values do not affect surjectivity.

Corollary 6.2.16. *At Most Countable Unions*
Let A be at most countable and let $(B_\alpha)_{\alpha\in A}$ be an indexed family of at most countable sets. Then $\bigcup_{\alpha\in A} B_\alpha$ is at most countable.

Proof. Set $T := \bigcup_{\alpha\in A} B_\alpha$. If $T = \emptyset$, then the conclusion holds. Suppose $T \neq \emptyset$ and set $I := \{\alpha \in A : B_\alpha \neq \emptyset\}$.
The set I is nonempty and at most countable by Proposition 6.2.12(i).
Choose a surjection $e : \mathbb{N} \to I$ by Proposition 6.2.13, and put $\alpha_j := e(j)$. For $j \in \mathbb{N}$, let $\mathcal{S}_j$ be the set of surjections $\mathbb{N} \to B_{\alpha_j}$.
Then $(\forall j \in \mathbb{N})\, \mathcal{S}_j \neq \emptyset$ by Proposition 6.2.13. Choose $(b_j)_{j\in\mathbb{N}}$ satisfying $(\forall j \in \mathbb{N})\, b_j \in \mathcal{S}_j$ by the axiom of choice, and write $b_{j,k} := b_j(k)$.
Choose a bijection $d : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by Proposition 6.2.14. For $t \in \mathbb{N}$, write $d(t) = (j_t, k_t)$ and define $a : \mathbb{N} \to T$ by $a(t) := b_{j_t,k_t}$.
Then $a(\mathbb{N}) = T$, so T is at most countable by Proposition 6.2.13.
Structural Commentary: Surjections index the nonempty pieces and their elements before a pairing map flattens both indices.
Key Insight: Empty pieces can be discarded; all remaining rows admit sequence images. □

Corollary 6.2.17. *Countable Cartesian Products*
Let A, B be sets.

(i) If A and B are at most countable, then $A \times B$ is at most countable.
(ii) If A and B are countable, then $A \times B$ is countable.

Proof.

(i) For $a \in A$, set $S_a := \{a\} \times B$. The map $b \mapsto (a,b)$ is a bijection $B \to S_a$, so $(S_a)_{a\in A}$ is an indexed family of at-most-countable sets. Since $A \times B = \bigcup_{a\in A} S_a$, the product is at most countable by Corollary 6.2.16.

(ii) By (i), $A \times B$ is at most countable. Choose $b_0 \in B$. The map $a \mapsto (a, b_0)$ is a bijection $A \to A \times \{b_0\}$, so $A \times B$ contains the countable subset $A \times \{b_0\}$. Thus $A \times B$ is not finite by Proposition 6.2.5(iv), and it is therefore countable.

Structural Commentary: Slicing proves at-most-countability; one fixed-coordinate slice rules out finiteness.
Key Insight: A product is a union of copies of either factor. □

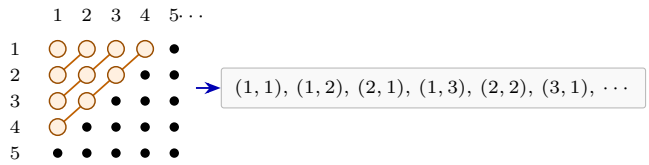

Figure 8: Diagonal enumeration of $\mathbb{N} \times \mathbb{N}$ by finite $i + j$ diagonals.

Figure 8 depicts the $\mathbb{N} \times \mathbb{N}$ pairing in Proposition 6.2.14.
Each pair lies on one finite coordinate-sum diagonal, which determines its interval of indices.

Theorem 6.2.18. *Finite Tuples over a Countable Set*
Let A be countable. For $n \in \mathbb{N}$, set $T_n := \{(a_1, \dots, a_n) : (\forall k \in J_n)\, a_k \in A\}$. Then T_n is countable.

Proof. For $r \in \mathbb{N}$, let P_r assert that T_r is countable. The map $a \mapsto (a)$ is a bijection $A \to T_1$, so P_1 holds. Assume P_r.
The product $T_r \times A$ is countable by Corollary 6.2.17(ii), and concatenation bijects $T_r \times A$ with T_{r+1}.
Thus P_{r+1} holds, so $(\forall r \in \mathbb{N})\, P_r$ by Theorem 2.6.2.
Structural Commentary: Induction appends one coordinate and transfers countability across a binary product bijection.
Key Insight: Every fixed finite product is built one coordinate at a time. □

Corollary 6.2.19. *The Rationals Are Countable*
The set $\mathbb{Q}$ is countable.

Proof. Set $D := \{(p, q) \in \mathbb{Z} \times \mathbb{Z} : q \neq 0\}$ and define $\pi : D \to \mathbb{Q}$ by $\pi(p, q) := \frac{p}{q}$.
The set $\mathbb{Z} \times \mathbb{Z}$ is countable by Proposition 6.2.10 and Corollary 6.2.17(ii), so D is at most countable by Proposition 6.2.12(i).
Since $\mathbb{Q} = \pi(D)$, the set $\mathbb{Q}$ is at most countable by Proposition 6.2.12(ii).
The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ identifies a countable subset, so $\mathbb{Q}$ is not finite by Proposition 6.2.5(iv), and therefore $\mathbb{Q}$ is countable.
Structural Commentary: Integer pairs form an at-most-countable parameter set whose quotient map covers $\mathbb{Q}$.
Key Insight: Fractions are images of nonzero-denominator integer pairs. □

Proof Method 6.2.20. *Cardinality Proof Patterns*
Let A, B be sets. Choose the route matching the target cardinality claim.

(i) *Same cardinality*: Construct $f : A \to B$, then verify that f is injective and surjective.
(ii) *Countable*: Construct a bijection $\mathbb{N} \to A$. Alternatively, prove that A is at most countable and not finite.
(iii) *Uncountable*: Prove $A \neq \emptyset$. Fix an arbitrary map $s : \mathbb{N} \to A$ and construct $a_s \in A \setminus s(\mathbb{N})$.
Then no sequence surjects onto A, so A is not at most countable by Proposition 6.2.13.

Key Idea: Bijections certify equal cardinality; diagonal witnesses obstruct every proposed enumeration.

6.3 Ordinals, Transfinite Recursion, and Well-Foundedness

Definition 6.3.1. *Ordinals and Order Types*

(i) A set T is *transitive* iff $(\forall x \in T)\, x \subseteq T$.
(ii) An *ordinal* is a transitive set α on which $\in$ is a strict well-order. The relation $\in$ is irreflexive and transitive on α, distinct elements are $\in$-comparable, and each nonempty $A \subseteq \alpha$ contains m satisfying $(\forall a \in A)\, a \notin m$.
(iii) For ordinals α, β, write $\alpha < \beta$ iff $\alpha \in \beta$ and $\alpha \leq \beta$ iff $\alpha \subseteq \beta$.
(iv) Set $0 := \emptyset$. The *successor* of α is $\alpha + 1 := \alpha \cup \{\alpha\}$. The *finite ordinals* are 0 and its finite iterated successors.
An ordinal is *infinite* iff it is not finite. It is a *limit ordinal* iff it is nonzero and is not a successor.
(v) The least infinite ordinal is ω; its elements are the finite ordinals $0, 1, 2, \dots$.
If a well-ordered set $(S, \leq_S)$ is order-isomorphic to an ordinal δ in the sense of Definition 5.4.7, then δ is the *order type* of $(S, \leq_S)$.

Principle 6.3.2. *Ordinal Foundation Principles*
The following standard properties of von Neumann ordinals form the ordinal framework used below.

(i) *Ordinal comparison*: two ordinals are equal or one belongs to the other, and each nonempty set of ordinals has a least member.
(ii) *Strong transfinite induction*: if P is a property of ordinals satisfying $(\forall \alpha)\, [((\forall \beta < \alpha)\, P(\beta)) \Rightarrow P(\alpha)]$, then $(\forall \alpha)\, P(\alpha)$.
(iii) *Order types*: a well-ordered set is order-isomorphic to a unique ordinal.
(iv) *Well-ordering theorem*: under Principle 6.1.2, each set admits a well-order and is therefore bijective with an ordinal by (iii).

These principles concern ordinal comparison and recursive construction; do not prescribe a particular well-order on an arbitrary set.

Definition 6.3.3. *Well-Founded Relation*
Let $\prec$ be a relation on a set X.

(i) The *predecessor set* of $x \in X$ is $\mathrm{Pred}_{\prec}(x) := \{y \in X : y \prec x\}$.
(ii) For $A \subseteq X$, an element $m \in A$ is *$\prec$-minimal in A* iff $\mathrm{Pred}_{\prec}(m) \cap A = \emptyset$.
(iii) The relation $\prec$ is *well-founded* iff $(\forall A \subseteq X)\, [A \neq \emptyset \Rightarrow (\exists m \in A)\ \mathrm{Pred}_{\prec}(m) \cap A = \emptyset]$.

Neither totality nor transitivity is required.

Theorem 6.3.4. *Well-Founded Induction*
Let $\prec$ be a well-founded relation on X, and let P be a property of elements of X.
If $(\forall x \in X)\, [((\forall y \in \mathrm{Pred}_{\prec}(x))\, P(y)) \Rightarrow P(x)]$, then $(\forall x \in X)\, P(x)$.

Proof. Set $B := \{x \in X : \neg P(x)\}$. Suppose $B \neq \emptyset$, and choose a $\prec$-minimal $b \in B$. If $y \in \mathrm{Pred}_{\prec}(b)$, then $y \notin B$ by minimality.
Thus $(\forall y \in \mathrm{Pred}_{\prec}(b))\, P(y)$. The induction premise gives $P(b)$, contradicting $b \in B$. Hence $B = \emptyset$ and $(\forall x \in X)\, P(x)$.
Structural Commentary: A least-counterexample contradiction applies the induction premise at one minimal failure.
Key Insight: A minimal failure has only successful predecessors. □

Principle 6.3.5. *Transfinite and Well-Founded Recursion*
Recursive values may depend on the full restriction of the desired function to earlier arguments.

(i) *Ordinal recursion*: let θ be an ordinal, and suppose $G(\alpha, g)$ is assigned whenever $\alpha < \theta$ and $\operatorname{dom} g = \alpha$.
Then $(\exists! F)\,[\operatorname{dom} F = \theta \wedge (\forall \alpha < \theta)\, F(\alpha) = G(\alpha, F|_\alpha)]$.

(ii) *Well-founded recursion*: let $\prec$ be well-founded on X, and suppose $H(x, g)$ is assigned whenever $x \in X$ and $\operatorname{dom} g = \operatorname{Pred}_\prec(x)$.
Then $(\exists! F)\,[\operatorname{dom} F = X \wedge (\forall x \in X)\, F(x) = H(x, F|_{\operatorname{Pred}_\prec(x)})]$.

In (i), the least disagreement proves uniqueness; in (ii), uniqueness follows by Theorem 6.3.4 for the property that two candidates agree.
The rules apply only when the value at an argument depends on values at its predecessors.

Definition 6.3.6. *Ordinal Arithmetic*
For a set A of ordinals, set $\sup A := \bigcup A$. Fix an ordinal α.
Transfinite recursion in the second argument defines the following operations.
Each limit clause assumes that λ is a nonzero limit ordinal.

(i) *Addition*: $\alpha + 0 := \alpha$ and $\alpha + (\beta + 1) := (\alpha + \beta) + 1$. For a limit λ, $\alpha + \lambda := \sup\{\alpha + \beta : \beta < \lambda\}$.
(ii) *Multiplication*: $\alpha \cdot 0 := 0$ and $\alpha \cdot (\beta + 1) := \alpha \cdot \beta + \alpha$. For a limit λ, $\alpha \cdot \lambda := \sup\{\alpha \cdot \beta : \beta < \lambda\}$.
(iii) *Exponentiation*: $\alpha^0 := 1$ and $\alpha^{\beta+1} := \alpha^\beta \cdot \alpha$. For a limit λ, $\alpha^\lambda := \sup\{\alpha^\beta : \beta < \lambda\}$.

The successor clause for addition agrees with Definition 6.3.1.

Example 6.3.7. *Ordinal Arithmetic Remembers Order*
The limit clauses give $1 + \omega = \sup\{1 + n : n < \omega\} = \omega$ and $2 \cdot \omega = \sup\{2 \cdot n : n < \omega\} = \omega$.
The successor clauses give $\omega + 1 > \omega$ and $\omega \cdot 2 = \omega + \omega > \omega$. Thus $1 + \omega < \omega + 1$ and $2 \cdot \omega < \omega \cdot 2$.
Ordinal addition and multiplication retain the order of concatenation and are not commutative.

Definition 6.3.8. *Cardinals as Initial Ordinals*

(i) An ordinal κ is a *cardinal*, or *initial ordinal*, iff $(\forall \beta < \kappa)\, \beta \not\sim_\# \kappa$.
(ii) If a set X is bijective with an ordinal, its *cardinality* $|X|$ is the least such ordinal.
The minimum exists by Principle 6.3.2(i), applied to the nonempty set of candidates no larger than a fixed witness.
Under Principle 6.1.2, every set has a cardinality by Principle 6.3.2(iv).
(iii) A cardinal is *infinite* iff it is not a finite ordinal, and $\aleph_0 := |\omega|$.

Definition 6.3.9. *Cofinal Subsets and Cofinality*
Let α be an ordinal.

(i) A subset $C \subseteq \alpha$ is *cofinal in* α, or *unbounded in* α, iff $(\forall \beta < \alpha)(\exists \gamma \in C)\, \beta \le \gamma$.
(ii) A function $f : \delta \to \alpha$ is *cofinal* iff $f(\delta)$ is cofinal in α.
(iii) The *cofinality* of α is $\operatorname{cf}(\alpha) := \min\{\delta : (\exists f : \delta \to \alpha)(\forall \beta < \alpha)(\exists \xi < \delta)\, \beta \le f(\xi)\}$.

The minimum exists because id_α is cofinal and nonempty sets of ordinals have least elements.
If α is a limit ordinal, then C is cofinal iff $\sup C = \alpha$. Indeed, if C is not cofinal, then some member of α strictly bounds C from above.
Conversely, cofinality above each successor below α forces $\bigcup C = \alpha$. The empty function is cofinal in 0, while $0 \mapsto \alpha$ is cofinal from 1 to $\alpha + 1$. Since an empty range is not cofinal in a successor, $\operatorname{cf}(0) = 0$ and $\operatorname{cf}(\alpha + 1) = 1$.

Lemma 6.3.10. *Nondecreasing Cofinal Witness*
Let λ be a nonzero limit ordinal and set $\delta := \operatorname{cf}(\lambda)$.

(i) The ordinal δ is nonzero and is a limit ordinal.
(ii) There is a cofinal function $f : \delta \to \lambda$ satisfying $(\forall \xi \le \eta < \delta)\, f(\xi) \le f(\eta)$.

Proof.

(i) Choose a cofinal function $h : \delta \to \lambda$. Since an empty range is not cofinal in λ, $\delta \neq 0$.
Suppose $\delta = \varepsilon + 1$. The restriction $h|_\varepsilon$ is not cofinal by the minimality of δ, so $(\exists \beta < \lambda)(\forall \eta < \varepsilon)\, h(\eta) < \beta$.
Set $\rho := \max\{\beta, h(\varepsilon)\} + 1$. Since λ is a limit ordinal, $\rho < \lambda$ and $(\forall \eta < \delta)\, h(\eta) < \rho$, contradicting the cofinality of h. QED.
(ii) For $\xi < \delta$, set $f(\xi) := \sup\{h(\eta) : \eta \le \xi\}$. Since $\xi + 1 < \delta$, range of $h|_{\xi+1}$ is not cofinal and is bounded below λ; hence $f(\xi) < \lambda$.
If $\xi \le \eta < \delta$, then the set defining $f(\xi)$ is contained in the set defining $f(\eta)$, so $f(\xi) \le f(\eta)$.
Finally, $h(\xi) \le f(\xi)$ for $\xi < \delta$, and therefore f is cofinal.

Structural Commentary: Minimality first excludes successor domains, then successive suprema monotone-ize a cofinal witness.
Key Insight: No proper initial segment of a minimal cofinal domain is cofinal. □

Definition 6.3.11. *Regular and Singular Cardinals*
An infinite cardinal κ is *regular* iff $\operatorname{cf}(\kappa) = \kappa$, and it is *singular* iff $\operatorname{cf}(\kappa) < \kappa$.

Theorem 6.3.12. *Idempotence and Cardinality of Cofinality*
Let α be an ordinal and set $\delta := \operatorname{cf}(\alpha)$.

(i) $\operatorname{cf}(\delta) = \delta$.
(ii) The ordinal δ is a cardinal.
(iii) If α is a nonzero limit ordinal, then δ is an infinite regular cardinal.

Proof.

(i) If $\alpha = 0$, then $\delta = 0$ and $\operatorname{cf}(\delta) = 0$. If α is a successor, then $\delta = 1$ and $\operatorname{cf}(\delta) = 1$.
Suppose α is a nonzero limit ordinal. A nondecreasing cofinal function $f : \delta \to \alpha$ exists by Lemma 6.3.10(ii); fix one.
The ordinal δ is a nonzero limit ordinal by Lemma 6.3.10(i).
A nondecreasing cofinal function $g : \operatorname{cf}(\delta) \to \delta$ therefore exists by Lemma 6.3.10(ii).
For $\beta < \alpha$, choose $\xi < \delta$ with $\beta \le f(\xi)$ and then $\eta < \operatorname{cf}(\delta)$ with $\xi \le g(\eta)$. Monotonicity gives $\beta \le f(g(\eta))$, so $f \circ g$ is cofinal in α.
The minimality of δ gives $\delta \le \operatorname{cf}(\delta)$, while id_δ gives $\operatorname{cf}(\delta) \le \delta$. Hence $\operatorname{cf}(\delta) = \delta$.
(ii) If δ were not a cardinal, choose $\beta < \delta$ and a bijection $b : \beta \to \delta$. Its range is cofinal, so $\operatorname{cf}(\delta) \le \beta < \delta$, contradicting (i). QED.
(iii) Let α be a nonzero limit ordinal. The ordinal δ is nonzero and limit by Lemma 6.3.10(i), hence it is infinite.
It is a cardinal by (ii), and $\operatorname{cf}(\delta) = \delta$ by (i), so it is regular.

Structural Commentary: Composition forbids cofinality from shrinking twice; initiality and regularity then follow.
Key Insight: Minimal cofinal domains are fixed points of cofinality. □

Example 6.3.13. *Basic Ordinal Cofinalities*
An empty range is bounded. For a nonempty finite range in a limit ordinal, recursion produces a maximum; its successor still lies below the limit and bounds the range. Hence no finite ordinal is a cofinal domain for a nonzero limit ordinal.
The maps $n \mapsto n$, $n \mapsto \omega + n$, and $n \mapsto \omega^n$ from ω are cofinal in ω, $\omega \cdot 2$, and ω^ω, respectively.
Each target therefore has cofinality at most ω and no finite cofinality. Thus $\mathrm{cf}(\omega) = \mathrm{cf}(\omega \cdot 2) = \mathrm{cf}(\omega^\omega) = \omega$.
Cofinality measures the shortest unbounded approach, not the order type or cardinality of its target.

Proposition 6.3.14. *Cardinality Is Canonical and Bijection-Invariant*
Assume Principle 6.1.2. For sets X, Y:

(i) $X \sim_\# |X|$, *and* $|X|$ *is a cardinal.*
(ii) $|X| = |Y| \iff X \sim_\# Y$.

Proof.

(*i*) The relation $X \sim_\# |X|$ holds by the definition of $|X|$. If $\beta < |X|$ and $\beta \sim_\# |X|$, then $\beta \sim_\# X$ by Proposition 6.2.2, contradicting the minimality of $|X|$. Thus $|X|$ is a cardinal.
(*ii*) ($\Rightarrow$) If $|X| = |Y|$, then $X \sim_\# |X| = |Y| \sim_\# Y$, so $X \sim_\# Y$ by Proposition 6.2.2.
(*ii*) ($\Leftarrow$) If $X \sim_\# Y$, then $(\forall \gamma)\,[\gamma \sim_\# X \iff \gamma \sim_\# Y]$ by Proposition 6.2.2.
The least ordinals satisfying these equivalent conditions are equal, so $|X| = |Y|$.

Structural Commentary: Minimal ordinal representatives identify exactly the equivalence classes of bijections.
Key Insight: Cardinality retains only bijection type. □

Proposition 6.3.15. *Finite Cardinality and $\aleph_0$ Agree with Earlier Notation*
Assume Principle 6.1.2.

(i) If X is finite and $\#X = n$, then $|X| = n$, where n on the right is the corresponding finite ordinal.
(ii) $\aleph_0 = \omega$.
(iii) A set X is countable iff $|X| = \aleph_0$.
(iv) The cardinal $\aleph_0$ is the least infinite cardinal.

Proof.

(*i*) The unique map $J_0 \to 0$ is bijective, and for $n \in \mathbb{N}$ the map $q_n : J_n \to n$ defined by $q_n(k) := k - 1$ is bijective.
Since $X \sim_\# J_n$ and $J_n \sim_\# n$, $X \sim_\# n$ by Proposition 6.2.2.
If $\beta < n$ and $\beta \sim_\# X$, choose $m < n$ with $\beta = m$. Then $J_m \sim_\# J_n$ by Proposition 6.2.2.
Hence $m = n$ by Proposition 4.2.2, contradicting $m < n$. Thus n is the least ordinal bijective with X, so $|X| = n$.
(*ii*) The map $q : \mathbb{N} \to \omega$ defined by $q(k) := k - 1$ is bijective. If $\beta < \omega$, choose $m \in \mathbb{N}_0$ with $\beta = m$.
If $\beta \sim_\# \omega$, then a witnessing bijection, q, and the bijection $J_m \to m$ from (i) compose to give an injection $\mathbb{N} \to J_m$, which is impossible by Proposition 6.2.5(i). Hence ω is the least ordinal bijective with itself, so $\aleph_0 = |\omega| = \omega$.
(*iii*) ($\Rightarrow$) If X is countable, then $X \sim_\# \mathbb{N}$ by Definition 6.2.4. Since $\mathbb{N} \sim_\# \omega$ by (ii), $X \sim_\# \omega$ by Proposition 6.2.2.
Hence $|X| = \aleph_0$ by Proposition 6.3.14(ii) and (ii).
(*iii*) ($\Leftarrow$) If $|X| = \aleph_0$, then $X \sim_\# \omega$ by Proposition 6.3.14(ii) and (ii). Since $\omega \sim_\# \mathbb{N}$ by (ii), $X \sim_\# \mathbb{N}$ by Proposition 6.2.2.
Thus X is countable by Definition 6.2.4.
(*iv*) An infinite cardinal is an infinite ordinal and is therefore at least the least infinite ordinal $\omega = \aleph_0$.

Structural Commentary: Explicit finite and countable bijections identify the initial ordinals selected by cardinality.
Key Insight: General cardinality extends both finite size and countability. □

Proposition 6.3.16. *Cardinality Comparison by Injections*
Assume Principle 6.1.2. For sets X, Y, $|X| \le |Y|$ iff an injection $X \to Y$ exists.

Proof. Set $\kappa := |X|$ and $\lambda := |Y|$. The ordinals κ, λ are cardinals and satisfy $X \sim_\# \kappa$ and $Y \sim_\# \lambda$ by Proposition 6.3.14(i).
Choose bijections $u : X \to \kappa$ and $v : Y \to \lambda$.
($\Rightarrow$) If $\kappa \le \lambda$, then $v^{-1} \circ \iota \circ u : X \to Y$ is injective, where $\iota : \kappa \hookrightarrow \lambda$ is inclusion.
($\Leftarrow$) Suppose $e : X \to Y$ is injective. Then $v \circ e \circ u^{-1} : \kappa \to \lambda$ is injective.
If $\lambda < \kappa$, inclusion injects λ into κ, so $\kappa \sim_\# \lambda$ by Theorem 6.2.3.
This contradicts the initiality of κ. Hence ordinal comparison gives $\kappa \le \lambda$.

Structural Commentary: Conjugate an injection to initial ordinals and exclude the reverse strict comparison.
Key Insight: Schroeder–Bernstein prevents a cardinal from embedding into a smaller cardinal. □

Lemma 6.3.17. *Infinite Cardinals Are Limit Ordinals*
Every infinite cardinal is a limit ordinal.

Proof. Suppose an infinite cardinal has the form $\kappa = \alpha + 1$. The ordinal α is infinite, otherwise its successor would be finite, so $\omega \le \alpha$.
Define $f : \alpha + 1 \to \alpha$ by $f(\alpha) := 0$, by $f(\xi) := \xi + 1$ when $\xi < \omega$, and by $f(\xi) := \xi$ when $\omega \le \xi < \alpha$.
The three domain pieces are disjoint, and their images are $\{0\}$, $\omega \setminus \{0\}$, and $\{\xi : \omega \le \xi < \alpha\}$.
These images are disjoint and cover α, so f is bijective. Thus κ is bijective with the smaller ordinal α, contradicting that κ is a cardinal.
Hence κ is neither zero nor a successor ordinal, and therefore it is a limit ordinal.
Structural Commentary: A direct folding bijection contradicts initiality for any alleged infinite successor cardinal.
Key Insight: An infinite successor ordinal absorbs its final endpoint. □

6.4 Diagonalization and Power Sets

Theorem 6.4.1. *Cantor's Theorem*
Let X be a set. No function $f : X \to \mathcal{P}(X)$ is surjective; hence $X \not\sim_{\#} \mathcal{P}(X)$.

Proof. Fix $f : X \to \mathcal{P}(X)$ and set $D_f := \{x \in X : x \notin f(x)\}$. Then $D_f \subseteq X$, so $D_f \in \mathcal{P}(X)$.
If $a \in X$ and $D_f = f(a)$, then $a \in D_f \iff a \in f(a)$, contrary to $a \in D_f \iff a \notin f(a)$.
Hence $(\forall a \in X)\, D_f \neq f(a)$, so $D_f \notin f(X)$ and f is not surjective. Since a bijection must be surjective, $X \not\sim_{\#} \mathcal{P}(X)$.
Structural Commentary: The diagonal subset differs from each image set at its own indexing element.
Key Insight: Diagonal self-membership forces $(\forall a \in X)\, D_f \neq f(a)$. □

Proposition 6.4.2. *Binary Sequences Model a Power Set*
Let $A := \{0,1\}^{\mathbb{N}}$. Define $\Phi : A \to \mathcal{P}(\mathbb{N})$ by $\Phi(s) := \{n \in \mathbb{N} : s(n) = 1\}$. Then Φ is bijective.

Proof. For $E \subseteq \mathbb{N}$, define $\chi_E : \mathbb{N} \to \{0,1\}$ by $\chi_E(n) := 1$ when $n \in E$ and $\chi_E(n) := 0$ otherwise. Define $\Psi : \mathcal{P}(\mathbb{N}) \to A$ by $\Psi(E) := \chi_E$.
If $E \subseteq \mathbb{N}$ and $n \in \mathbb{N}$, then $n \in \Phi(\Psi(E)) \iff \chi_E(n) = 1 \iff n \in E$. Hence $\Phi(\Psi(E)) = E$ by Assumption 1.3.2.
If $s \in A$ and $n \in \mathbb{N}$, then $\chi_{\Phi(s)}(n) = 1 \iff s(n) = 1$. Both values lie in $\{0,1\}$, so $\chi_{\Phi(s)}(n) = s(n)$. Thus $(\forall n \in \mathbb{N})\, \chi_{\Phi(s)}(n) = s(n)$.
Hence $\Psi(\Phi(s)) = s$ by Assumption 3.1.2. The maps Φ and Ψ are inverse, so Φ is bijective by Theorem 3.3.2.
Structural Commentary: Characteristic functions construct a two-sided inverse to the subset-extraction map.
Key Insight: A membership bit records exactly one subset decision. □

Theorem 6.4.3. *Binary Sequences Are Uncountable*
Let $A := \{0,1\}^{\mathbb{N}}$. Then A is uncountable.

Proof. Define $j : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ by $j(n) := \{n\}$, and define $\Phi : A \to \mathcal{P}(\mathbb{N})$ by $\Phi(s) := \{n \in \mathbb{N} : s(n) = 1\}$.
The map j is injective because $j(m) = j(n)$ implies $m = n$. The map Φ is bijective by Proposition 6.4.2, so $\Phi^{-1} \circ j : \mathbb{N} \to A$ is injective.
If A were finite, choose $m \in \mathbb{N}_0$ and a bijection $q : A \to J_m$.
Then $q \circ \Phi^{-1} \circ j : \mathbb{N} \to J_m$ would be injective. No such injection exists by Proposition 6.2.5(i), a contradiction.
Thus A is not finite. Suppose A were countable. Choose a bijection $e : \mathbb{N} \to A$, and write $s_n := e(n)$.
Define $s : \mathbb{N} \to \{0,1\}$ by $s(k) := 1 - s_k(k)$. Then $s \in A$ and $(\forall n \in \mathbb{N})\, s(n) \neq s_n(n)$, so $(\forall n \in \mathbb{N})\, s \neq e(n)$.
This contradicts the surjectivity of e. Hence A is neither finite nor countable, so it is uncountable.
Structural Commentary: An embedded copy of $\mathbb{N}$ excludes finiteness; a diagonal flip excludes enumerability.
Key Insight: The nth flipped bit separates the new sequence from row n. □

$$
\begin{array}{lcl}
s_1 & = & \boxed{a_{11}}\, a_{12}\ a_{13}\ a_{14}\ a_{15} \\
s_2 & = & a_{21}\ \boxed{a_{22}}\, a_{23}\ a_{24}\ a_{25} \\
s_3 & = & a_{31}\ a_{32}\ \boxed{a_{33}}\, a_{34}\ a_{35} \quad \cdots \\
s_4 & = & a_{41}\ a_{42}\ a_{43}\ \boxed{a_{44}}\, a_{45} \\
s_5 & = & a_{51}\ a_{52}\ a_{53}\ a_{54}\ \boxed{a_{55}}
\end{array}
$$

Figure 9: Boxed entries $a_{nn} = s_n(n)$ whose bit flips define a sequence distinct from every listed row.

Figure 9 marks the diagonal entries used in Theorem 6.4.3.
Flipping the nth boxed bit forces $s(n) \neq s_n(n)$, so the constructed sequence cannot equal row n.

6.5 Cardinal Comparison and the Continuum

Throughout this subsection, assume Principle 6.1.2 and use the initial-ordinal cardinality $|X|$ from Definition 6.3.8.

Definition 6.5.1. *Cardinal Comparison and Arithmetic*
For sets A, B, define the *function set* by $A^B := \{f \mid f : B \to A\}$. Let κ, λ be cardinals.

(i) *Comparison*: the relations $\kappa \leq \lambda$ and $\kappa < \lambda$ are the corresponding ordinal comparisons.
(ii) The *cardinal sum* is $\kappa + \lambda := |(\{0\} \times \kappa) \cup (\{1\} \times \lambda)|$.
(iii) The *cardinal product* is $\kappa\lambda := |\kappa \times \lambda|$.
(iv) The *cardinal power* is $\kappa^\lambda := |\{f \mid f : \lambda \to \kappa\}|$.

These are cardinal operations. Cardinal arguments distinguish them from the ordinal operations of Definition 6.3.6.

Proposition 6.5.2. *Cardinal Comparison and Operations on Representatives*
Let A, B be sets, and set $\kappa := |A|$ and $\lambda := |B|$.

(i) $\kappa \leq \lambda$ *iff an injection $A \to B$ exists.*
(ii) $\kappa < \lambda$ *iff an injection $A \to B$ exists and $A \not\sim_{\#} B$.*
(iii) $\kappa + \lambda = |(\{0\} \times A) \cup (\{1\} \times B)|$. *If $A \cap B = \emptyset$, then $\kappa + \lambda = |A \cup B|$.*
(iv) $\kappa\lambda = |A \times B|$.
(v) $\kappa^\lambda = |A^B|$.

Proof. Choose bijections $u : A \to \kappa$ and $v : B \to \lambda$ by Proposition 6.3.14(i).

(i) ($\Rightarrow$) If $\kappa \leq \lambda$, then an injection $A \to B$ exists by Proposition 6.3.16.
(i) ($\Leftarrow$) If an injection $A \to B$ exists, then $\kappa \leq \lambda$ by Proposition 6.3.16.
(ii) ($\Rightarrow$) Suppose $\kappa < \lambda$. Then $\kappa \leq \lambda$ and $\kappa \neq \lambda$. An injection $A \to B$ exists by (i), and $A \not\sim_{\#} B$ by Proposition 6.3.14(ii).
(ii) ($\Leftarrow$) Suppose an injection $A \to B$ exists and $A \not\sim_{\#} B$. Then $\kappa \leq \lambda$ by (i) and $\kappa \neq \lambda$ by Proposition 6.3.14(ii), so $\kappa < \lambda$.
(iii) Set $U := (\{0\} \times A) \cup (\{1\} \times B)$ and $V := (\{0\} \times \kappa) \cup (\{1\} \times \lambda)$. Define $F : U \to V$ by $F(0,a) := (0, u(a))$ and $F(1,b) := (1, v(b))$.
The inverse sends $(0,\xi)$ to $(0, u^{-1}(\xi))$ and $(1,\eta)$ to $(1, v^{-1}(\eta))$, so F is bijective by Theorem 3.3.2.
Hence $|U| = |V|$ by Proposition 6.3.14(ii), proving the first equality.
If $A \cap B = \emptyset$, define $G : U \to A \cup B$ by $G(0,a) := a$ and $G(1,b) := b$.
Its inverse sends $x \in A$ to $(0,x)$ and $x \in B$ to $(1,x)$, so G is bijective. Thus $\kappa + \lambda = |A \cup B|$ by Proposition 6.3.14(ii).

(iv) Define $P : A \times B \to \kappa \times \lambda$ by $P(a,b) := (u(a), v(b))$. Its inverse is $(\xi,\eta) \mapsto (u^{-1}(\xi), v^{-1}(\eta))$, so P is bijective by Theorem 3.3.2. Therefore $\kappa\lambda = |A \times B|$ by Proposition 6.3.14(ii).
(v) Define $T : A^B \to \{g \mid g : \lambda \to \kappa\}$ by $T(f) := u \circ f \circ v^{-1}$. Its inverse sends $g : \lambda \to \kappa$ to $u^{-1} \circ g \circ v$, so T is bijective by Theorem 3.3.2.
Hence $\kappa^\lambda = \left|A^B\right|$ by Proposition 6.3.14(ii).

Structural Commentary: Chosen bijections transport comparison and all three cardinal operations.
Key Insight: Bijections act coordinatewise on tags, pairs, and functions. □

Definition 6.5.3. *Continuum Cardinal*
The *continuum cardinal* is $\mathfrak{c} := 2^{\aleph_0}$.

Proposition 6.5.4. *Models of the Continuum Cardinal*
$\mathfrak{c} = \left|\{0,1\}^{\mathbb{N}}\right| = |\mathcal{P}(\mathbb{N})|$.

Proof. The map $q : J_2 \to \{0,1\}$ defined by $q(k) := k - 1$ is bijective. Thus $|\{0,1\}| = 2$ and $|\mathbb{N}| = \aleph_0$ by Proposition 6.3.15(i),(iii).
Hence $\mathfrak{c} = \left|\{0,1\}^{\mathbb{N}}\right|$ by Proposition 6.5.2(v). The relation $\{0,1\}^{\mathbb{N}} \sim_{\#} \mathcal{P}(\mathbb{N})$ holds by Proposition 6.4.2.
Hence $\left|\{0,1\}^{\mathbb{N}}\right| = |\mathcal{P}(\mathbb{N})|$ by Proposition 6.3.14(ii).
Structural Commentary: Finite and countable representatives reduce cardinal exponentiation to binary functions and subsets.
Key Insight: A binary function is the characteristic function of one subset. □

Corollary 6.5.5. *The Continuum Strictly Exceeds $\aleph_0$*
$\aleph_0 < \mathfrak{c}$.

Proof. The map $j : \mathbb{N} \to \mathcal{P}(\mathbb{N})$, $j(n) := \{n\}$, is injective since $j(m) = j(n) \Rightarrow m = n$.
No bijection $\mathbb{N} \to \mathcal{P}(\mathbb{N})$ exists by Theorem 6.4.1. Therefore $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$ by Proposition 6.5.2(ii).
The identities $|\mathbb{N}| = \aleph_0$ and $|\mathcal{P}(\mathbb{N})| = \mathfrak{c}$ hold by Proposition 6.3.15(iii) and Proposition 6.5.4, respectively. Hence $\aleph_0 < \mathfrak{c}$.
Structural Commentary: Singleton embedding gives weak comparison, while Cantor's theorem rules out equality.
Key Insight: The diagonal obstruction makes the continuum strictly larger. □

6.6 Cardinal Arithmetic

Throughout this subsection, assume Principle 6.1.2 and use the cardinal operations from Definition 6.5.1.

Proposition 6.6.1. *Cardinal Arithmetic Reference Laws*
Let κ, λ, μ be cardinals.
(i) $\kappa + \lambda = \lambda + \kappa$ *and* $\kappa\lambda = \lambda\kappa$.
(ii) $\kappa + (\lambda + \mu) = (\kappa + \lambda) + \mu$ *and* $\kappa(\lambda\mu) = (\kappa\lambda)\mu$.
(iii) $\kappa(\lambda + \mu) = \kappa\lambda + \kappa\mu$.
(iv) $\kappa^{\lambda+\mu} = \kappa^\lambda \kappa^\mu$, $(\kappa^\lambda)^\mu = \kappa^{\lambda\mu}$, *and* $(\kappa\lambda)^\mu = \kappa^\mu \lambda^\mu$.

Proof. Set $A := \{0\} \times \kappa$, $B := \{1\} \times \lambda$, and $C := \{2\} \times \mu$.
The sets are pairwise disjoint, and their coordinate projections are bijections onto κ, λ, μ, respectively.
Thus $|A| = \kappa$, $|B| = \lambda$, and $|C| = \mu$ by Proposition 6.3.14(ii) and Definition 6.3.8.
(i) The identity is a bijection $A \cup B \to B \cup A$. The map $(a,b) \mapsto (b,a)$ is a bijection $A \times B \to B \times A$ with itself as inverse.
(ii) The equality $A \cup (B \cup C) = (A \cup B) \cup C$ holds.
The map $(a,(b,c)) \mapsto ((a,b),c)$ is a bijection $A \times (B \times C) \to (A \times B) \times C$ with inverse $((a,b),c) \mapsto (a,(b,c))$.
(iii) The equality $A \times (B \cup C) = (A \times B) \cup (A \times C)$ holds. The union on the right is disjoint because $B \cap C = \emptyset$.
(iv) Define $R : A^{B \cup C} \to A^B \times A^C$ by $R(f) := (f|_B, f|_C)$. Its inverse is $R^{-1}(g,h) := g \cup h$; disjointness makes this union a function.
Define $Q : (A^B)^C \to A^{B \times C}$ by $Q(F)(b,c) := F(c)(b)$. Its inverse sends h to $c \mapsto (b \mapsto h(b,c))$. Let $\pi_A(a,b) := a$ and $\pi_B(a,b) := b$.
The map $D : (A \times B)^C \to A^C \times B^C$ defined by $D(h) := (\pi_A \circ h, \pi_B \circ h)$ has inverse $D^{-1}(f,g)(c) := (f(c), g(c))$.
Claims (i)–(iv) follow from these set equalities and bijections by Proposition 6.5.2(iii)–(v).
Structural Commentary: Set identities and explicit bijections transport representative arithmetic to cardinal laws.
Key Insight: Restriction, currying, and projections control the exponent laws. □

Theorem 6.6.2. *Infinite Cardinal Square Law*
If κ is an infinite cardinal, then $|\kappa \times \kappa| = \kappa$.

Proof.
(i) *Induction setup*: For an ordinal α, let $P(\alpha)$ mean that, if α is an infinite cardinal, then $|\alpha \times \alpha| = \alpha$.
Fix an ordinal κ and assume $(\forall \beta < \kappa)\, P(\beta)$. It suffices by Principle 6.3.2(ii) to prove $P(\kappa)$.
The assertion $P(\kappa)$ is vacuous unless κ is an infinite cardinal, so assume that it is.
(ii) *Well-order construction*: Define $r : \kappa \times \kappa \to \kappa$ by $r(\xi,\eta) := \max\{\xi,\eta\}$.
For $p = (\beta,\beta')$ and $q = (\gamma,\gamma')$ in $\kappa \times \kappa$, declare $p \prec q$ when the first applicable condition holds:
  (a) $r(p) < r(q)$;
  (b) $r(p) = r(q)$ and $\beta < \gamma$;
  (c) $r(p) = r(q)$, $\beta = \gamma$, and $\beta' < \gamma'$.

Put $p \preceq q$ iff $p = q$ or $p \prec q$. Ordinal comparison makes these clauses exhaustive for distinct pairs.
Comparison by the first differing entry of $(r(p), \beta, \beta')$ is transitive and antisymmetric; hence $\preceq$ is a total order.
If $\emptyset \neq S \subseteq \kappa \times \kappa$, set $R_S := \{r(p) : p \in S\}$. The nonempty set R_S has a least element ρ by Principle 6.3.2(i).
Set $I_S := \{\xi < \kappa : (\exists \eta < \kappa)\,[(\xi,\eta) \in S \wedge r(\xi,\eta) = \rho]\}$; it is nonempty by the definition of ρ.
The minimum $\xi_0 := \min I_S$ exists by Principle 6.3.2(i).
Set $J_S := \{\eta < \kappa : (\xi_0,\eta) \in S \wedge r(\xi_0,\eta) = \rho\}$; it is nonempty by the definition of I_S.
The minimum $\eta_0 := \min J_S$ exists by Principle 6.3.2(i). Then (ξ_0,η_0) is least in S, so $\preceq$ is a well-order.
(iii) *Ordinal comparison*: Choose an ordinal δ and an order isomorphism $f : (\delta, \leq) \to (\kappa \times \kappa, \preceq)$ by Principle 6.3.2(iii).
The map $\xi \mapsto (\xi, 0)$ injects κ into $\kappa \times \kappa$. Since $|\kappa| = \kappa$ by Definition 6.3.8, $\kappa \leq |\kappa \times \kappa|$ by Proposition 6.3.16.
The equality $|\kappa \times \kappa| = |\delta|$ holds by Proposition 6.3.14(ii), and $|\delta| \leq \delta$ by Definition 6.3.8. Thus $\kappa \leq |\kappa \times \kappa| = |\delta| \leq \delta$.

(iv) *Initial-segment contradiction*: Suppose $\kappa < \delta$. Write $f(\kappa) = (\beta, \beta')$ and set $X := \{q \in \kappa \times \kappa : q \prec (\beta, \beta')\}$.
The restriction $f|_\kappa : \kappa \to X$ is bijective because f preserves and reflects order.
Hence $|X| = \kappa$ by Proposition 6.3.14(ii) and Definition 6.3.8.
Set $\alpha := \max\{\beta, \beta'\} + 1$. Since $\beta, \beta' < \kappa$ and κ is a limit ordinal by Lemma 6.3.17, $\alpha < \kappa$.
Set $\lambda := |\alpha|$. Then λ is a cardinal, and a bijection $u : \alpha \to \lambda$ exists by Proposition 6.3.14(i).
Since $\lambda \leq \alpha$ by Definition 6.3.8, $\lambda < \kappa$. If $q \in X$, then $r(q) \leq \max\{\beta, \beta'\} < \alpha$, so $q \in \alpha \times \alpha$.
The coordinate map $(\xi, \eta) \mapsto (u(\xi), u(\eta))$ is a bijection $\alpha \times \alpha \to \lambda \times \lambda$, whose restriction to X is injective.
Thus $\kappa = |X| \leq |\lambda \times \lambda|$ by Proposition 6.3.16. If λ is finite, then $|\lambda \times \lambda| < \aleph_0 \leq \kappa$ by Theorem 4.2.5 and Proposition 6.3.15.
If λ is infinite, then $|\lambda \times \lambda| = \lambda < \kappa$ by induction hypothesis. In either case, $|\lambda \times \lambda| < \kappa$, contradicting $\kappa \leq |\lambda \times \lambda|$; hence $\kappa \not< \delta$.

(v) *Conclusion*: Since $\kappa \leq \delta$, $\delta = \kappa$ by Principle 6.3.2(i). Thus f is a bijection $\kappa \to \kappa \times \kappa$, so $|\kappa \times \kappa| = \kappa$ by Proposition 6.3.14(ii) and Definition 6.3.8. Hence $P(\kappa)$ holds, completing the induction.

Structural Commentary: Strong induction bounds max-ranked initial segments by smaller cardinal squares.
Key Insight: Every predecessor lies in a square of smaller cardinality. □

Theorem 6.6.3. *Infinite Cardinal Absorption*
Let κ, λ be cardinals, and let $\mu := \max\{\kappa, \lambda\}$ be infinite.

(i) $\kappa + \lambda = \mu$.
(ii) If $\kappa, \lambda > 0$, *then* $\kappa\lambda = \mu$.
(iii) $\aleph_0 + \aleph_0 = \aleph_0\aleph_0 = \aleph_0$.

Proof. The equality $|\mu \times \mu| = \mu$ holds by Theorem 6.6.2. Set $U := (\{0\} \times \mu) \cup (\{1\} \times \mu)$.
Since $0, 1 \in \mu$, the maps $\xi \mapsto (0, \xi)$ and $(i, \xi) \mapsto (\xi, i)$ inject μ into U and U into $\mu \times \mu$, respectively.
The identities $|U| = \mu + \mu$ and $|\mu| = \mu$ hold by Definition 6.5.1 and Definition 6.3.8, respectively.
Thus $\mu \leq \mu + \mu \leq |\mu \times \mu| = \mu$ by Proposition 6.3.16, so $\mu + \mu = \mu$.

(i) Set $V := (\{0\} \times \kappa) \cup (\{1\} \times \lambda)$. If $\mu = \kappa$, then $\xi \mapsto (0, \xi)$ injects μ into V; if $\mu = \lambda$, then $\xi \mapsto (1, \xi)$ does so.
The inclusions $\kappa \hookrightarrow \mu$ and $\lambda \hookrightarrow \mu$ induce an injection $V \to U$, and $|V| = \kappa + \lambda$ by Definition 6.5.1.
Thus $\mu \leq \kappa + \lambda \leq \mu + \mu = \mu$ by Proposition 6.3.16, so $\kappa + \lambda = \mu$.

(ii) Assume $\kappa, \lambda > 0$, so $0 \in \kappa \cap \lambda$. If $\mu = \kappa$, then $\xi \mapsto (\xi, 0)$ injects μ into $\kappa \times \lambda$; if $\mu = \lambda$, then $\xi \mapsto (0, \xi)$ does so.
The coordinate inclusions induce an injection $\kappa \times \lambda \to \mu \times \mu$, and $\kappa\lambda = |\kappa \times \lambda|$ by Definition 6.5.1.
Thus $\mu \leq \kappa\lambda \leq |\mu \times \mu| = \mu$ by Proposition 6.3.16, so $\kappa\lambda = \mu$.

(iii) The cardinal $\aleph_0$ is infinite by Proposition 6.3.15(iv). The asserted equalities follow by (i),(ii) with $\kappa = \lambda = \aleph_0$.

Structural Commentary: The square law sandwiches each sum and nonzero product between μ and itself.
Key Insight: An infinite maximum absorbs every bounded factor. □

7 Axiomatic and Advanced Set Theory

This section reserves space for deeper axiomatic and metatheoretic material: formal set axioms, models, forcing, large cardinals, and sets of reals.

7.1 Extensionality, Specification, and Replacement

Variables range over sets; formulas use first-order logic with equality and the sole nonlogical symbol $\in$.
For y' fresh in $\varphi(y)$, use $(\exists! y)\,\varphi(y) := (\exists y)[\varphi(y) \wedge (\forall y')(\varphi(y') \Rightarrow y' = y)]$.

Assumption 7.1.1. *Axiom of Extensionality*
Assume $(\forall A)(\forall B)[(\forall x)(x \in A \iff x \in B) \Rightarrow A = B]$.
The converse is equality substitution, so this axiom is the formal source of Assumption 1.3.2.

Assumption 7.1.2. *Separation/Specification Schema*
Fix $n \in \mathbb{N}_0$, put $\mathbf{z} = (z_1, \ldots, z_n)$, and let $\varphi(x, \mathbf{z})$ have no other free variables.
Require B not free in φ; the associated axiom is the universal closure in $A, \mathbf{z}$ of the next formula. $(\exists B)(\forall x)[x \in B \iff (x \in A \wedge \varphi(x, \mathbf{z}))]$.
In set-builder notation, its witness is $B = \{x \in A : \varphi(x, \mathbf{z})\} \subseteq A$.

Assumption 7.1.3. *Replacement Schema*
Fix $n \in \mathbb{N}_0$, put $\mathbf{z} = (z_1, \ldots, z_n)$, and let $\varphi(x, y, \mathbf{z})$ have no other free variables.
Require B not free in φ; the associated axiom is the universal closure in $A, \mathbf{z}$ of the next implication.
$[(\forall x \in A)(\exists! y)\,\varphi(x, y, \mathbf{z})] \Rightarrow (\exists B)(\forall y)[y \in B \iff (\exists x \in A)\,\varphi(x, y, \mathbf{z})]$.
The antecedent makes φ functional on A; the consequent collects its full range into a set.

Perspective 7.1.4. *Selection versus Functional Image*
Assumption 7.1.2 selects elements of a bounding set A, so its witness satisfies $B \subseteq A$.
Assumption 7.1.3 collects the range of a definable function on A, whose values may lie outside A.
An axiom schema comprises one first-order axiom per permitted formula and finite parameter list.

7.2 Models, Constructibility, and Independence

Fix a classical proof calculus with assumption discharge and double-negation elimination.
Use a ZFC metatheory formalizing first-order syntax, satisfaction, and the required set constructions.
Write Ord for the class of ordinals. The theories ZF, ZFC, and their extensions are *object theories* interpreted in models.
No claim below assumes that the metatheory proves its own consistency.

Definition 7.2.1. *Models and Inner Models of Set Theory*
Fix an object theory T in the first-order language of membership.
Let $\mathcal{M} = (M, E)$ satisfy $\emptyset \neq M$ and $E \subseteq M \times M$.

(i) The structure $\mathcal{M}$ is a *set model* of T iff $(\forall \sigma \in T)\,\mathcal{M} \models \sigma$.
(ii) It is a *transitive set model* of T iff it is a set model, $(\forall x \in M)\,x \subseteq M$, and $E = \{(x, y) \in M \times M : x \in y\}$.
(iii) A class N is an *inner model* of T iff $(\forall x \in N)\,x \subseteq N$, $\mathrm{Ord} \subseteq N$, and $(N, \in) \models T$.

Countability of a set model is an external metatheoretic property.
Thus it can regard one of its sets as uncountable when the relevant bijection is external.

Definition 7.2.2. *Relative Consistency, Consistency Strength, and Independence*
Let T, U be object theories in the fixed proof calculus; for a sentence φ, set $T + \varphi := T \cup \{\varphi\}$.

(i) $\mathrm{Con}(T)$ abbreviates $T \nvdash \bot$.
(ii) A *relative-consistency result from T to U* is a metatheoretic proof of $\mathrm{Con}(T) \Longrightarrow \mathrm{Con}(U)$.
(iii) The theories T and U are *equiconsistent* iff $[\mathrm{Con}(T) \Longrightarrow \mathrm{Con}(U)] \wedge [\mathrm{Con}(U) \Longrightarrow \mathrm{Con}(T)]$ is metatheoretically provable.
(iv) The theory U has at least the *consistency strength* of T iff the metatheory proves $\mathrm{Con}(U) \Longrightarrow \mathrm{Con}(T)$.
(v) A sentence φ is *independent of T* iff $T \nvdash \varphi$ and $T \nvdash \neg\varphi$.

Relative consistency does not assert its antecedent; independence selects no ambient truth.
The sentence CH asserts $2^{\aleph_0} = \aleph_1$. For a cardinal κ, let κ^+ be the least cardinal strictly greater than κ.
The sentence GCH asserts $(\forall \kappa \geq \aleph_0)\, 2^\kappa = \kappa^+$, with the quantifier restricted to cardinals.

Proposition 7.2.3. *Independence via Two Consistent Extensions*
Let T be a classical object theory and let φ be a sentence.
Then φ is independent of T iff $\mathrm{Con}(T + \varphi) \wedge \mathrm{Con}(T + \neg\varphi)$.

Proof.

($\Rightarrow$) Assume $T \nvdash \varphi$ and $T \nvdash \neg\varphi$. By assumption discharge, $T + \varphi \vdash \bot \Longrightarrow T \vdash \neg\varphi$.
By assumption discharge and double-negation elimination, $T + \neg\varphi \vdash \bot \Longrightarrow T \vdash \neg\neg\varphi \Longrightarrow T \vdash \varphi$.
Both consequents contradict the hypothesis, so $\mathrm{Con}(T + \varphi)$ and $\mathrm{Con}(T + \neg\varphi)$.
($\Leftarrow$) Assume $\mathrm{Con}(T + \varphi)$ and $\mathrm{Con}(T + \neg\varphi)$. Then $T \vdash \varphi \Longrightarrow T + \neg\varphi \vdash \bot$ and $T \vdash \neg\varphi \Longrightarrow T + \varphi \vdash \bot$.
Thus $T \nvdash \varphi$ and $T \nvdash \neg\varphi$, so φ is independent of T.

Structural Commentary: The two directions translate extension consistency into the two nonprovability obligations.
Key Insight: Opposite consistent extensions encode both nonprovability conditions. □

Definition 7.2.4. *Constructible Hierarchy*
For a set M, fix $n \in \mathbb{N}_0$ and $\mathbf{p} = (p_1, \ldots, p_n) \in M^n$. Let $\varphi(x, z_1, \ldots, z_n)$ be a formula with no other free variables.
For these data, set $A := \{x \in M : (M, \in) \models \varphi(x, p_1, \ldots, p_n)\}$. The family $\mathrm{Def}(M)$ consists exactly of these sets as the data vary.
Transfinite recursion sets $L_0 := \emptyset$ and $L_{\alpha+1} := \mathrm{Def}(L_\alpha)$. At a nonzero limit ordinal λ, set $L_\lambda := \bigcup_{\beta<\lambda} L_\beta$.
The *constructible universe* is $L := \bigcup_{\alpha \in \mathrm{Ord}} L_\alpha$; a set is *constructible* iff it belongs to L.

Proposition 7.2.5. *Basic Properties of the Constructible Hierarchy*
For ordinals $\alpha \leq \beta$:

(i) L_α is transitive;
(ii) $L_\alpha \subseteq L_\beta$;
(iii) L is transitive;
(iv) $L_\alpha \cap \mathrm{Ord} = \alpha$;
(v) $\mathrm{Ord} \subseteq L$.

Proof.

(i) Set $\mathrm{Trans}(M) := (\forall x \in M)\, x \subseteq M$ and $P(\gamma) := [\mathrm{Trans}(L_\gamma) \wedge L_\gamma \subseteq L_{\gamma+1}]$. To prove $(\forall \gamma \in \mathrm{Ord})\, P(\gamma)$ by transfinite induction.
Since $L_0 = \emptyset$ is transitive and $L_0 \subseteq L_1$, the assertion $P(0)$ holds. Assume $P(\gamma)$.
If $x \in y \in L_{\gamma+1}$, then $y \subseteq L_\gamma$ and $x \in L_\gamma \subseteq L_{\gamma+1}$; hence $L_{\gamma+1}$ is transitive.
For $x \in L_{\gamma+1}$, the parameter x and formula $z \in x$ define x over $L_{\gamma+1}$. Thus $x \in L_{\gamma+2}$ and $P(\gamma+1)$ holds.
At a nonzero limit ordinal λ, assume $(\forall \gamma < \lambda)\, P(\gamma)$. If $x \in y \in L_\lambda$, choose $\gamma < \lambda$ with $y \in L_\gamma$.
Then $x \in L_\gamma \subseteq L_\lambda$ by $P(\gamma)$, so L_λ is transitive. For $x \in L_\lambda$, the parameter x and formula $z \in x$ define x over L_λ.
Thus $L_\lambda \subseteq L_{\lambda+1}$ and $P(\lambda)$ holds. Thus $(\forall \alpha \in \mathrm{Ord})\ \mathrm{Trans}(L_\alpha)$.
(ii) Fix α and prove $(\forall \beta \in \mathrm{Ord})\, [\alpha \leq \beta \Rightarrow L_\alpha \subseteq L_\beta]$ by transfinite induction.
At $\beta = 0$, the premise $\alpha \leq 0$ gives $\alpha = 0$, so the inclusion is equality. If $\beta = \gamma + 1$ and $\alpha \leq \beta$, then either $\alpha = \beta$ or $\alpha \leq \gamma$.
In the latter case, $L_\alpha \subseteq L_\gamma \subseteq L_{\gamma+1}$ by the induction hypothesis and the simultaneous induction in (i).
At a nonzero limit β, the case $\alpha = \beta$ is trivial.
If $\alpha < \beta$, then $\alpha + 1 < \beta$ and $L_\alpha \subseteq L_{\alpha+1} \subseteq L_\beta$ by the simultaneous induction in (i) and the definition of L_β.
(iii) If $x \in y \in L$, choose $\gamma \in \mathrm{Ord}$ with $y \in L_\gamma$. Then $x \in L_\gamma \subseteq L$ by (i), so L is transitive.
(iv) Set $Q(\gamma) := [L_\gamma \cap \mathrm{Ord} = \gamma]$ and proceed by transfinite induction. At 0, $L_0 \cap \mathrm{Ord} = 0$.
At a nonzero limit ordinal λ, $L_\lambda \cap \mathrm{Ord} = \bigcup_{\xi<\lambda} (L_\xi \cap \mathrm{Ord}) = \bigcup_{\xi<\lambda} \xi = \lambda$ by the induction hypothesis.
Assume $Q(\gamma)$. If $\delta \in L_{\gamma+1} \cap \mathrm{Ord}$, then $\delta \subseteq L_\gamma \cap \mathrm{Ord} = \gamma$, so $\delta \leq \gamma$. Conversely, $\delta < \gamma \Longrightarrow \delta \in \gamma = L_\gamma \cap \mathrm{Ord} \subseteq L_{\gamma+1}$.
Ordinalhood is bounded and defines $\gamma = L_\gamma \cap \mathrm{Ord}$ over the transitive set L_γ; hence $\gamma \in L_{\gamma+1}$.
Therefore $L_{\gamma+1} \cap \mathrm{Ord} = \gamma + 1$, which proves $Q(\gamma + 1)$.
(v) $(\forall \alpha \in \mathrm{Ord})\, \alpha \in L_{\alpha+1}$ by (iv); hence $\mathrm{Ord} \subseteq L$.

Structural Commentary: Simultaneous induction gives transitivity and nesting; a second induction identifies the stage ordinals.
Key Insight: Definability carries each stage and its index forward. □

Principle 7.2.6. *Constructible Inner Model Result*

(i) The theory ZF proves that L is an inner model of ZFC + GCH.
(ii) The metatheory proves $\mathrm{Con}(\mathrm{ZF}) \Longrightarrow \mathrm{Con}(\mathrm{ZFC} + \mathrm{GCH})$.
(iii) The reverse consistency implication holds because ZFC + GCH contains ZF; hence the theories are equiconsistent.

Reflection and condensation are not developed here, so this principle records framework and supplies no proof authority.

Definition 7.2.7. *Forcing Posets, Generic Filters, and Extensions*
Let M be a transitive set. Let $\mathbb{P} = (P, \leq, 1_\mathbb{P}) \in M$ be a partial order whose greatest element is $1_\mathbb{P}$.
The element $1_\mathbb{P}$ is the weakest condition, and $p \leq q$ means that p is stronger than q.

(i) A set D is *dense in P*, written $\mathrm{Dense}_P(D)$, iff $D \subseteq P$ and $(\forall p \in P)(\exists q \in D)\, q \leq p$.

(ii) A *filter on P* is a set $\emptyset \neq G \subseteq P$ satisfying $(\forall p \in G)(\forall q \in P)\,[p \leq q \Rightarrow q \in G]$.
Its directedness condition is $(\forall p, q \in G)(\exists r \in G)\,[r \leq p \wedge r \leq q]$.
(iii) The filter G is *M-generic* iff $(\forall D \in M)\,[\mathrm{Dense}_P(D) \Rightarrow G \cap D \neq \emptyset]$.
(iv) Transfinite recursion defines $\mathcal{N}_0^P := \emptyset$ and $\mathcal{N}_{\alpha+1}^P := \mathcal{P}(\mathcal{N}_\alpha^P \times P)$.
At a nonzero limit ordinal λ, set $\mathcal{N}_\lambda^P := \bigcup_{\beta<\lambda} \mathcal{N}_\beta^P$. Set $\mathcal{N}^P := \bigcup_{\alpha \in \mathrm{Ord}} \mathcal{N}_\alpha^P$; its elements are the *$P$-names*.
(v) For a filter G and $\tau \in \mathcal{N}^P$, recursion on name rank defines $\tau^G := \{\sigma^G : (\exists p \in G)\,[(\sigma, p) \in \tau]\}$.
(vi) For an M-generic filter G, the *forcing extension* is $M[G] := \{\tau^G : \tau \in M \cap \mathcal{N}^P\}$.

Theorem 7.2.8. *Existence of Generic Filters over Countable Transitive Sets*
In the metatheory, let M be a countable transitive set. Let $\mathbb{P} = (P, \leq, 1_{\mathbb{P}}) \in M$ be a partial order, and fix $p \in P$. Then an M-generic filter $G \subseteq P$ satisfying $p \in G$ exists.

Proof.
(i) *Enumeration*: External countability gives $(D_n)_{n<\omega}$ enumerating $\{D \in M : \mathrm{Dense}_P(D)\}$. Repetitions are allowed; set $p_0 := p$.
(ii) *Descending construction*: Recursively choose $p_{n+1} \in D_n$ with $p_{n+1} \leq p_n$ by the density of D_n.
(iii) *Filter verification*: Set $G := \{q \in P : (\exists n < \omega)\, p_n \leq q\}$. Then $p = p_0 \in G$.
If $q \in G$, $q \leq r$, and $r \in P$, choose $n < \omega$ with $p_n \leq q$; transitivity gives $p_n \leq r$, so $r \in G$.
If $q_1, q_2 \in G$, choose $i, j < \omega$ with $p_i \leq q_1$ and $p_j \leq q_2$.
Set $k := \max\{i, j\}$. Then $p_k \in G$, $p_k \leq q_1$, and $p_k \leq q_2$ by the descending construction. Thus G is a filter on P.
(iv) *Genericity*: The construction gives $(\forall n < \omega)\, p_{n+1} \in G \cap D_n$.
If $D \in M$ and $\mathrm{Dense}_P(D)$, then $(\exists n < \omega)\, D = D_n$; hence $G \cap D \neq \emptyset$. Therefore G is M-generic and contains p.

Structural Commentary: A descending chain turns enumerated dense sets into one filter.
Key Insight: Later conditions strengthen all earlier choices. □

Principle 7.2.9. *Forcing Extension Framework*
Let M be a countable transitive model of ZFC. Let $\mathbb{P} = (P, \leq, 1_{\mathbb{P}}) \in M$, and let $G \subseteq P$ be M-generic.
The forcing theorem has the following conclusions.

(i) $M \subseteq M[G]$.
(ii) $M \cap \mathrm{Ord} = M[G] \cap \mathrm{Ord}$.
(iii) $M[G] \models \mathrm{ZFC}$.

Its proof recursively defines the forcing relation on formulas and names before proving the Truth Lemma.
That machinery is not developed here, so this principle supplies no proof authority.

Principle 7.2.10. *Relative Consistency and Independence of the Continuum Hypothesis*
The constructible-inner-model and forcing constructions establish the following relative-consistency implications.

(i) $\mathrm{Con}(\mathrm{ZFC}) \Longrightarrow \mathrm{Con}(\mathrm{ZFC} + \mathrm{CH})$.
(ii) $\mathrm{Con}(\mathrm{ZFC}) \Longrightarrow \mathrm{Con}(\mathrm{ZFC} + \neg\mathrm{CH})$.

Under Con(ZFC), the sentence CH is independent of ZFC by Proposition 7.2.3.
Each extension contains ZFC, so both are equiconsistent with ZFC. The constructible universe supplies (i).
For (ii), conditions are finite partial functions from $\omega \times \omega_2$ to 2.
Reverse inclusion orders them, and the resulting poset has the countable chain condition.
It preserves ω_1 and ω_2 and adds ω_2 distinct subsets of ω.
The relative-consistency proof is syntactic, or equivalently Boolean-valued. It does not assume a transitive model of all ZFC.
The required constructibility and forcing theorems are not proved here, so this principle supplies no proof authority.

7.3 Baire and Cantor Spaces

Definition 7.3.1. *Sequence Spaces and Cylinders*
Let $A \in \{2, \omega\}$, where $2 = \{0, 1\}$.

(i) For $n < \omega$, let A^n be the set of functions $n \to A$, and let A^ω be the set of functions $\omega \to A$. Set $A^{<\omega} := \bigcup_{n<\omega} A^n$.
(ii) For $s \in A^{<\omega}$, its *length* is $|s| := \mathrm{dom}\, s$. For $u \in A^{<\omega} \cup A^\omega$, write $s \sqsubseteq u$ iff $s = u|_{|s|}$; then s is an *initial segment of u*.
(iii) Strings $s, t \in A^{<\omega}$ are *compatible* iff $s \sqsubseteq t$ or $t \sqsubseteq s$; otherwise they are *incompatible*.
The *basic cylinder* determined by s is $N_s^A := \{x \in A^\omega : s \sqsubseteq x\}$.
(iv) A set $U \subseteq A^\omega$ is *open* iff it is a union of basic cylinders. It is *closed* iff $A^\omega \setminus U$ is open and *clopen* iff it is both open and closed.
(v) The spaces 2^ω and ω^ω are *Cantor space* and *Baire space*, respectively. Write N_s for N_s^A when A is fixed.

Proposition 7.3.2. *Basic Cylinders Form a Clopen Basis*
Let $A \in \{2, \omega\}$, let $s, t \in A^{<\omega}$, and let $U \subseteq A^\omega$ be open.

(i) If $s \sqsubseteq t$, then $N_s^A \cap N_t^A = N_t^A$. If $t \sqsubseteq s$, then $N_s^A \cap N_t^A = N_s^A$. If s, t are incompatible, then $N_s^A \cap N_t^A = \emptyset$.
(ii) The cylinder N_s^A is clopen.
(iii) $(\forall x \in U)(\exists n < \omega)\, x \in N_{x|_n}^A \subseteq U$.

Proof.
(i) If $s \sqsubseteq t$, then $(\forall x \in N_t^A)\, s = t|_{|s|} = x|_{|s|}$; hence $N_t^A \subseteq N_s^A$ and $N_s^A \cap N_t^A = N_t^A$.
If $t \sqsubseteq s$, then $(\forall x \in N_s^A)\, t = s|_{|t|} = x|_{|t|}$; hence $N_s^A \subseteq N_t^A$ and $N_s^A \cap N_t^A = N_s^A$.
If $x \in N_s^A \cap N_t^A$, then s, t are finite initial segments of x and are comparable by length.
Thus incompatible s, t have disjoint cylinders.
(ii) The cylinder N_s^A is open because it is a union of itself. Fix $x \notin N_s^A$. Since $s \neq x|_{|s|}$, choose $j < |s|$ with $x(j) \neq s(j)$.
If $y \in N_{x|_{j+1}}^A$, then $y(j) = x(j) \neq s(j)$. Therefore $N_{x|_{j+1}}^A \subseteq A^\omega \setminus N_s^A$.
The complement is a union of cylinders and is open, so N_s^A is clopen.
(iii) Fix $x \in U$. Since U is a union of cylinders, choose $r \in A^{<\omega}$ with $x \in N_r^A \subseteq U$. Then $r = x|_{|r|}$, so $n := |r|$ has required property.

Structural Commentary: Comparability controls intersections; first disagreements construct complementary cylinders.
Key Insight: Finite prefixes determine all basic neighborhoods. □

Definition 7.3.3. *Prefix Metric and Polish Space*
Let $A \in \{2, \omega\}$.

(i) For distinct $x, y \in A^\omega$, set $\Delta_A(x,y) := \min\{n < \omega : x(n) \neq y(n)\}$. Define $d_A(x,x) := 0$ and $d_A(x,y) := 2^{-(\Delta_A(x,y)+1)}$ for $x \neq y$.
(ii) A metric d on X is an *ultrametric* iff $(\forall x, y, z \in X)\, d(x,z) \leq \max\{d(x,y), d(y,z)\}$.
(iii) A sequence $(x_k)_{k<\omega} \in X^\omega$ is *d-Cauchy* iff $(\forall \varepsilon > 0)(\exists K < \omega)(\forall k, \ell \geq K)\, d(x_k, x_\ell) < \varepsilon$.
It *converges to $x \in X$* iff $(\forall \varepsilon > 0)(\exists K < \omega)(\forall k \geq K)\, d(x_k, x) < \varepsilon$.
The metric d is *complete* iff each d-Cauchy sequence in X converges to a point of X.
(iv) A set $D \subseteq A^\omega$ is *dense* iff $(\forall s \in A^{<\omega})\, D \cap N_s^A \neq \emptyset$.
(v) A topological space is *Polish* iff its topology is induced by some complete metric and it has a countable dense subset.

Theorem 7.3.4. *Baire and Cantor Spaces Are Polish*
For $A \in \{2, \omega\}$, set $E_A := \{x \in A^\omega : (\exists n < \omega)(\forall k \in \omega \setminus n)\, x(k) = 0\}$.

(i) The prefix metric d_A is an ultrametric.
(ii) The metric d_A is complete.
(iii) The metric d_A induces the cylinder topology on A^ω.
(iv) The eventually-zero set E_A is countable.
(v) The set E_A is dense in A^ω.
(vi) The space A^ω is Polish.

Proof.

(i) Symmetry of the disagreement set makes d_A symmetric. By definition, $(\forall x, y \in A^\omega)\, [d_A(x,y) \geq 0 \wedge (d_A(x,y) = 0 \iff x = y)]$.
Fix $x, y, z \in A^\omega$ and set $r := \max\{d_A(x,y), d_A(y,z)\}$. If $r = 0$, then $x = y = z$. If $r > 0$, choose $m < \omega$ with $r = 2^{-(m+1)}$.
Both pairs x, y and y, z agree below m, so $d_A(x,z) \leq r$. Thus the strong triangle inequality holds.
The ordinary triangle inequality follows from $r \leq d_A(x,y) + d_A(y,z)$, so d_A is an ultrametric.
(ii) Let $(x_k)_{k<\omega}$ be d_A-Cauchy. For $m < \omega$, choose $K_m < \omega$ such that $(\forall k, \ell \geq K_m)\, d_A(x_k, x_\ell) < 2^{-m}$.
Set $s_m := x_{K_m}|_m \in A^m$. If $k \geq K_m$, then $x_k|_m = s_m$. If $m \leq n$, comparison at an index $k \geq \max\{K_m, K_n\}$ gives $s_m = s_n|_m$.
Hence $x := \bigcup_{m<\omega} s_m$ is a function $\omega \to A$. Given $\varepsilon > 0$, choose $m < \omega$ with $2^{-m} < \varepsilon$.
For $k \geq K_m$, the sequences x_k, x agree below m, so $d_A(x_k, x) < 2^{-m} < \varepsilon$. Thus $x_k \to x$, and d_A is complete.
(iii) For $x \in A^\omega$ and $m < \omega$, $N^A_{x|_m} = \{y \in A^\omega : d_A(x,y) < 2^{-m}\}$.
Both sides require agreement below m, so each cylinder containing x is a metric ball.
Conversely, given $\varepsilon > 0$, choose $m < \omega$ with $2^{-m} < \varepsilon$.
Then $N^A_{x|_m} \subseteq \{y \in A^\omega : d_A(x,y) < \varepsilon\}$, so the metric and cylinder topologies coincide.
(iv) The map $k \mapsto k - 1$ is a bijection $\mathbb{N} \to \omega$, so ω is countable.
For $n \in \mathbb{N}$, the coordinate map $f \mapsto (f(0), \ldots, f(n-1))$ bijects ω^n with the n-tuples over ω.
Hence ω^n is countable by Theorem 6.2.18. Since $A^n \subseteq \omega^n$, the set A^n is at most countable by Proposition 6.2.12(i).
The set $A^0 = \{\emptyset\}$ is finite, so $A^{<\omega}$ is at most countable by Corollary 6.2.16.
For $s \in A^{<\omega}$, let $\widehat{s} \in A^\omega$ extend s and equal 0 on $\omega \setminus |s|$. Then $E_A = \{\widehat{s} : s \in A^{<\omega}\}$.
Thus E_A is at most countable by Proposition 6.2.12(ii), applied to the image of $s \mapsto \widehat{s}$.
For $j < \omega$, let $e_j \in A^\omega$ equal 1 at j and 0 elsewhere.
Distinct indices give distinct $e_j \in E_A$, so E_A is infinite and therefore countable.
(v) The relation $(\forall s \in A^{<\omega})\, \widehat{s} \in E_A \cap N_s^A$ holds, so E_A is dense.
(vi) The complete metric d_A induces the cylinder topology by (ii) and (iii).
The set E_A is a countable dense subset by (iv) and (v). Hence A^ω is Polish.

Structural Commentary: Prefix stabilization proves completeness; finite supports supply a countable dense set.
Key Insight: Finite information controls both distance and density. □

Definition 7.3.5. *Products, Projections, and Continuity*
Let $A, B \in \{2, \omega\}$.

(i) The *product topology* on $A^\omega \times B^\omega$ has basis $\{N_s^A \times N_t^B : s \in A^{<\omega},\ t \in B^{<\omega}\}$. Its open sets are the unions of these rectangles.
(ii) For $E \subseteq A^\omega \times B^\omega$, its first-coordinate projection is $\pi_A(E) := \{x \in A^\omega : (\exists y \in B^\omega)\, (x,y) \in E\}$.
(iii) A function $f : A^\omega \to B^\omega$ is *continuous* iff $(\forall t \in B^{<\omega})\, f^{-1}(N_t^B)$ is open.

Proposition 7.3.6. *Continuity by Finite Prefixes*
Let $A, B \in \{2, \omega\}$ and let $f : A^\omega \to B^\omega$. Then f is continuous iff the following finite-prefix condition holds.
$(\forall x \in A^\omega)(\forall m < \omega)(\exists n < \omega)(\forall y \in A^\omega)\, [y|_n = x|_n \implies f(y)|_m = f(x)|_m]$.

Proof.

($\Rightarrow$) Assume f is continuous, and fix $x \in A^\omega$ and $m < \omega$. Set $V := f^{-1}(N^B_{f(x)|_m})$. Then V is open and $x \in V$.
Choose $n < \omega$ with $x \in N^A_{x|_n} \subseteq V$ by Proposition 7.3.2(iii). If $y|_n = x|_n$, then $y \in V$ and $f(y)|_m = f(x)|_m$.
($\Leftarrow$) Assume the prefix condition, fix $t \in B^{<\omega}$, and set $m := |t|$.
For $x \in f^{-1}(N_t^B)$, choose $n < \omega$ such that $(\forall y \in A^\omega)\, [y|_n = x|_n \Rightarrow f(y)|_m = f(x)|_m]$ by the prefix condition.
If $y \in N^A_{x|_n}$, then $f(y)|_m = f(x)|_m = t$, so $N^A_{x|_n} \subseteq f^{-1}(N_t^B)$.
Thus $f^{-1}(N_t^B)$ is a union of cylinders and is open; hence f is continuous.

Structural Commentary: Each direction translates target prefixes into source-cylinder neighborhoods.
Key Insight: Finite output information depends on finite input information. □

Definition 7.3.7. *Trees and Infinite Branches*
Let $A \in \{2, \omega\}$, let T be a set, and let $C \subseteq A^\omega$.

(i) A *tree on A* is a possibly empty set $T \subseteq A^{<\omega}$ satisfying the next condition; write $\mathrm{Tree}_A(T)$. $(\forall t \in T)(\forall s \in A^{<\omega})\, [s \sqsubseteq t \implies s \in T]$.
(ii) The *body* of T is $[T] := \{x \in A^\omega : (\forall n < \omega)\, x|_n \in T\}$.
(iii) The *tree of initial segments* of C is $T_C := \{x|_n : x \in C,\ n < \omega\}$.

Theorem 7.3.8. *Closed Sets Are Exactly Tree Bodies*
Let $A \in \{2, \omega\}$, let $T \subseteq A^{<\omega}$, and let $C \subseteq A^{\omega}$.

(i) If $\mathrm{Tree}_A(T)$, then $[T]$ is closed.
(ii) $\mathrm{Tree}_A(T_C)$.
(iii) If C is closed, then $C = [T_C]$.
(iv) The set C is closed iff $(\exists S \subseteq A^{<\omega})\,[\mathrm{Tree}_A(S) \wedge C = [S]]$.

Proof.

(i) Fix $x \notin [T]$ and choose $n < \omega$ with $x|_n \notin T$. The cylinder $N^A_{x|_n}$ lies in $A^{\omega} \setminus [T]$ because each $y \in N^A_{x|_n}$ satisfies $y|_n = x|_n \notin T$. Thus the complement of $[T]$ is open.
(ii) By definition, $T_C \subseteq A^{<\omega}$. If $t = x|_n \in T_C$ and $s \in A^{<\omega}$ satisfies $s \sqsubseteq t$, then $s = x|_{|s|} \in T_C$; hence $\mathrm{Tree}_A(T_C)$.
(iii) By definition, $C \subseteq [T_C]$. Fix $x \notin C$ and choose $n < \omega$ with $x \in N^A_{x|_n} \subseteq A^{\omega} \setminus C$ by Proposition 7.3.2(iii).
If $x|_n \in T_C$, then some $y \in C$ extends $x|_n$, contradicting $N^A_{x|_n} \cap C = \emptyset$. Hence $x \notin [T_C]$, so $[T_C] \subseteq C$.
(iv) ($\Rightarrow$) If C is closed, then $\mathrm{Tree}_A(T_C)$ by (ii) and $C = [T_C]$ by (iii).
(iv) ($\Leftarrow$) If $\mathrm{Tree}_A(S)$ and $C = [S]$, then C is closed by (i) with $T = S$.

Structural Commentary: Complement cylinders prove closure; realized prefixes construct the representing tree.
Key Insight: Closed membership is determined by all finite approximations. □

7.4 Filters, Ultrafilters, and Sets of Reals

Definition 7.4.1. *Filters, Ideals, and Duality*
Let X be a set, let $\mathcal{F}, \mathcal{I} \subseteq \mathcal{P}(X)$, and let $A \subseteq X$.

(i) The family $\mathcal{F}$ is a *filter* on X iff $X \in \mathcal{F}$, $\emptyset \notin \mathcal{F}$, $(\forall B, C \in \mathcal{F})\, B \cap C \in \mathcal{F}$, and $(\forall B \in \mathcal{F})(\forall C \subseteq X)\,[B \subseteq C \Rightarrow C \in \mathcal{F}]$.
(ii) The family $\mathcal{I}$ is an *ideal* on X iff $\emptyset \in \mathcal{I}$, $X \notin \mathcal{I}$, $(\forall B, C \in \mathcal{I})\, B \cup C \in \mathcal{I}$, and $(\forall B \in \mathcal{I})(\forall C \subseteq B)\, C \in \mathcal{I}$.
(iii) If $\mathcal{F}$ is a filter, its *dual ideal* is $\mathcal{I}_{\mathcal{F}} := \{X \setminus B : B \in \mathcal{F}\}$. If $\mathcal{I}$ is an ideal, its *dual filter* is $\mathcal{F}_{\mathcal{I}} := \{X \setminus B : B \in \mathcal{I}\}$.
(iv) If $\mathcal{I}$ is an ideal, then A is *$\mathcal{I}$-positive* iff $A \notin \mathcal{I}$.

Definition 7.4.2. *Generated, Principal, and Fréchet Filters*
Let X be a set and let $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfy $\mathcal{B} \neq \emptyset$, $\emptyset \notin \mathcal{B}$, and $(\forall B_0, B_1 \in \mathcal{B})\, B_0 \cap B_1 \in \mathcal{B}$.

(i) The *filter generated by $\mathcal{B}$* is $\langle \mathcal{B} \rangle := \{A \subseteq X : (\exists B \in \mathcal{B})\, B \subseteq A\}$.
(ii) For $\emptyset \neq D \subseteq X$, the *principal filter generated by D* is $\mathcal{F}_D := \{A \subseteq X : D \subseteq A\}$.
(iii) For infinite X, the *Fréchet filter*, or *cofinite filter*, is $\mathrm{Fr}(X) := \{A \subseteq X : X \setminus A \text{ is finite}\}$.

Definition 7.4.3. *κ-Complete Filter*
Let X be a set, let κ be an infinite cardinal, and let $\mathcal{F}$ be a filter on X.
The filter $\mathcal{F}$ is *κ-complete* iff $(\forall \lambda < \kappa)(\forall A \in \mathcal{F}^{\lambda})\ \bigcap_{\xi<\lambda} A(\xi) \in \mathcal{F}$.

Proposition 7.4.4. *Basic Filter and Ideal Constructions*
Let X be a set, let $\mathcal{F}$ be a filter on X, let $\mathcal{I}$ be an ideal on X, and let $\emptyset \neq D \subseteq X$.
Let $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfy $\mathcal{B} \neq \emptyset$, $\emptyset \notin \mathcal{B}$, and $(\forall B_0, B_1 \in \mathcal{B})\, B_0 \cap B_1 \in \mathcal{B}$.

(i) $\mathcal{I}_{\mathcal{F}}$ is an ideal and $\mathcal{F}_{\mathcal{I}}$ is a filter on X.
(ii) $\langle \mathcal{B} \rangle$ is the least filter containing $\mathcal{B}$.
(iii) $\mathcal{F}_D$ is a filter; if X is infinite, then $\mathrm{Fr}(X)$ is a filter whose dual ideal is the family of finite subsets of X.
(iv) The filter $\mathcal{F}$ is $\aleph_0$-complete.

Proof.

(i) The equalities $X \setminus X = \emptyset$ and $X \setminus \emptyset = X$ show that $\mathcal{I}_{\mathcal{F}}$ contains $\emptyset$ but not X. If $A, B \in \mathcal{F}$, then $(X \setminus A) \cup (X \setminus B) = X \setminus (A \cap B) \in \mathcal{I}_{\mathcal{F}}$.
Let $A \in \mathcal{F}$ and $E \subseteq X \setminus A$. Then $A \subseteq X \setminus E$, so $X \setminus E \in \mathcal{F}$ and $E \in \mathcal{I}_{\mathcal{F}}$. Thus $\mathcal{I}_{\mathcal{F}}$ is an ideal.
The same two complement equalities give $X \in \mathcal{F}_{\mathcal{I}}$ and $\emptyset \notin \mathcal{F}_{\mathcal{I}}$. If $A, B \in \mathcal{I}$, then $(X \setminus A) \cap (X \setminus B) = X \setminus (A \cup B) \in \mathcal{F}_{\mathcal{I}}$.
Let $A \in \mathcal{I}$ and $X \setminus A \subseteq E \subseteq X$. Then $X \setminus E \subseteq A$, so $X \setminus E \in \mathcal{I}$ and $E \in \mathcal{F}_{\mathcal{I}}$. Hence $\mathcal{F}_{\mathcal{I}}$ is a filter.
(ii) Choose $B_0 \in \mathcal{B}$. Then $B_0 \subseteq X$ gives $X \in \langle \mathcal{B} \rangle$. Since no member of $\mathcal{B}$ is contained in $\emptyset$, one has $\emptyset \notin \langle \mathcal{B} \rangle$.
If $A, A' \in \langle \mathcal{B} \rangle$, choose $B, B' \in \mathcal{B}$ with $B \subseteq A$ and $B' \subseteq A'$. Then $B \cap B' \in \mathcal{B}$ and $B \cap B' \subseteq A \cap A'$.
Thus $\langle \mathcal{B} \rangle$ is closed under intersections, and upward closure follows from transitivity of inclusion. The inclusion $\mathcal{B} \subseteq \langle \mathcal{B} \rangle$ holds.
If a filter $\mathcal{G}$ contains $\mathcal{B}$ and $A \in \langle \mathcal{B} \rangle$, choose $B \in \mathcal{B}$ with $B \subseteq A$. Then $A \in \mathcal{G}$, so $\langle \mathcal{B} \rangle \subseteq \mathcal{G}$.
(iii) Since $\mathcal{F}_D = \langle \{D\} \rangle$, the family $\mathcal{F}_D$ is a filter by (ii). Suppose X is infinite. The set X is cofinite, whereas $\emptyset$ is not. The intersection of two cofinite sets is cofinite because its complement is the union of two finite sets, and every superset of a cofinite set is cofinite. Thus $\mathrm{Fr}(X)$ is a filter. Its complement transform is $\{A \subseteq X : A \text{ is finite}\}$.
(iv) Let $\lambda < \aleph_0$ and $A \in \mathcal{F}^{\lambda}$. The ordinal λ is finite because $\aleph_0$ is the least infinite cardinal. The empty intersection is $X \in \mathcal{F}$, and adjoining one $A(\xi)$ preserves membership by binary closure. Hence the required intersection belongs to $\mathcal{F}$ by induction on λ.

Structural Commentary: Complementation proves duality; base closure and finite induction handle the filter constructions.
Key Insight: Upward closure turns finite-intersection bases into filters. □

Definition 7.4.5. *Ultrafilters and Principal Ultrafilters*
Let X be a set and let $\mathcal{U}$ be a filter on X.

(i) The filter $\mathcal{U}$ is an *ultrafilter* iff $(\forall A \subseteq X)\, \#(\mathcal{U} \cap \{A, X \setminus A\}) = 1$.
(ii) The ultrafilter $\mathcal{U}$ is *principal* iff $(\exists x \in X)\, \mathcal{U} = \{A \subseteq X : x \in A\}$; otherwise it is *nonprincipal*.

Theorem 7.4.6. *Ultrafilters Are Exactly the Maximal Filters*
Let X be a set and let $\mathcal{F}$ be a filter on X. Then $\mathcal{F}$ is an ultrafilter iff $(\forall \mathcal{G} \subseteq \mathcal{P}(X))\,[\mathcal{G}$ is a filter on $X \wedge \mathcal{F} \subseteq \mathcal{G} \Rightarrow \mathcal{G} = \mathcal{F}]$.

Proof.

($\Rightarrow$) Suppose $\mathcal{F}$ is an ultrafilter and let $\mathcal{G}$ be a filter with $\mathcal{F} \subseteq \mathcal{G}$
If $A \in \mathcal{G} \setminus \mathcal{F}$, then $X \setminus A \in \mathcal{F} \subseteq \mathcal{G}$, so $\emptyset = A \cap (X \setminus A) \in \mathcal{G}$, contradicting properness. Thus $\mathcal{G} = \mathcal{F}$.

(⇐) Suppose the displayed maximality condition holds, and fix $A \subseteq X$. Assume $A, X \setminus A \notin \mathcal{F}$, and set $\mathcal{B}_A := \{B \cap A : B \in \mathcal{F}\}$.
The relation $A = X \cap A$ gives $\mathcal{B}_A \neq \emptyset$, and $(B \cap A) \cap (C \cap A) = (B \cap C) \cap A \in \mathcal{B}_A$ for $B, C \in \mathcal{F}$.
If $\emptyset \in \mathcal{B}_A$, choose $B \in \mathcal{F}$ with $B \cap A = \emptyset$. Then $B \subseteq X \setminus A$, so $X \setminus A \in \mathcal{F}$, a contradiction.
Hence $\langle \mathcal{B}_A \rangle$ is a filter by Proposition 7.4.4(ii). For $B \in \mathcal{F}$, the relation $B \cap A \subseteq B$ gives $B \in \langle \mathcal{B}_A \rangle$, so $\mathcal{F} \subseteq \langle \mathcal{B}_A \rangle$.
Since $A \in \mathcal{B}_A \subseteq \langle \mathcal{B}_A \rangle$ but $A \notin \mathcal{F}$, the containment is proper, contradicting maximality.
Thus $A \in \mathcal{F}$ or $X \setminus A \in \mathcal{F}$; properness prevents both, so $\mathcal{F}$ is an ultrafilter.

Structural Commentary: One direction excludes proper extensions; the other adjoins any undecided set through a filter base.
Key Insight: Maximality is equivalent to deciding every complementary pair. □

Principle 7.4.7. *Ultrafilter Lemma*
Let X be a set and let $\mathcal{F}$ be a filter on X. The Ultrafilter Lemma asserts $(\exists \mathcal{U} \subseteq \mathcal{P}(X))\,[\mathcal{U}$ is an ultrafilter on $X \wedge \mathcal{F} \subseteq \mathcal{U}]$.

Theorem 7.4.8. *Ultrafilter Extension under Zorn's Lemma*
Principle 6.1.3 implies Principle 7.4.7.

Proof.
(i) *Extension poset*: Assume Zorn's Lemma, let X be a set, and let $\mathcal{F}$ be a filter on X.
Let $\mathscr{P}$ be the set of filters on X containing $\mathcal{F}$, ordered by inclusion.
Then $\mathcal{F} \in \mathscr{P}$, and inclusion is a partial order on $\mathscr{P}$ by Proposition 6.1.5.
(ii) *Chain bounds*: The filter $\mathcal{F}$ bounds the empty chain. Let $\mathcal{C} \subseteq \mathscr{P}$ be a nonempty chain and set $\mathcal{G} := \bigcup_{\mathcal{H} \in \mathcal{C}} \mathcal{H}$.
Then $X \in \mathcal{G}$, $\emptyset \notin \mathcal{G}$, and $\mathcal{F} \subseteq \mathcal{G}$. If $A, B \in \mathcal{G}$, choose $\mathcal{H}_0, \mathcal{H}_1 \in \mathcal{C}$ with $A \in \mathcal{H}_0$ and $B \in \mathcal{H}_1$.
Since $\mathcal{C}$ is a chain, one of these filters contains both A and B, so $A \cap B \in \mathcal{G}$.
Upward closure passes from any member of $\mathcal{C}$ containing A. Thus $\mathcal{G} \in \mathscr{P}$ and it bounds $\mathcal{C}$.
(iii) *Maximal extension*: Every chain in $\mathscr{P}$ has an upper bound in $\mathscr{P}$, so a maximal $\mathcal{U} \in \mathscr{P}$ exists by Zorn's Lemma.
The filter $\mathcal{U}$ is an ultrafilter by Theorem 7.4.6, and $\mathcal{F} \subseteq \mathcal{U}$ by the definition of $\mathscr{P}$.

Structural Commentary: Unions preserve filters along chains, so Zorn produces a maximal extension.
Key Insight: Chain compatibility makes binary filter closure survive union. □

Remark 7.4.9. *Choice Strength of Ultrafilter Extension*
Over ZF, Principle 7.4.7 is equivalent to the Boolean Prime Ideal Theorem and is strictly weaker than the full Axiom of Choice. This metamathematical comparison is not used below as proof authority. Theorem 7.4.8 establishes only the implication from the stronger Zorn principle available earlier. No choice principle enters Theorem 7.4.6 or the definition of principal ultrafilters.

Corollary 7.4.10. *Nonprincipal Ultrafilters on Infinite Sets*
Assume Principle 7.4.7, and let X be infinite. Then $(\exists \mathcal{U} \subseteq \mathcal{P}(X))\,[\mathcal{U}$ is a nonprincipal ultrafilter on $X]$.

Proof. The family $\mathrm{Fr}(X)$ is a filter by Proposition 7.4.4(iii). Choose an ultrafilter $\mathcal{U} \supseteq \mathrm{Fr}(X)$ by the hypothesis of the corollary.
If $\mathcal{U}$ were principal at $x \in X$, then $\{x\} \in \mathcal{U}$. Since $X \setminus \{x\} \in \mathrm{Fr}(X) \subseteq \mathcal{U}$, closure under intersections would give $\emptyset \in \mathcal{U}$, contradicting properness. Hence $\mathcal{U}$ is nonprincipal.
Structural Commentary: Extend the cofinite filter and exclude principality by intersecting complementary singleton data.
Key Insight: Cofinite membership prevents concentration at one point. □

Definition 7.4.11. *Measurable Cardinal*
Let κ be an uncountable cardinal. In ZFC, the cardinal κ is *measurable* iff $(\exists \mathcal{U} \subseteq \mathcal{P}(\kappa))\,[\mathcal{U}$ is a nonprincipal κ-complete ultrafilter on $\kappa]$.

Remark 7.4.12. *Large-Cardinal Assumption Boundary*
The assertion that a measurable cardinal exists is an additional large-cardinal hypothesis; it is not assumed elsewhere in this part. The ordinary Ultrafilter Lemma does not supply such a cardinal, as extending a filter to an ultrafilter need not preserve κ-completeness. Thus the large-cardinal connection enters through the extra completeness requirement, not through ultrafilter maximality alone.

Assumption 7.4.13. *Club and Stationary Setting*
For the club-and-stationary results below, assume Principle 6.1.2 and fix a regular cardinal $\kappa > \omega$.

Definition 7.4.14. *Closed Unbounded Subset*
A set $C \subseteq \kappa$ is *closed in κ* iff $(\forall \nu < \kappa)\,[\nu$ is a nonzero limit ordinal $\wedge\, C \cap \nu$ is cofinal in $\nu \Rightarrow \nu \in C]$.
It is *unbounded in κ* iff it is cofinal in κ, and it is a *club subset of κ* iff it is both closed and unbounded.

Theorem 7.4.15. *Intersection of Fewer Than κ Clubs*
If $0 < \lambda < \kappa$ and $(C_i)_{i<\lambda}$ is a family of club subsets of κ, then $\bigcap_{i<\lambda} C_i$ is club in κ.

Proof. Set $C := \bigcap_{i<\lambda} C_i$. If $\nu < \kappa$ is a nonzero limit ordinal and $C \cap \nu$ is cofinal in ν, then $(\forall i < \lambda)\, C_i \cap \nu$ is cofinal in ν.
Hence $(\forall i < \lambda)\, \nu \in C_i$, so $\nu \in C$ and C is closed.
If $\theta < \kappa$ and $g : \theta \to \kappa$, then $\sup g(\theta) < \kappa$; otherwise g would be cofinal and $\mathrm{cf}(\kappa) \leq \theta < \kappa$, contradicting regularity.
Fix $\beta < \kappa$ and set $\beta_0 := \beta + 1 < \kappa$ by Lemma 6.3.17.
Recursively, after β_n is defined, choose $(\gamma_{n,i})_{i<\lambda}$ with $(\forall i < \lambda)\, \gamma_{n,i} \in C_i \setminus (\beta_n + 1)$ by the standing Choice Principle.
Set $\beta_{n+1} := \sup_{i<\lambda}(\gamma_{n,i} + 1)$. The limit-ordinal property of κ gives $(\forall i < \lambda)\, \gamma_{n,i} + 1 < \kappa$, so the preceding bound gives $\beta_{n+1} < \kappa$.
Set $\delta := \sup_{n<\omega} \beta_n$. The same observation with $\theta = \omega$ gives $\delta < \kappa$.
Fix $i < \lambda$. Since $(\forall n < \omega)\, \beta_n < \gamma_{n,i} < \beta_{n+1} \leq \delta$, the sequence $(\gamma_{n,i})_{n<\omega}$ is cofinal in the nonzero limit ordinal δ.
Closedness of C_i gives $\delta \in C_i$. As $i < \lambda$ was arbitrary, $\delta \in C$; moreover, $\beta < \delta$, so C is unbounded.
Structural Commentary: Regularity bounds a countable recursion whose stages visit every club.
Key Insight: Each club contributes a cofinal subsequence to one common limit. □

Definition 7.4.16. *Club Filter, Stationary Sets, and the Nonstationary Ideal*
(i) The *club filter* on κ is $\mathcal{F}_{\mathrm{club}}(\kappa) := \{A \subseteq \kappa : (\exists C \subseteq A)\, C$ is club in $\kappa\}$.
(ii) A set $S \subseteq \kappa$ is *stationary* iff $(\forall C \subseteq \kappa)\,[C$ is club in $\kappa \Rightarrow S \cap C \neq \emptyset]$.
(iii) The *nonstationary ideal* is $\mathrm{NS}_\kappa := \{A \subseteq \kappa : (\exists C \subseteq \kappa)\,[C$ is club in $\kappa \wedge A \cap C = \emptyset]\}$. Thus S is stationary iff $S \notin \mathrm{NS}_\kappa$.

Proposition 7.4.17. *Club-Filter Completeness and Stationary-Set Basics*
Let $C, S \subseteq \kappa$.

(i) The family $\mathcal{F}_{\mathrm{club}}(\kappa)$ is a κ-complete filter.
(ii) The ideal NS_κ is dual to $\mathcal{F}_{\mathrm{club}}(\kappa)$.
(iii) If C is club and S is stationary, then $C \cap S$ is stationary.
(iv) If S is stationary, then S is unbounded.
(v) If S is stationary, then $\{\nu \in S : \nu$ is a nonzero limit ordinal$\}$ is stationary.

Proof.

(i) The set κ is club in itself, so $\kappa \in \mathcal{F}_{\mathrm{club}}(\kappa)$. No club is empty, so $\emptyset \notin \mathcal{F}_{\mathrm{club}}(\kappa)$, and upward closure follows from the definition.
Let $\lambda < \kappa$ and $(A_i)_{i<\lambda} \in \mathcal{F}_{\mathrm{club}}(\kappa)^\lambda$. If $\lambda = 0$, then $\bigcap_{i<\lambda} A_i = \kappa$. If $\lambda > 0$, choose clubs $C_i \subseteq A_i$ by the standing Choice Principle.
The intersection $\bigcap_{i<\lambda} C_i$ is club by Theorem 7.4.15, and it is contained in $\bigcap_{i<\lambda} A_i$. Thus the club filter is κ-complete.

(ii) For $A \subseteq \kappa$, $\kappa \setminus A \in \mathcal{F}_{\mathrm{club}}(\kappa)$ iff $(\exists C \subseteq \kappa \setminus A)\,[C$ is club$]$, equivalently iff $A \in \mathrm{NS}_\kappa$.
Hence $\mathrm{NS}_\kappa = \mathcal{I}_{\mathcal{F}_{\mathrm{club}}(\kappa)}$, which is an ideal by (i) and Proposition 7.4.4(i).

(iii) Fix a club $D \subseteq \kappa$. The intersection $C \cap D$ is club by Theorem 7.4.15 with $\lambda = 2$, so $(C \cap S) \cap D \neq \emptyset$.
As D was arbitrary, $C \cap S$ is stationary.

(iv) Suppose $S \subseteq \kappa$ is bounded, and choose $\beta < \kappa$ with $S \subseteq \beta + 1$. Set $D_\beta := \{\xi < \kappa : \beta < \xi\}$.
Since κ is a limit ordinal, $(\forall \eta < \kappa)\ \max\{\eta, \beta\} + 1 \in D_\beta$, so D_β is unbounded.
If $D_\beta \cap \nu$ is cofinal in a nonzero limit ordinal $\nu < \kappa$, then $D_\beta \cap \nu \neq \emptyset$, so $\beta < \nu$ and $\nu \in D_\beta$.
Hence D_β is closed and therefore club. Since $S \cap D_\beta = \emptyset$, the set S is not stationary.

(v) Set $L := \{\nu < \kappa : \nu$ is a nonzero limit ordinal$\}$. If $L \cap \nu$ is cofinal in a nonzero limit ordinal $\nu < \kappa$, then $\nu \in L$, so L is closed.
For $\beta < \kappa$, the sequence $(\beta + n)_{n<\omega}$ is bounded below κ by regularity, so $\beta + \omega < \kappa$.
Since $\beta + \omega \in L$ lies above β, the set L is unbounded and therefore club. The set $S \cap L$ is stationary by (iii).

Structural Commentary: Club intersections prove completeness; closed tails and limit ordinals test stationarity.
Key Insight: Stationary sets survive every additional club restriction. □

Definition 7.4.18. *Diagonal Intersection and Normality*

(i) For $(X_\xi)_{\xi<\kappa} \in \mathcal{P}(\kappa)^\kappa$, its *diagonal intersection* is $\triangle_{\xi<\kappa} X_\xi := \{\beta < \kappa : (\forall \xi < \beta)\ \beta \in X_\xi\}$.

(ii) A filter $\mathcal{F}$ on κ is *normal* iff $(\forall (X_\xi)_{\xi<\kappa} \in \mathcal{F}^\kappa)\ \triangle_{\xi<\kappa} X_\xi \in \mathcal{F}$.

Theorem 7.4.19. *The Club Filter Is Normal*

(i) If $(C_\xi)_{\xi<\kappa}$ is a sequence of clubs in κ, then $\triangle_{\xi<\kappa} C_\xi$ is club.
(ii) The club filter on κ is normal.

Proof.

(i) We prove clubness through decreasing refinement, closure, and recursive unboundedness.

(a) *Decreasing refinement*: Define $(C'_\xi)_{\xi<\kappa}$ by $C'_\xi := \bigcap_{\eta\le\xi} C_\eta$.
Each C'_ξ is club by Theorem 7.4.15 with $\lambda = \xi + 1$, since $0 < \xi + 1 < \kappa$; moreover, $(C'_\xi)_{\xi<\kappa}$ is decreasing.
Fix $\rho < \kappa$. Then $(\forall \xi < \rho)\ \rho \in C'_\xi$ iff $(\forall \xi < \rho)(\forall \eta \le \xi)\ \rho \in C_\eta$ iff $(\forall \eta < \rho)\ \rho \in C_\eta$.
As $\rho < \kappa$ was arbitrary, the diagonal intersections agree. Hence assume $(C_\xi)_{\xi<\kappa}$ is decreasing and set $D := \triangle_{\xi<\kappa} C_\xi$.

(b) *Closedness*: Suppose $\nu < \kappa$ is a nonzero limit ordinal and $D \cap \nu$ is cofinal in ν. Fix $\xi < \nu$.
Then $\{\rho \in D \cap \nu : \xi < \rho\} \subseteq C_\xi$ is cofinal in ν, so $\nu \in C_\xi$ by closedness.
As $\xi < \nu$ was arbitrary, $\nu \in D$; thus D is closed.

(c) *Unboundedness*: Fix $\eta < \kappa$ and set $\beta_0 := \eta + 1 < \kappa$.
Recursively set $\beta_{n+1} := \min(C_{\beta_n} \setminus (\beta_n + 1))$; unboundedness of C_{β_n} makes the defining set nonempty.
Set $\beta := \sup_{n<\omega} \beta_n < \kappa$; otherwise $n \mapsto \beta_n$ would be cofinal in κ, contrary to $\mathrm{cf}(\kappa) = \kappa > \omega$.
The strict increase makes β a nonzero limit ordinal. Fix $\xi < \beta$ and choose $n < \omega$ with $\xi < \beta_n$.
Decreasingness gives $(\forall m \ge n)\ \beta_{m+1} \in C_{\beta_m} \subseteq C_\xi$. This tail is cofinal in β, so $\beta \in C_\xi$ by closedness.
As $\xi < \beta$ was arbitrary, $\beta \in D$; since $\eta < \beta$, the set D is unbounded and therefore club.

(ii) Let $(A_\xi)_{\xi<\kappa} \in \mathcal{F}_{\mathrm{club}}(\kappa)^\kappa$, and choose clubs $C_\xi \subseteq A_\xi$ by the standing Choice Principle.
The set $\triangle_{\xi<\kappa} C_\xi$ is club by (i), and $\triangle_{\xi<\kappa} C_\xi \subseteq \triangle_{\xi<\kappa} A_\xi$. Hence $\triangle_{\xi<\kappa} A_\xi \in \mathcal{F}_{\mathrm{club}}(\kappa)$.

Structural Commentary: Decreasing refinement separates closure from recursive unboundedness; contained clubs establish normality.
Key Insight: Each point satisfies only club conditions indexed below it. □

Theorem 7.4.20. *Fodor's Pressing-Down Lemma*
Let $S \subseteq \kappa$ be stationary and let $f : S \to \kappa$ be regressive, meaning $(\forall \alpha \in S)\ f(\alpha) < \alpha$. Then $(\exists \gamma < \kappa)\,[\{\alpha \in S : f(\alpha) = \gamma\}$ is stationary$]$.

Proof. For $\gamma < \kappa$, set $S_\gamma := \{\alpha \in S : f(\alpha) = \gamma\}$. Suppose $(\forall \gamma < \kappa)\,[S_\gamma$ is nonstationary$]$.
Choose $(C_\gamma)_{\gamma<\kappa}$ satisfying $(\forall \gamma < \kappa)\,[C_\gamma$ is club $\wedge\ C_\gamma \cap S_\gamma = \emptyset]$ by the standing Choice Principle.
The set $D := \triangle_{\gamma<\kappa} C_\gamma$ is club by Theorem 7.4.19(i). Choose $\alpha \in S \cap D$ and set $\gamma := f(\alpha)$. Then $\gamma < \alpha$ and $\alpha \in D$, so $\alpha \in C_\gamma$.
Since $\alpha \in S_\gamma$, this contradicts $C_\gamma \cap S_\gamma = \emptyset$. Hence $(\exists \gamma < \kappa)\,[S_\gamma$ is stationary$]$.
Structural Commentary: Avoid every fiber by a club, combine the clubs diagonally, and use regressiveness.
Key Insight: The regressive value indexes a club already tested at its argument. □

Assumption 7.4.21. *Descriptive-Set-Theory Setting*
The cardinal fixed in Assumption 7.4.13 is no longer in force. Work in ZFC, and let X be Baire space, Cantor space, or a finite product of these spaces with its product topology.

Definition 7.4.22. *Borel Sets and the Borel Pointclasses*
The *Borel σ-algebra* $\mathcal{B}(X)$ is the least family of subsets of X containing all open sets and closed under complements and countable unions. Its members are the *Borel sets*.
Let ω_1 be the least uncountable ordinal. The *Borel pointclasses* are defined recursively for $0 < \alpha < \omega_1$.

(i) $\mathbf{\Sigma}^0_1(X) := \{A \subseteq X : A \text{ is open}\}$ and $\mathbf{\Pi}^0_1(X) := \{X \setminus A : A \in \mathbf{\Sigma}^0_1(X)\}$.
(ii) Fix $1 < \alpha < \omega_1$ after all lower levels have been defined.
Set $\mathbf{\Sigma}^0_\alpha(X) := \{ \bigcup_{n<\omega} A(n) : A \in (\bigcup_{0<\beta<\alpha} \mathbf{\Pi}^0_\beta(X))^\omega\}$. Set $\mathbf{\Pi}^0_\alpha(X) := \{X \setminus A : A \in \mathbf{\Sigma}^0_\alpha(X)\}$.
(iii) $\mathbf{\Delta}^0_\alpha(X) := \mathbf{\Sigma}^0_\alpha(X) \cap \mathbf{\Pi}^0_\alpha(X)$.

Thus $\mathbf{\Sigma}^0_1(X)$, $\mathbf{\Pi}^0_1(X)$, and $\mathbf{\Delta}^0_1(X)$ are the open, closed, and clopen subsets of X, respectively.

Theorem 7.4.23. *The Borel Hierarchy Exhausts the Borel Sets*
A set $A \subseteq X$ is Borel iff $(\exists \alpha < \omega_1)\,[0 < \alpha \wedge A \in \mathbf{\Sigma}^0_\alpha(X)]$.

Proof.
(⇒) Set $\mathcal{H} := \bigcup_{0<\alpha<\omega_1} \mathbf{\Sigma}^0_\alpha(X)$. Then $\mathbf{\Sigma}^0_1(X) \subseteq \mathcal{H}$.
The relation $(\forall \alpha < \omega_1)\,[0 < \alpha \Rightarrow \mathbf{\Pi}^0_\alpha(X) \subseteq \mathbf{\Sigma}^0_{\alpha+1}(X)]$ holds by constant unions, so $\mathcal{H}$ is complement-closed.
Fix $B \in \mathcal{H}^\omega$ and choose $\alpha \in (\omega_1 \setminus \{0\})^\omega$ satisfying $(\forall n < \omega)\, B(n) \in \mathbf{\Sigma}^0_{\alpha(n)}(X)$ by Choice.
One has $(\forall n < \omega)\, B(n) \in \mathbf{\Pi}^0_{\alpha(n)+1}(X)$ by constant unions and complementation.
Set $\gamma := \sup_{n<\omega}(\alpha(n)+1) + 1 < \omega_1$ by Corollary 6.2.16. Since $(\forall n < \omega)\,\alpha(n) + 1 < \gamma$, one has $\bigcup_{n<\omega} B(n) \in \mathbf{\Sigma}^0_\gamma(X) \subseteq \mathcal{H}$.
Thus $\mathcal{H}$ is a σ-algebra containing $\mathbf{\Sigma}^0_1(X)$, so $\mathcal{B}(X) \subseteq \mathcal{H}$ by minimality.
(⇐) Set $P(\alpha) := \mathbf{\Sigma}^0_\alpha(X) \cup \mathbf{\Pi}^0_\alpha(X) \subseteq \mathcal{B}(X)$. We prove $(\forall \alpha < \omega_1)\,[0 < \alpha \Rightarrow P(\alpha)]$ by strong transfinite induction.
The property $P(1)$ holds by the definition of $\mathcal{B}(X)$. Fix $1 < \alpha < \omega_1$ and assume $(\forall \beta < \alpha)\,[0 < \beta \Rightarrow P(\beta)]$.
The inclusion $\mathbf{\Sigma}^0_\alpha(X) \subseteq \mathcal{B}(X)$ holds by the recursive definition, induction hypothesis, and countable-union closure.
Thus $\mathbf{\Pi}^0_\alpha(X) \subseteq \mathcal{B}(X)$ by complementation. Hence $P(\alpha)$.

Structural Commentary: Forward closure identifies the generated σ-algebra; transfinite induction proves the reverse inclusion.
Key Insight: Countable Borel operations have ranks below ω_1. □

Definition 7.4.24. *The Projective Pointclasses*
For $E \subseteq X \times \omega^\omega$, set $\pi_X(E) := \{x \in X : (\exists y \in \omega^\omega)\,(x, y) \in E\}$. Define the projective pointclasses recursively.

(i) $\mathbf{\Sigma}^1_1(X) := \{\pi_X(E) : E \in \mathcal{B}(X \times \omega^\omega)\}$; its members are the *analytic sets*.
(ii) For $n \in \mathbb{N}$, $\mathbf{\Pi}^1_n(X) := \{X \setminus A : A \in \mathbf{\Sigma}^1_n(X)\}$.
(iii) For $n \in \mathbb{N}$, $\mathbf{\Sigma}^1_{n+1}(X) := \{\pi_X(E) : E \in \mathbf{\Pi}^1_n(X \times \omega^\omega)\}$.
(iv) For $n \in \mathbb{N}$, $\mathbf{\Delta}^1_n(X) := \mathbf{\Sigma}^1_n(X) \cap \mathbf{\Pi}^1_n(X)$.

The members of $\mathbf{\Pi}^1_1(X)$ are the *coanalytic sets*. A set $A \subseteq X$ is *projective* iff $(\exists n \in \mathbb{N})\,[A \in \mathbf{\Sigma}^1_n(X) \vee A \in \mathbf{\Pi}^1_n(X)]$.

Proposition 7.4.25. *Borel Sets Are Analytic and Coanalytic*
(i) $\mathcal{B}(X) \subseteq \mathbf{\Delta}^1_1(X)$.
(ii) The class $\bigcup_{n\in\mathbb{N}} (\mathbf{\Sigma}^1_n(X) \cup \mathbf{\Pi}^1_n(X))$ *is closed under complements.*

Proof.
(i) Set $Y := \omega^\omega$ and $\mathscr{S} := \{C \subseteq X : C \times Y \in \mathcal{B}(X \times Y)\}$.
Then $(\forall C \in \mathscr{S})\,[(X \setminus C) \times Y = (X \times Y) \setminus (C \times Y) \in \mathcal{B}(X \times Y)]$, so $\mathscr{S}$ is complement-closed.
One has $(\forall C \in \mathscr{S}^\omega)\,[(\bigcup_{n<\omega} C(n)) \times Y = \bigcup_{n<\omega} (C(n) \times Y) \in \mathcal{B}(X \times Y)]$. Thus $\mathscr{S}$ is countable-union-closed.
The inclusion $\mathbf{\Sigma}^0_1(X) \subseteq \mathscr{S}$ holds because products of open sets are open. Thus $\mathscr{S}$ is a σ-algebra and $\mathcal{B}(X) \subseteq \mathscr{S}$ by minimality.
Since $Y \neq \emptyset$, $(\forall C \in \mathcal{B}(X))\, C = \pi_X(C \times Y) \in \mathbf{\Sigma}^1_1(X)$ by the definition of $\mathbf{\Sigma}^1_1(X)$.
Consequently, $(\forall A \in \mathcal{B}(X))\, A \in \mathbf{\Sigma}^1_1(X) \cap \mathbf{\Pi}^1_1(X) = \mathbf{\Delta}^1_1(X)$ by the pointclass definitions.
(ii) Set $\mathscr{P} := (\mathbf{\Sigma}^1_n(X) \cup \mathbf{\Pi}^1_n(X))_{n\in\mathbb{N}}$.
Then $(\forall n \in \mathbb{N})(\forall A \in \mathscr{P}(n))\, X \setminus A \in \mathscr{P}(n)$ by the definition of $\mathbf{\Pi}^1_n(X)$. Thus $\bigcup_{n\in\mathbb{N}} \mathscr{P}(n)$ is complement-closed.

Structural Commentary: Product closure proves the level-one inclusion; pointclass duality proves complement closure.
Key Insight: A constant witness realizes every Borel set as a projection. □

Remark 7.4.26. *Determinacy and Large-Cardinal Boundary*
The material from Definition 7.4.22 through Proposition 7.4.25 is developed in ZFC alone, without measurable-cardinal or determinacy hypotheses. No projective-regularity theorem is asserted; any later theorem of that kind must state its additional hypothesis explicitly.

8 Mathematical Writing and Research Practice

This section fixes the communication, evidence, integrity, and reading practices used to create and evaluate mathematical and machine-learning research.

8.1 Theorem-Style Results and Formal Statements

Formal environments advertise the role and authority of a block before the reader examines its details.

Convention 8.1.1. *Theorem-Style Result Hierarchy*
The theorem-style category contains exactly the following proved results.

(i) *Theorem*: a major result that advances a principal objective of the surrounding work.
(ii) *Proposition*: an independently useful result of smaller local weight.
(iii) *Lemma*: a subsidiary result introduced primarily to support a later proof.
(iv) *Corollary*: a result requiring little additional argument after an earlier result.

The classification communicates structural role rather than a universal rank of importance. Definitions introduce objects or terminology, while conjectures record unproved claims. This hierarchy follows standard mathematical-writing guidance (MIT Department of Mathematics, 1998).

Process 8.1.2. *Theorem-Statement Construction and Audit*
Given a candidate result and its supporting argument, produce a statement that can be cited and proved without hidden context.

(i) *Purpose*: identify the reusable conclusion and its role in the dependency chain.
(ii) *Objects*: bind every ambient set, structure, parameter, map, and index before its first use.
(iii) *Hypotheses*: state the exact domain, regularity, finiteness, independence, or boundary conditions needed.
(iv) *Conclusion*: state existence, uniqueness, characterization, bound, convergence, or other claim explicitly.
(v) *Claim census*: separate conclusions whose removal would erase a distinct proof obligation.
(vi) *Role*: choose theorem, proposition, lemma, or corollary by Convention 8.1.1.
(vii) *Locality*: remove motivation, examples, proof machinery, and unused assumptions from the statement.
(viii) *Traceability*: attach a stable title and label, and cite external ownership of a nonoriginal result.

The final statement should remain intelligible when read apart from the preceding paragraph. Self-contained statements and first-use symbol definitions are central expository checks (Knuth et al., 1988).

Principle 8.1.3. *Claim–Authority Separation*
A formal block should display its actual epistemic status. A definition stipulates meaning; an assumption declares a condition; a conjecture remains unproved; a theorem-style result requires a proof; and an empirical statement requires data and a protocol. A citation identifies prior ownership but does not replace a proof when the present work claims to establish the result.

Warning 8.1.4. *Statement–Proof Contamination*
Do not place a proof plan, auxiliary construction, example, or claim of obviousness inside a theorem-style statement. Such material obscures the logical contract, creates unused hypotheses, and makes later citation unreliable. Move proof-specific objects into the proof and explanatory material after the formal owner.

8.2 Proof Construction and Revision

A proof is a checkable argument from declared hypotheses, definitions, and earlier proved results to the exact stated conclusion.

Proof Method 8.2.1. *Proof Construction and Audit*
Given a theorem-style statement, construct and review its proof as follows.

(i) *Parse*: expose the objects, hypotheses, quantifier dependencies, and conclusion by Proof Method 2.4.1.
(ii) *Plan*: select a direct argument, cases, two inclusions, induction, contradiction, construction, or another fitting architecture.
(iii) *Dependencies*: list only earlier results whose hypotheses can be verified at the point of citation.
(iv) *Obligations*: map each numbered claim, implication direction, existence witness, and uniqueness assertion to a proof unit.
(v) *Execution*: introduce variables before use and justify each inference by a definition, calculation, hypothesis, or earlier result.
(vi) *Boundaries*: handle empty, zero, singular, endpoint, and other degenerate cases that affect an argument.
(vii) *Closure*: verify that the final sentence discharges the original conclusion rather than a nearby surrogate.
(viii) *Cold read*: reread without relying on memory; test notation, citations, and transitions as a skeptical reader would.

The proof should expose its plan when the route is not apparent, yet retain every step needed for verification. Mathematical-writing guidance likewise emphasizes status, motivation, and conceptual structure (MIT Department of Mathematics, 1998; Knuth et al., 1988).

Process 8.2.2. *Proof Revision Passes*
Revise a completed proof in three passes.

(i) *Validity*: check every dependency, implication, case split, witness, index, limit, and exceptional case.
(ii) *Architecture*: align proof units with statement claims and make long operative stages visible.
(iii) *Exposition*: remove repetition, bind notation earlier, compress routine algebra, and expand decisive unfamiliar steps.

Typography is reviewed only after the mathematical and logical structure is stable.

Warning 8.2.3. *Evidence Is Not Proof*
Finite testing, a diagram, numerical agreement, or a successful program run supports only the cases and conditions examined. None closes an unrestricted quantified claim without an exhaustive argument and a proof that the enumeration is complete. Computation may discover a conjecture, locate a counterexample, or verify a formally specified finite case.

Principle 8.2.4. *Proof Detail Proportionality*
Detail belongs where correctness or understanding depends on it. Expand novel constructions, fragile estimates, changes of quantifier, and boundary arguments; compress familiar algebra only while the remaining chain is checkable. The goal is neither maximal formality nor persuasive rhetoric, but a reproducible mathematical argument.

8.3 Methods, Algorithms, and Computational Artifacts

Procedural blocks must distinguish conceptual guidance, executable steps, and claims about correctness or performance.

Convention 8.3.1. *Procedural Environment Roles*
Use the following roles throughout these notes.

(i) *Proof method*: a reusable proof route, reduction, recognition criterion, or obligation checklist.
(ii) *Computation method*: a reusable construction, calculation, approximation, diagnostic, or test.
(iii) *Method*: a short convention, classification workflow, or inseparable hybrid with no clearer subtype.
(iv) *Process*: an organizational workflow for research, writing, review, or documentation.
(v) *Algorithm*: executable pseudocode with declared inputs, outputs, control flow, and termination conditions.

A worked instance belongs in an example, an actual proof in a proof, and implementation-specific behavior in code or an artifact record.

Process 8.3.2. *Algorithm Specification and Audit*
Before publishing pseudocode, record its contract and verification obligations.

(i) *Problem*: state the task and the abstraction level at which the algorithm operates.
(ii) *Input*: specify types, dimensions, admissibility, representation, and malformed-input behavior.

(iii) *Output*: specify the returned object and a postcondition that characterizes success.
(iv) *State*: define initialization, mutable state, randomness, and any invariant needed for correctness.
(v) *Steps*: write one unambiguous action per step with readable branches and loops.
(vi) *Termination*: state the stopping rule and justify termination when it is claimed.
(vii) *Correctness*: separate the proof of the postcondition from the pseudocode itself.
(viii) *Resources*: state complexity only with an input-size parameter, cost model, resource, and case type.
(ix) *Limits*: record numerical, memory, precision, platform, approximation, and applicability boundaries.

Labeled steps and explanatory invariants clarify the distinction between an algorithm and program text (Knuth et al., 1988).

Principle 8.3.3. *Formal and Informal Procedural Views*
Complex procedures often need complementary views. The formal view fixes inputs, outputs, updates, and stopping conditions; the informal view explains the invariant, decomposition, or design choice. Neither should duplicate the other, and only a separate proof can establish mathematical correctness.

Process 8.3.4. *Computational Artifact Record*
For code supporting a reported result, archive the source revision, dependencies, runtime, platform, configuration, random seeds, data identifiers, licenses, commands, and expected outputs. Map each reported figure, table, or metric to the command that creates it. State unavailable components and access restrictions explicitly. Current ML reporting guidance requests exact execution instructions and enough experimental detail to interpret the main results (Neural Information Processing Systems Foundation, 2026).

Warning 8.3.5. *Algorithm Evidence Boundaries*
Passing tests does not establish correctness outside the tested cases, and an asymptotic bound does not predict wall-clock performance without a cost model and operating conditions. Keep specification, proof, implementation, benchmark, and interpretation as distinct evidence types.

8.4 Mathematical Research and Integrity

Mathematical research alternates discovery, attempted refutation, proof, verification, and communication rather than proceeding linearly from question to theorem.

Process 8.4.1. *Mathematical Research Cycle*
For a proposed research problem, maintain a revisable cycle with the following stages.

(i) *Question*: state the object, scope, known assumptions, and a falsifiable or provable target.
(ii) *Literature*: identify definitions, known results, counterexamples, adjacent formulations, and open status.
(iii) *Exploration*: compute examples, boundary cases, and nonexamples while recording the tested domain.
(iv) *Conjecture*: state the strongest precise claim supported by exploration, with all suspected hypotheses.
(v) *Attack*: search independently for a proof and a counterexample; record failed routes and new obstructions.
(vi) *Revision*: weaken, strengthen, split, or reframe the conjecture when an example or proof obligation demands it.
(vii) *Verification*: check every proof dependency, compare with prior results, and seek independent criticism.
(viii) *Communication*: state novelty accurately, write a complete proof, cite provenance, and record limitations.

MIT's project laboratory similarly presents mathematical discovery as data generation, conjecture, experiment, proof or counterexample, and writing (MIT OpenCourseWare, 2013).

Process 8.4.2. *Research Claim Ledger*
Maintain one row per substantive claim with its exact wording, status, evidence, dependencies, source locator, and unresolved checks. Distinguish proved, conjectured, computed, observed, reproduced, replicated, and interpreted claims. Update the row when a hypothesis, dataset, code revision, or conclusion changes. This record prevents an exploratory observation from silently becoming a theorem or empirical conclusion.

Convention 8.4.3. *Reproducibility, Replicability, and Generalizability*
Use the National Academies terminology (National Academies of Sciences, Engineering, and Medicine, 2019).

(i) *Reproducibility*: consistent computational results from the same data, code, methods, and analysis conditions.
(ii) *Replicability*: consistent results from a new study addressing the same question with newly obtained data.
(iii) *Generalizability*: applicability of a result to contexts or populations beyond those directly studied.

These are separate targets. A reproducible computation need not replicate on new data or generalize to a new setting.

Principle 8.4.4. *Research Integrity and Stewardship*
Research records should permit scrutiny, correction, and appropriate reuse. Report methods and findings honestly; preserve accurate records; credit contributions and prior work; disclose conflicts; respect confidentiality, consent, licenses, and governing rules; and correct the record when an error is found. These duties reflect the honesty, accountability, fairness, and stewardship principles of the Singapore Statement (Second World Conference on Research Integrity, 2010).

Warning 8.4.5. *Novelty and Negative Results*
Do not equate novelty with truth or publication with correctness. Search for prior and concurrent work before claiming priority, and report a failed proof, null result, or nonreplication with its actual protocol and uncertainty. A negative outcome may diagnose a false conjecture, an inadequate method, or a boundary of generality; the evidence must distinguish these possibilities.

8.5 Machine Learning and AI Research

Machine-learning and AI claims combine statistical, computational, system, and social evidence, so the study design must identify which authority supports each conclusion.

Process 8.5.1. *Machine-Learning and AI Study Protocol*
Design an empirical study before interpreting its scores.

(i) *Question*: state the population, task, context of use, intervention or comparison, and claimed outcome.
(ii) *Claim type*: classify each target as theoretical, empirical, causal, computational, system, or normative.
(iii) *Data*: record provenance, version, license, collection, consent or access status, labels, exclusions, and preprocessing.
(iv) *Evaluation boundary*: protect test data from training, model selection, prompt design, preprocessing, and human iteration.
(v) *Comparators*: choose relevant baselines, ablations, and resource-matched alternatives before inspecting final results.

(vi) *Protocol*: fix splits, metrics, hyperparameter selection, seeds, stopping rules, hardware, software, and compute budget.
(vii) *Uncertainty*: repeat stochastic runs and report the source and calculation of intervals, variation, or significance measures.
(viii) *Stress tests*: examine subgroups, shift, robustness, failure modes, sensitivity, and plausible misuse within scope.
(ix) *Artifacts*: preserve code, data handles, checkpoints, configurations, logs, and command-to-result mappings.
(x) *Report*: separate observed results from interpretation and state limitations, costs, risks, and unsupported generalizations.

The NeurIPS checklist and its reproducibility program emphasize assumptions, complete proofs, experimental details, uncertainty, artifacts, ethics, and limitations (Neural Information Processing Systems Foundation, 2026; Pineau et al., 2021).

Process 8.5.2. *Dataset and Model Documentation*
Release documentation with each durable ML artifact.

(i) *Dataset*: record motivation, composition, collection, preprocessing, labeling, intended uses, restrictions, and maintenance.
(ii) *Model*: record identity, version, intended and excluded uses, training data, evaluation protocol, subgroup results, and limitations.
(iii) *System*: record interfaces, human roles, deployment assumptions, monitoring, update policy, safeguards, and rollback conditions.

Dataset datasheets and model cards establish complementary documentation patterns for provenance, intended use, evaluation, and limitations (Gebru et al., 2021; Mitchell et al., 2019).

Principle 8.5.3. *Contextual AI Risk Assessment*
Accuracy alone is not a complete system claim. For the intended context, select and justify tests for validity and reliability, safety, security and resilience, accountability and transparency, explainability and interpretability, privacy, and harmful-bias management. These characteristics can conflict and require documented tradeoffs. This versioned vocabulary follows the voluntary NIST AI RMF 1.0 (Tabassi, 2023).

Warning 8.5.4. *Adaptive Evaluation and Leakage*
Any choice influenced by protected evaluation data makes the final score adaptive. Repeated leaderboard queries, test-set debugging, manual example selection, and unreported prompt tuning can all leak evaluation information. Reserve a final assessment boundary or disclose the adaptation and narrow the claim.

Warning 8.5.5. *Benchmark Overclaiming*
A higher score supports a claim only under the reported dataset, metric, resources, selection procedure, and uncertainty. It does not by itself establish practical usefulness, causal benefit, robustness, fairness, safety, or superiority in another operating environment.

8.6 Reading and Synthesizing Research Papers

Reading depth should match the decision at hand: triage, comprehension, verification, reproduction, or synthesis.

Process 8.6.1. *Three-Pass Paper Reading*
Adapt the three-pass method of Keshav (2007) to the paper's evidence type.

(i) *Triage*: inspect title, abstract, introduction, headings, conclusion, figures, and references. Record category, context, claimed contribution, apparent assumptions, and relevance.
(ii) *Comprehension*: read the body carefully, map each main claim to its stated evidence, inspect figures and tables, and mark unfamiliar prerequisites or missing details.
(iii) *Reconstruction*: independently rebuild the proof, derivation, algorithm, or experiment closely enough to expose hidden assumptions, fragile steps, and reproducibility barriers.

Stop after the earliest pass sufficient for the decision, and resume only after acquiring a missing prerequisite when further reading would otherwise be guesswork.

Process 8.6.2. *Paper Claim–Evidence Ledger*
Create one entry per central claim and record its locator, exact scope, assumptions, evidence type, dependency, result, uncertainty, and limitation. For a mathematical paper, reconstruct definitions and proof obligations. For an empirical paper, reconstruct the sample, protocol, comparator, metric, variability, and analysis. For a systems paper, reconstruct the boundary, workload, implementation, platform, and failure model. Mark each claim as supported, partially supported, unsupported, or not assessed; reputation and venue are metadata rather than evidence.

Process 8.6.3. *Literature Search and Synthesis*
Build a literature map rather than a pile of summaries.

(i) *Seed*: start from precise keywords, a recent survey, and several directly relevant papers.
(ii) *Backward search*: follow repeated foundational citations to original results, data, or methods.
(iii) *Forward search*: locate corrections, replications, extensions, and critiques that cite each key work.
(iv) *Cluster*: group papers by question, assumptions, method, evidence, and operating context rather than by author.
(v) *Compare*: normalize terminology and record agreements, contradictions, gaps, and incomparable protocols.
(vi) *Saturate*: stop when new searches cease changing the conceptual map, while retaining the search date and boundary.

Keshav's survey procedure likewise expands seed papers through repeated authors, shared references, and recent proceedings (Keshav, 2007).

Warning 8.6.4. *Version and Source Status*
Before relying on a paper, check its version, venue, peer-review status, corrections, retractions, supplements, artifact links, and conflicts of interest. Cite the original source for a result or dataset when available. A preprint may be valuable, and peer review may improve reliability, but neither status settles the truth of a claim.

Principle 8.6.5. *Read Against the Claim*
Evaluate the strongest wording in the title, abstract, or conclusion against the narrowest evidence in the body. The decisive questions are what was established, under which assumptions, by which evidence, with what uncertainty, and within which boundary.

Part II
Structural Linear Algebra

1 Vector Spaces and Linear Structure

1.1 Vector Spaces

Vector spaces encode addition and scalar multiplication. Subspaces, spans, bases, dimension, and coordinates describe how linear structure is generated and measured.

Definition 1.1.1. *Vector Space*
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set $\mathcal{V}$ with the following structure.
The operations are $+\colon \mathcal{V}\times\mathcal{V}\to\mathcal{V}$ and scalar multiplication $\mathbb{F}\times\mathcal{V}\to\mathcal{V}$. The scalar action is written $(a,\mathbf{v})\mapsto a\mathbf{v}$, and the axioms are:

(i) $(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{V})\;\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$;
(ii) $(\forall\,\mathbf{u},\mathbf{v},\mathbf{w}\in\mathcal{V})\;(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$;
(iii) an element $\mathbf{0}\in\mathcal{V}$ satisfies $(\forall\,\mathbf{v}\in\mathcal{V})\;\mathbf{0}+\mathbf{v}=\mathbf{v}$;
(iv) $(\forall\,\mathbf{v}\in\mathcal{V})(\exists\,\mathbf{w}\in\mathcal{V})\;\mathbf{v}+\mathbf{w}=\mathbf{0}$;
(v) $(\forall\,\mathbf{v}\in\mathcal{V})\;1\mathbf{v}=\mathbf{v}$;
(vi) $(\forall\,a,b\in\mathbb{F})(\forall\,\mathbf{v}\in\mathcal{V})\;(ab)\mathbf{v}=a(b\mathbf{v})$;
(vii) $(\forall\,a\in\mathbb{F})(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{V})\;a(\mathbf{u}+\mathbf{v})=a\mathbf{u}+a\mathbf{v}$;
(viii) $(\forall\,a,b\in\mathbb{F})(\forall\,\mathbf{v}\in\mathcal{V})\;(a+b)\mathbf{v}=a\mathbf{v}+b\mathbf{v}$.

If $\mathcal{V}=\{\mathbf{0}\}$, then $\mathcal{V}$ is a *zero vector space*; otherwise $\mathcal{V}$ is a *nonzero vector space*.

Theorem 1.1.2. *Basic Consequences of Vector Space Axioms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then:

(i) the additive identity is unique;
(ii) each $\mathbf{v}\in\mathcal{V}$ has a unique additive inverse, denoted $-\mathbf{v}$;
(iii) $(\forall\,\mathbf{v}\in\mathcal{V})\;0\mathbf{v}=\mathbf{0}$;
(iv) $(\forall\,a\in\mathbb{F})\;a\mathbf{0}=\mathbf{0}$;
(v) $(\forall\,a\in\mathbb{F})(\forall\,\mathbf{v}\in\mathcal{V})\;[a\mathbf{v}=\mathbf{0}\iff(a=0\vee\mathbf{v}=\mathbf{0})]$.

Proof.

(i) If $\mathbf{0}$ and $\mathbf{0}'$ are additive identities, then $\mathbf{0}=\mathbf{0}+\mathbf{0}'=\mathbf{0}'$.
(ii) If $\mathbf{w},\mathbf{w}'$ are additive inverses of $\mathbf{v}$, then $\mathbf{w}=\mathbf{w}+\mathbf{0}=\mathbf{w}+(\mathbf{v}+\mathbf{w}')=(\mathbf{w}+\mathbf{v})+\mathbf{w}'=\mathbf{0}+\mathbf{w}'=\mathbf{w}'$.
(iii) $0\mathbf{v}=(0+0)\mathbf{v}=0\mathbf{v}+0\mathbf{v}$; after adding $-(0\mathbf{v})$ to both sides, $0\mathbf{v}=\mathbf{0}$.
(iv) $a\mathbf{0}=a(\mathbf{0}+\mathbf{0})=a\mathbf{0}+a\mathbf{0}$; after adding $-(a\mathbf{0})$ to both sides, $a\mathbf{0}=\mathbf{0}$.
(v) ($\Rightarrow$) Suppose $a\mathbf{v}=\mathbf{0}$. If $a=0$, then $a=0\vee\mathbf{v}=\mathbf{0}$. If $a\neq0$, then $\mathbf{v}=1\mathbf{v}=(a^{-1}a)\mathbf{v}=a^{-1}(a\mathbf{v})=a^{-1}\mathbf{0}=\mathbf{0}$ by (iv).
(v) ($\Leftarrow$) If $a=0$ or $\mathbf{v}=\mathbf{0}$, then the conclusion follows from (iii) or (iv).

Structural Commentary: The proof derives cancellation, zero actions, and inverse uniqueness from the axioms.
Key Insight: Zero and inverse behavior is forced by the axioms. □

Corollary 1.1.3. *Scalar Multiplication by -1 Is Negation*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then $(\forall\,\mathbf{v}\in\mathcal{V})\;(-1)\mathbf{v}=-\mathbf{v}$.

Proof. For $\mathbf{v}\in\mathcal{V}$, $\mathbf{v}+(-1)\mathbf{v}=1\mathbf{v}+(-1)\mathbf{v}=(1-1)\mathbf{v}=0\mathbf{v}$.
By Theorem 1.1.2, $\mathbf{v}+(-1)\mathbf{v}=\mathbf{0}$ and additive inverses are unique, so $(-1)\mathbf{v}=-\mathbf{v}$.
Structural Commentary: The proof identifies $(-1)\mathbf{v}$ as the unique additive inverse.
Key Insight: Negation is scalar action by -1. □

Example 1.1.4. *Function-Space Model*
Let $\mathbb{F}$ be a field and let X be a set. Set $\mathbb{F}^X:=\{f:X\to\mathbb{F}\}$. For $f,g\in\mathbb{F}^X$ and $a\in\mathbb{F}$, define $f+g$ and af pointwise.
Thus $(f+g)(x)=f(x)+g(x)$ and $(af)(x)=af(x)$ for $x\in X$. Then $\mathbb{F}^X$ is an $\mathbb{F}$-vector space.
Evaluating each axiom at $x\in X$ reduces it to the corresponding field identity, and $\mathbf{0}$ is the constant 0 function.
Key Insight: Pointwise operations inherit linear structure from the scalar field.

1.2 Subspaces and Spans

Definition 1.2.1. *Subspace*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$.
A *subspace* of $\mathcal{V}$ is a subset $\mathcal{U}\subseteq\mathcal{V}$ that is an $\mathbb{F}$-vector space under the inherited operations; write $\mathcal{U}\leq\mathcal{V}$.

Theorem 1.2.2. *Subspace Test*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $\emptyset\neq\mathcal{U}\subseteq\mathcal{V}$. Then $\mathcal{U}\leq\mathcal{V}\iff(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{U})(\forall\,c\in\mathbb{F})\;c\mathbf{u}+\mathbf{v}\in\mathcal{U}$.

Proof.

($\Rightarrow$) $(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{U})(\forall\,c\in\mathbb{F})\;c\mathbf{u}+\mathbf{v}\in\mathcal{U}$ by inherited scalar and additive closure.
($\Leftarrow$) Let C denote the stated closure condition, and choose $\mathbf{u}_0\in\mathcal{U}$. The identity $\mathbf{0}=(-1)\mathbf{u}_0+\mathbf{u}_0\in\mathcal{U}$ holds by C and Corollary 1.1.3.
$(\forall\,\mathbf{u}\in\mathcal{U})\;-\mathbf{u}=(-1)\mathbf{u}+\mathbf{0}\in\mathcal{U}$ by C and Corollary 1.1.3. $(\forall\,\mathbf{u}\in\mathcal{U})(\forall\,c\in\mathbb{F})\;c\mathbf{u}=c\mathbf{u}+\mathbf{0}\in\mathcal{U}$ by C.
$(\forall\,\mathbf{u},\mathbf{v}\in\mathcal{U})\;\mathbf{u}+\mathbf{v}=1\mathbf{u}+\mathbf{v}\in\mathcal{U}$ by C. The remaining vector-space axioms are inherited from $\mathcal{V}$, so $\mathcal{U}\leq\mathcal{V}$.

Structural Commentary: Nonemptiness supplies a zero; the single closure condition supplies inverses, scalar multiples, and sums.
Key Insight: One affine-looking closure test captures all subspace closure. □

Proposition 1.2.3. *Intersections of Subspaces*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$, let $\emptyset\neq I$, and let $\{\mathcal{U}_i\}_{i\in I}$ be a family of subspaces of $\mathcal{V}$. Then $\bigcap_{i\in I}\mathcal{U}_i\leq\mathcal{V}$.

Proof. Set $M := \bigcap_{i \in I} \mathcal{U}_i$. Then $\mathbf{0} \in M$ by $(\forall\, i \in I)\, \mathbf{0} \in \mathcal{U}_i$, so $M \neq \emptyset$.
$(\forall\, \mathbf{u}, \mathbf{v} \in M)(\forall\, c \in \mathbb{F})(\forall\, i \in I)\, c\mathbf{u} + \mathbf{v} \in \mathcal{U}_i$ by the definition of M and $\mathcal{U}_i \leq \mathcal{V}$.
Hence $(\forall\, \mathbf{u}, \mathbf{v} \in M)(\forall\, c \in \mathbb{F})\, c\mathbf{u} + \mathbf{v} \in M$, and $M \leq \mathcal{V}$ by Theorem 1.2.2.
Structural Commentary: The proof verifies nonemptiness and closure pointwise across the indexed family.
Key Insight: Shared closure survives intersection. □

Definition 1.2.4. *Lists, Linear Combinations, and Span*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

(i) For $m \in \mathbb{N}_0$, a *list* in $\mathcal{V}$ is $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$; the case $m = 0$ is the empty list $()$.

(ii) For $m \in \mathbb{N}_0$, $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, and $(a_1, \ldots, a_m) \in \mathbb{F}^m$, a *linear combination* of β is $\sum_{j=1}^{m} a_j \mathbf{v}_j$.
The case $m = 0$ is the empty linear combination, whose value is $\mathbf{0}$.

(iii) For $E \subseteq \mathcal{V}$, its *span* is the set of all finite linear combinations of entries from E.
Precisely, $\operatorname{span}(E) := \left\{ \sum_{j=1}^{m} a_j \mathbf{v}_j : m \in \mathbb{N}_0,\ (a_1, \ldots, a_m) \in \mathbb{F}^m,\ (\mathbf{v}_1, \ldots, \mathbf{v}_m) \in E^m \right\}$; hence $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.

(iv) For $m \in \mathbb{N}_0$ and $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, define $\operatorname{span}(\beta) := \operatorname{span}(\{\mathbf{v}_j : 1 \leq j \leq m\})$. For the empty list, $\operatorname{span}(()) = \{\mathbf{0}\}$.

(v) $E \subseteq \mathcal{V}$ *spans* $\mathcal{V}$ iff $\operatorname{span}(E) = \mathcal{V}$. A list β *spans* $\mathcal{V}$ iff $\operatorname{span}(\beta) = \mathcal{V}$.

Theorem 1.2.5. *Span is the Smallest Containing Subspace*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $E \subseteq \mathcal{V}$. Then:

(i) $\operatorname{span}(E) \leq \mathcal{V}$;
(ii) $E \subseteq \operatorname{span}(E)$;
(iii) $(\forall\, \mathcal{U} \leq \mathcal{V})\, [E \subseteq \mathcal{U} \Rightarrow \operatorname{span}(E) \subseteq \mathcal{U}]$.

Proof.

(i) The empty combination has value $\mathbf{0}$, so $\mathbf{0} \in \operatorname{span}(E)$. Let $\mathbf{x}, \mathbf{y} \in \operatorname{span}(E)$ and $c \in \mathbb{F}$.
Choose $m, n \in \mathbb{N}_0$ and finite linear-combination representations from E, $\mathbf{x} = \sum_{j=1}^{m} a_j \mathbf{e}_j$ and $\mathbf{y} = \sum_{k=1}^{n} b_k \mathbf{f}_k$.
Then $c\mathbf{x} + \mathbf{y} = \sum_{j=1}^{m} (ca_j)\mathbf{e}_j + \sum_{k=1}^{n} b_k \mathbf{f}_k \in \operatorname{span}(E)$ by concatenating the two lists. Thus $\operatorname{span}(E) \leq \mathcal{V}$ by Theorem 1.2.2.

(ii) $(\forall\, \mathbf{v} \in E)\, \mathbf{v} = 1\mathbf{v} \in \operatorname{span}(E)$, so $E \subseteq \operatorname{span}(E)$.

(iii) Let $\mathcal{U} \leq \mathcal{V}$ and $E \subseteq \mathcal{U}$. Set $P(m) := [(\forall\, (a_1, \ldots, a_m) \in \mathbb{F}^m)(\forall\, (\mathbf{e}_1, \ldots, \mathbf{e}_m) \in E^m)\ \sum_{j=1}^{m} a_j \mathbf{e}_j \in \mathcal{U}]$.
The base case $P(0)$ holds by $\mathbf{0} \in \mathcal{U}$. Assume $m \in \mathbb{N}_0$, $P(m)$, $(a_1, \ldots, a_{m+1}) \in \mathbb{F}^{m+1}$, and $(\mathbf{e}_1, \ldots, \mathbf{e}_{m+1}) \in E^{m+1}$.
Set $\mathbf{s} := \sum_{j=1}^{m} a_j \mathbf{e}_j$. Then $\mathbf{s} \in \mathcal{U}$ by $P(m)$. $a_{m+1}\mathbf{e}_{m+1} \in \mathcal{U}$ by $E \subseteq \mathcal{U}$ and scalar closure in $\mathcal{U}$.
Hence $\sum_{j=1}^{m+1} a_j \mathbf{e}_j = \mathbf{s} + a_{m+1}\mathbf{e}_{m+1} \in \mathcal{U}$ by additive closure, so $P(m+1)$.
$(\forall\, m \in \mathbb{N}_0)\, P(m)$ by induction. Hence $\operatorname{span}(E) \subseteq \mathcal{U}$ by the definition of span.

Structural Commentary: The proof verifies closure, generator inclusion, and minimality in statement order.
Key Insight: Finite linear combinations are exactly the closure forced by E. □

Proposition 1.2.6. *Monotonicity of Span*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$, and let $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$. Then $\mathcal{U} \subseteq \mathcal{W} \Rightarrow \operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.

Proof. Assume $\mathcal{U} \subseteq \mathcal{W}$. Then $\operatorname{span}(\mathcal{W}) \leq \mathcal{V}$ and $\mathcal{W} \subseteq \operatorname{span}(\mathcal{W})$ by Theorem 1.2.5(i)–(ii).
Thus $\mathcal{U} \subseteq \operatorname{span}(\mathcal{W})$, and $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$ by Theorem 1.2.5(iii).
Structural Commentary: The proof compares both generating sets inside the same containing subspace.
Key Insight: Minimality converts generator inclusion into span inclusion. □

Computation Method 1.2.7. *Subspace and Span Diagnostic*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $S \subseteq \mathcal{V}$. Return the output matching the task:

(i) *Subspace certificate*: establish $S \neq \emptyset$ and $(\forall\, \mathbf{u}, \mathbf{v} \in S)(\forall\, c \in \mathbb{F})\, c\mathbf{u} + \mathbf{v} \in S$. Then $S \leq \mathcal{V}$ by Theorem 1.2.2.

(ii) *Failure certificate*: if $S = \emptyset$, then $S \not\leq \mathcal{V}$. Otherwise exhibit $\mathbf{u}, \mathbf{v} \in S$ and $c \in \mathbb{F}$ with $c\mathbf{u} + \mathbf{v} \notin S$.
Then $S \not\leq \mathcal{V}$ by Theorem 1.2.2. The condition $\mathbf{0} \in S$ alone is insufficient.

(iii) *Generated subspace*: return $\operatorname{span}(S)$, the set of all finite linear combinations from S.
Precisely, $\operatorname{span}(S) = \left\{ \sum_{i=1}^{n} a_i \mathbf{s}_i : n \in \mathbb{N}_0,\ (a_1, \ldots, a_n) \in \mathbb{F}^n,\ (\mathbf{s}_1, \ldots, \mathbf{s}_n) \in S^n \right\}$, with the empty sum $\mathbf{0}$ at $n = 0$.

1.3 Linear Independence and Bases

Throughout this subsection, let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

Definition 1.3.1. *Linear Independence, Bases, and Finite-Dimensionality*

(i) A nonempty finite list $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *linearly independent* iff $(\forall\, (a_1, \ldots, a_m) \in \mathbb{F}^m)\, [\sum_{j=1}^{m} a_j \mathbf{v}_j = \mathbf{0} \Rightarrow a_1 = \cdots = a_m = 0]$.
The empty list is linearly independent. A finite list that is not linearly independent is *linearly dependent*.

(ii) A list β is a *basis* of $\mathcal{V}$ iff β is linearly independent and $\operatorname{span}(\beta) = \mathcal{V}$.

(iii) $\mathcal{V}$ is *finite-dimensional* iff $(\exists\, m \in \mathbb{N}_0)(\exists\, \beta \in \mathcal{V}^m)\ \operatorname{span}(\beta) = \mathcal{V}$. Otherwise $\mathcal{V}$ is *infinite-dimensional*.

Proposition 1.3.2. *Unique Coefficients for Independent Lists*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, $\mathbf{a} := (a_1, \ldots, a_m) \in \mathbb{F}^m$, $\mathbf{b} := (b_1, \ldots, b_m) \in \mathbb{F}^m$; $\sum_{j=1}^{m} a_j \mathbf{v}_j = \sum_{j=1}^{m} b_j \mathbf{v}_j$ iff $\mathbf{a} = \mathbf{b}$.

Proof.

($\Rightarrow$) $\sum_{j=1}^{m}(a_j - b_j)\mathbf{v}_j = \mathbf{0}$ by subtracting the two sums. Hence $(\forall\, j \in \{1,\dots,m\})\, a_j - b_j = 0$ by linear independence of β, so $\mathbf{a} = \mathbf{b}$.

($\Leftarrow$) The two sums are equal by equality of the coefficient tuples.

Structural Commentary: Subtracting two representations converts coefficient comparison into the defining zero-relation test.
Key Insight: Independence detects equality coefficient by coefficient. □

Proposition 1.3.3. *Unique Coordinate Representation*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Then β is a basis of $\mathcal{V}$ iff $(\forall\, \mathbf{v} \in \mathcal{V})(\exists!\,(a_1,\dots,a_n) \in \mathbb{F}^n)\, \mathbf{v} = \sum_{j=1}^{n} a_j\mathbf{v}_j$.

Proof.

($\Rightarrow$) Every $\mathbf{v} \in \mathcal{V}$ has a coefficient tuple by $\operatorname{span}(\beta) = \mathcal{V}$, and that tuple is unique by Proposition 1.3.2.

($\Leftarrow$) $\operatorname{span}(\beta) = \mathcal{V}$ by the existence clause. If $\sum_{j=1}^{n} a_j\mathbf{v}_j = \mathbf{0}$, then $\sum_{j=1}^{n} a_j\mathbf{v}_j = \sum_{j=1}^{n} 0\mathbf{v}_j$, so $(\forall\, j \in \{1,\dots,n\})\, a_j = 0$ by the uniqueness clause. Thus β is linearly independent and hence a basis.

Structural Commentary: Spanning supplies coordinate tuples, while independence separates any two such tuples.
Key Insight: A basis is precisely a list with universal and unique coordinates. □

Proposition 1.3.4. *Appending to a Dependent List*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ be linearly dependent in $\mathcal{V}$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\,(\mathbf{v}_1,\dots,\mathbf{v}_m,\mathbf{v})$ is linearly dependent.

Proof. Choose $(a_1,\dots,a_m) \in \mathbb{F}^m$ such that $\sum_{j=1}^{m} a_j\mathbf{v}_j = \mathbf{0}$ and $(\exists\, j \in \{1,\dots,m\})\, a_j \neq 0$ by dependence of β.

For $\mathbf{v} \in \mathcal{V}$, $\sum_{j=1}^{m} a_j\mathbf{v}_j + 0\mathbf{v} = \mathbf{0}$ is a nontrivial relation on $(\beta, \mathbf{v})$.

Structural Commentary: The original relation is embedded into the longer coefficient tuple by assigning zero to the appended entry.
Key Insight: An existing dependence relation survives every append. □

Definition 1.3.5. *Omitting a Vector from a List*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ be a nonempty list in $\mathcal{V}$. For $j \in \{1,\dots,m\}$, write $\beta_{\widehat{j}} := (\mathbf{v}_1,\dots,\widehat{\mathbf{v}_j},\dots,\mathbf{v}_m)$; if $m = 1$, then $\beta_{\widehat{1}} = ()$.

Proposition 1.3.6. *Omitting from an Independent List*

If $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ is a nonempty linearly independent list, then $(\forall\, j \in \{1,\dots,m\})$ $\beta_{\widehat{j}}$ is linearly independent.

Proof. Fix $j \in \{1,\dots,m\}$, and let $(a_i)_{i\neq j}$ satisfy $\sum_{i\neq j} a_i\mathbf{v}_i = \mathbf{0}$. Set $a_j := 0$.

Then $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$, so $(\forall\, i \in \{1,\dots,m\})\, a_i = 0$ by independence of β. Hence every coefficient in the original relation is zero.

Structural Commentary: Padding a sublist relation with a zero coefficient turns it into a relation on the original list.
Key Insight: Deletion cannot introduce a new cancellation. □

Proposition 1.3.7. *Adjoining and Removing Vectors from Lists*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_m)$ be a list in $\mathcal{V}$, and write $(\beta, \mathbf{v}) := (\mathbf{v}_1,\dots,\mathbf{v}_m,\mathbf{v})$.

(i) If β is linearly independent and $\mathbf{v} \notin \operatorname{span}(\beta)$, then $(\beta, \mathbf{v})$ is linearly independent.

(ii) If $(a_1,\dots,a_m) \in \mathbb{F}^m$, $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$, and $j \in \{1,\dots,m\}$ satisfies $a_j \neq 0$, then $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.
Hence $\operatorname{span}(\beta) = \mathcal{V}$ implies $\operatorname{span}(\beta_{\widehat{j}}) = \mathcal{V}$.

Proof.

(i) Let $\sum_{i=1}^{m} a_i\mathbf{v}_i + a\mathbf{v} = \mathbf{0}$. If $a \neq 0$, then $\mathbf{v} = -a^{-1}\sum_{i=1}^{m} a_i\mathbf{v}_i \in \operatorname{span}(\beta)$, contrary to hypothesis.
Hence $a = 0$, and then $(\forall\, i \in \{1,\dots,m\})\, a_i = 0$ by independence of β.

(ii) $\mathbf{v}_j = -a_j^{-1}\sum_{i\neq j} a_i\mathbf{v}_i \in \operatorname{span}(\beta_{\widehat{j}})$ by $a_j \neq 0$ and the stated relation.
Thus every vector of β lies in $\operatorname{span}(\beta_{\widehat{j}})$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$. The reverse inclusion is immediate because $\beta_{\widehat{j}}$ is a sublist of β.

Structural Commentary: The first argument isolates the appended coefficient; the second solves for a nonzero-coefficient entry.
Key Insight: Append outside the current span, or delete an entry already controlled by the others. □

Theorem 1.3.8. *Basis Characterizations*

Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Then the following are equivalent:

(i) β is a basis of $\mathcal{V}$;

(ii) $(\forall\, \mathbf{x} \in \mathcal{V})(\exists!\,(a_1,\dots,a_n) \in \mathbb{F}^n)\, \mathbf{x} = \sum_{j=1}^{n} a_j\mathbf{v}_j$;

(iii) $\operatorname{span}(\beta) = \mathcal{V}$, and no proper sublist of β spans $\mathcal{V}$;

(iv) β is linearly independent, and $(\forall\, \mathbf{x} \in \mathcal{V})$ $(\beta, \mathbf{x})$ is linearly dependent.

Proof.

(*i*) $\Rightarrow$ (*ii*) by Proposition 1.3.3.

(*ii*) $\Rightarrow$ (*iii*). The equality $\operatorname{span}(\beta) = \mathcal{V}$ holds by the existence clause.
Suppose a proper sublist γ of β spans $\mathcal{V}$, and choose an occurrence $\mathbf{v}_j$ omitted from γ.
A representation of $\mathbf{v}_j$ by γ extends to a β-coefficient tuple whose jth entry is 0, whereas the standard coefficient tuple for $\mathbf{v}_j$ has jth entry 1. This contradicts the uniqueness clause.

(*iii*) $\Rightarrow$ (*iv*). If β were linearly dependent, choose a nontrivial relation $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$ and an index j with $a_j \neq 0$.
Then $\beta_{\widehat{j}}$ would span $\mathcal{V}$ by Proposition 1.3.7(ii), contrary to minimality. Hence β is linearly independent.
For $\mathbf{x} \in \mathcal{V}$, choose $\mathbf{x} = \sum_{j=1}^{n} a_j \mathbf{v}_j$ by $\operatorname{span}(\beta) = \mathcal{V}$. Then $(a_1, \ldots, a_n, -1)$ is a nonzero relation tuple for $(\beta, \mathbf{x})$.

(*iv*) $\Rightarrow$ (*i*). If $\operatorname{span}(\beta) \neq \mathcal{V}$, choose $\mathbf{x} \in \mathcal{V} \setminus \operatorname{span}(\beta)$. Then $(\beta, \mathbf{x})$ is linearly independent by Proposition 1.3.7(i), contrary to (iv). Thus $\operatorname{span}(\beta) = \mathcal{V}$, and β is a basis by the independence clause in (iv).

Structural Commentary: The implication cycle passes through unique coordinates, minimal spanning, and maximal independence.
Key Insight: A basis is both an irredundant spanning list and an independence-maximal list. □

Lemma 1.3.9. *Replacement Lemma*
Let $\mathcal{V} \neq \{\mathbf{0}\}$, let $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_r)$ span $\mathcal{V}$, and let $\mathbf{0} \neq \mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i$. For $j \in \{1, \ldots, r\}$, set $\beta_{j \leftarrow \mathbf{v}} := (\mathbf{u}_1, \ldots, \mathbf{u}_{j-1}, \mathbf{v}, \mathbf{u}_{j+1}, \ldots, \mathbf{u}_r)$.

(i) $(\exists\, j \in \{1, \ldots, r\})\, c_j \neq 0$.
(ii) If $c_j \neq 0$, then $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$.
(iii) If $c_j \neq 0$ and β is a basis, then $\beta_{j \leftarrow \mathbf{v}}$ is a basis.
(iv) If $r \geq 2$, β is a basis, $k \in \{1, \ldots, r-1\}$, and $\mathbf{v} \notin \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, then $(\exists\, j \in \{k+1, \ldots, r\})$ $\beta_{j \leftarrow \mathbf{v}}$ is a basis.

Proof.

(i) If $(\forall\, j \in \{1, \ldots, r\})\, c_j = 0$, then $\mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i = \mathbf{0}$, contrary to hypothesis.

(ii) Assume $c_j \neq 0$. Then $\mathbf{u}_j = c_j^{-1} \left(\mathbf{v} - \sum_{i \neq j} c_i \mathbf{u}_i \right) \in \operatorname{span}(\beta_{j \leftarrow \mathbf{v}})$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{j \leftarrow \mathbf{v}})$.
Conversely, $\mathbf{v} \in \operatorname{span}(\beta)$, so $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) \subseteq \operatorname{span}(\beta)$. Hence $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$.

(iii) Assume $c_j \neq 0$ and β is a basis. Let $\sum_{i<j} b_i \mathbf{u}_i + a\mathbf{v} + \sum_{i>j} b_i \mathbf{u}_i = \mathbf{0}$. Then $\sum_{i \neq j} (b_i + a c_i) \mathbf{u}_i + a c_j \mathbf{u}_j = \mathbf{0}$ by substituting $\mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i$.
Thus $a c_j = 0$ and $(\forall\, i \neq j)\, b_i + a c_i = 0$ by independence of β. Hence $a = 0$ by $c_j \neq 0$, and then $(\forall\, i \neq j)\, b_i = 0$.
Therefore $\beta_{j \leftarrow \mathbf{v}}$ is linearly independent; it spans $\mathcal{V}$ by (ii), so it is a basis.

(iv) If $(\forall\, j \in \{k+1, \ldots, r\})\, c_j = 0$, then $\mathbf{v} = \sum_{i=1}^{k} c_i \mathbf{u}_i \in \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, contrary to hypothesis.

(v) Choose $j \in \{k+1, \ldots, r\}$ with $c_j \neq 0$. Then $\beta_{j \leftarrow \mathbf{v}}$ is a basis by (iii).

Structural Commentary: Solving for a nonzero-coefficient entry preserves span; comparing coefficients preserves independence.
Key Insight: A vector may replace precisely an entry on which its expansion genuinely depends. □

Lemma 1.3.10. *Linear Dependence Lemma*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly dependent with $\mathbf{v}_1 \neq \mathbf{0}$. Then $(\exists\, j \in \{2, \ldots, m\})\, [\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}) \wedge \operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)]$.

Proof.

(i) *Terminal coefficient*: Choose a nontrivial relation $\sum_{i=1}^{m} a_i \mathbf{v}_i = \mathbf{0}$, and set $j := \max\{i : a_i \neq 0\}$.
The equality $j = 1$ would imply $\mathbf{v}_1 = \mathbf{0}$ by $a_1 \mathbf{v}_1 = \mathbf{0}$ and $a_1 \neq 0$, contrary to hypothesis. Hence $j \in \{2, \ldots, m\}$.

(ii) *Earlier-span representation*: $(\forall\, i > j)\, a_i = 0$ by the choice of j, and therefore $\mathbf{v}_j = -a_j^{-1} \sum_{i=1}^{j-1} a_i \mathbf{v}_i \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$.

(iii) *Span preservation*: This representation rewrites every linear combination of β without $\mathbf{v}_j$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$.
The reverse inclusion holds because $\beta_{\widehat{j}}$ is a sublist of β. Hence $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.

Structural Commentary: The last nonzero relation coefficient identifies a later vector in the earlier span that can be removed.
Key Insight: Choose the final active coefficient to preserve both order and span. □

Theorem 1.3.11. *Independent Lists Are No Longer Than Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is linearly independent, and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$, then $m \leq n$.

Proof.

(i) *Contradictory regime*: Suppose $m > n$. Then $n \geq 1$: if $n = 0$, then $\mathbf{u}_1 \in \mathcal{V} = \operatorname{span}(\gamma) = \{\mathbf{0}\}$, contrary to independence of α.

(ii) *Successive replacement*: Construct spanning lists γ_k of length n with designated occurrences of $\mathbf{u}_1, \ldots, \mathbf{u}_k$ for $0 \leq k \leq n$.
Set $\gamma_0 := \gamma$. Suppose $k < n$ and γ_k has been constructed. Then $\mathbf{u}_{k+1} \notin \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ by independence of α.
Expand $\mathbf{u}_{k+1}$ in the spanning list γ_k. Some coefficient at a nondesignated position is nonzero; otherwise the expansion would place $\mathbf{u}_{k+1}$ in $\operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. The vector $\mathbf{u}_{k+1}$ is nonzero by independence of α. Replace the identified position by $\mathbf{u}_{k+1}$.
The resulting list γ_{k+1} spans $\mathcal{V}$ by Lemma 1.3.9(ii), has length n, and retains every earlier designated occurrence.

(iii) *Terminal list*: The designated occurrences of $\mathbf{u}_1, \ldots, \mathbf{u}_n$ occupy all n positions of γ_n. Hence $\operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n) = \operatorname{span}(\gamma_n) = \mathcal{V}$.

(iv) *Contradiction*: Since $m > n$, $\mathbf{u}_{n+1} \in \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$ by the terminal equality, contrary to independence of α.

Structural Commentary: Each replacement reserves one independent vector while keeping the spanning-list length and span fixed.
Key Insight: After all spanning positions are reserved, no further independent direction can remain. □

Theorem 1.3.12. *Reducing Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, then every spanning list in $\mathcal{V}$ contains a basis sublist.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a spanning list of $\mathcal{V}$. If β is independent, it is a basis. Otherwise remove one vector without changing span: omit $\mathbf{v}_1$ if $\mathbf{v}_1 = \mathbf{0}$; if $\mathbf{v}_1 \neq \mathbf{0}$, remove a suitable $\mathbf{v}_j$ by Lemma 1.3.10. The resulting sublist still spans $\mathcal{V}$ and has length one less. Repeating at most n times reaches a spanning independent sublist, hence a basis.
Structural Commentary: Each deletion shortens the list without changing span, so finite descent reaches an independent sublist.
Key Insight: Repeatedly remove a certified redundancy until only a basis remains. □

Theorem 1.3.13. *Extending Independent Lists*
If $\mathcal{V}$ is finite-dimensional, then every linearly independent list in $\mathcal{V}$ extends to a basis of $\mathcal{V}$.

Proof.

(i) *Setup*: Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, and choose a finite spanning list $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ of $\mathcal{V}$.
(ii) *Construction*: Starting from $\delta_0 := \beta$, inspect $\mathbf{w}_1, \dots, \mathbf{w}_n$ in order
At stage j, append $\mathbf{w}_j$ exactly when $\mathbf{w}_j \notin \operatorname{span}(\delta_{j-1})$, and call the resulting list δ_j.
Every δ_j is linearly independent by induction using Proposition 1.3.7(i) at each append.
(iii) *Verification*: $(\forall\, j \in \{1, \dots, n\})\ \mathbf{w}_j \in \operatorname{span}(\delta_n)$ because each $\mathbf{w}_j$ was either already spanned when inspected or appended, and subsequent stages only append. Thus $\operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_n) \subseteq \operatorname{span}(\delta_n)$ by Theorem 1.2.5(iii).
Hence $\operatorname{span}(\delta_n) = \mathcal{V}$, and the independent list δ_n is a basis extending β.

Structural Commentary: The finite scan preserves independence while absorbing each chosen spanning generator.
Key Insight: Append exactly the generators not already controlled by the current span. □

Corollary 1.3.14. *Bases and Subspaces in Finite Dimension*
If $\mathcal{V}$ is finite-dimensional, then:

(i) $\mathcal{V}$ has a basis;
(ii) $(\forall \mathcal{U} \leq \mathcal{V})$ $\mathcal{U}$ is finite-dimensional.

Proof. Choose a finite spanning list $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ of $\mathcal{V}$.

(i) A basis of $\mathcal{V}$ is contained in γ by Theorem 1.3.12.
(ii) Fix $\mathcal{U} \leq \mathcal{V}$, and set $\beta_0 := ()$. Whenever $\operatorname{span}(\beta_k) \neq \mathcal{U}$, choose $\mathbf{u}_{k+1} \in \mathcal{U} \setminus \operatorname{span}(\beta_k)$ and set $\beta_{k+1} := (\beta_k, \mathbf{u}_{k+1})$.
Every constructed list is linearly independent by induction using Proposition 1.3.7(i).
The construction cannot reach β_{n+1} by Theorem 1.3.11, because γ is a spanning list of length n.
Hence it stops at some $k \leq n$ with $\operatorname{span}(\beta_k) = \mathcal{U}$. Thus β_k is a finite basis of $\mathcal{U}$, so $\mathcal{U}$ is finite-dimensional.

Structural Commentary: Basis reduction handles $\mathcal{V}$; the ambient length bound forces every subspace construction to terminate.
Key Insight: Ambient finite spanning length bounds the size of every independent subspace list. □

Definition 1.3.15. *Independent Sets and Hamel Bases*
A subset $I \subseteq \mathcal{V}$ is *linearly independent* iff every finite list of distinct vectors in I is linearly independent.
A subset $\mathcal{B} \subseteq \mathcal{V}$ is a *Hamel basis* of $\mathcal{V}$ iff $\mathcal{B}$ is linearly independent and $\operatorname{span}(\mathcal{B}) = \mathcal{V}$.
For $a : \mathcal{B} \to \mathbb{F}$, define $\operatorname{supp}_{\mathcal{B}}(a) := \{\mathbf{b} \in \mathcal{B} : a(\mathbf{b}) \neq 0\}$. A *finite $\mathcal{B}$-expansion* of $\mathbf{v} \in \mathcal{V}$ is a function $a : \mathcal{B} \to \mathbb{F}$ with finite support such that $\mathbf{v} = \sum_{\mathbf{b} \in \operatorname{supp}_{\mathcal{B}}(a)} a(\mathbf{b})\mathbf{b}$. For a finite Hamel basis, any ordering of $\mathcal{B}$ is an ordered basis.

Proposition 1.3.16. *Hamel Coordinate Representation*
If $\mathcal{B}$ is a Hamel basis of $\mathcal{V}$, then every $\mathbf{v} \in \mathcal{V}$ has a unique finite $\mathcal{B}$-expansion.

Proof. Fix $\mathbf{v} \in \mathcal{V}$. Choose $m \in \mathbb{N}_0$, $(c_1, \dots, c_m) \in \mathbb{F}^m$, and $(\mathbf{b}_1, \dots, \mathbf{b}_m) \in \mathcal{B}^m$ such that $\mathbf{v} = \sum_{j=1}^{m} c_j \mathbf{b}_j$ by $\operatorname{span}(\mathcal{B}) = \mathcal{V}$.
For $\mathbf{b} \in \mathcal{B}$, set $J_{\mathbf{b}} := \{j \in \{1, \dots, m\} : \mathbf{b}_j = \mathbf{b}\}$, and define $a : \mathcal{B} \to \mathbb{F}$ by $a(\mathbf{b}) := \sum_{j \in J_{\mathbf{b}}} c_j$.
Then $\operatorname{supp}_{\mathcal{B}}(a) \subseteq \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$, and $\mathbf{v} = \sum_{\mathbf{b} \in \operatorname{supp}_{\mathcal{B}}(a)} a(\mathbf{b})\mathbf{b}$ by regrouping equal entries. Hence a is a finite $\mathcal{B}$-expansion of $\mathbf{v}$.
For uniqueness, let $a, a' : \mathcal{B} \to \mathbb{F}$ be finite $\mathcal{B}$-expansions of $\mathbf{v}$, and set $F := \operatorname{supp}_{\mathcal{B}}(a) \cup \operatorname{supp}_{\mathcal{B}}(a')$.
Then F is finite and $\sum_{\mathbf{b} \in F} (a(\mathbf{b}) - a'(\mathbf{b}))\mathbf{b} = \mathbf{0}$. Thus $(\forall\, \mathbf{b} \in F)\, a(\mathbf{b}) = a'(\mathbf{b})$ by independence of $\mathcal{B}$. Both vanish on $\mathcal{B} \setminus F$, so $a = a'$.
Structural Commentary: Existence consolidates repeated generators; uniqueness compares their combined finite support.
Key Insight: Hamel coordinates are ordinary independent-list coefficients indexed by finite support. □

Theorem 1.3.17. *Hamel Basis Extension*
Assume Zorn's lemma. Then:

(i) if $I \subseteq S \subseteq \mathcal{V}$, I is linearly independent, and $\operatorname{span}(S) = \mathcal{V}$, then $(\exists\, \mathcal{B})\, [I \subseteq \mathcal{B} \subseteq S \wedge \mathcal{B}$ is a Hamel basis of $\mathcal{V}]$;
(ii) $\mathcal{V}$ has a Hamel basis;
(iii) every linearly independent subset of $\mathcal{V}$ extends to a Hamel basis of $\mathcal{V}$;
(iv) every spanning subset of $\mathcal{V}$ contains a Hamel basis of $\mathcal{V}$.

Proof.

(i) Let $\mathscr{P} := \{E : I \subseteq E \subseteq S,\ E \text{ is linearly independent}\}$, ordered by inclusion.
Then $I \in \mathscr{P}$. Let $\mathcal{C} \subseteq \mathscr{P}$ be a nonempty chain, and set $U := \bigcup_{E \in \mathcal{C}} E$. Then $I \subseteq U \subseteq S$.
Let $m \in \mathbb{N}$, and consider a relation $\sum_{k=1}^{m} a_k \mathbf{u}_k = \mathbf{0}$ among distinct vectors of U.
Choose $E_1, \dots, E_m \in \mathcal{C}$ such that $(\forall\, k \in \{1, \dots, m\})\, \mathbf{u}_k \in E_k$.
Some $E_* \in \mathcal{C}$ contains $E_1 \cup \cdots \cup E_m$ by finite induction on the chain order.
Thus $(\forall\, k \in \{1, \dots, m\})\, a_k = 0$ by independence of E_*, so $U \in \mathscr{P}$.
Hence every nonempty chain has an upper bound in $\mathscr{P}$; the empty chain is bounded by I.
Choose a maximal $\mathcal{B} \in \mathscr{P}$ by Zorn's lemma. Suppose $s \in S \setminus \operatorname{span}(\mathcal{B})$.
In a finite relation among distinct vectors of $\mathcal{B} \cup \{s\}$, a nonzero coefficient of s would imply $s \in \operatorname{span}(\mathcal{B})$, while a zero coefficient leaves a relation in $\mathcal{B}$. Hence $\mathcal{B} \cup \{s\}$ is linearly independent, contrary to maximality of $\mathcal{B}$.
Therefore $S \subseteq \operatorname{span}(\mathcal{B})$, so $\operatorname{span}(\mathcal{B}) = \mathcal{V}$ by $\operatorname{span}(S) = \mathcal{V}$. Thus $I \subseteq \mathcal{B} \subseteq S$, and $\mathcal{B}$ is a Hamel basis.
(ii) $\mathcal{V}$ has a Hamel basis by (i) with the linearly independent set $I = \emptyset$ and the spanning set $S = \mathcal{V}$.
(iii) Every prescribed linearly independent subset extends to a Hamel basis by (i) with $S = \mathcal{V}$.
(iv) Every prescribed spanning subset contains a Hamel basis by (i) with $I = \emptyset$.

Structural Commentary: Chain unions preserve finite independence, and maximality rules out every unspanned element of S.
Key Insight: Finitary independence is stable under chain unions, so Zorn maximality forces a Hamel basis. □

1.4 Dimension and Coordinates

Throughout this subsection, let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

Theorem 1.4.1. *Basis Length Is Well-Defined*
If $\mathcal{V}$ is finite-dimensional and β_1, β_2 are bases of $\mathcal{V}$ of lengths m, n, respectively, then $m = n$.

Proof. The inequality $m \leq n$ holds by Theorem 1.3.11, with β_1 independent and β_2 spanning.
After interchanging the bases, $n \leq m$ holds by the same theorem. Hence $m = n$.
Structural Commentary: Using both basis roles proves the opposite length inequalities.
Key Insight: Symmetric comparison makes basis length intrinsic. □

Definition 1.4.2. *Dimension of a Finite-Dimensional Space*
Assume $\mathcal{V}$ is finite-dimensional. Choose a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ by Corollary 1.3.14, and define $\dim(\mathcal{V}) := n$.
This value is independent of β by Theorem 1.4.1, and $\mathcal{V}$ is called *n-dimensional*. Since $()$ is a basis of $\{\mathbf{0}\}$, $\dim(\{\mathbf{0}\}) = 0$.

Proposition 1.4.3. *Subspace Dimension Criterion*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U} \leq \mathcal{V}$. Then:
(i) $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$;
(ii) $\dim(\mathcal{U}) = \dim(\mathcal{V})$ *iff* $\mathcal{U} = \mathcal{V}$.

Proof. Choose a basis $\delta = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$ by Corollary 1.3.14.
(*i*) The list δ is independent in $\mathcal{V}$, while β spans $\mathcal{V}$. Hence $m \leq n$ by Theorem 1.3.11, so $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$.
(*ii*) ($\Rightarrow$) Suppose $\dim(\mathcal{U}) = \dim(\mathcal{V})$. Then $m = n$. Extend δ to a basis of $\mathcal{V}$ by Theorem 1.3.13.
The extension adds no entry because its length is n by Theorem 1.4.1. Thus $\mathcal{U} = \operatorname{span}(\delta) = \mathcal{V}$.
(*ii*) ($\Leftarrow$) If $\mathcal{U} = \mathcal{V}$, then δ is a basis of $\mathcal{V}$, so $m = n$ by Theorem 1.4.1. Hence $\dim(\mathcal{U}) = \dim(\mathcal{V})$.
Structural Commentary: Length comparison proves the bound; basis extension tests equality against a new ambient direction.
Key Insight: A subspace of full ambient dimension cannot be proper. □

Proposition 1.4.4. *Full-Dimension Basis Criteria*
Let $\mathcal{V}$ be finite-dimensional and set $n := \dim(\mathcal{V})$. Then:
(i) if γ is a spanning list in $\mathcal{V}$ of length n, then γ is a basis;
(ii) if α is a linearly independent list in $\mathcal{V}$ of length n, then α is a basis.

Proof.
(*i*) A basis sublist β of γ exists by Theorem 1.3.12. Its length is n by Definition 1.4.2 and Theorem 1.4.1.
Since γ also has length n, the sublist β equals γ. Hence γ is a basis.
(*ii*) Extend α to a basis β of $\mathcal{V}$ by Theorem 1.3.13. The length of β is n by Definition 1.4.2.
Hence the extension adds no entry to the length-n list α. Hence $\alpha = \beta$ is a basis.
Structural Commentary: At full length, basis reduction cannot delete and basis extension cannot append.
Key Insight: At the ambient dimension, either half of the basis condition forces the other. □

Theorem 1.4.5. *Dimension as Basis Cardinality*
Assume the Choice Principle 6.1.2. If $\mathcal{B}$ and $\mathcal{C}$ are Hamel bases of $\mathcal{V}$, then $|\mathcal{B}| = |\mathcal{C}|$.

Proof.
(i) *Finite case*: Assume $|\mathcal{B}| = n \in \mathbb{N}_0$, and order $\mathcal{B}$ as a spanning list β. Then $\mathcal{V}$ is finite-dimensional.
The set $\mathcal{C}$ cannot contain $n+1$ distinct vectors by Theorem 1.3.11, because they would form an independent list longer than β.
Thus $\mathcal{C}$ is finite; otherwise finite recursion would select such vectors. If γ orders $\mathcal{C}$ and has length k, then $k = n$ by Theorem 1.4.1, and $|\mathcal{C}| = k = n$ by Proposition 6.3.15(i). Interchanging the bases handles finite $\mathcal{C}$.
(ii) *Support incidence*: Assume $\mathcal{B}$ and $\mathcal{C}$ are infinite. For $\mathbf{c} \in \mathcal{C}$, let $a_{\mathbf{c}} : \mathcal{B} \to \mathbb{F}$ be its unique finite $\mathcal{B}$-expansion by Proposition 1.3.16, and set $S_{\mathbf{c}} := \operatorname{supp}_{\mathcal{B}}(a_{\mathbf{c}})$. Fix $\mathbf{b} \in \mathcal{B}$ and write $\mathbf{b} = \sum_{j=1}^{m} d_j \mathbf{c}_j$ with $\mathbf{c}_j \in \mathcal{C}$.
If $\mathbf{b} \notin \bigcup_{\mathbf{c} \in \mathcal{C}} S_{\mathbf{c}}$, then substituting the $\mathcal{B}$-expansions makes the coefficient at $\mathbf{b}$ equal to $\sum_{j=1}^{m} d_j a_{\mathbf{c}_j}(\mathbf{b}) = 0$, whereas the standard expansion of $\mathbf{b}$ has coefficient 1. This discrepancy is impossible by uniqueness in Proposition 1.3.16.
Hence $(\forall\, \mathbf{b} \in \mathcal{B})(\exists\, \mathbf{c} \in \mathcal{C})\, \mathbf{b} \in S_{\mathbf{c}}$. Set $D := \{(\mathbf{c}, \mathbf{b}) \in \mathcal{C} \times \mathcal{B} : \mathbf{b} \in S_{\mathbf{c}}\}$.
(iii) *Cardinal comparison*: Choose $(\mathbf{c}_{\mathbf{b}})_{\mathbf{b} \in \mathcal{B}}$ such that $(\forall\, \mathbf{b} \in \mathcal{B})\, [\mathbf{c}_{\mathbf{b}} \in \mathcal{C} \wedge \mathbf{b} \in S_{\mathbf{c}_{\mathbf{b}}}]$ by Principle 6.1.2.
Then $\mathbf{b} \mapsto (\mathbf{c}_{\mathbf{b}}, \mathbf{b})$ injects $\mathcal{B}$ into D. Fix a well-order of $\mathcal{B}$ by Principle 6.3.2(iv), let $p_{\mathbf{c}}(\mathbf{b}) \in \mathbb{N}$ be position of $\mathbf{b}$ in the induced ordering of the finite set $S_{\mathbf{c}}$. The map $(\mathbf{c}, \mathbf{b}) \mapsto (\mathbf{c}, p_{\mathbf{c}}(\mathbf{b}))$ injects D into $\mathcal{C} \times \mathbb{N}$. Hence $|\mathcal{B}| \leq |D| \leq |\mathcal{C} \times \mathbb{N}|$ by Proposition 6.5.2(i).
By Proposition 6.5.2(iv) and Proposition 6.3.15(iii), $|\mathcal{C} \times \mathbb{N}| = |\mathcal{C}|\, \aleph_0$.
Since $\mathcal{C}$ is infinite, $\aleph_0 \leq |\mathcal{C}|$ by Proposition 6.3.15(iv), so $|\mathcal{C}|\, \aleph_0 = |\mathcal{C}|$ by Theorem 6.6.3(ii). Therefore $|\mathcal{B}| \leq |\mathcal{C}|$.
(iv) *Symmetry*: Interchanging $\mathcal{B}$ and $\mathcal{C}$ proves $|\mathcal{C}| \leq |\mathcal{B}|$. These inequalities correspond to injections in both directions by Proposition 6.5.2(i), so $\mathcal{B} \sim_{\#} \mathcal{C}$ by Theorem 6.2.3. Hence $|\mathcal{B}| = |\mathcal{C}|$ by Proposition 6.3.14(ii).
Structural Commentary: Finite supports form an incidence relation that choice and well-ordering encode by cardinal injections.
Key Insight: Record each support occurrence by one basis vector and one finite position. □

Definition 1.4.6. *Hamel Dimension*
Assume Zorn's lemma and the Choice Principle 6.1.2. Choose a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.3.17, and define $\dim(\mathcal{V}) := |\mathcal{B}|$.
The value is independent of $\mathcal{B}$ by Theorem 1.4.5. If $\mathcal{V}$ has a finite ordered basis of length n, its underlying set is a Hamel basis of cardinality n by Proposition 6.3.15(i); hence this definition extends Definition 1.4.2.

1.5 Linear Maps and Operators

Linear maps preserve vector-space operations. Pointwise operations and composition form their coordinate-free algebra, while basis values determine finite-dimensional maps.

Throughout this subsection, $\mathcal{U}, \mathcal{V}, \mathcal{W}$ are vector spaces over the same fixed field $\mathbb{F}$.

Definition 1.5.1. *Linear Map*
A function $T : \mathcal{V} \to \mathcal{W}$ is *linear* iff $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})(\forall\, a, b \in \mathbb{F})\, T(a\mathbf{u} + b\mathbf{v}) = aT\mathbf{u} + bT\mathbf{v}$.
The collection of linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted $\mathcal{L}(\mathcal{V}, \mathcal{W})$. Write $\mathcal{L}(\mathcal{V}) := \mathcal{L}(\mathcal{V}, \mathcal{V})$; its elements are *operators* on $\mathcal{V}$.

Definition 1.5.2. *Zero and Identity Maps*
The *zero map* from $\mathcal{V}$ to $\mathcal{W}$ is $0_{\mathcal{V},\mathcal{W}} : \mathcal{V} \to \mathcal{W}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\, 0_{\mathcal{V},\mathcal{W}}\mathbf{v} = \mathbf{0}_{\mathcal{W}}$.
The *identity map* on $\mathcal{V}$ is $I_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\, I_{\mathcal{V}}\mathbf{v} = \mathbf{v}$.

Proposition 1.5.3. *Linear Maps Send Zero to Zero*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$.

Proof. By linearity, $T\mathbf{0}_{\mathcal{V}} = T(\mathbf{0}_{\mathcal{V}} + \mathbf{0}_{\mathcal{V}}) = T\mathbf{0}_{\mathcal{V}} + T\mathbf{0}_{\mathcal{V}}$. Hence $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$ by cancellation.
Structural Commentary: Additivity converts zero preservation into a cancellation identity.
Key Insight: The additive identity is fixed by every linear map. □

Definition 1.5.4. *Operations on Linear Maps*
For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, define pointwise $(S + T)\mathbf{v} := S\mathbf{v} + T\mathbf{v}$ and $(aT)\mathbf{v} := a(T\mathbf{v})$ for $\mathbf{v} \in \mathcal{V}$.

Proposition 1.5.5. *Zero, Identity, and the Linear-Map Space*
(i) $0_{\mathcal{V},\mathcal{W}} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(ii) $I_{\mathcal{V}} \in \mathcal{L}(\mathcal{V})$.
(iii) The operations above make $\mathcal{L}(\mathcal{V}, \mathcal{W})$ a vector space over $\mathbb{F}$.

Proof.
(i) For $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $0_{\mathcal{V},\mathcal{W}}(a\mathbf{u} + b\mathbf{v}) = \mathbf{0}_{\mathcal{W}} = a(0_{\mathcal{V},\mathcal{W}}\mathbf{u}) + b(0_{\mathcal{V},\mathcal{W}}\mathbf{v})$ by Theorem 1.1.2(iv). Hence $0_{\mathcal{V},\mathcal{W}} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(ii) For $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $I_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v}) = a\mathbf{u} + b\mathbf{v} = aI_{\mathcal{V}}\mathbf{u} + bI_{\mathcal{V}}\mathbf{v}$, so $I_{\mathcal{V}} \in \mathcal{L}(\mathcal{V})$.
(iii) For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a, b \in \mathbb{F}$, set $A := aS + bT$. For $c, d \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $A(c\mathbf{u} + d\mathbf{v}) = a(cS\mathbf{u} + dS\mathbf{v}) + b(cT\mathbf{u} + dT\mathbf{v}) = cA\mathbf{u} + dA\mathbf{v}$, so $A \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Hence $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is closed under addition and scalar multiplication.
For $\mathbf{v} \in \mathcal{V}$, $(T + 0_{\mathcal{V},\mathcal{W}})\mathbf{v} = T\mathbf{v} = (0_{\mathcal{V},\mathcal{W}} + T)\mathbf{v}$, and $(T + (-1)T)\mathbf{v} = \mathbf{0}_{\mathcal{W}}$ by Corollary 1.1.3.
Thus $T + 0_{\mathcal{V},\mathcal{W}} = T = 0_{\mathcal{V},\mathcal{W}} + T$ and $T + (-1)T = 0_{\mathcal{V},\mathcal{W}}$ by Assumption 3.1.2.
The remaining axioms hold pointwise in $\mathcal{W}$ and hence for the maps by Assumption 3.1.2.
Structural Commentary: Closure is algebraic; function extensionality lifts codomain axioms pointwise.
Key Insight: Function spaces inherit vector-space structure from their codomain. □

Definition 1.5.6. *Product of Linear Maps*
For $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define their *product* as the function $ST : \mathcal{U} \to \mathcal{W}$ satisfying $(\forall\, \mathbf{u} \in \mathcal{U})\, (ST)\mathbf{u} = S(T\mathbf{u})$.

Proposition 1.5.7. *Algebra of Linear Map Products*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
(i) $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$.
(ii) If $\mathcal{X}$ is a vector space over $\mathbb{F}$ and $R \in \mathcal{L}(\mathcal{W}, \mathcal{X})$, then $(RS)T = R(ST)$.
(iii) $TI_{\mathcal{U}} = T = I_{\mathcal{V}}T$.
(iv) If $S_1, S_2 \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $(S_1 + S_2)T = S_1T + S_2T$.
(v) If $T_1, T_2 \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, then $S(T_1 + T_2) = ST_1 + ST_2$.

Proof.
(i) For $\mathbf{u}, \mathbf{x} \in \mathcal{U}$ and $a, b \in \mathbb{F}$, $(ST)(a\mathbf{u} + b\mathbf{x}) = S(aT\mathbf{u} + bT\mathbf{x}) = aS(T\mathbf{u}) + bS(T\mathbf{x}) = a(ST)\mathbf{u} + b(ST)\mathbf{x}$, so $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$.
(ii) For $\mathbf{u} \in \mathcal{U}$, $((RS)T)\mathbf{u} = R(S(T\mathbf{u})) = (R(ST))\mathbf{u}$. Hence $(RS)T = R(ST)$ by Assumption 3.1.2.
(iii) For $\mathbf{u} \in \mathcal{U}$, $(TI_{\mathcal{U}})\mathbf{u} = T\mathbf{u} = (I_{\mathcal{V}}T)\mathbf{u}$. Hence $TI_{\mathcal{U}} = T = I_{\mathcal{V}}T$ by Assumption 3.1.2.
(iv) For $\mathbf{u} \in \mathcal{U}$, $((S_1 + S_2)T)\mathbf{u} = S_1(T\mathbf{u}) + S_2(T\mathbf{u}) = (S_1T + S_2T)\mathbf{u}$. Hence $(S_1 + S_2)T = S_1T + S_2T$ by Assumption 3.1.2.
(v) For $\mathbf{u} \in \mathcal{U}$, $(S(T_1 + T_2))\mathbf{u} = S(T_1\mathbf{u} + T_2\mathbf{u}) = S(T_1\mathbf{u}) + S(T_2\mathbf{u}) = (ST_1 + ST_2)\mathbf{u}$.
Hence $S(T_1 + T_2) = ST_1 + ST_2$ by Assumption 3.1.2.
Structural Commentary: Pointwise evaluation reduces composition identities to codomain equalities.
Key Insight: Composition and pointwise addition form the algebra of linear maps. □

Proposition 1.5.8. *Linear Maps Are Determined by a Finite Basis*
If $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ and $\mathbf{w}_1, \ldots, \mathbf{w}_n \in \mathcal{W}$, then $(\exists!\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}))(\forall\, j \in \{1, \ldots, n\})\, T(\mathbf{v}_j) = \mathbf{w}_j$.

Proof. By Proposition 1.3.3, each $\mathbf{v} \in \mathcal{V}$ has a unique expression $\mathbf{v} = \sum_{j=1}^{n} a_j\mathbf{v}_j$. Define $T(\mathbf{v}) := \sum_{j=1}^{n} a_j\mathbf{w}_j$.
The map T is well-defined by coordinate uniqueness, and $(\forall\, j \in \{1, \ldots, n\})\, T\mathbf{v}_j = \mathbf{w}_j$ by the standard coordinate tuples.
For $\mathbf{x} = \sum_{j=1}^{n} a_j\mathbf{v}_j$, $\mathbf{y} = \sum_{j=1}^{n} b_j\mathbf{v}_j$, and $c, d \in \mathbb{F}$, $c\mathbf{x} + d\mathbf{y} = \sum_{j=1}^{n} (ca_j + db_j)\mathbf{v}_j$, so $T(c\mathbf{x} + d\mathbf{y}) = cT\mathbf{x} + dT\mathbf{y}$. Thus $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
For $A \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ with the prescribed basis values, $A\mathbf{v} = \sum_{j=1}^{n} a_jA\mathbf{v}_j = \sum_{j=1}^{n} a_j\mathbf{w}_j = T\mathbf{v}$ for $\mathbf{v} \in \mathcal{V}$. Hence $A = T$ by Assumption 3.1.2.
Structural Commentary: Coordinate uniqueness supports construction; function extensionality completes uniqueness.
Key Insight: Arbitrary basis images extend to exactly one linear map. □

Principle 1.5.9. *Coordinate-Free Principle*
By Proposition 1.5.8, basis images determine a linear map before coordinates encode it.
This viewpoint organizes finite-dimensional representations but does not make the map depend on the chosen basis.

1.6 Kernels, Ranges, and Map Properties

Throughout this subsection, $\mathcal{U}, \mathcal{V}, \mathcal{W}$ are vector spaces over the same fixed field $\mathbb{F}$.

Definition 1.6.1. *Kernel and Range of a Linear Map*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define its *kernel*, or *nullspace*, by $\ker(T) := \{\mathbf{v} \in \mathcal{V} : T\mathbf{v} = \mathbf{0}_{\mathcal{W}}\}$ and its range by $\text{range}(T) := \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\}$.

Proposition 1.6.2. *Injectivity and the Kernel*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is injective iff $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$.

Proof.
($\Rightarrow$) Suppose T is injective. For $\mathbf{v} \in \ker(T)$, $T\mathbf{v} = \mathbf{0}_{\mathcal{W}} = T\mathbf{0}_{\mathcal{V}}$ by Proposition 1.5.3, so $\mathbf{v} = \mathbf{0}_{\mathcal{V}}$ by injectivity.
Moreover, $\mathbf{0}_{\mathcal{V}} \in \ker(T)$ by the displayed equality; hence $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$.
($\Leftarrow$) Suppose $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$. For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ with $T\mathbf{v} = T\mathbf{w}$, $T(\mathbf{v} - \mathbf{w}) = \mathbf{0}_{\mathcal{W}}$ by linearity, so $\mathbf{v} - \mathbf{w} \in \ker(T)$ and $\mathbf{v} = \mathbf{w}$.
Structural Commentary: The forward direction compares kernel vectors with zero; the reverse moves equal images into the kernel.
Key Insight: Differences convert image collisions into kernel vectors. □

Proposition 1.6.3. *Kernel and Range Are Subspaces*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then:
(i) $\ker(T)$ is a subspace of $\mathcal{V}$.
(ii) $\text{range}(T)$ is a subspace of $\mathcal{W}$.

Proof.
(i) The kernel contains $\mathbf{0}_{\mathcal{V}}$ by Proposition 1.5.3. For $\mathbf{u}, \mathbf{v} \in \ker(T)$ and $a \in \mathbb{F}$, $T(a\mathbf{u} + \mathbf{v}) = aT\mathbf{u} + T\mathbf{v} = \mathbf{0}_{\mathcal{W}}$, so $a\mathbf{u} + \mathbf{v} \in \ker(T)$. Therefore $\ker(T) \leq \mathcal{V}$ by Theorem 1.2.2.
(ii) The range contains $\mathbf{0}_{\mathcal{W}} = T\mathbf{0}_{\mathcal{V}}$ by Proposition 1.5.3. For $T\mathbf{u}, T\mathbf{v} \in \text{range}(T)$ and $a \in \mathbb{F}$, $aT\mathbf{u} + T\mathbf{v} = T(a\mathbf{u} + \mathbf{v}) \in \text{range}(T)$. Therefore $\text{range}(T) \leq \mathcal{W}$ by Theorem 1.2.2.
Structural Commentary: Each proof verifies nonemptiness and linear-combination closure before invoking the subspace test.
Key Insight: Linearity stabilizes the zero fiber and the image. □

Definition 1.6.4. *Invertible Linear Map and Isomorphism*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is *invertible* iff $(\exists\, S \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\, [ST = I_{\mathcal{V}} \wedge TS = I_{\mathcal{W}}]$.
Such an S is an *inverse* of T. An invertible linear map is an *isomorphism*; if an isomorphism $\mathcal{V} \to \mathcal{W}$ exists, write $\mathcal{V} \cong \mathcal{W}$.

Proposition 1.6.5. *Inverse Is Unique*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is invertible, then $(\exists!\, S \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\, [ST = I_{\mathcal{V}} \wedge TS = I_{\mathcal{W}}]$. Denote this map by T^{-1}.

Proof. An inverse exists by invertibility of T. Suppose $S_1, S_2 \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ satisfy $S_1T = I_{\mathcal{V}} = S_2T$ and $TS_1 = I_{\mathcal{W}} = TS_2$.
Then $S_1 = S_1I_{\mathcal{W}} = S_1(TS_2) = (S_1T)S_2 = I_{\mathcal{V}}S_2 = S_2$ by Proposition 1.5.7(ii)–(iii) and the candidate inverse identities.
Structural Commentary: Existence is definitional; associativity compares any two inverse candidates.
Key Insight: A two-sided inverse has no freedom. □

Proposition 1.6.6. *Inverse of a Product Reverses Order*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ be invertible.
(i) ST is invertible.
(ii) $(ST)^{-1} = T^{-1}S^{-1}$.

Proof.
(i) Set $R := T^{-1}S^{-1}$. Then $R \in \mathcal{L}(\mathcal{W}, \mathcal{U})$ by Proposition 1.5.7(i).
The identity $R(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I_{\mathcal{U}}$ holds by Proposition 1.5.7(ii)–(iii) and the inverse identities.
Similarly, $(ST)R = S(TT^{-1})S^{-1} = SS^{-1} = I_{\mathcal{W}}$ by Proposition 1.5.7(ii)–(iii) and the inverse identities. Hence ST is invertible.
(ii) Since R is a two-sided inverse of ST, $(ST)^{-1} = T^{-1}S^{-1}$ by Proposition 1.6.5.
Structural Commentary: The reversed product is checked against both inverse identities, then identified by uniqueness.
Key Insight: Composition order reverses under inversion. □

Proposition 1.6.7. *Invertibility and Bijectivity*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the map T is invertible iff it is both injective and surjective.

Proof.
($\Rightarrow$) Let $S = T^{-1}$. If $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $T\mathbf{v} = T\mathbf{w}$, then $\mathbf{v} = (ST)\mathbf{v} = S(T\mathbf{v}) = S(T\mathbf{w}) = (ST)\mathbf{w} = \mathbf{w}$, so T is injective.
If $\mathbf{y} \in \mathcal{W}$, then $\mathbf{y} = (TS)\mathbf{y} = T(S\mathbf{y})$, so T is surjective.
($\Leftarrow$) Suppose T is injective and surjective. For $\mathbf{y} \in \mathcal{W}$, a unique $\mathbf{v} \in \mathcal{V}$ satisfies $T\mathbf{v} = \mathbf{y}$ by surjectivity and injectivity; define $S\mathbf{y} := \mathbf{v}$.
For $a, b \in \mathbb{F}$ and $\mathbf{y}_1, \mathbf{y}_2 \in \mathcal{W}$, $S(a\mathbf{y}_1 + b\mathbf{y}_2) = aS\mathbf{y}_1 + bS\mathbf{y}_2$ by injectivity because both sides have T-image $a\mathbf{y}_1 + b\mathbf{y}_2$.
Hence $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$. By construction, $(\forall\, \mathbf{y} \in \mathcal{W})\, (TS)\mathbf{y} = \mathbf{y}$, so $TS = I_{\mathcal{W}}$ by Assumption 3.1.2.
For $\mathbf{v} \in \mathcal{V}$, $T((ST)\mathbf{v}) = T\mathbf{v}$, whence $(ST)\mathbf{v} = \mathbf{v}$ by injectivity. Therefore $ST = I_{\mathcal{V}}$ by Assumption 3.1.2, and T is invertible.
Structural Commentary: Surjectivity constructs the reverse map; injectivity makes it linear and enforces both inverse identities.
Key Insight: Unique preimages turn a bijection into a linear inverse. □

Definition 1.6.8. *Morphism Terminology for Linear Maps*
Within vector spaces, the following terminology applies to $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$:
(i) T is a *homomorphism*.
(ii) T is an *endomorphism* iff $\mathcal{V} = \mathcal{W}$.
(iii) T is a *monomorphism*, or *embedding*, iff T is injective; equivalently, $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ by Proposition 1.6.2.
(iv) T is an *epimorphism* iff T is surjective, equivalently $\text{range}(T) = \mathcal{W}$.
(v) T is an *isomorphism* iff T is invertible; equivalently, T is injective and surjective by Proposition 1.6.7.
(vi) T is an *automorphism* iff $\mathcal{V} = \mathcal{W}$ and T is invertible.

1.7 Rank-Nullity and Dimension Criteria

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Theorem 1.7.1. *Range from Basis Images*
Let $\mathcal{V}$ be finite-dimensional with basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) $\operatorname{range}(T) = \operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$.
(ii) $\operatorname{range}(T)$ *is finite-dimensional.*
(iii) $\dim(\operatorname{range}(T)) \leq n$.

Proof.

(i) ($\Rightarrow$) For $\mathbf{y} \in \operatorname{range}(T)$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{y} = T\mathbf{v}$ and write $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j$.

Then $\mathbf{y} = \sum_{j=1}^{n} c_j T\mathbf{v}_j$ by linearity, so $\operatorname{range}(T) \subseteq \operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$.

(i) ($\Leftarrow$) $\sum_{j=1}^{n} a_j T\mathbf{v}_j = T(\sum_{j=1}^{n} a_j \mathbf{v}_j) \in \operatorname{range}(T)$ for $a_1, \dots, a_n \in \mathbb{F}$, so the reverse inclusion holds.

(ii) The finite list in (i) spans $\operatorname{range}(T)$, so the range is finite-dimensional.

(i) A basis sublist of the list in (i) exists by Theorem 1.3.12. Its length is at most n, and therefore $\dim(\operatorname{range}(T)) \leq n$.

Structural Commentary: Two inclusions identify the range; basis reduction then converts its generating list into a dimension bound.
Key Insight: A linear map transports every basis expansion to its image. □

Definition 1.7.2. *Rank and Nullity*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Define $\operatorname{rank}(T) := \dim(\operatorname{range}(T))$ and $\operatorname{nullity}(T) := \dim(\ker(T))$.
The rank is finite by Theorem 1.7.1(ii); the nullity is finite by Proposition 1.6.3 and Corollary 1.3.14.

Theorem 1.7.3. *Rank-Nullity Theorem*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then:

(i) $\operatorname{range}(T)$ *is finite-dimensional.*
(ii) $\dim(\mathcal{V}) = \operatorname{nullity}(T) + \operatorname{rank}(T)$.

Proof. The kernel has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ by Proposition 1.6.3 and Corollary 1.3.14.
Extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r)$ of $\mathcal{V}$ by Theorem 1.3.13.
For $\mathbf{v} = \sum_{i=1}^{m} a_i \mathbf{u}_i + \sum_{j=1}^{r} b_j \mathbf{w}_j$, $T\mathbf{v} = \sum_{j=1}^{r} b_j T\mathbf{w}_j$ by linearity, so $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ spans $\operatorname{range}(T)$.
If $\sum_{j=1}^{r} c_j T\mathbf{w}_j = \mathbf{0}_{\mathcal{W}}$, then $\mathbf{x} := \sum_{j=1}^{r} c_j \mathbf{w}_j \in \ker(T) = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_m)$.
Thus $\mathbf{x}$ also has an extended-basis representation with every $\mathbf{w}_j$-coefficient zero, so $c_1 = \cdots = c_r = 0$ by Proposition 1.3.3.
Hence $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is a basis of $\operatorname{range}(T)$.

(i) The basis $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is finite, so $\operatorname{range}(T)$ is finite-dimensional.

(ii) The three bases have lengths m, r, and $m + r$, respectively. Hence $\operatorname{nullity}(T) = m$, $\operatorname{rank}(T) = r$, and $\dim(\mathcal{V}) = m + r$.

Structural Commentary: Extending a kernel basis produces complementary vectors whose images form a basis of the range.
Key Insight: Every domain basis direction is either killed or detected. □

Corollary 1.7.4. *Maps to Smaller Spaces Are Not Injective*
If finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ satisfy $\dim(\mathcal{V}) > \dim(\mathcal{W})$, then no $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is injective.

Proof. Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Proposition 1.6.3 and Proposition 1.4.3, $\operatorname{rank}(T) \leq \dim(\mathcal{W})$.
By Theorem 1.7.3, $\operatorname{nullity}(T) = \dim(\mathcal{V}) - \operatorname{rank}(T) \geq \dim(\mathcal{V}) - \dim(\mathcal{W}) > 0$.
Thus $\ker(T) \neq \{\mathbf{0}_{\mathcal{V}}\}$, so T is not injective by Proposition 1.6.2.
Structural Commentary: The range bound feeds rank-nullity, forcing a nontrivial kernel.
Key Insight: Excess domain dimension must collapse. □

Corollary 1.7.5. *Maps to Larger Spaces Are Not Surjective*
If finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ satisfy $\dim(\mathcal{V}) < \dim(\mathcal{W})$, then no $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is surjective.

Proof. Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 1.7.3, $\operatorname{rank}(T) = \dim(\mathcal{V}) - \operatorname{nullity}(T) \leq \dim(\mathcal{V}) < \dim(\mathcal{W})$. Since $\operatorname{range}(T) \leq \mathcal{W}$ by Proposition 1.6.3, $\operatorname{range}(T) \neq \mathcal{W}$ by Proposition 1.4.3. Hence T is not surjective.
Structural Commentary: Rank-nullity bounds the range below the codomain dimension.
Key Insight: A smaller domain cannot cover a larger codomain. □

Corollary 1.7.6. *Equal-Dimension Kernel-Range Criterion*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) = \dim(\mathcal{W})$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ *iff* $\operatorname{range}(T) = \mathcal{W}$.
(ii) T *is injective iff T is surjective.*

Proof.

(i) ($\Rightarrow$) If $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$, then $\operatorname{nullity}(T) = 0$ and $\operatorname{rank}(T) = \dim(\mathcal{V}) = \dim(\mathcal{W})$ by Theorem 1.7.3.
Since $\operatorname{range}(T) \leq \mathcal{W}$ by Proposition 1.6.3, $\operatorname{range}(T) = \mathcal{W}$ by Proposition 1.4.3.

(i) ($\Leftarrow$) If $\operatorname{range}(T) = \mathcal{W}$, then $\operatorname{rank}(T) = \dim(\mathcal{W}) = \dim(\mathcal{V})$ and $\operatorname{nullity}(T) = 0$ by Theorem 1.7.3.
The zero subspace and $\ker(T)$ both have dimension zero, so they are equal by Proposition 1.4.3.

(ii) ($\Rightarrow$) If T is injective, then $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$ by Proposition 1.6.2. By (i), $\operatorname{range}(T) = \mathcal{W}$, so T is surjective.

(ii) ($\Leftarrow$) If T is surjective, then $\operatorname{range}(T) = \mathcal{W}$. By (i), $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$, so T is injective by Proposition 1.6.2.

Structural Commentary: Rank-nullity exchanges kernel loss with range deficit; the kernel criterion translates this to maps.
Key Insight: Equal dimensions leave no room for one-sided failure. □

Theorem 1.7.7. *Finite-Dimensional Invertibility Criteria*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then the following are equivalent:

(i) T *is invertible;*
(ii) T *is injective;*
(iii) T *is surjective.*

Proof.

$(i) \Rightarrow (ii)$ The map T is injective by Proposition 1.6.7.
$(ii) \Rightarrow (iii)$ The map T is surjective by Corollary 1.7.6, applied with the same domain and codomain.
$(iii) \Rightarrow (i)$ The map T is injective by Corollary 1.7.6, hence bijective and therefore invertible by Proposition 1.6.7.

Structural Commentary: Bijectivity supplies the outer implications; equal dimension links injectivity and surjectivity.
Key Insight: A nonzero kernel is the sole finite-dimensional obstruction to operator invertibility. □

Proposition 1.7.8. *Isomorphisms Preserve Bases*
Let $T : \mathcal{V} \to \mathcal{W}$ be an isomorphism and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Set $T\beta = (T\mathbf{v}_1, \dots, T\mathbf{v}_n)$. The following are equivalent:

(i) β *is a basis of* $\mathcal{V}$.
(ii) $T\beta$ *is a basis of* $\mathcal{W}$.

Proof. The map T is injective and surjective by Proposition 1.6.7.

$(i) \Rightarrow (ii)$ For $\mathbf{w} \in \mathcal{W}$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{w} = T\mathbf{v}$ and write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$. Then $\mathbf{w} = \sum_{j=1}^{n} a_j T\mathbf{v}_j$ by linearity, so $T\beta$ spans $\mathcal{W}$.
If $\sum_{j=1}^{n} a_j T\mathbf{v}_j = \mathbf{0}_{\mathcal{W}}$, then $T(\sum_{j=1}^{n} a_j \mathbf{v}_j) = \sum_{j=1}^{n} a_j T\mathbf{v}_j = \mathbf{0}_{\mathcal{W}} = T\mathbf{0}_{\mathcal{V}}$ by linearity and Proposition 1.5.3.
Injectivity and independence of β imply $a_1 = \cdots = a_n = 0$. Hence $T\beta$ is a basis of $\mathcal{W}$.

$(ii) \Rightarrow (i)$ For $\mathbf{v} \in \mathcal{V}$, write $T\mathbf{v} = \sum_{j=1}^{n} a_j T\mathbf{v}_j = T(\sum_{j=1}^{n} a_j \mathbf{v}_j)$ by linearity. By injectivity, $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$, so β spans $\mathcal{V}$.
If $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}_{\mathcal{V}}$, then $\sum_{j=1}^{n} a_j T\mathbf{v}_j = \mathbf{0}_{\mathcal{W}}$ by linearity and Proposition 1.5.3.
Independence of $T\beta$ implies $a_1 = \cdots = a_n = 0$. Hence β is a basis of $\mathcal{V}$.

Structural Commentary: Surjectivity transports spanning representations; injectivity preserves and reflects linear relations.
Key Insight: Isomorphisms preserve both halves of the basis condition. □

Theorem 1.7.9. *Isomorphism and Dimension*
Finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ are isomorphic iff $\dim(\mathcal{V}) = \dim(\mathcal{W})$.

Proof.

$(\Rightarrow)$ Let $T : \mathcal{V} \to \mathcal{W}$ be an isomorphism. Choose a basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of $\mathcal{V}$ by Corollary 1.3.14.
Then $T\beta$ is a basis of $\mathcal{W}$ by Proposition 1.7.8, so $\dim(\mathcal{V}) = n = \dim(\mathcal{W})$.
$(\Leftarrow)$ Suppose $\dim(\mathcal{V}) = \dim(\mathcal{W}) = n$, and choose bases $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ by Corollary 1.3.14.
Choose $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfying $(\forall\, j \in \{1, \dots, n\})\, T\mathbf{v}_j = \mathbf{w}_j$ by Proposition 1.5.8. By Theorem 1.7.1(i), $\operatorname{range}(T) = \operatorname{span}(\gamma) = \mathcal{W}$.
or $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j \in \ker(T)$, $\mathbf{0}_{\mathcal{W}} = T\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{w}_j$, so $a_1 = \cdots = a_n = 0$ by independence of γ.
Thus $\ker(T) = \{\mathbf{0}_{\mathcal{V}}\}$, and T is injective by Proposition 1.6.2.
It is surjective by its range identity, hence invertible by Proposition 1.6.7 and therefore an isomorphism.

Structural Commentary: The forward direction transports a basis; the reverse prescribes an isomorphism on matching bases.
Key Insight: Dimension counts the coordinates preserved by an isomorphism. □

2 Direct Sums, Quotients, and Decomposition

2.1 Sums and Direct Sums

Sums combine subspaces inside a common ambient space, while products and external sums assemble separate spaces. Directness records when these decompositions are unique.

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 2.1.1. *Sums of Subspaces*
Let $m \in \mathbb{N}$ and let $\mathcal{U}_1, \dots, \mathcal{U}_m \leq \mathcal{V}$.

(i) Their *sum* is $\mathcal{U}_1 + \cdots + \mathcal{U}_m := \left\{ \sum_{j=1}^{m} \mathbf{u}_j : (\forall\, j \in \{1, \dots, m\})\, \mathbf{u}_j \in \mathcal{U}_j \right\}$.
(ii) For $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$, $\mathcal{U} + \mathcal{W} = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in \mathcal{U},\ \mathbf{w} \in \mathcal{W}\}$.
(iii) The sum is *direct* iff $(\forall\, \mathbf{v} \in \mathcal{U}_1 + \cdots + \mathcal{U}_m)(\exists!\, (\mathbf{u}_1, \dots, \mathbf{u}_m) \in \mathcal{U}_1 \times \cdots \times \mathcal{U}_m)\, \mathbf{v} = \sum_{j=1}^{m} \mathbf{u}_j$. A direct sum is denoted $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Proposition 2.1.2. *Sum of Subspaces Is the Smallest Containing Subspace*
Let $m \in \mathbb{N}$, let $\mathcal{U}_1, \dots, \mathcal{U}_m \leq \mathcal{V}$, and set $S := \sum_{j=1}^{m} \mathcal{U}_j$.

(i) $S \leq \mathcal{V}$, $(\forall\, j \in \{1, \dots, m\})\, \mathcal{U}_j \subseteq S$, *and* $(\forall\, \mathcal{W} \leq \mathcal{V})[(\forall\, j \in \{1, \dots, m\})\, \mathcal{U}_j \subseteq \mathcal{W} \Rightarrow S \subseteq \mathcal{W}]$.
(ii) $S = \operatorname{span}(\bigcup_{j=1}^{m} \mathcal{U}_j)$.

Proof.

(i) $(\forall\, j \in \{1,\dots,m\})\, \mathbf{0}_{\mathcal{V}} \in \mathcal{U}_j$, so $\mathbf{0}_{\mathcal{V}} \in S$. Let $\mathbf{x}, \mathbf{y} \in S$ and $c \in \mathbb{F}$.

Choose $(\mathbf{u}_1,\dots,\mathbf{u}_m), (\mathbf{v}_1,\dots,\mathbf{v}_m) \in \mathcal{U}_1 \times \cdots \times \mathcal{U}_m$ with $\mathbf{x} = \sum_{j=1}^{m} \mathbf{u}_j$ and $\mathbf{y} = \sum_{j=1}^{m} \mathbf{v}_j$.

Then $c\mathbf{x} + \mathbf{y} = \sum_{j=1}^{m} (c\mathbf{u}_j + \mathbf{v}_j) \in S$ by $(\forall\, j \in \{1,\dots,m\})\, c\mathbf{u}_j + \mathbf{v}_j \in \mathcal{U}_j$. Hence $S \leq \mathcal{V}$ by Theorem 1.2.2.

For $j \in \{1,\dots,m\}$ and $\mathbf{u} \in \mathcal{U}_j$, the sum with jth term $\mathbf{u}$ and all other terms $\mathbf{0}_{\mathcal{V}}$ belongs to S, so $\mathcal{U}_j \subseteq S$.
If $\mathcal{W} \leq \mathcal{V}$ and $(\forall\, j \in \{1,\dots,m\})\, \mathcal{U}_j \subseteq \mathcal{W}$, then each defining finite sum for S belongs to $\mathcal{W}$ by additive closure; hence $S \subseteq \mathcal{W}$.

(ii) Set $E := \bigcup_{j=1}^{m} \mathcal{U}_j$. Then $\operatorname{span}(E) \subseteq S$ by Theorem 1.2.5(iii) and part (i).

The inclusions $E \subseteq \operatorname{span}(E) \leq \mathcal{V}$ hold by Theorem 1.2.5(i)–(ii), so $S \subseteq \operatorname{span}(E)$ by part (i). Thus $S = \operatorname{span}(E)$.

Structural Commentary: Closure and minimality establish the sum; span minimality identifies the same subspace.
Key Insight: Both constructions are forced by the same generators. □

Perspective 2.1.3. *Sum Is Not Union*
Proposition 2.1.2 identifies $\mathcal{U} + \mathcal{W}$ with the linear closure of $\mathcal{U} \cup \mathcal{W}$. The union itself contains only vectors already in one summand.
If neither summand contains the other, choose $\mathbf{u} \in \mathcal{U} \setminus \mathcal{W}$ and $\mathbf{w} \in \mathcal{W} \setminus \mathcal{U}$.
Then $\mathbf{u} + \mathbf{w} \notin \mathcal{U} \cup \mathcal{W}$, since membership in either summand would place the other chosen vector there.
Thus the sum supplies exactly the mixed vectors missing from the union.

Theorem 2.1.4. *Finite Unions of Proper Subspaces*

Let $n \in \mathbb{N}$ and let $\mathcal{V} \neq \{\mathbf{0}_{\mathcal{V}}\}$ be over an infinite field $\mathbb{F}$. If $\mathcal{U}_1,\dots,\mathcal{U}_n \subsetneq \mathcal{V}$ are subspaces, then $\mathcal{V} \neq \bigcup_{j=1}^{n} \mathcal{U}_j$.

Proof.

(i) *Minimal cover*: Assume $\mathcal{V} = \bigcup_{j=1}^{n} \mathcal{U}_j$. Choose a minimal subcover $\mathcal{W}_1,\dots,\mathcal{W}_k$ and set $K := \{1,\dots,k\}$.

Then $(\forall\, i \in K)\, \mathcal{W}_i \not\subseteq \bigcup_{j \in K\setminus\{i\}} \mathcal{W}_j$ by minimality. The case $k = 1$ contradicts $\mathcal{W}_1 \subsetneq \mathcal{V}$; hence $k \geq 2$.

(ii) *Affine line*: Choose $\mathbf{w} \in \mathcal{W}_1 \setminus \bigcup_{j=2}^{k} \mathcal{W}_j$ and $\mathbf{v} \in \mathcal{V} \setminus \mathcal{W}_1$. Since $k \geq 2$ and $\mathbf{0}_{\mathcal{V}} \in \mathcal{W}_2$, $\mathbf{w} \neq \mathbf{0}_{\mathcal{V}}$. Set $A := \{\mathbf{v} + r\mathbf{w} : r \in \mathbb{F}\}$.

The parametrization $r \mapsto \mathbf{v} + r\mathbf{w}$ is injective, so A is infinite by the infinitude of $\mathbb{F}$.

(iii) *Intersections*: If $\mathbf{v} + r\mathbf{w} \in \mathcal{W}_1$, then $\mathbf{v} = (\mathbf{v} + r\mathbf{w}) - r\mathbf{w} \in \mathcal{W}_1$, a contradiction; hence $A \cap \mathcal{W}_1 = \emptyset$.
For $j \in \{2,\dots,k\}$, if $r_1 \neq r_2$ and $\mathbf{v} + r_1\mathbf{w}, \mathbf{v} + r_2\mathbf{w} \in \mathcal{W}_j$, then $(r_1 - r_2)\mathbf{w} \in \mathcal{W}_j$, so $\mathbf{w} \in \mathcal{W}_j$, a contradiction. Thus $|A \cap \mathcal{W}_j| \leq 1$.

(iv) *Contradiction*: The set $A \cap \bigcup_{j=1}^{k} \mathcal{W}_j$ is finite by part (iii), whereas $A \subseteq \mathcal{V} = \bigcup_{j=1}^{k} \mathcal{W}_j$ would make it equal to the infinite set A.

Therefore the assumed cover does not exist.

Structural Commentary: A minimal cover selects a direction whose affine line meets every summand finitely.
Key Insight: An infinite affine line defeats a finite cover. □

Definition 2.1.5. *Subspace Poset, Meet, and Join*

(i) The *subspace poset* is $\operatorname{Sub}(\mathcal{V}) := \{\mathcal{U} \subseteq \mathcal{V} : \mathcal{U} \leq \mathcal{V}\}$, ordered by inclusion.
(ii) For $\{\mathcal{U}_i\}_{i\in I} \subseteq \operatorname{Sub}(\mathcal{V})$, its *meet* is $\bigwedge_{i\in I} \mathcal{U}_i := \bigcap_{i\in I} \mathcal{U}_i$.
(iii) For $\{\mathcal{U}_i\}_{i\in I} \subseteq \operatorname{Sub}(\mathcal{V})$, its *join* is $\bigvee_{i\in I} \mathcal{U}_i := \sum_{i\in I} \mathcal{U}_i := \operatorname{span}(\bigcup_{i\in I} \mathcal{U}_i)$.
(iv) The empty meet is $\mathcal{V}$, and the empty join is $\{\mathbf{0}_{\mathcal{V}}\}$.

Proposition 2.1.6. *Subspace Meet and Join*
Let $\{\mathcal{U}_i\}_{i\in I} \subseteq \operatorname{Sub}(\mathcal{V})$.

(i) $\bigwedge_{i\in I} \mathcal{U}_i = \inf\{\mathcal{U}_i : i \in I\}$.

(ii) $\bigvee_{i\in I} \mathcal{U}_i = \sup\{\mathcal{U}_i : i \in I\}$.

(iii) $\bigvee_{i\in I} \mathcal{U}_i = \left\{ \sum_{i\in F} \mathbf{u}_i : F \subseteq I,\ |F| < \infty,\ (\forall\, i \in F)\, \mathbf{u}_i \in \mathcal{U}_i \right\}$, *where the empty sum is $\mathbf{0}_{\mathcal{V}}$.*

Proof.

(i) If $I = \emptyset$, the empty meet is $\mathcal{V}$, the greatest lower bound of the empty family. Assume $I \neq \emptyset$ and set $M := \bigcap_{i\in I} \mathcal{U}_i$.

Then $M \leq \mathcal{V}$ by Proposition 1.2.3, $(\forall\, i \in I)\, M \subseteq \mathcal{U}_i$, and $(\forall \mathcal{W} \leq \mathcal{V})[(\forall\, i \in I)\, \mathcal{W} \subseteq \mathcal{U}_i \Rightarrow \mathcal{W} \subseteq M]$. Hence $M = \inf\{\mathcal{U}_i : i \in I\}$.

(ii) Set $J := \operatorname{span}(\bigcup_{i\in I} \mathcal{U}_i)$. Then $J \leq \mathcal{V}$ and $(\forall\, i \in I)\, \mathcal{U}_i \subseteq J$ by Theorem 1.2.5(i)–(ii).

If $\mathcal{W} \leq \mathcal{V}$ and $(\forall\, i \in I)\, \mathcal{U}_i \subseteq \mathcal{W}$, then $\bigcup_{i\in I} \mathcal{U}_i \subseteq \mathcal{W}$, so $J \subseteq \mathcal{W}$ by Theorem 1.2.5(iii).

Thus J is the least upper bound, including $J = \{\mathbf{0}_{\mathcal{V}}\}$ when $I = \emptyset$.

(iii) Every stated finite sum is a linear combination of vectors in $\bigcup_{i\in I} \mathcal{U}_i$, so it belongs to J.

Conversely, each $\mathbf{x} \in J$ is a finite linear combination of vectors from that union.
Group terms assigned to the same index; linear closure of $\mathcal{U}_i$ replaces each nonempty group by one vector in $\mathcal{U}_i$.
The empty combination represents $\mathbf{0}_{\mathcal{V}}$, so the displayed representation set equals J.

Structural Commentary: Order bounds identify meet and join; grouping span terms establishes finite representation.
Key Insight: Intersection preserves constraints; span combines directions. □

Theorem 2.1.7. *Subspace Lattice*
The poset $\mathrm{Sub}(\mathcal{V})$ *has the following properties.*

(i) It is a complete lattice. For $\{\mathcal{U}_i\}_{i\in I} \subseteq \mathrm{Sub}(\mathcal{V})$, $\bot = \{\mathbf{0}_\mathcal{V}\}$, $\top = \mathcal{V}$, $\bigwedge_{i\in I} \mathcal{U}_i = \bigcap_{i\in I} \mathcal{U}_i$, *and* $\bigvee_{i\in I} \mathcal{U}_i = \sum_{i\in I} \mathcal{U}_i = \mathrm{span}(\bigcup_{i\in I} \mathcal{U}_i)$.

(ii) It is modular: $(\forall \mathcal{U},\mathcal{S},\mathcal{T} \leq \mathcal{V})[\mathcal{U} \leq \mathcal{S} \Rightarrow \mathcal{S} \cap (\mathcal{T}+\mathcal{U}) = (\mathcal{S}\cap\mathcal{T})+\mathcal{U}]$.

Proof.

(i) The arbitrary meet and join formulas hold by Proposition 2.1.6(i)–(ii).
The empty formulas identify $\top = \mathcal{V}$ and $\bot = \{\mathbf{0}_\mathcal{V}\}$, so $\mathrm{Sub}(\mathcal{V})$ is complete.

(ii) Let $\mathcal{U} \leq \mathcal{S}$. If $\mathbf{x} \in (\mathcal{S}\cap\mathcal{T})+\mathcal{U}$, write $\mathbf{x} = \mathbf{y}+\mathbf{u}$ with $\mathbf{y} \in \mathcal{S}\cap\mathcal{T}$ and $\mathbf{u} \in \mathcal{U}$. Then $\mathbf{x} \in \mathcal{S}\cap(\mathcal{T}+\mathcal{U})$.
Conversely, if $\mathbf{x} \in \mathcal{S}\cap(\mathcal{T}+\mathcal{U})$, write $\mathbf{x} = \mathbf{t}+\mathbf{u}$ with $\mathbf{t} \in \mathcal{T}$ and $\mathbf{u} \in \mathcal{U}$. Then $\mathbf{t} = \mathbf{x}-\mathbf{u} \in \mathcal{S}\cap\mathcal{T}$ by $\mathbf{x},\mathbf{u} \in \mathcal{S}$, so $\mathbf{x} \in (\mathcal{S}\cap\mathcal{T})+\mathcal{U}$.

Structural Commentary: Completeness follows from the bound formulas; two inclusions establish modularity.
Key Insight: A contained summand survives intersection unchanged. □

Proposition 2.1.8. *Two-Subspace Directness Criterion*
Let $\mathcal{U}$ *and* $\mathcal{W}$ *be subspaces of* $\mathcal{V}$. *Then* $\mathcal{U}+\mathcal{W}$ *is direct iff* $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$.

Proof.

(⇒) Assume $\mathcal{U}+\mathcal{W}$ is direct. For $\mathbf{x} \in \mathcal{U}\cap\mathcal{W}$, directness of $\mathbf{x} = \mathbf{x}+\mathbf{0}_\mathcal{V} = \mathbf{0}_\mathcal{V}+\mathbf{x}$ forces $\mathbf{x} = \mathbf{0}_\mathcal{V}$. Hence $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$.

(⇐) Assume $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$ and let $\mathbf{u}_1+\mathbf{w}_1 = \mathbf{u}_2+\mathbf{w}_2$, where $\mathbf{u}_1,\mathbf{u}_2 \in \mathcal{U}$ and $\mathbf{w}_1,\mathbf{w}_2 \in \mathcal{W}$.
Then $\mathbf{u}_1-\mathbf{u}_2 = \mathbf{w}_2-\mathbf{w}_1 \in \mathcal{U}\cap\mathcal{W}$, so $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$.
Every vector in the sum has a decomposition by its definition; therefore the sum is direct.

Structural Commentary: The two directions translate decomposition equality into intersection and back.
Key Insight: Trivial intersection prevents cross-summand cancellation. □

Perspective 2.1.9. *Sums Always Exist; Directness Is Extra*
For $\mathcal{U},\mathcal{W} \leq \mathcal{V}$, the sum $\mathcal{U}+\mathcal{W}$ exists as the smallest subspace containing both by Proposition 2.1.2.
The notation $\mathcal{U}\oplus\mathcal{W}$ records the additional equivalent conditions in Proposition 2.1.8: $\mathcal{U}\cap\mathcal{W} = \{\mathbf{0}_\mathcal{V}\}$ and uniqueness of $\mathbf{u}+\mathbf{w}$.
Thus $+$ records generation, whereas $\oplus$ records generation without ambiguity; this comparison is not proof authority.

Proposition 2.1.10. *Equivalent Conditions for a Direct Sum*
For $m \in \mathbb{N}$ *and subspaces* $\mathcal{U}_1,\ldots,\mathcal{U}_m \leq \mathcal{V}$, *the following are equivalent:*

(i) $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$;

(ii) $(\forall \mathbf{v} \in \mathcal{V})(\exists!\,(\mathbf{u}_1,\ldots,\mathbf{u}_m) \in \mathcal{U}_1\times\cdots\times\mathcal{U}_m)\,\mathbf{v} = \sum_{j=1}^{m} \mathbf{u}_j$;

(iii) $\mathcal{V} = \sum_{j=1}^{m} \mathcal{U}_j$ *and* $(\forall\,(\mathbf{u}_1,\ldots,\mathbf{u}_m) \in \mathcal{U}_1\times\cdots\times\mathcal{U}_m)[\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}_\mathcal{V} \Rightarrow (\forall\, j \in \{1,\ldots,m\})\,\mathbf{u}_j = \mathbf{0}_\mathcal{V}]$.

Proof.

(i) ⇒ *(ii)* Condition (i) asserts that the sum equals $\mathcal{V}$ and that its decompositions are unique, which is condition (ii).

(ii) ⇒ *(iii)* Existence in (ii) implies $\mathcal{V} = \sum_{j=1}^{m} \mathcal{U}_j$.
If $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}_\mathcal{V}$, uniqueness identifies this representation with $\sum_{j=1}^{m} \mathbf{0}_\mathcal{V}$, so $(\forall\, j \in \{1,\ldots,m\})\,\mathbf{u}_j = \mathbf{0}_\mathcal{V}$.

(iii) ⇒ *(i)* The equality in (iii) supplies a decomposition of every $\mathbf{v} \in \mathcal{V}$. If $\sum_{j=1}^{m} \mathbf{u}_j = \sum_{j=1}^{m} \mathbf{w}_j$, then $\sum_{j=1}^{m} (\mathbf{u}_j-\mathbf{w}_j) = \mathbf{0}_\mathcal{V}$.
Hence $(\forall\, j \in \{1,\ldots,m\})\,\mathbf{u}_j = \mathbf{w}_j$ by the zero-relation condition, so the sum is direct and equals $\mathcal{V}$.

Structural Commentary: The implication cycle separates existence of decompositions from the zero-relation obstruction.
Key Insight: Uniqueness is equivalent to trivial zero relations. □

Definition 2.1.11. *Product of Vector Spaces*
Let $m \in \mathbb{N}$ and let $\mathcal{V}_1,\ldots,\mathcal{V}_m$ be $\mathbb{F}$-vector spaces.

(i) Their *product* is $\mathcal{V}_1\times\cdots\times\mathcal{V}_m := \{(\mathbf{v}_1,\ldots,\mathbf{v}_m) : (\forall\, j \in \{1,\ldots,m\})\,\mathbf{v}_j \in \mathcal{V}_j\}$.
(ii) Addition is $(\mathbf{u}_1,\ldots,\mathbf{u}_m)+(\mathbf{v}_1,\ldots,\mathbf{v}_m) := (\mathbf{u}_1+\mathbf{v}_1,\ldots,\mathbf{u}_m+\mathbf{v}_m)$.
(iii) Scalar multiplication is $a(\mathbf{v}_1,\ldots,\mathbf{v}_m) := (a\mathbf{v}_1,\ldots,a\mathbf{v}_m)$ for $a \in \mathbb{F}$.

Proposition 2.1.12. *Products Are Vector Spaces*
If $m \in \mathbb{N}$ *and* $\mathcal{V}_1,\ldots,\mathcal{V}_m$ *are vector spaces over* $\mathbb{F}$, *then* $\mathcal{V}_1\times\cdots\times\mathcal{V}_m$ *is a vector space under componentwise operations.*

Proof. Set $P := \mathcal{V}_1\times\cdots\times\mathcal{V}_m$ and let $\mathbf{0}_j$ be the zero vector of $\mathcal{V}_j$. The operations are closed on P by componentwise closure.
The tuple $(\mathbf{0}_1,\ldots,\mathbf{0}_m)$ is an additive identity, and $(-\mathbf{v}_1,\ldots,-\mathbf{v}_m)$ is the additive inverse of $(\mathbf{v}_1,\ldots,\mathbf{v}_m)$.
For arbitrary tuples $\mathbf{u},\mathbf{v},\mathbf{w} \in P$, scalars $a,b \in \mathbb{F}$, and $j \in \{1,\ldots,m\}$, the jth coordinates on both sides of each commutativity, associativity, scalar-identity, scalar-associativity, and distributivity axiom are equal by the corresponding axiom in $\mathcal{V}_j$.
Coordinatewise equality proves each tuple identity, so P is an $\mathbb{F}$-vector space.
Structural Commentary: Closure, identities, and all algebraic laws are verified coordinatewise.
Key Insight: Tuple equality reduces every axiom to its components. □

Theorem 2.1.13. *Dimension of a Product*
Let $m \in \mathbb{N}$ *and let* $\mathcal{V}_1,\ldots,\mathcal{V}_m$ *be finite-dimensional* $\mathbb{F}$*-vector spaces.*
Then $P := \mathcal{V}_1\times\cdots\times\mathcal{V}_m$ *is finite-dimensional and* $\dim(P) = \sum_{j=1}^{m} \dim(\mathcal{V}_j)$.

Proof. For $j \in \{1,\dots,m\}$, let $(\mathbf{v}_{j,1},\dots,\mathbf{v}_{j,n_j})$ be a basis of $\mathcal{V}_j$.
For $j \in \{1,\dots,m\}$ and $k \in \{1,\dots,n_j\}$, let $\mathbf{e}_{j,k} \in P$ have jth component $\mathbf{v}_{j,k}$ and zero in every other component.
The list of all $\mathbf{e}_{j,k}$ spans P because each component has an expansion in its chosen basis.
If $\sum_{j=1}^{m}\sum_{k=1}^{n_j} a_{j,k}\mathbf{e}_{j,k} = \mathbf{0}$, then $(\forall\, j \in \{1,\dots,m\})\ \sum_{k=1}^{n_j} a_{j,k}\mathbf{v}_{j,k} = \mathbf{0}_{\mathcal{V}_j}$.
Hence $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,n_j\})\, a_{j,k} = 0$ by linear independence of each factor basis.
Thus the list is a basis, and its length $\sum_{j=1}^{m} n_j = \sum_{j=1}^{m} \dim(\mathcal{V}_j)$ proves the formula.
Structural Commentary: Embedded factor bases span componentwise and are independent componentwise.
Key Insight: Each factor contributes its own coordinate directions. □

Definition 2.1.14. *Graph of a Function*
For $T : \mathcal{V} \to \mathcal{W}$, its *graph* is $\operatorname{graph}(T) := \{(\mathbf{v}, T\mathbf{v}) : \mathbf{v} \in \mathcal{V}\} \subseteq \mathcal{V} \times \mathcal{W}$.

Proposition 2.1.15. *Linearity and Graphs*
A function $T : \mathcal{V} \to \mathcal{W}$ is linear iff $\operatorname{graph}(T) \le \mathcal{V} \times \mathcal{W}$.

Proof.
(⇒) Assume T is linear. Then $(\mathbf{0}_{\mathcal{V}}, \mathbf{0}_{\mathcal{W}}) \in \operatorname{graph}(T)$ by Proposition 1.5.3.
For $(\mathbf{u}, T\mathbf{u}), (\mathbf{v}, T\mathbf{v}) \in \operatorname{graph}(T)$ and $a, b \in \mathbb{F}$, $a(\mathbf{u}, T\mathbf{u}) + b(\mathbf{v}, T\mathbf{v}) = (a\mathbf{u} + b\mathbf{v}, T(a\mathbf{u} + b\mathbf{v})) \in \operatorname{graph}(T)$ by linearity.
Hence $\operatorname{graph}(T) \le \mathcal{V} \times \mathcal{W}$ by Theorem 1.2.2.
(⇐) Assume $\operatorname{graph}(T) \le \mathcal{V} \times \mathcal{W}$. For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $(a\mathbf{u} + b\mathbf{v}, aT\mathbf{u} + bT\mathbf{v}) = a(\mathbf{u}, T\mathbf{u}) + b(\mathbf{v}, T\mathbf{v}) \in \operatorname{graph}(T)$.
Membership in the graph forces $T(a\mathbf{u} + b\mathbf{v}) = aT\mathbf{u} + bT\mathbf{v}$, so T is linear.
Structural Commentary: Each direction translates the linear-combination law into graph closure.
Key Insight: Graph closure is the linearity identity in product form. □

Definition 2.1.16. *External Product*
Let $\{\mathcal{V}_i : i \in I\}$ be $\mathbb{F}$-vector spaces.
(i) The *external direct product* is $\prod_{i\in I} \mathcal{V}_i := \{(\mathbf{v}_i)_{i\in I} : (\forall i \in I)\, \mathbf{v}_i \in \mathcal{V}_i\}$.
(ii) Addition is defined by $(\forall\, i \in I)\, (\mathbf{u} + \mathbf{v})_i := \mathbf{u}_i + \mathbf{v}_i$.
(iii) Scalar multiplication is defined by $(\forall\, i \in I)\, (a\mathbf{v})_i := a\mathbf{v}_i$ for $a \in \mathbb{F}$.

Definition 2.1.17. *Support and Finite Sums of a Family*
Let I be an index set, and let $\mathcal{X}_i$ be a vector space with zero vector $\mathbf{0}_i$ for $i \in I$.
(i) The *support* of $\mathbf{x} = (\mathbf{x}_i)_{i\in I} \in \prod_{i\in I} \mathcal{X}_i$ is $\operatorname{supp}(\mathbf{x}) := \{i \in I : \mathbf{x}_i \neq \mathbf{0}_i\}$.
(ii) The family $\mathbf{x}$ is *finitely supported* iff $|\operatorname{supp}(\mathbf{x})| < \infty$.
(iii) If $(\forall\, i \in I)\, \mathcal{X}_i \le \mathcal{V}$ and $\mathbf{x}$ is finitely supported, its *finite sum* is $\sum_{i\in I} \mathbf{x}_i := \sum_{i\in\operatorname{supp}(\mathbf{x})} \mathbf{x}_i$; the empty sum is $\mathbf{0}_{\mathcal{V}}$.

Definition 2.1.18. *External Direct Sum*
Let $\{\mathcal{V}_i : i \in I\}$ be $\mathbb{F}$-vector spaces. The *external direct sum* is $\bigoplus_{i\in I}^{\text{ext}} \mathcal{V}_i := \{\mathbf{v} \in \prod_{i\in I} \mathcal{V}_i : |\operatorname{supp}(\mathbf{v})| < \infty\}$.

Proposition 2.1.19. *External Sums Are Finite-Support Products*
Let $\{\mathcal{V}_i\}_{i\in I}$ be $\mathbb{F}$-vector spaces and set $P := \prod_{i\in I} \mathcal{V}_i$ and $S := \bigoplus_{i\in I}^{\text{ext}} \mathcal{V}_i$.
(i) P is an $\mathbb{F}$-vector space under componentwise operations.
(ii) $S \le P$.
(iii) If $|I| < \infty$, then $S = P$.

Proof.
(i) Componentwise closure keeps addition and scalar multiplication in P.
The family $(\mathbf{0}_i)_{i\in I}$ is its zero, and $(-\mathbf{v}_i)_{i\in I}$ is the inverse of $\mathbf{v}$. For arbitrary elements, scalars, and $i \in I$, the relevant identity at coordinate i holds by the corresponding identity in $\mathcal{V}_i$; coordinatewise equality proves the identity in P.
(ii) The zero family has empty support. For $\mathbf{u}, \mathbf{v} \in S$ and $a \in \mathbb{F}$, $\operatorname{supp}(a\mathbf{u} + \mathbf{v}) \subseteq \operatorname{supp}(\mathbf{u}) \cup \operatorname{supp}(\mathbf{v})$, a finite set.
Hence $a\mathbf{u} + \mathbf{v} \in S$, so $S \le P$ by Theorem 1.2.2.
(iii) If $|I| < \infty$ and $\mathbf{v} \in P$, then $\operatorname{supp}(\mathbf{v}) \subseteq I$ is finite. Thus $P \subseteq S$, while $S \subseteq P$ by construction, so $S = P$.
Structural Commentary: Coordinatewise axioms build the product; support containment proves the remaining claims.
Key Insight: Finite linear combinations preserve finite support. □

Definition 2.1.20. *Finitely Nonzero Sequence Space*
The *finitely nonzero sequence space* is $\mathbb{F}^{(\mathbb{N})} := \bigoplus_{j\in\mathbb{N}}^{\text{ext}} \mathbb{F} = \{(x_j)_{j\in\mathbb{N}} \in \mathbb{F}^{\mathbb{N}} : |\{j \in \mathbb{N} : x_j \neq 0\}| < \infty\}$, with componentwise operations.

Proposition 2.1.21. *Finitely Nonzero Sequences Form a Vector Space*
The set $\mathbb{F}^{(\mathbb{N})}$ is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication.

Proof. The space $\mathbb{F}^{(\mathbb{N})}$ is the external direct sum of the constant family $\mathcal{V}_j = \mathbb{F}$ by its definition, so it is a subspace of the product $\mathbb{F}^{\mathbb{N}}$ by Proposition 2.1.19(ii). Hence it is an $\mathbb{F}$-vector space under the inherited componentwise operations.
Structural Commentary: The result specializes the external direct-sum subspace construction to a constant countable family.
Key Insight: Finitely supported sequences are a canonical external sum. □

Set $\operatorname{Seq}(\mathbb{F}) := \mathbb{F}^{\mathbb{N}}$. Then $\mathbb{F}^{(\mathbb{N})} \le \operatorname{Seq}(\mathbb{F})$ by Proposition 2.1.21.

Definition 2.1.22. *Internal Direct Sum of a Family*

For $\{\mathcal{U}_i\}_{i\in I} \subseteq \mathrm{Sub}(\mathcal{V})$, the internal sum $S := \sum_{i\in I} \mathcal{U}_i$ is *direct* iff $(\forall\, \mathbf{v} \in S)(\exists!\, \mathbf{u} \in \bigoplus_{i\in I}^{\text{ext}} \mathcal{U}_i)\, \mathbf{v} = \sum_{i\in I} \mathbf{u}_i$.
A direct internal sum is denoted $\bigoplus_{i\in I} \mathcal{U}_i$; the equality $\mathcal{V} = \bigoplus_{i\in I} \mathcal{U}_i$ additionally asserts $\mathcal{V} = S$.

Proposition 2.1.23. *Family Direct-Sum Criterion*
For $\{\mathcal{U}_i\}_{i\in I} \subseteq \mathrm{Sub}(\mathcal{V})$, the following are equivalent:

(i) $\sum_{i\in I} \mathcal{U}_i$ *is direct;*
(ii) $(\forall\, F \subseteq I)[|F| < \infty \Rightarrow (\forall\, (\mathbf{u}_i)_{i\in F} \in \prod_{i\in F} \mathcal{U}_i)(\sum_{i\in F} \mathbf{u}_i = \mathbf{0}_{\mathcal{V}} \Rightarrow (\forall\, i \in F)\, \mathbf{u}_i = \mathbf{0}_{\mathcal{V}})]$;
(iii) $(\forall\, i \in I)\mathcal{U}_i \cap \sum_{j\in I\setminus\{i\}} \mathcal{U}_j = \{\mathbf{0}_{\mathcal{V}}\}$.

Proof.

(i) $\Rightarrow$ *(ii)* Let $F \subseteq I$ be finite and let $(\mathbf{u}_i)_{i\in F} \in \prod_{i\in F} \mathcal{U}_i$ satisfy $\sum_{i\in F} \mathbf{u}_i = \mathbf{0}_{\mathcal{V}}$. Extend this family by zero outside F.
Directness identifies the resulting decomposition with the all-zero decomposition, so $(\forall\, i \in F)\, \mathbf{u}_i = \mathbf{0}_{\mathcal{V}}$.
(ii) $\Rightarrow$ *(iii)* Fix $i \in I$ and let $\mathbf{x} \in \mathcal{U}_i \cap \sum_{j\in I\setminus\{i\}} \mathcal{U}_j$.
Then $\mathbf{x} = \sum_{j\in F} \mathbf{u}_j$ for a finite $F \subseteq I \setminus \{i\}$ with $(\forall\, j \in F)\, \mathbf{u}_j \in \mathcal{U}_j$ by Proposition 2.1.6(iii).
The relation on $F \cup \{i\}$ with ith term $\mathbf{x}$ and jth term $-\mathbf{u}_j$ is zero. It is trivial by (ii), so $\mathbf{x} = \mathbf{0}_{\mathcal{V}}$.
(iii) $\Rightarrow$ *(i)* Let $F \subseteq I$ be finite and suppose $\sum_{i\in F} \mathbf{u}_i = \mathbf{0}_{\mathcal{V}}$ with $(\forall\, i \in F)\, \mathbf{u}_i \in \mathcal{U}_i$.
For $k \in F$, $\mathbf{u}_k = -\sum_{i\in F\setminus\{k\}} \mathbf{u}_i \in \mathcal{U}_k \cap \sum_{j\in I\setminus\{k\}} \mathcal{U}_j = \{\mathbf{0}_{\mathcal{V}}\}$ by (iii), so every finite zero relation is trivial.
Each vector in the sum has a finite decomposition by Proposition 2.1.6(iii).
The difference of two decompositions is a finite zero relation, so they coincide and the sum is direct.

Structural Commentary: The cycle converts unique decompositions into zero relations and collective intersection tests.
Key Insight: Directness forbids cancellation across distinct summands. □

Proposition 2.1.24. *Direct Sum Map*

Let $m \in \mathbb{N}$ and let $\mathcal{U}_1, \ldots, \mathcal{U}_m \leq \mathcal{V}$. Define $\Gamma : \prod_{j=1}^{m} \mathcal{U}_j \to \sum_{j=1}^{m} \mathcal{U}_j$ by $\Gamma((\mathbf{u}_j)_{j=1}^{m}) := \sum_{j=1}^{m} \mathbf{u}_j$. The following hold:

(i) $\Gamma \in \mathcal{L}(\mathcal{U}_1 \times \cdots \times \mathcal{U}_m, \mathcal{U}_1 + \cdots + \mathcal{U}_m)$;
(ii) $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ *is direct iff* Γ *is injective.*

Proof.

(i) For $\mathbf{u}, \mathbf{v} \in \mathcal{U}_1 \times \cdots \times \mathcal{U}_m$ and $a, b \in \mathbb{F}$, $\Gamma(a\mathbf{u} + b\mathbf{v}) = \sum_{j=1}^{m} (a\mathbf{u}_j + b\mathbf{v}_j) = a\Gamma(\mathbf{u}) + b\Gamma(\mathbf{v})$. Hence Γ is linear.
(ii) $(\Rightarrow)$ Assume sum is direct. If $\Gamma(\mathbf{u}) = \Gamma(\mathbf{v})$, then $\mathbf{u}$ and $\mathbf{v}$ are decompositions of same vector, so $\mathbf{u} = \mathbf{v}$ by directness. QED.
(ii) $(\Leftarrow)$ Assume Γ is injective. If $\sum_{j=1}^{m} \mathbf{u}_j = \sum_{j=1}^{m} \mathbf{v}_j$, then $\Gamma(\mathbf{u}) = \Gamma(\mathbf{v})$, so $\mathbf{u} = \mathbf{v}$ by injectivity.
Thus every vector in the sum has a unique decomposition, and the sum is direct.

Structural Commentary: Linearity is coordinatewise; injectivity translates equality of sums into equality of decompositions.
Key Insight: Directness is injectivity of the addition map. □

2.2 Quotient Spaces

Quotients collapse a prescribed subspace while retaining the linear structure that is independent of the collapsed directions.
Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 2.2.1. *Affine Subset Parallel to a Subspace*
Let $\mathcal{U} \leq \mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, the subset $\mathbf{v} + \mathcal{U} := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in \mathcal{U}\}$ is the *affine subset* through $\mathbf{v}$ parallel to $\mathcal{U}$.

Proposition 2.2.2. *Parallel Affine Subsets Are Equal or Disjoint*
Let $\mathcal{U} \leq \mathcal{V}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The following conditions are equivalent:

(i) $\mathbf{v} - \mathbf{w} \in \mathcal{U}$;
(ii) $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$;
(iii) $(\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U}) \neq \emptyset$.

Proof.

(i) $\Rightarrow$ *(ii)* Suppose $\mathbf{v} - \mathbf{w} \in \mathcal{U}$. For $\mathbf{u} \in \mathcal{U}$, $\mathbf{v} + \mathbf{u} = \mathbf{w} + ((\mathbf{v} - \mathbf{w}) + \mathbf{u}) \in \mathbf{w} + \mathcal{U}$, so $\mathbf{v} + \mathcal{U} \subseteq \mathbf{w} + \mathcal{U}$.
Since $\mathbf{w} - \mathbf{v} = -(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$, exchanging $\mathbf{v}$ and $\mathbf{w}$ establishes the reverse inclusion.
(ii) $\Rightarrow$ *(iii)* If $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$, then $\mathbf{v} \in (\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U})$.
(iii) $\Rightarrow$ *(i)* Choose $\mathbf{x} \in (\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U})$ and $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$ with $\mathbf{x} = \mathbf{v} + \mathbf{u}_1 = \mathbf{w} + \mathbf{u}_2$. Then $\mathbf{v} - \mathbf{w} = \mathbf{u}_2 - \mathbf{u}_1 \in \mathcal{U}$.

Structural Commentary: A cyclic implication proof connects differences, equality, and intersection.
Key Insight: A common point exposes the difference of representatives. □

Definition 2.2.3. *Underlying Quotient Set*
For $\mathcal{U} \leq \mathcal{V}$, the *quotient set* is $\mathcal{V}/\mathcal{U} := \{\mathbf{v} + \mathcal{U} : \mathbf{v} \in \mathcal{V}\}$.

Proposition 2.2.4. *Quotient Relation Is a Linear Congruence*
Let $\mathcal{U} \leq \mathcal{V}$ and define a relation on $\mathcal{V}$ by $\mathbf{v} \equiv_{\mathcal{U}} \mathbf{w} \iff \mathbf{v} - \mathbf{w} \in \mathcal{U}$.

(i) $(\forall\, \mathbf{v} \in \mathcal{V})\, \mathbf{v} \equiv_{\mathcal{U}} \mathbf{v}$.
(ii) $(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})[\mathbf{v} \equiv_{\mathcal{U}} \mathbf{w} \Rightarrow \mathbf{w} \equiv_{\mathcal{U}} \mathbf{v}]$.

(iii) $(\forall\, \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V})[(\mathbf{u} \equiv_{\mathcal{U}} \mathbf{v} \wedge \mathbf{v} \equiv_{\mathcal{U}} \mathbf{w}) \Rightarrow \mathbf{u} \equiv_{\mathcal{U}} \mathbf{w}]$.
(iv) $(\forall\, a, b \in \mathbb{F})(\forall\, \mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2 \in \mathcal{V})[(\mathbf{v}_1 \equiv_{\mathcal{U}} \mathbf{v}_2 \wedge \mathbf{w}_1 \equiv_{\mathcal{U}} \mathbf{w}_2) \Rightarrow a\mathbf{v}_1 + b\mathbf{w}_1 \equiv_{\mathcal{U}} a\mathbf{v}_2 + b\mathbf{w}_2]$.
Thus $\equiv_{\mathcal{U}}$ is an equivalence relation compatible with linear combinations.

Proof.
(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} - \mathbf{v} = \mathbf{0} \in \mathcal{U}$.
(ii) If $\mathbf{v} - \mathbf{w} \in \mathcal{U}$, then $\mathbf{w} - \mathbf{v} = -(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$.
(iii) If $\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w} \in \mathcal{U}$, then $\mathbf{u} - \mathbf{w} = (\mathbf{u} - \mathbf{v}) + (\mathbf{v} - \mathbf{w}) \in \mathcal{U}$.
(iv) If $\mathbf{v}_1 - \mathbf{v}_2, \mathbf{w}_1 - \mathbf{w}_2 \in \mathcal{U}$, then $(a\mathbf{v}_1 + b\mathbf{w}_1) - (a\mathbf{v}_2 + b\mathbf{w}_2) = a(\mathbf{v}_1 - \mathbf{v}_2) + b(\mathbf{w}_1 - \mathbf{w}_2) \in \mathcal{U}$ for $a, b \in \mathbb{F}$.
Structural Commentary: The four items verify the equivalence axioms and linear compatibility.
Key Insight: Linear combinations preserve differences lying in the subspace. □

Perspective 2.2.5. *Cosets as Equivalence Classes*
The equivalence class of $\mathbf{v}$ under $\equiv_{\mathcal{U}}$ is $\mathbf{v} + \mathcal{U}$ by Proposition 2.2.2. Hence the quotient set in Definition 2.2.3 is the abstract quotient set from Definition 5.2.5. It remembers a vector only up to addition by an element of $\mathcal{U}$.

Proposition 2.2.6. *Quotient-Space Structure*
Let $\mathcal{U} \leq \mathcal{V}$. For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a \in \mathbb{F}$, set $(\mathbf{v} + \mathcal{U}) + (\mathbf{w} + \mathcal{U}) := (\mathbf{v} + \mathbf{w}) + \mathcal{U}$ and $a(\mathbf{v} + \mathcal{U}) := a\mathbf{v} + \mathcal{U}$.
(i) These operations are independent of the chosen representatives.
(ii) With these operations, $\mathcal{V}/\mathcal{U}$ is a vector space, called the quotient space of $\mathcal{V}$ by $\mathcal{U}$.

Proof.
(i) Suppose $\mathbf{v} + \mathcal{U} = \mathbf{v}' + \mathcal{U}$ and $\mathbf{w} + \mathcal{U} = \mathbf{w}' + \mathcal{U}$. Then $\mathbf{v} - \mathbf{v}', \mathbf{w} - \mathbf{w}' \in \mathcal{U}$ by Proposition 2.2.2.
Hence $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}') \in \mathcal{U}$ by Proposition 2.2.4(iv), and $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in \mathcal{U}$.
The resulting sum and scalar-multiple cosets are equal by Proposition 2.2.2.
(ii) Write $[\mathbf{v}] := \mathbf{v} + \mathcal{U}$. For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $([\mathbf{u}] + [\mathbf{v}]) + [\mathbf{w}] = [\mathbf{u} + \mathbf{v} + \mathbf{w}] = [\mathbf{u}] + ([\mathbf{v}] + [\mathbf{w}])$, $[\mathbf{v}] + [\mathbf{w}] = [\mathbf{w}] + [\mathbf{v}]$, $[\mathbf{0}] = \mathcal{U}$ is an additive identity, and $[-\mathbf{v}]$ is the additive inverse of $[\mathbf{v}]$.
The scalar identities $1[\mathbf{v}] = [\mathbf{v}]$, $a(b[\mathbf{v}]) = (ab)[\mathbf{v}]$, $a([\mathbf{v}] + [\mathbf{w}]) = a[\mathbf{v}] + a[\mathbf{w}]$, and $(a + b)[\mathbf{v}] = a[\mathbf{v}] + b[\mathbf{v}]$ hold by the vector-space identities in $\mathcal{V}$. Thus all vector-space axioms hold.
Structural Commentary: Representative independence precedes direct verification of the vector-space axioms.
Key Insight: Coset arithmetic inherits every identity from representatives. □

Definition 2.2.7. *Quotient Map*
For $\mathcal{U} \leq \mathcal{V}$, the *quotient map* is $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ defined by $\pi(\mathbf{v}) := \mathbf{v} + \mathcal{U}$.

Proposition 2.2.8. *Properties of the Quotient Map*
Let $\mathcal{U} \leq \mathcal{V}$ and define $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ by $\pi(\mathbf{v}) = \mathbf{v} + \mathcal{U}$.
(i) π is linear.
(ii) π is surjective.
(iii) $\ker(\pi) = \mathcal{U}$.

Proof.
(i) For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $\pi(a\mathbf{v} + b\mathbf{w}) = (a\mathbf{v} + b\mathbf{w}) + \mathcal{U} = a\pi(\mathbf{v}) + b\pi(\mathbf{w})$.
(ii) If $\mathbf{x} \in \mathcal{V}/\mathcal{U}$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{x} = \mathbf{v} + \mathcal{U} = \pi(\mathbf{v})$ by the definition of $\mathcal{V}/\mathcal{U}$.
(iii) For $\mathbf{v} \in \mathcal{V}$, $\pi(\mathbf{v}) = \mathcal{U} \iff \mathbf{v} \in \mathcal{U}$ by Proposition 2.2.2. Therefore $\ker(\pi) = \mathcal{U}$.
Structural Commentary: The proof verifies linearity, surjectivity, and the kernel in statement order.
Key Insight: The zero coset is exactly the collapsed subspace. □

Theorem 2.2.9. *Dimension of a Quotient*
Let $\mathcal{V}$ be finite-dimensional and $\mathcal{U} \leq \mathcal{V}$.
(i) $\mathcal{V}/\mathcal{U}$ is finite-dimensional.
(ii) $\dim(\mathcal{V}/\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map. Its range is $\mathcal{V}/\mathcal{U}$ and its kernel is $\mathcal{U}$ by Proposition 2.2.8(ii)–(iii).
(i) The range $\mathcal{V}/\mathcal{U}$ is finite-dimensional by Theorem 1.7.3(i).
(ii) The formula in (ii) holds by Theorem 1.7.3(ii) with $\ker(\pi) = \mathcal{U}$ and $\operatorname{range}(\pi) = \mathcal{V}/\mathcal{U}$.
Structural Commentary: Quotient-map properties followed by rank-nullity handle finiteness and dimension.
Key Insight: Quotient dimension is the rank of the quotient map. □

Theorem 2.2.10. *Subspace Correspondence for Quotients*
Let $\mathcal{U} \leq \mathcal{V}$, let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map, and set $\mathfrak{A} := \{\mathcal{T} \leq \mathcal{V} : \mathcal{U} \leq \mathcal{T}\}$ and $\mathfrak{B} := \{\mathcal{Y} : \mathcal{Y} \leq \mathcal{V}/\mathcal{U}\}$. For $\mathcal{T} \in \mathfrak{A}$ and $\mathcal{Y} \in \mathfrak{B}$, set $\Phi(\mathcal{T}) := \mathcal{T}/\mathcal{U} = \pi(\mathcal{T})$ and $\Psi(\mathcal{Y}) := \pi^{-1}(\mathcal{Y})$.
(i) $\Phi : \mathfrak{A} \to \mathfrak{B}$ and $\Psi : \mathfrak{B} \to \mathfrak{A}$ are well-defined.
(ii) Φ and Ψ are inverse bijections: $\Psi\Phi = I_{\mathfrak{A}}$ and $\Phi\Psi = I_{\mathfrak{B}}$.
(iii) Both maps preserve inclusion.

Proof.
(i) Let $\mathcal{T} \in \mathfrak{A}$. The set $\mathcal{T}/\mathcal{U}$ contains $\mathcal{U} = \mathbf{0} + \mathcal{U}$, and $a(\mathbf{t}_1 + \mathcal{U}) + b(\mathbf{t}_2 + \mathcal{U}) = (a\mathbf{t}_1 + b\mathbf{t}_2) + \mathcal{U} \in \mathcal{T}/\mathcal{U}$ for $\mathbf{t}_1, \mathbf{t}_2 \in \mathcal{T}$ and $a, b \in \mathbb{F}$.
Hence $\Phi(\mathcal{T}) \leq \mathcal{V}/\mathcal{U}$ by Theorem 1.2.2. Let $\mathcal{Y} \in \mathfrak{B}$. The set $\pi^{-1}(\mathcal{Y})$ contains $\mathbf{0}_{\mathcal{V}}$, and $\pi(a\mathbf{v} + b\mathbf{w}) = a\pi(\mathbf{v}) + b\pi(\mathbf{w}) \in \mathcal{Y}$ for $\mathbf{v}, \mathbf{w} \in \pi^{-1}(\mathcal{Y})$ and $a, b \in \mathbb{F}$; hence $\pi^{-1}(\mathcal{Y}) \leq \mathcal{V}$ by Theorem 1.2.2. Moreover, $\mathcal{U} = \ker(\pi) \subseteq \pi^{-1}(\mathcal{Y})$ by Proposition 2.2.8(iii).
(ii) Let $\mathcal{T} \in \mathfrak{A}$. For $\mathbf{t} \in \mathcal{T}$, $\pi(\mathbf{t}) \in \pi(\mathcal{T})$, so $\mathcal{T} \subseteq \pi^{-1}(\pi(\mathcal{T}))$. If $\mathbf{v} \in \pi^{-1}(\pi(\mathcal{T}))$, choose $\mathbf{t} \in \mathcal{T}$ with $\mathbf{v} + \mathcal{U} = \mathbf{t} + \mathcal{U}$.
Then $\mathbf{v} - \mathbf{t} \in \mathcal{U} \subseteq \mathcal{T}$ by Proposition 2.2.2, so $\mathbf{v} \in \mathcal{T}$. Thus $\Psi(\Phi(\mathcal{T})) = \mathcal{T}$. Let $\mathcal{Y} \in \mathfrak{B}$.
If $\mathbf{v} \in \pi^{-1}(\mathcal{Y})$, then $\pi(\mathbf{v}) \in \mathcal{Y}$, so $\pi(\pi^{-1}(\mathcal{Y})) \subseteq \mathcal{Y}$. If $\mathbf{y} \in \mathcal{Y}$, choose $\mathbf{v} \in \mathcal{V}$ with $\pi(\mathbf{v}) = \mathbf{y}$ by Proposition 2.2.8(ii).
Then $\mathbf{v} \in \pi^{-1}(\mathcal{Y})$, so $\mathbf{y} \in \pi(\pi^{-1}(\mathcal{Y}))$. Therefore $\Phi(\Psi(\mathcal{Y})) = \mathcal{Y}$.

(iii) If $\mathcal{T}_1 \subseteq \mathcal{T}_2$ in $\mathfrak{A}$, then $\pi(\mathcal{T}_1) \subseteq \pi(\mathcal{T}_2)$. If $\mathcal{Y}_1 \subseteq \mathcal{Y}_2$ in $\mathfrak{B}$, then $\pi^{-1}(\mathcal{Y}_1) \subseteq \pi^{-1}(\mathcal{Y}_2)$.

Structural Commentary: Well-defined assignments, inverse identities, and monotonicity establish the correspondence.
Key Insight: Images and preimages undo the quotient construction exactly. □

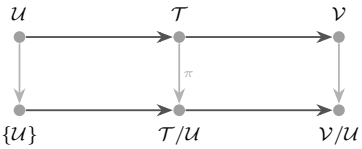

Figure 10: Intermediate subspaces above $\mathcal{U}$ correspond to subspaces of the quotient; horizontal arrows denote inclusions and vertical arrows denote quotient images.

Figure 10 illustrates Theorem 2.2.10: $\mathcal{U} \subseteq \mathcal{T} \subseteq \mathcal{V}$ descends to $\{\mathcal{U}\} \subseteq \mathcal{T}/\mathcal{U} \subseteq \mathcal{V}/\mathcal{U}$, while inverse images recover the original subspaces. The theorem, rather than the diagram, owns the correspondence.

Theorem 2.2.11. *Universal Property of the Quotient Map*
Let $\mathcal{U} \leq \mathcal{V}$, let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfy $\mathcal{U} \leq \ker(T)$.

(i) $(\exists!\, \overline{T} \in \mathcal{L}(\mathcal{V}/\mathcal{U}, \mathcal{W}))[\overline{T}\pi = T \wedge (\forall\, \mathbf{v} \in \mathcal{V})\, \overline{T}(\mathbf{v}+\mathcal{U}) = T\mathbf{v}]$.
(ii) The map $\overline{T}$ from (i) satisfies $\ker(\overline{T}) = \ker(T)/\mathcal{U}$.
(iii) The map $\overline{T}$ from (i) satisfies $\operatorname{range}(\overline{T}) = \operatorname{range}(T)$.

Proof.

(i) Define $\overline{T}(\mathbf{v}+\mathcal{U}) := T\mathbf{v}$. If $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$, then $\mathbf{v}-\mathbf{w} \in \mathcal{U} \leq \ker(T)$ by Proposition 2.2.2, so $T\mathbf{v} - T\mathbf{w} = T(\mathbf{v}-\mathbf{w}) = \mathbf{0}$. Hence $\overline{T}$ is well-defined. For $a, b \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\overline{T}(a(\mathbf{v}+\mathcal{U}) + b(\mathbf{w}+\mathcal{U})) = T(a\mathbf{v}+b\mathbf{w}) = aT\mathbf{v} + bT\mathbf{w}$, so $\overline{T}$ is linear. Also, $(\overline{T}\pi)(\mathbf{v}) = \overline{T}(\mathbf{v}+\mathcal{U}) = T\mathbf{v}$ for $\mathbf{v} \in \mathcal{V}$, so $\overline{T}\pi = T$. If $S \in \mathcal{L}(\mathcal{V}/\mathcal{U}, \mathcal{W})$ satisfies $S\pi = T$, then $(\forall\, \mathbf{v} \in \mathcal{V})\, S(\mathbf{v}+\mathcal{U}) = S\pi(\mathbf{v}) = T\mathbf{v} = \overline{T}(\mathbf{v}+\mathcal{U})$. Since $\mathcal{V}/\mathcal{U} = \{\mathbf{v}+\mathcal{U} : \mathbf{v} \in \mathcal{V}\}$, $S = \overline{T}$.
(ii) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v}+\mathcal{U} \in \ker(\overline{T}) \iff T\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(T)$. Hence $\ker(\overline{T}) = \{\mathbf{v}+\mathcal{U} : \mathbf{v} \in \ker(T)\} = \ker(T)/\mathcal{U}$.
(iii) The equality $\operatorname{range}(\overline{T}) = \{\overline{T}(\mathbf{v}+\mathcal{U}) : \mathbf{v} \in \mathcal{V}\} = \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\} = \operatorname{range}(T)$ is immediate from the defining formula.

Structural Commentary: Construction establishes factorisation; set calculations identify kernel and range.
Key Insight: Killing $\mathcal{U}$ is precisely the condition for descent. □

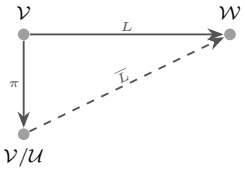

Figure 11: If $\mathcal{U} \subseteq \ker(L)$, the map L factors uniquely through the quotient map π; the dashed arrow is the induced map $\overline{L}$.

Figure 11 depicts Theorem 2.2.11, using L in place of T: collapsing $\mathcal{U}$ first does not change a map that already vanishes on $\mathcal{U}$. The theorem, rather than the diagram, owns existence and uniqueness.

Theorem 2.2.12. *Quotient by the Kernel*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and define $\widetilde{T}(\mathbf{v} + \ker(T)) := T\mathbf{v}$.

(i) The assignment defines a well-defined linear map $\widetilde{T} : \mathcal{V}/\ker(T) \to \mathcal{W}$.
(ii) $\widetilde{T}$ is injective.
(iii) $\operatorname{range}(\widetilde{T}) = \operatorname{range}(T)$.
(iv) The codomain restriction $\widehat{T} : \mathcal{V}/\ker(T) \to \operatorname{range}(T)$ is an isomorphism; hence $\mathcal{V}/\ker(T) \cong \operatorname{range}(T)$.

Proof.

(i) The claimed well-defined linear map exists by Theorem 2.2.11(i) with $\mathcal{U} = \ker(T)$.
(ii) The identity $\ker(\widetilde{T}) = \ker(T)/\ker(T) = \{\ker(T)\}$ holds by Theorem 2.2.11(ii). Thus $\widetilde{T}$ is injective by Proposition 1.6.2.
(iii) The range identity holds by Theorem 2.2.11(iii).
(iv) The codomain restriction $\widehat{T}$ is linear and injective by parts (i)–(ii), and it is surjective by part (iii). Therefore $\widehat{T}$ is invertible by Proposition 1.6.7, and $\mathcal{V}/\ker(T) \cong \operatorname{range}(T)$.

Structural Commentary: The universal property constructs the map; kernel and range identities produce the isomorphism.
Key Insight: Removing the kernel makes the induced map injective. □

Theorem 2.2.13. *Second Isomorphism Theorem*
If $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, then $\mathcal{S}/(\mathcal{S} \cap \mathcal{T}) \cong (\mathcal{S}+\mathcal{T})/\mathcal{T}$.

Proof.

(i) *Construction*: The sum $\mathcal{S}+\mathcal{T}$ is a subspace containing $\mathcal{T}$ by Proposition 2.1.2(i). Define $L : \mathcal{S} \to (\mathcal{S}+\mathcal{T})/\mathcal{T}$ by $L\mathbf{s} := \mathbf{s}+\mathcal{T}$. For $\mathbf{s}, \mathbf{r} \in \mathcal{S}$ and $a, b \in \mathbb{F}$, $L(a\mathbf{s}+b\mathbf{r}) = (a\mathbf{s}+b\mathbf{r}) + \mathcal{T} = aL\mathbf{s} + bL\mathbf{r}$, so L is linear.
(ii) *Kernel*: For $\mathbf{s} \in \mathcal{S}$, $L\mathbf{s} = \mathcal{T} \iff \mathbf{s} \in \mathcal{T}$ by Proposition 2.2.2. Hence $\ker(L) = \mathcal{S} \cap \mathcal{T}$.
(iii) *Surjectivity*: Each coset equals $(\mathbf{s}+\mathbf{t}) + \mathcal{T} = \mathbf{s} + \mathcal{T} = L\mathbf{s}$ for suitable $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$; hence L is surjective.
(iv) *Conclusion*: The equality $\operatorname{range}(L) = (\mathcal{S}+\mathcal{T})/\mathcal{T}$ holds by part (iii). Therefore $\mathcal{S}/(\mathcal{S} \cap \mathcal{T}) \cong (\mathcal{S}+\mathcal{T})/\mathcal{T}$ by Theorem 2.2.12 with the linearity and kernel identity from parts (i)–(ii).

Structural Commentary: Construction, kernel, surjectivity, and the first isomorphism theorem form four operative stages.
Key Insight: The target forgets exactly $\mathcal{S} \cap \mathcal{T}$. □

Theorem 2.2.14. *Third Isomorphism Theorem*
If $\mathcal{U} \leq \mathcal{T} \leq \mathcal{V}$, then $(\mathcal{V}/\mathcal{U})/(\mathcal{T}/\mathcal{U}) \cong \mathcal{V}/\mathcal{T}$.

Proof.

(i) *Construction*: The inclusion $\mathcal{T}/\mathcal{U} \leq \mathcal{V}/\mathcal{U}$ holds by Theorem 2.2.10(i). Define $L : \mathcal{V}/\mathcal{U} \to \mathcal{V}/\mathcal{T}$ by $L(\mathbf{v}+\mathcal{U}) := \mathbf{v}+\mathcal{T}$.
(ii) *Well-Definedness and Linearity*: If $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$, then $\mathbf{v}-\mathbf{w} \in \mathcal{U} \subseteq \mathcal{T}$ by Proposition 2.2.2, so $\mathbf{v}+\mathcal{T} = \mathbf{w}+\mathcal{T}$ by the same proposition. For $a, b \in \mathbb{F}$, $L(a(\mathbf{v}+\mathcal{U}) + b(\mathbf{w}+\mathcal{U})) = (a\mathbf{v}+b\mathbf{w})+\mathcal{T} = aL(\mathbf{v}+\mathcal{U}) + bL(\mathbf{w}+\mathcal{U})$.
(iii) *Kernel and Range*: For $\mathbf{v} \in \mathcal{V}$, $L(\mathbf{v}+\mathcal{U}) = \mathcal{T} \iff \mathbf{v} \in \mathcal{T}$ by Proposition 2.2.2; hence $\ker(L) = \mathcal{T}/\mathcal{U}$.
Every $\mathbf{v}+\mathcal{T} \in \mathcal{V}/\mathcal{T}$ equals $L(\mathbf{v}+\mathcal{U})$, so L is surjective.
(iv) *Conclusion*: The equality $\operatorname{range}(L) = \mathcal{V}/\mathcal{T}$ holds by part (iii).
Therefore $(\mathcal{V}/\mathcal{U})/(\mathcal{T}/\mathcal{U}) \cong \mathcal{V}/\mathcal{T}$ by Theorem 2.2.12 with the linearity and kernel identity from parts (ii)–(iii).

Structural Commentary: Construction, kernel computation, and the first isomorphism theorem identify the iterated quotient.
Key Insight: The nested collapse has kernel $\mathcal{T}/\mathcal{U}$. □

Quotient spaces retain precisely the structure invariant under translation by a chosen subspace. Their correspondence, universal property, and isomorphism theorems formalise which data survive that collapse.

2.3 Decomposition and Complementation

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$. Dimension counts first detect directness; complements then realize quotients as concrete subspaces.

Theorem 2.3.1. *Dimension Formula for a Sum*
If $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ are finite-dimensional, then $\dim(\mathcal{U}+\mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} \cap \mathcal{W})$.

Proof. The subspace $\mathcal{U} \cap \mathcal{W}$ is finite-dimensional by Proposition 1.4.3. Choose basis $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_r)$ of intersection by Corollary 1.3.14.
Extend ϵ to bases $\beta_{\mathcal{U}} = (\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$ and $\beta_{\mathcal{W}} = (\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$ by Theorem 1.3.13.
Set $\gamma := (\mathbf{e}_1, \ldots, \mathbf{e}_r, \mathbf{u}_1, \ldots, \mathbf{u}_p, \mathbf{w}_1, \ldots, \mathbf{w}_q)$. The list γ spans $\mathcal{U}+\mathcal{W}$.
Suppose $\sum_{i=1}^{r} a_i\mathbf{e}_i + \sum_{j=1}^{p} b_j\mathbf{u}_j + \sum_{k=1}^{q} c_k\mathbf{w}_k = \mathbf{0}$. Then $\sum_{i=1}^{r} a_i\mathbf{e}_i + \sum_{j=1}^{p} b_j\mathbf{u}_j = -\sum_{k=1}^{q} c_k\mathbf{w}_k \in \mathcal{U} \cap \mathcal{W}$.
Its representation in $\beta_{\mathcal{U}}$ therefore has $b_1 = \cdots = b_p = 0$.
The remaining relation belongs to $\beta_{\mathcal{W}}$, so $a_1 = \cdots = a_r = c_1 = \cdots = c_q = 0$.
Thus γ is a basis of $\mathcal{U}+\mathcal{W}$, and $\dim(\mathcal{U}+\mathcal{W}) = r+p+q = (r+p)+(r+q)-r$.
Structural Commentary: An intersection basis is extended twice; concatenation then separates shared from new directions.
Key Insight: Subtract the dimension counted in both summands. □

Corollary 2.3.2. *Dimension of a Direct Sum*
If $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ are finite-dimensional and $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, then $\dim(\mathcal{U} \oplus \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W})$.

Proof. The sum is direct by Proposition 2.1.8. Since $\dim(\mathcal{U} \cap \mathcal{W}) = 0$, the asserted equality holds by Theorem 2.3.1.
Structural Commentary: The directness criterion removes the intersection term from the sum formula.
Key Insight: Trivial overlap makes dimensions additive. □

Corollary 2.3.3. *Large Subspaces Must Intersect*
Let $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$, where $\mathcal{V}$ is finite-dimensional, and let $k \in \mathbb{N}$.

(i) If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, then $\mathcal{U} \cap \mathcal{W}$ contains a nonzero vector.
(ii) If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\mathcal{U} \cap \mathcal{W}$ contains k linearly independent vectors.

Proof. The inequality $\dim(\mathcal{U}+\mathcal{W}) \leq \dim(\mathcal{V})$ holds by Proposition 1.4.3.
Hence $\dim(\mathcal{U} \cap \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U}+\mathcal{W}) \geq \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{V})$ by Theorem 2.3.1.

(i) Under strict inequality, $\dim(\mathcal{U} \cap \mathcal{W}) \geq 1$. Choose basis of intersection by Corollary 1.3.14; any one of its vectors is nonzero.
(ii) Under second inequality, $\dim(\mathcal{U} \cap \mathcal{W}) \geq k$.
A basis of the intersection exists by Corollary 1.3.14, and any k of its vectors are linearly independent.

Structural Commentary: The sum formula converts excess ambient dimension into a lower bound on overlap.
Key Insight: Excess dimension must reappear in the intersection. □

Theorem 2.3.4. *Direct Sums and Dimension*
Let $m \in \mathbb{N}$, $\mathcal{V}$ be finite-dimensional, $\mathcal{U}_1, \ldots, \mathcal{U}_m \leq \mathcal{V}$. For $1 \leq j \leq m$, let β_j be an ordered basis of $\mathcal{U}_j$, $\beta = (\beta_1, \ldots, \beta_m)$ be their concatenation.

(i) The sum $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is direct iff $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m) = \sum_{j=1}^{m} \dim(\mathcal{U}_j)$.
(ii) If the sum is direct, then β is an ordered basis of $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Proof. The list β spans $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ and has length $\sum_{j=1}^{m} \dim(\mathcal{U}_j)$.

(i) ($\Rightarrow$) Suppose the sum is direct. A linear relation among the vectors of β determines vectors $\mathbf{u}_j \in \mathcal{U}_j$ satisfying $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}$.
Every $\mathbf{u}_j$ is zero by Proposition 2.1.23, and then every coefficient is zero by the independence of β_j.
Thus β is a basis, and its length proves the dimension equality.
(i) ($\Leftarrow$) Suppose dimension equality holds. The spanning list β has length $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m)$, so it is a basis by Proposition 1.4.4.
If $\sum_{j=1}^{m} \mathbf{u}_j = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$, expanding each $\mathbf{u}_j$ in β_j produces a relation in β.
Its coefficients, and hence every $\mathbf{u}_j$, are zero. The sum is therefore direct by Proposition 2.1.23.

(*ii*) Under directness, the linear-independence argument in the forward direction of part (i) applies to β.
Since β already spans the sum, it is an ordered basis of $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Structural Commentary: Concatenation converts directness into independence and dimension equality into full basis length.
Key Insight: Dimensions add exactly when concatenation has no redundancy. □

Definition 2.3.5. *Basis Adapted to a Direct Sum*
Let $m \in \mathbb{N}$, let $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$ be finite-dimensional, and let β be an ordered basis of $\mathcal{V}$.
The basis β is *adapted to the direct-sum decomposition* iff there are ordered bases β_j of $\mathcal{U}_j$ such that $\beta = (\beta_1, \ldots, \beta_m)$.

Proposition 2.3.6. *Complements Exist in Finite Dimensions*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \leq \mathcal{V}$, then $(\exists\, \mathcal{W} \leq \mathcal{V})\, \mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ by Corollary 1.3.14, and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$ by Theorem 1.3.13.
Set $\mathcal{W} := \operatorname{span}(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. The extended basis spans $\mathcal{V}$, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$.
If $\mathbf{x} \in \mathcal{U} \cap \mathcal{W}$, subtracting its two representations produces a relation in the extended basis.
Linear independence forces every coefficient to vanish, so $\mathbf{x} = \mathbf{0}$. Therefore $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ by Proposition 2.1.8.
Structural Commentary: Basis extension isolates the old directions from the newly appended directions.
Key Insight: The appended basis vectors span a complement. □

Theorem 2.3.7. *Complements Exist*
Assume Zorn's lemma. If $\mathcal{U} \leq \mathcal{V}$, then $(\exists\, \mathcal{W} \leq \mathcal{V})\, \mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a Hamel basis $\mathcal{B}_\mathcal{U}$ of $\mathcal{U}$ and extend it to a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.3.17. Set $\mathcal{C} := \mathcal{B} \setminus \mathcal{B}_\mathcal{U}$ and $\mathcal{W} := \operatorname{span}(\mathcal{C})$.
Every finite $\mathcal{B}$-expansion separates into terms indexed by $\mathcal{B}_\mathcal{U}$ and terms indexed by $\mathcal{C}$, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$.
If $\mathbf{x} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{x}$ has one $\mathcal{B}$-expansion supported in $\mathcal{B}_\mathcal{U}$ and another supported in the disjoint set $\mathcal{C}$.
These expansions agree by Proposition 1.3.16, so both have empty support and $\mathbf{x} = \mathbf{0}$. Therefore $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ by Proposition 2.1.8.
Structural Commentary: Hamel basis extension partitions all coordinates into subspace and complementary supports.
Key Insight: Disjoint coordinate supports produce a complement. □

Proposition 2.3.8. *Rank-Nullity as a Complement Isomorphism*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and suppose $\mathcal{V} = \ker(T) \oplus \mathcal{M}$.

(i) The restriction $T|_\mathcal{M} : \mathcal{M} \to \operatorname{range}(T)$ is an isomorphism.
(ii) If $\mathcal{V}$ is finite-dimensional, then $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\operatorname{range}(T))$.

Proof.

(i) The restriction is linear and $\ker(T|_\mathcal{M}) = \mathcal{M} \cap \ker(T) = \{\mathbf{0}\}$, so it is injective by Proposition 1.6.2.
If $T\mathbf{v} \in \operatorname{range}(T)$, write $\mathbf{v} = \mathbf{k} + \mathbf{m}$ with $\mathbf{k} \in \ker(T)$ and $\mathbf{m} \in \mathcal{M}$. Then $T\mathbf{v} = T\mathbf{m}$, so the restriction is surjective onto $\operatorname{range}(T)$.
It is therefore an isomorphism by Proposition 1.6.7.

(ii) Suppose $\mathcal{V}$ is finite-dimensional. The subspaces $\ker(T)$ and $\mathcal{M}$ are finite-dimensional by Proposition 1.4.3, and $\operatorname{range}(T)$ is finite-dimensional by Theorem 1.7.1. Thus $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\mathcal{M})$ by Corollary 2.3.2, while $\dim(\mathcal{M}) = \dim(\operatorname{range}(T))$ by part (i) and Theorem 1.7.9. Substitution proves the formula.

Structural Commentary: The restriction removes precisely the invisible kernel directions before dimensions are counted.
Key Insight: A kernel complement models the range exactly. □

Proposition 2.3.9. *One-Sided Inverses Split in Finite Dimensions*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) If T is injective, then $(\exists\, L \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\, LT = I_\mathcal{V}$.
(ii) If T is surjective, then $(\exists\, R \in \mathcal{L}(\mathcal{W}, \mathcal{V}))\, TR = I_\mathcal{W}$.

Proof.

(i) Suppose T is injective. Choose $\mathcal{H} \leq \mathcal{W}$ with $\mathcal{W} = \operatorname{range}(T) \oplus \mathcal{H}$ by Proposition 2.3.6.
For the unique decomposition $\mathbf{w} = T\mathbf{v} + \mathbf{h}$, define $L\mathbf{w} := \mathbf{v}$.
This is well-defined: directness determines $T\mathbf{v}$ and $\mathbf{h}$, while injectivity determines $\mathbf{v}$.
If $\mathbf{w}_i = T\mathbf{v}_i + \mathbf{h}_i$ for $i = 1, 2$ and $a, b \in \mathbb{F}$, then $L(a\mathbf{w}_1 + b\mathbf{w}_2) = L(T(a\mathbf{v}_1 + b\mathbf{v}_2) + (a\mathbf{h}_1 + b\mathbf{h}_2)) = a\mathbf{v}_1 + b\mathbf{v}_2 = aL\mathbf{w}_1 + bL\mathbf{w}_2$.
Hence L is linear, and $LT\mathbf{v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.

(ii) Suppose T is surjective. Choose $\mathcal{H} \leq \mathcal{V}$ with $\mathcal{V} = \ker(T) \oplus \mathcal{H}$ by Proposition 2.3.6.
Since $\operatorname{range}(T) = \mathcal{W}$, the restriction $T|_\mathcal{H} : \mathcal{H} \to \mathcal{W}$ is an isomorphism by Proposition 2.3.8(i).
Let $R := (T|_\mathcal{H})^{-1}$, regarded as a linear map from $\mathcal{W}$ to $\mathcal{V}$. Then $TR = I_\mathcal{W}$.

Structural Commentary: A range complement extends a left inverse; a kernel complement supports a right inverse.
Key Insight: Complements turn injectivity or surjectivity into a splitting. □

Proposition 2.3.10. *Quotient by a Complemented Subspace*
Let $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ with $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$. Then the map $J : \mathcal{W} \to \mathcal{V}/\mathcal{U}$ defined by $J(\mathbf{w}) := \mathbf{w} + \mathcal{U}$ is an isomorphism.
Consequently, every complement of $\mathcal{U}$ is isomorphic to $\mathcal{V}/\mathcal{U}$.

Proof.

(i) *Linearity*: For $a, b \in \mathbb{F}$, $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, equality $J(a\mathbf{w}_1 + b\mathbf{w}_2) = aJ(\mathbf{w}_1) + bJ(\mathbf{w}_2)$ holds by Proposition 2.2.6. Thus J is linear.

(ii) *Injectivity*: The equivalence $J(\mathbf{w}) = \mathcal{U} \Leftrightarrow \mathbf{w} \in \mathcal{U}$ holds by Proposition 2.2.2.
Hence $\ker(J) = \mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, and J is injective by Proposition 1.6.2.

(iii) *Surjectivity*: Let $\mathbf{v} + \mathcal{U} \in \mathcal{V}/\mathcal{U}$. Write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.
Then $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U} = J(\mathbf{w})$ by Proposition 2.2.2, so J is surjective.

(iv) *Conclusion*: The map J is an isomorphism by Proposition 1.6.7.
Since $\mathcal{W}$ was an arbitrary complement of $\mathcal{U}$, this proves the consequent.

Structural Commentary: The proof checks that each coset has exactly one representative in the complement.
Key Insight: A complement is a concrete model of the quotient. □

Definition 2.3.11. *Codimension*
If $\mathcal{U} \leq \mathcal{V}$ and $\mathcal{V}/\mathcal{U}$ is finite-dimensional, then the *codimension of $\mathcal{U}$ in $\mathcal{V}$* is $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) := \dim(\mathcal{V}/\mathcal{U})$.

If $\mathcal{V}$ is finite-dimensional, then $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$ by Theorem 2.2.9. If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ and $\mathcal{V}/\mathcal{U}$ is finite-dimensional, then $\mathcal{W} \cong \mathcal{V}/\mathcal{U}$ by Proposition 2.3.10, so $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) = \dim(\mathcal{W})$ by Proposition 1.7.8.

Perspective 2.3.12. *Quotients Split over Fields*
In finite dimensions, a complement $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ exists by Proposition 2.3.6, and $\mathcal{V}/\mathcal{U} \cong \mathcal{W}$ by Proposition 2.3.10.
Thus the quotient forgets the $\mathcal{U}$-component and retains the complementary coordinate.

Proposition 2.3.13. *Quotient of a Direct Sum*
If $\mathcal{V} = \mathcal{V}_1 \oplus \mathcal{V}_2$ and $\mathcal{U}_j \leq \mathcal{V}_j$ for $j = 1, 2$, then $\mathcal{V}/(\mathcal{U}_1 \oplus \mathcal{U}_2) \cong (\mathcal{V}_1/\mathcal{U}_1) \times (\mathcal{V}_2/\mathcal{U}_2)$.

Proof.

(i) *Denominator*: Since $\mathcal{U}_1 \cap \mathcal{U}_2 \subseteq \mathcal{V}_1 \cap \mathcal{V}_2 = \{\mathbf{0}\}$, the sum $\mathcal{U}_1 + \mathcal{U}_2$ is direct by Proposition 2.1.8.
(ii) *Construction*: For the unique decomposition $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$, define $\Phi : \mathcal{V} \to (\mathcal{V}_1/\mathcal{U}_1) \times (\mathcal{V}_2/\mathcal{U}_2)$ by $\Phi(\mathbf{v}_1 + \mathbf{v}_2) := (\mathbf{v}_1 + \mathcal{U}_1, \mathbf{v}_2 + \mathcal{U}_2)$. Componentwise quotient operations show that Φ is linear by Proposition 2.2.6.
(iii) *Range and kernel*: Every pair $(\mathbf{x}_1 + \mathcal{U}_1, \mathbf{x}_2 + \mathcal{U}_2)$ equals $\Phi(\mathbf{x}_1 + \mathbf{x}_2)$, so Φ is surjective. Moreover, $\Phi(\mathbf{v}_1 + \mathbf{v}_2) = (\mathcal{U}_1, \mathcal{U}_2)$ iff $\mathbf{v}_j \in \mathcal{U}_j$ for $j = 1, 2$ by Proposition 2.2.2. Hence $\ker(\Phi) = \mathcal{U}_1 \oplus \mathcal{U}_2$.
(iv) *Conclusion*: The claimed isomorphism holds by Theorem 2.2.12, because $\operatorname{range}(\Phi)$ is the whole product.

Structural Commentary: Direct-sum coordinates define the map; its componentwise kernel identifies the quotient.
Key Insight: Quotienting a direct sum acts coordinatewise. □

Perspective 2.3.14. *Quotients Depend on the Embedding*
A complement realizes a quotient, but the result depends on how the subspace is embedded.
Let $\mathcal{V}$ have countable basis $(\mathbf{e}_n)_{n \geq 1}$ and set $\mathcal{S} := \operatorname{span}(\mathbf{e}_n : n \geq 2)$.
The forward shift $F : \mathcal{V} \to \mathcal{S}$ defined by $F(\mathbf{e}_n) = \mathbf{e}_{n+1}$ is an isomorphism: unique finite basis expansions make it linear, and the backward shift $\mathbf{e}_{n+1} \mapsto \mathbf{e}_n$ is its inverse. Thus $\mathcal{S} \cong \mathcal{V}$.
Nevertheless, $\mathcal{V} = \mathcal{V} \oplus \{\mathbf{0}\}$ and $\mathcal{V} = \mathcal{S} \oplus \mathbb{F}\mathbf{e}_1$, so $\mathcal{V}/\mathcal{V} \cong \{\mathbf{0}\}$ while $\mathcal{V}/\mathcal{S} \cong \mathbb{F}\mathbf{e}_1$ by Proposition 2.3.10.
Hence isomorphic subspaces can have nonisomorphic quotients when embedded differently.

2.4 Algebraic Projections and Resolutions

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$. Projections encode direct-sum coordinates as operators, while resolutions package finite decompositions into mutually annihilating idempotents.

Definition 2.4.1. *Projection Along a Complement*

(i) An operator $P \in \mathcal{L}(\mathcal{V})$ is an *algebraic projection*, or *idempotent operator*, iff $P^2 = P$.
(ii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, the *projection onto $\mathcal{U}$ along $\mathcal{W}$* is the map $\rho_{\mathcal{U},\mathcal{W}} : \mathcal{V} \to \mathcal{V}$ defined by $(\forall\, (\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W})\, \rho_{\mathcal{U},\mathcal{W}}(\mathbf{u} + \mathbf{w}) := \mathbf{u}$.

Proposition 2.4.2. *Algebraic Projections and Direct-Sum Splittings*
The following hold.

(i) For $P \in \mathcal{L}(\mathcal{V})$, $P^2 = P$ iff $\mathcal{V} = \operatorname{range}(P) \oplus \ker(P)$ and $P = \rho_{\operatorname{range}(P),\ker(P)}$.
(ii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then $\rho_{\mathcal{U},\mathcal{W}} \in \mathcal{L}(\mathcal{V})$ and $\rho_{\mathcal{U},\mathcal{W}}^2 = \rho_{\mathcal{U},\mathcal{W}}$.
(iii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then $\operatorname{range}(\rho_{\mathcal{U},\mathcal{W}}) = \mathcal{U}$ and $\ker(\rho_{\mathcal{U},\mathcal{W}}) = \mathcal{W}$.

Proof.

(i) ($\Rightarrow$) Suppose $P^2 = P$. For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$, where $P\mathbf{v} \in \operatorname{range}(P)$ and $\mathbf{v} - P\mathbf{v} \in \ker(P)$.
If $\mathbf{x} \in \operatorname{range}(P) \cap \ker(P)$, choose $\mathbf{y} \in \mathcal{V}$ with $\mathbf{x} = P\mathbf{y}$. Then $\mathbf{x} = P^2\mathbf{y} = P\mathbf{x} = \mathbf{0}$, so the sum is direct by Proposition 2.1.8.
The displayed decomposition satisfies $(\forall\, \mathbf{v} \in \mathcal{V})\, \rho_{\operatorname{range}(P),\ker(P)}\mathbf{v} = P\mathbf{v}$; hence $P = \rho_{\operatorname{range}(P),\ker(P)}$.
(i) ($\Leftarrow$) Suppose $\mathcal{V} = \operatorname{range}(P) \oplus \ker(P)$ and $P = \rho_{\operatorname{range}(P),\ker(P)}$.
If $\mathbf{v} = \mathbf{x} + \mathbf{y}$ with $\mathbf{x} \in \operatorname{range}(P)$ and $\mathbf{y} \in \ker(P)$, then $P^2\mathbf{v} = P\mathbf{x} = \mathbf{x} = P\mathbf{v}$; hence $P^2 = P$.
(ii) Set $R := \rho_{\mathcal{U},\mathcal{W}}$ and write $\mathbf{v}_j = \mathbf{u}_j + \mathbf{w}_j$ with $(\mathbf{u}_j, \mathbf{w}_j) \in \mathcal{U} \times \mathcal{W}$ for $j = 1, 2$.
For $a, b \in \mathbb{F}$, $R(a\mathbf{v}_1 + b\mathbf{v}_2) = a\mathbf{u}_1 + b\mathbf{u}_2 = aR\mathbf{v}_1 + bR\mathbf{v}_2$ by uniqueness of direct-sum decompositions, so R is linear.
Also $R^2(\mathbf{u} + \mathbf{w}) = R\mathbf{u} = \mathbf{u} = R(\mathbf{u} + \mathbf{w})$, so R is idempotent.
(iii) The inclusion $\operatorname{range}(\rho_{\mathcal{U},\mathcal{W}}) \subseteq \mathcal{U}$ holds by the defining formula, while $\rho_{\mathcal{U},\mathcal{W}}\mathbf{u} = \mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$; hence the range is $\mathcal{U}$.
For $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\rho_{\mathcal{U},\mathcal{W}}\mathbf{v} = \mathbf{0}$ iff $\mathbf{u} = \mathbf{0}$ iff $\mathbf{v} \in \mathcal{W}$, so the kernel is $\mathcal{W}$.

Structural Commentary: Decomposition handles characterization; coordinates verify linearity, idempotence, range, and kernel.
Key Insight: Every idempotent keeps its range and kills its kernel. □

Proposition 2.4.3. *Complementary Algebraic Projections*
If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then the following hold.

(i) $\rho_{\mathcal{U},\mathcal{W}} + \rho_{\mathcal{W},\mathcal{U}} = I_{\mathcal{V}}$.
(ii) $\rho_{\mathcal{U},\mathcal{W}}\rho_{\mathcal{W},\mathcal{U}} = \rho_{\mathcal{W},\mathcal{U}}\rho_{\mathcal{U},\mathcal{W}} = 0$.

Proof. Let $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $(\mathbf{u}, \mathbf{w}) \in \mathcal{U} \times \mathcal{W}$.

(i) $(\rho_{\mathcal{U},\mathcal{W}} + \rho_{\mathcal{W},\mathcal{U}})\mathbf{v} = \mathbf{u} + \mathbf{w} = \mathbf{v}$. Since this equality holds on $\mathcal{V}$, the sum is $I_{\mathcal{V}}$.
(ii) $\rho_{\mathcal{U},\mathcal{W}}\rho_{\mathcal{W},\mathcal{U}}\mathbf{v} = \rho_{\mathcal{U},\mathcal{W}}\mathbf{w} = \mathbf{0}$ and $\rho_{\mathcal{W},\mathcal{U}}\rho_{\mathcal{U},\mathcal{W}}\mathbf{v} = \rho_{\mathcal{W},\mathcal{U}}\mathbf{u} = \mathbf{0}$. Since both equalities hold on $\mathcal{V}$, both products are zero.

Structural Commentary: The two coordinate projections are evaluated on the same unique decomposition.
Key Insight: Complementary projections partition identity and annihilate each other. □

Definition 2.4.4. *Algebraic Orthogonality and Resolutions*

(i) Algebraic projections $\rho, \sigma \in \mathcal{L}(\mathcal{V})$ are *algebraically orthogonal*, written $\rho \perp \sigma$, iff $\rho\sigma = \sigma\rho = 0$.
(ii) A *resolution of identity* on $\mathcal{V}$ is a finite family of algebraic projections $(\rho_i)_{i=1}^k$ such that $k \in \mathbb{N}$,

$$(\forall\, i, j \in \{1, \ldots, k\})(i \neq j \Rightarrow \rho_i \perp \rho_j), \text{ and } \sum_{i=1}^{k} \rho_i = I_{\mathcal{V}}.$$

Theorem 2.4.5. *Resolution Ranges Form a Direct Sum*
Let $(\rho_i)_{i=1}^k$ be a resolution of identity on $\mathcal{V}$, and set $\mathcal{R}_i := \operatorname{range}(\rho_i)$ for $i \in \{1,\dots,k\}$. Then the following hold.

(i) $\mathcal{V} = \bigoplus_{i=1}^{k} \mathcal{R}_i$.

(ii) $(\forall\, i \in \{1,\dots,k\})\ \ker(\rho_i) = \bigoplus_{j\in\{1,\dots,k\}\setminus\{i\}} \mathcal{R}_j$.

Proof.

(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \sum_{i=1}^{k} \rho_i \mathbf{v} \in \sum_{i=1}^{k} \mathcal{R}_i$. Suppose $\sum_{i=1}^{k} \mathbf{x}_i = \mathbf{0}$ with $\mathbf{x}_i \in \mathcal{R}_i$.
For $m \in \{1,\dots,k\}$, idempotence and algebraic orthogonality imply $\rho_m \mathbf{x}_m = \mathbf{x}_m$ and $(\forall\, i \in \{1,\dots,k\} \setminus \{m\})\ \rho_m \mathbf{x}_i = \mathbf{0}$.
Hence $\mathbf{x}_m = \rho_m \sum_{i=1}^{k} \mathbf{x}_i = \mathbf{0}$. The sum is direct by Proposition 2.1.23.

(ii) By part (i), write $\mathbf{v} = \sum_{j=1}^{k} \mathbf{x}_j$ uniquely with $\mathbf{x}_j \in \mathcal{R}_j$. The equality $\rho_i \mathbf{v} = \mathbf{x}_i$ holds by idempotence and algebraic orthogonality.
Hence $\mathbf{v} \in \ker(\rho_i)$ iff $\mathbf{x}_i = \mathbf{0}$. These are exactly the vectors in $\bigoplus_{j\in\{1,\dots,k\}\setminus\{i\}} \mathcal{R}_j$.

Structural Commentary: The identity establishes spanning; orthogonal idempotents isolate each component and identify every kernel.
Key Insight: Mutually annihilating projections recover unique direct-sum coordinates. □

Theorem 2.4.6. *Direct Sums Give Resolutions*
Let $k \in \mathbb{N}$, set $K := \{1,\dots,k\}$, and suppose $\mathcal{V} = \bigoplus_{i\in K} \mathcal{U}_i$. For $i \in K$, set $\rho_i := \rho_{\mathcal{U}_i,\ \sum_{j\in K\setminus\{i\}} \mathcal{U}_j}$. Then $(\rho_i)_{i\in K}$ is a resolution of identity.

Proof. For $i \in K$, set $\mathcal{W}_i := \sum_{j\in K\setminus\{i\}} \mathcal{U}_j$. Then $\mathcal{U}_i \cap \mathcal{W}_i = \{\mathbf{0}\}$ by Proposition 2.1.23(iii), while $\mathcal{V} = \mathcal{U}_i + \mathcal{W}_i$ by the assumed sum.
Hence $\mathcal{V} = \mathcal{U}_i \oplus \mathcal{W}_i$ by Proposition 2.1.8, and ρ_i is an algebraic projection by Proposition 2.4.2(ii).
For the unique decomposition $\mathbf{v} = \sum_{i\in K} \mathbf{u}_i$, $\rho_i \mathbf{v} = \mathbf{u}_i$; hence $(\sum_{i\in K} \rho_i)\mathbf{v} = \mathbf{v}$. This pointwise identity proves $\sum_{i\in K} \rho_i = I_{\mathcal{V}}$.
If $i \neq j$, then $\rho_i \rho_j \mathbf{v} = \rho_i \mathbf{u}_j = \mathbf{0}$ and $\rho_j \rho_i \mathbf{v} = \rho_j \mathbf{u}_i = \mathbf{0}$. Thus $\rho_i \perp \rho_j$, and the family is a resolution of identity.
Structural Commentary: Direct-sum uniqueness defines coordinate projections; pointwise identities verify the resolution conditions.
Key Insight: Coordinate projections isolate summands and reconstruct every vector. □

Theorem 2.4.7. *Algebra of Projections*
Let $\rho, \sigma \in \mathcal{L}(\mathcal{V})$ be algebraic projections, and assume $2 \neq 0$ in $\mathbb{F}$.

(i) The operator $\rho + \sigma$ is an algebraic projection iff $\rho \perp \sigma$.
Under these conditions, $\operatorname{range}(\rho+\sigma) = \operatorname{range}(\rho) \oplus \operatorname{range}(\sigma)$ *and* $\ker(\rho+\sigma) = \ker(\rho) \cap \ker(\sigma)$.

(ii) The operator $\rho - \sigma$ is an algebraic projection iff $\rho\sigma = \sigma\rho = \sigma$.
Under these conditions, $\operatorname{range}(\rho-\sigma) = \operatorname{range}(\rho) \cap \ker(\sigma)$ *and* $\ker(\rho-\sigma) = \ker(\rho) \oplus \operatorname{range}(\sigma)$.

(iii) If $\rho\sigma = \sigma\rho$, then $\rho\sigma$ is an algebraic projection, $\operatorname{range}(\rho\sigma) = \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$, *and* $\ker(\rho\sigma) = \ker(\rho) + \ker(\sigma)$.

Proof.

(i) ($\Rightarrow$) Set $A := \rho\sigma$, $B := \sigma\rho$, and $C := \rho\sigma\rho$. The identity $(\rho+\sigma)^2 = \rho+\sigma$ is equivalent to $A + B = 0$.
Then $A + C = \rho(A+B) = 0$ and $C + B = (A+B)\rho = 0$, so $A = B$ and $2A = 0$.
The scalar 2 is invertible by $2 \neq 0$ in the field $\mathbb{F}$; hence $A = B = 0$ and $\rho \perp \sigma$.

(i) ($\Leftarrow$) Suppose $\rho \perp \sigma$. Then $(\rho+\sigma)^2 = \rho+\sigma$, so the sum is an algebraic projection.
Its range is contained in $\operatorname{range}(\rho) + \operatorname{range}(\sigma)$, while $(\rho+\sigma)(\mathbf{x}+\mathbf{y}) = \mathbf{x}+\mathbf{y}$ for $(\mathbf{x},\mathbf{y}) \in \operatorname{range}(\rho) \times \operatorname{range}(\sigma)$.
If $\mathbf{z} \in \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$ and $\mathbf{z} = \sigma\mathbf{w}$, then $\mathbf{z} = \rho\mathbf{z} = \rho\sigma\mathbf{w} = \mathbf{0}$; hence the range sum is direct by Proposition 2.1.8.
$(\rho+\sigma)\mathbf{v} = \mathbf{0} \Rightarrow \rho\mathbf{v} = \rho(\rho+\sigma)\mathbf{v} = \mathbf{0}$ and $\sigma\mathbf{v} = \sigma(\rho+\sigma)\mathbf{v} = \mathbf{0}$. Reverse implication is immediate. Thus $\ker(\rho+\sigma) = \ker(\rho) \cap \ker(\sigma)$.

(ii) ($\Rightarrow$) Suppose $Q := \rho - \sigma$ is algebraic projection. Then $(I_{\mathcal{V}} - Q)^2 = I_{\mathcal{V}} - Q$ and $(I_{\mathcal{V}} - \rho)^2 = I_{\mathcal{V}} - \rho$ hold by $Q^2 = Q$ and $\rho^2 = \rho$.
Since $I_{\mathcal{V}} - Q = (I_{\mathcal{V}} - \rho) + \sigma$, $(I_{\mathcal{V}} - \rho) \perp \sigma$ by part (i). Therefore $(I_{\mathcal{V}} - \rho)\sigma = \sigma(I_{\mathcal{V}} - \rho) = 0$, which is equivalent to $\rho\sigma = \sigma\rho = \sigma$.

(ii) ($\Leftarrow$) Suppose $\rho\sigma = \sigma\rho = \sigma$ and set $Q := \rho - \sigma$. The operator $I_{\mathcal{V}} - \rho$ is an algebraic projection and is algebraically orthogonal to σ, so $I_{\mathcal{V}} - Q = (I_{\mathcal{V}} - \rho) + \sigma$ is an algebraic projection by part (i). Since $I_{\mathcal{V}} - Q$ is idempotent, $Q^2 = (I_{\mathcal{V}} - (I_{\mathcal{V}} - Q))^2 = I_{\mathcal{V}} - (I_{\mathcal{V}} - Q) = Q$.
Since $\rho Q = Q$ and $\sigma Q = 0$, $\operatorname{range}(Q) \subseteq \operatorname{range}(\rho) \cap \ker(\sigma)$; the reverse inclusion holds because $Q\mathbf{x} = \mathbf{x}$ when $\rho\mathbf{x} = \mathbf{x}$ and $\sigma\mathbf{x} = \mathbf{0}$.
The operator Q kills both $\ker(\rho)$ and $\operatorname{range}(\sigma)$. Their intersection is trivial: if $\mathbf{z} = \sigma\mathbf{w} \in \ker(\rho)$, then $\mathbf{z} = \rho\sigma\mathbf{w} = \rho\mathbf{z} = \mathbf{0}$.
Their sum is direct by Proposition 2.1.8, and $\ker(\rho) \oplus \operatorname{range}(\sigma) \subseteq \ker(Q)$ because Q kills both summands.
Conversely, if $\mathbf{v} \in \ker(Q)$, then $\rho\mathbf{v} = \sigma\mathbf{v}$ and $\mathbf{v} = (\mathbf{v} - \rho\mathbf{v}) + \sigma\mathbf{v} \in \ker(\rho) \oplus \operatorname{range}(\sigma)$. Hence $\ker(Q) = \ker(\rho) \oplus \operatorname{range}(\sigma)$.

(iii) Suppose $\rho\sigma = \sigma\rho$. Then $(\rho\sigma)^2 = \rho^2\sigma^2 = \rho\sigma$, so $\rho\sigma$ is an algebraic projection.
If $\mathbf{x} = \rho\sigma\mathbf{y}$, then $\rho\mathbf{x} = \mathbf{x} = \sigma\mathbf{x}$; conversely, $\rho\mathbf{x} = \mathbf{x} = \sigma\mathbf{x}$ implies $\mathbf{x} = \rho\sigma\mathbf{x}$. Thus $\operatorname{range}(\rho\sigma) = \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$.
If $\mathbf{x} \in \ker(\rho)$, then $\rho\sigma\mathbf{x} = \sigma\rho\mathbf{x} = \mathbf{0}$; if $\mathbf{x} \in \ker(\sigma)$, then $\rho\sigma\mathbf{x} = \mathbf{0}$. Hence $\ker(\rho) + \ker(\sigma) \subseteq \ker(\rho\sigma)$.
If $\rho\sigma\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = (\mathbf{v} - \sigma\mathbf{v}) + \sigma\mathbf{v}$ with $\mathbf{v} - \sigma\mathbf{v} \in \ker(\sigma)$ and $\sigma\mathbf{v} \in \ker(\rho)$; hence $\ker(\rho\sigma) = \ker(\rho) + \ker(\sigma)$.

Structural Commentary: Expanding idempotence isolates cross terms; fixed and killed subspaces determine each range and kernel.
Key Insight: Commuting idempotents encode intersections and sums. □

3 Matrix Representations, Change of Basis, Block Matrices, and Operations

3.1 Matrix Algebra and Linear Systems

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every size parameter is a positive integer.

Definition 3.1.1. *Matrices and Matrix Operations*

(i) *Matrix space and equality*: $\operatorname{Mat}(m,n,\mathbb{F}) := \{[A_{j,k}]_{1\le j\le m,\ 1\le k\le n} : A_{j,k} \in \mathbb{F}\}$.
For A, C in this space, $(A = C) \iff (\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,n\})\ A_{j,k} = C_{j,k}$.

(ii) *Entries, rows, and columns*: if $A = [A_{j,k}] \in \operatorname{Mat}(m,n,\mathbb{F})$, then $A_{j,k}$ is its (j,k)-entry, $A_{j,\cdot}$ is row j, and $A_{\cdot,k}$ is column k.

(iii) *Transpose*: for $A \in \operatorname{Mat}(m,n,\mathbb{F})$, the transpose $A^t \in \operatorname{Mat}(n,m,\mathbb{F})$ satisfies $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,n\})\ (A^t)_{k,j} := A_{j,k}$.

(iv) *Main diagonal and diagonal matrix*: if $A = [A_{j,k}] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $r = \min(m,n)$, its main diagonal is $(A_{1,1},\dots,A_{r,r})$. For $\lambda_1,\dots,\lambda_n \in \mathbb{F}$, $\mathrm{diag}(\lambda_1,\dots,\lambda_n) \in \mathrm{Mat}(n,n,\mathbb{F})$ has (j,j)-entry λ_j and all off-diagonal entries zero.
(v) *Symmetry*: a matrix $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is symmetric iff $A^t = A$, and skew-symmetric iff $A^t = -A$.
(vi) *Triangularity*: a matrix $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is upper triangular iff $(\forall\, j,k \in \{1,\dots,n\})(j > k \Rightarrow A_{j,k} = 0)$. Lower triangular iff $(\forall\, j,k \in \{1,\dots,n\})(j < k \Rightarrow A_{j,k} = 0)$. Diagonal iff $(\forall\, j,k \in \{1,\dots,n\})(j \neq k \Rightarrow A_{j,k} = 0)$.
(vii) *Linear operations*: for $A, C \in \mathrm{Mat}(m,n,\mathbb{F})$ and $a \in \mathbb{F}$, $(A+C)_{j,k} := A_{j,k} + C_{j,k}$ and $(aA)_{j,k} := aA_{j,k}$ for admissible j,k.
(viii) *Row-column pairing*: if $r_1,\dots,r_n,c_1,\dots,c_n \in \mathbb{F}$, $\mathbf{r} = (r_1,\dots,r_n)$, and $\mathbf{c} = (c_1,\dots,c_n)^t$, then $\mathbf{rc} := \sum_{s=1}^{n} r_s c_s$.
(ix) *Matrix product*: for $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(n,p,\mathbb{F})$, the product $AB \in \mathrm{Mat}(m,p,\mathbb{F})$ satisfies $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,p\})\,(AB)_{j,k} := \sum_{s=1}^{n} A_{j,s}B_{s,k}$.
(x) *Matrix-vector product*: identify $\mathbf{x} = (x_1,\dots,x_n)^t \in \mathbb{F}^n$ with its column matrix. For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, define $A\mathbf{x} \in \mathbb{F}^m$ by $(\forall\, j \in \{1,\dots,m\})\,(A\mathbf{x})_j := \sum_{k=1}^{n} A_{j,k}x_k$.
(xi) *Matrix units*: for $j \in \{1,\dots,m\}$ and $k \in \{1,\dots,n\}$, $E_{j,k} \in \mathrm{Mat}(m,n,\mathbb{F})$ has (j,k)-entry 1 and all other entries zero.
(xii) *Identity and inverse*: $I_n := \mathrm{diag}(1,\dots,1) = [\delta_{j,k}] \in \mathrm{Mat}(n,n,\mathbb{F})$, where $\delta_{j,k} = 1$ iff $j = k$. A matrix $B \in \mathrm{Mat}(n,n,\mathbb{F})$ is a two-sided inverse of $A \in \mathrm{Mat}(n,n,\mathbb{F})$ iff $AB = I_n = BA$; the matrix A is invertible iff it has a two-sided inverse.

Proposition 3.1.2. *Basic Matrix Product Laws*
Let $A, A' \in \mathrm{Mat}(m,n,\mathbb{F})$, $B, B' \in \mathrm{Mat}(n,p,\mathbb{F})$, $C \in \mathrm{Mat}(p,q,\mathbb{F})$, and $a \in \mathbb{F}$. Then:
(i) $(AB)C = A(BC)$;
(ii) $I_m A = A = AI_n$;
(iii) $(A+A')B = AB + A'B$ *and* $A(B+B') = AB + AB'$;
(iv) $(aA)B = a(AB) = A(aB)$.

Proof.
(i) For admissible j,k, $((AB)C)_{j,k} = \sum_{s=1}^{p}\sum_{r=1}^{n} A_{j,r}B_{r,s}C_{s,k} = \sum_{r=1}^{n}\sum_{s=1}^{p} A_{j,r}B_{r,s}C_{s,k} = (A(BC))_{j,k}$ by finite distributivity. Thus $(AB)C = A(BC)$ by entrywise equality.
(ii) For admissible j,k, $(I_m A)_{j,k} = \sum_{r=1}^{m} \delta_{j,r}A_{r,k} = A_{j,k} = \sum_{r=1}^{n} A_{j,r}\delta_{r,k} = (AI_n)_{j,k}$. Hence $I_m A = A = AI_n$.
(iii) For admissible j,k, finite distributivity implies $((A+A')B)_{j,k} = \sum_{r=1}^{n}(A_{j,r}+A'_{j,r})B_{r,k} = (AB+A'B)_{j,k}$. Also implies $(A(B+B'))_{j,k} = \sum_{r=1}^{n} A_{j,r}(B_{r,k}+B'_{r,k}) = (AB+AB')_{j,k}$. Entrywise equality proves both laws.
(iv) For admissible j,k, $((aA)B)_{j,k} = a\sum_{r=1}^{n} A_{j,r}B_{r,k} = (a(AB))_{j,k} = (A(aB))_{j,k}$; hence the matrices are equal.

Structural Commentary: Each law reduces to corresponding identities among finite scalar sums.
Key Insight: Entrywise equality transfers field arithmetic to compatible matrices. □

Lemma 3.1.3. *Uniqueness of Two-Sided Inverses*
Let $A, B, C \in \mathrm{Mat}(n,n,\mathbb{F})$. If $AB = I_n$ and $BC = I_n$, then $A = C$.

Proof. By Proposition 3.1.2 and the hypotheses, $A = AI_n = A(BC) = (AB)C = I_n C = C$.
Structural Commentary: A common middle inverse forces the left and right inverses to coincide.
Key Insight: Associativity turns $AB = I_n = BC$ into $A = C$. □

Definition 3.1.4. *Inverse Matrix Notation*
If $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, its unique two-sided inverse by Lemma 3.1.3 is denoted A^{-1}.
The set of invertible matrices in $\mathrm{Mat}(n,n,\mathbb{F})$ is denoted $\mathrm{Mat}(n,n,\mathbb{F})^{\times}$.

Proposition 3.1.5. *Matrix Spaces*
For $m,n \in \mathbb{Z}_{>0}$, the following hold.
(i) The entrywise operations make $\mathrm{Mat}(m,n,\mathbb{F})$ *a vector space over* $\mathbb{F}$.
(ii) The family $(E_{j,k})_{1\le j\le m,\ 1\le k\le n}$ *is a basis; hence* $\dim(\mathrm{Mat}(m,n,\mathbb{F})) = mn$.

Proof.
(i) The zero matrix and entrywise negatives supply the additive identity and inverses. Each remaining vector-space axiom holds in every entry by the corresponding field law, so it holds by matrix equality.
(ii) Every $A = [A_{j,k}]$ has the expansion $A = \sum_{j=1}^{m}\sum_{k=1}^{n} A_{j,k}E_{j,k}$. If $\sum_{j=1}^{m}\sum_{k=1}^{n} c_{j,k}E_{j,k} = 0$, then its (j,k)-entry is $c_{j,k} = 0$. Hence the family spans and is linearly independent. Its mn members form a basis, so the stated dimension is mn.

Structural Commentary: Entrywise field laws establish the space, and matrix units establish its basis.
Key Insight: Each matrix unit isolates one independent scalar coordinate. □

Proposition 3.1.6. *Matrix Product Entries Are Row Times Column*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(n,p,\mathbb{F})$, then $(\forall\, j \in \{1,\dots,m\})(\forall\, k \in \{1,\dots,p\})\,(AB)_{j,k} = A_{j,\cdot}B_{\cdot,k}$.

Proof. Both sides equal $\sum_{r=1}^{n} A_{j,r}B_{r,k}$ by the definitions of matrix multiplication and row-column pairing.
Structural Commentary: Matrix multiplication is assembled entry-by-entry from row-column pairings.
Key Insight: Each entry of AB sees one row of A and one column of B. □

Definition 3.1.7. *Map Induced by a Matrix*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, the *map induced by A* is $T_A : \mathbb{F}^n \to \mathbb{F}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 3.1.8. *Matrix-Induced Maps Are Linear*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$, then $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear.

Proof. For $\mathbf{x},\mathbf{y} \in \mathbb{F}^n$, $a,b \in \mathbb{F}$, and $j \in \{1,\dots,m\}$, $(T_A(a\mathbf{x}+b\mathbf{y}))_j = \sum_{k=1}^{n} A_{j,k}(ax_k + by_k) = a(T_A\mathbf{x})_j + b(T_A\mathbf{y})_j$.
Coordinatewise equality proves $T_A(a\mathbf{x}+b\mathbf{y}) = aT_A\mathbf{x} + bT_A\mathbf{y}$.
Structural Commentary: Matrix-vector multiplication preserves linear combinations.
Key Insight: Every matrix defines a canonical linear map between coordinate spaces. □

Proposition 3.1.9. *Matrix-Vector Products as Column Combinations*
If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{x} = (x_1,\dots,x_n)^t \in \mathbb{F}^n$, then $A\mathbf{x} = \sum_{k=1}^{n} x_k\mathbf{a}_k$.

Proof. For $j \in \{1,\dots,m\}$, both vectors have jth entry $\sum_{k=1}^{n} A_{j,k}x_k$.
Structural Commentary: Multiplying by a column selects a linear combination of the columns of A.
Key Insight: The coordinates of $\mathbf{x}$ are precisely the column coefficients. □

Proposition 3.1.10. *Matrix Products by Columns*
If $A \in \mathrm{Mat}(m,r,\mathbb{F})$ and $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \mathrm{Mat}(r,n,\mathbb{F})$, then $AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_n]$.

Proof. For $j \in \{1,\dots,m\}$ and $k \in \{1,\dots,n\}$, the jth entries of the kth column of AB and of $A\mathbf{b}_k$ both equal $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k}$.
Structural Commentary: Matrix products apply A to columns of B.
Key Insight: The columns of AB are obtained by applying A to the columns of B. □

Proposition 3.1.11. *Matrix Products as Sums of Outer Products*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_r] \in \mathrm{Mat}(m,r,\mathbb{F})$ and $B \in \mathrm{Mat}(r,n,\mathbb{F})$. If the ℓth row of B is $\mathbf{b}_\ell^t$ with $\mathbf{b}_\ell \in \mathbb{F}^n$, then $AB = \sum_{\ell=1}^{r} \mathbf{a}_\ell\mathbf{b}_\ell^t$.

Proof. For admissible j,k, the (j,k)-entry of $\mathbf{a}_\ell\mathbf{b}_\ell^t$ is $A_{j,\ell}B_{\ell,k}$, so the corresponding entry of the stated sum is $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k} = (AB)_{j,k}$.
Structural Commentary: Column-row products decompose a matrix product into single column-row pieces.
Key Insight: Each shared index ℓ contributes one outer product. □

Definition 3.1.12. *Column Space, Matrix Rank, and Nullity*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$.
(i) $\mathrm{col}(A) := \mathrm{span}(\mathbf{a}_1,\dots,\mathbf{a}_n) \subseteq \mathbb{F}^m$, and $\mathrm{row}(A) := \mathrm{col}(A^t) \subseteq \mathbb{F}^n$.
(ii) $\ker(A) := \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{0}\}$ is the *nullspace*, and $\mathrm{nullity}(A) := \dim(\ker(A))$.
(iii) $\mathrm{rank}(A) := \dim(\mathrm{col}(A))$ is the column rank, while $\dim(\mathrm{row}(A))$ is the row rank.
(iv) The matrix A has *full rank* iff $\mathrm{rank}(A) = \min(m,n)$, *full column rank* iff $\mathrm{rank}(A) = n$, and *full row rank* iff $\mathrm{rank}(A) = m$.
$\mathrm{range}(T_A) = \mathrm{col}(A)$ and $\mathrm{range}(T_{A^t}) = \mathrm{row}(A)$ by Proposition 3.1.9, while $\ker(T_A) = \ker(A)$ by definition.
These three sets are subspaces by Proposition 3.1.8 and Proposition 1.6.3, so the stated dimensions are defined.

Proposition 3.1.13. *Solvability and Complete Solution Sets*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{b} \in \mathbb{F}^m$. Then:
(i) The system $A\mathbf{x} = \mathbf{b}$ with $\mathbf{x} \in \mathbb{F}^n$ is solvable iff $\mathbf{b} \in \mathrm{col}(A)$.
(ii) If $A\mathbf{x}_p = \mathbf{b}$, its solution set is $\mathbf{x}_p + \ker(A) = \{\mathbf{x}_p + \mathbf{x}_0 : \mathbf{x}_0 \in \ker(A)\}$.

Proof.
(i) ($\Rightarrow$) If $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x} = (x_1,\dots,x_n)^t$, then $\mathbf{b} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j \in \mathrm{col}(A)$ by Proposition 3.1.9.
(i) ($\Leftarrow$) If $\mathbf{b} \in \mathrm{col}(A)$, then $(\exists\, x_1,\dots,x_n \in \mathbb{F})\, \mathbf{b} = \sum_{j=1}^{n} x_j\mathbf{a}_j$. For $\mathbf{x} = (x_1,\dots,x_n)^t$, $A\mathbf{x} = \mathbf{b}$ by Proposition 3.1.9.
(ii) If $A\mathbf{x}_p = \mathbf{b}$, then $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = \mathbf{b} \iff A(\mathbf{x}-\mathbf{x}_p) = \mathbf{0} \iff \mathbf{x}-\mathbf{x}_p \in \ker(A) \iff \mathbf{x} \in \mathbf{x}_p + \ker(A)$ by Proposition 3.1.8.
Structural Commentary: Column expansion proves existence, while subtraction identifies the complete affine fiber.
Key Insight: Column space decides existence; kernel decides nonuniqueness. □

Proposition 3.1.14. *Elementary Transpose Rules*
Let $A, C \in \mathrm{Mat}(m,n,\mathbb{F})$ and $a \in \mathbb{F}$. Then:
(i) $(A^t)^t = A$;
(ii) $(A+C)^t = A^t + C^t$ and $(aA)^t = aA^t$;
(iii) $I_n^t = I_n$.

Proof.
(i) For admissible j,k, $((A^t)^t)_{j,k} = (A^t)_{k,j} = A_{j,k}$.
(ii) For admissible j,k, $((A+C)^t)_{j,k} = A_{k,j} + C_{k,j} = (A^t + C^t)_{j,k}$ and $((aA)^t)_{j,k} = aA_{k,j} = (aA^t)_{j,k}$.
(iii) For $j,k \in \{1,\dots,n\}$, $(I_n^t)_{j,k} = \delta_{k,j} = \delta_{j,k} = (I_n)_{j,k}$.
Structural Commentary: Each identity follows by swapping the two entry indices.
Key Insight: Transposition preserves entrywise operations and reverses index order. □

Example 3.1.15. *Symmetric and Skew-Symmetric Decomposition*
Assume $2 \neq 0$ in $\mathbb{F}$ and set $M := \mathrm{Mat}(n, n, \mathbb{F})$. Define $\mathrm{Sym}_n(\mathbb{F}) := \{A \in M : A^t = A\}$ and $\mathrm{Skew}_n(\mathbb{F}) := \{A \in M : A^t = -A\}$.
The map $\tau : M \to M$, $\tau(A) := A^t$, is linear by Proposition 3.1.14(ii). Thus $\tau - I_M, \tau + I_M \in \mathcal{L}(M)$ by Proposition 1.5.5.
Since $\mathrm{Sym}_n(\mathbb{F}) = \ker(\tau - I_M)$ and $\mathrm{Skew}_n(\mathbb{F}) = \ker(\tau + I_M)$, both are subspaces of M by Proposition 1.6.3(i).
For $A \in M$, set $S_A := \frac{1}{2}(A + A^t)$ and $K_A := \frac{1}{2}(A - A^t)$. Then $S_A^t = S_A$, $K_A^t = -K_A$, and $A = S_A + K_A$ by Proposition 3.1.14.
Hence $M = \mathrm{Sym}_n(\mathbb{F}) + \mathrm{Skew}_n(\mathbb{F})$. If $A \in \mathrm{Sym}_n(\mathbb{F}) \cap \mathrm{Skew}_n(\mathbb{F})$, then $A = A^t = -A$ and $2A = 0$, whence $A = 0$ because $2 \neq 0$ is invertible in $\mathbb{F}$. Therefore $M = \mathrm{Sym}_n(\mathbb{F}) \oplus \mathrm{Skew}_n(\mathbb{F})$ by Proposition 2.1.8.
Structural Commentary: Transpose splits a matrix into its even and odd parts.
Key Insight: The directness of the split uses the ability to divide by 2.

Proposition 3.1.16. *Transpose Reverses Products*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(n, p, \mathbb{F})$, then $(AB)^t = B^t A^t$.

Proof. For $j \in \{1, \ldots, p\}$ and $k \in \{1, \ldots, m\}$, $((AB)^t)_{j,k} = (AB)_{k,j} = \sum_{\ell=1}^{n} A_{k,\ell} B_{\ell,j}$.
The same sum equals $\sum_{\ell=1}^{n} (B^t)_{j,\ell} (A^t)_{\ell,k} = (B^t A^t)_{j,k}$, so the matrices are equal.
Structural Commentary: Transposition swaps the order of composition encoded by matrix multiplication.
Key Insight: Rows become columns, so the product order reverses. □

Corollary 3.1.17. *Matrix Inverses and Transposes Reverse Products*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible. Then:
(i) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$;
(ii) A^t is invertible and $(A^t)^{-1} = (A^{-1})^t$.

Proof.
(i) For $D := B^{-1}A^{-1}$, $D(AB) = I_n = (AB)D$ by Proposition 3.1.2 and the inverse equations; hence $(AB)^{-1} = D$.
(ii) For $C := (A^{-1})^t$, $CA^t = I_n = A^t C$ by Proposition 3.1.16 and Proposition 3.1.14(iii); hence $(A^t)^{-1} = C$.
Structural Commentary: Each claim verifies the two inverse equations after reversing multiplication order.
Key Insight: Both inversion and transposition reverse products. □

3.2 Coordinates and Matrix Representations

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 3.2.1. *Coordinate Vector*
Let $n \geq 1$, let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of a vector space $\mathcal{V}$, and let $\mathbf{v} \in \mathcal{V}$. By Proposition 1.3.3, $(\exists!\, (c_1, \ldots, c_n) \in \mathbb{F}^n)\, \mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j$.
For this tuple, the *β-coordinate vector* of $\mathbf{v}$ is $[\mathbf{v}]_\beta := (c_1, \ldots, c_n)^t \in \mathbb{F}^n$.
The function $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is the *β-coordinate map*. The order of the list β fixes the order of the coordinates.

Proposition 3.2.2. *Coordinate Maps Are Isomorphisms*
Let $n \geq 1$, and let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of a vector space $\mathcal{V}$. The coordinate map $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ has the following properties:
(i) It is linear.
(ii) It is injective.
(iii) It is surjective and hence an isomorphism.

Proof.
(i) Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $c, d \in \mathbb{F}$, and write $[\mathbf{u}]_\beta = (a_1, \ldots, a_n)^t$ and $[\mathbf{v}]_\beta = (b_1, \ldots, b_n)^t$.
Then $[c\mathbf{u} + d\mathbf{v}]_\beta = (ca_1 + db_1, \ldots, ca_n + db_n)^t = c[\mathbf{u}]_\beta + d[\mathbf{v}]_\beta$, so $[\cdot]_\beta$ is linear.
(ii) If $[\mathbf{u}]_\beta = [\mathbf{v}]_\beta$, then $\mathbf{u}$ and $\mathbf{v}$ have the same coefficients in β, so $\mathbf{u} = \mathbf{v}$ by Proposition 1.3.3. Thus $[\cdot]_\beta$ is injective.
(iii) For $(c_1, \ldots, c_n)^t \in \mathbb{F}^n$, set $\mathbf{v} := \sum_{j=1}^{n} c_j \mathbf{v}_j$. Then $[\mathbf{v}]_\beta = (c_1, \ldots, c_n)^t$, so $[\cdot]_\beta$ is surjective.
It is therefore an isomorphism by Proposition 1.6.7.
Structural Commentary: The three checks compare vectors with their unique coefficient tuples.
Key Insight: An ordered basis identifies the vector space with its coordinate space without losing linear structure. □

Definition 3.2.3. *Matrix Representation*
Let $n, m \geq 1$, let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathcal{W}$.
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *matrix of T from β-coordinates to γ-coordinates* is $[T]_{\gamma \leftarrow \beta} := \big[[T(\mathbf{v}_1)]_\gamma \ \cdots \ [T(\mathbf{v}_n)]_\gamma\big] \in \mathrm{Mat}(m, n, \mathbb{F})$.
Thus its jth column is the uniquely determined coordinate vector $[T(\mathbf{v}_j)]_\gamma$.

Proposition 3.2.4. *Matrices Respect Linear Combinations of Maps*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathcal{W}$.
If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $[aS + T]_{\gamma \leftarrow \beta} = a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.

Proof. By Proposition 1.5.5, $aS + T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. For $j \in \{1, \ldots, n\}$, $[(aS + T)\mathbf{v}_j]_\gamma = [aS\mathbf{v}_j + T\mathbf{v}_j]_\gamma = a[S\mathbf{v}_j]_\gamma + [T\mathbf{v}_j]_\gamma$ by Proposition 3.2.2(i). These are the respective jth columns of the two matrices, so the matrices are equal.
Structural Commentary: The proof compares the coordinate images of each domain basis vector.
Key Insight: Taking a matrix representation preserves the vector-space operations on linear maps. □

Proposition 3.2.5. *Matrix Representation Acts on Coordinates*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be a basis of $\mathcal{W}$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\, [T(\mathbf{v})]_\gamma = [T]_{\gamma \leftarrow \beta} [\mathbf{v}]_\beta$.

Proof. Let $\mathbf{v} \in \mathcal{V}$, and write $[\mathbf{v}]_\beta = (a_1, \ldots, a_n)^t$, so $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$. By linearity of T and Proposition 3.2.2(i), $[T(\mathbf{v})]_\gamma = \sum_{j=1}^{n} a_j [T(\mathbf{v}_j)]_\gamma$.
The summands are the columns of $[T]_{\gamma \leftarrow \beta}$, and this sum equals $[T]_{\gamma \leftarrow \beta} [\mathbf{v}]_\beta$ by Proposition 3.1.9.
Structural Commentary: The proof expands a vector in the domain basis and tracks that expansion through T.
Key Insight: Applying a linear map becomes matrix multiplication after both bases are fixed. □

3.3 Matrix-Induced Maps and Rank

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Corollary 3.3.1. *Matrix Rank-Nullity*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:
(i) $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$; *equivalently,* $\mathrm{nullity}(A) + \mathrm{rank}(A) = n$.
(ii) If $m = n$, then $\ker(A) = \{\mathbf{0}\}$ iff $\mathrm{col}(A) = \mathbb{F}^n$.

Proof.
(i) Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. The map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear by Proposition 3.1.8.
For $\mathbf{x} = (x_1, \ldots, x_n)^t$, $T_A\mathbf{x} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j$ by Proposition 3.1.9.
Hence $\mathrm{range}(T_A) = \mathrm{col}(A)$, while $\ker(T_A) = \ker(A)$ holds by $T_A\mathbf{x} = A\mathbf{x}$; $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$ by Theorem 1.7.3(ii).
The equivalent identity is the same equality under $\mathrm{nullity}(A) = \dim(\ker(A))$ and $\mathrm{rank}(A) = \dim(\mathrm{col}(A))$.
(ii) ($\Rightarrow$) Suppose $m = n$ and $\ker(A) = \{\mathbf{0}\}$. Then $\dim(\mathrm{col}(A)) = n = \dim(\mathbb{F}^n)$ by part (i).
Since $\mathrm{col}(A) \subseteq \mathbb{F}^n$, $\mathrm{col}(A) = \mathbb{F}^n$ by Proposition 1.4.3(ii).
(ii) ($\Leftarrow$) Suppose $m = n$ and $\mathrm{col}(A) = \mathbb{F}^n$. Then $\dim(\ker(A)) = 0$ by part (i).
The zero subspace and $\ker(A)$ have the same dimension, so $\ker(A) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii).
Structural Commentary: Rank-nullity transfers through the induced map; the square case compares full-dimensional subspaces.
Key Insight: A matrix inherits rank-nullity from its coordinate-space map. □

Theorem 3.3.2. *Row Rank Equals Column Rank*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\dim(\mathrm{row}(A)) = \dim(\mathrm{col}(A))$. Equivalently, $\mathrm{rank}(A^t) = \mathrm{rank}(A)$.

Proof. Let $C = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_q] \in \mathrm{Mat}(p,q,\mathbb{F})$ be arbitrary, and set $s := \mathrm{rank}(C)$.
If $s = 0$, then $\dim(\mathrm{col}(C)) = 0 = \dim(\{\mathbf{0}\})$, so $\mathrm{col}(C) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii). Hence $C = 0$ and $\mathrm{rank}(C^t) = 0$.
Suppose $s > 0$. A basis sublist $(\mathbf{c}_{j_1}, \ldots, \mathbf{c}_{j_s})$ of the columns exists by Theorem 1.3.12; set $B := [\mathbf{c}_{j_1} \ \cdots \ \mathbf{c}_{j_s}] \in \mathrm{Mat}(p,s,\mathbb{F})$.
The relation $(\forall\, k \in \{1,\ldots,q\})(\exists!\, \mathbf{x}_k \in \mathbb{F}^s)\, \mathbf{c}_k = B\mathbf{x}_k$ holds by Proposition 1.3.3.
Set $X := [\mathbf{x}_1 \ \cdots \ \mathbf{x}_q] \in \mathrm{Mat}(s,q,\mathbb{F})$. Then $C = BX$ by Proposition 3.1.10, and $C^t = X^tB^t$ by Proposition 3.1.16.
Each column of C^t lies in the span of the s columns of X^t by Proposition 3.1.10 and Proposition 3.1.9.
Hence $\mathrm{col}(C^t)$ has a basis of at most s members by Theorem 1.3.12, and $\mathrm{rank}(C^t) \le s = \mathrm{rank}(C)$.
Thus $\mathrm{rank}(C^t) \le \mathrm{rank}(C)$ for the arbitrary matrix C above. For $C = A$, $\mathrm{rank}(A^t) \le \mathrm{rank}(A)$.
For $C = A^t$, $\mathrm{rank}(A) \le \mathrm{rank}(A^t)$ because $(A^t)^t = A$ by Proposition 3.1.14(i).
Structural Commentary: A column-basis factorization proves one rank inequality; transposition supplies the reverse inequality.
Key Insight: Transposition preserves the number of independent matrix directions. □

Perspective 3.3.3. *Row and Column Spaces Have Different Hosts*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\mathrm{row}(A) = \mathrm{col}(A^t) \subseteq \mathbb{F}^n$ and $\mathrm{col}(A) \subseteq \mathbb{F}^m$.
When $m \neq n$, Theorem 3.3.2 identifies their dimensions, not the subspaces in their different ambient spaces.

Theorem 3.3.4. *Four Fundamental Matrix Subspaces*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and let $r := \mathrm{rank}(A)$. Then:
(i) $\mathrm{row}(A), \ker(A) \subseteq \mathbb{F}^n$ *and* $\mathrm{col}(A), \ker(A^t) \subseteq \mathbb{F}^m$.
(ii) $\dim(\mathrm{col}(A)) = \dim(\mathrm{row}(A)) = r$, $\dim(\ker(A)) = n - r$, *and* $\dim(\ker(A^t)) = m - r$.

Proof.
(i) The row and column inclusions hold by their entry lengths, while the kernel inclusions hold by the domains of T_A and T_{A^t}.
(ii) The equality $\dim(\mathrm{col}(A)) = r$ holds by the definition of rank, and $\dim(\mathrm{row}(A)) = r = \mathrm{rank}(A^t)$ by Theorem 3.3.2.
Hence $\dim(\ker(A)) = n - r$ and $\dim(\ker(A^t)) = m - r$ by Corollary 3.3.1(i) for A and A^t, respectively.
Structural Commentary: Ambient sizes place the subspaces; row-column equality and rank-nullity determine their dimensions.
Key Insight: One rank parameter controls both image and kernel dimensions. □

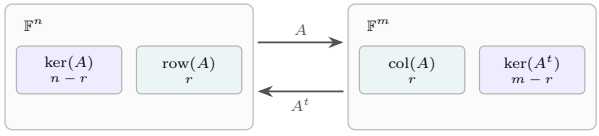

Figure 12: Hosts and dimensions of the four fundamental subspaces, with arrows for $A : \mathbb{F}^n \to \mathbb{F}^m$ and $A^t : \mathbb{F}^m \to \mathbb{F}^n$.

Figure 12 illustrates Theorem 3.3.4. Its boxes encode only ambient hosts and dimensions, not direct sums or orthogonality.

Corollary 3.3.5. *Homogeneous Systems with More Variables Than Equations*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $n > m$, then $(\exists\, \mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\, A\mathbf{x} = \mathbf{0}$.

Proof. The map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear by Proposition 3.1.8 and is not injective by Corollary 1.7.4.
Thus $\ker(T_A) \neq \{\mathbf{0}\}$ by Proposition 1.6.2. Choose $\mathbf{x} \in \ker(T_A) \setminus \{\mathbf{0}\}$. Then $A\mathbf{x} = T_A\mathbf{x} = \mathbf{0}$.
Structural Commentary: The dimension obstruction transfers from the induced map to the homogeneous system.
Key Insight: More variables force a nontrivial nullspace. □

Corollary 3.3.6. *Inhomogeneous Systems with More Equations Than Variables*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $m > n$, then $(\exists\, \mathbf{b} \in \mathbb{F}^m)(\forall\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} \neq \mathbf{b}$.

Proof. The map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear by Proposition 3.1.8 and is not surjective by Corollary 1.7.5.
Choose $\mathbf{b} \in \mathbb{F}^m \setminus \mathrm{range}(T_A)$. Then $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = T_A\mathbf{x} \neq \mathbf{b}$.
Structural Commentary: The dimension obstruction leaves a right-hand side outside the induced map's range.
Key Insight: More equations force an unreachable right-hand side. □

3.4 Invertibility and the Matrix-Map Dictionary

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Theorem 3.4.1. *Columns of an Invertible Matrix Form a Basis*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ *be invertible. Then* $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ *is a basis of* $\mathbb{F}^n$.

Proof. Fix $\mathbf{y} \in \mathbb{F}^n$ and set $\mathbf{x} = (x_1, \ldots, x_n)^t := A^{-1}\mathbf{y}$.
Then $\mathbf{y} = I_n\mathbf{y} = (AA^{-1})\mathbf{y} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j$ by Proposition 3.1.2, the inverse relation $AA^{-1} = I_n$, and Proposition 3.1.9.
If $\mathbf{c} = (c_1, \ldots, c_n)^t$ satisfies $\mathbf{y} = \sum_{j=1}^{n} c_j\mathbf{a}_j = A\mathbf{c}$, then $\mathbf{c} = I_n\mathbf{c} = (A^{-1}A)\mathbf{c} = A^{-1}\mathbf{y} = \mathbf{x}$ by Proposition 3.1.2 and $A^{-1}A = I_n$.
Hence $(\forall\, \mathbf{y} \in \mathbb{F}^n)(\exists!\, \mathbf{x} \in \mathbb{F}^n)\, \mathbf{y} = \sum_{j=1}^{n} x_j\mathbf{a}_j$, so $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis by Proposition 1.3.3.
Structural Commentary: The inverse constructs the unique coefficient column required by the basis criterion.
Key Insight: Invertibility turns matrix multiplication into unique coordinate representation. □

Theorem 3.4.2. *One-Sided Inverses of Square Matrices*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. *If* $AB = I_n$, *then* $BA = I_n$.

Proof. The induced map $T_B : \mathbb{F}^n \to \mathbb{F}^n$ is linear by Proposition 3.1.8.
If $\mathbf{x} \in \ker(T_B)$, then $B\mathbf{x} = \mathbf{0}$ and $\mathbf{x} = I_n\mathbf{x} = (AB)\mathbf{x} = A(B\mathbf{x}) = \mathbf{0}$ by Proposition 3.1.2; hence T_B is injective by Proposition 1.6.2.
It is surjective by Theorem 1.7.7, $(ii) \Rightarrow (iii)$. Fix $\mathbf{y} \in \mathbb{F}^n$ and choose $\mathbf{x} \in \mathbb{F}^n$ satisfying $\mathbf{y} = B\mathbf{x}$.
Then $BA\mathbf{y} = BA(B\mathbf{x}) = B((AB)\mathbf{x}) = B\mathbf{x} = \mathbf{y}$ by Proposition 3.1.2.
Substituting the columns of I_n for $\mathbf{y}$, $BA = I_n$ by Proposition 3.1.10 and Proposition 3.1.2.
Structural Commentary: Kernel triviality and equal dimension convert a left inverse into the missing right inverse.
Key Insight: Surjectivity lets the known inverse relation reach every vector. □

Motivation 3.4.3. *Coordinate Encoding*
Fixing bases encodes a linear map by a matrix. The next results verify composition compatibility and exact matrix recovery.

Proposition 3.4.4. *Composition and Matrix Multiplication*
Let $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_\ell)$, $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$, *and* $\delta = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ *be bases of* $\mathcal{U}$, $\mathcal{V}$, *and* $\mathcal{W}$, *respectively.*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ *and* $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, *then* $[ST]_{\delta\leftarrow\alpha} = [S]_{\delta\leftarrow\beta}[T]_{\beta\leftarrow\alpha}$.

Proof. $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$ by Proposition 1.5.7(i). Fix $j \in \{1, \ldots, \ell\}$.
Then $[(ST)\mathbf{u}_j]_\delta = [S(T\mathbf{u}_j)]_\delta = [S]_{\delta\leftarrow\beta}[T\mathbf{u}_j]_\beta$ by Proposition 3.2.5, with input $T\mathbf{u}_j$.
The first and last terms are jth columns of $[ST]_{\delta\leftarrow\alpha}$ and $[S]_{\delta\leftarrow\beta}[T]_{\beta\leftarrow\alpha}$, respectively, by Definition 3.2.3 and Proposition 3.1.10.
Since j was arbitrary, the matrices are equal by Definition 3.1.1(i).
Structural Commentary: Comparing basis-vector images identifies the corresponding columns of the composition and product matrices.
Key Insight: Composition order becomes matrix multiplication order. □

Theorem 3.4.5. *Linear Maps and Matrices Are the Same Data*
Let $\dim(\mathcal{V}) = n \geq 1$ *and* $\dim(\mathcal{W}) = m \geq 1$, *and fix bases* $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *and* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$.
Define $\Phi : \mathcal{L}(\mathcal{V}, \mathcal{W}) \to \mathrm{Mat}(m, n, \mathbb{F})$ *by* $\Phi(T) := [T]_{\gamma\leftarrow\beta}$. *Then:*

(i) Φ *is an isomorphism.*
(ii) $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Proof.

(i) Both spaces are vector spaces by Proposition 1.5.5 and Proposition 3.1.5(i), and Φ is linear by Proposition 3.2.4.
Suppose $\Phi(S) = \Phi(T)$. Then $(\forall\, j \in \{1, \ldots, n\})\, [S\mathbf{v}_j]_\gamma = [T\mathbf{v}_j]_\gamma$ by Definition 3.2.3, so $(\forall\, j \in \{1, \ldots, n\})\, S\mathbf{v}_j = T\mathbf{v}_j$ by Proposition 3.2.2(ii). Hence $S = T$ by Proposition 1.5.8, and Φ is injective.
Fix $A = [a_{i,j}] \in \mathrm{Mat}(m, n, \mathbb{F})$ and set $\mathbf{z}_j := \sum_{i=1}^{m} a_{i,j}\mathbf{w}_i$ $(j \in \{1, \ldots, n\})$.
The relation $(\exists!\, R \in \mathcal{L}(\mathcal{V}, \mathcal{W}))(\forall\, j \in \{1, \ldots, n\})\, R\mathbf{v}_j = \mathbf{z}_j$ holds by Proposition 1.5.8; fix this R.
Then $(\forall\, j \in \{1, \ldots, n\})\, [R\mathbf{v}_j]_\gamma = (a_{1,j}, \ldots, a_{m,j})^t$ by Definition 3.2.1, so $\Phi(R) = A$ by Definition 3.2.3.
Thus Φ is surjective and therefore an isomorphism by Proposition 1.6.7.

(ii) The matrix units form a basis of $\mathrm{Mat}(m, n, \mathbb{F})$ with mn members by Proposition 3.1.5(ii).
Their inverse images under Φ form a basis of $\mathcal{L}(\mathcal{V}, \mathcal{W})$ by part (i) and Proposition 1.7.8. Hence $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Structural Commentary: Basis images establish the isomorphism; transporting matrix units determines the dimension.
Key Insight: A matrix records exactly the images of a domain basis. □

Perspective 3.4.6. *Finite-Support Coordinate Model*
Proposition 3.2.2 and Theorem 3.4.5 identify finite-dimensional spaces and their maps with coordinate columns and matrices after bases are chosen. Definition 2.1.20 supplies the countable finite-support model $\mathbb{F}^{(\mathbb{N})}$; the analogy stops before infinite linear combinations.

3.5 Change of Basis and Coordinate Transformations

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 3.5.1. *Change-of-Basis Matrix*
Let $n \geq 1$, and let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be bases of the same vector space $\mathcal{V}$.
The *change-of-basis matrix from β-coordinates to γ-coordinates* is $C_{\gamma\leftarrow\beta} := [I_\mathcal{V}]_{\gamma\leftarrow\beta} \in \mathrm{Mat}(n, n, \mathbb{F})$.
The identity $(\forall\, \mathbf{v} \in \mathcal{V})\, [\mathbf{v}]_\gamma = C_{\gamma\leftarrow\beta}[\mathbf{v}]_\beta$ holds by Proposition 3.2.5, with $T = I_\mathcal{V}$.

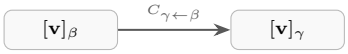

Figure 13: Direction of the coordinate rewrite implemented by $C_{\gamma\leftarrow\beta}$.

Figure 13 depicts $C_{\gamma\leftarrow\beta}$ carrying $[\mathbf{v}]_\beta$ to $[\mathbf{v}]_\gamma$; Definition 3.5.1 owns the exact identity.

Theorem 3.5.2. *Change-of-Basis Matrices*
Let $\mathcal{V}$ be n-dimensional, where $n \geq 1$, and let β be a basis of $\mathcal{V}$.
(i) If γ is a basis of $\mathcal{V}$, then $C_{\gamma\leftarrow\beta}$ is invertible with $(C_{\gamma\leftarrow\beta})^{-1} = C_{\beta\leftarrow\gamma}$.
(ii) If $S \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, then $S = C_{\gamma\leftarrow\beta}$ for a basis γ of $\mathcal{V}$.

Proof.
(i) The identity $[I_\mathcal{V}]_{\beta\leftarrow\beta} = I_n = [I_\mathcal{V}]_{\gamma\leftarrow\gamma}$ holds by Definition 3.2.1 and Definition 3.2.3.
Hence $C_{\beta\leftarrow\gamma}C_{\gamma\leftarrow\beta} = [I_\mathcal{V} I_\mathcal{V}]_{\beta\leftarrow\beta} = I_n$ and $C_{\gamma\leftarrow\beta}C_{\beta\leftarrow\gamma} = [I_\mathcal{V} I_\mathcal{V}]_{\gamma\leftarrow\gamma} = I_n$ by Definition 3.5.1 and Proposition 3.4.4.
Thus $C_{\gamma\leftarrow\beta}$ is invertible with inverse $C_{\beta\leftarrow\gamma}$.
(ii) Write $S^{-1} = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$. The relation $(\forall\, j \in \{1, \ldots, n\})(\exists!\, \mathbf{w}_j \in \mathcal{V})\, [\mathbf{w}_j]_\beta = \mathbf{s}_j$ holds by Proposition 3.2.2(ii),(iii).
Set $\gamma := (\mathbf{w}_1, \ldots, \mathbf{w}_n)$. The matrix S^{-1} is invertible with inverse S, so $(\mathbf{s}_1, \ldots, \mathbf{s}_n)$ is a basis of $\mathbb{F}^n$ by Theorem 3.4.1.
The map $[\cdot]_\beta^{-1} : \mathbb{F}^n \to \mathcal{V}$ is an isomorphism by Proposition 3.2.2(iii), so γ is a basis of $\mathcal{V}$ by Proposition 1.7.8.
The relation $(\forall\, j \in \{1, \ldots, n\})\, (C_{\beta\leftarrow\gamma})_{\cdot,j} = [\mathbf{w}_j]_\beta = \mathbf{s}_j$ holds by Definition 3.5.1 and Definition 3.2.3; hence $C_{\beta\leftarrow\gamma} = S^{-1}$ by Definition 3.1.1(i). Therefore $C_{\gamma\leftarrow\beta} = S$ by part (i) and Lemma 3.1.3.

Structural Commentary: Composition proves reciprocity; inverse-coordinate columns realize an arbitrary invertible matrix.
Key Insight: Coordinate isomorphisms transport bases in both directions. □

Theorem 3.5.3. *Change of Bases for a Linear Map*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\beta' = (\mathbf{v}'_1, \ldots, \mathbf{v}'_n)$ be bases of $\mathcal{V}$, and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ and $\gamma' = (\mathbf{w}'_1, \ldots, \mathbf{w}'_m)$ be bases of $\mathcal{W}$, where $n, m \geq 1$. Then $[T]_{\gamma'\leftarrow\beta'} = C_{\gamma'\leftarrow\gamma}[T]_{\gamma\leftarrow\beta}C_{\beta\leftarrow\beta'}$.

Proof. $T = I_\mathcal{W} T I_\mathcal{V}$ by Proposition 1.5.7(ii),(iii). Hence $[T]_{\gamma'\leftarrow\beta'} = [I_\mathcal{W} T I_\mathcal{V}]_{\gamma'\leftarrow\beta'} = [I_\mathcal{W}]_{\gamma'\leftarrow\gamma}[T]_{\gamma\leftarrow\beta}[I_\mathcal{V}]_{\beta\leftarrow\beta'}$ by Proposition 3.4.4 twice. The outer matrices are $C_{\gamma'\leftarrow\gamma}$ and $C_{\beta\leftarrow\beta'}$ by Definition 3.5.1.
Structural Commentary: Identity maps insert the codomain and domain coordinate transitions around the fixed representation.
Key Insight: Output coordinates transform on the left; input coordinates transform on the right. □

3.6 Rank and Invertibility Consequences

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Corollary 3.6.1. *Basis Columns Form an Invertible Matrix*
Let $n \geq 1$. If $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$, then $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible.

Proof. Let $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the ordered list of columns of I_n, and set $\alpha := (\mathbf{a}_1, \ldots, \mathbf{a}_n)$.
The matrix I_n is invertible with inverse I_n by Proposition 3.1.2(ii), so ϵ is a basis of $\mathbb{F}^n$ by Theorem 3.4.1.
The relation $(\forall\, j \in \{1, \ldots, n\})\, [\mathbf{a}_j]_\epsilon = \mathbf{a}_j$ holds by Definition 3.2.1, Proposition 3.1.9, and $I_n \mathbf{a}_j = \mathbf{a}_j$.
Thus $C_{\epsilon\leftarrow\alpha} = A$ by Definition 3.5.1 and Definition 3.2.3, and A is invertible by Theorem 3.5.2(i).
Structural Commentary: Standard coordinates identify the column matrix with a change-of-basis matrix.
Key Insight: A basis-to-basis coordinate rewrite is invertible. □

Corollary 3.6.2. *Invertibility and Full Rank*
Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is invertible $\iff \mathrm{rank}(A) = n$.

Proof.
($\Rightarrow$) Suppose A is invertible. Its columns form basis of $\mathbb{F}^n$ by Theorem 3.4.1; hence $\mathrm{col}(A) = \mathbb{F}^n$, $\mathrm{rank}(A) = n$ by Definition 3.1.12.
($\Leftarrow$) Suppose $\mathrm{rank}(A) = n$. Then $\dim(\mathrm{col}(A)) = n = \dim(\mathbb{F}^n)$ by Definition 3.1.12, so $\mathrm{col}(A) = \mathbb{F}^n$ by Proposition 1.4.3(ii).
The n columns of A therefore form a basis of $\mathbb{F}^n$ by Proposition 1.4.4(i), and A is invertible by Corollary 3.6.1.

Structural Commentary: One direction reads rank from basis columns; the converse upgrades full dimension to a basis.
Key Insight: Square-matrix invertibility is equivalent to full column rank. □

Corollary 3.6.3. *Transpose Preserves Invertibility*
Let $n \geq 1$ and $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is invertible $\iff A^t$ is invertible.

Proof.
($\Rightarrow$) Suppose A is invertible. Then $\mathrm{rank}(A) = n$ by Corollary 3.6.2, and $\mathrm{rank}(A^t) = \mathrm{rank}(A) = n$ by Theorem 3.3.2.
Hence A^t is invertible by Corollary 3.6.2.
($\Leftarrow$) Suppose A^t is invertible. The matrix $(A^t)^t$ is invertible by the forward direction with A^t in place of A.
Since $(A^t)^t = A$ by Proposition 3.1.14(i), A is invertible.

Structural Commentary: Rank invariance proves the forward direction; double transposition reverses it.
Key Insight: Transposition preserves full rank. □

Theorem 3.6.4. *Completing a Full-Column-Rank Matrix*
Let $1 \leq n < m$ and $X \in \mathrm{Mat}(m, n, \mathbb{F})$. If $\mathrm{rank}(X) = n$, then $(\exists\, X' \in \mathrm{Mat}(m, m-n, \mathbb{F}))\ [\, X \ X' \,] \in \mathrm{Mat}(m, m, \mathbb{F})^\times$.

Proof. Write $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$. Then $\dim(\mathrm{col}(X)) = n$ by Definition 3.1.12. The columns form a spanning list of length n in $\mathrm{col}(X)$.
Hence they form a basis of $\mathrm{col}(X)$ by Proposition 1.4.4(i) and are linearly independent in $\mathbb{F}^m$.
Extend them to a basis $(\mathbf{x}_1, \ldots, \mathbf{x}_m)$ of $\mathbb{F}^m$ by Theorem 1.3.13, and set $X' := [\mathbf{x}_{n+1} \ \cdots \ \mathbf{x}_m] \in \mathrm{Mat}(m, m-n, \mathbb{F})$.
The matrix $[\, X \ X' \,]$ is invertible by Corollary 3.6.1.
Structural Commentary: Column rank identifies an independent list; basis extension supplies the missing columns.
Key Insight: Full-column-rank matrices are initial blocks of invertible matrices. □

Theorem 3.6.5. *Full-Rank Products Preserve Rank*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:
(i) If $X \in \mathrm{Mat}(p,m,\mathbb{F})$ and $Y \in \mathrm{Mat}(n,q,\mathbb{F})$ satisfy $\mathrm{rank}(X) = m$ and $\mathrm{rank}(Y) = n$, then $\mathrm{rank}(XAY) = \mathrm{rank}(A)$.
(ii) If $P \in \mathrm{Mat}(m,m,\mathbb{F})$ and $Q \in \mathrm{Mat}(n,n,\mathbb{F})$ are invertible, then $\mathrm{rank}(PAQ) = \mathrm{rank}(A)$.

Proof.
(i) The equality $\dim(\ker(X)) = 0$ holds by Corollary 3.3.1(i), so $\ker(X) = \{\mathbf{0}\}$ by Proposition 1.4.3(ii).
Hence $\ker(XA) = \{\mathbf{z} \in \mathbb{F}^n : X(A\mathbf{z}) = \mathbf{0}\} = \{\mathbf{z} \in \mathbb{F}^n : A\mathbf{z} = \mathbf{0}\} = \ker(A)$, and $\mathrm{rank}(XA) = \mathrm{rank}(A)$ by Corollary 3.3.1(i).
Since $\mathrm{rank}(Y^t) = \mathrm{rank}(Y) = n$ by Theorem 3.3.2, the equality $\mathrm{rank}(Y^tA^t) = \mathrm{rank}(A^t)$ holds by the preceding kernel argument with (X, A) replaced by (Y^t, A^t). The identity $(AY)^t = Y^tA^t$ holds by Proposition 3.1.16.
Hence $\mathrm{rank}(AY) = \mathrm{rank}(Y^tA^t) = \mathrm{rank}(A^t) = \mathrm{rank}(A)$ by Theorem 3.3.2 and the preceding equality.
Finally, $\mathrm{rank}(XAY) = \mathrm{rank}(AY) = \mathrm{rank}(A)$ by the preceding kernel argument with A replaced by AY.
(ii) Set $X = P$ and $Y = Q$. Then $\mathrm{rank}(P) = m$ and $\mathrm{rank}(Q) = n$ by Corollary 3.6.2, so $\mathrm{rank}(PAQ) = \mathrm{rank}(A)$ by part (i).
Structural Commentary: Kernel preservation handles left factors; transposition converts right factors to the same argument.
Key Insight: Full column and row rank prevent dimensional collapse. □

3.7 Block Matrix Algebra

Throughout this subsection, all vector spaces and matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Definition 3.7.1. *Block Partitions and Conformality*
Let $r, s \geq 1$, let $m_1, \ldots, m_r$ and $n_1, \ldots, n_s$ be positive integers, and let $A \in \mathrm{Mat}\left(\sum_{i=1}^{r} m_i, \sum_{j=1}^{s} n_j, \mathbb{F}\right)$.
An *$r \times s$ block partition* $A = [A_{ij}]$ groups consecutive rows according to $(m_1, \ldots, m_r)$ and consecutive columns according to $(n_1, \ldots, n_s)$, so $(\forall\, i \in \{1, \ldots, r\})(\forall\, j \in \{1, \ldots, s\})\, A_{ij} \in \mathrm{Mat}(m_i, n_j, \mathbb{F})$. The submatrices A_{ij} are the *blocks*.
(i) Two $r \times s$ block partitions $A = [A_{ij}]$ and $C = [C_{ij}]$ are *conformal for addition* if corresponding blocks have the same size.
(ii) An $r \times s$ block partition $A = [A_{ik}]$ and an $s \times t$ block partition $B = [B_{kj}]$, where $t \geq 1$, are *conformal for multiplication* if $\exists\, p_1, \ldots, p_t \in \mathbb{Z}_{>0}$ such that $A_{ik} \in \mathrm{Mat}(m_i, n_k, \mathbb{F})$ and $B_{kj} \in \mathrm{Mat}(n_k, p_j, \mathbb{F})$ $\forall i \in \{1, \ldots, r\}$, $k \in \{1, \ldots, s\}$, $j \in \{1, \ldots, t\}$.

Definition 3.7.2. *Block Matrix of a Linear Map*
Suppose $\mathcal{V} = \mathcal{U}_1 \oplus \mathcal{U}_2$ and $\mathcal{W} = \mathcal{Z}_1 \oplus \mathcal{Z}_2$, where $\dim(\mathcal{U}_j) = n_j \geq 1$ and $\dim(\mathcal{Z}_i) = m_i \geq 1$ for $i, j \in \{1, 2\}$. Choose ordered bases α_j of $\mathcal{U}_j$ and β_i of $\mathcal{Z}_i$. By Definition 2.3.5, $\alpha = (\alpha_1, \alpha_2)$ and $\beta = (\beta_1, \beta_2)$ are bases of $\mathcal{V}$ and $\mathcal{W}$ adapted to the decompositions.
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, its representation matrix has conformal partition $[T]_{\beta \leftarrow \alpha} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where $A_{ij} \in \mathrm{Mat}(m_i, n_j, \mathbb{F})$ for $i, j \in \{1, 2\}$.
The jth block column contains coordinate columns of images of basis α_j, while the ith block row retains their β_i-coordinate entries.

Proposition 3.7.3. *Block Matrix Arithmetic*
The following identities hold.
(i) If $A = [A_{ij}]$ and $C = [C_{ij}]$ are conformal $r \times s$ block partitions, then $A + C = [A_{ij} + C_{ij}]$.
In particular, conformal 2×2 partitions satisfy $A + C = \begin{bmatrix} A_{11}+C_{11} & A_{12}+C_{12} \\ A_{21}+C_{21} & A_{22}+C_{22} \end{bmatrix}$.
(ii) If $A = [A_{ik}]$ and $B = [B_{kj}]$ are conformal for multiplication, then $(AB)_{ij} = \sum_{k=1}^{s} A_{ik}B_{kj}$ for every admissible i, j.
In particular, conformal 2×2 partitions satisfy $AB = \begin{bmatrix} A_{11}B_{11}+A_{12}B_{21} & A_{11}B_{12}+A_{12}B_{22} \\ A_{21}B_{11}+A_{22}B_{21} & A_{21}B_{12}+A_{22}B_{22} \end{bmatrix}$.

Proof.
(i) By Definition 3.1.1, matrix addition is entrywise. Because conformality assigns the same entry positions to corresponding blocks, addition may be performed blockwise, so $A + C = [A_{ij} + C_{ij}]$. The case $r = s = 2$ is the stated specialization.
(ii) By Proposition 3.1.6, each entry of AB is the row-column sum over the shared index. Group that index into the s consecutive intervals determined by the common block sizes. The contribution from the kth interval is the corresponding entry of $A_{ik}B_{kj}$.
Therefore $(AB)_{ij} = \sum_{k=1}^{s} A_{ik}B_{kj}$. The case of two block rows and two block columns is the stated specialization.
Structural Commentary: Addition is entrywise; multiplication groups the shared index.
Key Insight: Block arithmetic is ordinary matrix arithmetic performed at a coarser scale. □

Proposition 3.7.4. *Stacks Intersect Kernels, Concatenations Sum Columns*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(p,n,\mathbb{F})$, $C \in \mathrm{Mat}(m,r,\mathbb{F})$, and $D \in \mathrm{Mat}(m,s,\mathbb{F})$. Then:
(i) $\ker\left(\begin{bmatrix} A \\ B \end{bmatrix}\right) = \ker(A) \cap \ker(B)$.
(ii) $\mathrm{col}([\,C \;\; D\,]) = \mathrm{col}(C) + \mathrm{col}(D)$.

Proof.
(i) By Proposition 3.7.3, $\begin{bmatrix} A \\ B \end{bmatrix}\mathbf{x} = \begin{bmatrix} A\mathbf{x} \\ B\mathbf{x} \end{bmatrix}$ for every $\mathbf{x} \in \mathbb{F}^n$.
By Definition 3.1.12, $\mathbf{x} \in \ker\left(\begin{bmatrix} A \\ B \end{bmatrix}\right)$ if and only if $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$, which is equivalent to $\mathbf{x} \in \ker(A) \cap \ker(B)$.
(ii) By Proposition 3.7.3, $[\,C \;\; D\,]\begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix} = C\mathbf{u} + D\mathbf{v}$ for every $\mathbf{u} \in \mathbb{F}^r$ and $\mathbf{v} \in \mathbb{F}^s$.
By Definition 3.1.12, $\mathrm{col}([\,C \;\; D\,]) = \{C\mathbf{u} + D\mathbf{v} : \mathbf{u} \in \mathbb{F}^r,\ \mathbf{v} \in \mathbb{F}^s\} = \mathrm{col}(C) + \mathrm{col}(D)$.
Structural Commentary: The kernel and column-space definitions translate both block actions.
Key Insight: Vertical blocks intersect kernels; horizontal blocks add column spaces. □

Proposition 3.7.5. *Block Centrosymmetry and Commutation*
Let $n \geq 1$, let $A, B, C, D \in \mathrm{Mat}(n,n,\mathbb{F})$, and set $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ and $L_{2n} = \begin{bmatrix} 0 & I_n \\ I_n & 0 \end{bmatrix}$ in $\mathrm{Mat}(2n, 2n, \mathbb{F})$.
Then M has block centrosymmetric form, meaning $A = D$ and $B = C$, if and only if $L_{2n}M = ML_{2n}$.

Proof. By Proposition 3.7.3, $L_{2n}M = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$ and $ML_{2n} = \begin{bmatrix} B & A \\ D & C \end{bmatrix}$.
($\Rightarrow$) If M has block centrosymmetric form, then $A = D$ and $B = C$, so the two block products agree. Therefore $L_{2n}M = ML_{2n}$.

($\Leftarrow$) If $L_{2n}M = ML_{2n}$, then corresponding blocks are equal by Definition 3.1.1.
Thus $C = B$ and $D = A$, so M has block centrosymmetric form.

Structural Commentary: Explicit block multiplication reduces commutation with L_{2n} to equality of opposite blocks.
Key Insight: The exchange matrix detects block centrosymmetry. □

Definition 3.7.6. *Direct Sum of Matrices*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(p,q,\mathbb{F})$, the *direct sum* of A and B is $A \oplus B := \left[\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right] \in \mathrm{Mat}(m+p, n+q, \mathbb{F})$, where the upper-right and lower-left zero blocks have sizes $m \times q$ and $p \times n$, respectively.

3.8 Matrix Operations and Identities

Throughout this subsection, all matrices are over the same fixed field $\mathbb{F}$, and every matrix size is a positive integer.

Definition 3.8.1. *Polynomial Evaluation on a Square Matrix*
Let $n \geq 1$ and $A \in \mathrm{Mat}(n,n,\mathbb{F})$. Define $A^0 := I_n$ and $(\forall\, j \in \mathbb{Z}_{\geq 0})\, A^{j+1} := A^j A$.
For $p(z) = \sum_{j=0}^{d} c_j z^j \in \mathbb{F}[z]$, where $d \geq 0$, the *evaluation of p at A* is $p(A) := \sum_{j=0}^{d} c_j A^j \in \mathrm{Mat}(n,n,\mathbb{F})$.

Proposition 3.8.2. *Algebra of Matrix Direct Sums*
(i) If $A, C \in \mathrm{Mat}(m,n,\mathbb{F})$, $B, D \in \mathrm{Mat}(r,s,\mathbb{F})$, and $a, b \in \mathbb{F}$, then $a(A \oplus B) + b(C \oplus D) = (aA + bC) \oplus (aB + bD)$.
(ii) If $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $C \in \mathrm{Mat}(n,\ell,\mathbb{F})$, $B \in \mathrm{Mat}(r,s,\mathbb{F})$, and $D \in \mathrm{Mat}(s,t,\mathbb{F})$, then $(A \oplus B)(C \oplus D) = AC \oplus BD$.
(iii) If $A \in \mathrm{Mat}(m,m,\mathbb{F})$, $B \in \mathrm{Mat}(n,n,\mathbb{F})$, and $k \in \mathbb{Z}_{>0}$, then $(A \oplus B)^k = A^k \oplus B^k$.
(iv) If $A \in \mathrm{Mat}(m,m,\mathbb{F})$, $B \in \mathrm{Mat}(n,n,\mathbb{F})$, and $p \in \mathbb{F}[z]$, then $p(A \oplus B) = p(A) \oplus p(B)$.

Proof.
(i) By Definition 3.7.6, Definition 3.1.1, and Proposition 3.7.3, $a(A \oplus B) + b(C \oplus D) = \left[\begin{smallmatrix} aA+bC & 0 \\ 0 & aB+bD \end{smallmatrix}\right] = (aA + bC) \oplus (aB + bD)$.
(ii) By Definition 3.7.6 and Proposition 3.7.3, $(A \oplus B)(C \oplus D) = \left[\begin{smallmatrix} AC & 0 \\ 0 & BD \end{smallmatrix}\right] = AC \oplus BD$.
(iii) Set $P_k := ((A \oplus B)^k = A^k \oplus B^k)$. By Definition 3.8.1 and Proposition 3.1.2, P_1 holds.
If P_k holds, then $(A \oplus B)^{k+1} = (A \oplus B)^k (A \oplus B) = (A^k \oplus B^k)(A \oplus B) = A^{k+1} \oplus B^{k+1}$ by the recurrence defining powers, the induction hypothesis, and (ii). Therefore $(\forall\, k \in \mathbb{Z}_{>0})\, P_k$ by Theorem 2.6.2.
(iv) Write $p(z) = \sum_{j=0}^{d} c_j z^j$. The identity $(A \oplus B)^j = A^j \oplus B^j$ holds for $j = 0$ by Definition 3.8.1, Definition 3.7.6, for $j \geq 1$ by (iii).
By (i), scalar multiples, finite sums of these direct sums are componentwise. Hence $p(A \oplus B) = p(A) \oplus p(B)$ by Definition 3.8.1.

Structural Commentary: Block formulas extend by induction and linearity.
Key Insight: Direct-sum blocks evolve independently under matrix algebra. □

Proposition 3.8.3. *Diagonal Multiplication Scales Rows and Columns*
Let $A = [a_{ij}] \in \mathrm{Mat}(m,n,\mathbb{F})$, let $\lambda_1, \ldots, \lambda_n, \mu_1, \ldots, \mu_m \in \mathbb{F}$, and set $\Lambda := \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and $M := \mathrm{diag}(\mu_1, \ldots, \mu_m)$. Then:
(i) $A\Lambda = [\lambda_j a_{ij}]$; right multiplication by Λ scales column j by λ_j.
(ii) $MA = [\mu_i a_{ij}]$; left multiplication by M scales row i by μ_i.

Proof.
(i) For admissible i, j, $(A\Lambda)_{ij} = \sum_{r=1}^{n} a_{ir}\Lambda_{rj} = a_{ij}\lambda_j = \lambda_j a_{ij}$ by Definition 3.1.1 and commutativity in $\mathbb{F}$.
(ii) For admissible i, j, $(MA)_{ij} = \sum_{r=1}^{m} M_{ir} a_{rj} = \mu_i a_{ij}$ by Definition 3.1.1.

Structural Commentary: Diagonal sparsity isolates one summand in each entry formula.
Key Insight: The multiplication side determines whether rows or columns are scaled. □

Proposition 3.8.4. *Inverse of an Upper Block-Triangular Matrix*
Let $Y \in \mathrm{Mat}(m,m,\mathbb{F})$ and $Z \in \mathrm{Mat}(n,n,\mathbb{F})$ be invertible, and let $X \in \mathrm{Mat}(m,n,\mathbb{F})$.
Then $\left[\begin{smallmatrix} Y & X \\ 0 & Z \end{smallmatrix}\right]^{-1} = \left[\begin{smallmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{smallmatrix}\right] \in \mathrm{Mat}(m+n, m+n, \mathbb{F})$, where each lower-left zero block has size $n \times m$.

Proof. Set $M := \left[\begin{smallmatrix} Y & X \\ 0 & Z \end{smallmatrix}\right]$ and $N := \left[\begin{smallmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{smallmatrix}\right]$.
By Proposition 3.7.3 and Proposition 3.1.2, $MN = \left[\begin{smallmatrix} I_m & -XZ^{-1}+XZ^{-1} \\ 0 & I_n \end{smallmatrix}\right] = I_{m+n}$ and $NM = \left[\begin{smallmatrix} I_m & Y^{-1}X-Y^{-1}X \\ 0 & I_n \end{smallmatrix}\right] = I_{m+n}$.
Thus $N = M^{-1}$ by Definition 3.1.1 and Lemma 3.1.3.
Structural Commentary: Two block products verify both inverse identities.
Key Insight: The off-diagonal correction cancels the coupling block. □

Theorem 3.8.5. *Inverse of an Upper Triangular Matrix*
Let $n \geq 1$ and let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{F})$ be upper triangular with $(\forall\, i \in \{1, \ldots, n\})\, a_{ii} \neq 0$. Then:
(i) A is invertible.
(ii) A^{-1} is upper triangular.
(iii) The diagonal of A^{-1} is $(a_{11}^{-1}, \ldots, a_{nn}^{-1})$.

Proof. Let P_n denote the conjunction of (i)–(iii) in size n under the stated hypotheses.
For $n = 1$, $A^{-1} = [a_{11}^{-1}]$ by $[a_{11}][a_{11}^{-1}] = I_1 = [a_{11}^{-1}][a_{11}]$, Definition 3.1.1, Lemma 3.1.3.
This inverse is upper triangular with diagonal (a_{11}^{-1}), so P_1 holds. Suppose $n > 1$ and P_{n-1} holds.
Write $A = \left[\begin{smallmatrix} B & \mathbf{x} \\ 0 & a_{nn} \end{smallmatrix}\right]$, where $B \in \mathrm{Mat}(n-1, n-1, \mathbb{F})$ is upper triangular with nonzero diagonal and $\mathbf{x} \in \mathbb{F}^{n-1}$.
(i) The blocks B, $[a_{nn}]$ invertible by induction hypothesis, Definition 3.1.1, respectively. Hence A invertible by Proposition 3.8.4.

(ii) The matrix B^{-1} is upper triangular by the induction hypothesis.
By Proposition 3.8.4, $A^{-1} = \begin{bmatrix} B^{-1} & -B^{-1}\mathbf{x}a_{nn}^{-1} \\ 0 & a_{nn}^{-1} \end{bmatrix}$, which is upper triangular by Definition 3.1.1.

(iii) By the induction hypothesis, the diagonal of B^{-1} is $(a_{11}^{-1}, \dots, a_{n-1,n-1}^{-1})$.
By the block formula in (ii), the diagonal of A^{-1} is $(a_{11}^{-1}, \dots, a_{n-1,n-1}^{-1}, a_{nn}^{-1})$.

Thus P_n holds, so $(\forall\, n \in \mathbb{Z}_{>0})\, P_n$ by Theorem 2.6.2.
Structural Commentary: Induction transfers all three claims through the block inverse formula.
Key Insight: Triangular inversion reduces dimension while preserving the zero pattern. □

4 Duality and Multilinear Structure

4.1 Dual Spaces and Dual Bases

Duality studies vectors through scalar-valued tests. This subsection develops algebraic duals, dual bases, coordinate functionals, separation by functionals, and their dimension consequences.

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.1.1. *Linear Functional and Dual Space*

(i) A *linear functional* on $\mathcal{V}$ is an element $\varphi \in \mathcal{L}(\mathcal{V}, \mathbb{F})$.
(ii) The *algebraic dual space* of $\mathcal{V}$ is $\mathcal{V}' := \mathcal{L}(\mathcal{V}, \mathbb{F})$.

Pointwise addition and scalar multiplication make $\mathcal{V}'$ a vector space over $\mathbb{F}$ by Proposition 1.5.5.

Example 4.1.2. *Evaluation and Integral Functionals*

(i) Define $\mathrm{ev}_0 : \mathbb{F}[x] \to \mathbb{F}$ by $\mathrm{ev}_0(p) := p(0)$.
The identity $(\forall\, \alpha, \beta \in \mathbb{F})(\forall\, p, q \in \mathbb{F}[x])\ \mathrm{ev}_0(\alpha p + \beta q) = \alpha\, \mathrm{ev}_0(p) + \beta\, \mathrm{ev}_0(q)$ proves $\mathrm{ev}_0 \in \mathcal{V}'$ for $\mathcal{V} = \mathbb{F}[x]$.

(ii) For $\mathbb{F} = \mathbb{R}$, let $C[a,b]$ be the real vector space of continuous functions on $[a,b]$ and define $I : C[a,b] \to \mathbb{R}$ by $I(f) := \int_a^b f(x)\, dx$.
By linearity of the integral, $(\forall\, \alpha, \beta \in \mathbb{R})(\forall\, f, g \in C[a,b])\, I(\alpha f + \beta g) = \alpha I(f) + \beta I(g)$, so $I \in C[a,b]'$.

Definition 4.1.3. *Dual Basis*

Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an ordered basis of $\mathcal{V}$, where $n \in \mathbb{N}_0$.
Its *dual basis* is the list $\beta' = (\varphi_1, \dots, \varphi_n)$ in $\mathcal{V}'$ characterized by $(\forall\, j, k \in \{1, \dots, n\})\, \varphi_j(\mathbf{v}_k) = \delta_{j,k}$.
If $n \geq 1$, each φ_j exists uniquely by Proposition 1.5.8; if $n = 0$, then $\beta' := ()$.

Theorem 4.1.4. *Dual Basis and Dimension of the Dual Space*

Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of the finite-dimensional space $\mathcal{V}$, and let $\beta' = (\varphi_1, \dots, \varphi_n)$ be its dual basis. Then:

(i) β' is a basis of $\mathcal{V}'$;
(ii) $\dim(\mathcal{V}') = \dim(\mathcal{V})$.

Proof.

(i) If $n = 0$, then $\mathcal{V} = \{\mathbf{0}\}$, and $\mathcal{V}' = \{0\}$ by Proposition 1.5.3; thus $()$ is a basis of $\mathcal{V}'$.
Suppose $n \geq 1$. If $\sum_{j=1}^{n} a_j \varphi_j = 0$, then $(\forall\, k \in \{1, \dots, n\})\, a_k = \sum_{j=1}^{n} a_j \varphi_j(\mathbf{v}_k) = 0$, so β' is linearly independent.
For $\varphi \in \mathcal{V}'$ and $\mathbf{v} = \sum_{k=1}^{n} c_k \mathbf{v}_k$, one has $\big(\sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j\big)(\mathbf{v}) = \sum_{j=1}^{n} \varphi(\mathbf{v}_j) c_j = \varphi(\mathbf{v})$ by Definition 4.1.3.
Hence $\varphi = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j$ by Assumption 3.1.2, and β' spans $\mathcal{V}'$.

(ii) Both bases have length n by (i), so $\dim(\mathcal{V}') = n = \dim(\mathcal{V})$ by Definition 1.4.2.

Structural Commentary: Evaluation on the original basis proves independence and reconstructs every functional.
Key Insight: Basis values are precisely the coordinates of a functional. □

Proposition 4.1.5. *Coordinate Functionals for a Hamel Basis*

Let $\mathcal{B} = \{\mathbf{b}_i\}_{i \in I}$ be Hamel basis of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, let $a_{\mathbf{v}} : I \to \mathbb{F}$ be unique finitely supported function satisfying $\mathbf{v} = \sum_{j \in \mathrm{supp}(a_{\mathbf{v}})} a_{\mathbf{v}}(j)\mathbf{b}_j$, and define $\varepsilon_i(\mathbf{v}) := a_{\mathbf{v}}(i)$ for $i \in I$. Then:

(i) $(\forall\, i \in I)\, \varepsilon_i \in \mathcal{V}'$;
(ii) $\{\varepsilon_i\}_{i \in I}$ is linearly independent;
(iii) $\mathrm{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$ iff $|I| < \infty$;
(iv) $|I| < \infty$ iff $\mathcal{V}$ is finite-dimensional.

Proof.

(i) For $\alpha, \beta \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, the identity $a_{\alpha\mathbf{v}+\beta\mathbf{w}} = \alpha a_{\mathbf{v}} + \beta a_{\mathbf{w}}$ holds by uniqueness in Proposition 1.3.16.
Hence $(\forall\, i \in I)\, \varepsilon_i(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha\varepsilon_i(\mathbf{v}) + \beta\varepsilon_i(\mathbf{w})$, so $(\forall\, i \in I)\, \varepsilon_i \in \mathcal{V}'$ by Definition 4.1.1.

(ii) If $i_1, \dots, i_N \in I$ are distinct and $\sum_{r=1}^{N} c_r \varepsilon_{i_r} = 0$, then $(\forall\, s \in \{1, \dots, N\})\, c_s = \sum_{r=1}^{N} c_r \varepsilon_{i_r}(\mathbf{b}_{i_s}) = 0$.
Thus every finite relation is trivial, so $\{\varepsilon_i\}_{i \in I}$ is linearly independent.

(iii) ($\Rightarrow$) Suppose $\mathrm{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$ and I is infinite. Define $\varphi(\mathbf{v}) := \sum_{j \in \mathrm{supp}(a_{\mathbf{v}})} a_{\mathbf{v}}(j)$.
By coordinate uniqueness, $(\forall\, \alpha, \beta \in \mathbb{F})(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\, \varphi(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha\varphi(\mathbf{v}) + \beta\varphi(\mathbf{w})$, so $\varphi \in \mathcal{V}'$
The equality $(\forall\, i \in I)\, \varphi(\mathbf{b}_i) = 1$ holds, whereas every element of $\mathrm{span}\{\varepsilon_i : i \in I\}$ vanishes on all but finitely many $\mathbf{b}_i$.
Thus $\varphi \notin \mathrm{span}\{\varepsilon_i : i \in I\}$, a contradiction, and $|I| < \infty$.

(iii) ($\Leftarrow$) If I is finite, order $\mathcal{B}$. Then $\{\varepsilon_i\}_{i \in I}$ is its dual basis, so it spans $\mathcal{V}'$ by Theorem 4.1.4(i).

(iv) ($\Rightarrow$) If I is finite, then any ordering of $\mathcal{B}$ is a finite basis, so $\mathcal{V}$ is finite-dimensional.

(*iv*) (⇐) Suppose $\mathcal{V}$ has a spanning list of length n and I is infinite. Choose distinct $i_1, \dots, i_{n+1} \in I$.
The list $(\mathbf{b}_{i_1}, \dots, \mathbf{b}_{i_{n+1}})$ is linearly independent, so $n+1 \le n$ by Theorem 1.3.11, a contradiction. Hence I is finite.

Structural Commentary: Coefficient uniqueness proves linearity and independence; finite support detects finite-dimensionality.
Key Insight: Finite coordinate support cannot represent a functional nonzero on every basis vector. □

Theorem 4.1.6. *Dimension of the Algebraic Dual*
Assume Zorn's lemma and the Choice Principle 6.1.2, and let $\mathcal{V}$ be an $\mathbb{F}$-vector space. Then:
(*i*) $\dim(\mathcal{V}) \le \dim(\mathcal{V}')$;
(*ii*) $\dim(\mathcal{V}) = \dim(\mathcal{V}')$ *iff* $\dim(\mathcal{V}) < \infty$.

Proof. Choose a Hamel basis $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ by Theorem 1.3.17(ii), set $\kappa := |I| = \dim(\mathcal{V})$ by Definition 1.4.6, and let $\{\varepsilon_i\}_{i\in I}$ denote the coordinate functionals associated with $\mathcal{B}$.

(*i*) The family $\{\varepsilon_i\}_{i\in I} \subseteq \mathcal{V}'$ is linearly independent by Proposition 4.1.5(ii).
Extend it to a Hamel basis $\mathcal{D}$ of $\mathcal{V}'$ by Theorem 1.3.17(iii).
The map $i \mapsto \varepsilon_i$ injects I into $\mathcal{D}$, so $\kappa = |I| \le |\mathcal{D}| = \dim(\mathcal{V}')$ by Proposition 6.5.2(i) and Definition 1.4.6.

(*ii*) (⇒) Suppose $\dim(\mathcal{V}) = \dim(\mathcal{V}')$. Since $\kappa = \dim(\mathcal{V})$, assume for contradiction that κ is infinite.

(a) *Countable scalar core*: Set $D := \{(m,n) \in \mathbb{Z}^2 : n \cdot 1 \neq 0\}$ and $\mathbb{F}_0 := \{(m\cdot 1)(n\cdot 1)^{-1} : (m,n) \in D\} \subseteq \mathbb{F}$.
The elements $0, 1$ belong to $\mathbb{F}_0$. If $x = (m\cdot 1)(n\cdot 1)^{-1}$ and $y = (r\cdot 1)(s\cdot 1)^{-1}$, then $x - y = ((ms - rn)\cdot 1)((ns)\cdot 1)^{-1}$ and $xy = ((mr)\cdot 1)((ns)\cdot 1)^{-1}$; if $x \neq 0$, then $x^{-1} = (n\cdot 1)(m\cdot 1)^{-1}$.
Thus $\mathbb{F}_0$ is a subfield. The set D is at most countable by Proposition 6.2.10, Corollary 6.2.17(ii), and Proposition 6.2.12(i), so $\mathbb{F}_0$ is at most countable by Proposition 6.2.12(ii). The map $E \mapsto \mathbf{1}_E$ injects $\mathcal{P}(I)$ into $\mathbb{F}_0^I$.
The map $i \mapsto \{i\}$ injects I into $\mathcal{P}(I)$, while $I \not\sim_{\#} \mathcal{P}(I)$ by Theorem 6.4.1.
Hence $\kappa < |\mathcal{P}(I)| \le |\mathbb{F}_0^I|$ by Proposition 6.5.2(i),(ii).

(b) *Core-space dimension*: Suppose $\dim_{\mathbb{F}_0}(\mathbb{F}_0^I) \le \kappa$. Choose a Hamel basis $\mathcal{C}$ of $\mathbb{F}_0^I$ by Theorem 1.3.17(ii).
Since $|\mathcal{C}| = \dim_{\mathbb{F}_0}(\mathbb{F}_0^I) \le \kappa$, an injection $\iota : \mathcal{C} \to \kappa$ exists by Proposition 6.5.2(i).
By Proposition 1.3.16, the increasing list of ι-images of the support, paired with its nonzero coefficients, injectively encodes each $f \in \mathbb{F}_0^I$ in $\bigcup_{n<\omega} \big(\{n\} \times (\kappa \times \mathbb{F}_0)^n\big)$. Since $\mathbb{F}_0$ is at most countable and κ is infinite, each finite power injects into κ, and the tagged union injects into $\omega \times \kappa$, by Proposition 6.3.15(iv) and Theorem 6.6.3(ii).
Therefore $|\mathbb{F}_0^I| \le \kappa$ by Proposition 6.5.2(i), contrary to (a). Thus $\dim_{\mathbb{F}_0}(\mathbb{F}_0^I) > \kappa$.

(c) *Scalar transfer*: Choose an $\mathbb{F}_0$-basis $\mathcal{C}$ of $\mathbb{F}_0^I$ by Theorem 1.3.17(ii). For $\mathbf{v} \in \mathcal{V}$, a unique finitely supported $\mathcal{B}$-coefficient function $a_{\mathbf{v}} : I \to \mathbb{F}$ exists by Proposition 1.3.16. For $g \in \mathcal{C}$, define $\Phi_g : \mathcal{V} \to \mathbb{F}$ by $\Phi_g(\mathbf{v}) := \sum_{i \in \operatorname{supp}(a_{\mathbf{v}})} a_{\mathbf{v}}(i) g(i)$.
The identity $a_{\alpha\mathbf{v}+\beta\mathbf{w}} = \alpha a_{\mathbf{v}} + \beta a_{\mathbf{w}}$ holds by coordinate uniqueness.
Hence $(\forall\, \alpha, \beta \in \mathbb{F})(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\, \Phi_g(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha\Phi_g(\mathbf{v}) + \beta\Phi_g(\mathbf{w})$ and $\Phi_g \in \mathcal{V}'$.
Let $N \in \mathbb{N}$, let $g_1, \dots, g_N \in \mathcal{C}$ be distinct, and suppose $\sum_{r=1}^{N} c_r \Phi_{g_r} = 0$.
Equality $(\forall\, i \in I) \sum_{r=1}^{N} c_r g_r(i) = 0$ holds by evaluation at $\mathbf{b}_i$. Choose $\mathbb{F}_0$-basis $(e_1, \dots, e_s)$ of $\operatorname{span}_{\mathbb{F}_0}\{c_1, \dots, c_N\}$ by Corollary 1.3.14, and $d_{\ell r} \in \mathbb{F}_0$ satisfying $(\forall\, r \in \{1, \dots, N\})\, c_r = \sum_{\ell=1}^{s} d_{\ell r} e_\ell$. For $i \in I$, $0 = \sum_{r=1}^{N} c_r g_r(i) = \sum_{\ell=1}^{s} e_\ell \big(\sum_{r=1}^{N} d_{\ell r} g_r(i) \big)$.
Hence $(\forall\, \ell \in \{1, \dots, s\})(\forall\, i \in I) \sum_{r=1}^{N} d_{\ell r} g_r(i) = 0$ by independence of $(e_1, \dots, e_s)$.
$(\forall\, \ell \in \{1, \dots, s\}) \sum_{r=1}^{N} d_{\ell r} g_r = 0$ by Assumption 3.1.2, $(\forall\, \ell \in \{1, \dots, s\})(\forall\, r \in \{1, \dots, N\})\, d_{\ell r} = 0$ by independence of $\mathcal{C}$.
Hence $(\forall\, r \in \{1, \dots, N\})\, c_r = 0$, so $g \mapsto \Phi_g$ is injective and $\{\Phi_g : g \in \mathcal{C}\}$ is $\mathbb{F}$-linearly independent.
Extend this set to a Hamel basis $\mathcal{E}$ of $\mathcal{V}'$ by Theorem 1.3.17(iii). By (b), $\dim_{\mathbb{F}}(\mathcal{V}') = |\mathcal{E}| \ge |\mathcal{C}| = \dim_{\mathbb{F}_0}(\mathbb{F}_0^I) > \kappa = \dim_{\mathbb{F}}(\mathcal{V})$ by Proposition 6.5.2(i) and Definition 1.4.6, contrary to the assumed equality.

(*ii*) (⇐) If $\dim(\mathcal{V}) < \infty$, then $\dim(\mathcal{V}') = \dim(\mathcal{V})$ by Theorem 4.1.4(ii).

Structural Commentary: A countable scalar core transfers Cantor-sized independence from functions to functionals.
Key Insight: Scalar extension preserves independence of core-field coordinate functions. □

Proposition 4.1.7. *Functionals Separate Points*
Assume Zorn's lemma, and let $\mathcal{V}$ be a vector space.
(*i*) $(\forall\, \mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\})(\exists\, \varphi \in \mathcal{V}')\, \varphi(\mathbf{v}) \neq 0$.
(*ii*) $(\forall\, \mathbf{v} \in \mathcal{V})\, [\mathbf{v} = \mathbf{0} \iff (\forall\, \varphi \in \mathcal{V}')\, \varphi(\mathbf{v}) = 0]$.
(*iii*) *If $\mathcal{U} \le \mathcal{V}$ and $\mathbf{x} \in \mathcal{V} \setminus \mathcal{U}$, then* $(\exists\, \varphi \in \mathcal{V}')\, [\varphi|_{\mathcal{U}} = 0 \wedge \varphi(\mathbf{x}) = 1]$.
(*iv*) *If $\varphi \in \mathcal{V}'$ and $\mathbf{x} \in \mathcal{V}$ satisfy $\varphi(\mathbf{x}) \neq 0$, then $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \ker(\varphi)$.*
(*v*) *If $0 \neq \varphi, \psi \in \mathcal{V}'$ and $\ker(\varphi) = \ker(\psi)$, then $(\exists\, a \in \mathbb{F} \setminus \{0\})\, \varphi = a\psi$.*

Proof.

(*i*) Fix $\mathbf{v} \neq \mathbf{0}$. The set $\{\mathbf{v}\}$ is linearly independent, so it extends to a Hamel basis $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ by Theorem 1.3.17(iii).
Let $\{\varepsilon_i\}_{i\in I}$ denote its coordinate functionals, choose $i_0 \in I$ with $\mathbf{b}_{i_0} = \mathbf{v}$. Then $\varepsilon_{i_0} \in \mathcal{V}'$ by Proposition 4.1.5(i), and $\varepsilon_{i_0}(\mathbf{v}) = 1$.

(*ii*) (⇒) If $\mathbf{v} = \mathbf{0}$, then $(\forall\, \varphi \in \mathcal{V}')\, \varphi(\mathbf{v}) = 0$ by Proposition 1.5.3.

(*ii*) (⇐) If $\mathbf{v} \neq \mathbf{0}$, then $(\exists\, \psi \in \mathcal{V}')\, \psi(\mathbf{v}) \neq 0$ by (i), contrary to the assumed vanishing. Hence $\mathbf{v} = \mathbf{0}$.

(*iii*) Choose a Hamel basis $\mathcal{B}_{\mathcal{U}}$ of $\mathcal{U}$ by Theorem 1.3.17(ii).
Since $\mathbf{x} \notin \operatorname{span}(\mathcal{B}_{\mathcal{U}})$, every finite relation on $\mathcal{B}_{\mathcal{U}} \cup \{\mathbf{x}\}$ has zero $\mathbf{x}$-coefficient and is then trivial by independence of $\mathcal{B}_{\mathcal{U}}$.
Thus $\mathcal{B}_{\mathcal{U}} \cup \{\mathbf{x}\}$ is linearly independent and extends to a Hamel basis $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ of $\mathcal{V}$ by Theorem 1.3.17(iii).
Let $\{\varepsilon_i\}_{i\in I}$ denote its coordinate functionals, choose $i_0 \in I$ with $\mathbf{b}_{i_0} = \mathbf{x}$, and set $\varphi := \varepsilon_{i_0}$.
Then $\varphi \in \mathcal{V}'$ by Proposition 4.1.5(i), $\varphi(\mathbf{x}) = 1$, and $\varphi|_{\mathcal{U}} = 0$ by $\mathcal{U} = \operatorname{span}(\mathcal{B}_{\mathcal{U}})$.

(*iv*) If $c\mathbf{x} \in \ker(\varphi)$, then $c\varphi(\mathbf{x}) = 0$, so $c = 0$; hence $\operatorname{span}(\mathbf{x}) \cap \ker(\varphi) = \{\mathbf{0}\}$. For $\mathbf{v} \in \mathcal{V}$, set $c_{\mathbf{v}} := \varphi(\mathbf{v})\varphi(\mathbf{x})^{-1}$.
Then $\mathbf{v} = c_{\mathbf{v}}\mathbf{x} + (\mathbf{v} - c_{\mathbf{v}}\mathbf{x})$, with $\mathbf{v} - c_{\mathbf{v}}\mathbf{x} \in \ker(\varphi)$. Thus $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \ker(\varphi)$ by Proposition 2.1.8.

(*v*) Set $\mathcal{K} := \ker(\varphi) = \ker(\psi)$. Since $\varphi \neq 0$, choose $\mathbf{x} \in \mathcal{V} \setminus \mathcal{K}$, set $a := \varphi(\mathbf{x})\psi(\mathbf{x})^{-1} \neq 0$. The identity $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \mathcal{K}$ holds by (iv).
Fix $\mathbf{v} = c\mathbf{x} + \mathbf{k}$ with $c \in \mathbb{F}$ and $\mathbf{k} \in \mathcal{K}$. Then $\varphi(\mathbf{v}) = c\varphi(\mathbf{x}) = ac\psi(\mathbf{x}) = a\psi(\mathbf{v})$. Hence $\varphi = a\psi$ by Assumption 3.1.2.

Structural Commentary: Basis extension constructs separating coordinates; kernel decomposition then compares nonzero functionals.
Key Insight: One normalized coordinate detects any direction outside a prescribed subspace. □

Corollary 4.1.8. *Kernels of Nonzero Functionals*
If $0 \neq \varphi \in \mathcal{V}'$, then:

(i) φ *is surjective;*
(ii) $\mathcal{V}/\ker(\varphi) \cong \mathbb{F}$*;*
(iii) $\operatorname{codim}_{\mathcal{V}}(\ker(\varphi)) = 1$.

Proof. Choose $\mathbf{x} \in \mathcal{V}$ with $\varphi(\mathbf{x}) \neq 0$.

(i) $(\forall\, a \in \mathbb{F})\, \varphi(a\varphi(\mathbf{x})^{-1}\mathbf{x}) = a$, so $\operatorname{range}(\varphi) = \mathbb{F}$.
(ii) The isomorphism $\mathcal{V}/\ker(\varphi) \cong \operatorname{range}(\varphi) = \mathbb{F}$ holds by Theorem 2.2.12 and (i).
(iii) Choose an isomorphism $T : \mathcal{V}/\ker(\varphi) \to \mathbb{F}$ by (ii). Since (1) is a basis of $\mathbb{F}$ over itself, $(T^{-1}(1))$ is a basis of $\mathcal{V}/\ker(\varphi)$ by Proposition 1.7.8. Hence $\operatorname{codim}_{\mathcal{V}}(\ker(\varphi)) = \dim(\mathcal{V}/\ker(\varphi)) = 1$ by Definition 2.3.11 and Definition 1.4.2.

Structural Commentary: Surjectivity identifies the quotient with the one-dimensional scalar field.
Key Insight: A nonzero functional removes exactly one scalar direction. □

4.2 Double Duals and Dual Maps

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.2.1. *Algebraic Double Dual and Natural Map*

(i) The *algebraic double dual* of $\mathcal{V}$ is $\mathcal{V}'' := (\mathcal{V}')'$.
(ii) The *natural evaluation map* $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ is defined by $(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \varphi \in \mathcal{V}')\, \Lambda_{\mathcal{V}}(\mathbf{v})(\varphi) := \varphi(\mathbf{v})$.
For $\mathbf{v} \in \mathcal{V}$, $(\forall\, a, b \in \mathbb{F})(\forall\, \varphi, \psi \in \mathcal{V}')\, \Lambda_{\mathcal{V}}(\mathbf{v})(a\varphi + b\psi) = a\varphi(\mathbf{v}) + b\psi(\mathbf{v})$ by pointwise operations in Proposition 1.5.5, so $\Lambda_{\mathcal{V}}(\mathbf{v}) \in \mathcal{V}''$.
(iii) The space $\mathcal{V}$ is *algebraically reflexive* iff $\Lambda_{\mathcal{V}}$ is surjective.

Theorem 4.2.2. *Algebraic Reflexivity*
Let $\mathcal{V}$ be a vector space. Claims (iii)–(v) assume Zorn's lemma, and (v) additionally assumes the Choice Principle 6.1.2. Then:

(i) $\Lambda_{\mathcal{V}} \in \mathcal{L}(\mathcal{V}, \mathcal{V}'')$*;*
(ii) if $\dim(\mathcal{V}) < \infty$*, then* $\Lambda_{\mathcal{V}}$ *is an isomorphism;*
(iii) $\Lambda_{\mathcal{V}}$ *is injective;*
(iv) $\Lambda_{\mathcal{V}}$ *is an isomorphism iff* $\mathcal{V}$ *is algebraically reflexive;*
(v) $\mathcal{V}$ *is algebraically reflexive iff* $\dim(\mathcal{V}) < \infty$.

Proof.

(i) For $a, b \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, and $\varphi \in \mathcal{V}'$, one has $\Lambda_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v})(\varphi) = \varphi(a\mathbf{u} + b\mathbf{v}) = \big(a\Lambda_{\mathcal{V}}(\mathbf{u}) + b\Lambda_{\mathcal{V}}(\mathbf{v})\big)(\varphi)$.
Hence $\Lambda_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v}) = a\Lambda_{\mathcal{V}}(\mathbf{u}) + b\Lambda_{\mathcal{V}}(\mathbf{v})$ by Assumption 3.1.2, so $\Lambda_{\mathcal{V}} \in \mathcal{L}(\mathcal{V}, \mathcal{V}'')$.
(ii) Suppose $\dim(\mathcal{V}) < \infty$, let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis, and let $\beta' = (\varphi_1, \ldots, \varphi_n)$ be its dual basis.
If $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{v}_j \in \ker(\Lambda_{\mathcal{V}})$, then $(\forall\, k \in \{1, \ldots, n\})\, c_k = \Lambda_{\mathcal{V}}(\mathbf{v})(\varphi_k) = 0$; hence $\mathbf{v} = \mathbf{0}$, also for $n = 0$. Thus $\Lambda_{\mathcal{V}}$ is injective by Proposition 1.6.2. By Theorem 4.1.4(ii) for $\mathcal{V}$, $\dim(\mathcal{V}') = \dim(\mathcal{V}) < \infty$; by the same theorem for $\mathcal{V}'$, $\dim(\mathcal{V}'') = \dim(\mathcal{V}')$.
Therefore $\Lambda_{\mathcal{V}}$ is surjective by Corollary 1.7.6(ii) and is an isomorphism by Proposition 1.6.7.
(iii) Assume Zorn's lemma. By Proposition 1.6.2, it suffices to prove $\ker(\Lambda_{\mathcal{V}}) = \{\mathbf{0}\}$.
If $\mathbf{v} \in \ker(\Lambda_{\mathcal{V}})$, then $(\forall\, \varphi \in \mathcal{V}')\, \varphi(\mathbf{v}) = 0$, so $\mathbf{v} = \mathbf{0}$ by Proposition 4.1.7(ii).
(iv) ($\Rightarrow$) If $\Lambda_{\mathcal{V}}$ is an isomorphism, then it is surjective by Proposition 1.6.7; hence $\mathcal{V}$ is algebraically reflexive by Definition 4.2.1(iii).
(iv) ($\Leftarrow$) By Proposition 1.6.7, it suffices to prove that $\Lambda_{\mathcal{V}}$ is bijective. It is surjective by algebraic reflexivity and injective by (iii).
(v) ($\Rightarrow$) Suppose $\mathcal{V}$ is algebraically reflexive. Then $\Lambda_{\mathcal{V}}$ is an isomorphism by (iv).
If $\dim(\mathcal{V})$ were infinite, then $\dim(\mathcal{V}) < \dim(\mathcal{V}') < \dim(\mathcal{V}'')$ by Theorem 4.1.6(i),(ii), first for $\mathcal{V}$ and then for $\mathcal{V}'$.
Choose a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.3.17(ii). The set $\Lambda_{\mathcal{V}}(\mathcal{B})$ is a Hamel basis of $\mathcal{V}''$ by linearity, injectivity, and surjectivity of $\Lambda_{\mathcal{V}}$: linearity and injectivity preserve finite independence, while linearity and surjectivity preserve spanning.
Thus $\dim(\mathcal{V}) = \dim(\mathcal{V}'')$ by Definition 1.4.6, a contradiction. Hence $\dim(\mathcal{V}) < \infty$.
(v) ($\Leftarrow$) If $\dim(\mathcal{V}) < \infty$, then $\Lambda_{\mathcal{V}}$ is surjective by (ii) and Proposition 1.6.7; hence $\mathcal{V}$ is algebraically reflexive by Definition 4.2.1(iii).

Structural Commentary: Evaluation proves linearity, separation proves injectivity, and dual dimensions obstruct infinite reflexivity.
Key Insight: Canonical evaluation is surjective exactly before dual dimension grows. □

Perspective 4.2.3. *Why the Double Dual Is Natural*
An isomorphism $\mathcal{V} \cong \mathcal{V}'$ for finite-dimensional $\mathcal{V}$ generally depends on a chosen basis. In contrast, $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ is defined solely by evaluation and requires no basis.

Definition 4.2.4. *Dual Map*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, its *dual map* is $T' : \mathcal{W}' \to \mathcal{V}'$ defined by $(\forall\, \varphi \in \mathcal{W}')\, T'\varphi := \varphi T$.
The relation $(\forall\, \varphi \in \mathcal{W}')\, \varphi T \in \mathcal{L}(\mathcal{V}, \mathbb{F}) = \mathcal{V}'$ holds by Proposition 1.5.7(i). Equivalently, $(\forall\, \varphi \in \mathcal{W}')(\forall\, \mathbf{v} \in \mathcal{V})\, (T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v})$.

Proposition 4.2.5. *Dual Maps Are Linear*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$.

Proof. For $a, b \in \mathbb{F}$, $\varphi, \psi \in \mathcal{W}'$, and $\mathbf{v} \in \mathcal{V}$, one has $\big(T'(a\varphi + b\psi)\big)(\mathbf{v}) = (a\varphi + b\psi)(T\mathbf{v}) = \big(aT'\varphi + bT'\psi\big)(\mathbf{v})$.
Hence $T'(a\varphi + b\psi) = aT'\varphi + bT'\psi$ by Assumption 3.1.2, so $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$ by Definition 1.5.1.
Structural Commentary: Pointwise evaluation reduces linearity of the pullback to linearity of each functional.
Key Insight: Precomposition preserves linear combinations while reversing map direction. □

Theorem 4.2.6. *Algebra of Dual Maps*

(i) If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ *and* $a, b \in \mathbb{F}$*, then* $(aS + bT)' = aS' + bT'$.
(ii) If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ *and* $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$*, then* $(ST)' = T'S'$.
(iii) $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Proof.

(i) For $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, $\big((aS+bT)'\varphi\big)(\mathbf{v}) = \varphi(aS\mathbf{v}+bT\mathbf{v}) = \big(aS'\varphi + bT'\varphi\big)(\mathbf{v})$.
Hence $(aS+bT)' = aS' + bT'$ by Assumption 3.1.2 on $\mathcal{V}$ and $\mathcal{W}'$.

(ii) For $\varphi \in \mathcal{W}'$ and $\mathbf{u} \in \mathcal{U}$, $\big((ST)'\varphi\big)(\mathbf{u}) = \varphi(S(T\mathbf{u})) = \big(T'(S'\varphi)\big)(\mathbf{u})$. Hence $(ST)' = T'S'$ by Assumption 3.1.2 on $\mathcal{U}$ and $\mathcal{W}'$.

(iii) For $\varphi \in \mathcal{V}'$ and $\mathbf{v} \in \mathcal{V}$, $\big((I_\mathcal{V})'\varphi\big)(\mathbf{v}) = \varphi(I_\mathcal{V}\mathbf{v}) = \varphi(\mathbf{v})$. Hence $(I_\mathcal{V})' = I_{\mathcal{V}'}$ by Assumption 3.1.2 on $\mathcal{V}$ and $\mathcal{V}'$.

Structural Commentary: Function extensionality lifts pointwise linearity, composition, and identity calculations to dual maps.
Key Insight: Precomposition reverses products but preserves linear structure. □

Corollary 4.2.7. *Dual Map of an Inverse*
If $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ is an isomorphism, then:

(i) T' is an isomorphism;
(ii) $(T^{-1})' = (T')^{-1}$.

Proof. Set $S := (T^{-1})'$. Then $S \in \mathcal{L}(\mathcal{V}',\mathcal{W}')$ by Proposition 4.2.5.

(i) By Theorem 4.2.6(ii)–(iii), $T'S = (T^{-1}T)' = (I_\mathcal{V})' = I_{\mathcal{V}'}$ and $ST' = (TT^{-1})' = (I_\mathcal{W})' = I_{\mathcal{W}'}$.
Thus S is a two-sided inverse of T', and T' is an isomorphism by Definition 1.6.4.

(ii) The identity $S = (T')^{-1}$ holds by Proposition 1.6.5; hence $(T^{-1})' = (T')^{-1}$.

Structural Commentary: Reversed product identities construct the dual inverse; uniqueness identifies it.
Key Insight: Contravariance turns both inverse identities into dual identities. □

Corollary 4.2.8. *Double Dual Maps Identify with Original Maps*
Let $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ and set $T'' := (T')'$. Then:

(i) $T''\Lambda_\mathcal{V} = \Lambda_\mathcal{W} T$;
(ii) if $\dim(\mathcal{V}), \dim(\mathcal{W}) < \infty$, then $T'' = \Lambda_\mathcal{W} T \Lambda_\mathcal{V}^{-1}$.

Proof. $T' \in \mathcal{L}(\mathcal{W}',\mathcal{V}')$ and $T'' \in \mathcal{L}(\mathcal{V}'',\mathcal{W}'')$ by Proposition 4.2.5, first for T and then for T'.

(i) For $\mathbf{v} \in \mathcal{V}$ and $\psi \in \mathcal{W}'$, $\big(T''\Lambda_\mathcal{V}(\mathbf{v})\big)(\psi) = \Lambda_\mathcal{V}(\mathbf{v})(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \Lambda_\mathcal{W}(T\mathbf{v})(\psi)$.
Hence $T''\Lambda_\mathcal{V} = \Lambda_\mathcal{W} T$ by Assumption 3.1.2 on $\mathcal{W}'$ and $\mathcal{V}$.

(ii) If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional, then $\Lambda_\mathcal{V}$ and $\Lambda_\mathcal{W}$ are isomorphisms by Theorem 4.2.2(ii).
Thus $T'' = T''I_{\mathcal{V}''} = T''\Lambda_\mathcal{V}\Lambda_\mathcal{V}^{-1} = \Lambda_\mathcal{W} T \Lambda_\mathcal{V}^{-1}$ by (i) and Proposition 1.5.7(ii)–(iii).

Structural Commentary: Pointwise evaluation proves naturality; finite reflexivity transports T to its double dual.
Key Insight: Both routes evaluate each functional at the same image vector. □

4.3 Annihilators and Transposes

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.3.1. *Annihilator*
If $\mathcal{U} \subseteq \mathcal{V}$, the *annihilator of $\mathcal{U}$ in $\mathcal{V}'$* is $\mathcal{U}^0 := \{\varphi \in \mathcal{V}' : (\forall\, \mathbf{u} \in \mathcal{U})\, \varphi(\mathbf{u}) = 0\}$.
The ambient space $\mathcal{V}$ is understood from context; changing it changes the dual space containing $\mathcal{U}^0$.

Proposition 4.3.2. *Annihilators Are Subspaces*
If $\mathcal{U} \subseteq \mathcal{V}$, then $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.

Proof. The zero functional belongs to $\mathcal{U}^0$. If $\varphi, \psi \in \mathcal{U}^0$, $a \in \mathbb{F}$, and $\mathbf{u} \in \mathcal{U}$, then $(a\varphi+\psi)(\mathbf{u}) = a\varphi(\mathbf{u}) + \psi(\mathbf{u}) = 0$.
Hence $a\varphi + \psi \in \mathcal{U}^0$, and $\mathcal{U}^0 \leq \mathcal{V}'$ by Theorem 1.2.2.
Structural Commentary: The subspace test reduces closure to the preservation of pointwise vanishing.
Key Insight: Linear combinations preserve every annihilating constraint. □

Theorem 4.3.3. *Dimension of the Annihilator*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Then $\dim(\mathcal{U}) + \dim(\mathcal{U}^0) = \dim(\mathcal{V})$.

Proof. Choose basis $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ of $\mathcal{U}$ by Corollary 1.3.14, extend to basis $\beta = (\mathbf{u}_1,\dots,\mathbf{u}_m,\mathbf{w}_1,\dots,\mathbf{w}_k)$ of $\mathcal{V}$ by Theorem 1.3.13.
Let $\beta' = (\varphi_1,\dots,\varphi_m,\psi_1,\dots,\psi_k)$ be the dual basis.

By Theorem 4.1.4(i) and Definition 4.1.3, every $\varphi \in \mathcal{V}'$ satisfies $\varphi = \sum_{j=1}^{m} \varphi(\mathbf{u}_j)\varphi_j + \sum_{\ell=1}^{k} \varphi(\mathbf{w}_\ell)\psi_\ell$. Thus $\varphi \in \mathcal{U}^0$ implies $\varphi = \sum_{\ell=1}^{k} \varphi(\mathbf{w}_\ell)\psi_\ell$.
Conversely, each ψ_ℓ vanishes on $\mathcal{U} = \operatorname{span}(\mathbf{u}_1,\dots,\mathbf{u}_m)$ by Definition 4.1.3.
Hence $(\psi_1,\dots,\psi_k)$ spans $\mathcal{U}^0$ and is linearly independent as a sublist of β', so it is a basis of $\mathcal{U}^0$.
The adapted and annihilator bases have lengths $m+k$ and k, respectively, so $\dim(\mathcal{U}) + \dim(\mathcal{U}^0) = m + k = \dim(\mathcal{V})$.
Structural Commentary: An adapted basis identifies the annihilator with the dual coordinates outside $\mathcal{U}$.
Key Insight: Vanishing deletes precisely the dual coordinates indexed by $\mathcal{U}$. □

Theorem 4.3.4. *Annihilator Lattice Identities*
Let $\mathcal{V}$ be finite-dimensional, and let $\mathcal{S},\mathcal{T}$ be subspaces of $\mathcal{V}$.

(i) $\mathcal{S} \subseteq \mathcal{T} \Rightarrow \mathcal{T}^0 \subseteq \mathcal{S}^0$.
(ii) $(\mathcal{S}+\mathcal{T})^0 = \mathcal{S}^0 \cap \mathcal{T}^0$.
(iii) $(\mathcal{S}\cap\mathcal{T})^0 = \mathcal{S}^0 + \mathcal{T}^0$.

Proof.

(i) If $\mathcal{S} \subseteq \mathcal{T}$ and $\varphi \in \mathcal{T}^0$, then $(\forall\, \mathbf{s} \in \mathcal{S})\, \varphi(\mathbf{s}) = 0$, so $\varphi \in \mathcal{S}^0$.

(ii) If $\varphi \in (\mathcal{S}+\mathcal{T})^0$, then $\varphi|_\mathcal{S} = 0$ and $\varphi|_\mathcal{T} = 0$, so $\varphi \in \mathcal{S}^0 \cap \mathcal{T}^0$.
Conversely, if $\varphi \in \mathcal{S}^0 \cap \mathcal{T}^0$, then $(\forall\, \mathbf{s} \in \mathcal{S})(\forall\, \mathbf{t} \in \mathcal{T})\, \varphi(\mathbf{s}+\mathbf{t}) = \varphi(\mathbf{s}) + \varphi(\mathbf{t}) = 0$, so $\varphi \in (\mathcal{S}+\mathcal{T})^0$.

(iii) By (i), $\mathcal{S}^0 + \mathcal{T}^0 \subseteq (\mathcal{S}\cap\mathcal{T})^0$. By (ii) and Theorem 2.3.1, $\dim(\mathcal{S}^0+\mathcal{T}^0) = \dim(\mathcal{S}^0) + \dim(\mathcal{T}^0) - \dim((\mathcal{S}+\mathcal{T})^0)$.
The identity $\dim(\mathcal{S}^0+\mathcal{T}^0) = \dim(\mathcal{V}) - \dim(\mathcal{S}\cap\mathcal{T}) = \dim((\mathcal{S}\cap\mathcal{T})^0)$ holds by Theorem 4.3.3 and Theorem 2.3.1.
Hence the inclusion is an equality by Proposition 1.4.3.

Structural Commentary: Two direct inclusions establish order reversal and sums; a dimension comparison recovers intersections.
Key Insight: Annihilation reverses the finite-dimensional subspace lattice. □

Theorem 4.3.5. *Double Annihilator*
Let $\dim(\mathcal{V}) < \infty$ *and* $M \subseteq \mathcal{V}$, *and set* $M^{00} := (M^0)^0 \subseteq \mathcal{V}''$. *Then:*
(i) $M^{00} = \Lambda_{\mathcal{V}}(\operatorname{span}(M))$;
(ii) if $\mathcal{S} \leq \mathcal{V}$, *then* $\mathcal{S}^{00} = \Lambda_{\mathcal{V}}(\mathcal{S})$, *and the corestriction* $\Lambda_{\mathcal{V}}|_{\mathcal{S}} : \mathcal{S} \to \mathcal{S}^{00}$ *is an isomorphism.*

Proof.
(i) If $\mathbf{v} \in \operatorname{span}(M)$ and $\varphi \in M^0$, then $\Lambda_{\mathcal{V}}(\mathbf{v})(\varphi) = \varphi(\mathbf{v}) = 0$ by linearity of φ, so $\Lambda_{\mathcal{V}}(\operatorname{span}(M)) \subseteq M^{00}$.
For the reverse inclusion, let $\eta \in M^{00}$. The relation $(\exists!\, \mathbf{v} \in \mathcal{V})\, \eta = \Lambda_{\mathcal{V}}(\mathbf{v})$ holds by Theorem 4.2.2(ii); fix this $\mathbf{v}$.
Suppose $\mathbf{v} \notin \operatorname{span}(M)$. Choose a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ of $\operatorname{span}(M)$ by Corollary 1.3.14.
The list $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v})$ is linearly independent by Proposition 1.3.7 and extends to a basis of $\mathcal{V}$ by Theorem 1.3.13.
Its dual-basis functional φ corresponding to $\mathbf{v}$ satisfies $\varphi|_{\operatorname{span}(M)} = 0$ and $\varphi(\mathbf{v}) = 1$.
Thus $\varphi \in M^0$ but $\eta(\varphi) = 1$, contrary to $\eta \in M^{00}$. Hence $M^{00} \subseteq \Lambda_{\mathcal{V}}(\operatorname{span}(M))$.
(ii) The identity $\mathcal{S}^{00} = \Lambda_{\mathcal{V}}(\mathcal{S})$ holds by (i) with $M = \mathcal{S}$. The corestriction is linear and injective by Theorem 4.2.2(i)–(ii), and it is surjective onto $\mathcal{S}^{00}$. Therefore it is an isomorphism by Proposition 1.6.7.

Structural Commentary: One inclusion extends vanishing; an adapted dual basis separates every vector outside the span.
Key Insight: Double annihilation remembers exactly the generated subspace. □

Theorem 4.3.6. *Direct Sums and Annihilators*
Suppose $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$, *and define* $E : \mathcal{T}' \to \mathcal{S}^0$ *by* $(\forall\, \varphi \in \mathcal{T}')(\forall\, \mathbf{s} \in \mathcal{S})(\forall\, \mathbf{t} \in \mathcal{T})\, (E\varphi)(\mathbf{s}+\mathbf{t}) := \varphi(\mathbf{t})$. *Then:*
(i) E *is an isomorphism;*
(ii) $\mathcal{V}' = \mathcal{S}^0 \oplus \mathcal{T}^0$.

Proof.
(i) Each $\mathbf{v} \in \mathcal{V}$ has a unique representation $\mathbf{v} = \mathbf{s} + \mathbf{t}$ by Proposition 2.1.8, so E is well-defined.
If $\mathbf{v}_i = \mathbf{s}_i + \mathbf{t}_i$ for $i = 1, 2$ and $a, b \in \mathbb{F}$, then $(E\varphi)(a\mathbf{v}_1 + b\mathbf{v}_2) = \varphi(a\mathbf{t}_1 + b\mathbf{t}_2) = a(E\varphi)(\mathbf{v}_1) + b(E\varphi)(\mathbf{v}_2)$, so $E\varphi \in \mathcal{V}'$.
The relation $(\forall\, \mathbf{s} \in \mathcal{S})\, (E\varphi)(\mathbf{s}) = 0$ places $E\varphi$ in $\mathcal{S}^0$. For $a, b \in \mathbb{F}$ and $\varphi, \psi \in \mathcal{T}'$, the identity $E(a\varphi + b\psi) = aE\varphi + bE\psi$ holds pointwise and hence by Assumption 3.1.2, so $E \in \mathcal{L}(\mathcal{T}', \mathcal{S}^0)$. Define $R : \mathcal{S}^0 \to \mathcal{T}'$ by $(\forall\, \eta \in \mathcal{S}^0)\, R\eta := \eta|_{\mathcal{T}}$.
If $\eta \in \mathcal{S}^0$, $a, b \in \mathbb{F}$, and $\mathbf{t}_1, \mathbf{t}_2 \in \mathcal{T}$, then $(R\eta)(a\mathbf{t}_1 + b\mathbf{t}_2) = a(R\eta)(\mathbf{t}_1) + b(R\eta)(\mathbf{t}_2)$ by linearity of η, so $R\eta \in \mathcal{T}'$.
For $a, b \in \mathbb{F}$ and $\eta, \zeta \in \mathcal{S}^0$, the identity $R(a\eta + b\zeta) = aR\eta + bR\zeta$ holds pointwise, so $R \in \mathcal{L}(\mathcal{S}^0, \mathcal{T}')$ by Assumption 3.1.2.
The identities $(\forall\, \varphi \in \mathcal{T}')\, RE\varphi = \varphi$ and $(\forall\, \eta \in \mathcal{S}^0)(\forall\, \mathbf{s} \in \mathcal{S})(\forall\, \mathbf{t} \in \mathcal{T})\, (ER\eta)(\mathbf{s}+\mathbf{t}) = \eta(\mathbf{t}) = \eta(\mathbf{s}+\mathbf{t})$ hold by the definitions of E, R, and $\mathcal{S}^0$. Thus R is a two-sided inverse of E by Assumption 3.1.2, and E is an isomorphism.
(ii) For $\varphi \in \mathcal{V}'$, set $\varphi_{\mathcal{S}} := \varphi\rho_{\mathcal{S},\mathcal{T}}$ and $\varphi_{\mathcal{T}} := \varphi\rho_{\mathcal{T},\mathcal{S}}$. These functionals belong to $\mathcal{V}'$ by Proposition 2.4.2(ii) and Proposition 1.5.7(i).
Their respective membership in $\mathcal{T}^0$ and $\mathcal{S}^0$ holds by Proposition 2.4.2(iii).
The identity $\varphi = \varphi_{\mathcal{S}} + \varphi_{\mathcal{T}}$ holds by Proposition 2.4.3(i), so $\mathcal{V}' = \mathcal{S}^0 + \mathcal{T}^0$.
If $\eta \in \mathcal{S}^0 \cap \mathcal{T}^0$, then $(\forall\, \mathbf{s} \in \mathcal{S})(\forall\, \mathbf{t} \in \mathcal{T})\, \eta(\mathbf{s}+\mathbf{t}) = 0$, so $\eta = 0_{\mathcal{V}'}$ by Assumption 3.1.2. Thus $\mathcal{V}' = \mathcal{S}^0 \oplus \mathcal{T}^0$.

Structural Commentary: Extension and restriction identify one annihilator; component functionals split every element of the dual.
Key Insight: Each dual component ignores the opposite primal summand. □

Theorem 4.3.7. *Kernel and Range of the Dual Map*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. *The following statements hold, with finite-dimensionality of* $\mathcal{V}$ *and* $\mathcal{W}$ *required only in (ii)–(iv):*
(i) $\ker(T') = (\operatorname{range}(T))^0$;
(ii) $\dim(\ker(T')) = \dim(\ker(T)) + \dim(\mathcal{W}) - \dim(\mathcal{V})$;
(iii) $\operatorname{range}(T') = (\ker(T))^0$;
(iv) $\dim(\operatorname{range}(T')) = \dim(\operatorname{range}(T))$.

Proof. For (ii)–(iv), assume that $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional.
(i) For $\varphi \in \mathcal{W}'$, $T'\varphi = 0$ iff $(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(T\mathbf{v}) = 0$ iff $\varphi \in (\operatorname{range}(T))^0$.
(ii) By (i), Theorem 4.3.3, and Theorem 1.7.3(ii), $\dim(\ker(T')) = \dim(\mathcal{W}) - \dim(\operatorname{range}(T)) = \dim(\ker(T)) + \dim(\mathcal{W}) - \dim(\mathcal{V})$.
(iii) If $\varphi = T'\psi$ and $\mathbf{v} \in \ker(T)$, then $\varphi(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\operatorname{range}(T') \subseteq (\ker(T))^0$.
By Theorem 1.7.3(ii), (ii), and Theorem 4.1.4(ii), $\dim(\operatorname{range}(T')) = \dim(\mathcal{W}') - \dim(\ker(T')) = \dim(\mathcal{V}) - \dim(\ker(T))$.
The latter dimension equals $\dim((\ker(T))^0)$ by Theorem 4.3.3, so the inclusion is an equality by Proposition 1.4.3.
(iv) By (iii), Theorem 4.3.3, and Theorem 1.7.3(ii), $\dim(\operatorname{range}(T')) = \dim(\mathcal{V}) - \dim(\ker(T)) = \dim(\operatorname{range}(T))$.

Structural Commentary: A direct kernel identity and finite-dimensional counts determine the dual range and both dimensions.
Key Insight: Pullback exchanges unreachable vectors with annihilating functionals. □

Corollary 4.3.8. *Surjectivity and Dual Injectivity*
Let $\mathcal{V}$ *and* $\mathcal{W}$ *be finite-dimensional, and let* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. *Then* T *is surjective iff* T' *is injective.*

Proof.
(⇒) If T is surjective, then $\operatorname{range}(T) = \mathcal{W}$, so $\ker(T') = \mathcal{W}^0 = \{0_{\mathcal{W}'}\}$ by Theorem 4.3.7(i). Hence T' is injective by Proposition 1.6.2.
(⇐) If T' is injective, then $(\operatorname{range}(T))^0 = \ker(T') = \{0_{\mathcal{W}'}\}$ by Proposition 1.6.2 and Theorem 4.3.7(i).
Thus $\dim(\operatorname{range}(T)) = \dim(\mathcal{W})$ by Theorem 4.3.3, so $\operatorname{range}(T) = \mathcal{W}$ by Proposition 1.4.3.

Structural Commentary: The dual-kernel identity translates surjectivity into a zero-kernel criterion.
Key Insight: A full range has no nonzero annihilator. □

Corollary 4.3.9. *Injectivity and Dual Surjectivity*
Let $\mathcal{V}$ *and* $\mathcal{W}$ *be finite-dimensional, and let* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. *Then* T *is injective iff* T' *is surjective.*

Proof.
(⇒) If T is injective, then T' is surjective by Proposition 1.6.2, Theorem 4.3.7(iii), and Definition 4.3.1.

($\Leftarrow$) If T' is surjective, then $(\ker(T))^0 = \operatorname{range}(T') = \mathcal{V}'$ by Theorem 4.3.7(iii).
Thus $\dim(\ker(T)) = 0$ by Theorem 4.3.3, so $\ker(T) = \{\mathbf{0}_\mathcal{V}\}$ by Proposition 1.4.3. Therefore T is injective by Proposition 1.6.2.

Structural Commentary: The dual-range identity converts injectivity into surjectivity of the pullback.
Key Insight: A trivial kernel leaves no functional-extension obstruction. □

Corollary 4.3.10. *Dual of a Quotient as an Annihilator*
Let $\mathcal{V}$ be finite-dimensional, let $\mathcal{U} \leq \mathcal{V}$, and let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map. Then:

(i) $\ker(\pi') = \{0_{(\mathcal{V}/\mathcal{U})'}\}$;
(ii) $\operatorname{range}(\pi') = \mathcal{U}^0$;
(iii) the corestriction $\pi' : (\mathcal{V}/\mathcal{U})' \to \mathcal{U}^0$ is an isomorphism, so $(\mathcal{V}/\mathcal{U})' \cong \mathcal{U}^0$.

Proof. The quotient is finite-dimensional by Theorem 2.2.9.

(i) The map π is surjective, so π' is injective by Corollary 4.3.8. Hence $\ker(\pi') = \{0_{(\mathcal{V}/\mathcal{U})'}\}$ by Proposition 1.6.2.
(ii) Since $\ker(\pi) = \mathcal{U}$, $\operatorname{range}(\pi') = (\ker(\pi))^0 = \mathcal{U}^0$ by Theorem 4.3.7(iii).
(iii) By (i)–(ii), the corestriction is a linear bijection and hence an isomorphism by Proposition 1.6.7.

Structural Commentary: Dual kernel and range identities make the quotient pullback a bijection onto the annihilator.
Key Insight: Quotient functionals are exactly functionals vanishing on the collapsed subspace. □

Theorem 4.3.11. *Range Membership Dual Alternative*
Let $\mathcal{V}, \mathcal{W}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and $\mathbf{w} \in \mathcal{W}$. Exactly one holds:

(i) $\mathbf{w} \in \operatorname{range}(T)$;
(ii) $(\exists\, \psi \in \mathcal{W}')[\psi T = 0 \wedge \psi(\mathbf{w}) \neq 0]$.

Proof.

(i) If (i) and (ii) hold, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{w} = T\mathbf{v}$ and $\psi \in \mathcal{W}'$ satisfying (ii).
Then $\psi(\mathbf{w}) = \psi(T\mathbf{v}) = 0$, contrary to $\psi(\mathbf{w}) \neq 0$. Hence (i) and (ii) cannot both hold.
(ii) Suppose (i) fails. The map $\Lambda_\mathcal{W}$ is injective by Theorem 4.2.2(ii), so $\Lambda_\mathcal{W}(\mathbf{w}) \notin \Lambda_\mathcal{W}(\operatorname{range}(T)) = (\operatorname{range}(T))^{00}$ by Theorem 4.3.5(i).
Hence $(\exists\, \psi \in (\operatorname{range}(T))^0)\, \Lambda_\mathcal{W}(\mathbf{w})(\psi) \neq 0$ by Definition 4.3.1. For such ψ, one has $\psi(\mathbf{w}) \neq 0$ and $(\forall\, \mathbf{v} \in \mathcal{V})\, (\psi T)(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\psi T = 0$ by Assumption 3.1.2. Thus (ii) holds whenever (i) fails.

Structural Commentary: Direct evaluation proves exclusion; double annihilation constructs every missing-range certificate.
Key Insight: A separating functional certifies every missing range vector. □

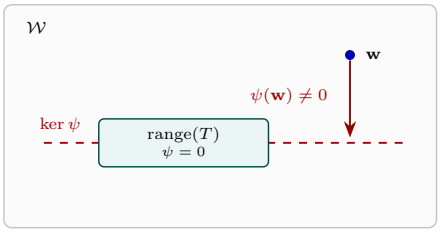

Figure 14: The dual certificate ψ vanishes on $\operatorname{range}(T)$ but detects $\mathbf{w}$ outside that range.

Figure 14 illustrates Theorem 4.3.11: $\operatorname{range}(T) \subseteq \ker(\psi)$ while $\mathbf{w} \notin \ker(\psi)$. It supplies no proof authority.

Proposition 4.3.12. *Standard Coordinate Dual Identification*
For $m \geq 1$, define $\Theta : \mathbb{F}^m \to (\mathbb{F}^m)'$ by $(\forall\, \mathbf{y}, \mathbf{z} \in \mathbb{F}^m)\, \Theta(\mathbf{y})(\mathbf{z}) := \mathbf{y}^t\mathbf{z}$. Then:

(i) Θ is an isomorphism;
(ii) $(\forall\, \psi \in (\mathbb{F}^m)')(\exists!\, \mathbf{y} \in \mathbb{F}^m)(\forall\, \mathbf{z} \in \mathbb{F}^m)\, \psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$.

Proof.

(i) For $a, b \in \mathbb{F}$ and $\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{w} \in \mathbb{F}^m$, the identity $\Theta(\mathbf{y})(a\mathbf{z} + b\mathbf{w}) = a\Theta(\mathbf{y})(\mathbf{z}) + b\Theta(\mathbf{y})(\mathbf{w})$ holds by Proposition 3.1.2, so $\Theta(\mathbf{y}) \in (\mathbb{F}^m)'$.
The identity $\Theta(a\mathbf{x} + b\mathbf{y})(\mathbf{z}) = a\Theta(\mathbf{x})(\mathbf{z}) + b\Theta(\mathbf{y})(\mathbf{z})$ holds by Proposition 3.1.14 and Proposition 3.1.2; hence Θ is linear by Assumption 3.1.2. If $\Theta(\mathbf{y}) = 0$, then $(\forall\, j \in \{1, \ldots, m\})\, 0 = \Theta(\mathbf{y})(\mathbf{e}_j) = y_j$, so $\mathbf{y} = \mathbf{0}$ and Θ is injective by Proposition 1.6.2.
Since (1) is a basis of $\mathbb{F}$ over itself, $\dim(\mathbb{F}^m) = m = \dim((\mathbb{F}^m)')$ by Theorem 2.1.13 and Theorem 4.1.4(ii).
Thus Θ is surjective by Corollary 1.7.6(ii) and is an isomorphism by Proposition 1.6.7.
(ii) Every $\psi \in (\mathbb{F}^m)'$ has a unique preimage $\mathbf{y} \in \mathbb{F}^m$ under Θ by (i), and $\psi(\mathbf{z}) = \Theta(\mathbf{y})(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$ by the defining formula.

Structural Commentary: Matrix identities prove linearity; coordinate evaluations and equal dimensions prove bijectivity.
Key Insight: Every coordinate functional is multiplication by one transposed column. □

Corollary 4.3.13. *Matrix Equation Alternative*
Let $m, n \geq 1$, $A \in \operatorname{Mat}(m, n, \mathbb{F})$, and $\mathbf{b} \in \mathbb{F}^m$. Exactly one holds:

(i) $(\exists\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = \mathbf{b}$;
(ii) $(\exists\, \mathbf{y} \in \mathbb{F}^m)[\mathbf{y}^t A = \mathbf{0}^t \wedge \mathbf{y}^t\mathbf{b} \neq 0]$.

Proof. Let $T_A : \mathbb{F}^n \to \mathbb{F}^m$ be the matrix-induced map.
Then $T_A \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ by Proposition 3.1.8, and both coordinate spaces are finite-dimensional by Theorem 2.1.13.

(i) The equivalence $\mathbf{b} \in \operatorname{range}(T_A) \iff (\exists\, \mathbf{x} \in \mathbb{F}^n)\, A\mathbf{x} = \mathbf{b}$ holds by $T_A\mathbf{x} = A\mathbf{x}$.
(ii) By Proposition 4.3.12(ii), each $\psi \in (\mathbb{F}^m)'$ has the unique form $\psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$. For this $\mathbf{y}$, $\psi T_A = 0$ iff $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, \mathbf{y}^t A\mathbf{x} = 0$ iff $\mathbf{y}^t A = \mathbf{0}^t$ by evaluation on the standard basis of $\mathbb{F}^n$; also, $\psi(\mathbf{b}) \neq 0$ iff $\mathbf{y}^t\mathbf{b} \neq 0$. Exactly one holds by Theorem 4.3.11.

Structural Commentary: Coordinate identification converts the abstract separating functional into a left-null vector.
Key Insight: Left-null vectors are coordinate dual certificates. □

Theorem 4.3.14. *Matrix of the Dual Map Is the Transpose*
Let $\dim(\mathcal{V}) = n \geq 1$ *and* $\dim(\mathcal{W}) = m \geq 1$, *let* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, *and let* $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *and* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ *be bases.*
If β' *and* γ' *are the corresponding dual bases, then* $[T']_{\beta' \leftarrow \gamma'} = ([T]_{\gamma \leftarrow \beta})^t$.

Proof. Set $A := [T]_{\gamma \leftarrow \beta}$ and $C := [T']_{\beta' \leftarrow \gamma'}$. Write $\beta' = (\varphi_1, \ldots, \varphi_n)$ and $\gamma' = (\psi_1, \ldots, \psi_m)$. By Definition 3.2.3 and Definition 4.1.3, $(\forall\, k \in \{1, \ldots, n\})(\forall\, j \in \{1, \ldots, m\})\, C_{k,j} = (T'\psi_j)(\mathbf{v}_k) = \psi_j(T\mathbf{v}_k) = A_{j,k}$, where $T\mathbf{v}_k = \sum_{r=1}^{m} A_{r,k}\mathbf{w}_r$. Hence $C = A^t$.
Structural Commentary: Evaluating dual-basis expansions identifies each dual-matrix entry with the transposed primal entry.
Key Insight: Dual evaluation swaps the row and column indices. □

4.4 Multilinear Structure

Throughout this subsection, all vector spaces are over the same fixed field $\mathbb{F}$.

Definition 4.4.1. *Bilinear Pairings, Matrices, and Induced Functions*
(i) A map $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ is *bilinear* if $(\forall\, a, b \in \mathbb{F})(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\, B(a\mathbf{u} + b\mathbf{v}, \mathbf{w}) = aB(\mathbf{u}, \mathbf{w}) + bB(\mathbf{v}, \mathbf{w})$, and $(\forall\, a, b \in \mathbb{F})(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{W})\, B(\mathbf{v}, a\mathbf{x} + b\mathbf{y}) = aB(\mathbf{v}, \mathbf{x}) + bB(\mathbf{v}, \mathbf{y})$.
(ii) Suppose $\dim(\mathcal{V}) = m \geq 1$ and $\dim(\mathcal{W}) = n \geq 1$, with ordered bases $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$. The *matrix of* B *in* β, γ is $[B]_{\beta,\gamma} := [B(\mathbf{v}_i, \mathbf{w}_j)]_{1 \leq i \leq m,\ 1 \leq j \leq n} \in \mathrm{Mat}(m, n, \mathbb{F})$.
(iii) For bilinear B, define Φ_B by $(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\, (\Phi_B\mathbf{v})(\mathbf{w}) := B(\mathbf{v}, \mathbf{w})$. Define Ψ_B by $(\forall\, \mathbf{w} \in \mathcal{W})(\forall\, \mathbf{v} \in \mathcal{V})\, (\Psi_B\mathbf{w})(\mathbf{v}) := B(\mathbf{v}, \mathbf{w})$.

Proposition 4.4.2. *Coordinate Evaluation of a Bilinear Pairing*
Let $\dim(\mathcal{V}) = m \geq 1$ *and* $\dim(\mathcal{W}) = n \geq 1$, *let* $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ *be bilinear.*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ *and* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ *be ordered bases of* $\mathcal{V}$ *and* $\mathcal{W}$, *respectively, and set* $M := [B]_{\beta,\gamma}$.
For $\mathbf{v} \in \mathcal{V}$ *and* $\mathbf{w} \in \mathcal{W}$, *set* $\mathbf{x} := [\mathbf{v}]_\beta$ *and* $\mathbf{y} := [\mathbf{w}]_\gamma$. *Then* $B(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i B(\mathbf{v}_i, \mathbf{w}_j) y_j = \mathbf{x}^t M \mathbf{y}$.

Proof. The identities $\mathbf{v} = \sum_{i=1}^{m} x_i \mathbf{v}_i$ and $\mathbf{w} = \sum_{j=1}^{n} y_j \mathbf{w}_j$ hold by Definition 3.2.1.
Hence $B(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j B(\mathbf{v}_i, \mathbf{w}_j)$ by Definition 4.4.1(i). The sum equals $\mathbf{x}^t M \mathbf{y}$ by Definition 4.4.1(ii) and Proposition 3.1.6.
Structural Commentary: Two slotwise expansions convert basis coefficients into a matrix product.
Key Insight: A pairing matrix records every basis-vector interaction. □

Proposition 4.4.3. *Bilinear Pairings Linearize through Duals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ *be bilinear. Then:*
(i) $\Phi_B : \mathcal{V} \to \mathcal{W}'$ *and* $\Psi_B : \mathcal{W} \to \mathcal{V}'$ *are well-defined linear maps;*
(ii) if $\dim(\mathcal{V}) = m \geq 1$ *and* $\dim(\mathcal{W}) = n \geq 1$, $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ *and* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ *are ordered bases of* $\mathcal{V}$ *and* $\mathcal{W}$ *with dual bases* β', γ', *and* $M := [B]_{\beta,\gamma}$, *then* $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$ *and* $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.

Proof.
(i) By Definition 4.4.1(i),(iii), $(\forall\, \mathbf{v} \in \mathcal{V})\, \Phi_B\mathbf{v} \in \mathcal{W}'$ and $(\forall\, \mathbf{w} \in \mathcal{W})\, \Psi_B\mathbf{w} \in \mathcal{V}'$.
For $a, b \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, and $\mathbf{w} \in \mathcal{W}$, $(\Phi_B(a\mathbf{u} + b\mathbf{v}))(\mathbf{w}) = aB(\mathbf{u}, \mathbf{w}) + bB(\mathbf{v}, \mathbf{w}) = (a\Phi_B\mathbf{u} + b\Phi_B\mathbf{v})(\mathbf{w})$.
Hence $\Phi_B(a\mathbf{u} + b\mathbf{v}) = a\Phi_B\mathbf{u} + b\Phi_B\mathbf{v}$ by Assumption 3.1.2. For $\mathbf{x}, \mathbf{y} \in \mathcal{W}$, the identity $\Psi_B(a\mathbf{x} + b\mathbf{y}) = a\Psi_B\mathbf{x} + b\Psi_B\mathbf{y}$ holds by Definition 4.4.1(i) and Assumption 3.1.2. Thus both maps are linear.
(ii) By Definition 3.2.3, 4.1.3, and 4.4.1(ii), $(\forall\, i \in \{1, \ldots, m\})(\forall\, j \in \{1, \ldots, n\})\, ([\Phi_B]_{\gamma' \leftarrow \beta})_{j,i} = (\Phi_B\mathbf{v}_i)(\mathbf{w}_j) = B(\mathbf{v}_i, \mathbf{w}_j) = M_{i,j}$.
Hence $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$. Likewise, $(\forall\, i \in \{1, \ldots, m\})(\forall\, j \in \{1, \ldots, n\})\, ([\Psi_B]_{\beta' \leftarrow \gamma})_{i,j} = (\Psi_B\mathbf{w}_j)(\mathbf{v}_i) = M_{i,j}$ by Definition 3.2.3, 4.1.3, and 4.4.1(ii), so $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.
Structural Commentary: Frozen-slot linearity establishes the maps; dual-basis evaluation identifies their matrices.
Key Insight: Freezing one slot turns bilinearity into dual-valued linearity. □

Definition 4.4.4. *Radicals and Nonsingularity of a Bilinear Pairing*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear.
(i) The *left* and *right radicals* are $\mathrm{rad}_L(B) := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{w} \in \mathcal{W})\, B(\mathbf{v}, \mathbf{w}) = 0\}$ and $\mathrm{rad}_R(B) := \{\mathbf{w} \in \mathcal{W} : (\forall\, \mathbf{v} \in \mathcal{V})\, B(\mathbf{v}, \mathbf{w}) = 0\}$.
(ii) For $\mathcal{S} \subseteq \mathcal{V}$ and $\mathcal{T} \subseteq \mathcal{W}$, the *right* and *left orthogonals* are $\mathcal{S}^{\perp_R} := \{\mathbf{w} \in \mathcal{W} : (\forall\, \mathbf{s} \in \mathcal{S})\, B(\mathbf{s}, \mathbf{w}) = 0\}$ and ${}^{\perp_L}\mathcal{T} := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{t} \in \mathcal{T})\, B(\mathbf{v}, \mathbf{t}) = 0\}$.
(iii) B is *left nondegenerate* if $\mathrm{rad}_L(B) = \{\mathbf{0}_\mathcal{V}\}$ and *right nondegenerate* if $\mathrm{rad}_R(B) = \{\mathbf{0}_\mathcal{W}\}$.
(iv) B is *nonsingular* or *perfect* if $\Phi_B : \mathcal{V} \to \mathcal{W}'$ and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ are isomorphisms.

Proposition 4.4.5. *Radicals, Matrices, and Dual Maps*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ *be bilinear. Then:*
(i) $\ker(\Phi_B) = \mathrm{rad}_L(B)$ *and* $\ker(\Psi_B) = \mathrm{rad}_R(B)$;
(ii) if $\dim(\mathcal{V}) = m \geq 1$ *and* $\dim(\mathcal{W}) = n \geq 1$, *with ordered bases* β *of* $\mathcal{V}$ *and* γ *of* $\mathcal{W}$ *and with* $M := [B]_{\beta,\gamma}$, *then* $\{[\mathbf{v}]_\beta : \mathbf{v} \in \mathrm{rad}_L(B)\} = \ker(M^t)$ *and* $\{[\mathbf{w}]_\gamma : \mathbf{w} \in \mathrm{rad}_R(B)\} = \ker(M)$;
(iii) under the hypotheses of (ii), $\dim(\mathrm{rad}_L(B)) = m - \mathrm{rank}(M)$ *and* $\dim(\mathrm{rad}_R(B)) = n - \mathrm{rank}(M)$.

Proof.
(i) By Definition 4.4.4(i), 4.4.1(iii), Assumption 3.1.2, $(\forall\, \mathbf{v} \in \mathcal{V})\, \mathbf{v} \in \ker(\Phi_B) \iff (\forall\, \mathbf{w} \in \mathcal{W})\, B(\mathbf{v}, \mathbf{w}) = 0 \iff \mathbf{v} \in \mathrm{rad}_L(B)$.
By Definition 4.4.4(i), 4.4.1(iii), Assumption 3.1.2, $(\forall\, \mathbf{w} \in \mathcal{W})\, \mathbf{w} \in \ker(\Psi_B) \iff (\forall\, \mathbf{v} \in \mathcal{V})\, B(\mathbf{v}, \mathbf{w}) = 0 \iff \mathbf{w} \in \mathrm{rad}_R(B)$
(ii) For $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$, set $\mathbf{x} := [\mathbf{v}]_\beta$ and $\mathbf{y} := [\mathbf{w}]_\gamma$. By Proposition 3.2.5 and 4.4.3(ii), $[\Phi_B\mathbf{v}]_{\gamma'} = M^t\mathbf{x}$ and $[\Psi_B\mathbf{w}]_{\beta'} = M\mathbf{y}$.
Hence $\mathbf{v} \in \mathrm{rad}_L(B) \iff \mathbf{x} \in \ker(M^t)$ and $\mathbf{w} \in \mathrm{rad}_R(B) \iff \mathbf{y} \in \ker(M)$ by (i) and Proposition 3.2.2(ii).
These equivalences prove both coordinate-set identities.

(iii) The coordinate-map restrictions to the two radicals are linear and injective by Proposition 3.2.2(i)–(ii), and they are surjective onto $\ker(M^t)$ and $\ker(M)$ by (ii). Thus their corestrictions are isomorphisms by Proposition 1.6.7, so the radical dimensions equal the corresponding kernel dimensions.
By Corollary 3.3.1(i) and Theorem 3.3.2, $\dim(\ker(M^t)) = m - \operatorname{rank}(M)$ and $\dim(\ker(M)) = n - \operatorname{rank}(M)$.

Structural Commentary: Kernel identities pass through coordinate isomorphisms; rank-nullity then determines both dimensions.
Key Insight: Radicals are the two matrix nullspaces in intrinsic form. □

Proposition 4.4.6. *Square Pairing Nonsingularity Criteria*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear with $\dim(\mathcal{V}) = \dim(\mathcal{W}) = n \geq 1$, let β and γ be ordered bases of $\mathcal{V}$ and $\mathcal{W}$, respectively. Set $M := [B]_{\beta,\gamma}$. The following are equivalent:

(i) B is left nondegenerate;
(ii) B is right nondegenerate;
(iii) M is invertible;
(iv) B is perfect.

Proof.

(i) $\Rightarrow$ (ii) If $\operatorname{rad}_L(B) = \{\mathbf{0}_\mathcal{V}\}$, then $\dim(\operatorname{rad}_R(B)) = n - \operatorname{rank}(M) = \dim(\operatorname{rad}_L(B)) = 0$ by Proposition 4.4.5(iii). Hence $\operatorname{rad}_R(B) = \{\mathbf{0}_\mathcal{W}\}$ by Proposition 1.4.3(ii).
(ii) $\Rightarrow$ (iii) If $\operatorname{rad}_R(B) = \{\mathbf{0}_\mathcal{W}\}$, then $0 = \dim(\operatorname{rad}_R(B)) = n - \operatorname{rank}(M)$ by Proposition 4.4.5(iii); M invertible by Corollary 3.6.2.
(iii) $\Rightarrow$ (iv) If M is invertible, then $\operatorname{rank}(M) = n$ by Corollary 3.6.2. Hence both radicals have dimension zero by Proposition 4.4.5(iii), so they are zero subspaces by Proposition 1.4.3(ii). Therefore $\ker(\Phi_B) = \{\mathbf{0}_\mathcal{V}\}$ and $\ker(\Psi_B) = \{\mathbf{0}_\mathcal{W}\}$ by Proposition 4.4.5(i). Their domains have dimension n by hypothesis, and their codomains have dimension n by Theorem 4.1.4(ii), so Φ_B and Ψ_B are isomorphisms by Corollary 1.7.6(i), Proposition 1.6.2, and Proposition 1.6.7. Thus B is perfect by Definition 4.4.4(iv).
(iv) $\Rightarrow$ (i) If B is perfect, then Φ_B is an isomorphism by Definition 4.4.4(iv) and hence injective by Proposition 1.6.7. Thus $\ker(\Phi_B) = \{\mathbf{0}_\mathcal{V}\}$ by Proposition 1.6.2, and $\operatorname{rad}_L(B) = \{\mathbf{0}_\mathcal{V}\}$ by Proposition 4.4.5(i).

Structural Commentary: Equal nullities link the radicals; full rank then identifies perfectness.
Key Insight: Square pairings have one shared rank obstruction. □

Corollary 4.4.7. *Perfect Pairings Represent Functionals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be perfect. Then:

(i) $(\forall \psi \in \mathcal{W}')(\exists! \, \mathbf{v} \in \mathcal{V})(\forall \mathbf{w} \in \mathcal{W}) \, \psi(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$;
(ii) $(\forall \varphi \in \mathcal{V}')(\exists! \, \mathbf{w} \in \mathcal{W})(\forall \mathbf{v} \in \mathcal{V}) \, \varphi(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$.

Proof.

(i) The map $\Phi_B : \mathcal{V} \to \mathcal{W}'$ is an isomorphism by Definition 4.4.4(iv), and hence a bijection by Proposition 1.6.7. Thus $(\forall \psi \in \mathcal{W}')(\exists! \, \mathbf{v} \in \mathcal{V}) \, \psi = \Phi_B \mathbf{v}$. For this $\mathbf{v}$, $(\forall \mathbf{w} \in \mathcal{W}) \, \psi(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$ by Definition 4.4.1(iii).
(ii) The map $\Psi_B : \mathcal{W} \to \mathcal{V}'$ is an isomorphism by Definition 4.4.4(iv), and hence a bijection by Proposition 1.6.7. Thus $(\forall \varphi \in \mathcal{V}')(\exists! \, \mathbf{w} \in \mathcal{W}) \, \varphi = \Psi_B \mathbf{w}$. For this $\mathbf{w}$, $(\forall \mathbf{v} \in \mathcal{V}) \, \varphi(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$ by Definition 4.4.1(iii).

Structural Commentary: Bijectivity of the two induced maps supplies the two unique representatives.
Key Insight: Perfect pairings turn functionals into uniquely paired vectors. □

Definition 4.4.8. *Matrix Congruence*
For $n \geq 1$, matrices $M, N \in \operatorname{Mat}(n, n, \mathbb{F})$ are *congruent* if $(\exists \, P \in \operatorname{Mat}(n, n, \mathbb{F}))[P \text{ is invertible} \wedge N = P^t M P]$.

Proposition 4.4.9. *Two-Sided Basis Change for Pairings*
Let $\dim(\mathcal{U}) = p \geq 1$ and $\dim(\mathcal{W}) = q \geq 1$, let $B : \mathcal{U} \times \mathcal{W} \to \mathbb{F}$ be bilinear, let $\alpha, \widetilde{\alpha}$ be ordered bases of $\mathcal{U}$, let $\beta, \widetilde{\beta}$ be ordered bases of $\mathcal{W}$. Set $M := [B]_{\alpha,\beta}$, $\widetilde{M} := [B]_{\widetilde{\alpha},\widetilde{\beta}}$, $P := C_{\alpha \leftarrow \widetilde{\alpha}}$, and $Q := C_{\beta \leftarrow \widetilde{\beta}}$. Then:

(i) $\widetilde{M} = P^t M Q$;
(ii) if $\mathcal{U} = \mathcal{W}$, $\alpha = \beta$, and $\widetilde{\alpha} = \widetilde{\beta}$, then $\widetilde{M}$ is congruent to M.

Proof.

(i) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, set $\mathbf{x} := [\mathbf{u}]_{\widetilde{\alpha}}$ and $\mathbf{y} := [\mathbf{w}]_{\widetilde{\beta}}$. Then $[\mathbf{u}]_\alpha = P\mathbf{x}$ and $[\mathbf{w}]_\beta = Q\mathbf{y}$ by Definition 3.5.1.
By Proposition 4.4.2 and Proposition 3.1.16, $\mathbf{x}^t \widetilde{M} \mathbf{y} = B(\mathbf{u}, \mathbf{w}) = (P\mathbf{x})^t M (Q\mathbf{y}) = \mathbf{x}^t P^t M Q \mathbf{y}$.
The coordinate maps are surjective by Proposition 3.2.2(iii), so $(\forall \mathbf{x} \in \mathbb{F}^p)(\forall \mathbf{y} \in \mathbb{F}^q) \, \mathbf{x}^t \widetilde{M} \mathbf{y} = \mathbf{x}^t P^t M Q \mathbf{y}$.
The relation $(\forall i \in \{1, \ldots, p\})(\forall j \in \{1, \ldots, q\}) \, \widetilde{M}_{i,j} = (P^t M Q)_{i,j}$ holds by substituting the standard coordinate columns.
Hence $\widetilde{M} = P^t M Q$ by Definition 3.1.1(i).
(ii) Under the stated equalities of spaces and bases, $P = Q$ and P is invertible by Theorem 3.5.2(i).
Thus $\widetilde{M} = P^t M P$ by (i), so $\widetilde{M}$ is congruent to M by Definition 4.4.8.

Structural Commentary: Coordinate evaluation transforms each slot; entry comparison proves the two-sided formula.
Key Insight: Each input slot contributes its own basis-change factor. □

Proposition 4.4.10. *Basis Change for a Square Bilinear Form*
Let $\dim(\mathcal{V}) = n \geq 1$, let $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ be bilinear, and let β, γ be ordered bases of $\mathcal{V}$. Set $M_\beta := [B]_{\beta,\beta}$, $M_\gamma := [B]_{\gamma,\gamma}$, and $C := C_{\beta \leftarrow \gamma}$. Then:

(i) $M_\gamma = C^t M_\beta C$;
(ii) M_β and M_γ are congruent.

Proof.

(i) With $\mathcal{U} = \mathcal{W} = \mathcal{V}$, old basis β in both slots, and new basis γ in both slots, $M_\gamma = C^t M_\beta C$ by Proposition 4.4.9(i).
(ii) The matrix C is invertible by Theorem 3.5.2(i), and $M_\gamma = C^t M_\beta C$ by (i). Hence M_β and M_γ are congruent by Definition 4.4.8.

Structural Commentary: The square case specializes both slot changes to one invertible matrix.
Key Insight: Reusing one basis change produces congruence. □

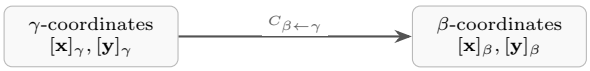

Figure 15: The same coordinate transition acts in both slots, producing $M_\gamma = C_{\beta\leftarrow\gamma}^{t} M_\beta C_{\beta\leftarrow\gamma}$.

Figure 15 depicts $C_{\beta\leftarrow\gamma}$ acting in both input slots; the formula belongs to Proposition 4.4.10, not to the illustration. The induced maps Φ_B and Ψ_B encode bilinear pairings in dual spaces. Part 4 develops real bilinear metric forms and complex sesquilinear inner products, while Part 8 constructs tensor products; neither later theory is used here.

Part III
Linear Maps and Matrix Representations

1 Linear Maps and Operations

1.1 Further Linear Map Operations and Examples

Operator polynomials encode powers and polynomial equations satisfied by a linear operator.

Definition 1.1.1. *Operator Powers*
Let $T \in \mathcal{L}(\mathcal{V})$. The *powers* of T are $T^m := \underbrace{T \cdots T}_{m \text{ times}}$ for $m \in \mathbb{N}$, with $T^0 := I_{\mathcal{V}}$. If T is invertible, set $T^{-m} := (T^{-1})^m$.

Definition 1.1.2. *Polynomial Applied to an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$. For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, the *polynomial p applied to T* is $p(T) := \sum_{j=0}^{m} a_j T^j$.

Definition 1.1.3. *Annihilating Polynomial*
Let $T \in \mathcal{L}(\mathcal{V})$. A polynomial $p \in \mathbb{F}[z]$ is an *annihilating polynomial* for T if $p(T) = 0_{\mathcal{L}(\mathcal{V})}$; equivalently, p *annihilates* T.

Definition 1.1.4. *Product of Polynomials*
If $p, q \in \mathbb{F}[z]$, their product $pq \in \mathbb{F}[z]$ is defined by $(pq)(z) = p(z)q(z)$ for $z \in \mathbb{F}$.

Definition 1.1.5. *Monic Polynomial*
A nonzero polynomial is *monic* if its highest-degree coefficient is 1.

Proposition 1.1.6. *Multiplicative Properties of Operator Polynomials*
Let $p, q \in \mathbb{F}[z]$ and $T \in \mathcal{L}(\mathcal{V})$. Then
(i) $(pq)(T) = p(T)q(T)$;
(ii) $p(T)q(T) = q(T)p(T)$.

Proof.
(i) Write $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$. Then $(pq)(z) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k z^{j+k}$.
Substituting T gives $(pq)(T) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} = \left(\sum_{j=0}^{m} a_j T^j \right) \left(\sum_{k=0}^{n} b_k T^k \right) = p(T)q(T)$.
(ii) Since polynomial multiplication is commutative, $pq = qp$. By part (i), $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$.
Structural Commentary: Polynomial evaluation preserves products.
Key Insight: Expanding a polynomial product works the same with T as with z. □

Proposition 1.1.7. *Existence of Annihilating Polynomials*
Let $\mathcal{V}$ be finite-dimensional with $n = \dim(\mathcal{V}) > 0$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then T has a nonconstant annihilating polynomial of degree at most n^2.

Proof. By Theorem 3.4.5, the vector space $\mathcal{L}(\mathcal{V})$ has dimension n^2, so the $n^2 + 1$ operators $I_{\mathcal{V}}, T, T^2, \ldots, T^{n^2}$ are linearly dependent.
Hence there are scalars $a_0, \ldots, a_{n^2}$, not all zero, such that $\sum_{j=0}^{n^2} a_j T^j = 0$.
If $a_1 = \cdots = a_{n^2} = 0$, then $a_0 I_{\mathcal{V}} = 0$, forcing $a_0 = 0$ because $\mathcal{V} \neq \{\mathbf{0}\}$, a contradiction.
Thus $p(z) = \sum_{j=0}^{n^2} a_j z^j$ is a nonconstant annihilating polynomial.
Structural Commentary: Finite-dimensionality makes the powers of one operator linearly dependent inside $\mathcal{L}(\mathcal{V})$.
Key Insight: A dependence among $I_{\mathcal{V}}, T, \ldots, T^{n^2}$ is exactly a polynomial equation satisfied by T. □

Proposition 1.1.8. *Polynomial Evaluation on Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, let $p \in \mathbb{F}[z]$, and suppose $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $p(T)\mathbf{v} = p(\lambda)\mathbf{v}$.

Proof. Induct on j to get $T^j\mathbf{v} = \lambda^j\mathbf{v}$ $\forall j \geq 0$. If $p(z) = \sum_{j=0}^{m} a_j z^j$, then $p(T)\mathbf{v} = \sum_{j=0}^{m} a_j T^j\mathbf{v} = \sum_{j=0}^{m} a_j \lambda^j \mathbf{v} = p(\lambda)\mathbf{v}$.
Structural Commentary: On a one-dimensional eigenspace, T acts like the scalar λ.
Key Insight: Polynomial expressions in T reduce to the same polynomial evaluated at λ. □

Theorem 1.1.9. *Annihilating Polynomials Control the Spectrum*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let p be a nonconstant annihilating polynomial for T.
(i) Every eigenvalue of T is a root of p.
(ii) If $\mathbb{F} = \mathbb{C}$, then at least one root of p is an eigenvalue of T.

Proof.
(i) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then Proposition 1.1.8 gives $0 = p(T)\mathbf{v} = p(\lambda)\mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, $p(\lambda) = 0$.
(ii) Factor $p(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$ over $\mathbb{C}$, with $c \neq 0$. Then $0 = p(T) = c(T - \lambda_1 I_{\mathcal{V}}) \cdots (T - \lambda_m I_{\mathcal{V}})$. At least one factor $T - \lambda_j I_{\mathcal{V}}$ is not injective; otherwise product would be injective. Hence λ_j is an eigenvalue.
Structural Commentary: Annihilating polynomial restricts spectrum to its roots; over $\mathbb{C}$, one root must appear.
Key Insight: Eigenvectors turn $p(T)$ into $p(\lambda)$. □

2 Kernels, Images, Rank, and Nullity

2.1 Shift Operators and Kernel-Range Examples

Fix a field $\mathbb{F}$, let $\mathcal{E} := \mathbb{F}^{(\mathbb{N})}$ be its finitely supported sequence space, and set $\mathbf{e}_1 := (1, 0, 0, \ldots)$ and $\mathcal{E}_0 := \{\mathbf{x} \in \mathcal{E} : x_1 = 0\}$.

Proposition 2.1.1. *Shift Operators on Finitely Supported Sequences*
Define $R, S : \mathcal{E} \to \mathcal{E}$ by $R(x_1, x_2, \ldots) := (0, x_1, x_2, \ldots)$ and $S(x_1, x_2, \ldots) := (x_2, x_3, \ldots)$. Then:

(i) $R, S \in \mathcal{L}(\mathcal{E})$.
(ii) $\ker(R) = \{\mathbf{0}_\mathcal{E}\}$.
(iii) $\text{range}(R) = \mathcal{E}_0$.
(iv) $\ker(S) = \text{span}(\mathbf{e}_1)$.
(v) $\text{range}(S) = \mathcal{E}$.
(vi) *R is injective but not surjective, whereas S is surjective but not injective.*
(vii) *$\mathcal{E}$ is infinite-dimensional.*

Proof.

(i) For $\mathbf{x} \in \mathcal{E}$, $|\text{supp}(R\mathbf{x})| = |\text{supp}(\mathbf{x})| < \infty$ and $|\text{supp}(S\mathbf{x})| \leq |\text{supp}(\mathbf{x})|$, so R and S are well-defined.
Coordinatewise operations establish $(\forall T \in \{R, S\})(\forall a, b \in \mathbb{F})(\forall \mathbf{x}, \mathbf{y} \in \mathcal{E})\, T(a\mathbf{x} + b\mathbf{y}) = aT\mathbf{x} + bT\mathbf{y}$; hence $R, S \in \mathcal{L}(\mathcal{E})$.
(ii) For $\mathbf{x} = (x_1, x_2, \ldots) \in \mathcal{E}$, $R\mathbf{x} = \mathbf{0}_\mathcal{E} \iff (\forall j \in \mathbb{N})\, x_j = 0 \iff \mathbf{x} = \mathbf{0}_\mathcal{E}$. Thus $\ker(R) = \{\mathbf{0}_\mathcal{E}\}$.
(iii) By definition, $\text{range}(R) \subseteq \mathcal{E}_0$. Conversely, $\mathbf{y} = (0, y_2, y_3, \ldots) \in \mathcal{E}_0$ has $\mathbf{x} := (y_2, y_3, \ldots) \in \mathcal{E}$ with $R\mathbf{x} = \mathbf{y}$; hence $\text{range}(R) = \mathcal{E}_0$.
(iv) For $\mathbf{x} = (x_1, x_2, \ldots) \in \mathcal{E}$, $S\mathbf{x} = \mathbf{0}_\mathcal{E} \iff (\forall j \in \mathbb{N} \setminus \{1\})\, x_j = 0 \iff \mathbf{x} = x_1\mathbf{e}_1$. Thus $\ker(S) = \text{span}(\mathbf{e}_1)$.
(v) The identity $(\forall \mathbf{y} \in \mathcal{E})\, S(R\mathbf{y}) = \mathbf{y}$ holds by the definitions, with $R\mathbf{y} \in \mathcal{E}$ by (i); hence $\text{range}(S) = \mathcal{E}$.
(vi) The map R is injective by Proposition 1.6.2 with (ii), and R is not surjective by (iii), since $\mathbf{e}_1 \notin \mathcal{E}_0$.
The map S is surjective by (v) and not injective by Proposition 1.6.2 with (iv), since $\mathbf{e}_1 \neq \mathbf{0}_\mathcal{E}$.
(vii) The assumption $\dim(\mathcal{E}) < \infty$ would make R surjective by Theorem 1.7.7 with (i) and injectivity from (vi), contradicting (vi).

Structural Commentary: Matched claim items verify linearity, fibers, images, map properties, and the dimension obstruction.
Key Insight: The shifts exchange a missing coordinate with a nontrivial kernel. □

Example 2.1.2. *Shift Operators Show Noncommutativity*
For the shifts above, $SR, RS \in \mathcal{L}(\mathcal{E})$ by Proposition 1.5.7(i). For $\mathbf{x} = (x_1, x_2, \ldots) \in \mathcal{E}$, $SR\mathbf{x} = \mathbf{x}$ and $RS\mathbf{x} = (0, x_2, x_3, \ldots)$.
Hence $SR = I_\mathcal{E}$ by Assumption 3.1.2. Since $RS\mathbf{e}_1 = \mathbf{0}_\mathcal{E} \neq \mathbf{e}_1 = SR\mathbf{e}_1$, $RS \neq SR$ and $\mathcal{L}(\mathcal{E})$ is noncommutative.

2.2 Further Rank-Nullity and Invertibility Results

2.3 Further Quotients and Isomorphism Theorems

2.4 Further Finite-Dimensional Splitting and Isomorphism Tests

3 Matrix Representations

3.1 Further Matrix Algebra and Linear Systems

3.2 Further Coordinates and Matrix Representations

3.3 Further Matrix-Induced Maps and Rank

3.4 Further Invertibility and the Matrix-Map Dictionary

4 Row Reduction and Elementary Matrices

4.1 Row Reduction and Elementary Matrices

Definition 4.1.1. *Elementary Row Operations and Elementary Matrices*
Let $m \geq 1$. For $i, j \in \{1, \ldots, m\}$, the *elementary row operations* are:

(i) *Row scaling*: $R_i \leftarrow cR_i$, where $c \in \mathbb{F}^\times$;
(ii) *Row swap*: $R_i \leftrightarrow R_j$, where $i \neq j$;
(iii) *Row replacement*: $R_i \leftarrow R_i + cR_j$, where $i \neq j$ and $c \in \mathbb{F}$.

For an elementary row operation ϵ on m rows, its *elementary matrix* is $E_\epsilon := \epsilon(I_m)$.

Proposition 4.1.2. *Elementary Matrices Perform Row Operations*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, let ϵ be an elementary operation on m rows, and set $E_\epsilon := \epsilon(I_m)$. Then:

(i) $E_\epsilon A = \epsilon(A)$.
(ii) $E_\epsilon^{-1} = E_{\epsilon^{-1}}$*; in particular, every elementary matrix has an elementary inverse.*
(iii) *If η is an elementary operation on n rows and $F_\eta := \eta(I_n)^t$, then $AF_\eta = (\eta(A^t))^t$.*
Thus right multiplication by F_η performs the corresponding column operation.

Proof.

(i) The rows of E_ϵ identify the rows of $E_\epsilon A$ by matrix multiplication:
  (a) For scaling, $(E_\epsilon)_{i,\cdot} = c\mathbf{e}_i^t$, while every other row is standard; hence $(E_\epsilon A)_{i,\cdot} = cA_{i,\cdot}$ and the other rows are unchanged.
  (b) For a swap, $(E_\epsilon)_{i,\cdot} = \mathbf{e}_j^t$ and $(E_\epsilon)_{j,\cdot} = \mathbf{e}_i^t$; hence $E_\epsilon A$ exchanges rows i and j, with every other row fixed.
  (c) For replacement, $(E_\epsilon)_{i,\cdot} = \mathbf{e}_i^t + c\mathbf{e}_j^t$, so $(E_\epsilon A)_{i,\cdot} = A_{i,\cdot} + cA_{j,\cdot}$; the other coordinate rows preserve their rows.
(ii) The inverse of a scaling by c is scaling by c^{-1}, a swap is self-inverse, and the inverse of replacement by c is replacement by $-c$.
Therefore $E_{\epsilon^{-1}} E_\epsilon = \epsilon^{-1}(\epsilon(I_m)) = I_m$ and $E_\epsilon E_{\epsilon^{-1}} = \epsilon(\epsilon^{-1}(I_m)) = I_m$ by part (i).
(iii) $(AF_\eta)^t = F_\eta^t A^t = \eta(I_n)A^t = \eta(A^t)$ by Proposition 3.1.16, Proposition 3.1.14(i), and part (i).
After transposition, $(AF_\eta)^{tt} = AF_\eta$, so $AF_\eta = (\eta(A^t))^t$.

Structural Commentary: The proof translates row operations, their inverses, and transposition into multiplication identities.
Key Insight: Elementary matrices store row operations as left factors. □

Definition 4.1.3. *Row Equivalence and Reduced Row Echelon Form*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, $k=0$ interpreted as I_m; $A \sim_{\mathrm{row}} B \iff B = E_k \cdots E_1 A$ for $k \in \mathbb{N}_0$, elementary $E_1, \dots, E_k \in \mathrm{Mat}(m,m,\mathbb{F})$.
A matrix $R = [r_{i,j}] \in \mathrm{Mat}(m,n,\mathbb{F})$ is in *reduced row echelon form* if $(\exists s \in \{0,\dots,m\})(\exists p_1,\dots,p_s \in \{1,\dots,n\})$ such that:
(i) $(\forall i \in \{1,\dots,s\})\, R_{i,\cdot} \neq \mathbf{0}^t$ and $(\forall i \in \{s+1,\dots,m\})\, R_{i,\cdot} = \mathbf{0}^t$;
(ii) $(\forall i \in \{1,\dots,s\})\, [r_{i,p_i} = 1 \wedge (\forall j \in \{1,\dots,p_i-1\})\, r_{i,j} = 0]$;
(iii) $p_1 < \cdots < p_s$;
(iv) $(\forall i \in \{1,\dots,s\})(\forall \ell \in \{1,\dots,m\} \setminus \{i\})\, r_{\ell,p_i} = 0$.
The indices $p_1,\dots,p_s$ are the *pivot columns* of R.

Theorem 4.1.4. *Reduced Row Echelon Form*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then $(\exists!\, R \in \mathrm{Mat}(m,n,\mathbb{F}))\, [R$ is in RREF $\wedge\, A \sim_{\mathrm{row}} R]$. Write $\mathrm{rref}(A) := R$.

Proof.
(i) *Construction*: Set $i=1$ and $p_0 = 0$. Suppose rows $1,\dots,i-1$ have pivot columns $p_1 < \cdots < p_{i-1}$, these pivot columns are coordinate columns, and the remaining rows vanish in columns $1,\dots,p_{i-1}$.
If the remaining rows are zero, stop. Otherwise let p_i be the least column containing a nonzero entry in rows $i,\dots,m$.
Swap such an entry into row i, scale it to 1, and clear column p_i by row replacement.
The new pivot row vanishes before p_i, so these operations preserve the preceding pivot columns and extend the stated invariant.
(ii) *Verification*: Each stage increases both the pivot-row index and the pivot-column index, so the construction stops after at most $\min(m,n)$ stages. At termination, the invariant is exactly the four RREF conditions.
Each step is elementary, and hence the resulting matrix R satisfies $R = E_k \cdots E_1 A$ by Proposition 4.1.2. Thus $A \sim_{\mathrm{row}} R$.
(iii) *Leading Indices*: Let R and S be RREF matrices row-equivalent to A. Every elementary row operation replaces rows by linear combinations of rows, and its elementary inverse reverses the inclusion by Proposition 4.1.2(i)–(ii).
Hence $\mathrm{row}(R) = \mathrm{row}(A) = \mathrm{row}(S) := \mathcal{W}$. If $\mathcal{W} = \{\mathbf{0}\}$, then $R = S = 0$; assume $\mathcal{W} \neq \{\mathbf{0}\}$.
For $\mathbf{v} \in \mathcal{W} \setminus \{\mathbf{0}\}$, set $\ell(\mathbf{v}) := \min\{j : v_j \neq 0\}$. Let $X \in \{R, S\}$ have nonzero rows $X_{1,\cdot}, \dots, X_{h,\cdot}$ and pivot columns $q_1 < \cdots < q_h$.
$\mathbf{v} = \sum_{i=1}^{h} c_i X_{i,\cdot}^t$, set $a := \min\{i : c_i \neq 0\}$. The RREF conditions imply $(\forall j \in \{1,\dots,q_a-1\})\, v_j = 0$ and $v_{q_a} = c_a \neq 0$, so $\ell(\mathbf{v}) = q_a$.
Conversely, $\ell(X_{i,\cdot}^t) = q_i$. Therefore $\{\ell(\mathbf{v}) : \mathbf{v} \in \mathcal{W} \setminus \{\mathbf{0}\}\}$ is the pivot set of both R and S.
(iv) *Uniqueness*: Let $p_1 < \cdots < p_s$ be the common pivot columns.
For $i \in \{1,\dots,s\}$, the vector $\mathbf{d}_i := R_{i,\cdot}^t - S_{i,\cdot}^t$ belongs to $\mathcal{W}$ and $(\forall j \in \{1,\dots,s\})\, (\mathbf{d}_i)_{p_j} = 0$.
If $\mathbf{d}_i \neq \mathbf{0}$, then its leading index is a pivot column by part (iii), contradicting these zero pivot coordinates.
Thus $R_{i,\cdot} = S_{i,\cdot}$ for $i \leq s$, while every row with index greater than s is zero. Hence $R = S$.

Structural Commentary: Elimination constructs a representative; leading-index invariance proves uniqueness in the common row space.
Key Insight: A reduced row space determines every pivot row. □

Lemma 4.1.5. *Invertible Matrices Row-Reduce to the Identity*
Let $P \in \mathrm{Mat}(m,m,\mathbb{F})$ be invertible. Then:
(i) $\mathrm{rref}(P) = I_m$.
(ii) $I_m = E_k \cdots E_1 P$ *for a* $k \in \mathbb{N}_0$ *and elementary* $E_1,\dots,E_k \in \mathrm{Mat}(m,m,\mathbb{F})$.

Proof.
(i) Set $R = [r_{i,j}] := \mathrm{rref}(P)$. Then $R = E_k \cdots E_1 P$ by Theorem 4.1.4 and the definition of row equivalence.
Each E_i is invertible by Proposition 4.1.2(ii), so R is invertible by repeated use of Corollary 3.1.17(i). Consequently, $\ker(R) = \{\mathbf{0}\}$.
Let $p_1 < \cdots < p_s$ be the pivot columns of R. If $s < m$, choose $j \in \{1,\dots,m\} \setminus \{p_1,\dots,p_s\}$ and define $\mathbf{x} \in \mathbb{F}^m$ by $x_j = 1$, $(\forall i \in \{1,\dots,s\})\, x_{p_i} = -r_{i,j}$, and $(\forall \ell \in \{1,\dots,m\} \setminus \{j, p_1,\dots,p_s\})\, x_\ell = 0$.
Then $(\forall i \in \{1,\dots,s\})\, (R\mathbf{x})_i = x_{p_i} + r_{i,j} x_j = 0$ and $(\forall i \in \{s+1,\dots,m\})\, (R\mathbf{x})_i = 0$. This contradicts $\ker(R) = \{\mathbf{0}\}$, so $s = m$.
The strictly increasing pivot indices $p_1,\dots,p_m \in \{1,\dots,m\}$ satisfy $p_i = i$.
Every pivot column is therefore the corresponding coordinate column, so $R = I_m$.
(ii) By part (i), $P \sim_{\mathrm{row}} \mathrm{rref}(P) = I_m$. Thus $I_m = E_k \cdots E_1 P$ by the definition of row equivalence.

Structural Commentary: Kernel triviality excludes free columns; the square pivot set then forces the identity.
Key Insight: An invertible reduced matrix has no free variable. □

Corollary 4.1.6. *Row Equivalence Characterization*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, the following are equivalent:
(i) $A \sim_{\mathrm{row}} B$.
(ii) $\mathrm{rref}(A) = \mathrm{rref}(B)$.
(iii) $(\exists\, P \in \mathrm{Mat}(m,m,\mathbb{F})^{\times})\, B = PA$.

Proof.
(i) $\Rightarrow$ *(ii)* Write $B = G_q \cdots G_1 A$ and set $G := G_q \cdots G_1$.
The matrix G is invertible, $G^{-1} = G_1^{-1} \cdots G_q^{-1}$, and each G_j^{-1} is elementary by Proposition 4.1.2(ii) and Corollary 3.1.17(i).
For $R := \mathrm{rref}(A)$, write $R = E_k \cdots E_1 A$. Then $R = E_k \cdots E_1 G^{-1} B$, so $B \sim_{\mathrm{row}} R$ and $R = \mathrm{rref}(B)$ by Theorem 4.1.4.
(ii) $\Rightarrow$ *(iii)* Let $R := \mathrm{rref}(A) = \mathrm{rref}(B)$ and write $R = E_k \cdots E_1 A = F_\ell \cdots F_1 B$. For $E := E_k \cdots E_1$ and $F := F_\ell \cdots F_1$, the matrix $P := F^{-1}E$ is invertible by Proposition 4.1.2(ii) and Corollary 3.1.17(i), and $B = PA$.
(iii) $\Rightarrow$ *(i)* Let $B = PA$ with P invertible. Choose elementary matrices satisfying $E_k \cdots E_1 P = I_m$ by Lemma 4.1.5(ii).
Then $P = E_1^{-1} \cdots E_k^{-1}$, and each E_i^{-1} is elementary by Proposition 4.1.2(ii). Hence $A \sim_{\mathrm{row}} B$.

Structural Commentary: The implication cycle passes among canonical representatives, elementary products, and invertible left factors.
Key Insight: Row equivalence is exactly invertible left multiplication. □

Computation Method 4.1.7. *Inverse Computation by Gauss–Jordan Reduction*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, this procedure returns A^{-1} or certifies that A is not invertible.

(i) Form $[\,A \mid I_n\,]$ and perform each Gauss–Jordan row operation on the entire augmented matrix until the left block is $\mathrm{rref}(A)$.
(ii) If the result is $[\,I_n \mid B\,]$, choose $P \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ such that $[\,I_n \mid B\,] = P[\,A \mid I_n\,] = [\,PA \mid P\,]$ by Corollary 4.1.6. Then $P = B$, $BA = I_n$, and $AB = I_n$ by Theorem 3.4.2; return $B = A^{-1}$.
(iii) If $\mathrm{rref}(A) \neq I_n$, then A is not invertible by Lemma 4.1.5(i).

5 Change of Basis

5.1 Further Change of Basis and Coordinate Transformations

Theorem 5.1.1. *Change of Basis Formula*
Let $n \geq 1$, let $\dim(\mathcal{V}) = n$, let $T \in \mathcal{L}(\mathcal{V})$, and fix a basis β of $\mathcal{V}$. Set $A := [T]_{\beta \leftarrow \beta}$.

(i) If γ is a basis of $\mathcal{V}$ and $S = C_{\gamma \leftarrow \beta}$, then $[T]_{\gamma \leftarrow \gamma} = SAS^{-1}$.
(ii) If $S \in \mathrm{Mat}(n,n,\mathbb{F})^\times$, then some basis γ of $\mathcal{V}$ satisfies $[T]_{\gamma \leftarrow \gamma} = SAS^{-1}$.

Proof.

(i) Suppose γ is a basis and $S = C_{\gamma \leftarrow \beta}$. Then $S^{-1} = C_{\beta \leftarrow \gamma}$ by Theorem 3.5.2(i). Hence $[T]_{\gamma \leftarrow \gamma} = C_{\gamma \leftarrow \beta} A C_{\beta \leftarrow \gamma} = SAS^{-1}$ by Theorem 3.5.3, with β as both original bases and γ as both new bases.
(ii) Let $S \in \mathrm{Mat}(n,n,\mathbb{F})^\times$. Choose a basis γ satisfying $S = C_{\gamma \leftarrow \beta}$ by Theorem 3.5.2(ii). Then $[T]_{\gamma \leftarrow \gamma} = SAS^{-1}$ by part (i).

Structural Commentary: The proof specializes two-sided transport, then realizes an arbitrary invertible coordinate map.
Key Insight: Conjugation records one operator after a simultaneous coordinate change. □

6 Matrix Equivalence and Similarity

6.1 Matrix Equivalence and Rank Normal Forms

Throughout this subsection, $m, n \geq 1$, and all vector spaces and matrices are over the same fixed field $\mathbb{F}$.

Definition 6.1.1. *Matrix Equivalence*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, write $A \sim_{\mathrm{eq}} B$ iff $(\exists\, P \in \mathrm{Mat}(m,m,\mathbb{F})^\times)(\exists\, Q \in \mathrm{Mat}(n,n,\mathbb{F})^\times)\; B = PAQ$.

Proposition 6.1.2. *Matrix Equivalence and Elementary Operations*
Let $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:

(i) The relation $\sim_{\mathrm{eq}}$ is an equivalence relation on $\mathrm{Mat}(m,n,\mathbb{F})$.
(ii) $A \sim_{\mathrm{eq}} B$ iff B can be obtained from A by finitely many elementary row and column operations.

Proof.

(i) The relation is reflexive by Definition 6.1.1, with $A = I_m A I_n$.
If $B = PAQ$ with P, Q invertible, then $A = P^{-1}BQ^{-1}$, so symmetry holds by Definition 6.1.1.
If $B = P_1 A Q_1$ and $C = P_2 B Q_2$, then $C = (P_2 P_1) A (Q_1 Q_2)$, and both products are invertible by Corollary 3.1.17(i).
Hence transitivity holds by Definition 6.1.1.
(ii) ($\Rightarrow$) Suppose $A \sim_{\mathrm{eq}} B$. Choose $P \in \mathrm{Mat}(m,m,\mathbb{F})^\times$ and $Q \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ satisfying $B = PAQ$ by Definition 6.1.1.
The matrix Q^t is invertible by Corollary 3.1.17(ii). The matrices P and Q^t are products of elementary matrices by Lemma 4.1.5(ii) and Proposition 4.1.2(ii). Hence left multiplication by the factors of P performs elementary row operations, while right multiplication by the transposes of the factors of Q^t performs elementary column operations by Proposition 4.1.2(i),(iii).
Thus these operations transform A into $PAQ = B$.
(ii) ($\Leftarrow$) Suppose B is obtained from A by finitely many elementary row and column operations.
Each elementary row operation is left multiplication by an invertible elementary matrix by Proposition 4.1.2(i),(ii).
Each elementary column operation is right multiplication by the transpose of an invertible elementary matrix by Proposition 4.1.2(ii),(iii) and Corollary 3.1.17(ii). Associativity collects the left factors into P and the right factors into Q, so the operation sequence has the form $B = PAQ$ with P, Q invertible by Corollary 3.1.17(i). Hence $A \sim_{\mathrm{eq}} B$ by Definition 6.1.1.

Structural Commentary: Part (i) uses invertible-factor algebra; part (ii) translates elementary operations into factors.
Key Insight: Two-sided factors are exactly row and column operations. □

Theorem 6.1.3. *Equivalent Matrices Represent the Same Linear Map*
Let $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$, and $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$.
Then $A \sim_{\mathrm{eq}} B$ iff some $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ and bases β, β' of $\mathcal{V}$ and γ, γ' of $\mathcal{W}$ satisfy $A = [T]_{\gamma \leftarrow \beta}$ and $B = [T]_{\gamma' \leftarrow \beta'}$.

Proof.

($\Rightarrow$) Suppose $A \sim_{\mathrm{eq}} B$. Choose $P \in \mathrm{Mat}(m,m,\mathbb{F})^\times$ and $Q \in \mathrm{Mat}(n,n,\mathbb{F})^\times$ satisfying $B = PAQ$ by Definition 6.1.1.
Choose bases β, γ by Corollary 1.3.14, and choose $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ satisfying $[T]_{\gamma \leftarrow \beta} = A$ by Theorem 3.4.5(i).
Choose bases γ' and β' satisfying $P = C_{\gamma' \leftarrow \gamma}$ and $Q^{-1} = C_{\beta' \leftarrow \beta}$ by Theorem 3.5.2(ii).
Then $Q = C_{\beta \leftarrow \beta'}$ by Theorem 3.5.2(i), and $[T]_{\gamma' \leftarrow \beta'} = PAQ = B$ by Theorem 3.5.3.
($\Leftarrow$) Choose $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$ and bases β, β' of $\mathcal{V}$ and γ, γ' of $\mathcal{W}$ satisfying $A = [T]_{\gamma \leftarrow \beta}$ and $B = [T]_{\gamma' \leftarrow \beta'}$ by the hypothesis.
Then $B = C_{\gamma' \leftarrow \gamma} A C_{\beta \leftarrow \beta'}$ by Theorem 3.5.3, so $A \sim_{\mathrm{eq}} B$ by Definition 6.1.1.

Structural Commentary: One direction realizes invertible factors as basis changes; the converse reads coordinate transport.
Key Insight: Domain and codomain coordinates account for both factors. □

Definition 6.1.4. *Matrix Rank Normal Form*
For $0 \leq r \leq \min(m,n)$, define $J_r^{m,n} \in \mathrm{Mat}(m,n,\mathbb{F})$ as follows:

(i) $J_0^{m,n} := 0$.
(ii) If $r \geq 1$, then $J_r^{m,n} := \begin{pmatrix} I_r & 0_{r \times (n-r)} \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{pmatrix}$.

Theorem 6.1.5. *Rank Classifies Matrix Equivalence*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$:
(i) $A \sim_{\mathrm{eq}} J^{m,n}_{\mathrm{rank}(A)}$.
(ii) $A \sim_{\mathrm{eq}} B$ *iff* $\mathrm{rank}(A) = \mathrm{rank}(B)$.

Proof.
(*i*) Set $r := \mathrm{rank}(A)$ and $T_A(\mathbf{x}) := A\mathbf{x}$ for $\mathbf{x} \in \mathbb{F}^n$. Then $T_A \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ by Proposition 3.1.8.
Choose $\mathcal{M}_A \leq \mathbb{F}^n$ satisfying $\mathbb{F}^n = \ker(T_A) \oplus \mathcal{M}_A$ by Proposition 2.3.6.
The restriction $T_A|_{\mathcal{M}_A} : \mathcal{M}_A \to \mathrm{range}(T_A)$ is an isomorphism by Proposition 2.3.8(i). The equality $\mathrm{range}(T_A) = \mathrm{col}(A)$ holds by Proposition 3.1.9 and Definition 3.1.12; hence $\dim(\mathcal{M}_A) = r$ by Theorem 1.7.9 and Definition 3.1.12.
Choose a basis $\mathbf{u} := (\mathbf{u}_1, \ldots, \mathbf{u}_r)$ of $\mathcal{M}_A$ and a basis κ of $\ker(T_A)$ by Corollary 1.3.14, where $\mathbf{u}$ is empty if $r = 0$.
The list $(T_A\mathbf{u}_1, \ldots, T_A\mathbf{u}_r)$ is a basis of $\mathrm{range}(T_A)$ by Proposition 1.7.8; extend it to a basis γ_A of $\mathbb{F}^m$ by Theorem 1.3.13.
Equality $\mathbb{F}^n = \mathcal{M}_A \oplus \ker(T_A)$ holds by Proposition 2.1.8. Hence concatenation $\beta_A := (\mathbf{u}, \kappa)$ is basis of $\mathbb{F}^n$ by Theorem 2.3.4(ii).
The first r basis vectors map to the first r vectors of γ_A, while the remaining basis vectors lie in $\ker(T_A)$.
Thus $[T_A]_{\gamma_A \leftarrow \beta_A} = J^{m,n}_r$ by Definition 3.2.3 and Definition 6.1.4.
The standard-basis matrix of T_A is A by Definition 3.1.7 and Definition 3.2.3. Therefore $A \sim_{\mathrm{eq}} J^{m,n}_r$ by Theorem 6.1.3.
(*ii*) ($\Rightarrow$) If $A \sim_{\mathrm{eq}} B$, then $B = PAQ$ for invertible P, Q by Definition 6.1.1, and $\mathrm{rank}(B) = \mathrm{rank}(A)$ by Theorem 3.6.5(ii).
(*ii*) ($\Leftarrow$) If $\mathrm{rank}(A) = \mathrm{rank}(B) = r$, then $A \sim_{\mathrm{eq}} J^{m,n}_r \sim_{\mathrm{eq}} B$ and $A \sim_{\mathrm{eq}} B$ by part (i) and Proposition 6.1.2(i).
Structural Commentary: Complement bases construct the normal form; rank preservation proves classification.
Key Insight: Kernel and range dimensions determine the two-sided orbit. □

Corollary 6.1.6. *Rank Normal Form for Linear Maps*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$, and $\mathrm{rank}(T) = r$. Then some bases β of $\mathcal{V}$ and γ of $\mathcal{W}$ satisfy $[T]_{\gamma \leftarrow \beta} = J^{m,n}_r$.

Proof. Choose initial bases β_0, γ_0 by Corollary 1.3.14, and set $A := [T]_{\gamma_0 \leftarrow \beta_0}$.
Then $\{[\mathbf{y}]_{\gamma_0} : \mathbf{y} \in \mathrm{range}(T)\} = \{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathrm{col}(A)$ by Proposition 3.2.5, 3.2.2(iii), 3.1.9, and Definition 3.1.12.
The coordinate map is linear and injective by Proposition 3.2.2(i),(ii), so its restriction is an isomorphism from $\mathrm{range}(T)$ onto $\mathrm{col}(A)$.
Hence $\mathrm{rank}(A) = \mathrm{rank}(T) = r$ by Theorem 1.7.9 and the definitions of matrix and operator rank.
Thus $A \sim_{\mathrm{eq}} J^{m,n}_r$ by Theorem 6.1.5(i); choose invertible P, Q satisfying $J^{m,n}_r = PAQ$ by Definition 6.1.1.
Choose bases γ, β satisfying $P = C_{\gamma \leftarrow \gamma_0}$ and $Q^{-1} = C_{\beta \leftarrow \beta_0}$ by Theorem 3.5.2(ii).
Then $Q = C_{\beta_0 \leftarrow \beta}$ by Theorem 3.5.2(i), and $[T]_{\gamma \leftarrow \beta} = J^{m,n}_r$ by Theorem 3.5.3.
Structural Commentary: Matrix normal form supplies factors that are realized as coordinate changes.
Key Insight: Rank fixes the canonical coordinate matrix. □

Corollary 6.1.7. *Linear Maps Classified by Rank*
Let $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$. For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $S \sim_{\mathrm{bc}} T$ iff some bases β_S, β_T of $\mathcal{V}$ and γ_S, γ_T of $\mathcal{W}$ satisfy $[S]_{\gamma_S \leftarrow \beta_S} = [T]_{\gamma_T \leftarrow \beta_T}$. Then $S \sim_{\mathrm{bc}} T$ iff $\mathrm{rank}(S) = \mathrm{rank}(T)$.

Proof.
($\Rightarrow$) Suppose $S \sim_{\mathrm{bc}} T$. Choose bases β_S, β_T of $\mathcal{V}$ and γ_S, γ_T of $\mathcal{W}$ satisfying $[S]_{\gamma_S \leftarrow \beta_S} = [T]_{\gamma_T \leftarrow \beta_T}$ by the definition of $\sim_{\mathrm{bc}}$, and set $A := [S]_{\gamma_S \leftarrow \beta_S}$. For $R \in \{S, T\}$, interpret β_R and γ_R accordingly.
Then $\{[\mathbf{y}]_{\gamma_R} : \mathbf{y} \in \mathrm{range}(R)\} = \{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathrm{col}(A)$ by Proposition 3.2.5, 3.2.2(iii), 3.1.9, and Definition 3.1.12.
The restricted coordinate map is an isomorphism by Proposition 3.2.2(i),(ii).
Thus $(\forall R \in \{S, T\})$ $\mathrm{rank}(R) = \dim(\mathrm{col}(A))$ by Theorem 1.7.9 and Definition 1.7.2, so $\mathrm{rank}(S) = \mathrm{rank}(T)$.
($\Leftarrow$) If $\mathrm{rank}(S) = \mathrm{rank}(T) = r$, choose bases satisfying $[S]_{\gamma_S \leftarrow \beta_S} = J^{m,n}_r = [T]_{\gamma_T \leftarrow \beta_T}$ by Corollary 6.1.6.
Structural Commentary: Both directions compare operator ranks with the shared rank normal form.
Key Insight: Equal ranks select the same canonical matrix. □

6.2 Similarity and Operator Coordinates

Throughout this subsection, $m, n \geq 1$, and all vector spaces and matrices are over the same fixed field $\mathbb{F}$.

Definition 6.2.1. *Similarity*
Matrices $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are similar iff $(\exists\, P \in \mathrm{Mat}(n,n,\mathbb{F})^{\times})\ B = PAP^{-1}$.

Proposition 6.2.2. *Similarity Is an Equivalence Relation*
Similarity is an equivalence relation on $\mathrm{Mat}(n,n,\mathbb{F})$.

Proof. Let $A, B, C \in \mathrm{Mat}(n,n,\mathbb{F})$. Similarity is reflexive by Definition 6.2.1, with $P = I_n$.
If $B = PAP^{-1}$ for invertible P, then $A = P^{-1}BP$, so symmetry holds by Definition 6.2.1. If $B = PAP^{-1}$ and $C = QBQ^{-1}$ for invertible P, Q, then $C = (QP)A(QP)^{-1}$ and QP is invertible by Corollary 3.1.17(i), so transitivity holds by Definition 6.2.1.
Structural Commentary: Identity, inverse, and product witnesses verify reflexivity, symmetry, and transitivity.
Key Insight: Conjugation is an orbit relation under invertible matrices. □

Remark 6.2.3. *Similarity Convention*
The forms PAP^{-1} and $M^{-1}AM$ are the same convention with $P = M^{-1}$. In Theorem 5.1.1, this means choosing opposite coordinate-change directions.

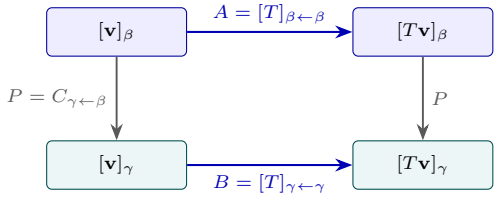

Figure 16: One operator in two bases: $PA = BP$, so $B = PAP^{-1}$.

In Figure 16, $A = [T]_{\beta\leftarrow\beta}$, $B = [T]_{\gamma\leftarrow\gamma}$, and $P = C_{\gamma\leftarrow\beta}$, so the coordinate square commutes as $PA = BP$. Theorem 5.1.1 owns the exact conjugation identity.

Corollary 6.2.4. *Similarity as One Operator in Two Bases*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) A and B are similar.
(ii) Some n-dimensional vector space $\mathcal{V}$, bases β, γ of $\mathcal{V}$, and $T \in \mathcal{L}(\mathcal{V})$ satisfy $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Suppose A and B are similar. Choose $P \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B = PAP^{-1}$ by Definition 6.2.1.
Let $\mathcal{V} := \mathbb{F}^n$, let β be its standard basis, and set $T_A(\mathbf{x}) := A\mathbf{x}$, which is linear by Proposition 3.1.8.
Then $[T_A]_{\beta\leftarrow\beta} = A$ by Definition 3.1.7 and Definition 3.2.3. Choose a basis γ satisfying $P = C_{\gamma\leftarrow\beta}$ by Theorem 3.5.2(ii).
Hence $[T_A]_{\gamma\leftarrow\gamma} = PAP^{-1} = B$ by Theorem 5.1.1(i).

(*ii*) $\Rightarrow$ (*i*) Suppose $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$. Set $P := C_{\gamma\leftarrow\beta}$. Then P is invertible with $P^{-1} = C_{\beta\leftarrow\gamma}$ by Theorem 3.5.2(i), and $B = PAP^{-1}$ by Theorem 5.1.1(i). Thus A and B are similar by Definition 6.2.1.

Structural Commentary: One direction realizes conjugation as basis change; the converse reads the transport formula.
Key Insight: Similarity is one operator in two coordinate systems. □

Perspective 6.2.5. *Similarity as Same Operator, New Coordinates*
For matrices, similarity means that they represent the same operator in two bases, not merely that their entries are related.
This is the structural content behind change of basis.

Definition 6.2.6. *Similarity of Operators*
For $T \in \mathcal{L}(\mathcal{V})$ and $S \in \mathcal{L}(\mathcal{W})$, $T \sim S$ iff $(\exists\, \Phi : \mathcal{V} \xrightarrow{\sim} \mathcal{W})\, S = \Phi T \Phi^{-1}$.
When $\mathcal{V} = \mathcal{W}$, the witnesses lie in $\mathrm{Aut}(\mathcal{V})$, so similarity is conjugacy in $\mathcal{L}(\mathcal{V})$.

Proposition 6.2.7. *Operator Similarity and Common Matrix Representatives*
Let $\dim \mathcal{V} = \dim \mathcal{W} = n$, $T \in \mathcal{L}(\mathcal{V})$, and $S \in \mathcal{L}(\mathcal{W})$.
Then $T \sim S$ iff some bases β of $\mathcal{V}$ and γ of $\mathcal{W}$ satisfy $[T]_{\beta\leftarrow\beta} = [S]_{\gamma\leftarrow\gamma}$.

Proof.

($\Rightarrow$) Suppose $T \sim S$. Choose an isomorphism $\Phi : \mathcal{V} \to \mathcal{W}$ satisfying $S = \Phi T \Phi^{-1}$ by Definition 6.2.6.
Choose a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$ by Corollary 1.3.14. Set $\gamma := (\Phi\mathbf{v}_1, \ldots, \Phi\mathbf{v}_n)$, which is a basis of $\mathcal{W}$ by Proposition 1.7.8.
The identities $(\forall\, \mathbf{v} \in \mathcal{V})\, [\Phi\mathbf{v}]_\gamma = [\mathbf{v}]_\beta$ and $S\Phi = \Phi T$ hold by Definition 3.2.1 and Proposition 1.5.7(ii),(iii), respectively.
Consequently, each $j \in \{1, \ldots, n\}$ satisfies $[S(\Phi\mathbf{v}_j)]_\gamma = [T\mathbf{v}_j]_\beta$. These are the respective jth columns of $[S]_{\gamma\leftarrow\gamma}$ and $[T]_{\beta\leftarrow\beta}$ by Definition 3.2.3; hence the matrices are equal by Definition 3.1.1(i).

($\Leftarrow$) Suppose bases $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ satisfy $[T]_{\beta\leftarrow\beta} = [S]_{\gamma\leftarrow\gamma}$.
Choose $\Phi \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ satisfying $(\forall\, j \in \{1, \ldots, n\})\, \Phi\mathbf{v}_j = \mathbf{w}_j$ by Proposition 1.5.8. Its range is $\mathrm{span}(\gamma) = \mathcal{W}$ by Theorem 1.7.1(i), so Φ is surjective and hence injective by Corollary 1.7.6(ii). Therefore Φ is an isomorphism by Proposition 1.6.7.
The identity $(\forall\, \mathbf{v} \in \mathcal{V})\, [\Phi\mathbf{v}]_\gamma = [\mathbf{v}]_\beta$ holds by Definition 3.2.1.
Thus $(\forall\, j \in \{1, \ldots, n\})\, [\Phi T\mathbf{v}_j]_\gamma = [S\Phi\mathbf{v}_j]_\gamma$ by Definition 3.2.3, and $(\forall\, j \in \{1, \ldots, n\})\, \Phi T\mathbf{v}_j = S\Phi\mathbf{v}_j$ by Proposition 3.2.2(ii).
The compositions are linear by Proposition 1.5.7(i), so $\Phi T = S\Phi$ by Proposition 1.5.8.
Thus $S = \Phi T \Phi^{-1}$ by Proposition 1.5.7(ii),(iii), so $T \sim S$ by Definition 6.2.6.

Structural Commentary: A conjugating isomorphism transports a basis; a shared matrix reconstructs the isomorphism.
Key Insight: Matching basis actions force operator conjugacy. □

Theorem 6.2.8. *Similarity and Scalar Shifts*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If A and B are similar, then $(\forall\, \lambda \in \mathbb{F})\, [A - \lambda I_n$ is similar to $B - \lambda I_n]$.
(ii) If $(\exists\, \lambda \in \mathbb{F})\, [A - \lambda I_n$ is similar to $B - \lambda I_n]$, then A and B are similar.

Proof.

(i) Choose $P \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B = PAP^{-1}$ by the hypothesis and Definition 6.2.1.
For $\lambda \in \mathbb{F}$, $B - \lambda I_n = PAP^{-1} - \lambda PP^{-1} = P(A - \lambda I_n)P^{-1}$, so the shifted matrices are similar by Definition 6.2.1.

(ii) Choose $\lambda \in \mathbb{F}$ and $P \in \mathrm{Mat}(n, n, \mathbb{F})^\times$ satisfying $B - \lambda I_n = P(A - \lambda I_n)P^{-1}$ by the hypothesis and Definition 6.2.1.
After adding $\lambda I_n = \lambda PP^{-1}$, the equality becomes $B = PAP^{-1}$, so A and B are similar by Definition 6.2.1.

Structural Commentary: Both directions add or remove the central term λI_n inside a conjugation.
Key Insight: Scalar identity shifts commute with every basis change. □

Proposition 6.2.9. *Upper Block-Triangular Similarity Shift*
Let $B \in \mathrm{Mat}(m, m, \mathbb{F})$, $D \in \mathrm{Mat}(n, n, \mathbb{F})$, $C, X \in \mathrm{Mat}(m, n, \mathbb{F})$. Set $M := \left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$, $N := \left(\begin{smallmatrix} B & C+XD-BX \\ 0 & D \end{smallmatrix}\right)$. Then M and N are similar.

Proof. Set $U := \left(\begin{smallmatrix} I_m & X \\ 0 & I_n \end{smallmatrix}\right)$. By block multiplication, $U^{-1} = \left(\begin{smallmatrix} I_m & -X \\ 0 & I_n \end{smallmatrix}\right)$ and $UMU^{-1} = \left(\begin{smallmatrix} B & -BX+C+XD \\ 0 & D \end{smallmatrix}\right) = N$.
Thus M and N are similar by Definition 6.2.1.
Structural Commentary: Block multiplication verifies the inverse and computes the conjugated coupling block.
Key Insight: Unit upper conjugation preserves diagonal blocks and shears the coupling. □

Part IV
Geometric and Spectral Linear Algebra

1 Inner Product Geometry

1.1 Inner Products and Norms

Throughout, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. This subsection develops the norm and metric geometry determined by an inner product.

Definition 1.1.1. *Preliminary Terminology*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$.

(i) For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the *dot product* is $\mathbf{x} \cdot \mathbf{y} := x_1 y_1 + \cdots + x_n y_n$.
(ii) For $z = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$, its *complex conjugate* and *modulus* are $\overline{z} := a - bi$ and $|z| := \sqrt{a^2 + b^2}$.
(iii) A map $f : \mathcal{V} \to \mathbb{F}$ is *conjugate linear* if $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})(\forall\, \lambda \in \mathbb{F})\ \big(f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \wedge f(\lambda \mathbf{v}) = \overline{\lambda} f(\mathbf{v})\big)$.
(iv) A map $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is *sesquilinear* if it is linear in its first slot and conjugate linear in its second slot.

Motivation 1.1.2. *From Dot Products to Inner Products*
For $\mathbf{x} \in \mathbb{R}^n$, Euclidean squared length is $\mathbf{x} \cdot \mathbf{x} = \sum_{j=1}^{n} x_j^2$.

For $z \in \mathbb{C}$, the scalar z^2 need not be real, whereas $z\overline{z} = |z|^2 \geq 0$.
Conjugating one factor therefore preserves nonnegative squared length and motivates the following axioms.

Definition 1.1.3. *Inner Product*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. An *inner product* on $\mathcal{V}$ is a map $\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ satisfying, for $(\mathbf{u}, \mathbf{v}, \mathbf{w}, \lambda) \in \mathcal{V}^3 \times \mathbb{F}$:

(i) *positivity*: $\langle \mathbf{v}, \mathbf{v} \rangle \in [0, \infty)$;
(ii) *definiteness*: $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$;
(iii) *additivity in the first slot*: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;
(iv) *homogeneity in the first slot*: $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$;
(v) *conjugate symmetry*: $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.

Definition 1.1.4. *Inner-Product Space*
An *inner-product space* is a vector space $\mathcal{V}$ together with an inner product on $\mathcal{V}$.

Henceforth, unless stated otherwise, $\mathcal{V}$ denotes an inner-product space over $\mathbb{F}$.

Definition 1.1.5. *Norm Induced by an Inner Product*
For $\mathbf{v} \in \mathcal{V}$, the *norm* of $\mathbf{v}$ is $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

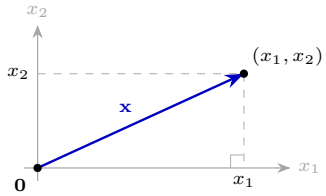

Figure 17: The norm is vector length, with $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$.

Figure 17 draws $\mathbf{x} \in \mathbb{R}^2$ with coordinate legs x_1, x_2, so its length realizes $\|\mathbf{x}\|^2 = x_1^2 + x_2^2$.

Example 1.1.6. *Standard Inner Product on $\mathbb{F}^n$*
For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, set $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{j=1}^{n} x_j \overline{y_j}$. Sesquilinearity and conjugate symmetry hold componentwise, while $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{j=1}^{n} |x_j|^2$ verifies positivity and definiteness. For $\mathbb{F} = \mathbb{R}$, this is the dot product.

Example 1.1.7. *Function and Matrix Inner Products*

(i) If $a < b$, then $\langle f, g \rangle := \int_a^b f(t)\overline{g(t)}\, dt$ is an inner product on $C_{\mathbb{F}}[a, b]$.
Positivity and definiteness follow from $\langle f, f \rangle = \int_a^b |f(t)|^2\, dt \geq 0$, with equality only for $f = 0$ by continuity.

(ii) On $\mathrm{Mat}(m, n, \mathbb{F})$, the *Frobenius inner product* is $\langle A, B \rangle_F := \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} \overline{b_{ij}}$.
Sesquilinearity holds entrywise, and $\langle A, A \rangle_F = \sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2$ is nonnegative and vanishes exactly when $A = 0$.

Example 1.1.8. *Square-Summable Sequences*
Set $\ell^2(\mathbb{F}) := \left\{ (x_j)_{j \geq 1} : \sum_{j=1}^{\infty} |x_j|^2 < \infty \right\}$ and $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{j=1}^{\infty} x_j \overline{y_j}$. The bound $|x_j \overline{y_j}| \leq \frac{1}{2}(|x_j|^2 + |y_j|^2)$ ensures absolute convergence.
Termwise algebra verifies sesquilinearity and conjugate symmetry, while $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{j=1}^{\infty} |x_j|^2$ verifies positivity and definiteness.
Thus $\ell^2(\mathbb{F})$ is an inner-product space containing $\mathbb{F}^{(\mathbb{N})}$.

Proposition 1.1.9. *Basic Inner-Product Identities*
For $(\mathbf{u}, \mathbf{v}, \mathbf{w}, \lambda) \in \mathcal{V}^3 \times \mathbb{F}$:

(i) for fixed $\mathbf{u}$, the map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle$ is linear;
(ii) $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ and $\langle \mathbf{u}, \mathbf{0} \rangle = 0$;

(iii) $\langle \mathbf{u}, \mathbf{v}+\mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$;
(iv) $\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \overline{\lambda}\,\langle \mathbf{u}, \mathbf{v}\rangle$.

Proof.

(i) The map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u}\rangle$ is linear by first-slot additivity and homogeneity.
(ii) $\langle \mathbf{0}, \mathbf{u}\rangle = 0$ by (i), and $\langle \mathbf{u}, \mathbf{0}\rangle = \overline{\langle \mathbf{0}, \mathbf{u}\rangle} = 0$ by conjugate symmetry.
(iii) $\langle \mathbf{u}, \mathbf{v}+\mathbf{w}\rangle = \overline{\langle \mathbf{v}+\mathbf{w}, \mathbf{u}\rangle} = \overline{\langle \mathbf{v}, \mathbf{u}\rangle} + \overline{\langle \mathbf{w}, \mathbf{u}\rangle} = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$ by conjugate symmetry and additivity.
(iv) $\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \overline{\langle \lambda\mathbf{v}, \mathbf{u}\rangle} = \overline{\lambda}\,\overline{\langle \mathbf{v}, \mathbf{u}\rangle} = \overline{\lambda}\,\langle \mathbf{u}, \mathbf{v}\rangle$ by conjugate symmetry and homogeneity.

Structural Commentary: Conjugate symmetry transfers the first-slot axioms to the second slot.
Key Insight: One linear slot determines conjugate linearity in the other. □

Lemma 1.1.10. *Inner Products Separate Vectors*
For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$:
(i) $\big((\forall\, \mathbf{x} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{u}\rangle = \langle \mathbf{x}, \mathbf{v}\rangle\big) \Longrightarrow \mathbf{u} = \mathbf{v}$;
(ii) $\big((\forall\, \mathbf{x} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{w}\rangle = 0\big) \Longleftrightarrow \mathbf{w} = \mathbf{0}$.

Proof.

(i) With $\mathbf{x} = \mathbf{u} - \mathbf{v}$, $\|\mathbf{u}-\mathbf{v}\|^2 = \langle \mathbf{u}-\mathbf{v}, \mathbf{u}\rangle - \langle \mathbf{u}-\mathbf{v}, \mathbf{v}\rangle = 0$ by Proposition 1.1.9. The equality $\mathbf{u} = \mathbf{v}$ holds by definiteness.
(ii) $(\Rightarrow)$ $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w}\rangle = 0$ by the hypothesis with $\mathbf{x} = \mathbf{w}$, and $\mathbf{w} = \mathbf{0}$ by definiteness.
(ii) $(\Leftarrow)$ If $\mathbf{w} = \mathbf{0}$, then $(\forall\, \mathbf{x} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{w}\rangle = 0$ by Proposition 1.1.9(ii).

Structural Commentary: Each claim tests a difference vector against a strategically chosen argument.
Key Insight: Testing a vector against itself detects whether it vanishes. □

Example 1.1.11. *Norms from Standard Inner Products*

(i) On $\mathbb{F}^n$, the standard inner product induces the *Euclidean norm* $\|(x_1, \ldots, x_n)\|_2 = \sqrt{\sum_{j=1}^{n} |x_j|^2}$.

(ii) On $C_{\mathbb{F}}[a,b]$ with $a < b$, the integral inner product induces the *L^2 norm* $\|f\|_2 = \sqrt{\int_a^b |f(t)|^2\, dt}$.

(iii) On $\mathrm{Mat}(m,n,\mathbb{F})$, the Frobenius inner product induces $\|A\|_F = \sqrt{\langle A, A\rangle_F} = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2}$ for $A = [a_{ij}]$.

This is the *Frobenius*, *Schur*, or *Hilbert–Schmidt norm*.

Proposition 1.1.12. *Basic Properties of the Derived Norm*
For $(\mathbf{v}, \lambda) \in \mathcal{V} \times \mathbb{F}$:
(i) $\|\mathbf{v}\|$ *is real and nonnegative;*
(ii) $\|\mathbf{v}\| = 0 \Longleftrightarrow \mathbf{v} = \mathbf{0}$;
(iii) $\|\lambda\mathbf{v}\| = |\lambda|\,\|\mathbf{v}\|$.

Proof.

(i) The scalar $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$ is real and nonnegative by positivity.
(ii) $(\Rightarrow)$ If $\|\mathbf{v}\| = 0$, then $\langle \mathbf{v}, \mathbf{v}\rangle = 0$, and $\mathbf{v} = \mathbf{0}$ by definiteness.
(ii) $(\Leftarrow)$ If $\mathbf{v} = \mathbf{0}$, then $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ by definiteness and $\|\mathbf{v}\| = 0$ by the norm definition.
(iii) $\|\lambda\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \lambda\mathbf{v}\rangle = \lambda\overline{\lambda}\,\langle \mathbf{v}, \mathbf{v}\rangle = |\lambda|^2\,\|\mathbf{v}\|^2$ by sesquilinearity. The equality $\|\lambda\mathbf{v}\| = |\lambda|\,\|\mathbf{v}\|$ holds by nonnegative square roots.

Structural Commentary: Positivity controls the square root, while sesquilinearity controls scalar rescaling.
Key Insight: Squared length is easier to manipulate than length itself. □

Theorem 1.1.13. *Cauchy–Schwarz Inequality*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:
(i) $\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$;
(ii) equality in (i) holds if and only if $\mathbf{u}, \mathbf{v}$ *are linearly dependent.*

Proof.

(i) If $\mathbf{v} = \mathbf{0}$, the inequality is an equality. Suppose $\mathbf{v} \neq \mathbf{0}$ and set $\lambda := \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{v}\|^2}$.

Then $0 \le \|\mathbf{u} - \lambda\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{\|\mathbf{v}\|^2}$ by positivity, sesquilinearity, and conjugate symmetry.

The inequality $\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$ follows by multiplication by $\|\mathbf{v}\|^2$ and nonnegative square roots.
(ii) $(\Rightarrow)$ If equality holds and $\mathbf{v} = \mathbf{0}$, then $\mathbf{u}, \mathbf{v}$ are linearly dependent.
If $\mathbf{v} \neq \mathbf{0}$, then $\|\mathbf{u} - \lambda\mathbf{v}\| = 0$ by equality in (i) and the preceding identity, and $\mathbf{u} = \lambda\mathbf{v}$ by Proposition 1.1.12(ii).
(ii) $(\Leftarrow)$ If $\mathbf{v} = \mathbf{0}$, equality holds. Else linear dependence has form $\mathbf{u} = \lambda\mathbf{v}$; $\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| = |\lambda|\,\|\mathbf{v}\|^2 = \|\mathbf{u}\|\,\|\mathbf{v}\|$ by Proposition 1.1.12(iii).

Structural Commentary: A least-residual scalar supplies the bound and characterizes equality.
Key Insight: Positivity controls the mixed term through optimal scalar comparison. □

Theorem 1.1.14. *Triangle Inequality*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:
(i) $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$;
(ii) equality in (i) holds if and only if one of $\mathbf{u}, \mathbf{v}$ *is a nonnegative real multiple of the other.*

Proof.

(i) $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\,\mathrm{Re}\,\langle \mathbf{u}, \mathbf{v}\rangle$ by sesquilinearity and conjugate symmetry.
By Theorem 1.1.13(i), this is at most $\big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2$. The inequality $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ follows by nonnegative square roots.

(*ii*) (⇒) If either vector is $\mathbf{0}$, it is a nonnegative multiple of the other.
Otherwise equality requires $\operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle = |\langle \mathbf{u}, \mathbf{v}\rangle|$ and equality in Cauchy–Schwarz.
Choose $a \in \mathbb{F}$ with $\mathbf{u} = a\mathbf{v}$ by Theorem 1.1.13(ii). Then $a \in [0, \infty)$ by $a\|\mathbf{v}\|^2 = \langle \mathbf{u}, \mathbf{v}\rangle \in [0, \infty)$.
(*ii*) (⇐) If $\mathbf{u} = a\mathbf{v}$ for $a \in [0, \infty)$, then $\|\mathbf{u} + \mathbf{v}\| = (a + 1)\|\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$. Interchanging $\mathbf{u}, \mathbf{v}$ in this calculation covers $\mathbf{v} = a\mathbf{u}$.

Structural Commentary: Cauchy–Schwarz bounds the cross term, and its equality case fixes the direction.
Key Insight: Equality requires aligned vectors with nonnegative relative scale. □

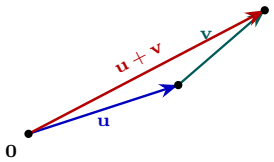

Figure 18: The direct displacement $\mathbf{u} + \mathbf{v}$ is no longer than the two-step path with legs $\mathbf{u}$ and $\mathbf{v}$.

Figure 18 identifies the three vectors compared by the triangle inequality; it does not supply the proof.

Proposition 1.1.15. *Reverse Triangle Inequality*
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ |\|\mathbf{u}\| - \|\mathbf{v}\|| \le \|\mathbf{u} - \mathbf{v}\|$.

Proof. $\|\mathbf{u}\| - \|\mathbf{v}\| \le \|\mathbf{u} - \mathbf{v}\|$ by Theorem 1.1.14(i) applied to $\mathbf{u} = (\mathbf{u} - \mathbf{v}) + \mathbf{v}$.
Likewise, $\|\mathbf{v}\| - \|\mathbf{u}\| \le \|\mathbf{u} - \mathbf{v}\|$ by the same result with $\mathbf{u}, \mathbf{v}$ interchanged.
The two bounds are equivalent to the asserted absolute-value inequality.
Structural Commentary: Two triangle-inequality bounds control both possible signs of the norm difference.
Key Insight: Distance bounds the change in length in either direction. □

Definition 1.1.16. *Norm and Normed Vector Space*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. A *norm* on $\mathcal{V}$ is a map $\|\cdot\| : \mathcal{V} \to [0, \infty)$ satisfying, for $(\mathbf{u}, \mathbf{v}, \lambda) \in \mathcal{V}^2 \times \mathbb{F}$:

(i) $\|\mathbf{u}\| \ge 0$;
(ii) $\|\mathbf{u}\| = 0 \iff \mathbf{u} = \mathbf{0}$;
(iii) $\|\lambda\mathbf{u}\| = |\lambda|\,\|\mathbf{u}\|$;
(iv) $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.

A *normed vector space* is a vector space equipped with a norm. Its *unit ball* is $\{\mathbf{v} \in \mathcal{V} : \|\mathbf{v}\| \le 1\}$.

Definition 1.1.17. *Distance Induced by an Inner Product*
The *distance induced by the inner product* is $d : \mathcal{V} \times \mathcal{V} \to [0, \infty)$ defined by $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$.

Proposition 1.1.18. *Inner-Product Distance Is a Metric*
For $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$ and $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$:

(i) $d(\mathbf{u}, \mathbf{v}) \ge 0$;
(ii) $d(\mathbf{u}, \mathbf{v}) = 0 \iff \mathbf{u} = \mathbf{v}$;
(iii) $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$;
(iv) $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$.

Proof.

(*i*) $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| \ge 0$ by Proposition 1.1.12(i).
(*ii*) (⇒) If $d(\mathbf{u}, \mathbf{v}) = 0$, then $\mathbf{u} - \mathbf{v} = \mathbf{0}$ by Proposition 1.1.12(ii); equivalently, $\mathbf{u} = \mathbf{v}$.
(*ii*) (⇐) If $\mathbf{u} = \mathbf{v}$, then $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{0}\| = 0$ by Proposition 1.1.12(ii).
(*iii*) $d(\mathbf{u}, \mathbf{v}) = \|-(\mathbf{v} - \mathbf{u})\| = \|\mathbf{v} - \mathbf{u}\| = d(\mathbf{v}, \mathbf{u})$ by Proposition 1.1.12(iii) with $\lambda = -1$.
(*iv*) Since $\mathbf{u} - \mathbf{v} = (\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})$, $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ by Theorem 1.1.14(i).

Structural Commentary: Each metric axiom reduces to the corresponding norm property on a difference vector.
Key Insight: Distances are norms of displacements. □

Perspective 1.1.19. *Metric Viewpoint*
Because $d(\mathbf{v}_n, \mathbf{v}) = \|\mathbf{v}_n - \mathbf{v}\|$, metric convergence is exactly norm convergence.
Closedness, continuity, and completeness require topological results developed later; no such conclusion is used as authority here.

Theorem 1.1.20. *Parallelogram Identity*
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\big(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\big)$.

Proof. For $\varepsilon \in \{1, -1\}$, $\|\mathbf{u} + \varepsilon\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\varepsilon \operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle$ by sesquilinearity and conjugate symmetry.
The asserted identity is the sum of the cases $\varepsilon = 1, -1$.
Structural Commentary: Paired expansions cancel the cross terms while preserving both squared norms.
Key Insight: Opposite diagonals erase the mixed inner-product contribution. □

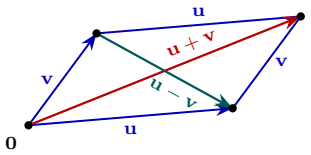

Figure 19: A parallelogram with side vectors $\mathbf{u}, \mathbf{v}$ and diagonal vectors $\mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v}$.

Figure 19 identifies the sides and diagonals whose squared lengths occur in the parallelogram identity.

Proposition 1.1.21. *Coordinate Norms on $\mathbb{F}^n$*
For $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$, each of the following defines a norm on $\mathbb{F}^n$:

(i) $\|\mathbf{x}\|_1 := \sum_{j=1}^{n} |x_j|$;

(ii) $\|\mathbf{x}\|_2 := \left(\sum_{j=1}^{n} |x_j|^2 \right)^{1/2}$;

(iii) $\|\mathbf{x}\|_\infty := \max_{1 \le j \le n} |x_j|$.

Proof.

(i) Nonnegativity, definiteness, and absolute homogeneity of $\|\cdot\|_1$ hold by the corresponding scalar properties.
Its triangle inequality is $\|\mathbf{x}+\mathbf{y}\|_1 = \sum_{j=1}^{n} |x_j + y_j| \le \sum_{j=1}^{n} (|x_j| + |y_j|) = \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$ by the scalar triangle inequality.

(ii) Standard inner product induces $\|\cdot\|_2$, first 3 norm axioms hold (Proposition 1.1.12), triangle inequality hold (Theorem 1.1.14(i)).

(iii) Nonnegativity and absolute homogeneity of $\|\cdot\|_\infty$ hold by modulus, and $\|\mathbf{x}\|_\infty = 0 \iff (\forall\, j \in \{1, \dots, n\})\, x_j = 0 \iff \mathbf{x} = \mathbf{0}$.
Choose $k \in \{1, \dots, n\}$ with $\|\mathbf{x}+\mathbf{y}\|_\infty = |x_k + y_k|$.
Then $\|\mathbf{x}+\mathbf{y}\|_\infty \le |x_k| + |y_k| \le \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$ by the scalar triangle inequality and the definition of the maximum.

Structural Commentary: Scalar modulus handles the ℓ_1 and ℓ_∞ cases; inner-product geometry handles the ℓ_2 case.
Key Insight: Distinct coordinate aggregations satisfy the same norm axioms. □

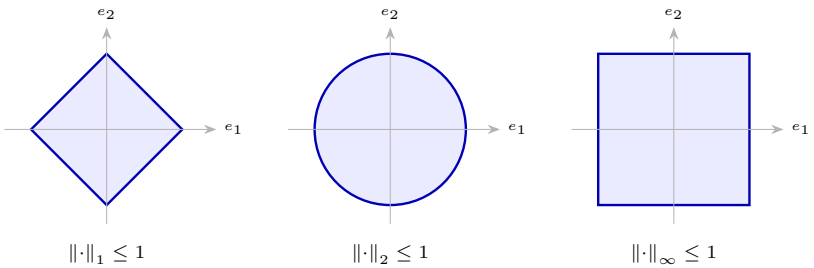

Figure 20: The ℓ_1, ℓ_2, and ℓ_∞ unit balls in $\mathbb{R}^2$ are respectively a diamond, a disk, and a square.

Figure 20 contrasts the three unit-ball shapes without serving as proof of the norm axioms.

Corollary 1.1.22. *ℓ_1 and ℓ_∞ Are Not Inner-Product Norms*
If $n \ge 2$, then the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $\mathbb{F}^n$ are not induced by any inner product.

Proof. For the first two coordinate vectors $\mathbf{e}_1, \mathbf{e}_2$, direct evaluation makes the two sides of the parallelogram identity 8 and 4 under $\|\cdot\|_1$, and 2 and 4 under $\|\cdot\|_\infty$. Neither norm is induced by an inner product by Theorem 1.1.20.
Structural Commentary: Two coordinate tests contradict the necessary parallelogram identity.
Key Insight: A failed parallelogram identity excludes every inducing inner product. □

Theorem 1.1.23. *Polarization Identities*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:

(i) If $\mathbb{F} = \mathbb{R}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 \right)$.

(ii) If $\mathbb{F} = \mathbb{C}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 + i\,\|\mathbf{u}+i\mathbf{v}\|^2 - i\,\|\mathbf{u}-i\mathbf{v}\|^2 \right)$.

Proof. Let $z := \langle \mathbf{u}, \mathbf{v} \rangle$.

(i) If $\mathbb{F} = \mathbb{R}$, then $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4 \operatorname{Re} z = 4z$ by sesquilinearity, which is equivalent to (i).

(ii) If $\mathbb{F} = \mathbb{C}$, then $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4 \operatorname{Re} z$ and $\|\mathbf{u}+i\mathbf{v}\|^2 - \|\mathbf{u}-i\mathbf{v}\|^2 = 4 \operatorname{Im} z$ by sesquilinearity.
Substitution into $z = \operatorname{Re} z + i \operatorname{Im} z$ is exactly (ii).

Structural Commentary: Norm differences recover the real part directly and the imaginary part after rotation.
Key Insight: The norm determines both components of every complex inner product. □

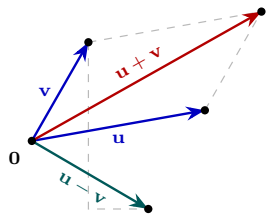

Figure 21: The sum and difference vectors isolate $4 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle$ through their squared-length difference.

Figure 21 illustrates the real-part identity; the complex identity also compares $\mathbf{u} \pm i\mathbf{v}$.

Proposition 1.1.24. *Operator Zero Tests from Inner Products*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) $\big((\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{w} \rangle = 0 \big) \Longrightarrow T = 0$.

(ii) *If* $\mathbb{F} = \mathbb{C}$, *then* $\big((\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = 0\big) \Longrightarrow T = 0$.
(iii) *If* $\mathbb{F} = \mathbb{R}$, *the implication in (ii) can fail.*

Proof.

(i) $(\forall\, \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\|^2 = \langle T\mathbf{v}, T\mathbf{v}\rangle = 0$ by the hypothesis with $\mathbf{w} = T\mathbf{v}$; $T = 0$ by definiteness.
(ii) Set $Q(\mathbf{x}) := \langle T\mathbf{x}, \mathbf{x}\rangle$. For fixed $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, set $a := \langle T\mathbf{x}, \mathbf{y}\rangle$ and $b := \langle T\mathbf{y}, \mathbf{x}\rangle$.
For $r \in \mathbb{C}$, $0 = Q(r\mathbf{x} + \mathbf{y}) = ra + \overline{r}b$ by the hypothesis and sesquilinearity.
The choices $r = 1, i$ establish $a + b = a - b = 0$, and $a = b = 0$ by these equations.
The identity $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y}\rangle = 0$ holds by the arbitrary choice of $\mathbf{x}, \mathbf{y}$, and $T = 0$ by (i).
(iii) On standard $\mathbb{R}^2$, define $T(x_1, x_2) := (-x_2, x_1)$. Then $(\forall\, \mathbf{v} \in \mathbb{R}^2)\ \langle T\mathbf{v}, \mathbf{v}\rangle = -v_2v_1 + v_1v_2 = 0$, while $T(1, 0) = (0, 1) \neq \mathbf{0}$.

Structural Commentary: Self-testing handles (i), complex scalar probes handle (ii), and rotation refutes the real analogue.
Key Insight: Complex phases recover mixed terms that real diagonal tests miss. □

Principle 1.1.25. *Geometry from Algebra*
For inner-product norms, expanding $\|\mathbf{u} \pm \mathbf{v}\|^2$ exposes the cross terms that control angles, inequalities, and polarization.
This heuristic does not apply to arbitrary norms, as the ℓ_1 and ℓ_∞ countertests demonstrate.

1.2 Unit Vectors and Orthogonality

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Definition 1.2.1. *Unit Vector and Normalization*
A vector $\mathbf{u} \in \mathcal{V}$ is a *unit vector* if $\|\mathbf{u}\| = 1$. If $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the *normalization* of $\mathbf{v}$.

Proposition 1.2.2. *Properties of Normalization*
For $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$:
(i) $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ *is a unit vector;*
(ii) $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ *is a positive real scalar multiple of* $\mathbf{v}$.

Proof.

(i) Since $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\| > 0$ by Proposition 1.1.12(i)–(ii). The identity $\left\|\frac{\mathbf{v}}{\|\mathbf{v}\|}\right\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$ holds by Proposition 1.1.12(iii).
(ii) $\frac{\mathbf{v}}{\|\mathbf{v}\|} = \lambda\mathbf{v}$ for $\lambda := \frac{1}{\|\mathbf{v}\|} \in \mathbb{R}_{>0}$ by Proposition 1.1.12(i)–(ii).

Structural Commentary: Normalization changes scale but preserves the generated one-dimensional subspace.
Key Insight: Divide by the length to make the length equal to one. □

Definition 1.2.3. *Orthogonal Vectors and Finite Lists*
Vectors $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ are *orthogonal*, written $\mathbf{u} \perp \mathbf{v}$, if $\langle \mathbf{u}, \mathbf{v}\rangle = 0$.
A finite list $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *mutually orthogonal* if $(\forall\, j, k \in \{1, \ldots, m\})[j \neq k \Longrightarrow \langle \mathbf{v}_j, \mathbf{v}_k\rangle = 0]$.

Proposition 1.2.4. *Orthogonality and the Zero Vector*
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$:
(i) $\mathbf{u} \perp \mathbf{v} \Longleftrightarrow \mathbf{v} \perp \mathbf{u}$.
(ii) $(\forall\, \mathbf{x} \in \mathcal{V})\,(\mathbf{0} \perp \mathbf{x} \wedge \mathbf{x} \perp \mathbf{0})$.
(iii) $\mathbf{v} \perp \mathbf{v} \Longrightarrow \mathbf{v} = \mathbf{0}$.

Proof.

(i) $(\Rightarrow)$ If $\mathbf{u} \perp \mathbf{v}$, then $\langle \mathbf{u}, \mathbf{v}\rangle = 0$ by Definition 1.2.3. Conjugate symmetry implies $\langle \mathbf{v}, \mathbf{u}\rangle = \overline{\langle \mathbf{u}, \mathbf{v}\rangle} = 0$, so $\mathbf{v} \perp \mathbf{u}$.
(i) $(\Leftarrow)$ If $\mathbf{v} \perp \mathbf{u}$, then $\langle \mathbf{v}, \mathbf{u}\rangle = 0$ by Definition 1.2.3. Conjugate symmetry implies $\langle \mathbf{u}, \mathbf{v}\rangle = \overline{\langle \mathbf{v}, \mathbf{u}\rangle} = 0$, so $\mathbf{u} \perp \mathbf{v}$.
(ii) $(\forall\, \mathbf{x} \in \mathcal{V})\ \big(\langle \mathbf{0}, \mathbf{x}\rangle = 0 \wedge \langle \mathbf{x}, \mathbf{0}\rangle = 0\big)$ by Proposition 1.1.9(ii).
(iii) If $\mathbf{v} \perp \mathbf{v}$, then $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ by Definition 1.2.3, and $\mathbf{v} = \mathbf{0}$ by definiteness.

Structural Commentary: Conjugate symmetry makes orthogonality symmetric, and definiteness isolates $\mathbf{0}$.
Key Insight: Conjugate symmetry preserves the equation "equals zero." □

Proposition 1.2.5. *Mutually Orthogonal Nonzero Vectors Are Independent*
Mutually orthogonal nonzero vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m \in \mathcal{V}$ *are linearly independent.*

Proof. Suppose $\sum_{j=1}^{m} c_j\mathbf{v}_j = \mathbf{0}$ for $c_1, \ldots, c_m \in \mathbb{F}$, and fix $k \in \{1, \ldots, m\}$.

By Proposition 1.1.9(i)–(ii) and mutual orthogonality, $0 = \langle \mathbf{0}, \mathbf{v}_k\rangle = \left\langle \sum_{j=1}^{m} c_j\mathbf{v}_j, \mathbf{v}_k \right\rangle = c_k \langle \mathbf{v}_k, \mathbf{v}_k\rangle = c_k \|\mathbf{v}_k\|^2$.

Since $\mathbf{v}_k \neq \mathbf{0}$, $\|\mathbf{v}_k\| > 0$ by Proposition 1.1.12(i)–(ii), so $c_k = 0$. The index k was arbitrary, hence every coefficient vanishes.
Structural Commentary: Orthogonality isolates one coefficient at a time.
Key Insight: Pair a dependence relation with one vector from the list; every cross term disappears. □

1.3 Orthonormal Bases and Gram–Schmidt

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Definition 1.3.1. *Orthogonal and Orthonormal Families*
For an index set I, a family $(\mathbf{u}_i)_{i\in I}$ in $\mathcal{V}$ is *orthogonal* if $(\forall\, i,j\in I)[i\neq j \Longrightarrow \langle \mathbf{u}_i,\mathbf{u}_j\rangle = 0]$. It is *orthonormal* if $(\forall\, i,j\in I)\ \langle \mathbf{u}_i,\mathbf{u}_j\rangle = \delta_{ij}$.
Thus a finite list $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ is orthonormal exactly when $(\forall\, j,k\in\{1,\dots,m\})\ \langle \mathbf{e}_j,\mathbf{e}_k\rangle = \delta_{jk}$.
Equivalently, the vectors in the list are unit and mutually orthogonal.
If $(\mathbf{u}_i)_{i\in I}$ is orthogonal and $(\forall\, i\in I)\,\mathbf{u}_i\neq\mathbf{0}$, then $\left(\frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}\right)_{i\in I}$ is orthonormal.

Proposition 1.3.2. *Norm of an Orthonormal Linear Combination*
If $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ is orthonormal, then $(\forall\,(a_1,\dots,a_m)\in\mathbb{F}^m)\left[\left\|\sum_{j=1}^m a_j\mathbf{e}_j\right\|^2 = \sum_{j=1}^m |a_j|^2\right]$.

Proof. $\left\|\sum_{j=1}^m a_j\mathbf{e}_j\right\|^2 = \left\langle \sum_{j=1}^m a_j\mathbf{e}_j, \sum_{k=1}^m a_k\mathbf{e}_k\right\rangle = \sum_{j=1}^m\sum_{k=1}^m a_j\overline{a_k}\,\langle \mathbf{e}_j,\mathbf{e}_k\rangle = \sum_{j=1}^m |a_j|^2$ by sesquilinearity and orthonormality.
Structural Commentary: Orthonormality deletes every cross term in the squared-norm expansion.
Key Insight: In orthonormal coordinates, length is computed coefficient-by-coefficient. □

Corollary 1.3.3. *Orthonormal Lists Are Linearly Independent*
If $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ is orthonormal in $\mathcal{V}$, then it is linearly independent.

Proof. Suppose $\sum_{j=1}^m a_j\mathbf{e}_j = \mathbf{0}$ for $(a_1,\dots,a_m)\in\mathbb{F}^m$. Then $0 = \|\mathbf{0}\|^2 = \sum_{j=1}^m |a_j|^2$ by Proposition 1.3.2.
Each $a_j = 0$ by nonnegativity of the summands, so the list is linearly independent.
Structural Commentary: The norm identity converts a linear relation into a sum of nonnegative scalars.
Key Insight: A zero orthonormal linear combination can only have zero coefficients. □

Definition 1.3.4. *Orthonormal Basis*
An *orthonormal basis* of $\mathcal{V}$ is a basis of $\mathcal{V}$ that is an orthonormal list.

Proposition 1.3.5. *Right-Length Orthonormal Lists Are Bases*
If $\dim(\mathcal{V}) = n < \infty$ and $(\mathbf{e}_1,\dots,\mathbf{e}_n)$ is orthonormal, then it is an orthonormal basis of $\mathcal{V}$.

Proof. The list is linearly independent by Corollary 1.3.3. It is a basis by Proposition 1.4.4 because its length is $\dim(\mathcal{V})$.
Structural Commentary: Orthonormality supplies independence; the dimension criterion supplies spanning.
Key Insight: In finite dimensions, n independent vectors in an n-dimensional space already form a basis. □

Theorem 1.3.6. *Gram–Schmidt Process*
Let $m\geq 1$ and let $(\mathbf{v}_1,\dots,\mathbf{v}_m)$ be linearly independent in $\mathcal{V}$. Define $\mathbf{u}_1 := \mathbf{v}_1$ and $\mathbf{e}_1 := \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$.
For $2\leq j\leq m$, define $\mathbf{u}_j := \mathbf{v}_j - \sum_{k=1}^{j-1}\langle \mathbf{v}_j,\mathbf{e}_k\rangle\,\mathbf{e}_k$.
For $2\leq j\leq m$, set $\mathbf{e}_j := \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}$. Then:
(i) $(\forall\, j\in\{1,\dots,m\})[\operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j) = \operatorname{span}(\mathbf{e}_1,\dots,\mathbf{e}_j)]$;
(ii) $(\mathbf{e}_1,\dots,\mathbf{e}_m)$ *is orthonormal.*

Proof.
(*i*) Put $V_j := \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$ and $E_j := \operatorname{span}(\mathbf{e}_1,\dots,\mathbf{e}_j)$. Induct on j.
Since $\mathbf{v}_1\neq\mathbf{0}$, $\mathbf{e}_1$ is a nonzero scalar multiple of $\mathbf{v}_1$, so $V_1 = E_1$. Suppose $V_{j-1} = E_{j-1}$.
The correction term in $\mathbf{u}_j$ lies in $E_{j-1} = V_{j-1}$, so $\mathbf{u}_j\in V_j$.
If $\mathbf{u}_j = \mathbf{0}$, then $\mathbf{v}_j\in V_{j-1}$, contrary to linear independence; hence $\mathbf{u}_j\neq\mathbf{0}$ and $\mathbf{e}_j\in V_j$, so $E_j\subseteq V_j$.
Conversely, $\mathbf{v}_j = \|\mathbf{u}_j\|\,\mathbf{e}_j + \sum_{k=1}^{j-1}\langle \mathbf{v}_j,\mathbf{e}_k\rangle\,\mathbf{e}_k\in E_j$, so $V_j\subseteq E_j$.

(*ii*) $(\forall\, j\in\{1,\dots,m\})\left[\|\mathbf{e}_j\| = \left\|\frac{1}{\|\mathbf{u}_j\|}\mathbf{u}_j\right\| = 1\right]$ by Proposition 1.1.12(iii) and nonvanishing established in (i). Induct on j to verify mutual orthogonality. If $(\mathbf{e}_1,\dots,\mathbf{e}_{j-1})$ is orthonormal and $k<j$, then $\langle \mathbf{u}_j,\mathbf{e}_k\rangle = \langle \mathbf{v}_j,\mathbf{e}_k\rangle - \sum_{i=1}^{j-1}\langle \mathbf{v}_j,\mathbf{e}_i\rangle\,\langle \mathbf{e}_i,\mathbf{e}_k\rangle = 0$.
Hence $\mathbf{e}_j\perp\mathbf{e}_k$ by first-slot homogeneity, and the reverse relation holds by Proposition 1.2.4(i). Thus the full list is orthonormal.
Structural Commentary: Prefix-span induction establishes well-definedness before orthogonality is verified.
Key Insight: Each remainder removes precisely the earlier orthonormal components. □

Theorem 1.3.7. *Gram–Schmidt for Arbitrary Finite Lists*
Let $m\geq 1$ and let $(\mathbf{v}_1,\dots,\mathbf{v}_m)$ be a finite list in $\mathcal{V}$. Set $\mathbf{u}_1 := \mathbf{v}_1$. For $2\leq k\leq m$, define $\mathcal{J}_{k-1} := \{j\in\{1,\dots,k-1\} : \mathbf{u}_j\neq\mathbf{0}\}$.
For $2\leq k\leq m$, set $\mathbf{u}_k := \mathbf{v}_k - \sum_{j\in\mathcal{J}_{k-1}}\frac{\langle \mathbf{v}_k,\mathbf{u}_j\rangle}{\langle \mathbf{u}_j,\mathbf{u}_j\rangle}\mathbf{u}_j$. Set $U_k := \operatorname{span}(\mathbf{u}_1,\dots,\mathbf{u}_k)$, $V_k := \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_k)$, with $U_0 = V_0 := \{\mathbf{0}\}$. Then:
(i) $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ *is orthogonal;*
(ii) $(\forall\, k\in\{1,\dots,m\})[U_k = V_k]$.

Proof.
(*i*) Induct on k. A one-vector list is orthogonal. Suppose $(\mathbf{u}_1,\dots,\mathbf{u}_{k-1})$ is orthogonal and fix $i<k$.
If $\mathbf{u}_i = \mathbf{0}$, then $\mathbf{u}_k\perp\mathbf{u}_i$ by Proposition 1.2.4(ii). If $\mathbf{u}_i\neq\mathbf{0}$, then $i\in\mathcal{J}_{k-1}$ and $\langle \mathbf{u}_k,\mathbf{u}_i\rangle = \langle \mathbf{v}_k,\mathbf{u}_i\rangle - \sum_{j\in\mathcal{J}_{k-1}}\frac{\langle \mathbf{v}_k,\mathbf{u}_j\rangle}{\langle \mathbf{u}_j,\mathbf{u}_j\rangle}\langle \mathbf{u}_j,\mathbf{u}_i\rangle = 0$
by the induction hypothesis: all terms with $j\neq i$ vanish, and the $j=i$ term cancels the first term.
The reverse orthogonality relation holds by Proposition 1.2.4(i), so the full list is orthogonal.

(*ii*) Induct on k, starting from $U_0 = V_0$. Suppose $U_{k-1} = V_{k-1}$.
The correction term in $\mathbf{u}_k$ lies in $U_{k-1} = V_{k-1}$, so $\mathbf{u}_k \in V_k$ and $U_k \subseteq V_k$.
The recursion also expresses $\mathbf{v}_k$ as $\mathbf{u}_k$ plus a vector in U_{k-1}, so $\mathbf{v}_k \in U_k$ and $V_k \subseteq U_k$. Therefore $U_k = V_k$.

Structural Commentary: Orthogonality and prefix-span preservation follow by separate inductions.
Key Insight: Subtracting all nonzero earlier directions preserves the generated span. □

Corollary 1.3.8. *Zero Remainders Detect Dependence*
For the construction in Theorem 1.3.7, $(\forall\, k \in \{1, \dots, m\})[\mathbf{u}_k = \mathbf{0} \iff \mathbf{v}_k \in V_{k-1}]$.

Proof. Fix $k \in \{1, \dots, m\}$.
(⇒) Suppose $\mathbf{u}_k = \mathbf{0}$. If $k = 1$, then $\mathbf{v}_1 = \mathbf{0} \in V_0$. If $k \geq 2$, the recursion places $\mathbf{v}_k$ in $U_{k-1} = V_{k-1}$ by Theorem 1.3.7(ii).
(⇐) Suppose $\mathbf{v}_k \in V_{k-1}$. If $k = 1$, then $\mathbf{v}_1 = \mathbf{0} = \mathbf{u}_1$.
If $k \geq 2$, then $V_{k-1} = U_{k-1}$ by Theorem 1.3.7(ii), so write $\mathbf{v}_k = \sum_{j \in \mathcal{J}_{k-1}} a_j \mathbf{u}_j$ for a coefficient family $(a_j)_{j \in \mathcal{J}_{k-1}}$.
The nonzero vectors in this sum are orthogonal by Theorem 1.3.7(i), and $(\forall\, i \in \mathcal{J}_{k-1})\ \langle \mathbf{v}_k, \mathbf{u}_i \rangle = a_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ by orthogonality.
Thus $(\forall\, i \in \mathcal{J}_{k-1})\, a_i = \frac{\langle \mathbf{v}_k, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle}$, so the correction term equals $\mathbf{v}_k$ and $\mathbf{u}_k = \mathbf{0}$ by the recursion.

Structural Commentary: Each direction compares the recursion with the preserved prefix span.
Key Insight: A zero remainder means the new vector adds no direction. □

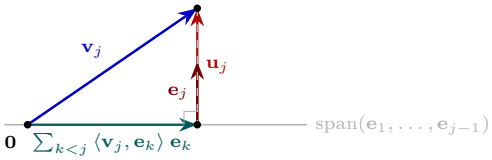

Figure 22: One Gram–Schmidt step: subtract the previous-span component, then normalize.

Figure 22 depicts the residual after removal of the previous-span component; normalization changes its length while preserving its generated one-dimensional subspace.

Example 1.3.9. *Legendre Polynomials from Gram–Schmidt*
Equip $\mathbb{R}[x]$ with $\langle p, q \rangle := \int_{-1}^{1} p(x) q(x)\, dx$ and perform Gram–Schmidt on $(1, x, x^2, x^3, \dots)$.
Parity annihilates mixed even–odd inner products. Since $\langle 1, 1 \rangle = 2$, $\langle x^2, 1 \rangle = \frac{2}{3}$, $\langle x, x \rangle = \frac{2}{3}$, and $\langle x^3, x \rangle = \frac{2}{5}$, the first remainders are $u_1(x) = 1$, $u_2(x) = x$, $u_3(x) = x^2 - \frac{1}{3}$, and $u_4(x) = x^3 - \frac{3}{5}x$.
These are nonzero scalar multiples of the first four Legendre polynomials.
Structural Commentary: Orthogonal polynomials are orthogonalized monomials.
Key Insight: Changing the inner product changes the polynomial family produced by Gram–Schmidt.

Lemma 1.3.10. *Gram–Schmidt on Orthogonal Lists*
Let $m \geq 1$ and let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a mutually orthogonal list of nonzero vectors. Then:
(i) $(\forall\, j \in \{1, \dots, m\})\, \mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$;
(ii) if the original list is orthonormal, then $(\forall\, j \in \{1, \dots, m\})\, \mathbf{e}_j = \mathbf{v}_j$.

Proof.
(*i*) The list is linearly independent by Proposition 1.2.5, so the construction is defined by Theorem 1.3.6. Induct on j.
The claim holds for $j = 1$. Suppose $(\forall\, i < j)\, \mathbf{e}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$. Then $(\forall\, i < j)\ \langle \mathbf{v}_j, \mathbf{e}_i \rangle = \frac{1}{\|\mathbf{v}_i\|} \langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0$ by mutual orthogonality.
Thus $\mathbf{u}_j = \mathbf{v}_j$ in the recursion and $\mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$.
(*ii*) If the input is orthonormal, then $(\forall\, j \in \{1, \dots, m\})\ \|\mathbf{v}_j\| = 1$ by orthonormality, so $\mathbf{e}_j = \mathbf{v}_j$ by (i).

Structural Commentary: Induction verifies that every orthogonal correction term vanishes.
Key Insight: Orthogonal input already has no earlier-direction component. □

Corollary 1.3.11. *Existence of Orthonormal Bases*
If $\dim(\mathcal{V}) < \infty$, then $\mathcal{V}$ has an orthonormal basis.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, the empty list is an orthonormal basis. Suppose $\dim(\mathcal{V}) = n \geq 1$ and choose a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$.
By Theorem 1.3.6(i)–(ii), its Gram–Schmidt output is orthonormal and has span $\mathcal{V}$, so it is an orthonormal basis.
Structural Commentary: Gram–Schmidt orthonormalizes a basis without changing its span.
Key Insight: Finite-dimensional inner-product spaces always admit geometry-friendly coordinates. □

Corollary 1.3.12. *Extending an Orthonormal List*
If $\dim(\mathcal{V}) < \infty$ and $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ is orthonormal, then it extends to an orthonormal basis of $\mathcal{V}$.

Proof. If $m = 0$, an orthonormal basis exists by Corollary 1.3.11 and extends the empty list.
Suppose $m \geq 1$. The list is linearly independent by Corollary 1.3.3. $\exists\, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n$ such that $(\mathbf{e}_1, \dots, \mathbf{e}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ is a basis by Theorem 1.3.13. Let $(\mathbf{f}_1, \dots, \mathbf{f}_n)$ be its Gram–Schmidt output.
The first m steps depend only on the orthonormal prefix, so $(\forall\, j \in \{1, \dots, m\})\, \mathbf{f}_j = \mathbf{e}_j$ by Lemma 1.3.10(ii). The output is orthonormal by Theorem 1.3.6(ii), and it spans $\mathcal{V}$ by Theorem 1.3.6(i) with $j = n$. Thus $(\mathbf{f}_1, \dots, \mathbf{f}_n)$ is the required extension.
Structural Commentary: Extend linearly first, then orthonormalize while preserving the original prefix.
Key Insight: Gram–Schmidt leaves an orthonormal prefix fixed. □

1.4 Orthonormal Coordinates and Inner-Product Identities

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Theorem 1.4.1. *Coordinates in an Orthonormal Basis*
Let $\dim(\mathcal{V}) = n \geq 1$ *and let* $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *be an orthonormal basis. For* $\mathbf{v} \in \mathcal{V}$*:*

(i) $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \, \mathbf{e}_j$*;*

(ii) $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2$*;*

(iii) $[\mathbf{v}]_\beta = \big(\langle \mathbf{v}, \mathbf{e}_1 \rangle, \ldots, \langle \mathbf{v}, \mathbf{e}_n \rangle \big)^t$.

Proof.

(i) Write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{e}_j$. For $k \in \{1, \ldots, n\}$, $\langle \mathbf{v}, \mathbf{e}_k \rangle = \sum_{j=1}^{n} a_j \langle \mathbf{e}_j, \mathbf{e}_k \rangle = a_k$ by orthonormality, which proves the expansion.

(ii) By (i) and Proposition 1.3.2, $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2$.

(iii) The coefficient computation in (i) means $[\mathbf{v}]_\beta = \big(\langle \mathbf{v}, \mathbf{e}_1 \rangle, \ldots, \langle \mathbf{v}, \mathbf{e}_n \rangle \big)^t$ by the coordinate-vector definition.

Structural Commentary: Pairing with each basis vector extracts the coefficients before the norm and coordinate formulas follow.
Key Insight: Orthonormality turns coefficient extraction into an inner-product calculation. □

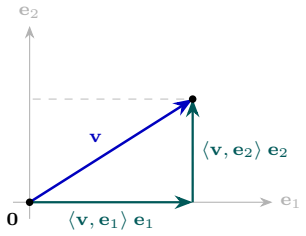

Figure 23: An orthonormal basis resolves a vector into perpendicular coordinate components.

Figure 23 illustrates Theorem 1.4.1(i) for $n = 2$: the two perpendicular coordinate components sum to $\mathbf{v}$.

Remark 1.4.2. *Cross Terms*
The mixed terms $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$ in $\|\mathbf{u} + \mathbf{v}\|^2$ are called *cross terms*; $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ are the pure squared-length terms.

Theorem 1.4.3. *Pythagorean Theorem*
$(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V}) \big[\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \big]$.

Proof. Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{u} \perp \mathbf{v}$. Then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ by orthogonality and $\langle \mathbf{v}, \mathbf{u} \rangle = 0$ by Proposition 1.2.4(i).
Hence $\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
Structural Commentary: Orthogonality cancels the cross terms in the squared-norm expansion.
Key Insight: Perpendicular components add in squared length, not in length. □

Proposition 1.4.4. *Pythagorean Converse over* $\mathbb{R}$
If $\mathbb{F} = \mathbb{R}$*, then* $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V}) \big[\mathbf{u} \perp \mathbf{v} \iff \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \big]$.

Proof. Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$. Conjugate symmetry over $\mathbb{R}$ implies $\langle \mathbf{v}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle$, so $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2 \langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$.
($\Rightarrow$) If $\mathbf{u} \perp \mathbf{v}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ by orthogonality, so the identity follows from the expansion.
($\Leftarrow$) If the identity holds, then $2 \langle \mathbf{u}, \mathbf{v} \rangle = 0$ by the expansion, so $\mathbf{u} \perp \mathbf{v}$ by orthogonality.
Structural Commentary: The two directions compare the norm identity with its single real cross term.
Key Insight: Over $\mathbb{R}$, vanishing length excess is equivalent to orthogonality. □

Theorem 1.4.5. *Bessel's Inequality*
Let $m \geq 1$ *and let* $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *be orthonormal in* $\mathcal{V}$*. Then* $(\forall\, \mathbf{v} \in \mathcal{V}) \sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2 \leq \|\mathbf{v}\|^2$.

Proof. Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{p} := \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \, \mathbf{e}_j$. For $k \in \{1, \ldots, m\}$, $\langle \mathbf{v} - \mathbf{p}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$ by orthonormality.
Hence $\langle \mathbf{v} - \mathbf{p}, \mathbf{p} \rangle = \sum_{j=1}^{m} \overline{\langle \mathbf{v}, \mathbf{e}_j \rangle} \langle \mathbf{v} - \mathbf{p}, \mathbf{e}_j \rangle = 0$, so $\mathbf{p} \perp (\mathbf{v} - \mathbf{p})$ by Proposition 1.2.4(i).
By Theorem 1.4.3 and Proposition 1.3.2, respectively, $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{v} - \mathbf{p}\|^2$ and $\|\mathbf{p}\|^2 = \sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2$. Thus $\sum_{j=1}^{m} \left| \langle \mathbf{v}, \mathbf{e}_j \rangle \right|^2 \leq \|\mathbf{v}\|^2$.
Structural Commentary: The coordinate sum and its perpendicular residual split the squared norm by Pythagoras.
Key Insight: The residual norm is the nonnegative energy missing from selected coordinates. □

Corollary 1.4.6. *Coefficients Against an Orthonormal Sequence Vanish*
If $(\mathbf{e}_n)_{n \geq 1}$ *is orthonormal in* $\mathcal{V}$*, then* $(\forall\, \mathbf{v} \in \mathcal{V}) \lim_{n \to \infty} \langle \mathbf{v}, \mathbf{e}_n \rangle = 0$.

Proof. Fix $\mathbf{v} \in \mathcal{V}$ and $\epsilon > 0$, and set $S_\epsilon := \{j \in \mathbb{N}_{\geq 1} : |\langle \mathbf{v}, \mathbf{e}_j \rangle| \geq \epsilon\}$.
If S_ϵ were infinite, choose $N \in \mathbb{N}_{\geq 1}$ with $N\epsilon^2 > \|\mathbf{v}\|^2$ and distinct $j_1, \ldots, j_N \in S_\epsilon$.
For this orthonormal sublist, $N\epsilon^2 \leq \sum_{k=1}^{N} |\langle \mathbf{v}, \mathbf{e}_{j_k} \rangle|^2 \leq \|\mathbf{v}\|^2$ by Theorem 1.4.5, contradicting $N\epsilon^2 > \|\mathbf{v}\|^2$. Thus S_ϵ is finite.
Choose $N_0 \in \mathbb{N}_{\geq 1}$ greater than every member of S_ϵ, with $N_0 = 1$ when $S_\epsilon = \varnothing$. Then $(\forall\, n \geq N_0)\ |\langle \mathbf{v}, \mathbf{e}_n \rangle| < \epsilon$, so $\langle \mathbf{v}, \mathbf{e}_n \rangle \to 0$.
Structural Commentary: A contradiction argument applies Bessel to arbitrarily many allegedly large coefficients.
Key Insight: Finite total energy permits only finitely many coefficients above each threshold. □

Theorem 1.4.7. *Parseval's Identity*
Let $\dim(\mathcal{V}) = n \geq 1$ *and let* $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *be an orthonormal basis. For* $\mathbf{v}, \mathbf{w} \in \mathcal{V}$*:*

(i) $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle}$*;*

(ii) $\langle \mathbf{v}, \mathbf{w} \rangle = \langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$*.*

Proof.

(i) Set $c_j := \langle \mathbf{v}, \mathbf{e}_j \rangle$ and $d_j := \langle \mathbf{w}, \mathbf{e}_j \rangle$ $(j \in \{1, \ldots, n\})$. Then $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{n} d_k \mathbf{e}_k$ by Theorem 1.4.1(i).
Hence $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \sum_{k=1}^{n} c_j \overline{d_k} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{n} c_j \overline{d_j}$ by sesquilinearity and orthonormality.

(ii) By Theorem 1.4.1(iii), $[\mathbf{v}]_\beta = (c_1, \ldots, c_n)^t$ and $[\mathbf{w}]_\beta = (d_1, \ldots, d_n)^t$. Therefore $\langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n} = \sum_{j=1}^{n} c_j \overline{d_j} = \langle \mathbf{v}, \mathbf{w} \rangle$ by (i).

Structural Commentary: Basis expansions remove cross terms before the coordinate formula identifies the remaining sum.
Key Insight: Orthonormal coordinates preserve the inner product itself. □

Theorem 1.4.8. *Matrix Entries from an Orthonormal Codomain Basis*
Let $\dim(\mathcal{V}) = n \geq 1$ *and* $\dim(\mathcal{W}) = m \geq 1$*, with* $\mathcal{W}$ *an inner-product space.*
Fix a basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *of* $\mathcal{V}$ *and an orthonormal basis* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ *of* $\mathcal{W}$*. Then:*

(i) $(\forall\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}),\ i \in \{1, \ldots, m\},\ j \in \{1, \ldots, n\})\ ([T]_{\gamma \leftarrow \beta})_{i,j} = \langle T\mathbf{v}_j, \mathbf{w}_i \rangle$*;*

(ii) $(\forall\, A \in \mathrm{Mat}(m, n, \mathbb{F}))(\exists!\, T \in \mathcal{L}(\mathcal{V}, \mathcal{W}))\ [T]_{\gamma \leftarrow \beta} = A$*.*

Proof.

(i) Fix $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $i \in \{1, \ldots, m\}$, and $j \in \{1, \ldots, n\}$. The jth column of $[T]_{\gamma \leftarrow \beta}$ is $[T\mathbf{v}_j]_\gamma = \big(\langle T\mathbf{v}_j, \mathbf{w}_1 \rangle, \ldots, \langle T\mathbf{v}_j, \mathbf{w}_m \rangle\big)^t$ by Theorem 1.4.1(iii). Therefore its ith entry is $\langle T\mathbf{v}_j, \mathbf{w}_i \rangle$.

(ii) The map $\Phi : \mathcal{L}(\mathcal{V}, \mathcal{W}) \to \mathrm{Mat}(m, n, \mathbb{F})$ defined by $\Phi(T) := [T]_{\gamma \leftarrow \beta}$ is an isomorphism by Theorem 3.4.5(i).
Hence every $A \in \mathrm{Mat}(m, n, \mathbb{F})$ has a unique preimage T under Φ.

Structural Commentary: Coordinate extraction identifies entries; the earlier encoding isomorphism supplies existence and uniqueness.
Key Insight: Entry (i, j) measures the jth basis image along output direction i. □

Figure 24 visualizes Theorem 1.4.8(i): applying T to $\mathbf{v}_j$ forms column j, and pairing with $\mathbf{w}_i$ reads entry (i, j).

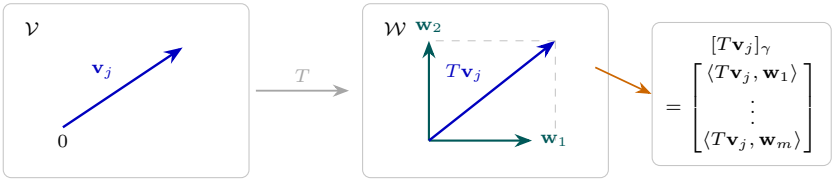

Figure 24: The jth column of $[T]_{\gamma \leftarrow \beta}$ is obtained by resolving $T\mathbf{v}_j$ against the orthonormal basis of $\mathcal{W}$.

Corollary 1.4.9. *Inner Products Through an Operator Matrix*
Let $\dim(\mathcal{V}) = n \geq 1$ *and let* $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *be an orthonormal basis.*
For $T \in \mathcal{L}(\mathcal{V})$*, set* $A := [T]_{\beta \leftarrow \beta}$*. Then* $(\forall\, \mathbf{v}, \mathbf{w} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{w} \rangle = \langle A[\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$*.*

Proof. Fix $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The identity $[T\mathbf{v}]_\beta = A[\mathbf{v}]_\beta$ holds by Proposition 3.2.5. Hence $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle A[\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$ by Theorem 1.4.7(ii).
Structural Commentary: Matrix action computes the first coordinate vector before Parseval transfers the inner product.
Key Insight: Orthonormal coordinates convert operator geometry into matrix algebra. □

1.5 Gram Matrices

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Definition 1.5.1. *Gram Matrix*
For $n \geq 1$ and $\mathbf{u}_1, \ldots, \mathbf{u}_n \in \mathcal{V}$, the *Gram matrix* is $\mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := [\langle \mathbf{u}_j, \mathbf{u}_i \rangle]_{i,j=1}^{n} \in \mathrm{Mat}(n, n, \mathbb{F})$.

Proposition 1.5.2. *Gram Matrix Quadratic Form*
Let $n \geq 1$*, let* $\mathbf{u}_1, \ldots, \mathbf{u}_n \in \mathcal{V}$*, and set* $G = [g_{ij}] := \mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$*. Then:*

(i) $(\forall\, i, j \in \{1, \ldots, n\})\ g_{ij} = \overline{g_{ji}}$*, so* G *is Hermitian;*

(ii) $(\forall\, \mathbf{c} = (c_1, \ldots, c_n)^t \in \mathbb{F}^n)\ \mathbf{c}^* G \mathbf{c} = \left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2 \geq 0$*;*

(iii) if $\mathbf{u}_1, \ldots, \mathbf{u}_n$ *are linearly independent, then* $(\forall\, \mathbf{c} \in \mathbb{F}^n \setminus \{\mathbf{0}\})\ \mathbf{c}^* G \mathbf{c} > 0$*.*

Proof.

(i) For $i, j \in \{1, \dots, n\}$, $g_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle = \overline{\langle \mathbf{u}_i, \mathbf{u}_j \rangle} = \overline{g_{ji}}$ by conjugate symmetry.

(ii) Fix $\mathbf{c} = (c_1, \dots, c_n)^t \in \mathbb{F}^n$ and set $\mathbf{x} := \sum_{j=1}^{n} c_j \mathbf{u}_j$.

By sesquilinearity and definiteness, $\mathbf{c}^* G \mathbf{c} = \sum_{i=1}^{n} \sum_{j=1}^{n} \overline{c_i} \langle \mathbf{u}_j, \mathbf{u}_i \rangle c_j = \langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2 \geq 0$.

(iii) Suppose $\mathbf{u}_1, \dots, \mathbf{u}_n$ are linearly independent and fix $\mathbf{c} = (c_1, \dots, c_n)^t \neq \mathbf{0}$.

Then $\sum_{j=1}^{n} c_j \mathbf{u}_j \neq \mathbf{0}$ by linear independence, so $\mathbf{c}^* G \mathbf{c} > 0$ by (ii) and definiteness.

Structural Commentary: Entry symmetry and a norm-square expansion establish Hermitian positivity; independence makes it strict.
Key Insight: Gram quadratic forms measure norms of coefficient-weighted vector sums. □

1.6 Finite-Dimensional Riesz Representation

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Theorem 1.6.1. *Riesz Representation Theorem*
Suppose $\dim(\mathcal{V}) < \infty$ *and let* $\varphi \in \mathcal{V}'$. *Then:*
(i) $(\exists\, \mathbf{u} \in \mathcal{V})(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$;
(ii) $(\forall\, \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{V}) \big[(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle \implies \mathbf{u}_1 = \mathbf{u}_2 \big]$.

Proof.

(i) If $\mathcal{V} = \{\mathbf{0}\}$, then $\varphi(\mathbf{0}) = 0$ by linearity, so $\mathbf{u} = \mathbf{0}$ represents φ.

Suppose $\dim(\mathcal{V}) = n \geq 1$ and choose an orthonormal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ by Corollary 1.3.11. Set $\mathbf{u} := \sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)} \mathbf{e}_j$.

Fix $\mathbf{v} \in \mathcal{V}$. By Theorem 1.4.1(i), linearity of φ, and Proposition 1.1.9(iii)–(iv), $\varphi(\mathbf{v}) = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \varphi(\mathbf{e}_j) = \langle \mathbf{v}, \mathbf{u} \rangle$.

(ii) Suppose $(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle$. Then $\mathbf{u}_1 = \mathbf{u}_2$ by Lemma 1.1.10(i).

Structural Commentary: An orthonormal expansion constructs the representative; inner-product separation proves uniqueness.
Key Insight: Functional values become conjugated coordinates of one vector. □

Definition 1.6.2. *Riesz Vector*
If $\dim(\mathcal{V}) < \infty$ and $\varphi \in \mathcal{V}'$, the unique $\mathbf{u} \in \mathcal{V}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\, \varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ is the *Riesz vector* representing φ.

Theorem 1.6.3. *Riesz Map*
Suppose $\dim(\mathcal{V}) < \infty$ *and define* $\tau : \mathcal{V} \to \mathcal{V}'$ *by* $\tau(\mathbf{u}) := \langle \cdot, \mathbf{u} \rangle$. *Then:*
(i) τ *is conjugate linear:* $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{u}, \mathbf{w} \in \mathcal{V})\, \tau(a\mathbf{u} + \mathbf{w}) = \overline{a}\, \tau(\mathbf{u}) + \tau(\mathbf{w})$;
(ii) τ *is bijective;*
(iii) $R := \tau^{-1}$ *is conjugate linear:* $(\forall\, a \in \mathbb{F})(\forall\, \varphi, \psi \in \mathcal{V}')\, R(a\varphi + \psi) = \overline{a}\, R(\varphi) + R(\psi)$;
(iv) if $\mathbb{F} = \mathbb{R}$, *then* τ *and* R *are linear.*

Proof.

(i) $(\forall\, \mathbf{u} \in \mathcal{V})\, \tau(\mathbf{u}) \in \mathcal{V}'$ by Proposition 1.1.9(i). Fix $a \in \mathbb{F}$, $\mathbf{u}, \mathbf{w}, \mathbf{v} \in \mathcal{V}$. Then $\tau(a\mathbf{u} + \mathbf{w})(\mathbf{v}) = \overline{a}\, \tau(\mathbf{u})(\mathbf{v}) + \tau(\mathbf{w})(\mathbf{v})$ by Proposition 1.1.9(iii)–(iv), so $\tau(a\mathbf{u} + \mathbf{w}) = \overline{a}\, \tau(\mathbf{u}) + \tau(\mathbf{w})$ by equality of functions.

(ii) If $\tau(\mathbf{u}) = \tau(\mathbf{w})$, then $\|\mathbf{u} - \mathbf{w}\|^2 = \tau(\mathbf{u} - \mathbf{w})(\mathbf{u} - \mathbf{w}) = 0$ by (i), so $\mathbf{u} = \mathbf{w}$ by definiteness.
Thus τ is injective. For $\varphi \in \mathcal{V}'$, $(\exists\, \mathbf{u} \in \mathcal{V})\, \tau(\mathbf{u}) = \varphi$ by Theorem 1.6.1(i), so τ is surjective.

(iii) Fix $a \in \mathbb{F}$ and $\varphi, \psi \in \mathcal{V}'$. By (i) and the definition of R, $\tau\big(\overline{a}\, R(\varphi) + R(\psi)\big) = a\varphi + \psi = \tau\big(R(a\varphi + \psi)\big)$.
Hence $R(a\varphi + \psi) = \overline{a}\, R(\varphi) + R(\psi)$ by injectivity from (ii).

(iv) If $\mathbb{F} = \mathbb{R}$, then $\overline{a} = a$ in (i) and (iii), so both maps are linear.

Structural Commentary: Conjugate-linearity, separation, and representation establish the bijection and its inverse law.
Key Insight: Conjugating the representing slot conjugates scalar multiplication. □

Figure 25 depicts Theorem 1.6.3: the vector $\mathbf{u}$ corresponds to the functional $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle$.

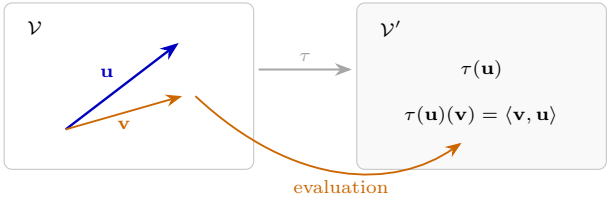

Figure 25: The map τ sends each vector to the functional it represents.

Perspective 1.6.4. *Riesz as Coordinate Duality*

For an orthonormal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$, Theorem 1.6.1(i) constructs the Riesz vector of $\varphi \in \mathcal{V}'$ as $\sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)} \mathbf{e}_j$.

Although this formula uses the basis, uniqueness in part (ii) makes its value basis-independent.

2 Orthogonality, Projections, and Least Squares

2.1 Orthogonal Complements

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Orthogonal complements turn orthogonality into a subspace operation and organize canonical direct-sum decompositions.

Definition 2.1.1. *Orthogonality and Orthogonal Direct Sums*
Let $\mathcal{X}, \mathcal{Y} \subseteq \mathcal{V}$ and let $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$.

(i) $\mathcal{X}$ and $\mathcal{Y}$ are *orthogonal*, written $\mathcal{X} \perp \mathcal{Y}$, if $(\forall\, \mathbf{x} \in \mathcal{X})(\forall\, \mathbf{y} \in \mathcal{Y})\ \langle \mathbf{x}, \mathbf{y} \rangle = 0$.
(ii) $\mathcal{V}$ is the *orthogonal direct sum* of $\mathcal{S}$ and $\mathcal{T}$, written $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, if $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{S} \perp \mathcal{T}$.

Definition 2.1.2. *Orthogonal Complement*
For $\mathcal{U} \subseteq \mathcal{V}$, its *orthogonal complement* is $\mathcal{U}^{\perp} := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{u} \in \mathcal{U})\ \langle \mathbf{v}, \mathbf{u} \rangle = 0\}$.

Proposition 2.1.3. *Basic Properties of Orthogonal Complements*
For subsets $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$,

(i) $\mathcal{U}^{\perp}$ is a subspace of $\mathcal{V}$;
(ii) $\{\mathbf{0}\}^{\perp} = \mathcal{V}$;
(iii) $\mathcal{V}^{\perp} = \{\mathbf{0}\}$;
(iv) $\mathcal{U} \cap \mathcal{U}^{\perp} \subseteq \{\mathbf{0}\}$;
(v) if $\mathcal{U} \subseteq \mathcal{W}$, then $\mathcal{W}^{\perp} \subseteq \mathcal{U}^{\perp}$.

Proof.

(i) $\mathbf{0} \in \mathcal{U}^{\perp}$ by Proposition 1.1.9(ii). Fix $\mathbf{v}, \mathbf{w} \in \mathcal{U}^{\perp}$ and $a \in \mathbb{F}$.
Then $(\forall\, \mathbf{u} \in \mathcal{U})\ \langle a\mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = a\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle = 0$ by linearity, so $a\mathbf{v} + \mathbf{w} \in \mathcal{U}^{\perp}$. Hence $\mathcal{U}^{\perp} \leq \mathcal{V}$ by Theorem 1.2.2.
(ii) $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{v}, \mathbf{0} \rangle = 0$ by Proposition 1.1.9(ii), so $\{\mathbf{0}\}^{\perp} = \mathcal{V}$.
(iii) $\mathbf{0} \in \mathcal{V}^{\perp}$ by (i). If $\mathbf{v} \in \mathcal{V}^{\perp}$, then $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$ by definiteness.
(iv) If $\mathbf{v} \in \mathcal{U} \cap \mathcal{U}^{\perp}$, then $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$ by definiteness.
(v) Suppose $\mathcal{U} \subseteq \mathcal{W}$ and fix $\mathbf{v} \in \mathcal{W}^{\perp}$. Then $(\forall\, \mathbf{u} \in \mathcal{U})\ \langle \mathbf{v}, \mathbf{u} \rangle = 0$, so $\mathbf{v} \in \mathcal{U}^{\perp}$.

Structural Commentary: Subspace closure, definiteness, and reversed constraint inclusion establish the five properties.
Key Insight: Larger tested sets impose stronger orthogonality constraints. □

Remark 2.1.4. *Definiteness Matters*
For a bilinear form $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ and $\mathcal{U} \subseteq \mathcal{V}$, set $\mathcal{U}^{\perp_B} := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{u} \in \mathcal{U})\, B(\mathbf{v}, \mathbf{u}) = 0\}$.
Bilinearity makes $\mathcal{U}^{\perp_B}$ a subspace, but without definiteness one may have $\mathcal{V}^{\perp_B} \neq \{\mathbf{0}\}$ or $\mathcal{U} \cap \mathcal{U}^{\perp_B} \neq \{\mathbf{0}\}$.
Thus $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp_B}$ is an inner-product conclusion, not a consequence of bilinearity alone.

Theorem 2.1.5. *Direct Sum with the Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$.

Proof. If $\mathcal{U} = \{\mathbf{0}\}$, then $\mathcal{U}^{\perp} = \mathcal{V}$ by Proposition 2.1.3(ii), and hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$.
Suppose $\dim(\mathcal{U}) = m \geq 1$ and choose an orthonormal basis $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ by Corollary 1.3.11.
Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{u} := \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \mathbf{e}_j \in \mathcal{U}$ and $\mathbf{w} := \mathbf{v} - \mathbf{u}$.
For $k \in \{1, \ldots, m\}$, $\langle \mathbf{w}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$ by orthonormality.
Fix $\mathbf{x} \in \mathcal{U}$ and write $\mathbf{x} = \sum_{k=1}^{m} a_k \mathbf{e}_k$. Then $\langle \mathbf{w}, \mathbf{x} \rangle = \sum_{k=1}^{m} \overline{a_k} \langle \mathbf{w}, \mathbf{e}_k \rangle = 0$ by Proposition 1.1.9(iii)–(iv), so $\mathbf{w} \in \mathcal{U}^{\perp}$ and $\mathbf{v} = \mathbf{u} + \mathbf{w}$.
Hence $\mathcal{V} = \mathcal{U} + \mathcal{U}^{\perp}$. The identity $\mathcal{U} \cap \mathcal{U}^{\perp} = \{\mathbf{0}\}$ holds by Proposition 2.1.3(i),(iv), so the sum is direct.
Structural Commentary: An orthonormal expansion constructs the component; definiteness makes the decomposition direct.
Key Insight: Every vector splits into a subspace component and orthogonal residual. □

Proposition 2.1.6. *Orthogonal Direct Sum Characterization*
For $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, the following are equivalent:

(i) $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$.
(ii) $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$.

Proof.

(i) $\Rightarrow$ (ii) Suppose $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$. Then $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ by orthogonal-direct-sum definition, and $\mathcal{T} \subseteq \mathcal{S}^{\perp}$ by Proposition 1.2.4(i).
Fix $\mathbf{v} \in \mathcal{S}^{\perp}$ and write $\mathbf{v} = \mathbf{s} + \mathbf{t}$ with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$.
Then $\mathbf{s} = \mathbf{v} - \mathbf{t} \in \mathcal{S}^{\perp}$, so $\mathbf{s} = \mathbf{0}$ by Proposition 2.1.3(iv). Hence $\mathbf{v} = \mathbf{t} \in \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$.
(ii) $\Rightarrow$ (i) Suppose $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$. Then $\mathcal{S} \perp \mathcal{T}$ by Proposition 1.2.4(i), so $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$ by orthogonal-direct-sum definition.

Structural Commentary: Direct-sum uniqueness forces an orthogonal partner to be maximal.
Key Insight: A full orthogonal summand is exactly the complement. □

Corollary 2.1.7. *Dimension of the Orthogonal Complement*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$, then $\dim(\mathcal{U}^{\perp}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. The subspace $\mathcal{U}$ is finite-dimensional by Corollary 1.3.14.
By Theorem 2.1.5, $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$, so $\dim(\mathcal{V}) = \dim(\mathcal{U}) + \dim(\mathcal{U}^{\perp})$ by Corollary 2.3.2. Rearranging proves the formula.
Structural Commentary: The direct-sum dimension formula converts decomposition into subtraction.
Key Insight: The complement contains precisely the unused dimensions. □

Corollary 2.1.8. *Double Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{U} = (\mathcal{U}^{\perp})^{\perp}$.

Proof. Fix $\mathbf{u} \in \mathcal{U}$. For $\mathbf{w} \in \mathcal{U}^\perp$, $\langle \mathbf{w}, \mathbf{u} \rangle = 0$ by membership, so $\langle \mathbf{u}, \mathbf{w} \rangle = 0$ by Proposition 1.2.4(i). Hence $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$.
Conversely, fix $\mathbf{v} \in (\mathcal{U}^\perp)^\perp$. By Theorem 2.1.5, write $\mathbf{v} = \mathbf{u}_0 + \mathbf{w}$ with $\mathbf{u}_0 \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$.
Since $(\mathcal{U}^\perp)^\perp$ is a subspace by Proposition 2.1.3(i), the established inclusion implies $\mathbf{w} = \mathbf{v} - \mathbf{u}_0 \in (\mathcal{U}^\perp)^\perp$.
Thus $\mathbf{w} \in \mathcal{U}^\perp \cap (\mathcal{U}^\perp)^\perp$, so $\mathbf{w} = \mathbf{0}$ by Proposition 2.1.3(iv), and $\mathbf{v} = \mathbf{u}_0 \in \mathcal{U}$.
Structural Commentary: Symmetry establishes one inclusion; the orthogonal decomposition eliminates every excess vector.
Key Insight: Double orthogonality adds no new finite-dimensional direction. □

Corollary 2.1.9. *Double Complement of a Finite Span*
If $\mathcal{X} \subseteq \mathcal{V}$ and $\operatorname{span}(\mathcal{X})$ is finite-dimensional, then $(\mathcal{X}^\perp)^\perp = \operatorname{span}(\mathcal{X})$.

Proof. Let $\mathcal{W} = \operatorname{span}(\mathcal{X})$. Since $\mathcal{X} \subseteq \mathcal{W}$, $\mathcal{W}^\perp \subseteq \mathcal{X}^\perp$ by Proposition 2.1.3(v).
Conversely, fix $\mathbf{v} \in \mathcal{X}^\perp$ and $\mathbf{w} \in \mathcal{W}$. Write $\mathbf{w} = \sum_{j=1}^{r} a_j \mathbf{x}_j$ with $\mathbf{x}_j \in \mathcal{X}$.
Then $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{r} \overline{a_j} \langle \mathbf{v}, \mathbf{x}_j \rangle = 0$ by Proposition 1.1.9(iii)–(iv), so $\mathbf{v} \in \mathcal{W}^\perp$.
Therefore $\mathcal{X}^\perp = \mathcal{W}^\perp$, and $(\mathcal{X}^\perp)^\perp = (\mathcal{W}^\perp)^\perp = \mathcal{W}$ by Corollary 2.1.8.
Structural Commentary: Span invariance reduces the double complement to the preceding corollary.
Key Insight: Orthogonality depends on the span of the tested set. □

Perspective 2.1.10. *Orthogonal Versus Complementary*
Let $\mathcal{U}, \mathcal{W} \leq \mathcal{V}$ with $\mathcal{U} \perp \mathcal{W}$. By Definitions 2.1.1 and 2.1.2, $\mathcal{W} \subseteq \mathcal{U}^\perp$ and $\mathcal{U} \subseteq \mathcal{W}^\perp$, not necessarily equality.
If $\mathcal{V}$ is finite-dimensional and $\dim(\mathcal{U}) + \dim(\mathcal{W}) = \dim(\mathcal{V})$, then $\dim(\mathcal{U}^\perp) = \dim(\mathcal{W})$ and $\dim(\mathcal{W}^\perp) = \dim(\mathcal{U})$ by Corollary 2.1.7.
The two inclusions are therefore equalities by Proposition 1.4.3(ii).

Figure 26 contrasts a proper inclusion $\mathcal{W} \subsetneq \mathcal{U}^\perp$ with the saturated decomposition $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$.

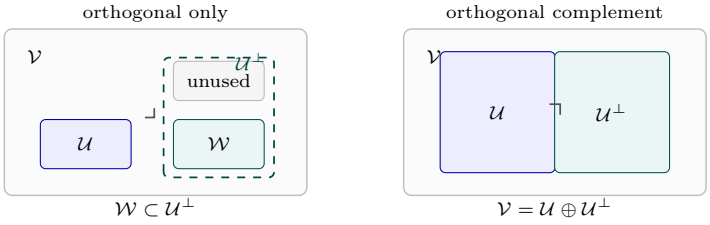

Figure 26: Orthogonal subspaces are complements only when their sum fills $\mathcal{V}$.

2.2 Orthogonal Projections

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ and $\mathcal{V}$ is an inner-product space over $\mathbb{F}$.

Proposition 2.2.1. *Orthogonal Projection onto a Line*
Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$, and set $c := \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$ and $\mathbf{w} := \mathbf{u} - c\mathbf{v}$. Then $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ and $\mathbf{w} \perp \mathbf{v}$.
The vector $\operatorname{proj}_{\mathbf{v}}(\mathbf{u}) := c\mathbf{v} = \left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|}$ is the orthogonal projection of $\mathbf{u}$ onto $\operatorname{span}(\mathbf{v})$.

Proof. Since $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\|^2 > 0$ by definiteness, so c is well-defined. The identity $\mathbf{w} = \mathbf{u} - c\mathbf{v}$ implies $\mathbf{u} = c\mathbf{v} + \mathbf{w}$.
By linearity, $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - c \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \|\mathbf{v}\|^2 = 0$, so $\mathbf{w} \perp \mathbf{v}$.
Since $\|\mathbf{v}\| \in \mathbb{R}_{>0}$, $\left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} = c\mathbf{v}$ by Proposition 1.1.9(iv).
Structural Commentary: The chosen coefficient makes the residual orthogonal.
Key Insight: Projection isolates the parallel component and leaves an orthogonal residual. □

Figure 27 depicts the decomposition $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ with $c\mathbf{v} \in \operatorname{span}(\mathbf{v})$ and $\mathbf{w} \perp \mathbf{v}$.

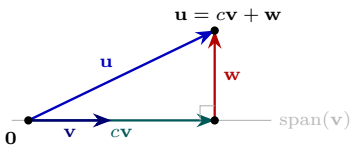

Figure 27: Writing $\mathbf{u}$ as a component parallel to $\mathbf{v}$ plus a component orthogonal to $\mathbf{v}$.

Definition 2.2.2. *Orthogonal Projection onto a Subspace*
Let $\mathcal{U}$ be finite-dimensional subspace of $\mathcal{V}$. By Theorem 2.1.5, every $\mathbf{v} \in \mathcal{V}$ has unique decomposition $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$.
The *orthogonal projection* of $\mathcal{V}$ onto $\mathcal{U}$ is the map $P_{\mathcal{U}} : \mathcal{V} \to \mathcal{V}$ defined by $P_{\mathcal{U}}\mathbf{v} := \mathbf{u}$.

Figure 28 illustrates the $\mathcal{U}$-component $P_{\mathcal{U}}\mathbf{v}$ and the residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^\perp$.

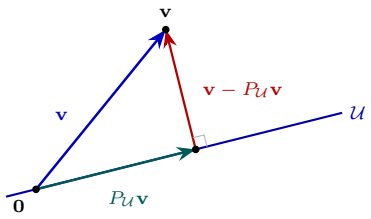

Figure 28: Orthogonal projection decomposes $\mathbf{v}$ into $P_{\mathcal{U}}\mathbf{v}$ plus an orthogonal residual.

Proposition 2.2.3. *Properties of Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. Then:

(i) $P_{\mathcal{U}} \in \mathcal{L}(\mathcal{V})$;
(ii) $(\forall\, \mathbf{u} \in \mathcal{U})\, P_{\mathcal{U}}\mathbf{u} = \mathbf{u}$;
(iii) $(\forall\, \mathbf{w} \in \mathcal{U}^{\perp})\, P_{\mathcal{U}}\mathbf{w} = \mathbf{0}$;
(iv) $\operatorname{range}(P_{\mathcal{U}}) = \mathcal{U}$;
(v) $(\forall\, \mathbf{v} \in \mathcal{V})\, \mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$;
(vi) $\ker(P_{\mathcal{U}}) = \mathcal{U}^{\perp}$;
(vii) $P_{\mathcal{U}}^2 = P_{\mathcal{U}}$;
(viii) $(\forall\, \mathbf{v} \in \mathcal{V})\, \|P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v}\|$;
(ix) if $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ is an orthonormal basis of $\mathcal{U}$, then $(\forall\, \mathbf{v} \in \mathcal{V})\, P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \mathbf{e}_j$.

Proof.

(i) Fix $a \in \mathbb{F}$ and $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$. By Theorem 2.1.5, write $\mathbf{v}_r = \mathbf{u}_r + \mathbf{w}_r$ with $\mathbf{u}_r \in \mathcal{U}$ and $\mathbf{w}_r \in \mathcal{U}^{\perp}$ for $r \in \{1, 2\}$.
Since $\mathcal{U}$ and $\mathcal{U}^{\perp}$ are subspaces, $a\mathbf{v}_1 + \mathbf{v}_2 = (a\mathbf{u}_1 + \mathbf{u}_2) + (a\mathbf{w}_1 + \mathbf{w}_2)$ is its orthogonal decomposition.
Uniqueness therefore implies $P_{\mathcal{U}}(a\mathbf{v}_1 + \mathbf{v}_2) = aP_{\mathcal{U}}\mathbf{v}_1 + P_{\mathcal{U}}\mathbf{v}_2$, so $P_{\mathcal{U}} \in \mathcal{L}(\mathcal{V})$.
(ii) Fix $\mathbf{u} \in \mathcal{U}$. The decomposition $\mathbf{u} = \mathbf{u} + \mathbf{0}$ has $\mathbf{0} \in \mathcal{U}^{\perp}$ by Proposition 2.1.3(i), so $P_{\mathcal{U}}\mathbf{u} = \mathbf{u}$.
(iii) Fix $\mathbf{w} \in \mathcal{U}^{\perp}$. The decomposition $\mathbf{w} = \mathbf{0} + \mathbf{w}$ has $\mathbf{0} \in \mathcal{U}$, so $P_{\mathcal{U}}\mathbf{w} = \mathbf{0}$.
(iv) The defining decomposition places every value of $P_{\mathcal{U}}$ in $\mathcal{U}$, so $\operatorname{range}(P_{\mathcal{U}}) \subseteq \mathcal{U}$. The reverse inclusion holds by (ii).
(v) Fix $\mathbf{v} \in \mathcal{V}$. The defining decomposition is $\mathbf{v} = P_{\mathcal{U}}\mathbf{v} + (\mathbf{v} - P_{\mathcal{U}}\mathbf{v})$ with $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$.
(vi) The inclusion $\mathcal{U}^{\perp} \subseteq \ker(P_{\mathcal{U}})$ holds by (iii). Conversely, if $P_{\mathcal{U}}\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = \mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$ by (v).
(vii) Fix $\mathbf{v} \in \mathcal{V}$. By (iv), $P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}$, so $P_{\mathcal{U}}(P_{\mathcal{U}}\mathbf{v}) = P_{\mathcal{U}}\mathbf{v}$ by (ii). Hence $P_{\mathcal{U}}^2 = P_{\mathcal{U}}$.
(viii) Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{p} := P_{\mathcal{U}}\mathbf{v}$ and $\mathbf{r} := \mathbf{v} - \mathbf{p}$. Then $\mathbf{p} \in \mathcal{U}$ by (iv) and $\mathbf{r} \in \mathcal{U}^{\perp}$ by (v).
Hence $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{r}\|^2 \ge \|\mathbf{p}\|^2$ by Theorem 1.4.3. Nonnegativity of norms implies $\|P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v}\|$.
(ix) Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{p} := \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \mathbf{e}_j \in \mathcal{U}$. For $k \in \{1, \dots, m\}$, $\langle \mathbf{v} - \mathbf{p}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$ by orthonormality.
If $\mathbf{x} \in \mathcal{U}$, write $\mathbf{x} = \sum_{k=1}^{m} a_k \mathbf{e}_k$; then $\langle \mathbf{v} - \mathbf{p}, \mathbf{x} \rangle = \sum_{k=1}^{m} \overline{a_k}\, \langle \mathbf{v} - \mathbf{p}, \mathbf{e}_k \rangle = 0$ by Proposition 1.1.9(iii)–(iv).
Thus $\mathbf{v} - \mathbf{p} \in \mathcal{U}^{\perp}$, and $P_{\mathcal{U}}\mathbf{v} = \mathbf{p}$ by uniqueness in Theorem 2.1.5.

Structural Commentary: The orthogonal decomposition organizes all nine arguments.
Key Insight: Projection preserves the $\mathcal{U}$-component and annihilates the $\mathcal{U}^{\perp}$-component. □

Definition 2.2.4. *Rank-One Projection Matrix*
Let $\mathbf{u} = (u_1, \dots, u_n)^t \in \mathbb{F}^n$ be a unit vector. The *rank-one projection matrix* onto $\operatorname{span}(\mathbf{u})$ is $P_{\mathbf{u}} := [u_i \overline{u_j}]_{i,j=1}^{n} \in \operatorname{Mat}(n, n, \mathbb{F})$.
It satisfies $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u}$.

Proposition 2.2.5. *Properties of Rank-One Projection Matrices*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. Then:

(i) $\operatorname{range}(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})$;
(ii) $\ker(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})^{\perp}$;
(iii) $(\forall, i, j \in \{1, \dots, n\})\, (P_{\mathbf{u}})_{ij} = \overline{(P_{\mathbf{u}})_{ji}}$;
(iv) $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.

Proof.

(i) For $\mathbf{x} \in \mathbb{F}^n$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u} \in \operatorname{span}(\mathbf{u})$, so $\operatorname{range}(P_{\mathbf{u}}) \subseteq \operatorname{span}(\mathbf{u})$. Since $P_{\mathbf{u}}\mathbf{u} = \langle \mathbf{u}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u} = \mathbf{u}$, the reverse inclusion also holds.
(ii) Fix $\mathbf{x} \in \mathbb{F}^n$. Since $\mathbf{u} \neq \mathbf{0}$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n}\, \mathbf{u} = \mathbf{0}$ if and only if $\langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} = 0$, which is equivalent to $\mathbf{x} \in \operatorname{span}(\mathbf{u})^{\perp}$.
(iii) For $i, j \in \{1, \dots, n\}$, $(P_{\mathbf{u}})_{ij} = u_i \overline{u_j} = \overline{u_j \overline{u_i}} = \overline{(P_{\mathbf{u}})_{ji}}$.
(iv) For $i, k \in \{1, \dots, n\}$, $(P_{\mathbf{u}}^2)_{ik} = \sum_{j=1}^{n} u_i \overline{u_j} u_j \overline{u_k} = u_i \overline{u_k} \sum_{j=1}^{n} |u_j|^2 = u_i \overline{u_k} = (P_{\mathbf{u}})_{ik}$ because $\mathbf{u}$ is a unit vector.

Structural Commentary: The projection formula proves all four matrix identities.
Key Insight: The matrix preserves the $\mathbf{u}$ direction and annihilates its orthogonal complement. □

Theorem 2.2.6. *Orthogonal Projection Minimizes Distance*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. If $\mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \in \mathcal{U}$, then $\|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v} - \mathbf{u}\|$, with equality if and only if $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$.

Proof. Set $\mathbf{p} := P_{\mathcal{U}}\mathbf{v}$, $\mathbf{r} := \mathbf{v} - \mathbf{p}$, and $\mathbf{h} := \mathbf{p} - \mathbf{u}$. By Proposition 2.2.3(iv)–(v), $\mathbf{p}, \mathbf{h} \in \mathcal{U}$ and $\mathbf{r} \in \mathcal{U}^{\perp}$, so $\mathbf{r} \perp \mathbf{h}$.
Since $\mathbf{v} - \mathbf{u} = \mathbf{r} + \mathbf{h}$, $\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{r}\|^2 + \|\mathbf{h}\|^2 \ge \|\mathbf{r}\|^2 = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|^2$ by Theorem 1.4.3. Nonnegativity of norms proves the inequality.
($\Rightarrow$) If equality holds, then the preceding identity forces $\|\mathbf{h}\| = 0$, so $\mathbf{h} = \mathbf{0}$ by definiteness and $\mathbf{u} = \mathbf{p} = P_{\mathcal{U}}\mathbf{v}$.
($\Leftarrow$) If $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$, then $\mathbf{h} = \mathbf{0}$ and equality holds in the same identity.

Structural Commentary: Pythagoras isolates the nonnegative correction term.
Key Insight: The orthogonal projection is the unique nearest point in the subspace. □

Definition 2.2.7. *Distance to a Subspace*
Let $\mathcal{U}$ be a nonempty subset of $\mathcal{V}$. The *distance* from $\mathbf{v} \in \mathcal{V}$ to $\mathcal{U}$ is $d(\mathbf{v},\mathcal{U}) := \inf_{\mathbf{u}\in\mathcal{U}} \|\mathbf{v}-\mathbf{u}\|$.
If $\mathcal{U}$ is a finite-dimensional subspace, then $d(\mathbf{v},\mathcal{U}) = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|$ by Theorem 2.2.6.

Definition 2.2.8. *Fourier Coefficients and Expansion*
Let $\mathcal{O} = (\mathbf{e}_1,\dots,\mathbf{e}_m)$ be orthonormal list, $\mathbf{v} \in \mathcal{V}$. The *Fourier coefficients* of $\mathbf{v}$ relative to $\mathcal{O}$ are $c_j(\mathbf{v}) := \langle \mathbf{v}, \mathbf{e}_j\rangle$ for $j \in \{1,\dots,m\}$.
The *Fourier expansion* of $\mathbf{v}$ onto $\mathcal{O}$ is $\widehat{\mathbf{v}}_{\mathcal{O}} := \sum_{j=1}^{m} c_j(\mathbf{v})\mathbf{e}_j$.

Theorem 2.2.9. *Fourier Expansion onto a Finite Orthonormal Set*
Let $\mathcal{O} = (\mathbf{e}_1,\dots,\mathbf{e}_m)$ be orthonormal, let $\mathcal{S} = \operatorname{span}(\mathcal{O})$, and let $\mathbf{v} \in \mathcal{V}$. Then:
(i) $\widehat{\mathbf{v}}_{\mathcal{O}}$ *is the unique* $\mathbf{s} \in \mathcal{S}$ *with* $\mathbf{v} - \mathbf{s} \in \mathcal{S}^{\perp}$;
(ii) $(\forall\, \mathbf{s} \in \mathcal{S})\ \|\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}\| \le \|\mathbf{v} - \mathbf{s}\|$, *and* $\widehat{\mathbf{v}}_{\mathcal{O}}$ *is the only* $\mathbf{s} \in \mathcal{S}$ *for which equality holds;*
(iii) $\|\widehat{\mathbf{v}}_{\mathcal{O}}\| \le \|\mathbf{v}\|$.

Proof.
(i) By construction, $\widehat{\mathbf{v}}_{\mathcal{O}} \in \mathcal{S}$. For $k \in \{1,\dots,m\}$, $\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{e}_k\rangle = c_k(\mathbf{v}) - \sum_{j=1}^{m} c_j(\mathbf{v})\delta_{jk} = 0$.
If $\mathbf{x} \in \mathcal{S}$, write $\mathbf{x} = \sum_{k=1}^{m} a_k\mathbf{e}_k$; then $\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{x}\rangle = \sum_{k=1}^{m} \overline{a_k}\,\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{e}_k\rangle = 0$ by Proposition 1.1.9(iii)–(iv). Hence $\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}} \in \mathcal{S}^{\perp}$.
If $\mathbf{s} \in \mathcal{S}$ also satisfies $\mathbf{v} - \mathbf{s} \in \mathcal{S}^{\perp}$, then $\mathbf{s} - \widehat{\mathbf{v}}_{\mathcal{O}} = (\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}) - (\mathbf{v} - \mathbf{s}) \in \mathcal{S} \cap \mathcal{S}^{\perp}$, so $\mathbf{s} = \widehat{\mathbf{v}}_{\mathcal{O}}$ by Proposition 2.1.3(iv).
(ii) By (i), $\widehat{\mathbf{v}}_{\mathcal{O}} = P_{\mathcal{S}}\mathbf{v}$. The inequality and equality condition hold by Theorem 2.2.6.
(iii) If $m = 0$, then $\widehat{\mathbf{v}}_{\mathcal{O}} = \mathbf{0}$ and the claim is immediate. Suppose $m \ge 1$.
By Proposition 1.3.2 and Theorem 1.4.5, $\|\widehat{\mathbf{v}}_{\mathcal{O}}\|^2 = \sum_{j=1}^{m} |c_j(\mathbf{v})|^2 \le \|\mathbf{v}\|^2$, so $\|\widehat{\mathbf{v}}_{\mathcal{O}}\| \le \|\mathbf{v}\|$ by nonnegativity of norms.

Structural Commentary: Coordinate construction and nearest-point optimality prove all three claims.
Key Insight: Fourier expansion is orthogonal projection expressed in orthonormal coordinates. □

Theorem 2.2.10. *Characterizing Finite Orthonormal Bases*
Let $\mathcal{O} = (\mathbf{e}_1,\dots,\mathbf{e}_m)$ be orthonormal, $\mathcal{S} = \operatorname{span}(\mathcal{O})$. Write $c_j(\mathbf{v})$, $\widehat{\mathbf{v}} := \widehat{\mathbf{v}}_{\mathcal{O}}$ as in Definition 2.2.8. The following are equivalent:
(i) $\mathcal{S} = \mathcal{V}$.
(ii) $\mathcal{S}^{\perp} = \{\mathbf{0}\}$.
(iii) $(\forall\, \mathbf{v} \in \mathcal{V})\ \widehat{\mathbf{v}} = \mathbf{v}$.
(iv) $(\forall\, \mathbf{v},\mathbf{w} \in \mathcal{V})\ \langle \mathbf{v},\mathbf{w}\rangle = \sum_{j=1}^{m} c_j(\mathbf{v})\overline{c_j(\mathbf{w})}$.
(v) $(\forall\, \mathbf{v} \in \mathcal{V})\ \|\widehat{\mathbf{v}}\| = \|\mathbf{v}\|$.

Proof.
(i) $\Rightarrow$ (ii) If $\mathcal{S} = \mathcal{V}$, then $\mathcal{S}^{\perp} = \mathcal{V}^{\perp} = \{\mathbf{0}\}$ by Proposition 2.1.3(iii).
(ii) $\Rightarrow$ (iii) Fix $\mathbf{v} \in \mathcal{V}$. By Theorem 2.2.9(i), $\mathbf{v} - \widehat{\mathbf{v}} \in \mathcal{S}^{\perp} = \{\mathbf{0}\}$, so $\widehat{\mathbf{v}} = \mathbf{v}$.
(iii) $\Rightarrow$ (iv) Fix $\mathbf{v},\mathbf{w} \in \mathcal{V}$. By (iii), $\mathbf{v} = \sum_{j=1}^{m} c_j(\mathbf{v})\mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{m} c_k(\mathbf{w})\mathbf{e}_k$.
Then $\langle \mathbf{v},\mathbf{w}\rangle = \sum_{j=1}^{m} c_j(\mathbf{v})\overline{c_j(\mathbf{w})}$ by Proposition 1.1.9(iii)–(iv) and orthonormality.
(iv) $\Rightarrow$ (v) Fix $\mathbf{v} \in \mathcal{V}$ and set $\mathbf{w} = \mathbf{v}$ in (iv). Then $\|\mathbf{v}\|^2 = \sum_{j=1}^{m} |c_j(\mathbf{v})|^2 = \|\widehat{\mathbf{v}}\|^2$ by Proposition 1.3.2, so $\|\widehat{\mathbf{v}}\| = \|\mathbf{v}\|$.
(v) $\Rightarrow$ (i) Fix $\mathbf{v} \in \mathcal{V}$. By Theorem 2.2.9(i), $\widehat{\mathbf{v}} \in \mathcal{S}$ and $\mathbf{v} - \widehat{\mathbf{v}} \in \mathcal{S}^{\perp}$. Hence $\|\mathbf{v}\|^2 = \|\widehat{\mathbf{v}}\|^2 + \|\mathbf{v} - \widehat{\mathbf{v}}\|^2$ by Theorem 1.4.3.
Condition (v) implies $\|\mathbf{v} - \widehat{\mathbf{v}}\| = 0$, so $\mathbf{v} = \widehat{\mathbf{v}} \in \mathcal{S}$ by definiteness. Thus $\mathcal{V} \subseteq \mathcal{S}$, while $\mathcal{S} \subseteq \mathcal{V}$ by construction, therefore $\mathcal{S} = \mathcal{V}$.

Structural Commentary: A five-step implication cycle proves equivalence.
Key Insight: An orthonormal list is a basis exactly when its projection loses no component. □

Perspective 2.2.11. *Why the Finite-Dimensional Hypothesis Matters*
In $\ell^2(\mathbb{F})$, let $\mathcal{S} = \mathbb{F}^{(\mathbb{N})} = \operatorname{span}\{\mathbf{e}_j : j \ge 1\}$. If $\mathbf{x} \in \mathcal{S}^{\perp}$, then $(\forall\, j \in \mathbb{N})\, x_j = \langle \mathbf{x}, \mathbf{e}_j\rangle = 0$, so $\mathcal{S}^{\perp} = \{\mathbf{0}\}$.
The vector $(2^{-j})_{j\ge1}$ belongs to $\ell^2(\mathbb{F})$ but not to $\mathcal{S}$, hence $\mathcal{S} \subsetneq \ell^2(\mathbb{F})$ and $\mathcal{S} \oplus \mathcal{S}^{\perp} \neq \ell^2(\mathbb{F})$.

2.3 Adjoints and Fundamental Subspaces

Adjoints express dual action in inner-product language. They become conjugate transposes in orthonormal coordinates and exchange kernels, ranges, and orthogonal complements.

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.3.1. *Adjoint of a Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$.
A function $S : \mathcal{W} \to \mathcal{V}$ is an *adjoint* of T if $(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\ \langle T\mathbf{v},\mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, S\mathbf{w}\rangle_{\mathcal{V}}$.

Perspective 2.3.2. *Dual Map and Inner-Product Adjoint*
For $T : \mathcal{V} \to \mathcal{W}$, the pullback $\varphi \mapsto \varphi \circ T$ is the *dual map* $T' : \mathcal{W}' \to \mathcal{V}'$, also called the *operator adjoint* and written $T^{\times}$.
This algebraic map is distinct from the inner-product adjoint T^*, whose construction requires inner products.

Lemma 2.3.3. *Uniqueness of Adjoints*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V},\mathcal{W})$. Then T has at most one adjoint.

Proof. Suppose $S_1, S_2 : \mathcal{W} \to \mathcal{V}$ are adjoints of T, $\mathbf{w} \in \mathcal{W}$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{v}, S_1\mathbf{w}\rangle_{\mathcal{V}} = \langle \mathbf{v}, S_2\mathbf{w}\rangle_{\mathcal{V}}$ by the two adjoint identities.
Hence $S_1\mathbf{w} = S_2\mathbf{w}$ by Lemma 1.1.10(i). Since $\mathbf{w}$ was arbitrary, $S_1 = S_2$.
Structural Commentary: Pointwise comparison reduces uniqueness to separation by inner products.
Key Insight: Distinct vectors cannot agree against every inner-product test. □

Theorem 2.3.4. *Existence of the Adjoint*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $\mathcal{W}$ be an inner-product space, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) T has an adjoint $T^* : \mathcal{W} \to \mathcal{V}$;
(ii) the adjoint T^* is unique;
(iii) $T^* \in \mathcal{L}(\mathcal{W}, \mathcal{V})$.

Proof.

(i) Fix $\mathbf{w} \in \mathcal{W}$. The map $\varphi_{\mathbf{w}} := \langle T(\cdot), \mathbf{w} \rangle_{\mathcal{W}}$ belongs to $\mathcal{V}'$ by linearity of T and the first inner-product slot.
The unique $T^*\mathbf{w} \in \mathcal{V}$ satisfying $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w} \rangle_{\mathcal{V}}$ exists by Theorem 1.6.1, and resulting map T^* is adjoint.
(ii) The adjoint T^* is unique by Lemma 2.3.3.
(iii) Fix $a \in \mathbb{F}$, $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$. Then $\langle \mathbf{v}, T^*(a\mathbf{w}_1 + \mathbf{w}_2) \rangle_{\mathcal{V}} = \langle T\mathbf{v}, a\mathbf{w}_1 + \mathbf{w}_2 \rangle_{\mathcal{W}} = \overline{a}\, \langle T\mathbf{v}, \mathbf{w}_1 \rangle_{\mathcal{W}} + \langle T\mathbf{v}, \mathbf{w}_2 \rangle_{\mathcal{W}} = \langle \mathbf{v}, aT^*\mathbf{w}_1 + T^*\mathbf{w}_2 \rangle_{\mathcal{V}}$ for $\mathbf{v} \in \mathcal{V}$ by conjugate linearity in the second slot and the adjoint identity.
Hence $T^*(a\mathbf{w}_1 + \mathbf{w}_2) = aT^*\mathbf{w}_1 + T^*\mathbf{w}_2$ by Lemma 1.1.10(i), so T^* is linear.

Structural Commentary: Riesz representation constructs codomain-indexed vectors; separation verifies that their assignment is linear.
Key Insight: Riesz vectors assemble pointwise into the adjoint. □

Figure 29 records the inner-product identity that transfers T from the first slot to T^* in the second slot.

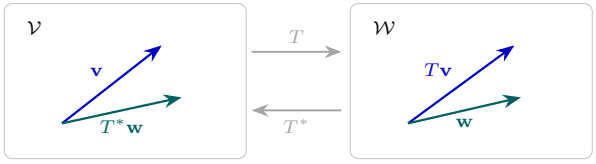

Figure 29: Adjoint relation: $\langle T\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w} \rangle_{\mathcal{V}}$.

Perspective 2.3.5. *Inner-Product Adjoints as Riesz-Transported Dual Maps*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces, $R_{\mathcal{V}} : \mathcal{V}' \to \mathcal{V}$ and $R_{\mathcal{W}} : \mathcal{W}' \to \mathcal{W}$ be Riesz maps from Theorem 1.6.3.
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the dual map $T' : \mathcal{W}' \to \mathcal{V}'$ and the inner-product adjoint T^* satisfy $T' = (R_{\mathcal{V}})^{-1} \circ T^* \circ R_{\mathcal{W}}$.
Indeed, if $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, then $(T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v}) = \langle T\mathbf{v}, R_{\mathcal{W}}\varphi \rangle_{\mathcal{W}} = \langle \mathbf{v}, T^* R_{\mathcal{W}}\varphi \rangle_{\mathcal{V}}$.
Thus T' pulls back functionals by moving their Riesz vectors with T^*.
Structural Commentary: Riesz representation conjugates the algebraic dual map into an inner-product operator.
Key Insight: Dual maps and adjoints encode the same reversed action.

Figure 30 displays the commuting square expressed by $R_{\mathcal{V}} T' = T^* R_{\mathcal{W}}$.

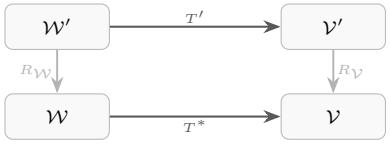

Figure 30: Riesz transports the dual map to the adjoint: $R_{\mathcal{V}} T' = T^* R_{\mathcal{W}}$.

Definition 2.3.6. *Conjugate Transpose*
For $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, the *conjugate transpose* of A is $A^* = [\overline{a_{ji}}] \in \mathrm{Mat}(n, m, \mathbb{F})$.
Over $\mathbb{R}$, the conjugate transpose is the ordinary transpose.

Theorem 2.3.7. *Matrix of the Adjoint*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces.
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and $\gamma = (\mathbf{f}_1, \ldots, \mathbf{f}_m)$ be orthonormal bases of $\mathcal{V}$ and $\mathcal{W}$, respectively. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[T^*]_{\beta \leftarrow \gamma} = ([T]_{\gamma \leftarrow \beta})^*$.

Proof. Set $A := [T]_{\gamma \leftarrow \beta}$ and $B := [T^*]_{\beta \leftarrow \gamma}$.
For $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, m\}$, $A_{j,i} = \langle T\mathbf{e}_i, \mathbf{f}_j \rangle_{\mathcal{W}}$ and $B_{i,j} = \langle T^*\mathbf{f}_j, \mathbf{e}_i \rangle_{\mathcal{V}}$ by Theorem 1.4.8.
Hence $B_{i,j} = \overline{\langle \mathbf{e}_i, T^*\mathbf{f}_j \rangle_{\mathcal{V}}} = \overline{\langle T\mathbf{e}_i, \mathbf{f}_j \rangle_{\mathcal{W}}} = \overline{A_{j,i}}$ by conjugate symmetry and the adjoint identity, so $B = A^*$.
Structural Commentary: Orthonormal coordinates convert the adjoint identity into entrywise conjugate transposition.
Key Insight: Moving an operator across an inner product becomes conjugate transposition in orthonormal bases. □

Proposition 2.3.8. *Basic Adjoint Identities*
Let $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ be finite-dimensional inner-product spaces. Then:

(i) $(aS + T)^* = \overline{a}S^* + T^*$ for $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$;
(ii) $(SR)^* = R^*S^*$ for $R \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
(iii) $(T^*)^* = T$ for $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
(iv) $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Proof. Each candidate is identified by Lemma 2.3.3 after verification of the adjoint identity.

(i) For $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$, $\langle (aS + T)\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = a\, \langle S\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} + \langle T\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = \langle \mathbf{v}, (\overline{a}S^* + T^*)\mathbf{w} \rangle_{\mathcal{V}}$. Hence $(aS + T)^* = \overline{a}S^* + T^*$.
(ii) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, $\langle SR\mathbf{u}, \mathbf{w} \rangle_{\mathcal{W}} = \langle R\mathbf{u}, S^*\mathbf{w} \rangle_{\mathcal{V}} = \langle \mathbf{u}, R^*S^*\mathbf{w} \rangle_{\mathcal{U}}$. Hence $(SR)^* = R^*S^*$.
(iii) Conjugate symmetry transforms $(\forall\, \mathbf{v} \in \mathcal{V})(\forall\, \mathbf{w} \in \mathcal{W})\ \langle T\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w} \rangle_{\mathcal{V}}$ into $(\forall\, \mathbf{w} \in \mathcal{W})(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T^*\mathbf{w}, \mathbf{v} \rangle_{\mathcal{V}} = \langle \mathbf{w}, T\mathbf{v} \rangle_{\mathcal{W}}$.
Thus T is the adjoint of T^*, so $(T^*)^* = T$.
(iv) For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle I_{\mathcal{V}}\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, I_{\mathcal{V}}\mathbf{w} \rangle$. Thus $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Structural Commentary: Each algebraic operation is transported through the two inner-product slots and then settled by uniqueness.
Key Insight: Adjoint formation conjugates scalars, reverses products, and is involutive. □

Proposition 2.3.9. *Adjoints of Inverses and Polynomials*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space. For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, set $p^{\#}(z) := \sum_{j=0}^{m} \overline{a_j} z^j$.

(i) If T is invertible, then $(T^*)^{-1} = (T^{-1})^*$.
(ii) $p(T)^* = p^{\#}(T^*)$.
(iii) $p \in \mathbb{R}[z] \Longrightarrow p(T)^* = p(T^*)$.

Proof.
(i) Identities $(T^{-1})^* T^* = (TT^{-1})^* = I_{\mathcal{V}}$ and $T^*(T^{-1})^* = (T^{-1}T)^* = I_{\mathcal{V}}$ hold by Proposition 2.3.8(ii),(iv); $(T^*)^{-1} = (T^{-1})^*$.
(ii) The base identity $(T^0)^* = I_{\mathcal{V}} = (T^*)^0$ holds by Proposition 2.3.8(iv).
If $(T^j)^* = (T^*)^j$, then $(T^{j+1})^* = (T^j T)^* = T^*(T^j)^* = (T^*)^{j+1}$ by Proposition 2.3.8(ii).
Thus $(\forall\, j \in \{0, \dots, m\})\, (T^j)^* = (T^*)^j$ by induction.
The identity $p(T)^* = (\sum_{j=0}^{m} a_j T^j)^* = \sum_{j=0}^{m} \overline{a_j} (T^*)^j = p^{\#}(T^*)$ holds by repeated application of Proposition 2.3.8(i).
(iii) If $p \in \mathbb{R}[z]$, then $p^{\#} = p$, so $p(T)^* = p(T^*)$ by (ii).

Structural Commentary: The product law controls inverses and powers, while conjugate additivity controls polynomial coefficients.
Key Insight: Adjoint formation applies coefficientwise conjugation to polynomial expressions in an operator. □

Proposition 2.3.10. *Adjoint Kernel-Range Orthogonality*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then:

(i) $\ker(T^*) = (\operatorname{range}(T))^{\perp}$;
(ii) $\ker(T) = (\operatorname{range}(T^*))^{\perp}$;
(iii) $\operatorname{range}(T^*) = (\ker(T))^{\perp}$;
(iv) $\operatorname{range}(T) = (\ker(T^*))^{\perp}$.

Equivalently, an adjoint annihilates exactly the vectors orthogonal to the range of the original map.

Proof.
(i) For $\mathbf{w} \in \mathcal{W}$, $\mathbf{w} \in \ker(T^*) \iff (\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{v}, T^*\mathbf{w} \rangle_{\mathcal{V}} = 0 \iff (\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{w}, T\mathbf{v} \rangle_{\mathcal{W}} = 0 \iff \mathbf{w} \in (\operatorname{range}(T))^{\perp}$ by Lemma 1.1.10(ii), the adjoint identity, and conjugate symmetry.
(ii) The identity $\ker(T) = \ker((T^*)^*) = (\operatorname{range}(T^*))^{\perp}$ holds by (i) for T^* and Proposition 2.3.8(iii).
(iii) Taking orthogonal complements in (ii), $(\ker(T))^{\perp} = ((\operatorname{range}(T^*))^{\perp})^{\perp} = \operatorname{range}(T^*)$ by Corollary 2.1.8.
(iv) Taking orthogonal complements in (i), $(\ker(T^*))^{\perp} = ((\operatorname{range}(T))^{\perp})^{\perp} = \operatorname{range}(T)$ by Corollary 2.1.8.

Structural Commentary: The kernel formula, involution, and double complementation establish all four identities.
Key Insight: Kernels and ranges are exchanged by taking an adjoint and an orthogonal complement. □

Proposition 2.3.11. *Adjoint Products and Algebraic Projections*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) $(\ker(T^*T), \ker(TT^*)) = (\ker(T), \ker(T^*))$.
(ii) $(\operatorname{range}(T^*T), \operatorname{range}(TT^*)) = (\operatorname{range}(T^*), \operatorname{range}(T))$.
(iii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$, then $(\rho_{\mathcal{U},\mathcal{Z}})^* = \rho_{\mathcal{Z}^{\perp},\mathcal{U}^{\perp}}$.

Proof.
(i) Fix $\mathbf{v} \in \mathcal{V}$. Then $\mathbf{v} \in \ker(T^*T) \iff 0 = \langle \mathbf{v}, T^*T\mathbf{v} \rangle_{\mathcal{V}} = \langle T\mathbf{v}, T\mathbf{v} \rangle_{\mathcal{W}} = \|T\mathbf{v}\|^2 \iff \mathbf{v} \in \ker(T)$ by adjointness and definiteness.
The equality $\ker(TT^*) = \ker(T^*)$ follows by this result for T^* and Proposition 2.3.8(iii).
(ii) The identities $(T^*T)^* = T^*T$ and $(TT^*)^* = TT^*$ hold by Proposition 2.3.8(ii),(iii).
The identity $\operatorname{range}(T^*T) = (\ker(T^*T))^{\perp} = (\ker(T))^{\perp} = \operatorname{range}(T^*)$ holds by (i) and Proposition 2.3.10(iii).
The identity $\operatorname{range}(TT^*) = (\ker(TT^*))^{\perp} = (\ker(T^*))^{\perp} = \operatorname{range}(T)$ holds by (i) and Proposition 2.3.10(iv).
(iii) Suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$ and set $P := \rho_{\mathcal{U},\mathcal{Z}}$. Fix $\mathbf{c} \in \mathcal{Z}^{\perp} \cap \mathcal{U}^{\perp}$ and $\mathbf{x} \in \mathcal{V}$. Write $\mathbf{x} = \mathbf{u} + \mathbf{z}$ with $(\mathbf{u}, \mathbf{z}) \in \mathcal{U} \times \mathcal{Z}$.
Then $\langle \mathbf{c}, \mathbf{x} \rangle = \langle \mathbf{c}, \mathbf{u} \rangle + \langle \mathbf{c}, \mathbf{z} \rangle = 0$, so $\mathbf{c} = \mathbf{0}$ by Proposition 2.1.3(iii). Hence $\mathcal{Z}^{\perp} \cap \mathcal{U}^{\perp} = \{\mathbf{0}\}$.
The identity $\dim(\mathcal{Z}^{\perp} \oplus \mathcal{U}^{\perp}) = \dim(\mathcal{V})$ follows by Corollaries 2.1.7 and 2.3.2, because $\dim(\mathcal{U}) + \dim(\mathcal{Z}) = \dim(\mathcal{V})$.
Hence $\mathcal{V} = \mathcal{Z}^{\perp} \oplus \mathcal{U}^{\perp}$ by Proposition 1.4.3(ii), and $Q := \rho_{\mathcal{Z}^{\perp},\mathcal{U}^{\perp}}$ is defined.
Fix $\mathbf{x}, \mathbf{y} \in \mathcal{V}$ and write $\mathbf{x} = \mathbf{u} + \mathbf{z}$ and $\mathbf{y} = \mathbf{a} + \mathbf{b}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{z} \in \mathcal{Z}$, $\mathbf{a} \in \mathcal{Z}^{\perp}$, and $\mathbf{b} \in \mathcal{U}^{\perp}$.
By the projection formulas, $P\mathbf{x} = \mathbf{u}$ and $Q\mathbf{y} = \mathbf{a}$; by conjugate symmetry, $\langle \mathbf{u}, \mathbf{b} \rangle = \langle \mathbf{z}, \mathbf{a} \rangle = 0$.
Consequently, $\langle P\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{u}, \mathbf{a} + \mathbf{b} \rangle = \langle \mathbf{u}, \mathbf{a} \rangle = \langle \mathbf{u} + \mathbf{z}, \mathbf{a} \rangle = \langle \mathbf{x}, Q\mathbf{y} \rangle$. Thus Q is the adjoint of P, and $P^* = Q$ by Lemma 2.3.3.

Structural Commentary: Positivity fixes kernels, complementation fixes ranges, and decomposition fixes projection adjoints.
Key Insight: Adjoints preserve effective directions and orthogonally exchange projection complements. □

Corollary 2.3.12. *Matrix Fundamental Subspaces as Orthogonal Complements*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ act as $A : \mathbb{F}^n \to \mathbb{F}^m$ with the standard inner products. Then:

(i) $\ker(A) = \operatorname{col}(A^*)^{\perp}$;
(ii) $\operatorname{col}(A^*) = \ker(A)^{\perp}$;
(iii) $\ker(A^*) = \operatorname{col}(A)^{\perp}$;
(iv) $\operatorname{col}(A) = \ker(A^*)^{\perp}$.
(v) $\mathbb{F} = \mathbb{R} \Longrightarrow \mathbb{R}^n = \ker(A) \odot \operatorname{row}(A)$.
(vi) $\mathbb{F} = \mathbb{R} \Longrightarrow \mathbb{R}^m = \ker(A^t) \odot \operatorname{col}(A)$.

Proof. The induced map and its adjoint have ranges $\operatorname{col}(A)$ and $\operatorname{col}(A^*)$, respectively, by Theorem 2.3.7.
(i) The identity $\ker(A) = \operatorname{col}(A^*)^{\perp}$ holds by Proposition 2.3.10(ii).
(ii) The identity $\operatorname{col}(A^*) = \ker(A)^{\perp}$ holds by Proposition 2.3.10(iii).
(iii) The identity $\ker(A^*) = \operatorname{col}(A)^{\perp}$ holds by Proposition 2.3.10(i).
(iv) The identity $\operatorname{col}(A) = \ker(A^*)^{\perp}$ holds by Proposition 2.3.10(iv).
(v) If $\mathbb{F} = \mathbb{R}$, then $A^* = A^t$ and $\operatorname{col}(A^t) = \operatorname{row}(A)$. Hence $\mathbb{R}^n = \ker(A) \odot \operatorname{row}(A)$ by (i),(ii) and Theorem 2.1.5.
(vi) If $\mathbb{F} = \mathbb{R}$, then $A^* = A^t$. Hence $\mathbb{R}^m = \ker(A^t) \odot \operatorname{col}(A)$ by (iii),(iv) and Theorem 2.1.5.

Structural Commentary: Standard coordinates convert operator formulas into matrix fundamental-subspace identities.
Key Insight: Row and column spaces are orthogonal complements of their nullspaces. □

2.4 Least Squares and Metric Optimization

Least squares is projection theory as optimization: normal equations, full-rank formulas, and minimum-norm solutions.

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Principle 2.4.1. *Least Squares as Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of an inner-product space $\mathcal{V}$, and let $\mathbf{v} \in \mathcal{V}$.
The best approximation to $\mathbf{v}$ from $\mathcal{U}$ is $P_{\mathcal{U}}\mathbf{v}$ by Theorem 2.2.6. The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ belongs to $\mathcal{U}^{\perp}$ by Proposition 2.2.3(v).

Definition 2.4.2. *Matrix Least-Squares Solution*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$. A vector $\mathbf{x}_0 \in \mathbb{F}^n$ is a *least-squares solution* of $A\mathbf{x} = \mathbf{y}$ if $(\forall\, \mathbf{x} \in \mathbb{F}^n)\ \|A\mathbf{x}_0 - \mathbf{y}\|_2 \le \|A\mathbf{x} - \mathbf{y}\|_2$.
The vector $A\mathbf{x}_0$ is the *fitted vector*, and $\mathbf{y} - A\mathbf{x}_0$ is the *residual*.

Theorem 2.4.3. *Normal Equations for Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of an inner-product space $\mathcal{V}$, suppose $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_n)$, and fix $\mathbf{v} \in \mathcal{V}$.
For $\mathbf{c} = (c_1, \dots, c_n) \in \mathbb{F}^n$, write $\mathsf{N}(\mathbf{c})$ for $(\forall\, i \in \{1, \dots, n\})\ \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i \rangle = \langle \mathbf{v}, \mathbf{u}_i \rangle$. Then:

(i) $(\forall\, \mathbf{c} = (c_1, \dots, c_n) \in \mathbb{F}^n)\ [P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j \iff \mathsf{N}(\mathbf{c})]$;
(ii) $(\exists\, \mathbf{c} \in \mathbb{F}^n)\ \mathsf{N}(\mathbf{c})$;
(iii) *if* $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ *is linearly independent, then* $(\exists!\, \mathbf{c} \in \mathbb{F}^n)\ \mathsf{N}(\mathbf{c})$.

Proof.

(i) $(\Rightarrow)$ Fix $\mathbf{c} = (c_1, \dots, c_n) \in \mathbb{F}^n$ and suppose $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$. The residual belongs to $\mathcal{U}^{\perp}$ by Proposition 2.2.3(v).

Hence $(\forall\, i \in \{1, \dots, n\})\ 0 = \langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i \rangle = \langle \mathbf{v}, \mathbf{u}_i \rangle - \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i \rangle$, so $\mathsf{N}(\mathbf{c})$ holds.

(i) $(\Leftarrow)$ Fix $\mathbf{c} = (c_1, \dots, c_n) \in \mathbb{F}^n$, suppose $\mathsf{N}(\mathbf{c})$, and set $\mathbf{q} := \sum_{j=1}^{n} c_j \mathbf{u}_j \in \mathcal{U}$.

For $\mathbf{u} = \sum_{i=1}^{n} a_i \mathbf{u}_i \in \mathcal{U}$, $\langle \mathbf{v} - \mathbf{q}, \mathbf{u} \rangle = \sum_{i=1}^{n} \overline{a_i} \langle \mathbf{v} - \mathbf{q}, \mathbf{u}_i \rangle = 0$ by $\mathsf{N}(\mathbf{c})$ and Proposition 1.1.9(iii)–(iv).
Thus $\mathbf{v} - \mathbf{q} \in \mathcal{U}^{\perp}$, and $\mathbf{q} = P_{\mathcal{U}}\mathbf{v}$ by uniqueness in Theorem 2.1.5.

(ii) Choose $\mathbf{c} = (c_1, \dots, c_n) \in \mathbb{F}^n$ with $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$.

This choice is possible because $P_{\mathcal{U}}\mathbf{v} \in \mathcal{U} = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_n)$ by Proposition 2.2.3(iv). Then $\mathsf{N}(\mathbf{c})$ holds by (i).

(iii) Suppose $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ is linearly independent. Existence holds by (ii).

If $\mathsf{N}(\mathbf{c})$ and $\mathsf{N}(\mathbf{d})$, then $\sum_{j=1}^{n} c_j \mathbf{u}_j = P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} d_j \mathbf{u}_j$ by (i), so $\mathbf{c} = \mathbf{d}$ by linear independence.

Structural Commentary: Projection membership supplies coefficients; residual orthogonality supplies the equations and converse.
Key Insight: Normal equations are coordinates of an orthogonal residual. □

Figure 31 illustrates the residual orthogonality encoded by the component equations.

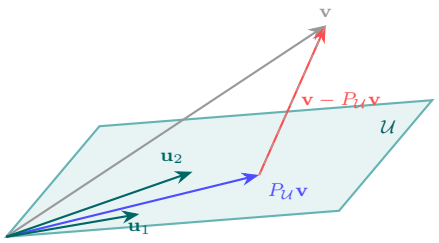

Figure 31: Normal equations impose $(\forall\, i \in \{1, \dots, n\})\ \langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i \rangle = 0$.

Theorem 2.4.4. *Matrix Least Squares and Normal Equations*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, and set $\mathrm{LS}(A, \mathbf{y}) := \arg\min_{\mathbf{x} \in \mathbb{F}^n} \|A\mathbf{x} - \mathbf{y}\|_2$. Then:

(i) $(\forall\, \mathbf{x}_0 \in \mathbb{F}^n)\ [\mathbf{x}_0 \in \mathrm{LS}(A, \mathbf{y}) \iff A^*A\mathbf{x}_0 = A^*\mathbf{y}]$;
(ii) $(\exists\, \mathbf{x}_0 \in \mathbb{F}^n)\ A^*A\mathbf{x}_0 = A^*\mathbf{y}$;
(iii) $[(\exists!\, \mathbf{x}_0 \in \mathbb{F}^n)\ A^*A\mathbf{x}_0 = A^*\mathbf{y}] \iff \mathrm{rank}(A) = n$.

Proof. Set $\mathcal{U} := \mathrm{col}(A)$.

(i) $(\Rightarrow)$ Suppose $\mathbf{x}_0$ is a least-squares solution. Then $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$ by Theorem 2.2.6, so $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^{\perp} = \ker(A^*)$ by Proposition 2.2.3(v) and Corollary 2.3.12(iii). Hence $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
(i) $(\Leftarrow)$ Suppose $A^*A\mathbf{x}_0 = A^*\mathbf{y}$. Then $\mathbf{y} - A\mathbf{x}_0 \in \ker(A^*) = \mathcal{U}^{\perp}$ by Corollary 2.3.12(iii). Since $A\mathbf{x}_0 \in \mathcal{U}$, the identity $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$ holds by uniqueness in Theorem 2.1.5. Thus $\mathbf{x}_0$ is a least-squares solution by Theorem 2.2.6.
(ii) Since $P_{\mathcal{U}}\mathbf{y} \in \mathcal{U} = \mathrm{col}(A)$, choose $\mathbf{x}_0 \in \mathbb{F}^n$ with $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$. Then $A^*A\mathbf{x}_0 = A^*\mathbf{y}$ by (i).
(iii) $(\Rightarrow)$ Suppose the normal equations have a unique solution $\mathbf{x}_0$. If $\mathbf{z} \in \ker(A)$, then $\mathbf{x}_0 + \mathbf{z}$ also solves them, so $\mathbf{z} = \mathbf{0}$ by uniqueness. Hence $\dim(\ker(A)) = 0$ and $\mathrm{rank}(A) = n$ by Corollary 3.3.1(i).

(*iii*) ($\Leftarrow$) Suppose $\operatorname{rank}(A) = n$. Existence holds by (ii), and $\ker(A) = \{\mathbf{0}\}$ by Corollary 3.3.1(i).
If $\mathbf{x}_0$ and $\mathbf{x}_1$ solve the normal equations, then $\mathbf{x}_0 - \mathbf{x}_1 \in \ker(A^*A) = \ker(A) = \{\mathbf{0}\}$ by Proposition 2.3.11(i). Thus $\mathbf{x}_0 = \mathbf{x}_1$.

Structural Commentary: Projection characterizes optimality; kernel identities establish consistency and uniqueness.
Key Insight: Normal equations annihilate the column-space residual. □

Figure 32 shows the fitted vector and its orthogonal residual in the codomain.

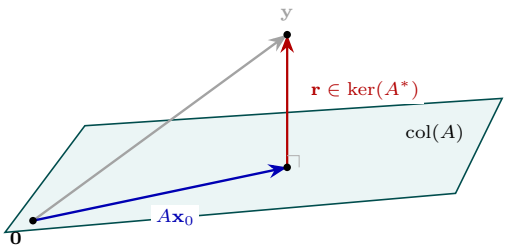

Figure 32: Fitted vector and residual: $A\mathbf{x}_0 = P_{\operatorname{col}(A)}\mathbf{y}$ and $\mathbf{y} - A\mathbf{x}_0 \in \operatorname{col}(A)^\perp = \ker(A^*)$.

Proposition 2.4.5. *Least-Squares Solution Set and Minimum Norm*
Let $A \in \operatorname{Mat}(m,n,\mathbb{F})$, let $\mathbf{y} \in \mathbb{F}^m$, and set $\operatorname{LS}(A,\mathbf{y}) := \operatorname{arg\,min}_{\mathbf{x}\in\mathbb{F}^n} \|A\mathbf{x}-\mathbf{y}\|_2$. Fix $\mathbf{x}_0 \in \operatorname{LS}(A,\mathbf{y})$ and set $\mathbf{s} := P_{(\ker(A))^\perp}\mathbf{x}_0$. Then:

(i) $\operatorname{LS}(A,\mathbf{y}) = \mathbf{x}_0 + \ker(A)$;
(ii) $\operatorname{LS}(A,\mathbf{y}) \cap (\ker(A))^\perp = \{\mathbf{s}\}$;
(iii) $\operatorname{arg\,min}_{\mathbf{x}\in\operatorname{LS}(A,\mathbf{y})} \|\mathbf{x}\|_2 = \{\mathbf{s}\}$.

Proof.
(i) For $\mathbf{x} \in \mathbb{F}^n$, $\mathbf{x} \in \operatorname{LS}(A,\mathbf{y}) \iff A\mathbf{x} = P_{\operatorname{col}(A)}\mathbf{y} \iff A\mathbf{x} = A\mathbf{x}_0 \iff \mathbf{x} - \mathbf{x}_0 \in \ker(A)$ by Theorem 2.2.6; $\operatorname{LS}(A,\mathbf{y}) = \mathbf{x}_0 + \ker(A)$.
(ii) The relations $\mathbf{s} \in (\ker(A))^\perp$ and $\mathbf{x}_0 - \mathbf{s} \in ((\ker(A))^\perp)^\perp = \ker(A)$ hold by Proposition 2.2.3(iv)–(v), Corollary 2.1.8.
Thus $\mathbf{s} \in \operatorname{LS}(A,\mathbf{y})$ by (i). If $\mathbf{u} \in \operatorname{LS}(A,\mathbf{y}) \cap (\ker(A))^\perp$, then $\mathbf{u} - \mathbf{s} \in \ker(A) \cap (\ker(A))^\perp = \{\mathbf{0}\}$ by (i), Proposition 2.1.3(iv); $\mathbf{u} = \mathbf{s}$.
(iii) Fix $\mathbf{x} \in \operatorname{LS}(A,\mathbf{y})$ and choose $\mathbf{z} \in \ker(A)$ with $\mathbf{x} = \mathbf{s} + \mathbf{z}$ by (i)–(ii).
Then $\mathbf{s} \perp \mathbf{z}$, so $\|\mathbf{x}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{z}\|_2^2 \geq \|\mathbf{s}\|_2^2$ by Theorem 1.4.3, with equality iff $\mathbf{z} = \mathbf{0}$, equivalently $\mathbf{x} = \mathbf{s}$.

Structural Commentary: An affine kernel description and orthogonal decomposition isolate the minimum-norm point.
Key Insight: Removing the kernel component uniquely minimizes norm. □

Corollary 2.4.6. *Projection Formula for Full Column Rank*
Let $A \in \operatorname{Mat}(m,n,\mathbb{F})$ satisfy $\operatorname{rank}(A) = n$. Then:

(i) A^*A *is invertible;*
(ii) $P_{\operatorname{col}(A)} = A(A^*A)^{-1}A^*$.

Proof.
(i) The identities $\ker(A^*A) = \ker(A) = \{\mathbf{0}\}$ hold by Proposition 2.3.11(i) and Corollary 3.3.1(i).
Hence A^*A is injective and therefore invertible by Theorem 1.7.7.
(ii) Fix $\mathbf{y} \in \mathbb{F}^m$ and set $\mathbf{x}_0 := (A^*A)^{-1}A^*\mathbf{y}$. Then $A^*A\mathbf{x}_0 = A^*\mathbf{y}$, so $\mathbf{x}_0$ is a least-squares solution by Theorem 2.4.4(i).
Consequently, $P_{\operatorname{col}(A)}\mathbf{y} = A\mathbf{x}_0 = A(A^*A)^{-1}A^*\mathbf{y}$ by Theorem 2.2.6. Since $\mathbf{y}$ was arbitrary, the operator identity follows.

Structural Commentary: Kernel equality proves invertibility; the normal-equation solution determines the projection matrix.
Key Insight: Full column rank makes normal-equation coordinates explicit. □

Definition 2.4.7. *Minimum-Norm Solution*
Let $A \in \operatorname{Mat}(m,n,\mathbb{F})$, $\mathbf{y} \in \operatorname{col}(A)$. A solution $\mathbf{s} \in \mathbb{F}^n$ of $A\mathbf{x} = \mathbf{y}$ is a *minimum-norm solution* if $(\forall\, \mathbf{x} \in \mathbb{F}^n)\,[A\mathbf{x} = \mathbf{y} \implies \|\mathbf{s}\|_2 \leq \|\mathbf{x}\|_2]$.

Theorem 2.4.8. *Minimum-Norm Solution of a Consistent Linear System*
Let $A \in \operatorname{Mat}(m,n,\mathbb{F})$, let $\mathbf{y} \in \mathbb{F}^m$, and suppose $A\mathbf{x} = \mathbf{y}$ is consistent. Then:

(i) $(\exists!\, \mathbf{s} \in \mathbb{F}^n)\,[A\mathbf{s} = \mathbf{y} \wedge (\forall\, \mathbf{x} \in \mathbb{F}^n)\,(A\mathbf{x} = \mathbf{y} \implies \|\mathbf{s}\|_2 \leq \|\mathbf{x}\|_2)]$;
(ii) for the vector $\mathbf{s}$ *in (i),* $\{\mathbf{x} \in \operatorname{col}(A^*) : A\mathbf{x} = \mathbf{y}\} = \{\mathbf{s}\}$;
(iii) $(\exists\, \mathbf{u} \in \mathbb{F}^m)\, AA^*\mathbf{u} = \mathbf{y}$;
(iv) for the vector $\mathbf{s}$ *in (i),* $(\forall\, \mathbf{u}_0 \in \mathbb{F}^m)\,[AA^*\mathbf{u}_0 = \mathbf{y} \implies \mathbf{s} = A^*\mathbf{u}_0]$.

Proof. The identities $\operatorname{col}(A^*) = (\ker(A))^\perp$ and $\ker(A) = (\operatorname{col}(A^*))^\perp$ hold by Corollary 2.3.12(ii) and (i), respectively.
Hence $\mathbb{F}^n = \operatorname{col}(A^*) \odot \ker(A)$ by Theorem 2.1.5.
(i) Choose $\mathbf{x}_0 \in \mathbb{F}^n$ with $A\mathbf{x}_0 = \mathbf{y}$ and write $\mathbf{x}_0 = \mathbf{s} + \mathbf{v}$ with $\mathbf{s} \in \operatorname{col}(A^*)$ and $\mathbf{v} \in \ker(A)$. Then $A\mathbf{s} = \mathbf{y}$.
Fix $\mathbf{x} \in \mathbb{F}^n$ with $A\mathbf{x} = \mathbf{y}$ and set $\mathbf{w} := \mathbf{x} - \mathbf{s} \in \ker(A)$. Then $\|\mathbf{x}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{w}\|_2^2 \geq \|\mathbf{s}\|_2^2$ by Theorem 1.4.3.
Nonnegativity of norms proves minimality, and equality holds iff $\mathbf{w} = \mathbf{0}$, which is equivalent to $\mathbf{x} = \mathbf{s}$ and proves uniqueness.
(ii) The constructed $\mathbf{s}$ is a solution in $\operatorname{col}(A^*)$. If $\mathbf{s}'$ is another such solution, then $\mathbf{s}' - \mathbf{s} \in \ker(A) \cap \operatorname{col}(A^*) = \{\mathbf{0}\}$, so $\mathbf{s}' = \mathbf{s}$.
(iii) Since $\mathbf{s} \in \operatorname{col}(A^*)$, choose $\mathbf{u} \in \mathbb{F}^m$ with $\mathbf{s} = A^*\mathbf{u}$. Then $AA^*\mathbf{u} = A\mathbf{s} = \mathbf{y}$.
(iv) If $AA^*\mathbf{u}_0 = \mathbf{y}$, then $A^*\mathbf{u}_0$ is a solution in $\operatorname{col}(A^*)$, so $A^*\mathbf{u}_0 = \mathbf{s}$ by (ii).

Structural Commentary: Orthogonal splitting constructs the minimizer; the column-space characterization proves representation.
Key Insight: Minimum norm means zero nullspace component. □

Figure 33 depicts the orthogonal representative of the affine solution set.

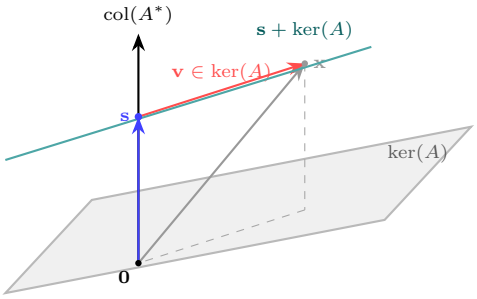

Figure 33: The unique solution in $\mathrm{col}(A^*) = (\ker(A))^\perp$ is the minimum-norm element of $\mathbf{s} + \ker(A)$.

Perspective 2.4.9. *Roadmap to the Pseudoinverse*
Theorem 2.4.4 projects $\mathbf{y}$ onto $\mathrm{col}(A)$; Proposition 2.4.5 selects the minimum-norm preimage in $(\ker(A))^\perp$, identified with $\mathrm{col}(A^*)$ in Theorem 2.4.8. Later, Definition 5.2.4 and Theorem 5.2.9 package both choices as $A^\dagger\mathbf{y}$.

2.5 Self-Adjoint Operators and Orthogonal Projections

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.5.1. *Self-Adjoint and Hermitian*

(i) Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F}$. An operator $T \in \mathcal{L}(\mathcal{V})$ is *self-adjoint* if $T^* = T$.
(ii) Let $n \geq 1$. A matrix $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is *Hermitian* if $A^* = A$.

In orthonormal coordinates, these conditions agree by Theorem 2.3.7.

Lemma 2.5.2. *Vanishing Complex Diagonal Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
If $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = 0$, then $T = 0$.

Proof. Set $Q : \mathcal{V} \to \mathbb{C}$ by $Q(\mathbf{x}) := \langle T\mathbf{x}, \mathbf{x}\rangle$ for $\mathbf{x} \in \mathcal{V}$.
For $(\mathbf{u}, \mathbf{w}) \in \mathcal{V} \times \mathcal{V}$, the identity $4\langle T\mathbf{u}, \mathbf{w}\rangle = Q(\mathbf{u}+\mathbf{w}) - Q(\mathbf{u}-\mathbf{w}) + iQ(\mathbf{u}+i\mathbf{w}) - iQ(\mathbf{u}-i\mathbf{w})$ holds by direct expansion.
Thus $(\forall\, (\mathbf{u}, \mathbf{w}) \in \mathcal{V} \times \mathcal{V})\ \langle T\mathbf{u}, \mathbf{w}\rangle = 0$ by the hypothesis.
Taking $\mathbf{w} = T\mathbf{u}$, $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = 0$, and hence $T = 0$ by positive definiteness.
Structural Commentary: Over $\mathbb{C}$, the diagonal values determine all mixed values.
Key Insight: Polarization turns the one-vector condition into a two-vector condition. □

Proposition 2.5.3. *Self-Adjointness from Real Diagonal Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
Then $T = T^* \iff (\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$.

Proof.
($\Rightarrow$) Suppose $T = T^*$ and fix $\mathbf{v} \in \mathcal{V}$. Then $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \overline{\langle T\mathbf{v}, \mathbf{v}\rangle}$, so $\langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$.
($\Leftarrow$) Suppose $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$. Then $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle (T - T^*)\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle - \overline{\langle T\mathbf{v}, \mathbf{v}\rangle} = 0$. Hence $T - T^* = 0$ by Lemma 2.5.2.
Structural Commentary: Adjoint symmetry proves reality; polarization eliminates the skew-adjoint difference.
Key Insight: Real diagonal values annihilate $T - T^*$. □

Proposition 2.5.4. *Vanishing Diagonal Values for Self-Adjoint Operators*
Let T be a self-adjoint operator on a finite-dimensional inner-product space $\mathcal{V}$. If $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{v}, \mathbf{v}\rangle = 0$, then $T = 0$.

Proof. If $\mathbb{F} = \mathbb{C}$, then $T = 0$ by Lemma 2.5.2. Suppose $\mathbb{F} = \mathbb{R}$.
For $(\mathbf{u}, \mathbf{w}) \in \mathcal{V} \times \mathcal{V}$, $\langle T\mathbf{w}, \mathbf{u}\rangle = \langle \mathbf{w}, T\mathbf{u}\rangle = \langle T\mathbf{u}, \mathbf{w}\rangle$ by self-adjointness and symmetry.
Hence $4\langle T\mathbf{u}, \mathbf{w}\rangle = \langle T(\mathbf{u}+\mathbf{w}), \mathbf{u}+\mathbf{w}\rangle - \langle T(\mathbf{u}-\mathbf{w}), \mathbf{u}-\mathbf{w}\rangle = 0$ by direct expansion and the hypothesis.
Taking $\mathbf{w} = T\mathbf{u}$, $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = 0$, and hence $T = 0$ by positive definiteness.
Structural Commentary: Self-adjointness supplies the missing symmetry needed for real polarization.
Key Insight: A self-adjoint operator is determined by its diagonal values. □

Definition 2.5.5. *Orthogonal Projection Matrix*
Let $n \geq 1$. A matrix $P \in \mathrm{Mat}(n, n, \mathbb{F})$ is an *orthogonal projection matrix* if $P^2 = P$ and $P^* = P$.

Theorem 2.5.6. *Orthogonal Projections as Self-Adjoint Idempotents*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $P \in \mathcal{L}(\mathcal{V})$.
Then $P = P_{\mathrm{range}(P)} \iff [P^2 = P \wedge P^* = P]$.

Proof.
($\Rightarrow$) Set $\mathcal{U} := \mathrm{range}(P)$ and suppose $P = P_{\mathcal{U}}$. Then $P^2 = P$ by Proposition 2.2.3(vii).
The identity $P = \rho_{\mathcal{U},\mathcal{U}^\perp}$ holds by the defining decompositions of the two projections.
Hence $P^* = \rho_{(\mathcal{U}^\perp)^\perp,\mathcal{U}^\perp} = \rho_{\mathcal{U},\mathcal{U}^\perp} = P$ by Proposition 2.3.11(iii) and Corollary 2.1.8.
($\Leftarrow$) Suppose $P^2 = P$ and $P^* = P$. Then $\mathcal{V} = \mathrm{range}(P) \oplus \ker(P)$ and $P = \rho_{\mathrm{range}(P),\ker(P)}$ by Proposition 2.4.2(i).
The identity $\ker(P) = \ker(P^*) = (\mathrm{range}(P))^\perp$ holds by Proposition 2.3.10(i).
Therefore $P = \rho_{\mathrm{range}(P),(\mathrm{range}(P))^\perp} = P_{\mathrm{range}(P)}$ by the defining decompositions of the two projections.
Structural Commentary: Algebraic splitting identifies the kernel; adjoint orthogonality makes the decomposition metric.
Key Insight: Orthogonality is the kernel identity $\ker(P) = (\mathrm{range}(P))^\perp$. □

2.6 Isometries and Unitary Matrices

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.6.1. *Isometry and Isometric Isomorphism*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) T is an *isometry* if $(\forall\, \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) T is an *isometric isomorphism* if it is bijective and isometric.
The spaces $\mathcal{V}$ and $\mathcal{W}$ are *isometrically isomorphic* if an isometric isomorphism from $\mathcal{V}$ to $\mathcal{W}$ exists.

Proposition 2.6.2. *Basic Properties of Isometries*
Let $\mathcal{V}$ be an inner-product space and let $S, T \in \mathcal{L}(\mathcal{V})$ be isometries.

(i) ST is an isometry;
(ii) T is injective;
(iii) if $\mathcal{V}$ is finite-dimensional, then T is invertible;
(iv) if $\mathcal{V}$ is finite-dimensional, then T^{-1} is an isometry.

Proof.
(i) $(\forall\, \mathbf{v} \in \mathcal{V})\ \|ST\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) Fix $\mathbf{v} \in \ker(T)$. Then $\|\mathbf{v}\| = \|T\mathbf{v}\| = 0$, so $\mathbf{v} = \mathbf{0}$ by positive definiteness. Hence $\ker(T) = \{\mathbf{0}\}$ and T is injective.
(iii) If $\mathcal{V}$ is finite-dimensional, then T is invertible by (ii) and Theorem 1.7.7.
(iv) Suppose $\mathcal{V}$ is finite-dimensional. By (iii), T^{-1} exists, and $(\forall\, \mathbf{u} \in \mathcal{V})\ \left\|T^{-1}\mathbf{u}\right\| = \left\|T(T^{-1}\mathbf{u})\right\| = \|\mathbf{u}\|$.

Structural Commentary: Norm identities prove closure and injectivity; finite-dimensional invertibility controls the inverse.
Key Insight: Length preservation prevents a nonzero vector from entering the kernel. □

Example 2.6.3. *An Isometry Need Not Be Onto*
Define $T : \ell^2(\mathbb{F}) \to \ell^2(\mathbb{F})$ by $T(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$. For $\mathbf{x} = (x_j)_{j \geq 1}$, $\|T\mathbf{x}\|_2^2 = \sum_{j \geq 1} |x_j|^2 = \|\mathbf{x}\|_2^2$, so T is an isometry.
Since every vector in $\operatorname{range}(T)$ has first coordinate 0, $(1, 0, 0, \ldots) \notin \operatorname{range}(T)$; hence T is not surjective.
Thus norm preservation need not imply surjectivity in infinite-dimensional inner-product spaces.

Theorem 2.6.4. *Isometries Preserve Inner Products*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $[(\forall\, \mathbf{x} \in \mathcal{V})\ \|T\mathbf{x}\| = \|\mathbf{x}\|] \iff [(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \langle T\mathbf{u}, T\mathbf{v}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}}]$.

Proof.
($\Rightarrow$) Suppose T preserves norms and fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$. Each norm in the polarization formula for $\langle T\mathbf{u}, T\mathbf{v}\rangle_{\mathcal{W}}$ equals its counterpart for $\langle \mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}}$ by linearity of T and the hypothesis. Hence $\langle T\mathbf{u}, T\mathbf{v}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}}$ by Theorem 1.1.23.
($\Leftarrow$) Suppose T preserves inner products. Then $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{u}\rangle_{\mathcal{V}} = \|\mathbf{u}\|^2$; T preserves norms by nonnegativity.

Structural Commentary: Polarization converts preservation of norms into preservation of inner products.
Key Insight: Universal length preservation also preserves angle information. □

Remark 2.6.5. *Isometry Conventions*
For linear maps between inner-product spaces, norm preservation and inner-product preservation are equivalent by Theorem 2.6.4, so either convention defines the same isometries. For general bilinear forms, however, form preservation need not imply injectivity; an isometry is therefore often required to be bijective and form-preserving.

Theorem 2.6.6. *Isometries and the Adjoint Product*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
Then $[(\forall\, \mathbf{v} \in \mathcal{V})\ \|T\mathbf{v}\| = \|\mathbf{v}\|] \iff T^*T = I_{\mathcal{V}}$.

Proof.
($\Rightarrow$) Suppose T preserves norms. Then $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{u}, T^*T\mathbf{v}\rangle = \langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$ by Theorem 2.6.4 and the adjoint relation.
For $\mathbf{v} \in \mathcal{V}$, set $\mathbf{u} := (T^*T - I_{\mathcal{V}})\mathbf{v}$. Then $\|\mathbf{u}\|^2 = \langle \mathbf{u}, (T^*T - I_{\mathcal{V}})\mathbf{v}\rangle = 0$, so $\mathbf{u} = \mathbf{0}$ by positive definiteness. Hence $T^*T = I_{\mathcal{V}}$.
($\Leftarrow$) Suppose $T^*T = I_{\mathcal{V}}$; $(\forall\, \mathbf{u} \in \mathcal{V})\ \|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle \mathbf{u}, T^*T\mathbf{u}\rangle = \|\mathbf{u}\|^2$ by adjoint relation; T preserves norms by nonnegativity.

Structural Commentary: The operator T^*T records inner products after applying T.
Key Insight: T is an isometry exactly when this Gram operator is the identity. □

Definition 2.6.7. *Unitary and Orthogonal*

(i) Let $\mathcal{V}$ be a finite-dimensional inner-product space. An operator $U \in \mathcal{L}(\mathcal{V})$ is *unitary* if $U^*U = UU^* = I_{\mathcal{V}}$.
(ii) Let $n \geq 1$. A matrix $U \in \operatorname{Mat}(n, n, \mathbb{F})$ is *unitary* if $U^*U = I_n$. If $\mathbb{F} = \mathbb{R}$, a unitary matrix is *orthogonal*.

Theorem 2.6.8. *Equivalent Conditions for a Unitary Matrix*
Let $n \geq 1$ and let $U = [\mathbf{u}_1\ \cdots\ \mathbf{u}_n] \in \operatorname{Mat}(n, n, \mathbb{F})$. The following are equivalent:

(i) $U^*U = I_n$;
(ii) the columns $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are orthonormal in $\mathbb{F}^n$;
(iii) $UU^* = I_n$;
(iv) U^* is unitary;
(v) the rows of U are orthonormal;
(vi) U is invertible and $U^{-1} = U^*$;
(vii) $(\forall\, \mathbf{x} \in \mathbb{F}^n)\ \|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$;
(viii) $(\forall\, \mathbf{x}, \mathbf{y} \in \mathbb{F}^n)\ \langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} = \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}$.

Proof.
(*i*) $\Rightarrow$ (*ii*) Since $(U^*U)_{ij} = \mathbf{u}_i^*\mathbf{u}_j = \langle \mathbf{u}_j, \mathbf{u}_i\rangle_{\mathbb{F}^n}$, the identity $U^*U = I_n$ is equivalent to orthonormality of the columns.
(*ii*) $\Rightarrow$ (*iii*) Orthonormality of the columns implies $U^*U = I_n$. Hence $UU^* = I_n$ by Theorem 3.4.2.
(*iii*) $\Rightarrow$ (*iv*) The identity $UU^* = I_n$ is $(U^*)^*U^* = I_n$ by conjugate-transpose involution, so U^* is unitary.

(*iv*) $\Rightarrow$ (*v*) Since U^* is unitary, its columns are orthonormal by the calculation in (i).
These columns are the conjugates of the rows of U, so the rows of U are orthonormal.
(*v*) $\Rightarrow$ (*vi*) Orthonormality of the rows implies $UU^* = I_n$. Hence $U^{-1} = U^*$ by Theorem 3.4.2.
(*vi*) $\Rightarrow$ (*vii*) The relation $U^{-1} = U^*$ implies $U^*U = I_n$. Thus $(\forall\, \mathbf{x} \in \mathbb{F}^n)\ \|U\mathbf{x}\|_2^2 = \langle U\mathbf{x}, U\mathbf{x}\rangle_{\mathbb{F}^n} = \langle \mathbf{x}, U^*U\mathbf{x}\rangle_{\mathbb{F}^n} = \|\mathbf{x}\|_2^2$ by the conjugate-transpose adjoint relation, so U preserves norms by nonnegativity.
(*vii*) $\Rightarrow$ (*viii*) Condition (viii) holds by Theorem 2.6.4 for the map $\mathbf{x} \mapsto U\mathbf{x}$.
(*viii*) $\Rightarrow$ (*i*) The relation $(\forall\, \mathbf{x}, \mathbf{y} \in \mathbb{F}^n)\ \langle \mathbf{x}, (U^*U - I_n)\mathbf{y}\rangle_{\mathbb{F}^n} = \langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} - \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = 0$ holds by the conjugate-transpose adjoint relation and (viii). Taking $\mathbf{x} = (U^*U - I_n)\mathbf{y}$, $(\forall\, \mathbf{y} \in \mathbb{F}^n)\ \|(U^*U - I_n)\mathbf{y}\|_2^2 = 0$, so $U^*U = I_n$ by positive definiteness.

Structural Commentary: A cyclic chain links Gram identities, inverse identities, and geometric invariance.
Key Insight: Unitarity is orthonormality expressed in coordinates. □

Figure 34 illustrates the preservation of lengths and inner products established by Theorem 2.6.8.

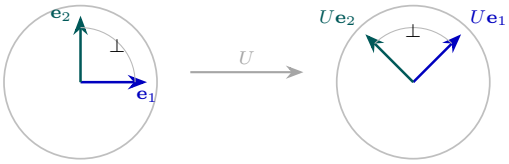

Figure 34: Unitary invariance: $\|U\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle U\mathbf{x}, U\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.

Definition 2.6.9. *Unitary and Permutation Similarity*
Let $n \geq 1$ and let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) The matrices A and B are *unitarily similar* if $A = UBU^*$ for a unitary $U \in \mathrm{Mat}(n, n, \mathbb{F})$.
(ii) They are *orthogonally similar* if $\mathbb{F} = \mathbb{R}$ and $A = UBU^*$ for a real orthogonal $U \in \mathrm{Mat}(n, n, \mathbb{R})$.
(iii) Matrix $P = [p_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is *permutation matrix* if $(\forall\, i, j \in \{1, \ldots, n\})\ p_{ij} \in \{0, 1\}$, $(\forall\, i \in \{1, \ldots, n\})\ \sum_{j=1}^{n} p_{ij} = \sum_{j=1}^{n} p_{ji} = 1$.
(iv) The matrices A and B are *permutation similar* if $A = PBP^*$ for a permutation matrix P.

Proposition 2.6.10. *Permutation Matrices Are Orthogonal*
Let $n \geq 1$, let $\sigma \in \mathfrak{S}_n$, and set $P := [\mathbf{e}_{\sigma(1)} \ \cdots \ \mathbf{e}_{\sigma(n)}] \in \mathrm{Mat}(n, n, \mathbb{R})$. Then $P^*P = PP^* = I_n$, so P is orthogonal.

Proof. The columns of P are a rearrangement of the standard orthonormal basis of $\mathbb{R}^n$.
Hence $P^*P = PP^* = I_n$ by Theorem 2.6.8(ii)–(iii), and P is orthogonal because it is real.
Structural Commentary: A permutation matrix only reorders the standard basis.
Key Insight: Reordering orthonormal vectors preserves orthonormality. □

Theorem 2.6.11. *Unitary Coordinate Form of an Orthogonal Projection*
Let $n \geq 1$, let $P \in \mathrm{Mat}(n, n, \mathbb{F})$ be an orthogonal projection matrix, and set $r := \mathrm{rank}(P)$.
Then $(\exists\, U \in \mathrm{Mat}(n, n, \mathbb{F}))\ [U^*U = I_n \wedge U^*PU = I_r \oplus 0_{n-r}]$. Equivalently, P is unitarily similar to $I_r \oplus 0_{n-r}$.

Proof. Set $\mathcal{U} := \mathrm{col}(P)$. The operator induced by P is $P_{\mathcal{U}}$ by Theorem 2.5.6.
Hence $(\forall\, \mathbf{u} \in \mathcal{U})\ P\mathbf{u} = \mathbf{u}$ and $(\forall\, \mathbf{w} \in \mathcal{U}^{\perp})\ P\mathbf{w} = \mathbf{0}$ by Proposition 2.2.3(ii)–(iii).
Choose an orthonormal basis $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ of $\mathcal{U}$ by Corollary 1.3.11, and extend it to an orthonormal basis $(\mathbf{u}_1, \ldots, \mathbf{u}_n)$ of $\mathbb{F}^n$ by Corollary 1.3.12. Set $U := [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$. Then $U^*U = I_n$ by Theorem 2.6.8(i)–(ii).
For $j \in \{1, \ldots, r\}$, $U^*PU\mathbf{e}_j = U^*P\mathbf{u}_j = U^*\mathbf{u}_j = \mathbf{e}_j$; for $j \in \{r+1, \ldots, n\}$, $U^*PU\mathbf{e}_j = \mathbf{0}$.
Therefore $U^*PU = I_r \oplus 0_{n-r}$ by equality on the standard basis.
Finally, $P = U(I_r \oplus 0_{n-r})U^*$ by multiplication on the left by U and on the right by U^*.
Structural Commentary: An adapted orthonormal basis converts the range–kernel split into block coordinates.
Key Insight: Projection becomes coordinate truncation in an adapted orthonormal basis. □

Figure 35 depicts the range–kernel coordinates constructed in Theorem 2.6.11.

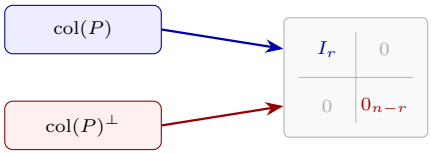

Figure 35: Adapted basis: $U^*PU = I_r \oplus 0_{n-r}$, identity on $\mathrm{col}(P)$ and zero on $\mathrm{col}(P)^{\perp}$.

Proposition 2.6.12. *Algebra of Unitary Matrices*
Let $m, n \geq 1$, let $U, V \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $W \in \mathrm{Mat}(m, m, \mathbb{F})$.
Let $V \oplus W$ denote the block diagonal matrix with blocks V and W.

(i) $[U^*U = I_n \wedge V^*V = I_n] \Longrightarrow (UV)^*(UV) = I_n$;
(ii) $[(V \oplus W)^*(V \oplus W) = I_{n+m}] \Longleftrightarrow [V^*V = I_n \wedge W^*W = I_m]$.

Proof.

(*i*) The identity $(UV)^*(UV) = V^*U^*UV = V^*V = I_n$ holds by the hypotheses.
(*ii*) ($\Rightarrow$) Suppose $(V \oplus W)^*(V \oplus W) = I_{n+m}$.
Then $V^*V \oplus W^*W = I_n \oplus I_m$, so $V^*V = I_n$ and $W^*W = I_m$ by equality of the diagonal blocks.
(*ii*) ($\Leftarrow$) If $V^*V = I_n$ and $W^*W = I_m$, then $(V \oplus W)^*(V \oplus W) = I_n \oplus I_m = I_{n+m}$.

Structural Commentary: Unitary structure is stable under composition and blockwise orthogonal sums.
Key Insight: The equation $U^*U = I$ makes every verification a short adjoint calculation. □

Theorem 2.6.13. *Extending Isometric Columns to a Unitary Matrix*
Let $m, n \geq 1$ and let $X \in \mathrm{Mat}(m, n, \mathbb{F})$ satisfy $X^*X = I_n$. Then:
(i) $n \leq m$;
(ii) $(\exists\, X' \in \mathrm{Mat}(m, m-n, \mathbb{F}))\, [X\ X']^*[X\ X'] = I_m$.

Proof. Write $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_n]$. The list $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is orthonormal in $\mathbb{F}^m$ because $(X^*X)_{ij} = \langle \mathbf{x}_j, \mathbf{x}_i \rangle_{\mathbb{F}^m} = \delta_{ij}$.
(i) The list is linearly independent by Corollary 1.3.3, so $n \leq m$ by Theorem 1.3.11.
(ii) Extend the list to an orthonormal basis $(\mathbf{x}_1, \ldots, \mathbf{x}_m)$ of $\mathbb{F}^m$ by Corollary 1.3.12, and set $X' := [\mathbf{x}_{n+1}\ \cdots\ \mathbf{x}_m]$. Then $[X\ X']^*[X\ X'] = I_m$ by Theorem 2.6.8(i)–(ii).

Structural Commentary: Column orthonormality proves the dimension bound and supplies an extendable partial basis.
Key Insight: Complete an isometric column family by extending its orthonormal list. □

2.7 Householder Reflections

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.7.1. *Householder Matrix*
Let $n \geq 1$ and $\mathbf{w} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$, and set $\mathbf{u} := \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$. The *Householder matrix* associated with $\mathbf{w}$ is $U_{\mathbf{w}} := I_n - 2P_{\mathbf{u}} = I_n - 2\frac{\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|_2^2}$.
Equivalently, $(\forall\, \mathbf{x} \in \mathbb{F}^n)\, U_{\mathbf{w}}\mathbf{x} = \mathbf{x} - 2\langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

Theorem 2.7.2. *Householder Matrices Are Unitary Reflections*
Let $n \geq 1$ and let $\mathbf{w} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$. Then:
(i) $U_{\mathbf{w}}^* = U_{\mathbf{w}}$;
(ii) $U_{\mathbf{w}}^2 = I_n$;
(iii) $U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$, *so $U_{\mathbf{w}}$ is unitary;*
(iv) $(\forall\, \mathbf{x} \in \mathbf{w}^{\perp})\, U_{\mathbf{w}}\mathbf{x} = \mathbf{x}$;
(v) $(\forall\, \mathbf{x} \in \mathrm{span}(\mathbf{w}))\, U_{\mathbf{w}}\mathbf{x} = -\mathbf{x}$;
(vi) $\mathbf{w} \in \mathbb{R}^n \Longrightarrow U_{\mathbf{w}}$ *is real symmetric and orthogonal.*

Proof. Set $\mathbf{u} := \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$ and $P := P_{\mathbf{u}}$.
(i) $(\forall\, i, j \in \{1, \ldots, n\})\, (P^*)_{ij} = \overline{P_{ji}} = P_{ij}$ by Proposition 2.2.5(iii). Hence $P^* = P$ and $U_{\mathbf{w}}^* = I_n - 2P = U_{\mathbf{w}}$.
(ii) The identity $P^2 = P$ holds by Proposition 2.2.5(iv). Hence $U_{\mathbf{w}}^2 = (I_n - 2P)^2 = I_n - 4P + 4P^2 = I_n$.
(iii) The identity $U_{\mathbf{w}}^* U_{\mathbf{w}} = U_{\mathbf{w}}^2 = I_n$ holds by (i)–(ii), so $U_{\mathbf{w}}$ is unitary.
(iv) Fix $\mathbf{x} \in \mathbf{w}^{\perp} = \mathrm{span}(\mathbf{u})^{\perp}$. Then $P\mathbf{x} = \mathbf{0}$ by Proposition 2.2.5(ii), so $U_{\mathbf{w}}\mathbf{x} = \mathbf{x}$.
(v) Fix $\mathbf{x} \in \mathrm{span}(\mathbf{w}) = \mathrm{span}(\mathbf{u})$. Then $(\exists\, a \in \mathbb{F})\, \mathbf{x} = a\mathbf{u}$. Thus $P\mathbf{x} = \langle a\mathbf{u}, \mathbf{u} \rangle\, \mathbf{u} = a\mathbf{u} = \mathbf{x}$ by $\|\mathbf{u}\|_2 = 1$, and $U_{\mathbf{w}}\mathbf{x} = -\mathbf{x}$.
(vi) If $\mathbf{w} \in \mathbb{R}^n$, then $\mathbf{u}$ and P have real entries. Thus $U_{\mathbf{w}}$ is real symmetric and orthogonal by (i) and (iii).

Structural Commentary: Self-adjoint idempotence of $P_{\mathbf{u}}$ separates the algebraic and fixed-space claims.
Key Insight: $I - 2P_{\mathbf{u}}$ flips exactly the projected component. □

Figure 36 depicts the hyperplane fixed and normal line reversed by Theorem 2.7.2(iv)–(v).

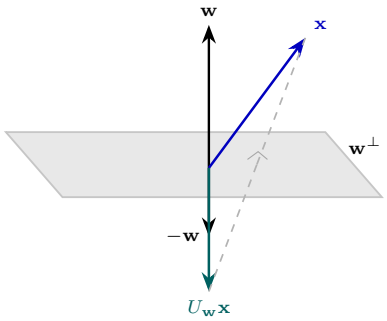

Figure 36: Householder reflection: $U_{\mathbf{w}}$ fixes the $\mathbf{w}^{\perp}$ component of $\mathbf{x}$ and reverses its $\mathrm{span}(\mathbf{w})$ component.

Theorem 2.7.3. *Householder Map Between Equal-Norm Vectors*
Let $n \geq 1$ and let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ satisfy $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 \neq 0$.
Set $z := \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n}$ and choose $\sigma \in \mathbb{F}$ so that $(z = 0 \Longrightarrow \sigma = 1) \wedge (z \neq 0 \Longrightarrow \sigma = -\frac{\overline{z}}{|z|})$. Set $\mathbf{w} := \mathbf{y} - \sigma\mathbf{x}$. Then:
(i) $|\sigma| = 1$;
(ii) $\mathbf{w} \neq \mathbf{0}$;
(iii) $\sigma U_{\mathbf{w}}\mathbf{x} = \mathbf{y}$.

Proof. Set $r := \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2$.
(i) The identity $|\sigma| = 1$ holds by the defining alternatives.
(ii) The identity $\sigma z = -|z|$ holds by the defining alternatives.
By (i) and conjugate symmetry, $\|\mathbf{w}\|_2^2 = 2r^2 - \sigma z - \overline{\sigma z} = 2(r^2 + |z|) > 0$, so $\mathbf{w} \neq \mathbf{0}$ by positive definiteness.
(iii) By $\mathbf{y} = \mathbf{w} + \sigma\mathbf{x}$, $\sigma U_{\mathbf{w}}\mathbf{x} = \mathbf{y} - \left(1 + 2\sigma\frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|_2^2}\right)\mathbf{w}$. The identity $\langle \mathbf{x}, \mathbf{w} \rangle = z - \overline{\sigma} r^2$ holds by the definition of $\mathbf{w}$. Thus $2\sigma\langle \mathbf{x}, \mathbf{w} \rangle = 2\sigma z - 2r^2 = -2(r^2 + |z|) = -\|\mathbf{w}\|_2^2$ by (i)–(ii), and $\sigma U_{\mathbf{w}}\mathbf{x} = \mathbf{y}$.

Structural Commentary: The phase choice makes the reflection direction nonzero and cancels its coefficient.
Key Insight: A unit phase aligns equal-norm vectors through one reflection. □

Corollary 2.7.4. *Householder Unitary Annihilator*
Let $n \geq 1$, $\mathbf{x} = [x_1 \ \cdots \ x_n]^t \in \mathbb{F}^n \setminus \{\mathbf{0}\}$. *Choose* $\sigma \in \mathbb{F}$, $(x_1 = 0 \implies \sigma = 1) \wedge (x_1 \neq 0 \implies \sigma = -\frac{\overline{x_1}}{|x_1|})$. *Set* $\mathbf{w} := \|\mathbf{x}\|_2 \mathbf{e}_1 - \sigma\mathbf{x}$. *Then:*

(i) $\mathbf{w} \neq \mathbf{0}$;
(ii) $\sigma U_{\mathbf{w}}\mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$;
(iii) $\|\mathbf{x}\|_2 = 1 \implies \overline{\sigma}U_{\mathbf{w}}$ *is unitary with first column* $\mathbf{x}$.

Proof. Set $\mathbf{y} := \|\mathbf{x}\|_2 \mathbf{e}_1$. Then $\|\mathbf{y}\|_2 = \|\mathbf{x}\|_2$ and $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \|\mathbf{x}\|_2$.
Since $\|\mathbf{x}\|_2 > 0$, the defining alternatives for σ coincide with those determined by $\langle \mathbf{x}, \mathbf{y} \rangle$.

(i) The relation $\mathbf{w} \neq \mathbf{0}$ holds by Theorem 2.7.3(ii) with this $\mathbf{y}$.
(ii) The identity $\sigma U_{\mathbf{w}}\mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$ holds by Theorem 2.7.3(iii) with this $\mathbf{y}$.
(iii) Suppose $\|\mathbf{x}\|_2 = 1$ and set $A := \sigma U_{\mathbf{w}}$. Then $|\sigma| = 1$ by Theorem 2.7.3(i), and $U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$ by Theorem 2.7.2(iii).
Hence $A^*A = |\sigma|^2 U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$, and A^* is unitary by Theorem 2.6.8(iv). By (ii), $A\mathbf{x} = \mathbf{e}_1$, so $A^*\mathbf{e}_1 = A^*A\mathbf{x} = \mathbf{x}$.
The identity $A^* = \overline{\sigma}U_{\mathbf{w}}^* = \overline{\sigma}U_{\mathbf{w}}$ holds by Theorem 2.7.2(i), so $\overline{\sigma}U_{\mathbf{w}}$ is unitary with first column $\mathbf{x}$.

Structural Commentary: Equal-norm reflection constructs the annihilator; taking its adjoint prescribes a column.
Key Insight: Reflect to $\mathbf{e}_1$, then adjoint to install $\mathbf{x}$. □

3 Bilinear and Quadratic Forms

3.1 Bilinear Form Types and Congruence Invariants

Metric vector spaces retain the algebraic structure of inner-product geometry without imposing positivity. Throughout this subsection, $\mathbb{F}$ is an arbitrary field and all vector spaces are finite-dimensional unless stated otherwise.
General bilinear pairings, their left and right radicals, their representing matrices, and the corresponding basis-change law were established earlier. Here the two input spaces coincide, and a fixed bilinear form B is written $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$.

Definition 3.1.1. *Symmetric, Skew-Symmetric, and Alternating Forms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ be bilinear.

(i) B is a *bilinear form* on $\mathcal{V}$; when fixed, write $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$.
(ii) $\mathrm{Sym}(B)$ means $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = B(\mathbf{y}, \mathbf{x})$; then B is *symmetric*.
(iii) $\mathrm{Skew}(B)$ means $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = -B(\mathbf{y}, \mathbf{x})$; then B is *skew-symmetric*.
(iv) $\mathrm{Alt}(B)$ means $(\forall\, \mathbf{x} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{x}) = 0$; then B is *alternating*.
(v) A *metric vector space* is a pair $(\mathcal{V}, B)$ such that $\mathrm{Sym}(B)$, $\mathrm{Skew}(B)$, or $\mathrm{Alt}(B)$ holds.
The symmetric and alternating cases are called *orthogonal geometry* and *symplectic geometry*, respectively.

Proposition 3.1.2. *Characteristic Controls the Form Types*
Let $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$ *be a bilinear form on* $\mathcal{V}$.

(i) $\mathrm{Alt}(B) \Rightarrow \mathrm{Skew}(B)$.
(ii) *If* $\mathrm{char}(\mathbb{F}) \neq 2$, *then* $\mathrm{Sym}(B) \wedge \mathrm{Alt}(B) \Rightarrow B = 0$.
(iii) *If* $\mathrm{char}(\mathbb{F}) \neq 2$, *then* $\mathrm{Alt}(B) \iff \mathrm{Skew}(B)$.
(iv) *If* B *is symmetric, skew-symmetric, or alternating, then* $\mathrm{rad}_L(B) = \mathrm{rad}_R(B)$.
(v) *If* $\mathrm{char}(\mathbb{F}) = 2$, *then* $\mathrm{Sym}(B) \iff \mathrm{Skew}(B)$.
(vi) *If* $\mathrm{char}(\mathbb{F}) = 2$, *then* $\mathrm{Alt}(B) \Rightarrow \mathrm{Sym}(B)$.

Proof.

(i) Assume $\mathrm{Alt}(B)$. By bilinearity and alternation, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) + B(\mathbf{y}, \mathbf{x}) = B(\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) = 0$. Hence $\mathrm{Skew}(B)$.
(ii) Assume $\mathrm{char}(\mathbb{F}) \neq 2$, $\mathrm{Sym}(B)$, and $\mathrm{Alt}(B)$. By (i), B is also skew-symmetric.
Thus $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, 2B(\mathbf{x}, \mathbf{y}) = 0$. Since $2 \neq 0$ in $\mathbb{F}$, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = 0$, and therefore $B = 0$.
(iii) ($\Rightarrow$) If $\mathrm{Alt}(B)$, then $\mathrm{Skew}(B)$ by (i).
(iii) ($\Leftarrow$) Assume $\mathrm{char}(\mathbb{F}) \neq 2$ and $\mathrm{Skew}(B)$. Then $(\forall\, \mathbf{x} \in \mathcal{V})\, 2B(\mathbf{x}, \mathbf{x}) = 0$. Since $2 \neq 0$ in $\mathbb{F}$, $(\forall\, \mathbf{x} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{x}) = 0$, so $\mathrm{Alt}(B)$.
(iv) If B is symmetric or skew-symmetric, then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, [B(\mathbf{x}, \mathbf{y}) = 0 \iff B(\mathbf{y}, \mathbf{x}) = 0]$, so $\mathrm{rad}_L(B) = \mathrm{rad}_R(B)$.
If B is alternating, the same conclusion holds because B is skew-symmetric by (i).
(v) ($\Rightarrow$) Assume $\mathrm{char}(\mathbb{F}) = 2$ and $\mathrm{Sym}(B)$. Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = B(\mathbf{y}, \mathbf{x}) = -B(\mathbf{y}, \mathbf{x})$, so $\mathrm{Skew}(B)$.
(v) ($\Leftarrow$) Assume $\mathrm{char}(\mathbb{F}) = 2$ and $\mathrm{Skew}(B)$. Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\, B(\mathbf{x}, \mathbf{y}) = -B(\mathbf{y}, \mathbf{x}) = B(\mathbf{y}, \mathbf{x})$, so $\mathrm{Sym}(B)$.
(vi) Assume $\mathrm{char}(\mathbb{F}) = 2$ and $\mathrm{Alt}(B)$. Then $\mathrm{Skew}(B)$ by (i), and hence $\mathrm{Sym}(B)$ by (v).

Structural Commentary: Items (ii), (iii), (v), and (vi) isolate the characteristic-dependent relations among the form types.
Key Insight: Expanding $B(\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) = 0$ converts alternation into the skew-symmetry identity without dividing by 2. □

Henceforth, $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is usually symmetric, skew-symmetric, or alternating. By Proposition 3.1.2(iv), its left and right radicals agree, so a single radical and a single orthogonality notation suffice.

Definition 3.1.3. *Matrix, Rank, and Discriminant of a Form*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an ordered basis of a metric vector space $\mathcal{V}$.

(i) The *matrix of the form* in β is $M_\beta := [\langle \mathbf{v}_i, \mathbf{v}_j \rangle]_{i,j}$.
(ii) The *rank* of the form is $\mathrm{rank}(M_\beta)$.
(iii) The *discriminant set* is $\Delta(\mathcal{V}) := \{r^2 \det(M_\beta) : r \in \mathbb{F}^\times\}$.
(iv) If $\det(M_\beta) \neq 0$, then $\Delta(\mathcal{V}) = \det(M_\beta)(\mathbb{F}^\times)^2$ is the square class of $\det(M_\beta)$ in $\mathbb{F}^\times/(\mathbb{F}^\times)^2$.

Proposition 3.1.4. *Matrices and Invariants under Congruence*
Let $n = \dim(\mathcal{V})$, *let* β, γ *be ordered bases of* $\mathcal{V}$, *and put* $C = C_{\beta \leftarrow \gamma}$.

(i) $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta$.
(ii) $M_\gamma = C^t M_\beta C$.
(iii) *If* $A = M_\beta$ *and* $N \in \mathrm{Mat}(n, n, \mathbb{F})$, *then* $(\exists\, \delta$ *an ordered basis of* $\mathcal{V})\, N = M_\delta \iff (\exists\, P \in \mathrm{Mat}(n, n, \mathbb{F}),\ P$ *invertible*$)\, N = P^t A P$.

(iv) $\operatorname{rank}(M_\gamma) = \operatorname{rank}(M_\beta)$.
(v) $\Delta(\mathcal{V})$ *is independent of the ordered basis used in its definition.*

Proof.
(i) By Proposition 4.4.2, $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta$.
(ii) By Proposition 4.4.10(i), $M_\gamma = C^t M_\beta C$.
(iii) ($\Rightarrow$) If $N = M_\delta$ for an ordered basis δ, then by (ii), $N = C_{\beta\leftarrow\delta}^t A C_{\beta\leftarrow\delta}$, where $C_{\beta\leftarrow\delta}$ is invertible by Theorem 3.5.2(i).
(iii) ($\Leftarrow$) Suppose $N = P^t A P$ with $P = [p_{ij}]$ invertible, and write $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$. Define $\delta = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ by $\mathbf{w}_j = \sum_i p_{ij}\mathbf{v}_i$.
The columns of P form a basis of $\mathbb{F}^n$ by Theorem 3.4.1. The inverse coordinate map $[\cdot]_\beta^{-1} : \mathbb{F}^n \to \mathcal{V}$ is an isomorphism by Proposition 3.2.2(iii), so δ is a basis by Proposition 1.7.8. By construction, $C_{\beta\leftarrow\delta} = P$. Hence $M_\delta = P^t A P = N$ by (ii).
(iv) By Theorem 3.5.2(i), C is invertible. By Corollary 3.6.3, C^t is invertible. Hence $\operatorname{rank}(M_\gamma) = \operatorname{rank}(M_\beta)$ by (ii), Theorem 3.6.5(ii).
(v) By (ii) and the determinant identities, $\det(M_\gamma) = \det(C)^2 \det(M_\beta)$. Since C is invertible, put $c := \det(C) \in \mathbb{F}^\times$.
The bijection $r \mapsto rc$ of $\mathbb{F}^\times$ identifies $\{r^2 \det(M_\gamma) : r \in \mathbb{F}^\times\}$ with $\{s^2 \det(M_\beta) : s \in \mathbb{F}^\times\}$, proving basis independence.

Structural Commentary: Items (ii)–(v) identify congruence with basis change and establish invariance of rank and discriminant.
Key Insight: Because basis change acts in both arguments, $M \mapsto P^t M P$; the determinant consequently changes by a nonzero square. □

3.2 Quadratic Forms and Polar Forms

Throughout this subsection, $\mathcal{V}$ is a vector space over an arbitrary field $\mathbb{F}$.

Definition 3.2.1. *Quadratic Forms and Polar Forms*
A *quadratic form* on $\mathcal{V}$ is a function $Q : \mathcal{V} \to \mathbb{F}$ such that:
(i) $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{v} \in \mathcal{V})\, Q(a\mathbf{v}) = a^2 Q(\mathbf{v})$;
(ii) the *polar form* $B_Q : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$, defined by $B_Q(\mathbf{u}, \mathbf{v}) := Q(\mathbf{u} + \mathbf{v}) - Q(\mathbf{u}) - Q(\mathbf{v})$, is bilinear.

Proposition 3.2.2. *Quadratic Forms and Symmetric Bilinear Forms*
(i) If $Q : \mathcal{V} \to \mathbb{F}$ is quadratic, then $\operatorname{Sym}(B_Q)$.
(ii) Suppose $\operatorname{char}(\mathbb{F}) \neq 2$ and $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is symmetric and bilinear. If $Q_B(\mathbf{x}) := \frac{1}{2} B(\mathbf{x}, \mathbf{x})$, then Q_B is quadratic and $B_{Q_B} = B$.
(iii) Suppose $\operatorname{char}(\mathbb{F}) \neq 2$ and $Q : \mathcal{V} \to \mathbb{F}$ is quadratic. If $Q_{B_Q}(\mathbf{x}) := \frac{1}{2} B_Q(\mathbf{x}, \mathbf{x})$, then $Q_{B_Q} = Q$.

Proof.
(i) By commutativity of vector addition, $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\, B_Q(\mathbf{u}, \mathbf{v}) = Q(\mathbf{u} + \mathbf{v}) - Q(\mathbf{u}) - Q(\mathbf{v}) = B_Q(\mathbf{v}, \mathbf{u})$. Hence $\operatorname{Sym}(B_Q)$.
(ii) By bilinearity, $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{x} \in \mathcal{V})\, Q_B(a\mathbf{x}) = a^2 Q_B(\mathbf{x})$. Fix $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.
Then $B_{Q_B}(\mathbf{u}, \mathbf{v}) = Q_B(\mathbf{u} + \mathbf{v}) - Q_B(\mathbf{u}) - Q_B(\mathbf{v}) = \frac{1}{2}\big(B(\mathbf{u}, \mathbf{v}) + B(\mathbf{v}, \mathbf{u})\big) = B(\mathbf{u}, \mathbf{v})$.
Thus $B_{Q_B} = B$ is bilinear, and Q_B is quadratic.
(iii) Fix $\mathbf{x} \in \mathcal{V}$. Then $Q_{B_Q}(\mathbf{x}) = \frac{1}{2} B_Q(\mathbf{x}, \mathbf{x}) = \frac{1}{2}\big(Q(2\mathbf{x}) - 2Q(\mathbf{x})\big)$. Since $Q(2\mathbf{x}) = 4Q(\mathbf{x})$, $Q_{B_Q}(\mathbf{x}) = Q(\mathbf{x})$. Hence $Q_{B_Q} = Q$.

Structural Commentary: Items (ii) and (iii) are the two reconstruction identities underlying the odd-characteristic correspondence.
Key Insight: When 2 is invertible, diagonal evaluation and polarization recover each other. □

Proposition 3.2.3. *Polar Forms in Characteristic Two*
Let $Q : \mathcal{V} \to \mathbb{F}$ be quadratic. If $\operatorname{char}(\mathbb{F}) = 2$, then $\operatorname{Alt}(B_Q)$.

Proof. By homogeneity with scalar 0, $Q(\mathbf{0}) = 0$. Since $\operatorname{char}(\mathbb{F}) = 2$, $(\forall\, \mathbf{x} \in \mathcal{V})\, B_Q(\mathbf{x}, \mathbf{x}) = Q(2\mathbf{x}) - 2Q(\mathbf{x}) = Q(\mathbf{0}) = 0$. Thus $\operatorname{Alt}(B_Q)$.
Structural Commentary: Diagonal evaluation makes every polar form in characteristic 2 alternating.
Key Insight: The identities $2\mathbf{x} = \mathbf{0}$ and $2Q(\mathbf{x}) = 0$ force the polar diagonal to vanish. □

Example 3.2.4. *Characteristic Two Loses Diagonal Data*
Assume $\operatorname{char}(\mathbb{F}) = 2$. On $\mathbb{F}^2$, let $Q_0(x, y) = xy$ and $Q_1(x, y) = xy + x^2$.
The identity $(\forall\, i \in \{0, 1\})(\forall\, a, x, y \in \mathbb{F})\, Q_i(ax, ay) = a^2 Q_i(x, y)$ verifies homogeneity. Define $B((x, y), (u, v)) = xv + uy$.
Then $B_{Q_0} = B_{Q_1} = B$, and B is bilinear. Hence Q_0 and Q_1 are quadratic with the same polar form, but $Q_0(1, 0) = 0 \neq 1 = Q_1(1, 0)$.
More generally, for quadratic Q and linear $L : \mathcal{V} \to \mathbb{F}$, set $L^2(\mathbf{v}) := L(\mathbf{v})^2$. Then $(\forall\, a \in \mathbb{F})(\forall\, \mathbf{v} \in \mathcal{V})\, (Q + L^2)(a\mathbf{v}) = a^2 (Q + L^2)(\mathbf{v})$, while $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{V})\, B_{Q+L^2}(\mathbf{u}, \mathbf{v}) = B_Q(\mathbf{u}, \mathbf{v}) + 2L(\mathbf{u})L(\mathbf{v}) = B_Q(\mathbf{u}, \mathbf{v})$. Hence $Q + L^2$ is quadratic.
Structural Commentary: The pair Q_0, Q_1 exemplifies the general square perturbations that polarization cannot detect.
Key Insight: In characteristic 2, a polar form does not determine the diagonal quadratic data.

Example 3.2.5. *Alternating versus Nonalternating Symmetric Forms*
Assume $\operatorname{char}(\mathbb{F}) = 2$. The matrix $H = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ defines the symmetric bilinear form $B_H((x, y), (u, v)) = xv + yu$ on $\mathbb{F}^2$.
Since $(\forall\, x, y \in \mathbb{F})\, B_H((x, y), (x, y)) = 2xy = 0$, the form is alternating.
Since $H^2 = I_2$, the matrix is invertible, so the form is nonsingular by Proposition 4.4.6.
Suppose a diagonal matrix D is congruent to H. By Proposition 3.1.4(iii), D represents B_H in some ordered basis.
Alternation forces every diagonal entry of D to vanish, so $D = 0$. Thus H is not congruent to a nonzero diagonal matrix.
By contrast, $[1]$ is symmetric and nonalternating because its basis vector has self-pairing 1.
Structural Commentary: The forms represented by H and $[1]$ separate the alternating and nonalternating symmetric cases.
Key Insight: A nonsingular alternating form has zero diagonal entries in every basis.

3.3 Orthogonality, Radicals, and Nonsingular Metric Spaces

Throughout this subsection, metric vector spaces are finite-dimensional over an arbitrary field $\mathbb{F}$.

Definition 3.3.1. *Orthogonality, Isotropy, and Radical*
Let $\mathcal{V}$ be a metric vector space. By Definition 3.1.1(ii)–(iv) and Proposition 3.1.2(i), vanishing of the form is symmetric in its two arguments, so left and right orthogonality agree.
(i) $\mathbf{x} \perp \mathbf{y}$ means $\langle \mathbf{x}, \mathbf{y} \rangle = 0$; then $\mathbf{x}, \mathbf{y}$ are *orthogonal*. For $S \subseteq \mathcal{V}$, $S^\perp := \{\mathbf{v} \in \mathcal{V} : (\forall\, \mathbf{s} \in S)\ \langle \mathbf{v}, \mathbf{s} \rangle = 0\}$.
(ii) $\mathcal{S} \leq \mathcal{V}$ is *totally isotropic* or *totally degenerate* if $\mathcal{S} \subseteq \mathcal{S}^\perp$, and *nonsingular as a subspace* if $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$.

(iii) $\mathbf{x} \neq \mathbf{0}$ is *isotropic* or *null* if $\langle \mathbf{x}, \mathbf{x} \rangle = 0$, and *nonisotropic* otherwise.
The space $\mathcal{V}$ is *isotropic* if $(\exists\, \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\})\ \langle \mathbf{x}, \mathbf{x} \rangle = 0$, and *anisotropic* if $(\forall\, \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\})\ \langle \mathbf{x}, \mathbf{x} \rangle \neq 0$.
(iv) $\mathbf{x}$ is *degenerate* if $(\forall\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{v} \rangle = 0$. The *radical* is $\operatorname{rad}(\mathcal{V}) := \mathcal{V}^{\perp}$.
(v) $\mathcal{V}$ is *nonsingular* or *nondegenerate* if $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, *singular* otherwise.
(vi) $\mathcal{V}$ is *totally singular* or *totally degenerate* if $\operatorname{rad}(\mathcal{V}) = \mathcal{V}$.

Proposition 3.3.2. *Metric Radical as a Bilinear Radical*
Let $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$ be the form of a metric vector space $\mathcal{V}$.
(i) $\operatorname{rad}_L(B) = \operatorname{rad}_R(B) = \operatorname{rad}(\mathcal{V})$.
(ii) $\mathcal{V}$ is nonsingular $\iff$ B is perfect.

Proof.
(i) By the definitions of $\mathcal{V}^{\perp}$ and $\operatorname{rad}_L(B)$, $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}_L(B)$. The equality $\operatorname{rad}_L(B) = \operatorname{rad}_R(B)$ holds by Proposition 3.1.2(iv).
(ii) ($\Rightarrow$) If $\mathcal{V}$ is nonsingular, then $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, so $\operatorname{rad}_L(B) = \{\mathbf{0}\}$ by (i). Hence B is perfect by Proposition 4.4.6.
(ii) ($\Leftarrow$) If B is perfect, then $\operatorname{rad}_L(B) = \{\mathbf{0}\}$ by Proposition 4.4.6. Hence $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ by (i), so $\mathcal{V}$ is nonsingular.
Structural Commentary: Item (i) identifies the intrinsic and bilinear radicals; item (ii) transfers nonsingularity to the square pairing.
Key Insight: A metric form is perfect exactly when its common radical vanishes. □

Let β be an ordered basis of $\mathcal{V}$. By Proposition 3.3.2(i) and Proposition 4.4.5, the coordinate radical is $\ker(M_\beta^t)$ and $\dim \operatorname{rad}(\mathcal{V}) = \dim \mathcal{V} - \operatorname{rank}(M_\beta)$. By Proposition 3.3.2(ii) and Proposition 4.4.6, $\mathcal{V}$ is nonsingular $\iff$ M_β is invertible.

Remark 3.3.3. *Subspace Radicals*
For $\mathcal{S} \leq \mathcal{V}$, the restricted form satisfies $\operatorname{rad}(\mathcal{S}) = \{\mathbf{s} \in \mathcal{S} : (\forall\, \mathbf{t} \in \mathcal{S})\ \langle \mathbf{s}, \mathbf{t} \rangle = 0\} = \mathcal{S} \cap \mathcal{S}^{\perp}$.
In general, $\operatorname{rad}(\mathcal{S}) \neq \mathcal{S}^{\perp}$ because the latter is computed in the ambient space.

Proposition 3.3.4. *Riesz Map Isomorphism Criterion*
Let $\mathcal{V}$ be a metric vector space and define $\tau : \mathcal{V} \to \mathcal{V}'$ by $\tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$.
(i) $\tau \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$ and $\ker(\tau) = \operatorname{rad}(\mathcal{V})$.
(ii) τ is an isomorphism $\iff$ $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Proof.
(i) For the metric form B, $\tau = \Psi_B$, so $\tau \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$ by Proposition 4.4.3(i).
The identity $\ker(\tau) = \operatorname{rad}_R(B) = \operatorname{rad}(\mathcal{V})$ follows by Proposition 4.4.5(i) and Proposition 3.3.2(i).
(ii) ($\Rightarrow$) If τ is isomorphism, then it is injective by Proposition 1.6.7, so $\ker(\tau) = \{\mathbf{0}\}$ by Proposition 1.6.2; $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ by (i).
(ii) ($\Leftarrow$) If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, then $\ker(\tau) = \{\mathbf{0}\}$ by (i), so τ is injective by Proposition 1.6.2. By Theorem 4.1.4(ii), $\dim(\mathcal{V}) = \dim(\mathcal{V}')$, so τ is surjective by Corollary 1.7.6(ii) and is an isomorphism by Proposition 1.6.7.
Structural Commentary: Item (i) locates the obstruction in the radical; item (ii) removes it by equal finite dimensions.
Key Insight: The Riesz map is invertible exactly when the metric form is nonsingular. □

Theorem 3.3.5. *Riesz Map for Nonsingular Metric Spaces*
Let $\mathcal{V}$ be a nonsingular metric vector space. Then $(\forall\, f \in \mathcal{V}')(\exists!\, \mathbf{x} \in \mathcal{V})(\forall\, \mathbf{v} \in \mathcal{V})\ f(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$.

Proof. Let $\tau : \mathcal{V} \to \mathcal{V}'$ be defined by $\tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$. By Proposition 3.3.4(ii), τ is an isomorphism.
Its bijectivity is precisely the stated existence and uniqueness.
Structural Commentary: The Riesz isomorphism converts the representation equation into a unique inverse-image problem.
Key Insight: Nonsingularity, rather than positivity, is the exact hypothesis for algebraic Riesz representation. □

Definition 3.3.6. *Isometries and Classical Groups*
Let $\mathcal{V}, \mathcal{W}$ be metric vector spaces.
(i) An *isometry* $\sigma : \mathcal{V} \to \mathcal{W}$ is a bijective linear map satisfying $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle \sigma\mathbf{x}, \sigma\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$; write $\mathcal{V} \cong \mathcal{W}$ if one exists.
(ii) The group of isometries $\mathcal{V} \to \mathcal{V}$ is denoted $\operatorname{Iso}(\mathcal{V})$. If $\mathcal{V}$ is nonsingular orthogonal, this is the *orthogonal group* $\mathcal{O}(\mathcal{V})$.
If $\mathcal{V}$ is nonsingular symplectic, this is the *symplectic group* $\operatorname{Sp}(\mathcal{V})$.

Proposition 3.3.7. *Matrix Criterion for Isometries*
Let β be an ordered basis of a metric vector space $\mathcal{V}$, let $\sigma \in \mathcal{L}(\mathcal{V})$, let $M = M_\beta$, and let $A = [\sigma]_{\beta \leftarrow \beta}$.
(i) $\sigma \in \operatorname{Iso}(\mathcal{V})$ $\iff$ σ is bijective and $A^t M A = M$.
(ii) If $\sigma \in \operatorname{Iso}(\mathcal{V})$ and M is invertible, then $\det A = \pm 1$.

Proof.
(i) ($\Rightarrow$) Suppose that $\sigma \in \operatorname{Iso}(\mathcal{V})$. Then σ is bijective.
By Proposition 3.1.4(i), metric preservation implies $(\forall\, X, Y \in \mathbb{F}^n)\ X^t A^t M A Y = X^t M Y$, where $n = \dim \mathcal{V}$.
Taking X, Y to be standard coordinate columns proves $A^t M A = M$.
(i) ($\Leftarrow$) Suppose that σ is bijective and $A^t M A = M$. By Proposition 3.1.4(i), $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle \sigma\mathbf{x}, \sigma\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$. Thus $\sigma \in \operatorname{Iso}(\mathcal{V})$.
(ii) Suppose that $\sigma \in \operatorname{Iso}(\mathcal{V})$ and M is invertible. By (i), $A^t M A = M$. Hence $\det(M) = \det(A^t M A) = (\det A)^2 \det(M)$.
Since $\det(M) \neq 0$, $(\det A)^2 = 1$, so $\det A = \pm 1$.
Structural Commentary: Item (i) states the matrix criterion; item (ii) records its determinant consequence.
Key Insight: The equation $A^t M A = M$ realizes an isometry group as a congruence stabilizer. □

Definition 3.3.8. *Adjoint for a Nonsingular Bilinear Form*
Let $\mathcal{V}$ be nonsingular. For $T \in \mathcal{L}(\mathcal{V})$, an operator $S \in \mathcal{L}(\mathcal{V})$ is a *metric adjoint* of T if $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, S\mathbf{y} \rangle$.
When this operator is unique, it is denoted by T^*.

Proposition 3.3.9. *Metric Adjoints and Isometries*
Let $\mathcal{V}$ be nonsingular, let $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle$ be its form, and let $T \in \mathcal{L}(\mathcal{V})$.
(i) $(\exists!\, T^* \in \mathcal{L}(\mathcal{V}))(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, T^*\mathbf{y} \rangle$.
(ii) If β is an ordered basis, $M = M_\beta$, and $A = [T]_\beta$, then $[T^*]_\beta = M^{-1} A^t M$.

(iii) T *is an isometry* $\iff$ T *is bijective and* $T^*T = I$.
(iv) *If* T *is bijective, then* $T^*T = I \iff T^* = T^{-1}$.

Proof.

(*i*) Define $\tau : \mathcal{V} \to \mathcal{V}'$ by $\tau(\mathbf{y})(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle$ and $F : \mathcal{V} \to \mathcal{V}'$ by $F(\mathbf{y})(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{y} \rangle$.
By Proposition 3.3.4(ii), τ is an isomorphism, while bilinearity implies $F \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$. Set $T^* = \tau^{-1} \circ F \in \mathcal{L}(\mathcal{V})$.
Then $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, T^*\mathbf{y} \rangle$. If $S \in \mathcal{L}(\mathcal{V})$ satisfies this identity, then $\tau \circ S = F = \tau \circ T^*$, so $S = T^*$.
(*ii*) By Proposition 3.3.2(ii) and Proposition 4.4.6, M is invertible. Put $D = [T^*]_\beta$.
By the adjoint identity and Proposition 3.1.4(i), $(\forall\, X, Y \in \mathbb{F}^n)\ X^t A^t M Y = X^t M D Y$.
Taking X, Y to be standard coordinate columns proves $A^t M = MD$, so $D = M^{-1} A^t M$.
(*iii*) ($\Rightarrow$) Suppose that T is an isometry. Then T is bijective. Fix $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.
The adjoint identity and metric preservation imply $\langle \mathbf{x}, T^*T\mathbf{y} \rangle = \langle T\mathbf{x}, T\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$.
Hence $(T^*T - I)\mathbf{y} \in \operatorname{rad}_R(B) = \{\mathbf{0}\}$ by Proposition 3.3.2(i), and therefore $T^*T = I$.
(*iii*) ($\Leftarrow$) Suppose that T is bijective and $T^*T = I$. For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, the adjoint identity implies $\langle T\mathbf{x}, T\mathbf{y} \rangle = \langle \mathbf{x}, T^*T\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$. QED.
(*iv*) ($\Rightarrow$) If $T^*T = I$, then right-composition with T^{-1} proves $T^* = T^{-1}$.
(*iv*) ($\Leftarrow$) If $T^* = T^{-1}$, then $T^*T = T^{-1}T = I$.

Structural Commentary: Items (i)–(ii) construct and compute the adjoint; items (iii)–(iv) characterize isometries by $T^*T = I$.
Key Insight: Nonsingularity makes the adjoint unique, and a bijective operator is an isometry exactly when its adjoint is its inverse. □

Proposition 3.3.10. *Isometry Is Matrix Congruence*
Let $(\mathcal{V}, B_\mathcal{V})$ *and* $(\mathcal{W}, B_\mathcal{W})$ *be* n*-dimensional metric vector spaces with ordered bases* $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *and* $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ *and representing matrices* M_β *and* N_γ. $\mathcal{V} \cong \mathcal{W} \iff M_\beta$ *and* N_γ *are congruent.*

Proof.

($\Rightarrow$) Let $\sigma : \mathcal{V} \to \mathcal{W}$ be an isometry. It is a bijective linear map and hence an isomorphism by Proposition 1.6.7.
Thus $\delta = (\sigma^{-1}\mathbf{w}_1, \ldots, \sigma^{-1}\mathbf{w}_n)$ is a basis of $\mathcal{V}$ by Proposition 1.7.8.
Metric preservation implies $(\forall\, i, j \in \{1, \ldots, n\})\, (M_\delta)_{ij} = (N_\gamma)_{ij}$, so $M_\delta = N_\gamma$.
Hence N_γ is congruent to M_β by Proposition 3.1.4(iii).
($\Leftarrow$) Suppose that M_β and N_γ are congruent. By Proposition 3.1.4(iii), fix an ordered basis $\delta = (\mathbf{d}_1, \ldots, \mathbf{d}_n)$ of $\mathcal{V}$ satisfying $M_\delta = N_\gamma$.
By Proposition 1.5.8, fix linear maps $\sigma : \mathcal{V} \to \mathcal{W}$ and $\rho : \mathcal{W} \to \mathcal{V}$ satisfying $(\forall\, j \in \{1, \ldots, n\})\, \sigma\mathbf{d}_j = \mathbf{w}_j$ and $\rho\mathbf{w}_j = \mathbf{d}_j$.
The compositions $\rho\sigma$ and $\sigma\rho$ agree with respective identity maps on δ and γ, so $\rho\sigma = I_\mathcal{V}$ and $\sigma\rho = I_\mathcal{W}$ by Proposition 1.5.8.
Hence σ is invertible with inverse ρ and is bijective by Proposition 1.6.7. Fix $\mathbf{x}, \mathbf{y} \in \mathcal{V}$ with δ-coordinate columns X, Y.
These are the γ-coordinate columns of $\sigma\mathbf{x}, \sigma\mathbf{y}$. By Proposition 3.1.4(i), the form values are $X^t N_\gamma Y$ and $X^t M_\delta Y$, respectively.
They agree because $M_\delta = N_\gamma$, so σ is an isometry.

Structural Commentary: Basis transport proves both directions of the congruence criterion.
Key Insight: Isometry classes of metric spaces coincide with congruence classes of their representing matrices. □

Perspective 3.3.11. *Congruence versus Similarity*
Similarity $N = P^{-1}MP$ governs operator matrices. Congruence $N = P^tMP$ governs bilinear-form matrices because the same basis change acts in both slots. Over $\mathbb{R}$, transpose equals conjugate transpose, so real form classification uses congruence.

3.4 Radicals, Hyperbolic Spaces, and Witt Theory

Throughout this subsection, metric vector spaces are finite-dimensional over a fixed arbitrary field $\mathbb{F}$.

Definition 3.4.1. *Orthogonal Direct Sum*
For $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, the notation $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$ means $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{S} \perp \mathcal{T}$; then $\mathcal{V}$ is the *orthogonal direct sum* of $\mathcal{S}$ and $\mathcal{T}$.

Theorem 3.4.2. *Dimensions of Orthogonal Complements*
Let $\mathcal{S} \leq \mathcal{V}$.

(i) *If* $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ *or* $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, *then* $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = \dim(\mathcal{V})$.
(ii) *If* $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, *then* $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$.
(iii) *If* $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, *then* $\mathcal{S}^{\perp\perp} = \mathcal{S}$.
(iv) *If* $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, *then* $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\} \iff \operatorname{rad}(\mathcal{S}^\perp) = \{\mathbf{0}\}$.

Proof.

(*i*) Define $\rho : \mathcal{V} \to \mathcal{S}'$ by $\rho(\mathbf{x})(\mathbf{s}) = \langle \mathbf{s}, \mathbf{x} \rangle$. Bilinearity implies $\rho \in \mathcal{L}(\mathcal{V}, \mathcal{S}')$, and $\ker(\rho) = \mathcal{S}^\perp$.
Fix $f \in \mathcal{S}'$. If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, choose $\mathcal{V} = \mathcal{S} \oplus \mathcal{C}$ by Proposition 2.3.6 and define $\widetilde{f}(\mathbf{s} + \mathbf{c}) = f(\mathbf{s})$.
By Theorem 3.3.5, fix $\mathbf{x} \in \mathcal{V}$ satisfying $(\forall\, \mathbf{s} \in \mathcal{S})\ f(\mathbf{s}) = \widetilde{f}(\mathbf{s}) = \langle \mathbf{s}, \mathbf{x} \rangle$. If $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, fix such an $\mathbf{x} \in \mathcal{S}$ by Theorem 3.3.5 for $\mathcal{S}$.
Thus ρ is surjective in either case. By Theorem 1.7.3 and Theorem 4.1.4(ii), $\dim(\mathcal{V}) = \dim(\ker\rho) + \dim(\mathcal{S}') = \dim(\mathcal{S}^\perp) + \dim(\mathcal{S})$.
(*ii*) If $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, then $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$. This trivial intersection and (i) imply $\mathcal{V} = \mathcal{S} \oplus \mathcal{S}^\perp$.
Also $\mathcal{S} \perp \mathcal{S}^\perp$ holds by the definition of $\mathcal{S}^\perp$. Hence $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$.
(*iii*) Suppose that $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$. The inclusion $\mathcal{S} \subseteq \mathcal{S}^{\perp\perp}$ is immediate.
By (i) for $\mathcal{S}$ and $\mathcal{S}^\perp$, $\dim(\mathcal{S}^{\perp\perp}) = \dim(\mathcal{S})$, so $\mathcal{S}^{\perp\perp} = \mathcal{S}$ by Proposition 1.4.3(ii).
(*iv*) ($\Rightarrow$) If $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, then $\operatorname{rad}(\mathcal{S}^\perp) = \mathcal{S}^\perp \cap \mathcal{S}^{\perp\perp} = \mathcal{S}^\perp \cap \mathcal{S} = \{\mathbf{0}\}$ by (iii).
(*iv*) ($\Leftarrow$) If $\operatorname{rad}(\mathcal{S}^\perp) = \{\mathbf{0}\}$, then $\operatorname{rad}(\mathcal{S}) = \mathcal{S} \cap \mathcal{S}^\perp = \mathcal{S}^{\perp\perp} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$ by (iii).

Structural Commentary: A restriction pairing gives the dimension formula; dimension counting gives the rest.
Key Insight: Nonsingularity turns orthogonal complementation into a dimension-reversing involution. □

Theorem 3.4.3. *Splitting off the Radical*
Let $R = \operatorname{rad}(\mathcal{V})$. *Then* $\exists\, \mathcal{S} \leq \mathcal{V}$ *such that* $\mathcal{V} = R \odot \mathcal{S}$, $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, *and* R *is totally singular.*

Proof. By Proposition 2.3.6, choose $\mathcal{S}$ satisfying $\mathcal{V} = R \oplus \mathcal{S}$. Since $R \perp \mathcal{V}$, $\mathcal{V} = R \odot \mathcal{S}$ and the restricted form on R vanishes.
Thus R is totally singular. If $\mathbf{s} \in \operatorname{rad}(\mathcal{S})$, then $\mathbf{s} \perp \mathcal{S}$ and $\mathbf{s} \perp R$, so $\mathbf{s} \in R \cap \mathcal{S} = \{\mathbf{0}\}$.
Structural Commentary: A vector-space complement isolates the radical and inherits a nonsingular restricted form.
Key Insight: Every singular direction is confined to an orthogonal zero-form summand. □

Proposition 3.4.4. *Metric Quotients*
Let $\mathcal{U} \leq \mathcal{V}$ and consider the rule $\langle \mathbf{x}+\mathcal{U}, \mathbf{y}+\mathcal{U}\rangle_{\mathcal{V}/\mathcal{U}} := \langle \mathbf{x}, \mathbf{y}\rangle$.

(i) The rule is well-defined $\iff \mathcal{U} \leq \operatorname{rad}(\mathcal{V})$.
(ii) If $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$, then $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\mathcal{U}$.
(iii) The metric space $\mathcal{V}/\operatorname{rad}(\mathcal{V})$ is nonsingular.

Proof.

(i) ($\Rightarrow$) Suppose that the rule is well-defined. Then $(\forall\, \mathbf{u} \in \mathcal{U})(\forall\, \mathbf{y} \in \mathcal{V})\, 0 = \langle \mathbf{u}+\mathcal{U}, \mathbf{y}+\mathcal{U}\rangle_{\mathcal{V}/\mathcal{U}} = \langle \mathbf{u}, \mathbf{y}\rangle$, so $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$.
(i) ($\Leftarrow$) Suppose that $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$. By Proposition 3.3.2(i), $(\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{U})(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle \mathbf{x}+\mathbf{u}, \mathbf{y}+\mathbf{v}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$. QED.
(ii) Suppose that $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$. Then $\mathbf{x}+\mathcal{U} \in \operatorname{rad}(\mathcal{V}/\mathcal{U}) \iff (\forall\, \mathbf{y} \in \mathcal{V})\ \langle \mathbf{x}, \mathbf{y}\rangle = 0 \iff \mathbf{x} \in \operatorname{rad}(\mathcal{V})$. Thus $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\mathcal{U}$.
(iii) Set $\mathcal{U} = \operatorname{rad}(\mathcal{V})$ in (ii). Then $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Structural Commentary: Item (i) tests representative independence; items (ii)–(iii) compute and remove the residual radical.
Key Insight: Exactly radical directions may be collapsed without changing pairings. □

Proposition 3.4.5. *Radicals and Orthogonal Direct Sums*
Suppose that $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$.

(i) $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}(\mathcal{S}) \odot \operatorname{rad}(\mathcal{T})$.
(ii) $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong (\mathcal{S}/\operatorname{rad}(\mathcal{S})) \odot (\mathcal{T}/\operatorname{rad}(\mathcal{T}))$.
(iii) $\mathcal{V}$ is nonsingular $\iff \mathcal{S}$ and $\mathcal{T}$ are nonsingular.

Proof.

(i) Fix $\mathbf{v} = \mathbf{s}+\mathbf{t}$ with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$. Orthogonality implies $\mathbf{v} \in \operatorname{rad}(\mathcal{V}) \iff \mathbf{s} \in \operatorname{rad}(\mathcal{S})$ and $\mathbf{t} \in \operatorname{rad}(\mathcal{T})$.
Uniqueness of the decomposition proves the radical identity.
(ii) Define $\Phi((\mathbf{s}+\mathbf{t})+\operatorname{rad}(\mathcal{V})) = (\mathbf{s}+\operatorname{rad}(\mathcal{S}), \mathbf{t}+\operatorname{rad}(\mathcal{T}))$. By (i), Φ well-defined, has inverse $\Psi(\mathbf{s}+\operatorname{rad}(\mathcal{S}), \mathbf{t}+\operatorname{rad}(\mathcal{T})) = (\mathbf{s}+\mathbf{t})+\operatorname{rad}(\mathcal{V})$.
Hence Φ is linear bijection. By orthogonality, $(\forall\, \mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S})(\forall\, \mathbf{t}_1, \mathbf{t}_2 \in \mathcal{T})\ \langle \mathbf{s}_1+\mathbf{t}_1, \mathbf{s}_2+\mathbf{t}_2\rangle = \langle \mathbf{s}_1, \mathbf{s}_2\rangle + \langle \mathbf{t}_1, \mathbf{t}_2\rangle$; Φ is isometry.
(iii) ($\Rightarrow$) If $\mathcal{V}$ is nonsingular, then $\operatorname{rad}(\mathcal{S}) \odot \operatorname{rad}(\mathcal{T}) = \{\mathbf{0}\}$ by (i), so $\mathcal{S}$ and $\mathcal{T}$ are nonsingular.
(iii) ($\Leftarrow$) If $\mathcal{S}$ and $\mathcal{T}$ are nonsingular, then $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ by (i).

Structural Commentary: Item (i) splits the radical; item (ii) descends the decomposition; item (iii) detects nonsingularity.
Key Insight: Orthogonality separates both the form and its invisible directions. □

Corollary 3.4.6. *Recovering a Singular Space from Its Regular Quotient*
If $\dim \mathcal{V} = \dim \mathcal{W}$ and $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong \mathcal{W}/\operatorname{rad}(\mathcal{W})$, then $\mathcal{V} \cong \mathcal{W}$ as metric vector spaces.

Proof.

(i) *Split*: By Theorem 3.4.3, choose $\mathcal{V} = \operatorname{rad}(\mathcal{V}) \odot \mathcal{S}$ and $\mathcal{W} = \operatorname{rad}(\mathcal{W}) \odot \mathcal{T}$ with $\mathcal{S}, \mathcal{T}$ nonsingular.
(ii) *Regular Parts*: By Proposition 3.4.5(ii), $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong \mathcal{S}$ and $\mathcal{W}/\operatorname{rad}(\mathcal{W}) \cong \mathcal{T}$. Hence $\mathcal{S} \cong \mathcal{T}$ by the assumed quotient isometry.
(iii) *Radicals*: The dimension hypotheses imply $\dim \operatorname{rad}(\mathcal{V}) = \dim \operatorname{rad}(\mathcal{W})$.
Fix a vector-space isomorphism between these radicals; it is an isometry because both restricted forms vanish.
(iv) *Assembly*: The orthogonal direct sum of the radical isometry and an isometry $\mathcal{S} \to \mathcal{T}$ is an isometry $\mathcal{V} \to \mathcal{W}$.

Structural Commentary: Split both radicals, identify the regular parts, match dimensions, and reassemble orthogonally.
Key Insight: A radical carries only its dimension once its form vanishes. □

Definition 3.4.7. *Hyperbolic Pairs and Hyperbolic Spaces*

(i) A *hyperbolic pair* is $(\mathbf{u}, \mathbf{v})$ with $\langle \mathbf{u}, \mathbf{u}\rangle = \langle \mathbf{v}, \mathbf{v}\rangle = 0$ and $\langle \mathbf{u}, \mathbf{v}\rangle = 1$.
(ii) The plane $\operatorname{span}(\mathbf{u}, \mathbf{v})$ of a hyperbolic pair is a *hyperbolic plane*.
(iii) A *hyperbolic space* is an orthogonal direct sum of hyperbolic planes.
(iv) A basis $(\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_k, \mathbf{v}_k)$ is a *hyperbolic basis* if each $(\mathbf{u}_i, \mathbf{v}_i)$ is hyperbolic and the planes $\operatorname{span}(\mathbf{u}_i, \mathbf{v}_i)$ are mutually orthogonal.
In the symplectic case it is a *symplectic basis*.

Theorem 3.4.8. *Capturing Isotropic Vectors*
Let $\mathcal{V}$ be nonsingular. If $\mathcal{V}$ is orthogonal, assume $\operatorname{char}(\mathbb{F}) \neq 2$.
Then $(\forall\, \mathbf{u} \in \mathcal{V} \setminus \{\mathbf{0}\})(\langle \mathbf{u}, \mathbf{u}\rangle = 0 \Longrightarrow (\exists\, \mathbf{v} \in \mathcal{V})\ \langle \mathbf{u}, \mathbf{v}\rangle = 1 \wedge \langle \mathbf{v}, \mathbf{v}\rangle = 0)$.

Proof. Fix $\mathbf{u} \neq \mathbf{0}$ with $\langle \mathbf{u}, \mathbf{u}\rangle = 0$. Since $\mathbf{u} \notin \operatorname{rad}(\mathcal{V})$, fix $\mathbf{x} \in \mathcal{V}$ with $c := \langle \mathbf{u}, \mathbf{x}\rangle \neq 0$.
In alternating case, set $\mathbf{v} = c^{-1}\mathbf{x}$; then $\langle \mathbf{u}, \mathbf{v}\rangle = 1$ and $\langle \mathbf{v}, \mathbf{v}\rangle = 0$. In symmetric case, set $\mathbf{v} = c^{-1}\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{x}\rangle}{2c^2}\mathbf{u}$.
Symmetry and $\langle \mathbf{u}, \mathbf{u}\rangle = 0$ imply $\langle \mathbf{u}, \mathbf{v}\rangle = 1$ and $\langle \mathbf{v}, \mathbf{v}\rangle = c^{-2}\langle \mathbf{x}, \mathbf{x}\rangle - c^{-2}\langle \mathbf{x}, \mathbf{x}\rangle = 0$. Thus $(\mathbf{u}, \mathbf{v})$ is hyperbolic.
Structural Commentary: Choose a nonorthogonal partner and remove its self-pairing in the symmetric case.
Key Insight: One scalar correction converts a partner into an isotropic partner. □

Theorem 3.4.9. *Nonsingular Completions*
Let $\mathcal{U}$ be a subspace of a nonsingular metric vector space $\mathcal{V}$, and assume $\operatorname{char}(\mathbb{F}) \neq 2$ in the orthogonal case.
Write $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_r) \odot \mathcal{W}$, where $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ is a basis of $\operatorname{rad}(\mathcal{U})$ and $\mathcal{W}$ is nonsingular.

(i) $(\exists\, \mathbf{v}_1, \ldots, \mathbf{v}_r \in \mathcal{V})$ such that $\mathcal{H}_r = \operatorname{span}(\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_r, \mathbf{v}_r)$ is hyperbolic and $\widehat{\mathcal{U}} = \mathcal{H}_r \odot \mathcal{W}$ contains $\mathcal{U}$ with $\dim \widehat{\mathcal{U}} = \dim \mathcal{U} + r$.
(ii) If $\mathcal{X} \supseteq \mathcal{U}$ is nonsingular, then $\dim \mathcal{X} \geq \dim \mathcal{U} + r$; equality is attained by $\widehat{\mathcal{U}}$, and all equality-case extensions are isometric.
(iii) If $\sigma : \mathcal{U} \to \mathcal{T}$ is an isometry between subspaces of nonsingular metric spaces of the same kind, equality-case extensions can be chosen so that σ extends to an isometry between them.

Proof.

(i) By Theorem 3.4.2(ii), $\mathcal{V} = \mathcal{W} \odot \mathcal{W}^{\perp}$. Thus $(\forall\, i \in \{1, \ldots, r\})\, \mathbf{u}_i \in \mathcal{W}^{\perp}$. Set $\mathcal{H}_0 = \{\mathbf{0}\}$ and proceed inductively.
After constructing $\mathcal{H}_{j-1}$, set $\mathcal{S}_j = \operatorname{span}(\mathbf{u}_{j+1}, \ldots, \mathbf{u}_r) \odot \mathcal{W} \odot \mathcal{H}_{j-1}$. Pairing a relation with $\mathbf{u}_1, \ldots, \mathbf{u}_{j-1}$ eliminates its $\mathbf{v}_i$-coefficients, and independence of the radical basis modulo $\mathcal{W}$ then proves $\mathbf{u}_j \notin \mathcal{S}_j$.
By Theorem 3.4.2(iii), $\mathcal{S}_j = \mathcal{S}_j^{\perp\perp}$, so fix $\mathbf{x} \in \mathcal{S}_j^{\perp}$ with $c := \langle \mathbf{u}_j, \mathbf{x}\rangle \neq 0$.
Set $\mathbf{v}_j = c^{-1}\mathbf{x}$ in the alternating case and $\mathbf{v}_j = c^{-1}\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{x}\rangle}{2c^2}\mathbf{u}_j$ in the orthogonal case.

Then $\mathbf{v}_j \in \mathcal{S}_j^\perp$, $\langle \mathbf{u}_j, \mathbf{v}_j \rangle = 1$, and $\langle \mathbf{v}_j, \mathbf{v}_j \rangle = 0$ by direct calculation. Put $\mathcal{H}_j = \mathcal{H}_{j-1} \odot \operatorname{span}(\mathbf{u}_j, \mathbf{v}_j)$.
After r steps, $\mathcal{H}_r$ is hyperbolic and $\widehat{\mathcal{U}} = \mathcal{H}_r \odot \mathcal{W}$ has the stated containment and dimension.

(*ii*) Let $\mathcal{X} \supseteq \mathcal{U}$ be nonsingular and choose $\mathcal{X} = \mathcal{U} \oplus \mathcal{Y}$. The pairing $\operatorname{rad}(\mathcal{U}) \times \mathcal{Y} \to \mathbb{F}$ has zero left radical: a vector in that radical pairs trivially with both $\mathcal{U}$ and $\mathcal{Y}$ and hence lies in $\operatorname{rad}(\mathcal{X})$.
By Proposition 4.4.5(iii), the pairing matrix has rank r, so $\dim \mathcal{Y} \geq r$ and $\dim \mathcal{X} \geq \dim \mathcal{U} + r$. Item (i) attains equality.
If equality holds, the construction inside $\mathcal{X}$ produces a subspace of equal dimension, hence all of $\mathcal{X}$.
Any two equality-case extensions are isometric because both are $\mathcal{W}\odot$ a hyperbolic r-space.

(*iii*) Put $\mathbf{u}_i' = \sigma \mathbf{u}_i$ and $\mathcal{W}' = \sigma\mathcal{W}$. By (i) for $\mathcal{T} = \operatorname{span}(\mathbf{u}_1', \dots, \mathbf{u}_r') \odot \mathcal{W}'$, choose partners $\mathbf{v}_1', \dots, \mathbf{v}_r'$.
Extend σ by $(\forall\, i \in \{1, \dots, r\})\, \mathbf{v}_i \mapsto \mathbf{v}_i'$. The resulting basis-to-basis linear map preserves the form on the hyperbolic and nonsingular summands, so it is an isometry between the equality-case extensions.

Structural Commentary: Item (i) pairs radical directions; item (ii) proves minimality; item (iii) transports the construction.
Key Insight: Each radical dimension requires exactly one hyperbolic partner. □

Definition 3.4.10. *Witt Extension and Cancellation Properties*
Let $\mathcal{V} \cong \mathcal{V}'$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. Write $\tau : \mathcal{A} \xrightarrow{\sim} \mathcal{B}$ when τ is an isometry. The pair $(\mathcal{V}, \mathcal{V}')$ has:

(i) the *Witt extension property* if $(\forall\, \mathcal{S} \leq \mathcal{V})(\forall\, \mathcal{T} \leq \mathcal{V}')(\forall\, \tau : \mathcal{S} \xrightarrow{\sim} \mathcal{T})(\exists\, \overline{\tau} : \mathcal{V} \xrightarrow{\sim} \mathcal{V}')\, \overline{\tau}|_{\mathcal{S}} = \tau$;
(ii) the *Witt cancellation property* if $(\forall\, \mathcal{S} \leq \mathcal{V})(\forall\, \mathcal{T} \leq \mathcal{V}')$, $\operatorname{rad}(\mathcal{S}) = \operatorname{rad}(\mathcal{T}) = \{\mathbf{0}\}$ and $\mathcal{S} \cong \mathcal{T}$ imply $\mathcal{S}^\perp \cong \mathcal{T}^\perp$.

Proposition 3.4.11. *Witt Extension–Cancellation Bridge*
Let $\mathcal{V} \cong \mathcal{V}'$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. *Then the pair has the Witt extension property $\iff$ it has the Witt cancellation property.*

Proof.

($\Rightarrow$) Let $\mathcal{S} \leq \mathcal{V}$ and $\mathcal{T} \leq \mathcal{V}'$ be nonsingular, and fix an isometry $\sigma : \mathcal{S} \to \mathcal{T}$. By Theorem 3.4.2(ii), $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$ and $\mathcal{V}' = \mathcal{T} \odot \mathcal{T}^\perp$.
By Witt extension property, σ extends to isometry $\widehat{\sigma} : \mathcal{V} \to \mathcal{V}'$. Form preservation, bijectivity imply $\widehat{\sigma}(\mathcal{S}^\perp) = \mathcal{T}^\perp$, so $\mathcal{S}^\perp \cong \mathcal{T}^\perp$.

($\Leftarrow$) Fix an isometry $\sigma : \mathcal{S} \to \mathcal{T}$ between arbitrary subspaces.
By Theorem 3.4.9(iii), choose equality-case nonsingular extensions $\widehat{\mathcal{S}} \supseteq \mathcal{S}$ and $\widehat{\mathcal{T}} \supseteq \mathcal{T}$ and an isometry $\widehat{\sigma} : \widehat{\mathcal{S}} \to \widehat{\mathcal{T}}$ extending σ.
By Theorem 3.4.2(ii), $\mathcal{V} = \widehat{\mathcal{S}} \odot \widehat{\mathcal{S}}^\perp$ and $\mathcal{V}' = \widehat{\mathcal{T}} \odot \widehat{\mathcal{T}}^\perp$. By the Witt cancellation property, fix an isometry $\eta : \widehat{\mathcal{S}}^\perp \to \widehat{\mathcal{T}}^\perp$.
Then $\widehat{\sigma} \odot \eta : \mathcal{V} \to \mathcal{V}'$ is an isometry extending σ.

Structural Commentary: Extension restricts to complements; cancellation extends maps after nonsingular completion.
Key Insight: Completion reduces arbitrary subspaces to an orthogonal complement problem. □

Theorem 3.4.12. *Witt Extension Theorem*
Let $\mathcal{V} \cong \mathcal{W}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
If $\mathcal{S} \leq \mathcal{V}$, $\mathcal{T} \leq \mathcal{W}$, and $\sigma : \mathcal{S} \xrightarrow{\sim} \mathcal{T}$, then $(\exists\, \widehat{\sigma} : \mathcal{V} \xrightarrow{\sim} \mathcal{W})\, \widehat{\sigma}|_{\mathcal{S}} = \sigma$.

Proof.

(i) *Symplectic Preparation*: Every two nonsingular symplectic spaces of equal dimension are isometric.
Indeed, a nonzero vector in such a space is isotropic and belongs to a hyperbolic plane by Theorem 3.4.8.
The plane splits off by Theorem 3.4.2(ii), and its orthogonal complement is nonsingular by Theorem 3.4.2(iv).
Induction on dimension produces a hyperbolic basis, so equal-dimensional spaces have isometric hyperbolic bases.

(ii) *Orthogonal Line Extension*: Let $\mathcal{V}, \mathcal{W}$ be orthogonal, fix an isometry $\theta : \mathcal{V} \to \mathcal{W}$, and suppose $\langle \mathbf{s}, \mathbf{s} \rangle = \langle \mathbf{t}, \mathbf{t} \rangle \neq 0$.
Put $\mathbf{x} = \theta\mathbf{s}$ and $\mathbf{y} = \mathbf{t}$. For nonisotropic $\mathbf{a} \in \mathcal{W}$, define $r_{\mathbf{a}}(\mathbf{z}) = \mathbf{z} - \frac{2\langle \mathbf{z}, \mathbf{a} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle}\mathbf{a}$.
Direct expansion proves that $r_{\mathbf{a}}$ is an isometry and $r_{\mathbf{a}}\mathbf{a} = -\mathbf{a}$. If $\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle \neq 0$, then $r_{\mathbf{x}-\mathbf{y}}\mathbf{x} = \mathbf{y}$.
Otherwise $\langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle = 4\langle \mathbf{y}, \mathbf{y} \rangle \neq 0$ and $r_{\mathbf{y}} r_{\mathbf{x}+\mathbf{y}} \mathbf{x} = \mathbf{y}$. Thus $r_{\mathbf{x}-\mathbf{y}}\theta$ or $r_{\mathbf{y}} r_{\mathbf{x}+\mathbf{y}}\theta$ extends $\mathbf{s} \mapsto \mathbf{t}$.

(iii) *Nonsingular Subspaces*: Assume that $\mathcal{S}, \mathcal{T}$ are nonsingular and induct on $\dim \mathcal{S}$. The zero-dimensional case follows from $\mathcal{V} \cong \mathcal{W}$.
In the symplectic case, choose a hyperbolic plane $\mathcal{A} \leq \mathcal{S}$ by Theorem 3.4.8.
In the orthogonal case, fix $\mathbf{s} \in \mathcal{S}$ with $\langle \mathbf{s}, \mathbf{s} \rangle \neq 0$ by Proposition 3.1.2(ii) and nonsingularity, and set $\mathcal{A} = \operatorname{span}(\mathbf{s})$. Put $\mathcal{B} = \sigma\mathcal{A}$.
In the symplectic case, fix an isometry $\beta : \mathcal{A}^\perp \to \mathcal{B}^\perp$ by (i) and set $\alpha = (\sigma|_{\mathcal{A}}) \odot \beta$.
In the orthogonal case, fix an ambient isometry α extending $\sigma|_{\mathcal{A}}$ by (ii). Set $\mathcal{S}_0 = \mathcal{S} \cap \mathcal{A}^\perp$ and $\mathcal{T}_0 = \mathcal{T} \cap \mathcal{B}^\perp$.
By Theorem 3.4.2(ii), $\mathcal{S} = \mathcal{A} \odot \mathcal{S}_0$, $\mathcal{T} = \mathcal{B} \odot \mathcal{T}_0$, $\mathcal{V} = \mathcal{A} \odot \mathcal{A}^\perp$, and $\mathcal{W} = \mathcal{B} \odot \mathcal{B}^\perp$.
The restriction $\alpha|_{\mathcal{A}^\perp}$ identifies the ambient complements, while $\dim \mathcal{S}_0 < \dim \mathcal{S}$.
By induction, fix an isometry $\eta : \mathcal{A}^\perp \to \mathcal{B}^\perp$ satisfying $\eta|_{\mathcal{S}_0} = \sigma|_{\mathcal{S}_0}$. Then $(\sigma|_{\mathcal{A}}) \odot \eta : \mathcal{V} \to \mathcal{W}$ extends σ.

(iv) *General Subspaces*: By Theorem 3.4.9(iii), choose nonsingular extensions $\widehat{\mathcal{S}} \supseteq \mathcal{S}$ and $\widehat{\mathcal{T}} \supseteq \mathcal{T}$, isometry $\widehat{\sigma} : \widehat{\mathcal{S}} \to \widehat{\mathcal{T}}$ extending σ.
Item (iii) extends $\widehat{\sigma}$ to an isometry $\mathcal{V} \to \mathcal{W}$ and hence extends σ.

Structural Commentary: Prepare elementary summands, induct through nonsingular subspaces, then complete arbitrary radicals.
Key Insight: Reflections handle orthogonal lines; hyperbolic planes handle symplectic pairs. □

Corollary 3.4.13. *Witt Cancellation*
Let $\mathcal{V} = \mathcal{S} \odot \mathcal{A}$, $\mathcal{W} = \mathcal{T} \odot \mathcal{B}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$.
If $\mathcal{V} \cong \mathcal{W}$ and $\mathcal{S} \cong \mathcal{T}$, then $\mathcal{A} \cong \mathcal{B}$.

Proof. Fix an isometry $\sigma : \mathcal{S} \to \mathcal{T}$. By Theorem 3.4.12, fix an isometry $\widehat{\sigma} : \mathcal{V} \to \mathcal{W}$ satisfying $\widehat{\sigma}|_{\mathcal{S}} = \sigma$.
Since $\mathcal{A} = \mathcal{S}^\perp$ and $\mathcal{B} = \mathcal{T}^\perp$, $\widehat{\sigma}(\mathcal{A}) = \widehat{\sigma}(\mathcal{S}^\perp) = (\widehat{\sigma}\mathcal{S})^\perp = \mathcal{T}^\perp = \mathcal{B}$. Hence $\widehat{\sigma}|_{\mathcal{A}} : \mathcal{A} \to \mathcal{B}$ is an isometry.
Structural Commentary: Extend the summand isometry and restrict the extension to its orthogonal complement.
Key Insight: Ambient extension converts cancellation into complement preservation. □

Definition 3.4.14. *Witt Index*
Let $\mathcal{V}$ be nonsingular orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. The *Witt index* is $w(\mathcal{V}) := \max\{\dim \mathcal{L} : \mathcal{L} \leq \mathcal{V},\ \mathcal{L} \subseteq \mathcal{L}^\perp\}$.

Theorem 3.4.15. *Witt Index and Anisotropic Decomposition*
Let $\mathcal{V}$ be orthogonal over $\mathbb{F}$ with $\operatorname{char}(\mathbb{F}) \neq 2$. *Then:*

(i) If $\mathcal{V}$ is nonsingular, all maximal totally isotropic subspaces have dimension $w(\mathcal{V})$.
(ii) If $\mathcal{V}$ is nonsingular, all maximal hyperbolic subspaces have dimension $2w(\mathcal{V})$.

(iii) If $R = \operatorname{rad}(\mathcal{V})$ and $\mathcal{U}$ is a nonsingular complement of R, then $(\exists\, \mathcal{H}, \mathcal{S} \le \mathcal{U})\ \mathcal{H}$ is hyperbolic, $\mathcal{S}$ is anisotropic, $\mathcal{V} = \mathcal{S} \odot \mathcal{H} \odot R$, and $\dim \mathcal{H} = 2w(\mathcal{U})$.

Proof.

(i) Let $\mathcal{L}$ be maximal totally isotropic and choose $\mathcal{M}$ totally isotropic with $\dim \mathcal{M} = w(\mathcal{V})$.
Since $\dim \mathcal{L} \le w(\mathcal{V})$, fix $\mathcal{M}_0 \le \mathcal{M}$ with $\dim \mathcal{M}_0 = \dim \mathcal{L}$.
A vector-space isomorphism $\mathcal{L} \to \mathcal{M}_0$ is an isometry because both restricted forms vanish.
By Theorem 3.4.12, extend it to $\Phi \in \mathcal{O}(\mathcal{V})$. Then $\mathcal{L} \subseteq \Phi^{-1}\mathcal{M}$, so maximality proves $\mathcal{L} = \Phi^{-1}\mathcal{M}$ and $\dim \mathcal{L} = w(\mathcal{V})$.

(ii) A hyperbolic subspace with k planes contains the k-dimensional totally isotropic span of one vector from each pair, so $k \le w(\mathcal{V})$.
By Theorem 3.4.9(i), a $w(\mathcal{V})$-dimensional totally isotropic subspace has a hyperbolic completion $\mathcal{H}_0$ of dimension $2w(\mathcal{V})$.
Suppose that a maximal hyperbolic $\mathcal{H}$ has dimension $2k < 2w(\mathcal{V})$. Choose a $2k$-dimensional hyperbolic $\mathcal{H}_1 \le \mathcal{H}_0$.
By Theorem 3.4.12, extend an isometry $\mathcal{H} \to \mathcal{H}_1$ to $\Phi \in \mathcal{O}(\mathcal{V})$.
hen $\Phi^{-1}\mathcal{H}_0$ is a larger hyperbolic subspace containing $\mathcal{H}$, a contradiction.

(iii) By Theorem 3.4.3, $\mathcal{V} = R \odot \mathcal{U}$ with $\mathcal{U}$ nonsingular. Choose a maximal hyperbolic $\mathcal{H} \le \mathcal{U}$ and put $\mathcal{S} = \mathcal{H}^{\perp} \cap \mathcal{U}$.
By (ii) and Theorem 3.4.2(ii), $\dim \mathcal{H} = 2w(\mathcal{U})$ and $\mathcal{U} = \mathcal{H} \odot \mathcal{S}$. If $\mathcal{S}$ contained a nonzero isotropic vector, that vector would belong to a hyperbolic plane in $\mathcal{S}$ by Theorem 3.4.8, contradicting maximality of $\mathcal{H}$. Hence $\mathcal{S}$ is anisotropic and $\mathcal{V} = \mathcal{S} \odot \mathcal{H} \odot R$.

Structural Commentary: Items (i)–(ii) identify the index geometrically; item (iii) splits the anisotropic residue.
Key Insight: The Witt index is the maximal number of hyperbolic planes. □

3.5 Symplectic Classification and Transvections

Throughout this subsection, symplectic geometries are finite-dimensional over a fixed arbitrary field $\mathbb{F}$.

Theorem 3.5.1. *Classification of Symplectic Geometries*
Let $\mathcal{V}$ be a symplectic geometry, $R = \operatorname{rad}(\mathcal{V})$, $n = \dim(\mathcal{V})$, and $r = \operatorname{rank}(\mathcal{V})$.

(i) If $R = \{\mathbf{0}\}$, then $\mathcal{V}$ is hyperbolic and n is even.
(ii) $\mathcal{V} = R \odot \mathcal{H}$ for a hyperbolic subspace $\mathcal{H} \le \mathcal{V}$, and R is totally singular.
(iii) $r \in 2\mathbb{Z}_{\ge 0}$, and if $r = 2k$, then $\mathcal{V}$ admits a basis β satisfying $M_\beta = X_{2k,n-2k} := \operatorname{diag}(J_2^{\oplus k}, 0_{n-2k})$, where $J_2 := \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.
(iv) For symplectic geometries $\mathcal{V}, \mathcal{W}$, $\mathcal{V} \cong \mathcal{W} \iff (\dim \mathcal{V}, \operatorname{rank}(\mathcal{V})) = (\dim \mathcal{W}, \operatorname{rank}(\mathcal{W}))$.

Proof.

(i) The case $n = 0$ is immediate. Assume $n > 0$ and fix $\mathbf{x} \neq \mathbf{0}$. By alternation and Theorem 3.4.8, $\mathbf{x}$ lies in a hyperbolic plane $\mathcal{H}_1$.
By Theorem 3.4.2(ii),(iv), $\mathcal{V} = \mathcal{H}_1 \odot \mathcal{H}_1^{\perp}$ with $\mathcal{H}_1^{\perp}$ nonsingular. Thus $\mathcal{V}$ is hyperbolic by induction on n, so n is even.

(ii) Fix $\mathcal{V} = R \odot \mathcal{S}$ with $\mathcal{S}$ nonsingular and R totally singular by Theorem 3.4.3. The subspace $\mathcal{S}$ is hyperbolic by (i), so take $\mathcal{H} = \mathcal{S}$.

(iii) In (ii), $\operatorname{rank}(\mathcal{V}) = \dim(\mathcal{H}) = 2k$ and $\dim R = n - 2k$ by the zero form on R and nonsingularity of $\mathcal{H}$.
A hyperbolic basis of $\mathcal{H}$ followed by any basis of R is a basis β of $\mathcal{V}$ with $M_\beta = X_{2k,n-2k}$.

(iv) ($\Rightarrow$) An isometry preserves dimension and rank by its invertibility and form preservation.

(iv) ($\Leftarrow$) Then $\mathcal{V} \cong \mathcal{W}$ by (iii) and Proposition 3.1.4(i) for the normal-form bases.

Structural Commentary: Split the radical, decompose the regular part into planes, and read off the normal form.
Key Insight: Hyperbolic planes are the unique nonsingular symplectic blocks. □

Definition 3.5.2. *Symplectic Transvection*
Let $\mathcal{V}$ be a nonsingular symplectic geometry.
For $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$ and $a \in \mathbb{F}$, the *symplectic transvection* with direction $\mathbf{v}$ and parameter a is $\tau_{\mathbf{v},a} : \mathcal{V} \to \mathcal{V}$, $\mathbf{x} \mapsto \mathbf{x} + a\langle \mathbf{x}, \mathbf{v}\rangle \mathbf{v}$.

Proposition 3.5.3. *Basic Properties of Symplectic Transvections*
Let $\mathcal{V}$ be a nonsingular symplectic geometry, $0 \neq \mathbf{v} \in \mathcal{V}$, $a, b \in \mathbb{F}$, $c \in \mathbb{F}^{\times}$, and $\sigma \in \operatorname{Sp}(\mathcal{V})$. Then:

(i) $\tau_{\mathbf{v},a} \in \operatorname{Sp}(\mathcal{V})$;
(ii) $\tau_{\mathbf{v},a} = I_{\mathcal{V}} \iff a = 0$;
(iii) $\tau_{\mathbf{v},a}\tau_{\mathbf{v},b} = \tau_{\mathbf{v},a+b}$ and $\tau_{\mathbf{v},a}^{-1} = \tau_{\mathbf{v},-a}$;
(iv) $\sigma\tau_{\mathbf{v},a}\sigma^{-1} = \tau_{\sigma\mathbf{v},a}$;
(v) $\tau_{c\mathbf{v},a} = \tau_{\mathbf{v},ac^2}$.

Proof.

(i) Set $T = \tau_{\mathbf{v},a}$ and $\lambda(\mathbf{x}) = \langle \mathbf{x}, \mathbf{v}\rangle$. By bilinearity and alternation, $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})\ \langle T\mathbf{x}, T\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle + a\lambda(\mathbf{y})\lambda(\mathbf{x}) - a\lambda(\mathbf{x})\lambda(\mathbf{y}) = \langle \mathbf{x}, \mathbf{y}\rangle$.
The map T is invertible by $(\forall\, \mathbf{x} \in \mathcal{V})\ \tau_{\mathbf{v},-a}\tau_{\mathbf{v},a}(\mathbf{x}) = \mathbf{x} = \tau_{\mathbf{v},a}\tau_{\mathbf{v},-a}(\mathbf{x})$. Hence $T \in \operatorname{Sp}(\mathcal{V})$.

(ii) ($\Rightarrow$) If $\tau_{\mathbf{v},a} = I_{\mathcal{V}}$ and $a \neq 0$, fix $\mathbf{x}$ with $\langle \mathbf{x}, \mathbf{v}\rangle \neq 0$ by nonsingularity. Then $\tau_{\mathbf{v},a}\mathbf{x} - \mathbf{x} = a\langle \mathbf{x}, \mathbf{v}\rangle \mathbf{v} \neq \mathbf{0}$, a contradiction.

(ii) ($\Leftarrow$) If $a = 0$, then $\tau_{\mathbf{v},a} = I_{\mathcal{V}}$.

(iii) By alternation, $(\forall\, \mathbf{x} \in \mathcal{V})\ \tau_{\mathbf{v},a}\tau_{\mathbf{v},b}(\mathbf{x}) = \mathbf{x} + (a+b)\langle \mathbf{x}, \mathbf{v}\rangle \mathbf{v} = \tau_{\mathbf{v},a+b}(\mathbf{x})$. Hence $\tau_{\mathbf{v},a}\tau_{\mathbf{v},b} = \tau_{\mathbf{v},a+b}$, and $\tau_{\mathbf{v},a}^{-1} = \tau_{\mathbf{v},-a}$ by $b = -a$.

(iv) By form preservation, $(\forall\, \mathbf{x} \in \mathcal{V})\ \sigma\tau_{\mathbf{v},a}\sigma^{-1}(\mathbf{x}) = \mathbf{x} + a\langle \sigma^{-1}\mathbf{x}, \mathbf{v}\rangle \sigma\mathbf{v} = \mathbf{x} + a\langle \mathbf{x}, \sigma\mathbf{v}\rangle \sigma\mathbf{v} = \tau_{\sigma\mathbf{v},a}(\mathbf{x})$.

(v) By bilinearity, $(\forall\, \mathbf{x} \in \mathcal{V})\ \tau_{c\mathbf{v},a}(\mathbf{x}) = \mathbf{x} + ac^2\langle \mathbf{x}, \mathbf{v}\rangle \mathbf{v} = \tau_{\mathbf{v},ac^2}(\mathbf{x})$.

Structural Commentary: Item (i) verifies the form; items (ii)–(v) compute the parameter laws.
Key Insight: Alternation cancels every quadratic correction term. □

Theorem 3.5.4. *Symplectic Transvections Generate*
If $\mathcal{V}$ is nonsingular symplectic, then $\operatorname{Sp}(\mathcal{V}) = \langle \tau_{\mathbf{v},a} : \mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\},\ a \in \mathbb{F}\rangle$.

Proof.

(i) *Generated Subgroup*: Set $G := \langle \tau_{\mathbf{v},a} : \mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\},\ a \in \mathbb{F}\rangle$. Then $G \le \operatorname{Sp}(\mathcal{V})$ by Proposition 3.5.3(i),(iii).

(ii) *Vector Transitivity*: Fix $\mathbf{x}, \mathbf{y} \in \mathcal{V} \setminus \{\mathbf{0}\}$. If $\mathbf{x} = \mathbf{y}$, then $I_{\mathcal{V}}\mathbf{x} = \mathbf{y}$. If $\langle \mathbf{x}, \mathbf{y}\rangle \neq 0$, then $\tau_{\mathbf{y}-\mathbf{x},\langle \mathbf{x},\mathbf{y}\rangle^{-1}}(\mathbf{x}) = \mathbf{y}$.
Suppose that $\langle \mathbf{x}, \mathbf{y}\rangle = 0$. If $\mathbf{y} = \lambda\mathbf{x}$ for $\lambda \in \mathbb{F}^{\times}$, fix $\mathbf{z}$ with $\langle \mathbf{x}, \mathbf{z}\rangle \neq 0$ by nonsingularity; then $\langle \mathbf{z}, \mathbf{y}\rangle = -\lambda\langle \mathbf{x}, \mathbf{z}\rangle \neq 0$.
If $\mathbf{x}, \mathbf{y}$ are independent, fix $\mathbf{z}$ with $\langle \mathbf{x}, \mathbf{z}\rangle = \langle \mathbf{z}, \mathbf{y}\rangle = 1$ by Proposition 2.3.6 for extending the functional $f(\mathbf{x}) = 1$, $f(\mathbf{y}) = -1$ and Theorem 3.3.5. In either case, set $c = \langle \mathbf{x}, \mathbf{z}\rangle$ and $d = \langle \mathbf{z}, \mathbf{y}\rangle$. Then $c, d \neq 0$, $\tau_{\mathbf{z}-\mathbf{x},c^{-1}}(\mathbf{x}) = \mathbf{z}$, and $\tau_{\mathbf{y}-\mathbf{z},d^{-1}}(\mathbf{z}) = \mathbf{y}$.
Thus G is transitive on $\mathcal{V} \setminus \{\mathbf{0}\}$.

(iii) *Hyperbolic-Pair Transitivity*: Fix hyperbolic pairs $(\mathbf{u},\mathbf{v})$ and $(\mathbf{u}',\mathbf{v}')$. Fix $\gamma\in G$ with $\gamma\mathbf{u}=\mathbf{u}'$ by (ii), and set $\mathbf{w}=\gamma\mathbf{v}$. Then $(\mathbf{u}',\mathbf{w})$ is hyperbolic. If $\mathbf{w}=\mathbf{v}'$, take $\delta=I_{\mathcal{V}}$. If $\langle\mathbf{w},\mathbf{v}'\rangle\neq 0$, take $\delta=\tau_{\mathbf{v}'-\mathbf{w},\langle\mathbf{w},\mathbf{v}'\rangle^{-1}}$; then $\delta\mathbf{u}'=\mathbf{u}'$ and $\delta\mathbf{w}=\mathbf{v}'$. Suppose finally that $\mathbf{w}\neq\mathbf{v}'$ and $\langle\mathbf{w},\mathbf{v}'\rangle=0$. The vectors $\mathbf{u}',\mathbf{w},\mathbf{v}'$ are independent by pairing a relation with $\mathbf{u}'$ and $\mathbf{w}$. Fix $\mathbf{z}$ with $\langle\mathbf{u}',\mathbf{z}\rangle=\langle\mathbf{w},\mathbf{z}\rangle=\langle\mathbf{z},\mathbf{v}'\rangle=1$ by Proposition 2.3.6 for extending the functional satisfying $(f(\mathbf{u}'),f(\mathbf{w}),f(\mathbf{v}'))=(1,1,-1)$ and Theorem 3.3.5. Then $\delta=\tau_{\mathbf{v}'-\mathbf{z},1}\tau_{\mathbf{z}-\mathbf{w},1}$ fixes $\mathbf{u}'$ and maps $\mathbf{w}$ to $\mathbf{v}'$. In all cases, $\delta\gamma\in G$ maps the first pair to the second.

(iv) *Dimension Induction*: Fix $\sigma\in\mathrm{Sp}(\mathcal{V})$ and induct on $n=\dim\mathcal{V}$. If $n=0$, then $\sigma=I_{\mathcal{V}}\in G$. For $n>0$, fix a hyperbolic plane $\mathcal{H}=\mathrm{span}(\mathbf{u},\mathbf{v})$ by Theorem 3.5.1(i), and fix $\gamma\in G$ satisfying $\gamma\sigma|_{\mathcal{H}}=I_{\mathcal{H}}$ by (iii). The restriction of $\gamma\sigma$ to $\mathcal{H}^{\perp}$ is a product of transvections by induction. Each such transvection extends to $\mathcal{V}$ by fixing $\mathcal{H}$ pointwise, so $\gamma\sigma\in G$ and $\sigma\in G$.

Structural Commentary: Establish vector and pair transitivity, then remove one hyperbolic plane inductively.
Key Insight: Transvections move hyperbolic frames while fixing their orthogonal complements. □

Proposition 3.5.5. *Symplectic Transformations Have Determinant One*
If $\mathcal{V}$ is nonsingular symplectic, then $(\forall\,\sigma\in\mathrm{Sp}(\mathcal{V}))\ \det(\sigma)=1$.

Proof. By Theorem 3.5.4, it suffices to prove $\det(\tau_{\mathbf{v},a})=1$. The map $\tau_{\mathbf{v},a}$ fixes $\mathbb{F}\mathbf{v}$ and induces $I_{\mathcal{V}/\mathbb{F}\mathbf{v}}$ by alternation. Its matrix in a basis beginning with $\mathbf{v}$ is therefore triangular with unit diagonal, so $\det(\tau_{\mathbf{v},a})=1$.
Structural Commentary: The generation theorem reduces the determinant to one triangular rank-one map.
Key Insight: Every transvection has unit diagonal. □

3.6 Orthogonal Classification over Common Fields

Throughout this subsection, orthogonal geometries are finite-dimensional over a field $\mathbb{F}$; additional hypotheses are stated locally.

Theorem 3.6.1. *Orthogonal Bases and Diagonal Congruence*
Let $(\mathcal{V},B)$ be an orthogonal geometry, $q(\mathbf{x}):=B(\mathbf{x},\mathbf{x})$, $n=\dim\mathcal{V}$, and $r=\operatorname{rank}B$.

(i) An orthogonal basis β with $M_\beta=\mathrm{diag}(a_1,\dots,a_r,0_{n-r})$ and $a_1,\dots,a_r\in\mathbb{F}^\times$ exists if and only if $\mathrm{char}(\mathbb{F})\neq 2$, $B=0$, or $q\not\equiv 0$.
(ii) A symmetric $A\in\mathrm{Mat}(n,n,\mathbb{F})$ is congruent to a diagonal matrix if and only if $\mathrm{char}(\mathbb{F})\neq 2$, $A=0$, or $(\exists\,\mathbf{x}\in\mathbb{F}^n)\,\mathbf{x}^tA\mathbf{x}\neq 0$.

Proof.

(i) ($\Rightarrow$) Suppose that such a basis exists. If $\mathrm{char}(\mathbb{F})=2$ and $q\equiv 0$, every diagonal entry $q(\mathbf{v})$ of M_β vanishes, so $M_\beta=0$ and $B=0$.
(i) ($\Leftarrow$) If $B=0$, any basis works. Assume $B\neq 0$. Then $q\not\equiv 0$ by hypothesis in characteristic 2, Proposition 3.1.2(ii) otherwise. Successive splittings by Theorem 3.4.2(ii) terminate with $\mathcal{V}=\mathbb{F}\mathbf{u}_1\odot\cdots\odot\mathbb{F}\mathbf{u}_s\odot\mathcal{W}$, where $q(\mathbf{u}_i)=a_i\in\mathbb{F}^\times$ and $q|_{\mathcal{W}}\equiv 0$. If $\mathrm{char}(\mathbb{F})\neq 2$, then $B|_{\mathcal{W}}=0$ by Proposition 3.1.2(ii), so append any basis of $\mathcal{W}$. Assume $\mathrm{char}(\mathbb{F})=2$. By Theorem 3.5.1(ii), $\mathcal{W}=R\odot\mathcal{H}_1\odot\cdots\odot\mathcal{H}_t$ with zero form on R and hyperbolic planes $\mathcal{H}_j$. Fix a current $\mathbf{u}$ with $q(\mathbf{u})=a\neq 0$. In each $\mathcal{H}_j$, rescale a hyperbolic basis $(\mathbf{e},\mathbf{f})$ so $B(\mathbf{e},\mathbf{f})=a$. The vectors $\mathbf{u}+\mathbf{e}$, $\mathbf{u}+\mathbf{f}$, and $\mathbf{u}+\mathbf{e}+\mathbf{f}$ are independent by coefficient comparison, have norm a, and are pairwise orthogonal because every cross-pairing is $a+a=0$. Replacing $\mathbb{F}\mathbf{u}\odot\mathcal{H}_j$ by these three lines successively and appending a basis of R completes the construction.
(ii) ($\Rightarrow$) Let $D=P^tAP$ be diagonal with P invertible. If $\mathrm{char}(\mathbb{F})=2$ and $(\forall\,\mathbf{x}\in\mathbb{F}^n)\,\mathbf{x}^tA\mathbf{x}=0$, then every diagonal entry of D vanishes. Thus $D=0$ and $A=0$.
(ii) ($\Leftarrow$) Let $B_A(\mathbf{x},\mathbf{y})=\mathbf{x}^tA\mathbf{y}$ on $\mathbb{F}^n$. The hypothesis for A is the hypothesis in (i) for B_A, so B_A has an orthogonal basis by (i). Its change-of-basis matrix diagonalizes A by Proposition 3.1.4(ii).

Structural Commentary: Split nonisotropic lines, absorb any alternating residue, and translate bases into matrix congruence.
Key Insight: Only a nonzero alternating characteristic-2 form resists diagonalization. □

Proposition 3.6.2. *Recognizing Orthogonal Hyperbolic Planes*
Let $\mathcal{H}$ be a 2-dimensional nonsingular orthogonal geometry with $\mathrm{char}(\mathbb{F})\neq 2$. The following are equivalent:

(i) $\mathcal{H}$ is a hyperbolic plane;
(ii) $(\exists\,\mathbf{u}\in\mathcal{H}\setminus\{\mathbf{0}\})\ \langle\mathbf{u},\mathbf{u}\rangle=0$;
(iii) $\Delta(\mathcal{H})=-(\mathbb{F}^\times)^2$.

Proof.

(i) $\Rightarrow$ (ii) A hyperbolic pair consists of nonzero isotropic vectors.
(ii) $\Rightarrow$ (iii) By Theorem 3.4.8, fix $\mathbf{v}$ such that $(\mathbf{u},\mathbf{v})$ is hyperbolic. $(\mathbf{u},\mathbf{v})$ is a basis as $\dim\mathcal{H}=2$, and its matrix is $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$. QED.
(iii) $\Rightarrow$ (i) By Theorem 3.6.1(i), write $M=\mathrm{diag}(a,b)$ with $a,b\in\mathbb{F}^\times$. Fix $c\in\mathbb{F}^\times$ satisfying $ab=-c^2$ by the discriminant hypothesis. Then $a\left(\frac{c}{a}\right)^2+b=0$, so $\left(\frac{c}{a},1\right)$ is isotropic. It lies in a hyperbolic plane by Theorem 3.4.8; that plane equals $\mathcal{H}$ by dimension.

Structural Commentary: Cycle from a hyperbolic basis to its discriminant, then reconstruct an isotropic vector.
Key Insight: A nonsingular orthogonal plane is hyperbolic exactly when it represents zero. □

Example 3.6.3. *Characteristic Two Orthogonal-Symplectic Plane*
Assume $\mathrm{char}(\mathbb{F})=2$ and let $(\mathcal{H},B)$ have basis $(\mathbf{e},\mathbf{f})$ with Gram matrix $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$.
Its matrix is symmetric with determinant 1, while $(\forall\,a,b\in\mathbb{F})\,B(a\mathbf{e}+b\mathbf{f},a\mathbf{e}+b\mathbf{f})=2ab=0$ makes the form alternating.
For $\mathbf{0}\neq\mathbf{x}\in\mathcal{H}$, $\mathbb{F}\mathbf{x}\leq(\mathbb{F}\mathbf{x})^{\perp}$ by alternation, and both spaces are one-dimensional by Theorem 3.4.2(i).
Hence they are equal, so no orthogonal basis exists, exactly the obstruction in Theorem 3.6.1.

Computation Method 3.6.4. *Lagrange Orthogonalization*
Let $(\mathcal{V},B)$ be an orthogonal geometry with $\mathrm{char}(\mathbb{F})\neq 2$ and $q(\mathbf{x}):=B(\mathbf{x},\mathbf{x})$. Input: a spanning list L. Output: an orthogonal basis.

(i) Initialize $\mathcal{W}_0=\mathcal{V}$ and $L_0=L$. At stage k, stop with $R=\mathcal{W}_k$ if $q|_{\mathcal{W}_k}\equiv 0$; otherwise fix $\mathbf{u}_{k+1}\in\mathcal{W}_k$ with $q(\mathbf{u}_{k+1})\neq 0$. If $(\forall\,\mathbf{x}\in L_k)\,q(\mathbf{x})=0$, fix i,j with $B(\mathbf{x}_i,\mathbf{x}_j)\neq 0$ and take $\mathbf{u}_{k+1}=\mathbf{x}_i+\mathbf{x}_j$; then $q(\mathbf{u}_{k+1})=2B(\mathbf{x}_i,\mathbf{x}_j)\neq 0$.
(ii) Put $\mathcal{W}_{k+1}=\mathcal{W}_k\cap(\mathbb{F}\mathbf{u}_{k+1})^{\perp}$. Replace each $\mathbf{x}\in L_k$ by $\mathbf{x}-\frac{B(\mathbf{x},\mathbf{u}_{k+1})}{q(\mathbf{u}_{k+1})}\mathbf{u}_{k+1}$ and discard zero vectors. The resulting list L_{k+1} spans $\mathcal{W}_{k+1}$ by the orthogonal splitting $\mathcal{W}_k=\mathbb{F}\mathbf{u}_{k+1}\odot\mathcal{W}_{k+1}$.
(iii) The process stops because $\dim\mathcal{W}_{k+1}=\dim\mathcal{W}_k-1$. If $(\mathbf{z}_1,\dots,\mathbf{z}_m)$ is any basis of R, output $\beta=(\mathbf{u}_1,\dots,\mathbf{u}_s,\mathbf{z}_1,\dots,\mathbf{z}_m)$. Then $q|_R\equiv 0$ and $B|_R=0$ by Proposition 3.1.2(ii), so β is orthogonal.

Definition 3.6.5. *Universal Orthogonal Geometry*
An orthogonal geometry $\mathcal{V}$ over $\mathbb{F}$ is *universal* if its norm map $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$ satisfies $q(\mathcal{V}) = \mathbb{F}$.

Proposition 3.6.6. *Finite-Field Square Classes and Universality*
Let $\mathbb{F}_q$ be a finite field and let $S = (\mathbb{F}_q^\times)^2$. For an orthogonal geometry $\mathcal{V}$, write $q_{\mathcal{V}}(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$.

(i) If $\operatorname{char}(\mathbb{F}_q) = 2$, then $(\forall\, c \in \mathbb{F}_q)(\exists\, x \in \mathbb{F}_q)\, c = x^2$.
(ii) If $\operatorname{char}(\mathbb{F}_q) \neq 2$, then $|S| = \frac{q-1}{2}$ and $\mathbb{F}_q^\times \setminus S = dS$ for any nonsquare d.
(iii) If $\mathcal{V}$ is orthogonal over odd $\mathbb{F}_q$ and contains a nonsingular subspace of dimension at least 2, then $q_{\mathcal{V}}(\mathcal{V}) = \mathbb{F}_q$.

Proof.

(i) In characteristic 2, $x^2 = y^2 \Rightarrow (x-y)^2 = 0 \Rightarrow x = y$. Thus $x \mapsto x^2$ is injective, hence bijective on the finite set $\mathbb{F}_q$.
(ii) Suppose $\operatorname{char}(\mathbb{F}_q) \neq 2$. Then $(\forall\, x, y \in \mathbb{F}_q^\times)\, x^2 = y^2 \iff x = \pm y$, so the $\frac{q-1}{2}$ pairs $\{x, -x\}$ determine the elements of S.
If d is nonsquare, then $dS \cap S = \varnothing$ and $|dS| = |S| = \frac{q-1}{2}$, so $\mathbb{F}_q^\times \setminus S = dS$.
(iii) Set $q = q_{\mathcal{V}}$. By Theorem 3.6.1(i), the nonsingular subspace contains orthogonal $\mathbf{u}, \mathbf{v}$ with $a := q(\mathbf{u}) \neq 0$ and $b := q(\mathbf{v}) \neq 0$.
Fix $c \in \mathbb{F}_q$. The sets $a\{\alpha^2 : \alpha \in \mathbb{F}_q\}$ and $c - b\{\beta^2 : \beta \in \mathbb{F}_q\}$ each have $\frac{q+1}{2}$ elements and therefore intersect.
Fix $\alpha, \beta \in \mathbb{F}_q$ satisfying $a\alpha^2 = c - b\beta^2$; then $c = q(\alpha\mathbf{u} + \beta\mathbf{v})$.

Structural Commentary: Count square fibers, identify the nonsquare coset, and force two quadratic-value sets to intersect.
Key Insight: Two subsets of size $\frac{q+1}{2}$ cannot be disjoint. □

Theorem 3.6.7. *Canonical Orthogonal Forms over Common Fields*
Let $M = M^t \in \operatorname{Mat}(n, n, \mathbb{F})$, let $r = \operatorname{rank} M$, and write $M \sim N$ for congruence. If $\operatorname{char}(\mathbb{F}) = 2$, assume M is not nonzero alternating.

(i) If $\mathbb{F}$ is algebraically closed, then $M \sim I_r \oplus 0_{n-r}$, and rank is a complete congruence invariant.
(ii) If $\mathbb{F} = \mathbb{R}$, then $M \sim I_p \oplus (-I_m) \oplus 0_k$ for unique $p, m, k \in \mathbb{Z}_{\geq 0}$ satisfying $p + m + k = n$.
(iii) If $\mathbb{F} = \mathbb{F}_q$ has characteristic 2, then $M \sim I_r \oplus 0_{n-r}$, and rank is a complete congruence invariant among the admitted matrices.
(iv) If $\mathbb{F} = \mathbb{F}_q$ has odd characteristic and $d \in \mathbb{F}_q^\times$ is a fixed nonsquare, then $r = 0$ implies $M \sim 0_n$; for $r > 0$, exactly one of $M \sim I_r \oplus 0_{n-r}$ and $M \sim I_{r-1} \oplus [d] \oplus 0_{n-r}$ holds.

Proof.

(i) By Theorem 3.6.1(ii), write $M \sim \operatorname{diag}(a_1, \ldots, a_r, 0_{n-r})$. Fix $s_i \in \mathbb{F}^\times$ satisfying $a_i = s_i^2$ by algebraic closure.
Rescaling produces $I_r \oplus 0_{n-r}$. Rank is congruence-invariant by Proposition 3.1.4(iv), hence complete.
(ii) By Theorem 3.6.1(ii), rescaling produces $D = I_p \oplus (-I_m) \oplus 0_k$. The positive coordinate subspace has dimension p, while every positive-definite subspace intersects the nonpositive coordinate subspace trivially and therefore has dimension at most p.
Thus p is its maximal possible dimension and is congruence-invariant. The same argument for $-D$ fixes m, and $k = n - p - m$.
(iii) By Theorem 3.6.1(ii), $M \sim \operatorname{diag}(a_1, \ldots, a_r, 0_{n-r})$. Each a_i is a square by Proposition 3.6.6(i), so rescaling produces $I_r \oplus 0_{n-r}$.
Rank invariance proves completeness.
(iv) By Theorem 3.6.1(ii), diagonalize M and let $\mathcal{U}$ be its nonsingular block, with norm map q. While $\dim \mathcal{U} \geq 2$, fix $\mathbf{u} \in \mathcal{U}$ with $q(\mathbf{u}) = 1$ by Proposition 3.6.6(iii), then split its line with nonsingular complement by Theorem 3.4.2(ii),(iv).
Thus regular block becomes $I_{r-1} \oplus [a]$. By Proposition 3.6.6(ii), rescale so $a \in \{1, d\}$. Congruence induces an isometry of regular quotients, whose two discriminants are distinct by Proposition 3.1.4(v). If $r = 0$, then $M = 0_n$.

Structural Commentary: Diagonalize first, normalize coefficients field by field, and separate the remaining forms by invariant data.
Key Insight: Square classes and signs control the coefficients that scaling cannot remove. □

Definition 3.6.8. *Minkowski Space*
Minkowski space $\mathbb{R}^{3,1}$ is $\mathbb{R}^4$ with standard Gram matrix $\operatorname{diag}(1, 1, 1, -1)$.

Example 3.6.9. *Null Vectors in Minkowski Space*
In $\mathbb{R}^{3,1}$, the null cone is $\mathcal{N} = \{\mathbf{x} \in \mathbb{R}^4 \setminus \{\mathbf{0}\} : x_1^2 + x_2^2 + x_3^2 = x_4^2\}$, and $\mathbf{e}_1 + \mathbf{e}_4 \in \mathcal{N}$. The space is not isometric to Euclidean $\mathbb{R}^4$: an isometry matrix A would satisfy $A^t I_4 A = \operatorname{diag}(1, 1, 1, -1)$, whose determinant identity $(\det A)^2 = -1$ is impossible over $\mathbb{R}$.

Example 3.6.10. *Two Rank-Two Orthogonal Planes over $\mathbb{F}_5$*
In $\mathbb{F}_5$, $(\mathbb{F}_5^\times)^2 = \{1, 4\}$ and the nonsquare class is $\{2, 3\}$. Let $q_1(x, y) = x^2 + y^2$ and $q_2(x, y) = x^2 + 2y^2$.

(i) The form q_1 has matrix I_2 and is hyperbolic by $q_1(1, 2) = 0$ and Proposition 3.6.2.
(ii) The form q_2 is anisotropic: $q_2(x, y) = 0$ and $y = 0$ imply $x = 0$, while $y \neq 0$ would make the nonsquare $3 = \left(\frac{x}{y}\right)^2$.

Thus the two forms realize the nonsingular rank-2 classes in Theorem 3.6.7(iv).

Lemma 3.6.11. *Odd Finite-Field Isotropy Bound*
Let $\mathcal{V}$ be nonsingular orthogonal over odd $\mathbb{F}_q$, put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$, and write $\mathcal{H}^{\odot s}$ for s hyperbolic planes.

(i) If $\dim \mathcal{V} \geq 3$, then $(\exists\, \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\})\, q(\mathbf{x}) = 0$.
(ii) $\mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$, where $\mathcal{R}$ is nonsingular anisotropic and $\dim \mathcal{R} \leq 2$.

Proof.

(i) Fix a nonsingular plane $\mathcal{P} \leq \mathcal{V}$ and $\mathbf{0} \neq \mathbf{z} \in \mathcal{P}^\perp$ by Theorems 3.6.1(i) and 3.4.2(i). If $q(\mathbf{z}) = 0$, take $\mathbf{x} = \mathbf{z}$.
Otherwise fix $\mathbf{p} \in \mathcal{P}$ with $q(\mathbf{p}) = -q(\mathbf{z})$ by Proposition 3.6.6(iii); then $\mathbf{0} \neq \mathbf{p} + \mathbf{z}$ is isotropic by $\mathcal{P} \cap \mathcal{P}^\perp = \{\mathbf{0}\}$ and $\mathcal{P} \perp \mathcal{P}^\perp$.
(ii) Induct on $\dim \mathcal{V}$. If $\mathcal{V}$ is anisotropic, take $(s, \mathcal{R}) = (0, \mathcal{V})$; then $\dim \mathcal{R} \leq 2$ by (i).
Otherwise an isotropic vector lies in a hyperbolic plane $\mathcal{H}$ by Theorem 3.4.8. By Theorem 3.4.2(ii),(iv), $\mathcal{V} = \mathcal{H} \odot \mathcal{H}^\perp$ with $\mathcal{H}^\perp$ nonsingular and two dimensions smaller, so induction completes the decomposition.

Structural Commentary: Force an isotropic vector by universality, then split hyperbolic planes inductively.
Key Insight: An anisotropic space over an odd finite field has dimension at most two. □

Theorem 3.6.12. *Odd Finite-Field Orthogonal Decomposition*
Let $\mathbb{F}_q$ have odd characteristic, fix a nonsquare $d \in \mathbb{F}_q^\times$, and let $\mathcal{V} \neq \{\mathbf{0}\}$ be nonsingular orthogonal.
Write $\mathcal{H}^{\odot s}$ for an orthogonal sum of s hyperbolic planes, $\langle a \rangle$ for a line of value a, and $\mathcal{A}_d := \langle 1 \rangle \odot \langle -d \rangle$.

(i) If $\dim \mathcal{V} = 2m + 1$, then $(\exists\, a \in \mathbb{F}_q^\times)\, \mathcal{V} \cong \mathcal{H}^{\odot m} \odot \langle a \rangle \;\wedge\; a(\mathbb{F}_q^\times)^2 = (-1)^m \Delta(\mathcal{V})$.

(ii) If $\dim \mathcal{V} = 2m$, then $m \geq 1$ and exactly one of $\mathcal{V} \cong \mathcal{H}^{\odot m}$ and $\mathcal{V} \cong \mathcal{H}^{\odot(m-1)} \odot \mathcal{A}_d$ holds. The first occurs if and only if $\Delta(\mathcal{V}) = (-1)^m(\mathbb{F}_q^\times)^2$; the second occurs if and only if $\Delta(\mathcal{V}) = (-1)^m d(\mathbb{F}_q^\times)^2$.

Proof.

(*i*) By Lemma 3.6.11(ii), $\mathcal{V} \cong \mathcal{H}^{\odot m} \odot \mathcal{R}$ with $\dim \mathcal{R} = 1$. Fix $a \in \mathbb{F}_q^\times$ satisfying $\mathcal{R} \cong \langle a \rangle$.
Since $\Delta(\mathcal{H}) = -(\mathbb{F}_q^\times)^2$, $\Delta(\mathcal{V}) = (-1)^m a(\mathbb{F}_q^\times)^2$, and multiplication by $(-1)^m$ proves the stated square class for a.

(*ii*) ($\Rightarrow$) Since $\mathcal{V} \neq \{\mathbf{0}\}$, $m \geq 1$. If $\mathcal{V} \cong \mathcal{H}^{\odot m}$, then $\Delta(\mathcal{V}) = (-1)^m(\mathbb{F}_q^\times)^2$. If $\mathcal{V} \cong \mathcal{H}^{\odot(m-1)} \odot \mathcal{A}_d$, then $\Delta(\mathcal{A}_d) = -d(\mathbb{F}_q^\times)^2$ and $\Delta(\mathcal{V}) = (-1)^m d(\mathbb{F}_q^\times)^2$. These square classes are distinct, so at most one alternative holds.

(*ii*) ($\Leftarrow$) By Lemma 3.6.11(ii), $\mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$ with $\dim \mathcal{R} \in \{0, 2\}$. If $\dim \mathcal{R} = 0$, then $s = m$ and $\mathcal{V} \cong \mathcal{H}^{\odot m}$.
Otherwise $s = m - 1$, while $\mathcal{R}$ and $\mathcal{A}_d$ are anisotropic planes because d is nonsquare.
They lie in the unique nonhyperbolic rank-2 class by Theorem 3.6.7(iv) and Proposition 3.6.2, so $\mathcal{R} \cong \mathcal{A}_d$.
The forward discriminant calculations select the unique alternative.

Structural Commentary: Split to the anisotropic residue, use parity for its dimension, and use discriminant for its class.
Key Insight: Over an odd finite field, parity and discriminant determine the Witt residue. □

3.7 Quadratic Refinements over $\mathbb{F}_2$

Throughout this subsection, vector spaces are finite-dimensional over $\mathbb{F}_2$.

Definition 3.7.1. *Quadratic Isometries and Arf Sums over $\mathbb{F}_2$*
Let $Q : \mathcal{V} \to \mathbb{F}_2$ be quadratic with nonsingular polar form B_Q.

(i) If $Q' : \mathcal{W} \to \mathbb{F}_2$ is quadratic, a *quadratic isometry* $Q \to Q'$ is a linear isomorphism $\sigma : \mathcal{V} \to \mathcal{W}$ satisfying $(\forall\, \mathbf{x} \in \mathcal{V})\, Q'(\sigma\mathbf{x}) = Q(\mathbf{x})$. Write $Q \cong Q'$ if such a map exists.

(ii) Given a symplectic basis $\beta = (\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_m, \mathbf{v}_m)$ of B_Q, its *Arf sum* is $\mathrm{Arf}_\beta(Q) := \sum_{i=1}^{m} Q(\mathbf{u}_i)Q(\mathbf{v}_i) \in \mathbb{F}_2$.

Example 3.7.2. *One Polar Form with Two Refinements*
On $\mathcal{V} = \mathbb{F}_2^2$, put $H(x, y) = xy$ and $A(x, y) = x^2 + xy + y^2$.
Both have polar form $B((x, y), (u, v)) = xv + yu$, whose radical is zero because $B((x, y), (0, 1)) = x$ and $B((x, y), (1, 0)) = y$.
However, $H^{-1}(0) \setminus \{(0, 0)\} = \{(1, 0), (0, 1)\}$, whereas $A^{-1}(0) = \{(0, 0)\}$.
Hence $H \ncong A$ because a quadratic isometry preserves the zero set.

Theorem 3.7.3. *Arf Classification over $\mathbb{F}_2$*
Let $Q : \mathcal{V} \to \mathbb{F}_2$ be quadratic with nonsingular polar form B_Q, and let $\mathscr{B}_Q$ be the set of symplectic bases of B_Q.
For $m \geq 1$, $a \in \mathbb{F}_2$, define $Q_a^{(m)}(x_1, y_1, \ldots, x_m, y_m) := x_1y_1 + a(x_1^2 + y_1^2) + \sum_{i=2}^{m} x_iy_i$. Let $Q_0^{(0)}$ be zero form on $\{\mathbf{0}\}$.

(i) $(\exists!\, m \in \mathbb{Z}_{\geq 0})\ \dim \mathcal{V} = 2m$.

(ii) For this m, $(\forall\, \beta \in \mathscr{B}_Q)\, Q \cong Q^{(m)}_{\mathrm{Arf}_\beta(Q)}$.

(iii) If $m \geq 1$, then $(\forall\, a \in \mathbb{F}_2)\, |(Q_a^{(m)})^{-1}(0)| = 2^{2m-1} + (-1)^a 2^{m-1}$; *in particular,* $Q_0^{(m)} \ncong Q_1^{(m)}$.

(iv) $(\forall\, \beta, \gamma \in \mathscr{B}_Q)\ \mathrm{Arf}_\beta(Q) = \mathrm{Arf}_\gamma(Q)$; *denote this common value by* $\mathrm{Arf}(Q)$.

(v) $Q \cong Q^{(m)}_{\mathrm{Arf}(Q)}$. If $m \geq 1$, then $(\forall\, a \in \mathbb{F}_2)\ \mathrm{Arf}(Q_a^{(m)}) = a$.

Proof.

(i) Form B_Q is alternating by Proposition 3.2.3. By Theorem 3.5.1(i), it has symplectic basis with $2m$ vectors for unique $m \in \mathbb{Z}_{\geq 0}$.

(ii) If $m = 0$, then $Q = Q_0^{(0)}$ by $Q(\mathbf{0}) = 0$. Assume $m \geq 1$ and fix $\beta = (\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_m, \mathbf{v}_m) \in \mathscr{B}_Q$.
The form Q is the orthogonal sum of its restrictions to the basis planes by their polar orthogonality.
On one pair $(\mathbf{u}, \mathbf{v})$, put $r = Q(\mathbf{u})$ and $s = Q(\mathbf{v})$; then $(\forall\, x, y \in \mathbb{F}_2)\, Q(x\mathbf{u} + y\mathbf{v}) = rx^2 + xy + sy^2$.
If $(r, s) = (0, 0)$, this is $H(x, y) = xy$. If $(r, s) = (0, 1)$, replace $\mathbf{v}$ by $\mathbf{u} + \mathbf{v}$; if $(r, s) = (1, 0)$, replace $\mathbf{u}$ by $\mathbf{u} + \mathbf{v}$.
Each replacement is a hyperbolic pair of Q-value zero, so these cases are also H.
The remaining case $(r, s) = (1, 1)$ is $A(x, y) = x^2 + xy + y^2$. Thus the parity of the A-summands is $\mathrm{Arf}_\beta(Q)$.
To cancel two A-summands with symplectic bases $(\mathbf{e}_1, \mathbf{f}_1)$ and $(\mathbf{e}_2, \mathbf{f}_2)$, put $\mathbf{p}_1 = \mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{q}_1 = \mathbf{f}_1 + \mathbf{e}_2$, $\mathbf{p}_2 = \mathbf{e}_1 + \mathbf{f}_1 + \mathbf{f}_2$, and $\mathbf{q}_2 = \mathbf{e}_1 + \mathbf{f}_1 + \mathbf{e}_2 + \mathbf{f}_2$. Direct evaluation has $Q(\mathbf{p}_j) = Q(\mathbf{q}_j) = 0$, $B_Q(\mathbf{p}_j, \mathbf{q}_j) = 1$, and every cross-pairing zero.
Hence $(\mathbf{p}_1, \mathbf{q}_1, \mathbf{p}_2, \mathbf{q}_2)$ is a symplectic basis for two H-summands, so cancelling pairs of A-summands proves the claim.

(iii) For $P : \mathcal{U} \to \mathbb{F}_2$, put $D(P) := \sum_{\mathbf{x} \in \mathcal{U}} (-1)^{P(\mathbf{x})}$. Then $D(P \odot P') = D(P)D(P')$ by factoring the sum over pairs, while direct evaluation has $D(H) = 1 + 1 + 1 - 1 = 2$ and $D(A) = 1 - 1 - 1 - 1 = -2$. Hence $D(Q_a^{(m)}) = (-1)^a 2^m$.
Since $|(Q_a^{(m)})^{-1}(0)| + |(Q_a^{(m)})^{-1}(1)| = 4^m$, solving the sum and difference identities proves the formula.
Its two values are distinct, and zero-set cardinality is invariant under quadratic isometry.

(iv) Let $\beta, \gamma \in \mathscr{B}_Q$. By (ii), $Q \cong Q^{(m)}_{\mathrm{Arf}_\beta(Q)}$ and $Q \cong Q^{(m)}_{\mathrm{Arf}_\gamma(Q)}$. If $m = 0$, both Arf sums are zero; if $m \geq 1$, their values agree by the nonisometry in (iii). Thus $\mathrm{Arf}(Q)$ is well-defined.

(v) The classification is (ii) with the basis-independent Arf sum. If $m \geq 1$, the first pair in the standard symplectic basis of $Q_a^{(m)}$ contributes $a^2 = a$ and every other pair contributes zero; hence $\mathrm{Arf}(Q_a^{(m)}) = a$.

Structural Commentary: Reduce the planes, cancel anisotropic pairs, and distinguish parity by zero counts.
Key Insight: The Arf invariant is the one-bit quadratic datum invisible to the common polar form. □

3.8 Orthogonal Isometries and Symmetry Groups

Throughout, orthogonal geometries are finite-dimensional over a fixed field $\mathbb{F}$; characteristic restrictions are local.

Proposition 3.8.1. *Isometries from Quadratic Values*
Let $\mathcal{V}$ be orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, put $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$, and let $T \in \mathrm{GL}(\mathcal{V})$. Then $T \in \mathcal{O}(\mathcal{V})$ if and only if $(\forall\, \mathbf{x} \in \mathcal{V})\, q(T\mathbf{x}) = q(\mathbf{x})$.

Proof.
($\Rightarrow$) If T is an isometry, then $(\forall\,\mathbf{x}\in\mathcal{V})\,q(T\mathbf{x})=\langle T\mathbf{x},T\mathbf{x}\rangle=\langle\mathbf{x},\mathbf{x}\rangle=q(\mathbf{x})$.
($\Leftarrow$) Suppose that $q\circ T=q$. By linearity of T and polarization, $(\forall\,\mathbf{x},\mathbf{y}\in\mathcal{V})$
$2\langle T\mathbf{x},T\mathbf{y}\rangle=q(T(\mathbf{x}+\mathbf{y}))-q(T\mathbf{x})-q(T\mathbf{y})=q(\mathbf{x}+\mathbf{y})-q(\mathbf{x})-q(\mathbf{y})=2\langle\mathbf{x},\mathbf{y}\rangle$. Cancellation of 2 proves the form.

Structural Commentary: Evaluate the form on the diagonal, then recover cross-pairings by polarization.
Key Insight: When 2 is invertible, preserving every quadratic value is equivalent to preserving the bilinear form. □

Definition 3.8.2. *Special Orthogonal Group and Symmetries*
Let $\mathcal{V}$ be nonsingular orthogonal with $\operatorname{char}(\mathbb{F})\neq 2$, and put $q(\mathbf{x}):=\langle\mathbf{x},\mathbf{x}\rangle$.

(i) The *special orthogonal group* is $\mathrm{SO}(\mathcal{V}):=\{\sigma\in\mathcal{O}(\mathcal{V}):\det\sigma=1\}$; its elements are *rotations*.

(ii) For $q(\mathbf{u})\neq 0$, the *symmetry* with normal line $\mathbb{F}\mathbf{u}$ is the map $\sigma_{\mathbf{u}}(\mathbf{x}):=\mathbf{x}-2\frac{\langle\mathbf{x},\mathbf{u}\rangle}{q(\mathbf{u})}\mathbf{u}$.

Proposition 3.8.3. *Symmetries and Equal Quadratic Values*
Let $\mathcal{V}$ be nonsingular orthogonal with $\operatorname{char}(\mathbb{F})\neq 2$, and put $q(\mathbf{x}):=\langle\mathbf{x},\mathbf{x}\rangle$.

(i) If $q(\mathbf{w})\neq 0$, then $\sigma_{\mathbf{w}}$ is a linear involutive isometry, $\sigma_{\mathbf{w}}\mathbf{w}=-\mathbf{w}$, and $\sigma_{\mathbf{w}}|_{(\mathbb{F}\mathbf{w})^{\perp}}=I$.

(ii) If $q(\mathbf{u})=q(\mathbf{v})\neq 0$, then at least one of $\mathbf{u}-\mathbf{v}$ and $\mathbf{u}+\mathbf{v}$ is nonisotropic; its symmetry maps $\mathbf{u}$ to $\mathbf{v}$ or $-\mathbf{v}$, respectively.

Proof.
(i) Fix $\mathbf{w}$ with $q(\mathbf{w})\neq 0$. The formula is linear, sends $\mathbf{w}$ to $-\mathbf{w}$, and fixes $(\mathbb{F}\mathbf{w})^{\perp}$ pointwise.
By Theorem 3.4.2(ii), $\mathcal{V}=\mathbb{F}\mathbf{w}\odot(\mathbb{F}\mathbf{w})^{\perp}$, so $\sigma_{\mathbf{w}}^2=I$ and $q\circ\sigma_{\mathbf{w}}=q$. Hence $\sigma_{\mathbf{w}}$ is bijective and isometric by Proposition 3.8.1.
(ii) Put $\mathbf{a}=\mathbf{u}-\mathbf{v}$ and $\mathbf{b}=\mathbf{u}+\mathbf{v}$. Then $\langle\mathbf{a},\mathbf{b}\rangle=q(\mathbf{u})-q(\mathbf{v})=0$.
If $q(\mathbf{a})=q(\mathbf{b})=0$, then $0=q(\mathbf{a}+\mathbf{b})=q(2\mathbf{u})=4q(\mathbf{u})$, a contradiction. Hence $q(\mathbf{a})\neq 0$ or $q(\mathbf{b})\neq 0$.
In the first case, $q(\mathbf{a})=2\langle\mathbf{u},\mathbf{a}\rangle$ and $\sigma_{\mathbf{a}}\mathbf{u}=\mathbf{u}-\mathbf{a}=\mathbf{v}$; in the second, $q(\mathbf{b})=2\langle\mathbf{u},\mathbf{b}\rangle$ and $\sigma_{\mathbf{b}}\mathbf{u}=\mathbf{u}-\mathbf{b}=-\mathbf{v}$.

Structural Commentary: Verify one symmetry directly, then test the orthogonal directions $\mathbf{u}\pm\mathbf{v}$.
Key Insight: Equal nonzero quadratic values make at least one of the two difference directions reflective. □

Theorem 3.8.4. *Orthogonal Transformations Are Products of Symmetries*
Let $\mathcal{V}$ be nonsingular orthogonal with $\operatorname{char}(\mathbb{F})\neq 2$, and put $q(\mathbf{x}):=\langle\mathbf{x},\mathbf{x}\rangle$. Then $\mathcal{O}(\mathcal{V})=\langle\sigma_{\mathbf{u}}:q(\mathbf{u})\neq 0\rangle$.

Proof. Induct on $n=\dim\mathcal{V}$. Case $n=0$ is immediate. If $n=1$, every isometry acts by some λ with $\lambda^2=1$, hence as I or $\sigma_{\mathbf{u}}$.
Assume $n>1$. Nonsingularity provides $\mathbf{u}$ with $q(\mathbf{u})\neq 0$; otherwise the form is alternating and zero by Proposition 3.1.2(ii).
For $\tau\in\mathcal{O}(\mathcal{V})$, choose σ by Proposition 3.8.3(ii) with $\sigma\tau\mathbf{u}=\varepsilon\mathbf{u}$, where $\varepsilon\in\{1,-1\}$. Put $\rho=\sigma$ for $\varepsilon=1$ and $\rho=\sigma_{\mathbf{u}}\sigma$ for $\varepsilon=-1$; then $\rho\tau\mathbf{u}=\mathbf{u}$. If $\mathbf{x}\in(\mathbb{F}\mathbf{u})^{\perp}$, then $\langle\rho\tau\mathbf{x},\mathbf{u}\rangle=\langle\rho\tau\mathbf{x},\rho\tau\mathbf{u}\rangle=\langle\mathbf{x},\mathbf{u}\rangle=0$, so $\rho\tau$ preserves this complement. It is nonsingular of dimension $n-1$ by Theorem 3.4.2(i),(iv). The inductive factors of the restriction extend by I on $\mathbb{F}\mathbf{u}$ to same-normal symmetries of $\mathcal{V}$. Thus $\rho\tau$ is a product of symmetries; so is ρ^{-1} because each symmetry is involutive by Proposition 3.8.3(i). Hence $\tau=\rho^{-1}(\rho\tau)$ is such a product.
Structural Commentary: Fix one nonisotropic line by at most two symmetries, then induct on its orthogonal complement.
Key Insight: Symmetries play the role of elementary generators for the orthogonal group. □

Proposition 3.8.5. *Fixed Spaces of Isometries*
Let $\mathcal{V}$ be nonsingular and let $\tau\in\mathrm{Iso}(\mathcal{V})$. Then $\ker(I-\tau)=\operatorname{im}(I-\tau)^{\perp}$.

Proof. If $\mathbf{x}\in\ker(I-\tau)$, then $(\forall\,\mathbf{y}\in\mathcal{V})\ \langle\mathbf{x},(I-\tau)\mathbf{y}\rangle=\langle\mathbf{x},\mathbf{y}\rangle-\langle\tau\mathbf{x},\tau\mathbf{y}\rangle=0$, so $\ker(I-\tau)\subseteq\operatorname{im}(I-\tau)^{\perp}$. These spaces have equal dimension by Theorem 1.7.3 and Theorem 3.4.2(i), so they are equal.
Structural Commentary: Establish orthogonality of fixed vectors to displacements, then close the inclusion by dimension.
Key Insight: The fixed space is exactly the orthogonal complement of the displacement space. □

Proposition 3.8.6. *Symmetries and Fixed Hyperplanes*
Let $\mathcal{V}$ be nonsingular orthogonal with $\operatorname{char}(\mathbb{F})\neq 2$, put $q(\mathbf{x}):=\langle\mathbf{x},\mathbf{x}\rangle$, and let $\tau\in\mathcal{O}(\mathcal{V})$ with $\tau\neq I$.

(i) For nonisotropic $\mathbf{u}$, $\ker(I-\sigma_{\mathbf{u}})=(\mathbb{F}\mathbf{u})^{\perp}$ is a nonsingular hyperplane.

(ii) The condition $(\exists\,\mathbf{u}\in\mathcal{V})\,(q(\mathbf{u})\neq 0\wedge\tau=\sigma_{\mathbf{u}})$ holds if and only if $\ker(I-\tau)$ is a nonsingular hyperplane.

Proof.
(i) By Proposition 3.8.3(i), $\sigma_{\mathbf{u}}$ fixes $(\mathbb{F}\mathbf{u})^{\perp}$ pointwise and sends $\mathbf{u}$ to $-\mathbf{u}$. Since $\operatorname{char}(\mathbb{F})\neq 2$, its fixed space is exactly $(\mathbb{F}\mathbf{u})^{\perp}$, which is a nonsingular hyperplane by Theorem 3.4.2(i),(iv).
(ii) ($\Rightarrow$) This is (i).
(ii) ($\Leftarrow$) Put $\mathcal{H}=\ker(I-\tau)$ and $\mathcal{L}=\mathcal{H}^{\perp}$. By Theorem 3.4.2(ii),(iv), $\mathcal{V}=\mathcal{H}\odot\mathcal{L}$ with $\mathcal{L}$ a nonsingular line. For $\mathbf{x}\in\mathcal{L}$ and $\mathbf{h}\in\mathcal{H}$, $\langle\tau\mathbf{x},\mathbf{h}\rangle=\langle\tau\mathbf{x},\tau\mathbf{h}\rangle=0$, so $\tau\mathcal{L}=\mathcal{L}$. Thus $\tau|_{\mathcal{L}}=\lambda I$ with $\lambda^2=1$. Since $\tau\neq I$ and $\operatorname{char}(\mathbb{F})\neq 2$, $\lambda=-1$. Every $\mathbf{0}\neq\mathbf{u}\in\mathcal{L}$ is nonisotropic, and $\tau=\sigma_{\mathbf{u}}$ by Proposition 3.8.3(i).

Structural Commentary: Read the fixed hyperplane from the symmetry formula, then recover the flipped normal line.
Key Insight: A nonidentity isometry is one symmetry exactly when its fixed space is a nonsingular hyperplane. □

Proposition 3.8.7. *No Unipotent Isometries in Anisotropic Spaces*
Let $\mathcal{V}$ be nonsingular anisotropic and let $\tau\in\mathrm{Iso}(\mathcal{V})$. If $(\exists\,m\in\mathbb{Z}_{>0})\,(\tau-I)^m=0$, then $\tau=I$.

Proof. Put $N=\tau-I$ and suppose $N\neq 0$. Fix maximal $k\geq 1$ with $N^k\neq 0$, so $N^{k+1}=0$, and choose $\mathbf{x}$ with $\mathbf{u}:=N^k\mathbf{x}\neq\mathbf{0}$. By Proposition 3.3.9, $N^*=\tau^{-1}-I=-N+N^2-\cdots+(-1)^kN^k$. Every monomial in $(N^*)^kN^k$ has N-degree at least $2k\geq k+1$, so this operator is zero. Hence $q(\mathbf{u})=\langle N^k\mathbf{x},N^k\mathbf{x}\rangle=\langle\mathbf{x},(N^*)^kN^k\mathbf{x}\rangle=0$, contrary to anisotropy. Therefore $N=0$ and $\tau=I$.
Structural Commentary: Pass the top nilpotent displacement through the adjoint and force its quadratic value to vanish.
Key Insight: An anisotropic geometry has no nonzero isotropic displacement on which unipotence can terminate. □

Witt extension and cancellation apply here by Theorem 3.4.12 and Corollary 3.4.13.

4 Affine, Projective, and Convex Geometry

4.1 Affine Flats, Hulls, and Incidence

Throughout, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$; affine geometry treats translated subspaces without choosing an origin.

Definition 4.1.1. *Flats*
Let $\mathcal{S} \le \mathcal{V}$ and $\mathbf{x} \in \mathcal{V}$, and put $\mathbf{x}+\mathcal{S} := \{\mathbf{x}+\mathbf{s} : \mathbf{s} \in \mathcal{S}\}$.

(i) The affine subset $\mathbf{x}+\mathcal{S}$ from Definition 2.2.1 is a *flat* in $\mathcal{V}$; $\mathcal{S}$ is its *base*, and $\mathbf{x}$ is a *flat representative*.
(ii) The *affine geometry* on $\mathcal{V}$ is $\mathcal{A}(\mathcal{V}) := \{\mathbf{y}+\mathcal{T} : \mathbf{y} \in \mathcal{V},\ \mathcal{T} \le \mathcal{V}\}$, with $\dim(\mathcal{A}(\mathcal{V})) := \dim(\mathcal{V})$.
(iii) The *dimension* of $\mathbf{x}+\mathcal{S}$ is $\dim(\mathcal{S})$.
A 0-flat is a *point*, a 1-flat is a *line*, a 2-flat is a *plane*, and a codimension-one flat is a *hyperplane*.

Proposition 4.1.2. *Flat Bases and Representatives*
Let $\mathcal{S}, \mathcal{T} \le \mathcal{V}$ and $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.

(i) $\mathbf{x}+\mathcal{S} = \mathbf{y}+\mathcal{T} \Rightarrow \mathcal{S} = \mathcal{T}$.
(ii) If $X = \mathbf{x}+\mathcal{S}$ and $\mathbf{z} \in X$, then $X = \mathbf{z}+\mathcal{S}$.

Proof.

(i) Suppose $\mathbf{x}+\mathcal{S} = \mathbf{y}+\mathcal{T}$ and put $\mathbf{t}_0 = \mathbf{x}-\mathbf{y} \in \mathcal{T}$. Then $(\forall\, \mathbf{s} \in \mathcal{S})\, \mathbf{t}_0+\mathbf{s} = (\mathbf{x}+\mathbf{s})-\mathbf{y} \in \mathcal{T}$, so $\mathcal{S} \subseteq \mathcal{T}$. Interchanging the representations proves $\mathcal{T} \subseteq \mathcal{S}$.
(ii) Write $\mathbf{z} = \mathbf{x}+\mathbf{s}_0$ with $\mathbf{s}_0 \in \mathcal{S}$. Then $\mathbf{z}+\mathcal{S} = \mathbf{x}+(\mathbf{s}_0+\mathcal{S}) = \mathbf{x}+\mathcal{S} = X$.

Structural Commentary: Identify directions by two inclusions, then absorb a base displacement into the representative.
Key Insight: A flat has one base but many representatives. □

Definition 4.1.3. *Parallel Flats*
For flats $X = \mathbf{x}+\mathcal{S}$ and $Y = \mathbf{y}+\mathcal{T}$, define $X \parallel Y \iff (\mathcal{S} \subseteq \mathcal{T} \vee \mathcal{T} \subseteq \mathcal{S})$. The relation is representation-independent by Proposition 4.1.2(i).

Proposition 4.1.4. *Basic Incidence of Flats*
Let $X = \mathbf{x}+\mathcal{S}$ and $Y = \mathbf{y}+\mathcal{T}$ be flats in $\mathcal{V}$, and write $\mathbf{w}+X := \{\mathbf{w}+\mathbf{u} : \mathbf{u} \in X\}$.

(i) $(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+X \subseteq Y) \iff \mathcal{S} \subseteq \mathcal{T}$.
(ii) $(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+X = Y) \iff \mathcal{S} = \mathcal{T}$.
(iii) $X \cap Y \neq \emptyset \Rightarrow (X \subseteq Y \iff \mathcal{S} \subseteq \mathcal{T})$.
(iv) $X \cap Y \neq \emptyset \Rightarrow (X = Y \iff \mathcal{S} = \mathcal{T})$.
(v) $X \parallel Y \Rightarrow (X \subseteq Y \vee Y \subseteq X \vee X \cap Y = \emptyset)$.
(vi) $X \parallel Y \iff [(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+X \subseteq Y)] \vee [(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+Y \subseteq X)]$.

Proof.

(i) (⇒) Fix $\mathbf{w}$ with $\mathbf{w}+X \subseteq Y$ and choose $\mathbf{z} \in \mathbf{w}+X$. Then $\mathbf{w}+X = \mathbf{z}+\mathcal{S}$ and $Y = \mathbf{z}+\mathcal{T}$ by Proposition 4.1.2(ii). Hence $(\forall\, \mathbf{s} \in \mathcal{S})\, \mathbf{z}+\mathbf{s} \in \mathbf{z}+\mathcal{T}$, so $\mathcal{S} \subseteq \mathcal{T}$.
(i) (⇐) If $\mathcal{S} \subseteq \mathcal{T}$, then $(\mathbf{y}-\mathbf{x})+X = \mathbf{y}+\mathcal{S} \subseteq Y$.
(ii) (⇒) If $\mathbf{w}+X = Y$, then $\mathcal{S} = \mathcal{T}$ by Proposition 4.1.2(i).
(ii) (⇐) If $\mathcal{S} = \mathcal{T}$, then $(\mathbf{y}-\mathbf{x})+X = Y$.
(iii) (⇒) Fix $\mathbf{z} \in X \cap Y$. Then $X = \mathbf{z}+\mathcal{S}$ and $Y = \mathbf{z}+\mathcal{T}$ by Proposition 4.1.2(ii). Thus $X \subseteq Y \Rightarrow (\forall\, \mathbf{s} \in \mathcal{S})\, \mathbf{z}+\mathbf{s} \in \mathbf{z}+\mathcal{T}$, so $\mathcal{S} \subseteq \mathcal{T}$.
(iii) (⇐) Fix $\mathbf{z} \in X \cap Y$. By Proposition 4.1.2(ii), $X = \mathbf{z}+\mathcal{S} \subseteq \mathbf{z}+\mathcal{T} = Y$ whenever $\mathcal{S} \subseteq \mathcal{T}$.
(iv) (⇒) If $X = Y$, then $\mathcal{S} = \mathcal{T}$ by Proposition 4.1.2(i).
(iv) (⇐) Fix $\mathbf{z} \in X \cap Y$. If $\mathcal{S} = \mathcal{T}$, then $X = \mathbf{z}+\mathcal{S} = \mathbf{z}+\mathcal{T} = Y$ by Proposition 4.1.2(ii).
(v) By Definition 4.1.3, $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$. If $X \cap Y \neq \emptyset$, then $X \subseteq Y$ or $Y \subseteq X$ by (iii); otherwise $X \cap Y = \emptyset$.
(vi) (⇒) By Definition 4.1.3 and (i), a translate of X lies in Y or a translate of Y lies in X.
(vi) (⇐) By (i), either translate-containment condition implies $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$, so $X \parallel Y$ by Definition 4.1.3.

Structural Commentary: Translate incidence to base comparison in (i)–(iv), then recover parallelism in (v)–(vi).
Key Insight: A common representative converts affine incidence into subspace incidence. □

Definition 4.1.5. *Affine Combinations*

(i) Let $n \in \mathbb{Z}_{>0}$, $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathcal{V}$, and $r_1, \dots, r_n \in \mathbb{F}$. The vector $\sum_{j=1}^{n} r_j \mathbf{x}_j$ is an *affine combination* iff $\sum_{j=1}^{n} r_j = 1$.
(ii) A nonempty $E \subseteq \mathcal{V}$ is *affine closed* iff $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, \mathbf{x}_1, \dots, \mathbf{x}_n \in E)(\forall\, r_1, \dots, r_n \in \mathbb{F})\, [\sum_{j=1}^{n} r_j = 1 \Rightarrow \sum_{j=1}^{n} r_j \mathbf{x}_j \in E]$.
(iii) A nonempty $E \subseteq \mathcal{V}$ is *two-affine closed* iff $(\forall\, \mathbf{x}, \mathbf{y} \in E)(\forall\, r \in \mathbb{F})\, r\mathbf{x}+(1-r)\mathbf{y} \in E$.
(iv) If $\mathbf{x} \neq \mathbf{y}$, their affine line is $\overline{\mathbf{xy}} := \{r\mathbf{x}+(1-r)\mathbf{y} : r \in \mathbb{F}\} = \mathbf{y}+\operatorname{span}(\mathbf{x}-\mathbf{y})$.
Thus a nonempty $E \subseteq \mathcal{V}$ is two-affine closed iff $(\forall\, \mathbf{x}, \mathbf{y} \in E)\, [\mathbf{x} \neq \mathbf{y} \Rightarrow \overline{\mathbf{xy}} \subseteq E]$.

Theorem 4.1.6. *Two-Point Affine Closure*
Let $\emptyset \neq E \subseteq \mathcal{V}$ and assume $\operatorname{char}(\mathbb{F}) \neq 2$. Then E is affine closed iff E is two-affine closed.

Proof.

(⇒) Two-affine closure is the $n = 2$ instance of affine closure.
(⇐) Assume that E is two-affine closed, fix $\mathbf{x}_0 \in E$, and put $\mathcal{S} = E-\mathbf{x}_0$. Fix $\mathbf{s}, \mathbf{t} \in \mathcal{S}$ and $c \in \mathbb{F}$. Put $\mathbf{a} := \mathbf{x}_0+c\mathbf{s} = c(\mathbf{x}_0+\mathbf{s})+(1-c)\mathbf{x}_0$; then $\mathbf{a} \in E$ by two-affine closure. Put $\mathbf{b} := \frac{1}{2}\mathbf{a}+\frac{1}{2}(\mathbf{x}_0+\mathbf{t}) = \mathbf{x}_0+\frac{1}{2}(c\mathbf{s}+\mathbf{t})$; then $\mathbf{b} \in E$ by two-affine closure. The identity $2\mathbf{b}-\mathbf{x}_0 = \mathbf{x}_0+c\mathbf{s}+\mathbf{t} \in E$ holds by two-affine closure with coefficient 2. Hence $(\forall\, \mathbf{s}, \mathbf{t} \in \mathcal{S})(\forall\, c \in \mathbb{F})\, c\mathbf{s}+\mathbf{t} \in \mathcal{S}$, so $\mathcal{S} \le \mathcal{V}$ by Theorem 1.2.2. Fix $n \in \mathbb{Z}_{>0}$, $\mathbf{x}_j = \mathbf{x}_0+\mathbf{s}_j \in E$, and $r_j \in \mathbb{F}$ with $\sum_{j=1}^{n} r_j = 1$. Then $\sum_{j=1}^{n} r_j \mathbf{x}_j = \mathbf{x}_0+\sum_{j=1}^{n} r_j \mathbf{s}_j \in \mathbf{x}_0+\mathcal{S} = E$. Thus E is affine closed.

Structural Commentary: Translate one point to zero, derive subspace closure, then reconstruct every affine combination.
Key Insight: Midpoints and dilation by 2 recover vector addition. □

Remark 4.1.7. *Characteristic Two Warning*
The characteristic restriction is necessary. In $\mathbb{F}_2^2$, put $E := \{(0,0),(1,0),(0,1)\}$. Since $r \in \{0,1\}$, every two-point affine combination equals one of its endpoints, so E is two-affine closed. However, $(1,1) = (0,0)+(1,0)+(0,1) \notin E$ is an affine combination because $1+1+1=1$ in $\mathbb{F}_2$.

Theorem 4.1.8. *Flats Are the Affine-Closed Sets*
Let $\emptyset \neq E \subseteq \mathcal{V}$. Then $E \in \mathcal{A}(\mathcal{V})$ iff E is affine closed.

Proof.

($\Rightarrow$) Write $E = \mathbf{x}+\mathcal{S}$ with $\mathcal{S} \leq \mathcal{V}$. Fix $n \in \mathbb{Z}_{>0}$, $\mathbf{x}_j = \mathbf{x}+\mathbf{s}_j \in E$, and $r_j \in \mathbb{F}$ with $\sum_{j=1}^{n} r_j = 1$. Then $\sum_{j=1}^{n} r_j\mathbf{x}_j = \mathbf{x} + \sum_{j=1}^{n} r_j\mathbf{s}_j \in E$, so E is affine closed.

($\Leftarrow$) Assume that E is affine closed, fix $\mathbf{x} \in E$, and put $\mathcal{S} = E - \mathbf{x}$. Fix $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}$ and $c \in \mathbb{F}$, and choose $\mathbf{x}_1, \mathbf{x}_2 \in E$ with $\mathbf{s}_j = \mathbf{x}_j - \mathbf{x}$. Then $c\mathbf{s}_1 + \mathbf{s}_2 + \mathbf{x} = c\mathbf{x}_1 + \mathbf{x}_2 - c\mathbf{x} \in E$ by affine closure, so $c\mathbf{s}_1 + \mathbf{s}_2 \in \mathcal{S}$. Hence $\mathcal{S} \leq \mathcal{V}$ by Theorem 1.2.2, and $E = \mathbf{x}+\mathcal{S} \in \mathcal{A}(\mathcal{V})$.

Structural Commentary: Check translated-subspace closure forward, then recover a base by shifting one point to zero.
Key Insight: Affine closure becomes linear closure after translation. □

Theorem 4.1.9. *Flats Are the Two-Affine-Closed Sets*
Let $\emptyset \neq E \subseteq \mathcal{V}$ and assume $\operatorname{char}(\mathbb{F}) \neq 2$. Then $E \in \mathcal{A}(\mathcal{V})$ iff E is two-affine closed.

Proof.

($\Rightarrow$) If $E \in \mathcal{A}(\mathcal{V})$, then E is affine closed by Theorem 4.1.8 and two-affine closed by Theorem 4.1.6.

($\Leftarrow$) If E is two-affine closed, then E is affine closed by Theorem 4.1.6 and $E \in \mathcal{A}(\mathcal{V})$ by Theorem 4.1.8.

Structural Commentary: Compose the two preceding characterizations in each direction.
Key Insight: Binary affine closure detects translated subspaces when 2 is invertible. □

Definition 4.1.10. *Affine Hull and Affine Span*
Let $\emptyset \neq E \subseteq \mathcal{V}$.

(i) The *affine hull* is $\operatorname{affhull}(E) := \bigcap\{X \in \mathcal{A}(\mathcal{V}) : E \subseteq X\}$; the defining family contains $\mathcal{V}$.

(ii) The *affine span* is $\operatorname{affspan}(E) := \{\sum_{j=1}^{n} r_j\mathbf{e}_j : n \in \mathbb{Z}_{>0},\ \mathbf{e}_j \in E,\ r_j \in \mathbb{F},\ \sum_{j=1}^{n} r_j = 1\}$.

Theorem 4.1.11. *Affine Hull Formula*
Let $\emptyset \neq E \subseteq \mathcal{V}$, fix $\mathbf{x} \in E$, and put $\mathcal{S}_{\mathbf{x}} := \operatorname{span}(E - \mathbf{x})$.

(i) $\operatorname{affhull}(E) = \operatorname{affspan}(E) = \mathbf{x} + \mathcal{S}_{\mathbf{x}}$.

(ii) $\dim(\operatorname{affhull}(E)) = \dim(\mathcal{S}_{\mathbf{x}})$.

Proof.

(i) If $\mathbf{z} = \sum_{j=1}^{n} r_j\mathbf{e}_j \in \operatorname{affspan}(E)$, then $\mathbf{z} = \mathbf{x} + \sum_{j=1}^{n} r_j(\mathbf{e}_j - \mathbf{x}) \in \mathbf{x} + \mathcal{S}_{\mathbf{x}}$. Conversely, if $\mathbf{z} = \mathbf{x} + \sum_{j=1}^{n} a_j(\mathbf{e}_j - \mathbf{x}) \in \mathbf{x} + \mathcal{S}_{\mathbf{x}}$, then $\mathbf{z} = (1 - \sum_{j=1}^{n} a_j)\mathbf{x} + \sum_{j=1}^{n} a_j\mathbf{e}_j \in \operatorname{affspan}(E)$. Hence $\operatorname{affspan}(E) = \mathbf{x} + \mathcal{S}_{\mathbf{x}}$, a flat containing E. Every flat containing E contains $\operatorname{affspan}(E)$ by Theorem 4.1.8, so this flat equals $\operatorname{affhull}(E)$ by Definition 4.1.10(i).

(ii) By (i) and Definition 4.1.1(iii), $\dim(\operatorname{affhull}(E)) = \dim(\mathbf{x} + \mathcal{S}_{\mathbf{x}}) = \dim(\mathcal{S}_{\mathbf{x}})$.

Structural Commentary: Prove the translated-span formula by two inclusions, then read dimension from its base.
Key Insight: Differences from one point encode every affine direction. □

Lemma 4.1.12. *Intersections of Flats*
Let $\emptyset \neq I$ and let $X_i = \mathbf{x}_i + \mathcal{S}_i$ be flats indexed by $i \in I$. If $\mathbf{z} \in \bigcap_{i\in I} X_i$, then $\bigcap_{i\in I} X_i = \mathbf{z} + \bigcap_{i\in I} \mathcal{S}_i$.

Proof. By Proposition 4.1.2(ii), $(\forall\, i \in I)\, X_i = \mathbf{z} + \mathcal{S}_i$. Hence $(\forall\, \mathbf{u} \in \mathcal{V})\,[\mathbf{u} \in \bigcap_{i\in I} X_i \iff \mathbf{u} - \mathbf{z} \in \bigcap_{i\in I} \mathcal{S}_i]$, which proves the formula.

Structural Commentary: Replace every representative by the common point, then intersect the remaining bases.
Key Insight: Common incidence synchronizes all affine origins. □

Theorem 4.1.13. *Lattice of Flats*
Let $\mathcal{A}_0(\mathcal{V}) := \mathcal{A}(\mathcal{V}) \cup \{\emptyset\}$, ordered by inclusion.

(i) $(\forall\, \mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V}))\ \bigwedge \mathcal{C} = \bigcap_{X\in\mathcal{C}} X$.

(ii) If $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$ and $U := \bigcup_{X\in\mathcal{C}} X$, then $U \neq \emptyset \Rightarrow \bigvee \mathcal{C} = \operatorname{affhull}(U)$, while $U = \emptyset \Rightarrow \bigvee \mathcal{C} = \emptyset$.

(iii) $\mathcal{A}_0(\mathcal{V})$ is a complete lattice.

Proof.

(i) Fix $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$ and put $M = \bigcap_{X\in\mathcal{C}} X$. If $\mathcal{C} = \emptyset$, then $M = \mathcal{V}$. If $\mathcal{C} \neq \emptyset$ and $M = \emptyset$, then $M \in \mathcal{A}_0(\mathcal{V})$. If $\mathcal{C} \neq \emptyset$ and $M \neq \emptyset$, choose $\mathbf{z} \in M$ and let $\mathcal{S}_X$ be the unique base of X; then $M = \mathbf{z} + \bigcap_{X\in\mathcal{C}} \mathcal{S}_X$ by Lemma 4.1.12, so M is a flat. Thus $M \in \mathcal{A}_0(\mathcal{V})$. It is a lower bound of $\mathcal{C}$, and any lower bound is contained in M, so $M = \bigwedge \mathcal{C}$.

(ii) Fix $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$ and put $U = \bigcup_{X\in\mathcal{C}} X$. If $U = \emptyset$, then $\emptyset$ is the least upper bound of $\mathcal{C}$. If $U \neq \emptyset$, then $\operatorname{affhull}(U)$ is a flat containing each $X \in \mathcal{C}$ by Theorem 4.1.11. If B is an upper bound of $\mathcal{C}$, then $U \subseteq B$ and $\operatorname{affhull}(U) \subseteq B$, so $\operatorname{affhull}(U) = \bigvee \mathcal{C}$.

(iii) The arbitrary meets and joins in (i)–(ii) prove completeness by Definition 5.3.4(iv).

Structural Commentary: Construct arbitrary meets by intersection and joins by affine hull, then invoke lattice completeness.
Key Insight: Empty incidence must be adjoined to close arbitrary meets. □

Remark 4.1.14. *Empty Affine Hull Convention*
Within $\mathcal{A}_0(\mathcal{V})$, set affhull$(\emptyset) := \emptyset$; then Theorem 4.1.13(ii) has the uniform form $\bigvee \mathcal{C} = \text{affhull}(\bigcup_{X \in \mathcal{C}} X)$.

Corollary 4.1.15. *Join of Two Flats*
Let $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$ be flats.

(i) $X \vee Y = \mathbf{x} + \text{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$.
(ii) $X \cap Y \neq \emptyset \Rightarrow X \vee Y = \mathbf{x} + (\mathcal{S} + \mathcal{T})$.

Proof.

(i) By Theorem 4.1.13(ii) and Theorem 4.1.11(i), $X \vee Y = \mathbf{x} + \text{span}((X \cup Y) - \mathbf{x})$. Since $(X \cup Y) - \mathbf{x} = \mathcal{S} \cup ((\mathbf{y} - \mathbf{x}) + \mathcal{T})$, its span is $\text{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$.
(ii) Fix $\mathbf{z} = \mathbf{x} + \mathbf{s} = \mathbf{y} + \mathbf{t} \in X \cap Y$. Then $\mathbf{x} - \mathbf{y} = \mathbf{t} - \mathbf{s} \in \mathcal{S} + \mathcal{T}$, so the span in (i) equals $\mathcal{S} + \mathcal{T}$.

Structural Commentary: Expand the affine hull around $\mathbf{x}$, then test whether the displacement adds a direction.
Key Insight: Intersection absorbs the displacement into the two bases. □

Corollary 4.1.16. *Dimension of the Join of Two Flats*
Let $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$ be flats.

(i) $X \cap Y \neq \emptyset \Rightarrow \dim(X \vee Y) = \dim(X) + \dim(Y) - \dim(X \cap Y)$.
(ii) $X \cap Y = \emptyset \Rightarrow \dim(X \vee Y) = \dim(\mathcal{S} + \mathcal{T}) + 1$.

Proof.

(i) Fix $\mathbf{z} \in X \cap Y$. By Corollary 4.1.15(ii) and Lemma 4.1.12, the bases of $X \vee Y$ and $X \cap Y$ are $\mathcal{S} + \mathcal{T}$ and $\mathcal{S} \cap \mathcal{T}$, respectively. Hence $\dim(X \vee Y) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$ by Theorem 2.3.1, which is the stated identity by Definition 4.1.1(iii).
(ii) By Corollary 4.1.15(i), the base of $X \vee Y$ is $\mathcal{U} := \text{span}(\{\mathbf{x} - \mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$. If $\mathbf{x} - \mathbf{y} \in \mathcal{S} + \mathcal{T}$, choose $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$ with $\mathbf{x} - \mathbf{y} = \mathbf{s} + \mathbf{t}$; then $\mathbf{x} - \mathbf{s} = \mathbf{y} + \mathbf{t} \in X \cap Y$, a contradiction. Thus $\mathcal{U} = (\mathcal{S} + \mathcal{T}) \oplus \text{span}(\mathbf{x} - \mathbf{y})$, and $\dim(X \vee Y) = \dim(\mathcal{S} + \mathcal{T}) + 1$ by Corollary 2.3.2.

Structural Commentary: Apply the subspace dimension formula when flats meet; isolate one new direction otherwise.
Key Insight: Disjointness makes the representative displacement linearly independent. □

4.2 Affine Independence and Barycentric Coordinates

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$.

Definition 4.2.1. *Affinely Independent Sets*
A nonempty set $E \subseteq \mathcal{V}$ is *affinely independent* iff $(\forall\, \mathbf{x} \in E)\,(E - \mathbf{x}) \setminus \{\mathbf{0}\}$ is linearly independent.

Theorem 4.2.2. *Affine Independence Criteria*
Let $\emptyset \neq E \subseteq \mathcal{V}$. For $F \subseteq \mathcal{V}$ and $n \in \mathbb{Z}_{>0}$, put $F^n_{\neq} := \{(\mathbf{x}_1, \dots, \mathbf{x}_n) \in F^n : \mathbf{x}_j \neq \mathbf{x}_k \ (j \neq k)\}$. The following conditions are equivalent.

(i) *E is affinely independent.*
(ii) $(\forall\, \mathbf{x} \in E)\, \mathbf{x} \notin \text{affhull}(E \setminus \{\mathbf{x}\})$, *where* affhull$(\emptyset) = \emptyset$.
(iii) $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \dots, \mathbf{x}_n) \in E^n_{\neq})(\forall\, \mathbf{r} \in \mathbb{F}^n)\,[\sum_{j=1}^n r_j \mathbf{x}_j = \mathbf{0} \wedge \sum_{j=1}^n r_j = 0 \Rightarrow \mathbf{r} = \mathbf{0}]$.
(iv) $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \dots, \mathbf{x}_n) \in E^n_{\neq})(\forall\, \mathbf{r}, \mathbf{s} \in \mathbb{F}^n)\,[\sum_{j=1}^n r_j \mathbf{x}_j = \sum_{j=1}^n s_j \mathbf{x}_j \wedge \sum_{j=1}^n r_j = \sum_{j=1}^n s_j = 1 \Rightarrow \mathbf{r} = \mathbf{s}]$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Fix $\mathbf{x} \in E$. The claim holds by the stated convention when $E = \{\mathbf{x}\}$. Otherwise, suppose $\mathbf{x} \in \text{affhull}(E \setminus \{\mathbf{x}\})$. By Theorem 4.1.11(i), after collecting equal points, $(\exists\, m \in \mathbb{Z}_{>0})(\exists\, (\mathbf{y}_1, \dots, \mathbf{y}_m) \in (E \setminus \{\mathbf{x}\})^m_{\neq})(\exists\, \mathbf{r} \in \mathbb{F}^m)\,[\mathbf{x} = \sum_{j=1}^m r_j \mathbf{y}_j \wedge \sum_{j=1}^m r_j = 1]$. Thus $\sum_{j=1}^m r_j(\mathbf{y}_j - \mathbf{x}) = \mathbf{0}$ with $\mathbf{r} \neq \mathbf{0}$, contrary to (i).

(*ii*) $\Rightarrow$ (*iii*) Fix $n \in \mathbb{Z}_{>0}$, $(\mathbf{x}_1, \dots, \mathbf{x}_n) \in E^n_{\neq}$, and $\mathbf{r} \in \mathbb{F}^n$ with $\sum_{j=1}^n r_j \mathbf{x}_j = \mathbf{0}$ and $\sum_{j=1}^n r_j = 0$. If $\mathbf{r} \neq \mathbf{0}$, choose k with $r_k \neq 0$. Then $n \geq 2$, $\mathbf{x}_k = \sum_{j \neq k} (-\frac{r_j}{r_k})\mathbf{x}_j$, and $\sum_{j \neq k} -\frac{r_j}{r_k} = 1$, so $\mathbf{x}_k \in \text{affhull}(E \setminus \{\mathbf{x}_k\})$ by Theorem 4.1.11(i), contrary to (ii). Hence $\mathbf{r} = \mathbf{0}$.

(*iii*) $\Rightarrow$ (*iv*) After subtraction, $\sum_{j=1}^n (r_j - s_j)\mathbf{x}_j = \mathbf{0}$ and $\sum_{j=1}^n (r_j - s_j) = 0$, so $\mathbf{r} = \mathbf{s}$ by (iii).

(*iv*) $\Rightarrow$ (*i*) Fix $\mathbf{x} \in E$. The empty list is linearly independent. Fix $m \in \mathbb{Z}_{>0}$, $(\mathbf{y}_1, \dots, \mathbf{y}_m) \in (E \setminus \{\mathbf{x}\})^m_{\neq}$, and $\mathbf{a} \in \mathbb{F}^m$ with $\sum_{j=1}^m a_j(\mathbf{y}_j - \mathbf{x}) = \mathbf{0}$. On $(\mathbf{y}_1, \dots, \mathbf{y}_m, \mathbf{x}) \in E^{m+1}_{\neq}$, put $\mathbf{c} := (a_1, \dots, a_m, -\sum_{j=1}^m a_j)$ and $\mathbf{b} := (0, \dots, 0, 1)$. Both $\mathbf{b}$ and $\mathbf{b} + \mathbf{c}$ have coefficient sum 1 and represent $\mathbf{x}$, so $\mathbf{c} = \mathbf{0}$ by (iv). Hence $\mathbf{a} = \mathbf{0}$, and $(E - \mathbf{x}) \setminus \{\mathbf{0}\}$ is linearly independent.

Structural Commentary: Cycle through excluded hulls, zero-sum relations, coordinate uniqueness, and translated independence.
Key Insight: Affine dependence is exactly a nonzero coefficient relation with total sum zero. □

Corollary 4.2.3. *Dimension Criterion for Affine Independence*
Let $E \subseteq \mathcal{V}$ be finite and nonempty with $|E| = n$. Then E is affinely independent iff $\dim(\text{affhull}(E)) = n - 1$.

Proof.

($\Rightarrow$) Fix $\mathbf{x} \in E$ and put $B_{\mathbf{x}} := (E - \mathbf{x}) \setminus \{\mathbf{0}\}$. The set $B_{\mathbf{x}}$ has $n - 1$ elements and is linearly independent by Definition 4.2.1; hence any ordering is a basis of $\text{span}(E - \mathbf{x})$. Thus $\dim(\text{affhull}(E)) = \dim(\text{span}(E - \mathbf{x})) = n - 1$ by Theorem 4.1.11(ii) and Definition 1.4.2.
($\Leftarrow$) Fix $\mathbf{x} \in E$ and put $B_{\mathbf{x}} := (E - \mathbf{x}) \setminus \{\mathbf{0}\}$. By Theorem 4.1.11(ii), $\dim(\text{span}(E - \mathbf{x})) = n - 1$. Any ordering of the $n - 1$ elements of $B_{\mathbf{x}}$ spans this space, so it is a basis by Proposition 1.4.4(i). Thus $B_{\mathbf{x}}$ is linearly independent, and E is affinely independent by Definition 4.2.1.

Structural Commentary: Translate one point to zero, then compare the remaining spanning list with the hull dimension.
Key Insight: Removing one affine origin leaves exactly $n-1$ direction vectors. □

Definition 4.2.4. *Affine Bases and Barycentric Coordinates*
Let $A = \mathbf{x} + \mathcal{S}$ be an n-flat.

(i) A finite subset $\mathcal{B} \subseteq A$ is an *affine basis* of A iff $\mathcal{B}$ is affinely independent and $\operatorname{affspan}(\mathcal{B}) = A$.
(ii) A tuple $(\mathbf{p}_1, \dots, \mathbf{p}_m)$ is an *ordered affine basis* of A iff its entries are distinct and $\{\mathbf{p}_1, \dots, \mathbf{p}_m\}$ is an affine basis of A.
(iii) If $(\mathbf{p}_1, \dots, \mathbf{p}_m)$ is an ordered affine basis of A and $\mathbf{v} \in A$, then $\mathbf{r} = (r_1, \dots, r_m) \in \mathbb{F}^m$ is a *barycentric coordinate vector* of $\mathbf{v}$ iff $\mathbf{v} = \sum_{j=1}^{m} r_j \mathbf{p}_j$ and $\sum_{j=1}^{m} r_j = 1$.

Theorem 4.2.5. *Affine Bases and Barycentric Coordinates*
Let $A = \mathbf{x} + \mathcal{S}$ be an n-flat.

(i) If $(\mathbf{s}_1, \dots, \mathbf{s}_n)$ is an ordered basis of $\mathcal{S}$, then $(\mathbf{x}+\mathbf{s}_1, \dots, \mathbf{x}+\mathbf{s}_n, \mathbf{x})$ is an ordered affine basis of A.
(ii) If $\mathcal{B}$ is an affine basis of A, then $|\mathcal{B}| = n+1$.
(iii) If $(\mathbf{p}_1, \dots, \mathbf{p}_{n+1})$ is an ordered affine basis of A, then $(\forall\, \mathbf{v} \in A)(\exists!\, \mathbf{r} \in \mathbb{F}^{n+1})\, [\mathbf{v} = \sum_{j=1}^{n+1} r_j \mathbf{p}_j \wedge \sum_{j=1}^{n+1} r_j = 1]$.

Proof.

(i) Put $\mathbf{p}_j := \mathbf{x} + \mathbf{s}_j$ for $1 \le j \le n$ and $\mathbf{p}_{n+1} := \mathbf{x}$. Every zero-sum relation on a subtuple of $(\mathbf{p}_1, \dots, \mathbf{p}_{n+1})$ extends by zero coefficients to the full tuple. If $\mathbf{a} \in \mathbb{F}^{n+1}$ satisfies $\sum_{j=1}^{n+1} a_j \mathbf{p}_j = \mathbf{0}$ and $\sum_{j=1}^{n+1} a_j = 0$, then $\sum_{j=1}^{n} a_j \mathbf{s}_j = \mathbf{0}$. Hence $a_1 = \cdots = a_n = 0$ by independence of the $\mathbf{s}_j$, and $a_{n+1} = 0$ by the coefficient sum. Thus $\{\mathbf{p}_1, \dots, \mathbf{p}_{n+1}\}$ is affinely independent by Theorem 4.2.2(iii). The identity $\operatorname{affspan}(\{\mathbf{p}_1, \dots, \mathbf{p}_{n+1}\}) = \mathbf{x} + \operatorname{span}(\mathbf{s}_1, \dots, \mathbf{s}_n) = A$ holds by Theorem 4.1.11(i), so the tuple is an ordered affine basis.
(ii) Put $m := |\mathcal{B}|$. Then $\dim(\operatorname{affhull}(\mathcal{B})) = m - 1$ by Corollary 4.2.3. By Definition 4.2.4(i) and Theorem 4.1.11(i), $\operatorname{affhull}(\mathcal{B}) = \operatorname{affspan}(\mathcal{B}) = A$. Thus $n = m - 1$, so $m = n + 1$.
(iii) Fix $\mathbf{v} \in A$. Since $\operatorname{affspan}(\{\mathbf{p}_1, \dots, \mathbf{p}_{n+1}\}) = A$, an affine combination of these basis points represents $\mathbf{v}$ by Definition 4.1.10(ii). Collecting coefficients at equal basis points and inserting zero coefficients for omitted points produces $\mathbf{r} \in \mathbb{F}^{n+1}$ with $\mathbf{v} = \sum_{j=1}^{n+1} r_j \mathbf{p}_j$ and $\sum_{j=1}^{n+1} r_j = 1$. This vector is unique by Theorem 4.2.2(iv).

Structural Commentary: Adjoin an affine origin to a linear basis, compare dimensions, then invoke affine-representation uniqueness.
Key Insight: One extra basis point records the missing affine origin. □

Example 4.2.6. *Barycentric Coordinates on a Plane*
Let $A = \mathbf{x} + \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2) \subseteq \mathbb{R}^3$, where $(\mathbf{v}_1, \mathbf{v}_2)$ is linearly independent.
Put $\mathbf{p}_1 := \mathbf{x} + \mathbf{v}_1$, $\mathbf{p}_2 := \mathbf{x} + \mathbf{v}_2$, and $\mathbf{p}_3 := \mathbf{x}$. Then $(\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3)$ is an ordered affine basis of A by Theorem 4.2.5(i).
If $\mathbf{a} = \mathbf{x} + t_1\mathbf{v}_1 + t_2\mathbf{v}_2$, then $\mathbf{a} = t_1\mathbf{p}_1 + t_2\mathbf{p}_2 + (1 - t_1 - t_2)\mathbf{p}_3$ and $t_1 + t_2 + (1 - t_1 - t_2) = 1$. Hence the barycentric coordinate vector of $\mathbf{a}$ is $(t_1, t_2, 1 - t_1 - t_2)$ by Theorem 4.2.5(iii).

4.3 Affine Maps and Affine Groups

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$.

Definition 4.3.1. *Affine Transformation*
A map $f : \mathcal{V} \to \mathcal{V}$ is *affine* iff $(\forall\, n \in \mathbb{Z}_{>0})(\forall\, (\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathcal{V}^n)(\forall\, \mathbf{r} \in \mathbb{F}^n)\, [\sum_{j=1}^{n} r_j = 1 \Rightarrow f(\sum_{j=1}^{n} r_j \mathbf{x}_j) = \sum_{j=1}^{n} r_j f(\mathbf{x}_j)]$.

Theorem 4.3.2. *Two-Point Test for Affine Maps*
Assume $\operatorname{char}(\mathbb{F}) \neq 2$. A map $f : \mathcal{V} \to \mathcal{V}$ is affine iff $(\forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V})(\forall\, r \in \mathbb{F})\, f(r\mathbf{x} + (1-r)\mathbf{y}) = rf(\mathbf{x}) + (1-r)f(\mathbf{y})$.

Proof.

(⇒) The identity holds by Definition 4.3.1 with $n = 2$.
(⇐) Let $\Gamma_f := \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in \mathcal{V}\} \subseteq \mathcal{V} \times \mathcal{V}$. The assumed identity makes Γ_f two-affine closed because $r(\mathbf{x}, f(\mathbf{x})) + (1-r)(\mathbf{y}, f(\mathbf{y})) = (r\mathbf{x} + (1-r)\mathbf{y}, f(r\mathbf{x} + (1-r)\mathbf{y}))$. The nonempty set Γ_f is affine closed by Theorem 4.1.6 in $\mathcal{V} \times \mathcal{V}$. Fix $n \in \mathbb{Z}_{>0}$, $(\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathcal{V}^n$, and $\mathbf{r} \in \mathbb{F}^n$ with $\sum_{j=1}^{n} r_j = 1$. Then $(\sum_{j=1}^{n} r_j \mathbf{x}_j, \sum_{j=1}^{n} r_j f(\mathbf{x}_j)) \in \Gamma_f$, so $f(\sum_{j=1}^{n} r_j \mathbf{x}_j) = \sum_{j=1}^{n} r_j f(\mathbf{x}_j)$. Thus f is affine.

Structural Commentary: Convert binary preservation into two-affine closure of the graph, then invoke affine closure.
Key Insight: Affine maps are exactly the maps with affine-closed graphs. □

Definition 4.3.3. *Translations*
For $\mathbf{a} \in \mathcal{V}$, the *translation by* $\mathbf{a}$ is the map $T_{\mathbf{a}} : \mathcal{V} \to \mathcal{V}$ defined by $T_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$.

Theorem 4.3.4. *Affine Maps Are Linear Maps Followed by Translations*
A map $f : \mathcal{V} \to \mathcal{V}$ is affine iff $(\exists!\, (\mathbf{a}, \tau) \in \mathcal{V} \times \mathcal{L}(\mathcal{V}))\, f = T_{\mathbf{a}} \circ \tau$. The unique pair satisfies $\mathbf{a} = f(\mathbf{0})$ and $\tau = T_{-\mathbf{a}} \circ f$.

Proof.

(⇒) Put $\mathbf{a} := f(\mathbf{0})$ and $\tau := T_{-\mathbf{a}} \circ f$. For $r, s \in \mathbb{F}$ and $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, the identity $\tau(r\mathbf{x} + s\mathbf{y}) = rf(\mathbf{x}) + sf(\mathbf{y}) + (1-r-s)\mathbf{a} - \mathbf{a} = r\tau(\mathbf{x}) + s\tau(\mathbf{y})$ holds by affine preservation with coefficients $r, s, 1-r-s$. Hence $\tau \in \mathcal{L}(\mathcal{V})$ and $f = T_{\mathbf{a}} \circ \tau$. If $f = T_{\mathbf{b}} \circ \sigma$ with $(\mathbf{b}, \sigma) \in \mathcal{V} \times \mathcal{L}(\mathcal{V})$, then $\mathbf{b} = f(\mathbf{0}) = \mathbf{a}$ and $\sigma = T_{-\mathbf{b}} \circ f = \tau$, proving uniqueness.
(⇐) Suppose $f = T_{\mathbf{a}} \circ \tau$ with $\tau \in \mathcal{L}(\mathcal{V})$. Fix $n \in \mathbb{Z}_{>0}$, $(\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathcal{V}^n$, and $\mathbf{r} \in \mathbb{F}^n$ with $\sum_{j=1}^{n} r_j = 1$. Then $f(\sum_{j=1}^{n} r_j \mathbf{x}_j) = \mathbf{a} + \sum_{j=1}^{n} r_j \tau(\mathbf{x}_j) = \sum_{j=1}^{n} r_j (\mathbf{a} + \tau(\mathbf{x}_j)) = \sum_{j=1}^{n} r_j f(\mathbf{x}_j)$, so f is affine.

Structural Commentary: Translate $f(\mathbf{0})$ to zero, verify linearity, then recover and identify the factorization.
Key Insight: The image of zero determines the entire translation part. □

Definition 4.3.5. *Affine Group Data*

(i) The *general linear group* is $\mathrm{GL}(\mathcal{V}) := \{\tau \in \mathcal{L}(\mathcal{V}) : (\exists\, \sigma \in \mathcal{L}(\mathcal{V}))\,[\sigma\tau = I_{\mathcal{V}} = \tau\sigma]\}$.
(ii) The *affine group* is $\mathrm{Aff}(\mathcal{V}) := \{T_{\mathbf{a}} \circ \tau : \mathbf{a} \in \mathcal{V},\ \tau \in \mathrm{GL}(\mathcal{V})\}$.
(iii) The *translation group* is $\mathrm{Trans}(\mathcal{V}) := \{T_{\mathbf{a}} : \mathbf{a} \in \mathcal{V}\}$.
(iv) The *linear-part map* is $\Lambda : \mathrm{Aff}(\mathcal{V}) \to \mathrm{GL}(\mathcal{V})$, defined by $\Lambda(T_{\mathbf{a}} \circ \tau) = \tau$; it is well defined by uniqueness in Theorem 4.3.4.

Corollary 4.3.6. *Bijective Affine Maps*
Let $\mathbf{a} \in \mathcal{V}$ and $\tau \in \mathcal{L}(\mathcal{V})$. Then $T_{\mathbf{a}} \circ \tau$ is bijective iff $\tau \in \mathrm{GL}(\mathcal{V})$.

Proof.
($\Rightarrow$) The translation $T_{-\mathbf{a}}$ is bijective with inverse $T_{\mathbf{a}}$. Hence $\tau = T_{-\mathbf{a}} \circ (T_{\mathbf{a}} \circ \tau)$ is bijective, so $\tau \in \mathrm{GL}(\mathcal{V})$ by Proposition 1.6.7.
($\Leftarrow$) If $\tau \in \mathrm{GL}(\mathcal{V})$, then $T_{\mathbf{a}} \circ \tau$ is bijective with inverse $\tau^{-1} \circ T_{-\mathbf{a}}$.
Structural Commentary: Cancel the translation in one direction and compose the two explicit inverses in the other.
Key Insight: Translation does not affect bijectivity. □

Corollary 4.3.7. *Affine Group and Translations*

(i) $(\forall\, \mathbf{a}, \mathbf{b} \in \mathcal{V})(\forall\, \tau, \sigma \in \mathcal{L}(\mathcal{V}))\ (T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma) = T_{\mathbf{a}+\tau(\mathbf{b})} \circ (\tau\sigma)$.
(ii) $\mathrm{GL}(\mathcal{V})$ *is a group under composition.*
(iii) $\mathrm{Aff}(\mathcal{V})$ *is a group under composition, with identity* $T_{\mathbf{0}} \circ I_{\mathcal{V}}$ *and* $(T_{\mathbf{a}} \circ \tau)^{-1} = T_{-\tau^{-1}(\mathbf{a})} \circ \tau^{-1}$.
(iv) $\mathrm{Trans}(\mathcal{V}) \trianglelefteq \mathrm{Aff}(\mathcal{V})$.
(v) Λ *is a surjective group homomorphism and* $\ker\Lambda = \mathrm{Trans}(\mathcal{V})$.
(vi) $\mathrm{Aff}(\mathcal{V})/\mathrm{Trans}(\mathcal{V}) \cong \mathrm{GL}(\mathcal{V})$.

Proof.
(i) For $\mathbf{x} \in \mathcal{V}$, $((T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma))(\mathbf{x}) = \mathbf{a} + \tau(\mathbf{b}) + (\tau\sigma)(\mathbf{x})$, which proves the map identity.
(ii) Composition is associative and $I_{\mathcal{V}} \in \mathrm{GL}(\mathcal{V})$. Closure under products holds by Proposition 1.6.6(i). If $\tau \in \mathrm{GL}(\mathcal{V})$, then $\tau^{-1} \in \mathcal{L}(\mathcal{V})$ with inverse τ by Definition 1.6.4 and Proposition 1.6.5, so $\tau^{-1} \in \mathrm{GL}(\mathcal{V})$.
(iii) Closure holds by (i)–(ii), and composition is associative. The identity is $T_{\mathbf{0}} \circ I_{\mathcal{V}}$. By (i), $T_{-\tau^{-1}(\mathbf{a})} \circ \tau^{-1}$ composes with $T_{\mathbf{a}} \circ \tau$ in both orders to $T_{\mathbf{0}} \circ I_{\mathcal{V}}$, proving the inverse formula and the group claim.
(iv) The identities $T_{\mathbf{a}}T_{\mathbf{b}} = T_{\mathbf{a}+\mathbf{b}}$, $T_{\mathbf{0}} = I_{\mathcal{V}}$, and $T_{\mathbf{a}}^{-1} = T_{-\mathbf{a}}$ hold by (i), so $\mathrm{Trans}(\mathcal{V})$ is a subgroup. If $g = T_{\mathbf{a}} \circ \tau$, then $gT_{\mathbf{b}}g^{-1} = T_{\tau(\mathbf{b})}$ by (i). Since τ is bijective by Proposition 1.6.7, $g\,\mathrm{Trans}(\mathcal{V})g^{-1} = \mathrm{Trans}(\mathcal{V})$.
(v) By (i), $\Lambda((T_{\mathbf{a}} \circ \tau)(T_{\mathbf{b}} \circ \sigma)) = \tau\sigma = \Lambda(T_{\mathbf{a}} \circ \tau)\Lambda(T_{\mathbf{b}} \circ \sigma)$, so Λ is a homomorphism. It is surjective because $\Lambda(T_{\mathbf{0}} \circ \tau) = \tau$, and $\ker\Lambda = \{T_{\mathbf{a}} \circ I_{\mathcal{V}} : \mathbf{a} \in \mathcal{V}\} = \mathrm{Trans}(\mathcal{V})$.
(vi) Put $N := \mathrm{Trans}(\mathcal{V})$. The rule $\overline{\Lambda}(gN) = \Lambda(g)$ defines $\overline{\Lambda} : \mathrm{Aff}(\mathcal{V})/N \to \mathrm{GL}(\mathcal{V})$ because $gN = hN \Rightarrow h^{-1}g \in N = \ker\Lambda \Rightarrow \Lambda(g) = \Lambda(h)$. It is a surjective homomorphism by (v), and $\ker\overline{\Lambda} = \{N\}$ because $\overline{\Lambda}(gN) = I_{\mathcal{V}} \iff g \in \ker\Lambda = N \iff gN = N$. Hence $\overline{\Lambda}$ is an isomorphism.

Structural Commentary: Derive product and inverse laws, prove normality, compute the kernel, then descend Λ to the quotient.
Key Insight: The linear-part map forgets exactly the translation subgroup. □

4.4 Linear Projective Geometry and Affine Completion

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over a fixed field $\mathbb{F}$.

Definition 4.4.1. *Projective Geometry of a Vector Space*

(i) The *projective geometry* of $\mathcal{V}$ is the poset $\mathrm{Sub}(\mathcal{V})$ of linear subspaces, ordered by inclusion.
Its elements are *projective subspaces*, and its dimension map is $\mathrm{pdim} : \mathrm{Sub}(\mathcal{V}) \to \mathbb{Z}$, $\mathrm{pdim}(\mathcal{U}) := \dim(\mathcal{U}) - 1$.
(ii) A *projective point* is a 1-dimensional subspace; a *projective line* is a 2-dimensional subspace.

Theorem 4.4.2. *Linear Span Map from Affine to Projective Geometry*
Let $H = \mathbf{x} + \mathcal{K}$ be an affine hyperplane with $\mathbf{0} \notin H$, and put $\mathcal{A}(H) := \{X \in \mathcal{A}(\mathcal{V}) : X \subseteq H\}$.
Define $P : \mathcal{A}(H) \to \mathrm{Sub}(\mathcal{V})$ by $P(X) := \mathrm{span}(X)$. Then:

(i) P *is injective, and* $(\forall\, \mathcal{U} \in \mathrm{im}\, P)\ P^{-1}(\mathcal{U}) = \mathcal{U} \cap H$.
(ii) $\mathrm{im}\, P = \{\mathcal{U} \leq \mathcal{V} : \mathcal{U} \not\subseteq \mathcal{K}\}$.
(iii) $(\forall\, X, Y \in \mathcal{A}(H))\,[X \subseteq Y \iff P(X) \subseteq P(Y)]$.
(iv) $(\forall\, \mathcal{C} \subseteq \mathcal{A}(H))\,[\mathcal{C} \neq \emptyset \wedge \bigcap \mathcal{C} \neq \emptyset \Rightarrow P(\bigcap \mathcal{C}) = \bigcap_{X \in \mathcal{C}} P(X)]$.
(v) $(\forall\, \mathcal{C} \subseteq \mathcal{A}(H))\,[\mathcal{C} \neq \emptyset \Rightarrow P(\bigvee \mathcal{C}) = \bigvee_{X \in \mathcal{C}} P(X) = \sum_{X \in \mathcal{C}} P(X)]$, *where* $\bigvee \mathcal{C}$ *is computed in* $\mathcal{A}_0(\mathcal{V})$.
(vi) $(\forall\, X \in \mathcal{A}(H))\ \mathrm{pdim}(P(X)) = \dim(X)$.
(vii) $(\forall\, X, Y \in \mathcal{A}(H))\,[X \parallel Y \iff (P(X) \cap \mathcal{K} \subseteq P(Y) \cap \mathcal{K} \vee P(Y) \cap \mathcal{K} \subseteq P(X) \cap \mathcal{K})]$.

Proof.
(i) Fix $X = \mathbf{u} + \mathcal{S} \in \mathcal{A}(H)$. Then $\mathcal{S} \leq \mathcal{K}$ by Proposition 4.1.4(iii). If $\mathbf{u} \in \mathcal{K}$, then $\mathbf{u} - \mathbf{x} \in \mathcal{K}$ because $\mathbf{u} \in H$, so $\mathbf{x} \in \mathcal{K}$ and $\mathbf{0} \in H$, a contradiction. Thus $\mathbf{u} \notin \mathcal{K}$. Since $\mathbf{u} \in X$ and $(\forall\, \mathbf{s} \in \mathcal{S})\,\mathbf{s} = (\mathbf{u} + \mathbf{s}) - \mathbf{u}$, $\{\mathbf{u}\} \cup \mathcal{S} \subseteq P(X)$. Together with $X \subseteq \mathbb{F}\mathbf{u} + \mathcal{S}$, this proves $P(X) = \mathbb{F}\mathbf{u} \oplus \mathcal{S}$; the sum is direct because $\mathcal{S} \leq \mathcal{K}$ and $\mathbf{u} \notin \mathcal{K}$. If $\mathbf{z} \in P(X) \cap H$, write $\mathbf{z} = r\mathbf{u} + \mathbf{s} = \mathbf{u} + \mathbf{k}$ with $(r, \mathbf{s}, \mathbf{k}) \in \mathbb{F} \times \mathcal{S} \times \mathcal{K}$. Then $(1 - r)\mathbf{u} = \mathbf{s} - \mathbf{k} \in \mathcal{K}$, whence $r = 1$ and $\mathbf{z} \in X$. Since $X \subseteq P(X) \cap H$, $P(X) \cap H = X$. Therefore $(\forall\, X, Y \in \mathcal{A}(H))\,[P(X) = P(Y) \Rightarrow X = P(X) \cap H = P(Y) \cap H = Y]$, and $(\forall\, \mathcal{U} \in \mathrm{im}\, P)\ P^{-1}(\mathcal{U}) = \mathcal{U} \cap H$.
(ii) By the representation in (i), $(\forall\, X \in \mathcal{A}(H))\ P(X) \not\subseteq \mathcal{K}$. Conversely, fix $\mathcal{U} \leq \mathcal{V}$ with $\mathcal{U} \not\subseteq \mathcal{K}$ and choose $\mathbf{u} \in \mathcal{U} \setminus \mathcal{K}$. Since $\mathcal{K}$ has codimension one, $\dim(\mathcal{V}/\mathcal{K}) = 1$ by Theorem 2.2.9(ii). The cosets $\mathbf{x} + \mathcal{K}$ and $\mathbf{u} + \mathcal{K}$ are nonzero; choose $r \in \mathbb{F}^{\times}$ with $\mathbf{x} + \mathcal{K} = r(\mathbf{u} + \mathcal{K})$, and put $\mathbf{h} := r\mathbf{u} \in \mathcal{U} \cap H$. Then $(\forall\, \mathbf{z} \in \mathcal{V})\,[\mathbf{z} \in \mathcal{U} \cap H \iff \mathbf{z} - \mathbf{h} \in \mathcal{U} \cap \mathcal{K}]$, so $\mathcal{U} \cap H = \mathbf{h} + (\mathcal{U} \cap \mathcal{K}) \in \mathcal{A}(H)$. By $\mathrm{span}(\mathbf{h} + \mathcal{K}) = \mathcal{V}/\mathcal{K}$, $(\forall\, \mathbf{w} \in \mathcal{U})(\exists\, a \in \mathbb{F})\,\mathbf{w} - a\mathbf{h} \in \mathcal{U} \cap \mathcal{K}$. Therefore $P(\mathcal{U} \cap H) = \mathbb{F}\mathbf{h} + (\mathcal{U} \cap \mathcal{K}) = \mathcal{U}$.
(iii) ($\Rightarrow$) Fix $X, Y \in \mathcal{A}(H)$. Then $X \subseteq Y \Rightarrow \mathrm{span}(X) \subseteq \mathrm{span}(Y)$.
(iii) ($\Leftarrow$) Fix $X, Y \in \mathcal{A}(H)$. By (i), $P(X) \subseteq P(Y) \Rightarrow X = P(X) \cap H \subseteq P(Y) \cap H = Y$.

(iv) Fix $\emptyset \neq \mathcal{C} \subseteq \mathcal{A}(H)$ with $\bigcap \mathcal{C} \neq \emptyset$, and choose $\mathbf{z} \in \bigcap \mathcal{C}$. For $X \in \mathcal{C}$, let $\mathcal{S}_X$ be its base; then $X = \mathbf{z} + \mathcal{S}_X$ by Proposition 4.1.2(ii). Hence $\bigcap \mathcal{C} = \mathbf{z} + \bigcap_{X \in \mathcal{C}} \mathcal{S}_X$ by Lemma 4.1.12. Since $\mathbf{z} \notin \mathcal{K}$ and $(\forall\, X \in \mathcal{C})\, \mathcal{S}_X \leq \mathcal{K}$, $\bigcap_{X \in \mathcal{C}} P(X) = \bigcap_{X \in \mathcal{C}} (\mathbb{F}\mathbf{z} \oplus \mathcal{S}_X) = \mathbb{F}\mathbf{z} \oplus \bigcap_{X \in \mathcal{C}} \mathcal{S}_X = P(\bigcap \mathcal{C})$.

(v) Fix $\emptyset \neq \mathcal{C} \subseteq \mathcal{A}(H)$ and put $E := \bigcup \mathcal{C}$. By Theorem 4.1.13(ii), $\bigvee \mathcal{C} = \operatorname{affhull}(E) \subseteq H$ because H is an upper bound, so $\bigvee \mathcal{C} \in \mathcal{A}(H)$. By Theorem 4.1.11(i), $\operatorname{affhull}(E) = \operatorname{affspan}(E) \subseteq \operatorname{span}(E)$. By $E \subseteq \operatorname{affhull}(E)$, $\operatorname{span}(E) \subseteq \operatorname{span}(\operatorname{affhull}(E))$. Hence $P(\bigvee \mathcal{C}) = \operatorname{span}(E) = \sum_{X \in \mathcal{C}} \operatorname{span}(X) = \sum_{X \in \mathcal{C}} P(X) = \bigvee_{X \in \mathcal{C}} P(X)$.

(vi) If $X = \mathbf{u} + \mathcal{S} \in \mathcal{A}(H)$, then $P(X) = \mathbb{F}\mathbf{u} \oplus \mathcal{S}$ by (i), so $\operatorname{pdim}(P(X)) = \dim(\mathbb{F}\mathbf{u} \oplus \mathcal{S}) - 1 = \dim \mathcal{S} = \dim X$.

(vii) ($\Rightarrow$) Fix $X = \mathbf{u} + \mathcal{S}, Y = \mathbf{v} + \mathcal{T} \in \mathcal{A}(H)$. By the calculation in (i), $P(X) \cap \mathcal{K} = \mathcal{S}$ and $P(Y) \cap \mathcal{K} = \mathcal{T}$. By $X \parallel Y$ and the definition of parallel flats, $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$, so the two intersections with $\mathcal{K}$ are comparable.

(vii) ($\Leftarrow$) Fix $X = \mathbf{u} + \mathcal{S}, Y = \mathbf{v} + \mathcal{T} \in \mathcal{A}(H)$. By the identities in the forward direction, comparability of $P(X) \cap \mathcal{K}$ and $P(Y) \cap \mathcal{K}$ is comparability of $\mathcal{S}$ and $\mathcal{T}$. Hence $X \parallel Y$ by the definition of parallel flats.

Structural Commentary: Compute spans and traces on H, then transport order, lattice operations, dimension, and direction.
Key Insight: Transversality makes intersection with H inverse to taking linear span. □

Perspective 4.4.3. *Projective Completion of an Affine Hyperplane*
Let $H = \mathbf{x} + \mathcal{K}$ be an affine hyperplane with $\mathbf{0} \notin H$. Theorem 4.4.2 identifies each affine flat $X \subseteq H$ with $\operatorname{span}(X)$ and records its direction as $\operatorname{span}(X) \cap \mathcal{K}$. If $\dim H = 2$, distinct parallel affine lines therefore determine the same one-dimensional subspace of $\mathcal{K}$, their common projective point at infinity. This viewpoint explains the completion; the theorem owns the incidence claims.

Figure 37 illustrates the preceding plane case: parallel affine lines are disjoint in H, whereas their projective spans meet in a shared direction contained in $\mathcal{K}$.

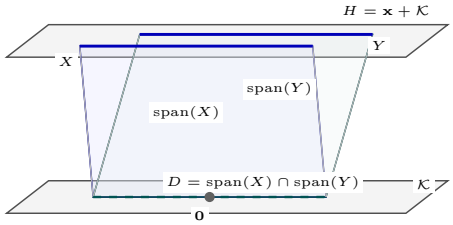

Figure 37: The affine plane $H = \mathbf{x} + \mathcal{K}$, direction plane $\mathcal{K}$, parallel lines X, Y, and shaded spans $\operatorname{span}(X), \operatorname{span}(Y)$. The spans meet along the dashed subspace $D = \operatorname{span}(X) \cap \mathcal{K} = \operatorname{span}(Y) \cap \mathcal{K}$, the common projective point at infinity.

4.5 Convex Combinations, Cones, and Hulls

Definition 4.5.1. *Convex Combinations, Cones, and Hulls*
Let $k \geq 1$ and $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$.

(i) A *convex combination* of these vectors is a vector $\sum_{i=1}^{k} t_i \mathbf{x}_i$, where $t_i \geq 0$ for every i and $\sum_{i=1}^{k} t_i = 1$.
(ii) A set $C \subseteq \mathbb{R}^n$ is *convex* if $t\mathbf{x} + (1-t)\mathbf{y} \in C$ for all $\mathbf{x}, \mathbf{y} \in C$ and $0 \leq t \leq 1$.
(iii) A set $K \subseteq \mathbb{R}^n$ is a *cone* if $\mathbf{x} \in K$ and $a \geq 0$ imply $a\mathbf{x} \in K$.
(iv) A *convex cone* is a subset of $\mathbb{R}^n$ that is both convex and a cone.
(v) For the nonempty finite set $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, its *convex hull* is

$$\operatorname{conv}(S) := \bigcap \{C \subseteq \mathbb{R}^n : C \text{ is convex and } S \subseteq C\}.$$

The defining family is nonempty because $\mathbb{R}^n$ is convex and contains S.

The closedness and compactness claims below use the Euclidean metric conventions from Definitions 3.1.2 and 4.1.1.

Proposition 4.5.2. *Finite Convex Hulls*
Let $k \geq 1$, $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\} \subseteq \mathbb{R}^n$, and $\Delta_{k-1} := \left\{ (t_1, \ldots, t_k) \in \mathbb{R}^k : t_i \geq 0,\ \sum_{i=1}^{k} t_i = 1 \right\}$.
Then $\operatorname{conv}(S) = \left\{ \sum_{i=1}^{k} t_i \mathbf{x}_i : (t_1, \ldots, t_k) \in \Delta_{k-1} \right\}$. *Moreover, $\operatorname{conv}(S)$ is compact.*

Proof. Define the linear map $\widetilde{T} : \mathbb{R}^k \to \mathbb{R}^n$ by $\widetilde{T}(t_1, \ldots, t_k) = \sum_{i=1}^{k} t_i \mathbf{x}_i$, and set $C := \widetilde{T}(\Delta_{k-1})$. For the ith standard basis vector $\mathbf{e}_i \in \Delta_{k-1}$, $\widetilde{T}(\mathbf{e}_i) = \mathbf{x}_i$, so $S \subseteq C$.
Let $K \subseteq \mathbb{R}^n$ be convex with $S \subseteq K$. Prove $C \subseteq K$ by induction on k. The case $k = 1$ is immediate. For $k \geq 2$ and $(t_1, \ldots, t_k) \in \Delta_{k-1}$, if $t_k = 1$, then $\sum_i t_i \mathbf{x}_i = \mathbf{x}_k \in K$. If $t_k < 1$, set $s := 1 - t_k > 0$. Then $\sum_{i=1}^{k-1} (t_i/s) = 1$, so $\mathbf{y} := \sum_{i=1}^{k-1} (t_i/s)\mathbf{x}_i \in K$ by induction, and $\sum_i t_i \mathbf{x}_i = s\mathbf{y} + t_k \mathbf{x}_k \in K$ by convexity. Hence C lies in every convex set containing S, so $C \subseteq \operatorname{conv}(S)$ by Definition 4.5.1(v).
Conversely, if $\mathbf{r}, \mathbf{s} \in \Delta_{k-1}$ and $0 \leq t \leq 1$, then $t\mathbf{r} + (1-t)\mathbf{s} \in \Delta_{k-1}$ and $\widetilde{T}(t\mathbf{r} + (1-t)\mathbf{s}) = t\widetilde{T}(\mathbf{r}) + (1-t)\widetilde{T}(\mathbf{s})$. Thus C is convex. Since $S \subseteq C$, $\operatorname{conv}(S) \subseteq C$ by Definition 4.5.1(v), and therefore $\operatorname{conv}(S) = C$.
The simplex Δ_{k-1} is closed and bounded, hence compact by Heine–Borel, Theorem 4.2.5.
The map $\widetilde{T}$ is continuous by Corollary 7.1.7, so $\operatorname{conv}(S) = \widetilde{T}(\Delta_{k-1})$ is compact by Theorem 5.1.35.
Structural Commentary: A finite convex hull is the continuous image of a compact coefficient simplex.
Key Insight: Convexity is carried by the coefficients t_i, not by the ambient coordinates. □

Proposition 4.5.3. *Sums of Convex and Closed Sets*
For $X, Y \subseteq \mathbb{R}^n$, set $X + Y := \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in X,\ \mathbf{y} \in Y\}$. Then:

(i) X, Y convex $\implies$ $X + Y$ convex.
(ii) X compact and Y closed $\implies$ $X + Y$ closed.

Proof.

(i) Assume X, Y are convex. Let $\mathbf{z}_r = \mathbf{x}_r + \mathbf{y}_r \in X + Y$ for $r = 0, 1$, and let $0 \le t \le 1$. Then $t\mathbf{z}_0 + (1-t)\mathbf{z}_1 = \big(t\mathbf{x}_0 + (1-t)\mathbf{x}_1\big) + \big(t\mathbf{y}_0 + (1-t)\mathbf{y}_1\big) \in X + Y$. Thus $X + Y$ is convex.

(ii) Assume X is compact and Y is closed. By Theorem 3.4.13(ii), it is enough to consider $\mathbf{z}_j \in X + Y$ with $\mathbf{z}_j \to \mathbf{z}$ and prove $\mathbf{z} \in X + Y$.
Write $\mathbf{z}_j = \mathbf{x}_j + \mathbf{y}_j$ with $\mathbf{x}_j \in X$ and $\mathbf{y}_j \in Y$. By Theorem 4.1.12, a subsequence satisfies $\mathbf{x}_{j_\ell} \to \mathbf{x} \in X$.
Then $\mathbf{y}_{j_\ell} = \mathbf{z}_{j_\ell} - \mathbf{x}_{j_\ell} \to \mathbf{z} - \mathbf{x}$. Since Y is closed, $\mathbf{z} - \mathbf{x} \in Y$ by Theorem 3.4.13(ii). Hence $\mathbf{z} = \mathbf{x} + (\mathbf{z} - \mathbf{x}) \in X + Y$, so $X + Y$ is closed.

Structural Commentary: Compactness lets one hold the X-part still enough for closedness of Y to apply.
Key Insight: A limit in a sum can be split by passing to a convergent compact subsequence. □

4.6 Hyperplanes, Nearest Points, and Separation

Definition 4.6.1. *Hyperplanes, Half-Spaces, and Separation*
Let $\mathbf{N} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$.

(i) The *affine hyperplane* with normal $\mathbf{N}$ and level b is
$$\begin{aligned}\mathcal{H}(\mathbf{N}, b) &:= \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x} \rangle = b\} \\ &= \frac{b}{\|\mathbf{N}\|^2}\mathbf{N} + \operatorname{span}(\mathbf{N})^\perp.\end{aligned}$$
(ii) The case $b = 0$ gives the *linear hyperplane* $\mathcal{H}(\mathbf{N}, 0) = \operatorname{span}(\mathbf{N})^\perp$.
(iii) The *closed half-spaces* determined by $\mathcal{H}(\mathbf{N}, b)$ are
$\mathcal{H}_+(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x} \rangle \ge b\}$ and $\mathcal{H}_-(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x} \rangle \le b\}$.
The *open half-spaces* are obtained by replacing $\ge, \le$ with $>, <$.
(iv) For $X \subseteq \mathbb{R}^n$ and $\star \in \{<, \le, >, \ge\}$, use the *set-inequality convention* $\langle \mathbf{N}, X \rangle \star b \iff \langle \mathbf{N}, \mathbf{x} \rangle \star b \ \forall \mathbf{x} \in X$.
(v) An *orientation* of $\mathcal{H}(\mathbf{N}, b)$ is $\sigma(\mathbf{N}, b) := (\sigma\mathbf{N}, \sigma b)$ for $\sigma \in \{\pm 1\}$.
(vi) Sets $X, Y \subseteq \mathbb{R}^n$ are *strictly separated* by $\mathcal{H}(\mathbf{N}, b)$ if $\exists \sigma \in \{\pm 1\}$ such that $\langle \sigma\mathbf{N}, X \rangle < \sigma b < \langle \sigma\mathbf{N}, Y \rangle$.
(vii) Sets $X, Y \subseteq \mathbb{R}^n$ are *strongly separated* by $\mathcal{H}(\mathbf{N}, b)$ if $\exists \sigma \in \{\pm 1\}$ and $\varepsilon > 0$ such that
$\langle \sigma\mathbf{N}, X \rangle < \sigma b - \varepsilon < \sigma b + \varepsilon < \langle \sigma\mathbf{N}, Y \rangle$.

Remark 4.6.2. *Hyperplane Geometry*
The half-spaces satisfy $\mathcal{H}_+(\mathbf{N}, b) \cap \mathcal{H}_-(\mathbf{N}, b) = \mathcal{H}(\mathbf{N}, b)$, and two open half-spaces with $\mathcal{H}(\mathbf{N}, b)$ partition $\mathbb{R}^n$.
Moreover, $\mathbf{N} \in \mathcal{H}(\mathbf{N}, \|\mathbf{N}\|^2)$. If $\mathbf{x}$ belongs to this hyperplane, then $\|\mathbf{N}\|^2 = \langle \mathbf{N}, \mathbf{x} \rangle \le \|\mathbf{N}\| \, \|\mathbf{x}\|$ by Theorem 1.1.13; hence $\|\mathbf{N}\| \le \|\mathbf{x}\|$, so $\mathbf{N}$ is a closest point of the hyperplane to $\mathbf{0}$.

Theorem 4.6.3. *Nearest Points and Separating a Point*
Let $C \subseteq \mathbb{R}^n$ be nonempty, closed, and convex.

(i) There exists a unique $\mathbf{N}_C \in C$ such that $\|\mathbf{N}_C\| = \min_{\mathbf{x} \in C} \|\mathbf{x}\|$.
(ii) If $\mathbf{b} \notin C$, set $D := C - \mathbf{b} := \{\mathbf{x} - \mathbf{b} : \mathbf{x} \in C\}$ and let $\mathbf{N} \in D$ satisfy $\|\mathbf{N}\| = \min_{\mathbf{z} \in D} \|\mathbf{z}\|$.
Then $\mathbf{N} \neq \mathbf{0}$ and $\langle \mathbf{N}, C \rangle \ge \|\mathbf{N}\|^2 + \langle \mathbf{N}, \mathbf{b} \rangle > \langle \mathbf{N}, \mathbf{b} \rangle$.
With $\alpha := \langle \mathbf{N}, \mathbf{b} \rangle + \frac{1}{2}\|\mathbf{N}\|^2$, the hyperplane $\mathcal{H}(\mathbf{N}, \alpha)$ strongly separates $\{\mathbf{b}\}$ and C.

Proof.

(i) Choose $\mathbf{x}_0 \in C$ and $s \ge \|\mathbf{x}_0\|$. Set $K := C \cap \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \le s\}$. The norm is 1-Lipschitz and hence continuous by Proposition 1.1.15, so its sublevel set is closed by Corollary 5.1.19. Thus K is nonempty, closed, and bounded, and it is compact by Theorem 4.2.5.
By Theorem 5.1.37, choose $\mathbf{N}_C \in K$ minimizing $\|\cdot\|$ on K.
If $\mathbf{x} \in C$ and $\|\mathbf{x}\| < \|\mathbf{N}_C\|$, then $\mathbf{x} \in K$, a contradiction; hence $\mathbf{N}_C$ minimizes on C.
If $\mathbf{u} \neq \mathbf{v}$ are minimizers of norm m, then $\mathbf{w} := (\mathbf{u} + \mathbf{v})/2 \in C$ and, by the parallelogram identity, Theorem 1.1.20, $\|\mathbf{w}\|^2 = m^2 - \frac{1}{4}\|\mathbf{u} - \mathbf{v}\|^2 < m^2$, a contradiction.

(ii) The singleton $\{-\mathbf{b}\}$ is convex and is compact by Proposition 4.5.2. Since $D = \{-\mathbf{b}\} + C$, it is nonempty, closed, and convex by Proposition 4.5.3. By part (i), let $\mathbf{N}$ be its unique minimum-norm vector.
Since $\mathbf{b} \notin C$, $\mathbf{0} \notin D$, hence $\mathbf{N} \neq \mathbf{0}$. Fix $\mathbf{z} \in D$. Convexity places $\mathbf{N} + t(\mathbf{z} - \mathbf{N})$ in D for $0 \le t \le 1$, so minimality implies, for $0 < t \le 1$,
$$\begin{aligned}0 &\le \|\mathbf{N} + t(\mathbf{z} - \mathbf{N})\|^2 - \|\mathbf{N}\|^2 \\ &= 2t\,\langle \mathbf{N}, \mathbf{z} - \mathbf{N} \rangle + t^2 \|\mathbf{z} - \mathbf{N}\|^2 .\end{aligned}$$
If $\langle \mathbf{N}, \mathbf{z} - \mathbf{N} \rangle < 0$, choosing sufficiently small $t > 0$ makes the last expression negative, a contradiction. Thus $\langle \mathbf{N}, \mathbf{z} \rangle \ge \|\mathbf{N}\|^2$.
Taking $\mathbf{z} = \mathbf{x} - \mathbf{b}$ proves $\langle \mathbf{N}, \mathbf{x} \rangle \ge \|\mathbf{N}\|^2 + \langle \mathbf{N}, \mathbf{b} \rangle$ for every $\mathbf{x} \in C$.
Set $\alpha := \langle \mathbf{N}, \mathbf{b} \rangle + \|\mathbf{N}\|^2/2$ and $\varepsilon := \|\mathbf{N}\|^2/4$. Then
$$\langle \mathbf{N}, \mathbf{b} \rangle < \alpha - \varepsilon < \alpha + \varepsilon < \langle \mathbf{N}, \mathbf{b} \rangle + \|\mathbf{N}\|^2 \le \langle \mathbf{N}, C \rangle .$$
Therefore $\mathcal{H}(\mathbf{N}, \alpha)$ strongly separates $\{\mathbf{b}\}$ and C by Definition 4.6.1(vi).

Structural Commentary: The nearest point in $C - \mathbf{b}$ supplies the normal vector to a separating hyperplane.
Key Insight: Minimal norm turns a metric problem into the linear inequality $\langle \mathbf{N}, \mathbf{z} \rangle \ge \|\mathbf{N}\|^2$. □

Figure 38 illustrates Theorem 4.6.3(ii): $\mathbf{p} = \mathbf{b} + \mathbf{N}$ is the nearest point in C, and $\mathbf{N}$ is normal to a separating hyperplane. The theorem, rather than the diagram, owns these claims. The schematic is redrawn after the separation geometry in (Roman, 2008, Chapter 15).

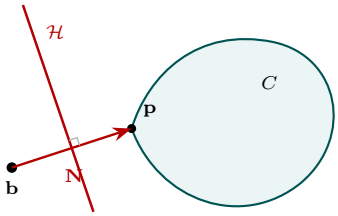

Figure 38: For $\mathbf{b} \notin C$, the labeled point $\mathbf{p} = \mathbf{b} + \mathbf{N} \in C$ is nearest to $\mathbf{b}$, the arrow $\mathbf{N}$ is the displacement from $\mathbf{b}$ to $\mathbf{p}$, and the labeled hyperplane lies strictly between $\mathbf{b}$ and the closed convex set C.

Theorem 4.6.4. *Separating a Compact Convex Set from a Subspace*
Let $C \subseteq \mathbb{R}^n$ be nonempty, compact, and convex, and let $S \leq \mathbb{R}^n$ satisfy $C \cap S = \emptyset$.
Then there exists $\mathbf{N} \in S^\perp \setminus \{\mathbf{0}\}$ such that $\langle \mathbf{N}, C \rangle \geq \|\mathbf{N}\|^2 > 0$.
Consequently, $\mathcal{H}(\mathbf{N}, \|\mathbf{N}\|^2/2)$ strongly separates S and C.

Proof.

(i) The subspace S is convex. To verify its Euclidean closedness, let $Q := P_{S^\perp}$. By Proposition 2.2.3(i),(vi) and Corollary 2.1.8, Q is linear and $\ker(Q) = S$. Thus Q is continuous by Corollary 7.1.7, and $S = Q^{-1}(\{\mathbf{0}\})$ is closed by Corollary 5.1.19.
Set $D := S + C = C + S$. By Proposition 4.5.3, D is closed and convex.
Also $\mathbf{0} \notin D$, for $\mathbf{0} = \mathbf{s} + \mathbf{c}$ would imply $\mathbf{c} = -\mathbf{s} \in C \cap S$.

(ii) By Theorem 4.6.3(ii), applied to D and $\mathbf{0}$, choose $\mathbf{N} \neq \mathbf{0}$ such that $\langle \mathbf{N}, \mathbf{s} + \mathbf{c} \rangle \geq \|\mathbf{N}\|^2$ for every $\mathbf{s} \in S$ and $\mathbf{c} \in C$.
Fix $\mathbf{c}_0 \in C$. If $a := \langle \mathbf{N}, \mathbf{s} \rangle \neq 0$ for some $\mathbf{s} \in S$, choose $\lambda \in \mathbb{R}$ with $\lambda a + \langle \mathbf{N}, \mathbf{c}_0 \rangle < \|\mathbf{N}\|^2$. Then $\lambda \mathbf{s} \in S$, contradicting the preceding inequality. Hence $\mathbf{N} \in S^\perp$. Taking $\mathbf{s} = \mathbf{0}$ proves $\langle \mathbf{N}, C \rangle \geq \|\mathbf{N}\|^2 > 0$.
Set $\alpha := \|\mathbf{N}\|^2/2$ and $\varepsilon := \|\mathbf{N}\|^2/4$. Since $\langle \mathbf{N}, \mathbf{s} \rangle = 0$ for every $\mathbf{s} \in S$,

$$0 < \alpha - \varepsilon < \alpha + \varepsilon < \|\mathbf{N}\|^2 \leq \langle \mathbf{N}, C \rangle .$$

Thus $\mathcal{H}(\mathbf{N}, \alpha)$ strongly separates S and C by Definition 4.6.1(vi).

Structural Commentary: The sum $S + C$ is a closed convex set disjoint from $\mathbf{0}$, so point separation applies.
Key Insight: Since the separating inequality holds for all translates by S, scaling inside S forces $\mathbf{N}$ to annihilate S. □

4.7 Positive Orthants, Alternatives, and Farkas's Lemma

This subsection uses finite-dimensional real inner-product geometry to decide when linear systems have positive solutions. The guiding theme is that a missing positive solution is often detected by a separating hyperplane, hence by a dual vector certificate.

Definition 4.7.1. *Positive Orthants*
For $n \geq 1$ and $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, use the following order and orthant notation:

(i) $\mathbf{x} \geq \mathbf{0}$ iff $\mathbf{x}$ is *coordinatewise nonnegative*: $(\forall i \in \{1, \ldots, n\})\, x_i \geq 0$. Write $\mathbf{x} \leq \mathbf{0}$ iff $-\mathbf{x} \geq \mathbf{0}$.
(ii) $\mathbf{x} > \mathbf{0}$ iff $\mathbf{x}$ is *nonzero nonnegative*: $\mathbf{x} \geq \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$.
(iii) $\mathbf{x} \gg \mathbf{0}$ iff $\mathbf{x}$ is *coordinatewise positive*: $(\forall i \in \{1, \ldots, n\})\, x_i > 0$.
(iv) $\mathbb{R}^n_{\geq 0} := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}\}$ is the *nonnegative orthant*.
(v) $\mathbb{R}^n_{+} := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} > \mathbf{0}\} = \mathbb{R}^n_{\geq 0} \setminus \{\mathbf{0}\}$ is the *nonzero nonnegative orthant*.
(vi) $\mathbb{R}^n_{++} := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \gg \mathbf{0}\}$ is the *coordinatewise positive orthant*.

Motivation 4.7.2. *Positive Null Vectors as Probabilities*
Let $m, n \geq 1$, $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{R})$, and $\mathbf{x} = (x_1, \ldots, x_n) > \mathbf{0}$ satisfy $A\mathbf{x} = \mathbf{0}$.

(i) Set $s := x_1 + \cdots + x_n > 0$ and $\boldsymbol{\pi} := s^{-1}\mathbf{x}$.
(ii) Then $\pi_i \geq 0$ and $\sum_{i=1}^n \pi_i = 1$, so $\boldsymbol{\pi}$ is a probability vector.
(iii) Since $A\boldsymbol{\pi} = s^{-1}A\mathbf{x} = \mathbf{0}$, equivalently $\sum_{i=1}^{n} \pi_i \mathbf{a}_i = \mathbf{0}$.
(iv) If $\mathbf{x} \gg \mathbf{0}$, then $(\forall i \in \{1, \ldots, n\})\, \pi_i > 0$.

Thus a positive null vector normalizes to probabilities under which the columns of A have zero expected value.

Principle 4.7.3. *Positive Solutions Are Orthant Intersections*
For $m, n \geq 1$ and $A \in \mathrm{Mat}(m, n, \mathbb{R})$, $\ker(A) = \mathrm{row}(A)^\perp$ by Corollary 2.3.12.
Hence $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0},\ \mathbf{x} > \mathbf{0}\} = \mathrm{row}(A)^\perp \cap \mathbb{R}^n_{+}$; and $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0},\ \mathbf{x} \gg \mathbf{0}\} = \mathrm{row}(A)^\perp \cap \mathbb{R}^n_{++}$.
Thus *positive null vectors* are precisely orthant intersections with $\mathrm{row}(A)^\perp$.

Theorem 4.7.4. *Positive Orthant Alternatives for Subspaces*
Let $n \geq 1$ and $S \leq \mathbb{R}^n$. Then:

(i) $S \cap \mathbb{R}^n_{+} = \emptyset \iff S^\perp \cap \mathbb{R}^n_{++} \neq \emptyset$.
(ii) $S \cap \mathbb{R}^n_{++} = \emptyset \iff S^\perp \cap \mathbb{R}^n_{+} \neq \emptyset$.

Proof.

(i) ($\Rightarrow$) Let $\Delta := \mathrm{conv}\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$. Then Δ is compact and convex by Proposition 4.5.2, and $\Delta \subseteq \mathbb{R}^n_{+}$. Hence $\Delta \cap S = \emptyset$. Choose $\mathbf{N} \in S^\perp \setminus \{\mathbf{0}\}$ by Theorem 4.6.4 such that $(\forall i \in \{1, \ldots, n\})\, N_i = \langle \mathbf{N}, \mathbf{e}_i \rangle \geq \|\mathbf{N}\|^2 > 0$. Thus $\mathbf{N} \in S^\perp \cap \mathbb{R}^n_{++}$.
(i) ($\Leftarrow$) Suppose $\mathbf{N} \in S^\perp \cap \mathbb{R}^n_{++}$. If $\mathbf{x} \in S \cap \mathbb{R}^n_{+}$, then $\langle \mathbf{N}, \mathbf{x} \rangle = \sum_{i=1}^n N_i x_i > 0$, since $\mathbf{x} \geq \mathbf{0}$ is nonzero and $\mathbf{N} \gg \mathbf{0}$. This contradicts $\mathbf{N} \perp S$, so $S \cap \mathbb{R}^n_{+} = \emptyset$.

(*ii*) ($\Rightarrow$) If $S = \{\mathbf{0}\}$, then $\mathbf{e}_1 \in S^\perp \cap \mathbb{R}^n_+$, so the claim holds. Assume $S \neq \{\mathbf{0}\}$, choose a basis $(\mathbf{B}_1, \dots, \mathbf{B}_k)$ of S, and set $M = [\mathbf{B}_1 \ \cdots \ \mathbf{B}_k] \in \mathrm{Mat}(n, k, \mathbb{R})$. Write the row vectors of M as $\mathbf{R}_1, \dots, \mathbf{R}_n \in \mathbb{R}^k$. For $\mathbf{a} = (a_1, \dots, a_n)$, $\mathbf{a} \in S^\perp \iff M^t\mathbf{a} = \mathbf{0} \iff \sum_{i=1}^n a_i\mathbf{R}_i = \mathbf{0}$. Set $K := \mathrm{conv}\{\mathbf{R}_1, \dots, \mathbf{R}_n\}$. If $\mathbf{a} \in S^\perp \cap \mathbb{R}^n_+$ and $s := \sum_{i=1}^n a_i$, then $s > 0$ and $\mathbf{0} = \sum_{i=1}^n (a_i/s)\mathbf{R}_i \in K$. Conversely, if $\mathbf{0} = \sum_{i=1}^n t_i\mathbf{R}_i$ with $(t_1, \dots, t_n) \in \Delta_{n-1}$, then $(t_1, \dots, t_n) \in S^\perp \cap \mathbb{R}^n_+$. Therefore $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset \iff \mathbf{0} \in K$. The set K is compact and convex by Proposition 4.5.2. If $\mathbf{0} \notin K$, choose $\mathbf{u} \in \mathbb{R}^k \setminus \{\mathbf{0}\}$ by Theorem 4.6.4 with the subspace $\{\mathbf{0}\}$ such that $(\forall i \in \{1, \dots, n\})\ \langle \mathbf{u}, \mathbf{R}_i \rangle \geq \|\mathbf{u}\|^2 > 0$. Thus $\mathbf{v} := M\mathbf{u} \in S \cap \mathbb{R}^n_{++}$, contradicting the hypothesis. Hence $\mathbf{0} \in K$, so $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset$.

(*ii*) ($\Leftarrow$) Suppose $\mathbf{N} \in S^\perp \cap \mathbb{R}^n_+$. If $\mathbf{x} \in S \cap \mathbb{R}^n_{++}$, then $\langle \mathbf{N}, \mathbf{x} \rangle = \sum_{i=1}^n N_i x_i > 0$, since $\mathbf{N} \geq \mathbf{0}$ is nonzero and $\mathbf{x} \gg \mathbf{0}$. This contradicts $\mathbf{N} \perp S$, so $S \cap \mathbb{R}^n_{++} = \emptyset$.

Structural Commentary: Use the simplex and row hull as compact witnesses.
Key Insight: Orthant avoidance is equivalent to an orthogonal positive certificate. □

Figure 39 illustrates Theorem 4.7.4(i) in $\mathbb{R}^2$: the displayed $\mathbf{y} \in S^\perp \cap \mathbb{R}^2_{++}$ certifies $S \cap \mathbb{R}^2_+ = \emptyset$. The theorem, rather than the illustration, owns this implication. The schematic is redrawn after the orthant geometry in (Roman, 2008, Chapter 15).

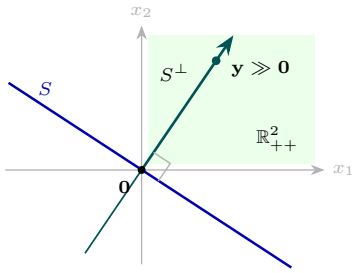

Figure 39: In $\mathbb{R}^2$, the negative-slope line S contains no nonzero nonnegative vector, while the labeled vector $\mathbf{y} \gg \mathbf{0}$ lies in $S^\perp$.

Theorem 4.7.5. *Homogeneous Positive Alternatives*
Let $m, n \geq 1$ and $A \in \mathrm{Mat}(m, n, \mathbb{R})$. Exactly one alternative holds in each pair:

(i) $\ker(A) \cap \mathbb{R}^n_{++} \neq \emptyset$ *or* $\mathrm{col}(A^t) \cap \mathbb{R}^n_+ \neq \emptyset$.
(ii) $\ker(A) \cap \mathbb{R}^n_+ \neq \emptyset$ *or* $\mathrm{col}(A^t) \cap \mathbb{R}^n_{++} \neq \emptyset$.

Proof.

(i) Set $S := \mathrm{row}(A) = \mathrm{col}(A^t) \subseteq \mathbb{R}^n$. Then $\ker(A) = S^\perp$ by Corollary 2.3.12(i). Hence the first pair is equivalent to $S^\perp \cap \mathbb{R}^n_{++} \neq \emptyset$ or $S \cap \mathbb{R}^n_+ \neq \emptyset$, and exactly one holds by Theorem 4.7.4(i).

(ii) With the same S, the second pair is equivalent to $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset$ or $S \cap \mathbb{R}^n_{++} \neq \emptyset$, and exactly one holds by Theorem 4.7.4(ii).

Structural Commentary: The row space carries certificates; its orthogonal complement carries null solutions.
Key Insight: Positive solvability is an orthogonal-complement alternative. □

Method 4.7.6. *Certifying Positive Homogeneous Systems*
Let $m, n \geq 1$, $A \in \mathrm{Mat}(m, n, \mathbb{R})$, and $S := \mathrm{row}(A) = \mathrm{col}(A^t)$. By Theorem 4.7.5, the following certificates and solvability tests are exact:

(i) $\ker(A) \cap \mathbb{R}^n_{++} = \emptyset \iff (\exists\, \mathbf{v} \in \mathbb{R}^m)\ A^t\mathbf{v} > \mathbf{0}$.
(ii) $\ker(A) \cap \mathbb{R}^n_+ = \emptyset \iff (\exists\, \mathbf{v} \in \mathbb{R}^m)\ A^t\mathbf{v} \gg \mathbf{0}$.
(iii) $S \cap \mathbb{R}^n_+ = \emptyset \iff \ker(A) \cap \mathbb{R}^n_{++} \neq \emptyset$, and $S \cap \mathbb{R}^n_{++} = \emptyset \iff \ker(A) \cap \mathbb{R}^n_+ \neq \emptyset$.

Lemma 4.7.7. *Nonnegative Matrix Cone*
Let $m, n \geq 1$, $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{R})$, and $C_A := A(\mathbb{R}^n_{\geq 0}) = \left\{ \sum_{i=1}^n t_i\mathbf{a}_i : t_i \geq 0 \right\}$.

Then:

(i) C_A is a convex cone in $\mathbb{R}^m$.
(ii) C_A is closed in $\mathbb{R}^m$.

Proof.

(i) Let $\mathbf{x}_r = A\mathbf{y}_r \in C_A$ with $\mathbf{y}_r \geq \mathbf{0}$ for $r \in \{1, 2\}$. If $0 \leq t \leq 1$ and $a \geq 0$, then $t\mathbf{x}_1 + (1-t)\mathbf{x}_2 = A(t\mathbf{y}_1 + (1-t)\mathbf{y}_2) \in C_A$ and $a\mathbf{x}_1 = A(a\mathbf{y}_1) \in C_A$.

(ii) Let $\mathcal{I}$ be the family of $I \subseteq \{1, \dots, n\}$ for which $\{\mathbf{a}_i : i \in I\}$ is linearly independent, including $I = \emptyset$, and set $K_I := \left\{ \sum_{i \in I} t_i\mathbf{a}_i : (\forall i \in I)\, t_i \geq 0 \right\}$. For $\mathbf{x} \in C_A \setminus \{\mathbf{0}\}$, choose a representation $\mathbf{x} = \sum_{i \in I} t_i\mathbf{a}_i$ with $(\forall i \in I)\, t_i > 0$ and minimal support I. If the active columns were dependent, choose $(d_i)_{i \in I} \neq \mathbf{0}$ with $\sum_{i \in I} d_i\mathbf{a}_i = \mathbf{0}$ and, after replacing (d_i) by its negative if necessary, $(\exists i \in I)\, d_i < 0$. Then $\lambda := \min_{d_i < 0} t_i/(-d_i) > 0$ satisfies $(\forall i \in I)\, t_i + \lambda d_i \geq 0$ and $(\exists i \in I)\, t_i + \lambda d_i = 0$, while $\mathbf{x} = \sum_{i \in I}(t_i + \lambda d_i)\mathbf{a}_i$. This contradicts minimality, so $I \in \mathcal{I}$. Since $\mathbf{0} \in K_\emptyset$,

$$C_A = \bigcup_{I \in \mathcal{I}} K_I.$$

Fix $I \in \mathcal{I}$ and set $U_I := \mathrm{span}\{\mathbf{a}_i : i \in I\}$. Let $(\varphi_i)_{i \in I}$ be the dual basis of $(\mathbf{a}_i)_{i \in I}$. By Theorem 1.6.1(i), choose $(\mathbf{d}_i)_{i \in I} \subseteq U_I$ such that $(\forall\, \mathbf{u} \in U_I)(\forall i \in I)\, \varphi_i(\mathbf{u}) = \langle \mathbf{u}, \mathbf{d}_i \rangle$. Thus $K_I = \{\mathbf{u} \in U_I : (\forall i \in I)\ \langle \mathbf{u}, \mathbf{d}_i \rangle \geq 0\}$. Let $\mathbf{x} \notin K_I$. If $\mathbf{x} \notin U_I$, then $\mathbf{q} := P_{U_I^\perp}\mathbf{x} \neq \mathbf{0}$ by Proposition 2.2.3(vi) and Corollary 2.1.8. For $\|\mathbf{y} - \mathbf{x}\| < \|\mathbf{q}\|/2$, $\left\|P_{U_I^\perp}\mathbf{y}\right\| \geq \|\mathbf{q}\| - \left\|P_{U_I^\perp}(\mathbf{y} - \mathbf{x})\right\| > \|\mathbf{q}\|/2 > 0$ by Proposition 2.2.3(i),(viii) and Proposition 1.1.15, so $\mathbf{y} \notin U_I$. If $\mathbf{x} \in U_I \setminus K_I$, choose $j \in I$ with $\langle \mathbf{x}, \mathbf{d}_j \rangle < 0$. Since $\langle \mathbf{a}_j, \mathbf{d}_j \rangle = 1$, $\mathbf{d}_j \neq \mathbf{0}$; and if $\|\mathbf{y} - \mathbf{x}\| < -\langle \mathbf{x}, \mathbf{d}_j \rangle / (2\|\mathbf{d}_j\|)$, then $\langle \mathbf{y}, \mathbf{d}_j \rangle \leq \langle \mathbf{x}, \mathbf{d}_j \rangle + \|\mathbf{y} - \mathbf{x}\|\,\|\mathbf{d}_j\| < \frac{1}{2}\langle \mathbf{x}, \mathbf{d}_j \rangle < 0$ by Theorem 1.1.13, so $\mathbf{y} \notin K_I$. Hence $\mathbb{R}^m \setminus K_I$ is open by Definition 3.1.2, and K_I is closed by Corollary 3.1.17. The family $\mathcal{I}$ is finite, so C_A is closed by Theorem 3.1.19(iv).

Structural Commentary: Minimal supports reduce the cone to finitely many closed independent-generator cones.
Key Insight: Dual coordinates and orthogonal projection expose every point outside each cone. □

Theorem 4.7.8. *Farkas's Lemma*
Let $m, n \geq 1$, write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{R})$, let $\mathbf{b} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$, and set $C_A := \{A\mathbf{y} : \mathbf{y} \in \mathbb{R}^n,\ \mathbf{y} \geq \mathbf{0}\}$.
Exactly one of the following holds:
(i) $\mathbf{b} \in C_A$, *equivalently* $(\exists\, \mathbf{u} \in \mathbb{R}^n)\,[\mathbf{u} > \mathbf{0}$ *and* $A\mathbf{u} = \mathbf{b}]$.
(ii) $(\exists\, \mathbf{v} \in \mathbb{R}^m)\,[A^t\mathbf{v} \leq \mathbf{0}$ *and* $\langle \mathbf{v}, \mathbf{b} \rangle > 0]$.

Proof.
(i) By the definition of C_A, $\mathbf{b} \in C_A \iff (\exists\, \mathbf{u} \in \mathbb{R}^n)\,[\mathbf{u} \geq \mathbf{0}$ and $A\mathbf{u} = \mathbf{b}]$; since $\mathbf{b} \neq \mathbf{0}$, $A\mathbf{u} = \mathbf{b} \implies \mathbf{u} \neq \mathbf{0}$, proving the stated equivalence in (i). If $\mathbf{b} = A\mathbf{u}$ with $\mathbf{u} \geq \mathbf{0}$ and $A^t\mathbf{v} \leq \mathbf{0}$, then $\langle \mathbf{v}, \mathbf{b} \rangle = \langle \mathbf{v}, A\mathbf{u} \rangle = \langle A^t\mathbf{v}, \mathbf{u} \rangle \leq 0$, so (i) and (ii) cannot both hold.
(ii) Suppose $\mathbf{b} \notin C_A$. Since $\mathbf{0} \in C_A$, the cone is nonempty; it is closed and convex by Lemma 4.7.7(i),(ii). Choose $\mathbf{N} \neq \mathbf{0}$ by Theorem 4.6.3(ii) such that $(\forall\, \mathbf{c} \in C_A)\ \langle \mathbf{N}, \mathbf{c} \rangle \geq \|\mathbf{N}\|^2 + \langle \mathbf{N}, \mathbf{b} \rangle$. Set $\mathbf{v} := -\mathbf{N}$. Then $(\forall\, \mathbf{c} \in C_A)\ \langle \mathbf{v}, \mathbf{c} \rangle \leq \langle \mathbf{v}, \mathbf{b} \rangle - \|\mathbf{v}\|^2$. Taking $\mathbf{c} = \mathbf{0}$ proves $\langle \mathbf{v}, \mathbf{b} \rangle \geq \|\mathbf{v}\|^2 > 0$. For $i \in \{1, \ldots, n\}$ and $\lambda \geq 0$, $\lambda\mathbf{a}_i \in C_A$, so $\lambda \langle \mathbf{v}, \mathbf{a}_i \rangle \leq \langle \mathbf{v}, \mathbf{b} \rangle - \|\mathbf{v}\|^2$. If $\langle \mathbf{v}, \mathbf{a}_i \rangle > 0$, a sufficiently large λ contradicts this inequality. Thus $(\forall i \in \{1, \ldots, n\})\ \langle \mathbf{v}, \mathbf{a}_i \rangle \leq 0$, equivalently $A^t\mathbf{v} \leq \mathbf{0}$.

Structural Commentary: Cone closedness permits nearest-point separation; scaling tests each generating column.
Key Insight: A cone certificate is nonpositive on every generator. □

Example 4.7.9. *Why the Weak Inequality Matters*
The conclusion $A^t\mathbf{v} \leq \mathbf{0}$ cannot generally be strengthened to $-A^t\mathbf{v} \gg \mathbf{0}$. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.
If $\mathbf{u} \geq \mathbf{0}$, then $(A\mathbf{u})_1 = u_1 \geq 0$, so $A\mathbf{u} \neq \mathbf{b}$.
The certificate $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ satisfies $A^t\mathbf{v} = (-1, 0, 0)^t \leq \mathbf{0}$ and $\langle \mathbf{v}, \mathbf{b} \rangle = 1 > 0$.
For $\mathbf{w} = (r, s)^t \in \mathbb{R}^2$, $A^t\mathbf{w} = (r, s, -s)^t$, so $-A^t\mathbf{w} \gg \mathbf{0}$ would require both $s < 0$ and $s > 0$, impossible.
Structural Commentary: Opposite columns force a zero in every certificate.
Key Insight: Boundary separation is sometimes the best possible separation.

5 Eigenstructure and Invariant Subspaces

5.1 Eigenvalues, Eigenspaces, and Spectral Criteria

Eigenstructure begins by identifying the scalars for which a shifted operator has a nontrivial kernel.
Throughout this subsection, $\mathcal{V}$ is a vector space over $\mathbb{F}$, and all matrix sizes are positive integers.

Definition 5.1.1. *Eigenvalue and Eigenvector*
Let $T \in \mathcal{L}(\mathcal{V})$. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T iff $(\exists\, \mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\})\, T\mathbf{v} = \lambda\mathbf{v}$. Such a vector $\mathbf{v}$ is an *eigenvector* of T corresponding to λ.

Definition 5.1.2. *Eigenspace*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\lambda \in \mathbb{F}$. The *eigenspace* of T corresponding to λ is $E_\lambda(T) := \ker(T - \lambda I_\mathcal{V})$. Thus λ is an eigenvalue of T if and only if $E_\lambda(T) \neq \{\mathbf{0}\}$. Equivalently, $E_\lambda(T)$ is the nullspace of the shifted operator $T - \lambda I_\mathcal{V}$.

Definition 5.1.3. *Spectrum and Geometric Multiplicity*
Let $T \in \mathcal{L}(\mathcal{V})$. The *spectrum* of T is $\mathrm{spec}(T) := \{\lambda \in \mathbb{F} : E_\lambda(T) \neq \{\mathbf{0}\}\}$.
The *geometric multiplicity* of $\lambda \in \mathrm{spec}(T)$ is $\dim E_\lambda(T)$. For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, set $E_\lambda(A) := \ker(A - \lambda I_n)$; $\mathrm{spec}(A)$ is the spectrum of the induced operator on $\mathbb{F}^n$.

Theorem 5.1.4. *Equivalent Conditions for an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{F}$. The following are equivalent:
(i) λ *is an eigenvalue of* T;
(ii) $T - \lambda I_\mathcal{V}$ *is not injective;*
(iii) $T - \lambda I_\mathcal{V}$ *is not surjective;*
(iv) $T - \lambda I_\mathcal{V}$ *is not invertible.*

Proof.
(i) $\Rightarrow$ *(ii)* The kernel $\ker(T - \lambda I_\mathcal{V}) = E_\lambda(T)$ is nontrivial by Definition 5.1.2, so $T - \lambda I_\mathcal{V}$ is not injective.
(ii) $\Rightarrow$ *(iii)* The operator $T - \lambda I_\mathcal{V}$ is not surjective by Theorem 1.7.7(ii),(iii).
(iii) $\Rightarrow$ *(iv)* The operator $T - \lambda I_\mathcal{V}$ is not invertible by Theorem 1.7.7(i),(iii).
(iv) $\Rightarrow$ *(i)* The operator $T - \lambda I_\mathcal{V}$ is not injective by Theorem 1.7.7(i),(ii), so $E_\lambda(T) \neq \{\mathbf{0}\}$ and λ is an eigenvalue by Definition 5.1.2.

Structural Commentary: Kernel nontriviality enters first; finite-dimensional invertibility cycles through the remaining conditions.
Key Insight: A scalar λ is spectral precisely when $T - \lambda I_\mathcal{V}$ has a nontrivial kernel. □

Theorem 5.1.5. *Eigenvectors for Distinct Eigenvalues Are Independent*
Let $m \geq 1$ and $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Suppose the list is linearly dependent, and let $k \geq 2$ be minimal such that $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is dependent. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is independent and $\mathbf{v}_k \in \mathrm{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$; choose $a_1, \ldots, a_{k-1} \in \mathbb{F}$ satisfying $\mathbf{v}_k = \sum_{i=1}^{k-1} a_i\mathbf{v}_i$. By linearity and $T\mathbf{v}_i = \lambda_i\mathbf{v}_i$, $\mathbf{0} = (T - \lambda_k I_\mathcal{V})\mathbf{v}_k = \sum_{i=1}^{k-1} a_i(\lambda_i - \lambda_k)\mathbf{v}_i$. Independence implies $(\forall i \in \{1, \ldots, k-1\})\, a_i(\lambda_i - \lambda_k) = 0$. Since the eigenvalues are distinct, $a_1 = \cdots = a_{k-1} = 0$, so $\mathbf{v}_k = \mathbf{0}$, contradicting that $\mathbf{v}_k$ is an eigenvector.
Structural Commentary: A minimal dependent prefix isolates one eigenvector; its shifted operator separates the earlier eigendirections.
Key Insight: Distinct shifts force every coefficient in the minimal dependence to vanish. □

Corollary 5.1.6. *Number of Distinct Eigenvalues*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then T has at most $\dim(\mathcal{V})$ distinct eigenvalues.

Proof. Set $d := \dim(\mathcal{V})$ and choose a basis β of $\mathcal{V}$ by Corollary 1.3.14; its length is d by Definition 1.4.2. If T had $d+1$ distinct eigenvalues, choose corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_{d+1}$. The list $(\mathbf{v}_1, \dots, \mathbf{v}_{d+1})$ is independent by Theorem 5.1.5, while β spans $\mathcal{V}$, contradicting $d+1 \leq d$ by Theorem 1.3.11.
Structural Commentary: A basis supplies the spanning-length bound; distinct eigenvalues supply independent vectors.
Key Insight: Count one eigenvector per eigenspace. □

Corollary 5.1.7. *Invertibility and the Zero Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is invertible if and only if $0 \notin \operatorname{spec}(T)$.

Proof.
(⇒) If T is invertible, then 0 is not an eigenvalue by Theorem 5.1.4(i),(iv), so $0 \notin \operatorname{spec}(T)$ by Definition 5.1.3.
(⇐) If $0 \notin \operatorname{spec}(T)$, then 0 is not an eigenvalue by Definition 5.1.3, so $T = T - 0I_{\mathcal{V}}$ is invertible by Theorem 5.1.4(i),(iv).
Structural Commentary: The zero eigenvalue is the spectral form of singularity.
Key Insight: Invertibility is equivalent to having trivial kernel, and $E_0(T) = \ker T$. □

Corollary 5.1.8. *Matrix Invertibility and the Spectrum*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$. Then A is invertible if and only if $0 \notin \operatorname{spec}(A)$.

Proof.
(⇒) If A is invertible, then $\operatorname{rank}(A) = n$ by Corollary 3.6.2, so $\dim(\ker(A)) = 0$ by Corollary 3.3.1(i). Since $\dim(\{\mathbf{0}\}) = 0$ by Definition 1.4.2, Proposition 1.4.3(ii) implies $\ker(A) = \{\mathbf{0}\}$. Hence $0 \notin \operatorname{spec}(A)$ by Definitions 5.1.2 and 5.1.3.
(⇐) If $0 \notin \operatorname{spec}(A)$, then $\ker(A) = \{\mathbf{0}\}$ by Definitions 5.1.2 and 5.1.3, so $\dim(\ker(A)) = 0$ by Definition 1.4.2. Thus $\operatorname{rank}(A) = n$ by Corollary 3.3.1(i), and A is invertible by Corollary 3.6.2.
Structural Commentary: Matrix invertibility is the coordinate form of trivial kernel.
Key Insight: A zero eigenvalue is exactly a nonzero solution of $A\mathbf{x} = \mathbf{0}$. □

5.2 Invariant Subspaces, Reducing Pairs, and Quotient Operators

Throughout this subsection, $\mathcal{V}$ is a vector space over $\mathbb{F}$. In statements involving inner products, adjoints, or unitary matrices, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 5.2.1. *Invariant Subspace*
Let $T \in \mathcal{L}(\mathcal{V})$. A subspace $\mathcal{U}$ of $\mathcal{V}$ is *invariant under T* iff $T\mathbf{u} \in \mathcal{U}$ for every $\mathbf{u} \in \mathcal{U}$.
The set of all T-invariant subspaces is $\operatorname{Inv}(T) := \{\mathcal{U} \leq \mathcal{V} : T\mathcal{U} \subseteq \mathcal{U}\}$.
Key Insight: Invariance means that T sends every vector of $\mathcal{U}$ back into $\mathcal{U}$.

Definition 5.2.2. *Restriction and Quotient Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T.
(i) The *restriction operator* $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is defined by $T|_{\mathcal{U}}(\mathbf{u}) = T\mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$.
(ii) The *quotient operator* $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ is defined by $(T/\mathcal{U})(\mathbf{v} + \mathcal{U}) = T\mathbf{v} + \mathcal{U}$, $\mathbf{v} \in \mathcal{V}$.

Proposition 5.2.3. *Restriction and Quotient Operators Are Well-Defined*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T. Then $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ and $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$.

Proof. Since $\mathcal{U}$ is invariant under T, $T|_{\mathcal{U}}$ maps $\mathcal{U}$ into itself, and its linearity follows from the linearity of T. For the quotient operator, suppose $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$. Then $\mathbf{v} - \mathbf{w} \in \mathcal{U}$ by Proposition 2.2.2.
Therefore $T(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$ by invariance, so $T\mathbf{v} + \mathcal{U} = T\mathbf{w} + \mathcal{U}$ by Proposition 2.2.2. Thus $T/\mathcal{U}$ is well-defined.
For $a, b \in \mathbb{F}$ and $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $(T/\mathcal{U})(a(\mathbf{v} + \mathcal{U}) + b(\mathbf{w} + \mathcal{U})) = T(a\mathbf{v} + b\mathbf{w}) + \mathcal{U} = a(T\mathbf{v} + \mathcal{U}) + b(T\mathbf{w} + \mathcal{U})$ by Proposition 2.2.6(ii) and the linearity of T. Hence $T/\mathcal{U}$ is linear.
Structural Commentary: Restriction stays inside $\mathcal{U}$; quotient collapses $\mathcal{U}$.
Key Insight: Invariance makes both operators well-defined. □

Proposition 5.2.4. *One-Dimensional Invariant Subspaces and Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$. Then $\operatorname{span}(\mathbf{v})$ is invariant under T if and only if $\mathbf{v}$ is an eigenvector of T.

Proof.
(⇒) If $\operatorname{span}(\mathbf{v})$ is invariant under T, then $T\mathbf{v} \in \operatorname{span}(\mathbf{v})$, so $(\exists\, \lambda \in \mathbb{F})\, T\mathbf{v} = \lambda\mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, it is an eigenvector by Definition 5.1.1.
(⇐) If $\mathbf{v}$ is an eigenvector, choose $\lambda \in \mathbb{F}$ satisfying $T\mathbf{v} = \lambda\mathbf{v}$ by Definition 5.1.1. For $a \in \mathbb{F}$, $T(a\mathbf{v}) = a\lambda\mathbf{v} \in \operatorname{span}(\mathbf{v})$, so $\operatorname{span}(\mathbf{v})$ is invariant.
Structural Commentary: One-dimensional invariance means rescaling one direction.
Key Insight: In dimension one, staying inside the subspace means being a scalar multiple of the same vector. □

Figure 40 illustrates Proposition 5.2.4: T rescales the nonzero vector $\mathbf{v}$ without leaving $\operatorname{span}(\mathbf{v})$. The proposition, rather than the diagram, owns the equivalence.

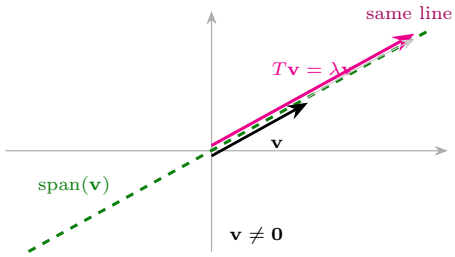

Figure 40: An eigendirection is a one-dimensional invariant subspace.

Proposition 5.2.5. *Invariant Subspaces and Block Form*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

(i) If $\mathcal{U}$ is invariant under T, then in a basis adapted to $\mathcal{U} \oplus \mathcal{W}$, the matrix of T has upper block-triangular form

$$\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}.$$

(ii) If both $\mathcal{U}$ and $\mathcal{W}$ are invariant under T, then the matrix of T has the block-diagonal form

$$\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}.$$

Proof. Choose bases $\beta_{\mathcal{U}} = (\mathbf{u}_1, \dots, \mathbf{u}_p)$ of $\mathcal{U}$ and $\beta_{\mathcal{W}} = (\mathbf{w}_1, \dots, \mathbf{w}_q)$ of $\mathcal{W}$ by Corollary 1.3.14. Their concatenation is an adapted basis of $\mathcal{V}$ by Theorem 2.3.4(ii).

(i) If $\mathcal{U}$ is invariant, then each $T\mathbf{u}_j$ lies in $\mathcal{U}$, so its coordinates along $\mathbf{w}_1, \dots, \mathbf{w}_q$ are zero. These entries form the lower-left block.
(ii) If $\mathcal{W}$ is also invariant, then each $T\mathbf{w}_j$ lies in $\mathcal{W}$, so its coordinates along $\mathbf{u}_1, \dots, \mathbf{u}_p$ are zero. These entries form the upper-right block.

Structural Commentary: Block form records how an operator moves vectors between chosen summands. Invariant summands force the corresponding off-diagonal movement to vanish.
Key Insight: Zero blocks are coordinate evidence of invariant subspaces. □

Definition 5.2.6. *Reducing Pair*
Let $T \in \mathcal{L}(\mathcal{V})$. A pair of subspaces $(\mathcal{U}, \mathcal{W})$ *reduces* T if $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$; $T(\mathcal{U}) \subseteq \mathcal{U}$; and $T(\mathcal{W}) \subseteq \mathcal{W}$.
Equivalently, both summands are invariant under T, and every $\mathbf{v} = \mathbf{u}+\mathbf{w}$ in the direct-sum decomposition satisfies $T\mathbf{v} = T|_{\mathcal{U}}\mathbf{u}+T|_{\mathcal{W}}\mathbf{w}$.

Theorem 5.2.7. *Adjoints and Invariant Orthogonal Complements*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space and let $\mathcal{U} \leq \mathcal{V}$.

(i) $\mathcal{U} \in \mathrm{Inv}(T) \iff \mathcal{U}^\perp \in \mathrm{Inv}(T^*)$.
(ii) $(\mathcal{U},\mathcal{U}^\perp)$ reduces $T \iff \mathcal{U} \in \mathrm{Inv}(T) \cap \mathrm{Inv}(T^*)$.
(iii) If $(\mathcal{U},\mathcal{U}^\perp)$ reduces T, then $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$.

Proof.

(i) ($\Rightarrow$) Suppose $\mathcal{U} \in \mathrm{Inv}(T)$. If $\mathbf{z} \in \mathcal{U}^\perp$ and $\mathbf{u} \in \mathcal{U}$, then $\langle T^*\mathbf{z}, \mathbf{u}\rangle = \overline{\langle \mathbf{u}, T^*\mathbf{z}\rangle} = \overline{\langle T\mathbf{u}, \mathbf{z}\rangle} = \langle \mathbf{z}, T\mathbf{u}\rangle = 0$ by conjugate symmetry, Definition 2.3.1, and invariance. Hence $T^*\mathbf{z} \in \mathcal{U}^\perp$, so $\mathcal{U}^\perp \in \mathrm{Inv}(T^*)$.
(i) ($\Leftarrow$) Suppose $\mathcal{U}^\perp \in \mathrm{Inv}(T^*)$. If $\mathbf{u} \in \mathcal{U}$ and $\mathbf{z} \in \mathcal{U}^\perp$, then $\langle T\mathbf{u}, \mathbf{z}\rangle = \langle \mathbf{u}, T^*\mathbf{z}\rangle = 0$ by Definition 2.3.1. Thus $T\mathbf{u} \in (\mathcal{U}^\perp)^\perp = \mathcal{U}$ by Corollary 2.1.8, and $\mathcal{U} \in \mathrm{Inv}(T)$.
(ii) ($\Rightarrow$) Suppose $(\mathcal{U},\mathcal{U}^\perp)$ reduces T. Then $\mathcal{U} \in \mathrm{Inv}(T)$ and $\mathcal{U}^\perp \in \mathrm{Inv}(T)$ by Definition 5.2.6. Thus $\mathcal{U} \in \mathrm{Inv}(T^*)$ by (i) for T^* and Proposition 2.3.8(iii).
(ii) ($\Leftarrow$) Suppose $\mathcal{U} \in \mathrm{Inv}(T) \cap \mathrm{Inv}(T^*)$. Then $\mathcal{U}^\perp \in \mathrm{Inv}(T)$ by (i) for T^* and Proposition 2.3.8(iii). Moreover, $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$ by Theorem 2.1.5, so $(\mathcal{U},\mathcal{U}^\perp)$ reduces T by Definition 5.2.6.
(iii) By (ii), $\mathcal{U}$ is invariant under T and T^*, so both restrictions are well-defined by Proposition 5.2.3. For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, $\langle T|_{\mathcal{U}}\mathbf{u}, \mathbf{v}\rangle_{\mathcal{U}} = \langle T\mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}} = \langle \mathbf{u}, T^*\mathbf{v}\rangle_{\mathcal{V}} = \langle \mathbf{u}, (T^*)|_{\mathcal{U}}\mathbf{v}\rangle_{\mathcal{U}}$ by Definition 2.3.1. Therefore $(T^*)|_{\mathcal{U}}$ is the adjoint of $T|_{\mathcal{U}}$, and $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$ by Lemma 2.3.3.

Structural Commentary: Adjoints trade invariance with orthogonal-complement invariance.
Key Insight: A subspace reduces T precisely when it is stable under both T and T^*. □

Proposition 5.2.8. *Reducing Pairs and Projections*
Let $T \in \mathcal{L}(\mathcal{V})$, let $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and let $\rho = \rho_{\mathcal{U},\mathcal{W}}$. Then $(\mathcal{U}, \mathcal{W})$ reduces $T \iff T\rho = \rho T$.

Proof.

($\Rightarrow$) Let $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.
Since both summands are invariant, $T\rho\mathbf{v} = T\mathbf{u}$ and $\rho T\mathbf{v} = \rho(T\mathbf{u} + T\mathbf{w}) = T\mathbf{u}$. Thus $T\rho = \rho T$.
($\Leftarrow$) Suppose $T\rho = \rho T$. If $\mathbf{u} \in \mathcal{U}$, then $\rho(T\mathbf{u}) = T(\rho\mathbf{u}) = T\mathbf{u}$, so $T\mathbf{u} \in \mathrm{range}(\rho) = \mathcal{U}$ by Proposition 2.4.2(iii).
If $\mathbf{w} \in \mathcal{W}$, then $\rho(T\mathbf{w}) = T(\rho\mathbf{w}) = \mathbf{0}$, so $T\mathbf{w} \in \ker(\rho) = \mathcal{W}$ by Proposition 2.4.2(iii). Hence $(\mathcal{U}, \mathcal{W})$ reduces T by Definition 5.2.6.

Structural Commentary: Commutation with ρ means T preserves the two projection components.
Key Insight: Reducing pairs are exactly decompositions whose projection commutes with T. □

Proposition 5.2.9. *Invariant Subspaces and Algebraic Projections*
Let $T \in \mathcal{L}(\mathcal{V})$, let $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and let $\rho = \rho_{\mathcal{U},\mathcal{W}}$. Then $\mathcal{U} \in \mathrm{Inv}(T) \iff \rho T\rho = T\rho$.
Consequently, if $\mathcal{U} \in \mathrm{Inv}(T)$, then $\rho_{\mathcal{U},\mathcal{Z}} T \rho_{\mathcal{U},\mathcal{Z}} = T\rho_{\mathcal{U},\mathcal{Z}}$ for every complement $\mathcal{Z}$ of $\mathcal{U}$.

Proof.

($\Rightarrow$) Suppose $\mathcal{U} \in \mathrm{Inv}(T)$, and write $\mathbf{v} = \mathbf{u}+\mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$. Then $T\mathbf{u} \in \mathcal{U}$, so $(\rho T\rho)\mathbf{v} = \rho T\mathbf{u} = T\mathbf{u} = (T\rho)\mathbf{v}$ by Definition 2.4.1. Hence $\rho T\rho = T\rho$.
($\Leftarrow$) Suppose $\rho T\rho = T\rho$. If $\mathbf{u} \in \mathcal{U}$, then $\rho T\mathbf{u} = \rho T\rho\mathbf{u} = T\rho\mathbf{u} = T\mathbf{u}$, so $T\mathbf{u} \in \mathrm{range}(\rho) = \mathcal{U}$ by Proposition 2.4.2(iii). Hence $\mathcal{U} \in \mathrm{Inv}(T)$.

The consequent is the forward implication for the decomposition $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$.
Structural Commentary: A projection onto $\mathcal{U}$ tests whether the $\mathcal{W}$-component of $T\mathbf{u}$ vanishes.
Key Insight: Invariance says applying T after projecting into $\mathcal{U}$ needs no further correction by ρ. □

Perspective 5.2.10. *Matrix Viewpoint on Structure*
Basic block matrices already capture important geometry. Upper block-triangular form records the presence of an invariant subspace; block diagonal form records a genuine decomposition into independent invariant pieces.

Theorem 5.2.11. *Invariant Subspaces, Blocks, and Projections*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, let $\mathcal{U}$ be an r-dimensional subspace of $\mathbb{F}^n$ with $1 \leq r \leq n-1$, and let $Q = [Q_1\ Q_2]$ be unitary, where the columns of Q_1 and Q_2 are orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$, respectively. Let $P = Q_1Q_1^*$ be the orthogonal projection onto $\mathcal{U}$. The following are equivalent:

(i) $\mathcal{U}$ is invariant under A;

(ii) $Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$ *for some compatible matrices* B, X, C;
(iii) $PAP = AP$.

Proof. Since Q is unitary, $Q_1^*Q_1 = I_r$ and $Q_2^*Q_1 = 0$. Thus $P^* = P$, $P^2 = P$, and $\operatorname{range}(P) = \operatorname{col}(Q_1) = \mathcal{U}$, so P is the orthogonal projection matrix onto $\mathcal{U}$ by Definition 2.5.5. Set $M := Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E := Q^*PQ = I_r \oplus 0_{n-r}$. Also $Y = Q_2^*AQ_1$ and $\ker(Q_2^*) = (\operatorname{col}(Q_2))^\perp = (\mathcal{U}^\perp)^\perp = \mathcal{U}$ by Proposition 2.3.10(i) and Corollary 2.1.8.

(i) $\Rightarrow$ *(ii)* If $A\mathcal{U} \subseteq \mathcal{U}$, then each column of AQ_1 lies in $\mathcal{U} = \ker(Q_2^*)$, so $Y = Q_2^*AQ_1 = 0$.

(ii) $\Rightarrow$ *(iii)* If $Y = 0$, then $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$.
Thus $Q^*(PAP)Q = EME = ME = Q^*(AP)Q$, so $PAP = AP$.

(iii) $\Rightarrow$ *(i)* If $PAP = AP$, then $EME = ME$. Since $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $Y = 0$.
For $\mathbf{u} \in \mathcal{U}$, choose $\mathbf{c} \in \mathbb{F}^r$ satisfying $\mathbf{u} = Q_1\mathbf{c}$. Then $Q_2^*A\mathbf{u} = Y\mathbf{c} = \mathbf{0}$, so $A\mathbf{u} \in \ker(Q_2^*) = \mathcal{U}$. Hence $A\mathcal{U} \subseteq \mathcal{U}$.

Structural Commentary: In adapted coordinates, the lower-left block is the component of $A\mathcal{U}$ in $\mathcal{U}^\perp$.
Key Insight: $PAP = AP$ exactly kills this component. □

Corollary 5.2.12. *Reducing Subspaces and Commuting Projections*
Maintain the notation of Theorem 5.2.11. The following are equivalent:
(i) $\mathcal{U}$ *is invariant under both* A *and* A^*;
(ii) $Q^*AQ = B \oplus C$ *for some* $B \in \operatorname{Mat}(r, r, \mathbb{F})$ *and* $C \in \operatorname{Mat}(n-r, n-r, \mathbb{F})$;
(iii) $PA = AP$.

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E = Q^*PQ = I_r \oplus 0_{n-r}$.
Then $Q^*A^*Q = M^*$, $Q^*(PA)Q = EM$, and $Q^*(AP)Q = ME$.

(i) $\Rightarrow$ *(ii)* The equality $Y = 0$ holds by Theorem 5.2.11(i),(ii) for A. The lower-left block X^* of M^* is zero by the same theorem for A^*. Hence $X = Y = 0$ and $M = B \oplus C$.

(ii) $\Rightarrow$ *(iii)* If $M = B \oplus C$, then $EM = ME$, hence $PA = AP$.

(iii) $\Rightarrow$ *(i)* If $PA = AP$, then $EM = ME$. Since $EM = \begin{bmatrix} B & X \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $X = Y = 0$.
The equality $Y = 0$ establishes A-invariance by Theorem 5.2.11(i),(ii). Since the lower-left block of M^* is $X^* = 0$, A^*-invariance holds by the same theorem for A^*.

Structural Commentary: Reducing subspaces are exactly block diagonal decompositions in adapted coordinates.
Key Insight: Commutation with P kills both off-diagonal blocks. □

Figure 41 summarizes the preceding formal results: invariance removes the lower-left block, while projection commutation removes both off-diagonal blocks. Proposition 5.2.5, Theorem 5.2.11, and Corollary 5.2.12 own these claims.

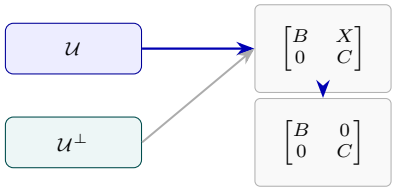

Figure 41: Adapted coordinates: invariance gives upper block form; $PA = AP$ gives block diagonal form.

Perspective 5.2.13. *Two Smaller Views of an Operator*
When $\mathcal{U}$ is invariant under T, the operators $T|_{\mathcal{U}}$ and $T/\mathcal{U}$ split attention between what happens inside $\mathcal{U}$ and what remains after $\mathcal{U}$ is collapsed. These two views often simplify an operator, even though they need not determine the whole operator by themselves.

5.3 Triangularization over Complex and Real Fields

Definition 5.3.1. *Diagonal and Upper-Triangular Matrices*
Let $n \geq 1$ and let $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$. The *diagonal* of A is $(a_{11}, \ldots, a_{nn})$. The matrix A is *diagonal* iff $(\forall\, i, j \in \{1, \ldots, n\})(i \neq j \Rightarrow a_{ij} = 0)$, and *upper triangular* iff $(\forall\, i, j \in \{1, \ldots, n\})(i > j \Rightarrow a_{ij} = 0)$.

Theorem 5.3.2. *Complex Operators Have Eigenvalues*
Let $\mathcal{V}$ *be a nonzero finite-dimensional complex vector space. Every* $T \in \mathcal{L}(\mathcal{V})$ *has an eigenvalue.*

Proof. Fix $T \in \mathcal{L}(\mathcal{V})$. A nonconstant annihilating polynomial for T exists by Proposition 1.1.7, and one of its roots is an eigenvalue of T by Theorem 1.1.9(ii).
Structural Commentary: Finite dimension supplies an annihilating polynomial; over $\mathbb{C}$, it factors linearly.
Key Insight: One noninjective factor produces an eigenvector. □

Theorem 5.3.3. *Criteria for an Upper-Triangular Operator Matrix*
Let $n \geq 1$, *let* $T \in \mathcal{L}(\mathcal{V})$, *and let* $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ *be a basis of* $\mathcal{V}$. *The following are equivalent:*
(i) $[T]_{\beta \leftarrow \beta}$ *is upper triangular;*
(ii) $(\forall\, j \in \{1, \ldots, n\})\, T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$;
(iii) $(\forall\, j \in \{1, \ldots, n\})\, \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j) \in \operatorname{Inv}(T)$.

Proof.

(*i*) $\Rightarrow$ (*ii*) The jth column of $[T]_{\beta\leftarrow\beta}$ is the coordinate column of $T\mathbf{v}_j$ in β. Upper triangularity makes every entry below position j zero, so $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$ for $j \in \{1,\dots,n\}$.

(*ii*) $\Rightarrow$ (*iii*) Fix $j \in \{1,\dots,n\}$ and write $\mathbf{u} = \sum_{k=1}^{j} a_k\mathbf{v}_k$. Then $T\mathbf{u} = \sum_{k=1}^{j} a_k T\mathbf{v}_k \in \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$ by (ii), so this span is invariant under T.

(*iii*) $\Rightarrow$ (*i*) For $j \in \{1,\dots,n\}$, invariance of $\operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$ implies $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$. Hence the jth column of $[T]_{\beta\leftarrow\beta}$ has zero entries below position j, and the matrix is upper triangular.

Structural Commentary: Upper triangularity records an invariant flag.
Key Insight: Columns below the diagonal vanish exactly when $T\mathbf{v}_j$ uses only earlier basis vectors and itself. □

Corollary 5.3.4. *Upper-Triangular Form in an Orthonormal Basis*
Let $\mathcal{V}$ be a nonzero finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. If T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$, then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a basis for which $[T]_{\beta\leftarrow\beta}$ is upper triangular. By Theorem 5.3.3, each subspace

$$\mathcal{U}_j = \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j), \qquad j = 1,\dots,n,$$

is invariant under T. Let $\gamma = (\mathbf{e}_1,\dots,\mathbf{e}_n)$ be the Gram–Schmidt output for β. This is an orthonormal basis and satisfies

$$\operatorname{span}(\mathbf{e}_1,\dots,\mathbf{e}_j) = \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j) = \mathcal{U}_j,$$

by Theorem 1.3.6(i)–(ii). Thus each $\operatorname{span}(\mathbf{e}_1,\dots,\mathbf{e}_j)$ is invariant under T, and $[T]_{\gamma\leftarrow\gamma}$ is upper triangular by Theorem 5.3.3(iii),(i).
Structural Commentary: Upper triangularity is preservation of a nested flag of subspaces. Gram–Schmidt preserves that flag while making the basis orthonormal.
Key Insight: Triangular form depends on the flag, not on the particular basis used inside the flag. □

Theorem 5.3.5. *Complex Upper Triangularisation*
Let $\mathcal{V}$ be a nonzero finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$.

Proof. Induct on $n = \dim(\mathcal{V})$. If $n = 1$, every operator matrix is upper triangular.
Suppose $n > 1$ and the result holds in smaller dimensions. Choose an eigenvector $\mathbf{v}_1$ of T by Theorem 5.3.2, and set $\mathcal{U} := \operatorname{span}(\mathbf{v}_1)$. The subspace $\mathcal{U}$ is invariant by Proposition 5.2.4; hence $T/\mathcal{U}$ is well-defined and $\dim(\mathcal{V}/\mathcal{U}) = n-1$ by Proposition 5.2.3 and Theorem 2.2.9. By induction, choose a basis $(\mathbf{v}_2+\mathcal{U},\dots,\mathbf{v}_n+\mathcal{U})$ of $\mathcal{V}/\mathcal{U}$ with respect to which $T/\mathcal{U}$ is upper triangular, and choose representatives $\mathbf{v}_2,\dots,\mathbf{v}_n \in \mathcal{V}$.
The lifted list $(\mathbf{v}_1,\dots,\mathbf{v}_n)$ is independent: a relation descends to the quotient basis, so the coefficients of $\mathbf{v}_2,\dots,\mathbf{v}_n$ vanish, and then the coefficient of $\mathbf{v}_1$ vanishes. Its length is n, so it is a basis by Proposition 1.4.4. Also $T\mathbf{v}_1 \in \operatorname{span}(\mathbf{v}_1)$.
For $j \in \{2,\dots,n\}$, $(T/\mathcal{U})(\mathbf{v}_j+\mathcal{U}) \in \operatorname{span}(\mathbf{v}_2+\mathcal{U},\dots,\mathbf{v}_j+\mathcal{U})$ by Theorem 5.3.3(i),(ii) for $T/\mathcal{U}$. By Definition 5.2.2 and Proposition 2.2.2, $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_j)$. Therefore $[T]_{\beta\leftarrow\beta}$ is upper triangular for $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ by Theorem 5.3.3(ii),(i).
Structural Commentary: Lift an upper-triangular quotient basis above an eigenline.
Key Insight: The quotient builds the remaining invariant flag. □

Definition 5.3.6. *Complexification of a Real Vector Space*
Let $\mathcal{V}$ be a real vector space.

(i) The *complexification* of $\mathcal{V}$ is $\mathcal{V}_{\mathbb{C}} := \mathcal{V}\times\mathcal{V}$, with $(\mathbf{u},\mathbf{v})$ denoted $\mathbf{u}+i\mathbf{v}$.

(ii) Addition and $\mathbb{C}$-scalar multiplication are

$$(\mathbf{u}_1+i\mathbf{v}_1)+(\mathbf{u}_2+i\mathbf{v}_2) = (\mathbf{u}_1+\mathbf{u}_2)+i(\mathbf{v}_1+\mathbf{v}_2),$$
$$(a+bi)(\mathbf{u}+i\mathbf{v}) = (a\mathbf{u}-b\mathbf{v})+i(b\mathbf{u}+a\mathbf{v}), \qquad a,b\in\mathbb{R}.$$

(iii) The *complexification map* is $\operatorname{cpx}: \mathcal{V}\to\mathcal{V}_{\mathbb{C}}$ defined by $\operatorname{cpx}(\mathbf{u}) := \mathbf{u}+i\mathbf{0}$; its image is $\{\mathbf{u}+i\mathbf{0} : \mathbf{u}\in\mathcal{V}\}$.

(iv) For $\mathbf{z} = \mathbf{u}+i\mathbf{v} \in \mathcal{V}_{\mathbb{C}}$, define $\operatorname{Re}\mathbf{z} := \mathbf{u}$ and $\operatorname{Im}\mathbf{z} := \mathbf{v}$. Hence $\mathbf{z} = \mathbf{0}$ iff $\operatorname{Re}\mathbf{z} = \operatorname{Im}\mathbf{z} = \mathbf{0}$.

Proposition 5.3.7. *Complexification Preserves Bases*
Let $\mathcal{V}$ be a finite-dimensional real vector space, let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a real basis of $\mathcal{V}$, and set $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$. Then $\mathcal{V}_{\mathbb{C}}$ is a complex vector space and $\beta_{\mathbb{C}}$ is a complex basis of $\mathcal{V}_{\mathbb{C}}$.
Thus $\dim_{\mathbb{C}}(\mathcal{V}_{\mathbb{C}}) = \dim_{\mathbb{R}}(\mathcal{V})$.

Proof. The vector-space axioms follow componentwise from those of $\mathcal{V}$.
For $\mathbf{u},\mathbf{v}\in\mathcal{V}$, write $\mathbf{u} = \sum_{j=1}^{n} a_j\mathbf{v}_j$ and $\mathbf{v} = \sum_{j=1}^{n} b_j\mathbf{v}_j$, $a_j,b_j\in\mathbb{R}$. Then $\mathbf{u}+i\mathbf{v} = \sum_{j=1}^{n}(a_j+ib_j)\operatorname{cpx}(\mathbf{v}_j)$, so $\beta_{\mathbb{C}}$ spans.
If $\sum_{j=1}^{n}\lambda_j\operatorname{cpx}(\mathbf{v}_j) = \mathbf{0}$ and $\lambda_j = a_j+ib_j$, then $\sum_{j=1}^{n} a_j\mathbf{v}_j = \mathbf{0}$ and $\sum_{j=1}^{n} b_j\mathbf{v}_j = \mathbf{0}$ by Definition 5.3.6(ii)–(iv).
Real linear independence implies $a_j = b_j = 0$ for all j, hence $\lambda_j = 0$ for all j.
Structural Commentary: Complexification keeps the basis length unchanged.
Key Insight: Separate real and imaginary parts. □

Corollary 5.3.8. *Coordinates Survive Complexification*
Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be a real basis of $\mathcal{V}$ and set $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$. Then, for all $\mathbf{v}\in\mathcal{V}$, $[\operatorname{cpx}(\mathbf{v})]_{\beta_{\mathbb{C}}} = [\mathbf{v}]_{\beta} \in \mathbb{R}^n \subseteq \mathbb{C}^n$.

Proof. If $[\mathbf{v}]_\beta = (a_1,\dots,a_n)^t$, then $a_j\in\mathbb{R}$ and $\operatorname{cpx}(\mathbf{v}) = \mathbf{v}+i\mathbf{0} = \sum_{j=1}^{n} a_j(\mathbf{v}_j+i\mathbf{0}) = \sum_{j=1}^{n} a_j\operatorname{cpx}(\mathbf{v}_j)$.
Thus $[\operatorname{cpx}(\mathbf{v})]_{\beta_{\mathbb{C}}} = (a_1,\dots,a_n)^t = [\mathbf{v}]_\beta$.
Structural Commentary: The embedded real copy keeps exactly the same coordinate columns.
Key Insight: Complexification enlarges the scalar field, not the coordinates of real vectors. □

Proposition 5.3.9. *Complexification as a Real Space*
If $\mathcal{V}$ has real basis $(\mathbf{v}_1,\dots,\mathbf{v}_n)$, then $\mathcal{V}_{\mathbb{C}}$, as a real vector space, has basis $(\operatorname{cpx}(\mathbf{v}_1),\dots,\operatorname{cpx}(\mathbf{v}_n), i\operatorname{cpx}(\mathbf{v}_1),\dots,i\operatorname{cpx}(\mathbf{v}_n))$.
Hence $\dim_{\mathbb{R}}(\mathcal{V}_{\mathbb{C}}) = 2\dim_{\mathbb{R}}(\mathcal{V})$, and $\operatorname{cpx}:\mathcal{V}\to\mathcal{V}_{\mathbb{C}}$, $\operatorname{cpx}(\mathbf{v}) = \mathbf{v}+i\mathbf{0}$, is real-linear and injective.

Proof. For $\mathbf{u} = \sum a_j \mathbf{v}_j$ and $\mathbf{w} = \sum b_j \mathbf{v}_j$, $\mathbf{u} + i\mathbf{w} = \sum a_j \operatorname{cpx}(\mathbf{v}_j) + \sum b_j (i \operatorname{cpx}(\mathbf{v}_j))$, so the displayed list spans $\mathcal{V}_{\mathbb{C}}$ over $\mathbb{R}$.
If $\sum a_j \operatorname{cpx}(\mathbf{v}_j) + \sum b_j (i \operatorname{cpx}(\mathbf{v}_j)) = \mathbf{0}$, then $\sum a_j \mathbf{v}_j = \sum b_j \mathbf{v}_j = \mathbf{0}$ by Definition 5.3.6(iv). Thus all a_j, b_j vanish by independence of $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, so the displayed list is a real basis.
For $r \in \mathbb{R}$, $\operatorname{cpx}(r\mathbf{v} + \mathbf{w}) = r \operatorname{cpx}(\mathbf{v}) + \operatorname{cpx}(\mathbf{w})$. Moreover, $\operatorname{cpx}(\mathbf{v}) = \mathbf{0}$ iff $\mathbf{v} = \mathbf{0}$ by Definition 5.3.6(iii)–(iv), so $\ker(\operatorname{cpx}) = \{\mathbf{0}\}$ and cpx is real-linear and injective.
Structural Commentary: Complexification is two real copies of $\mathcal{V}$.
Key Insight: The real and imaginary parts are independent real copies of the same basis. □

Definition 5.3.10. *Complexification of a Real Operator*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
The *complexification* of T is the map $T_{\mathbb{C}} : \mathcal{V}_{\mathbb{C}} \to \mathcal{V}_{\mathbb{C}}$ defined by $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) := T\mathbf{u} + iT\mathbf{v}$.

Proposition 5.3.11. *Complexification Preserves Operator Data*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.

(i) $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$.
(ii) $(T_{\mathbb{C}})^k(\mathbf{u} + i\mathbf{v}) = T^k\mathbf{u} + iT^k\mathbf{v}$ *for every integer* $k \geq 0$.
(iii) If β is a real basis of $\mathcal{V}$ and $\beta_{\mathbb{C}} := \operatorname{cpx}(\beta)$, then $[T_{\mathbb{C}}]_{\beta_{\mathbb{C}} \leftarrow \beta_{\mathbb{C}}} = [T]_{\beta \leftarrow \beta}$, now regarded as a complex matrix.

Proof.

(i) Additivity is immediate. For $\alpha = a + bi \in \mathbb{C}$,
$$T_{\mathbb{C}}\big(\alpha(\mathbf{u} + i\mathbf{v})\big) = T(a\mathbf{u} - b\mathbf{v}) + iT(b\mathbf{u} + a\mathbf{v}) = (aT\mathbf{u} - bT\mathbf{v}) + i(bT\mathbf{u} + aT\mathbf{v}) = \alpha T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}),$$
so $T_{\mathbb{C}}$ is complex-linear.

(ii) For $k = 0$, the claim is the identity. If it holds for k, then
$$(T_{\mathbb{C}})^{k+1}(\mathbf{u} + i\mathbf{v}) = T_{\mathbb{C}}(T^k\mathbf{u} + iT^k\mathbf{v}) = T^{k+1}\mathbf{u} + iT^{k+1}\mathbf{v}.$$
Thus the formula holds for all $k \geq 0$.

(iii) The list $\beta_{\mathbb{C}}$ is a complex basis by Proposition 5.3.7. If $T\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$ with $a_{ij} \in \mathbb{R}$, then $T_{\mathbb{C}} \operatorname{cpx}(\mathbf{v}_j) = \sum_i a_{ij} \operatorname{cpx}(\mathbf{v}_i)$. Hence the coordinate matrix is unchanged.

Structural Commentary: Complexification keeps the coordinate matrix unchanged.
Key Insight: Same real entries; larger scalar field. □

Figure 42 illustrates the defining action of $T_{\mathbb{C}}$ on the two real components established in Proposition 5.3.11. The proposition, rather than the diagram, owns complex linearity and coordinate compatibility.

Figure 42: Complexification packages two real vectors: $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = T\mathbf{u} + iT\mathbf{v}$.

Proposition 5.3.12. *Real Eigenvalues of a Complexification*
Let $\mathcal{V}$ be a finite-dimensional real vector space, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{R}$. Then λ is an eigenvalue of T if and only if it is an eigenvalue of $T_{\mathbb{C}}$.

Proof.

($\Rightarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\operatorname{cpx}(\mathbf{v}) \neq \mathbf{0}$ by Proposition 5.3.9, and $T_{\mathbb{C}} \operatorname{cpx}(\mathbf{v}) = \lambda \operatorname{cpx}(\mathbf{v})$ by Definition 5.3.10. Thus λ is an eigenvalue of $T_{\mathbb{C}}$ by Definition 5.1.1.

($\Leftarrow$) If $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = \lambda(\mathbf{u} + i\mathbf{v})$ and $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$, then $T\mathbf{u} = \lambda\mathbf{u}$ and $T\mathbf{v} = \lambda\mathbf{v}$ by Definitions 5.3.6(ii),(iv) and 5.3.10. At least one of $\mathbf{u}, \mathbf{v}$ is nonzero by Definition 5.3.6(iv), so λ is an eigenvalue of T by Definition 5.1.1.

Structural Commentary: Real eigenvalues survive complexification exactly.
Key Insight: For real λ, real and imaginary parts decouple. □

Method 5.3.13. *Complexifying a Real Matrix*
To use complex eigenvalues for a real matrix $A \in \operatorname{Mat}(n, n, \mathbb{R})$:

(i) Regard the same real matrix A as a complex matrix on $\mathbb{C}^n$.
(ii) Find complex eigenpairs or invariant data for this complexified operator.
(iii) If $A(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$, separate parts: $A\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $A\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.
(iv) Conclude that $\operatorname{span}(\mathbf{u}, \mathbf{v}) \subseteq \mathbb{R}^n$ is an invariant line or plane.

Theorem 5.3.14. *Invariant Line or Plane over $\mathbb{R}$*
Every operator on a nonzero finite-dimensional real vector space has an invariant subspace of dimension 1 *or* 2.

Proof. Let $\mathcal{V}$ be a nonzero finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
The space $\mathcal{V}_{\mathbb{C}}$ is a nonzero finite-dimensional complex vector space and $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$ by Proposition 5.3.7, Proposition 5.3.9, and Proposition 5.3.11(i). Choose an eigenvalue $a + bi$ of $T_{\mathbb{C}}$ and a corresponding eigenvector $\mathbf{u} + i\mathbf{v}$ by Theorem 5.3.2.
The eigenvector equation and Definitions 5.3.6(ii) and 5.3.10 imply $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.
Thus $\mathcal{U} = \operatorname{span}(\mathbf{u}, \mathbf{v})$ is nonzero, has dimension 1 or 2, and is invariant under T.
Structural Commentary: A complex eigenvector decomposes into two real directions that are closed under T.
Key Insight: Nonreal eigenvalues become real invariant planes; real eigenvalues become invariant lines. □

Figure 43 summarizes Theorem 5.3.14: separating a complex eigenvector equation produces a nonzero real span preserved by T. The theorem owns the invariant-subspace conclusion.

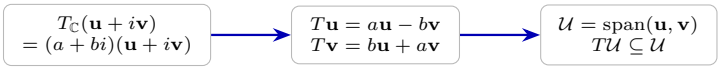

Figure 43: Separate a complex eigenpair into a real invariant plane.

Definition 5.3.15. *Almost Upper-Triangular Real Matrices*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{R})$. The matrix A is *almost upper triangular* if there exist $m \geq 1$ and $d_1, \ldots, d_m \in \{1, 2\}$ with $\sum_{j=1}^{m} d_j = n$ such that, in the corresponding block form $A = [A_{ij}]_{i,j=1}^{m}$,

(i) $A_{ij} = 0$ for $i > j$;

(ii) each diagonal block is either $A_{jj} = [\lambda_j]$, $\lambda_j \in \mathbb{R}$, or $A_{jj} = \begin{bmatrix} a_j & -b_j \\ b_j & a_j \end{bmatrix}$, $a_j, b_j \in \mathbb{R}$, $b_j \neq 0$.

The matrix A is in *real Schur block form* if condition (i) holds for some such block sizes $d_j \in \{1, 2\}$, without the additional restriction in (ii).

Theorem 5.3.16. *Real Almost Upper Triangularisation*
Let $\mathcal{V}$ be a nonzero finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then there exists a basis β of $\mathcal{V}$ such that $[T]_{\beta \leftarrow \beta}$ is almost upper triangular.

Proof. Induct on $n = \dim_{\mathbb{R}} \mathcal{V}$. If $n = 1$, every operator matrix is a real 1×1 diagonal block.
For $n \geq 2$, the complexification $T_{\mathbb{C}}$ is an operator on the nonzero n-dimensional complex vector space $\mathcal{V}_{\mathbb{C}}$ by Proposition 5.3.7 and Proposition 5.3.11(i). Choose $a + bi \in \mathbb{C}$ and $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$ satisfying $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$ by Theorem 5.3.2.
Thus $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.
If $b = 0$, choose $\mathbf{0} \neq \mathbf{x} \in \{\mathbf{u}, \mathbf{v}\}$, set $\mathcal{U} = \mathrm{span}(\mathbf{x})$, $\alpha = (\mathbf{x})$, so $[T|_{\mathcal{U}}]_{\alpha \leftarrow \alpha} = [a]$.
If $b \neq 0$, then $\mathbf{u} \neq \mathbf{0}$ because $\mathbf{u} = \mathbf{0}$ would force $\mathbf{v} = \mathbf{0}$ in $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$. If $\mathbf{v} = c\mathbf{u}$ for some $c \in \mathbb{R}$, the two displayed identities imply $c(a - bc) = b + ac$, hence $c^2 = -1$, a contradiction. Thus $\mathbf{u}, \mathbf{v}$ are independent. Set $\mathcal{U} = \mathrm{span}(\mathbf{u}, \mathbf{v})$ and $\alpha = (\mathbf{u}, -\mathbf{v})$; then $[T|_{\mathcal{U}}]_{\alpha \leftarrow \alpha} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.
In either case, $T(\mathcal{U}) \subseteq \mathcal{U}$, $\dim \mathcal{U} \in \{1, 2\}$, and the first diagonal block is admissible.
The quotient operator $T/\mathcal{U}$ is well-defined and $\dim(\mathcal{V}/\mathcal{U}) = n - \dim(\mathcal{U}) < n$ by Proposition 5.2.3 and Theorem 2.2.9. By induction, $T/\mathcal{U}$ has an almost upper-triangular matrix in a quotient basis $\overline{\gamma} = (\mathbf{w}_1 + \mathcal{U}, \ldots, \mathbf{w}_{n-\dim(\mathcal{U})} + \mathcal{U})$.
Let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_{n-\dim(\mathcal{U})})$ and $\beta = (\alpha, \gamma)$. A relation in β descends to the quotient basis, forcing the coefficients of γ to vanish; the remaining coefficients vanish because α is a basis of $\mathcal{U}$. Thus β is independent, and its length is n, so it is a basis by Proposition 1.4.4.
In this basis, $[T]_{\beta \leftarrow \beta}$ is block upper triangular with first diagonal block $[T|_{\mathcal{U}}]_{\alpha \leftarrow \alpha}$ and remaining block $[T/\mathcal{U}]_{\overline{\gamma} \leftarrow \overline{\gamma}}$ by Definition 5.2.2. Hence it is almost upper triangular.
Structural Commentary: Real triangularisation starts with an invariant line or plane.
Key Insight: Complex eigenvectors split into real and imaginary parts; those parts form the first real block. □

Corollary 5.3.17. *Orthonormal Real Schur Block Form*
Let $\mathcal{V}$ be a nonzero finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then there exists an orthonormal basis γ of $\mathcal{V}$ such that $[T]_{\gamma \leftarrow \gamma}$ is in real Schur block form.

Proof. By Theorem 5.3.16, choose $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ with block sizes $d_j \in \{1, 2\}$.
Put $s_k = d_1 + \cdots + d_k$ and $\mathcal{U}_k = \mathrm{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{s_k})$. Then $T(\mathcal{U}_k) \subseteq \mathcal{U}_k$ for $k \in \{1, \ldots, m\}$ by Definition 5.3.15(i).
Let $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the Gram–Schmidt output for β. By Theorem 1.3.6(i)–(ii), this is an orthonormal basis with $\mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{s_k}) = \mathcal{U}_k$ for $k \in \{1, \ldots, m\}$. Thus $[T]_{\gamma \leftarrow \gamma}$ is block upper triangular for the same $d_j \in \{1, 2\}$ and is in real Schur block form by Definition 5.3.15.
Structural Commentary: Gram–Schmidt preserves the invariant block flag.
Key Insight: Orthonormalization changes blocks, not the flag. □

Proposition 5.3.18. *Invertibility from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then T is invertible if and only if $\lambda_j \neq 0$ for every j.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be such a basis.

($\Rightarrow$) Suppose T is invertible and $\lambda_j = 0$ for some $j \in \{1, \ldots, n\}$. Set $\mathcal{U}_j := \mathrm{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$ and $\mathcal{U}_0 := \{\mathbf{0}\}$. For $i < j$, $T\mathbf{v}_i \in \mathcal{U}_{j-1}$ by Theorem 5.3.3(i),(ii), while $T\mathbf{v}_j \in \mathcal{U}_{j-1}$ because its $\mathbf{v}_j$-coefficient is $\lambda_j = 0$. Hence $T(\mathcal{U}_j) \subseteq \mathcal{U}_{j-1}$, and $S : \mathcal{U}_j \to \mathcal{U}_{j-1}$ defined by $S\mathbf{u} := T\mathbf{u}$ is linear. Since $\dim(\mathcal{U}_j) = j > j - 1 = \dim(\mathcal{U}_{j-1})$, there exists a nonzero $\mathbf{u} \in \mathcal{U}_j$ with $T\mathbf{u} = S\mathbf{u} = \mathbf{0}$ by Corollary 1.7.4. This contradicts that T is injective by Theorem 1.7.7(i),(ii). Therefore $\lambda_j \neq 0$ for every $j \in \{1, \ldots, n\}$.

($\Leftarrow$) Suppose $\lambda_j \neq 0$ for every $j \in \{1, \ldots, n\}$. The matrix $M := [T]_{\beta \leftarrow \beta}$ is invertible by Theorem 3.8.5(i), so its columns form a basis of $\mathbb{F}^n$ by Theorem 3.4.1. These columns are $[T\mathbf{v}_1]_\beta, \ldots, [T\mathbf{v}_n]_\beta$ by Definition 3.2.3. The coordinate map $[\cdot]_\beta$ is an isomorphism by Proposition 3.2.2(iii), so $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is a basis of $\mathcal{V}$ by Proposition 1.7.8. Thus T is surjective by Theorem 1.7.1(i), and it is invertible by Theorem 1.7.7(i),(iii).

Structural Commentary: Diagonal entries control each flag layer.
Key Insight: A zero diagonal entry forces a kernel vector. □

Proposition 5.3.19. *Eigenvalues from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then $\mathrm{spec}(T) = \{\lambda_1, \ldots, \lambda_n\}$.

Proof. For $\lambda \in \mathbb{F}$, the matrix of $T - \lambda I_{\mathcal{V}}$ in the same basis is upper triangular with diagonal entries $\lambda_1 - \lambda, \ldots, \lambda_n - \lambda$. By Proposition 5.3.18, $T - \lambda I_{\mathcal{V}}$ is not invertible if and only if $\lambda = \lambda_j$ for some j. By Theorem 5.1.4(i),(iv), this is equivalent to λ being an eigenvalue of T, or equivalently to $\lambda \in \mathrm{spec}(T)$ by Definition 5.1.3.
Structural Commentary: Scalar shifts preserve triangular form.
Key Insight: Eigenvalues occur exactly when one shifted diagonal entry becomes zero. □

Perspective 5.3.20. *Why Triangular Form Matters*
Upper-triangular form is the first useful simplification beyond finding one eigenvector. It preserves enough of the operator to read eigenvalues from the diagonal while still allowing nontrivial coupling above the diagonal.

5.4 Diagonalization, Commutants, and Simultaneous Forms

Definition 5.4.1. *Diagonalizable Operator*
Let $\mathcal{V}$ be finite-dimensional. An operator $T \in \mathcal{L}(\mathcal{V})$ is *diagonalizable* iff T has a diagonal matrix with respect to some basis of $\mathcal{V}$.

Proposition 5.4.2. *Eigenspaces for Distinct Eigenvalues Form a Direct Sum*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues of T, where $m \geq 1$.
Then $E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)$ is a direct sum. Consequently, $\dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T) \leq \dim(\mathcal{V})$.

Proof. Suppose $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. If some $\mathbf{u}_j$ were nonzero, then the nonzero vectors among $\mathbf{u}_1, \dots, \mathbf{u}_m$ would be linearly independent by Theorem 5.1.5, contrary to their displayed relation. Hence every $\mathbf{u}_j = \mathbf{0}$, and the sum is direct by Proposition 2.1.23(ii),(i).
The dimension identity $\dim(E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)) = \sum_{j=1}^{m} \dim E_{\lambda_j}(T)$ holds by Theorem 2.3.4(i). Its left side is at most $\dim(\mathcal{V})$ by Proposition 1.4.3(i).
Structural Commentary: Distinct eigenspaces form independent pieces.
Key Insight: A zero sum across them would contradict eigenvector independence. □

Theorem 5.4.3. *Conditions Equivalent to Diagonalizability*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T. Set $n = \dim(\mathcal{V})$. The following are equivalent:

(i) T is diagonalizable;
(ii) $\mathcal{V}$ has a basis consisting of eigenvectors of T;
(iii) There exist one-dimensional subspaces $\mathcal{U}_1, \dots, \mathcal{U}_n$ of $\mathcal{V}$, each invariant under T, such that $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$;
(iv) $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$;
(v) $\dim(\mathcal{V}) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.

Proof.

(i) $\Rightarrow$ (ii) If T is diagonalizable, choose a basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ for which $[T]_{\beta \leftarrow \beta}$ is diagonal by Definition 5.4.1. For each j, the jth matrix column has the form $a_j \mathbf{e}_j$, so $T\mathbf{v}_j = a_j \mathbf{v}_j$ by Definition 3.2.3. Thus β is an eigenbasis.
(ii) $\Rightarrow$ (iii) Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an eigenbasis, and set $\mathcal{U}_j := \operatorname{span}(\mathbf{v}_j)$. Each $\mathcal{U}_j$ is one-dimensional and invariant under T by Proposition 5.2.4. A zero relation among vectors chosen from the $\mathcal{U}_j$ is a relation in the basis, so $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$ by Proposition 2.1.23(ii),(i).
(iii) $\Rightarrow$ (iv) Choose $\mathbf{0} \neq \mathbf{v}_j \in \mathcal{U}_j$ for each j. Each $\mathbf{v}_j$ is an eigenvector by Proposition 5.2.4, so every $\mathcal{U}_j$ lies in one of $E_{\lambda_1}(T), \dots, E_{\lambda_m}(T)$. Hence these eigenspaces span $\mathcal{V}$, and their sum is direct by Proposition 5.4.2.
(iv) $\Rightarrow$ (v) The dimensions add by Theorem 2.3.4(i).
(v) $\Rightarrow$ (i) For each j, choose a basis β_j of $E_{\lambda_j}(T)$ and let β be their concatenation. The eigenspace sum is direct by Proposition 5.4.2, so β is a basis of that sum by Theorem 2.3.4(ii). In particular, β is independent, and its length is $\dim(\mathcal{V})$ by (v), so it is a basis of $\mathcal{V}$ by Proposition 1.4.4(ii). Every member of β is an eigenvector, and therefore $[T]_{\beta \leftarrow \beta}$ is diagonal by Definition 3.2.3. Thus T is diagonalizable by Definition 5.4.1.

Structural Commentary: Diagonalizability means the eigenspaces exhaust $\mathcal{V}$. Directness is automatic for distinct eigenspaces, so the real test is spanning, or equivalently the dimension count.
Key Insight: A diagonal basis is exactly an eigenbasis. □

Corollary 5.4.4. *Enough Eigenvalues Imply Diagonalizability*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$ has $\dim(\mathcal{V})$ distinct eigenvalues, then T is diagonalizable.

Proof. Let $n = \dim(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_n$ be distinct eigenvalues of T. For each j, choose an eigenvector $\mathbf{v}_j$ corresponding to λ_j. The list $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is linearly independent by Theorem 5.1.5, and its length is n, so it is a basis by Proposition 1.4.4(ii). Thus T is diagonalizable by Theorem 5.4.3(ii),(i).
Structural Commentary: One eigenvector from each distinct eigenvalue already supplies a full-size independent list.
Key Insight: Once the independent eigenvectors reach $\dim(\mathcal{V})$, they must form a basis. □

Perspective 5.4.5. *Diagonalization as Exhaustion by Eigenspaces*
Distinct eigenspaces are already separated by Proposition 5.4.2. Thus diagonalizability is the stronger statement that these separated pieces account for every direction in $\mathcal{V}$.

Theorem 5.4.6. *Algebraic Spectral Resolution*
Let $\mathcal{V}$ be nonzero and finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\Lambda = \{\lambda_1, \dots, \lambda_m\} \subseteq \mathbb{F}$, where $m \geq 1$ and the λ_i are distinct. The following are equivalent:

(i) T is diagonalizable and $\operatorname{spec}(T) = \Lambda$.
(ii) There exist nonzero algebraic projections $\rho_1, \dots, \rho_m \in \mathcal{L}(\mathcal{V})$ such that $\rho_i \rho_j = 0$ for $i \neq j$, $\sum_{i=1}^{m} \rho_i = I_{\mathcal{V}}$, and $T = \sum_{i=1}^{m} \lambda_i \rho_i$.

In this case, for each i, $\operatorname{range}(\rho_i) = E_{\lambda_i}(T)$, $\ker(\rho_i) = \bigoplus_{j \neq i} E_{\lambda_j}(T)$, and

$$\rho_i = \prod_{j \neq i} \frac{T - \lambda_j I_{\mathcal{V}}}{\lambda_i - \lambda_j}.$$

Proof.

(i) $\Rightarrow$ (ii) The decomposition $\mathcal{V} = \bigoplus_{i=1}^{m} E_{\lambda_i}(T)$ holds by Theorem 5.4.3(i),(iv). Set $\rho_i := \rho_{E_{\lambda_i}(T), \bigoplus_{j \neq i} E_{\lambda_j}(T)}$. Each ρ_i is an algebraic projection with range $E_{\lambda_i}(T)$ by Proposition 2.4.2(ii)–(iii), and it is nonzero because $E_{\lambda_i}(T) \neq \{\mathbf{0}\}$. The defining coordinate action of these projections satisfies $\rho_i \rho_j = 0$ for $i \neq j$ and $\sum_{i=1}^{m} \rho_i = I_{\mathcal{V}}$. For $\mathbf{x} = \sum_{i=1}^{m} \mathbf{x}_i$ with $\mathbf{x}_i \in E_{\lambda_i}(T)$, $T\mathbf{x} = \sum_{i=1}^{m} \lambda_i \mathbf{x}_i = \sum_{i=1}^{m} \lambda_i \rho_i \mathbf{x}$, so $T = \sum_{i=1}^{m} \lambda_i \rho_i$.
(ii) $\Rightarrow$ (i) Set $\mathcal{V}_i := \operatorname{range}(\rho_i)$. Every $\mathbf{x} \in \mathcal{V}$ satisfies $\mathbf{x} = \sum_{i=1}^{m} \rho_i \mathbf{x}$, so the $\mathcal{V}_i$ span $\mathcal{V}$. If $\sum_{i=1}^{m} \mathbf{x}_i = \mathbf{0}$ with $\mathbf{x}_i \in \mathcal{V}_i$, applying ρ_k and using idempotence and pairwise annihilation shows $\mathbf{x}_k = \mathbf{0}$ for every k. Thus $\mathcal{V} = \bigoplus_{i=1}^{m} \mathcal{V}_i$ by Proposition 2.1.23(ii),(i). If $\mathbf{x} \in \mathcal{V}_i$, then $\rho_i \mathbf{x} = \mathbf{x}$ by idempotence and $\rho_j \mathbf{x} = \mathbf{0}$ for $j \neq i$ by the hypotheses; hence $T\mathbf{x} = \lambda_i \mathbf{x}$ and $\mathcal{V}_i \subseteq E_{\lambda_i}(T)$. Conversely, if $T\mathbf{x} = \lambda_i \mathbf{x}$ and $\mathbf{x} = \sum_{j=1}^{m} \mathbf{x}_j$ with $\mathbf{x}_j \in \mathcal{V}_j$, then $\sum_{j=1}^{m} (\lambda_j - \lambda_i)\mathbf{x}_j = \mathbf{0}$. Directness and distinctness imply $\mathbf{x}_j = \mathbf{0}$ for $j \neq i$, so $\mathbf{x} \in \mathcal{V}_i$. Therefore $\mathcal{V}_i = E_{\lambda_i}(T)$. Since every $\mathcal{V}_i$ is nonzero, $\operatorname{spec}(T) = \Lambda$, and T is diagonalizable by Theorem 5.4.3(iv),(i).

In either case, every $\mathbf{x} \in \mathcal{V}$ has a unique decomposition $\mathbf{x} = \sum_{j=1}^{m} \mathbf{x}_j$ with $\mathbf{x}_j \in E_{\lambda_j}(T)$ and $\rho_i \mathbf{x} = \mathbf{x}_i$. Therefore $\ker(\rho_i) = \bigoplus_{j \neq i} E_{\lambda_j}(T)$. Define $\ell_i(z) := \prod_{j \neq i}(z - \lambda_j)/(\lambda_i - \lambda_j)$. For $\mathbf{x} \in E_{\lambda_k}(T)$, $\ell_i(T)\mathbf{x} = \ell_i(\lambda_k)\mathbf{x} = \delta_{ik}\mathbf{x} = \rho_i \mathbf{x}$. Since the eigenspaces span $\mathcal{V}$, $\ell_i(T) = \rho_i$, which is the displayed product formula.
Structural Commentary: A diagonalizable operator is a weighted sum of the projections onto its eigenspaces.
Key Insight: Spectral projections are Lagrange polynomials in the operator. □

Lemma 5.4.7. *Diagonalizable Restrictions to Invariant Subspaces*
Let $\mathcal{V}$ be a nonzero finite-dimensional vector space, let $T \in \mathcal{L}(\mathcal{V})$ be diagonalizable, and let $\mathcal{U} \in \mathrm{Inv}(T)$. For $\lambda \in \mathrm{spec}(T)$, set $\mathcal{U}_\lambda := \mathcal{U} \cap E_\lambda(T)$.
Then $\mathcal{U} = \bigoplus_{\lambda \in \mathrm{spec}(T)} \mathcal{U}_\lambda$, and $T|_{\mathcal{U}}$ is diagonalizable.

Proof. For $\lambda \in \mathrm{spec}(T)$, choose the corresponding polynomial spectral projection ρ_λ by Theorem 5.4.6. Invariance of $\mathcal{U}$ under T implies $T^k(\mathcal{U}) \subseteq \mathcal{U}$ for every integer $k \geq 0$ by induction. Since ρ_λ is a polynomial in T, $\rho_\lambda(\mathcal{U}) \subseteq \mathcal{U}$.
For $\mathbf{u} \in \mathcal{U}$, the spectral resolution satisfies $\mathbf{u} = \sum_{\lambda \in \mathrm{spec}(T)} \rho_\lambda \mathbf{u}$, with $\rho_\lambda \mathbf{u} \in \mathcal{U}_\lambda$. Hence $\mathcal{U} = \sum_{\lambda \in \mathrm{spec}(T)} \mathcal{U}_\lambda$, and the sum is direct by Proposition 5.4.2. Concatenating bases of its nonzero summands produces a basis of $\mathcal{U}$ by Theorem 2.3.4(ii). Every member is an eigenvector of $T|_{\mathcal{U}}$, so the restriction is diagonalizable by Theorem 5.4.3(ii),(i).
Structural Commentary: Invariant subspaces inherit spectral projections.
Key Insight: Polynomial spectral projections cannot move vectors out of a T-invariant subspace. □

Theorem 5.4.8. *Simultaneous Diagonalisation of Commuting Diagonalizable Operators*
Let $\mathcal{V}$ be a nonzero finite-dimensional vector space, and let $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be finite, with every member diagonalizable. Then the members of $\mathcal{F}$ commute pairwise if and only if there is a basis β of $\mathcal{V}$ such that $[T]_{\beta \leftarrow \beta}$ is diagonal for every $T \in \mathcal{F}$.

Proof.
(⇒) Induct on $n = \dim \mathcal{V}$. If every $T \in \mathcal{F}$ is scalar, any basis works. Otherwise choose nonscalar $S \in \mathcal{F}$.
The decomposition $\mathcal{V} = \bigoplus_{\lambda \in \mathrm{spec}(S)} E_\lambda(S)$ holds by Theorem 5.4.6. It has at least two nonzero summands, because the one-summand spectral formula in that theorem would make S a scalar multiple of $I_{\mathcal{V}}$. Hence each summand has dimension less than n.
If $T \in \mathcal{F}$ and $\mathbf{x} \in E_\lambda(S)$, then $S(T\mathbf{x}) = T(S\mathbf{x}) = \lambda T\mathbf{x}$, so $E_\lambda(S) \in \mathrm{Inv}(T)$.
By Lemma 5.4.7, each $T|_{E_\lambda(S)}$ is diagonalizable; the restricted family still commutes.
By induction, each $E_\lambda(S)$ has a common eigenbasis for the restricted family. Their concatenation is a basis of $\mathcal{V}$ by Theorem 2.3.4(ii), and it diagonalizes every member of $\mathcal{F}$.
(⇐) For $S, T \in \mathcal{F}$, set $D_S = [S]_{\beta \leftarrow \beta}$ and $D_T = [T]_{\beta \leftarrow \beta}$.
Since D_S, D_T are diagonal, $[ST]_{\beta \leftarrow \beta} = D_S D_T = D_T D_S = [TS]_{\beta \leftarrow \beta}$ by Proposition 3.4.4. Hence $ST = TS$ by Theorem 3.4.5(i).
Structural Commentary: Commutation preserves eigenspaces.
Key Insight: Diagonalize one operator, then restrict. □

Definition 5.4.9. *Commutant*
Let $n \geq 1$. For $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n, n, \mathbb{F})$, the *commutant* of $\mathcal{F}$ is $\mathrm{Comm}(\mathcal{F}) := \{X \in \mathrm{Mat}(n, n, \mathbb{F}) : (\forall\, A \in \mathcal{F})\, XA = AX\}$.
For a single matrix A, write $\mathrm{Comm}(A)$.

Lemma 5.4.10. *Commuting with a Block Scalar Diagonal Matrix*
Let $k, n_1, \ldots, n_k \geq 1$, set $n := \sum_{i=1}^{k} n_i$, let $\lambda_1, \ldots, \lambda_k \in \mathbb{F}$, and set $\Lambda := \lambda_1 I_{n_1} \oplus \cdots \oplus \lambda_k I_{n_k}$.
Partition $A = [A_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ conformally with Λ, so $A_{ij} \in \mathrm{Mat}(n_i, n_j, \mathbb{F})$.
(i) If A is block diagonal, then $A\Lambda = \Lambda A$.
(ii) If $A\Lambda = \Lambda A$ and $\lambda_1, \ldots, \lambda_k$ are distinct, then A is block diagonal.

Proof. By block multiplication, $(A\Lambda)_{ij} = \lambda_j A_{ij}$ and $(\Lambda A)_{ij} = \lambda_i A_{ij}$.
(i) If A is block diagonal, then $A_{ij} = 0$ for $i \neq j$, while the two diagonal-block expressions coincide. Hence $A\Lambda = \Lambda A$.
(ii) If $A\Lambda = \Lambda A$, then $(\lambda_j - \lambda_i)A_{ij} = 0$ for every i, j. Distinctness of the λ_i implies $A_{ij} = 0$ whenever $i \neq j$, so A is block diagonal.
Structural Commentary: Distinct scalar blocks force commuting matrices to respect the block decomposition.
Key Insight: Distinct block scalars make every off-diagonal block visible and force it to vanish. □

Example 5.4.11. *Distinctness Is Necessary*
The distinctness hypothesis in Lemma 5.4.10 cannot be dropped. If $\lambda_1 = \cdots = \lambda_k = 1$, then $\Lambda = I_n$ commutes with every matrix in $\mathrm{Mat}(n, n, \mathbb{F})$, including matrices that are not block diagonal for the chosen partition.

Theorem 5.4.12. *Commutant of a Diagonalizable Matrix*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonalizable. Let $\mu_1, \ldots, \mu_d$ be its distinct eigenvalues, and set $n_i := \dim_{\mathbb{C}} E_{\mu_i}(A)$. Choose $S \in \mathrm{Mat}(n, n, \mathbb{C})^\times$ satisfying $A = S(\mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d})S^{-1}$. Then

$$\mathrm{Comm}(A) = \{S(X_1 \oplus \cdots \oplus X_d)S^{-1} : X_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})\}.$$

In particular, $\dim_{\mathbb{C}} \mathrm{Comm}(A) = n_1^2 + \cdots + n_d^2$.

Proof. Set $\Lambda := \mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d}$. For $B \in \mathrm{Mat}(n, n, \mathbb{C})$, the equality $BA = AB$ is equivalent, after multiplication by S^{-1} on the left and S on the right, to $(S^{-1}BS)\Lambda = \Lambda(S^{-1}BS)$. By Lemma 5.4.10(ii), this holds exactly when $S^{-1}BS = X_1 \oplus \cdots \oplus X_d$ with $X_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})$, which proves the displayed description.
Conjugation by S is a linear bijection, with inverse given by conjugation by S^{-1}. The block-diagonal subspace has a basis formed by embedding the standard matrix-unit bases of its d diagonal blocks, so its dimension is $\sum_{i=1}^{d} n_i^2$ by Proposition 3.1.5(ii).
Structural Commentary: Diagonalizing A reduces commutation to respecting equal-eigenvalue blocks.
Key Insight: Distinct eigenvalue blocks cannot mix under a commuting matrix. □

Theorem 5.4.13. *Common Eigenvector for a Commuting Family*
Let $\mathcal{V}$ be a nonzero finite-dimensional complex vector space, and let $\mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a nonempty family satisfying $(\forall\, S, T \in \mathcal{F})\, ST = TS$. Then some $\mathbf{v} \neq \mathbf{0}$ is an eigenvector of every operator in $\mathcal{F}$.

Proof. First suppose $\mathcal{F} = \{T_1, \dots, T_m\}$ is finite, and induct on $m \geq 1$. If $m = 1$, the conclusion holds by Theorem 5.3.2. Suppose $m > 1$. Choose $\lambda \in \operatorname{spec}(T_1)$ by Theorem 5.3.2, and set $\mathcal{U} := E_\lambda(T_1)$. Then $\mathcal{U} \neq \{\mathbf{0}\}$.
If $\mathbf{u} \in \mathcal{U}$ and $j \in \{2, \dots, m\}$, then $(T_1 - \lambda I_\mathcal{V})T_j\mathbf{u} = T_j(T_1 - \lambda I_\mathcal{V})\mathbf{u} = \mathbf{0}$ by commutation. Hence $\mathcal{U}$ is invariant under $T_2, \dots, T_m$.
Their restrictions to $\mathcal{U}$ still commute, and by induction have a common eigenvector $\mathbf{v} \in \mathcal{U}$, $\mathbf{v} \neq \mathbf{0}$.
Since $\mathbf{v} \in E_\lambda(T_1)$, it is also an eigenvector of T_1. This proves the finite case.
For general $\mathcal{F}$, $\operatorname{span}(\mathcal{F})$ is finite-dimensional by Theorem 3.4.5(ii). If $\operatorname{span}(\mathcal{F}) = \{0\}$, every nonzero vector is a common eigenvector with eigenvalue 0 for each member. Otherwise, choose a finite basis $(R_1, \dots, R_q)$ of $\operatorname{span}(\mathcal{F})$ by Corollary 1.3.14. Each R_j is a finite linear combination of members of $\mathcal{F}$, so the union of the finitely many operators occurring in these combinations is a finite list from $\mathcal{F}$ that spans $\operatorname{span}(\mathcal{F})$. Choose a basis sublist $(T_1, \dots, T_m)$ by Theorem 1.3.12.
By the finite case, choose a nonzero $\mathbf{v}$ and scalars μ_j satisfying $T_j\mathbf{v} = \mu_j\mathbf{v}$. If $S \in \mathcal{F}$, write $S = \sum_j a_j T_j$.
Then $S\mathbf{v} = (\sum_j a_j\mu_j)\mathbf{v}$, so $\mathbf{v}$ is an eigenvector of every member of $\mathcal{F}$.
Structural Commentary: Commutation makes one eigenspace invariant, reducing the family by one.
Key Insight: Find the common eigenvector inside an eigenspace. □

Theorem 5.4.14. *Simultaneous Schur Triangularisation*
Let $n \geq 1$ and let $\emptyset \neq \mathcal{F} \subseteq \operatorname{Mat}(n, n, \mathbb{C})$ be a commuting family.
Then there is a unitary $U \in \operatorname{Mat}(n, n, \mathbb{C})$ such that U^*AU is upper triangular for every $A \in \mathcal{F}$.
If every matrix in $\mathcal{F}$ is real and has only real eigenvalues, then U can be chosen real orthogonal.

Proof. Proceed by strong induction on n; both conclusions are immediate for $n = 1$. Assume $n > 1$ and both conclusions in every smaller positive dimension.
For the complex conclusion, choose a common eigenvector by Theorem 5.4.13, normalize it to a unit vector $\mathbf{x}$, and extend its column to a unitary matrix $V = [\mathbf{x}\ V_2]$ by Theorem 2.6.13(ii). For each $A \in \mathcal{F}$, write

$$M_A := V^*AV = \begin{bmatrix} \lambda_A & \mathbf{r}_A \\ 0 & A' \end{bmatrix},$$

where the zero lower-left block follows from $A\mathbf{x} = \lambda_A\mathbf{x}$. If $A, B \in \mathcal{F}$, then $M_AM_B = V^*ABV = V^*BAV = M_BM_A$, so their lower-right blocks satisfy $A'B' = B'A'$. By induction, choose a unitary W that simultaneously upper triangularizes all lower-right blocks. The matrix $U := V(I_1 \oplus W)$ is unitary by Proposition 2.6.12(i)–(ii), and U^*AU is upper triangular for every $A \in \mathcal{F}$.
Now assume every member of $\mathcal{F}$ is real and has only real eigenvalues. First construct a common real unit eigenvector $\mathbf{x}$. If every member is scalar, fix any real unit vector $\mathbf{x}$. Otherwise, choose a nonscalar $A_0 \in \mathcal{F}$. Its complexification has the same matrix by Proposition 5.3.11(iii). By Theorem 5.3.2 and the real-spectrum hypothesis, it has an eigenvalue $\lambda \in \mathbb{R}$; hence the real operator induced by A_0 has a nonzero real eigenspace $\mathcal{U} := E_\lambda(A_0)$ by Proposition 5.3.12. Since A_0 is nonscalar, $\mathcal{U} \neq \mathbb{R}^n$.
In this nonscalar case, for $B \in \mathcal{F}$ and $\mathbf{u} \in \mathcal{U}$, $(A_0 - \lambda I_n)B\mathbf{u} = B(A_0 - \lambda I_n)\mathbf{u} = \mathbf{0}$ by commutation, so $\mathcal{U}$ is invariant under every B. The restricted family still commutes. It also has only real eigenvalues: if $\mu \in \mathbb{C}$ and $\mathbf{0} \neq \mathbf{z} \in \mathcal{U}_\mathbb{C}$ satisfy $(B|_\mathcal{U})_\mathbb{C}\mathbf{z} = \mu\mathbf{z}$, then the defining formula for complexification also implies $B_\mathbb{C}\mathbf{z} = \mu\mathbf{z}$, so μ is an eigenvalue of B and is real by hypothesis. Choose an orthonormal basis of $\mathcal{U}$ by Corollary 1.3.11. In these coordinates, induction supplies a simultaneous real upper-triangular form for the restricted family. The first column of each such matrix is supported in its first entry, so the first basis vector is a common real eigenvector; normalize it and denote it by $\mathbf{x}$.
Extend $\mathbf{x}$ to a real orthogonal matrix $V = [\mathbf{x}\ V_2]$ by Theorem 2.6.13(ii), and write $M_A = V^tAV = \left[\begin{smallmatrix} \lambda_A & \mathbf{r}_A \\ 0 & A' \end{smallmatrix}\right]$ for $A \in \mathcal{F}$. The lower-right blocks commute by the same block-product calculation as in the complex case. They are real and have only real eigenvalues. Indeed, if $A'\mathbf{y} = \mu\mathbf{y}$ for $\mathbf{0} \neq \mathbf{y} \in \mathbb{C}^{n-1}$, then either $\mu = \lambda_A \in \mathbb{R}$ or, with $c := \mathbf{r}_A\mathbf{y}/(\mu - \lambda_A)$,

$$M_A \begin{bmatrix} c \\ \mathbf{y} \end{bmatrix} = \mu \begin{bmatrix} c \\ \mathbf{y} \end{bmatrix}.$$

In the second case, $V[c\ \mathbf{y}^t]^t$ is an eigenvector of A for μ, so $\mu \in \mathbb{R}$ by hypothesis. By induction, choose a real orthogonal W that simultaneously upper triangularizes all lower-right blocks. Then $U := V(I_1 \oplus W)$ is real orthogonal by Proposition 2.6.12(i)–(ii), and every U^tAU is upper triangular.
Structural Commentary: A common eigenline creates the first block; commutation descends to the compression.
Key Insight: Simultaneous triangularisation is Schur induction with a common eigenvector at every step. □

Figure 44 summarizes the complex induction in Theorem 5.4.14: a common eigenline creates the first block, and pairwise commutation passes to the lower-right blocks. The theorem, rather than the diagram, owns the simultaneous triangularization conclusion.

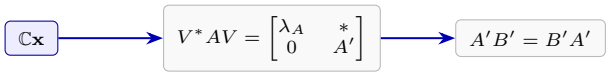

Figure 44: A shared eigenline gives a simultaneous first block; lower-right compressions still commute.

Corollary 5.4.15. *Eigenvalues of Sums and Products of Commuting Matrices*
Let $n \geq 1$ and let $A, B \in \operatorname{Mat}(n, n, \mathbb{C})$ satisfy $AB = BA$. There exist $\lambda_1, \dots, \lambda_n, \mu_1, \dots, \mu_n \in \mathbb{C}$ such that

$$\operatorname{spec}(A) = \{\lambda_1, \dots, \lambda_n\}, \qquad \operatorname{spec}(B) = \{\mu_1, \dots, \mu_n\},$$
$$\operatorname{spec}(A + B) = \{\lambda_i + \mu_i : 1 \leq i \leq n\}, \qquad \operatorname{spec}(AB) = \{\lambda_i\mu_i : 1 \leq i \leq n\}.$$

Proof. Choose a unitary U by Theorem 5.4.14, and set $T := U^*AU$, $R := U^*BU$, $\lambda_i := t_{ii}$, and $\mu_i := r_{ii}$. The matrices $T + R = U^*(A + B)U$ and $TR = U^*ABU$ are upper triangular with respective diagonal entries $\lambda_i + \mu_i$ and $\lambda_i\mu_i$. By Proposition 5.3.19, $\operatorname{spec}(T) = \{\lambda_1, \dots, \lambda_n\}$, $\operatorname{spec}(R) = \{\mu_1, \dots, \mu_n\}$, $\operatorname{spec}(T + R) = \{\lambda_i + \mu_i : 1 \leq i \leq n\}$, and $\operatorname{spec}(TR) = \{\lambda_i\mu_i : 1 \leq i \leq n\}$.
For any $M \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\theta \in \mathbb{C}$, a nonzero $\mathbf{x}$ satisfies $(U^*MU)\mathbf{x} = \theta\mathbf{x}$ iff the nonzero vector $U\mathbf{x}$ satisfies $M(U\mathbf{x}) = \theta(U\mathbf{x})$. Thus unitary similarity preserves the spectrum, and the displayed identities follow.
Structural Commentary: Simultaneous Schur form aligns both spectra on one diagonal.
Key Insight: In common triangular coordinates, diagonals add and multiply. □

6 Spectral Theorems and Operator Functions

6.1 Normal, Self-Adjoint, and Spectral Theorems

Structured spectral theory studies operators whose geometry and spectrum align. Normal, self-adjoint, positive, and unitary maps admit orthogonal decompositions and spectral resolutions.

Throughout this subsection, inner-product spaces are finite-dimensional over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ unless the field is specified explicitly.

Definition 6.1.1. *Normal Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ is *normal* iff $TT^* = T^*T$. Let $n \geq 1$. A matrix $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is *normal* iff $A^*A = AA^*$.

Theorem 6.1.2. *Normal Operators and Equal Adjoint Norms*
Let $T \in \mathcal{L}(\mathcal{V})$. Then T is normal if and only if $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Set $C := T^*T - TT^*$. The operator C is self-adjoint by Proposition 2.3.8(ii)–(iii). For every $\mathbf{v} \in \mathcal{V}$,

$$\langle C\mathbf{v}, \mathbf{v}\rangle = \|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2 .$$

($\Rightarrow$) If T is normal, then $C = 0$ by Definition 6.1.1, so the displayed identity implies $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v}$.
($\Leftarrow$) If the two norms agree for every $\mathbf{v}$, then $\langle C\mathbf{v}, \mathbf{v}\rangle = 0$ for every $\mathbf{v}$. Hence $C = 0$ by Proposition 2.5.4, and T is normal by Definition 6.1.1.

Structural Commentary: Normality is measured by the self-adjoint commutator $T^*T - TT^*$.
Key Insight: A normal operator and its adjoint move every vector the same distance. □

Theorem 6.1.3. *Normal Operators Share Eigenvectors with Their Adjoints*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. If $T\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$, then $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Proof. Set $S := T - \lambda I_{\mathcal{V}}$. Then $S^* = T^* - \overline{\lambda} I_{\mathcal{V}}$ by Proposition 2.3.8(i),(iv), and direct expansion using $TT^* = T^*T$ shows $SS^* = S^*S$. Thus S is normal. Since $S\mathbf{v} = \mathbf{0}$, $\|S^*\mathbf{v}\| = \|S\mathbf{v}\| = 0$ by Theorem 6.1.2. Hence $S^*\mathbf{v} = \mathbf{0}$ by positive definiteness, or equivalently $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.
Structural Commentary: For normal operators, kernels of $T - \lambda I$ and $T^* - \overline{\lambda} I$ coincide.
Key Insight: Normality makes eigenvector information visible from both T and T^*. □

Figure 45 illustrates Theorem 6.1.3: the eigenline of a normal operator is also an eigenline of its adjoint. The theorem, rather than the diagram, owns the conjugate-eigenvalue conclusion.

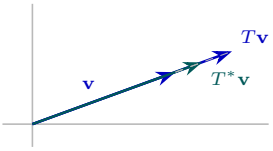

Figure 45: For normal T, $T\mathbf{v} = \lambda\mathbf{v}$ implies $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Theorem 6.1.4. *Orthogonal Eigenvectors for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. Eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose $T\mathbf{u} = \alpha\mathbf{u}$ and $T\mathbf{v} = \beta\mathbf{v}$ with $\alpha \neq \beta$. By Theorem 6.1.3, $T^*\mathbf{v} = \overline{\beta}\mathbf{v}$.
Hence $\alpha \langle \mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T^*\mathbf{v}\rangle = \left\langle \mathbf{u}, \overline{\beta}\mathbf{v}\right\rangle = \beta \langle \mathbf{u}, \mathbf{v}\rangle$. Since $\alpha - \beta \neq 0$, $\langle \mathbf{u}, \mathbf{v}\rangle = 0$.
Structural Commentary: Normality lets the operator move from one eigenvector to the other through the adjoint.
Key Insight: Distinct eigenvalue scalings are incompatible unless the inner product is zero. □

Theorem 6.1.5. *Eigenvalues of Self-Adjoint Operators Are Real*
Every eigenvalue of a self-adjoint operator on $\mathcal{V}$ lies in $\mathbb{R}$.

Proof. Let $T = T^*$ and $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $\lambda \|\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \langle \mathbf{v}, \lambda\mathbf{v}\rangle = \overline{\lambda} \|\mathbf{v}\|^2$.
Since $\|\mathbf{v}\|^2 > 0$, $\lambda = \overline{\lambda}$, so $\lambda \in \mathbb{R}$.
Structural Commentary: Self-adjointness moves T between inner-product slots without changing the value.
Key Insight: An eigenvalue must equal its conjugate because the eigenvector has positive squared norm. □

Proposition 6.1.6. *Invertible Quadratic Expressions*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint on a finite-dimensional inner-product space.
If $b, c \in \mathbb{R}$ and $b^2 < 4c$, then $T^2 + bT + cI_{\mathcal{V}}$ is invertible.

Proof. Set $S := T^2 + bT + cI_{\mathcal{V}}$. Self-adjointness implies $\langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$ and $\langle T^2\mathbf{v}, \mathbf{v}\rangle = \|T\mathbf{v}\|^2$. Thus, for $\mathbf{v} \neq \mathbf{0}$,

$$\begin{aligned}\langle S\mathbf{v}, \mathbf{v}\rangle &= \|T\mathbf{v}\|^2 + b \langle T\mathbf{v}, \mathbf{v}\rangle + c \|\mathbf{v}\|^2 \geq \|T\mathbf{v}\|^2 - |b| \, \|T\mathbf{v}\| \, \|\mathbf{v}\| + c \|\mathbf{v}\|^2 \\ &= \left(\|T\mathbf{v}\| - \frac{|b|}{2} \|\mathbf{v}\|\right)^2 + \left(c - \frac{b^2}{4}\right) \|\mathbf{v}\|^2 > 0\end{aligned}$$

where the inequality holds by Theorem 1.1.13. Thus $S\mathbf{v} \neq \mathbf{0}$ for every nonzero $\mathbf{v}$, so S is injective and therefore invertible by Theorem 1.7.7(ii),(i).
Structural Commentary: Negative discriminant keeps this self-adjoint quadratic positive.
Key Insight: Self-adjointness converts $\langle T^2\mathbf{v}, \mathbf{v}\rangle$ into $\|T\mathbf{v}\|^2$, so completing the square controls the kernel. □

Lemma 6.1.7. *Self-Adjoint Operators Have Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real inner-product space. Every self-adjoint $T \in \mathcal{L}(\mathcal{V})$ has an eigenvalue.

Proof. On the complexification $\mathcal{V}_{\mathbb{C}}$, define

$$\langle \mathbf{u}+i\mathbf{v}, \mathbf{x}+i\mathbf{y}\rangle_{\mathcal{V}_{\mathbb{C}}} := \langle \mathbf{u},\mathbf{x}\rangle_{\mathcal{V}} + \langle \mathbf{v},\mathbf{y}\rangle_{\mathcal{V}} + i\big(\langle \mathbf{v},\mathbf{x}\rangle_{\mathcal{V}} - \langle \mathbf{u},\mathbf{y}\rangle_{\mathcal{V}}\big).$$

This is a complex inner product by the real inner-product axioms and Definition 5.3.6(ii); in particular, $\langle \mathbf{u}+i\mathbf{v}, \mathbf{u}+i\mathbf{v}\rangle_{\mathcal{V}_{\mathbb{C}}} = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. The space $\mathcal{V}_{\mathbb{C}}$ is nonzero and finite-dimensional over $\mathbb{C}$ by Propositions 5.3.7 and 5.3.9.
The complexification $T_{\mathbb{C}}$ is complex-linear by Proposition 5.3.11(i). For $\mathbf{z} = \mathbf{u}+i\mathbf{v}$ and $\mathbf{w} = \mathbf{x}+i\mathbf{y}$, self-adjointness of T in each real component implies

$$\langle T_{\mathbb{C}}\mathbf{z}, \mathbf{w}\rangle_{\mathcal{V}_{\mathbb{C}}} = \langle \mathbf{z}, T_{\mathbb{C}}\mathbf{w}\rangle_{\mathcal{V}_{\mathbb{C}}}.$$

Hence $T_{\mathbb{C}}$ is self-adjoint. Choose an eigenvalue $\lambda \in \mathbb{C}$ of $T_{\mathbb{C}}$ by Theorem 5.3.2. Then $\lambda \in \mathbb{R}$ by Theorem 6.1.5, and λ is an eigenvalue of T by Proposition 5.3.12.
Structural Commentary: Complexification supplies an eigenvalue without losing the real self-adjoint structure.
Key Insight: The Hermitian extension forces the complex eigenvalue to be real, after which it descends to the original operator. □

Proposition 6.1.8. *Self-Adjoint Operators and Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint, and let $\mathcal{U}$ be a subspace of $\mathcal{V}$ invariant under T. Then:

(i) $\mathcal{U}^{\perp}$ is invariant under T;
(ii) $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is self-adjoint;
(iii) $T|_{\mathcal{U}^{\perp}} \in \mathcal{L}(\mathcal{U}^{\perp})$ is self-adjoint.

Proof.

(i) Since $\mathcal{U}$ is invariant under $T = T^*$, the subspace $\mathcal{U}^{\perp}$ is invariant under $T^* = T$ by Theorem 5.2.7(i).
(ii) Parts (i) and the hypothesis show that $(\mathcal{U},\mathcal{U}^{\perp})$ reduces T by Theorem 2.1.5 and Definition 5.2.6. Therefore $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}} = T|_{\mathcal{U}}$ by Theorem 5.2.7(iii).
(iii) The pair $(\mathcal{U}^{\perp},\mathcal{U})$ also reduces T by part (i), the hypothesis, Theorem 2.1.5, and the identity $(\mathcal{U}^{\perp})^{\perp} = \mathcal{U}$ from Corollary 2.1.8. Hence $(T|_{\mathcal{U}^{\perp}})^* = (T^*)|_{\mathcal{U}^{\perp}} = T|_{\mathcal{U}^{\perp}}$ by Theorem 5.2.7(iii), applied to $\mathcal{U}^{\perp}$.

Structural Commentary: Self-adjointness preserves orthogonal complements of invariant subspaces.
Key Insight: Move T across the inner product; invariance of $\mathcal{U}$ then forces orthogonality to persist. □

Theorem 6.1.9. *Schur Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, then the empty orthonormal basis has the required property. Assume $n := \dim(\mathcal{V}) \geq 1$. Choose an orthonormal basis β of $\mathcal{V}$ by Corollary 1.3.11, and set $A := [T]_{\beta\leftarrow\beta}$. For the nonempty commuting family $\{A\}$, fix a unitary U such that U^*AU is upper triangular by Theorem 5.4.14.
Let γ be the ordered list of vectors whose β-coordinate columns are the columns of U. The columns of U are orthonormal by Theorem 2.6.8(i),(ii), and orthonormal coordinates preserve inner products by Theorem 1.4.7(ii), so γ is an orthonormal basis of $\mathcal{V}$. Moreover, $C_{\beta\leftarrow\gamma} = U$ and $C_{\gamma\leftarrow\beta} = U^{-1} = U^*$ by Definition 3.5.1 and Theorem 2.6.8(i),(vi). Therefore $[T]_{\gamma\leftarrow\gamma} = U^*AU$ by Theorem 3.5.3, and this matrix is upper triangular.
Structural Commentary: The simultaneous theorem specializes to a singleton matrix family.
Key Insight: Schur triangularisation is the one-operator case of simultaneous triangularisation. □

Lemma 6.1.10. *Upper-Triangular Normal Matrices Are Diagonal*
Let $n \geq 1$. If $A \in \mathrm{Mat}(n,n,\mathbb{C})$ is upper triangular and normal, then A is diagonal.

Proof. Induct on n. The case $n = 1$ is immediate.
Since A is upper triangular, $A\mathbf{e}_1 = a_{11}\mathbf{e}_1$, while $A^*\mathbf{e}_1 = \overline{a_{11}}\mathbf{e}_1 + \overline{a_{12}}\mathbf{e}_2 + \cdots + \overline{a_{1n}}\mathbf{e}_n$.
The $(1,1)$ entries of the normality identity $A^*A = AA^*$ are $|a_{11}|^2$ and $|a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2$, respectively. Hence $a_{12} = \cdots = a_{1n} = 0$. Thus $A = a_{11} \oplus B$ with B upper triangular. The lower-right blocks in $A^*A = AA^*$ satisfy $B^*B = BB^*$, so B is normal. The induction hypothesis makes B diagonal.
Structural Commentary: Normality forbids upper-triangular row tails.
Key Insight: Compare the lengths of $A\mathbf{e}_1$ and $A^*\mathbf{e}_1$ to kill the first row, then induct. □

Theorem 6.1.11. *Complex Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

(i) T is normal;
(ii) $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
(iii) T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, all three conditions hold for the empty orthonormal basis. Assume $n := \dim(\mathcal{V}) \geq 1$.

(*i*) $\Rightarrow$ (*ii*) Fix an orthonormal basis γ such that $A := [T]_{\gamma\leftarrow\gamma}$ is upper triangular by Theorem 6.1.9. The identities $[T^*]_{\gamma\leftarrow\gamma} = A^*$ and $[ST]_{\gamma\leftarrow\gamma} = [S]_{\gamma\leftarrow\gamma}[T]_{\gamma\leftarrow\gamma}$ hold by Theorem 2.3.7 and Proposition 3.4.4. Thus $A^*A = AA^*$ because $T^*T = TT^*$, so A is normal. The matrix A is diagonal by Lemma 6.1.10; hence γ consists of eigenvectors of T.
(*ii*) $\Rightarrow$ (*iii*) The matrix of T in an orthonormal eigenbasis is diagonal by Definition 3.2.3.
(*iii*) $\Rightarrow$ (*i*) Fix an orthonormal basis γ in which $A := [T]_{\gamma\leftarrow\gamma}$ is diagonal. Then $[T^*]_{\gamma\leftarrow\gamma} = A^*$ by Theorem 2.3.7. Since the diagonal matrices A and A^* commute, $[TT^*]_{\gamma\leftarrow\gamma} = [T^*T]_{\gamma\leftarrow\gamma}$ by Proposition 3.4.4. Therefore $TT^* = T^*T$ by Theorem 3.4.5(i), and T is normal.

Structural Commentary: Schur supplies triangular form; normality kills the upper entries.
Key Insight: Equal norms of $T\mathbf{e}_j$ and $T^*\mathbf{e}_j$ force each row tail to vanish. □

Figure 46 summarizes the implication from normality to orthonormal diagonalization in Theorem 6.1.11. The theorem, rather than the diagram, owns the diagonalization conclusion.

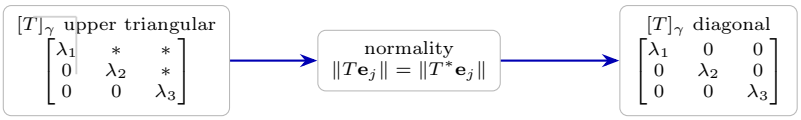

Figure 46: Normality turns Schur triangular form diagonal by forcing row tails to vanish.

Theorem 6.1.12. *Real Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

(i) T is self-adjoint;
(ii) $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
(iii) T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Assume $T = T^*$ and induct on $n := \dim(\mathcal{V})$. For $n = 0$, the empty basis suffices. For $n = 1$, any nonzero vector is an eigenvector by Proposition 5.2.4, and normalization produces an orthonormal eigenbasis.
Let $n \geq 2$. Fix an eigenvalue $\lambda \in \mathbb{R}$ and a corresponding unit eigenvector $\mathbf{u}$ by Lemma 6.1.7 and normalization, and set $\mathcal{U} := \operatorname{span}(\mathbf{u})$. The subspace $\mathcal{U}$ is invariant under T by Proposition 5.2.4. The subspace $\mathcal{U}^\perp$ is invariant under T, and $T|_{\mathcal{U}^\perp}$ is self-adjoint, by Proposition 6.1.8(i),(iii). Moreover, $\dim(\mathcal{U}^\perp) = n - 1$ by Corollary 2.1.7. The induction hypothesis supplies an orthonormal eigenbasis of $\mathcal{U}^\perp$. Together with $\mathbf{u}$, it is an orthonormal eigenbasis of $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$ by Theorem 2.1.5.

(*ii*) $\Rightarrow$ (*iii*) The matrix of T in an orthonormal eigenbasis is diagonal by Definition 3.2.3.

(*iii*) $\Rightarrow$ (*i*) Fix an orthonormal basis γ in which $A := [T]_{\gamma \leftarrow \gamma}$ is real diagonal. Then $[T^*]_{\gamma \leftarrow \gamma} = A^t = A$ by Theorem 2.3.7. If $\mathcal{V} \neq \{\mathbf{0}\}$, then $T^* = T$ by Theorem 3.4.5(i); the same equality is immediate when $\mathcal{V} = \{\mathbf{0}\}$.

Structural Commentary: Split off one eigendirection, then restrict to its orthogonal complement.
Key Insight: One eigenvector reduces the dimension. □

Theorem 6.1.13. *Spectral Resolution of a Normal Operator*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $d \geq 1$, and let $\mu_1, \ldots, \mu_d \in \mathbb{C}$ be distinct. For $T \in \mathcal{L}(\mathcal{V})$, the following are equivalent:

(i) T is normal and $\operatorname{spec}(T) \subseteq \{\mu_1, \ldots, \mu_d\}$.
(ii) There exist orthogonal projections $P_1, \ldots, P_d \in \mathcal{L}(\mathcal{V})$ such that $P_i P_j = 0$ for $i \neq j$, $P_1 + \cdots + P_d = I_\mathcal{V}$, and $T = \mu_1 P_1 + \cdots + \mu_d P_d$.

In this case, every family of projections satisfying (ii) obeys $\operatorname{range}(P_j) = E_{\mu_j}(T)$ for every $j \in \{1, \ldots, d\}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Fix an orthonormal eigenbasis of T by Theorem 6.1.11(i),(ii). Grouping its vectors by eigenvalue produces

$$\mathcal{V} = \bigoplus_{j=1}^{d} {}^{\perp} E_{\mu_j}(T),$$

where an absent eigenvalue contributes the zero subspace; the summands are orthogonal by Theorem 6.1.4. Set $P_j := P_{E_{\mu_j}(T)}$ by Definition 2.2.2. For the displayed orthogonal decomposition, P_j selects the $E_{\mu_j}(T)$ component. Consequently, $P_i P_j = 0$ when $i \neq j$, $\sum_{j=1}^{d} P_j = I_\mathcal{V}$, and $T = \sum_{j=1}^{d} \mu_j P_j$.

(*ii*) $\Rightarrow$ (*i*) Suppose (ii) holds. Each P_j is self-adjoint and idempotent by Theorem 2.5.6. By Proposition 2.3.8(i) and the pairwise product relations,

$$T^* = \sum_{j=1}^{d} \overline{\mu_j} P_j, \qquad T^* T = \sum_{j=1}^{d} |\mu_j|^2 P_j = T T^*.$$

Thus T is normal. If $\lambda \in \operatorname{spec}(T)$ and $\mathbf{0} \neq \mathbf{x} \in E_\lambda(T)$, then

$$\mathbf{0} = (T - \lambda I_\mathcal{V})\mathbf{x} = \sum_{j=1}^{d} (\mu_j - \lambda) P_j \mathbf{x}.$$

The vectors $P_j \mathbf{x}$ are pairwise orthogonal because the projections are self-adjoint and $P_i P_j = 0$ for $i \neq j$. Therefore

$$0 = \sum_{j=1}^{d} |\mu_j - \lambda|^2 \, \|P_j \mathbf{x}\|^2 .$$

At least one $P_k \mathbf{x}$ is nonzero because $\sum_j P_j = I_\mathcal{V}$ and $\mathbf{x} \neq \mathbf{0}$, so $\lambda = \mu_k$. Hence $\operatorname{spec}(T) \subseteq \{\mu_1, \ldots, \mu_d\}$.

Now fix any projections satisfying (ii) and any $j \in \{1, \ldots, d\}$. For $\mathbf{x} \in \mathcal{V}$,

$$\mathbf{x} = \sum_{i=1}^{d} P_i \mathbf{x}, \qquad (T - \mu_j I_\mathcal{V})\mathbf{x} = \sum_{i \neq j} (\mu_i - \mu_j) P_i \mathbf{x}.$$

By pairwise orthogonality,

$$\|(T - \mu_j I_\mathcal{V})\mathbf{x}\|^2 = \sum_{i \neq j} |\mu_i - \mu_j|^2 \, \|P_i \mathbf{x}\|^2 .$$

Since the μ_i are distinct, this norm vanishes exactly when $P_i \mathbf{x} = \mathbf{0}$ for all $i \neq j$, equivalently when $\mathbf{x} = P_j \mathbf{x}$. The latter condition is equivalent to $\mathbf{x} \in \operatorname{range}(P_j)$ because $P_j^2 = P_j$. Thus $E_{\mu_j}(T) = \operatorname{range}(P_j)$.

Structural Commentary: A normal operator decomposes into scalar actions on orthogonal eigenspaces.
Key Insight: Spectral projections isolate eigenspaces; the operator is the weighted sum of those projections. □

Figure 47 summarizes the projection identities established in Theorem 6.1.13. The theorem, rather than the diagram, owns the eigenspace decomposition and weighted-sum formula.

Figure 47: Spectral resolution: $\mathcal{V} = \bigoplus_j \operatorname{range}(P_j)$, $P_iP_j = 0$ for $i \neq j$, $\sum_j P_j = I_{\mathcal{V}}$, and $T = \sum_j \mu_j P_j$.

Perspective 6.1.14. *Spectral Resolutions Are Orthogonal Resolutions*
For the projections P_j in Theorem 6.1.13, $P_j = P_j^* = P_j^2$; $P_iP_j = 0$ $(i \neq j)$; $\sum_j P_j = I_{\mathcal{V}}$.
Hence $\mathcal{V} = \bigoplus_j^{\perp} \operatorname{range}(P_j)$ is the orthogonal decomposition into T-invariant spectral subspaces.
Structural Commentary: Each P_j is forced by its eigenspace: it is the orthogonal projection onto it.
Key Insight: Normality turns diagonalization into a projection-valued decomposition.

Corollary 6.1.15. *Functional Calculus from Spectral Resolution*
Let $d \geq 1$, let $\mu_1, \ldots, \mu_d \in \mathbb{C}$ be distinct, let $T = \sum_{j=1}^d \mu_j P_j$ be a spectral resolution from Theorem 6.1.13, and let $f \in \mathbb{C}[z]$. Then $f(T) = \sum_{j=1}^{d} f(\mu_j) P_j$.
In particular, for each j there is a polynomial p_j such that $p_j(T) = P_j$.

Proof. The identities $P_iP_j = 0$ for $i \neq j$, $P_j^2 = P_j$, and $\sum_j P_j = I_{\mathcal{V}}$ imply by induction that $T^m = \sum_j \mu_j^m P_j$ for every $m \geq 0$.
The displayed formula follows by expanding the polynomial f and collecting the P_j terms.
For the final assertion, choose the Lagrange polynomial p_j satisfying $p_j(\mu_i) = \delta_{ij}$. Then $p_j(T) = P_j$.
Structural Commentary: Orthogonal spectral projections multiply like coordinate idempotents.
Key Insight: Polynomial functions of a normal operator act by scalar functions on each eigenspace. □

6.2 Normal Operator Consequences

Throughout this subsection, spaces supporting operators are finite-dimensional inner-product spaces over the stated field.

Perspective 6.2.1. *Normal Means Diagonal in Orthogonal Coordinates*
Theorem 6.1.11 states that normal operators are precisely the complex operators whose full structure is visible in an orthonormal eigenbasis. This is why normality is the natural coordinate-free version of being diagonal.

Proposition 6.2.2. *Basic Spectral Properties of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, and let $p \in \mathbb{C}[z]$.

(i) $p(T)$ is normal and commutes with T.
(ii) T^* is normal and $E_\lambda(T) = E_{\overline{\lambda}}(T^*)$ for every $\lambda \in \mathbb{C}$.
(iii) $\ker T = \ker T^*$ and $\operatorname{range}(T) = \operatorname{range}(T^*)$.

Proof. Fix an orthonormal eigenbasis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ with $T\mathbf{e}_j = \mu_j \mathbf{e}_j$ by Theorem 6.1.11(i),(ii).

(i) The relation $p(T)\mathbf{e}_j = p(\mu_j)\mathbf{e}_j$ follows from the definition of an operator polynomial. Thus $p(T)$ is diagonal in this orthonormal basis and therefore normal by Theorem 6.1.11(iii),(i). It commutes with T by Proposition 1.1.6.

(ii) The identity $T^*\mathbf{e}_j = \overline{\mu_j}\mathbf{e}_j$ holds by Theorem 6.1.3, so T^* is diagonal in the same orthonormal basis and is normal by Theorem 6.1.11(iii),(i). The inclusion $E_\lambda(T) \subseteq E_{\overline{\lambda}}(T^*)$ holds by Theorem 6.1.3. The reverse inclusion follows by that theorem for the normal operator T^* and Proposition 2.3.8(iii), which identifies $(T^*)^*$ with T.

(iii) The kernel identity is part (ii) with $\lambda = 0$. Moreover, $\operatorname{range}(T) = \operatorname{span}\{\mathbf{e}_j : \mu_j \neq 0\} = \operatorname{range}(T^*)$ because T and T^* have respective diagonal entries μ_j and $\overline{\mu_j}$ in the same basis.

Structural Commentary: One orthonormal basis diagonalises both T and T^*.
Key Insight: Kernel and range are read from the nonzero diagonal entries in that basis. □

Lemma 6.2.3. *Adjoints of Normal Matrices as One Polynomial*
Let $k \geq 1$, let $n_1, \ldots, n_k \geq 1$, and let $A_j \in \operatorname{Mat}(n_j, n_j, \mathbb{C})$ be normal for every $j \in \{1, \ldots, k\}$. Then there is a polynomial $p \in \mathbb{C}[z]$ such that $p(A_j) = A_j^*$ for every $j \in \{1, \ldots, k\}$.

Proof. Let $\Omega := \bigcup_{j=1}^{k} \operatorname{spec}(A_j)$. This set is finite by Corollary 5.1.6 and nonempty by Theorem 5.3.2, with each A_j regarded as an operator on $\mathbb{C}^{n_j}$. Define

$$p(z) := \sum_{\lambda \in \Omega} \overline{\lambda} \prod_{\substack{\nu \in \Omega \\ \nu \neq \lambda}} \frac{z - \nu}{\lambda - \nu}.$$

Then $p(\lambda) = \overline{\lambda}$ for every $\lambda \in \Omega$. Fix j and regard A_j as an operator on $\mathbb{C}^{n_j}$. This operator is normal by Theorem 2.3.7 and Proposition 3.4.4. Write its spectral resolution as $A_j = \sum_{\mu \in \operatorname{spec}(A_j)} \mu P_{j,\mu}$ by Theorem 6.1.13. By Corollary 6.1.15, Proposition 2.3.8(i), Theorem 2.5.6, and Theorem 2.3.7,

$$p(A_j) = \sum_{\mu \in \operatorname{spec}(A_j)} \overline{\mu} P_{j,\mu} = A_j^*.$$

Since j was arbitrary, the same polynomial works for every matrix.
Structural Commentary: One polynomial interpolates conjugation on all spectra.
Key Insight: For normal matrices, A^* is $p(A)$ with $p(\lambda) = \overline{\lambda}$. □

Theorem 6.2.4. *Fuglede–Putnam Theorem*
Let $m, n \geq 1$, let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$ be normal, and let $X \in \operatorname{Mat}(m, n, \mathbb{C})$.
Then $AX = XB \iff A^*X = XB^*$.

Proof.

(⇒) Assume $AX = XB$. By induction, $A^rX = XB^r$ for every $r \geq 0$, so $p(A)X = Xp(B)$ for every polynomial p. Fix p satisfying $p(A) = A^*$ and $p(B) = B^*$ by Lemma 6.2.3. Thus $A^*X = XB^*$.

(⇐) Assume $A^*X = XB^*$. The matrices A^* and B^* are normal because $(A^*)^*A^* = AA^* = A^*A = A^*(A^*)^*$ and similarly for B^*. The same polynomial argument, with a polynomial q fixed by Lemma 6.2.3 such that $q(A^*) = A$ and $q(B^*) = B$, proves $AX = XB$.

Structural Commentary: An intertwining relation passes through every polynomial in the two normal matrices.
Key Insight: Normality makes adjoints polynomial, so intertwining transfers to adjoints. □

Method 6.2.5. *Finding a Spectral Decomposition of a Normal Matrix*
Let $n \geq 1$. For a normal matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$:

i. Find the distinct eigenvalues $\mu_1, \ldots, \mu_d$.
ii. For each j, solve $(A - \mu_j I_n)\mathbf{x} = 0$ to get $E_{\mu_j}(A)$.
iii. Orthonormalise a basis of each eigenspace.
iv. Concatenate these orthonormal bases as the columns of U; then U is unitary and $A = U\Lambda U^*$, where Λ repeats each μ_j according to $\dim E_{\mu_j}(A)$.

Theorem 6.2.6. *Spectral Classification of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.

(i) T is unitary if and only if $|\lambda| = 1$ for every $\lambda \in \mathrm{spec}(T)$.
(ii) T is self-adjoint if and only if $\mathrm{spec}(T) \subseteq \mathbb{R}$.
(iii) T is an orthogonal projection if and only if $\mathrm{spec}(T) \subseteq \{0, 1\}$.
(iv) T is skew-adjoint, meaning $T^* = -T$, if and only if every $\lambda \in \mathrm{spec}(T)$ is purely imaginary.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, its unique operator has all four properties on the left, while its spectrum is empty; hence all four equivalences hold. Assume $\mathcal{V} \neq \{\mathbf{0}\}$. List the distinct eigenvalues as $\mu_1, \ldots, \mu_d$, and fix the spectral resolution $T = \sum_{j=1}^{d} \mu_j P_j$ by Theorem 6.1.13. Then

$$I_{\mathcal{V}} = \sum_{j=1}^{d} P_j, \qquad T^* = \sum_{j=1}^{d} \overline{\mu_j} P_j, \qquad T^*T = \sum_{j=1}^{d} |\mu_j|^2 P_j, \qquad T^2 = \sum_{j=1}^{d} \mu_j^2 P_j,$$

where the adjoint identity holds by Proposition 2.3.8(i) and Theorem 2.5.6. If $\sum_j \alpha_j P_j = 0$, then applying this operator to a nonzero vector in $\mathrm{range}(P_k) = E_{\mu_k}(T)$ proves $\alpha_k = 0$. Thus the P_j are linearly independent.

(i) (⇒) If T is unitary, then $T^*T = I_{\mathcal{V}}$ by Definition 2.6.7. Comparing coefficients in the linearly independent family $(P_j)_{j=1}^d$ shows $|\mu_j|^2 = 1$ for every j.
(i) (⇐) If $|\mu_j| = 1$ for every j, then $T^*T = TT^* = \sum_j P_j = I_{\mathcal{V}}$, so T is unitary by Definition 2.6.7.
(ii) (⇒) If $T = T^*$, then $\sum_j (\mu_j - \overline{\mu_j})P_j = 0$. Linear independence forces $\mu_j = \overline{\mu_j}$ for every j, so $\mathrm{spec}(T) \subseteq \mathbb{R}$.
(ii) (⇐) If every μ_j is real, then the displayed resolutions of T and T^* agree, so $T = T^*$.
(iii) (⇒) If T is an orthogonal projection, then $T^* = T$ and $T^2 = T$ by Theorem 2.5.6. Comparing coefficients shows $\mu_j = \overline{\mu_j}$ and $\mu_j^2 = \mu_j$, so $\mu_j \in \{0, 1\}$ for every j.
(iii) (⇐) If every μ_j belongs to $\{0, 1\}$, then $T^* = T$ and $T^2 = T$ by the displayed resolutions. Hence T is an orthogonal projection by Theorem 2.5.6.
(iv) (⇒) If $T^* = -T$, then $\sum_j (\overline{\mu_j} + \mu_j)P_j = 0$. Linear independence forces $\overline{\mu_j} = -\mu_j$ for every j, so every eigenvalue is purely imaginary.
(iv) (⇐) If $\overline{\mu_j} = -\mu_j$ for every j, then the displayed resolutions satisfy $T^* = -T$.

Structural Commentary: For normal operators, structural classes are scalar conditions on spectral blocks.
Key Insight: For normal T, these properties reduce to scalar tests. □

Proposition 6.2.7. *Cartesian Decomposition and Normality*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space.
For every $T \in \mathcal{L}(\mathcal{V})$, there are unique self-adjoint operators $H, K \in \mathcal{L}(\mathcal{V})$ such that $T = H + iK$.
They are $H = \frac{1}{2}(T + T^*)$ and $K = \frac{1}{2i}(T - T^*)$. Moreover, T is normal if and only if $HK = KH$.

Proof. Set $H := \frac{1}{2}(T+T^*)$ and $K := \frac{1}{2i}(T-T^*)$. Then $H^* = H$, $K^* = K$, and $T = H+iK$ by Proposition 2.3.8(i),(iii). If $T = X+iY$ with X, Y self-adjoint, then $T^* = X - iY$ by Proposition 2.3.8(i), so $X = H$ and $Y = K$; this proves uniqueness.

(⇒) Suppose T is normal. The identities

$$T^*T = H^2 + K^2 + i(HK - KH), \qquad TT^* = H^2 + K^2 - i(HK - KH)$$

and $T^*T = TT^*$ imply $HK = KH$.

(⇐) If $HK = KH$, the two displayed products are equal, so T is normal.

Structural Commentary: Every complex operator splits into Hermitian and skew-Hermitian parts.
Key Insight: Normality says the two Hermitian coordinates of T commute. □

Figure 48 summarizes Proposition 6.2.7. The proposition, rather than the diagram, owns the equivalence between normality and commutation of the self-adjoint parts.

Figure 48: Cartesian decomposition: T is normal exactly when $HK = KH$.

Proposition 6.2.8. *Normal Block Upper-Triangular Matrices Collapse*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $r, s \geq 1$, and let

$$A := \begin{bmatrix} B & X \\ 0 & C \end{bmatrix} \in \mathrm{Mat}(r+s, r+s, \mathbb{F}),$$

where $B \in \mathrm{Mat}(r, r, \mathbb{F})$, $C \in \mathrm{Mat}(s, s, \mathbb{F})$, and $X \in \mathrm{Mat}(r, s, \mathbb{F})$.
Then A is normal if and only if $X = 0$ and both B and C are normal.

Proof.
($\Rightarrow$) Suppose A is normal, regarding the matrices as complex when $\mathbb{F} = \mathbb{R}$. Fix unitary matrices U_B, U_C such that $U_B^* B U_B$ and $U_C^* C U_C$ are upper triangular by Theorem 6.1.9, applied to the induced operators. Set $Q := U_B \oplus U_C$, which is unitary by Proposition 2.6.12(ii), with $Q^*Q = QQ^* = I$ by Theorem 2.6.8(i),(iii). Then

$$Q^* A Q = \begin{bmatrix} U_B^* B U_B & U_B^* X U_C \\ 0 & U_C^* C U_C \end{bmatrix}$$

is upper triangular. It is normal because $Q^*Q = QQ^* = I$ and

$$(Q^*AQ)^*(Q^*AQ) = Q^*A^*AQ, \qquad (Q^*AQ)(Q^*AQ)^* = Q^*AA^*Q.$$

Thus Q^*AQ is diagonal by Lemma 6.1.10. In particular, $U_B^* X U_C = 0$, and multiplication by U_B and U_C^* proves $X = 0$. The normality equation for $A = B \oplus C$ is blockwise, so $B^*B = BB^*$ and $C^*C = CC^*$.
($\Leftarrow$) If $X = 0$ and B, C are normal, then $A = B \oplus C$ and $A^*A = AA^*$ by blockwise multiplication.

Structural Commentary: Block-diagonal Schur coordinates make the whole matrix upper triangular.
Key Insight: An upper-triangular normal matrix is diagonal, so the coupling block vanishes. □

Figure 49 summarizes Proposition 6.2.8: block-diagonal Schur coordinates reduce the claim to the earlier upper-triangular normal-matrix lemma. The proposition, rather than the diagram, owns the vanishing of X.

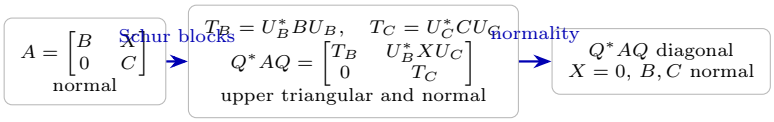

Figure 49: Normality removes the upper-right block in a block upper-triangular matrix.

Theorem 6.2.9. *Invariant Subspaces of Normal Matrices Reduce*
Let $n \geq 1$, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be normal, and let $\mathcal{U} \leq \mathbb{F}^n$ be invariant under A. Then $\mathcal{U}$ is invariant under A^*. If P is the orthogonal projection onto $\mathcal{U}$, then $AP = PA$.

Proof. The cases $\mathcal{U} = \{\mathbf{0}\}$ and $\mathcal{U} = \mathbb{F}^n$ are immediate. Assume $r := \dim(\mathcal{U}) \in \{1, \ldots, n-1\}$. Choose orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$ by Corollary 1.3.11, concatenate them, and let $Q = [Q_1 \; Q_2]$ be the resulting unitary matrix. Invariance of $\mathcal{U}$ implies $Q_2^* A Q_1 = 0$, so

$$Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}.$$

Since $Q^*Q = QQ^* = I$, the products $(Q^*AQ)^*(Q^*AQ) = Q^*A^*AQ$ and $(Q^*AQ)(Q^*AQ)^* = Q^*AA^*Q$ are equal, so Q^*AQ is normal. Hence $X = 0$ by Proposition 6.2.8. The identity $Q^*A^*Q = (Q^*AQ)^* = B^* \oplus C^*$ proves that $\mathcal{U}$ is invariant under A^*. Moreover, $Q^*PQ = I_r \oplus 0$ by Proposition 2.2.3(ii)–(iii). Both Q^*AQ and Q^*PQ are block diagonal, so they commute. Multiplication by Q on the left and Q^* on the right proves $AP = PA$.
Structural Commentary: In adapted coordinates, normal invariance becomes block diagonality.
Key Insight: Normal invariant subspaces are automatically reducing subspaces. □

Corollary 6.2.10. *Invariant Subspaces of Normal Operators Reduce*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, let $\mathcal{U}$ be invariant under T. Then:
(i) $\mathcal{U}^\perp$ is invariant under T;
(ii) $\mathcal{U}$ is invariant under T^*;
(iii) $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$;
(iv) $T|_{\mathcal{U}}$ and $T|_{\mathcal{U}^\perp}$ are normal.

Proof. If $\mathcal{U} = \{\mathbf{0}\}$ or $\mathcal{U} = \mathcal{V}$, all four conclusions are immediate. Assume $0 < r := \dim(\mathcal{U}) < n := \dim(\mathcal{V})$. Choose an orthonormal basis β adapted to $\mathcal{U} \oplus \mathcal{U}^\perp$ by Corollary 1.3.11 and Theorem 2.1.5. Set $A := [T]_{\beta \leftarrow \beta}$ and $R := I_r \oplus 0_{n-r}$. The matrix A is normal by Theorem 2.3.7, Proposition 3.4.4, and normality of T. The coordinate subspace $\mathbb{F}^r \oplus \{\mathbf{0}\}$ is invariant under A, and R is its orthogonal projection. Thus $AR = RA$ by Theorem 6.2.9, so $A = B \oplus C$ for square blocks B, C.

(i) The block C shows that $\mathcal{U}^\perp$ is invariant under T.
(ii) The identity $[T^*]_{\beta \leftarrow \beta} = A^* = B^* \oplus C^*$ holds by Theorem 2.3.7, so $\mathcal{U}$ is invariant under T^*.
(iii) The subspaces $\mathcal{U}$ and $\mathcal{U}^\perp$ are invariant under T by the hypothesis and part (i), while $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$ by Theorem 2.1.5. Thus $(\mathcal{U}, \mathcal{U}^\perp)$ reduces T by Definition 5.2.6, and $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$ by Theorem 5.2.7(iii).
(iv) The normality identity $A^*A = AA^*$ implies $B^*B = BB^*$ and $C^*C = CC^*$ blockwise. These are the orthonormal-basis matrices of the adjoint-product identities for $T|_{\mathcal{U}}$ and $T|_{\mathcal{U}^\perp}$ by Theorem 2.3.7 and Proposition 3.4.4. Hence both restrictions are normal by Theorem 3.4.5(i).

Structural Commentary: Normal operator turns every invariant subspace into an orthogonal reducing summand.
Key Insight: Normality kills the one-way coupling between $\mathcal{U}$ and $\mathcal{U}^\perp$. □

Theorem 6.2.11. *Simultaneous Unitary Diagonalisation of Commuting Normal Operators*
Let $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a family of normal operators on a finite-dimensional complex inner-product space. Then $\mathcal{F}$ is pairwise commuting if and only if $\mathcal{V}$ has an orthonormal basis of common eigenvectors for $\mathcal{F}$.

Proof.

(⇒) Assume $\mathcal{F}$ is pairwise commuting and induct on $\dim(\mathcal{V})$. If every $A \in \mathcal{F}$ is scalar, fix any orthonormal basis by Corollary 1.3.11; it is a common eigenbasis, including when $\mathcal{V} = \{\mathbf{0}\}$. Otherwise choose a nonscalar $A_0 \in \mathcal{F}$ and write its spectral resolution as $A_0 = \sum_{j=1}^{d} \mu_j P_j$ with nonzero projections P_j by Theorem 6.1.13. Since A_0 is nonscalar, $d \geq 2$, so every $\operatorname{range}(P_j)$ has dimension strictly between 0 and $\dim(\mathcal{V})$. Each P_j is a polynomial in A_0 by Corollary 6.1.15.
Fix $B \in \mathcal{F}$. Since $BA_0 = A_0B$, induction on powers and linearity show that B commutes with every polynomial in A_0; hence $BP_j = P_jB$. If $\mathbf{x} \in \operatorname{range}(P_j)$, then $P_jB\mathbf{x} = BP_j\mathbf{x} = B\mathbf{x}$, so $\operatorname{range}(P_j)$ is invariant under B. The restriction $B|_{\operatorname{range}(P_j)}$ is normal by Corollary 6.2.10(iv). Thus the restricted family on each $\operatorname{range}(P_j)$ is pairwise commuting and normal. The induction hypothesis supplies a common orthonormal eigenbasis on every range. Concatenating these bases produces a common orthonormal eigenbasis of $\mathcal{V} = \bigoplus_{j=1}^{d} {}^{\perp} \operatorname{range}(P_j)$ by Theorem 6.1.13.

(⇐) If $\mathcal{V} = \{\mathbf{0}\}$, all operators commute. Otherwise, in a common orthonormal eigenbasis, every $A \in \mathcal{F}$ is diagonal. Diagonal matrices commute, so the operators commute by Proposition 3.4.4 and Theorem 3.4.5(i).

Structural Commentary: Spectral projections reduce the whole commuting family.
Key Insight: Diagonalise one nonscalar member, then induct on its eigenspaces. □

Figure 50 illustrates the reduction to the spectral subspaces of one nonscalar family member in Theorem 6.2.11. The theorem, rather than the diagram, owns the common-eigenbasis conclusion.

Figure 50: For $A_0 = \sum_j \mu_j P_j$, commutativity gives $BP_j = P_jB$; the ranges reduce $\mathcal{F}$, and induction gives a common orthonormal eigenbasis.

Corollary 6.2.12. *Sums and Products of Commuting Normal Operators*
Let $S, T \in \mathcal{L}(\mathcal{V})$ be commuting normal operators on a finite-dimensional complex inner-product space. Then $S + T$ and ST are normal.

Proof. Fix a common orthonormal eigenbasis for S and T by Theorem 6.2.11. The operators $S + T$ and ST are diagonal in that basis and therefore normal by Theorem 6.1.11(iii),(i).
Structural Commentary: Commuting normal operators share orthonormal coordinates.
Key Insight: In common diagonal form, sums and products remain diagonal. □

Perspective 6.2.13. *Real Normal Operators Use Planes*
Over $\mathbb{R}$, normal operators need not have enough eigenvectors. The replacement for a missing conjugate eigenpair is a two-dimensional invariant plane carrying a rotation-dilation block.

Proposition 6.2.14. *Two-Dimensional Real Normal Blocks*
Let $\mathcal{V}$ be a two-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.

Then T is normal but not self-adjoint if and only if, in every orthonormal basis, $[T] = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $b \neq 0$.

Moreover, some orthonormal basis makes $b > 0$.

Proof.

(⇒) Fix an arbitrary orthonormal basis and write $A = [T] = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$.
The matrix satisfies $A^tA = AA^t$ and $A \neq A^t$ by Theorem 2.3.7, Proposition 3.4.4, and Theorem 3.4.5(i). Comparing the $(1,1)$ entries of the normality equation shows $b^2 = c^2$. If $c = b$, then $A = A^t$, a contradiction. Thus $c = -b$ and $b \neq 0$. The off-diagonal entries now satisfy $-ab + bd = ab - bd$, so $d = a$. Since the basis was arbitrary, every orthonormal basis has the displayed form.

(⇐) Choose an orthonormal basis. By hypothesis, its matrix $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ has $b \neq 0$. Direct multiplication shows $A^tA = AA^t$, while $A \neq A^t$. Therefore T is normal and not self-adjoint by Theorem 2.3.7, Proposition 3.4.4, and Theorem 3.4.5(i).

For the final assertion, start with any orthonormal basis and its nonzero parameter b. If $b < 0$, replacing the second basis vector by its negative replaces b by $-b > 0$; if $b > 0$, keep the basis.
Structural Commentary: In real dimension 2, non-self-adjoint normality is exactly a scalar plus a rotation block.
Key Insight: Normality forces $c = -b$ and $d = a$. □

Theorem 6.2.15. *Real Normal Operator Block Form*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then T is normal if and only if some orthonormal basis makes $[T]$ block diagonal, with each diagonal block either 1-by-1 or of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $b > 0$.

Proof.

(⇒) Induct on $n = \dim(\mathcal{V})$. The cases $n \leq 1$ are immediate.
For $n \geq 2$, fix an invariant subspace $\mathcal{U}$ with $\dim(\mathcal{U}) \in \{1, 2\}$ by Theorem 5.3.14. The subspace $\mathcal{U}^{\perp}$ is invariant under T, and both restrictions are normal, by Corollary 6.2.10(i),(iv). If $\dim(\mathcal{U}) = 1$, a unit vector in $\mathcal{U}$ produces a scalar block. If $\dim(\mathcal{U}) = 2$ and $T|_{\mathcal{U}}$ is self-adjoint, fix an orthonormal eigenbasis of $\mathcal{U}$ by Theorem 6.1.12(i),(ii), producing two scalar blocks. If the restriction is not self-adjoint, fix the required two-dimensional block basis by Proposition 6.2.14. The induction hypothesis supplies the required orthonormal block basis on $\mathcal{U}^{\perp}$. Concatenating the two bases produces an orthonormal basis of $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$ by Theorem 2.1.5.

(⇐) If $\mathcal{V} = \{\mathbf{0}\}$, normality is immediate. Otherwise fix the stated orthonormal basis and set $A := [T]$. Each allowed block D satisfies $D^*D = DD^*$, so the block diagonal matrix satisfies $A^*A = AA^*$. The corresponding operator identity $T^*T = TT^*$ follows by Theorem 2.3.7, Proposition 3.4.4, and Theorem 3.4.5(i). Hence T is normal.

Structural Commentary: Normality lets invariant lines or planes split orthogonally.
Key Insight: Real normal operators split into scalar lines and rotation-dilation planes. □

Figure 51 summarizes Theorem 6.2.15. The theorem, rather than the diagram, owns the orthogonal line–plane decomposition.

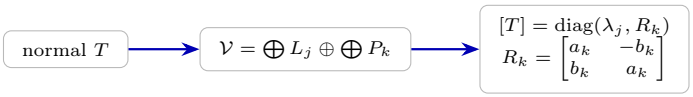

Figure 51: Orthogonal line/plane split into scalar and rotation blocks.

Theorem 6.2.16. *Description of Complex Isometries*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $S \in \mathcal{L}(\mathcal{V})$. Then S is an isometry if and only if $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of S whose eigenvalues all have absolute value 1.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, both conditions hold for its unique operator and the empty orthonormal basis. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

($\Rightarrow$) Suppose S is an isometry. Then $S^*S = I_{\mathcal{V}}$ by Theorem 2.6.6. The map S is injective by norm preservation and therefore invertible by Theorem 1.7.7(ii),(i). Thus $S^* = S^{-1}$ and $SS^* = I_{\mathcal{V}}$, so S is normal. Fix an orthonormal eigenbasis $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$ by Theorem 6.1.11(i),(ii). Since each $\mathbf{e}_j$ is unit and S preserves norms, $|\lambda_j| = \|S\mathbf{e}_j\| = 1$.

($\Leftarrow$) Suppose $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$ in an orthonormal basis with $|\lambda_j| = 1$. If $\mathbf{v} = \sum_j c_j \mathbf{e}_j$, then

$$\|S\mathbf{v}\|^2 = \sum_j |\lambda_j c_j|^2 = \sum_j |c_j|^2 = \|\mathbf{v}\|^2$$

by Theorem 1.4.7(i). Hence S is an isometry.

Structural Commentary: Complex isometries are normal with unit-circle spectrum.
Key Insight: In an eigenbasis, isometry means $|\lambda_j| = 1$. □

Example 6.2.17. *Plane Rotations*
For $\theta \in \mathbb{R}$, the real matrix $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is an isometry of $\mathbb{R}^2$. If θ is not an integer multiple of π, then R_θ has no real eigenvalue, which is why real isometries require two-dimensional rotation blocks.

Theorem 6.2.18. *Real Isometry Block Form*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $S \in \mathcal{L}(\mathcal{V})$. Then S is an isometry if and only if some orthonormal basis makes $[S]$ block diagonal, with each diagonal block one of $[1]$, $[-1]$, $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ with $0 < \theta < \pi$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, both conditions hold for its unique operator and the empty orthonormal basis. Assume $\dim(\mathcal{V}) \geq 1$.

($\Rightarrow$) Suppose S is an isometry. Then $S^*S = I_{\mathcal{V}}$ by Theorem 2.6.6. The map S is injective by norm preservation and therefore invertible by Theorem 1.7.7(ii),(i). Hence $S^* = S^{-1}$, so $SS^* = I_{\mathcal{V}}$ and S is normal. Fix a block basis by Theorem 6.2.15. In this basis, $[S]^t[S] = I$ by Theorem 2.3.7 and Proposition 3.4.4. Thus every scalar block $[\lambda]$ satisfies $\lambda^2 = 1$, so $\lambda = \pm 1$, and every two-dimensional block $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ satisfies $a^2 + b^2 = 1$ with $b > 0$. Therefore $a = \cos\theta$ and $b = \sin\theta$ for a unique $\theta \in (0, \pi)$.

($\Leftarrow$) Fix a basis with the stated block form. Each displayed block D satisfies $D^tD = I$ on its coordinate subspace, so $[S]^t[S] = I$. Hence $S^*S = I_{\mathcal{V}}$ by Theorem 2.3.7, Proposition 3.4.4, and Theorem 3.4.5(i). Thus S is an isometry by Theorem 2.6.6.

Structural Commentary: Isometries are the unit-length special case of real normal blocks.
Key Insight: Scalar blocks reduce to ± 1, and planar normal blocks reduce to rotations. □

Figure 52 summarizes Theorem 6.2.18. The theorem, rather than the diagram, owns the classification by fixed lines, flipped lines, and rotation planes.

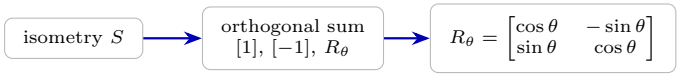

Figure 52: Normal form: fixed/flipped lines and rotation planes.

6.3 Operator Functions

This subsection develops finite spectral functions of normal operators, positive square roots, and commutativity through spectral projections.

Definition 6.3.1. *Spectral Resolution Notation*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space.
Set $\Omega_T := \operatorname{spec}(T)$. If $\mathcal{V} = \{\mathbf{0}\}$, use the empty family of spectral projections. Otherwise, T has an orthogonal spectral resolution indexed by Ω_T by Theorem 6.1.13. We use the following notation:

(i) *Spectral Set*: $\Omega_T = \operatorname{spec}(T)$.
(ii) *Spectral Projection*: P_λ is the orthogonal projection onto $E_\lambda(T)$.
(iii) *Spectral Resolution*: $\{P_\lambda\}_{\lambda \in \Omega_T}$, with $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $I_{\mathcal{V}} = \sum_{\lambda \in \Omega_T} P_\lambda$.

Theorem 6.3.2. *Finite Spectral Functional Calculus for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space, with spectral resolution $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$.

The finite spectral functional calculus is the map $\mathbb{C}^{\Omega_T} \to \mathcal{L}(\mathcal{V})$ defined by $f \mapsto f(T) := \sum_{\lambda \in \Omega_T} f(\lambda) P_\lambda$.

If $p \in \mathbb{C}[z]$ and $p|_{\Omega_T} = f$, then $p(T) = f(T)$. Hence every $f \in \mathbb{C}^{\Omega_T}$ has a polynomial representative in T.

Proof. If $\Omega_T = \emptyset$, then $\mathcal{V} = \{\mathbf{0}\}$ by Theorem 6.1.11(i),(ii). Hence $p(T) = f(T)$ is the unique operator on $\mathcal{V}$ for every polynomial p, and the zero polynomial represents the unique function on Ω_T. Assume $\Omega_T \neq \emptyset$. If $p|_{\Omega_T} = f$, then

$$p(T) = \sum_{\lambda \in \Omega_T} p(\lambda) P_\lambda = \sum_{\lambda \in \Omega_T} f(\lambda) P_\lambda = f(T)$$

by Corollary 6.1.15. Define

$$p_f(z) := \sum_{\lambda \in \Omega_T} f(\lambda) \prod_{\substack{\mu \in \Omega_T \\ \mu \neq \lambda}} \frac{z - \mu}{\lambda - \mu}.$$

Then $p_f(\lambda) = f(\lambda)$ for every $\lambda \in \Omega_T$, so $p_f(T) = f(T)$ by the first part.
Structural Commentary: Spectral sums reduce finite functions to polynomial evaluation.
Key Insight: Finite spectral data is interpolated by one polynomial. □

Theorem 6.3.3. *Rules for Normal Spectral Functions*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space, with spectral resolution $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$. For $f, g : \Omega_T \to \mathbb{C}$:

(i) $(f+g)(T) = f(T) + g(T)$ *and* $(fg)(T) = f(T)g(T)$;
(ii) $f(T)^* = \overline{f}(T)$, *where* $\overline{f}(\lambda) = \overline{f(\lambda)}$;
(iii) $f(T)$ *is normal and* $\operatorname{spec}(f(T)) = f(\Omega_T)$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, all three assertions hold because every operator equality is automatic and both spectra in (iii) are empty. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(i) By Theorem 6.1.13,

$$f(T)g(T) = \sum_{\lambda \in \Omega_T} f(\lambda) g(\lambda) P_\lambda = (fg)(T), \qquad (f+g)(T) = f(T) + g(T).$$

(ii) Since $P_\lambda^* = P_\lambda$ by Theorem 2.5.6,

$$f(T)^* = \sum_{\lambda \in \Omega_T} \overline{f(\lambda)} P_\lambda = \overline{f}(T).$$

(iii) For each $\lambda \in \Omega_T$, the restriction of $f(T)$ to $\operatorname{range}(P_\lambda)$ is $f(\lambda) I$. The orthogonal direct sum of these ranges is $\mathcal{V}$ by Theorem 6.1.13. Thus $f(T)$ is diagonal in an orthonormal basis and therefore normal by Theorem 6.1.11(iii),(i). Every $f(\lambda)$ is an eigenvalue because $\operatorname{range}(P_\lambda) \neq \{\mathbf{0}\}$, and the direct-sum decomposition leaves no other eigenvalues. Hence $\operatorname{spec}(f(T)) = f(\Omega_T)$.

Structural Commentary: Spectral projections make operator rules pointwise.
Key Insight: Spectral functions act as scalars on spectral subspaces. □

Definition 6.3.4. *Positive Operator*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An operator $T \in \mathcal{L}(\mathcal{V})$ is *positive* if $T = T^*$ and $\langle T\mathbf{v}, \mathbf{v} \rangle \geq 0$ for all $\mathbf{v} \in \mathcal{V}$.

Definition 6.3.5. *Square Root of an Operator*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
An operator $R \in \mathcal{L}(\mathcal{V})$ is a *square root* of $T \in \mathcal{L}(\mathcal{V})$ if $R^2 = T$.

Theorem 6.3.6. *Characterization of Positive Operators*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

(i) T *is positive;*
(ii) T *is self-adjoint and every eigenvalue of* T *is nonnegative;*
(iii) T *has a positive square root;*
(iv) T *has a self-adjoint square root;*
(v) there exists $R \in \mathcal{L}(\mathcal{V})$ *such that* $T = R^*R$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, all five conditions hold, so their equivalence is immediate. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(i) $\Rightarrow$ *(ii)* Positivity includes self-adjointness by Definition 6.3.4. If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\lambda \in \mathbb{R}$ by Theorem 6.1.5, and $0 \leq \langle T\mathbf{v}, \mathbf{v} \rangle = \lambda \|\mathbf{v}\|^2$, so $\lambda \geq 0$.
(ii) $\Rightarrow$ *(iii)* In the complex case, $T = T^*$ makes T normal. Fix an orthonormal eigenbasis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ by Theorem 6.1.12(i),(ii) when $\mathbb{F} = \mathbb{R}$ and by Theorem 6.1.11(i),(ii) when $\mathbb{F} = \mathbb{C}$. Write $T\mathbf{e}_j = \lambda_j \mathbf{e}_j$ with $\lambda_j \geq 0$, and define $S\mathbf{e}_j := \sqrt{\lambda_j}\mathbf{e}_j$. For $\mathbf{u} = \sum_j c_j \mathbf{e}_j$ and $\mathbf{v} = \sum_j d_j \mathbf{e}_j$,

$$\langle S\mathbf{u}, \mathbf{v} \rangle = \sum_j \sqrt{\lambda_j} c_j \overline{d_j} = \langle \mathbf{u}, S\mathbf{v} \rangle, \qquad \langle S\mathbf{v}, \mathbf{v} \rangle = \sum_j \sqrt{\lambda_j} |d_j|^2 \geq 0.$$

Thus S is positive, and $S^2\mathbf{e}_j = T\mathbf{e}_j$ for every j. Hence $S^2 = T$ by Proposition 1.5.8, so S is a positive square root.
(iii) $\Rightarrow$ *(iv)* A positive square root is self-adjoint by Definition 6.3.4.
(iv) $\Rightarrow$ *(v)* If $T = R^2$ with $R = R^*$, then $T = R^*R$.
(v) $\Rightarrow$ *(i)* If $T = R^*R$, then $T^* = T$ by Proposition 2.3.8(ii)–(iii), and $\langle T\mathbf{v}, \mathbf{v} \rangle = \langle R\mathbf{v}, R\mathbf{v} \rangle = \|R\mathbf{v}\|^2 \geq 0$ for every $\mathbf{v}$. Thus T is positive by Definition 6.3.4.

Structural Commentary: Positivity equals nonnegative spectrum and square-root factorization.
Key Insight: In an eigenbasis, the proof is scalar: $\lambda \geq 0$ iff $\sqrt{\lambda}$ exists. □

Proposition 6.3.7. *Unique Positive Square Root*
Let $\mathcal{V}$ be a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Every positive operator on $\mathcal{V}$ has a unique positive square root.

Proof. Existence holds by Theorem 6.3.6(i),(iii). If $\mathcal{V} = \{\mathbf{0}\}$, its unique operator is the unique square root. Assume $\mathcal{V} \neq \{\mathbf{0}\}$ and let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. They are nonnegative by Theorem 6.3.6(i),(ii); in the complex case, self-adjointness makes T normal. By Theorem 6.1.12(i),(ii) when $\mathbb{F} = \mathbb{R}$ and Theorem 6.1.11(i),(ii) when $\mathbb{F} = \mathbb{C}$,

$$\mathcal{V} = E_{\lambda_1}(T) \oplus^{\perp} \cdots \oplus^{\perp} E_{\lambda_m}(T).$$

Let R be a positive square root of T. Since $R^2 = T$, $RT = R^3 = TR$, so every $E_{\lambda_j}(T)$ is invariant under R. Set $R_j := R|_{E_{\lambda_j}(T)}$. The restriction R_j is self-adjoint by Proposition 6.1.8(ii), is positive by restriction of the defining quadratic inequality, and satisfies $R_j^2 = \lambda_j I$. In the complex case, self-adjointness makes R_j normal. Fix an orthonormal eigenbasis of R_j by Theorem 6.1.12(i),(ii) over $\mathbb{R}$ or Theorem 6.1.11(i),(ii) over $\mathbb{C}$. Every corresponding eigenvalue ρ is nonnegative by Theorem 6.3.6(i),(ii), while $R_j^2 = \lambda_j I$ forces $\rho^2 = \lambda_j$. Hence $\rho = \sqrt{\lambda_j}$, and $R_j = \sqrt{\lambda_j} I$ by Proposition 1.5.8. Thus every positive square root agrees on every summand in the displayed decomposition and is unique.
Structural Commentary: Each eigenspace of T determines its own scalar root.
Key Insight: Positivity selects the nonnegative root. □

Definition 6.3.8. *Positive Square Root Notation*
If T is positive, then $\sqrt{T}$ denotes the unique positive square root of T.

Example 6.3.9. *Basic Functions from the Spectral Calculus*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space, with spectral resolution $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$. The adjoint rule in Theorem 6.3.3(ii) gives $T^* = \sum_{\lambda \in \Omega_T} \overline{\lambda} P_\lambda$.
If $0 \notin \Omega_T$, then the function $f(\lambda) = \lambda^{-1}$ satisfies $Tf(T) = f(T)T = I_{\mathcal{V}}$ by Theorem 6.3.3(i), so $T^{-1} = \sum_{\lambda \in \Omega_T} \lambda^{-1} P_\lambda$. If T is positive, then $\lambda \geq 0$ on Ω_T by Theorem 6.3.6(i),(ii), and the function $g(\lambda) = \sqrt{\lambda}$ produces the unique positive square root

$$\sqrt{T} = \sum_{\lambda \in \Omega_T} \sqrt{\lambda} P_\lambda$$

by Theorem 6.3.3 and Proposition 6.3.7.

Theorem 6.3.10. *Commutativity via Spectral Projections*
Let $T, S \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space, with spectral resolutions $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $S = \sum_{\mu \in \Omega_S} \mu Q_\mu$.

(i) For $R \in \mathcal{L}(\mathcal{V})$, $RT = TR$ if and only if $RP_\lambda = P_\lambda R$ for all $\lambda \in \Omega_T$.
(ii) $TS = ST$ if and only if $P_\lambda Q_\mu = Q_\mu P_\lambda$ for all $\lambda \in \Omega_T$ and $\mu \in \Omega_S$.
(iii) If $f : \Omega_T \to \mathbb{C}$ and $g : \Omega_S \to \mathbb{C}$ are injective, then $f(T)g(S) = g(S)f(T)$ if and only if $TS = ST$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, all three equivalences are immediate. Assume $\mathcal{V} \neq \{\mathbf{0}\}$.

(*i*) ($\Rightarrow$) If $RT = TR$, induction on powers and linearity show that $Rp(T) = p(T)R$ for every $p \in \mathbb{C}[z]$. Each P_λ is a polynomial in T by Corollary 6.1.15, so $RP_\lambda = P_\lambda R$.
(*i*) ($\Leftarrow$) If $RP_\lambda = P_\lambda R$ for every $\lambda \in \Omega_T$, then

$$RT = \sum_{\lambda \in \Omega_T} \lambda R P_\lambda = \sum_{\lambda \in \Omega_T} \lambda P_\lambda R = TR.$$

(*ii*) ($\Rightarrow$) If $TS = ST$, then $SP_\lambda = P_\lambda S$ by part (i) with $R = S$. The equality $P_\lambda Q_\mu = Q_\mu P_\lambda$ follows from part (i) again, now for the normal operator S and $R = P_\lambda$.
(*ii*) ($\Leftarrow$) If all spectral projections commute pairwise, then

$$TS = \sum_{\lambda,\mu} \lambda\mu P_\lambda Q_\mu = \sum_{\lambda,\mu} \lambda\mu Q_\mu P_\lambda = ST.$$

(*iii*) ($\Rightarrow$) Suppose f and g are injective and $f(T)g(S) = g(S)f(T)$. The operators $f(T)$ and $g(S)$ are normal by Theorem 6.3.3(iii). Injectivity makes their spectral resolutions

$$f(T) = \sum_{\lambda \in \Omega_T} f(\lambda) P_\lambda, \qquad g(S) = \sum_{\mu \in \Omega_S} g(\mu) Q_\mu$$

have distinct coefficients, so their spectral projections are precisely the P_λ and Q_μ. Part (ii) for $f(T)$ and $g(S)$ makes all these projections commute, and part (ii) for T and S proves $TS = ST$.
(*iii*) ($\Leftarrow$) If $TS = ST$, then all P_λ commute with all Q_μ by part (ii). Therefore

$$f(T)g(S) = \sum_{\lambda,\mu} f(\lambda)g(\mu) P_\lambda Q_\mu = \sum_{\lambda,\mu} f(\lambda)g(\mu) Q_\mu P_\lambda = g(S)f(T).$$

Structural Commentary: Spectral projections are the coordinate idempotents of a normal operator.
Key Insight: To commute with a normal operator is to preserve each of its eigenspaces. □

Corollary 6.3.11. *A Commuting Normal Pair Comes from One Normal Operator*
Let $T, S \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space.
Then $TS = ST \iff \exists\, H \in \mathcal{L}(\mathcal{V})$ normal, $p, q \in \mathbb{C}[z]$ such that $T = p(H)$, $S = q(H)$.
If $TS = ST$, then H may be chosen in the unital algebra generated by T and S, as a finite sum of operators $c_{a,b} T^a S^b$.

Proof.
($\Rightarrow$) Suppose $TS = ST$. If $\mathcal{V} = \{\mathbf{0}\}$, choose $H = T = S = 0$ and $p = q = 0$. Assume $\mathcal{V} \neq \{\mathbf{0}\}$, and write $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $S = \sum_{\mu \in \Omega_S} \mu Q_\mu$. The projections commute by Theorem 6.3.10(ii). Set

$$\mathcal{I} := \{(\lambda, \mu) \in \Omega_T \times \Omega_S : P_\lambda Q_\mu \neq 0\}.$$

For $(\lambda,\mu) \in \mathcal{I}$, the relations $(P_\lambda Q_\mu)^2 = P_\lambda Q_\mu$ and $(P_\lambda Q_\mu)^* = P_\lambda Q_\mu$ follow from commutation and self-adjoint idempotence. Thus $P_\lambda Q_\mu$ is an orthogonal projection by Theorem 2.5.6. These projections are pairwise orthogonal, and their sum is $I_\mathcal{V}$ because

$$\sum_{\lambda,\mu} P_\lambda Q_\mu = \left(\sum_\lambda P_\lambda\right)\left(\sum_\mu Q_\mu\right) = I_\mathcal{V}.$$

Choose distinct scalars $\alpha_{\lambda,\mu}$ for $(\lambda,\mu) \in \mathcal{I}$ and set

$$H := \sum_{(\lambda,\mu)\in\mathcal{I}} \alpha_{\lambda,\mu} P_\lambda Q_\mu.$$

Choose an orthonormal basis in every joint projection range by Corollary 1.3.11. Their concatenation is an orthonormal eigenbasis for H, so H is normal by Theorem 6.1.11(ii),(i), with spectrum $\{\alpha_{\lambda,\mu} : (\lambda,\mu) \in \mathcal{I}\}$. Each P_λ is a polynomial in T and each Q_μ is a polynomial in S by Corollary 6.1.15; hence H is a finite sum of terms $c_{a,b}T^aS^b$. Define functions on $\operatorname{spec}(H)$ by $f(\alpha_{\lambda,\mu}) := \lambda$ and $g(\alpha_{\lambda,\mu}) := \mu$. Fix polynomial representatives p, q for f, g by Theorem 6.3.2. Then

$$p(H) = \sum_{(\lambda,\mu)\in\mathcal{I}} \lambda P_\lambda Q_\mu = T, \qquad q(H) = \sum_{(\lambda,\mu)\in\mathcal{I}} \mu P_\lambda Q_\mu = S.$$

($\Leftarrow$) If $T = p(H)$ and $S = q(H)$, then $TS = ST$ by Proposition 1.1.6(ii).

Structural Commentary: One scalar labels each common spectral block.
Key Insight: Joint spectral projections reduce a pair of commuting normal operators to finite scalar data. □

Part V
Polynomial Structure and Canonical Forms

1 Operator Polynomials

1.1 Polynomial Functional Calculus

Theorem 1.1.1. *Polynomial Functional Calculus for Diagonalizable Matrices*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, and suppose
$A = S\Lambda S^{-1}$ for some $S \in \mathrm{Mat}(n,n,\mathbb{F})^{\times}$ and $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$.
If $p \in \mathbb{F}[z]$, then $p(A) = S\,\mathrm{diag}(p(\lambda_1),\dots,p(\lambda_n))S^{-1}$.

Proof. For every $j \geq 0$, $A^j = S\Lambda^j S^{-1}$ by induction and the matrix product laws in Proposition 3.1.2.
If $p(z) = \sum_{j=0}^{d} c_j z^j$, then

$$p(A) = \sum_{j=0}^{d} c_j A^j = S\left(\sum_{j=0}^{d} c_j \Lambda^j\right) S^{-1} = Sp(\Lambda)S^{-1}.$$

by Definition 3.8.1 and the displayed power identity.
Since Λ is diagonal, $p(\Lambda) = \mathrm{diag}(p(\lambda_1),\dots,p(\lambda_n))$.
Structural Commentary: Diagonalization reduces $p(A)$ to scalar evaluation on eigenvalues.
Key Insight: Compute functions of A by applying the function to the eigenvalues. □

2 Characteristic and Minimal Polynomials

2.1 Characteristic Polynomial Foundations

Motivation 2.1.1. *Two Spectral Polynomials*
The characteristic polynomial records eigenvalues with algebraic multiplicity.
The minimal polynomial records the shortest polynomial relation satisfied by the operator.

Definition 2.1.2. *Characteristic Polynomial*
Let $n \geq 1$. For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, the *characteristic polynomial* of A is $p_A(z) := \det(zI_n - A)$.

Theorem 2.1.3. *Basic Form of the Characteristic Polynomial*
Let $n \geq 1$ and let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{F})$. Then p_A is monic of degree n and $p_A(z) = z^n - (\mathrm{tr}\,A)z^{n-1} + \cdots + (-1)^n \det A$.
Moreover, $\mathrm{spec}(A) = \{\lambda \in \mathbb{F} : p_A(\lambda) = 0\}$.

Proof. The only determinant term of degree greater than $n-2$ comes from the identity permutation:

$$(z-a_{11})(z-a_{22})\cdots(z-a_{nn}) = z^n - (a_{11}+\cdots+a_{nn})z^{n-1} + \cdots.$$

Thus p_A is monic and the coefficient of z^{n-1} is $-\mathrm{tr}\,A$; also $p_A(0) = \det(-A) = (-1)^n \det A$. Finally, $p_A(\lambda) = 0$ if and only if $\lambda I_n - A$ is singular, equivalently $\ker(A - \lambda I_n) \neq \{\mathbf{0}\}$.
Structural Commentary: The diagonal product determines the top two coefficients.
Key Insight: Eigenvalues are precisely the roots of $\det(zI_n - A)$. □

Definition 2.1.4. *Algebraic Multiplicity*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. For $\lambda \in \mathbb{F}$, let $m_A(\lambda)$ be the multiplicity of λ as a root of p_A, with $m_A(\lambda) = 0$ if $\lambda \notin \mathrm{spec}(A)$.
For $\lambda \in \mathrm{spec}(A)$, this is the *algebraic multiplicity* of λ. Unqualified multiplicity means algebraic multiplicity.

Figure 53 summarizes the passage from a matrix to its characteristic polynomial and roots.

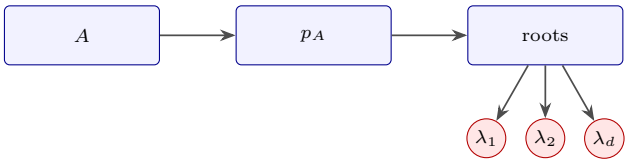

Figure 53: Form $p_A(z) = \det(zI_n - A)$ and factor it; each root λ is counted by $m_A(\lambda)$.

Method 2.1.5. *Characteristic Equation and Eigenspace Computation*
For $n \geq 1$, compute eigenvalues and eigenspaces of $A \in \mathrm{Mat}(n,n,\mathbb{F})$ as follows:

(i) Form the characteristic polynomial $p_A(z) = \det(zI_n - A)$ from Definition 2.1.2.
(ii) Solve the characteristic equation $p_A(\lambda) = 0$. By Theorem 2.1.3, the solutions are exactly the eigenvalues in $\mathbb{F}$.
(iii) For each eigenvalue λ, row-reduce $A - \lambda I_n$ and compute

$$E_\lambda(A) = \ker(A - \lambda I_n).$$

(iv) Choose a basis of each eigenspace. The dimension of this basis is the geometric multiplicity of λ.

Algebraic multiplicity comes from the factorisation of p_A, while geometric multiplicity comes from the nullspace computation.

Proposition 2.1.6. *Upper-Triangular Eigenvalues*
Let $n \geq 1$. If $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{F})$ is upper triangular, then $p_A(z) = \prod_{j=1}^{n}(z - a_{jj})$.
Thus $\mathrm{spec}(A) = \{a_{jj} : 1 \leq j \leq n\}$, and for $\lambda \in \mathrm{spec}(A)$, $m_A(\lambda) = \#\{j : a_{jj} = \lambda\}$.

Proof. Since $zI_n - A$ is upper triangular with diagonal entries $z - a_{jj}$, $p_A(z) = \det(zI_n - A) = \prod_{j=1}^{n} (z - a_{jj})$.
Thus the roots of p_A are exactly the diagonal entries, and the exponent of $(z - \lambda)$ is $\#\{j : a_{jj} = \lambda\}$.
Structural Commentary: Triangular form removes all off-diagonal determinant contributions.
Key Insight: Once a matrix is triangular, the spectrum is visible on the diagonal. □

Theorem 2.1.7. *Similarity Preserves Characteristic Data*
Let $n \geq 1$, let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, and suppose $B = SAS^{-1}$ for some $S \in \mathrm{Mat}(n, n, \mathbb{F})^{\times}$. Then $p_B = p_A$. In particular, for each $\lambda \in \mathrm{spec}(A)$, $m_B(\lambda) = m_A(\lambda)$ and $\dim E_\lambda(B) = \dim E_\lambda(A)$.

Proof. As an identity in $\mathbb{F}[z]$,

$$p_B(z) = \det(zI_n - SAS^{-1}) = \det(S(zI_n - A)S^{-1}) = \det(S)\det(zI_n - A)\det(S)^{-1} = p_A(z).$$

Thus roots and root multiplicities agree, so $m_B(\lambda) = m_A(\lambda)$. Also $B - \lambda I_n = S(A - \lambda I_n)S^{-1}$.
Hence $\mathbf{x} \in \ker(A - \lambda I_n)$ if and only if $S\mathbf{x} \in \ker(B - \lambda I_n)$, so S restricts to an isomorphism between the two eigenspaces.
Structural Commentary: Similarity changes coordinates, not the shifted determinant or shifted kernel dimension.
Key Insight: Conjugation inserts S and S^{-1}, which cancel in determinants and identify kernels. □

Proposition 2.1.8. *Block Upper-Triangular Characteristic Polynomial*
Let $r, s \geq 1$, let $B \in \mathrm{Mat}(r, r, \mathbb{F})$, $C \in \mathrm{Mat}(r, s, \mathbb{F})$, and $D \in \mathrm{Mat}(s, s, \mathbb{F})$, and set $A := \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$. Then $p_A(z) = p_B(z)p_D(z)$, so $m_A(\lambda) = m_B(\lambda) + m_D(\lambda)$ for every $\lambda \in \mathbb{F}$.

Proof. The shifted matrix is block upper triangular, $zI - A = \begin{bmatrix} zI - B & -C \\ 0 & zI - D \end{bmatrix}$.
$p_A(z) = \det(zI - B)\det(zI - D) = p_B(z)p_D(z)$ by Theorem 1.1.11.
Roots of a product are the roots of the factors, with multiplicity added.
Structural Commentary: The off-diagonal block couples vectors but not determinant factors.
Key Insight: In block upper-triangular form, spectral data lives on the diagonal blocks. □

Theorem 2.1.9. *Matrix Schur Triangularisation with Ordered Diagonal*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity and listed in a prescribed order. If $A\mathbf{x} = \lambda_1\mathbf{x}$ and $\|\mathbf{x}\|_2 = 1$, then there is a unitary $U = [\mathbf{x}\ U_2]$ such that U^*AU is upper triangular with diagonal $\lambda_1, \ldots, \lambda_n$.
If A is real, all $\lambda_i \in \mathbb{R}$, and $\mathbf{x} \in \mathbb{R}^n$, then U may be chosen real orthogonal.

Proof. Induct on n. The case $n = 1$ is immediate. Extend $\mathbf{x}$ to an orthonormal basis by Corollary 1.3.12, and set $V = [\mathbf{x}\ V_2]$.
Since $A\mathbf{x} = \lambda_1\mathbf{x}$, the first column of V^*AV is $\lambda_1\mathbf{e}_1$, so $V^*AV = \begin{bmatrix} \lambda_1 & * \\ 0 & A' \end{bmatrix}$.
By Theorem 2.1.7 and Proposition 2.1.8, $p_A(z) = p_{V^*AV}(z) = (z - \lambda_1)p_{A'}(z)$. Thus A' has eigenvalues $\lambda_2, \ldots, \lambda_n$, counted with algebraic multiplicity. By induction, choose unitary W with $W^*A'W = T'$ upper triangular and diagonal $\lambda_2, \ldots, \lambda_n$.
Then $U := V(I_1 \oplus W)$ has first column $\mathbf{x}$ and $U^*AU = \begin{bmatrix} \lambda_1 & * \\ 0 & T' \end{bmatrix}$.
The real case uses real orthonormal extensions; the block A' is real, and the same characteristic-polynomial factorization leaves the real eigenvalue list $\lambda_2, \ldots, \lambda_n$ for the induction step.
Structural Commentary: Isolate the chosen eigenline, then repeat on the lower-right block.
Key Insight: The diagonal order is controlled by which eigenline is peeled off first. □

Figure 54 summarizes the induction by isolating the prescribed first eigenline.

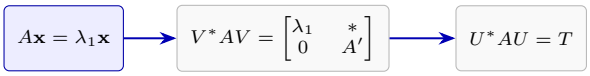

Figure 54: Schur induction: isolate an eigenline, then triangularise the compressed block.

Corollary 2.1.10. *Eigenvalues of the Inverse*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be invertible with eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity. Then A^{-1} has eigenvalues $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$, counted with algebraic multiplicity.

Proof. Choose an eigenvector $\mathbf{y}$ corresponding to λ_1 and set $\mathbf{x} := \mathbf{y}/\|\mathbf{y}\|_2$. Then $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 / \|\mathbf{y}\|_2 = 1$.
By Theorem 2.1.9, $A = UTU^*$ for a unitary U and an upper-triangular T with diagonal $\lambda_1, \ldots, \lambda_n$.
Then $A^{-1} = UT^{-1}U^*$, and T^{-1} is upper triangular with diagonal $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$ by Theorem 3.8.5. The eigenvalues of A^{-1}, counted with algebraic multiplicity, are therefore the displayed reciprocals by Proposition 2.1.6 and Theorem 2.1.7.
Structural Commentary: Invert the Schur form and read the diagonal.
Key Insight: The inverse spectrum is the reciprocal spectrum. □

Theorem 2.1.11. *Geometric Multiplicity Is at Most Algebraic Multiplicity*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $\lambda \in \mathrm{spec}(A)$. Then $\dim E_\lambda(A) \leq m_A(\lambda)$.

Proof. Let $k = \dim E_\lambda(A)$ and choose a basis $\mathbf{x}_1, \ldots, \mathbf{x}_k$ of $E_\lambda(A)$. Extend it to a basis of $\mathbb{F}^n$ by Theorem 1.3.17, and let S be the matrix whose columns are this basis. In the new basis,

$$S^{-1}AS = \begin{bmatrix} \lambda I_k & * \\ 0 & C \end{bmatrix}.$$

By Theorem 2.1.7 and Proposition 2.1.8, $p_A(z) = p_{S^{-1}AS}(z) = (z - \lambda)^k p_C(z)$. Thus $m_A(\lambda) \geq k = \dim E_\lambda(A)$.
Structural Commentary: An eigenspace-first basis exposes a λI_k block.
Key Insight: Each independent eigenvector forces one visible factor $z - \lambda$ in the characteristic polynomial. □

Theorem 2.1.12. *Diagonalizability and Multiplicity*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then A is similar to a diagonal matrix if and only if $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$.

Proof.
(⇒) Suppose A is similar to a diagonal matrix D. Then $m_A(\lambda) = m_D(\lambda)$ and $\dim E_\lambda(A) = \dim E_\lambda(D)$ by Theorem 2.1.7. For diagonal D, both numbers count diagonal entries equal to λ.
(⇐) Assume $\dim E_\lambda(A) = m_A(\lambda)$ for every eigenvalue.

If $\mu_1, \ldots, \mu_d$ are distinct eigenvalues, then $\sum_{i=1}^{d} m_A(\mu_i) = n$. Hence $\dim E_{\mu_1}(A) + \cdots + \dim E_{\mu_d}(A) = n$.

The induced operator on $\mathbb{C}^n$ has an eigenbasis by Theorem 5.4.3(v),(ii). Its matrix D in that eigenbasis is diagonal, and A is similar to D by Corollary 6.2.4.

Structural Commentary: Diagonalizability means root counts match eigenspace dimensions.
Key Insight: No missing eigendirections means an eigenbasis. □

Proposition 2.1.13. *Permutation Similarity Reorders Diagonal Entries*
Let $n \geq 1$, let $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, let $\sigma \in \mathfrak{S}_n$, set $\Lambda := \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, and let $P := [\mathbf{e}_{\sigma(1)} \ \cdots \ \mathbf{e}_{\sigma(n)}]$. Then $P^*\Lambda P = \mathrm{diag}(\lambda_{\sigma(1)}, \ldots, \lambda_{\sigma(n)})$.

Proof. The (i, j) entry of $P^*\Lambda P$ is $\mathbf{e}_{\sigma(i)}^* \Lambda \mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)} \mathbf{e}_{\sigma(i)}^* \mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)} \delta_{ij}$.
Thus all off-diagonal entries are zero, and the jth diagonal entry is $\lambda_{\sigma(j)}$.
Structural Commentary: Permutation similarity applies the same basis reordering to rows and columns.
Key Insight: Diagonal entries move exactly as the standard basis vectors are rearranged. □

Corollary 2.1.14. *Normal Matrices with the Same Spectrum Are Unitarily Similar*
Let $n \geq 1$. If $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are normal and have the same eigenvalues counted with algebraic multiplicity, then A and B are unitarily similar.

Proof. By Theorem 6.1.11(i),(iii), write $A = U\Lambda U^*$ and $B = V\Gamma V^*$ with U, V unitary and Λ, Γ diagonal.
Since the eigenvalue lists agree with multiplicity, choose $\sigma \in \mathfrak{S}_n$ that matches the diagonal of Λ to the diagonal of Γ. For $P = [\mathbf{e}_{\sigma(1)} \ \cdots \ \mathbf{e}_{\sigma(n)}]$, the identity $\Gamma = P^*\Lambda P$ holds by Proposition 2.1.13.
Hence $B = (VP^*U^*)A(UPV^*)$, and UPV^* is unitary by Proposition 2.6.12.
Structural Commentary: Normal matrices are classified up to unitary similarity by their eigenvalue multiset.
Key Insight: After diagonalising both matrices, only a permutation of diagonal entries remains. □

Definition 2.1.15. *Characteristic Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$, let $n = \dim(\mathcal{V})$, and let $T \in \mathcal{L}(\mathcal{V})$. For a basis β of $\mathcal{V}$, the *characteristic polynomial* of T is $q_T(z) := p_{[T]_{\beta \leftarrow \beta}}(z) = \det(zI_n - [T]_{\beta \leftarrow \beta})$.
This is independent of β by Theorems 5.1.1 and 2.1.7.

Proposition 2.1.16. *Degree and Roots of q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T.
(i) $\deg q_T = \dim(\mathcal{V})$.
(ii) The roots of q_T are precisely the eigenvalues of T.

Proof. Choose a basis β of $\mathcal{V}$ and set $A = [T]_{\beta \leftarrow \beta}$.
(i) The identity $q_T = p_A$ holds by Definition 2.1.15, so $\deg q_T = \dim(\mathcal{V})$ by Theorem 2.1.3.
(ii) For $\mathbf{v} \in \mathcal{V}$, $T\mathbf{v} = \lambda\mathbf{v}$ if and only if $A[\mathbf{v}]_\beta = \lambda[\mathbf{v}]_\beta$. Since the coordinate map is bijective, $\mathrm{spec}(T) = \mathrm{spec}(A)$. This set is the set of roots of $p_A = q_T$ by Theorem 2.1.3.
Structural Commentary: Coordinates transfer degree and eigenvalue data from A to T.
Key Insight: The characteristic polynomial is independent of the chosen basis. □

2.2 Minimal Polynomial Foundations

Theorem 2.2.1. *Existence of the Minimal Polynomial*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then T has a unique least-degree monic annihilator.

Proof. A nonconstant annihilator exists by Proposition 1.1.7. Scaling it by the inverse of its leading coefficient produces a monic annihilator, so one of least degree exists. If p and q are two such polynomials, then $(p - q)(T) = 0$ and $\deg(p - q) < \deg p$.
If $p \neq q$, the normalization of $p - q$ is a smaller-degree monic annihilator, which contradicts minimality. Hence $p = q$.
Structural Commentary: Existence comes from finite-dimensional dependence; uniqueness from minimality.
Key Insight: Two least monic annihilators cannot differ. □

Definition 2.2.2. *Minimal Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. The *minimal polynomial* of T, denoted μ_T, is the unique monic polynomial of smallest degree such that $\mu_T(T) = 0$, whose existence and uniqueness hold by Theorem 2.2.1.

Definition 2.2.3. *Minimal Polynomial of a Matrix*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $T_A \in \mathcal{L}(\mathbb{F}^n)$ be the operator induced by A from Definition 3.1.7 and Proposition 3.1.8. The *minimal polynomial* of A is $\mu_A := \mu_{T_A}$.
For every $p \in \mathbb{F}[z]$, the identity $p(T_A) = T_{p(A)}$ holds by Definition 3.8.1 and Proposition 3.4.4. Hence μ_A is equivalently the unique monic polynomial of smallest positive degree such that $\mu_A(A) = 0$.

Theorem 2.2.4. *Annihilators Are Multiples of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{F}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $f \in \mathbb{F}[z]$ be nonzero. Then $f(T) = 0$ if and only if μ_T divides f.

Proof.
(⇒) Suppose $f(T) = 0$. By Theorem 3.9.2, write $f = s\mu_T + r$, where $r = 0$ or $\deg r < \deg \mu_T$.
Then $r(T) = 0$, so minimality forces $r = 0$. Hence μ_T divides f.

($\Leftarrow$) If $f = \mu_T s$, then $f(T) = \mu_T(T)s(T) = 0$ by Proposition 1.1.6(i).
Structural Commentary: Polynomial division reduces every annihilator to its remainder modulo μ_T.
Key Insight: Minimality forces that remainder to be zero. □

Theorem 2.2.5. *Similarity Preserves the Minimal Polynomial*
Let $n \geq 1$. If $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are similar, then $p_A = p_B$ and $\mu_A = \mu_B$.

Proof. The identity $p_A = p_B$ holds by Theorem 2.1.7. Choose $S \in \mathrm{Mat}(n, n, \mathbb{C})^\times$ satisfying $A = SBS^{-1}$ by Definition 6.2.1.
For every $j \geq 0$, $A^j = SB^jS^{-1}$ by induction. Therefore $f(A) = Sf(B)S^{-1}$ for every $f \in \mathbb{C}[z]$ by Definition 3.8.1. Thus A and B have the same annihilating polynomials, and $\mu_A = \mu_B$ by Definition 2.2.3.
Structural Commentary: Similarity preserves polynomial equations.
Key Insight: Conjugation commutes with polynomial evaluation. □

3 Cayley-Hamilton and Annihilators

3.1 Cayley-Hamilton and Minimal Polynomials

Theorem 3.1.1. *Cayley–Hamilton Theorem*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have characteristic polynomial $p_A(z) = \det(zI_n - A)$. Then $p_A(A) = 0$.

Proof. List the eigenvalues of A as $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity, choose an eigenvector $\mathbf{y}$ for λ_1, and set $\mathbf{x} := \mathbf{y}/\|\mathbf{y}\|_2$. Then $\|\mathbf{x}\|_2 = 1$. By Theorem 2.1.9, write $A = UTU^*$ with U unitary and T upper triangular with diagonal $\lambda_1, \ldots, \lambda_n$.
The identity $p_A(z) = p_T(z) = \prod_{i=1}^n (z - \lambda_i)$ holds by Theorem 2.1.7 and Proposition 2.1.6. For every $k \geq 0$, $A^k = UT^kU^*$ by induction, so $p_A(A) = Up_A(T)U^*$ by Definition 3.8.1. It therefore suffices to prove $(T - \lambda_1 I_n)(T - \lambda_2 I_n) \cdots (T - \lambda_n I_n) = 0$.
Let $Q_j := \prod_{i=1}^{j} (T - \lambda_i I_n)$. We show that the first j columns of Q_j are zero. For $j = 1$, the first column of $T - \lambda_1 I_n$ is zero as T is upper triangular with first diagonal entry λ_1. Assume the first j columns of Q_j are zero.
Since $T - \lambda_{j+1} I_n$ is upper triangular and its $(j+1, j+1)$ entry is zero, its first $j+1$ columns lie in $\mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j)$. By the induction hypothesis, Q_j kills this span, so $Q_{j+1} = Q_j(T - \lambda_{j+1} I_n)$ has first $j+1$ zero columns.
At $j = n$, all columns of Q_n are zero, so $Q_n = 0$.
Structural Commentary: Schur form reduces the theorem to a triangular column-killing induction.
Key Insight: Each factor $T - \lambda_i I_n$ kills the next leading column. □

Figure 55 illustrates the leading-column induction in the Schur form.

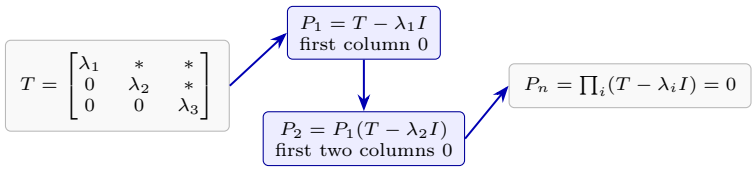

Figure 55: In Schur form, the factors $T - \lambda_i I$ kill the leading columns one layer at a time.

Theorem 3.1.2. *Cayley–Hamilton Theorem for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T. Then $q_T(T) = 0$.

Proof. Choose a basis β of $\mathcal{V}$ and set $A = [T]_{\beta \leftarrow \beta}$. The identity $q_T = p_A$ holds by Definition 2.1.15.
By Proposition 3.2.4 and Proposition 3.4.4,

$$[q_T(T)]_{\beta \leftarrow \beta} = q_T(A) = p_A(A) = 0,$$

where the final equality holds by Theorem 3.1.1. Therefore $q_T(T) = 0$ by Theorem 3.4.5.
Structural Commentary: Coordinates reduce the operator theorem to the matrix theorem.
Key Insight: Polynomial evaluation commutes with taking a matrix representation. □

Corollary 3.1.3. *Reduction Modulo the Characteristic Polynomial*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, and let $f \in \mathbb{C}[z]$. If $f = p_A q + r$ with $r = 0$ or $\deg r < n$, then $f(A) = r(A)$.

Proof. The identity $p_A(A) = 0$ holds by Theorem 3.1.1. Therefore,

$$f(A) = p_A(A)q(A) + r(A) = r(A).$$

Structural Commentary: Cayley–Hamilton makes polynomial expressions in A reducible modulo p_A.
Key Insight: Every high power of A can be rewritten using powers below n. □

Corollary 3.1.4. *Powers Span at Most n Dimensions*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then $\mathrm{span}(I_n, A, A^2, \ldots) = \mathrm{span}(I_n, A, \ldots, A^{n-1})$, so this has dimension at most n.

Proof. For each $k \geq 0$, write $z^{n+k} = p_A q_k + r_k$, where $r_k = 0$ or $\deg r_k < n$, by Theorem 3.9.2. Then $A^{n+k} = r_k(A)$ by Corollary 3.1.3. Thus every power of A lies in $\mathrm{span}(I_n, A, \ldots, A^{n-1})$; the reverse inclusion is immediate.
Structural Commentary: Polynomial reduction collapses the infinite power list to its first n terms.
Key Insight: Cayley–Hamilton encodes the recurrence relation for powers of A. □

Corollary 3.1.5. *Inverse from the Characteristic Polynomial*
Let $n \geq 1$ and let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have $p_A(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$.
If A is invertible, then

$$A^{-1} = -c_0^{-1} \left(A^{n-1} + \sum_{j=1}^{n-1} c_j A^{j-1} \right).$$

Proof. Since A is invertible, $0 \notin \operatorname{spec}(A)$ by Corollary 5.1.8. Hence $c_0 = p_A(0) \neq 0$ by Theorem 2.1.3.
By Theorem 3.1.1,

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

Move $c_0 I_n$ to the other side and multiply by $-c_0^{-1}A^{-1}$.
Structural Commentary: The characteristic equation contains an identity term exactly when A is invertible.
Key Insight: Solve Cayley–Hamilton for A^{-1} after factoring out one A. □

Theorem 3.1.6. *Basic Properties of the Minimal Polynomial*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $\lambda_1, \ldots, \lambda_d$ be the distinct eigenvalues of A.

(i) $\deg \mu_A \leq n$.

(ii) There are integers q_i satisfying $1 \leq q_i \leq m_A(\lambda_i)$ such that $\mu_A(z) = \prod_{i=1}^{d} (z - \lambda_i)^{q_i}$; in particular, $\mu_A \mid p_A$.

Proof.

(i) The polynomial p_A is a monic annihilator of degree n by Theorem 3.1.1 and Theorem 2.1.3. Hence $\deg \mu_A \leq n$ by Definition 2.2.3.

(ii) The divisibility

$$\mu_A \mid p_A = \prod_{i=1}^{d} (z - \lambda_i)^{m_A(\lambda_i)}$$

holds by Theorem 3.1.1, Theorem 2.2.4, and Definition 2.2.3. Thus every root of μ_A is an eigenvalue and its exponent is at most the corresponding algebraic multiplicity. If $A\mathbf{v} = \lambda_i \mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\mu_A(A)\mathbf{v} = \mu_A(\lambda_i)\mathbf{v} = \mathbf{0}$ by Proposition 1.1.8, so $\mu_A(\lambda_i) = 0$ and every λ_i occurs.

Structural Commentary: Cayley–Hamilton bounds degree and divisibility; eigenvectors identify the roots.
Key Insight: The least monic annihilator retains every eigenvalue with minimal exponents. □

Figure 56 summarizes the divisibility relations owned by the minimal polynomial.

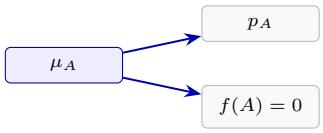

Figure 56: μ_A is the least monic annihilator; arrows indicate that μ_A divides p_A and every annihilator f.

3.2 Diagonalizability and Minimal Polynomial Tests

Theorem 3.2.1. *Diagonalizability and the Minimal Polynomial*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_d$.
Then A is diagonalizable if and only if $\mu_A(z) = (z - \lambda_1) \cdots (z - \lambda_d)$.
Equivalently, A is diagonalizable if and only if μ_A is square-free.

Proof. Set $p(z) := \prod_{i=1}^{d}(z - \lambda_i)$. By Theorem 3.1.6, $\mu_A(z) = \prod_{i=1}^{d}(z - \lambda_i)^{q_i}$ with $q_i \geq 1$, so μ_A is square-free if and only if $\mu_A = p$.

($\Rightarrow$) If $A = S\Lambda S^{-1}$ is diagonalizable, then $p(\Lambda) = 0$, so $p(A) = 0$.
Theorem 3.1.6 gives $\mu_A \mid p$; since every factor $z - \lambda_i$ occurs in μ_A, we have $\mu_A = p$.

($\Leftarrow$) Suppose $\mu_A = p$. For $i \in \{1, \ldots, d\}$, set $\ell_i(z) := \prod_{j \neq i} \frac{z - \lambda_j}{\lambda_i - \lambda_j}$ and $\rho_i := \ell_i(A)$.

The polynomials $\ell_i \ell_j$ for $i \neq j$, $\sum_i \ell_i - 1$, and $z - \sum_i \lambda_i \ell_i$ vanish at $\lambda_1, \ldots, \lambda_d$, so each is divisible by p. Since $p(A) = 0$, $\rho_i \rho_j = 0$ for $i \neq j$, $\sum_i \rho_i = I_n$, and $A = \sum_i \lambda_i \rho_i$. Thus $\rho_i^2 = \rho_i$.
If $\rho_i = 0$, then $\prod_{j \neq i}(A - \lambda_j I_n) = 0$, contradicting the minimality of the degree-d polynomial $\mu_A = p$. Hence every $\rho_i \neq 0$.
Theorem 5.4.6, applied to the operator induced by A, gives that A is diagonalizable.

Structural Commentary: Lagrange polynomials turn square-free annihilation into a spectral resolution.
Key Insight: Distinct roots produce complementary eigenspace projections. □

Corollary 3.2.2. *Square-Free Annihilator Criterion*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. If a polynomial f with no repeated roots satisfies $f(A) = 0$, then A is diagonalizable.

Proof. Theorem 2.2.4 gives $\mu_A \mid f$. Thus μ_A is square-free, so Theorem 3.2.1 applies.
Structural Commentary: A square-free annihilator forces square-free μ_A.
Key Insight: Square-free μ_A means diagonalizable. □

Method 3.2.3. *Computing μ_T from a Matrix*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be the matrix of T in some basis. Propositions 3.2.4 and 3.4.4 give $[p(T)] = p(A)$ for every $p \in \mathbb{F}[z]$; Theorem 3.4.5 then gives $\mu_T = \mu_A$.

(i) For $m = 1, \ldots, n^2$, test the monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1} z^{m-1} + z^m$.
(ii) Solve linear system by comparing matrix entries in $a_0 I + a_1 A + \cdots + a_{m-1} A^{m-1} = -A^m$ for $a_0, \ldots, a_{m-1}$.
(iii) If no solution exists, increase m. The first solvable candidate is μ_T.

Proposition 1.1.7 guarantees termination by $m = n^2$.

Figure 57 summarizes the linear-system search for the least-degree monic annihilator.

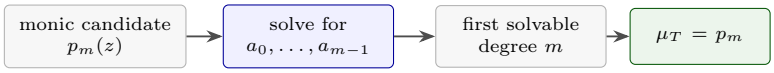

Figure 57: Solve $a_0 I + \cdots + a_{m-1}A^{m-1} = -A^m$; the first solvable monic candidate is μ_T.

Method 3.2.4. *Testing a Minimal Polynomial from p_A*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, and suppose $p_A(z) = \prod_i (z - \lambda_i)^{a_i}$ is known.

i. Write each candidate as $m_{\mathbf{q}}(z) = \prod_i (z - \lambda_i)^{q_i}$, where $1 \le q_i \le a_i$.
ii. Test exponent vectors $\mathbf{q} = (q_i)$ in increasing total degree $\sum_i q_i$.
iii. For each $\mathbf{q}$, compute $m_{\mathbf{q}}(A) = \prod_i (A - \lambda_i I)^{q_i}$.
iv. The first candidate with $m_{\mathbf{q}}(A) = 0$ is μ_A, by Theorem 3.1.6.

3.3 Annihilators and Spectral Control

Corollary 3.3.1. *q_T Is a Multiple of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then the characteristic polynomial q_T is a polynomial multiple of the minimal polynomial μ_T.

Proof. Theorem 3.1.2 gives $q_T(T) = 0$. Theorem 2.2.4 then implies that μ_T divides q_T.
Structural Commentary: Cayley–Hamilton turns the characteristic polynomial into an annihilator.
Key Insight: The minimal polynomial divides every annihilator, hence divides q_T. □

Theorem 3.3.2. *Roots of μ_T Are Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then the roots of μ_T are precisely the eigenvalues of T.

Proof.
($\Rightarrow$) If $\mu_T(\lambda) = 0$, write $\mu_T(z) = (z - \lambda)g(z)$. Since $\deg g < \deg \mu_T$, minimality gives $g(T) \neq 0$; choose $\mathbf{v}$ with $\mathbf{u} = g(T)\mathbf{v} \neq \mathbf{0}$. Then $(T - \lambda I_{\mathcal{V}})\mathbf{u} = \mu_T(T)\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $0 = \mu_T(T)\mathbf{v} = \mu_T(\lambda)\mathbf{v}$, so $\mu_T(\lambda) = 0$.

Structural Commentary: A root splits off one shifted operator.
Key Insight: The quotient cannot already annihilate T. □

Theorem 3.3.3. *Minimal Polynomial of a Spectrally Disjoint Direct Sum*
Let $A_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})$ for $i \in \{1, \dots, d\}$, let $A = A_1 \oplus \cdots \oplus A_d$, and suppose the spectra of $A_1, \dots, A_d$ are pairwise disjoint. Then $\mu_A = \mu_{A_1}\mu_{A_2}\cdots\mu_{A_d}$.

Proof. Let $f = \prod_i \mu_{A_i}$. Since $f(A_i) = 0$ for all i, blockwise evaluation gives $f(A) = 0$, and therefore $\mu_A \mid f$.

Conversely, $\mu_A(A_i) = 0$ for each i, so $\mu_{A_i} \mid \mu_A$ for each i. The roots of μ_{A_i} are exactly $\mathrm{spec}(A_i)$ by Theorem 3.3.2, so the disjointness of the spectra makes the polynomials μ_{A_i} pairwise relatively prime.
Hence their product f divides μ_A. Since $\mu_A \mid f$ and $f \mid \mu_A$, and both are monic, $\mu_A = f$.
Structural Commentary: Direct sums are annihilated blockwise; disjoint spectra make the block factors coprime.
Key Insight: Each spectral component contributes its own minimal-polynomial factor. □

Corollary 3.3.4. *Minimal and Characteristic Polynomials Have the Same Roots*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then $\{\lambda \in \mathbb{C} : \mu_T(\lambda) = 0\} = \{\lambda \in \mathbb{C} : q_T(\lambda) = 0\} = \mathrm{spec}(T)$.

Proof. By Theorem 3.3.2, $\{\lambda \in \mathbb{C} : \mu_T(\lambda) = 0\} = \mathrm{spec}(T)$.
By Proposition 2.1.16, $\{\lambda \in \mathbb{C} : q_T(\lambda) = 0\} = \mathrm{spec}(T)$.
Structural Commentary: Different multiplicities, same spectral scalars.
Key Insight: Minimal polynomial roots and characteristic polynomial roots are both just eigenvalues. □

Perspective 3.3.5. *Characteristic vs. Minimal Polynomial*
The characteristic polynomial records eigenvalues with algebraic multiplicity. The minimal polynomial has the same roots, with the least exponents needed to annihilate the operator.

3.4 Polynomial Spectral Mapping

Proposition 3.4.1. *Eigenvalues of a Polynomial in a Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, let $p \in \mathbb{C}[z]$, and let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity. Then the eigenvalues of $p(A)$ are $p(\lambda_1), \dots, p(\lambda_n)$, counted with algebraic multiplicity.

Proof. Choose a Schur form $U^*AU = T$ with diagonal $\lambda_1, \dots, \lambda_n$. Then $p(A) = Up(T)U^*$, and $p(T)$ is upper triangular with diagonal $p(\lambda_1), \dots, p(\lambda_n)$. Hence $p(A)$ has precisely these eigenvalues.
Structural Commentary: Schur form reduces the claim to the diagonal.
Key Insight: A polynomial preserves triangular form and applies p to each diagonal entry. □

Corollary 3.4.2. *Polynomial Spectral Mapping*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $p \in \mathbb{C}[z]$. Then $\mathrm{spec}(p(A)) = \{p(\lambda) : \lambda \in \mathrm{spec}(A)\}$.

Proof. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity.
By Proposition 3.4.1, the eigenvalues of $p(A)$, counted with algebraic multiplicity, are $p(\lambda_1), \dots, p(\lambda_n)$. Taking underlying sets gives the claim.
Structural Commentary: Polynomial evaluation pushes the spectrum forward through the scalar polynomial.
Key Insight: Once Schur form has exposed the diagonal, p acts entry by entry there. □

Corollary 3.4.3. *Polynomial Spectral Mapping for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $p \in \mathbb{C}[z]$.
Then $\operatorname{spec}(p(T)) = \{p(\lambda) : \lambda \in \operatorname{spec}(T)\}$.

Proof. Choose a basis β and set $A = [T]_{\beta \leftarrow \beta}$.
Then $[p(T)]_{\beta \leftarrow \beta} = p(A)$, and coordinate isomorphism gives $\operatorname{spec}(T) = \operatorname{spec}(A)$ and $\operatorname{spec}(p(T)) = \operatorname{spec}(p(A))$.
By Corollary 3.4.2, $\operatorname{spec}(p(T)) = \operatorname{spec}(p(A)) = \{p(\lambda) : \lambda \in \operatorname{spec}(A)\} = \{p(\lambda) : \lambda \in \operatorname{spec}(T)\}$.
Structural Commentary: Coordinates reduce the operator statement to the matrix statement.
Key Insight: Polynomial spectral mapping is an operator fact seen through any convenient basis. □

Corollary 3.4.4. *Polynomial Invertibility Test*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ and $p \in \mathbb{C}[z]$. If $p(\lambda) \neq 0$ for every $\lambda \in \operatorname{spec}(A)$, then $p(A)$ is invertible.

Proof. By Corollary 3.4.2, $0 \notin \operatorname{spec}(p(A))$. By Corollary 5.1.8, $p(A)$ is invertible.
Structural Commentary: Invertibility of $p(A)$ is checked on spectral values of A.
Key Insight: Avoiding zero on the spectrum prevents $p(A)$ from having a kernel. □

3.5 Polynomial Invariance and Block Separation

Proposition 3.5.1. *Kernel and Range of a Polynomial in an Operator Are Invariant*
Let $T \in \mathcal{L}(\mathcal{V})$ and $p \in \mathbb{F}[z]$. Then $\ker(p(T))$ and $\operatorname{range}(p(T))$ are invariant under T.

Proof. If $\mathbf{v} \in \ker(p(T))$, then $p(T)\mathbf{v} = \mathbf{0}$.
Since $p(T)$ commutes with T by Proposition 1.1.6, $p(T)(T\mathbf{v}) = T(p(T)\mathbf{v}) = T\mathbf{0} = \mathbf{0}$.
If $\mathbf{v} \in \operatorname{range}(p(T))$, write $\mathbf{v} = p(T)\mathbf{u}$. Then $T\mathbf{v} = Tp(T)\mathbf{u} = p(T)(T\mathbf{u})$, so $T\mathbf{v} \in \operatorname{range}(p(T))$.
Structural Commentary: Polynomial kernels and ranges are stable under the original operator.
Key Insight: Since T commutes with $p(T)$, it preserves both $\ker p(T)$ and $\operatorname{range} p(T)$. □

Theorem 3.5.2. *Sylvester Linear Matrix Equation*
Let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$. If $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$, then for every $C \in \operatorname{Mat}(m, n, \mathbb{C})$ the equation $AX - XB = C$ has a unique solution $X \in \operatorname{Mat}(m, n, \mathbb{C})$.

Proof. Let $L(X) = AX - XB$. Since L has equal finite-dimensional domain and codomain, it suffices to show $\ker L = \{0\}$. If $AX = XB$, then $p(A)X = Xp(B)$ for every polynomial p.
With $p = p_B$, $p_B(B) = 0$ by Theorem 3.1.1, so $p_B(A)X = Xp_B(B) = 0$. For every $\lambda \in \operatorname{spec}(A)$, disjointness gives $p_B(\lambda) \neq 0$; hence $p_B(A)$ is invertible by Corollary 3.4.4. Thus $X = 0$, so L is bijective.
Structural Commentary: Polynomial calculus passes through homogeneous solutions; disjoint spectra invert $p_B(A)$.
Key Insight: Spectral separation turns an intertwining relation into the zero relation. □

Figure 58 summarizes the spectral-separation argument behind invertibility of the Sylvester map.

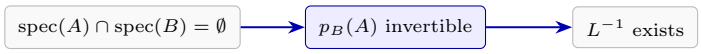

Figure 58: Disjoint spectra make $p_B(A)$ invertible, so the Sylvester map $L(X) = AX - XB$ is invertible.

Corollary 3.5.3. *Removing a Spectrally Separated Off-Diagonal Block*
Let $A \in \operatorname{Mat}(m, m, \mathbb{C})$, $B \in \operatorname{Mat}(n, n, \mathbb{C})$, $C \in \operatorname{Mat}(m, n, \mathbb{C})$, and $M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$. If $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$, then M is similar to $A \oplus B$.

Proof. By Theorem 3.5.2, choose X such that $AX - XB = -C$. Then

$$\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix} \begin{bmatrix} A & C \\ 0 & B \end{bmatrix} \begin{bmatrix} I & X \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & AX + C - XB \\ 0 & B \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}.$$

Structural Commentary: A block shear cancels the off-diagonal block via Sylvester's equation.
Key Insight: Spectrally separated diagonal blocks cannot carry essential upper-right coupling. □

Theorem 3.5.4. *Schur Form Grouped by Eigenvalues*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_d$ with algebraic multiplicities $n_1, \ldots, n_d$.
Then A is unitarily similar to a block upper-triangular matrix

$$T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1d} \\ 0 & T_{22} & \cdots & T_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T_{dd} \end{bmatrix}, \qquad T_{ii} \in \operatorname{Mat}(n_i, n_i, \mathbb{C}),$$

where each T_{ii} is upper triangular with all diagonal entries equal to λ_i. Moreover, A is similar to $T_{11} \oplus \cdots \oplus T_{dd}$.

Proof. Order the Schur diagonal as $\lambda_1, \ldots, \lambda_1, \lambda_2, \ldots, \lambda_2, \ldots, \lambda_d, \ldots, \lambda_d$.
By Theorem 2.1.9, A has the displayed block Schur form. Since $\operatorname{spec}(T_{11})$ is disjoint from the trailing block spectrum, Corollary 3.5.3 splits off T_{11}. Iterating splits off every diagonal block.
Structural Commentary: Grouped Schur form isolates spectral blocks; Sylvester shears split disjoint groups.
Key Insight: Cross-group upper blocks are coordinate artefacts. □

Corollary 3.5.5. *Commuting with Disjoint Blocks*
Let $A_{ii} \in \operatorname{Mat}(n_i, n_i, \mathbb{C})$ for $i \in \{1, \ldots, d\}$, set $n = \sum_{i=1}^{d} n_i$ and $A = A_{11} \oplus \cdots \oplus A_{dd}$, and suppose the spectra of $A_{11}, \ldots, A_{dd}$ are pairwise disjoint.
Let $B = [B_{ij}] \in \operatorname{Mat}(n, n, \mathbb{C})$ be partitioned conformally with A. If $AB = BA$, then $B_{ij} = 0$ for all $i \neq j$.

Proof. Equating (i,j) blocks in $AB = BA$ gives $A_{ii}B_{ij} - B_{ij}A_{jj} = 0$.
For $i \neq j$, the spectra of A_{ii} and A_{jj} are disjoint. By Theorem 3.5.2, the homogeneous equation has the unique solution 0, so $B_{ij} = 0$.
Structural Commentary: A commuting matrix cannot mix direct-sum blocks with disjoint spectra.
Key Insight: The off-diagonal block solves a homogeneous Sylvester equation, whose only solution is zero. □

4 Generalized Eigenspaces

4.1 Generalized Eigenspaces and Nilpotent Structure

Generalized eigenvectors repair the main defect of ordinary eigenvectors: an operator may not have enough eigendirections to form a basis. The replacement is to study vectors killed by powers of shifted operators.

Definition 4.1.1. *Generalized Eigenvectors and Generalized Eigenspaces*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. Nonzero vector $\mathbf{v} \in \mathcal{V}$ is a *generalized eigenvector* of T to λ if $(T - \lambda I_{\mathcal{V}})^j \mathbf{v} = \mathbf{0}$ for some positive integer j. The *generalized eigenspace* corresponding to λ is

$$\mathcal{G}_\lambda(T) := \{\mathbf{0}\} \cup \{\mathbf{v} \neq \mathbf{0} : \mathbf{v} \text{ is a generalized eigenvector of } T \text{ corresponding to } \lambda\}.$$

Every eigenvector is a generalized eigenvector by taking $j = 1$. The point is that higher powers of $T - \lambda I_{\mathcal{V}}$ may reveal directions that are not eigenvectors but are eventually pushed into the eigenspace.

Proposition 4.1.2. *Kernels of Powers Increase*
Let $T \in \mathcal{L}(\mathcal{V})$. Then $\{\mathbf{0}\} = \ker(T^0) \subseteq \ker(T) \subseteq \ker(T^2) \subseteq \cdots \subseteq \ker(T^k) \subseteq \ker(T^{k+1}) \subseteq \cdots$.

Proof. If $\mathbf{v} \in \ker(T^k)$, then $T^k\mathbf{v} = \mathbf{0}$, hence $T^{k+1}\mathbf{v} = T(T^k\mathbf{v}) = \mathbf{0}$. Thus $\ker(T^k) \subseteq \ker(T^{k+1})$ for every $k \geq 0$.
Structural Commentary: Powers of T form an increasing chain of kernels.
Key Insight: Once a vector is killed by T^k, applying one more T cannot revive it. □

Proposition 4.1.3. *Equality Forces Stabilization*
Let $T \in \mathcal{L}(\mathcal{V})$ and $m \geq 0$. If $\ker(T^m) = \ker(T^{m+1})$, then $\ker(T^m) = \ker(T^{m+1}) = \ker(T^{m+2}) = \cdots$.

Proof. By Proposition 4.1.2, it suffices to prove $\ker(T^{m+k+1}) \subseteq \ker(T^{m+k})$ for every $k \geq 0$.
Take $\mathbf{v} \in \ker(T^{m+k+1})$. Then $T^{m+1}(T^k\mathbf{v}) = T^{m+k+1}\mathbf{v} = \mathbf{0}$, so $T^k\mathbf{v} \in \ker(T^{m+1}) = \ker(T^m)$.
Hence $T^{m+k}\mathbf{v} = T^m(T^k\mathbf{v}) = \mathbf{0}$, proving the reverse inclusion.
Structural Commentary: Stabilization at one step propagates through the whole tail.
Key Insight: A later kernel vector can be shifted down into the already-stable kernel. □

Theorem 4.1.4. *Kernels Stop Growing by $\dim(\mathcal{V})$*
If $\mathcal{V}$ is finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and $n = \dim(\mathcal{V})$, then $\ker(T^n) = \ker(T^{n+1}) = \ker(T^{n+2}) = \cdots$.

Proof. If $\ker(T^n) \neq \ker(T^{n+1})$, then all inclusions $\{\mathbf{0}\} = \ker(T^0) \subsetneq \ker(T) \subsetneq \cdots \subsetneq \ker(T^{n+1})$ are strict by Proposition 4.1.3. Each strict inclusion raises dimension by at least 1, so $\dim \ker(T^{n+1}) \geq n+1$, impossible because $\ker(T^{n+1}) \subseteq \mathcal{V}$. Thus $\ker(T^n) = \ker(T^{n+1})$, and the tail equality follows from Proposition 4.1.3.
Structural Commentary: Strict kernel growth is dimension-limited.
Key Insight: Dimension bounds force the kernel chain to stabilize no later than step $\dim(\mathcal{V})$. □

Figure 59 records the dimension-limited growth and eventual stabilization of the kernel filtration.

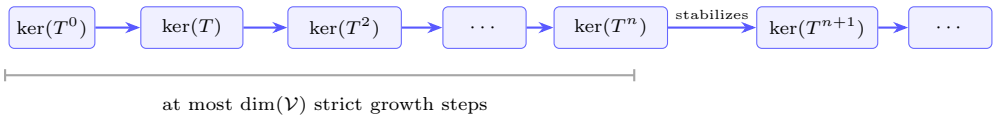

Figure 59: The kernel filtration grows until it stabilizes; after equality appears once, the tail is constant.

Theorem 4.1.5. *Stable Kernel-Range Decomposition*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and set $n = \dim(\mathcal{V})$. Then $\mathcal{V} = \ker(T^n) \oplus \operatorname{range}(T^n)$.

Proof. First show the intersection is trivial. If $\mathbf{v} \in \ker(T^n) \cap \operatorname{range}(T^n)$, then $\mathbf{v} = T^n\mathbf{u}$ and $T^n\mathbf{v} = \mathbf{0}$. Hence $T^{2n}\mathbf{u} = \mathbf{0}$, so $\mathbf{u} \in \ker(T^{2n}) = \ker(T^n)$ by Theorem 4.1.4. Thus $\mathbf{v} = T^n\mathbf{u} = \mathbf{0}$.
By Rank-Nullity Theorem 1.7.3, $\dim \ker(T^n) + \dim \operatorname{range}(T^n) = \dim(\mathcal{V})$.
The zero intersection and the dimension count imply the sum is direct and equals $\mathcal{V}$.
Structural Commentary: At the stable power, killed directions and surviving image directions separate cleanly.
Key Insight: Stabilization turns the usual rank-nullity count into an actual direct-sum decomposition. □

Theorem 4.1.6. *Description of Generalized Eigenspaces*
Let $\mathcal{V}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$, and $n = \dim(\mathcal{V})$. Then $\mathcal{G}_\lambda(T) = \ker\big((T - \lambda I_{\mathcal{V}})^n\big)$.
In particular, $\mathcal{G}_\lambda(T)$ is a subspace of $\mathcal{V}$.

Proof. Set $S = T - \lambda I_{\mathcal{V}}$.
($\Rightarrow$) If $\mathbf{v} \in \mathcal{G}_\lambda(T)$, then $S^j\mathbf{v} = \mathbf{0}$ for some $j \geq 1$. If $j \leq n$, then $\mathbf{v} \in \ker(S^n)$ by Proposition 4.1.2; if $j > n$, then $\ker(S^j) = \ker(S^n)$ by Theorem 4.1.4.
($\Leftarrow$) If $\mathbf{v} \in \ker(S^n)$, then either $\mathbf{v} = \mathbf{0}$ or $\mathbf{v}$ is a generalized eigenvector of T corresponding to λ.
Thus $\mathcal{G}_\lambda(T) = \ker(S^n)$, which is a subspace.
Structural Commentary: Generalized eigenspaces are stable shifted kernels.
Key Insight: Finite dimension makes exponent $\dim(\mathcal{V})$ enough. □

Proposition 4.1.7. *Generalized Eigenspaces Are Invariant*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{G}_\lambda(T)$ is invariant under T for every $\lambda \in \mathbb{F}$.

Proof. By Theorem 4.1.6, $\mathcal{G}_\lambda(T) = \ker((T - \lambda I_\mathcal{V})^n)$, where $n = \dim(\mathcal{V})$.
This is the kernel of a polynomial in T, so it is invariant under T by Proposition 3.5.1.
Structural Commentary: A generalized eigenspace is cut out by a polynomial equation in T.
Key Insight: Polynomial kernels commute with the operator that defines them. □

Definition 4.1.8. *Nilpotent Operator*
An operator $N \in \mathcal{L}(\mathcal{V})$ is *nilpotent* if $N^m = 0$ for some positive integer m.

Corollary 4.1.9. *Nilpotent Power Bound*
If $\mathcal{V}$ is finite-dimensional, $N \in \mathcal{L}(\mathcal{V})$ is nilpotent, and $n = \dim(\mathcal{V})$, then $N^n = 0$.

Proof. Since N is nilpotent, $\ker(N^m) = \mathcal{V}$ for some m.
The kernel chain has stabilized by the nth step by Theorem 4.1.4, so $\ker(N^n) = \mathcal{V}$. Hence $N^n = 0$.
Structural Commentary: Nilpotence says some kernel in the chain is already the whole space.
Key Insight: The stabilization bound pulls that full-kernel conclusion back to exponent $\dim(\mathcal{V})$. □

Lemma 4.1.10. *Nonzero Scalar Plus Nilpotent Is Invertible*
Let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent and let $a \in \mathbb{F}$ with $a \neq 0$. Then $aI_\mathcal{V} + N$ is invertible.

Proof. Choose m such that $N^m = 0$. Then $(aI_\mathcal{V} + N)\left(a^{-1}\sum_{r=0}^{m-1}(-a^{-1}N)^r\right) = I_\mathcal{V} - (-a^{-1}N)^m = I_\mathcal{V}$, and the same computation in the reverse order gives a two-sided inverse.
Structural Commentary: A nilpotent perturbation has a finite geometric-series inverse.
Key Insight: Nilpotence truncates the Neumann series exactly. □

Theorem 4.1.11. *Generalized Eigenvectors for Distinct Eigenvalues Are Independent*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding generalized eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Let $n = \dim(\mathcal{V})$ and suppose $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
Fix $r \in \{1, \ldots, m\}$ and apply $Q_r(T) := \prod_{i \neq r}(T - \lambda_i I_\mathcal{V})^n$ to the relation.
For $i \neq r$, $Q_r(T)\mathbf{v}_i = \mathbf{0}$ by Theorem 4.1.6, so $a_r Q_r(T)\mathbf{v}_r = \mathbf{0}$.
On $\mathcal{G}_{\lambda_r}(T)$, the operator $T - \lambda_i I_\mathcal{V}$ equals $(T - \lambda_r I_\mathcal{V}) + (\lambda_r - \lambda_i)I_\mathcal{V}$. The first term is nilpotent on $\mathcal{G}_{\lambda_r}(T)$ and $\lambda_r - \lambda_i \neq 0$, so each factor is invertible on $\mathcal{G}_{\lambda_r}(T)$ by Lemma 4.1.10. Thus $Q_r(T)\mathbf{v}_r \neq \mathbf{0}$, forcing $a_r = 0$. Since r was arbitrary, all coefficients vanish.
Structural Commentary: Polynomial filters isolate one generalized eigenspace at a time.
Key Insight: Distinct eigenvalues let polynomial filters kill the other components and preserve the target one. □

Theorem 4.1.12. *Nilpotent Operators and Strictly Upper-Triangular Matrices*
Let $\mathcal{V}$ be finite-dimensional and $N \in \mathcal{L}(\mathcal{V})$. Then N is nilpotent if and only if N has a strictly upper-triangular matrix in some basis.

Proof. Let $n = \dim(\mathcal{V})$. By Corollary 4.1.9, $\{\mathbf{0}\} \subseteq \ker(N) \subseteq \ker(N^2) \subseteq \cdots \subseteq \ker(N^n) = \mathcal{V}$.

(⇒) Choose a basis of $\ker(N)$, extend it to a basis of $\ker(N^2)$, and continue until obtaining a basis of $\mathcal{V}$.
If $\mathbf{v}$ first appears in $\ker(N^k)$, then $N\mathbf{v} \in \ker(N^{k-1})$, which is spanned by earlier basis vectors. Hence the matrix of N is strictly upper triangular.

(⇐) If the matrix of N is $n \times n$ strictly upper triangular, then N sends each basis vector into the span of earlier basis vectors. After n applications every basis vector is sent to $\mathbf{0}$, so $N^n = 0$.

Structural Commentary: Kernel filtration gives adapted basis; triangular form records motion to earlier layers.
Key Insight: Nilpotence is strict upper-triangularity in a suitable basis. □

Figure 60 illustrates how a basis adapted to the kernel filtration makes nilpotent action move toward earlier layers.

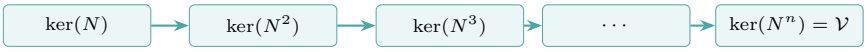

Figure 60: Extending bases through the kernel filtration makes N move vectors to earlier layers.

Perspective 4.1.13. *Eigenvalue Plus Nilpotent Part*
On $\mathcal{G}_\lambda(T)$, the operator T behaves as $\lambda I_\mathcal{V}$ plus the nilpotent operator $T - \lambda I_\mathcal{V}$. Generalized eigenvectors are therefore ordinary eigenvectors enriched by nilpotent chain data.

5 Primary Decomposition

5.1 Primary Decomposition and Spectral Splitting

Theorem 5.1.1. *Primary Decomposition over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. Then
(i) $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$;
(ii) each $(T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ is nilpotent.

Proof. Let $n = \dim(\mathcal{V})$.

(i) Proceed by induction on n. If $n = 1$, then T has single eigenvalue λ_1 and $\mathcal{V} = \mathcal{G}_{\lambda_1}(T)$.
Assume $n > 1$. Let $S = T - \lambda_1 I_{\mathcal{V}}$, $\mathcal{U} := \operatorname{range}(S^n)$.
By Theorems 4.1.5 and 4.1.6, $\mathcal{V} = \ker(S^n) \oplus \operatorname{range}(S^n) = \mathcal{G}_{\lambda_1}(T) \oplus \mathcal{U}$. The subspace $\mathcal{U}$ is T-invariant by Proposition 3.5.1, and $\dim(\mathcal{U}) < n$ because $\mathcal{G}_{\lambda_1}(T) \neq \{\mathbf{0}\}$. If $\mathcal{U} = \{\mathbf{0}\}$, then (i) follows.
Otherwise distinct eigenvalues of $T|_{\mathcal{U}}$ are $\lambda_2, \dots, \lambda_m$: every eigenvalue of $T|_{\mathcal{U}}$ is an eigenvalue of T but not λ_1, while an eigenvector $\mathbf{v}$ of T for λ_j, $j \geq 2$, satisfies $S^n\mathbf{v} = (\lambda_j - \lambda_1)^n\mathbf{v}$ and hence lies in $\mathcal{U}$.
Induction gives $\mathcal{U} = \mathcal{G}_{\lambda_2}(T|_{\mathcal{U}}) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T|_{\mathcal{U}})$. For $j \geq 2$, $\mathcal{G}_{\lambda_j}(T|_{\mathcal{U}}) = \mathcal{G}_{\lambda_j}(T)$.
The inclusion $\subseteq$ follows from the definition. Conversely, if $\mathbf{v} \in \mathcal{G}_{\lambda_j}(T)$, write $\mathbf{v} = \mathbf{g} + \mathbf{u}$ with $\mathbf{g} \in \mathcal{G}_{\lambda_1}(T)$ and $\mathbf{u} \in \mathcal{U}$. For some r, $(T - \lambda_j I_{\mathcal{V}})^r\mathbf{v} = \mathbf{0}$. Since both summands are T-invariant, the two terms $(T - \lambda_j I_{\mathcal{V}})^r\mathbf{g}$ and $(T - \lambda_j I_{\mathcal{V}})^r\mathbf{u}$ remain in the two direct summands and have sum zero, so both are zero.
Thus $\mathbf{g} \in \mathcal{G}_{\lambda_1}(T) \cap \mathcal{G}_{\lambda_j}(T) = \{\mathbf{0}\}$ by Theorem 4.1.11, and $\mathbf{v} = \mathbf{u} \in \mathcal{G}_{\lambda_j}(T|_{\mathcal{U}})$. Hence (i) follows.

(ii) By Proposition 4.1.7, the restriction is an operator on $\mathcal{G}_{\lambda_j}(T)$. By Theorem 4.1.6, $(T - \lambda_j I_{\mathcal{V}})^n\mathbf{v} = \mathbf{0}$ $\forall \mathbf{v} \in \mathcal{G}_{\lambda_j}(T)$. Hence nth power of $(T - \lambda_j I_{\mathcal{V}})|_{\mathcal{G}_{\lambda_j}(T)}$ is zero.

Structural Commentary: Peel off one generalized eigenspace, then induct on the invariant range.
Key Insight: Stable kernel-range splitting isolates one eigenvalue at a time. □

Figure 61 summarizes the primary summands and the nilpotent part of T on each one.

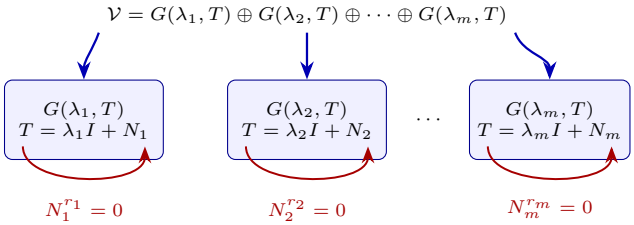

Figure 61: Primary decomposition: on $\mathcal{G}_{\lambda_j}(T)$, $T = \lambda_j I$ plus nilpotent.

Corollary 5.1.2. *Basis of Generalized Eigenvectors*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a generalized-eigenvector basis for T.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 5.1.1, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$. Choose a basis of each generalized eigenspace and concatenate the bases. Directness makes the concatenated list a basis of $\mathcal{V}$, and every nonzero vector in each list is a generalized eigenvector.
Structural Commentary: A direct sum basis is obtained by choosing bases in each summand.
Key Insight: Bases of the direct-sum pieces concatenate. □

Theorem 5.1.3. *Block Diagonal Form with Upper-Triangular Primary Blocks*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$, and set $d_j = \dim \mathcal{G}_{\lambda_j}(T)$. Then T has a matrix of the form

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j \in \operatorname{Mat}(d_j, d_j, \mathbb{C})$ is upper triangular with diagonal $\lambda_j, \dots, \lambda_j$.

Proof. Set $\mathcal{U}_j = \mathcal{G}_{\lambda_j}(T)$ and $N_j = (T - \lambda_j I_{\mathcal{V}})|_{\mathcal{U}_j}$.
By Theorem 5.1.1, the $\mathcal{U}_j$ form a T-invariant direct-sum decomposition and each N_j is nilpotent.
By Theorem 4.1.12, choose a basis of each $\mathcal{U}_j$ in which N_j is strictly upper triangular. Then $T|_{\mathcal{U}_j} = \lambda_j I_{\mathcal{U}_j} + N_j$ has an upper triangular matrix with diagonal $\lambda_j, \dots, \lambda_j$. Concatenating these bases gives the desired block diagonal matrix.
Structural Commentary: Primary decomposition gives the blocks; nilpotent triangularisation shapes them.
Key Insight: On each generalized eigenspace, T is eigenvalue plus nilpotent. □

Figure 62 illustrates the block form obtained from bases adapted to the primary summands.

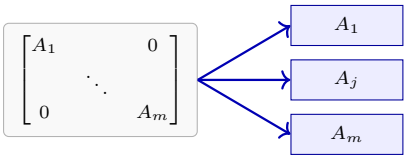

Figure 62: Primary blocks: $A_j = \lambda_j I + N_j$, with N_j strictly upper triangular.

Proposition 5.1.4. *Algebraic Multiplicity from the Generalized Eigenspace*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
The algebraic multiplicity of an eigenvalue λ of T equals $\dim \mathcal{G}_\lambda(T) = \dim \ker\big((T - \lambda I_{\mathcal{V}})^{\dim(\mathcal{V})}\big)$.

Proof. By Theorem 5.1.3 and Proposition 2.1.6, the multiplicity of λ as a root of q_T equals $\dim \mathcal{G}_\lambda(T)$.
By Theorem 4.1.6, $\mathcal{G}_\lambda(T) = \ker\big((T - \lambda I_{\mathcal{V}})^{\dim(\mathcal{V})}\big)$.
Structural Commentary: Each primary block contributes one characteristic factor whose exponent is the block dimension.
Key Insight: Algebraic multiplicity is the dimension of the generalized eigenspace. □

Figure 63 records how the primary-summand dimensions determine the characteristic-polynomial exponents.

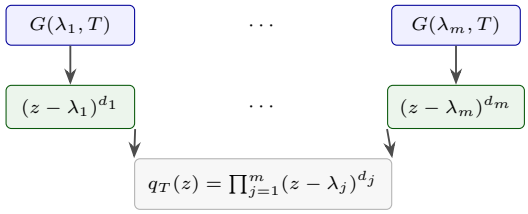

Figure 63: Each primary summand contributes $(z-\lambda_j)^{d_j}$, where $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.

Proposition 5.1.5. *Sum of Multiplicities*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then the sum of the multiplicities of all eigenvalues of T equals $\dim(\mathcal{V})$.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 5.1.1, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$.
Taking dimensions, by Corollary 2.3.2, $\dim(\mathcal{V}) = \sum_{j=1}^{m} \dim \mathcal{G}_{\lambda_j}(T)$, exactly the sum of the multiplicities.
Structural Commentary: Multiplicity is dimension of the generalized-eigenspace summand.
Key Insight: Multiplicities are dimensions of the direct-sum pieces. □

5.2 Real Spectral Data via Complexification

Lemma 5.2.1. *Conjugation Transfers Generalized Eigenspaces*
Let $\mathcal{V}$ be a real vector space, let $T \in \mathcal{L}(\mathcal{V})$, let $\lambda \in \mathbb{C}$, let $k \geq 0$, and let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$. Then $(T_{\mathbb{C}} - \lambda I_{\mathcal{V}_{\mathbb{C}}})^k(\mathbf{u}+i\mathbf{v}) = \mathbf{0} \iff (T_{\mathbb{C}} - \overline{\lambda} I_{\mathcal{V}_{\mathbb{C}}})^k(\mathbf{u}-i\mathbf{v}) = \mathbf{0}$.

Proof. Let $\Gamma(\mathbf{u}+i\mathbf{v}) = \mathbf{u} - i\mathbf{v}$.
Then Γ is a conjugate-linear bijection, $\Gamma T_{\mathbb{C}} = T_{\mathbb{C}}\Gamma$, and $\Gamma\big((T_{\mathbb{C}} - \lambda I)^k \mathbf{x}\big) = (T_{\mathbb{C}} - \overline{\lambda} I)^k \Gamma \mathbf{x}$.
($\Rightarrow$) Apply the displayed identity to $\mathbf{x} = \mathbf{u} + i\mathbf{v}$.
($\Leftarrow$) Apply the first direction with $\overline{\lambda}$ in place of λ and $\mathbf{u} - i\mathbf{v}$ in place of $\mathbf{u} + i\mathbf{v}$.
Structural Commentary: Conjugation commutes with $T_{\mathbb{C}}$ and conjugates scalars.
Key Insight: Conjugation pairs generalized eigenspaces. □

Corollary 5.2.2. *Conjugate Eigenvalues Have Equal Multiplicity*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$.
If λ is an eigenvalue of $T_{\mathbb{C}}$, then $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$, and $\dim_{\mathbb{C}} \mathcal{G}_{\lambda}(T_{\mathbb{C}}) = \dim_{\mathbb{C}} \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$.

Proof. By Lemma 5.2.1 with $k = 1$, conjugation sends a nonzero eigenvector for λ to a nonzero eigenvector for $\overline{\lambda}$. Thus $\overline{\lambda}$ is an eigenvalue. With $k = \dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$, the generalized eigenspaces are the corresponding stabilized kernels by Theorem 4.1.6. By Lemma 5.2.1, $\Gamma(\mathbf{u}+i\mathbf{v}) = \mathbf{u} - i\mathbf{v}$ restricts to conjugate-linear bijection $\mathcal{G}_{\lambda}(T_{\mathbb{C}}) \longrightarrow \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$.
Conjugate-linear bijections preserve complex dimension, so the multiplicities agree.
Structural Commentary: Conjugation mirrors generalized eigenspaces.
Key Insight: Nonreal spectral data of a real operator arrives in equal-sized conjugate pairs. □

Corollary 5.2.3. *Odd-Dimensional Real Operators Have Eigenvalues*
Every operator on a nonzero odd-dimensional real vector space has a real eigenvalue.

Proof. Let $T \in \mathcal{L}(\mathcal{V})$, where $\dim_{\mathbb{R}} \mathcal{V}$ is odd. By Proposition 5.3.7, $\dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$ is odd.
The sum of the multiplicities of all eigenvalues of $T_{\mathbb{C}}$ equals this odd dimension by Proposition 5.1.5. Nonreal eigenvalues contribute in conjugate pairs of equal multiplicity by Corollary 5.2.2, hence contribute an even total. Therefore $T_{\mathbb{C}}$ has at least one real eigenvalue, which is an eigenvalue of T by Proposition 5.3.12.
Structural Commentary: Odd dimension forces a real eigenvalue.
Key Insight: Parity forces at least one real spectral direction. □

Theorem 5.2.4. *Characteristic Polynomial over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be its characteristic polynomial.
(i) $q_T \in \mathbb{R}[z]$.
(ii) $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) The real zeros of q_T are precisely the eigenvalues of T.
(iv) $q_T(T) = 0$.

Proof. Choose a basis β of $\mathcal{V}$ and its complexification $\beta_{\mathbb{C}}$. Since $[T_{\mathbb{C}}]_{\beta_{\mathbb{C}} \leftarrow \beta_{\mathbb{C}}} = [T]_{\beta \leftarrow \beta}$, Definition 2.1.15 gives $q_{T_{\mathbb{C}}} = q_T$.

(i) Factor $q_{T_{\mathbb{C}}}$ over $\mathbb{C}$. Real eigenvalues give real linear factors. If $\lambda \notin \mathbb{R}$ has multiplicity m, then $\overline{\lambda}$ has multiplicity m by Corollary 5.2.2, and $(z-\lambda)^m(z-\overline{\lambda})^m = \big(z^2 - 2\operatorname{Re}(\lambda) z + |\lambda|^2\big)^m$, which is real.
(ii) By Propositions 2.1.16 and 5.3.7, $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) By Proposition 2.1.16, the real zeros of $q_T = q_{T_{\mathbb{C}}}$ are the real eigenvalues of $T_{\mathbb{C}}$; these are the eigenvalues of T by Proposition 5.3.12.
(iv) By Theorem 3.1.2, $q_T(T_{\mathbb{C}}) = 0$. By (i) and Proposition 5.3.11, $q_T(T_{\mathbb{C}}) = (q_T(T))_{\mathbb{C}}$.
If $q_T(T)$ were nonzero on some $\mathbf{u} \in \mathcal{V}$, then $(q_T(T))_{\mathbb{C}}(\mathbf{u}+i\mathbf{0}) = q_T(T)\mathbf{u} + i\mathbf{0}$ would be nonzero, a contradiction. Hence $q_T(T) = 0$.
Structural Commentary: Nonreal factors pair into real quadratics.
Key Insight: Complexify, then pair conjugates. □

Figure 64 shows how each conjugate factor pair becomes a real quadratic contribution to q_T.

Figure 64: Conjugate factors pair into real quadratics.

Corollary 5.2.5. *q_T Is a Multiple of μ_T over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
Then $\deg \mu_T \leq \dim_{\mathbb{R}} \mathcal{V}$, and q_T is a polynomial multiple of μ_T in $\mathbb{R}[z]$.

Proof. By Theorem 5.2.4, $q_T(T) = 0$.
By Theorem 2.2.4, μ_T divides q_T in $\mathbb{R}[z]$. The degree bound follows from $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
Structural Commentary: Cayley–Hamilton makes q_T a real annihilator of T.
Key Insight: The minimal polynomial divides every real annihilator. □

Proposition 5.2.6. *Minimal Polynomial Survives Complexification*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\mu_{T_{\mathbb{C}}} = \mu_T$ in $\mathbb{C}[z]$.

Proof.
($\Rightarrow$) Since $\mu_T(T) = 0$, $\mu_T(T_{\mathbb{C}}) = (\mu_T(T))_{\mathbb{C}} = 0$ by Proposition 5.3.11. Hence $\mu_{T_{\mathbb{C}}}$ divides μ_T in $\mathbb{C}[z]$.
($\Leftarrow$) Let $q = \mu_{T_{\mathbb{C}}} = r + is$, with $r, s \in \mathbb{R}[z]$. Since q is monic, r is monic and $\deg r = \deg q$.
For every $\mathbf{u} \in \mathcal{V}$, $0 = q(T_{\mathbb{C}})(\mathbf{u} + i\mathbf{0}) = r(T)\mathbf{u} + is(T)\mathbf{u}$. Thus $r(T) = 0$, so $\deg \mu_T \leq \deg r = \deg q$. The first part gives opposite degree inequality. As q divides μ_T and both are monic with equal degree, $q = \mu_T$.
Structural Commentary: Annihilation passes to complexification and back by real-part extraction.
Key Insight: Evaluate $q(T_{\mathbb{C}})$ on $\mathbf{u} + i\mathbf{0}$. □

5.3 Normal Primary Structure

Corollary 5.3.1. *Normal Operators Have No Nilpotent Primary Part*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.
(i) If $S = p(T)$ for some $p \in \mathbb{C}[z]$, then $\ker(S^k) = \ker(S)$ for every $k \geq 1$.
(ii) The minimal polynomial of T is square-free, where $\mu_T(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$.

Proof. By Theorem 6.1.11, choose an orthonormal eigenbasis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ with $T\mathbf{e}_j = \lambda_j \mathbf{e}_j$.
(i) For $S = p(T)$, $k \geq 1$, $S^k \mathbf{e}_j = p(\lambda_j)^k \mathbf{e}_j$; $\ker(S^k) = \operatorname{span}\{\mathbf{e}_j : p(\lambda_j)^k = 0\} = \operatorname{span}\{\mathbf{e}_j : p(\lambda_j) = 0\} = \ker(S)$.
(ii) Let $m(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$. Since $m(T)\mathbf{e}_j = 0$ for all j, $m(T) = 0$, so $\mu_T \mid m$. Conversely, every $\lambda \in \operatorname{spec}(T)$ is a root of μ_T by Theorem 3.3.2, hence $m \mid \mu_T$. Since both polynomials are monic, $\mu_T = m$.
Structural Commentary: Normality makes every primary component an eigenspace.
Key Insight: A diagonal operator has no hidden nilpotent part. □

Proposition 5.3.2. *Coprime Normal Primary Pieces Are Orthogonal*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, and let $\mathcal{S}, \mathcal{R} \leq \mathcal{V}$ be T-invariant.
If $p, q \in \mathbb{C}[z]$, $(p, q) = 1$, $p(T)|_{\mathcal{S}} = 0$, and $q(T)|_{\mathcal{R}} = 0$, then $\mathcal{S} \perp \mathcal{R}$.

Proof. Choose $a, b \in \mathbb{C}[z]$ with $ap + bq = 1$. For $\mathbf{s} \in \mathcal{S}$, $\mathbf{s} = a(T)p(T)\mathbf{s} + b(T)q(T)\mathbf{s} = b(T)q(T)\mathbf{s}$.
By Corollary 6.2.10, $\mathcal{R}$ reduces T, so it reduces $b(T)$ and $q(T)$. Hence $b(T)^*\mathcal{R} \subseteq \mathcal{R}$ and $q(T)^*|_{\mathcal{R}} = (q(T)|_{\mathcal{R}})^* = 0$.
Thus, for $\mathbf{r} \in \mathcal{R}$, $\langle \mathbf{s}, \mathbf{r} \rangle = \langle b(T)q(T)\mathbf{s}, \mathbf{r} \rangle = \langle \mathbf{s}, q(T)^* b(T)^* \mathbf{r} \rangle = 0$.
Structural Commentary: Coprime annihilators orthogonally separate normal invariant pieces.
Key Insight: Normality makes invariant subspaces reduce, so annihilation also holds after taking adjoints. □

6 Jordan and Canonical Forms

6.1 Jordan Chains and Nilpotent Blocks

Jordan and canonical forms turn primary decomposition into explicit blocks. Jordan chains, complexification, and real block forms organize the nilpotent and real spectral structure. Each Jordan block records one eigenvalue together with the length of a nilpotent chain.

Definition 6.1.1. *Jordan Chains, Blocks, and Matrices*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, let $\lambda \in \mathbb{C}$, and let $k \geq 1$. A *Jordan chain* for A at λ is a list $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ such that $(A - \lambda I_n)\mathbf{x}_1 = \mathbf{0}$ and $(A - \lambda I_n)\mathbf{x}_j = \mathbf{x}_{j-1}$ for $2 \leq j \leq k$.
The *nilpotent Jordan block* $J_k(0)$ is the $k \times k$ matrix with 1 on the first superdiagonal and 0 elsewhere:

$$J_k(0) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

Set $J_k(\lambda) := \lambda I_k + J_k(0)$. A *Jordan matrix* is a direct sum of Jordan blocks. A basis is a *Jordan basis* for an operator T if the matrix of T in that basis is a Jordan matrix.

Perspective 6.1.2. *Jordan Chains as Columns*
For a Jordan chain $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ and $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$, $AX = XJ_k(\lambda)$.
If all chains form the columns of a basis matrix M, then $AM = MJ$ and $M^{-1}AM = J$.

Figure 65 depicts the shifted action along a Jordan chain and its matrix-column interpretation.

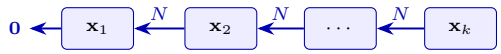

Figure 65: Jordan chain for $N = A - \lambda I$; columns give $AX = XJ_k(\lambda)$.

Theorem 6.1.3. *Nilpotent Jordan Chain Basis*
Let $\mathcal{V}$ be finite-dim, $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $\exists$ vectors $\mathbf{v}_1, \ldots, \mathbf{v}_r \in \mathcal{V}$ and nonnegative integers $m_1, \ldots, m_r$ such that $\mathcal{B}_N = \left(N^{m_1}\mathbf{v}_1, \ldots, N\mathbf{v}_1, \mathbf{v}_1, \ldots, N^{m_r}\mathbf{v}_r, \ldots, N\mathbf{v}_r, \mathbf{v}_r\right)$ is a basis of $\mathcal{V}$ and $N^{m_j+1}\mathbf{v}_j = \mathbf{0}$ for every j.

Proof. Induct on $n = \dim(\mathcal{V})$. If $N = 0$, any basis works with all $m_j = 0$.
Now assume $N \neq 0$. Then N is not injective, so range(N) is a proper subspace, and $N|_{\text{range}(N)}$ is nilpotent.
By induction, range(N) has a chain basis $\left(N^{m_1}\mathbf{u}_1, \ldots, N\mathbf{u}_1, \mathbf{u}_1, \ldots, N^{m_r}\mathbf{u}_r, \ldots, N\mathbf{u}_r, \mathbf{u}_r\right)$ with $N^{m_j+1}\mathbf{u}_j = \mathbf{0}$.
Choose $\mathbf{v}_j \in \mathcal{V}$ such that $N\mathbf{v}_j = \mathbf{u}_j$, and lift the chains to $\mathcal{L} = \left(N^{m_1+1}\mathbf{v}_1, \ldots, N\mathbf{v}_1, \mathbf{v}_1, \ldots, N^{m_r+1}\mathbf{v}_r, \ldots, N\mathbf{v}_r, \mathbf{v}_r\right)$.
The list $\mathcal{L}$ is linearly independent. Indeed, applying N to a dependence among vectors in $\mathcal{L}$ gives a dependence among the range-chain basis, so all coefficients except possibly those on the leading vectors $N^{m_j+1}\mathbf{v}_j$ vanish.
The remaining relation is a relation among $N^{m_j+1}\mathbf{v}_j = N^{m_j}\mathbf{u}_j$, which are members of the range-chain basis, so those coefficients vanish too. Also $N(\text{span}(\mathcal{L})) = \text{range}(N)$: each range-chain basis vector $N^q\mathbf{u}_j$ is $N(N^q\mathbf{v}_j)$ for the corresponding lifted vector. Extend $\mathcal{L}$ to a basis by adding $\mathbf{w}_1, \ldots, \mathbf{w}_s$. For each i, choose $\mathbf{x}_i \in \text{span}(\mathcal{L})$ such that $N\mathbf{x}_i = N\mathbf{w}_i$, and set $\mathbf{z}_i = \mathbf{w}_i - \mathbf{x}_i$. Then $N\mathbf{z}_i = \mathbf{0}$. Since each $\mathbf{x}_i$ lies in span$(\mathcal{L})$, replacing $\mathbf{w}_i$ by $\mathbf{z}_i$ preserves the span and linear independence of the full basis. Thus $\mathcal{L}, \mathbf{z}_1, \ldots, \mathbf{z}_s$ is the desired chain basis.
Structural Commentary: Build chains in range(N), lift them once, then add kernel chains.
Key Insight: Applying N turns lifted-chain dependences into known range-chain dependences. □

Figure 66 illustrates the nilpotent shift along one chain in the adapted basis order.

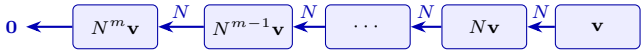

Figure 66: A nilpotent chain shifts left under N; in this basis order it becomes one Jordan block.

Method 6.1.4. *Filippov Column-Space Jordan Chains*
Let $A \in \text{Mat}(n, n, \mathbb{C})$, let $\lambda \in \text{spec}(A)$, and set $N = A - \lambda I_n$ on the generalized eigenspace $\mathcal{G}_\lambda(A) = \ker(N^n)$.

(i) Work first inside range$(N) \cap \mathcal{G}_\lambda(A)$, where N is again nilpotent.
(ii) Build Jordan chains for this smaller nilpotent operator.
(iii) For each chain ending in $\mathbf{u}$, solve $N\mathbf{v} = \mathbf{u}$ in $\mathcal{G}_\lambda(A)$ and lift the chain by appending $\mathbf{v}$.
(iv) Add enough vectors from $\ker(N)$ to complete a basis of $\mathcal{G}_\lambda(A)$; these added vectors are length-one chains.
(v) Repeat the process recursively until all of $\mathcal{G}_\lambda(A)$ is covered, then concatenate the chains for all eigenvalues.

The lifting steps are valid by Theorem 6.1.3; the chain relations give $AM = MJ$ for the resulting basis matrix.

Proposition 6.1.5. *Powers of a Jordan Block*
Let $k \geq 1$ and $\lambda \in \mathbb{C}$. For $p \geq 0$, $J_k(\lambda)^p = \sum_{r=0}^{\min(p,k-1)} \binom{p}{r} \lambda^{p-r} J_k(0)^r$. Thus $J_k(\lambda)^p \to 0$ if and only if $|\lambda| < 1$.

Proof.
($\Rightarrow$) If $J_k(\lambda)^p \to 0$, then its diagonal entries λ^p tend to 0, so $|\lambda| < 1$.
($\Leftarrow$) If $|\lambda| < 1$, then for each fixed $0 \leq r < k$ the scalar $\binom{p}{r}\lambda^{p-r}$ tends to 0, because a polynomial in p is dominated by the exponential decay of $|\lambda|^p$.

Structural Commentary: Jordan powers are exponential-polynomial terms.
Key Insight: Polynomial growth cannot defeat exponential decay when $|\lambda| < 1$. □

6.2 Complex and Real Jordan Forms and Block Sizes

Theorem 6.2.1. *Jordan Form over $\mathbb{C}$*
Let $\dim_{\mathbb{C}} \mathcal{V} = n \geq 1$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\exists$ a basis $\mathcal{B}$ of $\mathcal{V}$ and a Jordan matrix J with $[T]_{\mathcal{B}} = J$.

Proof. Let spec$(T) = \{\lambda_1, \ldots, \lambda_m\}$, set $G_j = \mathcal{G}_{\lambda_j}(T)$, $N_j = (T - \lambda_j I_{\mathcal{V}})|_{G_j}$, and $T_j = T|_{G_j}$.
By Theorem 5.1.1, $\mathcal{V} = \bigoplus_{j=1}^{m} G_j$ and each N_j is nilpotent. By Theorem 6.1.3, choose an N_j-chain basis $\mathcal{B}_j$ of G_j.
For an N_j-chain $\mathcal{C} \subset \mathcal{B}_j$ of length r, $[N_j]_{\mathcal{C}} = J_r(0)$ and $[T_j]_{\mathcal{C}} = \lambda_j I_r + J_r(0) = J_r(\lambda_j)$. Let $\mathcal{B} = \mathcal{B}_1 \sqcup \cdots \sqcup \mathcal{B}_m$.
By directness, $\mathcal{B}$ is a basis of $\mathcal{V}$, and $[T]_{\mathcal{B}}$ is a direct sum of Jordan blocks.
Structural Commentary: Primary decomposition reduces T to nilpotent chain bases.
Key Insight: On G_j, $T = \lambda_j I + N_j$. □

Figure 67 summarizes the passage from primary summands to nilpotent chains and Jordan blocks.

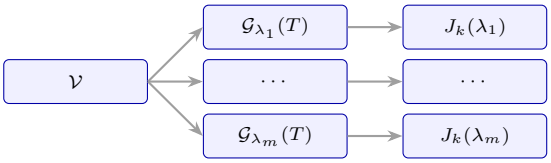

Figure 67: Primary pieces $\mathcal{G}_{\lambda_j}(T)$ give nilpotent chains, hence Jordan blocks.

Theorem 6.2.2. *Real Jordan Block Form*
Let $n \in \mathbb{N}$, $A \in \mathrm{Mat}(n,n,\mathbb{R})$. For $a \in \mathbb{R}$, $b > 0$, set $R(a,b) := \left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$.
Let $\mathcal{J}_k(a,b) = [C_{rs}]_{r,s=1}^{k}$ with $C_{rr} = R(a,b)$, $C_{r,r+1} = I_2$ for $r < k$, and $C_{rs} = 0_2$ otherwise.
Then $\exists S \in \mathrm{GL}_n(\mathbb{R})$ such that $S^{-1}AS$ is a direct sum of blocks of the following types:
(i) $J_k(\lambda)$, where $k \in \mathbb{N}$ and $\lambda \in \mathbb{R}$;
(ii) $\mathcal{J}_k(a,b)$, where $k \in \mathbb{N}$, $a \in \mathbb{R}$, and $b > 0$.

Proof. Complexify A to $A_{\mathbb{C}}$. By Theorem 6.2.1, choose Jordan chains for $A_{\mathbb{C}}$.
By $p_A \in \mathbb{R}[z]$, $\lambda \notin \mathbb{R}$ implies $\overline{\lambda} \in \mathrm{spec}(A_{\mathbb{C}})$.
(i) For $\lambda \in \mathbb{R}$, let $G_\lambda^{\mathbb{R}} = \ker(A - \lambda I_n)^n$ and $N_\lambda = (A - \lambda I_n)|_{G_\lambda^{\mathbb{R}}}$.
Then $(G_\lambda^{\mathbb{R}})_{\mathbb{C}} = \ker(A_{\mathbb{C}} - \lambda I_n)^n$ and N_λ is nilpotent.
By Theorem 6.1.3, take an N_λ-chain basis $\mathcal{B}_\lambda$; $[A]_{\mathcal{B}_\lambda}$ is a direct sum of $J_k(\lambda)$ blocks.
(ii) Let $\lambda = a + ib$, $b > 0$, and let $v = (v_1, \dots, v_k)$ be a λ-chain. By Lemma 5.2.1, $\overline{v}$ is a $\overline{\lambda}$-chain.
Let $v_j = u_j + iw_j$; $U_j = [u_j \ w_j]$, $U_0 = 0$, $\mathcal{U}_v = (u_1, w_1, \dots, u_k, w_k)$, $W_v = \mathrm{span}_{\mathbb{C}}(v, \overline{v}) \cap \mathbb{R}^n$.
By $(v, \overline{v}) \leftrightarrow \mathcal{U}_v$, $\mathcal{U}_v$ is real basis of W_v. By chain relations, $AU_j = U_j R(a,b) + U_{j-1}$; $[A]_{\mathcal{U}_v} = \mathcal{J}_k(a,b)$.
Let $\mathcal{B}$ concatenate all $\mathcal{B}_\lambda$ and one $\mathcal{U}_v$ for each pair $(\lambda, \overline{\lambda})$, $\mathrm{Im}\,\lambda > 0$.
By primary directness, $\mathcal{B}$ is a real basis of $\mathbb{R}^n$. With $S = [\mathcal{B}]$, $S^{-1}AS$ has the asserted blocks.
Structural Commentary: Real chains and conjugate-pair chains exhaust the primary decomposition.
Key Insight: A nonreal chain becomes one real block chain. □

Corollary 6.2.3. *Diagonalizability in Jordan Form*
Let $n \in \mathbb{N}$, $A \in \mathrm{Mat}(n,n,\mathbb{C})$, $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\mu_\alpha)$ be Jordan form of A.
Set $\delta_{\lambda\alpha} := \delta_{\lambda,\mu_\alpha}$ and $b_J(\lambda) = \sum_{\alpha=1}^{s} \delta_{\lambda\alpha}$ for $\lambda \in \mathrm{spec}(A)$. Then:
(i) $\dim E_\lambda(A) = b_J(\lambda)$, $\lambda \in \mathrm{spec}(A)$;
(ii) A is diagonalizable if and only if every Jordan block in J has size 1.

Proof. Choose $S \in \mathrm{GL}_n(\mathbb{C})$ such that $A = SJS^{-1}$.
(i) For $\lambda \in \mathrm{spec}(A)$, $S : E_\lambda(J) \cong E_\lambda(A)$ and $\dim E_\lambda(J_{k_\alpha}(\mu_\alpha)) = \delta_{\lambda\alpha}$. Thus $\dim E_\lambda(A) = \sum_{\alpha=1}^{s} \delta_{\lambda\alpha} = b_J(\lambda)$.
(ii) Put $a_J(\lambda) = \sum_{\alpha=1}^{s} k_\alpha \delta_{\lambda\alpha}$. By Theorem 2.1.12 and (i), A is diagonalizable $\iff a_J = b_J$ on $\mathrm{spec}(A)$.
Since $a_J(\lambda) - b_J(\lambda) = \sum_{\alpha=1}^{s} (k_\alpha - 1)\delta_{\lambda\alpha}$ and $k_\alpha \geq 1$, this is equivalent to $k_\alpha = 1$ for every α.
Structural Commentary: Similarity transfers eigenspaces; block kernels count directions.
Key Insight: Algebraic multiplicity counts block lengths. □

Lemma 6.2.4. *Powers of a Nilpotent Jordan Block*
Let $k \in \mathbb{N}$, let $p \in \mathbb{N}_0$, and set $N = J_k(0) \in \mathrm{Mat}(k,k,\mathbb{C})$. Then:
(i) $\mathrm{rank}(N^p) = \max(k - p, 0)$;
(ii) $\dim \ker(N^p) = \min(p, k)$.

Proof. Let $(\mathbf{e}_j)_{j=1}^{k}$ be the standard basis. Set $(q)_+ := \max(q, 0)$ and $\mathcal{E}_m = \mathrm{span}(\mathbf{e}_1, \dots, \mathbf{e}_m)$ for $0 \leq m \leq k$, with $\mathcal{E}_0 = \{\mathbf{0}\}$. By $N\mathbf{e}_j = \mathbf{e}_{j-1}$ for $j > 1$ and $N\mathbf{e}_1 = \mathbf{0}$, $N^p\mathbf{e}_j = \mathbf{e}_{j-p}$ for $j > p$ and $\mathbf{0}$ otherwise.
(i) $\mathrm{im}(N^p) = \mathcal{E}_{(k-p)_+}$, hence $\mathrm{rank}(N^p) = \dim \mathcal{E}_{(k-p)_+} = (k-p)_+ = \max(k-p, 0)$.
(ii) By rank-nullity and (i), $\dim \ker(N^p) = k - (k-p)_+ = \min(p, k)$.
Structural Commentary: Powers shift the chain and truncate its span.
Key Insight: Rank counts surviving positions. □

Definition 6.2.5. *Weyr Characteristic*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and let $\lambda \in \mathrm{spec}(A)$.
(i) For $p \in \mathbb{N}_0$, define the shifted-power rank $r_p(\lambda, A) := \mathrm{rank}\big((A - \lambda I_n)^p\big)$.
(ii) For $p \in \mathbb{N}$, define the Weyr number $w_p(\lambda, A) := r_{p-1}(\lambda, A) - r_p(\lambda, A)$. If $s := \max\{p \in \mathbb{N} : w_p(\lambda, A) > 0\}$, then the finite sequence $\big(w_1(\lambda, A), \dots, w_s(\lambda, A)\big)$ is the *Weyr characteristic* of A associated with λ.

Theorem 6.2.6. *Jordan Block Sizes from Rank Drops*
Let $n \in \mathbb{N}$, $A \in \mathrm{Mat}(n,n,\mathbb{C})$, $\lambda \in \mathrm{spec}(A)$, and $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\mu_\alpha)$ be a Jordan form of A.
For $q \in \mathbb{N}_0$, set $r_q = \mathrm{rank}\big((A - \lambda I_n)^q\big)$; for $q \in \mathbb{N}$, set $w_q = r_{q-1} - r_q$.
For $p \in \mathbb{N}$, set $I_p = \{\alpha : \mu_\alpha = \lambda,\ k_\alpha \geq p\}$. Then:
(i) $w_p = \#I_p$;
(ii) $w_p - w_{p+1} = \#(I_p \setminus I_{p+1}) = \#\{\alpha : \mu_\alpha = \lambda,\ k_\alpha = p\}$.

Proof. By similarity-invariance of rank, assume $A = J$. Let $\mathbf{1}_E$ be the indicator of E.
Set $B_\alpha = J_{k_\alpha}(\mu_\alpha)$ and $\rho_q(B) = \mathrm{rank}\big((B - \lambda I)^q\big)$.
Put $\Delta_{p,\alpha} = \rho_{p-1}(B_\alpha) - \rho_p(B_\alpha)$. By rank additivity, $w_p = \sum_{\alpha=1}^{s} \Delta_{p,\alpha}$.
(i) By invertibility for $\mu_\alpha \neq \lambda$ and by Lemma 6.2.4 for $\mu_\alpha = \lambda$, $\Delta_{p,\alpha} = \mathbf{1}_{I_p}(\alpha)$. Hence $w_p = \sum_{\alpha=1}^{s} \mathbf{1}_{I_p}(\alpha) = \#I_p$.
(ii) By (i), $w_p - w_{p+1} = \#I_p - \#I_{p+1} = \#(I_p \setminus I_{p+1}) = \#\{\alpha : \mu_\alpha = \lambda,\ k_\alpha = p\}$.

Structural Commentary: Rank drops split into blockwise indicators.
Key Insight: A block contributes until its length. □

Figure 68 records how successive shifted-rank drops recover the Jordan-block counts.

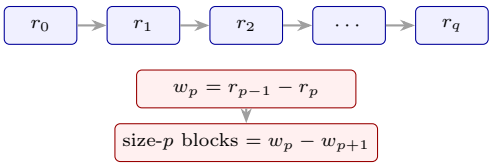

Figure 68: For $r_p = \operatorname{rank}((A-\lambda I)^p)$, drops count surviving chains; $w_p - w_{p+1}$ counts size-p blocks.

Corollary 6.2.7. *Uniqueness of Jordan Form*
Let $n \in \mathbb{N}$, $A \in \operatorname{Mat}(n,n,\mathbb{C})$. Let J, J' be Jordan matrices similar to A.
For $M \in \{J, J'\}$, let $\mathfrak{B}(M)$ be its multiset of Jordan blocks, set $m_M(\lambda,k) = \#\{B \in \mathfrak{B}(M) : B = J_k(\lambda)\}$. Then:
(i) $m_J(\lambda,k) = m_{J'}(\lambda,k)$ for all $(\lambda,k) \in \operatorname{spec}(A) \times \mathbb{N}$;
(ii) $\exists$ a block-permutation matrix P such that $J' = P^{-1}JP$.

Proof. Set $d_k^\lambda(M) = w_k(\lambda, M) - w_{k+1}(\lambda, M)$. By similarity invariance, $d_k^\lambda(J) = d_k^\lambda(J')\ \forall (\lambda,k) \in \operatorname{spec}(A) \times \mathbb{N}$.
(i) For $(\lambda,k) \in \operatorname{spec}(A) \times \mathbb{N}$, by Theorem 6.2.6, $m_J(\lambda,k) = d_k^\lambda(J) = d_k^\lambda(J') = m_{J'}(\lambda,k)$.
(ii) By (i), $\mathfrak{B}(J) = \mathfrak{B}(J')$, so $\exists$ a block-permutation matrix P with $J' = P^{-1}JP$.
Structural Commentary: Shifted ranks determine block multiplicities.
Key Insight: Weyr data fixes Jordan blocks. □

Method 6.2.8. *Computing Jordan Blocks by Rank Drops*
Let $A \in \operatorname{Mat}(n,n,\mathbb{C})$ and let $\lambda \in \operatorname{spec}(A)$. To determine the λ-Jordan blocks:
(i) Compute $r_p(\lambda, A) = \operatorname{rank}\big((A-\lambda I_n)^p\big)$ for $p = 0, 1, 2, \ldots$ until $r_{p-1}(\lambda, A) = r_p(\lambda, A)$.
(ii) Set $w_p(\lambda, A) = r_{p-1}(\lambda, A) - r_p(\lambda, A)$. This is the number of λ-blocks of size at least p.
(iii) For $p \in \mathbb{N}$, take $w_p(\lambda, A) - w_{p+1}(\lambda, A)$ copies of $J_p(\lambda)$.
(iv) Repeat (i)–(iii) as λ ranges over $\operatorname{spec}(A)$, and take the direct sum of all resulting blocks.
The validity of steps (ii)–(iii) follows from Theorem 6.2.6.

6.3 Products AB and BA and Jordan Structure

Proposition 6.3.1. *AB and BA Have the Same Characteristic Polynomial*
For $A, B \in \operatorname{Mat}(n,n,\mathbb{F})$, $p_{AB} = p_{BA}$.
If $A \in \operatorname{GL}(n,\mathbb{F})$ or $B \in \operatorname{GL}(n,\mathbb{F})$, then $AB \sim_{\mathbb{F}} BA$; otherwise not necessarily.

Proof. In $\mathbb{F}(z)$, by Sylvester's identity $\det(zI - AB) = z^n \det(I - z^{-1}AB) = z^n \det(I - z^{-1}BA) = \det(zI - BA)$, hence $p_{AB} = p_{BA}$.
If A is invertible, $BA = A^{-1}(AB)A$; if B is invertible, $AB = B^{-1}(BA)B$.
Without invertibility, take $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.
Then $AB = B$, $BA = 0$, $p_{AB} = p_{BA} = z^2$, but $\operatorname{rank}(AB) = 1 \neq 0 = \operatorname{rank}(BA)$, so $AB \not\sim_{\mathbb{F}} BA$.
Structural Commentary: Characteristic polynomials see less than similarity.
Key Insight: AB and BA share determinant data; invertibility yields similarity. □

Theorem 6.3.2. *Nonzero Jordan Blocks of AB and BA*
Let $A, B \in \operatorname{Mat}(n,n,\mathbb{C})$. For every $\lambda \neq 0$, the Jordan blocks of AB and BA at λ coincide.

Proof. Set $X = \begin{bmatrix} AB & A \\ 0 & 0 \end{bmatrix}$ and $Y = \begin{bmatrix} 0 & A \\ 0 & BA \end{bmatrix}$. Then $\begin{bmatrix} I_n & 0 \\ B & I_n \end{bmatrix} X \begin{bmatrix} I_n & 0 \\ -B & I_n \end{bmatrix} = Y$, so X and Y are similar.
For $\lambda \neq 0$ and $p \geq 1$, the shifted powers have block triangular forms

$$(X - \lambda I_{2n})^p = \begin{bmatrix} (AB - \lambda I_n)^p & * \\ 0 & (-\lambda)^p I_n \end{bmatrix}, \qquad (Y - \lambda I_{2n})^p = \begin{bmatrix} (-\lambda)^p I_n & * \\ 0 & (BA - \lambda I_n)^p \end{bmatrix}$$

The diagonal block $(-\lambda)^p I_n$ is invertible.
Block Gaussian elimination by invertible row and column operations removes the starred block, so

$$\operatorname{rank}\big((X - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((AB - \lambda I_n)^p\big),$$
$$\operatorname{rank}\big((Y - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((BA - \lambda I_n)^p\big).$$

Since X and Y are similar, these left sides agree for all p.
Hence the shifted rank drops of AB and BA at λ agree. By Theorem 6.2.6, their Jordan blocks at λ coincide.
Structural Commentary: Similar block embeddings make the nonzero shifted ranks match.
Key Insight: At $\lambda \neq 0$, the extra block is invertible and splits off. □

Figure 69 summarizes the auxiliary similarity that identifies the nonzero shifted-rank data of AB and BA.

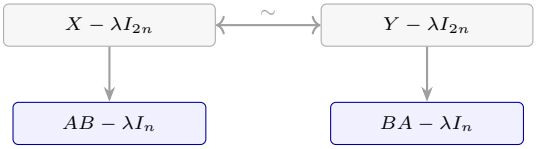

Figure 69: For $\lambda \neq 0$, the auxiliary similarity reduces the comparison to shifted ranks of AB and BA.

Corollary 6.3.3. *Similarity Criterion for AB and BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. AB is similar to BA if and only if $\operatorname{rank}\big((AB)^p\big) = \operatorname{rank}\big((BA)^p\big)$, $p = 1, \dots, n$.

Proof.
($\Rightarrow$) If $AB \sim BA$, then $(AB)^p \sim (BA)^p$ for all p, so their ranks are equal.
($\Leftarrow$) Assume the rank equalities hold for $p = 1, \dots, n$. By Theorem 6.3.2, the nonzero Jordan blocks coincide. Together with the equality at $p = 0$, the rank equalities give the same Weyr characteristic at $\lambda = 0$. By Theorem 6.2.6, the nilpotent blocks also agree, so the Jordan forms coincide up to block order.

Structural Commentary: Nonzero blocks agree automatically; only the nilpotent blocks must be tested.
Key Insight: Powers of AB and BA measure exactly the zero-Jordan-block data. □

Method 6.3.4. *Checking Whether AB and BA Are Similar*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$:

(i) The nonzero Jordan blocks of AB and BA already match.
(ii) Compute $\operatorname{rank}\big((AB)^p\big)$ and $\operatorname{rank}\big((BA)^p\big)$ for $p = 1, \dots, n$.
(iii) If all ranks agree, then $AB \sim BA$; if some rank differs, then only the nilpotent Jordan structure differs.

Step (i) follows from Theorem 6.3.2, and step (iii) is Corollary 6.3.3.

6.4 Nonderogatory Matrices and Commutants

Definition 6.4.1. *Nonderogatory Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *nonderogatory* if its Jordan form has exactly one Jordan block for each distinct eigenvalue. Equivalently, $\dim E_\lambda(A) = 1$ for every $\lambda \in \operatorname{spec}(A)$. The equivalence follows from Corollary 6.2.3.

Lemma 6.4.2. *Commutant of One Jordan Block*
If $J = J_m(\lambda)$ and $XJ = JX$, then $X = p(J)$ for some polynomial p with $\deg p < m$.

Proof. It suffices to treat $N = J_m(0)$, since $J_m(\lambda) = \lambda I_m + N$.
The vector $\mathbf{e}_m$ is cyclic for N: $(\mathbf{e}_m, N\mathbf{e}_m, \dots, N^{m-1}\mathbf{e}_m)$ is the standard basis in reverse order.
Choose the unique polynomial p with $\deg p < m$ such that $p(N)\mathbf{e}_m = X\mathbf{e}_m$.
For every $r = 0, \dots, m-1$, $p(N)N^r\mathbf{e}_m = N^r p(N)\mathbf{e}_m = N^r X\mathbf{e}_m = XN^r\mathbf{e}_m$.
Thus $p(N)$ and X agree on a basis, so $X = p(N) = q(J)$ for $q(z) = p(z - \lambda)$.
Structural Commentary: A single Jordan block is generated by one cyclic vector.
Key Insight: The cyclic vector determines the commutant. □

Theorem 6.4.3. *Nonderogatory Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is nonderogatory, then $\operatorname{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$.

Proof. The inclusion $\{p(A) : p \in \mathbb{C}[z]\} \subseteq \operatorname{Comm}(A)$ follows from Proposition 1.1.6.
For reverse inclusion, by Theorem 6.2.1, $A = SJS^{-1}$ for a Jordan matrix $J = J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_d}(\lambda_d)$ with distinct λ_i. Conjugation identifies $\operatorname{Comm}(A)$ with $\operatorname{Comm}(J)$ and satisfies $p(A) = Sp(J)S^{-1}$, so it suffices to prove the claim for J.
If $XJ = JX$ and $X = [X_{ij}]$ is partitioned conformally with J, the block equation is $X_{ij}J_{m_j}(\lambda_j) = J_{m_i}(\lambda_i)X_{ij}$. By Corollary 3.5.5, $X_{ij} = 0$ for $i \neq j$. Each diagonal block X_{ii} commutes with $J_{m_i}(\lambda_i)$, so by Lemma 6.4.2, $X_{ii} = p_i(J_{m_i}(\lambda_i))$ for some polynomial p_i with $\deg p_i < m_i$. The ideals generated by $(z - \lambda_i)^{m_i}$ are pairwise coprime, so Hermite interpolation gives a polynomial p with $p \equiv p_i \pmod{(z - \lambda_i)^{m_i}}$ for every i.
On the ith block, $p - p_i = (z - \lambda_i)^{m_i} q_i$ for some q_i, and $(J_{m_i}(\lambda_i) - \lambda_i I)^{m_i} = 0$. Hence $p(J_{m_i}(\lambda_i)) = p_i(J_{m_i}(\lambda_i)) = X_{ii}$ for every i, so $X = p(J)$. Conjugating back to A proves the claim.
Structural Commentary: Disjoint eigenvalue blocks force block diagonality; one Jordan block per eigenvalue makes each diagonal commutant polynomial.
Key Insight: Hermite interpolation glues the separate block polynomials into one polynomial in A. □

Figure 70 summarizes how the blockwise commutant polynomials are combined into one polynomial on the full Jordan matrix.

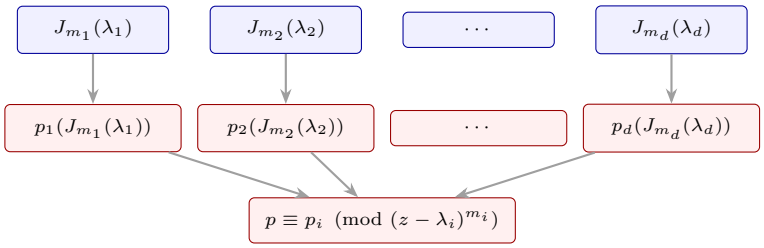

Figure 70: Blockwise commutant polynomials glue by interpolation.

Corollary 6.4.4. *Simple Spectrum Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ has n distinct eigenvalues, then $\operatorname{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$, $\dim \operatorname{Comm}(A) = n$.

Proof. Write $A = S \operatorname{diag}(\lambda_1, \dots, \lambda_n)S^{-1}$ with distinct λ_i. By Theorem 5.4.12, every $B \in \operatorname{Comm}(A)$ has the form $S \operatorname{diag}(\xi_1, \dots, \xi_n)S^{-1}$.
Choose the Lagrange interpolation polynomial $p(z) = \sum_{i=1}^{n} \xi_i \prod_{j \neq i} \frac{z - \lambda_j}{\lambda_i - \lambda_j}$.
Then $p(\lambda_i) = \xi_i$ for every i. By Theorem 1.1.1, $B = p(A)$.
The dimension statement is the case $n_1 = \cdots = n_n = 1$ of Theorem 5.4.12.
Structural Commentary: Simple spectrum leaves one scalar choice per eigendirection.
Key Insight: Interpolation converts arbitrary diagonal data into a polynomial in A. □

6.5 Jordan Operator Functions and Square Roots

Proposition 6.5.1. *Exponential of a Jordan Block*

For $k \in \mathbb{N}$ with $k \geq 1$, $\lambda \in \mathbb{C}$, and $t \in \mathbb{C}$, one has $e^{tJ_k(\lambda)} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} J_k(0)^r$.

Proof. Set $N = J_k(0)$, so $J_k(\lambda) = \lambda I_k + N$ and $N^k = 0$.
By binomial theorem, $(\lambda I_k + N)^m = \sum_{r=0}^{\min(m,k-1)} \binom{m}{r} \lambda^{m-r} N^r$.
Absolute convergence permits regrouping by powers of N. Thus $e^{tJ_k(\lambda)} = \sum_{r=0}^{k-1} \frac{t^r}{r!} N^r \sum_{s=0}^{\infty} \frac{(\lambda t)^s}{s!} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} N^r$.
Structural Commentary: Expand the scalar and nilpotent parts together, then regroup by nilpotent degree.
Key Insight: Nilpotence truncates the matrix directions while the scalar series remains exponential. □

Proposition 6.5.2. *Commuting and Similarity Laws for the Exponential*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ and $S \in GL_n(\mathbb{C})$. Then:
(i) $AB = BA \implies e^{A+B} = e^A e^B$.
(ii) $e^{SAS^{-1}} = Se^A S^{-1}$.

Proof.

(i) Since A and B commute, $(A+B)^m = \sum_{r=0}^{m} \binom{m}{r} A^r B^{m-r}$.

Absolute convergence permits the Cauchy product, $e^A e^B = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{r=0}^{m} \binom{m}{r} A^r B^{m-r} = e^{A+B}$.

(ii) Put $X = SAS^{-1}$ and $H_m \colon X^m = SA^m S^{-1}$. Then H_0 is $I_n = SI_n S^{-1}$.
If H_m holds, then $X^{m+1} = SA^m S^{-1} SAS^{-1} = SA^{m+1} S^{-1}$, so H_{m+1} holds.

By induction, H_m holds for all $m \in \mathbb{N}_0$, and $e^X = S \left(\sum_{m=0}^{\infty} \frac{A^m}{m!} \right) S^{-1} = Se^A S^{-1}$.

Structural Commentary: Apply power-series algebra only after verifying the required matrix identities.
Key Insight: Commutativity enables products; similarity passes through powers. □

Theorem 6.5.3. *Analytic Power Series on Jordan Blocks*
Let $n \in \mathbb{N}$, $A \in \mathrm{Mat}(n, n, \mathbb{C})$, and $f(z) = \sum_{m=0}^{\infty} c_m z^m$ have radius $R > 0$ with $\rho(A) < R$.
Set $D_R := \{z \in \mathbb{C} : |z| < R\}$ and $N_k := J_k(0)$. For $(k, \lambda) \in \mathbb{N} \times D_R$, set $F_k(\lambda) := \sum_{r=0}^{k-1} \frac{f^{(r)}(\lambda)}{r!} N_k^r$.
For $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$ with $\lambda_\alpha \in D_R$, set $f(J) := \bigoplus_{\alpha=1}^{s} F_{k_\alpha}(\lambda_\alpha)$. Then:
(i) $\sum_{m=0}^{\infty} c_m J_k(\lambda)^m$ *converges to* $F_k(\lambda)$ *for every* $(k, \lambda) \in \mathbb{N} \times D_R$.
(ii) The series $f(A) := \sum_{m=0}^{\infty} c_m A^m$ *converges.*
(iii) If $A = SJS^{-1}$ *is a Jordan decomposition, then* $f(A) = Sf(J)S^{-1}$.

Proof. For $M \in \mathbb{N}_0$, set $P_M(X) := \sum_{m=0}^{M} c_m X^m$, and use $\binom{m}{r} = 0$ for $r > m$.

(i) Fix $(k, \lambda) \in \mathbb{N} \times D_R$ and set $N = N_k$. Since $N^k = 0$, $J_k(\lambda)^m = \sum_{r=0}^{k-1} \binom{m}{r} \lambda^{m-r} N^r$ for $m \in \mathbb{N}_0$.

For $0 \leq r < k$, $\sum_{m=r}^{\infty} c_m \binom{m}{r} \lambda^{m-r} = \frac{f^{(r)}(\lambda)}{r!}$ by termwise differentiation.

Absolute convergence permits regrouping over $0 \leq r < k$, so $P_M(J_k(\lambda)) \to F_k(\lambda)$.

(ii) By Theorem 6.2.1, $A = SJS^{-1}$ with $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$. Since $\lambda_\alpha \in \mathrm{spec}(A)$ and $\rho(A) < R$, $\lambda_\alpha \in D_R$.

By (i), $P_M(J) \to f(J)$, and $P_M(A) = SP_M(J)S^{-1} \to Sf(J)S^{-1}$.

(iii) By (ii), $f(A) = \lim_{M \to \infty} P_M(A) = S \left(\lim_{M \to \infty} P_M(J) \right) S^{-1} = Sf(J)S^{-1}$.

Structural Commentary: Blockwise binomial expansion reduces $f(J_k(\lambda))$ to nilpotent powers.
Key Insight: Derivatives supply the chain coefficients. □

Example 6.5.4. *Closed Form for a 2×2 Exponential*
Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and set $\delta := \frac{a+d}{2}$ and $B := A - \delta I_2$. Choose $\Delta \in \mathbb{C}$ with $\Delta^2 = \left(\frac{a-d}{2}\right)^2 + bc$.
By Theorem 3.1.1, $B^2 = \Delta^2 I_2$. Since $A = \delta I_2 + B$, by Proposition 6.5.2(i) and even–odd decomposition of exponential series, $e^A = e^\delta \left(\cosh(\Delta) I_2 + \frac{\sinh(\Delta)}{\Delta} B \right)$. At $\Delta = 0$, $\frac{\sinh(\Delta)}{\Delta}$ is interpreted by continuity as 1.

Theorem 6.5.5. *Spectral Mapping and Jordan Structure of the Exponential*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, let $A = SJS^{-1}$ be a Jordan decomposition, and set $E := \exp(A)$.
Write $J = \bigoplus_{\alpha=1}^{s} B_\alpha$, $B_\alpha = J_{k_\alpha}(\lambda_\alpha)$. For $\rho \in \mathbb{C}$, set $I_\rho := \{\alpha : e^{\lambda_\alpha} = \rho\}$, $\Lambda_\rho(A) := \{\lambda \in \mathrm{spec}(A) : e^\lambda = \rho\}$.
Let $\mathcal{S}_\rho(M)$ denote the multiset of ρ-Jordan block sizes of M. Then:
(i) $\mathrm{spec}(E) = \{e^\lambda : \lambda \in \mathrm{spec}(A)\}$.
(ii) $\forall \rho \in \mathrm{spec}(E)$, $\mathcal{G}_\rho(E) = \bigoplus_{\lambda \in \Lambda_\rho(A)} \mathcal{G}_\lambda(A)$.

(iii) $\mathcal{S}_\rho(E) = \{k_\alpha : \alpha \in I_\rho\}$ *for every* $\rho \in \operatorname{spec}(E)$.
(iv) $\det(E) = e^{\operatorname{tr}(A)}$.

Proof. Set $\sigma(M) := \operatorname{spec}(M)$, $E_J := e^J$, $E_\alpha := e^{B_\alpha}$, and $\tau := \sum_{\alpha=1}^{s} k_\alpha \lambda_\alpha = \operatorname{tr}(A)$.

By Proposition 6.5.2(ii), $E = SE_JS^{-1}$; work with J. Let V_α be coordinate space of B_α; $K_\rho := \bigoplus_{\alpha \in I_\rho} V_\alpha$.

Write $B_\alpha = \lambda_\alpha I_{k_\alpha} + N_\alpha$, $q_\alpha(z) := \sum_{r=0}^{k_\alpha - 1} \frac{z^r}{(r+1)!}$, $Q_\alpha := q_\alpha(N_\alpha)$. By Proposition 6.5.1, $E_\alpha - e^{\lambda_\alpha} I_{k_\alpha} = e^{\lambda_\alpha} N_\alpha Q_\alpha$.

With $M_\alpha := Q_\alpha - I_{k_\alpha} \in N_\alpha \mathbb{C}[N_\alpha]$, $M_\alpha^{k_\alpha} = 0$ and $Q_\alpha^{-1} = \sum_{j=0}^{k_\alpha - 1} (-M_\alpha)^j$. Also $Q_\alpha N_\alpha = N_\alpha Q_\alpha$.

(i) $\operatorname{diag}(E_\alpha) = e^{\lambda_\alpha} I_{k_\alpha}$, so $\sigma(E_\alpha) = \{e^{\lambda_\alpha}\}$.
By $E = SE_JS^{-1}$ and $E_J = \bigoplus_{\alpha=1}^{s} E_\alpha$, $\sigma(E) = \{e^{\lambda_\alpha} : 1 \le \alpha \le s\} = \{e^\lambda : \lambda \in \sigma(A)\}$.
(ii) If $\alpha \in I_\rho$, then $(E_\alpha - \rho I_{k_\alpha})^r = \rho^r N_\alpha^r Q_\alpha^r$ for $r \in \mathbb{N}$, so $\mathcal{G}_\rho(E_J|_{V_\alpha}) = V_\alpha$.
If $\alpha \notin I_\rho$, then $E_\alpha - \rho I_{k_\alpha} \in GL_{k_\alpha}(\mathbb{C})$. Thus $\mathcal{G}_\rho(E) = S\mathcal{G}_\rho(E_J) = SK_\rho = \bigoplus_{\lambda \in \Lambda_\rho(A)} \mathcal{G}_\lambda(A)$.
(iii) For $\alpha \in I_\rho$ and $r \in \mathbb{N}$, $\operatorname{rank}((E_\alpha - \rho I_{k_\alpha})^r) = \operatorname{rank}(N_\alpha^r)$. By Theorem 6.2.6, $\mathcal{S}_\rho(E_\alpha) = \{k_\alpha\}$.
For $\alpha \notin I_\rho$, $\mathcal{S}_\rho(E_\alpha) = \varnothing$. By similarity and direct sums, (iii) follows.
(iv) $\det(E) = \det(E_J) = \prod_{\alpha=1}^{s} e^{k_\alpha \lambda_\alpha} = e^\tau = e^{\operatorname{tr}(A)}$.

Structural Commentary: Exponentials merge block labels with equal e^λ.
Key Insight: Q_α is invertible, so nilpotent ranks are unchanged. □

Definition 6.5.6. *Matrix Logarithm*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. A matrix $B \in \operatorname{Mat}(n, n, \mathbb{C})$ is a *matrix logarithm* of A if $e^B = A$.

Theorem 6.5.7. *Existence of Complex Matrix Logarithms*
Let $n \in \mathbb{N}$ *and* $A \in \operatorname{Mat}(n, n, \mathbb{C})$. *Then* $\exists B \in \operatorname{Mat}(n, n, \mathbb{C})$ *with* $e^B = A$ *iff* $A \in GL_n(\mathbb{C})$.

Proof.
(⇒) Suppose $A = e^B$. Since B and $-B$ commute, by Proposition 6.5.2(i), $Ae^{-B} = I_n = e^{-B}A$; $A \in GL_n(\mathbb{C})$.
(⇐) Suppose $A \in GL_n(\mathbb{C})$. Choose $A = SJS^{-1}$ with $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$. Then $\lambda_\alpha \neq 0$ for all α.

Choose $\ell_\alpha \in \mathbb{C}$, $e^{\ell_\alpha} = \lambda_\alpha$. Set $N_\alpha := J_{k_\alpha}(0)$, $X_\alpha := \lambda_\alpha^{-1} N_\alpha$. Set $Y_\alpha := \sum_{r=1}^{k_\alpha - 1} \frac{(-1)^{r+1}}{r} X_\alpha^r$, $L_\alpha := \ell_\alpha I_{k_\alpha} + Y_\alpha$.
Since $X_\alpha^{k_\alpha} = 0$, by $e^{\log(1+x)} = 1 + x$ in $\mathbb{C}[x]/(x^{k_\alpha})$, $e^{Y_\alpha} = I_{k_\alpha} + X_\alpha$.
Since $\ell_\alpha I_{k_\alpha}$ commutes with Y_α, by Proposition 6.5.2(i), $e^{L_\alpha} = e^{\ell_\alpha}(I_{k_\alpha} + X_\alpha) = J_{k_\alpha}(\lambda_\alpha)$.
Set $L_J := \bigoplus_{\alpha=1}^{s} L_\alpha$ and $B := SL_JS^{-1}$. By Proposition 6.5.2(ii), $e^B = Se^{L_J}S^{-1} = SJS^{-1} = A$.

Structural Commentary: Combine scalar logs with nilpotent log polynomials.
Key Insight: Nonzero block eigenvalues permit logarithms. □

Perspective 6.5.8. *Real Matrix Logarithms*
For $A \in \operatorname{Mat}(n, n, \mathbb{C})$, by Theorem 6.5.7, $A = e^B$ for some complex B iff $A \in GL_n(\mathbb{C})$.
Over $\mathbb{R}$, $A \in GL_n(\mathbb{R})$ is not sufficient.
If $A \in GL_n(\mathbb{R})$ and $\operatorname{spec}(A) \cap \mathbb{R} \subset (0, \infty)$, choose $\ell_\lambda = \log \lambda \in \mathbb{R}$ for real λ and $\ell_{\overline{\lambda}} = \overline{\ell_\lambda}$ for nonreal pairs.
By conjugate pairing of real Jordan blocks, the block logarithms assemble to $B \in \operatorname{Mat}(n, n, \mathbb{R})$ with $e^B = A$.
For $A \in GL_n(\mathbb{R})$, A^2 has a real logarithm: $\mu < 0$ and $\mu \in \operatorname{spec}(A^2)$ imply $\{\lambda, \overline{\lambda}\} = \{\pm i\sqrt{-\mu}\}$ for some $\lambda \in \operatorname{spec}(A)$. By conjugate pairing, the negative real blocks of A^2 occur in pairs.
These are sufficient facts, not the full real criterion.

Proposition 6.5.9. *Jordan Resolvent and Primary Spectral Projectors*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, *set* $\Sigma := \operatorname{spec}(A)$, *and set* $G_\nu := \mathcal{G}_\nu(A)$.
For $m \in \mathbb{N}$, $M \in \operatorname{Mat}(m, m, \mathbb{C})$, *and* $z \notin \operatorname{spec}(M)$, *set* $R_M(z) := (zI_m - M)^{-1}$.
By primary decomposition, $\mathbb{C}^n = \bigoplus_{\nu \in \Sigma} G_\nu$. *For* $\mu \in \Sigma$, *define* P_μ *by* $P_\mu|_{G_\nu} = \delta_{\mu\nu} I_{G_\nu}$.
For meromorphic F *near* μ, *let* $[F]_{\mu,-1}$ *be its* $(z-\mu)^{-1}$ *coefficient. Then:*

(i) *For* $k \in \mathbb{N}$, $\lambda \in \mathbb{C}$, $B = J_k(\lambda) = \lambda I_k + N$, *and* $z \neq \lambda$, $R_B(z) = \sum_{r=0}^{k-1} \frac{N^r}{(z-\lambda)^{r+1}}$.
(ii) $[R_A]_{\mu,-1} = P_\mu$ *for every* $\mu \in \Sigma$.
(iii) $P_\lambda P_\mu = \delta_{\lambda\mu} P_\mu$ *for all* $\lambda, \mu \in \Sigma$, *and* $AP_\mu = P_\mu A$ *for all* $\mu \in \Sigma$.
(iv) $\sum_{\mu \in \Sigma} P_\mu = I_n$.

Proof.
(i) Let $B = J_k(\lambda) = \lambda I_k + N$ and set $a := (z - \lambda)^{-1}$.

By $zI_k - B = (z - \lambda)(I_k - aN)$, $N^k = 0$, and the finite geometric identity, $(I_k - aN)^{-1} = \sum_{r=0}^{k-1} a^r N^r$.

By $R_B(z) = a(I_k - aN)^{-1}$, $R_B(z) = a \sum_{r=0}^{k-1} a^r N^r = \sum_{r=0}^{k-1} \frac{N^r}{(z-\lambda)^{r+1}}$.

(ii) Choose $A = SJS^{-1}$ by Theorem 6.2.1, with $J = \bigoplus_{\alpha=1}^{s} J_{k_\alpha}(\lambda_\alpha)$. By similarity, $R_A(z) = SR_J(z)S^{-1}$.

Set $H_\nu := \mathcal{G}_\nu(J)$ and $C_\mu := [R_J]_{\mu,-1}$. By (i), $C_\mu|_{H_\nu} = \delta_{\mu\nu} I_{H_\nu}$. By conjugation, $[R_A]_{\mu,-1} = SC_\mu S^{-1} = P_\mu$.

(iii) Let $v = \sum_{\nu\in\Sigma} v_\nu$, $v_\nu \in G_\nu$. Then $P_\mu v = v_\mu$, $P_\lambda P_\mu v = \delta_{\lambda\mu} v_\mu = \delta_{\lambda\mu} P_\mu v$; $AG_\nu \subseteq G_\nu$, $AP_\mu v = Av_\mu = P_\mu Av$.

(iv) For $v = \sum_{\nu\in\Sigma} v_\nu$ as in (iii), $\left(\sum_{\mu\in\Sigma} P_\mu\right) v = \sum_{\mu\in\Sigma} v_\mu = v$. Since v is arbitrary, (iv) follows.

Structural Commentary: Laurent coefficients isolate primary summands.
Key Insight: The simple-pole coefficient is the primary projection. □

Definition 6.5.10. *Contracting, Neutral, and Expanding Primary Subspaces*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, and let P_λ be the primary projection onto $\mathcal{G}_\lambda(A)$.
Set $\Sigma_- := \{\lambda \in \mathrm{spec}(A) : |\lambda| < 1\}$, $\Sigma_0 := \{\lambda \in \mathrm{spec}(A) : |\lambda| = 1\}$, and $\Sigma_+ := \{\lambda \in \mathrm{spec}(A) : |\lambda| > 1\}$. Define:

(i) $\mathcal{G}^-(A) := \bigoplus_{\lambda\in\Sigma_-} \mathcal{G}_\lambda(A)$ and $P_A^- := \sum_{\lambda\in\Sigma_-} P_\lambda$, the *contracting* part.

(ii) $\mathcal{G}^0(A) := \bigoplus_{\lambda\in\Sigma_0} \mathcal{G}_\lambda(A)$ and $P_A^0 := \sum_{\lambda\in\Sigma_0} P_\lambda$, the *neutral* part.

(iii) $\mathcal{G}^+(A) := \bigoplus_{\lambda\in\Sigma_+} \mathcal{G}_\lambda(A)$ and $P_A^+ := \sum_{\lambda\in\Sigma_+} P_\lambda$, the *expanding* part.

By primary decomposition, $\mathbb{C}^n = \mathcal{G}^-(A) \oplus \mathcal{G}^0(A) \oplus \mathcal{G}^+(A)$, and Proposition 6.5.9 identifies P_A^-, P_A^0, and P_A^+ as the corresponding projections.

Lemma 6.5.11. *Identity Plus Nilpotent Has a Square Root*
Let $\mathcal{V}$ be a complex vector space and let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $\exists\, R \in \mathcal{L}(\mathcal{V})$ such that $R^2 = I_\mathcal{V} + N$.

Proof. Choose $m \in \mathbb{N}$ with $N^m = 0$. Set $a_0 = 1$. For $1 \le k < m$, set $s_k := \sum_{r=1}^{k-1} a_r a_{k-r}$ and $a_k := \frac{1}{2}(\delta_{k1} - s_k)$.
Set $R := \sum_{k=0}^{m-1} a_k N^k$. For $0 \le k < m$, set $c_k := \sum_{r=0}^{k} a_r a_{k-r}$.
By recursion, $c_0 = 1$, $c_1 = 1$ if $m \ge 2$, and $c_k = 0$ for $2 \le k < m$. By $N^m = 0$, $R^2 = \sum_{k=0}^{m-1} c_k N^k = I_\mathcal{V} + N$.
Structural Commentary: Coefficient matching truncates at nilpotency.
Key Insight: Choose a_k so R^2 has only degrees 0 and 1. □

Figure 71 summarizes the finite polynomial construction used to square-root an identity-plus-nilpotent operator.

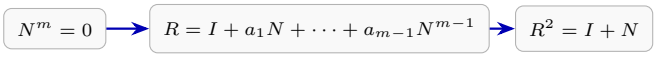

Figure 71: Nilpotence truncates the square-root series; coefficients are chosen so $R^2 = I + N$.

Theorem 6.5.12. *Invertible Complex Operators Have Square Roots*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$ be invertible. Then $\exists\, R \in \mathcal{L}(\mathcal{V})$ such that $R^2 = T$.

Proof. If $\mathcal{V} = \{\mathbf{0}\}$, take $R = I_\mathcal{V}$. Assume $\mathcal{V} \neq \{\mathbf{0}\}$. Let $\{\lambda_1, \ldots, \lambda_m\} = \mathrm{spec}(T)$ and set $\mathcal{U}_j := \mathcal{G}_{\lambda_j}(T)$.
By Theorem 5.1.1, $\mathcal{V} = \bigoplus_{j=1}^{m} \mathcal{U}_j$. By $T \in GL(\mathcal{V})$, $\lambda_j \neq 0$. Set $N_j := (T - \lambda_j I_\mathcal{V})|_{\mathcal{U}_j}$ and $M_j := \lambda_j^{-1} N_j$.
Then M_j is nilpotent and $T|_{\mathcal{U}_j} = \lambda_j(I_{\mathcal{U}_j} + M_j)$. By Lemma 6.5.11, choose $B_j \in \mathcal{L}(\mathcal{U}_j)$ with $B_j^2 = I_{\mathcal{U}_j} + M_j$.
Choose $\alpha_j \in \mathbb{C}$ with $\alpha_j^2 = \lambda_j$ and set $R_j := \alpha_j B_j$. By these choices, $R_j^2 = \lambda_j B_j^2 = \lambda_j(I_{\mathcal{U}_j} + M_j) = T|_{\mathcal{U}_j}$.
Define $R := \bigoplus_{j=1}^{m} R_j$. By $(R^2)|_{\mathcal{U}_j} = R_j^2 = T|_{\mathcal{U}_j}$ for every j, $R^2 = T$.
Structural Commentary: Square-root each primary restriction, then sum.
Key Insight: Nonzero eigenvalues separate scalar roots from nilpotent roots. □

7 Transpose and Conjugate Similarity

7.1 Transpose Similarity and Conjugate Similarity

Theorem 7.1.1. *Conjugate Similarity Criterion*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then A is similar to $\overline{A}$ if and only if each $J_k(\lambda)$ occurs in the Jordan form of A with the same multiplicity as $J_k(\overline{\lambda})$.

Proof. By Theorem 6.2.1, choose a Jordan form J of A. Then $\overline{J}$ is a Jordan form of $\overline{A}$, obtained by replacing each $J_k(\lambda)$ by $J_k(\overline{\lambda})$.
($\Rightarrow$) If A is similar to $\overline{A}$, then J and $\overline{J}$ are similar.
By Corollary 6.2.7, their Jordan blocks have the same multiplicities.
($\Leftarrow$) If the multiplicities agree, then J and $\overline{J}$ differ only by a permutation of blocks. Hence A is similar to $\overline{A}$.
Structural Commentary: Conjugation changes eigenvalue labels but not block sizes.
Key Insight: Non-real Jordan blocks must occur in conjugate pairs. □

Definition 7.1.2. *Reversal Matrix*
For $k \ge 1$, define the *reversal matrix* $R_k \in \mathrm{Mat}(k,k,\mathbb{C})$ by $(R_k)_{ij} = \begin{cases} 1, & i+j = k+1, \\ 0, & i+j \neq k+1 \end{cases}$. Then $R_k^t = R_k$ and $R_k^2 = I_k$.

Lemma 7.1.3. *Reversal Conjugates a Jordan Block to Its Transpose*
For every $k \ge 1$ and $\lambda \in \mathbb{C}$, $J_k(\lambda)^t = R_k J_k(\lambda) R_k$. Moreover, $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.

Proof. Let $N = J_k(0)$. Since R_k reverses standard basis order, $R_k N R_k = N^t$.
Adding scalar gives $R_k J_k(\lambda) R_k = J_k(\lambda)^t$. As $R_k = R_k^t = R_k^{-1}$, this identity gives $J_k(\lambda)^t R_k = R_k J_k(\lambda)$ and $R_k J_k(\lambda)^t = J_k(\lambda) R_k$, so $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.
Structural Commentary: Reversal changes the superdiagonal shift into the subdiagonal shift.
Key Insight: Transposition reads the same Jordan chain backwards. □

Theorem 7.1.4. *Transpose Similarity and Symmetric Factorisations*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then:

(i) A is similar to A^t via a symmetric similarity matrix.
(ii) $A = BC = DE$, where B, C, D, E are symmetric and B, E are invertible.

Proof. By Theorem 6.2.1, write $A = SJS^{-1}$ for a Jordan matrix J, and let R be the direct sum of the reversal matrices matching the Jordan blocks of J. By Lemma 7.1.3, $J^t = RJR$, $R^t = R$, $R^2 = I_n$, and both RJ and JR are symmetric.

(i) Set $P = (S^{-1})^t R S^{-1}$. Then $P^t = P$, $P^{-1} = SRS^t$, and $PAP^{-1} = (S^{-1})^t RJRS^t = (S^{-1})^t J^t S^t = A^t$.
(ii) Using $R^2 = I_n$, $A = (SRS^t)\big((S^{-1})^t RJS^{-1}\big)$ and $A = (SJRS^t)\big((S^{-1})^t RS^{-1}\big)$.
The products equal SJS^{-1} because $S^t(S^{-1})^t = I_n$ and $R^2 = I_n$. All four factors are symmetric because R, RJ, JR are symmetric; SRS^t and $(S^{-1})^t RS^{-1}$ are invertible.

Structural Commentary: Reduce to Jordan blocks, reverse chains, then conjugate back.
Key Insight: Symmetric involutions produce both transpose similarity and symmetric factors. □

Figure 72 illustrates how reversing a Jordan chain converts the superdiagonal shift into its transpose.

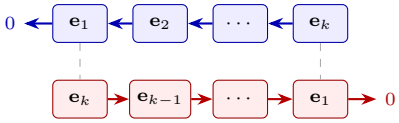

Figure 72: Reversal gives $R_k J_k(0) R_k = J_k(0)^t$: the same chain read backwards.

Corollary 7.1.5. *Hermitian Products and Conjugate Products*
Let $A, H, K \in \mathrm{Mat}(n, n, \mathbb{C})$, and suppose H and K are Hermitian.
Then HK is similar to KH, and $A\overline{A}$ is similar to $\overline{A}A$.

Proof. For each $p = 1, \ldots, n$, adjoints preserve rank, and $((HK)^p)^* = (KH)^p$. Hence $\mathrm{rank}\big((HK)^p\big) = \mathrm{rank}\big((KH)^p\big)$.
Similarly, entrywise conjugation preserves rank and gives $\overline{(A\overline{A})^p} = (\overline{A}A)^p$, so $\mathrm{rank}\big((A\overline{A})^p\big) = \mathrm{rank}\big((\overline{A}A)^p\big)$.
Both similarities follow from Corollary 6.3.3.
Structural Commentary: Adjoint and conjugation preserve rank and reverse product order.
Key Insight: The rank-power test turns order reversal into similarity. □

Theorem 7.1.6. *Conjugate Similarity and Hermitian Products*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is similar to $\overline{A}$.
(ii) A is similar to A^*.
(iii) A is similar to A^* via a Hermitian similarity matrix.
(iv) $A = HK = LM$, where H, K, L, M are Hermitian, and H, M are invertible.
(v) $A = HK$ for some Hermitian matrices H, K.

Proof.

(i) ⇒ (ii) By Theorem 7.1.4 applied to $\overline{A}$, $\overline{A} \sim (\overline{A})^t = A^*$. Thus $A \sim A^*$.
(ii) ⇒ (iii) Choose invertible S such that $A = SA^*S^{-1}$, so $AS = SA^*$. For $\theta \in \mathbb{R}$, put $S_\theta = e^{i\theta}S$.
Then $AS_\theta = S_\theta A^*$ and, by taking adjoints, $AS_\theta^* = S_\theta^* A^*$. Hence $A(S_\theta + S_\theta^*) = (S_\theta + S_\theta^*)A^*$.
The matrix $H_\theta = S_\theta + S_\theta^*$ is Hermitian. Also $S_\theta^{-1} H_\theta = I_n + e^{-2i\theta} S^{-1}S^*$. Thus H_θ is singular exactly when $-e^{2i\theta}$ is an eigenvalue of $S^{-1}S^*$. There are only finitely many such values, so choose θ for which H_θ is invertible. Then $A = H_\theta A^* H_\theta^{-1}$.
(iii) ⇒ (iv) Let R be invertible Hermitian and satisfy $A = RA^*R^{-1}$. Then $AR = RA^*$ and $R^{-1}A = A^*R^{-1}$.
Thus A^*R^{-1} and RA^* are Hermitian, and $A = R(A^*R^{-1}) = (RA^*)R^{-1}$.
This gives the two asserted Hermitian factorizations, with the first factor in the first product and the second factor in the second product invertible.
(iv) ⇒ (v) The first factorization in (iv) already has the form $A = HK$ with H and K Hermitian.
(v) ⇒ (i) If $A = HK$ with H, K Hermitian, then by Corollary 7.1.5, $HK \sim KH$.
By Theorem 7.1.4, $KH \sim (KH)^t = H^t K^t = \overline{H}\,\overline{K} = \overline{A}$. Hence $A \sim \overline{A}$.

Structural Commentary: The equivalences move between conjugation, adjoints, and Hermitian products by combining transpose similarity with a Hermitianized similarity matrix.
Key Insight: A phase-rotated average $e^{i\theta}S + e^{-i\theta}S^*$ turns an arbitrary similarity to A^* into a Hermitian one. □

Figure 73 summarizes the implication cycle connecting conjugate similarity, adjoint similarity, and Hermitian factorization.

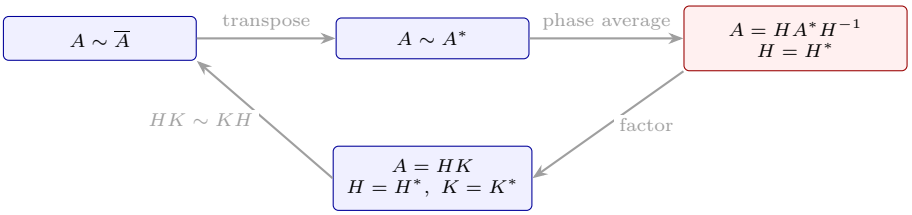

Figure 73: Conjugate similarity, adjoint similarity, and Hermitian product factorisations are equivalent.

Part VI
Matrix Analysis and Numerical Linear Algebra

Matrix classes package operator structure in coordinates. Trace, determinant, rank, positivity, idempotents, nilpotents, spectral radius, and structured products give computable invariants.

1 Trace, Determinant, Rank, and Matrix Invariants

1.1 Trace and Determinant

Definition 1.1.1. *Trace and Determinant of a Matrix*
For $A = [A_{j,k}] \in \mathrm{Mat}(n, n, \mathbb{F})$, the *trace* of A is $\mathrm{tr}(A) := A_{1,1} + \cdots + A_{n,n}$.
The *determinant* of A is $\det A := \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$.

Definition 1.1.2. *Gram Determinant*
For vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in an inner-product space, define their *Gram determinant* by $\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \det \mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.

Example 1.1.3. *Trace Functional and the Frobenius Inner Product*
On $\mathrm{Mat}(n, n, \mathbb{F})$ with the Frobenius inner product from Example 1.1.7, the Riesz vector of the trace functional $A \mapsto \mathrm{tr}(A)$ is I_n, since $\langle A, I_n \rangle_F = \sum_{j=1}^{n} a_{jj} = \mathrm{tr}(A)$.

Proposition 1.1.4. *Trace Arithmetic*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ and $r \in \mathbb{F}$. Then
(i) $\mathrm{tr}(rA) = r\,\mathrm{tr}(A)$;
(ii) $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$.

Proof. We prove the two identities separately.

(i) Summing the diagonal entries of rA gives $\mathrm{tr}(rA) = \sum_{j=1}^{n} r a_{jj} = r \sum_{j=1}^{n} a_{jj} = r\,\mathrm{tr}(A)$.

(ii) Summing the diagonal entries of $A + B$ gives $\mathrm{tr}(A + B) = \sum_{j=1}^{n} (a_{jj} + b_{jj}) = \sum_{j=1}^{n} a_{jj} + \sum_{j=1}^{n} b_{jj} = \mathrm{tr}(A) + \mathrm{tr}(B)$.

Structural Commentary: Trace is linear because it is the sum of diagonal entries.
Key Insight: Scalar multiplication and addition pass through the diagonal sum. □

Proposition 1.1.5. *Trace Is Cyclic for Two Factors*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(n, m, \mathbb{F})$, then $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.

Proof. Expand both traces by entries: $\mathrm{tr}(AB) = \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k} B_{k,j} = \sum_{k=1}^{n} \sum_{j=1}^{m} B_{k,j} A_{j,k} = \mathrm{tr}(BA)$.
Structural Commentary: Cyclicity is just reindexing the same double sum.
Key Insight: The two double sums contain exactly the same scalar terms. □

Proposition 1.1.6. *Cyclic Products*
Let $m \geq 2$, $A_j \in \mathrm{Mat}(n_j, n_{j+1}, \mathbb{F})$ for $j = 1, \ldots, m$, $n_{m+1} = n_1$, $P_{p,q} := A_p A_{p+1} \cdots A_q$.
Then $\mathrm{tr}(P_{1,m}) = \mathrm{tr}(A_m P_{1,m-1})$. *Consequently, for $A, B, C \in \mathrm{Mat}(n, n, \mathbb{F})$,* $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$.

Proof. Since $P_{1,m-1} \in \mathrm{Mat}(n_1, n_m, \mathbb{F})$ and $A_m \in \mathrm{Mat}(n_m, n_1, \mathbb{F})$, by Proposition 1.1.5, $\mathrm{tr}(P_{1,m}) = \mathrm{tr}(P_{1,m-1} A_m) = \mathrm{tr}(A_m P_{1,m-1})$.
Applying this cyclic move twice to ABC gives the three-factor identity.
Structural Commentary: Trace preserves cyclic order, not arbitrary order.
Key Insight: One cyclic move is $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ with $X = P_{1,m-1}$ and $Y = A_m$. □

Example 1.1.7. *Cyclicity Is Not Arbitrary Commutativity*
Let $A = E_{1,2}$, $B = E_{2,1}$, and $C = E_{1,1}$ in $\mathrm{Mat}(2, 2, \mathbb{F})$. Then $\mathrm{tr}(ABC) = 1$, while $\mathrm{tr}(ACB) = 0$.
Thus trace permits cyclic rotations of factors, but not arbitrary rearrangements.

Theorem 1.1.8. *Basic Determinant Laws*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) $\det I_n = 1$.
(ii) $\det(A)$ *is linear as a function of each column of A when the other columns are fixed.*
(iii) Swapping two columns negates the determinant; equal columns force $\det A = 0$.
(iv) If A is triangular, then $\det A = A_{1,1} \cdots A_{n,n}$.
(v) $\det(AB) = \det(A) \det(B)$.
(vi) A is invertible if and only if $\det A \neq 0$. *In that case* $\det(A^{-1}) = (\det A)^{-1}$.
(vii) $\det(A^t) = \det(A)$.

Proof.

(i) $\det I_n = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) (I_n)_{\sigma(1),1} \cdots (I_n)_{\sigma(n),n} = 1$ as product is nonzero only when $\sigma(k) = k\ \forall k$.

(ii) For fixed k, $\det(A_{\cdot,1}, \ldots, \mathbf{x}, \ldots, A_{\cdot,n}) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) x_{\sigma(k)} \prod_{\ell \neq k} A_{\sigma(\ell),\ell}$, which is linear in $\mathbf{x}$.

(iii) If $B_{\cdot,k} = A_{\cdot,\tau(k)}$ for transposition τ, then $\det B = \sum_{\rho \in \mathfrak{S}_n} \mathrm{sign}(\rho\tau) \prod_{k=1}^{n} A_{\rho(k),k} = -\det A$.
If the swapped columns are equal, then $B = A$, so $\det A = -\det A$ and $\det A = 0$.

(iv) If A is upper triangular and $\prod_{k=1}^{n} A_{\sigma(k),k} \neq 0$, then $\sigma(k) \leq k$ for all k, hence $\sigma(k) = k$ for all k.
Thus $\det A = A_{1,1} \cdots A_{n,n}$. If A is lower triangular, nonzero products require $\sigma(k) \geq k$ for all k, again forcing $\sigma(k) = k$ for all k, so the same diagonal product formula holds.
(v) If F is alternating and multilinear in columns, then expanding each column in the standard basis gives
$F(A_{\cdot,1}, \ldots, A_{\cdot,n}) = \sum_{i_1,\ldots,i_n} A_{i_1,1} \cdots A_{i_n,n} F(\mathbf{e}_{i_1}, \ldots, \mathbf{e}_{i_n}) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) F(I_n) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$.
The terms with repeated i_j vanish by alternation, and the surviving ordered bases differ from $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ by sign of permutation.
For $H(B) = \det(AB)$, map H is alternating multilinear and $H(I_n) = \det A$.
Hence $\det(AB) = H(B) = H(I_n) \det B = \det(A) \det(B)$.
(vi) If A is invertible, then $1 = \det I_n = \det(AA^{-1}) = \det(A) \det(A^{-1})$.
If $A_{\cdot,k} = \sum_{\ell \neq k} c_\ell A_{\cdot,\ell}$, then $\det A = \sum_{\ell \neq k} c_\ell \det(A_{\cdot,1}, \ldots, A_{\cdot,\ell}, \ldots, A_{\cdot,\ell}, \ldots, A_{\cdot,n}) = 0$.
Thus $\det A \neq 0$ iff the columns of A are independent, iff A is invertible.
(vii) Using the permutation definition and replacing σ by σ^{-1},
$\det(A^t) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) A_{1,\sigma(1)} \cdots A_{n,\sigma(n)} = \sum_{\tau \in \mathfrak{S}_n} \operatorname{sign}(\tau) A_{\tau(1),1} \cdots A_{\tau(n),n} = \det(A)$; $\operatorname{sign}(\tau^{-1}) = \operatorname{sign}(\tau)$.

Structural Commentary: The determinant is the normalized alternating multilinear measure of columns.
Key Insight: Multilinearity expands columns; alternation kills repeated directions. □

Figure 74 illustrates the alternating column geometry encoded by the basic determinant laws.

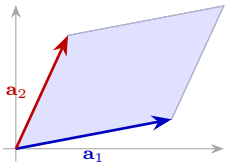

Figure 74: Determinant is signed column-volume; swaps reverse orientation and dependence gives zero volume.

Method 1.1.9. *Determinant by Structure*
To compute or simplify a determinant:
(i) first look for triangular or block triangular form;
(ii) use column linearity to split complicated columns only when it reduces structure;
(iii) use column swaps to move useful pivots into position, tracking a factor of -1 for each swap;
(iv) detect zero determinants from dependent columns or repeated columns;
(v) for products or changes of basis, use multiplicativity instead of expanding permutations.
These steps use Theorem 1.1.8.

Figure 75 summarizes the geometric scaling interpretation used by the determinant-by-structure method.

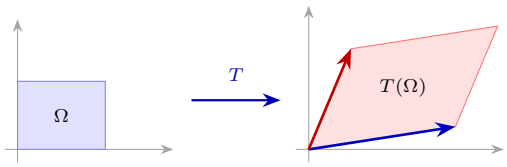

Figure 75: A linear map scales oriented volume by its determinant; singular maps collapse volume to zero.

Lemma 1.1.10. *Identity Blocks Do Not Change Determinant*
Let $A \in \operatorname{Mat}(m, m, \mathbb{F})$ and let $n \geq 0$. Then $\det(I_n \oplus A) = \det(A) = \det(A \oplus I_n)$.

Proof. In the permutation formula for $I_n \oplus A$, a nonzero term must satisfy $\sigma(k) = k$ for $1 \leq k \leq n$. Remaining choices are exactly permutations of $\{n+1, \ldots, n+m\}$, after relabeling, gives $\det(A)$. Hence $\det(I_n \oplus A) = \det(A)$.
The same argument applied to the last n coordinates gives $\det(A \oplus I_n) = \det(A)$.
The case $n = 0$ is the empty identity block convention.
Structural Commentary: Identity blocks add directions without changing volume.
Key Insight: In the determinant expansion, the identity block forces the matching coordinates to be fixed. □

Theorem 1.1.11. *Determinant of an Upper Block-Triangular Matrix*
Let $A \in \operatorname{Mat}(r, r, \mathbb{F})$, $B \in \operatorname{Mat}(r, s, \mathbb{F})$, and $D \in \operatorname{Mat}(s, s, \mathbb{F})$. Then $\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det(A) \det(D)$.

Proof. Factor $\begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} I_r & B \\ 0 & I_s \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & I_s \end{bmatrix}$.
The first and last factors have determinants $\det(D)$ and $\det(A)$ by Lemma 1.1.10.
The middle factor is upper triangular with every diagonal entry equal to 1, so its determinant is 1.
Multiplicativity of determinant gives the formula.
Structural Commentary: Block upper triangular matrices separate determinant contribution into diagonal blocks. The off-diagonal block does not affect the determinant.
Key Insight: Factor the matrix into two block diagonal pieces and one unit upper triangular piece. □

Proposition 1.1.12. *Schur Complement Determinant Formula*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \operatorname{Mat}(r, r, \mathbb{F})$, $D \in \operatorname{Mat}(s, s, \mathbb{F})$.
If A is invertible, then the Schur complement of A in M is $M/A := D - CA^{-1}B$, and $\det(M) = \det(A) \det(M/A)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & 0 \\ -CA^{-1} & I_s \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$.
The elimination matrix is lower block-triangular with identity diagonal blocks, so its determinant is 1. Take determinants, apply Theorem 1.1.11 to right-hand block triangular matrix, $\det(M) = \det(A)\det(D - CA^{-1}B)$.
Structural Commentary: Schur complementation is block Gaussian elimination.
Key Insight: Eliminating C splits the determinant into $\det(A)$ times the remaining block determinant. □

Proposition 1.1.13. *Schur Complement of the Lower-Right Block*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$.
If D is invertible, then the Schur complement of D in M is $M/D := A - BD^{-1}C$, and $\det(M) = \det(D)\det(M/D)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & -BD^{-1} \\ 0 & I_s \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A - BD^{-1}C & 0 \\ C & D \end{bmatrix}$.
The elimination matrix has determinant 1. The transpose of the matrix on the right is upper block-triangular, and $\det(X^t) = \det(X)$ by Theorem 1.1.8. By Theorem 1.1.11, $\det(M) = \det(A - BD^{-1}C)\det(D)$.
Structural Commentary: This is the lower-right version of Schur complementation. Eliminating the upper-right block isolates the determinant contribution of D.
Key Insight: Eliminate by blocks, then read the determinant from the diagonal blocks. □

Proposition 1.1.14. *Bordered Determinant Formulas*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, *let* $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, *and let* $c \in \mathbb{F}$.

(i) If $c \neq 0$, *then* $\det \begin{bmatrix} c & \mathbf{x}^t \\ \mathbf{y} & A \end{bmatrix} = c \det\left(A - \frac{1}{c}\mathbf{y}\mathbf{x}^t\right)$.

(ii) If A is invertible, then $\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = \left(c - \mathbf{y}^t A^{-1}\mathbf{x}\right)\det(A)$

Proof.
(i) This is Proposition 1.1.12 with the scalar block c in the upper-left corner.
(ii) This is Proposition 1.1.12 with A as the upper-left block and $D = c$.
Structural Commentary: Bordered determinants are Schur complements with scalar borders.
Key Insight: Treat the border as a 1×1 block and apply the Schur complement formula. □

Theorem 1.1.15. *Sylvester Determinant Identity*
Let $A \in \mathrm{Mat}(n, m, \mathbb{F})$ *and* $B \in \mathrm{Mat}(m, n, \mathbb{F})$. *Then* $\det(I_n + AB) = \det(I_m + BA)$.

Proof. Let $M = \begin{bmatrix} I_n & -A \\ B & I_m \end{bmatrix}$.
By Proposition 1.1.13 with lower-right block I_m, $\det(M) = \det(I_n + AB)$. By Proposition 1.1.12 with upper-left block I_n, $\det(M) = \det(I_m + BA)$. Hence the two determinants are equal.
Structural Commentary: Two Schur complements of one block matrix give Sylvester's identity.
Key Insight: Compare the same determinant after eliminating different identity blocks. □

Theorem 1.1.16. *Nonzero Eigenvalues of AB and BA*
Let $A \in \mathrm{Mat}(m, n, \mathbb{C})$, $B \in \mathrm{Mat}(n, m, \mathbb{C})$, *and suppose* $n \geq m$. *Then* $p_{BA}(z) = z^{n-m}p_{AB}(z)$.
Consequently, for $\lambda \neq 0$, $m_{AB}(\lambda) = m_{BA}(\lambda)$. *If* $m_{AB}(0) = k$, *then* $m_{BA}(0) = k + n - m$.

Proof. For $z \neq 0$, by Theorem 1.1.15 applied to $-z^{-1}B$ and A,

$$\det(zI_n - BA) = z^n \det(I_n - z^{-1}BA) = z^n \det(I_m - z^{-1}AB) = z^{n-m}\det(zI_m - AB).$$

Both sides are polynomials in z, so the identity holds for all z. The claimed multiplicities follow by factoring p_{AB} and comparing with $p_{BA} = z^{n-m}p_{AB}$.
Structural Commentary: Sylvester's determinant identity moves the rectangular product across the determinant.
Key Insight: Passing from AB to BA only adds the necessary extra zero roots when the sizes differ. □

Theorem 1.1.17. *Trace and Determinant Are Similarity Invariants*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ *are similar, then* $\mathrm{tr}(A) = \mathrm{tr}(B)$, $\det(A) = \det(B)$.

Proof. Write $B = PAP^{-1}$ with P invertible.
Proposition 1.1.5 gives $\mathrm{tr}(B) = \mathrm{tr}\big(P(AP^{-1})\big) = \mathrm{tr}\big((AP^{-1})P\big) = \mathrm{tr}(A)$.
Theorem 1.1.8 gives $\det(B) = \det(P)\det(A)\det(P^{-1}) = \det(A)$.
Structural Commentary: Trace and determinant ignore the coordinate system used to write an operator.
Key Insight: Conjugating by P inserts P and P^{-1}, which cancel inside trace and determinant. □

Proposition 1.1.18. *Trace and Determinant from Eigenvalues*
Let $\lambda_1, \ldots, \lambda_n$ *list the eigenvalues of* $A \in \mathrm{Mat}(n, n, \mathbb{C})$ *according to* m_A. *Then* $\mathrm{tr}\, A = \sum_{j=1}^{n} \lambda_j$ *and* $\det A = \prod_{j=1}^{n} \lambda_j$.

Proof. By Fundamental Theorem of Algebra, $p_A(z) = \prod_{j=1}^{n}(z - \lambda_j) = z^n - \left(\sum_{j=1}^{n}\lambda_j\right)z^{n-1} + \cdots + (-1)^n \prod_{j=1}^{n}\lambda_j$.
Theorem 2.1.3 gives $p_A(z) = z^n - (\mathrm{tr}\, A)z^{n-1} + \cdots + (-1)^n \det A$. Uniqueness of coefficients gives the identities.
Structural Commentary: Coefficients of p_A encode symmetric functions of the eigenvalues.
Key Insight: Trace is the first root sum and determinant is the full root product. □

Proposition 1.1.19. *Transpose, Conjugate, and Adjoint Spectra*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $p_{A^t}(z) = p_A(z)$, $p_{\overline{A}}(z) = \overline{p_A(\overline{z})}$, *and* $p_{A^*}(z) = \overline{p_A(\overline{z})}$.
Consequently, if $\lambda \in \mathrm{spec}(A)$, *then* $m_{A^t}(\lambda) = m_A(\lambda)$ *and* $m_{\overline{A}}(\overline{\lambda}) = m_{A^*}(\overline{\lambda}) = m_A(\lambda)$.

Proof. Use determinant identities:

$$p_{A^t}(z) = \det(zI_n - A^t) = \det((zI_n - A)^t) = p_A(z),$$
$$p_{\overline{A}}(z) = \det(zI_n - \overline{A}) = \det(\overline{\overline{z}I_n - A}) = \overline{\det(\overline{z}I_n - A)} = \overline{p_A(\overline{z})}.$$

Since $A^* = \overline{A}^t$, the first identity applied to $\overline{A}$ gives the formula for A^*.
Multiplicity is preserved because conjugating a polynomial sends a factor $(z-\lambda)^k$ to $(z-\overline{\lambda})^k$.
Structural Commentary: Transpose preserves p_A; conjugation conjugates its roots.
Key Insight: The adjoint combines conjugation with transposition, so its spectrum is conjugated. □

Perspective 1.1.20. *Similarity Invariants and Canonical Forms*
Similarity is equality up to change of basis. Trace, determinant, characteristic polynomials, and minimal polynomials are similarity invariants by Theorems 1.1.17, 2.1.7, and 2.2.5, but they are not generally complete.
Canonical forms, such as the Jordan form of Theorem 6.2.1, choose one structured representative of each similarity class when possible.

Principle 1.1.21. *Trace and Determinant Are Coordinate-Free*
Trace and determinant are first computed from a matrix, but their real structural meaning is operator-level: changing basis conjugates the matrix, and conjugation preserves both quantities.

Theorem 1.1.22. *Trace and Determinant of an Operator Are Basis-Independent*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β, γ be bases of $\mathcal{V}$.
Then $\operatorname{tr}([T]_{\beta\leftarrow\beta}) = \operatorname{tr}([T]_{\gamma\leftarrow\gamma})$, $\det([T]_{\beta\leftarrow\beta}) = \det([T]_{\gamma\leftarrow\gamma})$.

Proof. By Theorem 5.1.1, the two matrices are similar: $[T]_{\gamma\leftarrow\gamma} = C_{\gamma\leftarrow\beta}[T]_{\beta\leftarrow\beta}C_{\beta\leftarrow\gamma}$.
Theorem 1.1.17 now gives equality of traces and determinants.
Structural Commentary: Basis changes conjugate matrices; trace and determinant remain operator invariants.
Key Insight: Similar matrices have the same trace and determinant. □

Figure 76 summarizes the conjugation relation that makes operator trace and determinant basis-independent.

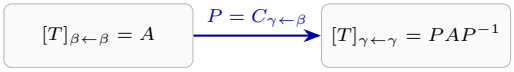

Figure 76: Basis changes conjugate matrices; trace and determinant are unchanged.

Definition 1.1.23. *Trace and Determinant of an Operator*
Let $\mathcal{V}$ be finite-dim, let $T \in \mathcal{L}(\mathcal{V})$. Choose any basis β of $\mathcal{V}$. Define $\operatorname{tr} T := \operatorname{tr}([T]_{\beta\leftarrow\beta})$, $\det T := \det([T]_{\beta\leftarrow\beta})$.
Theorem 1.1.22 shows these definitions do not depend on chosen basis.

Proposition 1.1.24. *Unitary Matrices and Isometries Have Determinant of Modulus One*
Let $U \in \operatorname{Mat}(n, n, \mathbb{F})$ be unitary. Then $|\det U| = 1$. More generally, if $S \in \mathcal{L}(\mathcal{V})$ is an isometry on a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $|\det S| = 1$.

Proof. The permutation definition gives $\det(U^*) = \overline{\det U}$. By Theorem 1.1.8, taking determinants in $U^*U = I_n$ gives $1 = \det(U^*)\det(U) = \overline{\det U}\det U = |\det U|^2$. Hence $|\det U| = 1$.
For an isometry S, choose an orthonormal basis and let U be the matrix of S in that basis. Then U is unitary by Theorem 2.6.6, and $\det S = \det U$.
Structural Commentary: In orthonormal coordinates, isometries are unitary or orthogonal, so the determinant calculation is the matrix one.
Key Insight: Determinants turn $U^*U = I$ into $|\det U|^2 = 1$. □

Proposition 1.1.25. *Trace and Determinant from q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $T \in \mathcal{L}(\mathcal{V})$, $q_T(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$, $n = \dim \mathcal{V}$.
Then $\operatorname{tr} T = -c_{n-1}$ and $\det T = (-1)^n c_0$. If $\mathbb{F} = \mathbb{C}$ and $\lambda_1, \ldots, \lambda_n$ list the eigenvalues of T with algebraic multiplicity, then $\operatorname{tr} T = \lambda_1 + \cdots + \lambda_n$, $\det T = \lambda_1 \cdots \lambda_n$.

Proof. Choose a basis β and set $A = [T]_{\beta\leftarrow\beta}$.
Then $q_T = p_A$, and by Definition 1.1.23, $\operatorname{tr} T = \operatorname{tr} A$ and $\det T = \det A$.
Theorem 2.1.3 gives $\operatorname{tr} T = -c_{n-1}$ and $\det T = (-1)^n c_0$.
If $\mathbb{F} = \mathbb{C}$, then comparing $q_T(z) = (z-\lambda_1)\cdots(z-\lambda_n)$ with the displayed form of q_T gives the formulas.
Structural Commentary: Characteristic polynomial stores trace and determinant in first and last coefficients.
Key Insight: Compare coefficients of $\det(zI - T)$. □

1.2 Minors, Cofactors, and Rank

Definition 1.2.1. *Minors, Cofactors, and Adjugate*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, let M_{ij} delete row i and column j from A; set $\det(\emptyset_{0\times 0}) = 1$. Define $C_{ij} := (-1)^{i+j}\det(M_{ij})$.
Then $C = [C_{ij}]$ is the *cofactor matrix*, and $\operatorname{adj}(A) := C^t$ is the *adjugate*.

Lemma 1.2.2. *Cofactor Expansion*
For $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$, $\det A = \sum_{k=1}^{n} a_{ik}C_{ik}$ and $\det A = \sum_{k=1}^{n} a_{kj}C_{kj}$.

Proof. Using $\det A = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{r=1}^{n} a_{r,\sigma(r)}$, group terms by $\sigma(i) = k$.
For fixed k, the remaining permutation sum equals $(-1)^{i+k}\det(M_{ik}) = C_{ik}$, giving the row formula. Applying this to A^t and using Theorem 1.1.8 gives the column formula.
Structural Commentary: Cofactor expansion is the big determinant formula collected around one row or column.
Key Insight: Fix one entry; the rest is its minor. □

Theorem 1.2.3. *Adjugate Identity and Cramer's Rule*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and let $\mathbf{b} \in \mathbb{F}^n$. Then $A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = (\det A)I_n$. If $\det A \neq 0$, then, for $A\mathbf{x} = \mathbf{b}$, $A^{-1} = \frac{1}{\det A}\mathrm{adj}(A)$ and $x_j = \frac{\det A_j(\mathbf{b})}{\det A}$ for $1 \leq j \leq n$, where $A_j(\mathbf{b})$ replaces column j of A by $\mathbf{b}$.

Proof. Since $(\mathrm{adj}\,A)_{kj} = C_{jk}$, $(A\,\mathrm{adj}\,A)_{ij} = \sum_k a_{ik}C_{jk}$.
For $i = j$ this is $\det A$ by Lemma 1.2.2; for $i \neq j$ it is the cofactor expansion of a matrix with rows i and j equal, hence 0. Thus $A\,\mathrm{adj}(A) = (\det A)I_n$. The same argument applied to A^t gives $\mathrm{adj}(A)A = (\det A)I_n$. If $\det A \neq 0$, division gives the inverse formula. Finally, $(\mathrm{adj}\,A\,\mathbf{b})_j = \sum_i C_{ij}b_i = \det A_j(\mathbf{b})$, by expansion down column j, so $x = A^{-1}\mathbf{b}$ gives Cramer's rule.
Structural Commentary: Cofactors assemble all one-column replacements of A into a single inverse formula.
Key Insight: Off-diagonal products vanish because they are determinants with two equal rows or columns. □

Remark 1.2.4. *Adjugate Form of the Bordered Formula*
Proposition 1.1.14 gives the adjugate form $\det\begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = c\det(A) - \mathbf{y}^t\,\mathrm{adj}(A)\mathbf{x}$. No invertibility of A is required.

Theorem 1.2.5. *Cauchy–Binet Formula*
Let $A \in \mathrm{Mat}(k,n,\mathbb{F})$, $B \in \mathrm{Mat}(n,k,\mathbb{F})$, and $k \leq n$. For $J = \{j_1 < \cdots < j_k\} \subseteq \{1,\ldots,n\}$, let A^J use the columns of A in J and B_J use the rows of B in J.
Then $\det(AB) = \sum_J \det(A^J)\det(B_J)$, where J ranges over all k-subsets.

Proof. Put $S(A,B) := \sum_J \det(A^J)\det(B_J)$. For an elementary $n \times n$ matrix E, determinant multilinearity and row/column operation rules give $S(A,B) = S(AE, E^{-1}B)$, and $AB = (AE)(E^{-1}B)$.
If $\mathrm{rank}\,A < k$, both sides vanish. Otherwise reduce A by elementary column operations to $A' = [C\ 0]$ with C invertible, and transform B oppositely to B'.
Then $AB = A'B'$ and $\det(A'B') = \det(C)\det(B'_{J_0}) = S(A',B')$, $J_0 = \{1,\ldots,k\}$, as every other k-minor of A' is zero. Hence $\det(AB) = S(A,B)$.
Structural Commentary: Cauchy–Binet is determinant multiplicativity for rectangular middle dimensions.
Key Insight: The middle space contributes one k-coordinate shadow at a time. □

Proposition 1.2.6. *Transpose Preserves Spectrum and Geometric Multiplicity*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then $\lambda \in \mathrm{spec}(A) \iff \lambda \in \mathrm{spec}(A^t)$, and $\dim E_\lambda(A) = \dim E_\lambda(A^t)$.

Proof. Since $(A - \lambda I_n)^t = A^t - \lambda I_n$, Theorem 3.3.2 gives $\mathrm{rank}(A - \lambda I_n) = \mathrm{rank}(A^t - \lambda I_n)$.
Matrix rank-nullity then gives $\dim E_\lambda(A) = n - \mathrm{rank}(A - \lambda I_n) = n - \mathrm{rank}(A^t - \lambda I_n) = \dim E_\lambda(A^t)$.
($\Rightarrow$) If $\lambda \in \mathrm{spec}(A)$, then $E_\lambda(A) \neq \{\mathbf{0}\}$, so $E_\lambda(A^t) \neq \{\mathbf{0}\}$ and $\lambda \in \mathrm{spec}(A^t)$.
($\Leftarrow$) The same argument with A and A^t interchanged gives the converse.
Structural Commentary: Transposition preserves rank, so shifted kernels have the same dimension.
Key Insight: Spectrum and geometric multiplicity are kernel facts for $A - \lambda I_n$. □

Corollary 1.2.7. *Square Matrix Algebraic Invertibility Dictionary*
Let $A = [\mathbf{a}_1\ \cdots\ \mathbf{a}_n] \in \mathrm{Mat}(n,n,\mathbb{F})$. The following are equivalent:
(i) A is invertible;
(ii) $\mathbf{a}_1,\ldots,\mathbf{a}_n$ form a basis of $\mathbb{F}^n$;
(iii) $\mathbf{a}_1,\ldots,\mathbf{a}_n$ are linearly independent;
(iv) $\mathbf{a}_1,\ldots,\mathbf{a}_n$ span $\mathbb{F}^n$;
(v) for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution;
(vi) the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the zero solution.

Proof.
(*i*) $\Rightarrow$ (*ii*) By Theorem 3.4.1, the columns of an invertible matrix form a basis.
(*ii*) $\Rightarrow$ (*iii*) A basis is linearly independent.
(*iii*) $\Rightarrow$ (*iv*) In $\mathbb{F}^n$, n linearly independent vectors form a basis, hence span.
(*iv*) $\Rightarrow$ (*v*) In $\mathbb{F}^n$, n spanning vectors form a basis, so every $\mathbf{b}$ has unique coordinates $\mathbf{x}$ in $A\mathbf{x} = \mathbf{b}$.
(*v*) $\Rightarrow$ (*vi*) Apply (v) to $\mathbf{b} = \mathbf{0}$; since $\mathbf{x} = \mathbf{0}$ is a solution, it is the only one.
(*vi*) $\Rightarrow$ (*i*) Condition (vi) says the columns are independent.
Hence they form a basis of $\mathbb{F}^n$, and Corollary 3.6.1 gives invertibility.
Structural Commentary: Square invertibility is column basishood.
Key Insight: Invertible columns give unique coordinates. □

Corollary 1.2.8. *Nonsingularity by Rank, Determinant, and Spectrum*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. The following are equivalent:
(i) A is invertible;
(ii) $\mathrm{rank}(A) = n$;
(iii) $\det A \neq 0$;
(iv) $0 \notin \mathrm{spec}(A)$.
Over $\mathbb{C}$, this is also equivalent to all eigenvalues of A being nonzero.

Proof.
(*i*) $\Rightarrow$ (*ii*) This is Corollary 3.6.2.
(*ii*) $\Rightarrow$ (*iii*) Full rank gives invertibility by Corollary 3.6.2, hence $\det A \neq 0$ by Theorem 1.1.8.
(*iii*) $\Rightarrow$ (*iv*) If $\det A \neq 0$, then A is invertible by Theorem 1.1.8; hence $0 \notin \mathrm{spec}(A)$ by Corollary 5.1.8.
(*iv*) $\Rightarrow$ (*i*) This is Corollary 5.1.8.

Over $\mathbb{C}$, Theorem 2.1.3 identifies $\lambda_1, \dots, \lambda_n$ as the roots of p_A, so (iv) is equivalent to $\lambda_j \neq 0$ for all j.
Structural Commentary: Rank, determinant, and spectrum detect singularity.
Key Insight: Singularity can be seen as rank loss, zero determinant, or the zero eigenvalue. □

Theorem 1.2.9. *Rank Bounds for Products and Concatenations*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$, $B \in \mathrm{Mat}(k,n,\mathbb{F})$, *and* $C \in \mathrm{Mat}(m,p,\mathbb{F})$. *Then* $\mathrm{rank}(AB) \leq \min(\mathrm{rank}(A), \mathrm{rank}(B))$ *and* $\mathrm{rank}(AB) = \mathrm{rank}(A)$ *if and only if* $\mathrm{col}(AB) = \mathrm{col}(A)$.
Also, $\max(\mathrm{rank}(A), \mathrm{rank}(C)) \leq \mathrm{rank}\begin{bmatrix} A & C \end{bmatrix}$ *and* $\mathrm{rank}\begin{bmatrix} A & C \end{bmatrix} = \mathrm{rank}(A)$ *if and only if* $\mathrm{col}(A) = \mathrm{col}(A) + \mathrm{col}(C)$.

Proof. Each column of AB is a linear combination of the columns of A, so $\mathrm{col}(AB) \subseteq \mathrm{col}(A)$. Therefore $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$; since both spaces are finite-dimensional, equality holds exactly when $\mathrm{col}(AB) = \mathrm{col}(A)$. Applying this same argument to $(AB)^t = B^t A^t$ and using Theorem 3.3.2 gives $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$.
For the block matrix, its columns are precisely the columns of A and C, hence $\mathrm{col}\begin{bmatrix} A & C \end{bmatrix} = \mathrm{col}(A) + \mathrm{col}(C)$.
This space contains both $\mathrm{col}(A)$ and $\mathrm{col}(C)$, giving the lower bound. Equality with $\mathrm{rank}(A)$ occurs exactly when the containing subspace has the same dimension as $\mathrm{col}(A)$, equivalently when $\mathrm{col}(A) + \mathrm{col}(C) = \mathrm{col}(A)$.
Structural Commentary: Products can only reduce accessible column directions, while horizontal concatenation can only add column directions.
Key Insight: Rank tracks whether a matrix operation preserves, loses, or adds column-space dimension. □

Theorem 1.2.10. *Sylvester Rank Inequality*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$ *and* $B \in \mathrm{Mat}(k,n,\mathbb{F})$. *Then* $\mathrm{rank}(A) + \mathrm{rank}(B) - k \leq \mathrm{rank}(AB)$.

Proof. Let $R = \mathrm{col}(B) \subseteq \mathbb{F}^k$. The columns of AB are A applied to the columns of B, so $\mathrm{col}(AB) = A(R)$.
Apply rank-nullity to the restriction $T_A|_R : R \to \mathbb{F}^m$. Its range is $A(R) = \mathrm{col}(AB)$ and its kernel is $R \cap \ker(A)$.
Hence $\mathrm{rank}(AB) = \dim(A(R)) = \dim(R) - \dim(R \cap \ker(A)) \geq \mathrm{rank}(B) - \dim(\ker(A))$.
Since A has k columns, Corollary 3.3.1 gives $\dim(\ker(A)) = k - \mathrm{rank}(A)$. Substitution gives desired inequality.
Structural Commentary: Multiplication by A can lose only the part of $\mathrm{col}(B)$ lying in $\ker(A)$. Sylvester's inequality is this loss estimate in dimensions.
Key Insight: The shared middle dimension k bounds how much rank can disappear in the product. □

Proposition 1.2.11. *Sylvester's Law of Nullity*
Use the matrix nullity $\mathrm{nullity}(X) = \dim(\ker(X))$ *from Definition 3.1.12.*
(i) If $A \in \mathrm{Mat}(m,k,\mathbb{F})$ *and* $B \in \mathrm{Mat}(k,n,\mathbb{F})$, *then* $\mathrm{nullity}(AB) \leq \mathrm{nullity}(A) + \mathrm{nullity}(B)$.
(ii) If $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, *then* $\max(\mathrm{nullity}(A), \mathrm{nullity}(B)) \leq \mathrm{nullity}(AB)$.

Proof.
(i) By rank-nullity and Theorem 1.2.10,

$$\begin{aligned} \mathrm{nullity}(AB) &= n - \mathrm{rank}(AB) \leq n - \mathrm{rank}(A) - \mathrm{rank}(B) + k \\ &= (k - \mathrm{rank}(A)) + (n - \mathrm{rank}(B)) = \mathrm{nullity}(A) + \mathrm{nullity}(B). \end{aligned}$$

(ii) For square A, B, $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$, $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$. Subtract from n gives desired lower bounds.
Structural Commentary: Nullity is rank loss from the domain.
Key Insight: Use rank-nullity to translate rank bounds into kernel bounds. □

Theorem 1.2.12. *Frobenius Rank Inequality*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$, $B \in \mathrm{Mat}(k,p,\mathbb{F})$, *and* $C \in \mathrm{Mat}(p,n,\mathbb{F})$.
Then $\mathrm{rank}(AB) + \mathrm{rank}(BC) \leq \mathrm{rank}(B) + \mathrm{rank}(ABC)$.

Proof. Let $R = \mathrm{col}(B)$ and $S = \mathrm{col}(BC)$. Since every column of BC is a linear combination of columns of B, we have $S \subseteq R$. The induced map T_A maps R onto $\mathrm{col}(AB)$ and maps S onto $\mathrm{col}(ABC)$.
Rank-nullity on the restrictions $T_A|_R$ and $T_A|_S$ gives

$$\mathrm{rank}(AB) = \dim R - \dim(R \cap \ker(A)), \quad \mathrm{rank}(ABC) = \dim S - \dim(S \cap \ker(A)).$$

Since $S \cap \ker(A) \subseteq R \cap \ker(A)$, we have $\dim(S \cap \ker(A)) \leq \dim(R \cap \ker(A))$.
Thus $\dim R - \dim(R \cap \ker(A)) + \dim S \leq \dim R + \dim S - \dim(S \cap \ker(A))$.
Substituting $\dim R = \mathrm{rank}(B)$ and $\dim S = \mathrm{rank}(BC)$ gives the stated inequality.
Structural Commentary: Frobenius tracks A on nested column spaces.
Key Insight: The smaller column space cannot lose more dimensions to $\ker(A)$ than the larger one. □

Corollary 1.2.13. *Rank Inequalities from Frobenius*
The following are special cases of Theorem 1.2.12:
(i) $\mathrm{rank}(AB) \leq \min(\mathrm{rank}(A), \mathrm{rank}(B))$;
(ii) $\mathrm{rank}(A) + \mathrm{rank}(B) - k \leq \mathrm{rank}(AB)$.

Proof.
(i) Apply Frobenius to conformable factors X, Y, Z. Taking $Z = 0$ gives $\mathrm{rank}(XY) \leq \mathrm{rank}(Y)$, because $\mathrm{rank}(Y0) = \mathrm{rank}(XY0) = 0$.
Taking $X = 0$ gives $\mathrm{rank}(YZ) \leq \mathrm{rank}(Y)$.
Renaming the factors gives $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$ and $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$.
(ii) Taking the middle matrix to be I_k gives $\mathrm{rank}(A) + \mathrm{rank}(B) \leq k + \mathrm{rank}(AB)$, which is Sylvester's inequality.
Structural Commentary: Frobenius is the three-factor parent inequality. The usual upper and lower product-rank bounds appear by making one factor zero or identity.
Key Insight: Zero factors isolate upper bounds; an identity middle factor yields the Sylvester lower bound. □

Theorem 1.2.14. *Rank Triangle Inequality*
If $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, *then* $|\mathrm{rank}(A) - \mathrm{rank}(B)| \leq \mathrm{rank}(A+B) \leq \mathrm{rank}(A) + \mathrm{rank}(B)$.

Proof. Since $\operatorname{col}(A+B) \subseteq \operatorname{col}(A) + \operatorname{col}(B)$, Theorem 2.3.1 gives $\operatorname{rank}(A+B) \leq \dim(\operatorname{col}(A) + \operatorname{col}(B)) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$. Applying this upper bound to $A = (A+B) - B$ gives $\operatorname{rank}(A) \leq \operatorname{rank}(A+B) + \operatorname{rank}(B)$, so $\operatorname{rank}(A) - \operatorname{rank}(B) \leq \operatorname{rank}(A+B)$. Applying it to $B = (A+B) - A$ gives $\operatorname{rank}(B) - \operatorname{rank}(A) \leq \operatorname{rank}(A+B)$.
Together these are exactly $|\operatorname{rank}(A) - \operatorname{rank}(B)| \leq \operatorname{rank}(A+B)$.
Structural Commentary: The rank of a sum is controlled by the sum of the two column spaces. The reverse bounds come by solving for either summand.
Key Insight: Rank behaves like a dimension-valued size under addition. □

1.3 Commutators and Special Matrix Classes

Definition 1.3.1. *Commutator*
For $S, T \in \mathcal{L}(\mathcal{V})$, the *commutator* of S and T is $[S,T] := ST - TS$.
For $A, B \in \operatorname{Mat}(n,n,\mathbb{F})$, the *matrix commutator* is $[A,B] := AB - BA$.

Proposition 1.3.2. *Trace Obstruction for Commutators*
If $A, B \in \operatorname{Mat}(n,n,\mathbb{F})$, then $\operatorname{tr}(AB - BA) = 0$. Hence any matrix similar to a commutator has trace zero.

Proof. By linearity of trace and Proposition 1.1.5, $\operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0$.
If C is similar to $AB - BA$, then Theorem 1.1.17 gives $\operatorname{tr}(C) = \operatorname{tr}(AB - BA) = 0$.
Structural Commentary: Trace kills commutators because it is cyclic.
Key Insight: AB and BA have the same trace, so their difference has trace zero. □

Corollary 1.3.3. *The Identity Is Not a Commutator*
Assume $\dim \mathcal{V} \cdot 1_{\mathbb{F}} \neq 0$. There do not exist $S, T \in \mathcal{L}(\mathcal{V})$ such that $ST - TS = I_{\mathcal{V}}$.

Proof. If $ST - TS = I_{\mathcal{V}}$, choose a basis and let $A = [S]$ and $B = [T]$ in that basis. Then $AB - BA = I_n$.
Taking traces and using Proposition 1.1.5, $0 = \operatorname{tr}(AB) - \operatorname{tr}(BA) = \operatorname{tr}(I_n) = n \cdot 1_{\mathbb{F}}$, contradicting the hypothesis.
Structural Commentary: Commutators have zero trace, while the identity has dimension as its trace.
Key Insight: Trace detects a scalar obstruction invisible from multiplication alone. □

Lemma 1.3.4. *Scalar Matrices and One-Step Orbits*
Let $A \in \operatorname{Mat}(n,n,\mathbb{F})$ with $n \geq 2$.
Then $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x} \in \mathbb{F}^n$ if and only if A is a scalar matrix.

Proof.
($\Rightarrow$) Suppose $A = cI_n$. Then $A\mathbf{x} = c\mathbf{x}$ for every $\mathbf{x}$, so $(\mathbf{x}, A\mathbf{x})$ is linearly dependent.
($\Leftarrow$) Suppose $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x}$.
For each standard basis vector $\mathbf{e}_i$, $\exists a_i \in \mathbb{F}$ such that $A\mathbf{e}_i = a_i\mathbf{e}_i$. For $i \geq 2$, dependence of $\mathbf{e}_1 + \mathbf{e}_i$ and $A(\mathbf{e}_1 + \mathbf{e}_i) = a_1\mathbf{e}_1 + a_i\mathbf{e}_i$ means $a_1\mathbf{e}_1 + a_i\mathbf{e}_i = b(\mathbf{e}_1 + \mathbf{e}_i)$ for some $b \in \mathbb{F}$. Comparing the independent vectors $\mathbf{e}_1, \mathbf{e}_i$ gives $a_1 = b = a_i$. Hence every a_i equals a_1, so $A = a_1 I_n$.
Structural Commentary: A scalar matrix preserves every one-dimensional coordinate direction and every diagonal line between two coordinates. These diagonal lines force all coordinate scalars to agree.
Key Insight: If every vector is an eigenvector, then there can be only one eigenvalue. □

Lemma 1.3.5. *Nonscalar Matrices Have a Zero Diagonal Entry After Similarity*
Let $A \in \operatorname{Mat}(n,n,\mathbb{F})$ with $n \geq 2$.
If A is not a scalar matrix, then A is similar over $\mathbb{F}$ to a matrix whose $(1,1)$ entry is 0.

Proof. By Lemma 1.3.4, there is $\mathbf{x} \in \mathbb{F}^n$ such that $(\mathbf{x}, A\mathbf{x})$ is linearly independent. Extend $(\mathbf{x}, A\mathbf{x})$ to a basis β of $\mathbb{F}^n$. In the basis β, the first column of the matrix of A is the coordinate vector of $A\mathbf{x}$, namely $(0, 1, 0, \ldots, 0)^t$. Thus the $(1,1)$ entry is 0, and this matrix is similar to A.
Structural Commentary: Nonscalarity produces a vector whose first iterate points in a genuinely new direction. Using these two vectors as the start of a basis forces a zero in the first diagonal position.
Key Insight: Put $\mathbf{x}$ first and $A\mathbf{x}$ second in the basis. □

Theorem 1.3.6. *Similarity to a Constant-Diagonal Matrix*
Let $n \geq 1$, let $A \in \operatorname{Mat}(n,n,\mathbb{F})$, and assume $k \cdot 1_{\mathbb{F}} \neq 0$ for every $1 \leq k \leq n$. Then A is similar over $\mathbb{F}$ to a matrix whose diagonal entries are all equal to $\frac{1}{n}\operatorname{tr}(A)$.

Proof. Set $\tau = \frac{1}{n}\operatorname{tr}(A)$ and $B = A - \tau I_n$. Then $\operatorname{tr}(B) = 0$. It is enough to show that every trace-zero matrix is similar to a matrix with zero diagonal, because Theorem 6.2.8 then adds τI_n back to the diagonal.
To prove the trace-zero claim by induction on n. If $n = 1$, a trace-zero matrix is $[0]$. Let $n \geq 2$ and assume the claim holds for smaller sizes. Let $\operatorname{tr}(B) = 0$. If B is scalar, then $B = cI_n$ and $0 = \operatorname{tr}(B) = nc$.
Since $n \cdot 1_{\mathbb{F}} \neq 0$, we get $c = 0$, so B already has zero diagonal.
If B is not scalar, Lemma 1.3.5 makes B similar to $\begin{bmatrix} 0 & * \\ * & B_1 \end{bmatrix}$, and $\operatorname{tr}(B_1) = 0$.
By induction, choose an invertible S_1 such that $S_1^{-1} B_1 S_1$ has zero diagonal. Conjugating the displayed block matrix by $1 \oplus S_1$ leaves the first diagonal entry equal to 0 and replaces the lower-right block by $S_1^{-1} B_1 S_1$.
Hence the resulting similar matrix has every diagonal entry equal to 0.
Structural Commentary: The proof removes one zero diagonal entry at a time.
Key Insight: Similarity creates the first zero; trace zero handles the smaller block. □

Theorem 1.3.7. *Shoda's Theorem in the Distinct-Diagonal Case*
Let $n \geq 1$ and assume $k \cdot 1_{\mathbb{F}} \neq 0$ for every $1 \leq k \leq n$. For $A \in \operatorname{Mat}(n,n,\mathbb{F})$, the matrix A is a commutator in $\operatorname{Mat}(n,n,\mathbb{F})$ if and only if $\operatorname{tr}(A) = 0$. In particular, no nonzero scalar matrix is a commutator.

Proof.

($\Rightarrow$) Proposition 1.3.2 gives $\operatorname{tr}(A) = 0$.
Thus cI_n can be a commutator only when $nc = \operatorname{tr}(cI_n) = 0$, hence only when $c = 0$.

($\Leftarrow$) Suppose $\operatorname{tr}(A) = 0$. By Theorem 1.3.6, there is an invertible S such that $B = S^{-1}AS = [b_{ij}]$ has zero diagonal. Let $X = \operatorname{diag}(1, 2, \ldots, n)$, viewing the integers as scalars in $\mathbb{F}$, and define $Y = [y_{ij}]$ by

$$y_{ij} = \begin{cases} (i-j)^{-1}b_{ij}, & i \neq j, \\ 0, & i = j. \end{cases}$$

The hypothesis ensures that $i - j \neq 0$ in $\mathbb{F}$ whenever $i \neq j$. Thus $XY - YX = [(i-j)y_{ij}] = B$.
Hence $A = (SXS^{-1})(SYS^{-1}) - (SYS^{-1})(SXS^{-1})$, so A is a commutator.

Structural Commentary: Trace is the obstruction; similarity exposes zero diagonal form.
Key Insight: Cyclicity forces trace zero; distinct diagonals realize it. □

Remark 1.3.8. *Field Hypothesis in the Diagonal Proof*
The proof above uses that the integers $1, \ldots, n$ are distinct and that each smaller block size remains nonzero in $\mathbb{F}$. In small characteristic, this diagonal proof must be replaced by a more delicate argument.

Method 1.3.9. *Trace–Determinant Invariant Method*
To get eigenvalue data without solving for eigenvalues:

(i) choose a basis in which $[T]_{\beta \leftarrow \beta}$ is easiest to compute;
(ii) compute $\operatorname{tr} T$ and $\det T$ from this matrix;
(iii) if the eigenvalues are $\lambda_1, \ldots, \lambda_n$, conclude $\lambda_1 + \cdots + \lambda_n = \operatorname{tr} T$, and $\lambda_1 \cdots \lambda_n = \det T$;
(iv) for equations $ST - TS = A$, first test $\operatorname{tr} A = 0$; otherwise no solution exists.

Definition 1.3.10. *Idempotent and Nilpotent Matrices*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$.

(i) A is *idempotent* if $A^2 = A$.
(ii) A is *nilpotent* if $A^k = 0$ for some positive integer k.

An idempotent matrix is an algebraic projection: it fixes its range and kills its kernel.

Proposition 1.3.11. *Basic Invariants of Idempotents*
If $P \in \operatorname{Mat}(n, n, \mathbb{F})$ satisfies $P^2 = P$, then $\mathbb{F}^n = \operatorname{range}(P) \oplus \ker(P)$, P is diagonalizable, and its only possible eigenvalues are 0 and 1. Moreover $\operatorname{tr}(P) = \operatorname{rank}(P)$.
If $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then P is the orthogonal projection onto $\operatorname{range}(P)$ if and only if $P = P^*$.

Proof. For every $\mathbf{x}$, write $\mathbf{x} = P\mathbf{x} + (\mathbf{x} - P\mathbf{x})$. The first term lies in $\operatorname{range}(P)$, while $P(\mathbf{x} - P\mathbf{x}) = P\mathbf{x} - P^2\mathbf{x} = 0$, so the second term lies in $\ker(P)$. The intersection is zero because $\mathbf{x} = P\mathbf{y}$ and $P\mathbf{x} = 0$ imply $\mathbf{x} = P^2\mathbf{y} = P\mathbf{x} = 0$.
In a basis adapted to $\operatorname{range}(P) \oplus \ker(P)$, the matrix of P is $I_r \oplus 0$, where $r = \operatorname{rank}(P)$.
Hence P is diagonalizable, its eigenvalues lie in $\{0, 1\}$, and $\operatorname{tr}(P) = r$.
For the final claim, self-adjoint idempotents are exactly orthogonal projections by Theorem 2.5.6.
Structural Commentary: Idempotents split the space into fixed directions and killed directions.
Key Insight: In adapted coordinates, every idempotent is just $I_r \oplus 0$. □

2 Positive Definite and Structured Matrices

2.1 Positive Semidefinite and Positive Definite Matrices

Definition 2.1.1. *Positive Semidefinite and Positive Definite Matrices*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $A \in \operatorname{Mat}(n, n, \mathbb{F})$.
The matrix A is *positive semidefinite*, written $A \succeq 0$, if $A = A^*$ and $\mathbf{x}^*A\mathbf{x} \geq 0$ for every $\mathbf{x} \in \mathbb{F}^n$.
It is *positive definite*, written $A \succ 0$, if $A = A^*$ and $\mathbf{x}^*A\mathbf{x} > 0$ for every nonzero $\mathbf{x} \in \mathbb{F}^n$.
Negative semidefinite and negative definite mean $-A \succeq 0$ and $-A \succ 0$, respectively.

Proposition 2.1.2. *Rectangular Gram Products Are Positive Semidefinite*
Let $R \in \operatorname{Mat}(m, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Then $R^*R \succeq 0$, $RR^* \succeq 0$, and $(R^*R)_{j,j} = \|R_{\cdot,j}\|_2^2$; $(RR^*)_{i,i} = \|R_{i,\cdot}\|_2^2$.
Full column rank gives $R^*R \succ 0$; full row rank gives $RR^* \succ 0$.

Proof. The matrices are self-adjoint by Proposition 2.3.8.
For $\mathbf{x} \in \mathbb{F}^n$ and $\mathbf{y} \in \mathbb{F}^m$, $\mathbf{x}^*R^*R\mathbf{x} = \|R\mathbf{x}\|_2^2 \geq 0$ and $\mathbf{y}^*RR^*\mathbf{y} = \|R^*\mathbf{y}\|_2^2 \geq 0$. Hence both products are positive semidefinite. Taking $\mathbf{x} = \mathbf{e}_j$ and $\mathbf{y} = \mathbf{e}_i$ gives the diagonal formulas. If R has full column rank, then $R\mathbf{x} \neq 0$ for $\mathbf{x} \neq 0$, so $R^*R \succ 0$. If R has full row rank, apply the same argument to R^*.
Structural Commentary: Multiplying a matrix by its adjoint produces a Gram matrix of columns or rows.
Key Insight: The quadratic form is a squared norm, so no cancellation can make it negative. □

Theorem 2.1.3. *Equivalent Forms of Positive Semidefiniteness*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. The following are equivalent:

(i) $A \succeq 0$.
(ii) *A is Hermitian and all eigenvalues of A are nonnegative.*
(iii) *$A = B^*B$ for some $B \in \operatorname{Mat}(m, n, \mathbb{F})$.*
(iv) *A has a positive semidefinite square root.*

In this case the positive semidefinite square root is unique.
If $A = U\Lambda U^*$ with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $\lambda_i \geq 0$, then $A^{1/2} = U \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})U^*$.

Proof.

(*i*) $\Rightarrow$ (*ii*) This is the matrix form of Theorem 6.3.6.
(*ii*) $\Rightarrow$ (*iv*) By the spectral theorem, write $A = U\Lambda U^*$, where U is unitary and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_i \geq 0$. Then $A^{1/2} = U \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})U^*$ is positive semidefinite and satisfies $(A^{1/2})^2 = A$.

(*iv*) $\Rightarrow$ (*iii*) If $C \succeq 0$ and $C^2 = A$, then $C = C^*$ and $A = C^*C$, so (iii) holds with $m = n$.
(*iii*) $\Rightarrow$ (*i*) If $A = B^*B$, then $A^* = A$ and $\mathbf{x}^*A\mathbf{x} = \mathbf{x}^*B^*B\mathbf{x} = \|B\mathbf{x}\|_2^2 \geq 0$, so $A \succeq 0$.

The positive semidefinite square root is unique by Proposition 6.3.7.
Structural Commentary: Positive semidefinite matrices are Gram-type matrices with nonnegative spectrum.
Key Insight: Positivity is best checked either on eigenvalues or by finding a factor B with $A = B^*B$. □

Theorem 2.1.4. *Kernel, Rank, and Columns of B^*B*
Let $B \in \mathrm{Mat}(m, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and set $A = B^*B$. Then:

(i) $A \succeq 0$;
(ii) $\ker(A) = \ker(B)$;
(iii) $\operatorname{rank}(A) = \operatorname{rank}(B)$;
(iv) $\operatorname{col}(A) = \operatorname{col}(B^*)$;
(v) $A = 0$ *if and only if* $B = 0$.

Proof.
(i) Positive semidefiniteness follows from Theorem 2.1.3.
(ii) If $B\mathbf{x} = 0$, then $A\mathbf{x} = 0$. Conversely, if $A\mathbf{x} = 0$, then $0 = \mathbf{x}^*A\mathbf{x} = \|B\mathbf{x}\|_2^2$ so $B\mathbf{x} = 0$.
(iii) Since A and B both have n columns, rank-nullity and (ii) give $\operatorname{rank}(A) = \operatorname{rank}(B)$.
(iv) Since $A = A^*$, Proposition 2.3.10 and (ii) give $\operatorname{col}(A) = \ker(A)^\perp = \ker(B)^\perp = \operatorname{col}(B^*)$.
(v) By (iii), $\operatorname{rank}(A) = 0$ if and only if $\operatorname{rank}(B) = 0$.
A matrix has rank zero exactly when it is the zero matrix, so $A = 0$ if and only if $B = 0$.

Structural Commentary: The Gram product B^*B preserves the kernel of B.
Key Insight: The identity $\mathbf{x}^*B^*B\mathbf{x} = \|B\mathbf{x}\|_2^2$ is the whole mechanism. □

Figure 77 summarizes the squared-norm argument that transfers kernel data from B^*B to B.

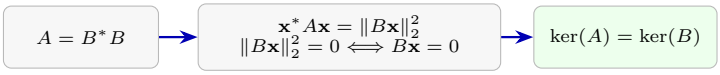

Figure 77: For $A = B^*B$, the squared-norm identity gives $\ker A = \ker B$.

Corollary 2.1.5. *Zero Quadratic Form Detects the Kernel*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $\mathbf{x}^*A\mathbf{x} = 0$ if and only if $A\mathbf{x} = 0$.

Proof. Write $A = B^*B$.
($\Rightarrow$) If $\mathbf{x}^*A\mathbf{x} = 0$, then $\|B\mathbf{x}\|_2^2 = 0$, so $B\mathbf{x} = 0$ and hence $A\mathbf{x} = B^*B\mathbf{x} = 0$.
($\Leftarrow$) If $A\mathbf{x} = 0$, then $\mathbf{x}^*A\mathbf{x} = 0$.

Structural Commentary: A zero PSD quadratic value has no hidden cancellation.
Key Insight: Zero length-square means the factored vector itself is zero. □

Method 2.1.6. *PSD Spectral Factorisation Method*
To analyse a Hermitian matrix A:

(i) diagonalise $A = U\Lambda U^*$;
(ii) check the signs of the diagonal entries of Λ;
(iii) if $\Lambda \geq 0$, set $A^{1/2} = U\Lambda^{1/2}U^*$ and use $A = (A^{1/2})^*A^{1/2}$;
(iv) translate kernel, rank, and zero-quadratic-form questions into the factor $A^{1/2}$.

Theorem 2.1.7. *Equivalent Forms of Positive Definiteness*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. The following are equivalent:

(i) $A \succ 0$.
(ii) *A is Hermitian and all eigenvalues of A are positive.*
(iii) $A \succeq 0$ *and A is invertible.*
(iv) $A = B^*B$ *for some invertible* $B \in \mathrm{Mat}(n, n, \mathbb{F})$.
(v) *A is invertible and* $A^{-1} \succ 0$.

Proof.
(*i*) $\Rightarrow$ (*ii*) By definition A is Hermitian. If $(\lambda, \mathbf{v})$ is an eigenpair with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\|\mathbf{v}\|_2^2 = \mathbf{v}^*A\mathbf{v} > 0$, so every eigenvalue is positive.
(*ii*) $\Rightarrow$ (*iii*) Positive eigenvalues imply $A \succeq 0$ by Theorem 2.1.3; they also exclude 0, so A is invertible.
(*iii*) $\Rightarrow$ (*iv*) By Theorem 2.1.3, A has a positive semidefinite square root $A^{1/2}$.
Since A is invertible, $A^{1/2}$ is invertible, and $A = (A^{1/2})^*A^{1/2}$.
(*iv*) $\Rightarrow$ (*i*) If $A = B^*B$ and B is invertible, then for every nonzero $\mathbf{x}$, $\mathbf{x}^*A\mathbf{x} = \|B\mathbf{x}\|_2^2 > 0$.
(*ii*) $\Leftrightarrow$ (*v*) Hermitian-ness is preserved by inversion, and eigenvalues of A^{-1} are reciprocal eigenvalues of A. Thus positive eigenvalues of A are equivalent to positive eigenvalues of A^{-1}.

Structural Commentary: Positive definite matrices are the invertible part of the positive semidefinite cone.
Key Insight: Positive definite means every eigenvalue is strictly on the positive side of zero. □

Proposition 2.1.8. *Trace and Determinant of Positive Semidefinite Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Then $\operatorname{tr} A \geq 0$, with equality if and only if $A = 0$, and $\det A \geq 0$, with strict inequality if and only if A is positive definite.

Proof. List the eigenvalues of A as $\lambda_1, \dots, \lambda_n$. By Theorem 2.1.3, each $\lambda_j \geq 0$.
Hence $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j \geq 0$ and $\det A = \prod_{j=1}^{n} \lambda_j \geq 0$.
The trace is zero exactly when every $\lambda_j = 0$, which for a Hermitian matrix means $A = 0$. The determinant is positive exactly when every $\lambda_j > 0$, which is positive definiteness by Theorem 2.1.7.
Structural Commentary: Trace and determinant aggregate the nonnegative eigenvalues by sum and product.
Key Insight: Equality is controlled by which eigenvalues vanish. □

Theorem 2.1.9. *Stability Rules for Positive Semidefinite Matrices*
Let $A, B \succeq 0$ over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Whenever the expressions are defined:

(i) $aA + bB \succeq 0$ for $a, b \geq 0$; if $a, b > 0$ and one of A, B is positive definite, then $aA + bB \succ 0$.
(ii) $S^*AS \succeq 0$; moreover, $S^*AS \succ 0$ if and only if $S\mathbf{x} \notin \ker(A)$ for all $\mathbf{x} \neq \mathbf{0}$.
(iii) $A \oplus B \succeq 0$, with $A \oplus B \succ 0$ if and only if $A \succ 0$ and $B \succ 0$.

Proof.

(i) $\mathbf{x}^*(aA + bB)\mathbf{x} = a\mathbf{x}^*A\mathbf{x} + b\mathbf{x}^*B\mathbf{x} \geq 0$. If, for example, $A \succ 0$ and $a > 0$, then for every nonzero $\mathbf{x}$ the first summand is positive and the second is nonnegative, so $aA + bB \succ 0$; the case $B \succ 0$ is identical.
(ii) $\mathbf{x}^*S^*ASx = (S\mathbf{x})^*A(S\mathbf{x}) \geq 0$. By Corollary 2.1.5, equality occurs exactly when $S\mathbf{x} \in \ker(A)$.
(iii) $(\mathbf{x} \oplus \mathbf{y})^*(A \oplus B)(\mathbf{x} \oplus \mathbf{y}) = \mathbf{x}^*A\mathbf{x} + \mathbf{y}^*B\mathbf{y} \geq 0$. This expression is positive for every nonzero $\mathbf{x} \oplus \mathbf{y}$ exactly when each summand is positive on its own nonzero vectors, namely when $A \succ 0$ and $B \succ 0$.

Structural Commentary: The PSD cone is stable under sums, pullbacks, and direct sums.
Key Insight: Test the quadratic form after each construction. □

Proposition 2.1.10. *Explicit Square Root for a 2×2 Positive Semidefinite Matrix*
Let $A \in \operatorname{Mat}(2, 2, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, be positive semidefinite and nonzero. Set $\tau = \left(\operatorname{tr} A + 2\sqrt{\det A}\right)^{1/2}$.
Then $\tau > 0$ and $A^{1/2} = \frac{1}{\tau}\left(A + \sqrt{\det A}\, I_2\right)$.

Proof. Let $C = \frac{1}{\tau}\left(A + \sqrt{\det A}\, I_2\right)$. By Proposition 2.1.8, $\det A \geq 0$ and $\operatorname{tr} A > 0$, hence $\tau > 0$; by Theorem 2.1.9, $C \succeq 0$. Cayley–Hamilton gives $A^2 - (\operatorname{tr} A)A + (\det A)I_2 = 0$, so

$$C^2 = \frac{1}{\tau^2}\left(A^2 + 2\sqrt{\det A}\, A + (\det A)I_2\right) = \frac{1}{\tau^2}\left((\operatorname{tr} A)A + 2\sqrt{\det A}\, A\right) = A.$$

Thus $C = A^{1/2}$ by uniqueness of the positive semidefinite square root.
Structural Commentary: In dimension 2, Cayley–Hamilton makes $A^{1/2}$ linear in A.
Key Insight: Square the linear candidate and use Cayley–Hamilton. □

Definition 2.1.11. *Directed Incidence Matrix*
For edges $e = u \to v$ on vertices $1, \dots, n$, $B \in \operatorname{Mat}(m, n, \mathbb{R})$ has row e equal to $-\mathbf{e}_u^t + \mathbf{e}_v^t$.

Figure 78 records the chosen edge orientations and their corresponding signed incidence rows.

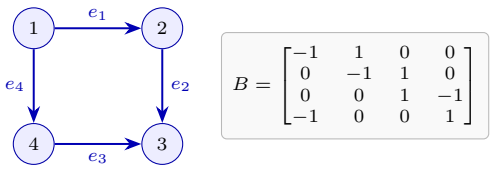

Figure 78: A directed incidence matrix records oriented edge differences from vertices to edges.

Example 2.1.12. *A Directed Incidence Matrix*
For Figure 78, $B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$. For node potentials $\mathbf{x} = (x_1, x_2, x_3, x_4)^t$, $B\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_4 \\ x_4 - x_1 \end{bmatrix}$.
Hence $\ker(B) = \operatorname{span}\{(1, 1, 1, 1)^t\}$, and $\operatorname{rank}(B) = 3$. The loop vector $\mathbf{y} = (1, 1, -1, -1)^t$ spans $\ker(B^t)$.
Thus $\mathbf{b} \in \operatorname{col}(B)$ if and only if $\mathbf{y}^t\mathbf{b} = b_1 + b_2 - b_3 - b_4 = 0$. This is Kirchhoff loop compatibility.
Structural Commentary: Incidence subspaces encode graph constraints.
Key Insight: A sparse matrix can encode both the algebra of a linear map and the combinatorics of a graph.

Perspective 2.1.13. *Incidence Data and Least Squares*
For an incidence matrix B as in Example 2.1.12, $B\mathbf{x} = \mathbf{b}$ seeks node potentials matching edge differences. If loop consistency fails, least squares projects $\mathbf{b}$ onto $\operatorname{col}(B)$, where $B^*(B\mathbf{x}_0 - \mathbf{b}) = \mathbf{0}$ (Theorem 2.4.4 on B). The constant-vector kernel means potentials are unique only after fixing one node or choosing minimum norm.

Proposition 2.1.14. *Incidence Laplacians Are Positive Semidefinite*
Let B be a directed incidence matrix and set $L = B^tB$. Then $L \succeq 0$ and $\mathbf{x}^tL\mathbf{x} = \sum_{e=(u\to v)} (x_v - x_u)^2$, with $\ker(L)$ equal to the vectors constant on each connected component.

Proof. Since $L = B^tB$, Proposition 2.1.2 gives $L \succeq 0$.
For $e = (u \to v)$, $(B\mathbf{x})_e = x_v - x_u$, so $\mathbf{x}^tL\mathbf{x} = \|B\mathbf{x}\|_2^2 = \sum_{e=(u\to v)} (x_v - x_u)^2$. Thus $\mathbf{x} \in \ker(L)$ if and only if $B\mathbf{x} = \mathbf{0}$, equivalently $x_u = x_v$ on every edge, which means $\mathbf{x}$ is constant on each connected component.
Structural Commentary: Laplacians sum squared edge differences.
Key Insight: A graph Laplacian is a Gram product, so positivity is automatic. □

2.2 Principal Minors and Inertia

Definition 2.2.1. *Principal and Leading Principal Minors*
For $A \in \mathrm{Mat}(n,n,\mathbb{C})$, a *principal minor* is the determinant of a principal submatrix of A.
A *leading principal minor* is the determinant of the leading $k \times k$ principal submatrix for some $k = 1, \dots, n$.

Corollary 2.2.2. *Principal Blocks and Column Inclusion*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ be PSD, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, and partition $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{bmatrix}$, $A_{11} \in \mathrm{Mat}(k,k,\mathbb{F})$.
Then $A_{11} \succeq 0$, $\mathrm{col}(A_{12}) \subseteq \mathrm{col}(A_{11})$, and $\mathrm{rank}(A_{12}) \leq \mathrm{rank}(A_{11})$. If $A \succ 0$, then $A_{11} \succ 0$.

Proof. Write $A = B^*B$ and partition $B = [B_1 \ B_2]$ conformally. Then $A_{11} = B_1^*B_1$ and $A_{12} = B_1^*B_2$.
Hence $A_{11} \succeq 0$ and $\mathrm{col}(A_{12}) = \mathrm{col}(B_1^*B_2) \subseteq \mathrm{col}(B_1^*) = \mathrm{col}(B_1^*B_1) = \mathrm{col}(A_{11})$, equality $\mathrm{col}(B_1^*) = \mathrm{col}(B_1^*B_1)$ follows from Theorem 2.1.4. The rank inequality follows by taking dimensions
If $A \succ 0$ and $\mathbf{x} \neq 0$ in $\mathbb{F}^k$, then $\mathbf{x}^*A_{11}\mathbf{x} = \begin{bmatrix}\mathbf{x}\\0\end{bmatrix}^* A \begin{bmatrix}\mathbf{x}\\0\end{bmatrix} > 0$, so $A_{11} \succ 0$.
Structural Commentary: In a positive block matrix, cross-block columns lie in principal block's column space.
Key Insight: Factor first; then column inclusion is immediate from $A_{12} = B_1^*B_2$. □

Figure 79 highlights the principal and cross blocks involved in the PSD column-inclusion result.

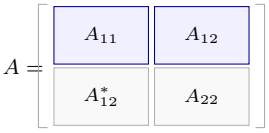

Figure 79: For $A \succeq 0$, factorisation forces $\mathrm{col}(A_{12}) \subseteq \mathrm{col}(A_{11})$.

Theorem 2.2.3. *Sylvester's Criterion*
A Hermitian matrix $A \in \mathrm{Mat}(n,n,\mathbb{C})$ is positive definite if and only if all leading principal minors are positive.

Proof.
(⇒) If $A \succ 0$, then each leading principal submatrix is positive definite by Corollary 2.2.2. Its eigenvalues are positive, so its determinant is positive.
(⇐) Induct on n. If $n = 1$, the single leading principal minor is the only diagonal entry, so positivity of that minor is exactly positive definiteness. Assume $n \geq 2$ and write $A = \begin{bmatrix} B & \mathbf{y} \\ \mathbf{y}^* & a \end{bmatrix}$, $B \in \mathrm{Mat}(n-1,n-1,\mathbb{C})$
The leading principal minors of B are positive, so $B \succ 0$.
Since $\det(B) > 0$ and $\det(A) > 0$, Proposition 1.1.12 gives $a - \mathbf{y}^*B^{-1}\mathbf{y} = \frac{\det(A)}{\det(B)} > 0$.
For $\mathbf{u} \in \mathbb{C}^{n-1}$ and $t \in \mathbb{C}$, $\begin{bmatrix}\mathbf{u}\\t\end{bmatrix}^* A \begin{bmatrix}\mathbf{u}\\t\end{bmatrix} = (\mathbf{u} + B^{-1}\mathbf{y}t)^*B(\mathbf{u} + B^{-1}\mathbf{y}t) + \left(a - \mathbf{y}^*B^{-1}\mathbf{y}\right)|t|^2$.
First term is nonnegative as $B \succ 0$, and second coefficient is positive. If displayed vector is nonzero, then either $t \neq 0$ and second term is positive, or $t = 0$ and then $\mathbf{u} \neq 0$, making first term positive. Thus $A \succ 0$.
Structural Commentary: Leading minors make each leading block positive and final Schur complement positive.
Key Insight: Complete the square with the leading block. □

Method 2.2.4. *2×2 Quadratic Form Test*
Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ be real symmetric, $\Delta = ac - b^2$, and $q(x,y) = \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2$.
For $a \neq 0$, $q(x,y) = a\left(x + \frac{b}{a}y\right)^2 + \frac{\Delta}{a}y^2$.
If $a = 0$, swap variables when $c \neq 0$; if $a = c = 0$, use $q = 2bxy$. Thus:
(i) $q \succ 0$ if and only if $a > 0$ and $\Delta > 0$;
(ii) $q \prec 0$ if and only if $a < 0$ and $\Delta > 0$;
(iii) q is indefinite if and only if $\Delta < 0$;
(iv) if $\Delta = 0$, then $q \succeq 0$ iff $a, c \geq 0$, and $q \preceq 0$ iff $a, c \leq 0$.
This is the 2×2 form of Sylvester's criterion and inertia.
Structural Commentary: Completing the square is the 2×2 shadow of the LDL^* and Schur-complement tests.
Key Insight: Δ and the leading sign classify the form.

Theorem 2.2.5. *Principal Axes for Real Quadratic Forms*
Let $A \in \mathrm{Mat}(n,n,\mathbb{R})$ be symmetric. Then:

(i) there are a real orthogonal Q and real $\lambda_1, \dots, \lambda_n$ such that, with $\mathbf{y} = Q^t\mathbf{x}$, $\mathbf{x}^tA\mathbf{x} = \sum_{j=1}^{n} \lambda_j y_j^2$.
(ii) $A \succeq 0$ if and only if every $\lambda_j \geq 0$;
(iii) $A \succ 0$ if and only if every $\lambda_j > 0$;
(iv) A is indefinite if and only if some $\lambda_j > 0$ and some $\lambda_k < 0$.

Proof.
(i) By the real spectral theorem, write $A = Q\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)Q^t$.
Then $\mathbf{x}^tA\mathbf{x} = \mathbf{x}^tQ\,\mathrm{diag}(\lambda_j)Q^t\mathbf{x} = \mathbf{y}^t\,\mathrm{diag}(\lambda_j)\mathbf{y} = \sum_{j=1}^{n} \lambda_j y_j^2$.
(ii) The displayed sum is nonnegative for all $\mathbf{y}$ exactly when each coefficient λ_j is nonnegative.
(iii) The displayed sum is positive for all nonzero $\mathbf{y}$ exactly when each coefficient λ_j is positive.
(iv) The diagonal form takes both signs exactly when its coefficients include both signs.

Structural Commentary: Principal axes turn the form into signed squares.
Key Insight: Eigenvalue signs classify the quadratic form. □

Method 2.2.6. *Quadratic Form Principal-Axes Method*
To analyse a real quadratic form:

(i) write it as $\mathbf{x}^t A\mathbf{x}$ with $A = A^t$; split each cross coefficient equally between a_{ij} and a_{ji};
(ii) diagonalise $A = Q\Lambda Q^t$;
(iii) change variables by $\mathbf{y} = Q^t\mathbf{x}$;
(iv) classify the form from the signs of the eigenvalues in Λ.

Figure 80 illustrates the orthogonal change of coordinates underlying the principal-axes method.

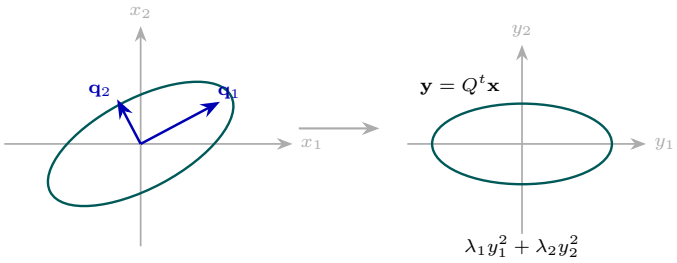

Figure 80: Orthogonal coordinates reveal the principal axes of a quadratic form.

Definition 2.2.7. *$*$-Congruence and Hermitian Inertia*
Two matrices $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are *$*$-congruent* if $A = SBS^*$ for some invertible $S \in \mathrm{Mat}(n, n, \mathbb{C})$.
For Hermitian A, its *inertia* is $\mathrm{in}(A) = (n_+(A), n_-(A), n_0(A))$, where these are the numbers of positive, negative, and zero eigenvalues of A, counted with multiplicity.

Lemma 2.2.8. *Positive Subspaces and Inertia*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian. The maximal dimensions of subspaces on which $\mathbf{x}^*A\mathbf{x}$ is positive or negative for every nonzero $\mathbf{x}$ are $n_+(A)$ and $n_-(A)$, respectively.

Proof. Let P_+, $P_{\leq 0}$ be positive, nonpositive spectral subspaces. Then $\dim P_+ = n_+(A)$, P_+ is positive for A.
If $\dim \mathcal{U} > n_+(A)$, then $\dim \mathcal{U} + \dim P_{\leq 0} > n_+(A) + n_-(A) + n_0(A) = n$, so $\mathcal{U} \cap P_{\leq 0}$ contains a nonzero vector.
On this vector quadratic form is nonpositive, so $\mathcal{U}$ cannot be positive. Apply this to $-A$ for negative case.
Structural Commentary: The positive spectral subspace is maximal; the nonpositive part blocks enlargement.
Key Insight: Inertia counts maximal sign-definite dimensions. □

Theorem 2.2.9. *Sylvester's Inertia Theorem*
Hermitian matrices A, B of the same size are $*$-congruent if and only if $n_+(A) = n_+(B)$ and $n_-(A) = n_-(B)$.

Proof.
(⇒) Suppose $A = SBS^*$ with S invertible. Then $\mathbf{x}^*A\mathbf{x} = (S^*\mathbf{x})^*B(S^*\mathbf{x})$.
Thus S^* transports positive and negative subspaces, so Lemma 2.2.8 gives $n_\pm(A) = n_\pm(B)$.
(⇐) Let the common inertia be (p, q, r) and set $D = I_p \oplus (-I_q) \oplus 0_r$.
By the spectral theorem, each of A and B is $*$-congruent to D, hence they are $*$-congruent to each other.
Structural Commentary: $*$-congruence is a coordinate change for Hermitian quadratic forms.
Key Insight: Spectral decomposition reduces the form to signed squares and zero directions. □

Figure 81 summarizes the $*$-congruence reduction to positive, negative, and zero coordinate directions.

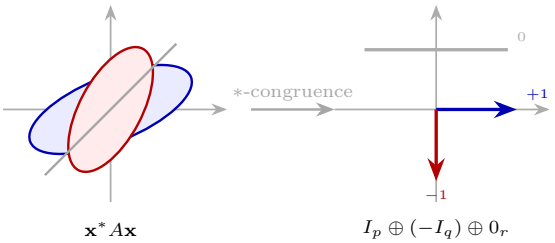

Figure 81: $*$-congruence reduces a Hermitian form to signed coordinate axes.

2.3 Hermitian and Normal Matrix Structure

Theorem 2.3.1. *Simultaneous Congruence with One Positive Definite Form*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be Hermitian, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \succ 0$.
Then, for some invertible S and real diagonal Λ, $A = SS^*$ and $B = S\Lambda S^*$.
The diagonal entries of Λ are the eigenvalues of $A^{-1}B$.

Proof. Let $C = A^{-1/2}BA^{-1/2} = U\Lambda U^*$ be spectral decomposition, $S = A^{1/2}U$. Then $SS^* = A$ and $S\Lambda S^* = B$.
Also $A^{-1}B = A^{-1/2}CA^{1/2}$, so $A^{-1}B$ is similar to C and has eigenvalues given by the diagonal entries of Λ.
Structural Commentary: The positive definite form A normalizes B to a Hermitian matrix.
Key Insight: Normalize, diagonalise, then transform back. □

Corollary 2.3.2. *Products with One Positive Definite Factor*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \succ 0$.

(i) If B is Hermitian, then AB is diagonalizable with real spectrum.
(ii) If $B \succeq 0$, then AB is diagonalizable with nonnegative real spectrum; if $B \neq 0$, one eigenvalue is positive.
(iii) If $B \succ 0$, then AB is diagonalizable with positive real spectrum.

Proof. Let $C = A^{1/2}BA^{1/2}$. Since $A^{-1/2}(AB)A^{1/2} = C$, the matrices AB and C are similar.

(i) If B is Hermitian, then C is Hermitian, hence diagonalizable with real spectrum.
(ii) If $B \succeq 0$, then $C \succeq 0$ by Theorem 2.1.9, so its spectrum is nonnegative.
If $B \neq 0$, then $C \neq 0$, hence some eigenvalue is positive.
(iii) If $B \succ 0$, then $C \succ 0$ by Theorem 2.1.9, so all eigenvalues are positive.

Structural Commentary: A positive definite factor makes AB similar to a Hermitian congruence.
Key Insight: Use $AB \sim A^{1/2}BA^{1/2}$. □

Definition 2.3.3. *Phase Inertia Matrix*
A *phase inertia matrix* is a diagonal complex matrix whose nonzero diagonal entries have modulus 1.

Proposition 2.3.4. *Phase Inertia Form for Normal Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be normal.
Then A is $*$-congruent to a phase inertia matrix D, whose entries record the spectral rays of A.

Proof. Write $A = U\,\mathrm{diag}(\lambda_j)U^*$. Write $\lambda_j = r_j d_j$, where $r_j > 0$ and $|d_j| = 1$ if $\lambda_j \neq 0$, while $r_j = 1$ and $d_j = 0$ if $\lambda_j = 0$. Set $D = \mathrm{diag}(d_j)$, $S = U\,\mathrm{diag}(\sqrt{r_j})$. Then S is invertible and $A = SDS^*$.
The entries $d_j = e^{i\theta}$ count exactly the eigenvalues on the ray $\{\rho e^{i\theta} : \rho > 0\}$.
Structural Commentary: Normal diagonalization separates modulus from phase; $*$-congruence absorbs moduli.
Key Insight: Only spectral phases survive. □

Perspective 2.3.5. *Hermitian Inertia as Real Phase Inertia*
For Hermitian matrices, the only nonzero spectral rays are the positive and negative real axes.
Theorem 2.2.9 is therefore the real-ray version of phase inertia.

3 Matrix Products, Vectorization, and Spectral Radius

3.1 Kronecker Products and Vectorization

Definition 3.1.1. *Kronecker Product*
Let $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$. The *Kronecker product* of A and B is the block matrix

$$A \otimes B := \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \in \mathrm{Mat}(mp, nq, \mathbb{F}).$$

The Kronecker product is also called the tensor product of matrices.

Proposition 3.1.2. *Basic Kronecker Identities*
Whenever the expressions are defined,

(i) $c(A \otimes B) = (cA) \otimes B = A \otimes (cB)$;
(ii) $(A + B) \otimes C = A \otimes C + B \otimes C$ and $A \otimes (B + C) = A \otimes B + A \otimes C$;
(iii) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$;
(iv) $(A \otimes B)^t = A^t \otimes B^t$;
(v) $I_m \otimes I_n = I_{mn}$.

If $\mathbb{F} = \mathbb{C}$, then also $\overline{A \otimes B} = \overline{A} \otimes \overline{B}$ and $(A \otimes B)^* = A^* \otimes B^*$.

Proof.

(i) The (i, j) block of $c(A \otimes B)$ is $c(a_{ij}B) = (ca_{ij})B = a_{ij}(cB)$, which is (i, j) block of $(cA) \otimes B$ and $A \otimes (cB)$.
(ii) The (i, j) block of $(A + B) \otimes C$ is $(a_{ij} + b_{ij})C = a_{ij}C + b_{ij}C$, so $(A + B) \otimes C = A \otimes C + B \otimes C$.
The proof of $A \otimes (B + C) = A \otimes B + A \otimes C$ is identical inside each block.
(iii) If $B = [b_{rs}]$, then both $(A \otimes B) \otimes C$ and $A \otimes (B \otimes C)$ have blocks $a_{ij}b_{rs}C$ in the same relative positions.
(iv) The (j, i) block of $(A \otimes B)^t$ is $(a_{ij}B)^t = a_{ij}B^t$, which is the (j, i) block of $A^t \otimes B^t$.
(v) The block matrix $I_m \otimes I_n$ has I_n on its diagonal blocks and zero elsewhere, hence equals I_{mn}.

If $\mathbb{F} = \mathbb{C}$, conjugation is checked blockwise, adjoint identity follows from conjugation together with part (iv).
Structural Commentary: Kronecker products replace scalar entries by whole scaled blocks. Standard algebraic rules survive because each block is still controlled by scalar distributivity and transposition.
Key Insight: Check identities block-by-block; each block reduces to an ordinary scalar-matrix identity. □

Theorem 3.1.3. *Mixed Product Property*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $B \in \mathrm{Mat}(p, q, \mathbb{F})$, $C \in \mathrm{Mat}(n, r, \mathbb{F})$, $D \in \mathrm{Mat}(q, s, \mathbb{F})$. $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

Proof. Write $A = [a_{ik}]$ and $C = [c_{kj}]$. The (i, j) block of $(A \otimes B)(C \otimes D)$ is
$\sum_{k=1}^{n} (a_{ik}B)(c_{kj}D) = \left(\sum_{k=1}^{n} a_{ik}c_{kj}\right) BD = (AC)_{ij}BD$, which is exactly the (i, j) block of $(AC) \otimes (BD)$.
Structural Commentary: The outer indices multiply as AC, while the inner block action multiplies as BD.
Key Insight: Kronecker multiplication separates into ordinary multiplication in each factor. □

Corollary 3.1.4. *Inverse of a Kronecker Product*
If $A \in \mathrm{Mat}(m, m, \mathbb{F})$ and $B \in \mathrm{Mat}(n, n, \mathbb{F})$ are invertible, then $A \otimes B$ is invertible and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Proof. By Theorem 3.1.3, $(A \otimes B)(A^{-1} \otimes B^{-1}) = I_m \otimes I_n = I_{mn}$ and the reverse product is identical.
Structural Commentary: Inverting a Kronecker product inverts each independent factor.
Key Insight: The mixed product property turns the inverse check into two ordinary inverse checks. □

Proposition 3.1.5. *Kronecker Product and Sum Spectra in the Diagonalizable Case*
Let $A = P\Lambda P^{-1}$, $B = QMQ^{-1}$, where $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and $M = \operatorname{diag}(\mu_1, \dots, \mu_q)$.
Then $A \otimes B \sim \Lambda \otimes M$; and $A \otimes I_q + I_n \otimes B \sim \Lambda \otimes I_q + I_n \otimes M$.
Their diagonal entries are respectively $\lambda_i \mu_j$ and $\lambda_i + \mu_j$ for $1 \le i \le n$, $1 \le j \le q$.

Proof. By Theorem 3.1.3 and $(P \otimes Q)^{-1} = P^{-1} \otimes Q^{-1}$, $A \otimes B = (P \otimes Q)(\Lambda \otimes M)(P^{-1} \otimes Q^{-1})$.
Similarly, $A \otimes I_q + I_n \otimes B = (P \otimes Q)(\Lambda \otimes I_q + I_n \otimes M)(P^{-1} \otimes Q^{-1})$.
The middle matrices are diagonal with the stated (i, j)-indexed entries.
Structural Commentary: Eigenvectors tensor together, and Kronecker sums add the two scalar eigenvalues.
Key Insight: Diagonalize each factor first; the Kronecker operation becomes entrywise on eigenvalue pairs. □

Let T_r denote the $r \times r$ tridiagonal matrix with diagonal entries 2 and first off-diagonal entries -1. Figure 82 illustrates how $T_m \otimes I_n + I_m \otimes T_n$ combines the two coordinate directions into a five-point stencil.

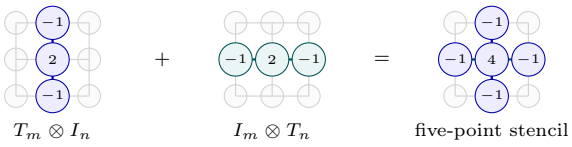

Figure 82: Vertical plus horizontal second differences give the five-point stencil.

Definition 3.1.6. *Vectorisation*
Let $X = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_n] \in \operatorname{Mat}(m, n, \mathbb{F})$. The *vectorisation* of X is

$$\operatorname{vec}(X) := \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \in \mathbb{F}^{mn}.$$

Thus vec stacks the columns of X vertically.

Theorem 3.1.7. *Vectorisation of a Triple Product*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $X \in \operatorname{Mat}(n, p, \mathbb{F})$, and $B \in \operatorname{Mat}(p, q, \mathbb{F})$. Then $\operatorname{vec}(AXB) = (B^t \otimes A)\operatorname{vec}(X)$.

Proof. Write $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_q]$ and $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_p]$.
The kth column of AXB is $AX\mathbf{b}_k = A \sum_{i=1}^{p} b_{ik}\mathbf{x}_i = \sum_{i=1}^{p} b_{ik} A\mathbf{x}_i = (\mathbf{b}_k^t \otimes A)\operatorname{vec}(X)$.
Stacking these q columns gives $(B^t \otimes A)\operatorname{vec}(X)$.
Structural Commentary: Vectorisation rewrites two-sided multiplication as one matrix action.
Key Insight: Left multiplication contributes A; right multiplication contributes B^t. □

Perspective 3.1.8. *Vectorising a Minimal-Polynomial Test*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$ and a monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1} z^{m-1} + z^m$,
the matrix equation $a_0 I + a_1 A + \cdots + a_{m-1} A^{m-1} = -A^m$ can be written as the ordinary linear system

$$\begin{bmatrix} \operatorname{vec}(I) & \operatorname{vec}(A) & \cdots & \operatorname{vec}(A^{m-1}) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{m-1} \end{bmatrix} = -\operatorname{vec}(A^m)$$

Structural Commentary: Vectorisation turns matrix identities into linear systems.
Key Insight: Polynomial membership becomes a coefficient system in vectorized powers.

3.2 Hadamard Products

Definition 3.2.1. *Hadamard Product*
Let $A = [a_{ij}], B = [b_{ij}] \in \operatorname{Mat}(m, n, \mathbb{F})$.
The *Hadamard product*, or *entrywise product*, of A and B is $A \circ B := [a_{ij} b_{ij}] \in \operatorname{Mat}(m, n, \mathbb{F})$.

Proposition 3.2.2. *Basic Hadamard Identities*
For matrices of compatible sizes:

(i) $A \circ B = B \circ A$;
(ii) $A \circ (\gamma B + C) = \gamma(A \circ B) + A \circ C$;
(iii) $(A \circ B)^* = A^* \circ B^*$;
(iv) if E is the all-ones matrix, then $A \circ E = A$;
(v) if $D = \operatorname{diag}(d_1, \dots, d_n)$, then $A \circ D = \operatorname{diag}(a_{11}d_1, \dots, a_{nn}d_n)$.

Proof.

(i) $(A \circ B)_{ij} = a_{ij} b_{ij} = b_{ij} a_{ij} = (B \circ A)_{ij}$.
(ii) $\big(A \circ (\gamma B + C)\big)_{ij} = a_{ij}(\gamma b_{ij} + c_{ij}) = \gamma a_{ij} b_{ij} + a_{ij} c_{ij}$.
(iii) $\big((A \circ B)^*\big)_{ij} = \overline{a_{ji} b_{ji}} = \overline{a_{ji}}\,\overline{b_{ji}} = (A^* \circ B^*)_{ij}$.
(iv) $(A \circ E)_{ij} = a_{ij}$.

(v) Since D has zero off-diagonal entries, $A \circ D$ is diagonal with entries $a_{ii}d_i$.

Structural Commentary: Hadamard multiplication is entrywise, with no row-column summation.
Key Insight: Prove Hadamard identities entry by entry. □

Theorem 3.2.3. *Schur Product Theorem*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ are positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $A \circ B$ is positive semidefinite.

Proof. By Theorem 2.1.3, regard A and B as Gram matrices: $a_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle$, $b_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$ for suitable vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Then $(A \circ B)_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle \langle \mathbf{v}_j, \mathbf{v}_i \rangle = \langle \mathbf{u}_j \otimes \mathbf{v}_j, \mathbf{u}_i \otimes \mathbf{v}_i \rangle$.
Thus $A \circ B$ is the Gram matrix of the vectors $\mathbf{u}_j \otimes \mathbf{v}_j$, hence is positive semidefinite by Proposition 1.5.2.
Structural Commentary: The Schur product is the Gram matrix of paired tensor vectors.
Key Insight: Pair the jth vector from each Gram representation and take tensor products. □

Figure 83 summarizes how paired tensor vectors turn two Gram representations into the Hadamard-product Gram matrix.

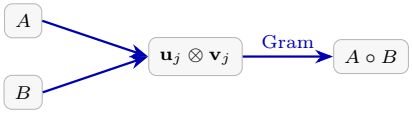

Figure 83: For Gram matrices, entrywise products become tensor inner products, so $A \circ B$ is Gram.

Corollary 3.2.4. *Positive Definite Hadamard Products*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) If A is positive definite and every diagonal entry of B is positive, then $A \circ B$ is positive definite.
(ii) If A and B are positive definite, then $A \circ B$ is positive definite.

Proof.

(i) Let $\lambda_{\min} > 0$ be the smallest eigenvalue of A. Then $A - \lambda_{\min} I_n \succeq 0$, so $(A - \lambda_{\min} I_n) \circ B \succeq 0$ by Theorem 3.2.3. Thus, for $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^*(A \circ B)\mathbf{x} = \mathbf{x}^*((A - \lambda_{\min} I_n) \circ B)\mathbf{x} \lambda_{\min} \sum_{j=1}^{n} b_{jj} |x_j|^2 > 0$.

(ii) If $B \succ 0$, then $b_{jj} = \mathbf{e}_j^* B \mathbf{e}_j > 0$ for every j, so (i) applies.

Structural Commentary: Subtract the spectral floor of A; the diagonal of B gives strictness.
Key Insight: Use $I_n \circ B = \mathrm{diag}(b_{11}, \ldots, b_{nn})$. □

Method 3.2.5. *Schur Product Positivity Method*
To prove an entrywise product is positive semidefinite:

(i) factor or interpret each matrix as a Gram matrix;
(ii) pair the corresponding vectors;
(iii) take tensor products of the paired vectors;
(iv) identify the Hadamard product as the Gram matrix of those tensor products.

3.3 Spectral Radius and Matrix Powers

Theorem 3.3.1. *Ranks of Matrix Powers Stabilize*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then:

(i) $\mathrm{rank}(A^k) \geq \mathrm{rank}(A^{k+1})$ for $k = 1, 2, \ldots$;
(ii) if $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+1})$, then $\mathrm{col}(A^k) = \mathrm{col}(A^{k+p})$ and $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+p})$ for every $p = 1, 2, \ldots$;
(iii) if A is not invertible, then there is a least positive integer $q \leq n$ such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Proof.

(i) Since $A^{k+1} = A^k A$, each column of A^{k+1} is a linear combination of the columns of A^k. Hence $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$, so $\mathrm{rank}(A^{k+1}) \leq \mathrm{rank}(A^k)$.

(ii) The inclusion $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$ and equality of dimensions give $\mathrm{col}(A^{k+1}) = \mathrm{col}(A^k)$. Let $U = \mathrm{col}(A^k)$. Then multiplication by A maps U onto U, so induction gives $\mathrm{col}(A^{k+p}) = U$ for every $p \geq 1$.

(iii) If A is not invertible, then Corollary 3.6.2 gives $\mathrm{rank}(A) \leq n - 1$. If all inequalities $\mathrm{rank}(A^k) > \mathrm{rank}(A^{k+1})$ held for $k = 1, \ldots, n$, then $\mathrm{rank}(A^{n+1}) \leq \mathrm{rank}(A) - n < 0$, impossible. Hence equality occurs for some $q \leq n$; choose the least such q.

Structural Commentary: The column spaces of powers form a decreasing chain. In finite dimensions, a decreasing chain cannot keep losing dimension forever.
Key Insight: Once two consecutive ranks agree, the image has stabilized under multiplication by A. □

Definition 3.3.2. *Index of a Matrix*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, define its *index* to be 0. If A is not invertible, its index is the least positive integer q such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Definition 3.3.3. *Spectral Radius and Convergent Matrix*
For $A \in \mathrm{Mat}(n, n, \mathbb{C})$, the *spectral radius* is $\rho(A) := \max\{|\lambda| : \lambda \in \mathrm{spec}(A)\}$.
We say $A^p \to 0$ if every entry of A^p converges to 0 as $p \to \infty$.

Proposition 3.3.4. *Basic Invariants of Nilpotents*
For $N \in \mathrm{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) N is nilpotent;
(ii) the characteristic polynomial of N is z^n;
(iii) $N^n = 0$.

In this case, $\mathrm{tr}(N) = 0$, $\det(N) = 0$ for $n \geq 1$, and the spectral radius of N over $\mathbb{C}$ is 0.

Proof. If $N^k = 0$, then every eigenvalue λ of N over an algebraic closure satisfies $\lambda^k = 0$, hence $\lambda = 0$. Thus the characteristic polynomial is z^n. If the characteristic polynomial is z^n, Cayley–Hamilton gives $N^n = 0$. The implication $N^n = 0 \Rightarrow$ nilpotent is immediate. The trace, determinant, and spectral-radius statements follow from the eigenvalue data and from Definitions 1.1.1 and 3.3.3.
Structural Commentary: Nilpotent matrices have no spectral mass away from zero.
Key Insight: Cayley–Hamilton turns the characteristic polynomial z^n into the equation $N^n = 0$. □

Theorem 3.3.5. *Matrix Powers Converge to Zero*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then $A^p \to 0$ if and only if $\rho(A) < 1$.

Proof. By Theorem 6.2.1, choose a Jordan form $A = SJS^{-1}$.
Since S and S^{-1} are fixed, $A^p \to 0$ if and only if every Jordan block power in J^p converges to 0.
($\Rightarrow$) Each block power tends to 0, so Proposition 6.1.5 gives $|\lambda| < 1$ for every eigenvalue. Thus $\rho(A) < 1$.
($\Leftarrow$) If $\rho(A) < 1$, then every eigenvalue satisfies $|\lambda| < 1$. Proposition 6.1.5 makes each block power tend to 0.
Structural Commentary: Similarity reduces matrix-power convergence to Jordan block-power convergence.
Key Insight: The largest eigenvalue modulus controls the asymptotic size of all blocks. □

Figure 84 summarizes how the unit circle separates decaying Jordan-block powers from nondecaying ones.

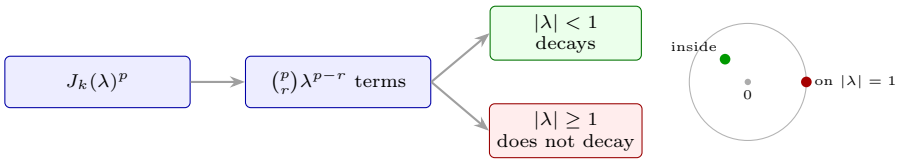

Figure 84: The gray circle is $|\lambda| = 1$: inside it, powers decay; on or outside it, they do not.

Definition 3.3.6. *Power Bounded Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *power bounded* if for each entry position (i, j), the sequence of entries of A^p in that position is bounded for $p = 1, 2, \ldots$.

Theorem 3.3.7. *Power Bounded Jordan Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is power bounded.
(ii) $\rho(A) \leq 1$, and every eigenvalue λ with $|\lambda| = 1$ has only 1×1 Jordan blocks.
(iii) $\rho(A) \leq 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. By Theorem 6.2.1, work in Jordan form.
(*i*) $\Rightarrow$ (*ii*): If $|\lambda| > 1$, the diagonal entries λ^p are unbounded.
If $|\lambda| = 1$ and $J_k(\lambda)$ has $k \geq 2$, then the first superdiagonal contains $p\lambda^{p-1}$, also unbounded.
(*ii*) $\Rightarrow$ (*i*): Blocks with $|\lambda| < 1$ converge to 0 by Proposition 6.1.5; 1×1 blocks with $|\lambda| = 1$ are bounded. Finite direct sums and fixed similarities preserve boundedness.
(*ii*) $\Leftrightarrow$ (*iii*): For fixed λ, $\dim E_\lambda(A)$ counts Jordan blocks, while $m_A(\lambda)$ counts their total size.
Equality holds exactly when every block has size 1.
Structural Commentary: Unit-circle eigenvalues are harmless only when their nilpotent parts vanish.
Key Insight: Nontrivial Jordan blocks on the unit circle create polynomial growth. □

4 Matrix Decompositions

Decompositions expose usable structure in a matrix. LU, QR, Cholesky, SVD, compact SVD, and polar decomposition separate elimination, geometry, positivity, rank, and stretching.

Perspective 4.0.1. *Matrix Factorisation Roadmap*
Factorisations expose structure by task: elimination gives LU and LDL^t (Theorem 4.1.4, Corollary 4.1.5); positivity gives Cholesky (Theorem 4.1.11); geometry gives QR (Theorem 4.1.7); singular directions give SVD, compact SVD, and the pseudoinverse (Theorems 4.2.8, 4.2.13, Definition 5.2.4); polar decomposition separates motion from stretch (Theorem 4.2.18). Spectral factorisations appeared earlier: diagonalisation, spectral theorems, Jordan form, and Schur form (Theorems 5.4.3, 6.1.11, 6.1.12, 6.2.1, 2.1.9).
Key Insight: Choose the factorisation that exposes the needed structure.

4.1 LU, QR, and Cholesky-Type Decompositions

Definition 4.1.1. *Full-Rank Factorization*
A factorization $A = XY$ with X full column rank and Y full row rank is called a *full-rank factorization* of A.

Theorem 4.1.2. *Full-Rank Factorization*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have rank $r \geq 1$, and let the columns of $X \in \mathrm{Mat}(m, r, \mathbb{F})$ be a basis of $\mathrm{col}(A)$. Then there is a unique $Y \in \mathrm{Mat}(r, n, \mathbb{F})$ such that $A = XY$, and $\mathrm{rank}(X) = \mathrm{rank}(Y) = r$.

Proof. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Since the columns of X form a basis of $\mathrm{col}(A)$, for each j there is a unique $\mathbf{y}_j \in \mathbb{F}^r$ such that $\mathbf{a}_j = X\mathbf{y}_j$.
Set $Y = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_n]$. Then $A = [X\mathbf{y}_1 \ \cdots \ X\mathbf{y}_n] = XY$ and uniqueness of the coordinates $\mathbf{y}_j$ gives uniqueness of Y. The columns of X are independent, so $\mathrm{rank}(X) = r$.
By Theorem 1.2.9, $r = \mathrm{rank}(A) = \mathrm{rank}(XY) \leq \mathrm{rank}(Y) \leq r$ because Y has only r rows. Hence $\mathrm{rank}(Y) = r$.
Structural Commentary: A matrix factors through a basis of its column space.
Key Insight: Full-rank factorization separates column directions from their coordinates. □

Definition 4.1.3. *LU Factorisation*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, an *LU factorisation* is a product $A = LU$, where L is lower triangular and U is upper triangular. It is *unit lower triangular* if every diagonal entry of L is 1.

Theorem 4.1.4. *LU Factorisation from Nonzero Leading Principal Minors*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. If every leading principal minor of A is nonzero, then $A = LU$ for a unique unit lower triangular L and a unique upper triangular U.

Proof. Induct on n. The case $n = 1$ is immediate. Write $A = \begin{bmatrix} a & \mathbf{r} \\ \mathbf{c} & B \end{bmatrix}$, $a \neq 0$.
Set $\boldsymbol{\ell} = a^{-1}\mathbf{c}$ and $S = B - a^{-1}\mathbf{cr}$. For each leading $k \times k$ principal block of A, the block determinant formula gives determinant a times the corresponding leading $(k-1) \times (k-1)$ principal minor of S. Hence all leading principal minors of S are nonzero. By induction, $S = L_1U_1$ with L_1 unit lower triangular and U_1 upper triangular.
Then $A = \begin{bmatrix} 1 & 0 \\ \boldsymbol{\ell} & L_1 \end{bmatrix} \begin{bmatrix} a & \mathbf{r} \\ 0 & U_1 \end{bmatrix}$. For uniqueness, comparing the first row and first column of LU determines a, $\mathbf{r}$, and $\boldsymbol{\ell}$, then reduces uniqueness to the lower-right Schur complement.
Structural Commentary: Gaussian elimination peels off one pivot and passes to a Schur complement.
Key Insight: Nonzero leading principal minors are exactly the no-row-swap pivot condition. □

Corollary 4.1.5. *Symmetric LU Gives an LDL^t Factorisation*
Let $A = A^t \in \mathrm{Mat}(n,n,\mathbb{R})$ have nonzero leading principal minors.
Then $A = LDL^t$, with L unit lower triangular and D diagonal invertible.

Proof. By Theorem 4.1.4, write $A = LU$ uniquely with L unit lower triangular and U upper triangular.
Since the leading principal minors are nonzero, each $u_{ii} \neq 0$. Put $D = \mathrm{diag}(u_{11}, \ldots, u_{nn})$ and $U_0 = D^{-1}U$; then U_0 is unit upper triangular and $A = LDU_0$. Since $A = A^t$, also $A = U_0^t DL^t$. This is another LU factorisation with unit lower factor U_0^t, so uniqueness gives $L = U_0^t$. Hence $A = LDL^t$.
Structural Commentary: Symmetry makes the upper triangular unit factor the transpose of the lower one.
Key Insight: Once the pivots are separated into D, the two triangular halves mirror each other. □

Definition 4.1.6. *Wide and Narrow QR Factorisation*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $m \geq n$, factorisation $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, $V \in \mathrm{Mat}(m,m,\mathbb{F})$ unitary, $R \in \mathrm{Mat}(n,n,\mathbb{F})$ upper triangular is called a *wide QR factorisation*. Writing $V = [Q\ Q']$ gives the *narrow QR factorisation* $A = QR$.

Theorem 4.1.7. *QR Factorisation*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and suppose $m \geq n$.

(i) $\exists$ a unitary V and an upper triangular R with real nonnegative diagonal entries such that $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$.
Writing $V = [Q\ Q']$ with Q the first n columns gives $A = QR$, with Q orthonormal columns.
(ii) If $\mathrm{rank}(A) = n$, then narrow $A = QR$ is unique when R has positive diagonal entries.

Proof.
(i) At step k, after the first $k-1$ columns have been triangularized, apply Corollary 2.7.4 to the tail of the kth column. This gives a unitary $I_{k-1} \oplus U_k$ that sends that tail to $r_{kk}\mathbf{e}_1$ with $r_{kk} \geq 0$.
Since the first $k-1$ entries of every earlier column already contain all possible nonzero entries, acting only on rows $k, \ldots, m$ preserves the earlier triangular columns.
After n steps, $U_n \cdots U_1 A = \begin{bmatrix} R \\ 0 \end{bmatrix}$, where R is upper triangular with nonnegative diagonal entries.
Let $V = (U_n \cdots U_1)^*$. Then $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, and $V = [Q\ Q']$ gives $A = QR$ with Q orthonormal columns.
If $\mathrm{rank}(A) = n$, then $n = \mathrm{rank}(A) = \mathrm{rank}(R)$, so the triangular matrix R is invertible and its nonnegative diagonal entries are positive.
(ii) Suppose $A = Q_1R_1 = Q_2R_2$, where $Q_i^*Q_i = I_n$ and the R_i are upper triangular with positive diagonal entries. Then $A^*A = R_1^*R_1 = R_2^*R_2$. Let $D = R_1R_2^{-1}$. Substituting $R_1 = DR_2$ gives $D^*D = I_n$.
Thus the columns of D are orthonormal. Since D is upper triangular, the first column has only one possible nonzero entry; orthogonality with it forces the rest of the first row to vanish.
Inducting on the lower-right principal submatrix shows D is diagonal, and $D^*D = I_n$ gives diagonal entries of modulus 1. These diagonal entries are ratios of positive diagonal entries of R_1 and R_2, so they are positive. Hence $D = I_n$. Therefore $R_1 = R_2$, and then $Q_1 = Q_2$.
Structural Commentary: Householder steps triangularise columns; full rank makes the diagonal positive.
Key Insight: Positive diagonals remove the sign/phase ambiguity. □

Corollary 4.1.8. *Prefix Column Spaces in QR Factorisation*
Let $A = QR$ be the narrow QR factorisation of $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $\mathrm{rank}(A) = n$ and positive diagonal entries in R. For $k = 1, \ldots, n$, let A_k and Q_k be the first k columns of A and Q, respectively. Then $\mathrm{col}(A_k) = \mathrm{col}(Q_k)$.
Thus the columns of Q_k form an orthonormal basis of $\mathrm{col}(A_k)$.

Proof. Let R_k be the leading $k \times k$ principal submatrix of R. Since R is upper triangular with positive diagonal entries, R_k is invertible. The first k columns of $A = QR$ satisfy $A_k = Q_kR_k$, because upper triangularity prevents later columns of Q from contributing. Hence $\mathrm{col}(A_k) \subseteq \mathrm{col}(Q_k)$ and $Q_k = A_kR_k^{-1}$ gives the reverse inclusion. The columns of Q_k are orthonormal because Q has orthonormal columns.
Structural Commentary: QR preserves prefix column spaces while orthonormalising them.
Key Insight: Invertible R_k only changes coordinates within that space. □

Figure 85 summarizes how QR replaces the original columns by orthonormal columns while retaining upper-triangular coordinate data.

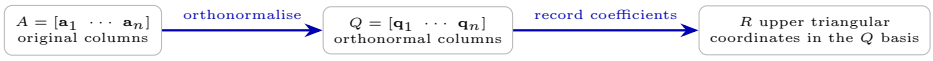

Figure 85: QR records the original columns in an orthonormal basis; upper triangularity preserves the prefix column spaces.

Perspective 4.1.9. *Gram–Schmidt and Householder QR*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ have full column rank.
Gram–Schmidt gives orthonormal columns $\mathbf{q}_1, \ldots, \mathbf{q}_n$ s.t. $\operatorname{span}(\mathbf{a}_1, \ldots, \mathbf{a}_k) = \operatorname{span}(\mathbf{q}_1, \ldots, \mathbf{q}_k)$, $k = 1, \ldots, n$.
Hence $A = QR$ with $Q^*Q = I$ and R upper triangular. The Householder construction gives the same factorization via unitary transformations, which is the numerically stable form.

Perspective 4.1.10. *Rank-Deficient QR from Gram–Schmidt*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \operatorname{Mat}(m, n, \mathbb{F})$, $m \geq n$, and let $\mathbf{u}_j$ be the arbitrary-list Gram–Schmidt remainders from Theorem 1.3.7. Set $\mathcal{J} = \{j : \mathbf{u}_j \neq \mathbf{0}\}$. For $j \in \mathcal{J}$ put $\mathbf{q}_j = \mathbf{u}_j / \|\mathbf{u}_j\|$; for $j \notin \mathcal{J}$ choose $\mathbf{q}_j \in \operatorname{col}(A)^\perp$ so that $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_n]$ has orthonormal columns. With $R = Q^*A$, one has $A = QR$, R upper triangular, and $r_{jj} = 0$ exactly when $\mathbf{u}_j = \mathbf{0}$. Thus uniqueness fails unless $\mathcal{J} = \{1, \ldots, n\}$ and $r_{jj} > 0$ for all j.
Structural Commentary: Rank defect becomes zero QR pivots.
Key Insight: The full-rank hypothesis in uniqueness is a statement about nonzero orthogonal remainders.

Theorem 4.1.11. *Cholesky Factorisation*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. If $A \succeq 0$, then $A = LL^*$ for some lower triangular L with nonnegative diagonal entries. If $A \succ 0$, then L is unique and has positive diagonal entries.

Proof. Let $B = A^{1/2}$ and take a QR factorisation $B = QR$ with R upper triangular and diagonal entries nonnegative. Then $A = B^*B = R^*R$, so $L = R^*$ works.
If $A \succ 0$, then L is invertible, so its diagonal entries are positive. For uniqueness, suppose $A = L_1L_1^* = L_2L_2^*$ with L_i lower triangular and positive diagonal. Set $R_i = L_i^*$. Then R_i is upper triangular with positive diagonal and $B^2 = A = R_i^*R_i$. Define $Q_i = BR_i^{-1}$. Then $Q_i^*Q_i = (R_i^{-1})^*B^2R_i^{-1} = (R_i^{-1})^*R_i^*R_iR_i^{-1} = I_n$, so $B = Q_iR_i$ are QR factorisations with positive diagonal. Uniqueness of QR gives $R_1 = R_2$, hence $L_1 = L_2$.
Structural Commentary: Cholesky is QR applied to $A^{1/2}$.
Key Insight: Triangularise the square root, then set $L = R^*$. □

Figure 86 summarizes the construction of a Cholesky factor from a QR factorization of the positive semidefinite square root.

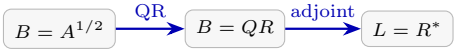

Figure 86: Apply QR to $B = A^{1/2}$; then $A = R^*R = LL^*$, with uniqueness in the positive definite case.

4.2 Polar and Singular Value Decompositions

Theorem 4.2.1. *Polar Decomposition*
For every $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $\exists$ an isometry $S \in \mathcal{L}(\mathcal{V})$ s.t. $T = S\sqrt{T^*T}$.

Proof. Let $P = \sqrt{T^*T}$. Since T^*T is positive, P is well defined by Proposition 6.3.7.
For every $\mathbf{v} \in \mathcal{V}$, $\|T\mathbf{v}\|^2 = \langle T^*T\mathbf{v}, \mathbf{v}\rangle = \langle P^2\mathbf{v}, \mathbf{v}\rangle = \|P\mathbf{v}\|^2$.
Define $S_1 : \operatorname{range}(P) \to \operatorname{range}(T)$ by $S_1(P\mathbf{v}) = T\mathbf{v}$. The norm identity shows this is well defined: if $P\mathbf{v}_1 = P\mathbf{v}_2$, then $\|T(\mathbf{v}_1 - \mathbf{v}_2)\| = \|P(\mathbf{v}_1 - \mathbf{v}_2)\| = 0$. It also shows S_1 is an isometry onto $\operatorname{range}(T)$.
The same norm identity gives $\ker(P) = \ker(T)$, so $\operatorname{rank}(P) = \operatorname{rank}(T)$ by rank-nullity. Hence $\operatorname{range}(P)^\perp$ and $\operatorname{range}(T)^\perp$ have the same dimension. Choose any isometry between these complements and combine it orthogonally with S_1 to obtain an isometry S on all of $\mathcal{V}$. Then $SP\mathbf{v} = S_1(P\mathbf{v}) = T\mathbf{v}$ for every $\mathbf{v}$, so $T = S\sqrt{T^*T}$.
Structural Commentary: The positive factor $\sqrt{T^*T}$ reproduces exactly the lengths produced by T.
Key Insight: Once the metric stretching has been extracted, the remaining map is forced to be an isometry. □

Corollary 4.2.2. *Uniqueness in Operator Polar Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, and put $|T| := \sqrt{T^*T}$ and $|T^*| := \sqrt{TT^*}$.

(i) If $T = SP$, $S^*S = I_{\mathcal{V}}$, and $P \succeq 0$, then $P = |T|$. Hence the positive right polar factor is unique.
(ii) $\exists$ left polar decomposition $T = |T^*|\,R$ with R an isometry, and $|T^*|$ is unique positive left polar factor.
(iii) If T is invertible, then both isometric factors are unique: $S = T\,|T|^{-1}$ and $R = |T^*|^{-1}\,T$.

Proof.

(i) If $T = SP$, then $T^*T = PS^*SP = P^2$. Since $P \succeq 0$, Proposition 6.3.7 gives $P = \sqrt{T^*T} = |T|$.
(ii) Apply Theorem 4.2.1 to T^*. There is an isometry S_0 such that $T^* = S_0\,|T^*|$.
Taking adjoints gives $T = |T^*|\,S_0^*$. Set $R = S_0^*$. If also $T = QR$ with $Q \succeq 0$ and R an isometry, then $RR^* = I_{\mathcal{V}}$ in finite dimensions, so $TT^* = QRR^*Q = Q^2$. Hence $Q = |T^*|$.
(iii) If T is invertible, then $|T|$ and $|T^*|$ are positive definite, hence invertible.
Therefore any right and left polar decompositions force $S = T\,|T|^{-1}$, $R = |T^*|^{-1}\,T$.

Structural Commentary: The positive polar factor is canonical.
Key Insight: Invertibility removes the kernel complement, so the whole factorisation becomes unique. □

Figure 87 summarizes the range isometry used to separate the positive stretch from the isometric factor.

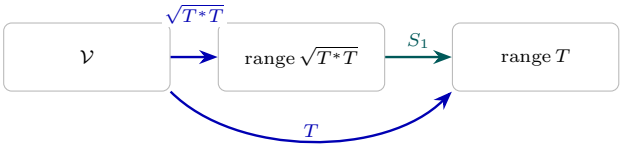

Figure 87: Polar decomposition builds S_1 using $\|T\mathbf{v}\| = \left\|\sqrt{T^*T}\mathbf{v}\right\|$.

Definition 4.2.3. *Singular Values*
For finite-dimensional inner-product spaces $\mathcal{U}, \mathcal{V}$ and $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, the *singular values* of T are the eigenvalues of $\sqrt{T^*T} \in \mathcal{L}(\mathcal{U})$, where each eigenvalue λ is repeated $\dim E_\lambda(\sqrt{T^*T})$.

Theorem 4.2.4. *Singular Value Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$ have singular values $s_1, \ldots, s_n$. Then there exist orthonormal bases $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and $(\mathbf{f}_1, \ldots, \mathbf{f}_n)$ of $\mathcal{V}$ such that $T\mathbf{v} = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_n \langle \mathbf{v}, \mathbf{e}_n \rangle \mathbf{f}_n$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $|T| = \sqrt{T^*T}$. Since $|T|$ is positive, the real or complex spectral theorem gives an orthonormal basis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $|T|\,\mathbf{e}_j = s_j \mathbf{e}_j$. Thus $|T|\,\mathbf{v} = \sum_{j=1}^{n} s_j \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$.
By Theorem 4.2.1, $T = S\,|T|$ for some isometry S. Put $\mathbf{f}_j = S\mathbf{e}_j$. The vectors $\mathbf{f}_1, \ldots, \mathbf{f}_n$ form an orthonormal basis, and applying S to the displayed expansion gives the desired formula.
Structural Commentary: SVD diagonalises $|T|$, then transports its eigenvectors by S.
Key Insight: The s_j are stretches; the $\mathbf{f}_j$ are their landing directions. □

Figure 88 illustrates how the positive polar factor stretches each right singular direction before the isometry supplies its left singular direction.

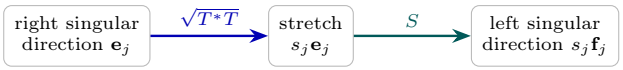

Figure 88: Since $T = S\sqrt{T^*T}$, each $\mathbf{e}_j$ maps to $s_j\mathbf{f}_j$.

Proposition 4.2.5. *Singular Values from T^*T*
The singular values of $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ are the nonnegative square roots of the eigenvalues of T^*T, with the same eigenspace multiplicities.

Proof. By the spectral theorem, choose an orthonormal basis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ and nonnegative scalars $\lambda_1, \ldots, \lambda_n$ such that $T^*T\mathbf{e}_j = \lambda_j \mathbf{e}_j$. The unique positive square root of T^*T sends $\mathbf{e}_j$ to $\sqrt{\lambda_j}\mathbf{e}_j$.
Thus the singular values are $\sqrt{\lambda_j}$, with the same eigenspace multiplicities.
Structural Commentary: The positive square root keeps eigenspaces and sends λ to $\sqrt{\lambda}$.
Key Insight: Singular values can be read from T^*T directly. □

Theorem 4.2.6. *Singular Vectors for a Linear Map*
Let $\mathcal{U}, \mathcal{V}$ be finite-dimensional inner-product spaces over $\mathbb{R}$ or $\mathbb{C}$, and let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ have rank r.
Put $n = \dim \mathcal{U}$, $m = \dim \mathcal{V}$, and $\mathcal{U}_+ = \text{range}(T^*)$, $\mathcal{U}_0 = \ker(T)$; $\mathcal{V}_+ = \text{range}(T)$, $\mathcal{V}_0 = \ker(T^*)$.
Then $\exists$ ordered orthonormal bases $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_r, \mathbf{u}_{r+1}, \ldots, \mathbf{u}_n)$; $\gamma = (\mathbf{v}_1, \ldots, \mathbf{v}_r, \mathbf{v}_{r+1}, \ldots, \mathbf{v}_m)$
with $\text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_r) = \mathcal{U}_+$; $\text{span}(\mathbf{u}_{r+1}, \ldots, \mathbf{u}_n) = \mathcal{U}_0$; $\text{span}(\mathbf{v}_1, \ldots, \mathbf{v}_r) = \mathcal{V}_+$; $\text{span}(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_m) = \mathcal{V}_0$,
and numbers $s_1, \ldots, s_r > 0$ such that $T\mathbf{u}_k = s_k \mathbf{v}_k$; $T^*\mathbf{v}_k = s_k \mathbf{u}_k$, for $1 \le k \le r$.
The s_k are the positive singular values; $\mathbf{u}_k$ and $\mathbf{v}_k$ are right and left singular vectors.

Proof. Let $A = T^*T$. Then $A \succeq 0$, $A = A^*$, and $\ker(A) = \ker(T)$. By spectral theorem, choose rthonormal eigenbasis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ s.t. $A\mathbf{u}_k = s_k^2 \mathbf{u}_k$ for $1 \le k \le r$; $A\mathbf{u}_k = 0$ for $k > r$; $s_k > 0$.
Thus $\text{span}(\mathbf{u}_{r+1}, \ldots, \mathbf{u}_n) = \mathcal{U}_0$, and Proposition 2.3.10 gives $\text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_r) = \mathcal{U}_+$. For $1 \le k \le r$,
set $\mathbf{v}_k = s_k^{-1} T\mathbf{u}_k$. Then $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \frac{1}{s_i s_j} \langle T\mathbf{u}_i, T\mathbf{u}_j \rangle = \frac{1}{s_i s_j} \langle T^*T\mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$; $T^*\mathbf{v}_k = s_k^{-1} A\mathbf{u}_k = s_k \mathbf{u}_k$.
Hence $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are orthonormal and span $\mathcal{V}_+ = T(\mathcal{U}_+)$. Complete them by an orthonormal basis of $\mathcal{V}_0 = \mathcal{V}_+^\perp$.
Structural Commentary: T^*T gives right directions; T gives left directions.
Key Insight: Away from the two kernels, T and T^* act symmetrically by the same positive scalars. □

Figure 89 summarizes the paired active subspaces and kernels for T and T^*.

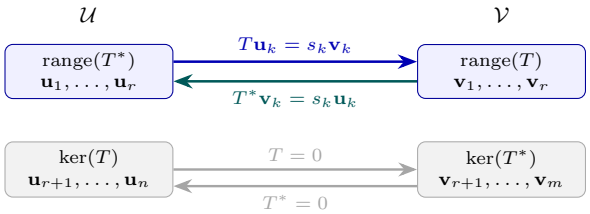

Figure 89: The singular-vector bases split the domain and codomain into active singular subspaces and kernels.

Definition 4.2.7. *Matrix Singular Values*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, let $q = \min(m,n)$, and let $r = \mathrm{rank}(A)$. The *positive singular values* of A are the positive eigenvalues of $|A| = (A^*A)^{1/2}$, written $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.
The full list of singular values is $\sigma_1, \ldots, \sigma_r, \sigma_{r+1}, \ldots, \sigma_q$, where $\sigma_{r+1} = \cdots = \sigma_q = 0$.

Theorem 4.2.8. *Matrix Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m,n)$. Then $\exists$ unitary matrices $V \in \mathrm{Mat}(m,m,\mathbb{F})$, $W \in \mathrm{Mat}(n,n,\mathbb{F})$ such that $A = V\Sigma W^*$, $\Sigma = \begin{bmatrix} \mathrm{diag}(\sigma_1,\ldots,\sigma_q) & 0 \\ 0 & 0 \end{bmatrix} \in \mathrm{Mat}(m,n,\mathbb{R})$ with the evident deletion of empty blocks.

Proof. If $A = 0$, take $\Sigma = 0$ and choose any unitary V, W. Assume $r = \mathrm{rank}(A) \geq 1$.
Diagonalise A^*A as $W^*A^*AW = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)$, where $\mathbf{w}_1, \ldots, \mathbf{w}_n$ are the columns of W.
For $1 \leq j \leq r$, set $\mathbf{v}_j = \sigma_j^{-1} A\mathbf{w}_j$. Then $\mathbf{v}_i^*\mathbf{v}_j = \frac{1}{\sigma_i\sigma_j}\mathbf{w}_i^*A^*A\mathbf{w}_j = \delta_{ij}$ so $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are orthonormal.
Complete them to an orthonormal basis of $\mathbb{F}^m$ and let V be the matrix with these columns.
For $j > r$, $\|A\mathbf{w}_j\|_2^2 = \mathbf{w}_j^*A^*A\mathbf{w}_j = 0$, hence $A\mathbf{w}_j = 0$. Thus $AW = V\Sigma$, equivalently $A = V\Sigma W^*$.
Structural Commentary: A^*A determines the right singular directions and singular values.
Key Insight: Apply A and normalize to get the left singular directions. □

Figure 90 illustrates the right-unitary motion, diagonal stretching, and left-unitary motion in the matrix SVD.

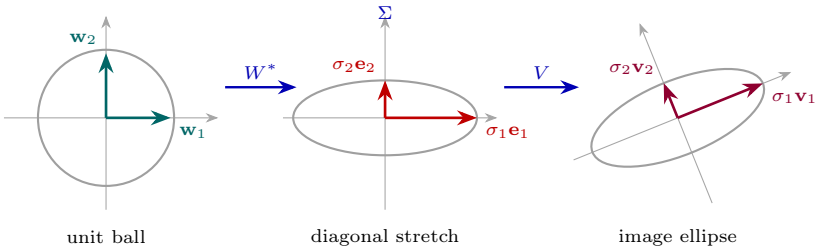

Figure 90: SVD: right-unitary motion, diagonal stretch, left-unitary motion.

Perspective 4.2.9. *Operator-to-Matrix SVD Bridge*
In the bases from Theorem 4.2.6, the matrix of T is rectangular diagonal with diagonal entries $s_1, \ldots, s_r$ and zeros elsewhere. Returning to standard orthonormal bases gives the matrix form $A = V\Sigma W^*$ in Theorem 4.2.8.

Proposition 4.2.10. *Basic Singular Value Identities*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m,n)$ and $r = \mathrm{rank}(A)$. Let $c \in \mathbb{F}$.
(i) The positive eigenvalues of A^*A and AA^* are $\sigma_1^2, \ldots, \sigma_r^2$.
(ii) $\sum_{j=1}^q \sigma_j^2 = \mathrm{tr}(A^*A) = \mathrm{tr}(AA^*) = \|A\|_F^2$, where $\|A\|_F^2 := \mathrm{tr}(A^*A)$.
(iii) A and A^* have the same singular values. Over $\mathbb{C}$, so do A^t and $\overline{A}$.
(iv) The singular values of cA are $|c|\,\sigma_1, \ldots, |c|\,\sigma_q$.

Proof. Let $A = V\Sigma W^*$ be an SVD. Then $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$.
(i) The nonzero diagonal entries of $\Sigma^*\Sigma$ and $\Sigma\Sigma^*$ are $\sigma_1^2, \ldots, \sigma_r^2$.
(ii) Since A^*A and $\Sigma^*\Sigma$ are unitarily similar, and AA^* and $\Sigma\Sigma^*$ are unitarily similar,
$\mathrm{tr}(A^*A) = \mathrm{tr}(\Sigma^*\Sigma) = \sum_{j=1}^q \sigma_j^2 = \mathrm{tr}(\Sigma\Sigma^*) = \mathrm{tr}(AA^*)$. The final equality is the definition of $\|A\|_F$.
(iii) For A^*, the two Gram matrices are interchanged. Over $\mathbb{C}$, $(A^t)^*A^t = (AA^*)^t$ and $(\overline{A})^*\overline{A} = (A^*A)^t$.
Since a matrix and its transpose have the same characteristic polynomial, the spectra agree.
(iv) Since $(cA)^*(cA) = |c|^2\, A^*A$, singular values scale by $|c|$.
Structural Commentary: Unitary factors do not change Gram spectra; Σ contains all singular data.
Key Insight: Reduce each claim to $\Sigma^*\Sigma$ and $\Sigma\Sigma^*$. □

Corollary 4.2.11. *Square Invertibility and Singular Values*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, have singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The following are equivalent:
(i) A is invertible;
(ii) $\mathrm{rank}(A) = n$;
(iii) $\sigma_n > 0$;
(iv) $A^*A \succ 0$.
If $r = \mathrm{rank}(A) < n$, then $\sigma_1, \ldots, \sigma_r > 0$ and $\sigma_{r+1} = \cdots = \sigma_n = 0$.

Proof.
(i) $\Rightarrow$ *(ii)* By Corollary 3.6.2, invertibility gives full rank.
(ii) $\Rightarrow$ *(iii)* By Definition 4.2.7, the number of positive singular values is $\mathrm{rank}(A)$.
(iii) $\Rightarrow$ *(iv)* If $\sigma_n > 0$, then A has full column rank, so Proposition 2.1.2 gives $A^*A \succ 0$.
(iv) $\Rightarrow$ *(i)* If $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}^*A^*A\mathbf{x} = 0$; PSD gives $\mathbf{x} = \mathbf{0}$. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.
The rank-defect statement follows again from Definition 4.2.7.
Structural Commentary: SVD sees nonsingularity as the absence of zero stretch directions.
Key Insight: A zero singular value is exactly a nonzero vector sent to zero. □

Definition 4.2.12. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.
A factorisation $A = X\Sigma_r Y^*$ is a *compact singular value decomposition* if $X \in \mathrm{Mat}(m,r,\mathbb{F})$ and $Y \in \mathrm{Mat}(n,r,\mathbb{F})$ have orthonormal columns and $\Sigma_r = \mathrm{diag}(\sigma_1,\dots,\sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

Theorem 4.2.13. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.

(i) If $A = V\Sigma W^*$ is an SVD and X, Y are the first r columns of V, W, then $A = X\Sigma_r Y^*$.
(ii) Conversely, if X, Y have orthonormal columns and $\Sigma_r = \mathrm{diag}(\sigma_1,\dots,\sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$, then $X\Sigma_r Y^*$ has positive singular values $\sigma_1,\dots,\sigma_r$.

Proof.

(i) Partition $V = [X\ X_0]$, $W = [Y\ Y_0]$, and $\Sigma = \begin{bmatrix}\Sigma_r & 0\\ 0 & 0\end{bmatrix}$. Multiplication gives $A = V\Sigma W^* = X\Sigma_r Y^*$.

(ii) Let $B = X\Sigma_r Y^*$. Since $X^*X = Y^*Y = I_r$, $B^*B = Y\Sigma_r^2 Y^*$.
Completing Y to a unitary matrix $\widetilde{Y} = [Y\ Y_0]$ gives $\widetilde{Y}^* B^* B \widetilde{Y} = \Sigma_r^2 \oplus 0$.
Thus the positive eigenvalues of B^*B are $\sigma_1^2,\dots,\sigma_r^2$, so the positive singular values of B are $\sigma_1,\dots,\sigma_r$.

Structural Commentary: Compact SVD keeps only the nonzero singular directions.
Key Insight: The rank-r data X, Σ_r, Y reconstructs A. □

Theorem 4.2.14. *Column and Kernel Spaces from Compact SVD*
Let $A = X\Sigma_r Y^*$ be a compact SVD. Then $\mathrm{col}(A) = \mathrm{col}(X) = (\ker A^*)^\perp$, and $\ker(A) = \mathrm{col}(Y)^\perp = (\mathrm{col}(A^*))^\perp$.

Proof. Since $A = X\Sigma_r Y^*$ and $AY\Sigma_r^{-1} = X$, $\mathrm{col}(A) = \mathrm{col}(X)$.
For $\mathbf{z} \in \mathbb{F}^n$, $A\mathbf{z} = 0 \iff X\Sigma_r Y^*\mathbf{z} = 0 \iff Y^*\mathbf{z} = 0 \iff \mathbf{z} \perp \mathrm{col}(Y)$, because X has independent columns and Σ_r is invertible. The orthogonal-complement identities follow from Proposition 2.3.10.
Structural Commentary: Compact SVD exposes image directions X and null directions $\mathrm{col}(Y)^\perp$.
Key Insight: A acts only on the right singular subspace $\mathrm{col}(Y)$. □

Figure 91 summarizes the active right singular subspace, its image, and the kernel annihilated by A.

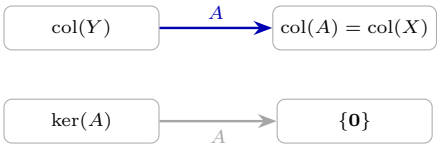

Figure 91: Compact SVD: A maps $\mathrm{col}(Y)$ onto $\mathrm{col}(X)$ and kills $\ker(A) = \mathrm{col}(Y)^\perp$.

Theorem 4.2.15. *Uniqueness of Compact SVD up to Singular Subspaces*
Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$ be compact SVDs with the same Σ_r. Then $X_1 = X_2U$, $Y_1 = Y_2U$ for some unitary U satisfying $U\Sigma_r = \Sigma_r U$. Hence $U = U_1 \oplus \cdots \oplus U_d$ along the equal singular-value blocks of Σ_r.

Proof. Theorem 4.2.14 gives $\mathrm{col}(X_1) = \mathrm{col}(X_2)$ and $\mathrm{col}(Y_1) = \mathrm{col}(Y_2)$. Since each X_i and Y_i has orthonormal columns, there are unitary $U, V \in \mathrm{Mat}(r,r,\mathbb{F})$ such that $X_1 = X_2U$ and $Y_1 = Y_2V$. Substitution and multiplication on the left by X_2^* and on the right by Y_2 give $\Sigma_r = U\Sigma_r V^*$, hence $U\Sigma_r = \Sigma_r V$.
Therefore $U\Sigma_r^2 U^* = \Sigma_r^2$, so U commutes with Σ_r^2. Lemma 5.4.10 makes U block diagonal on equal singular-value blocks. Hence $U\Sigma_r = \Sigma_r U$, and $U\Sigma_r = \Sigma_r V$ forces $V = U$ because Σ_r is invertible.
Structural Commentary: Compact SVD is unique up to unitary changes inside equal singular-value blocks.
Key Insight: Distinct singular values fix subspaces; repeated ones only fix the subspace, not its basis. □

Definition 4.2.16. *Matrix Modulus*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, the *modulus* of A is $|A| := (A^*A)^{1/2} \in \mathrm{Mat}(n,n,\mathbb{F})$.
Thus $|A^*| = (AA^*)^{1/2} \in \mathrm{Mat}(m,m,\mathbb{F})$.

Proposition 4.2.17. *Modulus from SVD*
Let $A = V\Sigma W^*$ be an SVD. Then $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$ and $|A^*| = V(\Sigma\Sigma^*)^{1/2}V^*$.

Proof. Since $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$, the displayed matrices are positive semidefinite square roots of A^*A and AA^*, respectively. Uniqueness of the positive semidefinite square root gives the claim.
Structural Commentary: The modulus keeps only the singular-value stretch.
Key Insight: In an SVD, $|A|$ is the right singular-vector version of the diagonal stretch. □

Theorem 4.2.18. *Rectangular Polar Decomposition*
Let $A = V\Sigma W^* \in \mathrm{Mat}(m,n,\mathbb{F})$ be an SVD.

(i) If $m \geq n$, let V_n be the first n columns of V and set $U = V_n W^*$. Then $U^*U = I_n$ and $A = U\,|A|$.
(ii) If $m \leq n$, let W_m be the first m columns of W and set $U = VW_m^*$. Then $UU^* = I_m$ and $A = |A^*|\,U$.
(iii) If $m = n$, then $U = VW^*$ is unitary and $A = U\,|A| = |A^*|\,U$.

Proof.

(i) Proposition 4.2.17 gives $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$.
Since $V_n(\Sigma^*\Sigma)^{1/2} = V\Sigma$, $U\,|A| = V_nW^*W(\Sigma^*\Sigma)^{1/2}W^* = V\Sigma W^* = A$. Also $U^*U = WV_n^*V_nW^* = I_n$.

(ii) Apply (i) to A^* with the SVD $A^* = W\Sigma^*V^*$, obtaining $A^* = U_0\,|A^*|$ with $U_0 = W_mV^*$ and $U_0^*U_0 = I_m$. Taking adjoints gives $A = |A^*|\,U_0^* = |A^*|\,VW_m^*$, and $U_0^*(U_0^*)^* = U_0^*U_0 = I_m$. Thus $U = VW_m^*$ works.

(iii) This is the common case of (i) and (ii), with $U = VW^*$ unitary.

Structural Commentary: Polar decomposition compresses the two SVD unitary factors into one partial isometry.
Key Insight: SVD records every singular direction; polar form records stretch plus motion. □

Method 4.2.19. *SVD and Polar Computation Method*
To compute an SVD and the associated polar factors of $A \in \mathrm{Mat}(m,n,\mathbb{F})$:

(i) Diagonalise A^*A; the square roots of its positive eigenvalues are the positive singular values.
(ii) Choose orthonormal right singular vectors $\mathbf{w}_j$ for the positive eigenvalues and set $\mathbf{v}_j = \sigma_j^{-1} A\mathbf{w}_j$.
(iii) Complete the $\mathbf{v}_j$ and $\mathbf{w}_j$ to unitary matrices V, W to obtain $A = V\Sigma W^*$.
Omit zero-singular directions for compact SVD.
(iv) Use $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$ and merge V, W as in Theorem 4.2.18 to obtain the polar factor.

Theorem 4.2.20. *Uniqueness of the Positive Polar Factor*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$.

(i) If $m \geq n$ and $A = UB$ with $U^*U = I_n$ and $B \succeq 0$, then $B = |A|$.
If $\mathrm{rank}(A) = n$, then $U = A\,|A|^{-1}$ is unique.
(ii) If $m \leq n$ and $A = CU$ with $UU^* = I_m$ and $C \succeq 0$, then $C = |A^*|$.
If $\mathrm{rank}(A) = m$, then $U = |A^*|^{-1}\,A$ is unique.

Proof.

(i) $A^*A = BU^*UB = B^2$, $B = |A|$ by uniqueness of positive square root. If $\mathrm{rank}(A) = n$, then $\ker(A) = \{\mathbf{0}\}$. Since $\mathbf{x}^*A^*A\mathbf{x} = \|A\mathbf{x}\|_2^2$, this gives $\ker(A^*A) = \{\mathbf{0}\}$ and $|A|$ is invertible. Thus $U = A\,|A|^{-1}$.
(ii) $AA^* = CUU^*C = C^2$, so $C = |A^*|$. If $\mathrm{rank}(A) = m$, then A^* has full column rank, so $\ker(A^*) = \{\mathbf{0}\}$. Since $\mathbf{y}^*AA^*\mathbf{y} = \|A^*\mathbf{y}\|_2^2$, this gives $\ker(AA^*) = \{\mathbf{0}\}$ and $|A^*|$ is invertible. Thus $U = |A^*|^{-1}\,A$.

Structural Commentary: The positive factor is fixed by the Gram matrix.
Key Insight: Full rank makes the partial isometry unique. □

Theorem 4.2.21. *Normality and Polar Commutation*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and suppose $A = U\,|A|$ with U unitary. Then A is normal if and only if $U\,|A| = |A|\,U$.

Proof.

(⇒) If A is normal, then $|A|^2 = A^*A = AA^* = U\,|A|^2\,U^*$. Thus $U\,|A|^2\,U^* = |A|^2$. The matrix $U\,|A|\,U^*$ is PSD and squares to $|A|^2$, so uniqueness of positive square root gives $U\,|A|\,U^* = |A|$. Hence $U\,|A| = |A|\,U$.
(⇐) If $U\,|A| = |A|\,U$, then $AA^* = U\,|A|^2\,U^* = |A|^2 = A^*A$.

Structural Commentary: Normality says that the polar motion preserves the stretching eigenspaces.
Key Insight: Normality is commutation of polar motion and stretch. □

Perspective 4.2.22. *Polar Decomposition Separates Stretch and Direction*
In $T = S\sqrt{T^*T}$, the positive factor $\sqrt{T^*T}$ determines the metric stretch and the isometry S determines the direction. Thus polar decomposition is the operator analogue of modulus times phase.
Structural Commentary: For normal T, the polar factors commute and share spectral data.
Key Insight: Polar decomposition factors an operator into positive geometry and isometric motion.

Proposition 4.2.23. *Absolute Determinant Comes from the Positive Polar Factor*
For $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $|\det T| = \det\sqrt{T^*T}$.

Proof. By Theorem 4.2.1, $T = S\sqrt{T^*T}$ for some isometry S. Hence $|\det T| = |\det S|\left|\det\sqrt{T^*T}\right| = \det\sqrt{T^*T}$, where Proposition 1.1.24 gives $|\det S| = 1$, and positivity of $\sqrt{T^*T}$ gives nonnegative determinant.
Structural Commentary: Polar decomposition isolates stretch; only the positive factor changes volume.
Key Insight: Determinant sign or phase belongs to the isometric part; volume belongs to the positive part. □

5 Norms, Conditioning, Perturbation, and Stability

Matrix norms measure stretch, energy, approximation quality, and sensitivity. Singular values organize norms, low-rank approximation, pseudoinverses, conditioning, and error bounds.

Perturbation theory studies how matrix information changes under small errors. Bauer–Fike, Gersgorin, interlacing, Weyl inequalities, and pseudospectra measure robustness.

5.1 Matrix Norms

Definition 5.1.1. *Spectral Norm and Extreme Singular Values*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, define the *spectral norm* $\|A\|_2 := \max_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2$.

If the singular values of A are $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, $q = \min(m,n)$, set $\sigma_{\max}(A) := \sigma_1$.
When A is square, $\sigma_{\min}(A) := \sigma_n$.

Theorem 5.1.2. *Spectral Norm is the Largest Singular Value*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$. Then $\|A\|_2 = \sigma_{\max}(A)$.
Moreover, if $A = V\Sigma W^*$ is an SVD and $\mathbf{w}_1$ is the first column of W, then $\|A\mathbf{w}_1\|_2 = \sigma_1$.

Proof. Let $\mathbf{w}_1, \ldots, \mathbf{w}_n$ be the columns of W. For $\mathbf{x} = \sum_{j=1}^n c_j\mathbf{w}_j$ with $\|\mathbf{x}\|_2 = 1$, unitary invariance of the Euclidean norm gives

$$\|A\mathbf{x}\|_2^2 = \|\Sigma W^*\mathbf{x}\|_2^2 = \sum_{j=1}^{q} \sigma_j^2\,|c_j|^2 \leq \sigma_1^2 \sum_{j=1}^{n} |c_j|^2 = \sigma_1^2.$$

Thus $\|A\|_2 \leq \sigma_1$. Taking $\mathbf{x} = \mathbf{w}_1$ gives equality.
Structural Commentary: The right singular basis diagonalises the stretching computation.
Key Insight: The largest stretch occurs in a top right singular direction. □

Proposition 5.1.3. *Basic Properties of the Spectral Norm*
For compatible matrices A, B and scalar c:

(i) $\|A\|_2 \geq 0$, *and* $\|A\|_2 = 0$ *if and only if* $A = 0$;
(ii) $\|cA\|_2 = |c|\,\|A\|_2$;
(iii) $\|A + B\|_2 \leq \|A\|_2 + \|B\|_2$;
(iv) $\|AB\|_2 \leq \|A\|_2\,\|B\|_2$.

Proof.

(i) Since $\|A\mathbf{x}\|_2 \geq 0$, $\|A\|_2 \geq 0$. Also $\|A\|_2 = 0$ iff $A\mathbf{x} = 0$ for every unit $\mathbf{x}$, iff $A = 0$.
(ii) $\|cA\|_2 = \max_{\|\mathbf{x}\|_2=1} \|cA\mathbf{x}\|_2 = |c|\,\|A\|_2$.
(iii) For $\|\mathbf{x}\|_2 = 1$, $\|(A + B)\mathbf{x}\|_2 \leq \|A\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \leq \|A\|_2 + \|B\|_2$. Maximize over unit $\mathbf{x}$.
(iv) For $\|\mathbf{x}\|_2 = 1$, $\|AB\mathbf{x}\|_2 \leq \|A\|_2\,\|B\mathbf{x}\|_2 \leq \|A\|_2\,\|B\|_2$. Maximize over unit $\mathbf{x}$.

Structural Commentary: The spectral norm is the operator norm induced by Euclidean geometry.
Key Insight: Bound each output vector first, then take the worst case. □

Definition 5.1.4. *Operator Norm on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$*
For $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, define $\|A\|_{\text{op}} := \sup_{\|\mathbf{h}\|_2 \leq 1} \|A\mathbf{h}\|_2$.
Equivalently, after standard bases identify A with its matrix, $\|A\|_{\text{op}}$ is the spectral norm from Definition 5.1.1.
Thus $\|A\mathbf{h}\|_2 \leq \|A\|_{\text{op}}\,\|\mathbf{h}\|_2$, and sums and products obey the norm estimates in Proposition 5.1.3.
The induced metric on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is $d(A, B) := \|A - B\|_{\text{op}}$; this is the metric used for continuity of derivative maps in Definition 3.2.2.

Proposition 5.1.5. *Frobenius and Spectral Norm Comparison*
Let $A \in \text{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$. Then $\|A\|_F^2 = \sum_{j=1}^q \sigma_j^2$ and $\|A\|_2 \leq \|A\|_F$.
If $A \neq 0$, equality $\|A\|_2 = \|A\|_F$ holds if and only if $\text{rank}(A) = 1$.

Proof. The Frobenius identity is Proposition 4.2.10. Hence $\|A\|_2^2 = \sigma_1^2 \leq \sum_{j=1}^q \sigma_j^2 = \|A\|_F^2$. Assume $A \neq 0$.

($\Rightarrow$) If $\|A\|_2 = \|A\|_F$, then $\sum_{j=2}^q \sigma_j^2 = 0$, so only σ_1 is nonzero and $\text{rank}(A) = 1$.
($\Leftarrow$) If $\text{rank}(A) = 1$, then $\sigma_2 = \cdots = \sigma_q = 0$, so $\|A\|_2 = \|A\|_F$.

Structural Commentary: Frobenius counts all singular energy; spectral keeps the largest part.
Key Insight: Equality means exactly one nonzero singular value. □

Theorem 5.1.6. *Submultiplicativity of Frobenius and Spectral Norms*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, $B \in \text{Mat}(n, k, \mathbb{F})$. Then $\|AB\|_F \leq \|A\|_2\,\|B\|_F \leq \|A\|_F\,\|B\|_F$ and $\|AB\|_2 \leq \|A\|_2\,\|B\|_2$.

Proof. Write $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_k]$. Then $\|AB\|_F^2 = \sum_{j=1}^k \|A\mathbf{b}_j\|_2^2 \leq \|A\|_2^2 \sum_{j=1}^k \|\mathbf{b}_j\|_2^2 = \|A\|_2^2\,\|B\|_F^2$.

Proposition 5.1.5 gives $\|A\|_2 \leq \|A\|_F$. The spectral-norm inequality is Proposition 5.1.3.
Structural Commentary: Frobenius norm is column energy; each column of AB is A applied to a column of B.
Key Insight: Control every column by the spectral norm of A. □

Proposition 5.1.7. *Unitary Invariance of Singular Values*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, and let $U \in \text{Mat}(m, m, \mathbb{F})$ and $V \in \text{Mat}(n, n, \mathbb{F})$ be unitary. Then A and UAV have the same singular values. In particular, $\|UAV\|_2 = \|A\|_2$ and $\|UAV\|_F = \|A\|_F$.

Proof. Since $(UAV)^*(UAV) = V^*A^*AV$, the Gram matrices of A and UAV are unitarily similar, hence have the same eigenvalues. Singular values are the nonnegative square roots of these eigenvalues. The two norm identities follow from Theorem 5.1.2 and Proposition 5.1.5.
Structural Commentary: Left and right unitary factors change coordinates, not singular-value data.
Key Insight: Singular values are spectral data of A^*A, and unitary conjugation preserves that spectrum. □

Example 5.1.8. *Similarity Need Not Preserve Singular Values*
Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Then $B = DAD^{-1}$ with $D = \text{diag}(2, 1)$, so A and B are similar.
However $A^*A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $B^*B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ have different eigenvalues, so A and B have different singular values.
Singular values are preserved by unitary left-right equivalence, not by arbitrary similarity.

Theorem 5.1.9. *Schur's Inequality and Normality*
Let $A \in \text{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted algebraically.
Then $\sum_{j=1}^n |\lambda_j|^2 \leq \|A\|_F^2$ with equality if and only if A is normal.

Proof. Let $A = UTU^*$ be a Schur form, where $T = [t_{ij}]$ is upper triangular and $t_{jj} = \lambda_j$.
Then $\|A\|_F^2 = \text{tr}(T^*T) = \sum_{j=1}^n |\lambda_j|^2 + \sum_{i<j} |t_{ij}|^2$. Inequality holds. Equality holds exactly when T is diagonal.

($\Rightarrow$) If equality holds, then T is diagonal, so A is unitarily diagonalizable and hence normal.
($\Leftarrow$) If A is normal, then T is normal and upper triangular, hence diagonal by Lemma 6.1.10.

Structural Commentary: Schur form splits Frobenius norm into eigenvalue mass plus off-diagonal mass.
Key Insight: Normality means the Schur off-diagonal part vanishes. □

Definition 5.1.10. *Defect from Normality*
Let $A \in \text{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity.
The *defect from normality* of A is $\Delta(A) := \|A\|_F^2 - \sum_{j=1}^n |\lambda_j|^2$.
By Theorem 5.1.9, $\Delta(A) \geq 0$, with equality if and only if A is normal.

Theorem 5.1.11. *Normality Transfers from AB to BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. If A, B, and AB are normal, then BA is normal.

Proof. By Theorem 1.1.16, AB and BA have the same eigenvalues counted with algebraic multiplicity. Also,

$$\begin{aligned}\operatorname{tr}\big((BA)^*(BA)\big) &= \operatorname{tr}(A^*B^*BA) = \operatorname{tr}(AA^*B^*B)\\ &= \operatorname{tr}(A^*AB^*B) = \operatorname{tr}(A^*ABB^*) = \operatorname{tr}(B^*A^*AB) = \operatorname{tr}\big((AB)^*(AB)\big),\end{aligned}$$

using cyclicity of trace and normality of A and B. Thus $\|BA\|_F^2 = \|AB\|_F^2$, so $\Delta(BA) = \Delta(AB)$.
Since AB is normal, $\Delta(AB) = 0$; hence $\Delta(BA) = 0$, and Theorem 5.1.9 gives that BA is normal.
Structural Commentary: Same spectrum and Frobenius norm give the same Schur defect.
Key Insight: Schur defect turns normality of AB into normality of BA. □

Theorem 5.1.12. *Eigenvalues Versus Singular Values*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ ordered so that $|\lambda_1| \geq \cdots \geq |\lambda_n|$, and singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Then:

(i) $|\det A| = \prod_{j=1}^n |\lambda_j| = \prod_{j=1}^n \sigma_j$;
(ii) $\rho(A) \leq \|A\|_2$;
(iii) $\sigma_n \leq |\lambda_n|$;
(iv) *A is normal if and only if $|\lambda_j| = \sigma_j$ for every j;*
(v) *A is positive semidefinite if and only if $\lambda_j = \sigma_j$ for every j.*

Proof.

(i) The eigenvalue product gives $|\det A| = \prod_j |\lambda_j|$.
If $A = V\Sigma W^*$ is an SVD, then $|\det V| = |\det W| = 1$, so $|\det A| = |\det \Sigma| = \prod_j \sigma_j$.
(ii) If $A\mathbf{x} = \lambda\mathbf{x}$ with $\|\mathbf{x}\|_2 = 1$, then $|\lambda| = \|A\mathbf{x}\|_2 \leq \|A\|_2$. Maximize over eigenvalues.
(iii) If A is singular, both sides are 0 or the inequality is immediate.
If A is invertible, apply (ii) to A^{-1}: $\frac{1}{|\lambda_n|} = \rho(A^{-1}) \leq \|A^{-1}\|_2 = \frac{1}{\sigma_n}$.
(iv) If A is normal, the spectral theorem gives $\sigma_j = |\lambda_j|$. Conversely, equality gives $\sum_j |\lambda_j|^2 = \sum_j \sigma_j^2 = \|A\|_F^2$, so Theorem 5.1.9 gives normality.
(v) If $A \succeq 0$, then A is normal and $\lambda_j \geq 0$, so (iv) gives $\lambda_j = \sigma_j$. Conversely, equality gives normality with nonnegative real eigenvalues, hence $A \succeq 0$ by Theorem 2.1.3.

Structural Commentary: Normality aligns eigenvalue moduli with singular values; positivity removes the moduli.
Key Insight: Normality is the bridge that turns spectral data into singular-value data. □

Theorem 5.1.13. *Trace Bound by Singular Values*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$, then $|\operatorname{tr} A| \leq \sum_{j=1}^n \sigma_j(A)$. Consequently, $\sum_{j=1}^n |\lambda_j(A)| \leq \sum_{j=1}^n \sigma_j(A)$ and $\sum_{j=1}^n |a_{jj}| \leq \sum_{j=1}^n \sigma_j(A)$.

Proof. Let $A = V\Sigma W^*$ be an SVD.
Writing $\mathbf{v}_k, \mathbf{w}_k$ for the columns of V, W, $\operatorname{tr} A = \sum_{k=1}^n \sigma_k \mathbf{w}_k^* \mathbf{v}_k$, $|\mathbf{w}_k^* \mathbf{v}_k| \leq 1$, so $|\operatorname{tr} A| \leq \sum_k \sigma_k$.
For eigenvalues, take a Schur form T of A and a diagonal unitary D with $\operatorname{tr}(DT) = \sum_j |\lambda_j|$.
For diagonal entries, take diagonal unitary D with $\operatorname{tr}(DA) = \sum_j |a_{jj}|$.
In both cases, unitary invariance of singular values and the trace bound give the result.
Structural Commentary: Phase rotations turn trace estimates into eigenvalue and diagonal estimates.
Key Insight: Prove $|\operatorname{tr} A| \leq \sum_j \sigma_j$ once, then rotate phases. □

5.2 Low-Rank Approximation and Pseudoinverses

Proposition 5.2.1. *Rank-One SVD Terms and Truncation Error*
Let $A = X\Sigma_r Y^* = \sum_{j=1}^r \sigma_j \mathbf{x}_j \mathbf{y}_j^*$ be a compact SVD, with $\sigma_1 \geq \cdots \geq \sigma_r > 0$.
For $0 \leq k \leq r$, set $A_k := \sum_{j=1}^k \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, $A_0 := 0$.
Then the rank-one summands are mutually orthogonal in the Frobenius inner product, and

$$\|A\|_F^2 = \sum_{j=1}^r \sigma_j^2, \qquad \|A - A_k\|_F^2 = \sum_{j=k+1}^r \sigma_j^2, \qquad \|A - A_k\|_2 = \begin{cases}\sigma_{k+1}, & k < r,\\ 0, & k = r.\end{cases}$$

Proof. For $i, j \in \{1, \dots, r\}$, $\left\langle \sigma_i \mathbf{x}_i \mathbf{y}_i^*, \sigma_j \mathbf{x}_j \mathbf{y}_j^* \right\rangle_F = \sigma_i \sigma_j \operatorname{tr}(\mathbf{y}_i \mathbf{x}_i^* \mathbf{x}_j \mathbf{y}_j^*) = \sigma_i \sigma_j \delta_{ij}$.
Pythagoras in the Frobenius inner product gives the two Frobenius formulas. The residual $A - A_k$ has compact SVD $\sum_{j=k+1}^r \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, so Theorem 5.1.2 gives the spectral-norm formula.
Structural Commentary: Compact SVD decomposes a matrix into orthogonal rank-one energy packets.
Key Insight: Truncation error is exactly the energy left in the discarded singular directions. □

Figure 92 separates the retained singular-value terms $1, \dots, k$ from the discarded tail $k+1, \dots, r$ whose squared entries determine the Frobenius error.

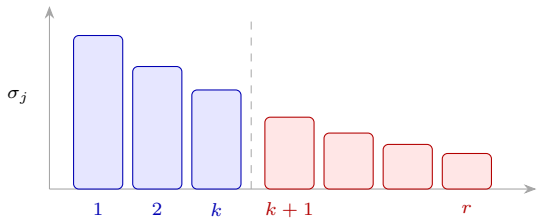

Figure 92: Blue bars form A_k; red bars are discarded, with $\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2$.

Theorem 5.2.2. *Eckart–Young Low-Rank Approximation*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m,n)$ and $\sigma_j = 0$ for $j > q$. If A_k is the rank-k truncated SVD, $0 \leq k < q$, and $B \in \mathrm{Mat}(m,n,\mathbb{F})$ has $\mathrm{rank}(B) \leq k$, then

$$\|A - A_k\|_2 \leq \|A - B\|_2\,, \qquad \|A - A_k\|_F \leq \|A - B\|_F\,.$$

Proof. Let $\mathbf{w}_1, \ldots, \mathbf{w}_n$ be right singular vectors for A.

i. Since $\mathrm{rank}(B) \leq k$, rank-nullity gives $\dim \ker(B) \geq n - k$. The subspaces $\ker(B)$ and $\mathrm{span}(\mathbf{w}_1, \ldots, \mathbf{w}_{k+1})$ in $\mathbb{F}^n$ have dimensions summing at least $n+1$, so their intersection contains a nonzero vector.
Choose unit $\mathbf{x}$ in this intersection. Writing $\mathbf{x} = \sum_{i=1}^{k+1} c_i \mathbf{w}_i$ gives $\sum_i |c_i|^2 = 1$ and $\|A\mathbf{x}\|_2^2 = \sum_{i=1}^{k+1} \sigma_i^2 |c_i|^2 \geq \sigma_{k+1}^2$.
Thus $\|A - B\|_2 \geq \|(A-B)\mathbf{x}\|_2 = \|A\mathbf{x}\|_2 \geq \sigma_{k+1} = \|A - A_k\|_2$.

ii. Choose an $(n-k)$-dimensional subspace $\mathcal{U} \subseteq \ker(B)$, with projection P and orthonormal basis $\mathbf{z}_1, \ldots, \mathbf{z}_{n-k}$.
Since $B\mathbf{z}_j = 0$, $\|A - B\|_F^2 \geq \sum_{j=1}^{n-k} \|(A-B)\mathbf{z}_j\|_2^2 = \sum_{j=1}^{n-k} \|A\mathbf{z}_j\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 \|P\mathbf{w}_i\|_2^2$.
Put $\alpha_i = \|P\mathbf{w}_i\|_2^2$. Then $0 \leq \alpha_i \leq 1$ and $\sum_i \alpha_i = \mathrm{tr}(P) = n - k$.
Among all such weights, the sum $\sum_i \sigma_i^2 \alpha_i$ is minimized by placing weight 1 on the $n-k$ smallest values of the decreasing sequence σ_i^2, hence $\sum_i \sigma_i^2 \alpha_i \geq \sum_{i=k+1}^{n} \sigma_i^2 = \|A - A_k\|_F^2$.

Structural Commentary: Rank-k competitors vanish on at least $n-k$ directions.
Key Insight: The optimal loss is precisely the singular-value tail. □

Method 5.2.3. *Truncated SVD Approximation Method*
To compress or approximate A by rank k:

(i) compute or estimate the leading singular triples $(\sigma_j, \mathbf{x}_j, \mathbf{y}_j)$;
(ii) keep $A_k = \sum_{j=1}^{k} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$;
(iii) use $\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2$ and $\|A - A_k\|_2 = \sigma_{k+1}$ to quantify the lost information.

The same compact SVD data also defines a canonical inverse on the nonzero singular subspaces.

Definition 5.2.4. *Moore–Penrose Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. If $A = 0$, define $A^\dagger = 0_{n,m}$.
If $\mathrm{rank}(A) = r \geq 1$ and $A = X\Sigma_r Y^*$ is a compact SVD, define $A^\dagger := Y\Sigma_r^{-1}X^* \in \mathrm{Mat}(n,m,\mathbb{F})$.
This is the *pseudoinverse* or *Moore–Penrose pseudoinverse* of A.

Proposition 5.2.5. *Pseudoinverse is Well Defined*
The definition of $A^\dagger$ does not depend on the chosen compact SVD of A.

Proof. Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$. By Theorem 4.2.15, $X_1 = X_2 U$, $Y_1 = Y_2 U$, and $U\Sigma_r^{-1}U^* = \Sigma_r^{-1}$.
Hence $Y_1\Sigma_r^{-1}X_1^* = Y_2 U\Sigma_r^{-1}U^*X_2^* = Y_2\Sigma_r^{-1}X_2^*$.
Structural Commentary: Compact SVD ambiguity cancels in $Y\Sigma_r^{-1}X^*$.
Key Insight: Inverting the singular values preserves the same block symmetries. □

Corollary 5.2.6. *Pseudoinverse from a Full SVD*
Let $A = V\Sigma W^*$ be a full SVD of $A \in \mathrm{Mat}(m,n,\mathbb{F})$, with rank r and positive singular values $\sigma_1, \ldots, \sigma_r$.
Write $V = [X\ X_0]$, $W = [Y\ Y_0]$, and $\Sigma_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$.
In conformal blocks, $\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}$, $\Sigma^\dagger := \begin{bmatrix} \Sigma_r^{-1} & 0 \\ 0 & 0 \end{bmatrix} \in \mathrm{Mat}(n,m,\mathbb{F})$. Then $A^\dagger = W\Sigma^\dagger V^*$.

Proof. If $r = 0$, then $A = 0$ and $W\Sigma^\dagger V^* = 0$.
If $r > 0$, the block form gives $A = X\Sigma_r Y^*$; $A^\dagger = Y\Sigma_r^{-1}X^*$; and $W\Sigma^\dagger V^* = Y\Sigma_r^{-1}X^*$. Thus $A^\dagger = W\Sigma^\dagger V^*$.
Structural Commentary: Full SVD pads compact SVD by zero blocks.
Key Insight: The pseudoinverse flips V and W and inverts only the positive singular values. □

Theorem 5.2.7. *Basic Properties of the Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:

(i) $\mathrm{col}(A^\dagger) = \mathrm{col}(A^*)$ *and* $\ker(A^\dagger) = \ker(A^*)$;
(ii) $AA^\dagger$ *is the orthogonal projection of* $\mathbb{F}^m$ *onto* $\mathrm{col}(A)$;
(iii) $A^\dagger A$ *is the orthogonal projection of* $\mathbb{F}^n$ *onto* $\mathrm{col}(A^*)$;
(iv) $AA^\dagger A = A$ *and* $A^\dagger AA^\dagger = A^\dagger$;
(v) $(A^\dagger)^\dagger = A$;
(vi) if $m = n$ and A is invertible, then $A^\dagger = A^{-1}$.

Proof. The zero case is immediate. Let $A = X\Sigma_r Y^*$ be a compact SVD; then $A^\dagger = Y\Sigma_r^{-1}X^*$.

(i) $\operatorname{col}(A^\dagger) = \operatorname{col}(Y) = \operatorname{col}(A^*)$ and $\ker(A^\dagger) = \ker(X^*) = \ker(A^*)$.
(ii) $AA^\dagger = XX^*$, the orthogonal projection onto $\operatorname{col}(X) = \operatorname{col}(A)$.
(iii) $A^\dagger A = YY^*$, the orthogonal projection onto $\operatorname{col}(Y) = \operatorname{col}(A^*)$.
(iv) These follow from $XX^*X = X$ and $YY^*Y = Y$.
(v) The compact SVD of $A^\dagger$ is $Y\Sigma_r^{-1}X^*$, so $(A^\dagger)^\dagger = A$.
(vi) If A is square and invertible, then $AA^\dagger = A^\dagger A = I_n$, hence $A^\dagger = A^{-1}$.

Structural Commentary: The pseudoinverse inverts nonzero singular directions and projects away null directions.
Key Insight: Replace each $\sigma_j > 0$ by σ_j^{-1} on the same singular subspaces. □

Figure 93 shows that A and $A^\dagger$ reverse one another between $\operatorname{col}(A^*)$ and $\operatorname{col}(A)$, while their respective kernel directions are annihilated.

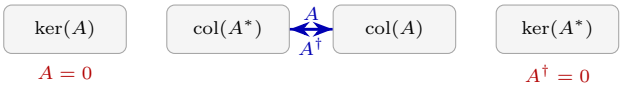

Figure 93: A and $A^\dagger$ invert the nonzero singular subspaces; they vanish on $\ker(A)$ and $\ker(A^*)$, respectively.

Theorem 5.2.8. *Penrose Characterization of the Pseudoinverse*
Let $A \in \operatorname{Mat}(m,n,\mathbb{F})$, $\mathcal{C} = \operatorname{col}(A)$, and $\mathcal{D} = \operatorname{col}(A^*) = (\ker A)^\perp$. Then $A^\dagger$ is unique $B \in \operatorname{Mat}(n,m,\mathbb{F})$ satisfying

(i) $ABA = A$;
(ii) $BAB = B$;
(iii) AB is Hermitian;
(iv) BA is Hermitian.

Proof.

(i) Theorem 5.2.7 gives $AA^\dagger A = A$; $A^\dagger AA^\dagger = A^\dagger$, and $AA^\dagger$, $A^\dagger A$ are orthogonal projections. Hence $A^\dagger$ satisfies the four equations.
(ii) Let B satisfy the four equations. Then $(AB)^2 = ABAB = AB$; $(AB)^* = AB$. Also $\operatorname{range}(AB) = \mathcal{C}$, since $\operatorname{range}(AB) \subseteq \mathcal{C}$ and $AB(A\mathbf{x}) = ABA\mathbf{x} = A\mathbf{x}$. By Theorem 2.5.6, $AB = P_{\mathcal{C}}$.
(iii) $(BA)^2 = BABA = BA$; $(BA)^* = BA$. Also $\ker(BA) = \ker(A)$, as $BA\mathbf{x} = 0 \Rightarrow A\mathbf{x} = ABA\mathbf{x} = 0$ and the reverse inclusion is immediate. By Theorem 2.5.6, $BA = P_{\mathcal{D}}$.
(iv) For $\mathbf{y} \in \mathbb{F}^m$, write $\mathbf{y} = \mathbf{p} + \mathbf{z}$ with $\mathbf{p} \in \mathcal{C}$ and $\mathbf{z} \in \mathcal{C}^\perp = \ker(A^*)$. Since $AB\mathbf{z} = 0$, $B\mathbf{z} = BAB\mathbf{z} = 0$. If $\mathbf{p} = A\mathbf{x}$, then $B\mathbf{p} = BA\mathbf{x} = P_{\mathcal{D}}\mathbf{x}$, independent of the choice of $\mathbf{x}$ because $\ker(P_{\mathcal{D}}) = \mathcal{D}^\perp = \ker(A)$. Thus the four equations determine B uniquely. Since $A^\dagger$ satisfies them, $B = A^\dagger$.

Structural Commentary: Penrose equations force the two projection factors.
Key Insight: Once AB and BA are projections, B is forced. □

Theorem 5.2.9. *Pseudoinverse and Minimum-Norm Least Squares*
Let $A \in \operatorname{Mat}(m,n,\mathbb{F})$, let $\mathbf{y} \in \mathbb{F}^m$, and set $\mathbf{x}_0 = A^\dagger\mathbf{y}$.

(i) If $A\mathbf{x} = \mathbf{y}$ is consistent, then $A\mathbf{x}_0 = \mathbf{y}$ and $\mathbf{x}_0$ is the unique solution of smallest Euclidean norm.
(ii) For arbitrary $\mathbf{y}$, $\mathbf{x}_0$ is a least-squares solution of $A\mathbf{x} \approx \mathbf{y}$, and it is the unique least-squares solution of smallest Euclidean norm.

Proof. Let $P = AA^\dagger = P_{\operatorname{col}(A)}$ and note that $\mathbf{x}_0 \in \operatorname{col}(A^\dagger) = \operatorname{col}(A^*) = (\ker A)^\perp$.

(i) If $\mathbf{y} \in \operatorname{col}(A)$, then $A\mathbf{x}_0 = P\mathbf{y} = \mathbf{y}$. For any solution $\mathbf{x}$, $\mathbf{x} - \mathbf{x}_0 \in \ker(A)$, so $\|\mathbf{x}\|_2^2 = \|\mathbf{x}_0 + (\mathbf{x} - \mathbf{x}_0)\|_2^2 = \|\mathbf{x}_0\|_2^2 + \|\mathbf{x} - \mathbf{x}_0\|_2^2$ with equality only when $\mathbf{x} = \mathbf{x}_0$.
(ii) For arbitrary $\mathbf{y}$, $A\mathbf{x}_0 = P\mathbf{y}$, so $\mathbf{y} - A\mathbf{x}_0 \perp \operatorname{col}(A)$ and $\mathbf{x}_0$ is least-squares optimal by Theorem 2.4.4. Any least-squares solution $\mathbf{x}$ satisfies $A\mathbf{x} = P\mathbf{y} = A\mathbf{x}_0$, hence the same orthogonal decomposition gives $\|\mathbf{x}\|_2 \geq \|\mathbf{x}_0\|_2$, with equality only for $\mathbf{x} = \mathbf{x}_0$.

Structural Commentary: Least squares projects onto $\operatorname{col}(A)$, then removes the $\ker(A)$ component.
Key Insight: $A^\dagger\mathbf{y}$ is the fitted solution orthogonal to $\ker(A)$. □

Method 5.2.10. *Pseudoinverse Solution Method*
For a system $A\mathbf{x} \approx \mathbf{y}$:

(i) compute a compact SVD $A = X\Sigma_r Y^*$;
(ii) form $A^\dagger = Y\Sigma_r^{-1}X^*$;
(iii) set $\mathbf{x}_0 = A^\dagger\mathbf{y}$;
(iv) interpret $A\mathbf{x}_0 = P_{\operatorname{col}(A)}\mathbf{y}$ and $\mathbf{x}_0 \in (\ker A)^\perp$.

5.3 Conditioning

Definition 5.3.1. *Spectral Condition Number*
If $A \in \operatorname{Mat}(n,n,\mathbb{F})$ is invertible, its *spectral condition number* is $\kappa_2(A) := \|A\|_2 \left\|A^{-1}\right\|_2$.
Equivalently, $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$. If A is singular, $\kappa_2(A)$ is undefined.

Proposition 5.3.2. *Properties of the Spectral Condition Number*
Let $A, B \in \operatorname{Mat}(n,n,\mathbb{F})$ be invertible.

(i) $\kappa_2(AB) \leq \kappa_2(A)\kappa_2(B)$;
(ii) $\kappa_2(A) \geq 1$, with equality if and only if A is a nonzero scalar multiple of a unitary matrix;
(iii) $\kappa_2(cA) = \kappa_2(A)$ for every nonzero scalar c;
(iv) $\kappa_2(A) = \kappa_2(A^*) = \kappa_2(\overline{A}) = \kappa_2(A^t) = \kappa_2(A^{-1})$ over $\mathbb{C}$, with the evident real special cases;
(v) $\kappa_2(A^*A) = \kappa_2(A)^2$;
(vi) if U, V are unitary, then $\kappa_2(UAV) = \kappa_2(A)$.

Proof.

(i) Use submultiplicativity: $\kappa_2(AB) = \|AB\|_2 \left\|B^{-1}A^{-1}\right\|_2 \le \|A\|_2 \|B\|_2 \left\|B^{-1}\right\|_2 \left\|A^{-1}\right\|_2$.
(ii) Since $I_n = AA^{-1}$, $1 = \|I_n\|_2 \le \kappa_2(A)$. Equality means $\sigma_{\max}(A) = \sigma_{\min}(A)$, equivalently every singular value is the same positive number s. If $A = V\Sigma W^*$, then $\Sigma = sI_n$, so $A = sVW^*$ is a nonzero scalar multiple of a unitary matrix. The converse is immediate.
(iii) This follows from $\|cA\|_2 = |c| \|A\|_2$ and $(cA)^{-1} = c^{-1}A^{-1}$.
(iv) These matrices have the same singular values as A, except A^{-1} has singular values $\sigma_n^{-1} \ge \cdots \ge \sigma_1^{-1}$.
(v) The singular values of A^*A are $\sigma_1(A)^2, \ldots, \sigma_n(A)^2$.
(vi) This follows from Proposition 5.1.7.

Structural Commentary: Condition number compares largest and smallest singular stretches.
Key Insight: A well-conditioned matrix stretches every direction by nearly the same factor. □

Corollary 5.3.3. *Positive Definite Condition Number by Eigenvalues*
Let $A \succ 0$ have eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_n$. Then $\|A\|_2 = \lambda_n$; $\left\|A^{-1}\right\|_2 = \frac{1}{\lambda_1}$; and $\kappa_2(A) = \frac{\lambda_n}{\lambda_1}$.

Proof. By Theorems 2.1.7 and 5.1.12, the singular values of A are $\lambda_1, \ldots, \lambda_n$.
Hence Theorem 5.1.2 gives $\|A\|_2 = \lambda_n$. Since $\operatorname{spec}(A^{-1}) = \{\lambda_1^{-1}, \ldots, \lambda_n^{-1}\}$, same argument gives $\left\|A^{-1}\right\|_2 = \lambda_1^{-1}$, and multiplying yields $\kappa_2(A) = \lambda_n/\lambda_1$.
Structural Commentary: For $A \succ 0$, singular and eigenvalue stretches coincide.
Key Insight: Conditioning compares the largest and smallest positive eigendirections. □

Perspective 5.3.4. *Determinant is Not Conditioning*
Determinants measure volume, not sensitivity.
$\kappa_2(cA) = \kappa_2(A)$ for $c \neq 0$ (Proposition 5.3.2), but $\det(cA) = c^n \det(A)$.

Theorem 5.3.5. *Condition Number and Linear-System Error*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible, let $\mathbf{y} \neq \mathbf{0}$, and let $A\mathbf{x} = \mathbf{y}$.
If $A(\Delta\mathbf{x}) = \Delta\mathbf{y}$, then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta\mathbf{y}\|_2}{\|\mathbf{y}\|_2}$

Proof. From $\mathbf{y} = A\mathbf{x}$, we have $\|\mathbf{y}\|_2 \le \|A\|_2 \|\mathbf{x}\|_2$, hence $1/\|\mathbf{x}\|_2 \le \|A\|_2 / \|\mathbf{y}\|_2$. Also $\Delta\mathbf{x} = A^{-1}\Delta\mathbf{y}$, so $\|\Delta\mathbf{x}\|_2 \le \left\|A^{-1}\right\|_2 \|\Delta\mathbf{y}\|_2$.
Multiplying the two estimates gives the displayed bound.
Structural Commentary: Solution error is controlled by the ratio of extreme singular stretches.
Key Insight: $\kappa_2(A)$ measures worst-case error magnification. □

Theorem 5.3.6. *Condition Number and Matrix Perturbation*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible. Suppose $A\mathbf{x} = \mathbf{y}$; $(A + \Delta A)(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{y}$, and $\mathbf{x} + \Delta\mathbf{x} \neq \mathbf{0}$.
Then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}+\Delta\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2}$.

Proof. Subtracting gives $A\Delta\mathbf{x} + \Delta A(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{0}$, so $\Delta\mathbf{x} = -A^{-1}\Delta A(\mathbf{x} + \Delta\mathbf{x})$.
Thus $\|\Delta\mathbf{x}\|_2 \le \left\|A^{-1}\right\|_2 \|\Delta A\|_2 \|\mathbf{x} + \Delta\mathbf{x}\|_2 = \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2} \|\mathbf{x} + \Delta\mathbf{x}\|_2$. Divide by $\|\mathbf{x} + \Delta\mathbf{x}\|_2 \neq 0$.
Structural Commentary: Perturbing the matrix changes the operator applied to the perturbed solution.
Key Insight: The same condition number controls matrix errors and right-hand-side errors. □

Figure 94 gives the geometry behind both error estimates: A sends the unit ball to an ellipse with longest and shortest semiaxes $\sigma_{\max}(A)$ and $\sigma_{\min}(A)$.

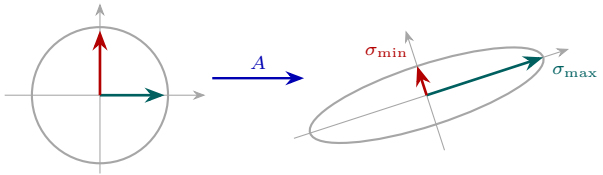

Figure 94: A maps a unit error ball to an ellipse; $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$ is the long/short semiaxis ratio.

5.4 Perturbation Bounds

Lemma 5.4.1. *Small Spectral-Norm Perturbations of the Identity*
If $E \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\|E\|_2 < 1$, then $I_n + E$ is invertible.

Proof. Theorem 5.1.12 gives $\rho(E) \le \|E\|_2 < 1$, so $-1 \notin \operatorname{spec}(E)$.
Hence $0 \notin \operatorname{spec}(I_n + E)$, and $I_n + E$ is invertible.
Structural Commentary: Spectral-norm smallness traps every eigenvalue inside the unit disk.
Key Insight: If -1 is not an eigenvalue of E, then $I + E$ has no kernel. □

Theorem 5.4.2. *Bauer–Fike Eigenvalue Perturbation Bound*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ be diagonalizable, with $A = S\Lambda S^{-1}$ and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.
If $\Delta A \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\mu \in \operatorname{spec}(A + \Delta A)$, then $\min_{1 \le j \le n} |\mu - \lambda_j| \le \kappa_2(S) \|\Delta A\|_2$.

Proof. If $\mu = \lambda_j$ for some j, the claim is immediate. Otherwise $\Lambda - \mu I_n$ is invertible.
Since μ is an eigenvalue of $A + \Delta A$, the matrix $A + \Delta A - \mu I_n$ is singular.
Conjugating gives $S^{-1}(A + \Delta A - \mu I_n)S = (\Lambda - \mu I_n)\left(I_n + (\Lambda - \mu I_n)^{-1} S^{-1}\Delta A S\right)$ so the second factor is singular.
Lemma 5.4.1 implies $1 \le \left\|(\Lambda - \mu I_n)^{-1} S^{-1}\Delta A S\right\|_2 \le \frac{\|S^{-1}\|_2 \|\Delta A\|_2 \|S\|_2}{\min_j |\mu - \lambda_j|}$. Rearranging gives the bound.
Structural Commentary: Diagonalize, then test how perturbation breaks shifted invertibility.
Key Insight: Eigenvalue sensitivity is controlled by how ill-conditioned the eigenvector matrix S is. □

Method 5.4.3. *Bauer–Fike Perturbation Method*
To bound eigenvalue movement for a diagonalizable matrix:

(i) diagonalize $A = S\Lambda S^{-1}$;
(ii) estimate $\|\Delta A\|_2$;
(iii) compute or bound $\kappa_2(S)$;
(iv) place every eigenvalue of $A + \Delta A$ inside the union of disks $|z - \lambda_j| \le \kappa_2(S)\,\|\Delta A\|_2$.

Definition 5.4.4. *Gersgorin Disks*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. The *deleted absolute row sum* in row k is $R_k^{\mathrm{off}}(A) := \sum_{j \neq k} |a_{kj}|$.

The *kth Gersgorin disk* of A is $\mathrm{Ger}_k(A) := \{z \in \mathbb{C} : |z - a_{kk}| \le R_k^{\mathrm{off}}(A)\}$, where $\mathrm{Ger}(A) := \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.

Theorem 5.4.5. *Gersgorin Disk Theorem*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) = \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.

Proof. Let $\lambda \in \mathrm{spec}(A)$ and choose an eigenvector $\mathbf{x} = [x_1 \ \cdots \ x_n]^t$.
Pick k such that $|x_k| = \max_j |x_j|$; then $x_k \neq 0$. The kth row of $A\mathbf{x} = \lambda\mathbf{x}$ gives $(\lambda - a_{kk})x_k = \sum_{j \neq k} a_{kj} x_j$.

Divide by x_k and use $|x_j/x_k| \le 1$: $|\lambda - a_{kk}| = \left|\sum_{j \neq k} a_{kj} \frac{x_j}{x_k}\right| \le \sum_{j \neq k} |a_{kj}| \left|\frac{x_j}{x_k}\right| \le R_k^{\mathrm{off}}(A)$.

Thus $\lambda \in \mathrm{Ger}_k(A) \subseteq \mathrm{Ger}(A)$.
Structural Commentary: Normalize by a largest component; all row ratios have modulus at most one.
Key Insight: Row k traps λ in the disk centered at a_{kk} with radius $R_k^{\mathrm{off}}(A)$. □

Corollary 5.4.6. *Row-Column Gersgorin Refinement*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$.
If $R_k(A) = \sum_{j=1}^{n} |a_{kj}|$ *and* $R_{\max}(A) = \max_{1 \le k \le n} R_k(A)$, *then* $\mathrm{spec}(A) \subseteq \{z \in \mathbb{C} : |z| \le \min(R_{\max}(A), R_{\max}(A^t))\}$.

Proof. Theorem 5.4.5 gives $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A)$.
Proposition 1.2.6 gives $\mathrm{spec}(A) = \mathrm{spec}(A^t)$, so applying Gersgorin to A^t gives $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A^t)$.
Also $\mathrm{Ger}_k(A) \subseteq \{z : |z| \le |a_{kk}| + R_k^{\mathrm{off}}(A) = R_k(A)\}$, and similarly for A^t.
Structural Commentary: Row disks and column disks are both valid because A and A^t have the same spectrum.
Key Insight: Intersecting independent enclosures usually sharpens the eigenvalue search region. □

Method 5.4.7. *Gersgorin Localisation Method*
For $A = [a_{ij}]$, Gersgorin localisation gives a disk enclosure for the spectrum.

(i) For each row k, place a disk at the diagonal entry a_{kk} with radius $R_k^{\mathrm{off}}(A) = \sum_{j \neq k} |a_{kj}|$.
The row enclosure is $\mathrm{Ger}(A) = \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.
(ii) Theorem 5.4.5 guarantees that every eigenvalue of A lies in $\mathrm{Ger}(A)$.
(iii) For a sharper enclosure, repeat the same construction for A^t. The centers are still a_{kk}, while the radii become deleted column sums of A. Then use $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$. This is the intersection of two unions of disks; an eigenvalue need not lie in the overlap of each individual row-column disk pair.

Figure 95 overlays the solid row disks with the dashed column disks; their intersection is the refined enclosure from Corollary 5.4.6, and the marked points illustrate possible eigenvalue locations.

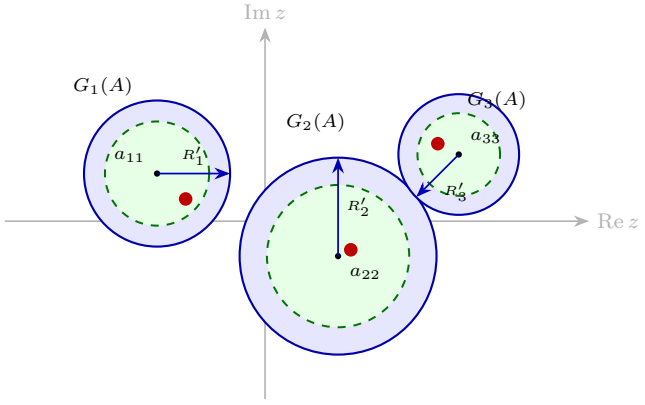

Figure 95: Blue row disks contain the spectrum; green dashed column disks give the sharper row-column enclosure $\mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$. Red points illustrate possible eigenvalue locations.

Perspective 5.4.8. *Pseudospectral Intuition*
For a matrix A, the ε-pseudospectrum records complex numbers z for which $A - zI$ is singular or nearly singular. Equivalently, it asks where a perturbation of size at most ε can move the spectrum. Bauer–Fike gives a clean enclosure for diagonalizable matrices, while Gersgorin and norm bounds give cruder but often cheaper enclosures. The main lesson is that eigenvalues of nonnormal matrices can be much more sensitive than their locations alone suggest.

5.5 Diagonal Dominance and Definiteness

Definition 5.5.1. *Diagonal Dominance*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. Matrix A is *diagonally dominant* if $|a_{kk}| \geq R_k^{\mathrm{off}}(A)$ for every k, and *strictly diagonally dominant* if $|a_{kk}| > R_k^{\mathrm{off}}(A)$ for every k.

Corollary 5.5.2. *Strict Diagonal Dominance Implies Invertibility*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$.

(i) If A is strictly diagonally dominant, then A is invertible.
(ii) If A has real spectrum and nonnegative diagonal entries, then diagonal dominance gives $\mathrm{spec}(A) \subseteq [0, \infty)$; strict diagonal dominance gives $\mathrm{spec}(A) \subseteq (0, \infty)$.

Proof.
(i) Strict dominance gives $0 \notin \mathrm{Ger}_k(A)$ for every k, since $|a_{kk}| > R_k^{\mathrm{off}}(A)$. Thus $0 \notin \mathrm{spec}(A)$ by Gersgorin, so A is invertible by Corollary 5.1.8.
(ii) If $a_{kk} \geq 0$ and $a_{kk} \geq R_k^{\mathrm{off}}(A)$, then each $\mathrm{Ger}_k(A)$ lies in the closed right half-plane. In the strict case, each disk lies in the open right half-plane. Gersgorin and the real-spectrum hypothesis give the claim.

Structural Commentary: Gersgorin converts diagonal dominance into disk placement.
Key Insight: The disks control whether eigenvalues can reach 0 or the left half-plane. □

Theorem 5.5.3. *Weak Diagonal Dominance Criterion*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonally dominant with no zero entries.
If $|a_{kk}| > R_k^{\mathrm{off}}(A)$ for some k, then A is invertible.

Proof. Assume $A\mathbf{x} = 0$ with $\mathbf{x} \neq \mathbf{0}$, and choose p with $|x_p| = \|\mathbf{x}\|_\infty$.

Row p gives $|a_{pp}|\,\|\mathbf{x}\|_\infty = \left|-\sum_{j \neq p} a_{pj} x_j\right| \leq \sum_{j \neq p} |a_{pj}|\,|x_j| \leq R_p^{\mathrm{off}}(A)\,\|\mathbf{x}\|_\infty \leq |a_{pp}|\,\|\mathbf{x}\|_\infty$.

Hence all inequalities are equalities. Since no a_{pj} vanishes, $|x_j| = \|\mathbf{x}\|_\infty$ for every j. Applying the same chain to any row i gives $|a_{ii}| = R_i^{\mathrm{off}}(A)$ for every i, contradicting the strict row. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.
Structural Commentary: A zero vector in the kernel would force equality in every Gersgorin row.
Key Insight: One strict row breaks that equality pattern. □

Corollary 5.5.4. *Diagonal Dominance and Positive Semidefiniteness*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian with nonnegative diagonal entries.

(i) If A is diagonally dominant, then A is positive semidefinite.
(ii) If A is diagonally dominant and invertible, then A is positive definite.
(iii) If A is strictly diagonally dominant, then A is positive definite.

Proof. Hermitian matrices have real spectrum.
(i) Corollary 5.5.2 gives $\mathrm{spec}(A) \subseteq [0, \infty)$; hence $A \succeq 0$ by Theorem 2.1.3.
(ii) If A is invertible, then $0 \notin \mathrm{spec}(A)$, so every eigenvalue is positive; hence $A \succ 0$.
(iii) Strict diagonal dominance implies invertibility by Corollary 5.5.2; apply (ii).

Structural Commentary: Gersgorin traps the real spectrum in $[0, \infty)$; invertibility removes 0.
Key Insight: For Hermitian matrices, positivity is spectral location. □

5.6 Rayleigh Quotients and Minimax

Definition 5.6.1. *Increasing Hermitian Eigenvalue Order*
For a Hermitian matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$, write $\lambda_1(A) \leq \lambda_2(A) \leq \cdots \leq \lambda_n(A)$ for its eigenvalues counted with algebraic multiplicity. If only one Hermitian matrix is under discussion, we write λ_i instead of $\lambda_i(A)$.

Theorem 5.6.2. *Rayleigh Quotient on a Spectral Interval*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian with increasing eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$ and corresponding orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$. For a unit vector, write $\mathrm{Ray}_A(\mathbf{x}) := \mathbf{x}^* A \mathbf{x}$. If $1 \leq p \leq q \leq n$ and $\mathbf{x} \in \mathrm{span}(\mathbf{u}_p, \ldots, \mathbf{u}_q)$ is a unit vector, then $\lambda_p \leq \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_q$.

Proof. Write $\mathbf{x} = \sum_{i=p}^{q} c_i \mathbf{u}_i$. Orthonormality gives $\sum_{i=p}^{q} |c_i|^2 = 1$, and $\mathbf{x}^* A \mathbf{x} = \sum_{i=p}^{q} \lambda_i\,|c_i|^2$.

This is a convex combination of $\lambda_p, \ldots, \lambda_q$, so it lies between the smallest and largest of them.
Structural Commentary: In an eigenbasis, the Rayleigh quotient is an eigenvalue average.
Key Insight: Spectral coordinates turn a quadratic form into weighted scalar data. □

Corollary 5.6.3. *Rayleigh Extrema and Diagonal Entry Bounds*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian.

(i) If $\mathcal{U}_{p,q} = \mathrm{span}(\mathbf{u}_p, \ldots, \mathbf{u}_q)$, then $\min_{\substack{\mathbf{x} \in \mathcal{U}_{p,q} \\ \|\mathbf{x}\|_2 = 1}} \mathbf{x}^* A \mathbf{x} = \lambda_p(A)$, and $\max_{\substack{\mathbf{x} \in \mathcal{U}_{p,q} \\ \|\mathbf{x}\|_2 = 1}} \mathbf{x}^* A \mathbf{x} = \lambda_q(A)$.
(ii) For every unit vector $\mathbf{x}$, $\lambda_1(A) \leq \mathbf{x}^* A \mathbf{x} \leq \lambda_n(A)$.
(iii) If $A = [a_{ij}]$, then $\lambda_1(A) \leq a_{ii} \leq \lambda_n(A)$ for $i = 1, \ldots, n$.

Proof.
(i) Theorem 5.6.2 gives the inequalities on $\mathcal{U}_{p,q}$, with equality at $\mathbf{u}_p$ and $\mathbf{u}_q$.
(ii) This is (i) with $p = 1$ and $q = n$.
(iii) Apply (ii) to $\mathbf{x} = \mathbf{e}_i$, since $\mathbf{e}_i^* A \mathbf{e}_i = a_{ii}$.

Structural Commentary: Rayleigh extrema on extreme eigendirections; diagonal entries are coordinate quotients.
Key Insight: The whole numerical range of a Hermitian matrix sits inside its spectral interval. □

Figure 96 places the Rayleigh value inside the eigenvalue interval allowed by the spectral support of $\mathbf{x}$.

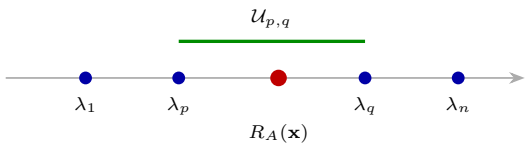

Figure 96: For unit $\mathbf{x} \in \mathcal{U}_{p,q}$, $R_A(\mathbf{x}) = \mathrm{Ray}_A(\mathbf{x})$ is a convex combination of $\lambda_p, \dots, \lambda_q$.

Method 5.6.4. *Rayleigh Eigenbasis Method*
To bound a Hermitian quadratic form:

(i) diagonalise A with an orthonormal eigenbasis;
(ii) expand the unit vector in that basis;
(iii) rewrite $\mathbf{x}^*A\mathbf{x}$ as $\sum_i \lambda_i |c_i|^2$;
(iv) use the eigenvalue range allowed by the support of $\mathbf{x}$.

Theorem 5.6.5. *Rayleigh Minimax Principle*
Let $A = A^* \in \mathrm{Mat}(n,n,\mathbb{C})$ have eigenvalues $\lambda_1 \le \cdots \le \lambda_n$, and set $R_A(\mathbf{x}) = \mathbf{x}^*A\mathbf{x}/(\mathbf{x}^*\mathbf{x})$ for $\mathbf{x} \neq \mathbf{0}$.
For $1 \le k \le n$, $\lambda_k = \min\limits_{\substack{\mathcal{S}\subseteq\mathbb{C}^n \\ \dim\mathcal{S}=k}} \max\limits_{\mathbf{0}\neq\mathbf{x}\in\mathcal{S}} R_A(\mathbf{x}) = \max\limits_{\substack{\mathcal{S}\subseteq\mathbb{C}^n \\ \dim\mathcal{S}=n-k+1}} \min\limits_{\mathbf{0}\neq\mathbf{x}\in\mathcal{S}} R_A(\mathbf{x})$.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_n$ be an orthonormal eigenbasis in increasing order.
Set $E_k = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$; $F_k = \mathrm{span}(\mathbf{u}_k, \dots, \mathbf{u}_n)$.
By Theorem 5.6.2, $\max_{\mathbf{0}\neq\mathbf{x}\in E_k} R_A(\mathbf{x}) = \lambda_k$ and $\min_{\mathbf{0}\neq\mathbf{x}\in F_k} R_A(\mathbf{x}) = \lambda_k$.
If $\dim\mathcal{S} = k$, then $\mathcal{S} \cap F_k \neq \{\mathbf{0}\}$, so $\max_{\mathbf{0}\neq\mathbf{x}\in\mathcal{S}} R_A(\mathbf{x}) \ge \lambda_k$; equality is attained by E_k.
If $\dim\mathcal{T} = n-k+1$, then $\mathcal{T} \cap E_k \neq \{\mathbf{0}\}$, so $\min_{\mathbf{0}\neq\mathbf{x}\in\mathcal{T}} R_A(\mathbf{x}) \le \lambda_k$; equality is attained by F_k.
Structural Commentary: Dimension forces intersections with spectral flags.
Key Insight: Minimax is Rayleigh's quotient plus dimension counting. □

5.7 Eigenvalue and Singular-Value Interlacing

Theorem 5.7.1. *Rank-One Positive Perturbation Interlacing*
Let $A, E \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian, with $E \succeq 0$ and $\mathrm{rank}(E) \le 1$.
Then $\lambda_i(A) \le \lambda_i(A+E) \le \lambda_{i+1}(A)$ $(1 \le i < n)$, $\lambda_n(A) \le \lambda_n(A+E)$
Equivalently, $\lambda_i(A-E) \le \lambda_i(A) \le \lambda_{i+1}(A-E)$ $(1 \le i < n)$, $\lambda_n(A-E) \le \lambda_n(A)$.

Proof. Let $\mathbf{u}_j$ and $\mathbf{v}_j$ be orthonormal eigenvectors of A and $A+E$, respectively, in increasing eigenvalue order.
For the lower bound, set $\mathcal{U} = \mathrm{span}(\mathbf{u}_i, \dots, \mathbf{u}_n)$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_i)$. Since $\dim\mathcal{U} + \dim\mathcal{V} = n+1$, choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V}$. Then $\lambda_i(A) \le \mathbf{x}^*A\mathbf{x} \le \mathbf{x}^*(A+E)\mathbf{x} \le \lambda_i(A+E)$ by Theorem 5.6.2 and $E \succeq 0$.
For the upper bound, set $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$, $\mathcal{W} = \ker(E)$. Here $\dim(\mathcal{U} \cap \mathcal{V}) \ge 2$ and $\dim\mathcal{W} \ge n-1$, so choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$. Thus $\lambda_i(A+E) \le \mathbf{x}^*(A+E)\mathbf{x} = \mathbf{x}^*A\mathbf{x} \le \lambda_{i+1}(A)$.
The statements for $A-E$ follow by applying the proved result to $A-E$ and E.
Structural Commentary: Interlacing comes from test vectors in overlapping spectral subspaces.
Key Insight: Rank one leaves a large kernel, so the upper-bound vector can avoid the perturbation. □

Corollary 5.7.2. *Rank-Two Indefinite Interlacing*
Let $A, H \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian, $n \ge 2$. Suppose $\mathrm{rank}(H) \le 2$ and H has at most one positive and one negative eigenvalue. Then

$$\lambda_1(A+H) \le \lambda_2(A), \qquad \lambda_{i-1}(A) \le \lambda_i(A+H) \le \lambda_{i+1}(A) \quad (2 \le i < n), \qquad \lambda_{n-1}(A) \le \lambda_n(A+H).$$

Proof. Write $H = E_+ - E_-$ with $E_\pm \succeq 0$ and $\mathrm{rank}(E_\pm) \le 1$. Set $B = A - E_-$; then $A = B + E_-$ and $A + H = B + E_+$. For $2 \le i < n$, rank-one interlacing gives $\lambda_{i-1}(A) \le \lambda_i(B) \le \lambda_i(A+H) \le \lambda_{i+1}(B) \le \lambda_{i+1}(A)$.
The endpoints follow similarly: $\lambda_1(A+H) \le \lambda_2(B) \le \lambda_2(A)$, and $\lambda_{n-1}(A) \le \lambda_n(B) \le \lambda_n(A+H)$.
Structural Commentary: Such an H is a difference of two rank-one PSD perturbations.
Key Insight: Interlace once with E_-, then once with E_+. □

Theorem 5.7.3. *Cauchy Interlacing: One Principal Deletion*
Let $A = \begin{bmatrix} B & \mathbf{y} \\ \mathbf{y}^* & c \end{bmatrix} \in \mathrm{Mat}(n+1,n+1,\mathbb{C})$ be Hermitian, with $B \in \mathrm{Mat}(n,n,\mathbb{C})$.
Then, for $1 \le i \le n$, $\lambda_i(A) \le \lambda_i(B) \le \lambda_{i+1}(A)$.

Proof. Let $\mathbf{u}_j$ and $\mathbf{v}_j$ be orthonormal eigenvectors of A and B, respectively, in increasing eigenvalue order. Identify $\mathbb{C}^n$ with $\{(\mathbf{z},0)\} \subseteq \mathbb{C}^{n+1}$, so $(\mathbf{z},0)^*A(\mathbf{z},0) = \mathbf{z}^*B\mathbf{z}$.
For the lower bound, set $\mathcal{S} = \mathrm{span}((\mathbf{v}_1,0), \dots, (\mathbf{v}_i,0))$ and $\mathcal{T} = \mathrm{span}(\mathbf{u}_i, \dots, \mathbf{u}_{n+1})$. Their dimensions are i and $n+2-i$, so $\dim\mathcal{S} + \dim\mathcal{T} > n+1$ and $\mathcal{S} \cap \mathcal{T}$ contains a nonzero vector.
Choose a unit $\mathbf{x} = (\mathbf{z},0)$ in the intersection. The spectral expansion in the $\mathbf{u}_j$ gives $\mathbf{x}^*A\mathbf{x} \ge \lambda_i(A)$, while the expansion in the $\mathbf{v}_j$ gives $\mathbf{z}^*B\mathbf{z} \le \lambda_i(B)$. Hence $\lambda_i(A) \le \lambda_i(B)$.
For the upper bound, take $\mathcal{S}' = \mathrm{span}((\mathbf{v}_i,0), \dots, (\mathbf{v}_n,0))$ and $\mathcal{T}' = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$. Again the dimensions sum to $n+2$, so choose a unit $\mathbf{x} = (\mathbf{z},0) \in \mathcal{S}' \cap \mathcal{T}'$.
Now $\mathbf{z}^*B\mathbf{z} \ge \lambda_i(B)$ and $\mathbf{x}^*A\mathbf{x} \le \lambda_{i+1}(A)$, giving $\lambda_i(B) \le \lambda_{i+1}(A)$.
Structural Commentary: A principal submatrix is a coordinate slice of the larger quadratic form.
Key Insight: Deleting one coordinate costs one dimension, so eigenvalues shift by at most one index. □

Corollary 5.7.4. *Cauchy Interlacing for Principal Submatrices*
Let $B \in \mathrm{Mat}(n,n,\mathbb{C})$ be a principal submatrix of Hermitian $A \in \mathrm{Mat}(n+m,n+m,\mathbb{C})$.
Then, for $1 \le i \le n$, $\lambda_i(A) \le \lambda_i(B) \le \lambda_{i+m}(A)$.

Proof. After a permutation similarity, B is leading.
Delete the other m coordinates one at a time and apply Theorem 5.7.3 at each step.
Structural Commentary: Principal submatrices come from repeated coordinate deletion.
Key Insight: Each deletion shifts eigenvalue indices by at most one. □

Corollary 5.7.5. *Singular Values Interlace After Deleting a Row or Column*
Let B be obtained from $A \in \mathrm{Mat}(n, n, \mathbb{C})$ by deleting one row or column.
With singular values ordered decreasingly, $\sigma_i(A) \geq \sigma_i(B) \geq \sigma_{i+1}(A) \quad (1 \leq i < n)$.

Proof. For a row deletion, permute rows so that $PA = \begin{bmatrix} B \\ \mathbf{y}^* \end{bmatrix}$.
Then PA has the same singular values as A, and BB^* is a principal submatrix of $(PA)(PA)^*$.
Cauchy interlacing applied to these Gram matrices gives the result after taking square roots and reversing the increasing eigenvalue order. A column deletion follows by applying the row case to A^*.
Structural Commentary: Row deletion makes BB^* a principal submatrix of a permuted AA^*.
Key Insight: Singular-value interlacing is Cauchy interlacing applied to Gram matrices. □

Figure 97 compares the two index windows: a rank-one positive perturbation places $\lambda_i(A+E)$ between consecutive eigenvalues of A, while an $n \times n$ principal submatrix of an $(n+m) \times (n+m)$ matrix places $\lambda_i(B)$ between $\lambda_i(A)$ and $\lambda_{i+m}(A)$.

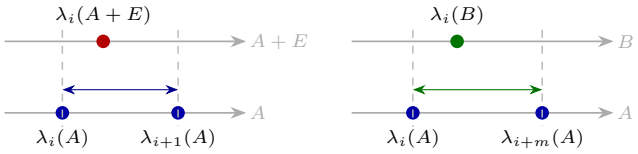

Figure 97: Rank-one and principal-submatrix interlacing place new eigenvalues between old ones.

Theorem 5.7.6. *Schur Majorization of the Diagonal*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian, with increasing eigenvalues $\lambda_i(A)$ and increasing diagonal entries d_i.
Then $\sum_{i=1}^{k} \lambda_i(A) \leq \sum_{i=1}^{k} d_i$, $(1 \leq k \leq n)$ with equality for $k = n$.

Proof. Permute coordinates so that $a_{ii} = d_i$, and let A_k be the leading $k \times k$ principal submatrix.
Cauchy interlacing gives $\lambda_i(A) \leq \lambda_i(A_k)$ for $1 \leq i \leq k$, hence $\sum_{i=1}^{k} \lambda_i(A) \leq \sum_{i=1}^{k} \lambda_i(A_k) = \mathrm{tr}(A_k) = \sum_{i=1}^{k} d_i$.
For $k = n$, this is $\mathrm{tr}(A) = \mathrm{tr}(A)$.
Structural Commentary: Ordered diagonal sums are traces of leading principal blocks.
Key Insight: Diagonal majorization is Cauchy interlacing plus trace. □

Method 5.7.7. *Eigenvalue Interlacing Method*
To prove or use an interlacing estimate:

(i) order Hermitian eigenvalues increasingly;
(ii) identify dimension lost: rank-one perturbation, one deleted coordinate, m-coordinate principal submatrix;
(iii) choose test vectors by intersecting spectral subspaces with the unaffected subspace;
(iv) convert the Rayleigh quotient bounds into eigenvalue inequalities.

5.8 Weyl Eigenvalue Bounds

Theorem 5.8.1. *Weyl Eigenvalue Inequalities*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian. For $1 \leq i \leq n$,

$$\lambda_i(A+B) \leq \lambda_{i+j}(A) + \lambda_{n-j}(B), \qquad 0 \leq j \leq n-i,$$
$$\lambda_{i-j+1}(A) + \lambda_j(B) \leq \lambda_i(A+B), \qquad 1 \leq j \leq i.$$

Proof. Let $\mathbf{u}_k, \mathbf{v}_k, \mathbf{w}_k$ be increasing eigenbases for $A, A+B, B$.
Fix i and $0 \leq j \leq n-i$, and set $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_{i+j})$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_i, \ldots, \mathbf{v}_n)$, $\mathcal{W} = \mathrm{span}(\mathbf{w}_1, \ldots, \mathbf{w}_{n-j})$.
Their dimensions sum to $2n+1$, so choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$.
Then, by Theorem 5.6.2, $\lambda_i(A+B) \leq \mathbf{x}^*(A+B)\mathbf{x} = \mathbf{x}^*A\mathbf{x} + \mathbf{x}^*B\mathbf{x} \leq \lambda_{i+j}(A) + \lambda_{n-j}(B)$.
The lower bound follows by applying this upper bound to $-A$ and $-B$ and using $\lambda_k(-C) = -\lambda_{n-k+1}(C)$.
Structural Commentary: Weyl bounds come from intersecting the spectral flags of A, B, and $A+B$.
Key Insight: One shared test vector controls all three Rayleigh quotients. □

Method 5.8.2. *Weyl Bound Method*
To bound eigenvalues of $A+B$:

(i) choose the target index i;
(ii) use the upper Weyl family with $j = 0, \ldots, n-i$ to get upper bounds;
(iii) use the lower Weyl family with $j = 1, \ldots, i$ to get lower bounds;
(iv) for positive semidefinite or low-rank B, insert the known zero or nonnegative eigenvalues to recover monotonicity or interlacing.

5.9 Brauer Rank-One Spectral Shifts

Theorem 5.9.1. *Brauer Rank-One Eigenvalue Shift*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, and let $\mathbf{y} \in \mathbb{C}^n$. If $\lambda, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A with multiplicity, then the eigenvalues of $A + \mathbf{x}\mathbf{y}^*$ are $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \ldots, \lambda_n$.
Moreover, $\mathbf{x}$ is an eigenvector of $A + \mathbf{x}\mathbf{y}^*$ corresponding to $\lambda + \mathbf{y}^*\mathbf{x}$.

Proof. Put $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|_2$. By Schur triangularisation, choose a unitary $U = [\mathbf{u}\ U_2]$ such that $U^*AU = \begin{bmatrix} \lambda & * \\ 0 & T' \end{bmatrix}$, where the diagonal entries of T' are $\lambda_2, \ldots, \lambda_n$. Also $U^*(\mathbf{x}\mathbf{y}^*)U = \begin{bmatrix} \mathbf{y}^*\mathbf{x} & * \\ 0 & 0 \end{bmatrix}$. Therefore $U^*(A + \mathbf{x}\mathbf{y}^*)U$ is upper triangular with diagonal entries $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \ldots, \lambda_n$. Finally, $(A + \mathbf{x}\mathbf{y}^*)\mathbf{x} = A\mathbf{x} + \mathbf{x}(\mathbf{y}^*\mathbf{x}) = (\lambda + \mathbf{y}^*\mathbf{x})\mathbf{x}$. *Structural Commentary*: A rank-one eigenvector update changes only the first Schur diagonal entry.
Key Insight: Align the update with the first coordinate; only one diagonal term shifts. □

Figure 98 shows the Schur-coordinate mechanism: the aligned rank-one update changes the first diagonal entry from λ to $\lambda + \mathbf{y}^*\mathbf{x}$ while leaving the trailing block T' unchanged.

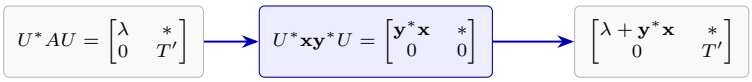

Figure 98: Brauer's update shifts one Schur diagonal entry.

Corollary 5.9.2. *Damped Rank-One Eigenvalue Reset*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have eigenpair $(\lambda, \mathbf{x})$, let $\mathbf{y}^*\mathbf{x} = 1$, and let $\tau \in \mathbb{C}$. If $\lambda, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A, then $A_\tau := \tau A + (1-\tau)\lambda\mathbf{x}\mathbf{y}^*$ has eigenvalues $\lambda, \tau\lambda_2, \ldots, \tau\lambda_n$.

Proof. The eigenvalues of τA are $\tau\lambda, \tau\lambda_2, \ldots, \tau\lambda_n$. Since $A_\tau = \tau A + \mathbf{x}((1-\overline{\tau})\overline{\lambda}\mathbf{y})^*$, Theorem 5.9.1 shifts only $\tau\lambda$ by $(1-\tau)\lambda\mathbf{y}^*\mathbf{x} = (1-\tau)\lambda$. Thus $\tau\lambda$ becomes λ, while the other eigenvalues stay $\tau\lambda_i$.
Structural Commentary: Damping scales the spectrum; Brauer's update restores one eigenvalue.
Key Insight: Brauer's theorem changes one spectral coordinate without moving the rest. □

6 Numerical Linear Algebra

Numerical linear algebra turns matrix structure into algorithms for direct solves, least squares, eigenvalues, SVD, structured transforms, Krylov methods, iteration, and preconditioning.

6.1 Direct Linear-System Solves

Perspective 6.1.1. *Solve with Factors Before Forming Inverses*
For $A\mathbf{x} = \mathbf{b}$, use available factors such as LU, QR, or Cholesky to solve triangular or orthogonal subproblems.
This avoids forming A^{-1} and preserves sparse or triangular structure.

Perspective 6.1.2. *Iterative Refinement by Residual Correction*
Given an approximate solution $\mathbf{x}_k$ of $A\mathbf{x} = \mathbf{b}$, set $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$.
Solve $A\mathbf{d}_k = \mathbf{r}_k$ with the same factors or preconditioner, then update $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{d}_k$.
Exact arithmetic corrects in one step; floating point helps when $\mathbf{r}_k$ is accurate and conditioning is moderate.

6.2 QR Least Squares

Proposition 6.2.1. *QR Route for Least Squares*
Let $A = QR \in \mathrm{Mat}(m,n,\mathbb{F})$ be a full-column-rank narrow QR factorisation. For $\mathbf{y} \in \mathbb{F}^m$, the least-squares solution solves $R\mathbf{x} = Q^*\mathbf{y}$. Moreover $P_{\mathrm{col}(A)} = QQ^*$.

Proof. The normal equations reduce by $A = QR$ and $Q^*Q = I_n$ to $R^*R\mathbf{x} = R^*Q^*\mathbf{y}$. Since R is invertible, this is equivalent to $R\mathbf{x} = Q^*\mathbf{y}$. Also $\mathrm{col}(A) = \mathrm{col}(Q)$, and the columns of Q are orthonormal, so $P_{\mathrm{col}(A)} = QQ^*$.
Structural Commentary: QR replaces $\mathrm{col}(A)$ by an orthonormal basis Q.
Key Insight: Project onto Q, then solve triangularly. □

Perspective 6.2.2. *Why Normal Equations Can Be Numerically Fragile*
For full-column-rank least squares, solving $A^*A\mathbf{x} = A^*\mathbf{y}$ squares the spectral condition number because $\kappa_2(A^*A) = \kappa_2(A)^2$ by Proposition 5.3.2. QR-based methods avoid this squaring and therefore preserve conditioning better.

Successive orthogonalization constructs an orthonormal basis for the same column space.

Algorithm 1: Least Squares by Successive Orthogonalization

Require: Full-column-rank design columns $x_0, \ldots, x_p$ and response vector y
Ensure: Least-squares coefficients $\widehat{\beta}$
for $j = 0, \ldots, p$ **do**
  $z_j \leftarrow x_j$
  for $\ell = 0, \ldots, j-1$ **do**
    $r_{\ell j} \leftarrow q_\ell^T x_j$
    $z_j \leftarrow z_j - r_{\ell j} q_\ell$
  end for
  $r_{jj} \leftarrow \|z_j\|_2$
  $q_j \leftarrow z_j / r_{jj}$
end for

Form $Q = [q_0, \dots, q_p]$ and the upper-triangular matrix $R = (r_{\ell j})$
Solve $R\widehat{\beta} = Q^T y$ by back-substitution

Perspective 6.2.3. *Solve Least Squares with Factors*
For $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, normal equations use $X^t X\widehat{\beta} = X^t Y$ but $\kappa_2(X^t X) = \kappa_2(X)^2$.
Prefer $X = QR$ with $Q^t Q = I_p$ and R nonsingular, then solve $R\widehat{\beta} = Q^t Y$ by back-substitution.
Use Cholesky on $X^t X$ only when $X^t X$ is well conditioned.

6.3 Banded and Sparse Direct Methods

Definition 6.3.1. *Band and Tridiagonal Matrices*
For $A = [a_{ij}] \in \operatorname{Mat}(n, n, \mathbb{F})$ and $1 \le w \le n$, say that A has *half-bandwidth* w if $a_{ij} = 0$ whenever $|i - j| \ge w$. Thus $w = 1$ gives a diagonal matrix, while $w = 2$ gives a *tridiagonal* matrix.

Method 6.3.2. *Banded Elimination Preview*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ have half-bandwidth w.

(i) Eliminate without row exchanges that leave the band.
(ii) At pivot k, update only entries with $|i - k| < w$ and $|j - k| < w$.
(iii) Obtain $A = LU$ with L and U of comparable bandwidth.

Cost: $O(nw^2)$ operations; for tridiagonal matrices ($w = 2$), $O(n)$.

Figure 99 shows the tridiagonal nonzero pattern splitting into a unit lower-bidiagonal factor and an upper-bidiagonal factor.

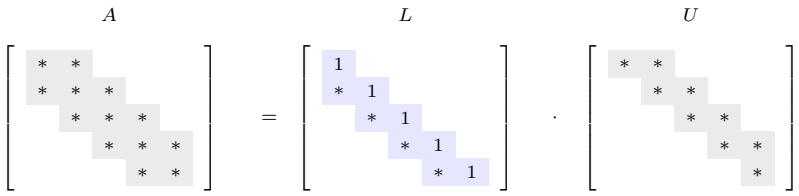

Figure 99: A tridiagonal matrix can factor as bidiagonal L times bidiagonal U, retaining local band structure.

Perspective 6.3.3. *Five-Point Stencils Are Banded but Less Local*
The 1D second-difference pattern $-1, 2, -1$ is tridiagonal; the row-ordered $N \times N$ five-point stencil is still sparse, but vertical neighbors create diagonals offset by about N. Thus bandwidth grows with grid width, and incomplete LU keeps a prescribed sparse pattern for a cheap approximate solve or preconditioner.

Figure 100 contrasts the main diagonal with the ± 1 offsets from horizontal grid neighbors and the $\pm N$ offsets from vertical grid neighbors.

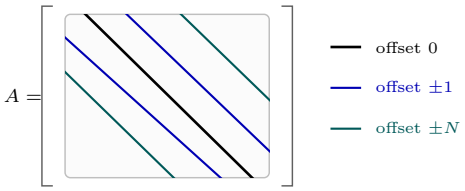

Figure 100: Row ordering gives matrix bands at offsets $0, \pm 1, \pm N$.

6.4 Krylov Subspace Methods

Definition 6.4.1. *Krylov Subspace*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$ and a vector $\mathbf{b} \in \mathbb{F}^n$, the mth *Krylov subspace* generated by $(A, \mathbf{b})$ is
$\mathcal{K}_m(A, \mathbf{b}) := \operatorname{span}\{\mathbf{b}, A\mathbf{b}, \dots, A^{m-1}\mathbf{b}\}$.

Perspective 6.4.2. *Krylov Methods Preview*
Krylov methods work in $\mathcal{K}_m(A, \mathbf{b})$: Arnoldi and Lanczos build bases, while CG and GMRES impose projection or residual-minimization conditions using only matrix-vector products.

Method 6.4.3. *Arnoldi, Lanczos, and CG Preview*
Let $\mathcal{K}_m = \mathcal{K}_m(A, \mathbf{b})$ and let $Q_m = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_m]$ be an orthonormal basis.

(i) *Arnoldi*: $AQ_m = Q_m H_m + h_{m+1,m}\mathbf{q}_{m+1}\mathbf{e}_m^t$, with H_m upper Hessenberg.
(ii) *Lanczos*: if $A = A^*$, then H_m is tridiagonal and $A\mathbf{q}_j = \beta_{j-1}\mathbf{q}_{j-1} + \alpha_j\mathbf{q}_j + \beta_j\mathbf{q}_{j+1}$.
(iii) *CG*: if $A \succ 0$, set $\mathbf{r}_0 = \mathbf{b} - A\mathbf{x}_0$ and choose $\mathbf{x}_k \in \mathbf{x}_0 + \mathcal{K}_k(A, \mathbf{r}_0)$ with $\mathbf{r}_k \perp \mathcal{K}_k(A, \mathbf{r}_0)$; search directions satisfy $\mathbf{p}_i^* A\mathbf{p}_j = 0$, $i \ne j$.

Each step uses one matrix-vector product.

6.5 Stationary Iterations and Splittings

Definition 6.5.1. *Matrix Splitting Iteration*
Let $A = S - T$ with S invertible, and set $B := S^{-1}T$.
The *splitting iteration* for $A\mathbf{x} = \mathbf{b}$ is $S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$, and $\mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$.
For a solution $\mathbf{x}_*$, the error $\mathbf{e}_k := \mathbf{x}_k - \mathbf{x}_*$ satisfies $\mathbf{e}_{k+1} = B\mathbf{e}_k$.

Theorem 6.5.2. *Splitting Convergence Criterion*
Let $A = S - T$, with S invertible, and set $B = S^{-1}T$. For $\mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$:
(i) if A is invertible, then $\mathbf{x}_k \to A^{-1}\mathbf{b}$ for all $\mathbf{x}_0, \mathbf{b}$ if and only if $\rho(B) < 1$;
(ii) $\rho(B) < 1$ implies A is invertible.

Proof.
(i) Let $\mathbf{x}_* = A^{-1}\mathbf{b}$ and $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_*$. Then $\mathbf{e}_k = B^k\mathbf{e}_0$, so convergence for all $\mathbf{x}_0, \mathbf{b}$ is equivalent to $B^k \to 0$, equivalently $\rho(B) < 1$ by Theorem 3.3.5.
(ii) If $\rho(B) < 1$, then $1 \notin \operatorname{spec}(B)$, so $I_n - B$ is invertible. Since $A = S - T = S(I_n - B)$, A is invertible.
Structural Commentary: A stationary method is a matrix-power problem in disguise.
Key Insight: The iteration matrix $B = S^{-1}T$ controls the errors. □

Corollary 6.5.3. *Singular Matrices Obstruct Splitting Convergence*
Let $A = S - T$, with S invertible. If A is singular, then $1 \in \operatorname{spec}(S^{-1}T)$.
Hence $\rho(S^{-1}T) \geq 1$, so $\rho(S^{-1}T) < 1$ cannot hold.

Proof. Choose $\mathbf{0} \neq \mathbf{x} \in \ker A$. Then $(S - T)\mathbf{x} = \mathbf{0}$, so $S\mathbf{x} = T\mathbf{x}$ and $S^{-1}T\mathbf{x} = \mathbf{x}$. Thus 1 is eigenvalue of $S^{-1}T$.
Structural Commentary: Singular A forces an undamped error mode.
Key Insight: A null vector of A becomes a fixed error mode. □

Method 6.5.4. *Stationary Iterative Method*
For $A\mathbf{x} = \mathbf{b}$, a stationary iteration is:
(i) Choose fixed M and $\mathbf{c}$.
(ii) Update $\mathbf{x}_{k+1} = M\mathbf{x}_k + \mathbf{c}$.
(iii) If $\mathbf{x}_*$ exists, use $\mathbf{x}_{k+1} - \mathbf{x}_* = M(\mathbf{x}_k - \mathbf{x}_*)$ and Theorem 3.3.5 for convergence.

Method 6.5.5. *Jacobi, Gauss–Seidel, and SOR Splittings*
Let $A = D + L + U$, with D diagonal, L strictly lower triangular, and U strictly upper triangular. For a splitting $A = S - T$, use $S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$ and $B = S^{-1}T$.
(i) *Jacobi*: $S = D$, $T = -(L + U)$, $B_J = -D^{-1}(L + U)$.
(ii) *Gauss–Seidel*: $S = D + L$, $T = -U$, $B_{\mathrm{GS}} = -(D + L)^{-1}U$.
(iii) *SOR*: for $\omega \neq 0$, $S_\omega = \frac{1}{\omega}D + L$; $T_\omega = \left(\frac{1}{\omega} - 1\right)D - U$; and $B_\omega = S_\omega^{-1}T_\omega$.
Convergence is controlled by $\rho(B)$; $\omega = 1$ gives Gauss–Seidel.

Corollary 6.5.6. *Strict Diagonal Dominance Gives Jacobi Convergence*
If $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ is strictly row diagonally dominant, then Jacobi converges for every right-hand side.

Proof. Here $a_{ii} \neq 0$ and $B_J = -D^{-1}(L + U) = I_n - D^{-1}A$. Thus $(B_J)_{ii} = 0$ and $\sum_{j \neq i} |(B_J)_{ij}| = \sum_{j \neq i} |a_{ij}| / |a_{ii}| < 1$.
Gersgorin gives $\operatorname{spec}(B_J) \subset \{z : |z| < 1\}$, so $\rho(B_J) < 1$; Theorem 6.5.2 applies.
Structural Commentary: Diagonal dominance puts the Jacobi spectrum in the unit disk.
Key Insight: Small off-diagonal row sums force the error matrix to have spectral radius below 1. □

Example 6.5.7. *Two-by-Two Jacobi and Gauss–Seidel*
For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} \in \mathbb{R}^2$, write $A = D + L + U$.
The Jacobi iteration matrix is $B_J = -D^{-1}(L + U) = \begin{bmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$, so $\rho(B_J) = \frac{1}{2}$.
The Gauss–Seidel iteration matrix is $B_{\mathrm{GS}} = -(D + L)^{-1}U = \begin{bmatrix} 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} \end{bmatrix}$, so $\rho(B_{\mathrm{GS}}) = \frac{1}{4}$.
Both converge by Theorem 6.5.2, and smaller spectral radius predicts faster asymptotic decay for Gauss–Seidel.

Example 6.5.8. *Poisson Tridiagonal Stationary Rates*
For $A_n = \operatorname{tridiag}(-1, 2, -1)$, Jacobi gives $B_J = I_n - \frac{1}{2}A_n$; $\lambda_k(B_J) = \cos\frac{k\pi}{n+1}$; and $\rho(B_J) = \cos\frac{\pi}{n+1}$.
For the same ordering, Gauss–Seidel satisfies $\rho(B_{\mathrm{GS}}) = \rho(B_J)^2 = \cos^2\frac{\pi}{n+1}$. As $n \to \infty$, both radii tend to 1; fine grids need acceleration or preconditioning.

Perspective 6.5.9. *Preconditioning Preview*
An invertible preconditioner M replaces $A\mathbf{x} = \mathbf{b}$ by an equivalent system such as $M^{-1}A\mathbf{x} = M^{-1}\mathbf{b}$ or $AM^{-1}\mathbf{y} = \mathbf{b}$, with $\mathbf{x} = M^{-1}\mathbf{y}$. It should improve conditioning or convergence while being much cheaper to apply than A^{-1}.

6.6 Hessenberg Reduction and QR Eigenvalue Methods

Definition 6.6.1. *Upper Hessenberg Matrix*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is *upper Hessenberg* if $a_{ij} = 0$ whenever $i > j + 1$.
Equivalently, all entries below the first subdiagonal are zero.

Numerical linear algebra also replaces a matrix by a unitarily similar one with many forced zero entries. Upper Hessenberg form is the standard preliminary reduction before QR-type eigenvalue algorithms.

Figure 101 separates the unrestricted diagonal-and-upper region, the retained first subdiagonal, and the entries below it that are forced to zero.

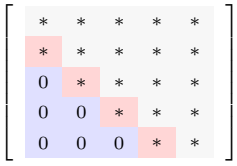

Figure 101: Upper Hessenberg pattern: red is the first subdiagonal; blue entries below it are forced zero.

Theorem 6.6.2. *Unitary Upper Hessenberg Reduction*
Every $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is unitarily similar to upper Hessenberg matrix with real nonnegative subdiagonal entries.

Proof. Set $A_0 = A$. For $k = 1, \ldots, n-2$, suppose columns $1, \ldots, k-1$ of A_{k-1} already have zeros below the first subdiagonal. Let $\mathbf{w}$ be the tail of column k in rows $k+1, \ldots, n$.
By Corollary 2.7.4, choose a unitary V_k with $V_k^* \mathbf{w} = \alpha \mathbf{e}_1$, $\alpha \geq 0$.
Put $U_k = I_k \oplus V_k$ and $A_k = U_k^* A_{k-1} U_k$. Right multiplication by U_k changes only columns $k+1, \ldots, n$, so columns $1, \ldots, k-1$ keep their existing Hessenberg zeros. Left multiplication by U_k^* acts only on rows $k+1, \ldots, n$; those rows are already zero in columns $1, \ldots, k-1$, so the zeros remain. In column k, the tail becomes $V_k^* \mathbf{w} = \alpha \mathbf{e}_1$, so entries below row $k+1$ are zero and the new subdiagonal entry is real nonnegative.
After $n-2$ steps, $A_{n-2} = U^* A U$ is upper Hessenberg, where $U = U_1 \cdots U_{n-2}$ is unitary.
Structural Commentary: Each trailing unitary creates one Hessenberg column and preserves earlier ones.
Key Insight: Kill one column tail at a time. □

Perspective 6.6.3. *Why Hessenberg Form Matters*
Hessenberg form preserves spectral data by unitary similarity, making QR-type eigenvalue algorithms cheaper.

Method 6.6.4. *QR Eigenvalue Algorithm Preview*
For a square matrix A_0, the basic QR eigenvalue iteration is:

(i) In practice, first reduce A_0 to Hessenberg form by Theorem 6.6.2.
(ii) Factor $A_k = Q_k R_k$, with Q_k unitary and R_k upper triangular.
(iii) Set $A_{k+1} = R_k Q_k = Q_k^* A_k Q_k$, using shifts to speed convergence.

Each step is a unitary similarity, so the spectrum is preserved.

Method 6.6.5. *Hessenberg QR Step by Plane Rotations*
Let $H \in \mathrm{Mat}(n, n, \mathbb{C})$ be upper Hessenberg and let $\mu \in \mathbb{C}$.

(i) Set $B_0 = H - \mu I_n$.
(ii) For $j = 1, \ldots, n-1$, choose a unitary plane rotation G_j acting on coordinates $j, j+1$, so that $(G_j^* B_{j-1})_{j+1,j} = 0$ and $B_j = G_j^* B_{j-1}$.
(iii) Put $Q = G_1 \cdots G_{n-1}$ and $R = B_{n-1}$; then $H - \mu I_n = QR$.
(iv) Define the shifted QR update $H_+ = RQ + \mu I_n = Q^* H Q$.

Adjacent rotations implement the Hessenberg step by local bulge chasing.

6.7 Power and Shift-Invert Iterations

Theorem 6.7.1. *Power Method Dominant-Eigenvector Preview*
Let $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$, where $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is a basis of $\mathbb{C}^n$, and assume $|\lambda_1| > \max_{j \geq 2} |\lambda_j|$.
If $\mathbf{u}_0 = \sum_{j=1}^{n} c_j \mathbf{v}_j$ with $c_1 \neq 0$, then $\lambda_1^{-k} A^k \mathbf{u}_0 \longrightarrow c_1 \mathbf{v}_1$. Thus the directions of $A^k \mathbf{u}_0$ converge to $\mathrm{span}\{\mathbf{v}_1\}$.

Proof. Set $\rho = \max_{j \geq 2} |\lambda_j / \lambda_1| < 1$. Since $A^k \mathbf{v}_j = \lambda_j^k \mathbf{v}_j$, $\lambda_1^{-k} A^k \mathbf{u}_0 = c_1 \mathbf{v}_1 + \sum_{j=2}^{n} c_j \left(\frac{\lambda_j}{\lambda_1}\right)^k \mathbf{v}_j$.

The remainder has norm at most $\rho^k \sum_{j=2}^{n} |c_j| \, \|\mathbf{v}_j\|$, hence tends to 0.

Structural Commentary: The power method is spectral filtering by repeated multiplication.
Key Insight: A spectral gap makes every non-dominant component decay relative to the dominant one. □

Method 6.7.2. *Inverse and Shifted Power Iteration Preview*
Let $\mathcal{N}(\mathbf{z}) := \mathbf{z} / \|\mathbf{z}\|_2$ for $\mathbf{z} \neq \mathbf{0}$.

(i) *Inverse*: if $0 \notin \mathrm{spec}(A)$, solve $A\mathbf{y}_{k+1} = \mathbf{x}_k$ and set $\mathbf{x}_{k+1} = \mathcal{N}(\mathbf{y}_{k+1})$. This is power iteration on A^{-1} and targets minimal $|\lambda|$.
(ii) *Shift-invert*: for $\sigma \notin \mathrm{spec}(A)$, $(A - \sigma I)\mathbf{y}_{k+1} = \mathbf{x}_k$, $\mathbf{x}_{k+1} = \mathcal{N}(\mathbf{y}_{k+1})$. Eigenvalues become $(\lambda - \sigma)^{-1}$, targeting λ nearest σ.
(iii) *Shifted power*: $\mathbf{x}_{k+1} = \mathcal{N}\big((A - \sigma I)\mathbf{x}_k\big)$; eigenvalues become $\lambda - \sigma$, emphasizing large $|\lambda - \sigma|$.

Use factors of A or $A - \sigma I$ for solves; do not form explicit inverses.

6.8 Numerical SVD Methods

Method 6.8.1. *SVD Numerics Preview*
Stable numerical SVD computation proceeds as:

(i) Avoid forming $A^* A$ when possible, since it can square the condition number.
(ii) Use orthogonal or unitary transformations to reduce A to bidiagonal form.
(iii) Compute singular values from the bidiagonal problem, then use them for low-rank approximation, pseudoinverse truncation, and least-squares diagnostics through Theorem 5.2.2 and Definition 5.3.1.

6.9 Discrete Fourier Transforms and Circulant Matrices

Definition 6.9.1. *Discrete Fourier Matrix*
For $n \geq 1$, set $\omega_n = e^{2\pi i/n}$ and $F_n := [\omega_n^{jk}]_{j,k=0}^{n-1}$.
Then $F_n\mathbf{c}$ is the unnormalised DFT of $\mathbf{c} \in \mathbb{C}^n$; $Q_n := n^{-1/2}F_n$ is the normalized Fourier matrix.

Definition 6.9.2. *Circulant Matrix*
On the 0-indexed basis of $\mathbb{C}^n$, define the *cyclic shift* $P_n\mathbf{e}_j = \mathbf{e}_{(j-1) \bmod n}$.
A matrix $C \in \mathrm{Mat}(n, n, \mathbb{C})$ is *circulant* if, for some $a_0, \ldots, a_{n-1} \in \mathbb{C}$,

$$C = a_0 I_n + a_1 P_n + \cdots + a_{n-1} P_n^{n-1} = p(P_n),$$
$$p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}$$

Theorem 6.9.3. *Fourier Basis for Cyclic Shifts and Circulants*
Let $n \geq 1$, let $\omega_n := e^{2\pi i/n}$, and let P_n be the cyclic shift.
For $0 \leq k < n$, set $\mathbf{f}_k := (\omega_n^{jk})_{j=0}^{n-1}$; $F_n := [\mathbf{f}_0 \ \cdots \ \mathbf{f}_{n-1}]$; and $\Omega_n := \mathrm{diag}(1, \omega_n, \ldots, \omega_n^{n-1})$.
Then $F_n^* F_n = nI_n$, so $Q_n := n^{-1/2}F_n$ is unitary, and $P_n F_n = F_n \Omega_n$.
If $C = p(P_n)$ is circulant, then $Q_n^* C Q_n = p(\Omega_n) = \mathrm{diag}\big(p(1), p(\omega_n), \ldots, p(\omega_n^{n-1})\big)$.

Proof. For $0 \leq k, \ell \leq n-1$, $\langle \mathbf{f}_k, \mathbf{f}_\ell \rangle = \sum_{r=0}^{n-1} (\omega_n^{k-\ell})^r = n\delta_{k\ell}$. For $k = \ell$ the sum is n; for $k \neq \ell$, $W = \omega_n^{k-\ell}$ has $W \neq 1$ and $W^n = 1$, so $(1-W)\sum_{r=0}^{n-1} W^r = 0$. Hence $F_n^* F_n = nI_n$, and Theorem 2.6.8 makes Q_n unitary.
Also $P_n \mathbf{f}_k = \omega_n^k \mathbf{f}_k$, so $Q_n^* P_n Q_n = \Omega_n$; applying p gives the formula.
Structural Commentary: Fourier coordinates diagonalize every circulant.
Key Insight: Orthogonality of roots of unity turns a cyclic matrix problem into scalar evaluation. □

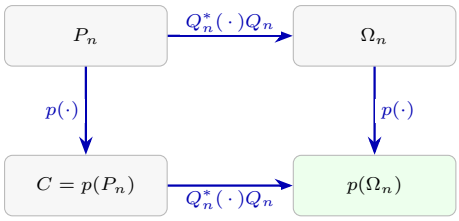

Figure 102: Fourier diagonalizes P_n; polynomial evaluation diagonalizes $C = p(P_n)$.

Proposition 6.9.4. *Inverse of the Fourier Matrix*
The Fourier matrix satisfies $F_n^* F_n = nI_n$, and therefore $F_n^{-1} = \frac{1}{n}F_n^*$.

Proof. Theorem 6.9.3 gives $Q_n^* Q_n = I_n$ for $Q_n = n^{-1/2}F_n$. Multiplying by n gives $F_n^* F_n = nI_n$, so $n^{-1}F_n^*$ is a left inverse of F_n. Since F_n is square, Theorem 3.4.2 makes it the inverse.
Structural Commentary: Fourier inversion is conjugate transpose plus one scalar normalization.
Key Insight: The inverse is simple because the Fourier columns are orthogonal. □

6.10 Fast Fourier Transform

Method 6.10.1. *Radix-Two FFT Split*
Let $n = 2m$ and $\mathbf{y} = F_n\mathbf{c}$.

(i) Split $\mathbf{c}' := (c_{2r})_{r=0}^{m-1}$ and $\mathbf{c}'' := (c_{2r+1})_{r=0}^{m-1}$.
(ii) Compute $\mathbf{y}' := F_m\mathbf{c}'$ and $\mathbf{y}'' := F_m\mathbf{c}''$.
(iii) With $d_j := \omega_n^j$, and using $\omega_n^2 = \omega_m$ and $\omega_n^m = -1$, set $y_j = y'_j + d_j y''_j$; and $y_{j+m} = y'_j - d_j y''_j$ for $0 \leq j < m$.
(iv) For $n = 2^\ell$, repeat (i)–(iii) for ℓ stages.
n^2 dense products become about $\frac{1}{2}n\log_2 n$ twiddle multiplications plus additions.

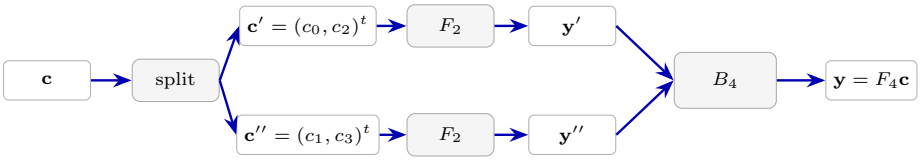

Figure 103: For $n = 4$: split, apply two F_2 transforms, then $B_4\binom{\mathbf{y}'}{\mathbf{y}''} = \binom{\mathbf{y}'+D\mathbf{y}''}{\mathbf{y}'-D\mathbf{y}''}$ with $D = \mathrm{diag}(1, i)$.

Part VII
Groups, Rings, and Fields

1 Algebraic Language and Categories

1.1 Universal Properties and Equivalence-Based Structure

The elementary set, order, choice, function, and bijection vocabulary used here is collected in the foundational prerequisites. This subsection continues with algebraic applications of universal properties and equivalence-based structure.

1.2 Categories and Functors

1.3 Algebraic Structures as Objects

1.4 Exactness and Diagrams

2 Group Theory

2.1 Groups, Subgroups, and Cyclic Groups

Groups are the algebraic language of reversible operations. They appear here as the additive structure underlying rings and modules, and later as symmetry groups acting on mathematical objects.

Definition 2.1.1. *Groups*
A *group* is a nonempty set G with a binary operation $(a, b) \mapsto ab$ such that:

(i) $(ab)c = a(bc)$ for all $a, b, c \in G$;
(ii) there is an element $e \in G$ with $eg = ge = g$ for all $g \in G$;
(iii) for each $g \in G$, there is $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

The group is *abelian* if $ab = ba$ for all $a, b \in G$.
In an abelian group the operation is often written additively, with identity 0 and inverse $-a$.

Example 2.1.2. *Basic Groups*

(i) The bijections $S \to S$ form a group under composition; it is usually nonabelian.
(ii) $\mathrm{Mat}(m, n, \mathbb{F})$ is an abelian group under matrix addition.
(iii) $\mathrm{GL}_n(\mathbb{F})$, the invertible $n \times n$ matrices, is a group under multiplication.
(iv) $\mathbb{Z}/n\mathbb{Z}$ is a finite abelian group under addition modulo n.

Definition 2.1.3. *Subgroup*
A *subgroup* of a group G is a nonempty subset $H \subseteq G$ that is itself a group under operation inherited from G.

Proposition 2.1.4. *Subgroup Test*
Let G be a group and let $H \subseteq G$ be nonempty. Then $H \leq G \iff ab^{-1} \in H \ \forall a, b \in H$.

Proof.
($\Rightarrow$) If $H \leq G$, then $b^{-1} \in H$ and $ab^{-1} \in H$ for all $a, b \in H$.
($\Leftarrow$) Assume $ab^{-1} \in H$ for all $a, b \in H$. Choose $h \in H$. Then $e_G = hh^{-1} \in H$. For $b \in H$, $b^{-1} = e_G b^{-1} \in H$.
For $a, c \in H$, since $c^{-1} \in H$, we have $ac = a(c^{-1})^{-1} \in H$. Hence $H \leq G$.

Structural Commentary: Closure under one expression packages identity, inverses, and products.
Key Insight: In a group, subtraction-style closure is enough. □

Definition 2.1.5. *Generated and Cyclic Groups*
Let G be a group.

(i) For $a \in G$, the *subgroup generated by a* is $\langle a \rangle := \{a^k : k \in \mathbb{Z}\} \leq G$.
(ii) G is *cyclic* if $G = \langle a \rangle$ for some $a \in G$.
If it exists, $\mathrm{ord}(a) := \min\{n \in \mathbb{N}_{>0} : a^n = e_G\}$; then $\langle a \rangle = \{e_G, a, \ldots, a^{\mathrm{ord}(a)-1}\}$.

2.2 Group Homomorphisms, Kernels, and Images

Definition 2.2.1. *Group Homomorphisms, Kernels, and Images*
Let G, H be groups.

(i) A *homomorphism* $\varphi : G \to H$ satisfies $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in G$.
(ii) For a homomorphism $\varphi : G \to H$, $\ker \varphi := \{g \in G : \varphi(g) = e_H\}$; and $\operatorname{im} \varphi := \varphi(G)$.
An *isomorphism* is a bijective homomorphism.

Proposition 2.2.2. *Basic Homomorphism Properties*
Let $\varphi : G \to H$ be a group homomorphism. Then:

(i) $\varphi(e_G) = e_H$;
(ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ *for all* $g \in G$;
(iii) $\ker \varphi \leq G$;
(iv) $\operatorname{im} \varphi \leq H$.

Proof.
(i) Since $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)^2$, cancellation gives $\varphi(e_G) = e_H$.
(ii) For $g \in G$, $\varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$, hence $\varphi(g^{-1}) = \varphi(g)^{-1}$.
(iii) If $a, b \in \ker \varphi$, then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H$, so $\ker \varphi \leq G$ by Proposition 2.1.4.
(iv) If $x = \varphi(a)$ and $y = \varphi(b)$ lie in $\operatorname{im} \varphi$, then $xy^{-1} = \varphi(ab^{-1}) \in \operatorname{im} \varphi$, so $\operatorname{im} \varphi \leq H$ by Proposition 2.1.4.

Structural Commentary: Homomorphisms preserve all group structure once they preserve multiplication.
Key Insight: $\varphi(ab) = \varphi(a)\varphi(b)$ preserves e, inverses, kernels, and images. □

Proposition 2.2.3. *Kernel Criterion for Injectivity*
Let $\varphi : G \to H$ be a group homomorphism. Then φ injective $\iff$ $\ker\varphi = \{e_G\}$.

Proof.
($\Rightarrow$) If φ injective, $g \in \ker\varphi$, then $\varphi(g) = e_H = \varphi(e_G)$ by Proposition 2.2.2, so $g = e_G$. Hence $\ker\varphi = \{e_G\}$.
($\Leftarrow$) If $\ker\varphi = \{e_G\}$ and $\varphi(a) = \varphi(b)$, then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H$, so $ab^{-1} = e_G$ and $a = b$.
Structural Commentary: Injectivity is detected by the identity fiber $\ker\varphi$.
Key Insight: The kernel measures exactly the failure of injectivity. □

2.3 Cosets, Normality, and Structure Theorems

2.4 Group Actions, Orbits, and Stabilizers

A group action packages the idea that elements of a group act as reversible motions of a set. The conjugation action is the finite-group ingredient used later in Wedderburn's theorem.

Definition 2.4.1. *Actions, Orbits, and Stabilizers*
Let G be a group and let $\mathcal{X}$ be a set.
(i) *Left action* of G on $\mathcal{X}$ is map $G \times \mathcal{X} \to \mathcal{X}$, $(g, x) \mapsto gx$, s.t. $e_G x = x$ and $(gh)x = g(hx)$ $\forall g, h \in G$, $x \in \mathcal{X}$.
(ii) The *orbit* of $x \in \mathcal{X}$ is $Gx := \{gx : g \in G\}$.
(iii) The *stabilizer* of $x \in \mathcal{X}$ is $G_x := \{g \in G : gx = x\}$.

Theorem 2.4.2. *Orbit-Stabilizer*
Let a finite group G act on $\mathcal{X}$. For each $x \in \mathcal{X}$, $|Gx| = [G : G_x] = \frac{|G|}{|G_x|}$.

Proof. Define $\Phi : G/G_x \to Gx$, $\Phi(gG_x) = gx$.
For $g, h \in G$, $gG_x = hG_x \iff h^{-1}g \in G_x \iff h^{-1}gx = x \iff gx = hx$. Thus Φ is well defined and injective. It is surjective by definition of Gx, hence bijective, so $|Gx| = |G/G_x| = [G : G_x]$.
Structural Commentary: Orbits are parametrized by cosets of the stabilizer.
Key Insight: Same orbit point $\iff$ same stabilizer coset. □

2.5 Conjugacy Classes and the Class Equation

Definition 2.5.1. *Conjugacy Classes and Centralizers*
Let G be a group.
(i) The *conjugation action* of G on itself is $g \cdot x := gxg^{-1}$.
(ii) The *conjugacy class* of $x \in G$ is $\{x\}^G := \{gxg^{-1} : g \in G\}$.
(iii) The *centralizer* of $x \in G$ is $C_G(x) := \{g \in G : gx = xg\}$.
(iv) The *center* of G is $Z(G) := \{g \in G : gx = xg \text{ for all } x \in G\}$.

Theorem 2.5.2. *Class Equation*
Let G be finite, $S \subseteq G$ contain one representative of each noncentral conjugacy class.
Then $|G| = |Z(G)| + \sum_{x \in S} [G : C_G(x)]$.

Proof. The conjugacy classes form a partition $G = Z(G) \sqcup \bigsqcup_{x \in S} \{x\}^G$.
Indeed, $\left|\{x\}^G\right| = 1 \iff gxg^{-1} = x$ for all $g \in G \iff x \in Z(G)$. For $x \in S$, Theorem 2.4.2 applied to conjugation gives $\left|\{x\}^G\right| = [G : C_G(x)]$. Taking cardinalities yields the formula.
Structural Commentary: G decomposes into central singleton classes and noncentral conjugacy classes.
Key Insight: Count conjugacy classes by orbit-stabilizer. □

2.6 Sylow Theory and Finite Groups

2.7 Representation-Theoretic Preview

3 Rings, Ideals, and Polynomial Rings

3.1 Rings, Identities, Fields, and Examples

Rings add multiplication to an abelian group.

Definition 3.1.1. *Rings, Identities, and Fields*
Let R be a set with operations $+$ and $\cdot$.
(i) R is a *ring* if $(R, +)$ is an abelian group, multiplication is associative, and $a(b + c) = ab + ac$, $(a + b)c = ac + bc$ for all $a, b, c \in R$.
(ii) A ring R is *commutative* if $ab = ba$ for all $a, b \in R$.
(iii) A ring R has *identity* if there is $1_R \in R$ such that $1_R a = a1_R = a$ for all $a \in R$.
(iv) A *field* is a commutative ring with identity $1_R \neq 0$ such that every $a \in R \setminus \{0\}$ has $a^{-1} \in R$.

Example 3.1.2. *Basic Rings*

(i) $(\mathbb{Z}, +, \cdot)$ and $(\mathbb{F}[x], +, \cdot)$ are commutative rings with identity 1.
(ii) $\mathrm{Mat}(n, n, \mathbb{F})$ is a unital ring under matrix addition and multiplication, with identity I_n; it is noncommutative for $n \geq 2$.
(iii) $2\mathbb{Z}$ is a commutative ring under inherited addition and multiplication, but has no identity.
(iv) $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring under addition and multiplication modulo n, with identity $1 + n\mathbb{Z}$.

3.2 Ring Homomorphisms and Subrings

Definition 3.2.1. *Ring Homomorphisms and Subrings*
Let R, S be rings with identity.

(i) A *ring homomorphism* $\varphi : R \to S$ preserves $+$, $\cdot$, and 1, where $\varphi(a+b) = \varphi(a) + \varphi(b)$, $\varphi(ab) = \varphi(a)\varphi(b)$, $\varphi(1_R) = 1_S$ for all $a, b \in R$.
(ii) A *subring* of R is a subset $T \subseteq R$ that is a ring under the inherited operations and has identity 1_R.

Proposition 3.2.2. *Subring Test*
Let R be a ring with identity and let $S \subseteq R$. Then $S \leq R \iff 1_R \in S$, $a - b \in S$, $ab \in S$ $\forall a, b \in S$.

Proof.
($\Rightarrow$) If $S \leq R$, then $1_R \in S$ and S is closed under subtraction and multiplication.
($\Leftarrow$) Assume $1_R \in S$, $a - b \in S$, and $ab \in S$ for all $a, b \in S$. Then S is nonempty, $(S, +) \leq (R, +)$ by Proposition 2.1.4, and multiplication restricts to S. All ring laws and the identity 1_R are inherited from R, so $S \leq R$.

Structural Commentary: A subring is a subset stable under the operations and containing the ambient unit.
Key Insight: Subtraction handles the additive group part in one stroke. □

3.3 Ideals and Generated Ideals

Ideals are the subsets by which quotient rings can be formed, just as subspaces are the subsets by which quotient vector spaces can be formed.

Definition 3.3.1. *Ideals and Generated Ideals*
Let R be a ring.

(i) A nonempty subset $\mathcal{I} \subseteq R$ is a *two-sided ideal* if $a - b \in \mathcal{I}$ $\forall a, b \in \mathcal{I}$, and $ra, ar \in \mathcal{I}$ $\forall a \in \mathcal{I}$, $r \in R$.
(ii) If R is commutative with identity, $S \subseteq R$, *ideal generated by S* is $\langle S \rangle := \left\{ \sum_{j=1}^{n} r_j s_j : n \geq 1,\ r_j \in R,\ s_j \in S \right\}$.
(iii) *Finite generated ideal notation*: if $S = \{s_1, \ldots, s_n\}$, write $\langle s_1, \ldots, s_n \rangle := \langle S \rangle$.

Example 3.3.2. *Principal Ideals in Polynomial Rings*
In $\mathbb{F}[x]$, $\langle p(x) \rangle = \{q(x)p(x) : q(x) \in \mathbb{F}[x]\}$ is the *principal* ideal generated by $p(x)$.

Proposition 3.3.3. *Generated Ideals Are Smallest*
Let R be a commutative ring with identity and let $S \subseteq R$.
Then $\langle S \rangle$ is an ideal, $S \subseteq \langle S \rangle$, and for every ideal $\mathcal{J} \subseteq R$, $S \subseteq \mathcal{J} \implies \langle S \rangle \subseteq \mathcal{J}$.

Proof. Let $x = \sum_{j=1}^{m} r_j s_j$ and $y = \sum_{k=1}^{n} t_k u_k$ lie in $\langle S \rangle$.
Then $x - y = \sum_{j=1}^{m} r_j s_j + \sum_{k=1}^{n} (-t_k) u_k \in \langle S \rangle$; $ax = \sum_{j=1}^{m} (ar_j) s_j \in \langle S \rangle$ for all $a \in R$, so $\langle S \rangle$ is an ideal.
Since $s = 1_R s \in \langle S \rangle$, $S \subseteq \langle S \rangle$. If $\mathcal{J}$ is an ideal with $S \subseteq \mathcal{J}$, then $r_j s_j \in \mathcal{J}$ and finite sums remain in $\mathcal{J}$, QED.
Structural Commentary: Generated ideals close a set under exactly the operations ideals must allow.
Key Insight: To force S into an ideal, one must include all ring-linear combinations of elements of S. □

Proposition 3.3.4. *Intersections and Chain Unions of Ideals*
Let R be a ring.

(i) If $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ is a nonempty family of ideals of R, then $\bigcap_{\alpha \in A} \mathcal{I}_\alpha$ is an ideal of R.
(ii) If $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ is a chain of ideals of R under inclusion, then $\bigcup_{\alpha \in A} \mathcal{I}_\alpha$ is an ideal of R.

Proof.
(i) Let $\mathcal{I} = \bigcap_{\alpha \in A} \mathcal{I}_\alpha$. If $a, b \in \mathcal{I}$ and $r \in R$, then $a - b, ra, ar \in \mathcal{I}_\alpha$ for every $\alpha \in A$, hence $a - b, ra, ar \in \mathcal{I}$.
(ii) Let $\mathcal{U} = \bigcup_{\alpha \in A} \mathcal{I}_\alpha$. If $a, b \in \mathcal{U}$, then $a \in \mathcal{I}_\alpha$ and $b \in \mathcal{I}_\beta$ for some α, β. Since the family is a chain, one of $\mathcal{I}_\alpha, \mathcal{I}_\beta$ contains both a, b; hence $a - b \in \mathcal{U}$. If $a \in \mathcal{U}$ and $r \in R$, then $a \in \mathcal{I}_\alpha$ for some α, so $ra, ar \in \mathcal{I}_\alpha \subseteq \mathcal{U}$.

Structural Commentary: Chains avoid the usual failure of unions to preserve closure.
Key Insight: Comparability places any finite test inside one ideal. □

3.4 Quotient Rings and Maximal Ideals

Definition 3.4.1. *Quotient Rings*
Let R be a ring and let $\mathcal{I} \trianglelefteq R$ be a two-sided ideal.

(i) The underlying set is the additive quotient group $R/\mathcal{I} = \{a + \mathcal{I} : a \in R\}$.
(ii) Addition and multiplication are $(a + \mathcal{I}) + (b + \mathcal{I}) = (a + b) + \mathcal{I}$ and $(a + \mathcal{I})(b + \mathcal{I}) = ab + \mathcal{I}$.
(iii) With these operations, $R/\mathcal{I}$ is the *quotient ring* of R by $\mathcal{I}$.

Proposition 3.4.2. *Ideals Make Quotient Multiplication Well Defined*
Let $\mathcal{I} \leq (R, +)$. The rule $(a + \mathcal{I})(b + \mathcal{I}) := ab + \mathcal{I}$ is well defined on $R/\mathcal{I}$ iff $\mathcal{I} \trianglelefteq R$.

Proof.
($\Rightarrow$) If the rule is well defined and $i \in \mathcal{I}$, then $i + \mathcal{I} = 0 + \mathcal{I}$.
For $r \in R$, $ri + \mathcal{I} = r0 + \mathcal{I}$, and $ir + \mathcal{I} = 0r + \mathcal{I}$, so $ri, ir \in \mathcal{I}$. Thus $\mathcal{I} \trianglelefteq R$.
($\Leftarrow$) If $\mathcal{I} \trianglelefteq R$ and $a' = a + i$, $b' = b + j$ with $i, j \in \mathcal{I}$, then $a'b' - ab = aj + ib + ij \in \mathcal{I}$.
Hence $a'b' + \mathcal{I} = ab + \mathcal{I}$, so the product is independent of representatives.

Structural Commentary: Ideals are exactly the additive subgroups compatible with multiplication of cosets.
Key Insight: Coset multiplication is well defined exactly for ideals. □

Definition 3.4.3. *Maximal Ideals*
Let R be a ring and let $\mathcal{M} \trianglelefteq R$.

(i) $\mathcal{M}$ is *proper* if $\mathcal{M} \neq R$.
(ii) $\mathcal{M}$ is *maximal* if $\mathcal{M} \neq R$ and, for every ideal $\mathcal{J} \trianglelefteq R$, $\mathcal{M} \subseteq \mathcal{J} \subseteq R \implies \mathcal{J} = \mathcal{M}$ or $\mathcal{J} = R$.

Theorem 3.4.4. *Maximal Ideals and Fields*
Let R be a commutative ring with identity and let $\mathcal{I} \trianglelefteq R$. Then $R/\mathcal{I}$ is a field $\iff$ $\mathcal{I}$ is maximal

Proof.
($\Rightarrow$) If $R/\mathcal{I}$ is a field and $\mathcal{I} \subsetneq \mathcal{J} \trianglelefteq R$, then $\mathcal{J}/\mathcal{I}$ is a nonzero ideal of $R/\mathcal{I}$.
Hence $\mathcal{J}/\mathcal{I} = R/\mathcal{I}$, so $\mathcal{J} = R$. Thus $\mathcal{I}$ is maximal.
($\Leftarrow$) If $\mathcal{I}$ is maximal and $0 \neq a + \mathcal{I} \in R/\mathcal{I}$, then $a \notin \mathcal{I}$.
The ideal $\mathcal{I} + \langle a \rangle$ strictly contains $\mathcal{I}$, so maximality gives $\mathcal{I} + \langle a \rangle = R$. Thus $1 = i + ra$ for some $i \in \mathcal{I}$, $r \in R$, and $(r + \mathcal{I})(a + \mathcal{I}) = 1 + \mathcal{I}$. Hence every nonzero coset is invertible.
Structural Commentary: Maximal ideals are exactly kernels whose quotients have no residual ideal structure.
Key Insight: Maximality makes every nonzero coset invertible. □

Theorem 3.4.5. *Existence of Maximal Ideals*
If R is a commutative ring with identity and $R \neq 0$, then R has a maximal ideal.

Proof. Let $\mathcal{P} := \{\mathcal{I} \trianglelefteq R : \mathcal{I} \neq R\}$, ordered by inclusion. Since $R \neq 0$, $\{0\} \in \mathcal{P}$.
If $\mathcal{C} \subseteq \mathcal{P}$ is a chain, then $\mathcal{U} := \bigcup_{\mathcal{I} \in \mathcal{C}} \mathcal{I}$ is an ideal by Proposition 3.3.4.
Also $\mathcal{U} \neq R$: if $1_R \in \mathcal{U}$, then $1_R \in \mathcal{I}$ for some $\mathcal{I} \in \mathcal{C}$, forcing $\mathcal{I} = R$, contradiction. Thus $\mathcal{U} \in \mathcal{P}$ is an upper bound for $\mathcal{C}$. By Zorn's lemma, Principle 6.1.3, $\mathcal{P}$ has a maximal element, i.e. R has a maximal ideal.
Structural Commentary: Maximal ideals are produced by maximality, not by an explicit formula in general.
Key Insight: Chain unions keep properness because containing 1 is a finite obstruction. □

3.5 Integral Domains and Fields of Quotients

Definition 3.5.1. *Zero Divisors and Integral Domains*
Let R be a ring.

(i) A *zero divisor* is a nonzero $a \in R$ such that $\exists\, 0 \neq b \in R$ with $ab = 0$ or $ba = 0$.
(ii) An *integral domain* is a commutative ring with identity $1_R \neq 0$ and no zero divisors.

Proposition 3.5.2. *Cancellation in Integral Domains*
Let R be a commutative ring with identity $1_R \neq 0$.
Then R is an integral domain $\iff$ $(\forall r, x, y \in R)(r \neq 0 \wedge rx = ry \Rightarrow x = y)$.

Proof.
($\Rightarrow$) If $rx = ry$ and $r \neq 0$, then $r(x - y) = 0$; no zero divisors gives $x - y = 0$.
($\Leftarrow$) If $ab = 0$ and $a \neq 0$, then $ab = a0$; cancellation gives $b = 0$, so no zero divisors exist.
Structural Commentary: Domains are exactly cancellative commutative rings.
Key Insight: Zero divisors obstruct cancellation. □

Definition 3.5.3. *Field of Quotients*
Let R be an integral domain. On $R \times (R \setminus \{0\})$, set $(a, b) \sim (c, d) \iff ad = bc$.
The *field of quotients* of R is $(R \times (R \setminus \{0\}))/\sim$; write $[(a, b)] = a/b$. For $b, d \neq 0$, $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$; $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}$.

Proposition 3.5.4. *Domains Embed in Fields*
Let R be an integral domain with field of quotients K.
Then $\iota : R \hookrightarrow K$; $\iota(a) = a/1$ is an injective ring homomorphism.

Proof. By Proposition 3.5.2, the relation and operations in Definition 3.5.3 are well defined.
For $a, b \in R$, $\iota(a + b) = \frac{a+b}{1} = \frac{a}{1} + \frac{b}{1}$; $\iota(ab) = \frac{ab}{1} = \frac{a}{1}\frac{b}{1}$; and $\iota(1_R) = \frac{1_R}{1}$..
Thus ι is a ring homomorphism. If $\iota(a) = \iota(b)$, then $a/1 = b/1$, so $a = b$; hence ι is injective.
Structural Commentary: The field of quotients formally adds denominators to a domain.
Key Insight: Integral domains are exactly the setting where fraction arithmetic is consistent. □

3.6 Ring Characteristic

Definition 3.6.1. *Characteristic*
Let R be a ring with identity. Its *characteristic* is $\operatorname{char}(R) := \begin{cases} \min\{n \in \mathbb{N}_{>0} : n1_R = 0_R\}, & \text{if this set is nonempty,} \\ 0, & \text{otherwise.} \end{cases}$.

Proposition 3.6.2. *Finite Domains Have Prime Characteristic*
Let R be a finite ring with identity. Then $\operatorname{char}(R) \neq 0$. If R is an integral domain, then $\operatorname{char}(R)$ is prime.

Proof. Since R is finite, the map $n \mapsto n1_R$ from $\mathbb{N}$ to R is not injective.
Thus $i1_R = j1_R$ for some $i < j$, so $(j - i)1_R = 0_R$ and $\operatorname{char}(R) > 0$.
Assume R is a domain and set $c = \operatorname{char}(R)$. Since $1_R \neq 0_R$, $c > 1$.
If $c = ab$ with $1 < a, b < c$, then $(a1_R)(b1_R) = ab1_R = c1_R = 0_R$.
By minimality of c, $a1_R \neq 0_R$ and $b1_R \neq 0_R$, contradicting that R has no zero divisors. Hence c is prime.
Structural Commentary: Characteristic records the additive order of the unit.
Key Insight: In a domain, a composite characteristic would factor zero nontrivially. □

3.7 Algebras over Fields

Definition 3.7.1. *Algebras over a Field*
Let $\mathbb{F}$ be a field.
An *$\mathbb{F}$-algebra* is an $\mathbb{F}$-vector space $\mathcal{A}$ with a ring product satisfying $\lambda(ab) = (\lambda a)b = a(\lambda b)$ $\forall \lambda \in \mathbb{F}$ and $a, b \in \mathcal{A}$.

Example 3.7.2. *Matrix and Polynomial Algebras*
$\mathrm{Mat}(n, n, \mathbb{F})$ and $\mathbb{F}[x]$ are algebras over $\mathbb{F}$. The first is generally noncommutative; the second is commutative.

3.8 Units, Divisibility, and Factorization Domains

Definition 3.8.1. *Units, Associates, Prime and Irreducible Elements*
Let R be an integral domain.

(i) A *unit* is an element $u \in R$ with $uv = 1$ for some $v \in R$.
(ii) Nonzero elements $a, b \in R$ are *associates*, written $a \sim b$, if $a = ub$ for some unit u.
(iii) A nonzero nonunit $p \in R$ is *prime* if $p \mid ab$ implies $p \mid a$ or $p \mid b$.
(iv) A nonzero nonunit $r \in R$ is *irreducible* if $r = ab$ implies a or b is a unit.

Proposition 3.8.2. *Divisibility and Principal Ideals*
Let R be an integral domain and let $r, s \in R$ be nonzero.

(i) $r \in R^\times \iff \langle r \rangle = R$.
(ii) $r \sim s \iff \langle r \rangle = \langle s \rangle$.
(iii) $r \mid s \iff \langle s \rangle \subseteq \langle r \rangle$.

Proof.

(i) ($\Rightarrow$) If $rv = 1$, then $1 \in \langle r \rangle$, so $\langle r \rangle = R$.
(i) ($\Leftarrow$) If $\langle r \rangle = R$, then $1 = ar$ for some $a \in R$, hence $r \in R^\times$.
(ii) ($\Rightarrow$) If $r = us$ with $u \in R^\times$, then $r \in \langle s \rangle$ and $s = u^{-1}r \in \langle r \rangle$, so $\langle r \rangle = \langle s \rangle$.
(ii) ($\Leftarrow$) If $\langle r \rangle = \langle s \rangle$, then $r = as$ and $s = br$ for some $a, b \in R$.
Thus $r = abr$, so $ab = 1$ by cancellation; hence $a \in R^\times$ and $r \sim s$.
(iii) ($\Rightarrow$) If $s = ar$, then every element of $\langle s \rangle$ is of the form $xs = xar \in \langle r \rangle$.
(iii) ($\Leftarrow$) If $\langle s \rangle \subseteq \langle r \rangle$, then $s \in \langle r \rangle$, so $s = ar$ for some $a \in R$; hence $r \mid s$.

Structural Commentary: Divisibility reverses inclusion of principal ideals.
Key Insight: Principal ideals translate multiplicative divisibility into order by containment. □

Definition 3.8.3. *PIDs and UFDs*
Let R be an integral domain.

(i) R is a *principal ideal domain* if every ideal $\mathcal{I} \trianglelefteq R$ has $\mathcal{I} = \langle a \rangle$ for some $a \in R$.
(ii) R is a *unique factorization domain* if every $0 \neq a \in R \setminus R^\times$ factors into irreducibles, uniquely up to reordering and associates.

3.9 Polynomial Rings, Division, and GCDs

Polynomial rings over fields behave like the integers in the ways linear algebra needs: division with remainder, gcds, Bezout identities, irreducible factorization, and principal ideals.

Definition 3.9.1. *Polynomials over a Field*
Let $\mathbb{F}$ be a field.

(i) $\mathbb{F}[x]$ denotes the ring of polynomials in one variable over $\mathbb{F}$.
(ii) If $0 \neq p \in \mathbb{F}[x]$ and $p(x) = a_0 + a_1 x + \cdots + a_n x^n$, $a_n \neq 0$, then $\mathrm{lc}(p) := a_n$, $\deg p := n$.
Also, p is *monic* iff $\mathrm{lc}(p) = 1$. Set $\deg 0 = -\infty$.
(iii) A nonzero $p \in \mathbb{F}[x]$ *splits over* $\mathbb{F}$ iff $p = \alpha \prod_{i=1}^{m} (x - \lambda_i)$ for some $m \geq 0$, $\alpha \in \mathbb{F}^\times$, and $\lambda_i \in \mathbb{F}$.

Theorem 3.9.2. *Polynomial Division Algorithm*
Let $f, g \in \mathbb{F}[x]$ with $g \neq 0$. Then $\exists!\, (q, r) \in \mathbb{F}[x]^2$ such that $f = qg + r$, $r = 0$ or $\deg r < \deg g$.

Proof. Let $d = \deg g$ and $\ell = \mathrm{lc}(g)$. Existence follows by induction on $\deg f$. If $f = 0$ or $\deg f < d$, take $(q, r) = (0, f)$. Otherwise, let $n = \deg f$, $a = \mathrm{lc}(f)$, and $f_1 = f - a\ell^{-1}x^{n-d}g$. Then $\deg f_1 < n$.
By induction, $f_1 = q_1 g + r$ with $r = 0$ or $\deg r < d$, hence $f = (q_1 + a\ell^{-1}x^{n-d})g + r$.
For uniqueness, if $qg + r = q'g + r'$ with admissible remainders, then $(q - q')g = r' - r$.
If $q \neq q'$, then $\deg((q - q')g) \geq d$, while $\deg(r' - r) < d$, contradiction. Hence $q = q'$ and then $r = r'$.
Structural Commentary: Division works because the leading coefficient of g is invertible in the field.
Key Insight: Cancel leading terms until the remainder is too small to divide again. □

Definition 3.9.3. *Divisibility, GCDs, and Coprimality*
For $p, q \in \mathbb{F}[x]$, write $p \mid q$ if $q = hp$ for some $h \in \mathbb{F}[x]$.
A *greatest common divisor* of $f, g \in \mathbb{F}[x]$, not both zero, is the unique monic polynomial $\gcd(f, g)$ such that:

(i) $\gcd(f, g) \mid f$ and $\gcd(f, g) \mid g$;
(ii) every common divisor of f and g divides $\gcd(f, g)$.

The polynomials f, g are *relatively prime* if $\gcd(f, g) = 1$.

Theorem 3.9.4. *Polynomial GCD and Bezout Identity*
For $f, g \in \mathbb{F}[x]$, not both zero, $\exists$ a unique monic $d \in \mathbb{F}[x]$ s.t. $d \mid f$, $d \mid g$, $h \mid f$, $h \mid g \Longrightarrow h \mid d$
Moreover, $d = \gcd(f, g)$ satisfies $d = af + bg$ for some $a, b \in \mathbb{F}[x]$.
Hence $\gcd(f, g) = 1$ iff $\exists\, a, b \in \mathbb{F}[x]$ with $af + bg = 1$.

Proof. If one of f, g is 0, the claim is immediate from the monic associate of the other.
Assume $f, g \neq 0$ and run the Euclidean algorithm: $r_{i-1} = q_i r_i + r_{i+1}$, where $r_{i+1} < \deg r_i$, starting with $(r_0, r_1) = (f, g)$ and ending with $r_N \neq 0$, $r_{N+1} = 0$. Then $r_N \mid r_{N-1}$, and upward substitution gives $r_N \mid f, g$.
Conversely, any common divisor of f, g divides every r_i, hence divides r_N. Thus $d := \mathrm{lc}(r_N)^{-1} r_N$ is a monic gcd; uniqueness follows since two monic gcds divide each other.
Back-substitution gives $d = af + bg$ for some $a, b \in \mathbb{F}[x]$. Finally, $d = 1$ gives $af + bg = 1$; conversely, if $af + bg = 1$, every common divisor divides 1, so $d = 1$.
Structural Commentary: The Euclidean algorithm produces both the gcd and its Bezout certificate.
Key Insight: GCD data is ideal data in $\mathbb{F}[x]$. □

3.10 Irreducible Polynomials and Unique Factorization

Definition 3.10.1. *Irreducible Polynomials*
A nonconstant polynomial $p \in \mathbb{F}[x]$ is *irreducible over* $\mathbb{F}$ if every factorization $p = ab$ in $\mathbb{F}[x]$ has a or b constant.

Proposition 3.10.2. *Irreducible Polynomials Are Prime in $\mathbb{F}[x]$*
Let $p \in \mathbb{F}[x]$ be nonconstant. Then p is irreducible over $\mathbb{F}$ iff $\forall f, g \in \mathbb{F}[x]$, $p \mid fg \implies p \mid f$ or $p \mid g$.

Proof.
(⇒) Assume p irreducible and $p \mid fg$. If $p \mid f$, done. Otherwise $\gcd(p, f) = 1$, so Theorem 3.9.4 gives $a, b \in \mathbb{F}[x]$ with $ap + bf = 1$. Multiplying by g gives $apg + bfg = g$; since $p \mid apg$ and $p \mid bfg$, we get $p \mid g$.
(⇐) Assume the divisibility property and write $p = ab$. Since $p \mid ab$, either $p \mid a$ or $p \mid b$. If $p \mid a$, then $a = ph$, so $p = phb$ and cancellation gives $1 = hb$, hence $b \in \mathbb{F}^\times$. Similarly, $p \mid b$ forces $a \in \mathbb{F}^\times$. Thus p is irreducible.
Structural Commentary: Over $\mathbb{F}[x]$, irreducible and prime are the same notion.
Key Insight: Bezout turns coprimality into a divisibility test. □

Theorem 3.10.3. *Unique Factorization in $\mathbb{F}[x]$*
For every nonconstant $f \in \mathbb{F}[x]$, $\exists\, c \in \mathbb{F}^\times$ and monic irreducibles $p_1, \ldots, p_m \in \mathbb{F}[x]$ such that $f = c \prod_{i=1}^{m} p_i$.
If $f = d \prod_{j=1}^{n} q_j$ is another such factorization, then $m = n$, $c = d$, and $p_i = q_{\sigma(i)}$ for some $\sigma \in S_m$.

Proof.
(i) *Existence*: Induct on $\deg f$. If f is irreducible, write $f = \mathrm{lc}(f)\big(\mathrm{lc}(f)^{-1} f\big)$.
If $f = uv$ with $0 < \deg u, \deg v < \deg f$, apply induction to u and v and concatenate the factors.
(ii) *Uniqueness*: Suppose $c \prod_{i=1}^{m} p_i = d \prod_{j=1}^{n} q_j$ with all p_i, q_j monic irreducible. Since $p_1 \mid \prod_{j=1}^{n} q_j$, iterating Proposition 3.10.2 gives $p_1 \mid q_j$ for some j. As p_1, q_j are monic irreducible, $p_1 = q_j$.
Cancel and continue by induction on m; the final scalar equality follows from leading coefficients.
Structural Commentary: Polynomial factorization over a field mirrors prime factorization in $\mathbb{Z}$.
Key Insight: The prime property of irreducibles drives uniqueness. □

3.11 Principal Ideal Domains and Unique Factorization

Theorem 3.11.1. *$\mathbb{Z}$ and $\mathbb{F}[x]$ Are Principal Ideal Domains*
(i) If $0 \neq \mathcal{I} \trianglelefteq \mathbb{Z}$, then $\mathcal{I} = \langle d \rangle$ for $d = \min(\mathcal{I} \cap \mathbb{N}_{>0})$.
(ii) If $0 \neq \mathcal{I} \trianglelefteq \mathbb{F}[x]$, then $\mathcal{I} = \langle m \rangle$ for the unique monic $m \in \mathcal{I}$ of minimal degree.

Proof.
(i) If $\mathcal{I} = 0$, then $\mathcal{I} = \langle 0 \rangle$. Otherwise choose $d = \min(\mathcal{I} \cap \mathbb{N}_{>0})$. For $a \in \mathcal{I}$, write $a = qd + r$ with $0 \leq r < d$. Since $r = a - qd \in \mathcal{I}$, minimality gives $r = 0$, so $\mathcal{I} = \langle d \rangle$.
(ii) If $\mathcal{I} = 0$, then $\mathcal{I} = \langle 0 \rangle$. Otherwise choose monic $m \in \mathcal{I}$ of minimal degree. For $p \in \mathcal{I}$, divide $p = qm + r$ with $r = 0$ or $\deg r < \deg m$. Since $r = p - qm \in \mathcal{I}$, minimality gives $r = 0$, so $\mathcal{I} = \langle m \rangle$.
If m' is another monic minimal generator, then $m \mid m'$ and $m' \mid m$, hence $m = m'$.
Structural Commentary: Euclidean division turns arbitrary ideals into one-generator ideals.
Key Insight: Choose the smallest positive integer or smallest-degree monic polynomial and divide by it. □

Corollary 3.11.2. *GCDs Generate Polynomial Ideals*
For $0 \neq p_1, \ldots, p_n \in \mathbb{F}[x]$ and monic $m = \gcd(p_1, \ldots, p_n)$, $\langle p_1, \ldots, p_n \rangle = \langle m \rangle$.

Proof. By Theorem 3.11.1, $\langle p_1, \ldots, p_n \rangle = \langle g \rangle$ for a unique monic g. Since $p_i \in \langle g \rangle$, $g \mid p_i$ for all i.
If $q \mid p_i$ for all i, then $\langle p_1, \ldots, p_n \rangle \subseteq \langle q \rangle$, so $g \in \langle q \rangle$ and $q \mid g$. Hence $g = m$, and the claim follows.
Structural Commentary: In $\mathbb{F}[x]$, gcds and finitely generated ideals are the same data.
Key Insight: The gcd is the monic generator of the ideal generated by the polynomials. □

Example 3.11.3. *A Polynomial Ring That Is Not a PID*
In $\mathbb{F}[x, y]$, the ideal $\langle x, y \rangle$ is not principal. If $\langle x, y \rangle = \langle p \rangle$, then $p \mid x$ and $p \mid y$, so $p \in \mathbb{F}^\times$. Hence $\langle x, y \rangle = \mathbb{F}[x, y]$, contradicting $1 \notin \langle x, y \rangle$ since every element of $\langle x, y \rangle$ has zero constant term.

Proposition 3.11.4. *PIDs Satisfy the Ascending Chain Condition*
If R is a PID and $\mathcal{I}_1 \subseteq \mathcal{I}_2 \subseteq \cdots$ is a chain of ideals, then $\exists N$ such that

$$\mathcal{I}_j = \mathcal{I}_N \qquad \forall j \geq N.$$

Proof. Let $\mathcal{I} = \bigcup_{j \geq 1} \mathcal{I}_j$. By Proposition 3.3.4, $\mathcal{I}$ is an ideal. Since R is a PID, $\mathcal{I} = \langle a \rangle$. Choose N with $a \in \mathcal{I}_N$. Then $\mathcal{I} = \langle a \rangle \subseteq \mathcal{I}_N \subseteq \mathcal{I}$, so $\mathcal{I}_j = \mathcal{I}_N$ for all $j \geq N$.
Structural Commentary: The generator appears at a finite stage.
Key Insight: Principal generation converts infinite ideal growth into a finite containment. □

Theorem 3.11.5. *Prime, Irreducible, and Bezout in a PID*
Let R be a principal ideal domain.

(i) For $0 \neq r \in R \setminus R^\times$, r is irreducible iff $\langle r \rangle$ is maximal among proper ideals.
(ii) Every irreducible $r \in R$ is prime.
(iii) For $a, b \in R$, $\gcd(a,b) = 1$ iff $\exists x, y \in R$ such that $xa + yb = 1$.

Proof.

(i) If r is irreducible and $\langle r \rangle \subseteq \mathcal{I} \subseteq R$, write $\mathcal{I} = \langle a \rangle$. Then $r = xa$, so irreducibility gives $\mathcal{I} = \langle r \rangle$ or $\mathcal{I} = R$. Conversely, if $r = ab$ with $a, b \notin R^\times$, then $\langle r \rangle \subsetneq \langle a \rangle \subsetneq R$.
(ii) Let r be irreducible and $r \mid ab$. By (i), $\langle r, a \rangle = \langle r \rangle$ or R. In the first case $r \mid a$. In the second, $1 = xr + ya$, so $b = xrb + yab$ and $r \mid b$.
(iii) If $xa + yb = 1$, every common divisor of a, b is a unit. Conversely, let $\langle a, b \rangle = \langle d \rangle$. If $\gcd(a,b) = 1$, then $d \in R^\times$, so $\langle a, b \rangle = R$ and $1 = xa + yb$ for some $x, y \in R$.

Structural Commentary: In a PID, factorization statements become ideal-containment statements.
Key Insight: Maximal principal ideals are the bridge from irreducible to prime. □

Theorem 3.11.6. *Every PID Is a UFD*
If R is a PID, then R is a UFD.

Proof. Let $0 \neq r \in R \setminus R^\times$. If factorization into nonunits never terminates, obtain $s_1, s_2, \ldots$ with $s_{j+1} \mid s_j$ properly. By Proposition 3.8.2,

$$\langle s_1 \rangle \subsetneq \langle s_2 \rangle \subsetneq \cdots,$$

contradicting Proposition 3.11.4; hence irreducible factorizations exist.
For uniqueness, let $p_1 \cdots p_m = q_1 \cdots q_n$ with all p_i, q_j irreducible. By Theorem 3.11.5, p_1 is prime, so $p_1 \mid q_j$ for some j; irreducibility gives $p_1 \sim q_j$. Cancel associates and iterate. Thus $m = n$ and the factors agree up to order and associates.
Structural Commentary: ACC gives existence; irreducible-is-prime gives uniqueness.
Key Insight: PIDs have enough ideal control to force unique factorization. □

3.12 Roots of Unity and Cyclotomic Polynomials

Definition 3.12.1. *Roots of Unity and Cyclotomic Polynomials*
For $n \geq 1$:

(i) The complex nth roots of unity are $U_n := \{z \in \mathbb{C} : z^n = 1\}$.
(ii) A root $\omega \in U_n$ is *primitive* iff $\operatorname{ord}(\omega) = n$; set $\Omega_n := \{\omega \in U_n : \operatorname{ord}(\omega) = n\}$.
(iii) The nth *cyclotomic polynomial* is $\Phi_n(x) := \prod_{\omega \in \Omega_n} (x - \omega)$.
(iv) The number of primitive nth roots is $\varphi(n) := |\Omega_n|$.

Proposition 3.12.2. *Roots of Unity Factorization*
Let $\zeta_n = e^{2\pi i/n}$. Then $U_n = \langle \zeta_n \rangle$; $|U_n| = n$; $\Omega_n = \{\zeta_n^k : 1 \leq k \leq n,\ \gcd(k,n) = 1\}$; $|\Omega_n| = \varphi(n)$.
Moreover, $U_n = \coprod_{d \mid n} \Omega_d$ and $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$.

Proof. The roots of $x^n - 1$ are precisely ζ_n^k for $0 \leq k < n$, hence $U_n = \langle \zeta_n \rangle$ and $|U_n| = n$.
In a cyclic group, $\operatorname{ord}(\zeta_n^k) = n/\gcd(k,n)$, so ζ_n^k is primitive iff $\gcd(k,n) = 1$; thus $|\Omega_n| = \varphi(n)$.
Each $z \in U_n$ has unique order $d \mid n$, giving $U_n = \coprod_{d \mid n} \Omega_d$.
Therefore $x^n - 1 = \prod_{z \in U_n} (x - z) = \prod_{d \mid n} \prod_{\omega \in \Omega_d} (x - \omega) = \prod_{d \mid n} \Phi_d(x)$ *Structural Commentary*: Orders partition roots of unity by divisors.
Key Insight: Cyclicity turns root counting into order counting. □

Proposition 3.12.3. *Cyclotomic Polynomials Have Integer Coefficients*
For every $n \geq 1$, $\Phi_n(x) \in \mathbb{Z}[x]$ is monic.

Proof. Induct on n. For $n = 1$, $\Phi_1(x) = x - 1$.
Assume $\Phi_d \in \mathbb{Z}[x]$ for every proper divisor $d \mid n$, and set $P_n(x) := \prod_{\substack{d \mid n \\ d < n}} \Phi_d(x) \in \mathbb{Z}[x]$.
Then P_n is monic, and Proposition 3.12.2 gives $x^n - 1 = \Phi_n(x) P_n(x)$.
Divide $x^n - 1$ by P_n in $\mathbb{Z}[x]$: $x^n - 1 = hP_n + r$, where $h, r \in \mathbb{Z}[x]$ and $r = 0$ or $\deg r < \deg P_n$.
Uniqueness of division over $\mathbb{C}[x]$ gives $h = \Phi_n$ and $r = 0$. Hence $\Phi_n \in \mathbb{Z}[x]$, and Φ_n is monic by definition.
Structural Commentary: Proper divisors isolate the next cyclotomic factor.
Key Insight: Monic division keeps the quotient in $\mathbb{Z}[x]$. □

Lemma 3.12.4. *A Cyclotomic Size Estimate*
For $n > 1$ and $z \in \mathbb{Z}_{\geq 2}$, $|\Phi_n(z)| > z - 1$.

Proof. For $\omega \in \Omega_n$, $\omega \neq 1$ and $|\omega| = 1$, hence $|z - \omega| > z - 1$. Thus $|\Phi_n(z)| = \prod_{\omega \in \Omega_n} |z - \omega| > (z-1)^{\varphi(n)} \geq z - 1$.
Structural Commentary: Primitive factors all lie outside the closest point 1.
Key Insight: Each primitive root contributes a factor larger than $z - 1$. □

4 Fields and Galois Theory

4.1 Field Extensions

4.2 Splitting Fields and Normality

4.3 Galois Groups

4.4 Finite Fields and Characteristic

Perspective 4.4.1. *Finite Fields and Characteristic*
If $\mathrm{char}(F) = 2$, then $(\forall a \in F)\ 2a = 0$, so $a = -a$. A finite field has $|F| = p^n$ with p prime and $n \geq 1$; conversely, each prime power $q = p^n$ determines a unique field $\mathbb{F}_q$ up to isomorphism.
Thus the classification is field-theoretic, while the characteristic constraint is ring-theoretic.

Definition 4.4.2. *Linear Code and Hamming Weight*
Let $n \in \mathbb{N}$ and let $\mathbb{F}_q$ be a finite field with q elements.

(i) The *ambient code space* is $V(n,q) := \mathbb{F}_q^n$.
(ii) A *linear code* of length n over $\mathbb{F}_q$ is a *subspace* $C \leq V(n,q)$.
(iii) For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{F}_q^n$, its *Hamming weight* is $\mathrm{wt}(\mathbf{x}) := \#\{j : 1 \leq j \leq n,\ x_j \neq 0\}$.

Example 4.4.3. *Even-Weight Code*
Let $n \in \mathbb{N}$. In $V(n,2) = \mathbb{F}_2^n$, let $E_n = \left\{ \mathbf{x} \in \mathbb{F}_2^n : \sum_{j=1}^{n} x_j = 0 \right\}$. The condition $\sum_{j=1}^{n} x_j = 0$ is equivalent to $\mathrm{wt}(\mathbf{x})$ being even. If $\mathbf{x}, \mathbf{y} \in E_n$ and $c \in \mathbb{F}_2$, then $\sum_{j=1}^{n} (cx_j + y_j) = c \sum_{j=1}^{n} x_j + \sum_{j=1}^{n} y_j = 0$, so $c\mathbf{x} + \mathbf{y} \in E_n$. Hence $E_n \leq V(n,2)$ by Theorem 1.2.2, and E_n is a linear code.
Structural Commentary: The even-weight code is the parity-zero subspace in $V(n,2)$.
Key Insight: Parity is linear over $\mathbb{F}_2$.

4.5 Classical Applications

Part VIII
Modules, Multilinear Algebra, and Homological Methods

1 Module Theory

1.1 Modules and Submodules

Throughout this subsection, R denotes a nonzero commutative ring with identity 1.
Modules are vector spaces with the scalar field replaced by a ring; the price is that nonzero scalars may fail to be invertible and may even kill nonzero vectors.

Perspective 1.1.1. *Polynomial Scalars from One Operator*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Every polynomial $p(z) \in \mathbb{F}[z]$ acts on $\mathcal{V}$ by $p \cdot \mathbf{v} := p(T)\mathbf{v}$.
Thus one operator turns $\mathcal{V}$ into an object with scalar ring $\mathbb{F}[z]$.

Definition 1.1.2. *R-Module*
An *R-module* is an abelian group $(M, +)$ equipped with a scalar action $R \times M \to M$, $(r, u) \mapsto ru$, such that for all $r, s \in R$ and $u, v \in M$,

(i) $r(u + v) = ru + rv$.
(ii) $(r + s)u = ru + su$.
(iii) $(rs)u = r(su)$.
(iv) $1u = u$.

The ring R is the *base ring* of M.

Example 1.1.3. *Basic Module Models*

(i) R^n is an R-module under componentwise addition and scalar multiplication.
(ii) $\mathrm{Mat}(m, n, R)$ is an R-module under entrywise operations.
(iii) R is an R-module over itself by ring multiplication.
(iv) Every vector space over a field $\mathbb{F}$ is an $\mathbb{F}$-module.

Structural Commentary: Modules are abelian groups with compatible scalar actions.
Key Insight: The new behavior comes from the scalar ring, not from changing addition.

Perspective 1.1.4. *The Base Ring Controls the Geometry*
The structure of an R-module is governed by the arithmetic of R. When R is a field, every nonzero scalar is invertible, giving the usual theory of bases, dimension, and complements. For a general ring, nonunits and zero divisors obstruct division and cancellation, so these notions need not retain their vector-space properties.

Definition 1.1.5. *Submodule*
A *submodule* of an R-module M is a nonempty subset $N \subseteq M$ that is an R-module under the operations inherited from M, written $N \leq M$.

Theorem 1.1.6. *Submodule Test*
Let M be an R-module and let $\emptyset \neq N \subseteq M$. Then $N \leq M \iff \forall r, s \in R,\ \forall u, v \in N,\ ru + sv \in N$.

Proof.

($\Rightarrow$) If $N \leq M$, then $ru, sv \in N$, hence $ru + sv \in N$.
($\Leftarrow$) Assume the displayed condition. Choose $n \in N$. Then $0 = 0n + 0n \in N$.
For $x, y \in N$ and $r \in R$, $x + y = 1x + 1y \in N$, $-x = (-1)x + 0x \in N$, $rx = rx + 0x \in N$.
Thus N is an additive subgroup of M closed under R-scalar multiplication, so the remaining module axioms are inherited from M and $N \leq M$.

Structural Commentary: One closure test packages addition, additive inverses, and scalar multiplication.
Key Insight: Submodules are exactly the subsets closed under all two-term R-linear combinations. □

Proposition 1.1.7. *Intersections and Sums of Submodules*
Let M be an R-module.
If $A, B \leq M$, then $A \cap B \leq M$; $A + B := \{a + b : a \in A,\ b \in B\} \leq M$, and $A + B = \bigcap_{\substack{N \leq M \\ A \cup B \subseteq N}} N$.

Proof. Since $0 \in A \cap B$ and $rx + sy \in A \cap B$ for all $x, y \in A \cap B$ and $r, s \in R$, the submodule test gives $A \cap B \leq M$. If $x = a + b$ and $y = a' + b'$ with $a, a' \in A$ and $b, b' \in B$, then $rx + sy = (ra + sa') + (rb + sb') \in A + B$.
Thus $A + B \leq M$. Moreover, $A \cup B \subseteq A + B$, and if $N \leq M$ contains $A \cup B$, then $a + b \in N$ for all $a \in A$, $b \in B$; hence $A + B \subseteq N$.
This proves the displayed intersection formula.
Structural Commentary: $A \cap B$ is the meet; $A + B$ is the join.
Key Insight: Sums are forced by closure under addition. □

Proposition 1.1.8. *Ideals Are Submodules of the Regular Module*
View R as the regular R-module ${}_RR$. For every $N \subseteq R$,

$$N \leq {}_RR \quad \iff \quad N \trianglelefteq R.$$

Proof. For $N \subseteq R$,

$$N \leq {}_RR \quad \iff \quad N \leq (R, +) \text{ and } RN \subseteq N \quad \iff \quad N \trianglelefteq R,$$

by Definition 3.3.1.
Structural Commentary: Ideals are the module-theoretic subobjects of the ring acting on itself.
Key Insight: Ring multiplication becomes scalar multiplication. □

1.2 Generated, Cyclic, and Finitely Generated Modules

Throughout this subsection, R denotes a nonzero commutative ring with identity and M denotes an R-module.

Definition 1.2.1. *Generated Submodule*

(i) For $S \subseteq M$, *submodule generated by S* is $\langle\langle S\rangle\rangle_R := \left\{ \sum_{j=1}^{n} r_j s_j : n \geq 1,\ r_j \in R,\ s_j \in S \right\}$; $\langle\langle \emptyset \rangle\rangle_R = \{0\}$.

(ii) A subset $S \subseteq M$ *spans* or *generates* M if $\langle\langle S\rangle\rangle_R = M$.

Proposition 1.2.2. *Generated Submodule Is Smallest*
For every $S \subseteq M$, $\langle\langle S\rangle\rangle_R = \bigcap_{\substack{N \leq M \\ S \subseteq N}} N$.

Proof. Let $G := \langle\langle S\rangle\rangle_R$. By construction, $S \subseteq G$ and the submodule test gives $G \leq M$.
If $N \leq M$ and $S \subseteq N$, then $\sum_{j=1}^{n} r_j s_j \in N$, where $r_j \in R,\ s_j \in S$, so $G \subseteq N$.
Structural Commentary: Generation means closing a set under the operations the module axioms require.
Key Insight: The generated submodule adds no elements except those forced by finite R-linear combination. □

Definition 1.2.3. *Cyclic and Finitely Generated Modules*

(i) For $v \in M$, the *cyclic submodule generated by v* is $\langle\langle v\rangle\rangle_R = Rv = \{rv : r \in R\}$.
(ii) M is *finitely generated* if $M = \langle\langle v_1, \ldots, v_n\rangle\rangle_R$ for some $v_1, \ldots, v_n \in M$.
(iii) M is *cyclic* if $M = Rv$ for some $v \in M$.

Example 1.2.4. *A Finitely Generated Module with a Non-Finitely Generated Submodule*
Let $R = \mathbb{F}[x_1, x_2, \ldots]$. Then $R = \langle\langle 1\rangle\rangle_R$ as an R-module.
Set $I := (x_1, x_2, \ldots) = \{f \in R : f(0, 0, \ldots) = 0\}$. Then $I \trianglelefteq R$, hence $I \leq R$, but I is not finitely generated.
If $I = \langle\langle p_1, \ldots, p_n\rangle\rangle_R$, choose k such that x_k occurs in no p_i. Let $\varphi_k : R \to \mathbb{F}[x_k]$ send $x_k \mapsto x_k$ and $x_j \mapsto 0$ for $j \neq k$. Since $\varphi_k(p_i) = 0$ for all i, every element of $\langle\langle p_1, \ldots, p_n\rangle\rangle_R$ maps to 0, but $\varphi_k(x_k) = x_k \neq 0$, contradiction.

1.3 Linear Independence in Modules

Throughout this subsection, R denotes a nonzero commutative ring with identity and M denotes an R-module.

Definition 1.3.1. *Linear Independence in a Module*

(i) Subset $S \subseteq M$ is *linearly independent over R* if,
$\forall$ distinct $v_1, \ldots, v_n \in S$, $r_1 v_1 + \cdots + r_n v_n = 0 \implies r_1 = \cdots = r_n = 0$.
(ii) A subset $S \subseteq M$ is *linearly dependent* if it is not linearly independent.

Example 1.3.2. *Dependence Without Solving for One Vector*
In the $\mathbb{Z}$-module $\mathbb{Z}$, $3 \cdot 2 - 2 \cdot 3 = 0$. Thus $\{2, 3\}$ is linearly dependent.
However, $2 \notin 3\mathbb{Z}$; $3 \notin 2\mathbb{Z}$, so dependence need not express one element as a scalar multiple of another.

Proposition 1.3.3. *Scaling Independent Sets over a Domain*
Assume R is an integral domain. If $\{v_1, \ldots, v_n\}$ is R-linearly independent and $0 \neq r \in R$, then $\{rv_1, \ldots, rv_n\}$ is R-linearly independent.

Proof. Let $a_1, \ldots, a_n \in R$ and suppose $a_1(rv_1) + \cdots + a_n(rv_n) = 0$. Then $(ra_1)v_1 + \cdots + (ra_n)v_n = 0$.
Independence gives $ra_i = 0$ for all i. Since R is a domain and $r \neq 0$, $a_i = 0$ for all i.
Structural Commentary: Domains preserve independence under nonzero scaling.
Key Insight: The failure of this statement is exactly zero-divisor behavior. □

1.4 Module Homomorphisms, Kernels, and Images

Throughout this subsection, R denotes a nonzero commutative ring with identity.

Definition 1.4.1. *Module Homomorphism*
Let M, N be R-modules and let $\tau : M \to N$.

(i) τ is *R-linear* if, $\forall r, s \in R$ and $u, v \in M$, $\tau(ru + sv) = r\tau(u) + s\tau(v)$, and is then called an *R-homomorphism*.
(ii) The set of all R-homomorphisms $M \to N$ is $\mathrm{Hom}_R(M, N)$.
(iii) An R-homomorphism $M \to M$ is an *endomorphism*.
(iv) An injective, surjective, or bijective R-homomorphism is a *monomorphism*, *epimorphism*, or *isomorphism*.

Proposition 1.4.2. *Hom, Kernel, and Image*
Let M, N be R-modules and $\tau \in \mathrm{Hom}_R(M, N)$.

(i) With $(a\sigma + b\rho)(m) := a\sigma(m) + b\rho(m)$, $\mathrm{Hom}_R(M, N)$ is an R-module.
(ii) $\ker \tau := \tau^{-1}(\{0\}) \leq M$.
(iii) $\mathrm{im}\, \tau := \tau(M) \leq N$.
(iv) τ is a monomorphism $\iff \ker \tau = \{0\}$.

Proof. By Theorem 1.1.6, for $A \subseteq P$ in an R-module P, $A \leq P \iff A \neq \emptyset$ and $\forall a, b \in R$, $\forall x, y \in A$, $ax + by \in A$.

(i) For $\sigma, \rho \in \mathrm{Hom}_R(M, N)$ and $a, b, r, s \in R$, $(a\sigma + b\rho)(ru + sv) = r(a\sigma + b\rho)(u) + s(a\sigma + b\rho)(v)$, where $u, v \in M$, using commutativity of R. Since $0 \in \mathrm{Hom}_R(M, N)$, the displayed criterion gives the claim inside the pointwise module of maps $M \to N$.
(ii) Since $0 \in \ker \tau$, and $x, y \in \ker \tau$, $a, b \in R$ imply $\tau(ax + by) = a\tau(x) + b\tau(y) = 0$,
so the displayed criterion gives $\ker \tau \leq M$.
(iii) Since $0 = \tau(0) \in \mathrm{im}\, \tau$, and $y_i = \tau(x_i) \in \mathrm{im}\, \tau$ $(i = 1, 2)$, $a, b \in R$ imply $ay_1 + by_2 = \tau(ax_1 + bx_2) \in \mathrm{im}\, \tau$, so the displayed criterion gives $\mathrm{im}\, \tau \leq N$.

(iv) If τ is injective, then $\tau(m) = 0 = \tau(0)$ implies $m = 0$, so $\ker \tau = \{0\}$.
Conversely, if $\ker \tau = \{0\}$ and $\tau(m) = \tau(m')$, then $\tau(m - m') = 0$, hence $m = m'$.

Structural Commentary: Kernels and images are the canonical submodules attached to a module map.
Key Insight: Homomorphisms preserve precisely the finite R-linear combinations that define submodules. □

1.5 Quotient Modules, Correspondence, and Isomorphism

Throughout this subsection, R denotes a nonzero commutative ring with identity and M denotes an R-module.

Definition 1.5.1. *Quotient Module*
Let $S \leq M$. Define $u \equiv v \pmod{S} \iff u - v \in S$.
The *quotient module* is $M/S := \{u + S : u \in M\}$; $u + S := \{u + s : s \in S\}$,
with operations $(u + S) + (v + S) := (u + v) + S$; $r(u + S) := ru + S$.

Proposition 1.5.2. *Quotient Operations Are Well-Defined*
Let $S \leq M$. The operations of Definition 1.5.1 are well-defined and make M/S an R-module.
Moreover, $0_{M/S} = S$, $\pi : M \twoheadrightarrow M/S$, $\pi(u) := u + S$, $\ker \pi = S$.

Proof. If $u \equiv u' \pmod{S}$ and $v \equiv v' \pmod{S}$, then $(u + v) - (u' + v') = (u - u') + (v - v') \in S$
and $ru - ru' = r(u - u') \in S$. Thus $+$ and R-scaling on cosets are representative-independent. The module axioms descend from M.
Finally, $0_{M/S} = 0 + S = S$, π is surjective, and $\ker \pi = \{u \in M : u + S = S\} = \{u \in M : u \in S\} = S$. *Structural Commentary*: Quotients collapse exactly the submodule S to zero.
Key Insight: Two representatives define the same coset precisely when their difference is invisible modulo S. □

Theorem 1.5.3. *Correspondence Theorem for Modules*
Let $S \leq M$ and let $\pi : M \twoheadrightarrow M/S$ be the quotient map. Then $\Phi : \{T \leq M : S \leq T\} \longrightarrow \{U \leq M/S\}$, where $\Phi(T) := T/S$ is an order isomorphism, with inverse $\Psi(U) := \pi^{-1}(U)$.

Proof. If $S \leq T \leq M$, then $\Phi(T) = \pi(T) \leq M/S$.
If $U \leq M/S$, then $\Psi(U) = \pi^{-1}(U) \leq M$ and $S = \ker \pi \leq \Psi(U)$.
Moreover, $\Psi\Phi(T) = \pi^{-1}(T/S) = T$ and $\Phi\Psi(U) = \pi(\pi^{-1}(U)) = U$, where the second equality uses surjectivity of π. Finally, $T_1 \leq T_2 \Longrightarrow T_1/S \leq T_2/S$, and $U_1 \leq U_2 \Longrightarrow \pi^{-1}(U_1) \leq \pi^{-1}(U_2)$. Thus Φ is an order isomorphism.
Structural Commentary: Submodules above S are exactly the submodules visible after S is collapsed.
Key Insight: Pull back along the quotient map to recover the original submodule. □

Theorem 1.5.4. *Module Isomorphism Theorems*
The following canonical maps are R-module isomorphisms.

(i) If $\tau \in \mathrm{Hom}_R(M, N)$, then $\overline{\tau} : M/\ker \tau \xrightarrow{\sim} \mathrm{im}\, \tau$ and $\overline{\tau}(m + \ker \tau) := \tau(m)$.
(ii) If $S, T \leq M$, then $\theta : S/(S \cap T) \xrightarrow{\sim} (S + T)/T$ and $\theta\big(s + (S \cap T)\big) := s + T$.
(iii) If $S \leq T \leq M$, then $\eta : (M/S)/(T/S) \xrightarrow{\sim} M/T$ and $\eta\big((m + S) + T/S\big) := m + T$.

Proof.

(i) If $m - m' \in \ker \tau$, then $\tau(m) = \tau(m')$, so $\overline{\tau}$ is well-defined and R-linear.
Also $\mathrm{im}\, \overline{\tau} = \mathrm{im}\, \tau$ and $\ker \overline{\tau} = \{m + \ker \tau : \tau(m) = 0\} = \{0_{M/\ker \tau}\}$.
(ii) Apply (i) to $\varphi \in \mathrm{Hom}_R(S, (S + T)/T)$, $\varphi(s) := s + T$.
Then $\mathrm{im}\, \varphi = (S + T)/T$ and $\ker \varphi = \{s \in S : s \in T\} = S \cap T$.
(iii) Apply (i) to $\psi \in \mathrm{Hom}_R(M/S, M/T)$, $\psi(m + S) := m + T$.
Then $\mathrm{im}\, \psi = M/T$ and $\ker \psi = \{m + S : m \in T\} = T/S$.

Structural Commentary: Each theorem identifies a quotient by the exact ambiguity in a natural map.
Key Insight: Image is domain modulo kernel, repeatedly. □

1.6 Torsion and Annihilators

Throughout this subsection, R denotes a nonzero commutative ring with identity and M denotes an R-module.

Definition 1.6.1. *Torsion*

(i) A nonzero element $v \in M$ is a *torsion element* if $\exists\, 0 \neq r \in R$ such that $rv = 0$.
(ii) M is *torsion-free* if it has no torsion elements.
(iii) M is a *torsion module* if every nonzero element of M is torsion.
(iv) The *torsion set* of M is $M_{\mathrm{tor}} := \{0\} \cup \{v \in M : v \text{ is torsion}\}$

Example 1.6.2. *$\mathbb{Z}/n\mathbb{Z}$ as a Torsion Module*
For $n > 1$, $\mathbb{Z}/n\mathbb{Z}$ is a $\mathbb{Z}$-module via $k \cdot \overline{a} = \overline{ka}$. For every $\overline{a} \in \mathbb{Z}/n\mathbb{Z}$, $n\overline{a} = \overline{0}$.
Hence $\mathbb{Z}/n\mathbb{Z}$ is a torsion module, and $\{\overline{a}\}$ is linearly dependent whenever $\overline{a} \neq \overline{0}$.

Proposition 1.6.3. *Torsion Submodule over an Integral Domain*
If R is an integral domain, then $M_{\mathrm{tor}} \leq M$, $(M/M_{\mathrm{tor}})_{\mathrm{tor}} = \{0\}$.

Proof. Let $x, y \in M_{\mathrm{tor}}$ and $r, s \in R$.
Choose $a, b \in R \setminus \{0\}$ with $ax = 0$ and $by = 0$, taking $a = 1$ if $x = 0$ and $b = 1$ if $y = 0$.
Since R is a domain, $ab \neq 0$, and $ab(rx + sy) = rb(ax) + sa(by) = 0$. Thus $rx + sy \in M_{\mathrm{tor}}$, so $M_{\mathrm{tor}} \leq M$.
Now let $0 \neq r \in R$ and suppose $r(m + M_{\mathrm{tor}}) = M_{\mathrm{tor}}$. Then $rm \in M_{\mathrm{tor}}$, so $srm = 0$ for some $0 \neq s \in R$; if $rm = 0$, take $s = 1$. Since $sr \neq 0$, $m \in M_{\mathrm{tor}}$, hence $m + M_{\mathrm{tor}} = M_{\mathrm{tor}}$.
Structural Commentary: Domains let annihilators be multiplied without becoming zero.
Key Insight: Domains make torsion closed and quotients torsion-free. □

Definition 1.6.4. *Annihilators*

(i) For $v \in M$, the *annihilator* of v is $\mathrm{Ann}_R(v) := \{r \in R : rv = 0\}$.

(ii) For $N \leq M$, the *annihilator* of N is $\mathrm{Ann}_R(N) := \{r \in R : rN = \{0\}\}$.

Proposition 1.6.5. *Annihilator Properties*
For $v \in M$ and $N \leq M$:
(i) $\mathrm{Ann}_R(v) \trianglelefteq R$ *and* $\mathrm{Ann}_R(N) \trianglelefteq R$.
(ii) v *is torsion* $\iff v \neq 0$ *and* $\mathrm{Ann}_R(v) \neq \{0\}$.
(iii) $A \leq B \leq M \implies \mathrm{Ann}_R(B) \subseteq \mathrm{Ann}_R(A)$.

Proof.
(i) Since $0 \in \mathrm{Ann}_R(v)$ and, for $a, b \in \mathrm{Ann}_R(v)$ and $r, s \in R$, $(ra + sb)v = r(av) + s(bv) = 0$, $\mathrm{Ann}_R(v) \trianglelefteq R$. The same computation for every $n \in N$ gives $\mathrm{Ann}_R(N) \trianglelefteq R$.
(ii) ($\Rightarrow$) If v is torsion, then $v \neq 0$ and $\exists\, 0 \neq r \in R$ with $rv = 0$, so $\mathrm{Ann}_R(v) \neq \{0\}$.
(ii) ($\Leftarrow$) If $v \neq 0$ and $\mathrm{Ann}_R(v) \neq \{0\}$, choose $0 \neq r \in \mathrm{Ann}_R(v)$. Then $rv = 0$, so v is torsion.
(iii) If $A \leq B \leq M$ and $r \in \mathrm{Ann}_R(B)$, then $rA = \{0\}$; hence $r \in \mathrm{Ann}_R(A)$.
Structural Commentary: Annihilators reverse inclusion.
Key Insight: Killing more vectors gives fewer scalars. □

Proposition 1.6.6. *Torsion Generators Give a Nonzero Annihilator over a Domain*
Assume R is an integral domain. If $M = \langle\langle u_1, \ldots, u_n \rangle\rangle_R$, $u_i \in M_{\mathrm{tor}}$ $\forall i$, then $\mathrm{Ann}_R(M) \neq \{0\}$.

Proof. Choose $0 \neq a_i \in R$ with $a_i u_i = 0$, and set $a := \prod_{i=1}^{n} a_i$. Since R is a domain, $a \neq 0$.
For each i, $au_i = 0$, hence $aM = \{0\}$ because the u_i generate M. Thus $0 \neq a \in \mathrm{Ann}_R(M)$.
Structural Commentary: Finite generation lets many annihilators be compressed into one product.
Key Insight: A finite torsion generating set gives a global order for the module. □

1.7 Direct Sums and Complements

Throughout this subsection, R denotes a nonzero commutative ring with identity and M denotes an R-module.

Definition 1.7.1. *Direct Sums and Direct Summands*
(i) For R-modules $M_1, \ldots, M_n$, their *external direct sum* is $\bigoplus_{i=1}^{n} M_i := \{(m_i)_{i=1}^n : m_i \in M_i\}$ with componentwise R-module operations.
(ii) For submodules $\{S_i : i \in I\}$ of M, their *internal direct sum* is denoted $M = \bigoplus_{i \in I} S_i$, which means $M = \sum_{i \in I} S_i$ and $\forall i \in I$, $S_i \cap \sum_{j \neq i} S_j = \{0\}$.
(iii) A submodule $S \leq M$ is a *direct summand* if $\exists\, T \leq M$ such that $M = S \oplus T$. Such a T is a *complement* of S in M.

Proposition 1.7.2. *Direct Sum Characterizations*
For a family $\{S_i : i \in I\}$ of distinct submodules of M, the following conditions are equivalent:
(i) $\sum_{i \in I} S_i$ *is direct, i.e.* $\forall k \in I$, $S_k \cap \sum_{i \neq k} S_i = \{0\}$.
(ii) For every $F \subseteq_{\mathrm{fin}} I$ *and* $x_i \in S_i$ $(i \in F)$, $\sum_{i \in F} x_i = 0 \implies x_i = 0\ \forall i \in F$.
(iii) Every $0 \neq x \in \sum_{i \in I} S_i$ *has a unique finite nonzero decomposition,* $\exists!\, (F, (x_i)_{i \in F})$ *where* $F \subseteq_{\mathrm{fin}} I$; $0 \neq x_i \in S_i$; $x = \sum_{i \in F} x_i$.

Proof.
(i) $\Rightarrow$ (ii). If (i) holds and $\sum_{i \in F} x_i = 0$, then any $x_k \neq 0$ gives $x_k = -\sum_{i \in F \setminus \{k\}} x_i \in S_k \cap \sum_{i \neq k} S_i$, a contradiction.
(ii) $\Rightarrow$ (i). If $x \in S_k \cap \sum_{i \neq k} S_i$, write $x = \sum_{i \in F} x_i$ with $F \subseteq_{\mathrm{fin}} I \setminus \{k\}$ and $x_i \in S_i$.
Then $x - \sum_{i \in F} x_i = 0$, so (ii) gives $x = 0$.
(ii) $\Rightarrow$ (iii). Existence follows from the definition of $\sum_i S_i$ after deleting zero terms.
If $\sum_{i \in F} x_i = \sum_{j \in G} y_j$, $0 \neq x_i \in S_i$, $0 \neq y_j \in S_j$, then, over $H = F \cup G$, missing terms set to 0, $\sum_{h \in H} (x_h - y_h) = 0$.
By (ii), $x_h = y_h$ for all $h \in H$, hence $F = G$ and the terms agree.
(iii) $\Rightarrow$ (ii). If $\sum_{i \in F} x_i = 0$, set $F_0 := \{i \in F : x_i \neq 0\}$.
If $F_0 \neq \emptyset$, choose $k \in F_0$. If $F_0 = \{k\}$, then $x_k = 0$, contradiction; otherwise $0 \neq x_k = -\sum_{i \in F_0 \setminus \{k\}} x_i$,
giving two finite nonzero decompositions of x_k, contradiction. Hence $F_0 = \emptyset$.
Structural Commentary: Directness is the absence of hidden cancellation among different summands.
Key Insight: Unique decomposition is controlled by the zero vector. □

Example 1.7.3. *Most Submodules of $\mathbb{Z}$ Are Not Complemented*
As a $\mathbb{Z}$-module, every submodule of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 0$. For $m, n > 0$, $m\mathbb{Z} \cap n\mathbb{Z} = \mathrm{lcm}(m, n)\mathbb{Z} \neq \{0\}$.
Thus $m\mathbb{Z} \oplus n\mathbb{Z} \neq \mathbb{Z}$ for all $m, n > 0$, so the only complemented submodules of $\mathbb{Z}$ are $\{0\}$ and $\mathbb{Z}$.

Theorem 1.7.4. *Complements and Quotients*
If $M = S \oplus T$, then $\pi|_T : T \xrightarrow{\sim} M/S$, $t \mapsto t + S$, where $\pi : M \twoheadrightarrow M/S$ is the quotient map.
Hence any two complements of S are isomorphic.

Proof. $\ker(\pi|_T) = T \cap S = \{0\}$. Also, for $m = s + t$ with $s \in S$, $t \in T$, one has $m + S = t + S$; hence $\mathrm{im}(\pi|_T) = M/S$. Thus $\pi|_T$ is an isomorphism, and all complements are isomorphic to M/S.
Structural Commentary: A complement gives a concrete copy of the quotient inside the original module.
Key Insight: Quotienting by S remembers exactly the complementary coordinate. □

1.8 Retracts, One-Sided Inverses, and Split Maps

Throughout this subsection, R denotes a nonzero commutative ring with identity.

Theorem 1.8.1. *Direct Summands Extend Epimorphisms*
Let $N \leq M$ and let M_1 be an R-module.

(i) If $M = N \oplus H$ and $\sigma : N \twoheadrightarrow M_1$, then $\exists$ a unique $\overline{\sigma} \in \mathrm{Hom}_R(M, M_1)$ such that $\overline{\sigma}(n+h) := \sigma(n)$, $n \in N$, $h \in H$, $\overline{\sigma}|_N = \sigma$, $\overline{\sigma} : M \twoheadrightarrow M_1$, and $\ker \overline{\sigma} = \ker \sigma \oplus H$.

(ii) If $\sigma : N \xrightarrow{\sim} M_1$, then $\mathcal{E}_\sigma := \{\overline{\sigma} : M \twoheadrightarrow M_1 : \overline{\sigma}|_N = \sigma\} \longleftrightarrow \mathcal{C}_N(M) := \{H \leq M : M = N \oplus H\}$ via $\overline{\sigma} \mapsto \ker \overline{\sigma}$ and $H \mapsto \overline{\sigma}_H$, where $\overline{\sigma}_H(n+h) := \sigma(n)$.

Proof.

(i) Direct-sum uniqueness makes $\overline{\sigma}$ well-defined, and componentwise computation gives R-linearity. Moreover, $\mathrm{im}\,\overline{\sigma} = \mathrm{im}\,\sigma = M_1$; $\ker \overline{\sigma} = \{n + h : \sigma(n) = 0\} = \ker \sigma \oplus H$. This shows uniqueness.

(ii) If $H \in \mathcal{C}_N(M)$, then (i) gives $\overline{\sigma}_H \in \mathcal{E}_\sigma$ and, since $\ker \sigma = \{0\}$, $\ker \overline{\sigma}_H = H$. Conversely, if $\overline{\sigma} \in \mathcal{E}_\sigma$, then $N \cap \ker \overline{\sigma} = \{0\}$, and for each $m \in M$, $n := \sigma^{-1}(\overline{\sigma}(m)) \in N$, $m - n \in \ker \overline{\sigma}$. Thus $M = N \oplus \ker \overline{\sigma}$, so the two displayed assignments are inverse.

Structural Commentary: A complement supplies the directions on which an extension may vanish.
Key Insight: Extending an isomorphism is equivalent to choosing its kernel complement. □

Definition 1.8.2. *Retract*
Let $N \leq M$.

(i) A *retraction* onto N is a map $\rho \in \mathrm{Hom}_R(M, N)$ such that $\rho|_N = \mathrm{id}_N$.
(ii) The submodule N is a *retract* of M iff $\exists\, \rho \in \mathrm{Hom}_R(M, N)$ with $\rho|_N = \mathrm{id}_N$.

Corollary 1.8.3. *Retracts Are Direct Summands*
For $N \leq M$, N is a retract of $M \iff \exists\, H \leq M$ such that $M = N \oplus H$.

Proof.

($\Rightarrow$) Let $\rho : M \to N$ satisfy $\rho|_N = \mathrm{id}_N$. For every $m \in M$, $m = \rho(m) + (m - \rho(m))$, $\rho(m) \in N$, $m - \rho(m) \in \ker \rho$. Moreover $N \cap \ker \rho = \{0\}$, so $M = N \oplus \ker \rho$.

($\Leftarrow$) If $M = N \oplus H$, define $\rho : M \to N$ by $\rho(n + h) := n$, $n \in N$, $h \in H$. Then $\rho \in \mathrm{Hom}_R(M, N)$ and $\rho|_N = \mathrm{id}_N$, so N is a retract.

Structural Commentary: Retractions are projections onto submodules.
Key Insight: A submodule is split exactly when it can be recovered by an idempotent map. □

Definition 1.8.4. *One-Sided Inverses*
Let $\tau \in \mathrm{Hom}_R(A, B)$.

(i) A *left inverse* of τ is $\lambda \in \mathrm{Hom}_R(B, A)$ such that $\lambda\tau = \mathrm{id}_A$.
(ii) A *right inverse* of τ is $\rho \in \mathrm{Hom}_R(B, A)$ such that $\tau\rho = \mathrm{id}_B$.
(iii) A *two-sided inverse* of τ is $\upsilon \in \mathrm{Hom}_R(B, A)$ such that $\upsilon\tau = \mathrm{id}_A$ and $\tau\upsilon = \mathrm{id}_B$.

Theorem 1.8.5. *One-Sided Inverses and Complements*
Let $\sigma : M \to M_1$ be an R-homomorphism.
For $S \leq N$, set $\mathcal{C}_S(N) := \{H \leq N : N = S \oplus H\}$, $\mathcal{L}_\sigma := \{\lambda \in \mathrm{Hom}_R(M_1, M) : \lambda\sigma = \mathrm{id}_M\}$,
and $\mathcal{R}_\sigma := \{\rho \in \mathrm{Hom}_R(M_1, M) : \sigma\rho = \mathrm{id}_{M_1}\}$.

(i) If σ is injective, then $\mathcal{L}_\sigma \xrightarrow{\sim} \mathcal{C}_{\mathrm{im}\,\sigma}(M_1)$ by $\lambda \mapsto \ker \lambda$.

(ii) If σ is surjective, then $\mathcal{R}_\sigma \xrightarrow{\sim} \mathcal{C}_{\ker \sigma}(M)$ by $\rho \mapsto \mathrm{im}\,\rho$.

Proof.

(i) For $H \in \mathcal{C}_{\mathrm{im}\,\sigma}(M_1)$, define $\lambda_H(\sigma m + h) := m$. Directness and injectivity give well-definedness; $\lambda_H\sigma = \mathrm{id}_M$ and $\ker \lambda_H = H$. Conversely, if $\lambda \in \mathcal{L}_\sigma$, then $\mathrm{im}\,\sigma \cap \ker \lambda = \{0\}$, since $\sigma m \in \ker \lambda \Rightarrow m = 0$, and $y = \sigma\lambda y + (y - \sigma\lambda y)$ with $y - \sigma\lambda y \in \ker \lambda$. Thus $M_1 = \mathrm{im}\,\sigma \oplus \ker \lambda$, and the constructions are inverse.

(ii) For $H \in \mathcal{C}_{\ker \sigma}(M)$, $\sigma|_H : H \to M_1$ is an isomorphism; let $\rho_H = (\sigma|_H)^{-1}$ followed by $H \hookrightarrow M$. Then $\rho_H \in \mathcal{R}_\sigma$ and $\mathrm{im}\,\rho_H = H$. Conversely, if $\rho \in \mathcal{R}_\sigma$, then ρ is injective, $\mathrm{im}\,\rho \cap \ker \sigma = \{0\}$, and $m = (m - \rho\sigma m) + \rho\sigma m$ with $m - \rho\sigma m \in \ker \sigma$, $\rho\sigma m \in \mathrm{im}\,\rho$. Thus $M = \ker \sigma \oplus \mathrm{im}\,\rho$, and the constructions are inverse.

Structural Commentary: One-sided inverses exist exactly when the relevant image or kernel splits.
Key Insight: Injective maps need a complemented image; surjective maps need a complemented kernel. □

Proposition 1.8.6. *Split Epimorphisms Split the Domain*
Let $\sigma : M \twoheadrightarrow N$ be an R-homomorphism and set $\mathcal{R}_\sigma := \{\rho \in \mathrm{Hom}_R(N, M) : \sigma\rho = \mathrm{id}_N\}$. These are equivalent:

(i) $\mathcal{R}_\sigma \neq \emptyset$.
(ii) $\exists\, H \leq M$ such that $M = \ker \sigma \oplus H$.

Moreover, $\rho \in \mathcal{R}_\sigma \Longrightarrow M = \ker \sigma \oplus \mathrm{im}\,\rho$.

Proof.

($\Rightarrow$) Let $\rho \in \mathcal{R}_\sigma$. If $\rho n = 0$, then $n = \sigma\rho n = 0$, so ρ is injective.
If $\rho n \in \ker \sigma$, then $n = \sigma\rho n = 0$, hence $\mathrm{im}\,\rho \cap \ker \sigma = \{0\}$.
For $m \in M$, $m = (m - \rho\sigma m) + \rho\sigma m$, with $m - \rho\sigma m \in \ker \sigma$ and $\rho\sigma m \in \mathrm{im}\,\rho$. Thus $M = \ker \sigma \oplus \mathrm{im}\,\rho$.

($\Leftarrow$) Let $M = \ker \sigma \oplus H$. Then $\ker(\sigma|_H) = H \cap \ker \sigma = \{0\}$, and $\mathrm{im}(\sigma|_H) = N$ since $n = \sigma m$, $m = k + h$, gives $n = \sigma h$. Thus $\sigma|_H : H \xrightarrow{\sim} N$, and $\rho := \iota_H(\sigma|_H)^{-1} \in \mathcal{R}_\sigma$, where $\iota_H : H \hookrightarrow M$.

Structural Commentary: A split epimorphism is a quotient map with complemented kernel.
Key Insight: The right inverse chooses canonical representatives for the cosets of $\ker \sigma$. □

2 Noetherian Modules and PID Structure Theorems

2.1 Free Modules, Bases, and Universal Property

This section begins with the structure facts that make free modules and finite generation usable: rank, splitting maps onto free modules, Noetherian modules. It then develops the classification of finitely generated modules over a principal ideal domain. The guiding model is the abelian-group theorem over $\mathbb{Z}$ and, later, the polynomial-module classification behind rational canonical form.

Definition 2.1.1. *Free Module and Basis*

(i) A subset $\mathcal{B} \subseteq M$ is a *basis* of M if $\mathcal{B}$ is R-linearly independent and $\langle\langle \mathcal{B} \rangle\rangle_R = M$.
(ii) An R-module M is *free* if $M = \{0\}$ or M has a basis.
(iii) If $\mathcal{B}$ is a basis of M, then M is *free on* $\mathcal{B}$.

Theorem 2.1.2. *Basis Means Unique Finite Coordinates*
A subset $\mathcal{B} \subseteq M$ is a basis if and only if $\forall v \in M,\ \exists!\, (F, (r_b)_{b\in F})$, with $F \subseteq_{\text{fin}} \mathcal{B}$, $r_b \in R \setminus \{0\}$, $v = \sum_{b\in F} r_b b$.

The case $F = \emptyset$ gives the empty sum 0.

Proof.

(⇒) Spanning gives a finite expression; collecting equal basis elements and deleting zero coefficients gives the displayed form. If $\sum_{b\in F} r_b b = \sum_{b\in G} s_b b$, then $\sum_{b\in F\cup G} (r_b - s_b)b = 0$, where missing coefficients are 0.
Independence gives $r_b = s_b$ for all b, hence $F = G$ and the coefficients agree.
(⇐) Existence gives $\langle\langle \mathcal{B} \rangle\rangle_R = M$. If $\sum_{b\in F} r_b b = 0$ and $F_0 := \{b \in F : r_b \neq 0\}$ is nonempty, then $0 = \sum_{b\in F_0} r_b b$ contradicts uniqueness of the empty expression. Thus all $r_b = 0$, so $\mathcal{B}$ is independent.

Structural Commentary: Bases give unique finite coordinates.
Key Insight: Existence is spanning; uniqueness is independence. □

Proposition 2.1.3. *Bases Are Minimal Spanning and Maximal Independent*
If $\mathcal{B}$ is a basis of M, then $\forall b \in \mathcal{B}$, $\langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R \neq M$, and $\forall v \in M \setminus \mathcal{B}$, $\mathcal{B} \cup \{v\}$ is R-linearly dependent.

Proof. If $b \in \mathcal{B}$ and $b \in \langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R$, then b has two finite coordinate expressions, contradicting Theorem 2.1.2.
Hence deleting b destroys spanning. If $v \notin \mathcal{B}$, write $v = \sum_{b\in F} r_b b$ with $F \subseteq_{\text{fin}} \mathcal{B}$.
Then $v - \sum_{b\in F} r_b b = 0$ is a nontrivial relation in $\mathcal{B} \cup \{v\}$, so $\mathcal{B} \cup \{v\}$ is dependent.
Structural Commentary: Basis still implies the familiar minimal/maximal properties.
Key Insight: The converses are vector-space privileges, not general module facts. □

Theorem 2.1.4. *Maps Out of a Free Module*
Let M be free on $\mathcal{B}$ and let N be an R-module.
For every function $f : \mathcal{B} \to N$, there exists a unique $\widetilde{f} \in \mathrm{Hom}_R(M, N)$ such that $\widetilde{f}|_{\mathcal{B}} = f$.

Proof. For $v = \sum_{b\in F} r_b b$ with $F \subseteq_{\text{fin}} \mathcal{B}$, define $\widetilde{f}(v) := \sum_{b\in F} r_b f(b)$.
Theorem 2.1.2 makes the expression of v unique, so $\widetilde{f}$ is well-defined.
For $x = \sum a_b b$ and $y = \sum c_b b$, combining supports gives $\widetilde{f}(rx + sy) = r\widetilde{f}(x) + s\widetilde{f}(y)$, so $\widetilde{f}$ is R-linear and clearly extends f. If $\phi \in \mathrm{Hom}_R(M, N)$ also extends f, then $\phi\left(\sum_{b\in F} r_b b\right) = \sum_{b\in F} r_b f(b) = \widetilde{f}\left(\sum_{b\in F} r_b b\right)$, hence $\phi = \widetilde{f}$.
Structural Commentary: A free module is controlled entirely by its basis values.
Key Insight: Freeness is the universal permission to choose images of basis elements arbitrarily. □

Example 2.1.5. *A Free Module with a Nonfree Submodule*
Let $R = \mathbb{Z} \times \mathbb{Z}$ with componentwise multiplication. As an R-module, $R = R(1, 1)$, so R is free on $\{(1, 1)\}$.
Set $N := \mathbb{Z} \times \{0\} \leq R$. For every $0 \neq (a, 0) \in N$, $(0, 1)(a, 0) = 0$.
Thus every nonzero element of N is torsion, so N has no nonempty independent subset and is not free.

Example 2.1.6. *A Quotient of a Free Module Need Not Be Free*
As a $\mathbb{Z}$-module, $\mathbb{Z}$ is free on $\{1\}$. For $n > 1$, the submodule $n\mathbb{Z}$ is cyclic, but the quotient $\mathbb{Z}/n\mathbb{Z}$ is torsion by Example 1.6.2. Hence $\mathbb{Z}/n\mathbb{Z}$ is not free as a $\mathbb{Z}$-module.

Principle 2.1.7. *Modules Are Not Just Vector Spaces with New Letters*
For a general ring R, the following vector-space implications fail:

(i) $N \leq M \not\Rightarrow \exists\, H \leq M$ with $M = N \oplus H$.
(ii) M finitely generated and $N \leq M \not\Rightarrow N$ finitely generated.
(iii) $M \neq 0 \not\Rightarrow \exists\, \emptyset \neq S \subseteq M$ linearly independent.
(iv) S minimal spanning or maximal independent $\not\Rightarrow S$ is a basis.
(v) M free and $N \leq M \not\Rightarrow N$ free.
(vi) F free and $S \leq F \not\Rightarrow F/S$ free.

Over a field, these obstructions vanish because $R \setminus \{0\} \subseteq R^\times$.

2.2 Rank and Coordinate Models of Free Modules

Perspective 2.2.1. *Dimension as the Field Case of Rank*
An $\mathbb{F}$-vector space is a free $\mathbb{F}$-module. By Theorem 1.4.1, finite bases have invariant cardinality; over general rings, rank invariance requires additional hypotheses.

Theorem 2.2.2. *Rank Is Well Defined fo r Free Modules*
Let A benonzero commutative ring with identity, M be free A-module, $\mathrm{Bas}_A(M)$ denote set of A-bases of M.

(i) $B, B' \in \mathrm{Bas}_A(M) \Longrightarrow |B| = |B'|$.
(ii) $B \in \mathrm{Bas}_A(M)$, $S \subseteq M$, *and* $\langle\!\langle S \rangle\!\rangle_A = M \Longrightarrow |B| \leq |S|$.

Proof.

(i) Choose a maximal ideal $\mathcal{I} \lhd A$ by Theorem 3.4.5.
Put $K := A/\mathcal{I}$, a field by Theorem 3.4.4, $\mathcal{I}M := \langle\!\langle im : i \in \mathcal{I},\ m \in M \rangle\!\rangle_A$, and $V := M/\mathcal{I}M$.
Since $\mathcal{I}V = 0$, V is a K-vector space via $(a+\mathcal{I})(m+\mathcal{I}M) := am + \mathcal{I}M$.

(ii) Let $B \in \mathrm{Bas}_A(M)$ and set $\overline{B} := \{\overline{b} := b + \mathcal{I}M : b \in B\}$.
In B-coordinates, $\mathcal{I}M = \{\sum_{b\in F} i_b b : F \subseteq_{\mathrm{fin}} B,\ i_b \in \mathcal{I}\}$; hence $\overline{B}$ spans V.
If $\sum_{j=1}^{n} (a_j + \mathcal{I})\overline{b_j} = 0$ with distinct $b_j \in B$, then $\sum_{j=1}^{n} a_j b_j \in \mathcal{I}M$, so $a_j \in \mathcal{I}\ \forall j$; hence $\overline{B}$ is K-independent.
If $\overline{b} = \overline{b'}$ with $b \neq b'$, then $b - b' \in \mathcal{I}M$, forcing the B-coefficient $1 \in \mathcal{I}$, impossible. Thus $B \xrightarrow{\sim} \overline{B}$.

(iii) Therefore $|B| = |\overline{B}| = \dim_K V$ for every $B \in \mathrm{Bas}_A(M)$, so (i) follows from Theorem 1.4.5.
If $\langle\!\langle S \rangle\!\rangle_A = M$, then $\overline{S} := \{s + \mathcal{I}M : s \in S\}$ spans V, so $|B| = \dim_K V \leq |\overline{S}| \leq |S|$.

Structural Commentary: Modulo a maximal ideal, free module becomes a vector space with same basis size.
Key Insight: The field quotient $A/\mathcal{I}$ lets vector-space dimension certify module rank. □

Definition 2.2.3. *Rank of a Free Module*
Let A be a nonzero commutative ring with identity. The *rank* of a free A-module M, denoted $\mathrm{rk}_A(M)$, is the cardinality of any basis of M. The zero module has rank 0.

Example 2.2.4. *Why Commutativity Matters for Rank*
Let $V = \bigoplus_{k\geq 0} \mathbb{F} e_k$ and $E = \mathcal{L}(V)$. As a left E-module, $E = E\,\mathrm{id}_V$, so $\{\mathrm{id}_V\}$ is a basis.
For each $n \geq 1$, define $\beta_s \in E$, $0 \leq s < n$, by $\beta_s(e_{kn+t}) = \begin{cases} e_k, & t = s, \\ 0, & t \neq s, \end{cases}$ where $k \geq 0$, $0 \leq t < n$.
If $\sum_{s=0}^{n-1} \alpha_s \beta_s = 0$, then evaluating at e_{kn+t} gives $\alpha_t(e_k) = 0$ for all k, t, hence $\alpha_t = 0$ for all t.
Conversely, for $h \in E$, define $\alpha_s(e_k) := h(e_{kn+s})$; then $h = \sum_{s=0}^{n-1} \alpha_s \beta_s$.
Thus $\{\beta_0, \ldots, \beta_{n-1}\}$ is an E-basis of E, so E has bases of all positive finite cardinalities.
Structural Commentary: Noncommutative scalar rings can make a free module have incompatible basis sizes.
Key Insight: The rank theorem is a commutative-ring theorem, not a formal consequence of freeness alone.

Theorem 2.2.5. *Finite-Support Coordinate Modules*
Let A be a commutative ring with identity and let B be a set. Put $A^{(B)} := \{f \in A^B : |\mathrm{supp}(f)| < \infty\}$.
For $b \in B$, let $\delta_b \in A^{(B)}$ be given by $\delta_b(b) = 1$ and $\delta_b(x) = 0$ for $x \neq b$.
Then $A^{(B)} = \bigoplus_{b\in B} A\delta_b$, so $\{\delta_b : b \in B\}$ is an A-basis. If $A \neq 0$, then $\mathrm{rk}_A(A^{(B)}) = |B|$.

Proof. For $f, g \in A^{(B)}$ and $a \in A$, $\mathrm{supp}(f+g) \subseteq \mathrm{supp}(f) \cup \mathrm{supp}(g)$ and $\mathrm{supp}(af) \subseteq \mathrm{supp}(f)$, hence $A^{(B)} \leq A^B$.
For $f \in A^{(B)}$, $f = \sum_{b\in \mathrm{supp}(f)} f(b)\delta_b$, so the δ_b span. If $\sum_{j=1}^{m} a_j \delta_{b_j} = 0$ with $b_1, \ldots, b_m$ distinct, then evaluation at b_i gives $a_i = 0$ for every i. Thus the δ_b are independent, and the rank statement follows from the displayed basis.
Structural Commentary: Free modules are finite-support coordinate modules on their bases.
Key Insight: Finite support is the module-theoretic version of finite linear combination. □

Proposition 2.2.6. *Free Modules Are Coordinate Modules*
Let A be a commutative ring with identity and let M be free on a basis B.
Then the coordinate map $\Phi_B : M \to A^{(B)}$ is an isomorphism.
If $A \neq 0$ and M, N are free A-modules, then $M \cong N \Longleftrightarrow \mathrm{rk}_A(M) = \mathrm{rk}_A(N)$.

Proof. For $m = \sum_{b\in F} a_b b$, set $\Phi_B(m) = f_m$, with $f_m(b) = a_b$ for $b \in F$ and $f_m(b) = 0$ for $b \notin F$.
Unique coordinates give $\Phi_B : M \xrightarrow{\sim} A^{(B)}$, with inverse $f \mapsto \sum_{b\in \mathrm{supp}(f)} f(b)b$.

($\Rightarrow$) If $\psi : M \xrightarrow{\sim} N$ and C is a basis of M, then $\psi(C)$ is a basis of N, so $\mathrm{rk}_A(M) = |C| = |\psi(C)| = \mathrm{rk}_A(N)$.
($\Leftarrow$) If $\mathrm{rk}_A(M) = \mathrm{rk}_A(N)$, choose bases C, D with $|C| = |D|$.
Any bijection $C \to D$ extends, by Theorem 2.1.4, to $M \xrightarrow{\sim} N$.

Structural Commentary: A basis identifies a free module with a standard coordinate model.
Key Insight: Over a nonzero commutative ring, rank classifies free modules up to isomorphism. □

Theorem 2.2.7. *Independent Sets Are Bounded by Rank over a Domain*
Let A be integral domain with field of quotients Q, M free over A, $S \subseteq M$ A-independent. Then $|S| \leq \mathrm{rk}_A(M)$.

Proof. Choose an A-basis B and identify $M = A^{(B)} \subseteq Q^{(B)}$.
If $\sum_{i=1}^{n} q_i s_i = 0$ with $q_i \in Q$ and distinct $s_i \in S$, choose $d \in A \setminus \{0\}$ with $dq_i \in A$ for all i.
Then $\sum_{i=1}^{n} (dq_i) s_i = 0$ in M, so $dq_i = 0$ by A-independence, hence $q_i = 0$ for all i.
Thus S is Q-independent in $Q^{(B)}$, and $|S| \leq \dim_Q Q^{(B)} = |B| = \mathrm{rk}_A(M)$.
Structural Commentary: The field of quotients lets ordinary dimension bound module independence.
Key Insight: Domains allow denominator clearing without creating false zero relations. □

2.3 Free Quotients and Splittings

Theorem 2.3.1. *Free Quotients Split*
Let A be a commutative ring with identity.

(i) If $\sigma : M \twoheadrightarrow F$ and F is free, then $\exists N \leq M$ such that $M = \ker(\sigma) \oplus N$ and $N \cong F$.
(ii) If $S \leq M$ and M/S is free, then $\exists T \leq M$ such that $M = S \oplus T$ and $T \cong M/S$; hence $M \cong S \oplus M/S$.
(iii) If $A \neq 0$ and $M, S, M/S$ are free, then $\mathrm{rk}_A(M) = \mathrm{rk}_A(S) + \mathrm{rk}_A(M/S)$, with cardinal addition; in finite rank, $\mathrm{rk}_A(M/S) = \mathrm{rk}_A(M) - \mathrm{rk}_A(S)$.

Proof.

(i) Let B be a basis of F. For $b \in B$, choose $m_b \in M$ with $\sigma(m_b) = b$. By Theorem 2.1.4, $b \mapsto m_b$ extends to $\tau : F \to M$. Since $\sigma\tau = \mathrm{id}_F$, Theorem 1.8.5 gives $M = \ker(\sigma) \oplus \mathrm{im}(\tau)$ and $\mathrm{im}(\tau) \cong F$.
(ii) Apply (i) to $\pi : M \twoheadrightarrow M/S$. Then $M = S \oplus T$ for some $T \cong M/S$, so $M \cong S \oplus M/S$.
(iii) From (ii), $M = S \oplus T$ with $T \cong M/S$. If B_S, B_T are bases of S, T, then $B_S \sqcup B_T$ is a basis of M, hence $\mathrm{rk}_A(M) = |B_S| + |B_T| = \mathrm{rk}_A(S) + \mathrm{rk}_A(M/S)$. The finite formula follows.

Structural Commentary: Free modules are projective: maps onto them have linear sections.
Key Insight: Choosing representatives for basis vectors of the quotient is enough to split the quotient map. □

2.4 Noetherian Modules

Definition 2.4.1. *Noetherian Modules and Rings*
Let A be a commutative ring with identity.

(i) An A-module M satisfies *ACC* if every chain $M_1 \leq M_2 \leq \cdots$ of submodules satisfies $\exists n_0, \forall n \geq n_0, M_n = M_{n_0}$. Such M is *Noetherian*.
(ii) The ring A is *Noetherian* if the regular A-module A is Noetherian, equivalently if every ideal-chain $I_1 \subseteq I_2 \subseteq \cdots$ stabilizes.

Theorem 2.4.2. *Noetherian Means Finitely Generated Submodules*
Let M be an A-module.

(i) M is Noetherian $\iff$ every $N \leq M$ is f.g.
(ii) If A is commutative with identity, then A is Noetherian $\iff$ every $\mathcal{I} \lhd A$ is f.g.

Proof.

(i) ($\Rightarrow$) Let $N \leq M$. If N is not f.g., choose $u_{n+1} \in N \setminus \langle\langle u_1, \ldots, u_n \rangle\rangle_A$ inductively. Then $\langle\langle u_1 \rangle\rangle_A < \langle\langle u_1, u_2 \rangle\rangle_A < \cdots$, contradicting ACC.
(i) ($\Leftarrow$) Let $M_1 \leq M_2 \leq \cdots$ and put $U = \bigcup_{n \geq 1} M_n$. Then $U \leq M$, so $U = \langle\langle u_1, \ldots, u_r \rangle\rangle_A$. Choose k with $u_1, \ldots, u_r \in M_k$. Hence $U = M_k$ and $M_n = M_k$ for all $n \geq k$.
(ii) Apply (i) to the regular A-module A: its submodules are exactly the ideals of A.

Structural Commentary: ACC and finite generation are two faces of the same finiteness condition.
Key Insight: Failure of finite generation builds an infinite increasing chain one new generator at a time. □

2.5 Orders and Cyclic Modules over a PID

Definition 2.5.1. *Standing PID Notation*
In the PID-structure results, R denotes a principal ideal domain unless otherwise stated.
For $r \in R$, write $\langle r \rangle = Rr$; for an R-module M and $v \in M$, write $\langle\langle v \rangle\rangle_R = Rv$.

Perspective 2.5.2. *PID Arithmetic Controls Module Structure*
Fields have $F \setminus \{0\} \subseteq F^\times$. PIDs replace invertibility by factorization, gcds, and lcms.
This divisibility data classifies finitely generated modules.

Definition 2.5.3. *Order of a Submodule or Element*
Let M be an R-module and $N \leq M$. Since $\mathrm{Ann}_R(N) = \{r \in R : rN = 0\} = \langle \alpha \rangle$, any such α is an *order* of N; write $o(N) = \alpha$, up to associates. For $v \in M$, set $o(v) := o(\langle\langle v \rangle\rangle_R)$.

Proposition 2.5.4. *Orders and Direct Sums*
Let $M = A \oplus B$. If $o(A) = \alpha$ and $o(B) = \beta$, then $o(M) \sim \mathrm{lcm}(\alpha, \beta)$.
More generally, $o(\bigoplus_{i=1}^{n} M_i) \sim \mathrm{lcm}(o(M_1), \ldots, o(M_n))$.

Proof. $r(A \oplus B) = 0 \iff rA = 0$ and $rB = 0$.
Hence $\mathrm{Ann}_R(A \oplus B) = \mathrm{Ann}_R(A) \cap \mathrm{Ann}_R(B) = \langle \alpha \rangle \cap \langle \beta \rangle = \langle \mathrm{lcm}(\alpha, \beta) \rangle$.
Thus $o(A \oplus B) \sim \mathrm{lcm}(\alpha, \beta)$, and induction gives the finite case.
Structural Commentary: Direct sums intersect annihilator conditions.
Key Insight: Lcms appear because one scalar must kill every summand at once. □

Theorem 2.5.5. *Cyclic Modules over a PID*
Let M be an R-module and let $v \in M$.

(i) If $\mathrm{Ann}_R(v) = \langle\alpha\rangle$, then $R/\langle\alpha\rangle \xrightarrow{\sim} \langle\langle v\rangle\rangle_R$, $r + \langle\alpha\rangle \mapsto rv$.
(ii) If $C = \langle\langle v\rangle\rangle_R$ and $S \le C$, then $S = \langle\langle sv\rangle\rangle_R$ for some $s \in R$.
(iii) If $o(v) = \alpha$ and $\beta \in R$, then $o(\beta v) \sim \alpha/\gcd(\alpha,\beta)$, and $\langle\langle \beta v\rangle\rangle_R = \langle\langle v\rangle\rangle_R \iff \gcd(\alpha,\beta) \sim 1$.

Proof.

(i) The map $\phi : R \to \langle\langle v\rangle\rangle_R$, $\phi(r) = rv$, is onto and $\ker\phi = \mathrm{Ann}_R(v) = \langle\alpha\rangle$. Hence $R/\langle\alpha\rangle \xrightarrow{\sim} \langle\langle v\rangle\rangle_R$.
(ii) For $S \le C$, set $I = \{r \in R : rv \in S\} \lhd R$.
Since R is a PID, $I = \langle s\rangle$ for some $s \in R$, and $S = \{rv : r \in I\} = \langle\langle sv\rangle\rangle_R$.
(iii) Put $d = \gcd(\alpha,\beta)$, $\alpha = d\alpha'$, and $\beta = d\beta'$ with $\gcd(\alpha',\beta') \sim 1$.
For $r \in R$, $r\beta v = 0 \iff \alpha \mid r\beta \iff \alpha' \mid r\beta' \iff \alpha' \mid r$, the last step by Bezout.
Thus $\mathrm{Ann}_R(\beta v) = \langle\alpha'\rangle = \langle\alpha/\gcd(\alpha,\beta)\rangle$. Finally, $\langle\langle\beta v\rangle\rangle_R = \langle\langle v\rangle\rangle_R \iff (\exists c \in R)(v = c\beta v)$
$\iff (\exists c \in R)(\alpha \mid 1 - c\beta) \iff \gcd(\alpha,\beta) \sim 1$.

Structural Commentary: A cyclic module is the base ring modulo the first relation satisfied by its generator.
Key Insight: Multiplying a cyclic generator divides its order by the gcd of the multiplier and the old order. □

Theorem 2.5.6. *Composing and Decomposing Cyclic Modules*
Let M be an R-module.

(i) If $o(u_i) = \alpha_i \neq 0$ and $\gcd(\alpha_i,\alpha_j) \sim 1$ for $i \neq j$, then $o(\sum_{i=1}^n u_i) \sim \prod_{i=1}^n \alpha_i$ and $\bigoplus_{i=1}^n \langle\langle u_i\rangle\rangle_R = \langle\langle \sum_{i=1}^n u_i\rangle\rangle_R$.

(ii) If $o(v) = \alpha = \prod_{i=1}^n \alpha_i \neq 0$ and $\gcd(\alpha_i,\alpha_j) \sim 1$ for $i \neq j$,
then $\exists v_1,\dots,v_n \in M$ such that $v = \sum_{i=1}^n v_i$, $o(v_i) \sim \alpha_i$, and $\langle\langle v\rangle\rangle_R = \bigoplus_{i=1}^n \langle\langle v_i\rangle\rangle_R$.

Proof.

(i) It suffices to prove $n = 2$ and iterate. Let $o(u) = \alpha$, $o(w) = \beta$, and $\gcd(\alpha,\beta) \sim 1$.
If $x \in \langle\langle u\rangle\rangle_R \cap \langle\langle w\rangle\rangle_R$, then $\alpha x = \beta x = 0$, so Bezout gives $x = 0$. If $r(u+w) = 0$, then $ru = -rw$ lies in this intersection; hence $ru = rw = 0$, so $\alpha \mid r$ and $\beta \mid r$, hence $\alpha\beta \mid r$. Since $\alpha\beta(u+w) = 0$, $o(u+w) \sim \alpha\beta$. Choose $a,b \in R$ with $a\alpha + b\beta = 1$. Then $u = b\beta(u+w)$ and $w = a\alpha(u+w)$, so $\langle\langle u\rangle\rangle_R \oplus \langle\langle w\rangle\rangle_R = \langle\langle u+w\rangle\rangle_R$.
(ii) Put $\beta_i = \alpha/\alpha_i$. Since $\gcd(\beta_1,\dots,\beta_n) \sim 1$, choose $c_i \in R$ with $\sum_{i=1}^n c_i\beta_i = 1$, and set $v_i = c_i\beta_i v$.
Then $v = \sum_{i=1}^n v_i$. Also $\gcd(c_i,\alpha_i) \sim 1$ and $\gcd(\alpha,\beta_i) \sim \beta_i$, so Theorem 2.5.5 gives $o(v_i) \sim \alpha_i$.
Part (i) gives $\langle\langle v\rangle\rangle_R = \bigoplus_{i=1}^n \langle\langle v_i\rangle\rangle_R$.

Structural Commentary: Relatively prime cyclic pieces merge into one cyclic piece.
Key Insight: Bezout identities recover each component from the single sum. □

Proposition 2.5.7. *Prime Divisors Give Prime-Order Cyclic Submodules*
Let M be f.g. torsion with order μ. If $p \mid \mu$ is prime, then $\exists w \in M$ with $o(w) \sim p$; equivalently $Rw \cong R/\langle p\rangle$.

Proof. Write $\mu = pq$. Since $q \notin \mathrm{Ann}_R(M) = \langle\mu\rangle$, choose $v \in M$ with $w := qv \neq 0$.
Then $pw = \mu v = 0$, so $o(w) \mid p$. Since $w \neq 0$, $o(w)$ is nonunit; as p is prime, $o(w) \sim p$.
Structural Commentary: Every prime in the global order appears in some one-dimensional primary layer.
Key Insight: Divide the order by one prime and apply the remaining scalar to an element it does not kill. □

2.6 Submodules of Finite Free PID Modules

Perspective 2.6.1. *Noetherian Modules over a PID*
Over a PID, $N \le R^n$ implies N free with $\mathrm{rk}_R(N) \le n$; Noetherianity is the f.g. shadow.

Theorem 2.6.2. *Submodules of Finite Free PID Modules*
Let F be free over R, $\mathrm{rk}_R(F) = n < \infty$, and $N \le F$. Then $\exists m \le n$ such that $N \cong R^m$.
Consequently, for $\mu_R(M) := \min\{|X| : \langle\langle X\rangle\rangle_R = M\}$, $\mu_R(M) \le n$ implies $S \le M \Longrightarrow \mu_R(S) \le n$.

Proof. Identify $F \simeq R^n$. For $1 \le k \le n$, set $I_k := \{x_k : x = (x_1,\dots,x_k,0,\dots,0) \in N\} \lhd R$ and write $I_k = \langle r_k\rangle$..
If $r_k \neq 0$, choose $u_k \in N \cap (R^k \times \{0\}^{n-k})$ with $(u_k)_k = r_k$. Let $K = \{k : r_k \neq 0\}$.
If $\sum_{k\in K} c_k u_k = 0$ and $j = \max\{k : c_k \neq 0\}$, then the jth coordinate gives $c_j r_j = 0$, hence $c_j = 0$, contradiction.
Thus $\{u_k : k \in K\}$ is R-independent. Set $\ell(0) = 0$ and $\ell(x) = \max\mathrm{supp}(x)$ for $0 \neq x \in N$.
If $j = \ell(x) > 0$, then $x_j \in I_j = \langle r_j\rangle$, say $x_j = cr_j$, and $\ell(x - cu_j) < j$.
Induction on $\ell(x)$ gives $N = \bigoplus_{k\in K} Ru_k \cong R^{|K|}$, so $\mathrm{rk}_R(N) = |K| \le n$.
If $\mu_R(M) \le n$, choose $\pi : R^n \twoheadrightarrow M$. For $S \le M$, $P := \pi^{-1}(S) \le R^n$ has $P = \langle\langle v_1,\dots,v_m\rangle\rangle_R$ with $m \le n$.
Hence $S = \langle\langle \pi(v_1),\dots,\pi(v_m)\rangle\rangle_R$ and $\mu_R(S) \le m \le n$.
Structural Commentary: PIDs let one choose pivot generators in a submodule of R^n.
Key Insight: Pivot ideals I_k yield triangular free generators. □

Corollary 2.6.3. *Finitely Generated PID Modules Are Noetherian*
If $\mu_R(M) < \infty$, then M is Noetherian.

Proof. Let $n = \mu_R(M)$. By Theorem 2.6.2, $N \le M \Longrightarrow \mu_R(N) \le n < \infty$.
Thus every submodule of M is f.g.; Theorem 2.4.2 gives M Noetherian.
Structural Commentary: Finite generation of every submodule prevents infinite strict growth.
Key Insight: Generators of the union appear at a finite stage. □

Proposition 2.6.4. *A Full-Size Spanning Set Is a Basis*
Let $\mathrm{rk}_R(F) = n < \infty$. *If* $F = \langle\!\langle s_1, \dots, s_n \rangle\!\rangle_R$, *then* $(s_1, \dots, s_n)$ *is an* R*-basis of* F.

Proof. Choose a basis $(b_i)_{i=1}^n$ and define $\tau \in \mathrm{End}_R(F)$ by $\tau(b_i) = s_i$. Then τ is onto.
Since F is free, $\exists \rho : F \to F$ with $\tau\rho = \mathrm{id}_F$; hence $F = \ker\tau \oplus \mathrm{im}\,\rho$ and $\mathrm{im}\,\rho \cong F$.
By Theorem 2.6.2, $\ker\tau \cong R^r$ for some $r \le n$. Thus $R^n \cong F \cong \ker\tau \oplus F \cong R^{r+n}$, so $r = 0$ by Theorem 2.2.2. Hence $\ker\tau = 0$, τ is an isomorphism, and $(s_i) = \tau(b_i)$ is a basis.
Structural Commentary: Surjective finite-free endomorphisms over a PID are injective.
Key Insight: Rank leaves no room for a split kernel. □

2.7 Smith Normal Form and Module Presentations

Perspective 2.7.1. *Smith Form as Diagonalized Relations*
For a presentation $R^n \xrightarrow{A} R^m \to M \to 0$, Smith form replaces A by a diagonal relation matrix, decoupling finite relations into cyclic constraints.

Theorem 2.7.2. *Smith Normal Form over a PID*
Let $A \in \mathrm{Mat}(m, n, R)$, *viewed as* $R^n \to R^m$.
$\exists$ *invertible* $P \in \mathrm{Mat}(m, m, R)$ *and* $Q \in \mathrm{Mat}(n, n, R)$ *such that* $PAQ = \mathrm{diag}(d_1, \dots, d_r, 0, \dots, 0)$,
where $d_i \in R \setminus \{0\}$, $d_1 \mid \cdots \mid d_r$, *and* $r = \mathrm{rk}_R(\mathrm{im}\,A)$. *The ideals* $\langle d_i \rangle$ *are unique.*

Proof.
(i) *Existence*: If $A = 0$, done. Set $I(A) = \langle a_{ij} \rangle = \langle d \rangle$. Elementary operations move a Bezout combination equal to d to position $(1,1)$. If a pivot $p \nmid a$, Bezout on (p, a) replaces p by $\gcd(p, a)$, so $\langle p \rangle$ strictly increases; ACC in a PID forces termination. Then p divides every entry, so clear its row and column.
Induct on the complementary submatrix. Finally replace each bad adjacent pair (a, b) by $(\gcd(a, b), \mathrm{lcm}(a, b))$ until $d_1 \mid \cdots \mid d_r$.
(ii) *Uniqueness*: Let $\Delta_k(A)$ be the ideal of $k \times k$ minors and $\Delta_0(A) = R$. Matrix equivalence preserves each Δ_k. In Smith form, $\Delta_k(A) = \langle d_1 \cdots d_k \rangle$ for $1 \le k \le r$, and $\Delta_k(A) = 0$ for $k > r$. Since R is a domain, $(\Delta_k(A) : \Delta_{k-1}(A)) = \langle d_k \rangle$, so d_k is determined up to associates.
Structural Commentary: Smith form preserves rank plus divisibility data.
Key Insight: Determinantal ideals encode invariant factors. □

Theorem 2.7.3. *Adapted Bases for Submodules of Free Modules*
Let $N \le F$, $\mathrm{rk}_R(F) = n$, *and* $\mathrm{rk}_R(N) = k$.
Then $\exists (v_i)_{i=1}^n \in \mathrm{Bas}_R(F)$ *and* $r_1, \dots, r_k \in R \setminus \{0\}$ *such that* $r_1 \mid \cdots \mid r_k$ *and* $(r_i v_i)_{i=1}^k \in \mathrm{Bas}_R(N)$.

Proof. Choose bases of N and F, and let A represent $\iota : N \hookrightarrow F$.
Smith form gives $PAQ = \mathrm{diag}(r_1, \dots, r_k, 0, \dots, 0)$ with $r_i \neq 0$ and $r_1 \mid \cdots \mid r_k$, since ι is injective.
The row operations give a basis $(v_i)_{i=1}^n$ of F and the column operations give a basis $(w_i)_{i=1}^k$ of N with $\iota(w_i) = r_i v_i$. Hence $(r_i v_i)_{i=1}^k = (w_i)_{i=1}^k$ is a basis of N.
Structural Commentary: Inclusions diagonalize after choosing bases.
Key Insight: Diagonal entries are the required rescalings. □

Corollary 2.7.4. *Smith Form Presents Finitely Generated PID Modules*
Let $R^n \xrightarrow{A} R^m \to M \to 0$ *be a finite presentation.*
If $A \sim \mathrm{diag}(d_1, \dots, d_r, 0, \dots, 0)$ *in Smith form.* $I = \{i : 1 \le i \le r,\ d_i \notin R^\times\}$, *then* $M \cong R^{m-r} \oplus \bigoplus_{i \in I} R/\langle d_i \rangle$.

Proof. Equivalence $A \sim PAQ$ changes bases in R^m, R^n, so $\mathrm{coker}\,A \cong \mathrm{coker}(PAQ)$.
For the diagonal map, $d_i \in R^\times$ gives 0, $d_i \notin R^\times$ gives $R/\langle d_i \rangle$, and each zero row gives R.
Structural Commentary: Smith form turns presentations into direct sums.
Key Insight: Diagonal relations create cyclic torsion summands, and zero rows create free summands. □

Example 2.7.5. *Smith Form for a Finite Abelian Group*
Let $M = \mathbb{Z}^2/\langle\!\langle (6,0), (0,10), (2,4) \rangle\!\rangle_{\mathbb{Z}}$, with relation matrix $A = \begin{bmatrix} 6 & 0 & 2 \\ 0 & 10 & 4 \end{bmatrix}$. Then $\Delta_1(A) = \langle 6, 10, 2, 4 \rangle = \langle 2 \rangle$ and $\Delta_2(A) = \langle 60, 24, -20 \rangle = \langle 4 \rangle$. Hence $d_1 \sim 2$, $d_1 d_2 \sim 4$, so $d_2 \sim 2$ and $M \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.
Key Insight: Determinantal ideals read off cyclic summands.

2.8 Finitely Generated PID Modules

Theorem 2.8.1. *Finitely Generated Torsion-Free Modules Are Free*
For a f.g. R*-module* M, M *is free* $\iff$ M *is torsion-free.*

Proof.
($\Rightarrow$) If $M \cong R^n$, then $am = 0$ with $a \neq 0$ forces every coordinate of m to be 0, since R is a domain.
($\Leftarrow$) Let M be torsion-free and $G = \{u_1, \dots, u_n\}$ generate M.
Choose maximal R-independent $S = \{v_1, \dots, v_k\} \subseteq G$. For $w \in G \setminus S$, dependence of $S \cup \{w\}$ gives $a_w \neq 0$ and $b_{w,i} \in R$ with $a_w w + \sum_i b_{w,i} v_i = 0$. Put $a = \prod_{w \in G \setminus S} a_w$. Then $aM \subseteq \langle\!\langle S \rangle\!\rangle_R$, so aM is free by Theorem 2.6.2. Since M is torsion-free, $M \xrightarrow{\sim} aM$, $m \mapsto am$; hence M is free.
Structural Commentary: Clear denominators into a free submodule.
Key Insight: One scalar embeds M into a free sp an. □

Theorem 2.8.2. *Free Plus Torsion Decomposition*
For f.g. M, $M = M_{\mathrm{tor}} \oplus F$ *with* F *f.g. free. Moreover* M_{tor} *is intrinsic and* $F \cong M/M_{\mathrm{tor}}$.

Proof. By Proposition 1.6.3, $T := M_{\text{tor}} \leq M$ and $Q := M/T$ is f.g. torsion-free.
Thus Q is free by Theorem 2.8.1. Since Q is free, the quotient map $\pi : M \twoheadrightarrow Q$ splits; hence $M = T \oplus \operatorname{im}\sigma$ for any section $\sigma : Q \to M$, and $\operatorname{im}\sigma \cong Q$. If $M = T' \oplus G$ with T' torsion and G free, then $m = t' + g \in M_{\text{tor}}$ forces $g = 0$, so $T' = M_{\text{tor}}$. Every complement is isomorphic to M/M_{tor}.
Structural Commentary: F.g. PID modules split into torsion plus free.
Key Insight: The quotient by torsion is free, and freeness makes the quotient map split. □

2.9 Primary Decomposition and Cyclic Primary Modules

Definition 2.9.1. *Primary Module*
Let $p \in R$ be prime. A torsion R-module M is *p-primary* if $o(M) \sim p^e$ for some $e \geq 1$.

Theorem 2.9.2. *Primary Decomposition*
Let M be f.g. torsion with $o(M) = \mu \sim \prod_{i=1}^{n} p_i^{e_i}$, where the p_i are distinct nonassociate primes.
Put $q_i := p_i^{e_i}$, $\mu_i := \mu/q_i$, and $M_i := \mu_i M$. Then $M = \bigoplus_{i=1}^{n} M_i$, $M_i = \{v \in M : q_i v = 0\}$, and $o(M_i) \sim q_i$.
The decomposition is unique up to ordering.

Proof. Since $\gcd(\mu_1, \ldots, \mu_n) \sim 1$, choose $a_i \in R$ with $\sum_i a_i\mu_i = 1$. Hence $x = \sum_i a_i\mu_i x$, so $M = \sum_i M_i$.
Also $q_i M_i = \mu M = 0$. Conversely, if $q_i v = 0$, choose $a, b \in R$ with $a\mu_i + bq_i = 1$; then $v = a\mu_i v \in M_i$.
Thus $M_i = \{v : q_i v = 0\}$. If $\sum_i x_i = 0$ with $x_i \in M_i$, applying μ_i gives $\mu_i x_i = 0$ as $q_j \mid \mu_i$ for $j \neq i$.
Since $q_i x_i = 0$ and $\gcd(q_i, \mu_i) \sim 1$, Bezout gives $x_i = 0$; hence the sum is direct.
Now $o(M_i) \mid q_i$ and $\operatorname{lcm}_i o(M_i) \sim o(M) \sim \prod_i q_i$, so $o(M_i) \sim q_i$.
Finally each summand is the intrinsic set $\{v : q_i v = 0\}$, where q_i is the p_i-part of $o(M)$.
Structural Commentary: Primary decomposition separates a torsion module prime by prime.
Key Insight: Bezout idempotents built from the coprime factors of the order project onto the primary pieces. □

Lemma 2.9.3. *The First p-Torsion Layer*
Let $p \in R$ be prime and $M[p] := \ker(p : M \to M) = \{v \in M : pv = 0\}$.
Then $M[p]$ is an $R/\langle p\rangle$-vector space. If $M = S \oplus T$, then $M[p] = S[p] \oplus T[p]$.

Proof. $\langle p\rangle$ is maximal by Theorem 3.11.5 and Theorem 3.4.4; hence $\kappa := R/\langle p\rangle$ is a field. Define $(r + \langle p\rangle)v := rv$.
If $r - r' \in \langle p\rangle$ and $v \in M[p]$, then $(r - r')v = 0$, so the action is well-defined; the axioms are inherited from M.
If $v = s + t \in M[p]$ under $M = S \oplus T$, then $0 = pv = ps + pt$ gives $ps = pt = 0$ by directness.
Thus $v \in S[p] \oplus T[p]$, and the reverse inclusion is immediate.
Structural Commentary: Killing by p turns scalar multiplication modulo p into field scalar multiplication.
Key Insight: The layer $M[p]$ counts the number of cyclic primary summands. □

Theorem 2.9.4. *Cyclic Decomposition of a Primary Module*
Let M be f.g. p-primary torsion with $o(M) \sim p^e$.
Then $M = \bigoplus_{i=1}^{m} \langle\langle v_i\rangle\rangle_R$ with $\operatorname{Ann}_R(v_i) = \langle p^{e_i}\rangle$ and $e = e_1 \geq e_2 \geq \cdots \geq e_m \geq 1$.
The annihilator chain $\langle p^{e_1}\rangle \subseteq \cdots \subseteq \langle p^{e_m}\rangle$ is intrinsic.

Proof.

(i) *Existence*: Choose a finite presentation of M.
Corollary 2.7.4 gives $M \cong R^s \oplus \bigoplus_{\ell=1}^{t} R/\langle d_\ell\rangle$, with d_ℓ nonunit. Since M is torsion, $s = 0$.
Since M is p-primary, $d_\ell \sim p^{e_\ell}$ with $e_\ell \geq 1$. Reorder so $e_1 \geq \cdots \geq e_m$.
Proposition 2.5.4 gives $o(M) \sim p^{e_1}$, hence $e_1 = e$.

(ii) *Uniqueness*: Put $\kappa = R/\langle p\rangle$. By Lemma 2.9.3, each summand $R/\langle p^{e_i}\rangle$ contributes one κ-dimension to $M[p]$, so $\dim_\kappa M[p] = m$.
Passing from M to pM replaces each $e_i > 1$ by $e_i - 1$ and removes each $e_i = 1$.
Induction on e recovers the exponent multiset, hence the annihilator chain.

Structural Commentary: A primary module splits into cyclic strings whose lengths are the exponents e_i.
Key Insight: The dimensions of the layers $M[p], pM[p], p^2M[p], \ldots$ determine the string lengths. □

Proposition 2.9.5. *Layer Counts Recover Primary Exponents*
Let $\kappa = R/\langle p\rangle$ and $\lambda_k(M) := \dim_\kappa(p^{k-1}M/p^kM)$ for $k \geq 1$.
If $M \cong \bigoplus_{i=1}^{m} R/\langle p^{e_i}\rangle$ with $e_1 \geq \cdots \geq e_m \geq 1$, then $\lambda_k(M) = \#\{i : e_i \geq k\}$.
Hence the number of summands of exact exponent k is $\lambda_k(M) - \lambda_{k+1}(M)$.

Proof. For $C = R/\langle p^e\rangle$, $p^{k-1}C/p^kC$ is 0 if $k > e$ and is κ if $k \leq e$, generated by the class of p^{k-1}.
Since direct sums commute with $p^{k-1}M$ and p^kM, dimensions add, giving $\lambda_k(M) = \#\{i : e_i \geq k\}$.
Thus $\lambda_k - \lambda_{k+1}$ counts exactly the summands ending at exponent k.
Structural Commentary: Successive p-layers determine the exponent list.
Key Insight: A cyclic summand of order p^e contributes one dimension to exactly the first e layers. □

2.10 Elementary Divisors and Invariant Factors

Definition 2.10.1. *Elementary Divisors*
For finitely generated torsion M, if its primary cyclic summands have annihilators $\langle p_i^{e_{i,j}} \rangle$, set $\mathrm{ElemDiv}(M) := \{p_i^{e_{i,j}}\}_{i,j}$ as a multiset, up to associates.

Theorem 2.10.2. *Primary Cyclic Decomposition*
Every f.g. torsion R-module M admits $M = \bigoplus_i \bigoplus_{j=1}^{k_i} \langle\!\langle v_{i,j} \rangle\!\rangle_R$, where $\mathrm{Ann}_R(v_{i,j}) = \langle p_i^{e_{i,j}} \rangle$, p_i are distinct nonassociate primes, and $e_{i,1} \geq \cdots \geq e_{i,k_i} \geq 1$. For f.g. torsion M, N, $M \cong N \iff \mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$.

Proof.

($\Rightarrow$) Theorem 2.9.2 gives $M = \bigoplus_i M_{p_i}$ with each M_{p_i} f.g. p_i-primary. Theorem 2.9.4 gives $M_{p_i} = \bigoplus_{j=1}^{k_i} \langle\!\langle v_{i,j} \rangle\!\rangle_R$ with $\mathrm{Ann}_R(v_{i,j}) = \langle p_i^{e_{i,j}} \rangle$. If $M \cong N$, the same uniqueness statements identify the primary pieces and their exponent chains; hence $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$.

($\Leftarrow$) If $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$, then both modules are isomorphic to the same direct sum $\bigoplus_{q \in \mathrm{ElemDiv}(M)} R/\langle q \rangle$, since a cyclic summand with annihilator $\langle q \rangle$ is $R/\langle q \rangle$ by Theorem 2.5.5. Thus $M \cong N$.

Structural Commentary: A torsion module is a direct sum of prime-power cyclic blocks.
Key Insight: Elementary divisors are the atoms of the PID module classification. □

Proposition 2.10.3. *Elementary Divisors of a Direct Sum*
For f.g. torsion A, B, $\mathrm{ElemDiv}(A \oplus B) = \mathrm{ElemDiv}(A) \uplus \mathrm{ElemDiv}(B)$, where $\uplus$ is multiset union.

Proof. If $A = \bigoplus_i A_i$ and $B = \bigoplus_j B_j$ are primary cyclic decompositions, then $A \oplus B = (\bigoplus_i A_i) \oplus (\bigoplus_j B_j)$ is one for $A \oplus B$. Thus the prime-power annihilator multiset is the multiset union.
Structural Commentary: Direct sums concatenate cyclic primary blocks.
Key Insight: Elementary divisors count blocks with multiplicity. □

Definition 2.10.4. *Invariant Factors*
For f.g. torsion M, write $\mathrm{ElemDiv}(M) = \{p_i^{e_{i,j}} : 1 \leq j \leq k_i\}_i$ with $e_{i,1} \geq \cdots \geq e_{i,k_i} \geq 1$.
Let $m = \max_i k_i$, set $e_{i,j} = 0$ for $j > k_i$, and define $d_j := \prod_i p_i^{e_{i,j}}$.
The nonunit d_j, up to associates, are the *invariant factors*; they satisfy $d_m \mid \cdots \mid d_1$.

Method 2.10.5. *From Elementary Divisors to Invariant Factors*

(i) Group $\mathrm{ElemDiv}(M)$ by prime: $\{p_i^{e_{i,j}}\}_j$.
(ii) Reorder each list so $e_{i,1} \geq e_{i,2} \geq \cdots$ and pad by $e_{i,j} = 0$ to common length.
(iii) Form $d_j = \prod_i p_i^{e_{i,j}}$.
(iv) Discard unit d_j; the rest are the invariant factors from Definition 2.10.4.

Example 2.10.6. *Packing Elementary Divisors into Invariant Factors*
Over $R = \mathbb{F}[z]$, let $\mathrm{ElemDiv}(M) = \{(z-1)^3, z-1, (z^2+1)^2, z^2+1\}$. The exponent rows are $(3,1)$ for $z-1$ and $(2,1)$ for z^2+1, so $d_1 = (z-1)^3(z^2+1)^2$ and $d_2 = (z-1)(z^2+1)$, with $d_2 \mid d_1$. Hence $M \cong R/\langle d_1 \rangle \oplus R/\langle d_2 \rangle$.
Structural Commentary: Elementary divisors separate primes; invariant factors pack rows.
Key Insight: Line up equal-height primary strings and multiply across the row.

Example 2.10.7. *Recombining Elementary Divisors*
Let $G = \mathbb{Z}/12\mathbb{Z} \oplus \mathbb{Z}/18\mathbb{Z}$. Then $\mathrm{ElemDiv}(G) = \{2, 2^2, 3, 3^2\}$. Recombining equal-height primary blocks gives $2 \cdot 3 = 6$ and $2^2 \cdot 3^2 = 36$; hence $G \cong \mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/36\mathbb{Z}$, with $6 \mid 36$.
Structural Commentary: Elementary divisors split by primes; invariant factors recombine them.
Key Insight: Prime-wise data equals one divisibility chain.

Theorem 2.10.8. *Invariant Factor Decomposition*
For f.g. torsion M, $M \cong \bigoplus_{j=1}^m R/\langle d_j \rangle$ with $d_j \notin R^\times$ and $d_m \mid \cdots \mid d_1$.
Writing $\mathrm{Inv}(M) := (d_m, \ldots, d_1)$ up to associates, for f.g. torsion M, N, $M \cong N \iff \mathrm{Inv}(M) = \mathrm{Inv}(N)$.

Proof. By Theorem 2.10.2, $M \cong \bigoplus_{q \in \mathrm{ElemDiv}(M)} R/\langle q \rangle$.
Let $\mathrm{ElemDiv}(M) = \{p_i^{e_{i,j}} : 1 \leq j \leq k_i\}_i$ with $e_{i,1} \geq \cdots \geq e_{i,k_i}$, pad by $e_{i,j} = 0$, and set $d_j = \prod_i p_i^{e_{i,j}}$.
For fixed j, the nonunit $p_i^{e_{i,j}}$ are pairwise coprime, so Theorem 2.5.6 gives $\bigoplus_{e_{i,j} > 0} R/\langle p_i^{e_{i,j}} \rangle \cong R/\langle d_j \rangle$.
Hence $M \cong \bigoplus_j R/\langle d_j \rangle$ with $d_m \mid \cdots \mid d_1$.

($\Rightarrow$) If $M \cong N$, then $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$ by Theorem 2.10.2. Grouping by height gives $\mathrm{Inv}(M) = \mathrm{Inv}(N)$.
($\Leftarrow$) If $\mathrm{Inv}(M) = \mathrm{Inv}(N)$, by factorization of each d_j, $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$; QED (Theorem 2.10.2).

Structural Commentary: Invariant factors pack prime-power blocks.
Key Insight: Same data, different packaging. □

Corollary 2.10.9. *Classification of Finitely Generated PID Modules*
For f.g. M, $M \cong R^r \oplus \bigoplus_{j=1}^m R/\langle d_j \rangle$, where $r \geq 0$, $d_j \notin R^\times$, and $d_m \mid \cdots \mid d_1$.
The pair $(r, \mathrm{Inv}(M_{\mathrm{tor}}))$, equivalently $(r, \mathrm{ElemDiv}(M_{\mathrm{tor}}))$, is a complete invariant.

Proof. Let $T := M_{\mathrm{tor}}$. Theorem 2.8.2 gives $M = T \oplus F$ with $F \cong R^r$.
Theorem 2.10.8 gives $T \cong \bigoplus_{j=1}^m R/\langle d_j \rangle$ with $d_m \mid \cdots \mid d_1$. Since T is intrinsic and $F \cong M/T$, uniqueness and completeness follow from rank uniqueness and the invariant-factor theorem.
Structural Commentary: F.g. PID modules are free plus cyclic torsion.
Key Insight: The free rank records the zero-annihilator part; invariant factors record the torsion part. □

Method 2.10.10. *Reading a Module from Smith Form*
For finite presentation $R^n \xrightarrow{A} R^m \to M \to 0$ over PID, put $A \sim D = \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0)$, $d_1 \mid \cdots \mid d_r$:
(i) $d_i \in R^\times \implies$ no summand;
(ii) $d_i \notin R^\times \implies R/\langle d_i \rangle$;
(iii) each zero target coordinate contributes one R;
(iv) factor nonunit d_i into prime powers for elementary divisors.
Key Insight: Smith form changes arbitrary relations into independent one-coordinate relations.

Perspective 2.10.11. *Largest Invariant Factor and Product*
If $M \cong \bigoplus_{j=1}^{m} R/\langle d_j \rangle$ with $d_m \mid \cdots \mid d_1$, then $\operatorname{Ann}_R(M) = \langle d_1 \rangle$ and $\prod_{j=1}^{m} d_j \sim \prod_{q \in \operatorname{ElemDiv}(M)} q$.

Theorem 2.10.12. *Characterizing Cyclic Torsion Modules*
Let M be f.g. torsion with order $\mu \sim \prod_{i=1}^{n} p_i^{e_i}$, p_i distinct nonassociate primes. These are equivalent:
(i) M is cyclic.
(ii) $\forall i$, the primary component M_{p_i} is cyclic.
(iii) $\operatorname{ElemDiv}(M) = \{p_i^{e_i} : 1 \le i \le n\}$.
(iv) $M \cong R/\langle \mu \rangle$.

Proof.
(i) $\Rightarrow$ (ii) If $M = \langle\!\langle x \rangle\!\rangle_R$, Theorem 2.5.6 gives $M = \bigoplus_i \langle\!\langle x_i \rangle\!\rangle_R$ with $o(x_i) \sim p_i^{e_i}$; hence $M_{p_i} = \langle\!\langle x_i \rangle\!\rangle_R$.
(ii) $\Rightarrow$ (iii) By Theorem 2.10.2, cyclic M_{p_i} of order $p_i^{e_i}$ has sole elementary divisor $p_i^{e_i}$; use Proposition 2.10.3.
(iii) $\Rightarrow$ (iv) By Theorem 2.10.8, there is one invariant factor $d_1 \sim \prod_i p_i^{e_i} \sim \mu$.
(iv) $\Rightarrow$ (i) Since $R/\langle \mu \rangle = \langle\!\langle 1 + \langle \mu \rangle \rangle\!\rangle_R$, any isomorphism $R/\langle \mu \rangle \xrightarrow{\sim} M$ sends $1 + \langle \mu \rangle$ to a generator of M.
Structural Commentary: Cyclic torsion means one string per prime.
Key Insight: Multiple elementary divisors for the same prime are the obstruction to one generator. □

Definition 2.10.13. *Indecomposable Module*
A nonzero R-module M is *indecomposable* if $M = A \oplus B \implies A = 0$ or $B = 0$.

Theorem 2.10.14. *Indecomposable Finitely Generated Torsion Modules*
Let $0 \neq M$ be f.g. torsion. The following are equivalent:
(i) M is indecomposable.
(ii) M is primary cyclic.
(iii) $\operatorname{ElemDiv}(M) = \{p^e\}$ *for some prime p and $e \ge 1$.*

Proof.
(i) $\Rightarrow$ (ii) By Theorem 2.10.2, write $M \cong \bigoplus_{\alpha \in A} C_\alpha$, where $0 \neq C_\alpha \cong R/\langle p_\alpha^{e_\alpha} \rangle$.
If $|A| \ge 2$, then $M \cong C_{\alpha_0} \oplus \bigoplus_{\alpha \neq \alpha_0} C_\alpha$ is a nontrivial decomposition, contradiction.
Since $M \neq 0$, $|A| = 1$; hence M is primary cyclic.
(ii) $\Rightarrow$ (iii) If $M \cong R/\langle p^e \rangle$, then $\operatorname{ElemDiv}(M) = \{p^e\}$.
(iii) $\Rightarrow$ (i) If $M = A \oplus B$, Proposition 2.10.3 gives $\{p^e\} = \operatorname{ElemDiv}(A) \sqcup \operatorname{ElemDiv}(B)$.
Hence one multiset is empty, so one summand is 0.
Structural Commentary: Indecomposable torsion modules are precisely the single primary cyclic blocks.
Key Insight: Elementary divisors are the indecomposable pieces. □

3 Linear Operators as Modules

3.1 Polynomial Modules from Operators

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over $\mathbb{F}$ unless explicitly stated otherwise.

Perspective 3.1.1. *An Operator as a Polynomial Action*
Fix $T \in \mathcal{L}(\mathcal{V})$. Evaluation at T defines an $\mathbb{F}[z]$-action on $\mathcal{V}$ by $p \cdot \mathbf{v} := p(T)\mathbf{v}$.
Thus the orbit $\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \dots$ is encoded by the cyclic subspace $\{p(T)\mathbf{v} : p \in \mathbb{F}[z]\} = \operatorname{span}\{T^j\mathbf{v} : j \ge 0\}$.

Definition 3.1.2. *The Module Attached to an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$. The *operator module* attached to T is the $\mathbb{F}[z]$-module $\mathcal{V}_T$ with underlying additive group $\mathcal{V}$ and action $p(z) \cdot \mathbf{v} := p(T)\mathbf{v}$.
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, define $(\mathbb{F}^n)_A$ analogously by $p(z) \cdot \mathbf{x} := p(A)\mathbf{x}$.

Proposition 3.1.3. *Operator Modules Are Finitely Generated Torsion Modules*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}_T$ is a finitely generated torsion $\mathbb{F}[z]$-module.

Proof. Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an $\mathbb{F}$-basis of $\mathcal{V}$. Since $\mathbb{F} \subseteq \mathbb{F}[z]$, this basis generates $\mathcal{V}_T$.
By Proposition 1.1.7, there exists $0 \neq p \in \mathbb{F}[z]$ with $p(T) = 0$. Thus $p \cdot \mathbf{v} = p(T)\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathcal{V}$, so $\mathcal{V}_T$ is torsion. Finally, $\mathbb{F}[z]$ is a PID by Theorem 3.11.1.
Structural Commentary: Finite-dimensionality gives a global polynomial relation for the operator.
Key Insight: Rational canonical form is the coordinate form of the structure theorem for $\mathcal{V}_T$. □

Theorem 3.1.4. *Submodules Are Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{S} \subseteq \mathcal{V}$. Then $\mathcal{S} \le_{\mathbb{F}[z]} \mathcal{V}_T$ if and only if $\mathcal{S} \le_{\mathbb{F}} \mathcal{V}$ and $T\mathcal{S} \subseteq \mathcal{S}$.

Proof.
($\Rightarrow$) If $\mathcal{S} \le_{\mathbb{F}[z]} \mathcal{V}_T$, then $\mathbb{F} \subseteq \mathbb{F}[z]$ gives $\mathcal{S} \le_{\mathbb{F}} \mathcal{V}$. Since $z \cdot \mathbf{s} = T\mathbf{s}$ for $\mathbf{s} \in \mathcal{S}$, we have $T\mathcal{S} \subseteq \mathcal{S}$.
($\Leftarrow$) If $\mathcal{S} \le_{\mathbb{F}} \mathcal{V}$ and $T\mathcal{S} \subseteq \mathcal{S}$, then $T^j\mathcal{S} \subseteq \mathcal{S}$ for all $j \ge 0$.
Hence for $p(z) = \sum_j a_j z^j$ and $\mathbf{s} \in \mathcal{S}$, $p \cdot \mathbf{s} = p(T)\mathbf{s} = \sum_j a_j T^j \mathbf{s} \in \mathcal{S}$, so $\mathcal{S} \le_{\mathbb{F}[z]} \mathcal{V}_T$.

Structural Commentary: The single scalar z remembers the operator.
Key Insight: Invariance under T is exactly closure under all polynomial scalars. □

Theorem 3.1.5. *Similarity Is Module Isomorphism*
Let $T, S \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S \iff T \sim S$. More precisely, for $\Phi \in \mathrm{GL}(\mathcal{V})$, $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is $\mathbb{F}[z]$-linear if and only if $\Phi T = S\Phi$, equivalently $S = \Phi T \Phi^{-1}$.

Proof.
(⇒) If $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is an $\mathbb{F}[z]$-isomorphism, then $\Phi \in \mathrm{GL}(\mathcal{V})$, $\Phi(T\mathbf{v}) = \Phi(z \cdot \mathbf{v}) = z \cdot \Phi(\mathbf{v}) = S\Phi(\mathbf{v})$ $\forall \mathbf{v} \in \mathcal{V}$.
Hence $\Phi T = S\Phi$, so $S = \Phi T \Phi^{-1}$.
(⇐) If $\Phi \in \mathrm{GL}(\mathcal{V})$ and $S = \Phi T \Phi^{-1}$, then $\Phi T = S\Phi$, hence $\Phi T^j = S^j \Phi$ for all $j \geq 0$.
Thus for $p(z) = \sum_j a_j z^j$, $\Phi(p(T)\mathbf{v}) = p(S)\Phi(\mathbf{v})$, so $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is an $\mathbb{F}[z]$-isomorphism.
Structural Commentary: A module isomorphism is precisely an intertwiner for the action of z.
Key Insight: Similarity is isomorphism of the associated $\mathbb{F}[z]$-modules. □

Corollary 3.1.6. *Matrix Similarity Is Matrix-Module Isomorphism*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. Then $A \sim B \iff (\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$. More precisely, for $X \in \mathrm{GL}_n(\mathbb{F})$, $X : (\mathbb{F}^n)_A \to (\mathbb{F}^n)_B$ is $\mathbb{F}[z]$-linear if and only if $XA = BX$, equivalently $B = XAX^{-1}$.

Proof.
(⇒) If $X : (\mathbb{F}^n)_A \to (\mathbb{F}^n)_B$ is an $\mathbb{F}[z]$-isomorphism, then $X \in \mathrm{GL}_n(\mathbb{F})$ and $X(A\mathbf{x}) = X(z \cdot \mathbf{x}) = z \cdot X\mathbf{x} = BX\mathbf{x}$ for all $\mathbf{x} \in \mathbb{F}^n$. Hence $XA = BX$, so $B = XAX^{-1}$.
(⇐) If $X \in \mathrm{GL}_n(\mathbb{F})$ and $B = XAX^{-1}$, then $XA = BX$, hence $XA^j = B^j X$ for all $j \geq 0$. Thus $Xp(A) = p(B)X$ for all $p \in \mathbb{F}[z]$, so $X : (\mathbb{F}^n)_A \to (\mathbb{F}^n)_B$ is an $\mathbb{F}[z]$-isomorphism.
Structural Commentary: Similarity is invertible module intertwining.
Key Insight: The matrix is remembered by the scalar action of z. □

Definition 3.1.7. *Module Annihilator and Minimal Polynomial*
(i) The *annihilator* of $\mathcal{V}_T$ is $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) := \{p \in \mathbb{F}[z] : p(T)\mathcal{V} = \{\mathbf{0}\}\}$.
(ii) *Minimal polynomial* μ_T is unique monic generator of this nonzero principal ideal, so $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$.
(iii) For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, define $\mu_A := \mu_{T_A}$, where $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 3.1.8. *Minimal Polynomial as Module Order*
Let $T \in \mathcal{L}(\mathcal{V})$. Then μ_T is the unique monic polynomial of minimal degree in $\{p \in \mathbb{F}[z] : p(T) = 0\}$. If $S \sim T$, then $\mu_S = \mu_T$; hence similar matrices have the same minimal polynomial.

Proof. By Definition 3.1.7, $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$ and $p \in \mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T)$ iff $p(T) = 0$. Thus μ_T is the unique monic annihilator of minimal degree. If $S = \Phi T \Phi^{-1}$, then $p(S) = \Phi p(T) \Phi^{-1}$ for all $p \in \mathbb{F}[z]$, so $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_S) = \mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T)$ and $\mu_S = \mu_T$. The matrix case follows by applying this to the associated operators.
Structural Commentary: The minimal polynomial is the order of the whole operator module.
Key Insight: Similarity preserves the module and therefore preserves its annihilator ideal. □

Definition 3.1.9. *Cyclic Subspace and Vector Annihilator*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V}$.
(i) The *T-cyclic subspace generated by $\mathbf{v}$* is $\mathcal{C}_T(\mathbf{v}) := \mathrm{span}\{T^j\mathbf{v} : j \geq 0\}$
(ii) The vector $\mathbf{v}$ is a *cyclic vector* for T if $\mathcal{C}_T(\mathbf{v}) = \mathcal{V}$.
(iii) The *annihilator of $\mathbf{v}$ for T* is $\mathrm{ann}_T(\mathbf{v}) := \{p \in \mathbb{F}[z] : p(T)\mathbf{v} = \mathbf{0}\}$

Proposition 3.1.10. *Cyclic Subspaces Are Smallest Invariant Hosts*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V}$. Then $\mathcal{C}_T(\mathbf{v}) = \{p(T)\mathbf{v} : p \in \mathbb{F}[z]\}$.
Moreover, $\mathcal{C}_T(\mathbf{v})$ is the smallest T-invariant subspace containing $\mathbf{v}$, and $\mathrm{ann}_T(\mathbf{v})$ is an ideal of $\mathbb{F}[z]$.

Proof. Equality follows as $p(T)\mathbf{v}$ is a finite linear combination of the orbit vectors $T^j\mathbf{v}$, and every such finite linear combination equals $p(T)\mathbf{v}$ for some $p \in \mathbb{F}[z]$. Also $T(p(T)\mathbf{v}) = (zp)(T)\mathbf{v}$, so $\mathcal{C}_T(\mathbf{v})$ is T-invariant.
If $\mathcal{U}$ is T-invariant and contains $\mathbf{v}$, then $T^j\mathbf{v} \in \mathcal{U}$ for every $j \geq 0$, hence $\mathcal{C}_T(\mathbf{v}) \subseteq \mathcal{U}$. Finally, if $p, q \in \mathrm{ann}_T(\mathbf{v})$ and $r \in \mathbb{F}[z]$, then $(p+q)(T)\mathbf{v} = \mathbf{0}$ and $(rp)(T)\mathbf{v} = r(T)p(T)\mathbf{v} = \mathbf{0}$ by Proposition 1.1.6. $\mathrm{ann}_T(\mathbf{v})$ is an ideal.
Structural Commentary: A cyclic subspace is the span of the orbit generated by repeatedly applying T.
Key Insight: Invariance is exactly what lets the whole orbit stay inside the same host subspace. □

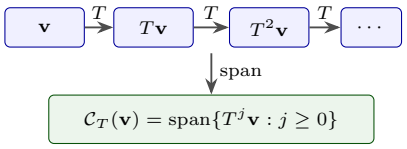

Figure 104: The T-orbit of $\mathbf{v}$ spans $\mathcal{C}_T(\mathbf{v})$.

Definition 3.1.11. *Order of a Vector under an Operator*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\mathbf{v} \neq \mathbf{0}$. The *order of $\mathbf{v}$ under T* is the unique monic polynomial $m_{\mathbf{v}} \in \mathrm{ann}_T(\mathbf{v})$ of least degree; equivalently, $\mathrm{ann}_T(\mathbf{v}) = (m_{\mathbf{v}})$

Definition 3.1.12. *Companion Matrix*
Let $f(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$ be monic. For $n \geq 2$, the *companion matrix* of f is

$$C_f := \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}.$$

For $n = 1$, set $C_f = [-c_0]$.

Theorem 3.1.13. *Companion Matrix Realises a Polynomial*
Let $f(z) = z^n + \sum_{i=0}^{n-1} c_i z^i$ be monic and let C_f be its companion matrix. Then $p_{C_f} = \mu_{C_f} = f$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the standard basis.
For $0 \leq j < n$, $C_f^j \mathbf{e}_1 = \mathbf{e}_{j+1}$, and last column gives $C_f^n \mathbf{e}_1 = -\sum_{i=0}^{n-1} c_i C_f^i \mathbf{e}_1$. Hence $f(C_f)\mathbf{e}_1 = 0$.
Since $\mathbf{e}_j = C_f^{j-1}\mathbf{e}_1$ and $f(C_f)C_f = C_f f(C_f)$, $f(C_f)\mathbf{e}_j = 0\ \forall j$, so $f(C_f) = 0$.
If $0 \neq g(z) = \sum_{i=0}^{m} b_i z^i$ with $m < n$, then $g(C_f)\mathbf{e}_1 = \sum_{i=0}^{m} b_i \mathbf{e}_{i+1} \neq \mathbf{0}$.
Thus no nonzero polynomial of degree $< n$ annihilates C_f, so $\deg \mu_{C_f} = n$.
Since $\mu_{C_f} \mid f$ and both are monic of degree n, $\mu_{C_f} = f$.
By Cayley–Hamilton, $\mu_{C_f} \mid p_{C_f}$; as p_{C_f} and f are monic of degree n, $p_{C_f} = f$.
Structural Commentary: The cyclic vector $\mathbf{e}_1$ generates the basis, and last column imposes relation $f(C_f) = 0$.
Key Insight: The companion matrix makes f the first possible annihilating relation. □

Theorem 3.1.14. *Cyclic Subspaces as Polynomial Quotients*
Let $\dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, $0 \neq \mathbf{v} \in \mathcal{V}$, and let $m_{\mathbf{v}}(z) = z^d + \sum_{i=0}^{d-1} c_i z^i$ be the order of $\mathbf{v}$.
Define $\Phi_{\mathbf{v}} : \mathbb{F}[z] \to \mathcal{C}_T(\mathbf{v})$ by $\Phi_{\mathbf{v}}(p) = p(T)\mathbf{v}$ and $\mathcal{B}_{\mathbf{v}} := (\mathbf{v}, T\mathbf{v}, \ldots, T^{d-1}\mathbf{v})$. Then
(i) $\operatorname{im}\Phi_{\mathbf{v}} = \mathcal{C}_T(\mathbf{v})$, $\ker \Phi_{\mathbf{v}} = \langle m_{\mathbf{v}} \rangle$, and $\mathcal{C}_T(\mathbf{v}) \cong \mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle$;
(ii) $\mathcal{B}_{\mathbf{v}}$ is a basis of $\mathcal{C}_T(\mathbf{v})$;
(iii) $[T|_{\mathcal{C}_T(\mathbf{v})}]_{\mathcal{B}_{\mathbf{v}}} = C_{m_{\mathbf{v}}}$.

Proof.
(i) By Proposition 3.1.10, $\operatorname{im}\Phi_{\mathbf{v}} = \mathcal{C}_T(\mathbf{v})$. $\ker \Phi_{\mathbf{v}} = \operatorname{ann}_T(\mathbf{v}) = \langle m_{\mathbf{v}} \rangle$, by Theorem 2.2.12 $\mathcal{C}_T(\mathbf{v}) \cong \mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle$.
(ii) If $\sum_{j=0}^{d-1} a_j T^j \mathbf{v} = \mathbf{0}$, then $q(z) = \sum_{j=0}^{d-1} a_j z^j \in \operatorname{ann}_T(\mathbf{v})$ and $\deg q < d$, hence $q = 0$; thus $\mathcal{B}_{\mathbf{v}}$ is independent.
For $p = hm_{\mathbf{v}} + r$ with $\deg r < d$ or $r = 0$, $p(T)\mathbf{v} = r(T)\mathbf{v} \in \operatorname{span}\mathcal{B}_{\mathbf{v}}$, so $\mathcal{B}_{\mathbf{v}}$ spans.
(iii) For $0 \leq j < d-1$, $T(T^j\mathbf{v}) = T^{j+1}\mathbf{v}$. Since $m_{\mathbf{v}}(T)\mathbf{v} = \mathbf{0}$, $T^d\mathbf{v} = -\sum_{i=0}^{d-1} c_i T^i \mathbf{v}$, last-column relation of $C_{m_{\mathbf{v}}}$.

Structural Commentary: The first relation cuts out the quotient and companion block.
Key Insight: One orbit carries the whole cyclic subspace. □

Perspective 3.1.15. *The Translation Dictionary*
The module vocabulary for $\mathcal{V}_T$ and the vector-space vocabulary for T match as follows:

$\mathbb{F}[z]$-module $\mathcal{V}_T$	$\mathbb{F}$-vector space $\mathcal{V}$
$p(z) \cdot \mathbf{v}$	$p(T)\mathbf{v}$
submodule	T-invariant subspace
$\operatorname{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$	minimal polynomial of T
$\langle\langle \mathbf{v} \rangle\rangle_{\mathbb{F}[z]}$	$\mathcal{C}_T(\mathbf{v})$
$\mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle$	cyclic block with companion matrix $C_{m_{\mathbf{v}}}$

Key Insight: Once z is interpreted as T, module structure becomes operator structure.

3.2 Invariant Factors

Definition 3.2.1. *Invariant Factors and Elementary Divisors*
Let $T \in \mathcal{L}(\mathcal{V})$.
(i) A *cyclic decomposition* is $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_j$, where each $\mathcal{C}_j$ is nonzero, T-invariant, and cyclic.
(ii) $\operatorname{InvFact}(T) := (f_1, \ldots, f_r)$, where f_j are monic nonconstant invariant factors of $\mathcal{V}_T$; $f_r \mid f_{r-1} \mid \cdots \mid f_1$.
(iii) $\operatorname{ElemDiv}(T)$ is the multiset of monic prime powers in the f_j, counted with multiplicity.
(iv) *Elementary divisor basis* is $\mathcal{B}_{\mathrm{ED}} := \bigsqcup_\alpha (\mathbf{v}_\alpha, T\mathbf{v}_\alpha, \ldots, T^{d_\alpha - 1}\mathbf{v}_\alpha)$, where $d_\alpha = \deg m_{\mathbf{v}_\alpha}$; $\{m_{\mathbf{v}_\alpha}\}_\alpha = \operatorname{ElemDiv}(T)$.
(v) For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, set $\operatorname{InvFact}(A) := \operatorname{InvFact}(T_A)$ and $\operatorname{ElemDiv}(A) := \operatorname{ElemDiv}(T_A)$.

Theorem 3.2.2. *Invariant-Factor Skeleton for Operators*
Let $0 < \dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, and $\operatorname{InvFact}(T) = (f_1, \ldots, f_r)$.
Then $f_r \mid \cdots \mid f_1$ and there are T-cyclic subspaces $\mathcal{C}_j$ such that $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_j$, $\mathcal{C}_j \cong \mathbb{F}[z]/\langle f_j \rangle$, $[T] = \bigoplus_{j=1}^{r} C_{f_j}$ in the concatenated cyclic bases, $\mu_T = f_1$, and $p_T = \prod_{j=1}^{r} f_j$.

Proof. By Proposition 3.1.3, $\mathcal{V}_T$ is finitely generated torsion over $\mathbb{F}[z]$.
Theorem 2.10.8 gives $\mathcal{V}_T \cong \bigoplus_{j=1}^{r} \mathbb{F}[z]/\langle f_j \rangle$ with $f_r \mid \cdots \mid f_1$, uniquely.
Transport cyclic generators to T-cyclic subspaces $\mathcal{C}_j$; Theorem 3.1.14 gives $\mathcal{C}_j \cong \mathbb{F}[z]/\langle f_j \rangle$ and block C_{f_j}.
Since $\operatorname{Ann}(\mathcal{V}_T) = \langle \operatorname{lcm}(f_1, \ldots, f_r) \rangle = \langle f_1 \rangle = \langle \mu_T \rangle$, $f_1 = \mu_T$.

Finally, Theorem 3.1.13 gives $p_{C_{f_j}} = f_j$, so $p_T = \prod_{j=1}^r f_j$.
Structural Commentary: Torsion module decomposition produces companion blocks.
Key Insight: Invariant factors are the polynomial-module version of choosing canonical blocks for similarity. □

3.3 Elementary Divisors

Theorem 3.3.1. *Primary Cyclic Decomposition for an Operator*
Let $\mu_T = \prod_{i=1}^s p_i^{e_i}$ with $p_i \in \mathbb{F}[z]$ distinct monic irreducible, and set $\mathcal{V}_{p_i} := \ker p_i(T)^{e_i}$. Then
(i) $\mathcal{V}_T = \bigoplus_{i=1}^s \mathcal{V}_{p_i}$;
(ii) $\mathcal{V}_{p_i} = \bigoplus_{j=1}^{k_i} \mathcal{C}_T(\mathbf{v}_{i,j})$, with $m_{\mathbf{v}_{i,j}} = p_i^{e_{i,j}}$ and $e_i = e_{i,1} \geq e_{i,2} \geq \cdots \geq e_{i,k_i}$;
(iii) $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}} : 1 \leq i \leq s,\ 1 \leq j \leq k_i\}$, corresponding cyclic bases form elementary divisor basis.

Proof.
(i) By Theorem 2.10.2, $\mathcal{V}_T = \bigoplus_i \mathcal{V}_T[p_i^\infty]$; $\mathcal{V}_T[p_i^\infty] := \{\mathbf{x} \in \mathcal{V} : \exists N \geq 1,\ p_i(T)^N \mathbf{x} = 0\} = \ker p_i(T)^{e_i} = \mathcal{V}_{p_i}$.
(ii) By Theorem 2.10.2, $\mathcal{V}_{p_i} = \bigoplus_{j=1}^{k_i} \mathcal{C}_T(\mathbf{v}_{i,j})$, $m_{\mathbf{v}_{i,j}} = p_i^{e_{i,j}}$ and $e_i = e_{i,1} \geq \cdots \geq e_{i,k_i}$.
(iii) Definition 3.2.1 gives $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}} : 1 \leq i \leq s,\ 1 \leq j \leq k_i\}$, and concatenating the bases of $\mathcal{C}_T(\mathbf{v}_{i,j})$ gives an elementary divisor basis.
Structural Commentary: Primary module pieces are polynomial-kernel pieces for the operator.
Key Insight: Elementary divisors are exactly the orders of the cyclic primary summands. □

Theorem 3.3.2. *Reducing Decompositions Combine the Data*
Let $\mathcal{V} = \mathcal{A} \oplus \mathcal{B}$ with $\mathcal{A}, \mathcal{B}$ T-invariant, and set $T_\mathcal{A} := T|_\mathcal{A}$, $T_\mathcal{B} := T|_\mathcal{B}$.
Then $\mu_T = \mathrm{lcm}(\mu_{T_\mathcal{A}}, \mu_{T_\mathcal{B}})$ and $\mathrm{ElemDiv}(T) = \mathrm{ElemDiv}(T_\mathcal{A}) \sqcup \mathrm{ElemDiv}(T_\mathcal{B})$, where $\sqcup$ is multiset union.
The invariant factors are obtained by recombining the elementary divisors into a divisibility chain.

Proof. By Theorem 3.1.4, $\mathcal{V}_T = \mathcal{A}_{T_\mathcal{A}} \oplus \mathcal{B}_{T_\mathcal{B}}$ as $\mathbb{F}[z]$-modules.
Hence $\mathrm{Ann}(\mathcal{V}_T) = \mathrm{Ann}(\mathcal{A}_{T_\mathcal{A}}) \cap \mathrm{Ann}(\mathcal{B}_{T_\mathcal{B}}) = \langle \mathrm{lcm}(\mu_{T_\mathcal{A}}, \mu_{T_\mathcal{B}}) \rangle$.
Proposition 2.10.3 gives elementary-divisor identity.
Structural Commentary: Reducing decompositions split the operator module.
Key Insight: Direct sums take lcm on global orders and multiset union on elementary divisors. □

Definition 3.3.3. *Characteristic Polynomial from Elementary Divisors*
For $T \in \mathcal{L}(\mathcal{V})$, define $\chi_T(z) := \prod_{q \in \mathrm{ElemDiv}(T)} q(z)$, with product over the multiset $\mathrm{ElemDiv}(T)$.
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, set $\chi_A := \chi_{T_A}$.

Theorem 3.3.4. *Minimal and Characteristic Polynomials from Elementary Divisors*
Let $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}} : 1 \leq i \leq s,\ 1 \leq j \leq k_i\}$.
Then $\mu_T = \prod_i p_i^{\max_j e_{i,j}}$, $\chi_T = \prod_{i,j} p_i^{e_{i,j}}$, and $\deg \chi_T = \dim \mathcal{V}$.
Hence $\mu_T \mid \chi_T$, and μ_T, χ_T have the same irreducible factors.

Proof. By Theorem 3.3.1, $\mathcal{V}_T = \bigoplus_{i,j} \mathcal{C}_T(\mathbf{v}_{i,j})$ with $m_{\mathbf{v}_{i,j}} = p_i^{e_{i,j}}$.
Hence $\mathrm{Ann}(\mathcal{V}_T) = \langle \mathrm{lcm}_{i,j}\, p_i^{e_{i,j}} \rangle = \langle \prod_i p_i^{\max_j e_{i,j}} \rangle$, so $\mu_T = \prod_i p_i^{\max_j e_{i,j}}$. Definition 3.3.3 gives $\chi_T = \prod_{i,j} p_i^{e_{i,j}}$.
By Theorem 3.1.14, $\dim \mathcal{C}_T(\mathbf{v}_{i,j}) = \deg p_i^{e_{i,j}}$; summing gives $\deg \chi_T = \dim \mathcal{V}$.
Divisibility and common irreducible factors are immediate.
Structural Commentary: μ_T records maximal exponents; χ_T records all exponents.
Key Insight: The pair (μ_T, χ_T) forgets the block distribution. □

Definition 3.3.5. *Nonderogatory Operators*
$T \in \mathcal{L}(\mathcal{V})$ is *nonderogatory* if $\mu_T = \chi_T$, equivalently $\deg \mu_T = \dim \mathcal{V}$.
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, A is nonderogatory if T_A is.

Theorem 3.3.6. *Cyclic and Nonderogatory Are the Same*
For $T \in \mathcal{L}(\mathcal{V})$, the following are equivalent:
(i) $\mathcal{V}_T$ is cyclic;
(ii) T has a cyclic vector;
(iii) $\mathrm{ElemDiv}(T) = \{p_1^{e_1}, \ldots, p_s^{e_s}\}$, one elementary divisor for each irreducible factor of μ_T;
(iv) T is nonderogatory.

Proof.
(i) $\Rightarrow$ *(ii)* If $\mathcal{V}_T = \mathbb{F}[z]\mathbf{v}$, then $\mathcal{V} = \{p(T)\mathbf{v} : p \in \mathbb{F}[z]\} = \mathcal{C}_T(\mathbf{v})$, so $\mathbf{v}$ is cyclic.
(ii) $\Rightarrow$ *(i)* If $\mathcal{C}_T(\mathbf{v}) = \mathcal{V}$, then $\mathcal{V}_T = \mathbb{F}[z]\mathbf{v}$.
(i) $\Leftrightarrow$ *(iii)* Theorem 2.10.12 on $\mathcal{V}_T$. Cyclic torsion module has one cyclic summand for each primary factor.
(iii) $\Rightarrow$ *(iv)* If $\mathrm{ElemDiv}(T) = \{p_i^{e_i}\}_{i=1}^s$, then $\chi_T = \prod_i p_i^{e_i} = \mu_T$, so T is nonderogatory.
(iv) $\Rightarrow$ *(iii)* Let $\mathrm{ElemDiv}(T) = \{p_i^{e_{i,j}}\}$. If $\mu_T = \chi_T$, then $\prod_i p_i^{\max_j e_{i,j}} = \prod_{i,j} p_i^{e_{i,j}}$, so $k_i = 1$ for all i.
Structural Commentary: Nonderogatory means all primary information can be generated by one vector.
Key Insight: Equality $\mu_T = \chi_T$ says that product and lcm of the cyclic orders coincide. □

Perspective 3.3.7. *Elementary Divisors and Jordan Blocks*
Over $\mathbb{C}$, elementary divisors are exactly $(z - \lambda)^k$, and $\mathbb{C}[z]/\langle (z-\lambda)^k \rangle$ corresponds to the Jordan block $J_k(\lambda)$.
Thus $\mathrm{ElemDiv}(T)$ is the multiset of Jordan blocks.

3.4 Operator Classification

Perspective 3.4.1. *Operator Modules and Similarity*
For $T \in \mathcal{L}(\mathcal{V})$, $\mathcal{V}_T$ has action $p \cdot \mathbf{v} = p(T)\mathbf{v}$. Then $S \sim T \iff \mathcal{V}_S \cong_{\mathbb{F}[z]} \mathcal{V}_T$; $\mathbb{F}[z]/\langle f \rangle \iff C_f$ (Theorem 3.1.14).

Theorem 3.4.2. *Complete Similarity Invariants*
Let $T, S \in \mathcal{L}(\mathcal{V})$ and $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) $T \sim S \iff \mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S \iff \mathrm{ElemDiv}(T) = \mathrm{ElemDiv}(S) \iff \mathrm{InvFact}(T) = \mathrm{InvFact}(S)$.
(ii) $A \sim B \iff (\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B \iff \mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B) \iff \mathrm{InvFact}(A) = \mathrm{InvFact}(B)$.
(iii) If $A = [T]_{\mathcal{B}}$ for some basis $\mathcal{B}$ of $\mathcal{V}$, then $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(T)$ and $\mathrm{InvFact}(A) = \mathrm{InvFact}(T)$.

Proof.

(*i*) ($\Rightarrow$) If $T \sim S$, then $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S$ by Theorem 3.1.5.
Module isomorphism preserves the PID decomposition, hence preserves ElemDiv and InvFact.
(*i*) ($\Leftarrow$) Equality of either ElemDiv or InvFact gives $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S$ by uniqueness of the PID decomposition.
Thus $T \sim S$ by Theorem 3.1.5.
(*ii*) ($\Rightarrow$) If $A \sim B$, then $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$ by Corollary 3.1.6; hence the divisor data agree.
(*ii*) ($\Leftarrow$) Equality of either matrix divisor datum gives $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$, so $A \sim B$ by Corollary 3.1.6.
(*iii*) If $A = [T]_{\mathcal{B}}$, then $\Phi(\mathbf{v}) = [\mathbf{v}]_{\mathcal{B}}$ satisfies $\Phi(p(T)\mathbf{v}) = p(A)\Phi(\mathbf{v})$.
Hence $\Phi : \mathcal{V}_T \to (\mathbb{F}^n)_A$ is an $\mathbb{F}[z]$-isomorphism, so the divisor data agree.

Structural Commentary: Similarity is module isomorphism plus PID data.
Key Insight: Elementary divisors classify operator modules. □

Lemma 3.4.3. *Determinant of a Companion Block*
Let $f(z) = z^n + \sum_{i=0}^{n-1} c_i z^i$ be monic. Then $\det(zI_n - C_f) = f(z)$.

Proof. Set $D_n(c_0, \ldots, c_{n-1}) := \det(zI_n - C_f)$. For $n = 1$, $D_1(c_0) = z + c_0$.
For $n \geq 2$, first-row expansion gives $D_n(c_0, \ldots, c_{n-1}) = zD_{n-1}(c_1, \ldots, c_{n-1}) + c_0$, since the $(1,1)$-minor is again companion-shaped and the last first-row cofactor contributes c_0. Induction gives $D_n = z^n + \sum_{i=0}^{n-1} c_i z^i = f(z)$.
Structural Commentary: The last column of C_f stores the coefficients of the polynomial relation.
Key Insight: The shifted determinant reads that stored relation back as f. □

Definition 3.4.4. *Rational Canonical Block Forms*

(i) *Invariant-factor RCF*: $A = \mathrm{diag}(C_{s_1}, \ldots, C_{s_r})$, where $s_j \in \mathbb{F}[z]$ are monic nonconstant and $s_r \mid \cdots \mid s_1$.
(ii) *Elementary-divisor RCF*: $A = \mathrm{diag}(C_{q_1}, \ldots, C_{q_m})$, where each q_j is power of monic irreducible polynomial.

Theorem 3.4.5. *Rational Canonical Form Skeleton*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, there is a unique tuple $\mathrm{InvFact}(A) = (s_1, \ldots, s_r)$ with $s_j \in \mathbb{F}[z]$ monic nonconstant, $s_r \mid \cdots \mid s_1$, and $A \sim_{\mathbb{F}} R_A := \mathrm{diag}(C_{s_1}, \ldots, C_{s_r})$. Moreover, $\mu_A = s_1$ and $p_A = \prod_{j=1}^{r} s_j$.
Equivalently, for $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, $A \sim_{\mathbb{F}} B \iff \mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$ as multisets.

Proof. Let $T_A : \mathbb{F}^n \to \mathbb{F}^n$, $\mathbf{x} \mapsto A\mathbf{x}$.
Theorem 3.2.2 gives $\mathbb{F}^n = \bigoplus_{j=1}^{r} \mathcal{C}_j$, $\mathcal{C}_j \cong_{\mathbb{F}[z]} \mathbb{F}[z]/\langle s_j \rangle$, $s_r \mid \cdots \mid s_1$, $[T_A] = R_A$, $\mu_{T_A} = s_1$, and $p_{T_A} = \prod_{j=1}^{r} s_j$.
Hence $A \sim_{\mathbb{F}} R_A$, $\mu_A = s_1$, and $p_A = \prod_{j=1}^{r} s_j$; uniqueness of $\mathrm{InvFact}(A)$ is uniqueness in the same theorem.

($\Rightarrow$) If $A \sim_{\mathbb{F}} B$, then Theorem 3.4.2 gives $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$.
($\Leftarrow$) If $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$, then Theorem 3.4.2 gives $A \sim_{\mathbb{F}} B$.

Structural Commentary: Companion blocks classify similarity without requiring p_A to split.
Key Insight: Similarity classes can be represented by polynomial data rather than by choosing roots. □

Theorem 3.4.6. *Rational Canonical Form: Two Versions*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) If $\mathrm{ElemDiv}(T) = \{q_1, \ldots, q_m\}$, then $\exists$ basis $\mathcal{R}$ with $[T]_{\mathcal{R}} = \mathrm{diag}(C_{q_1}, \ldots, C_{q_m})$.
The form is unique up to block permutation.
(ii) If $\mathrm{InvFact}(T) = (s_1, \ldots, s_r)$ and $s_r \mid \cdots \mid s_1$, then $\exists$ basis $\mathcal{B}$ with $[T]_{\mathcal{B}} = \mathrm{diag}(C_{s_1}, \ldots, C_{s_r})$.
The form is unique in this order.
(iii) If $A \in \mathrm{Mat}(n, n, \mathbb{F})$, then the similarity class $[A]_{\sim_{\mathbb{F}}}$ contains both forms.
Elementary-divisor form is unique up to block order, invariant-factor form by its divisibility chain.

Proof.

(i) Theorem 3.3.1 gives $\mathcal{V} = \bigoplus_{\alpha=1}^{m} \mathcal{C}_\alpha$ with orders q_α; Theorem 3.1.14 gives cyclic bases and blocks C_{q_α}.
Permuting summands conjugates by a permutation matrix.
(ii) Theorem 3.2.2 gives $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_j$, $\mathcal{C}_j \cong_{\mathbb{F}[z]} \mathbb{F}[z]/\langle s_j \rangle$, and cyclic bases with block matrix $\mathrm{diag}(C_{s_1}, \ldots, C_{s_r})$.
(iii) For A, apply (i),(ii) to $T_A : \mathbf{x} \mapsto A\mathbf{x}$; any chosen basis gives a similarity matrix.
Uniqueness follows from Theorem 3.4.2.

Structural Commentary: Elementary divisors split primary blocks; invariant factors recombine them.
Key Insight: Both forms classify the same similarity classes, but they expose different polynomial data. □

Method 3.4.7. *Constructing an Operator from Elementary Divisors*
Let $Q = \{q_1, \ldots, q_m\}$ be a finite multiset of monic prime powers in $\mathbb{F}[z]$, and set $n := \sum_{j=1}^{m} \deg q_j$.

(i) Set $R_Q := \mathrm{diag}(C_{q_1}, \ldots, C_{q_m}) \in \mathrm{Mat}(n, n, \mathbb{F})$.
(ii) Then $\mathrm{ElemDiv}(T_{R_Q}) = Q$; hence every such Q is realised on $\mathbb{F}^n$.
(iii) Apply Method 2.10.5 to recombine Q into $\mathrm{InvFact}(T_{R_Q}) = (s_1, \ldots, s_r)$, then $\mathrm{diag}(C_{s_1}, \ldots, C_{s_r}) \sim_{\mathbb{F}} R_Q$.

Structural Commentary: RCF constructs operators from divisor data.
Key Insight: Companion blocks realise cyclic summands; block sums realise direct sums.

Example 3.4.8. *Rational Canonical Possibilities from Degree Counts*
Let $T \in \mathcal{L}(\mathbb{R}^7)$ and $\mu_T = (z-1)(z^2+1)^2$. For $Q = \mathrm{ElemDiv}(T)$, $\operatorname{lcm} Q = \mu_T$ and $\sum_{q\in Q} \deg q = 7$.
Hence $Q_1 = \{z-1, (z^2+1)^2, z^2+1\}$ or $Q_2 = \{z-1, z-1, z-1, (z^2+1)^2\}$.
Thus the elementary-divisor RCF is $R_i = \bigoplus_{q\in Q_i} C_q$ for $i = 1, 2$, up to block order.

Theorem 3.4.9. *Determinant Form of the Characteristic Polynomial*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, $\chi_A(z) = \det(zI_n - A)$. If $A = [T]_{\mathcal{B}}$, then $\chi_T = \chi_A$.

Proof. Let $Q := \mathrm{ElemDiv}(A)$ and choose elementary-divisor RCF $R = \bigoplus_{q\in Q} C_q \sim_{\mathbb{F}} A$.
By Lemma 3.4.3, $\det(zI - R) = \prod_{q\in Q} \det(zI - C_q) = \prod_{q\in Q} q = \chi_R$.
Since similarity preserves both Q and $\det(zI - \cdot)$, $\chi_A = \chi_R = \det(zI - R) = \det(zI - A)$.
If $A = [T]_{\mathcal{B}}$, then $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(T)$, hence $\chi_A = \chi_T$.
Structural Commentary: Rational canonical form reduces the determinant computation to companion blocks.
Key Insight: The abstract product of elementary divisors is the concrete shifted determinant. □

Theorem 3.4.10. *Changing the Base Field*
Let $\mathbb{F} \subseteq \mathbb{K}$, $A \in \mathrm{Mat}(n, n, \mathbb{F})$, and $\mathrm{ElemDiv}_{\mathbb{F}}(A) = \{p_i^{e_{i,j}}\}_{i,j}$ with $p_i \in \mathbb{F}[z]$ monic irreducible.
If $p_i = \prod_{\ell=1}^{m_i} a_{i,\ell}^{d_{i,\ell}}$ in $\mathbb{K}[z]$, with $a_{i,\ell}$ pairwise distinct monic irreducible over $\mathbb{K}$, then $\mathrm{ElemDiv}_{\mathbb{K}}(A) = \{a_{i,\ell}^{d_{i,\ell} e_{i,j}}\}_{i,j,\ell}$.
Moreover, $\mathrm{InvFact}_{\mathbb{K}}(A) = \mathrm{InvFact}_{\mathbb{F}}(A)$.

Proof. Take elementary-divisor RCF over $\mathbb{F}$: $R = \bigoplus_{i,j} C_{p_i^{e_{i,j}}} \sim_{\mathbb{F}} A$.

Since $R \in \mathrm{Mat}(n, n, \mathbb{K})$ and $p_i^{e_{i,j}} = \prod_{\ell=1}^{m_i} a_{i,\ell}^{d_{i,\ell} e_{i,j}}$ with coprime factors in $\mathbb{K}[z]$,
Theorem 2.5.6 gives $C_{p_i^{e_{i,j}}} \sim_{\mathbb{K}} \bigoplus_{\ell=1}^{m_i} C_{a_{i,\ell}^{d_{i,\ell} e_{i,j}}}$. Summing over (i, j) yields $\mathrm{ElemDiv}_{\mathbb{K}}(A)$.

Recombination into the divisibility chain gives the original invariant factors.
Structural Commentary: Field extension refines elementary divisors.
Key Insight: Elementary divisors split; invariant factors persist. □

Corollary 3.4.11. *Nonderogatory Means One Companion Block*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, the following are equivalent:
(i) A is nonderogatory, i.e. $\mu_A = \chi_A$;
(ii) $(\mathbb{F}^n)_A$ is cyclic;
(iii) $A \sim_{\mathbb{F}} C_{\chi_A}$.

Proof.
(i) ⇒ (ii) This is Theorem 3.3.6.
(ii) ⇒ (iii) Cyclicity gives $\mathrm{InvFact}(A) = (s)$. By Theorem 3.2.2, $s = \mu_A = \chi_A$; hence $A \sim_{\mathbb{F}} C_{\chi_A}$.
(iii) ⇒ (i) Since $\mu_{C_f} = f$ by Theorem 3.1.13, similarity gives $\mu_A = \mu_{C_{\chi_A}} = \chi_A$.
Structural Commentary: Nonderogatory means one cyclic block.
Key Insight: One cyclic generator gives one companion block. □

Corollary 3.4.12. *When the Elementary-Divisor Form Is Diagonal*
Let $T \in \mathcal{L}(\mathcal{V})$, $Q = \mathrm{ElemDiv}(T)$, and $R_{\mathrm{ed}} = \bigoplus_{q\in Q} C_q$.
Then R_{ed} is diagonal iff $\deg q = 1$ for all $q \in Q$, iff $\mu_T = \prod_{\lambda\in\Lambda}(z - \lambda)$ for some finite $\Lambda \subseteq \mathbb{F}$.

Proof.
(⇒) If R_{ed} is diagonal, each block C_q has size 1, hence $\deg q = 1$ for all $q \in Q$.
Thus $\mu_T = \operatorname{lcm} Q$ is a product of distinct linear factors.
(⇐) If $\mu_T = \prod_{\lambda\in\Lambda}(z - \lambda)$, then each prime power $q \in Q$ divides μ_T, so $q = z - \lambda$ for some $\lambda \in \Lambda$.
Hence each C_q is 1×1, and R_{ed} is diagonal.
Structural Commentary: Diagonal RCF means all companion blocks are 1×1.
Key Insight: Nonlinear or repeated factors obstruct diagonal RCF. □

Example 3.4.13. *The Same Polynomial over Different Fields*
Let $\dim V = 4$ and $\mu_T = (z^2+1)(z^2-2)$. Since $\deg \mu_T = 4$, T is nonderogatory and $\mathrm{InvFact}_K(T) = (\mu_T)$ for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$; hence $R_{\mathrm{inv},K} = C_{\mu_T}$. Meanwhile

K	$\mathrm{ElemDiv}_K(T)$
$\mathbb{Q}$	$\{z^2+1,\ z^2-2\}$
$\mathbb{R}$	$\{z^2+1,\ z-\sqrt{2},\ z+\sqrt{2}\}$
$\mathbb{C}$	$\{z-i,\ z+i,\ z-\sqrt{2},\ z+\sqrt{2}\}$

Thus $R_{\mathrm{ed},K} = \bigoplus_{q\in\mathrm{ElemDiv}_K(T)} C_q$: scalar extension refines elementary divisors while preserving the invariant factor. Over $\mathbb{C}$, $R_{\mathrm{ed},\mathbb{C}}$ is diagonal.

Perspective 3.4.14. *Crosswalk with the Spectral Presentation*
The module and spectral presentations encode the same data at different splitting levels:
(i) diagonalizable $\iff$ $\mathrm{ElemDiv}(T)$ consists of linear factors, by Theorem 3.2.1;
(ii) $(z-\lambda)^k \iff J_k(\lambda)$ over an algebraically closed field, by Perspective 3.3.7 and Theorem 6.2.1;
(iii) rank drops recover the exponents k via Proposition 2.9.5, with $p = z - \lambda$ and $M = \mathcal{V}_T$.
Key Insight: Rational form is Jordan form before irreducibles split.

Method 3.4.15. *Moving Between Module and Spectral Data*
For $T \in \mathcal{L}(\mathcal{V})$, translate the $\mathbb{F}[z]$-module data of $\mathcal{V}_T$ into spectral data as follows.
(i) Factor $\mu_T = \prod_{i=1}^{s} p_i^{e_i}$ over $\mathbb{F}$; the primary components are $\ker p_i(T)^{e_i}$.

(ii) Decompose each $\ker p_i(T)^{e_i}$ into cyclic summands; their orders $p_i^{e_{i,j}}$ are the elementary divisors.
(iii) Over $\mathbb{F}$, each elementary divisor $p_i^{e_{i,j}}$ gives a companion block $C_{p_i^{e_{i,j}}}$ in rational canonical form.
(iv) Over a splitting field for the p_i, each elementary divisor refines into powers $(z-\lambda)^k$ by Theorem 3.4.10; each resulting cyclic block is similar to $J_k(\lambda)$ by Theorem 3.1.14 and Theorem 6.1.3.
(v) T is diagonalizable exactly when every elementary divisor is linear, i.e., Corollary 3.4.12 applies.

Key Insight: Rational and Jordan forms encode the same elementary-divisor data over different fields.

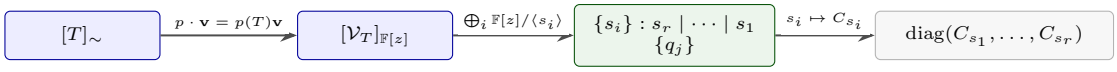

Figure 105: Rational-canonical dictionary: $[T]_\sim$, $[\mathcal{V}_T]_{\mathbb{F}[z]}$, $\{s_i\}$ or $\{q_j\}$, and $\operatorname{diag}(C_{s_i})$ are equivalent data.

4 Associative Algebras

4.1 Associative Algebra Structures

The module viewpoint in Definition 3.1.2 uses one operator at a time. An algebra keeps track of a whole vector space of operators or products at once. The guiding example is $\mathcal{L}(\mathcal{V})$, where addition and scalar multiplication are pointwise and multiplication is composition.

Definition 4.1.1. *Unital Associative Algebra*
Let $\mathbb{F}$ be a field.

(i) An *associative $\mathbb{F}$-algebra* is an $\mathbb{F}$-vector space $\mathcal{A}$ and unital ring $(\mathcal{A}, +, \cdot, 1_\mathcal{A})$ such that $r(ab) = (ra)b = a(rb)$ for all $r \in \mathbb{F}$ and $a, b \in \mathcal{A}$.
(ii) $\mathcal{A}$ is *finite-dimensional* iff $\dim_\mathbb{F} \mathcal{A} < \infty$, and *commutative* iff $ab = ba$ for all $a, b \in \mathcal{A}$.
(iii) $a \in \mathcal{A}$ is *invertible* iff $\exists b \in \mathcal{A}$ such that $ab = ba = 1_\mathcal{A}$.
(iv) In this subsection, *algebra* means unital associative $\mathbb{F}$-algebra unless stated otherwise.

Remark 4.1.2. *Convention*
Relative to Definition 3.7.1, assume multiplication is associative and unital.
Thus, *algebra* means unital associative $\mathbb{F}$-algebra.

Definition 4.1.3. *Center and Central Algebras*

(i) The *center* of an $\mathbb{F}$-algebra $\mathcal{A}$ is $Z(\mathcal{A}) := \{a \in \mathcal{A} : ax = xa \ \forall x \in \mathcal{A}\}$.
(ii) $\mathcal{A}$ is *central over $\mathbb{F}$* iff $Z(\mathcal{A}) = \mathbb{F}1_\mathcal{A} := \{r1_\mathcal{A} : r \in \mathbb{F}\}$.

Proposition 4.1.4. *The Scalar Field Sits in the Center*
If $\mathcal{A}$ is an $\mathbb{F}$-algebra with $1_\mathcal{A} \neq 0$, then $\iota_\mathcal{A} : \mathbb{F} \hookrightarrow Z(\mathcal{A}), \quad r \mapsto r1_\mathcal{A}$, is an injective unital ring homomorphism.
Conversely, if R is a unital ring and $\mathbb{F} \leq Z(R)$ is a subfield, then R is an $\mathbb{F}$-algebra via $r \cdot x := rx$.

Proof.
(⇒) Write $\iota = \iota_\mathcal{A}$. For $r, s \in \mathbb{F}$, $\iota(r+s) = (r+s)1_\mathcal{A} = \iota(r) + \iota(s)$, $\iota(rs) = (rs)1_\mathcal{A} = (r1_\mathcal{A})(s1_\mathcal{A})$, $\iota(1) = 1_\mathcal{A}$.
For $r \in \mathbb{F}$ and $a \in \mathcal{A}$, $\iota(r)a = (r1_\mathcal{A})a = r(1_\mathcal{A}a) = ra = r(a1_\mathcal{A}) = a(r1_\mathcal{A}) = a\iota(r)$, so $\operatorname{im}\iota \subseteq Z(\mathcal{A})$.
If $\iota(r) = 0$ and $r \neq 0$, then $1_\mathcal{A} = \iota(r^{-1})\iota(r) = 0$, contradiction; hence $\ker\iota = \{0\}$.
(⇐) Define $r \cdot x = rx$. The field laws in $\mathbb{F} \subseteq R$ give the vector-space axioms, and $1_\mathbb{F} \cdot x = x$.
For $r \in \mathbb{F}$ and $x, y \in R$, centrality gives $r \cdot (xy) = rxy = (rx)y = x(ry) = x(r \cdot y)$, so the scalar action is compatible with the ring product.

Structural Commentary: An algebra structure is a choice of central scalar copy $\mathbb{F}1_\mathcal{A} \subseteq Z(\mathcal{A})$.
Key Insight: Scalars are precisely central multipliers of the unit. □

Definition 4.1.5. *Subalgebras, Ideals, Quotients, Homomorphisms, and Regular Representation*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra.

(i) A *subalgebra* is a subset $\mathcal{B} \subseteq \mathcal{A}$ such that $1_\mathcal{A} \in \mathcal{B}$, $\mathcal{B} \leq_\mathbb{F} \mathcal{A}$, and $b, b' \in \mathcal{B} \Rightarrow bb' \in \mathcal{B}$.
(ii) For $X \subseteq \mathcal{A}$, the *subalgebra generated by X* is $\langle X \rangle_{\text{subalg}} := \bigcap\{\mathcal{B} : \mathcal{B} \text{ is a subalgebra of } \mathcal{A},\ X \subseteq \mathcal{B}\}$.
Equivalently, $\langle X \rangle_{\text{subalg}} = \operatorname{span}_\mathbb{F}\{1_\mathcal{A},\ x_1 \cdots x_m : m \geq 1,\ x_i \in X\}$.
(iii) A *two-sided ideal* is a subset $\mathcal{I} \subseteq \mathcal{A}$ such that $\mathcal{I} \neq \emptyset$, $u, v \in \mathcal{I} \Rightarrow u - v \in \mathcal{I}$, and $a \in \mathcal{A},\ u \in \mathcal{I} \Rightarrow au, ua \in \mathcal{I}$.
(iv) For nonempty $X \subseteq \mathcal{A}$, the *ideal generated by X* is $\langle X \rangle_{\text{ideal}} := \{\sum_{j=1}^n a_j x_j b_j : n \geq 1,\ x_j \in X,\ a_j, b_j \in \mathcal{A}\}$.
(v) $\mathcal{A}$ is *simple* iff $\mathcal{A}^2 \neq 0$ and its only two-sided ideals are 0 and $\mathcal{A}$.
(vi) If $\mathcal{I} \triangleleft \mathcal{A}$, then $\mathcal{A}/\mathcal{I}$ is an $\mathbb{F}$-algebra by $(a+\mathcal{I})(b+\mathcal{I}) := ab + \mathcal{I}$ and $r(a+\mathcal{I}) := ra + \mathcal{I}$.
(vii) For $\mathbb{F}$-algebras $\mathcal{A}, \mathcal{B}$, an *algebra homomorphism* is a map $\sigma : \mathcal{A} \to \mathcal{B}$ such that, for all $a, a' \in \mathcal{A}$ and $r \in \mathbb{F}$, $\sigma(a+a') = \sigma(a) + \sigma(a')$, $\sigma(aa') = \sigma(a)\sigma(a')$, $\sigma(1_\mathcal{A}) = 1_\mathcal{B}$, and $\sigma(ra) = r\sigma(a)$.
(viii) For $a \in \mathcal{A}$, the *left multiplication operator* is $L_a : \mathcal{A} \to \mathcal{A}$, $x \mapsto ax$.
The map $L_\mathcal{A} : \mathcal{A} \to \mathcal{L}(\mathcal{A})$, $a \mapsto L_a$, is the *left regular representation* of $\mathcal{A}$.

Proposition 4.1.6. *Kernel and First Isomorphism Theorem for Algebras*
If $\sigma : \mathcal{A} \to \mathcal{B}$ is an $\mathbb{F}$-algebra homomorphism,
then $\ker\sigma \triangleleft \mathcal{A}$ and $\overline{\sigma} : \mathcal{A}/\ker\sigma \to \operatorname{im}\sigma$, $a + \ker\sigma \mapsto \sigma(a)$, is an $\mathbb{F}$-algebra isomorphism.

Proof. Set $K := \ker\sigma$. Linearity gives $K \leq_\mathbb{F} \mathcal{A}$.
If $u \in K$ and $a \in \mathcal{A}$, then $\sigma(au) = \sigma(a)\sigma(u) = 0 = \sigma(u)\sigma(a) = \sigma(ua)$, hence $au, ua \in K$ and $K \triangleleft \mathcal{A}$.
Define $\overline{\sigma}(a+K) := \sigma(a)$. Since $a + K = a' + K \iff a - a' \in K \iff \sigma(a) = \sigma(a')$, $\overline{\sigma}$ is well defined and injective; it is surjective onto $\operatorname{im}\sigma$.
For $a, a' \in \mathcal{A}$ and $r \in \mathbb{F}$, $\overline{\sigma}((a+K)(a'+K)) = \sigma(aa') = \sigma(a)\sigma(a')$, $\overline{\sigma}(r(a+K)) = r\sigma(a)$, and $\overline{\sigma}(1_\mathcal{A} + K) = 1_\mathcal{B}$.
Thus $\overline{\sigma}$ is an $\mathbb{F}$-algebra isomorphism.
Structural Commentary: Quotienting by the kernel removes exactly the relations invisible to σ.
Key Insight: The image is the faithful algebra obtained after collapsing $\ker\sigma$. □

Theorem 4.1.7. *Left Regular Representation*
For every $\mathbb{F}$-algebra $\mathcal{A}$, $L_{\mathcal{A}} : \mathcal{A} \hookrightarrow \mathcal{L}(\mathcal{A})$, $a \mapsto L_a$, is an $\mathbb{F}$-algebra monomorphism.

Proof. For $a, b \in \mathcal{A}$, $r \in \mathbb{F}$, and $x \in \mathcal{A}$, $L_{a+b}(x) = (a+b)x = L_a(x) + L_b(x)$ and $L_{ra}(x) = (ra)x = rL_a(x)$, so $L_{\mathcal{A}}$ is $\mathbb{F}$-linear. Also $L_{ab}(x) = (ab)x = a(bx) = (L_aL_b)(x)$ and $L_{1_{\mathcal{A}}} = I_{\mathcal{A}}$, so products and units are preserved.
If $L_a = 0$, then $a = L_a(1_{\mathcal{A}}) = 0$, so $L_{\mathcal{A}}$ is injective.
Structural Commentary: Left multiplication realises $\mathcal{A}$ inside its own operator algebra.
Key Insight: The unit detects elements, since $L_a(1_{\mathcal{A}}) = a$. □

Figure 106: The left regular representation sends $a \in \mathcal{A}$ to $L_a : x \mapsto ax$; $L_{ab} = L_aL_b$ and $L_a(1_{\mathcal{A}}) = a$ give multiplicativity and injectivity.

Method 4.1.8. *Building Multiplication from Structure Constants*
Let $V = \bigoplus_{i \in I} \mathbb{F}b_i$. Choose $c_{ij}^k \in \mathbb{F}$ with $\{k : c_{ij}^k \neq 0\}$ finite for each (i, j).

(i) Define $b_ib_j := \sum_{k \in I} c_{ij}^k b_k$ and extend bilinearly: $(\sum_i r_ib_i)(\sum_j s_jb_j) := \sum_{i,j,k} r_is_jc_{ij}^k b_k$.
(ii) The product is associative iff $\sum_{\ell \in I} c_{ij}^{\ell} c_{\ell k}^m = \sum_{\ell \in I} c_{jk}^{\ell} c_{i\ell}^m$ for all $i, j, k, m \in I$.
(iii) The product is commutative iff $c_{ij}^k = c_{ji}^k$ for all $i, j, k \in I$.
(iv) A basis vector b_e is an identity iff $c_{ej}^m = \delta_j^m = c_{je}^m$ for all $j, m \in I$.

Example 4.1.9. *Basic Algebras*

(i) If $\mathbb{F} \subseteq \mathbb{E}$ are fields, then $\mathbb{E}$ is an $\mathbb{F}$-algebra by $r \cdot x = rx$ and multiplication in $\mathbb{E}$.
(ii) $\mathbb{F}[x]$ is a commutative $\mathbb{F}$-algebra with unit 1 and scalar embedding $r \mapsto r$.
(iii) $\mathrm{Mat}(n, n, \mathbb{F})$ is an $\mathbb{F}$-algebra with unit I_n and product AB.
(iv) For an $\mathbb{F}$-vector space $\mathcal{V}$, $\mathcal{L}(\mathcal{V}) = \mathrm{End}_{\mathbb{F}}(\mathcal{V})$ is an $\mathbb{F}$-algebra with unit $I_{\mathcal{V}}$ and product $S \circ T$.

Definition 4.1.10. *Group Algebra*
For a group G and field $\mathbb{F}$, the *group algebra* is $\mathbb{F}[G] := \{\sum_{g \in G} r_g g : r_g \in \mathbb{F},\ |\{g : r_g \neq 0\}| < \infty\}$.
It has coefficientwise vector-space operations and product $(\sum_g r_g g)(\sum_h s_h h) := \sum_{g,h} r_g s_h (gh)$.

Example 4.1.11. *Group Algebras*
In $\mathbb{F}[G]$, G is an $\mathbb{F}$-basis, e_G is the algebra identity, and associativity follows from $(gh)k = g(hk)$ by bilinearity.
If G is finite, then $\dim_{\mathbb{F}} \mathbb{F}[G] = |G|$. Moreover, $\mathbb{F}[G]$ is commutative iff G is abelian.

Example 4.1.12. *Cyclic Group Algebras and Circulant Shifts*
Let $G = \langle \alpha \rangle$, $|G| = n$, and let $\mathcal{B} = (1, \alpha, \ldots, \alpha^{n-1})$ be the ordered $\mathbb{F}$-basis of $\mathbb{F}[G]$.
For $0 \leq j, k < n$, $L_{\alpha^k}(\alpha^j) = \alpha^{j+k}$ with exponents read modulo n, so L_{α^k} is the k-fold cyclic shift of $\mathcal{B}$.
Let $P := [L_\alpha]_{\mathcal{B}}$, one has $[L_{\alpha^k}]_{\mathcal{B}} = P^k$, where

$$P = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$$

Hence $[L_{\sum_{k=0}^{n-1} c_k \alpha^k}]_{\mathcal{B}} = \sum_{k=0}^{n-1} c_k P^k$, precisely the circulant matrices.

Proposition 4.1.13. *Endomorphisms and Matrices Are the Same Algebra after Choosing a Basis*
Let $\dim \mathcal{V} = n$ and fix an ordered basis $\mathcal{B}$ of $\mathcal{V}$.
Then $\mu_{\mathcal{B}} : \mathcal{L}(\mathcal{V}) \to \mathrm{Mat}(n, n, \mathbb{F})$, $T \mapsto [T]_{\mathcal{B}}$, is an $\mathbb{F}$-algebra isomorphism.

Proof. By Theorem 3.4.5, $\mu_{\mathcal{B}}$ is an $\mathbb{F}$-linear bijection.
For $S, T \in \mathcal{L}(\mathcal{V})$ and $x \in \mathcal{V}$, $[STx]_{\mathcal{B}} = [S]_{\mathcal{B}}[Tx]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}[x]_{\mathcal{B}}$, hence $\mu_{\mathcal{B}}(ST) = \mu_{\mathcal{B}}(S)\mu_{\mathcal{B}}(T)$.
Also $\mu_{\mathcal{B}}(I_{\mathcal{V}}) = I_n$; therefore $\mu_{\mathcal{B}}$ is an $\mathbb{F}$-algebra isomorphism.
Structural Commentary: A basis is a coordinate bridge from abstract endomorphisms to concrete matrices.
Key Insight: Composition becomes matrix multiplication after taking coordinate columns. □

Theorem 4.1.14. *Center and Ideals of an Endomorphism Algebra*
Let $0 \neq \mathcal{V}$ be an $\mathbb{F}$-vector space, and set $\mathcal{F}(\mathcal{V}) := \{T \in \mathcal{L}(\mathcal{V}) : \dim \mathrm{im}\, T < \infty\}$.

(i) $Z(\mathcal{L}(\mathcal{V})) = \mathbb{F}I_{\mathcal{V}}$, so $\mathcal{L}(\mathcal{V})$ is central.
(ii) $\mathcal{F}(\mathcal{V}) \triangleleft \mathcal{L}(\mathcal{V})$, and $0 \neq \mathcal{J} \triangleleft \mathcal{L}(\mathcal{V}) \Rightarrow \mathcal{F}(\mathcal{V}) \subseteq \mathcal{J}$.
(iii) $\mathcal{L}(\mathcal{V})$ is simple iff $\dim \mathcal{V} < \infty$.

Proof.

(i) Let $T \in Z(\mathcal{L}(\mathcal{V}))$. For $0 \neq v \in \mathcal{V}$ and $\varphi \in \mathcal{V}^*$ with $\varphi(v) = 1$, commutation with $v \otimes \varphi : x \mapsto \varphi(x)v$ gives $Tv = T(v \otimes \varphi)(v) = (v \otimes \varphi)T(v) = \varphi(Tv)v$, so every line is T-stable. Thus $Tv = \lambda_v v$ for $v \neq 0$, and applying this to independent u, v and to $u + v$ gives $\lambda_u = \lambda_v$; hence $T = rI_{\mathcal{V}}$.
Scalars commute with all endomorphisms.
(ii) If $R, S \in \mathcal{L}(\mathcal{V})$ and $T \in \mathcal{F}(\mathcal{V})$, then $\mathrm{im}(RTS) \subseteq R(\mathrm{im}\, T)$, so $RTS \in \mathcal{F}(\mathcal{V})$ and $\mathcal{F}(\mathcal{V}) \triangleleft \mathcal{L}(\mathcal{V})$. If $0 \neq \mathcal{J} \triangleleft \mathcal{L}(\mathcal{V})$, choose $0 \neq S \in \mathcal{J}$, e with $Se \neq 0$, and $\eta \in \mathcal{V}^*$ with $\eta(Se) = 1$. For any $v \in \mathcal{V}$ and $\varphi \in \mathcal{V}^*$, $v \otimes \varphi = (v \otimes \eta)S(e \otimes \varphi) \in \mathcal{J}$, so $\mathcal{J}$ contains all rank-one maps, hence all finite-rank maps.

(iii) If $\dim \mathcal{V} < \infty$, then $\mathcal{F}(\mathcal{V}) = \mathcal{L}(\mathcal{V})$, so (ii) makes every nonzero two-sided ideal equal to $\mathcal{L}(\mathcal{V})$.
If $\dim \mathcal{V} = \infty$, then $0 \neq \mathcal{F}(\mathcal{V}) \neq \mathcal{L}(\mathcal{V})$, so $\mathcal{L}(\mathcal{V})$ is not simple.

Structural Commentary: Rank-one operators are the atoms controlling both centrality and ideals in $\mathcal{L}(\mathcal{V})$.
Key Insight: Sandwich one nonzero map to obtain every rank-one map. □

Perspective 4.1.15. *Operator Connection*
$T \in \mathcal{L}(\mathcal{V})$ gives an $\mathbb{F}[x]$-module by $x \cdot v = T(v)$. For any algebra $\mathcal{A}$, the left regular map $L_{\mathcal{A}} : \mathcal{A} \hookrightarrow \mathcal{L}(\mathcal{A})$ realizes $\mathcal{A}$ as a faithful endomorphism algebra.

4.2 Modules over Algebras

Definition 4.2.1. *Algebraic Elements and Minimal Polynomials*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and $a \in \mathcal{A}$.

(i) The *evaluation map at a* is $\mathrm{ev}_a : \mathbb{F}[x] \to \mathcal{A}$, $p \mapsto p(a)$.
(ii) a is *algebraic over $\mathbb{F}$* iff $\ker \mathrm{ev}_a \neq 0$.
(iii) If a is algebraic, its *minimal polynomial* m_a is the unique monic generator of $\ker \mathrm{ev}_a$, equivalently the monic polynomial of least degree with $m_a(a) = 0$.

Theorem 4.2.2. *Finite-Dimensional Algebras Are Algebraic*
If $\dim_{\mathbb{F}} \mathcal{A} < \infty$, then every $a \in \mathcal{A}$ is algebraic over $\mathbb{F}$. Moreover, the algebraic closure $\overline{\mathbb{Q}}_{\mathbb{C}}$ of $\mathbb{Q}$ inside $\mathbb{C}$ is an infinite-dimensional $\mathbb{Q}$-algebra whose elements are all algebraic.

Proof. Let $\dim_{\mathbb{F}} \mathcal{A} = n$ and $a \in \mathcal{A}$. The $n+1$ vectors $1, a, \dots, a^n$ are linearly dependent, so $\sum_{j=0}^{n} c_j a^j = 0$ for some $(c_0, \dots, c_n) \neq 0$; hence $p(x) = \sum_{j=0}^{n} c_j x^j \neq 0$ satisfies $p(a) = 0$.
If $\alpha, \beta \in \overline{\mathbb{Q}}_{\mathbb{C}}$, then $\mathbb{Q}(\alpha, \beta)/\mathbb{Q}$ is finite, so $\alpha + \beta$, $\alpha\beta$, and $r\alpha$ for $r \in \mathbb{Q}$ are algebraic; hence $\overline{\mathbb{Q}}_{\mathbb{C}}$ is a $\mathbb{Q}$-algebra.
For each $n \geq 1$, $x^n - 2$ is Eisenstein at 2, so it has a root of degree n in $\overline{\mathbb{Q}}_{\mathbb{C}}$; therefore $\dim_{\mathbb{Q}} \overline{\mathbb{Q}}_{\mathbb{C}} = \infty$.
Structural Commentary: Finite-dimensionality forces powers of each element into a linear relation.
Key Insight: Algebraic need not mean finite-dimensional as an algebra. □

Proposition 4.2.3. *The Algebra Generated by One Algebraic Element*
Let $a \in \mathcal{A}$ be algebraic with minimal polynomial m_a. Then $\mathbb{F}[a] := \mathrm{im}\, \mathrm{ev}_a = \{p(a) : \deg p < \deg m_a\}$ and $\mathbb{F}[a] \cong \mathbb{F}[x]/\langle m_a \rangle$.

Proof. $\mathrm{ev}_a : \mathbb{F}[x] \to \mathcal{A}$ is an $\mathbb{F}$-algebra homomorphism with image $\mathbb{F}[a]$.
Since $\ker \mathrm{ev}_a = \{p : p(a) = 0\}$ and m_a is the monic generator of this kernel, $\ker \mathrm{ev}_a = \langle m_a \rangle$.
Proposition 4.1.6 gives $\mathbb{F}[a] \cong \mathbb{F}[x]/\langle m_a \rangle$. $\forall p \in \mathbb{F}[x]$, by division $p = q m_a + r$, $\deg r < \deg m_a$, hence $p(a) = r(a)$.
Structural Commentary: One algebraic element gives one polynomial quotient.
Key Insight: Evaluation turns polynomial division modulo m_a into normal forms in $\mathbb{F}[a]$. □

Theorem 4.2.4. *Minimal Polynomial, Annihilator, and Invertibility*
Let $a \in \mathcal{A}$ be algebraic and write $m_a(x) = x^n + \sum_{j=0}^{n-1} \alpha_j x^j$.
Then $\mathrm{ann}_{\mathbb{F}[x]}(a) := \{f \in \mathbb{F}[x] : f(a) = 0\} = \langle m_a \rangle$, and $a \in \mathcal{A}^\times \iff \alpha_0 \neq 0$.

Proof. By Definition 4.2.1, $\mathrm{ann}_{\mathbb{F}[x]}(a) = \ker \mathrm{ev}_a = \langle m_a \rangle$.

(⇒) If $a \in \mathcal{A}^\times$ and $\alpha_0 = 0$, then $m_a = xq$ with $\deg q < \deg m_a$, and $0 = m_a(a) = aq(a)$ gives $q(a) = 0$, contradicting minimality of m_a; hence $\alpha_0 \neq 0$.
(⇐) If $\alpha_0 \neq 0$, then $0 = m_a(a) = a(a^{n-1} + \sum_{j=1}^{n-1} \alpha_j a^{j-1}) + \alpha_0$, so $b := -\alpha_0^{-1}(a^{n-1} + \sum_{j=1}^{n-1} \alpha_j a^{j-1})$ satisfies $ab = 1_{\mathcal{A}}$. Since $b \in \mathbb{F}[a]$, $ab = ba$, so $ba = 1_{\mathcal{A}}$ and $a \in \mathcal{A}^\times$.

Structural Commentary: Invertibility is read from the constant term of the minimal polynomial.
Key Insight: A nonzero constant term solves for a^{-1} as a polynomial in a. □

4.3 Zero Divisors and Spectra of Algebraic Elements

Definition 4.3.1. *Zero Divisors and Spectrum*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and $0 \neq a \in \mathcal{A}$.

(i) a is a *left zero divisor* iff $\exists b \in \mathcal{A} \setminus \{0\}$ with $ab = 0$.
(ii) a is a *right zero divisor* iff $\exists c \in \mathcal{A} \setminus \{0\}$ with $ca = 0$.
(iii) If a is algebraic, *spectrum* is $\mathrm{spec}_{\mathcal{A}}(a) := \{r \in \mathbb{F} : m_a(r) = 0\}$; elements are algebraic *eigenvalues* of a.

Theorem 4.3.2. *Invertible if and only if Not a Zero Divisor*
Let $a \in \mathcal{A}$ be algebraic. Then $a \in \mathcal{A}^\times$ iff a is neither a left nor a right zero divisor.

Proof.

(⇒) If $a \in \mathcal{A}^\times$ and $ab = 0$, then $b = a^{-1}ab = 0$; if $ca = 0$, then $c = caa^{-1} = 0$.
Hence a is neither a left nor a right zero divisor.
(⇐) Prove the contrapositive. If $a \notin \mathcal{A}^\times$, Theorem 4.2.4 gives $m_a(x) = xp(x)$ with $p \neq 0$ and $\deg p < \deg m_a$.
Then $0 = m_a(a) = ap(a) = p(a)a$, while minimality gives $p(a) \neq 0$. Thus a is both left, right zero divisor.

Structural Commentary: For algebraic elements, noninvertibility forces a polynomial kernel witness.
Key Insight: Factoring m_a by x manufactures the zero-divisor witness $p(a)$. □

Corollary 4.3.3. *One-Sided Inverses Become Two-Sided*
Let $a \in \mathcal{A}$ be algebraic. If $\exists b \in \mathcal{A}$ with $ba = 1_{\mathcal{A}}$ or $ab = 1_{\mathcal{A}}$, then $a \in \mathcal{A}^\times$.

Proof. Assume $ba = 1_{\mathcal{A}}$. If $ac = 0$, then $c = bac = 0$; if $ax = 0$, then $x = bax = 0$. Hence a is neither a right nor a left zero divisor, so Theorem 4.3.2 gives $a \in \mathcal{A}^\times$. The case $ab = 1_{\mathcal{A}}$ is symmetric.
Structural Commentary: Algebraicity upgrades one-sided cancellation to two-sided invertibility.
Key Insight: A one-sided inverse rules out both zero-divisor obstructions. □

Theorem 4.3.4. *Roots Detected by Invertibility*
Let $a \in \mathcal{A}$ be algebraic and $r \in \mathbb{F}$. Then $m_a(r) = 0 \iff a - r1_{\mathcal{A}} \notin \mathcal{A}^\times$. In particular, $a \in \mathcal{A}^\times \iff 0 \notin \operatorname{spec}_{\mathcal{A}}(a)$.

Proof.
($\Rightarrow$) If $m_a(r) = 0$, write $m_a = (x - r)q$ with $\deg q < \deg m_a$.
Then $0 = m_a(a) = (a - r1_{\mathcal{A}})q(a) = q(a)(a - r1_{\mathcal{A}})$ and $q(a) \neq 0$ by minimality, so $a - r1_{\mathcal{A}} \notin \mathcal{A}^\times$.
($\Leftarrow$) If $m_a(r) \neq 0$, then $\gcd(x - r, m_a) = 1$, so $\exists u, v \in \mathbb{F}[x]$ with $u(x)(x - r) + v(x)m_a(x) = 1$.
Evaluating at a gives $u(a)(a - r1_{\mathcal{A}}) = 1_{\mathcal{A}} = (a - r1_{\mathcal{A}})u(a)$, hence $a - r1_{\mathcal{A}} \in \mathcal{A}^\times$, contradiction.
Taking $r = 0$ gives $a \in \mathcal{A}^\times \iff m_a(0) \neq 0 \iff 0 \notin \operatorname{spec}_{\mathcal{A}}(a)$.
Structural Commentary: Roots of m_a are precisely the nonunit scalar shifts of a.
Key Insight: Bézout turns $\gcd(x - r, m_a) = 1$ into an explicit inverse. □

Theorem 4.3.5. *Polynomial Spectral Mapping for Algebraic Elements*
Let $\mathbb{F}$ be algebraically closed, $a \in \mathcal{A}$ algebraic, and $p \in \mathbb{F}[x]$.
Then $p(a)$ is algebraic and $\operatorname{spec}_{\mathcal{A}}(p(a)) = p(\operatorname{spec}_{\mathcal{A}}(a)) := \{p(r) : r \in \operatorname{spec}_{\mathcal{A}}(a)\}$.

Proof. Since $\mathbb{F}[p(a)] \subseteq \mathbb{F}[a]$ and $\dim_{\mathbb{F}} \mathbb{F}[a] < \infty$, $p(a)$ is algebraic. If p is constant, the claim is immediate.
If $\lambda \in \operatorname{spec}_{\mathcal{A}}(p(a))$, then $p(a) - \lambda 1_{\mathcal{A}} \notin \mathcal{A}^\times$. Factor $p(x) - \lambda = c\prod_{j=1}^{m}(x - r_j)^{e_j}$ over $\mathbb{F}$.
Since a product of units is a unit, $a - r_j 1_{\mathcal{A}} \notin \mathcal{A}^\times$ for some j. Theorem 4.3.4 gives $r_j \in \operatorname{spec}_{\mathcal{A}}(a)$ and $\lambda = p(r_j)$.
Conversely, if $r \in \operatorname{spec}_{\mathcal{A}}(a)$, then $p(x) - p(r) = (x - r)q(x)$, so $p(a) - p(r)1_{\mathcal{A}} = (a - r1_{\mathcal{A}})q(a)$ is noninvertible; otherwise $a - r1_{\mathcal{A}}$ has a one-sided inverse and Corollary 4.3.3 makes it invertible. Thus $p(r) \in \operatorname{spec}_{\mathcal{A}}(p(a))$.
Structural Commentary: Polynomial evaluation carries spectra by applying the same polynomial to roots.
Key Insight: Factor $p(x) - \lambda$ and detect a nonunit scalar shift of a. □

Proposition 4.3.6. *Spectra of ab and ba*
If $\dim_{\mathbb{F}} \mathcal{A} < \infty$, $\mathbb{F}$ is algebraically closed, and $a, b \in \mathcal{A}$, then $\operatorname{spec}_{\mathcal{A}}(ab) = \operatorname{spec}_{\mathcal{A}}(ba)$.

Proof. All elements are algebraic by Theorem 4.2.2. Let $r \in \mathbb{F}^\times$.
If $ba - r1_{\mathcal{A}} \in \mathcal{A}^\times$, set $u := a(ba - r1_{\mathcal{A}})^{-1}b - 1_{\mathcal{A}}$; then $(ab - r1_{\mathcal{A}})u = r1_{\mathcal{A}}$, so $ab - r1_{\mathcal{A}}$ has a right inverse, hence is invertible by Corollary 4.3.3. Thus $r \notin \operatorname{spec}_{\mathcal{A}}(ba) \Rightarrow r \notin \operatorname{spec}_{\mathcal{A}}(ab)$, and symmetry gives equality off 0.
If $ba \in \mathcal{A}^\times$ with inverse y, then $b(ay) = 1_{\mathcal{A}}$ and $(yb)a = 1_{\mathcal{A}}$, so $b, a \in \mathcal{A}^\times$ by Corollary 4.3.3; hence $ab \in \mathcal{A}^\times$. The converse is symmetric, so 0 belongs to both spectra or neither.
Structural Commentary: $\operatorname{spec}(ab) \setminus \{0\} = \operatorname{spec}(ba) \setminus \{0\}$, and algebraicity resolves the 0-case.
Key Insight: The resolvent identity transfers invertibility between $ba - r1_{\mathcal{A}}$ and $ab - r1_{\mathcal{A}}$ for $r \neq 0$. □

4.4 Division Algebras

Definition 4.4.1. *Division Algebras*
An associative $\mathbb{F}$-algebra $\mathcal{D}$ is a *division algebra* iff $\mathcal{D} \setminus \{0\} \subseteq \mathcal{D}^\times$.

Example 4.4.2. *Hamilton's Quaternions*
Hamilton's quaternion algebra is $\mathbb{H} := \mathbb{R}1 \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ with unit 1,
relations $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, and $ki = j = -ik$, extended $\mathbb{R}$-bilinearly.
It is noncommutative since $ij = -ji$. For $q = r_0 + r_1 i + r_2 j + r_3 k$, define $\overline{q} := r_0 - r_1 i - r_2 j - r_3 k$ and $N(q) := q\overline{q} = r_0^2 + r_1^2 + r_2^2 + r_3^2$.
If $q \neq 0$, then $N(q) > 0$ and $q^{-1} = N(q)^{-1}\overline{q}$, so $\mathbb{H}$ is a real division algebra.

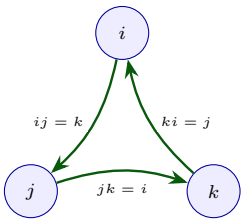

Figure 107: Quaternion basis products: the cycle $i \to j \to k \to i$ gives $ij = k$, $jk = i$, and $ki = j$; reversed order negates the product. Also $i^2 = j^2 = k^2 = -1$, and 1 is the unit.

Theorem 4.4.3. *Division Algebras over Algebraically Closed Fields*
If $\dim_{\mathbb{F}} \mathcal{D} < \infty$, $\mathcal{D}$ is a division $\mathbb{F}$-algebra, and $\mathbb{F}$ is algebraically closed, then $\mathcal{D} = \mathbb{F}1_{\mathcal{D}} \cong \mathbb{F}$.

Proof. Let $a \in \mathcal{D}$. By Theorem 4.2.2, a is algebraic with minimal polynomial m_a.
If $m_a = pq$ with $0 < \deg p, \deg q < \deg m_a$, then $0 = m_a(a) = p(a)q(a)$, while minimality gives $p(a), q(a) \neq 0$; this contradicts $\mathcal{D} \setminus \{0\} \subseteq \mathcal{D}^\times$. Thus m_a is irreducible, hence linear over algebraically closed $\mathbb{F}$: $m_a = x - r$.
Therefore $a = r1_{\mathcal{D}}$, so $\mathcal{D} \subseteq \mathbb{F}1_{\mathcal{D}}$, and the reverse inclusion is the scalar copy.
Structural Commentary: Algebraic closure leaves only scalars.
Key Insight: Irreducible minimal polynomials collapse to linear polynomials. □

Theorem 4.4.4. *Wedderburn's Theorem*
Every finite division algebra $\mathcal{D}$ is commutative; equivalently $\mathcal{D} = Z(\mathcal{D})$, hence $\mathcal{D}$ is a finite field.

Proof. Let $F := Z(\mathcal{D})$, $|F| = z$, $[\mathcal{D} : F] = n$, and $G := \mathcal{D}^\times$.
Suppose $\mathcal{D}$ is noncommutative, and let B be representatives of the noncentral conjugacy classes of G.
For $\beta \in B$, $C_{\mathcal{D}}(\beta)$ is a proper division subalgebra containing F, so $|C_{\mathcal{D}}(\beta)| = z^{k_\beta}$ with $1 < k_\beta < n$ and $k_\beta \mid n$.
By the class equation, Theorem 2.5.2, $z^n - 1 = z - 1 + \sum_{\beta \in B} (z^n - 1)/(z^{k_\beta} - 1)$.
Since $k_\beta \mid n$ and $k_\beta < n$, the primitive nth roots occur in $(x^n - 1)/(x^{k_\beta} - 1)$, so Propositions 3.12.2 and 3.12.3 give $Q_n(z) \mid (z^n - 1)/(z^{k_\beta} - 1)$ for each β, and also $Q_n(z) \mid z^n - 1$.
The class equation therefore gives $Q_n(z) \mid z - 1$, contradicting Lemma 3.12.4, which gives $|Q_n(z)| > z - 1$.
Thus G has no noncentral conjugacy classes, so G is abelian and $\mathcal{D}$ is commutative.
Structural Commentary: The class equation converts noncommutativity into cyclotomic divisibility.
Key Insight: The cyclotomic lower bound is too large to divide $z - 1$. □

Theorem 4.4.5. *Frobenius Classification over $\mathbb{R}$*
If $\dim_{\mathbb{R}} \mathcal{D} < \infty$ and $\mathcal{D}$ is a real division algebra, then $\mathcal{D} \cong \mathbb{R}$, $\mathcal{D} \cong \mathbb{C}$, or $\mathcal{D} \cong \mathbb{H}$.

Proof. For each $d \in \mathcal{D}$, m_d is linear or irreducible quadratic over $\mathbb{R}$; completing the square gives $d = r + u$ with $r \in \mathbb{R}$ and $u^2 \in \mathbb{R}_{\leq 0}$.
Set $\mathcal{D}_0 := \{u \in \mathcal{D} : u^2 \in \mathbb{R}_{\leq 0}\}$; the preceding decomposition and uniqueness give $\mathcal{D} = \mathbb{R} \oplus \mathcal{D}_0$. If $\mathcal{D}_0 = 0$, then $\mathcal{D} \cong \mathbb{R}$. If $\dim \mathcal{D}_0 = 1$, choose $i \in \mathcal{D}_0$ with $i^2 = -1$, so $\mathcal{D} = \mathbb{R} \oplus \mathbb{R}i \cong \mathbb{C}$.
If $\dim \mathcal{D}_0 > 1$, use $\langle u, v \rangle := -\frac{1}{2}(uv + vu)$ on $\mathcal{D}_0$ to choose orthonormal $i, j \in \mathcal{D}_0$ with $i^2 = j^2 = -1$; $ij + ji = 0$.
Then $k := ij$ satisfies $k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and reversed products change sign.
If $\ell \in \mathcal{D}_0$ is orthogonal to i, j, k, the same identities force $\ell = 0$, so $\mathcal{D} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k \cong \mathbb{H}$.
Structural Commentary: A real division algebra splits into scalars plus a Euclidean pure part.
Key Insight: Orthogonal imaginary units generate exactly $\mathbb{C}$ or $\mathbb{H}$. □

5 Tensor Products

5.1 Tensor Products and Universal Properties

Tensor products are the device that turns bilinear data into linear data. The guiding rule is the same one already used for quotient spaces in Theorem 2.2.11: a universal object is recognised by the unique way every compatible map factors through it.

Definition 5.1.1. *Bilinear Maps*
Let $\mathcal{U}, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces and $f : \mathcal{U} \times \mathcal{V} \to \mathcal{W}$.

(i) f is *bilinear* if $\forall r, s \in \mathbb{F}$, $\mathbf{u}, \mathbf{u}' \in \mathcal{U}$, $\mathbf{v}, \mathbf{v}' \in \mathcal{V}$, $f(r\mathbf{u} + s\mathbf{u}', \mathbf{v}) = rf(\mathbf{u}, \mathbf{v}) + sf(\mathbf{u}', \mathbf{v})$ and $f(\mathbf{u}, r\mathbf{v} + s\mathbf{v}') = rf(\mathbf{u}, \mathbf{v}) + sf(\mathbf{u}, \mathbf{v}')$.
(ii) $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W}) := \{f : \mathcal{U} \times \mathcal{V} \to \mathcal{W} : f \text{ bilinear}\}$.
(iii) If $\mathcal{W} = \mathbb{F}$, then $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathbb{F})$ is the space of *bilinear forms* from Definition 4.4.1.

Example 5.1.2. *Basic Bilinear Maps*

(i) If A is an $\mathbb{F}$-algebra, then $\mu : A \times A \to A$, $\mu(a, b) := ab$, is bilinear.
(ii) Evaluation $\mathrm{ev} : \mathcal{L}(\mathcal{V}, \mathcal{W}) \times \mathcal{V} \to \mathcal{W}$, $\mathrm{ev}(T, \mathbf{v}) := T\mathbf{v}$, is bilinear.
(iii) If $\varphi \in \mathcal{V}'$ and $\psi \in \mathcal{W}'$, then $B_{\varphi,\psi} : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$, $B_{\varphi,\psi}(\mathbf{v}, \mathbf{w}) := \varphi(\mathbf{v})\psi(\mathbf{w})$, is bilinear.

Definition 5.1.3. *Tensor Product by Universal Property*
Let $\mathcal{U}, \mathcal{V}$ be $\mathbb{F}$-vector spaces.

(i) A pair $(\mathcal{T}, t)$ is a *tensor product* of $\mathcal{U}$ and $\mathcal{V}$ if $\mathcal{T}$ is an $\mathbb{F}$-vector space, $t \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{T})$, and for every $\mathbb{F}$-vector space $\mathcal{W}$ and $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$, $\exists! \widetilde{f} \in \mathcal{L}(\mathcal{T}, \mathcal{W})$ such that $f = \widetilde{f} \circ t$.
(ii) The map $\widetilde{f}$ is the *mediating map* for f.
(iii) Write $\mathcal{T} = \mathcal{U} \otimes \mathcal{V}$ and $\mathbf{u} \otimes \mathbf{v} := t(\mathbf{u}, \mathbf{v})$; such tensors are *decomposable*.

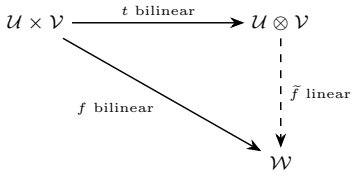

Figure 108: A tensor product is universal for bilinear maps out of $\mathcal{U} \times \mathcal{V}$.

Theorem 5.1.4. *Coordinate-Free Quotient Construction*
Let $\mathcal{U}, \mathcal{V}$ be $\mathbb{F}$-vector spaces and let $\mathbb{F}^{(\mathcal{U} \times \mathcal{V})}$ have basis $\{[\mathbf{u}, \mathbf{v}] : (\mathbf{u}, \mathbf{v}) \in \mathcal{U} \times \mathcal{V}\}$.
Let $\mathcal{R}$ be the span of all $r[\mathbf{u}_1, \mathbf{v}] + s[\mathbf{u}_2, \mathbf{v}] - [r\mathbf{u}_1 + s\mathbf{u}_2, \mathbf{v}]$; and $r[\mathbf{u}, \mathbf{v}_1] + s[\mathbf{u}, \mathbf{v}_2] - [\mathbf{u}, r\mathbf{v}_1 + s\mathbf{v}_2]$, where $r, s \in \mathbb{F}$, $\mathbf{u}, \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, and $\mathbf{v}, \mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$. Then $\mathcal{U} \otimes \mathcal{V} := \mathbb{F}^{(\mathcal{U} \times \mathcal{V})}/\mathcal{R}$, $t(\mathbf{u}, \mathbf{v}) = [\mathbf{u}, \mathbf{v}] + \mathcal{R}$ is a tensor product of $\mathcal{U}$ and $\mathcal{V}$.

Proof. Let $\pi : \mathbb{F}^{(\mathcal{U} \times \mathcal{V})} \to \mathcal{U} \otimes \mathcal{V}$ be the quotient map.
Since the generators of $\mathcal{R}$ vanish in the quotient, $t(\mathbf{u}, \mathbf{v}) = \pi[\mathbf{u}, \mathbf{v}]$ is bilinear.
Let $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ and define $\alpha : \mathbb{F}^{(\mathcal{U} \times \mathcal{V})} \to \mathcal{W}$ by $\alpha[\mathbf{u}, \mathbf{v}] := f(\mathbf{u}, \mathbf{v})$.
Bilinearity gives $\mathcal{R} \subseteq \ker \alpha$, so Theorem 2.2.11 gives a unique $\widetilde{f} \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$ with $\widetilde{f} \circ \pi = \alpha$. Hence $\widetilde{f} \circ t = f$.
If $L \circ t = f$, then L and $\widetilde{f}$ agree on the spanning set $\{[\mathbf{u}, \mathbf{v}] + \mathcal{R}\}$, so $L = \widetilde{f}$.
Structural Commentary: The quotient kills precisely the failure of the raw ordered-pair basis to be bilinear.
Key Insight: A tensor product is a free bilinear map, followed by no extra identifications. □

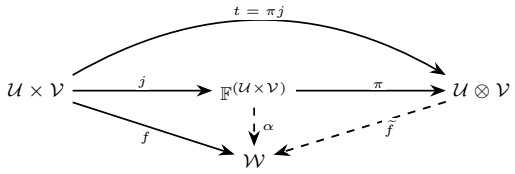

Figure 109: The free linear map α kills $\mathcal{R}$, so it descends uniquely to $\widetilde{f} : \mathcal{U} \otimes \mathcal{V} \to \mathcal{W}$.

Corollary 5.1.5. *Tensor Products Are Unique up to Isomorphism*
If $(\mathcal{T}, t)$ and $(\mathcal{S}, s)$ are tensor products of $\mathcal{U}$ and $\mathcal{V}$,
then there is a unique isomorphism $\Phi : \mathcal{T} \to \mathcal{S}$ satisfying $\Phi(t(\mathbf{u}, \mathbf{v})) = s(\mathbf{u}, \mathbf{v})$, where $\mathbf{u} \in \mathcal{U},\ \mathbf{v} \in \mathcal{V}$.

Proof. Apply universal property of $(\mathcal{T}, t)$ to bilinear map s and universal property of $(\mathcal{S}, s)$ to bilinear map t.
This gives linear maps $\Phi : \mathcal{T} \to \mathcal{S}$ and $\Psi : \mathcal{S} \to \mathcal{T}$. The composites $\Psi\Phi$ and $I_{\mathcal{T}}$ are both mediating maps for t, while $\Phi\Psi$ and $I_{\mathcal{S}}$ are both mediating maps for s. Uniqueness forces $\Psi\Phi = I_{\mathcal{T}}$ and $\Phi\Psi = I_{\mathcal{S}}$.
Structural Commentary: Universal objects are unique up to canonical isomorphism.
Key Insight: Two universal factorisations force the comparison maps to be inverse. □

Theorem 5.1.6. *Bilinearity Equals Linearity on a Tensor Product*
For $\mathbb{F}$-vector spaces $\mathcal{U}, \mathcal{V}, \mathcal{W}$, the map $\Phi : \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W}) \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$, $\Phi(f) = \widetilde{f}$,
where $\widetilde{f}(\mathbf{u} \otimes \mathbf{v}) = f(\mathbf{u}, \mathbf{v})$, is an isomorphism.

Proof. The universal property gives existence and uniqueness of $\widetilde{f}$.
Define $\Psi : \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W}) \to \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ by $\Psi(L) := L \circ t$.
Then $\Psi(\Phi(f)) = \widetilde{f} \circ t = f$, and $\Phi(\Psi(L)) = L$ by uniqueness of the mediating map for $L \circ t$. Hence Φ is bijective.
For $a \in \mathbb{F}$ and f, g bilinear, $\Phi(af + g)(\mathbf{u} \otimes \mathbf{v}) = af(\mathbf{u}, \mathbf{v}) + g(\mathbf{u}, \mathbf{v}) = (a\Phi(f) + \Phi(g))(\mathbf{u} \otimes \mathbf{v})$, so Φ is linear.
Structural Commentary: Tensor products package bilinear maps as ordinary linear maps.
Key Insight: To define a linear map out of $\mathcal{U} \otimes \mathcal{V}$, define a bilinear rule on $\mathcal{U} \times \mathcal{V}$. □

Method 5.1.7. *Defining Maps on Tensor Products*
To define $L \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$:

(i) define a candidate rule $f : \mathcal{U} \times \mathcal{V} \to \mathcal{W}$ by $f(\mathbf{u}, \mathbf{v})$;
(ii) prove $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$;
(iii) set $L := \Phi(f)$ from Theorem 5.1.6, so $L(\mathbf{u} \otimes \mathbf{v}) = f(\mathbf{u}, \mathbf{v})$.

Do not assign arbitrary values on decomposable tensors: they span $\mathcal{U} \otimes \mathcal{V}$ but are usually linearly dependent.

5.2 Tensor Product Bases and Maps

Definition 5.2.1. *Finite-Coordinate Tensor Product Model*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be ordered bases of $\mathcal{V}$ and $\mathcal{W}$.
Define the *coordinate tensor model* $\mathcal{V} \otimes_{\beta,\gamma} \mathcal{W} := \mathrm{Mat}(m, n, \mathbb{F})$ by $\mathbf{v}_i \otimes_{\beta,\gamma} \mathbf{w}_j := E_{i,j}$.
If $\mathbf{v} = \sum_{i=1}^{m} x_i \mathbf{v}_i$ and $\mathbf{w} = \sum_{j=1}^{n} y_j \mathbf{w}_j$, then $\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w} := \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j E_{i,j} \in \mathrm{Mat}(m, n, \mathbb{F})$.

Proposition 5.2.2. *Tensor Model Basis and Bilinear Evaluation*
In the coordinate tensor model associated to bases $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$:

(i) the matrices $E_{i,j} = \mathbf{v}_i \otimes_{\beta,\gamma} \mathbf{w}_j$ form a basis;
(ii) the map $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}$ is bilinear;
(iii) every bilinear form $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ determines a unique linear functional Θ_B on $\mathrm{Mat}(m, n, \mathbb{F})$ such that $\Theta_B(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) = B(\mathbf{v}, \mathbf{w})$ for all $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$.

Proof.

(i) This is the standard entry-matrix basis of $\mathrm{Mat}(m, n, \mathbb{F})$ from Proposition 3.1.5.
(ii) The coordinates of $a\mathbf{v}_1 + \mathbf{v}_2$ are the same linear combination of the coordinates of $\mathbf{v}_1$ and $\mathbf{v}_2$, and the displayed formula is linear in those coordinates. The same argument applies in the second variable.
(iii) Define $\Theta_B(E_{i,j}) = B(\mathbf{v}_i, \mathbf{w}_j)$ on the basis from (i), and extend linearly by Proposition 1.5.8. If $\mathbf{v} = \sum_i x_i \mathbf{v}_i$ and $\mathbf{w} = \sum_j y_j \mathbf{w}_j$, then $\Theta_B(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j B(\mathbf{v}_i, \mathbf{w}_j) = B(\mathbf{v}, \mathbf{w})$, by bilinearity of B.
Uniqueness follows since the values on basis matrices $E_{i,j}$ are forced by $\mathbf{v}_i, \mathbf{w}_j$.

Structural Commentary: The coordinate tensor model stores each pair of basis directions in its own matrix slot, so a bilinear form can be evaluated by a single linear functional on those slots.
Key Insight: Tensor coordinates turn products of coordinates into linear coordinates. □

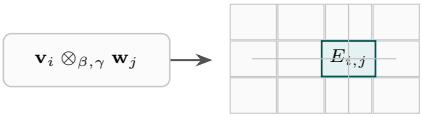

Figure 110: A basis pair occupies the (i, j) matrix slot.

Theorem 5.2.3. *Tensor Bases, Dimension, and Coordinate Rank*
Let $\mathcal{U}$ and $\mathcal{V}$ be finite-dimensional with bases $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_m)$ and $\gamma = (\mathbf{v}_1, \dots, \mathbf{v}_n)$.

(i) The tensors $\mathbf{u}_i \otimes \mathbf{v}_j$ form a basis of $\mathcal{U} \otimes \mathcal{V}$.
(ii) $\dim(\mathcal{U} \otimes \mathcal{V}) = \dim(\mathcal{U}) \dim(\mathcal{V})$.
(iii) Every $\mathbf{z} \in \mathcal{U} \otimes \mathcal{V}$ has a unique matrix $R_{\beta,\gamma}(\mathbf{z}) = [r_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$ such that $\mathbf{z} = \sum_{i=1}^{m} \sum_{j=1}^{n} r_{ij}(\mathbf{u}_i \otimes \mathbf{v}_j)$.
If β', γ' are other bases, then $\operatorname{rank} R_{\beta',\gamma'}(\mathbf{z}) = \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$.

Proof.

(i) The coordinate tensor model of Definition 5.2.1 has the universal property in Definition 5.1.3. Hence Corollary 5.1.5 identifies $\mathcal{U} \otimes \mathcal{V} \cong \mathrm{Mat}(m, n, \mathbb{F})$ with $\mathbf{u}_i \otimes \mathbf{v}_j \mapsto E_{i,j}$, and $\{E_{i,j}\}$ is a basis.
(ii) Counting the mn basis tensors gives $\dim(\mathcal{U} \otimes \mathcal{V}) = mn = \dim(\mathcal{U}) \dim(\mathcal{V})$.
(iii) The basis in (i) gives unique coefficients r_{ij}, hence $R_{\beta,\gamma}(\mathbf{z})$. If β', γ' are other bases, then $R_{\beta',\gamma'}(\mathbf{z}) = P R_{\beta,\gamma}(\mathbf{z}) Q$ for invertible change-of-basis matrices P, Q, so $\operatorname{rank} R_{\beta',\gamma'}(\mathbf{z}) = \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$.

Structural Commentary: Tensor rank is coordinate-free matrix rank.
Key Insight: Changing bases performs row and column changes, which preserve matrix rank. □

Definition 5.2.4. *Tensor Coordinate Matrix and Rank*
Let $\mathcal{U}, \mathcal{V}$ be finite-dimensional, with bases $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_m)$ and $\gamma = (\mathbf{v}_1, \dots, \mathbf{v}_n)$.

(i) For $\mathbf{z} = \sum_{i=1}^{m} \sum_{j=1}^{n} r_{ij}(\mathbf{u}_i \otimes \mathbf{v}_j)$, define the *coordinate matrix* $R_{\beta,\gamma}(\mathbf{z}) := [r_{ij}]$.
(ii) The *rank of the tensor* $\mathbf{z}$ is $\operatorname{rank}(\mathbf{z}) := \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$, independent of β, γ by Theorem 5.2.3.

Theorem 5.2.5. *Zero Tests and Decomposable Rank*
Let $\mathcal{U}$ and $\mathcal{V}$ be vector spaces.

(i) $\forall \mathbf{u} \in \mathcal{U}, \mathbf{v} \in \mathcal{V}$, $\mathbf{0} \otimes \mathbf{v} = \mathbf{u} \otimes \mathbf{0} = \mathbf{0}$.
(ii) If $\mathbf{u}_1, \dots, \mathbf{u}_r$ are linearly independent and $\sum_{i=1}^{r} \mathbf{u}_i \otimes \mathbf{v}_i = \mathbf{0}$, then $\mathbf{v}_1 = \cdots = \mathbf{v}_r = \mathbf{0}$.
(iii) If $\dim \mathcal{U}, \dim \mathcal{V} < \infty$, then $\mathbf{z} \neq \mathbf{0}$ is decomposable $\iff \operatorname{rank}(\mathbf{z}) = 1$.
More generally, $\operatorname{rank}(\mathbf{z}) = \min \left\{ r \geq 0 : \mathbf{z} = \sum_{i=1}^{r} \mathbf{x}_i \otimes \mathbf{y}_i,\ \mathbf{x}_i \in \mathcal{U}, \mathbf{y}_i \in \mathcal{V} \right\}$.

Proof.

(i) The tensor map $t : \mathcal{U} \times \mathcal{V} \to \mathcal{U} \otimes \mathcal{V}$ is bilinear, hence linear in each coordinate; therefore $t(\mathbf{0}, \mathbf{v}) = t(\mathbf{u}, \mathbf{0}) = \mathbf{0}$.
(ii) For each k, extend the dual basis functional on $\operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_r)$ to $\varphi_k \in \mathcal{U}'$ with $\varphi_k(\mathbf{u}_i) = \delta_{ki}$. The bilinear map $(\mathbf{u}, \mathbf{v}) \mapsto \varphi_k(\mathbf{u})\mathbf{v}$ induces $T_k \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{V})$.
Applying T_k gives $\mathbf{0} = T_k\left(\sum_{i=1}^{r} \mathbf{u}_i \otimes \mathbf{v}_i\right) = \sum_{i=1}^{r} \delta_{ki}\mathbf{v}_i = \mathbf{v}_k$.
(iii) Choose bases β, γ. If $\mathbf{z} = \mathbf{x} \otimes \mathbf{y} \neq \mathbf{0}$, then $R_{\beta,\gamma}(\mathbf{z}) = [\mathbf{x}]_\beta [\mathbf{y}]_\gamma^{\mathsf{T}}$, so $\operatorname{rank}(\mathbf{z}) = 1$.
Conversely, if $\operatorname{rank} R_{\beta,\gamma}(\mathbf{z}) = r$, write $R_{\beta,\gamma}(\mathbf{z}) = \sum_{i=1}^{r} a_i b_i^{\mathsf{T}}$ with columns a_i, b_i.
Translating a_i, b_i back to vectors $\mathbf{x}_i, \mathbf{y}_i$ gives $\mathbf{z} = \sum_{i=1}^{r} \mathbf{x}_i \otimes \mathbf{y}_i$ as a sum of r decomposable tensors.
Conversely, any sum of s decomposable tensors gives a sum of s rank-one coordinate matrices, so $\operatorname{rank}(\mathbf{z}) \leq s$; hence the displayed minimum equals $\operatorname{rank}(\mathbf{z})$.

Structural Commentary: Linear functionals can test one tensor coordinate at a time.
Key Insight: Tensor rank is matrix rank after choosing bases, but the number itself survives changing bases. □

Theorem 5.2.6. *Tensor Product of Linear Maps*
Let $A \in \mathcal{L}(\mathcal{U}, \mathcal{U}')$ and $B \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$.
There exists a unique $A \otimes B : \mathcal{U} \otimes \mathcal{V} \to \mathcal{U}' \otimes \mathcal{V}'$ such that $(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$.
Moreover, $(A_2 A_1) \otimes (B_2 B_1) = (A_2 \otimes B_2)(A_1 \otimes B_1)$, $I_{\mathcal{U}} \otimes I_{\mathcal{V}} = I_{\mathcal{U} \otimes \mathcal{V}}$.
In finite dimensions, $[A \otimes B]_{\beta \otimes \gamma}^{\beta' \otimes \gamma'} = [A]_\beta^{\beta'} \otimes [B]_\gamma^{\gamma'}$ for lexicographically ordered tensor-product bases, where the right-hand $\otimes$ is the matrix Kronecker product.

Proof. The map $(\mathbf{u}, \mathbf{v}) \mapsto A\mathbf{u} \otimes B\mathbf{v}$ is bilinear, so Theorem 5.1.6 gives the unique $A \otimes B$.
The two functorial identities hold on decomposable tensors $((A_2 \otimes B_2)(A_1 \otimes B_1))(\mathbf{u} \otimes \mathbf{v}) = A_2 A_1 \mathbf{u} \otimes B_2 B_1 \mathbf{v}$ and $(I_{\mathcal{U}} \otimes I_{\mathcal{V}})(\mathbf{u} \otimes \mathbf{v}) = \mathbf{u} \otimes \mathbf{v}$, and decomposable tensors span. If $A\mathbf{u}_j = \sum_i a_{ij}\mathbf{u}_i'$ and $B\mathbf{v}_\ell = \sum_k b_{k\ell}\mathbf{v}_k'$, then $(A \otimes B)(\mathbf{u}_j \otimes \mathbf{v}_\ell) = \sum_{i,k} a_{ij} b_{k\ell}(\mathbf{u}_i' \otimes \mathbf{v}_k')$, which is the Kronecker block-entry rule.
Structural Commentary: Tensoring maps acts independently in the two tensor coordinates.
Key Insight: The mixed product property for Kronecker products is the coordinate shadow of functoriality. □

Theorem 5.2.7. *Tensor Product of Spaces of Linear Maps*
Let $\mathcal{U}, \mathcal{V}, \mathcal{X}, \mathcal{Y}$ be $\mathbb{F}$-vector spaces. The assignment $(A, B) \mapsto A \otimes B$ induces a natural injective map
$\Theta : \mathcal{L}(\mathcal{U}, \mathcal{X}) \otimes \mathcal{L}(\mathcal{V}, \mathcal{Y}) \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{X} \otimes \mathcal{Y})$ with $\Theta(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$.
If $\mathcal{U}, \mathcal{V}, \mathcal{X}, \mathcal{Y}$ are finite-dimensional, then Θ is an isomorphism.
In particular, $\mathcal{U}' \otimes \mathcal{V}' \hookrightarrow (\mathcal{U} \otimes \mathcal{V})'$; $(\varphi \otimes \psi)(\mathbf{u} \otimes \mathbf{v}) = \varphi(\mathbf{u})\psi(\mathbf{v})$ is an isomorphism when $\dim \mathcal{U}, \dim \mathcal{V} < \infty$.

Proof.

(i) By Theorem 5.2.6, $(A, B) \mapsto A \otimes B$ is bilinear, hence induces Θ.

(ii) Let $0 \neq Z = \sum_{i=1}^{r} A_i \otimes B_i$ with r minimal. Then $A_i \neq 0$ and (B_i) is linearly independent.
Choose $\mathbf{u}$ with some $A_i\mathbf{u} \neq 0$, after reindexing let $A_1\mathbf{u}, \dots, A_m\mathbf{u}$ be basis of $\operatorname{span}\{A_i\mathbf{u} : 1 \leq i \leq r\}$.
Absorb remaining coefficients into B_i gives $C_1, \dots, C_m$, not all zero, s.t. $\Theta(Z)(\mathbf{u} \otimes \mathbf{v}) = \sum_{i=1}^{m} (A_i\mathbf{u}) \otimes C_i\mathbf{v}$.
If $\Theta(Z) = 0$, Theorem 5.2.5 gives $C_i\mathbf{v} = 0$ for all $i, \mathbf{v}$, contradiction. Thus $\ker\Theta = 0$.

(iii) If $\mathcal{U}, \mathcal{V}, \mathcal{X}, \mathcal{Y}$ are finite-dimensional, then the domain and codomain of Θ have equal dimension by Theorem 5.2.3, so injectivity implies bijectivity. Taking $\mathcal{X} = \mathcal{Y} = \mathbb{F}$, $\mathcal{L}(\mathcal{U}, \mathbb{F}) = \mathcal{U}'$, $\mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathbb{F}) = (\mathcal{U} \otimes \mathcal{V})'$ gives the dual embedding and finite-dimensional isomorphism.

Structural Commentary: Tensoring maps yields maps on tensor products.
Key Insight: The rule on decomposable tensors is enough, and finite dimensions make the embedding onto. □

Proposition 5.2.8. *Scalar Extension by Tensor Product*
Let $K/\mathbb{F}$ be a field extension and let $\mathcal{V}_\mathbb{F}$ be an $\mathbb{F}$-vector space.
Then $K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F}$ is a K-vector space via $\alpha \cdot (\beta \otimes \mathbf{v}) := (\alpha\beta) \otimes \mathbf{v}$. With $\mu : \mathcal{V}_\mathbb{F} \to K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F}$, $\mu(\mathbf{v}) := 1 \otimes \mathbf{v}$, every $\mathbb{F}$-linear $f : \mathcal{V}_\mathbb{F} \to \mathcal{Y}$ into a K-vector space $\mathcal{Y}$ admits a unique K-linear $\widetilde{f} : K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F} \to \mathcal{Y}$ such that $\widetilde{f} \circ \mu = f$.

Proof. For $\alpha \in K$, $(\beta, \mathbf{v}) \mapsto (\alpha\beta) \otimes \mathbf{v}$ is $\mathbb{F}$-bilinear, hence induces $M_\alpha \in \mathcal{L}_\mathbb{F}(K \otimes_\mathbb{F} \mathcal{V}_\mathbb{F})$.
On decomposable tensors, $M_\alpha M_\lambda = M_{\alpha\lambda}$, $M_1 = I$, and distributivity holds; since decomposable tensors span, these define a K-vector-space structure.
For f, set $g(\beta, \mathbf{v}) := \beta f(\mathbf{v})$. Then g is $\mathbb{F}$-bilinear, so it induces $\widetilde{f}$ with $\widetilde{f}(\beta \otimes \mathbf{v}) = \beta f(\mathbf{v})$. This is K-linear and gives $\widetilde{f} \circ \mu = f$. If h is K-linear and $h \circ \mu = f$, then $h(\beta \otimes \mathbf{v}) = \beta h(1 \otimes \mathbf{v}) = \beta f(\mathbf{v})$, so $h = \widetilde{f}$.
Structural Commentary: Scalar extension tensors in the larger field.
Key Insight: Tensoring with K upgrades which scalars are allowed to act. □

5.3 Iterated Tensor Products and Tensor Spaces

The tensor-product universal property extends from bilinear maps to maps with more variables, and then to tensor spaces and contractions.

Definition 5.3.1. *Multilinear Maps and Iterated Tensor Products*
Let $\mathcal{V}_1, \dots, \mathcal{V}_n, \mathcal{W}$ be $\mathbb{F}$-vector spaces.

(i) A map $f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{W}$ is *multilinear*, or *n-linear*, if for each k, the map $\mathbf{v}_k \mapsto f(\mathbf{v}_1, \dots, \mathbf{v}_k, \dots, \mathbf{v}_n)$ is linear with the other entries fixed.
(ii) $\operatorname{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{W}) := \{f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{W} : f \text{ multilinear}\}$.
(iii) If $\mathcal{W} = \mathbb{F}$, elements of $\operatorname{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathbb{F})$ are *multilinear forms*.
(iv) A pair $(\mathcal{T}, t)$ is an *iterated tensor product* of $\mathcal{V}_1, \dots, \mathcal{V}_n$ if $t \in \operatorname{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{T})$ and, for every $\mathbb{F}$-vector space $\mathcal{Y}$ and $f \in \operatorname{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{Y})$, $\exists! \widetilde{f} \in \mathcal{L}(\mathcal{T}, \mathcal{Y})$ such that $f = \widetilde{f} \circ t$.
(v) Write $\mathcal{T} = \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$ and $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n := t(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

Example 5.3.2. *Basic Multilinear Maps*

(i) If A is an associative $\mathbb{F}$-algebra, then $m_n : A^n \to A$, $m_n(a_1, \dots, a_n) := a_1 \cdots a_n$, is n-linear.
(ii) If $\dim \mathcal{V} = n$, then $\det : \mathcal{V}^n \to \mathbb{F}$, applied to coordinate columns in any basis, is alternating n-linear form.

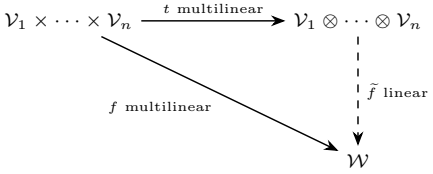

Figure 111: The iterated tensor product is universal for multilinear maps.

Theorem 5.3.3. *Basic Laws for Iterated Tensor Products*
Let $\mathcal{V}_1, \dots, \mathcal{V}_n, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces.

(i) $\operatorname{Mult}_\mathbb{F}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{W}) \cong \mathcal{L}(\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n, \mathcal{W})$ *naturally.*
(ii) For $1 \leq r < n$, the canonical α_r is $\alpha_r : (\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_r) \otimes (\mathcal{V}_{r+1} \otimes \cdots \otimes \mathcal{V}_n) \cong \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$, where $\alpha_r((\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_r) \otimes (\mathbf{v}_{r+1} \otimes \cdots \otimes \mathbf{v}_n)) = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n$.
(iii) For $\pi \in S_n$, the canonical τ_π is $\tau_\pi : \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n \cong \mathcal{V}_{\pi(1)} \otimes \cdots \otimes \mathcal{V}_{\pi(n)}$, where $\tau_\pi(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n) = \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}$.
(iv) $\lambda : \mathbb{F} \otimes \mathcal{V} \cong \mathcal{V}$, $\lambda(r \otimes \mathbf{v}) = r\mathbf{v}$, and $\rho : \mathcal{V} \otimes \mathbb{F} \cong \mathcal{V}$, $\rho(\mathbf{v} \otimes r) = r\mathbf{v}$.
(v) If $\dim \mathcal{V}_i < \infty$ for all i, then $\dim(\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n) = \prod_{i=1}^{n} \dim(\mathcal{V}_i)$.

Proof.

(i) Let $t : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$ be canonical. Define $\Phi(f) = \widetilde{f}$, where $\widetilde{f}$ is the unique linear map satisfying $\widetilde{f}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n) = f(\mathbf{v}_1, \dots, \mathbf{v}_n)$; conversely define $\Psi(T) = T \circ t$. Then $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are identity maps, so Φ is a linear bijection.
(ii) Set $A_r = (\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_r) \otimes (\mathcal{V}_{r+1} \otimes \cdots \otimes \mathcal{V}_n)$ and $B = \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$. The rules defining $\alpha_r : A_r \to B$ and $\beta_r : B \to A_r$, $\beta_r(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n) = (\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_r) \otimes (\mathbf{v}_{r+1} \otimes \cdots \otimes \mathbf{v}_n)$, are induced by multilinearity; $\alpha_r \circ \beta_r$ and $\beta_r \circ \alpha_r$ fix the spanning decomposable tensors, hence are identities.
(iii) Set $C_\pi = \mathcal{V}_{\pi(1)} \otimes \cdots \otimes \mathcal{V}_{\pi(n)}$. The map $p_\pi : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to C_\pi$, $p_\pi(\mathbf{v}_1, \dots, \mathbf{v}_n) = \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}$, is multilinear, so the universal property induces the unique τ_π with $\tau_\pi \circ t = p_\pi$.
The map $\sigma_\pi : C_\pi \to \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$, $\sigma_\pi(\mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}) = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n$, is induced similarly; $\sigma_\pi\tau_\pi$ and $\tau_\pi\sigma_\pi$ fix decomposable spanning tensors, hence are identities.

(iv) The maps $m_L : \mathbb{F} \times \mathcal{V} \to \mathcal{V}$, $m_L(r, \mathbf{v}) = r\mathbf{v}$, and $m_R : \mathcal{V} \times \mathbb{F} \to \mathcal{V}$, $m_R(\mathbf{v}, r) = r\mathbf{v}$, are bilinear, hence induce λ and ρ. Since $\lambda(1 \otimes \mathbf{v}) = \mathbf{v}$ and $1 \otimes r\mathbf{v} = r \otimes \mathbf{v}$, $\lambda^{-1}(\mathbf{v}) = 1 \otimes \mathbf{v}$.
Since $\rho(\mathbf{v} \otimes 1) = \mathbf{v}$ and $r\mathbf{v} \otimes 1 = \mathbf{v} \otimes r$, $\rho^{-1}(\mathbf{v}) = \mathbf{v} \otimes 1$.
(v) Choose finite bases B_i of $\mathcal{V}_i$ and set $\mathcal{B} = \{\mathbf{b}_1 \otimes \cdots \otimes \mathbf{b}_n : \mathbf{b}_i \in B_i\}$. By repeated application of Theorem 5.2.3, $\mathcal{B}$ is a basis; hence $\dim(\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n) = \#\mathcal{B} = \prod_{i=1}^n \#B_i = \prod_{i=1}^n \dim \mathcal{V}_i$.

Structural Commentary: Tensor parenthesising and ordering are canonical bookkeeping.
Key Insight: Universal properties extend pure-tensor rules uniquely. □

Definition 5.3.4. *Tensor Spaces of Type (p, q)*
Let $\dim \mathcal{V} < \infty$.
(i) For $(p, q) \in \mathbb{Z}^2_{\geq 0} \setminus \{(0, 0)\}$, with empty tensor powers equal to $\mathbb{F}$, $\mathcal{T}^p_q(\mathcal{V}) := \mathcal{V}^{\otimes p} \otimes (\mathcal{V}')^{\otimes q}$.
(ii) $\mathcal{T}^p_q(\mathcal{V})$ is the *tensor space of type (p, q)*; p is the *contravariant type*, q the *covariant type*.
(iii) $\mathcal{T}^0_0(\mathcal{V}) := \mathbb{F}$.

Proposition 5.3.5. *Tensor Spaces as Multilinear Functionals*
Let $\dim \mathcal{V} = n < \infty$ and $p, q \in \mathbb{Z}_{\geq 0}$.
(i) $\dim \mathcal{T}^p_q(\mathcal{V}) = n^{p+q}$.
(ii) Let $\iota : \mathcal{V} \to \mathcal{V}''$, $\iota(\mathbf{v})(\eta) = \eta(\mathbf{v})$. Via ι, natural isomorphism is $\Theta_{p,q} : \mathcal{T}^p_q(\mathcal{V}) \xrightarrow{\sim} \mathrm{Mult}_{\mathbb{F}}((\mathcal{V}')^{\times p} \times \mathcal{V}^{\times q}; \mathbb{F})$; where $\Theta_{p,q}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p \otimes \varphi_1 \otimes \cdots \otimes \varphi_q)(\eta_1, \ldots, \eta_p, \mathbf{x}_1, \ldots, \mathbf{x}_q) = \prod_{i=1}^p \eta_i(\mathbf{v}_i) \prod_{j=1}^q \varphi_j(\mathbf{x}_j)$, with empty products equal to 1.

Proof.
(i) By Theorem 5.3.3, $\dim \mathcal{V}' = n$, $\dim \mathcal{T}^p_q(\mathcal{V}) = \dim(\mathcal{V}^{\otimes p}) \dim((\mathcal{V}')^{\otimes q}) = n^p n^q = n^{p+q}$, with $n^0 = 1$.
(ii) Set $X = (\mathcal{V}')^{\otimes p} \otimes \mathcal{V}^{\otimes q}$. Since $\iota : \mathcal{V} \cong \mathcal{V}''$ and all factors are finite-dimensional,
$\mathcal{T}^p_q(\mathcal{V}) = \mathcal{V}^{\otimes p} \otimes (\mathcal{V}')^{\otimes q} \cong ((\mathcal{V}')')^{\otimes p} \otimes \mathcal{V}'^{\otimes q} \cong X' \cong \mathrm{Mult}_{\mathbb{F}}((\mathcal{V}')^{\times p} \times \mathcal{V}^{\times q}; \mathbb{F})$.
On decomposables, the composite is $\Theta_{p,q}$; spanning gives uniqueness.

Structural Commentary: Tensor elements are multilinear functionals after finite-dimensional duality.
Key Insight: Choosing which factors are active turns tensor factors into input slots. □

Definition 5.3.6. *Contraction*
Let $p, q \geq 1$. The *contraction* is the unique linear map $C : \mathcal{T}^p_q(\mathcal{V}) \to \mathcal{T}^{p-1}_{q-1}(\mathcal{V})$, where
$C(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p \otimes \varphi_1 \otimes \cdots \otimes \varphi_q) = \varphi_1(\mathbf{v}_1)\mathbf{v}_2 \otimes \cdots \otimes \mathbf{v}_p \otimes \varphi_2 \otimes \cdots \otimes \varphi_q$.
For (i, j), C_{ij} is C conjugated by the factor permutation placing $\mathbf{v}_i, \varphi_j$ first.

Example 5.3.7. *Rank-One Operators and Trace*
Under $\mathcal{V} \otimes \mathcal{V}' \cong \mathcal{L}(\mathcal{V})$, $\mathbf{v} \otimes \varphi \mapsto T_{\mathbf{v},\varphi}$, where $T_{\mathbf{v},\varphi}(\mathbf{x}) = \varphi(\mathbf{x})\mathbf{v}$. Hence $C(\mathbf{v} \otimes \varphi) = \varphi(\mathbf{v}) = \mathrm{tr}(T_{\mathbf{v},\varphi})$.

6 Symmetric and Exterior Algebras

6.1 Symmetric Algebras

Definition 6.1.1. *Symmetric, Skew-Symmetric, and Alternating Maps*
Let $f : \mathcal{V}^{\times n} \to \mathcal{W}$ be multilinear.
(i) f is *symmetric* if interchanging any two coordinate positions does not change its value.
(ii) f is *skew-symmetric* if interchanging any two coordinate positions multiplies its value by -1.
(iii) f is *alternating* if $f(\mathbf{v}_1, \ldots, \mathbf{v}_n) = 0$ whenever two coordinate entries are equal.

If $\mathrm{char}(\mathbb{F}) \neq 2$, alternating and skew-symmetric are equivalent.
If $\mathrm{char}(\mathbb{F}) = 2$, skew-symmetric is the same as symmetric, and alternating is the stronger condition.

Definition 6.1.2. *Permutation Action and Symmetric/Exterior Tensor Spaces*
Let $\dim \mathcal{V} < \infty$, $p \in \mathbb{Z}_{\geq 0}$, $\mathcal{V}^{\otimes 0} = \mathbb{F}$, $T_p := \mathcal{V}^{\otimes p}$, and $G_p := \mathfrak{S}_p$.
(i) For $\sigma \in G_p$, the *permutation action* is $\lambda_\sigma : T_p \to T_p$, determined by $\lambda_\sigma(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) = \mathbf{v}_{\sigma(1)} \otimes \cdots \otimes \mathbf{v}_{\sigma(p)}$.
(ii) The *symmetric tensors* are $\mathrm{ST}^p(\mathcal{V}) := \{\mathbf{z} \in T_p : \forall \sigma \in G_p, \lambda_\sigma \mathbf{z} = \mathbf{z}\}$.
(iii) The *antisymmetric tensors* are $\mathrm{AT}^p(\mathcal{V}) := \{\mathbf{z} \in T_p : \forall \sigma \in G_p, \lambda_\sigma \mathbf{z} = \mathrm{sign}(\sigma)\mathbf{z}\}$.
(iv) Set $R_\vee := \mathrm{span}\{\lambda_\sigma \mathbf{z} - \mathbf{z} : \sigma \in G_p, \mathbf{z} \in T_p\}$.
(v) The *symmetric power* is $\mathrm{Sym}^p(\mathcal{V}) := T_p / R_\vee$.
(vi) Set $A_p := \mathrm{span}\{\lambda_\sigma \mathbf{z} - \mathrm{sign}(\sigma)\mathbf{z} : \sigma \in G_p, \mathbf{z} \in T_p\}$ and $D_p := \mathrm{span}\{\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p : \exists i < j, \mathbf{v}_i = \mathbf{v}_j\}$.
The *exterior power* is $\bigwedge^p \mathcal{V} := T_p / (A_p + D_p)$.
(vii) For quotient map $q_\vee$, the *symmetric product* is $q_\vee(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) = \mathbf{v}_1 \vee \cdots \vee \mathbf{v}_p$
(viii) For quotient map $q_\wedge$, the *exterior product* is $q_\wedge(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_p$.

Theorem 6.1.3. *Universal Properties of Symmetric and Exterior Powers*
Let $\mathcal{U}$ be a vector space, $M = \mathrm{Mult}_{\mathbb{F}}(\mathcal{V}, \ldots, \mathcal{V}; \mathcal{U})$, and $f^\sigma(\mathbf{v}_1, \ldots, \mathbf{v}_p) := f(\mathbf{v}_{\sigma(1)}, \ldots, \mathbf{v}_{\sigma(p)})$.
(i) $\mathcal{L}(\mathrm{Sym}^p(\mathcal{V}), \mathcal{U}) \cong M_\vee$, where $M_\vee := \{f \in M : \forall \sigma \in \mathfrak{S}_p,\ f^\sigma = f\}$.
*(ii) $\mathcal{L}(\bigwedge^p \mathcal{V}, \mathcal{U}) \cong M_\wedge$, where $M_\wedge := \{f \in M : \forall \sigma \in \mathfrak{S}_p,\ f^\sigma = \mathrm{sign}(\sigma) f,\ \forall (\mathbf{v}_1, \ldots, \mathbf{v}_p),$
$(\exists i < j, \mathbf{v}_i = \mathbf{v}_j) \Rightarrow f(\mathbf{v}_1, \ldots, \mathbf{v}_p) = 0\}$.*

Proof.
(i) Let $t : \mathcal{V}^{\times p} \to \mathcal{V}^{\otimes p}$ and $q_\vee : \mathcal{V}^{\otimes p} \to \mathrm{Sym}^p(\mathcal{V})$ be canonical.
The map $\Phi_\vee(T) = T q_\vee t$ sends $\mathcal{L}(\mathrm{Sym}^p(\mathcal{V}), \mathcal{U})$ into $M_\vee$.
Conversely, for $f \in M_\vee$, there is a unique $\widetilde{f} : \mathcal{V}^{\otimes p} \to \mathcal{U}$ with $f = \widetilde{f} t$; since $\widetilde{f}(R_\vee) = 0$, it factors uniquely through $q_\vee$ as $\widetilde{f} = \overline{f} q_\vee$, where $\overline{f} : \mathrm{Sym}^p(\mathcal{V}) \to \mathcal{U}$. Thus $f \mapsto \overline{f}$ is $\Phi_\vee^{-1}$.
(ii) Let $t : \mathcal{V}^{\times p} \to \mathcal{V}^{\otimes p}$ and $q_\wedge : \mathcal{V}^{\otimes p} \to \bigwedge^p \mathcal{V}$ be canonical.
The map $\Phi_\wedge(T) = T q_\wedge t$ sends $\mathcal{L}(\bigwedge^p \mathcal{V}, \mathcal{U})$ into $M_\wedge$. Conversely, for $f \in M_\wedge$, the induced $\widetilde{f} : \mathcal{V}^{\otimes p} \to \mathcal{U}$ satisfies $\widetilde{f}(R_\wedge) = 0$, hence factors uniquely as $\widetilde{f} = \overline{f} q_\wedge$; thus $f \mapsto \overline{f}$ is $\Phi_\wedge^{-1}$.

Structural Commentary: Symmetric and exterior powers are tensor quotients by permutation relations.
Key Insight: Symmetric and alternating multilinear maps are ordinary linear maps out of the right quotient. □

Theorem 6.1.4. *Dimensions of Symmetric and Exterior Powers*
Let $\dim \mathcal{V} = n < \infty$ *and* $p \in \mathbb{Z}_{\geq 0}$. *Set* $I_\vee = \{1 \leq i_1 \leq \cdots \leq i_p \leq n\}$ *and* $I_\wedge = \{1 \leq i_1 < \cdots < i_p \leq n\}$.

(i) $\dim \mathrm{Sym}^p(\mathcal{V}) = \#I_\vee = \binom{n+p-1}{p}$.
(ii) $\dim \bigwedge^p \mathcal{V} = \#I_\wedge = \binom{n}{p}$, *with* $\binom{n}{p} = 0$ *for* $p > n$.

Proof. Let $E = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be a basis. For $I = (i_1, \ldots, i_p)$, set $\mathbf{e}_I^\vee = \mathbf{e}_{i_1} \vee \cdots \vee \mathbf{e}_{i_p}$ and $\mathbf{e}_I^\wedge = \mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_p}$.

(i) $R_\vee$ identifies each word with its nondecreasing representative. Thus $B_\vee = \{\mathbf{e}_I^\vee : I \in I_\vee\}$ is the quotient basis, and $\dim \mathrm{Sym}^p(\mathcal{V}) = \#B_\vee = \#I_\vee = \binom{n+p-1}{p}$.
(ii) $R_\wedge$ kills repeated words and identifies each remaining word with its increasing representative up to sign. Thus $B_\wedge = \{\mathbf{e}_I^\wedge : I \in I_\wedge\}$ is the quotient basis, and $\dim \bigwedge^p \mathcal{V} = \#B_\wedge = \#I_\wedge = \binom{n}{p}$.

Structural Commentary: Symmetric powers count repetitions; exterior powers forbid repetitions.
Key Insight: Multisets give symmetric dimensions, subsets give exterior dimensions. □

Proposition 6.1.5. *Symmetrization in Characteristic Zero*
Assume $\mathrm{char}(\mathbb{F}) = 0$. *Define* $S, A \in \mathcal{L}(\mathcal{V}^{\otimes p})$ *by* $S := \frac{1}{p!} \sum_{\sigma \in \mathfrak{S}_p} \lambda_\sigma$ *and* $A := \frac{1}{p!} \sum_{\sigma \in \mathfrak{S}_p} \mathrm{sign}(\sigma) \lambda_\sigma$.

(i) $S^2 = S$ *and* $\mathrm{im}\, S = \mathrm{ST}^p(\mathcal{V})$.
(ii) $A^2 = A$ *and* $\mathrm{im}\, A = \mathrm{AT}^p(\mathcal{V})$.

Proof.

(i) For $\tau \in \mathfrak{S}_p$, $\lambda_\tau S = p!^{-1} \sum_\sigma \lambda_{\tau\sigma} = S$, hence $\mathrm{im}\, S \subseteq \mathrm{ST}^p(\mathcal{V})$.
If $\mathbf{z} \in \mathrm{ST}^p(\mathcal{V})$, then $S\mathbf{z} = \mathbf{z}$, so $\mathrm{im}\, S = \mathrm{ST}^p(\mathcal{V})$ and $S^2 = S$.
(ii) For $\tau \in \mathfrak{S}_p$, $\lambda_\tau A = p!^{-1} \sum_\sigma \mathrm{sign}(\sigma) \lambda_{\tau\sigma} = \mathrm{sign}(\tau) A$, hence $\mathrm{im}\, A \subseteq \mathrm{AT}^p(\mathcal{V})$.
If $\mathbf{z} \in \mathrm{AT}^p(\mathcal{V})$, then $A\mathbf{z} = p!^{-1} \sum_\sigma \mathrm{sign}(\sigma)^2 \mathbf{z} = \mathbf{z}$, so $\mathrm{im}\, A = \mathrm{AT}^p(\mathcal{V})$ and $A^2 = A$.

Structural Commentary: Averaging gives symmetry projections.
Key Insight: The factor $1/p!$ makes the averaging map fix the tensors it projects onto. □

Definition 6.1.6. *Tensor, Symmetric, and Exterior Algebras*
Let $\dim \mathcal{V} < \infty$.

(i) The *tensor algebra* is $\mathcal{T}(\mathcal{V}) := \bigoplus_{p \geq 0} \mathcal{V}^{\otimes p}$, with $\mathcal{V}^{\otimes 0} := \mathbb{F}$ and product induced by tensor concatenation.
(ii) The *symmetric algebra* is $\mathrm{Sym}(\mathcal{V}) := \bigoplus_{p \geq 0} \mathrm{Sym}^p(\mathcal{V})$, with product induced by $\vee$.
(iii) The *exterior algebra* is $\bigwedge \mathcal{V} := \bigoplus_{0 \leq p \leq \dim \mathcal{V}} \bigwedge^p \mathcal{V}$, with product induced by $\wedge$.
(iv) Each is *graded*: degree p times degree q lies in degree $p + q$.

6.2 Exterior Algebras

Corollary 6.2.1. *Determinant from the Exterior Universal Property*
Let $\dim \mathcal{V} = n$, $E = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$, $\Omega_E := \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$, *and* $A_E(\mathbf{v}_1, \ldots, \mathbf{v}_n) = ([\mathbf{v}_j]_i)_{i,j}$.

(i) There is a unique $\ell_E \in (\bigwedge^n \mathcal{V})'$ *with* $\ell_E(\Omega_E) = 1$; *hence* $d_E(\mathbf{v}_1, \ldots, \mathbf{v}_n) := \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n)$ *is the unique alternating form satisfying* $d_E(\mathbf{e}_1, \ldots, \mathbf{e}_n) = 1$.
(ii) For all $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathcal{V}$, $d_E(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) \prod_{j=1}^n [\mathbf{v}_j]_{\sigma(j)} = \det A_E(\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

Proof.

(i) By Theorem 6.1.4, $\bigwedge^n \mathcal{V} = \mathbb{F}\Omega_E$, so $\ell_E(\Omega_E) = 1$ fixes a unique $\ell_E \in (\bigwedge^n \mathcal{V})'$; via Theorem 6.1.3, ℓ_E corresponds to $d_E(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n)$, hence d_E is unique.
(ii) Let $\mathbf{v}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$, $a_{ij} = [\mathbf{v}_j]_i$. Repeated index tuples have zero wedge; distinct tuples are $(\sigma(1), \ldots, \sigma(n))$, with wedge $\mathrm{sign}(\sigma)\Omega_E$. Hence $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) \prod_{j=1}^n a_{\sigma(j),j} \Omega_E$, and

$$d_E(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}$$
$$= \det(a_{ij}) = \det A_E(\mathbf{v}_1, \ldots, \mathbf{v}_n).$$

Structural Commentary: Determinant is the basis-normalised alternating form.
Key Insight: Top exterior dimension one turns normalisation into determinant. □

Corollary 6.2.2. *Determinant Properties from Exterior Powers*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, *and set* $d(A) := d_E(A_{\bullet 1}, \ldots, A_{\bullet n})$ *for* E *the standard basis of* $\mathbb{F}^n$.

(i) $d(A^t) = d(A)$.
(ii) $d(AB) = d(A)d(B)$.
(iii) $A \in \mathrm{GL}(n, \mathbb{F}) \iff d(A) \neq 0$.

Proof. Let $\omega = \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$ and let T_A denote the linear map with matrix A.

(i) By Corollary 6.2.1, $d(A^t) = \sum_\sigma \mathrm{sign}(\sigma) \prod_{j=1}^n A_{j,\sigma(j)}$. With $\tau = \sigma^{-1}$, $\sum_\tau \mathrm{sign}(\tau) \prod_{i=1}^n A_{\tau(i),i} = d(A)$.
(ii) Since $(\bigwedge^n T_A)\omega = d(A)\omega$ and $T_{AB} = T_A \circ T_B$, applying $\bigwedge^n (T_A \circ T_B) = (\bigwedge^n T_A)(\bigwedge^n T_B)$ to ω gives $d(AB)\omega = d(A)d(B)\omega$, hence $d(AB) = d(A)d(B)$.
(iii) ($\Rightarrow$) If $A \in \mathrm{GL}(n, \mathbb{F})$, then $I = AA^{-1}$, so $1 = d(I) = d(A)d(A^{-1})$ by (ii); hence $d(A) \neq 0$.

(*iii*) ($\Leftarrow$) Let $C_j = A_{\bullet j}$. Suppose $\sum_{j=1}^n c_j C_j = 0$ and $c_k \neq 0$. Then $C_k = -c_k^{-1} \sum_{j \neq k} c_j C_j$, so multilinearity in the kth slot gives $d(A) = d(C_1, \ldots, C_k, \ldots, C_n) = -c_k^{-1} \sum_{j \neq k} c_j d(C_1, \ldots, C_{k-1}, C_j, C_{k+1}, \ldots, C_n) = 0$, since each summand has equal jth and kth entries.
Thus $d(A) \neq 0$ implies $C_1, \ldots, C_n$ are independent, hence $A \in \mathrm{GL}(n, \mathbb{F})$.

Structural Commentary: Determinant laws are top-exterior scalar laws.
Key Insight: Top exterior dimension one makes determinants multiplicative. □

6.3 Multilinear Algebra Revisited

7 Chain Complexes and Homology

7.1 Chain Complexes

7.2 Homology

7.3 Derived Functors Preview

Part IX
Commutative Algebra

1 Prime Ideals and Spectra

1.1 Prime Ideals

1.2 Spectra

2 Localization and Local Rings

2.1 Localization

2.2 Local Rings

3 Noetherian Rings and Modules

3.1 Noetherian Rings

Theorem 3.1.1. *Hilbert Basis Theorem*
Let A be commutative with identity. If A is Noetherian, then $A[x]$ is Noetherian.

Proof. It suffices to show every ideal $\mathcal{J} \lhd A[x]$ is f.g. If $\mathcal{J} = 0$, done.
For $g \neq 0$, write $\mathrm{lc}(g)$ and $\mathrm{lt}(g)$. Set $L = \langle \mathrm{lc}(g) : g \in \mathcal{J}, g \neq 0 \rangle \lhd A$. Since A is Noetherian, choose $a_1, \dots, a_m$ generating L and $f_i \in \mathcal{J}$ with $\mathrm{lc}(f_i) = a_i$. Replacing f_i by powers of x times f_i, assume $\deg f_i = d$ for all i.
For $0 \leq k < d$, set $L_k = \langle \mathrm{lc}(g) : g \in \mathcal{J}, \deg g = k \rangle$ and choose finite $P_k \subseteq \mathcal{J}$ with $\deg p = k$ for $p \in P_k$ and $\{\mathrm{lc}(p) : p \in P_k\}$ generating L_k. Put $G = \{f_1, \dots, f_m\} \cup \bigcup_{k=0}^{d-1} P_k$ and $\mathcal{A} = \langle G \rangle \lhd A[x]$. Then $\mathcal{A} \subseteq \mathcal{J}$.
For $g \in \mathcal{J} \setminus \mathcal{A}$ of minimal degree, set $r = \deg g$. If $r < d$, write $\mathrm{lc}(g) = \sum_{p \in P_r} c_p \, \mathrm{lc}(p)$ and set $h = \sum_{p \in P_r} c_p p$.
If $r \geq d$, write $\mathrm{lc}(g) = \sum_{i=1}^{m} c_i a_i$ and set $h = \sum_{i=1}^{m} c_i x^{r-d} f_i$. In either case $h \in \mathcal{A}$ and $\mathrm{lt}(h) = \mathrm{lt}(g)$.
Then $g - h \in \mathcal{J} \setminus \mathcal{A}$ and $\deg(g - h) < \deg g$, contradiction. Thus $\mathcal{J} = \mathcal{A}$ is f.g.
Structural Commentary: Ideals in $A[x]$ are controlled by finitely many leading coefficients in A.
Key Insight: Match the leading term, subtract, and induct on degree. □

3.2 Noetherian Modules

Theorem 3.2.1. *Finitely Generated Modules over Noetherian Rings*
Let A be a commutative ring with identity. Then A is Noetherian $\iff$ every f.g. A-module is Noetherian.

Proof.
($\Rightarrow$) Assume A Noetherian. First prove by induction on n that every $S \leq A^n$ is f.g.
The case $n = 1$ is Theorem 2.4.2. For $n > 1$, let $\pi : S \to A$ be last-coordinate projection.
Then $\pi(S) \lhd A$ is f.g. and $\ker(\pi) \leq A^{n-1}$ is f.g. by induction. Lifts of generators of $\pi(S)$, together with generators of $\ker(\pi)$, generate S. Thus A^n is Noetherian. If M is f.g., choose $A^n \twoheadrightarrow M$; for $N \leq M$, its preimage in A^n is f.g., hence N is f.g. Therefore M is Noetherian.
($\Leftarrow$) Since $A = \langle\langle 1 \rangle\rangle_A$, the regular A-module A is f.g.; by hypothesis it is Noetherian. Hence A is Noetherian.
Structural Commentary: Noetherianity passes from rings to finite free modules to quotients.
Key Insight: Finitely generated modules are quotients of finite free modules. □

4 Integral Dependence and Dimension

4.1 Integral Dependence

4.2 Dimension Theory

5 Regularity and Singularities

5.1 Regularity

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Part X
Algebraic Geometry

1 Affine Algebraic Sets

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7 Divisors, Line Bundles, and Curves

7.1 Divisors

7.2 Line Bundles

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Part XI
Real and Topological Analysis

1 Number Systems and Completeness

1.1 Ordered Fields, Suprema, and Real Structure

Number systems supply the ordered and algebraic background for Analysis: $\mathbb{Q}$ has gaps, $\mathbb{R}$ repairs them by completeness, and $\mathbb{C}$ keeps field algebra without order.

Definition 1.1.1. *Natural, Rational, Irrational, and Dense Subsets*
A set $E \subseteq \mathbb{R}$ is *inductive* iff $1 \in E$ and, for all $x \in \mathbb{R}$, $x \in E$ implies $x + 1 \in E$.
Let $\mathcal{I}$ be the family of inductive subsets of $\mathbb{R}$.

(i) $\mathbb{N} := \bigcap_{E \in \mathcal{I}} E$, the least inductive subset of $\mathbb{R}$.
(ii) $\mathbb{Z} := \mathbb{N} \cup \{0\} \cup \{-n : n \in \mathbb{N}\}$.
(iii) $\mathbb{Q} := \{m/n : m, n \in \mathbb{Z},\ n \neq 0\}$; $x \in \mathbb{R}$ is *irrational* iff $x \notin \mathbb{Q}$.
(iv) $D \subseteq \mathbb{R}$ is *dense in* $\mathbb{R}$ iff for all $a, b \in \mathbb{R}$ with $a < b$, there exists $d \in D$ with $a < d < b$.

Definition 1.1.2. *Fields and Ordered Fields*
Let F be a set with operations $+, \cdot : F \times F \to F$ and elements $0, 1 \in F$ with $0 \neq 1$.

(i) A *field* has $(F, +, 0)$ and $(F \setminus \{0\}, \cdot, 1)$ abelian groups and $x(y + z) = xy + xz$ for all $x, y, z \in F$.
(ii) An *ordered field* is a field with total order $<$ satisfying $y < z \Rightarrow x + y < x + z$ and $0 < x, 0 < y \Rightarrow 0 < xy$.

Principle 1.1.3. *Completeness Axiom*
Every nonempty set $E \subseteq \mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$, denoted $\sup E$.
Equivalently, the real line has no missing cut between a nonempty bounded set and its upper bounds.

Remark 1.1.4. *Fields as Objects and Scalars*
In analysis, $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields: $\mathbb{Q}$ lacks the least-upper-bound property, while $\mathbb{R}$ has it.

Proposition 1.1.5. *Basic Consequences of the Field Axioms*
Let F be a field. For all $a, b, c, x, y \in F$:

(i) $a + c = b + c$ implies $a = b$; if $c \neq 0$, then $ac = bc$ implies $a = b$.
(ii) The identities $0, 1$ and inverses $-a$, a^{-1} $(a \neq 0)$ are unique.
(iii) $0x = 0$.
(iv) $xy = 0$ implies $x = 0$ or $y = 0$.
(v) $(-x)y = -(xy) = x(-y)$ and $(-x)(-y) = xy$.

Proof.

(i) Add $-c$ to $a + c = b + c$. If $c \neq 0$, multiply $ac = bc$ by c^{-1}.
(ii) In any group $(G, *, e)$, if e' is an identity, $e = e * e' = e'$.
If $u * a = e = a * v$, then $u = u * e = u * (a * v) = (u * a) * v = v$. Apply to $(F, +, 0)$ and $(F \setminus \{0\}, \cdot, 1)$.
(iii) Since $0x + 0x = (0 + 0)x = 0x$, cancellation gives $0x = 0$.
(iv) If $xy = 0$ and $x \neq 0$, multiply by x^{-1} to get $y = 0$.
(v) $xy + (-x)y = (x + (-x))y = 0$, so $(-x)y = -(xy)$; $x(-y) = -(xy)$. Also $(-x)(-y) = -(x(-y)) = xy$.

Structural Commentary: Inverses cancel terms.
Key Insight: Negatives are unique additive inverses. □

Proposition 1.1.6. *Basic Inequality Rules in Ordered Fields*
Let F be an ordered field. For all $c, x, y \in F$:

(i) $x > 0$ iff $-x < 0$.
(ii) If $c > 0$ and $x < y$, then $cx < cy$.
(iii) If $c < 0$ and $x < y$, then $cy < cx$.
(iv) If $x \neq 0$, then $x^2 > 0$; in particular, $1 > 0$.
(v) If $0 < x < y$, then $0 < 1/y < 1/x$.

Proof.

(i) If $x > 0$, add $-x$ to $0 < x$ to get $-x < 0$. If $-x < 0$, add x to get $0 < x$.
(ii) Since $x < y$, $y - x > 0$. With $c > 0$, $c(y - x) > 0$, so $cy - cx > 0$ and $cx < cy$.
(iii) Since $c < 0$, (i) gives $-c > 0$. By (ii), $(-c)x < (-c)y$; hence $-cx < -cy$, so $cy < cx$.
(iv) If $x > 0$, then $x^2 > 0$. If $x < 0$, then $-x > 0$ by (i), so $(-x)^2 > 0$ and $x^2 = (-x)^2$.
Since $x \neq 0$, totality gives one of these cases. Taking $x = 1$ gives $1 = 1^2 > 0$.
(v) By (iv), $(1/x)^2 > 0$ and $(1/y)^2 > 0$. Since $x, y > 0$, $1/x = x(1/x)^2 > 0$ and $1/y = y(1/y)^2 > 0$, so $(1/x)(1/y) > 0$. Apply (ii) to $x < y$ with multiplier $(1/x)(1/y)$ to get $1/y < 1/x$.

Structural Commentary: Positivity preserves order under multiplication.
Key Insight: Negative multiplication is positive multiplication by $-c$. □

Proposition 1.1.7. *The Rational Gap at $\sqrt{2}$*
There is no rational p such that $p^2 = 2$.

Proof. Let $p = m/n$. Then $m^2 = 2n^2$, so m is even; $m = 2k$. Hence $n^2 = 2k^2$, so n is even, a contradiction.
Structural Commentary: This exhibits a gap in $\mathbb{Q}$.
Key Insight: $\sqrt{2}$ is not rational. □

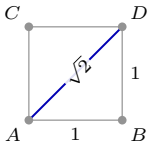

Figure 112: The diagonal of a unit square has length $\sqrt{2}$, so the rational gap is geometric as well as algebraic.

Example 1.1.8. *No Largest Lower Side and No Smallest Upper Side*
Let $A := \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$, and $B := \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$.
For every $p > 0$ define $q := p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2}$. Then $q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2}$. Hence if $p \in A$, then $q \in A$ and $q > p$; if $p \in B$, then $q \in B$ and $0 < q < p$. Thus A has no largest element in $\mathbb{Q}$, and B has no smallest element in $\mathbb{Q}$.

Remark 1.1.9. *Density Does Not Mean Completeness*
Between two rationals $r < s$ there is another rational, namely $(r + s)/2$. Nevertheless, $\mathbb{Q}$ is not complete.

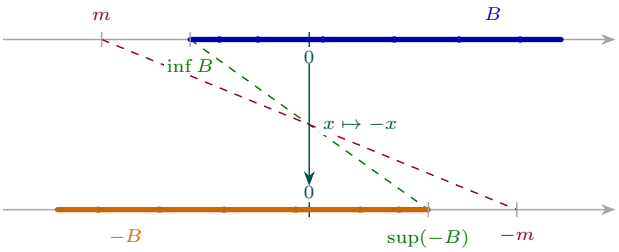

Figure 113: Reflecting B through 0 turns lower bounds of B into upper bounds of $-B$, so $\inf B = -\sup(-B)$.

Corollary 1.1.10. *Infimum by Reflection*
Let $B \subseteq \mathbb{R}$ be nonempty and bounded below, and define $-B := \{-b : b \in B\}$.
Then $-B$ is nonempty and bounded above, and $\inf B = -\sup(-B)$.

Proof. If m is a lower bound of B, then $-m$ is an upper bound of $-B$; hence $s := \sup(-B)$ exists.
For $b \in B$, $-b \le s$, so $-s \le b$; thus $-s$ is a lower bound of B.
If t is any lower bound of B, then $-t$ is an upper bound of $-B$, so $s \le -t$ and $t \le -s$. Therefore $-s = \inf B$.
Structural Commentary: Negation reverses bounds.
Key Insight: $\inf B = -\sup(-B)$. □

Definition 1.1.11. *Real Absolute Value and Distance*
For $x \in \mathbb{R}$, define $|x| := \begin{cases} x, & x \ge 0, \\ -x, & x < 0. \end{cases}$.
For $a, b \in \mathbb{R}$, the distance between a and b on the real line is $|a - b|$.

Proposition 1.1.12. *Real Absolute-Value Laws*
For $x, y, a, b, c \in \mathbb{R}$,
(i) $|x| \ge 0$, $|x| = 0$ *iff* $x = 0$, *and* $-|x| \le x \le |x|$;
(ii) $|xy| = |x|\,|y|$;
(iii) $|x + y| \le |x| + |y|$;
(iv) $|a - b| \le |a - c| + |c - b|$.

Proof.
(i) If $x \ge 0$, then $|x| = x$; if $x < 0$, then $|x| = -x > 0$. Hence $|x| \ge 0$, and $|x| = 0$ iff $x = 0$.
The bounds $-|x| \le x \le |x|$ follow from the same two cases.
(ii) If x, y have the same sign, then $xy \ge 0$ and $|xy| = xy = |x|\,|y|$.
If they have opposite signs, then $xy < 0$ and $|xy| = -xy = |x|\,|y|$.
(iii) By (i), $-|x| \le x \le |x|$ and $-|y| \le y \le |y|$, so $-(|x| + |y|) \le x + y \le |x| + |y|$.
Since $|x| + |y| \ge 0$, (i) applied to $x + y$ gives $|x + y| \le |x| + |y|$.
(iv) Apply (iii) to $a - b = (a - c) + (c - b)$.
Structural Commentary: Absolute value is two-sided order control.
Key Insight: Apply (iii) to $(a - c) + (c - b)$. □

Perspective 1.1.13. *Completeness Becomes a Theorem*
Dedekind cuts build completeness from $\mathbb{Q}$: real numbers are rational cuts, rational cuts embed $\mathbb{Q}$, and least upper bounds arise by taking unions of cuts.

Definition 1.1.14. *Dedekind Cuts and Their Order*
A *cut* is a set $\alpha \subseteq \mathbb{Q}$ such that
(i) α is nonempty and $\alpha \ne \mathbb{Q}$;
(ii) if $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$;
(iii) if $p \in \alpha$, then some $r \in \alpha$ satisfies $p < r$.
Let $\mathscr{D}$ be the set of all cuts. For $\alpha, \beta \in \mathscr{D}$, define $\alpha < \beta$ to mean $\alpha \subsetneq \beta$.
Rational letters $p, q, r, \ldots$ denote elements of $\mathbb{Q}$; Greek letters $\alpha, \beta, \gamma, \ldots$ denote cuts.
A cut is the rational left side of a real number: a proper initial segment with no last point.

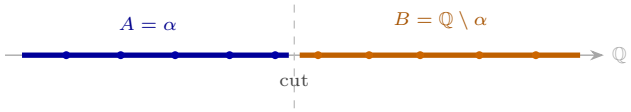

Figure 114: A Dedekind cut splits $\mathbb{Q}$ into α and $\mathbb{Q} \setminus \alpha$, with no last rational in α.

Remark 1.1.15. *Left Cuts Versus Two-Sided Cuts*
Two-sided notation writes a cut as $A|B$, with $A = \alpha$ and $B = \mathbb{Q} \setminus \alpha$; Definition 1.1.14 records only A.
Thus "no largest element" is the same left-side condition.

Definition 1.1.16. *Arithmetic of Cuts*
For cuts α, β, define $\alpha + \beta := \{r + s : r \in \alpha,\ s \in \beta\}$ and $0^* := \{q \in \mathbb{Q} : q < 0\}$.
Let candidate additive inverse of α be $-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q},\ r > 0,\ -p - r \notin \alpha\}$. Let $\mathscr{D}^+ := \{\alpha \in \mathscr{D} : \alpha > 0^*\}$.
For $\alpha, \beta \in \mathscr{D}^+$, define $\alpha\beta := \{p \in \mathbb{Q} : \exists r \in \alpha,\ \exists s \in \beta,\ r > 0,\ s > 0,\ p \le rs\}$ and $1^* := \{q \in \mathbb{Q} : q < 1\}$.
Extend multiplication to all cuts by setting $\alpha 0^* = 0^* \alpha = 0^*$ and, for nonzero signs,

$$\alpha\beta := \begin{cases} (-\alpha)(-\beta), & \alpha < 0^*,\ \beta < 0^*, \\ -\big((-\alpha)\beta\big), & \alpha < 0^*,\ \beta > 0^*, \\ -\big(\alpha(-\beta)\big), & \alpha > 0^*,\ \beta < 0^*. \end{cases}$$

Motivation: arithmetic uses rational approximations inside cuts; signs come after positive products.

Definition 1.1.17. *Rational Cuts*
For $r \in \mathbb{Q}$, define the *rational cut*, $r^* := \{p \in \mathbb{Q} : p < r\}$.
The map $r \mapsto r^*$ is intended to identify $\mathbb{Q}$ with a subfield of the cut field.

Theorem 1.1.18. *Existence of the Real Field*
There exists an ordered field $\mathbb{R}$ with the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Proof. Construct $\mathbb{R}$ from $\mathbb{Q}$ using Definitions 1.1.14–1.1.17.

(i) *Order.* The relation $\alpha < \beta$ is transitive, and at most one of $\alpha < \beta$, $\alpha = \beta$, and $\beta < \alpha$ can hold. For trichotomy, suppose $\alpha \not\subseteq \beta$ and $\alpha \neq \beta$. Choose $p \in \alpha$ with $p \notin \beta$. If $q \in \beta$, then $q < p$, since $p \notin \beta$ and β is downward closed. Hence $q \in \alpha$, so $\beta \subseteq \alpha$. Since $\alpha \neq \beta$, $\beta < \alpha$. Thus $\mathscr{D}$ is ordered.
Commentary: One missing rational decides containment.

(ii) *Least Upper Bounds.* Let $A \subseteq \mathscr{D}$ be nonempty and bounded above by a cut β. Define $\gamma := \bigcup_{\alpha \in A} \alpha$.
Since some $\alpha_0 \in A$ is nonempty, γ is nonempty. Since every $\alpha \in A$ lies in β, $\gamma \subseteq \beta$, so $\gamma \neq \mathbb{Q}$. If $p \in \gamma$ and $q < p$, then $p \in \alpha_1$ for some $\alpha_1 \in A$, hence $q \in \alpha_1 \subseteq \gamma$. If $p \in \gamma$, choose $\alpha_1 \in A$ with $p \in \alpha_1$ and then choose $r \in \alpha_1$ with $p < r$; then $r \in \gamma$. Thus γ is a cut.
Every $\alpha \in A$ satisfies $\alpha \le \gamma$. If $\delta < \gamma$, choose $s \in \gamma \setminus \delta$. Then $s \in \alpha$ for some $\alpha \in A$, so $\delta < \alpha$ and δ is not an upper bound of A. Hence $\gamma = \sup A$.
Motivation: Supremum is union.
Commentary: Bounded unions stay proper and give the smallest upper bound.

(iii) *Addition.* The set $\alpha + \beta$ is a cut. It is nonempty by choosing one element from each summand.
If $r' \notin \alpha$ and $s' \notin \beta$, then every $r \in \alpha$ and $s \in \beta$ satisfies $r < r'$ and $s < s'$, so $r + s < r' + s'$; hence $r' + s' \notin \alpha + \beta$, and $\alpha + \beta$ is proper. If $p = r + s \in \alpha + \beta$ and $q < p$, then $q = r + (s + q - p)$ with $s + q - p < s$, so $s + q - p \in \beta$ and $q \in \alpha + \beta$. If $p = r + s$, choose $r_1 \in \alpha$ with $r < r_1$; then $p < r_1 + s \in \alpha + \beta$.
Thus $\alpha + \beta$ is a cut. Commutativity and associativity follow by rewriting representatives in $\mathbb{Q}$.
The cut 0^* is the additive identity: $\alpha + 0^* \subseteq \alpha$ by downward closure, while if $p \in \alpha$, choose $r \in \alpha$ with $p < r$ and write $p = r + (p - r)$ with $p - r < 0$.
The inverse $-\alpha$ is a cut, and $\alpha + (-\alpha) = 0^*$. The inclusion $\alpha + (-\alpha) \subseteq 0^*$ follows because $s \in -\alpha$ forces $-s \notin \alpha$, so any $r \in \alpha$ satisfies $r < -s$. Conversely, if $v < 0$, put $w = -v/2 > 0$ and choose an integer n with $nw \in \alpha$ but $(n+1)w \notin \alpha$; then $p = -(n+2)w$ lies in $-\alpha$ and $v = nw + p \in \alpha + (-\alpha)$.
Commentary: Addition inherits its laws from $\mathbb{Q}$ once closure and inverses are checked.

(iv) *Order Compatibility with Addition.* The additive laws make $\mathscr{D}$ an additive group.
If $\beta < \gamma$, then $\alpha + \beta \subseteq \alpha + \gamma$. Equality would imply $\beta = \gamma$ after adding $-\alpha$ to both sides, so $\alpha + \beta < \alpha + \gamma$. Consequently, $\alpha > 0^*$ if and only if $-\alpha < 0^*$.
Commentary: Cancellation turns inclusion into strict inclusion.

(v) *Positive Multiplication.* If $\alpha, \beta \in \mathscr{D}^+$, then $\alpha\beta$ is a cut. Since each positive cut contains some positive rational, $\alpha\beta$ is nonempty. Choose positive $r' \notin \alpha$ and $s' \notin \beta$; then every positive $r \in \alpha$ and $s \in \beta$ satisfies $rs < r's'$, so $r's' \notin \alpha\beta$, and the product is proper. Downward closure is built into the condition $p \le rs$. If $p \le rs$ with $r, s > 0$, then either choose q with $p < q < rs$, or, when needed, choose $r_1 \in \alpha$ with $r < r_1$ and take q with $p < q \le r_1 s$; hence the product has no largest element.
Commutativity follows from $rs = sr$. Associativity and distributivity are proved by unfolding membership and using the rational identities $(rs)t = r(st)$ and $r(s + t) = rs + rt$, with a positive rational slack inserted whenever a representative must be made strictly larger inside a cut. The identity is 1^*: if $p \le rs$ with $r \in \alpha$, $s < 1$, and $r, s > 0$, then $p < r$, so $p \in \alpha$; conversely, if $p \in \alpha$, choose $r \in \alpha$ with $p < r$ and then choose $s \in \mathbb{Q}$ with $0 < s < 1$ and $p \le rs$. In particular, if $\alpha > 0^*$ and $\beta > 0^*$, then $\alpha\beta > 0^*$.
Commentary: Positivity avoids sign reversals.

(vi) *Signed Multiplication.* The definitions extend multiplication to all cuts by reducing to positive factors and additive inverses. Since $\delta = -(-\delta)$ and the sign rules for addition have already been proved, commutativity, associativity, and inverses reduce to the positive case.
For distributivity, split into signs of $\alpha, \beta, \gamma, \beta + \gamma$. For example, if $\alpha > 0^*$, $\beta < 0^*$, and $\beta + \gamma > 0^*$, then $\gamma = (\beta + \gamma) + (-\beta)$, so positive distributivity gives $\alpha\gamma = \alpha(\beta + \gamma) + \alpha(-\beta)$. Because $\alpha(-\beta) = -(\alpha\beta)$, adding $\alpha\beta$ to both sides gives $\alpha\beta + \alpha\gamma = \alpha(\beta + \gamma)$.
The other sign cases are identical reductions after replacing negative terms by their additive inverses.
Commentary: Distributivity reduces to positive cases plus signs.

(vii) *Embedding $\mathbb{Q}$.* Each r^* is a cut. For $r, s \in \mathbb{Q}$, $r^* + s^* = (r + s)^*$, $r^* s^* = (rs)^*$, and $r^* < s^* \iff r < s$.
For addition, if $p = u + v$ with $u < r$ and $v < s$, then $p < r + s$. Conversely, if $p < r + s$, choose $t > 0$ with $2t = r + s - p$ and put $r' = r - t$, $s' = s - t$; then $r' < r$, $s' < s$, and $p = r' + s'$. For positive products, if $p \le uv$ with $0 < u < r$ and $0 < v < s$,

then $p < rs$. Conversely, if $0 < p < rs$, choose positive $u < r$ and $v < s$ close enough to r and s that $p \le uv$; if $p \le 0$, use any positive $u < r$, $v < s$. The signed product cases follow from the definitions by additive inverses. The order equivalence follows because $r < s$ puts r in s^* but not in r^*, while $r^* < s^*$ gives some $p < s$ with $p \notin r^*$, hence $r \le p < s$.
Commentary: The map $r \mapsto r^*$ preserves arithmetic and order.

Identify $\mathbb{Q}$ with the subfield $\mathbb{Q}^* := \{r^* : r \in \mathbb{Q}\} \subseteq \mathscr{D}$, and write $\mathbb{R}$ for $\mathscr{D}$. Items (i), (iii)–(vi) give an ordered field; item (ii) gives the least-upper-bound property; item (vii) embeds $\mathbb{Q}$ as a subfield.
Any two ordered fields with the least-upper-bound property are isomorphic, so this construction characterizes the real field up to ordered-field structure. □

Perspective 1.1.19. *Number Systems from Representatives*
Choose representatives X, an equivalence relation $\sim$, and classes $[x] \in X/\!\sim$: $\mathbb{Z} : (a,b) \mapsto [(a,b)]$, $\mathbb{Q} : (a,b)$ with $b \neq 0 \mapsto [(a,b)]$, and $\mathbb{R}$ via cuts or Cauchy data. Prove each operation or order is well defined, e.g. $x \sim x'$ and $y \sim y'$ imply $x \star y \sim x' \star y'$ and $x < y \Leftrightarrow x' < y'$; then use $X/\!\sim$ as the number system.
Key Insight: Laws must descend to classes.

Theorem 1.1.20. *Archimedean Property and Density of $\mathbb{Q}$*
(i) If $x, y \in \mathbb{R}$ and $x > 0$, then there is a positive integer n such that $nx > y$.
(ii) If $x, y \in \mathbb{R}$ and $x < y$, then there exists $p \in \mathbb{Q}$ such that $x < p < y$.

Proof.
(i) Let $A = \{nx : n \in \mathbb{N}\}$. If no nx exceeded y, then A would be bounded above.
By Theorem 1.1.18, let $\alpha = \sup A$. Since $x > 0$, $\alpha - x < \alpha$, so $\alpha - x$ is not an upper bound.
Hence $\alpha - x < mx$ for some $m \in \mathbb{N}$, and $\alpha < (m+1)x \in A$, a contradiction.
(ii) Since $y - x > 0$, part (i) gives $n \in \mathbb{N}$ with $n(y-x) > 1$. Apply part (i) again to get $m_1, m_2 \in \mathbb{N}$ with $m_1 > nx$ and $m_2 > -nx$.
Then some $m \in \mathbb{Z}$ satisfies $m - 1 \le nx < m$.
Thus $nx < m \le nx + 1 < ny$, so $x < m/n < y$ and $m/n \in \mathbb{Q}$.

Structural Commentary: Completeness makes $\{nx : n \in \mathbb{N}\}$ unbounded.
Key Insight: Archimedean unboundedness gives rational density. □

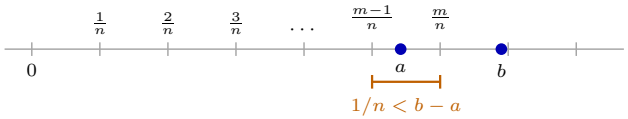

Figure 115: Choose n so the spacing $1/n$ is smaller than $b - a$; then one grid point m/n falls inside (a, b).

Theorem 1.1.21. *Existence and Uniqueness of Positive nth Roots*
For every $x \in \mathbb{R}$ with $x > 0$ and every positive integer n, $\exists$ a unique positive real number y such that $y^n = x$.

Proof.
(i) *Uniqueness.* If $0 < y_1 < y_2$, then $y_1^n < y_2^n$.
(ii) *Existence.* Let $E := \{t \in \mathbb{R} : t > 0,\ t^n < x\}$. $E \neq \varnothing$: for $t = x/(1+x)$, $0 < t < 1$ and $t^n \le t < x$.
Also E is bounded above by $1 + x$, since $t > 1 + x$ implies $t^n \ge t > x$. By Theorem 1.1.18, let $y = \sup E$.
For $0 < a < b$, $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1}) < (b-a)nb^{n-1}$.
If $y^n < x$, choose h with $0 < h < 1$ and $h < \frac{x - y^n}{n(y+1)^{n-1}}$.
Then $(y+h)^n - y^n < hn(y+h)^{n-1} < hn(y+1)^{n-1} < x - y^n$.
Thus $y + h \in E$, contradicting that y is an upper bound. If $y^n > x$, set $k := \frac{y^n - x}{ny^{n-1}}$. Then $0 < k < y$.
For $t \ge y - k$, $y^n - t^n \le y^n - (y-k)^n < kny^{n-1} = y^n - x$, so $t^n > x$ and $t \notin E$.
Hence $y - k$ is an upper bound for E, contradicting $y = \sup E$. Therefore $y^n = x$.

Structural Commentary: The root is $\sup\{t > 0 : t^n < x\}$.
Key Insight: Completeness supplies the missing root. □

Corollary 1.1.22. *Density of Rational and Irrational Points*
If $x < y$ in $\mathbb{R}$, then there exist $r \in \mathbb{Q}$ and $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < r < y$ and $x < s < y$.

Proof. The rational point r is supplied by Theorem 1.1.20. Let $\eta = \sqrt{2}$.
Choose $n \in \mathbb{N}$ with $0 < \eta/n < y - x$, and choose $q \in \mathbb{Q}$ with $x - \eta/n < q < y - \eta/n$.
Then $s := q + \eta/n$ lies in (x, y) and is irrational, since $s \in \mathbb{Q}$ would imply $\eta = n(s - q) \in \mathbb{Q}$.
Structural Commentary: Irrational density is rational density shifted by a small irrational amount.
Key Insight: Once one irrational is known, every interval contains irrational translates of rational points. □

Corollary 1.1.23. *Product Rule for Positive Roots*
If $a, b > 0$ and n is a positive integer, then $(ab)^{1/n} = a^{1/n} b^{1/n}$.

Proof. Let $\alpha = a^{1/n}$ and $\beta = b^{1/n}$. Then $ab = \alpha^n \beta^n = (\alpha\beta)^n$ by commutativity of multiplication.
The uniqueness part of Theorem 1.1.21 gives the identity.
Structural Commentary: Use uniqueness after checking nth powers.
Key Insight: Root identities are uniqueness arguments. □

Definition 1.1.24. *Extended Real Number System*
The *extended real number system* is $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$.
The order on $\mathbb{R}$ is preserved, and $-\infty < x < +\infty$, $(x \in \mathbb{R})$.
Every subset of $\mathbb{R}$ has a supremum and an infimum in $\overline{\mathbb{R}}$ after the conventions $\sup \emptyset = -\infty$, $\inf \emptyset = +\infty$, $\sup E = +\infty$ when E is unbounded above, and $\inf E = -\infty$ when E is unbounded below.

Remark 1.1.25. *Extended Arithmetic Conventions*
$\overline{\mathbb{R}}$ is not a field. For $x \in \mathbb{R}$:

(i) $\infty + \infty = \infty$ and $-\infty - \infty = -\infty$.
(ii) $x + \infty = \infty$ and $x - \infty = -\infty$.
(iii) If $x > 0$, then $x \cdot \infty = \infty$ and $x \cdot (-\infty) = -\infty$; if $x < 0$, the signs reverse.

Expressions $\infty - \infty$ and $0 \cdot \infty$ are undefined.

Remark 1.1.26. *Unbounded Intervals*
For $a, b \in \mathbb{R}$:

(i) $(a, \infty) := \{x \in \mathbb{R} : a < x\}$ and $[a, \infty) := \{x \in \mathbb{R} : a \leq x\}$.
(ii) $(-\infty, b) := \{x \in \mathbb{R} : x < b\}$ and $(-\infty, b] := \{x \in \mathbb{R} : x \leq b\}$.

Also $(-\infty, \infty) := \mathbb{R}$.

1.2 Complex Numbers and Absolute Value

Definition 1.2.1. *Complex Numbers*
Let $\mathbb{C} := \mathbb{R}^2$. For $x = (a, b)$ and $y = (c, d)$ in $\mathbb{C}$:

(i) $x = y$ iff $a = c$ and $b = d$.
(ii) $x + y := (a + c, b + d)$.
(iii) $xy := (ac - bd, ad + bc)$.

Elements of $\mathbb{C}$ are *complex numbers*.

Theorem 1.2.2. *The Complex Numbers Form a Field*
Let $\mathbb{C} = \mathbb{R}^2$ with $(a, b) + (c, d) = (a + c, b + d)$ and $(a, b)(c, d) = (ac - bd, ad + bc)$.

(i) $\mathbb{C}$ is a field.
(ii) The additive and multiplicative identities are $(0, 0)$ and $(1, 0)$.
(iii) If $x = (a, b) \neq (0, 0)$, then $x^{-1} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$.

Proof.

(i) For $x, y, z \in \mathbb{C}$, associativity, commutativity, and distributivity follow by expanding coordinates in $\mathbb{R}$; $-(a, b) = (-a, -b)$, and (iii) gives x^{-1} for $x \neq 0$.
(ii) Directly, $(a, b) + (0, 0) = (a, b)$ and $(a, b)(1, 0) = (a, b)$.
(iii) Since $(a, b) \neq (0, 0)$, $a^2 + b^2 > 0$. Multiplication gives $(a, b)\left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right) = (1, 0)$.

Structural Commentary: Inverses use conjugate-over-norm.
Key Insight: $a^2 + b^2 > 0$ for $(a, b) \neq (0, 0)$. □

Proposition 1.2.3. *Real Numbers Inside Complex Numbers*
Let $\iota : \mathbb{R} \to \mathbb{C}$ be $\iota(a) := (a, 0)$, and set $i := (0, 1)$.

(i) $\iota(a + b) = \iota(a) + \iota(b)$ and $\iota(ab) = \iota(a)\iota(b)$ for all $a, b \in \mathbb{R}$.
(ii) ι is injective, $\iota(0) = (0, 0)$, and $\iota(1) = (1, 0)$.

Where $a \in \mathbb{R}$ with $\iota(a) \in \mathbb{C}$; i is the complex unit.

Proof.

(i) $\iota(a) + \iota(b) = (a, 0) + (b, 0) = (a + b, 0) = \iota(a + b)$, and $\iota(a)\iota(b) = (a, 0)(b, 0) = (ab, 0) = \iota(ab)$.
(ii) If $\iota(a) = \iota(b)$, then $(a, 0) = (b, 0)$, so $a = b$. Also $\iota(0) = (0, 0)$ and $\iota(1) = (1, 0)$.

Structural Commentary: Real arithmetic embeds as the real axis.
Key Insight: ι preserves $+, \cdot, 0, 1$. □

Proposition 1.2.4. *Coordinate Form and i^2*
For $i = (0, 1) \in \mathbb{C}$:

(i) $i^2 = -1$.
(ii) For all $a, b \in \mathbb{R}$, $(a, b) = a + bi$.

Proof.

(i) $i^2 = (0, 1)(0, 1) = (-1, 0) = -1$.
(ii) $a + bi = (a, 0) + (b, 0)(0, 1) = (a, 0) + (0, b) = (a, b)$.

Structural Commentary: Pair coordinates become $a + bi$.
Key Insight: $i = (0, 1)$ encodes the second coordinate. □

Definition 1.2.5. *Conjugate, Real Part, and Imaginary Part*
Let $z = a + bi$ with $a, b \in \mathbb{R}$.

(i) The *complex conjugate* of z is $\overline{z} := a - bi$.
(ii) The *real part* and *imaginary part* of z are $\mathrm{Re}(z) := a$ and $\mathrm{Im}(z) := b$.

Proposition 1.2.6. *Basic Conjugation Identities*
If $z, w \in \mathbb{C}$, then

(i) $\overline{z + w} = \overline{z} + \overline{w}$;
(ii) $\overline{zw} = \overline{z}\,\overline{w}$;
(iii) $z + \overline{z} = 2\,\mathrm{Re}(z)$ and $z - \overline{z} = 2i\,\mathrm{Im}(z)$;
(iv) $z\overline{z}$ is real and positive unless $z = 0$.

Proof. Write $z = a + bi$ and $w = c + di$.

(i) $\overline{z + w} = (a + c) - (b + d)i = \overline{z} + \overline{w}$.

(ii) $\overline{zw} = (ac - bd) - (ad + bc)i = (a - bi)(c - di) = \overline{z}\,\overline{w}$.
(iii) $z + \overline{z} = 2a = 2\,\mathrm{Re}(z)$ and $z - \overline{z} = 2bi = 2i\,\mathrm{Im}(z)$.
(iv) $z\overline{z} = a^2 + b^2$, positive unless $a = b = 0$, i.e. unless $z = 0$.

Structural Commentary: Conjugation fixes Re and negates Im.
Key Insight: $z\overline{z} = a^2 + b^2 \geq 0$. □

Definition 1.2.7. *Absolute Value of a Complex Number*
For $z = a + bi \in \mathbb{C}$, define $|z| := \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}$, the unique $r \in [0, \infty)$ with $r^2 = z\overline{z} = a^2 + b^2$.

Theorem 1.2.8. *Absolute Value Identities*
For $z, w \in \mathbb{C}$,

(i) $|z| > 0$ *unless* $z = 0$, *and* $|0| = 0$;
(ii) $|\overline{z}| = |z|$;
(iii) $|zw| = |z|\,|w|$;
(iv) $|\mathrm{Re}(z)| \leq |z|$;
(v) $|z + w| \leq |z| + |w|$.

Proof.

(i) If $z = a + bi$, then $|z| = (a^2 + b^2)^{1/2}$; this is positive unless $a = b = 0$, and $|0| = 0$.
(ii) $|\overline{z}|^2 = \overline{z}z = z\overline{z} = |z|^2$, so nonnegative square roots give $|\overline{z}| = |z|$.
(iii) $|zw|^2 = zw\overline{zw} = z\overline{z}w\overline{w} = |z|^2\,|w|^2$; take nonnegative square roots.
(iv) If $z = a + bi$, then $|\mathrm{Re}(z)| = |a| \leq (a^2 + b^2)^{1/2} = |z|$.
(v) $z\overline{w}$ and $\overline{z}w$ are conjugates, $|z + w|^2 = |z|^2 + 2\,\mathrm{Re}(z\overline{w}) + |w|^2 \leq |z|^2 + 2\,|z\overline{w}| + |w|^2 = (|z| + |w|)^2$, using (iii) and (iv). Take nonnegative square roots.

Structural Commentary: The complex absolute value behaves like length because $z\overline{z}$ is a squared length.
Key Insight: Triangle inequality is a real-part estimate plus multiplicativity. □

2 Real Sequences and Completeness Methods

2.1 Sequences, Cauchy Criteria, and Completeness

Sequences are the first testing ground for limits, order, and completeness. The main tools are subsequences, Cauchy criteria, monotone convergence, limsup/liminf, and Bolzano–Weierstrass.

Definition 2.1.1. *Convergent Real and Complex Sequences*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, let (s_n) be a sequence in $\mathbb{K}$, and let $s \in \mathbb{K}$.

(i) (s_n) *converges* to s iff for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|s_n - s| < \varepsilon$.
(ii) Write $s_n \to s$ or $\lim_{n\to\infty} s_n = s$. If no $s \in \mathbb{K}$ satisfies (i), then (s_n) *diverges*.

Figure 116: For all $n \geq N$, the terms s_n stay inside $(s - \varepsilon, s + \varepsilon)$.

Definition 2.1.2. *Range and Boundedness of a Real Sequence*
Let (s_n) be a real sequence.

(i) The *range* of (s_n) is $\mathrm{range}(s_n) := \{s_n : n \in \mathbb{N}\}$.
(ii) (s_n) is *bounded* iff there exist $m, M \in \mathbb{R}$ such that $m \leq s_n \leq M$ for all $n \in \mathbb{N}$.

Proposition 2.1.3. *Basic Consequences of Sequence Convergence*
Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and let (s_n) and (t_n) be sequences in $\mathbb{K}$.

(i) If $s_n \to s$ and $s_n \to t$, then $s = t$.
(ii) If $s_n \to s$, then $\{|s_n| : n \in \mathbb{N}\}$ is bounded.
(iii) If $\mathbb{K} = \mathbb{R}$, $s_n \leq t_n$ eventually, $s_n \to s$, and $t_n \to t$, then $s \leq t$.
In particular, $s_n \leq c$ eventually implies $s \leq c$.

Proof.

(i) For $\varepsilon > 0$ and large n, $|s_n - s|, |s_n - t| < \varepsilon/2$, so $|s - t| < \varepsilon$; hence $s = t$.
(ii) For large n, $|s_n - s| < 1$, hence $|s_n| \leq |s| + 1$; the remaining finitely many $|s_n|$ are bounded.
(iii) If $s > t$, let $\varepsilon = (s - t)/3$. For large n, $s_n > s - \varepsilon > t + \varepsilon > t_n$, contradicting $s_n \leq t_n$.
Take $t_n = c$ for the final claim.

Structural Commentary: Limits are rigid: tails determine the value, stay bounded, and preserve eventual order.
Key Insight: The triangle inequality and an $\varepsilon/3$ gap control the basic consequences. □

Theorem 2.1.4. *Limit Algebra for Complex Sequences*
Suppose (s_n) and (t_n) are complex sequences with $s_n \to s$ and $t_n \to t$. Then, for every $c \in \mathbb{C}$,

(i) $s_n + t_n \to s + t$;
(ii) $cs_n \to cs$ *and* $c + s_n \to c + s$;
(iii) $s_n t_n \to st$;
(iv) if $s \neq 0$ *and* $s_n \neq 0$ *for all* n, *then* $1/s_n \to 1/s$.

Proof.

(i) Given $\varepsilon > 0$, choose N such that $|s_n - s| < \varepsilon/2$, $|t_n - t| < \varepsilon/2$ for $n \geq N$.
Then $|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \varepsilon$.

(ii) Scalar multiplication uses $|cs_n - cs| = |c|\,|s_n - s|$; translation is (i) with constant sequence c.
(iii) Since $s_n \to s$, choose $N \geq 2$ with $|s_n - s| < 1$ for $n \geq N$ and set $M := \max\{|s_1|, \ldots, |s_{N-1}|, |s| + 1\}$. Then $|s_n t_n - st| \leq |s_n|\,|t_n - t| + |t|\,|s_n - s| \leq M\,|t_n - t| + |t|\,|s_n - s| \to 0$.
(iv) If $s \neq 0$, then for all sufficiently large n, $|s_n| > |s|/2$. Hence $\left|\frac{1}{s_n} - \frac{1}{s}\right| = \frac{|s_n - s|}{|s_n||s|} \leq \frac{2}{|s|^2}\,|s_n - s| \to 0$.

Structural Commentary: Boundedness controls products; nonzero limits control reciprocals.
Key Insight: Product and reciprocal rules are estimates after the easy sum rule. □

Definition 2.1.5. *Subsequences and Subsequential Limits*
Let (p_n) be a sequence and let $\sigma : \mathbb{N} \to \mathbb{N}$ be strictly increasing, with $n_k := \sigma(k)$.
(i) The sequence $(p_{n_k})_{k \in \mathbb{N}}$ is a *subsequence* of (p_n).
(ii) A point p is a *subsequential limit* of (p_n) iff there exists such σ with $p_{\sigma(k)} \to p$.

Remark 2.1.6. *Subsequences as Selection Maps*
View (p_n) as a function $p : \mathbb{N} \to X$, $p(n) = p_n$.
For a strictly increasing $\sigma : \mathbb{N} \to \mathbb{N}$, the subsequence is $p \circ \sigma : \mathbb{N} \to X$, namely $k \mapsto p_{\sigma(k)}$.
Key Insight: A subsequence keeps the original order but is allowed to skip terms.

Remark 2.1.7. *Subsequences of Convergent Sequences*
If $p_n \to p$, then every subsequence of (p_n) also converges to p. Conversely, if every subsequence of (p_n) converges to p, then (p_n) itself converges to p, since (p_n) is one of its subsequences.

Definition 2.1.8. *Monotone Real Sequence*
Let (s_n) be a real sequence.
(i) (s_n) is *monotonically increasing* iff $s_n \leq s_{n+1}$ for all $n \in \mathbb{N}$.
(ii) (s_n) is *monotonically decreasing* iff $s_n \geq s_{n+1}$ for all $n \in \mathbb{N}$.
The sequence is *monotonic* iff it is increasing or decreasing.

Definition 2.1.9. *Dominant Index*
For a real sequence (s_n), an index $n \in \mathbb{N}$ is *dominant* iff $s_n \geq s_m$ for all $m > n$.

Theorem 2.1.10. *Monotone Subsequence Theorem*
Every real sequence has a monotone subsequence.

Proof. If the dominant indices $n_1 < n_2 < \cdots$ are infinite, then $s_{n_1} \geq s_{n_2} \geq \cdots$, giving a decreasing subsequence.
Otherwise choose N beyond all dominant indices and set $n_1 > N$.
Given $n_k > N$, nondominance gives $n_{k+1} > n_k$ with $s_{n_{k+1}} > s_{n_k}$; hence (s_{n_k}) is increasing.
Structural Commentary: Dominant indices give decreasing case; nondominance recursively gives increasing case.
Key Insight: The proof organizes arbitrary real behavior into one ordered thread. □

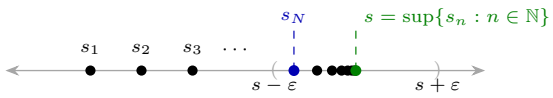

Figure 117: For an increasing bounded sequence, $s = \sup\{s_n\}$ is the limit.

Theorem 2.1.11. *Monotone Convergence Theorem for Sequences*
A monotonic real sequence converges if and only if it is bounded.

Proof.
(⇒) If $s_n \to s$, choose N with $|s_n - s| < 1$ for $n \geq N$.
The tail lies in $(s - 1, s + 1)$ and the initial set $\{s_1, \ldots, s_{N-1}\}$ is finite, so (s_n) is bounded.
(⇐) Let $E := \{s_n : n \in \mathbb{N}\}$. If (s_n) is increasing, set $s := \sup E$. Given $\varepsilon > 0$, choose N with $s - \varepsilon < s_N \leq s$; then $s - \varepsilon < s_n \leq s$ for $n \geq N$, so $s_n \to s$. If (s_n) is decreasing, set $s := \inf E$. Given $\varepsilon > 0$, choose N with $s \leq s_N < s + \varepsilon$; then $s \leq s_n \leq s_N < s + \varepsilon$ for $n \geq N$, so $s_n \to s$.
Structural Commentary: Monotonicity traps bounded tails at $\sup E$ or $\inf E$.
Key Insight: The least upper bound is forced to be the limit. □

Theorem 2.1.12. *Bolzano–Weierstrass for Real Sequences*
Every bounded real sequence has a convergent subsequence.

Proof. Let (s_{n_k}) be a monotone subsequence.
If $m \leq s_n \leq M$ for all n, then $m \leq s_{n_k} \leq M$ for all k; hence (s_{n_k}) is bounded and monotone, so it converges.
Structural Commentary: Boundedness prevents escape; monotone subsequences force one ordered thread.
Key Insight: Monotone subsequence plus monotone convergence is the real-line compactness mechanism. □

Definition 2.1.13. *Cauchy Real Sequence*
A real sequence (s_n) is *Cauchy* if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \geq N$ implies $|s_n - s_m| < \varepsilon$.

Lemma 2.1.14. *Cauchy Plus One Convergent Subsequence*
If a real Cauchy sequence has a subsequence converging to $s \in \mathbb{R}$, then the whole sequence converges to s.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m, n \geq N_0$ implies $|s_m - s_n| < \varepsilon/2$.
Choose j with $n_j \geq N_0$ and $|s_{n_j} - s| < \varepsilon/2$. Then for every $n \geq N_0$, $|s_n - s| \leq |s_n - s_{n_j}| + |s_{n_j} - s| < \varepsilon$.
Structural Commentary: Cauchy tails follow any convergent thread.
Key Insight: Mutual tail closeness turns one subsequential limit into the full limit. □

Theorem 2.1.15. *Real Cauchy Criterion*
A real sequence converges in $\mathbb{R}$ if and only if it is Cauchy.

Proof.
($\Rightarrow$) If $s_n \to s$, choose N with $|s_n - s| < \varepsilon/2$ for $n \geq N$. Then $m, n \geq N$ implies $|s_n - s_m| < \varepsilon$.
($\Leftarrow$) If (s_n) is Cauchy, choose N with $|s_m - s_n| < 1$ for $m, n \geq N$. Then the tail lies in $(s_N - 1, s_N + 1)$, so (s_n) is bounded. Bolzano–Weierstrass gives a convergent subsequence; Lemma 2.1.14 gives $s_n \to s$.

Structural Commentary: Completeness of $\mathbb{R}$ means Cauchy behavior has a real endpoint.
Key Insight: Cauchy boundedness plus a convergent subsequence forces the full limit. □

2.2 Divergence, Upper and Lower Limits, and Sequence Estimates

Definition 2.2.1. *Divergence to $\pm\infty$*
Let $s : \mathbb{N} \to \mathbb{R}$ and write $s_n := s(n)$.
(i) $s_n \to +\infty$ iff $\forall M \in \mathbb{R},\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N},\ n \geq N \Rightarrow s_n \geq M$.
(ii) $s_n \to -\infty$ iff $\forall M \in \mathbb{R},\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N},\ n \geq N \Rightarrow s_n \leq M$.

Proposition 2.2.2. *Monotone Sequences Have Extended Limits*
Let (s_n) be a real sequence.
(i) If (s_n) is increasing, then $s_n \to \sup\{s_n : n \in \mathbb{N}\}$ in $\overline{\mathbb{R}}$.
(ii) If (s_n) is decreasing, then $s_n \to \inf\{s_n : n \in \mathbb{N}\}$ in $\overline{\mathbb{R}}$.
Thus a monotone real sequence has a finite limit exactly when it is bounded.

Proof.
(i) Assume (s_n) is increasing and set $a := \sup\{s_n : n \in \mathbb{N}\} \in \overline{\mathbb{R}}$. If $a \in \mathbb{R}$, then for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ with $a - \varepsilon < s_N \leq a$, and $n \geq N$ implies $a - \varepsilon < s_n \leq a$, so $s_n \to a$. If $a = +\infty$, then for every $M \in \mathbb{R}$ there is $N \in \mathbb{N}$ with $s_N \geq M$, and $n \geq N$ implies $s_n \geq M$, so $s_n \to +\infty$.
(ii) Assume (s_n) is decreasing and set $b := \inf\{s_n : n \in \mathbb{N}\} \in \overline{\mathbb{R}}$. If $b \in \mathbb{R}$, then for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ with $b \leq s_N < b + \varepsilon$, and $n \geq N$ implies $b \leq s_n \leq s_N < b + \varepsilon$, so $s_n \to b$. If $b = -\infty$, then for every $M \in \mathbb{R}$ there is $N \in \mathbb{N}$ with $s_N \leq M$, and $n \geq N$ implies $s_n \leq M$, so $s_n \to -\infty$.

In (i), $a \in \mathbb{R}$ iff (s_n) is bounded; in (ii), $b \in \mathbb{R}$ iff (s_n) is bounded.
Thus the extended limit is finite exactly when (s_n) is bounded.
Structural Commentary: Monotone tails have one endpoint in $\overline{\mathbb{R}}$.
Key Insight: Finite endpoints give ε-traps; infinite endpoints give threshold tails. □

Proposition 2.2.3. *Unbounded Real Sequences Have Infinite Subsequences*
A real sequence is unbounded iff some subsequence tends to $+\infty$ or $-\infty$.

Proof.
($\Rightarrow$) If (s_n) is unbounded above, every tail is unbounded above; with $n_0 := 0$, choose $n_k > n_{k-1}$ with $s_{n_k} \geq k$, hence $s_{n_k} \to +\infty$. If (s_n) is unbounded below, every tail is unbounded below.
With $n_0 := 0$, choose $n_k > n_{k-1}$ with $s_{n_k} \leq -k$, hence $s_{n_k} \to -\infty$.
($\Leftarrow$) If $s_{n_k} \to +\infty$, then for every $M \in \mathbb{R}$ some k satisfies $s_{n_k} \geq M$, so (s_n) is unbounded above.
If $s_{n_k} \to -\infty$, then for every $M \in \mathbb{R}$ some k satisfies $s_{n_k} \leq M$, so (s_n) is unbounded below.

Structural Commentary: Unbounded tails supply the thresholds $\pm k$.
Key Insight: Large late terms form a divergent subsequence. □

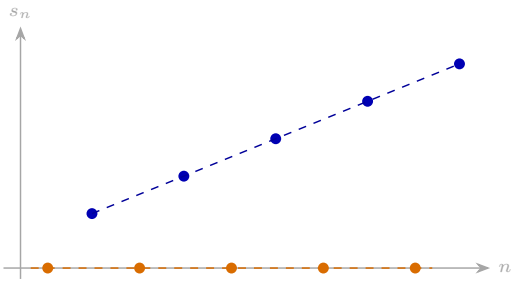

Figure 118: For $s_n = [(-1)^n + 1]n$, odd terms (orange) stay at 0, while even terms (blue) diverge to $+\infty$.

Definition 2.2.4. *Upper and Lower Limits*
Let (s_n) be a real sequence and set $E := E(s) := \{x \in \overline{\mathbb{R}} : \exists\, n_1 < n_2 < \cdots,\ s_{n_k} \to x\}$.
(i) The *upper limit* of (s_n) is $\limsup_{n\to\infty} s_n := s^* := \sup_{\overline{\mathbb{R}}} E$.
(ii) The *lower limit* of (s_n) is $\liminf_{n\to\infty} s_n := s_* := \inf_{\overline{\mathbb{R}}} E$.

Theorem 2.2.5. *Characterization of the Upper Limit*
Let (s_n) be real sequence. Set $E := \{x \in \overline{\mathbb{R}} : \exists\, n_1 < n_2 < \cdots,\ s_{n_k} \to x\}$, $s^* := \sup_{\overline{\mathbb{R}}} E$, $s_* := \inf_{\overline{\mathbb{R}}} E$. Then:
(i) $s^* \in E$.
(ii) If $x \in \mathbb{R}$ and $x > s^*$, then $s_n < x$ eventually.
(iii) s^* is the unique $y \in E$ such that $s_n < x$ eventually for all $x \in \mathbb{R}$ with $x > y$.
Dually, s_* is the unique $y \in E$ such that $s_n > x$ eventually for all $x \in \mathbb{R}$ with $x < y$.

Proof.

(i) Set $n_0 := 0$. If $s^* = +\infty$, choose $y_k \in E$ with $y_k > k$; diagonal selection gives $n_k > n_{k-1}$ with $s_{n_k} > k$, hence $+\infty \in E$. If $s^* \in \mathbb{R}$, choose $y_k \in E \cap (s^* - 1/k, s^*]$; diagonal selection gives $n_k > n_{k-1}$ with $|s_{n_k} - y_k| < 1/k$, hence $s_{n_k} \to s^*$. If $s^* = -\infty$ and $-\infty \notin E$, some tail is bounded below; that tail is unbounded above or bounded, giving respectively $+\infty \in E$ or a finite element of E, a contradiction. QED.

(ii) If $x > s^*$ and $s_n \geq x$ infinitely often, some subsequence has extended subsequential limit $y \in E \cap [x, +\infty]$, contradicting $s^* = \sup_{\overline{\mathbb{R}}} E < x \leq y$. Thus $s_n < x$ eventually.

(iii) By (i)–(ii), s^* satisfies the stated conditions. If $p < q$ both satisfy them, choose $x \in \mathbb{R}$ with $p < x < q$; eventual $s_n < x$ from p contradicts the subsequence tending to q.

Apply the upper-limit result to $(-s_n)$; its subsequential-limit set is $-E$ and its upper limit is $-s_*$.
Structural Commentary: Limsup is the maximal subsequential limit and eventual ceiling.
Key Insight: Limsup combines subsequences with eventual upper bounds. □

Proposition 2.2.6. *Tail Suprema Formula*
For a real sequence (s_n) and $N \in \mathbb{N}$, set $T_N := \{s_n : n \geq N\}$, $u_N := \sup_{\overline{\mathbb{R}}} T_N$, and $\ell_N := \inf_{\overline{\mathbb{R}}} T_N$.

(i) $u_{N+1} \leq u_N$ *and* $\ell_{N+1} \geq \ell_N$ *for all* $N \in \mathbb{N}$.
(ii) $\limsup_{n\to\infty} s_n = \inf_{N\in\mathbb{N}} u_N$ *and* $\liminf_{n\to\infty} s_n = \sup_{N\in\mathbb{N}} \ell_N$.

Proof.

(i) Since $T_{N+1} \subseteq T_N$, $\sup_{\overline{\mathbb{R}}} T_{N+1} \leq \sup_{\overline{\mathbb{R}}} T_N$ and $\inf_{\overline{\mathbb{R}}} T_{N+1} \geq \inf_{\overline{\mathbb{R}}} T_N$.

(ii) Set $s^* := \limsup s_n$, $s_* := \liminf s_n$, $u := \inf_{N\in\mathbb{N}} u_N$, and $\ell := \sup_{N\in\mathbb{N}} \ell_N$.
By Theorem 2.2.5, $x > s^* \Rightarrow u \leq x$ and $y < s^* \Rightarrow u \geq y$ for $x, y \in \mathbb{R}$, hence $u = s^*$.
Dually, $x < s_* \Rightarrow \ell \geq x$ and $y > s_* \Rightarrow \ell \leq y$ for $x, y \in \mathbb{R}$, hence $\ell = s_*$.

Structural Commentary: Tail envelopes converge to lim sup and lim inf.
Key Insight: Shrinking tails compute the extremal subsequential limits. □

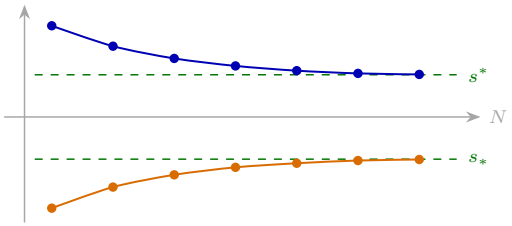

Figure 119: Blue tail suprema u_N decrease to $s^* = \limsup s_n$; orange tail infima ℓ_N increase to $s_* = \liminf s_n$.

Theorem 2.2.7. *Convergence and Equal Upper/Lower Limits*
A real sequence (s_n) converges to a finite number $s \in \mathbb{R}$ if and only if $\limsup_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = s$.

Proof.

($\Rightarrow$) If $s_n \to s$, every subsequence converges to s, and the tail is bounded, so no subsequence tends to $\pm\infty$. Thus $E = \{s\}$, hence $\limsup s_n = \liminf s_n = s$.

($\Leftarrow$) Suppose $\limsup s_n = \liminf s_n = s$. Given $\varepsilon > 0$, Theorem 2.2.5 gives $s_n < s + \varepsilon$ and $s_n > s - \varepsilon$ eventually. Hence $|s_n - s| < \varepsilon$ eventually, so $s_n \to s$.

Structural Commentary: Equality of the two extremal subsequential limits traps the whole tail.
Key Insight: Limsup supplies an eventual ceiling; liminf supplies an eventual floor. □

Theorem 2.2.8. *Order Comparison for Upper and Lower Limits*
If $s_n \leq t_n$ for all $n \geq N$, where N is fixed, then:

(i) $\limsup\limits_{n\to\infty} s_n \leq \limsup\limits_{n\to\infty} t_n$.
(ii) $\liminf\limits_{n\to\infty} s_n \leq \liminf\limits_{n\to\infty} t_n$.

Proof. Discard finitely many terms, so $s_n \leq t_n$ for all n.

(i) Put $a = \limsup s_n$ and $b = \limsup t_n$. If $x < a$, then $s_n > x$ infinitely often, hence $t_n > x$ infinitely often; Theorem 2.2.5 gives $x \leq b$. Thus $a \leq b$.

(ii) Apply (i) to $-t_n \leq -s_n$ to get $\limsup(-t_n) \leq \limsup(-s_n)$, equivalently $\liminf s_n \leq \liminf t_n$.

Structural Commentary: Eventual order is inherited by the extremal subsequential limits.
Key Insight: Limsup and liminf ignore finitely many initial terms. □

Theorem 2.2.9. *Pinching Theorem for Real Sequences*
Let (a_n), (s_n), and (b_n) be real sequences with $a_n \leq s_n \leq b_n$ for all sufficiently large n.
If $a_n \to s$ and $b_n \to s$, then $s_n \to s$.

Proof. Let $\varepsilon > 0$. Choose N_0 s.t. $a_n \leq s_n \leq b_n$ for $n \geq N_0$; N_1 s.t. $s - \varepsilon < a_n$ for $n \geq N_1$; and N_2 s.t. $b_n < s + \varepsilon$ for $n \geq N_2$. If $N = \max\{N_0, N_1, N_2\}$, then $s - \varepsilon < s_n < s + \varepsilon$ for all $n \geq N$. Hence $s_n \to s$.
Structural Commentary: Two convergent bounds squeeze the middle sequence to the same limit.
Key Insight: Once both walls are inside the ε-neighborhood, the middle term has no room to escape. □

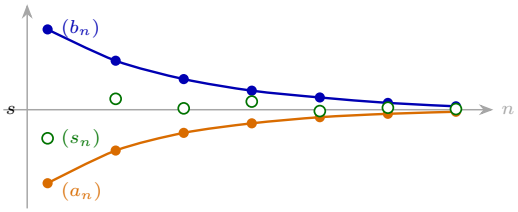

Figure 120: With $a_n \le s_n \le b_n$ and $a_n, b_n \to s$, the middle sequence also tends to s.

Theorem 2.2.10. *Special Sequence Limits*

(i) If $p > 0$, then $\lim_{n\to\infty} n^{-p} = 0$.
(ii) If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
(iii) $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
(iv) If $p > 0$ and $\alpha \in \mathbb{R}$, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
(v) If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

Proof.

(i) Given $\varepsilon > 0$, choose $N > (1/\varepsilon)^{1/p}$ by Theorem 1.1.20. Then $n \ge N$ gives $n^{-p} < \varepsilon$.
(ii) If $p > 1$, set $x_n := \sqrt[n]{p} - 1$. Then $x_n > 0$ and $1 + nx_n \le (1 + x_n)^n = p$, so $0 < x_n \le (p-1)/n \to 0$. By Theorem 2.2.9, $x_n \to 0$. If $p = 1$, then $\sqrt[n]{p} = 1$ for every n.
If $0 < p < 1$, then $1/p > 1$, so $\sqrt[n]{1/p} \to 1$ by the previous case.
Since $\sqrt[n]{p} = 1/\sqrt[n]{1/p}$ and limits preserve reciprocals away from 0, $\sqrt[n]{p} \to 1$.
(iii) Put $x_n := \sqrt[n]{n} - 1$. Then $x_n \ge 0$, and for $n \ge 2$ the binomial theorem gives $n = (1 + x_n)^n \ge \frac{n(n-1)}{2} x_n^2$.
Thus $0 \le x_n \le \sqrt{2/(n-1)}$, so Theorem 2.2.9 gives $x_n \to 0$.
(iv) Choose an integer $k > \alpha$ with $k > 0$. For $n > 2k$, $(1 + p)^n \ge \binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}$.
Hence $0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}$, and $n^{\alpha-k} \to 0$ by (i). Apply Theorem 2.2.9.
(v) If $x \ne 0$, write $|x| = (1 + p)^{-1}$ with $p > 0$ and apply (iv) with $\alpha = 0$ to get $|x|^n \to 0$. Since $-|x|^n \le x^n \le |x|^n$, Theorem 2.2.9 gives $x^n \to 0$; $x = 0$ is immediate.

Structural Commentary: Each limit is reduced to an elementary comparison sequence.
Key Insight: The binomial theorem supplies the right upper bounds for roots and exponential domination. □

Proposition 2.2.11. *Decimal Truncations Squeeze a Real*
Let $x \ge 0$. For $n \ge 0$, set $s_n := \max\{k \in \mathbb{Z}_{\ge 0} : k10^{-n} \le x\}$.
Then $s_n 10^{-n} \le x < (s_n + 1)10^{-n}$; and $0 \le x - s_n 10^{-n} < 10^{-n}$; so $s_n 10^{-n} \to x$.
Moreover $s_{n+1} = 10s_n + a_{n+1}$ for some $a_{n+1} \in \{0, \dots, 9\}$.

Proof. The maximum exists by the Archimedean property, Theorem 1.1.20, and Assumption 2.6.1. Maximality gives the displayed inequalities. The squeeze and Theorem 2.2.10 give $s_n 10^{-n} \to x$.
At levels n and $n + 1$, $10s_n \le s_{n+1} < 10s_n + 10$, so $a_{n+1} := s_{n+1} - 10s_n \in \{0, \dots, 9\}$.
Structural Commentary: Decimal existence is an Archimedean grid argument followed by a squeeze.
Key Insight: Refining the mesh by a factor of 10 adds exactly one digit. □

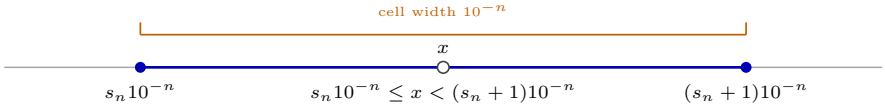

Figure 121: At precision 10^{-n}, the truncation $s_n 10^{-n}$ traps x in a cell of width 10^{-n}.

Method 2.2.12. *Sequence Proof Patterns*

(i) *Convergence*: fix $\varepsilon > 0$, then find N so that all later terms lie in the relevant ε-ball.
(ii) *Uniqueness*: compare two proposed limits with $\varepsilon/2$ and use the triangle inequality.
(iii) *Subsequences*: choose $n_k \uparrow$ recursively with distances $< 1/k$.
(iv) *Cauchy arguments*: estimate the diameter of tails rather than guessing the limit first.
(v) *Monotone sequences*: use sup or inf and then prove eventual trapping.

Key Insight: Most sequence proofs are bookkeeping for quantifiers, indices, and one controlling inequality.

3 Metric Spaces and Euclidean Topology

3.1 Metrics, Open Sets, and Euclidean Topology

Metric topology studies sets through balls: openness, closure, and limit points come from distance. Euclidean estimates rely on norms, Cauchy–Schwarz, and the triangle inequality.

Definition 3.1.1. *Metric Space*
A *metric space* is a set X equipped with a function $d : X \times X \to \mathbb{R}$ such that, for all $p, q, r \in X$,

i. $d(p, q) > 0$ if $p \ne q$, and $d(p, p) = 0$;

ii. $d(p,q) = d(q,p)$;
iii. $d(p,q) \le d(p,r) + d(r,q)$.

The function d is called a *distance function* or *metric*.

Definition 3.1.2. *Metric Topology Terminology*
Let X be a metric space, and let all points and sets below be understood as points of X and subsets of X.

(i) For $p \in X$ and $r > 0$, the *open ball* centered at p with radius r is $B_r(p) := \{q \in X : d(p,q) < r\}$.
The corresponding *closed ball* is $\overline{B}_r(p) := \{q \in X : d(p,q) \le r\}$.
The *sphere* of radius r about p is $\{q \in X : d(p,q) = r\}$.
(ii) A *neighborhood* of p is an open set $U \subseteq X$ with $p \in U$.
(iii) A point p is a *limit point* of E if every open ball $B_r(p)$ contains a point $q \neq p$ with $q \in E$.
(iv) If $p \in E$ and p is not a limit point of E, then p is an *isolated point* of E.
(v) A set E is *closed* if every limit point of E is a point of E.
(vi) A point p is an *interior point* of E if some neighborhood of p is contained in E.
(vii) A set E is *open* if every point of E is an interior point of E.
(viii) The *complement* of E is $E^c := X \setminus E$.
(ix) A set E is *dense* in X if every point of X is either a point of E, a limit point of E, or both.

For $X = \mathbb{R}$, this gives $B_\varepsilon(a) = (a - \varepsilon, a + \varepsilon)$.
Structural Commentary: Balls generate the local tests; neighborhoods are open sets containing the point.

Perspective 3.1.3. *ε-Closeness Becomes Metric Language*
On $\mathbb{R}$, ε-close means $|x - y| < \varepsilon$, i.e. $d(x,y) < \varepsilon$ with $d(x,y) = |x - y|$; may be abstracted to metric balls.
Key Insight: Absolute value is distance on $\mathbb{R}$.

Example 3.1.4. *Euclidean Metric and Subspaces*
The Euclidean space $\mathbb{R}^k$ is a metric space with $d(\mathbf{x},\mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_2, \qquad \mathbf{x},\mathbf{y} \in \mathbb{R}^k$.
Every subset Y of a metric space X is a metric space by restricting the same distance function to points of Y.

Example 3.1.5. *Discrete Metric*
On any set X, define $d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$.
This is the *discrete metric*: distinct points are all distance 1 apart.

Example 3.1.6. *Hamming Distance*
For $x, y \in \{0,1\}^n$, define $h(x,y) := |\{i : x_i \neq y_i\}|$. Then h is a metric.

Proof. $h \ge 0$, $h(x,y) = h(y,x)$, and $h(x,y) = 0 \iff x_i = y_i\ \forall i \iff x = y$.
For $x, y, z \in \{0,1\}^n$, $\{i : x_i \neq y_i\} \subseteq \{i : x_i \neq z_i\} \cup \{i : z_i \neq y_i\}$, so $h(x,y) \le h(x,z) + h(z,y)$.
Structural Commentary: Hamming distance counts coordinate disagreements.
Key Insight: Each disagreement between x and y must be paid for through z in at least one leg. □

Definition 3.1.7. *Euclidean Intervals, Cells, Balls, and Convexity*

(i) In $\mathbb{R}$, $(a,b) := \{x : a < x < b\}$, $[a,b] := \{x : a \le x \le b\}$, with $[a,b)$ and $(a,b]$ defined by mixed inequalities.
(ii) For $a_i < b_i$, a *k-cell* is $\prod_{i=1}^k [a_i, b_i] = \{\mathbf{x} \in \mathbb{R}^k : a_i \le x_i \le b_i,\ 1 \le i \le k\}$.
(iii) For $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, $B_r(\mathbf{x}) := \{\mathbf{y} : \|\mathbf{y} - \mathbf{x}\|_2 < r\}$ and $\overline{B}_r(\mathbf{x}) := \{\mathbf{y} : \|\mathbf{y} - \mathbf{x}\|_2 \le r\}$.
(iv) $E \subseteq \mathbb{R}^k$ is *convex* iff $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in E$ for all $\mathbf{x},\mathbf{y} \in E$ and $0 \le \lambda \le 1$.
(v) The *integer lattice* is $\mathbb{Z}^k$, and the *first orthant* is $\{\mathbf{x} \in \mathbb{R}^k : x_i \ge 0\ \forall i\}$.
(vi) The *unit ball* is $B^k := \{\mathbf{x} \in \mathbb{R}^k : \|\mathbf{x}\|_2 \le 1\}$, and the *unit sphere* is $S^{k-1} := \{\mathbf{x} \in \mathbb{R}^k : \|\mathbf{x}\|_2 = 1\}$.
(vii) A *convex combination* of $\mathbf{x},\mathbf{y}$ is $\lambda\mathbf{x} + (1-\lambda)\mathbf{y}$ with $0 \le \lambda \le 1$; these points form the line segment $[\mathbf{x},\mathbf{y}]$.

Proposition 3.1.8. *Balls and Cells Are Convex*
Open balls, closed balls, and k-cells in $\mathbb{R}^k$ are convex.

Proof. Let $\mathbf{y},\mathbf{z} \in B_r(\mathbf{x})$ and $0 < \lambda < 1$.
Then $\|\lambda\mathbf{y} + (1-\lambda)\mathbf{z} - \mathbf{x}\|_2 = \|\lambda(\mathbf{y} - \mathbf{x}) + (1-\lambda)(\mathbf{z} - \mathbf{x})\|_2 \le \lambda\|\mathbf{y} - \mathbf{x}\|_2 + (1-\lambda)\|\mathbf{z} - \mathbf{x}\|_2 < r$.
The same proof with $\le r$ gives closed balls. For a k-cell, each coordinate of $\lambda\mathbf{y} + (1-\lambda)\mathbf{z}$ lies between the corresponding coordinates of $\mathbf{y}$ and $\mathbf{z}$, hence remains in the same interval $[a_i, b_i]$.
Structural Commentary: Convexity is checked by following the line segment between two points.
Key Insight: Balls use the triangle inequality; cells use coordinate inequalities. □

Proposition 3.1.9. *Bounded Compression of a Metric*
If (X,d) is a metric space, then $\rho(x,y) := d(x,y)/(1 + d(x,y))$ is a bounded metric on X, and $\tau_\rho = \tau_d$.

Proof. $0 \le \rho < 1$, and definiteness and symmetry follow from d.
For $a, b \ge 0$, $(a+b)/(1+a+b) \le a/(1+a) + b/(1+b)$.
Since $t/(1+t)$ is increasing, $d(x,z) \le d(x,y) + d(y,z)$ gives $\rho(x,z) \le \rho(x,y) + \rho(y,z)$.
For balls, $B_d(x,\eta) \subseteq B_\rho(x,\eta)$ for $0 < \eta < 1$, and $B_\rho(x,\eta) \subseteq B_d(x, \eta/(1-\eta))$; hence $\tau_\rho = \tau_d$.
Structural Commentary: Compressing distances into $[0,1)$ changes boundedness but not local neighborhoods.
Key Insight: Topology depends on small-scale control, not on the large-scale size of distances. □

Definition 3.1.10. *Isometries and Pseudometrics*
Let (X,ρ) and (Y,σ) be metric spaces, and let $F : X \to Y$.

(i) F is an *isometry* from (X,ρ) onto (Y,σ) iff $F(X) = Y$ and $\forall x, y \in X$, $\sigma(Fx, Fy) = \rho(x,y)$.
(ii) (X,ρ) and (Y,σ) are *isometric*, written $(X,\rho) \cong (Y,\sigma)$, iff $\exists$ an isometry $F : (X,\rho) \to (Y,\sigma)$.

For a set Z, a *pseudometric* on Z is a map $d : Z \times Z \to [0,\infty)$ such that $\forall x,y,z \in Z$, $d(x,x) = 0$, $d(x,y) = d(y,x)$, and $d(x,z) \le d(x,y) + d(y,z)$; points $x \neq y$ may satisfy $d(x,y) = 0$.
Structural Commentary: Isometry preserves distance; pseudometrics permit zero-distance identification.

Proposition 3.1.11. *Metric Quotient of a Pseudometric*
Let X be a set and let ρ be a pseudometric on X.
Define $x \sim_\rho y$ iff $\rho(x,y) = 0$, and define $\widetilde{\rho}: X/\!\sim_\rho \times X/\!\sim_\rho \to [0,\infty)$ by $\widetilde{\rho}([x],[y]) := \rho(x,y)$.
(i) $\sim_\rho$ is an equivalence relation on X.
(ii) $\widetilde{\rho}$ is a well-defined metric on $X/\!\sim_\rho$.

Proof.
(i) For $x, y, z \in X$, $\rho(x,x) = 0$, $\rho(x,y) = 0 \Rightarrow \rho(y,x) = 0$;
$\rho(x,y) = \rho(y,z) = 0 \Rightarrow 0 \le \rho(x,z) \le \rho(x,y) + \rho(y,z) = 0$.
Hence $\sim_\rho$ is reflexive, symmetric, and transitive.
(ii) If $x \sim_\rho x'$ and $y \sim_\rho y'$, then $\rho(x,y) \le \rho(x,x') + \rho(x',y') + \rho(y',y) = \rho(x',y')$, and the reverse inequality is identical; hence $\widetilde{\rho}$ is well-defined. For $[x],[y],[z] \in X/\!\sim_\rho$, nonnegativity, symmetry, and
$\widetilde{\rho}([x],[z]) = \rho(x,z) \le \rho(x,y) + \rho(y,z) = \widetilde{\rho}([x],[y]) + \widetilde{\rho}([y],[z])$ follow from ρ.
Finally, $\widetilde{\rho}([x],[y]) = 0$ iff $\rho(x,y) = 0$ iff $[x] = [y]$.
Structural Commentary: Collapse exactly zero-distance classes.
Key Insight: The quotient restores definiteness. □

Example 3.1.12. *Open, Closed, Both, or Neither*
In $\mathbb{R}$, $\emptyset, \mathbb{R}$ are open and closed, while $[a,b)$, $a < b$, is neither. If d is discrete on X, then $B_{1/2}(x) = \{x\}$ for all $x \in X$, so every $E \subseteq X$ and E^c are open; hence every $E \subseteq X$ is closed.

Theorem 3.1.13. *Every Open Ball Is Open*
Every open ball in a metric space is an open set.

Proof. Let $E = B_r(p)$, and choose $q \in E$. Put $h := r - d(p,q) > 0$.
If $s \in B_h(q)$, then $d(p,s) \le d(p,q) + d(q,s) < d(p,q) + h = r$.
Thus $B_h(q) \subseteq E$, so q is an interior point of E. Since q was arbitrary, E is open.
Structural Commentary: Around any point inside a ball, the leftover distance to boundary gives a smaller ball.
Key Insight: The triangle inequality turns radius slack into openness. □

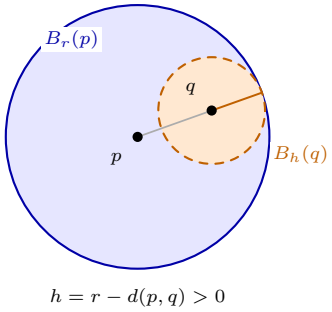

Figure 122: Every point inside a metric ball has a smaller ball still contained in the original ball.

Theorem 3.1.14. *Limit Points Force Infinitely Many Nearby Points*
Let (X,d) be a metric space, let $E \subseteq X$, and let $p \in X$. If p is a limit point of E, then $N \cap E$ is infinite for every neighborhood N of p.

Proof. Let N be a neighborhood of p and suppose $N \cap E$ is finite. Since N is open and $p \in N$, choose $\eta > 0$ with $B_\eta(p) \subseteq N$. Put $F := (N \cap E) \setminus \{p\}$. If $F = \emptyset$, then $B_\eta(p) \cap (E \setminus \{p\}) = \emptyset$, contradicting that p is a limit point. If $F \neq \emptyset$, set $\delta := \min(\{\eta\} \cup \{d(p,q) : q \in F\}) > 0$. If $q \in B_\delta(p) \cap (E \setminus \{p\})$, then $q \in F$ and $d(p,q) < \delta \le d(p,q)$, impossible. Thus $B_\delta(p) \cap (E \setminus \{p\}) = \emptyset$, again contradicting that p is a limit point.
Structural Commentary: A finite list of nearby exceptions has a smallest positive distance from p.
Key Insight: Limit points cannot be witnessed by only finitely many neighboring points. □

Corollary 3.1.15. *Finite Point Sets Have No Limit Points*
A finite point set in a metric space has no limit points.

Proof. Let $E \subseteq X$ be finite. If p were a limit point of E, then by Theorem 3.1.14, every neighborhood of p would contain infinitely many distinct points of E, impossible as E is finite. Hence E has no limit points.
Structural Commentary: This corollary is the finite obstruction to accumulation.
Key Insight: A limit point requires infinitely many nearby witnesses. □

Theorem 3.1.16. *Open Sets and Closed Complements*
A set E is open if and only if its complement E^c is closed.

Proof.
($\Rightarrow$) Suppose E is open. If x is a limit point of E^c and $x \in E$, then some neighborhood N of x satisfies $N \subseteq E$, contradicting $N \cap (E^c \setminus \{x\}) \neq \emptyset$. Hence $x \in E^c$, so E^c is closed.
($\Leftarrow$) Suppose E^c is closed. If $x \in E$, then $x \notin E^c$, hence x is not a limit point of E^c. Thus some neighborhood N of x has $N \cap E^c = \emptyset$, i.e. $N \subseteq E$. Hence E is open.
Structural Commentary: Openness of a set is equivalent to closure of the outside.
Key Insight: Interior points of E cannot be limit points of E^c. □

Corollary 3.1.17. *Closed Sets and Open Complements*
A set F is closed if and only if F^c is open.

Proof.
($\Rightarrow$) If F is closed, then $(F^c)^c = F$ is closed; by Theorem 3.1.16 with $E = F^c$, F^c is open.
($\Leftarrow$) If F^c is open, then Theorem 3.1.16 gives $(F^c)^c$ closed; since $(F^c)^c = F$, F is closed.
Structural Commentary: This is the same open–closed duality after taking complements.
Key Insight: Closedness of F is openness of the outside F^c. □

Corollary 3.1.18. *Closed Balls Are Closed*
For every $p \in X$ and $r > 0$, the closed ball $\overline{B}_r(p)$ is closed.

Proof. Let $q \notin \overline{B}_r(p)$, so $d(p,q) > r$. Put $h := (d(p,q) - r)/2$.
If $s \in B_h(q)$, then $d(p,s) \geq d(p,q) - d(q,s) > d(p,q) - h > r$, so $s \notin \overline{B}_r(p)$.
Hence the complement of $\overline{B}_r(p)$ is open, and $\overline{B}_r(p)$ is closed.
Structural Commentary: Outside a closed ball, positive distance remains to the boundary.
Key Insight: Reverse triangle estimates certify closedness. □

Theorem 3.1.19. *Basic Topological Calculus*
In a metric space:
(i) any union of open sets is open;
(ii) any intersection of closed sets is closed;
(iii) any finite intersection of open sets is open;
(iv) any finite union of closed sets is closed.

Proof.
(i) Let $G := \bigcup_\alpha G_\alpha$. If $x \in G$, then $x \in G_\alpha$ for some α.
Since G_α is open, some neighborhood of x lies in $G_\alpha \subseteq G$. Hence G is open.
(ii) Let $F := \bigcap_\alpha F_\alpha$. By Theorem 1.5.2, $F^c = \bigcup_\alpha F_\alpha^c$.
Each F_α^c is open by Corollary 3.1.17, so F^c is open by (i). Hence F is closed.
(iii) Let $H := \bigcap_{i=1}^n G_i$. If $x \in H$, let $r_i > 0$, $B_{r_i}(x) \subseteq G_i$, $r := \min_{1 \leq i \leq n} r_i$. Then $B_r(x) \subseteq H$, so H is open.
(iv) Let $F := \bigcup_{i=1}^n F_i$. By Theorem 1.5.2, $F^c = \bigcap_{i=1}^n F_i^c$.
Each F_i^c is open by Corollary 3.1.17, so F^c is open by (iii). Hence F is closed.
Structural Commentary: Arbitrary unions are easy for open sets; finite intersections require a minimum radius.
Key Insight: The finite minimum is the reason infinite intersections of open sets need not be open. □

Theorem 3.1.20. *Open Sets on the Real Line Split into Intervals*
Let $\emptyset \neq G \subseteq \mathbb{R}$ be open.
(i) $G = \bigsqcup_{\alpha \in A} I_\alpha$ for a family of pairwise disjoint open intervals $I_\alpha \subseteq \mathbb{R}$, possibly unbounded.
(ii) The index set A is at most countable.

Proof.
(i) For $a, b \in \mathbb{R}$, set $I(a,b) := [\min\{a,b\}, \max\{a,b\}]$.
For $x, y \in G$, set $x \sim y$ iff $I(x,y) \subseteq G$, and let $C_x := [x]_\sim$. Then $\sim$ is an equivalence relation.
If $y \in C_x$ and $z \in I(x,y)$, then $I(x,z) \subseteq I(x,y) \subseteq G$, so $z \in C_x$; hence C_x is an interval.
If $y \in C_x$, choose $r > 0$ with $(y - r, y + r) \subseteq G$; $z \in (y - r, y + r)$ implies $I(x,z) \subseteq I(x,y) \cup (y - r, y + r) \subseteq G$, so $z \in C_x$. Thus each C_x is open, and the classes give a disjoint union of G.
(ii) Let $\mathcal{C} := G/\sim$ and enumerate $\mathbb{Q} = \{q_j : j \in \mathbb{N}\}$. Since each $C \in \mathcal{C}$ contains a bounded nonempty open interval, $C \cap \mathbb{Q} \neq \emptyset$. Define $\phi(C) := q_{\min\{j : q_j \in C\}}$. Disjointness gives ϕ injective, so $\mathcal{C}$ is at most countable.
Structural Commentary: Maximal open intervals are counted by rationals.
Key Insight: Density of $\mathbb{Q}$ turns disjoint interval geometry into countability. □

Definition 3.1.21. *Closure*
If $E \subseteq X$ and E^{acc} denotes the set of all limit points of E in X, then the *closure* of E is $\overline{E} := E \cup E^{\text{acc}}$.

Theorem 3.1.22. *Closure Is the Smallest Closed Superset*
If X is a metric space and $E \subseteq X$, then:
(i) $\overline{E}$ is closed;
(ii) $\overline{E} = E$ if and only if E is closed;
(iii) $\overline{E} \subseteq F$ for every closed set $F \subseteq X$ such that $E \subseteq F$.
Thus $\overline{E}$ is the smallest closed subset of X that contains E.

Proof.
(i) Let $p \notin \overline{E}$. Then $p \notin E$ and p is not a limit point of E, so $B_r(p) \cap E = \emptyset$ for some $r > 0$.
If $q \in B_r(p)$, then $q \notin E$ and $B_{r - d(p,q)}(q) \subseteq B_r(p)$, so $q \notin E^{\text{acc}}$.
Hence $B_r(p) \subseteq (\overline{E})^c$, and $\overline{E}$ is closed by Corollary 3.1.17.
(ii) If $\overline{E} = E$, then E is closed by (i). Conversely, if E is closed, then $E^{\text{acc}} \subseteq E$, so $\overline{E} = E \cup E^{\text{acc}} = E$.
(iii) If F is closed and $E \subseteq F$, then $E^{\text{acc}} \subseteq F^{\text{acc}} \subseteq F$. Thus $\overline{E} = E \cup E^{\text{acc}} \subseteq F$.
Structural Commentary: Closure adds exactly the limit points forced by E.
Key Insight: Closed supersets must contain all limit points, so they must contain $\overline{E}$. □

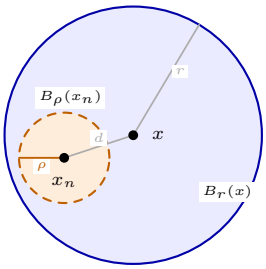

Figure 123: If $d = d(x, x_n) < r$, then any $0 < \rho < r - d$ gives $B_\rho(x_n) \subseteq B_r(x)$.

Definition 3.1.23. *Interior, Boundary, Adherence, and Distance*
Let (X, d) be a metric space and $E, F \subseteq X$.

(i) $E^\circ := \{p \in E : \exists r > 0,\ B_r(p) \subseteq E\}$ is the *interior*.
(ii) $\operatorname{ext}(E) := (E^c)^\circ = \{p \in X : \exists r > 0,\ B_r(p) \cap E = \emptyset\}$ is the *exterior*.
(iii) $\operatorname{bd}(E) := \overline{E} \setminus E^\circ = \{p \in X : \forall r > 0,\ B_r(p) \cap E \neq \emptyset,\ B_r(p) \cap E^c \neq \emptyset\}$ is the *boundary*.
(iv) $p \in X$ is *adherent* to E iff $\forall r > 0,\ B_r(p) \cap E \neq \emptyset$, equivalently $p \in \overline{E}$.
(v) For $p \in X$ and $\emptyset \neq E \subseteq X$, the *point-set distance* is $\operatorname{dist}(p, E) := \inf\{d(p, q) : q \in E\}$.
(vi) For nonempty $E, F \subseteq X$, the *set-set distance* is $\operatorname{dist}(E, F) := \inf\{d(x, y) : x \in E,\ y \in F\}$.

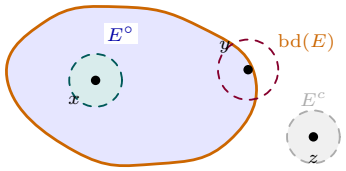

Figure 124: Interior, exterior, and boundary points are distinguished by how small balls meet E and E^c.

Proposition 3.1.24. *Distance-to-Set Closure Criterion*
Let $\emptyset \neq E \subseteq X$. For $p \in X$, $p \in \overline{E} \iff \operatorname{dist}(p, E) = 0$.

Proof.
($\Rightarrow$) If $p \in \overline{E}$, then $\forall \varepsilon > 0$, $B_\varepsilon(p) \cap E \neq \emptyset$; choose $q_\varepsilon \in E$ with $d(p, q_\varepsilon) < \varepsilon$.
Thus $0 \leq \operatorname{dist}(p, E) \leq \varepsilon$ for all $\varepsilon > 0$, so $\operatorname{dist}(p, E) = 0$.
($\Leftarrow$) If $\operatorname{dist}(p, E) = 0$, then $\forall r > 0$, $\exists q \in E$ with $d(p, q) < r$.
Hence $B_r(p) \cap E \neq \emptyset$ for all $r > 0$, so p is adherent to E, equivalently $p \in \overline{E}$.
Structural Commentary: Closure can be detected by vanishing distance to the set.
Key Insight: The infimum packages all small-ball tests into one number. □

Remark 3.1.25. *Set Distance Is Not a Metric on Subsets*
The function $\operatorname{dist}(E, F)$ is useful, but it is not a metric on nonempty subsets.
In $\mathbb{R}$, for example, $\operatorname{dist}((0, 1), (1, 2)) = 0$ although the sets are distinct and disjoint.

Figure 125: Adherent endpoints need not lie in E; isolated points are adherent but not limit points.

3.2 Relative Topology

Definition 3.2.1. *Relative Openness*
Suppose $E \subseteq Y \subseteq X$, where X is a metric space. The set E is *open relative to Y* if for each $p \in E$ there is an $r > 0$ such that $q \in E$ whenever $q \in Y$ and $d(p, q) < r$.

Theorem 3.2.2. *Relative Opens Are Intersections*
Suppose $Y \subseteq X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X.

Proof.
($\Rightarrow$) For each $p \in E$, choose $r_p > 0$ with $B_{r_p}(p) \cap Y \subseteq E$, and set $G := \bigcup_{p \in E} B_{r_p}(p)$. Then G is open in X, and $E \subseteq G \cap Y$. If $q \in G \cap Y$, then $q \in B_{r_p}(p) \cap Y$ for some $p \in E$, so $q \in E$. Thus $E = G \cap Y$.
($\Leftarrow$) If $E = G \cap Y$ with G open in X, then for each $p \in E$ choose $r > 0$ with $B_r(p) \subseteq G$.
Hence $B_r(p) \cap Y \subseteq G \cap Y = E$, so E is open relative to Y.
Structural Commentary: Relative openness is ordinary openness viewed through the ambient subset Y.
Key Insight: Intersect open sets in X with Y to get exactly the open sets of Y. □

Corollary 3.2.3. *Relative Closed Sets Are Intersections*
Let (X, d) be a metric space, let $Y \subseteq X$, and let $F \subseteq Y$.
Then F is closed relative to Y iff $\exists C \subseteq X$ such that C is closed in X and $F = Y \cap C$.

Proof.
(⇒) If F is closed relative to Y, then $Y \setminus F$ is open relative to Y. By Theorem 3.2.2, $\exists$ open $G \subseteq X$ such that $Y \setminus F = Y \cap G$. Put $C := X \setminus G$. Then C is closed in X, and $F = Y \setminus (Y \setminus F) = Y \setminus (Y \cap G) = Y \cap C$.
(⇐) If $F = Y \cap C$ with C closed in X, then $X \setminus C$ is open in X and $Y \setminus F = Y \setminus (Y \cap C) = Y \cap (X \setminus C)$. By Theorem 3.2.2, $Y \setminus F$ is open relative to Y, so F is closed relative to Y.

Structural Commentary: Relative closedness is ordinary closedness seen through the ambient subset.
Key Insight: The closed version is the complement of the relative-open intersection theorem. □

Example 3.2.4. *Relative Topology Depends on the Ambient Space*
Let $X = \{(x, 0) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the metric inherited from $\mathbb{R}^2$. The set $E = \{(x, 0) : -1 < x < 1\}$ is not open as a subset of $\mathbb{R}^2$, but it is open relative to X, since $E = X \cap \big((-1, 1) \times (-1, 1)\big)$.
Likewise, $[0, 1)$ is not closed in $\mathbb{R}$, but it is closed relative to $(-1, 1)$ because $[0, 1) = (-1, 1) \cap [0, \infty)$.

3.3 Density, Separability, and Closure Tests

Proposition 3.3.1. *Density and Closure*
Let X be a metric space and let $E \subseteq X$. Then E is dense in X if and only if $\overline{E} = X$.

Proof.
(⇒) Suppose E is dense in X. Let $p \in X$. By Definition 3.1.2, p is either a point of E or a limit point of E. Hence $p \in E \cup E^{\text{acc}} = \overline{E}$ by Definition 3.1.21. Thus $X \subseteq \overline{E}$. Since $\overline{E} \subseteq X$, $\overline{E} = X$.
(⇐) Suppose $\overline{E} = X$. Let $p \in X$. Then $p \in \overline{E} = E \cup E^{\text{acc}}$, so p is either a point of E or a limit point of E. Therefore E is dense in X by Definition 3.1.2.

Structural Commentary: Density says every point is already in E or is forced by nearby points of E.
Key Insight: The closure packages exactly those two possibilities. □

Proposition 3.3.2. *Real-Line Density Interval Test*
Let $E \subseteq \mathbb{R}$. Then E is dense in $\mathbb{R}$ if and only if every nonempty open interval (a, b) satisfies $(a, b) \cap E \neq \emptyset$.

Proof.
(⇒) Let $(a, b) \neq \emptyset$ and choose $p \in (a, b)$, $r := \frac{1}{2}\min\{p - a, b - p\}$. If $p \in E$, done. If $p \notin E$, density gives $p \in E^{\text{acc}}$, so choose $q \in (E \setminus \{p\}) \cap B_r(p)$. Since $B_r(p) \subseteq (a, b)$, $q \in (a, b) \cap E$.
(⇐) Suppose $(a, b) \cap E \neq \emptyset$ for every nonempty open interval (a, b).
For $p \in \mathbb{R}$, either $p \in E$, or $p \notin E$ and $\forall \varepsilon > 0$, choose $q \in (p - \varepsilon, p + \varepsilon) \cap E$.
Then $q \neq p$ and $q \in B_\varepsilon(p)$, so $p \in E^{\text{acc}}$. Hence $\mathbb{R} = E \cup E^{\text{acc}} = \overline{E}$, so E is dense in $\mathbb{R}$.

Structural Commentary: On $\mathbb{R}$, balls and open intervals test the same local condition.
Key Insight: Density means no interval can avoid the set. □

Definition 3.3.3. *Separable Metric Space*
A metric space X is *separable* if it contains a countable dense subset.

Example 3.3.4. *Separable and Nonseparable Metric Spaces*
(i) $\mathbb{R}$ is separable: $\mathbb{Q}$ is countable and dense. Hence $\mathbb{R}^k$ is separable via $\mathbb{Q}^k$, and $\mathbb{C}^k$ via $\mathbb{C}^k \simeq \mathbb{R}^{2k}$.
(ii) If X is discrete, then X is separable $\iff$ X is countable, since $B_{1/2}(x) = \{x\}$ forces every dense $D \subseteq X$ to contain x.

Structural Commentary: Dense rational grids handle Euclidean spaces; discrete balls force dense sets to contain every point.
Key Insight: Separability depends on whether countable data can meet every open ball.

Theorem 3.3.5. *Suprema Belong to Closures*
Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above and $y := \sup E$. Then $y \in \overline{E}$; if E is closed, then $y \in E$.

Proof. If $y \in E$, then $y \in \overline{E}$. If $y \notin E$, then $\forall h > 0$, $y - h$ is not an upper bound; choose $x_h \in E$ with $y - h < x_h \leq y$. Thus $x_h \neq y$, $|x_h - y| < h$, and $B_h(y) \cap (E \setminus \{y\}) \neq \emptyset$, so $y \in E^{\text{acc}} \subseteq \overline{E}$.
If E is closed, $\overline{E} = E$ by Theorem 3.1.22(ii).
Structural Commentary: LUB gives points of E arbitrarily close below $\sup E$.
Key Insight: A supremum is either attained or approached. □

3.4 Bounded Sets and Metric Sequences

Definition 3.4.1. *Bounded Sets and Metric Sequences*
Let (X, d) be a metric space, $E \subseteq X$, and (p_n) a sequence in X.
(i) E is *bounded* iff $\exists q \in X, \exists M > 0$ such that $E \subseteq B_M(q)$
(ii) E is *perfect* iff E is closed and every $p \in E$ is a limit point of E.
(iii) (p_n) *converges* to $p \in X$ iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $n \geq N \implies d(p_n, p) < \varepsilon$; write $p_n \to p$ or $\lim_{n\to\infty} p_n = p$. If no such p exists, (p_n) *diverges*.
(iv) The *range* of (p_n) is $\text{ran}(p_n) := \{p_n : n \in \mathbb{N}\}$; (p_n) is *bounded* iff $\text{ran}(p_n)$ is bounded.
(v) (p_n) is *eventually in E* iff $\exists N \in \mathbb{N}$ such that $n \geq N \implies p_n \in E$
(vi) (p_n) is *frequently in E* iff $\forall N \in \mathbb{N}, \exists n \geq N$ such that $p_n \in E$.

Structural Commentary: Boundedness, compactness, and completeness are intrinsic to restricted metrics; openness and closedness are ambient.
Key Insight: Convergence is tail control; frequent membership is arbitrarily late recurrence.

Theorem 3.4.2. *Basic Properties of Metric Convergence*
Let X be a metric space and $(p_n) \subseteq X$.
(i) $p_n \to p \iff \forall r > 0, \exists N \in \mathbb{N}$ such that $n \geq N \implies p_n \in B_r(p)$.
(ii) If $p_n \to p$ and $p_n \to p'$, then $p = p'$.

(iii) If $E \subseteq X$ and $p \in E^{\text{acc}}$, then $\exists (q_n) \subseteq E \setminus \{p\}$ such that $q_n \to p$.

Proof.

(i) ($\Rightarrow$) If $p_n \to p$ and $r > 0$, take $\varepsilon = r$ in the definition.
(i) ($\Leftarrow$) If the ball condition holds, then for every $\varepsilon > 0$ take $r = \varepsilon$, giving $d(p_n, p) < \varepsilon$ eventually.
(ii) Given $\varepsilon > 0$, choose N, N' with $n \geq N \implies d(p_n, p) < \varepsilon/2$ and $n \geq N' \implies d(p_n, p') < \varepsilon/2$.
If $n \geq \max\{N, N'\}$, then $d(p, p') \leq d(p, p_n) + d(p_n, p') < \varepsilon$, so $d(p, p') = 0$.
(iii) Since $p \in E^{\text{acc}}$, for each $n \in \mathbb{N}$ choose $q_n \in E \setminus \{p\}$ with $d(q_n, p) < 1/n$. Then $q_n \to p$.

Structural Commentary: Eventual containment gives convergence; triangle inequality gives uniqueness.
Key Insight: A limit point can always be approached by a sequence of points from the set. □

Corollary 3.4.3. *Subsequences Preserve Metric Limits*
If $p_n \to p$ in a metric space, then every subsequence (p_{n_k}) also converges to p.

Proof. Given $\varepsilon > 0$, choose N such that $n \geq N$ implies $d(p_n, p) < \varepsilon$.
Since $n_k \to \infty$, all sufficiently large k satisfy $n_k \geq N$, hence $d(p_{n_k}, p) < \varepsilon$.
Structural Commentary: A subsequence is only a delayed tail of the same sequence.
Key Insight: Eventual estimates survive deleting terms. □

Proposition 3.4.4. *Limit Points Are Subsequence Limits*
For $(p_n) \subseteq X$ and $p \in X$, the following are equivalent:

(i) $\forall r > 0, \forall N \in \mathbb{N}, \exists n \geq N$ *such that* $p_n \in B_r(p)$.
(ii) $\exists n_1 < n_2 < \cdots$ *such that* $p_{n_k} \to p$.

Proof.

(i) $\Rightarrow$ *(ii)* Set $n_0 = 0$. By (i), recurse $n_k > n_{k-1}$ with $p_{n_k} \in B_{1/k}(p)$; then $d(p_{n_k}, p) < 1/k$, so $p_{n_k} \to p$.
(ii) $\Rightarrow$ *(i)* If $p_{n_k} \to p$, then for any $r > 0$ and $N \in \mathbb{N}$, choose k with $n_k \geq N$ and $d(p_{n_k}, p) < r$.
Thus $p_{n_k} \in B_r(p)$, so (i) holds.

Structural Commentary: Frequent visits are exactly enough to extract a convergent subsequence.
Key Insight: Limits require all sufficiently late terms; limit points require arbitrarily late witnesses. □

Proposition 3.4.5. *Discrete-Metric Convergence*
Let X carry the discrete metric $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ if $x \neq y$.
For $(p_n) \subset X$ and $p \in X$, $p_n \to p$ if and only if $\exists N \in \mathbb{N}$ such that $n \geq N \implies p_n = p$.

Proof.

($\Rightarrow$) Suppose $p_n \to p$. For $\varepsilon = 1/2$, choose $N \in \mathbb{N}$ such that $n \geq N \implies d(p_n, p) < 1/2$.
Since d takes only the values $0, 1$, $d(p_n, p) = 0$, hence $p_n = p$ for all $n \geq N$.
($\Leftarrow$) Suppose $\exists N \in \mathbb{N}$ s.t. $n \geq N \implies p_n = p$. Then $\forall \varepsilon > 0$, $n \geq N$ gives $d(p_n, p) = 0 < \varepsilon$, so $p_n \to p$.

Structural Commentary: The discrete metric has no room for distinct points to approach one another.
Key Insight: In the discrete metric, $d(x, y) < 1 \iff x = y$. □

Example 3.4.6. *Coordinate Norm Metrics on $\mathbb{R}^k$*
By Proposition 1.1.21 and Theorem 1.1.13, $d_1(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_1$; $d_2(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_2$; $d_\infty(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_\infty$ are metrics on $\mathbb{R}^k$, and $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, $d_\infty(\mathbf{x}, \mathbf{y}) \leq d_2(\mathbf{x}, \mathbf{y}) \leq d_1(\mathbf{x}, \mathbf{y}) \leq \sqrt{k}\, d_2(\mathbf{x}, \mathbf{y})$; $d_2(\mathbf{x}, \mathbf{y}) \leq \sqrt{k}\, d_\infty(\mathbf{x}, \mathbf{y})$.
Hence d_1, d_2, d_∞ have the same convergent sequences.

Theorem 3.4.7. *Coordinatewise Convergence in $\mathbb{R}^k$*
Let $k \in \mathbb{N}$, $\mathbf{x}_n = (\alpha_{j,n})$, $\mathbf{x} = (\alpha_j)$, $\mathbf{y}_n = (\gamma_{j,n})$, $\mathbf{y} = (\gamma_j)$ in $\mathbb{R}^k$, and let $(\beta_n) \subset \mathbb{R}$, $\beta \in \mathbb{R}$.

(i) $\mathbf{x}_n \to \mathbf{x}$ *iff* $\alpha_{j,n} \to \alpha_j$ *for every* $1 \leq j \leq k$.
(ii) If $\mathbf{x}_n \to \mathbf{x}$ *and* $\mathbf{y}_n \to \mathbf{y}$, *then* $\mathbf{x}_n + \mathbf{y}_n \to \mathbf{x} + \mathbf{y}$.
(iii) If $\mathbf{x}_n \to \mathbf{x}$ *and* $\mathbf{y}_n \to \mathbf{y}$, *then* $\mathbf{x}_n \cdot \mathbf{y}_n \to \mathbf{x} \cdot \mathbf{y}$.
(iv) If $\beta_n \to \beta$ *and* $\mathbf{x}_n \to \mathbf{x}$, *then* $\beta_n \mathbf{x}_n \to \beta \mathbf{x}$.

Proof.

(i) ($\Rightarrow$) Since $|\alpha_{j,n} - \alpha_j| \leq \|\mathbf{x}_n - \mathbf{x}\|_2$ for each j, each coordinate converges.
(i) ($\Leftarrow$) For $\varepsilon > 0$, eventually $|\alpha_{j,n} - \alpha_j| < \varepsilon/\sqrt{k}$ for all j; hence $\|\mathbf{x}_n - \mathbf{x}\|_2^2 < k(\varepsilon^2/k) = \varepsilon^2$.
(ii) For each j, $\alpha_{j,n} + \gamma_{j,n} \to \alpha_j + \gamma_j$; apply (i).
(iii) Each product $\alpha_{j,n}\gamma_{j,n} \to \alpha_j\gamma_j$, hence $\sum_{j=1}^k \alpha_{j,n}\gamma_{j,n} \to \sum_{j=1}^k \alpha_j\gamma_j$.
(iv) For each j, $\beta_n\alpha_{j,n} \to \beta\alpha_j$; apply (i).

Structural Commentary: Euclidean convergence is exactly simultaneous coordinate convergence.
Key Insight: The norm packages finitely many coordinate estimates into one estimate. □

Definition 3.4.8. *Product Metrics*
For metric spaces (X, d_X) and (Y, d_Y), and $p = (x, y)$, $p' = (x', y')$ in $X \times Y$, define

$$d_E(p, p') := \sqrt{d_X(x, x')^2 + d_Y(y, y')^2},$$
$$d_{\max}(p, p') := \max\{d_X(x, x'), d_Y(y, y')\},$$
$$d_{\text{sum}}(p, p') := d_X(x, x') + d_Y(y, y')$$

These are the Euclidean, maximum, and sum product metrics; the sum metric is often simplest for estimates.

Proposition 3.4.9. *Comparing Product Metrics*
On $X \times Y$, $d_{\max} \leq d_E \leq d_{\text{sum}} \leq 2d_{\max}$. Hence the three product metrics have same convergent sequences.

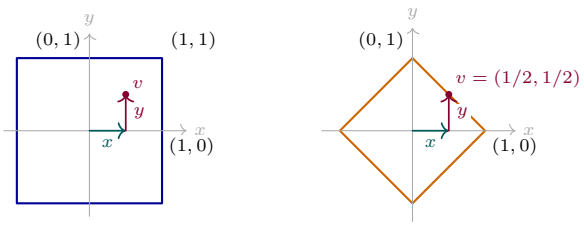

Figure 126: Unit balls in $\mathbb{R}^2$: $\max\{|x|, |y|\} < 1$ and $|x| + |y| < 1$.

Proof. Let $a = d_X(x, x')$ and $b = d_Y(y, y')$.
Then $\max\{a, b\} \le \sqrt{a^2 + b^2} \le a + b \le 2\max\{a, b\}$; convergence follows by squeezing.
Structural Commentary: The three product metrics differ by controlled constants.
Key Insight: Equivalent metric estimates preserve convergence. □

Definition 3.4.10. *Equivalent Metrics*
Let X be a set and let $\rho, \sigma : X \times X \to [0, \infty)$ be metrics.
They are *equivalent*, written $\rho \sim \sigma$, iff $\exists c_1, c_2 > 0$ with $\forall x, y \in X$, $c_1\sigma(x, y) \le \rho(x, y) \le c_2\sigma(x, y)$.

Proposition 3.4.11. *Equivalent Metrics Preserve Topological Tests*
Let X be a set, let $\rho, \sigma : X \times X \to [0, \infty)$ be metrics with $\rho \sim \sigma$, and let τ_ρ, τ_σ be the induced topologies.

(i) $\tau_\rho = \tau_\sigma$.
(ii) For every $E \subseteq X$, E is ρ-closed iff E is σ-closed.
(iii) For every sequence (x_n) in X and every $x \in X$, $\rho(x_n, x) \to 0$ iff $\sigma(x_n, x) \to 0$.

Proof. Choose $c_1, c_2 > 0$ such that $c_1\sigma(u, v) \le \rho(u, v) \le c_2\sigma(u, v)$ for all $u, v \in X$.
For $x \in X$ and $r > 0$, $B_\sigma(x, r/c_2) \subseteq B_\rho(x, r) \subseteq B_\sigma(x, r/c_1)$.

(i) If $U \in \tau_\rho$ and $x \in U$, choose $r > 0$ with $B_\rho(x, r) \subseteq U$; then $B_\sigma(x, r/c_2) \subseteq U$, so $U \in \tau_\sigma$.
If $U \in \tau_\sigma$ and $x \in U$, choose $s > 0$ with $B_\sigma(x, s) \subseteq U$; then $B_\rho(x, c_1 s) \subseteq U$, so $U \in \tau_\rho$.
(ii) Let $E \subseteq X$. By (i), $E^c \in \tau_\rho$ iff $E^c \in \tau_\sigma$; hence E is ρ-closed iff E is σ-closed.
(iii) For every $n \in \mathbb{N}$, $c_1\sigma(x_n, x) \le \rho(x_n, x) \le c_2\sigma(x_n, x)$; hence $\rho(x_n, x) \to 0$ iff $\sigma(x_n, x) \to 0$.

Structural Commentary: Ball inclusions transfer local tests between metrics.
Key Insight: Constant-comparison estimates let each metric ball contain a smaller ball for the other metric. □

Proposition 3.4.12. *Sequential Closure Criterion*
Let $E \subseteq X$, where X is a metric space, and let $p \in X$. Then $p \in \overline{E} \iff \exists (p_n)_{n \ge 1}$ with $p_n \in E$ $\forall n$, $p_n \to p$.

Proof.
($\Rightarrow$) If $p \in \overline{E}$, choose $p_n \in E \cap B_{1/n}(p)$ for $n \ge 1$ by Principle 6.1.2. Then $d(p_n, p) < 1/n$, so $p_n \to p$.
($\Leftarrow$) If $p_n \in E$ and $p_n \to p$, then every ball $B_r(p)$ contains some p_n.
Thus p is adherent to E by Definition 3.1.23, so $p \in \overline{E}$.

Structural Commentary: Closure can be read either by neighborhoods or by sequences.
Key Insight: Adherence supplies one witness at every scale. □

Theorem 3.4.13. *Closed Sets and Limit Points via Sequences*
Let X be a metric space and $E \subseteq X$.

(i) $p \in E^{\mathrm{acc}} \iff \exists (p_n) \subseteq E \setminus \{p\}$ *such that* $p_n \to p$.
(ii) E is closed $\iff$ every $(p_n) \subseteq E$ with $p_n \to p \in X$ satisfies $p \in E$.

Proof.
(i) ($\Rightarrow$) If $p \in E^{\mathrm{acc}}$, choose $p_n \in (E \setminus \{p\}) \cap B_{1/n}(p)$ for each n. Then $d(p_n, p) < 1/n$, so $p_n \to p$.
(i) ($\Leftarrow$) If $(p_n) \subseteq E \setminus \{p\}$ and $p_n \to p$, then every $B_r(p)$ contains some $p_n \ne p$. Hence $p \in E^{\mathrm{acc}}$.
(ii) ($\Rightarrow$) If E is closed, $(p_n) \subseteq E$, and $p_n \to p$, then Proposition 3.4.12 gives $p \in \overline{E} = E$.
(ii) ($\Leftarrow$) If every convergent sequence in E has limit in E, then Proposition 3.4.12 gives $\overline{E} \subseteq E$. QED.

Structural Commentary: In metric spaces, closedness can be tested by sequences.
Key Insight: Balls at radii $1/n$ convert local information into a sequence. □

Theorem 3.4.14. *Subsequential Limits Form a Closed Set*
For a sequence (p_n) in a metric space X, the set of all subsequential limits is closed.

Proof. Let E^* be the set of subsequential limits and let q be a limit point of E^*. Set $n_0 = 0$.
For each i, choose $x_i \in E^*$ with $d(x_i, q) < 2^{-i}$. Since x_i is a subsequential limit, choose $n_i > n_{i-1}$ such that $d(p_{n_i}, x_i) < 2^{-i}$. Then $d(p_{n_i}, q) < 2^{1-i}$, so $p_{n_i} \to q$, and $q \in E^*$.
Structural Commentary: A limit of subsequential limits is reached by diagonalizing through those subsequences.
Key Insight: Choose later and closer terms at the same time. □

Method 3.4.15. *Metric Proof Patterns*

(i) To prove G is open, fix $x \in G$ and find $r > 0$ with $B_r(x) \subseteq G$.
(ii) To prove F is closed, show $F^{\mathrm{acc}} \subseteq F$, or equivalently show F^c is open.
(iii) To prove $C = \overline{E}$, show $E \subseteq C$, C is closed, and $C \subseteq F$ for every closed $F \supseteq E$.
(iv) To prove $p \in \overline{E}$, show $\mathrm{dist}(p, E) = 0$, or build a sequence in E converging to p.

4 Compactness and Completeness

4.1 Compactness, Completeness, and Total Boundedness

Compactness extracts finite or convergent structure from infinite behavior; completeness asks whether Cauchy approximations converge internally.

Definition 4.1.1. *Open Covers, Compact Sets, and Finite Intersection Property*
Let (X, d) be a metric space.

(i) For $E \subseteq X$, an *open cover* of E is a family $(G_\alpha)_{\alpha\in A}$ such that each $G_\alpha \subseteq X$ is open and $E \subseteq \bigcup_{\alpha\in A} G_\alpha$.
(ii) A set $K \subseteq X$ is *compact* iff every open cover $(G_\alpha)_{\alpha\in A}$ of K admits $\alpha_1, \dots, \alpha_n \in A$ with $K \subseteq \bigcup_{j=1}^{n} G_{\alpha_j}$.
(iii) Family $\mathcal{F} \subseteq \mathcal{P}(X)$ has *finite intersection property* iff $\bigcap_{F\in\mathcal{F}_0} F \neq \emptyset$ $\forall$ finite nonempty subfamily $\mathcal{F}_0 \subseteq \mathcal{F}$.

Proposition 4.1.2. *Closed-Set Compactness Test*
Let (X, d) be a metric space. Then X is compact iff every nonempty family $\mathcal{F}$ of closed subsets of X with the finite intersection property satisfies $\bigcap_{F\in\mathcal{F}} F \neq \emptyset$.

Proof.
(⇒) Assume X is compact. Let $\emptyset \neq \mathcal{F} \subseteq \mathcal{P}(X)$ consist of closed sets and have the finite intersection property. If $\bigcap_{F\in\mathcal{F}} F = \emptyset$, then $\{X \setminus F : F \in \mathcal{F}\}$ is an open cover of X. Compactness gives $F_1, \dots, F_n \in \mathcal{F}$ with $X = \bigcup_{j=1}^{n}(X \setminus F_j)$; hence $\bigcap_{j=1}^{n} F_j = \emptyset$, contradicting the finite intersection property.
(⇐) Assume every nonempty closed family with the finite intersection property has nonempty intersection. Let $(G_\alpha)_{\alpha\in A}$ be an open cover of X. If no finite subcover exists, then $\mathcal{F} := \{X \setminus G_\alpha : \alpha \in A\}$ is a nonempty family of closed subsets of X with the finite intersection property. Thus $\exists x \in \bigcap_{\alpha\in A}(X \setminus G_\alpha)$, contradicting $X \subseteq \bigcup_{\alpha\in A} G_\alpha$. Hence every open cover has a finite subcover, so X is compact.

Structural Commentary: Complements turn finite subcovers into finite intersections.
Key Insight: Compactness can be read dually through closed sets. □

Figure 127: Compactness extracts a finite subcover.

Example 4.1.3. *An Open Cover Without a Finite Subcover*
For $x \in (0, 1)$, set $O_x := (x/2, 1)$. These sets cover $(0, 1)$, since $y \in O_x$ whenever $0 < x < y < 1$.
No finite subcollection covers $(0, 1)$: if $x' = \min\{x_1, \dots, x_n\}$, then $(0, x'/2]$ is missed by $O_{x_1} \cup \cdots \cup O_{x_n}$.

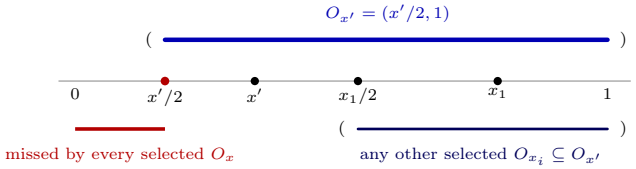

Figure 128: For finite choice, $x' = \min_i x_i$ makes $O_{x'}$ largest selected interval, but $(0, x'/2]$ remains uncovered.

Theorem 4.1.4. *Compactness Is Independent of the Ambient Space*
Suppose $K \subseteq Y \subseteq X$, where X is a metric space.
Then K is compact relative to X if and only if K is compact relative to Y.

Proof.
(⇒) Let $\{V_\alpha\}_{\alpha\in A}$ be a relative-open cover of K in Y.
By Theorem 3.2.2, write $V_\alpha = Y \cap G_\alpha$ with G_α open in X. Since $\{G_\alpha\}_{\alpha\in A}$ covers K, compactness in X gives a finite $F \subseteq A$ such that $K \subseteq \bigcup_{\alpha\in F} G_\alpha$. As $K \subseteq Y$, $K \subseteq \bigcup_{\alpha\in F} V_\alpha$.
(⇐) Let $\{G_\alpha\}_{\alpha\in A}$ be an open cover of K in X. Then $\{Y \cap G_\alpha\}_{\alpha\in A}$ is a relative-open cover of K in Y.
Compactness in Y gives a finite $F \subseteq A$ such that $K \subseteq \bigcup_{\alpha\in F}(Y \cap G_\alpha) \subseteq \bigcup_{\alpha\in F} G_\alpha$.

Structural Commentary: Relative covers are ambient covers intersected with Y.
Key Insight: Openness depends on the ambient space, but compactness does not. □

Theorem 4.1.5. *Compact Sets Are Closed*
Every compact subset of a metric space is closed.

Proof. Let K be compact in X. To prove K^c is open.
Fix $p \in K^c$. For each $q \in K$, put $V_q := B_{\frac{1}{2}d(p,q)}(p)$, $W_q := B_{\frac{1}{2}d(p,q)}(q)$.
The sets W_q form an open cover of K, so compactness gives $q_1, \dots, q_n \in K$ with $K \subseteq W_{q_1} \cup \cdots \cup W_{q_n}$.
Let $V := V_{q_1} \cap \cdots \cap V_{q_n}$, which is open by Theorem 3.1.19.
No point of V lies in any W_{q_i}, by the triangle inequality. Hence $V \cap K = \emptyset$, so p is an interior point of K^c.
Structural Commentary: Compactness makes the separation finite.
Key Insight: A finite subcover lets local point-separations become one open neighborhood of the complement. □

Theorem 4.1.6. *Closed Subsets of Compact Sets Are Compact*
Suppose $F \subseteq K \subseteq X$, where F is closed in X and K is compact. Then F is compact.

Proof. Let $\{V_\alpha\}_{\alpha\in A}$ cover F. Since F^c is open, $\{V_\alpha\}_{\alpha\in A} \cup \{F^c\}$ covers K.
Thus $K \subseteq F^c \cup \bigcup_{\alpha\in F_0} V_\alpha$ for some finite $F_0 \subseteq A$, so $F \subseteq \bigcup_{\alpha\in F_0} V_\alpha$.
Structural Commentary: The open complement of a closed set covers exactly the part of K not under discussion.
Key Insight: Cover K, then discard the complement piece. □

Corollary 4.1.7. *Closed Intersections with Compact Sets*
If F is closed in X and K is compact in X, then $F \cap K$ is compact.

Proof. By Theorem 4.1.5, K is closed in X. Since F is closed, Theorem 3.1.19 gives that $F \cap K$ is closed in X. Also $F \cap K \subseteq K$, so Theorem 4.1.6 implies $F \cap K$ is compact.
Structural Commentary: The intersection is closed inside a compact set.
Key Insight: Closed subsets of compact sets remain compact. □

Theorem 4.1.8. *Finite Intersection Property for Compact Sets*
Let $\{K_\alpha\}_{\alpha\in A}$ be compact subsets of a metric space X, with $A \neq \emptyset$.
If $\bigcap_{\alpha\in F} K_\alpha \neq \emptyset$ for every finite nonempty $F \subseteq A$, then $\bigcap_{\alpha\in A} K_\alpha \neq \emptyset$.

Proof. Choose $\alpha_0 \in A$ and suppose $\bigcap_{\alpha\in A} K_\alpha = \emptyset$. Then $K_{\alpha_0} \subseteq \bigcup_{\alpha\in A} K_\alpha^c$. Since each K_α^c is open by Theorem 4.1.5, compactness of K_{α_0} gives a finite $F \subseteq A$ with $K_{\alpha_0} \subseteq \bigcup_{\alpha\in F} K_\alpha^c$. Hence $\bigcap_{\alpha\in F\cup\{\alpha_0\}} K_\alpha = \emptyset$, a contradiction.
Structural Commentary: Complements turn failure into a finite cover.
Key Insight: For compact sets, failure of intersection is already witnessed finitely. □

Corollary 4.1.9. *Nested Nonempty Compact Sets Intersect*
If $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ is a sequence of nonempty compact sets, then $\bigcap_{n=1}^\infty K_n \neq \emptyset$.

Proof. Let $\emptyset \neq F \subseteq \mathbb{N}$ be finite and put $m := \max F$. Since the sequence is nested, $\bigcap_{n\in F} K_n = K_m \neq \emptyset$.
Thus $\{K_n\}_{n\in\mathbb{N}}$ has the finite intersection property, so Theorem 4.1.8 gives $\bigcap_{n=1}^\infty K_n \neq \emptyset$.
Structural Commentary: Nestedness reduces finite intersections to one term.
Key Insight: A nested compact sequence has the finite intersection property. □

Theorem 4.1.10. *Infinite Subsets of Compact Sets Have Limit Points*
Let $K \subseteq X$ be compact in a metric space X. If $E \subseteq K$ is infinite, then $E^{\text{acc}} \cap K \neq \emptyset$.

Proof. Suppose $E^{\text{acc}} \cap K = \emptyset$. $\forall q \in K$, choose $r_q > 0$ such that $B_{r_q}(q) \cap (E \setminus \{q\}) = \emptyset$, so $\#(B_{r_q}(q) \cap E) \leq 1$.
Since $\{B_{r_q}(q) : q \in K\}$ covers K, compactness gives $q_1, \ldots, q_n \in K$ with $K \subseteq \bigcup_{i=1}^n B_{r_{q_i}}(q_i)$.
As $E \subseteq K$, $E \subseteq \bigcup_{i=1}^n B_{r_{q_i}}(q_i)$, finite union of sets meeting E in at most one point. Thus E is finite, contradiction.
Structural Commentary: Compactness makes the local "at most one point" obstruction finite.
Key Insight: Infinite subsets of compact sets must accumulate somewhere inside the compact set. □

Definition 4.1.11. *Sequential Compactness*
A metric space is *sequentially compact* if every sequence has a convergent subsequence.

Theorem 4.1.12. *Compactness Produces Convergent Subsequences* *If (p_n) is a sequence in a compact metric space X, then $p_{n_i} \to p$ for some subsequence and some $p \in X$.*

Proof. Let $E = \{p_n : n \in \mathbb{N}\}$. If E is finite, choose a value p attained infinitely often; the corresponding constant subsequence converges. If E is infinite, Theorem 4.1.10 gives a limit point $p \in X$ of E. By Theorem 3.1.14, choose $n_i > n_{i-1}$ with $d(p_{n_i}, p) < 1/i$. Then $p_{n_i} \to p$. *Structural Commentary*: Compactness turns arbitrary infinite behavior into an accumulating subsequence.
Key Insight: A compact sequence range either repeats or accumulates. □

Definition 4.1.13. *Cauchy Sequence*
A sequence (p_n) in a metric space X is *Cauchy* if $\forall \varepsilon > 0$ $\exists$ integer N s.t. $n \geq N,\ m \geq N \implies d(p_n, p_m) < \varepsilon$.

Definition 4.1.14. *Diameter*
Let (X, d) be a metric space, $\emptyset \neq E \subseteq X$. The *diameter* of E is $\operatorname{diam} E := \sup_{\overline{\mathbb{R}}}\{d(p,q) : p, q \in E\} \in [0, +\infty]$.

Proposition 4.1.15. *Basic Diameter Properties*
Let $\emptyset \neq E, F \subseteq X$.

(i) E is bounded $\iff \operatorname{diam} E < \infty \iff \exists q \in X, \exists M > 0$ such that $E \subseteq B_M(q)$.
(ii) $\operatorname{diam} E = 0 \iff |E| = 1$.
(iii) $E \subseteq F \implies \operatorname{diam} E \leq \operatorname{diam} F$.
(iv) E, F bounded $\implies E \cup F$ bounded.

Proof.

(i) If $E \subseteq B_M(q)$, then $d(x,y) \leq d(x,q) + d(q,y) < 2M$ for $x, y \in E$, so $\operatorname{diam} E < \infty$.
If $\operatorname{diam} E < \infty$ and $q \in E$, then $E \subseteq B_{\operatorname{diam} E+1}(q)$.
(ii) $\operatorname{diam} E = 0 \iff d(x,y) = 0$ for all $x, y \in E \iff x = y$ for all $x, y \in E$; since $E \neq \emptyset$, this is $|E| = 1$.
(iii) $\{d(x,y) : x, y \in E\} \subseteq \{d(x,y) : x, y \in F\}$, so monotonicity of sup gives $\operatorname{diam} E \leq \operatorname{diam} F$.

(iv) Choose $e \in E$, $f \in F$, and set $D := \operatorname{diam} E + d(e,f) + \operatorname{diam} F < \infty$. For $x, y \in E \cup F$, either x, y lie in the same set, or after swapping, $x \in E$, $y \in F$, and $d(x,y) \le d(x,e) + d(e,f) + d(f,y) \le D$.
Thus $\operatorname{diam}(E \cup F) < \infty$, so $E \cup F$ is bounded by (i).

Structural Commentary: Diameter is a single-number form of boundedness for nonempty sets.
Key Insight: Cross-distances between two bounded sets are controlled by one bridge distance. □

Proposition 4.1.16. *Cauchy Sequences and Tail Diameters*
For (p_n) in a metric space X, set $E_N := \{p_j : j \ge N\}$. Then (p_n) is Cauchy if and only if $\operatorname{diam} E_N \to 0$.

Proof.

(⇒) Given $\varepsilon > 0$, choose N_0 so that $m, n \ge N_0$ implies $d(p_m, p_n) < \varepsilon/2$.
If $N \ge N_0$, then every pair in E_N has distance $< \varepsilon/2$, so $\operatorname{diam} E_N < \varepsilon$.
(⇐) Given $\varepsilon > 0$, choose N_0 with $\operatorname{diam} E_{N_0} < \varepsilon$.
If $m, n \ge N_0$, then $p_m, p_n \in E_{N_0}$, hence $d(p_m, p_n) \le \operatorname{diam} E_{N_0} < \varepsilon$.

Structural Commentary: Cauchy behavior is shrinking diameter of sequence tails.
Key Insight: The limit is not named; only mutual closeness is measured. □

Definition 4.1.17. *Contracting Sequence of Sets*
Let (X, d) be a metric space and let $(F_n)_{n \ge 1}$ satisfy $\emptyset \ne F_n \subseteq X$.

(i) (F_n) is *descending* iff $\forall n \ge 1$, $F_{n+1} \subseteq F_n$.
(ii) (F_n) is *contracting* iff it is descending and $\operatorname{diam} F_n \to 0$.

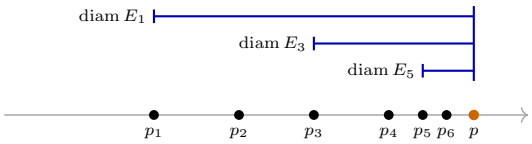

Figure 129: Tail diameters $\operatorname{diam} E_N$ shrink toward 0.

Theorem 4.1.18. *Nested Compact Sets with Shrinking Diameter*
Let X be a metric space.

(i) If $\emptyset \ne E \subseteq X$, then $\operatorname{diam} \overline{E} = \operatorname{diam} E$.
(ii) If $K_1 \supseteq K_2 \supseteq \cdots$ are nonempty compact subsets of X and $\operatorname{diam} K_n \to 0$, then $\left|\bigcap_{n=1}^{\infty} K_n\right| = 1$.

Proof.

(i) Since $E \subseteq \overline{E}$, $\operatorname{diam} E \le \operatorname{diam} \overline{E}$. For $p, q \in \overline{E}$ and $\varepsilon > 0$, choose $p', q' \in E$ with $d(p,p'), d(q,q') < \varepsilon$; then $d(p,q) \le \operatorname{diam} E + 2\varepsilon$. Hence $\operatorname{diam} \overline{E} \le \operatorname{diam} E$.
(ii) Set $I := \bigcap_{n \ge 1} K_n$. Corollary 4.1.9 gives $I \ne \emptyset$. If $x, y \in I$, then $d(x,y) \le \operatorname{diam} K_n$ for all n, so $d(x,y) = 0$ as $\operatorname{diam} K_n \to 0$. Thus $x = y$, hence $|I| = 1$.

Structural Commentary: Closure preserves diam; $K_n \downarrow$ with $\operatorname{diam} K_n \to 0$ forces a singleton.
Key Insight: Compactness gives existence; shrinking diameter gives uniqueness. □

Theorem 4.1.19. *Cauchy Criterion and Completeness Results*

(i) In any metric space, every convergent sequence is Cauchy.
(ii) If X is a compact metric space, then every Cauchy sequence in X converges to a point of X.

Proof.

(i) If $p_n \to p$, choose N with $d(p_n, p) < \varepsilon/2$ for $n \ge N$. For $m, n \ge N$, $d(p_n, p_m) < \varepsilon$.
(ii) Let $E_N := \{p_j : j \ge N\}$ and $F_N := \overline{E_N}$. Then $\operatorname{diam} F_N \to 0$, the F_N are compact and nested, and Theorem 4.1.18 gives $\bigcap_N F_N = \{p\}$. If $\operatorname{diam} F_N < \varepsilon$, then $p, p_n \in F_N$ for $n \ge N$, so $d(p_n, p) < \varepsilon$.

Structural Commentary: Cauchy controls $\operatorname{diam} E_N$; compactness turns $\bigcap_N F_N$ into the limit.
Key Insight: Compactness turns mutual tail closeness into an internal limit. □

Lemma 4.1.20. *A Cauchy Sequence with a Convergent Subsequence Converges*
Let (p_n) be a Cauchy sequence in a metric space X. If some subsequence (p_{n_j}) converges to $p \in X$, then $p_n \to p$.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m, n \ge N_0$ implies $d(p_m, p_n) < \varepsilon/2$.
Choose j with $n_j \ge N_0$ and $d(p_{n_j}, p) < \varepsilon/2$. Then for every $n \ge N_0$, $d(p_n, p) \le d(p_n, p_{n_j}) + d(p_{n_j}, p) < \varepsilon$.
Structural Commentary: The Cauchy tail follows any convergent subsequence to the same limit.
Key Insight: Mutual closeness plus one convergent thread forces the whole tail to converge. □

Proposition 4.1.21. *Boundedness and Distances of Cauchy Sequences*
Let $(p_n), (q_n)$ be Cauchy sequences in a metric space X.

(i) Cauchy boundedness: the range of $n \mapsto p_n$, $\operatorname{ran}(p_n) := \{p_n : n \in \mathbb{N}\}$ is bounded.
(ii) distance convergence: the sequence $n \mapsto d(p_n, q_n)$ converges in $\mathbb{R}$.
(iii) nonconverse: there exists a bounded sequence in $\mathbb{R}$ that is not Cauchy.

Proof.

(i) Choose N with $m, n \ge N \Longrightarrow d(p_m, p_n) < 1$.
Then $\{p_n : n \ge N\} \subseteq B_1(p_N)$, and $\{p_1, \ldots, p_{N-1}\}$ is finite, so $\operatorname{ran}(p_n)$ is bounded.
(ii) For $m, n \in \mathbb{N}$, $|d(p_n, q_n) - d(p_m, q_m)| \le d(p_n, p_m) + d(q_n, q_m)$. Hence $(d(p_n, q_n))$ Cauchy in $\mathbb{R}$, converges.

(iii) Let $a_n := (-1)^n$. Then $|a_n| \leq 1$ for all n, but with $\varepsilon = 1$, for every N there exist $m, n \geq N$ with $|a_m - a_n| = 2$; hence (a_n) is bounded but not Cauchy.

Structural Commentary: Cauchy tails are bounded; Cauchy-pair distances converge.
Key Insight: The four-point triangle estimate controls distance sequences. □

Definition 4.1.22. *Complete Metric Space*
A metric space X is *complete* if every Cauchy sequence in X converges to a point of X.
Compact metric spaces are complete by Theorem 4.1.19; $\mathbb{R}^k$ is complete by Corollary 4.2.8.

Proposition 4.1.23. *Completeness and Closed Subspaces*
Let (X, d) be a metric space, $E \subseteq X$, and $d_E := d|_{E\times E}$.

(i) If (E, d_E) is complete, then E is closed in X, equivalently $E = \overline{E}^X$.
(ii) If (X, d) is complete, then (E, d_E) is complete if and only if E is closed in X, equivalently $E = \overline{E}^X$.

Proof.

(i) Assume (E, d_E) complete. Let $p \in \overline{E}^X$. Choose $(p_n) \subseteq E$ with $d(p_n, p) < 1/n$.
Then $p_n \to p$ in X, hence (p_n) is Cauchy in (E, d_E). Thus $p_n \to q \in E$ in (E, d_E), hence also in X.
Uniqueness in X gives $p = q \in E$. Therefore $\overline{E}^X \subseteq E$, so $E = \overline{E}^X$.

(ii) Assume (X, d) complete. By (i), (E, d_E) complete $\Rightarrow E = \overline{E}^X$.
Conversely, if $E = \overline{E}^X$ and $(p_n) \subseteq E$ is Cauchy in (E, d_E), then (p_n) is Cauchy in X, so $p_n \to p \in X$.
Since $p \in \overline{E}^X = E$, every Cauchy sequence in (E, d_E) converges in (E, d_E).

Structural Commentary: Completeness gives the limit in X; closedness keeps it in E.
Key Insight: In a complete ambient space, E is complete exactly when $\overline{E} = E$. □

Theorem 4.1.24. *Cantor Intersection Theorem*
Let (X, d) be a metric space. Then X is complete iff every sequence $(F_n)_{n\geq 1}$ with $\emptyset \neq F_n \subseteq X$ closed, $F_{n+1} \subseteq F_n$, and $\operatorname{diam} F_n \to 0$ satisfies $\bigcap_{n=1}^{\infty} F_n = \{x\}$ for some $x \in X$.

Proof.

(⇒) Assume X is complete, and let (F_n) satisfy the stated hypotheses. Choose $x_n \in F_n$. For $m, n \geq N$, descendingness gives $x_m, x_n \in F_N$, so $d(x_m, x_n) \leq \operatorname{diam} F_N$; hence (x_n) is Cauchy. Let $x_n \to x \in X$. For each fixed N, the tail $(x_n)_{n\geq N}$ lies in closed F_N, so $x \in F_N$. Thus $x \in \bigcap_n F_n$. If $y \in \bigcap_n F_n$, then $d(x, y) \leq \operatorname{diam} F_N$ for all N, so $x = y$.

(⇐) Assume the stated intersection property. Let (x_n) be Cauchy, set $E_N := \{x_n : n \geq N\}$, and set $F_N := \overline{E_N}$. Then F_N is nonempty, closed, and descending. By Proposition 4.1.16 and Theorem 4.1.18(i), $\operatorname{diam} F_N = \operatorname{diam} E_N \to 0$, so $\bigcap_N F_N = \{x\}$ for some $x \in X$. Given $\varepsilon > 0$, choose N with $\operatorname{diam} F_N < \varepsilon$. If $n \geq N$, then $x_n, x \in F_N$, so $d(x_n, x) < \varepsilon$. Hence $x_n \to x$, and X is complete.

Structural Commentary: Contracting closed nests have one point exactly in complete spaces.
Key Insight: Cauchy tails are the canonical contracting closed sets. □

We now turn from contracting closed families to the finite-resolution side of compact metric spaces.

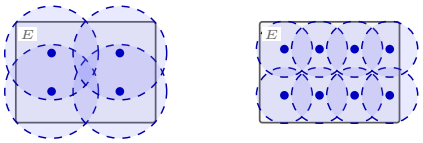

Figure 130: Finite ε- and δ-nets cover E, with $0 < \delta < \varepsilon$.

Definition 4.1.25. *Total Boundedness*
Let (X, d) be a metric space and let $E \subseteq X$.

(i) E is *totally bounded* iff $\forall \varepsilon > 0$, $\exists x_1, \ldots, x_N \in X$ with $E \subseteq \bigcup_{i=1}^{N} B_\varepsilon(x_i)$.
(ii) Such a finite family $\{B_\varepsilon(x_i)\}_{i=1}^{N}$ is a finite ε-*net* for E. If $E \neq \emptyset$, centers may equivalently be chosen in E.
(iii) The metric space X is *totally bounded* iff $E = X$ satisfies (i).

Theorem 4.1.26. *Total Boundedness Implies Boundedness*
Let X be a metric space. If X is totally bounded, then X is bounded.

Proof. If $X = \emptyset$, this is trivial. Otherwise cover X by finitely many 1-balls $B_1(x_i)$ and fix x_1.
Then $d(x, x_1) < 1 + \max_i d(x_i, x_1)$ for every $x \in X$.
Structural Commentary: One finite cover at scale 1 gives one global bound.
Key Insight: Total boundedness is boundedness with finite control at every scale. □

Example 4.1.27. *Bounded Need Not Mean Totally Bounded*
In $\ell^2(\mathbb{F})$, let $B := \{\mathbf{x} \in \ell^2 : \|\mathbf{x}\|_2 \leq 1\}$. Then B is bounded, since $B \subseteq B_2(0)$. It is not totally bounded: the standard unit vectors $e_n \in B$ satisfy $\|e_n - e_m\|_2 = \sqrt{2}$ for $n \neq m$. If $B \subseteq \bigcup_{i=1}^{N} B_{1/2}(x_i)$, then two distinct e_n, e_m lie in the same $B_{1/2}(x_i)$, so $\|e_n - e_m\|_2 < 1$, contradicting $\|e_n - e_m\|_2 = \sqrt{2}$.
Structural Commentary: Bounded balls can contain infinite separated sets.
Key Insight: Total boundedness forbids an infinite separated family at a fixed scale.

Definition 4.1.28. *Lebesgue Number*
Let (X, d) be a metric space and let $\mathcal{U}$ be an open cover of X.
A number $\delta > 0$ is a *Lebesgue number* for $\mathcal{U}$ iff $\forall x \in X$, $\exists U \in \mathcal{U}$ such that $B_\delta(x) \subseteq U$.

Lemma 4.1.29. *Lebesgue Covering Lemma*
Let (X, d) be a metric space.

(i) If X is sequentially compact, then every open cover of X has a Lebesgue number.
(ii) If X is compact, then every open cover of X has a Lebesgue number.

Proof.

(i) Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an open cover of sequentially compact X. If $\mathcal{U}$ has no Lebesgue number, then $\forall n \geq 1$, $\exists x_n \in X$ such that $B_{1/n}(x_n) \nsubseteq U_\alpha$ for all $\alpha \in A$. Choose a subsequence $x_{n_k} \to x \in X$. Since $\mathcal{U}$ covers X, choose α_0 and $r > 0$ with $x \in U_{\alpha_0}$ and $B_r(x) \subseteq U_{\alpha_0}$. For large k, $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$; hence $y \in B_{1/n_k}(x_{n_k})$ gives $d(y, x) < r$, so $B_{1/n_k}(x_{n_k}) \subseteq U_{\alpha_0}$, a contradiction.
(ii) If X is compact, then Theorem 4.1.12 gives sequential compactness.
(i) gives a Lebesgue number for every open cover of X.

Structural Commentary: Convergent subsequences force uniform cover radii.
Key Insight: Small balls near a convergent subsequence eventually stay in one cover set. □

Theorem 4.1.30. *Compactness as Complete Finite Resolution*
Let (X, d) be a metric space. The following are equivalent:

(i) X is compact;
(ii) X is complete and totally bounded;
(iii) X is sequentially compact.

Proof.

(i) $\Rightarrow$ *(ii)* Let (p_k) be Cauchy in X. By Theorem 4.1.19, (p_k) converges, so X is complete. For $\varepsilon > 0$, compactness applied to the cover $\{B_\varepsilon(x) : x \in X\}$ gives a finite ε-net, so X is totally bounded.
(ii) $\Rightarrow$ *(iii)* Let (x_n) be a sequence in X. Using finite $1/j$-ball covers, recursively choose nested infinite subsequences so that the jth subsequence lies in some ball $B_{1/j}(a_j)$. Let y_j be the jth term of the jth subsequence. If $k, \ell \geq j$, nestedness gives $y_k, y_\ell \in B_{1/j}(a_j)$, hence $d(y_k, y_\ell) < 2/j$. Thus (y_j) is Cauchy; completeness gives $y_j \to y \in X$, so (x_n) has a convergent subsequence.
(iii) $\Rightarrow$ *(ii)* If X is not totally bounded, then for some $\varepsilon > 0$ no finite ε-balls cover X. Choose $x_1 \in X$ and inductively $x_{n+1} \notin \bigcup_{j=1}^{n} B_\varepsilon(x_j)$. Then $d(x_n, x_j) \geq \varepsilon$ for $j < n$, so no subsequence is Cauchy or convergent, contradicting sequential compactness. If (p_n) is Cauchy, sequential compactness gives a convergent subsequence, and Lemma 4.1.20 gives convergence of (p_n). Hence X is complete.
(ii) $\Rightarrow$ *(i)* By *(ii)* $\Rightarrow$ *(iii)*, X is sequentially compact. Let $(U_\alpha)_{\alpha \in A}$ be an open cover of X, and let $\delta > 0$ be a Lebesgue number from Lemma 4.1.29. Choose a finite $\delta/2$-net $\{z_1, \ldots, z_m\}$ for X. For each i, choose α_i with $B_\delta(z_i) \subseteq U_{\alpha_i}$. Since every $x \in X$ lies in some $B_{\delta/2}(z_i)$, the finite family $U_{\alpha_1}, \ldots, U_{\alpha_m}$ covers X.

Structural Commentary: Compactness is complete finite-scale control.
Key Insight: Total boundedness gives Cauchy subsequences; completeness gives limits. □

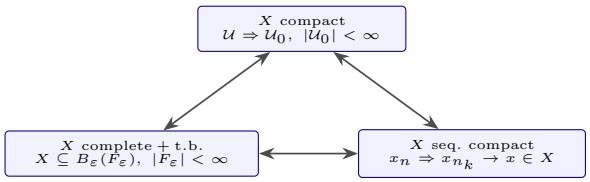

Figure 131: Equivalent compactness tests in a metric space X: finite subcovers for open covers, finite ε-nets with Cauchy limits in X, and convergent subsequences for every sequence.

Proposition 4.1.31. *Compact Metric Spaces Are Separable*
Let (X, d) be a compact metric space. Then $\exists$ countable $D \subseteq X$ such that $\overline{D}^X = X$.

Proof. By Theorem 4.1.30, X is totally bounded.
$\forall n \in \mathbb{N}$, choose finite $F_n \subseteq X$ such that $X \subseteq \bigcup_{a \in F_n} B_{1/n}(a)$, set $D := \bigcup_{n=1}^{\infty} F_n$. Then D is countable.
Let $x \in X$ and $r > 0$. Choose $n \in \mathbb{N}$ with $1/n < r$.
Since F_n is a $1/n$-net for X, $\exists a \in F_n$ such that $d(x, a) < 1/n < r$.
Thus $a \in B_r(x) \cap D$, so every X-open ball around x meets D. Hence $x \in \overline{D}^X$, and therefore $\overline{D}^X = X$.
Structural Commentary: Compactness gives finite nets at every scale; their countable union is dense.
Key Insight: Total boundedness converts compactness into countably many finite approximations. □

4.2 Heine–Borel and Euclidean Compactness

Theorem 4.2.1. *Nested Closed Intervals Intersect*
Let $(I_n)_{n \in \mathbb{N}}$ be nonempty compact intervals in $\mathbb{R}$, with $I_n = [a_n, b_n]$, such that $I_{n+1} \subseteq I_n$ $\forall n$. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Set $A := \{a_n : n \in \mathbb{N}\}$. Fix $m \in \mathbb{N}$. If $n \geq m$, then $a_n \in I_n \subseteq I_m$, so $a_n \leq b_m$.
If $n < m$, then $b_m \in I_m \subseteq I_n$, so $a_n \leq b_m$. Hence b_m is an upper bound for A.
Let $x := \sup A$. Since $a_m \in A$, $a_m \leq x$; since b_m is an upper bound for A, $x \leq b_m$.
Hence $x \in I_m$ for every m, and therefore $x \in \bigcap_{n=1}^{\infty} I_n$.
Structural Commentary: Nested endpoints form a LUB trap.
Key Insight: $\sup_n a_n$ lies in every interval. □

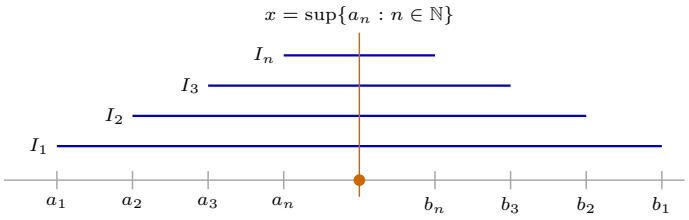

Figure 132: For nested closed intervals $[a_n, b_n]$, supremum of left endpoints lies between every a_n and b_n.

Theorem 4.2.2. *Nested k-Cells Intersect*
Let $k \in \mathbb{N}$, and for each $n \in \mathbb{N}$ let $I_n = \prod_{j=1}^{k} [a_{n,j}, b_{n,j}] \subseteq \mathbb{R}^k$ be a nonempty k-cell. Assume $I_{n+1} \subseteq I_n$ for every n. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. For each j, the jth coordinate projection of $I_{n+1} \subseteq I_n$ gives $[a_{n+1,j}, b_{n+1,j}] \subseteq [a_{n,j}, b_{n,j}]$.
By Theorem 4.2.1, choose $x_j^* \in \bigcap_{n=1}^{\infty} [a_{n,j}, b_{n,j}]$. Then $\mathbf{x}^* := (x_1^*, \ldots, x_k^*)$ satisfies $\mathbf{x}^* \in I_n$ $\forall n$, so $\mathbf{x}^* \in \bigcap_{n=1}^{\infty} I_n$.
Structural Commentary: Cells reduce coordinatewise to intervals.
Key Insight: Pick one persistent coordinate per axis. □

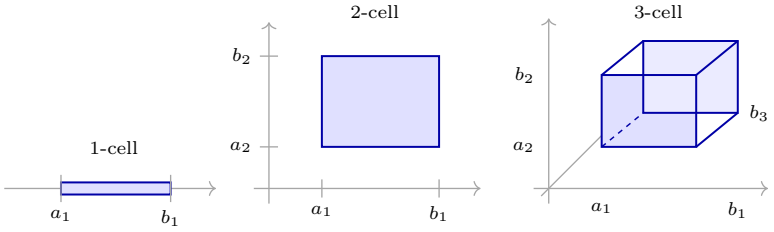

Figure 133: A 1-cell is a closed interval, a 2-cell is a coordinate rectangle, and a 3-cell is a coordinate box.

Theorem 4.2.3. *k-Cells Are Compact*
Let $k \in \mathbb{N}$, and let $I = \prod_{j=1}^{k} [a_j, b_j] \subseteq \mathbb{R}^k$ be a nonempty k-cell. Then I is compact.

Proof. Set $\delta := \left(\sum_{j=1}^{k} (b_j - a_j)^2\right)^{1/2} = \operatorname{diam} I$. Suppose open cover $\mathcal{G} = \{G_\alpha\}_{\alpha \in A}$ of I has no finite subcover.
Bisect each coordinate interval of I. If every resulting subcell had a finite $\mathcal{G}$-subcover, their union would finitely cover I, so choose a subcell I_1 with no finite $\mathcal{G}$-subcover. Repeating inside I_{n-1} gives nested k-cells $I \supseteq I_1 \supseteq I_2 \supseteq \cdots$ such that I_n has no finite $\mathcal{G}$-subcover and $\operatorname{diam} I_n \leq 2^{-n}\delta$.
By Theorem 4.2.2, choose $\mathbf{x}^* \in \bigcap_{n=1}^{\infty} I_n$. Since $\mathcal{G}$ covers I, choose α with $\mathbf{x}^* \in G_\alpha$. Openness gives $r > 0$ such that $B_r(\mathbf{x}^*) \subseteq G_\alpha$. Choose n with $2^{-n}\delta < r$. Since $\mathbf{x}^* \in I_n$ and $\operatorname{diam} I_n < r$, one has $I_n \subseteq B_r(\mathbf{x}^*) \subseteq G_\alpha$, contradicting the choice of I_n. Thus every open cover of I has a finite subcover.
Structural Commentary: Bisection gives shrinking nested cells.
Key Insight: One open set eventually contains the trapped cell. □

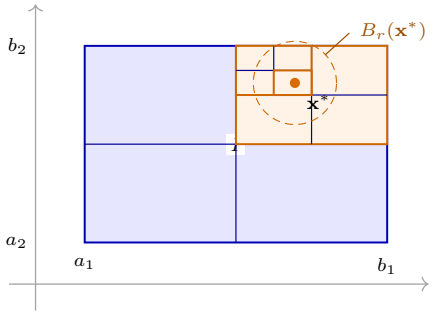

Figure 134: Repeated bisection traps a point.

Lemma 4.2.4. *Bounded Euclidean Sets Fit in k-Cells*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$ be bounded. Then there exists a k-cell $I \subseteq \mathbb{R}^k$ such that $E \subseteq I$.

Proof. If $E = \emptyset$, take $I = \prod_{j=1}^{k} [0, 0]$. Assume $E \neq \emptyset$.
Choose $\mathbf{q} = (q_1, \ldots, q_k) \in \mathbb{R}^k$ and $M > 0$ such that $E \subseteq B_M(\mathbf{q})$. Set $I := \prod_{j=1}^{k} [q_j - M, q_j + M]$.
For $\mathbf{p} = (p_1, \ldots, p_k) \in E$ and $1 \leq j \leq k$, $|p_j - q_j| \leq \|\mathbf{p} - \mathbf{q}\|_2 < M$, hence $\mathbf{p} \in I$. Thus $E \subseteq I$.
Structural Commentary: One ball fits inside one coordinate box.
Key Insight: A radius gives coordinate bounds. □

Theorem 4.2.5. *Heine–Borel in $\mathbb{R}^k$*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$. The following are equivalent:

(i) E is closed and bounded;

(ii) E *is compact;*
(iii) *every infinite subset of* E *has a limit point in* E.

Proof.

(*i*) $\Rightarrow$ (*ii*) Since E is bounded, Lemma 4.2.4 gives a k-cell I with $E \subseteq I$. By Theorem 4.2.3, I is compact. Since E is closed in $\mathbb{R}^k$, E is closed in I, so Theorem 4.1.6 implies that E is compact.
(*ii*) $\Rightarrow$ (*iii*) Theorem 4.1.10 applies directly to the compact set E.
(*iii*) $\Rightarrow$ (*i*) Suppose first that E is unbounded. Choose $\mathbf{x}_n \in E$ with $\|\mathbf{x}_n\|_2 > n$ and set $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$. For any $\mathbf{y} \in \mathbb{R}^k$, the set $S \cap B_1(\mathbf{y})$ is finite because $\mathbf{x}_n \in B_1(\mathbf{y})$ implies $n < \|\mathbf{y}\|_2 + 1$. Hence S has no limit point, contradicting (iii). Thus E is bounded. Suppose next that E is not closed. Choose $\mathbf{x}_0 \in \overline{E} \setminus E$ and distinct $\mathbf{x}_n \in E$ with $0 < \|\mathbf{x}_n - \mathbf{x}_0\|_2 < 1/n$. Then $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$ is infinite, and every limit point of S equals $\mathbf{x}_0 \notin E$, contradicting (iii). Thus E is closed.

Structural Commentary: Compactness equals closed boundedness in $\mathbb{R}^k$.
Key Insight: Cells convert bounded geometry into compactness. □

Theorem 4.2.6. *Weierstrass Limit-Point Theorem*
Let $k \in \mathbb{N}$ *and let* $E \subseteq \mathbb{R}^k$ *be bounded and infinite. Then* E *has a limit point in* $\mathbb{R}^k$.

Proof. By Lemma 4.2.4, choose a k-cell I with $E \subseteq I$. By Theorem 4.2.3, I is compact.
Since E is an infinite subset of I, Theorem 4.1.10 gives a limit point of E in $I \subseteq \mathbb{R}^k$.
Structural Commentary: Bounded sets sit in compact cells.
Key Insight: Infinite compact subsets accumulate. □

Corollary 4.2.7. *Bounded Euclidean Sequences Have Convergent Subsequences*
If $(\mathbf{x}_n)$ *is bounded in* $\mathbb{R}^k$, *then it has a convergent subsequence.*

Proof. The range of $(\mathbf{x}_n)$ has compact closure by Theorem 4.2.5; apply Theorem 4.1.12 in that closure.
Structural Commentary: Heine–Borel makes the range closure compact.
Key Insight: Bounded Euclidean sequences live inside compact closed sets. □

Corollary 4.2.8. *Euclidean Cauchy Convergence*
In $\mathbb{R}^k$, *every Cauchy sequence converges.*

Proof. For Cauchy $(\mathbf{x}_n) \subseteq \mathbb{R}^k$, choose N with $\operatorname{diam}\{\mathbf{x}_j : j \geq N\} < 1$.
Its range is bounded, hence has compact closure by Theorem 4.2.5; apply Theorem 4.1.19(ii).
Structural Commentary: A Cauchy tail has bounded compact closure.
Key Insight: In $\mathbb{R}^k$, Cauchy convergence follows from Heine–Borel. □

Corollary 4.2.9. *Euclidean Compactness Tests*
Let $k \in \mathbb{N}$ *and let* $E \subseteq \mathbb{R}^k$, *with the subspace metric. The following are equivalent:*

(i) E *is closed and bounded;*
(ii) E *is compact;*
(iii) *every sequence in* E *has a subsequence converging to a point of* E.

Proof.

(*i*) $\Rightarrow$ (*ii*) This is Theorem 4.2.5.
(*ii*) $\Rightarrow$ (*iii*) Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in E.
Since E is compact, Theorem 4.1.30 gives indices $n_1 < n_2 < \cdots$ and $x \in E$ such that $x_{n_j} \to x$.
(*iii*) $\Rightarrow$ (*i*) By (iii), E is sequentially compact. Theorem 4.1.30, applied to $X = E$, gives that E is compact.
Then Theorem 4.2.5 implies that E is closed and bounded.

Structural Commentary: Euclidean compactness has three tests.
Key Insight: Heine–Borel links geometry to sequences. □

Corollary 4.2.10. *Heine–Borel on the Real Line*
Let $E \subseteq \mathbb{R}$, *with the usual metric. The following are equivalent:*

(i) E *is closed and bounded;*
(ii) E *is compact;*
(iii) *every sequence in* E *has a subsequence converging to a point of* E.

Proof.

(*i*) $\Rightarrow$ (*ii*) By (i), E is closed, bounded as a subset of $\mathbb{R}^1$. Theorem 4.2.5 with $k = 1$ gives that E is compact.
(*ii*) $\Rightarrow$ (*iii*) Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in E.
By compactness of E, Theorem 4.1.30 gives indices $n_1 < n_2 < \cdots$ and $x \in E$ such that $x_{n_j} \to x$.
(*iii*) $\Rightarrow$ (*i*) Let $F \subseteq E$ be infinite, and choose distinct $x_n \in F$. By (iii), some subsequence satisfies $x_{n_j} \to x \in E$. Since the x_n are distinct, x is a limit point of F. Thus every infinite subset of E has a limit point in E, so Theorem 4.2.5 with $k = 1$ gives that E is closed and bounded.

Structural Commentary: On $\mathbb{R}$, compactness is sequential.
Key Insight: Closed bounded sets trap subsequences. □

Corollary 4.2.11. *Nested Closed Bounded Real Sets Intersect*
Let $(F_n)_{n\in\mathbb{N}}$ *be nonempty closed subsets of* $\mathbb{R}$ *with* $F_{n+1} \subseteq F_n$ *for all* n. *If* F_1 *is bounded, then* $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$.

Proof. By Corollary 4.2.10, F_1 is compact. For each n, F_n is a nonempty closed subset of the compact set F_1, so Theorem 4.1.6 gives that F_n is compact. Since $F_{n+1} \subseteq F_n$, Corollary 4.1.9 gives $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$.
Structural Commentary: Nested compact sets intersect.
Key Insight: Heine–Borel supplies compactness. □

4.3 Baire Category

Definition 4.3.1. *F_σ, G_δ, and Nowhere-Dense Sets*
For subsets of a metric space X:

(i) A is an *F_σ set* if $A = \bigcup_{n=1}^{\infty} F_n$ for closed sets F_n.
(ii) B is a *G_δ set* if $B = \bigcap_{n=1}^{\infty} G_n$ for open sets G_n.
(iii) E is *nowhere dense* if $(\overline{E})^\circ = \emptyset$; equivalently, no nonempty open set is contained in $\overline{E}$.

For $X = \mathbb{R}$, this says exactly that $\overline{E}$ contains no nonempty open interval.

Proposition 4.3.2. *Open Real Sets Are F_σ*
For subsets of $\mathbb{R}$:

(i) If $G \subseteq \mathbb{R}$ is open, then G is F_σ.
(ii) If $A \subseteq \mathbb{R}$ is closed, then A is G_δ.

Proof.

(i) If $G = \mathbb{R}$, then $G = \bigcup_{m \geq 1}[-m, m]$.
Otherwise set $C := \mathbb{R} \setminus G$ and $F_m := [-m, m] \cap \{x \in \mathbb{R} : \operatorname{dist}(x, C) \geq 1/m\}$ for $m \geq 1$.
The map $x \mapsto \operatorname{dist}(x, C)$ is continuous since $|\operatorname{dist}(x, C) - \operatorname{dist}(y, C)| \leq |x - y|$; hence each F_m is closed.
Moreover $F_m \subseteq G$. For $x \in G$, $r > 0$ with $(x - r, x + r) \subseteq G$, $m \geq |x|$; $1/m \leq r$ give $x \in F_m$. QED.
(ii) If A is closed, then $A^c = \bigcup_{m=1}^{\infty} F_m$ with F_m closed by (i), hence $A = \bigcap_{m=1}^{\infty} F_m^c$ is G_δ.

Structural Commentary: Positive distance from the complement gives closed exhaustion.
Key Insight: Distance to the complement converts local openness into a countable closed exhaustion. □

Definition 4.3.3. *First Category*
Let X be a metric space and let $E \subseteq X$. The set E is *of first category*, *meager*, or *thin* in X if there are nowhere-dense sets $N_n \subseteq X$ such that $E \subseteq \bigcup_{n=1}^{\infty} N_n$.
Structural Commentary: Meagerness is relative to the ambient space X.
Key Insight: The Baire Category Theorem says a nonempty complete metric space is not meager in itself.

Theorem 4.3.4. *Dense Open Intersections in $\mathbb{R}$*
If $G_1, G_2, G_3, \ldots$ are dense open subsets of $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$.

Proof. We construct nested closed intervals. Since G_1 is dense and open, $G_1 \cap (0, 1)$ contains a nonempty open interval; choose a closed interval $I_1 \subseteq G_1 \cap (0, 1)$ with nonempty interior.
Given $I_n = [a_n, b_n]$, density of G_{n+1} gives a point of G_{n+1} in (a_n, b_n). Since G_{n+1} is open, choose a closed interval $I_{n+1} \subseteq G_{n+1} \cap (a_n, b_n)$ with nonempty interior and length $< 1/(n + 1)$.
Then $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, so Theorem 4.2.1 gives $x \in \bigcap_n I_n$. Since $I_n \subseteq G_n$ for every n, this x lies in $\bigcap_n G_n$.
Structural Commentary: Dense openness builds nested intervals; completeness keeps a limit point.
Key Insight: A countable dense-open intersection cannot be destroyed one interval at a time. □

Theorem 4.3.5. *Dense Open Intersections in Complete Metric Spaces*
Let (X, d) be complete, let $O_1, O_2, \ldots$ be dense open subsets of X, and let $U \subseteq X$ be nonempty and open.
Then $U \cap \bigcap_{n=1}^{\infty} O_n \neq \emptyset$. Consequently, $\bigcap_{n=1}^{\infty} O_n$ is dense in X.

Proof. Choose $x_1 \in U \cap O_1$ and $0 < r_1 < 1$ with $C_1 := \overline{B}_{r_1}(x_1) \subseteq U \cap O_1$.
Inductively, if $C_n = \overline{B}_{r_n}(x_n) \subseteq U \cap O_1 \cap \cdots \cap O_n$ and $0 < r_n < 2^{-n}$, density and openness of O_{n+1} give $x_{n+1} \in B_{r_n}(x_n) \cap O_{n+1}$ and $0 < r_{n+1} < 2^{-(n+1)}$ such that $C_{n+1} := \overline{B}_{r_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1}$.
Then $C_{n+1} \subseteq C_n$. If $m > n$, then $x_m \in B_{r_n}(x_n)$, so $d(x_m, x_n) < 2^{-n}$; hence (x_n) is Cauchy. Let $x_n \to x$.
Since each C_n is closed and contains all sufficiently late x_m, we have $x \in C_n$ for every n. Thus $x \in U \cap \bigcap_n O_n$.
Since every nonempty open U meets $\bigcap_n O_n$, the intersection is dense.
Structural Commentary: The interval proof becomes a nested closed-ball proof.
Key Insight: Dense open sets choose the next ball; completeness supplies the limiting point. □

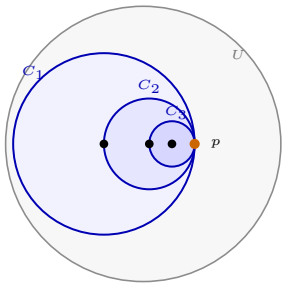

Figure 135: Nested closed balls shrink to p.

Definition 4.3.6. *Residual, Thick, and Generic Sets*
Let X be a metric space.

(i) A set $G \subseteq X$ is *residual*, or *thick*, in X if there are dense open sets $O_n \subseteq X$ such that $\bigcap_{n=1}^{\infty} O_n \subseteq G$.
(ii) A property P is *generic* in X if $\{x \in X : P(x)\}$ is residual in X.

If X is complete, then every residual subset of X is dense by Theorem 4.3.5.
Key Insight: Meager sets are topologically small; residual sets contain a countable dense-open core.

Theorem 4.3.7. *Baire Category Theorem for Complete Metric Spaces*
A nonempty complete metric space is not the union of a countable collection of nowhere-dense sets.

Proof. Suppose $X = \bigcup_n E_n$ with each E_n nowhere dense, and set $O_n := (\overline{E_n})^c$.
Then each O_n is dense open: it is open, and every nonempty open set meets it because no such set lies in $\overline{E_n}$.
By Theorem 4.3.5, choose $x \in \bigcap_n O_n$. Then $x \notin E_n$ for all n, contradicting $X = \bigcup_n E_n$.
Structural Commentary: Countably many nowhere-dense sets have dense open complements.
Key Insight: Completeness forbids countable assembly from thin pieces. □

Theorem 4.3.8. *Baire's Theorem for $\mathbb{R}$*
The real line $\mathbb{R}$ cannot be written as a countable union of nowhere-dense sets.

Proof. Assume $\mathbb{R} = \bigcup_n E_n$ with each E_n nowhere dense, and set $U_n := (\overline{E_n})^c$.
Each U_n is dense open: it is open, and every nonempty interval meets it because no such interval lies in $\overline{E_n}$.
By Theorem 4.3.4, choose $x \in \bigcap_n U_n$. Then $x \notin E_n$ for all n, contradicting $\mathbb{R} = \bigcup_n E_n$.
Structural Commentary: Nowhere-dense sets have dense open complements.
Key Insight: Countably many thin sets cannot exhaust the real line. □

Corollary 4.3.9. *Rationals and Irrationals by Category*
The set $\mathbb{Q}$ is F_σ but not G_δ. The set $\mathbb{R} \setminus \mathbb{Q}$ is G_δ but not F_σ.

Proof. Write $\mathbb{Q} = \{q_1, q_2, \ldots\}$. Since singletons are closed, $\mathbb{Q} = \bigcup_n \{q_n\}$ is F_σ, and $\mathbb{R} \setminus \mathbb{Q} = \bigcap_n (\{q_n\})^c$ is G_δ.
If $\mathbb{Q} = \bigcap_n G_n$ with each G_n open, then each G_n is dense because it contains the dense set $\mathbb{Q}$. Set $H_n := G_n \setminus \{q_n\}$.
Each H_n is dense open, so Theorem 4.3.4 gives a point of $\bigcap_n H_n$. But $\bigcap_n H_n = \emptyset$, since $\bigcap_n G_n = \mathbb{Q}$ and q_n is removed at stage n.
Hence $\mathbb{Q}$ is not G_δ. Therefore $\mathbb{R} \setminus \mathbb{Q}$ is not F_σ, or else its complement $\mathbb{Q}$ would be G_δ.
Structural Commentary: Category separates the familiar countable dense set from its dense complement.
Key Insight: Dense is not the same as large in the Baire-category sense. □

Corollary 4.3.10. *Closed Cover Interior Test*
Let X be a complete metric space and let $(F_n)_{n\in\mathbb{N}}$ be closed subsets of X.

(i) If $\emptyset \neq U \subseteq X$ is open and $U \subseteq \bigcup_{n=1}^{\infty} F_n$, then $F_N^\circ \neq \emptyset$ for some $N \in \mathbb{N}$.
(ii) If $X \neq \emptyset$ and $F_n^\circ = \emptyset$ for every $n \in \mathbb{N}$, then $X \neq \bigcup_{n=1}^{\infty} F_n$.

Proof.

(i) Assume $\emptyset \neq U \subseteq X$ is open and $U \subseteq \bigcup_{n=1}^{\infty} F_n$. Suppose $F_n^\circ = \emptyset$ for every n and set $O_n := X \setminus F_n$. Since F_n is closed, O_n is open. If $V \subseteq X$ nonempty and open, then $V \not\subseteq F_n$, so $V \cap O_n \neq \emptyset$; hence O_n dense.
Theorem 4.3.5 gives $x \in U \cap \bigcap_{n=1}^{\infty} O_n$. Then $x \in U \subseteq \bigcup_{n=1}^{\infty} F_n$ and $x \notin F_n$ $\forall n$, contradiction. QED.
(ii) Assume $X \neq \emptyset$ and $F_n^\circ = \emptyset$ for every n. If $X = \bigcup_{n=1}^{\infty} F_n$, then (i) with $U = X$ gives $F_N^\circ \neq \emptyset$ for some N, a contradiction. Hence $X \neq \bigcup_{n=1}^{\infty} F_n$.

Structural Commentary: Empty interiors give dense open complements.
Key Insight: Baire forces one complement-avoiding point. □

Theorem 4.3.11. *Pointwise Bounded Families Are Locally Uniformly Bounded*
Let (X, d) be a nonempty complete metric space, and let $\mathcal{F} \subseteq C(X, \mathbb{R})$.
Assume that for every $x \in X$ there is $M_x < \infty$ such that $|f(x)| \leq M_x$ for every $f \in \mathcal{F}$.
Then there exist a nonempty open set $O \subseteq X$ and $M < \infty$ such that $|f(x)| \leq M$ for all $x \in O$ and $f \in \mathcal{F}$.

Proof. For $n \in \mathbb{N}$, set $E_n := \{x \in X : \forall f \in \mathcal{F},\ |f(x)| \leq n\}$.
As $E_n = \bigcap_{f\in\mathcal{F}} \{x \in X : |f(x)| \leq n\}$ and each f is continuous, E_n is closed. The hypothesis gives $X = \bigcup_{n=1}^{\infty} E_n$.
By Corollary 4.3.10, $E_N^\circ \neq \emptyset$ for some $N \in \mathbb{N}$. Take $O := E_N^\circ$, $M := N$. Then $x \in O$ and $f \in \mathcal{F}$ imply $|f(x)| \leq M$.
Structural Commentary: Pointwise bounds form a closed cover.
Key Insight: Completeness turns pointwise control into uniform control somewhere. □

Theorem 4.3.12. *Dense Continuity Set for Pointwise Limits*
Let (X, d) be a complete metric space, let $f : X \to \mathbb{R}$, let $f_n : X \to \mathbb{R}$ be continuous for every $n \in \mathbb{N}$, and assume $f_n(x) \to f(x)$ for every $x \in X$. Then there exists $D \subseteq X$ such that:

(i) D is dense in X;
(ii) (f_n) is equicontinuous at every $x_0 \in D$;
(iii) f is continuous at every $x_0 \in D$.

Proof. For $m, n \in \mathbb{N}$, set $E_{m,n} := \{x \in X : \forall j, k \geq n,\ |f_j(x) - f_k(x)| \leq 1/m\}$.
Since $E_{m,n} = \bigcap_{j,k\geq n} \{x \in X : |f_j(x) - f_k(x)| \leq 1/m\}$, each $E_{m,n}$ is closed.
or fixed m, pointwise convergence makes $(f_n(x))$ Cauchy for every $x \in X$, hence $X = \bigcup_{n=1}^{\infty} E_{m,n}$.
For closed $E \subseteq X$, $\mathrm{bd}(E) = E \setminus E^\circ$ is closed and nowhere dense: if nonempty open $V \subseteq \mathrm{bd}(E)$, then $V \subseteq E^\circ$ and $V \subseteq E \setminus E^\circ$, a contradiction. Thus each $X \setminus \mathrm{bd}(E_{m,n})$ is dense open. Set $D := X \setminus \bigcup_{m,n\in\mathbb{N}} \mathrm{bd}(E_{m,n})$.

(i) Since $\mathbb{N}^2$ is countable, $D = \bigcap_{m,n\in\mathbb{N}} (X \setminus \mathrm{bd}(E_{m,n}))$ is a countable intersection of dense open sets.
By Theorem 4.3.5, D is dense in X.
(ii) Fix $x_0 \in D$ and $\varepsilon > 0$. Choose $m \in \mathbb{N}$ with $3/m < \varepsilon$. Since $(f_n(x_0))$ is Cauchy, choose N such that $x_0 \in E_{m,N}$. Since $x_0 \notin \mathrm{bd}(E_{m,N})$, actually $x_0 \in E_{m,N}^\circ$; choose $r > 0$ with $B_r(x_0) \subseteq E_{m,N}$. By continuity of $f_1, \ldots, f_N$ at x_0, choose $0 < \delta < r$ such that $d(x, x_0) < \delta$ implies $|f_i(x) - f_i(x_0)| < 1/m$ for $1 \leq i \leq N$.
If $d(x, x_0) < \delta$ and $j > N$, then $x, x_0 \in E_{m,N}$, so $|f_j(x) - f_N(x)| \leq 1/m$ and $|f_N(x_0) - f_j(x_0)| \leq 1/m$. Hence $|f_j(x) - f_j(x_0)| \leq |f_j(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f_j(x_0)| < 3/m < \varepsilon$.
For $1 \leq j \leq N$, the choice of δ gives $|f_j(x) - f_j(x_0)| < 1/m < \varepsilon$. Thus (f_n) is equicontinuous at x_0.
(iii) Fix $x_0 \in D$ and $\varepsilon > 0$, and choose m, N, δ as in (ii). If $d(x, x_0) < \delta$, the estimate in (ii) gives $|f_j(x) - f_j(x_0)| < 3/m < \varepsilon$ for all $j > N$. Letting $j \to \infty$ gives $|f(x) - f(x_0)| \leq 3/m < \varepsilon$. QED.

Structural Commentary: Baire avoids all layer boundaries.
Key Insight: Interior tail-Cauchy control gives equicontinuity. □

Perspective 4.3.13. *Category in Function Spaces*
Baire category is a theorem about complete metric spaces, not just $\mathbb{R}$.
It will be reused for complete function spaces under the supremum metric.

Method 4.3.14. *Baire-Category Proof Patterns*

(i) *Dense-open intersection*: build nested closed intervals $I_n \subseteq G_n$ and apply Theorem 4.2.1.
(ii) *Complete metric version*: build nested closed balls $C_{n+1} \subseteq C_n$ with $r_n \to 0$ and use completeness.
(iii) *Nowhere dense*: replace E by the dense open set $(\overline{E})^c$.
(iv) *First category*: show the exceptional set is contained in $\bigcup_n E_n$ with each E_n nowhere dense.
(v) *Residual/generic*: show the desired property contains $\bigcap_n O_n$ with each O_n dense open.
(vi) *F_σ/G_δ complements*: pass between countable unions and intersections by De Morgan's laws.

4.4 Perfect Sets and the Cantor Set

Theorem 4.4.1. *Perfect Sets Are Uncountable*
Every nonempty perfect subset of $\mathbb{R}^k$ is uncountable.

Proof. Assume $P = \{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$. Choose a neighborhood V_1 with $V_1 \cap P \neq \emptyset$.
Inductively, since $P \cap V_n$ contains infinitely many points by Theorem 3.1.14, choose $\mathbf{y}_n \in P \cap V_n$ with $\mathbf{y}_n \neq \mathbf{x}_n$ and then a neighborhood V_{n+1} of $\mathbf{y}_n$ such that $\overline{V}_{n+1} \subseteq V_n$ and $\mathbf{x}_n \notin \overline{V}_{n+1}$.
Put $K_n := \overline{V}_n \cap P$. Then K_n is nonempty, compact, and $K_{n+1} \subseteq K_n$. By Corollary 4.1.9, choose $\mathbf{z} \in \bigcap_{n=1}^{\infty} K_n$. For each n, $\mathbf{z} \in K_{n+1}$ but $\mathbf{x}_n \notin K_{n+1}$, so $\mathbf{z} \neq \mathbf{x}_n$ for all n, contradicting the enumeration.
Structural Commentary: Nested compact sets dodge the list.
Key Insight: Perfectness gives new points; compactness keeps one. □

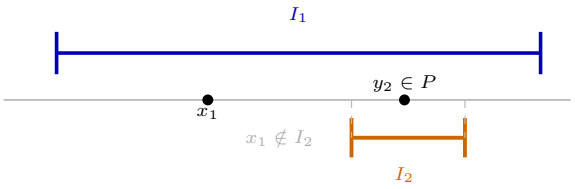

Figure 136: Choose $y_2 \in P \cap I_1$ with $y_2 \neq x_1$, then take a smaller interval $I_2 \subseteq I_1$ around y_2 that avoids x_1.

Corollary 4.4.2. *Nondegenerate Intervals and the Real Line Are Uncountable*
Let $J \subseteq \mathbb{R}$ be an interval.

(i) If J contains two distinct points, then J is uncountable.
(ii) $\mathbb{R}$ is uncountable.

Proof.

(i) Choose $u < v$ in J and then a, b with $u < a < b < v$. Since J is an interval, $I := [a, b] \subseteq J$.
The set I is a nonempty 1-cell, hence compact by Theorem 4.2.3 and closed by Theorem 4.1.5.
For every $x \in I$ and $\varepsilon > 0$, $I \cap (B_\varepsilon(x) \setminus \{x\}) \neq \emptyset$, so I is perfect.
Thus I is uncountable by Theorem 4.4.1; hence J is uncountable.
(ii) Since $\mathbb{R}$ is an interval and $0, 1 \in \mathbb{R}$ with $0 < 1$, (i) applied to $J = \mathbb{R}$ gives that $\mathbb{R}$ is uncountable.

Structural Commentary: Closed subintervals are perfect compact subsets.
Key Insight: One closed subinterval already forces uncountability. □

Example 4.4.3. *The Cantor Set*
Set $E_0 = [0, 1]$ and obtain E_{n+1} from E_n by deleting the open middle third of each component.
Then E_n is a union of 2^n compact intervals of length 3^{-n} and $E_{n+1} \subseteq E_n$.
The *Cantor set* is $P := \bigcap_{n=0}^{\infty} E_n$. Since (E_n) is nested nonempty compact, $P \neq \emptyset$ by Corollary 4.1.9.
Since P is closed in E_0, P is compact. Every nonempty open interval $J \subseteq [0, 1]$ contains a deleted middle third R with $R \cap P = \emptyset$, so P contains no nontrivial interval.If $x \in P$ and U is an interval neighborhood of x, choose n with the component $I_n(x) \subseteq U$; an endpoint $y \neq x$ of $I_n(x)$ lies in P, so $U \cap (P \setminus \{x\}) \neq \emptyset$. Hence P is perfect.
Structural Commentary: The Cantor set is built by nested compact deletion, not by listing points.
Key Insight: It is compact and perfect, but contains no interval.

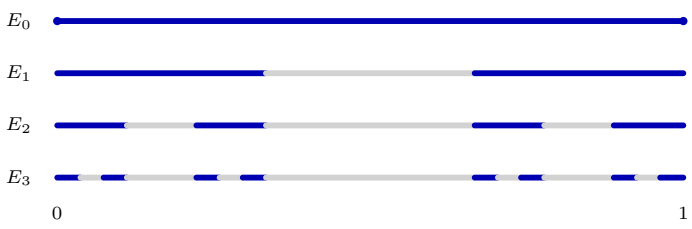

Figure 137: The first stages E_0, E_1, E_2, E_3 in the Cantor-set construction.

4.5 Complete Metric Spaces and Completions

Completeness turns Cauchy approximations into actual points. The examples below record the standard complete spaces used throughout analysis, then isolate the completion construction that adjoins missing Cauchy limits to an arbitrary metric space.

Example 4.5.1. *Finite-Dimensional Complete Metric Spaces*
For $k \in \mathbb{N}$, $(\mathbb{R}^k, d_2)$ and $(\mathbb{C}^k, d_2)$ are complete; in particular, so are $\mathbb{R}$ and $\mathbb{C}$.

Proof. Completeness of $\mathbb{R}$ is Theorem 2.1.15.
If $z_n = x_n + iy_n$ is Cauchy in $\mathbb{C}$, then (x_n) and (y_n) are Cauchy in $\mathbb{R}$, so $z_n \to x + iy \in \mathbb{C}$.
Corollary 4.2.8 gives completeness of $\mathbb{R}^k$, and the isometry $\mathbb{C}^k \cong \mathbb{R}^{2k}$ gives completeness of $\mathbb{C}^k$.
Structural Commentary: Finite-dimensional completeness is coordinatewise completeness.
Key Insight: A Cauchy sequence in finitely many coordinates converges coordinate by coordinate. □

Definition 4.5.2. *Contraction*
Let X be a metric space with metric d.
A map $\varphi : X \to X$ is a *contraction* if $\exists$ a constant $0 \le c < 1$ such that $d(\varphi(x), \varphi(y)) \le c\, d(x, y)$ for $x, y \in X$.
Structural Commentary: Bound $c < 1$ excludes nonexpansive maps and is the hypothesis in Theorem 4.5.3.

Theorem 4.5.3. *Contraction Principle*
If X is a complete metric space and $\varphi : X \to X$ is a contraction, then $\exists$ a unique $x \in X$ such that $\varphi(x) = x$.

Proof. Choose $x_0 \in X$ and define $x_{n+1} = \varphi(x_n)$.
If $c < 1$ is a contraction constant, induction gives $d(x_{n+1}, x_n) \le c^n d(x_1, x_0)$.
For $n < m$, the triangle inequality and the finite geometric sum give $d(x_n, x_m) \le \sum_{j=n}^{m-1} d(x_j, x_{j+1}) \le \frac{c^n}{1-c} d(x_1, x_0)$.
Thus (x_n) is Cauchy, so completeness gives $x_n \to x \in X$.
Since a contraction is continuous, $\varphi(x) = \lim \varphi(x_n) = \lim x_{n+1} = x$.
If $\varphi(x) = x$ and $\varphi(y) = y$, then $d(x, y) \le c\, d(x, y)$, hence $d(x, y) = 0$ and $x = y$.
Structural Commentary: Iteration turns the contraction estimate into a geometric Cauchy estimate.
Key Insight: Completeness supplies the limiting point; the strict constant $c < 1$ forces uniqueness. □

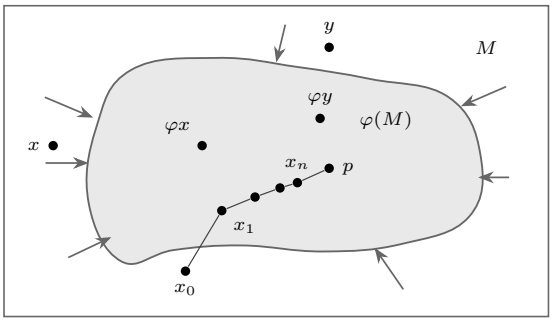

Figure 138: A contraction pulls iterates toward the unique fixed point.

Perspective 4.5.4. *Fixed Points as Existence Proofs*
A contraction proof solves $x = \varphi(x)$ by iteration in a complete metric space. This is the template behind the inverse function theorem, Theorem 4.1.4.

Example 4.5.5. *A Noncomplete Metric Space*
The rational numbers with $d(x, y) = |x - y|$ are not complete.
Let (r_n) be the decimal truncation sequence 1.4, 1.41, 1.414, 1.4142, ..., for $\sqrt{2}$.
This is a concrete instance of the truncation squeeze in Proposition 2.2.11. If $k, n \ge N$, then $|r_k - r_n| \le 10^{-N}$, so (r_n) is Cauchy in $\mathbb{Q}$. As a sequence in $\mathbb{R}$, it converges to $\sqrt{2}$.
If it converged to some $r \in \mathbb{Q}$ in the inherited metric, uniqueness of limits in $\mathbb{R}$ would force $r = \sqrt{2}$, contradicting Proposition 1.1.7.
Thus $\mathbb{Q}$ is not complete in the inherited metric.
Structural Commentary: The decimal approximants are Cauchy before their missing limit is available inside $\mathbb{Q}$.
Key Insight: Completeness asks whether every internally Cauchy approximation has an internal endpoint.

Perspective 4.5.6. *Cauchy Construction as a Parallel Model*
Cuts build $\mathbb{R}$ here; equivalence classes of rational Cauchy sequences give the parallel model.
Let $\mathcal{C}_{\mathbb{Q}}$ be the set of rational Cauchy sequences and set $(a_n) \sim (b_n)$ iff $|a_n - b_n| \to 0$; then $\mathbb{R} \simeq \mathcal{C}_{\mathbb{Q}}/\sim$.
Operations and order must be independent of representatives.

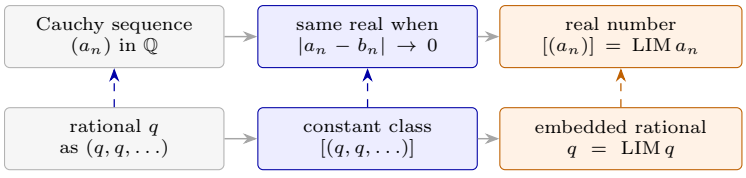

Figure 139: Real numbers as equivalence classes of rational Cauchy representatives.

Perspective 4.5.7. *Completion by Cauchy Sequences*
Complete by identifying Cauchy sequences with $d(p_n, q_n) \to 0$ and setting $D([(p_n)], [(q_n)]) = \lim_{n\to\infty} d(p_n, q_n)$.
Constant sequences embed the original space; triangle inequalities and diagonals give a complete metric.

Theorem 4.5.8. *Metric Completion*

For every metric space (X, d), $\exists$ complete metric space $(\widehat{X}, \widehat{d})$, isometry $\iota : X \to \widehat{X}$ with $\overline{\iota(X)}^{\widehat{X}} = \widehat{X}$.
If (Y, d_Y, τ), (Z, d_Z, σ) are two such completions, then $\exists!$ bijective isometry $\Phi : Y \to Z$ s.t. $\Phi \circ \tau = \sigma$.

Proof.

i. *Existence.* Let $\mathrm{CS}(X)$ be the Cauchy sequences in X.
For $a = (a_n), b = (b_n) \in \mathrm{CS}(X)$, $|d(a_n, b_n) - d(a_m, b_m)| \le d(a_n, a_m) + d(b_n, b_m)$, so $D(a, b) := \lim_{n\to\infty} d(a_n, b_n)$ exists. Put $a \sim b \Leftrightarrow D(a, b) = 0$, $\widehat{X} := \mathrm{CS}(X)/\sim$, and $\widehat{d}([a], [b]) := D(a, b)$.
The estimate gives well-definedness; metric axioms pass to limits. Map $\iota(x) := [(x, x, \ldots)]$ is an isometry.
If $a = (a_n)$ and $d(a_n, a_N) < \varepsilon$ for $n \ge N$, then $\widehat{d}(\iota(a_N), [a]) \le \varepsilon$, hence $\overline{\iota(X)}^{\widehat{X}} = \widehat{X}$.
Let $([a^{(k)}])$ be Cauchy in $\widehat{X}$. Choose $x_k \in X$ with $\widehat{d}(\iota(x_k), [a^{(k)}]) < 1/k$.
Since $d(x_k, x_l) \le 1/k + \widehat{d}([a^{(k)}], [a^{(l)}]) + 1/l$, (x_k) is Cauchy in X,
and $\widehat{d}([a^{(k)}], [(x_j)]) \le 1/k + \limsup_{j\to\infty} d(x_k, x_j) \to 0$. Thus $([a^{(k)}]) \to [(x_j)]$, so $(\widehat{X}, \widehat{d})$ is complete.

ii. *Uniqueness.* Let (Y, d_Y, τ) and (Z, d_Z, σ) be completions.
Define $\rho_0 : \tau(X) \to \sigma(X)$ by $\rho_0(\tau x) := \sigma x$; it is an isometry.
For $y \in Y$, choose $y_n \in \tau(X)$ with $y_n \to y$. Since $\rho_0(y_n)$ is Cauchy in Z, define $\Phi(y) := \lim_{n\to\infty} \rho_0(y_n)$.
If $z_n \in \tau(X)$ and $z_n \to y$, then $d_Z(\rho_0(y_n), \rho_0(z_n)) = d_Y(y_n, z_n) \to 0$, so Φ is well defined.
If $y'_n \to y'$, then $d_Z(\Phi(y), \Phi(y')) = \lim_{n\to\infty} d_Z(\rho_0(y_n), \rho_0(y'_n)) = \lim_{n\to\infty} d_Y(y_n, y'_n) = d_Y(y, y')$.
Hence Φ is an isometry extending ρ_0. Its image is complete, hence closed in Z, and contains the dense set $\sigma(X)$; therefore $\Phi(Y) = Z$. Uniqueness follows from continuity and density of $\tau(X)$.

Structural Commentary: Missing limits are installed as equivalence classes of Cauchy approximations.
Key Insight: Completion adds Cauchy limits and is unique over the dense copy of X. □

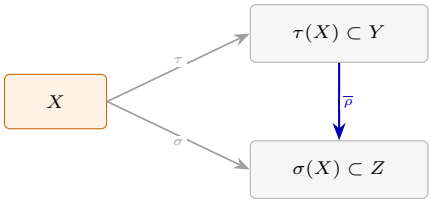

Figure 140: For completions Y, Z, the dense-copy isometry $\sigma\tau^{-1} : \tau(X) \to \sigma(X)$ extends uniquely to $\overline{\rho} : Y \cong Z$.

5 Connectedness and Continuity

5.1 Connectedness and Continuity

Connectedness prevents a space from splitting into separated pieces; continuity transports topological structure through maps. This subsection also records IVT/EVT and real-line refinements.

Definition 5.1.1. *Separated and Connected Sets*
Let X be a metric space.

(i) Subsets $A, B \subseteq X$ are *separated* if $A \cap \overline{B} = \overline{A} \cap B = \emptyset$.
(ii) A subset $E \subseteq X$ is *connected* if no nonempty separated subsets $A, B \subseteq E$ satisfy $E = A \cup B$.
(iii) A subset $U \subseteq X$ is *clopen* if it is both open and closed.

Structural Commentary: Separated sets avoid each other's closures.
Key Insight: Disconnected means a nontrivial clopen split; continuous preimages preserve such splits.

Theorem 5.1.2. *Connected Subsets of the Real Line*
A set $E \subseteq \mathbb{R}$ is connected if and only if $x, y \in E,\ x < y \implies [x, y] \subseteq E$.

Proof.

(⇒) If the interval property fails, choose $x, y \in E$ and $x < z < y$ with $z \notin E$. Then $A := E \cap (-\infty, z)$ and $B := E \cap (z, \infty)$ are nonempty separated sets with $E = A \cup B$, contradicting connectedness.
(⇐) Suppose $E = A \cup B$ with A, B nonempty separated. Choose $x \in A$, $y \in B$, $x < y$, and set $S := A \cap [x, y]$, $z := \sup S$. By Theorem 3.3.5, $z \in \overline{A}$; hence $z \notin B$ and $z < y$. If $z \notin A$, then $z \in [x, y] \setminus E$.
If $z \in A$, then $z \notin \overline{B}$, so choose $z_1 \in (z, y)$ with $z_1 \notin B$; by maximality of z, also $z_1 \notin A$.
Thus $z_1 \in [x, y] \setminus E$, contradicting $[x, y] \subseteq E$.

Structural Commentary: On the real line, connectedness is exactly having no gaps between any two of its points.
Key Insight: A separation creates a missing point by taking a supremum at the boundary of one side. □

Method 5.1.3. *Compactness and Connectedness Proof Patterns*

(i) *Open-cover compactness*: for every cover $\{G_\alpha\}_{\alpha\in A}$, find finite $F \subseteq A$ with $K \subseteq \bigcup_{\alpha\in F} G_\alpha$.
(ii) *Euclidean closed-bounded compactness*: if $E \subseteq \mathbb{R}^k$ is closed and bounded, choose a compact k-cell $I \supseteq E$ and apply Theorem 4.1.6.
(iii) *Connectedness on $\mathbb{R}$*: replace connectedness of $E \subseteq \mathbb{R}$ by $x, y \in E, x < y \implies [x, y] \subseteq E$.

Definition 5.1.4. *Limit of a Function Between Metric Spaces*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, and let p be a limit point of E. For $q \in Y$, $\lim_{x\to p} f(x) = q$ means that, for every $\varepsilon > 0$, there exists $\delta > 0$ such that $x \in E$, $0 < d_X(x,p) < \delta \implies d_Y(f(x),q) < \varepsilon$.
The point p need not belong to E, and $f(p)$ is irrelevant when it does.
Key Insight: Domain E controls approach: $J \setminus \{a\}$ gives two-sided limit, (a,b) or (c,a) gives one-sided limits.

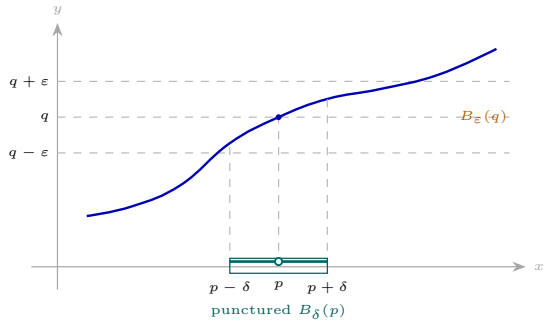

Figure 141: The epsilon–delta geometry for $\lim_{x\to p} f(x) = q$.

Theorem 5.1.5. *Sequential Characterization of Function Limits*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, and let p be a limit point of E.
For $q \in Y$, $\lim_{x\to p} f(x) = q$ if and only if $f(p_n) \to q$ for every sequence $(p_n) \subseteq E \setminus \{p\}$ with $p_n \to p$.

Proof.
($\Rightarrow$) If $p_n \to p$ with $p_n \in E \setminus \{p\}$, then eventually $0 < d_X(p_n,p) < \delta$; hence $d_Y(f(p_n),q) < \varepsilon$, so $f(p_n) \to q$.
($\Leftarrow$) If the limit fails, then for some $\varepsilon > 0$ and each n there is $x_n \in E$ with $0 < d_X(x_n,p) < 1/n$ and $d_Y(f(x_n),q) \geq \varepsilon$. Then $x_n \to p$, $x_n \neq p$, but $f(x_n) \not\to q$.
Structural Commentary: Function limits can be tested by all sequences approaching the input point.
Key Insight: The negation of an epsilon–delta limit manufactures a bad sequence. □

Corollary 5.1.6. *Uniqueness of Function Limits*
If a function has a limit at p, then that limit is unique.

Proof. Let q and r be limits of f at p. Choose $p_n \in E \setminus \{p\}$ with $d_X(p_n,p) < 1/n$.
Then $p_n \to p$, so Theorem 5.1.5 gives $f(p_n) \to q$ and $f(p_n) \to r$. By Theorem 3.4.2, $q = r$.
Structural Commentary: Limit uniqueness is inherited from metric sequence uniqueness.
Key Insight: One approaching sequence cannot have two different limits. □

Theorem 5.1.7. *Algebra of Function Limits*
Let X be a metric space, $E \subseteq X$, p be limit point of E.
For $f, g : E \to \mathbb{C}$, assume $\lim_{x\to p} f(x) = A$ and $\lim_{x\to p} g(x) = B$. Then:
(i) $\lim_{x\to p} (f+g)(x) = A + B$;
(ii) $\lim_{x\to p} (fg)(x) = AB$;
(iii) if $B \neq 0$ and f/g is defined near p, then $\lim_{x\to p} (f/g)(x) = A/B$.

For $f, g : E \to \mathbb{R}^k$ with limits a, b, the corresponding limits are $f + g \to a + b$, $\lambda f \to \lambda a$, and $f \cdot g \to a \cdot b$.

Proof.
(i) For any $p_n \in E \setminus \{p\}$ with $p_n \to p$, Theorem 5.1.5 gives $f(p_n) \to A$ and $g(p_n) \to B$.
Hence $(f+g)(p_n) \to A + B$ by Theorem 2.1.4, so Theorem 5.1.5 gives (i).
(ii) The same sequence gives $(fg)(p_n) \to AB$ by Theorem 2.1.4; hence (ii).
(iii) If $B \neq 0$ and f/g is defined near p, then every such sequence is eventually in the domain of f/g, and $(f/g)(p_n) \to A/B$ by Theorem 2.1.4; hence (iii).
The vector laws follow by the same sequential characterization and Theorem 3.4.7.
Structural Commentary: Limit algebra for functions is pulled back to sequence algebra.
Key Insight: Sequential characterization lets previous sequence theorems do the work. □

Proposition 5.1.8. *Function Limits Are Local*
Let X, Y be metric spaces, let $E \subseteq X$, let $f : E \to Y$, let p be a limit point of E, and let $q \in Y$.
For every $r > 0$, $\lim_{x\to p;\, x\in E} f(x) = q \iff \lim_{x\to p;\, x\in E\cap B_r(p)} f(x) = q$.

Proof.
($\Rightarrow$) Since $E \cap B_r(p) \subseteq E$, any δ witnessing the limit on E also witnesses the limit on $E \cap B_r(p)$.
($\Leftarrow$) Assume the restricted limit. Given $\varepsilon > 0$, choose $\delta_0 > 0$ such that $x \in E \cap B_r(p)$ and $0 < d_X(x,p) < \delta_0$ imply $d_Y(f(x),q) < \varepsilon$.
Let $\delta := \min\{\delta_0, r\}$.
If $x \in E$ and $0 < d_X(x,p) < \delta$, then $x \in E \cap B_r(p)$, so $d_Y(f(x),q) < \varepsilon$.
Structural Commentary: Values away from p cannot affect the limit at p.
Key Insight: Shrink δ until every tested point lies in the chosen local neighborhood. □

Definition 5.1.9. *Continuity at a Point and on a Set*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, $E \subseteq X$, and $f : E \to Y$.

(i) For $p \in E$, f is *continuous at p* if $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in E,\ d_X(x,p) < \delta \implies d_Y(f(x), f(p)) < \varepsilon$.
(ii) f is *continuous on E* if $\forall p \in E$, f is continuous at p.
(iii) If p is isolated in E, then f is continuous at p.

Corollary 5.1.10. *Discrete Domains Make Every Map Continuous*
If X carries the discrete metric, then every map $f : X \to Y$ into any metric space Y is continuous.

Proof. For $p \in X$, $B_{1/2}(p) = \{p\}$. Take $\delta = \frac{1}{2}$. If $d_X(x,p) < \delta$, then $x = p$, so $d_Y(f(x), f(p)) = 0 < \varepsilon$.
Structural Commentary: In the discrete metric, no distinct point can approach p.
Key Insight: Pointwise continuity is automatic when small enough balls are singletons. □

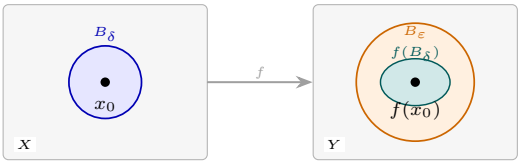

Figure 142: Continuity at x_0: for every $\varepsilon > 0$, choose $\delta > 0$ with $f(B_\delta(x_0)) \subset B_\varepsilon(f(x_0))$.

Theorem 5.1.11. *Continuity as Equality with the Function Limit*
Let X, Y be metric spaces, $E \subseteq X$, $p \in E$, and $f : E \to Y$.
If p is a limit point of E, then f is continuous at p if and only if $\lim_{x \to p} f(x) = f(p)$.

Proof.
($\Rightarrow$) If $0 < d_X(x,p) < \delta$, continuity at p gives $d_Y(f(x), f(p)) < \varepsilon$, hence $\lim_{x \to p} f(x) = f(p)$.
($\Leftarrow$) If $\lim_{x \to p} f(x) = f(p)$, the same δ proves continuity for $x \neq p$; for $x = p$, $d_Y(f(p), f(p)) = 0$.
Structural Commentary: Continuity is the compatibility of the value at p with nearby values.
Key Insight: Limits ignore the point; continuity ties the point back in. □

Remark 5.1.12. *Adherent-Point Limit Convention*
The adherent-point convention takes $p \in \overline{E}$ and allows $x = p$ in the δ-test; Definition 5.1.4 instead assumes p is a limit point and requires $0 < d_X(x,p)$. If $p \in E$, the unpunctured convention also tests $f(p)$, forcing $q = f(p)$; in the punctured convention this is exactly continuity at p by Theorem 5.1.11.
Key Insight: Punctured limits study nearby values; continuity adds compatibility with the value at the point.

Remark 5.1.13. *The Sequence-First View*
For functions on subsets of $\mathbb{R}$, continuity at limit points is equivalently sequential: $x_n \to p$ in the domain implies $f(x_n) \to f(p)$; isolated points are automatic by Theorems 5.1.5 and 5.1.11.

Theorem 5.1.14. *Sequential Characterization of Continuity*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, $f : X \to Y$, and $p \in X$.
Then f is continuous at p if and only if $\forall (x_n) \subseteq X$, $x_n \to p \implies f(x_n) \to f(p)$.

Proof.
($\Rightarrow$) Assume f is continuous at p and let $x_n \to p$.
Given $\varepsilon > 0$, choose $\delta > 0$ such that $d_X(x,p) < \delta \implies d_Y(f(x), f(p)) < \varepsilon$.
Since $x_n \to p$, eventually $d_X(x_n, p) < \delta$, hence eventually $d_Y(f(x_n), f(p)) < \varepsilon$. Thus $f(x_n) \to f(p)$.
($\Leftarrow$) Suppose the sequential condition holds. If f is not continuous at p, then $\exists \varepsilon_0 > 0$ such that $\forall \delta > 0$, $\exists x \in X$ with $d_X(x,p) < \delta$ and $d_Y(f(x), f(p)) \geq \varepsilon_0$. Taking $\delta = 1/n$ gives $x_n \in X$ with $d_X(x_n, p) < 1/n$ and $d_Y(f(x_n), f(p)) \geq \varepsilon_0$. Then $x_n \to p$, but $f(x_n) \not\to f(p)$, a contradiction.
Structural Commentary: Failure of epsilon–delta continuity manufactures a bad sequence.
Key Insight: In metric spaces, sequences detect continuity at points. □

Theorem 5.1.15. *Composition of Continuous Functions*
Let $E \subseteq X$ and let $f : E \to Y$, $g : f(E) \to Z$ be maps between metric spaces.
If f is continuous at $p \in E$ and g is continuous at $f(p)$, then $g \circ f$ is continuous at p.

Proof. Given $\varepsilon > 0$, choose $\eta > 0$ for g at $f(p)$, then choose $\delta > 0$ for f at p with tolerance η.
If $x \in E$ and $d_X(x,p) < \delta$, then $d_Y(f(x), f(p)) < \eta$, hence $d_Z(g(f(x)), g(f(p))) < \varepsilon$.
Structural Commentary: The output tolerance for g becomes the input tolerance for f.
Key Insight: Continuity composes by passing epsilon through an intermediate eta. □

Theorem 5.1.16. *The Distance Function Is Continuous*
Let (X, d) be a metric space and $D : X \times X \to \mathbb{R}$, $D(x,y) := d(x,y)$.
For $d_\times \in \{d_1, d_2, d_\infty\}$ on $X \times X$, $D : (X \times X, d_\times) \to \mathbb{R}$ is continuous.
Equivalently, $x_n \to x$ and $y_n \to y$ in X imply $d(x_n, y_n) \to d(x,y)$.

Proof. For $x, y, u, v \in X$, $|D(x,y) - D(u,v)| = |d(x,y) - d(u,v)| \leq d(x,u) + d(y,v)$. Hence $x_n \to x$ and $y_n \to y$ imply $D(x_n, y_n) \to D(x,y)$. Proposition 3.4.9 and Theorem 5.1.14 give continuity for each product metric.
Structural Commentary: Distance changes continuously when both endpoints move.
Key Insight: The triangle inequality controls distance error by endpoint errors. □

Theorem 5.1.17. *Continuity via Open Preimages*
For metric spaces X, Y, map $f : X \to Y$ is continuous if and only if $f^{-1}(V)$ is open in $X \forall$ open $V \subseteq Y$.

Proof.
($\Rightarrow$) If $p \in f^{-1}(V)$ and V is open, choose $\varepsilon > 0$ with $B_\varepsilon(f(p)) \subseteq V$.
Continuity gives $\delta > 0$ with $B_\delta(p) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open.
($\Leftarrow$) For $p \in X$ and $\varepsilon > 0$, the ball $V = B_\varepsilon(f(p))$ is open by Theorem 3.1.13.
Since $p \in f^{-1}(V)$ and $f^{-1}(V)$ is open, some $B_\delta(p) \subseteq f^{-1}(V)$; this is continuity at p.

Structural Commentary: Continuity is exactly preservation of openness under inverse image.
Key Insight: Images are not Boolean-exact, but preimages are; continuity belongs to preimages. □

Proposition 5.1.18. *Continuity on a Relative Domain*
Let (X, d_X) and (Y, d_Y) be metric spaces, let $E \subseteq X$, and let $f : E \to Y$. The following are equivalent:
(i) $f : (E, d_X|_{E\times E}) \to (Y, d_Y)$ is continuous.
(ii) For every open $V \subseteq Y$, $f^{-1}(V)$ is open relative to E.
(iii) For every open $V \subseteq Y$, there exists open $G \subseteq X$ such that $f^{-1}(V) = E \cap G$.

Proof.
(i) $\Rightarrow$ (ii) For open $V \subseteq Y$, Theorem 5.1.17 on the metric space E gives $f^{-1}(V)$ open in E.
(ii) $\Rightarrow$ (iii) For open $V \subseteq Y$, relative openness of $f^{-1}(V) \subseteq E$ gives $f^{-1}(V) = E \cap G$ for some open $G \subseteq X$.
(iii) $\Rightarrow$ (i) For open $V \subseteq Y$, $f^{-1}(V) = E \cap G$ with G open in X, hence $f^{-1}(V)$ is open in E.
Theorem 5.1.17 on E gives continuity.

Structural Commentary: Open preimages are relative opens in E.
Key Insight: Relative continuity says ordinary open sets in X are seen through intersection with E. □

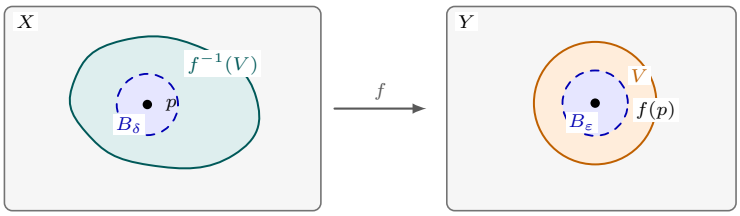

Figure 143: Continuity pulls open sets back to open sets.

Corollary 5.1.19. *Continuity via Closed Preimages*
Let X, Y be metric spaces, $f : X \to Y$. Then f is continuous iff $f^{-1}(C)$ is closed in X for every closed $C \subseteq Y$.

Proof.
($\Rightarrow$) For closed $C \subseteq Y$, $X \setminus f^{-1}(C) = f^{-1}(Y \setminus C)$ is open by Theorem 5.1.17; hence $f^{-1}(C)$ is closed.
($\Leftarrow$) For open $V \subseteq Y$, $X \setminus f^{-1}(V) = f^{-1}(Y \setminus V)$ is closed by hypothesis; hence $f^{-1}(V)$ is open, QED.

Structural Commentary: Closed-set continuity is the complement form of open-set continuity.
Key Insight: $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ transfers open and closed formulations. □

Perspective 5.1.20. *Three Views of Continuity*
For metric-space maps, continuity is equivalently ε–δ control, $x_n \to x \Rightarrow f(x_n) \to f(x)$,
and $V \subseteq Y$ open $\Rightarrow f^{-1}(V)$ open; see Definition 5.1.9 and Theorems 5.1.14, 5.1.17.
Key Insight: Continuity sends near points forward and pulls open sets back.

Proposition 5.1.21. *Distance-to-Set Is Continuous*
Let (X, d) be a metric space, let $\emptyset \neq E \subseteq X$, and define $f : X \to \mathbb{R}$ by $f(p) := \operatorname{dist}(p, E)$.
(i) f is 1-Lipschitz: $|f(p) - f(q)| \leq d(p, q)$ for all $p, q \in X$; in particular, f is continuous.
(ii) $f^{-1}(\{0\}) = \overline{E}$.

Proof.
(i) For $p, q \in X$ and $e \in E$, $d(p, e) \leq d(p, q) + d(q, e)$. Taking the infimum over $e \in E$ gives $f(p) \leq d(p, q) + f(q)$, hence $f(p) - f(q) \leq d(p, q)$. Hence $f(q) - f(p) \leq d(p, q)$, so $|f(p) - f(q)| \leq d(p, q)$, QED.
(ii) For $p \in X$, $p \in f^{-1}(\{0\})$ iff $\operatorname{dist}(p, E) = 0$ iff $p \in \overline{E}$ by Proposition 3.1.24.

Structural Commentary: Distance to a fixed set changes no faster than the point itself moves.
Key Insight: The triangle inequality turns nearest-set distance into a continuous detector of closure. □

Corollary 5.1.22. *Open and Closed Sets by Distance Functions*
Let X be a metric space.
(i) A set $O \subseteq X$ is open if and only if there is a continuous $f : X \to \mathbb{R}$ such that $O = \{x \in X : f(x) > 0\}$.
(ii) A set $F \subseteq X$ is closed if and only if there is a continuous $g : X \to \mathbb{R}$ such that $F = g^{-1}(\{0\})$.

Proof.
(i) If $O = X$, take $f \equiv 1$. Otherwise set $f(x) := \operatorname{dist}(x, O^c)$. Since O^c is closed, Proposition 5.1.21 gives $\{f = 0\} = O^c$, hence $O = \{f > 0\}$. Conversely, $\{f > 0\} = f^{-1}((0, \infty))$ is open by Theorem 5.1.17.
(ii) If $F = \emptyset$, take $g \equiv 1$. Otherwise set $g(x) := \operatorname{dist}(x, F)$.
If F closed, by Proposition 5.1.21, $g^{-1}(\{0\}) = \overline{F} = F$. Conversely, $g^{-1}(\{0\})$ closed by Corollary 5.1.19.

Structural Commentary: Metric spaces encode open and closed sets by continuous real-valued distance functions.
Key Insight: A set is closed exactly when its distance-zero locus has no extra points. □

Definition 5.1.23. *Homeomorphism*
A *homeomorphism* from X to Y is a bijection $f : X \to Y$ such that both f and f^{-1} are continuous.
If such a map exists, X and Y are *homeomorphic*, written $X \simeq Y$.

Proposition 5.1.24. *Homeomorphisms Preserve Topology*
If $f : X \to Y$ is a homeomorphism, then $U \mapsto f(U)$ bijects open subsets of X with open subsets of Y.

Proof. If $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open in Y since f^{-1} is continuous. If $V \subseteq Y$ is open, then $f^{-1}(V)$ is open in X since f is continuous. These two assignments are inverse because f is bijective.
Structural Commentary: A homeomorphism preserves open sets in both directions.
Key Insight: Topological sameness is continuity plus continuous reversibility. □

Definition 5.1.25. *Metric Isometry*
Let $(X, d_X), (Y, d_Y)$ be metric spaces.

(i) A map $f : X \to Y$ is an *isometry* if $\forall x, y \in X$, $d_Y(f(x), f(y)) = d_X(x, y)$.
(ii) X and Y are *isometric* if $\exists$ a bijective isometry $f : X \to Y$.

Theorem 5.1.26. *Basic Properties of Metric Isometries*
Let $f : (X, d_X) \to (Y, d_Y)$ be an isometry.

(i) f is injective.
(ii) f is continuous.
(iii) $f^{-1} : (f(X), d_Y|_{f(X)\times f(X)}) \to (X, d_X)$ is an isometry.

Proof.

(i) If $f(x) = f(y)$, then $d_X(x, y) = d_Y(f(x), f(y)) = 0$, hence $x = y$.
(ii) If $x_n \to x$ in X, then $d_Y(f(x_n), f(x)) = d_X(x_n, x) \to 0$; by Theorem 5.1.14, f is continuous.
(iii) For $u = f(x), v = f(y) \in f(X)$, $d_X(f^{-1}(u), f^{-1}(v)) = d_X(x, y) = d_Y(f(x), f(y)) = d_Y(u, v)$; QED.

Structural Commentary: An isometry preserves all distances, so injectivity and continuity are automatic.
Key Insight: Distance preservation is stronger than topological equivalence. □

Remark 5.1.27. *Metric Versus Linear Isometries*
Metric isometry is purely distance-theoretic. If $T : V \to W$ is linear between inner-product spaces, then $\|Tv\| = \|v\|\ \forall v \in V \iff d_W(Tu, Tv) = d_V(u, v)\ \forall u, v \in V$.

Example 5.1.28. *Continuous Bijection Need Not Be a Homeomorphism*
Let $f : [0, 2\pi) \to S^1$, and $f(t) = (\cos t, \sin t)$. This continuous bijection is not a homeomorphism: points of S^1 approaching $(1, 0)$ from below have preimages tending to 2π, not 0.

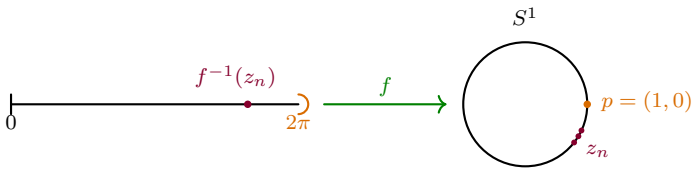

Figure 144: Parametrization $[0, 2\pi) \to S^1$ wraps continuously and bijectively, but inverse jumps at $(1, 0)$.

Remark 5.1.29. *Continuity Only Uses the Domain Metric*
For $f : E \to Y$ with $E \subseteq X$, continuity on E depends only on the restricted metric on E, not on $X \setminus E$.

Definition 5.1.30. *Product Map*
If $f : X \to Y$ and $g : X \to Z$ are maps between metric spaces, their *product map*, or direct sum, is $f \oplus g : X \to Y \times Z$, $(f \oplus g)(x) := (f(x), g(x))$, where $Y \times Z$ carries any of product metrics in Definition 3.4.8.

Proposition 5.1.31. *Continuity of Product Maps*
The product map $f \oplus g : X \to Y \times Z$ is continuous at x_0 iff f and g are continuous at x_0.
Hence it is continuous on X iff both components are.

Proof. By Proposition 3.4.9, it suffices to use d_{sum}.

($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ from continuity of $f \oplus g$ at x_0.
If $d_X(x, x_0) < \delta$, then $d_Y(f(x), f(x_0)), d_Z(g(x), g(x_0)) \le d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) < \varepsilon$.
($\Leftarrow$) Given $\varepsilon > 0$, choose δ_f, δ_g for f, g at tolerance $\varepsilon/2$ and set $\delta = \min\{\delta_f, \delta_g\}$.
Then $d_X(x, x_0) < \delta$ gives $d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) = d_Y(f(x), f(x_0)) + d_Z(g(x), g(x_0)) < \varepsilon$.

Structural Commentary: Product continuity is exactly componentwise continuity.
Key Insight: A product metric packages two output estimates into one. □

Theorem 5.1.32. *Continuous Algebra and Components*
Let X be a metric space.

(i) If $f, g : X \to \mathbb{C}$ are continuous, so are $f + g$, fg, and f/g where $g \neq 0$.
(ii) For $f = (f_1, \ldots, f_k) : X \to \mathbb{R}^k$, f is continuous if and only if each f_j is continuous.
(iii) If $f, g : X \to \mathbb{R}^k$ are continuous, so are $f + g$ and $f \cdot g$.

Proof.

(i) At isolated points continuity is automatic; at limit points, use Theorems 5.1.7 and 5.1.11.
(ii) By Theorem 3.4.7, convergence in $\mathbb{R}^k$ is exactly coordinatewise convergence; apply this to $f(x_n) \to f(x)$.
(iii) Apply (ii) to the coordinate functions of $f+g$ and $f \cdot g$, using the scalar algebra from (i).

Structural Commentary: Continuity is stable under the algebraic operations already stable for limits.
Key Insight: Componentwise control is equivalent to Euclidean control in finite dimensions. □

Example 5.1.33. *Coordinate, Polynomial, Rational, and Norm Maps*
Coordinate functions on $\mathbb{R}^k$ are continuous; hence polynomials are continuous, and rational functions are continuous where defined. Also $\mathbf{x} \mapsto \|\mathbf{x}\|_2$ is continuous since $\big|\|\mathbf{x}\|_2 - \|\mathbf{y}\|_2\big| \le \|\mathbf{x}-\mathbf{y}\|_2$.
Thus $p \mapsto \|f(p)\|_2$ is continuous whenever $f : X \to \mathbb{R}^k$ is, by Theorem 5.1.15.

Definition 5.1.34. *Bounded Mapping*
A mapping $f : E \to \mathbb{R}^k$ is *bounded* if there exists $M \in \mathbb{R}$ such that $\|f(x)\|_2 \le M$ for every $x \in E$.

Theorem 5.1.35. *Continuous Images of Compact Sets Are Compact*
If $f : X \to Y$ is continuous and X is compact, then $f(X)$ is compact.

Proof. Let $\{V_\alpha\}$ cover $f(X)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of X by Theorem 5.1.17.
Choose $\alpha_1, \dots, \alpha_n$ with $X \subseteq \bigcup_{i=1}^n f^{-1}(V_{\alpha_i})$; then $f(X) \subseteq \bigcup_{i=1}^n V_{\alpha_i}$.
Structural Commentary: Pull back, take a finite subcover, push forward.
Key Insight: Compactness is preserved by continuous images because inverse images preserve openness. □

Corollary 5.1.36. *Continuous Images into $\mathbb{R}^k$ Are Closed and Bounded*
If $f : X \to \mathbb{R}^k$ is continuous and X is compact, then $f(X)$ is closed and bounded. In particular, f is bounded.

Proof. By Theorem 5.1.35, $f(X)$ is compact.
By Theorem 4.2.5, compact subsets of $\mathbb{R}^k$ are closed and bounded; hence f is bounded.
Structural Commentary: Compactness in the domain becomes closed-bounded geometry in Euclidean range.
Key Insight: Continuous functions on compact spaces cannot escape to infinity. □

Theorem 5.1.37. *Extreme Value Theorem*
Let $f : X \to \mathbb{R}$ be continuous, let X be compact.
If $M := \sup_{p \in X} f(p)$, $m := \inf_{p \in X} f(p)$, then $\exists\ p, q \in X$ such that $f(p) = M$ and $f(q) = m$.

Proof. By Corollary 5.1.36, $f(X)$ is closed and bounded in $\mathbb{R}$. Thus $\sup f(X)$ and $\inf f(X)$ exist and belong to $f(X)$ by Theorems 3.3.5 and 5.3.3. Hence $M = f(p)$ and $m = f(q)$ for some $p, q \in X$.
Structural Commentary: Compactness turns suprema and infima into attained values.
Key Insight: Closed bounded images contain their endpoints. □

Theorem 5.1.38. *Compact Bijective Maps Have Continuous Inverses*
If $f : X \to Y$ is a continuous bijection from compact X to a metric space Y, then $f^{-1} : Y \to X$ is continuous.

Proof. It suffices, by Theorem 5.1.17, to show that $f(V)$ is open in Y for every open $V \subseteq X$. If V is open, then V^c is compact by Theorem 4.1.6, so $f(V^c)$ is compact, hence closed by Theorems 5.1.35 and 4.1.5.
Since f is bijective, $f(V) = Y \setminus f(V^c)$ is open.
Structural Commentary: Compactness turns closed complements into closed images.
Key Insight: A continuous bijection from compact domain has no room for inverse discontinuity. □

Remark 5.1.39. *Preimages, Not Forward Images*
Theorem 5.1.17 is about inverse images. A continuous map need not send open sets to open sets or closed sets to closed sets; compactness is one hypothesis that recovers closedness of images through Theorem 5.1.35.

Theorem 5.1.40. *Continuous Images of Connected Sets Are Connected*
If $f : X \to Y$ is continuous and $E \subseteq X$ is connected, then $f(E)$ is connected.

Proof. If $f(E) = A \cup B$ with A, B nonempty separated, set $G := E \cap f^{-1}(A)$ and $H := E \cap f^{-1}(B)$. Then $E = G \cup H$ and G, H are nonempty. Since $G \subseteq f^{-1}(A)$ and $f^{-1}(A)$ is closed, $\overline{G} \subseteq f^{-1}(A)$; hence $\overline{G} \cap H = \emptyset$. Similarly $G \cap \overline{H} = \emptyset$, so G, H separate E, a contradiction.
Structural Commentary: A separation of the image pulls back to a separation of the domain.
Key Insight: Continuous maps cannot create a gap in the image of a connected set. □

Theorem 5.1.41. *Intermediate Value Theorem*
If $f : [a,b] \to \mathbb{R}$ is continuous and c lies strictly between $f(a)$ and $f(b)$, then $f(x) = c$ for some $x \in (a,b)$.

Proof. The interval $[a,b]$ is connected by Theorem 5.1.2, so $f([a,b])$ is connected by Theorem 5.1.40.
Connected subsets of $\mathbb{R}$ contain every value between any two of their points; hence $c = f(x)$ for some $x \in [a,b]$. Since $c \neq f(a), f(b)$, $x \in (a,b)$.
Structural Commentary: The intermediate value theorem is connectedness of intervals transported by continuity.
Key Insight: Continuous real functions on intervals cannot jump over a value. □

Method 5.1.42. *Bisection Proof of the Intermediate Value Theorem*
Let $f : [a,b] \to \mathbb{R}$ be continuous and let c lie between $f(a)$ and $f(b)$.
Replacing (f, c) by $(-f, -c)$ if needed, assume $f(a) \le c \le f(b)$.
If $c = f(a)$ or $c = f(b)$, take the corresponding endpoint; otherwise $f(a) < c < f(b)$.

(i) Set $[a_1, b_1] := [a, b]$.
(ii) Given $[a_n, b_n]$ with $f(a_n) \le c \le f(b_n)$, set $m_n := (a_n + b_n)/2$.
If $f(m_n) \ge c$, set $[a_{n+1}, b_{n+1}] := [a_n, m_n]$; otherwise set $[a_{n+1}, b_{n+1}] := [m_n, b_n]$.

Then, for all n, $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$, $f(a_n) \le c \le f(b_n)$, and $b_n - a_n = (b-a)/2^{n-1}$.
By Theorem 4.2.1, choose $x_0 \in \bigcap_n [a_n, b_n]$. Since $b_n - a_n \to 0$, the intersection is $\{x_0\}$ and $a_n, b_n \to x_0$.
Continuity gives $f(a_n), f(b_n) \to f(x_0)$, so $f(x_0) \le c \le f(x_0)$; hence $f(x_0) = c$.
If $f(a) < c < f(b)$, then $x_0 \in (a, b)$.
Key Insight: Bisection keeps the target value bracketed while nested intervals collapse to one point.

Definition 5.1.43. *Path-Connected Set*
A set E in a metric space is *path-connected* if for every $p, q \in E$ there is a continuous map $\gamma : [a, b] \to E$ with $\gamma(a) = p$ and $\gamma(b) = q$. Such a map γ is a *path* from p to q.

Theorem 5.1.44. *Path-Connected Sets Are Connected*
Every path-connected set is connected.

Proof. Suppose $E = A \cup B$ with A, B nonempty separated.
Choose $p \in A$, $q \in B$, and a path $\gamma : [a, b] \to E$ from p to q. Since $[a, b]$ is connected, so is $\gamma([a, b])$.
But $\gamma([a, b])$ meets both A and B, which separate it, a contradiction.
Structural Commentary: A path transports connectedness from an interval to the set.
Key Insight: To connect two points by a path, the path image cannot be split into two separated pieces. □

Corollary 5.1.45. *Convex Sets Are Path-Connected*
Every convex subset of $\mathbb{R}^k$ is path-connected.

Proof. For $\mathbf{x}, \mathbf{y} \in E$, set $\gamma(t) = (1-t)\mathbf{x} + t\mathbf{y}$ on $[0, 1]$. Convexity gives $\gamma([0, 1]) \subseteq E$, with $\gamma(0) = \mathbf{x}$ and $\gamma(1) = \mathbf{y}$.
Since the coordinates of γ are affine in t, γ is continuous. Hence E is path-connected.
Structural Commentary: Convexity supplies the whole straight path between any two points.
Key Insight: In Euclidean space, line segments are continuous paths. □

5.2 Uniform Continuity and Compactness

Definition 5.2.1. *Uniform Continuity*
Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$.
The map f is *uniformly continuous* on X if $\forall \varepsilon > 0\, \exists \delta > 0\, \forall p, q \in X,\ d_X(p, q) < \delta \implies d_Y(f(p), f(q)) < \varepsilon$.
Structural Commentary: Uniform continuity is setwise: for each ε, one δ must work for all pairs in the set.

Principle 5.2.2. *Challenge–Response Order*
Limits and continuity are controlled by quantifier order. In Definition 5.1.4, the ε-challenge at the range point must be answered by a δ-restriction near the domain point. Uniform continuity, Definition 5.2.1, strengthens the same pattern by requiring one δ to work for all relevant domain points at once.

Proposition 5.2.3. *Sequential Test for Uniform Continuity*
Let X, Y be metric spaces and let $f : X \to Y$. The following are equivalent:

(i) f is uniformly continuous on X;
(ii) whenever (p_n) and (q_n) are sequences in X with $d_X(p_n, q_n) \to 0$, one has $d_Y(f(p_n), f(q_n)) \to 0$.

Proof.

(i) $\Rightarrow$ (ii) Given $\varepsilon > 0$, choose $\delta > 0$ such that $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$ for all $x, y \in X$.
Since $d_X(p_n, q_n) \to 0$, eventually $d_X(p_n, q_n) < \delta$, hence eventually $d_Y(f(p_n), f(q_n)) < \varepsilon$.

(ii) $\Rightarrow$ (i) Prove the contrapositive. If f is not uniformly continuous, then $\exists \varepsilon_0 > 0$ such that $\forall n \ge 1$, $\exists p_n, q_n \in X$ with $d_X(p_n, q_n) < 1/n$ and $d_Y(f(p_n), f(q_n)) \ge \varepsilon_0$.
Thus $d_X(p_n, q_n) \to 0$ but $d_Y(f(p_n), f(q_n)) \not\to 0$, contradicting (ii).

Structural Commentary: Uniform continuity preserves vanishing pairwise distances.
Key Insight: Failure of one global δ manufactures two close input sequences with separated outputs. □

Definition 5.2.4. *Lipschitz Map*
Let (X, d_X) and (Y, d_Y) be metric spaces, let $f : X \to Y$, and let $L \ge 0$.
The map f is *L-Lipschitz* if $\forall p, q \in X$, $d_Y(f(p), f(q)) \le L d_X(p, q)$.
It is *Lipschitz* if it is L-Lipschitz for some $L \ge 0$.
Key Insight: Lipschitz control is one global linear modulus of continuity.

Proposition 5.2.5. *Lipschitz Implies Uniform Continuity*
Every Lipschitz map between metric spaces is uniformly continuous, hence continuous.

Proof. Let f be L-Lipschitz and let $\varepsilon > 0$. If $L = 0$, then $d_Y(f(p), f(q)) = 0$ for all $p, q \in X$.
If $L > 0$, take $\delta := \varepsilon / L$; $d_X(p, q) < \delta$ implies $d_Y(f(p), f(q)) \le L d_X(p, q) < \varepsilon$.
Thus f is uniformly continuous. Fixing one point gives continuity.
Structural Commentary: A global Lipschitz bound gives a global modulus.
Key Insight: One constant controls all pairs. □

Example 5.2.6. *Continuous Need Not Be Lipschitz*
On $[0, 1]$, $f(x) = \sqrt{x}$ is continuous since $\left|\sqrt{x} - \sqrt{y}\right| \le \sqrt{|x - y|}$.
It is not Lipschitz: L-Lipschitz would imply $\sqrt{x} \le Lx$ for $0 < x \le 1$, i.e. $1/\sqrt{x} \le L$, impossible as $x \downarrow 0$.

Theorem 5.2.7. *Continuity on Compact Sets Is Uniform*
If $f : X \to Y$ is continuous and X is compact, then f is uniformly continuous.

Proof. Let $\varepsilon > 0$. For each $a \in X$, choose $r_a > 0$ such that $d_X(x, a) < r_a$ implies $d_Y(f(x), f(a)) < \varepsilon/2$.
Compactness gives a finite cover $X \subseteq \bigcup_{i=1}^{n} B_{r_{a_i}/2}(a_i)$. Set $\delta := \frac{1}{2} \min_i r_{a_i} > 0$.
If $d_X(p, q) < \delta$ and $p \in B_{r_{a_i}/2}(a_i)$, then $d_X(q, a_i) < r_{a_i}$ also; hence $f(p)$ and $f(q)$ within $\varepsilon/2$ of $f(a_i)$. QED.
Structural Commentary: Compactness converts pointwise radii into finitely many radii with positive minimum.
Key Insight: Uniform continuity is continuity with the point-dependence removed by a finite subcover. □

Example 5.2.8. *Separating Continuity, Uniform Continuity, and Lipschitz Control*

(i) Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$. Then f is continuous, but not uniformly continuous: with $p_n = n$ and $q_n = n + 1/n$, $|p_n - q_n| = 1/n \to 0$ while $|f(q_n) - f(p_n)| = 2 + 1/n^2 \not\to 0$.

(ii) Let $g : [0,1] \to \mathbb{R}$ be $g(x) = \sqrt{x}$. Then g is uniformly continuous by Theorem 5.2.7. If g were L-Lipschitz, then $1/n = |g(1/n^2) - g(0)| \le L/n^2$ $\forall n \in \mathbb{N}$, so $n \le L$ $\forall n$, impossible. Hence g is not Lipschitz.

Structural Commentary: Both reverse implications fail.
Key Insight: Compactness repairs uniform continuity, but it does not force a linear modulus.

Theorem 5.2.9. *Compactness Hypotheses Are Essential*
Let E be a noncompact subset of $\mathbb{R}$. Then:

(i) there exists a continuous function on E which is not bounded;
(ii) there exists a continuous bounded function on E which has no maximum;
(iii) if E is bounded, then there exists a continuous function on E which is not uniformly continuous.

Proof.

(i) If E is unbounded, $f(x) = x$ works. If E is bounded, Theorem 4.2.5 gives $x_0 \in \overline{E} \setminus E$, and $f(x) = 1/(x - x_0)$ is continuous on E and unbounded.

(ii) If E is unbounded, $h(x) = x^2/(1 + x^2)$ has $\sup_E h = 1$ but $h < 1$. If E is bounded, with x_0 as above, $g(x) = 1/(1 + (x - x_0)^2)$ is bounded by 1 and has supremum 1 but no maximum.

(iii) If E is bounded, let $f(x) = 1/(x - x_0)$. Given $\delta > 0$, let $x \in E$ with $|x - x_0| < \delta/2$, then $y \in E$ with $|y - x_0| < \min\{\delta/2, 1/(|f(x)| + 1)\}$. Then $|x - y| < \delta$, $|f(x) - f(y)| > 1$, so f is not uniformly continuous.

Structural Commentary: Compactness is essential in the preceding theorems.
Key Insight: Noncompact sets either escape to infinity or miss an accumulation point. □

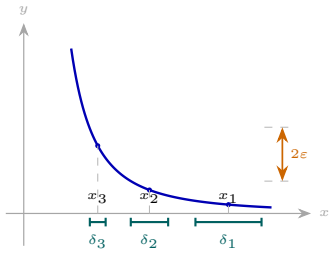

Figure 145: For $f(x) = 1/x^2$ near 0, a fixed vertical tolerance forces smaller pointwise horizontal tolerances.

Method 5.2.10. *Continuity Proof Patterns*

(i) *ε–δ at p*: prove $\forall \varepsilon > 0\, \exists \delta > 0$ such that $x \in E$ and $d_X(x,p) < \delta$ imply $d_Y(f(x), f(p)) < \varepsilon$.
(ii) *Limit at p*: prove same estimate with $0 < d_X(x,p) < \delta$, target q; equivalently test all $x_n \to p$, $x_n \neq p$.
(iii) *Topological continuity*: prove $V \subseteq Y$ open $\Rightarrow f^{-1}(V)$ open, or $C \subseteq Y$ closed $\Rightarrow f^{-1}(C)$ closed.
(iv) *Compactness*: from open cover $\{V_\alpha\}$ of $f(X)$, pulled-back cover $\{f^{-1}(V_\alpha)\}$ of X extract finite subcover.
(v) *Uniform continuity*: choose local radii r_p, cover compact X by $B_{r_p/2}(p)$, and set $\delta = \frac{1}{2}\min_i r_{p_i}$.

Key Insight: Continuity is local control; compactness turns local controls into finite control.

5.3 One-Sided Limits, Monotone Functions, and Extended Limits

Definition 5.3.1. *One-Sided Limits and Types of Discontinuity*
Let $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, $x_0 \in \mathbb{R}$, $q \in \mathbb{R}$, $X_+ := X \cap (x_0, \infty)$, and $X_- := X \cap (-\infty, x_0)$.

(i) If x_0 is a limit point of X_+, then $f(x_0+) = q$ means $\lim_{x \to x_0}(f|_{X_+})(x) = q$.
(ii) If x_0 is a limit point of X_-, then $f(x_0-) = q$ means $\lim_{x \to x_0}(f|_{X_-})(x) = q$.
(iii) If x_0 is a limit point of X_+ and X_-, then $\lim_{x \to x_0}(f|_{X \setminus \{x_0\}})(x)$ exists iff $f(x_0+)$ and $f(x_0-)$ exist with $f(x_0+) = f(x_0-)$.

For an interval I, $f : I \to \mathbb{R}$, and an interior discontinuity $x_0 \in I^\circ$:

(i) The discontinuity is of the *first kind*, or *simple*, iff both $f(x_0+)$ and $f(x_0-)$ exist.
(ii) A first-kind discontinuity is a *jump* iff $f(x_0+) \neq f(x_0-)$.
(iii) A first-kind discontinuity is *removable* iff $f(x_0+) = f(x_0-) \neq f(x_0)$.
(iv) The discontinuity is of the *second kind* iff at least one of $f(x_0+)$ and $f(x_0-)$ fails to exist.

Example 5.3.2. *Basic Discontinuity Types*
Dirichlet's function $\mathbf{1}_{\mathbb{Q}}$ has a discontinuity of the second kind at every point.
The function $f(x) = x$ on $\mathbb{Q}$ and $f(x) = 0$ off $\mathbb{Q}$ is continuous only at 0; elsewhere its discontinuities are of the second kind. A jump in a piecewise linear function is a simple discontinuity.
Use Corollary 1.1.22: every interval contains rational and irrational points.

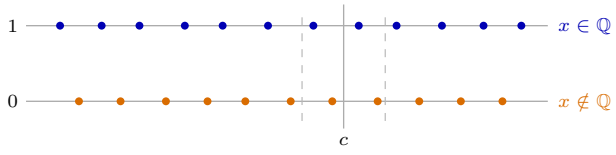

Figure 146: For $g = \mathbf{1}_{\mathbb{Q}}$, interval around c has rational inputs value 1 and irrational inputs value 0.

Definition 5.3.3. *Monotone Functions*
Let $f : (a, b) \to \mathbb{R}$.

(i) f is *monotonically increasing* iff $a < x < y < b$ implies $f(x) \le f(y)$.
(ii) f is *monotonically decreasing* iff $a < x < y < b$ implies $f(x) \ge f(y)$.
(iii) f is *monotone* iff f is monotonically increasing or monotonically decreasing.

Theorem 5.3.4. *One-Sided Limits of Monotone Functions*
If f is increasing on (a, b), then for every $x \in (a, b)$, $\sup_{a<t<x} f(t) = f(x-) \le f(x) \le f(x+) = \inf_{x<t<b} f(t)$.
Moreover, $a < x < y < b$ implies $f(x+) \le f(y-)$. For decreasing f, reverse the inequalities.

Proof. Let $A := \sup_{a<t<x} f(t)$.
For every $\varepsilon > 0$, choose $s < x$ with $A - \varepsilon < f(s) \le A$; then $s < t < x$ implies $A - \varepsilon < f(t) \le A$, so $f(x-) = A$.
Let $B := \inf_{x<t<b} f(t)$. For every $\varepsilon > 0$, choose $s > x$ with $B \le f(s) < B + \varepsilon$; then $x < t < s$ implies $B \le f(t) < B + \varepsilon$, so $f(x+) = B$.
If $x < y$, then $\inf_{x<t<y} f(t) \le \sup_{x<t<y} f(t)$ gives $f(x+) \le f(y-)$.
For decreasing f, apply the increasing case to $-f$ and multiply the resulting inequalities by -1.
Structural Commentary: Monotonicity converts one-sided limits into suprema and infima.
Key Insight: A monotone function has jumps, but not oscillatory one-sided failure. □

Corollary 5.3.5. *Monotone Functions Have No Second-Kind Discontinuities*
A monotone real function on an interval has no discontinuities of the second kind.

Proof. At every interior point, monotonicity gives both one-sided limits, so any discontinuity is of the first kind.
Structural Commentary: Monotonicity rules out oscillatory discontinuities.
Key Insight: All monotone discontinuities are jumps or removable value mismatches. □

Theorem 5.3.6. *Monotone Continuity and Interval Images*
Let $f : (a, b) \to \mathbb{R}$ be monotone. Then f is continuous iff $f((a, b))$ is an interval.

Proof. Put $I = (a, b)$.

($\Rightarrow$) If f is continuous, then $f(I)$ is connected, hence an interval.
($\Leftarrow$) Contrapositively, let f be discontinuous at $x_0 \in I$. Replace f by $-f$ if needed, assume f increasing.
Then $f(x_0-) \le f(x_0) \le f(x_0+)$, and discontinuity gives $f(x_0-) < f(x_0+)$.
Choose $y \in (f(x_0-), f(x_0+)) \setminus \{f(x_0)\}$. Then $x < x_0$ gives $f(x) < y$, $x > x_0$,$f(x) > y$, $f(x_0) \ne y$.
So $y \notin f(I)$ while $f(I)$ has values below and above y. Hence $f(I)$ is not an interval.

Structural Commentary: Monotone discontinuities create image gaps.
Key Insight: Interval images exclude jumps. □

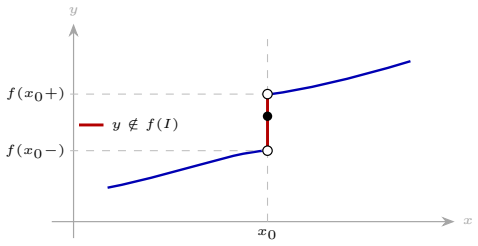

Figure 147: A jump of an increasing function leaves values between $f(x_0-)$ and $f(x_0+)$ out of the image; the red marker indicates such a $y \notin f(I)$, so $f(I)$ cannot be an interval.

Theorem 5.3.7. *Monotone Discontinuities Are Countable*
If f is monotone on (a, b), then the set of points of (a, b) at which f is discontinuous is at most countable.

Proof. Assume f is increasing, and let E be its discontinuity set.
For each $x \in E$, Theorem 5.3.4 gives $f(x-) < f(x+)$; choose $r(x) \in \mathbb{Q}$ with $f(x-) < r(x) < f(x+)$.
If $x_1 < x_2$, then $f(x_1+) \le f(x_2-)$, so $r(x_1) \ne r(x_2)$. Thus E injects into $\mathbb{Q}$, hence is countable by Corollary 6.2.19. If f is decreasing, apply the increasing case to $-f$, which has the same discontinuity set.
Structural Commentary: Each jump contains a rational marker, and monotonicity keeps the markers distinct.
Key Insight: Countability enters by injecting discontinuities into $\mathbb{Q}$. □

Remark 5.3.8. *Monotone Discontinuities Need Not Be Isolated*
For countable $E = \{x_n : n \in \mathbb{N}\} \subset (a, b)$ and positive summable (c_n), $f(x) = \sum_{x_n<x} c_n$ is increasing, has jumps of size c_n at x_n, and is continuous off E. Thus monotone discontinuities may be dense.

Definition 5.3.9. *Extended Limits and Limits at Infinity*
Neighborhoods of $+\infty$ are intervals $(c, +\infty)$; neighborhoods of $-\infty$ are $(-\infty, c)$.
For $f : E \to \mathbb{R}$ and extended real A, x, $f(t) \to A$ as $t \to x$ means: every neighborhood U of A contains $f(t)$ for all punctured $t \in E$ sufficiently near x, i.e. $\exists V \ni x$ such that $t \in V \cap E,\ t \ne x \implies f(t) \in U$.
For real A, x, this agrees with Definition 5.1.4.
Structural Commentary: Extended limits use same neighborhood test, $\pm\infty$ allowed as input or output limits.

Theorem 5.3.10. *Algebra of Extended Limits*
Let $f, g : E \to \mathbb{R}$ and $f(t) \to A$, $g(t) \to B$ as $t \to x$, with A, B, x extended real. Then, whenever RHS is defined:
(i) if $f(t) \to A'$ as $t \to x$, then $A' = A$;

(ii) $(f+g)(t) \to A+B$;
(iii) $(fg)(t) \to AB$;
(iv) $(f/g)(t) \to A/B$;
The forms $\infty - \infty$, $0 \cdot \infty$, ∞/∞, *and* $A/0$ *are undefined.*

Proof.

(i) If $A' \neq A$, choose disjoint neighborhoods U of A and U' of A'. The two limits give a common neighborhood V of x such that $t \in V \cap E$, $t \neq x$ implies $f(t) \in U \cap U'$, a contradiction.
(ii) For any neighborhood W of $A+B$, definedness gives neighborhoods U of A and V of B such that $u+v \in W$ for $u \in U$, $v \in V$. Apply the two limits in a common punctured neighborhood of x.
(iii) The same neighborhood argument applies to multiplication, excluding $0 \cdot \infty$.
(iv) Apply it to division; since $B \neq 0$, choose the neighborhood of B away from 0.

Structural Commentary: Extended limit algebra is ordinary limit algebra in neighborhood form.
Key Insight: The undefined forms are exactly where neighborhood control breaks. □

6 General Topology for Analysis

6.1 Topological Spaces, Bases, and Separation

General topology abstracts the open-set behavior of metric spaces. This selective pass records spaces, bases, subspaces, closure, and separation axioms for later use.

Definition 6.1.1. *Topological Space*
A *topology* on a set X is a collection $\mathcal{T} \subseteq 2^X$ such that:

(i) $\emptyset, X \in \mathcal{T}$;
(ii) if $\{\mathcal{O}_\alpha\}_{\alpha \in A} \subseteq \mathcal{T}$, then $\bigcup_{\alpha \in A} \mathcal{O}_\alpha \in \mathcal{T}$;
(iii) if $\mathcal{O}_1, \dots, \mathcal{O}_n \in \mathcal{T}$ with $n \geq 1$, then $\bigcap_{j=1}^n \mathcal{O}_j \in \mathcal{T}$.

Then $(X, \mathcal{T})$ is a *topological space*.

Definition 6.1.2. *Open, Closed, and Neighborhoods*
Let $(X, \mathcal{T})$ be a topological space.

(i) A set $U \subseteq X$ is *open* iff $U \in \mathcal{T}$.
(ii) A set $F \subseteq X$ is *closed* iff $X \setminus F \in \mathcal{T}$.
(iii) For $x \in X$, a *neighborhood* of x is an open set $U \subseteq X$ with $x \in U$.
(iv) For $A \subseteq X$, a *neighborhood* of A is an open set $U \subseteq X$ with $A \subseteq U$.

Proposition 6.1.3. *Open Sets by Neighborhoods*
Let $(X, \mathcal{T})$ *be a topological space and let* $E \subseteq X$. *Then* $E \in \mathcal{T}$ *iff for every* $x \in E$ *there exists an open set* $U_x \in \mathcal{T}$ *such that* $x \in U_x \subseteq E$.

Proof.

($\Rightarrow$) If $E \in \mathcal{T}$, take $U_x = E$ for each $x \in E$.
($\Leftarrow$) If such U_x exists for every $x \in E$, then $E = \bigcup_{x \in E} U_x$. Since arbitrary unions of open sets are open, $E \in \mathcal{T}$.

Structural Commentary: Openness is checked pointwise by open subsets.
Key Insight: Openness is exactly the local ability to fit an open neighborhood inside the set. □

Example 6.1.4. *Discrete and Trivial Topologies*
On a nonempty set X, the *discrete topology* is 2^X, so every subset is open and closed. It is induced by the discrete metric. The *trivial topology* is $\{\emptyset, X\}$, so the only neighborhood of any point is X.

Proposition 6.1.5. *Metric Open Sets Form a Topology*
Let (X, d) *be metric space,* $B_r(x) := \{y \in X : d(x,y) < r\}$, *and* $\tau_d := \{U \subseteq X : \forall x \in U,\ \exists r > 0,\ B_r(x) \subseteq U\}$.
Then τ_d *is a topology on* X.

Proof. Clearly $\emptyset, X \in \tau_d$. If $\{\mathcal{O}_\alpha\}_{\alpha \in A} \subseteq \tau_d$ and $x \in \bigcup_{\alpha \in A} \mathcal{O}_\alpha$, choose α with $x \in \mathcal{O}_\alpha$ and $r > 0$ with $B_r(x) \subseteq \mathcal{O}_\alpha$; hence $B_r(x) \subseteq \bigcup_{\alpha \in A} \mathcal{O}_\alpha$. If $\mathcal{O}_1, \dots, \mathcal{O}_n \in \tau_d$, $n \geq 1$, and $x \in \bigcap_{j=1}^n \mathcal{O}_j$, choose $r_j > 0$ with $B_{r_j}(x) \subseteq \mathcal{O}_j$ and set $r := \min_{1 \leq j \leq n} r_j$; then $B_r(x) \subseteq \bigcap_{j=1}^n \mathcal{O}_j$. Thus τ_d is a topology on X.
Structural Commentary: Metric balls generate an open-set system with the exact axioms of a topology.
Key Insight: General topology keeps the open-set calculus and forgets the distance function. □

Definition 6.1.6. *Closure, Interior, and Boundary*
Let $E \subseteq X$ in a topological space.

(i) The *closure* $\overline{E}$ is the set of all $x \in X$ such that every neighborhood of x meets E.
(ii) The *interior* $\operatorname{int}(E)$ is the union of all open sets contained in E.
(iii) The *exterior* of E is $\operatorname{int}(X \setminus E)$.
(iv) The *boundary* of E is $\operatorname{bd}(E) := \overline{E} \setminus \operatorname{int}(E)$.

Proposition 6.1.7. *Closure, Interior, and Boundary Tests*
Let $(X, \mathcal{T})$ *be a topological space and let* $E \subseteq X$.

(i) $\operatorname{int}(E)$ *is the largest open subset of* E, *and* E *is open iff* $E = \operatorname{int}(E)$;
(ii) $\overline{E}$ *is the smallest closed subset of* X *containing* E, *and* E *is closed iff* $E = \overline{E}$;
(iii) $x \in \operatorname{bd}(E)$ *iff every neighborhood of* x *meets both* E *and* $X \setminus E$;
(iv) $\operatorname{bd}(E) = \overline{E} \cap \overline{X \setminus E}$, *so* $\operatorname{bd}(E)$ *is closed.*

Proof. By definition, $\operatorname{int}(E) = \bigcup\{U \in \mathcal{T} : U \subseteq E\}$, hence it is the largest open subset of E. If $x \notin \overline{E}$, choose a neighborhood U of x with $U \cap E = \emptyset$; then $U \subseteq X \setminus \overline{E}$, so $\overline{E}$ is closed. If F is closed and $E \subseteq F$, then $X \setminus F$ is disjoint from E, so $\overline{E} \subseteq F$. Thus $\overline{E}$ is the smallest closed subset of X containing E.

(*i*) ($\Rightarrow$) If E is open, then $E \subseteq \operatorname{int}(E) \subseteq E$, so $E = \operatorname{int}(E)$.
(*i*) ($\Leftarrow$) If $E = \operatorname{int}(E)$, then E is open because $\operatorname{int}(E)$ is open.
(*ii*) ($\Rightarrow$) If E is closed, then the smallest-closed-superset property gives $\overline{E} \subseteq E$, while $E \subseteq \overline{E}$. Hence $E = \overline{E}$.
(*ii*) ($\Leftarrow$) If $E = \overline{E}$, then E is closed because $\overline{E}$ is closed.
(*iii*) ($\Rightarrow$) If $x \in \operatorname{bd}(E) = \overline{E} \setminus \operatorname{int}(E)$, then every neighborhood of x meets E. Since $x \notin \operatorname{int}(E)$, no neighborhood of x is contained in E, so every neighborhood of x also meets $X \setminus E$.
(*iii*) ($\Leftarrow$) If every neighborhood of x meets both E and $X \setminus E$, then $x \in \overline{E}$ and $x \notin \operatorname{int}(E)$. Hence $x \in \operatorname{bd}(E)$.
(*iv*) By (iii), $x \in \operatorname{bd}(E)$ iff $x \in \overline{E}$ and $x \in \overline{X \setminus E}$. Thus $\operatorname{bd}(E) = \overline{E} \cap \overline{X \setminus E}$, which is closed by (ii).

Structural Commentary: Interior, closure, and boundary are neighborhood tests.
Key Insight: $\operatorname{int}(E)$ is certified inside; $\overline{E}$ is neighborhood-forced; $\operatorname{bd}(E)$ is where both sides persist. □

Definition 6.1.8. *Bases and Subbases*
Let X be a set.

(i) If $\mathcal{T}$ is a topology on X and $x \in X$, then a collection $\mathcal{B}_x$ of neighborhoods of x is a *neighborhood basis at x* iff every neighborhood U of x contains some $B \in \mathcal{B}_x$ with $B \subseteq U$.
(ii) A collection $\mathcal{B} \subseteq 2^X$ is a *basis* for a topology on X iff $X = \bigcup_{B \in \mathcal{B}} B$ and, whenever $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.
(iii) If $\mathcal{B}$ is a basis on X, the *topology generated by $\mathcal{B}$* is $\mathcal{T}_{\mathcal{B}} := \{\bigcup_{B \in \mathcal{U}} B : \mathcal{U} \subseteq \mathcal{B}\}$.
(iv) A collection $\mathcal{S} \subseteq 2^X$ is a *subbasis* on X iff $\bigcup \mathcal{S} = X$ and its finite intersections form a basis on X.

Key Insight: Bases generate open sets by unions; subbases generate bases by finite intersections.

Proposition 6.1.9. *Basis Characterization*
Let X be a set and let $\mathcal{B} \subseteq 2^X$.

(i) If $\mathcal{B}$ is a basis on X, then $\mathcal{T}_{\mathcal{B}} := \{\bigcup \mathcal{U} : \mathcal{U} \subseteq \mathcal{B}\}$ is the unique topology on X of all $\mathcal{B}$-unions.
(ii) If $\mathcal{T}$ is a topology on X, $\mathcal{B} \subseteq \mathcal{T}$, and $\forall \emptyset \neq U \in \mathcal{T}$ there is $\mathcal{U} \subseteq \mathcal{B}$ with $U = \bigcup \mathcal{U}$, then $\mathcal{B}$ is a basis for $\mathcal{T}$.

Proof.

(i) Set $O_{\mathcal{U}} := \bigcup \mathcal{U}$ for $\mathcal{U} \subseteq \mathcal{B}$. Then $X = O_{\mathcal{B}}$, $\emptyset = O_{\emptyset}$, and $\bigcup_\alpha O_{\mathcal{U}_\alpha} = O_{\bigcup_\alpha \mathcal{U}_\alpha}$. If $x \in O_{\mathcal{U}} \cap O_{\mathcal{V}}$, choose $B_1 \in \mathcal{U}$ and $B_2 \in \mathcal{V}$ with $x \in B_1 \cap B_2$; by refinement, $x \in B_3 \subseteq B_1 \cap B_2$ for some $B_3 \in \mathcal{B}$. Hence $O_{\mathcal{U}} \cap O_{\mathcal{V}} = \bigcup\{B \in \mathcal{B} : B \subseteq O_{\mathcal{U}} \cap O_{\mathcal{V}}\}$, so $\mathcal{T}_{\mathcal{B}}$ is a topology. Uniqueness follows from the definition.
(ii) If $X \neq \emptyset$, apply the hypothesis to $U = X$ to get $X = \bigcup \mathcal{U}$ for some $\mathcal{U} \subseteq \mathcal{B}$; if $X = \emptyset$, the cover axiom is void. If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then $B_1 \cap B_2 \in \mathcal{T}$ and $B_1 \cap B_2 \neq \emptyset$, so $B_1 \cap B_2 = \bigcup \mathcal{V}$ for some $\mathcal{V} \subseteq \mathcal{B}$. Choose $B_3 \in \mathcal{V}$ with $x \in B_3 \subseteq B_1 \cap B_2$. Hence $\mathcal{B}$ is a basis and $\mathcal{T} = \mathcal{T}_{\mathcal{B}}$.

Structural Commentary: Local refinements make intersections open.
Key Insight: Basis pieces generate opens by unions. □

Example 6.1.10. *Metric Balls as a Basis*
In a metric space (X, d), $\{B_d(x, r) : x \in X,\ r > 0\}$ is a basis. On $\mathbb{R}^k$, $\{B(q, r) : q \in \mathbb{Q}^k,\ r \in \mathbb{Q}_{>0}\}$ is a countable basis. In $[a, b] \subseteq \mathbb{R}$, $\{[a, c) : a < c < b\} \cup \{(c, b] : a < c < b\}$ is a subbasis.

Definition 6.1.11. *Subspace Topology*
Let $(X, \mathcal{T})$ be a topological space and let $\emptyset \neq E \subseteq X$.
The *subspace topology* on E is $\mathcal{T}_E := \{E \cap U : U \in \mathcal{T}\}$, and $(E, \mathcal{T}_E)$ is a *subspace* of X.

Definition 6.1.12. *Separation Properties*
Let $(X, \mathcal{T})$ be a topological space. Disjoint sets $A, B \subseteq X$ are *separated by disjoint neighborhoods* if there exist $U, V \in \mathcal{T}$ such that $A \subseteq U$, $B \subseteq V$, and $U \cap V = \emptyset$. The space X is:

(i) T_1 iff $\{x\}$ is closed for every $x \in X$;
(ii) *Hausdorff* iff $\{x\}$ and $\{y\}$ can be separated by disjoint neighborhoods for all distinct $x, y \in X$;
(iii) *regular* iff X is T_1 and $\{x\}, F$ are separable by disjoint neighborhoods for closed $F \subseteq X$ and $x \notin F$;
(iv) *normal* iff X is T_1 and any disjoint closed $F, G \subseteq X$ can be separated by disjoint neighborhoods.

Key Insight: Stronger separation axioms separate larger closed objects.

Proposition 6.1.13. *T_1 Spaces and Closed Points*
Let $(X, \mathcal{T})$ be a topological space. Then X is T_1 iff $\forall$ distinct $x, y \in X$ $\exists U \in \mathcal{T}$ with $x \in U$ and $y \notin U$.

Proof.

($\Rightarrow$) Assume X is T_1. If $x \neq y$, then $\{y\}$ is closed, so $U = X \setminus \{y\} \in \mathcal{T}$ satisfies $x \in U$ and $y \notin U$.
($\Leftarrow$) Assume the stated condition. Fix $y \in X$. For each $x \in X \setminus \{y\}$, choose $U_x \in \mathcal{T}$ with $x \in U_x$ and $y \notin U_x$. Then $X \setminus \{y\} = \bigcup_{x \in X \setminus \{y\}} U_x \in \mathcal{T}$, so $\{y\}$ is closed. Hence X is T_1.

Structural Commentary: Point exclusion makes singleton complements open.
Key Insight: T_1 means each point can be omitted near every other point. □

Proposition 6.1.14. *Metric Spaces Are Normal*
Let (X, d) be a metric space with its metric topology.

(i) $\{x\}$ is closed for every $x \in X$.
(ii) If $F_1, F_2 \subseteq X$ closed and $F_1 \cap F_2 = \emptyset$, then $\exists$ disjoint open sets $O_1, O_2 \subseteq X$ with $F_i \subseteq O_i$ for $i = 1, 2$.

Hence X is normal.

Proof.

(i) Fix $x \in X$. For each $y \in X \setminus \{x\}$, $B_d(y, d(x, y)/2) \subseteq X \setminus \{x\}$, so $X \setminus \{x\}$ is open and $\{x\}$ is closed.

(ii) If one F_i is empty, take $O_i = \emptyset$ and $O_{3-i} = X$. Assume $F_1, F_2 \neq \emptyset$. Set $d_i(x) := \operatorname{dist}(x, F_i)$ and $O_i := \{x \in X : d_i(x) < d_{3-i}(x)\}$ for $i = 1, 2$. By Proposition 5.1.21, d_i is continuous and $d_i^{-1}(\{0\}) = F_i$.
Hence $O_i = (d_i - d_{3-i})^{-1}((-\infty, 0))$ is open, and if $x \in F_i$, then $d_i(x) = 0 < d_{3-i}(x)$, so $F_i \subseteq O_i$.
Finally, $O_1 \cap O_2 = \emptyset$, since $d_1(x) < d_2(x)$ and $d_2(x) < d_1(x)$ cannot both hold.

By Definition 6.1.12, (i) and (ii) imply that X is normal.
Structural Commentary: Distance inequalities separate closed sets.
Key Insight: Compare distances to build disjoint neighborhoods. □

Proposition 6.1.15. *Normality by Nested Neighborhoods*
Let X be a T_1 topological space. Then X is normal iff for every closed $F \subseteq X$ and every open $U \subseteq X$ with $F \subseteq U$, there exists an open $O \subseteq X$ such that $F \subseteq O \subseteq \overline{O} \subseteq U$.

Proof.
(⇒) Assume X is normal. Let $F \subseteq X$ be closed and let $U \subseteq X$ be open with $F \subseteq U$.
The closed sets F and $X \setminus U$ are disjoint, so choose disjoint open sets O, V with $F \subseteq O$ and $X \setminus U \subseteq V$.
Then $O \subseteq X \setminus V \subseteq U$, and since $X \setminus V$ is closed, $\overline{O} \subseteq X \setminus V \subseteq U$.
(⇐) Assume the nested-neighborhood property. Let $A, B \subseteq X$ be closed with $A \cap B = \emptyset$.
Apply the property to $F = A$ and $U = X \setminus B$ to get open O with $A \subseteq O \subseteq \overline{O} \subseteq X \setminus B$.
Then O and $X \setminus \overline{O}$ are disjoint open neighborhoods of A and B, respectively. Since X is T_1, X is normal.

Structural Commentary: Normality controls closures inside neighborhoods.
Key Insight: Thicken F without leaving U. □

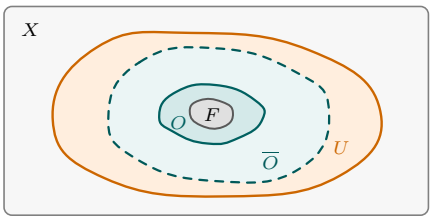

Figure 148: Normality thickens closed $F \subseteq U$ to an open O with $F \subseteq O \subseteq \overline{O} \subseteq U$.

6.2 Continuity and Topological Constructions

General topological constructions are organized by inverse images: homeomorphisms preserve open-set structure, while products, quotients, and weak topologies are characterized by which maps they make continuous.

Definition 6.2.1. *Continuity Between Topological Spaces*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, let $f : X \to Y$.
For $x \in X$, $y \in Y$, set $\mathcal{N}_X(x) := \{U \in \mathcal{T}_X : x \in U\}$, $\mathcal{N}_Y(y) := \{O \in \mathcal{T}_Y : y \in O\}$.
(i) For $x_0 \in X$, f is *continuous at x_0* iff $\forall O \in \mathcal{N}_Y(f(x_0))\ \exists U \in \mathcal{N}_X(x_0)$ with $f(U) \subseteq O$.
(ii) f is *continuous* iff $\forall x \in X$, f is continuous at x.

Proposition 6.2.2. *Continuity by Open Preimages*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and let $f : X \to Y$.
(i) f is continuous iff $\forall O \in \mathcal{T}_Y$, $f^{-1}(O) \in \mathcal{T}_X$.
(ii) If $A \subseteq X$, $\mathcal{T}_A = \{A \cap V : V \in \mathcal{T}_X\}$, and f is continuous, then $f|_A : (A, \mathcal{T}_A) \to (Y, \mathcal{T}_Y)$ is continuous.
(iii) If $(Z, \mathcal{T}_Z)$ is a topological space, $g : (Y, \mathcal{T}_Y) \to (Z, \mathcal{T}_Z)$ is continuous, and f is continuous, then $g \circ f : (X, \mathcal{T}_X) \to (Z, \mathcal{T}_Z)$ is continuous.

Proof.
(i) (⇒) Let $O \in \mathcal{T}_Y$ and $x \in f^{-1}(O)$. Since $O \in \mathcal{N}_Y(f(x))$ and f is continuous at x, choose $U_x \in \mathcal{N}_X(x)$ with $f(U_x) \subseteq O$. Then $x \in U_x \subseteq f^{-1}(O)$, so Proposition 6.1.3 gives $f^{-1}(O) \in \mathcal{T}_X$.
(i) (⇐) Assume $\forall O \in \mathcal{T}_Y$, $f^{-1}(O) \in \mathcal{T}_X$. Let $x_0 \in X$ and $O \in \mathcal{N}_Y(f(x_0))$. Then $U := f^{-1}(O) \in \mathcal{N}_X(x_0)$ and $f(U) \subseteq O$, so f is continuous at x_0. Since x_0 was arbitrary, f is continuous.
(ii) Let $O \in \mathcal{T}_Y$. Then $(f|_A)^{-1}(O) = A \cap f^{-1}(O) \in \mathcal{T}_A$, so $f|_A$ is continuous by *(i)*.
(iii) Let $W \in \mathcal{T}_Z$. Since g is continuous, $g^{-1}(W) \in \mathcal{T}_Y$.
Since f is continuous, $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)) \in \mathcal{T}_X$. Hence $g \circ f$ is continuous by *(i)*.

Structural Commentary: Subspaces and composites preserve open preimages by intersection and nesting.
Key Insight: Continuity is an inverse-image property. □

Proposition 6.2.3. *Continuity Tested on a Subbasis*
Let $(X, \mathcal{T}_X)$ be a topological space, let Y carry the topology $\mathcal{T}_Y$ generated by a subbasis $\mathcal{S}$, and let $f : X \to Y$. Then f is continuous iff $\forall S \in \mathcal{S}$, $f^{-1}(S) \in \mathcal{T}_X$.

Proof.
(⇒) If f is continuous and $S \in \mathcal{S}$, then $S \in \mathcal{T}_Y$, so Proposition 6.2.2 gives $f^{-1}(S) \in \mathcal{T}_X$.
(⇐) Assume $\forall S \in \mathcal{S}$, $f^{-1}(S) \in \mathcal{T}_X$. Let $\mathcal{B}$ be the basis of finite intersections from $\mathcal{S}$, including Y as the empty intersection. Then every $B \in \mathcal{B}$ has $f^{-1}(B) \in \mathcal{T}_X$, since $f^{-1}(\bigcap_{j=1}^{n} S_j) = \bigcap_{j=1}^{n} f^{-1}(S_j)$.
If $O = \bigcup_{\alpha \in A} B_\alpha \in \mathcal{T}_Y$, $B_\alpha \in \mathcal{B}$, then $f^{-1}(O) = \bigcup_{\alpha \in A} f^{-1}(B_\alpha) \in \mathcal{T}_X$. Proposition 6.2.2 gives continuity of f.

Structural Commentary: Subbasis tests extend through finite intersections and unions.
Key Insight: Inverse images preserve the operations generating the topology. □

Definition 6.2.4. *Homeomorphism*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces.

(i) A *homeomorphism* is a bijection $f : X \to Y$ such that both $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ and $f^{-1} : (Y, \mathcal{T}_Y) \to (X, \mathcal{T}_X)$ are continuous.
(ii) The spaces X and Y are *homeomorphic*, written $X \cong_{\text{top}} Y$, iff $\exists$ a homeomorphism $f : X \to Y$.

Proposition 6.2.5. *Homeomorphic Spaces Have the Same Open-Set Structure*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, let $f : X \to Y$ be a homeomorphism, and let $E \subseteq X$.

(i) $E \in \mathcal{T}_X$ iff $f(E) \in \mathcal{T}_Y$.
(ii) E is closed in X iff $f(E)$ is closed in Y.

Proof. Since f is a homeomorphism, f is bijective and both f and f^{-1} are continuous.

(i) ($\Rightarrow$) If $E \in \mathcal{T}_X$, then $f(E) = (f^{-1})^{-1}(E) \in \mathcal{T}_Y$.
(i) ($\Leftarrow$) If $f(E) \in \mathcal{T}_Y$, then $E = f^{-1}(f(E)) \in \mathcal{T}_X$.
(ii) ($\Rightarrow$) If E is closed in X, then $X \setminus E \in \mathcal{T}_X$. By *(i)*, $f(X \setminus E) \in \mathcal{T}_Y$. Since f is bijective, $f(X \setminus E) = Y \setminus f(E)$, so $f(E)$ is closed in Y.
(ii) ($\Leftarrow$) If $f(E)$ is closed in Y, then $Y \setminus f(E) \in \mathcal{T}_Y$. Since f is bijective, $Y \setminus f(E) = f(X \setminus E)$. By *(i)*, $X \setminus E \in \mathcal{T}_X$, so E is closed in X.

Structural Commentary: Openness transfers through inverse images of f and f^{-1}.
Key Insight: Homeomorphisms preserve and reflect open sets. □

Definition 6.2.6. *Product Topology*
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and let $\{(X_i, \mathcal{T}_i)\}_{i \in I}$ be a family of topological spaces.

(i) The *product topology* on $X \times Y$ is generated by $\{U \times V : U \in \mathcal{T}_X,\ V \in \mathcal{T}_Y\}$.
(ii) The *product topology* on $\prod_{i \in I} X_i$ is generated by $\{\prod_{i \in I} U_i : U_i \in \mathcal{T}_i\ \forall i \in I,\ |\{i \in I : U_i \neq X_i\}| < \infty\}$.

Proposition 6.2.7. *Product Topology and Coordinate Maps*
Let $(Z, \mathcal{T}_Z)$, $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be topological spaces. Let $\mathcal{T}_{X \times Y}$ be generated by $\mathcal{B} = \{U \times V : U \in \mathcal{T}_X,\ V \in \mathcal{T}_Y\}$.
For $f : Z \to X \times Y$, f is continuous iff $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous.

Proof.

($\Rightarrow$) If f is continuous and $U \in \mathcal{T}_X$, then $U \times Y \in \mathcal{B}$ and $(\pi_X \circ f)^{-1}(U) = f^{-1}(U \times Y) \in \mathcal{T}_Z$. Similarly, $(\pi_Y \circ f)^{-1}(V) = f^{-1}(X \times V) \in \mathcal{T}_Z$ for every $V \in \mathcal{T}_Y$.
($\Leftarrow$) Suppose $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous. For $U \times V \in \mathcal{B}$, $f^{-1}(U \times V) = (\pi_X \circ f)^{-1}(U) \cap (\pi_Y \circ f)^{-1}(V) \in \mathcal{T}_Z$. If $O \in \mathcal{T}_{X \times Y}$, write $O = \bigcup_{\alpha \in A} B_\alpha$ with $B_\alpha \in \mathcal{B}$; then $f^{-1}(O) = \bigcup_{\alpha \in A} f^{-1}(B_\alpha) \in \mathcal{T}_Z$. Hence f is continuous.

Structural Commentary: Basis rectangles pull back to intersections of coordinate preimages.
Key Insight: Product continuity is coordinatewise continuity. □

Proposition 6.2.8. *Product Convergence Is Pointwise*
Let $X^\Lambda = \prod_{\lambda \in \Lambda} X_\lambda$ carry the product topology from $\{(X_\lambda, \mathcal{T}_\lambda)\}_{\lambda \in \Lambda}$.
For $(x_n) \subseteq X^\Lambda$ and $x \in X^\Lambda$, $x_n \to x$ iff $x_n(\lambda) \to x(\lambda)$ in X_λ for every $\lambda \in \Lambda$.

Proof.

($\Rightarrow$) If $x_n \to x$, then each projection $\pi_\lambda : X^\Lambda \to X_\lambda$ is continuous, so $x_n(\lambda) = \pi_\lambda(x_n) \to \pi_\lambda(x) = x(\lambda)\ \forall \lambda \in \Lambda$.
($\Leftarrow$) Suppose $x_n(\lambda) \to x(\lambda)\ \forall \lambda \in \Lambda$. Let $W = \prod_{\lambda \in \Lambda} U_\lambda$ be a basic neighborhood of x, with $x(\lambda) \in U_\lambda \in \mathcal{T}_\lambda$ and finite $F = \{\lambda \in \Lambda : U_\lambda \neq X_\lambda\}$. Coordinatewise convergence on the finite set F gives n_0 such that $x_n(\lambda) \in U_\lambda$ for every $\lambda \in F$ and $n \geq n_0$. Hence $x_n \in W$ eventually, so $x_n \to x$.

Structural Commentary: Product neighborhoods impose finitely many coordinate constraints.
Key Insight: Coordinate convergence satisfies every finite constraint. □

Definition 6.2.9. *Quotient Topology*
Let $q : X \to Y$ be a surjection from a topological space X onto a set Y.

(i) The *quotient topology* on Y is $\{V \subseteq Y : q^{-1}(V)$ is open in $X\}$.
(ii) Equivalently, it is the strongest topology on Y for which q is continuous.

Proposition 6.2.10. *Quotient Universal Property*
Let $q : (X, \tau_X) \twoheadrightarrow Y$ and give Y the quotient topology $\tau_Y = \{V \subseteq Y : q^{-1}(V) \in \tau_X\}$.
For any map $g : Y \to (Z, \tau_Z)$, $g : (Y, \tau_Y) \to (Z, \tau_Z)$ is continuous $\iff$ $g \circ q : (X, \tau_X) \to (Z, \tau_Z)$ is continuous.

Proof.

($\Rightarrow$) If g is continuous, then $g \circ q$ is continuous by composition.
($\Leftarrow$) Suppose $g \circ q$ is continuous. For $W \in \tau_Z$, $q^{-1}(g^{-1}(W)) = (g \circ q)^{-1}(W) \in \tau_X$. Thus $g^{-1}(W) \in \tau_Y$ by the quotient topology, so g is continuous.

Structural Commentary: Continuity from a quotient is tested by precomposition with the quotient map.
Key Insight: Pull back through the quotient map and use the defining open-set test. □

Definition 6.2.11. *Weak Topology Induced by Maps*
Let X be a set.

(i) For topologies $\mathcal{T}_1, \mathcal{T}_2$ on X, $\mathcal{T}_2$ is *weaker* than $\mathcal{T}_1$ and $\mathcal{T}_1$ is *stronger* than $\mathcal{T}_2$ iff $\mathcal{T}_2 \subseteq \mathcal{T}_1$.
(ii) For topological spaces $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in A}$ and maps $f_\alpha : X \to X_\alpha$, the *weak topology induced by* $\{f_\alpha\}_{\alpha \in A}$ is generated by $\mathcal{S} := \{X\} \cup \{f_\alpha^{-1}(O) : \alpha \in A,\ O \in \mathcal{T}_\alpha\}$.

Proposition 6.2.12. *Weak Topology Universal Property*
Let $f_\alpha : X \to (X_\alpha, \mathcal{T}_\alpha)$ for $\alpha \in A$, and let $\mathcal{T}_w$ be generated by $\mathcal{S} := \{X\} \cup \{f_\alpha^{-1}(O) : \alpha \in A,\ O \in \mathcal{T}_\alpha\}$. Then $\mathcal{T}_w$ is the weakest topology on X making every f_α continuous.

Proof. Since $f_\alpha^{-1}(O) \in \mathcal{T}_w$ for all $\alpha \in A$ and $O \in \mathcal{T}_\alpha$, every $f_\alpha : (X, \mathcal{T}_w) \to (X_\alpha, \mathcal{T}_\alpha)$ is continuous.
If a topology $\mathcal{T}'$ on X makes every f_α continuous, then $\mathcal{S} \subseteq \mathcal{T}'$; hence $\mathcal{T}_w \subseteq \mathcal{T}'$.
Structural Commentary: Subbasic preimages are the forced open sets.
Key Insight: Weakest means contained in every topology passing the tests. □

6.3 Topological Compactness and Normality Tools

Compactness in general topology keeps the open-cover principle from metric spaces and pairs especially well with Hausdorff and normal separation. These tools also give useful continuous functions on normal spaces.

Definition 6.3.1. *Compactness in Topological Spaces*
Let X be a topological space.

(i) A family $\{O_\alpha\}_{\alpha \in A}$ is an *open cover* of $E \subseteq X$ if every O_α is open and $E \subseteq \bigcup_{\alpha \in A} O_\alpha$.
(ii) The space X is *compact* if every open cover of X has a finite subcover.
(iii) A subset $K \subseteq X$ is *compact* if it is compact with the subspace topology.
Equivalently, every cover of K by open subsets of X has a finite subcover.

Proposition 6.3.2. *Finite-Intersection Compactness Test*
Let $(X, \mathcal{T})$ be a topological space. Then X is compact if and only if every closed family $\{F_\alpha\}_{\alpha \in A}$ with the finite intersection property satisfies $\bigcap_{\alpha \in A} F_\alpha \neq \emptyset$.

Proof.
($\Rightarrow$) If X is compact and closed $\{F_\alpha\}_{\alpha \in A}$ has empty intersection, then $\{X \setminus F_\alpha\}_{\alpha \in A}$ covers X.
A finite subcover gives finite $J \subseteq A$ with $\bigcap_{\alpha \in J} F_\alpha = \emptyset$, so the family is not FIP.
($\Leftarrow$) Assume every closed FIP family intersects. If an open cover $\{O_\alpha\}_{\alpha \in A}$ has no finite subcover, then $\{X \setminus O_\alpha\}_{\alpha \in A}$ is closed and FIP, so some x lies in all $X \setminus O_\alpha$, contradicting that $\{O_\alpha\}_{\alpha \in A}$ covers X.
Structural Commentary: Complements exchange covers and intersections.
Key Insight: Compactness forbids closed FIP families from missing every point. □

Proposition 6.3.3. *Closed Subsets of Compact Spaces Are Compact*
Let $(X, \mathcal{T})$ be compact and let $K \subseteq X$ be closed. Then K is compact in the subspace topology.

Proof. Let $\{O_\alpha\}_{\alpha \in A}$ cover K, with $O_\alpha \in \mathcal{T}$. Since $X \setminus K \in \mathcal{T}$, the family $\{O_\alpha\}_{\alpha \in A} \cup \{X \setminus K\}$ covers X. Compactness gives a finite subcover; discarding $X \setminus K$ if present leaves a finite subcover of K.
Structural Commentary: Add the open complement to cover X.
Key Insight: Compactness passes to closed subspaces. □

Proposition 6.3.4. *Compact Subsets of Hausdorff Spaces Are Closed*
Let $(X, \mathcal{T})$ be Hausdorff and let $K \subseteq X$ be compact in the subspace topology. Then K is closed in X.

Proof. For $y \in X \setminus K$, choose disjoint open $O_x \ni x$ and $U_x \ni y$ for each $x \in K$.
Compactness gives $x_1, \ldots, x_n \in K$ with $K \subseteq \bigcup_{j=1}^{n} O_{x_j}$, so $U := \bigcap_{j=1}^{n} U_{x_j}$ is an open neighborhood of y disjoint from K. Thus $X \setminus K$ is open and K is closed.
Structural Commentary: Compactness turns pointwise Hausdorff separations into a finite one.
Key Insight: Every exterior point has an open neighborhood avoiding K. □

Theorem 6.3.5. *Compact Hausdorff Spaces Are Normal*
Let $(X, \mathcal{T})$ be compact and Hausdorff. Then X is normal.

Proof. Hausdorff implies T_1. Let $A, B \subseteq X$ be disjoint closed sets; then A, B are compact. For each $b \in B$, compact-set separation gives disjoint open $U_b \ni b$ and $V_b \supseteq A$. Choose $b_1, \ldots, b_n$ with $B \subseteq \bigcup_{j=1}^{n} U_{b_j}$.
Then $\bigcup_{j=1}^{n} U_{b_j}$ and $\bigcap_{j=1}^{n} V_{b_j}$ are disjoint open neighborhoods of B and A, so X is normal.
Structural Commentary: Compactness makes finitely many Hausdorff separations suffice.
Key Insight: Disjoint closed compact sets have disjoint neighborhoods. □

Theorem 6.3.6. *Continuous Images of Compact Spaces Are Compact*
Let $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ be continuous. If X is compact, then $f(X)$ is compact in the subspace topology.

Proof. Let $\{V_\alpha\}_{\alpha \in A}$ be an open cover of $f(X)$ by subsets open in Y. Then $\{f^{-1}(V_\alpha)\}_{\alpha \in A}$ covers X and each $f^{-1}(V_\alpha) \in \mathcal{T}_X$.
Compactness gives finite $J \subseteq A$ with $X = \bigcup_{\alpha \in J} f^{-1}(V_\alpha)$, hence $f(X) \subseteq \bigcup_{\alpha \in J} V_\alpha$.
Structural Commentary: Pull the cover back to the compact domain.
Key Insight: Finite subcovers descend through the image. □

Corollary 6.3.7. *Extreme Values on Compact Spaces*
If X is compact and $f : X \to \mathbb{R}$ is continuous, then f attains its minimum and maximum.

Proof. By Theorem 6.3.6, $f(X)$ is compact in $\mathbb{R}$. By Heine–Borel, Theorem 4.2.5, $f(X)$ is closed and bounded, so $\inf f(X), \sup f(X) \in f(X)$. Choose preimages $x_{\min}, x_{\max} \in X$.
Structural Commentary: Compact images in $\mathbb{R}$ are closed and bounded.
Key Insight: Closed bounded images contain their endpoints. □

Theorem 6.3.8. *Compact-to-Hausdorff Bijection Criterion*
Let $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ be continuous bijection. If X is compact, Y is Hausdorff, then f is a homeomorphism.

Proof. It remains to prove continuity of $f^{-1} : Y \to X$. If $F \subseteq X$ is closed, then F is compact by Proposition 6.3.3, so $f(F)$ is compact by Theorem 6.3.6. Since Y is Hausdorff, $f(F)$ is closed by Proposition 6.3.4.
Thus $(f^{-1})^{-1}(F) = f(F)$ is closed in Y, so f^{-1} is continuous.
Structural Commentary: Closed sets map compact, then closed.
Key Insight: Compact-to-Hausdorff bijections have continuous inverses. □

Proposition 6.3.9. *Compact Hausdorff Topologies Are Rigid*
Let X be a set and let $\mathcal{T}, \mathcal{T}_1, \mathcal{T}_2$ be topologies on X.

(i) If $(X, \mathcal{T})$ is compact and $\mathcal{T}_1 \subseteq \mathcal{T}$, then $(X, \mathcal{T}_1)$ is compact.
(ii) If $(X, \mathcal{T})$ is Hausdorff and $\mathcal{T} \subseteq \mathcal{T}_2$, then $(X, \mathcal{T}_2)$ is Hausdorff.
(iii) If $(X, \mathcal{T})$ is compact Hausdorff, then no strictly weaker topology is Hausdorff and no strictly stronger topology is compact.

Proof.

(*i*) Every $\mathcal{T}_1$-open cover is a $\mathcal{T}$-open cover, so compactness of $(X, \mathcal{T})$ gives a finite subcover.
(*ii*) If $x \neq y$, choose disjoint $\mathcal{T}$-open neighborhoods of x and y; they are also $\mathcal{T}_2$-open.
(*iii*) If $\mathcal{S} \subseteq \mathcal{T}$ is Hausdorff, Theorem 6.3.8 applied to $\mathrm{id} : (X, \mathcal{T}) \to (X, \mathcal{S})$ gives $\mathcal{T} \subseteq \mathcal{S}$, hence $\mathcal{S} = \mathcal{T}$.
If $\mathcal{T} \subseteq \mathcal{S}$ and $(X, \mathcal{S})$ is compact, apply it to $\mathrm{id} : (X, \mathcal{S}) \to (X, \mathcal{T})$ to get $\mathcal{S} \subseteq \mathcal{T}$, hence $\mathcal{S} = \mathcal{T}$.

Structural Commentary: Weakening preserves covers; strengthening preserves separations.
Key Insight: Compact Hausdorff topologies cannot be enlarged or shrunk compatibly. □

Lemma 6.3.10. *Maximal Finite-Intersection Families*
Let X be a set and let $\mathcal{A} \subseteq \mathcal{P}(X)$ have the finite intersection property. Then $\exists \mathcal{M} \subseteq \mathcal{P}(X)$ such that $\mathcal{A} \subseteq \mathcal{M}$, $\mathcal{M}$ has the finite intersection property, and no proper extension of $\mathcal{M}$ has it.

Proof. Let $\mathfrak{P} = \{\mathcal{B} \subseteq \mathcal{P}(X) : \mathcal{A} \subseteq \mathcal{B} \text{ and } \mathcal{B} \text{ has FIP}\}$, ordered by inclusion. If $\mathfrak{C} \subseteq \mathfrak{P}$ is a chain, then $\bigcup \mathfrak{C}$ has FIP: each finite subfamily lies in one $\mathcal{B} \in \mathfrak{C}$. Thus Zorn's lemma, Principle 6.1.3, gives a maximal $\mathcal{M} \in \mathfrak{P}$.
Structural Commentary: Chain unions preserve FIP because tests are finite.
Key Insight: Zorn extends any FIP family to a maximal one. □

Lemma 6.3.11. *Closure Properties of Maximal FIP Families*
Let X be a set and let $\mathcal{M} \subseteq \mathcal{P}(X)$ be maximal with respect to the finite intersection property.

(i) If $M_1, \ldots, M_n \in \mathcal{M}$, then $\bigcap_{j=1}^{n} M_j \in \mathcal{M}$.
(ii) If $C \subseteq X$ satisfies $C \cap \bigcap_{j=1}^{n} M_j \neq \emptyset$ for all $M_1, \ldots, M_n \in \mathcal{M}$, then $C \in \mathcal{M}$.

Proof.

(*i*) For $N = \bigcap_{j=1}^{n} M_j$, the family $\mathcal{M} \cup \{N\}$ still has the finite intersection property, so maximality gives $N \in \mathcal{M}$.
(*ii*) The hypothesis says $\mathcal{M} \cup \{C\}$ has the finite intersection property, so maximality gives $C \in \mathcal{M}$.

Structural Commentary: Maximality absorbs every FIP-compatible set.
Key Insight: Finite intersections are already forced inside $\mathcal{M}$. □

Theorem 6.3.12. *Tychonoff Product Theorem*
The product $X = \prod_{\lambda \in \Lambda} X_\lambda$ of compact topological spaces is compact in the product topology.

Proof. Let $\mathcal{F}$ be a closed FIP family in X, and extend it to maximal FIP $\mathcal{M}$ by Lemma 6.3.10.
For each λ, compactness of X_λ gives $x_\lambda \in \bigcap_{M \in \mathcal{M}} \overline{\pi_\lambda(M)}$; put $x = (x_\lambda)_{\lambda \in \Lambda}$.
If $N = \pi_\lambda^{-1}(O)$ is subbasic at x and $P = \bigcap_{j=1}^{n} M_j$ with $M_j \in \mathcal{M}$, then $P \in \mathcal{M}$ and $O \cap \pi_\lambda(P) \neq \emptyset$, so $N \cap P \neq \emptyset$.
Lemma 6.3.11 gives $N \in \mathcal{M}$, hence every basic neighborhood of x lies in $\mathcal{M}$.
Thus every basic neighborhood of x meets each $F \in \mathcal{F}$, so $x \in \overline{F} = F$.
Hence $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$, and Proposition 6.3.2 gives compactness.
Structural Commentary: Coordinate compactness selects a candidate point.
Key Insight: Maximal FIP forces all basic neighborhoods of that point to remain compatible. □

Definition 6.3.13. *Compact-Open Topology*
Let X, Y be topological spaces and let $Y^X = \{f : X \to Y\}$. The *compact-open topology* on Y^X is generated by the subbasis $\mathcal{U}_{K,O} := \{f \in Y^X : f(K) \subseteq O\}$, where $K \subseteq X$ is compact and $O \subseteq Y$ is open.
Structural Commentary: Subbasic opens prescribe uniform containment on compact subsets.

Remark 6.3.14. *Compact-Open Versus Product Topology*
On $Y^X = \prod_{x \in X} Y$, the product topology tests finitely many coordinates $f(x)$.
The compact-open topology tests compact sets at once through $\mathcal{U}_{K,O} = \{f \in Y^X : f(K) \subseteq O\}$
For metric Y, this encodes uniform control on compact subsets.

Definition 6.3.15. *Normally Ascending Families*
Let X be a topological space and let $\Lambda \subseteq \mathbb{R}$.
A family of open sets $\{O_\lambda\}_{\lambda\in\Lambda}$ is *normally ascending* iff $\overline{O_{\lambda_1}} \subseteq O_{\lambda_2}$ for all $\lambda_1, \lambda_2 \in \Lambda$ with $\lambda_1 < \lambda_2$.
Key Insight: Later open sets contain the closures of earlier ones.

Lemma 6.3.16. *Normally Ascending Families Define Continuous Functions*
Let X be a topological space, let $\Lambda \subseteq (a,b)$ be dense, and let $\{O_\lambda\}_{\lambda\in\Lambda}$ be normally ascending. Define $g : X \to [a,b]$ by $g(x) = \inf\{\lambda \in \Lambda : x \in O_\lambda\}$, with $\inf \emptyset = b$. Then g is continuous.

Proof. For $c \in (a,b)$, $\{g < c\} = \bigcup_{\lambda<c} O_\lambda$ and $\{g > c\} = \bigcup_{\lambda>c}(X \setminus \overline{O_\lambda})$. For the second identity, $x \notin \overline{O_\lambda}$ with $\lambda > c$ gives $g(x) \geq \lambda > c$; conversely, if $g(x) > c$, choose $\mu, \lambda \in \Lambda$ with $c < \mu < \lambda < g(x)$, so $x \notin O_\lambda \supseteq \overline{O_\mu}$. Both threshold sets are open, hence g is continuous.
Structural Commentary: Threshold preimages reduce to unions of nested open or closed-complement sets.
Key Insight: Normal ascent makes upper thresholds open. □

Lemma 6.3.17. *Dyadic Neighborhood Ladders in Normal Spaces*
Let X be normal, let $F \subseteq X$ be closed, and let U be an open neighborhood of F. For any open interval (a,b) there are dense $\Lambda \subseteq (a,b)$ and normally ascending open sets $\{O_\lambda\}_{\lambda\in\Lambda}$ with $F \subseteq O_\lambda \subseteq \overline{O_\lambda} \subseteq U$ for all $\lambda \in \Lambda$.

Proof. By affine reparametrization, take $(a,b) = (0,1)$ and $\Lambda = \{m2^{-n} : 0 < m < 2^n,\ n \in \mathbb{N}\}$.
Choose $O_{1/2}$ with $F \subseteq O_{1/2} \subseteq \overline{O_{1/2}} \subseteq U$ by Proposition 6.1.15.
Inductively, for each new dyadic $p < r < q$, choose O_r with $\overline{O_p} \subseteq O_r \subseteq \overline{O_r} \subseteq O_q$, interprete endpoints as F, U
This preserves normal ascent and containment in U.
Structural Commentary: Normality inserts closed-neighborhood buffers.
Key Insight: Dyadic induction builds the whole ascending ladder. □

Theorem 6.3.18. *Urysohn's Lemma*
Let X be normal, let $A, B \subseteq X$ be disjoint closed sets, and let $a < b$ in $\mathbb{R}$.
Then $\exists$ continuous $f : X \to [a,b]$ such that $f|_A = a$ and $f|_B = b$.

Proof. Apply Lemma 6.3.17 to $F = A$ and $U = X \setminus B$. Lemma 6.3.16 applied to the resulting ladder gives continuous $f : X \to [a,b]$. Since $A \subseteq O_\lambda$ for all λ and $B \cap \bigcup_\lambda O_\lambda = \emptyset$, the definition gives $f|_A = a$ and $f|_B = b$.
Structural Commentary: A nested open ladder becomes a continuous separator.
Key Insight: Normality separates closed sets by a real-valued function. □

Theorem 6.3.19. *Tietze Extension Theorem, Bounded Form*
Let X be normal, let $F \subseteq X$ be closed, let $a < b$, and let $f : F \to [a,b]$ be continuous.
Then $\exists$ continuous $g : X \to [a,b]$ with $g|_F = f$.

Proof. After an affine change of range, assume $[a,b] = [-1,1]$. The Urysohn step is: if $|h| \leq A$ on F, then some continuous $u : X \to [-A/3, A/3]$ satisfies $|h - u| \leq 2A/3$ on F. Starting with $s_0 = 0$, apply this to $h = f - s_{n-1}$ to get u_n with $|u_n| \leq \frac{1}{3}(2/3)^{n-1}$ on X and $|f - s_n| \leq (2/3)^n$ on F, where $s_n = s_{n-1} + u_n$.
Then $\sum_n u_n$ converges uniformly to continuous $g : X \to [-1,1]$ and $g|_F = f$. Undo the affine change of range.
Structural Commentary: Urysohn corrections shrink the residual geometrically.
Key Insight: Uniform convergence turns approximate extensions into an exact one. □

6.4 Sequences, Countability, and Metrization

Countability axioms control when topological questions can be reduced to sequences and countable tests. They also separate metric phenomena from genuinely non-metric topologies.

Definition 6.4.1. *Sequence Convergence and Sequential Compactness*
Let X be a topological space, and let $\mathcal{N}_X(x)$ denote the neighborhoods of $x \in X$.
(i) (x_n) *converges* to x, written $x_n \to x$, iff $\forall U \in \mathcal{N}_X(x)\ \exists N \in \mathbb{N}$ such that $n \geq N \Rightarrow x_n \in U$.
(ii) X is *sequentially compact* iff every sequence $(x_n) \subseteq X$ has a convergent subsequence $x_{n_k} \to x \in X$.
Key Insight: Sequential compactness is compactness tested by subsequences.

Proposition 6.4.2. *Hausdorff Sequence Limits Are Unique*
Let X be a topological space and let $(x_n) \subseteq X$.
(i) If X is Hausdorff, then (x_n) has at most one limit.
(ii) If X has the trivial topology $\{\emptyset, X\}$, then $x_n \to x$ for every $x \in X$.

Proof.
(*i*) If $x_n \to x$ and $x_n \to y$ with $x \neq y$, choose disjoint neighborhoods $U \in \mathcal{N}_X(x)$ and $V \in \mathcal{N}_X(y)$. Eventually $x_n \in U$ and $x_n \in V$, contradicting $U \cap V = \emptyset$.
(*ii*) In the trivial topology, $\mathcal{N}_X(x) = \{X\}$ for every $x \in X$, and every sequence is eventually in X.
Structural Commentary: Hausdorff neighborhoods separate competing limits.
Key Insight: A sequence tail cannot lie in two disjoint neighborhoods. □

Definition 6.4.3. *Countability Axioms*
Let (X, τ) be a topological space.
(i) X is *first countable* iff $\forall x \in X\ \exists$ countable $\mathcal{B}_x \subseteq \mathcal{N}_X(x)$ such that every $U \in \mathcal{N}_X(x)$ contains some $B \in \mathcal{B}_x$.
(ii) X is *second countable* iff τ has a countable basis.
(iii) X is *separable* iff $\exists$ countable $D \subseteq X$ with $\overline{D} = X$, equivalently every nonempty $U \in \tau$ meets D.
Key Insight: Countability turns topology into countable local, global, or dense tests.

Example 6.4.4. *Metric Spaces Are First Countable*
Every metric space is first countable: $\forall x$, balls $\{B_{1/n}(x) : n \geq 1\}$ form a countable neighborhood basis at x.

Proposition 6.4.5. *Second Countable Implies First Countable*
Let (X,τ) be second countable. Then X is first countable.

Proof. Let $\mathcal{B}$ be a countable basis for τ. For $x \in X$, set $\mathcal{B}_x := \{B \in \mathcal{B} : x \in B\}$.
Then $\mathcal{B}_x$ is countable, and for every $U \in \mathcal{N}_X(x)$, some $B \in \mathcal{B}_x$ satisfies $x \in B \subseteq U$.
Structural Commentary: Restrict the global basis to sets containing x.
Key Insight: Countable global tests give countable local tests. □

Proposition 6.4.6. *First Countable Sequential Closure*
Let X be first countable, let $E \subseteq X$, and let $x \in X$.
(i) $x \in \overline{E}$ iff $\exists (x_n) \subseteq E$ with $x_n \to x$.
(ii) E is closed iff every convergent sequence in E has all its limits in E.

Proof.
(i) ($\Rightarrow$) Let $\{B_n : n \geq 1\}$ be a countable neighborhood basis at x.
Since $x \in \overline{E}$ and $\bigcap_{j=1}^{n} B_j \in \mathcal{N}_X(x)$, choose $x_n \in E \cap \bigcap_{j=1}^{n} B_j$. Then $x_n \to x$.
(i) ($\Leftarrow$) If $(x_n) \subseteq E$ and $x_n \to x$, then every neighborhood of x contains some $x_n \in E$, so $x \in \overline{E}$.
(ii) ($\Rightarrow$) If E is closed and $(x_n) \subseteq E$ with $x_n \to x$, then $x \in \overline{E} = E$ by (i).
(ii) ($\Leftarrow$) If every convergent sequence in E has all its limits in E, then (i) gives $\overline{E} \subseteq E$. Hence $E = \overline{E}$, QED.
Structural Commentary: Countable local bases produce one testing sequence.
Key Insight: In first countable spaces, closure is sequential closure. □

Proposition 6.4.7. *Second Countable Consequences*
If (X,τ) is second countable, then there is a countable $D \subseteq X$ with $\overline{D} = X$.
Moreover, every subspace of a second countable space is second countable.

Proof. Let $\mathcal{B} = \{B_n : n \in I\}$ be a countable basis, with $I \subseteq \mathbb{N}$.
For each $n \in I$ with $B_n \neq \emptyset$, choose $x_n \in B_n$ and set $D = \{x_n : n \in I,\ B_n \neq \emptyset\}$. Then D is countable.
If $U \in \tau \setminus \{\emptyset\}$, choose $x \in U$; some $B_n \in \mathcal{B}$ satisfies $x \in B_n \subseteq U$, so $x_n \in D \cap U$. Hence $\overline{D} = X$.
If $Y \subseteq X$, then $\{B_n \cap Y : n \in I\}$ is a countable basis for the subspace topology on Y.
Structural Commentary: $\mathcal{B}$ countable gives countable D meeting every $U \in \tau \setminus \{\emptyset\}$.
Key Insight: D is dense iff $D \cap U \neq \emptyset$ for every $U \in \tau \setminus \{\emptyset\}$. □

Proposition 6.4.8. *Separable Metric Spaces and Countable Bases*
Let (X,d) be a metric space, and let τ_d be its metric topology. The following are equivalent:
(i) $\exists$ countable $D \subseteq X$ such that $\overline{D}^{\,X} = X$.
(ii) $\exists$ countable $\mathcal{B} \subseteq \tau_d$ such that $\mathcal{B}$ is a basis for τ_d.

Proof.
(i) $\Rightarrow$ (ii) Let $D \subseteq X$ be countable with $\overline{D}^{\,X} = X$, and set $\mathcal{B} := \{B_q(x) : x \in D,\ q \in \mathbb{Q},\ q > 0\}$.
Then $\mathcal{B}$ is countable. For $U \in \tau_d$ and $y \in U$, choose $r > 0$ with $B_r(y) \subseteq U$, then choose $q \in \mathbb{Q}$ with $0 < q < r/2$, $x \in D \cap B_q(y)$.
Then $y \in B_q(x) \subseteq B_r(y) \subseteq U$, since $z \in B_q(x)$ implies $d(z,y) < 2q < r$. QED.
(ii) $\Rightarrow$ (i) Let $\mathcal{B} = \{U_n : n \in I\}$ be a countable basis for τ_d, with $I \subseteq \mathbb{N}$.
Choose $x_n \in U_n$ for each nonempty U_n, and set $D := \{x_n : U_n \neq \emptyset\}$. Then D is countable.
If $\emptyset \neq U \in \tau_d$, choose $y \in U$ and $n \in I$ with $y \in U_n \subseteq U$; then $x_n \in D \cap U$.
Hence every nonempty open set meets D, so $\overline{D}^{\,X} = X$.
Structural Commentary: In metric spaces, one countable dense set produces all rational-radius local tests.
Key Insight: Separability and second countability coincide for metric spaces. □

Proposition 6.4.9. *Subspaces of Separable Metric Spaces*
Let (X,d) be a separable metric space and let $E \subseteq X$ with the subspace metric $d|_{E\times E}$. Then E is separable.

Proof. By Proposition 6.4.8, the metric topology τ_d has a countable basis $\mathcal{B}$. Set $\mathcal{B}_E := \{B \cap E : B \in \mathcal{B}\}$. Then $\mathcal{B}_E$ is countable. If U is open in E and $x \in U$, then $U = E \cap G$ for some $G \in \tau_d$; choose $B \in \mathcal{B}$ with $x \in B \subseteq G$. Hence $x \in B \cap E \subseteq U$, so $\mathcal{B}_E$ is a countable basis for E. By Proposition 6.4.8, E is separable.
Structural Commentary: A countable ambient basis restricts to a countable subspace basis.
Key Insight: Metric separability passes to all subspaces through countable bases. □

Corollary 6.4.10. *Euclidean Subsets Are Separable*
Let $k \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^k$ have the subspace Euclidean metric. Then E is separable.

Proof. The set $\mathbb{Q}^k$ is countable and dense in $\mathbb{R}^k$.
For $x \in \mathbb{R}^k$ and $r > 0$, choose $q = (q_1,\dots,q_k) \in \mathbb{Q}^k$ with $|x_i - q_i| < r/\sqrt{k}$ for each i, so $\|x - q\|_2 < r$.
Thus $\mathbb{R}^k$ is separable, and Proposition 6.4.9 gives that E is separable.
Structural Commentary: Rational grids are countable dense nets.
Key Insight: Euclidean separability passes to subspaces. □

Example 6.4.11. *Uncountable Cofinite Topology*
Let X be uncountable and let $\tau := \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$.
(i) τ is a topology on X.
(ii) (X,τ) is T_1.
(iii) (X,τ) is not first countable.

Proof.

(*i*) $\emptyset, X \in \tau$. Finite intersections and arbitrary unions of cofinite sets are cofinite, so τ is a topology.
(*ii*) For $x \in X$, $X \setminus \{x\} \in \tau$, so $\{x\}$ is closed. Hence (X, τ) is T_1.
(*iii*) Fix $x \in X$. If $\{B_n : n \geq 1\}$ were a countable neighborhood basis at x, then $C := \{x\} \cup \bigcup_n (X \setminus B_n)$ is countable. Choose $y \in X \setminus C$. Then $U := X \setminus \{y\} \in \mathcal{N}_X(x)$, but $y \in B_n$ for all n, so no $B_n \subseteq U$.

Key Insight: Cofinite local bases cannot be countable. □

Definition 6.4.12. *Lindelöf and Countably Compact Spaces*
Let (X, τ) be a topological space.

(i) X is *Lindelöf* iff every open cover $\{O_\alpha\}_{\alpha \in A} \subseteq \tau$ has a countable subcover $\{O_\alpha\}_{\alpha \in J}$, $J \subseteq A$.
(ii) X is *countably compact* iff every countable open cover $\{O_n\}_{n \in \mathbb{N}} \subseteq \tau$ has a finite subcover $\{O_n\}_{n \in J}$, $J \subseteq \mathbb{N}$.

Key Insight: Covers reduce to countable or finite subcovers.

Proposition 6.4.13. *Frechet Intersection Test for Countable Compactness*
Let X be a topological space. Then X is countably compact iff every descending sequence $F_1 \supseteq F_2 \supseteq \cdots$ of nonempty closed subsets of X satisfies $\bigcap_{n \geq 1} F_n \neq \emptyset$.

Proof.

($\Rightarrow$) Assume X is countably compact. If $\bigcap_{n \geq 1} F_n = \emptyset$, then $\{X \setminus F_n : n \geq 1\}$ is a countable open cover.
A finite subcover exists; since F_n descends, some $X \setminus F_N$ covers X, contradicting $F_N \neq \emptyset$.
($\Leftarrow$) Let $\{O_n : n \geq 1\}$ be a countable open cover with no finite subcover.
Then $F_n := X \setminus \bigcup_{j=1}^{n} O_j$ is nonempty, closed, and descending, while $\bigcap_{n \geq 1} F_n = \emptyset$, a contradiction.

Structural Commentary: Complements turn countable covers into descending closed sets.
Key Insight: Countable compactness is nested closed intersection. □

Theorem 6.4.14. *Dini's Theorem on Countably Compact Spaces*
Let X be countably compact, and let $f_n : X \to \mathbb{R}$ be continuous with $f_n(x) \downarrow 0$ for every $x \in X$. Then $f_n \to 0$ uniformly: $\forall \varepsilon > 0\ \exists N$ such that $0 \leq f_n(x) < \varepsilon$ for all $x \in X$ and $n \geq N$.

Proof. Fix $\varepsilon > 0$ and set $O_n := \{x \in X : f_n(x) < \varepsilon\}$. Then O_n is open, $O_n \subseteq O_{n+1}$, and $\bigcup_{n \geq 1} O_n = X$.

Countable compactness gives $X = \bigcup_{j=1}^{m} O_{n_j}$; with $N := \max_j n_j$, increasingness gives $O_N = X$.

Hence $0 \leq f_n(x) \leq f_N(x) < \varepsilon$ for all $x \in X$ and $n \geq N$.
Structural Commentary: Increasing sublevel covers reduce finite subcovers to one index.
Key Insight: Monotone pointwise convergence becomes uniform convergence. □

Proposition 6.4.15. *Second Countable Implies Lindelöf*
Let (X, τ) be second countable. Then X is Lindelöf: every open cover of X has a countable subcover.

Proof. Let $\mathcal{B} = \{B_n : n \in I\}$ be a countable basis, and let $\mathcal{U} \subseteq \tau$ cover X. For $I_0 := \{n \in I : \exists U \in \mathcal{U},\ B_n \subseteq U\}$, choose $U_n \in \mathcal{U}$ with $B_n \subseteq U_n$ for each $n \in I_0$. The family $\{U_n : n \in I_0\}$ is countable.
If $x \in X$, choose $U \in \mathcal{U}$ with $x \in U$ and $B_n \in \mathcal{B}$ with $x \in B_n \subseteq U$; then $n \in I_0$ and $x \in U_n$.
Structural Commentary: Basis elements refine arbitrary open covers.
Key Insight: A countable basis yields a countable subcover. □

Proposition 6.4.16. *Second Countable Countable Compactness Is Compactness*
Let X be second countable. Then X is compact iff X is countably compact.

Proof.

($\Rightarrow$) If $\{O_n\}_{n \in \mathbb{N}}$ covers X, compactness gives finite $J \subseteq \mathbb{N}$ with $X = \bigcup_{n \in J} O_n$.
($\Leftarrow$) Let $\mathcal{U} = \{O_\alpha\}_{\alpha \in A}$ cover X. By Proposition 6.4.15, $\exists$ countable $J \subseteq A$ with $X = \bigcup_{\alpha \in J} O_\alpha$.
Countable compactness gives finite $J_0 \subseteq J$ with $X = \bigcup_{\alpha \in J_0} O_\alpha$.

Structural Commentary: Second countability reduces covers to countable subcovers.
Key Insight: Countable compactness then makes the subcover finite. □

Theorem 6.4.17. *Second Countable Compactness and Sequential Compactness*
Let X be a second countable topological space. Then X is compact iff X is sequentially compact.

Proof.

($\Rightarrow$) Assume X is compact. Let $(x_n) \subseteq X$ and $F_n := \overline{\{x_k : k \geq n\}}$. Since $F_1 \supseteq F_2 \supseteq \cdots$ are nonempty closed, compactness gives $x \in \bigcap_{n \geq 1} F_n$. By first countability, choose a decreasing neighborhood basis (B_j) at x. With $n_0 = 0$, choose $n_j > n_{j-1}$ and $x_{n_j} \in B_j$ from $x \in F_{n_{j-1}+1}$. Then $x_{n_j} \to x$.
($\Leftarrow$) Assume X is sequentially compact. By Proposition 6.4.16, it is enough to prove countable compactness. If $\{O_n\}_{n \geq 1}$ is a countable open cover with no finite subcover, choose $x_n \in X \setminus \bigcup_{j=1}^{n} O_j$. Sequential compactness gives $x_{n_k} \to x \in O_N$ for some N. For k large, $n_k \geq N$ and $x_{n_k} \in O_N$, contradicting $x_{n_k} \notin O_N$.

Structural Commentary: Closed tails give a cluster point; local countability extracts a subsequence.
Key Insight: Countable covers and bases make compactness sequential. □

Definition 6.4.18. *Metrizable Space*
A topological space is *metrizable* if its topology is induced by some metric.

Definition 6.4.19. *Sorgenfrey Line*
Let $\mathcal{B}_S := \{[a,b) : a < b\}$ on $\mathbb{R}$. The *Sorgenfrey line* is $\mathbb{R}_S := (\mathbb{R}, \tau_S)$, where τ_S is the topology generated by $\mathcal{B}_S$.

Example 6.4.20. *Sorgenfrey Line Countability Tests*
The Sorgenfrey line $\mathbb{R}_S$ is first countable and separable, but not second countable; hence it is not metrizable.

Proof. For $x \in [a,b) \cap [c,d)$, choose e with $x < e \le \min\{b,d\}$; then $x \in [x,e) \subseteq [a,b) \cap [c,d)$, so $\mathcal{B}_S$ is a basis. Also $\{[x, x+1/n) : n \ge 1\} \subseteq \mathcal{N}_{\mathbb{R}_S}(x)$ is a local basis, and $\mathbb{Q} \cap [a,b) \neq \emptyset$ for every $[a,b) \neq \emptyset$.
Hence $\mathbb{R}_S$ is first countable and separable.
If $\mathcal{C}$ is a basis for $\mathbb{R}_S$, choose $C_x \in \mathcal{C}$ with $x \in C_x \subseteq [x, x+1)$. For $x < y$, $x \in C_x \setminus C_y$, so $C_x \neq C_y$.
Thus $x \mapsto C_x$ is injective, so no countable $\mathcal{C}$ exists. Hence $\mathbb{R}_S$ is not second countable.
If $\mathbb{R}_S$ were metrizable, separability and Proposition 6.4.8 would force second countability.
Structural Commentary: Half-open neighborhoods encode left endpoints.
Key Insight: Local countability does not imply one countable global basis. □

Definition 6.4.21. *Moore Plane*
Let $\mathbb{M} := \{(x,y) \in \mathbb{R}^2 : y \ge 0\}$ and $L := \mathbb{R} \times \{0\}$. The *Moore plane* is $\mathbb{M}$ with basis consisting of:
(i) $B_{\text{euc}}(p,r) \subseteq \mathbb{M}$ for $p \in \mathbb{M} \setminus L$ and $r > 0$;
(ii) $N(x,r) := \{(x,0)\} \cup B_{\text{euc}}((x,r), r)$ for $(x,0) \in L$ and $r > 0$.

Example 6.4.22. *Moore Plane Countability Tests*
The Moore plane $\mathbb{M}$ is Hausdorff and separable, but not second countable; hence it is not metrizable.

Proof. The basis and Hausdorff axioms follow by shrinking Euclidean disks and tangent neighborhoods $N(x,r)$. Let $D := \mathbb{Q} \times (\mathbb{Q} \cap (0,\infty))$; then every basic neighborhood meets D, so $\overline{D} = \mathbb{M}$. Thus $\mathbb{M}$ is separable.
For $(x,0) \in L$, $N(x,r) \cap L = \{(x,0)\}$, so L is an uncountable discrete subspace and hence not separable. If $\mathbb{M}$ were second countable, then L would be second countable and separable by Proposition 6.4.7, a contradiction. If $\mathbb{M}$ were metrizable, separability and Proposition 6.4.8 would force second countability.
Structural Commentary: Dense rational interior; discrete uncountable boundary.
Key Insight: The boundary is an uncountable discrete obstruction. □

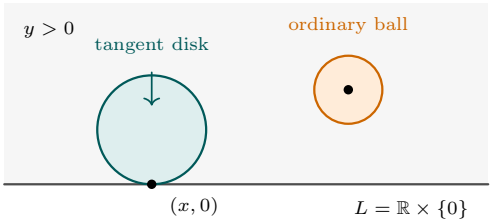

Figure 149: A Moore-plane boundary neighborhood is a tangent disk together with its tangent boundary point.

Theorem 6.4.23. *Urysohn Metrization Theorem*
Let (X,τ) be second countable. Then τ is induced by a metric iff (X,τ) is normal.

Proof.
($\Rightarrow$) Let d induce τ. Then X is T_1, since $X \setminus \{x\} = \bigcup_{y \neq x} B_d\left(y, \frac{d(x,y)}{2}\right)$. For disjoint nonempty closed $F, G \subseteq X$, set $u(x) := d(x,F)$ and $v(x) := d(x,G)$. Since u, v are continuous, $U := \{u < v\}$ and $V := \{v < u\}$ are disjoint open sets with $F \subseteq U$ and $G \subseteq V$. Empty cases are trivial, so (X,τ) is normal.
($\Leftarrow$) Assume (X,τ) is normal. Let $\{U_n : n \in \mathbb{N}\}$ be a basis and $A := \{(n,m) \in \mathbb{N}^2 : \overline{U_n} \subseteq U_m\}$.
For $(n,m) \in A$, choose $f_{n,m} : X \to [0,1]$ by Theorem 6.3.18, with $f_{n,m} = 0$ on $\overline{U_n}$, $f_{n,m} = 1$ on $X \setminus U_m$.
Set $\rho(x,y) := \sum_{(n,m) \in A} \frac{1}{2^{n+m}} |f_{n,m}(x) - f_{n,m}(y)|$, a τ-continuous pseudometric by uniform convergence.
For every $G \in \tau$ and $x \in G$, Proposition 6.1.15 and the basis give n, m with $x \in U_n \subseteq \overline{U_n} \subseteq U_m \subseteq G$.
If $y \notin U_m$, then $\rho(x,y) \ge \frac{1}{2^{n+m}}$, so $B_\rho\left(x, \frac{1}{2^{n+m+1}}\right) \subseteq U_m \subseteq G$.
Taking $G = X \setminus \{y\}$ shows $\rho(x,y) > 0$ for $x \neq y$. Hence ρ is a metric and its topology is τ.
Structural Commentary: Urysohn functions separate nested basis pairs.
Key Insight: Countably many separators recover the topology as a metric. □

6.5 Topological Connectedness and Paths

Connectedness is an open-set obstruction to splitting a space into two visible pieces. In general topology it is most naturally phrased by separations and then transported through continuous maps.

Definition 6.5.1. *Connected Topological Spaces*
Let (X,τ) be a topological space, and let $E \subseteq X$.
(i) A *separation* of X is a pair (U,V) with $U, V \in \tau$, $U, V \neq \emptyset$, $U \cap V = \emptyset$, and $X = U \cup V$.
(ii) X is *connected* iff no separation of X exists.
(iii) E is *connected* iff (E, τ_E) is connected, where $\tau_E := \{E \cap U : U \in \tau\}$.
Equivalently, E is disconnected iff $\exists U, V \in \tau$ with $U \cap E \neq \emptyset$, $V \cap E \neq \emptyset$, $E \subseteq U \cup V$, and $E \cap U \cap V = \emptyset$.

Key Insight: Connectedness forbids a nonempty disjoint open split.

Proposition 6.5.2. *Connectedness and Clopen Sets*
Let (X, τ) be a topological space. Then X is connected iff every clopen $A \subseteq X$ satisfies $A = \emptyset$ or $A = X$.

Proof.
($\Rightarrow$) If $A \subseteq X$ is clopen, nonempty, and proper, then $(A, X \setminus A)$ is a separation of X, impossible.
($\Leftarrow$) If (U, V) separates X, then $U = X \setminus V$ is clopen, nonempty, and proper, impossible.
Structural Commentary: Complements convert separations into clopen sets.
Key Insight: Connected means no nontrivial clopen subset. □

Proposition 6.5.3. *Continuous Images of Connected Spaces Are Connected*
Let $f : X \to Y$ be continuous. If X is connected, then $f(X)$ is connected in the subspace topology.

Proof. Let $g : X \to f(X)$ be f with codomain restricted to $f(X)$. If (U, V) separates $f(X)$, then $g^{-1}(U)$ and $g^{-1}(V)$ are nonempty disjoint open subsets of X with union X, contradicting connectedness.
Structural Commentary: Restrict the codomain to make the image map surjective.
Key Insight: Separations pull back by continuity. □

Definition 6.5.4. *Intermediate Value Property*
A space X has the *intermediate value property* iff $f(X)$ is an interval for every continuous $f : X \to \mathbb{R}$.

Theorem 6.5.5. *Connectedness and the Intermediate Value Property*
Let X be a topological space. Then X is connected iff X has the intermediate value property.

Proof.
($\Rightarrow$) If $f : X \to \mathbb{R}$ is continuous, then $f(X)$ is connected (Proposition 6.5.3), hence an interval (Theorem 5.1.2).
($\Leftarrow$) If (U, V) separates X, define $f : X \to \mathbb{R}$ by $f|_U = 0$ and $f|_V = 1$.
Since U, V are clopen, f is continuous, but $f(X) = \{0, 1\}$ is not an interval.
Structural Commentary: Separations produce continuous two-valued tests.
Key Insight: Intermediate values are connectedness seen by $\mathbb{R}$. □

Proposition 6.5.6. *Unions Through a Common Point*
Let $\{C_\alpha\}_{\alpha \in A}$ be connected subsets of a topological space X with $\bigcap_{\alpha \in A} C_\alpha \neq \emptyset$. Then $C := \bigcup_{\alpha \in A} C_\alpha$ is connected.

Proof. Choose $p \in \bigcap_{\alpha \in A} C_\alpha$. Suppose (U, V) separates C and $p \in U$. For each α, the connected set C_α meets U at p, so $C_\alpha \cap V = \emptyset$.
Hence $V \cap C = \emptyset$, contradicting that V is a nonempty side of the separation.
Structural Commentary: Each connected piece must stay on the side containing the shared point.
Key Insight: A common point blocks separation. □

Proposition 6.5.7. *Connectedness Between a Set and Its Closure*
Let $A \subseteq B \subseteq \overline{A}$ in a topological space X. If A is connected, then B and $\overline{A}$ are connected.

Proof. Suppose (P, Q) separates B. Since A is connected, $A \subseteq P$ or $A \subseteq Q$; assume $A \subseteq P$.
If $q \in Q$, choose $O \in \tau_X$ with $q \in O$ and $O \cap B \subseteq Q$.
Since $q \in B \subseteq \overline{A}$, $O \cap A \neq \emptyset$, contradicting $A \subseteq P$ and $P \cap Q = \emptyset$. Thus $Q = \emptyset$, impossible.
Structural Commentary: Relative open separation near $\overline{A}$ must meet A.
Key Insight: Closure points cannot form a new component. □

Definition 6.5.8. *Path-Connected and Arcwise Connected Spaces*
A topological space X is *path-connected*, or *arcwise connected*, iff for all $u, v \in X$ there exists continuous $\gamma : [0, 1] \to X$ with $\gamma(0) = u$ and $\gamma(1) = v$.

Theorem 6.5.9. *Path-Connected Spaces Are Connected*
Let X be path-connected. Then X is connected.

Proof. Suppose (U, V) separates X. Choose $u \in U$, $v \in V$, and a path $\gamma : [0, 1] \to X$ from u to v.
By Proposition 6.5.3, $\gamma([0, 1])$ is connected, but $(U \cap \gamma([0, 1]), V \cap \gamma([0, 1]))$ separates it, a contradiction.
Structural Commentary: A path imports interval connectedness into X.
Key Insight: Paths cannot cross a separation. □

Proposition 6.5.10. *Connected Open Sets in Euclidean Space Are Path-Connected*
Let $O \subseteq \mathbb{R}^n$ be open and connected. Then any $x, y \in O$ can be joined in O by a finite polygonal arc.

Proof. Fix $x_0 \in O$. Write $a \sim b$ if $a, b \in O$ joined in O by finite polygonal arc. Set $C := \{x \in O : x_0 \sim x\}$.
Then $x_0 \in C$. If $B(x, r) \subseteq O$, any two points of $B(x, r)$ are joined by a segment, so $B(x, r) \subseteq C$ when $x \in C$, and $B(x, r) \cap C = \emptyset$ when $x \notin C$. Thus C and $O \setminus C$ are open in O; connectedness gives $C = O$.
Structural Commentary: Polygonal reachability is both open and closed in an open set.
Key Insight: Local line segments make one component all of O. □

Example 6.5.11. *A Connected Space That Is Not Path-Connected*
Let $G := \{(x, \sin(1/x)) : 0 < x \leq 1\}$, $L := \{0\} \times [-1, 1]$, and $T := G \cup L \subseteq \mathbb{R}^2$. Then T is connected but not path-connected. Since G is the continuous image of connected $(0, 1]$ and $T = \overline{G}$, Proposition 6.5.7 gives connectedness. No path can join L to G without forcing an impossible one-sided limit of $\sin(1/x)$ as $x \to 0^+$.

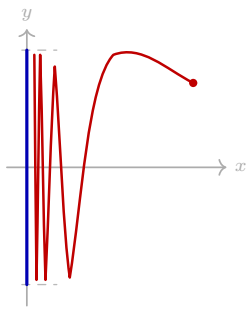

Figure 150: Topologist's sine curve: $y = \sin(1/x)$ accumulates on $\{0\} \times [-1, 1]$; connected, not path-connected.

7 Finite-Dimensional Topological Vector Spaces

7.1 Natural Topology and Continuity

Finite-dimensional real vector spaces carry a canonical topology: choose coordinates, use the standard Euclidean topology on $\mathbb{R}^n$, and then forget that coordinates were chosen. All coordinate changes are homeomorphisms.

Definition 7.1.1. *Topological Vector Space*
Let $\mathcal{V}$ be a real vector space and let $\mathcal{T}$ be a topology on its underlying set.
The pair $(\mathcal{V}, \mathcal{T})$ is a *topological vector space* if addition $A : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$, $A(\mathbf{u}, \mathbf{v}) = \mathbf{u} + \mathbf{v}$, and scalar multiplication $M : \mathbb{R} \times \mathcal{V} \to \mathcal{V}$, $M(a, \mathbf{v}) = a\mathbf{v}$ are continuous, where products carry the product topology.

Example 7.1.2. *The Standard Topology on $\mathbb{R}^n$*
With the Euclidean topology, $\mathbb{R}^n$ is a topological vector space: the maps $(\mathbf{x}, \mathbf{y}) \longmapsto \mathbf{x} + \mathbf{y}$, $(a, \mathbf{x}) \longmapsto a\mathbf{x}$ have polynomial coordinate functions $x_i + y_i$ and ax_i, hence are continuous.

Proposition 7.1.3. *Real Matrix Maps Are Continuous*
Let $A = [a_{j,k}] \in \mathrm{Mat}(m, n, \mathbb{R})$, and let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the induced map $T_A(\mathbf{x}) = A\mathbf{x}$. Then T_A is continuous for the Euclidean topologies. If $m = n$ and A is invertible, then T_A is a homeomorphism. In particular, every linear functional on $\mathbb{R}^n$ is continuous.

Proof. For $\mathbf{x} = (x_1, \dots, x_n)^t$, the jth coordinate of $T_A(\mathbf{x})$ is $\sum_{k=1}^{n} a_{j,k}x_k$, hence is continuous. Thus T_A is continuous. If A is invertible, then $(T_A)^{-1} = T_{A^{-1}}$, which is continuous by the first part. Finally, every $f \in (\mathbb{R}^n)'$ equals T_A for a unique $A \in \mathrm{Mat}(1, n, \mathbb{R})$, so f is continuous. *Structural Commentary*: Matrix maps on Euclidean coordinate spaces are continuous coordinate by coordinate.
Key Insight: Invertible real matrices give homeomorphisms of Euclidean space. □

Definition 7.1.4. *Natural Topology*
Let $\mathcal{V}$ be an n-dimensional real vector space, let β be a basis, and let $[\cdot]_\beta : \mathcal{V} \to \mathbb{R}^n$ be the coordinate isomorphism.
The *natural topology* on $\mathcal{V}$ is $\mathcal{T}_\beta := \{[\cdot]_\beta^{-1}(O) : O \subseteq \mathbb{R}^n \text{ is open}\}$.
Equivalently, $U \subseteq \mathcal{V}$ is open if and only if $[\cdot]_\beta(U)$ is open in $\mathbb{R}^n$.

Proposition 7.1.5. *Natural Topology Is Basis-Independent*
If β and γ are bases of $\mathcal{V}$, then $\mathcal{T}_\beta = \mathcal{T}_\gamma$.

Proof. Set $H_{\gamma\leftarrow\beta} := [\cdot]_\gamma \circ [\cdot]_\beta^{-1} : \mathbb{R}^n \to \mathbb{R}^n$.
Then $H_{\gamma\leftarrow\beta}\mathbf{x} = C_{\gamma\leftarrow\beta}\mathbf{x}$. Since $C_{\gamma\leftarrow\beta}$ is invertible with inverse $C_{\beta\leftarrow\gamma}$, Proposition 7.1.3 shows that $H_{\gamma\leftarrow\beta}$ is a homeomorphism with inverse $H_{\beta\leftarrow\gamma}$.
Hence, for $U \subseteq \mathcal{V}$, $U \in \mathcal{T}_\beta \iff [\cdot]_\beta(U)$ is open $\iff H_{\gamma\leftarrow\beta}([\cdot]_\beta(U)) = [\cdot]_\gamma(U)$ is open $\iff U \in \mathcal{T}_\gamma$. *Structural Commentary*: Basis changes are homeomorphisms.
Key Insight: The natural topology is carried by the vector-space structure, not by a preferred basis. □

Theorem 7.1.6. *Characterization of the Natural Topology*
Let $\dim_{\mathbb{R}} \mathcal{V} < \infty$. There is a unique topology $\mathcal{T}$ on $\mathcal{V}$ such that:
(i) $(\mathcal{V}, \mathcal{T})$ is a topological vector space;
(ii) $\mathcal{V}' \subseteq C((\mathcal{V}, \mathcal{T}), \mathbb{R})$.
Moreover, $\mathcal{T} = \mathcal{T}_\beta$ for every basis β.

Proof. Fix an arbitrary basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$.
(i) *Existence*: put $\mathcal{T} = \mathcal{T}_\beta$. Since $[\cdot]_\beta : (\mathcal{V}, \mathcal{T}_\beta) \to \mathbb{R}^n$ is a homeomorphism, the vector operations on $\mathcal{V}$ are transported from the continuous vector operations on $\mathbb{R}^n$.
Also, for $\varphi \in \mathcal{V}'$, $\varphi \circ [\cdot]_\beta^{-1} \in (\mathbb{R}^n)'$, hence is continuous by Proposition 7.1.3. Thus (i)–(ii) hold.
(ii) *Uniqueness*: suppose $\mathcal{T}$ satisfies (i)–(ii).
For $1 \le i \le n$, the coordinate functional $\pi_i \circ [\cdot]_\beta \in \mathcal{V}'$ is continuous, so $[\cdot]_\beta : (\mathcal{V}, \mathcal{T}) \to \mathbb{R}^n$ is continuous.
Its inverse $\Psi(a_1, \dots, a_n) = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ is continuous by (i), since it is built from scalar multiplication and finite sums.
Hence $[\cdot]_\beta$ is a homeomorphism, so $\mathcal{T} = \mathcal{T}_\beta$.
Structural Commentary: Functionals give coordinates; operations give inverse coordinates.
Key Insight: Functionals and vector operations determine finite-dimensional coordinates topologically. □

Corollary 7.1.7. *Finite-Dimensional Linear Maps Are Continuous*
Let $\dim_{\mathbb{R}} \mathcal{V}, \dim_{\mathbb{R}} \mathcal{W} < \infty$, with natural topologies. Then $\mathcal{L}(\mathcal{V}, \mathcal{W}) \subseteq C(\mathcal{V}, \mathcal{W})$.
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is an isomorphism, then T is a homeomorphism.

Proof. Choose bases β of $\mathcal{V}$ and γ of $\mathcal{W}$. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $[\cdot]_\gamma \circ T \circ [\cdot]_\beta^{-1}(\mathbf{x}) = [T]_{\gamma \leftarrow \beta}\mathbf{x}$, which is continuous by Proposition 7.1.3. Since $[\cdot]_\beta$ and $[\cdot]_\gamma$ are homeomorphisms, T is continuous.
If T is an isomorphism, apply the same argument to T^{-1}.
Structural Commentary: Finite-dimensional linear maps are just matrix maps in natural coordinates.
Key Insight: A basis turns topological questions about linear maps into continuity of matrix multiplication. □

Perspective 7.1.8. *Finite-Dimensional Topology Is Rigid*
For finite-dimensional real vector spaces, topology is algebraically forced.
Note that $\mathcal{T}_\beta = \mathcal{T}_\gamma$; $\mathcal{L}(\mathcal{V}, \mathcal{W}) \subseteq C(\mathcal{V}, \mathcal{W})$; $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ bijective $\Rightarrow T$ is a homeomorphism.
Infinite-dimensional topological vector spaces need not be rigid in this sense.

Part XII
Calculus, Series, and Approximation

1 One-Variable Differentiation and Riemann Integration

1.1 Differentiation and Basic Theorems

One-variable calculus studies functions on intervals through derivatives, mean-value theorems, Riemann integration, and the FTC, then points toward vector-valued estimates and linear approximation.

Definition 1.1.1. *Derivative of a Real Function*
Let f be real-valued on $[a,b]$ and let $x \in [a,b]$. For $a < t < b$ and $t \neq x$, set $\phi_x(t) := \frac{f(t)-f(x)}{t-x}$.
If $\lim_{t\to x} \phi_x(t)$ exists, denote *derivative* by $f'(x)$; then f is *differentiable at* x.
If this holds for every $x \in E \subseteq [a,b]$, then f is *differentiable on* E.

Figure 151: Domain matters: isolated points give no punctured approach; $[x_0, \infty)$ gives a right-hand one.

Perspective 1.1.2. *Derivative Depends on the Domain*
For $f : X \to \mathbb{R}$, $f'(x_0)$ is the limit of $\frac{f(x)-f(x_0)}{x-x_0}$ as $x \to x_0$ in $X \setminus \{x_0\}$; see Definition 5.1.4. Hence x_0 must be a limit point of X: isolated points give no derivative, while half-interval domains give one-sided derivatives.

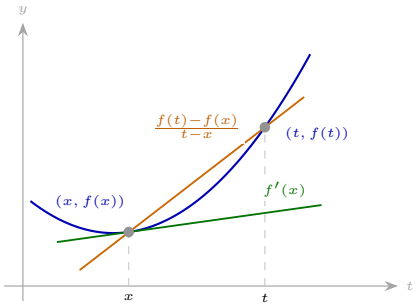

Figure 152: The derivative is the limiting slope of secant lines through $(x, f(x))$.

Proposition 1.1.3. *First-Order Approximation Form*
Let $X \subseteq \mathbb{R}$, let $x_0 \in X$ be a limit point, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. Then $f'(x_0) = L$ relative to X iff $\forall \varepsilon > 0\ \exists \delta > 0$ such that $|f(x) - f(x_0) - L(x - x_0)| \le \varepsilon\,|x - x_0|$ whenever $x \in X$ and $|x - x_0| < \delta$.

Proof. For $x \neq x_0$, $\frac{|f(x)-f(x_0)-L(x-x_0)|}{|x-x_0|} = \left|\frac{f(x)-f(x_0)}{x-x_0} - L\right|$, while $x = x_0$ makes approx. estimate automatic.

($\Rightarrow$) If $f'(x_0) = L$, then the displayed right-hand side is at most ε for all $x \in X \setminus \{x_0\}$ sufficiently near x_0. Multiplying by $|x - x_0|$ gives the approximation estimate.

($\Leftarrow$) If the approximation estimate holds, then division by $|x - x_0|$ gives $\left|\frac{f(x)-f(x_0)}{x-x_0} - L\right| \le \varepsilon$ for all $x \in X \setminus \{x_0\}$ sufficiently near x_0. Hence $f'(x_0) = L$.

Structural Commentary: Differentiability is first-order linear approximation.
Key Insight: Multiply or divide by $|x - x_0|$. □

Theorem 1.1.4. *Differentiability Implies Continuity*
If f is differentiable at x, then f is continuous at x.

Proof. For $t \neq x$, $f(t) - f(x) = \frac{f(t)-f(x)}{t-x}(t - x)$. As $t \to x$, the difference quotient tends to $f'(x)$ and $t - x \to 0$.
By Theorem 5.1.7, $f(t) - f(x) \to 0$, so $f(t) \to f(x)$. Theorem 5.1.11 gives continuity at x.
Structural Commentary: A differentiable function has first-order control near the point.
Key Insight: The quotient stays near a finite number while $t - x$ shrinks to zero. □

Theorem 1.1.5. *Algebra Rules for Derivatives*
Let f, g be differentiable at x.
(i) $f + g$ is differentiable at x, and $(f + g)'(x) = f'(x) + g'(x)$.
(ii) fg is differentiable at x, and $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$.
(iii) If $g(x) \neq 0$, then f/g is differentiable at x, and $\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x)-g'(x)f(x)}{g(x)^2}$.

Proof.
(i) Limit algebra applied to $\frac{(f+g)(t)-(f+g)(x)}{t-x} = \frac{f(t)-f(x)}{t-x} + \frac{g(t)-g(x)}{t-x}$ gives the sum rule.
(ii) For $h = fg$, $h(t) - h(x) = f(t)(g(t) - g(x)) + g(x)(f(t) - f(x))$. Divide by $t - x$ and let $t \to x$. Theorem 1.1.4 gives $f(t) \to f(x)$, hence the product rule.

(iii) If $h = f/g$, continuity of g gives $g(t) \neq 0$ near x, and $\frac{h(t)-h(x)}{t-x} = \frac{1}{g(t)g(x)}\left[g(x)\frac{f(t)-f(x)}{t-x} - f(x)\frac{g(t)-g(x)}{t-x}\right]$.
Let $t \to x$ to get the quotient rule.

Structural Commentary: Product and quotient rules are difference-quotient identities plus continuity.
Key Insight: Difference-quotient identities plus continuity give the algebra rules. □

Theorem 1.1.6. *Chain Rule*
Let f be differentiable at x and let g be differentiable at $f(x)$.
If $h = g \circ f$, then h is differentiable at x and $h'(x) = g'(f(x))f'(x)$.

Proof. Set $y = f(x)$. For $u(t) \to 0$, $f(t) - f(x) = (t - x)(f'(x) + u(t))$.
For $v(s) \to 0$, $g(s) - g(y) = (s - y)(g'(y) + v(s))$.
Since $f(t) \to y$, for $t \neq x$, $\frac{h(t)-h(x)}{t-x} = (g'(y) + v(f(t)))(f'(x) + u(t)) \to g'(y)f'(x)$.
Structural Commentary: Differentiate by composing first-order expansions.
Key Insight: First-order parts multiply; errors vanish. □

Example 1.1.7. *Oscillation at a Point*
Assuming the usual derivative of $\sin x$, define

$$f_1(x) = \begin{cases} x\sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases} \qquad f_2(x) = \begin{cases} x^2\sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

For f_1, the quotient at 0 is $\sin(1/t)$, so $f_1'(0)$ does not exist. For f_2, $|(f_2(t) - f_2(0))/t| \leq |t|$, so $f_2'(0) = 0$.
However, for $x \neq 0$, $f_2'(x) = 2x\sin(1/x) - \cos(1/x)$, so f_2' is not continuous at 0.

Definition 1.1.8. *Local Extrema*
Let f be a real-valued function on a metric space X. The function has a *local maximum* at $p \in X$ if there is $\delta > 0$ such that $f(q) \leq f(p)$ whenever $d(p, q) < \delta$. Local minima are defined by reversing the inequality.

Remark 1.1.9. *Local and Global Extrema*
Global extrema are local, but local extrema depend on domain: isolated points are automatic extrema. Derivative tests therefore need an interior interval point; compactness supplies global extrema (Theorem 5.1.37).

Theorem 1.1.10. *Interior Extremum Has Zero Derivative*
If f has a local extremum at $x \in (a, b)$ and $f'(x)$ exists, then $f'(x) = 0$.

Proof. For a local maximum, $f(t) \leq f(x)$ near x.
Thus the quotient $(f(t) - f(x))/(t - x)$ is nonnegative for $t < x$ and nonpositive for $t > x$. Since the two-sided limit exists, $f'(x) \geq 0$ and $f'(x) \leq 0$, so $f'(x) = 0$. For a local minimum, apply the local-maximum case to $-f$.
Since $(-f)'(x) = -f'(x)$, we get $-f'(x) = 0$, hence $f'(x) = 0$.
Structural Commentary: Interior extrema force opposite one-sided quotient signs.
Key Insight: A two-sided derivative compatible with both signs must be zero. □

Theorem 1.1.11. *Rolle's Theorem*
If $f : [a, b] \to \mathbb{R}$ continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $\exists x \in (a, b)$ s.t. $f'(x) = 0$.

Proof. If f is constant, any $x \in (a, b)$ works. Otherwise, by Theorem 5.1.37, f attains a nonendpoint extremum at some $x \in (a, b)$.
Theorem 1.1.10 gives $f'(x) = 0$.
Structural Commentary: Equal endpoints force any nonconstant height change to turn around inside.
Key Insight: Compactness gives the extremum; differentiability makes the interior extremum stationary. □

Theorem 1.1.12. *Cauchy Mean Value Theorem*
Let f, g be continuous on $[a, b]$, differentiable on (a, b). $\exists\, x \in (a, b)$, $\big(f(b) - f(a)\big)g'(x) = \big(g(b) - g(a)\big)f'(x)$.

Proof. Set $h(t) := \big(f(b) - f(a)\big)g(t) - \big(g(b) - g(a)\big)f(t)$.
Then $h(a) = h(b)$, h is continuous on $[a, b]$, and h is differentiable on (a, b). By Theorem 1.1.11, $h'(x) = 0$ for some $x \in (a, b)$. Since $h'(x) = \big(f(b) - f(a)\big)g'(x) - \big(g(b) - g(a)\big)f'(x)$, the identity follows.
Structural Commentary: Build one auxiliary function whose endpoint values agree.
Key Insight: Equal endpoints force an interior horizontal tangent. □

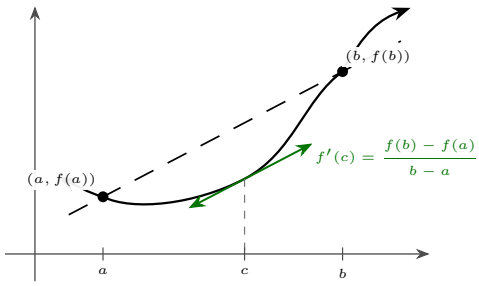

Figure 153: The Mean Value Theorem finds an interior tangent slope equal to the endpoint secant slope.

Corollary 1.1.13. *Mean Value Theorem*
If f is continuous on $[a, b]$, differentiable in (a, b), then $\exists\, x \in (a, b)$ such that $f(b) - f(a) = (b - a)f'(x)$.

Proof. Let $g(t) = t$. Then $g'(x) = 1$, $g(b) - g(a) = b - a$.
Hence Theorem 1.1.12 gives $f(b) - f(a) = (b - a)f'(x)$ for some $x \in (a, b)$.
Structural Commentary: This is Cauchy's theorem with $g(t) = t$.
Key Insight: A secant slope is achieved as a tangent slope somewhere inside. □

Corollary 1.1.14. *Bounded Derivative Gives Lipschitz Control*
If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $|f'| \le M$ on (a, b), then f is M-Lipschitz on $[a, b]$, hence uniformly continuous.

Proof. For $x < y$, Corollary 1.1.13 gives $f(y) - f(x) = (y - x)f'(c)$ for some $c \in (x, y)$,
so $|f(y) - f(x)| \le M(y - x)$. Symmetry handles $x > y$, and $x = y$ is trivial.
Structural Commentary: A derivative bound controls every secant slope.
Key Insight: Local slope bounds become global Lipschitz bounds through the Mean Value Theorem. □

Theorem 1.1.15. *Derivative Sign and Monotonicity*
Suppose f is differentiable in (a, b).

(i) If $f'(x) \ge 0$ for all $x \in (a, b)$, then f is monotonically increasing.
(ii) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.
(iii) If $f'(x) \le 0$ for all $x \in (a, b)$, then f is monotonically decreasing.

Proof. For $x_1 < x_2$ in (a, b), Corollary 1.1.13 on $[x_1, x_2]$ gives $f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$ for some $c \in (x_1, x_2)$.

(i) If $f' \ge 0$, then $f(x_2) - f(x_1) \ge 0$.
(ii) If $f' = 0$, then $f(x_2) - f(x_1) = 0$.
(iii) If $f' \le 0$, then $f(x_2) - f(x_1) \le 0$.

Structural Commentary: Monotonicity is a global reading of the mean-value identity.
Key Insight: The sign of every tangent slope controls every secant slope. □

Theorem 1.1.16. *Derivatives Have the Intermediate Value Property*
If f is differentiable on $[a, b]$ in the one-sided endpoint sense and $f'(a) < \lambda < f'(b)$,
Then $f'(x) = \lambda$ for some $x \in (a, b)$. The same conclusion holds when $f'(b) < \lambda < f'(a)$.

Proof. Set $g(t) = f(t) - \lambda t$. Since $g'(a) < 0 < g'(b)$, there are $t_1, t_2 \in (a, b)$ with $g(t_1) < g(a)$ and $g(t_2) < g(b)$. Hence the minimum of g on $[a, b]$ occurs at some $x \in (a, b)$. By Theorem 1.1.10, $0 = g'(x) = f'(x) - \lambda$.
Apply the same argument to $-f$ for the reversed case.
Structural Commentary: A derivative need not be continuous, but it cannot jump over values.
Key Insight: Tilt the graph by subtracting λt and force an interior extremum. □

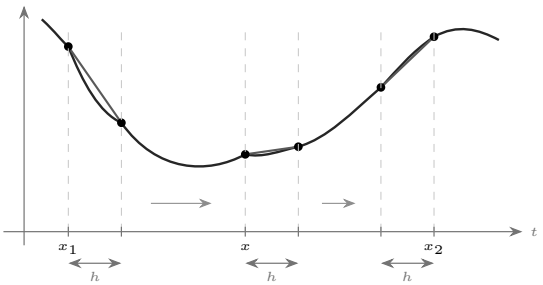

Figure 154: Sliding fixed-width secants in Darboux's theorem.

Perspective 1.1.17. *Sliding Secants*
For fixed small $h > 0$, the secant-slope function $\sigma_h(x) := \frac{f(x+h)-f(x)}{h}$ is continuous. Thus Theorem 5.1.41 finds intermediate secant slopes, and Corollary 1.1.13 converts them to derivative values.
Key Insight: Slide secants, then apply the Mean Value Theorem.

Corollary 1.1.18. *Derivatives Have No Simple Discontinuities*
If f is differentiable on $[a, b]$, then f' cannot have a simple discontinuity on $[a, b]$.

Proof. Suppose a simple discontinuity occurs at an interior point c; endpoint cases are one-sided.
If $f'(c-) \ne f'(c+)$, choose λ strictly between these limits, $\lambda \ne f'(c)$. For $x < c < y$ close to c, Theorem 1.1.16 gives $z \in (x, y)$ with $f'(z) = \lambda$, contradicting the one-sided limits. If $f'(c-) = f'(c+) = L \ne f'(c)$, choose λ between L and $f'(c)$ and apply the same theorem on a sufficiently small one-sided interval with endpoint c.
Structural Commentary: Darboux behavior rules out jump discontinuities for derivatives.
Key Insight: Derivatives may oscillate, but they do not jump. □

Theorem 1.1.19. *Derivative of an Inverse Function*
Let $f : I \to \mathbb{R}$ be one-to-one on an open interval I, let $x_0 \in I$, and put $y_0 = f(x_0)$. Suppose f is differentiable at x_0, $f'(x_0) \ne 0$, and $g = f^{-1} : f(I) \to I$ is continuous at y_0. Then g is differentiable at y_0 and $g'(y_0) = \frac{1}{f'(x_0)}$.

Proof. For $y \in f(I)$ with $y \ne y_0$, write $x = g(y)$. Then $x \ne x_0$ and $\frac{g(y)-g(y_0)}{y-y_0} = \frac{x-x_0}{f(x)-f(x_0)} = \left(\frac{f(x)-f(x_0)}{x-x_0}\right)^{-1}$.
As $y \to y_0$, the assumed continuity of g gives $x = g(y) \to x_0$.
The difference quotient for f tends to $f'(x_0) \ne 0$, so the reciprocal tends to $1/f'(x_0)$ by Theorem 5.1.7.
Structural Commentary: Matched inverse quotients are reciprocal.
Key Insight: Continuity of the inverse converts the limit $y \to y_0$ back into $x \to x_0$. □

Corollary 1.1.20. *One-Dimensional Inverse Function Theorem*
If $f:(a,b)\to(c,d)$ is a differentiable bijection and $f'\neq 0$, then f is a homeomorphism and f^{-1} is differentiable, with $(f^{-1})'(y)=\frac{1}{f'(f^{-1}(y))}$.

Proof. By Theorem 1.1.16, f' has constant sign. Corollary 1.1.13 makes f strictly monotone, hence its continuous interval inverse is continuous. Apply Theorem 1.1.19 pointwise.
Structural Commentary: Darboux gives constant sign; the inverse derivative formula is local.
Key Insight: Nonvanishing derivative is the one-dimensional obstruction to differentiating the inverse. □

Theorem 1.1.21. *L'Hospital's Rule*
Let $-\infty\le a<b\le+\infty$, let f,g be differentiable on (a,b), assume $g'\neq 0$, and suppose $\frac{f'}{g'}\to A\in\overline{\mathbb{R}}$ as $x\to a^+$.

(i) If $f,g\to 0$ as $x\to a^+$, then $\frac{f}{g}\to A$ as $x\to a^+$.

(ii) If $|g|\to\infty$ as $x\to a^+$, then $\frac{f}{g}\to A$ as $x\to a^+$.

The analogous statements hold as $x\to b^-$.

Proof. For $a<r<s<b$, Theorem 1.1.12 gives $t\in(r,s)$ with $\frac{f(s)-f(r)}{g(s)-g(r)}=\frac{f'(t)}{g'(t)}$, so eventual bounds on $\frac{f'}{g'}$ near a transfer to these difference quotients.

(i) Fix s and let $r\to a^+$; since $f(r),g(r)\to 0$, the same bounds hold for $\frac{f(s)}{g(s)}$.

(ii) Fix s near a and use $\frac{f(r)}{g(r)}=\frac{f(r)-f(s)}{g(r)-g(s)}\frac{g(r)-g(s)}{g(r)}+\frac{f(s)}{g(r)}$.
As $r\to a^+$, the second factor tends to 1 and the last term to 0, so the same bounds hold for $\frac{f(r)}{g(r)}$.

In both cases, the transferred eventual bounds force $\frac{f}{g}\to A$.
Structural Commentary: CMVT transfers derivative-quotient bounds to function quotients.
Key Insight: Bounds on f'/g' force the same limit for f/g. □

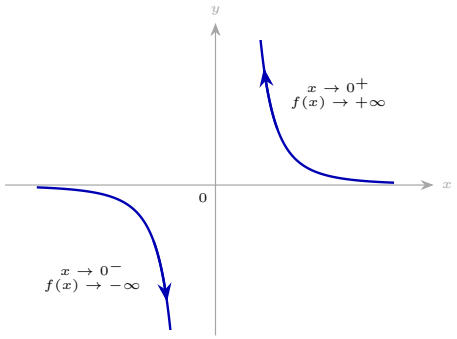

Figure 155: For $f(x)=x^{-3}$, the one-sided limits at 0 are opposite infinities.

Definition 1.1.22. *Higher Derivatives*
If f has derivative f' on an interval and f' is differentiable, then $(f')'$ is denoted f'' and called the *second derivative*. Continuing recursively gives $f^{(n)}$, the derivative of order n.

Definition 1.1.23. *Smoothness Classes*
On an interval, C^0 means continuous, C^r means $f^{(r)}$ exists and continuous, C^∞ means C^r for every finite r.

Theorem 1.1.24. *Higher-Order Leibniz Product Rule*
Let $r\ge 1$, and $f,g:(a,b)\to\mathbb{R}$ be r times differentiable. Then $\forall x\in(a,b)$, $(fg)^{(r)}(x)=\sum_{k=0}^{r}\binom{r}{k}f^{(k)}(x)g^{(r-k)}(x)$.

Proof. Induct on r. For $r=1$, this is the product rule in Theorem 1.1.5. Assume the formula holds at order r.
If f,g are $(r+1)$ times differentiable, differentiating gives $(fg)^{(r+1)}=\sum_{k=0}^{r}\binom{r}{k}f^{(k+1)}g^{(r-k)}+\sum_{k=0}^{r}\binom{r}{k}f^{(k)}g^{(r-k+1)}$.
Reindex the first sum by $j=k+1$; the coefficient of $f^{(j)}g^{(r+1-j)}$ is $\binom{r}{j-1}+\binom{r}{j}=\binom{r+1}{j}$, with the endpoint terms $j=0,r+1$ equal to 1.
Structural Commentary: Differentiating shifts one derivative from the total count onto f or g.
Key Insight: Pascal's identity produces the binomial coefficients. □

Theorem 1.1.25. *Taylor's Theorem with Lagrange Remainder*
Let $f:[a,b]\to\mathbb{R}$, let $n\ge 1$, suppose $f^{(n-1)}$ is continuous on $[a,b]$, and suppose $f^{(n)}$ exists on (a,b).
For distinct $\alpha,\beta\in[a,b]$, set $P(t):=\sum_{k=0}^{n-1}\frac{f^{(k)}(\alpha)}{k!}(t-\alpha)^k$.
Then for some x between α and β, $f(\beta)=P(\beta)+\frac{f^{(n)}(x)}{n!}(\beta-\alpha)^n$.

Proof. Choose M so that $f(\beta)=P(\beta)+M(\beta-\alpha)^n$, and set $G(t):=f(t)-P(t)-M(t-\alpha)^n$.
Then $G(\alpha)=G'(\alpha)=\cdots=G^{(n-1)}(\alpha)=G(\beta)=0$. Repeated applications of Corollary 1.1.13 give $G^{(n)}(x)=0$ for some x between α and β. Since $G^{(n)}(x)=f^{(n)}(x)-n!M$, the formula follows.
Structural Commentary: The auxiliary error vanishes at β and to order $n-1$ at α.
Key Insight: Repeated Rolle-style arguments turn endpoint cancellation into a precise remainder. □

Definition 1.1.26. *Vector-Valued Derivative*
For $\mathbf{f}:[a,b]\to\mathbb{R}^k$, set $\mathbf{f}'(x)=\mathbf{L}$ when $\lim_{t\to x}\left\|\frac{\mathbf{f}(t)-\mathbf{f}(x)}{t-x}-\mathbf{L}\right\|_2=0$.
For $\mathbf{f}=(f_1,\dots,f_k)$, this is equivalent to componentwise differentiability, with $\mathbf{f}'(x)=(f_1'(x),\dots,f_k'(x))$.

Theorem 1.1.27. *Vector-Valued Mean Value Inequality*
If $\mathbf{f}:[a,b]\to\mathbb{R}^k$ is continuous on $[a,b]$ and differentiable on (a,b).
Then for some $x\in(a,b)$, $\|\mathbf{f}(b)-\mathbf{f}(a)\|_2\le(b-a)\|\mathbf{f}'(x)\|_2$.

Proof. Set $\mathbf{z}:=\mathbf{f}(b)-\mathbf{f}(a)$ and $\varphi(t):=\mathbf{z}\cdot\mathbf{f}(t)$.
Corollary 1.1.13 gives $\varphi(b)-\varphi(a)=(b-a)\varphi'(x)=(b-a)\mathbf{z}\cdot\mathbf{f}'(x)$ for some $x\in(a,b)$. Since $\varphi(b)-\varphi(a)=\|\mathbf{z}\|_2^2$, Theorem 1.1.13 gives $\|\mathbf{z}\|_2^2\le(b-a)\|\mathbf{z}\|_2\|\mathbf{f}'(x)\|_2$, and the result follows by cancellation, with $\mathbf{z}=0$ trivial.
Structural Commentary: A scalar projection yields the vector norm bound.
Key Insight: Project onto the displacement vector, then use Schwarz to return to norms. □

Method 1.1.28. *Differentiation Proof Patterns*

(i) *Domain*: require x_0 to be a limit point of X; isolated points have no derivative.
(ii) *Definition*: factor $f(t)-f(x)=(t-x)Q_x(t)$ and take $Q_x(t)\to f'(x)$.
(iii) *Approximation*: use Proposition 1.1.3 to replace quotient limits by first-order error bounds.
(iv) *Algebra*: prove quotient identities first; use continuity to pass to limits.
(v) *Mean values*: choose h with equal endpoint values and apply Theorem 1.1.10.
(vi) *Taylor*: set $G=f-P-M(t-\alpha)^n$ and iterate the mean value theorem.

Key Insight: Difference quotients and auxiliary functions drive one-variable differentiation.

1.2 Riemann Integration

The integration side of one-variable calculus is organized by Riemann and Riemann–Stieltjes sums; the fundamental theorem connects the two halves.

This subsection develops Riemann and Riemann–Stieltjes integration by partitions, then links integrals to antiderivatives through the Fundamental Theorem of Calculus. Point partitions and interval-envelope partitions are equivalent via common refinement.

Definition 1.2.1. *Partitions and Riemann Upper/Lower Sums*

(i) A *partition* of $[a,b]$ is $\Pi=\{x_0,\dots,x_n\}$ with $a=x_0\le x_1\le\cdots\le x_n=b$.
Set $\Delta x_i:=x_i-x_{i-1}$ for $1\le i\le n$.
(ii) For bounded $f:[a,b]\to\mathbb{R}$, set $M_i:=\sup_{x\in[x_{i-1},x_i]}f(x)$ and $m_i:=\inf_{x\in[x_{i-1},x_i]}f(x)$.
(iii) The *upper* and *lower Darboux sums* are $U(\Pi,f):=\sum_{i=1}^n M_i\Delta x_i$ and $L(\Pi,f):=\sum_{i=1}^n m_i\Delta x_i$.
(iv) The *upper* and *lower Darboux integrals* are $\overline{\int_a^b f\,dx}:=\inf_\Pi U(\Pi,f)$ and $\underline{\int_a^b f\,dx}:=\sup_\Pi L(\Pi,f)$, where Π ranges over all partitions of $[a,b]$.
(v) If $\overline{\int_a^b f\,dx}=\underline{\int_a^b f\,dx}$, then f is *Riemann integrable* on $[a,b]$; $f\in\mathcal{R}[a,b]$, $\int_a^b f\,dx:=\overline{\int_a^b f\,dx}=\underline{\int_a^b f\,dx}$.

If $m\le f(x)\le M$ for $x\in[a,b]$, then $m(b-a)\le L(\Pi,f)\le U(\Pi,f)\le M(b-a)$ for every partition Π.
Hence the upper and lower Darboux integrals are finite for bounded f.
Key Insight: Riemann integrability means least upper Darboux area equals greatest lower Darboux area.

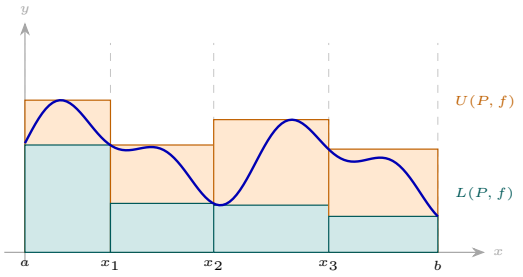

Figure 156: Upper and lower sums record the oscillation of f on each partition interval.

Definition 1.2.2. *Tagged Partitions and Riemann Sums*
Let $\Pi=\{x_0,\dots,x_n\}$ be a partition of $[a,b]$, and set $\Delta x_i:=x_i-x_{i-1}$.

(i) A *tagging* of Π is $c=(c_1,\dots,c_n)$ with $c_i\in[x_{i-1},x_i]$ for $1\le i\le n$; the pair (Π,c) is a *tagged partition*.
(ii) For $f:[a,b]\to\mathbb{R}$, the associated *Riemann sum* is $R(f,\Pi,c):=\sum_{i=1}^n f(c_i)\Delta x_i$.
(iii) If f is bounded, then $L(\Pi,f)\le R(f,\Pi,c)\le U(\Pi,f)$.

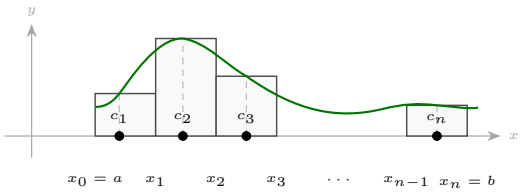

Figure 157: A tagged Riemann sum uses one sample value $f(c_i)$ on each subinterval $[x_{i-1},x_i]$.

Perspective 1.2.3. *Tagged Sums and Gauges*
Gauge integration keeps tagged sums but replaces one mesh bound δ with a local bound $\delta(c_i)$.
Thus fineness may vary by tag, giving a stronger framework for FTC-type results.

2 Riemann-Stieltjes and Improper Integration

2.1 Riemann-Stieltjes Integrability

Definition 2.1.1. *Riemann–Stieltjes Integral*
Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, let $\Pi = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$, and let $f : [a,b] \to \mathbb{R}$ be bounded. Set $M_i := \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i := \inf_{x \in [x_{i-1}, x_i]} f(x)$.

(i) The Stieltjes increment on $[x_{i-1}, x_i]$ is $\Delta\alpha_i := \alpha(x_i) - \alpha(x_{i-1}) \geq 0$ for $1 \leq i \leq n$.

(ii) The *upper* and *lower Riemann–Stieltjes sums* are $U(\Pi, f, \alpha) := \sum_{i=1}^{n} M_i \Delta\alpha_i$ and $L(\Pi, f, \alpha) := \sum_{i=1}^{n} m_i \Delta\alpha_i$.

(iii) The *upper* and *lower Riemann–Stieltjes integrals* are $\overline{\int_a^b f \, d\alpha} := \inf_{\Pi} U(\Pi, f, \alpha)$ and $\underline{\int_a^b f \, d\alpha} := \sup_{\Pi} L(\Pi, f, \alpha)$, where the infimum and supremum range over all partitions of $[a,b]$.

(iv) If $\overline{\int_a^b f \, d\alpha} = \underline{\int_a^b f \, d\alpha}$, then f is *Riemann integrable with respect to* α, written $f \in \mathcal{R}(\alpha)$, and $\int_a^b f \, d\alpha := \overline{\int_a^b f \, d\alpha} = \underline{\int_a^b f \, d\alpha}$, also written $\int_a^b f(x) \, d\alpha(x)$.

For $\alpha(x) = x$, this is the ordinary Riemann integral.

Remark 2.1.2. *Stieltjes Increments as α-Lengths*
Stieltjes sums replace ordinary length Δx_i by $\Delta\alpha_i$, the length measured by α.
Thus $\alpha(x) = x$ gives Riemann sums, while jumps of α place point mass as in Theorem 2.3.2.

Definition 2.1.3. *Refinements*
Let Π, Π^*, Π_1, Π_2 be partitions of $[a,b]$.

(i) Π^* is a *refinement* of Π iff $\Pi \subseteq \Pi^*$.

(ii) The *common refinement* of Π_1 and Π_2 is $\Pi_1 \cup \Pi_2$.

Theorem 2.1.4. *Refinement Monotonicity*
If Π^* is a refinement of Π, then $L(\Pi, f, \alpha) \leq L(\Pi^*, f, \alpha)$ and $U(\Pi^*, f, \alpha) \leq U(\Pi, f, \alpha)$.

Proof. It suffices to add one point $x^* \in (x_{i-1}, x_i)$. If w_1, w_2 are infima on $[x_{i-1}, x^*]$ and $[x^*, x_i]$, then $w_1, w_2 \geq m_i$ and $L(\Pi^*, f, \alpha) - L(\Pi, f, \alpha) = (w_1 - m_i)\big[\alpha(x^*) - \alpha(x_{i-1})\big] + (w_2 - m_i)\big[\alpha(x_i) - \alpha(x^*)\big] \geq 0$.
Iterate for additional points. The upper-sum inequality is the same argument with suprema.
Structural Commentary: Refinement raises lower sums and lowers upper sums.
Key Insight: Splitting an interval tightens the local oscillation bounds. □

Perspective 2.1.5. *Piecewise-Constant Envelopes*
A partition Π gives lower and upper step envelopes with values m_i and M_i; their integrals are exactly $L(\Pi, f)$ and $U(\Pi, f)$. Common refinements make these envelope sums compatible across partitions.

Theorem 2.1.6. *Lower Integral Below Upper Integral*
For every bounded real f and increasing α on $[a,b]$, $\underline{\int_a^b f \, d\alpha} \leq \overline{\int_a^b f \, d\alpha}$.

Proof. Given partitions Π_1, Π_2, let $\Pi^* = \Pi_1 \cup \Pi_2$.
By Theorem 2.1.4, $L(\Pi_1, f, \alpha) \leq L(\Pi^*, f, \alpha) \leq U(\Pi^*, f, \alpha) \leq U(\Pi_2, f, \alpha)$.
Take the supremum over Π_1 and then the infimum over Π_2.
Structural Commentary: Common refinements compare sums from different partitions.
Key Insight: The lower envelope never rises above the upper envelope. □

Theorem 2.1.7. *Upper–Lower Criterion*
A bounded real f is in $\mathcal{R}(\alpha)$ on $[a,b]$ if and only if $\forall \varepsilon > 0$ $\exists$ a partition Π with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.

Proof.

($\Rightarrow$) If $f \in \mathcal{R}(\alpha)$, choose Π_1, Π_2 whose lower and upper sums are within $\varepsilon/2$ of $\int_a^b f \, d\alpha$.
Their common refinement Π satisfies the displayed inequality by Theorem 2.1.4.

($\Leftarrow$) If such Π exist, then $L(\Pi, f, \alpha) \leq \underline{\int_a^b f \, d\alpha} \leq \overline{\int_a^b f \, d\alpha} \leq U(\Pi, f, \alpha)$ forces upper and lower integrals to agree.

Structural Commentary: Integrability is exactly the ability to make one partition squeeze the oscillation.
Key Insight: The gap $U - L$ is the quantitative defect of integrability. □

Perspective 2.1.8. *Darboux Gaps and Riemann Sums*
For $\alpha(x) = x$, a small Darboux gap $U(\Pi, f) - L(\Pi, f)$ traps every tagged sum for Π near the integral. This is the local reason the Darboux definition above and the tagged Riemann-sum definition give the same number. The same estimate works with Stieltjes weights $\Delta\alpha_i$ whenever the upper–lower gap is small.

Theorem 2.1.9. *Tagged Sums*
Let $\Pi = \{x_0, \dots, x_n\}$ satisfy $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$, and choose tags $t_i \in [x_{i-1}, x_i]$.

(i) If $s_i, t_i \in [x_{i-1}, x_i]$, then $\sum_{i=1}^{n} |f(s_i) - f(t_i)| \, \Delta\alpha_i < \varepsilon$.

(ii) If $f \in \mathcal{R}(\alpha)$, then $\left| \sum_{i=1}^{n} f(t_i) \Delta\alpha_i - \int_a^b f \, d\alpha \right| < \varepsilon$.

Proof.

(i) Since $|f(s_i) - f(t_i)| \le M_i - m_i$, summing gives $\sum_{i=1}^{n} |f(s_i) - f(t_i)|\,\Delta\alpha_i \le U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.

(ii) The tagged sum and $\int_a^b f\,d\alpha$ both lie between $L(\Pi, f, \alpha)$ and $U(\Pi, f, \alpha)$, so their distance is $< \varepsilon$.

Structural Commentary: A small Darboux gap makes tags and the integral close.
Key Insight: Tags are harmless after oscillation has been controlled. □

Theorem 2.1.10. *Continuous Functions Are Riemann–Stieltjes Integrable*
If f is continuous on $[a, b]$ and α is increasing on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. If $\alpha(b) = \alpha(a)$, the Darboux gap is zero.
Otherwise choose $\eta > 0$ with $\eta[\alpha(b) - \alpha(a)] < \varepsilon$. Uniform continuity gives partition Π with $M_i - m_i \le \eta\ \forall i$, so $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i \le \eta[\alpha(b) - \alpha(a)] < \varepsilon$. Apply Theorem 2.1.7.
Structural Commentary: Uniform continuity controls oscillation uniformly across the partition.
Key Insight: Continuity turns local small intervals into a small total Darboux gap. □

Theorem 2.1.11. *Monotone Functions and Continuous Integrators*
If f is monotone on $[a, b]$ and α is continuous and increasing on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. Assume first that f is increasing. If $\alpha(b) = \alpha(a)$, then every $\Delta\alpha_i = 0$, so $U(\Pi, f, \alpha) = L(\Pi, f, \alpha) = 0$ for every partition Π. Choose n and a partition Π such that $\Delta\alpha_i = [\alpha(b) - \alpha(a)]/n$ for all i.
Existence follows from continuity of α and Theorem 5.1.41.
Then $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \frac{\alpha(b)-\alpha(a)}{n} \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = \frac{\alpha(b)-\alpha(a)}{n}\big(f(b) - f(a)\big) < \varepsilon$ for n large.
If f is decreasing, use the same partition. Then the suprema and infima on $[x_{i-1}, x_i]$ are $f(x_{i-1})$ and $f(x_i)$, so the same computation gives $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \frac{\alpha(b)-\alpha(a)}{n}\big(f(a) - f(b)\big)$, which is $< \varepsilon$ for large n.
Structural Commentary: Equal α-increments convert the monotone oscillations into a telescoping sum.
Key Insight: A monotone function's total oscillation is finite and can be spread thinly. □

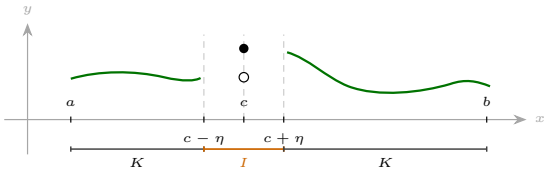

Figure 158: Isolate the discontinuity in a short interval; control the compact pieces outside it.

Theorem 2.1.12. *Finitely Many Discontinuities*
Suppose f is bounded on $[a, b]$, the discontinuity set $E := \{x \in [a, b] : f \text{ is discontinuous at } x\}$ is finite, and α is continuous at each point of E. Then $f \in \mathcal{R}(\alpha)$.

Proof. Let $M = \sup_{[a,b]} |f|$ and let $\eta > 0$. By continuity of α on the finite set E, choose disjoint intervals $I_j = [u_j, v_j]$ covering E, with each point of $E \cap (a, b)$ in some (u_j, v_j), such that $\sum_j \big(\alpha(v_j) - \alpha(u_j)\big) < \eta$.
Set $K = [a, b] \setminus \bigcup_j (u_j, v_j)$. Then $f|_K$ is uniformly continuous by Theorem 5.2.7.
Choose a partition Π containing all u_j, v_j, with no points inside the intervals (u_j, v_j), and fine enough on K that $M_i - m_i \le \eta$ whenever $[x_{i-1}, x_i] \subset K$.
Hence $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) \le \eta\big[\alpha(b) - \alpha(a)\big] + 2M \sum_j \big(\alpha(v_j) - \alpha(u_j)\big) < \eta\big[\alpha(b) - \alpha(a) + 2M\big]$.
Since η is arbitrary, Theorem 2.1.7 applies.
Structural Commentary: Isolate bad points in small α-intervals; use uniform continuity elsewhere.
Key Insight: Finitely many discontinuities do not matter unless α also jumps there. □

Example 2.1.13. *Dirichlet's Function Is Not Riemann Integrable*
Let $g = \mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$. Every nondegenerate subinterval contains rationals and irrationals by Corollary 1.1.22, so for every partition Π, $M_i = 1$ and $m_i = 0$ on each nonzero-width subinterval.
Thus $U(\Pi, g) = 1$ and $L(\Pi, g) = 0$, hence $g \notin \mathcal{R}$.

Remark 2.1.14. *Small Discontinuity Sets*
The same idea underlies Theorem 1.5.2: small discontinuity sets, later called measure-zero sets in Definition 4.1.13, do not obstruct integrability. Dirichlet's function fails because every point is discontinuous.

Theorem 2.1.15. *Continuous Composition Preserves Integrability*
Let α be increasing on $[a, b]$. Suppose $f \in \mathcal{R}(\alpha)$, $m \le f(x) \le M$ on $[a, b]$, and $\phi : [m, M] \to \mathbb{R}$ is continuous. Define $h : [a, b] \to \mathbb{R}$ by $h(x) := \phi(f(x))$. Then $h \in \mathcal{R}(\alpha)$.

Proof. Let $C := \sup_{y \in [m,M]} |\phi(y)|$ and let $\varepsilon > 0$. Choose $\eta > 0$ such that $\eta[\alpha(b) - \alpha(a) + 2C] < \varepsilon$. Since ϕ is uniformly continuous on $[m, M]$, choose $\gamma > 0$ such that $|s - t| < \gamma \Longrightarrow |\phi(s) - \phi(t)| < \eta$ $(s, t \in [m, M])$.
Choose a partition $\Pi = \{x_0, \dots, x_n\}$ with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$.
Let M_i, m_i denote upper and lower values for f on $[x_{i-1}, x_i]$, M_i^h, m_i^h denote corresponding values for h.
Split $A := \{i : M_i - m_i < \gamma\}$, $B := \{i : M_i - m_i \ge \gamma\}$. For $i \in A$, $M_i^h - m_i^h \le \eta$; for $i \in B$, $M_i^h - m_i^h \le 2C$.
Also $\gamma \sum_{i \in B} \Delta\alpha_i \le \sum_{i \in B} (M_i - m_i)\Delta\alpha_i \le U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$, so $\sum_{i \in B} \Delta\alpha_i < \eta$.

$U(\Pi, h, \alpha) - L(\Pi, h, \alpha) \le \eta \sum_{i\in A} \Delta\alpha_i + 2C \sum_{i\in B} \Delta\alpha_i < \eta[\alpha(b) - \alpha(a) + 2C] < \varepsilon$. Theorem 2.1.7 gives $h \in \mathcal{R}(\alpha)$.
Structural Commentary: Integrability survives continuous remeasurement of the values of f.
Key Insight: Large oscillation intervals have small total α-weight. □

Theorem 2.1.16. *Basic Properties of the Integral*
Let all functions be bounded and real on $[a,b]$, and let all integrators be increasing.

(i) If $f_1, f_2 \in \mathcal{R}(\alpha)$, $c \in \mathbb{R}$, then $f_1 + f_2, cf_1 \in \mathcal{R}(\alpha)$, and $\int_a^b (f_1 + f_2)\,d\alpha = \int_a^b f_1\,d\alpha + \int_a^b f_2\,d\alpha$; $\int_a^b cf_1\,d\alpha = c\int_a^b f_1\,d\alpha$.

(ii) If $f_1 \le f_2$ and both functions are in $\mathcal{R}(\alpha)$, then $\int_a^b f_1\,d\alpha \le \int_a^b f_2\,d\alpha$.

(iii) If $f \in \mathcal{R}(\alpha)$ and $a < c < b$, then f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^c f\,d\alpha + \int_c^b f\,d\alpha = \int_a^b f\,d\alpha$.

(iv) If $f \in \mathcal{R}(\alpha)$ and $|f| \le M$, then $\left|\int_a^b f\,d\alpha\right| \le M[\alpha(b) - \alpha(a)]$

(v) If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and $\int_a^b f\,d(\alpha_1 + \alpha_2) = \int_a^b f\,d\alpha_1 + \int_a^b f\,d\alpha_2$. If $c > 0$ and $f \in \mathcal{R}(\alpha)$, then $f \in \mathcal{R}(c\alpha)$ and $\int_a^b f\,d(c\alpha) = c\int_a^b f\,d\alpha$.

Proof.
(i) $L(\Pi, f_1, \alpha) + L(\Pi, f_2, \alpha) \le L(\Pi, f_1 + f_2, \alpha) \le U(\Pi, f_1 + f_2, \alpha) \le U(\Pi, f_1, \alpha) + U(\Pi, f_2, \alpha)$, where Π is common refinement, and Theorem 2.1.7; scalar multiplication is the same estimate with signs handled by swapping upper and lower sums when $c < 0$.
(ii) If $f_1 \le f_2$, then $L(\Pi, f_2 - f_1, \alpha) \ge 0$ for every Π; apply (i) to $f_2 - f_1$.
(iii) Refine so that $c \in \Pi$. Then the lower and upper sums split into the sums over $[a,c]$ and $[c,b]$.
(iv) Apply (ii) to $-M \le f \le M$ and use the integrals of the constant functions $\pm M$.
(v) Since $\Delta(\alpha_1 + \alpha_2)_i = \Delta\alpha_{1,i} + \Delta\alpha_{2,i}$ and $\Delta(c\alpha)_i = c\,\Delta\alpha_i$, corresponding lower and upper sums split exactly.
Structural Commentary: The algebra of integrals is inherited from the algebra of upper and lower sums.
Key Insight: Common refinements let several integrable functions be approximated by the same partition. □

Theorem 2.1.17. *Products and Absolute Values*
If $f, g \in \mathcal{R}(\alpha)$ on $[a,b]$, then $fg \in \mathcal{R}(\alpha)$, $|f| \in \mathcal{R}(\alpha)$, and $\left|\int_a^b f\,d\alpha\right| \le \int_a^b |f|\,d\alpha$.

Proof. By Theorem 2.1.16, $f \pm g \in \mathcal{R}(\alpha)$. Applying Theorem 2.1.15 to $\phi(t) = t^2$ gives $(f \pm g)^2 \in \mathcal{R}(\alpha)$, hence $fg \in \mathcal{R}(\alpha)$ from $4fg = (f+g)^2 - (f-g)^2$.
Applying the same theorem to $\phi(t) = |t|$ gives $|f| \in \mathcal{R}(\alpha)$, and $-|f| \le f \le |f|$ gives the inequality.
Structural Commentary: Products reduce to squares; absolute values to composition.
Key Insight: Once continuous composition is available, the usual algebra closes. □

2.2 Fundamental Theorems of Calculus

Theorem 2.2.1. *Integral Functions*
Let $f \in \mathcal{R}$ on $[a,b]$ and define $F(x) := \int_a^x f(t)\,dt$. Then F is continuous.
If f is continuous at x_0, then $F'(x_0) = f(x_0)$, with one-sided derivatives at endpoints.

Proof. Let $|f| \le M$. For $x < y$, Theorem 2.1.16 gives $|F(y) - F(x)| = \left|\int_x^y f(t)\,dt\right| \le M(y - x)$ so F continuous.
If f continuous at x_0, for $x \ne x_0$ near x_0, $\left|\frac{F(x)-F(x_0)}{x-x_0} - f(x_0)\right| \le \sup_{u\in[x,x_0]} |f(u) - f(x_0)| \to 0$. $F'(x_0) = f(x_0)$.
Structural Commentary: Boundedness controls increments; local continuity gives the derivative.
Key Insight: Averaging a continuous function over shrinking intervals returns its value. □

Theorem 2.2.2. *Fundamental Theorem of Calculus*
If $f \in \mathcal{R}$ on $[a,b]$ and $\exists$ a differentiable function F on $[a,b]$ such that $F' = f$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Proof. Choose $\Pi = \{x_0, \dots, x_n\}$ with $U(\Pi, f) - L(\Pi, f) < \varepsilon$.
By MVT, choose $t_i \in [x_{i-1}, x_i]$ so that $F(b) - F(a) = \sum_i \big(F(x_i) - F(x_{i-1})\big) = \sum_i f(t_i)\Delta x_i$.

Theorem 2.1.9 puts this tagged sum within ε of $\int_a^b f$. Let $\varepsilon \to 0$.
Structural Commentary: Mean-value increments convert antiderivatives into tagged Riemann sums.
Key Insight: The integral is the common limit of all sufficiently fine tagged sums. □

Theorem 2.2.3. *Integration by Parts*
Suppose F, G are differentiable on $[a,b]$, $F' = f \in \mathcal{R}$, and $G' = g \in \mathcal{R}$.
Then $\int_a^b F(x)g(x)\,dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)\,dx$.

Proof. Set $H := FG$. Since F, G are continuous, $F, G \in \mathcal{R}$; hence $Fg, fG \in \mathcal{R}$ by Theorem 2.1.17.
Thus $H' = Fg + fG \in \mathcal{R}$, and Theorem 2.2.2 gives $F(b)G(b) - F(a)G(a) = H(b) - H(a) = \int_a^b Fg + \int_a^b fG$.
Structural Commentary: Integration by parts is the product rule integrated over the interval.
Key Insight: Move one derivative from G onto F by applying the fundamental theorem to FG. □

2.3 Step Integrators and Change of Variable

Definition 2.3.1. *Unit Step Function*

The *unit step function* I is $I(x) = \begin{cases} 0, & x \le 0, \\ 1, & x > 0. \end{cases}$.

Theorem 2.3.2. *Single Jump Integrator*

If $a < s < b$, f is bounded on $[a,b]$ and continuous at s, and $\alpha(x) = I(x-s)$, then $\int_a^b f\,d\alpha = f(s)$.

Proof. Use partitions $\Pi = \{a, s, x_2, b\}$ with $s < x_2 < b$. The only nonzero $\Delta\alpha_i$ occurs on $[s, x_2]$, so $L(\Pi, f, \alpha)$ and $U(\Pi, f, \alpha)$ are the infimum and supremum of f on $[s, x_2]$. As $x_2 \downarrow s$, continuity at s squeezes both to $f(s)$.
Structural Commentary: A jump of α makes the integral sample f at the jump.
Key Insight: Stieltjes integration includes point masses without changing the Darboux framework. □

Theorem 2.3.3. *Pure Step Integrators Give Series*

Suppose $c_n \ge 0$, $\sum c_n$ converges, and (s_n) is a sequence of distinct points in (a,b).
Define $\alpha(x) := \sum_{n=1}^{\infty} c_n I(x - s_n)$. If f is continuous on $[a,b]$, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$.

Proof. By comparison, α is increasing. For $N \in \mathbb{N}$, put $\alpha_N(x) := \sum_{n=1}^{N} c_n I(x - s_n)$ and $\beta_N := \alpha - \alpha_N$..

Theorems 2.3.2 and 2.1.16 give $\int_a^b f\,d\alpha_N = \sum_{n=1}^{N} c_n f(s_n)$. Also $f \in \mathcal{R}(\beta_N)$ by Theorem 2.1.10.

If $|f| \le M$, then Theorem 2.1.16 gives $\left|\int_a^b f\,d\beta_N\right| \le M[\beta_N(b) - \beta_N(a)] = M \sum_{n>N} c_n$.

Letting $N \to \infty$ and using additivity in the integrator gives the formula.
Structural Commentary: Finite step integrators sum point samples; summable tails vanish by the integral bound.
Key Insight: A Stieltjes integral can be an ordinary series in disguise. □

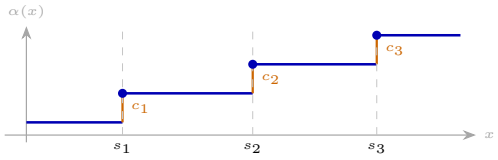

Figure 159: A step integrator concentrates mass at jumps: $\int f\,d\alpha = \sum_i c_i f(s_i)$.

Theorem 2.3.4. *Differentiable Integrators Reduce to Riemann Integrals*

Assume α increases monotonically, $\alpha' \in \mathcal{R}$ on $[a,b]$, and f is bounded and real on $[a,b]$.
Then $f \in \mathcal{R}(\alpha) \iff f\alpha' \in \mathcal{R}$, and in that case $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. Let $M = \sup_{[a,b]} |f|$. Choose Π with $U(\Pi, \alpha') - L(\Pi, \alpha') < \varepsilon$.
For any refinement and any tags $s_i \in [x_{i-1}, x_i]$, MVT gives points $t_i \in [x_{i-1}, x_i]$ with $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$.
Hence $\left|\sum_i f(s_i)\Delta\alpha_i - \sum_i f(s_i)\alpha'(s_i)\Delta x_i\right| \le M \sum_i |\alpha'(t_i) - \alpha'(s_i)|\,\Delta x_i \le M\varepsilon$. Thus every tagged Stieltjes sum for $f\,d\alpha$ is within $M\varepsilon$ of the corresponding tagged Riemann sum for $f\alpha'\,dx$, and conversely. By the tagged-sum criterion, one family has a limit exactly when the other does, and the two limits are equal.
Structural Commentary: The increment $d\alpha$ is replaced by $\alpha'(x)\,dx$ when α' is Riemann integrable.
Key Insight: The mean value theorem turns Stieltjes increments into ordinary Riemann weights. □

Theorem 2.3.5. *Change of Variable*

Let $\varphi : [A,B] \to [a,b]$ be strictly increasing, continuous, and onto.
Let α be increasing on $[a,b]$, $f \in \mathcal{R}(\alpha)$, and set $\beta(y) := \alpha(\varphi(y))$ and $g(y) := f(\varphi(y))$.
Then $g \in \mathcal{R}(\beta)$ and $\int_A^B g\,d\beta = \int_a^b f\,d\alpha$. If also $\alpha(x) = x$ and $\varphi' \in \mathcal{R}$, then $\int_a^b f(x)\,dx = \int_A^B f(\varphi(y))\varphi'(y)\,dy$.

Proof. Partitions $Q = \{y_i\}$ of $[A,B]$ correspond to partitions $\Pi = \{\varphi(y_i)\}$ of $[a,b]$. On corresponding intervals, g has same range as f and $\Delta\beta_i = \Delta\alpha_i$, so $U(Q, g, \beta) = U(\Pi, f, \alpha)$ and $L(Q, g, \beta) = L(\Pi, f, \alpha)$. The upper–lower criterion gives the Stieltjes formula. If $\alpha(x) = x$, then $\beta = \varphi$; apply Theorem 2.3.4 to the left side.
Structural Commentary: A monotone change of variable relabels the same partition intervals.
Key Insight: Stieltjes change of variable is equality of Darboux sums under reparametrization. □

2.4 Improper Integrals

Definition 2.4.1. *Improper Riemann Integrals*

Let $f \in \mathcal{R}$ on every compact subinterval of its domain.
Set $\int_a^b f(x)\,dx := \lim_{d \to b^-} \int_a^d f(x)\,dx$ at a right endpoint, and $\int_a^b f(x)\,dx := \lim_{c \to a^+} \int_c^b f(x)\,dx$ at a left endpoint.
If both endpoints are improper, choose $\xi \in (a,b)$ and set $\int_a^b f(x)\,dx := \int_a^\xi f(x)\,dx + \int_\xi^b f(x)\,dx$, provided both one-sided integrals exist and the sum is not indeterminate.
A finite value means the improper integral *converges*; otherwise it diverges to $+\infty$ or $-\infty$.

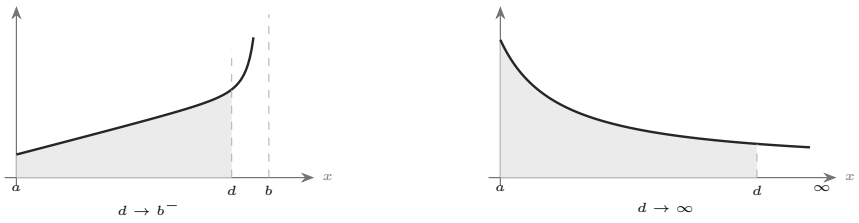

Figure 160: Improper integrals are limits of ordinary integrals over compact truncations.

Theorem 2.4.2. *Nonnegative Improper Integrals*
Let $f \geq 0$ and $f \in \mathcal{R}$ on every $[a,d]$, $a < d < b$. Then $\int_a^b f(x)\,dx = \sup\left\{\int_a^d f(x)\,dx : a < d < b\right\}$.
Thus $\int_a^b f$ exists in $[0,+\infty]$ and converges exactly when this supremum is finite.

Proof. For $a < d_1 < d_2 < b$, $\int_a^{d_2} f - \int_a^{d_1} f = \int_{d_1}^{d_2} f \geq 0$.
Hence $d \mapsto \int_a^d f$ is increasing, so its extended limit exists and equals its supremum.
Structural Commentary: Nonnegative area can only accumulate.
Key Insight: Improper convergence for $f \geq 0$ is boundedness of all finite truncations. □

Perspective 2.4.3. *Principal Value Is Different*
Improper integrals over $(-\infty,\infty)$ require separate one-sided limits.
A symmetric limit such as $\lim_{A\to\infty}\int_{-A}^{A}\sin x\,dx$ may still exist (Cauchy principal value, not Definition 2.4.1).

Definition 2.4.4. *Improper Riemann–Stieltjes Integrals on $\mathbb{R}$*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing and $f \in \mathcal{R}(\alpha)$ on every finite interval.
Define, when the limits exist, $\int_0^\infty f\,d\alpha := \lim_{b\to\infty}\int_0^b f\,d\alpha$; and $\int_{-\infty}^0 f\,d\alpha := \lim_{a\to-\infty}\int_a^0 f\,d\alpha$.
If their sum is not indeterminate, set $\int_{-\infty}^\infty f\,d\alpha := \int_{-\infty}^0 f\,d\alpha + \int_0^\infty f\,d\alpha$.
Finite value means f is *α-integrable on $\mathbb{R}$*. Write $\alpha(\pm\infty) := \lim_{t\to\pm\infty}\alpha(t)$ when these extended limits are used.

Theorem 2.4.5. *Nonnegative and Bounded-Mass Stieltjes Integrals*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing, and suppose $f \in \mathcal{R}(\alpha)$ on every finite interval.
(i) If $f \geq 0$, then either f is α-integrable on $\mathbb{R}$ or $\int_{-\infty}^\infty f\,d\alpha = +\infty$.
(ii) If $-\infty < \alpha(-\infty) < \alpha(\infty) < +\infty$ and f is bounded, then f is α-integrable on $\mathbb{R}$.

Proof.
(i) By additivity and order preservation, $\int_0^b f\,d\alpha$ and $\int_a^0 f\,d\alpha$ are monotone under outward truncation. Hence both extended one-sided limits exist, so their sum is finite or $+\infty$.
(ii) Let $|f| \leq M$. For finite $a < b$, Theorem 2.1.16 gives $\left|\int_a^b f\,d\alpha\right| \leq M\left[\alpha(b) - \alpha(a)\right]$. Since $\alpha(\infty)$ and $\alpha(-\infty)$ are finite, the right and left tails have vanishing α-increment. Thus the one-sided truncations are Cauchy and have finite limits.
Structural Commentary: Positivity gives monotone limits; finite mass bounds tails.
Key Insight: A bounded increasing integrator behaves like finite total mass. □

2.5 Vector-Valued Integrals and Curves

Definition 2.5.1. *Vector-Valued Riemann–Stieltjes Integral*
For $\mathbf{f} = (f_1,\dots,f_k) : [a,b] \to \mathbb{R}^k$, define $\mathbf{f} \in \mathcal{R}(\alpha)$ iff $f_j \in \mathcal{R}(\alpha)$ $\forall j$, and set $\int_a^b \mathbf{f}\,d\alpha := \left(\int_a^b f_1\,d\alpha,\dots,\int_a^b f_k\,d\alpha\right)$.

Theorem 2.5.2. *Vector-Valued Integral Results*
(i) If $\mathbf{f},\mathbf{F} : [a,b] \to \mathbb{R}^k$, $\mathbf{f} \in \mathcal{R}$, and $\mathbf{F}' = \mathbf{f}$, then $\int_a^b \mathbf{f}(t)\,dt = \mathbf{F}(b) - \mathbf{F}(a)$.
(ii) If $\mathbf{f} : [a,b] \to \mathbb{R}^k$ and $\mathbf{f} \in \mathcal{R}(\alpha)$, then $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$ and $\left\|\int_a^b \mathbf{f}\,d\alpha\right\| \leq \int_a^b \|\mathbf{f}\|\,d\alpha$

Proof.
(i) Write $\mathbf{f} = (f_1,\dots,f_k)$ and $\mathbf{F} = (F_1,\dots,F_k)$. For each j, Theorem 2.2.2 gives $\int_a^b f_j(t)\,dt = F_j(b) - F_j(a)$. The vector identity follows componentwise.
(ii) Write $\mathbf{f} = (f_1,\dots,f_k)$. Since $\|\mathbf{f}\| = (f_1^2 + \cdots + f_k^2)^{1/2}$, Theorems 2.1.17 and 2.1.15 give $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$.
Let $\mathbf{y} := \int_a^b \mathbf{f}\,d\alpha$. If $\mathbf{y} = 0$, the inequality is immediate.
Otherwise, $\|\mathbf{y}\|^2 = \sum_{j=1}^k y_j \int_a^b f_j\,d\alpha = \int_a^b \sum_{j=1}^k y_j f_j\,d\alpha \leq \int_a^b \|\mathbf{y}\|\,\|\mathbf{f}\|\,d\alpha = \|\mathbf{y}\|\int_a^b \|\mathbf{f}\|\,d\alpha$, by Theorem 1.1.13 and order preservation.
Divide by $\|\mathbf{y}\|$.
Structural Commentary: Coordinatewise integration; Schwarz controls norms.
Key Insight: Integrate the projection of $\mathbf{f}$ onto the resulting vector. □

Definition 2.5.3. *Curves and Rectifiability*
A *curve* in $\mathbb{R}^k$ is a continuous map $\gamma : [a,b] \to \mathbb{R}^k$; it is an *arc* if one-to-one and a *closed curve* if $\gamma(a) = \gamma(b)$.
Curves are maps, not just ranges. For a partition $\Pi = \{x_0,\dots,x_n\}$, set $\ell(\Pi,\gamma) := \sum_{i=1}^n \|\gamma(x_i) - \gamma(x_{i-1})\|$.
The *length* is $\ell(\gamma) := \sup_\Pi \ell(\Pi,\gamma)$; if $\ell(\gamma) < \infty$, γ is *rectifiable*.

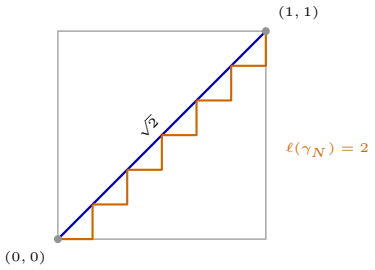

Figure 161: A staircase path can approach the diagonal in position while its polygonal length stays 2, not $\sqrt{2}$.

Perspective 2.5.4. *Length Is Not Just Shape*
Length is a supremum over partitions of one curve. Passing lengths through curve limits needs more than positional convergence; Theorem 2.5.5 uses C^1 speed control.

Theorem 2.5.5. *Length of a Continuously Differentiable Curve*
If $\gamma' : [a, b] \to \mathbb{R}^k$ is continuous, then γ is rectifiable and $\ell(\gamma) = \int_a^b \|\gamma'(t)\|\, dt$.

Proof. For $\Pi = \{x_0, \dots, x_n\}$, Theorem 2.5.2(i) gives $\|\gamma(x_i) - \gamma(x_{i-1})\| = \left\|\int_{x_{i-1}}^{x_i} \gamma'(t)\, dt\right\| \le \int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\, dt$.
Hence $\ell(\Pi, \gamma) \le \int_a^b \|\gamma'\|$. Conversely, choose Π so fine that $\|\gamma'(s) - \gamma'(t)\| < \varepsilon$ on each subinterval.
Choose $t_i \in [x_{i-1}, x_i]$. Then $\int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\, dt \le \|\gamma'(t_i)\|\, \Delta x_i + \varepsilon \Delta x_i$.
Also $\gamma(x_i) - \gamma(x_{i-1}) = \int_{x_{i-1}}^{x_i} \gamma'(t)\, dt$, so $\|\gamma'(t_i)\|\, \Delta x_i \le \|\gamma(x_i) - \gamma(x_{i-1})\| + \varepsilon \Delta x_i$.
Thus $\int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\, dt \le \|\gamma(x_i) - \gamma(x_{i-1})\| + 2\varepsilon \Delta x_i$.
Summing gives $\int_a^b \|\gamma'\| \le \ell(\gamma) + 2\varepsilon(b - a)$; let $\varepsilon \downarrow 0$.
Structural Commentary: Upper bounds are immediate; uniform continuity gives matching lower bounds.
Key Insight: For C^1 curves, length is accumulated speed. □

2.6 Further Topics in Integration

Method 2.6.1. *Integration Proof Patterns*

(i) *Definitions*: fix $\Pi = \{x_i\}_{i=0}^n$ before defining $M_i, m_i, \Delta x_i, \Delta \alpha_i$.
(ii) *Refinement*: prove the one-point insertion case; iterate.
(iii) *Integrability*: find Π with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.
(iv) *Bad sets*: cover E by $I_j = [u_j, v_j]$ with $\sum_j [\alpha(v_j) - \alpha(u_j)] < \varepsilon$; control $K = [a, b] \setminus \bigcup_j (u_j, v_j)$ uniformly.
(v) *Properties*: pass to a common refinement before comparing sums.
(vi) *Calculus*: use the mean value theorem to form tagged sums, then apply Theorem 2.1.9.
(vii) *Improper integrals*: truncate endpoints; use monotonicity or tail bounds.

Key Insight: Riemann–Stieltjes integration is Darboux approximation plus a choice of interval weights $\Delta \alpha_i$.

3 Several-Variable Differentiation

3.1 Partial and Total Derivatives

Several-variable differentiation studies best linear approximations and local geometry, using linear maps, operator norms, determinants, invertibility, and metric estimates.

Definition 3.1.1. *Total Derivative*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $\mathbf{x} \in E$, and $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ denote space of linear maps (Definition 1.5.1).
If $\exists A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ s.t. $\lim_{\mathbf{h} \to \mathbf{0}} \frac{\|\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - A\mathbf{h}\|_2}{\|\mathbf{h}\|_2} = 0$, then $\mathbf{f}$ is *differentiable at* $\mathbf{x}$, and A denoted $D\mathbf{f}(\mathbf{x})$ or $\mathbf{f}'(\mathbf{x})$.
Equivalently, $\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$, and $\frac{\|\mathbf{r}(\mathbf{h})\|_2}{\|\mathbf{h}\|_2} \to 0$.
If this holds for every $\mathbf{x} \in E$, then $\mathbf{f}$ is differentiable on E.

Perspective 3.1.2. *Derivative First, Matrix Second*
The total derivative is a linear map first; the Jacobian is its matrix in chosen bases. This keeps coordinate formulas subordinate to the intrinsic estimate $\|A\mathbf{h}\|_2 \le \|A\|_{\mathrm{op}} \|\mathbf{h}\|_2$ from Definition 5.1.4.

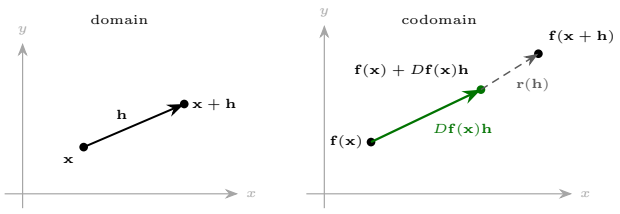

Figure 162: $D\mathbf{f}(\mathbf{x})\mathbf{h}$ is the linear part; $\mathbf{r}(\mathbf{h}) = o(\|\mathbf{h}\|_2)$ is the error.

Theorem 3.1.3. *Basic Consequences of Total Differentiability*
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable at $\mathbf{x} \in E$, then $D\mathbf{f}(\mathbf{x})$ is unique and $\mathbf{f}$ is continuous at $\mathbf{x}$.

Proof. If A_1, A_2 both satisfy Definition 3.1.1, set $B = A_1 - A_2$. Then $\|B\mathbf{h}\|_2 / \|\mathbf{h}\|_2 \to 0$ as $\mathbf{h} \to \mathbf{0}$.
For fixed $\mathbf{h} \neq \mathbf{0}$, applying this to $t\mathbf{h}$ and using linearity makes the quotient independent of t, so $B\mathbf{h} = \mathbf{0}$.
Hence $B = 0$. For continuity, write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$.
The operator-norm estimate gives $\|D\mathbf{f}(\mathbf{x})\mathbf{h}\|_2 \leq \|D\mathbf{f}(\mathbf{x})\|_{\text{op}} \|\mathbf{h}\|_2$, and remainder is $o(\|\mathbf{h}\|_2)$, so $\mathbf{f}(\mathbf{x}+\mathbf{h}) \to \mathbf{f}(\mathbf{x})$.
Structural Commentary: The derivative is a rigid first-order linear part.
Key Insight: A remainder smaller than $\|\mathbf{h}\|$ cannot hide a second linear map. □

Example 3.1.4. *Derivative of a Linear Map*
If $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$, then $D\mathbf{f}(\mathbf{x}) = A$ for every $\mathbf{x} \in \mathbb{R}^n$.
The remainder in Definition 3.1.1 is identically zero.

Theorem 3.1.5. *Chain Rule*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{x}_0$, $\mathbf{g} : G \subseteq \mathbb{R}^m \to \mathbb{R}^k$ is differentiable at $\mathbf{f}(\mathbf{x}_0)$, and $\mathbf{f}(E) \subseteq G$, then $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}_0$ and $D(\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0) = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))\, D\mathbf{f}(\mathbf{x}_0)$.

Proof. Let $A = D\mathbf{f}(\mathbf{x}_0)$, $B = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))$, $\mathbf{u}(\mathbf{h}) = \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0) - A\mathbf{h}$, and $\mathbf{v}(\mathbf{k}) = \mathbf{g}(\mathbf{f}(\mathbf{x}_0)+\mathbf{k}) - \mathbf{g}(\mathbf{f}(\mathbf{x}_0)) - B\mathbf{k}$.
For $\mathbf{k} = \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0)$, differentiability of $\mathbf{f}$ gives $\mathbf{k} \to \mathbf{0}$ and $(\mathbf{g}\circ\mathbf{f})(\mathbf{x}_0+\mathbf{h}) - (\mathbf{g}\circ\mathbf{f})(\mathbf{x}_0) - BA\mathbf{h} = B\mathbf{u}(\mathbf{h}) + \mathbf{v}(\mathbf{k})$, where $\|\mathbf{u}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ and $\|\mathbf{v}(\mathbf{k})\|_2 = o(\|\mathbf{k}\|_2) = o(\|\mathbf{h}\|_2)$. Divide by $\|\mathbf{h}\|_2$.
Structural Commentary: The first-order parts compose as linear maps.
Key Insight: The output increment of $\mathbf{f}$ is small enough to feed into the remainder estimate for $\mathbf{g}$. □

Theorem 3.1.6. *Componentwise Differentiability*
Let $\mathbf{f} = (f_1, \ldots, f_m) : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and let $\pi_i : \mathbb{R}^m \to \mathbb{R}$ be the ith coordinate projection. Then $\mathbf{f}$ is differentiable at $\mathbf{x} \in E$ iff each f_i is differentiable at $\mathbf{x}$, and $Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$ when these derivatives exist.

Proof.
(⇒) If $\mathbf{f}$ is differentiable, write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Applying π_i gives $f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) = \pi_i(D\mathbf{f}(\mathbf{x})\mathbf{h}) + \pi_i(\mathbf{r}(\mathbf{h}))$ and $|\pi_i(\mathbf{r}(\mathbf{h}))| \leq \|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Thus $Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$.
(⇐) Suppose $Df_i(\mathbf{x}) = A_i \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ for each i. Define $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ by $A\mathbf{h} = (A_1\mathbf{h}, \ldots, A_m\mathbf{h})$.
With $r_i(\mathbf{h}) = f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) - A_i\mathbf{h}$, each $r_i(\mathbf{h}) = o(\|\mathbf{h}\|_2)$.
So $\mathbf{r}(\mathbf{h}) = (r_1(\mathbf{h}), \ldots, r_m(\mathbf{h}))$ satisfies $\|\mathbf{r}(\mathbf{h})\|_2 \leq \sum_i |r_i(\mathbf{h})| = o(\|\mathbf{h}\|_2)$. Definition 3.1.1 gives $D\mathbf{f}(\mathbf{x}) = A$.
Structural Commentary: Component derivatives assemble into one linear map.
Key Insight: Components test the derivative; the total derivative packages the tests together. □

Definition 3.1.7. *Partial, Gradient, and Directional Derivatives*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} = (f_1, \ldots, f_m) : E \to \mathbb{R}^m$, $\mathbf{x} \in E$, and let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis of $\mathbb{R}^n$.
(i) For $1 \leq i \leq m$ and $1 \leq j \leq n$, the *partial derivative* of f_i in coordinate j at $\mathbf{x}$ is $(D_j f_i)(\mathbf{x}) := \lim_{t\to 0} \frac{f_i(\mathbf{x}+t\mathbf{e}_j) - f_i(\mathbf{x})}{t}$, when this limit exists.
(ii) For $\mathbf{u} \in \mathbb{R}^n$, the *directional derivative* of $\mathbf{f}$ at $\mathbf{x}$ in direction $\mathbf{u}$ is $(D_{\mathbf{u}}\mathbf{f})(\mathbf{x}) := \lim_{t\to 0} \frac{\mathbf{f}(\mathbf{x}+t\mathbf{u}) - \mathbf{f}(\mathbf{x})}{t}$, when this limit exists; if only direction matters, take $\|\mathbf{u}\|_2 = 1$.
(iii) If $m = 1$, write $f = f_1$; if all $(D_j f)(\mathbf{x})$ exist, the *gradient* of f at $\mathbf{x}$ is $(\nabla f)(\mathbf{x}) := \sum_{j=1}^{n} (D_j f)(\mathbf{x})\mathbf{e}_j$.

Perspective 3.1.8. *Partials Are Coordinate Tests*
Partials and directional derivatives test selected lines; the total derivative controls all small increments by one linear map. Theorem 3.1.9 extracts partials, and Theorem 3.2.4 gives a continuity-based converse.

Theorem 3.1.9. *Matrix of the Total Derivative*
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable at $\mathbf{x}$, then every $(D_j f_i)(\mathbf{x})$ exists and, in standard bases,

$$[D\mathbf{f}(\mathbf{x})] = \begin{bmatrix} (D_1 f_1)(\mathbf{x}) & \cdots & (D_n f_1)(\mathbf{x}) \\ \vdots & & \vdots \\ (D_1 f_m)(\mathbf{x}) & \cdots & (D_n f_m)(\mathbf{x}) \end{bmatrix}$$

If $m = 1$ and f is differentiable at $\mathbf{x}$, then $(D_{\mathbf{u}} f)(\mathbf{x}) = (\nabla f)(\mathbf{x}) \cdot \mathbf{u}$ for every $\mathbf{u} \in \mathbb{R}^n$.

Proof. Set $\mathbf{h} = t\mathbf{e}_j$ in Definition 3.1.1; limit is $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$, so components form column j (Proposition 3.2.5).
For $m = 1$, set $\mathbf{h} = t\mathbf{u}$, identify linear functional with dot product against $\nabla f(\mathbf{x})$ (Theorem 1.6.1).
Structural Commentary: Partials are coordinate columns of $D\mathbf{f}(\mathbf{x})$.
Key Insight: Differentiability controls all directions at once; partial derivatives record its coordinate columns. □

Remark 3.1.10. *Row and Column Conventions*
With the coordinate-column convention from Proposition 3.2.5, the Jacobian matrix above is the standard-basis matrix of the linear map $D\mathbf{f}(\mathbf{x})$, and its jth column is $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$.

Definition 3.1.11. *Jacobian*
Let $E \subseteq \mathbb{R}^n$, let $\mathbf{f} = (y_1, \ldots, y_n) : E \to \mathbb{R}^n$, and suppose $\mathbf{f}$ is differentiable at $\mathbf{x} \in E$.
The *Jacobian determinant* is $J_{\mathbf{f}}(\mathbf{x}) := \det D\mathbf{f}(\mathbf{x}) = \frac{\partial(y_1,\ldots,y_n)}{\partial(x_1,\ldots,x_n)}(\mathbf{x})$.
Then $D\mathbf{f}(\mathbf{x})$ is invertible iff $J_{\mathbf{f}}(\mathbf{x}) \neq 0$. Interpretation: $|J_{\mathbf{f}}(\mathbf{x})|$ is the infinitesimal volume-scaling factor.

3.2 Differentiability and Linear Approximation

Theorem 3.2.1. *Mean-Value Inequality on Convex Sets*
Let $E \subseteq \mathbb{R}^n$ be open and convex, and let $\mathbf{f} : E \to \mathbb{R}^m$ be differentiable.

(i) If $M \geq 0$ and $\|D\mathbf{f}(\mathbf{x})\|_{op} \leq M$ for all $\mathbf{x} \in E$, then $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 \leq M \|\mathbf{b} - \mathbf{a}\|_2$ for all $\mathbf{a}, \mathbf{b} \in E$.
(ii) If $D\mathbf{f}(\mathbf{x}) = 0$ for all $\mathbf{x} \in E$, then $\mathbf{f}$ is constant on E.

Proof.

(i) Fix $\mathbf{a}, \mathbf{b} \in E$, set $\gamma(t) := (1-t)\mathbf{a} + t\mathbf{b}$, and put $\mathbf{g} := \mathbf{f} \circ \gamma$. Convexity gives $\gamma([0,1]) \subseteq E$, and the chain rule gives $\mathbf{g}'(t) = D\mathbf{f}(\gamma(t))(\mathbf{b} - \mathbf{a})$, so $\|\mathbf{g}'(t)\|_2 \leq M \|\mathbf{b} - \mathbf{a}\|_2$.
Theorem 1.1.27 gives $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 = \|\mathbf{g}(1) - \mathbf{g}(0)\|_2 \leq M \|\mathbf{b} - \mathbf{a}\|_2$.
(ii) If $D\mathbf{f}(\mathbf{x}) = 0$ on E, apply (i) with $M = 0$: for all $\mathbf{a}, \mathbf{b} \in E$, $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 = 0$, hence $\mathbf{f}(\mathbf{a}) = \mathbf{f}(\mathbf{b})$.

Structural Commentary: Line restriction gives a one-variable estimate.
Key Insight: Derivative bounds control secants. □

Definition 3.2.2. *C^1 Mapping*
Let $E \subseteq \mathbb{R}^n$ be open. A differentiable map $\mathbf{f} : E \to \mathbb{R}^m$ is *continuously differentiable*, or C^1, if $\mathbf{x} \mapsto D\mathbf{f}(\mathbf{x})$ is continuous from E into $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ in operator norm.

Perspective 3.2.3. *C^1 Uses the Operator-Norm Metric*
C^1 continuity measures $\mathbf{x} \mapsto D\mathbf{f}(\mathbf{x})$ in operator norm. In coordinates this is spectral-norm continuity, equivalent here to continuity of the derivative matrix entries.

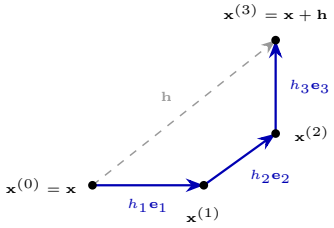

Figure 163: Coordinate increments telescope $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$.

Theorem 3.2.4. *C^1 Means Continuous Partials*
Let $E \subseteq \mathbb{R}^n$ be open and let $\mathbf{f} = (f_1, \ldots, f_m) : E \to \mathbb{R}^m$.
Then $\mathbf{f}$ is C^1 on E iff each $D_j f_i$ exists and is continuous on E, for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Proof.

(⇒) If $\mathbf{f}$ is C^1, then $D\mathbf{f} : E \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous. By Theorem 3.1.9, $(D_j f_i)(\mathbf{x})$ is the (i,j) matrix entry of $D\mathbf{f}(\mathbf{x})$; matrix-entry projections are continuous, so each $D_j f_i$ is continuous on E.
(⇐) By Theorem 3.1.6, it suffices to treat $m = 1$. Fix $\mathbf{x} \in E$ and small $\mathbf{h}$ with coordinate polygon in E.

Set $\mathbf{x}^{(0)} := \mathbf{x}$ and $\mathbf{x}^{(j)} := \mathbf{x} + \sum_{k=1}^{j} h_k \mathbf{e}_k$. Telescoping and the one-variable MVT give

$f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) = \sum_{j=1}^{n} h_j (D_j f)(\xi_j)$ for some $\xi_j \in [\mathbf{x}^{(j-1)}, \mathbf{x}^{(j)}]$. Let $A_{\mathbf{x}}\mathbf{h} := \sum_{j=1}^{n} h_j (D_j f)(\mathbf{x})$.

Since $\xi_j \to \mathbf{x}$ as $\mathbf{h} \to \mathbf{0}$, $|f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) - A_{\mathbf{x}}\mathbf{h}| \leq \sum_{j=1}^{n} |h_j| \, |(D_j f)(\xi_j) - (D_j f)(\mathbf{x})| = o(\|\mathbf{h}\|_2)$.

Thus $Df(\mathbf{x}) = A_{\mathbf{x}}$, i.e. $Df(\mathbf{x}) = [(D_1 f)(\mathbf{x}) \ \cdots \ (D_n f)(\mathbf{x})]$, and continuity of the entries gives $\mathbf{f} \in C^1$.

Structural Commentary: Continuous partials form the derivative matrix.
Key Insight: Coordinate telescoping gives the linear approximation. □

Higher derivatives repeat the same idea: once $D\mathbf{f}$ is itself a map into the finite-dimensional vector space $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, we differentiate it again. The linear-algebra viewpoint matters here because the second derivative is naturally bilinear, not merely an array of second partial derivatives.

Definition 3.2.5. *Higher Derivatives and C^r*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $r \geq 1$, and let $\mathcal{L}^r(\mathbb{R}^n, \mathbb{R}^m)$ denote the space of r-linear maps $(\mathbb{R}^n)^r \to \mathbb{R}^m$.

(i) The *rth derivative* is recursive: $D^1\mathbf{f} := D\mathbf{f}$.
For $r \geq 2$, if $D^{r-1}\mathbf{f}$ is differentiable at $\mathbf{x}$, then $D^r\mathbf{f}(\mathbf{x})(\mathbf{v}_1, \ldots, \mathbf{v}_r) := \big(D(D^{r-1}\mathbf{f})(\mathbf{x})\mathbf{v}_1\big)(\mathbf{v}_2, \ldots, \mathbf{v}_r)$.
(ii) For real-valued $f : E \to \mathbb{R}$, the mixed second partial is $D_{ij} f := D_i(D_j f)$ where defined.
(iii) $\mathbf{f}$ is C^r on E iff $D^r\mathbf{f} : E \to \mathcal{L}^r(\mathbb{R}^n, \mathbb{R}^m)$ exists and is continuous; $\mathbf{f}$ is C^∞ iff $\mathbf{f}$ is C^r for every $r \geq 1$.

Theorem 3.2.6. *Second Derivatives and Second Partials*
Let $\mathbf{f} = (f_1, \ldots, f_m) : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $p \in E$.
(⇒) If $D^2\mathbf{f}(p)$ exists, then each $D_{ij} f_\ell(p)$ exists and $D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j) = D_{ij} f_\ell(p)$.
(⇐) If all $D_{ij} f_\ell$ exist near p and are continuous at p, then $D^2\mathbf{f}(p)$ exists.

Proof.

(⇒) Apply Theorem 3.1.6 to $D\mathbf{f} : E \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. The entries of $D\mathbf{f}$ are $D_j f_\ell$ by Theorem 3.1.9; differentiating the (ℓ, j) entry in the $\mathbf{e}_i$ direction gives $D_{ij} f_\ell(p)$ and the displayed formula.
(⇐) Matrix entries of $D\mathbf{f}$ are functions $D_j f_\ell$. Since first partials $D_i(D_j f_\ell) = D_{ij} f_\ell$ exist near p and are continuous at p, coordinate criterion at p gives differentiability of $D\mathbf{f}$ there. Hence $D^2\mathbf{f}(p)$ exists.

Structural Commentary: Second partials are matrix entries of the derivative of the first-derivative matrix.
Key Insight: Continuous second partials make the first-derivative matrix differentiable. □

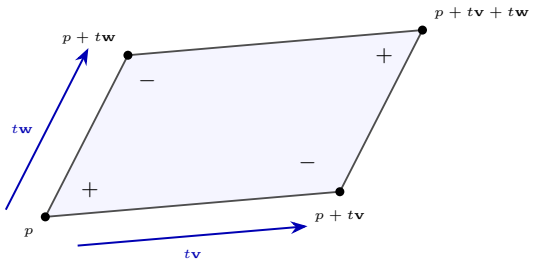

Figure 164: The signed vertex sum on a small parallelogram is symmetric in $\mathbf{v}$ and $\mathbf{w}$.

Theorem 3.2.7. *Symmetry of the Second Derivative*
If $D^2\mathbf{f}(p)$ exists, then $D^2\mathbf{f}(p)(\mathbf{v},\mathbf{w}) = D^2\mathbf{f}(p)(\mathbf{w},\mathbf{v})$ for all $\mathbf{v},\mathbf{w} \in \mathbb{R}^n$.

Proof. Reduce to scalar f by Theorem 3.1.6.
For small t, set $\Delta(t,\mathbf{v},\mathbf{w}) := f(p+t\mathbf{v}+t\mathbf{w}) - f(p+t\mathbf{v}) - f(p+t\mathbf{w}) + f(p)$.
Then $\Delta(t,\mathbf{v},\mathbf{w}) = \Delta(t,\mathbf{w},\mathbf{v})$. The one-variable mean value theorem and differentiability of Df at p give $D^2f(p)(\mathbf{v},\mathbf{w}) = \lim_{t\to 0} \frac{\Delta(t,\mathbf{v},\mathbf{w})}{t^2}$.
The same formula with $\mathbf{v},\mathbf{w}$ reversed proves symmetry.
Structural Commentary: A pointwise second derivative is already forced to be a symmetric bilinear form.
Key Insight: The parallelogram difference is symmetric in the two directions. □

Corollary 3.2.8. *Mixed Partials from D^2*
If $D^2\mathbf{f}(p)$ exists, then $D_{ij}f_\ell(p) = D_{ji}f_\ell(p)$ for all components and all i,j.

Proof. By Theorem 3.2.6, $D_{ij}f_\ell(p) = D^2f_\ell(p)(\mathbf{e}_i,\mathbf{e}_j)$; and $D_{ji}f_\ell(p) = D^2f_\ell(p)(\mathbf{e}_j,\mathbf{e}_i)$.
Theorem 3.2.7 makes the two right-hand sides equal.
Structural Commentary: Mixed partials are coordinate values of one bilinear map.
Key Insight: Symmetry of $D^2f_\ell(p)$ gives equality of mixed entries. □

Corollary 3.2.9. *Symmetry of Higher Derivatives*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f}: E \to \mathbb{R}^m$, $p \in E$, and $r \geq 2$.
If $D^r\mathbf{f}(p)$ exists, then for every $\sigma \in S_r$ and $\mathbf{v}_1,\dots,\mathbf{v}_r \in \mathbb{R}^n$, $D^r\mathbf{f}(p)(\mathbf{v}_{\sigma(1)},\dots,\mathbf{v}_{\sigma(r)}) = D^r\mathbf{f}(p)(\mathbf{v}_1,\dots,\mathbf{v}_r)$.

Proof. Induct on r; $r=2$ is Theorem 3.2.7. For $r>2$, the induction hypothesis gives symmetry in slots $2,\dots,r$ of $D^r\mathbf{f}(p)$. Applying Theorem 3.2.7 to $D^{r-2}\mathbf{f}: E \to \mathcal{L}^{r-2}(\mathbb{R}^n,\mathbb{R}^m)$ and evaluating at $\mathbf{v}_3,\dots,\mathbf{v}_r$ gives symmetry in slots $1,2$. Hence every adjacent transposition fixes $D^r\mathbf{f}(p)$, so every $\sigma \in S_r$ fixes it.
Structural Commentary: Symmetry propagates from D^2.
Key Insight: Adjacent swaps generate S_r. □

The next rectangle argument is the classical coordinate route to equality of mixed partials. It is weaker in hypothesis than global C^2, but it proves the same compatibility of coordinate second derivatives.

Theorem 3.2.10. *Rectangle Mean Value Formula*
Let $E \subseteq \mathbb{R}^2$ be open, let $f: E \to \mathbb{R}$, and assume D_1f and $D_{21}f = D_2(D_1f)$ exist on E. Let $h,k \neq 0$, let $Q \subseteq E$ be the closed coordinate rectangle with opposite vertices (a,b) and $(a+h,b+k)$, and set $\Delta(f,Q) := f(a+h,b+k) - f(a+h,b) - f(a,b+k) + f(a,b)$. Then there exists $(x,y) \in \operatorname{int} Q$ such that $\Delta(f,Q) = hk(D_{21}f)(x,y)$.

Proof. Let I,J be the open intervals with endpoints $a, a+h$ and $b, b+k$. Set $u(t) := f(t,b+k) - f(t,b)$.
By Corollary 1.1.13, $\Delta(f,Q) = u(a+h) - u(a) = h[(D_1f)(x,b+k) - (D_1f)(x,b)]$ for some $x \in I$. Applying the same result to $s \mapsto (D_1f)(x,s)$ gives $(D_1f)(x,b+k) - (D_1f)(x,b) = k(D_{21}f)(x,y)$ for some $y \in J$.
Thus $(x,y) \in I \times J = \operatorname{int} Q$ and $\Delta(f,Q) = hk(D_{21}f)(x,y)$.
Structural Commentary: Two MVT steps factor the rectangle increment.
Key Insight: $D_{21}f$ tracks vertical change of D_1f. □

Theorem 3.2.11. *Equality of Mixed Partials*
Let (a,b) lie in an open set $E \subseteq \mathbb{R}^2$. If D_1f, D_2f, and $D_{21}f$ exist on E, and $D_{21}f$ is continuous at (a,b), then $D_{12}f(a,b)$ exists and $(D_{12}f)(a,b) = (D_{21}f)(a,b)$. Hence $D_{12}f = D_{21}f$ when $f \in C^2(E)$.

Proof. Let $A = (D_{21}f)(a,b)$, $\varepsilon > 0$.
Continuity of $D_{21}f$, Theorem 3.2.10 gives $\forall$ small $h,k \neq 0$, $\left|\frac{\Delta(f,Q)}{hk} - A\right| < \varepsilon$.
Fix such an h and let $k \to 0$. Since D_2f exists at $(a+h,b)$ and (a,b), $\frac{\Delta(f,Q)}{hk} \to \frac{(D_2f)(a+h,b) - (D_2f)(a,b)}{h}$.
Thus this last quotient differs from A by at most ε $\forall$ sufficiently small $h \neq 0$.
Since ε was arbitrary, the quotient tends to A as $h \to 0$, so $D_{12}f(a,b)$ exists and equals A.
Structural Commentary: Continuity of $D_{21}f$ controls local rectangle quotients.
Key Insight: The missing mixed partial is recovered by taking the two one-variable limits in the reverse order. □

Perspective 3.2.12. *Mixed Partials Need Hypotheses*
$D_{12}f$ and $D_{21}f$ are different iterated limits. Theorems 3.2.7 and 3.2.11 give hypotheses for equality.

4 Inverse, Implicit, and Rank Theorems

4.1 Inverse Function Theorem

Theorem 4.1.1. *Finite-Dimensional Linear Maps Are Lipschitz*
Let $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$.

(i) For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\|A\mathbf{x} - A\mathbf{y}\|_2 \le \|A\|_{op} \|\mathbf{x} - \mathbf{y}\|_2$; hence A is $\|A\|_{op}$-Lipschitz.

(ii) If $m = n$ and A is invertible, then $\frac{1}{\|A^{-1}\|_{op}} \|\mathbf{x} - \mathbf{y}\|_2 \le \|A\mathbf{x} - A\mathbf{y}\|_2 \le \|A\|_{op} \|\mathbf{x} - \mathbf{y}\|_2$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$; hence $A : \mathbb{R}^n \to \mathbb{R}^n$ is a homeomorphism.

Proof.

(i) Definition 5.1.4 gives $\|A\mathbf{z}\|_2 \le \|A\|_{\text{op}} \|\mathbf{z}\|_2$ for all $\mathbf{z} \in \mathbb{R}^n$; set $\mathbf{z} = \mathbf{x} - \mathbf{y}$.

(ii) Apply (i) to A^{-1}: $\|\mathbf{x} - \mathbf{y}\|_2 = \|A^{-1}(A\mathbf{x} - A\mathbf{y})\|_2 \le \|A^{-1}\|_{\text{op}} \|A\mathbf{x} - A\mathbf{y}\|_2$.
This gives the lower bound; the upper bound is (i). Hence A and A^{-1} are continuous.

Structural Commentary: A^{-1} converts upper control into lower control.
Key Insight: Invertible linear maps are bi-Lipschitz. □

Theorem 4.1.2. *Invertible Operators Form an Open Set*
Let Ω be the set of invertible operators in $\mathcal{L}(\mathbb{R}^n)$.

(i) If $A \in \Omega$, $B \in \mathcal{L}(\mathbb{R}^n)$, and $\|B - A\|_{op} \|A^{-1}\|_{op} < 1$, then $B \in \Omega$.

(ii) Ω is open in $\mathcal{L}(\mathbb{R}^n)$.

(iii) The inversion map $\iota : \Omega \to \Omega$, $\iota(A) = A^{-1}$, is continuous.

Proof.

(i) Put $\alpha := 1/\|A^{-1}\|_{\text{op}}$ and $\beta := \|B - A\|_{\text{op}} < \alpha$.
For every $\mathbf{x} \in \mathbb{R}^n$, $\alpha\|\mathbf{x}\|_2 \le \|A\mathbf{x}\|_2 \le \beta\|\mathbf{x}\|_2 + \|B\mathbf{x}\|_2$, hence $(\alpha - \beta)\|\mathbf{x}\|_2 \le \|B\mathbf{x}\|_2$.
Thus $\ker B = \{\mathbf{0}\}$, so $B \in \Omega$ by Theorem 1.7.7; setting $\mathbf{x} = B^{-1}\mathbf{y}$ gives $\|B^{-1}\|_{\text{op}} \le (\alpha - \beta)^{-1}$.

(ii) For $A \in \Omega$, the ball $\{B : \|B - A\|_{\text{op}} < 1/\|A^{-1}\|_{\text{op}}\}$ lies in Ω by (i); hence Ω is open.

(iii) If $B \to A \in \Omega$, then eventually $\beta < \alpha/2$, so $B \in \Omega$ and $\|B^{-1}\|_{\text{op}} \le (\alpha - \beta)^{-1} \le 2/\alpha$.
Since $B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1}$, $\|B^{-1} - A^{-1}\|_{\text{op}} \le (2/\alpha)\|B - A\|_{\text{op}}\|A^{-1}\|_{\text{op}} \to 0$.

Structural Commentary: Perturbations below the lower stretch stay invertible.
Key Insight: $B^{-1} - A^{-1}$ factors through $A - B$. □

Perspective 4.1.3. *Linear Inverse Estimate*
For invertible $A \in \mathcal{L}(\mathbb{R}^n)$, Theorem 4.1.1 gives $c\|\mathbf{h}\|_2 \le \|A\mathbf{h}\|_2 \le \|A\|_{\text{op}}\|\mathbf{h}\|_2$; and $c := \frac{1}{\|A^{-1}\|_{\text{op}}}$.
The inverse function theorem preserves this lower stretch under small derivative changes.

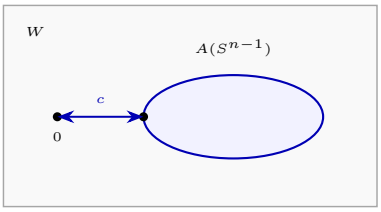

Figure 165: Invertibility keeps $A(S^{n-1})$ a positive distance from 0.

Theorem 4.1.4. *Inverse Function Theorem*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{f} : E \to \mathbb{R}^n$ be C^1, let $\mathbf{a} \in E$, put $\mathbf{b} := \mathbf{f}(\mathbf{a})$, and assume $A := D\mathbf{f}(\mathbf{a})$ is invertible. Then there exist open neighborhoods U of $\mathbf{a}$ and V of $\mathbf{b}$ such that:

(i) $\mathbf{f}|_U : U \to V$ is bijective; write $\mathbf{g} := (\mathbf{f}|_U)^{-1} : V \to U$.

(ii) $\mathbf{g}$ is differentiable on V and $D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{g}(\mathbf{y}))\big)^{-1}$ for all $\mathbf{y} \in V$.

(iii) $\mathbf{g} \in C^1(V, \mathbb{R}^n)$; equivalently, $\mathbf{f}|_U$ is a C^1 diffeomorphism.

Proof. Choose $0 < \theta < 1$ and $r > 0$ so that $\overline{B}_r(\mathbf{a}) \subset E$ and $\|I - A^{-1}D\mathbf{f}(\mathbf{x})\|_{\text{op}} \le \theta$ on $\overline{B}_r(\mathbf{a})$.
Let $V := \{\mathbf{y} : \|A^{-1}(\mathbf{y} - \mathbf{b})\|_2 < (1 - \theta)r\}$.

(i) For $\mathbf{y} \in V$, set $\varphi_{\mathbf{y}}(\mathbf{x}) := \mathbf{x} + A^{-1}(\mathbf{y} - \mathbf{f}(\mathbf{x}))$. Theorem 3.2.1 gives $\|\varphi_{\mathbf{y}}(\mathbf{x}) - \varphi_{\mathbf{y}}(\mathbf{z})\|_2 \le \theta\|\mathbf{x} - \mathbf{z}\|_2$ on $\overline{B}_r(\mathbf{a})$, and $\|\varphi_{\mathbf{y}}(\mathbf{x}) - \mathbf{a}\|_2 \le \theta r + \|A^{-1}(\mathbf{y} - \mathbf{b})\|_2 < r$. Since $\overline{B}_r(\mathbf{a})$ is complete, Theorem 4.5.3 gives a unique fixed point $\mathbf{x} \in \overline{B}_r(\mathbf{a})$, equivalently $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. With $U := B_r(\mathbf{a}) \cap \mathbf{f}^{-1}(V)$, $\mathbf{f}|_U : U \to V$ is bijective.

(ii) Let $\mathbf{y} = \mathbf{f}(\mathbf{x})$ and $\mathbf{y} + \mathbf{k} = \mathbf{f}(\mathbf{x} + \mathbf{h})$ lie in V. The derivative bound gives $\|\mathbf{h} - A^{-1}\mathbf{k}\|_2 \le \theta\|\mathbf{h}\|_2$, hence $\|\mathbf{h}\|_2 \le C\|\mathbf{k}\|_2$ locally. Since $\|I - A^{-1}D\mathbf{f}(\mathbf{x})\|_{\text{op}} < 1$, $D\mathbf{f}(\mathbf{x})$ is invertible, and
$\mathbf{g}(\mathbf{y} + \mathbf{k}) - \mathbf{g}(\mathbf{y}) - \big(D\mathbf{f}(\mathbf{x})\big)^{-1}\mathbf{k} = -\big(D\mathbf{f}(\mathbf{x})\big)^{-1}[\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}]$.
Since $\mathbf{f}$ is differentiable at $\mathbf{x}$, division by $\|\mathbf{k}\|_2$ gives $D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{x})\big)^{-1}$.

(iii) The estimate in (ii) makes $\mathbf{g}$ continuous. Since $D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{g}(\mathbf{y}))\big)^{-1}$, continuity of $D\mathbf{g}$ follows from continuity of $D\mathbf{f}$, $\mathbf{g}$, and Theorem 4.1.2.

Structural Commentary: A^{-1} turns solving $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ into a contraction.
Key Insight: Small derivative variation preserves invertibility locally. □

Corollary 4.1.5. *Local Diffeomorphisms Are Open*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is C^1 and $D\mathbf{f}$ is everywhere invertible, then $\mathbf{f}$ is an open map (locally one-to-one).

Proof. Let $W \subseteq E$ be open and let $\mathbf{y} \in \mathbf{f}(W)$. Choose $\mathbf{x} \in W$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. Applying Theorem 4.1.4 to $\mathbf{f}|_W$ at $\mathbf{x}$ gives open neighborhoods $U \subseteq W$ of $\mathbf{x}$ and V of $\mathbf{y}$ with $\mathbf{f}(U) = V$. Hence $V \subseteq \mathbf{f}(W)$, so $\mathbf{y}$ is interior to $\mathbf{f}(W)$. Since $\mathbf{y}$ was arbitrary, $\mathbf{f}(W)$ is open. Local one-to-one-ness is the same theorem applied pointwise.
Structural Commentary: Openness is checked one image point at a time.
Key Insight: A local inverse turns nearby domain points into a full image neighborhood. □

4.2 Implicit Function Theorem

Definition 4.2.1. *Block Splitting of a Linear Map*
For $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$ and $(\mathbf{h}, \mathbf{k}) \in \mathbb{R}^n \times \mathbb{R}^m$, define $A_x\mathbf{h} := A(\mathbf{h}, \mathbf{0})$ and $A_y\mathbf{k} := A(\mathbf{0}, \mathbf{k})$.
Then $A_x \in \mathcal{L}(\mathbb{R}^n)$, $A_y \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$, and $A(\mathbf{h}, \mathbf{k}) = A_x\mathbf{h} + A_y\mathbf{k}$.

Theorem 4.2.2. *Linear Implicit Function Theorem*
If $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$, A_x invertible, then $\forall \mathbf{k} \in \mathbb{R}^m$ $\exists$ unique $\mathbf{h} \in \mathbb{R}^n$ s.t. $A(\mathbf{h}, \mathbf{k}) = 0$, that $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k}$.

Proof. Fix $\mathbf{k} \in \mathbb{R}^m$ and set $\mathbf{h}_0 := -(A_x)^{-1}A_y\mathbf{k}$.
By Definition 4.2.1, $A(\mathbf{h}_0, \mathbf{k}) = A_x\mathbf{h}_0 + A_y\mathbf{k} = -A_y\mathbf{k} + A_y\mathbf{k} = 0$, so a solution exists.
If $\mathbf{h} \in \mathbb{R}^n$ also satisfies $A(\mathbf{h}, \mathbf{k}) = 0$, then $A_x\mathbf{h} + A_y\mathbf{k} = 0$, hence $A_x\mathbf{h} = -A_y\mathbf{k}$.
Applying $(A_x)^{-1}$ gives $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k} = \mathbf{h}_0$. Thus the solution exists and is unique.
Structural Commentary: Splitting A separates the unknown variable from the parameter.
Key Insight: Invertibility of A_x turns constraint $A(\mathbf{h}, \mathbf{k}) = 0$ into a uniquely solvable linear equation for $\mathbf{h}$. □

The implicit theorem below is the nonlinear counterpart of the block equation theorem. The invertible block is the set of variables one can solve for, just as in Theorem 4.2.2.

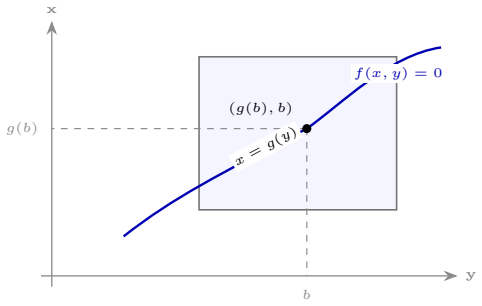

Figure 166: Near a regular point, the zero locus is the graph of an implicit function.

Theorem 4.2.3. *Implicit Function Theorem*
Let $E \subseteq \mathbb{R}^{n+m}$ be open, $\mathbf{f} \in C^1(E, \mathbb{R}^n)$, $(\mathbf{a}, \mathbf{b}) \in E$, and $\mathbf{f}(\mathbf{a}, \mathbf{b}) = \mathbf{0}$.
Put $A := D\mathbf{f}(\mathbf{a}, \mathbf{b})$ and write $A(\mathbf{h}, \mathbf{k}) = A_x\mathbf{h} + A_y\mathbf{k}$; assume A_x is invertible.
Then there exist open neighborhoods U of $\mathbf{a}$ and V of $\mathbf{b}$, and a unique map $\mathbf{g} : V \to U$, such that:

(i) $\{(\mathbf{x}, \mathbf{y}) \in U \times V : \mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{0}\} = \{(\mathbf{g}(\mathbf{y}), \mathbf{y}) : \mathbf{y} \in V\}$ *and* $\mathbf{g}(\mathbf{b}) = \mathbf{a}$.
(ii) $\mathbf{g} \in C^1(V, \mathbb{R}^n)$ *and* $D\mathbf{g}(\mathbf{y}) = -\big(D_1\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})\big)^{-1} D_2\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})$ *for $\mathbf{y} \in V$; in particular, $D\mathbf{g}(\mathbf{b}) = -(A_x)^{-1}A_y$.*

Proof. Let $\Phi(\mathbf{x}, \mathbf{y}) := (\mathbf{f}(\mathbf{x}, \mathbf{y}), \mathbf{y})$. Since $D\Phi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k}) = (A_x\mathbf{h} + A_y\mathbf{k}, \mathbf{k})$, $\ker D\Phi(\mathbf{a}, \mathbf{b}) = \{\mathbf{0}\}$; by Theorems 1.7.7 and 4.1.4, $\Phi : W \to Z$ is a local C^1 diffeomorphism. Shrink product neighborhoods $U \times V \subseteq W$ so that $\{\mathbf{0}\} \times V \subseteq Z$, $\Phi^{-1}(\{\mathbf{0}\} \times V) \subseteq U \times V$, and $D_1\mathbf{f}$ is invertible on $U \times V$.

(i) Define $\mathbf{g}(\mathbf{y})$ by $\Phi^{-1}(\mathbf{0}, \mathbf{y}) = (\mathbf{g}(\mathbf{y}), \mathbf{y})$. Since $\Phi(\mathbf{x}, \mathbf{y}) = (\mathbf{0}, \mathbf{y})$ iff $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$, the zero set in $U \times V$ is exactly the graph of $\mathbf{g}$, and $\mathbf{g}(\mathbf{b}) = \mathbf{a}$.
(ii) The map $\mathbf{g}$ is C^1 as a component of $\Phi^{-1}(\mathbf{0}, \mathbf{y})$.
Differentiating $\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y}) = \mathbf{0}$ gives $D_1\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y})D\mathbf{g}(\mathbf{y}) + D_2\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y}) = 0$, hence the formula.

Structural Commentary: Add $\mathbf{y}$ as a coordinate, then invert.
Key Insight: Invertible $D_x\mathbf{f}$ solves locally for $\mathbf{x}$. □

4.3 Rank Theorem

Perspective 4.3.1. *Rank Theorem Linear Model*
Theorem 2.2.12 reduces a linear map to its kernel–range split.
The constant-rank theorem is the local C^1 analogue after coordinate changes.

Theorem 4.3.2. *Rank Theorem*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{F} \in C^1(E, \mathbb{R}^m)$ have constant rank r, and fix $\mathbf{a} \in E$. Set $A := D\mathbf{F}(\mathbf{a})$, $Y := \operatorname{range} A$, and let $P : \mathbb{R}^m \to Y$ be the orthogonal projection. Then for some $\mathbf{x}_0 \in \mathbb{R}^n$, neighborhoods $V_0 \ni \mathbf{x}_0$ and $U \ni \mathbf{a}$, and C^1 diffeomorphism $H : V_0 \to U$, there exists $\varphi \in C^1(A(V_0), \ker P)$ such that $\mathbf{F} \circ H = A + \varphi \circ A$ on V_0.

Proof. If $r = 0$, Theorem 3.2.1 makes $\mathbf{F}$ locally constant; take $H = I$ and $\varphi(0) = \mathbf{F}(\mathbf{a})$.
Assume $r > 0$. Choose $S : Y \to \mathbb{R}^n$ with $AS = I_Y$ and set $G := I + SP(\mathbf{F} - A)$. Since $DG(\mathbf{a}) = I$, Theorem 4.1.4 gives a local C^1 diffeomorphism $G : U \to V_0$ with inverse H; shrink V_0 convex and put $\mathbf{x}_0 := G(\mathbf{a})$.
Let $\Psi := \mathbf{F} \circ H$. From $AG = P\mathbf{F}$, $P\Psi = A$ and $PD\Psi = A$.
Since $\operatorname{rank} D\Psi = \operatorname{rank} A = \operatorname{rank} PD\Psi = r$, P is injective on range $D\Psi$; hence $A\mathbf{v} = 0$ implies $D\Psi(\mathbf{x})\mathbf{v} = 0$.
By Theorem 3.2.1, Ψ is constant on each convex fiber $V_0 \cap (\mathbf{x} + \ker A)$.

Define $\varphi(A\mathbf{x}) := (I-P)\Psi(\mathbf{x})$. Fiber constancy makes φ well-defined, $\varphi(A\mathbf{x}) \in \ker P$, and $\Psi = A + \varphi \circ A$. After shrinking V_0 so $\sigma(\mathbf{u}) := \mathbf{x}_0 + S(\mathbf{u} - A\mathbf{x}_0)$ lies in V_0 for $\mathbf{u} \in A(V_0)$, $\varphi = (I-P)\Psi \circ \sigma$ is C^1.
Structural Commentary: Constant rank gives projection-plus-graph normal form.
Key Insight: Constant rank kills variation along A-fibers. □

Rank-nullity (Theorem 1.7.3) says that the domain directions split into rank directions and fiber directions. The rank theorem says a constant-rank map has, locally, the same split after C^1 changes of coordinates.

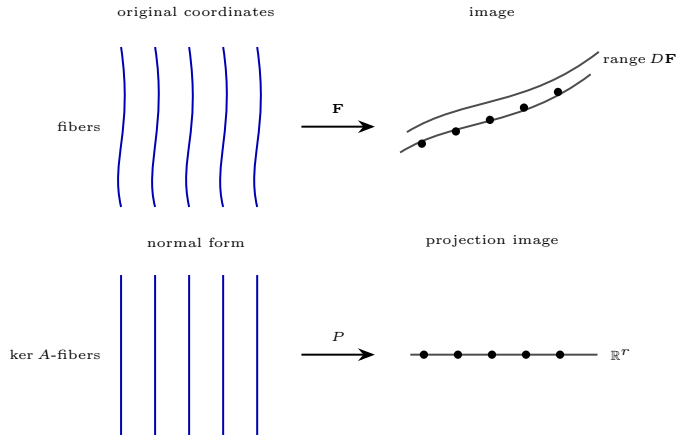

Figure 167: After local coordinate changes, a constant-rank map looks like projection onto the rank directions.

Theorem 4.3.3. *One-Constraint Lagrange Multiplier*
Let $E \subseteq \mathbb{R}^n$ be open, $f, g : E \to \mathbb{R}$ be C^1, and $S = \{\mathbf{x} : g(\mathbf{x}) = c\}$.
If $p \in S$, $\nabla g(p) \neq \mathbf{0}$, and $f|_S$ has a local extremum at p, then $\exists \lambda \in \mathbb{R}$ with $\nabla f(p) = \lambda \nabla g(p)$.

Proof. If $Df(p) = 0$, take $\lambda = 0$. Otherwise suppose $Df(p)$ is not a multiple of $Dg(p)$.
Complete $Dg(p), Df(p)$ to a dual basis $Dg(p)$, $Df(p)$, $\ell_3, \dots, \ell_n$ by Theorem 4.1.4,
and set $\Phi(\mathbf{x}) = \big(g(\mathbf{x}) - c,\ f(\mathbf{x}) - f(p),\ \ell_3(\mathbf{x}-p), \dots, \ell_n(\mathbf{x}-p)\big)$.
Then $D\Phi(p)$ is invertible, so Theorem 4.1.4 makes Φ locally invertible. In these coordinates, S is locally the slice with first coordinate 0, while the second coordinate is $f - f(p)$. Local invertibility gives nearby points of S with this second coordinate of both signs, contradicting extremality.
Hence $Df(p) = \lambda Dg(p)$; Riesz representation, Theorem 1.6.1, gives the gradient form.
Structural Commentary: The constraint kills first-order variation along S.
Key Insight: If Df were independent of Dg, f would be a free constraint coordinate. □

4.4 Further Topics in Local Differential Theory

Method 4.4.1. *Several-Variable Differentiation Proof Patterns*

(i) *Linear approximation*: write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
(ii) *Norm estimates*: use $\|A\mathbf{h}\|_2 \le \|A\|_{\text{op}} \|\mathbf{h}\|_2$ and the product estimate from Proposition 5.1.3.
(iii) *Line-segment reduction*: on convex sets, restrict to $\gamma(t) = (1-t)\mathbf{a} + t\mathbf{b}$ and apply one-variable results.
(iv) *Local inversion*: if $A = D\mathbf{f}(\mathbf{a})$ is invertible, write $D\mathbf{f}(\mathbf{x}) = A + E_{\mathbf{x}}$ with $\|E_{\mathbf{x}}\|_{\text{op}}$ small and solve by contraction (Theorem 4.5.3); linear block equations use Theorem 4.2.2.
(v) *Coordinates*: read partial derivatives as the entries of the matrix of $D\mathbf{f}(\mathbf{x})$, using Theorem 3.1.9.
(vi) *Higher derivatives*: regard $D^r\mathbf{f}(p)$ as an r-linear map; recover coordinate partials via Theorem 3.2.6.
(vii) *Rank normal forms*: for $A = D\mathbf{F}(\mathbf{a})$, split domain by $\ker A$ and image by range A; use the quotient-by-kernel picture from Theorem 2.2.12 and project with $P_{\text{range } A}$.

Key Insight: Several-variable calculus is one-variable limiting plus finite-dimensional linear algebra.

5 Differential Forms and Stokes Theory

5.1 Multiple Integration and Change of Variables

This subsection moves from Euclidean-region integration and change of variables to differential forms, pullbacks, and oriented parametrized surfaces.

Theorem 5.1.1. *Linear Maps Scale Volume by $|\det T|$*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ and let $\Omega \subseteq \mathbb{R}^n$ be bounded Jordan measurable.
(i) $T(\Omega)$ is Jordan measurable.
(ii) $\text{vol}(T(\Omega)) = |\det T| \,\text{vol}(\Omega)$.

Proof. If T invertible, let $T = SP$ with S orthogonal, $P = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_j > 0$, in orthonormal coordinates.
(i) If T is invertible, P and S preserve Jordan measurability, so $T(\Omega)$ is Jordan measurable.
If T is singular, $T(\Omega) \subseteq W$ for a proper affine subspace W, hence $T(\Omega)$ is Jordan null.
(ii) If T is invertible, rectangle scaling gives $\text{vol}(P\Omega) = (\det P)\,\text{vol}(\Omega)$ by Jordan approximation.
Hhence $\text{vol}(T\Omega) = \text{vol}(P\Omega) = |\det T|\,\text{vol}(\Omega)$. If T is singular, (i) gives $\text{vol}(T\Omega) = 0 = |\det T|\,\text{vol}(\Omega)$.
Structural Commentary: Polar decomposition leaves only diagonal stretching.
Key Insight: $|\det T|$ is total coordinate stretch. □

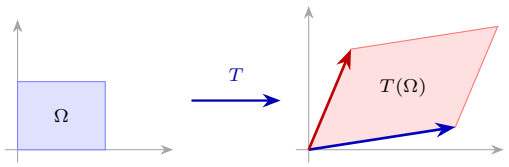

Figure 168: A linear map scales volume by $|\det T|$.

Definition 5.1.2. *k-Cells and Multiple Integrals*
Let $a_i \le b_i$.

(i) Set $I^j := \prod_{i=1}^{j} [a_i, b_i] \subseteq \mathbb{R}^j$ for $0 \le j \le k$, with I^0 a point. The *k-cell* is I^k.

(ii) For $f \in C(I^k)$, set $f_k := f$ and $f_{j-1}(x_1, \dots, x_{j-1}) := \int_{a_j}^{b_j} f_j(x_1, \dots, x_j)\, dx_j$, $j = k, \dots, 1$.
The *multiple integral* is $\int_{I^k} f(\mathbf{x})\, d\mathbf{x} := f_0$, also written $\int_{I^k} f$.

For bounded functions on rectangles, the Darboux picture from Definition 1.2.1 generalizes by replacing intervals with grid rectangles.

Definition 5.1.3. *Rectangular Grids and Riemann Sums*
Let $R = [a, b] \times [c, d] \subseteq \mathbb{R}^2$, let $P : a = x_0 < \cdots < x_m = b$, let $Q : c = y_0 < \cdots < y_n = d$, and set $G := P \times Q$.

(i) The *grid rectangles* are $R_{ij} := [x_{i-1}, x_i] \times [y_{j-1}, y_j]$, with $|R_{ij}| := (x_i - x_{i-1})(y_j - y_{j-1})$.

(ii) For tags $S = \{(s_{ij}, t_{ij}) \in R_{ij}\}$, the *Riemann sum* is $R(f, G, S) := \sum_{i=1}^{m} \sum_{j=1}^{n} f(s_{ij}, t_{ij})\, |R_{ij}|$.

(iii) For bounded $f : R \to \mathbb{R}$, set $(m_{ij}, M_{ij}) := (\inf_{R_{ij}} f, \sup_{R_{ij}} f)$ and $(L(f, G), U(f, G)) := \sum_{i,j} (m_{ij}, M_{ij})\, |R_{ij}|$.

(iv) The *lower* and *upper integrals* are $\underline{\int_R} f := \sup_G L(f, G)$ and $\overline{\int_R} f := \inf_G U(f, G)$.
If they agree, f is Riemann integrable on R, and $\int_R f$ denotes the common value.

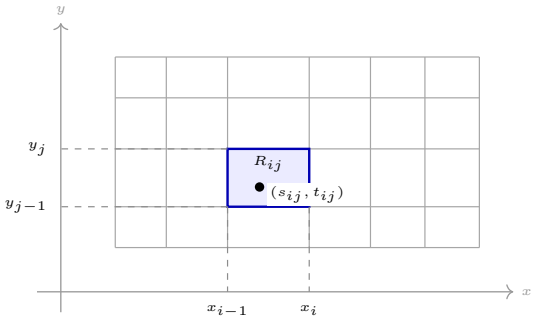

Figure 169: A grid rectangle R_{ij} and sample point (s_{ij}, t_{ij}).

Theorem 5.1.4. *Fubini on a Rectangle, Darboux Form*
Let $R = [a, b] \times [c, d]$, let $f : R \to \mathbb{R}$ be Riemann integrable, and set $f_y(x) := f(x, y)$.
For $y \in [c, d]$, define $\underline{F}(y) := \sup_P L(P, f_y)$ and $\overline{F}(y) := \inf_P U(P, f_y)$, with P ranging over partitions of $[a, b]$.

(i) $\underline{F}, \overline{F}$ are Riemann integrable on $[c, d]$.

(ii) $\int_R f = \int_c^d \underline{F}(y)\, dy = \int_c^d \overline{F}(y)\, dy$.

If $\int_a^b f(x, y)\, dx$ exists for every y, then $\underline{F}(y) = \overline{F}(y) = \int_a^b f(x, y)\, dx$.

Proof.

(i) For $G = P \times Q$, horizontal slice sums give $L(f, G) \le L(Q, \underline{F}) \le U(Q, \overline{F}) \le U(f, G)$.
Since f is integrable on R, choose G with $U(f, G) - L(f, G) < \varepsilon$; hence $U(Q, \overline{F}) - L(Q, \underline{F}) < \varepsilon$.
Thus $\underline{F}$ and $\overline{F}$ are integrable and have the same integral by Theorem 2.1.7.

(ii) The same inequalities give $L(f, G) \le \int_c^d \underline{F} \le \int_c^d \overline{F} \le U(f, G)$.
Letting $U(f, G) - L(f, G) \to 0$ gives both integrals equal to $\int_R f$.

Structural Commentary: Product grids become slice sums.
Key Insight: Fubini is a Darboux squeeze. □

Theorem 5.1.5. *Order Independence on Cells*
For $f \in C(I^k)$, all iterated orders defining $\int_{I^k} f$ give the same value.

Proof. Let L, L' be two iterated orders. For $h(\mathbf{x}) = h_1(x_1) \cdots h_k(x_k)$, $L(h) = L'(h) = \prod_{i=1}^{k} \int_{a_i}^{b_i} h_i(x_i)\, dx_i$, so $L(g) = L'(g)$ for every finite sum g of such products. These sums are dense in $C(I^k)$ by Theorem 9.2.8.

Choose g with $\|f - g\|_\infty < \varepsilon / V$, where $V = \prod_{i=1}^{k} (b_i - a_i)$.

Since $|L(r)|, |L'(r)| \le V \|r\|_\infty$, then $|L(f) - L'(f)| \le |L(f - g)| + |L'(f - g)| < 2\varepsilon$.
Structural Commentary: Product functions plus uniform approximation prove order independence.
Key Insight: Fubini-type order independence is first an algebraic fact, then a uniform approximation fact. □

Perspective 5.1.6. *Order Interchange Requires Hypotheses*
Finite sums are order-free; limiting processes are not.
Theorems 5.1.4 and 5.1.5 give hypotheses for interchanging integral order.

Definition 5.1.7. *Support and Simplex Integrals*
For $f : \mathbb{R}^k \to \mathbb{R}$ or $\mathbb{C}$, define $\operatorname{supp} f := \overline{\{\mathbf{x} \in \mathbb{R}^k : f(\mathbf{x}) \neq 0\}}$.

(i) For $f \in C_c(\mathbb{R}^k)$ and any cell $I^k \supseteq \operatorname{supp} f$, define $\int_{\mathbb{R}^k} f := \int_{I^k} f$; this is independent of I^k.

(ii) The *standard k-simplex* is $Q^k := \{\mathbf{x} \in \mathbb{R}^k : x_i \geq 0,\ \sum_{i=1}^{k} x_i \leq 1\}$.
For $f \in C(Q^k)$, extend f by 0 to $[0,1]^k$ and define $\int_{Q^k} f$ as the corresponding multiple integral.

Definition 5.1.8. *Primitive Mappings and Flips*
Let $E \subseteq \mathbb{R}^n$ be open.

(i) $G : E \to \mathbb{R}^n$ is *primitive in coordinate m* if $G(\mathbf{x}) = (x_1, \ldots, x_{m-1}, g(\mathbf{x}), x_{m+1}, \ldots, x_n)$.
(ii) If g is differentiable at $\mathbf{a} \in E$, then $DG(\mathbf{a})$ is I_n except row m is $\nabla g(\mathbf{a})$, and $J_G(\mathbf{a}) = (D_m g)(\mathbf{a})$.
(iii) A *flip* is a linear map $B : \mathbb{R}^n \to \mathbb{R}^n$ with $B\mathbf{e}_p = \mathbf{e}_q$, $B\mathbf{e}_q = \mathbf{e}_p$, and $B\mathbf{e}_i = \mathbf{e}_i$ for $i \notin \{p, q\}$, $p \neq q$.

Theorem 5.1.9. *Local Decomposition into Primitive Mappings*
Let $E \subseteq \mathbb{R}^n$ be open, let $\mathbf{F} \in C^1(E, \mathbb{R}^n)$, let $\mathbf{0} \in E$, and assume $\mathbf{F}(\mathbf{0}) = \mathbf{0}$ and $D\mathbf{F}(\mathbf{0})$ is invertible. Then there exist a neighborhood U of $\mathbf{0}$, primitive C^1 maps $G_1, \ldots, G_n$ near $\mathbf{0}$, and maps $B_1, \ldots, B_{n-1}$ such that:
(i) $G_i(\mathbf{0}) = \mathbf{0}$, $DG_i(\mathbf{0})$ is invertible, and each B_i is a flip or I.
(ii) $\mathbf{F} = B_1 \cdots B_{n-1} G_n \circ \cdots \circ G_1$ on U.

Proof. Let $P_m \mathbf{x} := \sum_{i=1}^{m} x_i \mathbf{e}_i$ and set $\mathbf{F}_1 := \mathbf{F}$.

(i) Suppose $P_{m-1}\mathbf{F}_m = P_{m-1}$, $\mathbf{F}_m(\mathbf{0}) = \mathbf{0}$, and $D\mathbf{F}_m(\mathbf{0})$ invertible, and write $\mathbf{F}_m = P_{m-1} + \sum_{i=m}^{n} \alpha_i \mathbf{e}_i$.
Choose $k \geq m$ with $(D_m \alpha_k)(\mathbf{0}) \neq 0$, let B_m swap m, k (or $B_m = I$), set $G_m(\mathbf{x}) := \mathbf{x} + (\alpha_k(\mathbf{x}) - x_m)\mathbf{e}_m$.
Then G_m is primitive, $G_m(\mathbf{0}) = \mathbf{0}$, $J_{G_m}(\mathbf{0}) \neq 0$, and G_m is locally invertible.
With $\mathbf{F}_{m+1} := B_m \mathbf{F}_m \circ G_m^{-1}$, one has $P_m \mathbf{F}_{m+1} = P_m$, $\mathbf{F}_{m+1}(\mathbf{0}) = \mathbf{0}$, and $D\mathbf{F}_{m+1}(\mathbf{0})$ invertible.
Iterate for $m = 1, \ldots, n-1$ and set $G_n := \mathbf{F}_n$.
(ii) From $\mathbf{F}_{m+1} = B_m \mathbf{F}_m \circ G_m^{-1}$ and $B_m^2 = I$, $\mathbf{F}_m = B_m \mathbf{F}_{m+1} \circ G_m$.
Iteration gives $\mathbf{F} = \mathbf{F}_1 = B_1 \cdots B_{n-1} G_n \circ G_{n-1} \circ \cdots \circ G_1$ near $\mathbf{0}$.

Structural Commentary: Local diffeomorphisms factor by coordinate elimination.
Key Insight: Nonlinear elimination follows $D\mathbf{F}(\mathbf{0})$. □

Theorem 5.1.10. *Partition of Unity on a Compact Set*
Let $K \subseteq \mathbb{R}^n$ be compact and let $\{V_\alpha\}$ cover K by open sets. Then there are $\psi_1, \ldots, \psi_s \in C^1(\mathbb{R}^n)$ such that
(i) $0 \leq \psi_i \leq 1$ for all i;
(ii) $\operatorname{supp}(\psi_i) \subseteq V_\alpha$ for some α;
(iii) $\sum_{i=1}^{s} \psi_i = 1$ on K.
If $f \in C(\mathbb{R}^n)$ and $\operatorname{supp}(f) \subseteq K$, then $f = \sum_{i=1}^{s} \psi_i f$, with each summand supported in one member of the cover.

Proof.

(i) For each $\mathbf{x} \in K$, choose balls with $\overline{B(\mathbf{x})} \subseteq W(\mathbf{x}) \subseteq \overline{W(\mathbf{x})} \subseteq V_{\alpha(\mathbf{x})}$.
By compactness, Definition 4.1.1, finitely many $B(\mathbf{x}_i)$ cover K.
(ii) Choose $\varphi_i \in C^1(\mathbb{R}^n)$ with $\varphi_i = 1$ on $B(\mathbf{x}_i)$ and $\operatorname{supp}(\varphi_i) \subseteq W(\mathbf{x}_i)$. Set $\psi_i = (1 - \varphi_1) \cdots (1 - \varphi_{i-1})\varphi_i$.
(iii) Then $0 \leq \psi_i \leq 1$, each ψ_i is supported in some V_α, and $\sum_{i=1}^{s} \psi_i = 1 - \prod_{i=1}^{s} (1 - \varphi_i)$, which equals 1 on K.
Multiplying by f gives the final assertion.

Structural Commentary: Compactness finite; product cutoffs sum to one.
Key Insight: A partition of unity turns local change-of-variable information into a global formula. □

The nonlinear change-of-variables theorem is the local volume-scaling formula from Theorem 5.1.1 applied cell by cell after a fine grid has been warped by a C^1 diffeomorphism.

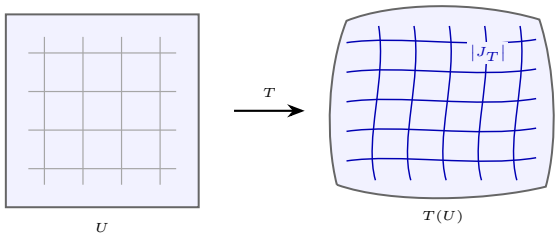

Figure 170: A C^1 change of variables turns a fine grid into curvilinear cells; $|J_T|$ records local volume scaling.

Theorem 5.1.11. *Change of Variables in $\mathbb{R}^k$*
Let $T : E \to \mathbb{R}^k$ be one-to-one and C^1 on open $E \subseteq \mathbb{R}^k$, with $J_T \neq 0$ on E.
If $f \in C(\mathbb{R}^k)$ and $\operatorname{supp}(f)$ is compact in $T(E)$, then $\int_{\mathbb{R}^k} f(\mathbf{y})\, d\mathbf{y} = \int_{\mathbb{R}^k} f(T(\mathbf{x}))\, |J_T(\mathbf{x})|\, d\mathbf{x}$.

Proof. Let $\mathcal{C}$ be the class of maps for which formula holds for all continuous functions with compact support in the image. Primitive maps lie in $\mathcal{C}$ by one-variable substitution, and flips lie in $\mathcal{C}$ by Theorem 5.1.5.
If $P, Q \in \mathcal{C}$, then $P \circ Q \in \mathcal{C}$, since Theorems 3.1.5 and 1.1.8 give $J_{P\circ Q}(\mathbf{x}) = J_P(Q(\mathbf{x}))J_Q(\mathbf{x})$.
After translating coordinates, Theorem 5.1.9 therefore gives the formula on a neighborhood of each point of E. By Theorem 4.1.4, these neighborhoods have open images covering $\operatorname{supp}(f)$.
Choose a partition of unity from Theorem 5.1.10, write $f = \sum_i \psi_i f$ with each $\psi_i f$ supported in one such image, apply the local formula to each summand, and sum.
Structural Commentary: Model cases plus partitions of unity prove the theorem.
Key Insight: The absolute Jacobian is local volume scaling, globalized by compact support. □

Method 5.1.12. *Change-of-Variables Jacobian Method*
For a differentiable map $\sigma : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$:

(i) compute the derivative matrix $D\sigma(\mathbf{x})$;
(ii) compute the local volume factor $|\det D\sigma(\mathbf{x})|$;
(iii) replace $d\mathbf{y}$ by $|\det D\sigma(\mathbf{x})|\, d\mathbf{x}$ when $\mathbf{y} = \sigma(\mathbf{x})$;
(iv) transform the domain bounds in the new coordinates.

Example 5.1.13. *Polar and Spherical Jacobian Factors*
For polar coordinates $\sigma(r,\theta) = (r\cos\theta, r\sin\theta)$, $D\sigma(r,\theta) = \begin{bmatrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{bmatrix}$, and $|\det D\sigma(r,\theta)| = r$.
Thus area integrals acquire the factor $r\, dr\, d\theta$.
For spherical coordinates $\sigma(\rho,\varphi,\theta) = (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)$, and $|\det D\sigma(\rho,\varphi,\theta)| = \rho^2\sin\varphi$.
Thus volume integrals acquire the factor $\rho^2\sin\varphi\, d\rho\, d\varphi\, d\theta$.

Method 5.1.14. *Multiple-Integration Proof Patterns*
For a multiple-integration identity:

(i) *Products*: prove product cases, then pass by uniform approximation;
(ii) *Grids*: compare product-grid upper and lower sums with one-variable slice sums;
(iii) *Support*: enclose compact support in a cell and check cell-independence;
(iv) *Local form*: reduce C^1 coordinate changes to primitive maps and flips;
(v) *Determinants*: use multiplicativity and Theorem 1.1.8;
(vi) *Globalize*: apply the partition of unity, Theorem 5.1.10.

Key Insight: Change of variables is local determinant geometry plus compactness.

5.2 Differential Forms and Manifolds

Differential forms package oriented integration on parametrized cells. They use determinant sign and laws (Definition 1.1.1, Theorems 1.1.8, 1.2.5) and change of variables (Theorem 5.1.11); conceptually, forms test oriented cells.

Perspective 5.2.1. *Forms as Functionals*
The notation $\int f\, dx$ treats dx as directed coordinate variation. Differential forms make this precise: a 1-form such as $f\, dx + g\, dy$ assigns a number to an oriented path; dx records net x-variation, and $f\, dx$ weights it by f.

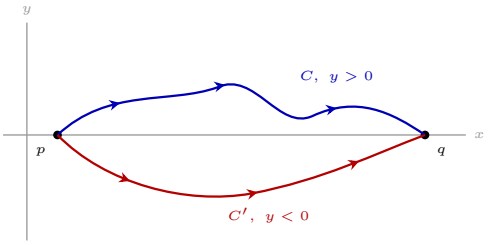

Figure 171: $y\, dx$ records signed, y-weighted x-variation.

Definition 5.2.2. *k-Surfaces and Differential Forms*
Let $E \subseteq \mathbb{R}^n$ be open.

(i) A compact *parameter domain* $D \subseteq \mathbb{R}^k$ is a k-cell or $Q^k = \{\mathbf{u} \in \mathbb{R}^k : u_i \geq 0,\ \sum_i u_i \leq 1\}$.
For compact D, $C^r(D)$ means extendable as C^r to an open neighborhood of D.
(ii) A *k-surface* in E is a C^1 map $\Phi : D \to E$ from a compact parameter domain $D \subseteq \mathbb{R}^k$.
(iii) For $k \geq 1$, a *k-form* on E is $\omega = \sum_I a_I(\mathbf{x})\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, where $I = (i_1, \ldots, i_k)$ and $a_I \in C(E, \mathbb{R})$.
(iv) For an increasing k-index $I = (i_1 < \cdots < i_k)$, set $dx_I := dx_{i_1} \wedge \cdots \wedge dx_{i_k}$.
A *standard presentation* is $\omega = \sum_I b_I(\mathbf{x})\, dx_I$, with I increasing.
(v) For a k-surface $\Phi : D \to E$, define $\int_\Phi \omega := \int_D \sum_I a_I(\Phi(\mathbf{u}))J_I\Phi(\mathbf{u})\, d\mathbf{u}$, where $J_I\Phi := \frac{\partial(x_{i_1},\ldots,x_{i_k})}{\partial(u_1,\ldots,u_k)}$.
(vi) A 0-form is a continuous function on E; a form is C^r iff all coefficients a_I are C^r.

Remark 5.2.3. *Parametrized Cells, Not Just Images*
Parametrizations may fail to be injective or regular. Forms therefore act on parametrized cells, not just images; this matters for orientation, multiplicity, and cancellation.

Proposition 5.2.4. *Basic Forms and Standard Presentation*

(i) Swapping neighboring factors in $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ changes sign; repeated indices give 0.
(ii) Every k-form has a unique standard presentation.

Proof.

(i) For $\Phi(\mathbf{u}) = (x_{i_1}, \dots, x_{i_k})$, the coefficient of $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ is the determinant of the corresponding $k \times k$ Jacobian minor. Row swaps change its sign and repeated rows make it 0.

(ii) By (i), each term either vanishes or sorts to an increasing index, giving existence.
For uniqueness, let $\omega = \sum_I b_I(\mathbf{x})\, dx_I$ be zero in standard presentation.

If $b_J(\mathbf{v}) \neq 0$ for increasing $J = (j_1, \dots, j_k)$, choose $D = [-h, h]^k$ so $b_J(\Phi(\mathbf{u}))$ has constant nonzero sign for $\Phi(\mathbf{u}) = \mathbf{v} + \sum_{r=1}^{k} u_r \mathbf{e}_{j_r}$.

Then the J-Jacobian is 1 and all other standard Jacobians are 0, so $0 = \int_\Phi \omega = \int_D b_J(\Phi(\mathbf{u}))\, d\mathbf{u}$, a contradiction. Hence all $b_I = 0$.

Structural Commentary: Wedge signs are determinant signs.
Key Insight: Standard form sorts indices once. □

Theorem 5.2.5. *Naturality Under Reparametrization*
Let $\Phi : D \to E \subseteq \mathbb{R}^n$ be a C^1 k-surface, and let $R : D' \to D$ be a C^1 diffeomorphism with $\epsilon := \operatorname{sgn} \det DR$ constant on D'. For every k-form ω on E, $\int_{\Phi \circ R} \omega = \epsilon \int_\Phi \omega$.

Proof. By linearity, take $\omega = f\, dx_I$. Cauchy-Binet gives $J_I(\Phi \circ R)(\mathbf{u}) = J_I\Phi(R(\mathbf{u})) \det DR(\mathbf{u})$.
Thus $\int_{\Phi\circ R} \omega = \int_{D'} f(\Phi(R\mathbf{u}))J_I\Phi(R\mathbf{u}) \det DR(\mathbf{u})\, d\mathbf{u} = \epsilon \int_D f(\Phi(\mathbf{v}))J_I\Phi(\mathbf{v})\, d\mathbf{v}$ by change of variables.
Structural Commentary: Reparametrization changes coordinates only.
Key Insight: Orientation contributes $\operatorname{sgn} \det DR$. □

Definition 5.2.6. *Product and Exterior Derivative*
For basic forms, $dx_I \wedge dx_J$ is obtained by concatenating I, J; it is 0 if an index repeats, and otherwise $\operatorname{sgn}(\sigma) dx_K$, where K is the increasing rearrangement and σ is the sorting permutation. Extend bilinearly.

(i) For $f \in C^1(E)$, define $df := \sum_{i=1}^{n} (D_i f)(\mathbf{x})\, dx_i$.

(ii) For a C^1 k-form $\omega = \sum_I b_I(\mathbf{x})\, dx_I$, define $d\omega := \sum_I (db_I) \wedge dx_I$.

Theorem 5.2.7. *Exterior Derivative Laws*
If ω, λ are C^1 forms in E and ω is a k-form, then $d(\omega \wedge \lambda) = (d\omega) \wedge \lambda + (-1)^k \omega \wedge d\lambda$. If ω is C^2, then $d^2\omega = 0$.

Proof. By bilinearity, it suffices to take $\omega = f\, dx_I$ and $\lambda = g\, dx_J$.
The ordinary product rule gives $d(fg) = f\, dg + g\, df$; moving dg past k factors in dx_I gives the displayed sign.
For a C^2 0-form f, $d^2 f = \sum_{i,j} (D_{ij} f)(\mathbf{x})\, dx_i \wedge dx_j = 0$, by Theorem 3.2.11 and antisymmetry.

The general case follows from $\omega = \sum_I b_I\, dx_I$ and the product rule just proved.
Structural Commentary: The exterior derivative is built to make mixed partial cancellation visible.
Key Insight: Symmetric second derivatives meet antisymmetric wedge products, so $d^2 = 0$. □

Definition 5.2.8. *Change of Variables for Forms*
Let $E \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$, let $T = (t_1, \dots, t_m) : E \to V$ be C^1, and let $\omega = \sum_I b_I(\mathbf{y})\, dy_I$ be a k-form on V.

(i) Set $dt_i := \sum_{j=1}^{n} (D_j t_i)(\mathbf{x})\, dx_j$ for $1 \leq i \leq m$.

(ii) The *pullback* of ω by T is $T^*\omega := \omega_T := \sum_I b_I(T(\mathbf{x}))\, dt_{i_1} \wedge \cdots \wedge dt_{i_k}$.

Pullback is the differential-form analogue of the dual map in Definition 4.2.4: cells move forward by composition with T, while forms move backward so they can be evaluated before or after that forward motion.

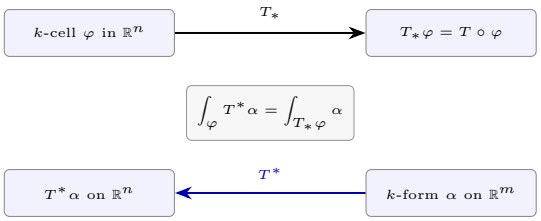

Figure 172: Cells push forward with T; forms pull back with T^*, so $T^*\alpha(\varphi) = \alpha(T_*\varphi)$.

Theorem 5.2.9. *Pullback Commutes with the Basic Operations*
Let $T = (t_1, \dots, t_m) : E \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ be C^1, and let $\omega_T := T^*\omega$.

(i) $(\omega + \lambda)_T = \omega_T + \lambda_T$ for forms of the same order;
(ii) $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$;
(iii) $d(\omega_T) = (d\omega)_T$ when ω is C^1 and T is C^2;
(iv) if $S : V \to W$ is C^1, then $(\omega_S)_T = \omega_{S\circ T}$.

Moreover, if Φ is a k-surface in E and ω is a k-form in V, then $\int_{T\circ\Phi} \omega = \int_\Phi \omega_T$.

Proof.

(i) Addition follows termwise from $b_I \mapsto b_I \circ T$ and $dy_i \mapsto dt_i$.
(ii) Pullback replaces each dy_i by dt_i; bilinearity gives $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$.
(iii) Let $\omega = \sum_I b_I\, dy_I$. Since $d(b_I \circ T) = (db_I)_T$ (Theorem 3.1.5) and $d(dt_i) = d^2 t_i = 0$, $d(\omega_T) = (d\omega)_T$.
(iv) If $S = (s_1, \dots, s_p)$, both $(\omega_S)_T$ and $\omega_{S\circ T}$ replace dz_q by $d(s_q \circ T)$, by the chain rule.

For the integral identity, expand over D and use Cauchy-Binet on $D(T \circ \Phi) = DT(\Phi)D\Phi$.
Structural Commentary: Pullback translates forms before integration.
Key Insight: Forms are defined so that composition of parameterizations becomes an algebraic substitution. □

5.3 Stokes Theory

Stokes theory relates exterior differentiation to oriented boundary, with Green's theorem, classical Stokes, and divergence as special cases.

Definition 5.3.1. *Chains and Boundaries*
Let $\mathbf{p}_0, \dots, \mathbf{p}_k \in \mathbb{R}^n$.

(i) The oriented affine *k-simplex* $[\mathbf{p}_0, \dots, \mathbf{p}_k]$ is the k-surface $\sigma : Q^k \to \mathbb{R}^n$, $\sigma(\mathbf{u}) := \mathbf{p}_0 + \sum_{i=1}^{k} u_i(\mathbf{p}_i - \mathbf{p}_0)$.

(ii) For a permutation π, $[\mathbf{p}_{\pi(0)}, \dots, \mathbf{p}_{\pi(k)}] := \operatorname{sgn}(\pi)[\mathbf{p}_0, \dots, \mathbf{p}_k]$.

(iii) An *affine k-chain* is $C = \sum_\ell c_\ell \sigma_\ell$; integration is linear: $\int_C \omega := \sum_\ell c_\ell \int_{\sigma_\ell} \omega$.

(iv) For $k \geq 1$, $\partial_{\mathrm{ch}}[\mathbf{p}_0, \dots, \mathbf{p}_k] := \sum_{j=0}^{k} (-1)^j [\mathbf{p}_0, \dots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k]$, extended linearly.

If $\Phi = T \circ \sigma$ is a C^2 oriented simplex, set $\partial_{\mathrm{ch}}\Phi := T(\partial_{\mathrm{ch}}\sigma)$, extended linearly to chains.

Theorem 5.3.2. *Stokes' Theorem*
Let $V \subseteq \mathbb{R}^m$ be open, let Ψ be a C^2 k-chain in V, and let ω be a C^1 $(k-1)$-form on V. Then $\int_\Psi d\omega = \int_{\partial_{\mathrm{ch}}\Psi} \omega$.

Proof. By linearity, take $\Psi = \Phi = T \circ \sigma$, where σ is the standard oriented k-simplex.
By Theorem 5.2.9, $\int_\Phi d\omega = \int_\sigma d(\omega_T)$ and $\int_{\partial_{\mathrm{ch}}\Phi} \omega = \int_{\partial_{\mathrm{ch}}\sigma} \omega_T$; it remains to prove the formula on σ.
By linearity, reduce to $\lambda = f\, dx_1 \wedge \cdots \wedge \widehat{dx_r} \wedge \cdots \wedge dx_k$.
For $\partial_{\mathrm{ch}}\sigma = \sum_{j=0}^{k} (-1)^j \sigma_j$, all faces vanish except the two with varying x_r endpoints.
The remaining equality is the one-variable FTC in the x_r integral.
Structural Commentary: Pullback reduces Stokes to the standard simplex.
Key Insight: Boundary signs match the missing wedge factor. □

Definition 5.3.3. *Closed and Exact Forms*
Let $E \subseteq \mathbb{R}^n$ be open and let ω be a k-form on E.

(i) ω is *exact* if $\omega = d\lambda$ for some $(k-1)$-form λ on E.

(ii) ω is *closed* if $\omega \in C^1$ and $d\omega = 0$.

Perspective 5.3.4. *Closed Is Local, Exact Is Global*
Exact $\Rightarrow$ closed is just $d^2 = 0$; the converse depends on the domain.
On $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ the standard warning example is $\omega = \frac{-y}{x^2+y^2}\, dx + \frac{x}{x^2+y^2}\, dy$, which is closed but not exact.
Holes can obstruct global antiderivatives even when local derivative tests vanish.

Theorem 5.3.5. *Poincare Lemma on Convex Sets*
Let $k \geq 1$.

(i) If $E \subseteq \mathbb{R}^n$ is open and convex, then every closed C^1 k-form on E is exact.

(ii) The same conclusion holds on any open set C^2-diffeomorphic to a convex open set.

Proof.

(i) Fix $\mathbf{a} \in E$ and set $H_t(\mathbf{x}) := \mathbf{a} + t(\mathbf{x} - \mathbf{a})$, $0 \leq t \leq 1$.
For a k-form ω, define $(K\omega)_{\mathbf{x}}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) := \int_0^1 t^{k-1} \omega_{H_t(\mathbf{x})}(\mathbf{x} - \mathbf{a}, \mathbf{v}_1, \dots, \mathbf{v}_{k-1})\, dt$.
Differentiating under the integral gives $\omega - H_0^*\omega = dK\omega + Kd\omega$.
Since $H_0^*\omega = 0$ for $k \geq 1$, $d\omega = 0$ implies $\omega = d(K\omega)$.

(ii) Let $\Phi : C \to E$ be a C^2 diffeomorphism from convex open C. If $d\omega = 0$, then $d(\Phi^*\omega) = \Phi^*(d\omega) = 0$.
By (i), $\Phi^*\omega = d\eta$ on C, hence $\omega = (\Phi^{-1})^* d\eta = d((\Phi^{-1})^*\eta)$ on E.

Structural Commentary: Convex contraction gives a homotopy operator.
Key Insight: Closed forms become derivatives along the contraction. □

Theorem 5.3.6. *Vector Analysis Dictionary in $\mathbb{R}^3$*
Let $E \subseteq \mathbb{R}^3$ be open and let $\mathbf{F} = F_1\mathbf{e}_1 + F_2\mathbf{e}_2 + F_3\mathbf{e}_3 \in C^1(E, \mathbb{R}^3)$.
Set $\lambda_{\mathbf{F}} := F_1\, dx + F_2\, dy + F_3\, dz$ and $\omega_{\mathbf{F}} := F_1\, dy \wedge dz + F_2\, dz \wedge dx + F_3\, dx \wedge dy$.

(i) For $u \in C^1(E)$, $\mathbf{F} = \nabla u$ iff $\lambda_{\mathbf{F}} = du$.

(ii) $d\lambda_{\mathbf{F}} = \omega_{\nabla \times \mathbf{F}}$; hence $\nabla \times \mathbf{F} = 0$ iff $d\lambda_{\mathbf{F}} = 0$.

(iii) For $\mathbf{G} \in C^1(E, \mathbb{R}^3)$, $\mathbf{F} = \nabla \times \mathbf{G}$ iff $\omega_{\mathbf{F}} = d\lambda_{\mathbf{G}}$.

(iv) $d\omega_{\mathbf{F}} = (\nabla \cdot \mathbf{F})\, dx \wedge dy \wedge dz$; hence $\nabla \cdot \mathbf{F} = 0$ iff $d\omega_{\mathbf{F}} = 0$.

(v) If E is C^2-diffeomorphic to a convex open set, then $\nabla \times \mathbf{F} = 0 \Rightarrow \exists u \in C^1(E) : \mathbf{F} = \nabla u$;
$\nabla \cdot \mathbf{F} = 0 \Rightarrow \exists \mathbf{G} \in C^1(E, \mathbb{R}^3) : \mathbf{F} = \nabla \times \mathbf{G}$.

Proof.

(i) This is $du = (D_1 u)\, dx + (D_2 u)\, dy + (D_3 u)\, dz$.

(ii) Direct expansion gives $d\lambda_{\mathbf{F}} = (D_2F_3 - D_3F_2)\, dy \wedge dz + (D_3F_1 - D_1F_3)\, dz \wedge dx + (D_1F_2 - D_2F_1)\, dx \wedge dy$.

(iii) By (ii), $d\lambda_{\mathbf{G}} = \omega_{\nabla \times \mathbf{G}}$; since $\mathbf{H} \mapsto \omega_{\mathbf{H}}$ is injective, $\omega_{\mathbf{F}} = d\lambda_{\mathbf{G}}$ iff $\mathbf{F} = \nabla \times \mathbf{G}$.

(iv) Direct expansion gives $d\omega_{\mathbf{F}} = (D_1F_1 + D_2F_2 + D_3F_3)\, dx \wedge dy \wedge dz$.

(v) If $\nabla \times \mathbf{F} = 0$, then $\lambda_{\mathbf{F}}$ is closed by (ii), so Poincare gives $\lambda_{\mathbf{F}} = du$, hence $\mathbf{F} = \nabla u$ by (i). If $\nabla \cdot \mathbf{F} = 0$, then $\omega_{\mathbf{F}}$ is closed by (iv), so Poincare gives $\omega_{\mathbf{F}} = d\eta$; writing $\eta = \lambda_{\mathbf{G}}$ gives $\mathbf{F} = \nabla \times \mathbf{G}$ by (iii).

Structural Commentary: Exterior derivative is grad–curl–div.
Key Insight: Vector identities are $d^2 = 0$ in $\mathbb{R}^3$. □

Theorem 5.3.7. *Green, Classical Stokes, and Divergence*
With the indicated positive orientations:

(i) For $\Omega \subseteq \mathbb{R}^2$ with positively oriented $\operatorname{bd}\Omega$ and $\alpha, \beta \in C^1$ near Ω, $\int_{\operatorname{bd}\Omega}(\alpha\, dx + \beta\, dy) = \int_\Omega (D_1\beta - D_2\alpha)\, dA$.

(ii) If Φ is a C^2 oriented 2-surface in $\mathbb{R}^3$ and $\mathbf{F} \in C^1$ near Φ, then $\int_{\partial_{\mathrm{ch}}\Phi} \mathbf{F}\cdot\mathbf{t}\,ds = \int_\Phi (\nabla\times\mathbf{F})\cdot\mathbf{n}\,dA$.
(iii) If $\Omega \subseteq \mathbb{R}^3$ has positively oriented boundary and $\mathbf{F} \in C^1$ near Ω, then $\int_{\mathrm{bd}\,\Omega} \mathbf{F}\cdot\mathbf{n}\,dA = \int_\Omega (\nabla\cdot\mathbf{F})\,dV$.

Proof. Use the form Stokes identity $\int_\Psi d\eta = \int_{\partial_{\mathrm{ch}}\Psi} \eta$.

(i) For $\lambda := \alpha\,dx + \beta\,dy$, $d\lambda = (D_1\beta - D_2\alpha)\,dx\wedge dy$; hence $\int_{\mathrm{bd}\,\Omega}\lambda = \int_\Omega d\lambda$ gives Green's formula.
(ii) For $\lambda := \lambda_{\mathbf{F}}$, $d\lambda = \omega_{\nabla\times\mathbf{F}}$; hence $\int_{\partial_{\mathrm{ch}}\Phi}\lambda = \int_\Phi d\lambda$ gives the classical Stokes formula.
(iii) For $\omega := \omega_{\mathbf{F}}$, $d\omega = (\nabla\cdot\mathbf{F})\,dx\wedge dy\wedge dz$; hence $\int_{\mathrm{bd}\,\Omega}\omega = \int_\Omega d\omega$ gives the divergence formula.

Structural Commentary: One form identity yields the three vector formulas.
Key Insight: Boundary plus d is invariant; notation changes. □

5.4 Further Topics in Differential Forms and Stokes Theory

Method 5.4.1. *Differential-Forms Proof Patterns*

(i) *Signs*: compute wedge signs by determinant swaps and permutation parity.
(ii) *Standard presentation*: remove repeated indices and sort the rest.
(iii) *Pullback*: substitute dt_i for dy_i and apply the chain rule.
(iv) *Stokes*: pull back to the standard simplex, prove the coordinate case, extend linearly.
(v) *Closed versus exact*: use $d^2 = 0$ for necessity; use convexity or C^2-equivalence for primitives.
(vi) *Vector analysis*: translate fields to forms, apply Stokes, translate back.

Key Insight: Forms package signs, orientation, and parameter changes.

6 Infinite Series and Summability

6.1 Series Tests

Series are sequences of partial sums. Comparison, condensation, root and ratio tests, rearrangements, products, and double sums all refine that viewpoint.

Definition 6.1.1. *Series and Partial Sums*
Let $(a_n)_{n\ge1} \subseteq \mathbb{C}$. Summation indices are bound variables and may be renamed within unchanged scope.

(i) For integers $1 \le p \le q$, $\sum_{n=p}^{q} a_n := a_p + a_{p+1} + \cdots + a_q$.
(ii) The *series* $\sum_{n=1}^{\infty} a_n$ is the *partial-sum sequence* (s_N), where $s_N := \sum_{n=1}^{N} a_n$.
(iii) The series *converges to* s, written $\sum_{n=1}^{\infty} a_n = s$, iff $s_N \to s$; otherwise it *diverges*. Also for $n = 0$.

Key Insight: A series is a sequence first; its sum is a limit when that limit exists.

Definition 6.1.2. *Absolute and Nonabsolute Convergence*
The series $\sum a_n$ *converges absolutely* if $\sum |a_n|$ converges.
If $\sum a_n$ converges but $\sum |a_n|$ diverges, then $\sum a_n$ converges *nonabsolutely*.

Motivation 6.1.3. *Partial Sums Remember Order*
An infinite sum is a limit of partial sums. For alternating harmonic series, odd and even partial sums approach the same value from opposite sides; see Corollary 6.1.25 and Theorem 6.1.34.

Perspective 6.1.4. *Formal Algebra Needs Convergence*
Finite sums are algebraic; infinite sums are partial-sum limits. Algebraic manipulations are justified only by convergence control, as in Theorem 6.1.5, Corollary 6.1.6, and Theorems 6.1.31, 6.1.34.
Key Insight: The symbol $\sum_{n=1}^{\infty}$ is a limit claim before it is an algebraic object.

Theorem 6.1.5. *Cauchy Criterion for Series*
The series $\sum a_n$ converges if and only if $\forall \varepsilon > 0$ $\exists$ an integer N s.t. $m \ge n \ge N$ implies $\left|\sum_{k=n}^{m} a_k\right| \le \varepsilon$.

Proof.
(⇒) If $s_n := \sum_{k=1}^{n} a_k$ converges, then (s_n) is Cauchy. Thus, for $m \ge n$ large enough, $\left|\sum_{k=n}^{m} a_k\right| = |s_m - s_{n-1}| \le \varepsilon$.
(⇐) The tail condition gives, for $m > n$ large enough, $|s_m - s_n| = \left|\sum_{k=n+1}^{m} a_k\right| \le \varepsilon$.
Hence (s_n) is Cauchy in $\mathbb{C} \cong \mathbb{R}^2$ and converges by Corollary 4.2.8.

Structural Commentary: A series converges exactly when all sufficiently late finite tails are small.
Key Insight: The Cauchy test measures tails, not the eventual value of the sum. □

Corollary 6.1.6. *Terms of a Convergent Series Tend to Zero*
If $\sum a_n$ converges, then $a_n \to 0$.

Proof. Let $\varepsilon > 0$. By Theorem 6.1.5, there is N such that $m \ge n \ge N$ implies $\left|\sum_{k=n}^{m} a_k\right| \le \varepsilon$.
Taking $m = n$ gives $|a_n| \le \varepsilon$ for all $n \ge N$, hence $a_n \to 0$.
Structural Commentary: A single term is the shortest possible tail.
Key Insight: The condition $a_n \to 0$ is necessary, but not sufficient for convergence. □

Theorem 6.1.7. *Nonnegative Series and Bounded Partial Sums*
A series of nonnegative real terms converges if and only if its partial sums form a bounded sequence.

Proof.
($\Rightarrow$) Every convergent sequence is bounded, so the partial sums are bounded.
($\Leftarrow$) Since $a_n \geq 0$, $s_n = \sum_{k=1}^{n} a_k$ increases; if bounded, it converges by Theorem 2.1.11.
Structural Commentary: Nonnegative series replace cancellation by monotonicity.
Key Insight: For nonnegative terms, convergence is exactly bounded accumulation. □

Theorem 6.1.8. *Comparison Test*
(i) If $|a_n| \leq c_n$ for $n \geq N_0$, and if $\sum c_n$ converges, then $\sum a_n$ converges.
(ii) If $a_n \geq d_n \geq 0$ for $n \geq N_0$, and if $\sum d_n$ diverges, then $\sum a_n$ diverges.

Proof.
(i) Given $\varepsilon > 0$, choose $N \geq N_0$ so that $m \geq n \geq N$ implies $\sum_{k=n}^{m} c_k \leq \varepsilon$.
Then $\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k| \leq \sum_{k=n}^{m} c_k \leq \varepsilon$, so Theorem 6.1.5 applies.
(ii) If $\sum a_n$ converged, then (i) applied to $d_n \leq a_n$ would force $\sum d_n$ to converge, a contradiction.
Structural Commentary: Absolute tail domination transfers Cauchy convergence.
Key Insight: Comparison is a tail test; finitely many early terms do not matter. □

Theorem 6.1.9. *Geometric Series*
If $0 \leq x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. If $x \geq 1$, the series diverges.

Proof. For $x \neq 1$, $\sum_{k=0}^{n} x^k = \frac{1-x^{n+1}}{1-x}$. Let $n \to \infty$ and use Theorem 2.2.10.
If $x = 1$, the partial sums are $n+1$, so they diverge.
Structural Commentary: The finite geometric identity converts convergence into the limit of x^{n+1}.
Key Insight: The geometric series is the model comparison series for ratio-like decay. □

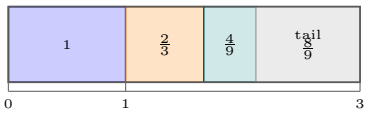

Figure 173: At $x = 2/3$, the strip widths sum to $1 + \frac{2}{3} + \frac{4}{9} + \cdots = 3$.

Definition 6.1.10. *Real Decimals as Series*
For digits $b_N, \ldots, b_0, a_1, a_2, \ldots \in \{0, \ldots, 9\}$, with $b_N \neq 0$ unless the integer part is 0,
$b_N \cdots b_0.a_1a_2\cdots := \sum_{i=0}^{N} b_i 10^i + \sum_{j=1}^{\infty} a_j 10^{-j}$. The fractional series converges absolutely since $0 \leq a_j 10^{-j} \leq 9 \cdot 10^{-j}$ and $\sum 9 \cdot 10^{-j}$ is geometric. A minus sign denotes the negative of this value.

Proposition 6.1.11. *Terminal Nines Are Nonunique*
The same real number can have two decimal representations: $1.000\ldots = 0.999\ldots$.

Proof. $1.000\ldots = 1 + \sum_{j=1}^{\infty} 0 \cdot 10^{-j} = 1$, while $0.999\ldots = \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 9 \cdot \frac{10^{-1}}{1-10^{-1}} = 1$.
Structural Commentary: The two digit strings have the same partial-sum limit.
Key Insight: Uniqueness requires excluding terminal 9 tails. □

Perspective 6.1.12. *Decimal Expansions Are Series*
Proposition 2.2.11 gives decimal truncations that converge to a real number, while Definition 6.1.10 interprets the infinite digit string as a series. Proposition 6.1.11 explains the standard terminal-9 ambiguity.

Theorem 6.1.13. *Cauchy Condensation Test*
Let $(a_n)_{n\geq 1}$ satisfy $a_1 \geq a_2 \geq \cdots \geq 0$. Then $\sum_{n=1}^{\infty} a_n$ converges iff $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Proof. Let $s_n := \sum_{j=1}^{n} a_j$ and $t_k := \sum_{j=0}^{k} 2^j a_{2^j}$. Both partial-sum sequences are nondecreasing.
($\Rightarrow$) If (s_n) is bounded, then $t_m \leq 2 \sum_{j=1}^{2^m} a_j = 2s_{2^m}$, so (t_m) is bounded and the condensed series converges.
($\Leftarrow$) If (t_k) is bounded, then for $2^m \leq n < 2^{m+1}$, $s_n \leq \sum_{j=0}^{m} 2^j a_{2^j} = t_m$; hence (s_n) is bounded, QED.
Structural Commentary: Dyadic blocks compare to endpoint terms.
Key Insight: Monotone tails are detected on dyadic scales. □

Corollary 6.1.14. *p-Series Test*
For $p \in \mathbb{R}$:
(i) If $p > 1$, then $\sum_{n=1}^{\infty} n^{-p}$ converges.
(ii) If $p \leq 1$, then $\sum_{n=1}^{\infty} n^{-p}$ diverges.

Proof.

(i) If $p > 1$, condensation reduces to $\sum_{k=0}^{\infty} 2^k (2^k)^{-p} = \sum_{k=0}^{\infty} 2^{(1-p)k}$, a geometric series with ratio $2^{1-p} < 1$.

(ii) If $p \le 0$, then $n^{-p} \not\to 0$; if $0 < p \le 1$, condensation gives a geometric series with ratio $2^{1-p} \ge 1$.

Structural Commentary: Condensation turns n-powers into a geometric series.
Key Insight: The threshold is the ratio $2^{1-p} = 1$. □

Theorem 6.1.15. *Integral Test*
Let $N \in \mathbb{N}$ and let $f : [N, \infty) \to [0, \infty)$ be decreasing and Riemann integrable on compact subintervals.

(i) For $M \ge N$, $\int_N^{M+1} f(x)\,dx \le \sum_{k=N}^{M} f(k) \le f(N) + \int_N^M f(x)\,dx$.

(ii) $\sum_{k=N}^{\infty} f(k)$ converges iff $\int_N^{\infty} f(x)\,dx$ converges.

Proof.

(i) For $x \in [k, k+1]$, $f(x) \le f(k)$, so $\int_N^{M+1} f \le \sum_{k=N}^{M} f(k)$.

For $k \ge N+1$ and $x \in [k-1, k]$, $f(k) \le f(x)$, so $\sum_{k=N}^{M} f(k) \le f(N) + \int_N^M f$.

(ii) The truncations $\sum_{k=N}^{M} f(k)$ and $\int_N^M f$ are nondecreasing in M. By (i), either both are bounded or both are unbounded; hence the series and improper integral converge or diverge together.

Structural Commentary: Unit intervals compare samples to area.
Key Insight: Same monotone tail, same boundedness. □

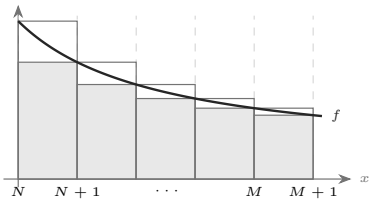

Figure 174: Left and right endpoint estimates for decreasing $f \ge 0$.

Definition 6.1.16. *The Number e*
Define $e := \sum_{n=0}^{\infty} \frac{1}{n!}$, where $0! := 1$; convergence follows by comparison with the geometric series.

Theorem 6.1.17. *A Limit Formula for e*
The limit formula is $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let $s_m := \sum_{k=0}^{m} \frac{1}{k!}$ and $t_n := (1 + \frac{1}{n})^n$. The binomial theorem gives $t_n \le s_n$, hence $\limsup t_n \le e$.
For fixed $m \le n$, first m binomial terms give $t_n \ge 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!}\left(1 - \frac{1}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right)$.
Letting $n \to \infty$ with m fixed gives $\liminf t_n \ge s_m$, and then $m \to \infty$ gives $e \le \liminf t_n$.
Structural Commentary: Bound above by the factorial series; bound below by fixed binomial terms.
Key Insight: Limsup and liminf squeeze the binomial limit to the series definition. □

Proposition 6.1.18. *Remainder Bound for the Series of e*
If $n \ge 1$ and $s_n = \sum_{k=0}^{n} 1/k!$, then $0 < e - s_n < \frac{1}{n!\,n}$.

Proof. Tail is $e - s_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots < \frac{1}{(n+1)!}\left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots\right) = \frac{1}{n!\,n}$
Structural Commentary: The factorial tail is bounded by a geometric tail.
Key Insight: Rapid convergence of $\sum \frac{1}{n!}$ gives arithmetic control of e. □

Theorem 6.1.19. *e Is Irrational*
The number e is irrational.

Proof. Suppose $e = p/q$ with p, q positive integers. By Proposition 6.1.18, $0 < q!(e - s_q) < \frac{1}{q}$.
But $q!e$ and $q!s_q$ are integers, so $q!(e - s_q)$ is an integer strictly between 0 and 1, impossible.
Structural Commentary: The sharp tail bound turns rationality into an impossible integer.
Key Insight: A positive integer cannot be smaller than 1. □

Theorem 6.1.20. *Root Test*
For a series $\sum a_n$, put $\alpha := \limsup_{n\to\infty} \sqrt[n]{|a_n|}$. Then:

(i) if $\alpha < 1$, then $\sum a_n$ converges absolutely;
(ii) if $\alpha > 1$, then $\sum a_n$ diverges;
(iii) if $\alpha = 1$, the test gives no information.

Proof.

(i) Choose $\alpha < \beta < 1$. Eventually $|a_n| < \beta^n$, so $\sum |a_n|$ converges by comparison with $\sum \beta^n$.
(ii) Since $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$; use Corollary 6.1.6.
(iii) $\sum 1/n$ and $\sum 1/n^2$ both have $\alpha = 1$, but behave differently by Corollary 6.1.14.

Structural Commentary: The root test compares term size directly to geometric decay.
Key Insight: Values below 1 force absolute convergence; values above 1 prevent terms from tending to 0. □

Theorem 6.1.21. *Ratio Test*
Assume $a_n \neq 0$ for all sufficiently large n.

(i) If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum a_n$ converges absolutely.
(ii) If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then $\sum a_n$ diverges.
(iii) In the remaining cases, the test gives no information.

Proof.

(i) Choose $\beta < 1$ with $|a_{n+1}/a_n| < \beta$ eventually.
Then $|a_n| = O(\beta^n)$, so $\sum |a_n|$ converges by comparison with a geometric series.
(ii) Choose $\beta > 1$ with $|a_{n+1}/a_n| > \beta$ eventually.
Then the tail $|a_n|$ is eventually increasing and bounded away from 0, so $a_n \not\to 0$.
(iii) The series $\sum 1/n$ and $\sum 1/n^2$ both have ratio limit 1, but behave differently by Corollary 6.1.14.

Structural Commentary: Ratios turn successive terms into a geometric majorant.
Key Insight: Ratios below 1 force geometric decay; ratios above 1 prevent terms from tending to zero. □

Theorem 6.1.22. *Root and Ratio Test Comparison*
For any sequence (c_n) of positive numbers, $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \le \liminf_{n\to\infty} \sqrt[n]{c_n} \le \limsup_{n\to\infty} \sqrt[n]{c_n} \le \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$.

Proof. Let $\alpha = \limsup c_{n+1}/c_n$. If $\alpha < +\infty$ and $\beta > \alpha$, then eventually $c_{n+1} \le \beta c_n$, hence $c_n \le c_N \beta^{n-N}$ for all large n. Taking nth roots gives $\limsup \sqrt[n]{c_n} \le \beta$; let $\beta \downarrow \alpha$. If $\alpha = +\infty$, the rightmost inequality is automatic.
Let $\ell = \liminf c_{n+1}/c_n$. If $\ell = 0$, the leftmost inequality is automatic. If $\gamma < \ell$ with $\gamma > 0$, then eventually $c_{n+1} \ge \gamma c_n$, hence $c_n \ge c_N \gamma^{n-N}$ for all large n. Taking nth roots gives $\liminf \sqrt[n]{c_n} \ge \gamma$; let $\gamma \uparrow \ell$.
The middle inequality is the general fact that $\liminf a_n \le \limsup a_n$.
Structural Commentary: A tail ratio bound multiplies into a root bound.
Key Insight: Whenever the ratio test proves convergence, the root test does too. □

Theorem 6.1.23. *Summation by Parts*
Given two sequences (a_n) and (b_n), put $A_n := \sum_{k=0}^{n} a_k$ and $A_{-1} := 0$.
If $0 \le p \le q$, then $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$.

Proof. Since $a_n = A_n - A_{n-1}$, $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p}^{q} A_{n-1} b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1}$.
Structural Commentary: This is the discrete analogue of integration by parts.
Key Insight: Move the difference from a_n onto b_n by rewriting $a_n = A_n - A_{n-1}$. □

Theorem 6.1.24. *Dirichlet Test*
If the partial sums A_n of $\sum a_n$ are bounded, $b_n \downarrow 0$, and $b_n \ge 0$, then $\sum a_n b_n$ converges.

Proof. Choose M with $|A_n| \le M$. If $M = 0$, the result is immediate. Otherwise choose N with $b_N \le \varepsilon/(2M)$.
For $N \le p \le q$, $\left|\sum_{n=p}^{q} a_n b_n\right| \le M\left(\sum_{n=p}^{q-1}(b_n - b_{n+1}) + b_q + b_p\right) = 2Mb_p \le 2Mb_N \le \varepsilon$. Theorem 6.1.5 applies.
Structural Commentary: Bounded oscillation in A_n is damped by a monotone scalar tending to 0.
Key Insight: Summation by parts converts cancellation into a total variation estimate for b_n. □

Corollary 6.1.25. *Alternating Series Test*
Suppose $|c_1| \ge |c_2| \ge |c_3| \ge \cdots$, the signs alternate, and $c_n \to 0$. Then $\sum c_n$ converges.

Proof. After shifting indices to start at 0 if desired, apply Theorem 6.1.24 with $a_n = (-1)^{n+1}$ and $b_n = |c_n|$.
Structural Commentary: Alternation gives bounded sums; decreasing magnitudes damp them.
Key Insight: Alternating convergence is Dirichlet convergence in its simplest form. □

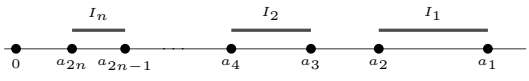

Figure 175: The intervals $I_j = [a_{2j}, a_{2j-1}]$ are disjoint inside $(0, a_1)$.

Perspective 6.1.26. *Alternating Partial Sums as Nested Control*
The interval picture explains the boundedness in Corollary 6.1.25: the even partial sums collect disjoint lengths inside a fixed interval, while the odd partial sums differ from the even ones by the next term, which tends to 0.

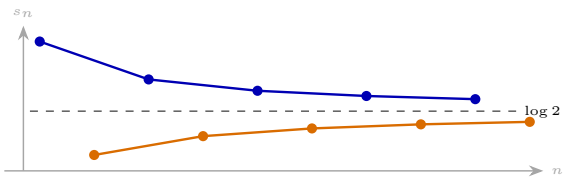

Figure 176: Odd partial sums (blue) decrease and even partial sums (orange) increase toward $\log 2$.

Theorem 6.1.27. *Absolute Convergence Implies Convergence*
If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

Proof. For $m \geq n$, $\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k|$. Apply Theorem 6.1.5 to $\sum |a_n|$.
Structural Commentary: Absolute convergence removes cancellation from the convergence question.
Key Insight: Small absolute tails force small signed or complex tails. □

Theorem 6.1.28. *Addition and Scalar Multiplication of Series*
If $\sum a_n = A$ and $\sum b_n = B$, then $\sum(a_n + b_n) = A + B$ and $\sum ca_n = cA$ for every fixed scalar c.

Proof. Apply Theorem 2.1.4 to the corresponding partial sums.
Structural Commentary: Linearity of convergent series is just limit algebra for partial sums.
Key Insight: Add series by adding their partial-sum sequences. □

Definition 6.1.29. *Cauchy Product*
Given $\sum a_n$, $\sum b_n$, define $c_n := \sum_{k=0}^{n} a_k b_{n-k}$, $(n = 0, 1, 2, \ldots)$. Series $\sum c_n$ is *Cauchy product* of given series.

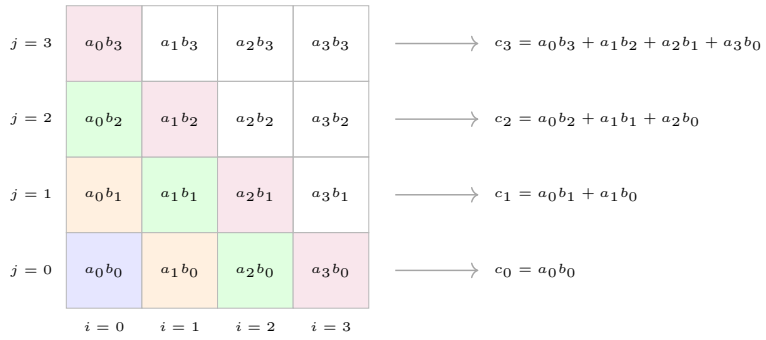

Figure 177: Diagonal sums of (a_ib_j) form the Cauchy product; finite rectangular sums are order-free.

Perspective 6.1.30. *Diagonal Ordering in a Product*
c_n is the diagonal sum of the products a_ib_j with $i + j = n$. Absolute convergence, as in Theorems 6.1.31 and 6.1.34, keeps the value independent of the accumulation order.

Theorem 6.1.31. *Mertens' Theorem for Cauchy Products*
Suppose $\sum a_n$ converges absolutely, $\sum a_n = A$, $\sum b_n = B$, and $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. Then $\sum_{n=0}^{\infty} c_n = AB$.

Proof. Let A_n, B_n, C_n be the partial sums and set $\beta_n := B_n - B$.
Then $C_n = A_nB + \gamma_n$, and $\gamma_n := a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$. Since $A_nB \to AB$, it suffices to show $\gamma_n \to 0$.
Put $\alpha := \sum |a_n|$. If $|\beta_j| \leq \varepsilon$ for $j \geq N$, then $|\gamma_n| \leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon\alpha$.
Let $n \to \infty$ with N fixed. The first term tends to 0, so $\limsup |\gamma_n| \leq \varepsilon\alpha$; let $\varepsilon \downarrow 0$.
Structural Commentary: Absolute convergence of one factor controls the convolution error.
Key Insight: One absolute factor makes the Cauchy product converge to AB. □

Definition 6.1.32. *Rearrangement of a Series*
If (k_n) lists each positive integer exactly once and $a'_n := a_{k_n}$, then $\sum a'_n$ is a *rearrangement* of $\sum a_n$.

Theorem 6.1.33. *Riemann Rearrangement Theorem*
Let $\sum a_n$ be a real series converging but not absolutely, and let $\alpha, \beta \in [-\infty, \infty]$ with $\alpha \leq \beta$.
Then there exists a bijection $\pi : \mathbb{N} \to \mathbb{N}$ such that, for $s'_N := \sum_{n=1}^{N} a_{\pi(n)}$:

(i) π rearranges every term of $\sum a_n$ exactly once.
(ii) $\liminf_{N\to\infty} s'_N = \alpha$.
(iii) $\limsup_{N\to\infty} s'_N = \beta$.

Proof. Write $a_n = p_n - q_n$, where $p_n := \max(a_n, 0)$ and $q_n := \max(-a_n, 0)$. Then $\sum p_n = \sum q_n = +\infty$ and $p_n, q_n \to 0$. List the nonzero p_n as $P_1, P_2, \dots$ and the nonzero q_n as $Q_1, Q_2, \dots$, preserving order.
Choose finite targets $\ell_N < u_N$ with $\ell_N \to \alpha$, $u_N \to \beta$, and $\ell_N < u_{N+1}$.

(i) Alternately add unused P terms until $s' > u_N$, then unused $-Q$ terms until $s' < \ell_N$.
Since $\sum P_j = \sum Q_j = +\infty$, all stops are finite and every nonzero term is used; insert zeros anywhere.
(ii) At the Nth down-crossing, $\ell_N - Q_j \le s'_k < \ell_N$ with $Q_j \to 0$. Thus lower crossing subsequences tend to α, and later sums stay above later ℓ_N up to $o(1)$; hence $\liminf s'_N = \alpha$.
(iii) At the Nth up-crossing, $u_N < s'_k \le u_N + P_j$ with $P_j \to 0$. Thus upper crossing subsequences tend to β, and later sums stay below later u_N up to $o(1)$; hence $\limsup s'_N = \beta$.

Structural Commentary: Divergent one-sided masses steer the partial sums.
Key Insight: Vanishing terms make the overshoot vanish. □

Theorem 6.1.34. *Absolutely Convergent Series Are Rearrangement-Invariant*
Every rearrangement of an absolutely convergent complex series converges to the same sum.

Proof. Let s_n, s'_n be the original and rearranged partial sums.
Given $\varepsilon > 0$, choose N so that $\sum_{i=N}^{m} |a_i| \le \varepsilon$ for $m \ge N$, then choose p so $a_1, \dots, a_N$ occur among $a'_1, \dots, a'_p$.
For $n > p$, $s_n - s'_n$ uses only tail terms, hence $|s_n - s'_n| \le \varepsilon$. Thus s'_n has the same limit as s_n.
Structural Commentary: Absolute tails control any finite reordering of the remaining terms.
Key Insight: Absolute convergence is exactly the protection needed against rearrangement. □

Definition 6.1.35. *Absolute Summation over Countable Sets*
Let X be countable, $f : X \to \mathbb{C}$, and $g : \mathbb{N} \to X$ be a bijection, $g_n := g(n)$.
If $\sum_{n=1}^{\infty} |f(g_n)| < \infty$, then $\sum_{x \in X} f(x)$ is *absolutely convergent*, and $\sum_{x \in X} f(x) := \sum_{n=1}^{\infty} f(g_n)$.

Theorem 6.1.36. *Countable Sums Are Order-Free Under Absolute Convergence*
Let X, Y be countable, let $f : X \to \mathbb{C}$ be absolutely summable, and let $h : Y \to X$ be a bijection.

(i) $f \circ h : Y \to \mathbb{C}$ *is absolutely summable.*
(ii) $\sum_{y \in Y} f(h(y)) = \sum_{x \in X} f(x)$.
(iii) Countable absolutely convergent sums are independent of enumeration.

Proof. Let $g : \mathbb{N} \to X$ and $k : \mathbb{N} \to Y$ be enumerations, and set $\sigma := g^{-1} \circ h \circ k : \mathbb{N} \to \mathbb{N}$.

(i) As σ is bijection, $h(k_n) = g_{\sigma(n)}$; $\sum_{n=1}^{\infty} |f(h(k_n))| = \sum_{n=1}^{\infty} \left|f(g_{\sigma(n)})\right| < \infty$ by absolute rearrangement invariance.
(ii) The same invariance gives $\sum_{y \in Y} f(h(y)) = \sum_{n=1}^{\infty} f(g_{\sigma(n)}) = \sum_{n=1}^{\infty} f(g_n) = \sum_{x \in X} f(x)$.
(iii) Take $Y = \mathbb{N}$ and let h be any enumeration of X.

Structural Commentary: Bijections only reorder an absolutely summable list.
Key Insight: Absolute tails make enumeration irrelevant. □

Theorem 6.1.37. *Fubini for Absolutely Summable Arrays*
Let $(a_{mn})_{m,n \ge 1} \subseteq \mathbb{C}$ and $L := \sum_{(m,n) \in \mathbb{N}^2} |a_{mn}| < \infty$. Define $S := \sum_{(m,n) \in \mathbb{N}^2} a_{mn}$ when summed absolutely over $\mathbb{N}^2$.

(i) Each row $\sum_{n=1}^{\infty} a_{mn}$ *and column* $\sum_{m=1}^{\infty} a_{mn}$ *converges absolutely.*
(ii) With $r_m := \sum_{n=1}^{\infty} a_{mn}$ *and* $c_n := \sum_{m=1}^{\infty} a_{mn}$, $\sum_m |r_m| \le L$ *and* $\sum_n |c_n| \le L$.
(iii) $\sum_{m=1}^{\infty} r_m = S = \sum_{n=1}^{\infty} c_n$.

Proof.

(i) For fixed m, N, $\sum_{n=1}^{N} |a_{mn}| \le L$; similarly $\sum_{m=1}^{M} |a_{mn}| \le L$. Hence every row, column absolutely summable.
(ii) For $M \ge 1$, $\sum_{m=1}^{M} |r_m| \le \sum_{m=1}^{M} \sum_{n=1}^{\infty} |a_{mn}| \le L$; let $M \to \infty$. The column estimate is identical.
(iii) Given $\varepsilon > 0$, choose finite $F \subseteq \mathbb{N}^2$ with $\sum_{\mathbb{N}^2 \setminus F} |a_{mn}| < \varepsilon$, then $R = \{1, \dots, M\} \times \{1, \dots, N\} \supseteq F$.
Each of S, $\sum_m r_m$, and $\sum_n c_n$ differs from $S_R := \sum_{(m,n) \in R} a_{mn}$ by at most $\sum_{\mathbb{N}^2 \setminus R} |a_{mn}| \le \varepsilon$. Let $\varepsilon \downarrow 0$.

Structural Commentary: Rectangles plus absolute tails justify Fubini.
Key Insight: Absolute summability makes order irrelevant. □

Method 6.1.38. *Series Proof Patterns*

(i) *Partial sums*: translate every assertion about $\sum a_n$ into an assertion about $s_n = \sum_{k=1}^{n} a_k$.
(ii) *Cauchy tails*: prove convergence by estimating $\left|\sum_{k=n}^{m} a_k\right|$ for $m \ge n$.
(iii) *Nonnegative terms*: replace convergence by boundedness of partial sums.
(iv) *Comparison*: dominate only the tail; finitely many early terms are irrelevant.
(v) *Absolute convergence*: use $\sum |a_n|$ when cancellation should not matter.
(vi) *Countable index sets*: choose a bijection $g : \mathbb{N} \to X$; use Theorem 6.1.36 so $\sum_{x \in X} a_x$ is independent of g.
(vii) *Double sums*: prove identities on $R_{M,N} := \{1, \dots, M\} \times \{1, \dots, N\}$; pass to ∞ arrays (Theorem 6.1.37).
(viii) *Conditional convergence*: order matters; use summation by parts or rearrangements.

Key Insight: Series arguments are sequence arguments with special notation for tails.

7 Uniform Convergence

7.1 Uniform Limits

Uniform convergence asks when limits of functions preserve structure. The key distinction is whether N may depend on x, which controls continuity, integration, differentiation, and the supremum metric.

Perspective 7.1.1. *Function Limits Before Function-Sequence Limits*
$\lim_{t\to x} f(t)$ denotes the metric-space function limit of Definition 5.1.4.
Uniform convergence is the hypothesis that lets the limits in n and t commute.

Definition 7.1.2. *Pointwise Limits of Functions*
Let $f_n : E \to (Y, d_Y)$.

(i) $f_n \to f$ *pointwise* on E if $d_Y(f_n(x), f(x)) \to 0$ for every $x \in E$.

(ii) If Y is scalar, $\sum f_n \to f$ *pointwise* on E if $s_N(x) := \sum_{n=1}^{N} f_n(x) \to f(x)$ for every $x \in E$.

No metric on E is required; topology matters only for preserving continuity, limits, or integrals.

Example 7.1.3. *Pointwise Limits Can Lose Structure*

(i) $s_{m,n} := m/(m+n)$ gives $\lim_n \lim_m s_{m,n} = 1$ but $\lim_m \lim_n s_{m,n} = 0$.
(ii) For $f_n(x) = x^2/(1+x^2)^n$, $\sum_{n=0}^{\infty} f_n$ equals 0 at 0 and $1 + x^2$ otherwise, by Theorem 6.1.9.
(iii) $f_n(x) = \sin(nx)/\sqrt{n}$ satisfies $f_n \to 0$ pointwise but $f_n'(0) = \sqrt{n} \to +\infty$.
(iv) $f_n(x) = n^2 x(1-x^2)^n$ on $[0,1]$ satisfies $f_n \to 0$ pointwise, while $\int_0^1 f_n\,dx = n^2/(2n+2) \to +\infty$.

Key Insight: Pointwise control can miss global spikes and oscillations.

Definition 7.1.4. *Uniform Convergence*
For $f_n, f : E \to (Y, d_Y)$, $f_n \rightrightarrows f$ on E if $\forall \varepsilon > 0\ \exists\ N \in \mathbb{N}$ such that $n \geq N,\ x \in E \implies d_Y(f_n(x), f(x)) \leq \varepsilon$.
In the scalar-valued case, take $d_Y(u,v) = |u - v|$; $\sum f_n$ converges uniformly on E if its partial sums do.

Perspective 7.1.5. *One Tail, All Points*
Pointwise convergence lets N depend on x; uniform convergence uses one N for all x.
Equivalently, late graphs lie in one ε-strip around f.

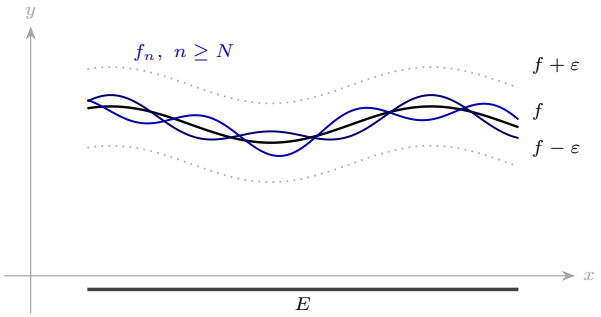

Figure 178: One index N puts all later graphs inside the same ε-strip around f on E.

Example 7.1.6. *x^n Converges Pointwise But Not Uniformly*
On $[0,1]$, let $g_n(x) = x^n$. Then $g_n \to g$ pointwise, where $g(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$.
The convergence is not uniform: for each n, some $x < 1$ has $|x^n - g(x)| = x^n > 1/2$.

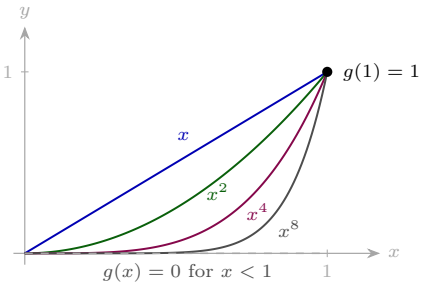

Figure 179: $x^n \to 0$ pointwise on $[0,1)$, but the endpoint prevents uniform convergence on $[0,1]$.

Theorem 7.1.7. *Cauchy Criterion for Uniform Convergence*
For complex-valued $f_n : E \to \mathbb{C}$, f_n converges uniformly on E if and only if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} : \quad m, n \geq N,\ x \in E \implies |f_n(x) - f_m(x)| \leq \varepsilon$$

Proof.
($\Rightarrow$) If $f_n \rightrightarrows f$, choose N so that $|f_k(x) - f(x)| \leq \varepsilon/2$ for $k \geq N$ and $x \in E$. Then $|f_n(x) - f_m(x)| \leq \varepsilon$.

($\Leftarrow$) For each x, $(f_n(x))$ is Cauchy in $\mathbb{C}$, so define $f(x) := \lim f_n(x)$.
Fix $n \geq N$ and let $m \to \infty$ in the criterion; then $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$, hence $f_n \rightrightarrows f$.

Structural Commentary: The same N gives Cauchy control simultaneously for every $x \in E$.
Key Insight: Uniform Cauchy tails are exactly uniform convergence. □

Theorem 7.1.8. *Supremum Test for Uniform Convergence*
Suppose $f_n(x) \to f(x)$ $\forall x \in E$, and $M_n := \sup_{x \in E} |f_n(x) - f(x)|$. Then $f_n \rightrightarrows f$ on E if and only if $M_n \to 0$.

Proof.
($\Rightarrow$) If $f_n \rightrightarrows f$, then $\forall \varepsilon > 0$ $\exists$ N such that $n \geq N$ implies $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$. Hence $M_n \leq \varepsilon$.
($\Leftarrow$) If $M_n \to 0$, then $\forall \varepsilon > 0$ $\exists$ N such that $n \geq N$ implies $M_n \leq \varepsilon$. Hence $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$.
Structural Commentary: Uniform convergence is convergence in the supremum error.
Key Insight: The pointwise errors must vanish as one global quantity. □

Theorem 7.1.9. *Weierstrass M-Test*
Let f_n be functions on E and suppose $|f_n(x)| \leq M_n$ for every $x \in E$.
If $\sum M_n$ converges, then $\sum f_n$ converges uniformly on E.

Proof. For $m \geq n$ and $x \in E$, $\left|\sum_{k=n}^{m} f_k(x)\right| \leq \sum_{k=n}^{m} M_k$.
Since $\sum M_k$ converges, Theorem 6.1.5 makes the right-hand side $< \varepsilon$ for all large m, n.
Thus the partial sums of $\sum f_k$ are uniformly Cauchy, so Theorem 7.1.7 applies.
Structural Commentary: Absolute numerical tails dominate functional tails.
Key Insight: A summable uniform majorant gives uniform convergence of a function series. □

Theorem 7.1.10. *Uniform Convergence Interchanges Limits*
Let E lie in a metric space, let x be a limit point of E, and suppose $f_n \rightrightarrows f$ on E.
If $A_n := \lim_{t \to x} f_n(t)$ exists for every n, then (A_n) converges and $\lim_{t \to x} f(t) = \lim_{n \to \infty} A_n$.

Proof. By Theorem 7.1.7, $\forall \varepsilon > 0$ $\exists$ N such that $m, n \geq N$ and $t \in E$ imply $|f_n(t) - f_m(t)| \leq \varepsilon$.
Letting $t \to x$ gives $|A_n - A_m| \leq \varepsilon$, hence $A_n \to A$ for some $A \in \mathbb{C}$.
For any n, $|f(t) - A| \leq |f(t) - f_n(t)| + |f_n(t) - A_n| + |A_n - A|$. Given $\varepsilon > 0$, choose n so $|f(t) - f_n(t)| < \varepsilon/3$ for all $t \in E$ and $|A_n - A| < \varepsilon/3$; then take t sufficiently close to x so $|f_n(t) - A_n| < \varepsilon/3$. Thus $\lim_{t \to x} f(t) = A$.
Structural Commentary: Uniform convergence controls the outer two terms in the three-term estimate.
Key Insight: Uniformity is precisely what allows the $n \to \infty$ limit to move past the $t \to x$ limit. □

Theorem 7.1.11. *Uniform Limits Preserve Continuity*
If (f_n) is a sequence of continuous functions on E and $f_n \rightrightarrows f$ on E, then f is continuous on E.

Proof. Let $x \in E$. If x is isolated, f is continuous at x.
If x is a limit point, then $\lim_{t \to x} f_n(t) = f_n(x)$ for all n, so Theorem 7.1.10 gives $\lim_{t \to x} f(t) = \lim_{n \to \infty} f_n(x) = f(x)$.
Structural Commentary: Uniform convergence passes continuity to the limit.
Key Insight: Uniform convergence protects continuity from the failure modes in Example 7.1.3. □

Theorem 7.1.12. *Dini's Monotone Uniform Convergence Theorem*
Let K be compact and let $f_n, f : K \to \mathbb{R}$ be continuous. If $f_n(x) \downarrow f(x)$ for every $x \in K$, then $f_n \rightrightarrows f$ on K.

Proof. Set $g_n := f_n - f$. Then g_n is continuous, $g_n \downarrow 0$, and $g_n \geq 0$. Given $\varepsilon > 0$, let $K_n := \{x \in K : g_n(x) \geq \varepsilon\}$. Then (K_n) is nested compact and $\bigcap_n K_n = \emptyset$, so Corollary 4.1.9 gives $K_N = \emptyset$ for some N.
Thus $0 \leq f_n(x) - f(x) < \varepsilon$ for all $x \in K$ and $n \geq N$.
Structural Commentary: Monotone convergence turns the bad sets into a nested compact sequence.
Key Insight: Compactness turns pointwise eventual escape from every bad set into a uniform escape. □

Definition 7.1.13. *The Supremum Metric on Bounded Continuous Functions*
Let X be a metric space.
(i) $C_b(X)$ is the set of bounded continuous maps $X \to \mathbb{C}$.
(ii) For $f \in C_b(X)$, $\|f\|_\infty := \sup_{x \in X} |f(x)|$.
(iii) The *supremum metric* on $C_b(X)$ is $d_\infty(f, g) := \|f - g\|_\infty$.
(iv) $f_n \to f$ in d_∞ iff $\|f_n - f\|_\infty \to 0$, i.e. iff $f_n \rightrightarrows f$ on X.
Key Insight: Uniform convergence is metric convergence in $C_b(X)$.

Perspective 7.1.14. *A Metric Space Whose Points Are Functions*
Uniform convergence is sequence convergence in the metric space $C_b(X)$.

Perspective 7.1.15. *Four Limit Notions*
The four limit notions are sequence limits, function limits, pointwise function limits, and uniform function limits. By Theorem 7.1.8, the last is ordinary sequence convergence in the supremum metric.

7.2 Interchange of Limits

Theorem 7.2.1. *Uniform Convergence and Riemann–Stieltjes Integration*
Let α be increasing on $[a,b]$. If $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ and $f_n \rightrightarrows f$, then $f \in \mathcal{R}(\alpha)$ and $\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha$.

Proof. Put $\varepsilon_n := \sup_{a\le x\le b} |f_n(x) - f(x)|$. Then $\varepsilon_n \to 0$ by Theorem 7.1.8, and $f_n - \varepsilon_n \le f \le f_n + \varepsilon_n$.
Upper and lower integrals of f are therefore squeezed between corresponding integrals of $f_n \pm \varepsilon_n$, and gap is at most $2\varepsilon_n[\alpha(b) - \alpha(a)]$.
Thus $f \in \mathcal{R}(\alpha)$. Same estimate gives $\left|\int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha\right| \le \varepsilon_n[\alpha(b) - \alpha(a)] \to 0$.
Structural Commentary: Uniform error becomes an integral error multiplied by the total α-variation.
Key Insight: Uniform convergence is strong enough to pass limits through the Riemann–Stieltjes integral. □

Corollary 7.2.2. *Termwise Integration of Uniformly Convergent Series*
If $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ and $\sum f_n$ converges uniformly to f, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.

Proof. Let $s_N := \sum_{n=1}^{N} f_n$. By linearity, $s_N \in \mathcal{R}(\alpha)$ and $\int_a^b s_N\,d\alpha = \sum_{n=1}^{N} \int_a^b f_n\,d\alpha$.
Since $s_N \rightrightarrows f$, Theorem 7.2.1 gives $\int_a^b f\,d\alpha = \lim_{N\to\infty} \int_a^b s_N\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.
Structural Commentary: Reduce the series to its uniformly convergent partial sums.
Key Insight: Uniform convergence lets integration be performed term by term. □

Theorem 7.2.3. *Differentiation Under a Riemann–Stieltjes Integral*
Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, let $\varphi : [a,b] \times [c,d] \to \mathbb{R}$, write $\varphi^t(x) := \varphi(x,t)$, and fix $s \in (c,d)$.
Assume $\varphi^t \in \mathcal{R}(\alpha)$ for every $t \in [c,d]$, and for some $\eta > 0$ with $(s-\eta, s+\eta) \subseteq (c,d)$, $D_2\varphi$ exists on $[a,b] \times (s-\eta, s+\eta)$ and $(D_2\varphi)^t \rightrightarrows (D_2\varphi)^s$ on $[a,b]$ as $t \to s$.
If $F(t) := \int_a^b \varphi(x,t)\,d\alpha(x)$, then $(D_2\varphi)^s \in \mathcal{R}(\alpha)$ and $F'(s) = \int_a^b (D_2\varphi)(x,s)\,d\alpha(x)$.

Proof. For $t \ne s$, set $\psi_t := (\varphi^t - \varphi^s)/(t-s) \in \mathcal{R}(\alpha)$.
By Corollary 1.1.13, $\psi_t(x) = D_2\varphi(x, \tau_x(t))$ for some $\tau_x(t)$ between s and t; hence $\psi_t \rightrightarrows (D_2\varphi)^s$.
Theorem 7.2.1 gives $(D_2\varphi)^s \in \mathcal{R}(\alpha)$ and $\frac{F(t)-F(s)}{t-s} = \int_a^b \psi_t\,d\alpha \to \int_a^b (D_2\varphi)^s\,d\alpha$, so $F'(s) = \int_a^b (D_2\varphi)(x,s)\,d\alpha(x)$.
Structural Commentary: Integrable difference quotients converge uniformly.
Key Insight: Uniform convergence permits $D_t \int = \int D_t$. □

Theorem 7.2.4. *Uniform Convergence and Differentiation*
Let f_n be differentiable on $[a,b]$, suppose $f_n(x_0)$ converges for some $x_0 \in [a,b]$, and suppose $f_n' \rightrightarrows g$ on $[a,b]$. Then $f_n \rightrightarrows f$ for some f, and $f' = g$ on $[a,b]$ (one-sided at endpoints).

Proof. For m, n, MVT 1.1.13 gives $|f_n(x) - f_m(x)| \le |f_n(x_0) - f_m(x_0)| + (b-a) \sup_{t\in[a,b]} |f_n'(t) - f_m'(t)|$.
Thus (f_n) is uniformly Cauchy, so $f_n \rightrightarrows f$. Fix x and set $\phi_n(t) := \frac{f_n(t)-f_n(x)}{t-x}$, $\phi(t) := \frac{f(t)-f(x)}{t-x}$.
For $t \ne x$, the same estimate gives $|\phi_n(t) - \phi_m(t)| \le \|f_n' - f_m'\|_\infty$, hence $\phi_n \rightrightarrows \phi$.
By Theorem 7.1.10, $f'(x) = \lim_{t\to x} \phi(t) = \lim_{n\to\infty} f_n'(x) = g(x)$.
Structural Commentary: Derivative control anchors functions, then quotients.
Key Insight: Uniform derivative convergence plus one anchor passes derivatives to the limit. □

7.3 Further Topics in Uniform Convergence

Method 7.3.1. *Uniform Convergence Proof Patterns*

(i) *Pointwise*: fix x, then choose $N = N(x,\varepsilon)$.
(ii) *Uniform*: choose $N = N(\varepsilon)$; equivalently $\sup_E |f_n - f| \to 0$.
(iii) *Uniform Cauchy*: prove $\sup_E |f_n - f_m| \to 0$ as $m, n \to \infty$.
(iv) *Preservation*: use a three-term estimate and choose the uniform index first.
(v) *Series of functions*: apply Weierstrass M-test when numerical absolute tails dominate all functional tails.
(vi) *Termwise operations*: require $f_n \to f$ locally uniformly, i.e. $\sup_{x\in K} |f_n - f| \to 0$ for each compact K, before $\lim_n \int_K f_n = \int_K f$; for derivatives, require $f_n' \to g$ locally uniformly and get $(\lim_n f_n)' = g$.

Key Insight: Uniformity means N is chosen before x.

8 Function-Space Compactness

8.1 Equicontinuity

Function-space compactness treats functions as points. Supremum metrics preview Banach spaces, Baire category describes generic behavior, and equicontinuity leads to Arzela–Ascoli.

Theorem 8.1.1. *Completeness of the Supremum Metric*
The metric space $C_b(X)$ with metric $d(f,g) = \|f - g\|_\infty$ is complete.

Proof. Let (f_n) be Cauchy in $C_b(X)$. Then Theorem 7.1.7 gives a function f with $f_n \rightrightarrows f$. By Theorem 7.1.11, f is continuous. For some n, $\|f - f_n\|_\infty < 1$, hence f_n is bounded, thus $f \in C_b(X)$ and $\|f_n - f\|_\infty \to 0$.
Structural Commentary: Uniform Cauchy convergence builds the limit inside the same function space.
Key Insight: Completeness of $\mathbb{C}$ plus uniform convergence makes $C_b(X)$ complete. □

Definition 8.1.2. *Bounded Target Metric for All Functions*
For metric spaces (X, d_X) and (Y, d_Y), let $\mathcal{F} = Y^X$ and set $\rho(y,z) := \frac{d_Y(y,z)}{1+d_Y(y,z)}$ and $d_\rho(f,g) := \sup_{x\in X} \rho(f(x), g(x))$.
Then $\rho < 1$, so d_ρ is finite on all of $\mathcal{F}$; d_ρ-convergence is uniform convergence measured by d_Y.

Theorem 8.1.3. *Completeness and Closed Subspaces for d_ρ*
If the target metric space (Y, d_Y) is complete, then:

(i) $(\mathcal{F}, d_\rho)$ *is complete;*
(ii) $\mathcal{F}_b := \{f \in \mathcal{F} : f(X)$ *is* d_Y*-bounded*$\}$ *is closed in* $\mathcal{F}$*;*
(iii) $C(X, Y)$ *is closed in* $\mathcal{F}$*.*

Proof.

(i) Let (f_n) be d_ρ-Cauchy. For each $x \in X$, $(f_n(x))$ is Cauchy in Y; define $f(x) := \lim_n f_n(x)$.
Given $\varepsilon > 0$, choose N with $d_\rho(f_n, f_m) < \varepsilon$ for $m, n \geq N$. Letting $m \to \infty$ gives $d_\rho(f_n, f) \leq \varepsilon$ for $n \geq N$.
(ii) If $f_n \in \mathcal{F}_b$ and $d_\rho(f_n, f) \to 0$, choose N with $d_\rho(f_N, f) < 1/2$.
Then $d_Y(f_N(x), f(x)) < 1$ for all x. Since $f_N(X)$ is bounded, so is $f(X)$.
(iii) If $f_n \in C(X, Y)$ and $d_\rho(f_n, f) \to 0$, then $f_n \to f$ uniformly in d_Y.
Given $x_0 \in X$ and $\varepsilon > 0$, choose N with $d_Y(f_N(x), f(x)) < \varepsilon/3$ for all x, then use continuity of f_N at x_0 and the triangle inequality. Thus f is continuous.

Structural Commentary: Compress the target metric; then close the useful subspaces.
Key Insight: Completeness and uniform control make boundedness and continuity closed. □

Definition 8.1.4. *Boundedness and Equicontinuity of Function Families*
Let $\mathcal{F}$ be a family of functions from E into a metric space (Y, d_Y); in scalar cases, take $d_Y(u, v) = |u - v|$.

(i) (f_n) is *pointwise bounded* on E if $(f_n(x))$ is bounded in Y for each $x \in E$.
(ii) A scalar sequence (f_n) is *uniformly bounded* on E if $\exists M$ such that $|f_n(x)| < M$ for all $x \in E, n \in \mathbb{N}$.
(iii) If E lies in a metric space (X, d_X), then $\mathcal{F}$ is *equicontinuous* on E if $\forall \varepsilon > 0\ \exists \delta > 0$, such that $d_X(x, y) < \delta,\ x, y \in E,\ f \in \mathcal{F} \implies d_Y(f(x), f(y)) < \varepsilon$.
(iv) The family is *pointwise equicontinuous* if for every $p \in E$ and $\varepsilon > 0$ there is $\delta = \delta(p, \varepsilon) > 0$ such that $d_X(p, y) < \delta,\ y \in E,\ f \in \mathcal{F}$ implies $d_Y(f(p), f(y)) < \varepsilon$.

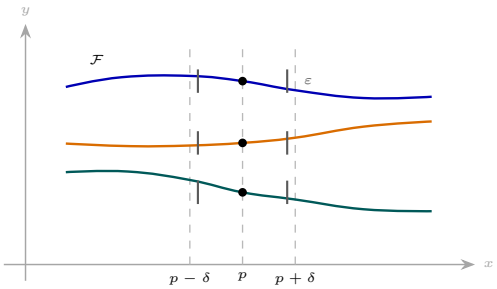

Figure 180: One input scale δ controls every curve in $\mathcal{F}$ within tolerance ε.

Theorem 8.1.5. *Pointwise Equicontinuity Becomes Uniform on Compact Sets*
Every pointwise equicontinuous family $\mathcal{F} \subset C(K, \mathbb{C})$ on compact K is equicontinuous.

Proof. Otherwise, choose $\varepsilon > 0$, $f_n \in \mathcal{F}$, and $x_n, t_n \in K$ with $d(x_n, t_n) < 1/n$ but $|f_n(x_n) - f_n(t_n)| \geq \varepsilon$.
By Theorem 4.1.12, pass to $x_n \to x_0$; then $t_n \to x_0$ too.
Pointwise equicontinuity at x_0 gives $|f_n(x_n) - f_n(t_n)| < \varepsilon$ for large n, a contradiction.
Structural Commentary: Compactness turns moving counterexamples into one base point.
Key Insight: The local $\delta(x, \varepsilon)$ becomes one uniform δ only after compactness prevents escape. □

Theorem 8.1.6. *Uniform Convergence Implies Equicontinuity on Compact Sets*
If K is compact, $f_n \in C_b(K)$, and $f_n \rightrightarrows f$ on K, then $\{f_n : n \in \mathbb{N}\}$ is equicontinuous.

Proof. Choose N so $\|f_n - f_N\|_\infty < \varepsilon$ for $n > N$.
By Theorem 5.2.7, choose one $\delta > 0$ giving ε-control for $f_1, \ldots, f_N$.
If $n > N$, $d(x, y) < \delta$, then $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < 3\varepsilon$.
Together with the finite initial family, $\{f_n\}$ is equicontinuous.
Structural Commentary: A uniformly convergent tail stays close to one uniformly continuous function.
Key Insight: Compactness handles finitely many functions; uniform convergence handles the tail. □

Example 8.1.7. *Uniform Boundedness Alone Is Not Enough*
On $[0, 1]$, let $f_n(x) := \frac{x^2}{x^2 + (1 - nx)^2}$. Then $|f_n| \leq 1$ and $f_n \to 0$ pointwise, but $f_n(1/n) = 1$; hence no uniformly convergent subsequence.
The missing hypothesis is equicontinuity.

Theorem 8.1.8. *Diagonal Selection on a Countable Set*
Every pointwise bounded sequence (f_n) on countable E has a subsequence converging pointwise on E.

Proof. List $E = \{x_1, x_2, \ldots\}$. Inductively choose nested subsequences S_i so that $S_i(x_i)$ converges.
The diagonal subsequence is eventually contained in each S_i, hence converges at every x_i.
Structural Commentary: The diagonal keeps the ith convergence property after the ith step.
Key Insight: Countability permits diagonal convergence. □

8.2 Arzela-Ascoli-Type Compactness

Theorem 8.2.1. *Arzela–Ascoli Selection Theorem*
Let K be a compact metric space and let $(f_n) \subset C_b(K)$.
If $\sup_n |f_n(x)| < \infty$ for each $x \in K$ and (f_n) is equicontinuous on K, then:

(i) (f_n) is uniformly bounded on K;
(ii) (f_n) contains a uniformly convergent subsequence.

Proof.

(i) Choose $\delta > 0$ so $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < 1$ $\forall n$. Let $K \subseteq \bigcup_{j=1}^{r} B_\delta(p_j)$, $M_j := \sup_n |f_n(p_j)| < \infty$.
If $x \in B_\delta(p_j)$, then $|f_n(x)| < 1 + M_j$; hence $\sup_{n,x\in K} |f_n(x)| \le 1 + \max_j M_j$.
(ii) Let $E \subset K$ be countable dense, obtained from finite $1/m$-nets. Pointwise boundedness on E and diagonal selection give a subsequence (g_i) converging pointwise on E. Given $\varepsilon > 0$, choose $\delta > 0$ for equicontinuity with tolerance $\varepsilon/3$ and $x_1, \dots, x_r \in E$ with $K \subseteq \bigcup_s B_\delta(x_s)$. For large i, j, $|g_i(x_s) - g_j(x_s)| < \varepsilon/3$ $\forall s$.
If $d(x, x_s) < \delta$, then $|g_i(x) - g_j(x)| < \varepsilon$, so (g_i) is uniformly Cauchy and hence uniformly convergent.

Structural Commentary: Dense convergence plus equicontinuity becomes uniform convergence.
Key Insight: Equicontinuity turns finite sampling into uniform control. □

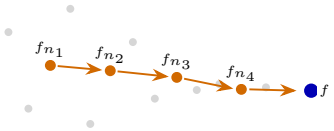

Figure 181: Arzela–Ascoli selects a uniformly convergent subsequence.

Theorem 8.2.2. *Compact Subsets of $C(K)$*
Let K be a compact metric space and let $C(K, \mathbb{R})$ carry the supremum metric.
For $\mathcal{F} \subseteq C(K, \mathbb{R})$, the following are equivalent:

(i) $\mathcal{F}$ is compact.
(ii) $\mathcal{F}$ is closed, uniformly bounded, and equicontinuous.

Proof.

(i) $\Rightarrow$ (ii) Compactness gives closedness and total boundedness in $\|\cdot\|_\infty$. If $\mathcal{F} \neq \emptyset$, choose $g \in \mathcal{F}$; boundedness follows from $\|f\|_\infty \le \|f - g\|_\infty + \|g\|_\infty$. Given $\varepsilon > 0$, take a finite $\varepsilon/3$-net $g_1, \dots, g_r \in \mathcal{F}$.
For $d(x, y) < \delta$, chosen so $|g_i(x) - g_i(y)| < \frac{\varepsilon}{3}$ $\forall i$, any f with $\|f - g_i\|_\infty < \frac{\varepsilon}{3}$ satisfies $|f(x) - f(y)| < \varepsilon$.
(ii) $\Rightarrow$ (i) Closedness makes $\mathcal{F}$ complete. For total boundedness, fix $\varepsilon > 0$, choose δ from equicontinuity at $\varepsilon/3$, and a finite δ-net $x_1, \dots, x_r \subset K$. With $|f| \le M$ on K for $f \in \mathcal{F}$, partition $[-M, M]$ into intervals of length $< \varepsilon/3$ and classify f by the interval data of $(f(x_1), \dots, f(x_r))$. For one representative g per class and $d(x, x_i) < \delta$, $|f(x) - g(x)| \le |f(x) - f(x_i)| + |f(x_i) - g(x_i)| + |g(x_i) - g(x)| < \varepsilon$. Hence $\mathcal{F}$ is totally bounded, so complete plus totally bounded implies compact.

Structural Commentary: Compactness in $C(K)$ is finite-resolution control plus closedness.
Key Insight: Equicontinuity lets finitely many sample values control whole functions. □

Example 8.2.3. *Hölder Balls Have Compact Closure*
For $0 < \alpha \le 1$, define $[f]_\alpha := \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}$ when finite.
Set $\mathcal{H}_\alpha := \{f \in C([0, 1], \mathbb{R}) : \|f\|_\infty + [f]_\alpha \le 1\}$, with closure taken in $C([0, 1], \mathbb{R})$.

(i) If $f \in \mathcal{H}_\alpha$, then $\|f\|_\infty \le 1$.
(ii) If $f \in \mathcal{H}_\alpha$, then $|f(x) - f(y)| \le |x - y|^\alpha$, so $\mathcal{H}_\alpha$ is equicontinuous.
(iii) Uniform limits preserve $\|f\|_\infty \le 1$ and $|f(x) - f(y)| \le |x - y|^\alpha$.

Thus $\overline{\mathcal{H}_\alpha}$ is closed, uniformly bounded, and equicontinuous; hence it is compact.

Definition 8.2.4. *σ-Compact Metric Space*
A metric space X is *σ-compact* if $X = \bigcup_{j=1}^{\infty} K_j$ for compact subsets $K_j \subseteq X$.

Theorem 8.2.5. *Arzela–Ascoli on σ-Compact Spaces*
Let X be a σ-compact metric space, Y a metric space, $f_n : X \to Y$ pointwise equicontinuous. If $\overline{\{f_n(x) : n \in \mathbb{N}\}}$ is compact in Y $\forall x \in X$, then some subsequence (g_m) converges uniformly on every compact $K \subseteq X$.

Proof. Choose an increasing compact exhaustion $X = \bigcup_j K_j$.

(i) On each K_j, the restrictions $(f_n|_{K_j})$ are uniformly equicontinuous and pointwise relatively compact in Y.
(ii) Choose subsequences successively so the jth one converges uniformly on K_j.
The diagonal g_m satisfies $g_m|_{K_j} \rightrightarrows g|_{K_j}$ for every j.
(iii) For compact $K \subseteq X$, $(g_m|_K)$ is uniformly equicontinuous and $g_m \to g$ pointwise.
Given $\varepsilon > 0$, take a finite δ-net $E \subset K$ with equicontinuity tolerance $\varepsilon/3$.
Convergence on E gives $\sup_{x\in K} d_Y(g_m(x), g_n(x)) < \varepsilon$ for large m, n, hence $g_m \rightrightarrows g$ on K.

Structural Commentary: Exhaust, select, diagonalize.
Key Insight: Compact-local control gives compact-local uniform convergence. □

Corollary 8.2.6. *Real-Line Local Uniform Subsequence*
Let $f_n : \mathbb{R} \to \mathbb{R}$ be pointwise equicontinuous and suppose $\sup_n |f_n(x_0)| < \infty$ for some $x_0 \in \mathbb{R}$.
Then some subsequence converges uniformly on every compact $K \subseteq \mathbb{R}$.

Proof. For $x \in \mathbb{R}$, take compact $I \supset \{x_0, x\}$. Uniform equicontinuity on I gives $\delta > 0$ with step-size tolerance 1.
Choose a finite δ-chain $x_0 = t_0, \dots, t_m = x$; then $|f_n(x)| \le |f_n(x_0)| + m$ for all n.
Thus each orbit $\{f_n(x) : n \in \mathbb{N}\}$ has compact closure in $\mathbb{R}$, and the σ-compact theorem applies.
Structural Commentary: One bounded orbit propagates by finite chains.
Key Insight: Equicontinuity turns one anchor into compact-local control. □

8.3 Further Topics in Function-Space Compactness

Method 8.3.1. *Function-Space Compactness Proof Patterns*

(i) *Completeness*: prove a function sequence is uniformly Cauchy, build the pointwise limit, and use uniform convergence to keep continuity or boundedness.
(ii) *Baire category*: show a desired property contains a countable intersection of dense open subsets of a complete function space.
(iii) *Compactness/equicontinuity*: pass from finite nets to all of K.
(iv) *Diagonal selection*: get pointwise convergence on a countable dense set, then upgrade using equicontinuity.
(v) *Locally uniform*: exhaust a σ-compact domain by compact sets and diagonalize.

Key Insight: Compactness in function spaces is finite-net control plus equicontinuity.

9 Power Series and Approximation

9.1 Power Series

Power series are controlled infinite polynomials, and approximation is uniform closure. This subsection covers radii, Taylor structure, Abel behavior, Weierstrass, Stone–Weierstrass, and elementary analytic functions.

Definition 9.1.1. *Core Objects: Power Series, Approximation, and Gamma*

(i) A real *power series about a* is $\sum_{n=0}^{\infty} c_n (x-a)^n$; a local representation of f by one makes f *analytic* at a.
(ii) A *polynomial approximant* to f on a compact set K is a polynomial P chosen so that $\|f - P\|_{\infty,K}$ is small.
(iii) For $x > 0$, the *gamma function* is $\Gamma(x) := \int_0^\infty t^{x-1} e^{-t}\, dt$ when the improper integral converges.

Key Insight: Power series are local analytic objects; approximation theorems turn algebraic families into dense function families; Γ extends factorials by integration.

Definition 9.1.2. *Power Series and Radius of Convergence*
Let $(c_n)_{n\ge 0} \subseteq \mathbb{C}$ and $\alpha := \limsup_{n\to\infty} \sqrt[n]{|c_n|} \in [0, +\infty]$.

(i) The series $\sum_{n=0}^{\infty} c_n z^n$ is a *power series*.
(ii) The *radius of convergence* is $R := \alpha^{-1} \in [0, +\infty]$, with $0^{-1} = +\infty$ and $(+\infty)^{-1} = 0$.

Theorem 9.1.3. *Convergence Inside the Circle*
For $\sum c_n z^n$, radius R: converge if $|z| < R$, diverge if $|z| > R$; boundary $|z| = R$ is undecided.

Proof. For $a_n = c_n z^n$, $\limsup_{n\to\infty} \sqrt[n]{|a_n|} = |z| \limsup_{n\to\infty} \sqrt[n]{|c_n|} = \frac{|z|}{R}$, with extended meaning if $R = 0$ or $R = +\infty$.
If $|z| < R$, this value is < 1; if $|z| > R$, it is > 1. Apply Theorem 6.1.20.
Structural Commentary: Radius controls the interior and exterior, not the boundary.
Key Insight: The root test is the radius-of-convergence theorem in disguise. □

Theorem 9.1.4. *A Boundary Test for Power Series*
If $\sum c_n z^n$ has radius 1, $c_n \downarrow 0$, and $c_n \ge 0$, then it converges for $|z| = 1$, except possibly at $z = 1$.

Proof. For $|z| = 1$ and $z \ne 1$, put $a_n = z^n$ and $b_n = c_n$.
The partial sums of $\sum z^n$ are bounded because $\left|\sum_{m=0}^{n} z^m\right| = \left|\frac{1-z^{n+1}}{1-z}\right| \le \frac{2}{|1-z|}$. Theorem 6.1.24 applies.
Structural Commentary: Boundary convergence away from $z = 1$ comes from bounded geometric partial sums.
Key Insight: Monotone coefficients can give conditional boundary convergence. □

Theorem 9.1.5. *Abel's Endpoint Theorem*
Let $s_n := \sum_{k=0}^{n} c_k \to s$. For $|x| < 1$, set $f(x) := \sum_{n\ge 0} c_n x^n$. Then $\lim_{x\to 1^-} f(x) = s$.

Proof. Set $s_{-1} := 0$. Since $c_n = s_n - s_{n-1}$, for $|x| < 1$, $f(x) = \sum_{n\ge 0} (s_n - s_{n-1}) x^n = (1-x) \sum_{n\ge 0} s_n x^n$.
Thus $f(x) - s = (1-x) \sum_{n\ge 0} (s_n - s) x^n$. Given $\varepsilon > 0$, choose N with $|s_n - s| < \varepsilon/2$ for $n > N$.
For $0 < x < 1$, $|f(x) - s| \le (1-x) \sum_{n=0}^{N} |s_n - s|\, x^n + \frac{\varepsilon}{2}(1-x) \sum_{n>N} x^n$.
The first term tends to 0 as $x \to 1^-$, and the second is $\le \varepsilon/2$; hence $f(x) \to s$.
Structural Commentary: Endpoint convergence is one-sided Abel continuity.
Key Insight: $(1-x)\sum s_n x^n$ is a geometric average of partial sums. □

Theorem 9.1.6. *Abel Product Theorem for Cauchy Products*
Let $\sum a_n = A$ and $\sum b_n = B$. Define $c_n := \sum_{k=0}^{n} a_k b_{n-k}$, and suppose $\sum c_n = C$. Then $C = AB$.

Proof. For $|x|<1$, set $f(x):=\sum_{n\geq 0}a_nx^n$, $g(x):=\sum_{n\geq 0}b_nx^n$, and $h(x):=\sum_{n\geq 0}c_nx^n$. Since $(a_n),(b_n),(c_n)$ are bounded, these series converge absolutely on $|x|<1$. Hence $f(x)g(x)=\sum_{n\geq 0}\big(\sum_{k=0}^{n}a_kb_{n-k}\big)x^n=\sum_{n\geq 0}c_nx^n=h(x)$.

By Abel's endpoint thm, $f(x)\to A$, $g(x)\to B$, $h(x)\to C$ as $x\to 1^-$;$C=\lim_{x\to 1^-}h(x)=\lim_{x\to 1^-}f(x)g(x)=AB$.

Structural Commentary: Regularize by x, multiply inside, pass to $x=1$.
Key Insight: Abel moves the safe product identity to the endpoint. □

Perspective 9.1.7. *Centered Series Are Translated Series*
$\sum c_n(x-a)^n$ is $\sum c_ny^n$ with $y=x-a$ and radius R: absolute convergence for $|x-a|<R$, divergence for $|x-a|>R$, and separate endpoint tests for $|x-a|=R$.

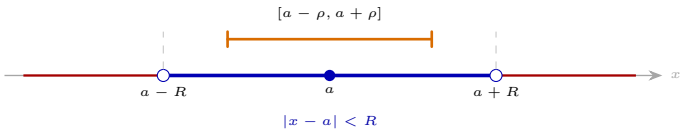

Figure 182: Inside $|x-a|<R$: absolute and locally uniform; outside divergence; endpoints undecided.

Definition 9.1.8. *Real Analytic at a Point*
Let $E\subseteq\mathbb{R}$, $f:E\to\mathbb{R}$, $a\in\operatorname{int}E$. The function f is *real analytic at a* if $\exists r>0$ and $(c_n)_{n\geq 0}\subseteq\mathbb{R}$ such that:

(i) $(a-r,a+r)\subseteq E$.
(ii) The power series $\sum c_n(x-a)^n$ has radius of convergence $\geq r$.
(iii) $f(x)=\sum_{n=0}^{\infty}c_n(x-a)^n$ for all $x\in(a-r,a+r)$.

If E is open and f is real analytic at every $a\in E$, then f is *real analytic on E*. Analytic functions form the class C^ω.

Theorem 9.1.9. *Power Series May Be Differentiated Term by Term*
Let $f(x)=\sum_{n\geq 0}c_nx^n$ have radius $R>0$, and set $I_R:=(-R,R)$, $K_\rho:=[-\rho,\rho]$ for $0<\rho<R$.

(i) $\sum c_nx^n$ converges uniformly on each K_ρ, and $f\in C^\infty(I_R)$.
(ii) For $k\geq 1$ and $x\in I_R$, $f^{(k)}(x)=\sum_{n\geq k}\frac{n!}{(n-k)!}c_nx^{n-k}$; in particular, $f^{(k)}(0)=k!c_k$.

Proof.

(i) If $0<\rho<R$, then $\sum|c_n|\,\rho^n<\infty$; hence $|c_nx^n|\leq|c_n|\,\rho^n$ on K_ρ, so $\sum c_nx^n$ converges uniformly there.
(ii) For fixed k, $\frac{n!}{(n-k)!}\sim n^k$, so $\limsup\sqrt[n]{\left|\frac{n!}{(n-k)!}c_n\right|}=\limsup\sqrt[n]{|c_n|}$.
Hence every derivative series has radius R and converges uniformly on each K_ρ.
Iterating termwise differentiation gives the formula and $f^{(k)}(0)=k!c_k$.

Structural Commentary: Work on $K_\rho\Subset I_R$ to justify termwise operations.
Key Insight: Polynomial coefficient growth does not change the radius. □

Corollary 9.1.10. *Termwise Calculus for Centered Power Series*
Let $f(x)=\sum_{n=0}^{\infty}c_n(x-a)^n$ have radius $R>0$.

(i) The series converges locally uniformly on $(a-R,a+R)$ and defines a continuous function.
(ii) For $|x-a|<R$, $f'(x)=\sum_{n=1}^{\infty}nc_n(x-a)^{n-1}$, locally uniformly.
(iii) If $[y,z]\subset(a-R,a+R)$, then $\int_y^z f(x)\,dx=\sum_{n=0}^{\infty}c_n\frac{(z-a)^{n+1}-(y-a)^{n+1}}{n+1}$.

Proof. Set $g(u):=\sum c_nu^n$ and $T(x):=x-a$, so $f=g\circ T$ on $(a-R,a+R)$.

(i) Local uniform convergence of $\sum c_nu^n$ on $(-R,R)$ implies local uniform convergence of $\sum c_n(Tx)^n$ on $(a-R,a+R)$; continuity follows from uniform limits.
(ii) Since $T'(x)=1$, $f'(x)=g'(Tx)=\sum_{n=1}^{\infty}nc_n(Tx)^{n-1}$, with local uniform convergence.
(iii) If $[y,z]\subset(a-R,a+R)$, choose $\rho<R$ with $[y,z]\subset[a-\rho,a+\rho]$.
Uniform convergence on $[a-\rho,a+\rho]$ gives $\int_y^z f=\lim_N\int_y^z\sum_{n=0}^{N}c_n(Tx)^n\,dx$, yielding the stated formula.

Structural Commentary: Work on compact intervals strictly inside the radius.
Key Insight: Translation reduces centered series to ordinary power series. □

Corollary 9.1.11. *Analytic Functions Are Their Taylor Series*
Suppose $f(x)=\sum_{n=0}^{\infty}c_n(x-a)^n$ for $|x-a|<r$, with $r>0$.

(i) $f\in C^\infty(a-r,a+r)$.
(ii) $c_k=f^{(k)}(a)/k!$ for every $k\geq 0$.
(iii) The power-series expansion of f about a is unique.
(iv) $f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$ for $|x-a|<r$.

Proof.

(i) Termwise differentiation gives $f \in C^\infty(a-r, a+r)$.
(ii) For $k \geq 0$, $f^{(k)}(x) = \sum_{n \geq k} \frac{n!}{(n-k)!} c_n (x-a)^{n-k}$; at $x = a$, $f^{(k)}(a) = k! c_k$.
(iii) If $f = \sum b_n (x-a)^n$ also, then $b_k = f^{(k)}(a)/k! = c_k$ for all k.
(iv) Replace c_n by $f^{(n)}(a)/n!$ in $f(x) = \sum c_n (x-a)^n$.

Structural Commentary: Derivatives determine every coefficient.
Key Insight: Analytic representation forces its own Taylor series. □

Theorem 9.1.12. *Re-Expansion and Identity for Power Series*
Let $f(x) = \sum_{n \geq 0} c_n x^n$ have radius $R > 0$, and set $S = (-R, R)$.

(i) If $a \in S$, then $f(x) = \sum_{m \geq 0} \frac{f^{(m)}(a)}{m!} (x-a)^m$ for $|x-a| < R - |a|$.
(ii) Let $F(x) = \sum_{n \geq 0} a_n x^n$ and $G(x) = \sum_{n \geq 0} b_n x^n$ converge on S.
If $\mathcal{Z}_{F,G} := \{x \in S : F(x) = G(x)\}$ has a limit point in S, then $a_n = b_n$ for all $n \geq 0$, hence $F = G$ on S.

Proof.

(i) Put $\rho := |x-a| + |a| < R$. Since $\sum_{n \geq 0} |c_n| \rho^n < \infty$, the binomial double sum is absolutely rearrangeable:
$f(x) = \sum_{m \geq 0} A_m (x-a)^m$, where $A_m := \sum_{n \geq m} \binom{n}{m} c_n a^{n-m}$. Also $f^{(m)}(a) = \sum_{n \geq m} \frac{n!}{(n-m)!} c_n a^{n-m} = m! A_m$.
(ii) Set $H := F - G = \sum_{n \geq 0} d_n x^n$, $d_n := a_n - b_n$, $Z := \{x \in S : H(x) = 0\}$, and $A := Z' \cap S$.
For $x_0 \in A$, (i) gives $H(x) = \sum e_n (x-x_0)^n$ near x_0.
If $k = \min\{n : e_n \neq 0\}$ existed, then $H = (x-x_0)^k g$ with $g(x_0) = e_k \neq 0$, so x_0 would be isolated in Z, contradiction. Thus all $e_n = 0$, so $H = 0$ near x_0 and A is open. Since $S \setminus A$ is open, $A \neq \emptyset$, and S is connected, $A = S$. As Z is closed in S, $Z = S$; hence $d_n = H^{(n)}(0)/n! = 0$.

Structural Commentary: Re-expand locally; connectedness propagates zeros.
Key Insight: Accumulating zeros force every local coefficient to vanish. □

Perspective 9.1.13. *Regularity Hierarchy*
The inclusions are $C^\omega \subseteq C^\infty \subseteq \cdots \subseteq C^2 \subseteq C^1 \subseteq C^0$.
The inclusions follow from Theorem 9.1.9 and Definition 1.1.23; they are generally proper.
Key Insight: Higher regularity is extra structure, not just a different notation for continuity.

Example 9.1.14. *Smooth but Not Analytic*
Define $e(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$. By repeated L'Hospital, Theorem 1.1.21, $e \in C^\infty$ and $e^{(r)}(0) = 0$ for all r.
Its Taylor series at 0 is zero, but $e(x) > 0$ for $x > 0$, so e is not analytic at 0 (Definition 9.1.8).
Key Insight: A function can be infinitely tangent to the x-axis without being locally equal to its Taylor series.

Perspective 9.1.15. *Taylor Coefficients Are Only the First Half*
Power-series coefficients are forced by derivatives (Corollary 9.1.11); for a function not already known to be analytic, proving the Taylor series represents the function needs a remainder estimate (Theorem 1.1.25).

Theorem 9.1.16. *Exponential and Logarithm from the Power Series*
Define $\mathsf{E}(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}$ for $z \in \mathbb{C}$.

(i) $\mathsf{E}(z+w) = \mathsf{E}(z)\mathsf{E}(w)$ for $z, w \in \mathbb{C}$.
(ii) On $\mathbb{R}$, $\mathsf{E} > 0$, $\mathsf{E}' = \mathsf{E}$, $\mathsf{E}(x) = e^x$, and E is strictly increasing.
(iii) The inverse $\mathsf{L} := \mathsf{E}^{-1} : (0, \infty) \to \mathbb{R}$ satisfies $\mathsf{L}'(y) = 1/y$ and $\mathsf{L}(uv) = \mathsf{L}(u) + \mathsf{L}(v)$ for $u, v > 0$.
(iv) For $x > 0$ and $\alpha \in \mathbb{R}$, set $x^\alpha := \mathsf{E}(\alpha \mathsf{L}(x))$; then $(x^\alpha)' = \alpha x^{\alpha-1}$.

Proof. The ratio test gives convergence for all $z \in \mathbb{C}$.

(i) Absolute convergence gives $\mathsf{E}(z)\mathsf{E}(w) = \sum_{n \geq 0} \sum_{j=0}^{n} \frac{z^j w^{n-j}}{j!(n-j)!} = \sum_{n \geq 0} \frac{(z+w)^n}{n!} = \mathsf{E}(z+w)$.
(ii) Termwise differentiation gives $\mathsf{E}' = \mathsf{E}$ on $\mathbb{R}$. By (i), $\mathsf{E}(x)\mathsf{E}(-x) = 1$, so continuity and $\mathsf{E}(0) = 1$ give $\mathsf{E} > 0$; hence $\mathsf{E}' = \mathsf{E} > 0$. Since $\mathsf{E}(1) = e$, (i) gives $\mathsf{E}(p) = e^p$ for $p \in \mathbb{Q}$, and continuity gives $\mathsf{E}(x) = e^x$.
(iii) For $y = \mathsf{E}(x)$, differentiating $\mathsf{L}(\mathsf{E}(x)) = x$ gives $\mathsf{L}'(y) = 1/\mathsf{E}'(x) = 1/y$.
If $u = \mathsf{E}(s)$ and $v = \mathsf{E}(t)$, then $\mathsf{L}(uv) = \mathsf{L}(\mathsf{E}(s+t)) = s + t = \mathsf{L}(u) + \mathsf{L}(v)$.
(iv) $\frac{d}{dx} \mathsf{E}(\alpha \mathsf{L}(x)) = \mathsf{E}(\alpha \mathsf{L}(x)) \alpha \mathsf{L}'(x) = x^\alpha \alpha / x = \alpha x^{\alpha-1}$.

Structural Commentary: Addition, inversion, powers.
Key Insight: $\mathsf{E}(z+w) = \mathsf{E}(z)\mathsf{E}(w)$ creates the logarithm law. □

Perspective 9.1.17. *Real Powers by Rational Approximation*
One route defines x^α for $x > 0$ by choosing $q_n \in \mathbb{Q}$ with $q_n \to \alpha$ and setting $x^\alpha = \lim_{n \to \infty} x^{q_n}$ after proving existence and independence of (q_n). The power-series construction sets $x^\alpha = \mathsf{E}(\alpha \mathsf{L}(x))$; both routes must agree with rational powers and be choice-independent.

Perspective 9.1.18. *Logarithmic Integral Route*
With the Riemann integral, set $L(y) := \int_1^y \frac{1}{t}\, dt$ for $y > 0$. Theorem 2.2.1 gives $L'(y) = 1/y$; endpoint limits make L invertible, and Theorem 1.1.6 then recovers $E' = E$ and the product laws.

Perspective 9.1.19. *From Exponentials to Factorials*
Theorems 9.1.16, 2.2.3 supply real-power laws and integration-by-parts pattern used for Γ in Theorem 9.1.22.

Theorem 9.1.20. *Trigonometric Functions from the Exponential*
For $x \in \mathbb{R}$, set $\mathsf{C}(x) := \frac{\mathsf{E}(ix)+\mathsf{E}(-ix)}{2}$, $\mathsf{S}(x) := \frac{\mathsf{E}(ix)-\mathsf{E}(-ix)}{2i}$, and $\gamma(t) := \mathsf{E}(it)$.

(i) $\mathsf{E}(ix) = \mathsf{C}(x) + i\mathsf{S}(x)$, $\mathsf{C}^2 + \mathsf{S}^2 = 1$, $\mathsf{C}' = -\mathsf{S}$, *and* $\mathsf{S}' = \mathsf{C}$.
(ii) $x_0 := \min\{x > 0 : \mathsf{C}(x) = 0\}$ *exists; with* $\pi := 2x_0$, $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, $\mathsf{E}(z + 2\pi i) = \mathsf{E}(z)$ *for* $z \in \mathbb{C}$.
(iii) C, S *are* 2π*-periodic,* $\gamma([0, 2\pi]) = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$, *and* γ *is injective on* $[0, 2\pi)$.

Proof.
(i) Real coefficients give $\overline{\mathsf{E}(ix)} = \mathsf{E}(-ix)$, so $\mathsf{E}(ix) = \mathsf{C} + i\mathsf{S}$ and $|\mathsf{E}(ix)|^2 = \mathsf{E}(ix)\mathsf{E}(-ix) = 1$.
Thus $\mathsf{C}^2 + \mathsf{S}^2 = 1$; differentiating the definitions gives $\mathsf{C}' = -\mathsf{S}$ and $\mathsf{S}' = \mathsf{C}$.
(ii) If C has no positive zero, then $\mathsf{C} > 0$ on $(0, \infty)$, hence $\mathsf{S}' > 0$ and $\mathsf{S}(x) > 0$ for $x > 0$.
For $0 < x < y$, $\mathsf{S}(x)(y - x) < \int_x^y \mathsf{S}(t)\,dt = \mathsf{C}(x) - \mathsf{C}(y) \le 2$, impossible as $y \to \infty$; continuity gives the first zero x_0. On $(0, x_0)$, $\mathsf{C} > 0$, so S increases from 0 to $\mathsf{S}(x_0) = 1$.
Since $x_0 = \pi/2$, $\mathsf{E}(\pi i/2) = i$, and $\mathsf{E}(z + w) = \mathsf{E}(z)\mathsf{E}(w)$ gives $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, and $\mathsf{E}(z + 2\pi i) = \mathsf{E}(z)$.
(iii) Periodicity of C, S follows from (ii). On $[0, \pi/2]$, C decreases from 1 to 0 and S increases from 0 to 1, so γ traverses the first quadrant once. Since $\gamma(t + m\pi/2) = \gamma(t)i^m$ for $m = 0, 1, 2, 3$, the four quadrants give $\gamma([0, 2\pi]) = \{\zeta : |\zeta| = 1\}$ and injectivity on $[0, 2\pi)$.

Structural Commentary: $\mathsf{E}(it)$ supplies trigonometry.
Key Insight: $\mathsf{E}(it)\mathsf{E}(-it) = 1$ puts the path on the unit circle. □

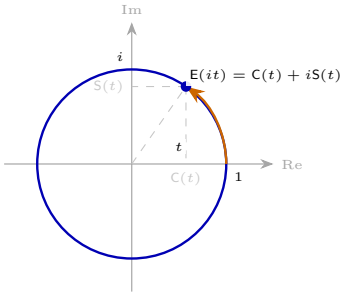

Figure 183: $\mathsf{E}(it) = \mathsf{C}(t) + i\mathsf{S}(t)$ traces the unit circle, defining sine and cosine analytically.

Theorem 9.1.21. *Algebraic Completeness of* $\mathbb{C}$
If $P \in \mathbb{C}[z]$ and $\deg P \ge 1$, then $\exists z_0 \in \mathbb{C}$ such that $P(z_0) = 0$.

Proof. Write $P(z) = \sum_{j=0}^n a_j z^j$, $n \ge 1$, $a_n = 1$.
Since $|P(z)| \to \infty$ as $|z| \to \infty$, choose z_0 with $\mu := |P(z_0)| = \min_{\mathbb{C}} |P|$. If $\mu = 0$, done.
Assume $\mu > 0$ and set $Q(w) := P(z_0 + w)/P(z_0)$. Then $Q(0) = 1$ and $|Q(w)| \ge 1$ for all w.
Write $Q(w) = 1 + b_k w^k + w^{k+1} H(w)$ with $k \ge 1$, $b_k \ne 0$, and H polynomial.
Choose $\theta \in \mathbb{R}$ with $b_k e^{ik\theta} = -|b_k|$. For $r > 0$, $Q(re^{i\theta}) = 1 - r^k A(r)$, where $A(r) := |b_k| - rH(re^{i\theta})$.
For small r, $\Re A(r) > |b_k|/2$, hence $\left|1 - r^k A(r)\right|^2 = 1 - 2r^k \Re A(r) + r^{2k} |A(r)|^2 < 1$, contradicting $|Q| \ge 1$.
Structural Commentary: Minimize $|P|$, translate, perturb.
Key Insight: A phase choice makes the first nonzero term decrease $|Q|$. □

Theorem 9.1.22. *Gamma Function, Beta Function, and Stirling Skeleton*
For $x > 0$, set $\Gamma(x) := \int_0^\infty t^{x-1} e^{-t}\,dt$; for $x, y > 0$, set $\mathrm{B}(x, y) := \int_0^1 t^{x-1}(1 - t)^{y-1}\,dt$.

(i) $\Gamma(x + 1) = x\Gamma(x)$ *for* $x > 0$, $\Gamma(n + 1) = n!$ *for* $n \in \mathbb{N}$, *and* $\log\Gamma$ *is convex on* $(0, \infty)$.
(ii) Γ *is the unique* $f : (0, \infty) \to (0, \infty)$ *with* $f(1) = 1$, $f(x + 1) = xf(x)$, *and* $\log f$ *convex.*
(iii) $\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$ *for* $x, y > 0$.
(iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, $\int_{-\infty}^\infty e^{-s^2}\,ds = \sqrt{\pi}$, $\Gamma(x) = \frac{2^{x-1}}{\sqrt{\pi}}\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{2}\right)$, *and* $\lim_{x\to\infty} \frac{\Gamma(x+1)}{(x/e)^x\sqrt{2\pi x}} = 1$.

Proof.
(i) Integration by parts gives $\Gamma(x + 1) = \int_0^\infty t^x e^{-t}\,dt = x\Gamma(x)$; $\Gamma(1) = 1$ gives $\Gamma(n + 1) = n!$.
For $0 < \lambda < 1$, Holder gives $\Gamma(\lambda x + (1 - \lambda)y) \le \Gamma(x)^\lambda \Gamma(y)^{1-\lambda}$, hence $\log\Gamma$ is convex.
(ii) Let $\varphi = \log f$. For $0 < x < 1$ and $n \ge 1$, convexity with $\varphi(t + 1) = \varphi(t) + \log t$ gives
$0 \le \varphi(x) - \log\frac{n!\,n^x}{x(x+1)\cdots(x+n)} \le x\log(1 + \frac{1}{n})$. Letting $n \to \infty$ fixes φ on $(0, 1)$; recurrence fixes φ on $(0, \infty)$.
(iii) Fix $y > 0$ and set $F_y(x) := \frac{\Gamma(x+y)}{\Gamma(y)}\mathrm{B}(x, y)$. Since $\mathrm{B}(1, y) = 1/y$ and $\mathrm{B}(x+1, y) = \frac{x}{x+y}\mathrm{B}(x, y)$, $F_y(1) = 1$ and $F_y(x+1) = xF_y(x)$; Holder gives log-convexity. Thus (ii) gives $F_y = \Gamma$.
(iv) $t = \sin^2\theta$ gives $\mathrm{B}(\frac{1}{2}, \frac{1}{2}) = \pi$, hence $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ by (iii). Then $t = s^2$ gives $\Gamma(\frac{1}{2}) = \int_{-\infty}^\infty e^{-s^2}\,ds$.
Beta duplication gives the duplication formula, and $t = x(1 + u)$ with Laplace asymptotics gives Stirling.

Structural Commentary: Recurrence plus log-convexity gives uniqueness.
Key Insight: Bohr–Mollerup turns beta identities into gamma identities. □

Method 9.1.23. *Gamma Function Proof Pattern*

(i) *Gamma function*: use $\Gamma(x + 1) = x\Gamma(x)$, $\Gamma(1) = 1$, and log-convexity for uniqueness.

9.2 Approximation Methods

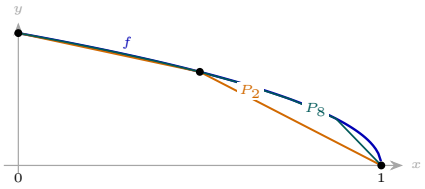

Figure 184: Finer piecewise-linear interpolants approach $f(x) = \sqrt{1-x}$ uniformly on $[0,1]$.

Theorem 9.2.1. *Weierstrass Polynomial Approximation*
Let $a < b$ and $f \in C([a,b],\mathbb{C})$. Then $\exists (P_n) \subset \mathbb{C}[x]$ with $P_n \rightrightarrows f$ on $[a,b]$. If $f([a,b]) \subset \mathbb{R}$, choose $(P_n) \subset \mathbb{R}[x]$.

Proof. By affine change and subtraction of the endpoint interpolant, reduce to $[0,1]$ with $f(0) = f(1) = 0$.
Extend $f = 0$ off $[0,1]$; then f is uniformly continuous on $[-1,2]$. Choose $c_n > 0$ so $Q_n(t) := c_n(1-t^2)^n$ satisfies $\int_{-1}^{1} Q_n = 1$. Use $c_n < \sqrt{n}$ and $Q_n(t) \le \sqrt{n}(1-\delta^2)^n$ for $0 < \delta \le |t| \le 1$. Set $P_n(x) := \int_{-1}^{1} f(x+t)Q_n(t)\,dt$.
Since $P_n(x) = \int_0^1 f(u)Q_n(u-x)\,du$, $P_n \in \mathbb{C}[x]$, and $P_n \in \mathbb{R}[x]$ if f is real-valued.
Given $\varepsilon > 0$, choose $0 < \delta \le 1$ with $|u-v| < \delta \Rightarrow |f(u) - f(v)| < \varepsilon/2$ on $[-1,2]$, and set $M = \sup_{[-1,2]} |f|$.
For $x \in [0,1]$, $Q_n \ge 0$, $\int_{-1}^{1} Q_n = 1$ give $|P_n(x) - f(x)| \le \int_{-1}^{1} |f(x+t) - f(x)|\, Q_n(t)\,dt \le \frac{\varepsilon}{2} + 4M\sqrt{n}(1-\delta^2)^n$.
Thus $\|P_n - f\|_\infty \to 0$ on $[0,1]$; undo the reductions.
Structural Commentary: Polynomial kernels approximate the identity.
Key Insight: Uniform continuity handles near mass; kernel concentration removes far mass. □

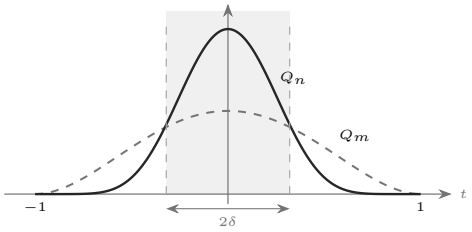

Figure 185: The normalized kernels $Q_n(t) = c_n(1-t^2)^n$ concentrate near 0 while retaining integral 1.

Corollary 9.2.2. *Polynomial Approximation of $|x|$*
Let $a > 0$. There exists $(P_n) \subset \mathbb{R}[x]$ such that $P_n(0) = 0$ and $P_n \rightrightarrows |x|$ on $[-a,a]$.

Proof. By Theorem 9.2.1, choose $Q_n \in \mathbb{R}[x]$ with $Q_n \rightrightarrows |x|$ on $[-a,a]$. Set $P_n(x) := Q_n(x) - Q_n(0)$.
Then $P_n \in \mathbb{R}[x]$, $P_n(0) = 0$, and $Q_n(0) \to 0$, so $\|P_n - |x|\|_\infty \le \|Q_n - |x|\|_\infty + |Q_n(0)| \to 0$.
Structural Commentary: Normalize at 0.
Key Insight: The constant correction vanishes uniformly. □

Definition 9.2.3. *Algebras and Uniform Closure*
Let $\mathcal{A}$ be a family of complex-valued functions on a set E.

(i) $\mathcal{A}$ is an *algebra* if it is closed under $+$, products, and scalar multiplication (real scalars in the real case).
(ii) $\mathcal{A}$ is *uniformly closed* if $f_n \in \mathcal{A}$ and $f_n \rightrightarrows f$ on E imply $f \in \mathcal{A}$.
(iii) The *uniform closure* of $\mathcal{A}$ is $\{f : f_n \in \mathcal{A},\ f_n \rightrightarrows f \text{ on } E\}$.

Theorem 9.2.4. *Uniform Closures Are Closed Algebras*
The uniform closure of an algebra of bounded functions is a uniformly closed algebra.

Proof. If $f, g \in \mathcal{B}$, choose $f_n, g_n \in \mathcal{A}$ with $f_n \rightrightarrows f$ and $g_n \rightrightarrows g$.
Boundedness and uniform convergence give $f_n + g_n \rightrightarrows f + g$; $f_n g_n \rightrightarrows fg$; and $cf_n \rightrightarrows cf$. Thus $\mathcal{B}$ is an algebra. If $h_n \in \mathcal{B}$ and $h_n \rightrightarrows h$, approximate each h_n uniformly by some $a_n \in \mathcal{A}$ within $1/n$; then $a_n \rightrightarrows h$, so $h \in \mathcal{B}$.
Structural Commentary: Uniform limits preserve algebraic operations for bounded functions.
Key Insight: The uniform closure of an algebra has already absorbed all further uniform limits. □

Definition 9.2.5. *Separation, Nonvanishing, and Self-Adjointness*
Let $\mathcal{A}$ be a family of functions on E.

(i) $\mathcal{A}$ *separates points* on E if for every distinct $x_1, x_2 \in E$ there is $f \in \mathcal{A}$ with $f(x_1) \ne f(x_2)$.
(ii) $\mathcal{A}$ *vanishes at no point* of E if for every $x \in E$ there is $g \in \mathcal{A}$ with $g(x) \ne 0$.
(iii) A complex algebra $\mathcal{A}$ is *self-adjoint* if $f \in \mathcal{A}$ implies $\overline{f} \in \mathcal{A}$, where $\overline{f}(x) := \overline{f(x)}$.

Theorem 9.2.6. *Two-Point Interpolation in a Separating Algebra*
Let $\mathcal{A}$ be a scalar algebra on E that separates points and vanishes nowhere.
For $x_1 \ne x_2$ in E and scalars c_1, c_2, $\exists f \in \mathcal{A}$ such that $f(x_i) = c_i$, $i = 1, 2$.

Proof. Choose $g, h, k \in \mathcal{A}$ with $g(x_1) \neq g(x_2)$, $h(x_1) \neq 0$, $k(x_2) \neq 0$. Set $u := gk - g(x_1)k$ and $v := gh - g(x_2)h$.
Then $u(x_1) = v(x_2) = 0$, while $u(x_2) = (g(x_2) - g(x_1))k(x_2) \neq 0$ and $v(x_1) = (g(x_1) - g(x_2))h(x_1) \neq 0$.
Hence $f := \frac{c_1 v}{v(x_1)} + \frac{c_2 u}{u(x_2)} \in \mathcal{A}$ and $f(x_i) = c_i$, $i = 1, 2$.
Structural Commentary: Build point-isolating functions.
Key Insight: Isolation reduces interpolation to linear combination. □

Perspective 9.2.7. *Interpolation Is Linear Algebra*
The two-point interpolation step can also be written as a 2×2 linear system. Its invertibility is exactly the finite-dimensional criterion from Theorem 1.7.7, with the determinant test supplied by Theorem 1.1.8. The proof above chooses explicit point-isolating functions instead of reopening the calculation.

Theorem 9.2.8. *Stone–Weierstrass Theorem, Real Form*
Let K be compact Hausdorff; $\mathcal{A} \subset C(K, \mathbb{R})$ point-separating, nowhere-vanishing real algebra. $\overline{\mathcal{A}}^{\|\cdot\|_\infty} = C(K, \mathbb{R})$.

Proof. Put $\mathcal{B} := \overline{\mathcal{A}}^{\|\cdot\|_\infty}$. By Theorem 9.2.4, $\mathcal{B}$ is a uniformly closed algebra and inherits separation/nonvanishing.
For $r \in \mathcal{B}$, if $a = \|r\|_\infty > 0$, Corollary 9.2.2 gives $p_n(0) = 0$ with $p_n \rightrightarrows |t|$ on $[-a, a]$, hence $p_n(r) \rightrightarrows |r|$.
$a = 0$ is trivial. Thus $|r| \in \mathcal{B}$, so for $r, s \in \mathcal{B}$, $r \vee s = \frac{r+s+|r-s|}{2}$ and $r \wedge s = \frac{r+s-|r-s|}{2}$ lie in $\mathcal{B}$.
Fix $f \in C(K, \mathbb{R})$ and $\varepsilon > 0$. For $x, y \in K$, choose $h_{x,y} \in \mathcal{B}$ with $h_{x,y}(x) = f(x)$, $h_{x,y}(y) = f(y)$, using Theorem 9.2.6 if $x \neq y$ and $f(x)q/q(x)$ with $q \in \mathcal{B}$, $q(x) \neq 0$, if $x = y$. For fixed x, set $V_y := \{h_{x,y} > f - \varepsilon\}$.
Since $y \in V_y$, choose y_i with $K = \bigcup_i V_{y_i}$ and put $g_x := \bigvee_i h_{x,y_i} \in \mathcal{B}$. Then $f - \varepsilon < g_x$ and $g_x(x) = f(x)$.
Set $U_x := \{g_x < f + \varepsilon\}$. Since $x \in U_x$, choose x_j with $K = \bigcup_j U_{x_j}$ and put $h := \bigwedge_j g_{x_j} \in \mathcal{B}$.
Then $f - \varepsilon < h < f + \varepsilon$, so $\|h - f\|_\infty < \varepsilon$. Since $\mathcal{B}$ is uniformly closed, $f \in \mathcal{B}$.
Structural Commentary: Finite joins/meets globalize interpolation.
Key Insight: Compactness upgrades pointwise to uniform. □

Theorem 9.2.9. *Stone–Weierstrass Theorem, Complex Form*
Let K be compact Hausdorff and let $\mathcal{A} \subset C(K, \mathbb{C})$ be a self-adjoint, point-separating, nowhere-vanishing complex algebra. Then $\overline{\mathcal{A}}^{\|\cdot\|_\infty} = C(K, \mathbb{C})$.

Proof. Set $\mathcal{A}_\mathbb{R} := \mathcal{A} \cap C(K, \mathbb{R})$. Self-adjointness gives $\operatorname{Re} f, \operatorname{Im} f \in \mathcal{A}_\mathbb{R}$ for all $f \in \mathcal{A}$. If $f(x) \neq f(y)$ or $f(x) \neq 0$, then one of $\operatorname{Re} f, \operatorname{Im} f$ witnesses the same condition; hence $\mathcal{A}_\mathbb{R}$ is point-separating and nowhere-vanishing.
By Theorem 9.2.8, $\overline{\mathcal{A}_\mathbb{R}} = C(K, \mathbb{R})$. For $F = u + iv \in C(K, \mathbb{C})$, take $u_n, v_n \in \mathcal{A}_\mathbb{R}$ with $u_n \rightrightarrows u$, $v_n \rightrightarrows v$.
Then $u_n + iv_n \in \mathcal{A}$ and $u_n + iv_n \rightrightarrows F$.
Structural Commentary: Self-adjointness yields a real subalgebra.
Key Insight: Approximate real and imaginary parts. □

Theorem 9.2.10. *Riesz Separability Criterion for $C(K)$*
Let K be a compact Hausdorff space. Then $(C(K, \mathbb{R}), \|\cdot\|_\infty)$ is separable iff K is metrizable.

Proof.
(⇒) Let $\{g_n\}$ be dense in $C(K, \mathbb{R})$ and $O_n := \{g_n > 1/2\}$. Each O_n is open. For open $O \ni x$, Urysohn gives $g \in C(K, \mathbb{R})$ with $0 \leq g \leq 1$, $g(x) = 1$, and $g|_{K \setminus O} = 0$. Pick n with $\|g - g_n\|_\infty < 1/3$; then $x \in O_n \subseteq O$.
Thus $\{O_n\}$ is a countable basis, so Theorem 6.4.23 makes K metrizable.
(⇐) Let d metrize K and choose dense $\{x_n : n \geq 1\} \subset K$. Put $f_0 = 1$ and $f_n(x) = d(x, x_n)$.
If $x \neq y$, choose x_n with $d(x, x_n) < d(x, y)/3$; then $f_n(x) \neq f_n(y)$. Hence $\{f_n : n \geq 0\}$ separates points and vanishes nowhere, so Theorem 9.2.8 makes $\operatorname{alg}(f_n : n \geq 0)$ dense in $C(K, \mathbb{R})$.
Its rational polynomial subalgebra is countable and dense.

Structural Commentary: Dense functions give a base; dense points give generators.
Key Insight: Countability moves between K and $C(K)$. □

Theorem 9.2.11. *A Continuous Nowhere Differentiable Function Exists*
There exists a real continuous function on $\mathbb{R}$ which is nowhere differentiable.

Proof. Let φ be the 2-periodic extension of $|x|$ on $[-1, 1]$. Then $0 \leq \varphi \leq 1$ and $|\varphi(s) - \varphi(t)| \leq |s - t|$.
Define $f(x) := \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$, converges uniformly (Theorem 7.1.9); hence f is continuous (Theorem 7.1.11).
Fix x and $m \in \mathbb{N}$. Choose $\delta_m = \pm\frac{1}{2}4^{-m}$ so that no integer lies between $4^m x$ and $4^m(x + \delta_m)$.
Set $\gamma_{n,m} := \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}$. If $n > m$, then $4^n \delta_m$ is an even integer, so $\gamma_{n,m} = 0$.
If $0 \leq n \leq m$, then $|\gamma_{n,m}| \leq 4^n$, and the choice of δ_m gives $|\gamma_{m,m}| = 4^m$.
Hence $\left|\frac{f(x+\delta_m) - f(x)}{\delta_m}\right| = \left|\sum_{n=0}^{m} \left(\frac{3}{4}\right)^n \gamma_{n,m}\right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1)$
Since $\delta_m \to 0$, the difference quotients are unbounded near x; since x was arbitrary, f is nowhere differentiable.
Structural Commentary: Uniform sums give continuity; scaled cusps force unbounded quotients.
Key Insight: Uniform convergence preserves continuity, not differentiability. □

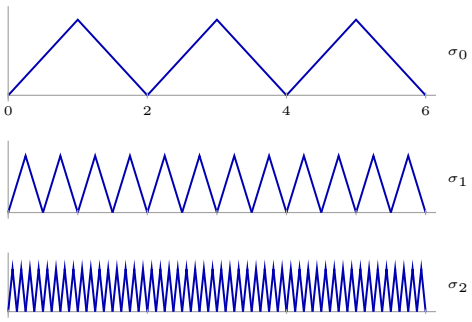

Figure 186: The first compressed sawtooth terms $\sigma_k(x) = (3/4)^k \varphi(4^k x)$: heights shrink, slopes grow.

Perspective 9.2.12. *Explicit Versus Generic Pathology*
The sawtooth series proves existence; Baire category shows nowhere differentiability is residual in $C([a,b],\mathbb{R})$.

Theorem 9.2.13. *Generic Nowhere Differentiability on Compact Intervals*
Let $a < b$, and let $C = C([a,b],\mathbb{R})$ with the supremum metric. There exists a residual set $\mathcal{G} \subseteq C$ such that:

(i) no $f \in \mathcal{G}$ has an available one-sided derivative at any point of $[a,b]$;
(ii) no $f \in \mathcal{G}$ is monotone on any nondegenerate subinterval of $[a,b]$;
(iii) the set $\{f \in C : f$ is differentiable at some point of $[a,b]\}$ is first category.

Proof.

(i) Write $\Delta_h f(x) := f(x+h) - f(x)$, and choose N_0 with $1/N_0 < b - a$. For $n \geq N_0$, define

$$R_n := \left\{ f \in C : \forall x \in [a, b - 1/n]\ \exists h > 0 \text{ with } \left| \frac{\Delta_h f(x)}{h} \right| > n \right\},$$

$$L_n := \left\{ f \in C : \forall x \in [a + 1/n, b]\ \exists h < 0 \text{ with } \left| \frac{\Delta_h f(x)}{h} \right| > n \right\},$$

$$G_n := \{ f \in C : f \text{ is nonmonotone on every subinterval of length } 1/n \}.$$

Each R_n, L_n, G_n is open and dense in C (Weierstrass Approximation Theorem 9.2.1), plus small high-slope sawtooth perturbation; openness follows from compactness and uniform perturbation estimates.

(ii) Set $\mathcal{G} := \bigcap_{n \geq N_0} (R_n \cap L_n \cap G_n)$. By Definition 4.3.6, $\mathcal{G}$ is residual. If $f \in \mathcal{G}$, then the R_n and L_n conditions give one-sided difference quotients of arbitrarily large absolute value; since f is bounded, the corresponding increments tend to 0. Thus no available one-sided derivative exists.

(iii) Let $f \in \mathcal{G}$. Every nondegenerate subinterval contains a subinterval of length $1/n$ for all sufficiently large n, so the G_n conditions make f nonmonotone on every nondegenerate subinterval. If f is differentiable somewhere, then $f \notin \mathcal{G}$; hence the differentiable-somewhere set lies in $C \setminus \mathcal{G}$, which is first category.

Structural Commentary: Baire category turns dense sawtooth perturbability into a generic regularity failure.
Key Insight: The explicit construction shows existence; the category proof shows typicality. □

Method 9.2.14. *Power-Series and Approximation Proof Patterns*

(i) *Power series*: work on $K_\rho \Subset I_R$; use uniform convergence before termwise limits or derivatives.
(ii) *Endpoint theorems*: prove identities inside the radius and pass to endpoints with Abel-type control.
(iii) *Approximate identities*: use kernels $Q_n \geq 0$, $\int Q_n = 1$, with tails $\to 0$ off 0.
(iv) *Uniform closure*: show $a_n \in \mathcal{A}$ and $a_n \rightrightarrows f$.
(v) *Lattice closure*: derive $|f|$, then $\max, \min$ from algebraic formulas.
(vi) *Stone–Weierstrass*: interpolate locally; use compactness and finite max/min to globalize.

Key Insight: Uniform convergence and compactness are what make infinite algebra behave like finite algebra.

9.3 The Umbral Calculus

The umbral calculus is the formal linear algebra of polynomial sequences. It identifies a formal power series with three related objects: an algebra element, a linear functional on $\mathbb{C}[x]$, and an operator on $\mathbb{C}[x]$. Refines duality of Definition 4.2.4 and polynomial operator calculus of Definition 1.1.2, without invoking convergence.

Definition 9.3.1. *Formal Power Series and Order*
Let $\mathscr{F} := \mathbb{C}[[t]]$.

(i) $f \in \mathscr{F}$ are *formal power series* $f(t) = \sum_{k \geq 0} a_k t^k$, w. coefficientwise addition, Cauchy product multiplication.
(ii) The *order* of f is $o(f) := \min\{k : a_k \neq 0\}$, where $o(0) := +\infty$.
(iii) A series is *invertible* if it has a multiplicative inverse in $\mathscr{F}$, and is a *delta series* if $o(f) = 1$.
(iv) If $o(f) = 1$, its *compositional inverse* is denoted $\overline{f}$ and satisfies $f(\overline{f}(t)) = \overline{f}(f(t)) = t$.
(v) The *lower factorial polynomials* are $x)_0 := 1$ and $(x)_n := x(x-1)\cdots(x-n+1)$, for $n \geq 1$

Proposition 9.3.2. *Formal Order Calculus*
For $f, g \in \mathscr{F}$:

(i) $f \in \mathscr{F}^\times \iff o(f) = 0$;
(ii) $o(fg) = o(f) + o(g)$ and $o(f+g) \geq \min\{o(f), o(g)\}$;
(iii) if $g(t) = \sum_{k \geq 0} b_k t^k$ and $o(f) \geq 1$, then $g \circ f := g(f(t)) := \sum_{k \geq 0} b_k f(t)^k \in \mathscr{F}$;
(iv) if $o(f) \geq 1$ and $g \neq 0$, then $o(g \circ f) = o(g) o(f)$;
(v) if $o(f) = 1$, then for every $g \in \mathscr{F}$ there are unique $c_k \in \mathbb{C}$ such that $g(t) = \sum_{k \geq 0} c_k f(t)^k$.

Proof.

(i) Let $[t^n] : \mathscr{F} \to \mathbb{C}$ denote coefficient extraction. Then $f \in \mathscr{F}^\times \iff [t^0]f \in \mathbb{C}^\times \iff o(f) = 0$.

(ii) Let $r = o(f)$ and $s = o(g)$. Then $[t^{r+s}]fg = [t^r]f\,[t^s]g \neq 0$, and all lower coefficients vanish; hence $o(fg) = r + s$. Also $[t^m](f+g) = 0$ for $m < \min\{r, s\}$, giving the stated lower bound.

(iii) Since $o(f) \geq 1$, $o(f^k) \to \infty$; hence $\forall n$, coefficient $[t^n] \sum_{k\geq 0} b_k f^k$ has only finitely many nonzero terms.

(iv) Let $s = o(g)$. Then $g \circ f = f^s \left(b_s + \sum_{k>s} b_k f^{k-s} \right)$, where second factor has order 0; hence $o(g \circ f) = s\, o(f)$.

(v) If $o(f) = 1$ and $f(t) = a_1 t + \cdots$ with $a_1 \neq 0$, then $f^k = a_1^k t^k +$ higher terms. Thus $\{t^k\}_{k\geq 0} \mapsto \{f^k\}_{k\geq 0}$ is triangular with nonzero diagonal, so recursive coefficient matching gives existence and uniqueness.

Structural Commentary: Delta series behave like a formal change of coordinate at $t = 0$.
Key Insight: Order replaces convergence as the bookkeeping device. □

Definition 9.3.3. *Derivation*
Let $\mathcal{A}$ be an algebra over a field. Linear operator $\partial : \mathcal{A} \to \mathcal{A}$ is a *derivation* if $\partial(ab) = (\partial a)b + a(\partial b)$ $\forall a, b \in \mathcal{A}$.

Proposition 9.3.4. *Formal Differentiation*
Define $\partial_t : \mathscr{F} \to \mathscr{F}$ *by* $\partial_t \left(\sum_{k\geq 0} a_k t^k \right) := \sum_{k\geq 1} k a_k t^{k-1}$. *Then:*

(i) ∂_t *is* $\mathbb{C}$*-linear.*
(ii) $\partial_t(fg) = (\partial_t f)g + f(\partial_t g)$ *for all* $f, g \in \mathscr{F}$*; hence* $\partial_t \in \mathrm{Der}_{\mathbb{C}}(\mathscr{F})$.

Proof.

(i) Let $f = \sum a_k t^k$, $g = \sum b_k t^k$, and $\alpha, \beta \in \mathbb{C}$. Then $\partial_t(\alpha f + \beta g) = \sum_{k\geq 1} k(\alpha a_k + \beta b_k) t^{k-1} = \alpha \partial_t f + \beta \partial_t g$.

(ii) Let $f(t) = \sum_{k\geq 0} a_k t^k$ and $g(t) = \sum_{k\geq 0} b_k t^k$. For $n \geq 0$, $[t^n]\partial_t(fg) = (n+1) \sum_{j=0}^{n+1} a_j b_{n+1-j}$.

Also $[t^n]\big((\partial_t f)g + f(\partial_t g)\big) = \sum_{j=1}^{n+1} j a_j b_{n+1-j} + \sum_{j=0}^{n} (n+1-j) a_j b_{n+1-j}$, equals preceding coefficient.

Structural Commentary: Formal differentiation is the canonical derivation of $\mathbb{C}[[t]]$.
Key Insight: Leibniz is a Cauchy-product coefficient identity. □

Definition 9.3.5. *The Umbral Algebra*

(i) *Polynomial algebra*: set $\mathscr{P} := \mathbb{C}[x]$ and let $\mathscr{P}' := \mathrm{Hom}_{\mathbb{C}}(\mathscr{P}, \mathbb{C})$.
(ii) *Umbral pairing*: for $L \in \mathscr{P}'$ and $p(x) \in \mathscr{P}$, write $\langle L \mid p(x) \rangle := L(p(x))$.
(iii) *Exponential normalization*: write each $f \in \mathscr{F}$ as $f(t) = \sum_{k\geq 0} \frac{a_k}{k!} t^k$.
(iv) *Associated functional*: set $\langle f(t) \mid x^n \rangle := a_n$ for $n \geq 0$, extending $\mathbb{C}$-linearly.

Theorem 9.3.6. *Formal Series Represent All Algebraic Functionals*
Define $\Phi : \mathscr{P}' \to \mathscr{F}$ *by* $\Phi(L) := \sum_{k\geq 0} \frac{\langle L | x^k \rangle}{k!} t^k$. *Then:*

(i) Φ *is a* $\mathbb{C}$*-linear isomorphism with inverse* $\Phi^{-1} \left(\sum_{k\geq 0} \frac{a_k}{k!} t^k \right) \left(\sum_{k=0}^{d} c_k x^k \right) := \sum_{k=0}^{d} c_k a_k$.
(ii) $\langle t^k \mid x^n \rangle = n! \delta_{n,k}$ *and* $\langle t^k \mid p \rangle = p^{(k)}(0)$ *for* $p \in \mathscr{P}$*; hence* $t^0 = \mathrm{ev}_0$ *and* $t^k = D^k|_0$.

Proof.

(i) Let $T : \mathscr{P}' \to \mathbb{C}^{\mathbb{N}}$ be $T(L) = (a_k)_{k\geq 0}$, $a_k = \langle L \mid x^k \rangle$.

Since $\{x^k\}_{k\geq 0}$ is a Hamel basis, T is a $\mathbb{C}$-linear bijection with $T^{-1}(a) \left(\sum_{k=0}^{d} c_k x^k \right) = \sum_{k=0}^{d} c_k a_k$.

The map $E : \mathbb{C}^{\mathbb{N}} \to \mathscr{F}$, $E(a) = \sum_{k\geq 0} \frac{a_k}{k!} t^k$, is a $\mathbb{C}$-linear bijection, hence $\Phi = E \circ T$ and $\Phi^{-1} = T^{-1} \circ E^{-1}$.

(ii) As $t^k = k!\, \frac{t^k}{k!}$, coefficient sequence is $(n!\delta_{n,k})_{n\geq 0}$. For $p = \sum_{n=0}^{d} c_n x^n$, by linearity, $\langle t^k \mid p \rangle = k! c_k = p^{(k)}(0)$.

Structural Commentary: Functionals are monomial-coordinate sequences.
Key Insight: Exponential normalization makes t^k equal $D^k|_0$. □

Definition 9.3.7. *Delta and Invertible Functionals*
Under the identification $\mathscr{F} \simeq \mathscr{P}'$:

(i) *Delta functional*: a functional represented by $f \in \mathscr{F}$ with $o(f) = 1$.
(ii) *Invertible functional*: a functional represented by $f \in \mathscr{F}^\times$, equivalently $o(f) = 0$.

Proposition 9.3.8. *Basic Umbral Identities*
For $f, g \in \mathscr{F}$, $p \in \mathscr{P}$, *and* $n \geq 0$*:*

(i) $f(t) = \sum_{k\geq 0} \frac{\langle f(t) \mid x^k \rangle}{k!} t^k$;

(ii) $p(x) = \sum_{k=0}^{\deg p} \frac{\langle t^k \mid p \rangle}{k!} x^k$;

(iii) $\langle fg \mid x^n \rangle = \sum_{k=0}^{n} \binom{n}{k} \langle f \mid x^k \rangle \langle g \mid x^{n-k} \rangle$;

(iv) *if* $o(f) > \deg p$, *then* $\langle f \mid p\rangle = 0$;

(v) *if* $o(f_k) = k$ *for all* $k \geq 0$, *then* $\left\langle \sum_{k\geq 0} a_k f_k \middle| p \right\rangle = \sum_{k=0}^{\deg p} a_k \langle f_k \mid p\rangle$, *with* $\sum_{k\geq 0} a_k f_k$ *well-defined in* $\mathscr{F}$.

Proof.

(i) If $f(t) = \sum_{k\geq 0} a_k t^k/k!$, then $a_k = \langle f \mid x^k\rangle$ by definition.

(ii) Since $\langle t^k \mid p\rangle = p^{(k)}(0)$, this is Taylor's formula for the polynomial p.

(iii) Write $f = \sum a_k t^k/k!$ and $g = \sum b_k t^k/k!$.

The Cauchy product is $fg = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \right) \frac{t^n}{n!}$, and $a_k = \langle f \mid x^k\rangle$, $b_k = \langle g \mid x^k\rangle$.

(iv) If $p = \sum_{j=0}^{d} c_j x^j$ and $o(f) > d$, then $\langle f \mid x^j\rangle = 0$ for $0 \leq j \leq d$.

(v) Since $o(f_k) = k$, each coefficient of $\sum_{k\geq 0} a_k f_k$ has only finitely many contributions.

With $d = \deg p$, (iv) gives $\langle f_k \mid p\rangle = 0$ for $k > d$, proving the finite formula.

Structural Commentary: The umbral pairing dualizes coefficient calculus.
Key Insight: Order makes infinite formal sums act finitely on each polynomial. □

Theorem 9.3.9. *Multiplication by x Moves to Formal Differentiation*
Let $M_x : \mathscr{P} \to \mathscr{P}$ *be* $M_x p = xp$, *and let* $\partial_t : \mathscr{F} \to \mathscr{F}$ *be formal differentiation. Then:*

(i) $\langle f \mid M_x p\rangle = \langle \partial_t f \mid p\rangle$ *for all* $f \in \mathscr{F}$ *and* $p \in \mathscr{P}$.
(ii) $M_x^* = \partial_t$ *under the umbral pairing.*

Proof.

(i) By linearity in p, take $p = x^n$.
For $f = \sum_{k\geq 0} \frac{a_k}{k!} t^k$, $\partial_t f = \sum_{k\geq 0} \frac{a_{k+1}}{k!} t^k$, hence $\langle \partial_t f \mid x^n\rangle = a_{n+1} = \langle f \mid x^{n+1}\rangle = \langle f \mid M_x x^n\rangle$.

(ii) For $f \in \mathscr{F}$, the adjoint satisfies $\langle M_x^* f \mid p\rangle = \langle f \mid M_x p\rangle$ for all $p \in \mathscr{P}$.
By (i), $\langle M_x^* f \mid p\rangle = \langle \partial_t f \mid p\rangle$ for all p, so nondegeneracy gives $M_x^* f = \partial_t f$. Hence $M_x^* = \partial_t$.

Structural Commentary: M_x and ∂_t are adjoint under the umbral pairing.
Key Insight: Exponential normalization makes multiplication by x a coefficient shift. □

Definition 9.3.10. *Basic Umbral Functionals*
For $a \in \mathbb{C}$ and $p \in \mathscr{P}$:

(i) *Evaluation*: $\epsilon_a = e^{at}$ and $\langle e^{at} \mid p\rangle = p(a)$; hence $\epsilon_a \epsilon_b = \epsilon_{a+b}$.
(ii) *Forward difference*: $\Delta_a = e^{at} - 1$ and $\langle \Delta_a \mid p\rangle = p(a) - p(0)$.
(iii) *Abel functional*: te^{at} and $\langle te^{at} \mid p\rangle = p'(a)$.
(iv) *Gamma functional*: $(1-t)^{-1}$ and $\left\langle (1-t)^{-1} \middle| p \right\rangle = \int_0^\infty p(u) e^{-u}\, du$.
(v) *Integral functional*: $\left\langle \frac{e^{at}-1}{t} \middle| p \right\rangle = \int_0^a p(u)\, du$
(vi) *Bernoulli numbers*: for $a = 1$, $B_n = \left\langle \frac{t}{e^t - 1} \middle| x^n \right\rangle$.

Definition 9.3.11. *Formal Series as Polynomial Operators*
For $f(t) = \sum_{k\geq 0} a_k t^k/k! \in \mathscr{F}$:

(i) *Operator action*: define $f(t) \in \mathrm{End}_{\mathbb{C}}(\mathscr{P})$ by $f(t)p(x) := \sum_{k\geq 0} \frac{a_k}{k!} p^{(k)}(x)$, a finite sum for each $p \in \mathscr{P}$.

(ii) *Monomial action*: for $n \geq 0$, $f(t)x^n = \sum_{k=0}^{n} \binom{n}{k} a_k x^{n-k}$.

Theorem 9.3.12. *Functional-Operator Compatibility*
For $f, g \in \mathscr{F}$ *and* $p \in \mathscr{P}$:

(i) $\langle fg \mid p\rangle = \langle f \mid g(t)p\rangle$.
(ii) $\langle f \mid p\rangle = \langle t^0 \mid f(t)p\rangle$.

Proof.

(i) By linearity in p, take $p = x^n$. Write $f = \sum_i \frac{a_i}{i!} t^i$ and $g = \sum_j \frac{b_j}{j!} t^j$.

Then $\langle fg \mid x^n\rangle = \sum_{j=0}^{n} \binom{n}{j} a_{n-j} b_j = \left\langle f \middle| \sum_{j=0}^{n} \binom{n}{j} b_j x^{n-j} \right\rangle = \langle f \mid g(t)x^n\rangle$.

(ii) In (i), set $(f, g) = (t^0, F)$ with $F \in \mathscr{F}$. Since $t^0 F = F$ in $\mathscr{F}$, $\langle F \mid p\rangle = \langle t^0 F \mid p\rangle = \langle t^0 \mid F(t)p\rangle$.

Structural Commentary: Multiplication is adjoint to action.
Key Insight: One finite coefficient sum appears on both sides. □

Definition 9.3.13. *Basic Umbral Operators*
For $a \in \mathbb{C}$ and $p \in \mathscr{P}$:

(i) *Translation*: $e^{at} p(x) = p(x + a)$.
(ii) *Forward difference*: $(e^{at} - 1)p(x) = p(x + a) - p(x)$.
(iii) *Abel operator*: $te^{at} p(x) = p'(x + a)$.
(iv) *Gamma operator*: $(1-t)^{-1} p(x) = \int_0^\infty p(x + u) e^{-u}\, du$
(v) *Integral operator*: $\frac{e^{at}-1}{t} p(x) = \int_x^{x+a} p(u)\, du$

Theorem 9.3.14. *Shift-Invariant Operators*
For $T \in \mathrm{End}_{\mathbb{C}}(\mathscr{P})$, the following are equivalent:

(i) $T = f(t)$ *for some* $f \in \mathscr{F}$;
(ii) $[T, t] = 0$;
(iii) $[T, h(t)] = 0$ *for every delta series* h;
(iv) $[T, e^{at}] = 0$ *for all* $a \in \mathbb{C}$.

Proof.

(i) $\Rightarrow$ *(ii), (iii), (iv)* If $T = f(t)$, then $\forall u(t) \in \mathscr{F}$, $[T, u(t)] = [f(t), u(t)] = 0$, since $\mathscr{F}$ is commutative.
(iii) $\Rightarrow$ *(ii)* Since $o(t) = 1$, $(\forall h \in \mathscr{F},\ o(h) = 1 \Rightarrow [T, h(t)] = 0) \Rightarrow [T, t] = 0$.
(iv) $\Rightarrow$ *(ii)* For each $p \in \mathscr{P}$, $0 = [a^1]\,[T, e^{at}]p = [T, t]p$, so $[T, t] = 0$.
(ii) $\Rightarrow$ *(i)* Set $g(t) := \sum_{k\geq 0} \frac{\langle t^0 \mid Tx^k\rangle}{k!} t^k$. Let $a_k := \langle t^0 \mid Tx^k\rangle$.

Since $[T, t] = 0$, for $m, k \geq 0$, $\langle t^m \mid Tx^k\rangle = \langle t^0 \mid t^m Tx^k\rangle = \langle t^0 \mid Tt^m x^k\rangle = (k)_m a_{k-m}$, with $a_{k-m} = 0$ if $m > k$. Also $g(t) = \sum_{r\geq 0} \frac{a_r}{r!} t^r$ and $t^r x^k = (k)_r x^{k-r}$ give $\langle t^m \mid g(t)x^k\rangle = \sum_{r=0}^{k} \binom{k}{r} a_r \langle t^m \mid x^{k-r}\rangle = (k)_m a_{k-m}$.

Thus $\langle t^m \mid Tx^k - g(t)x^k\rangle = 0$ for all m, k. Since $\{t^m : m \geq 0\}$ separates $\mathscr{P}$, $T = g(t)$.

Structural Commentary: The commutant of t on $\mathscr{P}$ is $\mathscr{F}$.
Key Insight: The derivative is enough to recover every coefficient of the operator. □

Definition 9.3.15. *Sheffer, Associated, and Appell Sequences*
Let f be a delta series, g an invertible series, and $\deg s_n = n$.

(i) *Sheffer sequence*: $(s_n)_{n\geq 0}$ is Sheffer for (g, f) if $\langle g(t)f(t)^k \mid s_n(x)\rangle = n!\delta_{n,k}$ for $n, k \geq 0$.
(ii) *Associated sequence*: a Sheffer sequence for $(1, f)$.
(iii) *Appell sequence*: a Sheffer sequence for (g, t).

Theorem 9.3.16. *Existence and Expansion Theorems*
Let f be a delta series, g an invertible series, and $q_k := g(t)f(t)^k$.

(i) There is a unique sequence $(s_n)_{n\geq 0}$ with $\deg s_n = n$ and $\langle q_k \mid s_n\rangle = n!\delta_{n,k}$.
(ii) If $p_n := g(t)s_n$, then (s_n) is Sheffer for (g, f) iff (p_n) is associated with f.
(iii) For all $h \in \mathscr{F}$, $h = \sum_{k\geq 0} \frac{\langle h|s_k\rangle}{k!} q_k$.
(iv) For all $p \in \mathscr{P}$, $p = \sum_{k=0}^{\deg p} \frac{\langle q_k|p\rangle}{k!} s_k$.

Proof.

(i) Since $o(q_k) = k$, the matrix $A_n = (\langle q_i \mid x^j\rangle)_{0\leq i,j\leq n}$ is triangular with nonzero diagonal.
Hence the equations $\langle q_k \mid s_n\rangle = n!\delta_{n,k}$ have a unique solution $s_n \in \mathscr{P}_{\leq n}$, necessarily with $\deg s_n = n$.
(ii) By Theorem 9.3.12, $\langle f(t)^k \mid p_n\rangle = \langle f(t)^k \mid g(t)s_n\rangle = \langle q_k \mid s_n\rangle$.
(iii) Since $o(q_k) = k$, every $h \in \mathscr{F}$ has a unique expansion $h = \sum c_k q_k$.
Pairing with s_n gives $\langle h \mid s_n\rangle = c_n n!$, so $c_n = \langle h \mid s_n\rangle/n!$.
(iv) Let $d = \deg p$ and $r := p - \sum_{k=0}^{d} \langle q_k \mid p\rangle s_k/k!$.

Then $\langle q_j \mid r\rangle = 0$ for all $j \leq d$, and also for $j > d$ by $o(q_j) > d$. Since the q_j separate $\mathscr{P}$, $r = 0$.

Structural Commentary: (s_n) is dual to $q_k = gf^k$.
Key Insight: The order filtration makes the orthogonality equations triangular. □

Theorem 9.3.17. *Generating and Operator Characterizations*
Let f be a delta series with compositional inverse $\overline{f}$, let $g \in \mathscr{F}^\times$, and let $(p_n), (s_n)$ be polynomial sequences with $\deg p_n = \deg s_n = n$. Set $P(y, u) := \sum_{n\geq 0} \frac{p_n(y)}{n!} u^n$ and $S(y, u) := \sum_{n\geq 0} \frac{s_n(y)}{n!} u^n$.

(i) (p_n) is associated with f iff $P(y, u) = e^{y\overline{f}(u)}$
(ii) (s_n) is Sheffer for (g, f) iff $S(y, u) = g(\overline{f}(u))^{-1} e^{y\overline{f}(u)}$
(iii) (p_n) is associated with f iff $p_n(0) = \delta_{n,0}$, $f(t)p_n = np_{n-1}$, where $p_{-1} := 0$.
(iv) (s_n) is Sheffer for (γ, f) for some $\gamma \in \mathscr{F}^\times$ iff $f(t)s_n = ns_{n-1}$, where $s_{-1} := 0$.
(v) If (p_n) is associated with f, then $p_n(x) = \sum_{k=0}^{n} \frac{\langle \overline{f}(t)^k|x^n\rangle}{k!} x^k$
(vi) If (s_n) is Sheffer for (g, f), then $s_n(x) = \sum_{k=0}^{n} \frac{1}{k!} \left\langle g(\overline{f}(t))^{-1}\overline{f}(t)^k \middle| x^n \right\rangle x^k$.

Proof.

(i) $(\Rightarrow)$ By Theorem 9.3.16, $e^{yt} = \sum_{n\geq 0} \frac{p_n(y)}{n!} f(t)^n$. Substitute $t = \overline{f}(u)$ to obtain $P(y, u) = e^{y\overline{f}(u)}$.

(i) $(\Leftarrow)$ If $P(x, u) = e^{x\overline{f}(u)}$, then for all $k \geq 0$, $\sum_{n\geq 0} \frac{\langle f(t)^k|p_n\rangle}{n!} u^n = \langle f(t)^k \mid P(x, u)\rangle = f(\overline{f}(u))^k = u^k$.

Thus $\langle f(t)^k \mid p_n\rangle = n!\delta_{n,k}$.

(ii) $(\Rightarrow)$ By Theorem 9.3.16, $e^{yt} = \sum_{n\geq 0} \frac{s_n(y)}{n!} g(t)f(t)^n$. Substitute $t = \overline{f}(u)$ and divide by $g(\overline{f}(u))$.

(ii) $(\Leftarrow)$ If $S(x, u) = g(\overline{f}(u))^{-1} e^{x\overline{f}(u)}$, then $\sum_{n\geq 0} \frac{\langle g(t)f(t)^k|s_n\rangle}{n!} u^n = \langle g(t)f(t)^k \mid S(x, u)\rangle = u^k$,

so $\langle g(t)f(t)^k \mid s_n\rangle = n!\delta_{n,k}$.

(*iii*) ($\Rightarrow$) If (p_n) is associated, then $p_n(0) = \langle 1 \mid p_n\rangle = \delta_{n,0}$ and, for all k, $\langle f(t)^k \mid f(t)p_n - np_{n-1}\rangle = 0$.
Since $\{f(t)^k : k \geq 0\}$ separates $\mathscr{P}$, $f(t)p_n = np_{n-1}$.
(*iii*) ($\Leftarrow$) The lowering relation gives $f(t)^k p_n = (n)_k p_{n-k}$.
Hence $\langle f(t)^k \mid p_n\rangle = \langle 1 \mid f(t)^k p_n\rangle = (n)_k p_{n-k}(0) = n!\delta_{n,k}$.
(*iv*) ($\Rightarrow$) If (s_n) is Sheffer for (γ, f), then for all k, $\langle \gamma(t)f(t)^k \mid f(t)s_n - ns_{n-1}\rangle = 0$.
Since $\{\gamma(t)f(t)^k : k \geq 0\}$ separates $\mathscr{P}$, $f(t)s_n = ns_{n-1}$.
(*iv*) ($\Leftarrow$) Define $L \in \mathscr{P}'$ by $L(s_0) = 1$ and $L(s_n) = 0$ for $n > 0$, and put $\gamma := \Phi(L)$. Then $\langle \gamma \mid 1\rangle = L(1) \neq 0$, so $\gamma \in \mathscr{F}^\times$, and $\langle \gamma(t)f(t)^k \mid s_n\rangle = \langle \gamma(t) \mid f(t)^k s_n\rangle = (n)_k L(s_{n-k}) = n!\delta_{n,k}$.
(*v*) From (i), $e^{x\overline{f}(u)} = \sum_{k\geq 0} \frac{\overline{f}(u)^k}{k!} x^k$. Applying $n![u^n]$ gives the stated formula for p_n.
(*vi*) From (ii), $g(\overline{f}(u))^{-1} e^{x\overline{f}(u)} = \sum_{k\geq 0} \frac{g(\overline{f}(u))^{-1}\overline{f}(u)^k}{k!} x^k$. Applying $n![u^n]$ gives the stated formula for s_n.

Structural Commentary: $\overline{f}$ converts f-lowering to exponential generating functions.
Key Insight: Generating functions package the whole triangular duality into one formal identity. □

Definition 9.3.18. *Binomial Type and Sheffer Identity*
Let (p_n) and (s_n) be polynomial sequences.

(i) *Binomial type*: (p_n) is of binomial type if $p_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} p_k(y) p_{n-k}(x)$.

(ii) *Sheffer identity*: (s_n) satisfies the Sheffer identity relative to (p_n) if $s_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} p_k(y) s_{n-k}(x)$.

Theorem 9.3.19. *Action and Binomial Identities*
Let (s_n) be Sheffer for (g, f), let (p_n) be associated with f, and write $E_y := e^{yt}$. Then:

(i) for $h \in \mathscr{F}$ and $n \geq 0$, $h(t)s_n = \sum_{k=0}^{n} \binom{n}{k} \langle h \mid s_k\rangle p_{n-k}$;
(ii) (p_n) is associated with a delta series iff (p_n) is of binomial type;
(iii) (s_n) is Sheffer for some $g \in \mathscr{F}^\times$ and same f iff (s_n) satisfies the Sheffer identity relative to (p_n).

Proof.

(i) By Theorem 9.3.16, $h = \sum_{k\geq 0} \frac{\langle h \mid s_k\rangle}{k!} g(t) f(t)^k$.

Since $g(t)s_n = p_n$ and $f(t)^k p_n = (n)_k p_{n-k}$, $h(t)s_n = \sum_{k=0}^{n} \frac{\langle h \mid s_k\rangle}{k!} (n)_k p_{n-k}$.
(*ii*) ($\Rightarrow$) If (p_n) is associated, apply (i) with $s_n = p_n$, $g = 1$, and $h = E_y$.
Since $\langle E_y \mid p_k\rangle = p_k(y)$ and $E_y p_n(x) = p_n(x+y)$, (p_n) is of binomial type.
(*ii*) ($\Leftarrow$) If (p_n) is of binomial type, then $p_n(0) = \delta_{n,0}$. Define $Qp_n := np_{n-1}$. The binomial identity gives $QE_y p_n = E_y Q p_n$ for all n, y, so $[Q, E_y] = 0$. By Theorem 9.3.14, $Q = f(t)$ for some $f \in \mathscr{F}$.
Since Q lowers degree by one, $o(f) = 1$; Theorem 9.3.17 makes (p_n) associated with f.
(*iii*) ($\Rightarrow$) If (s_n) is Sheffer for (g, f), apply (i) with $h = E_y$.

Since $\langle E_y \mid s_k\rangle = s_k(y)$ and $E_y s_n(x) = s_n(x+y)$, $s_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} s_k(y) p_{n-k}(x)$.

Relabel and swap x, y to get the Sheffer identity relative to (p_n).
(*iii*) ($\Leftarrow$) If the Sheffer identity holds, apply $f(t)$ in the y-variable and set $y = 0$.
Since $f(t)p_k(y)|_{y=0} = kp_{k-1}(0) = k\delta_{k,1}$, $f(t)s_n = ns_{n-1}$.
By Theorem 9.3.17, (s_n) is Sheffer for (g, f) for some $g \in \mathscr{F}^\times$.

Structural Commentary: Translation turns the operator formula into addition laws for polynomial sequences.
Key Insight: Binomial identities are the visible polynomial shadow of a delta operator. □

Example 9.3.20. *Standard Sheffer Examples*

(i) *Monomials*: $p_n(x) = x^n$ is associated with $f(t) = t$; the binomial identity is the usual binomial theorem.
(ii) *Lower factorials*: $(x)_n$ is associated with $f(t) = e^t - 1$, with $(1+t)^y = e^{y\log(1+t)} = \sum_{k\geq 0} \frac{(y)_k}{k!} t^k$,

and Vandermonde identity $(x+y)_n = \sum_{k=0}^{n} \binom{n}{k} (x)_k (y)_{n-k}$.

(iii) *Abel polynomials*: associated with $f(t) = te^{at}$, $A_n(x; a) = x(x - an)^{n-1}$.
(iv) *Hermite polynomials*: for $g(t) = e^{t^2/2}$, the polynomial is $H_n(x) = e^{-t^2/2} x^n$, where $e^{yt - t^2/2} = \sum_{k\geq 0} \frac{H_k(y)}{k!} t^k$.
(v) *Laguerre polynomials*: order α, Sheffer for $g(t) = (1-t)^{-\alpha-1}$, $f(t) = \frac{t}{t-1}$, $\frac{1}{(1-t)^{\alpha+1}} e^{yt/(t-1)} = \sum_{k\geq 0} \frac{L_k^{(\alpha)}(y)}{k!} t^k$.

Definition 9.3.21. *Umbral Operators, Continuity, and Adjoints*
Let (s_n) be Sheffer for (g, f), and let (p_n) be associated with f.

(i) *Sheffer operator*: $\lambda_{g,f} \in \mathcal{L}(\mathscr{P})$ is defined by $\lambda_{g,f}(x^n) = s_n(x)$.
(ii) *Umbral operator*: $\lambda_f \in \mathcal{L}(\mathscr{P})$ is defined by $\lambda_f(x^n) = p_n(x)$.
(iii) *Sheffer/umbral shift*: $\theta_{g,f} s_n = s_{n+1}$; in the associated case, $\theta_f p_n = p_{n+1}$.
(iv) *Continuous operator*: $T \in \mathcal{L}(\mathscr{F})$ is continuous iff $o(f_k) \to \infty \Longrightarrow o(Tf_k) \to \infty$.

Equivalently, for every defined formal sum with $o(f_k) \to \infty$, $T\left(\sum_{k\geq 0} a_k f_k\right) = \sum_{k\geq 0} a_k T(f_k)$

(v) *Algebraic adjoint*: for $\tau \in \mathcal{L}(\mathscr{P})$, $\tau^\times \in \mathcal{L}(\mathscr{F})$ is defined by $\langle \tau^\times h \mid p\rangle = \langle h \mid \tau p\rangle$ for $h \in \mathscr{F}$, $p \in \mathscr{P}$.
This is the dual-map adjoint of Definition 4.2.4, not an inner-product adjoint.

Theorem 9.3.22. *Continuous Operators Are Exactly Operator Adjoints*
Let $\mathcal{L}_{\text{cont}}(\mathscr{F})$ denote the continuous linear operators on $\mathscr{F}$.
Then $(-)^\times : \mathcal{L}(\mathscr{P}) \xrightarrow{\sim} \mathcal{L}_{\text{cont}}(\mathscr{F})$, $\tau \longmapsto \tau^\times$ is a vector-space isomorphism.

Proof.

($\Rightarrow$) Let $\tau \in \mathcal{L}(\mathscr{P})$ and $o(h_j) \to \infty$. For fixed n, $\tau x^n \in \mathscr{P}$, hence $\langle \tau^\times h_j \mid x^n \rangle = \langle h_j \mid \tau x^n \rangle = 0$ for all sufficiently large j. Thus $o(\tau^\times h_j) \to \infty$, so $\tau^\times \in \mathcal{L}_{\text{cont}}(\mathscr{F})$.
If $\tau^\times = 0$, then $\langle h \mid \tau p \rangle = 0$ for all h, p, hence $\tau = 0$; the map is injective.

($\Leftarrow$) Let $T \in \mathcal{L}_{\text{cont}}(\mathscr{F})$. Define $\tau_T x^n := \sum_{k \geq 0} \frac{\langle T t^k \mid x^n \rangle}{k!} x^k$. Since $o(Tt^k) \to \infty$, the sum is finite for each n.
For all m, n, $\langle \tau_T^\times t^m \mid x^n \rangle = \langle t^m \mid \tau_T x^n \rangle = \langle Tt^m \mid x^n \rangle$. Thus $\tau_T^\times t^m = Tt^m$ for every m.
By continuity, $\tau_T^\times = T$, so the map is surjective.

Structural Commentary: Continuity is the finiteness condition for dualising polynomial operators.
Key Insight: Finiteness on each polynomial degree is the hidden topology. □

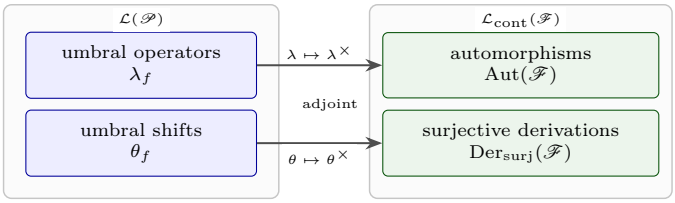

Figure 187: Adjoints identify $\mathcal{L}(\mathscr{P})$ with $\mathcal{L}_{\text{cont}}(\mathscr{F})$ and map the umbral classes shown.

Theorem 9.3.23. *Umbral Operators and Adjoint Automorphisms*
Let f be a delta series, let $\overline{f}$ be its compositional inverse, let (p_n) be associated with f, and set $\lambda_f(x^n) = p_n(x)$.
For $C_{\overline{f}}(h) := h(\overline{f})$ and $\lambda \in \mathcal{L}(\mathscr{P})$:

(i) $\lambda = \lambda_f$ *iff* $\lambda^\times = C_{\overline{f}}$; *in this case* $\lambda_f^\times f = t$.
(ii) λ *is umbral iff* $\lambda^\times \in \text{Aut}_{\mathbb{C}\text{-}alg}(\mathscr{F})$.

Proof.

(i) ($\Rightarrow$) If $\lambda = \lambda_f$, then $\langle \lambda^\times f^k \mid x^n \rangle = \langle f^k \mid p_n \rangle = n!\delta_{n,k} = \langle t^k \mid x^n \rangle$, so $\lambda^\times f^k = t^k$. For $h \in \mathscr{F}$, put $q := h(\overline{f})$, so $h = q(f)$. Continuity gives $\lambda^\times h = q(t) = h(\overline{f}) = C_{\overline{f}} h$. Hence $\lambda^\times = C_{\overline{f}}$ and $\lambda_f^\times f = t$.
(i) ($\Leftarrow$) If $\lambda^\times = C_{\overline{f}}$, set $r_n := \lambda x^n$. Then $\langle f^k \mid r_n \rangle = \langle \lambda^\times f^k \mid x^n \rangle = \langle t^k \mid x^n \rangle = n!\delta_{n,k}$.
Thus (r_n) is associated with f, so $r_n = p_n$ and $\lambda = \lambda_f$.
(ii) ($\Rightarrow$) If λ is umbral, then $\lambda = \lambda_u$ for some delta series u. By (i), $\lambda^\times = C_{\overline{u}} \in \text{Aut}_{\mathbb{C}\text{-alg}}(\mathscr{F})$.
(ii) ($\Leftarrow$) Let $A := \lambda^\times \in \text{Aut}_{\mathbb{C}\text{-alg}}(\mathscr{F})$ and set $u := A^{-1}t$. Since $A^{-1}((t)) = (t)$ and $A^{-1}((t)^2) = (t)^2$, $u \in (t) \setminus (t)^2$, so $o(u) = 1$. With $r_n := \lambda x^n$, $\langle u^k \mid r_n \rangle = \langle Au^k \mid x^n \rangle = \langle t^k \mid x^n \rangle = n!\delta_{n,k}$.
Hence (r_n) is associated with u, so $\lambda = \lambda_u$ is umbral.

Structural Commentary: Umbral adjoints are substitutions.
Key Insight: The delta series is $A^{-1}t$. □

Definition 9.3.24. *Umbral Composition*

For polynomial sequences (p_n) and (q_n) with $p_n(x) = \sum_{k=0}^n p_{n,k} x^k$, define $(p \circ_U q)_n(x) := \sum_{k=0}^n p_{n,k} q_k(x)$.

Theorem 9.3.25. *Group Law for Umbral Operators*
Let f, g be delta series, and let (p_n) and (q_n) be associated with f and g. Then:

(i) $\lambda_g \lambda_f = \lambda_{f \circ g}$.
(ii) $\lambda_f^{-1} = \lambda_{\overline{f}}$.
(iii) $(p \circ_U q)_n = (\lambda_g \lambda_f) x^n$; *hence associated sequences form a group under* $\circ_U$.

Proof.

(i) By $(AB)^\times = B^\times A^\times$ and Theorem 9.3.23, for $h \in \mathscr{F}$, $(\lambda_g \lambda_f)^\times h = \lambda_f^\times \lambda_g^\times h = h(\overline{g}(\overline{f})) = h(\overline{f \circ g})$. QED.

(ii) Put $g = \overline{f}$ in (i). Then $\lambda_{\overline{f}} \lambda_f = \lambda_t = I$ and $\lambda_f \lambda_{\overline{f}} = \lambda_t = I$, so $\lambda_f^{-1} = \lambda_{\overline{f}}$.

(iii) If $p_n = \sum_{k=0}^n p_{n,k} x^k$, then $(\lambda_g \lambda_f) x^n = \lambda_g p_n = \sum_{k=0}^n p_{n,k} q_k = (p \circ_U q)_n$.
By (i) and (ii), the operation inherits associativity, identity, and inverses from operator composition.

Structural Commentary: Umbral composition is substitution of delta series.
Key Insight: Adjoints reverse the operator composition. □

Proposition 9.3.26. *Sheffer Operators*
Let f, k be delta series, $g, h \in \mathscr{F}^\times$, (s_n) Sheffer for (g, f), and (p_n) associated with f. Then:

(i) $s_n = g(t)^{-1} p_n$ *and* $\lambda_{g,f} = g(t)^{-1} \lambda_f$.
(ii) $\lambda_{g,f} \lambda_{h,k} = \lambda_{g(h \circ f),\, k \circ f}$.
(iii) $\lambda_{g,f}^{-1} = \lambda_{g^{-1}(\overline{f}),\, \overline{f}}$.

Proof.

(i) Theorem 9.3.16 gives $g(t)s_n = p_n$, hence $s_n = g(t)^{-1}p_n$ and $\lambda_{g,f} = g(t)^{-1}\lambda_f$.
(ii) For $a \in \mathscr{F}$, Theorem 9.3.23 gives $\lambda_f a(t) = a(f(t))\lambda_f$.
Thus $\lambda_{g,f}\lambda_{h,k} = g(t)^{-1}\lambda_f h(t)^{-1}\lambda_k = g(t)^{-1}h(f(t))^{-1}\lambda_{k\circ f} = \lambda_{g(h\circ f),\,k\circ f}$.
(iii) In (ii), take $(h,k) = (g^{-1}(\overline{f}), \overline{f})$ to get $\lambda_{g,f}\lambda_{g^{-1}(\overline{f}),\,\overline{f}} = \lambda_{1,t} = I$.
Reversing the factors gives $\lambda_{g^{-1}(\overline{f}),\,\overline{f}}\lambda_{g,f} = \lambda_{g^{-1}(\overline{f})(g\circ\overline{f}),\,f\circ\overline{f}} = \lambda_{1,t} = I$.

Structural Commentary: Weight then umbral substitution.
Key Insight: $(g,f)(h,k) = (g(h\circ f), k\circ f)$. □

Theorem 9.3.27. *Umbral Shift Derivations*
Let f be a delta series and let θ_f be its umbral shift.
Define the continuous derivation $\partial_f : \mathscr{F} \to \mathscr{F}$ by $\partial_f(f^k) = kf^{k-1}$, $k \geq 0$. Then:

(i) $\theta_f^{\times} = \partial_f$.
(ii) $\partial_f f = 1$.
(iii) If g is a delta series, then $\partial_g = (\partial_g f)\partial_f$.

Proof.

(i) Let (p_n) be associated with f.
For $k,n \geq 0$, $\langle \theta_f^{\times} f^k \mid p_n\rangle = \langle f^k \mid p_{n+1}\rangle = (n+1)!\delta_{n+1,k} = \langle kf^{k-1} \mid p_n\rangle$.
Hence $\theta_f^{\times} f^k = kf^{k-1}$ for all k, so $\theta_f^{\times} = \partial_f$ by continuity.
(ii) By definition of ∂_f, $\partial_f f = \partial_f(f^1) = 1\cdot f^0 = 1$.
(iii) It suffices by continuity to test on f^k: $(\partial_g f)\partial_f(f^k) = k(\partial_g f)f^{k-1} = \partial_g(f^k)$.

Structural Commentary: Umbral shift is delta-coordinate differentiation.
Key Insight: The associated basis turns shifts into derivatives. □

Theorem 9.3.28. *Umbral Shifts and Surjective Derivations*
For $\theta \in \mathcal{L}(\mathscr{P})$, θ is an umbral shift $\iff$ $\theta^{\times}$ is a surjective derivation of $\mathscr{F}$.

Proof.

($\Rightarrow$) If $\theta = \theta_f$, then $\theta^{\times} = \partial_f$ by Theorem 9.3.27. Since $\partial_f\left(\frac{f^{k+1}}{k+1}\right) = f^k$, $k \geq 0$, ∂_f is a surjective derivation.
($\Leftarrow$) Let $D := \theta^{\times}$ be a surjective derivation. Choose $u \in \mathscr{F}$ with $Du = 1$ and set $f := u - [t^0]u$.
Then $Df = 1$ and, since D lowers order by one, $o(f) = 1$. Let (p_n) be associated with f.
For $k,n \geq 0$, $\langle f^k \mid \theta p_n\rangle = \langle Df^k \mid p_n\rangle = \langle kf^{k-1} \mid p_n\rangle = \langle f^k \mid p_{n+1}\rangle$.
The functionals f^k separate $\mathscr{P}$, so $\theta p_n = p_{n+1}$ and $\theta = \theta_f$.

Structural Commentary: Surjective derivations are derivatives in delta coordinates.
Key Insight: Surjectivity chooses the coordinate whose derivative is 1. □

9.4 Umbral Recurrences and Variants

Theorem 9.4.1. *Recurrence and Transfer Formulas*
Let (p_n) be associated with the delta series f, and set $u(t) := f(t)/t \in \mathscr{F}^{\times}$. Then:

(i) $p_{n+1} = x[f'(t)]^{-1}p_n$ *for* $n \geq 0$.
(ii) $p_n = f'(t)u(t)^{-n-1}x^n = xu(t)^{-n}x^{n-1}$ *for* $n \geq 1$.

Proof.

(i) By Theorem 9.3.27, $\theta_t = (\partial_t f)\theta_f = f'(t)\theta_f$. Since $\theta_t = x$ and $\theta_f p_n = p_{n+1}$, $p_{n+1} = x[f'(t)]^{-1}p_n$.
(ii) Put $q_n := f'u^{-n-1}x^n$. Then $q_0 = f'u^{-1}1 = 1 = p_0$ and, using $xa(t) = a(t)x - a'(t)$,
$q_{n+1} = f'u^{-n-2}x^{n+1} = xu^{-n-1}x^n = x(f')^{-1}q_n$. Thus $q_n = p_n$ by (i).
The same commutator gives $xu^{-n}x^{n-1} = u^{-n}x^n + nu^{-n-1}u'x^{n-1} = f'u^{-n-1}x^n$.

Structural Commentary: The recurrence determines (p_n); transfer solves it using $f = tu$.
Key Insight: The Pincherle commutator converts the shift recurrence into closed form. □

Corollary 9.4.2. *Sheffer Shift Formula*
Let (s_n) be Sheffer for (g,f), with f delta and $g \in \mathscr{F}^{\times}$. Write $f' = f'(t)$ and $g' = g'(t)$. Then:

(i) $\theta_{g,f} = g^{-1}\theta_f g = \left(x - \frac{g'}{g}\right)(f')^{-1}$.
(ii) $s_{n+1} = \left(x - \frac{g'}{g}\right)(f')^{-1}s_n$ *for* $n \geq 0$.

Proof.

(i) Since gs_n is associated with f, $\theta_{g,f} = g^{-1}\theta_f g$. By Theorem 9.4.1, $\theta_f = x(f')^{-1}$.
Since f' and g commute, and $g' = gx - xg$, $g^{-1}\theta_f g = g^{-1}x(f')^{-1}g = g^{-1}xg(f')^{-1} = \left(x - \frac{g'}{g}\right)(f')^{-1}$.
(ii) By definition of the Sheffer shift, $\theta_{g,f}s_n = s_{n+1}$ for $n \geq 0$. Using (i), $s_{n+1} = \theta_{g,f}s_n = \left(x - \frac{g'}{g}\right)(f')^{-1}s_n$.

Structural Commentary: Sheffer shifts are conjugated associated shifts.
Key Insight: Conjugation by g contributes the correction g'/g. □

Method 9.4.3. *Building Sheffer Sequences*
To construct the Sheffer sequence for (g,f):

(i) Verify $o(f) = 1$ and $o(g) = 0$.
(ii) If $\overline{f}$ is explicit, compute $S(y,t) := g(\overline{f}(t))^{-1}e^{y\overline{f}(t)}$, where $s_n(y) = n![t^n]S(y,t)$.
(iii) If $\overline{f}$ is unavailable, solve recursively from $f(t)s_n = ns_{n-1}$, where $\langle g(t)f(t)^k \mid s_n\rangle = n!\delta_{n,k}$.

(iv) For associated sequences, set $g = 1$ and use Theorem 9.4.1.

Definition 9.4.4. *Stirling and Exponential Polynomials*

(i) *Stirling numbers*: $S(n,k)$ are defined by $x^n = \sum_{k=0}^{n} S(n,k)(x)_k$.

(ii) *Exponential polynomials*: define $\phi_n(x) := \sum_{k=0}^{n} S(n,k)x^k$.

(iii) *Boundary convention*: $S(n,-1) = 0$ and $S(n,k) = 0$ for $k > n$.

Example 9.4.5. *Lower Factorials and Stirling Numbers*
For $f(t) = e^t - 1$, $f'(t) = e^t$, hence $p_{n+1}(x) = xe^{-t}p_n(x) = xp_n(x-1)$. With $p_0 = 1$, this gives $p_n = (x)_n$.
For $f(t) = \log(1+t)$, $\overline{f}(t) = e^t - 1$, and the associated sequence is (ϕ_n).
The inverse umbral relation is $x^n = \sum_{k=0}^{n} S(n,k)(x)_k$, and $\phi_n(x) = \sum_{k=0}^{n} S(n,k)x^k$.
The recurrence formula gives $\phi_{n+1}(x) = x(1+t)\phi_n(x) = x\big(\phi_n(x) + \phi_n'(x)\big)$,
so coefficient comparison yields $S(n+1,k) = S(n,k-1) + kS(n,k)$.

Definition 9.4.6. *q- and Logarithmic Umbral Variants*

(i) *Gaussian coefficient*: for $0 \le k \le n$, $\binom{n}{k}_q := \prod_{j=1}^{n}(1-q^j)/\left(\prod_{j=1}^{k}(1-q^j)\prod_{j=1}^{n-k}(1-q^j)\right)$.

(ii) *q-umbral calculus*: the deformation $\binom{n}{k} \rightsquigarrow \binom{n}{k}_q$.

(iii) *Logarithmic umbral calculus*: the logarithmic-type variant based on harmonic logarithms.

Remark 9.4.7. *Further Directions*
Classical umbral methods extend to many polynomial families and operations:

(i) *Families*: lower factorial, Stirling, exponential, Gould, central factorial, Abel, Mittag–Leffler, Bessel, Bell, Hermite, Bernoulli, Euler, Laguerre, Poisson–Charlier, actuarial, Meixner, Pidduck, Narumi, Boole, Peters, Mahler, and Mott polynomials.

(ii) *Operations*: connection constants, duplication formulas, Lagrange inversion, cross sequences, Steffensen sequences, operational formulas, inverse relations, recurrence solving, and binomial convolution.

(iii) *Variants*: q-umbral and logarithmic umbral calculi replace or modify the binomial structure through the objects in Definition 9.4.6.

10 Fourier Structure

10.1 Fourier Orthogonality

Fourier structure studies trigonometric series through orthogonality, projection, kernels, and convergence, with inner-product geometry reappearing throughout.

Definition 10.1.1. *Core Fourier Objects*

(i) A *trigonometric polynomial* is a finite sum $a_0 + \sum_{n=1}^{N}(a_n\cos nx + b_n \sin nx)$, equivalently $\sum_{n=-N}^{N}\alpha_n e^{inx}$.

(ii) If f is Riemann integrable on $[-\pi,\pi]$, its *Fourier coefficients* are $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-imx}\,dx$, $m \in \mathbb{Z}$, and $f(x) \sim_{\text{coef}} \sum_{m=-\infty}^{\infty}\widehat{f}(m)e^{imx}$ records coefficients, not convergence.

(iii) A sequence (ϕ_n) on $[a,b]$ is *orthonormal* if $\int_a^b \phi_n(x)\overline{\phi_m(x)}\,dx = 0 \quad (n \ne m)$ and $\int_a^b |\phi_n(x)|^2\,dx = 1$.
Its Fourier coefficients are $c_n := \int_a^b f(t)\overline{\phi_n(t)}\,dt$.

Key Insight: Fourier coefficients are orthonormal coordinates; convergence is a separate analytic question.

Definition 10.1.2. *Period-One Fourier Normalization*
Let $C(\mathbb{R}/\mathbb{Z};\mathbb{C})$ denote the continuous functions $f : \mathbb{R} \to \mathbb{C}$ satisfying $f(x+1) = f(x)$ for all x.
For $f, g \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, set $\langle f,g\rangle_{\mathbb{T}} := \int_0^1 f(x)\overline{g(x)}\,dx$; $\|f\|_{L^2(\mathbb{T})} := \sqrt{\langle f,f\rangle_{\mathbb{T}}}$; and $e_n(x) := e^{2\pi inx}$ for $n \in \mathbb{Z}$.
The *period-one Fourier coefficient* is $\widehat{f}_{\mathbb{T}}(n) := \langle f, e_n\rangle_{\mathbb{T}} = \int_0^1 f(x)e^{-2\pi inx}\,dx$.
A period-one trigonometric polynomial is a finite sum $\sum_{n=-N}^{N} c_n e_n$.

Perspective 10.1.3. *Two Equivalent Fourier Clocks*
The period-one and 2π-periodic conventions differ by $t = 2\pi x$; only normalizing constants change.
Key Insight: Changing Fourier clocks changes constants, not the projection geometry.

Proposition 10.1.4. *Characters Are an Orthonormal System*
For $m, n \in \mathbb{Z}$ and $N \ge 0$:

(i) $\langle e_n, e_m\rangle_{\mathbb{T}} = \delta_{n,m}$.

(ii) If $P = \sum_{k=-N}^{N} c_k e_k$, *then* $\langle P, e_n\rangle_{\mathbb{T}} = c_n$ *for* $-N \le n \le N$, *and* $\|P\|^2_{L^2(\mathbb{T})} = \sum_{k=-N}^{N}|c_k|^2$.

Proof.

(i) $\langle e_n, e_m\rangle_{\mathbb{T}} = \int_0^1 e^{2\pi i(n-m)x}\,dx$. This equals 1 if $n = m$ and $\frac{e^{2\pi i(n-m)}-1}{2\pi i(n-m)} = 0$ if $n \ne m$.

(ii) By (i), $\langle P, e_n\rangle_{\mathbb{T}} = \sum_{k=-N}^{N} c_k\delta_{k,n} = c_n$, and $\|P\|^2_{L^2(\mathbb{T})} = \langle P,P\rangle_{\mathbb{T}} = \sum_{j,k=-N}^{N} c_j\overline{c_k}\delta_{j,k} = \sum_{k=-N}^{N}|c_k|^2$.

Structural Commentary: The Gram matrix of $\{e_{-N},\dots,e_N\}$ is the identity.
Key Insight: Character orthogonality kills all cross terms. □

10.2 Fourier Approximation

Perspective 10.2.1. *Fourier and Plancherel Roadmap*
Path: orthonormal characters, Fejer density, best L^2 projection, then Parseval/Plancherel.
Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

Definition 10.2.2. *Periodic Convolution and Fejer Kernels*
For $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, define their *periodic convolution* as $(f * g)(x) := \int_0^1 f(y)g(x-y)\,dy$.
For $N \geq 1$, the *Fejer kernel* is $F_N(x) := \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e_n(x) = \frac{1}{N}\left|\sum_{n=0}^{N-1} e_n(x)\right|^2$.

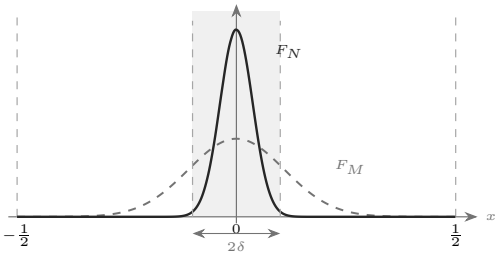

Figure 188: The Fejer kernels are nonnegative trigonometric polynomial approximate identities on $\mathbb{R}/\mathbb{Z}$.

Theorem 10.2.3. *Fejer Approximation on $\mathbb{R}/\mathbb{Z}$*
Let F_N be the period-one Fejer kernel, $N \geq 1$. Then:
(i) F_N is a nonnegative trigonometric polynomial and $\int_0^1 F_N(x)\,dx = 1$.
(ii) For each $0 < \delta < 1/2$, $\sup_{\delta \leq x \leq 1-\delta} F_N(x) \to 0$.
(iii) If $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then $f * F_N$ is a trigonometric polynomial for each N.
(iv) If $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then $f * F_N \rightrightarrows f$ on $\mathbb{R}/\mathbb{Z}$.

Proof.

(i) By definition, $F_N = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e_n$ is trigonometric, and $F_N = \frac{1}{N}\left|\sum_{n=0}^{N-1} e_n\right|^2 \geq 0$.
Since $\int_0^1 e_n(x)\,dx = 0$ for $n \neq 0$, the e_0 coefficient gives $\int_0^1 F_N(x)\,dx = 1$.

(ii) For $x \notin \mathbb{Z}$, $F_N(x) = \frac{\sin^2(\pi N x)}{N \sin^2(\pi x)}$. If $m_\delta := \min_{\delta \leq x \leq 1-\delta} \sin^2(\pi x) > 0$, then $0 \leq F_N(x) \leq \frac{1}{N m_\delta}$ on $[\delta, 1-\delta]$.

(iii) Since $f * e_n = \widehat{f}_{\mathbb{T}}(n) e_n$ and F_N is a finite linear combination of the e_n, $f * F_N$ is a trigonometric polynomial.

(iv) Write $(f * F_N)(x) = \int_0^1 f(x-t) F_N(t)\,dt$. Let $M = \|f\|_\infty$ and $\varepsilon > 0$.
Uniform continuity gives $0 < \delta < 1/2$ such that $|f(x-t) - f(x)| < \varepsilon$ for all x and $t \in [0, \delta] \cup [1-\delta, 1]$.
By (i), $F_N \geq 0$ and $\int_0^1 F_N(t)\,dt = 1$; hence $|(f * F_N)(x) - f(x)| \leq \varepsilon + 2M \sup_{\delta \leq t \leq 1-\delta} F_N(t)$.
By (ii), $\limsup_{N \to \infty} \|f * F_N - f\|_\infty \leq \varepsilon$. Since ε is arbitrary, $f * F_N \rightrightarrows f$.

Structural Commentary: Fejer kernels are positive approximate identities.
Key Insight: Normalized mass concentrating at 0 forces uniform convergence. □

Perspective 10.2.4. *Fourier Coefficients as Projections*
Fourier coefficients are projections onto orthonormal directions, modeled finitely by Theorems 2.1.5, 1.4.1, and 1.4.5; infinite directions make convergence separate.

Theorem 10.2.5. *Fourier Coefficients Give Best Mean-Square Approximation*
Let $\langle u, v \rangle := \int_a^b u\overline{v}$ and $\|u\|_2^2 := \langle u, u \rangle$. Let (ϕ_m) be orthonormal and let f satisfy $\|f\|_2 < \infty$.
For $n \geq 1$, set $c_m := \langle f, \phi_m \rangle$, $s_n := \sum_{m=1}^{n} c_m \phi_m$, and $t_n := \sum_{m=1}^{n} \gamma_m \phi_m$. Then:
(i) $\|f - s_n\|_2 \leq \|f - t_n\|_2$, with equality iff $\gamma_m = c_m$ for $m = 1, \ldots, n$.
(ii) $\sum_{m=1}^{\infty} |c_m|^2 \leq \|f\|_2^2$, hence $c_m \to 0$.

Proof.

(i) By orthonormality, $\|f - t_n\|_2^2 = \|f\|_2^2 - 2\,\mathrm{Re} \sum_{m=1}^{n} c_m \overline{\gamma_m} + \sum_{m=1}^{n} |\gamma_m|^2 = \|f\|_2^2 - \sum_{m=1}^{n} |c_m|^2 + \sum_{m=1}^{n} |\gamma_m - c_m|^2$. Thus $\|f - t_n\|_2^2 \geq \|f - s_n\|_2^2$, with equality iff $\gamma_m = c_m$ for all $m \leq n$.

(ii) Taking $\gamma_m = c_m$ in the identity from (i) gives $\|f - s_n\|_2^2 = \|f\|_2^2 - \sum_{m=1}^{n} |c_m|^2 \geq 0$.
Hence $\sum_{m=1}^{n} |c_m|^2 = \|s_n\|_2^2 \leq \|f\|_2^2$ for all n. Letting $n \to \infty$ gives Bessel's inequality.
Since $(|c_m|^2)$ is summable, $|c_m|^2 \to 0$, hence $c_m \to 0$.

Structural Commentary: Orthogonality diagonalizes the finite error.
Key Insight: Fourier coefficients are projection coordinates. □

Theorem 10.2.6. *Dirichlet Kernel and Local Fourier Convergence*
Let f be Riemann integrable and 2π-periodic, with $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-imt}\,dt$.
For $N \geq 0$, set $s_N(f;x) := \sum_{m=-N}^{N} \widehat{f}(m)e^{imx}$ and $D_N(t) := \sum_{m=-N}^{N} e^{imt}$. Then:

(i) $D_N(t) = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$ *for $t \notin 2\pi\mathbb{Z}$, with $D_N(0) = 2N+1$, and $s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt$.*
(ii) If $x \in \mathbb{R}$ and there exist $M, \delta > 0$ such that $|f(x+t) - f(x)| \leq M|t|$ for $|t| < \delta$, then $s_N(f;x) \to f(x)$.
(iii) If g is Riemann integrable and 2π-periodic and $f = g$ near x, then $s_N(f;x) - s_N(g;x) \to 0$.

Proof.

(i) For $z = e^{it} \neq 1$, $D_N(t) = z^{-N}\frac{1-z^{2N+1}}{1-z} = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$, and $D_N(0) = \sum_{m=-N}^{N} 1 = 2N+1$.
Substituting $\widehat{f}(m)$ and interchanging the finite sum with the integral gives
$s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(u)\sum_{m=-N}^{N} e^{im(x-u)}\,du = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt$.

(ii) Since $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N(t)\,dt = 1$, (i) gives $s_N(f;x) - f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi}(f(x-t) - f(x))D_N(t)\,dt$.
Set $q_x(t) := \frac{f(x-t)-f(x)}{\sin(t/2)}$ for $0 < |t| \leq \pi$ and $q_x(0) := 0$. The hypothesis makes q_x bounded near 0 and Riemann integrable on $[-\pi,\pi]$. Hence $s_N(f;x) - f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} q_x(t)\sin((N+\frac{1}{2})t)\,dt$.
Since $\sin((N+\frac{1}{2})t) = \sin Nt\cos(t/2) + \cos Nt\sin(t/2)$, Theorem 10.2.5 applied to $q_x(t)\cos(t/2)$ and $q_x(t)\sin(t/2)$ sends both coefficients to 0.

(iii) Apply (ii) to $h := f - g$. Then $h = 0$ near x, so $s_N(h;x) \to h(x) = 0$.
By linearity, $s_N(h;x) = s_N(f;x) - s_N(g;x)$.

Structural Commentary: Dirichlet kernels express partial sums as oscillatory local averages.
Key Insight: Local regularity makes the oscillatory error vanish. □

Corollary 10.2.7. *Uniform Approximation by Trigonometric Polynomials*
Let $f \in C(\mathbb{R};\mathbb{C})$ be 2π-periodic. For every $\varepsilon > 0$, there is $P(x) = \sum_{n=-N}^{N} c_n e^{inx}$ such that $\|f - P\|_\infty < \varepsilon$.

Proof. Let $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$ and define $F : \mathbb{T} \to \mathbb{C}$ by $F(e^{ix}) := f(x)$.
Periodicity makes F well-defined and continuous.
The algebra $\mathcal{T} := \operatorname{span}\{z^n : n \in \mathbb{Z}\} \subset C(\mathbb{T})$ is self-adjoint, separates points, and vanishes nowhere.
By Theorem 9.2.9, choose $Q(z) = \sum_{n=-N}^{N} c_n z^n \in \mathcal{T}$ with $\|F - Q\|_{C(\mathbb{T})} < \varepsilon$.
Set $P(x) := Q(e^{ix}) = \sum_{n=-N}^{N} c_n e^{inx}$; then $\|f - P\|_\infty = \|F - Q\|_{C(\mathbb{T})} < \varepsilon$.
Structural Commentary: Trigonometric polynomials are Laurent polynomials pulled back from $\mathbb{T}$.
Key Insight: Stone–Weierstrass on $\mathbb{T}$ gives uniform periodic approximation. □

Theorem 10.2.8. *Parseval's Theorem for Trigonometric Fourier Series*
Let f, g be Riemann integrable and 2π-periodic.
Set $\langle u, v\rangle_2 := \frac{1}{2\pi}\int_{-\pi}^{\pi} u(x)\overline{v(x)}\,dx$, $\|u\|_2^2 := \langle u, u\rangle_2$, and $s_N(f) := \sum_{|n|\leq N} \widehat{f}(n)e^{inx}$. Then:

(i) $\|f - s_N(f)\|_2 \to 0$.
(ii) $\langle f, g\rangle_2 = \sum_{n\in\mathbb{Z}} \widehat{f}(n)\overline{\widehat{g}(n)}$.
(iii) $\|f\|_2^2 = \sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\right|^2$.

Proof.

(i) Put $d_N(u) := \|u - s_N(u)\|_2$. If u is continuous, choose a trigonometric polynomial P with $\|u - P\|_2 < \varepsilon$; for $N \geq \deg P$, Theorem 10.2.5 gives $d_N(u) \leq \|u - P\|_2 < \varepsilon$, so $d_N(u) \to 0$.
For Riemann integrable f, choose continuous h with $\|f - h\|_2 < \varepsilon$.
Since $\|s_N(h - f)\|_2 \leq \|h - f\|_2$, $d_N(f) \leq 2\|f - h\|_2 + d_N(h)$; hence $\limsup_{N\to\infty} d_N(f) \leq 2\varepsilon$. Let $\varepsilon \downarrow 0$.

(ii) By orthogonality, $\langle e^{inx}, g\rangle_2 = \overline{\widehat{g}(n)}$. Hence $\langle s_N(f), g\rangle_2 = \sum_{|n|\leq N} \widehat{f}(n)\langle e^{inx}, g\rangle_2 = \sum_{|n|\leq N} \widehat{f}(n)\overline{\widehat{g}(n)}$.
By (i) and Cauchy–Schwarz, $\langle s_N(f), g\rangle_2 \to \langle f, g\rangle_2$.

(iii) Set $g = f$ in (ii). Then $\widehat{g}(n) = \widehat{f}(n)$ and $\langle f, f\rangle_2 = \|f\|_2^2$, so $\|f\|_2^2 = \sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\right|^2$.

Structural Commentary: Density turns finite projection identities into L^2 identities.
Key Insight: Partial Fourier projections converge in L^2. □

Method 10.2.9. *Fourier Proof Patterns*
(i) *Characters*: use orthogonality of e_n to compute coefficients and finite-dimensional projections.
(ii) *Kernels*: rewrite partial sums or averages as convolutions with Dirichlet or Fejer kernels.
(iii) *Approximation*: use Fejer positivity or Stone–Weierstrass to approximate continuous periodic functions by trigonometric polynomials.
(iv) *L^2 control*: project onto finite orthonormal spans, then pass to limits by density.
(v) *Localization*: express the error as an oscillatory integral and use local regularity near the point.
Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

10.3 Further Topics in Fourier Analysis

Example 10.3.1. *A Gaussian Fourier Integral*
Define $f(t) = \int_{-\infty}^{\infty} e^{-x^2} \cos(xt)\, dx$ and $g(t) = -\int_{-\infty}^{\infty} xe^{-x^2} \sin(xt)\, dx$.
Then $f'(t) = g(t)$, and integration by parts gives $2f'(t) + tf(t) = 0$. Since $f(0) = \sqrt{\pi}$, $f(t) = \sqrt{\pi} \exp\left(-\frac{t^2}{4}\right)$.

Part XIII
Measure Foundations and Extension

1 Sigma-Algebras and Measurable Spaces

1.1 Sigma-Algebras and Measurable Structures

Measure theory first builds measurable sets, then measurable functions and integrals. Outer measure covers from outside; measurability splits test sets without losing measure.

Perspective 1.1.1. *Why Abstract Measure Comes First*
Lebesgue integration is built so that, under hypotheses such as domination, $f_n \to f$ implies $\int f_n\,d\mu \to \int f\,d\mu$; see Theorem 1.3.7. The abstract data separate into a measurable space $(X, \mathcal{M})$, a countably additive measure $\mu : \mathcal{M} \to [0, +\infty]$, and an integral defined by approximation from simple functions.
Key Insight: Countable additivity is the set-level structure behind the convergence theorems.

Definition 1.1.2. *σ-Algebras and Measurable Spaces*
Let X be a set. A family $\mathcal{M} \subseteq \mathcal{P}(X)$ is a *σ-algebra* on X if:

(i) $\emptyset, X \in \mathcal{M}$;
(ii) $E \in \mathcal{M}$ implies $X \setminus E \in \mathcal{M}$;
(iii) $E_n \in \mathcal{M}$ for $n \in \mathbb{N}$ implies $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is a *measurable space*; the members of $\mathcal{M}$ are *measurable sets*.

Proposition 1.1.3. *Closure and Generation for σ-Algebras*
Let $\mathcal{M} \subseteq \mathcal{P}(X)$ be a σ-algebra.

(i) $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M} \Rightarrow \bigcap_{n=1}^{\infty} E_n \in \mathcal{M}$.
(ii) If $(\mathcal{M}_\alpha)_{\alpha\in A}$ is a nonempty family of σ-algebras on X, then $\bigcap_{\alpha\in A} \mathcal{M}_\alpha$ is a σ-algebra on X.
(iii) For $\mathcal{A} \subseteq \mathcal{P}(X)$, let $\mathscr{S}_{\mathcal{A}}$ be the family of σ-algebras $\mathcal{N}$ on X with $\mathcal{A} \subseteq \mathcal{N}$. Then $\sigma(\mathcal{A}) := \bigcap_{\mathcal{N}\in\mathscr{S}_{\mathcal{A}}} \mathcal{N}$ is the smallest σ-algebra containing $\mathcal{A}$.

Proof.

(i) Since $X \setminus E_n \in \mathcal{M}$, Theorem 1.5.2 gives $\bigcap_n E_n = X \setminus \bigcup_n (X \setminus E_n) \in \mathcal{M}$.
(ii) Set $\mathcal{N} := \bigcap_{\alpha\in A} \mathcal{M}_\alpha$. Then $\emptyset, X \in \mathcal{N}$, and $E \in \mathcal{N}$ implies $X \setminus E \in \mathcal{N}$. Likewise $(E_n) \subseteq \mathcal{N}$ implies $\bigcup_n E_n \in \mathcal{N}$.
(iii) The indexing class in the displayed definition is nonempty since $\mathcal{P}(X)$ belongs to it. By (ii), its intersection is a σ-algebra; it contains $\mathcal{A}$ and is contained in every σ-algebra containing $\mathcal{A}$.

Structural Commentary: Generation is intersection over all admissible σ-algebras.
Key Insight: Intersections preserve universal closure properties. □

Example 1.1.4. *Basic σ-Algebras*
Standard models:

(i) Extremes: $\{\emptyset, X\} \subseteq \mathcal{M} \subseteq \mathcal{P}(X)$ for every σ-algebra $\mathcal{M}$ on X.
(ii) For $E \subseteq X$, $\sigma(\{E\}) = \{\emptyset, E, X \setminus E, X\}$, with repetitions omitted.
(iii) On $\mathbb{N}$, if $O := \{2n-1 : n \in \mathbb{N}\}$, then $\sigma(\{O\}) = \{\emptyset, O, \mathbb{N} \setminus O, \mathbb{N}\}$.
(iv) If X is uncountable, $\mathcal{C}_X := \{E \subseteq X : E \text{ is countable or } X \setminus E \text{ is countable}\}$ is the countable–cocountable σ-algebra.

Key Insight: A σ-algebra records exactly which distinctions the theory is allowed to observe.

Definition 1.1.5. *Borel and Extended Borel σ-Algebras*

(i) For $n \in \mathbb{N}$, the *Borel σ-algebra* on $\mathbb{R}^n$ is $\mathcal{B}(\mathbb{R}^n) := \sigma(\{U \subseteq \mathbb{R}^n : U \text{ is open}\})$. On $\mathbb{R}$, it is also generated by open intervals, closed intervals, half-open intervals, or open half-rays $(a, +\infty)$.
(ii) The *extended Borel σ-algebra* on $\overline{\mathbb{R}}$ is $\overline{\mathcal{B}} := \{E \cup F : E \in \mathcal{B}(\mathbb{R}),\ F \subseteq \{-\infty, +\infty\}\}$.

Definition 1.1.6. *Monotone Classes*
A nonempty $\mathcal{C} \subseteq \mathcal{P}(X)$ is a *monotone class* iff $(E_n) \subseteq \mathcal{C},\ E_n \uparrow E \Rightarrow E \in \mathcal{C}$ and $(F_n) \subseteq \mathcal{C},\ F_n \downarrow F \Rightarrow F \in \mathcal{C}$.
For $\mathcal{A} \subseteq \mathcal{P}(X)$, define $\mathfrak{M}_{\mathcal{A}} := \{\mathcal{C} \subseteq \mathcal{P}(X) : \mathcal{C} \text{ is a monotone class, } \mathcal{A} \subseteq \mathcal{C}\}$; $\operatorname{mon}(\mathcal{A}) := \bigcap \mathfrak{M}_{\mathcal{A}}$.
Then $\operatorname{mon}(\mathcal{A})$ is the least monotone class containing $\mathcal{A}$, and $\operatorname{mon}(\mathcal{A}) \subseteq \sigma(\mathcal{A})$.
Key Insight: Monotone closure captures limits, not finite unions.

Definition 1.1.7. *Algebras of Sets and Premeasures*
Let X be a set.

(i) An *algebra of sets* (or *field of sets*) on X is $\mathcal{A} \subseteq \mathcal{P}(X)$ such that $\emptyset, X \in \mathcal{A}$; $E \in \mathcal{A} \Rightarrow X \setminus E \in \mathcal{A}$; and $E_1, \dots, E_n \in \mathcal{A} \Rightarrow \bigcup_{j=1}^{n} E_j \in \mathcal{A}$. Equivalently, $\mathcal{A}$ is a ring of sets containing X.
(ii) A *premeasure* (or *measure on an algebra*) on $\mathcal{A}$ is $\mu : \mathcal{A} \to [0, +\infty]$ such that $\mu(\emptyset) = 0$ and, whenever $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ is pairwise disjoint with $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$, then $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$.

Key Insight: Premeasures are countably additive where the algebra permits unions.

Theorem 1.1.8. *Monotone Class Theorem for Algebras*
Let X be a set, let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra, and set $\mathcal{C} := \operatorname{mon}(\mathcal{A})$. Then $\mathcal{C} = \sigma(\mathcal{A})$.

Proof. Since $\sigma(\mathcal{A})$ is monotone and contains $\mathcal{A}$, $\mathcal{C} \subseteq \sigma(\mathcal{A})$. It remains to prove that $\mathcal{C}$ is a σ-algebra.
Set $\mathcal{D} := \{E \in \mathcal{C} : E^c \in \mathcal{C}\}$. If $E_n \uparrow E$ in $\mathcal{D}$, then $E_n^c \downarrow E^c$; if $E_n \downarrow E$, then $E_n^c \uparrow E^c$. Thus $\mathcal{D}$ is monotone.
Since $\mathcal{A} \subseteq \mathcal{D}$, minimality gives $\mathcal{C} \subseteq \mathcal{D}$, so $\mathcal{C}$ is closed under complements.
For $B \in \mathcal{C}$, set $\mathcal{D}_B := \{E \in \mathcal{C} : E \cap B \in \mathcal{C}\}$. This class is monotone. If $B \in \mathcal{A}$, then $\mathcal{A} \subseteq \mathcal{D}_B$, hence $\mathcal{C} \subseteq \mathcal{D}_B$.
Therefore $E \cap B \in \mathcal{C}$ for $E \in \mathcal{C}$ and $B \in \mathcal{A}$. Fix $E \in \mathcal{C}$; the preceding sentence gives $\mathcal{A} \subseteq \mathcal{D}_E$, hence $\mathcal{C} \subseteq \mathcal{D}_E$.
Thus $\mathcal{C}$ is closed under finite intersections. Complements and finite intersections give finite unions.
If $(E_n) \subseteq \mathcal{C}$, then $F_N := \bigcup_{n=1}^N E_n \in \mathcal{C}$ and $F_N \uparrow \bigcup_{n=1}^\infty E_n$, so $\bigcup_{n=1}^\infty E_n \in \mathcal{C}$.
Hence $\mathcal{C}$ is a σ-algebra containing $\mathcal{A}$, so $\sigma(\mathcal{A}) \subseteq \mathcal{C}$.
Structural Commentary: Monotone lifting gives Boolean closure, then countable unions.
Key Insight: Show each closure property holds on a monotone class containing $\mathcal{A}$. □

Corollary 1.1.9. *Monotone-Class Test*
Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra and let $\mathcal{C} \subseteq \mathcal{P}(X)$ be a monotone class. If $\mathcal{A} \subseteq \mathcal{C}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{C}$.

Proof. Since $\mathcal{A} \subseteq \mathcal{C}$ and $\mathcal{C}$ is monotone, $\mathrm{mon}(\mathcal{A}) \subseteq \mathcal{C}$. By Theorem 1.1.8, $\mathrm{mon}(\mathcal{A}) = \sigma(\mathcal{A})$.
Structural Commentary: Algebra checks plus monotone stability control $\sigma(\mathcal{A})$.
Key Insight: Prove on the algebra; extend by monotone closure. □

2 Measures, Charges, and Basic Examples

2.1 Measures, Charges, and Basic Constructions

Definition 2.1.1. *Rings, σ-Rings, and Set Functions*
Let $\mathcal{A} \subseteq \mathcal{P}(X)$.

(i) $\mathcal{A}$ is a *ring of sets* iff $\emptyset \in \mathcal{A}$ and $A, B \in \mathcal{A} \Rightarrow A \cup B,\ A \setminus B \in \mathcal{A}$
(ii) $\mathcal{A}$ is a *σ-ring* iff it is a ring and $(A_n) \subseteq \mathcal{A} \Rightarrow \bigcup_{n=1}^\infty A_n \in \mathcal{A}$
(iii) A *set function* on $\mathcal{A}$ is $\phi : \mathcal{A} \to [-\infty, +\infty]$ with $\{+\infty, -\infty\} \not\subseteq \phi(\mathcal{A})$.
(iv) ϕ is *additive* iff $A \cap B = \emptyset \Rightarrow \phi(A \cup B) = \phi(A) + \phi(B)$
(v) ϕ is *countably additive* iff $(A_n) \subseteq \mathcal{A}$ pairwise disjoint, $\bigcup_n A_n \in \mathcal{A} \Rightarrow \phi\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{n=1}^\infty \phi(A_n)$.

Structural Commentary: Rings support finite operations; σ-rings support countable unions.

Definition 2.1.2. *Measures on Measurable Spaces*
Let $(X, \mathcal{M})$ be a measurable space. A *measure* on $(X, \mathcal{M})$ is a map $\mu : \mathcal{M} \to [0, +\infty]$ such that:

(i) $\mu(\emptyset) = 0$;
(ii) if $(E_n) \subseteq \mathcal{M}$ is pairwise disjoint, then $\mu\left(\bigcup_{n=1}^\infty E_n\right) = \sum_{n=1}^\infty \mu(E_n)$

The triple $(X, \mathcal{M}, \mu)$ is a *measure space*.
Key Insight: Measure is countably additive size on the chosen observable sets.

Definition 2.1.3. *Finite and σ-Finite Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) For nonnegative terms, $\sum_{n=1}^\infty a_n := \lim_{N\to\infty} \sum_{n=1}^N a_n \in [0, +\infty]$.
(ii) μ is *finite* iff $\mu(X) < +\infty$.
(iii) μ is *σ-finite* iff there exist $E_n \in \mathcal{M}$ such that $X = \bigcup_{n=1}^\infty E_n$ and $\mu(E_n) < +\infty$ for all n.
(iv) A measurable set $E \in \mathcal{M}$ has *finite measure* iff $\mu(E) < +\infty$.
(v) A measurable set $E \in \mathcal{M}$ is *σ-finite* iff $E = \bigcup_{n=1}^\infty E_n$ for some $E_n \in \mathcal{M}$ with $\mu(E_n) < +\infty$ for all n.

Key Insight: σ-finiteness replaces one finite global bound by a countable finite-measure exhaustion.

Example 2.1.4. *First Abstract Measures*

(i) The zero set function $\mu(E) = 0$ on $\mathcal{M}$ is a finite measure.
(ii) If $X \neq \emptyset$ and $\mathcal{M} = 2^X$, $\mu(\emptyset) = 0$ and $\mu(E) = +\infty$ for $E \neq \emptyset$ give a measure, neither finite nor σ-finite.
(iii) For $p \in X$, the *unit mass* $\delta_p(E) := \mathbf{1}_E(p)$ on 2^X is a finite measure.
(iv) On $2^{\mathbb{N}}$, *counting measure* is $\#(E)$ for finite E and $+\infty$ otherwise; it is σ-finite.
More generally, $(a_n) \subseteq [0, +\infty]$ gives $\mu(E) := \sum_{n\in E} a_n$, and every measure on $2^{\mathbb{N}}$ is uniquely of this form.
(v) If X is uncountable and $\mathcal{M} = 2^X$, $\mu(E) = 0$ for countable E and $\mu(E) = +\infty$ otherwise defines a measure.
(vi) If X is uncountable and $\mathcal{C}_X = \{E \subseteq X : E$ is countable or $X \setminus E$ is countable$\}$, then $\mu(E) = 0$ on countable sets and $\mu(E) = 1$ on cocountable sets defines a measure on $\mathcal{C}_X$.
(vii) On $2^{\mathbb{N}}$, $\mu(E) = 0$ for finite E and $\mu(E) = +\infty$ otherwise is not a measure, since $\mathbb{N} = \bigcup_{n=1}^\infty \{n\}$ gives $+\infty \neq 0$; the constant $+\infty$ rule fails at $\emptyset$.
(viii) Lebesgue measure on $\mathcal{B}(\mathbb{R})$ is σ-finite and gives length $b - a$ to standard intervals; Borel–Stieltjes measures come from increasing right-continuous functions as in Theorem 4.1.4.

Structural Commentary: Measures may be zero, atomic, counting-type, cardinality-based, or geometric.
Key Insight: Countable additivity turns disjoint decompositions into nonnegative series.

Proposition 2.1.5. *Elementary Measure Constructions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) If μ is a measure on $\mathcal{M}$ and $A \in \mathcal{M}$, then $\mu_A(E) := \mu(A \cap E)$ defines a measure μ_A on $\mathcal{M}$.
(ii) If $\mu_1, \ldots, \mu_m$ are measures on $\mathcal{M}$ and $a_1, \ldots, a_m \geq 0$, then $\lambda := \sum_{j=1}^m a_j \mu_j$ is a measure on $\mathcal{M}$.
(iii) If $(\mu_n)_{n\geq 1}$ are measures on $\mathcal{M}$ with $\mu_n(X) = 1$, then $\lambda := \sum_{n=1}^\infty 2^{-n}\mu_n$ is a measure on $\mathcal{M}$ and $\lambda(X) = 1$.

Proof.

(i) $\mu_A(\emptyset) = 0$. If (E_k) is disjoint in $\mathcal{M}$, then $A \cap \bigsqcup_k E_k = \bigsqcup_k (A \cap E_k)$, whence $\mu_A(\bigsqcup_k E_k) = \sum_k \mu_A(E_k)$.
(ii) $\lambda(\emptyset) = 0$. For disjoint (E_k), finite summation gives $\lambda(\bigsqcup_k E_k) = \sum_{j=1}^m a_j \sum_k \mu_j(E_k) = \sum_k \lambda(E_k)$.
(iii) $\lambda(\emptyset) = 0$. For disjoint (E_k), nonnegative reordering gives $\lambda(\bigsqcup_k E_k) = \sum_n 2^{-n} \sum_k \mu_n(E_k) = \sum_k \lambda(E_k)$.
Also $\lambda(X) = \sum_n 2^{-n}\mu_n(X) = \sum_n 2^{-n} = 1$.

Structural Commentary: New measures come from restriction, superposition, and countable convex mixtures.
Key Insight: Nonnegative series make the bookkeeping order-independent. □

Proposition 2.1.6. *Basic Laws for Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) For every $n \geq 1$, if $E_1, \ldots, E_n \in \mathcal{M}$ are pairwise disjoint, then $\mu\left(\bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} \mu(E_k)$.
(ii) If $A, B \in \mathcal{M}$ and $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
(iii) If $A, B \in \mathcal{M}$, $A \subseteq B$, and $\mu(A) < +\infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.
In particular, if $\mu(A) = 0$, then $\mu(B \setminus A) = \mu(B)$.
(iv) If $E \in \mathcal{M}$ and $(E_k)_{k\geq 1} \subseteq \mathcal{M}$ satisfy $E \subseteq \bigcup_{k=1}^{\infty} E_k$, then $\mu(E) \leq \sum_{k=1}^{\infty} \mu(E_k)$.

Proof.
(i) Define $G_k = E_k$ for $1 \leq k \leq n$ and $G_k = \emptyset$ for $k > n$. Then (G_k) is pairwise disjoint and countable additivity gives $\mu\left(\bigcup_{k=1}^{n} E_k\right) = \mu\left(\bigcup_{k=1}^{\infty} G_k\right) = \sum_{k=1}^{\infty} \mu(G_k) = \sum_{k=1}^{n} \mu(E_k)$.
(ii) Since $B = A \sqcup (B \setminus A)$, (i) gives $\mu(B) = \mu(A) + \mu(B \setminus A)$, and $\mu(B \setminus A) \geq 0$, hence $\mu(A) \leq \mu(B)$.
(iii) Since $B = A \sqcup (B \setminus A)$, (i) gives $\mu(B) = \mu(A) + \mu(B \setminus A)$. Because $\mu(A) < +\infty$, subtraction yields $\mu(B \setminus A) = \mu(B) - \mu(A)$. If $\mu(A) = 0$, this becomes $\mu(B \setminus A) = \mu(B)$.
(iv) Set $F_1 = E \cap E_1$ and $F_k = E \cap \left(E_k \setminus \bigcup_{j=1}^{k-1} E_j\right)$ for $k \geq 2$. Then $(F_k)_{k\geq 1} \subseteq \mathcal{M}$ is pairwise disjoint, $\bigcup_{k=1}^{\infty} F_k = E$, and $F_k \subseteq E_k$ for all k. Countable additivity and (ii) give $\mu(E) = \sum_{k=1}^{\infty} \mu(F_k) \leq \sum_{k=1}^{\infty} \mu(E_k)$.

Structural Commentary: Finite additivity yields order and excision; disjointization turns covers into sums.
Key Insight: Measure disjoint pieces, then compare. □

Theorem 2.1.7. *Continuity from Below*
Let $\mathcal{A}$ be a ring of subsets of X, and let ϕ be a countably additive set function on $\mathcal{A}$.
If $(A_n)_{n\geq 1} \subseteq \mathcal{A}$ and $A \in \mathcal{A}$ satisfy $A_n \subseteq A_{n+1}$ for all n and $A = \bigcup_{n=1}^{\infty} A_n$, then $\lim_{N\to\infty} \phi(A_N) = \phi(A)$.

Proof. Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Since $\mathcal{A}$ is a ring, each $B_n \in \mathcal{A}$.
The sets (B_n) are pairwise disjoint, $A_N = \bigsqcup_{n=1}^{N} B_n$, and $A = \bigsqcup_{n=1}^{\infty} B_n$.
Countable additivity, applied to the finite union by appending empty sets, gives $\phi(A_N) = \sum_{n=1}^{N} \phi(B_n)$, while countable additivity gives $\phi(A) = \sum_{n=1}^{\infty} \phi(B_n)$. Thus $\lim_{N\to\infty} \phi(A_N) = \phi(A)$.
Structural Commentary: Increasing chains split into disjoint increments.
Key Insight: Continuity is convergence of partial sums. □

Definition 2.1.8. *Charges*
Let $(X, \mathcal{M})$ be a measurable space. A *charge* is a map $\lambda : \mathcal{M} \to \mathbb{R}$ such that:
(i) $\lambda(\emptyset) = 0$.
(ii) For every pairwise disjoint $(E_n) \subseteq \mathcal{M}$, $\sum_{n=1}^{\infty} \lambda(E_n)$ converges in $\mathbb{R}$ and $\lambda\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \lambda(E_n)$.

Key Insight: A charge is a real-valued countably additive set function.

Definition 2.1.9. *Variation Candidates of a Charge*
Let λ be a charge on $\mathcal{M}$. For $E \in \mathcal{M}$, define
(i) $\mathfrak{P}_{\mathcal{M}}(E) := \{(A_j)_{j=1}^{n} : n \in \mathbb{N},\ E = \bigsqcup_{j=1}^{n} A_j,\ A_j \in \mathcal{M}\}$, the finite measurable partitions of E;
(ii) $\pi_\lambda(E) := \sup\{\lambda(A) : A \subseteq E,\ A \in \mathcal{M}\}$, the *positive variation* candidate;
(iii) $\nu_\lambda(E) := \sup\{\sum_{j=1}^{n} |\lambda(A_j)| : (A_j)_{j=1}^{n} \in \mathfrak{P}_{\mathcal{M}}(E)\}$, the *total variation* candidate.

For $E = \emptyset$, the empty partition is allowed and contributes 0.

Proposition 2.1.10. *Charge Operations and Variation Measures*
Let $(X, \mathcal{M})$ be a measurable space, let $m \geq 1$, and let $\lambda, \lambda_1, \ldots, \lambda_m$ be charges on $\mathcal{M}$.

(i) If $a_1, \ldots, a_m \in \mathbb{R}$ and $\Lambda : \mathcal{M} \to \mathbb{R}$ is defined by $\Lambda(E) := \sum_{j=1}^{m} a_j \lambda_j(E)$, then Λ is a charge.
(ii) If μ is a measure on $\mathcal{M}$, $f \in \mathscr{L}^1(\mu)$, and $\lambda_f : \mathcal{M} \to \mathbb{R}$ is defined by $\lambda_f(A) := \int_A f\, d\mu$, then λ_f is a charge.
(iii) $\pi_\lambda : \mathcal{M} \to [0, +\infty]$ is a measure.
(iv) $\nu_\lambda : \mathcal{M} \to [0, +\infty]$ is a measure.

Proof.
(i) $\Lambda(\emptyset) = 0$. For $E = \bigsqcup_n E_n$, $\sum_n \Lambda(E_n)$ converges as $\sum_{j=1}^{m} a_j \sum_n \lambda_j(E_n)$; $\Lambda(E) = \sum_{j=1}^{m} a_j \sum_n \lambda_j(E_n) = \sum_n \Lambda(E_n)$.
(ii) $\lambda_f(\emptyset) = 0$. For disjoint (A_n), $\sum_n |\lambda_f(A_n)| \leq \int_{\bigsqcup_n A_n} |f|\, d\mu < +\infty$
and $\lambda_f(\bigsqcup_n A_n) = \int_{\bigsqcup_n A_n} f\, d\mu = \sum_n \int_{A_n} f\, d\mu = \sum_n \lambda_f(A_n)$.
(iii) $\pi_\lambda(\emptyset) = 0$ and $\pi_\lambda \geq 0$. Let $E = \bigsqcup_n E_n$. For measurable $A \subseteq E$, $\lambda(A) = \sum_n \lambda(A \cap E_n) \leq \sum_n \pi_\lambda(E_n)$, so $\pi_\lambda(E) \leq \sum_n \pi_\lambda(E_n)$. Put $S_N = \sum_{n=1}^{N} \pi_\lambda(E_n)$. If $S_N < +\infty$, choose measurable $A_n \subseteq E_n$ for $n \leq N$ with $\lambda(A_n) > \pi_\lambda(E_n) - \frac{\varepsilon}{N}$; then $\pi_\lambda(E) > S_N - \varepsilon$. If $S_N = +\infty$, some E_ℓ has $\pi_\lambda(E_\ell) = +\infty$, hence $\pi_\lambda(E) = +\infty$. Thus $\pi_\lambda(E) \geq S_N$ for all N, and $N \to \infty$ gives equality.

(iv) $\nu_\lambda(\emptyset) = 0$ and $\nu_\lambda \geq 0$. Let $E = \bigsqcup_n E_n$ and $v(P) = \sum_{A\in P} |\lambda(A)|$. Refining $P = (A_j)_{j=1}^r \in \mathfrak{P}_{\mathcal{M}}(E)$ by E_n gives $v(P) \leq \sum_{j=1}^r \sum_n |\lambda(A_j \cap E_n)| \leq \sum_n \nu_\lambda(E_n)$, so $\nu_\lambda(E) \leq \sum_n \nu_\lambda(E_n)$. Put $T_N = \sum_{n=1}^N \nu_\lambda(E_n)$.
If $T_N < +\infty$, choose $P_n \in \mathfrak{P}_{\mathcal{M}}(E_n)$ for $n \leq N$ with $v(P_n) > \nu_\lambda(E_n) - \frac{\varepsilon}{N}$.
Adjoining the tail $\bigcup_{n>N} E_n$ gives $\nu_\lambda(E) > T_N - \varepsilon$. If $T_N = +\infty$, some E_ℓ has $\nu_\lambda(E_\ell) = +\infty$, and partitions of E_ℓ extend to E, so $\nu_\lambda(E) = +\infty$. Thus $\nu_\lambda(E) \geq T_N$ for all N, and $N \to \infty$ gives equality.

Structural Commentary: Suprema preserve additivity after refinement.
Key Insight: Refine first, then take suprema. □

3 Outer Measures and Caratheodory Extension

3.1 Outer Measures and Extension Theorems

Definition 3.1.1. *Abstract Outer Measure and Clean Division*
Let M be a set.

(i) An *abstract outer measure* on M is a map $\omega : 2^M \to [0, +\infty]$ such that $\omega(\emptyset) = 0$, $A \subseteq B \Rightarrow \omega(A) \leq \omega(B)$, and $\omega(\bigcup_{k=1}^\infty A_k) \leq \sum_{k=1}^\infty \omega(A_k)$ for all $A, B, A_k \subseteq M$.
(ii) For an outer measure ω, $E \subseteq M$ is *ω-measurable* iff $\omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$ for every $X \subseteq M$.

For $\omega = m_n^*$, ω-measurability is Caratheodory measurability.

Definition 3.1.2. *Outer Measure Induced by a Covering Family*
Let $\mathcal{S} \subseteq \mathcal{P}(X)$ and let $\rho : \mathcal{S} \to [0, +\infty]$.

(i) For $E \subseteq X$, $\mathcal{C}_{\mathcal{S}}(E) := \{(S_k)_{k\geq 1} \in \mathcal{S}^{\mathbb{N}} : E \subseteq \bigcup_{k=1}^\infty S_k\}$.
(ii) The induced map $\rho^* : \mathcal{P}(X) \to [0, +\infty]$ is defined by $\rho^*(\emptyset) = 0$ and, for $E \neq \emptyset$, $\rho^*(E) := \inf_{(S_k)\in\mathcal{C}_{\mathcal{S}}(E)} \sum_{k=1}^\infty \rho(S_k)$, with $\inf \emptyset = +\infty$.

The map ρ^* is the *outer measure induced by ρ and $\mathcal{S}$*.
Key Insight: Minimize total ρ-cost over countable $\mathcal{S}$-covers.

Proposition 3.1.3. *Cover-Induced Outer Measure*
For $\mathcal{S} \subseteq \mathcal{P}(X)$ and $\rho : \mathcal{S} \to [0, +\infty]$, the cover-induced map $\rho^* : \mathcal{P}(X) \to [0, +\infty]$ is an outer measure on X.

Proof. By construction, $\rho^*(\emptyset) = 0$ and $\rho^* \geq 0$. If $A \subseteq B$, then $\mathcal{C}_{\mathcal{S}}(B) \subseteq \mathcal{C}_{\mathcal{S}}(A)$, hence $\rho^*(A) \leq \rho^*(B)$.
Let $E = \bigcup_k E_k$, $I = \{k : E_k \neq \emptyset\}$, and assume $I \neq \emptyset$ and $\sum_k \rho^*(E_k) < +\infty$.
Choose $(S_{k,j})_j \in \mathcal{C}_{\mathcal{S}}(E_k)$ for $k \in I$ with $\sum_j \rho(S_{k,j}) < \rho^*(E_k) + \varepsilon 2^{-k}$.
Since $(S_{k,j})_{k\in I, j\geq 1}$ covers E, $\rho^*(E) \leq \sum_{k\in I} \sum_j \rho(S_{k,j}) \leq \sum_k \rho^*(E_k) + \varepsilon$; let $\varepsilon \downarrow 0$. Excluded cases are immediate.

Structural Commentary: Inclusion orders covers; regrouping covers sums.
Key Insight: Covers generate subadditivity. □

Lemma 3.1.4. *Finite-Measure Test for Caratheodory Measurability*
Let $\omega : 2^M \to [0, +\infty]$ be an outer measure. Then $E \subseteq M$ is ω-measurable iff, for all $X \subseteq M$, $\omega(X) < +\infty \Longrightarrow \omega(X) \geq \omega(X \cap E) + \omega(X \cap E^c)$.

Proof.
($\Rightarrow$) If E is ω-measurable, then $\omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$ $\forall X \subseteq M$, hence finite-test inequality holds.
($\Leftarrow$) For all $X \subseteq M$, $X = (X \cap E) \sqcup (X \cap E^c)$ gives $\omega(X) \leq \omega(X \cap E) + \omega(X \cap E^c)$. If $\omega(X) < +\infty$, the hypothesis gives $\geq$; if $\omega(X) = +\infty$, $\geq$ is automatic in $[0, +\infty]$. Thus equality holds for every X.

Structural Commentary: To verify a clean split, only finite outer-measure test sets need real work.
Key Insight: Subadditivity supplies one inequality for free. □

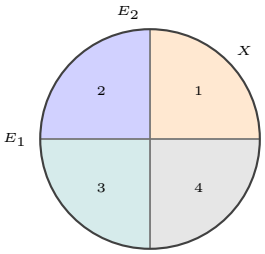

Figure 189: Two cuts split X into four regions; $E_1 \setminus E_2$ is region 2.

Definition 3.1.5. *Zero Sets*
For outer measure $\omega : 2^M \to [0, +\infty]$, *$\omega$-zero set* is $Z \subseteq M$ with $\omega(Z) = 0$; $\mathcal{Z}_\omega := \{Z \subseteq M : \omega(Z) = 0\}$.

Theorem 3.1.6. *Caratheodory Measurable Sets*
Let $\omega : 2^M \to [0, +\infty]$ be an outer measure.
Let $\mathcal{M}_\omega := \{E \subseteq M : \omega(X) = \omega(X \cap E) + \omega(X \cap E^c)\ \forall X \subseteq M\}$ and $\mathcal{Z}_\omega := \{Z \subseteq M : \omega(Z) = 0\}$. Then:
(i) $\mathcal{Z}_\omega \subseteq \mathcal{M}_\omega$;

(ii) $\mathcal{M}_\omega$ *is a* σ*-algebra;*
(iii) $\omega|_{\mathcal{M}_\omega}$ *is countably additive;*
(iv) if $E \in \mathcal{M}_\omega$ *and* $Z \in \mathcal{Z}_\omega$*, then* $E \cup Z, E \setminus Z, E \triangle Z \in \mathcal{M}_\omega$ *and* $\omega(E \cup Z) = \omega(E \setminus Z) = \omega(E \triangle Z) = \omega(E)$.

Proof.

(i) If $Z \in \mathcal{Z}_\omega$ and $X \subseteq M$, then $\omega(X) \le \omega(X \cap Z) + \omega(X \cap Z^c) \le 0 + \omega(X)$, so equality holds and $Z \in \mathcal{M}_\omega$.
(ii) Complements are immediate. If $E, F \in \mathcal{M}_\omega$, with $E^1 = E$, $E^0 = E^c$, $F^1 = F$, and $F^0 = F^c$, then $\omega(X) = \sum_{\epsilon,\eta\in\{0,1\}} \omega(X \cap E^\epsilon \cap F^\eta)$; hence finite Boolean combinations are measurable.
For pairwise disjoint $E_i \in \mathcal{M}_\omega$, set $E = \bigsqcup_i E_i$ and $F_N = (\bigcup_{i=1}^N E_i)^c$.
Finite splitting gives $\omega(X) \ge \sum_{i=1}^N \omega(X \cap E_i) + \omega(X \cap E^c)$ for every N.
Letting $N \to \infty$ and using subadditivity gives $\omega(X) = \sum_{i=1}^\infty \omega(X \cap E_i) + \omega(X \cap E^c)$.
Since $\omega(X \cap E) \le \sum_i \omega(X \cap E_i)$ and $\omega(X) \le \omega(X \cap E) + \omega(X \cap E^c)$, $E \in \mathcal{M}_\omega$.
General countable unions follow from $H_1 = E_1$ and $H_n = E_n \setminus \bigcup_{i<n} E_i$.
(iii) If $E = \bigsqcup_{i=1}^\infty E_i$ with $E_i \in \mathcal{M}_\omega$, then the equality in (ii) with $X = E$ gives $\omega(E) = \sum_{i=1}^\infty \omega(E_i)$.
(iv) By (i)–(ii), $E \cup Z, E \setminus Z, E \triangle Z \in \mathcal{M}_\omega$.
Since $E \cap Z$ and $Z \setminus E$ are null, additivity gives $\omega(E \setminus Z) = \omega(E \cup Z) = \omega(E \triangle Z) = \omega(E)$.

Structural Commentary: Measurable sets preserve countable additivity.
Key Insight: Measurable cuts turn subadditivity into equality. □

Corollary 3.1.7. *Disjointization of Countable Measurable Unions*
Let $\omega : 2^M \to [0, +\infty]$ *be an outer measure and let* $(E_k)_{k\ge1} \subseteq \mathcal{M}_\omega$.
Set $F_1 := E_1$ *and* $F_k := E_k \setminus \bigcup_{i=1}^{k-1} E_i$ *for* $k \ge 2$.
Then (F_k) *is pairwise disjoint in* $\mathcal{M}_\omega$, $\bigcup_{k\ge1} F_k = \bigcup_{k\ge1} E_k$, *and* $\omega(\bigcup_{k\ge1} E_k) = \sum_{k\ge1} \omega(F_k)$ *in* $[0, +\infty]$.

Proof. Since $\mathcal{M}_\omega$ is a σ-algebra, $F_k = E_k \cap (\bigcup_{i<k} E_i)^c \in \mathcal{M}_\omega$. If $i < k$, then $F_i \cap F_k \subseteq E_i \cap (\bigcup_{j<k} E_j)^c = \emptyset$.
For $x \in \bigcup_{k\ge1} E_k$, $x \in F_{\min\{j : x\in E_j\}}$; hence $\bigcup_{k\ge1} F_k = \bigcup_{k\ge1} E_k$, and countable additivity gives $\omega(\bigcup_{k\ge1} E_k) = \sum_{k\ge1} \omega(F_k)$.
Structural Commentary: Disjointization turns unions into sums.
Key Insight: Peel off earlier pieces so a countable union becomes a disjoint sum. □

Definition 3.1.8. *Countable Cover Classes*
Let $\mathcal{S} \subseteq \mathcal{P}(X)$.

(i) *Countable-union class*: $\mathcal{S}_\sigma := \left\{ \bigcup_{n=1}^\infty S_n : (S_n)_{n\ge1} \in \mathcal{S}^{\mathbb{N}} \right\}$.
(ii) *Countable-intersection class*: $\mathcal{S}_{\sigma\delta} := \left\{ \bigcap_{n=1}^\infty A_n : (A_n)_{n\ge1} \in (\mathcal{S}_\sigma)^{\mathbb{N}} \right\}$.

Key Insight: $\mathcal{S}_\sigma$ closes under countable unions; $\mathcal{S}_{\sigma\delta}$ then closes once under countable intersections.

Proposition 3.1.9. *Finite Outer-Measure Envelope*
Let $\mathcal{S} \subseteq \mathcal{P}(X)$, *let* $\rho : \mathcal{S} \to [0, +\infty]$, *and let* ρ^* *be the cover-induced map.*
If $\emptyset \ne E \subseteq X$ *and* $\rho^*(E) < +\infty$, *then there exists* $A \in \mathcal{S}_{\sigma\delta}$ *such that:*
(i) $E \subseteq A$ *and* $\rho^*(A) = \rho^*(E)$.
(ii) If E *and every* $S \in \mathcal{S}$ *are* ρ^**-measurable, then* A *is* ρ^**-measurable and* $\rho^*(A \setminus E) = 0$.

Proof. $\forall n \ge 1$, choose $(S_{n,k})_{k\ge1} \in \mathcal{C}_\mathcal{S}(E)$ with $\sum_{k=1}^\infty \rho(S_{n,k}) < \rho^*(E) + \frac{1}{n}$. Set $A_n := \bigcup_{k=1}^\infty S_{n,k}$, $A := \bigcap_{n=1}^\infty A_n$.
Then $A_n \in \mathcal{S}_\sigma$, $A \in \mathcal{S}_{\sigma\delta}$, $E \subseteq A \subseteq A_n$, and $\rho^*(A_n) < \rho^*(E) + \frac{1}{n}$.

(i) Monotonicity gives $\rho^*(E) \le \rho^*(A) \le \rho^*(A_n) < \rho^*(E) + \frac{1}{n}$ for all n, hence $\rho^*(A) = \rho^*(E)$.
(ii) If E and every $S \in \mathcal{S}$ are ρ^*-measurable, then the ρ^*-measurable sets form a σ-algebra, so A_n and A are ρ^*-measurable. Since $E \subseteq A$ and $\rho^*(E) < +\infty$, excision gives $\rho^*(A \setminus E) = \rho^*(A) - \rho^*(E) = 0$.

Structural Commentary: Finite cost gives a countable-intersection envelope.
Key Insight: Cover within $\frac{1}{n}$, then intersect. □

Definition 3.1.10. *Outer Measure Generated by a Premeasure*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$.

(i) *Algebra covers*: For $B \subseteq X$, set $\mathcal{C}_\mathcal{A}(B) := \{(E_j)_{j\ge1} \in \mathcal{A}^{\mathbb{N}} : B \subseteq \bigcup_{j=1}^\infty E_j\}$.
(ii) *Generated outer measure*: Define $\mu^* : \mathcal{P}(X) \to [0, +\infty]$ by $\mu^*(\emptyset) = 0$ and, for $B \ne \emptyset$, $\mu^*(B) := \inf_{(E_j)\in\mathcal{C}_\mathcal{A}(B)} \sum_{j=1}^\infty \mu(E_j)$, with $\inf \emptyset = +\infty$.

Key Insight: Cover arbitrary sets by algebra sets and minimize total premeasure.

Lemma 3.1.11. *Generated Outer-Measure Properties*
Let $\mu : \mathcal{A} \to [0, +\infty]$ *be a premeasure on an algebra* $\mathcal{A} \subseteq \mathcal{P}(X)$.
For $B \subseteq X$, *put* $\mathcal{C}_\mathcal{A}(B) := \{(E_j) \in \mathcal{A}^{\mathbb{N}} : B \subseteq \bigcup_j E_j\}$.
Set $\inf \emptyset = +\infty$, $\mu^*(\emptyset) = 0$, *and* $\mu^*(B) := \inf_{(E_j)\in\mathcal{C}_\mathcal{A}(B)} \sum_j \mu(E_j)$ *for* $B \ne \emptyset$. *Then:*
(i) $\mu^*(\emptyset) = 0$.
(ii) If $C \subseteq D \subseteq X$, *then* $\mu^*(C) \le \mu^*(D)$.
(iii) If $(B_n)_{n\ge1} \subseteq \mathcal{P}(X)$, *then* $\mu^*(\bigcup_{n=1}^\infty B_n) \le \sum_{n=1}^\infty \mu^*(B_n)$.

(iv) If $E \in \mathcal{A}$, then $\mu^*(E) = \mu(E)$.

Proof.

(i) This is the assigned value of $\mu^*(\emptyset)$.
(ii) If $C \subseteq D$, then $\mathcal{C}_{\mathcal{A}}(D) \subseteq \mathcal{C}_{\mathcal{A}}(C)$; taking infima gives $\mu^*(C) \le \mu^*(D)$.
(iii) Let $B = \bigcup_n B_n$ and $I = \{n : B_n \neq \emptyset\}$. If $I = \emptyset$ or $\sum_n \mu^*(B_n) = +\infty$, the claim is immediate.
Otherwise, for $\varepsilon > 0$ choose $(E_{n,k})_{k\ge 1} \in \mathcal{C}_{\mathcal{A}}(B_n)$ for $n \in I$ with $\sum_k \mu(E_{n,k}) < \mu^*(B_n) + \varepsilon 2^{-n}$.
Then $(E_{n,k})_{n\in I, k\ge 1}$ covers B, so $\mu^*(B) \le \sum_{n\in I}\sum_k \mu(E_{n,k}) \le \sum_n \mu^*(B_n) + \varepsilon$; let $\varepsilon \downarrow 0$.
(iv) Cover $E, \emptyset, \emptyset, \ldots$ gives $\mu^*(E) \le \mu(E)$. Conversely, for any cover $(F_n) \in \mathcal{C}_{\mathcal{A}}(E)$, set $G_n := E \cap \left(F_n \setminus \bigcup_{k<n} F_k\right)$.
Then (G_n) is disjoint in $\mathcal{A}$, $\bigcup_n G_n = E$, and $G_n \subseteq F_n$, so $\mu(E) = \sum_n \mu(G_n) \le \sum_n \mu(F_n)$.
Taking infima gives $\mu(E) \le \mu^*(E)$.

Structural Commentary: Cover containment and regrouping give the outer-measure axioms.
Key Insight: Disjointize algebra covers to recover μ. □

Proposition 3.1.12. *Single-Set Envelope Comparison*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, let μ^* be the generated outer measure, and define $\mu^\sharp : \mathcal{P}(X) \to [0, +\infty]$ by $\mu^\sharp(B) := \inf\{\mu(C) : C \in \mathcal{A},\ B \subseteq C\}$ with $\inf \emptyset = +\infty$.
(i) $\mu^\sharp(E) = \mu(E)$ for every $E \in \mathcal{A}$.
(ii) $\mu^*(B) \le \mu^\sharp(B)$ for every $B \subseteq X$.

Proof.

(i) If $E \in \mathcal{A}$, then E is admissible, so $\mu^\sharp(E) \le \mu(E)$.
Conversely, $E \subseteq C \in \mathcal{A}$ implies $\mu(E) \le \mu(C)$ by monotonicity of μ, hence $\mu(E) \le \mu^\sharp(E)$.
(ii) If no $C \in \mathcal{A}$ contains B, then $\mu^\sharp(B) = +\infty$. Otherwise, each $C \in \mathcal{A}$ with $B \subseteq C$ gives the algebra cover $C, \emptyset, \emptyset, \ldots$, so $\mu^*(B) \le \mu(C)$. Taking infima over such C gives $\mu^*(B) \le \mu^\sharp(B)$.

Structural Commentary: One algebra superset is a special countable cover.
Key Insight: The single-set envelope is larger. □

Example 3.1.13. *Single-Set Envelopes Need Not Be Outer Measures*
Let X be uncountable, let $\mathcal{A}$ be the finite–cofinite algebra on X, and set $\mu(E) := 0$ for finite E, $\mu(E) := +\infty$ for cofinite E. Then μ is a premeasure, $\mu^*(B) = 0$ iff B is countable, and $\mu^\sharp(B) = 0$ iff B is finite.
For any countably infinite $S \subseteq X$, $\mu^\sharp(S) = +\infty$ while $S = \bigcup_{x\in S}\{x\}$ and $\sum_{x\in S} \mu^\sharp(\{x\}) = 0$, QED.
Structural Commentary: Countable covers can be strictly smaller than one-set covers.
Key Insight: Outer measure is built from countable covers precisely to recover countable subadditivity.

Theorem 3.1.14. *Generated Caratheodory Extension*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$. For $B \subseteq X$, put $\mathcal{C}_{\mathcal{A}}(B) := \{(E_j) \in \mathcal{A}^{\mathbb{N}} : B \subseteq \bigcup_j E_j\}$. Define $\mu^*(\emptyset) = 0$, $\mu^*(B) := \inf_{(E_j)\in\mathcal{C}_{\mathcal{A}}(B)} \sum_j \mu(E_j)$ for $B \neq \emptyset$ with $\inf \emptyset = +\infty$, and $\mathcal{C}_\mu := \{E \subseteq X : (\forall B \subseteq X)\ \mu^*(B) = \mu^*(B \cap E) + \mu^*(B \setminus E)\}$. Then:
(i) $\mathcal{A} \subseteq \mathcal{C}_\mu$ and $\mu^*|_{\mathcal{A}} = \mu$.
(ii) $\mathcal{C}_\mu$ is a σ-algebra on X.
(iii) $\mu^*|_{\mathcal{C}_\mu}$ is a complete measure extending μ.

Proof. Lemma 3.1.11 makes μ^* an outer measure and gives $\mu^*|_{\mathcal{A}} = \mu$.

(i) Let $E \in \mathcal{A}$ and $B \subseteq X$. Subadditivity gives $\mu^*(B) \le \mu^*(B \cap E) + \mu^*(B \setminus E)$. If $(F_n) \in \mathcal{C}_{\mathcal{A}}(B)$, then $F_n \cap E, F_n \setminus E \in \mathcal{A}$ and $\mu(F_n) = \mu(F_n \cap E) + \mu(F_n \setminus E)$, so $\mu^*(B \cap E) + \mu^*(B \setminus E) \le \sum_n \mu(F_n)$.
Infimizing over (F_n) gives equality; hence $E \in \mathcal{C}_\mu$. The lemma gives $\mu^*|_{\mathcal{A}} = \mu$.
(ii) The class $\mathcal{C}_\mu$ is exactly the μ^*-measurable class $\mathcal{M}_{\mu^*}$.
Since μ^* is an outer measure, Theorem 3.1.6(ii) gives that $\mathcal{C}_\mu$ is a σ-algebra on X.
(iii) Theorem 3.1.6(iii) gives countable additivity of $\mu^*|_{\mathcal{C}_\mu}$. If $A \subseteq Z \in \mathcal{C}_\mu$ and $\mu^*(Z) = 0$, then monotonicity gives $\mu^*(A) = 0$, so Theorem 3.1.6(i) gives $A \in \mathcal{C}_\mu$. Thus $\mu^*|_{\mathcal{C}_\mu}$ is complete, and (i) shows it extends μ.

Structural Commentary: Caratheodory equality is the additivity domain.
Key Insight: Algebra sets split every algebra cover. □

Proposition 3.1.15. *Caratheodory Splitters for Finitely Subadditive Size*
Let $\alpha : \mathcal{P}(X) \to [0, +\infty)$ satisfy $0 \le \alpha(E) \le \alpha(E \cup F) \le \alpha(E) + \alpha(F)$ for all $E, F \subseteq X$.
Set $\mathcal{S} := \{E \subseteq X : (\forall A \subseteq X)\ \alpha(A) = \alpha(A \cap E) + \alpha(A \setminus E)\}$.
(i) If $\mathcal{S} \neq \emptyset$, then $\mathcal{S}$ is an algebra on X.
(ii) If $E, F \in \mathcal{S}$ and $E \cap F = \emptyset$, then $\alpha(E \cup F) = \alpha(E) + \alpha(F)$.
(iii) The hypothesis $\mathcal{S} \neq \emptyset$ in (i) is necessary: if $\alpha \equiv 1$, then $\mathcal{S} = \emptyset$.

Proof.

(i) Choose $E_0 \in \mathcal{S}$. Splitting $\emptyset$ gives $\alpha(\emptyset) = 0$, hence $\emptyset, X \in \mathcal{S}$.
Complement symmetry gives $E \in \mathcal{S} \Rightarrow X \setminus E \in \mathcal{S}$. For $E, F \in \mathcal{S}$ and $A \subseteq X$, put $A_{11} = A \cap E \cap F$, $A_{10} = A \cap E \setminus F$, $A_{01} = A \cap F \setminus E$, and $A_{00} = A \setminus (E \cup F)$. By successive splitting,
$\alpha(A) = \alpha(A_{11}) + \alpha(A_{10}) + \alpha(A_{01}) + \alpha(A_{00})$; $\alpha(A \setminus (E \cap F)) = \alpha(A_{10}) + \alpha(A_{01}) + \alpha(A_{00})$.
Thus $\alpha(A) = \alpha(A \cap E \cap F) + \alpha(A \setminus (E \cap F))$, so $E \cap F \in \mathcal{S}$. Complements give finite unions.
(ii) Since $F \subseteq X \setminus E$, the splitting identity for E applied to $A = E \cup F$ gives $\alpha(E \cup F) = \alpha(E) + \alpha(F)$.
(iii) If $\alpha \equiv 1$ and $E \in \mathcal{S}$, applying the splitting identity to $A = \emptyset$ gives $1 = 1 + 1$, impossible.

Structural Commentary: Splitters are Boolean once one splitter exists.
Key Insight: Empty-set splitting forces $\alpha(\emptyset) = 0$. □

Theorem 3.1.16. *σ-Finite Generated Extensions Are Unique*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, let μ^* be generated by μ, and put $\mathcal{C}_\mu := \{E \subseteq X : (\forall B \subseteq X)\ \mu^*(B) = \mu^*(B \cap E) + \mu^*(B \setminus E)\}$. Assume $X = \bigcup_n F_n$; $F_n \in \mathcal{A}$, $\mu(F_n) < +\infty$ $\forall n$.

(i) $\mu^*|_{\mathcal{C}_\mu}$ is a measure on $\mathcal{C}_\mu$ and extends μ.
(ii) If ν is a measure on $\mathcal{C}_\mu$ with $\nu|_{\mathcal{A}} = \mu$, then $\nu = \mu^*|_{\mathcal{C}_\mu}$.

Proof.
(i) By Theorem 3.1.14, $\mathcal{C}_\mu$ is complete σ-algebra, $\mathcal{A} \subseteq \mathcal{C}_\mu$, $\mu^*|_{\mathcal{C}_\mu}$ is complete measure with $(\mu^*|_{\mathcal{C}_\mu})|_{\mathcal{A}} = \mu$.
(ii) Let ν be a measure on $\mathcal{C}_\mu$ with $\nu|_{\mathcal{A}} = \mu$.
If $G \in \mathcal{C}_\mu$ and $G \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{A}$, then $\nu(G) \le \sum_n \mu(A_n)$; hence $\nu(G) \le \mu^*(G)$.
If $\mu(X) < +\infty$, apply to $G = E$, $G = X \setminus E$, while $\nu(E) + \nu(X \setminus E) = \nu(X) = \mu^*(X) = \mu^*(E) + \mu^*(X \setminus E)$, gives $\nu(E) = \mu^*(E)$.
In general, put $H_N := \bigcup_{j=1}^{N} F_j$. Then $H_N \in \mathcal{A}$ and $\mu(H_N) < +\infty$. Apply bound to
$E \cap H_N$, $H_N \setminus E$, while $\nu(E \cap H_N) + \nu(H_N \setminus E) = \nu(H_N) = \mu^*(H_N) = \mu^*(E \cap H_N) + \mu^*(H_N \setminus E)$, gives $\nu(E \cap H_N) = \mu^*(E \cap H_N)$.
Since $E \cap H_N \uparrow E$, continuity from below gives $\nu(E) = \mu^*(E)$.

Structural Commentary: Finite uniqueness localizes along a finite-mass exhaustion.
Key Insight: Compare on each finite piece, then pass upward. □

Corollary 3.1.17. *Uniqueness on the Generated σ-Algebra*
Let $\mu : \mathcal{A} \to [0, +\infty]$ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, and assume $X = \bigcup_n F_n$ with $F_n \in \mathcal{A}$ and $\mu(F_n) < +\infty$. If measures ν_1, ν_2 on $\sigma(\mathcal{A})$ extend μ, then $\nu_1 = \nu_2$.

Proof. Put $H_n := \bigcup_{j=1}^{n} F_j$. Then $H_n \in \mathcal{A}$, $H_n \uparrow X$, and $\nu_i(H_n) = \mu(H_n) < +\infty$.
For fixed n, $\mathcal{D}_n := \{E \in \sigma(\mathcal{A}) : \nu_1(E \cap H_n) = \nu_2(E \cap H_n)\}$ is a Dynkin class by finite mass on H_n and contains $\mathcal{A}$, since $A \cap H_n \in \mathcal{A}$.
As $\mathcal{A}$ is a π-system, the π-λ theorem gives $\mathcal{D}_n = \sigma(\mathcal{A})$.
Hence $\nu_1(E \cap H_n) = \nu_2(E \cap H_n)$ for all n; since $E \cap H_n \uparrow E$, continuity from below gives $\nu_1(E) = \nu_2(E)$.
Structural Commentary: Finite exhaustions reduce uniqueness to Dynkin closure.
Key Insight: Compare on H_n, then pass upward. □

Example 3.1.18. *Why σ-Finiteness Matters*
Let $X = \mathbb{Q} \cap (0, 1]$, and let $\mathcal{A}$ be the algebra of finite unions of sets $(a, b] \cap X$, where $0 \le a < b \le 1$ and $a, b \in \mathbb{Q}$.

(i) *Infinite algebra pieces*: Every nonempty $A \in \mathcal{A}$ is infinite.
(ii) *Generated σ-algebra*: For $r \in X$, choose $a_n \in \mathbb{Q} \cap [0, r)$ with $a_n \uparrow r$.
Then $\{r\} = \bigcap_{n=1}^{\infty} ((a_n, r] \cap X) \in \sigma(\mathcal{A})$, hence $\sigma(\mathcal{A}) = \mathcal{P}(X)$.
(iii) *Nonunique extensions*: Let μ_1 be counting measure and $\mu_2 = 2\mu_1$ on $\mathcal{P}(X)$.
Since every nonempty $A \in \mathcal{A}$ is infinite, $\mu_1|_{\mathcal{A}} = \mu_2|_{\mathcal{A}}$, but $\mu_1(\{r\}) = 1 \ne 2 = \mu_2(\{r\})$ for $r \in X$.

Thus uniqueness may fail without σ-finiteness.
Key Insight: Infinite algebra pieces cannot determine singleton scale.

3.2 Euclidean Cells and Cell Outer Measure

Definition 3.2.1. *Cells, Intervals, and Volume in $\mathbb{R}^p$*
Fix $p \ge 1$.

(i) *Cells in $\mathbb{R}$*. For $a \le b$, a cell with endpoints a, b is $I \in \{(a, b), [a, b], (a, b], [a, b)\}$.
If $a = b$, only $[a, a]$ is nonempty.
(ii) *Length*. If I is a cell with endpoints $a \le b$, then $\ell(I) := b - a$; unbounded intervals $(-\infty, a)$, $(-\infty, a]$, $[a, +\infty)$, $(a, +\infty)$, and $\mathbb{R}$ are not cells and have length $+\infty$.
(iii) *Cells in $\mathbb{R}^p$*. A cell is $I = \prod_{i=1}^{p} I_i$, each I_i a cell in $\mathbb{R}$; $\prod_i (a_i, b_i)$ is open, $\prod_i [a_i, b_i]$ is closed, $\prod_i [a_i, b_i)$ is *southwest*, $\prod_i (a_i, b_i]$ is *northeast*, and I is a *cube* iff $\ell(I_1) = \cdots = \ell(I_p)$.
(iv) *Volume*. For a cell $I = \prod_{i=1}^{p} I_i$ with endpoints $a_i \le b_i$, $\mathrm{vol}_p(I) := \prod_{i=1}^{p} \ell(I_i) = \prod_{i=1}^{p} (b_i - a_i)$.
Also $\mathrm{vol}_p(\emptyset) := 0$ and $\mathrm{vol}_p(I^\circ) = \mathrm{vol}_p(\overline{I}) = \mathrm{vol}_p(I)$.
(v) *Unbounded products*. For $I = \prod_{i=1}^{p} I_i$ with possibly unbounded intervals, $\mathrm{vol}_p(I) = 0$ if $\exists j : \ell(I_j) = 0$, $\mathrm{vol}_p(I) = +\infty$ if $\ell(I_i) > 0$ for all i and $\exists j : \ell(I_j) = +\infty$, and $\mathrm{vol}_p(I) = \prod_{i=1}^{p} \ell(I_i)$ otherwise.

Remark 3.2.2. *Cells versus Intervals*
Cells are precisely bounded intervals with finite endpoints, so every cell is an interval but not conversely.
Open cells are open in $\mathbb{R}^p$, closed cells are closed in $\mathbb{R}^p$, and nonempty half-open cells are neither.
In $\mathbb{R}$, $(-\infty, a)$ and $(b, +\infty)$ are open, $(-\infty, a]$ and $[b, +\infty)$ are closed, and $\mathbb{R} = (-\infty, +\infty)$ is clopen.
Key Insight: Topology likes open and closed endpoints; measure bookkeeping often likes half-open endpoints.

Proposition 3.2.3. *Half-Open Cell Calculus*
Fix $\sigma \in \{\text{southwest}, \text{northeast}\}$, and let $\mathcal{H}_\sigma$ be the half-open cells in $\mathbb{R}^p$ with orientation σ.

(i) Any finite Boolean combination of $H_1, \dots, H_n \in \mathcal{H}_\sigma$ is $\bigsqcup_{r=1}^{N} K_r$ with $K_r \in \mathcal{H}_\sigma$.
(ii) If $H \in \mathcal{H}_\sigma$, then $\mathbb{R}^p \setminus H = \bigsqcup_{r=1}^{\infty} K_r$ for some $K_r \in \mathcal{H}_\sigma$.
(iii) For $H \in \mathcal{H}_\sigma$ and $\delta > 0$, $H = \bigsqcup_{r=1}^{N} K_r$ for some $K_r \in \mathcal{H}_\sigma$ with $\operatorname{diam} K_r < \delta$.

Proof.

(i) For coordinate endpoints $E_j = \{c_{j,0} < \cdots < c_{j,m_j}\}$ of $H_1, \ldots, H_n$, set $[a,b)_\sigma = [a,b)$ if $\sigma =$ *southwest* and $[a,b)_\sigma = (a,b]$ if $\sigma =$ *northeast*, $\mathcal{P}_j = \{[c_{j,q-1}, c_{j,q})_\sigma : 1 \le q \le m_j\}$, and $\mathcal{G} = \{\prod_{j=1}^p P_j : P_j \in \mathcal{P}_j\}$.
Every finite Boolean combination is $\bigsqcup_{G \in \mathcal{G}'} G$, $\mathcal{G}' \subseteq \mathcal{G}$.

(ii) Write $H = \prod_{j=1}^p H_j$ and $D_s = (\prod_{j<s} H_j) \times (\mathbb{R} \setminus H_s) \times \mathbb{R}^{p-s}$; then $\mathbb{R}^p \setminus H = \bigsqcup_{s=1}^p D_s$. Decomposing each one-dimensional ray and $\mathbb{R}$ into countably many bounded σ-half-open intervals and taking products gives the required union; countability follows from Corollary 6.2.17.

(iii) Choose m with $2^{-m}(\sum_{j=1}^p \ell(H_j)^2)^{1/2} < \delta$. After m bisections per coordinate, each subcell has diameter $< \delta$; disjointness and volume preservation follow coordinatewise.

Structural Commentary: Half-open cells behave like grid atoms: finite set operations become finite grid selection.
Key Insight: Half-open endpoints prevent overlap while still partitioning boxes cleanly. □

Definition 3.2.4. *Translation of a Set*
If $A \subseteq \mathbb{R}^p$ and $t \in \mathbb{R}^p$, the translate of A by t is $t + A := \{t + a : a \in A\}$.

Proposition 3.2.5. *Cell Volume Is Translation Invariant*
Let $I = \prod_{i=1}^p I_i$ be a cell in $\mathbb{R}^p$ and $t = (t_i)_{i=1}^p \in \mathbb{R}^p$.
Then $t + I = \prod_{i=1}^p (t_i + I_i)$ is a cell with the same endpoint pattern as I, and $\mathrm{vol}_p(t + I) = \mathrm{vol}_p(I)$.

Proof. Let $a_i \le b_i$ be the endpoints of I_i. Then $t_i + I_i$ has endpoints $t_i + a_i \le t_i + b_i$, unchanged endpoint inclusion, and $\ell(t_i + I_i) = (t_i + b_i) - (t_i + a_i) = \ell(I_i)$. Hence $\mathrm{vol}_p(t + I) = \prod_{i=1}^p \ell(t_i + I_i) = \prod_{i=1}^p \ell(I_i) = \mathrm{vol}_p(I)$.
Structural Commentary: Translation moves location but not side lengths.
Key Insight: Coordinate volume depends only on side lengths. □

Definition 3.2.6. *Elementary Sets and Regularity in $\mathbb{R}^p$*

(i) *Elementary sets.* $\mathcal{E} := \{\bigcup_{r=1}^n I_r : n \in \mathbb{N}\}$, where each $I_r \subseteq \mathbb{R}^p$ is a bounded cell.

(ii) *Elementary volume.* If $A = \bigsqcup_{r=1}^n I_r$ with cells I_r, set $m(A) := \sum_{r=1}^n \mathrm{vol}_p(I_r)$.

(iii) *Regularity.* A nonnegative additive $\mu : \mathcal{E} \to [0, +\infty]$ is regular iff $\forall A \in \mathcal{E}$, $\forall \varepsilon > 0$, $\exists F, G \in \mathcal{E}$ with $F \subseteq A \subseteq G$, F closed, G open, and $\mu(G) - \varepsilon \le \mu(A) \le \mu(F) + \varepsilon$.

$\mathcal{E}$ is a ring; every $A \in \mathcal{E}$ admits a finite disjoint cell decomposition; m is well defined, additive, and regular.

Perspective 3.2.7. *Box Volume and Linear Algebra*
For a box $B = \prod_i (a_i, b_i)$ or $\prod_i [a_i, b_i]$, $m(B) = \prod_i (b_i - a_i)$, the diagonal case of determinant volume scaling.
Measure extends this product rule beyond boxes.
Key Insight: Lebesgue measure starts from determinant-style coordinate volume.

Proposition 3.2.8. *Lattice Counts Recover Cell Volume*
Fix $p \ge 1$. For $q \ge 1$ and $A \subseteq \mathbb{R}^q$, write $N_{\mathbb{Z}}(A) := \#(A \cap \mathbb{Z}^q)$, and for $\lambda > 0$ set $D_\lambda A := \{\lambda x : x \in A\}$.
If $I = \prod_{j=1}^p I_j$ is a bounded cell in $\mathbb{R}^p$, then $\lim_{\lambda \to +\infty} \lambda^{-p} N_{\mathbb{Z}}(D_\lambda I) = \mathrm{vol}_p(I)$.

Proof. For a bounded interval $J \subseteq \mathbb{R}$ with length L, $\max\{0, \lambda L - 1\} \le N_{\mathbb{Z}}(D_\lambda J) \le \lambda L + 1$.
Hence $\lambda^{-1} N_{\mathbb{Z}}(D_\lambda J) \to L$ if $L > 0$, and $N_{\mathbb{Z}}(D_\lambda J) \le 1$ if $L = 0$.
Let $L_j := \ell(I_j)$. Since $D_\lambda I = \prod_j D_\lambda I_j$, $N_{\mathbb{Z}}(D_\lambda I) = \prod_j N_{\mathbb{Z}}(D_\lambda I_j)$.
If every $L_j > 0$, then $\lambda^{-p} N_{\mathbb{Z}}(D_\lambda I) = \prod_j \lambda^{-1} N_{\mathbb{Z}}(D_\lambda I_j) \to \prod_j L_j = \mathrm{vol}_p(I)$.
If some $L_j = 0$, then $N_{\mathbb{Z}}(D_\lambda I) = O(\lambda^{p-1})$, so $\lambda^{-p} N_{\mathbb{Z}}(D_\lambda I) \to 0 = \mathrm{vol}_p(I)$.
Structural Commentary: Coordinate lattice density multiplies.
Key Insight: Endpoint effects are lower order. □

Perspective 3.2.9. *Outer Measure as Overestimation*
Elementary $A = \bigsqcup_{r=1}^n I_r$ have $m(A) = \sum_{r=1}^n \mathrm{vol}_p(I_r)$.
General $E \subseteq \mathbb{R}^p$, e.g. $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$, need not be a finite disjoint cell union.
Replace decompositions by countable cell covers $E \subseteq \bigcup_{k=1}^\infty I_k$ and minimize $\sum_{k=1}^\infty \mathrm{vol}_p(I_k)$.
Key Insight: Outer covers size every $E \subseteq \mathbb{R}^p$ before measurability.

Definition 3.2.10. *Cell-Cover Outer Measure*
For $E \subseteq \mathbb{R}^p$, let $\mathcal{C}(E)$ be the countable cell covers $(I_k)_{k \ge 1}$ with $E \subseteq \bigcup_{k=1}^\infty I_k$. Define $m_p^*(E) := \inf_{(I_k) \in \mathcal{C}(E)} \sum_{k=1}^\infty \mathrm{vol}_p(I_k)$. Here $\mathcal{C}(E) \neq \emptyset$ since $E \subseteq \bigcup_{j=1}^\infty [-j, j]^p$, and sum is nonnegative extended-real series, hence order-independent.

Proposition 3.2.11. *Equivalent Covering Families*
Let $E \subseteq \mathbb{R}^p$, $\delta > 0$, and fix a half-open orientation σ. Let $\mathcal{C}_{\mathsf{cl}}(E), \mathcal{C}_{\mathsf{op}}(E), \mathcal{C}_\sigma(E), \mathcal{C}_\delta(E) \subseteq \mathcal{C}(E)$ denote covers by closed cells, open cells, σ-half-open cells, and cells with diameter $< \delta$, respectively.
For $\mathcal{F} \subseteq \mathcal{C}(E)$, put $\alpha(\mathcal{F}) := \inf_{(I_k) \in \mathcal{F}} \sum_k \mathrm{vol}_p(I_k)$. Then $m_p^*(E) = \alpha(\mathcal{C}_{\mathsf{cl}}(E)) = \alpha(\mathcal{C}_{\mathsf{op}}(E)) = \alpha(\mathcal{C}_\sigma(E)) = \alpha(\mathcal{C}_\delta(E))$.

Proof. Let $\mathfrak{C} := \{\mathcal{C}_{\mathsf{cl}}(E), \mathcal{C}_{\mathsf{op}}(E), \mathcal{C}_\sigma(E), \mathcal{C}_\delta(E)\}$. Since each $\mathcal{F} \in \mathfrak{C}$ satisfies $\mathcal{F} \subseteq \mathcal{C}(E)$, $\alpha(\mathcal{F}) \ge m_p^*(E)$.
Conversely, fix $(I_k) \in \mathcal{C}(E)$ and $\varepsilon > 0$. The chain $I_k \subseteq \overline{I_k} \subseteq J_k$, with J_k open and $\sum_k \mathrm{vol}_p(J_k) \le \sum_k \mathrm{vol}_p(I_k) + \varepsilon$, followed by disjoint σ-half-open decomposition and finite δ-subdivision, gives covers in every $\mathcal{F} \in \mathfrak{C}$ of total volume $\le \sum_k \mathrm{vol}_p(I_k) + \varepsilon$. Infimize over (I_k) and let $\varepsilon \downarrow 0$.
Structural Commentary: Cover normalizations preserve the infimum.
Key Insight: Change endpoints, then refine. □

Theorem 3.2.12. *Basic Properties of Cell-Cover Outer Measure*
For $E, F, E_n \subseteq \mathbb{R}^p$,

(i) Null empty set. $0 \le m_p^*(E) \le +\infty$ *and* $m_p^*(\emptyset) = 0$.
(ii) Monotonicity. $E \subseteq F \Longrightarrow m_p^*(E) \le m_p^*(F)$.
(iii) Subadditivity. $m_p^*(\bigcup_{n=1}^{\infty} E_n) \le \sum_{n=1}^{\infty} m_p^*(E_n)$.

Proof.

(i) Since all cover sums lie in $[0, +\infty]$, $0 \le m_p^*(E) \le +\infty$; the cover $(\emptyset, \emptyset, \ldots)$ gives $m_p^*(\emptyset) = 0$.
(ii) If $E \subseteq F$, then $\mathcal{C}(F) \subseteq \mathcal{C}(E)$, hence $\inf_{\mathcal{C}(E)} \le \inf_{\mathcal{C}(F)}$.
(iii) If some $m_p^*(E_n) = +\infty$, the claim is trivial.
Otherwise, for $\varepsilon > 0$, choose $(I_k^n)_{k\ge1} \in \mathcal{C}(E_n)$ with $\sum_k \mathrm{vol}_p(I_k^n) \le m_p^*(E_n) + \varepsilon 2^{-n}$.
Then $(I_k^n)_{(n,k)\in\mathbb{N}^2}$ countable, covers $\bigcup_n E_n$, and $m_p^*(\bigcup_n E_n) \le \sum_{n,k} \mathrm{vol}_p(I_k^n) \le \sum_n m_p^*(E_n) + \varepsilon$.

Structural Commentary: Countable subadditivity is proved by merging near-optimal covers.
Key Insight: Nonnegative double sums may be scanned in one sequence. □

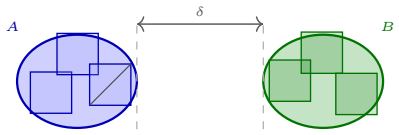

Figure 190: Positive distance separates cells with $\operatorname{diam} I_k < \delta$.

Theorem 3.2.13. *Additivity at Positive Distance*
Let $A, B \subseteq \mathbb{R}^p$ be nonempty and disjoint, and suppose $\operatorname{dist}(A, B) := \inf\{\|a - b\|_2 : a \in A,\ b \in B\} > 0$.
Then $m_p^*(A \cup B) = m_p^*(A) + m_p^*(B)$.

Proof. The inequality $\le$ follows from Theorem 3.2.12. For $\ge$, set $\delta := \operatorname{dist}(A, B)$. If $m_p^*(A \cup B) = +\infty$, done.
Otherwise, given $\varepsilon > 0$, choose $(I_n) \in \mathcal{C}(A \cup B)$ with $\operatorname{diam} I_n < \delta$ and $\sum_n \mathrm{vol}_p(I_n) \le m_p^*(A \cup B) + \varepsilon$.
Put $N_A := \{n : I_n \cap A \ne \emptyset\}$ and $N_B := \{n : I_n \cap B \ne \emptyset\}$. Since $N_A \cap N_B = \emptyset$, $(I_n)_{n\in N_A}$ covers A, $(I_n)_{n\in N_B}$ covers B, and $m_p^*(A) + m_p^*(B) \le \sum_{n\in N_A} \mathrm{vol}_p(I_n) + \sum_{n\in N_B} \mathrm{vol}_p(I_n) \le \sum_n \mathrm{vol}_p(I_n) \le m_p^*(A \cup B) + \varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: Positive separation lets a sufficiently fine cover be sorted unambiguously.
Key Insight: Countable subadditivity becomes equality when no small cell can touch both pieces. □

Theorem 3.2.14. *Outer Measure of a Cell*
If I is a cell in $\mathbb{R}^p$, then $m_p^*(I) = \mathrm{vol}_p(I)$.

Proof. The cover $(I, \emptyset, \ldots)$ gives $m_p^*(I) \le \mathrm{vol}_p(I)$. For $\ge$, the case $\mathrm{vol}_p(I) = 0$ follows by nonnegativity.
Assume $\mathrm{vol}_p(I) > 0$ and fix $\varepsilon > 0$. Choose a closed cell $J \subseteq I$ with $\mathrm{vol}_p(I) - \varepsilon < \mathrm{vol}_p(J)$, taking $J = I$ when I is closed. For any open-cell cover (G_k) of I, compactness gives $J \subseteq \bigcup_{k=1}^m G_k$.
Let $\mathcal{Q}$ be the finite rectangular grid determined by the faces of $J, \overline{G_1}, \ldots, \overline{G_m}$. Up to zero-volume boundary atoms, each $Q \in \mathcal{Q}$ with $Q \subseteq J$ lies in some G_k, hence $\mathrm{vol}_p(J) \le \sum_{k=1}^m \mathrm{vol}_p(G_k) \le \sum_{k=1}^{\infty} \mathrm{vol}_p(G_k)$.
Infimize over open-cell covers, use Proposition 3.2.11, and let $\varepsilon \downarrow 0$.
Structural Commentary: Compact inner cells reduce covers to finite grids.
Key Insight: Outer measure is calibrated so that cells keep their original volume. □

Corollary 3.2.15. *Outer Measure of Real Intervals*
Let $J \subseteq \mathbb{R}$ be a nonempty interval, and set $\ell(J) = +\infty$ when J is unbounded. Then $m_1^*(J) = \ell(J)$.

Proof. If J is bounded, then J is a one-dimensional cell, so $m_1^*(J) = \mathrm{vol}_1(J) = \ell(J)$ by Theorem 3.2.14.
If J is unbounded, then for every $N \in \mathbb{N}$ there exists a bounded interval $I_N \subseteq J$ with $\ell(I_N) = N$.
Hence monotonicity and the bounded case give $m_1^*(J) \ge m_1^*(I_N) = N$. Thus $m_1^*(J) = +\infty = \ell(J)$.
Structural Commentary: Cells calibrate bounded intervals; monotonicity handles unbounded ones.
Key Insight: Infinite length is forced by arbitrarily long bounded subintervals. □

Theorem 3.2.16. *Translation Invariance of Outer Measure*
For $E \subseteq \mathbb{R}^p$ and $t \in \mathbb{R}^p$, $m_p^*(t + E) = m_p^*(E)$.

Proof. The map $\mathcal{C}(E) \to \mathcal{C}(t + E)$, $(I_k) \mapsto (t + I_k)$, is bijective with inverse $(J_k) \mapsto (-t + J_k)$.
By Proposition 3.2.5, $\sum_k \mathrm{vol}_p(t + I_k) = \sum_k \mathrm{vol}_p(I_k)$, so the two infima are equal.
Structural Commentary: Translate covers forward and backward.
Key Insight: Translation invariance of cell volume lifts directly to translation invariance of outer measure. □

3.3 Caratheodory Measurability and Lebesgue Measure

Definition 3.3.1. *Outer Measure and μ-Measurable Sets*
Let $\mathcal{E}$ and $\mathcal{E}_{\mathrm{op}}$ denote the elementary and open elementary subsets of $\mathbb{R}^p$, let $\mu : \mathcal{E} \to [0, \infty)$ be additive and regular, and set $A \triangle B := (A \setminus B) \cup (B \setminus A)$.

(i) For $E \subseteq \mathbb{R}^p$, $\mu^*(E) := \inf\{\sum_{n=1}^{\infty} \mu(A_n) : E \subseteq \bigcup_{n=1}^{\infty} A_n,\ A_n \in \mathcal{E}_{\mathrm{op}}\}$.
(ii) $d_\mu(A, B) := \mu^*(A \triangle B)$.
(iii) $\mathcal{M}_F(\mu)$ is the class of *finitely μ-measurable* sets: $E \in \mathcal{M}_F(\mu)$ iff $\exists (E_k)_{k\ge1} \subseteq \mathcal{E}$ with $d_\mu(E_k, E) \to 0$.
(iv) $\mathcal{M}(\mu)$ is the class of *μ-measurable* sets: $E \in \mathcal{M}(\mu)$ iff $E = \bigcup_{k=1}^{\infty} E_k$ for some $E_k \in \mathcal{M}_F(\mu)$.

Perspective 3.3.2. *Outer Measure as Outside Approximation*
Outer measure covers A from outside; $d_\mu(A,B)=\mu^*(A\triangle B)$ measures closeness.
Key Insight: Countable subadditivity collects near-optimal covers.

Theorem 3.3.3. *Elementary Approximation Extension*
Let $\mathcal{E}$ be the elementary subsets of $\mathbb{R}^p$, let $\mathcal{E}_{\mathrm{op}}\subseteq\mathcal{E}$ be the open elementary sets, and let $\mu:\mathcal{E}\to[0,\infty)$ be additive and regular. For $E\subseteq\mathbb{R}^p$, set $\mu^*(E):=\inf\{\sum_{n=1}^\infty\mu(A_n):E\subseteq\bigcup_{n=1}^\infty A_n,\ A_n\in\mathcal{E}_{\mathrm{op}}\}$; set $d_\mu(A,B):=\mu^*(A\triangle B)$, let $\mathcal{M}_F(\mu)$ be the d_μ-closure of $\mathcal{E}$, and let $\mathcal{M}(\mu):=\{\bigcup_{k=1}^\infty E_k:E_k\in\mathcal{M}_F(\mu)\}$. Then:
(i) $\mu^*(A)=\mu(A)$ *for every* $A\in\mathcal{E}$;
(ii) μ^* *is countably subadditive on* $2^{\mathbb{R}^p}$;
(iii) $\mathcal{M}(\mu)$ *is a* σ*-ring;*
(iv) $\mu^*|_{\mathcal{M}(\mu)}$ *is a measure.*

Proof.
(i) Let $A\in\mathcal{E}$. For $\varepsilon>0$, regularity gives $G\in\mathcal{E}_{\mathrm{op}}$ with $A\subseteq G$ and $\mu(G)\le\mu(A)+\varepsilon$, so $\mu^*(A)\le\mu(A)$.
Conversely, choose closed elementary $F\subseteq A$ with $\mu(F)\ge\mu(A)-\varepsilon$. Every open elementary cover of A covers F; compactness gives a finite subcover, whose total μ-mass is at least $\mu(F)$.
Taking infima and letting $\varepsilon\downarrow 0$ gives $\mu(A)\le\mu^*(A)$.
(ii) Let $E_j\subseteq\mathbb{R}^p$ for $j\ge1$, and let $\varepsilon>0$. For each j, choose an open elementary cover $(A_{j,n})_{n\ge1}$ of E_j with $\sum_{n=1}^\infty\mu(A_{j,n})\le\mu^*(E_j)+\varepsilon2^{-j}$. Then $(A_{j,n})_{j,n}$ covers $\bigcup_{j=1}^\infty E_j$, so $\mu^*(\bigcup_j E_j)\le\sum_j\mu^*(E_j)+\varepsilon$; let $\varepsilon\downarrow0$.
(iii) If $A_k\to A$ and $B_k\to B$ in d_μ, with $A_k,B_k\in\mathcal{E}$, then $d_\mu(A_k\diamond B_k,A\diamond B)\le d_\mu(A_k,A)+d_\mu(B_k,B)$ for $\diamond\in\{\cup,\cap,\setminus\}$. Hence $\mathcal{M}_F(\mu)$ is ring. Countable unions give σ-ring since difference distributes over them.
(iv) If $A=\bigsqcup_{j=1}^\infty A_j$ with $A_j\in\mathcal{M}_F(\mu)$, subadditivity gives $\mu^*(A)\le\sum_j\mu^*(A_j)$.
Finite additivity on $\mathcal{M}_F(\mu)$ gives $\mu^*(A)\ge\sum_{j=1}^N\mu^*(A_j)$ for every N, hence $\mu^*(A)\ge\sum_j\mu^*(A_j)$.
Disjointize countable $\mathcal{M}_F(\mu)$-decompositions.
Structural Commentary: Elementary sets are completed under the distance measured by symmetric difference.
Key Insight: Approximate by elementary sets, prove identities there, then pass through $d_\mu(A,B)$. □

Definition 3.3.4. *Extended and Lebesgue Measure*
Let μ^* be the outer measure induced by μ and let $\mathcal{M}(\mu)$ be its measurable class.
(i) The *extended measure* of μ is $\mu_{\mathrm{ext}}:=\mu^*|_{\mathcal{M}(\mu)}$, usually denoted again by μ.
(ii) If $\mu=m$ is elementary volume on $\mathbb{R}^p$, then $m^*|_{\mathcal{M}(m)}$ is *Lebesgue measure* on $\mathbb{R}^p$.

Definition 3.3.5. *Box-Cover Outer Measure Model*
Fix $n\ge1$.
(i) $\mathcal{B}_n:=\{\prod_{i=1}^n(a_i,b_i):a_i,b_i\in\mathbb{R},\ a_i\le b_i\ \forall i\}$ is the class of *open boxes*.
For $B=\prod_{i=1}^n(a_i,b_i)$, $\mathrm{vol}_n(B):=\prod_{i=1}^n(b_i-a_i)$.
(ii) For $\Omega\subseteq\mathbb{R}^n$, $m_n^*(\Omega):=\inf\{\sum_{j\in J}\mathrm{vol}_n(B_j):\Omega\subseteq\bigcup_{j\in J}B_j,\ B_j\in\mathcal{B}_n,\ |J|\le\aleph_0\}$.

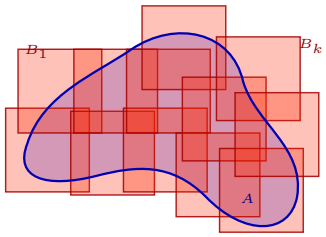

Figure 191: Cover $A\subseteq\bigcup_k B_k$ by squares; minimize $\sum_k|B_k|$.

Method 3.3.6. *Lebesgue Measure Construction*
(i) *Cover from outside*
For $A\subseteq\mathbb{R}^n$, define $m_n^*(A):=\inf\left\{\sum_{j\in J}\mathrm{vol}_n(B_j):(B_j)_{j\in J}\text{ covers }A,\ J\text{ at most countable}\right\}$.
This is defined for every set, but it is only countably subadditive in general.
(ii) *Keep the clean splitters*
Call A measurable when every test set $X\subseteq\mathbb{R}^n$ splits additively: $m_n^*(X)=m_n^*(X\cap A)+m_n^*(X\cap A^c)$.
These measurable sets form a σ-algebra.
(iii) *Restrict the outer measure*
Define Lebesgue measure on measurable sets by $m(A):=m_n^*(A)$.
On this restricted class, m is countably additive.
Key Insight: Outer measure sizes all sets; clean splitting restores additivity.

Definition 3.3.7. *Lebesgue σ-Algebra via Caratheodory*
(i) $\mathcal{L}^n:=\{E\subseteq\mathbb{R}^n:E\text{ is }m_n^*\text{-measurable}\}$ is the *Lebesgue σ-algebra*.
(ii) $m:=m_n^*|_{\mathcal{L}^n}$ is *Lebesgue measure*; for $E\in\mathcal{L}^n$, write $m(E)=m_n^*(E)$.

Proposition 3.3.8. *Outer-Null Sets Are Lebesgue Measurable*
If $Z\subseteq\mathbb{R}^n$ and $m_n^*(Z)=0$, then:
(i) $Z\in\mathcal{L}^n$;
(ii) $m(Z)=0$;
(iii) $\mathcal{P}(Z)\subseteq\mathcal{L}^n$ *and* $m(W)=0$ *for every* $W\subseteq Z$.

Proof.

(i) For $A \subseteq \mathbb{R}^n$, monotonicity gives $m_n^*(A \cap Z) = 0$, so $m_n^*(A) \leq m_n^*(A \cap Z) + m_n^*(A \cap Z^c) = m_n^*(A \cap Z^c) \leq m_n^*(A)$; hence Z satisfies Caratheodory's criterion.

(ii) Since $Z \in \mathcal{L}^n$, $m(Z) = m_n^*(Z) = 0$.

(iii) If $W \subseteq Z$, then $m_n^*(W) \leq m_n^*(Z) = 0$; applying (i)–(ii) to W gives $W \in \mathcal{L}^n$ and $m(W) = 0$.

Structural Commentary: Lebesgue measure is complete because outer-null subsets split every test set trivially.
Key Insight: A zero piece cannot disturb additivity. □

Example 3.3.9. *Countable Real Sets Are Outer-Null*
Let $C \subseteq \mathbb{R}$ be finite or countable, and choose $(c_k)_{k\geq 1}$ with $C \subseteq \{c_k : k \geq 1\}$. For $\varepsilon > 0$, choose open intervals $I_k \ni c_k$ with $\ell(I_k) < \varepsilon 2^{-k}$. Then $C \subseteq \bigcup_{k\geq 1} I_k$, so $0 \leq m_1^*(C) \leq \sum_{k=1}^{\infty} \ell(I_k) < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $m_1^*(C) = 0$. By Proposition 3.3.8, $C \in \mathcal{L}^1$ and $m(C) = 0$.
Key Insight: A countable list can spend a rapidly decreasing measure budget around its points.

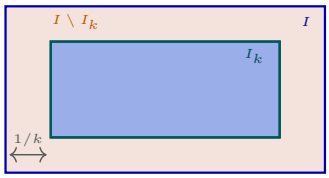

Figure 192: Inner cells I_k exhaust I; the collar $I \setminus I_k$ has vanishing volume.

Proposition 3.3.10. *Cells Are Lebesgue Measurable*
Let $I \subseteq \mathbb{R}^n$ be a coordinate cell. Then $I \in \mathcal{L}^n$ and $m(I) = \mathrm{vol}_n(I)$.

Proof. If $\mathrm{vol}_n(I) = 0$, then $m_n^*(I) = 0$ by Theorem 3.2.14, so Proposition 3.3.8 gives $I \in \mathcal{L}^n$ and $m(I) = 0$.
Assume $\mathrm{vol}_n(I) > 0$. For small $r > 0$, let I_r be the inward r-shrink of I.
Then $\mathrm{dist}(I_r, \mathbb{R}^n \setminus I) > 0$ and $\eta_r := m_n^*(I \setminus I_r) \to 0$ by finite slab covers.
Fix $A \subseteq \mathbb{R}^n$ with $m_n^*(A) < +\infty$. Positive-distance additivity gives $m_n^*(A) \geq m_n^*(A \cap I_r) + m_n^*(A \setminus I)$.
Since $A \cap I \subseteq (A \cap I_r) \cup (I \setminus I_r)$, $m_n^*(A \cap I) \leq m_n^*(A \cap I_r) + \eta_r$; hence $m_n^*(A) \geq m_n^*(A \cap I) + m_n^*(A \setminus I) - \eta_r$.
Letting $r \downarrow 0$ gives the finite-measure test inequality, so Lemma 3.1.4 gives $I \in \mathcal{L}^n$.
Theorem 3.2.14 gives $m(I) = m_n^*(I) = \mathrm{vol}_n(I)$.
Structural Commentary: Cells are the bridge between coordinate volume and Lebesgue measure.
Key Insight: Shrink, split, let the collar vanish. □

3.4 Caratheodory-Hahn Extension

Definition 3.4.1. *Premeasures on Set Families*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$ and $\mu : \mathcal{S} \to [0, +\infty]$.

(i) μ is *finitely additive on $\mathcal{S}$* if $\mu(\bigcup_{k=1}^{n} E_k) = \sum_{k=1}^{n} \mu(E_k)$ $\forall$ finite pairwise disjoint family $E_k \in \mathcal{S}$ with $\bigcup_{k=1}^{n} E_k \in \mathcal{S}$.

(ii) μ is *countably monotone on $\mathcal{S}$* if, for all $E, E_1, E_2, \ldots \in \mathcal{S}$ with $E \subseteq \bigcup_{k=1}^{\infty} E_k$, $\mu(E) \leq \sum_{k=1}^{\infty} \mu(E_k)$.

(iii) μ is a *premeasure on $\mathcal{S}$* if μ is finitely additive on $\mathcal{S}$, countably monotone on $\mathcal{S}$, and $\emptyset \in \mathcal{S} \Rightarrow \mu(\emptyset) = 0$.

Structural Commentary: No closure property of $\mathcal{S}$ is assumed; additivity is imposed only for finite disjoint unions which remain in $\mathcal{S}$, and countable monotonicity only for countable $\mathcal{S}$-covers of members of $\mathcal{S}$.

Proposition 3.4.2. *Necessary Conditions for Extension*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$, let $\mu : \mathcal{S} \to [0, +\infty]$, let μ^* be the outer measure induced by μ, and put $\overline{\mu} := \mu^*|_{\mathcal{M}_{\mu^*}}$.
If $\mathcal{S} \subseteq \mathcal{M}_{\mu^*}$ and $\overline{\mu}(E) = \mu(E)$ for all $E \in \mathcal{S}$, then the following hold:

(i) $\emptyset \in \mathcal{S} \Rightarrow \mu(\emptyset) = 0$.
(ii) μ *is finitely additive on* $\mathcal{S}$.
(iii) μ *is countably monotone on* $\mathcal{S}$.

Hence μ is a premeasure on $\mathcal{S}$.

Proof. By the extension hypotheses, $\mu(F) = \overline{\mu}(F) = \mu^*(F)$ for every $F \in \mathcal{S}$.

(i) If $\emptyset \in \mathcal{S}$, then $\mu(\emptyset) = \overline{\mu}(\emptyset) = 0$.

(ii) Let $n \geq 1$, let $E_1, \ldots, E_n \in \mathcal{S}$ be pairwise disjoint, and assume $F := \bigcup_{k=1}^{n} E_k \in \mathcal{S}$.

Since $\overline{\mu}$ is a measure, $\mu(F) = \overline{\mu}(F) = \sum_{k=1}^{n} \overline{\mu}(E_k) = \sum_{k=1}^{n} \mu(E_k)$.

(iii) Let $E, E_1, E_2, \ldots \in \mathcal{S}$ and $E \subseteq \bigcup_{k=1}^{\infty} E_k$. By monotonicity and countable subadditivity of μ^*,

$\mu(E) = \mu^*(E) \leq \mu^*(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} \mu^*(E_k) = \sum_{k=1}^{\infty} \mu(E_k)$.

Structural Commentary: The measure extension handles finite disjoint unions; μ^* handles countable covers.
Key Insight: Outer subadditivity supplies the inequality even when $\bigcup_k E_k \notin \mathcal{S}$. □

Definition 3.4.3. *Relative Complements and Semirings*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$.

(i) $\mathcal{S}$ is *closed under relative complements* if $A, B \in \mathcal{S} \Rightarrow A \setminus B \in \mathcal{S}$.

(ii) $\mathcal{S}$ is a *semiring of sets* if, for all $A, B \in \mathcal{S}$, $A \cap B \in \mathcal{S}$ and $A \setminus B = \bigsqcup_{j=1}^{n} C_j$ for some $n \geq 0$ and $C_j \in \mathcal{S}$.

Structural Commentary: Relative-complement closure gives $\emptyset = A \setminus A \in \mathcal{S}$ and $A \cap B = A \setminus (A \setminus B) \in \mathcal{S}$.

Theorem 3.4.4. *Extension from Relative-Complement Families*
Let $\emptyset \neq \mathcal{S} \subseteq \mathcal{P}(X)$ be closed under relative complements, let $\mu : \mathcal{S} \to [0, +\infty]$ be a premeasure, and let μ^* be the outer measure induced by μ. Then:

(i) $\mu^*(E) = \mu(E)$ for every $E \in \mathcal{S}$.
(ii) $\mathcal{S} \subseteq \mathcal{M}_{\mu^*}$.

Proof.

(i) Fix $E \in \mathcal{S}$. The one-set cover gives $\mu^*(E) \leq \mu(E)$. If $(E_k)_{k\geq 1} \subseteq \mathcal{S}$ covers E, then countable monotonicity gives $\mu(E) \leq \sum_{k=1}^{\infty} \mu(E_k)$; taking the infimum over all such covers gives $\mu(E) \leq \mu^*(E)$.

(ii) Fix $A \in \mathcal{S}$. By Lemma 3.1.4, it suffices to prove $\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \mu^*(B)$ for every $B \subseteq X$ with $\mu^*(B) < +\infty$. Let $\varepsilon > 0$ and choose an $\mathcal{S}$-cover $(E_k)_{k\geq 1}$ of B with $\sum_{k=1}^{\infty} \mu(E_k) < \mu^*(B) + \varepsilon$.

For each k, relative-complement closure gives $E_k \setminus A \in \mathcal{S}$ and $E_k \cap A = E_k \setminus (E_k \setminus A) \in \mathcal{S}$. By finite additivity, $\mu(E_k) = \mu(E_k \cap A) + \mu(E_k \setminus A)$. Since $(E_k \cap A)_{k\geq 1}$ covers $B \cap A$ and $(E_k \setminus A)_{k\geq 1}$ covers $B \setminus A$, $\mu^*(B \cap A) + \mu^*(B \setminus A) \leq \sum_{k=1}^{\infty} \mu(E_k \cap A) + \sum_{k=1}^{\infty} \mu(E_k \setminus A) = \sum_{k=1}^{\infty} \mu(E_k) < \mu^*(B) + \varepsilon$. Let $\varepsilon \downarrow 0$.

Structural Commentary: Countable monotonicity identifies μ with μ^*; finite additivity splits covers.
Key Insight: Relative complements turn one cover into inside/outside covers. □

Remark 3.4.5. *Two Roles of a Premeasure*
In Theorem 3.4.4, finite additivity is used to split each covering set across A and A^c, while countable monotonicity is used to force $\mu^*(E) = \mu(E)$ on the original family.
Key Insight: Measurability comes from finite splitting; extension equality comes from countable cover control.

Proposition 3.4.6. *Finite-Union Extension from a Semiring*
Let $\mathcal{S}$ be a semiring of subsets of X, and set $\mathcal{S}' := \{\bigsqcup_{k=1}^{n} A_k : n \geq 0,\ A_1, \ldots, A_n \in \mathcal{S}\}$. If $\mu : \mathcal{S} \to [0, +\infty]$ is a premeasure, define $\mu' : \mathcal{S}' \to [0, +\infty]$ by $\mu'(\bigsqcup_{k=1}^{n} A_k) := \sum_{k=1}^{n} \mu(A_k)$ for any disjoint representation. Then:

(i) $\mathcal{S}'$ is closed under relative complements.
(ii) μ' is well-defined and extends μ.
(iii) μ' is a premeasure on $\mathcal{S}'$.
(iv) μ' is the unique premeasure on $\mathcal{S}'$ extending μ.

Proof.

(i) Let $E = \bigsqcup_{i=1}^{r} A_i$ and $F = \bigsqcup_{j=1}^{s} B_j$ in $\mathcal{S}'$.
For each i, repeated semiring differences decompose $A_i \setminus F$ into finitely many disjoint $\mathcal{S}$-sets.
These decompositions, over $i = 1, \ldots, r$, form a finite disjoint $\mathcal{S}$-decomposition of $E \setminus F$, so $E \setminus F \in \mathcal{S}'$.

(ii) If $E = \bigsqcup_{i=1}^{r} A_i = \bigsqcup_{j=1}^{s} B_j$, then $A_i = \bigsqcup_{j=1}^{s} (A_i \cap B_j)$ and $B_j = \bigsqcup_{i=1}^{r} (A_i \cap B_j)$ with all intersections in $\mathcal{S}$.
Finite additivity of μ gives $\sum_{i=1}^{r} \mu(A_i) = \sum_{i=1}^{r} \sum_{j=1}^{s} \mu(A_i \cap B_j) = \sum_{j=1}^{s} \mu(B_j)$, so μ' is well-defined.
If $A \in \mathcal{S}$, the one-piece representation gives $\mu'(A) = \mu(A)$.

(iii) Since $\emptyset$ is the empty union, $\mu'(\emptyset) = 0$.
Finite additivity of μ' follows by refining finitely many disjoint $\mathcal{S}'$-sets to their disjoint $\mathcal{S}$-pieces.
For countable monotonicity, let $E = \bigsqcup_{i=1}^{r} A_i \in \mathcal{S}'$, let $F_m = \bigsqcup_{j=1}^{N_m} B_{mj} \in \mathcal{S}'$, assume $E \subseteq \bigcup_{m=1}^{\infty} F_m$.
For each i, $A_i \subseteq \bigcup_{m,j} (A_i \cap B_{mj})$ with $A_i \cap B_{mj} \in \mathcal{S}$, so $\mu(A_i) \leq \sum_{m,j} \mu(A_i \cap B_{mj})$.
Summing over i gives $\mu'(E) \leq \sum_m \sum_j \sum_i \mu(A_i \cap B_{mj})$.
For fixed m, j, repeated semiring differences decompose $B_{mj} \setminus \bigcup_i (A_i \cap B_{mj})$ into finitely many disjoint $\mathcal{S}$-sets, so finite additivity gives $\sum_i \mu(A_i \cap B_{mj}) \leq \mu(B_{mj})$. Thus $\mu'(E) \leq \sum_m \sum_j \mu(B_{mj}) = \sum_m \mu'(F_m)$.

(iv) If ν is a premeasure on $\mathcal{S}'$ extending μ, then finite additivity gives
$\nu(\bigsqcup_{k=1}^{n} A_k) = \sum_{k=1}^{n} \nu(A_k) = \sum_{k=1}^{n} \mu(A_k) = \mu'(\bigsqcup_{k=1}^{n} A_k)$ for every disjoint representation. Hence $\nu = \mu'$.

Structural Commentary: Finite disjoint unions close a semiring under relative complements.
Key Insight: All estimates reduce to intersections of semiring pieces. □

Theorem 3.4.7. *Caratheodory-Hahn Extension from a Semiring*
Let $\mathcal{S}$ be a semiring of subsets of X, let $\mu : \mathcal{S} \to [0, +\infty]$ be a premeasure, let μ^* be the outer measure induced by μ, and set $\overline{\mu} := \mu^*|_{\mathcal{M}_{\mu^*}}$. Then:

(i) $\mathcal{S} \subseteq \mathcal{M}_{\mu^*}$ and $\overline{\mu}(A) = \mu(A)$ for every $A \in \mathcal{S}$.
(ii) If $X = \bigcup_{k\geq 1} S_k$ with $S_k \in \mathcal{S}$ and $\mu(S_k) < +\infty$, then $\overline{\mu}$ is σ-finite.
(iii) Under the hypothesis in (ii), if ν is a measure on $\mathcal{M}_{\mu^*}$ with $\nu(A) = \mu(A)$ for every $A \in \mathcal{S}$, then $\nu = \overline{\mu}$.

Proof.

(i) Let $(\mathcal{S}', \mu')$ be the finite-union extension from Proposition 3.4.6. The $\mathcal{S}$- and $\mathcal{S}'$-cover infima coincide, so μ^* is also induced by μ'. Theorem 3.4.4 gives $\mathcal{S}' \subseteq \mathcal{M}_{\mu^*}$ and $\mu^*|_{\mathcal{S}'} = \mu'$. Since $\mu'|_{\mathcal{S}} = \mu$, the claim follows.

(ii) Assume $X = \bigcup_{k\geq 1} S_k$, $S_k \in \mathcal{S}$, $\mu(S_k) < +\infty$. By (i), each $S_k \in \mathcal{M}_{\mu^*}$, $\overline{\mu}(S_k) = \mu(S_k) < +\infty$, so $\overline{\mu}$ is σ-finite.

(iii) Assume (ii), and let ν be a measure on $\mathcal{M}_{\mu^*}$ extending μ. By finite additivity, $\nu = \overline{\mu} = \mu'$ on $\mathcal{S}'$.

Hence every $\mathcal{S}'$-cover of $E \in \mathcal{M}_{\mu^*}$ gives $\nu(E) \leq$ its μ'-mass, so $\nu(E) \leq \overline{\mu}(E)$. Put $H_n := \bigcup_{k=1}^{n} S_k \in \mathcal{S}'$.

Then $H_n \uparrow X$ and $\nu(H_n) = \overline{\mu}(H_n) = \mu'(H_n) < +\infty$.

Applying $\nu \leq \overline{\mu}$ to $E \cap H_n$ and $H_n \setminus E$ forces $\nu(E \cap H_n) = \overline{\mu}(E \cap H_n)$, because both measures split the same finite mass H_n.

Since $E \cap H_n \uparrow E$, continuity from below gives $\nu(E) = \overline{\mu}(E)$.

Structural Commentary: Pass to finite disjoint unions, then exhaust by finite-mass pieces.
Key Insight: $\mathcal{S}'$ supplies relative complements and the cover infimum for uniqueness. □

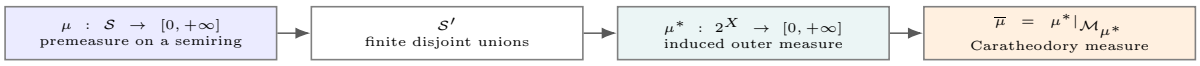

Figure 193: Caratheodory-Hahn: pass to $\mathcal{S}'$, form μ^* by cover infima, then $\overline{\mu} := \mu^*|_{\mathcal{M}_{\mu^*}}$ with $\overline{\mu}|_{\mathcal{S}} = \mu$.

Corollary 3.4.8. *Agreement on a σ-Finite Semiring*
Let $\mathcal{S}$ be a semiring of subsets of X, let $\mathcal{B} := \sigma(\mathcal{S})$, and let ν_1, ν_2 be measures on $\mathcal{B}$ with $\nu_1|_{\mathcal{S}} = \nu_2|_{\mathcal{S}}$.
If $X = \bigcup_{k\geq 1} S_k$ for some $S_k \in \mathcal{S}$ with $\nu_1(S_k) = \nu_2(S_k) < +\infty$ for all k, then $\nu_1 = \nu_2$.

Proof. Let $\mathcal{S}'$ be the finite-disjoint-union closure of $\mathcal{S}$. Finite additivity gives $\nu_1 = \nu_2$ on $\mathcal{S}'$.
Put $H_n := \bigcup_{k=1}^{n} S_k \in \mathcal{S}'$; then $H_n \uparrow X$ and $\nu_i(H_n) < +\infty$. For fixed n, $\mathcal{D}_n := \{E \in \mathcal{B} : \nu_1(E \cap H_n) = \nu_2(E \cap H_n)\}$ is a Dynkin class containing the π-system $\mathcal{S}$, since $A \cap H_n \in \mathcal{S}'$ for $A \in \mathcal{S}$. Hence $\mathcal{D}_n = \mathcal{B}$.
Thus $\nu_1(E \cap H_n) = \nu_2(E \cap H_n)$ $\forall E \in \mathcal{B}$ and $n \geq 1$; since $E \cap H_n \uparrow E$, continuity from below gives $\nu_1(E) = \nu_2(E)$.
Structural Commentary: Localize to finite-mass H_n, then apply π-λ.
Key Insight: Agreement on $\mathcal{S}$ propagates through $\mathcal{S}'$ and the exhaustion. □

Example 3.4.9. *Bounded Intervals and Rectangles Are Semirings*
Let $\mathcal{I}$ be the bounded intervals of $\mathbb{R}$, and let $\mathcal{R}_n := \{I_1 \times \cdots \times I_n : I_j \in \mathcal{I}\}$.

(i) $\mathcal{I}$ is a semiring: if $I, J \in \mathcal{I}$, then $I \cap J \in \mathcal{I}$ and $I \setminus J$ is a disjoint union of at most two members of $\mathcal{I}$.

(ii) $\mathcal{R}_n$ is a semiring in $\mathbb{R}^n$: intersections are coordinatewise products, and $R \setminus Q$ decomposes by coordinate cuts into finitely many disjoint rectangles in $\mathcal{R}_n$.

Thus length on $\mathcal{I}$ and rectangular volume on $\mathcal{R}_n$ fit the Caratheodory-Hahn extension pattern.

3.5 Metric Outer Measures and Hausdorff Measure

Metric outer measures are outer measures whose additivity recognizes positive separation. They abstract the positive-distance additivity of Euclidean cell-cover outer measure from Theorem 3.2.13.

Definition 3.5.1. *Metric/Caratheodory Outer Measure*
Let (X, d) be a metric space and let $\omega : 2^X \to [0, +\infty]$ be an outer measure.

(i) For nonempty $A, B \subseteq X$, $\operatorname{dist}(A, B) := \inf\{d(a, b) : a \in A,\ b \in B\}$.

(ii) ω is *metric outer measure* if $\operatorname{dist}(A, B) > 0 \Rightarrow \omega(A \cup B) = \omega(A) + \omega(B)$ $\forall$ nonempty $A, B \subseteq X$.

Key Insight: Positive metric separation forces outer-measure additivity.

Definition 3.5.2. *Separation by a Real-Valued Function*
Let X be a set and let $\varphi : X \to \mathbb{R}$. Sets $A, B \subseteq X$ are *φ-separated* if, for some $r < s$, one of A, B is contained in $\{\varphi \leq r\}$ and the other is contained in $\{\varphi \geq s\}$.

Proposition 3.5.3. *Separation by One Real-Valued Function*
Let X be a set, let $\omega : 2^X \to [0, +\infty]$ be an outer measure, and let $\varphi : X \to \mathbb{R}$. If $\omega(A \cup B) = \omega(A) + \omega(B)$ for every φ-separated pair $A, B \subseteq X$, then $\varphi : (X, \mathcal{M}_\omega) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable.

Proof. Fix $t \in \mathbb{R}$, put $E_t := \{\varphi > t\}$, and fix $A \subseteq X$ with $\omega(A) < +\infty$. Set $B := A \cap E_t$ and $C := A \setminus E_t$.
For $n \geq 1$, let $B_n := A \cap \{\varphi > t + \frac{1}{n}\}$, $B_0 := \emptyset$, and $R_n := B_n \setminus B_{n-1}$.
Same-parity strips are pairwise φ-separated, so $\sum_{n\geq 1} \omega(R_n) < +\infty$. Given $\varepsilon > 0$, choose n with $\sum_{k>n} \omega(R_k) < \varepsilon$.
Since $B \subseteq B_n \cup \bigcup_{k>n} R_k$ and B_n, C are φ-separated, $\omega(B) + \omega(C) \leq \omega(B_n \cup C) + \varepsilon \leq \omega(A) + \varepsilon$.
Let $\varepsilon \downarrow 0$, then $\omega(A) \geq \omega(A \cap E_t) + \omega(A \setminus E_t)$; Lemma 3.1.4 gives $E_t \in \mathcal{M}_\omega$ for every t, hence φ is measurable.
Structural Commentary: Measurability reduces to a finite-mass split of each level set.
Key Insight: Summable level strips leave an arbitrarily small boundary layer. □

Theorem 3.5.4. *Metric Outer Measures Measure Borel Sets*
Let (X, d) be a metric space and let $\omega : 2^X \to [0, +\infty]$ be a metric outer measure. Then:

(i) Every closed $F \subseteq X$ belongs to $\mathcal{M}_\omega$.

(ii) $\mathcal{B}(X) \subseteq \mathcal{M}_\omega$.

Proof.

(i) The case $F = \emptyset$ is immediate. For nonempty closed F, set $\varphi(x) := \operatorname{dist}(\{x\}, F)$. Proposition 5.1.21 gives φ 1-Lipschitz and $F = \varphi^{-1}(\{0\})$, so φ-separated pairs have positive distance. Metric additivity verifies Proposition 3.5.3; hence φ is $\mathcal{M}_\omega$-measurable and $F \in \mathcal{M}_\omega$.

(ii) By (i), all closed sets lie in the σ-algebra $\mathcal{M}_\omega$; since they generate $\mathcal{B}(X)$, $\mathcal{B}(X) \subseteq \mathcal{M}_\omega$.

Structural Commentary: Distance functions turn closed sets into real-valued separation tests.
Key Insight: Closed measurability plus generation gives Borel measurability. □

Definition 3.5.5. *Hausdorff Outer Measure*
Let (X, d) be a metric space, $\alpha > 0$, $\varepsilon > 0$.

(i) $\mathcal{H}^*_{\alpha,\varepsilon}(\emptyset) := 0$; $\mathcal{H}^*_{\alpha,\varepsilon}(E) := \inf\{\sum_{k=1}^{\infty} (\operatorname{diam} A_k)^\alpha : E \subseteq \bigcup_{k=1}^{\infty} A_k,\ \emptyset \neq A_k \subseteq X,\ \operatorname{diam} A_k < \varepsilon\}$, nonempty $E \subseteq X$
(ii) If $0 < \varepsilon_1 < \varepsilon_2$, then $\mathcal{H}^*_{\alpha,\varepsilon_1}(E) \geq \mathcal{H}^*_{\alpha,\varepsilon_2}(E)$.
(iii) $\mathcal{H}^*_\alpha(E) := \sup_{\varepsilon>0} \mathcal{H}^*_{\alpha,\varepsilon}(E) = \lim_{\varepsilon \downarrow 0} \mathcal{H}^*_{\alpha,\varepsilon}(E)$.

Key Insight: $\mathcal{H}^*_\alpha$ is the limiting cost of countable small-diameter covers.

Figure 194: $\operatorname{dist}(E, F) = \delta$ and $\operatorname{diam} A_k < \delta$: each A_k meets at most one of E, F.

Theorem 3.5.6. *Hausdorff Measure on Borel Sets*
Let (X, d) be a metric space and let $\alpha > 0$. Then:
(i) $\mathcal{H}^*_\alpha : 2^X \to [0, +\infty]$ is a metric outer measure.
(ii) $\mathcal{H}_\alpha := \mathcal{H}^*_\alpha|_{\mathcal{B}(X)}$ is a measure on $\mathcal{B}(X)$, called the α-dimensional Hausdorff measure on X.

Proof.

(i) For each $\varepsilon > 0$, $\mathcal{H}^*_{\alpha,\varepsilon}(\emptyset) = 0$, monotonicity is immediate from admissible covers, and countable subadditivity follows by joining admissible ε-covers; hence $\mathcal{H}^*_\alpha(\bigcup_i E_i) = \sup_{\varepsilon>0} \mathcal{H}^*_{\alpha,\varepsilon}(\bigcup_i E_i) \leq \sum_i \mathcal{H}^*_\alpha(E_i)$.
Thus $\mathcal{H}^*_\alpha$ is an outer measure. If $\operatorname{dist}(E, F) = \delta > 0$ and $0 < \varepsilon < \delta$, every admissible ε-cover of $E \cup F$ splits into disjoint subcovers of E and F, so $\mathcal{H}^*_{\alpha,\varepsilon}(E \cup F) \geq \mathcal{H}^*_{\alpha,\varepsilon}(E) + \mathcal{H}^*_{\alpha,\varepsilon}(F)$.
Letting $\varepsilon \downarrow 0$ and using subadditivity gives metric additivity.

(ii) By (i), $\mathcal{H}^*_\alpha$ is a metric outer measure; Theorem 3.5.4 gives $\mathcal{B}(X) \subseteq \mathcal{M}_{\mathcal{H}^*_\alpha}$.
Since $\mathcal{H}^*_\alpha|_{\mathcal{M}_{\mathcal{H}^*_\alpha}}$ is a measure, its restriction $\mathcal{H}^*_\alpha|_{\mathcal{B}(X)} = \mathcal{H}_\alpha$ is a measure on $\mathcal{B}(X)$.

Structural Commentary: Cover infima give the outer measure; separated cover splitting gives metric additivity.
Key Insight: Small-diameter covers cannot straddle positive gaps. □

Remark 3.5.7. *One-Dimensional Calibration*
On $\mathbb{R}$, interval covers give $\mathcal{H}^*_1((a, b]) \leq b - a$, and every diameter cover of $(a, b]$ enlarges to interval covers with arbitrarily small extra total length, giving $\mathcal{H}^*_1((a, b]) \geq b - a$. Hence $\mathcal{H}^*_1((a, b]) = b - a$, and uniqueness of the Borel extension from half-open intervals gives $\mathcal{H}_1 = m$ on $\mathcal{B}(\mathbb{R})$.

Example 3.5.8. *Finite Sets Have Zero Positive-Dimensional Hausdorff Measure*
If $E = \{x_1, \ldots, x_N\} \subseteq X$ and $\alpha > 0$, then $\mathcal{H}^*_\alpha(E) = 0$: for every $\varepsilon > 0$, the singleton cover $E \subseteq \bigcup_{j=1}^{N} \{x_j\}$ is admissible and has total α-content $\sum_{j=1}^{N} (\operatorname{diam}\{x_j\})^\alpha = 0$.
Key Insight: Finite sets are covered by zero-diameter singletons.

Proposition 3.5.9. *Hausdorff Measures Decrease Through Dimension*
Let (X, d) be a metric space, let $E \subseteq X$, and let $0 < \alpha < \beta$. If $\mathcal{H}^*_\alpha(E) < +\infty$, then $\mathcal{H}^*_\beta(E) = 0$.

Proof. Fix $\varepsilon > 0$. Choose admissible ε-cover (A_k) of E with $\sum_k (\operatorname{diam} A_k)^\alpha \leq \mathcal{H}^*_{\alpha,\varepsilon}(E) + 1 \leq \mathcal{H}^*_\alpha(E) + 1$.
Since $\operatorname{diam} A_k < \varepsilon$, $\sum_k (\operatorname{diam} A_k)^\beta = \sum_k (\operatorname{diam} A_k)^{\beta-\alpha} (\operatorname{diam} A_k)^\alpha \leq \varepsilon^{\beta-\alpha}(\mathcal{H}^*_\alpha(E) + 1)$.
Hence $\mathcal{H}^*_{\beta,\varepsilon}(E) \leq \varepsilon^{\beta-\alpha}(\mathcal{H}^*_\alpha(E) + 1)$. Letting $\varepsilon \downarrow 0$ gives $\mathcal{H}^*_\beta(E) = 0$.
Structural Commentary: Compare one small cover at dimensions α and β.
Key Insight: The factor $\varepsilon^{\beta-\alpha}$ kills finite α-content. □

Definition 3.5.10. *Hausdorff Dimension*
For $E \subseteq X$, the *Hausdorff dimension* is $\dim_H(E) := \inf\{\beta > 0 : \mathcal{H}^*_\beta(E) = 0\}$, with $\inf \emptyset := +\infty$.

Remark 3.5.11. *Dimension as a Critical Exponent*
For $E \subseteq X$, set $Z_E := \{\beta > 0 : \mathcal{H}^*_\beta(E) = 0\}$. Proposition 3.5.9 gives $Z_E = (\dim_H(E), +\infty)$ or $[\dim_H(E), +\infty)$ inside $(0, +\infty)$, so $\dim_H(E)$ is the zero-measure threshold for the Hausdorff scale.

4 Borel, Lebesgue, and Stieltjes Measures

4.1 Borel, Lebesgue, and Stieltjes Measure Constructions

Proposition 4.1.1. *Half-Open Interval Algebra on $\mathbb{R}$*
Let $\mathcal{F}$ be the algebra of finite disjoint unions of $(a,b]$, $(-\infty,b]$, $(a,+\infty)$ $(a,b\in\mathbb{R})$, and $\mathbb{R}$. Set $\ell((a,b])=b-a$ and $\ell((-\infty,b])=\ell((a,+\infty))=\ell(\mathbb{R})=+\infty$, extending additively over disjoint unions. Then

(i) $\mathcal{F}$ is an algebra on $\mathbb{R}$ and $\sigma(\mathcal{F})=\mathcal{B}(\mathbb{R})$.
(ii) ℓ is a premeasure on $\mathcal{F}$.

Proof.

(i) Generator complements lie in $\mathcal{F}$, hence $\mathcal{F}$ is an algebra.
Since $\mathcal{F}\subseteq\mathcal{B}(\mathbb{R})$ and $(a,b)=\bigcup_{n>1/(b-a)}(a,b-1/n]$, $\sigma(\mathcal{F})=\mathcal{B}(\mathbb{R})$.

(ii) Normal forms make ℓ well defined and finitely additive.

For countable additivity, reduce to $(a,b]=\bigsqcup_{j\geq1}(a_j,b_j]$: finite subunions give $\sum_{j=1}^{N}(b_j-a_j)\leq b-a$.

Conversely, choose $\varepsilon_j>0$ with $\sum_{j\geq1}\varepsilon_j<\varepsilon$; by compactness, $\exists$ finite $K\subseteq\mathbb{N}$ s.t. $[a+\delta,b]\subseteq\bigcup_{j\in K}(a_j,b_j+\varepsilon_j)$, hence $b-a-\delta\leq\sum_{j\in K}(b_j-a_j+\varepsilon_j)\leq\sum_{j\geq1}(b_j-a_j)+\varepsilon$.

Let $\delta,\varepsilon\downarrow0$. Truncation handles half-lines; finite disjoint unions follow componentwise.

Structural Commentary: Lebesgue measure starts as length on a finite interval algebra, not on all subsets.
Key Insight: Compactness supplies the reverse inequality in countable additivity of interval length. □

Remark 4.1.2. *Why the Half-Open Convention Matters*
Finite unions generated by (a,b), $(-\infty,b)$, $(a,+\infty)$, and $\mathbb{R}$ still generate $\mathcal{B}(\mathbb{R})$, but are not an algebra, where $(0,1)^c=(-\infty,0]\cup[1,+\infty)$. Half-open endpoints make complements remain in the starting family.

Definition 4.1.3. *Borel–Stieltjes and Lebesgue–Stieltjes Measures*
Let $g:\mathbb{R}\to\mathbb{R}$ be increasing and right-continuous, and set $g(\pm\infty):=\lim_{x\to\pm\infty}g(x)\in\overline{\mathbb{R}}$.
Let $\mathcal{I}:=\{(a,b]:a<b\}\cup\{(-\infty,b]:b\in\mathbb{R}\}\cup\{(a,+\infty):a\in\mathbb{R}\}\cup\{\mathbb{R}\}$ and $\mathcal{F}:=\operatorname{alg}(\mathcal{I})$.

(i) Define G_g on $\mathcal{I}$ by endpoint increments and set $\mu_g(\bigsqcup_{j=1}^{m}I_j):=\sum_{j=1}^{m}G_g(I_j)$ for disjoint $I_j\in\mathcal{I}$.
(ii) The unique extension of μ_g to $\mathcal{B}(\mathbb{R})$ is the *Borel–Stieltjes measure generated by g*.
(iii) The completion of this Borel measure is the *Lebesgue–Stieltjes measure generated by g*.

Theorem 4.1.4. *Borel–Stieltjes Measure Existence*
Let $g:\mathbb{R}\to\mathbb{R}$ be increasing and right-continuous, and put $g(\pm\infty):=\lim_{x\to\pm\infty}g(x)\in\overline{\mathbb{R}}$.
Set $\mathcal{I}:=\{(a,b]:a<b\}\cup\{(-\infty,b]:b\in\mathbb{R}\}\cup\{(a,+\infty):a\in\mathbb{R}\}\cup\{\mathbb{R}\}$ and $\mathcal{F}:=\operatorname{alg}(\mathcal{I})$.
Define G_g on $\mathcal{I}$ by endpoint increments and set $\mu_g(\bigsqcup_{j=1}^{m}I_j):=\sum_{j=1}^{m}G_g(I_j)$ for disjoint $I_j\in\mathcal{I}$. Then:

(i) μ_g is a σ-finite premeasure on $\mathcal{F}$ and has a unique extension to $\mathcal{B}(\mathbb{R})$, again denoted μ_g.
(ii) If $g=\mathrm{id}_{\mathbb{R}}$, then μ_g is Lebesgue measure.
(iii) If g is continuous, then $\mu_g(J)=g(b)-g(a)$ for $J\in\{(a,b),[a,b),(a,b],[a,b]\}$, $a<b$.

Proof.

(i) Normal forms give well-defined finite additivity: G_g-increments telescope and $G_g\geq0$.
If $(a,b]=\bigsqcup_{j\geq1}(a_j,b_j]$, finite subunions give $\sum_{j=1}^{N}G_g((a_j,b_j])\leq G_g((a,b])$.
Conversely, compactness gives finite K with $[a+\varepsilon,b]\subseteq\bigcup_{j\in K}(a_j,b_j+\varepsilon_j)$. Hence $g(b)-g(a+\varepsilon)\leq\sum_{j\in K}(g(b_j+\varepsilon_j)-g(a_j))$. Let $\varepsilon,\varepsilon_j\downarrow0$, hence countable additivity on bounded generators.
Half-lines follow by truncation, finite unions componentwise, so μ_g is a premeasure. Since $\mathbb{R}=\bigcup_{n\geq1}(-n,n]$ and $\mu_g((-n,n])=g(n)-g(-n)<+\infty$, μ_g is σ-finite; Theorem 3.1.16 gives uniqueness.

(ii) If $g=\mathrm{id}_{\mathbb{R}}$, then $G_g((a,b])=b-a$, so μ_g agrees with half-open length on $\mathcal{F}$.
Uniqueness identifies its Borel extension with Lebesgue measure.

(iii) If g is continuous, then $(x-1/n,x]\downarrow\{x\}$ gives $\mu_g(\{x\})=\lim_n(g(x)-g(x-1/n))=0$.

Structural Commentary: Stieltjes measure replaces ordinary length $b-a$ by the increment $g(b)-g(a)$.
Key Insight: Right-continuity aligns the half-open interval convention with countable covers. □

Proposition 4.1.5. *Finite Borel Measures Give Stieltjes Generators*
Let ν be finite on $\mathcal{B}(\mathbb{R})$ and set $g(x):=\nu((-\infty,x])$. Then:

(i) g is increasing and right-continuous;
(ii) $g(-\infty)=0$ and $g(+\infty)=\nu(\mathbb{R})$;
(iii) $\nu((a,b])=g(b)-g(a)$ for $a\leq b$;
(iv) $\nu=\mu_g$ on $\mathcal{B}(\mathbb{R})$.

Proof.

(i) If $a\leq b$, then $(-\infty,a]\subseteq(-\infty,b]$, so $g(a)\leq g(b)$. If $x_n\downarrow x$, then $(-\infty,x_n]\downarrow(-\infty,x]$.
Finite continuity from above gives $g(x_n)\downarrow g(x)$.

(ii) Since $(-\infty,-n]\downarrow\emptyset$ and $(-\infty,n]\uparrow\mathbb{R}$, continuity from above/below gives $g(-\infty)=0$ and $g(+\infty)=\nu(\mathbb{R})$.

(iii) For $a\leq b$, $(-\infty,b]=(-\infty,a]\sqcup(a,b]$, hence $\nu((a,b])=g(b)-g(a)$.

(iv) By (ii)–(iii), ν and μ_g agree on $\mathcal{I}$, hence on $\mathcal{F}$. Theorem 4.1.4 gives $\nu=\mu_g$ on $\mathcal{B}(\mathbb{R})$.

Structural Commentary: A finite real-line measure is encoded by its cumulative mass function.
Key Insight: Right-continuity of distribution functions is continuity from above for measures. □

Example 4.1.6. *Basic Borel–Stieltjes Generators*
The following generators illustrate the interval rule in Theorem 4.1.4.

(i) If $g(x) = 2x$, then $\mu_g = 2m$ on $\mathcal{B}(\mathbb{R})$.
(ii) If $g(x) = \arctan x$, then $\mu_g((a,b]) = \arctan b - \arctan a = \int_a^b (1+x^2)^{-1}\,dx$ and $\mu_g(\mathbb{R}) = \pi$.
(iii) If $g(x) = 0$ for $x < 0$ and $g(x) = 1$ for $x \geq 0$, then $\mu_g = \delta_0$.
(iv) If $g(x) = 0$ for $x < 0$ and $g(x) = x$ for $x \geq 0$, then $\mu_g(E) = m(E \cap [0,+\infty))$ for Borel E.

Lemma 4.1.7. *Open Sets Are Countable Unions of Rational Boxes*
For every open $G \subseteq \mathbb{R}^n$, there is a countable $\Sigma \subseteq \{\prod_{i=1}^n (a_i,b_i) : a_i, b_i \in \mathbb{Q},\ a_i < b_i\}$ such that $G = \bigcup_{B\in\Sigma} B$.

Proof. Let $\mathcal{B}_{\mathbb{Q}} := \{\prod_{i=1}^n (a_i,b_i) : a_i, b_i \in \mathbb{Q},\ a_i < b_i\}$; then $\mathcal{B}_{\mathbb{Q}}$ is countable since $\mathbb{Q}^{2n}$ is countable.
Set $\Sigma := \{B \in \mathcal{B}_{\mathbb{Q}} : B \subseteq G\}$. The inclusion $\bigcup_{B\in\Sigma} B \subseteq G$ is immediate.
If $x \in G$, choose $r > 0$ with $B_r(x) \subseteq G$, then choose $a_i, b_i \in \mathbb{Q}$ with $x_i - r/n < a_i < x_i < b_i < x_i + r/n$.
Thus $x \in B := \prod_i (a_i,b_i) \subseteq B_r(x) \subseteq G$, so $x \in \bigcup_{B\in\Sigma} B$.
Structural Commentary: Rational boxes give a countable Euclidean base.
Key Insight: Openness supplies a ball; density of $\mathbb{Q}$ supplies rational box endpoints inside it. □

Theorem 4.1.8. *Lebesgue Measure Basic Properties*
Let $m = m_n^*|_{\mathcal{L}^n}$.

(i) $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{L}^n$, and $\mathcal{L}^n$ is closed under $E \mapsto E^c$, countable $\cup$, and countable $\cap$.
(ii) For $E, F, E_k \in \mathcal{L}^n$, $0 \leq m(E)$, $E \subseteq F \Rightarrow m(E) \leq m(F)$, and $m(\bigcup_k E_k) \leq \sum_k m(E_k)$.
(iii) If $(E_k) \subseteq \mathcal{L}^n$ are pairwise disjoint, then $m(\bigsqcup_k E_k) = \sum_k m(E_k)$.
(iv) $m([0,1]^n) = 1$, and $E \in \mathcal{L}^n$, $t \in \mathbb{R}^n$ imply $E + t \in \mathcal{L}^n$ and $m(E+t) = m(E)$.

Proof.

(i) By Theorem 3.1.6, $\mathcal{L}^n$ is a σ-algebra, hence closed under complements and countable unions/intersections. Open boxes are cells, hence in $\mathcal{L}^n$ by Proposition 3.3.10; by Lemma 4.1.7, each open $G = \bigcup_j B_j$ with rational open boxes B_j, so $G \in \mathcal{L}^n$. Thus $\mathcal{B}(\mathbb{R}^n) = \sigma(\mathcal{O}_{\mathbb{R}^n}) \subseteq \mathcal{L}^n$.
(ii) Since $m = m_n^*|_{\mathcal{L}^n}$, nonnegativity follows from $m_n^* \geq 0$. If $E \subseteq F$, then $m(E) \leq m(F)$ by outer-measure monotonicity. Also $m(\bigcup_k E_k) = m_n^*(\bigcup_k E_k) \leq \sum_k m_n^*(E_k) = \sum_k m(E_k)$ by outer-measure subadditivity.
(iii) If $(E_k) \subseteq \mathcal{L}^n$ pairwise disjoint, by Theorem 3.1.6, m countably additive on $\mathcal{L}^n$, so $m(\bigsqcup_k E_k) = \sum_k m(E_k)$.
(iv) By Theorem 3.2.14, since $[0,1]^n$ is a cell, $m([0,1]^n) = \mathrm{vol}_n([0,1]^n) = 1$.
For translation, let $A \subseteq \mathbb{R}^n$. By Theorem 3.2.16 and
$E \in \mathcal{L}^n$, $m_n^*(A) = m_n^*(A-t) = m_n^*((A-t)\cap E) + m_n^*((A-t)\cap E^c) = m_n^*(A\cap(E+t)) + m_n^*(A\cap(E+t)^c)$; hence $E+t \in \mathcal{L}^n$, and $m(E+t) = m_n^*(E+t) = m_n^*(E) = m(E)$.

Structural Commentary: The wish list is recovered only after restricting outer measure to measurable sets.
Key Insight: The box-cover construction gives size; the measurable-set condition restores additivity. □

Theorem 4.1.9. *Uniqueness from Open Cells*
Let ν be a measure on $\mathcal{L}^n$ with $\nu(I) = \mathrm{vol}_n(I)$ for every open cell $I \subseteq \mathbb{R}^n$. Then $\nu = m$ on $\mathcal{L}^n$.

Proof. Fix $N \in \mathbb{N}$, $Q_N = (-N,N)^n$. If $E \in \mathcal{L}^n$ and $E \subseteq Q_N$, then any open-cell cover $E \subseteq \bigcup_j J_j$ gives $\nu(E) \leq \sum_j \nu(J_j) = \sum_j \mathrm{vol}_n(J_j)$, hence $\nu(E) \leq m(E)$ by infimum.
Applying this to E and $Q_N \setminus E$, and using $\nu(Q_N) = m(Q_N) < +\infty$, yields $\nu(E) + \nu(Q_N \setminus E) = m(E) + m(Q_N \setminus E)$ with each left summand $\leq$ its right counterpart; hence $\nu(E) = m(E)$.
For arbitrary $E \in \mathcal{L}^n$, set $A_1 = E \cap Q_1$, $A_N = E \cap (Q_N \setminus Q_{N-1})$ for $N > 1$.
Then $E = \bigsqcup_N A_N$, each $A_N \subseteq Q_N$, and countable additivity gives $\nu(E) = \sum_N \nu(A_N) = \sum_N m(A_N) = m(E)$.
Structural Commentary: Agreement on coordinate cells determines the whole Lebesgue measure.
Key Insight: Covers give $\nu \leq m$ locally; finite complements force equality. □

Method 4.1.10. *Translation Invariance by Moving the Test Set*

(i) For $t \in \mathbb{R}^n$ and $A \subseteq \mathbb{R}^n$, translated box covers give $m_n^*(A+t) = m_n^*(A)$.
(ii) If $E \in \mathcal{L}^n$, then for all $A \subseteq \mathbb{R}^n$, $(A\cap(E+t)) - t = (A-t)\cap E$ and $(A\cap(E+t)^c) - t = (A-t)\cap E^c$.
(iii) Apply the Caratheodory split for E to $A - t$ to get $E + t \in \mathcal{L}^n$ and $m(E+t) = m(E)$.

Key Insight: Translation invariance of outer measure transports the clean-splitting identity.

Proposition 4.1.11. *First Borel Hierarchy Examples*
In $\mathbb{R}^p$:

(i) If $G = \bigcap_{n\geq 1} U_n$ with U_n open or $F = \bigcup_{n\geq 1} C_n$ with C_n closed, then $G, F \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$.
(ii) Any set obtained from open or closed sets by countably many $\cup, \cap, {}^c$ operations is Borel.
(iii) In $\mathbb{R}$, $\{0\} = \bigcap_{n\geq 1} (-\frac{1}{n}, \frac{1}{n})$ is G_δ but not open.
(iv) In $\mathbb{R}$, $(0,1) = \bigcup_{n\geq 2} [\frac{1}{n}, 1 - \frac{1}{n}]$ is F_σ but not closed.

Proof.

(i) Since $\mathcal{B}(\mathbb{R}^p) = \sigma(\mathcal{O}_{\mathbb{R}^p})$, it contains open sets, closed sets by complements, and countable $\cap, \cup$ by σ-algebra closure; Theorem 4.1.8 gives $\mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$.
(ii) This is Proposition 1.1.3 applied to $\mathcal{B}(\mathbb{R}^p)$.
(iii) For $x \in \mathbb{R}$, $x = 0 \iff |x| < \frac{1}{n}$ for all $n \geq 1$; the singleton $\{0\}$ is not open.
(iv) For $x \in \mathbb{R}$, $0 < x < 1 \iff \exists n \geq 2 : \frac{1}{n} \leq x \leq 1 - \frac{1}{n}$; the interval $(0,1)$ is not closed.

Structural Commentary: Borel sets are generated by countable topological operations.
Key Insight: Countable intersections and unions quickly leave the class of open and closed sets. □

Perspective 4.1.12. *Why Not Measure Every Subset?*
Vitali-type sets show m_1^* is not additive on $2^{\mathbb{R}}$: disjoint unions and refinements can break additivity.
Thus measure is m^* on clean splitters E: $m^*(X) = m^*(X \cap E) + m^*(X \cap E^c)$ for all X.
Key Insight: Outer measure is universal; measure lives on clean splitters.

Definition 4.1.13. *Null Sets and Real-Line Measure Zero*
Let $(X, \mathcal{A}, \mu)$ be a measure space and let $A \subseteq \mathbb{R}$.

(i) $N \subseteq X$ is *null* iff $N \in \mathcal{A}$ and $\mu(N) = 0$.
(ii) A has *measure zero* iff $\forall \varepsilon > 0$, $\exists$ open intervals (O_n) with $A \subseteq \bigcup_{n=1}^{\infty} O_n$ and $\sum_{n=1}^{\infty} |O_n| \leq \varepsilon$.
(iii) If $A \subseteq [a, b]$, then A has *content zero* iff the condition in (ii) holds with finitely many intervals.

For Lebesgue measure on $\mathbb{R}$, null sets are also *zero sets*.

Proposition 4.1.14. *Basic Measure-Zero Examples on $\mathbb{R}$*

(i) Every finite $A \subseteq \mathbb{R}$ has content 0.
(ii) Every countable $A \subseteq \mathbb{R}$ has measure 0.
(iii) The Cantor set P has content 0.

Proof.

(i) For $A = \{a_1, \ldots, a_N\}$ and $\varepsilon > 0$, choose open $I_j \ni a_j$ with $|I_j| < \frac{\varepsilon}{N}$; then $A \subseteq \bigcup_{j=1}^{N} I_j$ and $\sum_{j=1}^{N} |I_j| < \varepsilon$. The case $A = \emptyset$ is trivial.
(ii) For $A = \{a_j : j \in \mathbb{N}\}$ and $\varepsilon > 0$, choose open $I_j \ni a_j$ with $|I_j| < \frac{\varepsilon}{2^j}$; then $A \subseteq \bigcup_{j \geq 1} I_j$ and $\sum_{j \geq 1} |I_j| < \varepsilon$.
(iii) Given $\varepsilon > 0$, choose N with $\left(\frac{2}{3}\right)^N < \frac{\varepsilon}{2}$. If $E_N = \bigcup_{k=1}^{2^N} J_k$, $|J_k| = 3^{-N}$, choose open $I_k \supseteq J_k$ with $|I_k| < 3^{-N} + \frac{\varepsilon}{2^{N+1}}$; then $P \subseteq \bigcup_{k=1}^{2^N} I_k$ and $\sum_{k=1}^{2^N} |I_k| < \left(\frac{2}{3}\right)^N + \frac{\varepsilon}{2} < \varepsilon$.

Structural Commentary: Zero size means arbitrarily small interval covers.
Key Insight: Lists split ε by terms; Cantor stages have length $(\frac{2}{3})^N \to 0$. □

Corollary 4.1.15. *Uncountable Null Sets Exist*
There exists $P \subseteq \mathbb{R}$ such that:

(i) $|P| = 2^{\aleph_0}$;
(ii) $P \in \mathcal{L}^1$ and $m(P) = 0$.

Proof. Let $P := \{\sum_{k=1}^{\infty} \varepsilon_k 3^{-k} : \varepsilon_k \in \{0, 2\}\}$.

(i) The map $(\varepsilon_k) \mapsto \sum_k \varepsilon_k 3^{-k}$ identifies P with $\{0, 2\}^{\mathbb{N}}$, so $|P| = 2^{\aleph_0}$.
(ii) At stage N, P is covered by 2^N closed intervals of length 3^{-N}, hence $m_1^*(P) \leq (2/3)^N$ for all N. Letting $N \to \infty$ gives $m_1^*(P) = 0$, so outer-null completeness gives $P \in \mathcal{L}^1$ and $m(P) = 0$.

Structural Commentary: Cantor coding separates cardinality from length.
Key Insight: Null need not mean countable. □

Corollary 4.1.16. *Real-Line Lebesgue Measure Checks*
Let m be Lebesgue measure on $\mathbb{R}$.

(i) $a < b \Rightarrow m((a, b)) = m((a, b]) = m([a, b)) = m([a, b]) = b - a$.
(ii) $A \subseteq \mathbb{R}$ countable $\Rightarrow A \in \mathcal{B}(\mathbb{R})$ and $m(A) = 0$.
(iii) $\emptyset \neq G \subseteq \mathbb{R}$ open $\Rightarrow m(G) > 0$.
(iv) $K \subseteq \mathbb{R}$ compact $\Rightarrow m(K) < +\infty$.

Proof.

(i) By cell volume, $m([a, b]) = b - a$; endpoint sets finite null; add or delete $\{a\}, \{b\}$ does not change measure.
(ii) If $A = \{a_k : k \in J\}$ with J finite or countable, then $A = \bigcup_{k \in J} \{a_k\} \in \mathcal{B}(\mathbb{R})$; $m(A) \leq \sum_{k \in J} m(\{a_k\}) = 0$.
(iii) Choose $(a, b) \subseteq G$ with $a < b$; by monotonicity and (i), $m(G) \geq m((a, b)) = b - a > 0$.
(iv) Compact K is bounded, so $K \subseteq [-R, R]$ for some $R > 0$; hence $m(K) \leq m([-R, R]) = 2R < +\infty$.

Structural Commentary: On $\mathbb{R}$, length, null endpoints, and compact boundedness give quick measure estimates.
Key Insight: Nonempty open sets contain intervals; compact sets fit inside one. □

Example 4.1.17. *A Positive-Measure Cantor-Type Set with No Interval*
Let $S_0 = [0, 1]$ and, for $n \geq 1$, let R_n be the union of the middle open intervals of length 2^{-2n} removed from the 2^{n-1} components of S_{n-1}; set $S_n := S_{n-1} \setminus R_n$ and $S := \bigcap_{n=0}^{\infty} S_n$.
Then $m(R_n) = 2^{-n-1}$, $\sum_{n \geq 1} m(R_n) = 1/2$, so $S = [0, 1] \setminus \bigcup_n R_n$ is compact with $m(S) = 1/2$.
Since every component of S_n has length $\leq 2^{-n}$, no interval (a, b) can lie in S; hence $\operatorname{int} S = \emptyset$.
Structural Commentary: Summable removals leave measure; repeated splitting kills interior.
Key Insight: Total removed length controls measure; repeated splitting controls interior.

5 Continuity and Regularity of Measures

5.1 Measure Continuity and Regularity

Corollary 5.1.1. *Measure Continuity*
Let μ be a measure, and let (E_k) and (F_k) be measurable sequences.

(i) If $E_k \uparrow E$, then $\mu(E_k) \uparrow \mu(E)$.
(ii) If $F_k \downarrow F$ and $\mu(F_1) < +\infty$, then $\mu(F_k) \downarrow \mu(F)$.

Proof.

(i) Set $G_1 = E_1$ and $G_k = E_k \setminus E_{k-1}$ for $k \geq 2$.
Then $E_k = \bigsqcup_{j=1}^{k} G_j$ and $E = \bigsqcup_{j\geq 1} G_j$, so $\mu(E_k) = \sum_{j=1}^{k} \mu(G_j) \uparrow \sum_{j\geq 1} \mu(G_j) = \mu(E)$.

(ii) Apply (i) to $H_k = F_1 \setminus F_k \uparrow F_1 \setminus F$. Since $\mu(F_1) < +\infty$, $\mu(F_k) = \mu(F_1) - \mu(H_k) \downarrow \mu(F_1) - \mu(F_1 \setminus F) = \mu(F)$.

Structural Commentary: Monotone set limits are disjoint series or complements of such series.
Key Insight: Downward continuity needs one finite anchor so subtraction is legal. □

Definition 5.1.2. *Limit Inferior and Superior of Sets*
For $(E_n) \subseteq \mathcal{P}(X)$, define $\limsup_n E_n := \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} E_n$ and $\liminf_n E_n := \bigcup_{m=1}^{\infty} \bigcap_{n\geq m} E_n$.

(i) *Infinitely often*: $x \in \limsup_n E_n$ iff $\{n : x \in E_n\}$ is infinite;

(ii) *Eventually always*: $x \in \liminf_n E_n$ iff $\{n : x \notin E_n\}$ is finite;

(iii) *Set limit*: if $\limsup_n E_n = \liminf_n E_n$, write $\lim_n E_n$ for the common set.

Proposition 5.1.3. *Basic Laws for Set Limits*
Let $(E_n) \subseteq \mathcal{P}(X)$.

(i) $\emptyset \subseteq \liminf_n E_n \subseteq \limsup_n E_n \subseteq X$.

(ii) $E_n \uparrow E \Longrightarrow \liminf_n E_n = \limsup_n E_n = E = \bigcup_n E_n$.

(iii) $E_n \downarrow E \Longrightarrow \liminf_n E_n = \limsup_n E_n = E = \bigcap_n E_n$.

(iv) $(E_n) \subseteq \mathcal{M} \Longrightarrow \liminf_n E_n, \limsup_n E_n \in \mathcal{M}$.

Proof.

(i) Since $\bigcap_m \bigcup_{n\geq m} E_n \subseteq \bigcup_{n\geq 1} E_n \subseteq X$ and $\bigcap_{n\geq m} E_n \subseteq \bigcup_{n\geq m} E_n$ for all m, the inclusions follow.

(ii) If $E_n \uparrow E$, then $\{n : x \in E_n\}$ is nonempty iff it is cofinite, hence $\liminf_n E_n = \limsup_n E_n = \bigcup_n E_n = E$.

(iii) If $E_n \downarrow E$, then $\{n : x \in E_n\}$ is infinite iff it is all of $\mathbb{N}$, hence $\liminf_n E_n = \limsup_n E_n = \bigcap_n E_n = E$.

(iv) If $(E_n) \subseteq \mathcal{M}$, then $\bigcup_{n\geq m} E_n$, $\bigcap_{n\geq m} E_n \in \mathcal{M}$ $\forall m$; countable closure gives $\limsup_n E_n, \liminf_n E_n \in \mathcal{M}$.

Structural Commentary: Set limsup means infinitely often; set liminf means eventually always.
Key Insight: Translate the definitions into pointwise membership statements. □

Proposition 5.1.4. *Measure Bounds for Set Limits*
Let (E_n) be measurable in a measure space $(X, \mathcal{M}, \mu)$.

(i) $\mu(\liminf_n E_n) \leq \liminf_n \mu(E_n)$.

(ii) $\mu(\bigcup_{n=1}^{\infty} E_n) < +\infty \Longrightarrow \limsup_n \mu(E_n) \leq \mu(\limsup_n E_n)$.

Proof.

(i) Set $A_m := \bigcap_{n\geq m} E_n$. Then $A_m \uparrow \liminf_n E_n$ and $A_m \subseteq E_n$ for all $n \geq m$, so $\mu(A_m) \leq \inf_{n\geq m} \mu(E_n)$.
By continuity from below, $\mu(\liminf_n E_n) = \lim_m \mu(A_m) \leq \sup_m \inf_{n\geq m} \mu(E_n) = \liminf_n \mu(E_n)$.

(ii) Set $B_m := \bigcup_{n\geq m} E_n$. $B_m \downarrow \limsup_n E_n$, $\mu(B_1) < +\infty$, $E_n \subseteq B_m$ $\forall n \geq m$, so $\sup_{n\geq m} \mu(E_n) \leq \mu(B_m)$.
By continuity from above, $\limsup_n \mu(E_n) = \inf_m \sup_{n\geq m} \mu(E_n) \leq \lim_m \mu(B_m) = \mu(\limsup_n E_n)$.

Structural Commentary: Tail intersections increase to liminf; tail unions decrease to limsup.
Key Insight: Compare each tail envelope to the corresponding tail bounds of the measures. □

Definition 5.1.5. *Almost Everywhere*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, and let $P(x)$ be a property on E.

(i) P holds *μ-almost everywhere on E* iff $\exists N \in \mathcal{M}$ with $N \subseteq E$, $\mu(N) = 0$, and P holds on $E \setminus N$.

(ii) For Lebesgue measure, write *almost everywhere*.

Theorem 5.1.6. *First Borel–Cantelli Lemma*
Let $(X, \mathcal{M}, \mu)$ be measure space, $(E_n) \subseteq \mathcal{M}$, set $L := \limsup_n E_n = \bigcap_{m\geq 1} \bigcup_{n\geq m} E_n$. Assume $\sum_{n=1}^{\infty} \mu(E_n) < +\infty$.

(i) $L \in \mathcal{M}$ and $\mu(L) = 0$.

(ii) μ-almost every $x \in X$ belongs to at most finitely many E_n.

Proof.

(i) Set $B_m := \bigcup_{n\geq m} E_n$. Then $B_m \downarrow L$ and $\mu(B_m) \leq \sum_{n\geq m} \mu(E_n)$.
Since $\mu(B_1) \leq \sum_{n\geq 1} \mu(E_n) < +\infty$, continuity from above gives $\mu(L) = \lim_{m\to\infty} \mu(B_m) \leq \lim_{m\to\infty} \sum_{n\geq m} \mu(E_n) = 0$.

(ii) Since $x \in L \iff \forall m\, \exists n \geq m$ with $x \in E_n$, L is the set of points lying in infinitely many E_n.
By (i), outside the null set L, membership occurs only finitely often.

Structural Commentary: Summable tails force the limsup set to have zero measure.
Key Insight: Points appearing infinitely often must lie in every tail union. □

Remark 5.1.7. *Borel Sets, Null Sets, and Regular Approximation*
Open, closed, and Borel subsets of $\mathbb{R}^p$ are measurable. Each $A \in \mathcal{M}(\mu)$ admits closed–open approximations $F \subseteq A \subseteq G$ with $\mu(G \setminus A)$ and $\mu(A \setminus F)$ arbitrarily small. In Lebesgue measure, one may take $F_\sigma \subseteq A \subseteq G_\delta$ with $\mu(G_\delta \setminus F_\sigma) = 0$. Null sets form a σ-ring, and countable subsets of $\mathbb{R}^p$ are null.
Structural Commentary: Measurable sets are Borel sets modified by null sets.
Key Insight: Regularity explains why open and closed approximations are enough for integration.

Proposition 5.1.8. *Local Measurability on Bounded Cubes*
For arbitrary $E \subseteq \mathbb{R}^p$, $E \in \mathcal{L}^p \iff E \cap [-n,n]^p \in \mathcal{L}^p$ for all $n \in \mathbb{N}$.

Proof.
($\Rightarrow$) If $E \in \mathcal{L}^p$, then $E \cap [-n,n]^p \in \mathcal{L}^p$ for all n, since $[-n,n]^p$ is Borel.
($\Leftarrow$) If $E \cap [-n,n]^p \in \mathcal{L}^p$ for all n, then $E = \bigcup_{n=1}^{\infty} (E \cap [-n,n]^p) \in \mathcal{L}^p$.
Structural Commentary: Nested Borel cubes exhaust $\mathbb{R}^p$.
Key Insight: Measurability is local along a countable exhaustion. □

Proposition 5.1.9. *Local Measurability on Unit Intervals*
Let $I_n := (n, n+1]$, $n \in \mathbb{Z}$. For $E \subseteq \mathbb{R}$, $E \in \mathcal{L}^1$ iff $E \cap I_n \in \mathcal{L}^1$ for all $n \in \mathbb{Z}$.

Proof.
($\Rightarrow$) Since $I_n \in \mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}^1$, $E \in \mathcal{L}^1$ implies $E \cap I_n \in \mathcal{L}^1$ for all n.
($\Leftarrow$) If $E \cap I_n \in \mathcal{L}^1$ for all n, then $E = \bigcup_{n \in \mathbb{Z}} (E \cap I_n) \in \mathcal{L}^1$.
Structural Commentary: A countable Borel exhaustion localizes measurability.
Key Insight: Local measurable pieces patch by countable unions. □

Proposition 5.1.10. *Finite Measure Meeting Every Unit Interval*
With $I_n := (n, n+1]$, there exists $E \in \mathcal{B}(\mathbb{R})$ such that $0 < m(E \cap I_n)$ for all $n \in \mathbb{Z}$ and $m(E) < +\infty$.

Proof. Set $\varepsilon_n := 2^{-|n|-3}$ and $E := \bigcup_{n \in \mathbb{Z}} (n, n+\varepsilon_n]$. Then $(n, n+\varepsilon_n] \subseteq I_n$ are pairwise disjoint Borel intervals, so $E \in \mathcal{B}(\mathbb{R})$ and $m(E \cap I_n) = \varepsilon_n > 0$. Countable additivity gives $m(E) = \sum_{n \in \mathbb{Z}} \varepsilon_n = \frac{3}{8} < +\infty$.
Structural Commentary: Finite measure can be spread over countably many cells.
Key Insight: Positive local masses can be summable globally. □

Proposition 5.1.11. *Null Sets Form a Sigma-Ideal*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$.
Then $\mathcal{Z}$ is a σ-ring, and $N \in \mathcal{Z}$, $E \in \mathcal{M}$ imply $N \cap E \in \mathcal{Z}$.

Proof. If $(N_k) \subseteq \mathcal{Z}$, then $\mu(\bigcup_k N_k) \le \sum_k \mu(N_k) = 0$, so $\bigcup_k N_k \in \mathcal{Z}$.
If $A, B \in \mathcal{Z}$, then $A \setminus B \subseteq A$, hence $A \setminus B \in \mathcal{Z}$. Finally, $N \cap E \subseteq N$ gives $N \cap E \in \mathcal{Z}$.
Structural Commentary: Null sets are stable under countable unions and measurable cuts.
Key Insight: Monotonicity makes every measurable piece of a null set null. □

Proposition 5.1.12. *Completed Sigma-Algebra*
Let $(X, \mathcal{M}, \mu)$ be measure space, $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$; $\overline{\mathcal{M}}_\mu := \{A \subseteq X : \exists E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ A \triangle E \subseteq N\}$.
(i) $\mathcal{M} \subseteq \overline{\mathcal{M}}_\mu$, and $\overline{\mathcal{M}}_\mu$ is a σ-algebra;
(ii) $A \subseteq N \in \mathcal{Z} \Longrightarrow A \in \overline{\mathcal{M}}_\mu$;
(iii) $\overline{\mathcal{M}}_\mu = \{E \cup Z : E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ Z \subseteq N\}$.

Proof.
(i) For $E \in \mathcal{M}$, $E \triangle E = \emptyset$, so $\mathcal{M} \subseteq \overline{\mathcal{M}}_\mu$. If $A \triangle E \subseteq N$, then $A^c \triangle E^c = A \triangle E \subseteq N$.
If $A_k \triangle E_k \subseteq N_k$, then $(\bigcup_k A_k) \triangle (\bigcup_k E_k) \subseteq \bigcup_k N_k \in \mathcal{Z}$. Hence $\overline{\mathcal{M}}_\mu$ is a σ-algebra.
(ii) If $A \subseteq N \in \mathcal{Z}$, then $A \triangle \emptyset \subseteq N$, so $A \in \overline{\mathcal{M}}_\mu$.
(iii) If $A \triangle E \subseteq N$, then $A = (E \setminus N) \cup (A \cap N)$, with $E \setminus N \in \mathcal{M}$ and $A \cap N \subseteq N$.
Conversely, if $A = E \cup Z$ and $Z \subseteq N \in \mathcal{Z}$, then $A \triangle E \subseteq Z \subseteq N$.
Structural Commentary: Completion adds all null modifications of measurable sets.
Key Insight: Equality modulo null sets is the membership criterion. □

Proposition 5.1.13. *Completed Measure*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$,
and $\overline{\mathcal{M}}_\mu := \{A \subseteq X : \exists E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ A \triangle E \subseteq N\}$. For $A \in \overline{\mathcal{M}}_\mu$, define $\overline{\mu}(A) := \mu(E)$ whenever $E \in \mathcal{M}$ and $A \triangle E \subseteq N \in \mathcal{Z}$.
Then $\overline{\mu}$ is a well-defined complete measure on $\overline{\mathcal{M}}_\mu$, and $\overline{\mu}|_{\mathcal{M}} = \mu$.

Proof. If $A \triangle E \subseteq N_1$ and $A \triangle F \subseteq N_2$, with $E, F \in \mathcal{M}$ and $N_1, N_2 \in \mathcal{Z}$, then $E \triangle F \subseteq N_1 \cup N_2$, so $\mu(E) = \mu(F)$. Hence $\overline{\mu}$ is well defined; since $E \triangle E = \emptyset$, $\overline{\mu}|_{\mathcal{M}} = \mu$.
For disjoint $(A_k) \subseteq \overline{\mathcal{M}}_\mu$, choose $E_k \in \mathcal{M}$ and $N_k \in \mathcal{Z}$ with $A_k \triangle E_k \subseteq N_k$, and set $N := \bigcup_k N_k$.
Then $E_i \cap E_j \subseteq N$ for $i \neq j$. Put $F_1 := E_1$ and $F_k := E_k \setminus \bigcup_{j<k} E_j$.
Then (F_k) disjoint, $A_k \triangle F_k \subseteq N$, $(\bigcup_k A_k) \triangle (\bigcup_k F_k) \subseteq N$, so $\overline{\mu}\left(\bigcup_k A_k\right) = \mu\left(\bigcup_k F_k\right) = \sum_k \mu(F_k) = \sum_k \overline{\mu}(A_k)$.
If $A \subseteq B \in \overline{\mathcal{M}}_\mu$ and $\overline{\mu}(B) = 0$, choose $E \in \mathcal{M}$ and $N \in \mathcal{Z}$ with $B \triangle E \subseteq N$.
Then $\mu(E) = 0$, so $A \subseteq B \subseteq E \cup N \in \mathcal{Z}$; hence $A \in \overline{\mathcal{M}}_\mu$ and $\overline{\mu}(A) = 0$.
Structural Commentary: The completed measure ignores null errors.
Key Insight: Null perturbations do not change countable additivity. □

5.2 Radon Measures on Locally Compact Spaces

Radon measure theory is where topology and measure start determining one another. The topological input is a locally compact Hausdorff space; the measure-theoretic output is a Borel measure that can be recovered from open neighborhoods, compact subsets, and compactly supported continuous test functions.

Definition 5.2.1. *Locally Compact Borel Setup*
Let $(X, \mathcal{T})$ be a topological space.

(i) X is *locally compact* if every $x \in X$ admits $U \in \mathcal{T}$ with $x \in U$ and $\overline{U}$ compact.
(ii) The *Borel σ-algebra* is $\mathcal{B}(X) := \sigma(\mathcal{T})$.
(iii) For $f : X \to \mathbb{R}$, define $\operatorname{supp} f := \overline{\{x \in X : f(x) \neq 0\}}$.
(iv) $C_c(X)$ is the subspace of $C(X, \mathbb{R})$ consisting of functions with compact support.

Proposition 5.2.2. *Compact Neighborhood Shrinking*
Let X be locally compact and Hausdorff, $K \subseteq X$ compact, and $O \subseteq X$ open with $K \subseteq O$. Then there is an open $V \subseteq X$ such that $K \subseteq V$, $\overline{V} \subseteq O$, and $\overline{V}$ is compact.

Proof. For each $x \in K$, choose open $U_x \ni x$ with $\overline{U_x}$ compact.
Normality of the compact Hausdorff space $\overline{U_x}$ gives a relatively open $W_x \ni x$ with $\overline{W_x}^{\overline{U_x}} \subseteq O \cap U_x$.
Choose open $A_x \subseteq X$ with $W_x = A_x \cap \overline{U_x}$ and set $V_x := A_x \cap U_x$. Then $x \in V_x$ and $\overline{V_x} \subseteq \overline{W_x}^{\overline{U_x}} \subseteq O \cap U_x$.
The family $(V_x)_{x \in K}$ covers K, so $K \subseteq \bigcup_{i=1}^n V_{x_i}$ for some $x_1, \dots, x_n \in K$.
Set $V := \bigcup_{i=1}^n V_{x_i}$. Then $\overline{V} = \bigcup_{i=1}^n \overline{V_{x_i}} \subseteq O$, and $\overline{V}$ is compact as a finite union of compact sets.
Structural Commentary: Shrink each point neighborhood inside O, then take a finite union.
Key Insight: Compactness converts local shrinkings into one global compact neighborhood. □

Proposition 5.2.3. *Compactly Supported Cutoff*
Let X be locally compact and Hausdorff, $K \subseteq X$ compact, and $O \subseteq X$ open with $K \subseteq O$. Then there is $f \in C_c(X)$ such that $0 \leq f \leq 1$ on X, $f|_K = 1$, and $\operatorname{supp} f \subseteq O$.

Proof. Choose open $V \subseteq X$ as in Proposition 5.2.2, so $K \subseteq V$, $\overline{V}$ is compact, and $\overline{V} \subseteq O$.
The disjoint sets K and $\overline{V} \setminus V$ are closed in the compact Hausdorff space $\overline{V}$.
By Urysohn's lemma, Theorem 6.3.18, there is $f_0 : \overline{V} \to [0,1]$ with $f_0|_K = 1$ and $f_0|_{\overline{V} \setminus V} = 0$.
Define $f : X \to [0,1]$ by $f|_{\overline{V}} = f_0$ and $f|_{X \setminus V} = 0$.
Closed-set pasting gives $f \in C_c(X)$ with $f|_K = 1$ and $\operatorname{supp} f \subseteq \overline{V} \subseteq O$.
Structural Commentary: Apply Urysohn separation on $\overline{V}$ and extend by zero.
Key Insight: The compact neighborhood forces the extension to have compact support. □

Proposition 5.2.4. *Finite Partition of Unity on a Compact Set*
Let X be locally compact and Hausdorff, $K \subseteq X$ compact, and $O_1, \dots, O_n \subseteq X$ open with $K \subseteq \bigcup_{i=1}^n O_i$. Then there are $\varphi_1, \dots, \varphi_n \in C_c(X)$ such that $0 \leq \varphi_i \leq 1$, $\operatorname{supp} \varphi_i \subseteq O_i$ for every i, and $(\sum_{i=1}^n \varphi_i)|_K = 1$.

Proof. For each $x \in K$, choose $i(x)$ with $x \in O_{i(x)}$ and, by Proposition 5.2.2, an open $V_x \ni x$ with $\overline{V_x}$ compact and $\overline{V_x} \subseteq O_{i(x)}$. Choose $x_1, \dots, x_m \in K$ with $K \subseteq \bigcup_{j=1}^m V_{x_j}$, write $i_j := i(x_j)$, and set $U_i := \bigcup_{\{j : i_j = i\}} V_{x_j}$.
Then $K \subseteq \bigcup_i U_i$, each U_i is open, and $\overline{U_i}$ is compact with $\overline{U_i} \subseteq O_i$.
For $U_i \neq \emptyset$, Proposition 5.2.3 gives $f_i \in C_c(X)$ with $0 \leq f_i \leq 1$, $f_i|_{\overline{U_i}} = 1$, $\operatorname{supp} f_i \subseteq O_i$; set $f_i = 0$ otherwise. Applying the same proposition to $K \subseteq \bigcup_i U_i$ gives $h \in C_c(X)$ with $0 \leq h \leq 1$, $h|_K = 1$, and $\operatorname{supp} h \subseteq \bigcup_i U_i$.
Set $g := \sum_{i=1}^n f_i$ and $d := g + 1 - h$. Since $g \geq 1$ on $\operatorname{supp} h$ and $h = 0$ off $\operatorname{supp} h$, $d \geq 1$. Hence $\varphi_i := f_i / d$ belongs to $C_c(X)$, satisfies $0 \leq \varphi_i \leq 1$, and has $\operatorname{supp} \varphi_i \subseteq O_i$. On K, $h = 1$ and $g \geq 1$, so $d = g$ and $(\sum_{i=1}^n \varphi_i)|_K = 1$.
Structural Commentary: Build finitely many local cutoffs, then normalize their sum.
Key Insight: The auxiliary cutoff keeps the denominator positive where local cutoffs vanish. □

Theorem 5.2.5. *Locally Compact Separation and Cutoff Package*
Let X be Hausdorff. The following are equivalent.

(i) X is locally compact.
(ii) For every compact $K \subseteq O \in \mathcal{T}$, there is open $V \supseteq K$ with $\overline{V}$ compact and $\overline{V} \subseteq O$.
(iii) For every compact $K \subseteq O \in \mathcal{T}$, there is $f \in C_c(X)$ with $0 \leq f \leq 1$, $f|_K = 1$, and $f|_{X \setminus O} = 0$.

Proof.
(*i*) $\Rightarrow$ (*ii*) Proposition 5.2.2.
(*i*) $\Rightarrow$ (*iii*) Proposition 5.2.3.
(*ii*) $\Rightarrow$ (*i*) For $x \in X$, apply (ii) to $K = \{x\}$ and $O = X$; the resulting open $V \ni x$ has compact closure.
(*iii*) $\Rightarrow$ (*i*) For $x \in X$, apply (iii) to $K = \{x\}$ and $O = X$, and set $U := f^{-1}((1/2, 3/2))$. Then $x \in U$ and $\overline{U} \subseteq f^{-1}([1/2, 3/2]) \subseteq \operatorname{supp} f$, so $\overline{U}$ is compact.
Structural Commentary: Compact cutoffs recover compact neighborhoods through level sets.
Key Insight: Compact-neighborhood shrinking and compactly supported cutoffs are equivalent local forms of local compactness. □

Method 5.2.6. *Metric Cutoff Formula*
Let X be locally compact and Hausdorff, assume that ρ induces its topology.
Let $K \subseteq O \subseteq X$ with K compact and O open, and choose open $U \subseteq X$ with $K \subseteq U$, $\overline{U}$ compact, and $\overline{U} \subseteq O$.
For every nonempty $A \subseteq X$, set $\operatorname{dist}_A(x) := \inf_{a \in A} \rho(x, a)$.

(i) *Empty Set*: If $K = \emptyset$, set $f := 0$.
(ii) *Trivial Complement*: If $K \neq \emptyset$ and $X \setminus U = \emptyset$, set $f := 1$.
(iii) *Distance Ratio*: Otherwise, set $f(x) := \frac{\operatorname{dist}_{X \setminus U}(x)}{\operatorname{dist}_{X \setminus U}(x) + \operatorname{dist}_K(x)}$.

In case (iii), K and $X \setminus U$ are disjoint closed sets, so the denominator is positive; continuity of distance functions gives $f \in C(X, [0,1])$. In all cases, $f \in C_c(X)$, $f|_K = 1$, $f|_{X\setminus U} = 0$, and $\operatorname{supp} f \subseteq \overline{U} \subseteq O$.
Key Insight: Metric distances give an explicit compactly supported cutoff.

Proposition 5.2.7. *Alexandroff One-Point Compactification*
Let $(X, \mathcal{T})$ be locally compact and Hausdorff, let $\omega \notin X$, and write $\mathcal{K}(X)$ for the compact subsets of X. Set $X^* := X \cup \{\omega\}$ and $\mathcal{T}^* := \mathcal{T} \cup \{X^* \setminus K : K \in \mathcal{K}(X)\}$. Let $\iota : X \hookrightarrow X^*$ be the inclusion.

(i) $\mathcal{T}^*$ is a topology on X^*, $(X^*, \mathcal{T}^*)$ is compact Hausdorff, and $\iota : X \to X^* \setminus \{\omega\}$ is a homeomorphism.
(ii) Every $h \in C(K, \mathbb{R})$ with $K \subseteq X$ compact has an extension $\widetilde{h} \in C(X, \mathbb{R})$ satisfying $\widetilde{h}|_K = h$.

Proof.

(i) Compact subsets of the Hausdorff space X are closed.
Thus $U \setminus C \in \mathcal{T}$ and $(X^* \setminus C) \cap (X^* \setminus D) = X^* \setminus (C \cup D)$ for $U \in \mathcal{T}$ and $C, D \in \mathcal{K}(X)$.
If $\omega \notin \bigcup_\alpha U_\alpha$, then $\bigcup_\alpha U_\alpha \in \mathcal{T}$; otherwise choose $U_{\alpha_0} = X^* \setminus C$, so $X^* \setminus \bigcup_\alpha U_\alpha$ is closed in C and hence compact. Thus $\mathcal{T}^*$ is a topology.
Any open cover contains $X^* \setminus C$ for some $C \in \mathcal{K}(X)$, and finitely many remaining members cover C; hence X^* is compact. Points of X are already separated.
For $x \in X$, choose open $V \ni x$ with $\overline{V}$ compact; then V and $X^* \setminus \overline{V}$ separate x and ω.
Finally, $\mathcal{T}^*|_X = \mathcal{T}$ since $X \setminus C \in \mathcal{T}$ for $C \in \mathcal{K}(X)$, so ι is a homeomorphism onto $X^* \setminus \{\omega\}$.

(ii) By (i), X^* is compact Hausdorff and hence normal. The compact set K is closed in X^*, and h is bounded.
Tietze extension, Theorem 6.3.19, gives $H \in C(X^*, \mathbb{R})$ with $H|_K = h$. Set $\widetilde{h} := H|_X$.

Structural Commentary: Neighborhoods of ω are complements of compact subsets of X.
Key Insight: Compactification transfers extension problems to a compact Hausdorff space. □

Definition 5.2.8. *Radon Measures and Radon Premeasures*
Let $(X, \mathcal{T})$ be a topological space, and write $\mathcal{K}(X)$ for its compact subsets.

(i) $\nu : \mathcal{B}(X) \to [0, +\infty]$ is a *Radon measure* if $\nu(K) < +\infty \ \forall K \in \mathcal{K}(X)$, $\nu(E) = \inf\{\nu(O) : E \subseteq O,\ O \in \mathcal{T}\} \ \forall E \in \mathcal{B}(X)$, and $\nu(O) = \sup\{\nu(K) : K \subseteq O,\ K \in \mathcal{K}(X)\} \ \forall O \in \mathcal{T}$.
(ii) $\mu : \mathcal{T} \to [0, +\infty]$ is a *Radon premeasure* if $\mu(U) < +\infty \ \forall U \in \mathcal{T}$ with $\overline{U} \in \mathcal{K}(X)$ and $\mu(O) = \sup\{\mu(U) : U \in \mathcal{T},\ \overline{U} \in \mathcal{K}(X),\ \overline{U} \subseteq O\} \ \forall O \in \mathcal{T}$.

Lemma 5.2.9. *Open Envelope for a Topological Premeasure*
Let $\mu : \mathcal{T} \to [0, +\infty]$ be a premeasure. For $E \subseteq X$, define $\mu^*(E) := \inf\{\sum_{j=1}^\infty \mu(U_j) : E \subseteq \bigcup_{j=1}^\infty U_j,\ U_j \in \mathcal{T}\}$. Then $\mu^*(E) = \inf\{\mu(U) : E \subseteq U,\ U \in \mathcal{T}\}$.

Proof.

($\Rightarrow$) Set $I(E) := \inf\{\mu(U) : E \subseteq U,\ U \in \mathcal{T}\}$. Every $U \in \mathcal{T}$ with $E \subseteq U$ is a one-set cover, so $\mu^*(E) \le I(E)$.
($\Leftarrow$) If $(U_j)_{j\ge1} \subseteq \mathcal{T}$ covers E, then $U := \bigcup_{j=1}^\infty U_j$ belongs to $\mathcal{T}$ and satisfies $\mu(U) \le \sum_{j=1}^\infty \mu(U_j)$.
Hence $I(E) \le \sum_{j=1}^\infty \mu(U_j)$; taking the infimum over covers gives $I(E) \le \mu^*(E)$.

Structural Commentary: Replace each countable cover by its union.
Key Insight: Subadditivity makes the union no more expensive than the cover. □

Lemma 5.2.10. *Open-Set Test for Caratheodory Measurability*
Let $\mu : \mathcal{T} \to [0, +\infty]$ be a premeasure, let μ^* be the outer measure induced by open covers, and let $E \subseteq X$. Assume $\mu(O) \ge \mu^*(O \cap E) + \mu^*(O \setminus E)$ for every $O \in \mathcal{T}$ with $\mu(O) < +\infty$. Then E is μ^*-measurable.

Proof. Fix $A \subseteq X$. Outer subadditivity gives $\mu^*(A) \le \mu^*(A \cap E) + \mu^*(A \setminus E)$.
If $\mu^*(A) = +\infty$, the reverse inequality is automatic. Otherwise, for $\varepsilon > 0$, Lemma 5.2.9 gives $O \in \mathcal{T}$ with $A \subseteq O$ and $\mu(O) < \mu^*(A) + \varepsilon$, so $\mu^*(A \cap E) + \mu^*(A \setminus E) \le \mu^*(O \cap E) + \mu^*(O \setminus E) \le \mu(O) < \mu^*(A) + \varepsilon$.
Letting $\varepsilon \downarrow 0$ gives the reverse inequality, hence the Caratheodory criterion.
Structural Commentary: Approximate A by an open envelope of nearly equal outer mass.
Key Insight: Open-set splitting propagates to all sets through outer regularity. □

Proposition 5.2.11. *Open Sets from a Radon Premeasure Are Measurable*
Let X be locally compact and Hausdorff, $\mu : \mathcal{T} \to [0, +\infty]$ Radon premeasure, μ^* open-cover outer measure.

(i) $\mu^*|_{\mathcal{T}} = \mu$.
(ii) Every $V \in \mathcal{T}$ is μ^*-measurable.

Proof.

(i) The one-set cover gives $\mu^*(V) \le \mu(V)$ for $V \in \mathcal{T}$.
Every $(U_j)_{j\ge1} \subseteq \mathcal{T}$ covering V satisfies $\mu(V) \le \mu(\bigcup_j U_j) \le \sum_j \mu(U_j)$, so $\mu(V) \le \mu^*(V)$.

(ii) Let $V, O \in \mathcal{T}$ with $\mu(O) < +\infty$.
For $\varepsilon > 0$, choose $U \in \mathcal{T}$ with $\overline{U}$ compact, $\overline{U} \subseteq O \cap V$, and $\mu(U) > \mu(O \cap V) - \varepsilon$.
Since U and $O \setminus \overline{U}$ are disjoint open subsets of O and $O \setminus V \subseteq O \setminus \overline{U}$,
$\mu(O) \ge \mu(U) + \mu(O \setminus \overline{U}) \ge \mu^*(O \cap V) - \varepsilon + \mu^*(O \setminus V)$ by (i). Let $\varepsilon \downarrow 0$ and apply Lemma 5.2.10.

Structural Commentary: Inner approximation leaves an open remainder containing $O \setminus V$.
Key Insight: Open inner regularity yields Caratheodory splitting. □

Theorem 5.2.12. *Radon Premeasure Extension*
Let X be locally compact and Hausdorff, let $\mu : \mathcal{T} \to [0, +\infty]$ be a Radon premeasure, let μ^* be its open-cover outer measure, write $\mathcal{K}(X)$ for the compact subsets of X, and set $\widehat{\mu} := \mu^*|_{\mathcal{B}(X)}$.

(i) $\widehat{\mu}|_{\mathcal{T}} = \mu$.
(ii) $\widehat{\mu}$ is a Radon measure on $\mathcal{B}(X)$.

Proof.

(i) For $O \in \mathcal{T} \subseteq \mathcal{B}(X)$, $\widehat{\mu}(O) = \mu^*(O) = \mu(O)$ by Proposition 5.2.11; hence $\widehat{\mu}|_{\mathcal{T}} = \mu$.

(ii) Let $\mathcal{C}_{\mu^*}$ be the Caratheodory σ-algebra of μ^*. Proposition 5.2.11 gives $\mathcal{T} \subseteq \mathcal{C}_{\mu^*}$, hence $\mathcal{B}(X) = \sigma(\mathcal{T}) \subseteq \mathcal{C}_{\mu^*}$.
As $\mu^*|_{\mathcal{C}_{\mu^*}}$ is measure, so is $\widehat{\mu}$. Lemma 5.2.9, (i) give $\widehat{\mu}(E) = \inf\{\widehat{\mu}(O) : E \subseteq O,\ O \in \mathcal{T}\}$ for $E \in \mathcal{B}(X)$.
For $K \in \mathcal{K}(X)$, Proposition 5.2.2 gives $K \subseteq U \in \mathcal{T}$ with $\overline{U} \in \mathcal{K}(X)$, so $\widehat{\mu}(K) \leq \widehat{\mu}(U) = \mu(U) < +\infty$.
For $O \in \mathcal{T}$, inner regularity and (i) give
$\widehat{\mu}(O) = \sup\{\mu(U) : U \in \mathcal{T},\ \overline{U} \in \mathcal{K}(X),\ \overline{U} \subseteq O\} \leq \sup\{\widehat{\mu}(K) : K \in \mathcal{K}(X),\ K \subseteq O\} \leq \widehat{\mu}(O)$.

Structural Commentary: $\mathcal{T} \subseteq \mathcal{C}_{\mu^*}$ yields $\widehat{\mu}$ on $\mathcal{B}(X)$; $K \in \mathcal{K}(X) \Rightarrow \widehat{\mu}(K) < +\infty$.
Key Insight: Open inner regularity becomes compact inner regularity after extension. □

Theorem 5.2.13. *Radon Measures Determined by $C_c(X)$*
Let X be locally compact and Hausdorff, and let μ_1, μ_2 be Radon measures on $\mathcal{B}(X)$.
If $\int_X f\, d\mu_1 = \int_X f\, d\mu_2$ for every $f \in C_c(X)$, then $\mu_1 = \mu_2$.

Proof. Write $\mathcal{K}(X)$ for the compact subsets of X. Let $K \in \mathcal{K}(X)$ and $\varepsilon > 0$.
By outer regularity, choose $O \in \mathcal{T}$ with $K \subseteq O$ and $\mu_i(O \setminus K) < \varepsilon$ for $i = 1, 2$.
Choose $f \in C_c(X)$ with $0 \leq f \leq 1$, $f|_K = 1$, and $\operatorname{supp} f \subseteq O$, and set $r_i := \int_{O \setminus K} f\, d\mu_i$.
Then $0 \leq r_i < \varepsilon$ and $\int_X f\, d\mu_i = \mu_i(K) + r_i$, so $|\mu_1(K) - \mu_2(K)| = |r_1 - r_2| < \varepsilon$.
Hence $\mu_1 = \mu_2$ on $\mathcal{K}(X)$, and inner then outer regularity give $\mu_1|_{\mathcal{T}} = \mu_2|_{\mathcal{T}}$ and $\mu_1 = \mu_2$ on $\mathcal{B}(X)$.
Structural Commentary: $C_c(X)$ identifies μ_1 and μ_2 on $\mathcal{K}(X)$.
Key Insight: $\mathcal{K}(X) \to \mathcal{T} \to \mathcal{B}(X)$ by Radon regularity. □

Figure 195: Riesz–Markov: local compactness supplies cutoffs; orange denotes Caratheodory extension.

Theorem 5.2.14. *Riesz–Markov Theorem on $C_c(X)$*
Let X be locally compact and Hausdorff, $I : C_c(X) \to \mathbb{R}$ linear with $I(f) \geq 0$ $\forall f \in C_c(X)$ satisfying $f \geq 0$.
Then $\exists$ a unique Radon measure $\widehat{\mu}$ on $\mathcal{B}(X)$ such that $I(f) = \int_X f\, d\widehat{\mu}$ $\forall f \in C_c(X)$.

Proof. For $O \in \mathcal{T}$, define $\mu(O) := \sup\{I(f) : f \in C_c(X),\ 0 \leq f \leq 1,\ \operatorname{supp} f \subseteq O\}$ and $\mu(\emptyset) := 0$.

(i) *Premeasure*: If $O \subseteq \bigcup_{j=1}^{\infty} O_j$ and f is admissible for $\mu(O)$, a finite partition $(\varphi_\ell)_{\ell=1}^m$ subordinate to a finite subcover of $\operatorname{supp} f$ gives $f = \sum_{\ell=1}^m f\varphi_\ell$ and $I(f) \leq \sum_{\ell=1}^m \mu(O_{j_\ell}) \leq \sum_{j=1}^{\infty} \mu(O_j)$, so μ is countably subadditive. For disjoint $O_1, O_2 \in \mathcal{T}$, countable subadditivity gives one inequality and admissible f_1, f_2 give
$\mu(O_1) + \mu(O_2) \leq \mu(O_1 \cup O_2)$, hence finite additivity.

(ii) *Radon Regularity*: For $a < \mu(O)$, choose admissible f with $I(f) > a$ and $U \in \mathcal{T}$ with $\operatorname{supp} f \subseteq U \subseteq \overline{U} \subseteq O$ and $\overline{U}$ compact. Then $\mu(U) > a$, proving inner regularity. If $\overline{U}$ is compact, a cutoff $g \in C_c(X)$ with $g|_{\overline{U}} = 1$ dominates every admissible f for $\mu(U)$, so $\mu(U) \leq I(g) < +\infty$. Thus μ is a Radon premeasure.

(iii) *Extension*: Stages (i)–(ii) make μ a Radon premeasure on $\mathcal{T}$. Theorem 5.2.12 applies to its open-cover outer measure and yields the Radon measure $\widehat{\mu} := \mu^*|_{\mathcal{B}(X)}$ with $\widehat{\mu}|_{\mathcal{T}} = \mu$.

(iv) *Identification*: Let $0 \leq f \leq 1$ belong to $C_c(X)$, choose $O \in \mathcal{T}$ with $\operatorname{supp} f \subseteq O$ and $\overline{O}$ compact, and set $\psi_k := \min\{1, \max\{0, nf - (k-1)\}\}$, $O_0 := O$, $O_{n+1} := \emptyset$, $O_k := \{x \in O : f(x) > \frac{k-1}{n}\}$ for $1 \leq k \leq n$.
Then $f = \frac{1}{n}\sum_{k=1}^n \psi_k$, $\operatorname{supp} \psi_k \subseteq O_{k-1}$, and $\psi_k|_{O_{k+1}} = 1$, so $\mu(O_{k+1}) \leq I(\psi_k) \leq \mu(O_{k-1})$
and $\mu(O_{k+1}) \leq \int_X \psi_k\, d\widehat{\mu} \leq \mu(O_{k-1})$. Therefore $|I(f) - \int_X f\, d\widehat{\mu}| \leq \frac{1}{n}\sum_{k=1}^n (\mu(O_{k-1}) - \mu(O_{k+1})) \leq \frac{2\mu(O)}{n}$.
Letting $n \to \infty$, scaling, and $f = f^+ - f^-$ give $I(f) = \int_X f\, d\widehat{\mu}$ for every $f \in C_c(X)$.
Uniqueness follows from Theorem 5.2.13.

Structural Commentary: Open-set contents and compactly supported cutoffs build the Radon measure before identifying its integral.
Key Insight: Layer functions recover I from $\widehat{\mu}$. □

Theorem 5.2.15. *Lusin Approximation for Radon Measures*
Let X be locally compact Hausdorff, let μ be Radon on $\mathcal{B}(X)$, let $f : X \to \mathbb{R}$ be Borel, and set $A := \{f \neq 0\}$ with $\mu(A) < +\infty$. For every $\varepsilon > 0$, there exist $X_0 \in \mathcal{B}(X)$ and $g \in C_c(X)$ such that:

(i) $g = f$ on X_0.
(ii) $\mu(X \setminus X_0) < \varepsilon$.

Proof. Set $\delta := \varepsilon/4$. Choose $M > 0$ with $\mu(A_M) < \delta$, where $A_M := A \cap \{|f| > M\}$, and put $B := A \setminus A_M$.
For $n \geq 1$, choose finite-valued Borel $s_n : B \to [-M, M]$ with $\|s_n - f\|_{\infty, B} \leq \frac{1}{n}$.
Let $s_n(B) = \{a_{n,1}, \ldots, a_{n,m_n}\}$ and $E_{n,j} := s_n^{-1}(\{a_{n,j}\})$. Inner regularity gives compact $K_{n,j} \subseteq E_{n,j}$, $\mu(E_{n,j} \setminus K_{n,j}) < \frac{\delta}{2^n m_n}$. Set $F_n := \bigcup_{j=1}^{m_n} K_{n,j}$ and $F := \bigcap_{n=1}^{\infty} F_n$.
Then F is compact, $B \setminus F \subseteq \bigcup_{n \geq 1} \bigcup_{j=1}^{m_n} (E_{n,j} \setminus K_{n,j})$, and $\mu(B \setminus F) \leq \delta$.
For each n, $F \cap E_{n,j} = F \cap K_{n,j}$ is closed in F, so $s_n|_F \in C(F)$; since $s_n|_F \to f|_F$ uniformly, $f|_F \in C(F)$.
Choose $W \in \mathcal{T}$ with $F \subseteq W$ and $\mu(W \setminus F) < \delta$, then choose $O \in \mathcal{T}$ by Proposition 5.2.2 with $F \subseteq O \subseteq \overline{O} \subseteq W$ and $\overline{O}$ compact.

Extend $f|_F$ to $H \in C(X)$ by Proposition 5.2.7, choose $q \in C_c(X)$ with $q|_F = 1$ and $\operatorname{supp} q \subseteq O$ by Proposition 5.2.3, and set $g := qH$, $X_0 := F \cup (X \setminus (A \cup O))$. Then $g \in C_c(X)$ and $X_0 \in \mathcal{B}(X)$.

(i) On F, $g = qH = f$; on $X \setminus (A \cup O)$, $f = q = 0$.
(ii) $X \setminus X_0 \subseteq (A \setminus F) \cup (O \setminus F)$, $\mu(A \setminus F) \le \mu(A_M) + \mu(B \setminus F) < 2\delta$, $\mu(O \setminus F) \le \mu(W \setminus F) < \delta$; $\mu(X \setminus X_0) < 3\delta < \varepsilon$.

Structural Commentary: Compact core plus cutoff extension.
Key Insight: Discard a small set to make Borel data continuous. □

5.3 Invariant Measures on Compact Groups

Compact groups are the meeting point of topological symmetry and Radon measure theory. The compactness supplies fixed points in dual spaces, and Riesz–Markov converts those fixed functionals into invariant Borel measures.

Definition 5.3.1. *Compact Groups and Haar Measures*
For a group G, $A \subseteq G$, and $g \in G$, set $gA := \{ga : a \in A\}$ and $Ag := \{ag : a \in A\}$.

(i) A *topological group* is G with Hausdorff topology such that $(g,h) \mapsto gh$ and $g \mapsto g^{-1}$ are continuous.
(ii) A *compact group* is a topological group whose underlying topological space is compact.
(iii) A Borel measure μ on $\mathcal{B}(G)$ is *left-invariant* if $\mu(gA) = \mu(A)$ for all $g \in G$ and $A \in \mathcal{B}(G)$.
(iv) A Borel measure μ on $\mathcal{B}(G)$ is *right-invariant* if $\mu(Ag) = \mu(A)$ for all $g \in G$ and $A \in \mathcal{B}(G)$.
(v) A *Haar measure* on a compact group G is a left-invariant Radon probability measure on $\mathcal{B}(G)$.

Definition 5.3.2. *Regular Actions on $C(G)$*
Let G be compact, $C(G)$ be continuous maps $G \to \mathbb{R}$ with $\|f\|_\infty := \max_{x \in G} |f(x)|$. For $g, x \in G$ and $f \in C(G)$:

(i) The *left regular action* is $\pi_{\mathrm{L}} : G \to \mathcal{L}(C(G))$, $[\pi_{\mathrm{L}}(g)f](x) := f(g^{-1}x)$.
(ii) The *right regular action* is $\pi_{\mathrm{R}} : G \to \mathcal{L}(C(G))$, $[\pi_{\mathrm{R}}(g)f](x) := f(xg^{-1})$.
(iii) A *probability functional* is $\psi \in C(G)^*$ with $\psi(\mathbf{1}_G) = 1$ and $f \ge 0 \Rightarrow \psi(f) \ge 0$ for all $f \in C(G)$.

Theorem 5.3.3. *Kakutani Fixed Probability Functional*
Let G be a compact group. Then there are probability functionals $\psi_{\mathrm{L}}, \psi_{\mathrm{R}} \in C(G)^*$ such that $\psi_{\mathrm{L}}(\pi_{\mathrm{L}}(g)f) = \psi_{\mathrm{L}}(f)$ and $\psi_{\mathrm{R}}(\pi_{\mathrm{R}}(g)f) = \psi_{\mathrm{R}}(f)$ for every $g \in G$ and $f \in C(G)$.

Proof. Let $\mathcal{P}(G) := \{\psi \in C(G)^* : \psi(\mathbf{1}_G) = 1,\ \forall f \in C(G)\ (f \ge 0 \Rightarrow \psi(f) \ge 0)\}$. Then $\delta_e \in \mathcal{P}(G)$, $\mathcal{P}(G)$ is convex and weak-* closed, and $\psi \in \mathcal{P}(G) \Rightarrow \|\psi\|_{\mathrm{op}} = 1$; hence $\mathcal{P}(G)$ is weak-* compact by Banach–Alaoglu.
For $g \in G$, set $T_{\mathrm{L}}(g)\psi := \psi \circ \pi_{\mathrm{L}}(g^{-1})$ and $T_{\mathrm{R}}(g)\psi := \psi \circ \pi_{\mathrm{R}}(g)$. Since $\pi_{\mathrm{L}}, \pi_{\mathrm{R}}$ preserve $\mathbf{1}_G$ and $C(G)_+$ and are norm-continuous in g, T_{L} and T_{R} are continuous affine G-actions on $\mathcal{P}(G)$.
Kakutani gives $\psi_{\mathrm{L}}, \psi_{\mathrm{R}} \in \mathcal{P}(G)$ with $T_{\mathrm{L}}(g)\psi_{\mathrm{L}} = \psi_{\mathrm{L}}$ and $T_{\mathrm{R}}(g)\psi_{\mathrm{R}} = \psi_{\mathrm{R}}$ for all $g \in G$.
Thus $\psi_{\mathrm{L}} \circ \pi_{\mathrm{L}}(g) = \psi_{\mathrm{L}}$ and $\psi_{\mathrm{R}} \circ \pi_{\mathrm{R}}(g) = \psi_{\mathrm{R}}$ for all $g \in G$.
Structural Commentary: Compactness lives in $\mathcal{P}(G)$; invariance comes from fixed points.
Key Insight: A fixed probability functional is invariant. □

Method 5.3.4. *Fixed Functional Pattern*
For a compact group G, set $\mathcal{P}(G) := \{\psi \in C(G)^* : \psi(\mathbf{1}_G) = 1,\ \forall f \in C(G)\ (f \ge 0 \Rightarrow \psi(f) \ge 0)\}$.

(i) Show $\mathcal{P}(G) \neq \varnothing$ via $\delta_x(f) := f(x)$, $x \in G$.
(ii) Show $\mathcal{P}(G)$ is weak-* compact convex: $\psi \in \mathcal{P}(G) \Rightarrow \|\psi\|_{\mathrm{op}} = 1$, Banach–Alaoglu, and weak-* closed defining conditions.
(iii) Define $T_{\mathrm{L}}(g)\psi := \psi \circ \pi_{\mathrm{L}}(g^{-1})$ or $T_{\mathrm{R}}(g)\psi := \psi \circ \pi_{\mathrm{R}}(g)$. For $T = T_{\mathrm{L}}, T_{\mathrm{R}}$, check $T(g)\mathcal{P}(G) \subseteq \mathcal{P}(G)$ and weak-* continuity.
(iv) Apply Kakutani: find $\psi \in \mathcal{P}(G)$ with $T(g)\psi = \psi$ for all $g \in G$.
(v) Apply Riesz–Markov, $\psi(f) = \int_G f \, d\mu$, and read $T(g)\psi = \psi$ as invariance of μ.

Structural Commentary: Compact state space plus adjoint action.
Key Insight: Haar measure is an invariant state represented by Riesz–Markov.

Proposition 5.3.5. *Translated Measures and Test Functions*
Let G be a compact group, let μ be a Borel measure on $\mathcal{B}(G)$, and fix $g \in G$.
Set $L_g(x) := gx$ and $R_g(x) := xg$ for $x \in G$.
For $A \in \mathcal{B}(G)$, define $\mu_g^{\mathrm{L}}(A) := \mu(L_g(A)) = \mu(gA)$ and $\mu_g^{\mathrm{R}}(A) := \mu(R_g(A)) = \mu(Ag)$.

(i) μ_g^{L} and μ_g^{R} are Borel measures on $\mathcal{B}(G)$.
(ii) If μ is Radon, then μ_g^{L} and μ_g^{R} are Radon.
(iii) For every $f \in C(G)$, $\int_G \pi_{\mathrm{L}}(g)f \, d\mu = \int_G f \, d\mu_g^{\mathrm{L}}$ and $\int_G \pi_{\mathrm{R}}(g)f \, d\mu = \int_G f \, d\mu_g^{\mathrm{R}}$.

Proof. For every $h \in G$, the maps L_h and R_h are homeomorphisms of G.

(i) Since L_g and R_g map Borel sets bijectively onto Borel sets, μ_g^{L} and μ_g^{R} are defined on $\mathcal{B}(G)$.
If $(A_n)_{n \ge 1}$ are pairwise disjoint Borel sets, then $L_g(\bigcup_n A_n) = \bigcup_n L_g(A_n)$ and $R_g(\bigcup_n A_n) = \bigcup_n R_g(A_n)$, with disjoint unions; countable additivity follows from countable additivity of μ.
(ii) Since $L_{g^{-1}}^{-1} = L_g$ and $R_{g^{-1}}^{-1} = R_g$, $\mu_g^{\mathrm{L}} = (L_{g^{-1}})_* \mu$ and $\mu_g^{\mathrm{R}} = (R_{g^{-1}})_* \mu$.
If μ is Radon, pushforward by a homeomorphism preserves compact sets, open sets, and Radon regularity.
Hence both translated measures are Radon.
(iii) For $f \in C(G)$, the pushforward identity gives $\int_G f \, d\mu_g^{\mathrm{L}} = \int_G f \circ L_{g^{-1}} \, d\mu = \int_G \pi_{\mathrm{L}}(g)f \, d\mu$
and $\int_G f \, d\mu_g^{\mathrm{R}} = \int_G f \circ R_{g^{-1}} \, d\mu = \int_G \pi_{\mathrm{R}}(g)f \, d\mu$.

Structural Commentary: Translated measures are inverse-translation pushforwards.
Key Insight: Measure translation is dual to test-function precomposition. □

Theorem 5.3.6. *Haar Measure on Compact Groups*
Let G be a compact group. Let $\mathcal{P}_{\mathrm{L}}(G)$ and $\mathcal{P}_{\mathrm{R}}(G)$ be the left- and right-invariant Radon probability measures on $\mathcal{B}(G)$, respectively.

(i) $\mathcal{P}_{\mathrm{L}}(G) \neq \emptyset$.
(ii) $\mathcal{P}_{\mathrm{R}}(G) \neq \emptyset$.

(iii) For every $(\lambda, \rho) \in \mathcal{P}_{\mathrm{L}}(G) \times \mathcal{P}_{\mathrm{R}}(G)$, $\lambda = \rho$.

Consequently, G admits a unique Haar measure, and this measure is right-invariant.

Proof. For a Radon probability measure σ on $\mathcal{B}(G)$, set $\Phi_\sigma(f) := \int_G f \, d\sigma$ for $f \in C(G)$.

(i) Choose ψ_{L} from Theorem 5.3.3. Riesz–Markov gives a Radon probability measure μ with $\psi_{\mathrm{L}} = \Phi_\mu$ on $C(G) = C_c(G)$. For $g \in G$, Proposition 5.3.5 gives $\Phi_{\mu_g^{\mathrm{L}}}(f) = \Phi_\mu(\pi_{\mathrm{L}}(g)f) = \Phi_\mu(f)$ for every $f \in C(G)$. Thus $\mu_g^{\mathrm{L}} = \mu$ by Theorem 5.2.13, so $\mu(gA) = \mu(A)$ for every $A \in \mathcal{B}(G)$ and $\mu \in \mathcal{P}_{\mathrm{L}}(G)$.

(ii) Choose ψ_{R} from Theorem 5.3.3. Riesz–Markov gives a Radon probability measure ν with $\psi_{\mathrm{R}} = \Phi_\nu$ on $C(G) = C_c(G)$. For $g \in G$, Proposition 5.3.5 gives $\Phi_{\nu_g^{\mathrm{R}}}(f) = \Phi_\nu(\pi_{\mathrm{R}}(g)f) = \Phi_\nu(f)$ for every $f \in C(G)$. Thus $\nu_g^{\mathrm{R}} = \nu$ by Theorem 5.2.13, so $\nu(Ag) = \nu(A)$ for every $A \in \mathcal{B}(G)$ and $\nu \in \mathcal{P}_{\mathrm{R}}(G)$.

(iii) Let $(\lambda, \rho) \in \mathcal{P}_{\mathrm{L}}(G) \times \mathcal{P}_{\mathrm{R}}(G)$, fix $f \in C(G)$, and set $h_f(x, y) := f(xy)$. Fubini applies to $h_f \in C(G \times G)$. Left-invariance gives $\int_G h_f(x, y)\, d\lambda(y) = \Phi_\lambda(f)$ for every x, and right-invariance gives $\int_G h_f(x, y)\, d\rho(x) = \Phi_\rho(f)$ for every y. Since $\lambda(G) = \rho(G) = 1$, Fubini gives $\Phi_\lambda(f) = \Phi_\rho(f)$. Hence $\lambda = \rho$ by Theorem 5.2.13.

Taking $\mu \in \mathcal{P}_{\mathrm{L}}(G)$ from (i) and $\nu \in \mathcal{P}_{\mathrm{R}}(G)$ from (ii), (iii) gives $\mu = \nu$, so μ is right-invariant. If η is Haar, then $\eta \in \mathcal{P}_{\mathrm{L}}(G)$ and (iii) gives $\eta = \nu$, so Haar measure is unique.
Structural Commentary: Fixed states give existence; Fubini gives uniqueness.
Key Insight: Invariance makes both iterated integrals constant. □

Proposition 5.3.7. *Point Masses of Haar Measure*
Let G be a compact group, and let μ be Haar measure on $\mathcal{B}(G)$.

(i) If G is finite, then every $A \subseteq G$ is Borel and $\mu(A) = \frac{|A|}{|G|}$.
(ii) If G is infinite, then $\mu(\{x\}) = 0$ for every $x \in G$.

Proof. Set $c := \mu(\{e\})$. For each $x \in G$, left-invariance gives $\mu(\{x\}) = \mu(x\{e\}) = c$.

(i) If G is finite, then G is discrete, so every $A \subseteq G$ is Borel. Since $1 = \mu(G) = \sum_{x \in G} \mu(\{x\}) = |G|\, c$, one has $c = \frac{1}{|G|}$, and finite additivity gives $\mu(A) = \sum_{a \in A} \mu(\{a\}) = \frac{|A|}{|G|}$.

(ii) If G is infinite and $c > 0$, choose $n \in \mathbb{N}$ with $n > \frac{1}{c}$ and distinct $x_1, \dots, x_n \in G$.
Finite additivity gives $1 = \mu(G) \geq \sum_{j=1}^{n} \mu(\{x_j\}) = nc > 1$, a contradiction; hence $c = 0$.

Structural Commentary: Translation reduces point masses to one scalar.
Key Insight: Normalization fixes or kills that scalar. □

Proposition 5.3.8. *Nonempty Open Sets Have Positive Haar Measure*
Let G be a compact group, μ be Haar measure on $\mathcal{B}(G)$, $O \subseteq G$ be nonempty and open. Then $\mu(O) > 0$.

Proof. Choose $a \in O$. For each $x \in G$, $x = (xa^{-1})a \in (xa^{-1})O$, so $\{gO : g \in G\}$ is an open cover of G.
Compactness gives $g_1, \dots, g_n \in G$ such that $G = \bigcup_{j=1}^{n} g_j O$. If $\mu(O) = 0$, then left-invariance and subadditivity give $1 = \mu(G) \leq \sum_{j=1}^{n} \mu(g_j O) = \sum_{j=1}^{n} \mu(O) = 0$, a contradiction. Hence $\mu(O) > 0$.
Structural Commentary: Compactness gives a finite translate cover.
Key Insight: Zero mass cannot cover total mass. □

6 Regular Approximation and Caratheodory Tests

6.1 Borel Approximation and Regular Measurable Sets

The Caratheodory definition is intrinsic to outer measure, but in $\mathbb{R}^p$ it has a topological shadow: measurable sets are exactly those that can be approximated by open, closed, compact, and cell-built sets up to arbitrarily small error. That bridge is also the route to the first nonmeasurable and non-Borel examples.

Lemma 6.1.1. *Outer Approximation by Open and G_δ Sets*
For $A \subseteq \mathbb{R}^p$:

(i) for every $\varepsilon > 0$, there exists open $G \subseteq \mathbb{R}^p$ with $A \subseteq G$ and $m(G) \leq m_p^*(A) + \varepsilon$;
(ii) $m_p^*(A) = \inf\{m(G) : A \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\}$;
(iii) there exists G_δ set $H \supseteq A$ with $m(H) = m_p^*(A)$.

Proof.

(i) If $m_p^*(A) = +\infty$, take $G = \mathbb{R}^p$.
If $m_p^*(A) < +\infty$, choose an open-cell cover $A \subseteq \bigcup_k I_k$ with $\sum_k \mathrm{vol}_p(I_k) \leq m_p^*(A) + \varepsilon$, and set $G := \bigcup_k I_k$.
Then G is open, $A \subseteq G$, and $m(G) \leq \sum_k m(I_k) = \sum_k \mathrm{vol}_p(I_k) \leq m_p^*(A) + \varepsilon$.

(ii) By (i), $\inf\{m(G) : A \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\} \leq m_p^*(A) + \varepsilon$ for all $\varepsilon > 0$; let $\varepsilon \downarrow 0$.
Conversely, $A \subseteq G$ gives $m_p^*(A) \leq m_p^*(G) = m(G)$ for open G; take the infimum over G.

(iii) If $m_p^*(A) = +\infty$, take $H = \mathbb{R}^p$.
If $m_p^*(A) < +\infty$, choose open $G_n \supseteq A$ with $m(G_n) \leq m_p^*(A) + 1/n$ and set $H := \bigcap_n G_n$.
Then H is G_δ, $A \subseteq H$, and $m_p^*(A) \leq m(H) \leq m(G_n) \leq m_p^*(A) + 1/n$ for all n, so $m(H) = m_p^*(A)$.

Structural Commentary: Outer measure can be realised by a Borel envelope from outside.
Key Insight: Infima over cell covers can be repackaged as open-set approximation. □

Corollary 6.1.2. *Null Sets Sit Inside Borel Null Sets*
Let $Z \subseteq \mathbb{R}^p$ with $m_p^*(Z) = 0$. Then there exists $B \subseteq \mathbb{R}^p$ such that:

(i) $Z \subseteq B$;
(ii) $B \in \mathcal{B}(\mathbb{R}^p)$;
(iii) $m(B) = 0$.

Proof. By Lemma 6.1.1, choose open sets $(G_n)_{n\geq 1}$ in $\mathbb{R}^p$ and set $B := \bigcap_{n=1}^{\infty} G_n$ so that $Z \subseteq B$ and $m(B) = m_p^*(Z)$.

(i) The chosen envelope satisfies $Z \subseteq B$.

(ii) Since each G_n is open, $B = \bigcap_{n=1}^{\infty} G_n \in \mathcal{B}(\mathbb{R}^p)$.

(iii) Since $m_p^*(Z) = 0$, $m(B) = m_p^*(Z) = 0$.

Structural Commentary: A null set has a G_δ null envelope.
Key Insight: Outer approximation gives a Borel null superset. □

Theorem 6.1.3. *Open and G_δ Characterizations of Measurability*
For $E \subseteq \mathbb{R}^p$, the following are equivalent:

(i) $E \in \mathcal{L}^p$;
(ii) $\forall \varepsilon > 0$, $\exists$ *open* $G \subseteq \mathbb{R}^p$ *such that* $E \subseteq G$ *and* $m_p^*(G \setminus E) < \varepsilon$;
(iii) $\exists$ G_δ *set* $H \subseteq \mathbb{R}^p$ *such that* $E \subseteq H$ *and* $m_p^*(H \setminus E) = 0$;
(iv) $\exists$ G_δ *set* $H \subseteq \mathbb{R}^p$ *and* $Z \in \mathcal{L}^p$ *such that* $Z \subseteq H$, $m(Z) = 0$, *and* $E = H \setminus Z$.

Proof.

(i) $\Rightarrow$ (ii). If $m(E) < +\infty$, choose open $G \supseteq E$ with $m(G) \leq m(E) + \varepsilon/2$.
Since $E \in \mathcal{L}^p$, $m_p^*(G \setminus E) = m(G) - m(E) < \varepsilon$. If $m(E) = +\infty$, set $E_1 := E \cap \{x : \|x\|_2 \leq 1\}$ and $E_n := E \cap \{x : n-1 < \|x\|_2 \leq n\}$ for $n \geq 2$. For each n, choose open $G_n \supseteq E_n$ with $m_p^*(G_n \setminus E_n) < \varepsilon 2^{-n}$; then $G := \bigcup_n G_n$ is open, $E \subseteq G$, and $G \setminus E \subseteq \bigcup_n (G_n \setminus E_n)$, so $m_p^*(G \setminus E) < \varepsilon$.

(ii) $\Rightarrow$ (iii). Choose open $G_n \supseteq E$ with $m_p^*(G_n \setminus E) < 1/n$, and set $H := \bigcap_n G_n$.
Then H is G_δ, $E \subseteq H$, and $H \setminus E \subseteq G_n \setminus E$ for all n; hence $m_p^*(H \setminus E) = 0$.

(iii) $\Rightarrow$ (i). Since H is G_δ, $H \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. Also $m_p^*(H \setminus E) = 0$ implies $H \setminus E \in \mathcal{L}^p$ by Proposition 3.3.8.
Thus $E = H \setminus (H \setminus E) \in \mathcal{L}^p$.

(iii) $\Leftrightarrow$ (iv). If (iii) holds, set $Z := H \setminus E$; then $Z \subseteq H$, $m(Z) = 0$, and $E = H \setminus Z$.
Conversely, if (iv) holds, then $E \subseteq H$ and $H \setminus E = Z$, so $m_p^*(H \setminus E) = 0$.

Structural Commentary: Lebesgue measurable sets are G_δ sets with null material removed.
Key Insight: Small open excess becomes a null G_δ excess after countably many approximations. □

Corollary 6.1.4. *Open Outer Regularity for Measurable Sets*
If $E \in \mathcal{L}^p$, then

(i) $\forall \varepsilon > 0$, $\exists$ *open* $G \supseteq E$ *such that* $m(G) \leq m(E) + \varepsilon$;
(ii) $m(E) = \inf\{m(G) : E \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\}$.

Proof.

(i) By Theorem 6.1.3, choose open $G \supseteq E$ with $m_p^*(G \setminus E) < \varepsilon$.
If $m(E) < +\infty$, then $m(G) = m(E) + m(G \setminus E) \leq m(E) + \varepsilon$; if $m(E) = +\infty$, the inequality is trivial.

(ii) From (i), $\inf_G m(G) \leq m(E)$. Conversely, $E \subseteq G$ gives $m(E) \leq m(G)$ by monotonicity; take $\inf_G$.

Structural Commentary: Measurable sets are measured from outside by open supersets.
Key Insight: Open excess controls the outer regular infimum. □

Theorem 6.1.5. *Closed and F_σ Characterizations of Measurability*
For $E \subseteq \mathbb{R}^p$, the following are equivalent:

(i) $E \in \mathcal{L}^p$;
(ii) $\forall \varepsilon > 0$, $\exists$ *closed* $F \subseteq E$ *such that* $m_p^*(E \setminus F) < \varepsilon$;
(iii) $\exists$ F_σ *set* $K \subseteq E$ *such that* $m_p^*(E \setminus K) = 0$;
(iv) $\exists$ F_σ *set* $K \subseteq E$ *and* $Z \in \mathcal{L}^p$ *such that* $Z \subseteq E$, $m(Z) = 0$, *and* $E = K \cup Z$.

Proof.

(i) $\Rightarrow$ (ii). Since $E^c \in \mathcal{L}^p$, Theorem 6.1.3 gives open $G \supseteq E^c$ with $m_p^*(G \setminus E^c) < \varepsilon$. Set $F := G^c$.
Then F is closed, $F \subseteq E$, and $E \setminus F = G \setminus E^c$, so $m_p^*(E \setminus F) < \varepsilon$.

(ii) $\Rightarrow$ (iii). Choose closed $F_n \subseteq E$ with $m_p^*(E \setminus F_n) < 1/n$, and set $K := \bigcup_n F_n$.
Then K is F_σ, $K \subseteq E$, and $E \setminus K \subseteq E \setminus F_n$ for all n; hence $m_p^*(E \setminus K) = 0$.

(iii) $\Rightarrow$ (i). Since K is F_σ, $K \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. Also $m_p^*(E \setminus K) = 0$ implies $E \setminus K \in \mathcal{L}^p$ by Proposition 3.3.8.
Thus $E = K \cup (E \setminus K) \in \mathcal{L}^p$.

(iii) $\Leftrightarrow$ (iv). If (iii) holds, set $Z := E \setminus K$; then $Z \subseteq E$, $m(Z) = 0$, and $E = K \cup Z$.
Conversely, if (iv) holds, then $E \setminus K \subseteq Z$, so $m_p^*(E \setminus K) = 0$.

Structural Commentary: Inside approximation is the complement of outside approximation.
Key Insight: Measurable sets are also F_σ sets with null material added. □

Corollary 6.1.6. *Closed Inner Regularity*
If $E \in \mathcal{L}^p$, then

(i) $\forall \varepsilon > 0$, $\exists$ *closed* $F \subseteq E$ *such that* $m(E) \leq m(F) + \varepsilon$;
(ii) $m(E) = \sup\{m(F) : F \subseteq E,\ F \subseteq \mathbb{R}^p \text{ closed}\}$.

Proof.

(i) By Theorem 6.1.5, choose closed $F \subseteq E$ with $m_p^*(E \setminus F) < \varepsilon$.
Since $F \in \mathcal{L}^p$ and $E = F \sqcup (E \setminus F)$, $m(E) = m(F) + m(E \setminus F) \leq m(F) + \varepsilon$.

(ii) From (i), $m(E) \leq \sup_F m(F)$. Conversely, $F \subseteq E$ gives $m(F) \leq m(E)$; take $\sup_F$.

Structural Commentary: Measurable sets are measured from inside by closed subsets.
Key Insight: Small discarded measure controls the inner regular supremum. □

Corollary 6.1.7. *Lebesgue Regularity and Borel Completion on $\mathbb{R}$*
For $A \in \mathcal{L}^1$,

(i) $m(A) = \inf\{m(G) : A \subseteq G,\ G \text{ open in } \mathbb{R}\}$.
(ii) $m(A) = \sup\{m(K) : K \subseteq A,\ K \text{ compact in } \mathbb{R}\}$, *finite-measure case applied to* $A \cap (-n, n]$; $m(A) = +\infty$.
(iii) There exists $B \in \mathcal{B}(\mathbb{R})$ *with* $A \subseteq B$ *and* $m(B \setminus A) = 0$; *equivalently,* $\mathcal{L}^1$ *is the completion of* $\mathcal{B}(\mathbb{R})$.

Proof.

(i) For $\varepsilon > 0$, regular approximation gives open $G \supseteq A$ with $m(G \setminus A) < \varepsilon$, hence $m(G) \leq m(A) + \varepsilon$. Monotonicity gives the reverse inequality.
(ii) If $m(A) < +\infty$, choose closed $F \subseteq A$ with $m(A \setminus F) < \varepsilon$. Since $F \cap [-n, n] \uparrow F$, continuity from below gives compact $K = F \cap [-n, n]$ with $m(F) - m(K) < \varepsilon$, so $m(A) - m(K) < 2\varepsilon$. If $m(A) = +\infty$, apply the finite case to $A_n = A \cap (-n, n]$ and let $n \to \infty$.
(iii) Choose open $G_k \supseteq A$ with $m(G_k \setminus A) < 1/k$, and set $B = \bigcap_k G_k$. Then $B \in \mathcal{B}(\mathbb{R})$, $A \subseteq B$, and $m(B \setminus A) = 0$. Thus every Lebesgue set is Borel plus a null subset.
Conversely, Borel sets and null subsets are Lebesgue measurable.

Structural Commentary: Lebesgue sets are Borel sets plus null-set ambiguity.
Key Insight: Open outer approximation and compact inner approximation are the usable form of regularity. □

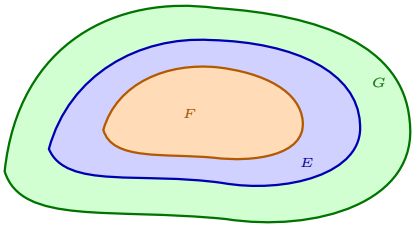

Figure 196: Regular approximation sandwiches E between F and G with small excess.

Theorem 6.1.8. *Lebesgue Measurable Non-Borel Sets Exist*

(i) There exists $E \subseteq \mathbb{R}$ *such that* $E \in \mathcal{L}^1$ *and* $E \notin \mathcal{B}(\mathbb{R})$.
(ii) $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}^1$.

Proof.

(i) Lemma 4.1.7 gives $|\mathcal{B}(\mathbb{R})| \leq \mathfrak{c}$. Let C be the Cantor set. Since $|C| = \mathfrak{c}$ and $m(C) = 0$, Proposition 3.3.8 gives $\mathcal{P}(C) \subseteq \mathcal{L}^1$, so $|\mathcal{L}^1| \geq 2^{\mathfrak{c}} > \mathfrak{c}$. Hence some $E \in \mathcal{L}^1$ is not Borel.
(ii) Proposition 4.1.11 gives $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}^1$, and (i) makes the containment proper.

Structural Commentary: Countable generation is smaller than completion by null subsets.
Key Insight: Completeness creates more measurable sets than topology can generate. □

6.2 Compact and Cell Approximation

Theorem 6.2.1. *Compact Approximation at Finite Outer Measure*
Let $E \subseteq \mathbb{R}^p$ *with* $m_p^*(E) < +\infty$. *Then* $E \in \mathcal{L}^p$ *iff* $\forall \varepsilon > 0$, $\exists$ *compact* $K \subseteq E$ *with* $m_p^*(E \setminus K) < \varepsilon$.

Proof.

($\Rightarrow$) Since $m(E) < +\infty$, choose closed $F \subseteq E$ with $m(E \setminus F) < \frac{\varepsilon}{2}$ by Theorem 6.1.5.
Let $Q_N := [-N, N]^p$. Since $F \cap Q_N \uparrow F$, continuity from below gives N with $m(F \setminus Q_N) < \frac{\varepsilon}{2}$.
Then $K := F \cap Q_N$ is compact, $K \subseteq E$, and $m_p^*(E \setminus K) = m(E \setminus K) \leq m(E \setminus F) + m(F \setminus Q_N) < \varepsilon$.
($\Leftarrow$) Choose compact $K_n \subseteq E$ with $m_p^*(E \setminus K_n) < \frac{1}{n}$ and set $K := \bigcup_{n \geq 1} K_n$.
Then K is F_σ, hence $K \in \mathcal{L}^p$, and $E \setminus K \subseteq E \setminus K_n$ for all n, so $m_p^*(E \setminus K) = 0$.
By Proposition 3.3.8, $E \setminus K \in \mathcal{L}^p$, hence $E = K \cup (E \setminus K) \in \mathcal{L}^p$.

Structural Commentary: Finite measure permits bounded truncation before inner closure.
Key Insight: Compact approximation is closed approximation plus a finite-measure tail estimate. □

Theorem 6.2.2. *Finite Cell Approximation in Symmetric Difference*
If $E \in \mathcal{L}^p$ *and* $m(E) < +\infty$, *then* $\forall \varepsilon > 0$, $\exists U = \bigcup_{j=1}^{N} I_j$ *with* I_j *bounded open cells and* $m(E \triangle U) < \varepsilon$.

Proof. Choose compact $K \subseteq E$ with $m(E \setminus K) < \frac{\varepsilon}{2}$ by Theorem 6.2.1, and open $G \supseteq E$ with $m_p^*(G \setminus E) < \frac{\varepsilon}{2}$ by Theorem 6.1.3. For each $x \in K$, choose a bounded open cell I_x with $x \in I_x$ and $\overline{I_x} \subseteq G$. Compactness gives $I_1, \ldots, I_N$ with $K \subseteq U := \bigcup_{j=1}^{N} I_j \subseteq G$. Then $E \triangle U \subseteq (E \setminus K) \cup (G \setminus E)$, so subadditivity gives $m(E \triangle U) < \varepsilon$.
Structural Commentary: Compactness turns a countable open cover into a finite geometric approximant.
Key Insight: Approximate from outside by cells and from inside by a compact core. □

Corollary 6.2.3. *Cell Approximation Variant Choices*
Let $E \in \mathcal{L}^p$ *with* $m(E) < +\infty$, *and let* $\varepsilon > 0$. *There exists* $A \subseteq \mathbb{R}^p$ *with* $m(E \triangle A) < \varepsilon$ *in each format:*

(i) A is a finite union of bounded open cells;
(ii) A is a finite union of bounded closed cells;

(iii) A is a finite union of bounded half-open cells of a fixed orientation.

Proof.

(i) By Theorem 6.2.2, choose $A = \bigcup_{j=1}^{N} I_j$ with I_j bounded open cells and $m(E \triangle A) < \varepsilon$.

(ii) Choose a finite open-cell union $U = \bigcup_{j=1}^{M} J_j$ with $m(E \triangle U) < \frac{\varepsilon}{2}$. If $M = 0$, take $A = \emptyset$; otherwise choose closed cells $C_j \subseteq J_j$ with $m(J_j \setminus C_j) < \frac{\varepsilon}{2M}$ and set $A := \bigcup_{j=1}^{M} C_j$. Then $m(U \setminus A) < \frac{\varepsilon}{2}$, so $m(E \triangle A) < \varepsilon$.

(iii) Choose a finite open-cell union U with $m(E \triangle U) < \varepsilon$. Refine by the finite coordinate grid from the endpoints of the open cells, and let A be the union of the half-open grid atoms contained in U.
Then $U \triangle A$ is contained in finitely many cell faces, each outer-null, so $m(E \triangle A) < \varepsilon$.

Structural Commentary: Cell format changes only boundary-null pieces.
Key Insight: A compact core inside an open envelope permits finite cell variants. □

Corollary 6.2.4. *Real-Line Finite Open-Interval Approximation*

If $E \in \mathcal{L}^1$ and $m(E) < +\infty$, then $\forall \varepsilon > 0$, $\exists O = \bigcup_{j=1}^{N} I_j$ with I_j pairwise disjoint bounded open intervals and $m(E \setminus O) + m(O \setminus E) = m(E \triangle O) < \varepsilon$.

Proof. By Theorem 6.2.2 with $p = 1$, choose a finite union U of bounded open intervals with $m(E \triangle U) < \varepsilon$. Write $U = O = \bigcup_{j=1}^{N} I_j$ as the disjoint union of its open interval components.
Since $E \setminus O$ and $O \setminus E$ are disjoint measurable sets, additivity gives $m(E \triangle O) = m(E \setminus O) + m(O \setminus E) < \varepsilon$.
Structural Commentary: Merge finite open intervals into disjoint components.
Key Insight: A compact core makes the countable cover finite; interval merging makes it disjoint. □

6.3 Caratheodory Tests and Inner Measure

Theorem 6.3.1. *Outer Measure Additivity with One Measurable Set*
Let $E \in \mathcal{L}^p$ and $F \subseteq \mathbb{R}^p$. Then $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.

Proof. Since $E \in \mathcal{L}^p$, Caratheodory splitting: $m_p^*(E \cup F) = m(E) + m_p^*(F \setminus E)$; $m_p^*(F) = m_p^*(E \cap F) + m_p^*(F \setminus E)$.
Hence $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.
Structural Commentary: One measurable cut splits arbitrary sets exactly.
Key Insight: Split both $E \cup F$ and F along the same measurable cut E. □

Corollary 6.3.2. *Disjoint Additivity and Difference*
Let $E \in \mathcal{L}^p$ and $F \subseteq \mathbb{R}^p$.
(i) $E \cap F = \emptyset \Longrightarrow m_p^*(E \cup F) = m(E) + m_p^*(F)$.
(ii) $E \subseteq F$, $m(E) < +\infty \Longrightarrow m_p^*(F \setminus E) = m_p^*(F) - m(E)$.

Proof.

(i) By Theorem 6.3.1, $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.
Since $E \cap F = \emptyset$ and $m_p^*(\emptyset) = 0$, $m_p^*(E \cup F) = m(E) + m_p^*(F)$.

(ii) Since $E \subseteq F$, Caratheodory splitting gives $m_p^*(F) = m(E) + m_p^*(F \setminus E)$; subtract $m(E) < +\infty$.

Structural Commentary: Exact splitting yields disjoint sums and finite differences.
Key Insight: Disjointness removes the overlap; containment isolates the remainder. □

Remark 6.3.3. *Excision Needs Measurability*
Let $E \subseteq \mathbb{R}^p$ with $m_p^*(E) < +\infty$. Open approximation may give, $\forall \varepsilon > 0$, an open $G \supseteq E$ with $m(G) - m_p^*(E) < \varepsilon$, but this does not imply $m_p^*(G \setminus E) < \varepsilon$. If $E \in \mathcal{L}^p$, Corollary 6.3.2 gives $m_p^*(G \setminus E) = m(G) - m(E)$.
Key Insight: Small open excess becomes small set difference only after E is a Caratheodory splitter.

Theorem 6.3.4. *Finite Outer Measure Inner Test*
If $m_p^*(E) < +\infty$, then $E \in \mathcal{L}^p \iff \exists B \in \mathcal{L}^p,\ B \subseteq E,\ m(B) = m_p^*(E)$.

Proof.

($\Rightarrow$) Set $B := E$. Then $B \in \mathcal{L}^p$, $B \subseteq E$, and $m(B) = m(E) = m_p^*(E)$.

($\Leftarrow$) Assume $B \in \mathcal{L}^p$, $B \subseteq E$, and $m(B) = m_p^*(E)$.
Since $m(B) < +\infty$, the difference formula gives $m_p^*(E \setminus B) = m_p^*(E) - m(B) = 0$.
Thus $E \setminus B \in \mathcal{L}^p$, and $E = B \cup (E \setminus B) \in \mathcal{L}^p$.

Structural Commentary: Exact inner mass leaves only a null remainder.
Key Insight: Equality of outer mass converts containment into measurability. □

Theorem 6.3.5. *Single Finite Container Test*
Let $A \in \mathcal{L}^p$, $m(A) < +\infty$, and $E \subseteq A$. Then $E \in \mathcal{L}^p \iff m(A) = m_p^*(E) + m_p^*(A \setminus E)$.

Proof.

($\Rightarrow$) Since $E \in \mathcal{L}^p$ and $E \subseteq A$,
Caratheodory splitting of A by E gives $m(A) = m(E) + m(A \setminus E) = m_p^*(E) + m_p^*(A \setminus E)$.

($\Leftarrow$) Choose G_δ set $H \supseteq A \setminus E$ with $m(H) = m_p^*(A \setminus E)$. Since $A \setminus E \subseteq A \cap H \subseteq H$, $m(A \cap H) = m_p^*(A \setminus E)$.
Set $B := A \setminus H$. $B \in \mathcal{L}^p$, $B \subseteq E$; $m(B) = m(A) - m(A \cap H) = m_p^*(E)$. By Theorem 6.3.4, $E \in \mathcal{L}^p$.

Structural Commentary: Inside a finite measurable container, Caratheodory only needs to be tested once.
Key Insight: Outer-approximate the complement; the remainder has full outer mass. □

Corollary 6.3.6. *Bounded Inner Measure Test*
For bounded cell $Q \subseteq \mathbb{R}^p$ and $E \subseteq Q$, define $m_{*,Q}(E) := m(Q) - m_p^*(Q \setminus E)$. Then $E \in \mathcal{L}^p \iff m_{*,Q}(E) = m_p^*(E)$.

Proof.
($\Rightarrow$) If $E \in \mathcal{L}^p$, then $m(Q) = m(E) + m_p^*(Q \setminus E)$, hence $m_{*,Q}(E) = m_p^*(E)$.
($\Leftarrow$) If $m_{*,Q}(E) = m_p^*(E)$, then $m(Q) = m_p^*(E) + m_p^*(Q \setminus E)$, so $E \in \mathcal{L}^p$ by the finite-container test.
Structural Commentary: Bounded inner measure tests one finite container.
Key Insight: Equality of inner and outer measure is a bounded version of Caratheodory. □

7 Vitali, Steinhaus, and Nonmeasurable Sets

7.1 Vitali, Steinhaus, and Nonmeasurability

Definition 7.1.1. *Difference Sets and Vitali Selectors*
(i) *Difference set*: for $A \subseteq \mathbb{R}^p$, $A - A := \{x - y : x, y \in A\}$; $A \subseteq B \Longrightarrow A - A \subseteq B - B$.
(ii) *Rational equivalence*: $x \sim_{\mathbb{Q}^p} y \iff x - y \in \mathbb{Q}^p$; equivalently, $[x]_{\mathbb{Q}^p} = x + \mathbb{Q}^p$.
(iii) *Vitali selector*: choose $\mathcal{V} \subseteq \mathbb{R}^p$ such that $|\mathcal{V} \cap (x + \mathbb{Q}^p)| = 1$ for every $x \in \mathbb{R}^p$.
(iv) *Rational translate*: for $q \in \mathbb{Q}^p$, $\mathcal{V}_q := q + \mathcal{V}$.

Lemma 7.1.2. *Steinhaus Lemma for Compact Sets*
If $K \subseteq \mathbb{R}^p$ is compact and $m(K) > 0$, then $\exists \delta > 0$ such that $B(0, \delta) \subseteq K - K$.

Proof. Choose open $G \supseteq K$ with $m(G) < 2m(K)$. Since K is compact and G^c is closed, set $\delta := \operatorname{dist}(K, G^c) > 0$. If $\|x\|_2 < \delta$, then $x + K \subseteq G$. If $(x + K) \cap K = \emptyset$, translation invariance and additivity give
$2m(K) = m(x + K) + m(K) = m((x + K) \cup K) \leq m(G) < 2m(K)$, a contradiction.
Hence $(x + K) \cap K \neq \emptyset$, so $x = k_1 - k_2 \in K - K$. Thus $B(0, \delta) \subseteq K - K$.
Structural Commentary: A positive compact set overlaps all sufficiently small translates of itself.
Key Insight: Disjoint small translates cannot fit inside a tight open envelope. □

Theorem 7.1.3. *Steinhaus Theorem*
If $E \in \mathcal{L}^p$ and $m(E) > 0$, then $\exists \delta > 0$ such that $B(0, \delta) \subseteq E - E$.

Proof. Set $E_n := E \cap B(0, n)$. Since $E_n \uparrow E$ and $m(E) > 0$, choose n with $0 < m(E_n) < +\infty$.
By compact inner approximation, choose compact $K \subseteq E_n$ with $m(K) > 0$.
Lemma 7.1.2 gives $\delta > 0$ with $B(0, \delta) \subseteq K - K \subseteq E - E$.
Structural Commentary: Positive measurable sets contain positive compact cores.
Key Insight: Difference-set thickness follows from compact inner regularity. □

Lemma 7.1.4. *Bounded Disjoint Translates Force Nullity*
Let $E \in \mathcal{L}^1$ and $\Lambda \subseteq \mathbb{R}$ be bounded, with E bounded and Λ countably infinite.
If $(\lambda + E)_{\lambda \in \Lambda}$ are pairwise disjoint, then $m(E) = 0$.

Proof. Since E and Λ are bounded, $U := \bigcup_{\lambda \in \Lambda} (\lambda + E)$ is bounded, so $m(U) < +\infty$.
By disjointness, countable additivity, and translation invariance, $m(U) = \sum_{\lambda \in \Lambda} m(\lambda + E) = \sum_{\lambda \in \Lambda} m(E)$.
Since Λ is countably infinite, finiteness forces $m(E) = 0$.
Structural Commentary: Infinitely many disjoint bounded translates cannot all carry positive equal mass.
Key Insight: Boundedness gives a finite container; translation invariance makes every summand the same. □

Proposition 7.1.5. *Real-Line Vitali Choice Sets*
Let $E \subseteq \mathbb{R}$ with $m_1^*(E) > 0$, and write $x \sim y$ iff $x - y \in \mathbb{Q}$.
(i) There exists bounded $E_0 \subseteq E$ with $m_1^*(E_0) > 0$.
(ii) If $C \subseteq E_0$ contains exactly one point from each $\sim$-class meeting E_0, then $C \notin \mathcal{L}^1$.
In particular, E contains a nonmeasurable subset.

Proof.
(i) Since $E = \bigcup_{n \geq 1} (E \cap [-n, n])$, countable subadditivity gives n with $E_0 := E \cap [-n, n]$ satisfying $m_1^*(E_0) > 0$.
(ii) Choose $b > 0$ with $E_0 \subseteq [-b, b]$ and set $\Lambda_0 := [-2b, 2b] \cap \mathbb{Q}$. Suppose $C \in \mathcal{L}^1$.
From $\lambda + c = \lambda' + c'$ with $\lambda, \lambda' \in \Lambda_0$ and $c, c' \in C$, we get $c - c' = \lambda' - \lambda \in \mathbb{Q}$; selector uniqueness gives $c = c'$ and $\lambda = \lambda'$. Thus $(\lambda + C)_{\lambda \in \Lambda_0}$ is pairwise disjoint, and Lemma 7.1.4 gives $m(C) = 0$.
For every $x \in E_0$, choose $c \in C$ with $x - c \in \mathbb{Q}$; since $x, c \in [-b, b]$, $x - c \in \Lambda_0$, so $E_0 \subseteq \bigcup_{\lambda \in \Lambda_0} (\lambda + C)$.
The right side is a countable union of null measurable sets, contradicting $m_1^*(E_0) > 0$.
Structural Commentary: Bounded rational shifts force the contradiction.
Key Insight: Selectors are either null under disjoint shifts or cover too much by null shifts. □

Lemma 7.1.6. *Rational Translates of a Vitali Selector Partition $\mathbb{R}^p$*
Let $(q_i)_{i \geq 1}$ enumerate $\mathbb{Q}^p$.
(i) $\mathbb{R}^p = \bigcup_{i=1}^{\infty} \mathcal{V}_{q_i}$.
(ii) Each $x \in \mathbb{R}^p$ has a unique representation $x = q_i + v$ with $i \in \mathbb{N}$ and $v \in \mathcal{V}$; hence $\mathbb{R}^p = \bigsqcup_{i=1}^{\infty} \mathcal{V}_{q_i}$.

Proof.

(i) For $x \in \mathbb{R}^p$, choose the unique $v \in \mathcal{V} \cap (x + \mathbb{Q}^p)$. Then $x - v = q_i$ for some i, so $x = q_i + v \in \mathcal{V}_{q_i}$.
(ii) If $q_i + v = q_j + w$ with $v, w \in \mathcal{V}$, then $v - w = q_j - q_i \in \mathbb{Q}^p$.
Selector uniqueness gives $v = w$, hence $q_i = q_j$ and $i = j$.

Structural Commentary: Rational translates enumerate the equivalence classes selected by $\mathcal{V}$.
Key Insight: One representative per class makes the rational translates disjoint and exhaustive. □

Theorem 7.1.7. *Vitali Selectors Are Nonmeasurable*
If $\mathcal{V} \subseteq \mathbb{R}^p$ is a Vitali selector, then $\mathcal{V} \notin \mathcal{L}^p$.

Proof. Assume $\mathcal{V} \in \mathcal{L}^p$.

(i) If $m(\mathcal{V}) > 0$, Steinhaus gives $\delta > 0$ with $B(0, \delta) \subseteq \mathcal{V} - \mathcal{V}$. Choose $0 \neq q \in \mathbb{Q}^p \cap B(0, \delta)$. Then $q = v_1 - v_2$ for some $v_1, v_2 \in \mathcal{V}$, so $v_1 \sim_{\mathbb{Q}^p} v_2$; selector uniqueness gives $v_1 = v_2$, hence $q = 0$, contradiction.
(ii) If $m(\mathcal{V}) = 0$, then each translate $\mathcal{V}_{q_i}$ is null. By Lemma 7.1.6, $\mathbb{R}^p = \bigsqcup_i \mathcal{V}_{q_i}$, so every bounded cell is contained in a countable union of null sets and has measure 0, contradicting positive cell volume.

Structural Commentary: Positive and zero measure are both impossible.
Key Insight: Steinhaus forbids positive measure; countable translated partition forbids zero measure. □

Theorem 7.1.8. *Nonmeasurable Subsets Are Everywhere*
If $E \subseteq \mathbb{R}^p$ and $m_p^*(E) > 0$, then $\exists N \subseteq E$ with $N \notin \mathcal{L}^p$.

Proof. Let (q_i) enumerate $\mathbb{Q}^p$ and set $E_i := E \cap \mathcal{V}_{q_i}$.
If $E_i \in \mathcal{L}^p$ and $m(E_i) > 0$, then $E_i - E_i$ contains a ball about 0, so $0 \neq q \in \mathbb{Q}^p \cap (E_i - E_i)$.
But $E_i \subseteq q_i + \mathcal{V}$, so $q = (q_i + v_1) - (q_i + v_2) = v_1 - v_2$ with $v_1, v_2 \in \mathcal{V}$, contradicting selector uniqueness.
Hence each measurable E_i is null. If every E_i were measurable, then $E = \bigcup_i E_i$ would satisfy $m_p^*(E) = 0$, contradiction. Thus some $E_i \subseteq E$ is nonmeasurable.
Structural Commentary: Any positive outer-measure set meets the Vitali partition in a nonmeasurable piece.
Key Insight: Vitali pieces cannot be measurable and positive. □

Corollary 7.1.9. *Nonadditive Decompositions*

(i) If $E \in \mathcal{L}^p$ and $0 < m(E) < +\infty$, then $\exists B, C \notin \mathcal{L}^p$ such that $E = B \sqcup C$ and $m(E) < m_p^*(B) + m_p^*(C)$.
(ii) If $B \notin \mathcal{L}^p$, $m_p^*(B) < +\infty$, and H is G_δ with $B \subseteq H$, $m(H) = m_p^*(B)$, then $C := H \setminus B \notin \mathcal{L}^p$ and $m(H) = m_p^*(B \cup C) < m_p^*(B) + m_p^*(C)$.

Proof.

(i) Pick $B \subseteq E$, $B \notin \mathcal{L}^p$, and put $C := E \setminus B$. Then $C \notin \mathcal{L}^p$, else $B = E \setminus C \in \mathcal{L}^p$.
Subadditivity gives $\leq$; equality would force $B \in \mathcal{L}^p$ by the finite-container test in E.
(ii) Since $H \in \mathcal{L}^p$, $C \in \mathcal{L}^p$ would give $B = H \setminus C \in \mathcal{L}^p$, contradiction. Also $m_p^*(C) > 0$, else $C \in \mathcal{L}^p$.
Thus $B \cup C = H$ and $m(H) = m_p^*(B)$ give $m_p^*(B \cup C) = m(H) < m_p^*(B) + m_p^*(C)$.

Structural Commentary: Nonmeasurability appears exactly as strict failure of additive splitting.
Key Insight: Equality in a finite measurable container is a measurability test. □

Theorem 7.1.10. *A Nonmeasurable Set with Only Null Measurable Pieces*
There exists $\mathcal{U} \subseteq \mathbb{R}^p$ such that:

(i) if $F \in \mathcal{L}^p$ and $F \subseteq \mathcal{U}$, then $m(F) = 0$;
(ii) if $F \in \mathcal{L}^p$ and $F \subseteq \mathcal{U}^c$, then $m(F) = 0$;
(iii) $\mathcal{U} \notin \mathcal{L}^p$.

Proof. Let $e := (1, \ldots, 1)$, $G_0 := \mathbb{Q}^p + (2\sqrt{2})\mathbb{Z}e$, $G_1 := \sqrt{2}e + G_0$, $G := G_0 \sqcup G_1 = \mathbb{Q}^p + \sqrt{2}\mathbb{Z}e$; G_0, G_1 dense.
Let $\mathcal{E}$ select one representative from each class modulo G, and set $\mathcal{U} := G_0 + \mathcal{E}$.
Then $\mathbb{R}^p = (G_0 + \mathcal{E}) \sqcup (G_1 + \mathcal{E})$ and $\mathcal{U}^c = G_1 + \mathcal{E} = \sqrt{2}e + \mathcal{U}$.

(i) Suppose $F \in \mathcal{L}^p$, $F \subseteq \mathcal{U}$, and $m(F) > 0$. By Steinhaus, $B(0, r) \subseteq F - F$ for some $r > 0$; choose $g_1 \in G_1 \cap B(0, r)$. Since $F \subseteq G_0 + \mathcal{E}$, $g_1 = (g_0 + s_1) - (g_0' + s_2)$ for some $g_0, g_0' \in G_0$ and $s_1, s_2 \in \mathcal{E}$. Thus $s_1 - s_2 = g_1 - g_0 + g_0' \in G$, so $s_1 = s_2$ and $g_1 = g_0 - g_0' \in G_0$, contradicting $G_0 \cap G_1 = \emptyset$.
(ii) If $F \in \mathcal{L}^p$ and $F \subseteq \mathcal{U}^c = \sqrt{2}\,e + \mathcal{U}$, then $F - \sqrt{2}\,e \subseteq \mathcal{U}$ and $m(F) = m(F - \sqrt{2}\,e) = 0$ by (i).
(iii) If $\mathcal{U} \in \mathcal{L}^p$, then (i)–(ii) give $m(\mathcal{U}) = m(\mathcal{U}^c) = 0$, so $m([0,1]^p) = 0$, contradiction.

Structural Commentary: This construction partitions representatives using two dense cosets.
Key Insight: Positive measurable pieces force forbidden coset differences. □

8 Cantor–Lebesgue Singular Function

8.1 Cantor-Lebesgue Singular Function Construction

Definition 8.1.1. *Cantor–Lebesgue Singular Function*
Let P be the Cantor set. For $k \geq 1$, let $I_{k,1}, \ldots, I_{k,2^k-1}$ list, left to right, the components of the stage-k removed set U_k. Set $U := \bigcup_{k \geq 1} U_k = [0,1] \setminus P$.

(i) For $x \in U$, choose $k \geq 1$ and $1 \leq j \leq 2^k - 1$ with $x \in I_{k,j}$, and set $\varphi(x) := \frac{j}{2^k}$.
(ii) This value is independent of the chosen stage: if $I_{k,j} = I_{\ell,m}$ with $\ell \geq k$, then $m = 2^{\ell-k} j$ and $\frac{m}{2^\ell} = \frac{j}{2^k}$.
(iii) On P, set $\varphi(0) := 0$ and $\varphi(x) := \sup\{\varphi(t) : t \in U \cap [0, x]\}$ for $x \in P \setminus \{0\}$.

The resulting $\varphi : [0,1] \to [0,1]$ is the *Cantor–Lebesgue singular function*.

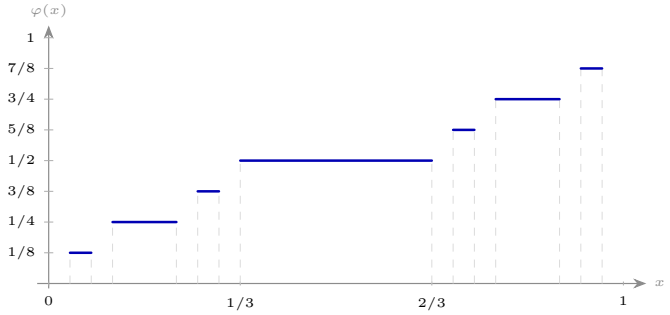

Figure 197: Horizontal segments show the Cantor–Lebesgue plateaus on the intervals removed through stage 3.

Proposition 8.1.2. *Properties of the Singular Function*
Let P be the Cantor set, $U := [0,1] \setminus P$, and $\varphi : [0,1] \to [0,1]$ the Cantor–Lebesgue singular function. Then:

(i) φ *is nondecreasing.*
(ii) φ *is continuous on* $[0,1]$.
(iii) $\varphi([0,1]) = [0,1]$.
(iv) $\varphi'(x) = 0$ *for every* $x \in U$.
(v) $m(U) = 1$.

Proof.

(i) At each stage k, the components $I_{k,1}, \dots, I_{k,2^k-1}$ occur left to right and carry heights $\frac{1}{2^k}, \dots, \frac{2^k-1}{2^k}$.
Thus the prescription on U is nondecreasing, and the supremum extension on P preserves order.
(ii) If $x_0 \in U$, then $x_0 \in J$ for some component J of U and $\varphi|_J$ is constant. If $x_0 \in P$, then for each k there are $a_k, b_k \in 2^{-k}\mathbb{Z} \cap [0,1]$ with $a_k \le \varphi(x_0) \le b_k$, $b_k - a_k \le 2^{-k}$, and $\varphi(y) \in [a_k, b_k]$ for all y near x_0.
Thus $\limsup_{y \to x_0} |\varphi(y) - \varphi(x_0)| \le 2^{-k}$ for every k, hence φ is continuous at x_0.
(iii) The definition gives $\varphi(0) = 0$. For each k, the rightmost component of U_k has height $1 - 2^{-k}$, so $\varphi(1) = 1$.
By (ii) and the intermediate value theorem, $\varphi([0,1]) = [0,1]$.
(iv) For each component J of U, $\varphi|_J$ is constant; hence $\varphi'(x) = 0$ for every $x \in U$.
(v) Proposition 4.1.14 gives $m(P) = 0$; since $[0,1] = P \sqcup U$, $m(U) = m([0,1]) - m(P) = 1$.

Structural Commentary: Plateaus on U force variation onto P.
Key Insight: At stage k, neighboring plateau heights differ by 2^{-k}. □

Theorem 8.1.3. *Lebesgue Measurable Non-Borel Sets Exist*
There exist $A \subseteq [0,1]$ and a homeomorphism $\psi : [0,1] \to [0,2]$ such that $A \in \mathcal{L}^1 \setminus \mathcal{B}(\mathbb{R})$ and $\psi(A) \notin \mathcal{L}^1$.

Proof. Let P be the Cantor set, $U := [0,1] \setminus P$, φ the Cantor–Lebesgue singular function, and $\psi := \mathrm{id}_{[0,1]} + \varphi$.
Proposition 8.1.2 gives φ continuous and nondecreasing; hence ψ is continuous and strictly increasing.
Since $\psi(0) = 0$ and $\psi(1) = 2$, $\psi : [0,1] \to [0,2]$ is a homeomorphism.
Let $(J_k)_{k \ge 1}$ be the components of U. Since $\varphi|_{J_k}$ is constant, $\psi(J_k)$ is a translate of J_k and $m(\psi(J_k)) = m(J_k)$.
Thus $m(\psi(U)) = \sum_{k \ge 1} m(\psi(J_k)) = \sum_{k \ge 1} m(J_k) = m(U) = 1$.
Since $\psi(P)$ is compact and $[0,2] = \psi(P) \sqcup \psi(U)$, $m(\psi(P)) = 1$.
Choose $W \subseteq \psi(P)$ with $W \notin \mathcal{L}^1$ by Theorem 7.1.8, and set $A := \psi^{-1}(W)$. Then $A \subseteq P$ and $m(P) = 0$, so $A \in \mathcal{L}^1$. If $A \in \mathcal{B}(\mathbb{R})$, then $W = \psi(A) \in \mathcal{B}([0,2]) \subseteq \mathcal{L}^1$, contradiction. Hence $A \in \mathcal{L}^1 \setminus \mathcal{B}(\mathbb{R})$ and $\psi(A) = W \notin \mathcal{L}^1$.
Structural Commentary: Completion by null sets creates Lebesgue sets invisible to the Borel hierarchy.
Key Insight: A null Cantor set can be homeomorphically stretched to positive measure. □

Part XIV
Measure, Integration, and Functional Analysis

1 Measurable Functions and Lebesgue Integration

1.1 Measurable Functions

Definition 1.1.1. *Borel, Lebesgue, and Function Measurability*

(i) *Borel*: $\mathcal{B}(\mathbb{R}^p) := \sigma(\mathcal{O}_{\mathbb{R}^p})$; its elements are *Borel sets*.
(ii) *Lebesgue*: $\mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$, and $\mathcal{L}^p$ is obtained by adjoining subsets of Lebesgue null sets.
(iii) *Maps*: for measurable spaces $(X,\mathcal{M})$, $(Y,\mathcal{N})$, $f : X \to Y$ is *$(\mathcal{M},\mathcal{N})$-measurable* iff $\forall E \in \mathcal{N},\ f^{-1}(E) \in \mathcal{M}$.
(iv) *Borel maps*: $f : \mathbb{R}^p \to \mathbb{R}^q$ is *Borel measurable* iff $\forall B \in \mathcal{B}(\mathbb{R}^q),\ f^{-1}(B) \in \mathcal{B}(\mathbb{R}^p)$.
(v) *Thresholds*: write $\{f \bowtie a\} := \{x \in X : f(x) \bowtie a\}$ for $\bowtie \in \{>, \geq, <, \leq\}$.
Extended-real f is *$\mathcal{M}$-measurable* iff $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M}$.

Structural Commentary: Borel measurability tracks topology; Lebesgue measurability adds null-set completion.
Key Insight: Measurability is inverse-image stability.

Proposition 1.1.2. *Generator Test for Measurable Maps*
Let $f : X \to Y$, $\mathcal{A} \subseteq \mathcal{P}(Y)$, and $\mathcal{N} := \sigma(\mathcal{A})$.
(i) If $\forall A \in \mathcal{A},\ f^{-1}(A) \in \mathcal{M}$, then $f : X \to (Y,\mathcal{N})$ is measurable.
(ii) For real-valued f, $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M} \iff \forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$.

Proof.
(i) Let $\mathcal{G} := \{E \subseteq Y : f^{-1}(E) \in \mathcal{M}\}$. Since $f^{-1}(E^c) = f^{-1}(E)^c$ and $f^{-1}(\bigcup_n E_n) = \bigcup_n f^{-1}(E_n)$, $\mathcal{G}$ is a σ-algebra. Thus $\mathcal{A} \subseteq \mathcal{G} \Longrightarrow \mathcal{N} = \sigma(\mathcal{A}) \subseteq \mathcal{G}$.
(ii) ($\Rightarrow$) Set $\mathcal{A} := \{(a,+\infty) : a \in \mathbb{R}\}$. Since $\forall a \in \mathbb{R},\ f^{-1}((a,+\infty)) = \{f > a\} \in \mathcal{M}$ and $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$, (i) gives $\forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$.
(ii) ($\Leftarrow$) If $\forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$, then $\forall a \in \mathbb{R},\ \{f > a\} = f^{-1}((a,+\infty)) \in \mathcal{M}$.

Structural Commentary: Measurability can be checked on any generating family of target sets.
Key Insight: Pullbacks turn generated σ-algebras into generated collections of preimages. □

Proposition 1.1.3. *Open-Set Test for Euclidean Targets*
Let $\Omega \subseteq \mathbb{R}^n$ be Lebesgue measurable and let $f : \Omega \to \mathbb{R}^m$. The following tests are equivalent:
(i) $\forall V \subseteq \mathbb{R}^m$ open, $f^{-1}(V)$ is measurable;
(ii) $\forall B \subseteq \mathbb{R}^m$ open boxes, $f^{-1}(B)$ is measurable;
(iii) $\forall j \in \{1,\dots,m\}$, $f_j : \Omega \to \mathbb{R}$ is threshold-measurable.
In particular, every continuous $f : \Omega \to \mathbb{R}^m$ is measurable.

Proof.
(i) $\Rightarrow$ *(ii)*. Open boxes are open.
(ii) $\Rightarrow$ *(i)*. By Lemma 4.1.7, $V = \bigcup_k B_k$ for rational boxes B_k, hence $f^{-1}(V) = \bigcup_k f^{-1}(B_k)$ is measurable.
(ii) $\Rightarrow$ *(iii)*. By *(ii)* $\Rightarrow$ *(i)*, $\{f_j > a\} = f^{-1}(\{y : y_j > a\})$ is measurable.
(iii) $\Rightarrow$ *(ii)*. If $B = \prod_{j=1}^m (a_j, b_j)$, then $f^{-1}(B) = \bigcap_{j=1}^m \big(\{f_j > a_j\} \cap \{f_j < b_j\}\big)$,
where $\{f_j < b_j\} = \Omega \setminus \bigcap_{\ell=1}^{\infty} \{f_j > b_j - \frac{1}{\ell}\}$. Thus each factor is measurable, hence so is $f^{-1}(B)$.

If f is continuous, $f^{-1}(V)$ is open relative to Ω by Theorem 5.1.17.
By Theorem 3.2.2, $f^{-1}(V) = W \cap \Omega$ for some open $W \subseteq \mathbb{R}^n$, hence it is measurable by Theorem 4.1.8.
Structural Commentary: Euclidean-valued measurability is just countable box algebra applied to inverse images.
Key Insight: Open sets reduce to rational boxes; vector functions reduce to coordinate threshold tests. □

Example 1.1.4. *Basic Measurable Functions*
Let $(X,\mathcal{M})$ be a measurable space.
(i) Constant real-valued functions are measurable, since each strict superlevel set is either X or $\emptyset$.
(ii) For $E \subseteq X$, the indicator $\mathbf{1}_E$ is measurable iff $E \in \mathcal{M}$.
(iii) Continuous maps between Euclidean measurable subsets are measurable by Proposition 1.1.3.
(iv) Monotone functions $\mathbb{R} \to \mathbb{R}$ are Borel measurable, since their strict superlevel sets are half-lines, $\mathbb{R}$, or $\emptyset$.

Key Insight: Constants, indicators of measurable sets, continuous functions, and monotone functions form the first reusable stock of measurable functions.

Theorem 1.1.5. *Closure Properties of Measurable Functions*
Let $(X,\mathcal{M})$ be a measurable space.
(i) The four tests $\forall a \in \mathbb{R},\ \{f > a\}, \{f \geq a\}, \{f < a\}, \{f \leq a\} \in \mathcal{M}$ are equivalent.
(ii) f measurable implies $|f|$ measurable.
(iii) If f_n are measurable, then $\sup_n f_n, \inf_n f_n, \limsup_n f_n, \liminf_n f_n$ are measurable.
(iv) If f, g real-valued measurable, $F : \mathbb{R}^2 \to \mathbb{R}$ continuous, then $F(f,g)$ measurable; so are $f+g$ and fg.

Proof.
(i) Use $\{f \geq a\} = \bigcap_{m \geq 1} \{f > a - \frac{1}{m}\}$; $\{f < a\} = X \setminus \{f \geq a\}$; $\{f \leq a\} = \bigcap_{m \geq 1} \{f < a + \frac{1}{m}\}$.
(ii) For $a > 0$, $\{|f| < a\} = \{f < a\} \cap \{f > -a\}$; for $a \leq 0$ the set is empty.

(iii) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$. Also $\limsup_n f_n = \inf_m \sup_{n\geq m} f_n$; $\liminf_n f_n = \sup_m \inf_{n\geq m} f_n$; inf follow from (i).

(iv) For $G_a = \{(u,v) : F(u,v) > a\}$, write $G_a = \bigcup_k (\alpha_k, \beta_k) \times (\gamma_k, \delta_k)$.

Then $\{F(f,g) > a\} = \bigcup_k \big(\{\alpha_k < f < \beta_k\} \cap \{\gamma_k < g < \delta_k\}\big) \in \mathcal{M}$.

Structural Commentary: Measurability is countable set algebra applied to level sets.
Key Insight: Once threshold sets are measurable, ordinary limiting and algebraic operations stay measurable. □

Corollary 1.1.6. *Continuous Composition and Finite Lattice Operations*
Let $(X, \mathcal{M})$ be a measurable space and let $n \in \mathbb{N}_{\geq 1}$.
For $f_1, \dots, f_n : X \to \mathbb{R}$, set $\bigvee_{k=1}^n f_k := \max\{f_1, \dots, f_n\}$ and $\bigwedge_{k=1}^n f_k := \min\{f_1, \dots, f_n\}$.
(i) If $g : X \to \mathbb{R}$ is measurable and $\varphi : \mathbb{R} \to \mathbb{R}$ is continuous, then $\varphi \circ g$ is measurable.
(ii) If $f_1, \dots, f_n : X \to \mathbb{R}$ are measurable, then $\bigvee_{k=1}^n f_k$ and $\bigwedge_{k=1}^n f_k$ are measurable.

Proof.

(i) For $a \in \mathbb{R}$, $\{\varphi \circ g > a\} = g^{-1}(\varphi^{-1}((a, +\infty))) \in \mathcal{M}$, since $\varphi^{-1}((a, +\infty)) \in \mathcal{B}(\mathbb{R})$ and g is measurable.

(ii) Set $h_k = f_k$ for $1 \leq k \leq n$ and $h_k = f_n$ for $k > n$. Then (h_k) is measurable, $\bigvee_{k=1}^n f_k = \sup_{k\geq 1} h_k$, and $\bigwedge_{k=1}^n f_k = \inf_{k\geq 1} h_k$, hence both are measurable by Theorem 1.1.5(iii).

Structural Commentary: Pullbacks preserve Borel thresholds; finite lattices are finite suprema and infima.
Key Insight: Continuity and finite lattice operations preserve measurable threshold structure. □

Example 1.1.7. *Measurable Composition Can Fail*
Choose $A \in \mathcal{L}^1$ with $A \subseteq [0,1]$ and an increasing homeomorphism $\psi : [0,1] \to [0,2]$ such that $\psi(A) \notin \mathcal{L}^1$. Define

$$\widetilde{\psi}(x) := \begin{cases} x, & x \leq 0, \\ \psi(x), & 0 \leq x \leq 1, \\ x+1, & x \geq 1. \end{cases}$$

Then $\widetilde{\psi} : \mathbb{R} \to \mathbb{R}$ is a homeomorphism. Put $h := \mathbf{1}_A \circ \widetilde{\psi}^{-1}$.
The maps $\mathbf{1}_A$ and $\widetilde{\psi}^{-1}$ are measurable, but $h^{-1}((\frac{1}{2}, \frac{3}{2})) = \widetilde{\psi}(A) = \psi(A) \notin \mathcal{L}^1$.
Hence h is not Lebesgue measurable, so Lebesgue measurable functions are not composition-closed.

Definition 1.1.8. *Positive and Negative Parts*
For an extended-real-valued function f, define $f^+(x) := \sup\{f(x), 0\}$; $f^-(x) := \sup\{-f(x), 0\}$.
When f is finite-valued, $f = f^+ - f^-$ and $|f| = f^+ + f^-$; equivalently $f^+ = \frac{|f|+f}{2}$; $f^- = \frac{|f|-f}{2}$.

Corollary 1.1.9. *Measurability of Positive and Negative Parts*
If $f : X \to [-\infty, +\infty]$ is measurable, then $f^+, f^-, |f|$ are measurable.
If $f : X \to \mathbb{R}$, then f measurable $\iff$ f^+, f^- measurable.

Proof.

($\Rightarrow$) Since $\{-f > a\} = \{f < -a\}$, $-f$ is measurable.
Thus $f^+ = f \vee 0$, $f^- = (-f) \vee 0$, and $|f| = f \vee (-f)$ are measurable by finite sup-closure.

($\Leftarrow$) If $f : X \to \mathbb{R}$ and f^+, f^- are measurable, then $f = f^+ - f^-$ is measurable.

Structural Commentary: Sign decomposition is compatible with threshold-set measurability.
Key Insight: Reduce signs to suprema and finite algebra. □

Lemma 1.1.10. *Extended-Valued Measurability via the Finite Part*
Let $f : X \to \overline{\mathbb{R}}$ and set $A := f^{-1}(\{+\infty\})$, and $B := f^{-1}(\{-\infty\})$. Define $f_0(x) := \begin{cases} f(x), & x \notin A \cup B, \\ 0, & x \in A \cup B. \end{cases}$
Then f is measurable iff $A, B \in \mathcal{M}$ and $f_0 : X \to \mathbb{R}$ is measurable.

Proof.

($\Rightarrow$) If f is measurable, then $A = \bigcap_{n=1}^{\infty} \{f > n\}$; and $B = X \setminus \bigcup_{n=1}^{\infty} \{f > -n\}$, so $A, B \in \mathcal{M}$.

Also, $\{f_0 > \alpha\} = \begin{cases} \{f > \alpha\} \setminus A, & \alpha \geq 0, \\ \{f > \alpha\} \cup B, & \alpha < 0, \end{cases}$ so f_0 is measurable.

($\Leftarrow$) If $A, B \in \mathcal{M}$, f_0 is measurable, then $\{f > \alpha\} = \begin{cases} \{f_0 > \alpha\} \cup A, & \alpha \geq 0, \\ \{f_0 > \alpha\} \setminus B, & \alpha < 0, \end{cases}$ so $\{f > \alpha\} \in \mathcal{M}\ \forall \alpha \in \mathbb{R}$.

Structural Commentary: The infinite-level sets are measurable exactly when the finite core is measurable.
Key Insight: Assign harmless finite values at $\pm\infty$, then put the infinite fibers back by set algebra. □

Corollary 1.1.11. *Extended-Valued Algebra Conventions*
Let $f, g : X \to \overline{\mathbb{R}}$ be measurable.
(i) For $c \in \mathbb{R}$, cf, $|f|$, f^+, and f^- are measurable, with $0(\pm\infty) = 0$.
(ii) Let $E_1 := \{f = -\infty,\ g = +\infty\}$, $E_2 := \{f = +\infty,\ g = -\infty\}$, and $F := E_1 \cup E_2$.
If $h = f + g$ on $X \setminus F$ and $h = 0$ on F, then h is measurable.
(iii) With $0(\pm\infty) = 0$, fg is measurable.

Proof.

(i) By Lemma 1.1.10, $A_\pm(f) := \{f = \pm\infty\} \in \mathcal{M}$ and f_0 is measurable.
For $c \in \mathbb{R}$, the infinite fibers of cf are among $A_\pm(f)$ and $(cf)_0 = cf_0$, so cf is measurable.
The functions $|f|, f^+, f^-$ are measurable by Corollary 1.1.9.

(ii) $E_1, E_2 \in \mathcal{M}$. Set $f_n := \max\{-n, \min(f, n)\}$ and $g_n := \max\{-n, \min(g, n)\}$.
Then f_n, g_n are real-valued measurable and $(f_n + g_n)\mathbf{1}_{X\setminus F} \to h$ pointwise; hence h is measurable.
(iii) With f_n, g_n as in (ii), $f_n g_n$ measurable, $f_n g_n \to fg$ pointwise under $0(\pm\infty) = 0$; hence fg measurable.

Structural Commentary: Extended algebra is finite algebra plus measurable exceptional fibers.
Key Insight: Truncate first, use real-valued closure, then pass to pointwise limits. □

Definition 1.1.12. *Simple Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) *Indicator*: for $E \subseteq X$, $\mathbf{1}_E(x) := 1$ on E and $\mathbf{1}_E(x) := 0$ on $X \setminus E$.
(ii) *Simple*: a real-valued s is simple iff $s(X)$ is finite.
(iii) *Standard representation*: if $s(X) = \{c_1, \dots, c_n\}$ and $E_i := \{s = c_i\} \neq \emptyset$, then $X = \bigsqcup_{i=1}^{n} E_i$; $s = \sum_{i=1}^{n} c_i \mathbf{1}_{E_i}$.
The pairs (c_i, E_i) are unique up to reordering.
(iv) *Measurability*: s measurable iff $E_i \in \mathcal{M}$ $\forall i$; step functions are special case $X \subseteq \mathbb{R}$ with interval level sets.

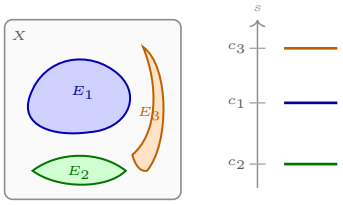

Figure 198: A simple function takes value c_i on each measurable piece E_i.

Proposition 1.1.13. *Simple Functions Form a Lattice Algebra*
Let $s, t : X \to \mathbb{R}$ be simple and $a \in \mathbb{R}$. Then as, $s + t$, st, $s \vee t$, and $s \wedge t$ are simple.
If s, t are measurable, so are these five functions.

Proof. Let $s(X) = \{c_i\}_{i=1}^m$ and $t(X) = \{d_j\}_{j=1}^n$. Each displayed range is contained in the corresponding finite set $\{ac_i\}$, $\{c_i + d_j\}$, $\{c_i d_j\}$, $\{c_i \vee d_j\}$, or $\{c_i \wedge d_j\}$, hence is finite.
If s, t are measurable, the atoms $P_{ij} := \{s = c_i\} \cap \{t = d_j\}$ are measurable and form a finite partition of X. Each displayed function is constant on every P_{ij}, so its level sets are finite unions of P_{ij}.
Structural Commentary: Algebra and lattice operations reduce to finite tables.
Key Insight: Refine two simple functions by the common partition of their level sets. □

Lemma 1.1.14. *Bounded Simple Approximation Brackets*
Let $(X, \mathcal{M})$ be a measurable space and let $f : X \to \mathbb{R}$ be bounded and $\mathcal{M}$-measurable. Then $\forall \varepsilon > 0$ $\exists$ simple $\mathcal{M}$-measurable functions $\varphi_\varepsilon, \psi_\varepsilon : X \to \mathbb{R}$ with $\varphi_\varepsilon \le f \le \psi_\varepsilon$ and $0 \le \psi_\varepsilon - \varphi_\varepsilon < \varepsilon$ on X.

Proof. Choose $c = y_0 < \cdots < y_N = d$ with $f(X) \subseteq (c, d)$ and $\max_k (y_k - y_{k-1}) < \varepsilon$.
Set $E_k := \{y_{k-1} \le f < y_k\} = \{f \ge y_{k-1}\} \cap \{f < y_k\} \in \mathcal{M}$; then $(E_k)_{k=1}^N$ partitions X.
Define $\varphi_\varepsilon := \sum_{k=1}^{N} y_{k-1} \mathbf{1}_{E_k}$ and $\psi_\varepsilon := \sum_{k=1}^{N} y_k \mathbf{1}_{E_k}$; these are $\mathcal{M}$-measurable simple functions.
For $x \in E_k$, $\varphi_\varepsilon(x) = y_{k-1} \le f(x) < y_k = \psi_\varepsilon(x)$ and $\psi_\varepsilon(x) - \varphi_\varepsilon(x) = y_k - y_{k-1} < \varepsilon$.
Structural Commentary: Pull back a finite range partition.
Key Insight: Mesh size controls the bracket gap. □

Theorem 1.1.15. *Approximation by Simple Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) $\forall f : X \to \mathbb{R}$ $\exists$ simple functions (s_n) with $s_n \to f$ pointwise.
(ii) For $f : X \to \overline{\mathbb{R}}$, f is $\mathcal{M}$-measurable iff f is the pointwise limit of $\mathcal{M}$-measurable simple functions.
(iii) If $f : X \to \overline{\mathbb{R}}$ is $\mathcal{M}$-measurable, the approximants in (ii) may be chosen with $|s_n| \le |f|$.
If $f \ge 0$, they may be chosen with $0 \le s_n \uparrow f$.

Proof. For $u : X \to [0, +\infty]$, set $Q_n u := 2^{-n} \lfloor 2^n (u \wedge n) \rfloor$.
Then $Q_n u(X) \subseteq 2^{-n}\{0, \dots, n2^n\}$, so $Q_n u$ is simple, and binary truncation gives $0 \le Q_n u \uparrow u$.
If u is measurable, the level sets of $Q_n u$ are threshold sets of $u \wedge n$, hence $Q_n u$ is measurable.

(i) For $f : X \to \mathbb{R}$, write $f = f^+ - f^-$ and set $s_n := Q_n f^+ - Q_n f^-$. Then s_n is simple and $s_n \to f$ pointwise.
(*ii*) ($\Rightarrow$) If $f : X \to \overline{\mathbb{R}}$ is measurable, then f^+, f^- are measurable and $s_n := Q_n f^+ - Q_n f^-$ is measurable simple. Since at each x at most one of $f^+(x), f^-(x)$ is infinite, $s_n(x) \to f(x)$ in $\overline{\mathbb{R}}$.
(*ii*) ($\Leftarrow$) If measurable simple $s_n \to f$ pointwise, then $f = \liminf_n s_n$ is measurable.
(*iii*) In (*ii*) $\Rightarrow$, $|s_n| \le Q_n f^+ + Q_n f^- \le |f|$. If $f \ge 0$, take $s_n := Q_n f$.

Structural Commentary: Simple functions quantize the range of f.
Key Insight: Dyadic range partitions give pointwise approximation while preserving measurability. □

Definition 1.1.16. *Complex-Valued Measurable Functions*
A map $f : X \to \mathbb{C}$ is *measurable* iff $\operatorname{Re} f, \operatorname{Im} f : X \to \mathbb{R}$ are measurable.
Equivalently, $f = \operatorname{Re} f + i \operatorname{Im} f$ with measurable real components.

Proposition 1.1.17. *Complex Measurable Closure*
If $f, g : X \to \mathbb{C}$ are measurable, $a \in \mathbb{C}$, and $h_n : X \to \mathbb{C}$ are measurable with $h_n \to h$ pointwise, then $f + g$, fg, af, and h are measurable.

Proof. Write $f = u + iv$, $g = r + is$, and $a = \alpha + i\beta$. Then $f + g = (u + r) + i(v + s)$, $af = (\alpha u - \beta v) + i(\alpha v + \beta u)$, and $fg = (ur - vs) + i(us + vr)$, so real closure gives measurability.
If $h_n \to h$, then $\operatorname{Re} h_n \to \operatorname{Re} h$ and $\operatorname{Im} h_n \to \operatorname{Im} h$, so pointwise-limit closure gives measurability.
Structural Commentary: Complex measurability is two real measurability statements.
Key Insight: Split into real and imaginary parts. □

Definition 1.1.18. *Almost Everywhere and In Measure*
Let E be measurable.

(i) *A.E. equality*: $f \sim_{\text{a.e.}} g$ on E iff $\mu(\{x \in E : f(x) \neq g(x)\}) = 0$.
(ii) *A.E. property*: a property holds a.e. on E iff its failure set in E is null.
(iii) *Null invariance*: $\sim_{\text{a.e.}}$ is an equivalence relation.
(iv) *A.E. convergence*: $f_n \to f$ a.e. on E iff $\exists N \subseteq E$ null such that $\forall x \in E \setminus N,\ f_n(x) \to f(x)$.
(v) *In measure*: $f_n \to f$ in measure on E iff $\forall \varepsilon > 0$, $\mu(\{x \in E : |f_n(x) - f(x)| > \varepsilon\}) \to 0$.

Remark 1.1.19. *Completion and Almost-Everywhere Limits*
If $E \subseteq N \in \mathcal{M}$, $\mu(N) = 0$, and $E \notin \mathcal{M}$, then $f_n \equiv 0$ is measurable and $f_n \to \mathbf{1}_E$ a.e.,
since $\{x : f_n(x) \not\to \mathbf{1}_E(x)\} \subseteq N$. Yet $\mathbf{1}_E$ is not $\mathcal{M}$-measurable.
In the completion, every $E \subseteq N$ is measurable, so null modifications preserve measurability.
Key Insight: Almost-everywhere statements are most stable on complete measure spaces.

Corollary 1.1.20. *Pointwise and Almost-Everywhere Limits*
Let (f_n) be $\overline{\mathbb{R}}$-valued measurable functions on $(X, \mathcal{M})$, and let $f : X \to \overline{\mathbb{R}}$.

(i) If $f_n \to f$ pointwise on X, then f is measurable.
(ii) If $(X, \mathcal{M}, \mu)$ is complete and $f_n \to f$ a.e. on X, then f is measurable.

Proof.

(i) Pointwise, $f = \liminf_n f_n = \limsup_n f_n$; closure under lim inf and lim sup gives measurability.

(ii) Choose $N \in \mathcal{M}$ with $\mu(N) = 0$ and $f_n \to f$ on $X \setminus N$.
Define $g = f$ and $g_n = f_n$ on $X \setminus N$, and $g = g_n = 0$ on N.
Then each g_n is measurable and $g_n \to g$ pointwise, so g is measurable by (i).
Since $\forall a \in \mathbb{R},\ \{f > a\} \triangle \{g > a\} \subseteq N$, completeness gives $\{f > a\} \in \mathcal{M}$; hence f is measurable.

Structural Commentary: Limits reduce to upper and lower limits; a.e. limits require completion.
Key Insight: Completion is exactly what lets null modifications preserve measurability. □

Proposition 1.1.21. *Completed Measurable Functions Have Original Representatives*
Let $(X, \mathcal{M}, \mu)$ be a measure space with completion $(X, \overline{\mathcal{M}}_\mu, \overline{\mu})$.
If $f : X \to \mathbb{R}$ is $\overline{\mathcal{M}}_\mu$-measurable, then $\exists g : X \to \mathbb{R}$ that is $\mathcal{M}$-measurable and satisfies $f = g$ μ-a.e.

Proof. For $q \in \mathbb{Q}$, let $A_q := \{f > q\} \in \overline{\mathcal{M}}_\mu$, let $A_q = E_q \cup Z_q$, with $E_q \in \mathcal{M}$ and $Z_q \subseteq N_q \in \mathcal{M}$, $\mu(N_q) = 0$.
Put $N := \bigcup_{q \in \mathbb{Q}} N_q$, so $N \in \mathcal{M}$ and $\mu(N) = 0$, and define $g(x) := \begin{cases} f(x), & x \notin N, \\ 0, & x \in N. \end{cases}$.
For $q \in \mathbb{Q}$, $\{g > q\} = (E_q \setminus N) \cup \begin{cases} N, & q < 0, \\ \emptyset, & q \geq 0, \end{cases}$ belongs to $\mathcal{M}$. Hence, for $a \in \mathbb{R}$, $\{g > a\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} \{g > q\} \in \mathcal{M}$.
Thus g is $\mathcal{M}$-measurable and $f = g$ on $X \setminus N$.
Structural Commentary: Completion only changes measurable functions on null sets.
Key Insight: Replace completed rational levels off one null set. □

Method 1.1.22. *Measure Proof Patterns*

(i) *Disjointize*. If $A_n \uparrow$, set $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$; then $A_n = \bigsqcup_{k \leq n} B_k$ and $\bigcup_n A_n = \bigsqcup_n B_n$.
(ii) *Approximate*. Prove on $\mathcal{E}$; use $d_\mu(C, D) := \mu^*(C \triangle D)$ and $\triangle$ estimates to pass limits through $\cup, \cap, \setminus$.
(iii) *Threshold*. For real f, show $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M}$; for $f : X \to \mathbb{R}^m$, use coordinate and open-set tests.
(iv) *Generators*. For $f : X \to (Y, \sigma(\mathcal{A}))$, check $\forall A \in \mathcal{A},\ f^{-1}(A) \in \mathcal{M}$.
(v) *Truncate*. For $f : X \to \overline{\mathbb{R}}$, prove results for $f_n := (-n) \vee (f \wedge n)$, then let $n \to \infty$.
(vi) *Topological bridge*. Use continuous open preimages.
Reduce Borel tests to countable interval or rectangle preimages.

Key Insight: Measure theory alternates between countable set algebra and approximation by elementary sets.

1.2 Lebesgue Integration

Lebesgue integration is calibrated on indicators, extended linearly to nonnegative simple functions, then extended by monotone approximation and positive–negative decomposition.

Perspective 1.2.1. *Simple Minorants Are the Right Primitive*
For Lebesgue measure on $\mathbb{R}$, bounded measurable functions supported on finite-measure sets form a useful intermediate class; arbitrary measure spaces need not admit that reduction.
Let $X \neq \emptyset$, $\mathcal{M} = \{\emptyset, X\}$, and $\mu(X) = +\infty$. If g is measurable and vanishes outside some $F \in \mathcal{M}$ with $\mu(F) < +\infty$, then $F = \emptyset$ and $g = 0$; hence finite-measure-support minorants of $\mathbf{1}_X$ have supremal integral 0, while the required normalization is $\int_X \mathbf{1}_X \, d\mu = \mu(X) = +\infty$.
Thus the defining minorants must include $\mathbf{1}_A$ $\forall A \in \mathcal{M}$, and more generally all nonnegative simple $s \leq f$.
Key Insight: The primitive calibration is $\int_X \mathbf{1}_A \, d\mu = \mu(A)$, with $\mu(A) = +\infty$ allowed.

Definition 1.2.2. *Lebesgue Integral*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$; let $\mathcal{S}^+$ be nonnegative measurable simple functions on X.
Write $s \in \mathcal{S}^+$ as $s = \sum_{i=1}^{n} c_i \mathbf{1}_{A_i}$ with $c_i \geq 0$ and pairwise disjoint $A_i \in \mathcal{M}$.
For measurable $f \geq 0$, put $\mathcal{S}^+(f) := \{s \in \mathcal{S}^+ : s \leq f\}$; for measurable $f : X \to \overline{\mathbb{R}}$, put $f^{\pm} := (\pm f) \vee 0$.

(i) *Simple*: $\mathcal{I}_E(s) := \sum_{i=1}^{n} c_i \mu(E \cap A_i)$, with $0 \cdot (+\infty) := 0$.
(ii) *Nonnegative*: $\int_E f \, d\mu := \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$; f integrable on $E \iff \int_E f \, d\mu < +\infty$.
(iii) *Signed*: if $\min\{\int_E f^+ \, d\mu, \int_E f^- \, d\mu\} < +\infty$, set $\int_E f \, d\mu := \int_E f^+ \, d\mu - \int_E f^- \, d\mu$.
(iv) *L^1*: $f \in \mathscr{L}^1(\mu)$ on E iff f is measurable and $\int_E |f| \, d\mu < +\infty$.
When $E = X$, write $\int f \, d\mu := \int_X f \, d\mu$ and drop "on E".

Simple functions are finite linear combinations of indicators of measurable sets: coefficients carry size, while the level sets carry measurability.

Lemma 1.2.3. *Refining a Simple Integral*
Let $s \geq 0$ be simple measurable and suppose $s = \sum_{k=1}^{m} b_k \mathbf{1}_{F_k}$ with $b_k \geq 0$ and $F_k \in \mathcal{M}$.
Then $\forall E \in \mathcal{M}$, $\mathcal{I}_E(s) = \sum_{k=1}^{m} b_k \mu(E \cap F_k)$.

Proof. For $\eta \in \{0,1\}^m$, set $F_k^1 := F_k$, $F_k^0 := F_k^c$, discard empty atoms $G_\eta := \bigcap_{k=1}^{m} F_k^{\eta_k}$. On G_η, $s = c_\eta := \sum_{\eta_k = 1} b_k$.
Group equal c_η values to recover the standard representation; finite additivity gives
$\mathcal{I}_E(s) = \sum_{\eta} c_\eta \mu(E \cap G_\eta) = \sum_{k=1}^{m} b_k \sum_{\eta : \eta_k = 1} \mu(E \cap G_\eta) = \sum_{k=1}^{m} b_k \mu(E \cap F_k)$.
Structural Commentary: Refine nonstandard sums to one finite partition.
Key Insight: Refine first, then regroup by equal function values. □

Proposition 1.2.4. *Simple Integral Algebra*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, let $c \geq 0$, and let $s, t \in \mathcal{S}^+$. For fixed $s \in \mathcal{S}^+$, define $\nu_s : \mathcal{M} \to [0, +\infty]$ by $\nu_s(A) := \mathcal{I}_A(s)$. Then:
(i) $\mathcal{I}_E(cs) = c\mathcal{I}_E(s)$.
(ii) $\mathcal{I}_E(s + t) = \mathcal{I}_E(s) + \mathcal{I}_E(t)$.
(iii) ν_s is a measure on $\mathcal{M}$.

Proof. Write $s = \sum_i a_i \mathbf{1}_{A_i}$ and $t = \sum_j b_j \mathbf{1}_{B_j}$ on finite measurable partitions of X, with $a_i, b_j \geq 0$.

(i) $\mathcal{I}_E(cs) = \sum_i (ca_i)\mu(E \cap A_i) = c \sum_i a_i \mu(E \cap A_i) = c\mathcal{I}_E(s)$, with $0 \cdot (+\infty) = 0$.

(ii) On the common refinement $A_i \cap B_j$, $s + t = \sum_{i,j} (a_i + b_j) \mathbf{1}_{A_i \cap B_j}$.
$\mathcal{I}_E(s + t) = \sum_{i,j} (a_i + b_j) \mu(E \cap A_i \cap B_j) = \sum_i a_i \mu(E \cap A_i) + \sum_j b_j \mu(E \cap B_j)$ by Lemma 1.2.3.

(iii) For $A \in \mathcal{M}$, $\nu_s(A) = \sum_i a_i \mu(A \cap A_i)$.
Each $A \mapsto \mu(A \cap A_i)$ is a measure, and finite nonnegative sums of measures are measures.

Structural Commentary: Simple integration is weighted measure summation on finite partitions.
Key Insight: Refine to common atoms, then use finite additivity. □

Proposition 1.2.5. *Monotonicity for Nonnegative Integrals*
Let $f, g \geq 0$ be measurable and let $E, F \in \mathcal{M}$.
(i) If $f \leq g$ on E, then $\int_E f \, d\mu \leq \int_E g \, d\mu$.
(ii) If $E \subseteq F$, then $\int_E f \, d\mu \leq \int_F f \, d\mu$.

Proof.
(i) If $s \in \mathcal{S}^+(f)$, then $s\mathbf{1}_E \in \mathcal{S}^+(g)$, $\mathcal{I}_E(s) = \mathcal{I}_E(s\mathbf{1}_E) \leq \int_E g \, d\mu$. Taking $\sup_{s \in \mathcal{S}^+(f)}$ gives $\int_E f \, d\mu \leq \int_E g \, d\mu$.
(ii) If $E \subseteq F$ and $s \in \mathcal{S}^+(f)$, then $\mathcal{I}_E(s) \leq \mathcal{I}_F(s)$. Taking $\sup_{s \in \mathcal{S}^+(f)}$ gives $\int_E f \, d\mu \leq \int_F f \, d\mu$.

Structural Commentary: The nonnegative integral is a supremum over simple functions below the integrand.
Key Insight: More room below the integrand, or over a larger set, can only increase the supremum. □

Proposition 1.2.6. *Chebyshev Inequality for Nonnegative Integrals*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, and let $f : X \to [0, +\infty]$ be measurable.
For $\lambda > 0$, set $E_\lambda := E \cap \{f \geq \lambda\}$. Then $E_\lambda \in \mathcal{M}$ and $\mu(E_\lambda) \leq \lambda^{-1} \int_E f \, d\mu$.

Proof. Measurability gives $E_\lambda \in \mathcal{M}$.
Since $\lambda \mathbf{1}_{E_\lambda} \leq f$ on E, monotonicity gives $\lambda \mu(E_\lambda) = \int_E \lambda \mathbf{1}_{E_\lambda} \, d\mu \leq \int_E f \, d\mu$; divide by $\lambda > 0$.
Structural Commentary: Level-set minorants convert mass to measure.
Key Insight: Use $\lambda \mathbf{1}_{E_\lambda} \leq f$ on E. □

Method 1.2.7. *Splitting a Simple Minorant Under a Sum*
Let $E \in \mathcal{M}$, $f, g \geq 0$ measurable, and $\omega \in \mathcal{S}^+(f + g)$ simple.
(i) *Grid*: for $n \geq 1$, use $D_n := \{0, \frac{1}{n}, \ldots, 1\}$.

(ii) *Split*: define $\varphi_n(x) := \max\{\theta\omega(x) : \theta \in D_n,\ \theta\omega(x) \le f(x)\}$; $\psi_n(x) := \big((1-\frac{1}{n})\,\omega(x) - \varphi_n(x)\big)^+$.
(iii) *Bounds*: φ_n, ψ_n are simple, $\varphi_n \le f$, $\psi_n \le g$, and $(1-\frac{1}{n})\omega \le \varphi_n + \psi_n$.
(iv) *Use*: simple additivity gives $(1-\frac{1}{n})\mathcal{I}_E(\omega) \le \int_E f\,d\mu + \int_E g\,d\mu$; then let $n \to \infty$ and take $\sup_{\omega \in \mathcal{S}^+(f+g)}$.

Key Insight: Grid-split ω between f and g, losing only $\frac{1}{n}$.

Proposition 1.2.8. *Homogeneity and Additivity for Nonnegative Integrals*
Let $f, g \ge 0$ be measurable, let $c \ge 0$, and let $E \in \mathcal{M}$. All identities are in $[0,+\infty]$, with $0 \cdot (+\infty) = 0$.
(i) Homogeneity: $\int_E cf\,d\mu = c\int_E f\,d\mu$.
(ii) Additivity: $\int_E (f+g)\,d\mu = \int_E f\,d\mu + \int_E g\,d\mu$.

Proof.
(i) If $c = 0$, both sides are 0; if $c > 0$, $s \mapsto cs$ bijects $\mathcal{S}^+(f)$ with $\mathcal{S}^+(cf)$ and satisfies $\mathcal{I}_E(cs) = c\mathcal{I}_E(s)$. Hence taking suprema gives the claim.
(ii) If $s \in \mathcal{S}^+(f)$ and $t \in \mathcal{S}^+(g)$, then $s+t \in \mathcal{S}^+(f+g)$ and $\mathcal{I}_E(s) + \mathcal{I}_E(t) = \mathcal{I}_E(s+t) \le \int_E (f+g)\,d\mu$, so taking $\sup_{s,t}$ gives $\int_E f\,d\mu + \int_E g\,d\mu \le \int_E (f+g)\,d\mu$.
Conversely, for $\omega \in \mathcal{S}^+(f+g)$, choose simple $\varphi_n \le f$ and $\psi_n \le g$ with $(1-\frac{1}{n})\omega \le \varphi_n + \psi_n$.
Then $\big(1-\frac{1}{n}\big)\,\mathcal{I}_E(\omega) \le \mathcal{I}_E(\varphi_n) + \mathcal{I}_E(\psi_n) \le \int_E f\,d\mu + \int_E g\,d\mu$. Let $n \to \infty$, then take $\sup_{\omega \in \mathcal{S}^+(f+g)}$.

Structural Commentary: The nonnegative integral is linear before signed integrability is introduced.
Key Insight: Scale minorants; split sum-minorants on a finite grid. □

Proposition 1.2.9. *Signed Decompositions Give the Same Integral*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $E \in \mathcal{M}$, and let $f, f_1, f_2 : X \to \overline{\mathbb{R}}$ be measurable. Assume $f_i \ge 0$, $\int_E f_i\,d\mu < +\infty$ for $i = 1, 2$, and $f = f_1 - f_2$ on E. Then:
(i) $f \in \mathscr{L}^1(\mu)$ on E.
(ii) $\int_E f\,d\mu = \int_E f_1\,d\mu - \int_E f_2\,d\mu$.

Proof.
(i) Since $-f_2 \le f \le f_1$ on E, $f^+ \le f_1$ and $f^- \le f_2$ on E; monotonicity gives $\int_E f^\pm\,d\mu < +\infty$.
(ii) Since $f^+ - f^- = f_1 - f_2$ on E, $f^+ + f_2 = f_1 + f^-$ on E.
Nonnegative additivity gives $\int_E f^+\,d\mu + \int_E f_2\,d\mu = \int_E f_1\,d\mu + \int_E f^-\,d\mu$; subtract the finite terms.

Structural Commentary: Finite positive decompositions determine one signed integral.
Key Insight: Move all nonnegative terms to one equality, integrate, then cancel finite terms. □

Perspective 1.2.10. *Step Functions versus Simple Functions*
Riemann: $\sup \sum_i m_i \ell(I_i)$ over interval partitions. Lebesgue: $\sup\{\mathcal{I}(s) : 0 \le s \le f,\ s = \sum_i c_i \mathbf{1}_{E_i},\ E_i \in \mathcal{M}\}$.

Structural Commentary: Riemann partitions intervals; Lebesgue partitions level sets.
Key Insight: Same finite constants, measurable pieces.

Perspective 1.2.11. *Undergraph Picture for Nonnegative Functions*
For measurable $f : \mathbb{R} \to [0,+\infty]$, set $\mathcal{U}_f := \{(x,y) \in \mathbb{R} \times [0,+\infty) : 0 \le y < f(x)\}$.
Geometrically, $\int f\,dm = m_2(\mathcal{U}_f)$; analytically, the integral is built as $\sup \mathcal{I}(s)$ over simple $0 \le s \le f$.

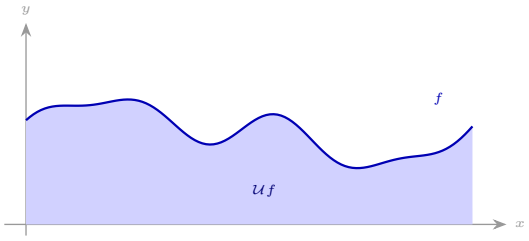

Figure 199: The integral of $f \ge 0$ is modeled by the measure of its undergraph.

Proposition 1.2.12. *Basic Integral Properties*
Let $E \in \mathcal{M}$.
(i) If f is measurable and bounded on E, and $\mu(E) < +\infty$, then $f \in \mathscr{L}^1(\mu)$ on E.
(ii) If $a \le f(x) \le b$ on E and $\mu(E) < +\infty$, then $a\mu(E) \le \int_E f\,d\mu \le b\mu(E)$.
(iii) If $f, g \in \mathscr{L}^1(\mu)$ on E and $f \le g$ on E, then $\int_E f\,d\mu \le \int_E g\,d\mu$.
(iv) If $f \in \mathscr{L}^1(\mu)$ on E, then $cf \in \mathscr{L}^1(\mu)$ on E for finite c, and $\int_E cf\,d\mu = c\int_E f\,d\mu$.
(v) If $\mu(E) = 0$ and f is measurable, then $\int_E f\,d\mu = 0$.
(vi) If $f \in \mathscr{L}^1(\mu)$ on E, $A \in \mathcal{M}$, and $A \subseteq E$, then $f \in \mathscr{L}^1(\mu)$ on A.

Proof.
(i) If $|f| \le M$ on E, then $0 \le f^\pm \le M\mathbf{1}_E$ on E, so $\int_E f^\pm\,d\mu \le M\mu(E) < +\infty$.
(ii) Since $a\mathbf{1}_E \le f \le b\mathbf{1}_E$ on E, order for the positive integral gives $a\mu(E) \le \int_E f\,d\mu \le b\mu(E)$.
(iii) Since $g - f \ge 0$ on E, $\int_E (g-f)\,d\mu \ge 0$, hence $\int_E f\,d\mu \le \int_E g\,d\mu$.
(iv) For $c \ge 0$, scale the simple approximants. For $c < 0$, use $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$.
(v) If $\mu(E) = 0$, then $\forall s \ge 0$ simple, $\mathcal{I}_E(s) = 0$, so $\int_E f^\pm\,d\mu = 0$.

(vi) Since $A \subseteq E$, $\mathcal{I}_A(s) \le \mathcal{I}_E(s)$ for $s \ge 0$ simple; hence $\int_A f^{\pm}\,d\mu \le \int_E f^{\pm}\,d\mu < +\infty$.

Structural Commentary: Estimate $f^{\pm}$ by simpler functions.
Key Insight: Monotonicity makes bounds, null sets, and restrictions immediate. □

Theorem 1.2.13. *The Integral as a Measure or Charge*
For $A \in \mathcal{M}$, define $\phi_f(A) := \int_A f\,d\mu$ when $f \ge 0$ is measurable, and define $\lambda_f(A) := \int_A f\,d\mu$ when $f \in \mathscr{L}^1(\mu)$.
(i) If $f \ge 0$ is measurable, then $\phi_f : \mathcal{M} \to [0,+\infty]$ is a measure.
(ii) If $f \in \mathscr{L}^1(\mu)$, then $\lambda_f : \mathcal{M} \to \mathbb{R}$ is finite-valued and countably additive.

Proof.
(i) For simple $s \ge 0$, $A \mapsto \mathcal{I}_A(s)$ is a measure, so $\phi_f(\emptyset) = 0$.
If $A = \bigsqcup_n A_n$, $s \in \mathcal{S}^+(f)$, then $\mathcal{I}_A(s) = \sum_n \mathcal{I}_{A_n}(s) \le \sum_n \phi_f(A_n)$; taking $\sup_s$ gives $\phi_f(A) \le \sum_n \phi_f(A_n)$. Conversely, $\forall N$, finite additivity gives $\sum_{n=1}^{N} \phi_f(A_n) = \phi_f(\bigcup_{n=1}^{N} A_n) \le \phi_f(A)$.
Let $N \to \infty$. Hence $\phi_f(A) = \sum_n \phi_f(A_n)$.
(ii) Since $f \in \mathscr{L}^1(\mu)$, $\phi_{f^+}(X), \phi_{f^-}(X) < +\infty$ and $\lambda_f = \phi_{f^+} - \phi_{f^-}$ is finite-valued.
If $A = \bigsqcup_n A_n$, then item (i) gives $\lambda_f(A) = \sum_n \phi_{f^+}(A_n) - \sum_n \phi_{f^-}(A_n) = \sum_n \lambda_f(A_n)$.

Structural Commentary: Setwise integration turns densities into measures.
Key Insight: Sup simple measures; subtract finite signs. □

Corollary 1.2.14. *Domain Additivity and Continuity for Integrable Functions*
Let $(X,\mathcal{M},\mu)$ be a measure space, let $E \in \mathcal{M}$, let $f \in \mathscr{L}^1(\mu)$ on E, and set $\mathcal{M}_E := \{A \in \mathcal{M} : A \subseteq E\}$.
(i) If $(A_n) \subseteq \mathcal{M}_E$ are pairwise disjoint, then $\int_{\bigsqcup_n A_n} f\,d\mu = \sum_n \int_{A_n} f\,d\mu$ and $\sum_n \left|\int_{A_n} f\,d\mu\right| < +\infty$.
(ii) If $A_n, A \in \mathcal{M}_E$ and $A_n \uparrow A$, then $\int_A f\,d\mu = \lim_{n\to\infty} \int_{A_n} f\,d\mu$.
(iii) If $A_n, A \in \mathcal{M}_E$ and $A_n \downarrow A$, then $\int_A f\,d\mu = \lim_{n\to\infty} \int_{A_n} f\,d\mu$.

Proof. Let $\mu_E := \mu|_{\mathcal{M}_E}$ and define $\lambda(A) := \int_A f\,d\mu$, $\nu(A) := \int_A |f|\,d\mu$ on $\mathcal{M}_E$.
By Theorem 1.2.13, λ is a finite signed measure, $\nu(E) < +\infty$, and $|\lambda(A)| \le \nu(A)$.
(i) Additivity gives $\lambda(\bigsqcup_n A_n) = \sum_n \lambda(A_n)$, and $\sum_n |\lambda(A_n)| \le \sum_n \nu(A_n) = \nu(\bigsqcup_n A_n) \le \nu(E) < +\infty$.
(ii) Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \ge 2$.
Since $A = \bigsqcup_n B_n$ and $A_N = \bigsqcup_{n\le N} B_n$, (i) gives $\lambda(A_N) = \sum_{n\le N} \lambda(B_n) \to \sum_n \lambda(B_n) = \lambda(A)$.
(iii) Set $C_n := A_1 \setminus A_n$ and $C := A_1 \setminus A$. Then $C_n \uparrow C$, so (ii) gives $\lambda(C_n) \to \lambda(C)$.
Hence $\lambda(A_n) = \lambda(A_1) - \lambda(C_n) \to \lambda(A_1) - \lambda(C) = \lambda(A)$.

Structural Commentary: Domain integrals form a finite signed measure.
Key Insight: Monotone domain limits are signed-measure continuity. □

Proposition 1.2.15. *Finite-Measure Support*
Let $(X,\mathcal{M},\mu)$ be a measure space, let $E \in \mathcal{M}$, and let $h : E \to \mathbb{R}$ be bounded and measurable. Suppose $h = 0$ on $E \setminus E_0$ for some $E_0 \in \mathcal{M}$ with $E_0 \subseteq E$ and $\mu(E_0) < +\infty$. Then:
(i) $h \in \mathscr{L}^1(\mu)$ on E.
(ii) $\int_E h\,d\mu = \int_{E_0} h\,d\mu$.
(iii) If $E_1 \in \mathcal{M}$, $E_1 \subseteq E$, $\mu(E_1) < +\infty$, and $h = 0$ on $E \setminus E_1$, then $\int_{E_0} h\,d\mu = \int_{E_1} h\,d\mu$.

Proof.
(i) Choose $M < +\infty$ with $|h| \le M$ on E.
Since $h = 0$ on $E \setminus E_0$, $0 \le h^{\pm} \le M\mathbf{1}_{E_0}$ on E, so $\int_E h^{\pm}\,d\mu \le M\mu(E_0) < +\infty$.
(ii) Since $\int_{E\setminus E_0} h\,d\mu = 0$, $\int_E h\,d\mu = \int_{E_0} h\,d\mu$ by Corollary 1.2.14.
(iii) Apply item (ii) with E_0 and with E_1: $\int_{E_0} h\,d\mu = \int_E h\,d\mu = \int_{E_1} h\,d\mu$.

Structural Commentary: Finite support turns boundedness into integrability.
Key Insight: The integral only sees where h can be nonzero. □

Corollary 1.2.16. *Compact Continuous Functions on Finite Measure Spaces*
Let X be a compact topological space, let $\mathcal{M}$ be a σ-algebra containing the topology of X, and let $\mu(X) < +\infty$. If $f : X \to \mathbb{R}$ is continuous, then:
(i) f is measurable.
(ii) $f \in \mathscr{L}^1(\mu)$.

Proof.
(i) $\forall O \subseteq \mathbb{R}$ open, $f^{-1}(O)$ is open in X by continuity, hence $f^{-1}(O) \in \mathcal{M}$.
(ii) The continuous image $f(X)$ is compact in $\mathbb{R}$, hence bounded.
Since $\mu(X) < +\infty$, $f \in \mathscr{L}^1(\mu)$ by Proposition 1.2.12(i).

Structural Commentary: Compactness supplies boundedness, while the measure hypothesis supplies finite mass.
Key Insight: Continuous on compact plus finite measure implies bounded measurable, hence integrable. □

Corollary 1.2.17. *Null and Finiteness Consequences for $f \ge 0$*
Let $f : X \to [0,+\infty]$ be measurable and $\phi_f(A) := \int_A f\,d\mu$.
(i) $\mu(A) = 0 \Longrightarrow \phi_f(A) = 0$ for $A \in \mathcal{M}$.
(ii) $\int_X f\,d\mu = 0 \Longleftrightarrow f = 0$ a.e.

(iii) $\int_X f\,d\mu < +\infty \Longrightarrow \mu\{f = +\infty\} = 0$ *and* $\{f > 0\}$ *is* σ*-finite.*

Proof.

(i) If $\mu(A) = 0$ and $0 \le s \le f$ is simple, then $\mathcal{I}_A(s) = 0$; hence $\phi_f(A) = \sup_s \mathcal{I}_A(s) = 0$.

(ii) ($\Rightarrow$) Set $E_n := \{f > \frac{1}{n}\}$. By Proposition 1.2.6, $\mu(E_n) \le n \int_X f\,d\mu = 0$.
Thus $\forall n,\ \mu(E_n) = 0$, and $\{f > 0\} = \bigcup_n E_n$ is null.

(ii) ($\Leftarrow$) If $N := \{f > 0\}$ null, every simple $0 \le s \le f$ vanishes on $X \setminus N$, so $\mathcal{I}_X(s) = \mathcal{I}_N(s) = 0$; take $\sup_s$.

(iii) Let $F_n := \{f \ge n\}$. By Proposition 1.2.6, $\mu(F_n) \le n^{-1} \int_X f\,d\mu$, so $\mu\{f = +\infty\} \le \mu(F_n) \to 0$.
Also $\{f > 0\} = \bigcup_n \{f > \frac{1}{n}\}$ and $\mu\{f > \frac{1}{n}\} \le n \int_X f\,d\mu < +\infty$.

Structural Commentary: Finite nonnegative integrals control threshold sets.
Key Insight: Threshold sets convert integral bounds into measure bounds. □

Theorem 1.2.18. *Absolute Integrability and Domination*
Let $(X, \mathcal{M}, \mu)$ *be a measure space and let* $E \in \mathcal{M}$. *Then:*

(i) If $f \in \mathscr{L}^1(\mu)$ *on* E, *then* $|f| \in \mathscr{L}^1(\mu)$ *on* E *and* $\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu$.

(ii) If $f : X \to \overline{\mathbb{R}}$ *is measurable,* $g \in \mathscr{L}^1(\mu)$ *on* E, *and* $|f| \le g$ *on* E, *then* $f \in \mathscr{L}^1(\mu)$ *on* E.

Proof.

(i) Set $a := \int_E f^+\,d\mu$ and $b := \int_E f^-\,d\mu$. Since $f \in \mathscr{L}^1(\mu)$ on E, $a, b < +\infty$.
Hence $\int_E |f|\,d\mu = \int_E (f^+ + f^-)\,d\mu = a + b < +\infty$, and $\left|\int_E f\,d\mu\right| = |a - b| \le a + b = \int_E |f|\,d\mu$.

(ii) Since $|f| \le g$ on E, $g \ge 0$ on E and $\int_E g\,d\mu < +\infty$.
Also $0 \le f^\pm \le |f| \le g$ on E, so monotonicity gives $\int_E f^\pm\,d\mu < +\infty$.

Structural Commentary: Both claims reduce to finite positive and negative parts.
Key Insight: Use $|f| = f^+ + f^-$ and $0 \le f^\pm \le g$. □

Proposition 1.2.19. *Linearity of the Lebesgue Integral*
Let $(X, \mathcal{M}, \mu)$ *be a measure space, let* $E \in \mathcal{M}$, *let* $f, g \in \mathscr{L}^1(\mu)$ *on* E, *and let* $a, b \in \mathbb{R}$.
Then:

(i) For $c \in \mathbb{R}$ *and* $u \in \mathscr{L}^1(\mu)$ *on* E, $cu \in \mathscr{L}^1(\mu)$ *on* E *and* $\int_E cu\,d\mu = c\int_E u\,d\mu$.

(ii) $f + g \in \mathscr{L}^1(\mu)$ *on* E *and* $\int_E (f+g)\,d\mu = \int_E f\,d\mu + \int_E g\,d\mu$.

(iii) $af + bg \in \mathscr{L}^1(\mu)$ *on* E *and* $\int_E (af+bg)\,d\mu = a\int_E f\,d\mu + b\int_E g\,d\mu$.

Proof.

(i) Since $(cu)^\pm = c^+u^\pm + c^-u^\mp$, both $(cu)^+$ and $(cu)^-$ have finite integral. Nonnegative linearity gives $\int_E cu\,d\mu = (c^+a_u + c^-b_u) - (c^+b_u + c^-a_u) = c(a_u - b_u)$, where $a_u := \int_E u^+\,d\mu$ and $b_u := \int_E u^-\,d\mu$.

(ii) Since $|f+g| \le |f| + |g|$ and $|f| + |g| \in \mathscr{L}^1(\mu)$ on E, $f + g \in \mathscr{L}^1(\mu)$ on E by Theorem 1.2.18(ii).
Integrate $(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+$ over E and cancel finite terms.

(iii) By (i), $af, bg \in \mathscr{L}^1(\mu)$ on E, $\int_E af\,d\mu = a\int_E f\,d\mu$, and $\int_E bg\,d\mu = b\int_E g\,d\mu$.
Applying (ii) to af, bg gives $af + bg \in \mathscr{L}^1(\mu)$ on E and $\int_E (af+bg)\,d\mu = a\int_E f\,d\mu + b\int_E g\,d\mu$.

Structural Commentary: Signed linearity follows from finite sign pieces.
Key Insight: Integrate one nonnegative equality, then cancel finite terms. □

Proposition 1.2.20. *L^1 Level Sets, Null Changes, and Indefinite Integrals*
Let $f, g \in \mathscr{L}^1(\mu)$ *on* X, *and write* $\lambda_u(A) := \int_A u\,d\mu$.

(i) For $a > 0$, $\mu\{|f| \ge a\} \le a^{-1}\int_X |f|\,d\mu < +\infty$; *hence* $\{f \ne 0\}$ *is* σ*-finite.*

(ii) If measurable $h = 0$ *a.e., then* $h \in \mathscr{L}^1(\mu)$ *and* $\lambda_h \equiv 0$.

(iii) $\lambda_f \ge 0$ *on* $\mathcal{M} \Longrightarrow f \ge 0$ *a.e.;* $\lambda_f \equiv 0 \Longrightarrow f = 0$ *a.e.*

(iv) $\lambda_f = \lambda_g$ *on* $\mathcal{M} \Longleftrightarrow f = g$ *a.e.*

Proof.

(i) Since $a\mathbf{1}_{\{|f| \ge a\}} \le |f|$, integration gives $a\mu\{|f| \ge a\} \le \int_X |f|\,d\mu$. Also $\{f \ne 0\} = \bigcup_{k \ge 1} \{|f| \ge \frac{1}{k}\}$.

(ii) Let $N := \{h \ne 0\}$. For $A \in \mathcal{M}$, $h^\pm \mathbf{1}_A$ vanish off $A \cap N$, a null set.
Hence $\int_A h^+\,d\mu = \int_A h^-\,d\mu = 0$, so $\lambda_h(A) = 0$.

(iii) For $k \ge 1$, set $N_k := \{f \le -\frac{1}{k}\}$. By (i), $\mu(N_k) < +\infty$, and $\lambda_f(N_k) \le -k^{-1}\mu(N_k)$.
If $\lambda_f \ge 0$, then $\mu(N_k) = 0$, so $\{f < 0\} = \bigcup_k N_k$ is null. If $\lambda_f \equiv 0$, apply this to f and $-f$.

(iv) If $\lambda_f = \lambda_g$, then $\lambda_{f-g} \equiv 0$, so $f - g = 0$ a.e. by (iii). Conversely, if $f = g$ a.e., apply (ii) to $h = f - g$.

Structural Commentary: An L^1 function is detected by its integrals over all measurable sets, up to null sets.
Key Insight: Test sign on the level sets where the function is bounded away from 0. □

Proposition 1.2.21. *Finite-Measure Indicators by Intervals and Continuous Functions*
Let $A \subseteq \mathbb{R}$ *be Lebesgue measurable with* $m(A) < +\infty$. *Then* $\forall \varepsilon > 0$*:*

(i) $\exists G = \bigcup_{j=1}^{N} I_j$, I_j *open intervals, such that* $m(A \triangle G) < \varepsilon$.

(ii) $\exists f \in C(\mathbb{R})$ *such that* $0 \le f \le 1$ *and* $\int_{\mathbb{R}} |\mathbf{1}_A - f|\,dm < \varepsilon$.

Proof.

(i) Choose open $O \supseteq A$ with $m(O \setminus A) < \frac{\varepsilon}{2}$ and $m(O) < +\infty$.
Write $O = \bigsqcup_{j \geq 1} I_j$ with open intervals I_j, choose N with $m(O \setminus G) < \frac{\varepsilon}{2}$ for $G := \bigcup_{j=1}^{N} I_j$.
Then $A \setminus G \subseteq O \setminus G$ and $G \setminus A \subseteq O \setminus A$, so $m(A \triangle G) < \varepsilon$.

(ii) Apply (i) with $\frac{\varepsilon}{2}$ to get finite open G. Choose compact $K \subseteq G$ with $m(G \setminus K) < \frac{\varepsilon}{2}$, and choose piecewise-linear f with $0 \leq f \leq 1$, $f = 1$ on K, $f = 0$ on $\mathbb{R} \setminus G$.
Then $\int_{\mathbb{R}} |\mathbf{1}_G - f| \, dm \leq m(G \setminus K) < \frac{\varepsilon}{2}$, hence $\int_{\mathbb{R}} |\mathbf{1}_A - f| \, dm \leq m(A \triangle G) + \int_{\mathbb{R}} |\mathbf{1}_G - f| \, dm < \varepsilon$.

Structural Commentary: Finite-measure sets can be made geometrically simple up to small L^1 error.
Key Insight: Approximate by intervals, then smooth the indicator. □

Example 1.2.22. *Counting and Unit-Mass Integrals*

(i) *Counting*: for counting measure $\#$ on $(\mathbb{N}, 2^{\mathbb{N}})$ and $f : \mathbb{N} \to \mathbb{R}$,
$f \in \mathscr{L}^1(\#) \iff \sum_{n=1}^{\infty} |f(n)| < +\infty$, and then $\int_{\mathbb{N}} f \, d\# = \sum_{n=1}^{\infty} f(n)$.

(ii) *Unit mass*: for $(X, \mathcal{M})$, $p \in X$, and $\delta_p(E) := \mathbf{1}_E(p)$, $\int_X f \, d\delta_p = f(p)$ for measurable $f : X \to [0, +\infty]$.
If $f : X \to \mathbb{R}$ is measurable, then $f \in \mathscr{L}^1(\delta_p)$ and the same formula holds.

For δ_p, verify first on indicators, then extend to simple functions by finite linearity, to nonnegative functions by monotone simple approximation, and to real-valued functions by positive–negative splitting.

Example 1.2.23. *L^1 Is Not Closed Under Squaring*
On $X = \mathbb{N}$ with $\mathcal{M} = 2^{\mathbb{N}}$, define $\mu(E) := \sum_{n \in E} 4^{-n}$ and $f(n) := \frac{2^n}{\sqrt{n}}$.

Then $\int_X |f| \, d\mu = \sum_{n=1}^{\infty} \frac{1}{2^n \sqrt{n}} < +\infty$; $\int_X f^2 \, d\mu = \sum_{n=1}^{\infty} \frac{1}{n} = +\infty$.
The first series is dominated by the geometric series $\sum 2^{-n}$, while the second diverges by Corollary 6.1.14.
Thus $f \in \mathscr{L}^1(\mu)$ need not imply $f^2 \in \mathscr{L}^1(\mu)$.

1.3 Convergence Theorems

This subsection collects the main limit-interchange tools: monotone convergence, Fatou, and dominated convergence.

Theorem 1.3.1. *Monotone Convergence*
Let $E \in \mathcal{M}$, let f_n, f be measurable on E, and assume $0 \leq f_n \uparrow f$. Then $\int_E f_n \, d\mu \uparrow \int_E f \, d\mu$.

Proof. By monotonicity, $\int_E f_n \, d\mu \uparrow L \leq \int_E f \, d\mu$.
For $s \in \mathcal{S}^+(f)$ and $0 < c < 1$, set $E_n := \{x \in E : f_n(x) \geq c s(x)\}$.
Since $f_n \uparrow f$ and $s \leq f$, $E_n \uparrow E$, and $\int_E f_n \, d\mu \geq \int_{E_n} f_n \, d\mu \geq c \int_{E_n} s \, d\mu$.
Continuity from below for $\phi_s(A) := \int_A s \, d\mu$ gives $L \geq c \int_E s \, d\mu$. Let $c \uparrow 1$, then take $\sup_{s \in \mathcal{S}^+(f)}$.

Structural Commentary: Nonnegative integration commutes with monotone limits.
Key Insight: Simple minorants of the limit are eventually captured below the sequence. □

Theorem 1.3.2. *Series Form of Monotone Convergence*
Let $E \in \mathcal{M}$, let $f_n \geq 0$ be measurable, and set $f := \sum_{n=1}^{\infty} f_n$ on E. Then $\int_E f \, d\mu = \sum_{n=1}^{\infty} \int_E f_n \, d\mu$.

Proof. Set $S_N := \sum_{n=1}^{N} f_n$. Then $0 \leq S_N \uparrow f$.

By Theorem 1.3.1, $\int_E f \, d\mu = \lim_{N \to \infty} \int_E S_N \, d\mu = \lim_{N \to \infty} \sum_{n=1}^{N} \int_E f_n \, d\mu = \sum_{n=1}^{\infty} \int_E f_n \, d\mu$.

Structural Commentary: Nonnegative series are monotone partial sums.
Key Insight: Integrate partial sums, then pass upward. □

Corollary 1.3.3. *Beppo Levi Bounded-Integral Criterion*
Let $E \in \mathcal{M}$, let (f_n) be measurable on E, and assume $0 \leq f_n \uparrow f$ with $\sup_n \int_E f_n \, d\mu < +\infty$.
Then $f < +\infty$ a.e. on E and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu < +\infty$.

Proof. The monotone limit f is measurable, and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu \leq \sup_n \int_E f_n \, d\mu < +\infty$ by Theorem 1.3.1. Thus $f < +\infty$ a.e. on E by Corollary 1.2.17.
Structural Commentary: Bounded mass survives monotone limits.
Key Insight: Finite nonnegative integral forces a.e. finiteness. □

Proposition 1.3.4. *L^1 Approximation and Bounded Multipliers*
Let $f \in \mathscr{L}^1(\mu)$ on X.
(i) $\forall \varepsilon > 0$ $\exists$ a measurable simple function φ with $\int_X |f - \varphi| \, d\mu < \varepsilon$.
(ii) If g is bounded measurable, then $fg \in \mathscr{L}^1(\mu)$.

Proof.

(i) Choose simple $0 \leq s_n \uparrow f^+$ and $0 \leq t_n \uparrow f^-$. Then $\int_X (f^+ - s_n) \, d\mu \to 0$ and $\int_X (f^- - t_n) \, d\mu \to 0$.
For large n, $\varphi := s_n - t_n$ satisfies $\int_X |f - \varphi| \, d\mu \leq \int_X (f^+ - s_n) \, d\mu + \int_X (f^- - t_n) \, d\mu < \varepsilon$.

(ii) If $|g| \leq M$, then fg is measurable and $\int_X |fg| \, d\mu \leq M \int_X |f| \, d\mu < +\infty$.

Structural Commentary: L^1 is simple-closed and bounded-multiplier stable.
Key Insight: Approximate signs separately; dominate products by $M|f|$. □

Theorem 1.3.5. *Fatou's Lemma*
Let $E \in \mathcal{M}$ and let $f_n : E \to [0, +\infty]$ be measurable. Then $\int_E \liminf_n f_n \, d\mu \le \liminf_n \int_E f_n \, d\mu$.

Proof. Set $g_n := \inf_{i \ge n} f_i$. Then $0 \le g_n \uparrow \liminf_i f_i$ and $\forall i \ge n$, $g_n \le f_i$, so $\int_E g_n \, d\mu \le \inf_{i \ge n} \int_E f_i \, d\mu$.
By Theorem 1.3.1, $\int_E \liminf_i f_i \, d\mu = \lim_n \int_E g_n \, d\mu \le \lim_n \inf_{i \ge n} \int_E f_i \, d\mu = \liminf_n \int_E f_n \, d\mu$.
Structural Commentary: Replace lim inf by increasing tail infima.
Key Insight: Tail infima are monotone minorants. □

Example 1.3.6. *Strictness in Fatou's Lemma*
Fatou's inequality can be strict by concentration or escape of mass.

(i) *Concentration*: On $E = (0, 1)$ with Lebesgue measure m, set $f_n := n\mathbf{1}_{(0, \frac{1}{n})}$.
Then $f_n \to 0$ pointwise and $\forall n$, $\int_E f_n \, dm = 1$.
(ii) *Escape*: On $E = \mathbb{R}$ with Lebesgue measure m, set $g_n := \mathbf{1}_{(n, n+1)}$.
Then $g_n \to 0$ pointwise and $\forall n$, $\int_E g_n \, dm = 1$.

In either case, with $h_n = f_n$ or $h_n = g_n$, $\int_E \liminf_n h_n \, dm = 0 < 1 = \liminf_n \int_E h_n \, dm$.

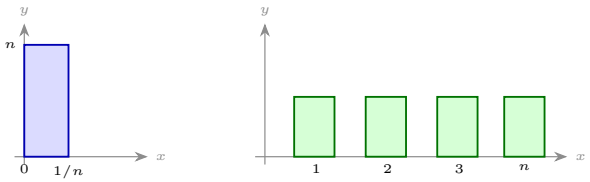

Figure 200: Strict Fatou behavior: $f_n = n\mathbf{1}_{(0, \frac{1}{n})}$ and $g_n = \mathbf{1}_{(n, n+1)}$ have integral 1 but vanish pointwise.

Theorem 1.3.7. *Dominated Convergence*
Let $E \in \mathcal{M}$, let f_n, f be measurable on E, and let $g \in \mathscr{L}^1(\mu)$ on E.
Suppose $f_n \to f$ a.e. on E and $\forall n$, $|f_n| \le g$ on E. Then:
(i) $f \in \mathscr{L}^1(\mu)$ on E.
(ii) $\lim_{n \to \infty} \int_E f_n \, d\mu = \int_E f \, d\mu$.

Proof. Choose null $N \subseteq E$ such that $f_n \to f$ on $E \setminus N$.
Changing values on N does not change integrals, so assume $f_n \to f$ everywhere on E.

(i) The pointwise limit f is measurable on E, and $|f| \le g$ on E; hence $f \in \mathscr{L}^1(\mu)$ on E by Theorem 1.2.18(ii).
(ii) Put $a_n := \int_E f_n \, d\mu$. Fatou applied to $g + f_n \ge 0$ gives $\int_E (g + f) \, d\mu \le \int_E g \, d\mu + \liminf_n a_n$.
Fatou applied to $g - f_n \ge 0$ gives $\int_E (g - f) \, d\mu \le \int_E g \, d\mu - \limsup_n a_n$.
Subtracting $\int_E g \, d\mu < +\infty$ gives $\int_E f \, d\mu \le \liminf_n a_n \le \limsup_n a_n \le \int_E f \, d\mu$.

Structural Commentary: Domination reduces signed convergence to two Fatou bounds.
Key Insight: Apply Fatou to $g + f_n$ and $g - f_n$. □

Theorem 1.3.8. *General Dominated Convergence*
Let $E \in \mathcal{M}$, let f_n, f, g be measurable on E, and let $g_n \ge 0$ be integrable on E.
Suppose $f_n \to f$ and $g_n \to g$ a.e. on E, with $\forall n$, $|f_n| \le g_n$ a.e. on E.
Assume $\lim_n \int_E g_n \, d\mu = \int_E g \, d\mu < +\infty$. Then:
(i) $\forall n$, $f_n \in \mathscr{L}^1(\mu)$ on E, and $f \in \mathscr{L}^1(\mu)$ on E.
(ii) $\lim_{n \to \infty} \int_E f_n \, d\mu = \int_E f \, d\mu$.

Proof. Choose null $N \subseteq E$ so that $f_n \to f$, $g_n \to g$, and $\forall n$, $|f_n| \le g_n$ on $E \setminus N$.
Changing values on N does not change integrals, so assume these relations hold on E.

(i) Since $|f_n| \le g_n \in \mathscr{L}^1(\mu)$, $f_n \in \mathscr{L}^1(\mu)$ on E by Theorem 1.2.18(ii).
Passing to the limit gives $|f| \le g$, and $\int_E g \, d\mu < +\infty$, hence $f \in \mathscr{L}^1(\mu)$ on E.
(ii) Put $a_n := \int_E f_n \, d\mu$, $b_n := \int_E g_n \, d\mu$, and $b := \int_E g \, d\mu$; then $b_n \to b$.
Fatou applied to $g_n + f_n \ge 0$ gives $b + \int_E f \, d\mu \le \liminf_n (b_n + a_n) = b + \liminf_n a_n$.
Fatou applied to $g_n - f_n \ge 0$ gives $b - \int_E f \, d\mu \le \liminf_n (b_n - a_n) = b - \limsup_n a_n$.
Subtracting $b < +\infty$ gives $\limsup_n a_n \le \int_E f \, d\mu \le \liminf_n a_n$, hence $a_n \to \int_E f \, d\mu$.

Structural Commentary: Varying domination works when masses converge.
Key Insight: Fatou on $g_n \pm f_n$ traps the limit. □

Corollary 1.3.9. *Monotone Convergence Almost Everywhere*
Let $E \in \mathcal{M}$ and $0 \le f_n \uparrow f$ a.e. on E. Then $\int_E f_n \, d\mu \uparrow \int_E f \, d\mu$.

Proof. Choose $N \subseteq E$ with $\mu(N) = 0$ and $f_n \uparrow f$ on $E \setminus N$. Set $g_n := f_n \mathbf{1}_{E \setminus N}$ and $g := f\mathbf{1}_{E \setminus N}$.
Then $0 \le g_n \uparrow g$, so $\int_E g_n \, d\mu \uparrow \int_E g \, d\mu$. Since $f_n = g_n$ and $f = g$ off N, the integrals over N vanish.
Structural Commentary: Monotone convergence ignores null exceptional sets.
Key Insight: Remove the exceptional set, prove the theorem there, then add back zero. □

Example 1.3.10. *Counting Measure Integrals Are Series*

Let μ be counting measure on $(\mathbb{N}, 2^{\mathbb{N}})$. For $f : \mathbb{N} \to [0, +\infty]$, set $s_N := \sum_{n=1}^{N} f(n)\mathbf{1}_{\{n\}}$; then $0 \le s_N \uparrow f$, with the case $f(k) = +\infty$ giving both sides $+\infty$. Hence $\int_{\mathbb{N}} f\, d\mu = \lim_N \int_{\mathbb{N}} s_N\, d\mu = \lim_N \sum_{n=1}^{N} f(n) = \sum_{n=1}^{\infty} f(n)$.

Structural Commentary: Counting measure turns integration into summation.
Key Insight: Finite-support truncations are the simple-function partial sums.

Example 1.3.11. *Boundary Tests for Monotone Convergence and Fatou*

On $([0, +\infty), \mathcal{B}, m)$:

(i) $f_n := \mathbf{1}_{[0,n]} \uparrow \mathbf{1}_{[0,+\infty)}$ and $\int f_n\, dm = n \uparrow +\infty$; monotone convergence allows $+\infty$.
(ii) $g_n := n^{-1}\mathbf{1}_{[n,+\infty)} \downarrow 0$ uniformly, but $\int g_n\, dm = +\infty\ \forall n$; decreasing convergence needs finite mass.
(iii) $w_n := -n^{-1}\mathbf{1}_{[0,n]} \rightrightarrows 0$, but $\int w_n\, dm = -1$; Fatou can fail without $f_n \ge 0$ or $f_n \ge -h \in L^1$.

Theorem 1.3.12. *Fatou with an Integrable Lower Bound*

Let $h \ge 0$, $h \in \mathscr{L}^1(\mu)$, and let f_n be measurable with $f_n \ge -h$.
Then, whenever the signed integrals are defined in $\overline{\mathbb{R}}$, $\int \liminf_{n\to\infty} f_n\, d\mu \le \liminf_{n\to\infty} \int f_n\, d\mu$.

Proof. Set $F_n := f_n + h \ge 0$. Fatou gives $\int \liminf_n F_n\, d\mu \le \liminf_n \int F_n\, d\mu$.
Since $\liminf_n F_n = h + \liminf_n f_n$ and $\int F_n\, d\mu = \int f_n\, d\mu + \int h\, d\mu$, subtract the finite term $\int h\, d\mu$.
Structural Commentary: Fatou survives controlled negative parts.
Key Insight: Add an integrable floor to make the sequence nonnegative. □

Remark 1.3.13. *Why the Negative Example Has No Floor*

For $w_n := -n^{-1}\mathbf{1}_{[0,n]}$, any $h \in L^1$ with $-h \le w_n\ \forall n$ must satisfy $h \ge n^{-1}$ on $[0, n]$.
Thus $h \ge k^{-1}$ on $[k-1, k]$, and $\int_0^N h\, dm \ge H_N := \sum_{k=1}^{N} \frac{1}{k} \to +\infty$, contradicting $h \in L^1$.

Perspective 1.3.14. *Domination Is the Safeguard*

$f_n \to f$ a.e. need not imply $\int f_n \to \int f$; domination $|f_n| \le g \in L^1$ restores the limit interchange.

Example 1.3.15. *A Nonuniform Limit with Vanishing Integrals*

For $x > 0$, set $f_n(x) := e^{-nx}x^{-\frac{1}{2}}$. Then $f_n \to 0$ pointwise, but $\forall \delta > 0$, $\sup_{0<x<\delta} f_n(x) = +\infty$.
For $n \ge 1$, $0 \le f_n \le g := e^{-x}x^{-\frac{1}{2}}$; $g \le x^{-\frac{1}{2}}\mathbf{1}_{(0,1]} + e^{-x}\mathbf{1}_{[1,\infty)} \in \mathscr{L}^1(0, \infty)$.
Thus dominated convergence gives $I_n := \int_0^\infty e^{-nx}x^{-\frac{1}{2}}\, dx \to 0$.
Structural Commentary: Uniform convergence is unnecessary once one integrable envelope controls the sequence.
Key Insight: Domination handles the singular behavior at 0 and the tail at infinity simultaneously.

Proposition 1.3.16. *Uniform Convergence on Finite Measure Spaces*

Assume $\mu(X) < +\infty$.

(i) If $0 \le f_n$ are measurable and $f_n \rightrightarrows f$, then $f \ge 0$ is measurable and $\int_X f\, d\mu = \lim_n \int_X f_n\, d\mu$ in $[0, +\infty]$.
(ii) If $f_n \in \mathscr{L}^1(\mu)$ and $f_n \rightrightarrows f$, then $f \in \mathscr{L}^1(\mu)$ and $\int_X f_n\, d\mu \to \int_X f\, d\mu$ in $\mathbb{R}$.

Proof.

(i) Measurability and $f \ge 0$ pass to pointwise limits. For $\varepsilon > 0$ and $n \gg 1$, $f \le f_n + \varepsilon\mathbf{1}_X$ and $f_n \le f + \varepsilon\mathbf{1}_X$.
Thus $\int_X f\, d\mu \le \int_X f_n\, d\mu + \varepsilon\mu(X)$ and $\int_X f_n\, d\mu \le \int_X f\, d\mu + \varepsilon\mu(X)$. Let $\varepsilon \downarrow 0$.
(ii) Choose N with $|f - f_N| \le 1$. Then $f = f_N + (f - f_N) \in \mathscr{L}^1(\mu)$, and
$\left|\int_X f_n\, d\mu - \int_X f\, d\mu\right| \le \int_X |f_n - f|\, d\mu \le \mu(X) \sup_X |f_n - f| \to 0$.

Structural Commentary: On finite measure spaces, uniform error controls integral error.
Key Insight: Finiteness of $\mu(X)$ turns a supremum bound into an integral bound. □

Theorem 1.3.17. *Bounded Convergence on Finite Measure Spaces*

Let $\mu(X) < +\infty$, and let $f_n, f : X \to \mathbb{R}$ be measurable.
Suppose $f_n \to f$ a.e. and $\exists M < +\infty$ such that $\forall n$, $|f_n| \le M$ a.e. Then:

(i) $f \in \mathscr{L}^1(\mu)$.
(ii) $\lim_{n\to\infty} \int_X f_n\, d\mu = \int_X f\, d\mu$.

Proof. Since $|f_n| \le M$ a.e. and $f_n \to f$ a.e., $|f| \le M$ a.e.
Set $G := M\mathbf{1}_X$. Then $G \in \mathscr{L}^1(\mu)$, since $\int_X G\, d\mu = M\mu(X) < +\infty$.

(i) Since $|f| \le G$ a.e., $f \in \mathscr{L}^1(\mu)$ by Theorem 1.2.18(ii).
(ii) Since $\forall n$, $|f_n| \le G$ a.e., $\int_X f_n\, d\mu \to \int_X f\, d\mu$ by Theorem 1.3.7(ii).

Structural Commentary: On a finite-measure space, a uniform pointwise bound is an integrable dominator.
Key Insight: Bounded convergence is the finite-measure special case of dominated convergence. □

Example 1.3.18. *Why the Finite-Measure Hypothesis Matters*

On $(\mathbb{R}, m)$, let $f_n := n^{-1}\mathbf{1}_{[0,n]}$. Then $\|f_n\|_\infty = \frac{1}{n} \to 0$, but $\int_{\mathbb{R}} f_n\, dm = 1\ \forall n$.
Thus uniform convergence need not commute with integrals when $m(\mathbb{R}) = +\infty$; Fatou only yields $0 \le 1$.

Example 1.3.19. *Why Domination Cannot Be Dropped*

On $(\mathbb{R}, m)$, let $f_n := n\mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}$. Then $f_n \to 0$ pointwise, but $\int_{\mathbb{R}} f_n\, dm = 1\ \forall n$.
Thus pointwise convergence without domination need not commute with integrals; Fatou only yields $0 \le 1$.

Theorem 1.3.20. *Absolutely Summable L^1 Series*
Let $f_n \in \mathscr{L}^1(\mu)$ and $\sum_{n\geq 1} \|f_n\|_1 < +\infty$.
Set $S_N := \sum_{n=1}^{N} f_n$. Then $\exists f : X \to \mathbb{R}$ measurable such that:
(i) $S_N \to f$ absolutely a.e.
(ii) $f \in \mathscr{L}^1(\mu)$.
(iii) $\int_X f\,d\mu = \sum_{n\geq 1} \int_X f_n\,d\mu$.

Proof. Set $G := \sum_{n\geq 1} |f_n|$. By monotone convergence, $\int_X G\,d\mu = \sum_{n\geq 1} \|f_n\|_1 < +\infty$, so $G < +\infty$ a.e. and $G \in \mathscr{L}^1(\mu)$. Define $f := \lim_N S_N$ on $\{G < +\infty\}$ and $f := 0$ elsewhere.

(i) Since $G < +\infty$ a.e., $\sum_{n\geq 1} |f_n| < +\infty$ a.e.; hence $S_N \to f$ absolutely a.e.

(ii) Since $|f| \leq G$ a.e. and $G \in \mathscr{L}^1(\mu)$, $f \in \mathscr{L}^1(\mu)$ by Theorem 1.2.18(ii).

(iii) Since $|S_N| \leq G$ and $S_N \to f$ a.e., dominated convergence gives

$$\int_X f\,d\mu = \lim_N \int_X S_N\,d\mu = \lim_N \sum_{n=1}^{N} \int_X f_n\,d\mu = \sum_{n\geq 1} \int_X f_n\,d\mu.$$

Structural Commentary: Absolute summability in L^1 gives a.e. absolute convergence and termwise integration.
Key Insight: First integrate the nonnegative majorant $\sum |f_n|$. □

Proposition 1.3.21. *L^1 Convergence Controls Absolute Integrals*
If $f_n, f \in \mathscr{L}^1(\mu)$ and $\|f_n - f\|_1 \to 0$, then:
(i) $\|f_n\|_1 \to \|f\|_1$.
(ii) $\int_X |f|\,d\mu = \lim_n \int_X |f_n|\,d\mu$.

Proof.

(i) The reverse triangle inequality gives $||f_n| - |f|| \leq |f_n - f|$; hence $\big|\|f_n\|_1 - \|f\|_1\big| \leq \|f_n - f\|_1 \to 0$.

(ii) Since $\|h\|_1 = \int_X |h|\,d\mu$, item (i) gives $\int_X |f_n|\,d\mu \to \int_X |f|\,d\mu$.

Structural Commentary: L^1 convergence is strong enough to converge the L^1 sizes.
Key Insight: The reverse triangle inequality converts function convergence into norm convergence. □

Proposition 1.3.22. *Finite Integral Localizes to Finite Measure*
Let $f : X \to [0, +\infty]$ be measurable and $\int_X f\,d\mu < +\infty$. For $n \geq 1$, set $E_n := \{\frac{1}{n} < f \leq n\}$. Then:
(i) $\forall n,\ \mu(E_n) < +\infty$.
(ii) $\int_{X\setminus E_n} f\,d\mu \downarrow 0$.
Consequently, $\forall \varepsilon > 0\ \exists E \in \mathcal{M}$ with $\mu(E) < +\infty$ and $\int_{X\setminus E} f\,d\mu < \varepsilon$.

Proof.

(i) Since $f > \frac{1}{n}$ on E_n, $\mu(E_n) \leq n \int_{E_n} f\,d\mu \leq n \int_X f\,d\mu < +\infty$.

(ii) Since $f\mathbf{1}_{E_n} \uparrow f$ a.e., monotone convergence gives $\int_{E_n} f\,d\mu \uparrow \int_X f\,d\mu$.
Thus $\int_{X\setminus E_n} f\,d\mu = \int_X f\,d\mu - \int_{E_n} f\,d\mu \downarrow 0$.

Given $\varepsilon > 0$, choose n with $\int_{X\setminus E_n} f\,d\mu < \varepsilon$ and set $E = E_n$.
Structural Commentary: Finite integral mass can be captured on a finite-measure part of the space.
Key Insight: Keep the levels where f is bounded above and away from zero. □

Proposition 1.3.23. *Setwise Convergence from Total Mass Convergence*
Let $f_n, f : X \to [0, +\infty]$ be measurable, $f_n \to f$ pointwise, and $\int_X f\,d\mu = \lim_{n\to\infty} \int_X f_n\,d\mu < +\infty$.
Then $\forall E \in \mathcal{M}$, $\int_E f\,d\mu = \lim_{n\to\infty} \int_E f_n\,d\mu$.

Proof. Fix $E \in \mathcal{M}$ and set $E^c := X \setminus E$. By Fatou, $\int_E f\,d\mu \leq \liminf_n \int_E f_n\,d\mu$ and $\int_{E^c} f\,d\mu \leq \liminf_n \int_{E^c} f_n\,d\mu$.
Since $\int_X f_n\,d\mu = \int_E f_n\,d\mu + \int_{E^c} f_n\,d\mu \to \int_X f\,d\mu < +\infty$, $\limsup_n \int_E f_n\,d\mu \leq \int_X f\,d\mu - \int_{E^c} f\,d\mu = \int_E f\,d\mu$.
Thus $\liminf_n \int_E f_n\,d\mu = \limsup_n \int_E f_n\,d\mu = \int_E f\,d\mu$.
Structural Commentary: Total mass plus Fatou on complements fixes each part.
Key Insight: A lower bound on the complement is an upper bound on the chosen set. □

Example 1.3.24. *Why the Finite Total Mass Assumption Matters*
Let $X = \mathbb{N} \times \{0, 1\}$ with counting measure, set $E := \mathbb{N} \times \{1\}$.
Define $f := \mathbf{1}_{\mathbb{N}\times\{0\}}$; $f_n := \mathbf{1}_{\{1,\dots,n\}\times\{0\}} + n\mathbf{1}_{\{n\}\times\{1\}}$.
Then $f_n \to f$ pointwise and $\int_X f\,d\mu = \lim_n \int_X f_n\,d\mu = +\infty$; $\int_E f\,d\mu = 0$, $\int_E f_n\,d\mu = n$.
Thus Proposition 1.3.23 can fail when the common total mass is infinite.

Definition 1.3.25. *Function Truncates*
For $f : X \to \mathbb{R}$ and $n \in \mathbb{N}$, define the *nth truncate*

$$T_n f(x) := (-n) \vee (f(x) \wedge n) = \begin{cases} -n, & f(x) < -n, \\ f(x), & -n \leq f(x) \leq n, \\ n, & f(x) > n. \end{cases}$$

Proposition 1.3.26. *Truncates Characterize Integrability*
Let $f : X \to \mathbb{R}$ be measurable and set $f_n := T_n f$ for $n \in \mathbb{N}$.
(i) $f \in \mathscr{L}^1(\mu) \iff \sup_n \|f_n\|_1 < +\infty$.
(ii) If $f \in \mathscr{L}^1(\mu)$, then $\int_X f_n \, d\mu \to \int_X f \, d\mu$.

Proof. $\forall n$, $T_n : \mathbb{R} \to [-n, n]$ is continuous, so f_n is measurable; moreover $f_n \to f$, $|f_n| \le |f|$, and $|f_n| \uparrow |f|$.
(i) ($\Rightarrow$) If $f \in \mathscr{L}^1(\mu)$, then $\|f_n\|_1 \le \|f\|_1$ for all n, so $\sup_n \|f_n\|_1 < +\infty$.
(i) ($\Leftarrow$) If $\sup_n \|f_n\|_1 < +\infty$, then MCT gives $\|f\|_1 = \lim_{n\to\infty} \|f_n\|_1 = \sup_n \|f_n\|_1 < +\infty$, so $f \in \mathscr{L}^1(\mu)$.
(ii) If $f \in \mathscr{L}^1(\mu)$, then DCT applied to $f_n \to f$ with $|f_n| \le |f|$ gives $\int_X f_n \, d\mu \to \int_X f \, d\mu$.
Structural Commentary: Truncation reduces L^1 questions to bounded approximants.
Key Insight: MCT reads $\|f\|_1$ from $|f_n| \uparrow |f|$; DCT passes signed integrals. □

1.4 Parameter Integrals and Differentiation

Corollary 1.4.1. *Continuity Under a Lebesgue Integral*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in [a,b]$, and suppose $\forall t \in [a,b]$, $|f_t| \le g \in \mathscr{L}^1(\mu)$. Then:
(i) If $t_0 \in [a,b]$ and $f_t \to f_{t_0}$ a.e. as $t \to t_0$, then $\lim_{t\to t_0} \int_X f_t \, d\mu = \int_X f_{t_0} \, d\mu$.
(ii) If $t \mapsto f(x,t)$ is continuous on $[a,b]$ for a.e. x, then $F(t) := \int_X f_t \, d\mu$ is continuous on $[a,b]$.

Proof.
(i) For any $t_n \to t_0$, set $u_n := f_{t_n}$. Then $u_n \to f_{t_0}$ a.e. and $|u_n| \le g$, so dominated convergence gives $\int_X u_n \, d\mu \to \int_X f_{t_0} \, d\mu$. The sequential criterion gives the claim.
(ii) If $t \mapsto f(x,t)$ is continuous a.e., then $\forall t_0 \in [a,b]$, $f_t \to f_{t_0}$ a.e. as $t \to t_0$.
By item (i), $\forall t_0 \in [a,b]$, $F(t) \to F(t_0)$, so F is continuous on $[a,b]$.
Structural Commentary: Parameter continuity reduces to dominated convergence along sequences.
Key Insight: The parameter disappears once t_n is frozen into a sequence of functions. □

Theorem 1.4.2. *Differentiation Under a Lebesgue Integral*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in [a,b]$, and suppose $f_{t_0} \in \mathscr{L}^1(\mu)$ for some $t_0 \in [a,b]$.
Assume $t \mapsto f(x,t)$ is continuous on $[a,b]$ and differentiable on (a,b) $\forall x \in X$, and $\exists g \in \mathscr{L}^1(\mu)$ such that $\forall (x,t) \in X \times (a,b)$, $|\partial_t f(x,t)| \le g(x)$. For $t \in (a,b)$, set $D_t(x) := \partial_t f(x,t)$; set $F(t) := \int_X f_t \, d\mu$. Then:
(i) $\forall t \in (a,b)$, D_t is measurable.
(ii) $\forall t \in [a,b]$, $f_t \in \mathscr{L}^1(\mu)$.
(iii) F is differentiable on (a,b), and $\forall t \in (a,b)$, $F'(t) = \int_X D_t \, d\mu$.

Proof.
(i) Fix $t \in (a,b)$ and choose $h_k \to 0$ with $t + h_k \in [a,b]$, $h_k \neq 0$.
The quotients $\frac{f_{t+h_k} - f_t}{h_k}$ are measurable and converge pointwise to D_t, so D_t is measurable.
(ii) By the mean value theorem, $|f_t - f_{t_0}| \le |t - t_0| \, g$, so $|f_t| \le |f_{t_0}| + |t - t_0| \, g \in \mathscr{L}^1(\mu)$.
(iii) If $t_n \to t \in (a,b)$ and $t_n \neq t$, then $\frac{F(t_n) - F(t)}{t_n - t} = \int_X \frac{f_{t_n} - f_t}{t_n - t} \, d\mu$. The integrand converges pointwise to D_t and has modulus at most g, so dominated convergence gives $\frac{F(t_n) - F(t)}{t_n - t} \to \int_X D_t \, d\mu$.
Structural Commentary: Bounded parameter derivatives dominate the difference quotients.
Key Insight: Differentiating the integral is just applying dominated convergence to secant slopes. □

Corollary 1.4.3. *A Pointwise Difference-Quotient Test*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in [a,b]$, let $f_{t_0} \in \mathscr{L}^1(\mu)$ for some $t_0 \in [a,b]$, and fix $t_1 \in [a,b]$.
Assume $\partial_t f(x,t_1)$ exists $\forall x$. For $t \neq t_1$, set $Q_t(x) := \frac{f_t(x) - f_{t_1}(x)}{t - t_1}$, $|Q_t| \le g$ for some $g \in \mathscr{L}^1(\mu)$.
Set $F(t) := \int_X f_t \, d\mu$. Then:
(i) $\forall t \in [a,b]$, $f_t \in \mathscr{L}^1(\mu)$.
(ii) F is differentiable at t_1, and $F'(t_1) = \int_X \partial_t f(x,t_1) \, d\mu$.

Proof.
(i) If $t_0 \neq t_1$, then $|f_{t_1}| \le |f_{t_0}| + |t_0 - t_1| \, g$; if $t_0 = t_1$, this is given.
Hence $f_{t_1} \in \mathscr{L}^1(\mu)$. For $t \neq t_1$, $|f_t| \le |f_{t_1}| + |t - t_1| \, g$, so $f_t \in \mathscr{L}^1(\mu)$.
(ii) For any $t_n \to t_1$ with $t_n \neq t_1$, $Q_{t_n} \to \partial_t f(\cdot, t_1)$ pointwise and $|Q_{t_n}| \le g$.
Dominated convergence gives $\frac{F(t_n) - F(t_1)}{t_n - t_1} \to \int_X \partial_t f(x,t_1) \, d\mu$.
Structural Commentary: One parameter point needs only domination of the secants into that point.
Key Insight: A global derivative bound is sufficient, but not necessary, for a one-point formula. □

Example 1.4.4. *Differentiating Under the Integral Sign*
For $t > 0$, set $F(t) := \int_0^\infty x^2 e^{-tx} \, dx$. For $t_0 > 0$ and $0 < |h| < \frac{t_0}{2}$, put $Q_h(x) := \frac{x^2 e^{-(t_0+h)x} - x^2 e^{-t_0 x}}{h}$.
Then $Q_h \to -x^3 e^{-t_0 x}$ pointwise and $|Q_h| \le x^3 e^{-\frac{t_0 x}{2}} \in \mathscr{L}^1(0,\infty)$. Hence $F'(t_0) = -\int_0^\infty x^3 e^{-t_0 x} \, dx$.
Structural Commentary: Parameter differentiation is a limit-interchange problem.
Key Insight: Prove a local-in-parameter integrable majorant for the difference quotients.

Example 1.4.5. *Laplace Transform Powers and Factorials*
For $t > 0$, set $I(t) := \int_0^\infty e^{-tx}\,dx$. Then $I(t) = \lim_{b\to+\infty} \frac{1-e^{-tb}}{t} = t^{-1}$.

For $[a,b] \subset (0,+\infty)$ and $k \geq 0$, $0 \leq x^k e^{-tx} \leq x^k e^{-ax} \in \mathscr{L}^1(0,\infty)$, so repeated differentiation under the integral gives $I^{(n)}(t) = (-1)^n \int_0^\infty x^n e^{-tx}\,dx$. Since $I^{(n)}(t) = (-1)^n n! t^{-n-1}$, $\forall n \in \mathbb{N}\cup\{0\}$, $\int_0^\infty x^n e^{-x}\,dx = n!$.
Structural Commentary: A parameter integral turns powers of x into derivatives in t.
Key Insight: The exponential envelope e^{-ax} controls all nearby parameter values $t \geq a$.

Corollary 1.4.6. *One Lebesgue and One Riemann Integral*
Let $f_t(x) := f(x,t)$, $t \in [a,b]$, with f_t measurable, $t \mapsto f(x,t)$ continuous a.e., and $\forall t \in [a,b]$, $|f_t| \leq g \in \mathscr{L}^1(\mu)$. Then, with Riemann integrals in t, $\int_a^b \left[\int_X f_t\,d\mu\right] dt = \int_X \left[\int_a^b f(x,t)\,dt\right] d\mu(x)$.

Proof. Set $h_t(x) := \int_a^t f(x,s)\,ds$. Then h_t is measurable, $|h_t| \leq (b-a)g$, and $\partial_t h_t = f_t$.
By differentiation under the integral, $H(t) := \int_X h_t\,d\mu$ satisfies $H'(t) = \int_X f_t\,d\mu$.
Hence $\int_a^b \left[\int_X f_t\,d\mu\right] dt = H(b) - H(a) = \int_X \left[\int_a^b f(x,t)\,dt\right] d\mu(x)$.
Structural Commentary: Finite parameter integrals move through by differentiating primitives.
Key Insight: Integrate first in the parameter up to t, then differentiate the resulting Lebesgue integral. □

Corollary 1.4.7. *Improper Parameter Interchange*
Let $f_t(x) := f(x,t)$ be measurable $\forall t \in \mathbb{R}$, and let $t \mapsto f(x,t)$ be continuous for a.e. x. Suppose $\exists g, h \in \mathscr{L}^1(\mu)$ such that $|f_t| \leq g$ $\forall t$ and $\int_{\mathbb{R}} |f(x,t)|\,dt \leq h(x)$ for a.e. x. Put $G(t) := \int_X f_t\,d\mu$, $H(x) := \int_{\mathbb{R}} f(x,t)\,dt$, and $H_R(x) := \int_{-R}^R f(x,t)\,dt$. Then:

(i) $H \in \mathscr{L}^1(\mu)$.
(ii) $\lim_{R\to+\infty} \int_{-R}^R G(t)\,dt = \int_X H\,d\mu$.

Thus $\int_{\mathbb{R}} G := \lim_{R\to+\infty} \int_{-R}^R G(t)\,dt$ exists and equals $\int_X H\,d\mu$.

Proof.
(i) Since $|H| \leq \int_{\mathbb{R}} |f(x,t)|\,dt \leq h$ a.e. and $h \in \mathscr{L}^1(\mu)$, $H \in \mathscr{L}^1(\mu)$ by Theorem 1.2.18(ii).
(ii) For $R > 0$, finite-interval interchange gives $\int_{-R}^R G(t)\,dt = \int_X H_R\,d\mu$.
Since $H_R \to H$ a.e. and $|H_R| \leq h$, dominated convergence gives $\int_X H_R\,d\mu \to \int_X H\,d\mu$.
Structural Commentary: Finite interchanges pass to improper limits under an L^1 bound.
Key Insight: Truncate in t, interchange on finite intervals, then pass $R \to +\infty$ by domination in x. □

1.5 Riemann-Lebesgue Comparison

Lemma 1.5.1. *Monotone Integral Squeeze*
Let $E \in \mathcal{M}$, $(\varphi_n), (\psi_n) \subseteq \mathscr{L}^1(\mu)$ on E, and $f : E \to \mathbb{R}$. Assume $\varphi_n \uparrow$, $\psi_n \downarrow$, $\varphi_n \leq f \leq \psi_n$ on E, and $\int_E (\psi_n - \varphi_n)\,d\mu \to 0$. Put $\varphi := \lim_n \varphi_n$ and $\psi := \lim_n \psi_n$. Then:

(i) $\varphi = \psi$ *a.e. on* E.
(ii) For $h := \varphi$, $f = h$ a.e., $h \in \mathscr{L}^1(\mu)$ on E, and $\int_E \varphi_n\,d\mu, \int_E \psi_n\,d\mu \to \int_E h\,d\mu$.

Proof.
(i) The limits φ, ψ are measurable and $\varphi \leq f \leq \psi$.
Since $0 \leq \psi - \varphi \leq \psi_n - \varphi_n$, $0 \leq \int_E (\psi - \varphi)\,d\mu \leq \int_E (\psi_n - \varphi_n)\,d\mu \to 0$; hence $\psi = \varphi$ a.e.
(ii) Let h be the common a.e. value from (i). Then $f = h$ a.e. and $|h| \leq |\varphi_1| + |\psi_1|$ a.e., so $h \in \mathscr{L}^1(\mu)$ on E.
Monotone convergence applied to $\varphi_n - \varphi_1 \uparrow h - \varphi_1$ and $\psi_1 - \psi_n \uparrow \psi_1 - h$ gives $\int_E \varphi_n\,d\mu, \int_E \psi_n\,d\mu \to \int_E h\,d\mu$.
Structural Commentary: Vanishing gaps collapse monotone envelopes.
Key Insight: Integral gaps force the two monotone limits to coincide a.e. □

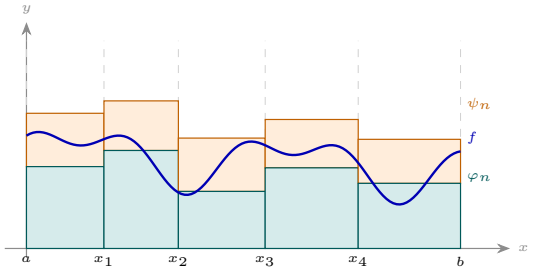

Figure 201: Step envelopes squeeze f to a common a.e. limit.

Theorem 1.5.2. *Comparison with the Riemann Integral*
Let m be Lebesgue measure on $[a,b]$ and let $f \in \mathcal{R}[a,b]$.

(i) $f \in \mathscr{L}^1(m)$.
(ii) $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.

Proof. Choose nested partitions P_n with $U(P_n,f) - L(P_n,f) \to 0$. Let φ_n, ψ_n be lower and upper step envelopes.
(i) Lemma 1.5.1 gives $f \in \mathscr{L}^1(m)$.

(ii) Since $\int \varphi_n\,dm = L(P_n,f)$ and $\int \psi_n\,dm = U(P_n,f)$, Lemma 1.5.1 gives $\int_{[a,b]} f\,dm = \lim_n L(P_n,f) = \lim_n U(P_n,f) = \int_a^b f(x)\,dx$.

Structural Commentary: Darboux envelopes squeeze the Lebesgue integral.
Key Insight: Vanishing Darboux gap identifies the two integral notions. □

Corollary 1.5.3. *Riemann Integrable Functions as Monotone Step Limits*
Let $f:[a,b]\to\mathbb{R}$ be Riemann integrable. Then $\exists(\varphi_n),(\psi_n)$ step-function sequences such that

(i) $\varphi_n\uparrow$, $\psi_n\downarrow$, and $\varphi_n\le f\le\psi_n$ on $[a,b]$;
(ii) $\lim_n \int_a^b \varphi_n\,dx = \lim_n \int_a^b \psi_n\,dx = \int_a^b f\,dx$;
(iii) $\varphi_n,\psi_n\to f$ a.e.; hence f is Lebesgue measurable and $\int_{[a,b]} f\,dm = \int_a^b f\,dx$.

Proof.

(i) Choose nested partitions P_n with $\max_{I\in P_n}\ell(I)\to 0$ and $U(P_n,f)-L(P_n,f)\to 0$.
Let φ_n,ψ_n be the lower and upper step envelopes on P_n; refinement gives $\varphi_n\uparrow$, $\psi_n\downarrow$, and $\varphi_n\le f\le\psi_n$.
(ii) Since $\int_a^b \varphi_n\,dx = L(P_n,f)$ and $\int_a^b \psi_n\,dx = U(P_n,f)$, by Darboux gap condition, both limits equal $\int_a^b f\,dx$.
(iii) By Lemma 1.5.1, $\exists h\in\mathscr{L}^1(m)$ with $h=f$ a.e., $\varphi_n,\psi_n\to h$ a.e., and $\int_{[a,b]} h\,dm = \lim_n \int_a^b \varphi_n\,dx = \int_a^b f\,dx$.

Structural Commentary: Riemann integrability is monotone step squeezing.
Key Insight: Refining partitions make Darboux sums into pointwise a.e. step approximations. □

Corollary 1.5.4. *Nonnegative Continuous Functions and Riemann Integrals*

(i) If $f\ge 0$ is continuous on $[a,b]$, then $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.
(ii) If $f\ge 0$ is continuous on $[0,+\infty)$, then $\int_{[0,+\infty)} f\,dm = \lim_{b\to+\infty}\int_0^b f(x)\,dx$ in $[0,+\infty]$.

Proof.

(i) Continuous f on $[a,b]$ is Riemann integrable; by Riemann–Lebesgue comparison, $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.
(ii) Put $f_b := f\mathbf{1}_{[0,b]}$. By (i), $\int_{[0,b]} f\,dm = \int_0^b f(x)\,dx$. As $b\uparrow+\infty$, $f_b\uparrow f$; monotone convergence gives the left limit, and the right limit is the nonnegative improper Riemann integral.

Structural Commentary: Compact comparison plus monotone truncation.
Key Insight: Truncate to compact intervals and pass upward. □

Theorem 1.5.5. *Lebesgue's Criterion for Riemann Integrability*
Let $f:[a,b]\to\mathbb{R}$ be bounded with discontinuity set D_f. Then $f\in\mathcal{R}[a,b]\iff m(D_f)=0$.

Proof. Set $\omega_f(x) := \lim_{r\downarrow 0}\operatorname{diam} f([a,b]\cap(x-r,x+r))$ and $D_k := \{x:\omega_f(x)\ge\frac{1}{k}\}$; then $D_f = \bigcup_{k\ge1} D_k$.

($\Rightarrow$) Fix k and $\varepsilon>0$. Choose P with $U(P,f)-L(P,f)<\frac{\varepsilon}{k}$. Set $\mathcal{B} := \{I\in P:\operatorname{int} I\cap D_k\neq\emptyset\}$.
Then $\frac{1}{k}\sum_{I\in\mathcal{B}}\ell(I)\le U(P,f)-L(P,f)$, so $\sum_{I\in\mathcal{B}}\ell(I)<\varepsilon$.
These intervals plus the partition points cover D_k; hence $m(D_k)=0$ and $m(D_f)=0$.
($\Leftarrow$) Suppose $m(D_f)=0$ and $|f|\le B$. If $B=0$, $f\in\mathcal{R}[a,b]$. For $B>0$ and $\varepsilon>0$, choose k with $\frac{b-a}{k}<\frac{\varepsilon}{2}$ and cover the compact null set D_k by finitely many intervals of total length $<\frac{\varepsilon}{4B}$.
Refining by this cover gives $U(P,f)-L(P,f)<\frac{b-a}{k}+\frac{\varepsilon}{2}<\varepsilon$, hence $f\in\mathcal{R}[a,b]$.

Structural Commentary: Darboux gaps measure oscillation.
Key Insight: Riemann failure is positive-measure discontinuity. □

Example 1.5.6. *Dirichlet Function: Lebesgue but Not Riemann*
Let $D := \mathbf{1}_{\mathbb{Q}\cap[0,1]}$. Since $\mathbb{Q}\cap[0,1]$ is countable, $D=0$ a.e. on $[0,1]$, so $\int_{[0,1]} D\,dm = 0$.
Every nondegenerate subinterval $I\subseteq[0,1]$ meets both $\mathbb{Q}$ and $\mathbb{R}\setminus\mathbb{Q}$, hence $\inf_I D = 0$ and $\sup_I D = 1$.
Thus all lower Darboux sums are 0, all upper Darboux sums are 1, and D is not Riemann integrable.
Structural Commentary: Null support, dense oscillation.
Key Insight: A dense null set can be harmless for Lebesgue integration and fatal for Riemann integration.

Perspective 1.5.7. *Lebesgue's Criterion as a Roadmap*
For bounded $f:[a,b]\to\mathbb{R}$ with discontinuity set D_f, $f\in\mathcal{R}[a,b]\iff m(D_f)=0$.
Thus finite, countable, and null Cantor-type D_f are allowed, while $D_f=[a,b]$ is not.

1.6 Littlewood and Lusin Approximation

The measurable world is often controlled by ordinary topological objects after discarding a small exceptional set. In the real-line finite-measure setting, this principle has three recurring forms: measurable sets are nearly elementary, measurable functions are nearly continuous, and a.e. convergence is nearly uniform.

Principle 1.6.1. *Littlewood's Three Principles, Finite-Measure Form*
For subsets of the real line with finite measure:

(i) measurable sets are nearly finite unions of intervals; see Corollary 6.2.4;
(ii) measurable functions are nearly continuous; this is Lusin's theorem below;
(iii) a.e. pointwise convergence is nearly uniform; see Egoroff's theorem, Theorem 1.6.3.

Key Insight: The word "nearly" means after changing or discarding a set of arbitrarily small measure.

Definition 1.6.2. *Almost Uniform Convergence*
Let (f_n) be measurable functions on X.

(i) $f_n\to f$ *almost uniformly* if $\forall\delta>0\ \exists E_\delta\in\mathcal{M}$ with $\mu(E_\delta)<\delta$ and $f_n\rightrightarrows f$ on $X\setminus E_\delta$.

(ii) (f_n) is *almost uniformly Cauchy* if $\forall \delta > 0\ \exists E_\delta \in \mathcal{M}$ with $\mu(E_\delta) < \delta$ s.t. (f_n) uniformly Cauchy on $X \setminus E_\delta$.

Theorem 1.6.3. *Egoroff's Theorem*
If $\mu(X) < +\infty$, $f_n \to f$ a.e., with all functions finite-valued off null set, then $f_n \to f$ almost uniformly.

Proof. Absorb the null exceptional set into every later E_δ, and assume $\forall x \in X,\ f_n(x) \to f(x)$.
For $m, n \in \mathbb{N}$, set $E_n(m) := \bigcup_{k=n}^{\infty} \{|f_k - f| \geq \frac{1}{m}\}$. Then $E_n(m) \downarrow \emptyset$, so $\mu(E_n(m)) \to 0$ by Corollary 5.1.1.
Given $\delta > 0$, choose k_m with $\mu(E_{k_m}(m)) < \delta 2^{-m}$ and set $E_\delta := \bigcup_{m \geq 1} E_{k_m}(m)$.
Then $\mu(E_\delta) < \delta$, and $x \notin E_\delta$, $k \geq k_m$ imply $|f_k(x) - f(x)| < \frac{1}{m}$; hence $f_n \rightrightarrows f$ on $X \setminus E_\delta$.
Structural Commentary: Finite measure turns decreasing bad sets with empty intersection into small bad sets.
Key Insight: A.e. convergence is almost uniform on finite measure spaces. □

Example 1.6.4. *Egoroff Needs a Finite-Valued Limit*
On $([0,1], \lambda)$, set $f_n := n\mathbf{1}_{[0,1]}$. Then $\forall x,\ f_n(x) \to +\infty$, so the pointwise limit is not finite-valued. If $f_n \to f$ almost uniformly for finite-valued measurable f, then for some E with $\lambda(E) < \frac{1}{2}$, $f_n \rightrightarrows f$ on $[0,1] \setminus E$; this nonempty set has $f_n(x) = n \to +\infty$, contradicting finite-valued pointwise convergence.

Lemma 1.6.5. *Continuous Extension from a Closed Subset of $\mathbb{R}$*
Let $F \subseteq \mathbb{R}$ be closed and let $h : F \to \mathbb{R}$ be continuous. Then $\exists g \in C(\mathbb{R})$ such that $g|_F = h$.

Proof. Cases $F = \emptyset$ and $F = \mathbb{R}$ are immediate. Otherwise $\mathbb{R} \setminus F = \bigsqcup_\alpha I_\alpha$, set $g = h$ on F; on each component:

(i) If $I_\alpha = (a, b)$, set $g(x) := \frac{b-x}{b-a}h(a) + \frac{x-a}{b-a}h(b)$.
(ii) If $I_\alpha = (-\infty, b)$, set $g(x) := h(b)$.
(iii) If $I_\alpha = (a, +\infty)$, set $g(x) := h(a)$.

The formulas give continuity off F; at $x \in F$, adjacent finite endpoints converge to x.
Hence continuity of h gives $g(y) \to h(x)$ as $y \to x$. Thus $g \in C(\mathbb{R})$ and $g|_F = h$.
Structural Commentary: Fill complementary intervals.
Key Insight: Interpolate between boundary values. □

Proposition 1.6.6. *Simple Functions Are Nearly Continuous*
Let $E \in \mathcal{L}^1$ with $m(E) < +\infty$, and $s : E \to \mathbb{R}$ measurable simple. $\forall \varepsilon > 0$, $\exists F \subseteq E$ and $g : \mathbb{R} \to \mathbb{R}$ such that:
(i) F is closed.
(ii) $g \in C(\mathbb{R})$ and $g = s$ on F.
(iii) $m(E \setminus F) < \varepsilon$.

Proof. If $E = \emptyset$, take $F = \emptyset$ and $g = 0$. Otherwise $s = \sum_{i=1}^{N} a_i \mathbf{1}_{E_i}$ with disjoint measurable E_i and $E = \bigsqcup_i E_i$. Choose closed $F_i \subseteq E_i$ with $m(E_i \setminus F_i) < \frac{\varepsilon}{N}$, set $F := \bigcup_i F_i$, and define $h : F \to \mathbb{R}$ by $h|_{F_i} := a_i$.

(i) Each F_i is closed and $N < +\infty$, so F is closed; also $F \subseteq E$.
(ii) For $x \in F_i$, the closed set $\bigcup_{j \neq i} F_j$ misses x, so h is locally constant at x in F; hence $h \in C(F)$.
$\exists g \in C(\mathbb{R})$ with $g|_F = h = s|_F$ by Lemma 1.6.5.
(iii) $m(E \setminus F) \leq \sum_i m(E_i \setminus F_i) < \varepsilon$.

Structural Commentary: Closed cores isolate simple levels.
Key Insight: Approximate each level set from inside, then extend. □

Theorem 1.6.7. *Lusin's Theorem*
Let $E \in \mathcal{L}^1$ with $m(E) < +\infty$, $f : E \to \mathbb{R}$ measurable. $\forall \varepsilon > 0$, $\exists F \subseteq E$ and $g : \mathbb{R} \to \mathbb{R}$ such that:
(i) F is closed.
(ii) $g \in C(\mathbb{R})$ and $g = f$ on F.
(iii) $m(E \setminus F) < \varepsilon$.

Proof. Take simple $s_n : E \to \mathbb{R}$ with $s_n \to f$ pointwise.
Pick $A \subseteq E$ with $s_n \to f$ uniformly on A and $m(E \setminus A) < \frac{\varepsilon}{4}$, and closed $F_0 \subseteq A$ with $m(A \setminus F_0) < \frac{\varepsilon}{4}$.
For $n \geq 1$, choose closed $F_n \subseteq E$ with $s_n|_{F_n} \in C(F_n)$ and $m(E \setminus F_n) < \varepsilon 2^{-(n+1)}$. Set $F := \bigcap_{n \geq 0} F_n$.

(i) F is an intersection of closed sets and $F \subseteq E$.
(ii) Since $F \subseteq F_0 \subseteq A$, the continuous functions $s_n|_F$ converge uniformly to $f|_F$, so $f|_F \in C(F)$.
Since F is closed and $f|_F \in C(F)$, Lemma 1.6.5 gives $\exists g \in C(\mathbb{R})$ with $g|_F = f|_F$.
(iii) $m(E \setminus F) \leq m(E \setminus F_0) + \sum_{n \geq 1} m(E \setminus F_n) < \varepsilon$.

Structural Commentary: Discard a small set to gain continuity.
Key Insight: Simple cores plus Egoroff give uniform continuous limits. □

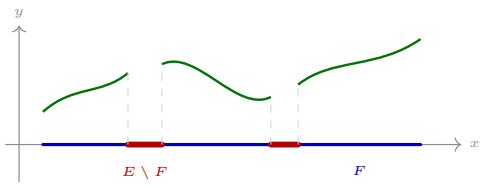

Figure 202: Lusin's theorem: keep F, discard $E \setminus F$, and extend with $g = f$ on F.

1.7 Convex Functions and Jensen's Inequality

Convexity is the geometric condition behind Jensen's inequality: averages move inside the argument, while supporting lines convert the nonlinear statement into a linear integral estimate.

Definition 1.7.1. *Convex Function on an Interval*
Let $I \subseteq \mathbb{R}$ be an interval and let $\varphi : I \to \mathbb{R}$.

(i) φ is *convex* iff $\forall x_1, x_2 \in I\ \forall \lambda \in [0,1]$, $\varphi(\lambda x_1 + (1-\lambda)x_2) \le \lambda\varphi(x_1) + (1-\lambda)\varphi(x_2)$.
(ii) Equivalently, $\forall x_1 < x < x_2$ in I, $\varphi(x) \le \frac{x_2 - x}{x_2 - x_1}\varphi(x_1) + \frac{x - x_1}{x_2 - x_1}\varphi(x_2)$.

Key Insight: Graphs lie below enclosing chords.

Lemma 1.7.2. *Chordal Slope Lemma*
Let $I \subseteq \mathbb{R}$ be an interval, let $\varphi : I \to \mathbb{R}$ be convex, and let $x_1, x, x_2 \in I$ satisfy $x_1 < x < x_2$.
Then $\frac{\varphi(x) - \varphi(x_1)}{x - x_1} \le \frac{\varphi(x_2) - \varphi(x_1)}{x_2 - x_1} \le \frac{\varphi(x_2) - \varphi(x)}{x_2 - x}$.

Proof. Since $x = \frac{x_2 - x}{x_2 - x_1}x_1 + \frac{x - x_1}{x_2 - x_1}x_2$, convexity gives $(x_2 - x_1)\varphi(x) \le (x_2 - x)\varphi(x_1) + (x - x_1)\varphi(x_2)$.
Rearrange, $(x_2 - x_1)(\varphi(x) - \varphi(x_1)) \le (x - x_1)(\varphi(x_2) - \varphi(x_1))$; $(x_2 - x)(\varphi(x_2) - \varphi(x_1)) \le (x_2 - x_1)(\varphi(x_2) - \varphi(x))$.
Divide by $(x - x_1)(x_2 - x_1) > 0$ and $(x_2 - x)(x_2 - x_1) > 0$, respectively.
Structural Commentary: One chord inequality, two rearrangements.
Key Insight: Convex secant slopes are nondecreasing. □

Proposition 1.7.3. *Increasing Derivative Gives Convexity*
Let $I \subseteq \mathbb{R}$ be open and let $\varphi : I \to \mathbb{R}$ be differentiable. If φ' is increasing on I, then φ is convex on I.
In particular, this holds if φ is twice differentiable and $\varphi'' \ge 0$ on I.

Proof. For $a < b$ in I, set $S(a,b) := \frac{\varphi(b) - \varphi(a)}{b - a}$. Let $u < v$ and $0 < \lambda < 1$, and put $x = \lambda u + (1-\lambda)v$.
By MVT, $S(u,x) = \varphi'(c_1)$, $S(x,v) = \varphi'(c_2)$ for some $c_1 \in (u,x)$, $c_2 \in (x,v)$. As $c_1 < c_2$, $S(u,x) \le S(x,v)$.
Clearing denominators, $\varphi(x) \le \frac{v - x}{v - u}\varphi(u) + \frac{x - u}{v - u}\varphi(v) = \lambda\varphi(u) + (1-\lambda)\varphi(v)$; the cases $\lambda = 0, 1$ are equality.
If $\varphi'' \ge 0$, for $a < b$, MVT gives $\varphi'(b) - \varphi'(a) = \varphi''(d)(b - a) \ge 0$ for some $d \in (a,b)$, so φ' is increasing.
Structural Commentary: MVT orders secant slopes.
Key Insight: Monotone slopes give convexity. □

Example 1.7.4. *Basic Convex Functions*
By Proposition 1.7.3, $x \mapsto x^p$ for $p \ge 1$, $x \mapsto e^{ax}$, and $x \mapsto \log(\frac{1}{x})$ are convex on $(0, +\infty)$, $\mathbb{R}$, and $(0, +\infty)$.

Proposition 1.7.5. *Finite Jensen Inequality*
Let $I \subseteq \mathbb{R}$ be an interval, $\varphi : I \to \mathbb{R}$ be convex, $x_1, \dots, x_n \in I$, $\lambda_1, \dots, \lambda_n \ge 0$ with $n \in \mathbb{N}$, $\sum_{k=1}^{n} \lambda_k = 1$.
Set $\bar{x} := \sum_{k=1}^{n} \lambda_k x_k$. Then $\bar{x} \in I$ and $\varphi(\bar{x}) \le \sum_{k=1}^{n} \lambda_k \varphi(x_k)$.

Proof. Induct on $m = \#\{k : \lambda_k > 0\}$. If $m = 1$, equality holds.
For $m \ge 2$, relabel so $\lambda_1, \dots, \lambda_m > 0$ and $\lambda_k = 0$ for $k > m$. Let $\Lambda = 1 - \lambda_m$ and $y = \Lambda^{-1} \sum_{k=1}^{m-1} \lambda_k x_k$.
By induction, $y \in I$ and $\varphi(y) \le \Lambda^{-1} \sum_{k=1}^{m-1} \lambda_k \varphi(x_k)$.
Since $\bar{x} = \Lambda y + \lambda_m x_m$, convexity gives $\bar{x} \in I$ and $\varphi(\bar{x}) \le \Lambda\varphi(y) + \lambda_m\varphi(x_m) \le \sum_{k=1}^{n} \lambda_k \varphi(x_k)$.
Structural Commentary: Induct by peeling off one weight.
Key Insight: Jensen iterates convexity. □

Lemma 1.7.6. *One-Sided Slopes and Supporting Lines*
Let $I \subseteq \mathbb{R}$ be open and let $\varphi : I \to \mathbb{R}$ be convex.
(i) $\forall x \in I$, $\varphi'(x-)$ and $\varphi'(x+)$ exist and are finite.
(ii) If $u < v$ in I, then $\varphi'(u-) \le \varphi'(u+) \le \frac{\varphi(v) - \varphi(u)}{v - u} \le \varphi'(v-) \le \varphi'(v+)$.
(iii) If $\alpha \in I$ and $\varphi'(\alpha-) \le m_0 \le \varphi'(\alpha+)$, then $\forall t \in I$, $\varphi(t) \ge \varphi(\alpha) + m_0(t - \alpha)$.

Proof. For $a < b$ in I, write $S(a,b) := \frac{\varphi(b) - \varphi(a)}{b - a}$.

(i) Fix $x \in I$ and choose $a < x < b$ in I.
If $a < r < s < x$, then $S(a,x) \le S(r,x) \le S(s,x) \le S(x,b)$ by Lemma 1.7.2, so $\lim_{r \uparrow x} S(r,x)$ exists in $\mathbb{R}$.
If $x < r < s < b$, the same lemma gives $S(a,x) \le S(x,r) \le S(x,s) \le S(x,b)$, so $\lim_{r \downarrow x} S(x,r)$ exists in $\mathbb{R}$.

(ii) Let $u < v$ in I. For $r < u < s < v$ and $u < q < v < t$, the chordal slope inequalities give
$S(r,u) \le S(u,s) \le S(u,v) \le S(q,v) \le S(v,t)$. Letting $r \uparrow u$, $s \downarrow u$, $q \uparrow v$, and $t \downarrow v$ gives asserted chain.

(iii) If $t = \alpha$, equality holds. If $t > \alpha$, then (ii) gives $m_0 \le \varphi'(\alpha+) \le S(\alpha, t)$, hence $\varphi(t) \ge \varphi(\alpha) + m_0(t-\alpha)$.
If $t < \alpha$, then (ii) gives $S(t,\alpha) \le \varphi'(\alpha-) \le m_0$, which rearranges to the same estimate.

Structural Commentary: Monotone secant slopes have finite one-sided limits.
Key Insight: One-sided slope bounds produce supporting lines. □

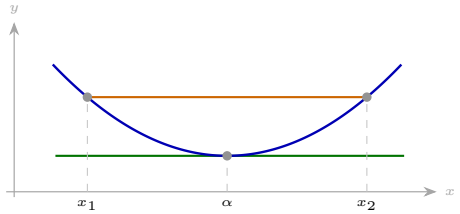

Figure 203: Color code: $y = \varphi(x)$ lies below the chord through x_1, x_2 and above the supporting line at α.

Theorem 1.7.7. *Jensen's Inequality on $[0,1]$*
Let m be Lebesgue measure on $[0,1]$, let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex, and let $f, \varphi \circ f \in L^1(m)$.
Then $\varphi\left(\int f\, dm\right) \le \int \varphi(f)\, dm$.

Proof. Set $\alpha := \int f\, dm$. By Lemma 1.7.6, choose $m_0 \in [\varphi'(\alpha-), \varphi'(\alpha+)]$; $\varphi(t) \ge \varphi(\alpha) + m_0(t-\alpha)\ \forall t \in \mathbb{R}$.
Substituting $t = f$ and integrating gives $\int \varphi(f)\, dm \ge \varphi(\alpha) + m_0\left(\int f\, dm - \alpha\right) = \varphi(\alpha)$.
Structural Commentary: Support at the mean.
Key Insight: The linear term vanishes. □

Remark 1.7.8. *Integrability in Jensen*
Linear growth, $|\varphi(t)| \le c_1 + c_2\,|t|$, makes $\varphi \circ f \in L^1([0,1])$ by Proposition 4.3.4.
If $\varphi \circ f \ge 0$, Jensen's right side may be extended-valued.

2 Product Measures and Fubini Theory

2.1 Product Measures

Product measure builds measurable rectangles and reads higher-dimensional integrals through slices. The abstract construction comes first; the Euclidean statements below are its geometric specialization.

Definition 2.1.1. *Product Measurable Spaces and Sections*
Let $(X, \mathcal{M})$, $(Y, \mathcal{N})$ be measurable spaces and $Z := X \times Y$.

(i) A *measurable rectangle* is $A \times B$ with $A \in \mathcal{M}$ and $B \in \mathcal{N}$.
Set $\mathcal{Z}_0 := \{ \bigcup_{j=1}^{n} (A_j \times B_j) : A_j \in \mathcal{M},\ B_j \in \mathcal{N} \}$ and $\mathcal{M} \otimes \mathcal{N} := \sigma\{A \times B : A \in \mathcal{M},\ B \in \mathcal{N}\}$.

(ii) For $E \subseteq Z$, the *sections* are $E_x := \{y \in Y : (x,y) \in E\}$ and $E^y := \{x \in X : (x,y) \in E\}$.

(iii) For $F : Z \to \overline{\mathbb{R}}$, the *function sections* are $F_x(y) := F(x,y)$ and $F^y(x) := F(x,y)$, where defined.

Lemma 2.1.2. *Finite Rectangle Algebra*
(i) If $R_j = A_j \times B_j$, then $R_1 \cap R_2 = (A_1 \cap A_2) \times (B_1 \cap B_2)$ and $R_1 \setminus R_2 = \left[(A_1 \cap A_2) \times (B_1 \setminus B_2)\right] \sqcup \left[(A_1 \setminus A_2) \times B_1\right]$.
(ii) $\mathcal{Z}_0$ is an algebra on Z.
(iii) $\forall E \in \mathcal{Z}_0\ \exists n \ge 1$, $A_j \in \mathcal{M}$, and $B_j \in \mathcal{N}$ such that $E = \bigsqcup_{j=1}^{n} (A_j \times B_j)$.

Proof.

(i) For $(x,y) \in Z$, $(x,y) \in R_1 \cap R_2 \iff x \in A_1 \cap A_2$ and $y \in B_1 \cap B_2$.
Also, $(x,y) \in R_1 \setminus R_2$ iff $x \in A_1$, $y \in B_1$, and either $x \notin A_2$ or $y \notin B_2$.
Splitting by $x \in A_2$ gives the disjoint union in (i).

(ii) If $E = \bigcup_j R_j$ and $F = \bigcup_k S_k$, then $E \cap F = \bigcup_{j,k} (R_j \cap S_k) \in \mathcal{Z}_0$ by (i).
Repeatedly applying (i) gives $R_j \setminus F \in \mathcal{Z}_0$, hence $E \setminus F = \bigcup_j (R_j \setminus F) \in \mathcal{Z}_0$.
Since $Z = X \times Y \in \mathcal{Z}_0$, complements are differences from Z.

(iii) If $E = \bigcup_{j=1}^{n} R_j$, set $S_j := R_j \setminus \bigcup_{i<j} R_i$. Then $S_j \in \mathcal{Z}_0$ by (ii), the S_j are disjoint, and $E = \bigsqcup_j S_j$.
Writing each S_j as a finite union of rectangles and relabeling gives required finite disjoint representation.

Structural Commentary: Product rectangles behave like elementary sets for the product space.
Key Insight: Before extending to a σ-algebra, make all finite overlaps into disjoint rectangle pieces. □

Theorem 2.1.3. *Product Measure Theorem*
For measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$, $\exists \pi : \mathcal{M} \otimes \mathcal{N} \to [0, +\infty]$ a measure such that $\forall (A,B) \in \mathcal{M} \times \mathcal{N}$, $\pi(A \times B) = \mu(A)\nu(B)$, where $0 \cdot (+\infty) = 0$. If μ, ν are σ-finite, π is unique; write $\pi = \mu \otimes \nu$.

Proof. For $E = \bigsqcup_{j=1}^{n} (A_j \times B_j) \in \mathcal{Z}_0$, set $\pi_0(E) := \sum_{j=1}^{n} \mu(A_j)\nu(B_j)$.
If $A \times B = \bigsqcup_{j \ge 1} (A_j \times B_j)$, then $\mathbf{1}_{A \times B} = \sum_{j \ge 1} \mathbf{1}_{A_j \times B_j}$, so monotone convergence in y, then x,
gives $\mu(A)\nu(B) = \sum_{j \ge 1} \mu(A_j)\nu(B_j)$. Refining two finite disjoint rectangle decompositions by intersections gives equal π_0-values; hence

π_0 is well defined. If $E = \bigsqcup_k E_k \in \mathcal{Z}_0$ with $E_k \in \mathcal{Z}_0$, write $E = \bigsqcup_i R_i$ and $E_k = \bigsqcup_\ell R_{k\ell}$.
Since $R_i = \bigsqcup_{k,\ell}(R_i \cap R_{k\ell})$, the rectangle comparison yields $\pi_0(E) = \sum_k \pi_0(E_k)$; hence π_0 is a premeasure.
Extend π_0 to $\sigma(\mathcal{Z}_0) = \mathcal{M} \otimes \mathcal{N}$. If μ, ν are σ-finite, then $\mathcal{Z}_0$ is π_0-σ-finite, so the extension is unique.
Structural Commentary: Rectangles fix the product measure.
Key Insight: Monotone convergence makes rectangle mass respect countable disjoint refinements. □

Lemma 2.1.4. *Measurable Sections*
(i) $E \in \mathcal{M} \otimes \mathcal{N}$ implies $\forall x \in X,\ E_x \in \mathcal{N}$ and $\forall y \in Y,\ E^y \in \mathcal{M}$.
(ii) If $F : Z \to \overline{\mathbb{R}}$ is $\mathcal{M} \otimes \mathcal{N}$-measurable, then F_x is $\mathcal{N}$-measurable $\forall x \in X$ and F^y is $\mathcal{M}$-measurable $\forall y \in Y$.

Proof.
(i) Fix $x \in X$ and set $\mathcal{C}_x := \{E \subseteq Z : E_x \in \mathcal{N}\}$. Since $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$, $\mathcal{C}_x$ is a σ-algebra. Also $(A \times B)_x = B$ if $x \in A$ and $\emptyset$ otherwise; hence $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{C}_x$. The E^y proof is identical.
(ii) For $a \in \mathbb{R}$, set $H_a := \{(u, v) \in Z : F(u, v) > a\} \in \mathcal{M} \otimes \mathcal{N}$. By (i), $(H_a)_x \in \mathcal{N}$ and $(H_a)^y \in \mathcal{M}$.
Since $(H_a)_x = \{y : F_x(y) > a\}$ and $(H_a)^y = \{x : F^y(x) > a\}$, all threshold sets are measurable.
Structural Commentary: Product measurability is stable under freezing one coordinate.
Key Insight: Section operations preserve the set algebra used to generate the product σ-algebra. □

Proposition 2.1.5. *Section Measures Integrate to Product Measure*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite, $\pi := \mu \otimes \nu$, and $E \in \mathcal{M} \otimes \mathcal{N}$. Set $f_E(x) := \nu(E_x)$ and $g_E(y) := \mu(E^y)$.
Then f_E, g_E are measurable and $\int_X f_E \, d\mu = \pi(E) = \int_Y g_E \, d\nu$.

Proof. Assume $\mu(X), \nu(Y) < +\infty$ and let $\mathcal{C}$ be the class of E satisfying the claim.
For $E = A \times B$, $f_E = \nu(B)\mathbf{1}_A$ and $g_E = \mu(A)\mathbf{1}_B$, so $A \times B \in \mathcal{C}$; finite additivity gives $\mathcal{Z}_0 \subseteq \mathcal{C}$.
If $E_n \uparrow E$ in $\mathcal{C}$, then $(E_n)_x \uparrow E_x$ and $(E_n)^y \uparrow E^y$; MCT gives $E \in \mathcal{C}$.
If $E_n \downarrow E$, finite total mass gives downward continuity and DCT, hence $E \in \mathcal{C}$.
Thus $\mathcal{C}$ is a monotone class containing $\mathcal{Z}_0$, so $\mathcal{C} = \mathcal{M} \otimes \mathcal{N}$. For σ-finite μ, ν, choose $X_n \uparrow X$, $Y_n \uparrow Y$ with $\mu(X_n), \nu(Y_n) < +\infty$, apply the finite case to $E_n := E \cap (X_n \times Y_n)$, then let $n \to \infty$ by MCT.
Structural Commentary: Cavalieri's principle is already true for abstract σ-finite products.
Key Insight: Prove rectangles, close under monotone limits, then exhaust by finite-measure rectangles. □

Perspective 2.1.6. *Rectangle Premeasure and the Completed Product*
Let $\mathcal{R} := \{A \times B : A \in \mathcal{M},\ B \in \mathcal{N}\}$ and define $\lambda(A \times B) := \mu(A)\nu(B)$ on $\mathcal{R}$.
(i) Lemma 2.1.2 and Theorem 2.1.3 imply that λ is a semiring premeasure.
(ii) Theorem 3.4.7 extends λ to a complete measure $\overline{\lambda}$ on the Caratheodory σ-algebra $\mathcal{C}_\lambda$.
(iii) The restriction $\overline{\lambda}|_{\mathcal{M}\otimes\mathcal{N}}$ is $\mu \otimes \nu$; the full space $(X \times Y, \mathcal{C}_\lambda, \overline{\lambda})$ is the completed product.
Thus $\mathcal{C}_\lambda$ may contain $N' \subseteq N$ with $N \in \mathcal{M} \otimes \mathcal{N}$, $(\mu \otimes \nu)(N) = 0$, and $N' \notin \mathcal{M} \otimes \mathcal{N}$.
This is why some section-measurability forms of Fubini require completeness.
Key Insight: Rectangles set mass; completion adds null subsets.

2.2 Fubini and Tonelli Theorems

Theorem 2.2.1. *Tonelli's Theorem for σ-Finite Products*
Let $(X, \mathcal{M}, \mu)$, $(Y, \mathcal{N}, \nu)$ be σ-finite, Let $Z := X \times Y$, $\pi := \mu \otimes \nu$.
Let $F : Z \to [0, +\infty]$ be $\mathcal{M} \otimes \mathcal{N}$-measurable. Put $I_F(x) := \int_Y F_x \, d\nu$ and $J_F(y) := \int_X F^y \, d\mu$.
(i) I_F is $\mathcal{M}$-measurable.
(ii) J_F is $\mathcal{N}$-measurable.
(iii) $\int_X I_F \, d\mu = \int_Z F \, d\pi = \int_Y J_F \, d\nu$.

Proof. For $F = \mathbf{1}_E$, the claims follow from Proposition 2.1.5.
Finite linearity gives them for nonnegative simple $\Phi = \sum_{j=1}^n a_j \mathbf{1}_{E_j}$. Choose simple $0 \le \Phi_n \uparrow F$.
(i) Since I_{Φ_n} is $\mathcal{M}$-measurable and $I_{\Phi_n}(x) = \int_Y (\Phi_n)_x \, d\nu \uparrow \int_Y F_x \, d\nu = I_F(x)$, I_F is $\mathcal{M}$-measurable.
(ii) Since J_{Φ_n} is $\mathcal{N}$-measurable and $J_{\Phi_n}(y) = \int_X (\Phi_n)^y \, d\mu \uparrow \int_X F^y \, d\mu = J_F(y)$, J_F is $\mathcal{N}$-measurable.
(iii) From $\int_X I_{\Phi_n} \, d\mu = \int_Z \Phi_n \, d\pi = \int_Y J_{\Phi_n} \, d\nu$, MCT in X, Z, and Y gives $\int_X I_F \, d\mu = \int_Z F \, d\pi = \int_Y J_F \, d\nu$.
Structural Commentary: Approximate by nonnegative simple functions.
Key Insight: MCT passes sections and integrals together. □

Theorem 2.2.2. *Fubini's Theorem for σ-Finite Products*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite, $\pi := \mu \otimes \nu$, and $F \in \mathscr{L}^1(\pi)$.
(i) $F_x \in \mathscr{L}^1(\nu)$ for μ-a.e. x and $F^y \in \mathscr{L}^1(\mu)$ for ν-a.e. y.
(ii) With $f(x) := \int_Y F_x \, d\nu$ and $g(y) := \int_X F^y \, d\mu$ on those full-measure sets, and 0 elsewhere, $f \in \mathscr{L}^1(\mu)$, $g \in \mathscr{L}^1(\nu)$, and $\int_X f \, d\mu = \int_Z F \, d\pi = \int_Y g \, d\nu$.

Proof.
(i) Tonelli on $|F|$ gives $\int_X (\int_Y |F_x| \, d\nu) d\mu = \int_Z |F| \, d\pi = \int_Y (\int_X |F^y| \, d\mu) d\nu < +\infty$, hence finite a.e.
(ii) Tonelli on F^+ and F^- gives finite identities; subtract them to obtain $\int_X f \, d\mu = \int_Z F \, d\pi = \int_Y g \, d\nu$.
Moreover $|f| \le \int_Y |F_x| \, d\nu$ and $|g| \le \int_X |F^y| \, d\mu$, so (i) gives $f, g \in \mathscr{L}^1$.
Structural Commentary: Fubini is Tonelli plus subtraction.
Key Insight: $\int |F| < +\infty$ makes the two nonnegative Tonelli identities finite enough to subtract. □

Proposition 2.2.3. *Product Borel and Measurable Building Blocks*
The following hold.

(i) $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$.
(ii) *If* $f : X \to \mathbb{R}$ *is* $\mathcal{M}$*-measurable and* $g : Y \to \mathbb{R}$ *is* $\mathcal{N}$*-measurable, then* $(x, y) \mapsto f(x)g(y)$ *is* $\mathcal{M} \otimes \mathcal{N}$*-measurable.*
(iii) *If* $u : \mathbb{R} \to \mathbb{R}$ *is Borel measurable, then* $(x, y) \mapsto u(x - y)$ *is* $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$*-measurable.*

Proof.

(i) Open rectangles give $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$.
Conversely, every open $O \subseteq \mathbb{R}^2$ is a countable union of rational rectangles, each in $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$.
(ii) For $\tilde{f}(x, y) := f(x)$ and $\tilde{g}(x, y) := g(y)$, $\{\tilde{f} > a\} = f^{-1}((a, +\infty)) \times Y$ and $\{\tilde{g} > a\} = X \times g^{-1}((a, +\infty))$.
Thus $\tilde{f}, \tilde{g}$ are measurable, so $\tilde{f}\tilde{g}$ is measurable.
(iii) The map $s(x, y) := x - y$ is continuous on $\mathbb{R}^2$, hence measurable by (i); therefore $u \circ s$ is measurable.

Structural Commentary: Product measurability supports two-variable formulas.
Key Insight: Check coordinate pullbacks on rectangles, then use ordinary measurable-function closure. □

Proposition 2.2.4. *Counting Factor Products*
Let $X = \mathbb{N}$, $\mathcal{M} = 2^{\mathbb{N}}$, μ *be counting measure, and* $(Y, \mathcal{N}, \nu)$ *be a measure space.*

(i) *For* $E \subseteq \mathbb{N} \times Y$, $E \in 2^{\mathbb{N}} \otimes \mathcal{N} \iff \forall n,\ E_n \in \mathcal{N}$, *and then* $E = \bigsqcup_{n \geq 1} (\{n\} \times E_n)$.
(ii) *The unique product measure is* $\pi = \mu \otimes \nu$ *with* $\pi(E) = \sum_{n \geq 1} \nu(E_n)$ *for* $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$.
(iii) *For* $F : \mathbb{N} \times Y \to \mathbb{R}$, F *is product-measurable* $\iff \forall n,\ F_n$ *is* $\mathcal{N}$*-measurable.*
(iv) *For product-measurable* F, $F \in \mathscr{L}^1(\pi) \iff \sum_{n \geq 1} \int_Y |F_n|\, d\nu < +\infty$;
then $\int_{\mathbb{N} \times Y} F\, d\pi = \sum_{n \geq 1} \int_Y F_n\, d\nu = \int_Y \sum_{n \geq 1} F_n\, d\nu$, *with* $\sum_n F_n$ *absolutely convergent* ν*-a.e.*

Proof.

(i) ($\Rightarrow$) If $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$, then $\forall n,\ E_n \in \mathcal{N}$ by Lemma 2.1.4.
(i) ($\Leftarrow$) If $\forall n,\ E_n \in \mathcal{N}$, then each $\{n\} \times E_n \in 2^{\mathbb{N}} \otimes \mathcal{N}$ and $E = \bigsqcup_{n \geq 1} (\{n\} \times E_n)$, hence $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$.
(ii) Define $\pi(E) := \sum_{n \geq 1} \nu(E_n)$.
If $(E^k)_k$ are disjoint, then $(\bigsqcup_k E^k)_n = \bigsqcup_k E^k_n$, so $\pi(\bigsqcup_k E^k) = \sum_n \sum_k \nu(E^k_n) = \sum_k \pi(E^k)$.
Also $\pi(\{n\} \times A) = \nu(A)$, and (i) forces uniqueness.
(iii) ($\Rightarrow$) If F is product-measurable, then $\forall n$, F_n is measurable by Lemma 2.1.4.
(iii) ($\Leftarrow$) If $\forall n$, F_n is measurable, then $\forall a \in \mathbb{R}$, $H_a := \{F > a\}$ satisfies $(H_a)_n = \{F_n > a\} \in \mathcal{N}$,
By (i), $H_a \in 2^{\mathbb{N}} \otimes \mathcal{N}$, so F is product-measurable.
(iv) For $G \geq 0$, (ii) gives the identity for $G = \mathbf{1}_E$; linearity and MCT give $\int_{\mathbb{N} \times Y} G\, d\pi = \sum_{n \geq 1} \int_Y G_n\, d\nu$.
(iv) Apply the preceding identity to $G = |F|$: $F \in \mathscr{L}^1(\pi) \iff \sum_{n \geq 1} \int_Y |F_n|\, d\nu < +\infty$.
If finite, then $\int_Y \sum_n |F_n|\, d\nu < +\infty$, so $\sum_n F_n$ converges absolutely a.e..
Applying the nonnegative identity to F^+ and F^- and subtracting gives the integral formulas.

Structural Commentary: A product with counting measure is a measurable sequence of fibers.
Key Insight: Counting atoms turn products into sums. □

Example 2.2.5. *A Diagonal Failure Without* σ*-Finiteness*
Let $X = Y = [0, 1]$, $\mathcal{M} = \mathcal{N} = \mathcal{B}([0, 1])$, $\mu = m$, $\nu = \#$, and $D := \{(x, y) \in [0, 1]^2 : x = y\}$.
Then $D \in \mathcal{M} \otimes \mathcal{N}$, $D_x = \{x\}$, and $D^y = \{y\}$, so $\nu(D_x) = 1$ and $\mu(D^y) = 0$.
Hence $\int_X \nu(D_x)\, d\mu = 1$ but $\int_Y \mu(D^y)\, d\nu = 0$, so Tonelli–Fubini can fail without σ-finite factors.

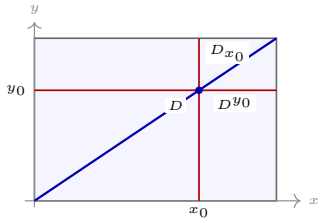

Figure 204: Diagonal sections: $\nu(D_x) = 1$ but $\mu(D^y) = 0$.

Example 2.2.6. *Nonunique Products Without* σ*-Finiteness*
Let $X = \mathbb{R}$, $\mathcal{M} = 2^{\mathbb{R}}$, and $\mu(A) = 0$ if A is countable, $\mu(A) = +\infty$ otherwise.
Define $\lambda_i : \mathcal{M} \otimes \mathcal{M} \to \{0, +\infty\}$ by $\lambda_i(E) = 0$ exactly on the following null ideals:
$\mathcal{I}_1 = \{E : E = G \cup H,\ \mathrm{pr}_x\, G$ countable, $\mathrm{pr}_y\, H$ countable$\}$
and $\mathcal{I}_2 = \{E : E = G \cup H \cup K,\ \mathrm{pr}_x\, G$ countable, $\mathrm{pr}_y\, H$ countable, $\mathrm{pr}_{y=x}\, K$ countable$\}$.
Both ideals are closed under countable unions, hence λ_i are measures.
For a rectangle $A \times B$, both measures equal $\mu(A)\mu(B)$: countable sides are null, and uncountable-by-uncountable rectangles are not in either ideal.
For $E = \{(x, y) : x + y = 0\}$, $\mathrm{pr}_{y=x}\, E$ is a point, so $\lambda_2(E) = 0$ but $\lambda_1(E) = +\infty$.

Structural Commentary: Without σ-finiteness, rectangle values need not determine a product measure.
Key Insight: Extra null lines preserve rectangles but change non-rectangles.

Example 2.2.7. *Double Series as Product Integrals*
With counting measure on $\mathbb{N}^2$, Tonelli gives, for $a_{mn} \geq 0$,
$\sum_m \sum_n a_{mn} = \sum_{(m,n)\in\mathbb{N}^2} a_{mn} = \sum_n \sum_m a_{mn} \in [0, +\infty]$.
Fubini needs integrability: set $a_{nn} = 1$, $a_{n,n+1} = -1$, and $a_{mn} = 0$ otherwise.
Then $\sum_m \sum_n a_{mn} = 0$, while $\sum_n \sum_m a_{mn} = 1$, and $\sum_{m,n} |a_{mn}| = +\infty$.

Proposition 2.2.8. *Separated Products*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite. Let $f \in \mathscr{L}^1(\mu)$, $g \in \mathscr{L}^1(\nu)$, and $H(x,y) := f(x)g(y)$.
(i) $H \in \mathscr{L}^1(\mu \otimes \nu)$.
(ii) $\int_{X\times Y} H\, d(\mu\otimes\nu) = (\int_X f\, d\mu)(\int_Y g\, d\nu)$.

Proof.
(i) Proposition 2.2.3 gives measurability. Tonelli gives $\int_{X\times Y} |H|\, d(\mu\otimes\nu) = (\int_X |f|\, d\mu)(\int_Y |g|\, d\nu) < +\infty$.
(ii) Since $H \in \mathscr{L}^1(\mu\otimes\nu)$, Fubini gives $\int_{X\times Y} H\, d(\mu\otimes\nu) = (\int_X f\, d\mu)(\int_Y g\, d\nu)$.
Structural Commentary: Tonelli handles size; Fubini handles sign.
Key Insight: Separated products integrate as products of integrals. □

Corollary 2.2.9. *Equal Section Measures*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite.
If $E, F \in \mathcal{M}\otimes\mathcal{N}$ and $\forall x \in X$, $\nu(E_x) = \nu(F_x)$, then $(\mu\otimes\nu)(E) = (\mu\otimes\nu)(F)$.

Proof. Set $f_E(x) := \nu(E_x)$ and $f_F(x) := \nu(F_x)$.
Since $f_E = f_F$, $(\mu\otimes\nu)(E) = \int_X f_E\, d\mu = \int_X f_F\, d\mu = (\mu\otimes\nu)(F)$ by Proposition 2.1.5.
Structural Commentary: Product mass is recovered from vertical section masses.
Key Insight: Equal fibers give equal product mass. □

Definition 2.2.10. *Convolution*
For Borel $f, g : \mathbb{R} \to \mathbb{R}$, if $y \mapsto f(x-y)g(y)$ is integrable for a.e. x,
define the *convolution* $(f * g)(x) := \int_{\mathbb{R}} f(x-y)g(y)\, dm(y)$ on that full-measure set and 0 elsewhere.

Proposition 2.2.11. *Convolution of L^1 Functions*
Let $f, g \in \mathscr{L}^1(m)$ be Borel.
(i) $f * g$ is defined for a.e. $x \in \mathbb{R}$.
(ii) $f * g \in \mathscr{L}^1(m)$.
(iii) $\int_{\mathbb{R}} |f * g|\, dm \leq (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm)$.

Proof. Set $\Phi(x,y) := f(x-y)g(y)$. By Proposition 2.2.3, Φ is product-measurable.
For fixed y, translation invariance gives $\int_{\mathbb{R}} |f(x-y)|\, dm(x) = \int_{\mathbb{R}} |f|\, dm$.
Hence Tonelli yields $\int_{\mathbb{R}^2} |\Phi|\, d(m\otimes m) = (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm) < +\infty$.
(i) Fubini gives $y \mapsto \Phi(x,y) \in \mathscr{L}^1(m)$ for a.e. x, so $f * g$ is a.e. defined.
(ii) Since $|f * g(x)| \leq \int_{\mathbb{R}} |\Phi(x,y)|\, dm(y)$ a.e., Fubini gives $\int_{\mathbb{R}} |f * g|\, dm < +\infty$.
(iii) The same estimate gives $\int_{\mathbb{R}} |f * g|\, dm \leq \int_{\mathbb{R}^2} |\Phi|\, d(m\otimes m) = (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm)$.
Structural Commentary: Apply Fubini to $\Phi(x,y) = f(x-y)g(y)$.
Key Insight: Translation invariance fixes the x-slice norm. □

Definition 2.2.12. *Products and Slices*
For $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^k$, set $A \times B \subseteq \mathbb{R}^{n+k}$.
For $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ define $E_x := \{y \in \mathbb{R}^k : (x,y) \in E\}$ as *x-slice*, and $E^y := \{x \in \mathbb{R}^n : (x,y) \in E\}$ as *y-slice*.
For $f : E \to \mathbb{R}$, write $f_x(y) = f^y(x) := f(x,y)$ where defined.

Theorem 2.2.13. *Measurable Products*
Let $A \in \mathcal{L}^n$ and $B \in \mathcal{L}^k$.
(i) $A \times B \in \mathcal{L}^{n+k}$.
(ii) $m_{n+k}(A\times B) = m_n(A)m_k(B)$, with $0 \cdot \infty = \infty \cdot 0 = 0$.

Proof. Boxes satisfy $m_{n+k}(I\times J) = m_n(I)m_k(J)$, hence so do open sets by disjoint box decompositions.
For bounded finite-measure A, B, choose $F_A \subseteq A \subseteq G_A$, $F_B \subseteq B \subseteq G_B$ with F_A, F_B closed, G_A, G_B bounded open, and $m_n(G_A \setminus F_A), m_k(G_B \setminus F_B) < \varepsilon$.
Then $F_A \times F_B \subseteq A\times B \subseteq G_A \times G_B$ and $m_n(G_A)m_k(G_B) - m_n(F_A)m_k(F_B) \to 0$.
(i) The squeeze proves $A\times B \in \mathcal{L}^{n+k}$ in the bounded finite-measure case.
For general A, B, with $A_N := A \cap [-N,N]^n$ and $B_N := B \cap [-N,N]^k$, $A\times B = \bigcup_N (A_N \times B_N) \in \mathcal{L}^{n+k}$.
(ii) The squeeze also gives $m_{n+k}(A\times B) = m_n(A)m_k(B)$ in the bounded finite-measure case.
Since $A_N \times B_N \uparrow A \times B$, $m_{n+k}(A\times B) = \lim_N m_n(A_N)m_k(B_N) = m_n(A)m_k(B)$, with $0\cdot\infty = \infty\cdot 0 = 0$.
Structural Commentary: Product measure comes from boxes and regularity.
Key Insight: Regularity turns rough measurable products into box products plus zero-set errors. □

Theorem 2.2.14. *Zero Slices*
If $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ is Lebesgue measurable, then E is null iff E_x is null for a.e. x.

Proof.
(⇒) If $m(E) = 0$, cover E by an open W with small measure.
For $X_\alpha(W) = \{x : m(W_x) > \alpha\}$, $m(W) \geq \alpha m(X_\alpha(W))$; letting $m(W) \to 0$ gives $m(E_x) = 0$ for a.e. x.

($\Leftarrow$) Remove the exceptional null base so every slice is null. For compact $K \subseteq E$, cover each K_x by a small-measure slice neighborhood; compactness gives finitely many product neighborhoods covering K, whose total measure is arbitrarily small by Theorem 2.2.13. Thus $m(K) = 0$, hence $m(E) = 0$ by regularity.

Structural Commentary: Nullity can be checked fiberwise after removing an exceptional base set.
Key Insight: Product measure turns small slice covers into small total measure. □

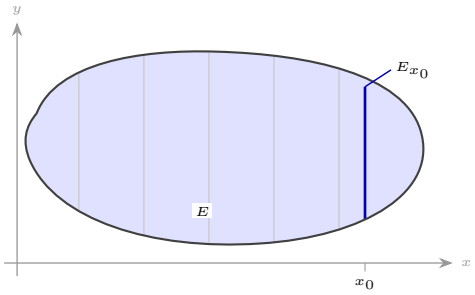

Figure 205: A planar set can be read by vertical slices; Fubini integrates the slice data over the base variable.

Theorem 2.2.15. *Cavalieri's Principle for Lebesgue Measure*
Let $n, k \in \mathbb{N}$ and $E \in \mathcal{L}^{n+k}$. For $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^k$, set $E_x := \{y : (x, y) \in E\}$ and $E^y := \{x : (x, y) \in E\}$.
(i) $E_x \in \mathcal{L}^k$ for a.e. x, and $f_E(x) := m_k(E_x)$ has a measurable representative with $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$.
(ii) $E^y \in \mathcal{L}^n$ for a.e. y, and $g_E(y) := m_n(E^y)$ has a measurable representative with $m_{n+k}(E) = \int_{\mathbb{R}^k} g_E \, dm_k$.

Proof. Let $\mathsf{C}_x(E)$ denote: $E_x \in \mathcal{L}^k$ for a.e. x, f_E has a measurable representative, and $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$.
(i) Rectangles satisfy $\mathsf{C}_x(A \times B)$ since $f_{A\times B} = m_k(B)\mathbf{1}_A$; finite additivity gives finite unions.
For bounded $Q = I \times J$, $\mathcal{C}_Q := \{E \subseteq Q : \mathsf{C}_x(E)\}$ is monotone:
$E_j \uparrow E$ gives $f_{E_j} \uparrow f_E$ and MCT, while $E_j \downarrow E$ gives $f_{E_j} \downarrow f_E \le m_k(J)\mathbf{1}_I$ and DCT.
Thus $\mathcal{C}_Q$ contains all Borel $E \subseteq Q$, and exhaustion by $Q_N = [-N, N]^{n+k}$ gives all Borel E.
For $E \in \mathcal{L}^{n+k}$, choose Borel B with $m_{n+k}(E \triangle B) = 0$; then $m_k((E \triangle B)_x) = 0$ a.e. by Theorem 2.2.14.
Hence $E_x \in \mathcal{L}^k$ a.e., $f_E = f_B$ a.e., and $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$.
(ii) Let $\tau : \mathbb{R}^k \times \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^k$ be $\tau(y, x) = (x, y)$ and put $\widetilde{E} := \tau^{-1}(E)$.
Apply (i) to $\widetilde{E} \subseteq \mathbb{R}^k \times \mathbb{R}^n$ gives $\widetilde{E}_y = E^y$ a.e., $g_E(y) = m_n(\widetilde{E}_y)$, $m_{n+k}(E) = m_{n+k}(\widetilde{E}) = \int_{\mathbb{R}^k} g_E \, dm_k$.

Structural Commentary: Slices reduce product measure to base integrals.
Key Insight: Extend from rectangles, then pass null differences. □

Perspective 2.2.16. *Fubini Proof Ladder*
Fubini ladder: $\mathbf{1}_E \to$ simple $\Phi \ge 0 \to f \ge 0$ by MCT, then $f = f^+ - f^-$ when $\int |f| < +\infty$.
Theorems 2.2.1 and 2.2.2 are the abstract form.
Key Insight: Fubini is a slice theorem, proof runs through indicators, simple functions, monotone convergence.

Proposition 2.2.17. *Undergraphs and Level-Set Measurability*
Let $f : \mathbb{R}^n \to [0, \infty]$, set $F_a := \{x : f(x) > a\}$, and let $\mathcal{U}f := \{(x, t) \in \mathbb{R}^n \times [0, \infty) : 0 < t < f(x)\}$.
(i) f is Lebesgue measurable iff $\forall q \in \mathbb{Q}_{>0}$, $F_q \in \mathcal{L}^n$.
(ii) If f is Lebesgue measurable, then $\mathcal{U}f \in \mathcal{L}^{n+1}$ and $\forall t > 0$, $(\mathcal{U}f)^t = F_t$.

Proof.
(i) ($\Rightarrow$) If f is measurable, each $F_q = f^{-1}((q, +\infty])$ is in $\mathcal{L}^n$.

(i) ($\Leftarrow$) If $\forall q \in \mathbb{Q}_{>0}$, $F_q \in \mathcal{L}^n$, then for $a \ge 0$, $F_a = \bigcup_{q\in\mathbb{Q},\ q>a} F_q \in \mathcal{L}^n$, while for $a < 0$, $F_a = \mathbb{R}^n$.
Thus all strict superlevel sets are measurable, so f is measurable.
(ii) By (i), every F_r with $r \in \mathbb{Q}_{>0}$ is measurable. Since $\mathcal{U}f = \bigcup_{q,r\in\mathbb{Q},\ 0<q<r} F_r \times (q, r)$, $\mathcal{U}f \in \mathcal{L}^{n+1}$.
For $t > 0$, $(\mathcal{U}f)^t = \{x : 0 < t < f(x)\} = F_t$.

Structural Commentary: Measurability is encoded by rational superlevels.
Key Insight: Undergraph slices are superlevel sets. □

Theorem 2.2.18. *Fubini–Tonelli for Nonnegative Functions*
Let $f : \mathbb{R}^n \times \mathbb{R}^k \to [0, \infty]$ be Lebesgue measurable.
Define $F(x) := \int_{\mathbb{R}^k} f_x \, dm_k$, f_x is measurable and arbitrarily elsewhere; define $G(y) := \int_{\mathbb{R}^n} f^y \, dm_n$ analogously.
(i) f_x is measurable for a.e. x, F is measurable, and $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^n} F \, dm_n$.
(ii) f^y is measurable for a.e. y, G is measurable, and $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^k} G \, dm_k$.

Proof. Set $U := \mathcal{U}f \subseteq \mathbb{R}^{n+k} \times [0, \infty)$. Then $U \in \mathcal{L}^{n+k+1}$ by Proposition 2.2.17.
(i) By Cavalieri, U_x is measurable for a.e. x, $H(x) := m_{k+1}(U_x)$ is measurable, and $m_{n+k+1}(U) = \int_{\mathbb{R}^n} H \, dm_n$.
For such x, $U_x = \mathcal{U}(f_x)$, so f_x is measurable and $H(x) = \int_{\mathbb{R}^k} f_x \, dm_k = F(x)$.
Since $m_{n+k+1}(U) = \int_{\mathbb{R}^{n+k}} f \, dm_{n+k}$, (i) follows.
(ii) Let $\tau(y, x) = (x, y)$ and $\tilde{f} := f \circ \tau$.
Applying (i) to $\tilde{f}$ gives $\tilde{f}_y = f^y$ a.e., $\tilde{F} = G$, and $\int_{\mathbb{R}^k} G \, dm_k = \int_{\mathbb{R}^{k+n}} \tilde{f} \, dm_{k+n} = \int_{\mathbb{R}^{n+k}} f \, dm_{n+k}$.

Structural Commentary: Cavalieri applied to the undergraph.
Key Insight: $(\mathcal{U}f)_x = \mathcal{U}(f_x)$. □

Theorem 2.2.19. *Fubini for L^1 Functions*
Let $f \in L^1(\mathbb{R}^2)$.

(i) $f(x,\cdot) \in L^1(\mathbb{R})$ for a.e. x; defining $F(x) := \int_{\mathbb{R}} f(x,y)\,dy$ there and $F = 0$ elsewhere gives $F \in L^1(\mathbb{R})$.
(ii) $f(\cdot,y) \in L^1(\mathbb{R})$ for a.e. y; defining $G(y) := \int_{\mathbb{R}} f(x,y)\,dx$ there and $G = 0$ elsewhere gives $G \in L^1(\mathbb{R})$.
(iii) $\int_{\mathbb{R}} F\,dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G\,dy$.

Proof. From Theorems 2.2.13–2.2.14, the identity holds for indicators.
By linearity and MCT, every measurable $h \geq 0$ satisfies $(*)$ $\int_{\mathbb{R}}(\int_{\mathbb{R}} h(x,y)\,dy)dx = \int_{\mathbb{R}^2} h = \int_{\mathbb{R}}(\int_{\mathbb{R}} h(x,y)\,dx)dy$.

(i) Applying $(*)$ to $|f|$ gives $\int_{\mathbb{R}}\int_{\mathbb{R}} |f(x,y)|\,dy\,dx = \int_{\mathbb{R}^2} |f| < +\infty$, so $f(x,\cdot) \in L^1(\mathbb{R})$ a.e.
Let $F^{\pm}(x) := \int_{\mathbb{R}} f^{\pm}(x,y)\,dy$; then $F = F^+ - F^-$ a.e. and $\int_{\mathbb{R}} |F|\,dx \leq \int_{\mathbb{R}}(F^+ + F^-)\,dx = \int_{\mathbb{R}^2} |f| < +\infty$.
(ii) Let $\tau(y,x) = (x,y)$ and $\tilde{f} := f \circ \tau$. Since τ preserves m_2, $\tilde{f} \in L^1(\mathbb{R}^2)$.
Applying (i) to $\tilde{f}$ gives $\tilde{f}(y,\cdot) = f(\cdot,y) \in L^1(\mathbb{R})$ a.e. and $\tilde{F} = G \in L^1(\mathbb{R})$.
(iii) Let $G^{\pm}(y) := \int_{\mathbb{R}} f^{\pm}(x,y)\,dx$. By $(*)$, $\int_{\mathbb{R}} F^{\pm}\,dx = \int_{\mathbb{R}^2} f^{\pm} = \int_{\mathbb{R}} G^{\pm}\,dy$. Since these quantities are finite and $F = F^+ - F^-$, $G = G^+ - G^-$ a.e., subtraction gives $\int_{\mathbb{R}} F\,dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G\,dy$.

Structural Commentary: Indicators, simple functions, MCT, then $f^{\pm}$.
Key Insight: Absolute integrability permits subtraction. □

2.3 Linear Changes of Variables and Euclidean Invariance

Perspective 2.3.1. *Coordinate Splitting as a Product Identification*
For $n = m + k$, let $\Phi : \mathbb{R}^n \to \mathbb{R}^m \times \mathbb{R}^k$ be $\Phi(x_1,\dots,x_n) = ((x_1,\dots,x_m),(x_{m+1},\dots,x_n))$. Then Φ sends rectangles to product rectangles and preserves rectangular volume, so m_n is the completion of $\Phi_{\#}^{-1}(m_m \otimes m_k)$. Equivalently, integration over $\mathbb{R}^{m+k}$ matches iterated integration over $\mathbb{R}^m \times \mathbb{R}^k$ whenever Fubini applies.
Key Insight: Coordinate splitting preserves Lebesgue mass.

Proposition 2.3.2. *Lipschitz Images of Lebesgue Sets*
Let $\Psi : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz and let $E \in \mathcal{L}^n$. Then $\Psi(E) \in \mathcal{L}^n$.
Hence every linear map $\mathbb{R}^n \to \mathbb{R}^n$ preserves Lebesgue measurability.

Proof. Let L be a Lipschitz constant.

(i) $\forall I \subseteq \mathbb{R}^n$ bounded cells, $m_n^*(\Psi(I)) \leq C_{n,L}\,\mathrm{vol}_n(I)$. Indeed, partition I into grid cells of side at most δ.
Each image has diameter at most $L\sqrt{n}\,\delta$, hence lies in a cube of volume at most $C_{n,L}\delta^n$. Summing and letting $\delta \downarrow 0$ gives the estimate.
(ii) If $m_n^*(N) = 0$ and $\varepsilon > 0$, cover N by cells (I_k) with $\sum_k \mathrm{vol}_n(I_k) < \varepsilon$.
Then $\Psi(N) \subseteq \bigcup_k \Psi(I_k)$, so (i) gives $m_n^*(\Psi(N)) \leq C_{n,L}\varepsilon$; hence $m_n^*(\Psi(N)) = 0$.
(iii) If E is bounded, choose F_σ set $F \subseteq E$ with $m_n^*(E \setminus F) = 0$. Write $F = \bigcup_j F_j$ with each F_j compact.
Then $\Psi(F) = \bigcup_j \Psi(F_j)$ is Borel, $\Psi(E \setminus F)$ is null by (ii), and $\Psi(E) = \Psi(F) \cup \Psi(E \setminus F) \in \mathcal{L}^n$.
(iv) For general E, use $E = \bigcup_{r \geq 1} (E \cap [-r,r]^n)$ and apply (iii) to each bounded term.

Structural Commentary: Reduce to compact pieces plus null error.
Key Insight: Lipschitz maps preserve null sets. □

Theorem 2.3.3. *Linear Change of Variables for Nonnegative Functions*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible and let $f : \mathbb{R}^n \to [0,+\infty]$ be Lebesgue measurable.

(i) $f \circ T$ is Lebesgue measurable.
(ii) $\int f \circ T\,dm_n = |\det T|^{-1} \int f\,dm_n$.

Proof.

(i) Since $(f \circ T)^{-1}((a,+\infty]) = T^{-1}(f^{-1}((a,+\infty]))$ and T^{-1} is Lipschitz, Proposition 2.3.2 gives measurability.
(ii) The $n = 1$ identity follows from $m_1(\alpha E + \beta) = |\alpha|\,m_1(E)$.
Let $\mathcal{M}_+$ denote the measurable maps $\mathbb{R}^n \to [0,+\infty]$, set $J_S(h) := \int h(Sx)\,dm_n(x)$.
Set $\mathcal{A}_n := \{S \in GL(n) : J_S(h) = |\det S|^{-1} \int h\,dm_n\ \forall h \in \mathcal{M}_+\}$.
If $S_1, S_2 \in \mathcal{A}_n$, then $J_{S_1S_2}(h) = |\det S_2|^{-1} J_{S_1}(h) = |\det(S_1S_2)|^{-1} \int h\,dm_n$, so $S_1S_2 \in \mathcal{A}_n$.
The generators $D_j(c) : x_j \mapsto cx_j$, $P_{j,j+1} : x_j \leftrightarrow x_{j+1}$, and $H_{jk}(c) : x_j \mapsto x_j + cx_k$ lie in $\mathcal{A}_n$ for $c \neq 0$ by the $n = 1$ identity, Fubini, and slice translation invariance, with determinant factors $|c|$, 1, and 1.
Row reduction gives $T \in \mathcal{A}_n$; hence $J_T(f) = |\det T|^{-1} \int f\,dm_n$.

Structural Commentary: Verify elementary row operations, then compose.
Key Insight: Only scalings change mass. □

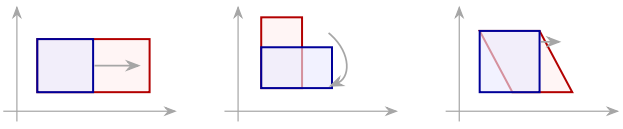

Figure 206: Original and image cells under scale, swap, and shear; Fubini checks each coordinatewise.

Corollary 2.3.4. *Invertible Linear Images of Lebesgue Sets*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible and let $E \subseteq \mathbb{R}^n$.

(i) $E \in \mathcal{L}^n \iff T(E) \in \mathcal{L}^n$.
(ii) If $E \in \mathcal{L}^n$, then $m_n(T(E)) = |\det T|\, m_n(E)$.

Proof.

(i) Since T and T^{-1} are Lipschitz, Proposition 2.3.2 gives $E \in \mathcal{L}^n \iff T(E) \in \mathcal{L}^n$.
(ii) For $E \in \mathcal{L}^n$, Theorem 2.3.3 applied to $\mathbf{1}_{T(E)}$ gives $m_n(E) = \int \mathbf{1}_{T(E)}(Tx)\, dm_n(x) = |\det T|^{-1}\, m_n(T(E))$.

Structural Commentary: Measurability is Lipschitz; mass follows from indicators.
Key Insight: Indicators turn integral change into set change. □

Corollary 2.3.5. *Linear Change of Variables for L^1 Functions*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ be invertible, let $f \in \mathscr{L}^1(m_n)$, and set $c_T := |\det T|^{-1}$.

(i) $f \circ T \in \mathscr{L}^1(m_n)$.
(ii) $\int f \circ T\, dm_n = c_T \int f\, dm_n$.

Proof.

(i) Apply Theorem 2.3.3 to $|f|$: $\int |f \circ T|\, dm_n = c_T \int |f|\, dm_n < +\infty$, hence $f \circ T \in \mathscr{L}^1(m_n)$.
(ii) Apply Theorem 2.3.3 to $f^{\pm}$: $\int f^{\pm} \circ T\, dm_n = c_T \int f^{\pm}\, dm_n < +\infty$.
Hence $\int f \circ T\, dm_n = \int f^+ \circ T\, dm_n - \int f^- \circ T\, dm_n = c_T \int f\, dm_n$.

Structural Commentary: Reduce signed integrals to finite positive parts.
Key Insight: Absolute integrability permits subtraction. □

Corollary 2.3.6. *Linear Images and Rigid Motions*
Let $E \in \mathcal{L}^n$.

(i) If $T \in \mathcal{L}(\mathbb{R}^n)$ is singular, then $T(E)$ is Lebesgue measurable and $m_n(T(E)) = 0$.
(ii) If $\Psi(x) = Ux + x_0$, where U is a linear isometry, then $\Psi(E) \in \mathcal{L}^n$ and $m_n(\Psi(E)) = m_n(E)$.

Proof.

(i) $T(E) \in \mathcal{L}^n$ by Proposition 2.3.2. Set $W := \operatorname{range} T$, $H := \mathbb{R}^{n-1} \times \{0\}$; choose invertible S with $S(W) \subseteq H$.
Since $H = \bigcup_N (H \cap [-N, N]^n)$ is null, $m_n(W) = 0$ by Corollary 2.3.4; hence $m_n(T(E)) = 0$.
(ii) Since U is an isometry, $|\det U| = 1$ by Proposition 1.1.24.
Thus $U(E) \in \mathcal{L}^n$ and $m_n(U(E)) = m_n(E)$ by Corollary 2.3.4.
By translation invariance, $\Psi(E) = U(E) + x_0 \in \mathcal{L}^n$ and $m_n(\Psi(E)) = m_n(U(E)) = m_n(E)$.

Structural Commentary: Singular maps land in null subspaces; rigid motions preserve determinant mass.
Key Insight: Lebesgue measure records determinant size, not the coordinate description of the set. □

3 Radon-Nikodym and Differentiation

3.1 Signed and Absolutely Continuous Measures

This subsection first decomposes finite signed set functions into measures, then compares measures by absolute continuity and singularity. Later subsections develop the local Euclidean counterpart of the Radon–Nikodym viewpoint.

Definition 3.1.1. *Positive, Negative, and Null Sets for a Charge*
Let λ be a charge on $(X, \mathcal{M})$.

(i) $\mathcal{P}_\lambda := \{P \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap P) \geq 0\}$, the *positive* sets.
(ii) $\mathcal{N}_\lambda := \{N \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap N) \leq 0\}$, the *negative* sets.
(iii) $\mathcal{Z}_\lambda := \{M \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap M) = 0\}$, the *null* sets.

Proposition 3.1.2. *Positive and Negative Set Closure*
For a charge λ:

(i) $A \in \mathcal{M}$ and $A \subseteq P \in \mathcal{P}_\lambda$ imply $A \in \mathcal{P}_\lambda$.
(ii) $A \in \mathcal{M}$ and $A \subseteq N \in \mathcal{N}_\lambda$ imply $A \in \mathcal{N}_\lambda$.
(iii) $\mathcal{P}_\lambda$ is closed under countable unions.
(iv) $\mathcal{N}_\lambda$ is closed under countable unions.

Proof.

(i) If $A \subseteq P \in \mathcal{P}_\lambda$, then $\lambda(E \cap A) = \lambda((E \cap A) \cap P) \geq 0\ \forall E \in \mathcal{M}$.
(ii) If $A \subseteq N \in \mathcal{N}_\lambda$, then $\lambda(E \cap A) = \lambda((E \cap A) \cap N) \leq 0\ \forall E \in \mathcal{M}$.
(iii) Let $(P_k) \subseteq \mathcal{P}_\lambda$ and set $P := \bigcup_{k \geq 1} P_k$. For $E \in \mathcal{M}$, set $F_1 := E \cap P_1$ and $F_k = E \cap P_k \setminus \bigcup_{j=1}^{k-1} P_j$ for $k \geq 2$.
Then (F_k) is pairwise disjoint, $\bigcup_k F_k = E \cap P$, and $F_k \subseteq P_k$, so $\lambda(F_k) \geq 0$.
Countable additivity gives $\lambda(E \cap P) = \sum_k \lambda(F_k) \geq 0$.
(iv) Apply the disjointization in (iii) to $(N_k) \subseteq \mathcal{N}_\lambda$; every disjoint piece has nonpositive charge, so $\bigcup_k N_k \in \mathcal{N}_\lambda$.

Structural Commentary: Sign sets are hereditary and countable-union stable.
Key Insight: Test sets can always be disjointized against a countable sign cover. □

Theorem 3.1.3. *Hahn Decomposition Theorem*
Let λ be a charge on $(X, \mathcal{M})$. Then $\exists P, N \in \mathcal{M}$ such that

(i) $P \cap N = \emptyset$ *and* $X = P \sqcup N$.

(ii) $P \in \mathcal{P}_\lambda$.
(iii) $N \in \mathcal{N}_\lambda$.
Such a pair (P, N) *is a Hahn decomposition of* X *for* λ.

Proof.

(i) Put $\alpha = \sup\{\lambda(A) : A \in \mathcal{P}_\lambda\}$. If $\alpha = +\infty$, choose $B_m \in \mathcal{P}_\lambda$, $\lambda(B_m) > m$, and set $C_m = \bigcup_{j=1}^{m} B_j$; then $C_m \in \mathcal{P}_\lambda$, $C_m \uparrow C$, and $\lambda(C_m) \geq \lambda(B_m) > m$, contradicting continuity from below and $\lambda(C) \in \mathbb{R}$.
Choose $A_n \in \mathcal{P}_\lambda$, $\lambda(A_n) \to \alpha$, replace A_n by $\bigcup_{j=1}^{n} A_j$, and set $P = \bigcup_{n\geq 1} A_n$, $N = X \setminus P$.
Hence $P, N \in \mathcal{M}$, $P \cap N = \emptyset$, and $X = P \sqcup N$.
(ii) For $E \in \mathcal{M}$, $E \cap A_n \uparrow E \cap P$ and $\lambda(E \cap A_n) \geq 0$, so continuity from below gives $\lambda(E \cap P) \geq 0$.
Thus $P \in \mathcal{P}_\lambda$, and $\lambda(P) = \alpha$.
(iii) Suppose $E \in \mathcal{M}$, $E \subseteq N$, and $\lambda(E) > 0$. Then $E \notin \mathcal{P}_\lambda$, otherwise $P \cup E \in \mathcal{P}_\lambda$ and $\lambda(P \cup E) > \alpha$.
Set $R_0 := E$. Inductively, if $R_{k-1} \notin \mathcal{P}_\lambda$, let $n_k := \min\{n \geq 1 : \exists H \in \mathcal{M},\ H \subseteq R_{k-1},\ \lambda(H) \leq -\frac{1}{n}\}$.
Choose such E_k, and set $R_k := R_{k-1} \setminus E_k$. If some $R_m \in \mathcal{P}_\lambda$, then $\lambda(R_m) = \lambda(E) - \sum_{k\leq m} \lambda(E_k) > 0$, so $P \cup R_m \in \mathcal{P}_\lambda$ and $\lambda(P \cup R_m) > \alpha$, impossible. Hence all $R_k \notin \mathcal{P}_\lambda$. Put $F_m := \bigcup_{k\leq m} E_k$ and $F := \bigcup_{k\geq 1} E_k$.
Since $\lambda(F_m) = \sum_{k\leq m} \lambda(E_k) \leq -\sum_{k\leq m} \frac{1}{n_k}$ and $F_m \uparrow F$, by finiteness of $\lambda(F)$, $\sum_k \frac{1}{n_k} < +\infty$, hence $n_k \to +\infty$.
If $G \in \mathcal{M}$ and $G \subseteq E \setminus F$, then $G \subseteq R_{k-1}$ $\forall k$, by minimality $\lambda(G) > -\frac{1}{n_k - 1}$ $\forall k \gg 1$; hence $\lambda(G) \geq 0$ and $E \setminus F \in \mathcal{P}_\lambda$. Since $\lambda(E \setminus F) = \lambda(E) - \lambda(F) > 0$, $P \cup (E \setminus F) \in \mathcal{P}_\lambda$ and $\lambda(P \cup (E \setminus F)) > \alpha$, contradiction.

Structural Commentary: A charge splits X into nonnegative-piece and nonpositive-piece regions.
Key Insight: Maximize positive mass; purify any leftover positive piece. □

Lemma 3.1.4. *Hahn Parts Are Unique up to Null Sets*
Let (P_i, N_i), $i = 1, 2$, *be Hahn decompositions for a charge* λ. $\forall E \in \mathcal{M}$*:*
(i) $\lambda(E \cap P_1) = \lambda(E \cap P_2)$.
(ii) $\lambda(E \cap N_1) = \lambda(E \cap N_2)$.

Proof. Fix $E \in \mathcal{M}$. For $\{i, j\} = \{1, 2\}$,

(i) $E \cap (P_i \setminus P_j) \subseteq P_i \cap N_j$, so $\lambda(E \cap (P_i \setminus P_j)) = 0$. Hence $\lambda(E \cap P_i) = \lambda(E \cap P_1 \cap P_2)$ for $i = 1, 2$.
(ii) $E \cap (N_i \setminus N_j) \subseteq N_i \cap P_j$, so $\lambda(E \cap (N_i \setminus N_j)) = 0$. Hence $\lambda(E \cap N_i) = \lambda(E \cap N_1 \cap N_2)$ for $i = 1, 2$.

Structural Commentary: Hahn decompositions are not unique as sets, but the measures they define are unique.
Key Insight: Positive-and-negative overlap is necessarily null. □

Definition 3.1.5. *Jordan Variations of a Charge*
Let λ be a charge, (P, N) a Hahn decomposition, and $E \in \mathcal{M}$.
(i) $\lambda^+(E) := \lambda(E \cap P)$, the *positive variation*.
(ii) $\lambda^-(E) := -\lambda(E \cap N)$, the *negative variation*.
(iii) $|\lambda|(E) := \lambda^+(E) + \lambda^-(E)$, the *total variation*.
These are independent of (P, N) by Lemma 3.1.4.

Theorem 3.1.6. *Jordan Decomposition Theorem*
Let λ *be a charge.*
(i) $\lambda = \lambda^+ - \lambda^-$.
(ii) If $\lambda = \mu - \nu$ *for finite measures* μ, ν*, then* $\lambda^+ \leq \mu$.
(iii) Under the same hypothesis, $\lambda^- \leq \nu$.

Proof. Let (P, N) be a Hahn decomposition and fix $E \in \mathcal{M}$.

(i) $\lambda(E) = \lambda(E \cap P) + \lambda(E \cap N) = \lambda^+(E) - \lambda^-(E)$.
(ii) $\lambda^+(E) = \mu(E \cap P) - \nu(E \cap P) \leq \mu(E)$.
(iii) $\lambda^-(E) = \nu(E \cap N) - \mu(E \cap N) \leq \nu(E)$.

Structural Commentary: Jordan decomposition is the minimal finite-measure difference.
Key Insight: Positive and negative Hahn pieces isolate the irreducible positive masses. □

Corollary 3.1.7. *Variation by Subsets and Null Sets*
For a charge λ *and* $E, M \in \mathcal{M}$*:*
(i) $\lambda^+(E) = \sup\{\lambda(F) : F \in \mathcal{M}, F \subseteq E\}$.
(ii) $\lambda^-(E) = -\inf\{\lambda(F) : F \in \mathcal{M}, F \subseteq E\}$.
(iii) $M \in \mathcal{Z}_\lambda \Leftrightarrow |\lambda|(M) = 0$.
(iv) $\pi_\lambda = \lambda^+$.
(v) $\nu_\lambda = |\lambda|$.

Proof.

(i) Let (P, N) be Hahn. If $F \in \mathcal{M}$ and $F \subseteq E$, then $\lambda(F) = \lambda^+(F) - \lambda^-(F) \leq \lambda^+(E)$, with equality at $F = E \cap P$.
(ii) Under the same hypotheses, $\lambda(F) \geq -\lambda^-(E)$, with equality at $F = E \cap N$.
(iii) ($\Rightarrow$) If $M \in \mathcal{Z}_\lambda$, then $\lambda^+(M) = \lambda^-(M) = 0$, so $|\lambda|(M) = 0$.

(iii) ($\Leftarrow$) If $|\lambda|(M) = 0$, then $\lambda^\pm(G \cap M) = 0$ $\forall G \in \mathcal{M}$, hence $M \in \mathcal{Z}_\lambda$.
(iv) This is (i) and the definition of π_λ.
(v) $\sum_j |\lambda(A_j)| \leq |\lambda|(E)$ $\forall$ finite measurable partitions (A_j) of E; the Hahn partition attains equality.

Structural Commentary: Variation is intrinsic to λ.
Key Insight: The positive part is the largest charge obtainable inside a set. □

Theorem 3.1.8. *Charges Form a Banach Space*
Let $\mathrm{ca}(X,\mathcal{M}) := \{\lambda : \mathcal{M} \to \mathbb{R} : \lambda \text{ is a charge}\}$ *and* $\|\lambda\|_{\mathrm{var}} := |\lambda|(X)$.
With pointwise operations, $(\mathrm{ca}(X,\mathcal{M}), \|\cdot\|_{\mathrm{var}})$ *is Banach.*

Proof.

(i) Charge linearity gives closure under pointwise operations. The variation axioms follow from $|\lambda|(X) = 0 \Rightarrow |\lambda|(E) = 0\ \forall E$, $|a\lambda| = |a|\,|\lambda|$, and $\sum_j |(\lambda+\eta)(A_j)| \le \sum_j |\lambda(A_j)| + \sum_j |\eta(A_j)|$ over finite measurable partitions (A_j).

(ii) Let (λ_n) be variation-Cauchy.
Since $|\lambda_n(E) - \lambda_m(E)| \le \|\lambda_n - \lambda_m\|_{\mathrm{var}}$, set $\lambda(E) := \lim_n \lambda_n(E)$; then λ is finitely additive.
For fixed n and finite disjoint (A_j), $\sum_j |(\lambda - \lambda_n)(A_j)| \le \limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}}$.

(iii) For disjoint (E_k), set $E := \bigcup_k E_k$, $R_K := \bigcup_{k>K} E_k$, and choose n with $\limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}} < \varepsilon$.
Since $\lambda_n(R_K) \to 0$, for large K, $|\lambda(R_K)| \le |\lambda_n(R_K)| + |(\lambda - \lambda_n)(R_K)| < 2\varepsilon$.
Hence $\lambda(R_K) \to 0$, so λ is countably additive.

(iv) Taking suprema in (ii) over finite partitions gives $\|\lambda - \lambda_n\|_{\mathrm{var}} \le \limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}} \to 0$.
Hence every variation-Cauchy sequence converges in variation to a charge.

Structural Commentary: Total variation is the signed-measure analogue of an L^1 norm.
Key Insight: Variation-Cauchy controls every finite partition. □

Theorem 3.1.9. *Variations of Integral Charges*
Let $f \in \mathscr{L}^1(\mu)$ *and* $\lambda(E) := \int_E f\,d\mu$. *For* $E \in \mathcal{M}$:
(i) $\lambda^+(E) = \int_E f^+\,d\mu$.
(ii) $\lambda^-(E) = \int_E f^-\,d\mu$.
(iii) $|\lambda|(E) = \int_E |f|\,d\mu$.

Proof. Set $P := \{f \ge 0\}$ and $N := \{f < 0\}$.
For $A \in \mathcal{M}$, $\lambda(A \cap P) = \int_{A\cap P} f\,d\mu \ge 0$ and $\lambda(A \cap N) = \int_{A\cap N} f\,d\mu \le 0$, so (P,N) is Hahn. Fix $E \in \mathcal{M}$.

(i) $\lambda^+(E) = \lambda(E \cap P) = \int_E f^+\,d\mu$.
(ii) $\lambda^-(E) = -\lambda(E \cap N) = \int_E f^-\,d\mu$.
(iii) $|\lambda|(E) = \lambda^+(E) + \lambda^-(E) = \int_E |f|\,d\mu$.

Structural Commentary: Measure variations match function parts.
Key Insight: Sign sets of the density are Hahn sets. □

Definition 3.1.10. *Absolute Continuity and Singularity*
Let λ, μ be measures and let η be a charge on $(X,\mathcal{M})$.

(i) *Absolute continuity for measures*: $\lambda \ll \mu \Leftrightarrow \forall E \in \mathcal{M}, \mu(E) = 0 \Rightarrow \lambda(E) = 0$.
(ii) *Absolute continuity for charges*: $\eta \ll \mu \Leftrightarrow |\eta| \ll \mu$.
(iii) *Mutual singularity*: $\lambda \perp \mu \Leftrightarrow \exists A, B \in \mathcal{M}$ with $A \cap B = \emptyset$, $X = A \cup B$, $\lambda(A) = 0$, and $\mu(B) = 0$.

Proposition 3.1.11. *Absolute Continuity Calculus*
Let λ, μ, ν *be measures and* η *a charge.*
(i) $\lambda \ll \lambda$.
(ii) $\lambda \ll \mu \ll \nu \Rightarrow \lambda \ll \nu$.
(iii) $\ll$ *is not symmetric.*
(iv) $(\forall E \in \mathcal{M}, \mu(E) = 0 \Rightarrow \eta(E) = 0) \Rightarrow \eta^+, \eta^-, |\eta| \ll \mu$.

Proof.

(i) Reflexivity follows directly from the null-set definition.
(ii) If $\lambda \ll \mu \ll \nu$ and $\nu(E) = 0$, then $\mu(E) = 0$, hence $\lambda(E) = 0$.
(iii) On $X = \{0,1\}$, $\mathcal{M} = 2^X$, $\delta_0 \ll \#$, but $\# \not\ll \delta_0$ since $\delta_0(\{1\}) = 0$ and $\#\{1\} = 1$.
(iv) If $\mu(E) = 0$ and $F \in \mathcal{M}$ satisfies $F \subseteq E$, then $\mu(F) = 0$ and $\eta(F) = 0$.
Hence $E \in \mathcal{Z}_\eta$ and $|\eta|(E) = 0$ by Corollary 3.1.7; therefore $\eta^{\pm}(E) \le |\eta|(E) = 0$.

Structural Commentary: Absolute continuity is null-set order; charges reduce to variation.
Key Insight: Null sets pass to measurable subsets, so all signed pieces vanish there. □

Corollary 3.1.12. *Absolutely Continuous and Singular Forces Zero*
Let λ, μ *be measures and let* η *be a charge.*
(i) $\lambda \ll \mu$ *and* $\lambda \perp \mu$ *imply* $\lambda = 0$.
(ii) $|\eta| \perp \mu$ *implies* $\eta^+ \perp \mu$.
(iii) $|\eta| \perp \mu$ *implies* $\eta^- \perp \mu$.

Proof.

(i) Choose $X = A \cup B$ with $\lambda(A) = 0$ and $\mu(B) = 0$. Since $\lambda \ll \mu$, $\lambda(B) = 0$, hence $\lambda(X) = 0$.
(ii) Choose $X = A \cup B$ with $|\eta|(A) = 0$ and $\mu(B) = 0$. Since $\eta^+ \le |\eta|$, $\eta^+(A) = 0$.
(iii) With the same $X = A \cup B$, $\eta^-(A) \le |\eta|(A) = 0$ and $\mu(B) = 0$; hence $\eta^- \perp \mu$.

Structural Commentary: Absolute continuity and singular support are incompatible except at zero.
Key Insight: A measure supported on a null set must vanish when it also respects that null ideal. □

Example 3.1.13. *Dominating a Countable Family of Measures*
Let (μ_n) be measures with $\mu_n(X) \le 1$, and set $\lambda(E) := \sum_{n\ge 1} 2^{-n}\mu_n(E)$.
Then λ is finite, since $\lambda(X) \le 1$, and countably additive by Tonelli for nonnegative series.
The bound $2^{-n}\mu_n(E) \le \lambda(E)$ implies $\lambda(E) = 0 \Rightarrow \mu_n(E) = 0$ and $\mu_n \ll \lambda\ \forall n$.
Key Insight: A summable positive mixture packages countably many null-set tests into one reference measure.

Lemma 3.1.14. *Epsilon–Delta Form of Absolute Continuity*
Let λ be finite and μ a measure on $(X, \mathcal{M})$. Then $\lambda \ll \mu \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0, \forall E \in \mathcal{M}, \mu(E) < \delta \Rightarrow \lambda(E) < \varepsilon$.

Proof.
($\Rightarrow$) Suppose the ε-δ condition fails. Then $\exists \varepsilon > 0$ and $E_n \in \mathcal{M}$ with $\mu(E_n) < 2^{-n}$ and $\lambda(E_n) \ge \varepsilon$.
Let $F_n = \bigcup_{k\ge n} E_k$ and $F = \bigcap_n F_n$. Then $\mu(F_n) \le 2^{-n+1}$, so $\mu(F) = 0$.
Since $F_n \downarrow F$, $\lambda(F_n) \ge \varepsilon$, and λ is finite, continuity from above gives $\lambda(F) \ge \varepsilon$, contradicting $\lambda \ll \mu$.
($\Leftarrow$) If $\mu(E) = 0$, then $\forall \delta > 0,\ \mu(E) < \delta$; hence $\forall \varepsilon > 0,\ \lambda(E) < \varepsilon$, so $\lambda(E) = 0$.
Structural Commentary: For finite target measure, absolute continuity is uniform smallness on small μ-sets.
Key Insight: A failing sequence is converted into a decreasing null tail with positive λ-mass. □

Example 3.1.15. *Finite Target Is Needed*
On $(\mathbb{N}, 2^{\mathbb{N}})$, let $\lambda = \#$ and $\mu(E) := \sum_{n\in E} 2^{-n}$. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$.
But for $E_n = \{n\}$, $\mu(E_n) = 2^{-n} \to 0$ while $\lambda(E_n) = 1$, so the ε-δ condition fails for $\varepsilon = 1$.

Example 3.1.16. *Mutual Absolute Continuity Need Not Preserve Finiteness*
On $(\mathbb{N}, 2^{\mathbb{N}})$, set $\mu(E) := \sum_{n\in E} 2^{-n}$ and $\lambda(\emptyset) = 0$, $\lambda(E) = +\infty$ for $E \ne \emptyset$.
Then $\mu(\mathbb{N}) = 1$, while λ is not σ-finite. Since both measures have null ideal $\{\emptyset\}$, $\lambda \ll \mu$ and $\mu \ll \lambda$.
Key Insight: Absolute continuity compares null sets; it does not compare finiteness, σ-finiteness, or size.

3.2 Radon-Nikodym Theorem

Theorem 3.2.1. *Radon–Nikodym Theorem*
Let λ, μ be σ-finite measures on $(X, \mathcal{M})$ with $\lambda \ll \mu$.
(i) Representation: $\exists$ measurable $f : X \to [0, +\infty)$ such that $\lambda(E) = \int_E f\, d\mu$ for $E \in \mathcal{M}$.
(ii) Uniqueness: Any two representing densities agree μ-a.e.
The μ-a.e. class of f is denoted by $\frac{d\lambda}{d\mu}$.

Proof.
(i) Assume first $\lambda(X), \mu(X) < +\infty$. For $c > 0$, let (P_k, N_k) be Hahn for $\lambda - kc\mu$, set $A_k := N_k \setminus \bigcup_{j<k} A_j$, and set $B := X \setminus \bigcup_{k\ge 1} A_k$.
If $k \ge 2$, then $A_k \subseteq N_k \cap P_{k-1}$.
Hence $E \subseteq A_k$ implies $(k-1)c\mu(E) \le \lambda(E) \le kc\mu(E)$, with the lower bound $0 \le \lambda(E)$ when $k = 1$.
Since $B \subseteq P_k$ for $k \ge 1$, $\lambda(B) \ge kc\mu(B)\ \forall k \ge 1$; finiteness and $\lambda \ll \mu$ give $\mu(B) = \lambda(B) = 0$. Set $f_c := \sum_{k\ge 1} (k-1)c\, \mathbf{1}_{A_k}$.
Countable additivity over (A_k) gives $0 \le \lambda(E) - \int_E f_c\, d\mu \le c\mu(E)$ for $E \in \mathcal{M}$.
Let $c_n := 2^{-n}$ and $f_n := f_{c_n}$. For $E \in \mathcal{M}$, $\left|\int_E (f_n - f_m)\, d\mu\right| \le (c_n + c_m)\mu(E)$; by Proposition 1.2.20, $|f_n - f_m| \le c_n + c_m$ μ-a.e. Thus $\sum_n \|f_{n+1} - f_n\|_1 < +\infty$. By Tonelli, f_n converges μ-a.e. and in $L^1(\mu)$ to a measurable $f \ge 0$. Letting $n \to \infty$ in the approximation inequality gives $\lambda(E) = \int_E f\, d\mu$.
For the σ-finite case, choose a measurable partition $X = \bigsqcup_r Y_r$ with $\lambda(Y_r), \mu(Y_r) < +\infty$. Apply the finite case on each Y_r, obtaining f_r, and set $f := \sum_r f_r \mathbf{1}_{Y_r}$.
Then $\lambda(E) = \sum_r \lambda(E \cap Y_r) = \int_E f\, d\mu$; also $f < +\infty$ μ-a.e. because $\int_{Y_r} f\, d\mu = \lambda(Y_r) < +\infty\ \forall r$.
(ii) If f, g represent λ, then $f, g \in L^1(\mu|_{Y_r})$ and $f = g$ μ-a.e. on Y_r by Proposition 1.2.20.
Taking the countable union of the exceptional sets gives $f = g$ μ-a.e. on X.
Structural Commentary: Hahn decompositions of $\lambda - c\mu$ locate density bands.
Key Insight: Summably refined band approximations converge without invoking later L^1 completeness. □

Example 3.2.2. *A Non-σ-Finite Failure*
Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}([0, 1])$, $\mu = \#$, and $\lambda = m$. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$.
If $\lambda(E) = \int_E f\, d\mu$ for some $f \ge 0$, then $\forall x$, $0 = \lambda(\{x\}) = \int_{\{x\}} f\, d\mu = f(x)$, so $\lambda([0, 1]) = 0$, contradiction.
Thus RN can fail when μ is not σ-finite.

Example 3.2.3. *Another Non-σ-Finite Failure*
Let X be uncountable, $\mathcal{M} := \{E \subseteq X : E \text{ or } X \setminus E \text{ countable}\}$, $\mu(E) := \#E$ if E is finite and $+\infty$ otherwise, and $\lambda(E) := 0$ if E is countable and $+\infty$ otherwise. Then $\mu(E) = 0 \Leftrightarrow E = \emptyset$, so $\lambda \ll \mu$.
If $\lambda(E) = \int_E f\, d\mu$ with $f \ge 0$, then $\forall x$, $0 = \lambda(\{x\}) = \int_{\{x\}} f\, d\mu = f(x)$, forcing $\lambda(X) = 0$, contradiction.
Structural Commentary: Singleton tests determine f, but not $\lambda(X)$ without σ-finiteness.
Key Insight: Non-σ-finite domination can lose global mass.

Remark 3.2.4. *When the Radon–Nikodym Derivative Is Integrable*
In Theorem 3.2.1, $\frac{d\lambda}{d\mu} \in L^1(\mu) \Leftrightarrow \lambda(X) < +\infty$, since $\int_X \frac{d\lambda}{d\mu}\, d\mu = \lambda(X)$.

Theorem 3.2.5. *Extended-Valued Radon–Nikodym Theorem*
Let μ be σ-finite, let λ be a measure on $(X, \mathcal{M})$, and assume $\lambda \ll \mu$.
(i) Representation: $\exists f : X \to [0, +\infty]$ measurable such that $\lambda(E) = \int_E f\, d\mu$ for $E \in \mathcal{M}$.
(ii) Uniqueness: Any two representing densities agree μ-a.e.

Proof.

(i) Assume first $\mu(X) < +\infty$ and set $\alpha := \sup\{\mu(A) : A \in \mathcal{M}, \lambda(A) < +\infty\}$. Choose $(A_n) \subseteq \mathcal{M}$ with $\lambda(A_n) < +\infty$ and $\mu(A_n) \to \alpha$, and set $C_n := \bigcup_{j \le n} A_j$, $N := \bigcup_n C_n$, and $P := X \setminus N$. Then $\lambda|_N$ is σ-finite.

For $E \subseteq P$, $\lambda(E) < +\infty \Rightarrow \mu(E) = 0$, since $\mu(E) > 0$ would give $\mu(C_n \cup E) = \mu(C_n) + \mu(E) > \alpha$ eventually. Hence $\mu(E) > 0 \Rightarrow \lambda(E) = +\infty$, while $\mu(E) = 0 \Rightarrow \lambda(E) = 0$ by $\lambda \ll \mu$.
By Theorem 3.2.1, $\exists g : N \to [0, +\infty)$ measurable such that $\lambda(E \cap N) = \int_{E \cap N} g \, d\mu$. Define $f = g$ on N and $f = +\infty$ on P; preceding dichotomy gives $\lambda(E) = \int_E f \, d\mu$. For σ-finite μ, $X = \bigsqcup_r Y_r$, $\mu(Y_r) < +\infty$.

Let f_r be the finite-μ density on Y_r and set $f := \sum_r f_r \mathbf{1}_{Y_r}$. Then $\lambda(E) = \sum_r \lambda(E \cap Y_r) = \int_E f \, d\mu$.

(ii) If f, h represent λ, fix $r, m, n \in \mathbb{N}$ and set $A^r_{m,n} := Y_r \cap \{f \le m\} \cap \{h \ge f + \frac{1}{n}\}$.
Since $\int_A f \, d\mu = \int_A h \, d\mu$ for measurable $A \subseteq Y_r$ and $\int_{A^r_{m,n}} f \, d\mu < +\infty$, $\mu(A^r_{m,n}) = 0$.

Taking the union over r, m, n gives $\mu\{f < h\} = 0$; symmetry gives $\mu\{h < f\} = 0$.

Structural Commentary: Maximal finite-λ regions separate ordinary density from infinite density.
Key Insight: Outside the maximal σ-finite part, every positive-μ set has infinite λ-mass. □

Theorem 3.2.6. *Lebesgue Decomposition Theorem*
Let λ, μ be σ-finite measures on $(X, \mathcal{M})$.

(i) $\exists$ measures λ_s, λ_a with $\lambda = \lambda_s + \lambda_a$, $\lambda_s \perp \mu$, and $\lambda_a \ll \mu$.
(ii) If $(\tilde\lambda_s, \tilde\lambda_a)$ is another such pair, then $\tilde\lambda_s = \lambda_s$ and $\tilde\lambda_a = \lambda_a$.

Proof.

(i) Put $\rho := \lambda + \mu$ and choose $f = \frac{d\lambda}{d\rho}$, $g = \frac{d\mu}{d\rho}$. With $A := \{g = 0\}$ and $B := \{g > 0\}$, set $\lambda_s(E) := \lambda(E \cap A)$ and $\lambda_a(E) := \lambda(E \cap B)$.
Then $\lambda = \lambda_s + \lambda_a$. Since $\mu(A) = \int_A g \, d\rho = 0$, $\lambda_s \perp \mu$.
If $\mu(E) = 0$, then $g = 0$ ρ-a.e. on E, so $\rho(E \cap B) = 0$ and $\lambda_a(E) = 0$; hence $\lambda_a \ll \mu$.

(ii) Let $(\tilde\lambda_s, \tilde\lambda_a)$ be another such pair. Since $\lambda_s \perp \mu$ and $\tilde\lambda_a \ll \mu$, $\lambda_s \perp \tilde\lambda_a$; choose $X = C \cup D$ with $\lambda_s(C) = 0$ and $\tilde\lambda_a(D) = 0$. Then $\lambda_s(E) = \lambda_s(E \cap D) \le \lambda(E \cap D) = \tilde\lambda_s(E \cap D) \le \tilde\lambda_s(E)$.
By symmetry, $\lambda_s = \tilde\lambda_s$, hence $\lambda_a = \tilde\lambda_a$.

Structural Commentary: Split λ into μ-visible and μ-singular parts.
Key Insight: Use $\rho = \lambda + \mu$ and split where $\frac{d\mu}{d\rho} = 0$. □

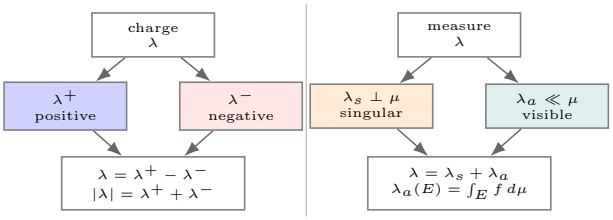

Figure 207: Charges split by sign; measures split into $\lambda_s \perp \mu$ and $\lambda_a \ll \mu$ parts.

Proposition 3.2.7. *Radon–Nikodym Calculus*
Let all measures below be σ-finite.

(i) $\lambda \ll \mu$, $g \ge 0$ measurable $\Rightarrow \int_X g \, d\lambda = \int_X g \frac{d\lambda}{d\mu} \, d\mu$.
(ii) $\nu \ll \lambda \ll \mu \Rightarrow \frac{d\nu}{d\mu} = \frac{d\nu}{d\lambda}\frac{d\lambda}{d\mu}$ μ-a.e.
(iii) $\lambda_1, \lambda_2 \ll \mu \Rightarrow \frac{d(\lambda_1+\lambda_2)}{d\mu} = \frac{d\lambda_1}{d\mu} + \frac{d\lambda_2}{d\mu}$ μ-a.e.
(iv) $\lambda \ll \mu \ll \lambda \Rightarrow \frac{d\lambda}{d\mu} = \left(\frac{d\mu}{d\lambda}\right)^{-1}$ μ-a.e.

Proof.

(i) For $g = \mathbf{1}_E$, $\int_X g \, d\lambda = \lambda(E) = \int_E \frac{d\lambda}{d\mu} \, d\mu = \int_X g \frac{d\lambda}{d\mu} \, d\mu$.
By linearity, identity holds for nonnegative simple g.
For measurable $g \ge 0$, take simple $g_n \uparrow g$ and apply monotone convergence to both sides.

(ii) Let $h := \frac{d\nu}{d\lambda}\frac{d\lambda}{d\mu}$. $\forall E \in \mathcal{M}$, (i) with $g = \mathbf{1}_E \frac{d\nu}{d\lambda}$ gives $\nu(E) = \int_E \frac{d\nu}{d\lambda} \, d\lambda = \int_E h \, d\mu$.
Hence h represents ν relative to μ, so $h = \frac{d\nu}{d\mu}$ μ-a.e.

(iii) Let $h := \frac{d\lambda_1}{d\mu} + \frac{d\lambda_2}{d\mu}$. $\forall E \in \mathcal{M}$, $\int_E h \, d\mu = \lambda_1(E) + \lambda_2(E) = (\lambda_1 + \lambda_2)(E)$.
Hence h represents $\lambda_1 + \lambda_2$ relative to μ, so uniqueness gives the identity.

(iv) Apply (ii) to $\mu \ll \lambda \ll \mu$. Since $\frac{d\mu}{d\mu} = 1$, $1 = \frac{d\mu}{d\lambda}\frac{d\lambda}{d\mu}$ μ-a.e.
Since $\mu \ll \lambda$, $\frac{d\mu}{d\lambda} > 0$ μ-a.e.; divide to obtain $\frac{d\lambda}{d\mu} = \left(\frac{d\mu}{d\lambda}\right)^{-1}$ μ-a.e.

Structural Commentary: Radon–Nikodym derivatives obey chain and linearity rules.
Key Insight: Prove identities by integrating over arbitrary measurable sets, then invoke uniqueness. □

Corollary 3.2.8. *Radon–Nikodym for Charges*
Let μ be σ-finite and let λ be a charge with $\lambda \ll \mu$. Then $\exists f \in \mathscr{L}^1(\mu)$ such that $\forall E \in \mathcal{M}$:

(i) $\lambda(E) = \int_E f \, d\mu$.
(ii) $\lambda^+(E) = \int_E f^+ \, d\mu$.
(iii) $\lambda^-(E) = \int_E f^- \, d\mu$.
(iv) $|\lambda|(E) = \int_E |f| \, d\mu$.

Proof. Since $|\lambda| \ll \mu$, $\lambda^{\pm} \ll \mu$. By Theorem 3.2.1, choose $f_{\pm} \geq 0$ with $\lambda^{\pm}(E) = \int_E f_{\pm}\, d\mu$.
Mutual singularity of λ^+ and λ^- gives $f_+ f_- = 0$ μ-a.e. Set $f := f_+ - f_-$.
Then $f^+ = f_+$ and $f^- = f_-$ μ-a.e., while $\int_X |f|\, d\mu = |\lambda|(X) < +\infty$.

(i) $\lambda(E) = \lambda^+(E) - \lambda^-(E) = \int_E f\, d\mu$.
(ii) $\lambda^+(E) = \int_E f_+\, d\mu = \int_E f^+\, d\mu$.
(iii) $\lambda^-(E) = \int_E f_-\, d\mu = \int_E f^-\, d\mu$.
(iv) $|\lambda|(E) = \lambda^+(E) + \lambda^-(E) = \int_E |f|\, d\mu$.

Structural Commentary: Signed absolute continuity is reduced to the two positive Jordan pieces.
Key Insight: Differentiate λ^+ and λ^- separately, then subtract. □

Example 3.2.9. *Radon–Nikodym Classification of Basic Stieltjes Generators*
Let m be Lebesgue measure on $\mathcal{B}(\mathbb{R})$, and let λ_i be the Borel–Stieltjes measure generated by g_i.

(i) $g_1(x) = 2x$: $\lambda_1 = 2m$, so $\lambda_1 \ll m$, $\frac{d\lambda_1}{dm} = 2$, $m \ll \lambda_1$, $\frac{dm}{d\lambda_1} = \frac{1}{2}$, and $\lambda_1(\mathbb{R}) = +\infty$.
(ii) $g_2(x) = \arctan x$: $\lambda_2(E) = \int_E (1+x^2)^{-1}\, dm$, so $\frac{d\lambda_2}{dm} = (1+x^2)^{-1}$, $\lambda_2(\mathbb{R}) = \pi$, $m \ll \lambda_2$, and $\frac{dm}{d\lambda_2} = 1 + x^2$.
(iii) $g_3 = \mathbf{1}_{[0,+\infty)}$: $\lambda_3 = \delta_0$, so $\lambda_3 \perp m$, $\lambda_3(\mathbb{R}) = 1$, $\lambda_3 \not\ll m$, and $m \not\ll \lambda_3$.
(iv) $g_4(x) = x\mathbf{1}_{[0,+\infty)}(x)$: $\lambda_4(E) = m(E \cap [0,+\infty))$, so $\lambda_4 \ll m$, $\frac{d\lambda_4}{dm} = \mathbf{1}_{[0,+\infty)}$, $m \not\ll \lambda_4$, and $\lambda_4(\mathbb{R}) = +\infty$.

Thus $\lambda_1, \lambda_2, \lambda_4 \ll m$, $\lambda_3 \perp m$, only λ_2, λ_3 are finite, and $m \ll \lambda_i$ exactly for $i = 1, 2$.
Structural Commentary: Stieltjes generators split naturally into density, atom, and restriction examples.
Key Insight: A positive density makes null sets equivalent; an atom lives on a Lebesgue-null set.

3.3 Differentiation of Measures

Definition 3.3.1. *Averages and Absolute Continuity*

(i) *Average*: for $f \in \mathscr{L}^1(E)$ and measurable $A \subseteq E$ with $m(A) > 0$, set $[f : A] := m(A)^{-1} \int_A f\, dm$.
(ii) *Density ratio*: for measurable B and cube or ball Q, $m(Q) > 0$, set $[B : Q] := \frac{m(B \cap Q)}{m(Q)}$.
(iii) *Indefinite integral*: for $f \in \mathscr{L}^1([a,b])$, $F(x) = F(a) + \int_a^x f(t)\, dt$.
(iv) *Absolute continuity*: $F : [a,b] \to \mathbb{R}$ is absolutely continuous iff $\forall \varepsilon > 0\ \exists \delta > 0$: if $(a_j, b_j)_{j=1}^N \subseteq [a,b]$ is a finite pairwise disjoint family, then $\sum_j (b_j - a_j) < \delta \Rightarrow \sum_j |F(b_j) - F(a_j)| < \varepsilon$.

Perspective 3.2.2. *Absolute Continuity as Interval Bookkeeping*
If $F(x) = \int_a^x f(t)\, dt$, absolute continuity controls total F-variation on disjoint intervals of small total length.

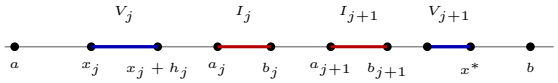

Figure 208: Interval bookkeeping for absolute continuity.

Perspective 3.3.3. *Vitali Covering Control*
Select balls disjointly by decreasing radius.
If B is rejected, then $B \cap B_j \neq \emptyset$ for some earlier B_j with $r(B) \leq r(B_j)$, hence $B \subseteq 5B_j$.
Key Insight: Disjoint balls add; fixed dilations cover misses.

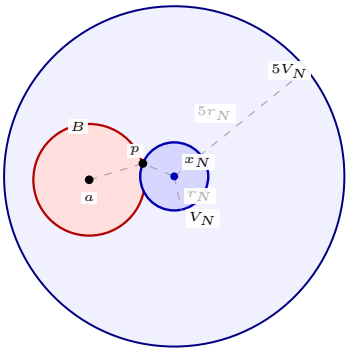

Figure 209: A comparable missed ball lies inside the fivefold dilation of the selected ball.

Definition 3.3.4. *Vitali Coverings and Density Points*
Let (M, d) be a metric space and let $A \subseteq M$.

(i) $\mathcal{V}$ is a *Vitali covering* of A iff $\forall p \in A\ \forall r > 0\ \exists V \in \mathcal{V}$ such that $p \in V$ and $\operatorname{diam}(V) < r$.
(ii) For measurable $E \subseteq \mathbb{R}^n$, its *density-point set* is $\operatorname{dp}(E) := \{p \in \mathbb{R}^n : \lim_{Q \downarrow p} [E : Q] = 1\}$, with Q cubes, equivalently balls, shrinking to p.

Theorem 3.3.5. *Vitali Covering Lemma for Balls and Cubes*
Let $E \subseteq \mathbb{R}^n$, $m^*(E) < +\infty$, and let $\mathcal{V}$ be a Vitali cover of E by closed balls or axis-parallel cubes.
Then $\exists (V_j)_{j \geq 1} \subseteq \mathcal{V}$ pairwise disjoint with $m^*(E \setminus \bigcup_{j \geq 1} V_j) = 0$.

Proof. Fix open $O \supseteq E$ with $m(O) < m^*(E) + \varepsilon$, and replace $\mathcal{V}$ by $\{V \in \mathcal{V} : V \subseteq O\}$.
Greedily choose a maximal disjoint (V_j) by dyadic diameter scales.
For a fixed dilation $V \mapsto V^*$, if $W \cap V \neq \emptyset$ and $\operatorname{diam} W \leq 2 \operatorname{diam} V$, then $W \subseteq V^*$ and $m(V^*) \leq C_n m(V)$.
Let $R = E \setminus \bigcup_j V_j$. Fix N and $p \in R$.

Since $V_1, \ldots, V_N$ are closed, choose $W \in \mathcal{V}$ with $p \in W$ and $W \cap \bigcup_{j \leq N} V_j = \emptyset$. Maximality gives some $j > N$ with $W \cap V_j \neq \emptyset$ and comparable diameter, hence $p \in W \subseteq V_j^*$. Thus $R \subseteq \bigcup_{j>N} V_j^*$.

Therefore $m^*(R) \leq C_n \sum_{j>N} m(V_j) \to 0$, because $\sum_j m(V_j) = m(\bigsqcup_j V_j) \leq m(O) < +\infty$. Hence $m^*(R) = 0$.
Structural Commentary: Vitali selection gives a disjoint cover modulo null sets.
Key Insight: $R \subseteq \bigcup_{j>N} V_j^*$ and disjoint tails vanish. □

Theorem 3.3.6. *Lebesgue Density Theorem*
For measurable $E \subseteq \mathbb{R}^n$, $\lim_{Q \downarrow p} [E : Q] = \mathbf{1}_E(p)$ for a.e. p, with $Q \downarrow p$ through cubes or balls.

Proof. Assume $m(E) < +\infty$. For $a \in (0,1)$ set $E_a := \{p \in E : \liminf_{Q \downarrow p} [E : Q] < a\}$.
Cubes $Q \ni p$ with $[E : Q] < a$ form a Vitali cover of E_a. For open $O \supseteq E_a$ with $m(O) < m(E_a) + \varepsilon$, choose disjoint $(Q_j) \subseteq O$ covering E_a modulo null sets by Theorem 3.3.5.
Hence $m(E_a) \leq \sum_j m(E_a \cap Q_j) \leq \sum_j m(E \cap Q_j) < a \sum_j m(Q_j) \leq a\, m(O) < a(m(E_a) + \varepsilon)$.
Let $\varepsilon \downarrow 0$; since $a < 1$, $m(E_a) = 0$. Thus $p \in E \setminus \bigcup_{a \in \mathbb{Q} \cap (0,1)} E_a$ has density 1. Apply same argument to bounded pieces of E^c to get density 0 a.e. on E^c; then exhaust arbitrary E by bounded pieces.
Structural Commentary: Measurable sets are locally 0 or 1 a.e.
Key Insight: Vitali turns bad density into $m(E_a) \leq a\, m(E_a)$. □

Corollary 3.3.7. *Density Points Represent a Measurable Set*
For measurable $E \subseteq \mathbb{R}^n$, $m(E \triangle \operatorname{dp}(E)) = 0$.

Proof. By Theorem 3.3.6, $[E : Q] \to 1$ at a.e. point of E, $[E : Q] \to 0$ at a.e. point of E^c; hence $E \triangle \operatorname{dp}(E)$ null.
Structural Commentary: The pointwise density dichotomy is converted into equality modulo null sets.
Key Insight: Density-one membership recovers the measurable set almost everywhere. □

Corollary 3.3.8. *Density Points Commute with Slicing Almost Everywhere*
Let $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ be measurable. For a.e. $x \in \mathbb{R}^n$, $\operatorname{dp}(E_x) = \{y : (x,y) \in \operatorname{dp}(E)\}$ up to a null subset of $\mathbb{R}^k$.

Proof. Set $D = \operatorname{dp}(E)$ and $N = E \triangle D$.
By Corollary 3.3.7, $m_{n+k}(N) = 0$; hence $m_k(N_x) = 0$ for a.e. x by Theorem 2.2.14.
Intersecting with the Fubini good set, E_x, D_x are measurable and $m_k(E_x \triangle D_x) = m_k((E \triangle D)_x) = 0$.
$m_k(E_x \triangle \operatorname{dp}(E_x)) = 0$ by Corollary 3.3.7 in $\mathbb{R}^k$. Therefore $m_k(\operatorname{dp}(E_x) \triangle D_x) = 0$.
Structural Commentary: Almost-everywhere local fullness is compatible with almost-everywhere slicing.
Key Insight: Null ambient errors have null slices for almost every base point. □

Lemma 3.3.9. *Large Averages Have Small Outer Measure*
Let $Q_0 \subseteq \mathbb{R}^n$ be a cube, let $0 \leq g \in \mathscr{L}^1(Q_0)$, and let $\lambda > 0$.
Set $A_\lambda := \{p \in Q_0 : \forall r > 0\ \exists$ a cube $Q \subseteq Q_0$, $p \in Q$, $\operatorname{diam} Q < r$, $[g : Q] > \lambda\}$. Then $m^*(A_\lambda) \leq \frac{1}{\lambda} \int_{Q_0} g\, dm$.

Proof. Let $\mathcal{V}_\lambda := \{Q \subseteq Q_0 : Q \text{ is a cube and } [g : Q] > \lambda\}$. This is a Vitali cover of A_λ; choose disjoint $(Q_j) \subseteq \mathcal{V}_\lambda$ covering A_λ modulo a null set. Then $\lambda m^*(A_\lambda) \leq \lambda \sum_j m(Q_j) < \sum_j \int_{Q_j} g\, dm \leq \int_{Q_0} g\, dm$.
Structural Commentary: Weak type for local averages.
Key Insight: Disjoint Vitali cubes turn pointwise large averages into one integral bound. □

Theorem 3.3.10. *Lebesgue Average Value Theorem*
For $f \in \mathscr{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, $\lim_{Q \downarrow p} [f : Q] = f(p)$ for a.e. $p \in \mathbb{R}^n$ with Q cubes, equivalently balls, shrinking to p.

Proof. Fix cube Q_0; to prove claim for a.e. $p \in Q_0^\circ$. For $\alpha > 0$, let $B_\alpha = \left\{ p \in Q_0^\circ : \limsup_{Q \downarrow p} |[f : Q] - f(p)| > \alpha \right\}$.
Choose $h \in C(Q_0)$ with $\|f - h\|_{L^1(Q_0)} < \eta$, and set $g = |f - h|$. Since $[h : Q] \to h(p)$, $B_\alpha \subseteq \{g > \frac{\alpha}{3}\} \cup A_{\frac{\alpha}{3}}(g)$.
By Markov and Lemma 3.3.9, $m^*(B_\alpha) \leq \frac{3}{\alpha} \int_{Q_0} g + \frac{3}{\alpha} \int_{Q_0} g < \frac{6\eta}{\alpha}$. Let $\eta \downarrow 0$; then $m(B_\alpha) = 0$.
Hence $B(Q_0) := \{p \in Q_0^\circ : \limsup_{Q \downarrow p} |[f : Q] - f(p)| > 0\} = \bigcup_{\alpha \in \mathbb{Q}_{>0}} B_\alpha$ is null.
With $Q_N := [-N, N]^n$, $\mathbb{R}^n = \bigcup_N Q_N^\circ$, so $\bigcup_N B(Q_N)$ is null.
Structural Commentary: Local averages recover the original integrable function almost everywhere.
Key Insight: Continuous approximation handles the good part; Vitali's estimate makes the bad part small. □

Theorem 3.3.11. *Lebesgue's Antiderivative Theorem*
For $f \in \mathscr{L}^1([a,b])$, define $F(x) := c + \int_a^x f(t)\, dt$.
(i) $F \in AC([a,b])$.
(ii) $F' = f$ a.e. on $[a,b]$.

Proof.

(i) Given $\varepsilon > 0$, choose $\delta > 0$ such that $m(A) < \delta$ implies $\int_A |f|\, dm < \varepsilon$.
If (α_j, β_j) are disjoint and $\sum_j (\beta_j - \alpha_j) < \delta$, then $\sum_j |F(\beta_j) - F(\alpha_j)| \le \int_{\bigcup_j (\alpha_j,\beta_j)} |f|\, dm < \varepsilon$.

(ii) For $x, x+h \in [a,b]$, let $I_h(x) := (\min\{x, x+h\}, \max\{x, x+h\})$. Then $\frac{F(x+h)-F(x)}{h} = [f : I_h(x)]$.
Extend f by 0 outside $[a,b]$; by Theorem 3.3.10 in one dimension, $[f : I_h(x)] \to f(x)$ for a.e. x.

Structural Commentary: The indefinite Lebesgue integral is the correct antiderivative for integrable functions.
Key Insight: Differentiating the integral is the same as shrinking the averaging interval. □

3.4 Monotone and Bounded-Variation Differentiation

Monotone functions are the one-dimensional model case for differentiation almost everywhere. Bounded-variation functions enter by Jordan decomposition: finite oscillation is exactly enough to write a function as the difference of two monotone functions.

Definition 3.4.1. *Bounded Variation*
Let $F : [a,b] \to \mathbb{R}$. For $a \le u < v \le b$, let $\mathcal{P}[u,v] := \{P = (x_0, \dots, x_N) : N \in \mathbb{N},\ u = x_0 < \cdots < x_N = v\}$.

(i) For $P = (x_0, \dots, x_N) \in \mathcal{P}[u,v]$, define $V(F,P) := \sum_{j=1}^{N} |F(x_j) - F(x_{j-1})|$.

(ii) Define $V_u^v(F) := \sup_{P \in \mathcal{P}[u,v]} V(F,P)$ for $u < v$, and $V_u^u(F) := 0$.

(iii) F has *bounded variation* on $[a,b]$ iff $F \in BV([a,b]) := \{G : [a,b] \to \mathbb{R} : V_a^b(G) < +\infty\}$.

Theorem 3.4.2. *Variation Function and Jordan Decomposition*
Let $F : [a,b] \to \mathbb{R}$. For $F \in BV([a,b])$, set $T_F : [a,b] \to \mathbb{R}$ by $T_F(a) := 0$ and $T_F(x) := V_a^x(F)$ for $a < x \le b$.

(i) Variation additivity: If $F \in BV([a,b])$, then $T_F(v) - T_F(u) = V_u^v(F)$ for $a \le u \le v \le b$.
(ii) Variation monotonicity: If $F \in BV([a,b])$, then T_F is nondecreasing.
(iii) Compensated monotonicity: If $F \in BV([a,b])$, then $F + T_F$ is nondecreasing.
(iv) Decomposition: If $F \in BV([a,b])$, then $F = (F + T_F) - T_F$ is a difference of nondecreasing functions.
(v) Converse: If $F = G - H$ for nondecreasing $G, H : [a,b] \to \mathbb{R}$,
then $F \in BV([a,b])$ and $V_a^b(F) \le G(b) - G(a) + H(b) - H(a)$.

Proof.

(i) Assume $F \in BV([a,b])$ and fix $a \le u \le v \le b$. The cases $u = a$ and $u = v$ are immediate; for $a < u < v$, concatenation gives $V_a^v(F) \ge V_a^u(F) + V_u^v(F)$, while inserting u into $P \in \mathcal{P}[a,v]$ and using the triangle inequality gives $V(F,P) \le V_a^u(F) + V_u^v(F)$. Taking suprema yields $V_a^v(F) = V_a^u(F) + V_u^v(F)$, QED.

(ii) By (i), $T_F(v) - T_F(u) = V_u^v(F) \ge 0$.

(iii) Since $|F(v) - F(u)| \le T_F(v) - T_F(u)$, $(F + T_F)(v) - (F + T_F)(u) \ge 0$.

(iv) The identity $F = (F + T_F) - T_F$ and (ii)–(iii) give the required monotone difference.

(v) Assume $F = G - H$ for nondecreasing $G, H : [a,b] \to \mathbb{R}$.
For $P = (x_0, \dots, x_N) \in \mathcal{P}[a,b]$, by monotonicity and triangle inequality,

$$V(F,P) \le \sum_{j=1}^{N} (G(x_j) - G(x_{j-1})) + \sum_{j=1}^{N} (H(x_j) - H(x_{j-1})) = G(b) - G(a) + H(b) - H(a).$$

Taking the supremum over P gives $V_a^b(F) \le G(b) - G(a) + H(b) - H(a) < +\infty$.

Structural Commentary: Bounded variation is exactly controlled monotone difference.
Key Insight: The variation function supplies the missing monotone part. □

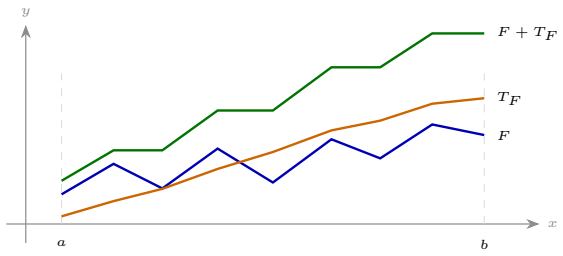

Figure 210: Variation T_F turns F into the increasing function $F + T_F$.

Example 3.4.3. *Basic Bounded-Variation Tests*

(i) If $F : [a,b] \to \mathbb{R}$ is nondecreasing, then $F \in BV([a,b])$ and $V_a^b(F) = F(b) - F(a)$.

(ii) If $F : [a,b] \to \mathbb{R}$ is L-Lipschitz, then $F \in BV([a,b])$ and $V_a^b(F) \le L(b-a)$.

(iii) Let $F : [0,1] \to \mathbb{R}$ satisfy $F(0) = 0$ and $F(x) = x\cos(\frac{\pi}{2x})$ for $0 < x \le 1$. Then F is continuous and $F \notin BV([0,1])$: for $P_n = (0, \frac{1}{2n}, \frac{1}{2n-1}, \dots, \frac{1}{2}, 1) \in \mathcal{P}[0,1]$, $V(F,P_n) = \sum_{k=1}^{n} \frac{1}{k} \to +\infty$.

Definition 3.4.4. *Upper and Lower Derivatives*
Let $I \subseteq \mathbb{R}$ be an interval, $F : I \to \mathbb{R}$, and $x \in I^\circ$.
For $h > 0$, set $Q_h(x) := \{t \in \mathbb{R} : 0 < |t| \le h,\ x + t \in I\}$ and $\Delta_t F(x) := \frac{F(x+t)-F(x)}{t}$.

(i) The *upper derivative* at x is $\overline{D}F(x) := \lim_{h \downarrow 0} \sup_{t \in Q_h(x)} \Delta_t F(x)$.

(ii) The *lower derivative* at x is $\underline{D}F(x) := \lim_{h \downarrow 0} \inf_{t \in Q_h(x)} \Delta_t F(x)$.

They exist in $\overline{\mathbb{R}}$, satisfy $\underline{D}F(x) \le \overline{D}F(x)$, and $F'(x)$ exists iff $\underline{D}F(x) = \overline{D}F(x) \in \mathbb{R}$.

Lemma 3.4.5. *Weak Bound for Upper Derivatives*
Let $F : [a,b] \to \mathbb{R}$ be nondecreasing. For $\alpha > 0$, set $E_\alpha := \{x \in (a,b) : \overline{D}F(x) \ge \alpha\}$.
Then:

(i) $m^*(E_\alpha) \le \frac{F(b)-F(a)}{\alpha}$ for every $\alpha > 0$.
(ii) $m^*\{x \in (a,b) : \overline{D}F(x) = +\infty\} = 0$.

Proof.

(i) Fix $\alpha > 0$ and $0 < \alpha' < \alpha$. For $x \in E_\alpha$ and $\delta > 0$, choose $[c,d] \subset (a,b)$ with $x \in [c,d]$, $d-c < \delta$, and $F(d) - F(c) > \alpha'(d-c)$. These intervals form a Vitali cover of E_α.
By Theorem 3.3.5, choose pairwise disjoint $I_k = [c_k, d_k]$ covering E_α modulo a null set.
Then $\alpha' m^*(E_\alpha) \le \alpha' \sum_k (d_k - c_k) < \sum_k (F(d_k) - F(c_k)) \le F(b) - F(a)$. Let $\alpha' \uparrow \alpha$.

(ii) Set $E_\infty := \{x \in (a,b) : \overline{D}F(x) = +\infty\}$. Since $E_\infty \subseteq E_n$ $\forall n \in \mathbb{N}$, (i) gives $m^*(E_\infty) \le \frac{F(b)-F(a)}{n} \to 0$.

Structural Commentary: Vitali selection converts local slope lower bounds into total rise.
Key Insight: Large local slopes are paid for by total monotone rise. □

Lemma 3.4.6. *Dini Gap Sets Are Null*
Let $F : [a,b] \to \mathbb{R}$ be nondecreasing and let $\alpha < \beta$.
Set $E_{\alpha,\beta} := \{x \in (a,b) : \underline{D}F(x) < \alpha < \beta < \overline{D}F(x)\}$. Then $m^*(E_{\alpha,\beta}) = 0$.

Proof. If $\alpha \le 0$, then $E_{\alpha,\beta} = \emptyset$ because every secant quotient of nondecreasing F is nonnegative.
Assume $0 < \alpha < \beta$, set $E = E_{\alpha,\beta}$, and fix $\varepsilon > 0$. Choose open O with $E \subseteq O \subseteq (a,b)$ and $m(O) < m^*(E) + \varepsilon$.
Since $\underline{D}F(x) < \alpha$ on E, intervals $[c,d] \subset O$ with $x \in [c,d]$, $F(d) - F(c) < \alpha(d-c)$ form Vitali cover of E.
Select pairwise disjoint $I_k = [c_k, d_k]$ covering E modulo a null set.
$\forall k$, modulo endpoints, $E \cap I_k \subseteq \{x \in (c_k, d_k) : \overline{D}(F|_{I_k})(x) \ge \beta\}$, so $m^*(E \cap I_k) \le \frac{F(d_k)-F(c_k)}{\beta}$ by Lemma 3.4.5.
Hence $m^*(E) \le \sum_k m^*(E \cap I_k) \le \frac{1}{\beta} \sum_k (F(d_k) - F(c_k)) < \frac{\alpha}{\beta} \sum_k (d_k - c_k) \le \frac{\alpha}{\beta} m(O)$.
Let $\varepsilon \downarrow 0$; then $m^*(E) \le \frac{\alpha}{\beta} m^*(E)$, and since $m^*(E) < +\infty$ and $\frac{\alpha}{\beta} < 1$, $m^*(E) = 0$.
Structural Commentary: Vitali intervals turn a slope gap into a contraction estimate.
Key Insight: Dini gaps force excess rise. □

Perspective 3.4.7. *Right Secants and Differentiation*
Right quotients $g_n(x) = n\big(F(x + \frac{1}{n}) - F(x)\big)$ encode local slope data.
Key Insight: Secants turn differentiation into slope estimates.

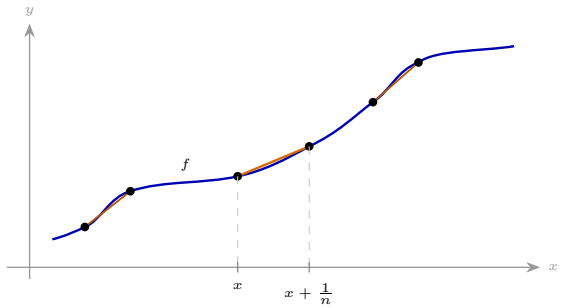

Figure 211: $g_n(x)$ is the slope of the short right secant from x to $x + \frac{1}{n}$.

Proposition 3.4.8. *Divided Differences and Moving Averages*
Let $f \in \mathscr{L}^1([a,b])$, extend f to $[a, b+1]$ by $f(x) = f(b)$ for $b < x \le b+1$, and fix $0 < h \le 1$.
For $x \in [a,b]$, set $\mathrm{Diff}_h f(x) := \frac{f(x+h)-f(x)}{h}$ and $\mathrm{Av}_h f(x) := h^{-1} \int_x^{x+h} f\, dm$.
Then $\forall a \le u < v \le b$, $\int_u^v \mathrm{Diff}_h f\, dm = \mathrm{Av}_h f(v) - \mathrm{Av}_h f(u)$.

Proof. For $a \le u < v \le b$, translation invariance gives $\int_u^v f(\cdot + h)\, dm = \int_{u+h}^{v+h} f\, dm$.
Hence $\int_u^v \mathrm{Diff}_h f\, dm = h^{-1}(\int_{u+h}^{v+h} f\, dm - \int_u^v f\, dm) = h^{-1}(\int_v^{v+h} f\, dm - \int_u^{u+h} f\, dm) = \mathrm{Av}_h f(v) - \mathrm{Av}_h f(u)$.
Structural Commentary: Difference quotients integrate to endpoint moving averages.
Key Insight: The interior contributions cancel after translation. □

Theorem 3.4.9. *Lebesgue Differentiability Theorem for Monotone Functions*
If $F : [a,b] \to \mathbb{R}$ is monotone, then F' exists a.e. on $[a,b]$.

Proof. Assume F is nondecreasing. Set $G_{\alpha,\beta} := \{\underline{D}F < \alpha < \beta < \overline{D}F\}$ for $\alpha, \beta \in \mathbb{Q}$, $\alpha < \beta$, and $E_\infty := \{\overline{D}F = +\infty\}$. Lemma 3.4.6 gives $m(G_{\alpha,\beta}) = 0$, and Lemma 3.4.5 gives $m(E_\infty) = 0$.
Outside $E_\infty \cup \bigcup_{\alpha<\beta} G_{\alpha,\beta}$, $\underline{D}F = \overline{D}F \in \mathbb{R}$, so F' exists. If F is nonincreasing, apply the result to $-F$.
Structural Commentary: Null Dini gaps plus finite upper derivative give differentiability.
Key Insight: Monotone rise controls exceptional slopes. □

Corollary 3.4.10. *Lipschitz and BV Differentiability*
Let $F : [a,b] \to \mathbb{R}$.

(i) If F is Lipschitz, then F' exists a.e. on $[a,b]$.

(ii) If $F \in BV([a,b])$, then F' exists a.e. on $[a,b]$.

Proof.

(i) If F is L-Lipschitz, then $G_+(x) := F(x) + Lx$ and $G_-(x) := Lx - F(x)$ are nondecreasing.
Their derivatives exist a.e. by Theorem 3.4.9, so $F' = \frac{G'_+ - G'_-}{2}$ a.e.

(ii) By Theorem 3.4.2, $F = G - H$ for nondecreasing G, H.
Their derivatives exist a.e. by Theorem 3.4.9, so $F' = G' - H'$ a.e.

Structural Commentary: Lipschitz and BV functions are reduced to finite combinations of monotone functions.
Key Insight: Almost-everywhere differentiability is stable under finite linear combinations. □

Corollary 3.4.11. *Absolute Continuity Implies BV Differentiability*
If $F \in AC([a,b])$, then:

(i) $F \in BV([a,b])$.
(ii) F' exists a.e. on $[a,b]$.

Proof.

(i) Choose $\delta > 0$ from absolute continuity with $\varepsilon = 1$, and choose $a = x_0 < \cdots < x_N = b$ with $x_j - x_{j-1} < \delta$. After inserting the x_j into a partition P, the increments within each $[x_{j-1}, x_j]$ have total length below δ and total F-variation below 1. Thus $V(F,P) \le N$ and $F \in BV([a,b])$.

(ii) By (i), $F \in BV([a,b])$; by Corollary 3.4.10 $\exists G \subseteq [a,b]$, $m([a,b] \setminus G) = 0$ and $F'(x) \in \mathbb{R}$ $\forall x \in G$.

Structural Commentary: Absolute continuity bounds variation; BV decomposition then gives differentiability.
Key Insight: One absolute-continuity scale controls every partition after a finite interval decomposition. □

Corollary 3.4.12. *Derivative Bounds for Monotone and BV Functions*
Let $F : [a,b] \to \mathbb{R}$.

(i) If F is nondecreasing, then $F' \ge 0$ a.e.
(ii) If F is nondecreasing, then $F' \in \mathscr{L}^1([a,b])$ and $\int_a^b F'\,dm \le F(b) - F(a)$.
(iii) If $F \in BV([a,b])$, then F' exists a.e.
(iv) If $F \in BV([a,b])$, then $F' \in \mathscr{L}^1([a,b])$.

Proof.

(i) Every secant quotient of a nondecreasing function is nonnegative; hence $F' \ge 0$ a.e. by Theorem 3.4.9.

(ii) Extend F by $F(b)$ on $(b, b+1]$, set $g_n(x) := n(F(x + \frac{1}{n}) - F(x))$. Then $g_n \ge 0$, $g_n \to F'$ a.e. on (a,b).
By Proposition 3.4.8, $\int_a^b g_n\,dm = F(b) - n\int_a^{a+\frac{1}{n}} F\,dm \le F(b) - F(a)$.
Fatou's lemma gives $\int_a^b F'\,dm \le F(b) - F(a) < +\infty$.

(iii) Corollary 3.4.10(ii) gives $\exists E \subseteq [a,b]$ with $m([a,b] \setminus E) = 0$ and $F'(x) \in \mathbb{R}$ $\forall x \in E$.

(iv) Write $F = G - H$ with G, H nondecreasing by Theorem 3.4.2. By (ii), $G', H' \in \mathscr{L}^1([a,b])$; on the full-measure set where G', H' exist, $F' = G' - H'$, so $F' \in \mathscr{L}^1([a,b])$.

Structural Commentary: BV inherits finite derivative mass from monotone parts.
Key Insight: Fatou turns right-secant bounds into an integral bound for the derivative. □

3.5 Absolute Continuity and BV Decomposition

Absolute continuity is the part of bounded variation detected by integration. The remaining bounded-variation behavior is singular: it may be monotone and continuous, but its derivative vanishes almost everywhere.

Proposition 3.5.1. *Absolutely Continuous Variation Function*
Let $F \in AC([a,b])$ and let $T_F(x) := V_a^x(F)$ be its total variation function.

(i) $T_F \in AC([a,b])$.
(ii) $F = (F + T_F) - T_F$ is a difference of two nondecreasing absolutely continuous functions.

Proof.

(i) Let $\varepsilon > 0$ and choose $\delta > 0$ from absolute continuity of F for the challenge $\frac{\varepsilon}{2}$.
Suppose $(c_k, d_k)_{k=1}^n$ are disjoint subintervals of (a,b) with $\sum_k (d_k - c_k) < \delta$.
Choose partitions P_k of $[c_k, d_k]$ such that $V_{c_k}^{d_k}(F) \le V(F, P_k) + \frac{\varepsilon}{2n}$.
Let $\mathcal{I}$ be all subintervals of all P_k; then $\mathcal{I}$ is disjoint up to endpoints and $\sum_{I \in \mathcal{I}} |I| = \sum_k (d_k - c_k)$.
Hence absolute continuity of F gives $\sum_k V(F, P_k) < \frac{\varepsilon}{2}$, thus $\sum_{k=1}^n |T_F(d_k) - T_F(c_k)| = \sum_{k=1}^n V_{c_k}^{d_k}(F) < \varepsilon$.

(ii) By (i), $T_F \in AC([a,b])$, hence $F + T_F \in AC([a,b])$.
For $x < y$, $T_F(y) - T_F(x) = V_x^y(F) \ge 0$ and $(F + T_F)(y) - (F + T_F)(x) = F(y) - F(x) + V_x^y(F) \ge 0$.
Thus T_F and $F + T_F$ are nondecreasing, and $F = (F + T_F) - T_F$.

Structural Commentary: Absolute continuity controls local variation, not just endpoint displacement.
Key Insight: Partition each local variation, then apply one absolute-continuity test to all pieces. □

Theorem 3.5.2. *Absolute Continuity via Divided Differences*
Let $F \in C([a,b])$, extend F constantly by $F(b)$ on $(b, b+1]$. Let $\operatorname{Diff}_h F(x) := \frac{F(x+h) - F(x)}{h}$, $0 < h \le 1$, $x \in [a,b]$.
Then $F \in AC([a,b])$ iff $\{\operatorname{Diff}_h F : 0 < h \le 1\}$ is uniformly integrable:
$\forall \varepsilon > 0\ \exists \delta > 0\ \forall E \in \mathcal{L}^1,\ E \subseteq [a,b]$ *and* $m(E) < \delta$ *imply* $\sup_{0<h\le 1} \int_E |\operatorname{Diff}_h F|\,dm < \varepsilon$.

Proof.

($\Leftarrow$) Assume the divided differences are uniformly integrable. Given $\varepsilon > 0$, choose $\delta > 0$ for $\frac{\varepsilon}{2}$.
For disjoint (c_k, d_k) with total length below δ, set $E := \bigcup_k (c_k, d_k)$.

By Proposition 3.4.8, $\sum_k |\mathrm{Av}_h F(d_k) - \mathrm{Av}_h F(c_k)| \le \int_E |\mathrm{Diff}_h F|\, dm < \frac{\varepsilon}{2}$.

Letting $h \downarrow 0$ gives $\sum_k |F(d_k) - F(c_k)| \le \frac{\varepsilon}{2} < \varepsilon$.

($\Rightarrow$) Suppose $F \in AC([a,b])$. By Proposition 3.5.1, write $F = U - V$ with U, V nondecreasing and absolutely continuous; since $|\mathrm{Diff}_h F| \le \mathrm{Diff}_h U + \mathrm{Diff}_h V$, assume F nondecreasing.
Let $\widetilde{F}$ be the constant extension to $[a, b+1]$.
Choose $\eta > 0$ for $\frac{\varepsilon}{2}$ in the absolute continuity of $\widetilde{F}$ and set $\delta := \frac{\eta}{2}$. If $m(E) < \delta$, take open $O \supseteq E \cap (a,b)$ with $O \cap (a,b) = \bigsqcup_j (\alpha_j, \beta_j)$ and $m(O \cap (a,b)) < \eta$; set $O_N := \bigcup_{j \le N} (\alpha_j, \beta_j)$.

For $0 < t \le h$, $\sum_{j \le N} ((\beta_j + t) - (\alpha_j + t)) < \eta$, so $\sum_{j \le N} (\widetilde{F}(\beta_j + t) - \widetilde{F}(\alpha_j + t)) < \frac{\varepsilon}{2}$.

Hence $\int_{O_N} \mathrm{Diff}_h F\, dm = \frac{1}{h} \int_0^h \sum_{j \le N} (\widetilde{F}(\beta_j + t) - \widetilde{F}(\alpha_j + t))\, dt < \frac{\varepsilon}{2}$.

Let $N \to \infty$ and use $E \subseteq O$ to get $\sup_{0<h\le 1} \int_E \mathrm{Diff}_h F\, dm < \varepsilon$.

Structural Commentary: Absolute continuity is uniform small-set control for translated secants.
Key Insight: Moving averages turn endpoint increments into integrals of divided differences. □

Theorem 3.5.3. *Fundamental Theorem for Absolutely Continuous Functions*
Let $F \in AC([a,b])$.

(i) F' exists a.e. on $[a,b]$.
(ii) $F' \in \mathscr{L}^1([a,b])$.
(iii) $F(y) - F(x) = \int_x^y F'\, dm$ for $a \le x \le y \le b$.

Proof.

(i) Corollary 3.4.11(ii) gives $\exists E \subseteq [a,b]$ with $m([a,b] \setminus E) = 0$ and $F'(x) \in \mathbb{R}$ $\forall x \in E$.
(ii) Corollary 3.4.11(i) gives $F \in BV([a,b])$, so Corollary 3.4.12(iv) gives $F' \in \mathscr{L}^1([a,b])$.
(iii) Set $f_n := \mathrm{Diff}_{\frac{1}{n}} F$. By Theorem 3.5.2, (f_n) is uniformly integrable; by (i), $f_n \to F'$ a.e.

Fix $a \le x < y \le b$ and $\varepsilon > 0$. Choose $M > 0$ with $\int_{\{|F'|>M\}} |F'|\, dm < \frac{\varepsilon}{6}$ and then $\delta > 0$ such that $M\delta < \frac{\varepsilon}{6}$ and $m(A) < \delta$ implies $\sup_n \int_A |f_n|\, dm < \frac{\varepsilon}{3}$. By Egorov's theorem, $f_n \to F'$ uniformly on $[x,y] \setminus A$ for some measurable A with $m(A) < \delta$. For large n, $\int_x^y |f_n - F'|\, dm < \varepsilon$.
Thus $\int_x^y f_n\, dm \to \int_x^y F'\, dm$. By Proposition 3.4.8, $\int_x^y f_n\, dm = \mathrm{Av}_{\frac{1}{n}} F(y) - \mathrm{Av}_{\frac{1}{n}} F(x) \to F(y) - F(x)$.

Structural Commentary: Uniform integrability is the condition that permits the discrete secant formula to pass to the derivative under the integral sign.
Key Insight: Uniform small-set control upgrades a.e. secant convergence to convergence of interval integrals. □

Corollary 3.5.4. *Indefinite Integral Characterization of Absolute Continuity*
A function $F : [a,b] \to \mathbb{R}$ satisfies:

(i) $F \in AC([a,b])$ iff $\exists g \in \mathscr{L}^1([a,b])$ such that $F(x) = F(a) + \int_a^x g\, dm$ for $x \in [a,b]$.
(ii) Any such g satisfies $g = F'$ a.e.

Proof.

(i) ($\Rightarrow$) If $F \in AC([a,b])$, Theorem 3.5.3 gives $F' \in \mathscr{L}^1([a,b])$ and $F(x) = F(a) + \int_a^x F'\, dm$; take $g = F'$.
(i) ($\Leftarrow$) If $F(x) = F(a) + \int_a^x g\, dm$ with $g \in \mathscr{L}^1([a,b])$, Theorem 3.3.11 gives $F \in AC([a,b])$.
(ii) If $F(x) = F(a) + \int_a^x g\, dm$, then Theorem 3.3.11 gives $F' = g$ a.e.

Structural Commentary: The two directions are the fundamental theorem and the antiderivative theorem.
Key Insight: Absolute continuity is exactly representability by an L^1 density. □

Proposition 3.5.5. *Radon–Nikodym on an Interval Recovers Absolute Continuity*
Let λ be a finite measure on $\mathcal{B}([a,b])$ with $\lambda \ll m$, let $f := \frac{d\lambda}{dm}$, and set $F(x) := \lambda([a,x])$.

(i) $F(y) - F(x) = \int_x^y f\, dm$ for $a \le x \le y \le b$.
(ii) $F \in AC([a,b])$.
(iii) $F' = f$ a.e. on $[a,b]$.

Proof. By Theorem 3.2.1, $f \in L^1(m)$, $\lambda(E) = \int_E f\, dm$ for $E \in \mathcal{B}([a,b])$. Since $\lambda \ll m$, $\lambda(\{x\}) = 0$ for $x \in [a,b]$.

(i) For $a \le x \le y \le b$, $\lambda(\{x\}) = 0$ gives $F(y) - F(x) = \lambda((x,y]) = \int_{(x,y]} f\, dm = \int_x^y f\, dm$.
(ii) Taking $x = a$ in (i), $F(y) = F(a) + \int_a^y f\, dm$ for $y \in [a,b]$; Corollary 3.5.4(i) gives $F \in AC([a,b])$.
(iii) Since $F = F(a) + \int_a^{(\cdot)} f\, dm$, Corollary 3.5.4(ii) gives $F' = f$ a.e. on $[a,b]$.

Structural Commentary: Interval RN representation differentiates cumulative mass.
Key Insight: No atoms, so interval increments equal density integrals. □

Remark 3.5.6. *Lebesgue–Stieltjes Integrals*
Let $g : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and let μ_g be its Borel–Stieltjes measure.
For $E \in \mathcal{B}(\mathbb{R})$ and bounded Borel $f : E \to \mathbb{R}$, define $\int_E f\, dg := \int_E f\, d\mu_g$.
If $J \subseteq \mathbb{R}$ is an interval, $\mu_g|_J \ll m|_J$, and $h := \frac{d\mu_g}{dm}$ on J, then $\int_E f\, dg = \int_E fh\, dm$ for every Borel $E \subseteq J$.
For $a < b$ with $[a,b] \subseteq J$, $F_a : [a,b] \to \mathbb{R}$, $F_a(x) := \mu_g([a,x])$, is absolutely continuous and satisfies $F_a' = h$ a.e. by Proposition 3.5.5. If $g|_J$ is absolutely continuous, then $h = g'$ a.e. on J.

Example 3.5.7. *Cantor–Lebesgue Function Is Not Absolutely Continuous*
Let $\varphi : [0,1] \to [0,1]$ be the Cantor–Lebesgue function: φ is continuous, nondecreasing, $\varphi(0) = 0$, $\varphi(1) = 1$, and constant on every complementary interval of the Cantor set.
At stage n, write the remaining intervals as $I_{n,k} = [c_{n,k}, d_{n,k}]$, $1 \le k \le 2^n$, so $\sum_k |I_{n,k}| = (\frac{2}{3})^n$.

Since φ is constant on the deleted gaps, $\sum_{k=1}^{2^n} (\varphi(d_{n,k}) - \varphi(c_{n,k})) = 1$.

By continuity, choose $c_{n,k} < \alpha_{n,k} < \beta_{n,k} < d_{n,k}$ such that $\sum_k (\beta_{n,k} - \alpha_{n,k}) < (\frac{2}{3})^n$ and $\sum_k (\varphi(\beta_{n,k}) - \varphi(\alpha_{n,k})) > \frac{1}{2}$. Since $(\frac{2}{3})^n \to 0$, absolute continuity fails for $\varepsilon = \frac{1}{2}$.
Key Insight: The increase of φ lies on intervals with arbitrarily small total length.

Proposition 3.5.8. *Lipschitz, Absolutely Continuous, and BV Classes*
Let $[a,b]$ be nondegenerate, and let $\mathcal{F}_{\mathrm{Lip}}$ denote the Lipschitz functions on $[a,b]$.
(i) $\mathcal{F}_{\mathrm{Lip}} \subsetneq AC([a,b]) \subsetneq BV([a,b])$.
(ii) For $\mathcal{C} \in \{\mathcal{F}_{\mathrm{Lip}}, AC([a,b]), BV([a,b])\}$, $F, G \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$ imply $\alpha F + \beta G \in \mathcal{C}$.
(iii) If $F \in \mathcal{C}$ for some $\mathcal{C} \in \{\mathcal{F}_{\mathrm{Lip}}, AC([a,b]), BV([a,b])\}$, then $T_F(x) := V_a^x(F)$ lies in $\mathcal{C}$.

Proof.
(i) If F is L-Lipschitz, then $\sum_j |F(d_j) - F(c_j)| \le L \sum_j (d_j - c_j)$, so $F \in AC([a,b])$.
Corollary 3.4.11(i) gives $AC([a,b]) \subseteq BV([a,b])$.
Strictness: $x \mapsto \sqrt{\frac{x-a}{b-a}}$ lies in $AC([a,b]) \setminus \mathcal{F}_{\mathrm{Lip}}$, and $x \mapsto \varphi(\frac{x-a}{b-a})$, with φ Cantor–Lebesgue, lies in $BV([a,b]) \setminus AC([a,b])$ because φ is nondecreasing and its Cantor-stage intervals have total length $(\frac{2}{3})^n \to 0$ but total φ-increase 1.
(ii) If F, G are Lipschitz with constants L_F, L_G, then $\alpha F + \beta G$ is Lipschitz with constant $|\alpha| L_F + |\beta| L_G$.
If $F, G \in AC([a,b])$, the interval criterion gives $\alpha F + \beta G \in AC([a,b])$.
If $F, G \in BV([a,b])$, then $V(\alpha F + \beta G, P) \le |\alpha| V(F,P) + |\beta| V(G,P)$, so $V_a^b(\alpha F + \beta G) < +\infty$.
(iii) If F is L-Lipschitz, then $T_F(v) - T_F(u) = V_u^v(F) \le L(v-u)$.
If $F \in AC([a,b])$, Proposition 3.5.1(i) gives $T_F \in AC([a,b])$.
If $F \in BV([a,b])$, then T_F is nondecreasing and $V_a^b(T_F) = T_F(b) - T_F(a) = V_a^b(F) < +\infty$.

Structural Commentary: $\mathcal{F}_{\mathrm{Lip}} \subsetneq AC \subsetneq BV$ is a strict control hierarchy.
Key Insight: $\sqrt{\cdot}$ breaks Lipschitz; Cantor mass breaks absolute continuity. □

Proposition 3.5.9. *Algebra and Safe Composition*
Let $F, G \in AC([a,b])$.
(i) For $\alpha, \beta \in \mathbb{R}$, $\alpha F + \beta G \in AC([a,b])$.
(ii) $FG \in AC([a,b])$.
(iii) If $\Phi : \mathbb{R} \to \mathbb{R}$ is Lipschitz, then $\Phi \circ G \in AC([a,b])$.
(iv) If G is strictly monotone and $\Phi \in AC(J)$ on a compact interval $J \supseteq G([a,b])$, then $\Phi \circ G \in AC([a,b])$.

Proof.
(i) For disjoint (c_k, d_k), $\sum_k |(\alpha F + \beta G)(d_k) - (\alpha F + \beta G)(c_k)| \le |\alpha| \sum_k |F(d_k) - F(c_k)| + |\beta| \sum_k |G(d_k) - G(c_k)|$. Thus the right side is $< \varepsilon$ whenever $\sum_k (d_k - c_k) < \delta$ for a common AC scale of F, G, QED.
(ii) Let $|F| \le M_F$ and $|G| \le M_G$. For $\Delta F_k := F(d_k) - F(c_k)$ and $\Delta G_k := G(d_k) - G(c_k)$, $|F(d_k)G(d_k) - F(c_k)G(c_k)| \le M_F |\Delta G_k| + M_G |\Delta F_k|$. Choose a common AC scale with $\sum_k |\Delta F_k| < \frac{\varepsilon}{2(1+M_G)}$ and $\sum_k |\Delta G_k| < \frac{\varepsilon}{2(1+M_F)}$; then $\sum_k |F(d_k)G(d_k) - F(c_k)G(c_k)| < \varepsilon$.
(iii) If Φ is L-Lipschitz, then $\sum_k |\Phi(G(d_k)) - \Phi(G(c_k))| \le L \sum_k |G(d_k) - G(c_k)|$.
Choose the AC scale of G for $\frac{\varepsilon}{L}$ if $L > 0$; the case $L = 0$ is trivial.
(iv) Suppose G is strictly increasing. Choose $\eta > 0$ from $AC(\Phi)$ for ε and $\delta > 0$ from $AC(G)$ for η.
If $\sum_k (d_k - c_k) < \delta$, then $\sum_k (G(d_k) - G(c_k)) < \eta$.
Since $(G(c_k), G(d_k))$ are disjoint, $\sum_k |\Phi(G(d_k)) - \Phi(G(c_k))| < \varepsilon$.
If G is strictly decreasing, replace $(G(c_k), G(d_k))$ by $(G(d_k), G(c_k))$.

Structural Commentary: AC is stable under algebra and controlled composition.
Key Insight: Lipschitz bounds sizes; monotonicity preserves disjoint intervals. □

Example 3.5.10. *Absolutely Continuous Composition Can Fail*
Define $f, g : [-1,1] \to \mathbb{R}$ by $f(x) := \operatorname{sgn}(x) |x|^{\frac{1}{3}}$, $g(0) := 0$, and $g(x) := x^2 \cos \frac{\pi}{2x}$ for $x \ne 0$. Then:
(i) $f, g \in AC([-1,1])$.
(ii) $f \circ g \notin BV([-1,1])$, hence $f \circ g \notin AC([-1,1])$.

Proof.
(i) Let $h(t) := \frac{1}{3} |t|^{-\frac{2}{3}} \mathbf{1}_{t \ne 0}$. Then $h \in L^1([-1,1])$ and $f(x) = \int_0^x h \, dm$, so $f \in AC([-1,1])$.
For $x \ne 0$, $g'(x) = 2x \cos \frac{\pi}{2x} + \frac{\pi}{2} \sin \frac{\pi}{2x}$, hence $|g'| \le 2 + \frac{\pi}{2}$ off 0.
Together with $|g(x)| \le x^2$, this gives g Lipschitz, so $g \in AC([-1,1])$.
(ii) For $k \ge 1$, $g(\frac{1}{2k-1}) = 0$ and $g(\frac{1}{2k}) = \frac{(-1)^k}{4k^2}$. With $P_n := \{0\} \cup \{\frac{1}{j} : 1 \le j \le 2n\}$,
$V(f \circ g, P_n) \ge \sum_{k=1}^{n} \left| f(g(\frac{1}{2k})) - f(g(\frac{1}{2k-1})) \right| = 4^{-\frac{1}{3}} \sum_{k=1}^{n} k^{-\frac{2}{3}} \to +\infty$. Thus $f \circ g \notin BV([-1,1])$.

Key Insight: The cube root amplifies the inner oscillations until variation diverges. □

Definition 3.5.11. *Singular Bounded-Variation Function*
A function $S \in BV([a,b])$ is *singular* if $S' = 0$ a.e. on $[a,b]$.

Theorem 3.5.12. *Lebesgue Decomposition of a BV Function*
Let $F \in BV([a,b])$. Define $G(x) := F(a) + \int_a^x F'\,dm$, $H(x) := F(x) - G(x)$.

(i) $G \in AC([a,b])$ and $G' = F'$ a.e.
(ii) $H \in BV([a,b])$, $H' = 0$ a.e., $H(a) = 0$, and $F = G + H$.
(iii) If $F = G_1 + H_1$ with $G_1 \in AC([a,b])$, $H_1' = 0$ a.e., and $H_1(a) = 0$, then $G_1 = G$ and $H_1 = H$.

Proof.

(i) By Corollary 3.4.12, $F' \in \mathscr{L}^1([a,b])$. By Theorem 3.3.11, $G \in AC([a,b])$ and $G' = F'$ a.e.
(ii) Since $F, G \in BV([a,b])$, $H = F - G \in BV([a,b])$.
Moreover, $H' = F' - G' = 0$ a.e., $H(a) = 0$, and $F = G + H$.
(iii) If $F = G_1 + H_1$ is another normalized decomposition, then $G_1 - G = H - H_1$ is absolutely continuous, has derivative 0 a.e., and vanishes at a. By Theorem 3.5.3, $G_1 - G = 0$, hence $G_1 = G$ and $H_1 = H$.

Structural Commentary: A BV function splits into an L^1-density part and a derivative-invisible singular part.
Key Insight: Integrating the a.e. derivative separates the absolutely continuous and singular parts. □

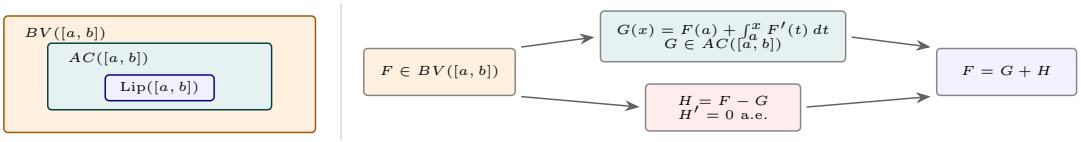

Figure 212: Lip $\subsetneq AC \subsetneq BV$, and each $F \in BV$ decomposes as $F = G + H$ with $G \in AC$ and $H' = 0$ a.e.

Corollary 3.5.13. *Absolutely Continuous Singular Functions Are Constant*

(i) If $F \in AC([a,b])$ and $F' = 0$ a.e., then F is constant.
(ii) The Cantor–Lebesgue function is nonconstant and singular.
(iii) The Cantor–Lebesgue function is not absolutely continuous.

Proof.

(i) By Theorem 3.5.3, $F(y) - F(x) = \int_x^y F'\,dm = 0$.
(ii) The function φ is nondecreasing, hence in $BV([0,1])$, and $\varphi' = 0$ on the full-measure set $[0,1] \setminus P$ by Proposition 8.1.2. Since $\varphi(0) = 0$ and $\varphi(1) = 1$, it is nonconstant.
(iii) If $\varphi \in AC([0,1])$, then $\varphi' = 0$ a.e. by (ii), so (i) gives φ constant, contradicting $\varphi(0) = 0 \neq 1 = \varphi(1)$.

Structural Commentary: The fundamental theorem makes absolute continuity exclude nonconstant singularity.
Key Insight: Derivative-invisible increase can occur in BV but not in absolute continuity. □

3.6 Nikodym Metric and Setwise Measure Limits

The Nikodym metric turns measurable sets modulo μ-null symmetric difference into a complete metric space. This lets Baire category control setwise limits of absolutely continuous measures.

Definition 3.6.1. *Setwise Convergence and Uniform Absolute Continuity*
Let $(X, \mathcal{M})$ be a measurable space.

(i) For $A, B \in \mathcal{M}$, $A \triangle B := (A \setminus B) \cup (B \setminus A)$.
(ii) Finite measures ν_n *converge setwise* to $\nu : \mathcal{M} \to [0, +\infty]$ iff $\nu_n(E) \to \nu(E)$ for every $E \in \mathcal{M}$.
(iii) For $\nu_n \ll \mu$ for all n, (ν_n) is *uniformly absolutely continuous* w.r.t. μ iff
$\forall \varepsilon > 0\ \exists \delta > 0$: $\mu(E) < \delta \Rightarrow \sup_n \nu_n(E) < \varepsilon$ for all $E \in \mathcal{M}$.

Lemma 3.6.2. *Nikodym Symmetric-Difference Calculus*
For $A, B, C \in \mathcal{M}$ and a finite measure ν:

(i) $A \triangle C \subseteq (A \triangle B) \cup (B \triangle C)$.
(ii) $|\nu(A) - \nu(B)| \leq \nu(A \triangle B)$.

Proof.

(i) If $x \in A \triangle C$, then $\mathbf{1}_A(x) \neq \mathbf{1}_C(x)$. Hence $\mathbf{1}_A(x) \neq \mathbf{1}_B(x)$ or $\mathbf{1}_B(x) \neq \mathbf{1}_C(x)$, so $x \in (A \triangle B) \cup (B \triangle C)$.
(ii) Since $\nu(A) - \nu(B) = \nu(A \setminus B) - \nu(B \setminus A)$, $|\nu(A) - \nu(B)| \leq \nu(A \setminus B) + \nu(B \setminus A) = \nu(A \triangle B)$.

Structural Commentary: Set inclusion gives the triangle inequality; disjoint differences give the mass bound.
Key Insight: Nikodym distance measures exactly the part where two measurable sets disagree. □

Definition 3.6.3. *Nikodym Metric Space*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space.

(i) *Null equivalence*: $A \sim_\mu B \Leftrightarrow \mu(A \triangle B) = 0$.
(ii) *Quotient space*: $\mathcal{M}_\mu := \mathcal{M}/\!\sim_\mu$, with class notation $[A] \in \mathcal{M}_\mu$.
(iii) *Metric*: $\rho_\mu([A],[B]) := \mu(A \triangle B)$.

By Lemma 3.6.2 and Proposition 3.1.11, ρ_μ is a well-defined metric on $\mathcal{M}_\mu$.

Proposition 3.6.4. *Nikodym Pseudometric*
Let $E \subseteq \mathbb{R}$ be Lebesgue measurable with $m(E) < \infty$, let X be the collection of Lebesgue measurable subsets of E, and for $A, B \in X$ set $A \triangle B := (A \setminus B) \cup (B \setminus A)$ and $\rho(A,B) := m(A \triangle B)$. Define $A \sim_\rho B$ iff $\rho(A,B) = 0$.

(i) ρ is a pseudometric on X.

(ii) $\widetilde{\rho}([A],[B]) := \rho(A,B)$ *is a metric on* $X/\!\sim_\rho$.
(iii) For all $A, B \in X$, $\rho(A,B) = \int_E |\mathbf{1}_A - \mathbf{1}_B|\, dm$.

Proof.

(i) For $A, B, C \in X$, $A\Delta B \subseteq E$, $A\Delta A = \emptyset$, $A\Delta B = B\Delta A$, and $A\Delta C \subseteq (A\Delta B) \cup (B\Delta C)$.
Hence $0 \le \rho(A,B) \le m(E) < \infty$, $\rho(A,A) = 0$, $\rho(A,B) = \rho(B,A)$, and $\rho(A,C) \le \rho(A,B) + \rho(B,C)$.
(ii) By (i), ρ is a pseudometric and $A \sim_\rho B \iff \rho(A,B) = 0$.
Proposition 3.1.11 gives that $\widetilde{\rho}([A],[B]) := \rho(A,B)$ is a metric on $X/\!\sim_\rho$.
(iii) For $t \in E$, $|\mathbf{1}_A(t) - \mathbf{1}_B(t)| = \mathbf{1}_{A\Delta B}(t)$; integrating gives $\int_E |\mathbf{1}_A - \mathbf{1}_B|\, dm = m(A\Delta B) = \rho(A,B)$.

Structural Commentary: Symmetric difference measures set disagreement.
Key Insight: Quotienting null disagreement gives definiteness. □

Theorem 3.6.5. *Completeness of the Nikodym Metric*
If $(X, \mathcal{M}, \mu)$ *is a finite measure space, then* $(\mathcal{M}_\mu, \rho_\mu)$ *is complete.*

Proof. Let (A_n) be ρ_μ-Cauchy. Choose $n_k \uparrow \infty$ with $\rho_\mu(A_{n_{k+1}}, A_{n_k}) < 2^{-k}$, and set $D_k := A_{n_{k+1}} \triangle A_{n_k}$.
Since $\mu(\bigcup_{k\ge N} D_k) \le \sum_{k\ge N} 2^{-k} \to 0$, $N_0 := \bigcap_N \bigcup_{k\ge N} D_k$ is null. Put $A := \bigcup_N \bigcap_{k\ge N} A_{n_k}$.
Then $A_{n_k} \triangle A \subseteq N_0 \cup \bigcup_{j\ge k} D_j$, so $\rho_\mu(A_{n_k}, A) \le \sum_{j\ge k} 2^{-j} \to 0$. Since (A_n) is Cauchy, $\rho_\mu(A_n, A) \to 0$.

Structural Commentary: Summable jumps give an a.e. eventual set limit.
Key Insight: Summable symmetric differences confine infinitely changing membership to a null set. □

Proposition 3.6.6. *Absolutely Continuous Measures Are Nikodym-Continuous*
Let $(X, \mathcal{M}, \mu)$ *be finite, let* ν *be a finite measure with* $\nu \ll \mu$, *and define* $\Phi_\nu([E]) := \nu(E)$.
(i) $\Phi_\nu : \mathcal{M}_\mu \to \mathbb{R}$ *is well-defined.*
(ii) Φ_ν *is uniformly continuous.*

Proof.

(i) If $A \sim_\mu B$, then $\mu(A\triangle B) = 0$, hence $\nu(A\triangle B) = 0$ and $\nu(A) = \nu(B)$.
(ii) By Lemma 3.6.2, $|\nu(A) - \nu(B)| \le \nu(A\triangle B)$. Given $\varepsilon > 0$, choose $\delta > 0$ by Lemma 3.1.14 such that $\mu(E) < \delta$ implies $\nu(E) < \varepsilon$.
Thus $\rho_\mu(A,B) < \delta$ implies $|\Phi_\nu(A) - \Phi_\nu(B)| < \varepsilon$.

Structural Commentary: Null-set identification defines the map; absolute continuity controls its modulus.
Key Insight: Absolute continuity converts small Nikodym distance into small change of mass. □

Proposition 3.6.7. *Uniform Absolute Continuity and Radon–Nikodym Derivatives*
Let $(X, \mathcal{M}, \mu)$ *be finite, let* (ν_n) *be finite measures with* $\nu_n \ll \mu$, *and set* $f_n = \frac{d\nu_n}{d\mu}$. *The following are equivalent.*

(i) (ν_n) *is uniformly absolutely continuous with respect to* μ.
(ii) The functions $\Phi_{\nu_n} : \mathcal{M}_\mu \to \mathbb{R}$ *are uniformly equicontinuous.*
(iii) The functions Φ_{ν_n} *are equicontinuous at* $\emptyset$.
(iv) $\{f_n : n \ge 1\}$ *is uniformly integrable:* $\forall \varepsilon > 0\ \exists \delta > 0\ \forall E \in \mathcal{M}$, $\mu(E) < \delta \Rightarrow \sup_n \int_E f_n\, d\mu < \varepsilon$.

Proof.

(i) $\Rightarrow$ (ii) Given $\varepsilon > 0$, choose $\delta > 0$ with $\mu(E) < \delta \Rightarrow \sup_n \nu_n(E) < \varepsilon$.
If $\rho_\mu(A,B) = \mu(A\triangle B) < \delta$, then $|\Phi_{\nu_n}(A) - \Phi_{\nu_n}(B)| = |\nu_n(A) - \nu_n(B)| \le \nu_n(A\triangle B) < \varepsilon$ uniformly in n.
(ii) $\Rightarrow$ (iii) Given $\varepsilon > 0$, uniform equicontinuity gives $\delta > 0$ such that
$\rho_\mu(A,B) < \delta \Rightarrow \sup_n |\Phi_{\nu_n}(A) - \Phi_{\nu_n}(B)| < \varepsilon$. With $B = \emptyset$, this is equicontinuity at $\emptyset$.
(iii) $\Rightarrow$ (i) Given $\varepsilon > 0$, choose $\delta > 0$ such that $\rho_\mu(E,\emptyset) < \delta \Rightarrow \sup_n |\Phi_{\nu_n}(E) - \Phi_{\nu_n}(\emptyset)| < \varepsilon$.
Since $\rho_\mu(E,\emptyset) = \mu(E)$ and $\Phi_{\nu_n}(\emptyset) = 0$, $\mu(E) < \delta \Rightarrow \sup_n \nu_n(E) < \varepsilon$.
(i) $\Leftrightarrow$ (iv) For $E \in \mathcal{M}$, $\nu_n(E) = \int_E f_n\, d\mu$. Thus $\sup_n \nu_n(E) < \varepsilon \Leftrightarrow \sup_n \int_E f_n\, d\mu < \varepsilon$ for every small-μ set E.

Structural Commentary: Uniform AC of ν_n equals uniform integrability of f_n.
Key Insight: Radon–Nikodym turns set bounds into integral bounds. □

Theorem 3.6.8. *Setwise Limits under Uniform Absolute Continuity*
Let $(X, \mathcal{M}, \mu)$ *be finite, let* ν_n *be finite measures with* $\nu_n \ll_{\mathrm{u}} \mu$, $\sup_n \nu_n(X) < +\infty$, *and* $\nu_n(E) \to \nu(E)\ \forall E \in \mathcal{M}$.

(i) ν *is a finite measure.*
(ii) $\nu \ll \mu$.

Proof.

(i) Since $\nu(X) = \lim_n \nu_n(X) \le \sup_n \nu_n(X)$, ν is finite-valued. Finite additivity passes to the pointwise limit.
For pairwise disjoint (E_k), set $E := \bigcup_k E_k$ and $R_N := \bigcup_{k>N} E_k$.
Given $\varepsilon > 0$, uniform absolute continuity and $\mu(R_N) \downarrow 0$ give $\nu_n(R_N) < \varepsilon\ \forall n$ and all sufficiently large N, hence $\nu(R_N) \le \varepsilon$.
Therefore $\nu(E) = \sum_{k=1}^{N} \nu(E_k) + \nu(R_N) \to \sum_k \nu(E_k)$.
(ii) If $\mu(A) = 0$, then for every $\varepsilon > 0$, uniform absolute continuity gives $\sup_n \nu_n(A) < \varepsilon$.
Passing to the setwise limit, $\nu(A) \le \varepsilon\ \forall \varepsilon > 0$, hence $\nu(A) = 0$.

Structural Commentary: Uniform tail control yields countable additivity and preserves the null ideal.
Key Insight: Uniform absolute continuity makes the tail of any disjoint union uniformly small. □

Lemma 3.6.9. *One-Point Equicontinuity Spreads*
Let $(X, \mathcal{M}, \mu)$ be finite, and let (ν_n) be finite measures with $\nu_n \ll \mu$.
If the maps Φ_{ν_n} are equicontinuous at one point $E_0 \in \mathcal{M}_\mu$, then they are uniformly equicontinuous on $\mathcal{M}_\mu$.

Proof. Fix $\varepsilon > 0$ and choose $\delta > 0$ so that $\rho_\mu(E, E_0) < \delta$ implies $\sup_n |\nu_n(E) - \nu_n(E_0)| < \frac{\varepsilon}{2}$.
If $\mu(C) < \delta$, then both $E_0 \setminus C$ and $E_0 \cup (C \setminus E_0)$ are within δ of E_0, so $\nu_n(C \cap E_0) < \frac{\varepsilon}{2}$, $\nu_n(C \setminus E_0) < \frac{\varepsilon}{2}$ $\forall n$.
Hence $\sup_n \nu_n(C) < \varepsilon$. Taking $C = A \triangle B$ gives $\sup_n |\nu_n(A) - \nu_n(B)| < \varepsilon$ whenever $\rho_\mu(A, B) < \delta$.
Structural Commentary: Split a small set into its portions inside and outside the equicontinuity base point.
Key Insight: Smallness near one set controls both the part inside it and the part outside it. □

Method 3.6.10. *Baire Mechanism for Vitali–Hahn–Saks*
Regard each absolutely continuous measure ν_n as $\Phi_{\nu_n} : \mathcal{M}_\mu \to \mathbb{R}$, $\Phi_{\nu_n}(E) := \nu_n(E)$.
(i) *Translate*: Setwise convergence becomes pointwise convergence of (Φ_{ν_n}).
(ii) *Apply Baire*: Completeness of $\mathcal{M}_\mu$ supplies a point of equicontinuity.
(iii) *Spread*: Lemma 3.6.9 upgrades one-point equicontinuity to uniform equicontinuity.
(iv) *Return to measures*: Proposition 3.6.7 converts uniform equicontinuity into uniform absolute continuity.

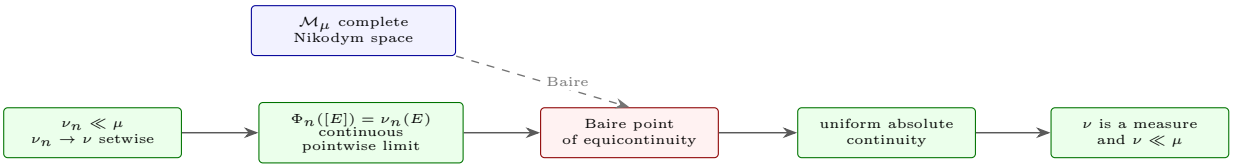

Figure 213: Vitali–Hahn–Saks mechanism: setwise convergence becomes pointwise convergence on $\mathcal{M}_\mu$; Baire gives uniform absolute continuity.

Theorem 3.6.11. *Vitali–Hahn–Saks Theorem*
Let $(X, \mathcal{M}, \mu)$ be finite, and let (ν_n) be finite measures with $\nu_n \ll \mu$, $\sup_n \nu_n(X) < +\infty$, and $\nu_n \to \nu$ setwise.

(i) (ν_n) is uniformly absolutely continuous with respect to μ.
(ii) ν is a finite measure.
(iii) $\nu \ll \mu$.

Proof.
(i) Set $\Phi_n([E]) := \nu_n(E)$. By Proposition 3.6.6, each Φ_n is continuous on complete $\mathcal{M}_\mu$.
If $A \sim_\mu B$, then $\nu_n(A) = \nu_n(B)$ $\forall n$, hence $\nu(A) = \nu(B)$.
Thus $\Phi([E]) := \nu(E)$ is well-defined and $\Phi_n \to \Phi$ pointwise. Since $|\Phi_n([E])| \le \sup_n \nu_n(X)$ and $\nu_n(E) \to \nu(E)$, Φ is real-valued.
Theorem 4.3.12 gives an equicontinuity point $E_0 \in \mathcal{M}_\mu$ for (Φ_n).
Lemma 3.6.9 gives uniform equicontinuity on $\mathcal{M}_\mu$, and Proposition 3.6.7 gives $\nu_n \ll_{\mathrm{u}} \mu$.
(ii) By (i), $\nu_n \ll_{\mathrm{u}} \mu$; Theorem 3.6.8(i) gives that ν is a finite measure.
(iii) By (i), $\nu_n \ll_{\mathrm{u}} \mu$; Theorem 3.6.8(ii) gives $\nu \ll \mu$.
Structural Commentary: Baire gives one equicontinuity point; Nikodym calculus spreads it.
Key Insight: Setwise convergence plus bounded total mass forces uniform null-set control. □

Corollary 3.6.12. *Nikodym Bounded Setwise Limit Theorem*
If finite measures (ν_n) on $(X, \mathcal{M})$ converge setwise to ν and $\sup_n \nu_n(X) < +\infty$, then ν is finite.

Proof. Define $\mu(E) := \sum_{n=1}^{\infty} 2^{-n} \nu_n(E)$. Then μ is a finite measure by countable additivity of nonnegative series, and $\nu_n \ll \mu$ $\forall n$ because $2^{-n}\nu_n \le \mu$. Thus ν is a finite measure by Theorem 3.6.11(ii).
Structural Commentary: A summable mixture places the sequence under one finite dominating measure.
Key Insight: A summable mixture builds one finite measure that dominates the whole sequence. □

4 L^p Spaces and Modes of Convergence

4.1 L^p Spaces

Lebesgue spaces turn integrability into normed linear geometry. The essential point is that an integral norm cannot distinguish functions that differ only on null sets, so the actual vectors are almost-everywhere equivalence classes. Their bounded linear functionals are themselves represented by integration against functions in the conjugate space.

Definition 4.1.1. *Norms and Seminorms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
(i) A *seminorm* is $N : \mathcal{V} \to [0, \infty)$ with $N(\alpha v) = |\alpha| N(v)$ and $N(u + v) \le N(u) + N(v)$ $\forall u, v \in \mathcal{V}$, $\alpha \in \mathbb{F}$.
(ii) A *norm* is a seminorm with $N(v) = 0 \iff v = 0$. A norm N induces $d_N(u, v) := N(u - v)$.

Example 4.1.2. *Norms and Seminorms Before Quotienting*
(i) On $\mathbb{F}^n$, $\|u\|_p := (\sum_{j=1}^{n} |u_j|^p)^{\frac{1}{p}}$ for $1 \le p < \infty$ and $\|u\|_\infty := \max_{1 \le j \le n} |u_j|$ are norms.
Minkowski gives $\|u + v\|_p \le \|u\|_p + \|v\|_p$.

(ii) On sequences, $\ell^p := \{a : \sum_n |a_n|^p < +\infty\}$ with $\|a\|_p := (\sum_n |a_n|^p)^{\frac{1}{p}}$.
$\ell^\infty := \{a : \sup_n |a_n| < +\infty\}$ with $\|a\|_\infty := \sup_n |a_n|$.

(iii) On $B(X)$, $\|f\|_\infty := \sup_{x\in X} |f(x)|$ is a norm. On $C[0,1]$, $N_0(f) := |f(0)|$ and $N_1(f) := \sup_{0\le x\le \frac{1}{2}} |f(x)|$ are seminorms. On $C^1[a,b]$, $N_D(f) := \sup_{a\le x\le b} |f'(x)|$ is a seminorm with kernel the constant functions.

Key Insight: Seminorms become norms after quotienting out their zero directions.

Remark 4.1.3. *Integral Norm on Continuous Functions*
On $C[0,1]$, $N(f) := \int_0^1 |f(x)|\,dx$ is a norm: if $f(x_0) \neq 0$, continuity gives $N(f) > 0$. It is not complete, since continuous ramps can be Cauchy in N and converge in L^1 to a discontinuous step function.

Definition 4.1.4. *L^p Spaces and Mean Convergence*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) For $1 \le p < \infty$, the *p-integrable class* is $\mathscr{L}^p(\mu) := \{f : X \to \mathbb{F} : f \text{ measurable}, \int_X |f|^p\,d\mu < +\infty\}$.

(ii) The *L^p space* and *L^p norm* are $L^p(\mu) := \mathscr{L}^p(\mu)/=_{\text{a.e.}}$ and $\|f\|_p := (\int_X |f|^p\,d\mu)^{\frac{1}{p}}$.

(iii) *L^p convergence*, or *mean convergence of order p*, means $f_n \to f \iff \|f_n - f\|_p \to 0$.
L^p-Cauchy means $\|f_m - f_n\|_p \to 0$ as $m, n \to \infty$.

(iv) *Essentially bounded* means $|f| \le M$ a.e. for some $M < +\infty$; $L^\infty(\mu)$ is the a.e.-quotient of such functions, with *essential sup norm* $\|f\|_\infty := \inf\{M \ge 0 : |f| \le M \text{ a.e.}\} = \inf_{\mu(N)=0} \sup_{x\in X\setminus N} |f(x)|$.

Proposition 4.1.5. *L^1 as a Quotient of a Seminorm*
On $\mathscr{L}^1(\mu)$ set $N_\mu(f) := \int_X |f|\,d\mu$.

(i) N_μ is a seminorm.
(ii) $N_\mu(f) = 0 \iff f = 0$ a.e.
(iii) $L^1(\mu) := \mathscr{L}^1(\mu)/=_{\text{a.e.}}$ is normed by $\|[f]\|_1 := \int_X |f|\,d\mu$; write f for $[f]$.

Proof.

(i) If $f, g \in \mathscr{L}^1(\mu)$ and $a, b \in \mathbb{R}$, then $|af + bg| \le |a|\,|f| + |b|\,|g| \in \mathscr{L}^1(\mu)$, so $\mathscr{L}^1(\mu)$ is linear. Also $N_\mu(af) = |a|\,N_\mu(f)$ and $N_\mu(f+g) \le N_\mu(f) + N_\mu(g)$.
(ii) ($\Rightarrow$) If $N_\mu(f) = 0$, set $E_n := \{|f| \ge \frac{1}{n}\}$. Then $\frac{1}{n}\mu(E_n) \le \int_{E_n} |f|\,d\mu \le N_\mu(f) = 0$, $\{f \neq 0\} = \bigcup_n E_n$ null.
(ii) ($\Leftarrow$) If $f = 0$ a.e., then $|f| = 0$ a.e., hence $N_\mu(f) = 0$.
(iii) If $f = g$ a.e., then $|f| = |g|$ a.e. and $\int_X |f|\,d\mu = \int_X |g|\,d\mu$, so $\|[f]\|_1$ is well defined. The seminorm axioms descend from (i), and $\|[f]\|_1 = 0 \Rightarrow f = 0$ a.e. by (ii), hence $[f] = [0]$.

Structural Commentary: The seminorm kernel is exactly the a.e.-zero functions.
Key Insight: Quotient by null differences before calling the integral a norm. □

Definition 4.1.6. *Complex Integrals and L^2*

(i) For $f = u + iv$, *complex measurability* means measurability of u, v.
(ii) On $E \in \mathcal{M}$, *complex integrability* means $u, v \in \mathscr{L}^1(\mu)$ on E; set $\int_E f\,d\mu := \int_E u\,d\mu + i\int_E v\,d\mu$.
(iii) $L^2(\mu)$: complex measurable f satisfying $\int_X |f|^2\,d\mu < +\infty$, modulo a.e. equality; $\|f\|_2^2 := \int_X |f|^2\,d\mu$.
(iv) The *L^2 inner product* is $\langle f, g\rangle_{L^2} := \int_X f\overline{g}\,d\mu$; it induces $\|f\|_2$.

Proposition 4.1.7. *Complex L^1 and Dominated Convergence*
The following statements hold.

(i) For complex measurable f on E, f is integrable $\iff |f| \in \mathscr{L}^1(\mu)$ on E; then $\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu$.
(ii) If $f_n \to f$ a.e. on E and $|f_n| \le g \in \mathscr{L}^1(\mu)$, then f is integrable and $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$.

Proof.

(i) ($\Rightarrow$) Write $f = u + iv$. If $u, v \in \mathscr{L}^1$, then $|f| \le |u| + |v|$, so $|f| \in \mathscr{L}^1$.
(i) ($\Leftarrow$) If $|f| \in \mathscr{L}^1$, then $|u|, |v| \le |f|$, so $u, v \in \mathscr{L}^1$ and f is integrable.
If $\int_E f\,d\mu = re^{i\theta}$ and $h = e^{-i\theta}f$, then $\left|\int_E f\,d\mu\right| = r = \operatorname{Re}\int_E h\,d\mu = \int_E \operatorname{Re} h\,d\mu \le \int_E |h|\,d\mu = \int_E |f|\,d\mu$.
(ii) Write $f_n = u_n + iv_n$ and $f = u + iv$. Since $|u_n|, |v_n| \le g$ and $u_n \to u$, $v_n \to v$ a.e., real DCT gives $\int_E u_n\,d\mu \to \int_E u\,d\mu$ and $\int_E v_n\,d\mu \to \int_E v\,d\mu$; hence the complex identity.

Structural Commentary: Complex integration is two real integrals.
Key Insight: Rotate the integral to make its value real and nonnegative before applying the real inequality. □

Theorem 4.1.8. *L^2 Geometry*
For $f, g \in L^2(\mu)$, the following hold.

(i) Schwarz: $fg \in L^1(\mu)$ and $\int_X |fg|\,d\mu \le \|f\|_2\,\|g\|_2$.
(ii) Minkowski: $f + g \in L^2(\mu)$ and $\|f+g\|_2 \le \|f\|_2 + \|g\|_2$.
(iii) Completeness: modulo equality a.e., $(L^2(\mu), \|\cdot\|_2)$ is complete.

Proof.

(i) Since $2|fg| \le |f|^2 + |g|^2$, $fg \in L^1(\mu)$.
For $\lambda \in \mathbb{R}$, $0 \le Q(\lambda) := \int_X (|f| + \lambda|g|)^2\,d\mu = \|f\|_2^2 + 2\lambda\int_X |fg|\,d\mu + \lambda^2\|g\|_2^2$; hence $\Delta(Q) \le 0$.
(ii) By (i), $\|f+g\|_2^2 \le \int_X (|f| + |g|)^2\,d\mu \le (\|f\|_2 + \|g\|_2)^2$.

(iii) Let (f_n) be Cauchy. Choose $n_k \uparrow \infty$ with $\|f_{n_{k+1}} - f_{n_k}\|_2 \le 2^{-k}$, and set $h_N := \sum_{k<N} |f_{n_{k+1}} - f_{n_k}|$.

Then $\|h_N\|_2 \le \sum_{k<N} 2^{-k}$, so Fatou gives $h := \lim_N h_N \in L^2$ and $f_{n_k} \to f$ a.e. for some f.

For fixed n, Fatou gives $\|f - f_n\|_2 \le \liminf_k \|f_{n_k} - f_n\|_2$; the Cauchy property sends the right side to 0.

Structural Commentary: Schwarz and Minkowski supply geometry; rapid subsequences prove completeness.
Key Insight: Positivity of the integral transfers Euclidean quadratic estimates to function classes. □

Example 4.1.9. *Mazur Homeomorphism Between L^1 and L^2*
Let $E \subseteq \mathbb{R}$ be Lebesgue measurable. Define $\Phi(f) := \operatorname{sgn}(f)\,|f|^{\frac{1}{2}}$ on $L^1(E)$ and $\Psi(g) := \operatorname{sgn}(g)\,|g|^2$ on $L^2(E)$.
Then $\Phi : L^1(E) \to L^2(E)$ is a homeomorphism with inverse $\Psi : L^2(E) \to L^1(E)$.

Proof. $|\Phi(f)|^2 = |f|$ and $|\Psi(g)| = |g|^2$, so the maps are well-defined; pointwise $\Psi\Phi = \mathrm{id}_{L^1}$ and $\Phi\Psi = \mathrm{id}_{L^2}$.
For $a, b \in \mathbb{R}$, $\left|\operatorname{sgn}(a)\,|a|^{\frac{1}{2}} - \operatorname{sgn}(b)\,|b|^{\frac{1}{2}}\right|^2 \le 2\,|a-b|$, hence $\|\Phi(f) - \Phi(h)\|_2^2 \le 2\,\|f-h\|_1$.
Also $\left|\operatorname{sgn}(a)\,|a|^2 - \operatorname{sgn}(b)\,|b|^2\right| \le 2\,|a-b|\,(|a|+|b|)$, so $\|\Psi(g) - \Psi(k)\|_1 \le 2\,\|g-k\|_2\,(\|g\|_2 + \|k\|_2)$ by Cauchy–Schwarz. Thus Φ and Ψ are continuous.
Key Insight: Nonlinear power maps can homeomorphically identify different L^p norm topologies. □

Definition 4.1.10. *Conjugate Indices*
For $p, q \in (1, \infty)$, p and q are *conjugate indices* if $p^{-1} + q^{-1} = 1$, equivalently $p + q = pq$.
The unique self-conjugate index is $p = q = 2$.

Lemma 4.1.11. *Young's Inequality*
If $A, B \ge 0$, $1 < p < \infty$, and $p^{-1} + q^{-1} = 1$, then:
(i) $AB \le \frac{A^p}{p} + \frac{B^q}{q}$;
(ii) equality holds iff $A^p = B^q$.

Proof.
(i) If $A, B > 0$, set $\alpha := \frac{1}{p}$ and $t := \frac{A^p}{B^q}$. Since $q^{-1} = 1 - \alpha$, weighted AM–GM gives $t^\alpha \le \alpha t + 1 - \alpha$.
Multiplying by B^q yields $AB \le \frac{A^p}{p} + \frac{B^q}{q}$. If $AB = 0$, the inequality is immediate.
(ii) For $A, B > 0$, equality in weighted AM–GM is $t = 1$, i.e. $A^p = B^q$.
If $AB = 0$, equality in (i) holds iff $A = B = 0$, again equivalent to $A^p = B^q$.
Structural Commentary: Young's inequality is weighted AM–GM in conjugate exponents.
Key Insight: Rescale to A^p and B^q, then use convexity. □

Theorem 4.1.12. *Hölder and Minkowski for Sequence Spaces*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
(i) If $p, q > 1$ and $p^{-1} + q^{-1} = 1$, then $\mathbf{x} \in \ell^p(\mathbb{F})$, $\mathbf{y} \in \ell^q(\mathbb{F})$ imply $\sum_{n\ge1} |x_n y_n| \le \|\mathbf{x}\|_p\,\|\mathbf{y}\|_q$.
(ii) If $1 \le p < \infty$ and $\mathbf{x}, \mathbf{y} \in \ell^p(\mathbb{F})$, then $\mathbf{x} + \mathbf{y} \in \ell^p(\mathbb{F})$ and $\|\mathbf{x}+\mathbf{y}\|_p \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$.

Proof.
(i) If $\|\mathbf{x}\|_p\,\|\mathbf{y}\|_q = 0$, the claim is immediate.
Otherwise set $X := \|\mathbf{x}\|_p$, $Y := \|\mathbf{y}\|_q$, $u_n := |x_n|/X$, and $v_n := |y_n|/Y$. Young's inequality gives $u_n v_n \le u_n^p/p + v_n^q/q$; summing over $1 \le n \le N$ and letting $N \to \infty$ gives $\sum_{n\ge1} |x_n y_n| \le XY$.
(ii) For $p = 1$, apply $|x_n + y_n| \le |x_n| + |y_n|$ and sum. For $p > 1$, put $s_N := (\sum_{n=1}^N |x_n + y_n|^p)^{1/p}$. If $s_N = 0$, the bound is trivial; if $s_N > 0$, then $\sum_{n=1}^N |x_n + y_n|^p \le \sum_{n=1}^N (|x_n| + |y_n|)\,|x_n + y_n|^{p-1} \le (\|\mathbf{x}\|_p + \|\mathbf{y}\|_p) s_N^{p-1}$ by (i), so $s_N \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$. Let $N \to \infty$.
Structural Commentary: Hölder controls products; Minkowski controls sums.
Key Insight: The metric property of ℓ^p is exactly Minkowski's inequality. □

Example 4.1.13. *Sequence Spaces as Vector Spaces*
Over $\mathbb{C}$, the sequence spaces $c_0 = \{(z_n) : z_n \to 0\}$, $\ell^\infty = \{(z_n) : \sup_n |z_n| < \infty\}$, and, for $1 \le p < \infty$, $\ell^p = \left\{(z_n) : \sum_{n=1}^{\infty} |z_n|^p < \infty\right\}$ are vector spaces under componentwise operations; for ℓ^p, closure under addition follows from Minkowski's inequality.

Example 4.1.14. *Function and Sequence Metric Spaces*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.
(i) On $C_{\mathbb{F}}[a,b]$, $d_\infty(f,g) := \sup_{t\in[a,b]} |f(t) - g(t)|$ and $d_1(f,g) := \int_a^b |f(t) - g(t)|\,dt$ are metrics.
Definiteness of d_1 uses continuity.
(ii) On $\ell^\infty(\mathbb{F})$, $d_\infty(\mathbf{x}, \mathbf{y}) := \sup_{n\ge1} |x_n - y_n|$ is a metric.
(iii) For $1 \le p < \infty$, $d_p(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_p = (\sum_{n\ge1} |x_n - y_n|^p)^{1/p}$ is a metric on $\ell^p(\mathbb{F})$ by Theorem 4.1.12.
Structural Commentary: Norm estimates produce metrics on function and sequence spaces.
Key Insight: Infinite-dimensional examples require estimates, not just coordinate formulas.

Theorem 4.1.15. *Hölder's Inequality*
Let $1 < p < \infty$, $p^{-1} + q^{-1} = 1$, $f \in L^p(\mu)$, and $g \in L^q(\mu)$.
(i) $fg \in L^1(\mu)$.
(ii) $\|fg\|_1 \le \|f\|_p\,\|g\|_q$.

Proof. If $\|f\|_p\,\|g\|_q = 0$, then $f = 0$ a.e. or $g = 0$ a.e., so both claims hold.
Assume $\|f\|_p\,\|g\|_q > 0$ and set $A := \frac{|f|}{\|f\|_p}$, $B := \frac{|g|}{\|g\|_q}$. Young gives $\frac{|fg|}{\|f\|_p\|g\|_q} \le \frac{A^p}{p} + \frac{B^q}{q}$.
(i) Since $\int A^p\,d\mu = \int B^q\,d\mu = 1$, the majorant $\frac{A^p}{p} + \frac{B^q}{q}$ lies in $L^1(\mu)$; hence $fg \in L^1(\mu)$.

(ii) Integrating the displayed inequality gives $\frac{\|fg\|_1}{\|f\|_p\|g\|_q} \le \frac{1}{p}\int A^p\,d\mu + \frac{1}{q}\int B^q\,d\mu = p^{-1}+q^{-1} = 1$.

Structural Commentary: Normalize both factors to unit L^p and L^q size.
Key Insight: Products are controlled by conjugate powers. □

Corollary 4.1.16. *Cauchy–Bunyakovsky–Schwarz in L^2*
If $f, g \in L^2(\mu)$, then:
(i) $fg \in L^1(\mu)$;
(ii) $\left|\int_X fg\,d\mu\right| \le \int_X |fg|\,d\mu \le \|f\|_2\,\|g\|_2$.

Proof.
(i) Since $2^{-1}+2^{-1}=1$, Theorem 4.1.15(i) gives $fg \in L^1(\mu)$.
(ii) Theorem 4.1.15(ii) gives $\int_X |fg|\,d\mu = \|fg\|_1 \le \|f\|_2\,\|g\|_2$, and $\left|\int_X fg\,d\mu\right| \le \int_X |fg|\,d\mu$.

Structural Commentary: Hölder at $p=q=2$ is Cauchy–Schwarz.
Key Insight: Pair two L^2 factors through $2^{-1}+2^{-1}=1$. □

Theorem 4.1.17. *Minkowski's Inequality*
If $1 \le p < \infty$ and $f, h \in L^p(\mu)$, then:
(i) $f+h \in L^p(\mu)$;
(ii) $\|f+h\|_p \le \|f\|_p + \|h\|_p$.

Proof. For $p=1$, integrate $|f+h| \le |f|+|h|$. Let $p>1$ and $q := \frac{p}{p-1}$.
(i) Since $|f+h|^p \le 2^{p-1}(|f|^p+|h|^p)$, $f+h \in L^p(\mu)$. Also $|f+h|^{p-1} \in L^q(\mu)$ because $(p-1)q = p$.
(ii) Pointwise, $|f+h|^p \le (|f|+|h|)\,|f+h|^{p-1}$.
By Hölder, $\|f+h\|_p^p \le (\|f\|_p + \|h\|_p)\left\||f+h|^{p-1}\right\|_q = (\|f\|_p+\|h\|_p)\,\|f+h\|_p^{p-1}$.
If $\|f+h\|_p > 0$, divide by $\|f+h\|_p^{p-1}$; otherwise the claim is trivial.

Structural Commentary: Hölder turns the pointwise triangle inequality into a norm triangle inequality.
Key Insight: Minkowski is exactly the closure and triangle estimate for L^p. □

Theorem 4.1.18. *L^p Is Normed*
For $1 \le p < \infty$, $(L^p(\mu), \|\cdot\|_p)$ is a normed linear space.

Proof.
(i) If $f = f'$ and $g = g'$ a.e., then $\alpha f + \beta g = \alpha f' + \beta g'$ a.e..
Hence pointwise operations descend to $L^p(\mu)$. Minkowski gives $\alpha f + \beta g \in L^p(\mu)$.
(ii) $\|\alpha f\|_p^p = \int_X |\alpha f|^p\,d\mu = |\alpha|^p\,\|f\|_p^p$, so $\|\alpha f\|_p = |\alpha|\,\|f\|_p$.
(iii) $\|f\|_p = 0 \Rightarrow \int_X |f|^p\,d\mu = 0 \Rightarrow f = 0$ a.e.; hence $[f] = 0$ in $L^p(\mu)$.
(iv) Minkowski gives $\|f+g\|_p \le \|f\|_p + \|g\|_p$.

Structural Commentary: Quotienting by a.e. equality turns pointwise operations into vector-space operations.
Key Insight: Minkowski supplies the nontrivial norm axiom. □

Definition 4.1.19. *Completeness Language*
Let $(V, \|\cdot\|)$ be a normed space.
(i) *Complete*: V is complete iff every Cauchy sequence $(x_n) \subset V$ has $x \in V$ with $\|x_n - x\| \to 0$.
(ii) *Banach*: A Banach space is a complete normed space.
(iii) *L^p criterion*: $L^p(\mu)$ is complete iff every L^p-Cauchy sequence converges in $\|\cdot\|_p$.

Lemma 4.1.20. *Convergent Implies Cauchy*
If $f_n \to f$ in $L^p(\mu)$, then (f_n) is Cauchy in $L^p(\mu)$.

Proof. Given $\varepsilon > 0$, choose N so $n \ge N$ implies $\|f_n - f\|_p < \frac{\varepsilon}{2}$.
Then $m, n \ge N$ gives $\|f_m - f_n\|_p \le \|f_m - f\|_p + \|f - f_n\|_p < \varepsilon$ by Minkowski.
Structural Commentary: A convergent sequence has uniformly small tails around its limit.
Key Insight: Use the common limit as the bridge between two tails. □

Definition 4.1.21. *Rapidly Cauchy Sequence*
Let $(V, \|\cdot\|)$ be a normed space. A sequence $(x_k) \subset V$ is *rapidly Cauchy* if $\exists (\varepsilon_k) \subset (0, +\infty)$ such that:
(i) $\sum_{k=1}^{\infty} \varepsilon_k < +\infty$.
(ii) $\|x_{k+1} - x_k\| \le \varepsilon_k^2$ for every $k \ge 1$.

Proposition 4.1.22. *Rapidly Cauchy Subsequences*
In a normed space, the following hold.
(i) Every rapidly Cauchy sequence is Cauchy.
(ii) Every Cauchy sequence has a rapidly Cauchy subsequence.

Proof.
(i) Let (x_k) be rapidly Cauchy with witnesses (ε_k). For $m > n$, $\|x_m - x_n\| \le \sum_{j=n}^{m-1} \|x_{j+1} - x_j\| \le \sum_{j=n}^{\infty} \varepsilon_j^2$.
Since $\sum_j \varepsilon_j < +\infty$, eventually $\varepsilon_j \le 1$, so $\sum_{j=n}^{\infty} \varepsilon_j^2 \le \sum_{j=n}^{\infty} \varepsilon_j \to 0$.
(ii) Let (x_n) be Cauchy. Choose $n_1 < n_2 < \cdots$ such that $m, n \ge n_k \Rightarrow \|x_m - x_n\| \le 4^{-k}$.
Set $y_k := x_{n_k}$ and $\varepsilon_k := 2^{-k}$. Then $\sum_k \varepsilon_k < +\infty$ and $\|y_{k+1} - y_k\| \le 4^{-k} = \varepsilon_k^2$, so (y_k) is rapidly Cauchy.

Structural Commentary: A Cauchy tail contains a summably controlled thread.
Key Insight: Replace arbitrary tail control by a geometric subsequence. □

Theorem 4.1.23. *Riesz–Fischer Completeness Theorem*
For $1 \le p < \infty$, the following hold.

(i) Completeness: $(L^p(\mu), \|\cdot\|_p)$ is complete.
(ii) A.E. subsequence: if $f_n \to f$ in $L^p(\mu)$, then some subsequence converges to a representative of f a.e.

Proof.

(i) Let (f_n) be Cauchy. Choose $g_k = f_{n_k}$ with $\|g_{k+1} - g_k\|_p \le 4^{-k}$. Set $G_N := |g_1| + \sum_{k<N} |g_{k+1} - g_k|$.
By Minkowski, $\|G_N\|_p \le \|g_1\|_p + \sum_{k<N} 4^{-k} \le \|g_1\|_p + \frac{1}{3}$; Fatou gives $G := \lim_N G_N \in L^p$.
The series $h := g_1 + \sum_k (g_{k+1} - g_k)$ converges a.e. with $|h| \le G$ and $g_k \to h$ a.e.
Since $|g_k - h| \le 2G$, DCT gives $\|g_k - h\|_p \to 0$. Given $\varepsilon > 0$, choose k with $\|g_k - h\|_p < \frac{\varepsilon}{2}$ and then N with $n \ge N \Rightarrow \|f_n - g_k\|_p < \frac{\varepsilon}{2}$; hence $\|f_n - h\|_p \to 0$.
(ii) If $f_n \to f$ in L^p, then (f_n) is Cauchy. Applying (i) gives a subsequence $f_{n_k} \to h$ a.e. and in L^p.
Since also $f_{n_k} \to f$ in L^p, uniqueness in normed space gives $[h] = f$ in L^p, so h is a representative of f.

Structural Commentary: A rapid subsequence converges a.e.; DCT gives norm convergence.
Key Insight: Fatou finds the pointwise limit, dominated convergence brings it back to L^p. □

Proposition 4.1.24. *Essential Supremum Test*
If $f \in L^\infty(\mu)$, then:

(i) $|f| \le \|f\|_\infty$ a.e.;
(ii) $A < \|f\|_\infty \Rightarrow \mu(\{|f| > A\}) > 0$.

Proof.

(i) Let $M := \|f\|_\infty$. For each k, $N_k := \{|f| > M + \frac{1}{k}\}$ is null. On $X \setminus \bigcup_k N_k$, $|f| \le M + \frac{1}{k}$ $\forall k$, hence $|f| \le M$.
(ii) If $A < M$ and $\mu(\{|f| > A\}) = 0$, then $|f| \le A$ a.e., so A is an essential upper bound. Contradiction.

Structural Commentary: The essential supremum is the least a.e. upper bound.
Key Insight: The sharp bound cannot be lowered on a full-measure set. □

Theorem 4.1.25. *L^∞ Is Banach*
$(L^\infty(\mu), \|\cdot\|_\infty)$ is a Banach space.

Proof.

(i) Pointwise operations descend modulo a.e. equality, and $\|\alpha f\|_\infty = |\alpha| \|f\|_\infty$. Proposition 4.1.24(i) gives definiteness; outside the union of two null sets, $|f + g| \le \|f\|_\infty + \|g\|_\infty$, hence $\|f + g\|_\infty \le \|f\|_\infty + \|g\|_\infty$.
(ii) Let (f_n) be L^∞-Cauchy. For each m, n, choose null N_{mn} with $|f_m - f_n| \le \|f_m - f_n\|_\infty$ on $X \setminus N_{mn}$, and set $N := \bigcup_{m,n} N_{mn}$.
On $X \setminus N$, (f_n) is uniformly Cauchy, hence converges uniformly to a measurable f.
(iii) The uniform convergence off N gives $\|f_n - f\|_\infty \to 0$.
For fixed n_0, $|f| \le |f_{n_0}| + \sup_{n \ge n_0} \|f_n - f_{n_0}\|_\infty$ a.e.; hence $f \in L^\infty(\mu)$.

Structural Commentary: Completeness is uniform convergence off one null set.
Key Insight: Essential bounds become ordinary uniform bounds off a common null set. □

Proposition 4.1.26. *Simple Approximation, Support, and Tails*
Let $1 \le p < \infty$ and $f \in L^p(\mu)$.

(i) Simple measurable functions are dense in $L^p(\mu)$; also in $L^\infty(\mu)$ for $\|\cdot\|_\infty$.
(ii) $\{|f| > 0\}$ is σ-finite.
(iii) For $E_n := \{|f| \ge n\}$, $\mu(E_n) \to 0$.
(iv) $\forall \varepsilon > 0$ $\exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset$ implies $\|f \mathbf{1}_F\|_p < \varepsilon$.

Proof.

(i) Choose simple $s_n \to f$ a.e. with $|s_n| \le |f|$ in the real case; treat real and imaginary parts separately in the complex case. Then $|f - s_n|^p \le 2^p |f|^p$, so DCT gives $\|f - s_n\|_p \to 0$.
For $f \in L^\infty$, discard null N with $|f| \le \|f\|_\infty + \eta$ on $X \setminus N$, then quantize that bounded range.
(ii) Put $A_k := \{|f| \ge \frac{1}{k}\}$. Then $\mu(A_k) \le k^p \|f\|_p^p < +\infty$ and $\{|f| > 0\} = \bigcup_k A_k$.
(iii) $n^p \mu(E_n) \le \int_{E_n} |f|^p \, d\mu \le \|f\|_p^p$, hence $\mu(E_n) \le n^{-p} \|f\|_p^p \to 0$.
(iv) With A_k as in (ii), $A_k \uparrow \{|f| > 0\}$ and $|f|^p \mathbf{1}_{X \setminus A_k} \to 0$ a.e. DCT gives $\int_{X \setminus A_k} |f|^p \, d\mu \to 0$.
Choose $E_\varepsilon = A_k$ with this integral $< \varepsilon^p$; if $F \cap E_\varepsilon = \emptyset$, then $\|f \mathbf{1}_F\|_p^p \le \int_{X \setminus E_\varepsilon} |f|^p \, d\mu < \varepsilon^p$.

Structural Commentary: L^p mass is simple-approximable and finite-piece capturable.
Key Insight: Integrability turns level sets into finite-measure exhaustions. □

Proposition 4.1.27. *Shell Criterion on Finite Measure Spaces*
Let $\mu(X) < +\infty$, f be measurable, and $E_n := \{n - 1 \le |f| < n\}$ for $n \ge 1$.
For $1 \le p < \infty$, $f \in L^p(\mu) \iff \sum_{n \ge 1} n^p \mu(E_n) < +\infty$.

Proof.

($\Rightarrow$) If $f \in L^p(\mu)$, then $\mu(E_1) \le \mu(X) < +\infty$ and, for $n \ge 2$, $n^p \mathbf{1}_{E_n} \le 2^p |f|^p \mathbf{1}_{E_n}$.
Thus $\sum_n n^p \mu(E_n) \le \mu(X) + 2^p \int_X |f|^p \, d\mu < +\infty$.

($\Leftarrow$) If $\sum_n n^p\mu(E_n) < +\infty$, then $|f|^p = \sum_n |f|^p \mathbf{1}_{E_n} \le \sum_n n^p \mathbf{1}_{E_n}$, hence $\int_X |f|^p \, d\mu < +\infty$.

Structural Commentary: The finite base shell handles small values; weighted shells control tails.
Key Insight: Shell weights characterize L^p mass. □

Proposition 4.1.28. *Inclusions, Interpolation, and Multipliers*

(i) If $\mu(X) < +\infty$ and $1 \le r \le p < \infty$, then $L^p(\mu) \subseteq L^r(\mu)$ and $\|f\|_r \le \|f\|_p \, \mu(X)^{\frac{1}{r}-\frac{1}{p}}$.
(ii) On $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, $1 \le p \le s < \infty$ implies $\ell^p \subseteq \ell^s$ and $\|a\|_s \le \|a\|_p$.
(iii) If $1 \le p_1 < p_2 < \infty$ and $f \in L^{p_1}(\mu) \cap L^{p_2}(\mu)$, then $f \in L^p(\mu)$ for $p_1 \le p \le p_2$.
(iv) If $1 \le p \le \infty$, then $L^p(\mu) \cdot L^\infty(\mu) \subseteq L^p(\mu)$ and $\|fg\|_p \le \|f\|_p \|g\|_\infty$.
(v) $L^\infty(\mu) \subseteq L^1(\mu) \iff \mu(X) < +\infty$. If $\mu(X) = 1$, then $\|f\|_\infty = \lim_{p\to\infty} \|f\|_p$ for $f \in L^\infty(\mu)$.

Proof.

(i) For $r = p$, $L^p \subseteq L^p$ and the estimate reads $\|f\|_p \le \|f\|_p$, i.e. equality.
For $r < p$, Hölder applied to $|f|^r \cdot 1$ gives $\int_X |f|^r \, d\mu \le \|f\|_p^r \, \mu(X)^{1-\frac{r}{p}}$; take rth roots.
(ii) For $a \in \ell^p$, $\forall n$, $|a_n| \le \|a\|_p$, hence $\sum_n |a_n|^s \le \|a\|_p^{s-p} \sum_n |a_n|^p \le \|a\|_p^s$.
(iii) For $p \in [p_1, p_2]$, $|f|^p \le |f|^{p_1} \mathbf{1}_{\{|f|\le 1\}} + |f|^{p_2} \mathbf{1}_{\{|f|>1\}}$.
(iv) For $p < \infty$, $|fg|^p \le \|g\|_\infty^p \, |f|^p$ a.e.; for $p = \infty$, $|fg| \le \|f\|_\infty \|g\|_\infty$ a.e.
(v) ($\Rightarrow$) If $L^\infty(\mu) \subseteq L^1(\mu)$, then $\mathbf{1}_X \in L^\infty(\mu)$ lies in $L^1(\mu)$, so $\mu(X) = \int_X \mathbf{1}_X \, d\mu < +\infty$.
(v) ($\Leftarrow$) If $\mu(X) < +\infty$ and $f \in L^\infty(\mu)$, then $\int_X |f| \, d\mu \le \|f\|_\infty \, \mu(X) < +\infty$. If $\mu(X) = 1$, then $\|f\|_p \le \|f\|_\infty$.
For $A < \|f\|_\infty$, $\mu(\{|f| > A\}) > 0$ and $\|f\|_p \ge A\mu(\{|f| > A\})^{\frac{1}{p}} \to A$. Let $A \uparrow \|f\|_\infty$.

Structural Commentary: Inclusions depend on domain size and pointwise bounds.
Key Insight: Check whether size comes from the function values or from the measure of the domain. □

Example 4.1.29. *Membership Models*

(i) On $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, $a_n = \frac{1}{n}$ satisfies $\|a\|_p^p = \sum_{n\ge 1} n^{-p}$; hence $a \in \ell^p \iff p > 1$, while $a \in \ell^\infty \setminus \ell^1$.
(ii) On $\mathbb{N}$ with $\lambda(\{n\}) = n^{-2}$, $\lambda(\mathbb{N}) < +\infty$ and $f(n) = n^{\frac{1}{p_0}}$ satisfies $\int_{\mathbb{N}} |f|^p \, d\lambda = \sum_{n\ge 1} n^{\frac{p}{p_0}-2}$.
Hence $f \in L^p(\lambda) \iff 1 \le p < p_0$. In particular, $f(n) = \sqrt{n}$ gives $p_0 = 2$.
(iii) On $(0, \infty)$, $f(x) = x^{-\frac{1}{2}}(1 + |\log x|)^{-1}$ gives $\int_0^\infty |f|^p \, dx = \int_{-\infty}^\infty e^{(1-\frac{p}{2})t}(1 + |t|)^{-p} \, dt$ under $x = e^t$.

Structural Commentary: L^p membership reduces to series and endpoint tests.
Key Insight: Different ends impose different p-conditions.

Proposition 4.1.30. *Hölder Dual Norm Attainment*
Let $1 < p < \infty$, $p^{-1} + q^{-1} = 1$, and let $f \in L^p(\mu)$ be real-valued.

(i) $\sup_{\|g\|_q \le 1} \left|\int_X fg \, d\mu\right| = \|f\|_p$.
(ii) If $f \ne 0$, the supremum is attained by $g_0 = \|f\|_p^{-\frac{p}{q}} \operatorname{sign}(f) \, |f|^{p-1}$.

Proof.

(i) For $\|g\|_q \le 1$, Hölder gives $\left|\int_X fg \, d\mu\right| \le \|f\|_p \|g\|_q \le \|f\|_p$.
If $f = 0$ this is sharp; if $f \ne 0$, item (ii) gives an admissible g_0 with $\int_X fg_0 \, d\mu = \|f\|_p$.
(ii) Since $(p-1)q = p$, $\|g_0\|_q^q = \|f\|_p^{-p} \int_X |f|^p \, d\mu = 1$ and $\int_X fg_0 \, d\mu = \|f\|_p^{-\frac{p}{q}} \int_X |f|^p \, d\mu = \|f\|_p$.

Structural Commentary: Equality in Hölder is obtained by matching the conjugate powers.
Key Insight: The L^q test function points in the power direction of f. □

Definition 4.1.31. *Bounded Linear Functionals on L^p*

(i) A *linear functional* on real $L^p(\mu)$ is a linear map $G : L^p(\mu) \to \mathbb{R}$.
(ii) G is *bounded* if $\exists M < +\infty$ such that $\forall f \in L^p(\mu)$, $|G(f)| \le M \|f\|_p$.
(iii) The *operator norm* is $\|G\| := \sup\{|G(f)| : f \in L^p(\mu), \ \|f\|_p \le 1\}$.
(iv) G is *positive* if $f \ge 0$ a.e. implies $G(f) \ge 0$.

Lemma 4.1.32. *Positive Split of a Bounded Functional*
Let $1 \le p < +\infty$ and $G \in (L^p(\mu))^*_{\mathbb{R}}$. Then $G = G^+ - G^-$ for positive bounded linear functionals $G^\pm$.

Proof. For $f \ge 0$, set $G^+(f) := \sup\{G(g) : 0 \le g \le f\}$.
This is finite since $|G(g)| \le \|G\| \|g\|_p \le \|G\| \|f\|_p$, and positive homogeneity is immediate.
For $f_1, f_2 \ge 0$, $G^+(f_1) + G^+(f_2) \le G^+(f_1 + f_2)$; conversely, if $0 \le h \le f_1 + f_2$, set $h_1 = (h - f_2)^+$ and $h_2 = h - h_1$, so $h = h_1 + h_2$ with $0 \le h_j \le f_j$, hence $G(h) \le G^+(f_1) + G^+(f_2)$.
Thus G^+ is additive on the positive cone. Extend by $G^+(f) = G^+(f^+) - G^+(f^-)$; then G^+ is positive, linear, and bounded. Let $G^- = G^+ - G$. Since $G^+(f) \ge G(f)$ for $f \ge 0$, G^- is positive and bounded.
Structural Commentary: Function-lattice functionals split into positive parts.
Key Insight: Split any competitor $h \le f_1 + f_2$ into the portions assigned to f_1 and f_2. □

Theorem 4.1.33. *Riesz Representation for L^1*
Let $(X, \mathcal{M}, \mu)$ be σ-finite and $G \in (L^1(\mu))^*_{\mathbb{R}}$.

(i) $\exists g \in L^\infty(\mu)$ such that $\forall f \in L^1(\mu)$, $G(f) = \int_X fg \, d\mu$.
(ii) $\|G\| = \|g\|_\infty$.

(iii) If G is positive, then $g \geq 0$ a.e.

Proof.

(i) First let $\mu(X) < +\infty$ and $G \geq 0$. Set $\lambda(E) := G(\mathbf{1}_E)$. Finite additivity is linearity; if $E_n \uparrow E$, then $\mathbf{1}_{E_n} \to \mathbf{1}_E$ in L^1, hence $\lambda(E_n) \to \lambda(E)$. Thus λ is finite, and $\lambda \ll \mu$ since $\mu(E) = 0$ gives $\mathbf{1}_E = 0$ in L^1. Radon–Nikodym gives $g \geq 0$ with $\lambda(E) = \int_E g \, d\mu$, so simple functions are represented.
By (ii), $g \in L^\infty$, hence L^1-density of simple functions extends the identity to all $f \in L^1$. For σ-finite μ, apply the finite case to $f \mapsto G(f\mathbf{1}_{X_n})$ on $X_n \uparrow X$, $\mu(X_n) < +\infty$, and patch densities on overlaps. For general G, use the positive split $G = G^+ - G^-$ and subtract the two densities.

(ii) The pairing estimate gives $\|G\| \leq \|g\|_\infty$. Conversely, if $A > \|G\|$ and $E \subseteq \{|g| > A\}$ has $\mu(E) < +\infty$, then with $f = \operatorname{sign}(g)\mathbf{1}_E$, $A\mu(E) < \int_E |g| \, d\mu = G(f) \leq \|G\| \, \mu(E)$, so $\mu(E) = 0$.
A σ-finite exhaustion gives $\forall A > \|G\|$, $\mu(\{|g| > A\}) = 0$, hence $\|g\|_\infty \leq \|G\|$.

(iii) If G is positive and $B = \{g < 0\}$, then $\forall E \subseteq B$ of finite measure, $0 \leq G(\mathbf{1}_E) = \int_E g \, d\mu \leq 0$ only if $\int_E (-g) \, d\mu = 0$. Hence $\mu(E) = 0$ on $\{g < -\frac{1}{k}\}$ $\forall k$, and a σ-finite exhaustion gives $\mu(B) = 0$.

Structural Commentary: A bounded functional on L^1 is integration against an essentially bounded density.
Key Insight: Indicators give a measure; Radon–Nikodym gives the density. □

Theorem 4.1.34. *Riesz Representation for L^p, $1 < p < +\infty$*
Let $(X, \mathcal{M}, \mu)$ be any measure space, $1 < p < +\infty$, $q = \frac{p}{p-1}$, and $G \in (L^p(\mu))^*_{\mathbb{R}}$.

(i) Representation: $\exists g \in L^q(\mu)$ such that $\forall f \in L^p(\mu)$, $G(f) = \int_X fg \, d\mu$.
(ii) Norm: every representing g satisfies $\|G\| = \|g\|_q$.
(iii) Uniqueness: if $g, h \in L^q(\mu)$ represent G, then $g = h$ a.e.

Proof.

(i) Finite μ: for $G \geq 0$, set $\lambda(E) := G(\mathbf{1}_E)$. Since $E_n \uparrow E \Rightarrow \mathbf{1}_{E_n} \to \mathbf{1}_E$ in L^p, λ is finite.
Since $\mu(E) = 0 \Rightarrow \mathbf{1}_E = 0$ in L^p, $\lambda \ll \mu$. Let $g = \frac{d\lambda}{d\mu}$.
By simple approximation, $G(\phi) = \int_X \phi g \, d\mu$ for bounded ϕ.
For $g_m := g \wedge m$, $\int_X g_m^q \, d\mu \leq G(g_m^{q-1}) \leq \|G\| \, (\int_X g_m^q \, d\mu)^{1/p}$; hence $g \in L^q$, by L^p-density $G(f) = \int_X fg \, d\mu$.
General finite G follows from $G = G^+ - G^-$. For σ-finite μ, choose $X_n \uparrow X$, $\mu(X_n) < +\infty$, represent $G_n : u \mapsto G(\tilde{u})$ by g_n, and patch the compatible g_n to g.
Since $\|g_n\|_q = \|G_n\| \leq \|G\|$, by MCT $g \in L^q$; $f\mathbf{1}_{X_n} \to f$ in $L^p \Rightarrow G(f) = \int_X fg \, d\mu$.
For arbitrary μ, choose $\|f_n\|_p = 1$, $G(f_n) \to \|G\|$, and $X_0 := \bigcup_{n,k} \{|f_n| > 1/k\}$.
If $E \subseteq X \setminus X_0$, $\mu(E) < +\infty$, and $a = G(\mathbf{1}_E)$, then $G(f_n) \pm ta \leq \|G\| \, (1 + t^p \mu(E))^{1/p}$; after $n \to \infty$ and $t \downarrow 0$, $a = 0$. Thus G vanishes off X_0; represent on X_0 and extend by 0.

(ii) For representing g, by Hölder, $\|G\| \leq \|g\|_q$; by Proposition 4.1.30, $\|g\|_q = \sup_{\|f\|_p \leq 1} \left| \int_X fg \, d\mu \right| \leq \|G\|$.

(iii) If g, h represent G, then $k := g - h$ satisfies $\int_X fk \, d\mu = 0$ $\forall f \in L^p(\mu)$.
By Proposition 4.1.30 applied to $k \in L^q(\mu)$ with exponents q, p, $\|k\|_q = 0$; hence $g = h$ a.e.

Structural Commentary: For $1 < p < \infty$, every bounded functional comes from the conjugate space L^q.
Key Insight: Localize to σ-finite support, then patch. □

Definition 4.1.35. *Bounded Finitely Additive Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.
For $E \in \mathcal{M}$, let $\Pi(E)$ be the finite families $(E_1, \ldots, E_n)$ with $E_k \in \mathcal{M}$, $E_k \subseteq E$, and $E_i \cap E_j = \emptyset$ for $i \neq j$.
For $\nu : \mathcal{M} \to \mathbb{R}$, set $|\nu|_{\text{fa}}(E) := \sup_{(E_k) \in \Pi(E)} \sum_k |\nu(E_k)|$ and $\|\nu\|_{\text{var}} := |\nu|_{\text{fa}}(X)$.

Define $\text{BFA}(X, \mathcal{M}, \mu)$ as the set of $\nu : \mathcal{M} \to \mathbb{R}$ satisfying:

(i) *Finite additivity*: $\nu(\emptyset) = 0$ and $\nu(\bigsqcup_{k=1}^n E_k) = \sum_{k=1}^n \nu(E_k)$.
(ii) *Null compatibility*: $\mu(E) = 0 \Rightarrow \nu(E) = 0$.
(iii) *Bounded variation*: $\|\nu\|_{\text{var}} < +\infty$.

Key Insight: BFA requires finite additivity, null compatibility, and finite variation, not countable additivity.

Theorem 4.1.36. *Kantorovitch Representation for $(L^\infty)^*$*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

For $\nu \in \text{BFA}(X, \mathcal{M}, \mu)$ and simple $\varphi = \sum_{k=1}^n c_k \mathbf{1}_{E_k}$ over a finite measurable partition, set $I_\nu(\varphi) := \sum_{k=1}^n c_k \nu(E_k)$.

(i) Extension: I_ν descends uniquely to $T_\nu \in (L^\infty(\mu))^*_{\mathbb{R}}$.
(ii) Isometry: $\|T_\nu\| = \|\nu\|_{\text{var}}$.
(iii) Surjectivity: if $S \in (L^\infty(\mu))^*_{\mathbb{R}}$ and $\nu_S(E) := S(\mathbf{1}_E)$, then $\nu_S \in \text{BFA}(X, \mathcal{M}, \mu)$ and $S = T_{\nu_S}$.

Thus $\nu \mapsto T_\nu$ is an isometric isomorphism $\text{BFA}(X, \mathcal{M}, \mu) \cong (L^\infty(\mu))^*_{\mathbb{R}}$.

Proof.

(i) Refinement and finite additivity make I_ν well defined.
For simple φ, discard $N = \{|\varphi| > \|\varphi\|_\infty\}$; since $|\nu|_{\text{fa}}(N) = 0$, $|I_\nu(\varphi)| \leq \|\varphi\|_\infty \|\nu\|_{\text{var}}$.
Hence uniform simple approximation extends I_ν to bounded measurable f.
If $\operatorname{supp} f \subseteq N$ and $\mu(N) = 0$, then $|\nu|_{\text{fa}}(N) = 0$, so $I_\nu(f) = 0$; therefore the extension depends only on the L^∞ class and defines T_ν with $\|T_\nu\| \leq \|\nu\|_{\text{var}}$.

(ii) For $(E_k) \in \Pi(X)$, set $c_k := \operatorname{sign}(\nu(E_k))$ and $\varphi := \sum_k c_k \mathbf{1}_{E_k}$.
Then $\|\varphi\|_\infty \leq 1$ and $\sum_k |\nu(E_k)| = T_\nu(\varphi) \leq \|T_\nu\|$.
Supremizing over $\Pi(X)$ gives $\|\nu\|_{\text{var}} \leq \|T_\nu\|$, and item (i) gives equality.

(iii) For $S \in (L^\infty(\mu))^*_{\mathbb{R}}$, $\nu_S(\emptyset) = 0$ and finite additivity follow from linearity.
If $\mu(E) = 0$, then $\mathbf{1}_E = 0$ in L^∞, so $\nu_S(E) = 0$.
For $(E_k) \in \Pi(X)$, with $c_k = \operatorname{sign}(\nu_S(E_k))$ and $\varphi = \sum_k c_k \mathbf{1}_{E_k}$, $\sum_k |\nu_S(E_k)| = S(\varphi) \le \|S\|$.
Hence $\nu_S \in \mathrm{BFA}(X, \mathcal{M}, \mu)$. Since S and T_{ν_S} agree on simple functions, uniform simple density gives $S = T_{\nu_S}$.

Structural Commentary: Simple extensions give T_ν; indicators recover ν.
Key Insight: Finite additivity supplies the full L^∞ dual. □

Proposition 4.1.37. *The Countably Additive Part*
For $g \in L^1(\mu)$, set $\nu_g(E) := \int_E g\, d\mu$.
(i) $\nu_g \in \mathrm{BFA}(X, \mathcal{M}, \mu)$.
(ii) $\|\nu_g\|_{\mathrm{var}} = \|g\|_1$.
(iii) $T_{\nu_g}(f) = \int_X fg\, d\mu$ for $f \in L^\infty(\mu)$; hence $L^1(\mu)$ embeds isometrically into $(L^\infty(\mu))^*$.

Proof.
(i) For disjoint $E_1, \dots, E_n$, $\nu_g(\bigsqcup_k E_k) = \sum_k \nu_g(E_k)$ by finite additivity of the integral, and
$\mu(E) = 0 \Rightarrow \nu_g(E) = 0$. Also, for $(E_k) \in \Pi(X)$, $\sum_k |\nu_g(E_k)| \le \sum_k \int_{E_k} |g|\, d\mu \le \|g\|_1$, so $\nu_g \in \mathrm{BFA}(X, \mathcal{M}, \mu)$.
(ii) Corollary 3.1.7 gives $\|\nu_g\|_{\mathrm{var}} = |\nu_g|(X)$, and Theorem 3.1.9 gives $|\nu_g|(X) = \int_X |g|\, d\mu = \|g\|_1$.
(iii) If $f = \sum_k c_k \mathbf{1}_{E_k}$ is simple, then $T_{\nu_g}(f) = \sum_k c_k \nu_g(E_k) = \int_X fg\, d\mu$.
For $f \in L^\infty(\mu)$, choose simple $f_n \to f$ uniformly; then $T_{\nu_g}(f_n) \to T_{\nu_g}(f)$ and $\int_X f_n g\, d\mu \to \int_X fg\, d\mu$.

Structural Commentary: Variation fixes the norm; simple approximation extends the pairing.
Key Insight: Countably additive representatives are exactly the familiar L^1 integral pairings. □

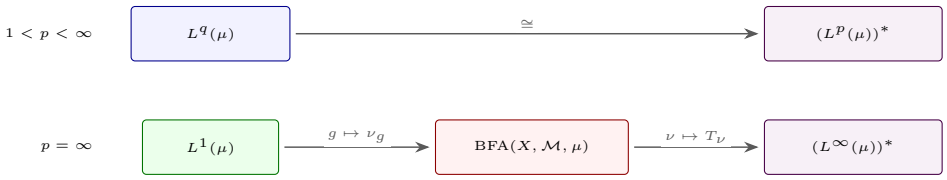

Figure 214: $L^q(\mu) \cong (L^p(\mu))^*$ for $1 < p < \infty$; $L^1(\mu)$ is the countably additive part of BFA $\cong (L^\infty(\mu))^*$.

Remark 4.1.38. *L^2 Riesz Gives Another Radon–Nikodym Route*
Let λ, μ be finite and $\lambda \ll \mu$. Put $\rho = \lambda + \mu$ and $G(h) := \int h\, d\lambda$ on $L^2(\rho)$.
If $G(h) = \int hg\, d\rho$, indicator tests give $0 \le g \le 1$ ρ-a.e. and $\mu(\{g = 1\}) = 0$.
Since $\int h(1-g)\, d\lambda = \int hg\, d\mu$, take $E_N = E \cap \{1 - g \ge \frac{1}{N}\}$ and $h = \frac{\mathbf{1}_{E_N}}{1-g} \in L^2(\rho)$ to get $\lambda(E_N) = \int_{E_N} \frac{g}{1-g}\, d\mu$. Let $N \to \infty$ by monotone convergence and $\lambda \ll \mu$: $\lambda(E) = \int_E \frac{g}{1-g}\, d\mu$.
Key Insight: Riesz representation on $L^2(\lambda + \mu)$ encodes the density in the ratio $\frac{g}{1-g}$.

Lemma 4.1.39. *Absolute Continuity of L^p Mass*
For $1 \le p < \infty$ and $f \in L^p(\mu)$, $\forall \varepsilon > 0\, \exists \delta > 0$ such that $\mu(E) < \delta$ implies $\|f\mathbf{1}_E\|_p < \varepsilon$.

Proof. Let $h = |f|^p \in L^1$. Choose $M > 0$ with $\int_{\{h > M\}} h\, d\mu < \frac{\varepsilon^p}{2}$, and set $\delta = \frac{\varepsilon^p}{2M}$.
If $\mu(E) < \delta$, then $\|f\mathbf{1}_E\|_p^p = \int_E h\, d\mu \le M\mu(E) + \int_{\{h > M\}} h\, d\mu < \varepsilon^p$.
Structural Commentary: L^p mass is absolutely continuous with respect to measure.
Key Insight: Truncate the tail; small measure controls the bounded part. □

Proposition 4.1.40. *Local Mass Limits for L^p Cauchy Sequences*
For $1 \le p < \infty$ and $f_n \in L^p(\mu)$, set $\beta_n(E) := \|f_n \mathbf{1}_E\|_p$ for $E \in \mathcal{M}$. The following hold.
(i) $|\beta_n(E) - \beta_m(E)| \le \|f_n - f_m\|_p$ for $E \in \mathcal{M}$.
(ii) If (f_n) is L^p-Cauchy, then $\forall E \in \mathcal{M}$, $\beta(E) := \lim_n \beta_n(E)$ exists.

Proof.
(i) Since $f_n \mathbf{1}_E = f_m \mathbf{1}_E + (f_n - f_m)\mathbf{1}_E$, $\beta_n(E) - \beta_m(E) \le \|f_n - f_m\|_p$.
Interchanging m, n gives $\beta_m(E) - \beta_n(E) \le \|f_n - f_m\|_p$; hence $|\beta_n(E) - \beta_m(E)| \le \|f_n - f_m\|_p$.
(ii) Fix $E \in \mathcal{M}$ and $\varepsilon > 0$. Choose N such that $m, n \ge N \Rightarrow \|f_n - f_m\|_p < \varepsilon$.
Then item (i) gives $|\beta_n(E) - \beta_m(E)| < \varepsilon$, so $(\beta_n(E))$ converges in $\mathbb{R}$.

Structural Commentary: Restricting to E is a 1-Lipschitz operation on L^p mass.
Key Insight: Local mass inherits global Cauchy control. □

Proposition 4.1.41. *Uniform Local Mass Control for L^p Cauchy Sequences*
Let $1 \le p < \infty$, f_n be L^p-Cauchy, and $\beta_n(E) := \|f_n \mathbf{1}_E\|_p$. Then $\forall \varepsilon > 0$:
(i) $\exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset$ implies $\sup_n \beta_n(F) < \varepsilon$;
(ii) $\exists \delta > 0$ such that $\mu(E) < \delta$ implies $\sup_n \beta_n(E) < \varepsilon$.

Proof. Choose N so $n \geq N$ implies $\|f_n - f_N\|_p < \frac{\varepsilon}{3}$.

(i) Choose finite-measure E_j for $1 \leq j \leq N$ with $F \cap E_j = \emptyset \Rightarrow \beta_j(F) < \frac{\varepsilon}{3}$, and set $E_\varepsilon = \bigcup_{j \leq N} E_j$.
For $n \leq N$ this works; for $n > N$, $\beta_n(F) \leq \|f_n - f_N\|_p + \beta_N(F) < \varepsilon$.

(ii) Choose $\delta_j > 0$ for $1 \leq j \leq N$ with $\mu(E) < \delta_j \Rightarrow \beta_j(E) < \frac{\varepsilon}{3}$, and set $\delta = \min_{j \leq N} \delta_j$.
For $n \leq N$ this works; for $n > N$, $\beta_n(E) \leq \|f_n - f_N\|_p + \beta_N(E) < \varepsilon$.

Structural Commentary: Cauchy families are uniformly tight and absolutely continuous.
Key Insight: Control finitely many early terms directly and the tail by Cauchy closeness. □

Method 4.1.42. *Lebesgue Proof Patterns*

(i) *Approximate*: for $f \geq 0$, use $s \in \mathcal{S}^+(f)$ and $\int_E f\, d\mu = \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$.
(ii) *Promote*: prove for $\mathbf{1}_A$, extend to $s = \sum_i c_i \mathbf{1}_{A_i}$, then pass through $s_n \uparrow f$.
(iii) *Ignore null sets*: if $f \sim_{\text{a.e.}} g$ on E, replace f by g inside integrals over measurable $A \subseteq E$.
(iv) *Split signs*: use $f = f^+ - f^-$ and $h = u + iv$.
(v) *Exchange limits*: reduce Fatou and dominated convergence to $0 \leq f_n \uparrow f$.
(vi) *Slice*: prove boxes first; use regularity and null slices to pass to products and Fubini.

Key Insight: Reduce to simple functions, monotone convergence, null sets, and product boxes.

4.2 Modes of Convergence

Throughout this subsection $(X, \mathcal{M}, \mu)$ is a fixed measure space and, unless otherwise stated, functions are real-valued and measurable. The point is not to rank convergence modes linearly, but to know which hypotheses convert one mode into another.

Definition 4.2.1. *Pointwise, Uniform, and Almost-Everywhere Convergence*
Let (f_n) and f be functions on X.

(i) $f_n \to f$ *pointwise* if $\forall x \in X$, $\varepsilon > 0$, $\exists N(\varepsilon, x)$ s.t. $n \geq N(\varepsilon, x)$ implies $|f_n(x) - f(x)| < \varepsilon$.
(ii) $f_n \to f$ *uniformly* if $\forall \varepsilon > 0$ $\exists N(\varepsilon)$ s.t. $n \geq N(\varepsilon)$ implies $|f_n(x) - f(x)| < \varepsilon$ $\forall x \in X$.
(iii) $f_n \to f$ *almost everywhere* if $\exists A \in \mathcal{M}$ with $\mu(A) = 0$ s.t. $\forall x \in X \setminus A$, $\varepsilon > 0$, $\exists N(\varepsilon, x)$ s.t. $n \geq N(\varepsilon, x)$ implies $|f_n(x) - f(x)| < \varepsilon$.

Proposition 4.2.2. *First Implications and First Failures*
Let $f_n, f : X \to \mathbb{R}$.

(i) $f_n \rightrightarrows f \Rightarrow f_n \to f$ pointwise $\Rightarrow f_n \to f$ a.e.; converses fail.
(ii) $|X| < +\infty$ and $f_n \to f$ pointwise $\Rightarrow f_n \rightrightarrows f$.
(iii) $\mu(A) = 0 \Rightarrow A = \emptyset$ and $f_n \to f$ a.e. $\Rightarrow f_n \to f$ pointwise.

Proof.

(i) If $f_n \rightrightarrows f$, then $\forall \varepsilon > 0$ $\exists N_\varepsilon$ such that $\forall n \geq N_\varepsilon$ $\forall x \in X$, $|f_n(x) - f(x)| < \varepsilon$; hence pointwise convergence with $N(\varepsilon, x) = N_\varepsilon$.
Pointwise convergence is a.e. convergence with $A = \emptyset$.
Conversely, on $([0,1], \lambda)$, $x^n \to \mathbf{1}_{\{1\}}$ pointwise but not uniformly, and $\mathbf{1}_{\{0\}} \to 0$ a.e. but not pointwise.
(ii) If X is finite and $f_n \to f$ pointwise, then $\forall \varepsilon > 0$ $\forall x \in X$, choose $N(\varepsilon, x)$.
Set $N_\varepsilon = \max_{x \in X} N(\varepsilon, x)$; then $\forall n \geq N_\varepsilon$ $\forall x \in X$, $|f_n(x) - f(x)| < \varepsilon$, so $f_n \rightrightarrows f$.
(iii) If $f_n \to f$ a.e., then $\exists A \in \mathcal{M}$ with $\mu(A) = 0$ and $\forall x \in X \setminus A$, $f_n(x) \to f(x)$.
Since every null set is empty, $A = \emptyset$, so $f_n \to f$ pointwise.

Structural Commentary: Uniformity is bounded x-dependence of $N(\varepsilon, x)$.
Key Insight: Uniformity controls indices; a.e. convergence controls null exceptions. □

Proposition 4.2.3. *Uniform Convergence Gives L^p Convergence on Finite Measure Spaces*
Let $1 \leq p < \infty$, $\mu(X) < +\infty$, and $f_n \in L^p(\mu)$. If $f_n \rightrightarrows f$ on X, then $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.

Proof. For $\varepsilon > 0$, choose N with $n \geq N \Rightarrow |f_n - f| \leq \varepsilon$ on X.
Then $n \geq N \Rightarrow \|f_n - f\|_p \leq (\int_X \varepsilon^p\, d\mu)^{\frac{1}{p}} = \varepsilon \mu(X)^{\frac{1}{p}}$, so $\|f_n - f\|_p \to 0$.
Taking $\varepsilon = 1$, choose n_0 with $|f_{n_0} - f| \leq 1$ on X; hence $f - f_{n_0} \in L^p(\mu)$ and $f = f_{n_0} + (f - f_{n_0}) \in L^p(\mu)$.

Structural Commentary: Uniform error becomes L^p error after multiplication by $\mu(X)^{\frac{1}{p}}$.
Key Insight: Finite measure converts a pointwise sup-bound into an integral bound. □

Theorem 4.2.4. *Dominated Almost-Everywhere Convergence Gives L^p Convergence*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and $f_n \to f$ a.e. Suppose $\exists g \in L^p(\mu)$ such that $\forall n$, $|f_n| \leq g$ a.e.

(i) $f \in L^p(\mu)$.
(ii) $\|f_n - f\|_p \to 0$.

Proof.

(i) Choose $N \in \mathcal{M}$ with $\mu(N) = 0$ such that on $X \setminus N$, $f_n \to f$ and $|f_n| \leq g$ $\forall n$.
Then $|f| \leq g$ a.e., so $|f|^p \leq g^p \in \mathscr{L}^1(\mu)$.
(ii) Since $|f_n - f|^p \to 0$ a.e. and $|f_n - f|^p \leq (2g)^p \in \mathscr{L}^1(\mu)$ a.e.,
Theorem 1.3.7 yields $\|f_n - f\|_p^p = \int_X |f_n - f|^p\, d\mu \to 0$.

Structural Commentary: Apply ordinary dominated convergence to the pth-power error.
Key Insight: A single L^p envelope controls both membership of the limit and convergence in mean. □

Corollary 4.2.5. *Uniform Boundedness on Finite Measure Spaces*
Let $\mu(X) < +\infty$, $0 \leq K < +\infty$, and $f_n \to f$ a.e. Assume $\forall n$, $|f_n| \leq K$ a.e. Then for every $1 \leq p < \infty$:

(i) $f \in L^p(\mu)$.

(ii) $\|f_n - f\|_p \to 0$.

Proof. Fix $1 \le p < \infty$ and set $g := K\mathbf{1}_X$. Then $\int_X g^p \, d\mu = K^p \mu(X) < +\infty$, so $g \in L^p(\mu)$ and $|f_n| \le g$ a.e.
(i) Since $f_n \to f$ a.e. and $|f_n| \le g \in L^p(\mu)$ a.e., Theorem 4.2.4(i) gives $f \in L^p(\mu)$.
(ii) The same hypotheses and Theorem 4.2.4(ii) give $\|f_n - f\|_p \to 0$.
Structural Commentary: Finite measure turns a uniform a.e. bound into an L^p envelope.
Key Insight: Bounded a.e. convergence on finite measure is dominated convergence. □

Example 4.2.6. *Uniform Need Not Imply L^p on Infinite Measure*
On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, fix $1 \le p < \infty$ and set $f_n := n^{-\frac{1}{p}} \mathbf{1}_{[0,n]}$. Then $\|f_n\|_\infty = n^{-\frac{1}{p}} \to 0$, so $f_n \rightrightarrows 0$.
However $\|f_n\|_p^p = \int_0^n n^{-1} \, d\lambda = 1$, hence $\|f_n\|_p \nrightarrow 0$.
Structural Commentary: The support grows exactly enough to preserve L^p mass.
Key Insight: Uniform smallness alone does not control integrals on infinite measure spaces.

Example 4.2.7. *Almost Everywhere and In-Measure Need Not Imply L^p*
On $[0,1]$ with Lebesgue measure, set $f_n := n\mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}$.
$\forall x \in [0,1]$, $x \in [\frac{1}{n}, \frac{2}{n}]$ for only finitely many n, hence $f_n(x) \to 0$.
If $\alpha > 0$ and $n \ge \alpha$, then $\lambda\{|f_n| \ge \alpha\} = \frac{1}{n} \to 0$, so $f_n \to 0$ in measure.
Yet for $1 \le p < \infty$, $\|f_n\|_p^p = n^p \lambda([\frac{1}{n}, \frac{2}{n}]) = n^{p-1} \nrightarrow 0$.
Given $\delta > 0$, choose $0 < \eta < \delta$; on $[0,1] \setminus [0, \eta]$, $\forall n > \frac{2}{\eta}$, $f_n = 0$, so $f_n \to 0$ almost uniformly.
Thus a.e., in-measure, and almost-uniform convergence do not imply L^p convergence.

Theorem 4.2.8. *Pointwise A.E. Convergence Plus Norms*
Let $1 \le p < \infty$, $f_n, f \in L^p(\mu)$, $f_n \to f$ a.e. Then $f_n \to f$ in $L^p(\mu)$ iff $\lim_{n\to\infty} \int_X |f_n|^p \, d\mu = \int_X |f|^p \, d\mu$.

Proof.
($\Rightarrow$) If $\|f_n - f\|_p \to 0$, then $\left| \|f_n\|_p - \|f\|_p \right| \le \|f_n - f\|_p$, hence $\int_X |f_n|^p \, d\mu = \|f_n\|_p^p \to \|f\|_p^p = \int_X |f|^p \, d\mu$.
($\Leftarrow$) Assume $\int_X |f_n|^p \, d\mu \to \int_X |f|^p \, d\mu$. Set $h_n := \frac{|f_n|^p + |f|^p}{2} - \left| \frac{f_n - f}{2} \right|^p \ge 0$ by convexity.
Since $h_n \to |f|^p$ a.e., Fatou gives $\int_X |f|^p \, d\mu \le \liminf_n \int_X h_n \, d\mu = \int_X |f|^p \, d\mu - 2^{-p} \limsup_n \int_X |f_n - f|^p \, d\mu$.
Thus $\limsup_n \|f_n - f\|_p^p \le 0$, so $f_n \to f$ in $L^p(\mu)$.
Structural Commentary: A.e. convergence identifies the limit; p-mass convergence prevents loss.
Key Insight: Fatou applied to the convexity defect forces the error norm to vanish. □

Definition 4.2.9. *Convergence and Cauchy Convergence in Measure*
Let (f_n) be measurable functions on X.
(i) $f_n \to f$ *in measure* if $\forall \alpha > 0$, $\mu\{x \in X : |f_n(x) - f(x)| \ge \alpha\} \to 0$.
(ii) (f_n) is *Cauchy in measure* if $\forall \alpha > 0$, $\mu\{x \in X : |f_m(x) - f_n(x)| \ge \alpha\} \to 0$ as $m, n \to \infty$.
The same definitions result with $>$ in place of $\ge$.

Proposition 4.2.10. *L^p Convergence Implies Convergence in Measure*
Let $1 \le p < \infty$. If $\|f_n - f\|_p \to 0$, then $f_n \to f$ in measure.

Proof. For $\alpha > 0$, set $E_n(\alpha) := \{|f_n - f| \ge \alpha\}$.
Since $\|f_n - f\|_p^p \ge \int_{E_n(\alpha)} |f_n - f|^p \, d\mu \ge \alpha^p \mu(E_n(\alpha))$, $\mu(E_n(\alpha)) \le \alpha^{-p} \|f_n - f\|_p^p \to 0$.
Structural Commentary: Large-error sets are controlled by the integral of the pth-power error.
Key Insight: This is Chebyshev's inequality in deterministic measure form. □

Example 4.2.11. *L^p Convergence Without Pointwise Convergence*
Let $X = [0,1]$ with Lebesgue measure and list $I_{m,k} := [\frac{k-1}{m}, \frac{k}{m}]$, $1 \le k \le m$, row by row in m.
Let $f_n := \mathbf{1}_{I_n}$, where I_n is the nth interval. If I_n lies in row $m(n)$, then $m(n) \to \infty$ and
$\|f_n\|_p^p = \lambda(I_n) = \frac{1}{m(n)} \to 0$; hence $f_n \to 0$ in L^p, therefore in measure by Proposition 4.2.10.
$\forall x \in [0,1]$, every row contains some $I_{m,k}$ with $x \in I_{m,k}$, giving a subsequence with value 1 at x; for infinitely many rows, some $I_{m,k}$ omits x, giving a subsequence with value 0 at x. Thus $(f_n(x))$ converges nowhere.
The subfamily $I_{m,2} = [\frac{1}{m}, \frac{2}{m}]$, $m \ge 2$, gives a subsequence converging pointwise to 0 on all of X.

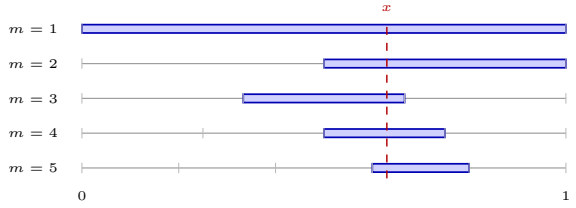

Figure 215: In each row, one shrinking typewriter interval contains the fixed point x.

Theorem 4.2.12. *Riesz Subsequence Principle for Convergence in Measure*
If (f_n) is Cauchy in measure, then there exist a subsequence (g_k) and a measurable f such that:
(i) $g_k \to f$ *a.e.*
(ii) $g_k \to f$ *in measure.*

(iii) $\forall \varepsilon > 0\ \exists E \in \mathcal{M}$ *with* $\mu(E) < \varepsilon$ *such that* $g_k \to f$ *uniformly on* $X \setminus E$.

Proof.

(i) Choose $n_1 < n_2 < \cdots$ so that $m, n \geq n_k \Rightarrow \mu\{|f_m - f_n| \geq \frac{1}{2^k}\} < \frac{1}{2^k}$, and set $g_k := f_{n_k}$.
For $E_k := \{|g_{k+1} - g_k| \geq \frac{1}{2^k}\}$, $\mu(E_k) < \frac{1}{2^k}$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$. Then $F_k \downarrow$, $\mu(F_k) \leq \sum_{j=k}^{\infty} \frac{1}{2^j} = \frac{1}{2^{k-1}}$.
For $i > j \geq k$, $x \notin F_k$, $|g_i(x) - g_j(x)| \leq \sum_{r=j}^{i-1} |g_{r+1}(x) - g_r(x)| < \frac{1}{2^{j-1}}$.
Hence (g_j) is uniformly Cauchy on $X \setminus F_k$ $\forall k$.
Let $F := \bigcap_k F_k$, so $\mu(F) = 0$. Define $f = \lim_j g_j$ on $X \setminus F$, $f = 0$ on F. f is measurable, $g_j \to f$ a.e.

(ii) Letting $i \to \infty$ gives $j \geq k$, $x \notin F_k \Rightarrow |f(x) - g_j(x)| \leq \frac{1}{2^{j-1}} \leq \frac{1}{2^{k-1}}$. Given $\alpha, \varepsilon > 0$, choose k with $\frac{1}{2^{k-1}} < \alpha$ and $\mu(F_k) < \varepsilon$; then $j \geq k \Rightarrow \{|f - g_j| \geq \alpha\} \subseteq F_k$, so $g_j \to f$ in measure.

(iii) Given $\varepsilon > 0$, choose k with $\mu(F_k) < \varepsilon$.
The estimate in (ii) gives $\sup_{x \in X \setminus F_k} |f(x) - g_j(x)| \leq \frac{1}{2^{j-1}} \to 0$, so $g_j \to f$ uniformly on $X \setminus F_k$.

Structural Commentary: Cauchy in measure lets us select summably small exceptional sets.
Key Insight: Outside the tail union of exceptional sets, the subsequence is uniformly Cauchy. □

Corollary 4.2.13. *Completeness for Convergence in Measure*
(f_n) *is Cauchy in measure iff* $\exists f$ *measurable with* $f_n \to f$ *in measure.*

Proof.

(⇒) If (f_n) is Cauchy in measure, then $f_{n_k} \to f$ in measure for some measurable f by Theorem 4.2.12.
For $\alpha > 0$, $\{|f_n - f| \geq \alpha\} \subseteq \{|f_n - f_{n_k}| \geq \frac{\alpha}{2}\} \cup \{|f_{n_k} - f| \geq \frac{\alpha}{2}\}$.
Choose k large, then n large, to get $f_n \to f$ in measure.

(⇐) For $\alpha > 0$, $\{|f_m - f_n| \geq \alpha\} \subseteq \{|f_m - f| \geq \frac{\alpha}{2}\} \cup \{|f_n - f| \geq \frac{\alpha}{2}\}$.
Hence $\mu\{|f_m - f_n| \geq \alpha\} \to 0$ as $m, n \to \infty$.

Structural Commentary: The subsequential in-measure limit attracts the whole Cauchy sequence.
Key Insight: Convergence in measure is complete after measurable limits are admitted. □

Corollary 4.2.14. *Uniqueness for Convergence in Measure*
If $f_n \to f$ *and* $f_n \to g$ *in measure, then* $f = g$ *a.e.*

Proof. For $\alpha > 0$, $\{|f - g| \geq \alpha\} \subseteq \{|f - f_n| \geq \frac{\alpha}{2}\} \cup \{|f_n - g| \geq \frac{\alpha}{2}\}$, so $\mu\{|f - g| \geq \alpha\} = 0$.
Hence $\mu\{|f - g| > 0\} = 0$ and $f = g$ a.e.
Structural Commentary: Two in-measure limits cannot differ on a positive-measure set.
Key Insight: In-measure limits are unique modulo null sets. □

Corollary 4.2.15. *In-Measure Limits Have A.E. Subsequences*
If $f_n \to f$ *in measure, then some subsequence* f_{n_k} *converges to* f *a.e.*

Proof. Since $f_n \to f$ in measure, (f_n) is Cauchy in measure.
Choose a subsequence $f_{n_k} \to h$ a.e. and in measure by Theorem 4.2.12.
Then $h = f$ a.e. by Corollary 4.2.14; after enlarging the exceptional null set, $f_{n_k} \to f$ a.e.
Structural Commentary: Convergence in measure contains an almost-everywhere convergent subsequence.
Key Insight: Select a Cauchy-in-measure subsequence, then identify its limit by uniqueness. □

Proposition 4.2.16. *Subsequence and Indicator Consequences*

(i) $f_n \to f$ *in measure* ⇒ *every subsequence* $f_{n_k} \to f$ *in measure; if* (f_n) *is Cauchy in measure, then every* (f_{n_k}) *is Cauchy in measure.*
(ii) If $\|f_n - f\|_p \to 0$ *and* $\|f_{n_k} - g\|_p \to 0$, *then* $f = g$ *a.e.*
(iii) If $f_n = \mathbf{1}_{E_n}$ *and* $\|f_n - f\|_p \to 0$, *then* $f = \mathbf{1}_E$ *a.e. for some* $E \in \mathcal{M}$.

Proof.

(i) For $\alpha > 0$, $\mu\{|f_{n_k} - f| \geq \alpha\} \to 0$ since $n_k \to \infty$; the Cauchy claim is identical with $m_k, n_\ell \to \infty$.

(ii) $f_n \to f$ and $f_{n_k} \to g$ in measure by Proposition 4.2.10; also $f_{n_k} \to f$ in measure by (i).
Thus $f = g$ a.e. by Corollary 4.2.14.

(iii) By Proposition 4.2.10 and Theorem 4.2.12, some $f_{n_k} \to f$ a.e.
Since $f_{n_k} \in \{0, 1\}$, $f \in \{0, 1\}$ a.e.; with $E := \{f = 1\}$, $E \in \mathcal{M}$ and $f = \mathbf{1}_E$ a.e.

Structural Commentary: Subsequence permanence and a.e. uniqueness transfer mode limits to representatives.
Key Insight: In-measure subsequences preserve limits; indicator limits remain indicators a.e. □

Proposition 4.2.17. *Algebra of Convergence in Measure on Finite Measure Spaces*
Let $(X, \mathcal{M}, \mu)$ *satisfy* $\mu(X) < +\infty$. *All functions below are measurable and finite-valued a.e.*

(i) If $f_n \to f$ *and* $g_n \to g$ *in measure, then* $\forall a, b \in \mathbb{R}$, $af_n + bg_n \to af + bg$ *in measure.*
(ii) If $u_n \to 0$ *in measure and* h *is measurable and finite-valued a.e., then* $hu_n \to 0$ *in measure.*
(iii) If $f_n \to f$ *and* $g_n \to g$ *in measure, then* $f_n g_n \to fg$ *in measure.*
(iv) If $f_n \to f$ *in measure, then* $f_n^2 \to f^2$ *in measure.*

Proof.

(i) Fix $\alpha > 0$. Since $\{|a(f_n - f) + b(g_n - g)| \geq \alpha\} \subseteq \{|a(f_n - f)| \geq \frac{\alpha}{2}\} \cup \{|b(g_n - g)| \geq \frac{\alpha}{2}\}$, the measure of the right-hand side tends to 0 by convergence in measure of f_n and g_n.

(ii) Fix $\alpha, \eta > 0$. Since h is finite a.e. and $\mu(X) < +\infty$, choose $M > 0$ with $\mu\{|h| > M\} < \eta$.
Then $\{|hu_n| \geq \alpha\} \subseteq \{|h| > M\} \cup \{|u_n| \geq \frac{\alpha}{M}\}$, so $\limsup_n \mu\{|hu_n| \geq \alpha\} \leq \eta$. Let $\eta \downarrow 0$.

(iii) Set $u_n := f_n - f$ and $v_n := g_n - g$. Then $u_n, v_n \to 0$ in measure and $f_n g_n - fg = fv_n + gu_n + u_n v_n$. By (ii), $fv_n \to 0$ and $gu_n \to 0$ in measure. For $\alpha > 0$, $\{|u_n v_n| \geq \alpha\} \subseteq \{|u_n| \geq \sqrt{\alpha}\} \cup \{|v_n| \geq \sqrt{\alpha}\}$, so $u_n v_n \to 0$ in measure. Apply (i) to the three summands.
(iv) Apply (iii) with $g_n = f_n$ and $g = f$.

Structural Commentary: Finite measure makes finite measurable multipliers harmless after truncation.
Key Insight: Products split into fixed-multiplier errors and a product of two small-in-measure errors. □

Theorem 4.2.18. *Dominated Convergence in Measure Gives L^p Convergence*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, $f_n \to f$ in measure, and $|f_n| \leq g$ a.e. for some $g \in L^p(\mu)$. Then:
(i) $f \in L^p(\mu)$;
(ii) $\|f_n - f\|_p \to 0$.

Proof. Since $f_n \to f$ in measure, (f_n) is Cauchy in measure.

(i) By Theorem 4.2.12, some subsequence (u_j) satisfies $u_j \to h$ a.e. and in measure; $h = f$ a.e. by Corollary 4.2.14. Since $|u_j| \leq g$ a.e., $f \in L^p(\mu)$ by Theorem 4.2.4.
(ii) If norm convergence fails, then $\exists \varepsilon > 0$ and a subsequence (g_k) with $\|g_k - f\|_p \geq \varepsilon$. The argument in (i) produces a subsubsequence (h_j) with $h_j \to f$ a.e.; then $\|h_j - f\|_p \to 0$ by Theorem 4.2.4, a contradiction.

Structural Commentary: Reduce convergence in measure to almost-everywhere convergence along a subsequence.
Key Insight: Domination turns every in-measure subsequence into an L^p-convergent subsubsequence. □

Lemma 4.2.19. *Almost Uniform Cauchy Sequences Have Almost Uniform Limits*
If (f_n) is almost uniformly Cauchy, then $\exists f$ measurable such that $f_n \to f$ almost uniformly and a.e.

Proof. Choose $E_k \in \mathcal{M}$ with $\mu(E_k) < 2^{-k}$ and (f_n) uniformly Cauchy on $X \setminus E_k$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$ and $F := \bigcap_k F_k$; then $F_k \downarrow$, $\mu(F_k) < 2^{-(k-1)}$, $\mu(F) = 0$, and (f_n) is uniformly Cauchy on $X \setminus F_k$. By Theorem 7.1.7, f_n has a uniform limit on each $X \setminus F_k$; these limits agree on overlaps, so define measurable f on $X \setminus F$, and set $f = 0$ on F. Hence $f_n \to f$ a.e. and uniformly on each $X \setminus F_k$; since $\mu(F_k) \to 0$, $f_n \to f$ almost uniformly.
Structural Commentary: Tail unions convert many small exceptional sets into nested uniform domains.
Key Insight: Almost uniform Cauchy means uniformly Cauchy after discarding a tail-small exceptional set. □

Theorem 4.2.20. *Almost Uniform Convergence and Convergence in Measure*
The following hold.
(i) If $f_n \to f$ almost uniformly, then $f_n \to f$ in measure.
(ii) If $f_n \to f$ in measure, then $\exists (f_{n_k})$ with $f_{n_k} \to f$ almost uniformly.

Proof.
(i) If $f_n \rightrightarrows f$ on $X \setminus E_\delta$ with $\mu(E_\delta) < \delta$, then $\forall \alpha > 0$ and sufficiently large n, $\{|f_n - f| \geq \alpha\} \subseteq E_\delta$. Thus $\limsup_n \mu\{|f_n - f| \geq \alpha\} \leq \delta$, and let $\delta \downarrow 0$.
(ii) Convergence in measure makes (f_n) Cauchy in measure. Choose $f_{n_k} \to h$ almost uniformly and in measure by Theorem 4.2.12. Uniqueness gives $h = f$ a.e.; absorb $\{h \neq f\}$ into the exceptional sets.

Structural Commentary: Almost uniform convergence is uniform convergence off arbitrarily small sets.
Key Insight: Convergence in measure always hides an almost-uniform subsequence. □

Corollary 4.2.21. *Finite-Measure Almost-Everywhere Convergence Implies In-Measure Convergence*
If $\mu(X) < +\infty$ and $f_n \to f$ a.e., with f_n, f finite-valued off a null set, then $f_n \to f$ in measure.

Proof. By Theorem 1.6.3, $f_n \to f$ almost uniformly; hence $f_n \to f$ in measure by Theorem 4.2.20.
Structural Commentary: Finite measure upgrades a.e. convergence through almost uniform convergence.
Key Insight: On finite measure spaces, a.e. convergence implies in-measure convergence. □

Corollary 4.2.22. *Finite-Measure Subsequence Test for Convergence in Measure*
Let $(X, \mathcal{M}, \mu)$ satisfy $\mu(X) < +\infty$, and let $f_n, f : X \to \overline{\mathbb{R}}$ be measurable and finite-valued a.e. Then $f_n \to f$ in measure iff every subsequence of (f_n) has a further subsequence converging to f a.e.

Proof.
($\Rightarrow$) If $f_n \to f$ in measure, then every subsequence also converges to f in measure by Proposition 4.2.16. Choose a further subsequence converging to f a.e. by Corollary 4.2.15.
($\Leftarrow$) Suppose $f_n \not\to f$ in measure. Then $\exists \alpha, \varepsilon > 0$ and subsequence (f_{n_k}) with $\mu\{|f_{n_k} - f| \geq \alpha\} \geq \varepsilon$ $\forall k$. But some $f_{n_{k_j}} \to f$ a.e., hence in measure by Corollary 4.2.21, contradicting the uniform lower bound.

Structural Commentary: Subsequence tests characterize in-measure convergence.
Key Insight: Nonconvergence leaves a bad subsequence. □

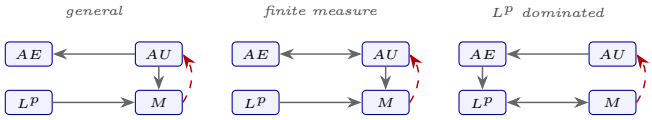

Figure 216: Modes of convergence in three settings: solid arrows are implications; dashed arrows yield convergent subsequences.

Method 4.2.23. *Counterexample Dictionary*

(i) On $(\mathbb{R},\lambda)$, $f_n = n^{-\frac{1}{p}}\mathbf{1}_{[0,n]}$ satisfies $f_n \rightrightarrows 0$ but $\|f_n\|_p = 1$.
(ii) On $([0,1],\lambda)$, $f_n = n\mathbf{1}_{[\frac{1}{n},\frac{2}{n}]}$ satisfies $f_n \to 0$ a.e., in measure, almost uniformly, but $\|f_n\|_p^p = n^{p-1} \nrightarrow 0$.
(iii) Typewriter intervals on $[0,1]$: if $f_n = \mathbf{1}_{I_n}$ and $\lambda(I_n) \to 0$ rowwise, then $\|f_n\|_p \to 0$ and $f_n \to 0$ in measure, but $(f_n(x))$ converges nowhere.
(iv) On $(\mathbb{R},\lambda)$, $f_n = \mathbf{1}_{[n,n+1]}$ satisfies $f_n \to 0$ pointwise but $\forall n,\ \lambda\{|f_n| \geq \frac{1}{2}\} = 1$, so $f_n \nrightarrow 0$ in measure.
(v) On $(\mathbb{R},\lambda)$, $f_n = \mathbf{1}_{[n,n+1]}$ is not almost uniformly convergent to 0: if $\lambda(E) < 1$ and $f_n \rightrightarrows 0$ on $\mathbb{R}\setminus E$, then $[n,n+1] \subseteq E$ for sufficiently large n, impossible.

Proposition 4.2.24. *An Integral Metric for Convergence in Measure on Finite Measure Spaces*
Let $\mu(X) < +\infty$ and $r(h) := \int_X |h|\,(1+|h|)^{-1}\,d\mu$. Then $f_n \to f$ in measure iff $r(f_n - f) \to 0$.

Proof.
($\Rightarrow$) Set $h_n := f_n - f$. For $\eta > 0$ and $A_n := \{|h_n| \geq \eta\}$, $r(h_n) \leq \mu(A_n) + \eta\mu(X)$. Let $n \to \infty$, then $\eta \downarrow 0$.
($\Leftarrow$) For $\alpha > 0$, $\alpha(1+\alpha)^{-1}\mathbf{1}_{\{|h_n|\geq\alpha\}} \leq |h_n|\,(1+|h_n|)^{-1}$, hence $\mu\{|h_n| \geq \alpha\} \leq (1+\alpha)\alpha^{-1} r(h_n) \to 0$.
Structural Commentary: The bounded error transform metrizes convergence in measure on finite measure spaces.
Key Insight: Small integral of the bounded error transform is equivalent to small large-error sets. □

Corollary 4.2.25. *Fatou and Dominated Convergence with In-Measure Hypotheses*
(i) $f_n \geq 0$ and $f_n \to f$ in measure $\Rightarrow \int_X f\,d\mu \leq \liminf_n \int_X f_n\,d\mu$.
(ii) $f_n \to f$ in measure and $|f_n| \leq g \in \mathscr{L}^1(\mu) \Rightarrow \int_X f_n\,d\mu \to \int_X f\,d\mu$.

Proof.
(i) Choose f_{n_k} with $\int f_{n_k}\,d\mu \to \liminf_n \int f_n\,d\mu$. By Theorem 4.2.12 and uniqueness, some $f_{n_{k_j}} \to f$ a.e.
Fatou gives $\int f\,d\mu \leq \liminf_j \int f_{n_{k_j}}\,d\mu = \liminf_n \int f_n\,d\mu$.
(ii) By Theorem 4.2.18 with $p=1$, $\|f_n - f\|_1 \to 0$, hence $\left|\int_X (f_n - f)\,d\mu\right| \leq \|f_n - f\|_1 \to 0$.
Structural Commentary: In-measure hypotheses reduce to a.e. subsequences or L^1 convergence.
Key Insight: Fatou and DCT survive convergence in measure through subsequences and domination. □

The next criterion packages the local mass controls needed to reverse convergence in measure into norm convergence.

Theorem 4.2.26. *Vitali Convergence Criterion for L^p*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and let f be measurable. Consider the following conditions.
(i) $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.
(ii) $f_n \to f$ in measure.
(iii) $\forall \varepsilon > 0\ \exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F |f_n|^p\,d\mu < \varepsilon^p$.
(iv) $\forall \varepsilon > 0\ \exists \delta(\varepsilon) > 0$ such that $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \int_E |f_n|^p\,d\mu < \varepsilon^p$.
Precisely, (i) is equivalent to the conjunction of (ii)–(iv).

Proof.
(i) Assume (i). Then (ii) holds by Proposition 4.2.10, and (f_n) is L^p-Cauchy. Conditions (iii)–(iv) follow from Proposition 4.1.41 after raising its norm bounds to the pth power.
(ii) Assume (ii)–(iv). Fix $\eta > 0$, choose E_η from (iii) and $\delta(\eta)$ from (iv), and put $F = X \setminus E_\eta$.
Then $\|(f_n - f_m)\mathbf{1}_F\|_p \leq \|f_n\mathbf{1}_F\|_p + \|f_m\mathbf{1}_F\|_p < 2\eta$.
If $\mu(E_\eta) > 0$, set $\alpha := \eta\mu(E_\eta)^{-\frac{1}{p}}$ and $H_{nm} := E_\eta \cap \{|f_n - f_m| \geq \alpha\}$.
By (ii), (f_n) is Cauchy in measure, so $\mu(H_{nm}) < \delta(\eta)$ for large m,n. On $E_\eta \setminus H_{nm}$, $|f_n - f_m| < \alpha$; on H_{nm}, use $|f_n - f_m| \leq |f_n| + |f_m|$. Hence $\|(f_n - f_m)\mathbf{1}_{E_\eta}\|_p \leq \alpha\mu(E_\eta)^{\frac{1}{p}} + \|f_n\mathbf{1}_{H_{nm}}\|_p + \|f_m\mathbf{1}_{H_{nm}}\|_p < 3\eta$ by (iv). If $\mu(E_\eta) = 0$, the E_η term is 0.
Thus $\|f_n - f_m\|_p < 5\eta$ eventually, so (f_n) is L^p-Cauchy.
By Riesz–Fischer, $f_n \to h$ in L^p; by Proposition 4.2.10 and Corollary 4.2.14, $h = f$ a.e., so (i) holds.
Structural Commentary: Vitali = in-measure convergence + tightness + uniform small-set control.
Key Insight: Tightness and small-set control convert in-measure convergence into an L^p Cauchy estimate. □

4.3 Composition and Convergence Modes

Proposition 4.3.1. *Composition and Pointwise Modes*
Let $\varphi : \mathbb{R} \to \mathbb{R}$.
(i) φ continuous, $f_n \to f$ a.e. $\Rightarrow \varphi \circ f_n \to \varphi \circ f$ a.e.
(ii) φ uniformly continuous preserves uniform, a.u., and in-measure convergence.
(iii) If φ is discontinuous, composition need not preserve a.e. convergence.
(iv) Non-uniform continuity can destroy uniform, even in-measure, convergence.

Proof.
(i) Outside a null set, $f_n(x) \to f(x)$; continuity gives $\varphi(f_n(x)) \to \varphi(f(x))$.
(ii) Given $\varepsilon > 0$, choose $\delta > 0$ with $|s-t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$. Uniform and almost-uniform convergence use $\sup|f_n - f| < \delta$ on X or $X \setminus E_\eta$; in measure, $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \delta\}$.
(iii) If φ is discontinuous at a, choose $t_n \to a$ and $\alpha > 0$ with $|\varphi(t_n) - \varphi(a)| \geq \alpha$.
On $X = \{*\}$, $f_n(*) = t_n \to a = f(*)$, but $\varphi(f_n(*)) \nrightarrow \varphi(f(*))$.
(iv) If φ is not uniformly continuous, choose a_n, b_n with $|a_n - b_n| \to 0$ and $|\varphi(a_n) - \varphi(b_n)| \geq \alpha > 0$.
On $(\mathbb{N},\#)$, set $f(k) = b_k$ and $f_n(k) = a_n$ for $k = n$, $f_n(k) = b_k$ for $k \neq n$.
Then $\|f_n - f\|_\infty \to 0$, but $\forall n,\ \#\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} = 1$.

Structural Commentary: Continuity is pointwise; uniform continuity gives one global δ.
Key Insight: Composition needs the same uniformity as the convergence mode. □

Proposition 4.3.2. *Composition and Convergence in Measure*
Let $f_n, f : X \to \mathbb{R}$ be measurable with $f_n \to f$ in measure, and let $\varphi : \mathbb{R} \to \mathbb{R}$.

(i) If φ is uniformly continuous, then $\forall \varepsilon > 0$, $\mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \to 0$.
(ii) If $\mu(X) < +\infty$ and φ is continuous, then $\forall \varepsilon > 0$, $\mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \to 0$.

Proof.

(i) Fix $\varepsilon > 0$ and choose $\delta > 0$ with $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$.
Then $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \delta\}$, so the measures tend to 0.

(ii) Fix $\varepsilon, \eta > 0$. Since $\mu(X) < +\infty$ and f is finite-valued, choose M with $\mu\{|f| > M\} < \eta$. Uniform continuity of φ on $[-M-1, M+1]$ gives $0 < \delta < 1$ such that $s, t \in [-M-1, M+1]$ and $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$. Hence $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f| > M\} \cup \{|f_n - f| \geq \delta\}$, so $\limsup_n \mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \leq \eta \downarrow 0$.

Structural Commentary: Use global uniformity, or truncate to a bounded range.
Key Insight: On finite measure spaces, measurable finite-valued functions are almost bounded. □

Example 4.3.3. *Continuity Alone Is Not Enough on Infinite Measure*
On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, set $f(x) = x$, $f_n(x) = x + \frac{1}{n}$, and $\varphi(t) = t^2$.
Then $\|f_n - f\|_\infty = \frac{1}{n} \to 0$, hence $f_n \rightrightarrows f$ and $f_n \to f$ in measure.
However $\varphi(f_n) - \varphi(f) = \frac{2x}{n} + \frac{1}{n^2}$; for $E_n := \{|\varphi(f_n) - \varphi(f)| \geq 1\}$, $[\frac{n}{2}, +\infty) \subseteq E_n$, so $\lambda(E_n) = +\infty$.
Thus $\varphi \circ f_n \not\to \varphi \circ f$ in measure.

Proposition 4.3.4. *Growth Conditions for L^p Composition*
Let $1 \leq p < \infty$, let $\varphi : \mathbb{R} \to \mathbb{R}$ be measurable, and let $f_n, f : X \to \mathbb{R}$ be measurable.

(i) If $\exists A, B \geq 0$ such that $\forall t$, $|\varphi(t)| \leq A|t| + B$, $\mu(X) < +\infty$, and $f \in L^p(\mu)$, then $\varphi \circ f \in L^p(\mu)$.
(ii) If $\exists A \geq 0$ such that $\forall t$, $|\varphi(t)| \leq A|t|$, and $f \in L^p(\mu)$, then $\varphi \circ f \in L^p(\mu)$.
(iii) If $\mu(X) < +\infty$, φ is uniformly continuous, $f_n, f \in L^p(\mu)$, and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iv) If φ is Lipschitz, $\varphi(0) = 0$, $f_n, f \in L^p(\mu)$, and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.

Proof.

(i) $|\varphi(f)|^p \leq 2^{p-1}(A^p |f|^p + B^p)$, so $\|\varphi \circ f\|_p^p \leq 2^{p-1}(A^p \|f\|_p^p + B^p \mu(X)) < +\infty$.

(ii) $|\varphi(f)|^p \leq A^p |f|^p$, so $\|\varphi \circ f\|_p \leq A \|f\|_p < +\infty$.

(iii) Uniform continuity gives constants $C, D \geq 0$ with $|\varphi(t)| \leq C|t| + D$ and $|\varphi(s) - \varphi(t)| \leq C|s - t| + D$.
Hence all compositions lie in L^p by (i). Let $h_n := |f_n - f|$.
Given $\eta > 0$, choose $\delta > 0$ with $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \eta$ and set $H_n := \{h_n \geq \delta\}$.
Then $\int_X |\varphi(f_n) - \varphi(f)|^p \, d\mu \leq \eta^p \mu(X) + 2^{p-1} \int_{H_n} (C^p h_n^p + D^p) \, d\mu$.
Since $\int h_n^p \, d\mu \to 0$ and $\mu(H_n) \leq \delta^{-p} \int h_n^p \, d\mu \to 0$, take $\limsup_n$ and then $\eta \downarrow 0$.

(iv) If L is Lipschitz, $|\varphi(f)| \leq L|f|$ and $\|\varphi \circ f_n - \varphi \circ f\|_p \leq L \|f_n - f\|_p \to 0$.

Structural Commentary: Growth gives membership; continuity estimates transfer L^p convergence.
Key Insight: Finite measure makes small uniform errors integrable; Lipschitz bounds give norm control. □

Example 4.3.5. *Uniform Continuity Does Not Suffice on Infinite Measure*
On $(\mathbb{R}, \lambda)$, fix $1 \leq p < \infty$, set $f = 0$, $f_n := n^{-1} \mathbf{1}_{[0, n^{\frac{p}{2}}]}$, and $\varphi(t) := \sqrt{|t|}$.
Then φ is uniformly continuous, $\varphi(0) = 0$, $\|f_n - f\|_p^p = n^{-\frac{p}{2}} \to 0$, but $\forall n$, $\|\varphi \circ f_n - \varphi \circ f\|_p^p = 1$.
Thus uniform continuity need not preserve L^p convergence on infinite measure spaces.

Lemma 4.3.6. *L^p Convergence Allows Continuous Composition in Measure*
Let $1 \leq p < \infty$, $f_n, f \in L^p(\mu)$, $\|f_n - f\|_p \to 0$, and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous.
Then $\varphi \circ f_n \to \varphi \circ f$ in measure.

Proof. Fix $\alpha, \eta > 0$. Choose M with $\mu\{|f| > M\} < \eta$. By uniform continuity of φ on $[-M-1, M+1]$, choose $0 < \delta < 1$ such that $s, t \in [-M-1, M+1]$ and $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \alpha$.
Then $\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} \subseteq \{|f| > M\} \cup \{|f_n - f| \geq \delta\}$, so $\limsup_n \mu\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} \leq \eta$. Let $\eta \downarrow 0$.

Structural Commentary: L^p convergence localizes to a large bounded range where continuity is uniform.
Key Insight: Norm convergence gives in-measure control; bounded-range continuity transfers it. □

Theorem 4.3.7. *Composition Growth Criteria in L^p*
Let $1 \leq p < \infty$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous.

(i) If $\mu(X) < +\infty$ and $\exists K > 0$ such that $|\varphi(t)| \leq K|t|$ for $|t| \geq K$, then $f \in L^p(\mu) \Rightarrow \varphi \circ f \in L^p(\mu)$.
(ii) Under (i), if $f_n, f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iii) If $\exists K \geq 0$ such that $\forall t$, $|\varphi(t)| \leq K|t|$, then $f \in L^p(\mu) \Rightarrow \varphi \circ f \in L^p(\mu)$ on any measure space.
(iv) Under (iii), if $f_n, f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ on any measure space.

Proof.

(i) Let $M := \sup_{|t| \leq K} |\varphi(t)|$, $C := K + M$. Then $\forall t$, $|\varphi(t)| \leq C(1 + |t|)$; hence $|\varphi(f)|^p \leq 2^{p-1} C^p (1 + |f|^p) \in L^1(\mu)$.

(ii) Let C be as in (i). Then $\varphi \circ f_n \to \varphi \circ f$ in measure by Lemma 4.3.6.
Since (f_n) is L^p-Cauchy, Proposition 4.1.41 and the growth bound from (i) supply the finite-tail and small-set estimates. Thus $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ by Theorem 4.2.26.

(iii) $|\varphi(f)|^p \leq K^p |f|^p \in L^1(\mu)$.

(iv) $\varphi \circ f_n \to \varphi \circ f$ in measure by Lemma 4.3.6. Since (f_n) is L^p-Cauchy and $|\varphi(f_n)|^p \le K^p |f_n|^p$, the finite-tail and small-set estimates for $(\varphi \circ f_n)$ follow from Proposition 4.1.41.
Thus $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ by Theorem 4.2.26.

Structural Commentary: Local growth needs finite measure; global growth does not.
Key Insight: Vitali converts in-measure composition plus inherited mass control into L^p convergence. □

Method 4.3.8. *Atomic Counterexamples for Composition Growth*

(i) If finite-measure tail growth fails, choose $|t_n| \ge n$ with $|\varphi(t_n)| > n^2 |t_n|$. On $\mathbb{N}$, set $\mu(\{n\}) = (n^2 |t_n|^p)^{-1}$ and $f(n) = t_n$. Then $\mu(\mathbb{N}) < +\infty$, $\int |f|^p \, d\mu = \sum_n n^{-2} < +\infty$, but $\int |\varphi(f)|^p \, d\mu \ge \sum_n n^{2p-2} = +\infty$.

(ii) If the same tail growth fails, choose $|t_n| \ge n$ with $|\varphi(t_n) - \varphi(0)| \ge n |t_n|$. With $\mu(\{n\}) = (n^p |t_n|^p)^{-1}$, $f = 0$, and $f_n = t_n \mathbf{1}_{\{n\}}$, one has $\mu(\mathbb{N}) < +\infty$, $\|f_n - f\|_p = \frac{1}{n} \to 0$, but $\|\varphi \circ f_n - \varphi \circ f\|_p \ge 1$.

(iii) If global linear growth fails and $\varphi(0) = 0$, the constructions in (i)–(ii), without requiring $\mu(\mathbb{N}) < +\infty$, give membership and convergence counterexamples on arbitrary measure spaces.

(iv) If $\varphi(0) \ne 0$, then on $(\mathbb{N}, \#)$, $0 \in L^p$ but $\varphi \circ 0$ is a nonzero constant and hence not in L^p; the constant sequence $f_n = 0$ gives the convergence failure.

4.4 Uniform Integrability and Vitali Convergence

Uniform integrability is the L^1 condition that rules out concentration of mass on tiny sets. On infinite-measure spaces it must be paired with tightness, which rules out mass escaping to remote finite-measure tails.

Lemma 4.4.1. *Finite-Measure Lebesgue Sets Have Small Pieces*
Let $E \in \mathcal{L}^1$, $m(E) < +\infty$, and $\delta > 0$. Then $\exists E_1, \dots, E_N \in \mathcal{L}^1$ such that:

(i) $E_i \cap E_j = \emptyset$ *for* $i \ne j$.

(ii) $E = \bigsqcup_{k=1}^{N} E_k$.

(iii) $m(E_k) < \delta$ *for* $1 \le k \le N$.

Proof. Choose n_0 with $T := E \setminus [-n_0, n_0]$ and $m(T) < \delta$. Partition $[-n_0, n_0]$ into finitely many disjoint intervals I_j with $m(I_j) < \delta$, and let (E_k) be the nonempty sets among T and $E \cap I_j$.

(i) Since $T \cap [-n_0, n_0] = \emptyset$ and the I_j are disjoint, the E_k are pairwise disjoint.

(ii) $E = T \bigsqcup (E \cap [-n_0, n_0]) = T \bigsqcup \bigsqcup_j (E \cap I_j) = \bigsqcup_{k=1}^{N} E_k$.

(iii) $m(T) < \delta$ and $m(E \cap I_j) \le m(I_j) < \delta$.

Structural Commentary: Finite measure gives a small tail T; a fine partition makes $E \setminus T$ small-piece.
Key Insight: One small tail plus a fine finite grid gives a finite small-piece decomposition. □

Proposition 4.4.2. *L^1 Absolute Continuity on Small Sets*
Let $E \in \mathcal{L}^1$, let $\mathcal{L}^1_E = \{A \in \mathcal{L}^1 : A \subseteq E\}$, let $m_E = m|_{\mathcal{L}^1_E}$, and let $f : E \to \mathbb{R}$ be measurable.
Consider the condition $(*)$: $\forall \varepsilon > 0 \; \exists \delta > 0$ such that $A \in \mathcal{L}^1_E$ and $m(A) < \delta$ imply $\int_A |f| \, dm < \varepsilon$.

(i) If $f \in L^1(m_E)$, then $(*)$ holds.
(ii) If $m(E) < +\infty$ and $(*)$ holds, then $f \in L^1(m_E)$.

Proof.

(i) Since $f \in L^1(m_E)$, Lemma 4.1.39 on $(E, \mathcal{L}^1_E, m_E)$ with $p = 1$ gives $\forall \varepsilon > 0 \; \exists \delta > 0$ such that $A \in \mathcal{L}^1_E$ and $m_E(A) < \delta$ imply $\int_A |f| \, dm_E < \varepsilon$. Since $m_E(A) = m(A)$ and $\int_A |f| \, dm_E = \int_A |f| \, dm$, $(*)$ holds.

(ii) Choose $\delta > 0$ from $(*)$ with $\varepsilon = 1$. By Lemma 4.4.1, write $E = \bigsqcup_{k=1}^{N} E_k$ with $E_k \in \mathcal{L}^1_E$ and $m(E_k) < \delta$. Then $(*)$ gives $\int_{E_k} |f| \, dm < 1$ for each k, hence $\int_E |f| \, dm = \sum_{k=1}^{N} \int_{E_k} |f| \, dm < N < +\infty$.

Structural Commentary: L^1 gives small-set control; finite partitions recover L^1.
Key Insight: A finite small-piece decomposition turns local integral bounds into one global finite integral. □

Definition 4.4.3. *Uniform Integrability and Tightness*
Let $\mathcal{F}$ be a family of measurable functions $h : X \to \mathbb{R}$ on $(X, \mathcal{M}, \mu)$.

(i) $\mathcal{F}$ is *uniformly integrable* on X if $\forall \varepsilon > 0 \, \exists \delta > 0$ such that $A \in \mathcal{M}$ and $\mu(A) < \delta$ imply $\sup_{h \in \mathcal{F}} \int_A |h| \, d\mu < \varepsilon$.

(ii) $\mathcal{F}$ is *tight* on X if $\forall \varepsilon > 0 \, \exists X_\varepsilon \in \mathcal{M}$ with $\mu(X_\varepsilon) < +\infty$ and $\sup_{h \in \mathcal{F}} \int_{X \setminus X_\varepsilon} |h| \, d\mu < \varepsilon$.

On finite measure spaces, tightness is automatic by taking $X_\varepsilon = X$.
Key Insight: UI controls mass on small A; tightness controls mass on $X \setminus X_\varepsilon$.

Example 4.4.4. *Concentration versus Escape*

(i) *Concentration*: on $(0, 1)$, $u_n := n \mathbf{1}_{(0, \frac{1}{n})}$. For every $\delta > 0$, choose $n > \delta^{-1}$ and $A_n = (0, \frac{1}{n})$.
Then $m(A_n) < \delta$ but $\int_{A_n} u_n \, dm = 1$, so (u_n) is not uniformly integrable.

(ii) *Escape*: on $\mathbb{R}$, let $I_n := (n, n+1)$ and $v_n := \mathbf{1}_{I_n}$. Since $\int_A v_n \, dm \le m(A)$, (v_n) is uniformly integrable.
If $m(E_0) < +\infty$ and $\sup_n \int_{\mathbb{R} \setminus E_0} v_n \, dm < \frac{1}{2}$, then $m(E_0 \cap I_n) > \frac{1}{2}$ for all n, contradiction.

Structural Commentary: Concentration occurs even on $m(X) < +\infty$; escape is an $m(X) = +\infty$ obstruction.
Key Insight: Uniform integrability controls small-set mass; tightness keeps mass in finite-measure cores.

Proposition 4.4.5. *Basic Sources of Uniform Integrability and Tightness*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) If $0 \le g \in L^1(\mu)$ and $\forall h \in \mathcal{F}$, $|h| \le g$ a.e., then $\mathcal{F}$ is uniformly integrable and tight.
(ii) If $f_1, \dots, f_N \in L^1(\mu)$, then $\{f_1, \dots, f_N\}$ is uniformly integrable and tight.

Proof. Let $\varepsilon > 0$.

(i) Choose $\delta > 0$ with $\mu(A) < \delta \Rightarrow \int_A g \, d\mu < \varepsilon$ by Lemma 4.1.39; hence $\sup_{h \in \mathcal{F}} \int_A |h| \, d\mu < \varepsilon$.
Choose finite-measure X_ε with $\int_{X \setminus X_\varepsilon} g \, d\mu < \varepsilon$ by Proposition 1.3.22; hence $\sup_{h \in \mathcal{F}} \int_{X \setminus X_\varepsilon} |h| \, d\mu < \varepsilon$.

(ii) For each j, apply (i) to $\{f_j\}$ with ε: choose $\delta_j > 0$ such that $\mu(A) < \delta_j \Rightarrow \int_A |f_j| \, d\mu < \varepsilon$, and choose $X_{\varepsilon,j}$ with $\mu(X_{\varepsilon,j}) < +\infty$ and $\int_{X \setminus X_{\varepsilon,j}} |f_j| \, d\mu < \varepsilon$. Set $\delta := \min_{1 \le j \le N} \delta_j$ and $X_\varepsilon := \bigcup_{j=1}^{N} X_{\varepsilon,j}$.
Then $\mu(A) < \delta \Rightarrow \sup_j \int_A |f_j| \, d\mu < \varepsilon$, while $\mu(X_\varepsilon) < +\infty$ and $\sup_j \int_{X \setminus X_\varepsilon} |f_j| \, d\mu < \varepsilon$.

Structural Commentary: An envelope gives common bounds; finite families use $\min_j \delta_j$ and $\bigcup_j X_{\varepsilon,j}$.

Key Insight: Domination gives uniform estimates; finite families allow minima and finite unions. □

Corollary 4.4.6. *L^p-Bounded Families Are Uniformly Integrable*
Let $1 < p < \infty$, q conjugate to p. If $\mathcal{F} \subseteq L^p(\mu)$ and $\forall h \in \mathcal{F}$, $\|h\|_p \le M$, then $\mathcal{F}$ is uniformly integrable.

Proof. If $M = 0$, every $h \in \mathcal{F}$ vanishes a.e. and the claim is immediate. Else, for measurable A with $\mu(A) < +\infty$,
by Hölder's inequality, $\int_A |h| \, d\mu = \int_X |h| \, \mathbf{1}_A \, d\mu \le \|h\|_p \, \|\mathbf{1}_A\|_q \le M \mu(A)^{\frac{1}{q}}$.
Given $\varepsilon > 0$, take $\delta = (\frac{\varepsilon}{M})^q$. Then $\mu(A) < \delta$ implies $\sup_{h \in \mathcal{F}} \int_A |h| \, d\mu < \varepsilon$.

Structural Commentary: Apply Hölder to $h\mathbf{1}_A$ and choose one measure threshold for the whole family.
Key Insight: Hölder converts a uniform L^p bound into uniform small-set L^1 control. □

Proposition 4.4.7. *Uniform Integrability by Large-Value Tails*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space and let (f_n) be bounded in $L^1(\mu)$. The following are equivalent.

(i) (f_n) is uniformly integrable.
(ii) $\forall \varepsilon > 0 \ \exists M > 0 : \ \sup_n \int_{\{|f_n| \ge M\}} |f_n| \, d\mu < \varepsilon$.

Proof. Let $C = \sup_n \|f_n\|_1 < +\infty$.

($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ from uniform integrability and $M > C/\delta$.
By Chebyshev, $\mu(\{|f_n| \ge M\}) \le M^{-1} \|f_n\|_1 \le C/M < \delta$, so $\sup_n \int_{\{|f_n| \ge M\}} |f_n| \, d\mu < \varepsilon$.

($\Leftarrow$) Given $\varepsilon > 0$, choose $M > 0$ with $\sup_n \int_{\{|f_n| \ge M\}} |f_n| \, d\mu < \frac{\varepsilon}{2}$ and set $\delta := \frac{\varepsilon}{2M}$.
If $\mu(A) < \delta$, then $\int_A |f_n| \, d\mu \le \int_{\{|f_n| \ge M\}} |f_n| \, d\mu + M\mu(A) < \varepsilon$.

Structural Commentary: Chebyshev shrinks tails; truncation recovers UI.
Key Insight: UI iff large tails vanish uniformly. □

Theorem 4.4.8. *Vitali Convergence for L^1 Families*
Let $(X, \mathcal{M}, \mu)$ be a measure space, let $f_n \in L^1(\mu)$, let $f : X \to \mathbb{R}$ be measurable, and assume $f_n \to f$ a.e.

(i) If $\mu(X) < +\infty$ and $\{f_n : n \ge 1\}$ is uniformly integrable, then $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.
(ii) If $\{f_n : n \ge 1\}$ is uniformly integrable and tight, then $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.
(iii) Under either hypothesis, $\int_X f_n \, d\mu \to \int_X f \, d\mu$.

Proof.

(i) Since $\mu(X) < +\infty$ and $f_n \to f$ a.e., $f_n \to f$ in measure by Corollary 4.2.21. Uniform integrability is the small-set hypothesis of Theorem 4.2.26 with $p = 1$; the finite-tail hypothesis holds with $E_\varepsilon = X$, QED.

(ii) Fix $\alpha, \eta > 0$ and set $B_n^\alpha := \{|f_n - f| \ge \alpha\}$.
By tightness, choose $X_0 \in \mathcal{M}$ with $\mu(X_0) < +\infty$, $\sup_n \int_{X \setminus X_0} |f_n| \, d\mu < \frac{\alpha\eta}{4}$. By Fatou, $\int_{X \setminus X_0} |f| \, d\mu \le \frac{\alpha\eta}{4}$.
Then $\alpha\mu(B_n^\alpha \cap (X \setminus X_0)) \le \int_{X \setminus X_0} (|f_n| + |f|) \, d\mu < \frac{\alpha\eta}{2}$. Since $f_n \to f$ a.e. on finite X_0, $\mu(B_n^\alpha \cap X_0) < \frac{\eta}{2}$ eventually by Corollary 4.2.21. Thus $\mu(B_n^\alpha) < \eta$ eventually, so $f_n \to f$ in measure.
By Theorem 4.2.26 with $p = 1$, using uniform integrability and tightness, $f \in L^1(\mu)$ and $\|f_n - f\|_1 \to 0$.

(iii) In either case, $\left| \int_X f_n \, d\mu - \int_X f \, d\mu \right| = \left| \int_X (f_n - f) \, d\mu \right| \le \|f_n - f\|_1 \to 0$.

Structural Commentary: Tightness gives finite cores; UI upgrades in-measure convergence to L^1.
Key Insight: Uniform integrability prevents concentration; tightness prevents escape. □

Corollary 4.4.9. *Nonnegative Integral Vanishing Criterion*
Let $f_n \in L^1(\mu)$ with $f_n \ge 0$ a.e. The following are equivalent.

(i) $\int_X f_n \, d\mu \to 0$.
(ii) $\|f_n\|_1 \to 0$.
(iii) $f_n \to 0$ in measure, and $\{f_n : n \ge 1\}$ is uniformly integrable and tight.

Proof.

(i) $\Rightarrow$ (ii) Since $f_n \ge 0$, $\|f_n\|_1 = \int_X f_n \, d\mu \to 0$.

(ii) $\Rightarrow$ (iii) By Theorem 4.2.26 with $p = 1$ and limit 0, $\|f_n\|_1 \to 0$ implies $f_n \to 0$ in measure,
$\forall \varepsilon > 0 \ \exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ and $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F f_n \, d\mu < \varepsilon$,
and $\forall \varepsilon > 0 \ \exists \delta > 0$ with $\mu(A) < \delta \Rightarrow \sup_n \int_A f_n \, d\mu < \varepsilon$; hence tightness and uniform integrability.

(iii) $\Rightarrow$ *(i)* Condition (iii) is hypotheses (ii)–(iv) of Theorem 4.2.26 with $p = 1$ and limit 0, so $\|f_n\|_1 \to 0$.
Since $f_n \geq 0$, $\int_X f_n \, d\mu = \|f_n\|_1 \to 0$.

Structural Commentary: For nonnegative functions, integral convergence to zero is L^1 convergence to zero.
Key Insight: Vitali's conditions characterize when nonnegative mass vanishes. □

Corollary 4.4.10. *Domination Gives Tail and Small-Set Control*
Let $1 \leq p < \infty$, $g \in L^p(\mu)$, and $\forall n$, $|f_n| \leq g$ a.e. Then $\forall \varepsilon > 0$:
(i) $\exists E_\varepsilon \in \mathcal{M}$ *with* $\mu(E_\varepsilon) < +\infty$ *such that* $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F |f_n|^p \, d\mu < \varepsilon^p$.
(ii) $\exists \delta(\varepsilon) > 0$ *such that* $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \int_E |f_n|^p \, d\mu < \varepsilon^p$.

Proof.
(i) Choose E_ε with $\mu(E_\varepsilon) < +\infty$ and $F \cap E_\varepsilon = \emptyset \Rightarrow \int_F |g|^p \, d\mu < \varepsilon^p$ by Proposition 4.1.26.
Since $|f_n|^p \leq |g|^p$ a.e., $\sup_n \int_F |f_n|^p \, d\mu < \varepsilon^p$.
(ii) Choose $\delta(\varepsilon) > 0$ with $\mu(E) < \delta(\varepsilon) \Rightarrow \int_E |g|^p \, d\mu < \varepsilon^p$ by Lemma 4.1.39.
Since $|f_n|^p \leq |g|^p$ a.e., $\sup_n \int_E |f_n|^p \, d\mu < \varepsilon^p$.

Structural Commentary: One L^p envelope supplies both tail tightness and small-set control.
Key Insight: Domination transfers both estimates from g to every f_n. □

4.5 Approximation and Separability in L^p

Approximation results explain why L^p spaces are large enough to contain rough measurable functions but still small enough to be controlled by countable data when $p < \infty$. The guiding pattern is to localize the L^p mass, approximate measurable atoms by elementary sets, and then code the elementary sets with rational parameters.

Definition 4.5.1. *Dense Subsets and Separable Normed Spaces*
Let $A \subseteq B$ be subsets of a normed space $(V, \|\cdot\|)$.
(i) A is *dense in* B if $\forall b \in B \; \forall \varepsilon > 0 \; \exists a \in A$ with $\|a - b\| < \varepsilon$.
(ii) V is *separable* if it has a countable dense subset.
Equivalently, A is dense in B iff every $b \in B$ is the norm-limit of a sequence from A.

Example 4.5.2. *Sequence-Space Separability*
(i) $\ell^\infty(\mathbf{F})$ is not separable. The set $\{0,1\}^{\mathbb{N}}$ is uncountable, and $s \neq t$ gives $d_\infty(s,t) = 1$; the balls $B_{1/3}(s)$ are pairwise disjoint, so no countable dense set can meet all of them.
(ii) For $1 \leq p < \infty$, $\ell^p(\mathbf{F})$ is separable. Let D be the countable set of finite-support sequences with rational coordinates, using rational real and imaginary parts if $\mathbf{F} = \mathbb{C}$.
For $x \in \ell^p$ and $\varepsilon > 0$, choose N with $\sum_{n>N} |x_n|^p < (\varepsilon/2)^p$, then choose $y \in D$ with support in $\{1, \ldots, N\}$ and $\sum_{n=1}^N |x_n - y_n|^p < (\varepsilon/2)^p$. Thus $\|x - y\|_p < \varepsilon$.

Structural Commentary: Separability is the existence of a countable set meeting every nonempty open ball.
Key Insight: Finite rational data are dense in ℓ^p, while ℓ^∞ contains an uncountable 1-separated set.

Theorem 4.5.3. *Separability and C_c Density in L^p*
Let $E \subseteq \mathbb{R}$ be Lebesgue measurable and let $1 \leq p < \infty$.
If $C_c(E)$ denotes restrictions to E of continuous functions on $\mathbb{R}$ with bounded support, then:
(i) $L^p(E)$ is separable;
(ii) $C_c(E)$ is dense in $L^p(E)$.

Proof.
(i) Let $\mathbb{F}_0 := \mathbb{Q}$ if real, $\mathbb{F}_0 := \mathbb{Q} + i\mathbb{Q}$ if complex. Let $\mathcal{D} := \operatorname{span}_{\mathbb{F}_0}\{\mathbf{1}_{E \cap (r,s)} : r, s \in \mathbb{Q}, \; r < s\}$.
Then $\mathcal{D}$ is countable. Extend $f \in L^p(E)$ by 0 to $\tilde{f} \in L^p(\mathbb{R})$.
By Proposition 4.1.26, it suffices to approximate $s = \sum_{\ell=1}^M a_\ell \mathbf{1}_{A_\ell}$ with $m(A_\ell) < +\infty$.
For each ℓ, Corollary 6.2.4 and rational endpoint perturbation give a finite union V_ℓ of rational open intervals with $m(A_\ell \triangle V_\ell)$ small, hence $\|\mathbf{1}_{E \cap A_\ell} - \mathbf{1}_{E \cap V_\ell}\|_p \leq m(A_\ell \triangle V_\ell)^{1/p}$. Approximate a_ℓ by $q_\ell \in \mathbb{F}_0$; then $\sum_{\ell=1}^M q_\ell \mathbf{1}_{E \cap V_\ell} \in \mathcal{D}$ approximates f arbitrarily well. Hence $L^p(E)$ is separable.
(ii) For rational $r < s$ and $0 < \eta < \frac{s-r}{2}$, choose $\psi_\eta \in C_c(\mathbb{R})$ with $0 \leq \psi_\eta \leq 1$, $\psi_\eta = 1$ on $[r + \eta, s - \eta]$, and $\psi_\eta = 0$ on $\mathbb{R} \setminus (r, s)$. Then $\|\psi_\eta|_E - \mathbf{1}_{E \cap (r,s)}\|_p^p \leq m((r, r+\eta) \cup (s - \eta, s)) \leq 2\eta \to 0$.
Thus each generator of $\mathcal{D}$ lies in $\overline{C_c(E)}^{L^p(E)}$; by finite linearity and (i), $C_c(E)$ is dense in $L^p(E)$.

Structural Commentary: Finite-measure atoms reduce to rational intervals, then to continuous cutoffs.
Key Insight: Countable rational steps lie in $\overline{C_c(E)}^{L^p(E)}$. □

Remark 4.5.4. *L^1 Does Not Exhaust the Dual of L^∞*
Let $I = [a,b]$ with $a < b$, and let $\iota : L^1(I) \to (L^\infty(I))^*$ be $\iota(g)(f) := \int_I fg \, dm$. The map ι from Proposition 4.1.37 is not onto. If ι were onto, then $(L^\infty(I))^*$ would be separable by Theorem 4.5.3.
Let (Φ_n) be dense in its unit sphere and choose $u_n \in L^\infty(I)$ with $\|u_n\|_\infty = 1$ and $|\Phi_n(u_n)| > \frac{1}{2}$.
If $Y := \overline{\operatorname{span}}\{u_n : n \geq 1\} \neq L^\infty(I)$, Hahn–Banach gives $\Phi \in (L^\infty(I))^*$ with $\|\Phi\| = 1$ and $\Phi|_Y = 0$.
Choosing n with $\|\Phi - \Phi_n\| < \frac{1}{4}$ gives $|\Phi_n(u_n)| < \frac{1}{4}$, a contradiction. Hence $L^\infty(I)$ would be separable.
But $v_t := \mathbf{1}_{[a,t]}$ satisfies $\|v_t - v_s\|_\infty = 1$ for $s < t$, an uncountable separated family, so $L^\infty(I)$ is not separable.

Example 4.5.5. *L^∞ on an Interval Need Not Be Separable*
Let $I=[a,b]$ with $a<b$. For $t\in I$, set $u_t:=\mathbf{1}_{[a,t]}$. If $s<t$, then $\|u_t-u_s\|_\infty=1$, since $u_t-u_s=\mathbf{1}_{(s,t]}$ a.e.
Thus $\{B(u_t,\frac{1}{3}):t\in I\}$ is pairwise disjoint. If countable $D\subseteq L^\infty(I)$ were dense, choose $d_t\in D\cap B(u_t,\frac{1}{3})$ for each $t\in I$; disjointness makes $t\mapsto d_t$ injective, impossible. Hence $L^\infty(I)$ is not separable.
Key Insight: Uncountably many distance-1 indicators preclude countable density.

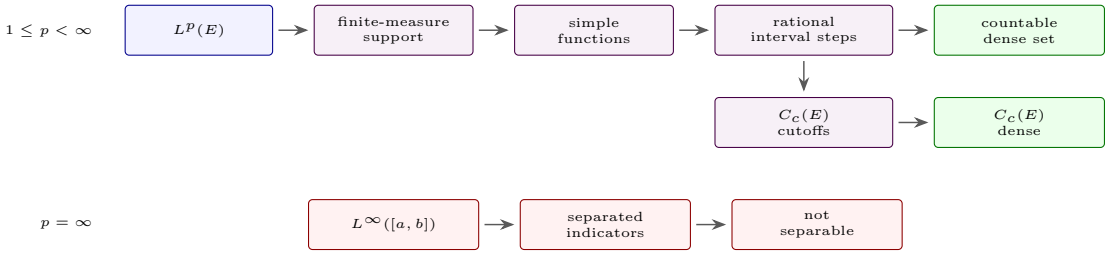

Figure 217: For $1\le p<\infty$, rational interval steps and $C_c(E)$ are dense in $L^p(E)$; $L^\infty([a,b])$ is nonseparable.

4.6 Weak Convergence in L^p

Weak convergence in L^p keeps only the scalar information detected by conjugate functions. For $1<p<+\infty$, reflexivity turns boundedness into weak subsequential compactness; at $p=1$, bounded mass can still concentrate and destroy weak compactness unless uniform integrability is imposed.

Theorem 4.6.1. *L^p Weak Convergence by Dual Tests*
Let $(X,\mathcal{M},\mu)$ be a measure space, $1<p<+\infty$, $q=\frac{p}{p-1}$, $(f_n)\subseteq L^p(\mu)$, and $f\in L^p(\mu)$, all real-valued.
Then $f_n\rightharpoonup f$ in $L^p(\mu)$ iff $\int_X f_n g\,d\mu\to\int_X fg\,d\mu$ for all $g\in L^q(\mu)$.

Proof.
($\Rightarrow$) For $g\in L^q(\mu)$, $\Lambda_g(h):=\int_X hg\,d\mu$ is in $(L^p(\mu))^*$ by Holder.
Hence $f_n\rightharpoonup f$ implies $\int_X f_n g\,d\mu=\Lambda_g(f_n)\to\Lambda_g(f)=\int_X fg\,d\mu$.
($\Leftarrow$) Let $\Lambda\in(L^p(\mu))^*$. By Theorem 4.1.34, choose $g\in L^q(\mu)$ with $\Lambda(h)=\int_X hg\,d\mu\ \forall h\in L^p(\mu)$.
The assumed integral test yields $\Lambda(f_n)\to\Lambda(f)$, hence $f_n\rightharpoonup f$.

Structural Commentary: Dual tests are exactly L^q pairings.
Key Insight: Weak convergence in L^p is conjugate-function testing. □

Proposition 4.6.2. *Weak Limits and Norm Lower Semicontinuity*
Let $(X,\mathcal{M},\mu)$ be a measure space, $1<p<+\infty$, $(f_n)\subseteq L^p(\mu)$, and $f_n\rightharpoonup f$ in real $L^p(\mu)$. Then:
(i) $\sup_n\|f_n\|_p<+\infty$;
(ii) if $f_n\rightharpoonup h$ in $L^p(\mu)$, then $h=f$ in $L^p(\mu)$;
(iii) $\|f\|_p\le\liminf_{n\to\infty}\|f_n\|_p$.

Proof.
(i) By Proposition 6.1.4 applied to $u_n=f_n$ in $L^p(\mu)$, $f_n\rightharpoonup f$ implies $\sup_n\|f_n\|_p<+\infty$.
(ii) Let $q=\frac{p}{p-1}$, $f_n\rightharpoonup h$. For $g\in L^q(\mu)$, by Theorem 4.6.1, $\int_X(f-h)g\,d\mu=\lim_n\int_X f_n g\,d\mu-\lim_n\int_X f_n g\,d\mu=0$.
By Proposition 4.1.30, $\|f-h\|_p=0$, so $h=f$ in $L^p(\mu)$.
(iii) By Proposition 6.1.4, $u\mapsto\|u\|_p$ is weakly lower semicontinuous on $L^p(\mu)$;
with $u_n=f_n$ and $u=f$, $\|f\|_p\le\liminf_{n\to\infty}\|f_n\|_p$.

Structural Commentary: General weak facts handle bounds; L^q tests force uniqueness.
Key Insight: Weak limits are norm-controlled and conjugate-tested. □

Theorem 4.6.3. *Reflexivity of L^p, $1<p<+\infty$*
Let $(X,\mathcal{M},\mu)$ be σ-finite, $1<p<+\infty$, $q=\frac{p}{p-1}$, and $Jf(\Phi):=\Phi(f)$.
For $r\in\{p,q\}$ with conjugate r', set $T_r u\in(L^{r'}(\mu))^*$ by $(T_r u)(v):=\int_X uv\,d\mu$. Then:
(i) T_p and T_q are isometric isomorphisms;
(ii) $J:L^p(\mu)\to(L^p(\mu))^{**}$ is onto; hence real $L^p(\mu)$ is reflexive.

Proof.
(i) By Theorem 4.1.34, $T_p:L^p(\mu)\to(L^q(\mu))^*$ and $T_q:L^q(\mu)\to(L^p(\mu))^*$ are isometric isomorphisms.
(ii) Let $S\in(L^p(\mu))^{**}$. Since $S\circ T_q\in(L^q(\mu))^*$, by (i), $f\in L^p(\mu)$, $S(T_q g)=(T_p f)(g)=\int_X fg\,d\mu\ \forall g\in L^q(\mu)$.
For $\Phi\in(L^p(\mu))^*$, by (i) $\Phi=T_q g$; $S(\Phi)=S(T_q g)=\int_X fg\,d\mu=\Phi(f)=Jf(\Phi)$. Hence $S=Jf$, J is onto.

Structural Commentary: Represent both dual layers by conjugate pairings.
Key Insight: Every bidual functional is evaluation at an L^p function. □

Definition 4.6.4. *Weak Sequential Compactness*
Let $\mathcal{X}$ be a normed space and $K\subseteq\mathcal{X}$.
Then K is *weakly sequentially compact* if every $(x_n)\subseteq K$ admits $n_k\uparrow\infty$ and $x\in K$ with $x_{n_k}\rightharpoonup x$ in $\mathcal{X}$.

Theorem 4.6.5. *Riesz Weak Compactness Theorem*
Let $(X, \mathcal{M}, \mu)$ be σ-finite, $1 < p < +\infty$, $q = \frac{p}{p-1}$, and $(f_n) \subseteq L^p(\mu)$ be real-valued with $M := \sup_n \|f_n\|_p < +\infty$.
Then there exist $n_k \uparrow \infty$ and $f \in L^p(\mu)$ such that:
(i) $f_{n_k} \rightharpoonup f$ in $L^p(\mu)$;
(ii) $\int_X f_{n_k} g \, d\mu \to \int_X fg \, d\mu$ for all $g \in L^q(\mu)$.

Proof. Set $B_M := \{h \in L^p(\mu) : \|h\|_p \le M\}$. By Theorem 4.6.3, $L^p(\mu)$ is reflexive; hence B_M is weakly compact. By Eberlein–Smulian, B_M is weakly sequentially compact.
(i) Since $(f_n) \subseteq B_M$ and B_M is weakly sequentially compact, $\exists n_k \uparrow \infty$ and $f \in B_M$ with $f_{n_k} \rightharpoonup f$ in $L^p(\mu)$.
(ii) From (i) and Theorem 4.6.1, $\forall g \in L^q(\mu)$, $\int_X f_{n_k} g \, d\mu \to \int_X fg \, d\mu$.
Structural Commentary: Reflexive bounded balls are weakly sequentially compact.
Key Insight: Bounded L^p sequences have conjugate-test convergent subsequences. □

Corollary 4.6.6. *The L^p Unit Ball Is Weakly Sequentially Compact*
Let $(X, \mathcal{M}, \mu)$ be σ-finite, $1 < p < +\infty$, and $B_p := \{f \in L^p(\mu) : \|f\|_p \le 1\}$.
Then B_p is weakly sequentially compact in $L^p(\mu)$.

Proof. Let $(f_n) \subseteq B_p$. By Theorem 4.6.5, $\exists n_k \uparrow \infty$ and $f \in L^p(\mu)$ with $f_{n_k} \rightharpoonup f$.
By Proposition 4.6.2, $\|f\|_p \le \liminf_{k\to\infty} \|f_{n_k}\|_p \le 1$, so $f \in B_p$.
Structural Commentary: Weak limits stay in B_p by lower semicontinuity.
Key Insight: Closed norm balls are weakly sequentially closed. □

Theorem 4.6.7. *Radon–Riesz Principle*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f \in L^p(\mu)$, all real-valued.
Then $f_n \rightharpoonup f$ and $\|f_n\|_p \to \|f\|_p$ iff $\|f_n - f\|_p \to 0$.

Proof.
($\Rightarrow$) If $\|f\|_p = 0$, then $\|f_n - f\|_p \le \|f_n\|_p + \|f\|_p \to 0$. Let $a := \|f\|_p > 0$.
Otherwise, after subsequences, $\|f_n - f\|_p \ge \varepsilon > 0$ and, with $a_n := \|f_n\|_p$, $u_n := \frac{f_n}{a_n}$, $u := \frac{f}{a}$, one has $u_n \rightharpoonup u$, $\|u_n\|_p = \|u\|_p = 1$, and $\|u_n - u\|_p \ge \eta > 0$. By uniform convexity, $\left\|\frac{u_n+u}{2}\right\|_p \le 1 - \delta$.
Since $\frac{u_n+u}{2} \rightharpoonup u$, weak lower semicontinuity implies $1 = \|u\|_p \le \liminf_n \left\|\frac{u_n+u}{2}\right\|_p \le 1 - \delta$, contradiction.
($\Leftarrow$) Assume $\|f_n - f\|_p \to 0$. Then $\left|\|f_n\|_p - \|f\|_p\right| \le \|f_n - f\|_p \to 0$.
For $\Lambda \in (L^p(\mu))^*$, $|\Lambda(f_n) - \Lambda(f)| \le \|\Lambda\| \, \|f_n - f\|_p \to 0$, hence $f_n \rightharpoonup f$.
Structural Commentary: Uniform convexity turns weak lower semicontinuity into strong convergence.
Key Insight: Weak direction plus norm size fixes the vector. □

Remark 4.6.8. *L^2 Proof of Radon–Riesz*
For $p = 2$, $f_n \rightharpoonup f$ in $L^2(\mu)$, $\|f_n\|_2 \to \|f\|_2$. Then $\int_X f_n f \, d\mu \to \|f\|_2^2$,
$\|f_n - f\|_2^2 = \|f_n\|_2^2 - 2\int_X f_n f \, d\mu + \|f\|_2^2 \to 0$. The converse follows from norm convergence in $L^2(\mu)$.
Key Insight: Weak convergence fixes the cross term.

Corollary 4.6.9. *Strong Subsequence Criterion*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f_n \rightharpoonup f$ in real $L^p(\mu)$.
Then $\exists n_k \uparrow \infty$ with $\|f_{n_k} - f\|_p \to 0$ iff $\|f\|_p = \liminf_{n\to\infty} \|f_n\|_p$.

Proof. Set $L := \liminf_{n\to\infty} \|f_n\|_p$.
($\Rightarrow$) If $\|f_{n_k} - f\|_p \to 0$, then $\|f_{n_k}\|_p \to \|f\|_p$.
Since $L \le \lim_k \|f_{n_k}\|_p = \|f\|_p$ and, by Proposition 4.6.2, $\|f\|_p \le L$, one has $L = \|f\|_p$.
($\Leftarrow$) Assume $L = \|f\|_p$. Choose $n_k \uparrow \infty$ with $\|f_{n_k}\|_p \to L$.
Since $f_{n_k} \rightharpoonup f$ and $\|f_{n_k}\|_p \to \|f\|_p$, Theorem 4.6.7 implies $\|f_{n_k} - f\|_p \to 0$.
Structural Commentary: Match the liminf subsequence to Radon–Riesz.
Key Insight: Strong subsequences occur exactly when no norm is lost. □

Theorem 4.6.10. *Dunford–Pettis Criterion in L^1*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space and let (f_n) be bounded in $L^1(\mu)$. The following are equivalent.
(i) (f_n) is uniformly integrable.
(ii) $\forall (f_{n_k}) \subseteq (f_n)$ $\exists (f_{n_{k_j}}) \subseteq (f_{n_k})$ converging weakly in $L^1(\mu)$.

Proof.
($\Rightarrow$) Let (u_j) be a subsequence of (f_n), set $C := \sup_j \|u_j\|_1$, and put $T_m(t) := (-m) \vee (t \wedge m)$.
By Proposition 4.4.7, $\sup_j \|u_j - T_m(u_j)\|_1 \to 0$. For fixed m and sign $\pm$, $\left\|T_m(u_j^{\pm})\right\|_2^2 \le m \left\|u_j^{\pm}\right\|_1 \le mC$.
By bounded Hilbert weak subsequence extraction and diagonalization over $(m, \pm)$, pass to a subsequence, still (u_j), such that $\int_E T_m(u_j^{\pm}) \, d\mu$ converges for every $E \in \mathcal{M}$ and $m \in \mathbb{N}$. Set $\nu_j^{\pm}(E) := \int_E u_j^{\pm} \, d\mu$.
For r, s, m and $E \in \mathcal{M}$, $\left|\nu_r^{\pm}(E) - \nu_s^{\pm}(E)\right| \le \left|\int_E (T_m(u_r^{\pm}) - T_m(u_s^{\pm})) \, d\mu\right| + 2\sup_j \left\|u_j^{\pm} - T_m(u_j^{\pm})\right\|_1$.
Hence $(\nu_j^{\pm}(E))_j$ is Cauchy. Put $\nu^{\pm}(E) := \lim_j \nu_j^{\pm}(E)$.
By Theorem 3.6.11, $\nu^{\pm}$ are finite measures and $\nu^{\pm} \ll \mu$. Write $h^{\pm} = \frac{d\nu^{\pm}}{d\mu}$ and $h := h^+ - h^- \in L^1(\mu)$.
Then $\int_X \varphi u_j \, d\mu \to \int_X \varphi h \, d\mu$ for simple φ; by Lemma 1.1.14 and $\sup_j \|u_j\|_1 + \|h\|_1 < +\infty$, the convergence holds for every $\varphi \in L^\infty(\mu)$. Thus $u_j \rightharpoonup h$ in $L^1(\mu)$.

($\Leftarrow$) Suppose (f_n) is not uniformly integrable. Since finite subfamilies are uniformly integrable by Proposition 4.4.5(ii), choose a subsequence (u_j) and sets $A_j \in \mathcal{M}$ with $\mu(A_j) \to 0$ and $\int_{A_j} |u_j|\, d\mu \geq \varepsilon_0 > 0$.

By (ii), some (u_{j_ℓ}) converges weakly in $L^1(\mu)$ to u. Set $\eta_\ell(E) := \int_E u_{j_\ell}\, d\mu$ and $\eta(E) := \int_E u\, d\mu$.
Then $\eta_\ell(E) \to \eta(E)$ for every $E \in \mathcal{M}$, and $\sup_\ell |\eta_\ell|(X) = \sup_\ell \|u_{j_\ell}\|_1 < +\infty$ by Theorem 3.1.9.
The signed form of Theorem 3.6.11 implies $\eta_\ell \ll_u \mu$ in variation:
$\forall \varepsilon > 0\ \exists \delta > 0$ such that $\mu(A) < \delta \Rightarrow \sup_\ell |\eta_\ell|(A) < \varepsilon$. Since $|\eta_\ell|(A) = \int_A |u_{j_\ell}|\, d\mu$, the subsequence (u_{j_ℓ}) is uniformly integrable, contradicting $\int_{A_{j_\ell}} |u_{j_\ell}|\, d\mu \geq \varepsilon_0$ and $\mu(A_{j_\ell}) \to 0$.

Structural Commentary: Truncate, pass to setwise limits, apply VHS.
Key Insight: UI is weak sequential compactness in finite L^1. □

Corollary 4.6.11. *Dominated L^1 Families Have Weak Subsequences*
Let $(X, \mathcal{M}, \mu)$ be finite. If $g \in L^1(\mu)$, $\forall n\, |f_n| \leq g$ a.e., then (f_n) has subsequence that converges weakly in $L^1(\mu)$.

Proof. Since $|f_n| \leq g$ a.e., $\|f_n\|_1 \leq \|g\|_1\ \forall n$, (f_n) is bounded in $L^1(\mu)$. By Proposition 4.4.5(i), $\{f_n : n \geq 1\}$ is uniformly integrable. Theorem 4.6.10, $(i) \Rightarrow (ii)$, yields a weakly convergent subsequence.
Structural Commentary: Domination verifies Dunford–Pettis through a common integrable envelope.
Key Insight: One L^1 envelope prevents concentration throughout the sequence. □

Corollary 4.6.12. *Bounded L^p Families Are Weakly Compact in L^1*
Let $(X, \mathcal{M}, \mu)$ be finite, let $1 < p < +\infty$, and let (f_n) be bounded in $L^p(\mu)$.
Then (f_n) has a subsequence that converges weakly in $L^1(\mu)$.

Proof. Let q be conjugate to p and $M := \sup_n \|f_n\|_p < +\infty$. By Hölder, $\|f_n\|_1 \leq M\mu(X)^{\frac{1}{q}}\ \forall n$, so (f_n) is bounded in $L^1(\mu)$. By Corollary 4.4.6, $\{f_n : n \geq 1\}$ is uniformly integrable.
Hence Theorem 4.6.10, $(i) \Rightarrow (ii)$, yields a weakly convergent subsequence in $L^1(\mu)$.
Structural Commentary: Hölder gives L^1 boundedness, uniform integrability before Dunford–Pettis is applied.
Key Insight: On finite measure spaces, an L^p bound with $p > 1$ rules out L^1 concentration. □

Example 4.6.13. *A Bounded L^1 Sequence with No Weakly Convergent Subsequence*
In $L^1[0,1]$, $I_n = [0, \frac{1}{n}]$ $f_n = n\mathbf{1}_{I_n}$. Then $\|f_n\|_1 = 1\ \forall n$, and no subsequence of (f_n) converges weakly in $L^1[0,1]$.

Proof. Assume $f_{n_k} \rightharpoonup f$ in $L^1[0,1]$. For $0 < c < d \leq 1$, $\mathbf{1}_{[c,d]} \in L^\infty[0,1]$ and $[c,d] \cap I_{n_k} = \varnothing$ for k large.
Hence $\int_c^d f\, dm = \lim_{k\to\infty} \int_c^d f_{n_k}\, dm = 0$. Thus $f = 0$ a.e. on $(0,1]$.
Testing by $\mathbf{1}_{[0,1]}$ yields $0 = \int_0^1 f\, dm = \lim_{k\to\infty} \int_0^1 f_{n_k}\, dm = 1$, contradiction.
Key Insight: Unit mass concentrates on shrinking supports. □

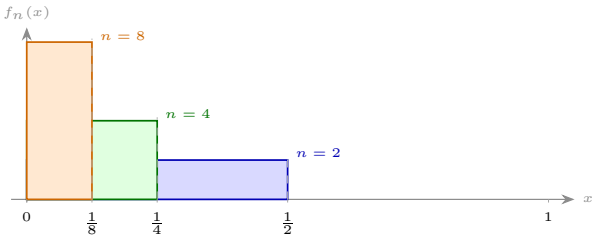

Figure 218: $f_n = n\mathbf{1}_{[0,\frac{1}{n}]}$: unit L^1 mass on shrinking support.

Remark 4.6.14. *Endpoint Failure and Uniform Integrability*
For $f_n = n\mathbf{1}_{[0,\frac{1}{n}]} \in L^1[0,1]$, $\|f_n\|_1 = 1$ but $\int_{\{f_n > R\}} f_n\, dm = 1$ whenever $n > R$.
Hence (f_n) is not uniformly integrable; the $p = 1$ failure is concentration, not lack of boundedness.

Theorem 4.6.15. *Banach–Saks Theorem*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 < p < +\infty$, $(f_n) \subseteq L^p(\mu)$, and $f_n \rightharpoonup f$ in real $L^p(\mu)$.
Then $\exists n_k \uparrow \infty$ such that, with $A_k := \frac{1}{k}\sum_{j=1}^k f_{n_j}$, $\|A_k - f\|_p \to 0$.

Proof. Set $h_n := f_n - f$. Then $h_n \rightharpoonup 0$ and $\sup_n \|h_n\|_p < +\infty$ by Proposition 4.6.2.
Since real $L^p(\mu)$ is uniformly convex, by the selection lemma there is $n_k \uparrow \infty$ with $\left\|\frac{1}{k}\sum_{j=1}^k h_{n_j}\right\|_p \to 0$.
Since $\frac{1}{k}\sum_{j=1}^k h_{n_j} = A_k - f$, $\|A_k - f\|_p \to 0$.
Structural Commentary: Reduce to weakly null; select Cesaro means by uniform convexity.
Key Insight: Convex averaging turns weak nullity into norm decay. □

Remark 4.6.16. *The L^2 Selection Estimate*
For $p = 2$, replace f_n by $f_n - f$ and assume $f_n \rightharpoonup 0$ in $L^2(\mu)$ with $\|f_n\|_2^2 \leq M$. Set $n_1 = 1$ and $S_k := \sum_{j=1}^k f_{n_j}$.

(i) Given S_k, choose $n_{k+1} > n_k$ so that $\left|\int_X S_k f_{n_{k+1}}\, d\mu\right| \leq 1$.
(ii) Then $\|S_{k+1}\|_2^2 = \|S_k\|_2^2 + 2\int_X S_k f_{n_{k+1}}\, d\mu + \|f_{n_{k+1}}\|_2^2 \leq \|S_k\|_2^2 + 2 + M$.
(iii) Hence $\|S_k/k\|_2^2 \leq \frac{2+M}{k}$ by induction.

Thus $S_k/k \to 0$ in $L^2(\mu)$.
Key Insight: Weak convergence makes new terms nearly orthogonal to old sums.

Definition 4.6.17. *Closed Convex Sets and Convex Functionals*
Let C be a subset of a normed space $\mathcal{X}$.

(i) C is *convex* if $\lambda f + (1-\lambda)g \in C$ whenever $f, g \in C$ and $0 \le \lambda \le 1$.
(ii) C is *closed* if every strongly convergent sequence in C has its limit in C.
(iii) A real-valued functional $T : C \to \mathbb{R}$ is *continuous* if $f_n \to f$ strongly in C implies $T(f_n) \to T(f)$.
(iv) If C is convex, then T is *convex* if $T(\lambda f + (1-\lambda)g) \le \lambda T(f) + (1-\lambda)T(g)$ $\forall f, g \in C$ and $0 \le \lambda \le 1$.

Example 4.6.18. *Basic Closed Convex Sets in L^p*
Let $E \subseteq \mathbb{R}$ be measurable and $1 \le p < +\infty$.

(i) If $g \in L^p(E)$ and $g \ge 0$ a.e., then $C_g := \{f \in L^p(E) : |f| \le g \text{ a.e.}\}$ is closed and convex.
(ii) $B := \{f \in L^p(E) : \|f\|_p \le 1\}$ is closed and convex.

Proof.

(i) If $f, h \in C_g$ and $0 \le \lambda \le 1$, then $|\lambda f + (1-\lambda)h| \le \lambda |f| + (1-\lambda)|h| \le g$ a.e., so C_g is convex.
If $f_n \in C_g$ and $f_n \to f$ in $L^p(E)$, choose $f_{n_k} \to f$ a.e. by Theorem 4.1.23; then $|f| \le g$ a.e., so $f \in C_g$.
(ii) If $f, h \in B$ and $0 \le \lambda \le 1$, then by Minkowski, $\|\lambda f + (1-\lambda)h\|_p \le \lambda \|f\|_p + (1-\lambda)\|h\|_p \le 1$.
If $f_n \in B$ and $f_n \to f$ in $L^p(E)$, then $\|f\|_p \le \|f - f_n\|_p + \|f_n\|_p \le \|f - f_n\|_p + 1$, hence $\|f\|_p \le 1$.

Key Insight: Norm limits preserve a.e. bounds and norm bounds. □

Lemma 4.6.19. *Weak Limits and Convex Lower Semicontinuity*
Let $E \subseteq \mathbb{R}$ be measurable, $1 < p < +\infty$, $C \subseteq L^p(E)$ closed, bounded, and convex.
Let $T : C \to \mathbb{R}$ be continuous and convex. If $(f_n) \subseteq C$ and $f_n \rightharpoonup f$ in $L^p(E)$, then, with $\alpha := \liminf_{n\to\infty} T(f_n)$:

(i) $f \in C$;
(ii) $T(f) \le \alpha$.

Proof.

(i) By Theorem 4.6.15, choose $n_k \uparrow \infty$ with $a_k := \frac{1}{k}\sum_{j=1}^{k} f_{n_j} \to f$ in $L^p(E)$.
Since $a_k \in \operatorname{co}\{f_{n_1}, \dots, f_{n_k}\} \subseteq C$ and C is closed, $f \in C$.
(ii) Choose $m_k \uparrow \infty$ with $T(f_{m_k}) \to \alpha$.

By Theorem 4.6.15, choose $g_j := f_{m_{k_j}}$ such that $b_k := \frac{1}{k}\sum_{j=1}^{k} g_j \to f$ in $L^p(E)$.

Then $b_k \in C$ and $T(g_j) \to \alpha$; by continuity and convexity, $T(f) = \lim_k T(b_k) \le \lim_k \frac{1}{k}\sum_{j=1}^{k} T(g_j) = \alpha$.

Structural Commentary: Banach–Saks supplies strong convex averages; convexity transfers the liminf.
Key Insight: Weak limits inherit closed convex constraints and convex lsc. □

Theorem 4.6.20. *Direct Method for Convex Functionals on L^p*
Let $E \subseteq \mathbb{R}$ be measurable, $1 < p < +\infty$, $C \subseteq L^p(E)$ nonempty, closed, bounded, and convex.
Let $T : C \to \mathbb{R}$ be continuous and convex. Then $\exists f_0 \in C$ such that $T(f_0) = \inf_{h\in C} T(h)$.

Proof.

(i) If $\inf_C T = -\infty$, choose $f_n \in C$ with $T(f_n) \to -\infty$.
Since C is bounded, by Theorem 4.6.5, $\exists n_k \uparrow \infty$ and $f \in L^p(E)$ with $f_{n_k} \rightharpoonup f$.
By Lemma 4.6.19, $f \in C$ and $T(f) \le \liminf_k T(f_{n_k}) = -\infty$, contradicting $T : C \to \mathbb{R}$. Thus $\inf_C T > -\infty$.
(ii) Let $c := \inf_C T$ and choose $f_n \in C$ with $T(f_n) \to c$.
Since C is bounded, by Theorem 4.6.5, $\exists n_k \uparrow \infty$ and $f_0 \in L^p(E)$ with $f_{n_k} \rightharpoonup f_0$.
By Lemma 4.6.19, $f_0 \in C$ and $T(f_0) \le \liminf_k T(f_{n_k}) = c$. Since $c \le T(f_0)$, $T(f_0) = c$.

Structural Commentary: Weak compactness supplies a limit; convex lsc keeps the infimum.
Key Insight: Direct method equals compactness plus lower semicontinuity. □

Corollary 4.6.21. *Integral Convex Functionals Have Unit-Ball Minimizers*
Let $\mu(E) < +\infty$, $1 < p < +\infty$, and $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous convex with $|\varphi(s)| \le c_1 + c_2|s|^p$ for some $c_1, c_2 \ge 0$. Set $B := \{f \in L^p(E) : \|f\|_p \le 1\}$ and $T(f) := \int_E \varphi(f)\,dm$. Then:

(i) $T : B \to \mathbb{R}$ is continuous and convex;
(ii) $\exists f_0 \in B$ such that $T(f_0) = \min_{f\in B} T(f)$.

Proof.

(i) For $f \in B$, $|\varphi(f)| \le c_1 + c_2|f|^p \in L^1(E)$, so $T(f) \in \mathbb{R}$.
Convexity of φ implies $T(\lambda f + (1-\lambda)h) \le \lambda T(f) + (1-\lambda)T(h)$ for $f, h \in B$, $0 \le \lambda \le 1$.
If $f_n \to f$ in $L^p(E)$, every subsequence admits $f_{n_k} \to f$ a.e. with $|f_{n_k}| \le g$ for some $g \in L^p(E)$.
Then $|\varphi(f_{n_k})| \le c_1 + c_2 g^p \in L^1(E)$, so by dominated convergence $T(f_{n_k}) \to T(f)$.
By the subsequence criterion, $T(f_n) \to T(f)$.
(ii) The set B is nonempty, bounded, convex by Minkowski, and closed since $f_n \in B$, $f_n \to f$ in $L^p(E)$ imply $\|f\|_p \le \liminf_n \|f_n\|_p \le 1$.
By Theorem 4.6.20, $\exists f_0 \in B$ with $T(f_0) = \min_{f\in B} T(f)$.

Structural Commentary: p-growth ⇒ continuity; direct method ⇒ attainment.
Key Insight: Growth converts strong L^p convergence to dominated convergence. □

5 Banach and Hilbert Spaces

5.1 Normed and Banach Spaces

Normed spaces make vector spaces metric; Banach spaces add completeness. This is the setting for infinite series, fixed points, bounded linear maps, and operator norms.

Definition 5.1.1. *Normed and Banach Spaces*
Let $\mathcal{X}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) *Norm*: $\|\cdot\| : \mathcal{X} \to [0,\infty)$; $\|x\| = 0 \iff x = 0$, $\|\lambda x\| = |\lambda| \, \|x\|$; $\|x + y\| \le \|x\| + \|y\|$ for $x, y \in \mathcal{X}$, $\lambda \in \mathbb{F}$.
(ii) *Normed space*: $(\mathcal{X}, \|\cdot\|)$ carries the metric $d(x, y) := \|x - y\|$.
(iii) *Banach space*: $(\mathcal{X}, \|\cdot\|)$ is complete in the norm metric.

Definition 5.1.2. *Basic Normed-Space Geometry*
Let $\mathcal{X}$ be a normed space.

(i) *Open unit ball*: $B_{\mathcal{X}} := \{x \in \mathcal{X} : \|x\| < 1\}$.
(ii) *Closed unit ball*: $\overline{B}_{\mathcal{X}} := \{x \in \mathcal{X} : \|x\| \le 1\}$.
(iii) *Unit vector*: $x \in \mathcal{X}$ satisfies $\|x\| = 1$.
(iv) *Minkowski sum*: $A + B := \{a + b : a \in A,\ b \in B\}$ for nonempty $A, B \subseteq \mathcal{X}$.
(v) *Scalar multiple*: $\lambda A := \{\lambda a : a \in A\}$ for nonempty $A \subseteq \mathcal{X}$ and $\lambda \in \mathbb{F}$.
(vi) *Closed linear span*: $\operatorname{cspan}(S) := \overline{\operatorname{span}(S)}$ for $S \subseteq \mathcal{X}$.
(vii) *Equivalent norms*: $\|\cdot\|_1 \sim \|\cdot\|_2$ iff $\exists c_1, c_2 > 0$ such that $\forall x \in \mathcal{X}$, $c_1 \|x\|_1 \le \|x\|_2 \le c_2 \|x\|_1$.

Proposition 5.1.3. *Linear Operations, Closed Spans, and Equivalent Norms*
Let $\mathcal{X}$ be a normed space.

(i) The norm map $x \mapsto \|x\|$, addition $\mathcal{X} \times \mathcal{X} \to \mathcal{X}$, and scalar multiplication $\mathbb{F} \times \mathcal{X} \to \mathcal{X}$ are continuous.
(ii) If $M \le \mathcal{X}$, then $\overline{M} \le \mathcal{X}$; for $S \subseteq \mathcal{X}$, $\operatorname{cspan}(S)$ is the least closed subspace containing S.
(iii) Equivalent norms have the same convergent and Cauchy sequences, topology, and compact sets.

Proof.

(i) $|\|x\| - \|y\|| \le \|x - y\|$, so $x \mapsto \|x\|$ is 1-Lipschitz.
Also $\|(x + h) + (y + k) - (x + y)\| \le \|h\| + \|k\|$, so addition is continuous.
If $\lambda_n \to \lambda$ and $x_n \to x$, then $\|\lambda_n x_n - \lambda x\| \le |\lambda_n| \, \|x_n - x\| + |\lambda_n - \lambda| \, \|x\| \to 0$, since (λ_n) is bounded.
(ii) Let $x, y \in \overline{M}$ and choose $x_n, y_n \in M$ with $x_n \to x$, $y_n \to y$.
For $\alpha, \beta \in \mathbb{F}$, $\alpha x_n + \beta y_n \in M$ and $\alpha x_n + \beta y_n \to \alpha x + \beta y$ by (i); hence $\overline{M} \le \mathcal{X}$.
If C is closed, $C \le \mathcal{X}$, and $S \subseteq C$, then $\operatorname{span}(S) \subseteq C$, so $\operatorname{cspan}(S) = \overline{\operatorname{span}(S)} \subseteq C$.
(iii) If $c \|z\|_1 \le \|z\|_2 \le C \|z\|_1$ for all $z \in \mathcal{X}$, then $B_1(x, \frac{\varepsilon}{C}) \subseteq B_2(x, \varepsilon) \subseteq B_1(x, \frac{\varepsilon}{c})$. Thus the topologies agree, the identity maps are bi-Lipschitz, convergent and Cauchy sequences agree, and compact sets agree.

Structural Commentary: Norms turn the algebraic operations into continuous operations.
Key Insight: Closing a subspace preserves linearity because linear combinations commute with limits. □

Definition 5.1.4. *Bounded Linear Maps and Operator Norm*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces and let $T : \mathcal{X} \to \mathcal{Y}$ be linear.

(i) *Bounded*: $\exists C \ge 0$ such that $\forall x \in \mathcal{X}$, $\|Tx\|_{\mathcal{Y}} \le C \|x\|_{\mathcal{X}}$.
(ii) *Operator norm*: $\|T\|_{\mathrm{op}} := \sup_{\|x\|_{\mathcal{X}} \le 1} \|Tx\|_{\mathcal{Y}} = \sup_{x \ne 0} \frac{\|Tx\|_{\mathcal{Y}}}{\|x\|_{\mathcal{X}}}$.
(iii) *Operator space*: $\mathcal{L}(\mathcal{X}, \mathcal{Y}) := \{T : \mathcal{X} \to \mathcal{Y} : T \text{ is bounded and linear}\}$ and $\mathcal{L}(\mathcal{X}) := \mathcal{L}(\mathcal{X}, \mathcal{X})$.

Theorem 5.1.5. *Bounded Equals Continuous for Linear Maps*
For a linear map $T : \mathcal{X} \to \mathcal{Y}$ between normed spaces, the following are equivalent:

(i) T is bounded.
(ii) T is continuous at 0.
(iii) T is continuous everywhere.

Proof.

(i) $\Rightarrow$ *(iii)* If T is bounded, then $\|Tx - Ty\| \le \|T\|_{\mathrm{op}} \|x - y\|$ for $x, y \in \mathcal{X}$; hence T is Lipschitz.
(iii) $\Rightarrow$ *(ii)* Since $0 \in \mathcal{X}$, continuity on $\mathcal{X}$ implies continuity at 0.
(ii) $\Rightarrow$ *(i)* Choose $\delta > 0$ with $\|z\| < \delta \Rightarrow \|Tz\| < 1$. For $x \ne 0$, set $z := \frac{\delta x}{2\|x\|}$.
Then $\|z\| < \delta$ and $\|Tx\| = \frac{2\|x\|}{\delta} \|Tz\| \le \frac{2}{\delta} \|x\|$. Thus $\|T\|_{\mathrm{op}} \le \frac{2}{\delta}$.

Structural Commentary: Continuity at 0 rescales to a global operator bound.
Key Insight: Linear continuity is boundedness. □

Theorem 5.1.6. *Bounded Operators Form a Banach Space*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces.

(i) $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is normed by $\|\cdot\|_{op}$ under pointwise operations.
(ii) If $\mathcal{Y}$ is Banach, then $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is Banach.

Proof.

(i) If $S, T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ and $\alpha, \beta \in \mathbb{F}$, then $\|(\alpha S + \beta T)x\| \le (|\alpha| \, \|S\|_{\mathrm{op}} + |\beta| \, \|T\|_{\mathrm{op}}) \|x\|$, so $\alpha S + \beta T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$.
Also $\|\alpha T\|_{\mathrm{op}} = |\alpha| \, \|T\|_{\mathrm{op}}$, $\|S + T\|_{\mathrm{op}} \le \|S\|_{\mathrm{op}} + \|T\|_{\mathrm{op}}$, and $\|T\|_{\mathrm{op}} = 0 \Rightarrow Tx = 0$ for all $x \in \mathcal{X}$.

(ii) Assume $\mathcal{Y}$ is Banach and let (T_n) be Cauchy in $\|\cdot\|_{\mathrm{op}}$.
For each $x \in \mathcal{X}$, $\|T_n x - T_m x\| \le \|T_n - T_m\|_{\mathrm{op}} \|x\|$, so define $Tx := \lim_n T_n x$.
Then $T(\alpha x + \beta y) = \alpha Tx + \beta Ty$. Given $\varepsilon > 0$, choose N with $\|T_n - T_m\|_{\mathrm{op}} < \varepsilon$ for $m, n \ge N$.
Letting $m \to \infty$ gives $\|(T_n - T)x\| \le \varepsilon \|x\|$, so $\|T_n - T\|_{\mathrm{op}} \le \varepsilon$.
Hence $T_N - T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, and $T_n \to T$.

Structural Commentary: Pointwise limits are linear; operator-norm Cauchy control is uniform in x.
Key Insight: Completeness of $\mathcal{Y}$ lifts to $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. □

Corollary 5.1.7. *Neumann Series*
Let $\mathcal{X}$ be Banach and let $T \in \mathcal{L}(\mathcal{X})$ satisfy $\|T\|_{op} < 1$.

Then $I - T$ has bounded inverse, with Neumann series converging in $\mathcal{L}(\mathcal{X})$: $(I - T)^{-1} = \sum_{n=0}^{\infty} T^n$.

Proof. Let $S_N := \sum_{n=0}^{N} T^n$. For $M > N$, $\|S_M - S_N\|_{\mathrm{op}} \le \sum_{n=N+1}^{M} \|T\|_{\mathrm{op}}^n \le \frac{\|T\|_{\mathrm{op}}^{N+1}}{1 - \|T\|_{\mathrm{op}}} \to 0$, so $S_N \to S \in \mathcal{L}(\mathcal{X})$.
Since $(I - T)S_N = I - T^{N+1} = S_N(I - T)$ and $\|T^{N+1}\|_{\mathrm{op}} \le \|T\|_{\mathrm{op}}^{N+1} \to 0$.
Passing to the limit gives $(I - T)S = S(I - T) = I$.
Structural Commentary: Operator-norm convergence turns the geometric identity into an inverse.
Key Insight: If $\|T\|_{\mathrm{op}} < 1$, then $\sum T^n = (I - T)^{-1}$. □

Proposition 5.1.8. *Quotient Norm and Quotient Map*
Let $M \le \mathcal{X}$ be closed, let $\mathcal{X}/M := \{x + M : x \in \mathcal{X}\}$, and let $q : \mathcal{X} \to \mathcal{X}/M$ be $q(x) := x + M$.
Set $\|x + M\|_{\mathcal{X}/M} := \inf_{m \in M} \|x - m\|$.

(i) $\|\cdot\|_{\mathcal{X}/M}$ is a norm on $\mathcal{X}/M$.
(ii) q is continuous with $\|q\|_{op} \le 1$.
(iii) $\forall x \in \mathcal{X}$ and $r > 0$, $q(B_r(x)) = B_r(qx)$.
(iv) q is open.

Proof.
(i) If $x' = x + m_0$ with $m_0 \in M$, then $\inf_{m \in M} \|x' - m\| = \inf_{m \in M} \|x - m\|$, so the formula is well defined.
Also $\|x + M\| = 0 \iff x \in \overline{M} = M$. For $\alpha \neq 0$,
$\|\alpha(x + M)\| = \inf_{m \in M} \|\alpha x - m\| = |\alpha| \inf_{m \in M} \|x - \alpha^{-1} m\| = |\alpha| \|x + M\|$; $\alpha = 0$ is trivial.
For $m, n \in M$, $\|(x + y) - (m + n)\| \le \|x - m\| + \|y - n\|$; take infima.
(ii) Since $\|q(x)\| \le \|x - 0\| = \|x\|$, $q \in \mathcal{L}(\mathcal{X}, \mathcal{X}/M)$ and $\|q\|_{\mathrm{op}} \le 1$.
(iii) If $z \in B_r(x)$, then $\|qz - qx\| \le \|z - x\| < r$, so $q(B_r(x)) \subseteq B_r(qx)$. Conversely, if $\eta = qz$ and $\|\eta - qx\| < r$, choose $m \in M$ with $\|z - x - m\| < r$; then $z - m \in B_r(x)$ and $q(z - m) = \eta$.
(iv) If $U \subseteq \mathcal{X}$ is open and $y \in q(U)$, choose $x \in U$ with $qx = y$ and $r > 0$ with $B_r(x) \subseteq U$.
By (iii), $B_r(y) = q(B_r(x)) \subseteq q(U)$; hence $q(U)$ is open.

Structural Commentary: Quotient norm is distance to the closed subspace M.
Key Insight: Closedness makes zero distance equivalent to membership. □

Theorem 5.1.9. *Banach Quotients*
If M is a closed linear subspace of a Banach space $\mathcal{X}$, then $\mathcal{X}/M$ is Banach in the quotient norm.

Proof. Let (ξ_n) be Cauchy in $\mathcal{X}/M$. Choose $n_1 < n_2 < \cdots$ with $\|\xi_{n_{k+1}} - \xi_{n_k}\| < 2^{-k}$ and choose $u_k \in \mathcal{X}$ with $q(u_k) = \xi_{n_{k+1}} - \xi_{n_k}$, $\|u_k\| < 2^{-k}$. Since $\sum_k \|u_k\| < +\infty$ and $\mathcal{X}$ is Banach, $u := \sum_{k \ge 1} u_k$ exists.
For $K \ge 1$, $\xi_{n_K} = \xi_{n_1} + q(\sum_{k<K} u_k) \to \xi_{n_1} + q(u)$.
Since (ξ_n) is Cauchy and has a convergent subsequence, $\xi_n \to \xi_{n_1} + q(u)$.
Structural Commentary: Rapid quotient increments lift to an absolutely convergent series.
Key Insight: A Cauchy sequence with a convergent subsequence converges. □

Proposition 5.1.10. *Operator Factor through the Kernel Quotient*
Let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, set $M := \ker T$, and let $\widetilde{T} : \mathcal{X}/M \to \mathcal{Y}$ be induced by Theorem 2.2.12.

(i) $\widetilde{T}$ is bounded.
(ii) $\|\widetilde{T}\|_{op} = \|T\|_{op}$.

Proof. Since $q(x) = q(y) \iff x - y \in M = \ker T$, the formula $\widetilde{T}(qx) := Tx$ is well defined and $T = \widetilde{T}q$.
For $m \in M$, $\|\widetilde{T}(qx)\| = \|Tx\| = \|T(x - m)\| \le \|T\|_{\mathrm{op}} \|x - m\|$; taking $\inf_{m \in M}$ gives $\|\widetilde{T}(qx)\| \le \|T\|_{\mathrm{op}} \|qx\|$.

(i) The estimate gives $\widetilde{T} \in \mathcal{L}(\mathcal{X}/M, \mathcal{Y})$.
(ii) The estimate gives $\|\widetilde{T}\|_{\mathrm{op}} \le \|T\|_{\mathrm{op}}$. Since $T = \widetilde{T}q$ and $\|q\|_{\mathrm{op}} \le 1$, $\|T\|_{\mathrm{op}} \le \|\widetilde{T}\|_{\mathrm{op}} \|q\|_{\mathrm{op}} \le \|\widetilde{T}\|_{\mathrm{op}}$.

Structural Commentary: Collapsing $\ker T$ makes T descend without changing its norm.
Key Insight: The quotient infimum is taken over exactly the invisible directions of T. □

Example 5.1.11. *Standard Banach Space Examples*
Standard Banach spaces include the following canonical examples.

(i) If S is a set and $B(S) := \{f : S \to \mathbb{F} : \|f\|_\infty < +\infty\}$, then $(B(S), \|\cdot\|_\infty)$ is Banach.
(ii) $\ell^\infty := \{x = (x_n) : \sup_n |x_n| < +\infty\}$ is Banach under $\|x\|_\infty := \sup_n |x_n|$.

(iii) For $1 \le p < \infty$, $\ell^p := \{x = (x_n) : \sum_n |x_n|^p < +\infty\}$ is Banach under $\|x\|_p := (\sum_n |x_n|^p)^{1/p}$.
(iv) For a measure space $(X, \mathcal{M}, \mu)$ and $1 \le p \le \infty$, $L^p(\mu)$ is Banach.
(v) $L^2(\mu)$ is Hilbert with $\langle f, g\rangle := \int_X f\overline{g}\, d\mu$.

Example 5.1.12. *Completeness of ℓ^∞ and ℓ^p*
For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $(\ell^\infty(\mathbb{F}), \|\cdot\|_\infty)$ and $(\ell^p(\mathbb{F}), \|\cdot\|_p)$, $1 \le p < \infty$, are complete.

Proof. For ℓ^∞, let $\mathbf{x}^{(n)} = (x_j^{(n)})_j$ be Cauchy. Then $x_j^{(n)} \to y_j \in \mathbb{F}$ for each j.
If $\|\mathbf{x}^{(n)} - \mathbf{x}^{(m)}\|_\infty < \varepsilon$ for $m, n \ge N$, then $|x_j^{(n)} - y_j| \le \varepsilon$ for all j, so $\|\mathbf{x}^{(n)} - \mathbf{y}\|_\infty \le \varepsilon$.
Taking $n = N$ gives $\mathbf{y} \in \ell^\infty$, hence $\mathbf{x}^{(n)} \to \mathbf{y}$.
For ℓ^p, let $\mathbf{x}^{(n)}$ be Cauchy and define $\mathbf{y} = (y_j)$ by coordinatewise limits.
If $\|\mathbf{x}^{(n)} - \mathbf{x}^{(m)}\|_p < \varepsilon$ for $m, n \ge N$, then for each r, $\sum_{j=1}^r |x_j^{(n)} - y_j|^p = \lim_{m\to\infty} \sum_{j=1}^r |x_j^{(n)} - x_j^{(m)}|^p \le \varepsilon^p$.
Letting $r \to \infty$ gives $\|\mathbf{x}^{(n)} - \mathbf{y}\|_p \le \varepsilon$ for $n \ge N$. Taking $n = N$ gives $\mathbf{y} \in \ell^p$, hence $\mathbf{x}^{(n)} \to \mathbf{y}$.
Structural Commentary: Completeness of scalar coordinates combines with uniform tail control.
Key Insight: Norm-Cauchy control keeps coordinatewise limits inside the space. □

Example 5.1.13. *Two Metrics on $C[0,1]$*
For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $C_\mathbb{F} := C([0,1], \mathbb{F})$ is complete under $d_\infty(f,g) := \sup_{x\in[0,1]} |f(x) - g(x)|$ by Theorem 8.1.1, but not under $d_1(f,g) := \int_0^1 |f(x) - g(x)|\, dx$.

Proof. For $n \ge 2$, define $f_n \in C_\mathbb{F}$ by

$$f_n(x) := \begin{cases} 0, & 0 \le x \le \frac{1}{2}, \\ n(x - \frac{1}{2}), & \frac{1}{2} \le x \le \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \le x \le 1. \end{cases}$$

Let $s := \mathbf{1}_{(1/2,1]}$. Then $d_1(f_n, s) = 1/(2n)$, hence (f_n) is d_1-Cauchy. If $f_n \to g \in C_\mathbb{F}$ in d_1, then $d_1(g, s) = 0$, so $g = s$ a.e. Continuity forces $g = 0$ on $[0, 1/2]$ and $g = 1$ on $(1/2, 1]$, impossible at $1/2$.
Structural Commentary: The integral metric permits a continuous ramp to collapse to a jump.
Key Insight: Supremum control preserves continuity; integral control can miss jumps. □

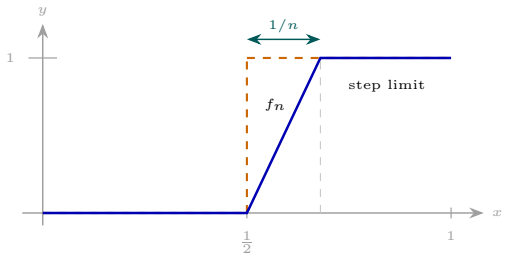

Figure 219: The continuous ramps f_n converge in the integral metric to a discontinuous step function.

Theorem 5.1.14. *Finite-Dimensional Normed Spaces*
Let $\mathcal{X}$ be a finite-dimensional normed space.
(i) $\mathcal{X}$ is Banach.
(ii) Every subspace of $\mathcal{X}$ is closed.
(iii) $\overline{B}_\mathcal{X}$ is compact.

Proof. Choose a basis $e_1, \dots, e_n$ and set $A : \mathbb{F}^n \to \mathcal{X}$, $Aa := \sum_{i=1}^n a_i e_i$.
Then A is linear bijective and $\|Aa\| \le \sum_i |a_i|\, \|e_i\| \le C\, \|a\|_2$.
Since $a \mapsto \|Aa\|$ is continuous, positive on $\{a : \|a\|_2 = 1\}$, $\exists c > 0$ with $c\, \|a\|_2 \le \|Aa\| \le C\, \|a\|_2$ $\forall a \in \mathbb{F}^n$.
(i) If (x_k) is Cauchy in $\mathcal{X}$, then $a_k := A^{-1}x_k$ is Cauchy in $\mathbb{F}^n$ by $c\, \|a_k - a_\ell\|_2 \le \|x_k - x_\ell\|$.
Since $\mathbb{F}^n$ is complete, $a_k \to a$, hence $x_k \to Aa$.
(ii) If $M \le \mathcal{X}$, then M is finite-dimensional, hence Banach by (i). If $x_k \in M$ and $x_k \to x$ in $\mathcal{X}$, then (x_k) converges in M to some $y \in M$; uniqueness of limits gives $x = y \in M$. Thus M is closed.
(iii) Since $\overline{B}_\mathcal{X} \subseteq A(\overline{B}_2(0, \frac{1}{c}))$ and $A^{-1}(\overline{B}_\mathcal{X})$ is closed in the compact Euclidean ball $\overline{B}_2(0, \frac{1}{c})$, $A^{-1}(\overline{B}_\mathcal{X})$ is compact; continuity of A gives compactness of $\overline{B}_\mathcal{X}$.
Structural Commentary: The coordinate map is a bi-Lipschitz linear isomorphism.
Key Insight: Finite-dimensional normed spaces inherit completeness and compactness from $\mathbb{F}^n$. □

Theorem 5.1.15. *Riesz Lemma*
Let M be a closed proper linear subspace of a normed space $\mathcal{X}$.
Then $\forall \varepsilon \in (0,1)$ $\exists u \in \mathcal{X}$ such that $\|u\| = 1$ and $\forall m \in M$, $\|u - m\| > 1 - \varepsilon$.

Proof. Choose $x \in \mathcal{X} \setminus M$ and set $d := \inf_{m\in M} \|x - m\|$. Since M is closed, $d > 0$.
Choose $m_0 \in M$ with $d \le \|x - m_0\| < \frac{d}{1-\varepsilon}$ and set $u := \frac{x - m_0}{\|x - m_0\|}$.
Then $\|u\| = 1$. For $m \in M$, $m_0 + \|x - m_0\|\, m \in M$, so $\|u - m\| = \frac{\|x - (m_0 + \|x - m_0\| m)\|}{\|x - m_0\|} \ge \frac{d}{\|x - m_0\|} > 1 - \varepsilon$.

Structural Commentary: A positive distance to M survives normalization.
Key Insight: Normalize an almost-minimizer of $\operatorname{dist}(x, M)$. □

Corollary 5.1.16. *Compact Unit Balls Characterize Finite Dimension*
For a normed space $\mathcal{X}$, $\overline{B}_{\mathcal{X}}$ is compact iff $\dim \mathcal{X} < \infty$.

Proof.
($\Leftarrow$) If $\dim \mathcal{X} < \infty$, then $\overline{B}_{\mathcal{X}}$ is compact by Theorem 5.1.14(iii).
($\Rightarrow$) Suppose $\dim \mathcal{X} = \infty$. Choose $x_1 \in \mathcal{X}$ with $\|x_1\| = 1$. Given $x_1, \dots, x_n$, set $M_n := \operatorname{span}\{x_1, \dots, x_n\}$.
Then M_n is finite-dimensional, closed, and proper, so Theorem 5.1.15 with $\varepsilon = \frac{1}{2}$ gives $x_{n+1} \in \mathcal{X}$ with $\|x_{n+1}\| = 1$ and $\operatorname{dist}(x_{n+1}, M_n) > \frac{1}{2}$. Since $x_i \in M_n$ for $i \le n$, $\|x_{n+1} - x_i\| > \frac{1}{2}$.
Thus $(x_n) \subseteq \overline{B}_{\mathcal{X}}$ has no Cauchy subsequence, so $\overline{B}_{\mathcal{X}}$ is not compact.
Structural Commentary: Riesz iteration builds a separated sequence in $\overline{B}_{\mathcal{X}}$.
Key Insight: Compact unit ball forces finite dimension. □

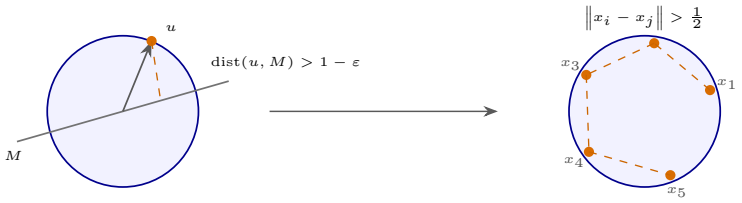

Figure 220: Riesz iteration: $M_n = \operatorname{span}\{x_1, \dots, x_n\}$, $\operatorname{dist}(x_{n+1}, M_n) > \frac{1}{2} \Rightarrow \|x_i - x_j\| > \frac{1}{2}$ $(i \neq j)$; hence no Cauchy subsequence in $\overline{B}_{\mathcal{X}}$.

5.2 Hilbert Spaces

Hilbert spaces are complete inner-product spaces. They extend the orthogonal geometry of Definition 1.1.3 to function and sequence spaces, supporting projection, Fourier expansion, and Riesz representation.

Definition 5.2.1. *Hilbert Space*
Let $\mathcal{H}$ be an inner-product space.
(i) *Hilbert space*: $\mathcal{H}$ is complete in the norm $\|x\| := \sqrt{\langle x, x \rangle}$.
(ii) *Orthonormal family*: $(e_i)_{i \in I} \subseteq \mathcal{H}$ satisfies $\forall i, j \in I$, $\langle e_i, e_j \rangle = \delta_{ij}$.

Proposition 5.2.2. *Continuity of the Inner Product*
Let $\mathcal{V}$ be an inner-product space, let $(x_n), (y_n) \subseteq \mathcal{V}$, and let $x, y \in \mathcal{V}$.
(i) $x_n \to x \Longrightarrow \|x_n\| \to \|x\|$.
(ii) $x_n \to x$ and $y_n \to y \Longrightarrow \langle x_n, y_n \rangle \to \langle x, y \rangle$.

Proof.
(i) $|\|x_n\| - \|x\|| \le \|x_n - x\| \to 0$, hence $\|x_n\| \to \|x\|$.
(ii) Since (y_n) is bounded, choose $M < \infty$ with $\|y_n\| \le M$.
Cauchy–Schwarz gives $\left|\langle x_n, y_n \rangle - \langle x, y \rangle\right| \le M \|x_n - x\| + \|x\| \|y_n - y\| \to 0$.
Structural Commentary: Cauchy–Schwarz turns norm convergence in either coordinate into scalar convergence.
Key Insight: Inner products are continuous because their errors are controlled by products of norms. □

Theorem 5.2.3. *Hilbert Completion*
Let $\mathcal{V}$ be an inner-product space.
(i) $\exists (\widehat{\mathcal{V}}, J)$ such that $\widehat{\mathcal{V}}$ is Hilbert, $J : \mathcal{V} \to \widehat{\mathcal{V}}$ is a linear isometry, and $\overline{J(\mathcal{V})} = \widehat{\mathcal{V}}$.
(ii) If $(\mathcal{K}, I)$ also satisfies (i), then $\exists! U : \widehat{\mathcal{V}} \to \mathcal{K}$ a surjective linear isometry with $U \circ J = I$.

Proof.
(i) Let $C(\mathcal{V})$ be the Cauchy sequences, set $(x_n) \sim (y_n) \iff \|x_n - y_n\| \to 0$, and define $\widehat{\mathcal{V}} := C(\mathcal{V})/\sim$.
Set $[x_n] + [y_n] := [x_n + y_n]$, $a[x_n] := [ax_n]$, $Jx := [(x)]$, and $\langle [x_n], [y_n] \rangle_{\widehat{\mathcal{V}}} := \lim_{n \to \infty} \langle x_n, y_n \rangle_{\mathcal{V}}$.
If $(x_n), (y_n)$ are bounded by M, then $\left|\langle x_n, y_n \rangle - \langle x_m, y_m \rangle\right| \le M \|y_n - y_m\| + M \|x_n - x_m\| \to 0$.
The same estimate for equivalent representatives makes the limit well defined.
Also $\|[x_n] - [y_n]\| = \lim_n \|x_n - y_n\|$, the metric-completion norm; hence $\widehat{\mathcal{V}}$ is Hilbert and $\overline{J(\mathcal{V})} = \widehat{\mathcal{V}}$.
(ii) Since $(\widehat{\mathcal{V}}, J)$ and $(\mathcal{K}, I)$ are metric completions of $\mathcal{V}$, there is a unique surjective isometry $U : \widehat{\mathcal{V}} \to \mathcal{K}$ with $U \circ J = I$. For $x, y \in \mathcal{V}$ and $a, b \in \mathbb{F}$, $U(aJx + bJy) = U(J(ax + by)) = I(ax + by) = aUJx + bUJy$; by density and continuity, U is linear. Since $\|Uz - Uw\| = \|z - w\|$, polarization gives $\langle Uz, Uw \rangle_{\mathcal{K}} = \langle z, w \rangle_{\widehat{\mathcal{V}}}$.
Structural Commentary: Complete the metric first, then recover the inner product by taking limits.
Key Insight: Cauchy–Schwarz makes the lifted inner product well defined. □

Theorem 5.2.4. *Complete, Closed, and Finite-Dimensional Subspaces*
Let $\mathcal{S}$ be a subspace of an inner-product space $\mathcal{V}$.
(i) If $\mathcal{S}$ is complete in the inherited metric, then $\mathcal{S}$ is closed in $\mathcal{V}$.
(ii) If $\mathcal{V}$ is Hilbert, then $\mathcal{S}$ is Hilbert if and only if $\mathcal{S}$ is closed.
(iii) If $\mathcal{S}$ is finite-dimensional, then $\mathcal{S}$ is complete and closed in $\mathcal{V}$.

Proof.

(i) Let $v \in \overline{\mathcal{S}}$ and choose $s_n \in \mathcal{S}$ with $s_n \to v$ in $\mathcal{V}$. Then (s_n) is Cauchy in $\mathcal{S}$, so $s_n \to s \in \mathcal{S}$ by completeness. Uniqueness of limits gives $v = s$, hence $\overline{\mathcal{S}} = \mathcal{S}$.

(ii) Assume $\mathcal{V}$ Hilbert. If $\mathcal{S}$ is closed and $(s_n) \subseteq \mathcal{S}$ is Cauchy, then $s_n \to v \in \mathcal{V}$ and $v \in \mathcal{S}$, so $\mathcal{S}$ is complete. Conversely, if $\mathcal{S}$ is Hilbert, then $\mathcal{S}$ is complete, hence closed by (i).

(iii) Let $\dim \mathcal{S} = m$ and choose an orthonormal basis $e_1, \dots, e_m$.

If (s_n) is Cauchy in $\mathcal{S}$ and $s_n = \sum_{j=1}^{m} a_{n,j} e_j$, then $|a_{n,j} - a_{k,j}| = |\langle s_n - s_k, e_j \rangle| \le \|s_n - s_k\|$.

Thus $a_{n,j} \to a_j \in \mathbb{F}$ for each j, and $s_n \to s := \sum_{j=1}^{m} a_j e_j \in \mathcal{S}$. Hence $\mathcal{S}$ is complete, so closed by (i).

Structural Commentary: In a complete ambient space, closed subspaces are exactly complete subspaces.
Key Insight: Finite orthonormal coordinates reduce completeness to $\mathbb{F}^m$. □

Definition 5.2.5. *Series in an Inner-Product Space*
Let $(x_k) \subseteq \mathcal{V}$.

(i) The *partial sums* are $s_n := \sum_{k=1}^{n} x_k$.

(ii) The *series* $\sum_{k=1}^{\infty} x_k$ *converges* to $s \in \mathcal{V}$ iff $s_n \to s$.

(iii) The series is *absolutely convergent* iff $\sum_{k=1}^{\infty} \|x_k\| < \infty$.

Theorem 5.2.6. *Completeness via Absolutely Convergent Series*
$\mathcal{V}$ is complete $\iff$ every series $\sum_{k=1}^{\infty} x_k$ in $\mathcal{V}$ with $\sum_{k=1}^{\infty} \|x_k\| < \infty$ converges in $\mathcal{V}$.

Proof.

($\Rightarrow$) Let $s_n := \sum_{k=1}^{n} x_k$ and assume $\sum_k \|x_k\| < +\infty$. For $n > m$, $\|s_n - s_m\| \le \sum_{k=m+1}^{n} \|x_k\| \to 0$ as $m, n \to \infty$. Thus (s_n) is Cauchy, hence convergent by completeness.

($\Leftarrow$) Let (x_n) be Cauchy. Let $N_1 < N_2 < \cdots$ such that $i, j \ge N_k \Rightarrow \|x_i - x_j\| < 2^{-k}$, $y_k := x_{N_{k+1}} - x_{N_k}$. Then $\sum_k \|y_k\| \le \sum_k 2^{-k} < +\infty$, so $\sum_k y_k$ converges. Since $x_{N_m} = x_{N_1} + \sum_{k<m} y_k$, the subsequence (x_{N_m}) converges; because (x_n) is Cauchy, (x_n) converges.

Structural Commentary: Absolute tails make partial sums Cauchy; Cauchy tails contain summable increments.
Key Insight: Completeness is summability of norm-controlled series. □

Example 5.2.7. *ℓ^2 and the Parallelogram Test*
Let $\ell^2 := \{x = (x_n) : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$. With $\langle x, y \rangle := \sum_{n=1}^{\infty} x_n \overline{y_n}$ and $\|x\|_2 = \langle x, x \rangle^{\frac{1}{2}}$, Cauchy coordinate limits and finite-set estimates give completeness, so ℓ^2 is Hilbert.
For $p \neq 2$, $\|\cdot\|_p$ is not inner-product-induced: with $x = (1, 1, 0, \dots)$ and $y = (1, -1, 0, \dots)$,

$\|x + y\|_p^2 + \|x - y\|_p^2 = 8$ while $2\|x\|_p^2 + 2\|y\|_p^2 = 4 \cdot 2^{\frac{2}{p}}$, so the parallelogram identity holds only for $p = 2$.
Structural Commentary: Completeness alone does not make a Banach space Hilbert.
Key Insight: The parallelogram identity detects whether a norm really came from an inner product.

Example 5.2.8. *The Supremum Norm Is Not Hilbertian*
For $a < b$, $C([a, b], \mathbb{F})$ with $\|h\|_\infty := \sup_{t \in [a,b]} |h(t)|$ is Banach but not Hilbertian: take $f(t) = 1$ and $g(t) = \frac{t-a}{b-a}$, so $\|f\|_\infty = \|g\|_\infty = 1$, $\|f + g\|_\infty = 2$, $\|f - g\|_\infty = 1$, and $\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = 5 \neq 4 = 2\|f\|_\infty^2 + 2\|g\|_\infty^2$, violating the parallelogram identity.
Structural Commentary: Banach need not mean Hilbert.
Key Insight: One failed parallelogram identity rules out every possible inner product inducing the norm.

Theorem 5.2.9. *Hilbert Projection Theorem*
Let $C \subseteq \mathcal{H}$ be nonempty, closed, and convex. Then $\forall x \in \mathcal{H}$ $\exists! p \in C$ such that $\|x - p\| = \inf_{q \in C} \|x - q\|$.

Proof. Let $d := \inf_{q \in C} \|x - q\|$, $q_n \in C$, $\|x - q_n\| \to d$. Since $\frac{q_n + q_m}{2} \in C$, by parallelogram identity,

$\|q_n - q_m\|^2 = 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4\left\|x - \frac{q_n + q_m}{2}\right\|^2 \le 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4d^2 \to 0.$

Thus (q_n) is Cauchy. Hence $q_n \to p \in \mathcal{H}$; closedness gives $p \in C$, and $\|x - p\| = d$.
If $p_1, p_2 \in C$ both minimize, the same identity gives $\|p_1 - p_2\|^2 \le 2d^2 + 2d^2 - 4d^2 = 0$.
Structural Commentary: Convexity and parallelogram force minimizing sequences to converge in C.
Key Insight: Midpoint minimality turns near minimizers into a Cauchy sequence. □

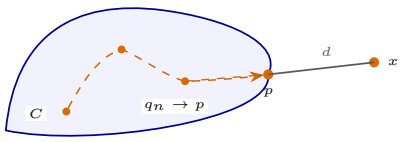

Figure 221: Hilbert projection: $q_n \in C$, $\|x - q_n\| \to d = \operatorname{dist}(x, C) \Rightarrow q_n \to p \in C$ and $\|x - p\| = d$.

Corollary 5.2.10. *Orthogonal Projection onto a Closed Subspace*
Let $M \leq \mathcal{H}$ be closed.

(i) $\forall x \in \mathcal{H}\ \exists! p \in M$ *such that* $x - p \in M^\perp$.
(ii) $\mathcal{H} = M \odot M^\perp$.

Proof.

(i) By Theorem 5.2.9, choose the unique $p \in M$ minimizing $\|x - p\|$, and set $r := x - p$.
For $y \in M$ and $t \in \mathbb{R}$, $p + ty \in M$, so $0 \leq \|r - ty\|^2 - \|r\|^2 = -2t \operatorname{Re}\langle r, y\rangle + t^2 \|y\|^2$.
Thus $\operatorname{Re}\langle r, y\rangle = 0$; over $\mathbb{C}$, replace y by iy to get $\operatorname{Im}\langle r, y\rangle = 0$. Hence $r \in M^\perp$.
If $x = p_1 + r_1 = p_2 + r_2$ with $p_i \in M$, $r_i \in M^\perp$, then $p_1 - p_2 = r_2 - r_1 \in M \cap M^\perp = \{0\}$.
(ii) By (i), each $x \in \mathcal{H}$ has $x = p + r$ with $p \in M$, $r \in M^\perp$, so $\mathcal{H} = M + M^\perp$.
Since $M \cap M^\perp = \{0\}$, the sum is direct: $\mathcal{H} = M \odot M^\perp$.

Structural Commentary: Nearest points are characterized by orthogonal residuals.
Key Insight: Every vector splits uniquely into M and $M^\perp$ parts. □

Definition 5.2.11. *Closed Linear Span and Total Sets*
For $A \subseteq \mathcal{H}$:

(i) *Closed linear span*: $\operatorname{cspan}(A) := \overline{\operatorname{span}(A)}$.
(ii) *Total*: A is total in $\mathcal{H}$ iff $\operatorname{cspan}(A) = \mathcal{H}$.

Theorem 5.2.12. *Double Orthogonal Complements in Hilbert Spaces*
Let $\mathcal{H}$ be a Hilbert space.

(i) $\forall A \subseteq \mathcal{H}$, $\operatorname{cspan}(A) = A^{\perp\perp}$.
(ii) $\forall \mathcal{S} \leq \mathcal{H}$, $\overline{\mathcal{S}} = \mathcal{S}^{\perp\perp}$.
(iii) $\forall \mathcal{K} \leq \mathcal{H}$, $\mathcal{K}$ *closed* $\iff \mathcal{K} = \mathcal{K}^{\perp\perp}$.
(iv) $\forall A \subseteq \mathcal{H}$, $\operatorname{span}(A)$ *dense in* $\mathcal{H} \iff A^\perp = \{\mathbf{0}\}$.

Proof.

(i) Let $B := \operatorname{cspan}(A)$. Since $A \subseteq B$, $B^\perp \subseteq A^\perp$. If $y \in A^\perp$, by $y \perp \operatorname{span}(A)$, continuity of $z \mapsto \langle z, y\rangle$, $y \perp B$, so $A^\perp = B^\perp$. By Corollary 5.2.10, $\mathcal{H} = B \odot B^\perp$; $B^{\perp\perp} = B$. Therefore $A^{\perp\perp} = B^{\perp\perp} = B = \operatorname{cspan}(A)$.
(ii) With $A = \mathcal{S}$ in (i), $\mathcal{S}^{\perp\perp} = \operatorname{cspan}(\mathcal{S}) = \overline{\operatorname{span}(\mathcal{S})} = \overline{\mathcal{S}}$.
(iii) By (ii), $\mathcal{K}^{\perp\perp} = \overline{\mathcal{K}}$; thus $\mathcal{K} = \mathcal{K}^{\perp\perp} \iff \mathcal{K} = \overline{\mathcal{K}} \iff \mathcal{K}$ is closed.
(iv) By (i), $\operatorname{span}(A)$ is dense $\iff A^{\perp\perp} = \mathcal{H}$. If $A^\perp = \{\mathbf{0}\}$, then $A^{\perp\perp} = \{\mathbf{0}\}^\perp = \mathcal{H}$.
Conversely, if $A^{\perp\perp} = \mathcal{H}$ and $y \in A^\perp$, then $y \in A^{\perp\perp}$, so $\|y\|^2 = \langle y, y\rangle = 0$; hence $A^\perp = \{\mathbf{0}\}$.

Structural Commentary: Double orthogonal complement equals closed linear span.
Key Insight: Missing closed-span directions are exactly nonzero orthogonal vectors. □

Example 5.2.13. *Completeness Is Needed for Double Complements*
Let $\mathcal{V} := c_{00}(\mathbb{N}; \mathbb{R}) \subseteq \ell^2$ with inherited inner product, $Lx := \sum_{n=1}^{\infty} \frac{x_n}{n}$, and $\mathcal{K} := \ker L$.

Then $|Lx| \leq \|x\|_2 \left(\sum_{n=1}^{\infty} n^{-2}\right)^{\frac{1}{2}}$, so L is continuous and $\mathcal{K}$ is a closed subspace of $\mathcal{V}$.

If $y \in \mathcal{K}^\perp$, then $e_1 - ne_n \in \mathcal{K}$ for $n \geq 2$, hence $0 = \langle y, e_1 - ne_n\rangle = y_1 - ny_n$.
Since $y \in c_{00}$, choose $n \geq 2$ with $y_n = 0$, so $y_1 = 0$ and then $\forall n,\ y_n = 0$.
Thus $\mathcal{K}^\perp = \{\mathbf{0}\}$ and $\mathcal{K}^{\perp\perp} = \mathcal{V}$. But $e_1 \notin \mathcal{K}$, hence $\mathcal{K} \neq \mathcal{V} = \mathcal{K}^{\perp\perp}$.
Structural Commentary: Closedness need not recover double complements without completeness.
Key Insight: The identity $\mathcal{K} = \mathcal{K}^{\perp\perp}$ is genuinely Hilbert-space geometry.

Perspective 5.2.14. *Hilbert and Hamel Basis Caution*
In Hilbert-space theory, an *orthonormal basis* usually means a complete orthonormal family: vectors are recovered by convergent Fourier expansion, not necessarily by finite linear combinations.
A *Hamel basis* is a vector-space basis and is much stronger in infinite dimension.
For example, the standard $(\mathbf{e}_j)$ in $\ell^2(\mathbb{F})$ is a complete orthonormal family, but $\operatorname{span}\{\mathbf{e}_j : j \geq 1\} = \mathbb{F}^{(\mathbb{N})} \subsetneq \ell^2(\mathbb{F})$.

Definition 5.2.15. *Hilbert Basis*

(i) A *Hilbert basis*, or complete orthonormal family, is a maximal orthonormal set under inclusion.
(ii) A *Hamel basis* is a vector-space basis.
(iii) Here *orthonormal basis* means Hilbert basis outside finite-dimensional Hamel context.

Proposition 5.2.16. *Finite Hilbert Bases Are Hamel Bases*
Let $\mathcal{O} = \{\mathbf{e}_1, \ldots, \mathbf{e}_m\}$ be orthonormal in $\mathcal{V}$ and set $W = \operatorname{span}(\mathcal{O})$.
Then $\mathcal{O}$ is a Hilbert basis $\iff W = \mathcal{V} \iff \mathcal{O}$ is an orthonormal Hamel basis.

Proof. Since $\mathcal{O}$ is orthonormal, it is linearly independent; hence $W = \mathcal{V} \iff \mathcal{O}$ is an orthonormal Hamel basis.
It remains to show maximality $\iff W = \mathcal{V}$.

($\Rightarrow$) Suppose $\mathcal{O}$ is maximal orthonormal and $W \neq \mathcal{V}$. Choose $\mathbf{v} \notin W$ and set $\mathbf{u} := \mathbf{v} - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j\rangle\, \mathbf{e}_j$.

Then $\langle \mathbf{u}, \mathbf{e}_i\rangle = 0$ for $1 \leq i \leq m$, and $\mathbf{u} \neq \mathbf{0}$ since $\mathbf{v} \notin W$.
Thus $\mathcal{O} \cup \{\mathbf{u}/\|\mathbf{u}\|\}$ is orthonormal, contradiction. Hence $W = \mathcal{V}$.

($\Leftarrow$) Suppose $W = \mathcal{V}$. If $\mathbf{w} \perp \mathcal{O}$, write $\mathbf{w} = \sum_{j=1}^{m} a_j \mathbf{e}_j$. Then $a_j = \langle \mathbf{w}, \mathbf{e}_j\rangle = 0$ for every j, so $\mathbf{w} = \mathbf{0}$.

Hence no nonzero vector can be adjoined orthogonally to $\mathcal{O}$; $\mathcal{O}$ is maximal orthonormal.

Structural Commentary: In finite orthonormal families, maximality equals spanning.
Key Insight: Any missing vector has a nonzero orthogonal residual. □

Example 5.2.17. *A Hilbert Basis Need Not Be a Hamel Basis*
In $\ell^2(\mathbb{F})$, let $\mathbf{e}_j = (0,\dots,0,1,0,\dots)$. Then $\{\mathbf{e}_j : j \geq 1\}$ is orthonormal and maximal, since $\forall j,\ \mathbf{x} \perp \mathbf{e}_j$ implies $\forall j,\ x_j = \langle \mathbf{x}, \mathbf{e}_j \rangle = 0$. Its finite span is $\mathbb{F}^{(\mathbb{N})} \subsetneq \ell^2(\mathbb{F})$, so it is a Hilbert basis but not a Hamel basis.

Theorem 5.2.18. *Hilbert Basis Characterizations*
Let $\mathcal{O} \subseteq \mathcal{H}$ be orthonormal. The following are equivalent:

(i) $\mathcal{O}$ *is maximal orthonormal.*
(ii) $\mathcal{O}^\perp = \{\mathbf{0}\}$.
(iii) $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$.

Proof.

(i) $\Rightarrow$ (ii) If $0 \neq v \in \mathcal{O}^\perp$, then $\mathcal{O} \cup \{v/\|v\|\}$ is orthonormal, contradicting maximality. Hence $\mathcal{O}^\perp = \{\mathbf{0}\}$.
(ii) $\Rightarrow$ (iii) By Theorem 5.2.12, $\operatorname{cspan}(\mathcal{O}) = \mathcal{O}^{\perp\perp} = (\mathcal{O}^\perp)^\perp = \{\mathbf{0}\}^\perp = \mathcal{H}$.
(iii) $\Rightarrow$ (i) If $\mathcal{O} \subsetneq \mathcal{O} \cup \{u\}$ is orthonormal with $\|u\| = 1$, then $u \in \mathcal{O}^\perp$.
By (iii) and Theorem 5.2.12(iv), $\mathcal{O}^\perp = \{\mathbf{0}\}$, contradiction. Thus $\mathcal{O}$ is maximal orthonormal.

Structural Commentary: Maximality, trivial orthogonal complement, and closed span are equivalent.
Key Insight: A Hilbert basis is total, not necessarily Hamel. □

Definition 5.2.19. *Arbitrary Sums in a Hilbert Space*
Let $(x_k)_{k\in K} \subseteq \mathcal{H}$.

(i) *Finite partial sums*: $\mathcal{F}(K) := \{J \subseteq K : |J| < \infty\}$, ordered by $\subseteq$, and $s_J := \sum_{k\in J} x_k$.
(ii) *Arbitrary convergence*: $\sum_{k\in K} x_k = x \iff \forall \varepsilon > 0\, \exists S \in \mathcal{F}(K)\, \forall T \in \mathcal{F}(K) : S \subseteq T \Rightarrow \|s_T - x\| \leq \varepsilon$.
(iii) *Nonnegative scalar sum*: for $(a_k)_{k\in K} \subseteq [0,\infty)$, $\sum_{k\in K} a_k < \infty \iff \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} a_k < \infty$.

Proposition 5.2.20. *Basic Properties of Arbitrary Sums*
Let $(x_k)_{k\in K}, (y_k)_{k\in K} \subseteq \mathcal{H}$ satisfy $\sum_{k\in K} x_k = x$ and $\sum_{k\in K} y_k = y$.

(i) $\forall a, b \in \mathbb{F},\ \sum_{k\in K} (ax_k + by_k) = ax + by$.
(ii) $\forall z \in \mathcal{H},\ \sum_{k\in K} \langle x_k, z\rangle = \langle x, z\rangle$.
(iii) $\forall z \in \mathcal{H},\ \sum_{k\in K} \langle z, x_k\rangle = \langle z, x\rangle$.

Proof. For $J \in \mathcal{F}(K)$, set $s_J := \sum_{k\in J} x_k$ and $t_J := \sum_{k\in J} y_k$.

(i) Since $s_J \to x$ and $t_J \to y$, $as_J + bt_J \to ax + by$.
(ii) For $z \in \mathcal{H}$ and $J \in \mathcal{F}(K)$, $\sum_{k\in J} \langle x_k, z\rangle = \langle s_J, z\rangle$; since $s_J \to x$, Proposition 5.2.2 gives $\langle s_J, z\rangle \to \langle x, z\rangle$.
(iii) For $z \in \mathcal{H}$ and $J \in \mathcal{F}(K)$, $\sum_{k\in J} \langle z, x_k\rangle = \langle z, s_J\rangle$; since $s_J \to x$, Proposition 5.2.2 gives $\langle z, s_J\rangle \to \langle z, x\rangle$.

Structural Commentary: Arbitrary sums are ordinary finite sums plus a convergence net.
Key Insight: Once the vector sum converges, continuous linear tests may pass through the sum. □

Theorem 5.2.21. *Arbitrary Orthonormal Sums*
Let $(u_k)_{k\in K} \subseteq \mathcal{H}$ be orthonormal.

(i) For $(r_k)_{k\in K} \subseteq \mathbb{F}$, $\sum_{k\in K} r_k u_k$ converges iff $\sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 < \infty$.
(ii) Under the conditions in (i), $\left\| \sum_{k\in K} r_k u_k \right\|^2 = \sum_{k\in K} |r_k|^2$.
(iii) $\forall (x_k)_{k\in K} \subseteq \mathcal{H}$, $\sum_{k\in K} x_k$ convergent implies $|\{k \in K : x_k \neq \mathbf{0}\}| \leq \aleph_0$.

Proof.

(i) ($\Rightarrow$) Let $s_J := \sum_{k\in J} r_k u_k$. If $s_J \to x$, choose $S \in \mathcal{F}(K)$ with $S \subseteq T \Rightarrow \|s_T - x\| \leq 1$.
For $J \in \mathcal{F}(K)$, $\sum_{k\in J} |r_k|^2 \leq \|s_{J\cup S}\|^2 \leq (\|x\| + 1)^2$.
(i) ($\Leftarrow$) Let $M := \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 < +\infty$. Given $\varepsilon > 0$, choose $S \in \mathcal{F}(K)$ with $M - \sum_{k\in S} |r_k|^2 < \varepsilon^2$.
If $S \subseteq J, L$, then $\|s_J - s_L\|^2 = \sum_{k\in J\triangle L} |r_k|^2 \leq \sum_{k\in (J\cup L)\setminus S} |r_k|^2 < \varepsilon^2$.
Thus (s_J) is Cauchy, hence converges in $\mathcal{H}$.
(ii) If $s_J \to s := \sum_{k\in K} r_k u_k$, then $\|s\|^2 = \lim_J \|s_J\|^2 = \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 = \sum_{k\in K} |r_k|^2$.
(iii) If $s_J := \sum_{k\in J} x_k$ converges, then (s_J) is Cauchy.
For each $n \geq 1$, choose $I_n \in \mathcal{F}(K)$ such that $I_n \subseteq J, L \Rightarrow \|s_J - s_L\| \leq \frac{1}{n}$.
Then $F \in \mathcal{F}(K \setminus I_n)$ implies $\left\| \sum_{k\in F} x_k \right\| = \|s_{I_n\cup F} - s_{I_n}\| \leq \frac{1}{n}$.
With $I := \bigcup_{n\geq 1} I_n$, I is countable; if $k \notin I$, then $\|x_k\| \leq \frac{1}{n}$ for all n, so $x_k = \mathbf{0}$.

Structural Commentary: Orthonormal finite sums turn net tails into square-sum tails.
Key Insight: Convergent arbitrary sums have countable support. □

Theorem 5.2.22. *Fourier Expansion in a Hilbert Space*
Let $\mathcal{O} = (u_k)_{k\in K} \subseteq \mathcal{H}$ be orthonormal. For $x \in \mathcal{H}$, set $\widehat{x}_{\mathcal{O}} := \sum_{k\in K} \langle x, u_k\rangle\, u_k$.

(i) $\widehat{x}_{\mathcal{O}}$ converges in $\mathcal{H}$.
(ii) $\|\widehat{x}_{\mathcal{O}}\|^2 = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2 \le \|x\|^2$.
(iii) $\widehat{x}_{\mathcal{O}}$ is the unique minimizer of $\|x - y\|$ over $y \in \operatorname{cspan}(\mathcal{O})$.
(iv) $\|\widehat{x}_{\mathcal{O}}\| = \|x\| \iff x \in \operatorname{cspan}(\mathcal{O})$.

Proof.
(i) For $J \in \mathcal{F}(K)$, finite Bessel gives $\sum_{k\in J} \left|\langle x, u_k\rangle\right|^2 \le \|x\|^2$.
Hence $\sup_J \sum_{k\in J} \left|\langle x, u_k\rangle\right|^2 < +\infty$, so $\sum_{k\in K} \langle x, u_k\rangle\, u_k$ converges by Theorem 5.2.21(i).
(ii) Theorem 5.2.21(ii) gives $\|\widehat{x}_{\mathcal{O}}\|^2 = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2$; finite Bessel gives the inequality $\sum_{k\in K} \left|\langle x, u_k\rangle\right|^2 \le \|x\|^2$.
(iii) For $i \in K$, Proposition 5.2.20 gives $\langle \widehat{x}_{\mathcal{O}}, u_i\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \langle u_k, u_i\rangle = \langle x, u_i\rangle$, so $r := x - \widehat{x}_{\mathcal{O}} \perp \mathcal{O}$.
By continuity, $r \perp M := \operatorname{cspan}(\mathcal{O})$. For $y \in M$, $x - y = r + (\widehat{x}_{\mathcal{O}} - y)$ orthogonally,
so $\|x - y\|^2 = \|r\|^2 + \|\widehat{x}_{\mathcal{O}} - y\|^2 \ge \|r\|^2 = \|x - \widehat{x}_{\mathcal{O}}\|^2$, with equality iff $y = \widehat{x}_{\mathcal{O}}$.
(iv) ($\Rightarrow$) As $x = \widehat{x}_{\mathcal{O}} + r$ orthogonally, $\|x\|^2 = \|\widehat{x}_{\mathcal{O}}\|^2 + \|r\|^2$. If $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$, then $r = 0$, $x = \widehat{x}_{\mathcal{O}} \in \operatorname{cspan}(\mathcal{O})$.
(iv) ($\Leftarrow$) If $x \in \operatorname{cspan}(\mathcal{O})$, the minimizer of $\|x - y\|$ over $y \in \operatorname{cspan}(\mathcal{O})$ is x; by (iii), $\widehat{x}_{\mathcal{O}} = x$, $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$.
Structural Commentary: Fourier expansion is orthogonal projection onto $\operatorname{cspan}(\mathcal{O})$.
Key Insight: Bessel bounds the coefficients; completeness sums them. □

Corollary 5.2.23. *Parseval Characterization of Hilbert Bases*
For an orthonormal family $\mathcal{O} = \{u_k : k \in K\}$ in $\mathcal{H}$, the following are equivalent:

(i) $\mathcal{O}$ is a Hilbert basis.
(ii) $\forall x \in \mathcal{H}$, $x = \sum_{k\in K} \langle x, u_k\rangle\, u_k$.
(iii) $\forall x, y \in \mathcal{H}$, Parseval's identity $\langle x, y\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \overline{\langle y, u_k\rangle}$ holds.
(iv) $\forall x \in \mathcal{H}$, $\|x\|^2 = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2$.

Proof.
(i) $\Rightarrow$ (ii) By Theorem 5.2.18, $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$. For $x \in \mathcal{H}$, both x and $\widehat{x}_{\mathcal{O}}$ minimize $\|x - y\|$ over $y \in \operatorname{cspan}(\mathcal{O})$.
By uniqueness in Theorem 5.2.22(iii), $x = \widehat{x}_{\mathcal{O}} = \sum_{k\in K} \langle x, u_k\rangle\, u_k$.
(ii) $\Rightarrow$ (iii) For $x, y \in \mathcal{H}$, (ii) and Proposition 5.2.20 give $\langle x, y\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \langle u_k, y\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \overline{\langle y, u_k\rangle}$.
(iii) $\Rightarrow$ (iv) Put $y = x$: $\|x\|^2 = \langle x, x\rangle = \sum_{k\in K} \langle x, u_k\rangle\, \overline{\langle x, u_k\rangle} = \sum_{k\in K} \left|\langle x, u_k\rangle\right|^2$.
(iv) $\Rightarrow$ (i) By Theorem 5.2.22(ii), (iv) gives $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$ for every $x \in \mathcal{H}$.
By Theorem 5.2.22(iv), $x \in \operatorname{cspan}(\mathcal{O})$ for every x, so $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$. Theorem 5.2.18 gives (i).
Structural Commentary: Totality, Fourier expansion, Parseval, and norm equality are equivalent.
Key Insight: Parseval preserves all Hilbert geometry. □

Theorem 5.2.24. *Existence and Cardinality of Hilbert Bases*
Let $\mathcal{H}$ be a Hilbert space.

(i) $\mathcal{H}$ admits a Hilbert basis.
(ii) If $\mathcal{B}, \mathcal{C}$ are Hilbert bases of $\mathcal{H}$, then $|\mathcal{B}| = |\mathcal{C}|$.

Proof.
(i) Let $\mathfrak{O}$ be the orthonormal subsets of $\mathcal{H}$, ordered by $\subseteq$. Chains in $\mathfrak{O}$ have orthonormal unions, so Principle 6.1.3 gives maximal $\mathcal{O} \in \mathfrak{O}$; by Theorem 5.2.18, $\mathcal{O}$ is a Hilbert basis.
(ii) Let $\mathcal{B} = \{b_i : i \in I\}$ and $\mathcal{C} = \{c_j : j \in J\}$ be Hilbert bases.
If $|I| < \infty$, Proposition 5.2.16 gives $\dim \mathcal{H} = |I|$; orthonormality gives $|J| \le \dim \mathcal{H}$, hence $|J| < \infty$, and Proposition 5.2.16 gives $|J| = \dim \mathcal{H}$. The case $|J| < \infty$ is symmetric.
Assume I, J infinite. For $i \in I$, set $J_i := \{j \in J : \langle b_i, c_j\rangle \neq 0\}$. By Theorem 5.2.21(iii), $|J_i| \le \aleph_0$.
Since $\mathcal{B}^\perp = \{\mathbf{0}\}$, $j \notin \bigcup_{i\in I} J_i$ is impossible; hence $J = \bigcup_{i\in I} J_i$ and $|J| \le \sum_{i\in I} |J_i| \le \aleph_0\, |I| = |I|$.
Symmetry gives $|I| \le |J|$; Theorem 6.2.3 gives $|I| = |J|$.
Structural Commentary: Zorn gives existence; countable coordinate supports compare cardinalities.
Key Insight: Hilbert-basis size is intrinsic. □

Definition 5.2.25. *Hilbert Dimension*
For a Hilbert space $\mathcal{H}$ with Hilbert basis $\mathcal{B}$, define $\operatorname{hdim}(\mathcal{H}) := |\mathcal{B}|$; this is well defined by Theorem 5.2.24.

Proposition 5.2.26. *The Hilbert Space $\ell^2(K)$*
Let K be a set and set $\ell^2(K) := \{f : K \to \mathbb{F} : \sum_{k\in K} |f(k)|^2 < \infty\}$.

(i) $\ell^2(K)$ is Hilbert under $\langle f, g\rangle := \sum_{k\in K} f(k)\overline{g(k)}$.
(ii) The coordinate family $(\delta_k)_{k\in K}$ is a Hilbert basis of $\ell^2(K)$.

Proof.

(i) For $f, g \in \ell^2(K)$ and finite $F \subseteq K$, $\sum_{k\in F} \left|f(k)\overline{g(k)}\right| \le (\sum_{k\in F} |f(k)|^2)^{1/2}(\sum_{k\in F} |g(k)|^2)^{1/2} \le \|f\|_2 \|g\|_2$, so $\langle f, g\rangle$ is well defined. If (f_n) is Cauchy, then $|f_m(k) - f_n(k)| \le \|f_m - f_n\|_2$ for each k, so set $f(k) := \lim_n f_n(k)$.
Given $\varepsilon > 0$, choose N with $\|f_m - f_n\|_2 < \varepsilon$ for $m, n \ge N$.
Then for finite $F \subseteq K$ and $n \ge N$, $\sum_{k\in F} |f(k) - f_n(k)|^2 = \lim_m \sum_{k\in F} |f_m(k) - f_n(k)|^2 \le \varepsilon^2$.
Hence $f - f_n \in \ell^2(K)$, $\|f - f_n\|_2 \le \varepsilon$, $f \in \ell^2(K)$, and $f_n \to f$.

(ii) $\langle \delta_j, \delta_k\rangle = \mathbf{1}_{\{j=k\}}$. If $f \perp \delta_k$ for every $k \in K$, then $f(k) = \langle f, \delta_k\rangle = 0$ for every k, so $(\delta_k)_{k\in K}^{\perp} = \{\mathbf{0}\}$.
By Theorem 5.2.18, $(\delta_k)_{k\in K}$ is a Hilbert basis.

Structural Commentary: Coordinatewise limits and uniform finite-subset bounds prove completeness.
Key Insight: Square-summability is precisely the norm condition for arbitrary orthogonal coordinates. □

Corollary 5.2.27. *Classification by Hilbert Dimension*
For Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, $\mathcal{H}_1 \cong_{\text{iso}} \mathcal{H}_2$ iff $\operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2)$.

Proof.

(⇒) Let $U : \mathcal{H}_1 \to \mathcal{H}_2$ be a surjective linear isometry and let $\mathcal{B}$ be a Hilbert basis of $\mathcal{H}_1$.
Then $U\mathcal{B}$ is orthonormal and $\operatorname{cspan}(U\mathcal{B}) = U(\operatorname{cspan}\mathcal{B}) = \mathcal{H}_2$.
Thus $U\mathcal{B}$ is a Hilbert basis and $\operatorname{hdim}(\mathcal{H}_1) = |\mathcal{B}| = |U\mathcal{B}| = \operatorname{hdim}(\mathcal{H}_2)$.

(⇐) Let $\operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2) = |I|$, and choose Hilbert bases $(b_i)_{i\in I}$ and $(c_i)_{i\in I}$.
Define $\Phi_1 x := (i \mapsto \langle x, b_i\rangle)$ and $\Phi_2 y := (i \mapsto \langle y, c_i\rangle)$. By Corollary 5.2.23, $\|\Phi_1 x\|_2 = \|x\|$, $\|\Phi_2 y\|_2 = \|y\|$.
If $a = (a_i) \in \ell^2(I)$, then $\sum_{i\in I} a_i b_i$ and $\sum_{i\in I} a_i c_i$ converge by Theorem 5.2.21, with images a under Φ_1 and Φ_2.
Hence $U := \Phi_2^{-1}\Phi_1 : \mathcal{H}_1 \to \mathcal{H}_2$ is a surjective linear isometry.

Structural Commentary: Map both spaces isometrically onto the same coordinate model.
Key Insight: Hilbert spaces are classified by the number of orthogonal coordinate directions. □

Corollary 5.2.28. *Separable Hilbert Spaces*
$\mathcal{H}$ separable $\iff \operatorname{hdim}(\mathcal{H}) \le \aleph_0$.

Proof.

(⇒) Let $D \subseteq \mathcal{H}$ be countable dense and let $\mathcal{O}$ be a Hilbert basis. $\forall u \in \mathcal{O}$, choose $d_u \in D$ with $\|u - d_u\| < \frac{1}{3}$.
If $u \ne v$ and $d_u = d_v$, then $\sqrt{2} = \|u - v\| \le \|u - d_u\| + \|d_v - v\| < \frac{2}{3}$, impossible.
Hence $u \mapsto d_u$ injects $\mathcal{O}$ into D, so $\operatorname{hdim}(\mathcal{H}) = |\mathcal{O}| \le \aleph_0$.

(⇐) Let $\mathcal{O}$ be a Hilbert basis with $|\mathcal{O}| \le \aleph_0$. The finite $\mathbb{Q}$-span of $\mathcal{O}$ over $\mathbb{R}$, respectively finite $(\mathbb{Q} + i\mathbb{Q})$-span over $\mathbb{C}$, is countable and dense in $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$.

Structural Commentary: Countable bases and countable dense sets force each other.
Key Insight: Separability of a Hilbert space is exactly countability of its orthogonal coordinate directions. □

Corollary 5.2.29. *Infinite Hilbert Bases Are Not Hamel Bases*
If $\operatorname{hdim}(\mathcal{H})$ is infinite, no Hilbert basis of $\mathcal{H}$ is a Hamel basis.

Proof. Let $\mathcal{O}$ be Hilbert basis, choose distinct $u_n \in \mathcal{O}$.
Since $\sum_n 4^{-n} < \infty$, $x := \sum_n 2^{-n} u_n \in \mathcal{H}$ by Theorem 5.2.21.

For every n, continuity of $\langle \cdot, u_n\rangle$ gives $\langle x, u_n\rangle = 2^{-n} \ne 0$. If $\mathcal{O}$ were Hamel, then $x = \sum_{j=1}^{N} a_j v_j$ with $v_j \in \mathcal{O}$; choosing n with $u_n \notin \{v_1, \dots, v_N\}$ gives $\langle x, u_n\rangle = 0$, contradiction.
Structural Commentary: Infinite Hilbert coordinates allow square-summable infinite expansions.
Key Insight: A Hilbert basis spans by limits, while a Hamel basis spans by finite sums. □

Perspective 5.2.30. *Bounded Functionals*
Here $\mathcal{H}^* = \mathcal{L}(\mathcal{H}, \mathbb{F})$ by Definition 5.1.4 and Theorem 5.1.5.

Proposition 5.2.31. *Operator Norm Algebra*
Let $S, T \in \mathcal{L}(\mathcal{H})$ and $a \in \mathbb{F}$.
(i) $\|aT\|_{op} = |a| \|T\|_{op}$.
(ii) $\|S + T\|_{op} \le \|S\|_{op} + \|T\|_{op}$.
(iii) $\|ST\|_{op} \le \|S\|_{op} \|T\|_{op}$.
(iv) $\mathcal{L}(\mathcal{H})$ is a normed algebra under composition.

Proof.

(i) $\|aT\|_{\text{op}} = \sup_{\|x\|\le 1} \|aTx\| = |a| \sup_{\|x\|\le 1} \|Tx\| = |a| \|T\|_{\text{op}}$.

(ii) For $\|x\| \le 1$, $\|(S + T)x\| \le \|Sx\| + \|Tx\| \le \|S\|_{\text{op}} + \|T\|_{\text{op}}$; taking sup gives the claim.

(iii) For $\|x\| \le 1$, $\|STx\| \le \|S\|_{\text{op}} \|Tx\| \le \|S\|_{\text{op}} \|T\|_{\text{op}}$; taking sup gives the claim.

(iv) Items (i)–(ii), with $\|T\|_{\text{op}} = 0 \Rightarrow T = 0$, give a norm on $\mathcal{L}(\mathcal{H})$.
Item (iii) gives closure and submultiplicativity for associative composition.

Structural Commentary: Composition multiplies stretch bounds.
Key Insight: Bounded operators behave like matrices with a compatible norm. □

Theorem 5.2.32. *Riesz Representation for Hilbert Spaces*
For $\varphi \in \mathcal{H}^*$:
(i) $\exists! y \in \mathcal{H}$ such that $\forall x \in \mathcal{H}$, $\varphi(x) = \langle x, y\rangle$.

(ii) The representing vector satisfies $\|\varphi\|_{op} = \|y\|$.

Proof.

(i) If $\varphi = 0$, take $y = 0$. If $\varphi \neq 0$, set $M := \ker\varphi$, $a \notin M$, $a = p + u$, $p \in M$, $u \in M^{\perp}$ by Corollary 5.2.10.
Then $u \neq 0$ and $\varphi(u) = \varphi(a) \neq 0$. For $x \in \mathcal{H}$, $x - \frac{\varphi(x)}{\varphi(u)}u \in M$, hence $\langle x, u\rangle = \frac{\varphi(x)}{\varphi(u)}\|u\|^2$.
Thus $y := \overline{\varphi(u)}\,\|u\|^{-2}\,u$ satisfies $\varphi(x) = \langle x, y\rangle$ for every x.
If z also represents φ, then $\langle x, y - z\rangle = 0$ for every x; with $x = y - z$, $z = y$.

(ii) For every x, $|\varphi(x)| = \big|\langle x, y\rangle\big| \leq \|x\|\,\|y\|$, so $\|\varphi\|_{\mathrm{op}} \leq \|y\|$.
If $y \neq 0$, then $\|\varphi\|_{\mathrm{op}} \geq \frac{|\varphi(y)|}{\|y\|} = \|y\|$; if $y = 0$, both norms vanish.

Structural Commentary: The kernel is a hyperplane; its orthogonal direction represents the functional.
Key Insight: Bounded functionals are exactly inner products with fixed vectors. □

Corollary 5.2.33. *Riesz Identifies the Continuous Dual*
Define $R : \mathcal{H} \to \mathcal{H}^*$ *by* $Ry := (x \mapsto \langle x, y\rangle)$.

(i) R *is an isometric bijection.*
(ii) R *is conjugate-linear over* $\mathbb{C}$ *and linear over* $\mathbb{R}$.

Proof.

(i) For $y \in \mathcal{H}$, $|Ry(x)| = \big|\langle x, y\rangle\big| \leq \|x\|\,\|y\|$, so $Ry \in \mathcal{H}^*$. By Theorem 5.2.32, $\mathcal{H}^* = R(\mathcal{H})$ and $\|Ry\|_{\mathrm{op}} = \|y\|$.
If $Ry = Rz$, then $0 = \|R(y - z)\|_{\mathrm{op}} = \|y - z\|$; hence R is bijective and isometric.

(ii) For $x, y, z \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{C}$, $R(\alpha y + \beta z)(x) = \langle x, \alpha y + \beta z\rangle = \overline{\alpha}Ry(x) + \overline{\beta}Rz(x)$.
Over $\mathbb{R}$, $R(\alpha y + \beta z) = \alpha Ry + \beta Rz$.

Structural Commentary: Continuous functionals are represented by vectors.
Key Insight: Riesz identifies $\mathcal{H}^*$ with $\mathcal{H}$, up to complex conjugation. □

Theorem 5.2.34. *Bounded Hilbert Adjoints*
Let $T \in \mathcal{L}(\mathcal{H})$.

(i) $\exists! T^* \in \mathcal{L}(\mathcal{H})$ *such that* $\forall u, v \in \mathcal{H}$, $\langle Tu, v\rangle = \langle u, T^*v\rangle$.
(ii) $\|T^*\|_{op} = \|T\|_{op}$.
(iii) $(T^*)^* = T$.

Proof.

(i) For $v \in \mathcal{H}$, set $\phi_v(u) := \langle Tu, v\rangle$. Since $|\phi_v(u)| \leq \|T\|_{\mathrm{op}}\,\|v\|\,\|u\|$, $\phi_v \in \mathcal{H}^*$.
By Theorem 5.2.32, $\exists! T^*v \in \mathcal{H}$ with $\phi_v(u) = \langle u, T^*v\rangle$ for every u.
For $\alpha, \beta \in \mathbb{F}$, $\phi_{\alpha v + \beta w} = \overline{\alpha}\phi_v + \overline{\beta}\phi_w$, so uniqueness gives $T^*(\alpha v + \beta w) = \alpha T^*v + \beta T^*w$.
Also $\|T^*v\| = \|\phi_v\|_{\mathrm{op}} \leq \|T\|_{\mathrm{op}}\,\|v\|$, hence $T^* \in \mathcal{L}(\mathcal{H})$.
If A also satisfies the adjoint identity, then $\langle u, (T^* - A)v\rangle = 0$ for every u, v, so $A = T^*$.

(ii) If $\mathcal{H} = \{\mathbf{0}\}$, immediate. Else $\|T\|_{\mathrm{op}} = \sup_{\|u\|=\|v\|=1}\big|\langle Tu, v\rangle\big| = \sup_{\|u\|=\|v\|=1}\big|\langle u, T^*v\rangle\big| = \|T^*\|_{\mathrm{op}}$.

(iii) For $u, v \in \mathcal{H}$, $\langle T^*u, v\rangle = \overline{\langle v, T^*u\rangle} = \overline{\langle Tv, u\rangle} = \langle u, Tv\rangle$.
Thus T satisfies the adjoint identity for T^*; by uniqueness, $(T^*)^* = T$.

Structural Commentary: Riesz turns the scalar tests induced by an operator back into vectors.
Key Insight: The adjoint is the dual action transported through the Hilbert-space Riesz map. □

Corollary 5.2.35. *Hilbert Adjoint Algebra Identities*
Let $S, T \in \mathcal{L}(\mathcal{H})$ *and* $a \in \mathbb{F}$.

(i) $(aS + T)^* = \overline{a}S^* + T^*$.
(ii) $(ST)^* = T^*S^*$.

Proof.

(i) For $u, v \in \mathcal{H}$, $\langle (aS + T)u, v\rangle = a\langle Su, v\rangle + \langle Tu, v\rangle = \langle u, \overline{a}S^*v + T^*v\rangle$.
Uniqueness of adjoints gives $(aS + T)^* = \overline{a}S^* + T^*$.

(ii) For $u, v \in \mathcal{H}$, $\langle STu, v\rangle = \langle Tu, S^*v\rangle = \langle u, T^*S^*v\rangle$; uniqueness gives $(ST)^* = T^*S^*$.

Structural Commentary: Move each operator across one inner product and invoke adjoint uniqueness.
Key Insight: Taking adjoints reverses composition and conjugates scalars. □

Definition 5.2.36. *Self-Adjoint, Nonnegative, and Coercive Operators*
Let $T \in \mathcal{L}(\mathcal{H})$.

(i) *Self-adjoint*: $T = T^*$.
(ii) *Nonnegative*: $T = T^*$ and $\forall h \in \mathcal{H}$, $\langle Th, h\rangle \geq 0$; write $T \geq 0$.
(iii) *Coercive*: if $\mathcal{H}$ is real, $\exists c > 0$ such that $\forall h \in \mathcal{H}$, $\langle Th, h\rangle \geq c\,\|h\|^2$.

Proposition 5.2.37. *Positive Order, Polarization, and Reducing Complements*
Let $\mathcal{H}$ *be a real Hilbert space.*

(i) If $T \geq 0$ *and* $S \in \mathcal{L}(\mathcal{H})$, *then* $S^*TS \geq 0$.
(ii) If $T \in \mathcal{L}(\mathcal{H})$ *is self-adjoint and* $Q_T(u) := \langle Tu, u\rangle$, *then* $\langle Tu, v\rangle = \frac{1}{4}\left[Q_T(u + v) - Q_T(u - v)\right]$.
(iii) If $T \in \mathcal{L}(\mathcal{H})$ *is self-adjoint and* $Q_T \equiv 0$, *then* $T = 0$.
(iv) If $T \in \mathcal{L}(\mathcal{H})$ *is self-adjoint and* $V \leq \mathcal{H}$ *satisfies* $T(V) \subseteq V$, *then* $T(V^{\perp}) \subseteq V^{\perp}$.

Proof.

(i) Since $T = T^*$, $(S^*TS)^* = S^*T^*S = S^*TS$. For $h \in \mathcal{H}$, $\langle S^*TSh, h\rangle = \langle TSh, Sh\rangle \geq 0$; hence $S^*TS \geq 0$.

(ii) For $u, v \in \mathcal{H}$, self-adjointness gives $Q_T(u \pm v) = Q_T(u) \pm 2\langle Tu, v\rangle + Q_T(v)$.
Subtracting gives $Q_T(u+v) - Q_T(u-v) = 4\langle Tu, v\rangle$.
(iii) By (ii), $\langle Tu, v\rangle = 0$ for all $u, v \in \mathcal{H}$; with $v = Tu$, $\|Tu\|^2 = 0$ for every u, so $T = 0$.
(iv) If $w \in V^\perp$ and $v \in V$, then $Tv \in V$ and $\langle Tw, v\rangle = \langle w, Tv\rangle = 0$; hence $Tw \in V^\perp$.

Structural Commentary: Adjoints transfer positivity and invariance; polarization recovers mixed terms.
Key Insight: For self-adjoint operators, the diagonal quadratic form controls the mixed inner products. □

Theorem 5.2.38. *Self-Adjoint Operator Norm from the Quadratic Form*
If $\mathcal{H}$ is real and $T \in \mathcal{L}(\mathcal{H})$ is self-adjoint, then $\|T\|_{op} = \sup_{\|u\|=1} \left|\langle Tu, u\rangle\right|$.

Proof. Let $\eta := \sup_{\|u\|=1} \left|\langle Tu, u\rangle\right|$. By Cauchy–Schwarz, $\eta \le \|T\|_{\mathrm{op}}$.
If $\eta = 0$, then $Q_T \equiv 0$ by scaling, so $T = 0$ by Proposition 5.2.37(iii).
Assume $\eta > 0$ and set $A_\pm := \eta I \pm T$. For $u \in \mathcal{H}$, $\left|\langle Tu, u\rangle\right| \le \eta \|u\|^2$, hence $\langle A_\pm u, u\rangle \ge 0$ and $A_\pm \ge 0$.
By Proposition 5.2.37(i), $A_+^* A_- A_+ \ge 0$ and $A_-^* A_+ A_- \ge 0$.
Since $A_\pm = A_\pm^*$ and $A_+A_- = A_-A_+ = \eta^2 I - T^2$, their sum is $2\eta(\eta^2 I - T^2) \ge 0$, so $\eta^2 I - T^2 \ge 0$.
Thus $\|Tu\|^2 = \langle T^2 u, u\rangle \le \eta^2 \|u\|^2$ for every u, hence $\|T\|_{\mathrm{op}} \le \eta$.
Structural Commentary: Self-adjointness lets order information about $\eta I \pm T$ control T^2.
Key Insight: A bounded self-adjoint operator is normed by its one-vector quadratic tests. □

Proposition 5.2.39. *Closed Range and the Adjoint Kernel*
If $T \in \mathcal{L}(\mathcal{H})$ has closed range, then $\mathcal{H} = \operatorname{range}(T) \odot \ker(T^*)$.

Proof. For $z \in \mathcal{H}$, $z \in \ker(T^*) \iff \forall u \in \mathcal{H},\ \langle u, T^* z\rangle = 0 \iff \forall u \in \mathcal{H},\ \langle Tu, z\rangle = 0 \iff z \in \operatorname{range}(T)^\perp$.
Thus $\ker(T^*) = \operatorname{range}(T)^\perp$, and Corollary 5.2.10 gives $\mathcal{H} = \operatorname{range}(T) \odot \ker(T^*)$ since $\operatorname{range}(T)$ is closed.
Structural Commentary: Identify the orthogonal complement of the range by moving T across the inner product.
Key Insight: The directions missed by T are exactly the vectors annihilated by T^*. □

Theorem 5.2.40. *Coercive Operators Are Invertible*
Let $\mathcal{H}$ be real and let $T \in \mathcal{L}(\mathcal{H})$ satisfy $\exists c > 0$ such that $\forall h \in \mathcal{H}$, $\langle Th, h\rangle \ge c\|h\|^2$.
(i) T is bijective.
(ii) $T^{-1} \in \mathcal{L}(\mathcal{H})$ and $\left\|T^{-1}\right\|_{op} \le c^{-1}$.

Proof. For $u \ne \mathbf{0}$, $c\|u\|^2 \le \langle Tu, u\rangle \le \|Tu\|\,\|u\|$, hence $\|Tu\| \ge c\|u\|$; the same holds for $u = \mathbf{0}$, so $\ker T = \{\mathbf{0}\}$.
If $Tu_n \to y$, then $\|u_n - u_m\| \le c^{-1}\|Tu_n - Tu_m\|$; hence $u_n \to u$ and $Tu = y$, so $\operatorname{range}(T)$ is closed.
Also $\langle T^* h, h\rangle = \langle Th, h\rangle \ge c\|h\|^2$, so $\ker(T^*) = \{\mathbf{0}\}$.
(i) By Proposition 5.2.39, $\mathcal{H} = \operatorname{range}(T) \odot \ker(T^*) = \operatorname{range}(T)$; with $\ker T = \{\mathbf{0}\}$, T is bijective.
(ii) For $y \in \mathcal{H}$, put $u = T^{-1}y$. Then $c\left\|T^{-1}y\right\| = c\|u\| \le \|Tu\| = \|y\|$, so $T^{-1} \in \mathcal{L}(\mathcal{H})$ and $\left\|T^{-1}\right\|_{\mathrm{op}} \le c^{-1}$.

Structural Commentary: Coercivity closes the range and kills both kernels; adjoint splitting gives surjectivity.
Key Insight: Coercivity supplies both a lower Lipschitz bound and, through the adjoint, surjectivity. □

Theorem 5.2.41. *Lax–Milgram Lemma*
Let $\mathcal{H}$ be a real Hilbert space and let $B : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ be bilinear.
Suppose there are constants $M, c > 0$ such that $|B(u,v)| \le M\|u\|\,\|v\|$ and $B(h,h) \ge c\|h\|^2$ $\forall u, v, h \in \mathcal{H}$.
Then $\forall \psi \in \mathcal{H}^*$ $\exists! h \in \mathcal{H}$ such that $\psi(u) = B(h,u)$ $(u \in \mathcal{H})$.

Proof. Let $R : \mathcal{H} \to \mathcal{H}^*$ be the real Riesz isometry, $Ry(u) = \langle u, y\rangle$, and define $S : \mathcal{H} \to \mathcal{H}^*$ by $Sh(u) = B(h,u)$. For $h, u \in \mathcal{H}$, $|Sh(u)| \le M\|h\|\,\|u\|$, hence $S \in \mathcal{L}(\mathcal{H}, \mathcal{H}^*)$.
Set $A := R^{-1}S \in \mathcal{L}(\mathcal{H})$. For $h \in \mathcal{H}$, $\langle Ah, h\rangle = \langle h, Ah\rangle = (RAh)(h) = Sh(h) = B(h,h) \ge c\|h\|^2$.
By Theorem 5.2.40, A is bijective, hence $S = RA$ is bijective.
For $\psi \in \mathcal{H}^*$, set $h := A^{-1}R^{-1}\psi$; then $\forall u \in \mathcal{H}$, $\psi(u) = (RAh)(u) = Sh(u) = B(h,u)$.
If h_1, h_2 solve this identity, then $S(h_1 - h_2) = 0$, so $h_1 = h_2$.
Structural Commentary: A coercive bilinear form is a Riesz-represented coercive operator in disguise.
Key Insight: Solving $B(h, \cdot) = \psi$ is solving an invertible operator equation after Riesz representation. □

Theorem 5.2.42. *Orthonormal Systems in L^2*
Let $(\phi_n)_{n\ge1} \subseteq L^2(\mu)$ be orthonormal, $\langle \phi_n, \phi_m\rangle = \delta_{nm}$, and $c_n(f) := \langle f, \phi_n\rangle = \int_X f\overline{\phi_n}\,d\mu$.

(i) $\forall f \in L^2(\mu)$, $\sum_{n=1}^{\infty} |c_n(f)|^2 \le \|f\|_2^2$.

(ii) If $(c_n) \in \ell^2$, then $s_N := \sum_{k=1}^{N} c_k \phi_k$ converges in $L^2(\mu)$.

(iii) If (ϕ_n) is complete, then $\forall f \in L^2(\mu)$, $\|f\|_2^2 = \sum_{n=1}^{\infty} |c_n(f)|^2$.

Proof.

(i) Set $s_N(f) := \sum_{k=1}^{N} c_k(f)\phi_k$. Finite orthonormality gives $\|f - s_N(f)\|_2^2 = \|f\|_2^2 - \sum_{k=1}^{N} |c_k(f)|^2 \ge 0$.
Monotone convergence of the partial sums yields $\sum_{n=1}^{\infty} |c_n(f)|^2 \le \|f\|_2^2$.

(ii) If $m < N$, then $\|s_N - s_m\|_2^2 = \sum_{k=m+1}^{N} |c_k|^2$. Since $(c_n) \in \ell^2$, (s_N) is Cauchy.
By completeness, $\exists g \in L^2(\mu)$ with $s_N \to g$.

(iii) Assume (ϕ_n) is complete. By (i), $(c_n(f)) \in \ell^2$; by (ii), $s_N(f) \to g$.
For fixed j, $\langle s_N(f), \phi_j\rangle = c_j(f)$ for $N \geq j$, hence $\langle f-g, \phi_j\rangle = c_j(f) - \lim_N \langle s_N(f), \phi_j\rangle = 0$.
Thus $f - g \perp \{\phi_j : j \geq 1\}$, so $f = g$; therefore $\|f\|_2^2 = \lim_N \|s_N(f)\|_2^2 = \sum_{n=1}^{\infty} |c_n(f)|^2$.

Structural Commentary: Complete orthonormal systems coordinatize L^2 by square-summable sequences.
Key Insight: Parseval is Pythagoras after passing from finite coordinates to an L^2 limit. □

Method 5.2.43. *Hilbert Space Proof Patterns*

(i) *Complete*: Cauchy tails, completions, and convergent subsequences force limits.
(ii) *Parallelogram*: test Hilbertian norms and make distance-minimizing sequences Cauchy.
(iii) *Project*: for closed M, write $x = p + r$, $p \in M$, $r \in M^\perp$.
(iv) *Separate*: density and equality via $M^\perp = \{\mathbf{0}\}$ and $M^{\perp\perp} = \overline{M}$.
(v) *Sum*: arbitrary convergence is net convergence over $\mathcal{F}(K)$; control finite tails.
(vi) *Expand*: orthonormal finite identities + Bessel bounds + completeness ⇒ Fourier expansions.
(vii) *Classify*: Hilbert bases give hdim, separability, and $\mathcal{H} \cong \ell^2(K)$.
(viii) *Dualize*: Riesz identifies $\mathcal{H}^*$ with $\mathcal{H}$, producing adjoints and coercive operator equations.

Key Insight: Hilbert-space arguments are finite orthogonal geometry plus completeness.

5.3 Core Functional Analytic Theorems

The core theorems explain why complete normed spaces behave rigidly under linear maps. Hahn–Banach supplies norm-preserving extensions, while Baire category turns pointwise control into uniform boundedness.

Theorem 5.3.1. *Hahn–Banach Extension Theorem*
Let $\mathcal{X}$ be a real normed space, $M \leq \mathcal{X}$, $p : \mathcal{X} \to \mathbb{R}$ sublinear, and $\varphi : M \to \mathbb{R}$ linear with $\varphi \leq p$ on M.

(i) If $x_0 \in \mathcal{X} \setminus M$ and $M_0 := M + \mathbb{R}x_0$, then $\exists \varphi_0 : M_0 \to \mathbb{R}$ linear such that $\varphi_0|_M = \varphi$ and $\varphi_0 \leq p$ on M_0.
(ii) $\exists \Phi : \mathcal{X} \to \mathbb{R}$ linear such that $\Phi|_M = \varphi$ and $\Phi \leq p$ on $\mathcal{X}$.

Proof.

(i) For $y, z \in M$, $\varphi(y) + \varphi(z) = \varphi(y+z) \leq p(y+z) \leq p(y - x_0) + p(z + x_0)$; $\varphi(y) - p(y - x_0) \leq p(z + x_0) - \varphi(z)$.
Set $a := \sup_{y \in M} (\varphi(y) - p(y - x_0))$, $b := \inf_{z \in M} (p(z + x_0) - \varphi(z))$. Then $-p(-x_0) \leq a \leq b \leq p(x_0)$; $\alpha \in [a, b]$.
Since $x_0 \notin M$, $y + tx_0$ has unique coordinates, so define $\varphi_0(y + tx_0) := \varphi(y) + t\alpha$.
If $t > 0$, then $\varphi_0(y + tx_0) = t(\varphi(y/t) + \alpha) \leq tp(y/t + x_0) = p(y + tx_0)$.
If $t < 0$, then $\varphi_0(y + tx_0) = (-t)(\varphi(y/(-t)) - \alpha) \leq (-t)p(y/(-t) - x_0) = p(y + tx_0)$.
If $t = 0$, domination is $\varphi \leq p$ on M.
(ii) Let $\mathscr{E}$ be (N, ψ), $M \leq N \leq \mathcal{X}$, $\psi : N \to \mathbb{R}$ linear, $\psi|_M = \varphi$, $\psi \leq p$ on N, ordered by extension.
Chains have upper bound $(\bigcup N_i, \psi_*)$, where $\psi_*|_{N_i} = \psi_i$, so Principle 6.1.3 gives maximal $(N_*, \psi_*) \in \mathscr{E}$.
If $N_* \neq \mathcal{X}$, choose $x_0 \in \mathcal{X} \setminus N_*$; by (i), ψ_* extends to $N_* + \mathbb{R}x_0$, contradiction. Thus $N_* = \mathcal{X}$ and $\Phi := \psi_*$.

Structural Commentary: Extend across one new direction while preserving domination, then use maximality.
Key Insight: Sublinearity leaves a nonempty interval of admissible values on each added direction. □

Corollary 5.3.2. *Norm-Preserving Extensions and Norming Functionals*
Let $\mathcal{X}$ be a normed space and let $M \subseteq \mathcal{X}$ be a linear subspace.

(i) A bounded linear functional $\varphi : M \to \mathbb{F}$ admits an extension $\Phi \in \mathcal{X}^*$ with $\|\Phi\|_{op} = \|\varphi\|_{op}$.
(ii) $\forall 0 \neq x \in \mathcal{X}$ $\exists \Phi \in \mathcal{X}^*$ such that $\|\Phi\|_{op} = 1$ and $\Phi(x) = \|x\|$.

Proof.

(i) If $\mathbb{F} = \mathbb{R}$, apply Theorem 5.3.1 with $p(u) := \|\varphi\|_{\mathrm{op}} \|u\|$ to get $\Phi|_M = \varphi$ and $\Phi \leq p$.
Since $-\Phi(u) = \Phi(-u) \leq p(u)$, $|\Phi(u)| \leq \|\varphi\|_{\mathrm{op}} \|u\| \forall u \in \mathcal{X}$, so $\|\Phi\|_{\mathrm{op}} \leq \|\varphi\|_{\mathrm{op}}$; restriction gives equality.
If $\mathbb{F} = \mathbb{C}$, apply the real case to $\operatorname{Re}\varphi$ and set $\Phi(u) := F(u) - iF(iu)$.
Then $\Phi|_M = \varphi$, Φ is complex-linear, and $|\Phi(u)| = \sup_{|\alpha|=1} F(\alpha u) \leq \|\varphi\|_{\mathrm{op}} \|u\|$, so again $\|\Phi\|_{\mathrm{op}} = \|\varphi\|_{\mathrm{op}}$.
(ii) Fix $0 \neq x \in \mathcal{X}$, put $M_x := \mathbb{F}x$, and define $\varphi_x(\lambda x) := \lambda \|x\|$.
Then $\|\varphi_x\|_{\mathrm{op}} = 1$ and $\varphi_x(x) = \|x\|$; by (i), choose $\Phi \in \mathcal{X}^*$ with $\Phi|_{M_x} = \varphi_x$ and $\|\Phi\|_{\mathrm{op}} = 1$.

Structural Commentary: Preserve the norm, then extend a one-dimensional norming functional.
Key Insight: Hahn–Banach continues a bounded scalar test without weakening its norm bound. □

Corollary 5.3.3. *Natural Embedding Is an Isometry*
For a normed space $\mathcal{X}$, the canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$, $Jx(\Phi) = \Phi(x)$, satisfies $\|Jx\|_{op} = \|x\|$ $(x \in \mathcal{X})$.

Proof. $\forall \Phi \in \mathcal{X}^*$, $|Jx(\Phi)| = |\Phi(x)| \leq \|\Phi\|_{\mathrm{op}} \|x\|$, so $\|Jx\|_{\mathrm{op}} \leq \|x\|$. If $x \neq 0$, choose $\Phi \in \mathcal{X}^*$ with $\|\Phi\|_{\mathrm{op}} = 1$ and $\Phi(x) = \|x\|$ by Corollary 5.3.2; hence $\|Jx\|_{\mathrm{op}} \geq \|x\|$. The case $x = 0$ is immediate.
Structural Commentary: The upper bound is duality; a norming functional attains the matching lower bound.
Key Insight: The double dual records every vector without changing its norm. □

Example 5.3.4. *A Dominated Extension of the Limit Functional*
On real ℓ^∞, set $p((x_n)) := \limsup_{n\to\infty} x_n$. This functional is subadditive and positively homogeneous.
If $c \subseteq \ell^\infty$ is the subspace of convergent sequences and $L((x_n)) := \lim_{n\to\infty} x_n$, then $L \leq p$ on c.
By Theorem 5.3.1, L extends to a linear functional $\Lambda : \ell^\infty \to \mathbb{R}$ with $\Lambda(x) \leq \limsup_{n\to\infty} x_n$ $(x \in \ell^\infty)$.
This is the Hahn–Banach domination step behind limit-extension arguments on ℓ^∞.

Corollary 5.3.5. *Positive Riesz Representation on $C([a,b])$*
Let $J=[a,b]$ and let $G\in C(J)^*$ be positive, i.e. $f\ge 0$ on J implies $G(f)\ge 0$.

(i) $\exists \gamma:\mathcal{B}(\mathbb{R})\to[0,+\infty]$ *a finite Borel measure with* $\operatorname{supp}\gamma\subseteq J$ *and* $\forall f\in C(J)$, $G(f)=\int_J f\,d\gamma$.
(ii) $\|G\|_{op}=G(\mathbf{1}_J)=\gamma(J)$.

Proof.

(i) By Theorem 5.2.14, $\exists$ finite Radon μ on J such that $G(f)=\int_J f\,d\mu$ for $f\in C(J)$.
Define $\gamma(E):=\mu(E\cap J)$ for $E\in\mathcal{B}(\mathbb{R})$. Then $\gamma(\mathbb{R}\setminus J)=0$, so $\operatorname{supp}\gamma\subseteq J$, and $\int_J f\,d\gamma=\int_J f\,d\mu=G(f)$.
(ii) If $\|f\|_\infty\le 1$, then $\mathbf{1}_J\pm f\ge 0$, hence $-G(\mathbf{1}_J)\le G(f)\le G(\mathbf{1}_J)$ and $|G(f)|\le G(\mathbf{1}_J)$.
Thus $\|G\|_{\mathrm{op}}\le G(\mathbf{1}_J)$; taking $f=\mathbf{1}_J$ gives $\|G\|_{\mathrm{op}}\ge G(\mathbf{1}_J)$. Also $G(\mathbf{1}_J)=\int_J \mathbf{1}_J\,d\gamma=\gamma(J)$.

Structural Commentary: Specialize Riesz–Markov to the interval, then extend the measure by zero.
Key Insight: Positivity converts the order unit $\mathbf{1}_J$ into the exact dual norm. □

Corollary 5.3.6. *Signed Riesz Representation by a Charge*
Let $J=[a,b]$, and suppose $G\in C(J)^*$ admits $G=G^+-G^-$ with positive $G^\pm\in C(J)^*$.
Then $\exists\lambda:\mathcal{B}(\mathbb{R})\to\mathbb{R}$ a charge with $\operatorname{supp}\lambda\subseteq J$ and $\forall f\in C(J)$, $G(f)=\int_J f\,d\lambda$.

Proof. By Corollary 5.3.5, choose finite Borel measures $\gamma^\pm$ on $\mathbb{R}$ such that $\gamma^\pm(\mathbb{R}\setminus J)=0$ and $G^\pm(f)=\int_J f\,d\gamma^\pm$ for $f\in C(J)$. Define $\lambda(E):=\gamma^+(E)-\gamma^-(E)$ for $E\in\mathcal{B}(\mathbb{R})$.
Then λ is a charge, $\lambda(\mathbb{R}\setminus J)=0$, and for $f\in C(J)$, $G(f)=G^+(f)-G^-(f)=\int_J f\,d\gamma^+-\int_J f\,d\gamma^-=\int_J f\,d\lambda$.
Structural Commentary: Subtract the positive representing measures.
Key Insight: Signed representation is positive representation plus subtraction. □

Theorem 5.3.7. *Uniform Boundedness Principle*
Let $\mathcal{X}$ be Banach, $\mathcal{Y}$ normed. If $\mathcal{F}\subseteq\mathcal{L}(\mathcal{X},\mathcal{Y})$ pointwise bounded, $\forall x\in\mathcal{X}$, $\sup_{T\in\mathcal{F}}\|Tx\|<\infty$, then $\sup_{T\in\mathcal{F}}\|T\|_{op}<\infty$.

Proof. For $T\in\mathcal{F}$, $f_T(x):=\|Tx\|$ is continuous. Pointwise boundedness and Theorem 4.3.11 give $x_0\in\mathcal{X}$, $r>0$, and $C<\infty$ with $\sup_{T\in\mathcal{F}}\|Tz\|\le C$ for $z\in B(x_0,r)$. Set $M_0:=\sup_{T\in\mathcal{F}}\|Tx_0\|<\infty$.
If $\|y\|<r$, then $\sup_{T\in\mathcal{F}}\|Ty\|\le\sup_{T\in\mathcal{F}}\|T(x_0+y)\|+\sup_{T\in\mathcal{F}}\|Tx_0\|\le C+M_0$.
For $\|x\|=1$ and $0<t<r$, $t\sup_{T\in\mathcal{F}}\|Tx\|=\sup_{T\in\mathcal{F}}\|T(tx)\|\le C+M_0$. By $t\uparrow r$, $\sup_{T\in\mathcal{F}}\|T\|_{\mathrm{op}}\le\frac{C+M_0}{r}<\infty$.
Structural Commentary: Baire controls one ball; translation and scaling transfer control to the unit ball.
Key Insight: Baire and linearity turn pointwise bounds into one uniform operator bound. □

Corollary 5.3.8. *Banach–Steinhaus Pointwise Limit Theorem*
Let $\mathcal{X}$ be Banach, $\mathcal{Y}$ normed, $(T_n)\subseteq\mathcal{L}(\mathcal{X},\mathcal{Y})$. Suppose $\forall x\in\mathcal{X}$, T_nx converges in $\mathcal{Y}$, $Tx:=\lim_{n\to\infty}T_nx$.

(i) $\sup_n\|T_n\|_{op}<\infty$.
(ii) T *is linear.*
(iii) $T\in\mathcal{L}(\mathcal{X},\mathcal{Y})$ *and* $\|T\|_{op}\le\liminf_{n\to\infty}\|T_n\|_{op}$.

Proof.

(i) For each $x\in\mathcal{X}$, convergence of (T_nx) gives $\sup_n\|T_nx\|<\infty$; hence Theorem 5.3.7 gives $\sup_n\|T_n\|_{\mathrm{op}}<\infty$.
(ii) For $\alpha,\beta\in\mathbb{F}$ and $x,y\in\mathcal{X}$, $T(\alpha x+\beta y)=\lim_n T_n(\alpha x+\beta y)=\alpha Tx+\beta Ty$.
(iii) Let $L:=\liminf_n\|T_n\|_{\mathrm{op}}<\infty$. For $x\in\mathcal{X}$, $\|Tx\|=\lim_n\|T_nx\|\le L\|x\|$, so $T\in\mathcal{L}(\mathcal{X},\mathcal{Y})$ and $\|T\|_{\mathrm{op}}\le L$.

Structural Commentary: Uniform boundedness controls the family before taking its pointwise limit.
Key Insight: Completeness prevents the norms of pointwise-convergent operators from escaping. □

5.4 Open Mapping, Closed Graph, and Complemented Subspaces

The Baire-category principle has three operator-theoretic consequences that should be read together: closed range gives uniform preimage estimates, open maps are detected at the unit ball, and closed graphs force boundedness.

Theorem 5.4.1. *Closed Range Estimate*
Let $\mathcal{X},\mathcal{Y}$ be Banach spaces, let $T\in\mathcal{L}(\mathcal{X},\mathcal{Y})$, and set $\mathcal{R}:=T(\mathcal{X})$. The following are equivalent.

(i) $\mathcal{R}$ *is closed in* $\mathcal{Y}$.
(ii) $\exists M>0$ *such that* $\forall y\in\mathcal{R}$ $\exists x\in\mathcal{X}$ *with* $Tx=y$ *and* $\|x\|\le M\|y\|$.

Proof.

(*ii*) $\Rightarrow$ (*i*) Let $y\in\overline{\mathcal{R}}$. Choose $y_n\in\mathcal{R}$ with $y_n\to y$ and $\|y_n-y_{n-1}\|\le 2^{-n}$ for $n\ge 2$.
Pick $x_1,u_n\in\mathcal{X}$ with $Tx_1=y_1$, $Tu_n=y_n-y_{n-1}$, and $\|u_n\|\le M2^{-n}$.
Since $\sum_{n\ge2}\|u_n\|<\infty$, $x:=x_1+\sum_{n\ge2}u_n$ exists and $Tx=\lim_N y_N=y$, so $y\in\mathcal{R}$.

(*i*) $\Rightarrow$ (*ii*) If $\mathcal{R}=\{\mathbf{0}\}$, take $M=1$. Otherwise $\mathcal{R}$ is Banach and $\mathcal{R}=\bigcup_{n\ge1}n\,\overline{T(B_{\mathcal{X}})}^{\mathcal{R}}$.

By Baire and balanced convexity, $\exists c>0$ with $\overline{B}_{\mathcal{R}}\subseteq c\,\overline{T(B_{\mathcal{X}})}^{\mathcal{R}}$.
Thus each $0\ne r\in\mathcal{R}$ has $u\in\mathcal{X}$ with $\|u\|\le c\|r\|$ and $\|r-Tu\|\le\frac12\|r\|$.
For $y\ne0$, iterate from $r_0=y$ to get $r_k=r_{k-1}-Tu_k$, $\|r_k\|\le 2^{-k}\|y\|$, and $\|u_k\|\le c2^{1-k}\|y\|$.
Then $x:=\sum_{k\ge1}u_k$ satisfies $Tx=y$ and $\|x\|\le 2c\|y\|$. For $y=0$, take $x=0$.

Structural Commentary: Baire category turns surjectivity onto a closed range into uniform preimage control.
Key Insight: Closed range is equivalent to solving $Tx=y$ with a norm bound proportional to $\|y\|$. □

Proposition 5.4.2. *Relative Openness and Closed Range*
Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$.
Then $T : \mathcal{X} \to T(\mathcal{X})$ is open, for subspace topology on range, iff $T(\mathcal{X})$ closed in $\mathcal{Y}$.

Proof. Let $\mathcal{R} := T(\mathcal{X})$.

($\Rightarrow$) If $T : \mathcal{X} \to \mathcal{R}$ is open, then $\exists r > 0$ with $B_{\mathcal{R}}(0, r) \subseteq T(B_{\mathcal{X}})$.
For $0 \neq y \in \mathcal{R}$, choose $u \in B_{\mathcal{X}}$ with $Tu = \frac{r}{2\|y\|} y$, and set $x := \frac{2\|y\|}{r} u$.
Then $Tx = y$ and $\|x\| \leq \frac{2}{r} \|y\|$; for $y = 0$, take $x = 0$. By Theorem 5.4.1, $\mathcal{R}$ is closed.

($\Leftarrow$) If $\mathcal{R}$ is closed, then by Theorem 5.4.1, $\exists M > 0$ such that every $y \in \mathcal{R}$ has $x \in \mathcal{X}$ with $Tx = y$ and $\|x\| \leq M \|y\|$. Let $U \subseteq \mathcal{X}$ be open and $x_0 \in U$. Choose $\rho > 0$ with $x_0 + \rho B_{\mathcal{X}} \subseteq U$. If $z \in B_{\mathcal{R}}(Tx_0, \frac{\rho}{M})$, choose $h \in \mathcal{X}$ with $Th = z - Tx_0$ and $\|h\| < \rho$; then $z = T(x_0 + h) \in T(U)$. Thus $T(U)$ is open in $\mathcal{R}$.

Structural Commentary: For linear maps, openness is detected at the unit ball around the origin.
Key Insight: Relative openness is the topological form of uniform preimage control. □

Theorem 5.4.3. *Open Mapping Theorem*
Every surjective $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ between Banach spaces is open.

Proof. Since T is surjective, $T(\mathcal{X}) = \mathcal{Y}$, $\mathcal{Y}$ is closed in itself. By Proposition 5.4.2, $T : \mathcal{X} \to T(\mathcal{X}) = \mathcal{Y}$ open.
Structural Commentary: Surjectivity identifies the relative range with the codomain.
Key Insight: A Banach-space surjection has uniformly controlled preimages near the origin. □

Corollary 5.4.4. *Bounded Inverse Theorem*
If $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is bijective between Banach spaces, then $T^{-1} \in \mathcal{L}(\mathcal{Y}, \mathcal{X})$.

Proof. Set $U := T^{-1} : \mathcal{Y} \to \mathcal{X}$. By Theorem 5.4.3, T is open: if $O \subseteq \mathcal{X}$ is open, then $U^{-1}(O) = T(O)$ is open in $\mathcal{Y}$. Thus U is continuous; since U is linear, Theorem 5.1.5 gives $U \in \mathcal{L}(\mathcal{Y}, \mathcal{X})$.
Structural Commentary: Openness of the forward bijection is continuity of the inverse.
Key Insight: Banach completeness prevents a bounded linear bijection from having an unbounded inverse. □

Corollary 5.4.5. *One-Sided Control of Banach Norms*
Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on $\mathcal{X}$ such that $(\mathcal{X}, \|\cdot\|_1)$ and $(\mathcal{X}, \|\cdot\|_2)$ are Banach.
Assume $\exists c > 0$ such that $\|x\|_2 \leq c \|x\|_1$ for all $x \in \mathcal{X}$.

(i) $\exists C > 0$ such that $\|x\|_1 \leq C \|x\|_2$ for all $x \in \mathcal{X}$.
(ii) $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.

Proof.

(i) Let $I : (\mathcal{X}, \|\cdot\|_1) \to (\mathcal{X}, \|\cdot\|_2)$ be the identity. Since $\|Ix\|_2 \leq c \|x\|_1$, $I \in \mathcal{L}$ and is bijective.
By Corollary 5.4.4, $I^{-1} \in \mathcal{L}$, hence $\exists C > 0$ such that $\|x\|_1 = \left\|I^{-1}x\right\|_1 \leq C \|x\|_2$.

(ii) For all $x \in \mathcal{X}$, $\|x\|_2 \leq c \|x\|_1$ by hypothesis and $\|x\|_1 \leq C \|x\|_2$ by (i).
Hence the identity maps between the two normed structures are Lipschitz.

Structural Commentary: Apply the bounded inverse theorem to the identity between the two normed structures.
Key Insight: Completeness upgrades one-sided norm control to two-sided equivalence. □

Definition 5.4.6. *Closed Operators and Closed Complements*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces.

(i) *Closed operator*: $T : \mathcal{X} \to \mathcal{Y}$ is linear and $\forall (x_n) \subseteq \mathcal{X}$, $x_n \to x$ and $Tx_n \to y$ imply $Tx = y$.
(ii) *Closed linear complement*: $W \leq \mathcal{X}$ satisfies $\mathcal{X} = V \oplus W$ and is closed.

Theorem 5.4.7. *Closed Graph Theorem*
Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces and let $T : \mathcal{X} \to \mathcal{Y}$ be linear. Set $\Gamma(T) := \{(x, Tx) : x \in \mathcal{X}\}$.

(i) T is bounded iff T is closed.
(ii) T is closed iff $\Gamma(T)$ is closed in $\mathcal{X} \times \mathcal{Y}$.

Proof. On $\mathcal{X} \times \mathcal{Y}$, use $\|(x, y)\| := \|x\| + \|y\|$.

(i) ($\Rightarrow$) If T is bounded, $x_n \to x$ and $Tx_n \to y$ imply $Tx_n \to Tx$; hence $y = Tx$, so T is closed.
(i) ($\Leftarrow$) Assume T is closed and set $\|x\|_* := \|x\| + \|Tx\|$. If (x_n) is $\|\cdot\|_*$-Cauchy, then $x_n \to x$ in $\mathcal{X}$ and $Tx_n \to y$ in $\mathcal{Y}$; by closedness, $y = Tx$, so $x_n \to x$ in $\|\cdot\|_*$. Thus $(\mathcal{X}, \|\cdot\|_*)$ is Banach.
Since $\|x\| \leq \|x\|_*$, by Corollary 5.4.5, $\exists C > 0$ such that $\|x\|_* \leq C \|x\|$; hence $\|Tx\| \leq C \|x\|$.
(ii) ($\Rightarrow$) If T is closed and $(x_n, Tx_n) \to (x, y)$, then $x_n \to x$ and $Tx_n \to y$, so $y = Tx$ and $(x, y) \in \Gamma(T)$.
(ii) ($\Leftarrow$) If $\Gamma(T)$ is closed, $x_n \to x$, and $Tx_n \to y$, then $(x_n, Tx_n) \to (x, y) \in \Gamma(T)$; hence $y = Tx$.

Structural Commentary: The graph norm turns closedness of the graph into completeness.
Key Insight: Graph-norm completeness and domination force equivalence with the original norm. □

Theorem 5.4.8. *Closed Complements and Continuous Projections*
Let V be a closed subspace of a Banach space $\mathcal{X}$. The following are equivalent.

(i) $\exists W \leq \mathcal{X}$ closed such that $\mathcal{X} = V \oplus W$.
(ii) $\exists P \in \mathcal{L}(\mathcal{X})$ such that $P^2 = P$ and $P(\mathcal{X}) = V$.

Proof.

(i) $\Rightarrow$ *(ii)* Let $\mathcal{X} = V \oplus W$ with W closed, and define $P(v + w) := v$. Then $P^2 = P$ and $P(\mathcal{X}) = V$.
If $x_n \to x$ and $Px_n \to v$, then $v \in V$ and $(I - P)x_n \to x - v \in W$; by uniqueness in $V \oplus W$, $Px = v$.
Thus P is closed, so $P \in \mathcal{L}(\mathcal{X})$ by Theorem 5.4.7.

(ii) $\Rightarrow$ *(i)* Let $P \in \mathcal{L}(\mathcal{X})$ with $P^2 = P$ and $P(\mathcal{X}) = V$. By Proposition 2.4.2, $\mathcal{X} = V \oplus \ker P$.
Since $\ker P = P^{-1}(\{\mathbf{0}\})$ and P is continuous, $\ker P$ is closed.

Structural Commentary: Closed complementary coordinates are exactly bounded projection coordinates.
Key Insight: The Closed Graph Theorem converts algebraic splitting into continuous splitting. □

Theorem 5.4.9. *Complemented Ranges Are Closed*
Let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$ act between Banach spaces. If $\operatorname{range}(T)$ has a closed linear complement in $\mathcal{Y}$, then it is closed.

Proof. Set $\mathcal{R} := T(\mathcal{X})$, $\mathcal{Y} = \mathcal{R} \oplus Y_0$, Y_0 closed. Put $Z := \mathcal{X} \times Y_0$, $\|(x, y_0)\|_Z := \|x\| + \|y_0\|$, and $S : Z \to \mathcal{Y}$, $S(x, y_0) := Tx + y_0$. Then Z is Banach, $S \in \mathcal{L}(Z, \mathcal{Y})$, and $S(Z) = \mathcal{Y}$.
By Theorem 5.4.1, $\exists M > 0$ such that $\forall y \in \mathcal{Y}\ \exists (x, y_0) \in Z$: $y = Tx + y_0$ and $\|x\| + \|y_0\| \le M \|y\|$. If $y \in \mathcal{R}$, by direct-sum uniqueness $y_0 = 0$; $\forall y \in \mathcal{R}\ \exists x \in \mathcal{X}$: $Tx = y$, $\|x\| \le M \|y\|$. By Theorem 5.4.1, $\mathcal{R}$ closed.
Structural Commentary: A closed complement transfers preimage control to the range summand.
Key Insight: Complementability supplies the missing uniform preimage estimate for closed range. □

Corollary 5.4.10. *Finite-Codimensional Operator Ranges Are Closed*
Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces and let $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$. If $\operatorname{codim}_{\mathcal{Y}} T(\mathcal{X}) < \infty$, then $T(\mathcal{X})$ is closed.

Proof. Set $\mathcal{R} := T(\mathcal{X})$. Since $\operatorname{codim}_{\mathcal{Y}} \mathcal{R} < \infty$, choose $Y_0 \le \mathcal{Y}$ with $\mathcal{Y} = \mathcal{R} \oplus Y_0$ and $\dim Y_0 < \infty$.
By Theorem 5.1.14, Y_0 is closed; by Theorem 5.4.9, $\mathcal{R}$ is closed.
Structural Commentary: Finite codimension produces a finite-dimensional closed complement.
Key Insight: Only finitely many missing codomain directions force the operator range to be closed. □

6 Weak Topologies, Compact Operators, and Spectral Theory

6.1 Weak Topologies and Duality

Weak topologies replace norm control by scalar tests. They are usually too weak to preserve distances, but strong enough to expose compactness phenomena that are invisible in norm topology.

Definition 6.1.1. *Dual Space, Weak Topology, and Weak-* Topology*
For a normed space $\mathcal{X}$ over $\mathbb{F}$, its continuous dual is $\mathcal{X}^* := \mathcal{L}(\mathcal{X}, \mathbb{F})$.

(i) The *weak topology* on $\mathcal{X}$ is the weakest topology making $\varphi : \mathcal{X} \to \mathbb{F}$ continuous $\forall \varphi \in \mathcal{X}^*$.
(ii) The *weak-* topology* on $\mathcal{X}^*$ is the weakest topology making the evaluation $\varphi \mapsto \varphi(x)$ continuous $\forall x \in \mathcal{X}$.
(iii) The *strong topology* is the norm topology.

For $m \in \mathbb{N}$, $\varepsilon > 0$, $\varphi_\bullet = (\varphi_1, \dots, \varphi_m) \in (\mathcal{X}^*)^m$, and $x_\bullet = (x_1, \dots, x_m) \in \mathcal{X}^m$:

(i) Weak: $N(x; \varphi_\bullet; \varepsilon) := \{y \in \mathcal{X} : |\varphi_j(y - x)| < \varepsilon\ \forall j\}$.
(ii) Weak-*: $N_*(\varphi; x_\bullet; \varepsilon) := \{\psi \in \mathcal{X}^* : |(\psi - \varphi)(x_j)| < \varepsilon\ \forall j\}$.

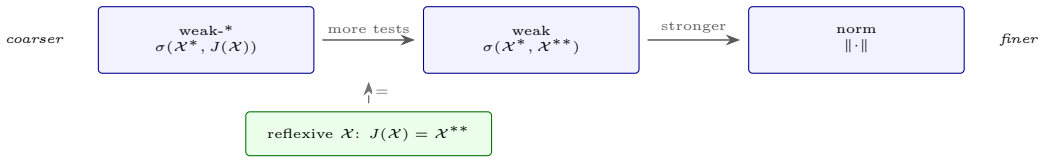

Figure 222: Topologies on $\mathcal{X}^*$: weak-* $\subseteq$ weak $\subseteq$ norm; reflexivity identifies the first two.

Definition 6.1.2. *Weak and Weak-* Convergence*

(i) $(x_n) \subseteq \mathcal{X}$ *converges weakly* to x, written $x_n \rightharpoonup x$, iff $\varphi(x_n) \to \varphi(x)\ \forall \varphi \in \mathcal{X}^*$.
(ii) $(\varphi_n) \subseteq \mathcal{X}^*$ *converges weak-** to $\varphi \in \mathcal{X}^*$ iff $\varphi_n(x) \to \varphi(x)\ \forall x \in \mathcal{X}$.

Proposition 6.1.3. *Strong Convergence Implies Weak Convergence*
If $x_n \to x$ in norm in a normed space $\mathcal{X}$, then $x_n \rightharpoonup x$.

Proof. $\forall \varphi \in \mathcal{X}^*$, $|\varphi(x_n) - \varphi(x)| = |\varphi(x_n - x)| \le \|\varphi\|_{\mathrm{op}} \|x_n - x\| \to 0$.
Structural Commentary: Apply an arbitrary bounded scalar test to the norm error.
Key Insight: Norm convergence controls every bounded linear test. □

Proposition 6.1.4. *Weakly Convergent Sequences Are Bounded*
If $x_n \rightharpoonup x$ in a normed space $\mathcal{X}$, then:

(i) $\sup_n \|x_n\| < +\infty$.
(ii) $\|x\| \le \liminf_{n \to \infty} \|x_n\|$.

Proof.

(i) $\forall \varphi \in \mathcal{X}^*$, the scalar sequence $(\varphi(x_n))$ is bounded. The space $\mathcal{X}^*$ is Banach by Theorem 5.1.6, so the Uniform Boundedness Principle, Theorem 5.3.7, applies to $Jx_n : \mathcal{X}^* \to \mathbb{F}$, $Jx_n(\varphi) := \varphi(x_n)$.
Since $\|Jx_n\|_{\mathrm{op}} = \|x_n\|$ by Corollary 5.3.3, the sequence is norm bounded.
(ii) If $x \neq 0$, choose $\varphi \in \mathcal{X}^*$ with $\|\varphi\|_{\mathrm{op}} = 1$ and $\varphi(x) = \|x\|$ by Corollary 5.3.2.
Then $\|x\| = \lim_{n \to \infty} |\varphi(x_n)| \le \liminf_{n \to \infty} \|x_n\|$. The case $x = 0$ is immediate.

Structural Commentary: Uniform boundedness proves (i); one norming functional proves (ii).
Key Insight: Scalar-test boundedness forces norm boundedness. □

Definition 6.1.5. *Reflexivity*
For a normed space $\mathcal{X}$, the canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$ is defined by $Jx(\varphi) := \varphi(x)$.
A Banach space $\mathcal{X}$ is *reflexive* if $J(\mathcal{X}) = \mathcal{X}^{**}$.

Theorem 6.1.6. *Banach–Alaoglu Theorem*
For a normed space $\mathcal{X}$, the closed unit ball $\overline{B}_{\mathcal{X}^*}$ is weak- compact.*

Proof. Let $B := \overline{B}_{\mathcal{X}}$, $B^* := \overline{B}_{\mathcal{X}^*}$, and $D := \{a \in \mathbb{F} : |a| \le 1\}$. By Theorem 6.3.12, D^B is compact.
The injection $R : B^* \hookrightarrow D^B$, $R\varphi = \varphi|_B$, identifies $\sigma(\mathcal{X}^*, \mathcal{X})|_{B^*}$ with pointwise convergence on B.
Let $f \in \overline{R(B^*)} \subseteq D^B$. If $x_j \in B$, $a_j \in \mathbb{F}$, and $\sum_j a_j x_j \in B$, then $f(\sum_j a_j x_j) = \sum_j a_j f(x_j)$ by passage from the identities for $R\varphi$.
Define $\Phi(0) := 0$ and $\Phi(x) := \|x\| f(x/\|x\|)$ for $x \neq 0$.
Then Φ is linear, $|\Phi(x)| \le \|x\|$, and $R\Phi = f$; hence $R(B^*)$ is closed in D^B. Therefore B^* is weak-* compact.
Structural Commentary: $B^* \hookrightarrow D^B$; closed linear constraints plus Tychonoff give compactness.
Key Insight: Weak-* topology records precisely the coordinates used in Tychonoff compactness. □

Remark 6.1.7. *Compactness and Sequences*
Weak-* compactness need not be weak-* sequential compactness. For $\mathcal{X} = \ell^\infty$, let $\varphi_n(x) := x_n$.
Then $\varphi_n \in \overline{B}_{\mathcal{X}^*}$, but for any subsequence (n_k), choosing $x \in \ell^\infty$ with $x_{n_k} = (-1)^k$ makes $\varphi_{n_k}(x) = (-1)^k$, so no subsequence is weak-* convergent. Helly's theorem gives a sequential substitute when $\mathcal{X}$ is separable.

Theorem 6.1.8. *Helly Selection for Bounded Dual Sequences*
Let $\mathcal{X}$ be a separable normed space. If $(T_n) \subseteq \mathcal{X}^*$ and $\sup_n \|T_n\|_{op} < +\infty$, then $\exists$ a subsequence (T_{n_k}) and $T \in \mathcal{X}^*$ such that $\forall x \in \mathcal{X}$, $T_{n_k}(x) \to T(x)$.

Proof. Let $D = \{x_j : j \ge 1\} \subseteq \mathcal{X}$ be dense and set $M := \sup_n \|T_n\|_{\mathrm{op}}$.
If $M = 0$, take $T = 0$; assume $M > 0$. By diagonal extraction, choose (n_k) such that $T_{n_k} x_j$ converges $\forall j$.
For $x \in \mathcal{X}$ and $\varepsilon > 0$, choose j with $\|x - x_j\| < \varepsilon/(4M)$ and then k, ℓ large with $|T_{n_k} x_j - T_{n_\ell} x_j| < \varepsilon/2$.
Then $|T_{n_k} x - T_{n_\ell} x| \le 2M \|x - x_j\| + |T_{n_k} x_j - T_{n_\ell} x_j| < \varepsilon$, so $(T_{n_k} x)$ is Cauchy. Define $Tx := \lim_k T_{n_k} x$.
Linearity passes to limits, and $|Tx| \le M \|x\|$; hence $T \in \mathcal{X}^*$ and $T_{n_k} x \to Tx$ $\forall x \in \mathcal{X}$.
Structural Commentary: Diagonalize on dense D; use $M = \sup_n \|T_n\|_{\mathrm{op}}$ to extend D-limits to $\mathcal{X}$.
Key Insight: A countable dense set reduces pointwise convergence to diagonal convergence. □

Corollary 6.1.9. *Embedding into a Continuous Function Space*
For every normed space $\mathcal{X}$, $\exists$ compact Hausdorff K and a linear isometry $\Phi : \mathcal{X} \to C(K)$.

Proof. Let $K := \overline{B}_{\mathcal{X}^*}$ with the weak-* topology. By Theorem 6.1.6, K is compact; since evaluations separate points of K, K is Hausdorff. Define $\Phi : \mathcal{X} \to C(K)$ by $\Phi(x)(\varphi) := \varphi(x)$. Each $\Phi(x)$ is weak-* continuous, Φ is linear, and by Corollary 5.3.3, $\|\Phi(x)\|_\infty = \|x\|$.
Structural Commentary: $K = \overline{B}^{w^*}_{\mathcal{X}^*}$ turns scalar tests into compact coordinates.
Key Insight: Evaluation turns each vector into a continuous function without changing its norm. □

Proposition 6.1.10. *Natural Embedding and Weak Topologies*
For a normed space $\mathcal{X}$, the canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$ is a homeomorphism $(\mathcal{X}, w) \to (J(\mathcal{X}), w^*|_{J(\mathcal{X})})$.

Proof. By Corollary 5.3.3, J is injective. Fix $x_0 \in \mathcal{X}$, $\varphi_\bullet = (\varphi_1, \dots, \varphi_m) \in (\mathcal{X}^*)^m$, and $\varepsilon > 0$.
Since $(Jx - Jx_0)(\varphi_j) = \varphi_j(x - x_0)$, $J^{-1}(N_*(Jx_0; \varphi_\bullet; \varepsilon) \cap J(\mathcal{X})) = N(x_0; \varphi_\bullet; \varepsilon)$.
Thus J and $J^{-1} : J(\mathcal{X}) \to \mathcal{X}$ are continuous.
Structural Commentary: Pullbacks of weak-* tests are weak tests.
Key Insight: J preserves precisely the scalar tests that define the weak topology. □

Theorem 6.1.11. *Kakutani's Weak Compactness Criterion*
For a Banach space $\mathcal{X}$, the following are equivalent.
(i) $\mathcal{X}$ is reflexive.
(ii) The closed unit ball $\overline{B}_{\mathcal{X}}$ is weakly compact.

Proof.
(*i*) $\Rightarrow$ (*ii*) If $J(\mathcal{X}) = \mathcal{X}^{**}$, then by Corollary 5.3.3, $J(\overline{B}_{\mathcal{X}}) = \overline{B}_{\mathcal{X}^{**}}$.
By Theorem 6.1.6 applied to $\mathcal{X}^*$, $\overline{B}_{\mathcal{X}^{**}}$ is weak-* compact. By Proposition 6.1.10, $\overline{B}_{\mathcal{X}}$ is weakly compact.
(*ii*) $\Rightarrow$ (*i*) By Proposition 6.1.10, $K := J(\overline{B}_{\mathcal{X}})$ is weak-* compact, hence weak-* closed, in $\mathcal{X}^{**}$.
If $F \in \overline{B}_{\mathcal{X}^{**}} \setminus K$, weak-* separation yields $\varphi \in \mathcal{X}^*$ with $\operatorname{Re} F(\varphi) > \sup_{\|x\| \le 1} \operatorname{Re} Jx(\varphi) = \|\varphi\|_{\mathrm{op}}$.
But $\operatorname{Re} F(\varphi) \le |F(\varphi)| \le \|F\|_{\mathrm{op}} \|\varphi\|_{\mathrm{op}} \le \|\varphi\|_{\mathrm{op}}$, a contradiction. Thus $\overline{B}_{\mathcal{X}^{**}} \subseteq K$.
For $F \in \mathcal{X}^{**}$ with $F \neq 0$, $F/\|F\|_{\mathrm{op}} \in K$, so $F \in J(\mathcal{X})$; hence $J(\mathcal{X}) = \mathcal{X}^{**}$.

Structural Commentary: Reflexivity identifies $J(\overline{B}_{\mathcal{X}})$ with $\overline{B}_{\mathcal{X}^{**}}$; weak-* separation forces the converse.
Key Insight: Reflexivity identifies the weak-* compact bidual ball with the embedded original ball. □

Corollary 6.1.12. *Closed Bounded Convex Sets in Reflexive Spaces*
Let $\mathcal{X}$ be reflexive Banach. If $C \subseteq \mathcal{X}$ is norm-closed, bounded, and convex, then C is weakly compact.

Proof. If $C = \emptyset$, the claim is trivial. Choose $r > 0$ with $C \subseteq r\overline{B}_{\mathcal{X}}$. By Theorem 6.1.11, $r\overline{B}_{\mathcal{X}}$ is weakly compact.
For $x_0 \notin C$, by Hahn–Banach separation, $\varphi \in \mathcal{X}^*$ and $\alpha \in \mathbb{R}$ with $\operatorname{Re}\varphi(x_0) > \alpha \ge \sup_{c \in C} \operatorname{Re}\varphi(c)$; hence $\{\operatorname{Re}\varphi > \alpha\}$ is weakly open, contains x_0, and misses C. Thus C is weakly closed, so C is weakly compact. *Structural Commentary*: $C \subseteq r\overline{B}_{\mathcal{X}}$ and C weakly closed inside a weakly compact ball.
Key Insight: Convex norm-closed sets remain closed when the topology is weakened to scalar tests. □

Theorem 6.1.13. *Goldstine's Theorem*
For a normed space $\mathcal{X}$, the weak- closure of $J(\overline{B}_{\mathcal{X}})$ in $\mathcal{X}^{**}$ is $\overline{B}_{\mathcal{X}^{**}}$.*

Proof. Set $C := \overline{J(\overline{B}_{\mathcal{X}})}^{\sigma(\mathcal{X}^{**},\mathcal{X}^{*})}$. J is isometric by Corollary 5.3.3, $\overline{B}_{\mathcal{X}^{**}} = \bigcap_{\varphi\in\overline{B}_{\mathcal{X}^{*}}} \{F \in \mathcal{X}^{**} : |F(\varphi)| \le 1\}$ is weak-* closed, hence $C \subseteq \overline{B}_{\mathcal{X}^{**}}$. Suppose $F \in \overline{B}_{\mathcal{X}^{**}} \setminus C$. By weak-* Hahn–Banach separation,
$\exists \varphi \in \mathcal{X}^{*}$ and $\alpha \in \mathbb{R}$ such that $\operatorname{Re} F(\varphi) > \alpha \ge \sup_{\|x\|\le 1} \operatorname{Re} Jx(\varphi) = \sup_{\|x\|\le 1} \operatorname{Re}\varphi(x) = \|\varphi\|_{\mathrm{op}}$.
But $\operatorname{Re} F(\varphi) \le |F(\varphi)| \le \|F\|_{\mathrm{op}}\|\varphi\|_{\mathrm{op}} \le \|\varphi\|_{\mathrm{op}}$, a contradiction. Hence $\overline{B}_{\mathcal{X}^{**}} \subseteq C$.
Structural Commentary: $J(\overline{B}_{\mathcal{X}}) \subseteq \overline{B}_{\mathcal{X}^{**}}$; weak-* separation excludes $\overline{B}_{\mathcal{X}^{**}} \setminus C$.
Key Insight: $J(\overline{B}_{\mathcal{X}})$ is weak-* dense in $\overline{B}_{\mathcal{X}^{**}}$ even when $J\mathcal{X} \neq \mathcal{X}^{**}$. □

Theorem 6.1.14. *Eberlein–Smulian Theorem for Unit Balls*
Let $\mathcal{X}$ be Banach and set $B := \overline{B}_{\mathcal{X}}$. The following are equivalent:
(i) B is compact in $(\mathcal{X}, w)$.
(ii) Every $(x_n) \subseteq B$ has a subsequence $x_{n_k} \rightharpoonup x$ for some $x \in B$.

Proof. By Eberlein–Smulian, for bounded $A \subseteq \mathcal{X}$, A is relatively weakly compact iff every sequence in A has a weakly convergent subsequence with limit in $\overline{A}^{w}$. By Hahn–Banach separation, B is weakly closed, so $\overline{B}^{w} = B$.
$(i) \Rightarrow (ii)$ Apply the criterion with $A = B$.
$(ii) \Rightarrow (i)$ The criterion makes B relatively weakly compact; since $\overline{B}^{w} = B$, B is weakly compact.
Structural Commentary: For $B = \overline{B}_{\mathcal{X}}$, weak compactness $\iff$ weak subsequential compactness.
Key Insight: Weak compactness of B is sequence-detectable. □

Corollary 6.1.15. *Reflexive Unit-Ball Characterization*
Let $\mathcal{X}$ be Banach, set $B := \overline{B}_{\mathcal{X}}$, and let $J : \mathcal{X} \to \mathcal{X}^{**}$ be canonical. The following are equivalent:
(i) $J(\mathcal{X}) = \mathcal{X}^{**}$.
(ii) B is compact in $(\mathcal{X}, w)$.
(iii) Every $(x_n) \subseteq B$ has a subsequence $x_{n_k} \rightharpoonup x$ for some $x \in B$.

Proof.
$(i) \Rightarrow (ii)$ $J(\mathcal{X}) = \mathcal{X}^{**} \Rightarrow B$ is compact in $(\mathcal{X}, w)$ by Theorem 6.1.11.
$(ii) \Rightarrow (iii)$ By Theorem 6.1.14, B weakly compact implies every $(x_n) \subseteq B$ has $x_{n_k} \rightharpoonup x \in B$.
$(iii) \Rightarrow (i)$ By Theorem 6.1.14, (iii) implies B weakly compact; by Theorem 6.1.11, $J(\mathcal{X}) = \mathcal{X}^{**}$.
Structural Commentary: $J(\mathcal{X}) = \mathcal{X}^{**} \iff B_w$ compact $\iff B_w$ subsequentially compact.
Key Insight: Reflexivity is unit-ball compactness. □

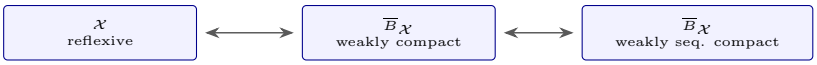

Figure 223: For Banach $\mathcal{X}$: $\mathcal{X}$ reflexive $\iff \overline{B}_{\mathcal{X}}$ weakly compact $\iff \overline{B}_{\mathcal{X}}$ weakly sequentially compact.

Proposition 6.1.16. *Hilbert Weak Convergence Tests*
Let $\mathcal{H}$ be a Hilbert space, let $(h_n), (u_n) \subseteq \mathcal{H}$, and let $h, u \in \mathcal{H}$.
(i) $h_n \rightharpoonup h$ if and only if $\forall u \in \mathcal{H}$, $\langle h_n, u\rangle \to \langle h, u\rangle$.
(ii) If $h_n \rightharpoonup h$ and $u_n \to u$ strongly, then $\langle h_n, u_n\rangle \to \langle h, u\rangle$.

Proof.
(i) $(\Rightarrow)$ For $u \in \mathcal{H}$, $\phi_u(x) := \langle x, u\rangle \in \mathcal{H}^{*}$, so $\langle h_n, u\rangle = \phi_u(h_n) \to \phi_u(h) = \langle h, u\rangle$.
(i) $(\Leftarrow)$ For $\phi \in \mathcal{H}^{*}$, $u_\phi \in \mathcal{H}$ with $\phi(x) = \langle x, u_\phi\rangle$ by Theorem 5.2.32; $\phi(h_n) = \langle h_n, u_\phi\rangle \to \langle h, u_\phi\rangle = \phi(h)$.
(ii) Let $C := \sup_n \|h_n\| < +\infty$ by Proposition 6.1.4(i).
Since $|\langle h_n, u_n\rangle - \langle h, u\rangle| \le C\|u_n - u\| + |\langle h_n - h, u\rangle|$, the limit is 0.
Structural Commentary: $\mathcal{H}^{*} = \{\langle \cdot, u\rangle : u \in \mathcal{H}\}$; boundedness controls $u_n - u$.
Key Insight: $h_n \rightharpoonup h \iff \langle h_n, u\rangle \to \langle h, u\rangle$ for all fixed $u \in \mathcal{H}$. □

Theorem 6.1.17. *Bounded Hilbert Sequences Have Weak Subsequences*
Let $\mathcal{H}$ be Hilbert and let $(h_n) \subseteq \mathcal{H}$ satisfy $C := \sup_n \|h_n\| < +\infty$. Then $\exists n_k \uparrow \infty$ and $h \in \mathcal{H}$ such that $h_{n_k} \rightharpoonup h$.

Proof. Set $M := \operatorname{cspan}\{h_n : n \ge 1\}$. Then M is a separable Hilbert space.
If $\dim M < +\infty$, then $C\overline{B}_M$ is compact by Theorem 5.1.14(iii); choose $h_{n_k} \to h \in M$ strongly, hence weakly.
Assume $\dim M = \infty$. Choose a Hilbert basis $(e_j)_{j\ge 1}$ of M by Corollary 5.2.28.
Diagonalize so $\langle h_{n_k}, e_j\rangle \to a_j$ $\forall j$. For $m \ge 1$, $\sum_{j=1}^{m} |a_j|^2 = \lim_k \sum_{j=1}^{m} |\langle h_{n_k}, e_j\rangle|^2 \le C^2$ by Theorem 5.2.22(ii).
Hence $a \in \ell^2$ and $h := \sum_j a_j e_j \in M$ by Theorem 5.2.21(i). If $v \in \operatorname{span}\{e_j : j \ge 1\}$, then $\langle h_{n_k} - h, v\rangle \to 0$.
For $u \in M$, $\varepsilon > 0$, $\|u - v\| < \varepsilon$ for v; then $\limsup_k |\langle h_{n_k} - h, u\rangle| \le (C + \|h\|)\varepsilon$, so $\langle h_{n_k} - h, u\rangle \to 0$.
For $u \in \mathcal{H}$, write $u = P_M u + u_\perp$, $u_\perp \in M^{\perp}$. Since $h_{n_k} - h \in M$, $\langle h_{n_k} - h, u\rangle = \langle h_{n_k} - h, P_M u\rangle \to 0$.
Thus $h_{n_k} \rightharpoonup h$ by Proposition 6.1.16.
Structural Commentary: Work in $M = \operatorname{cspan}\{h_n\}$; diagonal limits on (e_j) extend by $\overline{\operatorname{span}}\{e_j\} = M$ and $M^{\perp}$.
Key Insight: Bounded Hilbert sequences have countably coordinatized weak limits. □

Theorem 6.1.18. *Hilbert Spaces Are Reflexive*
For every Hilbert space $\mathcal{H}$, the canonical embedding $J : \mathcal{H} \to \mathcal{H}^{**}$ is onto.

Proof. Set $B := \overline{B}_{\mathcal{H}}$. If $(h_n) \subseteq B$, then by Theorem 6.1.17, $h_{n_k} \rightharpoonup h$ for some $h \in \mathcal{H}$.
By Proposition 6.1.4, $\|h\| \le \liminf_k \|h_{n_k}\| \le 1$, so $h \in B$.
Thus B is weakly subsequentially compact, hence $J(\mathcal{H}) = \mathcal{H}^{**}$ by Corollary 6.1.15.
Structural Commentary: B_w subsequentially compact $\Longrightarrow J(\mathcal{H}) = \mathcal{H}^{**}$.
Key Insight: Hilbert coordinates make B_w sequence-compact. □

Theorem 6.1.19. *Banach–Saks Theorem in Hilbert Spaces*

Let $\mathcal{H}$ be Hilbert and let $h_n \rightharpoonup h$. Then $\exists n_k \uparrow \infty$ such that $\frac{1}{k}\sum_{j=1}^{k} h_{n_j} \to h$ strongly.

Proof. Replace h_n by $h_n - h$; let $h = 0$. Put $C := \sup_n \|h_n\|^2 < +\infty$.

Let $n_1 = 1$, $n_1 < \cdots < n_k$, $s_k := \sum_{j=1}^{k} h_{n_j}$; $n_{k+1} > n_k$ with $\left|\langle s_k, h_{n_{k+1}}\rangle\right| \le 1$, since $\langle s_k, h_n\rangle \to 0$.

Then $\|s_{k+1}\|^2 = \|s_k\|^2 + 2\operatorname{Re}\langle s_k, h_{n_{k+1}}\rangle + \|h_{n_{k+1}}\|^2 \le \|s_k\|^2 + C + 2$, so $\|s_k\|^2 \le (C+2)k$.

Hence $\left\|\frac{1}{k}s_k\right\|^2 \le \frac{C+2}{k} \to 0$. For the original sequence, $\frac{1}{k}\sum_{j=1}^{k} h_{n_j} = h + \frac{1}{k}\sum_{j=1}^{k}(h_{n_j} - h) \to h$.

Structural Commentary: Choose n_{k+1} so $\left|\langle s_k, h_{n_{k+1}}\rangle\right| \le 1$; then $\|s_k\|^2 = O(k)$.
Key Insight: Weak-null subsequences can have $o(k)$ Cesàro norm. □

Theorem 6.1.20. *Radon–Riesz Theorem in Hilbert Spaces*
Let $\mathcal{H}$ be Hilbert and let $h_n \rightharpoonup h$. Then $h_n \to h$ in norm iff $\|h_n\| \to \|h\|$.

Proof.
($\Rightarrow$) If $\|h_n - h\| \to 0$, then $|\|h_n\| - \|h\|| \le \|h_n - h\| \to 0$.
($\Leftarrow$) Since $h_n \rightharpoonup h$, $\langle h_n, h\rangle \to \|h\|^2$. If also $\|h_n\| \to \|h\|$, then $\|h_n - h\|^2 = \|h_n\|^2 - 2\operatorname{Re}\langle h_n, h\rangle + \|h\|^2 \to 0$.

Structural Commentary: Weak convergence fixes $\langle h_n, h\rangle$; norm convergence fixes $\|h_n\|^2$.
Key Insight: $\|h_n - h\|^2$ has no remaining term. □

Theorem 6.1.21. *Metrizing Weak Topologies on the Unit Ball*
Let $\mathcal{X}$ be normed, $W \subseteq \mathcal{X}^*$ separable and point-separating $B := \overline{B}_{\mathcal{X}}$. Choose $(\varphi_k)_{k\ge1}$ dense in $W \cap \overline{B}_{\mathcal{X}^*}$. Then $\rho(u,v) := \sum_{k\ge1} 2^{-k}|\varphi_k(u-v)|$ is a metric on B inducing $\sigma(\mathcal{X}, W)|_B$.

Proof. For $u, v \in B$, $|\varphi_k(u-v)| \le 2$, hence $\rho(u,v) < +\infty$; symmetry and the triangle inequality are termwise.
If $\rho(u,v) = 0$, then $\varphi_k(u-v) = 0$ for all k; for $\psi \in W \setminus \{0\}$, density gives $\varphi_{k_j} \to \frac{\psi}{\|\psi\|}$, hence $\psi(u-v) = 0$.
Since W separates points, $u = v$, so ρ is a metric. Fix $u \in B$. It remains to compare neighborhoods at u.
($\Rightarrow$) Given $r > 0$, choose N with $\sum_{k>N} 2^{1-k} < \frac{r}{2}$ and $\delta > 0$ with $\delta \sum_{k\le N} 2^{-k} < \frac{r}{2}$.
Then $V := \{v \in B : |\varphi_k(v-u)| < \delta, 1 \le k \le N\}$ is a $\sigma(\mathcal{X}, W)|_B$-neighborhood of u.
By $v \in V$, $\rho(u,v) \le \delta \sum_{k\le N} 2^{-k} + \sum_{k>N} 2^{1-k} < r$, hence $V \subseteq B_\rho(u, r)$.
($\Leftarrow$) Let $U = \{v \in B : |\psi_j(v-u)| < \varepsilon, 1 \le j \le m\}$ with $\psi_j \in W$ and $\varepsilon > 0$. Put $K := \{j : \psi_j \ne 0\}$.
For $j \in K$, choose k_j with $\left\|\frac{\psi_j}{\|\psi_j\|} - \varphi_{k_j}\right\| < \frac{\varepsilon}{4\|\psi_j\|}$, then choose $r > 0$ with $2^{k_j} r < \frac{\varepsilon}{2\|\psi_j\|}$ $\forall j \in K$.
If $\rho(u,v) < r$, then $\left|\varphi_{k_j}(v-u)\right| < 2^{k_j} r$, $\|v-u\| \le 2$,
so $|\psi_j(v-u)| \le \|\psi_j\|\left|\varphi_{k_j}(v-u)\right| + \|\psi_j\|\left\|\frac{\psi_j}{\|\psi_j\|} - \varphi_{k_j}\right\|\|v-u\| < \varepsilon$ for $j \in K$.
For $j \notin K$, $\psi_j = 0$. Thus $B_\rho(u,r) \subseteq U$.
Therefore ρ induces $\sigma(\mathcal{X}, W)|_B$.
Structural Commentary: $\rho = \sum_{k\ge1} 2^{-k}|\varphi_k(\cdot)|$ metrizes the bounded test topology generated by W.
Key Insight: On B, dense countable tests determine all W-tests. □

Corollary 6.1.22. *Metrizability of Weak Unit Balls*
Let $\mathcal{X}$ be normed, write $B_{\mathcal{Y}} := \overline{B}_{\mathcal{Y}}$ for normed $\mathcal{Y}$, and let $J : \mathcal{X} \to \mathcal{X}^{**}$ be canonical.
(i) If $\mathcal{X}^*$ is separable, then $(B_{\mathcal{X}}, \sigma(\mathcal{X}, \mathcal{X}^*))$ is metrizable.
(ii) If $\mathcal{X}$ is separable, then $(B_{\mathcal{X}^*}, \sigma(\mathcal{X}^*, J(\mathcal{X})))$ is metrizable.
(iii) If $\mathcal{X}$ is reflexive Banach, then $(B_{\mathcal{X}}, \sigma(\mathcal{X}, \mathcal{X}^*))$ is metrizable iff $\mathcal{X}$ is separable.

Proof.
(i) $\mathcal{X}^*$ is separable and point-separating on $\mathcal{X}$: if $x \ne y$, then $\exists \varphi \in \mathcal{X}^*$ with $\varphi(x-y) \ne 0$.
Theorem 6.1.21 with $(\mathcal{Y}, W) = (\mathcal{X}, \mathcal{X}^*)$ metrizes $(B_{\mathcal{X}}, \sigma(\mathcal{X}, \mathcal{X}^*))$.
(ii) Since J is an isometry, $\mathcal{X}$ separable implies $J(\mathcal{X})$ separable.
If $\varphi \ne \psi$ in $\mathcal{X}^*$, choose $x \in \mathcal{X}$ with $(\varphi - \psi)(x) \ne 0$, hence $Jx(\varphi - \psi) \ne 0$, $J(\mathcal{X})$ point-separating on $\mathcal{X}^*$.
Theorem 6.1.21 with $(\mathcal{Y}, W) = (\mathcal{X}^*, J(\mathcal{X}))$ metrizes $(B_{\mathcal{X}^*}, \sigma(\mathcal{X}^*, J(\mathcal{X})))$.
(*iii*) ($\Rightarrow$) Let $B := B_{\mathcal{X}}$, B_w metrizable. By reflexivity, Theorem 6.1.11, B_w is compact metrizable, separable.
Choose countable $D \subseteq B$ with $\overline{D}^w = B$. Set $K := \overline{\operatorname{co} D}^{\|\cdot\|}$. If $K \ne B$, choose $x \in B \setminus K$.
Hahn–Banach gives $\varphi \in \mathcal{X}^*$ with $\operatorname{Re}\varphi(x) > \sup_{y\in K}\operatorname{Re}\varphi(y)$.
Since $D \subseteq K$ and $x \in \overline{D}^w$, weak continuity gives $\operatorname{Re}\varphi(x) \le \sup_{y\in K}\operatorname{Re}\varphi(y)$, contradiction.
Thus $B = K = \overline{\operatorname{co}_{\mathbb{Q}} D}^{\|\cdot\|}$; $\operatorname{co}_{\mathbb{Q}} D$ is countable, so B is norm separable, and $\mathcal{X} = \bigcup_{m\ge1} mB$ is separable.
(*iii*) ($\Leftarrow$) Assume $\mathcal{X}$ separable. By (ii), Theorem 6.1.6, $J(\mathcal{X}) = \mathcal{X}^{**}$, $B_{\mathcal{X}^*}$ compact metrizable in $\sigma(\mathcal{X}^*, \mathcal{X}^{**})$.
Pick countable dense $D \subseteq B_{\mathcal{X}^*}$. The ($\Rightarrow$) argument in $\mathcal{X}^*$ gives $B_{\mathcal{X}^*} = \overline{\operatorname{co}_{\mathbb{Q}} D}^{\|\cdot\|}$.
Since $\operatorname{co}_{\mathbb{Q}} D$ is countable, $B_{\mathcal{X}^*}$ is norm separable and $\mathcal{X}^* = \bigcup_{m\ge1} mB_{\mathcal{X}^*}$ is separable, so (i) applies.

Structural Commentary: W separable metrizes $B_{\sigma(\mathcal{X},W)}$; compact metrizable B_w yields $B = \overline{\operatorname{co}_{\mathbb{Q}} D}^{\|\cdot\|}$.
Key Insight: Reflexive $B_{\mathcal{X}}$ is weakly metrizable iff $\mathcal{X}$ is separable. □

Theorem 6.1.23. *Whole Weak Topologies Are Not Metrizable in Infinite Dimension*
Let $\mathcal{X}$ be Banach with $\dim \mathcal{X} = +\infty$.

(i) $(\mathcal{X}, \sigma(\mathcal{X}, \mathcal{X}^*))$ *is not metrizable.*
(ii) $(\mathcal{X}^*, \sigma(\mathcal{X}^*, J(\mathcal{X})))$ *is not metrizable.*

Proof.

(i) Suppose ρ metrizes $\sigma(\mathcal{X}, \mathcal{X}^*)$. $\forall n$, choose $U_n \subseteq B_\rho(0, \frac{1}{n})$ with $U_n = \{x : |\varphi_{n,j}(x)| < \varepsilon_n,\ 1 \le j \le m_n\}$.
Since $\bigcap_{j=1}^{m_n} \ker \varphi_{n,j} \neq \{0\}$, choose x_n in this kernel with $\|x_n\| = n$.
Then $x_n \in U_n$, so $\rho(x_n, 0) < \frac{1}{n}$ and $x_n \rightharpoonup 0$, contradicting Proposition 6.1.4(i).

(ii) Suppose ρ_* metrizes $\sigma(\mathcal{X}^*, J(\mathcal{X}))$. $\forall n$, choose finite-dim $M_n \le \mathcal{X}$ with $M_n^\perp := \{\varphi : \varphi|_{M_n} = 0\} \subseteq B_{\rho_*}(0, \frac{1}{n})$.
Choose $y_n \notin M_n$ and define ψ_n on $M_n + \operatorname{span}\{y_n\}$ by $\psi_n|_{M_n} = 0$ and $\psi_n(y_n) = 1$.
Extend ψ_n to $\mathcal{X}$ by Corollary 5.3.2(i), and rescale to $\varphi_n \in M_n^\perp$ with $\|\varphi_n\|_{\text{op}} = n$.
Then $\rho_*(\varphi_n, 0) < \frac{1}{n}$, so $\varphi_n \to 0$ weak-*. By Theorem 5.3.7, $(\|\varphi_n\|_{\text{op}})$ is bounded, a contradiction.

Structural Commentary: Finite tests leave nonzero kernels in $\mathcal{X}$ and nonzero annihilators in $\mathcal{X}^*$.
Key Insight: Metric convergence would force bounded norms; finite scalar tests cannot. □

Method 6.1.24. *Weak Compactness Proof Patterns*

(i) *Compactify*: replace $\|x\|$ by finite coordinates $(\varphi_1(x), \dots, \varphi_m(x))$.
(ii) *Embed*: use $J : \mathcal{X} \to \mathcal{X}^{**}$ and compare $\sigma(\mathcal{X}, \mathcal{X}^*)$ with $\sigma(\mathcal{X}^{**}, \mathcal{X}^*)|_{J(\mathcal{X})}$.
(iii) *Separate*: if $F \notin J(B)$, choose $\varphi \in \mathcal{X}^*$ separating F from $J(B)$.
(iv) *Metrize on balls*: for separable $W \le \mathcal{X}^*$, use $\rho(u, v) = \sum_{k \ge 1} 2^{-k} |\varphi_k(u - v)|$ on bounded B.
(v) *Hilbertize*: replace φ by $\langle \cdot, h \rangle$ and use orthogonal coordinates/projections.

Key Insight: Weak compactness is compactness for scalar coordinates $\varphi(x)$ on bounded sets.

6.2 Compact Operators

Compact operators carry bounded sets to relatively compact sets. Their spectral and solvability theory therefore resembles matrix theory inside infinite-dimensional spaces.

Definition 6.2.1. *Compact and Finite-Rank Operators*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces and let $T : \mathcal{X} \to \mathcal{Y}$ be linear.

(i) *Compact*: $\overline{T(B)}$ is compact in $\mathcal{Y}$ $\forall B \subseteq \mathcal{X}$ bounded.
(ii) *Finite rank*: $\dim \operatorname{range}(T) < \infty$.

Proposition 6.2.2. *Compactness Tests on Hilbert Spaces*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$, and set $B := \overline{B}_{\mathcal{H}}$. The following are equivalent:

(i) K *is compact.*
(ii) $K(B)$ *is totally bounded.*
(iii) Every bounded $(h_n) \subseteq \mathcal{H}$ has a subsequence (h_{n_j}) such that Kh_{n_j} converges in norm.

Proof. For bounded $E \subseteq \mathcal{H}$, choose $r > 0$ with $E \subseteq rB$; then $K(E) \subseteq K(rB) = rK(B)$.

(i) $\Rightarrow$ *(ii)* K compact $\Rightarrow$ $\overline{K(B)}$ compact $\Rightarrow$ $\overline{K(B)}$ totally bounded $\Rightarrow$ $K(B)$ totally bounded.
(ii) $\Rightarrow$ *(iii)* (h_n) bounded $\Rightarrow \exists r > 0$, $\{h_n : n \ge 1\} \subseteq rB$. Then $(Kh_n) \subseteq K(rB) = rK(B)$ with $rK(B)$ totally bounded, so $\exists (n_j)$ such that (Kh_{n_j}) is Cauchy; $\mathcal{H}$ complete $\Rightarrow \exists y \in \mathcal{H}$, $\|Kh_{n_j} - y\| \to 0$.
(iii) $\Rightarrow$ *(i)* Let $E \subseteq \mathcal{H}$ be bounded. If $K(E)$ is not totally bounded, then $\exists \varepsilon > 0$ and $(x_n) \subseteq E$ such that $n \neq m \Rightarrow \|Kx_n - Kx_m\| \ge \varepsilon$. By (iii), $\exists (n_j)$ and $y \in \mathcal{H}$ with $\|Kx_{n_j} - y\| \to 0$, contradiction. Hence $K(E)$ is totally bounded, so $\overline{K(E)}$ is totally bounded and complete; Theorem 4.1.30 gives $\overline{K(E)}$ compact.

Structural Commentary: $E \subseteq rB \Rightarrow K(E) \subseteq rK(B)$; $K(B)$ totally bounded $\Rightarrow (h_n) \subseteq rB \Rightarrow \exists (n_j) : (Kh_{n_j})$ Cauchy.
Key Insight: K compact iff K sends bounded sequences to norm-precompact sequences. □

Proposition 6.2.3. *Finite Rank Operators Are Compact*
Let $\mathcal{X}, \mathcal{Y}$ be normed, $T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$,$R := \operatorname{range}(T)$. If $\dim R < +\infty$, then T is compact.

Proof. Let $B \subseteq \mathcal{X}$ be bounded. Then $T(B) \subseteq R$ is bounded, so $\overline{T(B)}^{\,R}$ is compact by Theorem 5.1.14.
Since $R \le \mathcal{Y}$ is finite-dimensional, R is closed in $\mathcal{Y}$; hence $\overline{T(B)}^{\,\mathcal{Y}} = \overline{T(B)}^{\,R}$ is compact. Thus T is compact.
Structural Commentary: $T(B) \subseteq R$, $\dim R < +\infty$, and bounded subsets of R have compact closure.
Key Insight: Finite-dimensional range makes bounded images relatively compact. □

Theorem 6.2.4. *Norm Limits of Compact Operators*
Let $\mathcal{X}$ be normed, $\mathcal{Y}$ Banach, $T_n, T \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$. If each T_n compact and $\|T_n - T\|_{op} \to 0$, then T is compact.

Proof. Let $B \subseteq \mathcal{X}$ be bounded, $M > 0$, $\|x\| \le M$ for $x \in B$. For $\varepsilon > 0$, choose N with $\|T - T_N\|_{\text{op}} < \frac{\varepsilon}{2M}$.
Since $\overline{T_N(B)}$ is compact, choose $y_1, \dots, y_r \in \mathcal{Y}$ with $T_N(B) \subseteq \bigcup_{j=1}^r B_{\mathcal{Y}}(y_j, \frac{\varepsilon}{2})$.
Then $T(B) \subseteq \bigcup_{j=1}^r B_{\mathcal{Y}}(y_j, \varepsilon)$, so $T(B)$ is totally bounded. Since $\mathcal{Y}$ is complete, $\overline{T(B)}$ is compact; T is compact. *Structural Commentary*: $\|T - T_N\|_{\text{op}}$ transfers finite nets from $T_N(B)$ to $T(B)$ for bounded B.
Key Insight: Norm limits preserve relative compactness of bounded images. □

Proposition 6.2.5. *Compact Operators Form an Operator Ideal*
Let $\mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be normed spaces.

(i) If $K_1, \dots, K_m : \mathcal{X} \to \mathcal{Y}$ are compact, then $\sum_{j=1}^m K_j$ is compact.
(ii) If $K : \mathcal{X} \to \mathcal{Y}$ is compact, $A \in \mathcal{L}(\mathcal{Y}, \mathcal{Z})$, and $B \in \mathcal{L}(\mathcal{W}, \mathcal{X})$, then $AKB : \mathcal{W} \to \mathcal{Z}$ is compact.

Proof.

(i) Let $E \subseteq \mathcal{X}$ be bounded, $C_j := \overline{K_j(E)}$. Then each C_j is compact, and $\sum_{j=1}^{m} K_j(E) \subseteq \sigma(C_1 \times \cdots \times C_m)$ for the continuous map $\sigma(y_1, \ldots, y_m) := \sum_{j=1}^{m} y_j$. Thus $\sum_{j=1}^{m} K_j(E)$ is relatively compact.

(ii) Let $E \subseteq \mathcal{W}$ be bounded. Then $B(E)$ is bounded, $C := \overline{K(B(E))}$ is compact, and $AKB(E) \subseteq A(C)$. Since $A(C)$ is compact and closed in $\mathcal{Z}$, $\overline{AKB(E)} \subseteq A(C)$ is compact.

Structural Commentary: Products of compact image closures control sums; bounded precomposition and continuous postcomposition control AKB.
Key Insight: Compact operators form a two-sided ideal in $\mathcal{L}$. □

Example 6.2.6. *Diagonal Compact Operators*
Let $(e_k)_{k\geq 1}$ be a Hilbert basis of separable $\mathcal{H}$ and let $\lambda = (\lambda_k) \in \ell^\infty$ with $\lambda_k \to 0$.
For $h \in \mathcal{H}$, set $Th := \sum_{k=1}^{\infty} \lambda_k \langle h, e_k\rangle e_k$ and $T_N h := \sum_{k=1}^{N} \lambda_k \langle h, e_k\rangle e_k$.
By Parseval: $\sum_k |\lambda_k \langle h, e_k\rangle|^2 \leq \|\lambda\|_\infty^2 \|h\|^2 \Rightarrow T \in \mathcal{L}(\mathcal{H})$.
For $N \geq 1$, $\operatorname{range} T_N \subseteq \operatorname{span}\{e_1, \ldots, e_N\} \Rightarrow \operatorname{rank} T_N \leq N \Rightarrow T_N$ compact by Proposition 6.2.3.
Also $\|(T - T_N)h\|^2 = \sum_{k>N} |\lambda_k|^2 |\langle h, e_k\rangle|^2 \leq \sup_{k>N} |\lambda_k|^2 \|h\|^2 \Rightarrow \|T - T_N\|_{\text{op}} \leq \sup_{k>N} |\lambda_k| \to 0$.
Hence $T_N \to T$ in $\|\cdot\|_{\text{op}}$ and T is compact by Theorem 6.2.4.

Theorem 6.2.7. *Weak-to-Strong Characterization*
Let $\mathcal{H}$ be Hilbert, $K \in \mathcal{L}(\mathcal{H})$. Then K is compact iff $h_n \rightharpoonup h$ implies $\|Kh_n - Kh\| \to 0$ $\forall (h_n) \subseteq \mathcal{H}$ and $h \in \mathcal{H}$.

Proof.

($\Rightarrow$) Let $h_n \rightharpoonup h$. By Proposition 6.1.4(i), (h_n) bounded.
For every subsequence (n_j), By Proposition 6.2.2(iii), $\exists (j_\ell)$ and $y \in \mathcal{H}$ with $\left\|Kh_{n_{j_\ell}} - y\right\| \to 0$.
For $u \in \mathcal{H}$, $\langle y, u\rangle = \lim_\ell \left\langle Kh_{n_{j_\ell}}, u\right\rangle = \lim_\ell \left\langle h_{n_{j_\ell}}, K^* u\right\rangle = \langle h, K^* u\rangle = \langle Kh, u\rangle$, so $y = Kh$.
If $\|Kh_n - Kh\| \not\to 0$, then $\exists \varepsilon > 0$, (n_j) with $\left\|Kh_{n_j} - Kh\right\| \geq \varepsilon$, contradicting $\exists (j_\ell) : \left\|Kh_{n_{j_\ell}} - Kh\right\| \to 0$.

($\Leftarrow$) Let (h_n) be bounded. By Theorem 6.1.17, $\exists (n_j)$, $h \in \mathcal{H}$ with $h_{n_j} \rightharpoonup h$. By hypothesis, $\left\|Kh_{n_j} - Kh\right\| \to 0$.
Hence $\forall (h_n)$ bounded, $\exists (n_j)$ and $y \in \mathcal{H}$ with $\left\|Kh_{n_j} - y\right\| \to 0$; By Proposition 6.2.2(iii), K compact.

Structural Commentary: K compact, $h_n \rightharpoonup h \Rightarrow Kh_{n_{j_\ell}} \to y = Kh$; (h_n) bounded $\Rightarrow h_{n_j} \rightharpoonup h \Rightarrow Kh_{n_j} \to Kh$.
Key Insight: K compact iff $h_n \rightharpoonup h$ forces $Kh_n \to Kh$ in norm. □

Definition 6.2.8. *Orthogonal Projection Sequence*
Let $(e_k)_{k\geq 1}$ be a Hilbert basis for a separable Hilbert space $\mathcal{H}$.
The induced *orthogonal projection sequence* $(P_n)_{n\geq 1}$ is defined by:

(i) $P_n h := \sum_{k=1}^{n} \langle h, e_k\rangle e_k$ for $h \in \mathcal{H}$.
(ii) $\operatorname{range}(P_n) = \operatorname{span}\{e_1, \ldots, e_n\}$ and $\|P_n\|_{\text{op}} = 1$.
(iii) $P_n h \to h$ for every $h \in \mathcal{H}$.

Proposition 6.2.9. *Finite-Rank Approximation by Orthogonal Projections*
Let (P_n) be the orthogonal projection sequence induced by a Hilbert basis of a separable Hilbert space $\mathcal{H}$. For $T \in \mathcal{L}(\mathcal{H})$, T is compact iff $\|P_n T - T\|_{op} \to 0$.

Proof.

($\Rightarrow$) $B := \overline{B}_{\mathcal{H}}$, $C := \overline{T(B)}$; T compact $\Rightarrow C$ compact.
$\varepsilon > 0 \Rightarrow \exists y_1, \ldots, y_r \in C, N \in \mathbb{N} : C \subseteq \bigcup_{j=1}^{r} B_{\mathcal{H}}(y_j, \frac{\varepsilon}{3})$ and $n \geq N \Rightarrow \max_{1\leq j\leq r} \|(P_n - I)y_j\| < \frac{\varepsilon}{3}$.
Thus $n \geq N$, $x \in B \Rightarrow \exists j : \|Tx - y_j\| < \frac{\varepsilon}{3}$ and $\|(P_n T - T)x\| \leq \|Tx - y_j\| + \|(P_n - I)y_j\| + \|y_j - Tx\| < \varepsilon$.
Hence $\|P_n T - T\|_{\text{op}} = \sup_{x\in B} \|(P_n T - T)x\| \to 0$.

($\Leftarrow$) $\operatorname{range}(P_n T) \subseteq \operatorname{range}(P_n)$, $\dim \operatorname{range}(P_n) < \infty \Rightarrow P_n T$ compact by Proposition 6.2.3.
Also $P_n T \to T$ in $\|\cdot\|_{\text{op}}$; Theorem 6.2.4 $\Rightarrow T$ compact.

Structural Commentary: Compact $C = \overline{T(B)}$ upgrades $P_n y \to y$ to uniform convergence on C.
Key Insight: T compact iff finite-rank projections approximate T in operator norm. □

Corollary 6.2.10. *Finite-Rank Operators Are Dense in Compact Operators*
Let $\mathcal{H}$ be separable, and let $\mathcal{K}(\mathcal{H})$ and $\mathcal{F}(\mathcal{H})$ denote the compact and finite-rank operators in $\mathcal{L}(\mathcal{H})$.

(i) $\mathcal{K}(\mathcal{H})$ is a norm-closed linear subspace of $\mathcal{L}(\mathcal{H})$.
(ii) $\overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{op}} = \mathcal{K}(\mathcal{H})$.

Proof.

(i) Proposition 6.2.5: $K_1, K_2 \in \mathcal{K}(\mathcal{H})$, $\alpha, \beta \in \mathbb{F} \Rightarrow \alpha K_1 + \beta K_2 \in \mathcal{K}(\mathcal{H})$.
Theorem 6.2.4: $K_n \in \mathcal{K}(\mathcal{H})$, $\|K_n - K\|_{\text{op}} \to 0 \Rightarrow K \in \mathcal{K}(\mathcal{H})$.

(ii) ($\Rightarrow$) $\mathcal{F}(\mathcal{H}) \subseteq \mathcal{K}(\mathcal{H})$, (i) $\Rightarrow \overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{\text{op}}} \subseteq \mathcal{K}(\mathcal{H})$.

(ii) ($\Leftarrow$) $\dim \mathcal{H} < +\infty \Rightarrow \mathcal{F}(\mathcal{H}) = \mathcal{L}(\mathcal{H}) \supseteq \mathcal{K}(\mathcal{H})$. If $\dim \mathcal{H} = +\infty$,
take (P_n); $T \in \mathcal{K}(\mathcal{H}) \Rightarrow P_n T \in \mathcal{F}(\mathcal{H})$, $\|P_n T - T\|_{\text{op}} \to 0 \Rightarrow T \in \overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{\text{op}}}$ by Proposition 6.2.9.

Structural Commentary: $\mathcal{K}(\mathcal{H})$ is norm closed, and $P_nT \in \mathcal{F}(\mathcal{H})$ with $P_nT \to T$ for compact T.
Key Insight: On separable $\mathcal{H}$, $\mathcal{K}(\mathcal{H}) = \overline{\mathcal{F}(\mathcal{H})}^{\|\cdot\|_{\text{op}}}$. □

Theorem 6.2.11. *Schauder's Theorem*
Let $\mathcal{H}$ be Hilbert. If $K \in \mathcal{L}(\mathcal{H})$ is compact, then K^* is compact.

Proof. ByTheorem 5.2.34,$K^* \in \mathcal{L}(\mathcal{H})$. By Theorem 6.2.7, it suffices to prove $h_n \rightharpoonup h \Rightarrow \|K^*(h_n - h)\| \to 0$.
Put $u_n := h_n - h$. Then $u_n \rightharpoonup 0$, $M := \sup_n \|u_n\| < \infty$, and $\forall v \in \mathcal{H}$, $\langle K^*u_n, v\rangle = \langle u_n, Kv\rangle \to 0$, so $K^*u_n \rightharpoonup 0$.
Compactness of K gives $\|KK^*u_n\| \to 0$, hence $0 \le \|K^*u_n\|^2 = \langle KK^*u_n, u_n\rangle \le M\|KK^*u_n\| \to 0$.
Structural Commentary: $u_n \rightharpoonup 0 \Rightarrow K^*u_n \rightharpoonup 0$, K compact $\Rightarrow KK^*u_n \to 0$; $\|K^*u_n\|^2 \le M\|KK^*u_n\|$.
Key Insight: $\|K^*u_n\|^2 = \langle KK^*u_n, u_n\rangle$. □

Proposition 6.2.12. *Injective Compact Perturbations of the Identity Are Onto*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$ be compact, and set $A := I + K$. If $\ker A = \{0\}$, then $A(\mathcal{H}) = \mathcal{H}$.

Proof.
(i) If $\inf_{\|h\|=1}\|Ah\| = 0$, choose $\|h_n\| = 1$, $Ah_n \to 0$. By Proposition 6.2.2(iii), $Kh_{n_j} \to y$,
hence $h_{n_j} = Ah_{n_j} - Kh_{n_j} \to -y := h$, $\|h\| = 1$, and $Ah = 0$, contradiction.
Thus $\exists c > 0 : \|Ah\| \ge c\|h\|\ \forall h \in \mathcal{H}$.
(ii) Fix $g \in \mathcal{H}$ and put $M := \text{cspan}\{K^m g : m \ge 0\}$. Then $g \in M$, $K(M) \subseteq M$, and M is separable.
If $\dim M < +\infty$, then $A|_M : M \to M$ injective $\Rightarrow$ $A|_M$ surjective $\Rightarrow$ $g \in A(\mathcal{H})$.
(iii) Assume $\dim M = +\infty$. Let (P_n) be projection sequence of Hilbert basis of M,
$H_n := P_n(M)$, $\dim H_n < +\infty$. By Proposition 6.2.9, $\|P_nK - K\|_{\text{op}} \to 0$. Let N,
$n \ge N \Rightarrow \|P_nK - K\|_{\text{op}} < \frac{c}{2}$. Then $n \ge N$, $h \in H_n \Rightarrow \|(I + P_nK)h\| \ge \|Ah\| - \|(P_nK - K)h\| \ge \frac{c}{2}\|h\|$, so $I + P_nK : H_n \to H_n$
injective $\Rightarrow$ surjective.
(iv) Choose $u_n \in H_n$ with $(I + P_nK)u_n = P_ng$. Then $\|u_n\| \le \frac{2}{c}\|P_ng\| \le \frac{2}{c}\|g\|$.
By Theorem 6.1.17, $u_{n_j} \rightharpoonup u$ for some subsequence.
By Theorem 6.2.7, $Ku_{n_j} \to Ku$, $\|P_{n_j}Ku_{n_j} - Ku_{n_j}\| \le \|P_{n_j}K - K\|_{\text{op}}\|u_{n_j}\| \to 0$.
Since $P_{n_j}g \to g$, $u_{n_j} = P_{n_j}g - P_{n_j}Ku_{n_j} \to g - Ku$; by weak uniqueness $u = g - Ku$, hence $Au = g$.
Since $g \in \mathcal{H}$ was arbitrary, $A(\mathcal{H}) = \mathcal{H}$.
Structural Commentary: $\|Ah\| \ge c\|h\|$ and $(I + P_nK)u_n = P_ng \Rightarrow Au = g$.
Key Insight: Compactness removes approximate kernels and passes projected solutions to a true solution. □

Definition 6.2.13. *Fredholm Operator and Index*
Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{L}(\mathcal{H})$.
(i) *Fredholm*: $\text{range}(T)$ is closed, $\dim\ker T < \infty$, and $\text{codim}\,\text{range}(T) < \infty$.
(ii) *Index*: $\text{ind}\,T := \dim\ker T - \text{codim}\,\text{range}(T)$ when T is Fredholm.

Theorem 6.2.14. *Riesz–Schauder for Compact Perturbations of the Identity*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$ be compact, and set $A := I + K$. Then:
(i) $\dim\ker A < \infty$.
(ii) $\text{range}(A)$ *is closed.*
(iii) $\text{codim}\,\text{range}(A) = \dim\ker(A^*) = \dim\ker A < \infty$.
(iv) A is Fredholm with $\text{ind}\,A = 0$.
(v) A is injective iff it is surjective.

Proof.
(i) $\dim\ker A = +\infty \Rightarrow \exists(u_n) \subseteq \ker A$ with $\langle u_n, u_m\rangle = \delta_{nm}$.
Then $Ku_n = -u_n$ and $n \ne m \Rightarrow \|Ku_n - Ku_m\| = \sqrt{2}$, contradicting Proposition 6.2.2(iii).
(ii) Let $H_0 := (\ker A)^\perp$. If $\inf\{\|Ah\| : h \in H_0, \|h\| = 1\} = 0$, choose $h_n \in H_0$, $\|h_n\| = 1$, $Ah_n \to 0$.
By Proposition 6.2.2(iii), $Kh_{n_j} \to y$, so $h_{n_j} = Ah_{n_j} - Kh_{n_j} \to -y := h \in H_0$, $Ah = 0$.
Thus $h \in \ker A \cap H_0 = \{0\}$, contradiction. Hence $\exists c > 0 : \|Ah\| \ge c\|h\|$ on H_0.
Also $A(H_0) = A(\mathcal{H})$. If $Ah_n \to y$ with $h_n \in H_0$, then $\|h_n - h_m\| \le c^{-1}\|Ah_n - Ah_m\|$.
Thus $h_n \to h \in H_0$ and $Ah = y$; hence $\text{range}(A)$ is closed.
(iii) By Theorem 6.2.11, K^* compact, so (i)–(ii) apply to $A^* = I + K^*$.
By Proposition 5.2.39, $\mathcal{H} = \text{range}(A) \odot \ker(A^*)$, $\text{codim}\,\text{range}(A) = \dim\ker(A^*)$.
If $\dim\ker A < \dim\ker(A^*)$, choose injective $L : \ker A \to \ker(A^*)$ with $L(\ker A) \ne \ker(A^*)$, $F := LP_{\ker A}$.
Then $\text{rank}\,F < +\infty$, $K + F$ compact. For $x = z + n$, $z \in H_0$, $n \in \ker A$,
$(A + F)x = Az + Ln \in \text{range}(A) \odot L(\ker A)$; injectivity of L gives $\ker(A + F) = \{0\}$ and
$\text{range}(A + F) = \text{range}(A) \odot L(\ker A) \ne \mathcal{H}$, contradicting Proposition 6.2.12. Thus $\dim\ker A \ge \dim\ker(A^*)$.
Apply the same argument to A^* and use $(K^*)^* = K$ to get $\dim\ker A = \dim\ker(A^*)$.
(iv) (i)–(iii) give $\text{range}(A)$ closed, $\dim\ker A < +\infty$, $\text{codim}\,\text{range}(A) < +\infty$.
Also $\dim\ker A = \text{codim}\,\text{range}(A)$; hence A is Fredholm and $\text{ind}\,A = 0$.
(v) (iii) gives $\ker A = \{0\} \iff \text{codim}\,\text{range}(A) = 0 \iff \text{range}(A) = \mathcal{H}$.
Structural Commentary: Compactness $\Rightarrow \dim\ker A < \infty$, $\text{range}(A)$ closed; $F = LP_{\ker A} \Rightarrow \dim\ker A = \dim\ker(A^*)$.
Key Insight: For $A = I + K$ compactly perturbed, injectivity and surjectivity have the same finite defect. □

Corollary 6.2.15. *Fredholm Alternative*
Let $\mathcal{H}$ be Hilbert, let $K \in \mathcal{L}(\mathcal{H})$ be compact, let $\lambda \in \mathbb{F} \setminus \{0\}$, and set $T_\lambda := \lambda I - K$.
(i) $\text{range}(T_\lambda) = \mathcal{H}$ *iff* $\ker T_\lambda = \{0\}$.
(ii) If the conditions in (i) hold, then $T_\lambda^{-1} \in \mathcal{L}(\mathcal{H})$.

Proof. Let $\widetilde{K} := -\lambda^{-1}K$; $\widetilde{K}$ compact, $T_\lambda = \lambda(I + \widetilde{K})$, $\ker T_\lambda = \ker(I + \widetilde{K})$, $\text{range}(T_\lambda) = \mathcal{H}$ iff $\text{range}(I + \widetilde{K}) = \mathcal{H}$.
(i) ($\Rightarrow$) $\text{range}(T_\lambda) = \mathcal{H} \Rightarrow \text{range}(I + \widetilde{K}) = \mathcal{H} \Rightarrow \ker(I + \widetilde{K}) = \{0\}$ by Theorem 6.2.14(v), hence $\ker T_\lambda = \{0\}$.

(*i*) ($\Leftarrow$) $\ker T_\lambda = \{0\} \Rightarrow \ker(I + \widetilde{K}) = \{0\} \Rightarrow \operatorname{range}(I + \widetilde{K}) = \mathcal{H}$ by Theorem 6.2.14(v), hence $\operatorname{range}(T_\lambda) = \mathcal{H}$.
(*ii*) (i) $\Rightarrow T_\lambda$ bijective; Corollary 5.4.4 $\Rightarrow T_\lambda^{-1} \in \mathcal{L}(\mathcal{H})$.

Structural Commentary: $T_\lambda = \lambda(I + \widetilde{K})$ with $\widetilde{K}$ compact; bijective T_λ has bounded inverse.
Key Insight: $\forall f$, $T_\lambda u = f$ is solvable iff $\ker T_\lambda = \{0\}$. □

Theorem 6.2.16. *Fredholm Index Zero as an Invertible Plus Compact Operator*
Let $\mathcal{H}$ be Hilbert and let $T \in \mathcal{L}(\mathcal{H})$. The following are equivalent:

(i) T is Fredholm and $\operatorname{ind} T = 0$.
(ii) $\exists S, K \in \mathcal{L}(\mathcal{H})$ such that $T = S + K$, S is invertible, and K is compact.

Proof.

(*i*) $\Rightarrow$ (*ii*) $N := \ker T$, $R := \operatorname{range}(T)$, $Z := \ker(T^*)$.
By Proposition 5.2.39, $\mathcal{H} = R \odot Z$ and $\dim Z = \operatorname{codim} R$. $\operatorname{ind} T = 0 \Rightarrow \dim N = \dim Z < +\infty$.
$A : N \to Z$ isomorphism, P_N orthogonal projection, $K := AP_N$, $S := T - K$.
Then $\operatorname{rank} K < +\infty \Rightarrow K$ compact and $T = S + K$. For $h = v + u$, $v \in N^\perp$, $u \in N$, $Sh = Tv - Au \in R \odot Z$ and $Sh = 0 \Rightarrow Tv = 0$, $Au = 0 \Rightarrow v = 0$, $u = 0$; hence $\ker S = \{0\}$.
Also $T(N^\perp) = R$, $A(N) = Z$, and $y = r + z \in R \odot Z \Rightarrow \exists v \in N^\perp, u \in N : Tv = r, Au = -z, S(v + u) = y$; hence $\operatorname{range} S = \mathcal{H}$ and $S^{-1} \in \mathcal{L}(\mathcal{H})$ by Corollary 5.4.4.
(*ii*) $\Rightarrow$ (*i*) $C := S^{-1}K$, $A := I + C$. By Proposition 6.2.5(ii), C is compact and $T = SA$.
By Theorem 6.2.14(iv), A is Fredholm, $\operatorname{ind} A = 0$. S isomorphism $\Rightarrow \ker T = \ker A$, $\operatorname{range} T = S(\operatorname{range} A)$, $\operatorname{codim} \operatorname{range} T = \operatorname{codim} \operatorname{range} A$. Hence T is Fredholm and $\operatorname{ind} T = 0$.

Structural Commentary: $K = AP_{\ker T}$ makes $S = T - K$ invertible; $T = S(I + C)$ with C compact.
Key Insight: $\operatorname{ind} T = 0$ iff T is invertible modulo compact operators. □

Corollary 6.2.17. *Index-Zero Fredholm Operators Are Closed under Composition*
Let $\mathcal{H}$ be Hilbert, $S, T \in \mathcal{L}(\mathcal{H})$ be Fredholm with $\operatorname{ind} S = \operatorname{ind} T = 0$. Then ST is Fredholm and $\operatorname{ind}(ST) = 0$.

Proof. By Theorem 6.2.16, $S = S_0 + K_S$, $T = T_0 + K_T$, with S_0, T_0 invertible and K_S, K_T compact.
Let $U := S_0T_0$, $C := S_0K_T + K_ST_0 + K_SK_T$. Then $ST = U + C$, U invertible, by Proposition 6.2.5, C compact.
Theorem 6.2.16 $\Rightarrow ST$ Fredholm and $\operatorname{ind}(ST) = 0$.
Structural Commentary: $(S_0 + K_S)(T_0 + K_T) = S_0T_0 + C$, $C \in \mathcal{K}(\mathcal{H})$.
Key Insight: Invertible modulo compact operators are closed under products. □

Corollary 6.2.18. *Compact Perturbations Preserve Index-Zero Fredholm Operators*
Let $\mathcal{H}$ be Hilbert and let $T, K \in \mathcal{L}(\mathcal{H})$. If T is Fredholm, $\operatorname{ind} T = 0$, and K is compact, then:

(i) $T + K$ is Fredholm.
(ii) $\operatorname{ind}(T + K) = 0$.

Proof. By Theorem 6.2.16, $T = S + C$, S invertible, C compact. Set $\widetilde{C} := C + K$.
By Proposition 6.2.5, $\widetilde{C} \in \mathcal{K}(\mathcal{H})$ and $T + K = S + \widetilde{C}$.

(i) $T + K = S + \widetilde{C}$, S invertible, $\widetilde{C}$ compact $\Rightarrow T + K$ Fredholm by Theorem 6.2.16.
(ii) $T + K = S + \widetilde{C}$, S invertible, $\widetilde{C}$ compact $\Rightarrow \operatorname{ind}(T + K) = 0$ by Theorem 6.2.16.

Structural Commentary: $T = S + C$, $K \in \mathcal{K}(\mathcal{H}) \Rightarrow T + K = S + \widetilde{C}$, $\widetilde{C} = C + K \in \mathcal{K}(\mathcal{H})$.
Key Insight: ind = 0 is stable under compact perturbation. □

6.3 Spectral Theory of Operators

For bounded operators, spectrum replaces the finite-dimensional eigenvalue list. The resolvent tracks invertibility, while the spectrum records every obstruction to it.

Definition 6.3.1. *Spectrum and Resolvent*
Let $\mathcal{X}$ be a complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$.

(i) *Resolvent set*: $\rho(T) := \{\lambda \in \mathbb{C} : \lambda I - T$ is bijective and $(\lambda I - T)^{-1} \in \mathcal{L}(\mathcal{X})\}$.
(ii) *Spectrum*: $\sigma(T) := \mathbb{C} \setminus \rho(T)$.

Theorem 6.3.2. *Spectrum of a Complex Bounded Operator*
Let $\mathcal{X}$ be a nonzero complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$. Then:

(i) $\sigma(T) \neq \varnothing$;
(ii) $\sigma(T)$ is compact in $\mathbb{C}$;
(iii) $\sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq \|T\|_{op}\}$.

Proof. For $\lambda \in \rho(T)$, $R(\lambda) := (\lambda I - T)^{-1}$. If $\mu \in \rho(T)$, $|\lambda - \mu| \, \|R(\mu)\|_{\mathrm{op}} < 1$; $\lambda I - T = (I + (\lambda - \mu)R(\mu))(\mu I - T)$.
By Corollary 5.1.7, $\lambda \in \rho(T)$, $R(\lambda) = R(\mu)(I + (\lambda - \mu)R(\mu))^{-1}$. If $|\lambda| > \|T\|_{\mathrm{op}}$, then $\lambda I - T = \lambda(I - \lambda^{-1}T)$.
By Corollary 5.1.7, $\lambda \in \rho(T)$ and $\|R(\lambda)\|_{\mathrm{op}} \leq (|\lambda| - \|T\|_{\mathrm{op}})^{-1}$.

(i) Suppose $\sigma(T) = \varnothing$. By the local Neumann formula above, $R : \mathbb{C} \to \mathcal{L}(\mathcal{X})$ is entire.
For $x \in \mathcal{X}$ and $\Phi \in \mathcal{X}^*$, $f_{x,\Phi}(\lambda) := \Phi(R(\lambda)x)$ is an entire scalar function. By the estimate above, $f_{x,\Phi}(\lambda) \to 0$ as $|\lambda| \to +\infty$, and continuity bounds $f_{x,\Phi}$ on compact disks. By Liouville's theorem, $f_{x,\Phi} = 0$. By Corollary 5.3.3, bounded linear functionals separate points, so $R(0) = 0$, contradicting $(-T)R(0) = I$ on the nonzero space $\mathcal{X}$.
(ii) By the local Neumann formula, $\rho(T)$ is open, so $\sigma(T)$ is closed in $\mathbb{C}$.
By outside-disk formula, $\sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq \|T\|_{\mathrm{op}}\}$, hence $\sigma(T)$ is bounded. Thus $\sigma(T)$ is compact.
(iii) If $|\lambda| > \|T\|_{\mathrm{op}}$, then the outside-disk condition holds. By the outside-disk Neumann argument, $\lambda \in \rho(T)$.
Therefore every $\lambda \in \sigma(T)$ satisfies $|\lambda| \leq \|T\|_{\mathrm{op}}$.

Structural Commentary: By Neumann, $\rho(T)$ is open and $\sigma(T) \subseteq \{\lambda : |\lambda| \leq \|T\|_{\text{op}}\}$. By Liouville, $\sigma(T) \neq \varnothing$.
Key Insight: Complex Banach spectra are nonempty compact obstructions to invertibility. □

Definition 6.3.3. *Spectral Radius*
Let $\mathcal{X}$ be a nonzero complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$. The *spectral radius* of T is $r(T) := \sup_{\lambda \in \sigma(T)} |\lambda|$.

Theorem 6.3.4. *Spectral Radius Formula*
Let $\mathcal{X}$ be a nonzero complex Banach space and let $T \in \mathcal{L}(\mathcal{X})$.
Then the sequence $(\|T^n\|_{op}^{1/n})_{n\geq 1}$ converges in $[0,\infty)$, and $\lim_{n\to\infty} \|T^n\|_{op}^{1/n} = r(T)$.

Proof. Set $a_n := \|T^n\|_{\text{op}}$ for $n \geq 0$, so $a_0 = 1$ and $a_{m+n} \leq a_m a_n$. If $a_m = 0$ for some $m \geq 1$, then T is nilpotent.
Also $\|T^n\|_{\text{op}}^{1/n} \to 0$, $\sigma(T) = \{0\}$ by finite Neumann series and Theorem 6.3.2.
Hence assume $a_n > 0$ for all $n \geq 1$. For fixed $m \geq 1$ and $n = qm + r$ with $0 \leq r < m$, $a_n^{1/n} \leq a_m^{q/n} a_r^{1/n}$.
By $\limsup$ in n, $\limsup_n a_n^{1/n} \leq a_m^{1/m}$, hence $\limsup_n a_n^{1/n} \leq \inf_{m\geq 1} a_m^{1/m} \leq \liminf_n a_n^{1/n}$.
Thus $L := \lim_n a_n^{1/n} = \inf_{m\geq 1} a_m^{1/m}$. If $|\lambda| > L$, choose n with $a_n < |\lambda|^n$.
Then $\|T^n/\lambda^n\|_{\text{op}} < 1$, so Corollary 5.1.7 makes $\lambda^n I - T^n$ invertible.
With $Q_n := \sum_{j=0}^{n-1} \lambda^{n-1-j} T^j$, $\lambda^n I - T^n = (\lambda I - T)Q_n = Q_n(\lambda I - T)$; as these are polynomials in T,
$(\lambda I - T)^{-1} = Q_n(\lambda^n I - T^n)^{-1}$. Thus $\lambda \in \rho(T)$, so $\sigma(T) \subseteq \{z \in \mathbb{C} : |z| \leq L\}$ and $r(T) \leq L$.
Let $R > r(T)$, $R_0 > \max\{R, \|T\|_{\text{op}}\}$. For $|z| = R_0$, by Neumann expansion, $(zI - T)^{-1} = \sum_{k\geq 0} z^{-k-1} T^k$.
By Cauchy's coefficient formula $T^n = \frac{1}{2\pi i} \int_{|z|=R_0} z^n (zI - T)^{-1}\, dz$.
Since $\{z : R \leq |z| \leq R_0\} \subseteq \rho(T)$, Cauchy's theorem moves the contour to $|z| = R$.
With $M_R := \sup_{|z|=R} \left\|(zI - T)^{-1}\right\|_{\text{op}} < +\infty$, $a_n \leq M_R R^{n+1}$, hence $L \leq R$. By letting $R \downarrow r(T)$, $L \leq r(T)$.
Structural Commentary: By submultiplicativity, L exists; Neumann, $r(T) \leq L$; resolvent bounds, $L \leq r(T)$.
Key Insight: The spectral radius is asymptotic operator-power growth. □

Definition 6.3.5. *Quadratic Form and Rayleigh Quotient*
Let $\mathcal{H}$ be a real Hilbert space and let $T \in \mathcal{L}(\mathcal{H})$ be self-adjoint.

(i) *Quadratic form*: $Q_T(h) := \langle Th, h\rangle$ for $h \in \mathcal{H}$.
(ii) *Rayleigh quotient*: $R_T(h) := \frac{\langle Th,h\rangle}{\langle h,h\rangle}$ for $h \in \mathcal{H} \setminus \{0\}$.

A unit-sphere maximizer of Q_T is equivalently a maximizer of R_T on $\mathcal{H} \setminus \{0\}$.

Example 6.3.6. *A Self-Adjoint Operator without Eigenvectors*
Let $a < b$, set $H := L^2([a,b];\mathbb{C})$, and define $(Tf)(x) := xf(x)$.
With $M := \max\{|a|, |b|\}$, $\|Tf\|_2 \leq M \|f\|_2$, so $T \in \mathcal{L}(H)$.
For $u, v \in H$, $\langle Tu, v\rangle = \int_a^b xu(x)\overline{v(x)}\, dx = \int_a^b u(x)\overline{xv(x)}\, dx = \langle u, Tv\rangle$, hence $T = T^*$.
If $Tf = \lambda f$ for $\lambda \in \mathbb{C}$, then $(x - \lambda)f(x) = 0$ a.e. Let $E_\lambda := \{x \in [a,b] : x = \lambda\}$.
Since $m(E_\lambda) = 0$, $f = 0$ a.e. on $[a,b] \setminus E_\lambda$ and on E_λ, hence $f = 0$ in H.
Thus no nonzero $f \in H$ satisfies $Tf = \lambda f$, so T has no eigenvectors.

Lemma 6.3.7. *Hilbert–Schmidt Lemma*
Let $0 \neq \mathcal{H}$ be real Hilbert space, $T = T^* \in \mathcal{L}(\mathcal{H})$ compact, $Q_T(h) := \langle Th, h\rangle$. Then $\exists \lambda \in \mathbb{R}$, $h_* \in \mathcal{H}$ such that:

(i) $\|h_*\| = 1$;
(ii) $Th_* = \lambda h_*$;
(iii) $|\lambda| = \|T\|_{op} = \sup_{\|h\|=1} |Q_T(h)|$.

Proof. If $T = 0$, choose $\|h_*\| = 1$ and set $\lambda = 0$; then (i)–(iii) hold. Assume $T \neq 0$ and put $\eta := \|T\|_{\text{op}}$.
By Theorem 5.2.38, $\eta = \sup_{\|h\|=1} |Q_T(h)| > 0$. Choose $\|x_n\| = 1$ and $s_n \in \{-1, 1\}$ with $Q_{s_n T}(x_n) \to \eta$.
Passing to a subsequence, $s_n = s$; write $h_n := x_n$ and $A := sT$.
By Theorems 6.1.17 and 6.2.7, after passing to a further subsequence, $h_n \rightharpoonup h_*$ and $Ah_n \to Ah_*$ in norm.
Since $Q_A(h_n) - Q_A(h_*) = \langle Ah_n - Ah_*, h_n\rangle + \langle Ah_*, h_n - h_*\rangle \to 0$, $Q_A(h_*) = \eta$. Set $\lambda := s\eta$.

(i) By weak lower semicontinuity, $\|h_*\| \leq 1$.
Since $|Q_A(g)| \leq \eta \|g\|^2$ $\forall g \in \mathcal{H}$, $\eta = Q_A(h_*) \leq \eta \|h_*\|^2$. As $\eta > 0$, $\|h_*\| = 1$.
(ii) By (i), $R_A(h_*) = \eta$. For $\|h\| = 1$, $R_A(h) = Q_A(h) \leq |Q_T(h)| \leq \eta$, so h_* maximizes R_A.
For $h_0 \in \mathcal{H}$, $\phi(t) := R_A(h_* + th_0)$ has $\phi'(0) = 0$, hence $0 = 2\langle Ah_* - \eta h_*, h_0\rangle$.
Since h_0 is arbitrary, $Ah_* = \eta h_*$. Thus $Th_* = s\eta h_* = \lambda h_*$.
(iii) By the definition of λ and by Theorem 5.2.38, $|\lambda| = \eta = \|T\|_{\text{op}} = \sup_{\|h\|=1} |Q_T(h)|$.

Structural Commentary: By compactness, signed Q_T attains $\|T\|_{\text{op}}$; by stationarity, $Th_* = \lambda h_*$.
Key Insight: A compact self-adjoint norm maximizer is an eigenvector. □

Corollary 6.3.8. *Extremal Rayleigh Values Are Eigenvalues*
Let $\mathcal{H}$ be a nonzero real Hilbert space, let $T \in \mathcal{L}(\mathcal{H})$ be compact and self-adjoint, and set $Q_T(h) := \langle Th, h\rangle$, $\alpha := \inf_{\|h\|=1} Q_T(h)$, $\beta := \sup_{\|h\|=1} Q_T(h)$.

(i) If $\beta > 0$, then $\exists h_\beta \in \mathcal{H}$ such that $\|h_\beta\| = 1$ and $Th_\beta = \beta h_\beta$.
(ii) If $\alpha < 0$, then $\exists h_\alpha \in \mathcal{H}$ such that $\|h_\alpha\| = 1$ and $Th_\alpha = \alpha h_\alpha$.
(iii) If $\beta > 0$ and $(h_n) \subset \mathcal{H}$ satisfies $\|h_n\| = 1$, $Q_T(h_n) \to \beta$, then $\exists n_j \uparrow \infty$ and $h_\beta \in \mathcal{H}$ with $h_{n_j} \to h_\beta$ in norm, $\|h_\beta\| = 1$, and $Th_\beta = \beta h_\beta$.

Proof.

(i) Assume $\beta > 0$. Choose $\|u_n\| = 1$ with $Q_T(u_n) \to \beta$.
By Theorems 6.1.17 and 6.2.7, after passing to a subsequence, $u_n \rightharpoonup u_*$ and $Tu_n \to Tu_*$ in norm.
Since $Q_T(u_n) - Q_T(u_*) = \langle Tu_n - Tu_*, u_n\rangle + \langle Tu_*, u_n - u_*\rangle \to 0$, $Q_T(u_*) = \beta$.
By weak lower semicontinuity, $\|u_*\| \le 1$. As $Q_T(g) \le \beta\|g\|^2\ \forall g \in \mathcal{H}$, $\beta = Q_T(u_*) \le \beta\|u_*\|^2$; $\|u_*\| = 1$.
Thus u_* maximizes R_T. For $h \in \mathcal{H}$, $\phi(t) := R_T(u_* + th)$ has $\phi'(0) = 0$, so $0 = 2\langle Tu_* - \beta u_*, h\rangle$.
Hence $Tu_* = \beta u_*$; take $h_\beta := u_*$.

(ii) Assume $\alpha < 0$. Apply (i) to $-T$. Its upper Rayleigh value is $\beta_{-T} = \sup_{\|h\|=1}\langle -Th, h\rangle = -\alpha > 0$, so $\exists h_\alpha$ with $\|h_\alpha\| = 1$ and $(-T)h_\alpha = (-\alpha)h_\alpha$. Hence $Th_\alpha = \alpha h_\alpha$.

(iii) Assume $\beta > 0$, $\|h_n\| = 1$, $Q_T(h_n) \to \beta$. By compactness step in (i), $h_{n_j} \rightharpoonup h_\beta$, $Th_{n_j} \to Th_\beta$ in norm.
As in (i), $Q_T(h_\beta) = \beta$, $\|h_\beta\| = 1$, and $Th_\beta = \beta h_\beta$.
By Theorem 6.1.20, since $\|h_{n_j}\| = \|h_\beta\| = 1$, $h_{n_j} \to h_\beta$ in norm.

Structural Commentary: By compactness, attain β; by $-T$, attain α; by Radon–Riesz, $\rightharpoonup + \|\cdot\| \to 1 \Rightarrow \to$.
Key Insight: Compact self-adjoint Rayleigh extrema are eigenvalues. □

Theorem 6.3.9. *Hilbert–Schmidt Theorem for Compact Self-Adjoint Operators*
Let $\mathcal{H}$ be a real Hilbert space and let $K \in \mathcal{L}(\mathcal{H})$ be compact and self-adjoint.
If $\operatorname{rank} K = +\infty$, then $\exists$ orthonormal sequence $(\psi_k)_{k\ge1} \subset \mathcal{H}$ and nonzero scalars $(\lambda_k)_{k\ge1} \subset \mathbb{R}$ such that:

(i) $K\psi_k = \lambda_k\psi_k$ *for all* $k \ge 1$, $|\lambda_{k+1}| \le |\lambda_k|$, *and* $\lambda_k \to 0$;
(ii) $\operatorname{cspan}\{\psi_k : k \ge 1\} = (\ker K)^\perp$;
(iii) $Kh = \sum_{k\ge1} \lambda_k \langle h, \psi_k\rangle \psi_k$ *for every* $h \in \mathcal{H}$, *with norm convergence.*

If $\operatorname{rank} K < +\infty$, then:

(iv) $\exists m \ge 0$, *nonzero* $\lambda_1, \dots, \lambda_m \in \mathbb{R}$, *and an orthonormal basis* $(\psi_k)_{k=1}^m$ *of* $\operatorname{range} K = (\ker K)^\perp$ *such that* $K\psi_k = \lambda_k\psi_k$ *and* $Kh = \sum_{k=1}^m \lambda_k \langle h, \psi_k\rangle \psi_k$ *for every* $h \in \mathcal{H}$.

Proof.

(i) Put $M_0 := \{0\}$ and $H_0 := \mathcal{H}$. If $K|_{H_j} \ne 0$, then H_j is K-invariant and $K|_{H_j}$ is compact self-adjoint.
By Lemma 6.3.7, $\psi_{j+1} \in H_j$, $\lambda_{j+1} \ne 0$ with $\|\psi_{j+1}\| = 1$, $K\psi_{j+1} = \lambda_{j+1}\psi_{j+1}$, and $|\lambda_{j+1}| = \left\|K|_{H_j}\right\|_{\mathrm{op}}$.
Set $M_{j+1} := M_j \oplus \operatorname{span}\{\psi_{j+1}\}$, $H_{j+1} := M_{j+1}^\perp$. Since $K(M_{j+1}) \subseteq M_{j+1}$, $K = K^*$, $K(H_{j+1}) \subseteq H_{j+1}$.
Thus the construction continues while $K|_{H_j} \ne 0$, and $|\lambda_{j+1}| \le |\lambda_j|$.
If $\operatorname{rank} K = +\infty$, it cannot stop, since $K|_{H_m} = 0$ implies $\operatorname{range} K \subseteq M_m$.
If $\lambda_k \not\to 0$, then for some $\varepsilon > 0$, $\|K\psi_j - K\psi_k\|^2 = \lambda_j^2 + \lambda_k^2 \ge 2\varepsilon^2$ for infinitely many $j \ne k$, contradiction.

(ii) Set $M := \operatorname{cspan}\{\psi_k : k \ge 1\}$. Since $K(M) \subseteq M$ and $K = K^*$, $K(M^\perp) \subseteq M^\perp$.
If $K|_{M^\perp} \ne 0$, by Lemma 6.3.7 choose $\phi \in M^\perp$, $\mu \ne 0$ with $\|\phi\| = 1$ and $K\phi = \mu\phi$.
Then $\phi \in H_{k-1}\ \forall k$, so $|\mu| \le \left\|K|_{H_{k-1}}\right\|_{\mathrm{op}} = |\lambda_k| \to 0$, contradiction. Hence $M^\perp \subseteq \ker K$.
Conversely, $x \in \ker K$ implies $\langle x, \psi_k\rangle = \lambda_k^{-1}\langle Kx, \psi_k\rangle = 0\ \forall k$, so $\ker K \subseteq M^\perp$.
Thus $M^\perp = \ker K$ and $M = (\ker K)^\perp$ by Theorem 5.2.12.

(iii) For $h \in \mathcal{H}$, write $h = m + z$ with $m \in M = (\ker K)^\perp$ and $z \in \ker K$. By Theorem 5.2.22,
$m = \sum_{k\ge1} \langle h, \psi_k\rangle \psi_k$ in norm. By continuity of K, $Kh = Km = \sum_{k\ge1} \lambda_k \langle h, \psi_k\rangle \psi_k$ in norm.

(iv) If $\operatorname{rank} K < +\infty$, the construction stops at some $m \ge 0$; otherwise $\{\psi_k\} \subset \operatorname{range} K$ is an infinite orthonormal set in finite-dimensional $\operatorname{range} K$. Let $M_m := \operatorname{span}\{\psi_1, \dots, \psi_m\}$.
Since $K|_{M_m^\perp} = 0$, $\operatorname{range} K \subseteq M_m$; since $\psi_k = \lambda_k^{-1}K\psi_k$, $M_m \subseteq \operatorname{range} K$. Hence $M_m = \operatorname{range} K$.
Also $M_m^\perp = \ker K$ by argument in (ii), so $\operatorname{range} K = (\ker K)^\perp$. For $h = m_0 + z$ with $m_0 \in M_m$, $z \in \ker K$,
by the finite orthonormal expansion of m_0, $Kh = Km_0 = \sum_{k=1}^m \lambda_k \langle h, \psi_k\rangle \psi_k$.

Structural Commentary: Iterated extremal eigenvectors exhaust $(\ker K)^\perp$; Fourier expansion diagonalizes K.
Key Insight: Nonzero compact self-adjoint eigendirections are finite or countable, accumulating only at 0. □

Perspective 6.3.10. *Hilbert Operator Spectral Theory*
On a complex Hilbert space $\mathcal{H}$, normal $T \in \mathcal{L}(\mathcal{H})$ is diagonalized by spectral functional calculus.
By Theorem 6.3.9, compact self-adjoint $K \in \mathcal{L}(\mathcal{H})$ has nonzero eigenvalues (λ_k) and an orthonormal eigenfamily (ψ_k) spanning $(\ker K)^\perp$, so $\mathcal{H} = \ker K \oplus \operatorname{cspan}\{\psi_k\}$ and $Kh = \sum_k \lambda_k \langle h, \psi_k\rangle \psi_k\ \forall h \in \mathcal{H}$.
The sum is finite iff $\operatorname{rank} K < +\infty$; otherwise $\lambda_k \to 0$.

6.4 Measure Duality and Riesz Representation

On a compact Hausdorff space, Riesz–Markov turns positive functionals on $C(K)$ into Radon measures. The remaining step is order-theoretic: every bounded functional is the difference of two positive functionals, exactly as a signed measure is the difference of its Jordan variations.

Proposition 6.4.1. *Positive Functionals on $C(K)$ Are Continuous*
Let K be compact Hausdorff, $C(K) = C(K;\mathbb{R})$, $\|\cdot\|_\infty$, $L : C(K) \to \mathbb{R}$ linear with $L(C(K)_+) \subseteq [0, +\infty)$. Then:

(i) $L \in C(K)^*$.
(ii) $\|L\|_{op} = L(\mathbf{1}_K)$.

Proof.

(i) For $f \in C(K)$, $-\|f\|_\infty \mathbf{1}_K \le f \le \|f\|_\infty \mathbf{1}_K$; $-L(\mathbf{1}_K)\|f\|_\infty \le L(f) \le L(\mathbf{1}_K)\|f\|_\infty$; $|L(f)| \le L(\mathbf{1}_K)\|f\|_\infty$.
(ii) By (i), $\|L\|_{\mathrm{op}} \le L(\mathbf{1}_K)$. If $K = \varnothing$, both sides are 0; if $K \ne \varnothing$, then $\|\mathbf{1}_K\|_\infty = 1$, so $\|L\|_{\mathrm{op}} \ge L(\mathbf{1}_K)$.

Structural Commentary: $-\|f\|_\infty \mathbf{1}_K \le f \le \|f\|_\infty \mathbf{1}_K$ yields $|L(f)| \le L(\mathbf{1}_K)\,\|f\|_\infty$.
Key Insight: Positivity plus $\mathbf{1}_K$ controls $\|L\|_{\mathrm{op}}$. □

Proposition 6.4.2. *Positive Decomposition of Functionals on $C(K)$*
Let K be compact Hausdorff and let $L \in C(K)^*$. Then $\exists L_+, L_- \in C(K)^*$ positive such that:
(i) $L = L_+ - L_-$.
(ii) $\|L\|_{op} = L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K)$.

Proof.
(i) For $f \in C(K)_+$, set $P(f) := \sup_{0\le g\le f} L(g)$; boundedness of L implies $P(f) < +\infty$.
If $0 \le h \le f_1 + f_2$, then $h = (h \wedge f_1) + (h - h \wedge f_1)$, terms in $[0, f_1]$ and $[0, f_2]$; $P(f_1 + f_2) \le P(f_1) + P(f_2)$.
The reverse inequality follows from $0 \le g_i \le f_i \Rightarrow 0 \le g_1 + g_2 \le f_1 + f_2$; homogeneity is immediate.
Define $L_+(u) := P(u^+) - P(u^-)$ and $L_- := L_+ - L$. Then L_+ is positive linear, $L_-(f) = P(f) - L(f) \ge 0$ for $f \in C(K)_+$, and by Proposition 6.4.1(i), $L_\pm \in C(K)^*$.
(ii) Set $A := L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K)$. By (i) and Proposition 6.4.1(ii), $\|L\|_{\mathrm{op}} \le A$.
For $\varepsilon > 0$, choose $0 \le \varphi \le \mathbf{1}_K$ with $L(\varphi) > P(\mathbf{1}_K) - \varepsilon = L_+(\mathbf{1}_K) - \varepsilon$.
Since $\|2\varphi - \mathbf{1}_K\|_\infty \le 1$ and $L_-(\mathbf{1}_K) = L_+(\mathbf{1}_K) - L(\mathbf{1}_K)$, $\|L\|_{\mathrm{op}} \ge L(2\varphi - \mathbf{1}_K) > A - 2\varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: $P(f) = \sup_{0\le g\le f} L(g)$ is the positive part on $C(K)_+$; $2\varphi - \mathbf{1}_K$ tests $\|L\|_{\mathrm{op}}$.
Key Insight: $L = L_+ - L_-$ and $\|L\|_{\mathrm{op}} = L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K)$. □

Theorem 6.4.3. *Riesz Representation for $C(K)^*$*
Let K be compact Hausdorff. Let $\mathcal{M}_{\mathrm{Rad}}(K)$ be the finite signed measures ν on K with $\nu^\pm$ Radon.
Let $\|\nu\|_{\mathrm{var}} := |\nu|(K)$, and define $(T\nu)(f) := \int_K f\,d\nu$ for $f \in C(K)$. Then:
(i) T is linear and well defined.
(ii) $\|T(\nu)\|_{op} = \|\nu\|_{\mathrm{var}}$ for all $\nu \in \mathcal{M}_{\mathrm{Rad}}(K)$.
(iii) T is bijective.

Proof.
(i) For $\nu \in \mathcal{M}_{\mathrm{Rad}}(K)$ and $f \in C(K)$, $|T(\nu)(f)| \le \int_K |f|\,d\,|\nu| \le \|f\|_\infty \|\nu\|_{\mathrm{var}}$.
Hence $T(\nu) \in C(K)^*$ and $\|T(\nu)\|_{\mathrm{op}} \le \|\nu\|_{\mathrm{var}}$. Linearity is signed-integral linearity.
(ii) Set $L := T(\nu)$. By Proposition 6.4.2(i), Theorem 5.2.14, $L = L_+ - L_-$, $L_\pm = T(\mu_\pm)$ for finite Radon $\mu_\pm$.
Since $T(\nu^+ + \mu_-) = T(\nu^- + \mu_+)$, by uniqueness in Theorem 5.2.14, $\nu^+ + \mu_- = \nu^- + \mu_+$.
By Theorem 3.1.6(ii),(iii), $\nu^\pm \le \mu_\pm$. Therefore $\|\nu\|_{\mathrm{var}} \le \mu_+(K) + \mu_-(K) = L_+(\mathbf{1}_K) + L_-(\mathbf{1}_K) = \|L\|_{\mathrm{op}}$ by Proposition 6.4.2(ii).
Combine with (i).
(iii) Let $L \in C(K)^*$. Proposition 6.4.2(i), Theorem 5.2.14: $L = L_+ - L_-$, $L_\pm = T(\mu_\pm)$ for finite Radon $\mu_\pm$.
Set $\nu := \mu_+ - \mu_-$. By Theorem 3.1.6(ii),(iii), $\nu^\pm \le \mu_\pm$.
If $0 \le \theta \le \mu$ and μ is finite Radon, then θ is Radon since the same open and compact approximants work with no larger error.
Hence $\nu \in \mathcal{M}_{\mathrm{Rad}}(K)$, $T(\nu) = L$. Thus T is surjective; injectivity follows from (ii).
Structural Commentary: Positive Riesz–Markov: $L_\pm = T(\mu_\pm)$; Jordan minimality: $\|T(\nu)\|_{\mathrm{op}} = \|\nu\|_{\mathrm{var}}$.
Key Insight: $C(K)^* \cong \mathcal{M}_{\mathrm{Rad}}(K)$ isometrically via $L(f) = \int_K f\,d\nu$. □

Example 6.4.4. *Point Evaluation Is a Dirac Measure*
Let $K \neq \emptyset$ be compact Hausdorff and fix $x_0 \in K$. Define $L_{x_0} \in C(K)^*$ by $L_{x_0}(f) := f(x_0)$.
Then $L_{x_0} = T(\delta_{x_0})$, since $\forall f \in C(K)$, $T(\delta_{x_0})(f) = \int_K f\,d\delta_{x_0} = f(x_0)$. Hence $\|L_{x_0}\|_{\mathrm{op}} = \|\delta_{x_0}\|_{\mathrm{var}} = 1$.

Remark 6.4.5. *Interval Form and Stieltjes Integrators*
Let $a < b$ and let ν be a finite signed Radon measure on $[a, b]$, extended by 0 to $\mathbb{R}$. Set $g(x) := \nu((-\infty, x])$.
By Proposition 4.1.5 applied to $\nu^\pm$, $g \in BV(\mathbb{R})$ is right-continuous and $g(-\infty) = 0$.
For $a \le c \le d \le b$, $\nu((c, d]) = g(d) - g(c)$ and $\nu(\{a\}) = g(a) - g(a-)$ with $g(a-) = 0$.
Thus, in the Lebesgue–Stieltjes convention, $\forall f \in C([a, b])$, $T(\nu)(f) = \int_{[a,b]} f\,d\nu = \int_{[a,b]} f\,dg$.

Corollary 6.4.6. *Weak-* Compactness for Radon Measures*
Let K be compact Hausdorff and set $\mathcal{M} := \mathcal{M}_{\mathrm{Rad}}(K)$, the finite signed measures ν on K with $\nu^\pm$ Radon.
Let $\|\nu\|_{\mathrm{var}} := |\nu|(K)$ and let $\tau := \sigma(\mathcal{M}, C(K))$ be induced by $\langle \nu, f\rangle := \int_K f\,d\nu$.
If $A \subseteq \mathcal{M}$ is τ-closed and $\sup_{\nu\in A} \|\nu\|_{\mathrm{var}} < +\infty$, then A is τ-compact.

Proof. Let $(T\nu)(f) := \int_K f\,d\nu$. By Theorem 6.4.3, $T : \mathcal{M} \to C(K)^*$ is an isometric bijection identifying τ with $\sigma(C(K)^*, C(K))$.
Choose $R < +\infty$ with $A \subseteq R\overline{B}_{\mathcal{M}}$. Then $T(A)$ is weak-* closed and $T(A) \subseteq R\overline{B}_{C(K)^*}$.
By Theorem 6.1.6, $R\overline{B}_{C(K)^*}$ is weak-* compact; hence $T(A)$ and $A = T^{-1}(T(A))$ are weak-* compact.
Structural Commentary: Riesz isometry: reduce to weak-* closed $T(A) \subseteq R\overline{B}_{C(K)^*}$; Banach–Alaoglu: compact.
Key Insight: $T : \mathcal{M}_{\mathrm{Rad}}(K) \cong C(K)^*$ transfers Banach–Alaoglu compactness to Radon measures. □

Corollary 6.4.7. *Bounded Measures on Compact Metric Spaces Have Weak-* Subsequences*
Let K be compact metric and let $\mu_n \ge 0$ be Borel measures on K with $M := \sup_n \mu_n(K) < +\infty$.
Then $\exists n_k \uparrow \infty$ and a Radon measure $\mu \ge 0$ on K such that $\int_K f\,d\mu_{n_k} \to \int_K f\,d\mu$ for all $f \in C(K)$.

Proof. Set $T_n(f) := \int_K f\,d\mu_n$. By Proposition 6.4.1(ii), $T_n \in C(K)^*$, $T_n \ge 0$, and $\|T_n\|_{\mathrm{op}} = \mu_n(K) \le M$.
Since $C(K)$ is separable by Theorem 9.2.10, by Theorem 6.1.8, $\exists n_k \uparrow \infty$ and $T \in C(K)^*$ with $T_{n_k}(f) \to T(f)$ for all $f \in C(K)$.
Positivity passes to T, so by Theorem 5.2.14, $T(f) = \int_K f\,d\mu$ for some Radon $\mu \ge 0$.
Structural Commentary: Bound (T_n) in $C(K)^*$; use separability and Helly; represent the positive limit.
Key Insight: Bounded positive measures on compact metric K are weak-* sequentially compact on $C(K)$. □

Part XV
Probability Foundations and Distributions

1 Probability Spaces and Random Variables

1.1 Probability Spaces and Event Algebra

Probability assigns numbers to experimental outcomes; measure theory specifies which subsets are events.

Definition 1.1.1. *Sample Spaces, Events, and Probability Spaces*

(i) A *sample space* is a set Ω of possible outcomes.
(ii) An *outcome* or *realization* is a point $\omega \in \Omega$.
(iii) An *event* is a subset $A \subseteq \Omega$; the admissible events form a *σ-algebra* $\mathcal{F} \subseteq \mathcal{P}(\Omega)$.
(iv) A *probability measure* on $(\Omega, \mathcal{F})$ is a measure $\mathbb{P} : \mathcal{F} \to [0,1]$ with $\mathbb{P}(\Omega) = 1$.
(v) The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Key Insight: Probability is countably additive size normalized so that the whole sample space has mass 1.

Method 1.1.2. *Sample Space Construction Method*
Choose Ω as the set of complete outcomes and encode the target condition as an event $A \subseteq \Omega$:

(i) *Finite product method*: for m discrete trials with state set S, use $\Omega = S^m$ and coordinate constraints.
For two coin tosses, $S = \{H, T\}$, $\Omega = S^2$, and first toss heads gives $A = \{\omega \in \Omega : \omega_1 = H\} = \{HH, HT\}$.
(ii) *Interval method*: for real-valued measurements, use $\Omega = \mathbb{R}$ and translate inequalities into intervals.
$10 < \omega \leq 23$ gives $A = (10, 23]$.
(iii) *Cylinder method*: for infinite sequences with state set S, use $\Omega = S^{\mathbb{N}}$ and constrain finitely many coordinates; for coin tosses, $S = \{H, T\}$ and first head on toss 3 gives $A = \{\omega \in \Omega : \omega_1 = T, \omega_2 = T, \omega_3 = H\}$.

Example 1.1.3. *Introductory Sample Spaces and Events*

(i) If a commute passes three traffic lights, each with outcome stop s or continue c, then $\Omega = \{c, s\}^3$. The event that the first light is a stop is $\{sss, ssc, scc, scs\}$.
(ii) If a print queue length is modeled as random, one possible sample space is $\Omega = \{0, 1, 2, \ldots\}$; with a known capacity N, one may instead use $\Omega = \{0, 1, \ldots, N\}$.
(iii) If the measured waiting time between events is nonnegative, a natural sample space is $\Omega = [0, \infty)$.

Key Insight: Choose Ω at the event resolution needed.

Definition 1.1.4. *Event Algebra, Indicators, and Monotone Events*

(i) For $A \in \mathcal{F}$, the *complement* is $A^c := \Omega \setminus A$.
(ii) For $A, B \in \mathcal{F}$, the *union* and *intersection* are $A \cup B$ and $A \cap B$.
(iii) For $A, B \in \mathcal{F}$, the *difference* is $A \setminus B := A \cap B^c$.
(iv) *Commutativity*: $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
(v) *Associativity*: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.
(vi) *Distributivity*: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
(vii) *De Morgan laws*: $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.
(viii) Events $(A_i)_{i \in I}$ are *disjoint* or *mutually exclusive* if $A_i \cap A_j = \emptyset$ for $i \neq j$.
(ix) Events $(A_i)_{i \in I}$ form a *partition* of Ω if $A_i \cap A_j = \emptyset$ for $i \neq j$ and $\bigcup_{i \in I} A_i = \Omega$.
(x) The *indicator* of $A \in \mathcal{F}$ is $\mathbf{1}_A : \Omega \to \{0, 1\}$, where $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ for $\omega \notin A$.
(xi) A sequence (A_n) is *increasing* to A, written $A_n \uparrow A$, if $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_{n \geq 1} A_n$.
(xii) A sequence (A_n) is *decreasing* to A, written $A_n \downarrow A$, if $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_{n \geq 1} A_n$.

Proposition 1.1.5. *Indicator Algebra for Unions and Intersections*
For events $A, B \in \mathcal{F}$: $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B = \min\{\mathbf{1}_A, \mathbf{1}_B\}$, and $\mathbf{1}_{A \cup B} = \max\{\mathbf{1}_A, \mathbf{1}_B\}$.

Proof. Fix $\omega \in \Omega$ and set $a := \mathbf{1}_A(\omega)$, $b := \mathbf{1}_B(\omega)$. Then $a, b \in \{0, 1\}$, $\mathbf{1}_{A \cap B}(\omega) = ab = \min\{a, b\}$, and $\mathbf{1}_{A \cup B}(\omega) = \max\{a, b\}$. Since ω was arbitrary, the identities hold pointwise.
Structural Commentary: Boolean operations on events become arithmetic operations on 0-1 variables.
Key Insight: Indicators let event algebra be manipulated inside sums and expectations. □

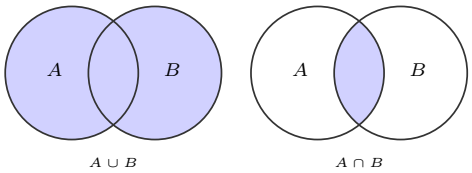

Figure 224: Set operations become event operations in a probability space.

Example 1.1.6. *Endpoint Sensitivity in Monotone Limits*
Let $\Omega = \mathbb{R}$.

(i) If $A_i = [0, \frac{1}{i}]$, then $\bigcup_{i=1}^{\infty} A_i = [0, 1]$ and $\bigcap_{i=1}^{\infty} A_i = \{0\}$.
(ii) If $A_i = (0, \frac{1}{i})$, then $\bigcup_{i=1}^{\infty} A_i = (0, 1)$ and $\bigcap_{i=1}^{\infty} A_i = \emptyset$.

Key Insight: Monotone set limits remember whether boundary points are included.

Proposition 1.1.7. *First Consequences of the Probability Axioms*
For events $A, B \in \mathcal{F}$:

(i) Null event: $\mathbb{P}(\emptyset) = 0$.
(ii) Monotonicity: $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.
(iii) Bounds: $0 \leq \mathbb{P}(A) \leq 1$.
(iv) Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(v) Disjoint additivity: $A \cap B = \emptyset \Rightarrow \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
(vi) Two-set inclusion-exclusion: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof.

(i) Countable additivity on $\Omega = \Omega \sqcup \emptyset \sqcup \emptyset \sqcup \cdots$ gives $1 = 1 + \sum_{n \geq 1} \mathbb{P}(\emptyset)$, hence $\mathbb{P}(\emptyset) = 0$.
(ii) If $A \subseteq B$, then $B = A \sqcup (B \setminus A)$, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.
(iii) Since $\emptyset \subseteq A \subseteq \Omega$, (i) and (ii) give $0 = \mathbb{P}(\emptyset) \leq \mathbb{P}(A) \leq \mathbb{P}(\Omega) = 1$.
(iv) Since $\Omega = A \sqcup A^c$, $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$, so $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(v) If $A \cap B = \emptyset$, then $A \cup B = A \sqcup B$, so $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
(vi) Decompose $A \cup B = (A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A)$, $A = (A \setminus B) \sqcup (A \cap B)$, and $B = (A \cap B) \sqcup (B \setminus A)$. Hence $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Structural Commentary: The algebraic identities are set partitions in disguise.
Key Insight: Split overlapping sets into disjoint pieces before adding probabilities. □

Example 1.1.8. *Two Coin Tosses and Inclusion-Exclusion*
Let $\Omega = \{HH, HT, TH, TT\}$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$ for all $\omega \in \Omega$. Set $H_1 = \{HH, HT\}$ and $H_2 = \{HH, TH\}$.
Since $H_1 \cap H_2 = \{HH\}$, $\mathbb{P}(H_1 \cup H_2) = \mathbb{P}(H_1) + \mathbb{P}(H_2) - \mathbb{P}(H_1 \cap H_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$.

Proposition 1.1.9. *Three-Set Inclusion-Exclusion and Bonferroni*
For events $A, B, C \in \mathcal{F}$:

(i) $\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$.
(ii) $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$.

Proof.

(i) Let $U = A \cup B$. Then $A \cup B \cup C = U \cup C$ and $U \cap C = (A \cap C) \cup (B \cap C)$.
By two-set inclusion-exclusion, $\mathbb{P}(U \cup C) = \mathbb{P}(U) + \mathbb{P}(C) - \mathbb{P}(U \cap C)$, $\mathbb{P}(U) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(U \cap C) = \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C)$. Substitution gives (i).
(ii) Since $\mathbb{P}(A \cup B) \leq 1$ and $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, rearrange.

Structural Commentary: Inclusion-exclusion alternates between overcounting and correcting overlaps.
Key Insight: Bonferroni inequalities are one-sided forms of inclusion-exclusion. □

Theorem 1.1.10. *Continuity of Probabilities*

(i) Continuity from below: if $A_n \uparrow A$, then $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.
(ii) Continuity from above: if $A_n \downarrow A$, then $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.

Proof.

(i) Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then (B_n) is disjoint, $A_n = \bigcup_{j=1}^{n} B_j$, and $A = \bigcup_{j=1}^{\infty} B_j$, so countable additivity gives $\mathbb{P}(A_n) = \sum_{j=1}^{n} \mathbb{P}(B_j) \to \sum_{j=1}^{\infty} \mathbb{P}(B_j) = \mathbb{P}(A)$.
(ii) Set $C_n = A_1 \setminus A_n$. Then $C_n \uparrow A_1 \setminus A$, so (i) gives $\mathbb{P}(C_n) \to \mathbb{P}(A_1 \setminus A)$; since $A_1 = A_n \sqcup C_n$ and $A_1 = A \sqcup (A_1 \setminus A)$, finite additivity gives $\mathbb{P}(A_n) = \mathbb{P}(A_1) - \mathbb{P}(C_n) \to \mathbb{P}(A_1) - \mathbb{P}(A_1 \setminus A) = \mathbb{P}(A)$.

Structural Commentary: Monotone limits convert countable additivity into limit rules.
Key Insight: Increasing uses increments; decreasing uses complements. □

Definition 1.1.11. *Events Occurring Infinitely Often and Eventually Always*
For events $(A_n)_{n \geq 1}$:

(i) *Tail union*: $B_n := \bigcup_{i=n}^{\infty} A_i$.
(ii) *Tail intersection*: $C_n := \bigcap_{i=n}^{\infty} A_i$.
(iii) *Limsup*: $\limsup_{n \to \infty} A_n := \bigcap_{n=1}^{\infty} B_n = \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i$.
(iv) *Liminf*: $\liminf_{n \to \infty} A_n := \bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i$.
(v) *Infinitely often*: $\omega \in \limsup_n A_n$ iff $\omega \in A_n$ for infinitely many n.
(vi) *Eventually always*: $\omega \in \liminf_n A_n$ iff $\omega \in A_n$ for all but finitely many n.

Proposition 1.1.12. *Tail Monotonicity*
For events $(A_n)_{n \geq 1}$ with tail union B_n and tail intersection C_n:

(i) Tail unions: $B_{n+1} \subseteq B_n$ for all n, hence $B_n \downarrow \bigcap_{n \geq 1} B_n$.
(ii) Tail intersections: $C_n \subseteq C_{n+1}$ for all n, hence $C_n \uparrow \bigcup_{n \geq 1} C_n$.

Proof.

(i) Since $\{i : i \geq n+1\} \subseteq \{i : i \geq n\}$, $B_{n+1} = \bigcup_{i=n+1}^{\infty} A_i \subseteq \bigcup_{i=n}^{\infty} A_i = B_n$.

(ii) Since $\{i : i \geq n+1\} \subseteq \{i : i \geq n\}$, $C_n = \bigcap_{i=n}^{\infty} A_i \subseteq \bigcap_{i=n+1}^{\infty} A_i = C_{n+1}$.

Structural Commentary: Tail shifts turn recurrence and permanence into monotone limits.
Key Insight: Tail unions track recurrence; tail intersections track permanence. □

Proposition 1.1.13. *Union Bound and Countable Intersections of Almost-Sure Events*
For events $(A_n)_{n\geq 1}$*:*

(i) Union bound: $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

(ii) Countable almost-sure intersection: if $\mathbb{P}(A_n) = 1$ *for all* n*, then* $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = 1$.

Proof.

(i) Set $B_n := A_n \setminus \bigcup_{j<n} A_j$. Then (B_n) is disjoint, $\bigcup_n B_n = \bigcup_n A_n$, and $B_n \subseteq A_n$.
Hence $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(B_n) \leq \sum_n \mathbb{P}(A_n)$.

(ii) If $\mathbb{P}(A_n) = 1$ for all n, then $\mathbb{P}(A_n^c) = 0$ for all n.
By (i), $\mathbb{P}((\bigcap_n A_n)^c) = \mathbb{P}(\bigcup_n A_n^c) \leq \sum_n \mathbb{P}(A_n^c) = 0$, so $\mathbb{P}(\bigcap_n A_n) = 1$.

Structural Commentary: Countable subadditivity is countable additivity after disjointization.
Key Insight: Almost sure statements survive countable intersections. □

1.2 Finite Probability, Counting, and Independence

Definition 1.2.1. *Uniform Probability on a Countable Space*
Let Ω be countable.

(i) A probability measure $\mathbb{P}$ on $(\Omega, \mathcal{P}(\Omega))$ is *uniform* if $\exists c \in [0,1]$ such that $\mathbb{P}(\{\omega\}) = c$ for all $\omega \in \Omega$.
(ii) The constant c is the *common atom mass*.

Proposition 1.2.2. *No Uniform Probability on a Countably Infinite Set*
Let $\Omega = \{0, 1, 2, \ldots\}$*. No uniform probability measure exists on* $(\Omega, \mathcal{P}(\Omega))$.

Proof. Assume uniform $\mathbb{P}$ exists, and let c be its common atom mass.
Since $\Omega = \bigsqcup_{n=0}^{\infty} \{n\}$, countable additivity gives $1 = \mathbb{P}(\Omega) = \sum_{n=0}^{\infty} \mathbb{P}(\{n\}) = \sum_{n=0}^{\infty} c$.
If $c = 0$, the sum is 0; if $c > 0$, the sum diverges. Both contradict $1 = \mathbb{P}(\Omega)$.
Structural Commentary: Countable additivity forces total mass to equal the sum of the atom masses.
Key Insight: Equal mass on countably many atoms is incompatible with total mass 1. □

Method 1.2.3. *Finite Uniform Probability by Counting*
Let $0 < |\Omega| < \infty$ and assume the *uniform model* $\mathbb{P}(\{\omega\}) = |\Omega|^{-1}$ for all $\omega \in \Omega$.
For $A \subseteq \Omega$, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) = \frac{|A|}{|\Omega|}$, so probability reduces to counting.

Method 1.2.4. *Counting Workflow for Uniform Finite Probability*
For a finite uniform sample space Ω and an event $A \subseteq \Omega$:

(i) choose complete outcomes so that each $\omega \in \Omega$ is equally likely;
(ii) count $|\Omega|$; and count $|A|$, usually by products, complements, or disjoint cases;
(iii) conclude $\mathbb{P}(A) = |A| \,/\, |\Omega|$.

Key Insight: The hard step is often choosing equally likely outcomes, not applying the ratio.

Example 1.2.5. *Simpson's Paradox in Two Urn Games*
Let $p(U)$ be the red probability in urn U, and let B_i, W_i be the black and white urns in game i.
Within games, $p(B_1) = 5/11 > 3/7 = p(W_1)$ and $p(B_2) = 6/9 > 9/14 = p(W_2)$.
After pooling by color into B and W, $p(B) = 11/20 < 12/21 = p(W)$, so the aggregate winner is white.
Key Insight: A within-group advantage can reverse after aggregation when the group sizes change.

Example 1.2.6. *Complement Counting in Uniform Models*

(i) For plates $\Omega = L^3 \times D^3$ with $|L| = 26$ and $|D| = 10$, $|\Omega| = 26^3 \cdot 10^3$. If E is the event of no repeated letters and no repeated digits, then $|E| = (26)_3(10)_3$, so $\mathbb{P}(E) = \frac{(26)_3(10)_3}{26^3 \cdot 10^3}$.
(ii) For $1 \leq n \leq 365$ iid uniform birthdays, take $\Omega = \{1, \ldots, 365\}^n$. If C is the event of a shared birthday, then $C^c = D_n(\{1, \ldots, 365\})$, so $\mathbb{P}(C) = 1 - \frac{(365)_n}{365^n}$; for $n = 23$, $\mathbb{P}(C) > 1/2$.

Key Insight: "At least one collision" is often easier to count by subtracting the no-collision event.

Definition 1.2.7. *Independent Events*

(i) Events $A, B \in \mathcal{F}$ are *independent*, written $A \perp\!\!\!\perp B$, if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
(ii) A family $(A_i)_{i \in I} \subseteq \mathcal{F}$ is *pairwise independent* if A_i and A_j are independent whenever $i \neq j$.
(iii) A family $(A_i)_{i \in I} \subseteq \mathcal{F}$ is *mutually independent* if $\mathbb{P}(\bigcap_{i \in J} A_i) = \prod_{i \in J} \mathbb{P}(A_i)$ for every finite $J \subseteq I$.

Key Insight: Mutual independence means multiplicativity for every finite subfamily, not just pairs.

Example 1.2.8. *Pairwise Independence Need Not Be Mutual*
Let $\Omega = \{HH, HT, TH, TT\}$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}\ \forall \omega \in \Omega$.
Set $A = \{HH, HT\}$, $B = \{HH, TH\}$, $C = \{HT, TH\}$.
Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{4}$, so A, B, C pairwise independent.
But $A \cap B \cap C = \emptyset$, so $\mathbb{P}(A \cap B \cap C) = 0 \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$.
Key Insight: Pairwise checks do not control higher-order intersections.

Proposition 1.2.9. *Basic Independence Tests*
Let $A, B \in \mathcal{F}$.

(i) If $A \cap B = \emptyset$ and $\mathbb{P}(A)\mathbb{P}(B) > 0$, then $A \not\perp\!\!\!\perp B$.
(ii) If $A \perp\!\!\!\perp B$, then $A^c \perp\!\!\!\perp B$, $A \perp\!\!\!\perp B^c$, and $A^c \perp\!\!\!\perp B^c$.
(iii) If $\mathbb{P}(A) \in \{0, 1\}$, then $A \perp\!\!\!\perp C$ for every $C \in \mathcal{F}$.
(iv) If $A \perp\!\!\!\perp A$, then $\mathbb{P}(A) \in \{0, 1\}$.

Proof.

(i) Since $A \cap B = \emptyset$, $\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B)$.
(ii) If $A \perp\!\!\!\perp B$, then $\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = (1 - \mathbb{P}(A))\mathbb{P}(B) = \mathbb{P}(A^c)\mathbb{P}(B)$
and $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(B^c)$.
Applying the first identity to $A \perp\!\!\!\perp B^c$ gives $A^c \perp\!\!\!\perp B^c$.
(iii) Let $C \in \mathcal{F}$. If $\mathbb{P}(A) = 0$, then $0 \leq \mathbb{P}(A \cap C) \leq \mathbb{P}(A) = 0$, so $\mathbb{P}(A \cap C) = 0 = \mathbb{P}(A)\mathbb{P}(C)$.
If $\mathbb{P}(A) = 1$, then $\mathbb{P}(A^c) = 0$, hence $A^c \perp\!\!\!\perp C$ by the previous case and $A \perp\!\!\!\perp C$ by (ii).
(iv) Let $p = \mathbb{P}(A)$. From $A \perp\!\!\!\perp A$, $p = \mathbb{P}(A \cap A) = p^2$, hence $p \in \{0, 1\}$.

Structural Commentary: Independence is probabilistic, not visual; disjointness usually means strong dependence.
Key Insight: Complements preserve independence because B splits into $A \cap B$ and $A^c \cap B$. □

Method 1.2.10. *Independence, Complement, and Decomposition Methods*

(i) *Verify independence*: compute $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$; conclude $A \perp\!\!\!\perp B$ iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
(ii) *Use complements*: if A^c is simpler, compute $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.
If $A^c = \bigcap_{j=1}^{n} C_j$ with independent C_j, then $\mathbb{P}(A) = 1 - \prod_{j=1}^{n} \mathbb{P}(C_j)$.
(iii) *Disjoint decomposition*: write $E = \bigsqcup_j A_j$ with disjoint A_j, then $\mathbb{P}(E) = \sum_j \mathbb{P}(A_j)$.
(iv) *Geometric summation*: if $\mathbb{P}(A_j) = ar^{j-1}$ for $j \geq 1$ and $|r| < 1$, then $\mathbb{P}(E) = \sum_{j \geq 1} ar^{j-1} = \frac{a}{1-r}$.

Example 1.2.11. *Independence, Complements, and First Success*

(i) *Independence by counting*: for a fair die, $\Omega = \{1, \ldots, 6\}$, $A = \{2, 4, 6\}$, and $B = \{1, 2, 3, 4\}$.
Then $\mathbb{P}(A \cap B) = \frac{2}{6} = (\frac{3}{6})(\frac{4}{6}) = \mathbb{P}(A)\mathbb{P}(B)$, hence $A \perp\!\!\!\perp B$.
(ii) *Complement method*: for independent fair tosses $X_1, \ldots, X_{10}$, let $A = \{\exists j \leq 10 : X_j = H\}$
and $T_j = \{X_j = T\}$. Then $A^c = \bigcap_{j=1}^{10} T_j$ and $\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \prod_{j=1}^{10} \mathbb{P}(T_j) = 1 - (\frac{1}{2})^{10}$.
(iii) *First-success decomposition*: let $p_1 = \frac{1}{3}$, $p_2 = \frac{1}{4}$.
Let A_j be player 1's first success on turn j before player 2 succeeds.
Then $E = \bigsqcup_{j \geq 1} A_j$, $\mathbb{P}(A_j) = ((1 - p_1)(1 - p_2))^{j-1} p_1 = (\frac{1}{2})^{j-1}/3$, and $\mathbb{P}(E) = \sum_{j \geq 1} \mathbb{P}(A_j) = \frac{2}{3}$.

Example 1.2.12. *Reliability and Match Probabilities*
Let A_i be the event that component i works, and assume all relevant events are mutually independent.

(i) Series: if $\mathbb{P}(A_i^c) = p$ for $1 \leq i \leq n$ and $W_s = \bigcap_{i=1}^{n} A_i$, then $\mathbb{P}(W_s) = (1 - p)^n$ and $\mathbb{P}(W_s^c) = 1 - (1 - p)^n$.
(ii) Parallel: if $W_p = \bigcup_{i=1}^{n} A_i$, then $\mathbb{P}(W_p^c) = \mathbb{P}(\bigcap_{i=1}^{n} A_i^c) = p^n$.
(iii) DNA match: let $N_j \in \{A, C, G, T\}$ with $\mathbb{P}(N_j = x) = p_{xj}$ for $j = 1, 2$.
If $N_1 \perp\!\!\!\perp N_2$, then $\mathbb{P}(N_1 = N_2) = \sum_{x \in \{A,C,G,T\}} p_{x1} p_{x2}$.

Key Insight: Independence multiplies intersections; complements handle failures; matches sum disjoint cases.

1.3 Random Variables, Laws, and Distribution Functions

Definition 1.3.1. *Random Variables, Laws, and Distribution Functions*

(i) A *real-valued random variable* on $(\Omega, \mathcal{F}, \mathbb{P})$ is map $X : \Omega \to \mathbb{R}$ s.t. $\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}\ \forall B \in \mathcal{B}(\mathbb{R})$.
(ii) Its *law* or *distribution* is $\mathcal{L}(X) : \mathcal{B}(\mathbb{R}) \to [0, 1]$ given by $\mathcal{L}(X)(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$.
(iii) Its *cumulative distribution function* is $F_X : \mathbb{R} \to [0, 1]$ given by $F_X(x) := \mathbb{P}(X \leq x) = \mathcal{L}(X)((-\infty, x])$.

Definition 1.3.2. *Distribution Functions and Densities*

(i) A *distribution function* is $F : \mathbb{R} \to [0, 1]$ such that $x \leq y \Rightarrow F(x) \leq F(y)$, $F(x) = \lim_{t \downarrow x} F(t)$ for all $x \in \mathbb{R}$, $\lim_{x \to -\infty} F(x) = 0$, and $\lim_{x \to \infty} F(x) = 1$.
(ii) A *density* for F is an integrable $g : \mathbb{R} \to [0, \infty)$ with $\int_{-\infty}^{\infty} g(x)\,dx = 1$ and $F(t) = \int_{-\infty}^{t} g(x)\,dx\ \forall t \in \mathbb{R}$.
If g is continuous, then $F'(t) = g(t)$.

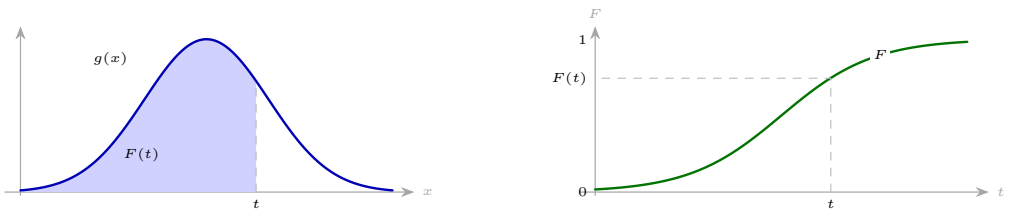

Figure 225: A normal density accumulates area into the distribution function $\Phi(t) = \int_{-\infty}^{t} g(x)\,dx$.

Definition 1.3.3. *Discrete and Continuous Random Variables*

(i) X is *discrete* if $X(\Omega) \subseteq S$ for some finite or countable $S = \{x_1, x_2, \ldots\} \subseteq \mathbb{R}$.
(ii) The *probability mass function* of discrete X is $f_X : S \to [0,1]$, $f_X(x) := \mathbb{P}(X = x)$, $\sum_{x \in S} f_X(x) = 1$.
(iii) The *discrete CDF* is $F_X(x) = \mathbb{P}(X \le x) = \sum_{x_i \le x} f_X(x_i)$.
(iv) X is *continuous* if its law has a density $f_X : \mathbb{R} \to [0, \infty)$.
(v) The *continuous probabilities* satisfy $\mathbb{P}(a < X < b) = \int_a^b f_X(t)\,dt$ and $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$.
(vi) The *point-mass distinction*: for continuous X, $\mathbb{P}(X = x) = 0$ for every fixed x, so $f_X(x) \neq \mathbb{P}(X = x)$.

A law may be mixed. It can have atoms and a continuous density on the remaining part.

Remark 1.3.4. *Mixed Laws Are Natural*
For a component lifetime X, one may have $\mathbb{P}(X = 0) = p > 0$ and, conditional on $X > 0$, a density g on $(0, \infty)$.
Then $\mathcal{L}(X)$ has an atom plus a continuous part, so it is mixed.

Example 1.3.5. *Counting Heads in Two Tosses*
Let $\Omega = \{H, T\}^2$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$, and set $X(\omega) = |\{j \in \{1,2\} : \omega_j = H\}|$.
Then $f_X(k) = \mathbb{P}(X = k) = \binom{2}{k} 2^{-2}$ for $k \in \{0,1,2\}$, so $(f_X(0), f_X(1), f_X(2)) = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$.
Hence $F_X(x) = 0$ for $x < 0$, $F_X(x) = \frac{1}{4}$ for $0 \le x < 1$, $F_X(x) = \frac{3}{4}$ for $1 \le x < 2$, $F_X(x) = 1$ for $x \ge 2$.
Key Insight: A CDF is defined for every real x, even when a discrete random variable takes only a few values.

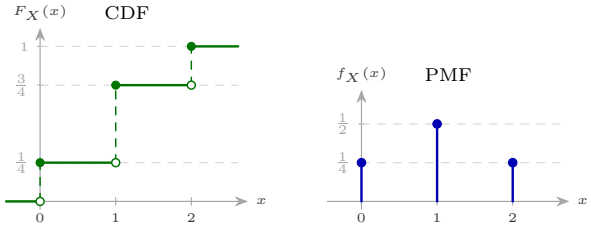

Figure 226: CDF and PMF for the number of heads in two independent fair coin tosses.

Proposition 1.3.6. *CDF Probability Calculus*
Let X have CDF F and define $F(x^-) := \lim_{y \uparrow x} F(y)$.

(i) Point mass: $\mathbb{P}(X = x) = F(x) - F(x^-)$.
(ii) Half-open interval: if $x < y$, then $\mathbb{P}(x < X \le y) = F(y) - F(x)$.
(iii) Right tail: $\mathbb{P}(X > x) = 1 - F(x)$.
(iv) Continuous endpoints: if X is continuous and $a < b$, then $\mathbb{P}(a < X < b) = \mathbb{P}(a \le X < b) = \mathbb{P}(a < X \le b) = \mathbb{P}(a \le X \le b)$.

Proof.

(i) Let $E_n = \{x - \frac{1}{n} < X \le x\}$. Then $E_n \downarrow \{X = x\}$ and $\mathbb{P}(E_n) = F(x) - F(x - \frac{1}{n})$.
Hence $\mathbb{P}(X = x) = \lim_n \mathbb{P}(E_n) = F(x) - F(x^-)$.
(ii) Since $\{X \le y\} = \{X \le x\} \sqcup \{x < X \le y\}$, $F(y) = F(x) + \mathbb{P}(x < X \le y)$.
(iii) Since $\{X > x\} = \{X \le x\}^c$, $\mathbb{P}(X > x) = 1 - \mathbb{P}(X \le x) = 1 - F(x)$.
(iv) If X is continuous, then $\mathbb{P}(X = t) = 0$ for all $t \in \mathbb{R}$. For $I_1 = (a,b)$, $I_2 = [a,b)$, $I_3 = (a,b]$, and $I_4 = [a,b]$, each $X^{-1}(I_j) \backslash X^{-1}(I_1) \subseteq \{X = a\} \cup \{X = b\}$, hence all four probabilities are equal.

Structural Commentary: CDF calculus subtracts nested lower-level events.
Key Insight: CDF jumps are point masses; continuous laws have none. □

Definition 1.3.7. *Survival and Hazard Functions*
Let $T \ge 0$ have CDF F and density f, and set $D := \{t \ge 0 : 1 - F(t) > 0\}$.

(i) *Survival Function*: $S : [0, \infty) \to [0,1]$, $S(t) := \mathbb{P}(T > t) = 1 - F(t)$.
(ii) *Hazard Function*: $h : D \to [0, \infty)$, $h(t) := \frac{f(t)}{1 - F(t)} = \frac{f(t)}{S(t)}$.

Structural Commentary: On intervals where S is differentiable and positive, $h(t) = -\frac{d}{dt} \log S(t)$.

Proposition 1.3.8. *Hazard Determines the Density*
Let $T \ge 0$ have survival $S(t) := \mathbb{P}(T > t)$ with $S \in C^1([0, \infty))$, $S(t) > 0$, $S(0) = 1$, density $f(t) := -S'(t)$, and hazard $h(t) := \frac{f(t)}{S(t)}$.
Put $H(t) := \int_0^t h(s)\,ds$. Then, for $t \ge 0$:

(i) Survival: $S(t) = e^{-H(t)}$.

(ii) Density: $f(t) = h(t)e^{-H(t)}$.

Proof. Fix $t \geq 0$.

(i) *Survival*: Since $(\log S)' = S'/S = -h$, $\log S(t) - \log S(0) = -H(t)$. Since $S(0) = 1$, $S(t) = e^{-H(t)}$.
(ii) *Density*: Since $h = f/S$, $f(t) = h(t)S(t) = h(t)e^{-H(t)}$.

Structural Commentary: $h = -S'/S = -(\log S)'$, so H is the cumulative log-decay of survival.
Key Insight: The hazard determines the law through $S = e^{-H}$ and $f = hS$. □

Example 1.3.9. *Constant and Power Hazards*
For $t \geq 0$:

(i) *Exponential*: If $F(t) = 1 - e^{-\lambda t}$ with $\lambda > 0$, then $S(t) = e^{-\lambda t}$, $f(t) = \lambda e^{-\lambda t}$, and $h(t) = \lambda$.
(ii) *Weibull-Type*: If $F(t) = 1 - e^{-\alpha t^\beta}$ with $\alpha, \beta > 0$, then $S(t) = e^{-\alpha t^\beta}$, $f(t) = \alpha\beta t^{\beta-1}e^{-\alpha t^\beta}$, $h(t) = \alpha\beta t^{\beta-1}$.

Key Insight: Power hazards increase for $\beta > 1$, decrease for $0 < \beta < 1$, and are constant for $\beta = 1$.

Definition 1.3.10. *Quantiles and Equality in Distribution*

(i) For a CDF F, the *quantile function* is $F^{-1}(q) := \inf\{x \in \mathbb{R} : F(x) \geq q\}$, $q \in (0, 1)$.
(ii) If F is continuous and strictly increasing, then F^{-1} is the ordinary inverse: $F(F^{-1}(q)) = q$ for $q \in (0, 1)$ and $F^{-1}(F(x)) = x$ for $x \in \mathbb{R}$.
(iii) The *pth quantile* is $x_p = F^{-1}(p)$; under (ii), $F(x_p) = p$ uniquely. Here $x_{\frac{1}{2}}$ is the median and $x_{\frac{1}{4}}, x_{\frac{3}{4}}$ are the lower and upper quartiles.
(iv) Random variables X, Y are *equal in distribution*, written $X \overset{\text{dist}}{=} Y$, iff $F_X = F_Y$ on $\mathbb{R}$.

Example 1.3.11. *Value at Risk as a Quantile*
Let $V_0 > 0$, $R = \frac{V_1 - V_0}{V_0}$, and $F_R(x) = \mathbb{P}(R \leq x)$.
The loss is $L = V_0 - V_1 = -V_0 R$, so $\{L \geq v\} = \{R \leq -\frac{v}{V_0}\}$ and $\mathbb{P}(L \geq v) = F_R(-\frac{v}{V_0})$.
If F_R is invertible and $\mathbb{P}(L \geq v_\alpha) = \alpha$, then $v_\alpha = -V_0 F_R^{-1}(\alpha)$.
Key Insight: VaR is a lower-tail return quantile converted back into a dollar loss.

1.4 Standard Distribution Families and Approximations

Definition 1.4.1. *Core Discrete Distributions*
Common discrete laws include:

(i) *Point mass*: $X \sim \delta_a$ for $a \in \mathbb{R}$ if $\mathbb{P}(X = a) = 1$.
(ii) *Discrete uniform*: $X \sim \text{Unif}\{1, \dots, k\}$ for $k \in \mathbb{N}$ if $\mathbb{P}(X = x) = \frac{1}{k}$ for $x = 1, \dots, k$.
(iii) *Bernoulli*: $X \sim \text{Bernoulli}(p)$ for $p \in [0, 1]$ if $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. Equivalently $\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$ for $x \in \{0, 1\}$.
(iv) *Binomial*: $X \sim \text{Binomial}(n, p)$ for $n \in \mathbb{N}$ and $p \in [0, 1]$ if $\mathbb{P}(X = x) = \binom{n}{x}p^x(1 - p)^{n-x}$ for $x = 0, \dots, n$.
(v) *Geometric*: $X \sim \text{Geom}(p)$, $p \in (0, 1]$, if $\mathbb{P}(X = k) = p(1 - p)^{k-1}$ for $k \geq 1$; X counts trials to first success.
(vi) *Negative binomial*: $X \sim \text{NegBin}(r, p)$, $r \in \mathbb{N}$, $p \in (0, 1]$, if $\mathbb{P}(X = k) = \binom{k-1}{r-1}p^r(1 - p)^{k-r}$ for $k \geq r$; X counts trials to the rth success.
(vii) *Hypergeometric*: $X \sim \text{Hypergeom}(n, r, m)$ if $\mathbb{P}(X = k) = \dfrac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}$ for $\max\{0, m - n + r\} \leq k \leq \min\{r, m\}$.
(viii) *Poisson*: $X \sim \text{Poisson}(\lambda)$ for $\lambda \geq 0$ if $\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, \dots$.

Definition 1.4.2. *Core Continuous Distributions*
Common continuous laws include:

(i) *Uniform*: $X \sim \text{Uniform}(a, b)$ for $a < b$ if $f_X(x) = (b - a)^{-1}\mathbf{1}_{[a,b]}(x)$.
(ii) *Normal*: $X \sim N(\mu, \sigma^2)$ for $\mu \in \mathbb{R}$, $\sigma > 0$, if $f_X(x) = (\sigma\sqrt{2\pi})^{-1}\exp(-\frac{(x-\mu)^2}{2\sigma^2})$ for $x \in \mathbb{R}$. $N(0, 1)$ has density ϕ and CDF Φ.
(iii) *Exponential*: $X \sim \text{Exp}(\lambda)$ for $\lambda > 0$ if $f_X(x) = \lambda e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x)$.
(iv) *Gamma*: $X \sim \text{Gamma}(\alpha, \lambda)$ for $\alpha, \lambda > 0$ if $f_X(x) = \lambda^\alpha\Gamma(\alpha)^{-1}x^{\alpha-1}e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x)$; $\text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$.
(v) *Beta*: $X \sim \text{Beta}(\alpha, \beta)$ for $\alpha, \beta > 0$ if $f_X(x) = \Gamma(\alpha + \beta)(\Gamma(\alpha)\Gamma(\beta))^{-1}x^{\alpha-1}(1 - x)^{\beta-1}\mathbf{1}_{(0,1)}(x)$.
(vi) *Cauchy*: $X \sim \text{Cauchy}(0, 1)$ if $f_X(x) = (\pi(1 + x^2))^{-1}$ for $x \in \mathbb{R}$.
(vii) χ^2: $X \sim \chi^2_p$ for $p > 0$ if $f_X(x) = (\Gamma(\frac{p}{2})2^{\frac{p}{2}})^{-1}x^{\frac{p}{2}-1}e^{-\frac{x}{2}}\mathbf{1}_{(0,\infty)}(x)$.

Definition 1.4.3. *Normal Right-Tail Quantile*
For $0 < \alpha < 1$ and $Z \sim N(0, 1)$, $z(\alpha) := \Phi^{-1}(1 - \alpha)$, so $\mathbb{P}(Z > z(\alpha)) = \alpha$.

Method 1.4.4. *Discrete Distribution Model Recipes*

(i) *Indicators*: for $A \in \mathcal{F}$, $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$.
(ii) *Binomial counts*: if $B_1, \dots, B_n$ are independent $\text{Bernoulli}(p)$ variables, then $\sum_{i=1}^{n} B_i \sim \text{Binomial}(n, p)$.
(iii) *Geometric waiting*: the probability of first success on trial k is $(1 - p)^{k-1}p$.
(iv) *Negative-binomial waiting*: for iid $B_i \sim \text{Bernoulli}(p)$ and $T_r = \min\{k : \sum_{i=1}^{k} B_i = r\}$, $\{T_r = k\} = \{B_k = 1, \sum_{i=1}^{k-1} B_i = r - 1\}$, so $\mathbb{P}(T_r = k) = \binom{k-1}{r-1}p^r(1 - p)^{k-r}$.
(v) *Hypergeometric sampling*: if $M \subseteq [n]$, $|M| = r$, and S is a uniform m-subset, then $X = |S \cap M|$ and $\mathbb{P}(X = k) = \dfrac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}$ for $\max\{0, m - n + r\} \leq k \leq \min\{r, m\}$.

Key Insight: Name the event; PMFs come from independence or subset counts.

Proposition 1.4.5. *Poisson Limit of Rare Binomial Counts*
Let $\lambda > 0$, $k \in \mathbb{N}_0$, and $X_n \sim \text{Binomial}(n, \frac{\lambda}{n})$ for all sufficiently large n. Then $\lim_{n\to\infty} \mathbb{P}(X_n = k) = \frac{e^{-\lambda}\lambda^k}{k!}$.

Proof. For large n, $\mathbb{P}(X_n = k) = \binom{n}{k}(\frac{\lambda}{n})^k(1-\frac{\lambda}{n})^{n-k} = \frac{\lambda^k}{k!}\prod_{j=0}^{k-1}(1-\frac{j}{n})(1-\frac{\lambda}{n})^n(1-\frac{\lambda}{n})^{-k}$.
Since $\prod_{j=0}^{k-1}(1-\frac{j}{n}) \to 1$, $(1-\frac{\lambda}{n})^n \to e^{-\lambda}$, and $(1-\frac{\lambda}{n})^{-k} \to 1$, the limit is $\frac{e^{-\lambda}\lambda^k}{k!}$.
Structural Commentary: Rare binomial masses split into fixed-k choices and an exponential no-event factor.
Key Insight: Large n, small p, fixed np turns binomial masses into Poisson masses. □

Method 1.4.6. *Recursive Binomial and Poisson Mass Computation*
Let $q := 1 - p$.

(i) If $X \sim \text{Binomial}(n, p)$, $p_k := \mathbb{P}(X = k)$, and $q > 0$, then $p_0 = q^n$ and $\frac{p_k}{p_{k-1}} = \frac{(n-k+1)p}{kq}$ for $1 \le k \le n$.

(ii) If $X \sim \text{Poisson}(\lambda)$ and $p_k := \mathbb{P}(X = k)$, then $p_0 = e^{-\lambda}$ and $\frac{p_k}{p_{k-1}} = \frac{\lambda}{k}$ for $k \ge 1$.

Key Insight: Successive mass ratios avoid repeatedly evaluating large factorials.

Example 1.4.7. *Majority Decoding as a Binomial Tail*
Let $n = 2m + 1$ and let $C_i \sim \text{Bernoulli}(p)$ be independent corruption indicators.
Then $X = \sum_{i=1}^{n} C_i \sim \text{Binomial}(n, p)$ and the majority decoder is correct iff $X \le m$.
For $n = 5$, $p = 0.1$, $\mathbb{P}(X \le 2) = 0.9^5 + 5(0.1)(0.9)^4 + 10(0.1)^2(0.9)^3 \approx 0.9914$.
Key Insight: Repetition coding converts reliability into a lower binomial tail.

Example 1.4.8. *Poisson Approximation for Rare Repeated Events*
Let $I_j = \mathbf{1}_{\{\text{double six on roll } j\}}$ for $1 \le j \le 100$.
Then $I_j \sim \text{Bernoulli}(\frac{1}{36})$ iid, $X = \sum_{j=1}^{100} I_j \sim \text{Binomial}(100, \frac{1}{36})$, and with $\lambda = \frac{100}{36} = \frac{25}{9}$, $\mathbb{P}(X = k) \approx \frac{e^{-\frac{25}{9}}(\frac{25}{9})^k}{k!}$.
Key Insight: Rare repeated event counts are often well approximated by a Poisson law with $\lambda = np$.

Proposition 1.4.9. *Exponential Memorylessness*
If $T \sim \text{Exp}(\lambda)$ with $\lambda > 0$, then $\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t)$ for all $s, t \ge 0$.

Proof. For $u \ge 0$, $\mathbb{P}(T > u) = e^{-\lambda u}$. Hence $\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T>s+t)}{\mathbb{P}(T>s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t)$.
Structural Commentary: Conditioning on survival to time s leaves the residual tail unchanged.
Key Insight: Exponential tails convert elapsed time into a cancellable factor. □

Remark 1.4.10. *Poisson Counts and Exponential Gaps*
In homogeneous Poisson process, counts satisfy $N(t) \sim \text{Poisson}(\lambda t)$; interarrival gaps satisfy $S_i \sim \text{Exp}(\lambda)$ iid.

Proposition 1.4.11. *Standardizing Normal Variables*
For $\sigma, \sigma_i > 0$:

(i) Standardization: if $X \sim N(\mu, \sigma^2)$, then $\frac{X-\mu}{\sigma} \sim N(0, 1)$.

(ii) Affine reconstruction: if $Z \sim N(0, 1)$, then $\mu + \sigma Z \sim N(\mu, \sigma^2)$.

(iii) Independent sums: if $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $\sum_{i=1}^{n} X_i \sim N(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2)$.

(iv) Interval probabilities: if $X \sim N(\mu, \sigma^2)$ and $a < b$, then $\mathbb{P}(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$.

Proof.

(i) Let $Y = \frac{X-\mu}{\sigma}$. By change of variables, $f_Y(z) = \sigma f_X(\mu + \sigma z) = (2\pi)^{-\frac{1}{2}} e^{-\frac{z^2}{2}}$, so $Y \sim N(0, 1)$.

(ii) Let $Y = \mu + \sigma Z$. Then $f_Y(y) = \sigma^{-1}\phi(\frac{y-\mu}{\sigma}) = (\sigma\sqrt{2\pi})^{-1}\exp(-\frac{(y-\mu)^2}{2\sigma^2})$, hence $Y \sim N(\mu, \sigma^2)$.

(iii) It suffices by induction to treat $n = 2$.
Let $X_i \sim N(\mu_i, v_i)$ be independent, $S = X_1 + X_2$, $v = v_1 + v_2$, $w = \frac{v_1 v_2}{v}$, and $a_s = \frac{v_2\mu_1 + v_1(s-\mu_2)}{v}$.
Then $f_S(s) = \int_{\mathbb{R}} f_1(u) f_2(s-u)\, du$, and completing the square gives
$f_1(u)f_2(s-u) = (2\pi v)^{-\frac{1}{2}} e^{-\frac{(s-\mu_1-\mu_2)^2}{2v}} (2\pi w)^{-\frac{1}{2}} e^{-\frac{(u-a_s)^2}{2w}}$.
The second factor integrates to 1, so $S \sim N(\mu_1 + \mu_2, v_1 + v_2)$; induction gives (iii).

(iv) Let $Z = \frac{X-\mu}{\sigma}$. Since $\sigma > 0$, $\{a < X < b\} = \{\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\}$.
Hence $\mathbb{P}(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$.

Structural Commentary: Normal laws are stable under affine maps and independent addition.
Key Insight: Standardize to $N(0, 1)$, compute with Φ, then transform back. □

Example 1.4.12. *Normal Tail and Quantile Standardization*
Let $X \sim N(3, 5)$ and $Z = \frac{X-3}{\sqrt{5}} \sim N(0, 1)$. Then $\mathbb{P}(X > 1) = \mathbb{P}(Z > -\frac{2}{\sqrt{5}}) = 1 - \Phi(-\frac{2}{\sqrt{5}}) \approx 0.81$.
If $\mathbb{P}(X < q) = 0.2$, then $\Phi(\frac{q-3}{\sqrt{5}}) = 0.2$, so $q = 3 + \sqrt{5}\,\Phi^{-1}(0.2) \approx 3 - 0.8416\sqrt{5} \approx 1.1181$.

2 Joint Distributions, Multivariate Laws, and Transformations

2.1 Joint Distributions and Independence

Definition 2.1.1. *Joint and Marginal Distributions*

(i) The *joint mass function* of discrete X, Y is $f_{X,Y}(x,y) := \mathbb{P}(X = x, Y = y)$.
(ii) The *discrete marginals* are $f_X(x) = \sum_y f_{X,Y}(x,y)$ and $f_Y(y) = \sum_x f_{X,Y}(x,y)$.
(iii) The *continuous marginals* are $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$, and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$.
(iv) The *joint CDF* is $F_{X,Y}(x,y) := \mathbb{P}(X \le x, Y \le y)$. (X,Y) has joint density $f_{X,Y}$.

Example 2.1.2. *A Three-Toss Joint Mass Table*

Let $\Omega = \{H,T\}^3$ with $\mathbb{P}(\{\omega\}) = 2^{-3}$, set $X = \mathbf{1}_{\{\omega_1 = H\}}$, $Y = \sum_{i=1}^{3} \mathbf{1}_{\{\omega_i = H\}}$, and write $p_{xy} := \mathbb{P}(X = x, Y = y)$.

	$Y=0$	$Y=1$	$Y=2$	$Y=3$
$X=0$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0
$X=1$	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$

Thus $p_{0,2} = \frac{1}{8}$, $\sum_y p_{0y} = \sum_y p_{1y} = \frac{1}{2}$, and $\sum_x p_{xj} = \binom{3}{j} 2^{-3}$ for $j = 0, \dots, 3$.

Proposition 2.1.3. *Rectangular Increments of a Joint CDF*
Let $F(x,y) := \mathbb{P}(X \le x, Y \le y)$.

(i) If $x_1 < x_2$ and $y_1 < y_2$, then $\mathbb{P}(x_1 < X \le x_2,\ y_1 < Y \le y_2) = F(x_2,y_2) - F(x_1,y_2) - F(x_2,y_1) + F(x_1,y_1)$.
(ii) If (X,Y) has density $f_{X,Y}$, $F \in C^2$, then $F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$; $f_{X,Y}(x,y) = \partial_x \partial_y F(x,y)$.

Proof.

(i) Set $A_{ij} := \{X \le x_i,\ Y \le y_j\}$ and $R := \{x_1 < X \le x_2,\ y_1 < Y \le y_2\}$.
Pointwise, $\mathbf{1}_R = \mathbf{1}_{A_{22}} - \mathbf{1}_{A_{12}} - \mathbf{1}_{A_{21}} + \mathbf{1}_{A_{11}}$; applying $\mathbb{E}$ gives the identity.
(ii) The density assumption gives $F(x,y) = \mathbb{P}(X \le x, Y \le y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$.
Since $F \in C^2$, the two-variable FTC gives $\partial_x \partial_y F = f_{X,Y}$.

Structural Commentary: Joint CDFs encode southwest rectangles; finite rectangles are four-corner increments.
Key Insight: Mixed derivatives are infinitesimal rectangular increments. □

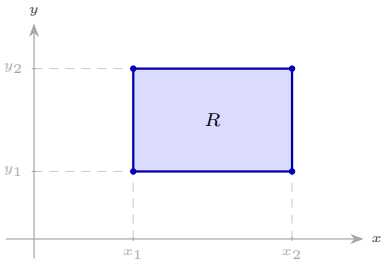

Figure 227: The rectangular probability is a four-term increment of the joint CDF.

Example 2.1.4. *A Non-Rectangular Bivariate Density*
Let $S = \{(x,y) : x^2 \le y \le 1\}$ and $f_{X,Y}(x,y) = cx^2 y \mathbf{1}_S(x,y)$.
Since $S = \{(x,y) : -1 \le x \le 1,\ x^2 \le y \le 1\}$, $1 = c\int_{-1}^{1}\int_{x^2}^{1} x^2 y\,dy\,dx = \frac{4c}{21}$, so $c = \frac{21}{4}$.
For $E = \{X \ge Y\}$, $S \cap E = \{(x,y) : 0 \le x \le 1,\ x^2 \le y \le x\}$, hence $\mathbb{P}(E) = (\frac{21}{4})\int_0^1\int_{x^2}^{x} x^2 y\,dy\,dx = \frac{3}{20}$.
Key Insight: In bivariate integration, pick one variable first and let the other range over the corresponding slice.

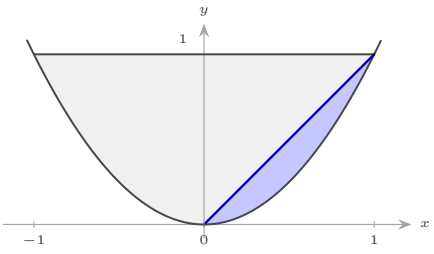

Figure 228: Gray: support $x^2 \le y \le 1$; blue: event $X \ge Y$ $(x^2 \le y \le x)$.

Definition 2.1.5. *Independent Random Variables and IID Samples*

(i) Random variables $X_1, \dots, X_n$ are *independent* if $\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i)\ \forall B_i \in \mathcal{B}(\mathbb{R})$.
(ii) They are *identically distributed* if $\mathcal{L}(X_1) = \cdots = \mathcal{L}(X_n)$.
(iii) They are *iid* if they are independent and identically distributed.
(iv) The *random vector* $X = (X_1, \dots, X_n)$ is a *random sample* when $X_1, \dots, X_n$ are iid from a common law.

Proposition 2.1.6. *Factorization Tests for Independence*

(i) Discrete factorization: discrete X, Y are independent iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all x, y.
(ii) Density factorization: if (X, Y) has joint density $f_{X,Y}$ and marginals f_X, f_Y, then X, Y are independent iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ a.e.
(iii) Rectangular support test: if $f_{X,Y}(x,y) = g(x)h(y)\mathbf{1}_{A\times B}(x,y)$ for measurable A, B, then X, Y are independent after normalization.

Proof.

(i) ($\Rightarrow$) If $X \perp\!\!\!\perp Y$, then $f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) = f_X(x)f_Y(y)$.
(i) ($\Leftarrow$) If $f_{X,Y} = f_X f_Y$, then for A, B, $\mathbb{P}(X \in A, Y \in B) = \sum_{x\in A}\sum_{y\in B} f_X(x)f_Y(y) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$.
(ii) ($\Rightarrow$) If $X \perp\!\!\!\perp Y$, the product law has density $f_X f_Y$, so uniqueness of densities gives $f_{X,Y} = f_X f_Y$ a.e.
(ii) ($\Leftarrow$) If $f_{X,Y} = f_X f_Y$ a.e., then for Borel A, B, $\mathbb{P}(X \in A, Y \in B) = \int_A \int_B f_X(x)f_Y(y)\,dy\,dx = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$.
(iii) Let $G = \int_A g$ and $H = \int_B h$. Normalization gives $GH = 1$, while $f_X(x) = g(x)H\mathbf{1}_A(x)$ and $f_Y(y) = h(y)G\mathbf{1}_B(y)$, hence $f_X(x)f_Y(y) = f_{X,Y}(x,y)$ a.e.; apply (ii).

Structural Commentary: Independence is product structure of the joint law.
Key Insight: Independence means the joint law factors. □

2.2 Multivariate Models and Copulas

Definition 2.2.1. *Two Multivariate Families*

(i) *Multinomial*: for $n \in \mathbb{N}$ and $p \in [0,1]^k$ with $\sum_{j=1}^k p_j = 1$, $X \sim \text{Multinomial}(n,p)$ if $\mathbb{P}(X = x) = n!(x_1!\cdots x_k!)^{-1}\prod_{j=1}^k p_j^{x_j}$ for $x \in \mathbb{N}_0^k$ with $\sum_{j=1}^k x_j = n$; then $X_j \sim \text{Binomial}(n, p_j)$.
(ii) *Multivariate normal*: for $\mu \in \mathbb{R}^k$ and symmetric positive definite $\Sigma \in \mathbb{R}^{k\times k}$, $X \sim N(\mu,\Sigma)$ if $f_X(x) = ((2\pi)^k \det\Sigma)^{-\frac{1}{2}}\exp(-\frac{(x-\mu)^t\Sigma^{-1}(x-\mu)}{2})$ for $x \in \mathbb{R}^k$. $N(0, I_k)$ has independent $N(0,1)$ coordinates.

Proposition 2.2.2. *Multivariate Normal Affine Standardization*

Let $k \in \mathbb{N}$, $\mu \in \mathbb{R}^k$, $\Sigma \in \mathbb{R}^{k\times k}$ with $\Sigma \succ 0$, and $A = \Sigma^{\frac{1}{2}}$.

(i) For $X \in \mathbb{R}^k$, $X \sim N(\mu,\Sigma) \iff A^{-1}(X-\mu) \sim N(0, I_k)$.
(ii) For $Z \in \mathbb{R}^k$, $Z \sim N(0, I_k) \iff \mu + AZ \sim N(\mu,\Sigma)$.

Proof.

(i) ($\Rightarrow$) Put $Z = A^{-1}(X-\mu)$. With $x = \mu + Az$, $|\det A| = (\det\Sigma)^{\frac{1}{2}}$ and $(Az)^t\Sigma^{-1}(Az) = z^t z$. By change of variables, $f_Z(z) = f_X(\mu+Az)|\det A| = (2\pi)^{-\frac{k}{2}}\exp(-\frac{z^t z}{2})$.
(i) ($\Leftarrow$) Put $Z = A^{-1}(X-\mu) \sim N(0, I_k)$, so $X = \mu + AZ$. With $z = A^{-1}(x-\mu)$ and $|\det A^{-1}| = (\det\Sigma)^{-\frac{1}{2}}$, $f_X(x) = f_Z(A^{-1}(x-\mu))|\det A^{-1}| = ((2\pi)^k\det\Sigma)^{-\frac{1}{2}}\exp(-\frac{(x-\mu)^t\Sigma^{-1}(x-\mu)}{2})$.
(ii) ($\Rightarrow$) If $Z \sim N(0, I_k)$ and $X := \mu + AZ$, then $A^{-1}(X-\mu) = Z$; by (i) ($\Leftarrow$), $X \sim N(\mu,\Sigma)$.
(ii) ($\Leftarrow$) If $X := \mu + AZ \sim N(\mu,\Sigma)$, then $Z = A^{-1}(X-\mu)$; by (i) ($\Rightarrow$), $Z \sim N(0, I_k)$.

Structural Commentary: Positive definite covariance is an affine deformation of the standard normal.
Key Insight: Standardize by $\Sigma^{-\frac{1}{2}}$; reconstruct by $\Sigma^{\frac{1}{2}}$. □

Proposition 2.2.3. *Basic Multivariate Normal Calculus*

Let $X = (X_a, X_b) \sim N(\mu,\Sigma)$ with $X_a \in \mathbb{R}^r$, $X_b \in \mathbb{R}^s$, $\mu = (\mu_a,\mu_b)$, and $\Sigma = \begin{pmatrix}\Sigma_{aa} & \Sigma_{ab}\\ \Sigma_{ba} & \Sigma_{bb}\end{pmatrix} \succ 0$.

(i) Marginal: $X_a \sim N(\mu_a, \Sigma_{aa})$.
(ii) Conditional: $X_b \mid X_a = x_a \sim N(\mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a-\mu_a), \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab})$.
(iii) Linear form: for any vector a, $a^tX \sim N(a^t\mu, a^t\Sigma a)$.

Proof.

(i) Let $P_a X = X_a$. By Proposition 2.2.2, $X = \mu + \Sigma^{\frac{1}{2}}Z$ with $Z \sim N(0, I)$. Hence $X_a = P_a\mu + P_a\Sigma^{\frac{1}{2}}Z \sim N(\mu_a, P_a\Sigma P_a^t) = N(\mu_a,\Sigma_{aa})$.
(ii) Set $S := \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}$ and $m(x_a) := \mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a-\mu_a)$. The block determinant and inverse identities give $\det\Sigma = \det\Sigma_{aa}\det S$ and $(x-\mu)^t\Sigma^{-1}(x-\mu) = (x_a-\mu_a)^t\Sigma_{aa}^{-1}(x_a-\mu_a) + (x_b-m(x_a))^tS^{-1}(x_b-m(x_a))$. Thus $f_{b|a}(x_b \mid x_a) = \frac{f_{a,b}(x_a,x_b)}{f_a(x_a)}$ is the $N(m(x_a), S)$ density.
(iii) With $X = \mu + \Sigma^{\frac{1}{2}}Z$, $a^tX = a^t\mu + (\Sigma^{\frac{1}{2}}a)^tZ$. Since Z has independent $N(0,1)$ coordinates, $(\Sigma^{\frac{1}{2}}a)^tZ \sim N(0, a^t\Sigma a)$, hence $a^tX \sim N(a^t\mu, a^t\Sigma a)$.

Structural Commentary: Marginals, conditionals, and linear projections of Gaussian vectors remain Gaussian.
Key Insight: Gaussian calculus is affine calculus plus Schur complements. □

Definition 2.2.4. *Copulas*

(i) A *copula* is a joint CDF $C : [0,1]^2 \to [0,1]$.
(ii) *Uniform marginals*: $C(u,1) = u$ and $C(1,v) = v$ for $u, v \in [0,1]$.
(iii) If $C \in C^2$, its *copula density* is $c(u,v) := \partial_u\partial_v C(u,v)$.

Proposition 2.2.5. *Constructing a Joint Law from Marginals and a Copula*
Let F_X, F_Y be continuous CDFs with densities f_X, f_Y, and let C be a copula.
Define $F_{X,Y}(x,y) := C(F_X(x), F_Y(y))$. Then $F_{X,Y}$ has marginals F_X, F_Y.
If $C \in C^2$ with density $c = \partial_u \partial_v C$, then $f_{X,Y}(x,y) = c(F_X(x), F_Y(y)) f_X(x) f_Y(y)$.

Proof. $F_{X,Y}(x,\infty) = C(F_X(x), 1) = F_X(x)$ and $F_{X,Y}(\infty, y) = C(1, F_Y(y)) = F_Y(y)$.
If densities exist, $\partial_x \partial_y C(F_X(x), F_Y(y)) = (\partial_u \partial_v C)(F_X(x), F_Y(y)) f_X(x) f_Y(y)$ by the chain rule.
Structural Commentary: Marginals and dependence are separated into F_X, F_Y and C.
Key Insight: Marginals alone do not determine a joint law; the copula supplies the dependence. □

Example 2.2.6. *Farlie–Morgenstern Copulas*
For $|\alpha| \le 1$, $C_\alpha(u,v) = uv\{1 + \alpha(1-u)(1-v)\}$ is a copula with density $c_\alpha(u,v) = 1 + \alpha(1-2u)(1-2v)$.
Here $\alpha = 0$ gives the independence copula uv, while $\alpha = -1$ and $\alpha = 1$ give densities $2u + 2v - 4uv$ and $2 - 2u - 2v + 4uv$ on $[0,1]^2$, respectively; all have uniform marginals.

2.3 Transformations of Random Variables

Method 2.3.1. *One-Variable Transformations*
Let $Y = r(X)$.
(i) *Discrete pushforward*: if X is discrete, then $f_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(X \in r^{-1}(\{y\})) = \sum_{x : r(x) = y} f_X(x)$.
(ii) *CDF method*: for $Y = r(X)$, compute $F_Y(y) = \int_{\{x : r(x) \le y\}} f_X(x)\,dx$; $f_Y(y) = F_Y'(y)$ where differentiable.
(iii) *Monotone change*: if r is strictly monotone on $\operatorname{supp} X$ with inverse s, then $f_Y(y) = f_X(s(y))\,|s'(y)|\,\mathbf{1}_{r(\operatorname{supp} X)}(y)$.
Key Insight: The CDF method works because it translates the transformed event back into an event about X.

Example 2.3.2. *Log Transform of an Exponential Variable*
Let $X \sim \mathrm{Exp}(1)$ and $Y = \log X$. Since $r(x) = \log x$ has inverse $s(y) = e^y$ with $s'(y) = e^y$, the change-of-variables formula gives $f_Y(y) = f_X(e^y) e^y = e^y e^{-e^y}$ for $y \in \mathbb{R}$.

Example 2.3.3. *A Nonmonotone Square Transform*
Let $X \sim \mathrm{Uniform}(-1, 3)$ and $Y = X^2$. Since $f_X = \frac{1}{4}$ on $[-1,3]$, $r^{-1}(\{y\}) \cap [-1,3] = \{-\sqrt{y}, \sqrt{y}\}$ for $0 < y < 1$ and $\{\sqrt{y}\}$ for $1 < y < 9$.
Hence $f_Y(y) = \frac{1}{4\sqrt{y}} \mathbf{1}_{(0,1)}(y) + \frac{1}{8\sqrt{y}} \mathbf{1}_{(1,9)}(y)$.
Key Insight: Non-one-to-one transformations require splitting the range into cases.

Example 2.3.4. *Squaring a Standard Normal*
Let $Z \sim N(0,1)$ and $X = Z^2$. For $x > 0$, $F_X(x) = \mathbb{P}(|Z| \le \sqrt{x}) = \Phi(\sqrt{x}) - \Phi(-\sqrt{x})$,
so $f_X(x) = x^{-\frac{1}{2}} \phi(\sqrt{x}) = (2\pi)^{-\frac{1}{2}} x^{-\frac{1}{2}} e^{-\frac{x}{2}}$. Hence $X \sim \chi^2_1 = \mathrm{Gamma}(\frac{1}{2}, \frac{1}{2})$ in the rate convention.

Example 2.3.5. *Reciprocal of a Uniform Variable*
Let $U \sim \mathrm{Uniform}(0,1)$ and $V = \frac{1}{U}$. Then $F_V(v) = 0$ for $v < 1$, while for $v \ge 1$, $F_V(v) = \mathbb{P}(U \ge \frac{1}{v}) = 1 - \frac{1}{v}$.
Hence $f_V(v) = v^{-2} \mathbf{1}_{[1,\infty)}(v)$.

Proposition 2.3.6. *Probability Integral Transform and Inverse Transform*
Let $F : I \to (0,1)$ be a continuous strictly increasing CDF with inverse $F^{-1} : (0,1) \to I$.
(i) If $X \sim F$, then $F(X) \sim \mathrm{Uniform}(0,1)$.
(ii) If $U \sim \mathrm{Uniform}(0,1)$ and $Y = F^{-1}(U)$, then $F_Y = F$.

Proof.
(i) For $0 < u < 1$, $\mathbb{P}(F(X) \le u) = \mathbb{P}(X \le F^{-1}(u)) = F(F^{-1}(u)) = u$, with endpoint values $0, 1$.
(ii) For $x \in I$, $\mathbb{P}(Y \le x) = \mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$.
Structural Commentary: F maps its law to uniform scale; F^{-1} maps uniform scale back.
Key Insight: Uniform random numbers become target laws by applying inverse CDFs. □

Method 2.3.7. *Several-Variable Transformations*
For $Z = r(X, Y)$ with joint density $f_{X,Y}$:
(i) *Sublevel set*: define $A_z := \{(x,y) : r(x,y) \le z\}$.
(ii) *CDF integral*: compute $F_Z(z) = \mathbb{P}(Z \le z) = \iint_{A_z} f_{X,Y}(x,y)\,dx\,dy$.
(iii) *Density*: differentiate where possible, $f_Z(z) = F_Z'(z)$.

Example 2.3.8. *Sum of Two Independent Uniforms*
Let $X_1, X_2 \sim \mathrm{Uniform}(0,1)$ be independent and $Y = X_1 + X_2$.
Since $f_{X_1,X_2} = \mathbf{1}_{[0,1]^2}$, $F_Y(y)$ is the area of $A_y = \{(x_1, x_2) \in [0,1]^2 : x_1 + x_2 \le y\}$,
so $F_Y(y) = \frac{y^2}{2} \mathbf{1}_{[0,1]}(y) + (1 - \frac{(2-y)^2}{2}) \mathbf{1}_{(1,2]}(y) + \mathbf{1}_{(2,\infty)}(y)$. Hence $f_Y(y) = y \mathbf{1}_{(0,1)}(y) + (2-y) \mathbf{1}_{(1,2)}(y)$.

Proposition 2.3.9. *Two-Dimensional Density Change of Variables*
Let (X,Y) have density $f_{X,Y}$ on $D \subseteq \mathbb{R}^2$, and let $T : D \xrightarrow{C^1} E$ be one-to-one with inverse h and $\det DT \neq 0$ on D. For $(U,V) = T(X,Y)$, $f_{U,V}(u,v) = f_{X,Y}(h(u,v))\,|\det Dh(u,v)|\,\mathbf{1}_E(u,v)$.

Proof. For Borel $B \subseteq E$, $\mathbb{P}((U,V) \in B) = \mathbb{P}((X,Y) \in h(B))$
$= \int_{h(B)} f_{X,Y}(x,y)\,dx\,dy = \int_B f_{X,Y}(h(u,v))\,|\det Dh(u,v)|\,du\,dv$ (by Theorem 5.1.11)
Structural Commentary: Transforming a density means preserving probability mass while rescaling area.
Key Insight: The absolute Jacobian accounts for local volume distortion. □

Example 2.3.10. *Sums and Quotients*
Let (X,Y) have joint density $f_{X,Y}$.
(i) Sum: for $S = X + Y$, $f_S(s) = \int_{\mathbb{R}} f_{X,Y}(x, s-x)\,dx$; if $X \perp\!\!\!\perp Y$, then $f_S(s) = \int_{\mathbb{R}} f_X(x) f_Y(s-x)\,dx$.

(ii) Quotient: for $Z = \frac{Y}{X}$, $f_Z(z) = \int_{\mathbb{R}} |x|\, f_{X,Y}(x, xz)\, dx$; if $X \perp\!\!\!\perp Y$, then $f_Z(z) = \int_{\mathbb{R}} |x|\, f_X(x) f_Y(xz)\, dx$.

Key Insight: Add one auxiliary coordinate, change variables, integrate it out.

Example 2.3.11. *Polar Transform of Two Standard Normals*
Let $X, Y \overset{\text{iid}}{\sim} N(0,1)$, $R = (X^2+Y^2)^{\frac{1}{2}}$, and $\Theta = \operatorname{atan2}(Y, X)$.

Polar change of variables gives $f_{R,\Theta}(r,\theta) = (2\pi)^{-1} r e^{-\frac{r^2}{2}} \mathbf{1}_{[0,\infty)}(r)\mathbf{1}_{[0,2\pi)}(\theta)$,
so $R \perp\!\!\!\perp \Theta$, $\Theta \sim \text{Uniform}(0, 2\pi)$, and $R^2 \sim \text{Exp}(\frac{1}{2})$.
Hence for iid $U_1, U_2 \sim \text{Uniform}(0,1)$, $(\sqrt{-2\log U_1}\cos(2\pi U_2), \sqrt{-2\log U_1}\sin(2\pi U_2)) \sim N(0, I_2)$.

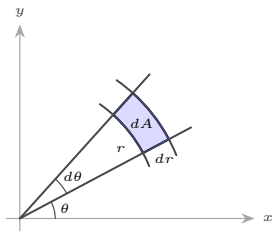

Figure 229: In polar coordinates, a small patch has area approximately $r\, dr\, d\theta$.

3 Expectation and Integration in Probability

3.1 Expectation, Variance, and LOTUS

Expectation is the averaging operation attached to a probability law. In the notation of Section 1.1.1, if X has law F_X, then integration against F_X is the common notation that covers both sums for discrete variables and ordinary integrals for variables with densities.

Definition 3.1.1. *Expectation Through the Law*
Let X be real-valued with distribution function F_X.

(i) *Expectation*: if $\int_{\mathbb{R}} |x|\, dF_X(x) < \infty$, then $\mathbb{E}(X) := \int_{\mathbb{R}} x\, dF_X(x)$.
(ii) *Discrete form*: if X has support S, mass function f_X, $\mathbb{E}(X) = \sum_{x\in S} x f_X(x)$ when $\sum_{x\in S} |x| f_X(x) < \infty$.
(iii) *Density form*: if X has density f_X, then $\mathbb{E}(X) = \int_{\mathbb{R}} x f_X(x)\, dx$ when $\int_{\mathbb{R}} |x| f_X(x)\, dx < \infty$.
(iv) *Function form*: for measurable $r : \mathbb{R} \to \mathbb{R}$, if $\int_{\mathbb{R}} |r(x)|\, dF_X(x) < \infty$, then $\mathbb{E}(r(X)) := \int_{\mathbb{R}} r(x)\, dF_X(x)$.

Proposition 3.1.2. *Indicators Turn Probabilities into Expectations*
For $A \in \mathcal{F}$, let $\mathbf{1}_A : \Omega \to \{0,1\}$ be $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ for $\omega \notin A$. Then $\mathbb{E}(\mathbf{1}_A) = \mathbb{P}(A)$.

Proof. Since $\Omega = A \sqcup A^c$ and $\mathbf{1}_A = 1$ on A, 0 on A^c, $\mathbb{E}(\mathbf{1}_A) = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A)$.
Structural Commentary: Indicators encode events as 0-1 random variables.
Key Insight: Probabilities are expectations of indicators. □

Example 3.1.3. *Expectation of the Two-Toss Head Count*
Let $\Omega = \{H,T\}^2$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$, and let $X(\omega)$ be the number of heads in ω.

Then $f_X(k) = \binom{2}{k} 2^{-2}$ for $k = 0,1,2$, so $\mathbb{E}(X) = \sum_{k=0}^{2} k f_X(k) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$.

Example 3.1.4. *First Expectation Computations*

(i) If $X \sim \text{Bernoulli}(p)$, then $\mathbb{E}X = 0(1-p) + 1p = p$.
(ii) If $X \sim \text{Uniform}(-1, 3)$, then $\mathbb{E}X = (\frac{1}{4})\int_{-1}^{3} x\, dx = 1$.
(iii) If $X \sim \text{Cauchy}(0,1)$, then $\mathbb{E}|X| = (\frac{2}{\pi})\int_0^\infty x(1+x^2)^{-1}\, dx = \infty$, so $\mathbb{E}X$ does not exist.

Theorem 3.1.5. *Expectation of a Function*
Let r be measurable and assume the displayed expectations exist.

(i) One variable: for real-valued X, $\mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x)\, dF_X(x)$.
(ii) Several variables: for (X,Y) with joint law $F_{X,Y}$, $\mathbb{E}(r(X,Y)) = \int_{\mathbb{R}^2} r(x,y)\, dF_{X,Y}(x,y)$.

Proof.

(i) Since $\mathcal{L}(X) = X_*\mathbb{P}$, $\mathbb{E}(r(X)) = \int_\Omega r(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} r(x)\, d\mathcal{L}(X)(x) = \int_{\mathbb{R}} r(x)\, dF_X(x)$.
(ii) Let $W = (X,Y)$. Since $\mathcal{L}(W) = W_*\mathbb{P} = F_{X,Y}$,
$\mathbb{E}(r(X,Y)) = \int_\Omega r(W(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}^2} r(x,y)\, d\mathcal{L}(W)(x,y) = \int_{\mathbb{R}^2} r(x,y)\, dF_{X,Y}(x,y)$.

Structural Commentary: Expectations of transformations integrate over the transformed law.
Key Insight: Pushforward laws let functions move from random variables to their values. □

Remark 3.1.6. *Law-Based Integration*
For measurable r, $\mathbb{E}(r(X))$ depends only on the law of X: $\mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x)\, dF_X(x)$ whenever integrable.
Discrete laws give $\sum_x r(x) f_X(x)$; density laws give $\int_{\mathbb{R}} r(x) f_X(x)\, dx$.

Example 3.1.7. *Using the Function Rule*

(i) If $X \sim \text{Uniform}(0,1)$ and $Y = e^X$, then $\mathbb{E}Y = \mathbb{E}(e^X) = \int_0^1 e^x\, dx = e - 1$.
(ii) If $X \sim \text{Uniform}(0,1)$ and $Y = \max\{X, 1-X\}$, then $\mathbb{E}Y = \int_0^{\frac{1}{2}} (1-x)\, dx + \int_{\frac{1}{2}}^1 x\, dx = \frac{3}{4}$.
(iii) If $(X,Y) \sim \text{Uniform}([0,1]^2)$ and $Z = X^2 + Y^2$, then $\mathbb{E}Z = \int_0^1 \int_0^1 (x^2+y^2)\, dx\, dy = \frac{2}{3}$.

Definition 3.1.8. *Variance and Covariance*

(i) The *variance* of X is $\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2]$, assuming $\mathbb{E}(X^2) < \infty$.
(ii) The *standard deviation* of X is $\mathrm{sd}(X) := \sqrt{\mathrm{Var}(X)}$.
(iii) The *covariance* of X, Y is $\mathrm{Cov}(X,Y) := \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, assuming $\mathbb{E}(X^2), \mathbb{E}(Y^2) < \infty$.

3.2 Moments, Sample Statistics, and Random Vectors

Definition 3.2.1. *Moments, Correlation, and Sample Statistics*

(i) The *kth moment* of X is $\mathbb{E}(X^k)$, assuming $\mathbb{E}|X|^k < \infty$.
(ii) The *correlation* of X, Y is $\mathrm{Corr}(X,Y) := \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$, assuming $\mathrm{Var}(X), \mathrm{Var}(Y) > 0$.
(iii) The *sample mean* of $X_1, \dots, X_n$ is $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$.
(iv) The *sample variance* of $X_1, \dots, X_n$ is $S_n^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$.

Theorem 3.2.2. *Expectation, Variance, and Covariance Algebra*
Assume all displayed quantities exist and $a_i, a, b \in \mathbb{R}$.

(i) Linearity: $\mathbb{E}(\sum_i a_i X_i) = \sum_i a_i \mathbb{E}X_i$.
(ii) Independent product: if $X_1, \dots, X_n$ are independent, then $\mathbb{E}(\prod_i X_i) = \prod_i \mathbb{E}X_i$.
(iii) Variance identities: $\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$ *and* $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$.
(iv) Covariance identity: $\mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y$.
(v) Variance of a linear combination: $\mathrm{Var}(\sum_i a_i X_i) = \sum_i a_i^2 \mathrm{Var}(X_i) + 2 \sum_{i<j} a_i a_j \mathrm{Cov}(X_i, X_j)$.

Proof.

(i) $\mathbb{E}(\sum_i a_i X_i) = \int_\Omega \sum_i a_i X_i \, d\mathbb{P} = \sum_i a_i \int_\Omega X_i \, d\mathbb{P} = \sum_i a_i \mathbb{E}X_i$.
(ii) By independence, $\mathcal{L}(X_1, \dots, X_n) = \prod_i F_{X_i}$; $\mathbb{E}(\prod_i X_i) = \int_{\mathbb{R}^n} \prod_i x_i \, d\prod_i F_{X_i}(x_i) = \prod_i \int_{\mathbb{R}} x_i \, dF_{X_i}(x_i) = \prod_i \mathbb{E}X_i$.
(iii) $\mathrm{Var}(X) = \mathbb{E}(X^2 - 2X\mathbb{E}X + (\mathbb{E}X)^2) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$.
Also $\mathrm{Var}(aX + b) = \mathbb{E}[(aX + b - a\mathbb{E}X - b)^2] = a^2 \mathrm{Var}(X)$.
(iv) $\mathrm{Cov}(X,Y) = \mathbb{E}(XY - X\mathbb{E}Y - Y\mathbb{E}X + \mathbb{E}X\mathbb{E}Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y$.
(v) With $S = \sum_i a_i X_i$, (iv) gives

$$\mathrm{Var}(S) = \mathrm{Cov}(S,S) = \sum_{i,j} a_i a_j \mathrm{Cov}(X_i, X_j) = \sum_i a_i^2 \mathrm{Var}(X_i) + 2 \sum_{i<j} a_i a_j \mathrm{Cov}(X_i, X_j).$$

Structural Commentary: Expectation is linear; variance and covariance are quadratic and bilinear consequences.
Key Insight: Expand products, then use linearity. □

Example 3.2.3. *Indicator Sums as a Method*

(i) If $X \sim \mathrm{Binomial}(n,p)$, write $X = \sum_{i=1}^{n} I_i$ with independent $I_i \sim \mathrm{Bernoulli}(p)$.
Then $\mathbb{E}X = np$ and $\mathrm{Var}(X) = np(1-p)$.
(ii) Let $\xi_i \in \{-1, 1\}$ with $\mathbb{P}(\xi_i = -1) = p$ and $\mathbb{P}(\xi_i = 1) = 1 - p$, independently, and $X_n = \sum_{i=1}^{n} \xi_i$.
Then $\mathbb{E}X_n = n(1-2p)$ and $\mathrm{Var}(X_n) = 4np(1-p)$.
(iii) Let $F_n = c \prod_{i=1}^{n} M_i$ with independent $M_i \in \{2, \frac{1}{2}\}$ and $\mathbb{P}(M_i = 2) = \mathbb{P}(M_i = \frac{1}{2}) = \frac{1}{2}$.
Then $\mathbb{E}F_n = c \prod_{i=1}^{n} \mathbb{E}M_i = c(\frac{5}{4})^n$.

Proposition 3.2.4. *Zero Variance Means Almost Surely Constant*
For $X \in L^2$ and $m := \mathbb{E}X$, $\mathrm{Var}(X) = 0$ iff $X = m$ a.s.

Proof.

($\Rightarrow$) If $\mathrm{Var}(X) = 0$, then $\mathbb{E}[(X-m)^2] = 0$. Since $(X-m)^2 \geq 0$, $\mathbb{P}((X-m)^2 > 0) = 0$, hence $X = m$ a.s.
($\Leftarrow$) If $X = m$ a.s., then $(X-m)^2 = 0$ a.s., so $\mathrm{Var}(X) = \mathbb{E}[(X-m)^2] = 0$.
Structural Commentary: Variance is the mean squared distance from the mean.
Key Insight: Zero nonnegative expectation forces zero almost surely. □

Proposition 3.2.5. *Tail Integral Formula*
If $X \geq 0$ has density f_X, distribution function F_X, and $\mathbb{E}X < \infty$,
then $\mathbb{E}X = \int_0^\infty \mathbb{P}(X > x)\,dx = \int_0^\infty (1 - F_X(x))\,dx$.

Proof. For $b > 0$, integration by parts gives $\int_0^b x f_X(x)\,dx = -b(1 - F_X(b)) + \int_0^b (1 - F_X(x))\,dx$.
Since $\mathbb{E}X < \infty$, $b(1 - F_X(b)) \leq \int_b^\infty x f_X(x)\,dx \to 0$.
Letting $b \to \infty$ yields $\mathbb{E}X = \int_0^\infty (1 - F_X(x))\,dx = \int_0^\infty \mathbb{P}(X > x)\,dx$.
Structural Commentary: A nonnegative variable is integrated by summing its survival probabilities.
Key Insight: Expectation equals area under the tail. □

Example 3.2.6. *Maximum of Uniform Samples*
Let $X_1, \dots, X_n$ be independent $\mathrm{Uniform}(0,1)$ variables and $Y_n = \max_i X_i$.
Then $F_{Y_n}(y) = \mathbb{P}(X_1 \leq y, \dots, X_n \leq y) = y^n$ for $0 \leq y \leq 1$.
So $\mathbb{E}Y_n = \int_0^1 (1 - F_{Y_n}(y))\,dy = \int_0^1 (1 - y^n)\,dy = \frac{n}{n+1}$.

Theorem 3.2.7. *Mean and Variance of the Basic Sample Statistics*
Let $X_1, \dots, X_n$ be iid with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$.
(i) Sample-mean expectation: $\mathbb{E}\overline{X}_n = \mu$.
(ii) Sample-mean variance: $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$.
(iii) Unbiased sample variance: $\mathbb{E}S_n^2 = \sigma^2$.

Proof.
(i) $\mathbb{E}\overline{X}_n = n^{-1}\sum_{i=1}^{n}\mathbb{E}X_i = n^{-1}n\mu = \mu$.
(ii) Independence gives $\mathrm{Var}(\overline{X}_n) = n^{-2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = n^{-2}n\sigma^2 = \frac{\sigma^2}{n}$.
(iii) Since $\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\overline{X}_n - \mu)^2$. Thus $\mathbb{E}[(n-1)S_n^2] = n\sigma^2 - n(\frac{\sigma^2}{n}) = (n-1)\sigma^2$.
Structural Commentary: Averaging preserves mean and scales variance.
Key Insight: Center at μ, then subtract the fluctuation of $\overline{X}_n$. □

Lemma 3.2.8. *Expected Squared Deviation from a Sample Mean*
Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = \mu_i$ and $\mathrm{Var}(X_i) = \sigma^2$ for $1 \le i \le n$.
Set $\overline{X} := \frac{1}{n}\sum_{i=1}^{n}X_i$ and $\overline{\mu} := \frac{1}{n}\sum_{i=1}^{n}\mu_i$. Then $\mathbb{E}(X_i - \overline{X})^2 = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2$ for $1 \le i \le n$.

Proof. Set $U_i := X_i - \overline{X}$. Then $\mathbb{E}U_i = \mu_i - \overline{\mu}$, $\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$, and $\mathrm{Cov}(X_i, \overline{X}) = \frac{\sigma^2}{n}$ by independence.
Hence $\mathrm{Var}(U_i) = \mathrm{Var}(X_i) + \mathrm{Var}(\overline{X}) - 2\,\mathrm{Cov}(X_i, \overline{X}) = \sigma^2 + \frac{\sigma^2}{n} - 2\frac{\sigma^2}{n} = \frac{n-1}{n}\sigma^2$.
Thus $\mathbb{E}U_i^2 = (\mathbb{E}U_i)^2 + \mathrm{Var}(U_i) = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2$.
Structural Commentary: $X_i - \overline{X}$ has mean $\mu_i - \overline{\mu}$ and variance $\mathrm{Var}(X_i - \overline{X}) = \frac{n-1}{n}\sigma^2$.
Key Insight: $\mathrm{Cov}(X_i, \overline{X}) = \frac{\sigma^2}{n}$ removes exactly one $\frac{\sigma^2}{n}$ from the raw variance σ^2. □

Definition 3.2.9. *Mean Vector and Covariance Matrix*
For a random vector $X = (X_1, \dots, X_k)^t$:
(i) *Mean vector*: $\mathbb{E}X := (\mathbb{E}X_1, \dots, \mathbb{E}X_k)^t$.
(ii) *Covariance matrix*: $\mathrm{Var}(X) := [\mathrm{Cov}(X_i, X_j)]_{i,j=1}^{k}$.

Proposition 3.2.10. *Linear Transformations of Random Vectors*
Let $X \in \mathbb{R}^k$ satisfy $\mathbb{E}X = \mu$ and $\mathrm{Var}(X) = \Sigma$.
i. Linear form: for $a \in \mathbb{R}^k$, $\mathbb{E}(a^tX) = a^t\mu$ and $\mathrm{Var}(a^tX) = a^t\Sigma a$.
ii. Matrix transform: for $A \in \mathbb{R}^{m\times k}$, $\mathbb{E}(AX) = A\mu$ and $\mathrm{Var}(AX) = A\Sigma A^t$.

Proof.
i. $\mathbb{E}(a^tX) = a^t\mathbb{E}X = a^t\mu$, and $\mathrm{Var}(a^tX) = \mathbb{E}[(a^t(X-\mu))^2] = a^t\mathbb{E}[(X-\mu)(X-\mu)^t]a = a^t\Sigma a$.
ii. $\mathbb{E}(AX) = A\mathbb{E}X = A\mu$, and $\mathrm{Var}(AX) = \mathbb{E}[(AX - A\mu)(AX - A\mu)^t] = A\mathbb{E}[(X-\mu)(X-\mu)^t]A^t = A\Sigma A^t$.
Structural Commentary: Linear maps push means forward and conjugate covariance matrices.
Key Insight: Center first; then move constants outside expectations. □

Example 3.2.11. *Multinomial Covariance Structure*
If $X \sim \mathrm{Multinomial}(n, p)$ with $p = (p_1, \dots, p_k)$, then $\mathbb{E}X = np$ and $\mathrm{Var}(X_i) = np_i(1-p_i)$.
For $i \neq j$, $X_i + X_j \sim \mathrm{Binomial}(n, p_i + p_j)$, so $n(p_i+p_j)(1-p_i-p_j) = np_i(1-p_i) + np_j(1-p_j) + 2\,\mathrm{Cov}(X_i, X_j)$.
Hence $\mathrm{Cov}(X_i, X_j) = -np_ip_j$.

3.3 Probability Inequalities

Theorem 3.3.1. *Markov and Chebyshev Inequalities*
i. Markov: if $X \ge 0$ and $\mathbb{E}X < \infty$, then for $t > 0$, $\mathbb{P}(X \ge t) \le \frac{\mathbb{E}X}{t}$.
ii. Chebyshev: if $\mu := \mathbb{E}X$ and $\sigma^2 := \mathrm{Var}(X) < \infty$, then for $t > 0$, $\mathbb{P}(|X - \mu| \ge t) \le \frac{\sigma^2}{t^2}$.

Proof.
i. Since $X \ge t\mathbf{1}_{\{X \ge t\}}$, taking expectations gives $\mathbb{E}X \ge t\mathbb{P}(X \ge t)$.
ii. Apply (i) to $Y := (X-\mu)^2 \ge 0$ with threshold t^2, then $\mathbb{P}(|X-\mu| \ge t) = \mathbb{P}(Y \ge t^2) \le \frac{\mathbb{E}Y}{t^2} = \frac{\sigma^2}{t^2}$.
Structural Commentary: Tail bounds convert moment control into probability control.
Key Insight: Dominate the tail indicator by a scaled nonnegative random variable. □

Example 3.3.2. *A Distribution-Free Error-Rate Bound*
Let $X_1, \dots, X_n$ be independent Bernoulli(p) error indicators and $\overline{X}_n := n^{-1}\sum_{i=1}^{n}X_i$. Since $\mathbb{E}\overline{X}_n = p$ and $\mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}$, Chebyshev gives, for $\varepsilon > 0$, $\mathbb{P}(|\overline{X}_n - p| > \varepsilon) \le \frac{p(1-p)}{n\varepsilon^2} \le \frac{1}{4n\varepsilon^2}$.

Theorem 3.3.3. *Cauchy–Schwarz and Jensen*
If $X, Y \in L^2$, then $\mathbb{E}|XY| \le \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}$.
If $g : I \to \mathbb{R}$ convex, $X \in I$ a.s., $X, g(X) \in L^1$, then $\mathbb{E}g(X) \ge g(\mathbb{E}X)$; for concave g, inequality reverses.

Proof. For Cauchy–Schwarz, apply the L^2 inner-product inequality to $|X|$ and $|Y|$.
For Jensen, set $m := \mathbb{E}X$ and take an affine support $\ell(x) := g(m) + s(x-m)$ with $\ell \le g$ on I.
Then $\mathbb{E}g(X) \ge \mathbb{E}\ell(X) = g(m) + s(\mathbb{E}X - m) = g(\mathbb{E}X)$. For concave g, apply the convex case to $-g$.
Structural Commentary: Cauchy–Schwarz controls products; Jensen moves convexity through expectation.
Key Insight: Use Hilbert-space geometry for products and supporting lines for convex functions. □

Theorem 3.3.4. *Hoeffding's Bounded-Sum Bound*
Let $Y_1, \dots, Y_n$ be independent, $\mathbb{E}Y_i = 0$, $a_i \le Y_i \le b_i$ a.s., and $V := \sum_{i=1}^{n} (b_i - a_i)^2$. For $S_n := \sum_{i=1}^{n} Y_i$ and $t, \varepsilon > 0$, $\mathbb{P}(S_n \ge \varepsilon) \le \exp(-t\varepsilon + \frac{t^2 V}{8})$; if $V > 0$, then $\mathbb{P}(S_n \ge \varepsilon) \le \exp(-\frac{2\varepsilon^2}{V})$.

Proof. For $t > 0$, Markov gives $\mathbb{P}(S_n \ge \varepsilon) = \mathbb{P}(e^{tS_n} \ge e^{t\varepsilon}) \le e^{-t\varepsilon}\mathbb{E}e^{tS_n}$.
Independence gives $\mathbb{E}e^{tS_n} = \prod_{i=1}^{n} \mathbb{E}e^{tY_i}$, and Hoeffding's lemma gives $\mathbb{E}e^{tY_i} \le \exp(\frac{t^2(b_i-a_i)^2}{8})$.
Hence $\mathbb{P}(S_n \ge \varepsilon) \le \exp(-t\varepsilon + \frac{t^2 V}{8})$. If $V > 0$, minimize the exponent at $t = \frac{4\varepsilon}{V}$ to obtain $\exp(-\frac{2\varepsilon^2}{V})$.
Structural Commentary: Hoeffding turns bounded independent increments into exponential tail decay.
Key Insight: Exponentiate the sum, factor the MGF, then optimize t. □

Corollary 3.3.5. *Hoeffding for Bernoulli Averages*
If $X_1, \dots, X_n$ independent Bernoulli(p) *variables, $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$, then $\mathbb{P}(|\overline{X}_n - p| > \varepsilon) \le 2e^{-2n\varepsilon^2}$.*

Proof. Set $Y_i^+ := \frac{X_i - p}{n}$.
Then $\mathbb{E}Y_i^+ = 0$, $Y_i^+ \in [-\frac{p}{n}, \frac{1-p}{n}]$, and $V = \sum_i (\frac{1}{n})^2 = \frac{1}{n}$, by Theorem 3.3.4, $\mathbb{P}(\overline{X}_n - p \ge \varepsilon) \le e^{-2n\varepsilon^2}$.
Apply same argument to $Y_i^- := \frac{p - X_i}{n}$, $\mathbb{P}(p - \overline{X}_n \ge \varepsilon) \le e^{-2n\varepsilon^2}$; the union bound yields the claim.
Structural Commentary: Bernoulli averages are sums of bounded centered increments.
Key Insight: Control both tails separately, then add them. □

3.4 Moment Generating Functions

Definition 3.4.1. *Moment Generating Function*
The *moment generating function* of X is $\psi_X(t) := \mathbb{E}(e^{tX})$ on $D_X := \{t \in \mathbb{R} : \mathbb{E}e^{tX} < \infty\}$.
If $(-\delta, \delta) \subseteq D_X$ for some $\delta > 0$, then $\psi_X^{(k)}(0) = \mathbb{E}(X^k)$ for all $k \ge 1$.

Proposition 3.4.2. *Basic MGF Properties*
Assume the MGFs below are finite on a common open interval about 0.

(i) Affine transform: if $a, b \in \mathbb{R}$ and $Y = aX + b$, then $\psi_Y(t) = e^{bt}\psi_X(at)$.

(ii) Independent sum: if $X_1, \dots, X_n$ are independent and $Y := \sum_{i=1}^{n} X_i$, then $\psi_Y(t) = \prod_{i=1}^{n} \psi_{X_i}(t)$.

(iii) Uniqueness: if $\psi_X(t) = \psi_Y(t)$ for all t in an open interval containing 0, then $\mathcal{L}(X) = \mathcal{L}(Y)$.

Proof.
(i) $\psi_Y(t) = \mathbb{E}e^{t(aX+b)} = e^{bt}\mathbb{E}e^{(at)X} = e^{bt}\psi_X(at)$.
(ii) For admissible t, $e^{tX_1}, \dots, e^{tX_n}$ are independent and integrable.
Hence $\psi_Y(t) = \mathbb{E}e^{t\sum_{i=1}^{n} X_i} = \mathbb{E}\prod_{i=1}^{n} e^{tX_i} = \prod_{i=1}^{n} \mathbb{E}e^{tX_i} = \prod_{i=1}^{n} \psi_{X_i}(t)$.
(iii) Let $\mu := \mathcal{L}(X)$ and $\nu := \mathcal{L}(Y)$. If $\psi_X = \psi_Y$ on $(-\delta, \delta)$, then $\int_{\mathbb{R}} e^{tx}\, d\mu(x) = \int_{\mathbb{R}} e^{tx}\, d\nu(x)$ for all $|t| < \delta$.
Uniqueness of the two-sided Laplace transform gives $\mu = \nu$.
Structural Commentary: MGFs turn affine maps and independent sums into algebraic operations.
Key Insight: Exponentials convert sums into products, and the MGF near 0 determines the law. □

Example 3.4.3. *MGF Computations*
(i) If $X \sim \text{Exp}(1)$, then $\psi_X(t) = \mathbb{E}e^{tX} = \int_0^\infty e^{(t-1)x}\, dx = (1-t)^{-1}$ for $t < 1$.
Hence $\mathbb{E}X = \psi_X'(0) = 1$ and $\text{Var}(X) = \psi_X''(0) - (\psi_X'(0))^2 = 1$.
(ii) If $X \sim \text{Binomial}(n, p)$, then $X = \sum_{i=1}^{n} B_i$ with independent $B_i \sim \text{Bernoulli}(p)$.
Hence $\psi_X(t) = \prod_{i=1}^{n} \psi_{B_i}(t) = (pe^t + 1 - p)^n$.

Method 3.4.4. *MGF Calculus for Classical Laws*
If $\psi_X(t) = \mathbb{E}e^{tX}$ exists near 0, then $\mathbb{E}X = \psi_X'(0)$ and $\text{Var}(X) = \psi_X''(0) - \psi_X'(0)^2$.
(i) If $X \sim \text{Poisson}(\lambda)$, then $\psi_X(t) = \exp(\lambda(e^t - 1))$, $\mathbb{E}X = \lambda$, and $\text{Var}(X) = \lambda$.
(ii) If $X \sim \text{Gamma}(\alpha, \lambda)$ with rate λ, then $\psi_X(t) = (\frac{\lambda}{\lambda - t})^\alpha$ for $t < \lambda$, $\mathbb{E}X = \frac{\alpha}{\lambda}$, and $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.
(iii) If $Z \sim N(0,1)$, then $\psi_Z(t) = e^{\frac{t^2}{2}}$; for $Y = \mu + \sigma Z$, $\psi_Y(t) = \exp(\mu t + \frac{\sigma^2 t^2}{2})$.

Corollary 3.4.5. *Independent Sums from MGFs*
Let $X \perp\!\!\!\perp Y$, and interpret all MGFs on a common neighborhood of 0.

(i) If $\lambda, \mu > 0$, $X \sim \text{Poisson}(\lambda)$, and $Y \sim \text{Poisson}(\mu)$, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.

(ii) If $X \sim \text{Gamma}(\alpha_1, \lambda)$ and $Y \sim \text{Gamma}(\alpha_2, \lambda)$ with $\alpha_1, \alpha_2, \lambda > 0$, then $X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$.

(iii) If $\mu, \nu \in \mathbb{R}$, $\sigma, \tau > 0$, $X \sim N(\mu, \sigma^2)$, and $Y \sim N(\nu, \tau^2)$, then $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$.

Proof.

(i) $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = \exp(\lambda(e^t-1))\exp(\mu(e^t-1)) = \exp((\lambda+\mu)(e^t-1))$, QED.

(ii) For $t < \lambda$, $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = (\frac{\lambda}{\lambda-t})^{\alpha_1+\alpha_2}$, QED.

(iii) $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = \exp(\mu t + \frac{\sigma^2t^2}{2})\exp(\nu t + \frac{\tau^2t^2}{2}) = \exp((\mu+\nu)t + \frac{(\sigma^2+\tau^2)t^2}{2})$, QED.

Structural Commentary: Independence turns convolution into multiplication of MGFs.
Key Insight: Closed distribution families are stable when their MGF parameters add. □

Method 3.4.6. *Common Means, Variances, and MGFs*
MGFs are stated on their natural neighborhood of 0; values at $t = 0$ are by continuity.
The geometric and negative-binomial rows use the total-trials convention.

Distribution	$\mathbb{E}X$	$\mathrm{Var}(X)$	$\psi_X(t)$
point mass at a	a	0	e^{at}
Bernoulli(p)	p	$p(1-p)$	$1-p+pe^t$
Binomial(n,p)	np	$np(1-p)$	$(1-p+pe^t)^n$
Geometric(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
NegBin(r,p)	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Poisson(λ)	λ	λ	$e^{\lambda(e^t-1)}$
Uniform(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$
$N(\mu,\sigma^2)$	μ	σ^2	$\exp(\mu t + \frac{\sigma^2t^2}{2})$
Exp(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Gamma(α,λ)	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$
Beta(α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	not useful
χ^2_p	p	$2p$	$(1-2t)^{-\frac{p}{2}}$

4 Order Statistics, Extrema, and Moment Methods

4.1 Order Statistics, Extrema, and Moment Methods

Order statistics describe the sorted values of an independent sample. They turn joint density calculations into formulas for minima, maxima, medians, ranges, and distribution-free tolerance intervals.

Definition 4.1.1. *Order Statistics*
For real-valued $X_1, \dots, X_n$,

(i) The *order statistics* are the pointwise rearrangement $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ of $X_1, \dots, X_n$.

(ii) $X_{(1)} = \min_{1\le i\le n} X_i$, $X_{(n)} = \max_{1\le i\le n} X_i$, and if $n = 2m+1$, then $X_{(m+1)}$ is the sample median.

Proposition 4.1.2. *Minimum and Maximum of an IID Continuous Sample*
Let $n \ge 1$ and $X_1, \dots, X_n \overset{\text{iid}}{\sim} F$, where F is differentiable with density $f = F'$.
Set $V = X_{(1)} = \min_i X_i$ and $U = X_{(n)} = \max_i X_i$; then for $u, v \in \mathbb{R}$, $F_U(u) = F(u)^n$, $f_U(u) = nf(u)F(u)^{n-1}$, $F_V(v) = 1-(1-F(v))^n$, and $f_V(v) = nf(v)(1-F(v))^{n-1}$.

Proof. Since $\{U \le u\} = \bigcap_{i=1}^n \{X_i \le u\}$, independence gives $F_U(u) = \prod_{i=1}^n F(u) = F(u)^n$.
Differentiation gives $f_U(u) = nf(u)F(u)^{n-1}$.
Since $\{V > v\} = \bigcap_{i=1}^n \{X_i > v\}$, independence gives $1 - F_V(v) = \prod_{i=1}^n (1-F(v)) = (1-F(v))^n$.
Differentiation of $F_V(v) = 1-(1-F(v))^n$ gives $f_V(v) = nf(v)(1-F(v))^{n-1}$.
Structural Commentary: Extremes convert order events into products over independent coordinates.
Key Insight: Maxima use all observations to the left; minima use all observations to the right. □

Example 4.1.3. *Series and Parallel Exponential Lifetimes*
Let $T_1, \dots, T_n \overset{\text{ind}}{\sim} \text{Exp}(\lambda)$ be component lifetimes.

(i) For a series system, all components must survive, so system failure occurs at the first component failure: $V = \min_i T_i = T_{(1)}$. Thus $F_V(v) = 1 - e^{-n\lambda v}$ and $f_V(v) = n\lambda e^{-n\lambda v}$ for $v \ge 0$; hence $V \sim \text{Exp}(n\lambda)$.

(ii) For a parallel system, one surviving component suffices, so system failure occurs only after all components fail: $U = \max_i T_i = T_{(n)}$. Thus $F_U(u) = (1-e^{-\lambda u})^n$ and $f_U(u) = n\lambda e^{-\lambda u}(1-e^{-\lambda u})^{n-1}$ for $u \ge 0$.

Theorem 4.1.4. *Density of the kth Order Statistic*
Let $X_1, \dots, X_n$ be iid real-valued random variables with differentiable CDF F and density $f = F'$.
For $1 \le k \le n$ and $x \in \mathbb{R}$, $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} f(x)F(x)^{k-1}(1-F(x))^{n-k}$.

Proof. $N_x = \sum_{i=1}^n \mathbf{1}_{\{X_i \le x\}}$; $N_x \sim \text{Bin}(n, F(x))$, $\{X_{(k)} \le x\} = \{N_x \ge k\}$; $F_{X_{(k)}}(x) = \sum_{j=k}^n \binom{n}{j} F(x)^j (1-F(x))^{n-j}$.
With $p = F(x)$, $\frac{d}{dp}\sum_{j=k}^n \binom{n}{j} p^j(1-p)^{n-j} = \frac{n!}{(k-1)!(n-k)!}p^{k-1}(1-p)^{n-k}$, and the chain rule gives stated density.
Structural Commentary: The multinomial count of left, middle, and right observations supplies the coefficient.
Key Insight: An order statistic is located by counting how many sample points fall on each side. □

Corollary 4.1.5. *Uniform Order Statistics Are Beta*
If $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0,1)$, *then for* $1 \le k \le n$, $X_{(k)} \sim \text{Beta}(k, n-k+1)$ *with density* $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k} \mathbf{1}_{(0,1)}(x)$.

Proof. On $(0,1)$, $F(x) = x$ and $f(x) = 1$, so Theorem 4.1.4 gives $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}$.
Also $B(k, n-k+1) = \frac{(k-1)!(n-k)!}{n!}$, hence this density is the Beta$(k, n-k+1)$ density.
Structural Commentary: Uniform order statistics reduce the general density to the beta normalizing constant.
Key Insight: Counting positions in a uniform sample produces beta powers of left and right mass. □

Proposition 4.1.6. *Joint Density of Minimum and Maximum*
Let $n \ge 2$, *let* $X_1, \dots, X_n \overset{\text{iid}}{\sim} F$ *with differentiable CDF* F, *density* $f = F'$; $V = X_{(1)}$, $U = X_{(n)}$, $R = U - V$.

(i) For $v < u$, $f_{V,U}(v,u) = n(n-1)f(v)f(u)(F(u) - F(v))^{n-2}$, *and* $f_{V,U}(v,u) = 0$ *for* $v \ge u$.
(ii) If $F(t) = t$ *on* $[0,1]$, *then* $f_{V,U}(v,u) = n(n-1)(u-v)^{n-2}\mathbf{1}_{\{0<v<u<1\}}$.
(iii) In the uniform case, $f_R(r) = n(n-1)r^{n-2}(1-r)\mathbf{1}_{(0,1)}(r)$, *with* $F_R(r) = 0$ *for* $r < 0$, $F_R(r) = nr^{n-1} - (n-1)r^n$ *for* $0 \le r \le 1$, *and* $F_R(r) = 1$ *for* $r \ge 1$.

Proof.
(i) For $v < u$, $G(v,u) = \mathbb{P}(V \ge v, U \le u) = \mathbb{P}(v \le X_i \le u,\ 1 \le i \le n) = (F(u) - F(v))^n$, so $f_{V,U}(v,u) = -\partial_v \partial_u G(v,u) = n(n-1)f(v)f(u)(F(u) - F(v))^{n-2}$.
(ii) If $F(t) = t$ and $f(t) = 1$ on $(0,1)$, by (i), $f_{V,U}(v,u) = n(n-1)(u-v)^{n-2}$ on $0 < v < u < 1$, else 0.
(iii) For $0 < r < 1$, $f_R(r) = \int_0^{1-r} f_{V,U}(v, v+r)\, dv = n(n-1)r^{n-2}(1-r)$. Integration gives $F_R(r) = \int_0^r f_R(s)\, ds = nr^{n-1} - (n-1)r^n$; outside $[0,1]$, the support gives $F_R(r) \in \{0,1\}$.
Structural Commentary: Mixed derivatives select the two boundary observations.
Key Insight: Fixed extremes leave only interior mass. □

Example 4.1.7. *Distribution-Free Tolerance Interval*
Let $X_1, \dots, X_n$ be iid with continuous CDF F, and set $Q = F(X_{(n)}) - F(X_{(1)})$.
By the probability integral transform, $U_i = F(X_i) \overset{\text{iid}}{\sim} \text{Uniform}(0,1)$, so $Q \overset{d}{=} U_{(n)} - U_{(1)}$.
Therefore, for $0 \le \alpha \le 1$, $\mathbb{P}(Q > \alpha) = 1 - F_R(\alpha) = 1 - n\alpha^{n-1} + (n-1)\alpha^n$. For $n = 100$ and $\alpha = .95$, $\mathbb{P}(Q > .95) \approx .96$: $[X_{(1)}, X_{(n)}]$ covers at least .95 of the F-mass with probability about .96.

The expectation calculus developed above also supports a useful collection of moment methods: identifying whether the mean exists, replacing difficult distributions of functions by direct expectation, controlling spread by covariance, and using generating functions or Taylor expansion when direct integration is cumbersome.

Example 4.1.8. *Existence and First Distributional Means*
(i) If $X \sim \text{Geom}(p)$ on $\{1, 2, \dots\}$ and $q = 1 - p$, then $\mathbb{E}X = p \sum_{k \ge 1} k q^{k-1} = p \frac{d}{dq} \frac{q}{1-q} = \frac{1}{p}$.
(ii) If $X \sim \text{Poisson}(\lambda)$, then $\mathbb{E}X = e^{-\lambda} \sum_{k \ge 1} \frac{k\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{j \ge 0} \frac{\lambda^j}{j!} = \lambda$.
(iii) If $X \sim \text{Cauchy}(0,1)$, then $\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty x(1+x^2)^{-1}\, dx = \infty$, so X is not integrable and $\mathbb{E}X$ is undefined.
(iv) If $\mathbb{P}(Y = 2^k) = 2^{-(k+1)}$ for $k \ge 0$, then $\mathbb{E}Y = \sum_{k \ge 0} 2^{-1} = \infty$; truncation is essential.
Key Insight: Expectation is a finite absolute average, not merely a formal symmetry or a plausible center.

Method 4.1.9. *Indicator Sums Without Independence*
(i) For a finite index set I and events $(A_i)_{i \in I}$, if $W = \sum_{i \in I} \mathbf{1}_{A_i}$, then $\mathbb{E}W = \sum_{i \in I} \mathbb{P}(A_i)$.
(ii) The identity uses only linearity of expectation; the events A_i need not be independent.

Example 4.1.10. *Counting Word Occurrences*
Let $S_1, \dots, S_N \overset{\text{iid}}{\sim} \text{Uniform}\{A, C, G, T\}$ and fix a word $w \in \{A, C, G, T\}^q$.
For $1 \le m \le N - q + 1$, set $I_m = \mathbf{1}_{\{(S_m, \dots, S_{m+q-1}) = w\}}$ and $W = \sum_{m=1}^{N-q+1} I_m$.
Then $\mathbb{P}(I_m = 1) = 4^{-q}$ and $\mathbb{E}W = (N - q + 1)4^{-q}$.
Overlaps may create dependence; linearity does not require independence.

Example 4.1.11. *Coupon Collection by Waiting Times*
Draw uniformly from n coupon types, and let X_r be the waiting time for the rth new type. Then $X_r \sim \text{Geom}(\frac{n-r+1}{n})$ on $\{1, 2, \dots\}$, so $\mathbb{E}X_r = \frac{n}{n-r+1}$.
For $T = \sum_{r=1}^n X_r$, linearity gives $\mathbb{E}T = \sum_{r=1}^n \frac{n}{n-r+1} = nH_n$, where $H_n = \sum_{j=1}^n \frac{1}{j} \sim \log n + \gamma$.

Theorem 4.1.12. *Correlation Bound and Equality Case*
Let $X, Y \in L^2$ *have means* μ_X, μ_Y *and standard deviations* $0 < \sigma_X, \sigma_Y < \infty$, *set* $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$, *and define* $Z_X = \frac{X - \mu_X}{\sigma_X}$, $Z_Y = \frac{Y - \mu_Y}{\sigma_Y}$.

(i) $\mathbb{E}Z_X = \mathbb{E}Z_Y = 0$, $\text{Var}(Z_X) = \text{Var}(Z_Y) = 1$, *and* $\text{Cov}(Z_X, Z_Y) = \rho$.
(ii) $-1 \le \rho \le 1$, *equivalently* $|\rho| \le 1$.
(iii) $\rho = \pm 1$ *iff* $Z_Y = \pm Z_X$ *a.s., equivalently* $Y = \mu_Y \pm (\frac{\sigma_Y}{\sigma_X})(X - \mu_X)$ *a.s., with the same sign.*

Proof.
(i) $\mathbb{E}Z_X = 0$, $\text{Var}(Z_X) = \sigma_X^{-2} \text{Var}(X) = 1$, and similarly for Z_Y. Moreover, $\text{Cov}(Z_X, Z_Y) = \sigma_X^{-1}\sigma_Y^{-1} \text{Cov}(X, Y) = \rho$.

(ii) Since $0 \le \mathrm{Var}(Z_X \pm Z_Y) = 2 \pm 2\rho$, both $1+\rho \ge 0$ and $1-\rho \ge 0$, hence $|\rho| \le 1$.
(*iii*) ($\Rightarrow$) If $\rho = 1$, then $\mathbb{E}(Z_X - Z_Y)^2 = 0$, so $Z_X = Z_Y$ a.s..
If $\rho = -1$, then $\mathbb{E}(Z_X + Z_Y)^2 = 0$, so $Z_X = -Z_Y$ a.s.
(*iii*) ($\Leftarrow$) If $Z_Y = \pm Z_X$, then $\rho = \mathrm{Cov}(Z_X, \pm Z_X) = \pm 1$, and $\frac{Y-\mu_Y}{\sigma_Y} = \pm\frac{X-\mu_X}{\sigma_X}$
iff $Y = \mu_Y \pm (\frac{\sigma_Y}{\sigma_X})(X - \mu_X)$ a.s.

Structural Commentary: Correlation is centred covariance after unit-variance normalization.
Key Insight: Perfect correlation means no residual scatter around an affine line. □

Example 4.1.13. *Bivariate Uniform Association*
For (X, Y) with uniform marginals on $[0,1]$, $\mathbb{E}X = \mathbb{E}Y = \frac{1}{2}$ and $\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{1}{12}$.
Hence $\mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \frac{1}{4}$ and $\rho = 12\,\mathrm{Cov}(X,Y)$.
If $\mathbb{E}(XY) = \frac{2}{9}$, then $\mathrm{Cov}(X,Y) = -\frac{1}{36}$, $\rho = -\frac{1}{3}$; if $\mathbb{E}(XY) = \frac{5}{18}$, then $\mathrm{Cov}(X,Y) = \frac{1}{36}$, $\rho = \frac{1}{3}$.
The sign of ρ records negative or positive association.

Method 4.1.14. *Moment Propagation by Taylor Expansion*
Let the relevant variables have finite second moments, and let g be C^2 near the relevant mean point.
(i) If $Y = g(X)$, $\mu = \mathbb{E}X$, and $\sigma^2 = \mathrm{Var}(X)$, then $\mathbb{E}Y \approx g(\mu)$ and $\mathrm{Var}(Y) \approx [g'(\mu)]^2\sigma^2$.
(ii) The second-order mean correction is $\mathbb{E}g(X) \approx g(\mu) + \frac{1}{2}g''(\mu)\sigma^2$.
(iii) For $Z = g(\mathbf{X})$, $\mu = \mathbb{E}\mathbf{X}$, and $\Sigma = \mathrm{Cov}(\mathbf{X})$, $\mathrm{Var}(Z) \approx \nabla g(\mu)^\top \Sigma \nabla g(\mu)$ and $\mathbb{E}Z \approx g(\mu) + \frac{1}{2}\,\mathrm{tr}(\Sigma H_g(\mu))$.

Structural Commentary: Taylor expansion propagates moments through local derivatives.
Key Insight: Smooth nonlinear functions are locally linear for variance and locally quadratic for mean correction.

Example 4.1.15. *Approximate Moments of a Ratio*
Let $Z = \frac{Y}{X}$, $\mu_X \neq 0$, $\mu = (\mu_X, \mu_Y)$, and write $\sigma_X^2 = \mathrm{Var}(X)$, $\sigma_Y^2 = \mathrm{Var}(Y)$, $\sigma_{XY} = \mathrm{Cov}(X,Y)$.
For $g(x,y) = \frac{y}{x}$, Taylor propagation gives $\mathbb{E}Z \approx \frac{\mu_Y}{\mu_X} + \frac{\sigma_X^2\mu_Y}{\mu_X^3} - \frac{\sigma_{XY}}{\mu_X^2}$
and $\mathrm{Var}(Z) \approx \frac{\sigma_X^2\mu_Y^2}{\mu_X^4} + \frac{\sigma_Y^2}{\mu_X^2} - \frac{2\sigma_{XY}\mu_Y}{\mu_X^3}$. The approximation is ill-conditioned as $|\mu_X| \downarrow 0$.

5 Normal-Derived Distributions

5.1 Normal-Derived Distribution Families

Independent $N(0,1)$ variables generate the normal-derived inference laws used below: χ^2 sums, Student t ratios, and F ratios.

Proposition 5.1.1. *Chi-Square Sums*
Let $n \ge 1$ and $Z_1, \dots, Z_n \overset{\text{ind}}{\sim} N(0,1)$.

(i) For $V = \sum_{i=1}^n Z_i^2$, $V \sim \chi_n^2 = \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$ in the rate convention,
$f_V(v) = 2^{-\frac{n}{2}}\Gamma(\frac{n}{2})^{-1}v^{\frac{n}{2}-1}e^{-\frac{v}{2}}\mathbf{1}_{(0,\infty)}(v)$, $\psi_V(t) = (1-2t)^{-\frac{n}{2}}$ for $t < \frac{1}{2}$, $\mathbb{E}V = n$, and $\mathrm{Var}(V) = 2n$.
(ii) If $U \sim \chi_m^2$, $W \sim \chi_n^2$, and $U \perp\!\!\!\perp W$, then $U + W \sim \chi_{m+n}^2$.

Proof.

(i) For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, $\mathbb{E}e^{tZ^2} = (2\pi)^{-\frac{1}{2}}\int_{\mathbb{R}} e^{-\frac{(1-2t)z^2}{2}}\,dz = (1-2t)^{-\frac{1}{2}}$.
By independence, $\psi_V(t) = \prod_{i=1}^n \mathbb{E}e^{tZ_i^2} = (1-2t)^{-\frac{n}{2}}$. By MGF uniqueness, $V \sim \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$.
Differentiating ψ_V at 0 gives $\mathbb{E}V = n$ and $\mathrm{Var}(V) = 2n$.
(ii) If $U \perp\!\!\!\perp W$, then $\psi_{U+W}(t) = (1-2t)^{-\frac{m}{2}}(1-2t)^{-\frac{n}{2}} = (1-2t)^{-\frac{m+n}{2}}$, hence $U + W \sim \chi_{m+n}^2$.

Structural Commentary: Chi-square degrees of freedom add by MGF exponents.
Key Insight: Squared independent standard normals form gamma sums. □

Definition 5.1.2. *Normal Sample Statistics*
(i) *Sample*: $X_1, \dots, X_n$ are independent with $X_i \sim N(\mu, \sigma^2)$, $n \ge 2$, and $\sigma^2 > 0$.
(ii) *Sample mean*: $\overline{X} := n^{-1}\sum_{i=1}^n X_i$ estimates the location μ.
(iii) *Sample variance*: $S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$ estimates the scale σ^2.

Proposition 5.1.3. *Normal Sample Mean*
Let $X_1, \dots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$, and set $\overline{X} := n^{-1}\sum_{i=1}^n X_i$. Then $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

Proof. Since independent normal sums are normal, $\overline{X}$ is normal.
Also $\mathbb{E}\overline{X} = n^{-1}\sum_i \mathbb{E}X_i = \mu$ and $\mathrm{Var}(\overline{X}) = n^{-2}\sum_i \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$. Thus $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

Structural Commentary: Averaging preserves normality and divides variance by n.
Key Insight: The sample mean is an exact normal pivot center, not an asymptotic approximation. □

Theorem 5.1.4. *Mean-Residual Independence*
Let $X_1, \dots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$, set $\overline{X} := n^{-1}\sum_i X_i$, $R := (X_1 - \overline{X}, \dots, X_n - \overline{X})$,
and $S^2 := (n-1)^{-1}\sum_i R_i^2$. Then $\overline{X} \perp\!\!\!\perp R$, hence $\overline{X} \perp\!\!\!\perp S^2$.

Proof. Let $Z_i := \frac{X_i-\mu}{\sigma}$, so $Z \sim N_n(0, I_n)$. Choose orthogonal Q with first row $n^{-\frac{1}{2}}\mathbf{1}^\top$, set $Y := QZ$.
Then $Y \sim N_n(0, I_n)$, $Y_1 \perp\!\!\!\perp (Y_2, \ldots, Y_n)$. Since $\overline{X} = \mu + \frac{\sigma Y_1}{\sqrt{n}}$ and $R = \sigma(I - n^{-1}\mathbf{1}\mathbf{1}^\top)Z = \sigma Q^\top(0, Y_2, \ldots, Y_n)^\top$, $\overline{X}$ is function of Y_1, R is function of $(Y_2, \ldots, Y_n)$. Therefore $\overline{X} \perp\!\!\!\perp R$, by $S^2 = (n-1)^{-1}\sum_i R_i^2$, $\overline{X} \perp\!\!\!\perp S^2$.

Structural Commentary: Mean and residuals are orthogonal projections.
Key Insight: For Gaussian vectors, orthogonal components are independent. □

Theorem 5.1.5. *Sample Variance Chi-Square Law*
Given independent $X_1, \ldots, X_n$; $X_i \sim N(\mu, \sigma^2)$; $S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$. *Then* $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Proof. Let $Z_i := \frac{X_i-\mu}{\sigma}$, so $Z = (Z_1, \ldots, Z_n) \sim N_n(0, I_n)$.
Choose orthogonal matrix Q with first row $n^{-\frac{1}{2}}(1, \ldots, 1)$, set $Y := QZ$.
Orthogonal invariance gives $Y \sim N_n(0, I_n)$ with independent coordinates.
Then $Y_1 = \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}$ and $\sum_{j=2}^n Y_j^2 = \sum_{i=1}^n (Z_i - \overline{Z})^2 = \frac{(n-1)S^2}{\sigma^2}$.
Thus $\frac{(n-1)S^2}{\sigma^2}$ is the sum of squares of $n-1$ independent $N(0,1)$ variables, hence χ^2_{n-1}.
Structural Commentary: Cantering leaves $n-1$ residual directions.
Key Insight: The sample variance is the squared Gaussian residual length scaled by σ^2. □

Definition 5.1.6. *Student t Distribution*
If $Z \sim N(0,1)$, $U \sim \chi^2_n$, $Z \perp\!\!\!\perp U$, then $T = Z/\sqrt{\frac{U}{n}}$ has *Student t distribution* with $df = n \geq 1$, $T \sim t_n$.

Proposition 5.1.7. *Student t Density*
Let $n \geq 1$ *and* $T = Z/\sqrt{\frac{U}{n}}$ *with* $Z \sim N(0,1)$, $U \sim \chi^2_n$, *and* $Z \perp\!\!\!\perp U$.

(i) For $t \in \mathbb{R}$, $f_T(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$.
(ii) For all $t \in \mathbb{R}$, $f_T(-t) = f_T(t)$.
(iii) For each fixed $t \in \mathbb{R}$, $\lim_{n\to\infty} f_T(t) = \phi(t) = (2\pi)^{-\frac{1}{2}}e^{-\frac{t^2}{2}}$.

Proof.

(i) For fixed $u > 0$, $z = t\sqrt{\frac{u}{n}}$ has Jacobian $\sqrt{\frac{u}{n}}$, so $f_T(t) = \int_0^\infty \phi(t\sqrt{\frac{u}{n}})\sqrt{\frac{u}{n}} f_U(u)\, du$.
Substituting $\phi(z) = (2\pi)^{-\frac{1}{2}}e^{-\frac{z^2}{2}}$ and $f_U(u) = 2^{-\frac{n}{2}}\Gamma(\frac{n}{2})^{-1}u^{\frac{n}{2}-1}e^{-\frac{u}{2}}\mathbf{1}_{(0,\infty)}(u)$ gives
$f_T(t) = \frac{1}{\sqrt{n\pi}2^{\frac{n+1}{2}}\Gamma(\frac{n}{2})}\int_0^\infty u^{\frac{n+1}{2}-1}e^{-\frac{(1+\frac{t^2}{n})u}{2}}\, du$. Then $F_T(t) = \Gamma(\frac{n+1}{2})2^{\frac{n+1}{2}}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$, yielding density.
(ii) By (i), $f_T(t) = c_n h_n(t^2)$ for $c_n > 0$, so $f_T(-t) = f_T(t)$ for all t.
(iii) For fixed t, $f_T(t) = (2\pi)^{-\frac{1}{2}}\frac{\Gamma(\frac{n+1}{2})}{\sqrt{\frac{n}{2}}\Gamma(\frac{n}{2})}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$, gamma ratio tends to 1, last factor tends to $e^{-\frac{t^2}{2}}$.

Structural Commentary: Student t is a normal scale mixture with random chi-square scale.
Key Insight: Random denominator uncertainty thickens the normal tails. □

Example 5.1.8. *t_1 is Cauchy*
If $T \sim t_1$, then $f_T(t) = \Gamma(1)(\sqrt{\pi}\Gamma(\frac{1}{2}))^{-1}(1+t^2)^{-1} = \frac{1}{\pi(1+t^2)}$, so $T \sim \text{Cauchy}(0,1)$.

Definition 5.1.9. *F Distribution*
If $U \sim \chi^2_m$, $V \sim \chi^2_n$, $U \perp\!\!\!\perp V$, then $W = (\frac{U}{m})/(\frac{V}{n})$ has *F distribution*, $df = (m,n)$, $W \sim F_{m,n}$.

Proposition 5.1.10. *F Density*
Let $m, n \geq 1$ *and* $W = (\frac{U}{m})/(\frac{V}{n})$, *where* $U \sim \chi^2_m$, $V \sim \chi^2_n$, *and* $U \perp\!\!\!\perp V$.

(i) $f_W(w) = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})}(\frac{m}{n})^{\frac{m}{2}}w^{\frac{m}{2}-1}(1 + \frac{mw}{n})^{-\frac{m+n}{2}}\mathbf{1}_{(0,\infty)}(w)$.
(ii) If $n > 2$, *then* $\mathbb{E}W = \frac{n}{n-2}$.

Proof.

(i) For $w > 0$, set $u = \frac{mwv}{n}$; the Jacobian is $\frac{mv}{n}$, so $f_W(w) = \int_0^\infty f_U(\frac{mwv}{n})f_V(v)(\frac{mv}{n})\, dv$. By substitution,
$f_W(w) = Cw^{\frac{m}{2}-1}\int_0^\infty v^{\frac{m+n}{2}-1}e^{-\frac{(1+\frac{mw}{n})v}{2}}\, dv$; $C = \frac{(\frac{m}{n})^{\frac{m}{2}}}{2^{\frac{m+n}{2}}\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})}$, by gamma integral, QED.
(ii) Since $U \perp\!\!\!\perp V$, $\mathbb{E}W = (\frac{n}{m})\mathbb{E}U\,\mathbb{E}(V^{-1}) = n\,\mathbb{E}(V^{-1}) = \frac{n}{n-2}$ for $n > 2$.

Structural Commentary: An F variable is a ratio of independent chi-square averages.
Key Insight: Denominator variability controls the right tail and the mean condition $n > 2$. □

Corollary 5.1.11. *Reciprocal F and Squared t*

(i) For $m, n \geq 1$, *if* $W \sim F_{m,n}$, *then* $W^{-1} \sim F_{n,m}$.
(ii) If $T \sim t_n$, *then* $T^2 \sim F_{1,n}$.

Proof.

(i) Write $W = (\frac{U}{m})/(\frac{V}{n})$ with $U \sim \chi^2_m$, $V \sim \chi^2_n$, and $U \perp\!\!\!\perp V$; then $W^{-1} = (\frac{V}{n})/(\frac{U}{m}) \sim F_{n,m}$.
(ii) Write $T = Z/\sqrt{\frac{V}{n}}$ with $Z \sim N(0,1)$, $V \sim \chi^2_n$, and $Z \perp\!\!\!\perp V$; since $Z^2 \sim \chi^2_1$, $T^2 = (\frac{Z^2}{1})/(\frac{V}{n}) \sim F_{1,n}$.

Structural Commentary: Reciprocals swap F degrees of freedom, and squaring t removes the normal sign.
Key Insight: Both identities are immediate from the same chi-square-ratio representation. □

Part XVI
Probability and Stochastic Systems

1 Conditional Probability and Conditional Expectation

1.1 Conditional Probability

Definition 1.1.1. *Conditional Probability*
For $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the *conditional probability* of A given B is $\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, i.e. the probability of A under the normalised restriction of $\mathbb{P}$ to B.

Example 1.1.2. *Diagnostic Table*
Let T be the event of a positive test and D the event of disease, with observed counts

	D	D^c	total
T	25	14	39
T^c	18	78	96
total	43	92	135

Let $N(\cdot)$ be counts, $\widehat{\mathbb{P}}(A \mid B) = \frac{N(A \cap B)}{N(B)}$.
Thus $\widehat{\mathbb{P}}(D \mid T) = \frac{25}{39}$, $\widehat{\mathbb{P}}(D^c \mid T^c) = \frac{78}{96}$, $\widehat{\mathbb{P}}(D^c \mid T) = \frac{14}{39}$, and $\widehat{\mathbb{P}}(D \mid T^c) = \frac{18}{96}$.

Proposition 1.1.3. *Conditioning on a Fixed Event Gives a Probability Measure*
For $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, define $\mathbb{P}_B : \mathcal{F} \to [0,1]$ by $\mathbb{P}_B(A) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.
Then $\mathbb{P}_B$ is a probability measure on $(\Omega, \mathcal{F})$.

Proof. For $A \in \mathcal{F}$, $\mathbb{P}_B(A) \geq 0$, and $\mathbb{P}_B(\Omega) = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1$.
If $(A_i)_{i \geq 1}$ are pairwise disjoint, then $(A_i \cap B)_{i \geq 1}$ are pairwise disjoint and $(\bigcup_i A_i) \cap B = \bigcup_i (A_i \cap B)$.
Thus $\mathbb{P}_B(\bigcup_i A_i) = \frac{\mathbb{P}(\bigcup_i (A_i \cap B))}{\mathbb{P}(B)} = \sum_i \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_i \mathbb{P}_B(A_i)$.
Structural Commentary: Conditioning normalizes the restriction of $\mathbb{P}$ to B.
Key Insight: Once B is fixed, conditional probability is ordinary probability on the restricted world. □

Remark 1.1.4. *Conditional Probability Is Not Symmetric*
In general, $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$; conditioning fixes the reference population. For disease event D and positive-test event T, large $\mathbb{P}(T \mid D)$ need not imply large $\mathbb{P}(D \mid T)$ when $\mathbb{P}(D)$ is small.

Proposition 1.1.5. *Multiplication and Chain Rules*
Let $A, B, C \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, $\mathbb{P}(B) > 0$, and $\mathbb{P}(B \cap C) > 0$.
(i) $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$.
(ii) $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C)$.

Proof.
(i) For $\mathbb{P}(F) > 0$, $\mathbb{P}(E \mid F)\mathbb{P}(F) = \mathbb{P}(E \cap F)$; apply this with $(E, F) = (A, B)$ and (B, A).
(ii) Since $\mathbb{P}(B \cap C) > 0$, $\mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C) = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(B \cap C)} \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} \mathbb{P}(C) = \mathbb{P}(A \cap B \cap C)$.
Structural Commentary: The chain rule orders information as C, then B within C, then A within $B \cap C$.
Key Insight: A joint probability factors into conditional increments along a chosen order. □

Example 1.1.6. *Cards Without Replacement*
Draw two cards from a standard deck without replacement. Let A be the event that the first card is the ace of clubs and B the event that the second card is the queen of diamonds. Since $\mathbb{P}(A) = \frac{1}{52}$ and $\mathbb{P}(B \mid A) = \frac{1}{51}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \frac{1}{52 \cdot 51}$. The factor $\mathbb{P}(B \mid A)$ uses the 51-card deck remaining after A occurs.

Proposition 1.1.7. *Independence Through Conditioning*
Let $A, B \in \mathcal{F}$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. The following are equivalent:
(i) A and B are independent.
(ii) $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.
(iii) $\mathbb{P}(B \mid A) = \mathbb{P}(B)$.

Proof.
(i) $\Rightarrow$ *(ii)* If A and B are independent, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A)$.
(ii) $\Rightarrow$ *(iii)* If $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \mathbb{P}(B)$.
(iii) $\Rightarrow$ *(i)* If $\mathbb{P}(B \mid A) = \mathbb{P}(B)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A) = \mathbb{P}(A)\mathbb{P}(B)$, hence A and B are independent.
Structural Commentary: Each condition is $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ normalized by a positive marginal.
Key Insight: Independence means conditioning on either event preserves the probability of the other. □

Theorem 1.1.8. *Law of Total Probability*
(i) Let $A_1, \ldots, A_k \in \mathcal{F}$ form a finite partition of Ω with $\mathbb{P}(A_i) > 0$ for all i.
(ii) For every $B \in \mathcal{F}$, $\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$.

Proof.

(i) The partition assumptions give $B = \bigsqcup_{i=1}^{k} (B \cap A_i)$, hence $\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \cap A_i)$.

(ii) For each i, $\mathbb{P}(B \cap A_i) = \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$; substituting in (i) gives the formula.

Structural Commentary: The partition $\{A_i\}_{i=1}^{k}$ decomposes B into disjoint cases $B \cap A_i$.
Key Insight: Total probability is weighted averaging over cases. □

Example 1.1.9. *Second Draw by Total Probability*
An urn has 3 red balls, 1 blue ball; draw twice without replacement, let R_i be the event that draw i is red.
By total probability, $\mathbb{P}(R_2) = \mathbb{P}(R_2 \mid R_1)\mathbb{P}(R_1) + \mathbb{P}(R_2 \mid R_1^c)\mathbb{P}(R_1^c) = \frac{2}{3} \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \frac{3}{4}$.
Key Insight: The second draw has the same marginal red probability as the first.

Definition 1.1.10. *Prior and Posterior Probabilities*
Let $A_1, \ldots, A_k \in \mathcal{F}$ be a partition with $\mathbb{P}(A_i) > 0$ for all i, and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$.

(i) $\mathbb{P}(A_i)$ is the *prior probability* of A_i.

(ii) $\mathbb{P}(A_i \mid B)$ is the *posterior probability* of A_i after conditioning on B.

Theorem 1.1.11. *Bayes' Theorem*

Let $A_1, \ldots, A_k \in \mathcal{F}$ satisfy $A_i \cap A_j = \varnothing$ for $i \neq j$, $\bigcup_{i=1}^{k} A_i = \Omega$, and $\mathbb{P}(A_i) > 0$ for all i.

If $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, then for each i, $\mathbb{P}(A_i \mid B) = \dfrac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B|A_j)\mathbb{P}(A_j)}$.

Proof. For fixed i, $\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\mathbb{P}(B)}$.
Since $B = \bigsqcup_{j=1}^{k} (B \cap A_j)$, $\mathbb{P}(B) = \sum_{j=1}^{k} \mathbb{P}(B \cap A_j) = \sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)$. Substitution gives the formula.
Structural Commentary: Bayes reweights priors by likelihoods, then normalises.
Key Insight: Posterior equals likelihood times prior divided by evidence. □

Corollary 1.1.12. *Posterior Mass Must Balance*
Let $A_1, \ldots, A_k \in \mathcal{F}$ be a partition of Ω with $\mathbb{P}(A_i) > 0$ for all i, and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$.
If $\mathbb{P}(A_1 \mid B) < \mathbb{P}(A_1)$, then there exists $i \in \{2, \ldots, k\}$ such that $\mathbb{P}(A_i \mid B) > \mathbb{P}(A_i)$.

Proof. Suppose $\mathbb{P}(A_i \mid B) \leq \mathbb{P}(A_i)$ for all $i \geq 2$
As $\sum_{i=1}^{k} \mathbb{P}(A_i \mid B) = \sum_{i=1}^{k} \mathbb{P}(A_i) = 1$, hence $1 = \sum_{i=1}^{k} \mathbb{P}(A_i \mid B) < \mathbb{P}(A_1) + \sum_{i=2}^{k} \mathbb{P}(A_i) = 1$, a contradiction.
Structural Commentary: Posterior mass is conserved across the partition.
Key Insight: A decrease in one cell must be offset by an increase in another. □

Example 1.1.13. *Transition Probabilities*
Let states U, M, L denote upper, middle, lower groups, and let $P = (p_{ij})$ with $p_{ij} = \mathbb{P}(j_2 \mid i_1)$:

	U_2	M_2	L_2
U_1	.45	.48	.07
M_1	.05	.70	.25
L_1	.01	.50	.49

For $\pi_1 = (.10, .40, .50)$, total probability gives $\pi_2 = \pi_1 P$ and $\mathbb{P}(U_2) = .45(.10) + .05(.40) + .01(.50) = .07$.
Bayes gives the inverse query $\mathbb{P}(U_1 \mid U_2) = \frac{\mathbb{P}(U_2|U_1)\mathbb{P}(U_1)}{\mathbb{P}(U_2)} = \frac{.45(.10)}{.07} \approx .64$.
Key Insight: Use πP forward and Bayes backward.

Example 1.1.14. *Base Rates in a Medical Test*
Let D be disease and T a positive test, with joint law

	D	D^c
T	0.0081	0.0900
T^c	0.0009	0.9010

Then $\mathbb{P}(T \mid D) = \frac{0.0081}{0.0081+0.0009} = 0.9$, $\mathbb{P}(T^c \mid D^c) = \frac{0.9010}{0.9010+0.0900} \approx 0.91$, $\mathbb{P}(D \mid T) = \frac{0.0081}{0.0081+0.0900} \approx 0.083$.
Key Insight: A rare condition can have small posterior probability after a positive test.

Example 1.1.15. *Rare Classes and Screening Errors*
Let L denote lying and $T = L^c$ truth, with $\mathbb{P}(+ \mid L) = .88$, $\mathbb{P}(+ \mid T) = .14$, $\mathbb{P}(L) = .01$, and $\mathbb{P}(T) = .99$.
Bayes gives $\mathbb{P}(T \mid +) = \frac{\mathbb{P}(+|T)\mathbb{P}(T)}{\mathbb{P}(+|T)\mathbb{P}(T)+\mathbb{P}(+|L)\mathbb{P}(L)} = \frac{(.14)(.99)}{(.14)(.99)+(.88)(.01)} \approx .94$.
Key Insight: Rare-class positives can mostly come from the common class.

Example 1.1.16. *Monty Hall*
The player chooses door 1. Let $\omega = (\omega_1, \omega_2)$, where ω_1 is the prize door and ω_2 is Monty's opened door.
Monty opens neither door 1 nor the prize door, and chooses uniformly from $\{2, 3\}$ when $\omega_1 = 1$.
The induced probabilities are $\mathbb{P}(1, 2) = \mathbb{P}(1, 3) = \frac{1}{6}$ and $\mathbb{P}(2, 3) = \mathbb{P}(3, 2) = \frac{1}{3}$.
Let $S = \{\omega_1 = 1\}$ be the event that staying wins.
Given $\omega_2 = 2$, the staying-win probability is $\mathbb{P}(S \mid \omega_2 = 2) = \frac{\frac{1}{6}}{\frac{1}{6}+\frac{1}{3}} = \frac{1}{3}$.
Thus switching wins with probability $\frac{2}{3}$, and the case $\omega_2 = 3$ is symmetric.
Key Insight: Once Monty's rule is part of the sample space, switching has conditional probability $\frac{2}{3}$.

Example 1.1.17. *Three Cards and Conditional Sampling*
Cards are GG, RR, and GR. Choose a card uniformly and then a side uniformly.
Let V be the event that the visible side is green and H the event that the hidden side is green.
Among the three equally likely green visible sides, two have green hidden side, so $\mathbb{P}(H \mid V) = \frac{2}{3}$.

Example 1.1.18. *Conditioning Changes the Reference Class*
Let $X \sim \text{Binomial}(5, \frac{1}{4})$ be number of blue-eyed children, Y be event that youngest child has blue eyes.

(i) Given $X \geq 1$, $\mathbb{P}(X \geq 3 \mid X \geq 1) = \frac{\binom{5}{3}(\frac{1}{4})^3(\frac{3}{4})^2 + \binom{5}{4}(\frac{1}{4})^4(\frac{3}{4}) + (\frac{1}{4})^5}{1-(\frac{3}{4})^5} = \frac{106}{781}$.

(ii) Given Y, $\mathbb{P}(X \geq 3 \mid Y) = 1 - \left(\frac{3}{4}\right)^4 - 4\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^3 = \frac{67}{256}$.

Key Insight: Similar information can determine different conditioning events.

Example 1.1.19. *Bayes with Several Categories*
Let $O \in \{M, W, L\}$ denote operating system with probabilities $(0.30, 0.50, 0.20)$.
Let I be infection, with $\mathbb{P}(I \mid O = M) = 0.65$, $\mathbb{P}(I \mid O = W) = 0.82$, and $\mathbb{P}(I \mid O = L) = 0.50$.
Bayes' theorem gives $\mathbb{P}(W \mid I) = \frac{0.82 \cdot 0.50}{0.65 \cdot 0.30 + 0.82 \cdot 0.50 + 0.50 \cdot 0.20} = \frac{82}{141}$.

Example 1.1.20. *Posterior and Predictive Probabilities*
Let C be uniform on $\{C_1, \ldots, C_5\}$, and suppose coin C_i has head probability p_i.
The head-probability vector is $(p_1, p_2, p_3, p_4, p_5) = \left(0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\right)$. Let H_n be the event that toss n is heads.
The posterior after one head is $\mathbb{P}(C_i \mid H_1) = \frac{p_i}{p_1+p_2+p_3+p_4+p_5} = \left(0, \frac{1}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{5}\right)_i$.
The predictive probability is $\mathbb{P}(H_2 \mid H_1) = \sum_{i=1}^{5} p_i \mathbb{P}(C_i \mid H_1) = \frac{3}{4}$.
If B_4 is the event that the first head occurs on toss 4,
the posterior is $\mathbb{P}(C_i \mid B_4) = \frac{(1-p_i)^3 p_i}{\sum_{j=1}^{5}(1-p_j)^3 p_j}$, and $\left(\mathbb{P}(C_i \mid B_4)\right)_{i=1}^{5} = \left(0, \frac{27}{46}, \frac{16}{46}, \frac{3}{46}, 0\right)$.

Key Insight: Bayes' theorem updates hidden parameters; prediction averages over the posterior.

Definition 1.1.21. *Conditional Mass and Density Functions*

(i) *Discrete*: if X, Y are discrete and $f_Y(y) > 0$, the conditional mass function is
$f_{X|Y}(x \mid y) := \mathbb{P}(X = x \mid Y = y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$.

(ii) *Continuous*: if X, Y have a joint density and $f_Y(y) > 0$, the conditional density is
$f_{X|Y}(x \mid y) := \frac{f_{X,Y}(x,y)}{f_Y(y)}$, so $\mathbb{P}(X \in A \mid Y = y) = \int_A f_{X|Y}(x \mid y)\, dx$ for measurable A.

Proposition 1.1.22. *Joint Factorizations and Total Probability for Conditional Laws*
Assume the displayed sums or integrals are well defined.

(i) If (X, Y) is discrete, $p_Y(y) > 0$, then $p_{X,Y}(x,y) = p_{X|Y}(x \mid y) p_Y(y)$, $p_X(x) = \sum_y p_{X|Y}(x \mid y) p_Y(y)$.

(ii) If (X, Y) has density, $f_X(x) > 0$, then $f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x)$, $f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x) f_X(x)\, dx$.

Proof.

(i) For $p_Y(y) > 0$, $p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$.
Hence $p_{X,Y}(x,y) = p_{X|Y}(x \mid y) p_Y(y)$ and $p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_{X|Y}(x \mid y) p_Y(y)$.

(ii) For $f_X(x) > 0$, $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$.
Hence $f_{X,Y}(x,y) = f_{Y|X}(y \mid x) f_X(x)$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x) f_X(x)\, dx$.

Structural Commentary: Conditional laws disintegrate a joint law into a marginal law and a fiber law.
Key Insight: Total probability is marginalization after conditional factorization. □

Example 1.1.23. *Conditional Masses from the Three-Toss Table*
Let Y be the number of heads in three fair tosses, and let $X = 1$ iff the first toss is heads.
Since $p_Y(1) = \frac{3}{8}$, $p_{X,Y}(0,1) = \frac{2}{8}$, and $p_{X,Y}(1,1) = \frac{1}{8}$, one has $p_{X|Y}(0 \mid 1) = \frac{2}{3}$ and $p_{X|Y}(1 \mid 1) = \frac{1}{3}$.
Thus, given $Y = 1$, the first toss is more likely tails.

Example 1.1.24. *Uniform Square*
If $(X, Y) \sim \text{Uniform}([0,1]^2)$, then for $0 \leq y \leq 1$, $0 \leq x \leq 1$, $f_{X|Y}(x \mid y) = 1$. Hence $X \mid Y = y \sim \text{Uniform}(0, 1)$.

Example 1.1.25. *Conditioning in the Parabolic Region*
Let (X, Y) have density $f(x,y) = \frac{21}{4}x^2 y$, $x^2 \leq y \leq 1$. Then $f_X(x) = \frac{21}{8}x^2(1 - x^4)$, $-1 \leq x \leq 1$.
For $x^2 \leq y \leq 1$, $f_{Y|X}(y \mid x) = \frac{\frac{21}{4}x^2 y}{\frac{21}{8}x^2(1-x^4)} = \frac{2y}{1-x^4}$.
At $x = \frac{1}{2}$, $f_{Y|X}(y \mid \frac{1}{2}) = \frac{32y}{15}$, so $\mathbb{P}(Y \geq \frac{3}{4} \mid X = \frac{1}{2}) = \int_{\frac{3}{4}}^{1} \frac{32y}{15}\, dy = \frac{7}{15}$.

Example 1.1.26. *Poisson Thinning*
Let $\lambda > 0$, $0 \leq p \leq 1$, $N \sim \text{Poisson}(\lambda)$, and $X \mid N = n \sim \text{Binomial}(n, p)$.
For $k \in \mathbb{N}_0$, by total probability, $\mathbb{P}(X = k) = \sum_{n=k}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \binom{n}{k} p^k (1-p)^{n-k} = e^{-\lambda p} \frac{(\lambda p)^k}{k!}$, thus $X \sim \text{Poisson}(\lambda p)$.

Example 1.1.27. *A Triangular Exponential Joint Density*
Let $\lambda > 0$ and $f_{X,Y}(x,y) = \lambda^2 e^{-\lambda y} \mathbf{1}_{\{0 \leq x \leq y\}}$.
For $x \geq 0$, $f_X(x) = \int_x^{\infty} \lambda^2 e^{-\lambda y}\, dy = \lambda e^{-\lambda x}$; for $y \geq 0$, $f_Y(y) = \int_0^y \lambda^2 e^{-\lambda y}\, dx = \lambda^2 y e^{-\lambda y}$.
Hence $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \lambda e^{-\lambda(y-x)} \mathbf{1}_{[x,\infty)}(y)$ and $f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = y^{-1} \mathbf{1}_{[0,y]}(x)$.
Thus $Y \mid X = x \sim x + \text{Exp}(\lambda)$ and $X \mid Y = y \sim \text{Uniform}(0, y)$.

Example 1.1.28. *Bivariate Normal Conditional Density*
Let (X, Y) be bivariate normal with means μ_X, μ_Y, standard deviations $\sigma_X, \sigma_Y > 0$, and correlation $\rho \in (-1, 1)$.
For fixed x, $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \{\sigma_Y \sqrt{2\pi(1-\rho^2)}\}^{-1} \exp\left[-\frac{(y-\mu_Y-\rho\sigma_Y\sigma_X^{-1}(x-\mu_X))^2}{2\sigma_Y^2(1-\rho^2)}\right]$.
Thus $Y \mid X = x \sim N(\mu_Y + \rho\sigma_Y\sigma_X^{-1}(x-\mu_X), \sigma_Y^2(1-\rho^2))$; if $\rho = 0$, this law is independent of x.

1.2 Conditional Expectation

Definition 1.2.1. *Conditional Expectation Given a Random Variable*
Let X, Y be random variables, and assume the relevant sums or integrals exist.

(i) *Discrete case*: If X, Y are discrete, then $\mathbb{E}(X \mid Y = y) := \sum_x x\, p_{X|Y}(x \mid y)$.

(ii) *Density case*: If X, Y have conditional density $f_{X|Y}$, then $\mathbb{E}(X \mid Y = y) := \int_{\mathbb{R}} x f_{X|Y}(x \mid y)\, dx$

(iii) *Measurable transforms*: For measurable r,

$$\mathbb{E}(r(X,Y) \mid Y = y) = \begin{cases} \sum_x r(x,y) p_{X|Y}(x \mid y), & \text{discrete case,} \\ \int_{\mathbb{R}} r(x,y) f_{X|Y}(x \mid y)\, dx, & \text{density case.} \end{cases}$$

(iv) *Random-variable form*: $\mathbb{E}(X \mid Y)$ is the function $g(Y)$ with $g(y) = \mathbb{E}(X \mid Y = y)$.

Example 1.2.2. *A Conditional Uniform Model*
Let $X \sim \text{Uniform}(0,1)$ and, conditionally on $X = x$, let $Y \sim \text{Uniform}(x, 1)$.
For $0 \le x < 1$ and $x < y < 1$, $f_{Y|X}(y \mid x) = \frac{1}{1-x}$, so $\mathbb{E}(Y \mid X = x) = \frac{1}{1-x}\int_x^1 y\, dy = \frac{1+x}{2}$; $\mathbb{E}(Y \mid X) = \frac{1+X}{2}$.

Theorem 1.2.3. *Tower Property*
Let X, Y be real-valued random variables whose joint law is either discrete or admits a density, and let $r : \mathbb{R}^2 \to \mathbb{R}$ be measurable. If $\mathbb{E}|Y| < \infty$ and $\mathbb{E}|r(X,Y)| < \infty$, then $\mathbb{E}[\mathbb{E}(Y \mid X)] = \mathbb{E}Y$ and $\mathbb{E}[\mathbb{E}(r(X,Y) \mid X)] = \mathbb{E}(r(X,Y))$.

Proof. Let $h(x,y)$ denote either y or $r(x,y)$, and set $g_h(x) := \mathbb{E}(h(X,Y) \mid X = x)$.
In the density case, $g_h(x) = \int h(x,y) f_{Y|X}(y \mid x)\, dy$ and $f(x,y) = f_X(x) f_{Y|X}(y \mid x)$,
so Fubini gives $\mathbb{E}[g_h(X)] = \int\int h(x,y) f(x,y)\, dy\, dx = \mathbb{E}h(X,Y)$.
In the discrete case, $\mathbb{E}[g_h(X)] = \sum_x \sum_y h(x,y) p_{Y|X}(y \mid x) p_X(x) = \sum_x \sum_y h(x,y) p_{X,Y}(x,y) = \mathbb{E}h(X,Y)$.
Taking $h(x,y) = y$ and $h(x,y) = r(x,y)$ gives the two identities.
Structural Commentary: Conditional expectation first averages over fibers $\{X = x\}$, and the outer expectation averages those fiber means over the law of X.
Key Insight: Iterated expectation removes the conditioning variable without changing the total mean. □

Example 1.2.4. *Using the Tower Property*
Let $X \sim \text{Uniform}(0,1)$ and, conditionally on $X = x$, let $Y \sim \text{Uniform}(x, 1)$.
Since $\mathbb{E}(Y \mid X) = \frac{1+X}{2}$ and $\mathbb{E}X = \frac{1}{2}$, the tower property gives $\mathbb{E}Y = \mathbb{E}[\mathbb{E}(Y \mid X)] = \mathbb{E}\left[\frac{1+X}{2}\right] = \frac{3}{4}$.

Proposition 1.2.5. *Pulling Out What Is Known*
Let X, Y be real-valued random variables whose joint law is either discrete or admits a density, and let $r, s : \mathbb{R} \to \mathbb{R}$ be measurable with $\mathbb{E}|s(Y)| < \infty$, $\mathbb{E}|r(X)s(Y)| < \infty$, and $\mathbb{E}|r(X)| < \infty$. Then $\mathbb{E}(r(X)s(Y) \mid X) = r(X)\mathbb{E}(s(Y) \mid X)$ and $\mathbb{E}(r(X) \mid X) = r(X)$.

Proof. Let K_x be the conditional law of Y given $X = x$, so $K_x(\{y\}) = p_{Y|X}(y \mid x)$ in the discrete case and $K_x(dy) = f_{Y|X}(y \mid x)\, dy$ in the density case. Set $m_s(x) := \int s(y) K_x(dy) = \mathbb{E}(s(Y) \mid X = x)$, with the integral read as a sum in the discrete case. For admissible x, $\mathbb{E}(r(X)s(Y) \mid X = x) = \int r(x)s(y) K_x(dy) = r(x) m_s(x)$, and $\mathbb{E}(r(X) \mid X = x) = \int r(x) K_x(dy) = r(x)$. Evaluating at $x = X$ gives the identities.
Structural Commentary: Conditioning on X makes every function of X fixed inside remaining average over Y.
Key Insight: Known factors can be treated as constants under conditional expectation. □

Corollary 1.2.6. *Constant Conditional Mean Implies Uncorrelated*
Let $X, Y \in L^2$ and suppose $\mathbb{E}(X \mid Y) = c$ a.s. for some constant $c \in \mathbb{R}$. Then $c = \mathbb{E}X$ and $\text{Cov}(X,Y) = 0$.

Proof. The tower property gives $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X \mid Y)] = c$.
Since Y is known when conditioning on Y, $\mathbb{E}(XY \mid Y) = Y\mathbb{E}(X \mid Y) = cY$.
Taking expectations gives $\mathbb{E}(XY) = c\mathbb{E}Y = \mathbb{E}X\,\mathbb{E}Y$, hence $\text{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y = 0$.
Structural Commentary: A constant conditional mean kills covariance with all functions of Y, hence with Y.
Key Insight: Mean independence implies uncorrelatedness, but not conversely. □

Definition 1.2.7. *Conditional Variance*
Let X, Y be real-valued random variables with finite second conditional moment for each x under consideration.

(i) The conditional mean is $\mu(x) := \mathbb{E}(Y \mid X = x)$.

(ii) The conditional variance at x is $v(x) := \text{Var}(Y \mid X = x) := \mathbb{E}((Y - \mu(x))^2 \mid X = x)$.

(iii) The conditional-variance random variable is $\text{Var}(Y \mid X) := v(X)$.

Theorem 1.2.8. *Law of Total Variance*
Let X, Y be real-valued random variables with $\mathbb{E}Y^2 < \infty$. Set $b(x) := \mathbb{E}(Y \mid X = x)$ and $v(x) := \text{Var}(Y \mid X = x)$, and write $b(X)$ and $v(X)$ for the induced random variables. Then $\text{Var}(Y) = \mathbb{E}[v(X)] + \text{Var}(b(X))$.

Proof. Let $m := \mathbb{E}Y$ and $e := Y - b(X)$.
The tower property gives $\mathbb{E}b(X) = m$, and $\mathbb{E}(e \mid X) = \mathbb{E}(Y \mid X) - b(X) = 0$.
Since $Y - m = e + (b(X) - m)$, expansion gives $\text{Var}(Y) = \mathbb{E}e^2 + 2\mathbb{E}[e(b(X) - m)] + \mathbb{E}(b(X) - m)^2$.
The cross term is zero because $\mathbb{E}[e(b(X) - m)] = \mathbb{E}[(b(X) - m)\mathbb{E}(e \mid X)] = 0$.
Also $\mathbb{E}e^2 = \mathbb{E}[\mathbb{E}(e^2 \mid X)] = \mathbb{E}[v(X)]$, $\mathbb{E}(b(X) - m)^2 = \text{Var}(b(X))$. Thus $\text{Var}(Y) = \mathbb{E}[v(X)] + \text{Var}(b(X))$.
Structural Commentary: Variance splits into within-fiber variance and between-fiber mean variation.
Key Insight: Total variance equals average conditional variance plus variance of conditional expectation. □

Example 1.2.9. *Random Binomial Proportion*
Let $Q \sim \text{Uniform}(0,1)$ and, conditionally on $Q = q$, let $X \sim \text{Binomial}(n, q)$.
Then $\mathbb{E}(X \mid Q) = nQ$, so $\mathbb{E}X = \mathbb{E}(nQ) = \frac{n}{2}$. Also $\text{Var}(X \mid Q) = nQ(1-Q)$.
Hence $\mathbb{E}[\text{Var}(X \mid Q)] = n\int_0^1 q(1-q)\,dq = \frac{n}{6}$, $\text{Var}(\mathbb{E}(X \mid Q)) = \text{Var}(nQ) = \frac{n^2}{12}$. Thus $\text{Var}(X) = \frac{n}{6} + \frac{n^2}{12}$.

1.3 Conditional Moment Methods and Prediction

Conditional expectation is not only a way to compute means. It is also the geometric projection rule behind random-sum formulas and least-squares prediction.

Proposition 1.3.1. *Random Sums*
Let $N \in \mathbb{N}_0$ a.s. with $\mathbb{E}N^2 < \infty$, and let $(X_i)_{i\geq 1}$ be iid, independent of N, with $\mathbb{E}X_1 = m$ and $\text{Var}(X_1) = v < \infty$.
Set $T = \sum_{i=1}^{N} X_i$, with $T = 0$ on $\{N = 0\}$.

(i) $\mathbb{E}T = m\mathbb{E}N$.
(ii) $\text{Var}(T) = v\mathbb{E}N + m^2\,\text{Var}(N)$.

Proof.

(i) On $\{N = n\}$, $T = \sum_{i=1}^{n} X_i$, so $\mathbb{E}(T \mid N = n) = nm$ and $\mathbb{E}(T \mid N) = mN$; hence $\mathbb{E}T = \mathbb{E}\mathbb{E}(T \mid N) = m\mathbb{E}N$.

(ii) On $\{N = n\}$, independence gives $\text{Var}(T \mid N = n) = nv$, so $\text{Var}(T \mid N) = vN$.
Total variance gives $\text{Var}(T) = \mathbb{E}(vN) + \text{Var}(mN) = v\mathbb{E}N + m^2\,\text{Var}(N)$.

Structural Commentary: The randomness of the number of summands contributes the second variance term.
Key Insight: First condition on how many summands exist; then average over that count. □

Proposition 1.3.2. *MGF of a Random Sum*
Let $T = \sum_{i=1}^{N} X_i$, where $N \in \mathbb{N}_0$ is independent of iid $(X_i)_{i\geq 1}$.
If ψ_N and ψ_X are finite at the relevant arguments, then $\psi_T(t) = \psi_N(\log \psi_X(t))$.

Proof. On $\{N = n\}$, independence gives $\mathbb{E}(e^{tT} \mid N = n) = \mathbb{E}e^{t\sum_{i=1}^{n} X_i} = \prod_{i=1}^{n} \psi_X(t) = [\psi_X(t)]^n$.
Hence $\psi_T(t) = \mathbb{E}[\psi_X(t)^N] = \mathbb{E}e^{N\log\psi_X(t)} = \psi_N(\log\psi_X(t))$.
Structural Commentary: Conditioning on N turns the random sum into a fixed product of MGFs.
Key Insight: Random counts compose MGFs by substituting $\log\psi_X(t)$ into ψ_N. □

Example 1.3.3. *Compound Poisson Count*
Let $N \sim \text{Poisson}(\mu)$ and, given N, let $T = \sum_{i=1}^{N} X_i$ with iid $X_i \sim \text{Poisson}(\lambda)$ independent of N.
Since $\psi_N(s) = \exp(\mu(e^s - 1))$ and $\psi_X(t) = \exp(\lambda(e^t - 1))$, one has $\psi_T(t) = \psi_N(\log\psi_X(t)) = \exp(\mu(e^{\lambda(e^t-1)} - 1))$.

Proposition 1.3.4. *Best Constant Predictor*
For $Y \in L^2$, $\arg\min_{c\in\mathbb{R}} \mathbb{E}(Y-c)^2 = \mathbb{E}Y$, and $\min_{c\in\mathbb{R}} \mathbb{E}(Y-c)^2 = \text{Var}(Y)$.

Proof. Let $\mu = \mathbb{E}Y$. Since $\mathbb{E}(Y-\mu) = 0$, $\mathbb{E}(Y-c)^2 = \mathbb{E}[(Y-\mu)+(\mu-c)]^2 = \text{Var}(Y) + (\mu-c)^2$.
Since $(\mu-c)^2 \geq 0$ with equality iff $c = \mu$, the unique minimizer is $c = \mu$.
Structural Commentary: The mean is the orthogonal projection of Y onto the constants.
Key Insight: Best constant prediction equals the average. □

Theorem 1.3.5. *Conditional Expectation as Best Predictor*
Let $X, Y \in L^2$ and $\mathcal{G} := \{h(X) : h(X) \in L^2\}$. Let $M := \mathbb{E}(Y \mid X)$, then $M \in \mathcal{G}$, $\mathbb{E}(Y-M)^2 \leq \mathbb{E}(Y-Z)^2 \forall Z \in \mathcal{G}$.

Proof. For $Z \in \mathcal{G}$, both M and Z are $\sigma(X)$-measurable.
Since $\mathbb{E}(Y - M \mid X) = 0$, $\mathbb{E}[(Y-M)(M-Z)] = \mathbb{E}[(M-Z)\mathbb{E}(Y-M \mid X)] = 0$.
Hence $\mathbb{E}(Y-Z)^2 = \mathbb{E}(Y-M)^2 + \mathbb{E}(M-Z)^2 \geq \mathbb{E}(Y-M)^2$.
Structural Commentary: Prediction from X is performed fiber by fiber over the conditional laws of Y.
Key Insight: The optimal nonlinear least-squares predictor is the conditional mean. □

Definition 1.3.6. *Best Linear Predictor*
For $X, Y \in L^2$, set $\mathcal{A}_X := \{\alpha + \beta X : \alpha, \beta \in \mathbb{R}\}$.
A *best linear predictor* of Y from X is any $\widehat{Y}^* \in \arg\min_{Z\in\mathcal{A}_X} \mathbb{E}(Y-Z)^2$.

Theorem 1.3.7. *Best Linear Predictor*
Let $X, Y \in L^2$, $\sigma_X^2 := \text{Var}(X) > 0$, $\mu_X := \mathbb{E}X$, and $\mu_Y := \mathbb{E}Y$.

(i) $\arg\min_{Z\in\mathcal{A}_X} \mathbb{E}(Y-Z)^2 = \{\alpha^* + \beta^* X\}$, *where* $\beta^* = \frac{\text{Cov}(X,Y)}{\sigma_X^2}$ *and* $\alpha^* = \mu_Y - \beta^*\mu_X$.

(ii) $\min_{Z\in\mathcal{A}_X} \mathbb{E}(Y-Z)^2 = \text{Var}(Y) - \frac{\text{Cov}(X,Y)^2}{\sigma_X^2}$.

(iii) If $\text{Var}(Y) > 0$ *and* $\rho = \text{Corr}(X,Y)$, *then the minimum error is* $\text{Var}(Y)(1-\rho^2)$.

Proof.

(i) Let $q(\alpha,\beta) := \mathbb{E}(Y-\alpha-\beta X)^2$. For fixed β, $q(\alpha,\beta) = \text{Var}(Y-\beta X) + (\mu_Y - \alpha - \beta\mu_X)^2$,
so $\alpha(\beta) = \mu_Y - \beta\mu_X$ uniquely minimizes over α.

(ii) Then $q(\alpha(\beta),\beta) = \mathrm{Var}(Y-\beta X) = \mathrm{Var}(Y) + \beta^2\sigma_X^2 - 2\beta\,\mathrm{Cov}(X,Y)$.
Since $\sigma_X^2 > 0$, this quadratic has unique minimizer $\beta^* = \frac{\mathrm{Cov}(X,Y)}{\sigma_X^2}$.

(iii) Substituting β^* gives $\mathrm{Var}(Y) - \frac{\mathrm{Cov}(X,Y)^2}{\sigma_X^2}$. If $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$, then this equals $\mathrm{Var}(Y)(1-\rho^2)$.

Structural Commentary: Linear prediction projects Y onto $\mathcal{A}_X$.
Key Insight: The optimal slope is normalized covariance. □

Example 1.3.8. *Bivariate Normal Prediction*
If (X,Y) is bivariate normal with means μ_X,μ_Y, standard deviations $\sigma_X,\sigma_Y > 0$, and correlation $\rho \in (-1,1)$, then $Y \mid X \sim N(\mu_Y + \rho\sigma_Y\sigma_X^{-1}(X-\mu_X), \sigma_Y^2(1-\rho^2))$.
Thus $\mathbb{E}(Y \mid X) = \mu_Y + \rho\sigma_Y\sigma_X^{-1}(X-\mu_X)$ is linear, with conditional variance $\sigma_Y^2(1-\rho^2)$.

Example 1.3.9. *Regression to the Mean*
Let $\mathbb{E}X = \mathbb{E}Y = \mu$, $\mathrm{Var}(X) = \mathrm{Var}(Y) = \sigma^2 > 0$, and $\mathrm{Corr}(X,Y) = \rho$. Best linear predictor is $\widehat{Y} = \mu + \rho(X-\mu)$.
For $0 < \rho < 1$, $|\widehat{Y}-\mu| = \rho|X-\mu| < |X-\mu|$, so predicted scores regress toward μ.

2 Probabilistic Convergence

2.1 Modes of Probabilistic Convergence

Pointwise convergence is too strict for most random sequences. The useful replacement is to ask whether probabilities, laws, or moments of the errors become small.

Definition 2.1.1. *Modes of Probabilistic Convergence*
Let $(X_n)_{n\ge1}$ and X be real-valued random variables with distribution functions F_n and F.

(i) *Convergence in probability*: $X_n \xrightarrow{\mathbb{P}} X$ iff $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$.
(ii) *Convergence in distribution*: $X_n \rightsquigarrow X$ iff $F_n(t) \to F(t)$ at every continuity point t of F.
(iii) *Convergence in L^p*: for $p \ge 1$, $X_n \xrightarrow{L^p} X$ iff $\mathbb{E}\,|X_n - X|^p \to 0$.
(iv) *Convergence in quadratic mean*: $X_n \xrightarrow{qm} X$ iff $X_n \xrightarrow{L^2} X$.
(v) *Almost-sure convergence*: $X_n \xrightarrow{\text{a.s.}} X$ iff $\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \to X(\omega)\}) = 1$.
(vi) *Constant limits*: if $X = c$ almost surely for a constant c, then $X_n \xrightarrow{\mathbb{P}} c$, $X_n \rightsquigarrow c$, $X_n \xrightarrow{L^p} c$, $X_n \xrightarrow{qm} c$, and $X_n \xrightarrow{\text{a.s.}} c$ denote convergence to that constant random variable in the corresponding modes.

Example 2.1.2. *A Shrinking Normal Law*
Let $X_n \sim N(0, \frac{1}{n})$, and let F be the distribution function of δ_0.
Since $\sqrt{n}X_n \sim N(0,1)$, $F_n(t) = \mathbb{P}(X_n \le t) = \Phi(\sqrt{n}t)$, so $F_n(t) \to 0$ for $t < 0$ and $F_n(t) \to 1$ for $t > 0$.
Hence $F_n(t) \to F(t)$ at every continuity point $t \ne 0$ of F, and therefore $X_n \rightsquigarrow 0$.

Theorem 2.1.3. *Basic Implications Between Convergence Modes*
Let X_n and X be real-valued random variables with distribution functions F_n and F, and let $c \in \mathbb{R}$.

(i) $X_n \xrightarrow{qm} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$.
(ii) $X_n \xrightarrow{\mathbb{P}} X \Rightarrow X_n \rightsquigarrow X$.
(iii) $X_n \rightsquigarrow c \Rightarrow X_n \xrightarrow{\mathbb{P}} c$.
(iv) $X_n \xrightarrow{\text{a.s.}} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$.
(v) $X_n \xrightarrow{L^1} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$, *and* $X_n \xrightarrow{qm} X \Rightarrow X_n \xrightarrow{L^1} X$.

Proof.

(i) If $X_n \xrightarrow{qm} X$, then for each $\varepsilon > 0$, Markov gives $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-2}\mathbb{E}\,|X_n - X|^2 \to 0$.
(ii) Let x be a continuity point of F.
For each $\varepsilon > 0$, $F(x-\varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon) \le F_n(x) \le F(x+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$,
so $F(x-\varepsilon) \le \liminf_n F_n(x) \le \limsup_n F_n(x) \le F(x+\varepsilon)$; letting $\varepsilon \downarrow 0$ gives $F_n(x) \to F(x)$.
(iii) For $\varepsilon > 0$, the distribution function of c is continuous at $c-\varepsilon$ and $c+\varepsilon$.
Hence $\mathbb{P}(|X_n - c| > \varepsilon) \le F_n(c-\varepsilon) + 1 - F_n(c+\varepsilon) \to 0$.
(iv) Fix $\varepsilon > 0$ and set $Y_n = \mathbf{1}_{\{|X_n - X| > \varepsilon\}}$. If $X_n \to X$ a.s., then $Y_n \to 0$ a.s. and $0 \le Y_n \le 1$.
Hence dominated convergence gives $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{E}Y_n \to 0$.
(v) If $X_n \xrightarrow{L^1} X$, then $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-1}\mathbb{E}\,|X_n - X| \to 0$ by Markov.
If $X_n \xrightarrow{qm} X$, then $\mathbb{E}\,|X_n - X| \le (\mathbb{E}\,|X_n - X|^2)^{\frac{1}{2}} \to 0$ by Cauchy–Schwarz.

Structural Commentary: Strong modes give tail bounds; distribution convergence gives c.d.f. control.
Key Insight: Convert to tail bounds; use c.d.f. squeezes when laws enter. □

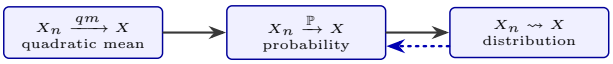

Figure 230: Solid: always; dotted: point-mass limit.

Example 2.1.4. *Why the Reverse Arrows Fail*

(i) Let $U \sim \mathrm{Uniform}(0,1)$ and $X_n = \sqrt{n}\mathbf{1}_{(0,\frac{1}{n})}(U)$. Then for each $\varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) \le \frac{1}{n} \to 0$,
so $X_n \xrightarrow{\mathbb{P}} 0$, but $\mathbb{E}X_n^2 = 1$ for all n; hence $X_n \not\to 0$ in quadratic mean.

(ii) Let $X \sim N(0,1)$ and set $X_n = -X$. Then $X_n \sim X$ for all n, so $X_n \rightsquigarrow X$,
but for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(2|X| > \varepsilon) > 0$ for all n; hence $X_n \not\to X$ in probability.

Remark 2.1.5. *A Warning About Expectations*
Convergence in probability does not imply convergence of expectations.
If $\mathbb{P}(X_n = n^2) = \frac{1}{n}$, $\mathbb{P}(X_n = 0) = 1 - \frac{1}{n}$, then $\forall \varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) = \frac{1}{n} \to 0$, so $X_n \xrightarrow{\mathbb{P}} 0$, $\mathbb{E}X_n = n \to +\infty$.

Definition 2.1.6. *Asymptotic Uniform Integrability*
A sequence (X_n) is *asymptotically uniformly integrable* if $\lim_{M\to\infty} \limsup_{n\to\infty} \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n|>M\}}) = 0$.

Proposition 2.1.7. *Recovering Expectations from Probability Convergence*
Let $b \in \mathbb{R}$. If $X_n \xrightarrow{\mathbb{P}} b$ and (X_n) is asymptotically uniformly integrable, then $\mathbb{E}X_n \to b$.

Proof. For $M > |b|$, set $\tau_M(x) = (-M) \vee (x \wedge M)$. As τ_M bounded, continuous; $\tau_M(b) = b$, $\tau_M(X_n) \xrightarrow{\mathbb{P}} b$.
For $\eta > 0$, $\mathbb{E}|\tau_M(X_n) - b| \le \eta + 2M\mathbb{P}(|\tau_M(X_n) - b| > \eta)$, so $\mathbb{E}\tau_M(X_n) \to b$ after $n \to \infty$ and $\eta \downarrow 0$.
Moreover, $|X_n - \tau_M(X_n)| \le |X_n| \mathbf{1}_{\{|X_n|>M\}}$, hence $\limsup_n |\mathbb{E}X_n - b| \le \limsup_n \mathbb{E}(|X_n| \mathbf{1}_{\{|X_n|>M\}})$.
Letting $M \to \infty$ and using asymptotic uniform integrability gives $\mathbb{E}X_n \to b$.
Structural Commentary: Truncation makes convergence bounded; asymptotic uniform integrability kills tails.
Key Insight: Probability convergence handles the truncations; tail control restores expectations. □

Theorem 2.1.8. *Continuous Mapping and Slutsky Rules*
Let $g : \mathbb{R} \to \mathbb{R}$ be continuous, and let $c \in \mathbb{R}$.

(i) $X_n \xrightarrow{\mathbb{P}} X \Rightarrow g(X_n) \xrightarrow{\mathbb{P}} g(X)$.
(ii) $X_n \rightsquigarrow X \Rightarrow g(X_n) \rightsquigarrow g(X)$.
(iii) $X_n \rightsquigarrow X$ *and* $Y_n \xrightarrow{\mathbb{P}} c \Rightarrow X_n + Y_n \rightsquigarrow X + c$ *and* $X_n Y_n \rightsquigarrow cX$.
(iv) $X_n \xrightarrow{\mathbb{P}} X$ *and* $Y_n \xrightarrow{\mathbb{P}} Y \Rightarrow X_n + Y_n \xrightarrow{\mathbb{P}} X + Y$ *and* $X_n Y_n \xrightarrow{\mathbb{P}} XY$.

Proof.
(i) Fix $\varepsilon, \eta > 0$. Choose K, $\mathbb{P}(|X| > K) < \eta$, $\delta \le 1$ s.t. $|x - y| < \delta$, $|x|, |y| \le K + 1$; $|g(x) - g(y)| < \varepsilon$.
Then $\mathbb{P}(|g(X_n) - g(X)| > \varepsilon) \le \mathbb{P}(|X| > K) + \mathbb{P}(|X_n - X| > \delta)$, so $\limsup$ at most η.
(ii) Let t be a continuity point of the law of $g(X)$ and set $A_t = \{x : g(x) \le t\}$.
Since $\partial A_t \subseteq \{x : g(x) = t\}$ and $\mathbb{P}(g(X) = t) = 0$, the continuity-set form of weak convergence gives
$\mathbb{P}(g(X_n) \le t) = \mathbb{P}(X_n \in A_t) \to \mathbb{P}(X \in A_t) = \mathbb{P}(g(X) \le t)$.
(iii) If $Z_n \rightsquigarrow Z$ and $R_n \xrightarrow{\mathbb{P}} 0$, then $Z_n + R_n \rightsquigarrow Z$ by the c.d.f. squeeze
$F_{Z_n}(t - \eta) - \mathbb{P}(|R_n| > \eta) \le F_{Z_n + R_n}(t) \le F_{Z_n}(t + \eta) + \mathbb{P}(|R_n| > \eta)$,
followed by $n \to \infty$ and $\eta \downarrow 0$ through continuity points of F_Z.
Since $X_n + c \rightsquigarrow X + c$ by (ii), applying this with $Z_n = X_n + c$ and $R_n = Y_n - c$ gives $X_n + Y_n \rightsquigarrow X + c$.
Also $X_n \rightsquigarrow X$ implies (X_n) is tight, so $\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \le \mathbb{P}(|X_n| > K) + \mathbb{P}(|Y_n - c| > \frac{\varepsilon}{K})$.
Letting $n \to \infty$, then $K \uparrow \infty$ gives $X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0$.
Since $cX_n \rightsquigarrow cX$ by (ii), the additive perturbation just proved gives $X_n Y_n = cX_n + X_n(Y_n - c) \rightsquigarrow cX$.
(iv) For sums, $\mathbb{P}(|(X_n + Y_n) - (X + Y)| > \varepsilon) \le \mathbb{P}(|X_n - X| > \frac{\varepsilon}{2}) + \mathbb{P}(|Y_n - Y| > \frac{\varepsilon}{2}) \to 0$.
For products, (X_n) and Y are tight, $\mathbb{P}(|X_n(Y_n - Y)| > \frac{\varepsilon}{2}) \le \mathbb{P}(|X_n| > K) + \mathbb{P}(|Y_n - Y| > \frac{\varepsilon}{2K})$,
while $\mathbb{P}(|Y(X_n - X)| > \frac{\varepsilon}{2}) \le \mathbb{P}(|Y| > K) + \mathbb{P}(|X_n - X| > \frac{\varepsilon}{2K})$; let $n \to \infty$ and then $K \uparrow \infty$.
Structural Commentary: Continuity transports convergence; Slutsky absorbs probability-small errors.
Key Insight: Control the bounded high-probability part; let the tails vanish. □

2.2 Probability on Metric Spaces

3 Limit Theorems

3.1 Laws of Large Numbers

The law of large numbers formalizes the idea that a large sample average is close to the mean of the distribution.

Perspective 3.1.1. *Law of the Sample Mean*
If $X_1, \ldots, X_n$ are IID, mean μ, variance σ^2, then $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$ is a random variable with law $\mathcal{L}(\overline{X}_n)$.
By Theorem 3.2.7, $\mathbb{E}\overline{X}_n = \mu$ and $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$, so $\mathcal{L}(\overline{X}_n)$ concentrates around μ as $n \to \infty$.

Theorem 3.1.2. *Weak Law of Large Numbers*
Let $(X_i)_{i\ge1}$ be IID real-valued random variables with $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 < +\infty$. Set $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$.
Then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$, equivalently $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \to 0$ for every $\varepsilon > 0$.

Proof. For $n \ge 1$, $\mathbb{E}\overline{X}_n = n^{-1} \sum_{i=1}^n \mathbb{E}X_i = \mu$ and, by independence, $\mathrm{Var}(\overline{X}_n) = n^{-2} \sum_{i=1}^n \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$.
For every $\varepsilon > 0$, Chebyshev gives $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$, hence $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.
Structural Commentary: Averaging keeps the mean fixed and divides variance by n.
Key Insight: Chebyshev turns vanishing variance into convergence in probability. □

Proposition 3.1.3. *Weak LLN with Vanishing Average Variance*
Let $(X_i)_{i\ge1}$ be independent, $X_i \in L^2$, with $\mu_i := \mathbb{E}X_i$ and $\sigma_i^2 := \mathrm{Var}(X_i)$.
Set $\overline{X}_n := n^{-1} \sum_{i=1}^n X_i$ and $m_n := n^{-1} \sum_{i=1}^n \mu_i$. If $m_n \to \mu$ and $n^{-2} \sum_{i=1}^n \sigma_i^2 \to 0$, then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. $\mathbb{E}\overline{X}_n = m_n$. By independence, $\mathrm{Var}(\overline{X}_n) = n^{-2}\sum_{i=1}^{n}\sigma_i^2 \to 0$. Fix $\varepsilon > 0$. $\forall$ large n, $|m_n - \mu| < \frac{\varepsilon}{2}$.

Hence $\{|\overline{X}_n - \mu| > \varepsilon\} \subseteq \{|\overline{X}_n - m_n| > \frac{\varepsilon}{2}\}$. By Chebyshev, $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{4\,\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} \to 0$.
Structural Commentary: Identical distribution is not essential for a weak law.
Key Insight: It is enough that the mean target stabilises and the average variance vanishes. □

Example 3.1.4. *Coin Proportions and Chebyshev Scale*

Let $X_i \sim \mathrm{Bernoulli}(p)$ be IID head indicators, $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$. Then $\mathbb{E}\overline{X}_n = p$, $\mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}$, $\overline{X}_n \xrightarrow{\mathbb{P}} p$.

For $p = \frac{1}{2}$, Chebyshev gives $\mathbb{P}(0.4 \le \overline{X}_n \le 0.6) \ge 1 - \frac{1}{4n(0.1)^2} = 1 - \frac{25}{n}$, at least 0.7 for $n \ge 84$.

If $n = 10000$ fair tosses produce 5067 heads, then $\overline{X}_n = 0.5067$, while $\mathrm{Var}(\overline{X}_n) = \frac{1}{4n} = 2.5 \cdot 10^{-5}$ and $\mathrm{sd}(\overline{X}_n) = 0.005$. The observed deviation is about 1.34 standard deviations from $\frac{1}{2}$.

Theorem 3.1.5. *Strong Law of Large Numbers*

Let $(X_i)_{i\ge1}$ be IID real-valued random variables with $X_1 \in L^1$, $\mu := \mathbb{E}X_1$, and $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$.

Then $\overline{X}_n \xrightarrow{\text{a.s.}} \mu$, equivalently $\mathbb{P}(\lim_{n\to\infty} \overline{X}_n = \mu) = 1$.

Proof. After replacing X_i by $X_i - \mu$, assume $\mu = 0$. Set $Y_i := X_i \mathbf{1}_{\{|X_i| \le i\}}$.

Since $\sum_{i=1}^{\infty}\mathbb{P}(X_i \ne Y_i) = \sum_{i=1}^{\infty}\mathbb{P}(|X_1| > i) < +\infty$, Borel–Cantelli gives $X_i = Y_i$ eventually a.s.

Hence $n^{-1}\sum_{i=1}^{n}(X_i - Y_i) \to 0$ a.s. For $Z_i := Y_i - \mathbb{E}Y_i$, independence gives

$\sum_{i=1}^{\infty}\frac{\mathrm{Var}(Z_i)}{i^2} \le \sum_{i=1}^{\infty}\frac{\mathbb{E}(X_1^2\mathbf{1}_{\{|X_1|\le i\}})}{i^2} < +\infty$, by Kolmogorov's variance criterion, $n^{-1}\sum_{i=1}^{n} Z_i \to 0$ a.s.

By dominated convergence, $\mathbb{E}Y_i \to 0$, hence Cesaro averaging gives $n^{-1}\sum_{i=1}^{n}\mathbb{E}Y_i \to 0$.

Combining $X_i - Y_i$, Z_i, and $\mathbb{E}Y_i$ gives $\overline{X}_n \to 0$ a.s.; adding back μ completes the proof.
Structural Commentary: Truncate tails; centered truncations have summable variance.
Key Insight: Replace by an eventually equal truncated sequence, then average the centered part. □

Corollary 3.1.6. *Strong Implies Weak*

Let $(X_i)_{i\ge1}$ be IID real-valued random variables, $X_1 \in L^1$, $\mu := \mathbb{E}X_1$, $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$. Then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. By Theorem 3.1.5, $\overline{X}_n \to \mu$ a.s. For $\varepsilon > 0$, set $Y_n := \mathbf{1}_{\{|\overline{X}_n - \mu| > \varepsilon\}}$.

Then $Y_n \to 0$ a.s. and $0 \le Y_n \le 1$, so dominated convergence gives $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) = \mathbb{E}Y_n \to 0$.
Structural Commentary: Almost-sure convergence gives pointwise tail indicators tending to zero.
Key Insight: Dominate the indicators by 1 and integrate. □

Proposition 3.1.7. *Consistency of Empirical Moments and Variance*
Let $(X_i)_{i\ge1}$ be iid, $X_1 \in L^2$, $\mu := \mathbb{E}X_1$, and $\sigma^2 := \mathrm{Var}(X_1)$.

For $n \ge 2$, set $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$ and $S_n^2 := (n-1)^{-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$. Then $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$.

More generally, if $X_1 \in L^r$ for an integer $r \ge 1$, then $n^{-1}\sum_{i=1}^{n} X_i^r \xrightarrow{\mathbb{P}} \mathbb{E}X_1^r$.

Proof. Since $(X_i^r)_{i\ge1}$ is iid and integrable, the strong law gives $n^{-1}\sum_{i=1}^{n} X_i^r \to \mathbb{E}X_1^r$ a.s., hence in probability.

Let $m_2 := \mathbb{E}X_1^2 = \sigma^2 + \mu^2$. The cases $r = 1, 2$ give $\overline{X}_n \to \mu$ and $n^{-1}\sum_{i=1}^{n} X_i^2 \to m_2$ a.s.

Since $S_n^2 = \frac{n}{n-1}(n^{-1}\sum_{i=1}^{n} X_i^2 - \overline{X}_n^2)$, continuous mapping gives $S_n^2 \to m_2 - \mu^2 = \sigma^2$ a.s., hence in probability.
Structural Commentary: Sample moments are averages of transformed observations.
Key Insight: Apply the strong law to X_i^r, then pass through the continuous variance formula. □

Remark 3.1.8. *Why Integrability Matters*
If $X_1 \notin L^1$, the mean $\mathbb{E}X_1$ need not exist and sample averages need not converge to any finite deterministic limit; integrability is the condition that makes a law of large numbers meaningful.

3.2 Central Limit Theory

The law of large numbers says that $\overline{X}_n$ is close to μ for large n. The central limit theorem describes the scale and shape of the remaining fluctuation.

Theorem 3.2.1. *Central Limit Theorem*
Let $(X_i)_{i\ge1}$ be IID real-valued random variables, $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$.

Let $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$, $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$.

Then $Z_n \rightsquigarrow N(0,1)$, equivalently $\mathbb{P}(Z_n \le z) \to \Phi(z) := \int_{-\infty}^{z}(2\pi)^{-\frac{1}{2}}e^{-\frac{x^2}{2}}\,dx$ $\forall z \in \mathbb{R}$.

Proof. Set $Y_i := \frac{X_i-\mu}{\sigma}$, so $\mathbb{E}Y_i = 0$, $\mathrm{Var}(Y_i) = 1$, and $Z_n = n^{-\frac{1}{2}} \sum_{i=1}^{n} Y_i$.

For $\varphi(t) := \mathbb{E}e^{\mathrm{i}tY_1}$, the finite second moment expansion gives $\varphi(u) = 1 - \frac{u^2}{2} + o(u^2)$ as $u \to 0$.

By independence, $\varphi_{Z_n}(t) = [\varphi(\frac{t}{\sqrt{n}})]^n = [1 - \frac{t^2}{2n} + o(\frac{1}{n})]^n \to e^{-\frac{t^2}{2}}$ for every $t \in \mathbb{R}$.

Since $e^{-\frac{t^2}{2}}$ is the characteristic function of $N(0,1)$, Levy's continuity theorem gives $Z_n \rightsquigarrow N(0,1)$; continuity of Φ gives the displayed c.d.f. convergence.

Structural Commentary: Centering and scaling turn the sum into unit-variance IID increments.

Key Insight: Characteristic functions turn normalized sums into powers, whose limit is Gaussian. □

Method 3.2.2. *Equivalent CLT Approximation Forms*

Let $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ with mean μ and variance $\sigma^2 \in (0, +\infty)$, and set $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$.

(i) Standardized form: $Z_n \approx N(0,1)$.
(ii) Centered form: $\sqrt{n}(\overline{X}_n - \mu) \approx N(0, \sigma^2)$.
(iii) Sample-mean form: $\overline{X}_n \approx N(\mu, \frac{\sigma^2}{n})$.

Use any equivalent form to approximate probabilities involving $\overline{X}_n$.

Example 3.2.3. *Measurement Error and Sample Size*

Let $X_1, \ldots, X_n$ be independent measurements with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$.
For $c > 0$, the CLT gives $\mathbb{P}(|\overline{X}_n - \mu| < c) \approx \Phi(\frac{c\sqrt{n}}{\sigma}) - \Phi(\frac{-c\sqrt{n}}{\sigma})$.
For $(n, \sigma, c) = (16, 1, 0.5)$, this is $\Phi(2) - \Phi(-2) \approx 0.954$.

Lemma 3.2.4. *MGF Continuity Theorem*

Let Z_n, Z be real-valued with MGFs $\psi_n(t) := \mathbb{E}e^{tZ_n}$ and $\psi(t) := \mathbb{E}e^{tZ}$ finite on $(-a, a)$ for some $a > 0$.
If $\psi_n(t) \to \psi(t)$ for all $t \in (-a, a)$, then $Z_n \rightsquigarrow Z$.

Proof. For each $s \in \mathbb{R}$, the characteristic functions satisfy $\varphi_n(s) = \mathbb{E}e^{\mathrm{i}sZ_n} = \psi_n(\mathrm{i}s)$ by analytic continuation on the strip $\{z : |\mathrm{Re}\, z| < a\}$.
Pointwise MGF convergence near 0 gives $\varphi_n(s) \to \varphi(s)$, the characteristic function of Z; Levy's continuity theorem yields $Z_n \rightsquigarrow Z$.

Structural Commentary: Local MGF convergence determines local characteristic-function convergence.

Key Insight: MGFs near 0 identify the weak limit. □

Proposition 3.2.5. *MGF Proof of the IID CLT*

Let $(X_i)_{i\geq 1}$ be iid with $\mathbb{E}X_1 = 0$, $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$, and MGF $M(t) := \mathbb{E}e^{tX_1}$ finite near 0.
Set $S_n := \sum_{i=1}^{n} X_i$ and $Z_n := \frac{S_n}{\sigma\sqrt{n}}$. Then $Z_n \rightsquigarrow N(0,1)$, equivalently $\mathbb{P}(Z_n \leq x) \to \Phi(x)$ for all $x \in \mathbb{R}$.

Proof. For fixed t near 0, independence gives $M_{Z_n}(t) = \mathbb{E}e^{\frac{tS_n}{\sigma\sqrt{n}}} = \prod_{i=1}^{n} \mathbb{E}e^{\frac{tX_i}{\sigma\sqrt{n}}} = [M(\frac{t}{\sigma\sqrt{n}})]^n$.

Since $M(0) = 1$, $M'(0) = 0$, and $M''(0) = \sigma^2$, Taylor expansion gives $M(s) = 1 + \frac{\sigma^2 s^2}{2} + o(s^2)$ as $s \to 0$.

Hence $M_{Z_n}(t) = (1 + \frac{t^2}{2n} + o(\frac{1}{n}))^n \to e^{\frac{t^2}{2}}$, the MGF of $N(0,1)$.
Lemma 3.2.4 gives $Z_n \rightsquigarrow N(0,1)$, and continuity of Φ gives the c.d.f. convergence.

Structural Commentary: The MGF proof needs the stronger near-zero MGF assumption.

Key Insight: Standardization gives $\frac{t^2}{2n}$; independence exponentiates. □

Example 3.2.6. *A Poisson Error-Count Approximation*

Let $X_1, \ldots, X_{125} \sim \mathrm{Poisson}(5)$ be IID and set $\overline{X}_{125} = 125^{-1} \sum_{i=1}^{125} X_i$.

Since $\mu = 5$ and $\sigma^2 = 5$, the CLT gives $\mathbb{P}(\overline{X}_{125} < 5.5) \approx \Phi(\frac{\sqrt{125}(5.5-5)}{\sqrt{5}}) = \Phi(2.5) \approx 0.9938$.

Example 3.2.7. *Normal Approximation to a Poisson Count*

Let $X_\lambda \sim \mathrm{Poisson}(\lambda)$, $Z_\lambda := \frac{X_\lambda - \lambda}{\sqrt{\lambda}}$. Then $M_{Z_\lambda}(t) = \exp(-t\sqrt{\lambda} + \lambda(e^{\frac{t}{\sqrt{\lambda}}} - 1))$ and $\log M_{Z_\lambda}(t) = \frac{t^2}{2} + o(1)$, so $Z_\lambda \rightsquigarrow N(0,1)$. If $X \sim \mathrm{Poisson}(900)$, then $\mathbb{P}(X > 950) = \mathbb{P}(\frac{X-900}{30} > \frac{5}{3}) \approx 1 - \Phi(\frac{5}{3}) \approx 0.04779$, close to 0.04712.

Example 3.2.8. *Normal Approximation to the Binomial*

If $X \sim \mathrm{Binomial}(n, p)$, $np > 5$, $n(1-p) > 5$, then $X \approx N(np, np(1-p))$; $\frac{X-np}{\sqrt{np(1-p)}} \approx N(0,1)$.

For $X \sim \mathrm{Binomial}(100, \frac{1}{2})$, $\mathbb{P}(X \geq 60) \approx 1 - \Phi(\frac{60-50}{5}) = 1 - \Phi(2) \approx 0.0228$.

Example 3.2.9. *Skewed Summands Need More Averaging*

Let $E_1, \ldots, E_n$ be iid $\mathrm{Exp}(1)$ and $G_n := \sum_{i=1}^{n} E_i \sim \mathrm{Gamma}(n, 1)$. Then $\frac{G_n - n}{\sqrt{n}} \rightsquigarrow N(0,1)$.

Skewness makes the normal approximation slower for small n.

Example 3.2.10. *Products and Lognormal Shape*

Let $Y_n = y_0 \prod_{i=1}^{n} X_i$ with iid $X_i > 0$, and suppose $\mu_\ell := \mathbb{E}\log X_1$ and $\sigma_\ell^2 := \mathrm{Var}(\log X_1) \in (0, +\infty)$.

Since $\log Y_n = \log y_0 + \sum_{i=1}^{n} \log X_i$, the CLT gives $\log Y_n \approx N(\log y_0 + n\mu_\ell, n\sigma_\ell^2)$, so Y_n is approximately lognormal. Likewise $V_n = v_0 \prod_{i=1}^{n} R_i$ when iid log-returns $\log R_i$ have finite positive variance.

Theorem 3.2.11. *Studentized Central Limit Theorem*

Let $(X_i)_{i\geq 1}$ be IID real-valued random variables with $\mathbb{E}X_1 = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$.
Set $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$, $S_n^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$, and $T_n := S_n^{-1}\sqrt{n}(\overline{X}_n - \mu)$. Then $T_n \rightsquigarrow N(0,1)$.

Proof. The CLT gives $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N(0,1)$, and Proposition 3.1.7 gives $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$, hence $S_n \xrightarrow{\mathbb{P}} \sigma$.
Thus $\frac{\sigma}{S_n} \xrightarrow{\mathbb{P}} 1$, and Slutsky's rule from Theorem 2.1.8 gives $T_n = Z_n(\frac{\sigma}{S_n}) \rightsquigarrow N(0,1)$.
Structural Commentary: Studentization replaces the unknown σ by the consistent estimate S_n.
Key Insight: Slutsky permits multiplying the CLT statistic by $\frac{\sigma}{S_n} \to 1$. □

Theorem 3.2.12. *Berry–Esseen Rate*
Let $(X_i)_{i\geq 1}$ be IID real-valued random variables, $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0,+\infty)$, $\rho_3 := \mathbb{E}\,|X_1-\mu|^3 < +\infty$.
Let $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$, $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$, $\Phi(z) := \int_{-\infty}^z (2\pi)^{-\frac{1}{2}} e^{-\frac{x^2}{2}}\,dx$; $\sup_{z\in\mathbb{R}} |\mathbb{P}(Z_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{\rho_3}{\sigma^3\sqrt{n}}$.

Proof. Set $Y_i := \frac{X_i-\mu}{\sigma}$, $\rho := \mathbb{E}\,|Y_1|^3 = \frac{\rho_3}{\sigma^3}$, and $W_n := n^{-\frac{1}{2}}\sum_{i=1}^n Y_i = Z_n$.
For $\varphi(t) := \mathbb{E}e^{itY_1}$, Taylor's formula with $\mathbb{E}Y_1 = 0$, $\mathbb{E}Y_1^2 = 1$, $\mathbb{E}\,|Y_1|^3 = \rho$ gives $\varphi(t) = 1 - \frac{t^2}{2} + O(\rho\,|t|^3)$ near 0.
Thus $\varphi_{W_n}(t) = [\varphi(\frac{t}{\sqrt{n}})]^n$ differs from $e^{-\frac{t^2}{2}}$ by $O(\rho n^{-\frac{1}{2}})$ in Esseen's smoothing integral.
Standard optimised smoothing bound gives $\sup_z |\mathbb{P}(W_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{\rho}{\sqrt{n}}$. Substitute $\rho = \frac{\rho_3}{\sigma^3}$, QED.
Structural Commentary: Berry–Esseen bounds the CLT error by standardised third absolute moment.
Key Insight: Normalise, compare characteristic functions, then apply Esseen smoothing. □

Theorem 3.2.13. *Multivariate Central Limit Theorem*
Let $(X_i)_{i\geq 1}$ be IID random vectors in $\mathbb{R}^k$ with $\mathbb{E}X_1 = \mu \in \mathbb{R}^k$ and $\mathrm{Cov}(X_1) = \Sigma$, and set $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$.
Then $\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N_k(0,\Sigma)$.

Proof. For $a \in \mathbb{R}^k$, set $Y_{i,a} := a^t(X_i - \mu)$ and $s_a^2 := a^t\Sigma a$.
Then $(Y_{i,a})_{i\geq 1}$ are IID, $\mathbb{E}Y_{1,a} = 0$, $\mathrm{Var}(Y_{1,a}) = s_a^2$, and $a^t\sqrt{n}(\overline{X}_n - \mu) = n^{-\frac{1}{2}}\sum_{i=1}^n Y_{i,a}$.
If $s_a^2 > 0$, the scalar CLT gives $a^t\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N(0,s_a^2)$; if $s_a^2 = 0$, then $\mathbb{E}Y_{1,a}^2 = 0$, so the same convergence holds with degenerate limit $N(0,0)$. Since $a^tZ \sim N(0,s_a^2)$ for $Z \sim N_k(0,\Sigma)$, by Cramer–Wold, QED.
Structural Commentary: Multivariate convergence is checked through one-dimensional projections.
Key Insight: Project first, apply the scalar CLT, then use Cramer–Wold. □

Theorem 3.2.14. *Delta Method*
Let $(Y_n)_{n\geq 1}$ be real-valued random variables, let $\mu \in \mathbb{R}$ and $\sigma > 0$, and assume $\sigma^{-1}\sqrt{n}(Y_n - \mu) \rightsquigarrow N(0,1)$.
If $g : \mathbb{R} \to \mathbb{R}$ is differentiable at μ and $g'(\mu) \neq 0$, then $(|g'(\mu)|\,\sigma)^{-1}\sqrt{n}(g(Y_n) - g(\mu)) \rightsquigarrow N(0,1)$.
Equivalently $g(Y_n) \approx N(g(\mu), \frac{(g'(\mu))^2\sigma^2}{n})$.

Proof. Set $A_n := \sigma^{-1}\sqrt{n}(Y_n - \mu)$ and define $\rho(\mu) := 0$, $\rho(y) := \frac{g(y)-g(\mu)}{y-\mu} - g'(\mu)$ for $y \neq \mu$.
Then $A_n \rightsquigarrow Z \sim N(0,1)$, $A_n = O_{\mathbb{P}}(1)$, and $Y_n - \mu = \frac{\sigma A_n}{\sqrt{n}} \xrightarrow{\mathbb{P}} 0$.
Differentiability gives $\rho(y) \to 0$ as $y \to \mu$, hence $\rho(Y_n) \xrightarrow{\mathbb{P}} 0$.
For all n, $g(Y_n) - g(\mu) = (g'(\mu) + \rho(Y_n))(Y_n - \mu)$. Thus, with $T_n := (|g'(\mu)|\,\sigma)^{-1}\sqrt{n}(g(Y_n) - g(\mu))$,
$T_n = \mathrm{sgn}(g'(\mu))A_n(1 + \frac{\rho(Y_n)}{g'(\mu)}) \rightsquigarrow \mathrm{sgn}(g'(\mu))Z \sim N(0,1)$ by Theorem 2.1.8.
Structural Commentary: The delta method is first-order Taylor expansion at the limit point.
Key Insight: The derivative rescales the fluctuation; the remainder is negligible in probability. □

Theorem 3.2.15. *Multivariate Delta Method*
Let $(Y_n)_{n\geq 1}$ be random vectors in $\mathbb{R}^k$ with $\sqrt{n}(Y_n - \mu) \rightsquigarrow N_k(0,\Sigma)$, and let $g : \mathbb{R}^k \to \mathbb{R}$ be differentiable at μ with gradient $\nabla g(\mu)$.
Then $\sqrt{n}(g(Y_n) - g(\mu)) \rightsquigarrow N(0, \nabla g(\mu)^t\Sigma\nabla g(\mu))$.

Proof. Set $A_n := \sqrt{n}(Y_n - \mu)$, so $A_n \rightsquigarrow N_k(0,\Sigma)$ and $Y_n \xrightarrow{\mathbb{P}} \mu$.
Differentiability gives $g(Y_n) - g(\mu) = \nabla g(\mu)^t(Y_n - \mu) + r(Y_n)\,\|Y_n - \mu\|$ with $r(Y_n) \xrightarrow{\mathbb{P}} 0$.
Since $A_n = O_{\mathbb{P}}(1)$, $\sqrt{n}r(Y_n)\,\|Y_n - \mu\| = r(Y_n)\,\|A_n\| \xrightarrow{\mathbb{P}} 0$.
Thus $\sqrt{n}(g(Y_n) - g(\mu)) = \nabla g(\mu)^tA_n + o_{\mathbb{P}}(1) \rightsquigarrow N(0, \nabla g(\mu)^t\Sigma\nabla g(\mu))$ by Theorem 2.1.8.
Structural Commentary: Multivariate delta method linearizes g by its gradient.
Key Insight: The gradient projection carries the Gaussian limit; the differentiability remainder vanishes. □

Proposition 3.2.16. *Delta Method*
If $\sqrt{n}(T_n - \theta) \Rightarrow N(0,\Sigma)$ and $g : \mathbb{R}^d \to \mathbb{R}^k$ is differentiable at θ with derivative $Dg(\theta)$,
then $\sqrt{n}(g(T_n) - g(\theta)) \Rightarrow N(0, Dg(\theta)\Sigma Dg(\theta)^\top)$.

Proof. Differentiability gives $g(T_n) - g(\theta) = Dg(\theta)(T_n - \theta) + r_n$ with $\frac{\|r_n\|}{\|T_n-\theta\|} \to 0$ in probability.
Since $\sqrt{n}(T_n - \theta) = O_P(1)$, $\sqrt{n}r_n = o_P(1)$; $\sqrt{n}(g(T_n) - g(\theta)) = Dg(\theta)\sqrt{n}(T_n - \theta) + o_P(1)$, by Slutsky QED.
Structural Commentary: The derivative transports the limiting normal fluctuation through g.
Key Insight: First-order linearisation determines asymptotic variance. □

Example 3.2.17. *Exponentiating a Sample Mean*
Let $(X_i)_{i=1}^n$ be IID with mean μ and variance $\sigma^2 < +\infty$, and set $W_n := e^{\overline{X}_n}$.
For $g(s) = e^s$, $g'(\mu) = e^\mu$, so the delta method gives $W_n \approx N(e^\mu, \frac{e^{2\mu}\sigma^2}{n})$.

Example 3.2.18. *Product and Ratio of Sample Means*

Let $X_i = (X_{1i}, X_{2i})^t$ be IID with mean $\mu = (\mu_1, \mu_2)^t$ and covariance $\Sigma = (\sigma_{ij})$, and set $\overline{X}_{jn} := n^{-1} \sum_{i=1}^{n} X_{ji}$.

For $g(x_1, x_2) = x_1 x_2$, $\nabla g(\mu) = (\mu_2, \mu_1)^t$, so $\sqrt{n}(\overline{X}_{1n}\overline{X}_{2n} - \mu_1\mu_2) \rightsquigarrow N(0, \mu_2^2\sigma_{11} + 2\mu_1\mu_2\sigma_{12} + \mu_1^2\sigma_{22})$.

If $\mu_2 \neq 0$, then for $h(x_1, x_2) = \frac{x_1}{x_2}$ and $a := \nabla h(\mu) = (\frac{1}{\mu_2}, -\frac{\mu_1}{\mu_2^2})^t$, $\sqrt{n}(\frac{\overline{X}_{1n}}{\overline{X}_{2n}} - \frac{\mu_1}{\mu_2}) \rightsquigarrow N(0, a^t\Sigma a)$.

4 Stochastic Processes

4.1 Process-Level Probability

Definition 4.1.1. *Stochastic Process, State Space, and Index Set*

(i) A *stochastic process* is a family $(X_t)_{t\in T}$ of random variables.
(ii) The set T is the *index set*, often representing time.
(iii) Each X_t takes values in a *state space* $\mathcal{X}$.

Typical choices are discrete time $T = \{0, 1, 2, \ldots\}$ and continuous time $T = [0, \infty)$.

Example 4.1.2. *IID Sequences as Processes*
An IID sequence $(X_n)_{n\geq 1}$ is a stochastic process indexed by $T = \{1, 2, \ldots\}$.

Example 4.1.3. *Discrete States and Discrete Time*
If $\mathcal{X} = \{\text{sunny}, \text{cloudy}\}$ and X_n is the weather on day n, then $(X_n)_{n\geq 1}$ is a stochastic process with discrete state space and discrete index set.

Example 4.1.4. *The Empirical CDF as a Process*

Let $X_1, \ldots, X_n$ be IID with CDF F on $[0, 1]$, and define $\widehat{F}_n(t) = n^{-1} \sum_{i=1}^{n} \mathbf{1}\{X_i \leq t\}$.

Then $(\widehat{F}_n(t))_{t\in[0,1]}$ is a stochastic process with continuous index set $[0, 1]$ and state space $[0, 1]$.

Remark 4.1.5. *Sample Paths*
A *sample path* is one realized function $t \mapsto X_t(\omega)$ of a stochastic process.
The state space and index set may each be discrete or continuous.

Proposition 4.1.6. *Finite-Dimensional Chain Rule*
Let $X_{1:n} = (X_1, \ldots, X_n)$ have joint mass function or density $f_{1:n}$.

At points where the needed marginals are positive, $f_{1:n}(x_{1:n}) = f_1(x_1) \prod_{i=2}^{n} f_{i|1:i-1}(x_i \mid x_{1:i-1})$.

Proof. For $i \geq 2$, the definition of conditional density gives $f_{1:i}(x_{1:i}) = f_{i|1:i-1}(x_i \mid x_{1:i-1}) f_{1:i-1}(x_{1:i-1})$.
Iterating this identity from $i = n$ down to 2 yields the product formula.
Structural Commentary: The joint law factors by revealing one coordinate at a time.
Key Insight: Conditional densities convert successive joint marginals into a telescoping product. □

4.2 Filtrations and Adapted Processes

4.3 Martingales in Discrete and Continuous Time

4.4 Markov Chains and Markov Processes

Definition 4.4.1. *Probability Vectors and Markov Matrices*
A vector $\mathbf{x} \in \mathbb{R}^n$ is a *probability vector* if its entries are nonnegative and sum to 1. Let $\mathbf{e} := (1, \ldots, 1)^t$. A matrix $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is a *Markov matrix* if $a_{ij} \geq 0$ for all i, j and $A\mathbf{e} = \mathbf{e}$; equivalently, every row sum is 1.

Lemma 4.4.2. *Basic Properties of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $1 \in \mathrm{spec}(A)$, and each A^p is Markov with entries in $[0, 1]$.

Proof. Since $A\mathbf{e} = \mathbf{e}$, 1 is an eigenvalue. Inductively, A^p has nonnegative entries and $A^p\mathbf{e} = \mathbf{e}$, so A^p is Markov. Each nonnegative row sums to 1, so all entries lie in $[0, 1]$.
Structural Commentary: Markov structure is closed under powers.
Key Insight: Nonnegative row sums of 1 bound every entry. □

Theorem 4.4.3. *Unit-Circle Spectrum of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $\rho(A) = 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Lemma 4.4.2 makes A power bounded. Theorem 3.3.7 gives $\rho(A) \leq 1$ and $\dim E_\lambda(A) = m_A(\lambda)$ on $|\lambda| = 1$. Since $A\mathbf{e} = \mathbf{e}$, $1 \in \mathrm{spec}(A)$, so $\rho(A) \geq 1$. Thus $\rho(A) = 1$.
Structural Commentary: Bounded Markov powers rule out unit-circle Jordan growth.
Key Insight: $A\mathbf{e} = \mathbf{e}$ forces the eigenvalue 1. □

Theorem 4.4.4. *Positive Markov Matrices Converge to a Stationary Distribution*
Let $n \geq 2$, $\mathbf{e} := (1, \ldots, 1)^t$, and $A = (a_{ij}) \in \mathrm{Mat}(n, n, \mathbb{R})$ satisfy $a_{ij} > 0$ and $A\mathbf{e} = \mathbf{e}$.
Let $\Delta_{n-1} := \{\mathbf{u} \in \mathbb{R}^n : \mathbf{u} \geq 0,\ \mathbf{e}^t\mathbf{u} = 1\}$. Then $\exists!\, \mathbf{x} \in \Delta_{n-1}$ s.t. $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x} > 0$, $\lim_{p\to\infty} A^p = \mathbf{e}\mathbf{x}^t$.

Proof. Let $A\mathbf{y} = \lambda\mathbf{y}$ with $\mathbf{y} \neq 0$ and $|\lambda| = 1$, and choose k with $|y_k| = \|\mathbf{y}\|_\infty$.

Since $\sum_j a_{kj} = 1$, then $\|\mathbf{y}\|_\infty = |\lambda y_k| = \left|\sum_{j=1}^{n} a_{kj} y_j\right| \leq \sum_{j=1}^{n} a_{kj} |y_j| \leq \|\mathbf{y}\|_\infty$. Thus all inequalities are equalities.
Since $a_{kj} > 0$, $|y_j| = \|\mathbf{y}\|_\infty$ $\forall j$; equality in the triangle inequality gives $\mathbf{y} = c\mathbf{e}$, $c \neq 0$.
Hence $\lambda = 1$, $E_1(A) = \mathrm{span}\{\mathbf{e}\}$, and $\mathrm{spec}(A) \cap \{z \in \mathbb{C} : |z| = 1\} = \{1\}$.
Theorem 4.4.3 makes 1 semisimple, so $A = S([1] \oplus J)S^{-1}$ with $S = [\mathbf{e}\ Y]$ and $\rho(J) < 1$.

First row of S^{-1} is $\mathbf{x}^t$. Then $\mathbf{x}^t A = \mathbf{x}^t$, $\mathbf{x}^t\mathbf{e} = 1$, and $A^p = S([1] \oplus J^p)S^{-1} \longrightarrow S([1] \oplus 0)S^{-1} = \mathbf{e}\mathbf{x}^t$.
Since $A^p \geq 0$, the limit gives $\mathbf{x} \geq 0$; since $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x}^t\mathbf{e} = 1$, and $A^t > 0$, we get $\mathbf{x} > 0$.
Finally, $\dim E_1(A^t) = n - \operatorname{rank}(A^t - I_n) = n - \operatorname{rank}(A - I_n) = \dim E_1(A) = 1$, so any $\mathbf{u} \in \Delta_{n-1}$ with $A^t\mathbf{u} = \mathbf{u}$ satisfies $\mathbf{u} = c\mathbf{x}$ and $\mathbf{e}^t\mathbf{u} = 1 = \mathbf{e}^t\mathbf{x}$, hence $\mathbf{u} = \mathbf{x}$.
Structural Commentary: Positivity leaves one simple unit-circle block; all other blocks decay.
Key Insight: The limit is the rank-one projection $\mathbf{e}\mathbf{x}^t$. □

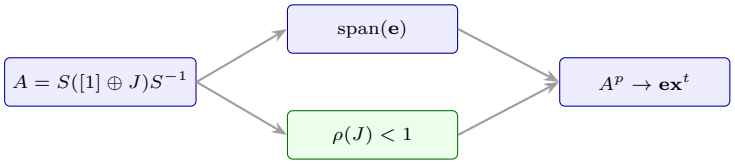

Figure 231: The line span($\mathbf{e}$) remains; all other Jordan directions decay, so $A^p \to \mathbf{e}\mathbf{x}^t$.

The preceding results are matrix facts. In probability language, a row-stochastic matrix becomes a transition matrix for a discrete-time chain.

Definition 4.4.5. *Markov Chain and Transition Matrix*
(i) Let $\mathcal{X}$ be countable and let $(X_n)_{n\geq 0}$ take values in $\mathcal{X}$.
(ii) Process is *Markov chain* if $\mathbb{P}(X_n = x \mid X_0, \ldots, X_{n-1}) = \mathbb{P}(X_n = x \mid X_{n-1})$ $\forall$ admissible n and $x \in \mathcal{X}$.
(iii) It is *homogeneous* if $p_{ij} := \mathbb{P}(X_{n+1} = j \mid X_n = i)$ is independent of n.
(iv) The matrix $P = [p_{ij}]$ is the *transition matrix*; $p_{ij} \geq 0$ and $\sum_{j\in\mathcal{X}} p_{ij} = 1$ for each i.

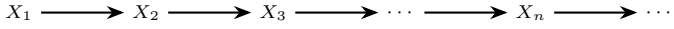

Figure 232: A Markov chain remembers the current state, not the full past.

Example 4.4.6. *Random Walk with Absorbing Barriers*
Let $\mathcal{X} = \{1, \ldots, N\}$ and $q = 1 - p$. The absorbing random walk has $p_{11} = p_{NN} = 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $2 \leq i \leq N - 1$, and all other transition probabilities equal to 0.

Definition 4.4.7. *n-Step Transition Probabilities*
For a homogeneous Markov chain, the *n-step transition probability* from i to j is
$p_{ij}(n) := \mathbb{P}(X_{m+n} = j \mid X_m = i)$; set $P_n = [p_{ij}(n)]$.

Theorem 4.4.8. *Chapman-Kolmogorov Equations*
Let $(X_r)_{r\geq 0}$ be homogeneous Markov chain on countable $\mathcal{X}$. Let $p_{ij}(r) := \mathbb{P}_i(X_r = j)$, $P_r = (p_{ij}(r))_{i,j\in\mathcal{X}}$.
$\forall i, j \in \mathcal{X}$ and $m, n \in \mathbb{N}_0$, $p_{ij}(m+n) = \sum_{k\in\mathcal{X}} p_{ik}(m)p_{kj}(n)$; equivalently, $P_{m+n} = P_m P_n$ and $P_n = P^n$ for $P = P_1$.

Proof. Fix $i, j \in \mathcal{X}$ and $m, n \in \mathbb{N}_0$. By total probability, the Markov property, and homogeneity,
$p_{ij}(m+n) = \mathbb{P}_i(X_{m+n} = j) = \sum_{k\in\mathcal{X}} \mathbb{P}_i(X_{m+n} = j \mid X_m = k)\mathbb{P}_i(X_m = k) = \sum_{k\in\mathcal{X}} p_{kj}(n)p_{ik}(m)$.
Thus $(P_m P_n)_{ij} = \sum_{k\in\mathcal{X}} p_{ik}(m)p_{kj}(n) = p_{ij}(m+n)$ for all i, j, so $P_{m+n} = P_m P_n$.
With $P_0 = I$ and $P = P_1$, induction gives $P_n = P^n$.
Structural Commentary: Transition over $m + n$ steps factors through the intermediate state after m steps.
Key Insight: Conditioning over intermediate states is matrix multiplication. □

Lemma 4.4.9. *Marginal Evolution*
Let $(X_n)_{n\geq 0}$ be a homogeneous Markov chain on countable $\mathcal{X}$ with $P = P_1 = (p_{ij})$, where $p_{ij}(n) := \mathbb{P}_i(X_n = j)$ and $P_n = (p_{ij}(n))$. Define row laws $\mu_n(j) := \mathbb{P}(X_n = j)$. Then $\mu_n = \mu_0 P^n$ for all $n \in \mathbb{N}_0$.

Proof. $\forall j \in \mathcal{X}$, by total probability $\mu_n(j) = \sum_{i\in\mathcal{X}} \mathbb{P}(X_n = j \mid X_0 = i)\mathbb{P}(X_0 = i) = \sum_{i\in\mathcal{X}} \mu_0(i)p_{ij}(n) = (\mu_0 P_n)_j$.
By Theorem 4.4.8, $P_n = P^n$, so $\mu_n(j) = (\mu_0 P^n)_j$ for every $j \in \mathcal{X}$; hence $\mu_n = \mu_0 P^n$.
Structural Commentary: Marginals evolve by multiplying the initial law by the transition semigroup.
Key Insight: Conditioning on X_0 turns distribution evolution into row-vector matrix multiplication. □

Definition 4.4.10. *Communication Classes*
(i) State i *reaches* j, written $i \to j$, if $p_{ij}(n) > 0$ for some $n \geq 0$.
(ii) States i, j *communicate*, written $i \leftrightarrow j$, if $i \to j$ and $j \to i$.
(iii) A *communication class* is an equivalence class under $\leftrightarrow$.
(iv) The chain is *irreducible* if all states lie in one communication class.
(v) A set C is *closed* if $p_{ij} = 0$ for all $i \in C$ and $j \notin C$; a singleton closed class is an *absorbing state*.

Proposition 4.4.11. *Communication Classes Partition the State Space*
Let $(X_n)_{n\geq 0}$ be a homogeneous Markov chain on countable $\mathcal{X}$ with n-step probabilities $p_{ij}(n)$.
Define $i \to j$ iff $\exists n \in \mathbb{N}_0$ such that $p_{ij}(n) > 0$, and $i \leftrightarrow j$ iff $i \to j$ and $j \to i$.
Then $\leftrightarrow$ is equivalence relation, and $\mathcal{X} = \bigsqcup_{C\in\mathcal{C}} C$ where $\mathcal{C} := \{[i]_\leftrightarrow : i \in \mathcal{X}\}$ is set of communication classes.

Proof. Since $p_{ii}(0) = 1$, $i \leftrightarrow i$ for all $i \in \mathcal{X}$; symmetry is immediate from the definition.
If $i \leftrightarrow j$ and $j \leftrightarrow k$, choose $m, n, r, s \in \mathbb{N}_0$ with $p_{ij}(m) > 0$, $p_{jk}(n) > 0$, $p_{kj}(r) > 0$, and $p_{ji}(s) > 0$.
By Theorem 4.4.8, $p_{ik}(m+n) \geq p_{ij}(m)p_{jk}(n) > 0$ and $p_{ki}(r+s) \geq p_{kj}(r)p_{ji}(s) > 0$, hence $i \leftrightarrow k$.
Thus $\leftrightarrow$ is reflexive, symmetric, and transitive.
Its equivalence classes are pairwise disjoint, cover $\mathcal{X}$, and are exactly the communication classes.
Structural Commentary: Communication classes are the equivalence classes generated by mutual reachability.
Key Insight: Positive transition paths concatenate by Chapman-Kolmogorov. □

Definition 4.4.12. *Recurrent and Transient States*

(i) Let $\tau_i^+ := \inf\{n \geq 1 : X_n = i\}$ be the first return time to i.
(ii) State i is *recurrent* or *persistent* if $\mathbb{P}_i(\tau_i^+ < \infty) = 1$.
(iii) State i is *transient* if $\mathbb{P}_i(\tau_i^+ < \infty) < 1$.

Theorem 4.4.13. *Recurrence Criterion*

(i) State i is recurrent iff $S_i := \sum_{n=0}^{\infty} p_{ii}(n) = \infty$.
(ii) State i is transient iff $S_i < \infty$.

Proof. Let $N_i := \sum_{n=0}^{\infty} \mathbf{1}\{X_n = i\}$ and $r_i := \mathbb{P}_i(\tau_i^+ < \infty)$. By Tonelli, $\mathbb{E}_i N_i = \sum_{n=0}^{\infty} p_{ii}(n) = S_i$.

By the Markov property after returns to i, $\mathbb{P}_i(N_i \geq k) = r_i^{k-1}$ for $k \geq 1$, hence $\mathbb{E}_i N_i = \sum_{k=1}^{\infty} r_i^{k-1}$.

(i) (⇒) If i is recurrent, then $r_i = 1$, so $S_i = \mathbb{E}_i N_i = \infty$.
(i) (⇐) If $S_i = \infty$, then $r_i < 1$ would give $\mathbb{E}_i N_i = (1 - r_i)^{-1} < \infty$, a contradiction; hence $r_i = 1$.
(ii) (⇒) If i is transient, then $r_i < 1$, so $S_i = \mathbb{E}_i N_i = (1 - r_i)^{-1} < \infty$.
(ii) (⇐) If $S_i < \infty$, then $r_i = 1$ would give $\mathbb{E}_i N_i = \infty$, a contradiction; hence $r_i < 1$.

Structural Commentary: Recurrence is detected by the expected total number of returns.
Key Insight: Visits to i repeat with geometric return probability r_i. □

Definition 4.4.14. *Period and Ergodicity*

(i) For recurrent i, it is *positive recurrent* if $\mathbb{E}_i \tau_i^+ < \infty$ and *null recurrent* otherwise.
(ii) The *period* of i is $d(i) := \gcd\{n \geq 1 : p_{ii}(n) > 0\}$.
(iii) State i is *aperiodic* if $d(i) = 1$.
(iv) State i is *ergodic* if it is recurrent, positive recurrent, and aperiodic.
(v) The chain is *ergodic* if every state is ergodic.

Definition 4.4.15. *Stationary and Limiting Distributions*

(i) A probability row vector π is *stationary* or *invariant* if $\pi = \pi P$.
(ii) A finite chain has *limiting distribution* π if $p_{ij}(n) \to \pi_j$ for all states i, j.

Equivalently, every row of P^n converges to π.

Remark 4.4.16. *Stationary Need Not Mean Convergent*
For the deterministic three-cycle on $\{1, 2, 3\}$, set $p_{12} = p_{23} = p_{31} = 1$ and all other $p_{ij} = 0$.
Then $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ satisfies $\pi P = \pi$, but $P^{n+3} = P^n$, so P^n does not converge.

Theorem 4.4.17. *Finite Ergodic Chain Convergence*
Let P be the transition matrix of a finite irreducible aperiodic Markov chain on $\mathcal{X}$.
There is a unique $\pi \in \Delta(\mathcal{X})$ with $\pi P = \pi$, and $P^n \to \mathbf{e}\pi$.
For every bounded $g : \mathcal{X} \to \mathbb{R}$, $N^{-1} \sum_{n=1}^{N} g(X_n) \to \pi(g) := \sum_{j \in \mathcal{X}} g(j)\pi_j$ almost surely.

Proof. Finite irreducibility and aperiodicity imply P is primitive: $P^m > 0$ for some $m \geq 1$.
Row-vector form of Theorem 4.4.4 on $A = P^m$, gives unique $\pi \in \Delta(\mathcal{X})$ with $\pi P^m = \pi$, $(P^m)^k \to \mathbf{e}\pi$.
Since $(\pi P)P^m = \pi P$ and $\pi P \in \Delta(\mathcal{X})$, uniqueness gives $\pi P = \pi$.
For $n = mk + r$ with $0 \leq r < m$, $P^n = (P^m)^k P^r \to \mathbf{e}\pi P^r = \mathbf{e}\pi$.
The finite-state Markov-chain strong law yields $N^{-1} \sum_{n=1}^{N} g(X_n) \to \pi(g)$ almost surely for bounded g.
Structural Commentary: Irreducibility gives one communicating class; aperiodicity removes cyclic obstruction.
Key Insight: Primitivity makes P^n converge to the rank-one projection $\mathbf{e}\pi$. □

Definition 4.4.18. *Detailed Balance*
Let $P = [p_{ij}]$ be a transition matrix on $\mathcal{X}$.
A probability row vector π satisfies *detailed balance* for P if $\pi_i p_{ij} = \pi_j p_{ji}$ for all $i, j \in \mathcal{X}$.

Theorem 4.4.19. *Detailed Balance Implies Stationarity*
Let $P = [p_{ij}]$ be row-stochastic on countable $\mathcal{X}$, and let $\pi \in \Delta(\mathcal{X})$. If $\pi_i p_{ij} = \pi_j p_{ji}$ $\forall i, j \in \mathcal{X}$, then $\pi P = \pi$.

Proof. $\forall j \in \mathcal{X}$, $(\pi P)_j = \sum_{i \in \mathcal{X}} \pi_i p_{ij} = \sum_{i \in \mathcal{X}} \pi_j p_{ji} = \pi_j \sum_{i \in \mathcal{X}} p_{ji} = \pi_j$, since row j of P sums to 1. QED.
Structural Commentary: Detailed balance gives pairwise cancellation of probability fluxes.
Key Insight: Reversibility is stronger than stationarity. □

Proposition 4.4.20. *Detailed Balance Implies Stationarity*
Let $\pi(dx) = f(x)\,dx$ and let P be a Markov kernel on state space X.
If $\pi(dx)P(x, dy) = \pi(dy)P(y, dx)$ as measures on $\mathsf{X} \times \mathsf{X}$, then $\pi P = \pi$.

Proof. For measurable $A \subseteq \mathsf{X}$, $\pi P(A) = \int P(x, A)\,\pi(dx) = \int \mathbf{1}_A(y)\,\pi(dx)P(x, dy)$.
By detailed balance, this equals $\int \mathbf{1}_A(y)\,\pi(dy)P(y, dx) = \int_A P(y, \mathsf{X})\,\pi(dy) = \pi(A)$, proving $\pi P = \pi$.
Structural Commentary: Detailed balance makes the joint law of consecutive states symmetric.
Key Insight: Marginalising the symmetric joint measure gives the same one-step density. □

Proposition 4.4.21. *Two-State Chain Limit*
Let $0 < a, b < 1$, $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$. *Set* $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$. *Then* $\pi P = \pi$, π *unique,* $P^n \to \mathbf{e}\pi = \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$.

Proof. Solving $\pi P = \pi$ with $\pi\mathbf{e} = 1$ gives $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$.
The characteristic polynomial is $(\lambda - 1)(\lambda - (1 - a - b))$, so $\operatorname{spec}(P) = \{1, 1 - a - b\}$ and $|1 - a - b| < 1$.
Thus stationary component is projected onto $\mathbf{e}\pi$, transverse component decays geometrically, giving $P^n \to \mathbf{e}\pi$.
Structural Commentary: The dynamics split into the stationary line and one transverse eigendirection.
Key Insight: The non-unit eigenvalue $1 - a - b$ controls convergence. □

4.5 Stationary Processes and Ergodicity

Definition 4.5.1. *Strict and Weak Stationarity*
Let $(X_t)_{t\in\mathbb{Z}}$ be a real-valued stochastic process.

(i) *Strict stationarity*: $\forall k \in \mathbb{N},\ t_1, \ldots, t_k, h \in \mathbb{Z}$, $(X_{t_1}, \ldots, X_{t_k}) \overset{\text{dist}}{=} (X_{t_1+h}, \ldots, X_{t_k+h})$.
(ii) *Weak stationarity*: $\exists \mu \in \mathbb{R}$ and $\gamma : \mathbb{Z} \to \mathbb{R}$ with $\forall t, h \in \mathbb{Z}$, $X_t \in L^2$, $\mathbb{E}X_t = \mu$, and $\operatorname{Cov}(X_t, X_{t-h}) = \gamma(h)$.

Key Insight: (i) plus L^2 implies (ii); (ii) fixes only μ and γ, not finite-dimensional laws.

4.6 Gaussian Processes

4.7 Point Processes and Renewal Structure

Counting processes model random event arrivals over time. The Poisson process is the basic model in which disjoint time intervals contribute independent counts and short intervals usually contain either zero or one event.

Definition 4.7.1. *Counting Process*
A *counting process* is a stochastic process $\{X(t) : t \geq 0\}$ with state space $\{0, 1, 2, \ldots\}$ such that $X(t)$ records the number of events observed up to and including time t.

Definition 4.7.2. *Poisson Process with Intensity Function*
Let $\lambda : [0, \infty) \to [0, \infty)$ be locally integrable.
A counting process $X = \{X(t) : t \geq 0\}$ is a *Poisson process* with *intensity function* λ if:

(i) $X(0) = 0$.
(ii) For $0 = t_0 < t_1 < \cdots < t_n$, the increments $\Delta_k X := X(t_k) - X(t_{k-1})$, $1 \leq k \leq n$, are independent.
(iii) For each $t \geq 0$, as $h \downarrow 0$, $\mathbb{P}(X(t + h) - X(t) = 1) = \lambda(t)h + o(h)$ and $\mathbb{P}(X(t + h) - X(t) \geq 2) = o(h)$.

Theorem 4.7.3. *Increment Distribution*
Let X *be a Poisson process with intensity* λ, *and set* $\Lambda(s, t) := \int_s^t \lambda(u)\,du$ *for* $0 \leq s \leq t$.
Then $X(t) - X(s) \sim \text{Poisson}(\Lambda(s, t))$. *In particular,* $X(t) \sim \text{Poisson}(\Lambda(0, t))$, *so* $\mathbb{E}X(t) = \operatorname{Var}(X(t)) = \Lambda(0, t)$.

Proof. Fix $s \geq 0$ and let $p_k(u) := \mathbb{P}(X(s + u) - X(s) = k)$ with $p_{-1} := 0$. By small-time conditions and independent increments, $p_k(u + h) = p_k(u)(1 - \lambda(s + u)h + o(h)) + p_{k-1}(u)\lambda(s + u)h + o(h)$.
Hence $p_k'(u) = -\lambda(s + u)p_k(u) + \lambda(s + u)p_{k-1}(u)$, with $p_0(0) = 1$ and $p_k(0) = 0$ for $k \geq 1$.
Writing $M(u) := \Lambda(s, s + u)$, the unique solution is $p_k(u) = e^{-M(u)}\frac{M(u)^k}{k!}$ by induction on k.
Taking $u = t - s$ yields $X(t) - X(s) \sim \text{Poisson}(\Lambda(s, t))$. Mean-variance identities follow from Poisson law.
Structural Commentary: The infinitesimal jump rules form a triangular system for the increment probabilities.
Key Insight: Integrating the local rate λ gives the Poisson mean over the interval. □

Definition 4.7.4. *Homogeneous Poisson Process*
If $\lambda(t) \equiv \lambda > 0$, then $X(t)$ is a *homogeneous Poisson process* with *rate* λ. In this case, $X(t) \sim \text{Poisson}(\lambda t)$.

Definition 4.7.5. *Waiting and Interarrival Times*
For a homogeneous Poisson process:

(i) Set $W_0 = 0$ and $W_n := \inf\{t \geq 0 : X(t) \geq n\}$ for $n \geq 1$; $W_1, W_2, \ldots$ are *waiting times*.
(ii) Set $S_n := W_{n+1} - W_n$ for $n \geq 0$; $S_0, S_1, \ldots$ are *sojourn* or *interarrival times*.

Theorem 4.7.6. *Poisson Interarrival Times*
Let X *be a homogeneous Poisson process with rate* $\lambda > 0$.
Then $S_0, S_1, \ldots \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, *with density* $f(s) = \lambda e^{-\lambda s}$ *for* $s \geq 0$.
For $n \geq 1$, $W_n = \sum_{r=0}^{n-1} S_r \sim \text{Gamma}(n, \lambda)$, *so* $\mathbb{E}W_n = \frac{n}{\lambda}$ *and* $\operatorname{Var}(W_n) = \frac{n}{\lambda^2}$.

Proof. For $t \geq 0$, $\mathbb{P}(S_0 > t) = \mathbb{P}(X(t) = 0) = e^{-\lambda t}$, so $S_0 \sim \text{Exp}(\lambda)$.
By independent stationary increments, $\mathbb{P}(S_n > t \mid \mathcal{F}_{W_n}) = \mathbb{P}(X(W_n + t) - X(W_n) = 0 \mid \mathcal{F}_{W_n}) = e^{-\lambda t}$.
Thus each $S_n \sim \text{Exp}(\lambda)$ and is independent of the past, so the S_n are IID.
Since $W_n = \sum_{r=0}^{n-1} S_r$, the Gamma law and its mean and variance follow.
Structural Commentary: Arrival times are cumulative sums of IID interarrival gaps.
Key Insight: No events in a length-t interval has probability $e^{-\lambda t}$. □

Proposition 4.7.7. *Conditional Binomial Count*
Let X be a homogeneous Poisson process with rate $\lambda > 0$.
For $s, t > 0$ and $n \in \mathbb{N}_0$, $X(t) \mid X(t+s) = n \sim \text{Binomial}\left(n, \frac{t}{t+s}\right)$.

Proof. Let $Y = X(t)$ and $Z = X(t+s) - X(t)$. Then $Y \sim \text{Poisson}(\lambda t)$, $Z \sim \text{Poisson}(\lambda s)$, and $Y \perp Z$.
For $0 \le k \le n$, $\mathbb{P}(Y = k \mid Y + Z = n) = \frac{\frac{(\lambda t)^k}{k!}\frac{(\lambda s)^{n-k}}{(n-k)!}}{\frac{(\lambda(t+s))^n}{n!}} = \binom{n}{k}\left(\frac{t}{t+s}\right)^k\left(\frac{s}{t+s}\right)^{n-k}$; QED.
Structural Commentary: Conditioning on the total allocates independent Poisson counts.
Key Insight: Independent Poisson counts become multinomial after conditioning on their sum. □

Remark 4.7.8. *Uniform Event-Time Diagnostic*
For a homogeneous Poisson process on $[0, T]$, conditional on $X(T) = n$, the event times $(W_1, \ldots, W_n)$ have the law of the order statistics of n IID Uniform$(0, T)$ variables.
Marked nonuniformity in the observed event times is evidence against homogeneity.

Proposition 4.7.9. *Odd Poisson Count*
Let X be a homogeneous Poisson process with rate $\lambda > 0$. For $t \ge 0$, $\mathbb{P}(X(t) \in 2\mathbb{N}_0 + 1) = \frac{1 - e^{-2\lambda t}}{2}$.

Proof. Let $\mu = \lambda t$. Since $X(t) \sim \text{Poisson}(\mu)$, $\mathbb{P}(X(t) \in 2\mathbb{N}_0 + 1) = e^{-\mu}\sum_{k=0}^{\infty}\frac{\mu^{2k+1}}{(2k+1)!} = e^{-\mu}\sinh\mu = \frac{1 - e^{-2\mu}}{2}$.
Structural Commentary: Split the Poisson mass by parity.
Key Insight: The odd terms of e^{μ} sum to $\sinh\mu$. □

Proposition 4.7.10. *Nearest Neighbor in a Planar Poisson Process*
Let Φ be a planar Poisson point process with intensity $\lambda > 0$, fix $x_0 \in \mathbb{R}^2$, and set $R := \inf_{Y \in \Phi}\|Y - x_0\|$.
Then for $t \ge 0$, $\mathbb{P}(R > t) = e^{-\lambda\pi t^2}$, and $\mathbb{E}R = \frac{1}{2\sqrt{\lambda}}$.

Proof. Let $B_t := B(x_0, t)$, $N_t := \#(\Phi \cap B_t)$. Then $N_t \sim \text{Poisson}(\lambda\pi t^2)$, $\{R > t\} = \{N_t = 0\}$,
so $\mathbb{P}(R > t) = e^{-\lambda\pi t^2}$. Since $R \ge 0$, $\mathbb{E}R = \int_0^{\infty}\mathbb{P}(R > t)\,dt = \int_0^{\infty} e^{-\lambda\pi t^2}\,dt = \frac{1}{2\sqrt{\lambda}}$.
Structural Commentary: Nearest-neighbor distance is encoded by empty disks around x_0.
Key Insight: The tail of R is the zero-count probability of a Poisson disk count. □

5 Stochastic Calculus and SDEs

5.1 Brownian Motion

5.2 Stochastic Integration

5.3 Itô Calculus

5.4 Stochastic Differential Equations

5.5 Change of Measure and Girsanov Theory

5.6 Diffusions and PDE Connections

5.7 Stochastic Flows and Advanced Dynamics

Part XVII
Dynamical and PDE Systems

1 Dynamical Systems and ODEs

1.1 ODE Foundations and Flows

Ordinary differential equations prescribe the evolution of an unknown curve through relations among the curve and its derivatives. Their order, dependence on time, and linearity determine the natural state variables and solution methods.

Definition 1.1.1. *Implicit Scalar ODE and Regular Point*
Let $k \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R}^{k+2}$ be open, and $F \in C(U, \mathbb{R})$.
For $I \subseteq \mathbb{R}$, $x \in C^k(I, \mathbb{R})$, set $J^k_x(t) := (t, x(t), x^{(1)}(t), \dots, x^{(k)}(t))$.

(i) $F(t, y_0, \dots, y_k) = 0$ is an *implicit scalar ODE written through order k*; its k-jet form is $F(J^k_x(t)) = 0$.
(ii) x is a *solution* on I iff $J^k_x(t) \in U$ and $F(J^k_x(t)) = 0$ $\forall t \in I$.
(iii) $F \in C^1(U, \mathbb{R})$; $p = (t_*, y_0^*, \dots, y_k^*) \in F^{-1}(0)$ is *regular w.r.t. highest derivative* iff $\partial_{y_k} F(p) \neq 0$.

Key Insight: $\partial_{y_k} F(p) \neq 0$ locally solves the relation for $x^{(k)}$.

Proposition 1.1.2. *Local Explicit Form of a Regular Implicit ODE*
Let $k \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R}^{k+2}$ be open, $F \in C^1(U, \mathbb{R})$, $u_* = (t_*, y_0^*, \dots, y_{k-1}^*)$, and $p = (u_*, y_k^*) \in U$ with $F(p) = 0$ and $\partial_{y_k} F(p) \neq 0$. Then $\exists$ open $V \subseteq \mathbb{R}^{k+1}$, $W \subseteq \mathbb{R}$ with $u_* \in V$, $y_k^* \in W$, and a unique $f \in C^1(V, \mathbb{R})$ such that $f(V) \subseteq W$, $f(u_*) = y_k^*$, $V \times W \subseteq U$, and:

(i) $F(u, v) = 0$ iff $v = f(u)$ $\forall (u, v) \in V \times W$.
(ii) If $x \in C^k(I, \mathbb{R})$ and $J^k_x(I) \subseteq V \times W$, then $F(J^k_x(t)) = 0$ $\forall t \in I$ iff $x^{(k)}(t) = f(t, x(t), \dots, x^{(k-1)}(t))$ $\forall t \in I$.

Proof. Let $u = (t, y_0, \dots, y_{k-1})$, $v = y_k$, $\widehat{U} := \{(v, u) : (u, v) \in U\}$, and $\widehat{F}(v, u) := F(u, v)$.
Since $\partial_v \widehat{F}(y_k^*, u_*) = \partial_{y_k} F(p) \neq 0$, by Theorem 4.2.3, there are W, V and a unique $f \in C^1(V, \mathbb{R})$ with $W \times V \subseteq \widehat{U}$, $f(V) \subseteq W$, $f(u_*) = y_k^*$, and $\widehat{F}(v, u) = 0$ iff $v = f(u)$.

(i) For $(u, v) \in V \times W$, $F(u, v) = \widehat{F}(v, u)$, so $F(u, v) = 0$ iff $v = f(u)$.
(ii) Set $u_x(t) := (t, x(t), \dots, x^{(k-1)}(t))$ and $v_x(t) := x^{(k)}(t)$. Since $(u_x(t), v_x(t)) \in V \times W$ $\forall t \in I$, (i) gives $F(J^k_x(t)) = 0$ iff $v_x(t) = f(u_x(t))$ $\forall t \in I$.

Structural Commentary: IFT solves $F(u, y_k) = 0$ for y_k.
Key Insight: $\partial_{y_k} F(p) \neq 0$ turns $F = 0$ into $y_k = f(u)$ locally. □

Definition 1.1.3. *Explicit ODE Systems and Classes*
Let $k, n \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R} \times (\mathbb{R}^n)^k$ open, and $f : U \to \mathbb{R}^n$.
Use $X_x(t) := (x(t), x^{(1)}(t), \dots, x^{(k-1)}(t))$, $\xi = (\xi_0, \dots, \xi_{k-1})$, and $\pi(U) := \{\xi \in (\mathbb{R}^n)^k : \exists t \in \mathbb{R},\ (t, \xi) \in U\}$.

(i) $x^{(k)}(t) = f(t, X_x(t))$ is an *explicit ODE system of order k*.
(ii) For an interval $I \subseteq \mathbb{R}$, $x \in C^k(I, \mathbb{R}^n)$ is a *solution* iff $\forall t \in I$, $(t, X_x(t)) \in U$ and $x^{(k)}(t) = f(t, X_x(t))$.
(iii) The system is *autonomous* iff $\exists F : \pi(U) \to \mathbb{R}^n$ with $f(t, \xi) = F(\xi)$ $\forall (t, \xi) \in U$.
(iv) On an interval I, the system is *linear* iff $\exists g \in C(I, \mathbb{R}^n)$ and $A_j \in C(I, \mathbb{R}^{n \times n})$, $0 \leq j < k$,
with $f(t, \xi) = g(t) + \sum_{j=0}^{k-1} A_j(t) \xi_j$ $\forall (t, \xi) \in U \cap (I \times (\mathbb{R}^n)^k)$.
(v) A linear system on I is *homogeneous* iff (iv) holds with $g = 0$.
(vi) A solution $x : I \to \mathbb{R}^n$ is *global* iff $I = \mathbb{R}$.
(vii) A solution $x : I \to \mathbb{R}^n$ is *maximal* iff $\nexists$ solution $\bar{x} : \bar{I} \to \mathbb{R}^n$ with $I \subsetneq \bar{I}$ and $\bar{x}|_I = x$.

Key Insight: The state lift X_x has dimension kn.

Proposition 1.1.4. *Reduction to First Order*
Let $k, n \in \mathbb{N}$, $k \geq 1$, $U \subseteq \mathbb{R} \times (\mathbb{R}^n)^k$ open, and $f : U \to \mathbb{R}^n$.
For $y = (y_1, \dots, y_k)$, define $G(t, y) := (y_2, \dots, y_k, f(t, y))$.
For an interval $I \subseteq \mathbb{R}$ and $x \in C^k(I, \mathbb{R}^n)$, set $y := (x, x^{(1)}, \dots, x^{(k-1)})$ and assume $(t, y(t)) \in U$ $\forall t \in I$.
Then $y \in C^1(I, (\mathbb{R}^n)^k)$, and $x^{(k)}(t) = f(t, y(t))$ $\forall t \in I$ iff $y'(t) = G(t, y(t))$ $\forall t \in I$.

Proof.
(⇒) Since $x \in C^k(I, \mathbb{R}^n)$, then $y \in C^1(I, (\mathbb{R}^n)^k)$. The identities $y_j' = x^{(j)} = y_{j+1}$, $1 \leq j < k$, and $y_k' = x^{(k)} = f(t, y(t))$ imply $y' = G(\,\cdot\,, y)$ on I.
(⇐) Since $y_k = x^{(k-1)}$, the kth coordinate gives $x^{(k)}(t) = y_k'(t) = G_k(t, y(t)) = f(t, y(t))$ $\forall t \in I$.

Structural Commentary: Coordinates shift; the last coordinate carries f.
Key Insight: Adjoin derivatives to get first order. □

Proposition 1.1.5. *Autonomization by Adjoining Time*
Let $m \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^m$ be open, $f : U \to \mathbb{R}^m$, and $\widetilde{f}(s, \xi) := (1, f(s, \xi))$. For an interval $I \subseteq \mathbb{R}$:

(i) If $y \in C^1(I, \mathbb{R}^m)$, $(t, y(t)) \in U$, and $y' = f(t, y)$, then $z(t) := (t, y(t))$ solves $z' = \widetilde{f}(z)$.
(ii) If $z = (s, y) \in C^1(I, \mathbb{R}^{m+1})$, $z(I) \subseteq U$, $z' = \widetilde{f}(z)$, and $s(t_0) = t_0$ for some $t_0 \in I$, then $s = \mathrm{id}_I$ and $y'(t) = f(t, y(t))$ on I.

Proof.
(i) For $z = (t, y)$, $z' = (1, y') = (1, f(t, y)) = \widetilde{f}(t, y) = \widetilde{f}(z)$.
(ii) Since $(s', y') = z' = \widetilde{f}(s, y) = (1, f(s, y))$, $s' = 1$ and $y' = f(s, y)$.
With $s(t_0) = t_0$, $s = \mathrm{id}_I$; hence $y'(t) = f(t, y(t))$.

Structural Commentary: The clock coordinate has $s' = 1$.
Key Insight: $y' = f(t, y)$ becomes $z' = \widetilde{f}(z)$ with $z = (t, y)$. □

Example 1.1.6. *Newtonian Motion as a First-Order System*
Let $q : \mathbb{R} \to \mathbb{R}^3$ be the position of a particle of mass $m > 0$, let $v := q'$, and let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be a position-dependent force. Newton's equation $mq''(t) = F(q(t))$ becomes $(q, v)' = (v, m^{-1}F(q))$ by Proposition 1.1.4. For uniform gravity $F(q) = -mge_3$, where $e_3 := (0, 0, 1)^{\mathsf{T}}$ and $g > 0$, the initial state $(q(0), v(0)) = (q_0, v_0)$ gives $q(t) = q_0 + v_0 t - \frac{g}{2}e_3 t^2$. Thus position and velocity together determine the mechanical state.

Proposition 1.1.7. *Scalar Autonomous Separation*
Let $D \subseteq \mathbb{R}$ be open, $f \in C(D, \mathbb{R})$, $D_f := \{x \in D : f(x) \neq 0\}$, and $x_0 \in D_f$.
Let C be the component of D_f containing x_0, and set $H(x) := \int_{x_0}^{x} \frac{du}{f(u)}$, $K := H(C)$. Then:

(i) K is an open interval and $H : C \to K$ is a differentiable bijection.
(ii) $\phi := H^{-1} : K \to C$ is the unique C-valued solution of $x' = f(x)$ with $x(0) = x_0$.
(iii) K is maximal among intervals of C-valued solutions through x_0, and ϕ is global iff $K = \mathbb{R}$.

Proof.

(i) Since D_f is open, C is an open interval. By Theorem 2.2.1, $H' = 1/f \neq 0$ on C.
Since f has constant sign on C by Theorem 5.1.41, H is strictly monotone.
Hence $H : C \to K = H(C)$ is bijective and K is an open interval.
(ii) By Corollary 1.1.20, $\phi \in C^1(K, C)$ and $\phi' = 1/(H' \circ \phi) = f \circ \phi$. Since $H(x_0) = 0$, $\phi(0) = x_0$.
If $u : L \to C$ solves $u' = f(u)$ with $0 \in L$ and $u(0) = x_0$, then $(H \circ u)' = H'(u)u' = 1$ and $(H \circ u)(0) = 0$.
Hence $H(u(t)) = t$, so $L \subseteq K$ and $u = \phi|_L$.
(iii) If $\bar{\phi} : \bar{K} \to C$ solves $x' = f(x)$, $K \subseteq \bar{K}$, and $\bar{\phi}|_K = \phi$, then (ii) gives $\bar{K} \subseteq K$; hence $\bar{K} = K$.
Thus K is maximal. Since $\operatorname{dom}\phi = K$, ϕ is global iff $K = \mathbb{R}$.

Structural Commentary: Monotone coordinate, inverse trajectory.
Key Insight: Time is accumulated reciprocal speed. □

Method 1.1.8. *Separation of Variables*
Let $D \subseteq \mathbb{R}$ be open, $J \subseteq \mathbb{R}$ open interval, $f \in C(D, \mathbb{R})$, $g \in C(J, \mathbb{R})$, $t_0 \in J$, $x_0 \in D_f := \{x \in D : f(x) \neq 0\}$.
Let $C \ni x_0$ be the component of D_f and $H(x) := \int_{x_0}^{x} \frac{du}{f(u)}$.
Since f has constant sign on C and $H' = 1/f$, $H : C \to H(C)$ is a differentiable bijection.

(i) Set $G(t) := \int_{t_0}^{t} g(s)\,ds$ and let L be the component of $\{t \in J : G(t) \in H(C)\}$ containing t_0.
(ii) Output $x := H^{-1} \circ G$ on L; then $x' = f(x)g$ on L and $x(t_0) = x_0$.

Example 1.1.9. *Global Growth, Finite-Time Blow-Up, and Nonuniqueness*
The separation coordinate distinguishes three basic behaviors.

(i) For $x' = x$ and $x(0) = x_0 > 0$, $H(x) = \log\left(\frac{x}{x_0}\right)$ and $x(t) = x_0 e^t$ on $\mathbb{R}$.
(ii) For $x' = x^2$ and $x(0) = x_0 > 0$, $H(x) = x_0^{-1} - x^{-1}$ and $x(t) = \frac{x_0}{1 - x_0 t}$ on $(-\infty, x_0^{-1})$; $x(t) \to +\infty$ as $t \uparrow x_0^{-1}$.
(iii) For $x' = \sqrt{|x|}$ and $x(0) = 0$, $a \in [0, +\infty)$ determines $x_a(t) := \begin{cases} 0, & t \leq a, \\ \frac{(t-a)^2}{4}, & t \geq a. \end{cases}$ Then $x_a \in C^1(\mathbb{R})$, $x_a' = \sqrt{|x_a|}$, $x_a(0) = 0$, and if $0 \leq a < b$, then $x_a = x_b$ on $(-\infty, a]$ while $x_a > x_b$ on $(a, b]$; hence the IVP is nonunique.

First-order equations can also be rewritten as fixed-point problems on complete function spaces.

Definition 1.1.10. *Scalar Initial-Value Problem*
Let $O \subseteq \mathbb{R}^2$ be open, $(x_0, y_0) \in O$, $g : O \to \mathbb{R}$, and $\mathrm{IVP}(O, g, x_0, y_0) := \{y' = g(x, y),\ y(x_0) = y_0\}$.
For $J \subseteq \mathbb{R}$ with $x_0 \in J$, $f : J \to \mathbb{R}$ solves $\mathrm{IVP}(O, g, x_0, y_0)$ iff

(i) J is open and f is differentiable.
(ii) $\Gamma_f := \{(x, f(x)) : x \in J\} \subseteq O$.
(iii) $f'(x) = g(x, f(x))\ \forall x \in J$, and $f(x_0) = y_0$.

Structural Commentary: $\Gamma_f \subseteq O$ makes $g(\,\cdot\,, f)$ defined; $f(x_0) = y_0$ selects the curve.

Proposition 1.1.11. *Differential and Integral Forms*
Let $O \subseteq \mathbb{R}^2$ be open, $(x_0, y_0) \in O$, $g \in C(O, \mathbb{R})$, $J \subseteq \mathbb{R}$ open with $x_0 \in J$, and $f \in C(J, \mathbb{R})$.
Let $\Gamma_f := \{(x, f(x)) : x \in J\} \subseteq O$. Then f solves $\mathrm{IVP}(O, g, x_0, y_0)$ iff $f(x) = y_0 + \int_{x_0}^{x} g(t, f(t))\,dt\ \forall x \in J$.

Proof.

($\Rightarrow$) If $f'(x) = g(x, f(x))$ on J, $f(x_0) = y_0$, by Theorem 2.2.2, $f(x) - y_0 = \int_{x_0}^{x} f'(t)\,dt = \int_{x_0}^{x} g(t, f(t))\,dt\ \forall x \in J$.
($\Leftarrow$) If $f(x) = y_0 + \int_{x_0}^{x} g(t, f(t))\,dt\ \forall x \in J$, then $g(\,\cdot\,, f(\,\cdot\,))$ is continuous on J.
By Theorem 2.2.1, $f'(x) = g(x, f(x))$ on J. Evaluating at $x = x_0$ gives $f(x_0) = y_0$.

Structural Commentary: Differentiation and integration are inverse along the continuous graph.
Key Insight: The IVP is equivalent to its Volterra integral equation. □

Theorem 1.1.12. *Picard Local Existence Theorem*
Let $O \subseteq \mathbb{R}^2$ be open, $(x_0, y_0) \in O$, and $g \in C(O, \mathbb{R})$.
Assume $|g(x, y_1) - g(x, y_2)| \leq M\,|y_1 - y_2|$ for some $M > 0$ whenever $(x, y_1), (x, y_2) \in O$.
Then $\exists a, b, \ell > 0$ with $Q := [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b] \subseteq O$ and $J_\ell := (x_0 - \ell, x_0 + \ell)$ such that:

(i) The IVP $y' = g(x, y)$, $y(x_0) = y_0$, has a differentiable solution $f : J_\ell \to \mathbb{R}$ with $\Gamma_f \subseteq Q$.
(ii) If $h : J_\ell \to \mathbb{R}$ is a differentiable solution with $\Gamma_h \subseteq Q$, then $h = f$.

Proof.

(i) Choose $a, b > 0$ with $Q := [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b] \subseteq O$.
Let $K := \sup_Q |g| < +\infty$, and choose $0 < \ell \le a$ with $\ell K \le b$ and $\ell M < 1$.
Put $I_\ell := [x_0 - \ell, x_0 + \ell]$, $\mathcal{X}_\ell := \{u \in C(I_\ell, \mathbb{R}) : \|u - y_0\|_\infty \le b\}$, and $(Tu)(x) := y_0 + \int_{x_0}^{x} g(t, u(t))\, dt$.
The space $\mathcal{X}_\ell$ is complete, and for $u, v \in \mathcal{X}_\ell$, $\|Tu - y_0\|_\infty \le \ell K \le b$ and $\|Tu - Tv\|_\infty \le \ell M \|u - v\|_\infty$.
Thus $T : \mathcal{X}_\ell \to \mathcal{X}_\ell$ is a contraction; by Theorem 4.5.3, $\exists! u_* \in \mathcal{X}_\ell$ with $Tu_* = u_*$.
For $f := u_*|_{J_\ell}$, $\Gamma_f \subseteq Q$; by Proposition 1.1.11, $f' = g(\cdot, f)$ and $f(x_0) = y_0$.

(ii) Let $h : J_\ell \to \mathbb{R}$ solve the IVP with $\Gamma_h \subseteq Q$. For $0 < r < \ell$, put $I_r := [x_0 - r, x_0 + r]$,
$\mathcal{X}_r := \{u \in C(I_r, \mathbb{R}) : \|u - y_0\|_\infty \le b\}$, and $(T_r u)(x) := y_0 + \int_{x_0}^{x} g(t, u(t))\, dt$.
Since $rM < 1$, T_r is a contraction. By Proposition 1.1.11, $T_r(h|_{I_r}) = h|_{I_r}$ and $T_r(u_*|_{I_r}) = u_*|_{I_r}$.
Uniqueness gives $h = f$ on I_r. Hence $h = f$ on $\bigcup_{0<r<\ell} I_r = J_\ell$.

Structural Commentary: Box control plus short-time contraction.
Key Insight: Picard iteration is Banach iteration. □

Method 1.1.13. *Picard Iteration*
Let $Q := [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b] \subseteq O$, assume $|g(x, y)| \le K$ for $(x, y) \in Q$,
$|g(x, y_1) - g(x, y_2)| \le M |y_1 - y_2|$ whenever $(x, y_1), (x, y_2) \in Q$. Let $\ell > 0$ with $\ell \le a$, $\ell K \le b$, and $\ell M < 1$.
Set $I_\ell := [x_0 - \ell, x_0 + \ell]$, $J_\ell := (x_0 - \ell, x_0 + \ell)$, $\mathcal{X}_\ell := \{u \in C(I_\ell, \mathbb{R}) : \|u - y_0\|_\infty \le b\}$,
$(Tu)(x) := y_0 + \int_{x_0}^{x} g(t, u(t))\, dt$. Then

(i) Initialize $f_0(x) \equiv y_0$ on I_ℓ.
(ii) Iterate $f_{n+1} := Tf_n$, i.e. $f_{n+1}(x) = y_0 + \int_{x_0}^{x} g(t, f_n(t))\, dt$ for $x \in I_\ell$.
(iii) Let $f_\infty := \lim_{n\to\infty} f_n$ uniformly on I_ℓ, and output $f := f_\infty|_{J_\ell}$.

By Theorem 4.5.3, the limit exists and is the unique fixed point of T; Proposition 1.1.11 identifies its restriction as the unique local solution with graph in Q.

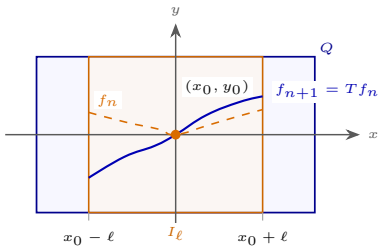

Figure 233: Picard setup: choose ℓ with $\ell K \le b$ so $T(\mathcal{X}_\ell) \subseteq \mathcal{X}_\ell$ and graphs stay in Q.

The same fixed-point formulation applies to vector fields and supports quantitative iteration estimates.

Definition 1.1.14. *Vector IVP and Local State Lipschitz Condition*
Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f : U \to \mathbb{R}^n$, and $(t_0, x_0) \in U$.

(i) The *initial-value problem* is $x' = f(t, x)$, $x(t_0) = x_0$; a solution on an interval $I \ni t_0$ is $x \in C^1(I, \mathbb{R}^n)$ with $(t, x(t)) \in U$ and $x'(t) = f(t, x(t))$ $\forall t \in I$.
(ii) f is *locally Lipschitz in the state, uniformly in time* iff $\forall$ compact $V \subseteq U$, $\exists L_V < +\infty$ such that $\|f(t, x) - f(t, y)\| \le L_V \|x - y\|$ whenever $(t, x), (t, y) \in V$.

Key Insight: State-Lipschitz control makes the integral operator asymptotically contractive under iteration.

Lemma 1.1.15. *C^1 Vector Fields Are Locally State-Lipschitz*
Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, and $f \in C^1(U, \mathbb{R}^n)$.
$\forall$ compact $K \subseteq U$, $\exists L_K < +\infty$ such that $\|f(t, x) - f(t, y)\| \le L_K \|x - y\|$ whenever $(t, x), (t, y) \in K$.

Proof. Fix compact $K \subseteq U$. If $K = \emptyset$, take $L_K = 0$. Otherwise choose $r > 0$, $K_r := \{z : \operatorname{dist}(z, K) \le r\} \subseteq U$.
Set $A := \max_{K_r} \|D_x f\|$, $M := \max_K \|f\|$, and $L_K := \max\{A, \frac{2M}{r}\}$. For $(t, x), (t, y) \in K$, put $\rho := \|x - y\|$.
If $\rho < r$, then $\sigma(s) := (t, x + s(y - x)) \in K_r$ for $s \in [0, 1]$; by Theorem 3.2.1, $\|f(t, x) - f(t, y)\| \le A\rho$.
If $\rho \ge r$, then $\|f(t, x) - f(t, y)\| \le 2M \le \frac{2M}{r}\rho$. Thus $\|f(t, x) - f(t, y)\| \le L_K \rho$.
Structural Commentary: Nearby pairs use $D_x f$; separated pairs use boundedness.
Key Insight: Compactness yields one state-Lipschitz constant. □

Theorem 1.1.16. *Quantitative Vector Picard–Lindelöf Theorem*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_0, x_0) \in U$. Choose $T, \delta > 0$ and set $V := [t_0, t_0 + T] \times \overline{B}_\delta(x_0) \subseteq U$.
Assume $L \ge 0$ and $\|f(t, x) - f(t, y)\| \le L \|x - y\|$ $\forall (t, x), (t, y) \in V$.
Set $M := \max_V \|f\|$, $T_0 := \min\{T, \delta/M\}$ with $\delta/0 := +\infty$, and $I := [t_0, t_0 + T_0]$.
Let $X := C(I, \mathbb{R}^n)$, $\|u\|_\infty := \max_{t\in I} \|u(t)\|$, and $\mathcal{C} := \{u \in X : \|u - x_0\|_\infty \le \delta\}$.
Define $Ku := x_0 + \int_{t_0}^{(\cdot)} f(s, u(s))\, ds$, $u_0 \equiv x_0$, and $u_{m+1} := Ku_m$.

(i) $(u_m) \subseteq \mathcal{C}$ and $u_m \to \phi$ uniformly for some $\phi \in \mathcal{C}$ satisfying $K\phi = \phi$; hence $\phi \in C^1(I, \mathbb{R}^n)$ solves the IVP.
(ii) ϕ is the unique solution on I whose graph lies in V.
(iii) If $M_0 := \max_{t\in I} \|f(t, x_0)\|$, then $\|\phi - u_m\|_\infty \le T_0 M_0 e^{LT_0} \frac{(LT_0)^m}{m!}$ $\forall m \ge 1$.

Proof.

(i) For $u \in \mathcal{C}$, $\|Ku - x_0\|_\infty \le MT_0 \le \delta$; hence $K(\mathcal{C}) \subseteq \mathcal{C}$.
For $u, v \in \mathcal{C}$, induction on $m \ge 1$ yields $\|K^m u(t) - K^m v(t)\| \le \frac{L^m (t - t_0)^m}{m!} \|u - v\|_\infty$ for $t \in I$.

Hence $\|u_{m+1}-u_m\|_\infty \le \frac{(LT_0)^m}{m!}\|Ku_0-u_0\|_\infty$, so (u_m) is uniformly Cauchy.
By Theorem 8.1.1, X is complete; since $\mathcal{C}$ is closed, $u_m \to \phi \in \mathcal{C}$.
The case $m=1$ implies K is continuous on $\mathcal{C}$, so $K\phi = \lim_{m\to\infty} Ku_m = \lim_{m\to\infty} u_{m+1} = \phi$.
Thus $\phi(t) = x_0 + \int_{t_0}^t f(s,\phi(s))\,ds$, so $\phi \in C^1(I,\mathbb{R}^n)$, $\phi(t_0)=x_0$, and $\phi' = f(\cdot,\phi)$.

(ii) If $Ku=u$ and $Kv=v$ in $\mathcal{C}$, then $\|u-v\|_\infty = \|K^m u - K^m v\|_\infty \le \frac{(LT_0)^m}{m!}\|u-v\|_\infty \ \forall m \ge 1$.
Choose m with $(LT_0)^m/m! < 1$; hence $u=v$.
If y solves the IVP on I with graph in V, then $y \in \mathcal{C}$ and integration implies $Ky = y$; hence $y=\phi$.

(iii) Since $\|Ku_0-u_0\|_\infty \le T_0 M_0$, (i) implies $\|\phi-u_m\|_\infty \le \sum_{j=m}^{\infty}\|u_{j+1}-u_j\|_\infty \le T_0M_0\sum_{j=m}^{\infty}\frac{(LT_0)^j}{j!}$.

For $a \ge 0$, $(m+k)! \ge m!k!$ implies $\sum_{j=m}^{\infty}\frac{a^j}{j!} \le \frac{a^m}{m!}e^a$; set $a = LT_0$.

Structural Commentary: Factorial iterate bounds replace the short-time contraction restriction.
Key Insight: Iterates contract after sufficiently many integrations. □

Theorem 1.1.17. *Picard–Lindelöf with an Integrable Lipschitz Bound*
Let $U \subseteq \mathbb{R}\times\mathbb{R}^n$ be open, $f \in C(U,\mathbb{R}^n)$, $(t_0,x_0)\in U$, and choose $T,\delta>0$ with $V := [t_0,t_0+T]\times\overline{B}_\delta(x_0) \subseteq U$.
Set $b(t) := \max_{x\in\overline{B}_\delta(x_0)}\|f(t,x)\|$. Let $L:[t_0,t_0+T]\to[0,+\infty)$ be integrable,
satisfy $\|f(t,x)-f(t,y)\| \le L(t)\|x-y\|$ for $(t,x),(t,y)\in V$. Choose $0<T_0\le T$ with $\int_{t_0}^{t_0+T_0} b(s)\,ds \le \delta$.
Put $I := [t_0,t_0+T_0]$, $A(t) := \int_{t_0}^t L(s)\,ds$, $A := A(t_0+T_0)$, and $B_0 := \int_{t_0}^{t_0+T_0}\|f(s,x_0)\|\,ds$.
Let $\mathcal{C} := \{u\in C(I,\mathbb{R}^n) : \|u-x_0\|_\infty \le \delta\}$, $(Ku)(t) := x_0 + \int_{t_0}^t f(s,u(s))\,ds$, $u_0 \equiv x_0$, and $u_{m+1} := Ku_m$. Then:
(i) $K(\mathcal{C}) \subseteq \mathcal{C}$ and $u_m \to \phi \in \mathcal{C}$ uniformly, where $K\phi=\phi$.
(ii) $\phi \in C^1(I,\mathbb{R}^n)$ is the unique solution of $x' = f(t,x)$, $x(t_0)=x_0$, whose graph lies in V.
(iii) $\|\phi-u_m\|_\infty \le B_0 e^A \frac{A^m}{m!}$ for $m \ge 0$.

Proof. For $u\in\mathcal{C}$ and $t\in I$, $\|Ku(t)-x_0\| \le \int_{t_0}^t b(s)\,ds \le \delta$; hence $K(\mathcal{C})\subseteq\mathcal{C}$.
Induction and $dA = L\,ds$ give $\|K^m u(t) - K^m v(t)\| \le \frac{A(t)^m}{m!}\|u-v\|_\infty$ for $u,v\in\mathcal{C}$ and $m\ge 1$.
With $u = Ku_0$ and $v=u_0$, $\|u_{j+1}-u_j\|_\infty \le B_0\frac{A^j}{j!}$ for $j\ge 0$. Thus (u_m) is uniformly Cauchy.

(i) By Theorem 8.1.1, $C(I,\mathbb{R}^n)$ is complete; closedness of $\mathcal{C}$ gives $u_m\to\phi\in\mathcal{C}$.
The case $m=1$ makes K continuous, so $K\phi = \lim_{m\to\infty}Ku_m = \lim_{m\to\infty}u_{m+1} = \phi$.

(ii) By Theorem 2.2.1, the fixed-point identity implies $\phi\in C^1(I,\mathbb{R}^n)$, $\phi(t_0)=x_0$, and $\phi' = f(\cdot,\phi)$.
If y is a solution with graph in V, then $y\in\mathcal{C}$ and $Ky=y$.
If $Ku=u$ and $Kv=v$, then $\|u-v\|_\infty \le \frac{A^m}{m!}\|u-v\|_\infty$ for $m\ge 1$; choose m with $\frac{A^m}{m!}<1$, hence $u=v$.

(iii) By telescoping, $\|\phi-u_m\|_\infty \le B_0\sum_{j=m}^{\infty}\frac{A^j}{j!}$. Since $(m+r)! \ge m!r!$, $\sum_{j=m}^{\infty}\frac{A^j}{j!} \le e^A\frac{A^m}{m!}$.

Structural Commentary: Integrated Lipschitz mass gives factorial iterate bounds.
Key Insight: Repeated integration contracts eventually. □

Remark 1.1.18. *Carathéodory Solutions*
Let $I\subseteq\mathbb{R}$, $t_0\in I$, $x_0\in\mathbb{R}^n$, $\delta>0$, $V := I\times\overline{B}_\delta(x_0)$, and $f: V\to\mathbb{R}^n$.
Suppose $f(\cdot,x)$ is measurable, $f(t,\cdot)$ is continuous for a.e. t, and $\exists b, L\in L^1(I,[0,+\infty))$ with $\|f(t,x)\|\le b(t)$ and $\|f(t,x)-f(t,y)\| \le L(t)\|x-y\|$ for a.e. t and $x,y\in\overline{B}_\delta(x_0)$.
Then a fixed point of $(Kx)(t) := x_0 + \int_{t_0}^t f(s,x(s))\,ds$ lies in $AC(I,\mathbb{R}^n)$ and satisfies $x' = f(\cdot,x)$ a.e.
Generally AC cannot be replaced by C^1. For $f(t,x)=\operatorname{sign}(t)x$, $x(t) := e^{|t|}$ satisfies $x(0)=1$,
$x(t) = 1+\int_0^t \operatorname{sign}(s)x(s)\,ds$, and $x' = \operatorname{sign}(t)x$ for $t\ne 0$, but $x\notin C^1((-1,1))$.

Lemma 1.1.19. *Time Regularity of ODE Solutions*
Let $I\subseteq\mathbb{R}$ be an interval, $k,n\in\mathbb{N}$, $k\ge 1$, $U\subseteq\mathbb{R}\times\mathbb{R}^n$ open, and $f\in C^k(U,\mathbb{R}^n)$.
Let $x\in C^1(I,\mathbb{R}^n)$ and $\gamma := (\mathrm{id}_I, x)$. If $\gamma(I)\subseteq U$ and $x' = f\circ\gamma$, then $x\in C^{k+1}(I,\mathbb{R}^n)$.

Proof. Let P_j mean $x\in C^j(I,\mathbb{R}^n)$. Since $x\in C^1$, P_1 holds.
If P_j holds and $1\le j\le k$, then $\gamma\in C^j(I,U)$, so $x' = f\circ\gamma\in C^j(I,\mathbb{R}^n)$; hence P_{j+1}. Thus P_{k+1}.
Structural Commentary: Bootstrap $x' = f\circ\gamma$.
Key Insight: Vector-field regularity lifts the trajectory by one derivative. □

Lemma 1.1.20. *Generalized Gronwall Inequality*
Let $\alpha,\beta,\psi\in C([0,T],\mathbb{R})$ with $\beta\ge 0$ and $\psi(t) \le \alpha(t) + \int_0^t \beta(s)\psi(s)\,ds \ \forall t\in[0,T]$.

(i) $\psi(t) \le \alpha(t) + \int_0^t \alpha(s)\beta(s)\exp\left(\int_s^t\beta(r)\,dr\right)ds \ \forall t\in[0,T]$.

(ii) If $\alpha(s)\le\alpha(t)$ whenever $0\le s\le t\le T$, then $\psi(t)\le\alpha(t)\exp\left(\int_0^t\beta(s)\,ds\right) \ \forall t\in[0,T]$.

Proof.

(i) Put $q(t) := \int_0^t\beta(s)\psi(s)\,ds$ and $w(t) := \exp\left(-\int_0^t\beta(s)\,ds\right)$. Then $(wq)' = w\beta(\psi-q) \le w\beta\alpha$.
Since $q(0)=0$, integration gives $q(t)\le\int_0^t\alpha(s)\beta(s)\exp\left(\int_s^t\beta(r)\,dr\right)ds$; combine this with $\psi\le\alpha+q$.

(ii) Under the monotonicity hypothesis, (i) gives $\psi(t)\le\alpha(t)\left(1+\int_0^t\beta(s)\exp\left(\int_s^t\beta(r)\,dr\right)ds\right)$.
For $q(s) := \exp\left(\int_s^t\beta(r)\,dr\right)$, $q' = -\beta q$, so the parenthesis equals $1+q(0)-q(t) = \exp\left(\int_0^t\beta(s)\,ds\right)$.

Structural Commentary: An integrating factor controls the accumulated integral term.
Key Insight: Exponential growth is the exact cost of repeated feedback. □

Theorem 1.1.21. *Continuous Dependence on Initial Data and Vector Fields*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f, g \in C(U, \mathbb{R}^n)$, $I := [t_0, T]$, and $x, y \in C^1(I, \mathbb{R}^n)$ satisfy $x' = f(\cdot, x)$, $y' = g(\cdot, y)$.
For $w : I \to \mathbb{R}^n$, set $\Gamma_w := \{(t, w(t)) : t \in I\}$. Assume $\Gamma_x \cup \Gamma_y \subseteq V \subseteq U$ with V compact.
Assume $\exists L \geq 0$ such that $\|f(t,p) - f(t,q)\| \leq L \|p - q\|$ whenever $(t,p), (t,q) \in V$.
Set $a := \|x(t_0) - y(t_0)\|$, $E := f - g$, $M := \max_V \|E\|$, $\rho_0(h) := h$, and $\rho_L(h) := L^{-1}(e^{Lh} - 1)$ for $L > 0$.
Then $\|x(t) - y(t)\| \leq e^{L(t-t_0)} a + M \rho_L(t - t_0)$ $\forall t \in I$.

Proof. Put $\Delta := x - y$, $\psi := \|\Delta\|$, and $R(s) := f(s, x(s)) - f(s, y(s)) + E(s, y(s))$.
Then $\Delta(t) = \Delta(t_0) + \int_{t_0}^t R(s)\, ds$ and $\|R(s)\| \leq L\psi(s) + M$. Hence $\psi(t) \leq a + \int_{t_0}^t (L\psi(s) + M)\, ds$.
If $L = 0$, then $\psi(t) \leq a + M(t - t_0) = a + M\rho_0(t - t_0)$.
If $L > 0$, set $\chi := \psi + M/L$. Then $\chi(t) \leq a + M/L + \int_{t_0}^t L\chi(s)\, ds$.
By Lemma 1.1.20(ii), $\chi(t) \leq (a + M/L)e^{L(t-t_0)}$, hence $\psi(t) \leq e^{L(t-t_0)} a + M\rho_L(t - t_0)$.
Structural Commentary: Split state error from field error; apply Gronwall.
Key Insight: Initial and model errors propagate exponentially. □

Theorem 1.1.22. *Local Lipschitz Dependence on the Initial Pair*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_*, x_*) \in U$. Assume f is locally state-Lipschitz:
$\forall$ compact $K \subseteq U$, $\exists L_K < +\infty$ such that $\|f(t,p) - f(t,q)\| \leq L_K \|p - q\|$ whenever $(t,p), (t,q) \in K$.
Then $\exists I, B, V, \Phi$ with $I \ni t_*$ compact, $B \ni x_*$ an open ball, $V \subseteq U$ compact, and $\Phi : I \times I \times B \to \mathbb{R}^n$.
Set $H := \sup I - \inf I$, $M := \max_V \|f\|$, choose a state-Lipschitz constant L for f on V.
Let $D((r,x),(q,y)) := \|x - y\| + M\,|r - q|$. Then:
(i) $\Phi(r, r, x) = x$, $(t, \Phi(t,r,x)) \in V$, and $\partial_t \Phi(t,r,x) = f(t, \Phi(t,r,x))$ for $(t,r,x) \in I \times I \times B$.
(ii) $\|\Phi(t,r,x) - \Phi(s,q,y)\| \leq e^{LH} D((r,x),(q,y)) + M\,|t - s|$ for $(t,r,x),(s,q,y) \in I \times I \times B$.

Proof. Choose $a, \delta > 0$ with $Q := [t_* - a, t_* + a] \times \overline{B}_\delta(x_*) \subseteq U$, and set $M_Q := \max_Q \|f\|$.
Choose $0 < h < a/2$ with $2hM_Q < \delta/2$, set $I := [t_* - h, t_* + h]$, $B := B_{\delta/2}(x_*)$, and $V := I \times \overline{B}_\delta(x_*)$.
Set $H := 2h$, $M := \max_V \|f\|$, and choose a state-Lipschitz constant L for f on V.

(i) Fix $(r,x) \in I \times B$, set $I_r^+ := [r, t_* + h]$ and $I_r^- := [t_* - h, r]$.
Since $I_r^\pm \times \overline{B}_{\delta/2}(x) \subseteq Q$ and $\left|I_r^\pm\right| \leq 2h$, Theorem 1.1.16, applied forward on I_r^+ and backward on I_r^-, gives a unique $u_{r,x} \in C^1(I, \mathbb{R}^n)$ with $u_{r,x}(r) = x$ and $u'_{r,x} = f(\cdot, u_{r,x})$.
For $t \in I$, $\|u_{r,x}(t) - x\| \leq M_Q\,|t - r| < \delta/2$, hence $\|u_{r,x}(t) - x_*\| < \delta$ and $(t, u_{r,x}(t)) \in V$.
Set $\Phi(t,r,x) := u_{r,x}(t)$; then $\Phi(r,r,x) = x$, $(t, \Phi(t,r,x)) \in V$, and $\partial_t \Phi(t,r,x) = f(t, \Phi(t,r,x))$.

(ii) Let $(t,r,x),(s,q,y) \in I \times I \times B$, $p := \max\{r,q\}$, and $D_0 := D((r,x),(q,y))$.
By (i) and $\|\partial_t \Phi\| \leq M$, $\|\Phi(p,r,x) - \Phi(p,q,y)\| \leq D_0$.
By Lemma 1.1.20(ii), after reversing parameter when $t < p$, $\|\Phi(t,r,x) - \Phi(t,q,y)\| \leq e^{L|t-p|} D_0 \leq e^{LH} D_0$.
The bound $\|\Phi(t,q,y) - \Phi(s,q,y)\| \leq M\,|t - s|$ and the triangle inequality give (ii).

Structural Commentary: Compare at the later start time, then apply Gronwall.
Key Insight: Initial-time error first becomes state error. □

Theorem 1.1.23. *Smooth Dependence and First Variational Equations*
Let $n, p \in \mathbb{N}$, $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^p$ be open, $f \in C^1(\Omega, \mathbb{R}^n)$, and $(t_0, x_0, \lambda_0) \in \Omega$.
Then $\exists I, B, \Lambda_0, \phi$ with $I \ni t_0$ compact, $B \ni x_0$ and $\Lambda_0 \ni \lambda_0$ open, and $\phi : I \times B \times \Lambda_0 \to \mathbb{R}^n$.
For $(x, \lambda) \in B \times \Lambda_0$, set $\gamma_{x,\lambda}(t) := (t, \phi(t,x,\lambda), \lambda)$.
Write $A_{x,\lambda} := D_x f \circ \gamma_{x,\lambda}$ and $G_{x,\lambda} := D_\lambda f \circ \gamma_{x,\lambda}$. The following hold:
(i) $\phi \in C^1(I \times B \times \Lambda_0, \mathbb{R}^n)$, $\phi(t_0, x, \lambda) = x$, and $\partial_t \phi(t,x,\lambda) = f(t, \phi(t,x,\lambda), \lambda)$ $\forall (t,x,\lambda) \in I \times B \times \Lambda_0$.
(ii) $D_x \phi(\cdot, x, \lambda)$ is the unique solution of $Y' = A_{x,\lambda} Y$, $Y(t_0) = I_n$, in $C^1(I, \mathbb{R}^{n \times n})$ $\forall (x,\lambda) \in B \times \Lambda_0$.
(iii) $D_\lambda \phi(\cdot, x, \lambda)$ is the unique solution of $Z' = A_{x,\lambda} Z + G_{x,\lambda}$, $Z(t_0) = 0_{n \times p}$, in $C^1(I, \mathbb{R}^{n \times p})$ $\forall (x,\lambda) \in B \times \Lambda_0$.

Proof. Put $z := (x, \lambda)$, $z_0 := (x_0, \lambda_0)$, $F : \Omega \to \mathbb{R}^{n+p}$ by $F(t,z) := (f(t,x,\lambda), 0)$; then $F \in C^1(\Omega, \mathbb{R}^{n+p})$.
Choose $a, r > 0$ with $W := [t_0 - a, t_0 + a] \times \overline{B}_{2r}(z_0) \subseteq \Omega$. Set $H := \max_W \|F\|$.
Let $\tau := a$ if $H = 0$, and $\tau := \min\{a, \frac{r}{H}\}$ if $H > 0$. For $z \in B_r(z_0)$, $[t_0 - \tau, t_0 + \tau] \times \overline{B}_r(z) \subseteq W$.
By Lemma 1.1.15, choose a state-Lipschitz constant L_W for F on W, and set $I := [t_0 - \tau, t_0 + \tau]$.
By Theorem 1.1.16, applied forward and backward, $\forall z \in B_r(z_0)$, $\exists! \Phi_z \in C^1(I, \mathbb{R}^{n+p})$ with $\Phi_z(t_0) = z$ and $\Phi'_z = F(\cdot, \Phi_z)$. Define $\Phi(t,z) := \Phi_z(t)$. Choose open B, Λ_0 with $x_0 \in B$, $\lambda_0 \in \Lambda_0$, and $B \times \Lambda_0 \subseteq B_r(z_0)$.

(i) Since the last p components of F vanish, $\Phi(t,x,\lambda) = (\phi(t,x,\lambda), \lambda)$.
Hence $\phi(t_0, x, \lambda) = x$ and $\partial_t \phi(t,x,\lambda) = f(t, \phi(t,x,\lambda), \lambda)$.
By Theorem 1.1.21, applied forward and backward, $\|\Phi_z - \Phi_w\|_\infty \leq e^{\tau L_W} \|z - w\|$ for $z, w \in B_r(z_0)$.
Hence Φ and $\partial_t \Phi = F(t, \Phi)$ are continuous.
Fix $z \in B_r(z_0)$ and set $\mathcal{L}_z(t) := D_z F(t, \Phi_z(t))$. Since $\mathcal{L}_z$ is bounded on I, apply Theorem 1.1.16 finitely many times forward and backward to obtain $P_z \in C^1(I, \mathbb{R}^{(n+p)\times(n+p)})$ with $P'_z = \mathcal{L}_z P_z$ and $P_z(t_0) = I_{n+p}$. Set $C_0 := \max_W \|D_z F\|$; By Lemma 1.1.20(ii) $\|P_z\|_\infty \leq e^{2\tau C_0}$.
For $h \to 0$ with $z + h \in B_r(z_0)$, put $\Delta_h := \Phi_{z+h} - \Phi_z$ and $e_h := \Delta_h - P_z h$.
Set $\eta_{h,s} := \Phi_z + s\Delta_h$, $\mathcal{R}_h(t,s) := D_z F(t, \eta_{h,s}(t)) - \mathcal{L}_z(t)$.
Then $R_h(t) := \int_0^1 \mathcal{R}_h(t,s)\, ds$, $\varepsilon_h := \|R_h\|_\infty$; $\varepsilon_h \to 0$; $e'_h = \mathcal{L}_z e_h + R_h \Delta_h$, $e_h(t_0) = 0$, $\|\Delta_h\|_\infty \leq e^{\tau L_W} \|h\|$.
By Lemma 1.1.20(ii), $\|e_h\|_\infty = O(\varepsilon_h \|h\|) = o(\|h\|)$; thus $D_z \Phi = P_z$.
For $z, w \in B_r(z_0)$, $P_z - P_w$ satisfies $(P_z - P_w)' = \mathcal{L}_z(P_z - P_w) + (\mathcal{L}_z - \mathcal{L}_w)P_w$ and $(P_z - P_w)(t_0) = 0$.
By Lemma 1.1.20(ii), $\|P_z - P_w\|_\infty \leq 2\tau e^{4\tau C_0} \|\mathcal{L}_z - \mathcal{L}_w\|_\infty \to 0$ as $z \to w$. Thus Φ, hence ϕ, is C^1.

(ii) Fix (x,λ) and set $z = (x,\lambda)$. Put $Y := D_x \phi(\cdot, x, \lambda)$, $A := A_{x,\lambda}$, and $G := G_{x,\lambda}$.
Here $\mathcal{L}_z(\xi, \mu) = (A\xi + G\mu, 0)$ and $D_x \Phi = (Y, 0)$. The x-columns of $P'_z = \mathcal{L}_z P_z$ give $Y' = AY$, $Y(t_0) = I_n$.
If $\widetilde{Y}$ also solves this IVP, $W := Y - \widetilde{Y}$ satisfies $W' = AW$, $W(t_0) = 0$; By Lemma 1.1.20(ii) $W = 0$.

(iii) Fix (x,λ) and set $z = (x,\lambda)$. Put $Z := D_\lambda \phi(\cdot, x, \lambda)$, $A := A_{x,\lambda}$, and $G := G_{x,\lambda}$.
Here $\mathcal{L}_z(\xi, \mu) = (A\xi + G\mu, 0)$. Since $D_\lambda \Phi = (Z, I_p)$, the λ-columns give $Z' = AZ + G$, $Z(t_0) = 0_{n \times p}$.
If $\widetilde{Z}$ also solves this IVP, $W := Z - \widetilde{Z}$ satisfies $W' = AW$, $W(t_0) = 0$; By Lemma 1.1.20(ii) $W = 0$.

Structural Commentary: Lift parameters; differentiate the extended flow.
Key Insight: Flow derivatives satisfy variational equations. □

Theorem 1.1.24. *Higher-Order Smooth Dependence*
Let $k, n, p \in \mathbb{N}$, $k \geq 1$, $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^p$ open, $f \in C^k(\Omega, \mathbb{R}^n)$, and $q_0 = (t_0, x_0, \lambda_0) \in \Omega$.
Then $\exists I \ni t_0$ compact, $B \ni x_0$ open in $\mathbb{R}^n$, $\Lambda_0 \ni \lambda_0$ open in $\mathbb{R}^p$, and $\phi : D \to \mathbb{R}^n$, with $D := I \times B \times \Lambda_0$.
For $\Gamma_\phi(t, x, \lambda) := (t, \phi(t, x, \lambda), \lambda)$:
(i) $\Gamma_\phi(D) \subseteq \Omega$, $\phi(t_0, x, \lambda) = x$, and $\partial_t \phi = f \circ \Gamma_\phi$ on D.
(ii) $\phi \in C^k(D, \mathbb{R}^n)$.

Proof. Let $\mathsf{H}_{r,d}$ mean: every $F \in C^r(\Omega_d, \mathbb{R}^d)$, $\Omega_d \subseteq \mathbb{R} \times \mathbb{R}^d$ open, has a local C^r flow near each $(\tau, z_*) \in \Omega_d$.
By Theorem 1.1.23(i) with a dummy parameter, $\mathsf{H}_{1,d}$ holds $\forall d$. Assume $\mathsf{H}_{j,d}$ for all d. Let $F \in C^{j+1}(\Omega_d, \mathbb{R}^d)$, $(\tau, z_*) \in \Omega_d$, and let Ψ be its local flow. By $\mathsf{H}_{j,d}$, $\Psi \in C^j$. Set $Q := D_z \Psi$ and $E_d := \mathrm{Mat}(d, d, \mathbb{R})$.
By Theorem 1.1.23(ii), $\partial_t Q = D_z F(t, \Psi) Q$ and $Q(\tau, z) = I_d$.
Set $\mathcal{H}(t, \zeta, P) := (F(t, \zeta), D_z F(t, \zeta) P)$ on $\Omega_d \times E_d$, and set $Y := (\Psi, Q)$. Then $\mathcal{H} \in C^j$ and $Y_t = \mathcal{H}(t, Y)$, $Y(\tau, z) = (z, I_d)$.
By $\mathsf{H}_{j,d+d^2}$, the flow Θ of $\mathcal{H}$ is C^j. By uniqueness, $Y(t, z) = \Theta(t, z, I_d)$, so $Q \in C^j$.
By $D_z \Psi = Q \in C^j$ and $\partial_t \Psi = F(t, \Psi) \in C^j$, $\Psi \in C^{j+1}$. By induction, $\mathsf{H}_{r,d}$ holds for all $r, d \in \mathbb{N}$.
Set $m := n + p$, $z := (x, \lambda)$, $z_0 := (x_0, \lambda_0)$, $\widetilde{\Omega} := \{(t, z) : (t, x, \lambda) \in \Omega\}$, and $\widetilde{F}(t, z) := (f(t, x, \lambda), 0_p)$.
By $\mathsf{H}_{k,m}$, after shrinking, choose compact $I \ni t_0$, open $U_0 \ni z_0$, and $\Phi \in C^k(I \times U_0, \mathbb{R}^m)$ with $\partial_t \Phi = \widetilde{F}(t, \Phi)$ and $\Phi(t_0, z) = z$.
Choose $B \ni x_0$ and $\Lambda_0 \ni \lambda_0$ with $B \times \Lambda_0 \subseteq U_0$. Set $D := I \times B \times \Lambda_0$, $\pi_x := \pi_{\mathbb{R}^n}$, $\pi_\lambda := \pi_{\mathbb{R}^p}$, and $\phi := \pi_x \circ \Phi$.
(i) By $\partial_t(\pi_\lambda \circ \Phi) = 0$ and $(\pi_\lambda \circ \Phi)(t_0, x, \lambda) = \lambda$, $\Phi = (\phi, \lambda)$ on D.
By this identity and $(t, \Phi) \in \widetilde{\Omega}$, $\Gamma_\phi(D) \subseteq \Omega$; by $\Phi(t_0, x, \lambda) = (x, \lambda)$, $\phi(t_0, x, \lambda) = x$.
By $\partial_t \Phi = \widetilde{F}(t, \Phi)$ and $\phi = \pi_x \circ \Phi$, $\partial_t \phi = f \circ \Gamma_\phi$ on D.
(ii) By $\Phi \in C^k(I \times U_0, \mathbb{R}^m)$ and linearity of π_x, $\phi \in C^k(D, \mathbb{R}^n)$.
Structural Commentary: Bootstrap through the variational flow.
Key Insight: Parameters are frozen coordinates. □

Proposition 1.1.25. *Regular Perturbation Expansion*
Let $k, n \in \mathbb{N}$, $k \geq 1$, $\Omega \subseteq \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}$ open, $f \in C^k(\Omega, \mathbb{R}^n)$, and $q_0 = (t_0, x_0, \varepsilon_0) \in \Omega$.
Let $D := I \times \Lambda_0$, where $I \ni t_0$ is compact and $\Lambda_0 \ni \varepsilon_0$ is open, and let $\phi \in C^k(D, \mathbb{R}^n)$.
Set $\Gamma_\phi(t, \varepsilon) := (t, \phi(t, \varepsilon), \varepsilon)$ and $h_\varepsilon := \varepsilon - \varepsilon_0$.
Assume $\Gamma_\phi(D) \subseteq \Omega$, $\phi(t_0, \varepsilon) = x_0$, and $\partial_t \phi = f \circ \Gamma_\phi$.
For $0 \leq j \leq k$, set $Z_j := (z_0, \ldots, z_j)$.
For $g = g(t, Z_j, \varepsilon)$ and $0 \leq j < k$, set $\mathcal{D}_j g := \partial_\varepsilon g + \sum_{r=0}^{j} D_{z_r} g \, z_{r+1}$.
Set $f_0(t, z_0, \varepsilon) := f(t, z_0, \varepsilon)$ and $f_{j+1} := \mathcal{D}_j f_j$ for $0 \leq j < k$.
For $0 \leq j \leq k$, set $\phi_j(t) := \partial_\varepsilon^j \phi(t, \varepsilon_0)$ and $\Phi_j(t) := (\phi_0(t), \ldots, \phi_j(t))$.
Set $T_k(t, \varepsilon) := \sum_{j=0}^{k} \frac{\phi_j(t)}{j!} h_\varepsilon^j$.
Set $r_k(t, \varepsilon) := \phi(t, \varepsilon) - T_k(t, \varepsilon)$ and $\rho_k(\varepsilon) := \sup_{t \in I} \|r_k(t, \varepsilon)\|$.
Then:
(i) For $0 \leq j \leq k$, $\phi_j' = f_j(t, \Phi_j, \varepsilon_0)$ and $\phi_j(t_0) = \delta_{j0} x_0$.
(ii) $\rho_k(\varepsilon) = o(|h_\varepsilon|^k)$.
(iii) If $f \in C^{k+1}(\Omega, \mathbb{R}^n)$, then, after shrinking I and Λ_0, $\rho_k(\varepsilon) = O(|h_\varepsilon|^{k+1})$.

Proof. For $0 \leq j \leq k$, set $z_j := \partial_\varepsilon^j \phi$, $\widehat{Z}_j := (z_0, \ldots, z_j)$, and $R_j := f_j(t, \widehat{Z}_j, \varepsilon)$.
For $g = g(t, \varepsilon)$, set $(\mathcal{I} g)(t, \varepsilon) := \int_{t_0}^{t} g(s, \varepsilon) \, ds$.
(i) Since $R_0 = f(t, \phi, \varepsilon)$, the IVP is $z_0 = x_0 + \mathcal{I} R_0$. By the chain rule, $\partial_\varepsilon R_j = R_{j+1}$ for $0 \leq j < k$.
Base $j = 0$ is the IVP. If $z_j = \delta_{j0} x_0 + \mathcal{I} R_j$, then by differentiation under $\mathcal{I}$, $z_{j+1} = \mathcal{I} R_{j+1}$.
By induction, $z_j = \delta_{j0} x_0 + \mathcal{I} R_j$ for $0 \leq j \leq k$.
At $\varepsilon = \varepsilon_0$, $z_j = \phi_j$ and $R_j = f_j(t, \Phi_j, \varepsilon_0)$.
By the preceding identity, $\phi_j = \delta_{j0} x_0 + \mathcal{I}(f_j(\,\cdot\,, \Phi_j, \varepsilon_0))$. Differentiating in t proves (i).
(ii) Put $h := h_\varepsilon$ and $G_t(\eta) := \partial_\varepsilon^k \phi(t, \eta) - \phi_k(t)$.
By integral Taylor, $r_k(t, \varepsilon) = \frac{h^k}{(k-1)!} \int_0^1 (1 - s)^{k-1} G_t(\varepsilon_0 + sh) \, ds$.
Set $\omega(h) := \sup_{(t,s) \in I \times [0,1]} \|G_t(\varepsilon_0 + sh)\|$. By uniform continuity, $\omega(h) \to 0$.
By the formula above, $\rho_k(\varepsilon) \leq \frac{|h|^k}{k!} \omega(h) = o(|h|^k)$.
(iii) If $f \in C^{k+1}$, by Theorem 1.1.24, choose $\widetilde{I} \ni t_0$ and $\widetilde{\Lambda} \ni \varepsilon_0$.
Let $\widetilde{\phi} \in C^{k+1}(\widetilde{I} \times \widetilde{\Lambda}, \mathbb{R}^n)$ satisfy the same IVP.
By local uniqueness from Theorem 1.1.16, after shrinking $I \times \Lambda_0$, $\widetilde{\phi} = \phi$, so $\phi \in C^{k+1}(I \times \Lambda_0, \mathbb{R}^n)$.
Choose compact $J \Subset \Lambda_0$ with $\varepsilon_0 \in J^\circ$.
Set $C_J := \max_{(t,\eta) \in I \times J} \left\| \partial_\varepsilon^{k+1} \phi(t, \eta) \right\|$.
For $\varepsilon \in J$, by the integral remainder, $\rho_k(\varepsilon) \leq \frac{C_J |h_\varepsilon|^{k+1}}{(k+1)!}$.
Structural Commentary: Differentiate, then Taylor expand.
Key Insight: Coefficients satisfy forced IVPs. □

Example 1.1.26. *Linear Drag as a Regular Perturbation*
Let $g > 0$, $\varepsilon \geq 0$, and $v' = -\varepsilon v - g$, $v(0) = 0$. Then $v(t, \varepsilon) = \frac{g(e^{-\varepsilon t} - 1)}{\varepsilon}$ if $\varepsilon > 0$, and $v_0(t) := v(t, 0) = -gt$.

At $\varepsilon_0 = 0$, $v_1' = -v_0 = gt$, $v_1(0) = 0$, hence $v_1(t) = \frac{gt^2}{2}$ and $v(t,\varepsilon) = v_0(t) + \varepsilon v_1(t) + o(\varepsilon)$ on compact t-intervals.
For fixed $\varepsilon > 0$, $t \to +\infty$ gives $v(t,\varepsilon) \to -\frac{g}{\varepsilon}$ but $v_0(t) + \varepsilon v_1(t) \to +\infty$; the expansion is local for bounded t.

Theorem 1.1.27. *Unique Maximal Solution*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_0, x_0) \in U$.
Assume $\forall$ compact $V \subseteq U$, $\exists L_V < +\infty$ such that $\|f(t,p) - f(t,q)\| \le L_V \|p - q\|$ whenever $(t,p),(t,q) \in V$.
For an open interval $J \ni t_0$, write $u \in \mathcal{A}(J)$ iff $u \in C^1(J, \mathbb{R}^n)$, $u(t_0) = x_0$, $(t, u(t)) \in U$, and $u' = f(\cdot, u)$.
Let $\mathcal{S}$ be all such pairs (J,u). For $(J,u),(K,v) \in \mathcal{S}$, write $(J,u) \preceq (K,v)$ iff $J \subseteq K$ and $v|_J = u$. Then:

(i) $\exists$ a maximal $(I_*, \phi_*) \in \mathcal{S}$ with $I_* = (T_-, T_+)$.
(ii) $(J,u) \preceq (I_*, \phi_*)$ $\forall (J,u) \in \mathcal{S}$; hence the maximal solution is unique.

Proof. By Theorem 1.1.16, applied forward and backward, $\mathcal{S} \neq \emptyset$. For $(J,u),(K,v) \in \mathcal{S}$, set $E := \{t \in J \cap K : u(t) = v(t)\}$.
Then $t_0 \in E$, E is closed, and local uniqueness at each $s \in E$ makes E open in $J \cap K$.
Since $J \cap K$ is an interval, $E = J \cap K$; thus $u = v$ on $J \cap K$ $\forall (J,u),(K,v) \in \mathcal{S}$.

(i) Set $I_* := \bigcup_{(J,u)\in\mathcal{S}} J$ and let $\phi_*(t) := u(t)$ for any $(J,u) \in \mathcal{S}$ with $t \in J$. Agreement on overlaps makes ϕ_* well defined.
Since all J are open intervals containing t_0, $I_* = (T_-, T_+)$. Locally $\phi_* = u$, so $(I_*, \phi_*) \in \mathcal{S}$.
If $(I_*, \phi_*) \preceq (\widehat{I}, \widehat{\phi}) \in \mathcal{S}$, then $I_* \subseteq \widehat{I} \subseteq I_*$; hence $\widehat{I} = I_*$.
(ii) If $(J,u) \in \mathcal{S}$, then $J \subseteq I_*$ and overlap agreement gives $u = \phi_*|_J$; hence $(J,u) \preceq (I_*, \phi_*)$.
If (K,ψ) is maximal, then $(K,\psi) \preceq (I_*, \phi_*)$; maximality gives $K = I_*$ and $\psi = \phi_*$.

Structural Commentary: Local uniqueness makes solution germs compatible.
Key Insight: The maximal solution is their union. □

Lemma 1.1.28. *Extension Criterion at a Finite Endpoint*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and assume f is locally Lipschitz in x on compact subsets of U.
Let $\phi \in C^1((T_-, T_+), \mathbb{R}^n)$ satisfy $\phi' = f(\cdot, \phi)$, $(t, \phi(t)) \in U$, and $T_+ < +\infty$.
Then $\exists \varepsilon > 0$ and a solution $\widehat{\phi} : (T_-, T_+ + \varepsilon) \to \mathbb{R}^n$ with $\widehat{\phi}|_{(T_-,T_+)} = \phi$ iff $\exists t_m \uparrow T_+$ and $y \in \mathbb{R}^n$ such that $\phi(t_m) \to y$ and $(T_+, y) \in U$.

Proof.
($\Rightarrow$) Let $\widehat{\phi}$ be such an extension, choose $t_m \uparrow T_+$, and put $y := \widehat{\phi}(T_+)$. Then $\phi(t_m) = \widehat{\phi}(t_m) \to y$ and $(T_+, y) \in U$.
($\Leftarrow$) Choose $r, \eta > 0$ with $W := [T_+ - \eta, T_+ + \eta] \times \overline{B}_r(y) \subseteq U$ and set $M := \max_W \|f\|$.
Choose m large enough that $t_m > T_+ - \eta$, $a := \|\phi(t_m) - y\| < \frac{r}{2}$, and $Mh < \frac{r}{2}$ for $h := T_+ - t_m$; put $J := [t_m, T_+)$ and $B := B_r(y)$.
If $\tau \in J$ is the first exit from B, then $r \le a + M(\tau - t_m) \le a + Mh < r$, impossible. Hence $\phi(J) \subseteq B$.
For $s,t \in J$, $\|\phi(t) - \phi(s)\| \le M|t - s|$, so $\phi(t)$ is Cauchy as $t \uparrow T_+$. Since $\phi(t_m) \to y$, $\lim_{t\uparrow T_+} \phi(t) = y$.
By Theorem 1.1.16, $\exists \varepsilon > 0$ and a solution $\widetilde{\phi}$ on $[T_+, T_+ + \varepsilon)$ with $\widetilde{\phi}(T_+) = y$. Define $\widehat{\phi} = \phi$ on (T_-, T_+) and $\widehat{\phi} = \widetilde{\phi}$ on $[T_+, T_+ + \varepsilon)$.
Continuity of f gives $\phi'(t) \to f(T_+, y) = \widetilde{\phi}'(T_+)$; hence $\widehat{\phi} \in C^1((T_-, T_+ + \varepsilon), \mathbb{R}^n)$ and is the required extension.

Structural Commentary: A convergent endpoint subsequence traps the tail for gluing.
Key Insight: Finite endpoint limits are exactly extendible states. □

Corollary 1.1.29. *Compact Escape at a Finite Maximal Endpoint*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and $f \in C(U, \mathbb{R}^n)$.
Assume $\forall C \Subset U$, $\exists L_C < +\infty$ such that $(t,p),(t,q) \in C \Rightarrow \|f(t,p) - f(t,q)\| \le L_C \|p - q\|$.
Let $J := (T_-, T_+)$ and let $\phi : J \to \mathbb{R}^n$ be maximal for $x' = f(t,x)$, with $\gamma(t) := (t, \phi(t)) \in U$. Then:

(i) $T_+ < +\infty \Rightarrow \forall K \Subset U$, $\exists \tau < T_+ : \gamma((\tau, T_+)) \cap K = \emptyset$.
(ii) $T_- > -\infty \Rightarrow \forall K \Subset U$, $\exists \tau > T_- : \gamma((T_-, \tau)) \cap K = \emptyset$.
(iii) If $U = \mathbb{R} \times \mathbb{R}^n$, then $T_+ < +\infty \Rightarrow \|\phi(t)\| \to +\infty$ as $t \uparrow T_+$ and $T_- > -\infty \Rightarrow \|\phi(t)\| \to +\infty$ as $t \downarrow T_-$.

Proof.
(i) If (i) fails for some $K \Subset U$, then $\exists t_m \uparrow T_+$ with $\gamma(t_m) \in K$. By compactness, after passing to a subsequence, $\gamma(t_m) \to (T_+, y) \in K \subseteq U$. By Lemma 1.1.28, ϕ admits a proper extension past T_+, contradiction.
(ii) Set $\widetilde{U} := \{(s,x) : (-s,x) \in U\}$, $\widetilde{f}(s,x) := -f(-s,x)$, $\widetilde{J} := (-T_+, -T_-)$, and $u(s) := \phi(-s)$.
Then $u : \widetilde{J} \to \mathbb{R}^n$ is maximal for $u' = \widetilde{f}(\cdot, u)$. For $K \Subset U$, put $\widetilde{K} := \{(-t,x) : (t,x) \in K\} \Subset \widetilde{U}$.
By (i) applied to u at $-T_-$, $\exists a < -T_-$ such that $(s, u(s)) \notin \widetilde{K}$ for $a < s < -T_-$.
With $\tau := -a$, $\gamma((T_-, \tau)) \cap K = \emptyset$.
(iii) Assume $U = \mathbb{R} \times \mathbb{R}^n$ and $T_+ < +\infty$. For $R > 0$, set $K_R := [T_+ - 1, T_+] \times \overline{B}_R(0)$.
By (i), $\exists \tau < T_+$ such that $\gamma((\tau, T_+)) \cap K_R = \emptyset$, hence $\|\phi(t)\| > R$ for $\max\{\tau, T_+ - 1\} < t < T_+$.
Since R is arbitrary, $\|\phi(t)\| \to +\infty$ as $t \uparrow T_+$. If $T_- > -\infty$, apply (ii) to $K_R^- := [T_-, T_- + 1] \times \overline{B}_R(0)$ to obtain $\|\phi(t)\| > R$ near T_-; hence $\|\phi(t)\| \to +\infty$ as $t \downarrow T_-$.

Structural Commentary: Compact recurrence creates an endpoint extension.
Key Insight: Finite maximal time forces compact escape. □

Corollary 1.1.30. *Global Existence under Linear Growth*
Let $f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ be continuous.
Assume (LL): $\forall$ compact $V \subseteq \mathbb{R} \times \mathbb{R}^n$, $\exists L_V < +\infty$ such that $\|f(t,p) - f(t,q)\| \le L_V \|p - q\|$ $\forall (t,p),(t,q) \in V$.
Assume (LG): $\forall T > 0$, $\exists M_T, L_T \ge 0$ such that $\|f(t,x)\| \le M_T + L_T \|x\|$ $\forall (t,x) \in [-T,T] \times \mathbb{R}^n$.
If $\phi : (T_-, T_+) \to \mathbb{R}^n$ is maximal and $\phi' = f(\cdot, \phi)$, then $(T_-, T_+) = \mathbb{R}$.

Proof. Fix $t_0 \in (T_-, T_+)$, $x_0 := \phi(t_0)$. If $T_+ < +\infty$, choose $T > \max\{|t_0|, |T_+|\}$ and M_T, L_T from (LG), and set $C_T := (\|x_0\| + M_T(T_+ - t_0))e^{L_T(T_+ - t_0)}$.
With $q(t) := \|\phi(t)\|$, $q(t) \le \|x_0\| + \int_{t_0}^t (M_T + L_T q(s))\, ds$ on $[t_0, T_+)$. Lemma 1.1.20(ii) gives $q(t) \le (\|x_0\| + M_T(t - t_0))e^{L_T(t - t_0)} \le C_T < +\infty$.
Thus $\Gamma_+ := \{(t, \phi(t)) : t \in [t_0, T_+)\} \subseteq K_T := [-T, T] \times \overline{B}_{C_T}(0)$. For $t_m \uparrow T_+$, compactness gives $\phi(t_{m_j}) \to y$ and $(T_+, y) \in K_T$. By (LL) and Lemma 1.1.28, ϕ extends past T_+, contradiction. Hence $T_+ = +\infty$.
Set $g(s,x) := -f(t_0 - s, x)$ and $u(s) := \phi(t_0 - s)$ on $(t_0 - T_+, t_0 - T_-)$. Then $u' = g(\cdot, u)$, u is maximal, and g satisfies (LL),(LG) on

bounded s-intervals. Apply right-endpoint result to u, then $t_0 - T_- = +\infty$, $T_- = -\infty$.
Structural Commentary: Gronwall bounds; compactness forces extension.
Key Insight: Linear growth rules out finite escape. □

Theorem 1.1.31. *Peano Local Existence Theorem*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, and $(t_0, x_0) \in U$. Choose $T, \delta > 0$ with $V := [t_0, t_0 + T] \times \overline{B}_\delta(x_0) \subseteq U$.
Set $M := \max_V \|f\|$, $T_0 := \min\{T, \delta/M\}$ with $\delta/0 := +\infty$, and $I := [t_0, t_0 + T_0]$.
For $y : I \to \mathbb{R}^n$, write $\Gamma_y := \{(t, y(t)) : t \in I\}$. Then:

(i) $\exists \phi \in C(I, \mathbb{R}^n)$ with $\Gamma_\phi \subseteq V$ and $\phi = x_0 + \int_{t_0}^{(\cdot)} f(s, \phi(s))\, ds$ on I.
(ii) This ϕ lies in $C^1(I, \mathbb{R}^n)$ and satisfies $\phi(t_0) = x_0$, $\phi' = f(\cdot\,, \phi)$.

Proof. For $N \geq 1$, set $h_N := T_0/N$, $t_j := t_0 + j h_N$, $\tau_N(t) := t_j$ on $[t_j, t_{j+1})$ for $0 \leq j < N$, and $\tau_N(t_N) := t_{N-1}$.
Construct x_N recursively by $x_N(t_0) := x_0$ and $x_N(t) := x_N(t_j) + (t - t_j) f(t_j, x_N(t_j))$ for $t \in [t_j, t_{j+1}]$, $0 \leq j < N$.
Write $(\mathcal{I}G)(t) := \int_{t_0}^t G(s)\, ds$, $Y_N := (\tau_N, x_N \circ \tau_N)$, and $F_N := f \circ Y_N$; then $x_N = x_0 + \mathcal{I} F_N$.
By induction, $\|x_N(t) - x_0\| \leq M(t - t_0) \leq \delta$ and $\|x_N(t) - x_N(s)\| \leq M\,|t - s|$.
Hence $\Gamma_{x_N} \subseteq V$ and (x_N) is uniformly bounded/equicontinuous.

(i) By Theorem 8.2.1, applied coordinatewise, $x_{N_k} \to \phi \in C(I, \mathbb{R}^n)$ uniformly.
Then $\Gamma_\phi \subseteq V$. Set $Y_\phi := (\mathrm{id}_I, \phi)$ and $f_\phi := f \circ Y_\phi$.
Since $\|\mathrm{id}_I - \tau_N\|_\infty \leq h_N$ and $\|x_N - x_N \circ \tau_N\|_\infty \leq M h_N$, $Y_{N_k}(I), Y_\phi(I) \subseteq V$ and $Y_{N_k} \to Y_\phi$ uniformly.
By uniform continuity of $f|_V$, $F_{N_k} \to f_\phi$ uniformly.
Passing to the limit in $x_{N_k} = x_0 + \mathcal{I} F_{N_k}$ gives $\phi = x_0 + \mathcal{I} f_\phi$.
(ii) Since $f_\phi \in C(I, \mathbb{R}^n)$, by Theorem 2.2.1 applied to (i), $\phi \in C^1(I, \mathbb{R}^n)$, $\phi(t_0) = x_0$, and $\phi' = f_\phi$.

Structural Commentary: Euler polygons plus Arzelà–Ascoli produce a convergent approximate trajectory.
Key Insight: Continuity yields existence through compactness, without uniqueness. □

Definition 1.1.32. *Osgood Modulus of Continuity*
Let $\rho \in C([0, +\infty), [0, +\infty))$.

(i) ρ is a *modulus of continuity* iff $\rho(0) = 0$ and ρ is nondecreasing.
(ii) A modulus ρ satisfies the *Osgood condition* iff $\int_0^1 \frac{dr}{\rho(r)} = +\infty$, interpreted as an improper extended integral; equivalently, $\int_0^a \frac{dr}{\rho(r)} = +\infty\ \forall a > 0$.

Key Insight: Osgood divergence prevents a zero separation from becoming positive in finite time.

Example 1.1.33. *Lipschitz, Hölder, and Almost-Lipschitz Moduli*
For $0 < \alpha < 1$, set $\rho_1(r) := r$, $\rho_\alpha(r) := r^\alpha$, $\rho_{\log}(0) := 0$, and $\rho_{\log}(r) := r(1 + |\log r|)$ for $r > 0$.
These functions are continuous, nondecreasing, and vanish at 0.
Since $\int_0^1 \frac{dr}{r} = +\infty$ and $\int_0^1 r^{-\alpha}\, dr = \frac{1}{1-\alpha} < +\infty$, ρ_1 is Osgood while the Hölder modulus ρ_α is not.
For the almost-Lipschitz modulus, $s = -\log r$ gives $\int_0^1 \frac{dr}{r(1+|\log r|)} = \int_0^{+\infty} \frac{ds}{1+s} = +\infty$, so $\rho_{\log}$ is Osgood.

Theorem 1.1.34. *Osgood Uniqueness Criterion*
Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, $(t_0, x_0) \in U$, $T > t_0$, and $\delta > 0$.
Put $J := [t_0, T]$, $V := J \times \overline{B}_\delta(x_0)$, and assume $V \subseteq U$.
Let $\rho \in C([0, +\infty), [0, +\infty))$ be nondecreasing, $\rho(0) = 0$, and $\int_0^a \frac{dr}{\rho(r)} = +\infty\ \forall a > 0$.
For $(t, p), (t, q) \in V$, set $A_t(p, q) := |\langle p - q, f(t, p) - f(t, q)\rangle|$.
Assume $\exists C \geq 0$ such that $A_t(p, q) \leq C\,\|p - q\|\,\rho(\|p - q\|)\ \forall (t, p), (t, q) \in V$.
For $w \in C(J, \mathbb{R}^n)$, set $\Gamma_w := \{(t, w(t)) : t \in J\}$.
If $u, v \in C^1(J, \mathbb{R}^n)$ satisfy $\Gamma_u \cup \Gamma_v \subseteq V$, $u(t_0) = v(t_0) = x_0$, $u' = f(\cdot\,, u)$, and $v' = f(\cdot\,, v)$, then $u = v$ on J.

Proof. Assume $S := \{t \in (t_0, T] : u(t) \neq v(t)\} \neq \emptyset$ and fix $t \in S$. Put $\Delta := u - v$ and $d := \|\Delta\|$.
For $s \in J$, set $E(s) := f(s, u(s)) - f(s, v(s))$. The set $Z_t := \{s \in [t_0, t] : d(s) = 0\}$ is nonempty and compact; set $a := \max Z_t$.
Then $a < t$, $d(a) = 0$, and $d > 0$ on $(a, t]$. On $(a, t]$, $d \in C^1$ and $d'(s) = \frac{\langle \Delta(s), E(s)\rangle}{d(s)}$. Hence $d'(s) \leq \frac{A_s(u(s), v(s))}{d(s)} \leq C\rho(d(s))$.
For $m \geq 1$, set $H_m(r) := \int_0^r \frac{dq}{\rho(q) + \frac{1}{m}}$. For $s \in (a, t]$, set $b_m(s) := \frac{\rho(d(s))}{\rho(d(s)) + \frac{1}{m}} \leq 1$.
If $a < r < t$, then on $[r, t]$, $(H_m \circ d)'(s) = \frac{d'(s)}{\rho(d(s)) + \frac{1}{m}} \leq C b_m(s) \leq C$.
Thus $H_m(d(t)) - H_m(d(r)) \leq C(t - r)$. Letting $r \downarrow a$ gives $H_m(d(t)) \leq C(t - a)$.
By Theorem 1.3.1, $H_m(d(t)) \uparrow \int_0^{d(t)} \frac{dq}{\rho(q)} = +\infty$, contradiction. Hence $S = \emptyset$ and $u = v$.
Structural Commentary: Regularize $1/\rho$; separate variables.
Key Insight: Divergence forbids first separation. □

Proposition 1.1.35. *Time Translation and the Local Flow Law*
Let $D \subseteq \mathbb{R}^n$ be open and $f : D \to \mathbb{R}^n$. For $(a, y) \in \mathbb{R} \times D$, let $\mathcal{S}(a, y)$ contain the pairs (I, u) such that $I \ni a$ is an open interval, $u \in C^1(I, D)$, $u(a) = y$, and $u' = f(u)$.
Assume $\mathcal{S}(a, y) \neq \emptyset\ \forall (a, y) \in \mathbb{R} \times D$, and $u = v$ on $I \cap J\ \forall (a, y) \in \mathbb{R} \times D$, $\forall (I, u), (J, v) \in \mathcal{S}(a, y)$.
For $x \in D$, let $\phi_x : I_x \to D$ be the maximal solution with $0 \in I_x$, $\phi_x(0) = x$, and $\phi_x' = f(\phi_x)$.
Define $\mathcal{D}_\Phi := \{(t, x) \in \mathbb{R} \times D : t \in I_x\}$ and $\Phi_t(x) := \phi_x(t)$ for $(t, x) \in \mathcal{D}_\Phi$.

(i) If $\phi : I \to D$, $0 \in I$, and $\phi' = f(\phi)$, then $\psi_\tau(t) := \phi(t - \tau)$ on $\tau + I$ satisfies $\psi_\tau' = f(\psi_\tau)\ \forall \tau \in \mathbb{R}$.
Its initial value is $\psi_\tau(\tau) = \phi(0)$.
(ii) $(0, x) \in \mathcal{D}_\Phi$ and $\Phi_0(x) = x\ \forall x \in D$; hence $\Phi_0 = \mathrm{id}_D$.
(iii) If $(s, x), (t, \Phi_s(x)), (t + s, x) \in \mathcal{D}_\Phi$, then $\Phi_{t+s}(x) = \Phi_t(\Phi_s(x))$.

Proof.

(i) For $t \in \tau + I$, $\psi_\tau'(t) = \phi'(t - \tau) = f(\phi(t - \tau)) = f(\psi_\tau(t))$ and $\psi_\tau(\tau) = \phi(0)$.
(ii) Since $0 \in I_x$, $(0, x) \in \mathcal{D}_\Phi$ and $\Phi_0(x) = \phi_x(0) = x\ \forall x \in D$.

(iii) Set $R := (I_x - s) \cap I_{\Phi_s(x)}$. Then $0, t \in R$ and $(r+s, x), (r, \Phi_s(x)) \in \mathcal{D}_\Phi$ for $r \in R$.
Define $u, v : R \to D$ by $u(r) := \Phi_{r+s}(x)$, $v(r) := \Phi_r(\Phi_s(x))$.
Then $u' = f(u)$ by (i), $v' = f(v)$, $u(0) = v(0) = \Phi_s(x)$.
Uniqueness gives $u = v$ on R, so $\Phi_{t+s}(x) = \Phi_t(\Phi_s(x))$.

Structural Commentary: Shift time; use uniqueness.
Key Insight: Autonomy makes elapsed time compose. □

1.2 Linear Systems and Stability

Scalar first-order linear equations admit an explicit evolution factor and a variation-of-constants representation.

Proposition 1.2.1. *Scalar Variation of Constants*
Let $I \subseteq \mathbb{R}$ be open, $a, g \in C(I, \mathbb{R})$, and $(t_0, x_0) \in I \times \mathbb{R}$.
Put $E(t,s) := \exp\left(\int_s^t a(r)\,dr\right)$ and $\phi(t) := E(t,t_0)x_0 + \int_{t_0}^t E(t,s)g(s)\,ds$. Then:

(i) $\phi \in C^1(I, \mathbb{R})$ and $(\phi' - a\phi, \phi(t_0)) = (g, x_0)$.
(ii) If $x \in C^1(I, \mathbb{R})$ and $(x' - ax, x(t_0)) = (g, x_0)$, then $x = \phi$.

Proof.
(i) Let $B(t) := x_0 + \int_{t_0}^t E(t_0, s)g(s)\,ds$. By additivity, $E(t,s) = E(t,t_0)E(t_0,s)$, so $\phi(t) = E(t,t_0)B(t)$.
By Theorem 2.2.1, $B \in C^1$, $\partial_t E(t,t_0) = a(t)E(t,t_0)$, and $B'(t) = E(t_0,t)g(t)$.
Hence $\phi \in C^1$, $\phi'(t) = a(t)\phi(t) + E(t,t_0)E(t_0,t)g(t) = a(t)\phi(t) + g(t)$, and $\phi(t_0) = x_0$.
(ii) If $(x' - ax, x(t_0)) = (g, x_0)$, then $\frac{d}{dt}(E(t_0,t)x(t)) = E(t_0,t)(x'(t) - a(t)x(t)) = E(t_0,t)g(t)$.
By Theorem 2.2.2, $E(t_0,t)x(t) = x_0 + \int_{t_0}^t E(t_0,s)g(s)\,ds$.
Multiplying by $E(t,t_0)$ and using $E(t,t_0)E(t_0,s) = E(t,s)$ gives $x(t) = \phi(t)$.

Structural Commentary: $E(t_0,t)$ removes $a(t)x$; integration accumulates g.
Key Insight: $(E(t_0,t)x)' = E(t_0,t)g$. □

Proposition 1.2.2. *Fiber-Preserving Change of Variables*
Let $\Omega, \widetilde{\Omega} \subseteq \mathbb{R}^2$ be open, $f : \Omega \to \mathbb{R}$, and $T : \Omega \to \widetilde{\Omega}$ be a C^1 diffeomorphism.
Assume $T(t,x) = (\sigma(t), \eta(t,x))$ and $T^{-1}(s,y) = (\tau(s), \xi(s,y))$. Put $G := \partial_t\eta + \partial_x\eta\, f$.
Define $\widetilde{f} : \widetilde{\Omega} \to \mathbb{R}$ by $\widetilde{f}(s,y) := \tau'(s)G(\tau(s), \xi(s,y))$.
Let $I \subseteq \mathbb{R}$ be an interval, $x \in C^1(I, \mathbb{R})$, $\Gamma_x := \{(t, x(t)) : t \in I\} \subseteq \Omega$, and $J := \sigma(I)$.
Set $y : J \to \mathbb{R}$ by $y(s) := \eta(\tau(s), x(\tau(s)))$. Then $y \in C^1(J, \mathbb{R})$, and $x' = f(\cdot\,, x)$ on I iff $y' = \widetilde{f}(\cdot\,, y)$ on J.

Proof. Since $\det DT(t,x) = \sigma'(t)\partial_x\eta(t,x) \neq 0$, $\sigma' \neq 0$, $\partial_x\eta \neq 0$, and $\tau'(\sigma(t))\sigma'(t) = 1$.

($\Rightarrow$) If $x' = f(\cdot\,, x)$ and $s \in J$, then $y'(s) = \tau'(s)G(\tau(s), x(\tau(s)))$. Since $\xi(s, y(s)) = x(\tau(s))$, $y'(s) = \widetilde{f}(s, y(s))$.
($\Leftarrow$) Let $t \in I$. Differentiating $y(\sigma(t)) = \eta(t, x(t))$ gives $\sigma'(t)y'(\sigma(t)) = \partial_t\eta(t,x(t)) + \partial_x\eta(t,x(t))x'(t)$.
Since $y' = \widetilde{f}(\cdot\,, y)$, $\sigma'(t)y'(\sigma(t)) = G(t, x(t))$. Hence $\partial_x\eta(t,x(t))[x'(t) - f(t,x(t))] = 0$, so $x' = f(\cdot\,, x)$.

Structural Commentary: Transform graphs; apply the chain rule.
Key Insight: $\widetilde{f}$ is the pushed-forward vector field. □

Method 1.2.3. *Homogeneous First-Order Reduction*
Let $D \subseteq \mathbb{R}$ be open, let $f \in C(D, \mathbb{R})$, and let $I \subseteq \mathbb{R} \setminus \{0\}$ be an interval. To solve $x' = f(x/t)$ on a branch for which $x(t)/t \in D$:

(i) Set $y := x/t$, so $x = ty$.
(ii) Differentiate to obtain $y' = (tx' - x)/t^2 = [f(y) - y]/t$.
(iii) Each $c \in D$ satisfying $f(c) = c$ gives the constant-ratio solution $x(t) = ct$.
(iv) On every subinterval where $f(y) \neq y$, separate $dy/(f(y) - y) = dt/t$, solve for y, and recover $x = ty$.

Method 1.2.4. *Affine-Fractional First-Order Reduction*
Let $D \subseteq \mathbb{R}$ be open, let $f \in C(D, \mathbb{R})$, put $A(t,x) := ax + bt + c$ and $B(t,x) := \alpha x + \beta t + \gamma$, and set $\Delta := a\beta - \alpha b$.
Work on a solution interval on which $B(t, x(t)) \neq 0$ and $A(t, x(t))/B(t, x(t)) \in D$.

(i) If $\Delta \neq 0$, there is a unique (t_*, x_*) with $A(t_*, x_*) = B(t_*, x_*) = 0$. Set $s := t - t_*$ and $y := x - x_*$.
Then $\frac{dy}{ds} = f(\frac{ay+bs}{\alpha y+\beta s})$; for $s \neq 0$, $z := y/s$ satisfies $s\frac{dz}{ds} = f(\frac{az+b}{\alpha z+\beta}) - z$.
(ii) If $\Delta = 0$ and $(a,b) \neq (0,0)$, choose the unique $\lambda \in \mathbb{R}$ with $(\alpha, \beta) = \lambda(a,b)$ and set $\widehat{f}(u) := f(\frac{u+c}{\lambda u+\gamma})$ wherever the quotient lies in D.
If $a \neq 0$, use $u := ax + bt$: $u' = a\widehat{f}(u) + b$, with $x = \frac{u-bt}{a}$. If $a = 0$, then $x' = \widehat{f}(bt)$.

Example 1.2.5. *Euler Equation Under Logarithmic Time*
For $t > 0$, set $s := \log t$ and $y(s) := x(e^s)$. Then $x'(t) = t^{-1}y'(s)$ and $x''(t) = t^{-2}(y''(s) - y'(s))$.
Hence Euler equation $t^2x'' + 3tx' + x = 2/t$ becomes constant-coefficient equation $y'' + 2y' + y = 2e^{-s}$.

Proposition 1.2.6. *Exact First-Order Equations*
Let $R := I \times J$ be an open rectangle, let $p, q \in C^1(R, \mathbb{R})$, and fix $(x_0, y_0) \in R$. Then:

(i) If $F \in C^2(R, \mathbb{R})$ satisfies $\partial_y F = p$ and $\partial_x F = q$, then $\partial_x p = \partial_y q$ on R.
(ii) If $\partial_x p = \partial_y q$ on R, then $F_0(x,y) := \int_{x_0}^x q(s, y_0)\,ds + \int_{y_0}^y p(x,r)\,dr$ belongs to $C^2(R, \mathbb{R})$ and satisfies $\partial_y F_0 = p$, $\partial_x F_0 = q$.
(iii) Under the hypothesis of (ii), every $F_1 \in C^2(R, \mathbb{R})$ satisfying $\partial_y F_1 = p$, $\partial_x F_1 = q$ has the form $F_1 = F_0 + c$ for a unique $c \in \mathbb{R}$.
(iv) Under the hypothesis of (ii), for every interval $L \subseteq I$ and every $y \in C^1(L, J)$, $p(\cdot\,, y)y' + q(\cdot\,, y) = 0$ iff $F_0(\cdot\,, y)$ is constant on L.

Proof.
(i) By Corollary 3.2.8, $\partial_x p = \partial_x\partial_y F = \partial_y\partial_x F = \partial_y q$.
(ii) The one-variable fundamental theorem gives $\partial_y F_0 = p$ and $\partial_x F_0 = q(x, y_0) + \int_{y_0}^y \partial_x p(x,r)\,dr$. By the hypothesis and the same theorem, $\partial_x F_0 = q(x,y_0) + \int_{y_0}^y \partial_y q(x,r)\,dr = q(x,y)$. Since these first partials are C^1, $F_0 \in C^2(R, \mathbb{R})$.

(iii) Set $H := F_1 - F_0$. Then $\partial_x H = \partial_y H = 0$. Applying the one-variable fundamental theorem first along the horizontal segment and then along the vertical segment gives $H(x,y) = H(x_0,y_0)$ for every $(x,y) \in R$. Thus $F_1 = F_0 + c$, and evaluation at (x_0,y_0) makes c unique.
(iv) The chain rule gives $(F_0(\,\cdot\,,y))' = q(\,\cdot\,,y) + p(\,\cdot\,,y)y'$. Therefore the differential equation holds exactly when this derivative vanishes, which on an interval is equivalent to constancy.

Structural Commentary: Build a potential; differentiate along graphs.
Key Insight: Mixed partials are the compatibility condition. □

Method 1.2.7. *Integrating Factor Depending Only on x*
Let $R := I \times J \subseteq \mathbb{R}^2$ be an open rectangle, let $p, q \in C^1(R,\mathbb{R})$, assume $p \neq 0$ on R, and fix $x_* \in I$. For $p(x,y)y' + q(x,y) = 0$:
(i) Set $r(x,y) := (\partial_y q - \partial_x p)/p$.
(ii) If $r(x,y) = r_0(x)$ on R, set $\mu(x) := \exp(\int_{x_*}^{x} r_0(s)\,ds)$.
(iii) Multiply by μ: $\partial_x(\mu p) = \mu(r_0 p + \partial_x p) = \mu \partial_y q = \partial_y(\mu q)$, so $\mu p\, y' + \mu q = 0$ is exact by Proposition 1.2.6.

For example, on either half-plane $x > 0$ or $x < 0$, $xy' + 3x - 2y = 0$ has $r_0(x) = -3/x$, so one may take the nonvanishing factor $\mu(x) = x^{-3}$.
A potential is $F(x,y) = y/x^2 - 3/x$, and the solutions are $y = 3x + Cx^2$.

Method 1.2.8. *Bernoulli Linearization*
Let $I \subseteq \mathbb{R}$ be open, let $a, b \in C(I,\mathbb{R})$, and let $n \in \mathbb{R} \setminus \{0,1\}$. On an interval $J \subseteq I$, consider a positive solution $x \in C^1(J,(0,+\infty))$ of $x' = a(t)x + b(t)x^n$.
(i) Set $y := x^{1-n}$.
(ii) Linearize: $y' = (1-n)x^{-n}x' = (1-n)a(t)y + (1-n)b(t)$ on J.
(iii) Solve $y' = \alpha y + \beta$ with $\alpha := (1-n)a$, $\beta := (1-n)b$ (Proposition 1.2.1); recover $x = y^{1/(1-n)}$ on $\{y > 0\}$.

Method 1.2.9. *Riccati Linearization from a Particular Solution*
Let $I \subseteq \mathbb{R}$ be open, $a, b, c \in C(I,\mathbb{R})$, and let $x_p \in C^1(I,\mathbb{R})$ solve $x_p' = ax_p + bx_p^2 + c$.
Let $x \in C^1(J,\mathbb{R})$ solve $x' = ax + bx^2 + c$ on an interval $J \subseteq I$ on which $x \neq x_p$.
(i) Set $u := x - x_p$ and $y := u^{-1}$ on J.
(ii) Subtract and invert: $u' = (a + 2bx_p)u + bu^2$, so $y' = -(a + 2bx_p)y - b$.
(iii) Solve $y' = \alpha y + \beta$ with $\alpha := -(a + 2bx_p)$, $\beta := -b$ (Proposition 1.2.1); recover $x = x_p + y^{-1}$ on $\{y \neq 0\}$.

Definition 1.2.10. *Matrix Exponential*
For $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and $t \in \mathbb{C}$, define $e^{tA} := \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}$.

Theorem 1.2.11. *Autonomous Matrix Evolution and Variation of Constants*
Let $A \in \mathbb{C}^{n\times n}$, $t_0 \in \mathbb{R}$, $x_0 \in \mathbb{C}^n$, and $g \in C(\mathbb{R},\mathbb{C}^n)$. For the matrix exponential from Definition 1.2.10, the following statements hold:
(i) $e^{0A} = I_n$, and $\forall s,t \in \mathbb{R}$, $e^{(t+s)A} = e^{tA}e^{sA}$, $(e^{tA})^{-1} = e^{-tA}$, and $\frac{d}{dt}e^{tA} = Ae^{tA} = e^{tA}A$.
(ii) $\exists!\, x \in C^1(\mathbb{R},\mathbb{C}^n)$ with $(x' - Ax, x(t_0)) = (0, x_0)$, namely $x(t) = e^{(t-t_0)A}x_0$.
(iii) $\exists!\, x \in C^1(\mathbb{R},\mathbb{C}^n)$ with $(x' - Ax, x(t_0)) = (g, x_0)$, namely $x(t) = e^{(t-t_0)A}x_0 + \int_{t_0}^{t} e^{(t-s)A}g(s)\,ds$.

Proof. Fix a submultiplicative matrix norm. Since $\|t^k A^k/k!\| \le |t|^k \|A\|^k/k!$, the defining series converges absolutely for every $t \in \mathbb{C}$. On bounded real t-intervals, it and its termwise derivative converge uniformly.
(i) The identities at $t = 0$ and for the derivative follow from the series. Absolute convergence permits the Cauchy product, and the binomial theorem gives $e^{tA}e^{sA} = \sum_{m=0}^{\infty} \frac{(t+s)^m A^m}{m!} = e^{(t+s)A}$. Taking $s = -t$ gives $e^{tA}e^{-tA} = I_n$.
(ii) Differentiation gives $x' = Ae^{(t-t_0)A}x_0 = Ax$ and $x(t_0) = x_0$.
If y is another solution, then $(e^{-(t-t_0)A}y(t))' = 0$, so $y(t) = e^{(t-t_0)A}x_0$.
(iii) Put $u(t) := e^{-(t-t_0)A}x(t)$. Then $u'(t) = e^{-(t-t_0)A}g(t)$, hence $u(t) = x_0 + \int_{t_0}^{t} e^{-(s-t_0)A}g(s)\,ds$. Multiplication by $e^{(t-t_0)A}$ and (i) give the formula. Conversely, differentiation verifies the equation and initial value; the difference of two solutions is zero by (ii).

Structural Commentary: The inverse exponential removes the homogeneous drift.
Key Insight: Matrix evolution transports forcing impulses from s to t. □

Proposition 1.2.12. *Pure Modes for Powers and Exponentials*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be diagonalizable with $A = S\Lambda S^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ and the columns of S are $\mathbf{v}_1,\dots,\mathbf{v}_n$. If $\mathbf{x}_0 = \sum_{j=1}^{n} c_j\mathbf{v}_j$, then $A^k\mathbf{x}_0 = \sum_{j=1}^{n} c_j\lambda_j^k\mathbf{v}_j$ for $k \ge 0$; and $e^{tA}\mathbf{x}_0 = \sum_{j=1}^{n} c_j e^{\lambda_j t}\mathbf{v}_j$ for $t \in \mathbb{C}$.

Proof. Let $c = (c_1,\dots,c_n)^t$, so $\mathbf{x}_0 = Sc$.
Then $A^k = S\Lambda^k S^{-1}$, and Definition 1.2.10 gives $e^{tA} = S\left(\sum_{r=0}^{\infty} \frac{t^r\Lambda^r}{r!}\right)S^{-1} = S\,\mathrm{diag}(e^{\lambda_1 t},\dots,e^{\lambda_n t})S^{-1}$.
Applying these two formulas to Sc gives the stated identities.
Structural Commentary: Eigenbasis coordinates decouple powers and exponentials.
Key Insight: Diagonalization turns coupled evolution into independent scalar modes. □

Theorem 1.2.13. *Principal Matrix, Liouville Formula, and Variation of Constants*
Let $A \in C(\mathbb{R},\mathbb{C}^{n\times n})$. Then $\exists!\,\Pi : \mathbb{R}^2 \to \mathbb{C}^{n\times n}$ such that $\forall s,t \in \mathbb{R}$, $\partial_t\Pi(t,s) = A(t)\Pi(t,s)$ and $\Pi(s,s) = I_n$. The following statements hold:
(i) $\forall r,s,t \in \mathbb{R}$, $\Pi(t,r)\Pi(r,s) = \Pi(t,s)$ and $\Pi(t,s)^{-1} = \Pi(s,t)$.
(ii) $\forall s,t \in \mathbb{R}$, $\det\Pi(t,s) = \exp\left(\int_s^t \mathrm{tr}\,A(r)\,dr\right)$.
(iii) If $g \in C(\mathbb{R},\mathbb{C}^n)$ and $x_s \in \mathbb{C}^n$, then $\exists!\, x \in C^1(\mathbb{R},\mathbb{C}^n)$ with $(x' - Ax, x(s)) = (g, x_s)$, namely $x(t) = \Pi(t,s)x_s + \int_s^t \Pi(t,r)g(r)\,dr$ for $t \in \mathbb{R}$.

The matrix $\Pi(t,s)$ is the principal matrix solution from s; any matrix solution $X' = AX$ with $\det X(s) \neq 0$ is a fundamental matrix solution.

Proof. Identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$. For the canonical basis $e_1, \dots, e_n$, Theorem 1.1.27 and Corollary 1.1.30 give a unique global solution of $x' = A(t)x$, $x(s) = e_j$; use these solutions as the columns of $\Pi(t,s)$. Existence and uniqueness of Π hold columnwise.

(i) For fixed r, s, both $t \mapsto \Pi(t,r)\Pi(r,s)$ and $t \mapsto \Pi(t,s)$ solve $X' = A(t)X$ and agree at $t = r$; hence they coincide. Applying the cocycle law to the triples (s,t,s) and (t,s,t) gives $I_n = \Pi(s,t)\Pi(t,s) = \Pi(t,s)\Pi(s,t)$, so $\Pi(t,s)^{-1} = \Pi(s,t)$. If $X(t) := \Pi(t,0)$, then $\Pi(t,s) = X(t)X(s)^{-1}$; hence Π is jointly continuous.

(ii) Let $X(t) := \Pi(t,s)$ and write its columns as $x_1(t), \dots, x_n(t)$. The Leibniz formula and $x_j' = Ax_j$ give $\frac{d}{dt}\det X(t) = \sum_{j=1}^{n} \det(x_1, \dots, Ax_j, \dots, x_n)$ $\operatorname{tr}(A(t))\det X(t)$. Multiplying by $\exp(-\int_s^t \operatorname{tr} A(r)\,dr)$ and using $\det X(s) = 1$ gives the claimed formula.

(iii) Differentiating the asserted formula gives $x' = A(t)x + g(t)$ and $x(s) = x_s$. If x, y have the same data, then $z := x - y$ solves $z' = A(t)z$, $z(s) = 0$, so the defining property of Π gives $z(t) = \Pi(t,s)z(s) = 0$.

Structural Commentary: Columnwise existence builds Π; uniqueness forces its cocycle law.
Key Insight: $\Pi(t,s)$ transports states and forcing between arbitrary times. □

Method 1.2.14. *Solving $\mathbf{x}'(t) = A\mathbf{x}(t)$ by Jordan Form*
Let $A \in \mathbb{C}^{n\times n}$ and $\mathbf{x}_0 \in \mathbb{C}^n$. To solve $\mathbf{x}'(t) = A\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{x}_0$:

(i) By Theorem 6.2.1, choose $S \in GL_n(\mathbb{C})$ with $A = SJS^{-1}$, where $J = \bigoplus_{i=1}^{r} J_{k_i}(\lambda_i)$.
(ii) Set $\mathbf{y}(t) = S^{-1}\mathbf{x}(t)$ and $\mathbf{y}_0 = S^{-1}\mathbf{x}_0$. Then $\mathbf{y}'(t) = J\mathbf{y}(t)$ and $\mathbf{x}(t) = S\mathbf{y}(t)$.
(iii) Split $\mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_r)$ according to the Jordan blocks. Then $\mathbf{y}_i'(t) = J_{k_i}(\lambda_i)\mathbf{y}_i(t)$.
(iv) By Proposition 6.5.1, $\mathbf{y}_i(t) = e^{tJ_{k_i}(\lambda_i)}\mathbf{y}_i(0) = e^{\lambda_i t}\sum_{q=0}^{k_i-1} \frac{t^q}{q!} J_{k_i}(0)^q \mathbf{y}_i(0)$ for $1 \le i \le r$.
Differentiating verifies the block equation.
(v) Reassemble $\mathbf{y}(t)$ and transform back: $\mathbf{x}(t) = S\mathbf{y}(t)$.

Thus a size-k block with eigenvalue λ contributes entries of the form $e^{\lambda t}p(t)$, where $\deg p < k$.

Perspective 1.2.15. *Jordan Functions and Polynomial Factors*

For a Jordan block $J_k(\lambda) = \lambda I_k + N$ with $N^k = 0$, any polynomial f satisfies $f(J_k(\lambda)) = \sum_{r=0}^{k-1} \frac{f^{(r)}(\lambda)}{r!} N^r$.

The nilpotent part contributes derivative terms; for the exponential these become polynomial factors in t by Proposition 6.5.1. As $t \to +\infty$, $\operatorname{Re}\lambda < 0$ gives blockwise decay and $\operatorname{Re}\lambda > 0$ gives growth.
If $\operatorname{Re}\lambda = 0$, the block evolution is bounded iff $N = 0$.

Example 1.2.16. *Nilpotent Jordan Block Evolution*

Let $N := J_3(0) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, so $N^3 = 0$. By Proposition 6.5.1, $e^{tN} = I_3 + tN + \frac{t^2}{2}N^2 = \left(\begin{smallmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{smallmatrix}\right)$. Nilpotent Jordan blocks create polynomial, rather than exponential, growth along generalized-eigenvector chains.

Theorem 1.2.17. *Spectral Criteria for Autonomous Linear Stability*
Let $A \in \mathbb{C}^{n\times n}$ and set $\alpha(A) := \max\{\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\}$. Then:

(i) $e^{tA} \to 0$ *as* $t \to +\infty$ *iff* $\alpha(A) < 0$.
(ii) $\sup_{t\ge 0} \|e^{tA}\| < +\infty$ *iff* $\alpha(A) \le 0$ *and* $\forall \lambda \in \operatorname{spec}(A)$, $\operatorname{Re}\lambda = 0 \implies \dim\ker(A - \lambda I_n) = m_A(\lambda)$.
(iii) If $\alpha(A) < 0$, *then* $\forall \gamma \in (0, -\alpha(A))$, $\exists C_\gamma > 0$ *such that* $\|e^{tA}\| \le C_\gamma e^{-\gamma t}$ $\forall t \ge 0$.

Proof. Choose $S \in GL_n(\mathbb{C})$ with $A = SJS^{-1}$ and use Proposition 6.5.1 on every block $J_k(\lambda)$. Fixed similarity changes all norms by at most the factor $\|S\|\,\|S^{-1}\|$.

(i) Each block exponential is $e^{\lambda t}\sum_{r=0}^{k-1} \frac{t^r N^r}{r!}$. It tends to zero exactly when $\operatorname{Re}\lambda < 0$; if $\operatorname{Re}\lambda \ge 0$, its diagonal entry $e^{\lambda t}$ does not tend to zero.
(ii) If $\operatorname{Re}\lambda < 0$, every polynomial factor is dominated by $e^{\operatorname{Re}\lambda t}$ and the block is bounded. If $\operatorname{Re}\lambda = 0$, the block is bounded exactly when $N = 0$, equivalently $k = 1$. If $\operatorname{Re}\lambda > 0$, its diagonal is unbounded.
(iii) Fix $0 < \gamma < -\alpha(A)$. For a block of size k, $e^{\gamma t}\|e^{tJ_k(\lambda)}\|$ is bounded because every $t^r e^{(\operatorname{Re}\lambda + \gamma)t}$ tends to zero. Take the maximum block bound and include the fixed similarity factor.

Structural Commentary: Jordan blocks separate exponential rates from nilpotent polynomial factors.
Key Insight: Boundary-spectrum nilpotence is the sole obstruction to boundedness. □

Definition 1.2.18. *Hurwitz Matrix*
A matrix $A \in \mathbb{C}^{n\times n}$ is *Hurwitz* iff $\max\{\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\} < 0$.

Perspective 1.2.19. *Planar Linear Phase Classification*
For $x' = Ax$ with $A \in \mathbb{R}^{2\times 2}$, two real eigenvalues of negative sign give a sink, two of positive sign give a source, and opposite signs give a saddle.
A nonreal pair $\alpha \pm i\beta$, $\beta \neq 0$, gives a spiral sink when $\alpha < 0$, a spiral source when $\alpha > 0$, and a center when $\alpha = 0$. A defective repeated real eigenvalue λ adds a polynomial shear along its generalized eigenvector; it still gives decay for $\lambda < 0$ and growth for $\lambda > 0$. For $\lambda = 0$, $e^{tA} = I_2 + tA$ in the defective case, so generalized-eigenvector directions grow linearly while the eigenspace consists of equilibria. These labels describe the origin; similarity changes the coordinates of the phase portrait without changing its spectral type.

Theorem 1.2.20. *Basis for Constant-Coefficient Linear ODEs*
Let $p(z) := z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0 = \prod_{j=1}^{m}(z - \lambda_j)^{a_j}$ over $\mathbb{C}$, where the λ_j are distinct and $\sum_{j=1}^{m} a_j = n$. Then:

(i) The functions $x_{j,k}(t) := t^k e^{\lambda_j t}$, $1 \le j \le m$, $0 \le k < a_j$, form a basis of the complex solutions of $p(D)x = 0$, where $D = d/dt$.

(ii) If $c_0, \dots, c_{n-1} \in \mathbb{R}$, write $\lambda_j = \alpha_j + i\beta_j$. The functions $t^k e^{\lambda_j t}$ for the real roots $\beta_j = 0$, together with $t^k e^{\alpha_j t}\cos(\beta_j t)$ and $t^k e^{\alpha_j t}\sin(\beta_j t)$ for one root from each conjugate pair with $\beta_j > 0$, form a real basis, where always $0 \le k < a_j$.

Proof. For $q \in \mathbb{C}[t]$, $(D-\lambda)(e^{\lambda t}q) = e^{\lambda t}q'$. Thus $(D-\lambda_j)^{a_j} x_{j,k} = 0$ for $k < a_j$, so every listed function solves $p(D)x = 0$.

(i) Suppose $\sum_{j=1}^{m} e^{\lambda_j t} q_j(t) = 0$ with $\deg q_j < a_j$. Fix j and apply $\prod_{i \ne j} (D-\lambda_i)^{a_i}$. All terms except the jth vanish, while the remaining term is $e^{\lambda_j t} R_j(D) q_j$, where $R_j(z) := \prod_{i \ne j} (z + \lambda_j - \lambda_i)^{a_i}$ and $R_j(0) \ne 0$.
On polynomials of degree below a_j, $R_j(D) - R_j(0)I$ is nilpotent because it is divisible by D. Thus $R_j(D)$ is invertible, so $q_j = 0$. The n listed functions are linearly independent.
Writing $y := (x, x', \dots, x^{(n-1)})$ converts $p(D)x = 0$ into a first-order companion system. Theorem 1.2.13(iii), with zero forcing, makes $y(0) \mapsto y$ a complex-linear bijection from $\mathbb{C}^n$ onto the global solution space. Thus that space has complex dimension n and the list is a basis.
(ii) Real coefficients pair nonreal roots by conjugation. Taking real and imaginary parts of the paired complex basis functions gives the stated real solutions; the change from each conjugate pair to its real and imaginary parts is invertible over $\mathbb{R}$.

Structural Commentary: Factor $p(D)$; isolate each exponential-polynomial root space.
Key Insight: Root multiplicity becomes polynomial degree. □

Method 1.2.21. *Exponential-Polynomial Forcing*
Let $p \in \mathbb{C}[z] \setminus \{0\}$, $q \in \mathbb{C}[t] \setminus \{0\}$, and $\beta \in \mathbb{C}$, and consider $p(D)x = q(t)e^{\beta t}$. Write $p(z) = (z-\beta)^a h(z)$ with $a \in \mathbb{N}_0$ and $h(\beta) \ne 0$. Set $d := \deg q$ and $\mathbb{C}_{\le d}[t] := \{u \in \mathbb{C}[t] : \deg u \le d\}$.

(i) Solve $h(D+\beta)u = q$ in $\mathbb{C}_{\le d}[t]$; the operator is triangular with nonzero diagonal $h(\beta)$ and hence invertible.
(ii) Choose the polynomial r with $r^{(a)} = u$ and vanishing coefficients in degrees $0, \dots, a-1$.
(iii) Set $x_p(t) := e^{\beta t} r(t)$. Then $p(D)x_p = e^{\beta t} h(D+\beta) D^a r = q(t)e^{\beta t}$ and $\deg r \le \deg q + a$.

The factor t^a in r is the resonance correction when β is a root of p.

Example 1.2.22. *Damping and Resonance*
For $x'' + 2\eta x' + \omega_0^2 x = 0$, with $\eta, \omega_0 > 0$, the roots are $-\eta \pm \sqrt{\eta^2 - \omega_0^2}$. Thus $\eta > \omega_0$, $\eta = \omega_0$, and $\eta < \omega_0$ give over-damping, critical damping, and under-damping, respectively; homogeneous solutions decay.
For the forced equation $x'' + 2\eta x' + \omega_0^2 x = \cos(\omega t)$, a complex particular solution is $z_p(t) = e^{i\omega t}/(\omega_0^2 - \omega^2 + 2i\eta\omega)$, and $x_p = \operatorname{Re} z_p$. If $\eta = 0$ and $\omega = \omega_0$, Method 1.2.21 instead gives $x_p(t) = t\sin(\omega_0 t)/(2\omega_0)$, whose amplitude grows linearly.
For a series RLC circuit, $\eta = R/(2L)$ and $\omega_0 = 1/\sqrt{LC}$; the same three regimes and resonance mechanism govern the current.

Method 1.2.23. *Reduction of Order for a Second Solution*
Let $I \subseteq \mathbb{R}$ be open, let $q_0, q_1 \in C(I, \mathbb{C})$, let $\phi_1 \in C^2(I, \mathbb{C})$ solve $x'' + q_1 x' + q_0 x = 0$, and restrict to an interval $J \subseteq I$ on which $\phi_1 \ne 0$. Fix $t_* \in J$.

(i) Set $x = c\phi_1$. Cancellation of the equation for ϕ_1 gives $\phi_1 c'' + (2\phi_1' + q_1\phi_1)c' = 0$.
(ii) With $d := c'$, solve $d' + (2\phi_1'/\phi_1 + q_1)d = 0$ to obtain $d(t) = C\phi_1(t)^{-2}\exp(-\int_{t_*}^{t} q_1(r)\,dr)$.
(iii) Taking $C = 1$ gives $\phi_2(t) := \phi_1(t)\int_{t_*}^{t} \phi_1(s)^{-2}\exp(-\int_{t_*}^{s} q_1(r)\,dr)\,ds$. Then $W(\phi_1, \phi_2)(t) = \phi_1\phi_2' - \phi_1'\phi_2 = \exp(-\int_{t_*}^{t} q_1(r)\,dr) \ne 0$. Thus ϕ_1, ϕ_2 are linearly independent; uniqueness for the associated two-dimensional first-order system makes them a basis of the solution space on J.

Theorem 1.2.24. *Floquet Normal Form*
Let $A \in C(\mathbb{R}, \mathbb{C}^{n\times n})$ satisfy $A(t+T) = A(t)$ for some $T > 0$, let Π be its principal matrix solution, and set $M(t_0) := \Pi(t_0 + T, t_0)$. Then:

(i) $\forall s, t \in \mathbb{R}$, $\Pi(t+T, s+T) = \Pi(t, s)$.
(ii) $M(t_0)$ is invertible.
(iii) $\exists Q(t_0) \in \mathbb{C}^{n\times n}$ and a T-periodic $P(\,\cdot\,, t_0) \in C^1(\mathbb{R}, GL_n(\mathbb{C}))$ such that $\Pi(t, t_0) = P(t, t_0)e^{(t-t_0)Q(t_0)}$ and $P(t_0, t_0) = I_n$.
(iv) $\forall t_0, t_1 \in \mathbb{R}$, $M(t_1) = \Pi(t_1, t_0)M(t_0)\Pi(t_1, t_0)^{-1}$; consequently, the spectrum and Jordan block sizes of $M(t_0)$ are independent of t_0.

Proof.

(i) Both $t \mapsto \Pi(t+T, s+T)$ and $t \mapsto \Pi(t, s)$ solve $X' = A(t)X$ and equal I_n at $t = s$, so uniqueness gives the identity.
(ii) Theorem 1.2.13(i) gives $M(t_0)^{-1} = \Pi(t_0, t_0 + T)$.
(iii) By (ii), every Jordan block of $M(t_0)$ has the form $J = \rho I + N$ with $\rho \ne 0$ and $N^k = 0$. Choose $\ell \in \mathbb{C}$ with $e^\ell = \rho$ and set $L := \ell I + \sum_{q=1}^{k-1} \frac{(-1)^{q+1}}{q}(\rho^{-1}N)^q$. In the quotient algebra $\mathbb{C}[z]/(z^k)$, the finite logarithm and exponential series satisfy $\exp(\log(1+z)) = 1 + z$; substituting $z = \rho^{-1}N$ gives $e^L = J$. Reassembling and conjugating these block logarithms produces $R \in \mathbb{C}^{n\times n}$ with $e^R = M(t_0)$.
Set $Q(t_0) := T^{-1}R$ and $P(t, t_0) := \Pi(t, t_0)e^{-(t-t_0)Q(t_0)}$. Then $P(t_0, t_0) = I_n$. By (i) and the cocycle law, $\Pi(t+T, t_0) = \Pi(t, t_0)M(t_0)$; because $M(t_0) = e^{TQ(t_0)}$, this identity gives $P(t+T, t_0) = P(t, t_0)$. Both factors defining P are C^1 and invertible.
(iv) Periodicity from (i) and the cocycle law give $\Pi(t_1 + T, t_0) = \Pi(t_1, t_0)M(t_0)$ and also $\Pi(t_1 + T, t_0) = M(t_1)\Pi(t_1, t_0)$. Right multiplication by $\Pi(t_1, t_0)^{-1}$ proves the similarity identity, and similarity preserves the spectrum and Jordan block sizes.

Structural Commentary: Take a logarithm of one-period transport; factor the remainder.
Key Insight: Periodic evolution is constant exponential growth in periodic coordinates. □

Definition 1.2.25. *Floquet Multipliers and Exponents*
Let $A \in C(\mathbb{R}, \mathbb{C}^{n\times n})$ be T-periodic, let Π be its principal matrix solution, set $M(t_0) := \Pi(t_0 + T, t_0)$, and choose $Q(t_0)$ with $e^{TQ(t_0)} = M(t_0)$.

(i) The eigenvalues ρ of $M(t_0)$ are the *Floquet multipliers*.

(ii) The eigenvalues γ of a Floquet matrix $Q(t_0)$ are *Floquet exponents*. Spectral mapping gives $\operatorname{spec}(M(t_0)) = \{e^{T\gamma} : \gamma \in \operatorname{spec}(Q(t_0))\}$ and $m_{M(t_0)}(\rho) = \sum_{\substack{\gamma \in \operatorname{spec}(Q(t_0)) \\ e^{T\gamma}=\rho}} m_{Q(t_0)}(\gamma)$. Thus every exponent determines a multiplier by $\rho = e^{T\gamma}$, and the scalar logarithm determines γ only modulo $2\pi i/T$.

Corollary 1.2.26. *Floquet Multiplier Stability Criterion*
Let $A \in C(\mathbb{R}, \mathbb{C}^{n\times n})$ be T-periodic, let Π be its principal matrix solution, and set $M(t_0) := \Pi(t_0 + T, t_0)$. Then:

(i) $\sup_{t\ge t_0} \|\Pi(t, t_0)\| < +\infty$ *iff* $|\rho| \le 1$ *for* $\rho \in \operatorname{spec}(M(t_0))$ *and* $\dim\ker(M(t_0) - \rho I_n) = m_{M(t_0)}(\rho)$ *whenever* $|\rho| = 1$.
(ii) $\Pi(t, t_0) \to 0$ *as* $t \to +\infty$ *iff* $\max\{|\rho| : \rho \in \operatorname{spec}(M(t_0))\} < 1$.

Proof. Write $t = t_0 + kT + \tau$, $k \in \mathbb{N}_0$, $0 \le \tau < T$. Periodicity and the cocycle law give by induction $\Pi(t, t_0) = \Pi(t_0 + \tau, t_0)M(t_0)^k$. The first factor and its inverse are uniformly bounded for $0 \le \tau \le T$.

(i) Hence $\Pi(\cdot\,, t_0)$ is bounded iff $(M(t_0)^k)$ is bounded. By Theorem 3.3.7, this is equivalent to the stated multiplier and multiplicity conditions.
(ii) Likewise, $\Pi(t, t_0) \to 0$ iff $M(t_0)^k \to 0$. By Theorem 3.3.5, this is equivalent to $\rho(M(t_0)) < 1$.

Structural Commentary: Sample at integer periods; reduce continuous evolution to powers of M.
Key Insight: Floquet stability is matrix-power stability of one-period transport. □

Example 1.2.27. *Hill Equation and Its Floquet Discriminant*
Let $q \in C(\mathbb{R}, \mathbb{R})$ be T-periodic and consider $x'' + q(t)x = 0$. Its first-order matrix is $A(t) = \left(\begin{smallmatrix} 0 & 1 \\ -q(t) & 0 \end{smallmatrix}\right)$. Let Π be the principal matrix solution and $M(t_0) := \Pi(t_0 + T, t_0)$. Since $\operatorname{tr} A(t) = 0$, by Theorem 1.2.13(ii), $\det M(t_0) = 1$.
With $\Delta := \frac{1}{2}\operatorname{tr} M(t_0)$, the multiplier equation is $\rho^2 - 2\Delta\rho + 1 = 0$. If $|\Delta| < 1$, the multipliers are distinct conjugates on the unit circle and all solutions are bounded. If $|\Delta| > 1$, one real multiplier has modulus above 1 and an unbounded solution exists. At $|\Delta| = 1$, boundedness holds exactly when $M(t_0) = \Delta I_2$; otherwise its nontrivial Jordan block produces linear growth in the period count.
The Mathieu equation $x'' + \omega^2(1 + \varepsilon\cos t)x = 0$ is the standard periodic-coefficient example; its stability regions are determined by $\Delta(\omega, \varepsilon)$.

Theorem 1.2.28. *Perturbation Bounds for Linear Evolution*
Let $A, B \in C(\mathbb{R}, \mathbb{C}^{n\times n})$, let Π_A, Π_{A+B} be the corresponding principal matrix solutions, and fix $t_0 \in \mathbb{R}$. Then:

(i) If $\|\Pi_A(t, s)\| \le Ce^{-\alpha(t-s)}$ *for all* $t \ge s \ge t_0$ *and* $\|B(t)\| \le b$ *for all* $t \ge t_0$, *where* $C, \alpha > 0$, $b \ge 0$, *and* $Cb < \alpha$, *then* $\|\Pi_{A+B}(t, s)\| \le Ce^{-(\alpha - Cb)(t-s)}$ *for all* $t \ge s \ge t_0$.
(ii) If $\|\Pi_A(t, s)\| \le C\ \forall t \ge s \ge t_0$ *and* $B_1 := \int_{t_0}^{\infty} \|B(r)\|\,dr < +\infty$, *then* $\|\Pi_{A+B}(t, s)\| \le Ce^{CB_1}\ \forall t \ge s \ge t_0$.

Proof. For fixed $s \ge t_0$, by Theorem 1.2.13(iii), applied columnwise with forcing $B(t)x(t)$, $\Pi_{A+B}(t, s) = \Pi_A(t, s) + \int_s^t \Pi_A(t, r)B(r)\Pi_{A+B}(r, s)\,dr$.

(i) Put $y(t) := e^{\alpha(t-s)}\,\|\Pi_{A+B}(t, s)\|$. The hypotheses give $y(t) \le C + Cb\int_s^t y(r)\,dr$. By Lemma 1.1.20(ii), after translating s to 0, $y(t) \le Ce^{Cb(t-s)}$, which is the asserted estimate.
(ii) Put $y(t) := \|\Pi_{A+B}(t, s)\|$. Then $y(t) \le C + C\int_s^t \|B(r)\|\,y(r)\,dr$. The same Gronwall estimate gives $y(t) \le C\exp(C\int_s^t \|B(r)\|\,dr) \le Ce^{CB_1}$.

Structural Commentary: Variation of constants converts perturbation into Gronwall feedback.
Key Insight: Perturbation mass subtracts from decay or multiplies boundedness. □

Corollary 1.2.29. *Vanishing Perturbations of a Hurwitz Matrix*
Let $A \in \mathbb{C}^{n\times n}$ be Hurwitz, let $B \in C(\mathbb{R}, \mathbb{C}^{n\times n})$, and suppose $\|B(t)\| \to 0$ as $t \to +\infty$. Set $\delta_A := \min\{-\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\}$. Then $\forall \gamma \in (0, \delta_A)$, $\exists K_\gamma > 0$ such that $\|\Pi_{A+B}(t, 0)\| \le K_\gamma e^{-\gamma t}\ \forall t \ge 0$.

Proof. Fix $0 < \gamma < \eta < \delta_A$. By Theorem 1.2.17(iii), $\exists C_\eta > 0$ with $\left\|e^{(t-s)A}\right\| \le C_\eta e^{-\eta(t-s)}$ for $t \ge s$.
Choose $t_0 \ge 0$ such that $\|B(t)\| \le (\eta - \gamma)/C_\eta$ for $t \ge t_0$. By Theorem 1.2.28(i), $\|\Pi_{A+B}(t, t_0)\| \le C_\eta e^{-\gamma(t-t_0)}$ for $t \ge t_0$.
The cocycle law yields $\Pi_{A+B}(t, 0) = \Pi_{A+B}(t, t_0)\Pi_{A+B}(t_0, 0)$.
Set $K_1 := C_\eta e^{\gamma t_0}\,\|\Pi_{A+B}(t_0, 0)\|$; then $\|\Pi_{A+B}(t, 0)\| \le K_1 e^{-\gamma t}$ for $t \ge t_0$.
Continuity gives $K_0 := \max_{0\le t\le t_0} e^{\gamma t}\,\|\Pi_{A+B}(t, 0)\| < +\infty$. Taking $K_\gamma := \max\{K_0, K_1\}$ proves the claim.
Structural Commentary: Wait until B is smaller than the spectral decay margin.
Key Insight: A vanishing perturbation eventually preserves any smaller decay rate. □

1.3 Nonlinear Stability Theory

Exact formulas are often unnecessary for proving that scalar trajectories remain ordered or trapped. Differential inequalities provide comparison curves for this purpose.

Definition 1.3.1. *Upper and Lower Solutions*
Let $U \subseteq \mathbb{R}^2$ be open, $f : U \to \mathbb{R}$, and $J \subseteq \mathbb{R}$ an open interval. For $u \in C^1(J, \mathbb{R})$ with $\Gamma_u := \{(t, u(t)) : t \in J\} \subseteq U$:

(i) *Upper solution*: $u'(t) \ge f(t, u(t))\ \forall t \in J$; it is *strict* iff $u'(t) > f(t, u(t))\ \forall t \in J$.
(ii) *Lower solution*: $u'(t) \le f(t, u(t))\ \forall t \in J$; it is *strict* iff $u'(t) < f(t, u(t))\ \forall t \in J$.

Key Insight: Differential inequalities encode one-sided barriers without solving the ODE.

Definition 1.3.2. *Scalar Isocline*
Let $U \subseteq \mathbb{R}^2$ be open, $f : U \to \mathbb{R}$, and $c \in \mathbb{R}$.
For $x' = f(t, x)$, the set $\mathcal{I}_c := \{(t, x) \in U : f(t, x) = c\}$ is the *isocline of slope c*.
Let $J \subseteq \mathbb{R}$ be open and $w \in C^1(J, \mathbb{R})$ satisfy $\Gamma_w := \{(t, w(t)) : t \in J\} \subseteq \mathcal{I}_c$.

(i) If $w'(t) > c\ \forall t \in J$, then w is a strict upper solution.
(ii) If $w'(t) < c\ \forall t \in J$, then w is a strict lower solution.

Key Insight: An isocline fixes the ODE slope, so only the graph slope remains to be compared.

Lemma 1.3.3. *Strict Scalar Comparison*
Let $U \subseteq \mathbb{R}^2$ be open, $f \in C(U,\mathbb{R})$, $J \subseteq \mathbb{R}$ open, and $[t_0,T] \subseteq J$ with $t_0 < T$.
For $z \in C^1(J,\mathbb{R})$, set $\Gamma_z := \{(t,z(t)) : t \in J\}$, and let $I_+ := (t_0,T]$. Suppose $x \in C^1(J,\mathbb{R})$, $\Gamma_x \subseteq U$, and $x' = f(\cdot\,,x)$.

(i) If $u \in C^1(J,\mathbb{R})$, $\Gamma_u \subseteq U$, $u'(t) > f(t,u(t))$ $\forall t \in J$, and $x(t_0) \le u(t_0)$, then $x(t) < u(t)$ $\forall t \in I_+$.
(ii) If $v \in C^1(J,\mathbb{R})$, $\Gamma_v \subseteq U$, $v'(t) < f(t,v(t))$ $\forall t \in J$, and $v(t_0) \le x(t_0)$, then $v(t) < x(t)$ $\forall t \in I_+$.

Proof.

(i) Set $\Delta := u - x$. At every $t \in [t_0,T]$ with $\Delta(t) = 0$, one has $x(t) = u(t)$ and $\Delta'(t) = u'(t) - f(t,u(t)) > 0$. Thus $\Delta(t_0) \ge 0$ implies $\Delta > 0$ on $(t_0, t_0+\varepsilon]$ for some $\varepsilon > 0$.
If $Z := \{t \in (t_0,T] : \Delta(t) = 0\} \neq \emptyset$, let $t_1 := \inf Z$. Continuity gives $\Delta(t_1) = 0$ and $\Delta > 0$ on (t_0,t_1), so the left derivative satisfies $\Delta'(t_1) \le 0$. This contradicts $\Delta'(t_1) > 0$. Hence $Z = \emptyset$ and $\Delta > 0$ on I_+.
(ii) Set $\Delta := x - v$. At every zero of Δ, $\Delta' = f(\cdot\,,v) - v' > 0$. The argument from (i), with $x - v$ in place of $u - x$, gives $\Delta > 0$ on I_+.

Structural Commentary: A first contact forces $\Delta' \le 0$, contradicting the strict residual sign.
Key Insight: Strict differential inequalities forbid barrier crossing. □

Definition 1.3.4. *Local State Lipschitz Condition*
Let $U \subseteq \mathbb{R}^2$ be open and $f : U \to \mathbb{R}$. Then f is *locally Lipschitz in the state, uniformly in time* iff, for every compact $K \subseteq U$, there exists $L_K < +\infty$ such that $|f(t,x) - f(t,y)| \le L_K |x - y|$ whenever $(t,x),(t,y) \in K$.

Theorem 1.3.5. *Scalar Lipschitz Comparison*
Let $U \subseteq \mathbb{R}^2$ be open and $f \in C(U,\mathbb{R})$. Assume that for every compact $K \subseteq U$, there exists $L_K < +\infty$ such that $|f(t,p) - f(t,q)| \le L_K |p - q|$ whenever $(t,p),(t,q) \in K$.
Let $I := [t_0,T] \subseteq J$, $J \subseteq \mathbb{R}$ open, $t_0 < T$. For $z : J \to \mathbb{R}$, set $\Gamma_z(A) := \{(t,z(t)) : t \in A\}$, $R_z := z' - f(\cdot\,,z)$.
Suppose $x, y \in C^1(J,\mathbb{R})$, $\Gamma_x(J) \cup \Gamma_y(J) \subseteq U$, and $R_x \le R_y$ on I. Then:

(i) If $x(t_0) \le y(t_0)$, then $x \le y$ on I.
(ii) If $t_1 \in I$ and $x(t_1) < y(t_1)$, then $x < y$ on $[t_1,T]$.

Proof.

(i) Set $\Delta := x - y$ and suppose $S := \{t \in I : \Delta(t) > 0\} \neq \emptyset$. Since $\Delta(t_0) \le 0$, every component C of S has a left endpoint $\alpha \in I$ with $\Delta(\alpha) = 0$. Choose $t_2 \in C$, so $\Delta > 0$ on $(\alpha,t_2]$.
With $K := \Gamma_x([\alpha,t_2]) \cup \Gamma_y([\alpha,t_2])$ and $L = L_K$, $R_x \le R_y$ gives $\Delta' \le f(\cdot\,,x) - f(\cdot\,,y) \le L\Delta$ on $(\alpha,t_2]$.
Thus $(e^{-L(t-\alpha)}\Delta)' \le 0$, so $\Delta(t_2) \le e^{L(t_2-\alpha)}\Delta(\alpha) = 0$, contradicting $t_2 \in S$.
(ii) Set $\delta := y - x$. If $Z := \{t \in (t_1,T] : \delta(t) = 0\} \neq \emptyset$, let $t_2 := \inf Z$. Then $\delta > 0$ on $[t_1,t_2)$ and $\delta(t_2) = 0$.
With $K := \Gamma_x([t_1,t_2]) \cup \Gamma_y([t_1,t_2])$ and $L = L_K$, by $R_x \le R_y$, $\delta' \ge f(\cdot\,,y) - f(\cdot\,,x) \ge -L\delta$ on $[t_1,t_2)$.
Hence $(e^{L(t-t_1)}\delta)' \ge 0$ and $0 = e^{L(t_2-t_1)}\delta(t_2) \ge \delta(t_1) > 0$, contradiction.

Structural Commentary: On a positive component, the integrating factor makes the order defect nonincreasing.
Key Insight: A nonpositive order defect cannot become positive. □

Lemma 1.3.6. *A Convergent Autonomous State Is an Equilibrium*
Let $f \in C(\mathbb{R},\mathbb{R})$ and let $x \in C^1([t_0,+\infty),\mathbb{R})$ satisfy $x' = f(x)$. If $x(t) \to x_* \in \mathbb{R}$ as $t \to +\infty$, then:

(i) $x'(t) \to 0$.
(ii) $f(x_*) = 0$.

Proof.

(i) By continuity, $x'(t) = f(x(t)) \to f(x_*) := L$. If $L > 0$, then $x'(t) \ge L/2$ eventually, so integration forces $x(t) \to +\infty$. If $L < 0$, the analogous estimate forces $x(t) \to -\infty$. Both conclusions contradict $x(t) \to x_*$, so $L = 0$ and $x'(t) \to 0$.
(ii) The identity $L = f(x_*)$ from (i) now gives $f(x_*) = 0$.

Structural Commentary: The differential equation converts state convergence into derivative convergence.
Key Insight: A nonzero limiting velocity is incompatible with a finite limiting state. □

Example 1.3.7. *A Trapping Wedge for $x' = x^2 - t^2$*
For $f(t,x) := x^2 - t^2$ on $(0,+\infty) \times \mathbb{R}$, set $u(t) := t$ and $v(t) := -t$.
Then $u' = 1 > 0 = f(t,u)$ and $v' = -1 < 0 = f(t,v)$.
If $t_0 > 0$, $x_0 \in [-t_0,t_0]$, and $x \in C^1([t_0,T_+),\mathbb{R})$ satisfies $x' = x^2 - t^2$ and $x(t_0) = x_0$, then Lemma 1.3.3 gives $-t < x(t) < t$ $\forall t \in (t_0,T_+)$.

Proposition 1.3.8. *Central-Wedge Convergence for $x' = x^2 - t^2$*
Let $t_0 > 0$, $-t_0 < x_0 < t_0$, and let $x : [t_0,T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = x^2 - t^2$, $x(t_0) = x_0$.

(i) $-t < x(t) < t$ for $t \in (t_0,T_+)$.
(ii) $T_+ = +\infty$.
(iii) $x(t) + t \to 0$ as $t \to +\infty$.

Proof.

(i) Let $u(t) := t$ and $v(t) := -t$. Since $u' - (u^2 - t^2) = 1 > 0$, $v' - (v^2 - t^2) = -1 < 0$, and $v(t_0) < x_0 < u(t_0)$, Lemma 1.3.3 implies $v(t) < x(t) < u(t)$ on every compact subinterval of $[t_0,T_+)$.
(ii) By (i), $x' = (x - t)(x + t) < 0$, so x is decreasing.
If $T_+ < +\infty$, then $-T_+ < x(t) \le x_0$ for $t < T_+$, hence $\ell := \lim_{t \uparrow T_+} x(t) \in [-T_+, x_0]$.
Lemma 1.1.28 extends x through (T_+,ℓ), contradicting maximality. Hence $T_+ = +\infty$.

(iii) Fix $T \geq \max\{t_0, 2\}$, and set $r(t) := \sqrt{t^2-2}$, $w(t) := -r(t)$ for $t \geq T$. Since $r^2 - t^2 = w^2 - t^2 = -2$, $r' = t/r > 0 > -2$, and $w' = -t/r \geq -\sqrt{2} > -2$, both r, w are strict upper solutions.
Let $d(t) := t - x(t)$. By (i), $d > 0$ and $t + x > 0$, so $d' = 1 + t^2 - x^2 = 1 + d(t+x) > 1$.
Since $d(t) \to +\infty$ and $t - r(t) \to 0$, choose $T_1 \geq T$ with $d(T_1) > T_1 - r(T_1)$, i.e. $x(T_1) < r(T_1)$.
By strict comparison, $x < r$ for $t > T_1$. If $x(T_1) \leq w(T_1)$, then by strict comparison $x < w$ for $t > T_1$.
If $w(T_1) < x(T_1) < r(T_1)$, then while $w < x < r$, one has $x' < -2$ and $w' \geq -\sqrt{2}$, hence $(x-w)' < -(2-\sqrt{2})$. Global existence forces a first contact $x(\tau) = w(\tau)$ in finite time, and strict comparison from τ gives $x < w$ thereafter.
Therefore eventually $0 < x(t) + t < t - r(t) = \frac{2}{t+\sqrt{t^2-2}} \to 0$.

Structural Commentary: Two nested isocline barriers squeeze the trajectory against $x = -t$.
Key Insight: Every trajectory entering the central wedge is asymptotic to its lower edge. □

Proposition 1.3.9. *Finite-Time Blow-Up Above the Upper Isocline*
Let $t_0 \geq 0$, $x_0 \geq \sqrt{t_0^2+2}$, and let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = x^2 - t^2$, $x(t_0) = x_0$.

(i) $x(t) > \sqrt{t^2+2}$ and $x'(t) > 2$ for $t \in (t_0, T_+)$.
(ii) $T_+ < +\infty$ and $x(t) \to +\infty$ as $t \uparrow T_+$.

Proof.

(i) Set $b(t) := \sqrt{t^2+2}$. Then $b^2 - t^2 = 2$ and $b' = \frac{t}{b} < 1 < 2$, so $b' < b^2 - t^2$.
Since $x_0 \geq b(t_0)$, Lemma 1.3.3 implies $b(t) < x(t)$ for $t > t_0$; hence $x' = x^2 - t^2 > b^2 - t^2 = 2$.

(ii) Suppose $T_+ = +\infty$. By (i), $x(t) > x_0 + 2(t - t_0)$; choose $t_1 \geq t_0$ with $x(t) > \sqrt{2}t > 0$ for $t \geq t_1$.
Then $x' = x^2 - t^2 > \frac{1}{2}x^2$ and $(\frac{1}{x})' = -\frac{x'}{x^2} < -\frac{1}{2}$ on $[t_1, +\infty)$, so $\frac{1}{x(t)} < \frac{1}{x(t_1)} - \frac{1}{2}(t - t_1)$.
This contradicts $x > 0$ for $t > t_1 + \frac{2}{x(t_1)}$. Thus $T_+ < +\infty$. Also $x' > 2$, so $\ell := \lim_{t \uparrow T_+} x(t) \in (x_0, +\infty]$.
If $\ell < +\infty$, Lemma 1.1.28 extends x through (T_+, ℓ), contradicting maximality. Hence $\ell = +\infty$.

Structural Commentary: Above $b(t) = \sqrt{t^2+2}$, linear escape enters the Riccati regime.
Key Insight: Once $x' \geq \frac{1}{2}x^2$, reciprocal decay forces finite-time escape. □

Theorem 1.3.10. *Separatrix Classification for $x' = x^2 - t^2$*
For $a \in \mathbb{R}$, let $x_a : [0, T_a) \to \mathbb{R}$ be the maximal forward solution of $x' = x^2 - t^2$, $x_a(0) = a$.

(i) $\exists!\, a_* \in [0, \sqrt{2}]$ such that $T_{a_*} = +\infty$, $t \leq x_{a_*}(t) \leq \sqrt{t^2+2}$ for $t \geq 0$, and $x_{a_*}(t) - t \to 0$ as $t \to +\infty$.
(ii) If $a > a_*$, then $T_a < +\infty$ and $x_a(t) \to +\infty$ as $t \uparrow T_a$.
(iii) If $a < a_*$, then $T_a = +\infty$ and $x_a(t) + t \to 0$ as $t \to +\infty$.

Proof. Set $\ell(t) := t$ and $b(t) := \sqrt{t^2+2}$. For $T > 0$, let p_T, q_T solve $x' = x^2 - t^2$ backward from $p_T(T) = \ell(T)$ and $q_T(T) = b(T)$. Under $s = T - t$, the reversed equation is $y' = (T-s)^2 - y^2$, while $\widehat{\ell}(s) := T - s$ and $\widehat{b}(s) := \sqrt{(T-s)^2+2}$ satisfy $\widehat{\ell}' < (T-s)^2 - \widehat{\ell}^2$ and $\widehat{b}' > (T-s)^2 - \widehat{b}^2$.
Lemma 1.3.3 and Theorem 1.3.5 trap and order the reversed solutions while they exist. Corollary 1.1.29 then gives existence on $[0, T]$, with $\ell \leq p_T < q_T \leq b$ on $[0, T]$ and strict outer inequalities on $[0, T)$.
Set $\alpha_T := p_T(0)$, $\beta_T := q_T(0)$, and $K_T := [\alpha_T, \beta_T]$. Order preservation gives $a \in K_T$ iff x_a exists on $[0, T]$ and $\ell \leq x_a \leq b$ there. Thus $S \geq T$ implies $K_S \subseteq K_T$, and $(K_n)_{n \geq 1}$ is a nested family of nonempty compact subintervals of $[0, \sqrt{2}]$.

(i) Choose $a_* \in \bigcap_{n \geq 1} K_n \subseteq [0, \sqrt{2}]$. Then x_{a_*} is global, $\ell \leq x_{a_*} \leq b$, and $0 \leq x_{a_*}(t) - t \leq b(t) - t = \frac{2}{b(t)+t} \to 0$.
If $a_1 < a_2$ lie in $\bigcap_{n \geq 1} K_n$, then $\delta := x_{a_2} - x_{a_1}$ satisfies $\delta(0) > 0$ and $\delta' = (x_{a_1} + x_{a_2})\delta \geq 0$.
Thus $0 < a_2 - a_1 \leq \delta(t) \leq b(t) - t \to 0$, impossible, so $\bigcap_{n \geq 1} K_n = \{a_*\}$.

(ii) Let $a > a_*$. Choose n with $a > \beta_n$. By comparison, $x_a > q_n > t$, so $x_a > 0$ and $x_a' > 0$ while both exist.
If $T_a \leq n$, then $\ell_a := \lim_{t \uparrow T_a} x_a(t) \in (a, +\infty]$. Lemma 1.1.28 excludes $\ell_a < +\infty$, so $x_a(t) \to +\infty$.
If $T_a > n$, then $x_a(n) > q_n(n) = \sqrt{n^2+2}$; Proposition 1.3.9 implies $T_a < +\infty$ and $x_a(t) \to +\infty$ as $t \uparrow T_a$.

(iii) Let $a < a_*$. Choose n with $a < \alpha_n$. By comparison, $x_a < p_n < t$.
On $[0, T_a \wedge n)$, whenever $x_a < -t$ one has $x_a' > 0$; after a contact with $-t$, strict comparison prevents a later downward crossing.
Hence $\min\{a, -n\} \leq x_a(t) < n$ on this interval. Corollary 1.1.29 therefore gives $T_a > n$.
Now $x_a(n) < n$. If $-n < x_a(n) < n$, Proposition 1.3.8 gives $T_a = +\infty$ and $x_a(t) + t \to 0$.
If $x_a(n) \leq -n$, set $h := x_a + t$. While $h < 0$, one has $h' = x_a^2 - t^2 + 1 > 1$ and $x_a(n) \leq x_a(t) < -t$. Compact escape gives existence through the finite time $n - h(n)$, so h has a first zero $\tau \in [n, n - h(n)]$. Strict comparison from τ gives $-t < x_a(t)$ for $t > \tau$, and continuity gives $x_a(t_2) < t_2$ for some $t_2 > \tau$.
Proposition 1.3.8 implies $T_a = +\infty$ and $x_a(t) + t \to 0$.

Structural Commentary: Nested corridors isolate the threshold separating blow-up from lower-edge attraction.
Key Insight: The orbit $x_{a_*} \sim t$ separates blow-up from attraction to $-t$. □

Theorem 1.3.11. *Asymptotic Stability by Linearization*
Let $U \subseteq \mathbb{R}^n$ be open with $0 \in U$, let $f \in C^1(U, \mathbb{R}^n)$ satisfy $f(0) = 0$, and set $A := Df(0)$. Assume $\operatorname{Re}\lambda < 0$ for every $\lambda \in \operatorname{spec}(A)$. Then $\exists r, K, \gamma > 0$ such that, for every $x_0 \in \overline{B}_r(0)$:

(i) The IVP $x' = f(x)$, $x(0) = x_0$, has a unique solution $x \in C^1([0, +\infty), U)$.
(ii) The solution satisfies $\|x(t)\| \leq Ke^{-\gamma t}\|x_0\|$ $\forall t \geq 0$.

Proof. Choose $\alpha \in (0, \min\{-\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(A)\})$. By Theorem 1.2.17(iii), $\exists C \geq 1$ with $\|e^{tA}\| \leq Ce^{-\alpha t}$ for $t \geq 0$. Set $g(x) := f(x) - Ax$. Since $Dg(0) = 0$, choose $b \in (0, \alpha/C)$ and $R > 0$ such that $\overline{B}_R(0) \subseteq U$ and $\|g(x)\| \leq b\|x\|$ for $\|x\| \leq R$.
Set $r := R/(2C)$, $K := C$, and $\gamma := \alpha - Cb > 0$. Fix $x_0 \in \overline{B}_r(0)$. By Lemma 1.1.15 and Theorem 1.1.27, the autonomous field $(t, x) \mapsto f(x)$ has a unique maximal solution $x : [0, T_+) \to U$ with $x(0) = x_0$.

(i) Work before the first exit from $B_R(0)$. By Theorem 1.2.11(iii), $x(t) = e^{tA}x_0 + \int_0^t e^{(t-s)A}g(x(s))\,ds$. For $y(t) := e^{\alpha t}\|x(t)\|$, this gives $y(t) \leq C\|x_0\| + Cb\int_0^t y(s)\,ds$.

Lemma 1.1.20(ii) yields $\|x(t)\| \le Ce^{-\gamma t}\|x_0\| \le R/2$, so no exit occurs. If $T_+ < +\infty$, the graph remains in $[0,T_+] \times \overline{B}_{R/2}(0) \Subset \mathbb{R} \times U$, contradicting Corollary 1.1.29. Hence $T_+ = +\infty$; uniqueness is part of Theorem 1.1.27.

(ii) The estimate obtained in (i) is valid for every $t \ge 0$ and is exactly $\|x(t)\| \le Ke^{-\gamma t}\|x_0\|$.

Structural Commentary: Split off Ax; absorb the differentiable remainder by Gronwall.
Key Insight: A Hurwitz linear part dominates its $o(\|x\|)$ remainder. □

1.4 Bifurcation Theory

A parameter can change both the position of equilibria and their number and stability. The harvested logistic equation is a complete scalar model of such a transition.

Definition 1.4.1. *Scalar Saddle-Node Pattern*
Let $\Lambda \subseteq \mathbb{R}$ be an interval, let $F : \mathbb{R} \times \Lambda \to \mathbb{R}$, and set $F_h(x) := F(x,h)$.
For $(x_c,h_c) \in \mathbb{R} \times \Lambda$, the family $(F_h)_{h\in\Lambda}$ has a *saddle-node pattern* at (x_c,h_c) iff $\exists \varepsilon > 0$, a neighborhood $X \subseteq \mathbb{R}$ of x_c, and $\sigma \in \{-1,1\}$ such that the sets $H_\pm := \{h \in \Lambda : 0 < \pm\sigma(h-h_c) < \varepsilon\}$ are nonempty and:

(i) *Critical section*: $F_{h_c}^{-1}(0) \cap X = \{x_c\}$.

(ii) *Two-equilibrium side*: $\exists x_-, x_+ : H_+ \to X$ such that $F_h^{-1}(0) \cap X = \{x_-(h), x_+(h)\}$ and $x_-(h) < x_+(h)\ \forall h \in H_+$, with $x_\pm(h) \to x_c$ as $h \to h_c$ through H_+.

(iii) *Zero-equilibrium side*: $F_h^{-1}(0) \cap X = \emptyset\ \forall h \in H_-$.

Key Insight: A saddle-node is local creation or annihilation of two equilibria.

Proposition 1.4.2. *Equilibrium Branches of the Harvested Logistic Equation*
For $h \in [0,+\infty)$, let $F_h : \mathbb{R} \to \mathbb{R}$ be given by $F_h(x) := (1-x)x - h$. Set $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, and $x_\pm(h) := \frac{1}{2}(1 \pm \sqrt{1-4h})$ for $h \in [0,h_c]$.

(i) $F_h^{-1}(0) = \{x_-(h), x_+(h)\}$ *and* $x_-(h) < x_+(h)\ \forall h \in [0,h_c)$.
(ii) $F_{h_c}^{-1}(0) = \{x_c\}$ *and* $F_{h_c}(x) = -(x-x_c)^2\ \forall x \in \mathbb{R}$.
(iii) $F_h^{-1}(0) = \emptyset\ \forall h \in (h_c,+\infty)$.
(iv) $x_\pm(h) \to x_c$ *as* $h \uparrow h_c$.
(v) The family $(F_h)_{h\in[0,+\infty)}$ has a saddle-node pattern at (x_c,h_c).

Proof. The identity $F_h(x) = h_c - h - (x-x_c)^2$ holds $\forall (x,h) \in \mathbb{R} \times [0,+\infty)$; hence $F_h(x) = 0$ iff $(x-x_c)^2 = h_c - h$.

(i) For $h \in [0,h_c)$, $h_c - h > 0$ and $x_c \pm \sqrt{h_c-h} = x_\pm(h)$, which gives the two distinct zeros.
(ii) Substitution of $h = h_c$ gives $F_{h_c}(x) = -(x-x_c)^2$ and therefore $F_{h_c}^{-1}(0) = \{x_c\}$.
(iii) For $h > h_c$, the relation $(x-x_c)^2 = h_c - h < 0$ has no solution in $\mathbb{R}$.
(iv) The identity $x_\pm(h) = x_c \pm \sqrt{h_c-h}$ and the limit $h_c - h \downarrow 0$ give $x_\pm(h) \to x_c$.
(v) Take $X = \mathbb{R}$, $\sigma = -1$, and $\varepsilon = \frac{1}{8}$. Then $H_+ = (\frac{1}{8},\frac{1}{4})$ and $H_- = (\frac{1}{4},\frac{3}{8})$; items (i)–(iv) establish the critical zero, the two coalescing branches on H_+, and the absence of zeros on H_-.

Structural Commentary: Completing the square controls the zero count on both parameter sides.
Key Insight: At h_c, two simple roots merge into one double root. □

Lemma 1.4.3. *Scalar Phase-Line Barriers and Limits*
Let $F \in C^1(\mathbb{R},\mathbb{R})$, and let $x : [t_0,T_+) \to \mathbb{R}$, $T_+ \in (t_0,+\infty]$, be a maximal forward solution of $x' = F(x)$.

(i) If $e \in F^{-1}(0)$ and $\exists \tau \in [t_0,T_+)$ such that $x(\tau) = e$, then $x(t) = e\ \forall t \in [t_0,T_+)$.
(ii) If x is monotone and $x([t_0,T_+))$ is bounded, then $T_+ = +\infty$ and $\exists \ell \in F^{-1}(0)$ such that $x(t) \to \ell$ as $t \to +\infty$.

Proof.

(i) The constant function $u \equiv e$ satisfies $u' = F(u)$ and $u(\tau) = x(\tau)$. Set $S := \{t \in [t_0,T_+) : x(t) = e\}$. Continuity makes S relatively closed, while local uniqueness makes S relatively open. Since $\tau \in S$ and $[t_0,T_+)$ is connected, $S = [t_0,T_+)$.

(ii) Monotone boundedness gives $\ell := \lim_{t\uparrow T_+} x(t) \in \mathbb{R}$. If $T_+ < +\infty$, by Lemma 1.1.28, x extends through (T_+,ℓ), contradicting maximality. Thus $T_+ = +\infty$.
Suppose $F(\ell) \ne 0$. By continuity and $x(t) \to \ell$, $\exists c > 0$ and $t_1 \ge t_0$ such that $F(x(t))$ has constant sign and $|F(x(t))| \ge c\ \forall t \ge t_1$.
Hence $|x(t) - x(s)| = \left|\int_s^t F(x(r))\,dr\right| \ge c(t-s)\ \forall t > s \ge t_1$, contradicting $x(t) \to \ell$. Therefore $F(\ell) = 0$.

Structural Commentary: Uniqueness makes equilibria barriers; maximal extension identifies bounded monotone limits.
Key Insight: A finite phase-line limit must be a zero of F. □

Theorem 1.4.4. *Autonomous Scalar Phase-Line Classification*
Let $F \in C^1(\mathbb{R},\mathbb{R})$ and let $x : [t_0,T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F(x)$ with $x(t_0) = x_0$.
Set $E := F^{-1}(0)$. If $x_0 \notin E$, set $\sigma := \operatorname{sgn}(F(x_0))$ and $Z_\sigma := \{e \in E : \sigma(e-x_0) > 0\}$.

(i) If $x_0 \in E$, then $x \equiv x_0$.
(ii) If $x_0 \notin E$, then $(\sigma x)'(t) > 0\ \forall t \in [t_0,T_+)$.
(iii) If $x_0 \notin E$ and $Z_\sigma \ne \emptyset$, then $\exists! e_\sigma \in Z_\sigma$ such that $\sigma e_\sigma = \inf\{\sigma e : e \in Z_\sigma\}$. In this case, $T_+ = +\infty$ and $\sigma(x(t) - e_\sigma) \uparrow 0$ as $t \to +\infty$.
(iv) If $x_0 \notin E$ and $Z_\sigma = \emptyset$, then $\sigma x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Proof.

(i) Since $x_0 \in E$, by Lemma 1.4.3(i), $x(t) = x_0\ \forall t \in [t_0,T_+)$.

(ii) Let C be the component of $\mathbb{R} \setminus E$ containing x_0. Continuity gives $\sigma F > 0$ on C. If x left C, continuity would give $x(\tau) \in E$ for some τ, and Lemma 1.4.3(i) would force $x \equiv x(\tau)$, contradicting $x_0 \notin E$. Thus $x([t_0,T_+)) \subseteq C$ and $(\sigma x)' = \sigma F(x) > 0$.

(iii) Since $F(x_0) \ne 0$, continuity gives $\delta > 0$ such that $E \cap [x_0-\delta, x_0+\delta] = \emptyset$. Let $m := \inf\{\sigma e : e \in Z_\sigma\}$ and choose $(e_n) \subseteq Z_\sigma$ with $\sigma e_n \to m$. Then (e_n) converges to $e_\sigma := \sigma m$. Since E is closed, $e_\sigma \in E$; the zero-free neighborhood gives $\sigma(e_\sigma - x_0) > 0$, so $e_\sigma \in Z_\sigma$. Its defining equality makes it unique.
No zero lies strictly between x_0 and e_σ. By (ii) and Lemma 1.4.3(i), σx is increasing and bounded above by σe_σ. Lemma 1.4.3(ii) gives $T_+ = +\infty$ and $x(t) \to \ell \in E$. Since $\sigma x_0 < \sigma\ell \le \sigma e_\sigma$, one has $\ell \in Z_\sigma$; minimality gives $\sigma e_\sigma \le \sigma\ell$, hence $\ell = e_\sigma$.

(iv) By (ii), σx is strictly increasing. If it were bounded above, Lemma 1.4.3(ii) would give $x(t) \to \ell \in E$ with $\sigma(\ell - x_0) > 0$, so $\ell \in Z_\sigma$, a contradiction. Therefore $\sigma x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Structural Commentary: Components of $\mathbb{R} \setminus E$ fix the sign; the nearest zero supplies the barrier.
Key Insight: A trajectory approaches its next zero or escapes in the sign direction. □

Theorem 1.4.5. *Subcritical Harvest Phase Line*
Let $h_c := \frac{1}{4}$, $F_h(x) := (1-x)x - h$, $0 \le h < h_c$, and $x_\pm := \frac{1}{2}(1 \pm \sqrt{1-4h})$.
Let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F_h(x)$ with $x(t_0) = x_0$.

(i) If $x_0 \in \{x_-, x_+\}$, then $T_+ = +\infty$ and $x \equiv x_0$.
(ii) If $x_0 < x_-$, then $x'(t) < 0$ $\forall t \in [t_0, T_+)$ and $x(t) \to -\infty$ as $t \uparrow T_+$.
(iii) If $x_- < x_0 < x_+$, then $T_+ = +\infty$, $x'(t) > 0$ $\forall t \ge t_0$, and $x(t) \to x_+$ as $t \to +\infty$.
(iv) If $x_0 > x_+$, then $T_+ = +\infty$, $x'(t) < 0$ $\forall t \ge t_0$, and $x(t) \to x_+$ as $t \to +\infty$.

Proof. The factorization $F_h = -(\cdot - x_-)(\cdot - x_+)$ gives $F_h < 0$ on $(-\infty, x_-) \cup (x_+, +\infty)$ and $F_h > 0$ on (x_-, x_+).

(i) Since $x_0 \in F_h^{-1}(0)$, by Theorem 1.4.4(i), $x \equiv x_0$. The constant solution is defined on $[t_0, +\infty)$, so maximality gives $T_+ = +\infty$.
(ii) Here $\operatorname{sgn}(F_h(x_0)) = -1$, and no zero lies below x_0. By Theorem 1.4.4(ii),(iv), $x' < 0$ and $-x(t) \uparrow +\infty$ as $t \uparrow T_+$.
(iii) Here $\operatorname{sgn}(F_h(x_0)) = 1$, and x_+ is the nearest zero above x_0. By Theorem 1.4.4(ii),(iii), $T_+ = +\infty$, $x' > 0$, and $x(t) \to x_+$.
(iv) Here $\operatorname{sgn}(F_h(x_0)) = -1$, and x_+ is the nearest zero below x_0. By Theorem 1.4.4(ii),(iii), $T_+ = +\infty$, $x' < 0$, and $x(t) \to x_+$.

Structural Commentary: The factorization fixes three sign intervals; the phase-line theorem handles each.
Key Insight: The upper root attracts every initial state above the lower threshold. □

Theorem 1.4.6. *Critical Harvest Phase Line*
Let $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, $F_h(x) := (1-x)x - h$, and $h = h_c$.
Let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F_{h_c}(x)$ with $x(t_0) = x_0$.

(i) If $x_0 = x_c$, then $T_+ = +\infty$ and $x \equiv x_c$.
(ii) If $x_0 > x_c$, then $T_+ = +\infty$, $x'(t) < 0$ $\forall t \ge t_0$, and $x(t) \to x_c$ as $t \to +\infty$.
(iii) If $x_0 < x_c$, then $x'(t) < 0$ $\forall t \in [t_0, T_+)$ and $x(t) \to -\infty$ as $t \uparrow T_+$.

Proof. The identity $F_{h_c}(x) = -(x - x_c)^2$ gives $F_{h_c}^{-1}(0) = \{x_c\}$ and $F_{h_c} < 0$ on $\mathbb{R} \setminus \{x_c\}$.

(i) Since $x_0 \in F_{h_c}^{-1}(0)$, by Theorem 1.4.4(i), $x \equiv x_c$. The constant solution is defined on $[t_0, +\infty)$, so maximality gives $T_+ = +\infty$.
(ii) Here $\operatorname{sgn}(F_{h_c}(x_0)) = -1$, and x_c is the nearest zero below x_0. By Theorem 1.4.4(ii),(iii), $T_+ = +\infty$, $x' < 0$, and $x(t) \to x_c$.
(iii) Here $\operatorname{sgn}(F_{h_c}(x_0)) = -1$, and no zero lies below x_0. By Theorem 1.4.4(ii),(iv), $x' < 0$ and $-x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Structural Commentary: The double zero is the sole barrier; the sign is negative on both sides.
Key Insight: The critical equilibrium attracts from above and repels toward $-\infty$ below. □

Theorem 1.4.7. *Supercritical Harvest Phase Line*
Let $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, $F_h(x) := (1-x)x - h$, and $h > h_c$.
Let $x : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $x' = F_h(x)$ with $x(t_0) = x_0$. Then:

(i) $x'(t) < 0$ $\forall t \in [t_0, T_+)$.
(ii) $x(t) \to -\infty$ as $t \uparrow T_+$.

Proof. For $h > h_c$, $F_h = -(\cdot - x_c)^2 - (h - h_c) < 0$ on $\mathbb{R}$ and $F_h^{-1}(0) = \emptyset$.

(i) By Theorem 1.4.4(ii), with $\sigma = -1$, $x' < 0$.
(ii) Since $F_h^{-1}(0) = \emptyset$, by Theorem 1.4.4(iv), $-x(t) \uparrow +\infty$ as $t \uparrow T_+$.

Structural Commentary: The completed square removes every zero, so the phase-line escape case applies.
Key Insight: Above critical harvest, every trajectory escapes toward $-\infty$. □

Corollary 1.4.8. *Critical Dimensional Harvest Rate*
Let $\kappa, N_{\max} > 0$, let $H \ge 0$, and set $G_H(N) := \kappa(1 - \frac{N}{N_{\max}})N - H$ and $H_c := \frac{\kappa N_{\max}}{4}$.
Let $N : [t_0, T_+) \to \mathbb{R}$ be the maximal forward solution of $N' = G_H(N)$ with $N(t_0) = N_0$.

(i) $G_H^{-1}(0) \cap (0, +\infty) \neq \emptyset$ iff $H \in [0, H_c]$, and $H_c = \max\{\eta \ge 0 : G_\eta^{-1}(0) \cap (0, +\infty) \neq \emptyset\}$.
(ii) For $H \in [0, H_c)$, set $N_\pm(H) := \frac{N_{\max}}{2}(1 \pm \sqrt{1 - \frac{4H}{\kappa N_{\max}}})$. Then $G_H^{-1}(0) = \{N_-(H), N_+(H)\}$ and $G_{H_c}^{-1}(0) = \{\frac{N_{\max}}{2}\}$.
(iii) If $H \in [0, H_c)$ and $N_0 > N_-(H)$, then $T_+ = +\infty$ and $N(t) \to N_+(H)$ as $t \to +\infty$.

Proof. Set $h := \frac{H}{\kappa N_{\max}}$, $h_c := \frac{1}{4}$, $x_c := \frac{1}{2}$, $x_\pm(h) := \frac{1}{2}(1 \pm \sqrt{1-4h})$ for $h \in [0, h_c]$, and $F_h(x) := (1-x)x - h$. Then $G_H(N) = \kappa N_{\max} F_h(\frac{N}{N_{\max}})$.
For the solution, set $x(\tau) := \frac{N(t_0 + \tau/\kappa)}{N_{\max}}$, $x_0 := \frac{N_0}{N_{\max}}$, and $\widehat{T}_+ := \kappa(T_+ - t_0)$. Thus $x : [0, \widehat{T}_+) \to \mathbb{R}$ is the maximal forward solution of $\frac{dx}{d\tau} = F_h(x)$ with $x(0) = x_0$.

(i) By Proposition 1.4.2(i)–(iii), $F_h^{-1}(0) \cap (0, +\infty) \neq \emptyset$ iff $h \in [0, h_c]$. Since $H = \kappa N_{\max} h$ and $H_c = \kappa N_{\max} h_c$, the asserted equivalence and maximum follow.
(ii) By Proposition 1.4.2(i),(ii), $F_h^{-1}(0) = \{x_-(h), x_+(h)\}$ for $h < h_c$ and $F_{h_c}^{-1}(0) = \{x_c\}$. Multiplication by $N_{\max}$ gives $N_\pm(H) = N_{\max} x_\pm(h)$ and $N_{\max} x_c = \frac{N_{\max}}{2}$.
(iii) The inequality $N_0 > N_-(H)$ is equivalent to $x_0 > x_-(h)$. By Theorem 1.4.5(i),(iii),(iv), $\widehat{T}_+ = +\infty$ and $x(\tau) \to x_+(h)$ as $\tau \to +\infty$. Hence $T_+ = +\infty$ and $N(t) \to N_{\max} x_+(h) = N_+(H)$.

Structural Commentary: State and time scaling reduce the dimensional model to the universal quadratic family.
Key Insight: The threshold is one quarter of the unharvested growth scale. □

1.5 Dynamical Systems Structure

For a time-periodic scalar equation, one period of continuous evolution can be compressed into a discrete map. Periodic solutions then become fixed points, and their long-time behavior can be studied through iteration.

Definition 1.5.1. *Scalar Poincaré Map*
Let $f \in C^1(\mathbb{R}^2, \mathbb{R})$ satisfy $f(t+1, x) = f(t, x)\ \forall (t, x) \in \mathbb{R}^2$.

(i) For $x_0 \in \mathbb{R}$, let $\phi_{x_0} : I_{x_0} \to \mathbb{R}$ be the unique maximal solution of $x' = f(t, x)$ with $0 \in I_{x_0}$ and $\phi_{x_0}(0) = x_0$; set $\phi(t, x_0) := \phi_{x_0}(t)$.
(ii) The *one-period domain* is $\mathcal{D} := \{x_0 \in \mathbb{R} : [0, 1] \subseteq I_{x_0}\}$.
(iii) The *Poincaré map* is $P : \mathcal{D} \to \mathbb{R}$, $P(x_0) := \phi(1, x_0)$.

Key Insight: P compresses one period of continuous evolution into one discrete step.

Proposition 1.5.2. *Periodic Orbits and Ordered Returns*
Let $f \in C^1(\mathbb{R}^2, \mathbb{R})$ satisfy $f(t+1, x) = f(t, x)$.
For $x_0 \in \mathbb{R}$, let $\phi_{x_0} : I_{x_0} \to \mathbb{R}$ be the unique maximal solution of $x' = f(t, x)$ with $0 \in I_{x_0}$ and $\phi_{x_0}(0) = x_0$.
Set $\mathcal{D} := \{x_0 \in \mathbb{R} : [0, 1] \subseteq I_{x_0}\}$ and $P : \mathcal{D} \to \mathbb{R}$, $P(x_0) := \phi_{x_0}(1)$.

(i) For $x_0 \in \mathcal{D}$, $P(x_0) = x_0 \iff I_{x_0} = \mathbb{R}$ and $\phi_{x_0}(t+1) = \phi_{x_0}(t)\ \forall t \in \mathbb{R}$.
(ii) For $x_0, y_0 \in \mathcal{D}$, $x_0 < y_0 \implies P(x_0) < P(y_0)$.

Proof.

(i) Suppose first that $P(x_0) = x_0$. For $n \in \mathbb{Z}$, define $u_n : [n, n+1] \to \mathbb{R}$ by $u_n(t) := \phi_{x_0}(t - n)$. Then $u_n' = f(t-n, u_n) = f(t, u_n)$, while $u_n(n+1) = \phi_{x_0}(1) = x_0 = u_{n+1}(n+1)$. The one-periodicity of f makes the one-sided derivatives agree at every integer, so the u_n glue to a one-periodic $u \in C^1(\mathbb{R}, \mathbb{R})$ satisfying $u' = f(t, u)$ and $u(0) = x_0$. Local uniqueness and maximality give $I_{x_0} = \mathbb{R}$ and $\phi_{x_0} = u$.
Conversely, if $I_{x_0} = \mathbb{R}$ and ϕ_{x_0} is one-periodic, then $P(x_0) = \phi_{x_0}(1) = \phi_{x_0}(0) = x_0$.
(ii) Choose an open interval J with $[0, 1] \subseteq J \subseteq I_{x_0} \cap I_{y_0}$. Since $f \in C^1$, it is state-Lipschitz on compact subsets of $\mathbb{R}^2$. By Theorem 1.3.5(ii), $\phi_{x_0}(t) < \phi_{y_0}(t)\ \forall t \in [0, 1]$; evaluation at $t = 1$ gives $P(x_0) < P(y_0)$.

Structural Commentary: Glue one-period segments; compare distinct initial states by scalar order preservation.
Key Insight: Periodic trajectories are fixed points of a strictly increasing return map. □

Proposition 1.5.3. *Periodic Scalar Linear Forcing*
Let $a \in \mathbb{R}$ and $g \in C(\mathbb{R}, \mathbb{R})$ satisfy $g(t+1) = g(t)$ and $g(t) \geq 0\ \forall t \in \mathbb{R}$. Set $I_a := \int_0^1 e^{-as} g(s)\, ds$.
For $x_0 \in \mathbb{R}$, let $x(\cdot\,; x_0)$ be the solution of $x' = ax + g(t)$ with $x(0; x_0) = x_0$, and set $P(x_0) := x(1; x_0)$.
If $a \neq 0$, put $x_{0,*} := -\frac{e^a}{e^a - 1} I_a$ and $x_*(t) := e^{at}\big(x_{0,*} + \int_0^t e^{-as} g(s)\, ds\big)$.

(i) $a \neq 0 \implies x_*$ is the unique one-periodic solution.
(ii) If $a = 0$ and $g \equiv 0$, the one-periodic solutions are $t \mapsto c$, $c \in \mathbb{R}$.
(iii) $a = 0$ and $g \not\equiv 0 \implies$ no one-periodic solution exists.

Proof. By Proposition 1.2.1, $x(t; x_0) = e^{at}\big(x_0 + \int_0^t e^{-as} g(s)\, ds\big)$ and $P(x_0) - x_0 = (e^a - 1)x_0 + e^a I_a$.
If $P(x_0) = x_0$, then $d(t) := x(t+1; x_0) - x(t; x_0)$ satisfies $d' = ad$ and $d(0) = 0$, so $d \equiv 0$. The converse is immediate; hence the one-periodic solutions correspond to $\mathrm{Fix}(P)$.

(i) If $a \neq 0$, then $e^a - 1 \neq 0$ and $\mathrm{Fix}(P) = \{x_{0,*}\}$. Substitution of $x_0 = x_{0,*}$ into the solution formula gives x_*.
(ii) If $a = 0$ and $g \equiv 0$, then $x(t; x_0) = x_0\ \forall (t, x_0) \in \mathbb{R}^2$.
(iii) If $a = 0$ and $g \not\equiv 0$, continuity and nonnegativity give $I_0 = \int_0^1 g(s)\, ds > 0$. Thus $P(x_0) - x_0 = I_0 \neq 0\ \forall x_0 \in \mathbb{R}$.

Structural Commentary: Variation of constants makes the return displacement affine in x_0.
Key Insight: $P(x_0) - x_0 = (e^a - 1)x_0 + e^a I_a$. □

Proposition 1.5.4. *Concave Periodic Return-Map Geometry*
Let $f, g \in C^2(\mathbb{R}, \mathbb{R})$ satisfy $f''(u) < 0\ \forall u \in \mathbb{R}$, $g(t+1) = g(t)$ and $g(t) \geq 0\ \forall t \in \mathbb{R}$, and $g \not\equiv 0$.
Let $W \subseteq \mathbb{R}^2$ be open and $(h, x) \in W$. For $(\eta, y) \in W$, let $\phi_\eta(\cdot\,, y)$ be the unique solution of $z' = f(z) + \eta g(t)$, $z(0) = y$, on $[0, 1]$.
Set $P_\eta(y) := \phi_\eta(1, y)$, $\phi(t) := \phi_h(t, x)$, $A(t) := f'(\phi(t))$, and $\theta(t) := \partial_x \phi_h(t, x)$.

(i) $P_h'(x) = \exp\Big(\int_0^1 A(s)\, ds\Big) > 0$.
(ii) $P_h''(x) = P_h'(x) \int_0^1 f''(\phi(s))\theta(s)\, ds < 0$.
(iii) $\partial_h P_h(x) = \int_0^1 \exp\Big(\int_s^1 A(r)\, dr\Big) g(s)\, ds > 0$.

Proof. Set $\mathcal{F}(t, z, \eta) := f(z) + \eta g(t)$. Since $\mathcal{F} \in C^2(\mathbb{R}^3, \mathbb{R})$, by Theorem 1.1.24, applied successively along the solution arcs over $[0, 1]$, the flow has the required two state derivatives and one parameter derivative.

(i) $\theta' = A\theta$ and $\theta(0) = 1$. Hence $\theta(t) = \exp\Big(\int_0^t A(s)\, ds\Big) > 0$, so $P_h'(x) = \theta(1)$.
(ii) Put $\zeta := \partial_x^2 \phi_h(\cdot\,, x)$. Then $\zeta' = A\zeta + f''(\phi)\theta^2$ and $\zeta(0) = 0$.
By Proposition 1.2.1, $\zeta(1) = \theta(1) \int_0^1 f''(\phi(s))\theta(s)\, ds$. Since $f'' < 0$ and $\theta > 0$, $P_h''(x) = \zeta(1) < 0$.
(iii) Put $\psi := \partial_h \phi_h(\cdot\,, x)$. Then $\psi' = A\psi + g$ and $\psi(0) = 0$.
By Proposition 1.2.1, $\psi(1) = \int_0^1 \exp\Big(\int_s^1 A(r)\, dr\Big) g(s)\, ds$. The integrand is nonnegative and not identically zero, so $\psi(1) > 0$ and $\partial_h P_h(x) = \psi(1)$.

Structural Commentary: Differentiate the flow; solve each scalar variational equation by an integrating factor.
Key Insight: $f'' < 0$ bends the return map; $g \geq 0$ moves it upward. □

Proposition 1.5.5. *Geometry of the Periodically Harvested Return Map*
Let $g(t) := 1 - \sin(2\pi t)$ and $F_\eta(t, z) := (1 - z)z - \eta g(t)$.
Let $W \subseteq \mathbb{R}^2$ be open and $(h, x) \in W$. For $(\eta, y) \in W$, let $\phi_\eta(\cdot\,, y)$ be the unique solution of $z' = F_\eta(t, z)$, $z(0) = y$, on $[0, 1]$.
Set $P_\eta(y) := \phi_\eta(1, y)$ and $\theta_h(t, x) := \partial_x \phi_h(t, x)$.

(i) $P_h'(x) = \exp\Big(\int_0^1 (1 - 2\phi_h(s, x))\, ds\Big) > 0$.

(ii) $P_h''(x) = -2P_h'(x)\int_0^1 \theta_h(s,x)\,ds < 0$.
(iii) $\partial_h P_h(x) = -\int_0^1 \exp\Big(\int_s^1 (1-2\phi_h(r,x))\,dr\Big)\,g(s)\,ds < 0$.

Proof. Put $f(z) := (1-z)z$, $\widetilde{W} := \{(-\eta, y) : (\eta, y) \in W\}$, and $Q_\lambda(y) := P_{-\lambda}(y)$ for $(\lambda, y) \in \widetilde{W}$. Then Q_λ is the return map for $z' = f(z) + \lambda g(t)$, and $f'' = -2$, while $g \geq 0$ and $g \not\equiv 0$. Set $E(s) := \exp\Big(\int_s^1 (1-2\phi_h(r,x))\,dr\Big)$.

(i) By Proposition 1.5.4(i), applied at $(-h, x)$, $P_h'(x) = Q'_{-h}(x) = \exp\Big(\int_0^1 (1-2\phi_h(s,x))\,ds\Big) > 0$.
(ii) By Proposition 1.5.4(ii), $P_h''(x) = Q''_{-h}(x) = -2P_h'(x)\int_0^1 \theta_h(s,x)\,ds < 0$.
(iii) By the chain rule and Proposition 1.5.4(iii), $\partial_h P_h(x) = -\partial_\lambda Q_{-h}(x) = -\int_0^1 E(s)g(s)\,ds < 0$.

Structural Commentary: Reflect the parameter; specialize the general concave return-map identities.
Key Insight: Harvesting shifts the increasing concave return map downward. □

Example 1.5.6. *The Unharvested Return Map*
For $x \in [0, +\infty)$, let $\phi_0(\,\cdot\,, x)$ solve $z' = (1-z)z$ with $z(0) = x$, and set $P_0(x) := \phi_0(1, x)$.
Direct differentiation verifies $\phi_0(t,x) = \frac{e^t x}{1+(e^t-1)x}$ for $t \geq 0$, so $P_0(x) = \frac{ex}{1+(e-1)x}$. Solving $P_0(x) = x$ gives $\mathrm{Fix}(P_0) = \{0, 1\}$. By Proposition 1.5.2(i), these fixed points correspond to the constant one-periodic solutions $z \equiv 0$ and $z \equiv 1$.

Theorem 1.5.7. *Iteration of an Increasing Concave Map*
Let $I \subseteq \mathbb{R}$ be an interval, $P : I \to I$ continuous, strictly increasing, strictly concave, and $\mathrm{Fix}(P) = \{x_1, x_2\}$ with $x_1 < x_2$. For $x \in I$, set $x_0 := x$ and $x_{n+1} := P(x_n)$.
(i) If $x > x_1$, then $x_n \to x_2$.
(ii) If $x = x_1$, then $x_n = x_1$ $\forall n \geq 0$.
(iii) If $x < x_1$, then $x_{n+1} < x_n$ $\forall n \geq 0$ and $x_n \downarrow \inf I$ in $\overline{\mathbb{R}}$; if $\inf I = -\infty$, then $x_n \to -\infty$.

Proof. For $a, b \in I$ with $a < b$, set $m(a,b) := \frac{P(b)-P(a)}{b-a}$. If $a < b < c$, strict concavity at $b = (1-\lambda)a + \lambda c$, $\lambda := \frac{b-a}{c-a} \in (0,1)$, gives $m(a,b) > m(a,c) > m(b,c)$. Since $P(x_j) = x_j$, $m(x_1, x_2) = 1$.

(i) If $x \in (x_1, x_2)$, then $m(x_1, x) > m(x_1, x_2) = 1$ and monotonicity gives $x < P(x) < x_2$. Hence $x_n \uparrow \ell \in (x_1, x_2]$, and continuity gives $P(\ell) = \ell$, so $\ell = x_2$. If $x = x_2$, the sequence is constant.
If $x > x_2$, then $m(x_2, x) < m(x_1, x_2) = 1$ and monotonicity gives $x_2 < P(x) < x$. Hence $x_n \downarrow \ell \in [x_2, x)$, and continuity again gives $\ell = x_2$.
(ii) Since $P(x_1) = x_1$, induction gives $x_n = x_1$ $\forall n \geq 0$.
(iii) If $x < x_1$, then $m(x, x_1) > m(x_1, x_2) = 1$, so $P(x) < x$. Induction gives $x_{n+1} < x_n < x_1$ $\forall n \geq 0$, and $x_n \downarrow \ell \in \overline{\mathbb{R}}$ with $\ell \geq \inf I$. If $\ell > \inf I$, then $\ell \in I$, continuity gives $P(\ell) = \ell < x_1$, and this contradicts $\mathrm{Fix}(P) = \{x_1, x_2\}$. Thus $\ell = \inf I$.

Structural Commentary: Nested secant slopes determine the sign of $P(x) - x$ on three intervals.
Key Insight: The lower fixed point is a threshold; the upper one attracts from above. □

Corollary 1.5.8. *Periodic Logistic Threshold Structure*
Let $g(t) := 1 - \sin(2\pi t)$. For $(\eta, y) \in \mathbb{R}^2$, let $\phi_{\eta,y} : I_{\eta,y} \to \mathbb{R}$ be the unique maximal solution of $z' = (1-z)z - \eta g(t)$ with $0 \in I_{\eta,y}$ and $\phi_{\eta,y}(0) = y$.
Set $\mathcal{D}_\eta := \{y \in \mathbb{R} : [0,1] \subseteq I_{\eta,y}\}$ and $P_\eta : \mathcal{D}_\eta \to \mathbb{R}$, $P_\eta(y) := \phi_{\eta,y}(1)$. Fix $h \geq 0$ and an interval $J \subseteq \mathcal{D}_h$.
(i) $\#\,\mathrm{Fix}(P_h|_J) \leq 2$; hence there are at most two one-periodic solutions with initial values in J.
(ii) If $P_h(J) \subseteq J$, $\mathrm{Fix}(P_h|_J) = \{x_1, x_2\}$ with $x_1 < x_2$, and $x \in J$ satisfies $x > x_1$, then $P_h^n(x) \to x_2$.
(iii) If $P_h(J) \subseteq J$, $\mathrm{Fix}(P_h|_J) = \{x_1, x_2\}$ with $x_1 < x_2$, and $x \in J$ satisfies $x < x_1$, then $P_h^n(x) \downarrow \inf J$ in $\overline{\mathbb{R}}$.
(iv) $\partial_h P_h(x) < 0$ $\forall x \in \mathcal{D}_h$.

Proof. Fix $x \in \mathcal{D}_h$. The solution arc $\{(t, \phi_{h,x}(t), h) : t \in [0,1]\}$ is compact. By Theorem 1.1.24, applied successively along a finite partition of this arc, $\exists$ an open neighborhood $W_x \subseteq \mathbb{R}^2$ of (h, x) on which the time-1 flow is defined and C^2. By Proposition 1.5.5, $P_h'(x) > 0$, $P_h''(x) < 0$, and $\partial_h P_h(x) < 0$. Thus P_h is strictly increasing and strictly concave on J.

(i) If $a < b < c$ were fixed points in J, strict concavity would give $P_h(b) > L_{a,c}(b) = b$, where $L_{a,c}$ is the chord through (a,a) and (c,c), contradicting $P_h(b) = b$. Hence $\#\,\mathrm{Fix}(P_h|_J) \leq 2$. By Proposition 1.5.2(i), these fixed points are exactly the initial values in J of one-periodic solutions.
(ii) Under the stated hypotheses, $P_h|_J : J \to J$ is continuous, strictly increasing, and strictly concave. By Theorem 1.5.7(i), $P_h^n(x) \to x_2$.
(iii) Under the stated hypotheses, by Theorem 1.5.7(iii), $P_h^n(x) \downarrow \inf J$ in $\overline{\mathbb{R}}$.
(iv) The local calculation preceding (i) gives $\partial_h P_h(x) < 0$ $\forall x \in \mathcal{D}_h$.

Structural Commentary: Smooth dependence supplies concavity; graph geometry and iteration classify fixed points.
Key Insight: Two periodic states form a threshold with an attracting upper branch. □

1.6 Semigroups and Evolution Viewpoint

1.7 Measure-Preserving Dynamics and Ergodicity

Measure-preserving dynamics study maps whose pullbacks keep mass fixed. The central question is whether time averages along one orbit recover space averages with respect to an invariant probability measure.

Definition 1.7.1. *Measure-Preserving and Ergodic Transformations*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $T : X \to X$.
For $E, A, B \in \mathcal{M}$, write $T^{-1}E := \{x \in X : T(x) \in E\}$ and $A \triangle B := (A \setminus B) \cup (B \setminus A)$.
(i) T is *measurable* iff $T^{-1}E \in \mathcal{M}$ $\forall E \in \mathcal{M}$.
(ii) If T is measurable, the *pushforward measure* is $(T_*\mu)(E) := \mu(T^{-1}E)$, $E \in \mathcal{M}$.
(iii) A measurable T is *measure preserving* iff $T_*\mu = \mu$.
(iv) For measurable T, $A \in \mathcal{M}$ is *invariant modulo μ* iff $\mu(A \triangle T^{-1}A) = 0$.
(v) If $0 < \mu(X) < +\infty$ and T is measure preserving, then T is *ergodic* iff $\mu(A \triangle T^{-1}A) = 0 \implies \mu(A) \in \{0, \mu(X)\}$ $\forall A \in \mathcal{M}$.

Key Insight: Pullbacks preserve mass; ergodic invariants are trivial.

Proposition 1.7.2. *Integral Test for Measure Preservation*
Let $(X, \mathcal{M}, \mu)$ be a finite measure space and $T : X \to X$ measurable. Then $T_*\mu = \mu$ iff $\forall g \in L^1(X,\mu)$, $g \circ T \in L^1(X,\mu)$ and $\int_X g \circ T \, d\mu = \int_X g \, d\mu$.

Proof.
($\Rightarrow$) If $T_*\mu = \mu$, then $\int_X \mathbf{1}_E \circ T \, d\mu = \mu(T^{-1}E) = \mu(E) = \int_X \mathbf{1}_E \, d\mu$ $\forall E \in \mathcal{M}$. Linearity gives the identity for nonnegative simple functions. For $0 \le g \in L^1(X,\mu)$, choose simple $0 \le s_n \uparrow g$; monotone convergence applied to s_n and $s_n \circ T$ gives the identity and $g \circ T \in L^1(X,\mu)$. Apply this conclusion to $g^{\pm}$ for general $g \in L^1(X,\mu)$.
($\Leftarrow$) Since $\mu(X) < +\infty$, $\mathbf{1}_E \in L^1(X,\mu)$ $\forall E \in \mathcal{M}$. Taking $g = \mathbf{1}_E$ gives $(T_*\mu)(E) = \int_X \mathbf{1}_E \circ T \, d\mu = \int_X \mathbf{1}_E \, d\mu = \mu(E)$.

Structural Commentary: Indicators encode sets; approximation extends the identity to observables.
Key Insight: Measure preservation is equality of integrals after pullback. □

Remark 1.7.3. *Invariant Sets as Invariant Indicators*
If $T_*\mu = \mu$ and $A \in \mathcal{M}$, then $\mu(A \triangle T^{-1}A) = 0 \iff \mathbf{1}_A \circ T = \mathbf{1}_A$ μ-a.e.

Proposition 1.7.4. *Invariant Functions Detect Ergodicity*
Let $(X, \mathcal{M}, \mu)$ satisfy $0 < \mu(X) < +\infty$, and let $T : X \to X$ be measurable with $T_*\mu = \mu$. Then T is ergodic iff $\forall$ measurable $g : X \to \overline{\mathbb{R}}$, $g \circ T = g$ μ-a.e. implies $\exists c \in \overline{\mathbb{R}}$ such that $g = c$ μ-a.e.

Proof.
($\Rightarrow$) Assume T is ergodic and $g \circ T = g$ μ-a.e. For $a \in \mathbb{Q}$, set $A_a := \{g < a\}$. Then $\mu(A_a \triangle T^{-1}A_a) = 0$, so $\mu(A_a) \in \{0, \mu(X)\}$. With $\inf \varnothing := +\infty$, set $c := \inf\{a \in \mathbb{Q} : \mu(A_a) = \mu(X)\} \in \overline{\mathbb{R}}$. If $a < c$, then $\mu(A_a) = 0$; if $b > c$, monotonicity and the definition of c give $\mu(X \setminus A_b) = 0$. Countable unions give $\mu(g < c) = \mu(g > c) = 0$, including $c = \pm\infty$; hence $g = c$ μ-a.e.
($\Leftarrow$) Assume every measurable T-invariant $g : X \to \overline{\mathbb{R}}$ is constant μ-a.e. If $A \in \mathcal{M}$ and $\mu(A \triangle T^{-1}A) = 0$, then $\mathbf{1}_A \circ T = \mathbf{1}_A$ μ-a.e. Thus $\mathbf{1}_A = c$ μ-a.e. for some $c \in \{0,1\}$, so $\mu(A) = c\mu(X) \in \{0, \mu(X)\}$.

Structural Commentary: Level sets convert invariant functions into invariant sets.
Key Insight: Ergodicity collapses invariant observables to constants. □

Method 1.7.5. *Krylov–Bogoliubov Averaging*
Let $X \neq \varnothing$ be compact metric, let $f \in C(X,X)$, and let $\mathcal{P}(X)$ denote the Borel probability measures on X.

(i) Choose $\eta \in \mathcal{P}(X)$ and set $\mu_n := \frac{1}{n}\sum_{k=0}^{n-1} (f^k)_*\eta \in \mathcal{P}(X)$.
(ii) For $g \in C(X)$, record $\int_X g \, d\mu_n = \frac{1}{n}\sum_{k=0}^{n-1} \int_X g \circ f^k \, d\eta$.
(iii) By Corollary 6.4.7, choose $n_j \uparrow \infty$ and a positive Radon measure μ with $\int_X g \, d\mu_{n_j} \to \int_X g \, d\mu$ $\forall g \in C(X)$; testing $g = 1$ gives $\mu \in \mathcal{P}(X)$.
(iv) Use $\int_X g \circ f \, d\mu_n - \int_X g \, d\mu_n = \frac{1}{n}\int_X (g \circ f^n - g)\, d\eta$ and the bound $2\|g\|_\infty / n$. Passing to n_j gives equality of the $C(X)$-integrals of the Radon measures $f_*\mu$ and μ. By Theorem 5.2.13, $f_*\mu = \mu$.

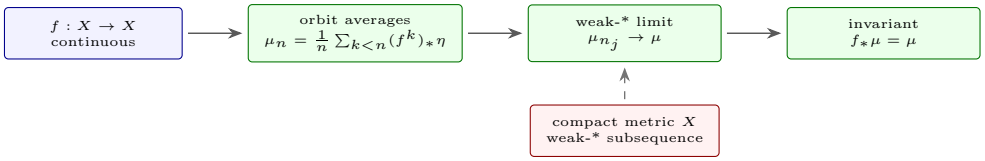

Figure 234: Krylov–Bogoliubov: $\mu_n = \frac{1}{n}\sum_{k<n}(f^k)_*\eta \xrightarrow{w^*} \mu$ and $\frac{1}{n}(g \circ f^n - g) \to 0$, hence $f_*\mu = \mu$.

Theorem 1.7.6. *Krylov–Bogoliubov Invariant Measure Theorem*
Let $X \neq \varnothing$ be compact metric, let $f \in C(X,X)$, and let $\mathcal{P}(X)$ denote the Borel probability measures on X. Then $\exists \mu \in \mathcal{P}(X)$ such that $f_*\mu = \mu$.

Proof. Choose $\eta \in \mathcal{P}(X)$ and set $\mu_n := \frac{1}{n}\sum_{k=0}^{n-1} (f^k)_*\eta \in \mathcal{P}(X)$. By Corollary 6.4.7, $\exists n_j \uparrow \infty$ and a positive Radon measure μ such that $\int_X g \, d\mu_{n_j} \to \int_X g \, d\mu$ $\forall g \in C(X)$. Taking $g = 1$ gives $\mu(X) = 1$, so $\mu \in \mathcal{P}(X)$.

For $g \in C(X)$, $\int_X g \circ f \, d\mu_n - \int_X g \, d\mu_n = \frac{1}{n}\int_X (g \circ f^n - g)\, d\eta$, whose absolute value is at most $2\|g\|_\infty / n$. Passing to n_j gives $\int_X g \, d(f_*\mu) = \int_X g \, d\mu$ $\forall g \in C(X)$. Both measures are Radon, and $C_c(X) = C(X)$; hence $f_*\mu = \mu$ by Theorem 5.2.13. *Structural Commentary*: Average pushforwards; extract a weak-* limit; cancel endpoints.
Key Insight: Compactness turns orbit averages into an invariant probability. □

Proposition 1.7.7. *Ergodic Measures Are Extreme Invariant Measures*
Let X be compact metric, let $f \in C(X,X)$, let $\mathcal{P}(X)$ denote the Borel probability measures on X, and set $\mathcal{I}_f(X) := \{\lambda \in \mathcal{P}(X) : f_*\lambda = \lambda\}$. For $\mu \in \mathcal{I}_f(X)$, the following are equivalent:

(i) $\mu \in \mathrm{Ext}(\mathcal{I}_f(X))$;
(ii) f is μ-ergodic.

Proof.
(i) $\Rightarrow$ (ii) If f is not μ-ergodic, choose $A \in \mathcal{B}(X)$ with $\mu(A \triangle f^{-1}A) = 0$ and $0 < \mu(A) < 1$. Set $\nu(E) := \mu(E \cap A)/\mu(A)$ and $\rho(E) := \mu(E \cap A^c)/\mu(A^c)$. For $E \in \mathcal{B}(X)$, invariance modulo μ gives $\mu(f^{-1}E \cap A) = \mu(f^{-1}(E \cap A)) = \mu(E \cap A)$, and the same identity holds with A^c. Thus $\nu, \rho \in \mathcal{I}_f(X)$, while $\mu = \mu(A)\nu + \mu(A^c)\rho$ and $\nu(A) \neq \rho(A)$, contradicting (i).

(*ii*) $\Rightarrow$ (*i*) Let $\mu = \theta\nu + (1-\theta)\rho$ with $0 < \theta < 1$ and $\nu, \rho \in \mathcal{I}_f(X)$. Since $\theta\nu \le \mu$, by Theorem 3.2.1, $h := d\nu/d\mu$ satisfies $0 \le h \le \theta^{-1}$ μ-a.e. Invariance of ν and μ gives $\int_X (h\circ f)h\,d\mu = \int_X h\circ f\,d\nu = \int_X h\,d\nu = \int_X h^2\,d\mu$ and $\int_X (h\circ f)^2\,d\mu = \int_X h^2\,d\mu$. Hence $\int_X (h\circ f - h)^2\,d\mu = 0$. By Proposition 1.7.4, $h = c$ μ-a.e.; since $1 = \int_X h\,d\mu$, $c = 1$ and $\nu = \mu$. The convex identity then gives $\rho = \mu$.

Structural Commentary: Invariant sets split measures; invariant densities test splits.
Key Insight: Extreme invariant measures are exactly ergodic measures. □

Corollary 1.7.8. *Existence of an Ergodic Invariant Measure*
Let $X \neq \varnothing$ be compact metric, let $f \in C(X,X)$, let $\mathcal{P}(X)$ denote the Borel probability measures on X, and set $\mathcal{I}_f(X) := \{\mu \in \mathcal{P}(X) : f_*\mu = \mu\}$. Then $\exists \mu \in \mathcal{I}_f(X)$ such that f is μ-ergodic.

Proof. By Theorem 1.7.6, $\mathcal{I}_f(X) \neq \varnothing$, and it is convex. Let a net $(\mu_\alpha) \subseteq \mathcal{I}_f(X)$ converge weak-* among finite signed Radon measures to μ. Positivity on $C(X)_+$ and the identity $\int_X 1\,d\mu = 1$ pass to the limit, so $\mu \in \mathcal{P}(X)$. For $g \in C(X)$, $\int_X g\circ f\,d\mu = \lim_\alpha \int_X g\circ f\,d\mu_\alpha = \lim_\alpha \int_X g\,d\mu_\alpha = \int_X g\,d\mu$; hence $f_*\mu = \mu$ by Theorem 5.2.13. Thus $\mathcal{I}_f(X)$ is weak-* closed and $\|\lambda\|_{\mathrm{var}} = 1$ $\forall \lambda \in \mathcal{I}_f(X)$, so it is weak-* compact by Corollary 6.4.6.

By Theorem 9.2.10, choose a dense sequence $(g_j)_{j\ge1}$ in $C(X,\mathbb{R})$. Set $K_0 := \mathcal{I}_f(X)$ and, recursively, $a_j := \max\{\int_X g_j\,d\lambda : \lambda \in K_{j-1}\}$ and $K_j := \{\lambda \in K_{j-1} : \int_X g_j\,d\lambda = a_j\}$. Weak-* continuity of $\lambda \mapsto \int_X g_j\,d\lambda$ makes K_j nonempty and compact, and affinity makes it a face of K_{j-1}. Hence $K_\infty := \bigcap_{j\ge0} K_j \neq \varnothing$. If $\lambda, \nu \in K_\infty$, their integrals agree on the dense family (g_j), hence on $C(X,\mathbb{R})$. By Theorem 5.2.13, $\lambda = \nu$; write $K_\infty = \{\mu\}$.

If $\mu = \theta\lambda + (1-\theta)\nu$ with $0 < \theta < 1$ and $\lambda, \nu \in \mathcal{I}_f(X)$, the face property applied successively to $K_1, K_2, \dots$ gives $\lambda, \nu \in K_\infty$, so $\lambda = \nu = \mu$. Thus $\mu \in \mathrm{Ext}(\mathcal{I}_f(X))$, and f is μ-ergodic by Proposition 1.7.7. *Structural Commentary*: Compact nested faces isolate one extreme invariant measure.
Key Insight: Countably many observables isolate an ergodic invariant measure. □

Theorem 1.7.9. *Asymptotic Ergodic Averages*
Let $(X, \mathcal{M}, \mu)$ satisfy $0 < \mu(X) < +\infty$, and let $T : X \to X$ be measurable with $T_*\mu = \mu$. For $g \in L^1(X,\mu)$, set $U_T g := g \circ T$ and $A_n g := \frac{1}{n}\sum_{k=0}^{n-1} U_T^k g$. Then T is ergodic iff $A_n g \to \mu(X)^{-1}\int_X g\,d\mu$ μ-a.e. $\forall g \in L^1(X,\mu)$.

Proof.

(⇒) First establish the maximal inequality. For $v \in L^1(X,\mu)$, set $S_0 v := 0$ and $S_n v := \sum_{k=0}^{n-1} v\circ T^k$. For $N \ge 1$, set $M_N v := \max_{0\le n\le N} S_n v$ and $E_N := \{M_N v > 0\}$. On E_N, $v + M_N v \circ T \ge M_N v$. Since $M_N v = 0$ on E_N^c, $\int_{E_N} M_N v\,d\mu = \int_X M_N v\,d\mu$. Since $M_N v \circ T \ge 0$ and $T_*\mu = \mu$, $\int_{E_N} M_N v \circ T\,d\mu \le \int_X M_N v\,d\mu$. Integrating the pointwise inequality therefore gives $\int_{E_N} v\,d\mu \ge 0$. Since $E_N \uparrow E := \{\sup_{n\ge1} S_n v > 0\}$, dominated convergence gives $\int_E v\,d\mu \ge 0$.

Fix $g \in L^1(X,\mu)$. For $\varepsilon > 0$, $\sum_{n\ge1} \mu(|g\circ T^n| > \varepsilon n) = \sum_{n\ge1} \mu(|g| > \varepsilon n) \le \varepsilon^{-1}\int_X |g|\,d\mu$. By Theorem 5.1.6, applied with $\varepsilon = 1/m$, $m \ge 1$, $g\circ T^n/n \to 0$ μ-a.e. Hence $A_n g \circ T - A_n g = (g\circ T^n - g)/n \to 0$ μ-a.e., so $\overline{g} := \limsup_n A_n g$ and $\underline{g} := \liminf_n A_n g$ are T-invariant. By Proposition 1.7.4, $\exists a, b \in \overline{\mathbb{R}}$ such that $(\overline{g}, \underline{g}) = (a,b)$ μ-a.e. and $b \le a$.

Set $m := \mu(X)^{-1}\int_X g\,d\mu$. If $a > m$, choose $m < c < a$. Then $\limsup_n A_n(g-c) = a - c > 0$ μ-a.e., so the maximal inequality applied to $v = g - c$ gives $\int_X (g-c)\,d\mu \ge 0$, contradicting $c > m$. Thus $a \le m$. If $b < m$, choose $b < c < m$ and apply the maximal inequality to $v = c - g$; this gives $\int_X (c-g)\,d\mu \ge 0$, contradicting $c < m$. Therefore $b \ge m$, and $b \le a \le m \le b$ proves $A_n g \to m$ μ-a.e.

(⇐) If $A \in \mathcal{M}$ and $\mu(A \triangle T^{-1}A) = 0$, then $A_n \mathbf{1}_A = \mathbf{1}_A$ μ-a.e. $\forall n \ge 1$. The assumed limit gives $\mathbf{1}_A = \mu(A)/\mu(X)$ μ-a.e., so $\mu(A) \in \{0, \mu(X)\}$.

Structural Commentary: Hopf's maximal inequality squeezes invariant limsup and liminf.
Key Insight: Ergodicity collapses both asymptotic envelopes to the space mean. □

Proposition 1.7.10. *Koopman Isometry Characterization*
Let $(X, \mathcal{M}, \mu)$ satisfy $\mu(X) < +\infty$, and let $T : X \to X$ be measurable. Then $T_*\mu = \mu$ iff $\forall p \in [1, +\infty)$ and $\forall g \in L^p(X,\mu)$, the pullback $U_T g := g \circ T$ belongs to $L^p(X,\mu)$ and $\|U_T g\|_p = \|g\|_p$.

Proof.

(⇒) If $T_*\mu = \mu$, Proposition 1.7.2 applied to $|g|^p$ gives $\|U_T g\|_p^p = \int_X |g\circ T|^p\,d\mu = \int_X |g|^p\,d\mu = \|g\|_p^p$ $\forall p \in [1,+\infty)$ and $g \in L^p(X,\mu)$.
(⇐) Apply the $p = 1$ identity to $g = \mathbf{1}_E$, $E \in \mathcal{M}$: $(T_*\mu)(E) = \|\mathbf{1}_E \circ T\|_1 = \|\mathbf{1}_E\|_1 = \mu(E)$.

Structural Commentary: Test pullback norms on powers and indicators.
Key Insight: Koopman isometries encode measure preservation. □

Proposition 1.7.11. *Linear Preservation of Lebesgue Measure*
Let $n \ge 1$ and $T \in \mathcal{L}(\mathbb{R}^n)$. Then:

(i) $T_* m_n = m_n \iff T \in GL_n(\mathbb{R})$ and $|\det T| = 1$.
(ii) For $D := \mathrm{diag}(2, \frac{1}{2}) \in GL_2(\mathbb{R})$, $D_* m_2 = m_2$.

Proof.

(i) If $T_* m_n = m_n$ and T were singular, then $L := T(\mathbb{R}^n)$ would satisfy $m_n(L) = 0$ by Corollary 2.3.6, whereas $(T_* m_n)(L) = m_n(T^{-1}L) = m_n(\mathbb{R}^n) = +\infty$. Thus $T \in GL_n(\mathbb{R})$. For $E \in \mathcal{L}^n$ with $0 < m_n(E) < +\infty$, by Corollary 2.3.4, $m_n(E) = m_n(T^{-1}E) = |\det T|^{-1} m_n(E)$, so $|\det T| = 1$. Conversely, if $T \in GL_n(\mathbb{R})$ and $|\det T| = 1$, then $(T_* m_n)(E) = m_n(T^{-1}E) = m_n(E)$ $\forall E \in \mathcal{L}^n$ by the same corollary.
(ii) Since $\det D = 1$, claim (i) gives $D_* m_2 = m_2$.

Structural Commentary: Exclude singular range; apply determinant scaling to preimages.
Key Insight: Linear volume preservation is exactly unit determinant. □

Theorem 1.7.12. *Poincaré Recurrence*
Let $(X, \mathcal{M}, \mu)$ satisfy $\mu(X) < +\infty$, let $T : X \to X$ be measurable with $T_*\mu = \mu$, and let $A \in \mathcal{M}$. Set $R_A := \{x \in A : \#\{n \geq 1 : T^n x \in A\} = +\infty\}$. Then $\mu(A \setminus R_A) = 0$.

Proof. Set $E := A \cap \bigcap_{n\geq 1} T^{-n}(A^c)$, so $E \cap T^{-m}A = \varnothing$ $\forall m \geq 1$. If $x \in T^{-i}E \cap T^{-j}E$ with $0 \leq i < j$, then $T^i x \in E \cap T^{-(j-i)}E \subseteq E \cap T^{-(j-i)}A$, a contradiction. Thus $(T^{-k}E)_{k\geq 0}$ is pairwise disjoint. For $N \geq 0$, measure preservation gives $(N+1)\mu(E) = \mu(\bigcup_{k=0}^{N} T^{-k}E) \leq \mu(X)$; letting $N \to \infty$ yields $\mu(E) = 0$.

If $x \in A \setminus R_A$, let $r \geq 0$ be its last visit time to A. Then $T^r x \in E$, so $A \setminus R_A \subseteq \bigcup_{r\geq 0} T^{-r}E$. Countable subadditivity and $\mu(T^{-r}E) = \mu(E) = 0$ give $\mu(A \setminus R_A) = 0$. *Structural Commentary*: The nonreturn set is wandering; finite-return points eventually enter it.
Key Insight: Finite measure forbids positive-measure wandering sets. □

2 Partial Differential Equations

2.1 First-Order PDE and Characteristics

2.2 Elliptic Equations

2.3 Parabolic Equations

2.4 Hyperbolic Equations

3 Weak Solutions and Evolution Equations

3.1 Sobolev Spaces for PDE

Distributions test against smooth compactly supported functions; Sobolev spaces package weak derivatives in normed or Hilbert spaces as a bridge to PDE.

Definition 3.1.1. *Test Functions and Distributions*
Let $\Omega \subseteq \mathbb{R}^n$ be open. A *test function* is a smooth function $\varphi \in C_c^\infty(\Omega)$. A *distribution* on Ω is a continuous linear functional on the test-function space, written $u \in \mathcal{D}'(\Omega)$ and paired as $\langle u, \varphi \rangle$.

Definition 3.1.2. *Weak Derivative*
Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let α be a multi-index. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak derivative* $D^\alpha u$ if

$$\int_\Omega u\, D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega v\, \varphi \, dx \qquad \text{for every } \varphi \in C_c^\infty(\Omega).$$

Definition 3.1.3. *Sobolev Spaces*
For $1 \leq p \leq \infty$ and an integer $k \geq 0$, the Sobolev space $W^{k,p}(\Omega)$ consists of functions $u \in L^p(\Omega)$ whose weak derivatives $D^\alpha u$ lie in $L^p(\Omega)$ for every $|\alpha| \leq k$. The norm is

$$\|u\|_{W^{k,p}} := \sum_{|\alpha|\leq k} \|D^\alpha u\|_{L^p}$$

with the usual supremum modification when $p = \infty$.

3.2 Weak Solutions and Variational Methods

3.3 Evolution Equations

Part XVIII
Numerical and Stochastic Computation

1 Numerical Analysis

1.1 Floating Point and Error Analysis

1.2 Nonlinear Equations and Optimization Basics

Method 1.2.1. *Newton and Secant Iterations*
To solve $f(x) = 0$:

(i) Newton uses tangent-line zero: $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$, $n \geq 1$ when $f'(x_{n-1}) \neq 0$.

(ii) Secant uses two-point slope: $x_n = x_{n-1} - \frac{f(x_{n-1})(x_{n-2}-x_{n-1})}{f(x_{n-2})-f(x_{n-1})}$, $n \geq 2$ when the denominator is nonzero.

These define iterates only; local convergence is proved with Taylor's theorem (Theorem 1.1.25).

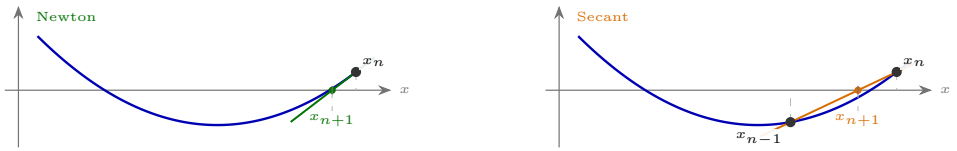

Figure 235: Newton: tangent-line zero. Secant: two-point line zero.

1.3 Interpolation and Approximation

1.4 Numerical Differentiation and Quadrature

1.5 Numerical Stability and Convergence

2 Numerical Differential Systems

2.1 Numerical ODEs

Numerical one-step methods replace the integral evolution of an initial-value problem by finitely many vector-field evaluations on a time grid.

Method 2.1.1. *Explicit Euler Method*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$, $f : U \to \mathbb{R}^n$, $(t_0, x_0) \in U$, and $T > 0$. For $x' = f(t, x)$, $x(t_0) = x_0$, choose $N \in \mathbb{N}$ with $N \geq 1$, set $h := T/N$ and $t_m := t_0 + mh$.

(i) Initialize $x_0^{(h)} := x_0$.

(ii) For $0 \leq m < N$ with $(t_m, x_m^{(h)}) \in U$, compute $x_{m+1}^{(h)} := x_m^{(h)} + hf(t_m, x_m^{(h)})$.

(iii) On each completed mesh interval, define the continuous Euler polygon by $x_h(t) := x_m^{(h)} + (t - t_m)f(t_m, x_m^{(h)})$ for $t_m \leq t \leq t_{m+1}$.

Thus $x_h' = f(t_m, x_m^{(h)})$ on each open mesh interval, while the nodes satisfy the explicit recurrence in (ii).

Theorem 2.1.2. *Uniform Error of the Euler Polygon*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C(U, \mathbb{R}^n)$, $(t_0, x_0) \in U$, and choose $T, \delta > 0$ with $V := [t_0, t_0 + T] \times \overline{B}_\delta(x_0) \subseteq U$.
Set $M := \max_V \|f\|$, assume $MT \leq \delta$, and assume $\exists L \geq 0$ such that $\|f(t, x) - f(t, y)\| \leq L \|x - y\|$ $\forall (t, x), (t, y) \in V$.
Let $\phi \in C^1([t_0, t_0 + T], \mathbb{R}^n)$ satisfy $\phi' = f(t, \phi)$, $\phi(t_0) = x_0$, and $(t, \phi(t)) \in V$ $\forall t \in [t_0, t_0 + T]$.
For $N \in \mathbb{N}$ with $N \geq 1$, set $h := T/N$, $t_m := t_0 + mh$, $x_0^{(h)} := x_0$, $x_{m+1}^{(h)} := x_m^{(h)} + hf(t_m, x_m^{(h)})$, and $x_h(t) := x_m^{(h)} + (t - t_m)f(t_m, x_m^{(h)})$ for $t_m \leq t \leq t_{m+1}$.
Define $\Delta(h) := \sup\{\|f(s, y) - f(t, x)\| : (s, y), (t, x) \in V,\ |s - t| \leq h,\ \|y - x\| \leq Mh\}$. Then:

(i) $(t, x_h(t)) \in V$ $\forall t \in [t_0, t_0 + T]$, and $\|x_h(t) - x_h(s)\| \leq M |t - s|$ $\forall s, t \in [t_m, t_{m+1}]$, $0 \leq m < N$.
(ii) With $h = T/N$, as $N \to \infty$, $\Delta(h) \to 0$ and $\|x_h - \phi\|_\infty \leq Te^{LT}\Delta(h)$; hence $x_h \to \phi$ uniformly.
(iii) If $\exists L_0 \geq 0$ such that $\|f(t, x) - f(s, x)\| \leq L_0 |t - s|$ $\forall (t, x), (s, x) \in V$, then $\Delta(h) \leq (L_0 + LM)h$ and $\|x_h - \phi\|_\infty \leq Te^{LT}(L_0 + LM)h$.

Proof.

(i) The initial graph point (t_0, x_0) lies in V. If $\left\|x_m^{(h)} - x_0\right\| \leq mhM$, then $(t_m, x_m^{(h)}) \in V$ and $\left\|x_{m+1}^{(h)} - x_0\right\| \leq \left\|x_m^{(h)} - x_0\right\| + h\left\|f(t_m, x_m^{(h)})\right\| \leq (m + 1)hM$.
Thus induction gives $\left\|x_m^{(h)} - x_0\right\| \leq mhM$. For $t \in [t_m, t_{m+1}]$, $\|x_h(t) - x_0\| \leq M(t - t_0) \leq MT \leq \delta$. Hence the graph lies in V, and the affine slope gives the meshwise Lipschitz bound.

(ii) Let $\tau_h(t) := t_m$ for $t \in [t_m, t_{m+1})$. The Euler polygon satisfies $x_h(t) = x_0 + \int_{t_0}^t f(\tau_h(s), x_h(\tau_h(s)))\, ds$.
Uniform continuity of $f|_V$ gives $\Delta(h) \to 0$. With $e(t) := \|x_h(t) - \phi(t)\|$, item (i) gives $\|x_h(s) - x_h(\tau_h(s))\| \leq Mh$, so $e(t) \leq \int_{t_0}^t (Le(s) + \Delta(h))\, ds$.
By Lemma 1.1.20(ii), $e(t) \leq (t - t_0)e^{L(t-t_0)}\Delta(h) \leq Te^{LT}\Delta(h)$.

(iii) For pairs in the definition of $\Delta(h)$, $\|f(s, y) - f(t, x)\| \leq L \|y - x\| + L_0 |s - t| \leq (LM + L_0)h$. Combine this with (ii).

Structural Commentary: Consistency supplies $\Delta(h)$; Gronwall propagates the local defect.
Key Insight: Uniform continuity is enough for convergence; time-Lipschitz control gives first order. □

Remark 2.1.3. *Euler Step-Doubling Control*
From a node (t_m, x_m), compare the full-step prediction $z_h := x_m + hf(t_m, x_m)$ with the two-half-step prediction $z_{1/2} := x_m + \frac{h}{2}f(t_m, x_m)$, $z_{2/2} := z_{1/2} + \frac{h}{2}f(t_m + h/2, z_{1/2})$. If $\|z_h - z_{2/2}\|$ exceeds a prescribed tolerance, reduce h and repeat the step. This is a practical error indicator rather than an a priori error bound.

Method 2.1.4. *Classical Fourth-Order Runge–Kutta Method*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$, $f : U \to \mathbb{R}^n$, $h > 0$, and $(t_m, x_m) \in U$, where x_m approximates a solution of $x' = f(t, x)$ at t_m. Set $t_{m+1} := t_m + h$ and assume every displayed argument of f below lies in U.

(i) Compute $k_{1,m} := f(t_m, x_m)$ and $k_{2,m} := f(t_m + h/2, x_m + hk_{1,m}/2)$.
(ii) Compute $k_{3,m} := f(t_m + h/2, x_m + hk_{2,m}/2)$ and $k_{4,m} := f(t_m + h, x_m + hk_{3,m})$.
(iii) Update $x_{m+1} := x_m + \frac{h}{6}(k_{1,m} + 2k_{2,m} + 2k_{3,m} + k_{4,m})$.

The four stages sample two midpoint predictions and one endpoint prediction before forming the weighted increment.

Method 2.1.5. *Heun's Improved Euler Method*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$, $f : U \to \mathbb{R}^n$, $h > 0$, and $(t_m, x_m) \in U$, where x_m approximates a solution of $x' = f(t, x)$ at t_m. Set $t_{m+1} := t_m + h$ and assume the predicted endpoint used below lies in U.

(i) Predict the endpoint by $y_m := x_m + hf(t_m, x_m)$.
(ii) Correct with the averaged slopes: $x_{m+1} := x_m + \frac{h}{2}[f(t_m, x_m) + f(t_{m+1}, y_m)]$.

Thus Heun replaces Euler's single left-endpoint slope by the average of the initial slope and its predicted endpoint value.

Proposition 2.1.6. *Heun-Type One-Step Error*
Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f \in C^2(U, \mathbb{R}^n)$, $I \subseteq \mathbb{R}$ an open interval, $t_0 \in I$, and $\phi \in C^1(I, \mathbb{R}^n)$ satisfy $(t, \phi(t)) \in U$ and $\phi'(t) = f(t, \phi(t))$ $\forall t \in I$, with $\phi(t_0) = x_0$.
Set $f_0 := f(t_0, x_0)$. Then $\exists h_* > 0$ such that, $\forall h \in (0, h_*)$, setting $t_1 := t_0 + h$ and $y_0 := x_0 + hf_0$ gives $t_1 \in I$ and $(t_1, y_0), (t_1, x_0), (t_0, y_0) \in U$. As $h \downarrow 0$:

(i) For $H_h := x_0 + \frac{h}{2}[f(t_0, x_0) + f(t_1, y_0)]$, $\|\phi(t_1) - H_h\| = O(h^3)$.
(ii) For $\widetilde{H}_h := x_0 + \frac{h}{2}[f(t_1, x_0) + f(t_0, y_0)]$, $\left\|\phi(t_1) - \widetilde{H}_h\right\| = O(h^3)$.

Proof. Set $G_0 := \partial_t f(t_0, x_0) + D_x f(t_0, x_0) f_0$. By Lemma 1.1.19, $\phi \in C^3$ locally, so $\phi(t_1) = x_0 + hf_0 + \frac{h^2}{2}G_0 + O(h^3)$.

(i) Taylor expansion of f at (t_0, x_0) gives $f(t_1, y_0) = f_0 + h\partial_t f(t_0, x_0) + hD_x f(t_0, x_0) f_0 + O(h^2) = f_0 + hG_0 + O(h^2)$.
Hence $H_h = x_0 + hf_0 + h^2 G_0/2 + O(h^3)$, which matches the expansion of $\phi(t_1)$.
(ii) Separately, $f(t_1, x_0) = f_0 + h\partial_t f(t_0, x_0) + O(h^2)$ and $f(t_0, y_0) = f_0 + hD_x f(t_0, x_0) f_0 + O(h^2)$.
Thus $\widetilde{H}_h = x_0 + hf_0 + h^2 G_0/2 + O(h^3)$, proving the second estimate.

Structural Commentary: Expand the exact flow and both predicted slopes through second order.
Key Insight: Averaging reconstructs the quadratic term of the exact solution. □

Example 2.1.7. *Euler and RK4 for Exponential Growth*
For $x' = x$, $x(0) = 1$, and $h = 0.1$, Euler gives $x_m = (1 + h)^m$, hence $x_{10} = 1.1^{10} \approx 2.5937424601$.
Classical RK4 gives $x_{m+1} = R(h)x_m$ with $R(h) = 1 + h + h^2/2 + h^3/6 + h^4/24$, so $x_{10} = R(0.1)^{10} \approx 2.7182797441$.
Since $x(1) = e \approx 2.7182818285$, the respective absolute errors at $t = 1$ are approximately 1.2454×10^{-1} and 2.0843×10^{-6}.

2.2 Geometric Numerical Integration

2.3 Finite Difference PDE Methods

2.4 Finite Element Methods

2.5 Numerical SDEs

2.6 Computational Stability Across ODE/PDE/SDE

3 Monte Carlo and Stochastic Computation

3.1 Monte Carlo Foundations

Monte Carlo methods turn integrals into averages of simulated random variables. This is especially useful when the integral is high-dimensional or when the target density is known only up to a normalizing constant.

Method 3.1.1. *Basic Monte Carlo Integration*

(i) Let $I = \int h(x) f(x)\, dx$, where f is a probability density.
(ii) For IID draws $X_1, \ldots, X_N \sim f$, define $\widehat{I}_N = N^{-1} \sum_{i=1}^{N} h(X_i)$; then $\widehat{I}_N \xrightarrow{p} I$ by the law of large numbers.
(iii) With $Y_i = h(X_i)$, set $s^2 = (N-1)^{-1} \sum_{i=1}^{N} (Y_i - \widehat{I}_N)^2$ and $\widehat{\mathrm{se}} = \frac{s}{\sqrt{N}}$.
(iv) An approximate $1 - \alpha$ Monte Carlo interval is $\widehat{I}_N \pm z_{\frac{\alpha}{2}} \widehat{\mathrm{se}}$.

Example 3.1.2. *Monte Carlo Integration from the LLN*
Let $X_1, \ldots, X_n$ be iid Uniform$(0, 1)$ and $f \in L^1([0, 1])$.
The estimator $\widehat{I}_n(f) := n^{-1} \sum_{i=1}^{n} f(X_i)$ satisfies $\widehat{I}_n(f) \xrightarrow{\mathbb{P}} \mathbb{E} f(X_1) = \int_0^1 f(x)\, dx$.
For $f(x) = (2\pi)^{-\frac{1}{2}} e^{-\frac{x^2}{2}}$ on $[0, 1]$, a 1000-point estimate near 0.3417 is consistent with $\int_0^1 f(x)\, dx \approx 0.3413$.

Example 3.1.3. *Integrating on an Interval*
For $X \sim$ Uniform(a, b), $I = \int_a^b h(x)\, dx = (b - a)\mathbb{E}[h(X)]$.
Given IID draws $X_1, \ldots, X_N \sim$ Uniform(a, b), estimate I by $\widehat{I}_N = (b - a)N^{-1} \sum_{i=1}^{N} h(X_i)$.

Example 3.1.4. *Estimating a Normal CDF*

Let $Z \sim N(0,1)$ and $\Phi(x) = \mathbb{P}(Z \le x)$. For IID draws $Z_1, \dots, Z_N \sim N(0,1)$, use $h(s) = \mathbf{1}\{s \le x\}$ and $\widehat{\Phi}(x) = N^{-1} \sum_{i=1}^{N} \mathbf{1}\{Z_i \le x\}$.

Thus $\widehat{\Phi}(x)$ is the empirical fraction of draws not exceeding x.

Method 3.1.5. *Inverse-Transform Generation*

(i) *Exponential*: if $U \sim \text{Uniform}(0,1)$ and $\lambda > 0$, then $T = -\lambda^{-1} \log(1-U) \sim \text{Exp}(\lambda)$. (or $T = -\lambda^{-1} \log U$)
(ii) *Weibull*: if $F(x) = 1 - \exp(-(\frac{x}{\alpha})^{\beta})$ for $x \ge 0$ and $\alpha, \beta > 0$,
then $F^{-1}(u) = \alpha[-\log(1-u)]^{\frac{1}{\beta}}$ and $W = \alpha[-\log(1-U)]^{\frac{1}{\beta}} \sim F$.
(iii) *Integer-valued inverse CDF*: if F is integer-valued and $U \sim \text{Uniform}(0,1)$, set $Y = \inf\{k \in \mathbb{Z} : F(k) \ge U\}$.
Equivalently $F(k-1) < U \le F(k)$. Then $F_Y = F$.

Key Insight: The simulation problem is reduced to solving $u = F(x)$ for x.

Method 3.1.6. *Rejection Sampling*

Let f be a target density on an interval $[a,b]$, possibly with infinite endpoints.
Choose an envelope M such that $M(x) \ge f(x)$ on $[a,b]$ and $A := \int_a^b M(t)\,dt < \infty$. Let $m(x) := \frac{M(x)}{A}$.

(i) Generate T with density m.
(ii) Generate $U \sim \text{Uniform}(0,1)$ independently of T.
(iii) Accept $X = T$ if $U \le \frac{f(T)}{M(T)}$; otherwise reject T and repeat.

Key Insight: Sample under an easy envelope and keep only points that fall below the target curve.

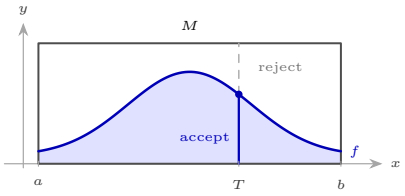

Figure 236: Rejection sample accepts when the vertical draw lies below $f(T)$.

Proposition 3.1.7. *Rejection Sampling Correctness*
Let f be a density, let $M \ge f$ with $A := \int M(x)\,dx < \infty$, and set $m := \frac{M}{A}$. If $T \sim m$, $U \sim \text{Uniform}(0,1)$ independently, and $\mathcal{A} := \{U \le \frac{f(T)}{M(T)}\}$, then $\mathbb{P}(\mathcal{A}) = \frac{1}{A}$ and $T \mid \mathcal{A}$ has density f.

Proof. For measurable B, $\mathbb{P}(T \in B, \mathcal{A}) = \int_B \frac{m(x)f(x)}{M(x)}\,dx = A^{-1} \int_B f(x)\,dx$.
Taking $B = \mathbb{R}$ gives $\mathbb{P}(\mathcal{A}) = A^{-1}$, and hence $\mathbb{P}(T \in B \mid \mathcal{A}) = \int_B f(x)\,dx$.
Structural Commentary: Acceptance reweights the envelope m back to f.
Key Insight: Conditional on acceptance, the normalized accepted density is exactly the target. □

3.2 Variance Reduction

Method 3.2.1. *Importance Sampling*

(i) For $I = \int h(x)f(x)\,dx$, choose a tractable proposal density g with $g(x) > 0$ wherever $h(x)f(x) \ne 0$.
(ii) Then $I = \int h(x)f(x)g(x)^{-1}g(x)\,dx = \mathbb{E}_g\left[\frac{h(X)f(X)}{g(X)}\right]$.
(iii) For IID draws $X_1, \dots, X_N \sim g$, use $\widehat{I}_N = N^{-1} \sum_{i=1}^{N} \frac{h(X_i)f(X_i)}{g(X_i)}$.

Remark 3.2.2. *Choosing the Proposal*
The importance weights have large, possibly infinite, variance when g is too small on regions where $|h|\,f$ is large.
Choose g to approximate the shape of $|h|\,f$ while giving it tails at least as thick as the target integrand.

Theorem 3.2.3. *Optimal Importance Density*
Let $I = \int h(x)f(x)\,dx$ and $A = \int |h(x)|\,f(x)\,dx \in (0,\infty)$.
For proposal densities g with $g(x) > 0$ wherever $|h(x)|\,f(x) > 0$, define $W_g = \frac{h(X)f(X)}{g(X)}$ for $X \sim g$.
The variance $\text{Var}_g(W_g)$ is minimised by $g^*(x) = \frac{|h(x)|f(x)}{A}$, and the minimum variance is $A^2 - I^2$.

Proof. For each admissible g, $\mathbb{E}_g W_g = I$, so $\text{Var}_g(W_g) = M(g) - I^2$, where $M(g) = \mathbb{E}_g(W_g^2) = \int \frac{h(x)^2 f(x)^2}{g(x)}\,dx$.
For any admissible g, Cauchy-Schwarz gives $A^2 = (\int |h(x)|\,f(x)g(x)^{-\frac{1}{2}} g(x)^{\frac{1}{2}}\,dx)^2 \le M(g) \int g(x)\,dx = M(g)$.
Equality holds precisely when $|h(x)|\,f(x)g(x)^{-\frac{1}{2}} = c\,g(x)^{\frac{1}{2}}$ a.e., equivalently $g(x) \propto |h(x)|\,f(x)$.
Normalization gives $g^*(x) = \frac{|h(x)|f(x)}{A}$, hence $M(g) \ge M(g^*) = A^2$ and $\text{Var}_g(W_g) \ge A^2 - I^2 = \text{Var}_{g^*}(W_{g^*})$.
Structural Commentary: Fixed mean reduces variance to second-moment minimization.
Key Insight: Cauchy-Schwarz is sharp exactly when the proposal is proportional to $|h|\,f$. □

3.3 Markov Chain Monte Carlo

Monte Carlo integration in Method 3.1.1 uses independent draws from the target density. Markov chain Monte Carlo instead constructs a Markov chain $X_0, X_1, \dots$ whose stationary density is the target f. When the chain mixes well, sample averages from the chain are used as approximations to expectations under f.

Method 3.3.1. *Metropolis-Hastings*
Let f be a target density available up to proportionality, and let $q(\cdot \mid x)$ be a tractable proposal density.
Choose X_0 arbitrarily; given $X_i = x$, define X_{i+1} as follows.

(i) Draw $Y \sim q(\cdot \mid x)$.
(ii) Set $r(x, Y) = \min\left\{\frac{f(Y)q(x|Y)}{f(x)q(Y|x)}, 1\right\}$.
(iii) Set $X_{i+1} = Y$ with probability $r(x, Y)$, and set $X_{i+1} = x$ otherwise.

Key Insight: Only ratios of f enter r, so unknown normalising constants cancel.

Proposition 3.3.2. *Detailed Balance for Metropolis-Hastings*
Let $\Phi(x, y) = f(x)q(y \mid x)$ and assume $\Phi(x, y) > 0$ iff $\Phi(y, x) > 0$.
For $x \neq y$, define $\alpha(x, y) = \min\left\{\frac{\Phi(y,x)}{\Phi(x,y)}, 1\right\}$ when $\Phi(x, y) > 0$ and $\alpha(x, y) = 0$ otherwise.
Set $p(x, y) = q(y \mid x)\alpha(x, y)$. Then $f(x)p(x, y) = f(y)p(y, x)$ for $x \neq y$.

Proof. If $\Phi(x, y) = 0$, the support condition gives $\Phi(y, x) = 0$, hence $p(x, y) = p(y, x) = 0$.
If $\Phi(x, y) > 0$, then $f(x)p(x, y) = \Phi(x, y)\min\left\{\frac{\Phi(y,x)}{\Phi(x,y)}, 1\right\} = \min\{\Phi(x, y), \Phi(y, x)\}$.
Interchanging x, y gives the same expression for $f(y)p(y, x)$, QED.
Structural Commentary: The acceptance rule symmetrizes the weighted proposal flow.
Key Insight: Both directions reduce to the same minimum of the two proposal fluxes. □

Method 3.3.3. *Random-Walk Metropolis-Hastings*

(i) Use the proposal $Y = X_i + \varepsilon_i$, where ε_i has a symmetric density.
(ii) Then $q(y \mid x) = q(x \mid y)$, so the acceptance probability is $r(x, y) = \min\left\{\frac{f(y)}{f(x)}, 1\right\}$.
(iii) For $\varepsilon_i \sim N(0, b^2)$, small b gives slow local moves, large b gives frequent rejection. Tune b for adequate mixing.

Example 3.3.4. *Cauchy Target*
For Cauchy target $f(x) = \frac{1}{\pi(1+x^2)}$ and symmetric random-walk proposal, $r(x, y) = \min\left\{\frac{1+x^2}{1+y^2}, 1\right\}$.
The proposal scale changes mixing efficiency, not the stationary target.

Method 3.3.5. *Independence Metropolis-Hastings*

(i) Use a state-independent proposal $Y \sim g$, so $q(y \mid x) = g(y)$.
(ii) The acceptance probability is $r(x, y) = \min\{\frac{f(y)g(x)}{f(x)g(y)}, 1\}$.
(iii) Choose g as a tractable approximation to the target with sufficient mass on important regions.

Method 3.3.6. *Gibbs Sampling*

(i) Let (X, Y) have joint density $f_{X,Y}$ with tractable conditionals $f_{X|Y}$ and $f_{Y|X}$.
(ii) Starting from (X_0, Y_0), iterate $X_{n+1} \sim f_{X|Y}(\cdot \mid Y_n)$ and $Y_{n+1} \sim f_{Y|X}(\cdot \mid X_{n+1})$.
(iii) In higher dimension, update one coordinate or block at a time from its full conditional.

Method 3.3.7. *Metropolis within Gibbs*

(i) If $f_{X_j|X_{-j}}$ is not directly simulable, update X_j by Metropolis-Hastings conditional on X_{-j}.
(ii) In the bivariate case, propose $Z \sim q(\cdot \mid X_n)$ and accept $X_{n+1} = Z$ with probability $\min\{\frac{f(Z,Y_n)q(X_n|Z)}{f(X_n,Y_n)q(Z|X_n)}, 1\}$.
(iii) Then update Y analogously, conditional on the newly updated X.

3.4 Sequential Monte Carlo

3.5 Rare-Event Simulation

Example 3.5.1. *Rare Tail Probability*
For $Z \sim N(0, 1)$, write $I = \mathbb{P}(Z > 3) = \int \mathbf{1}\{x > 3\}f(x)\,dx$.
If g is centered near the right tail and $X_1, \dots, X_N \sim g$, estimate I by $\widehat{I}_N = N^{-1}\sum_{i=1}^{N}\frac{\mathbf{1}\{X_i>3\}f(X_i)}{g(X_i)}$.

3.6 Stochastic Optimization

Definition 3.6.1. *Online Estimator*
Let $(X_t)_{t\geq 1}$ be observations in $\mathcal{X}$, let $\widehat{\theta}_0 \in \Theta \subseteq \mathbb{R}^d$, let $(a_t)_{t\geq 0}$ be positive step sizes, and let $H : \Theta \times \mathcal{X} \to \mathbb{R}^d$.
$\widehat{\theta}_{t+1} := \widehat{\theta}_t + a_t H(\widehat{\theta}_t, X_{t+1})$ for $t \geq 0$.
Structural Commentary: $(\widehat{\theta}_t)$ is adapted to $\mathcal{F}_t := \sigma(\widehat{\theta}_0, X_1, \dots, X_t)$.
Key Insight: Under stationarity, analyze the recursion through $h(\theta) := \mathbb{E}H(\theta, X)$.

Method 3.6.2. *Minibatch Stochastic Gradient Descent*
Let $J_n(\theta) = n^{-1}\sum_{i=1}^{n} L(f_\theta(x_i), y_i)$, $|B_k| = b$, and $\eta_k > 0$.

(i) Sample B_k uniformly.
(ii) Compute $g_k = b^{-1}\sum_{i\in B_k}\nabla_\theta L(f_{\theta_k}(x_i), y_i)$.
(iii) Update $\theta_{k+1} = \theta_k - \eta_k g_k$.

Uniform sampling gives $\mathbb{E}[g_k \mid \theta_k] = \nabla J_n(\theta_k)$; larger b trades lower noise for higher cost.

Definition 3.6.3. *Momentum and Adaptive Methods*
Let g_k denote the stochastic gradient at θ_k; $\odot$, division, and square roots are coordinatewise.

(i) *Momentum*: $v_k = \mu v_{k-1} + g_k$ and $\theta_{k+1} = \theta_k - \eta_k v_k$, with $0 \le \mu < 1$.
(ii) *AdaGrad*: $r_k = r_{k-1} + g_k \odot g_k$ and $\theta_{k+1} = \theta_k - \frac{\eta_k g_k}{\sqrt{r_k}+\varepsilon}$.
(iii) *RMSProp*: $r_k = \rho r_{k-1} + (1-\rho) g_k \odot g_k$ and $\theta_{k+1} = \theta_k - \frac{\eta_k g_k}{\sqrt{r_k}+\varepsilon}$, with $0 < \rho < 1$.
(iv) *Adam*: update (m_k, r_k) as exponential averages of $(g_k, g_k \odot g_k)$,
set $(\widehat{m}_k, \widehat{r}_k) = (\frac{m_k}{1-\beta_1^k}, \frac{r_k}{1-\beta_2^k})$, and $\theta_{k+1} = \theta_k - \frac{\eta_k \widehat{m}_k}{\sqrt{\widehat{r}_k}+\varepsilon}$.

Part XIX
Statistical Inference and Time Series

Statistical inference turns probabilistic models into procedures for learning from data. The probability, measure-theoretic, stochastic-process, optimization, and Monte Carlo material developed in Parts 11–14 is treated here as prerequisite structure.

1 Statistical Models and Experiments

1.1 Statistical Experiments and Likelihoods

A statistical problem starts with observable data and a family of probability laws that could have generated it. The model is not the data itself; it is the mathematical language in which uncertainty about the data-generating mechanism is expressed.

Definition 1.1.1. *Statistical Experiment and Model*

(i) A *statistical experiment* is $(\mathcal{X}, \mathcal{A}, \mathcal{P})$, where $(\mathcal{X}, \mathcal{A})$ is measurable and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is a family of probability measures on $(\mathcal{X}, \mathcal{A})$.
(ii) Θ is the *parameter space*, $X \in \mathcal{X}$ is the *sample*, and $\mathcal{P}$ is the *statistical model*.
(iii) The model is *well specified* if the true law $P_0 \in \mathcal{P}$.
(iv) The model is *misspecified* if $P_0 \notin \mathcal{P}$.

Definition 1.1.2. *Identifiability and Functionals*

(i) The parameterization is *identifiable* if $P_\theta = P_{\theta'}$ implies $\theta = \theta'$.
(ii) A *statistical functional* is a map $T : \mathcal{P} \to \mathcal{T}$, such as $T(P) = \int x\, dP(x)$ or $T(P) = \text{median}(P)$.
(iii) In a parametric model, a *target parameter* may be a function $\psi(\theta)$ rather than the full parameter θ.

Key Insight: Estimation should target a law or functional that the experiment can actually distinguish.

Definition 1.1.3. *Dominated Models, Likelihood, Score, and Information*

(i) A model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is *dominated* by μ if $P_\theta \ll \mu$ for all $\theta \in \Theta$; write $p_\theta := \frac{dP_\theta}{d\mu}$.
(ii) Given x, the *likelihood* and *log-likelihood* are $L_x(\theta) := p_\theta(x)$ and $\ell_x(\theta) := \log p_\theta(x)$ when $p_\theta(x) > 0$.
(iii) If $\Theta \subseteq \mathbb{R}^d$ and differentiation is valid, the *score* is $S_\theta(X) := \nabla_\theta \ell_X(\theta)$.
(iv) The *Fisher information* is $I(\theta) := \mathbb{E}_\theta[S_\theta(X) S_\theta(X)^\top]$.

Definition 1.1.4. *Maximum Likelihood Estimator*

For likelihood $L_X(\theta)$ and log-likelihood $\ell_X(\theta)$, a *maximum likelihood estimator* is any measurable $\widehat{\theta}_n(X) \in \arg\max_{\theta \in \Theta} L_X(\theta) = \arg\max_{\theta \in \Theta} \ell_X(\theta)$; equivalently, $\widehat{\theta}_n \in \arg\min_{\theta \in \Theta} [-\ell_X(\theta)]$.

Example 1.1.5. *Bernoulli and Normal Likelihoods*

Let $X_1, \dots, X_n$ be independent observations.

(i) If $X_i \sim \text{Bernoulli}(p)$, then $L_x(p) = p^s (1-p)^{n-s}$, where $s := \sum_{i=1}^n x_i$.

(ii) If $X_i \sim N(\mu, \sigma^2)$ and σ^2 is known, then $\ell_x(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2$, maximized at $\widehat{\mu} = \overline{x}$.

Example 1.1.6. *Multinomial Cell Probabilities*

Let $X = (X_1, \dots, X_m)$ have total count n.

(i) If $X \sim \text{Multinomial}(n; p_1, \dots, p_m)$ and $\sum_{j=1}^m p_j = 1$, then $L_x(p) \propto \prod_{j=1}^m p_j^{x_j}$ and $\widehat{p}_j = \frac{x_j}{n}$.

(ii) If $p_j = p_j(\theta)$, then $L_x(\theta) \propto \prod_{j=1}^m p_j(\theta)^{x_j}$ and $\widehat{\theta} \in \arg\max_\theta \sum_{j=1}^m x_j \log p_j(\theta)$.

Remark 1.1.7. *Likelihood as Relative Evidence*

For fixed data x, the likelihood is a function of θ, not a probability distribution on Θ.
Ratios $\frac{L_x(\theta_1)}{L_x(\theta_0)}$ compare how well two model laws explain the same observation.

1.2 Sufficiency, Exponential Families, and Local Asymptotics

The shape of a model determines which summaries of the data preserve information and which approximations become valid in large samples.

Definition 1.2.1. *Statistics, Sufficiency, and Ancillarity*

(i) A *statistic* is a measurable map $T : \mathcal{X} \to \mathcal{T}$.
(ii) T is *sufficient* if $\mathcal{L}_\theta(X \mid T)$ is independent of θ.
(iii) T is *ancillary* if $\mathcal{L}_\theta(T)$ is independent of θ.

Theorem 1.2.2. *Factorization Criterion*

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be dominated by μ, with $p_\theta := \frac{dP_\theta}{d\mu}$. A statistic $T : \mathcal{X} \to \mathcal{T}$ is sufficient iff there exist measurable $g_\theta : \mathcal{T} \to [0, \infty)$ and $h : \mathcal{X} \to [0, \infty)$ such that $p_\theta(x) = g_\theta(T(x))h(x)$ μ-a.e. for all θ.

Proof.

($\Rightarrow$) If T is sufficient, write $\nu_\theta := \mathcal{L}_\theta(T)$ and choose a θ-free kernel $K_t = \mathcal{L}_\theta(X \mid T = t)$, so $P_\theta(A) = \int K_t(A)\nu_\theta(dt)$. Choosing densities for ν_θ and for K_t on each fiber gives $p_\theta(x) = g_\theta(T(x))h(x)$.
($\Leftarrow$) If $p_\theta = g_\theta \circ T\, h$, then on the fiber $T = t$ the factor $g_\theta(t)$ cancels: for $c(t) := \int_{T^{-1}\{t\}} h\, d\mu_t > 0$, $\mathbb{P}_\theta(X \in A \mid T = t) = c(t)^{-1} \int_{A \cap T^{-1}\{t\}} h\, d\mu_t$, independent of θ.

Structural Commentary: Factorisation separates parameter information $g_\theta(T)$ from residual shape h.
Key Insight: Sufficiency keeps exactly the likelihood's θ-dependent part. □

Example 1.2.3. *Bernoulli and Normal Sufficient Statistics*
Let $X_1, \dots, X_n$ be independent and let $x = (x_1, \dots, x_n)$.

(i) If $X_i \sim \text{Bernoulli}(\theta)$, then $p_\theta(x) = \theta^s(1-\theta)^{n-s}$ with $s := \sum_i x_i$.
Hence $T = \sum_i X_i$ is sufficient for θ, and $\mathcal{L}_\theta(X \mid T = t)$ is uniform on $\{x : \sum_i x_i = t\}$.

(ii) If $X_i \sim N(\mu, \sigma^2)$ with (μ, σ^2) unknown, then $p_{\mu,\sigma^2}(x) \propto (\sigma^2)^{-\frac{n}{2}} \exp\{-\frac{\sum_i x_i^2 - 2\mu \sum_i x_i + n\mu^2}{2\sigma^2}\}$.
Hence $T = (\sum_i X_i, \sum_i X_i^2)$ is sufficient for (μ, σ^2).

Corollary 1.2.4. *Likelihood Estimators Use Sufficient Statistics*
Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be dominated with densities p_θ, and let T be sufficient with $p_\theta(x) = g_\theta(T(x))h(x)$.
If $a(t) \in \arg\max_{\theta \in \Theta} g_\theta(t)$ exists, then $\widehat{\theta}(X) := a(T(X))$ is an MLE.

Proof. For fixed x, $L_x(\theta) = p_\theta(x) = h(x)g_\theta(T(x))$. If $h(x) > 0$, then $\arg\max_\theta L_x(\theta) = \arg\max_\theta g_\theta(T(x))$.
If $h(x) = 0$, then $L_x(\theta) = 0$ for all θ. Thus an MLE may be chosen as $a(T(x))$.
Structural Commentary: The θ-free factor $h(x)$ is irrelevant to maximization.
Key Insight: MLEs can depend on data only through a sufficient statistic. □

Theorem 1.2.5. *Rao–Blackwell Improvement*
Let T be sufficient and let $\widehat{\psi} \in L^2(P_\theta)$ estimate a real target $\psi(\theta)$. Choose the θ-free version $\widetilde{\psi}(T) := \mathbb{E}_\theta(\widehat{\psi} \mid T)$.
Then $\mathbb{E}_\theta \widetilde{\psi} = \mathbb{E}_\theta \widehat{\psi}$ and $\mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2 \le \mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2$ for all θ, with equality iff $\widehat{\psi} = \widetilde{\psi}$ P_θ-a.s.

Proof. Sufficiency makes $\mathcal{L}_\theta(X \mid T)$ independent of θ, so $\widetilde{\psi}$ is a T-statistic. Also $\mathbb{E}_\theta \widetilde{\psi} = \mathbb{E}_\theta \widehat{\psi}$.
Since $\mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi} \mid T) = 0$ and $\widetilde{\psi} - \psi(\theta)$ is T-measurable, $\mathbb{E}_\theta[(\widehat{\psi} - \widetilde{\psi})(\widetilde{\psi} - \psi(\theta))] = 0$.
Hence $\mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2 = \mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi})^2 + \mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2 \ge \mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2$, with equality iff $\mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi})^2 = 0$.
Structural Commentary: Rao–Blackwellization projects an estimator onto the $\sigma(T)$-measurable statistics.
Key Insight: Conditioning on a sufficient statistic preserves bias and removes residual noise. □

Definition 1.2.6. *Exponential Family*

(i) A dominated model $\{P_\eta : \eta \in \mathcal{N}\}$ is a *k-parameter exponential family* if $\mathcal{N} \subseteq \mathbb{R}^k$ and $p_\eta(x) = \exp\{\eta^\top T(x) - A(\eta)\}h(x)$.
(ii) η is the *natural parameter*, $T(X)$ is the *natural statistic*, and h is the carrier density.
(iii) $A(\eta)$ is the *log-partition function*: $A(\eta) = \log \int \exp\{\eta^\top T(x)\}h(x)\, d\mu(x)$ when finite.

Proposition 1.2.7. *Moment Identities in Full Exponential Families*
Let $p_\eta(x) = \exp\{\eta^\top T(x) - A(\eta)\}h(x)$ be a full exponential family with open natural parameter space, and assume differentiation under the integral is valid. Then $\nabla A(\eta) = \mathbb{E}_\eta T(X)$ and $\nabla^2 A(\eta) = \text{Cov}_\eta(T(X))$.

Proof. From $1 = \int p_\eta\, d\mu$, differentiation gives $0 = \int (T - \nabla A(\eta)) p_\eta\, d\mu$, hence $\nabla A(\eta) = \mathbb{E}_\eta T(X)$.
For coordinates i, j, differentiating $\mathbb{E}_\eta T_i = \int T_i p_\eta\, d\mu$ gives $\partial_j \mathbb{E}_\eta T_i = \mathbb{E}_\eta[T_i(T_j - \partial_j A(\eta))] = \text{Cov}_\eta(T_i, T_j)$, QED.
Structural Commentary: The log-partition function cumulatively encodes the natural statistic.
Key Insight: First and second derivatives of A give mean and covariance. □

Definition 1.2.8. *Local Asymptotic Normality*
For experiments $\{P_\theta^{(n)} : \theta \in \Theta \subseteq \mathbb{R}^d\}$, set $\theta_{n,h} := \theta_0 + \frac{h}{\sqrt{n}}$.

(i) The sequence is *locally asymptotically normal* at θ_0 with information $I(\theta_0)$ if,
for each fixed $h \in \mathbb{R}^d$, $\log(\frac{dP^{(n)}_{\theta_{n,h}}}{dP^{(n)}_{\theta_0}}) = h^\top \Delta_n - \frac{1}{2} h^\top I(\theta_0) h + o_{P^{(n)}_{\theta_0}}(1)$.

(ii) The *central sequence* satisfies $\Delta_n \Rightarrow N(0, I(\theta_0))$ under $P^{(n)}_{\theta_0}$.

Remark 1.2.9. *Why Local Asymptotics Matter*
LAN replaces local likelihood ratios by the Gaussian quadratic experiment $h^\top \Delta - \frac{1}{2} h^\top I(\theta_0) h$, giving normal approximations for estimators, tests, and confidence regions.

1.3 Design, Randomization, and Observational Comparisons

Comparative inference is only as meaningful as the experiment or study that produced the groups. The same two-sample statistic can estimate a causal treatment effect in a randomised experiment, describe an association in an observational study, or amplify bias if the groups were formed by a hidden mechanism.

Definition 1.3.1. *Controls, Randomisation, and Blinding*

(i) A *controlled experiment* compares $Y \mid T = 1$ and $Y \mid T = 0$ under common conditions.
(ii) *Randomisation* assigns $T \in \{0, 1\}$ so that T is independent of pre-treatment factors C by design.
(iii) *Blinding* conceals T when knowledge of treatment could affect Y or its measurement.

Definition 1.3.2. *Placebo Effect and Confounding*

(i) A *placebo effect* is variation in Y caused by belief in treatment rather than the active mechanism.
(ii) Placebos and blinding separate expectation effects from the causal effect of T.
(iii) A pre-treatment variable C is *confounded* with treatment T for response Y if $\mathcal{L}(C \mid T = 1) \neq \mathcal{L}(C \mid T = 0)$ and C affects Y.

(iv) An *observational study* compares groups without assigning T.
Group differences may reflect confounding rather than the factor of interest.

Example 1.3.3. *Mammary Artery Ligation*
A blinded trial randomised patients after preliminary incisions to ligation ($T = 1$) or sham surgery ($T = 0$). Similar outcomes show that controls remove background improvement; blinding limits placebo, observer effects.

Principle 1.3.4. *Confounding, Blocking, and Pairing*
Randomisation balances unmeasured covariates C in probability; blocking, matching, and pairing condition on known influential C before randomisation. Pairing is one-factor blocking; blocked and factorial designs extend this control to multiple factors.

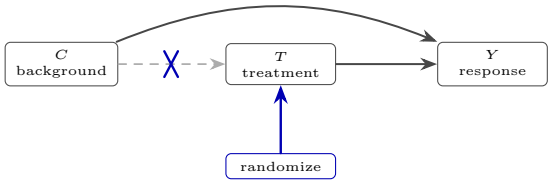

Figure 237: Randomization breaks the confounding path from C to T.

Example 1.3.5. *Uncontrolled Adjustments and Position Effects*
In the Lanarkshire milk experiment, post-randomisation movement between groups made T depend on perceived nourishment C, reintroducing selection bias; weighing children in clothes also confounded measurement with socioeconomic status. In the FD&C Red No. 40 study, cage location was systematic in sex and dose, so position effects confounded treatment comparisons; randomising or blocking cage location would protect inference.

Remark 1.3.6. *Comparative Study Checklist*
Before interpreting group comparisons, record sampling, treatment assignment T, controls, blinding, blocking or pairing factors C, dependence among repeated measurements, and the number of tested comparisons.
These determine whether the analysis supports causality, association, or hypothesis generation.

2 Estimation Theory

2.1 Estimators, Bias, Variance, and Risk

An estimator is a statistic interpreted as an answer to a target question. Its quality depends on the target, the loss function, and the model law under which performance is measured.

Definition 2.1.1. *Estimator, Loss, and Risk*
(i) A target parameter is a map $\psi : \Theta \to \Psi$.
(ii) An *estimator* of $\psi(\theta)$ is a statistic $\delta(X) \in \mathcal{D}$.
(iii) A *loss function* is $L : \Theta \times \mathcal{D} \to [0, \infty]$.
(iv) The *risk* of δ is $R(\theta, \delta) := \mathbb{E}_\theta L(\theta, \delta(X))$.
(v) For squared-error loss $L(\theta, d) = \|d - \psi(\theta)\|^2$, $R(\theta, \delta)$ is mean squared error.

Definition 2.1.2. *Sampling Distribution and Standard Error*
(i) For a statistic $T_n = T_n(X_1, \ldots, X_n)$ under P_θ, its *sampling distribution* is $\mathcal{L}_\theta(T_n)$.
(ii) If $T_n \in L^2(P_\theta)$, its *standard error* is $\mathrm{SE}_\theta(T_n) := \sqrt{\mathrm{Var}_\theta(T_n)}$.
(iii) An *estimated standard error* estimates $\mathrm{SE}_\theta(T_n)$, typically by plug-in, asymptotics, or resampling.

Definition 2.1.3. *Bias, Variance, and Consistency*
For real-valued $\widehat{\psi}$, $\mathrm{Bias}_\theta(\widehat{\psi}) := \mathbb{E}_\theta\widehat{\psi} - \psi(\theta)$, and $\mathrm{Var}_\theta(\widehat{\psi}) := \mathbb{E}_\theta(\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi})^2$.
(i) $\widehat{\psi}$ is *unbiased* if $\mathrm{Bias}_\theta(\widehat{\psi}) = 0$ for all θ.
(ii) $\widehat{\psi}_n$ is *consistent* at θ if $\widehat{\psi}_n \to \psi(\theta)$ in P_θ-probability.

Proposition 2.1.4. *Bias-Variance Decomposition*
Let $\widehat{\psi} \in L^2(P_\theta)$ be real-valued. Under squared-error loss, $\mathrm{MSE}_\theta(\widehat{\psi}) := \mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2 = \mathrm{Var}_\theta(\widehat{\psi}) + \mathrm{Bias}_\theta(\widehat{\psi})^2$.

Proof. Write $\widehat{\psi} - \psi(\theta) = (\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi}) + (\mathbb{E}_\theta\widehat{\psi} - \psi(\theta))$. Square, take $\mathbb{E}_\theta$ gives the identity since $\mathbb{E}_\theta(\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi}) = 0$.
Structural Commentary: MSE splits into random fluctuation plus squared systematic error.
Key Insight: Accuracy requires controlling both variance and bias. □

Example 2.1.5. *Sample Mean*
If $X_1, \ldots, X_n$ are independent with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$,
then $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ satisfies $\mathbb{E}\overline{X}_n = \mu$, $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$, and $\overline{X}_n \xrightarrow{P} \mu$.

Definition 2.1.6. *Plug-In and M-Estimators*
(i) A *plug-in estimator* estimates $T(P)$ by $T(\widehat{P}_n)$, where $\widehat{P}_n$ is empirical or fitted.
(ii) An *M-estimator* satisfies $\widehat{\theta}_n \in \arg\min_{\theta \in \Theta} M_n(\theta)$ for an empirical criterion M_n.
(iii) Common choices are $M_n =$ negative log-likelihood, squared loss, or absolute loss.

Definition 2.1.7. *Location M-Estimates*
For data $x_1, \ldots, x_n \in \mathbb{R}$, scale $\sigma > 0$, and loss $\Psi : \mathbb{R} \to \mathbb{R}$, $\widehat{\mu} \in \arg\min_{\mu \in \mathbb{R}} \sum_{i=1}^{n} \Psi(\frac{x_i - \mu}{\sigma})$.
Key Insight: Quadratic loss gives $\overline{x}$; absolute or subquadratic loss improves outlier resistance.

2.2 Information Bounds and Asymptotic Efficiency

Information inequalities describe the best possible local accuracy under regular models. Efficient procedures are those whose asymptotic error reaches these bounds.

Lemma 2.2.1. *Two Forms of Fisher Information*
For a regular one-parameter dominated model, let $\ell_\theta(x) := \log p_\theta(x)$ *and* $S_\theta(X) := \partial_\theta \ell_\theta(X)$.
Then $I(\theta) := \mathbb{E}_\theta S_\theta(X)^2 = -\mathbb{E}_\theta \partial_\theta^2 \ell_\theta(X)$.

Proof. Differentiating $\int p_\theta \, d\mu = 1$ gives $\mathbb{E}_\theta S_\theta = 0$.
Differentiating, $0 = \partial_\theta \mathbb{E}_\theta S_\theta = \mathbb{E}_\theta[\partial_\theta S_\theta + S_\theta^2] = \mathbb{E}_\theta[\partial_\theta^2 \ell_\theta + S_\theta^2]$, since $\partial_\theta p_\theta = S_\theta p_\theta$. Thus $I(\theta) = \mathbb{E}_\theta S_\theta^2 = -\mathbb{E}_\theta \partial_\theta^2 \ell_\theta$.
Structural Commentary: Fisher information is both score variance and expected log-likelihood curvature.
Key Insight: Normalisation forces the mean score to vanish, linking slope variance to curvature. □

Theorem 2.2.2. *Cramer–Rao Bound*
Let $X_1, \dots, X_n \sim P_\theta$ *be iid from a regular one-parameter dominated model with one-observation information* $I(\theta) > 0$. *If* $\widehat{\psi} \in L^2(P_\theta)$ *is unbiased for differentiable* $\psi(\theta)$, *then* $\mathrm{Var}_\theta(\widehat{\psi}) \geq \frac{\psi'(\theta)^2}{nI(\theta)}$.

Proof. Let $S_n(\theta) := \partial_\theta \log L_n(\theta) = \sum_{i=1}^n S_\theta(X_i)$. Regularity gives $\mathbb{E}_\theta S_n = 0$ and $\mathrm{Var}_\theta(S_n) = \mathbb{E}_\theta S_n^2 = nI(\theta)$.
Differentiating $\mathbb{E}_\theta \widehat{\psi} = \psi(\theta)$ gives $\psi'(\theta) = \mathbb{E}_\theta[\widehat{\psi} S_n] = \mathrm{Cov}_\theta(\widehat{\psi}, S_n)$.
Thus $\psi'(\theta)^2 \leq \mathrm{Var}_\theta(\widehat{\psi}) \, \mathrm{Var}_\theta(S_n) = \mathrm{Var}_\theta(\widehat{\psi}) nI(\theta)$ by Cauchy–Schwarz.
Structural Commentary: The score is the direction in which unbiased estimators must covary to track $\psi(\theta)$.
Key Insight: Information lower-bounds variance through Cauchy–Schwarz. □

Example 2.2.3. *Poisson Mean Attains the Bound*
For $X \sim \mathrm{Poisson}(\lambda)$, $\ell_\lambda(X) = X \log \lambda - \lambda - \log(X!)$ and $\partial_\lambda^2 \ell_\lambda(X) = -\frac{X}{\lambda^2}$, so $I(\lambda) = -\mathbb{E}_\lambda \partial_\lambda^2 \ell_\lambda(X) = \frac{1}{\lambda}$.
For iid $X_1, \dots, X_n$, any unbiased estimator T of λ satisfies $\mathrm{Var}_\lambda(T) \geq \frac{\lambda}{n}$.
$\overline{X}$ attains equality since $\mathbb{E}\overline{X} = \lambda$ and $\mathrm{Var}(\overline{X}) = \frac{\lambda}{n}$.

Definition 2.2.4. *Asymptotic Efficiency*
For a scalar regular parameter with information $I(\theta_0) > 0$, an estimator sequence $\widehat{\theta}_n$ is *asymptotically efficient* at θ_0 if $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$, i.e. it attains the inverse Fisher-information variance.

Theorem 2.2.5. *Regular MLE Asymptotic Normality*
Let $X_1, \dots, X_n \sim P_{\theta_0}$ *iid in a regular one-parameter model, with* θ_0 *interior,* $I(\theta_0) > 0$, *and a consistent interior MLE* $\widehat{\theta}_n$. *Then* $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$.

Proof. Let $S_n(\theta) := \partial_\theta \ell_n(\theta)$ and $J_n(\theta) := -n^{-1} \partial_\theta S_n(\theta)$.
Since $S_n(\widehat{\theta}_n) = 0$, Taylor expansion gives $0 = S_n(\theta_0) - nJ_n(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta_0)$ for some $\widetilde{\theta}_n$ between $\widehat{\theta}_n$ and θ_0.
Hence $\sqrt{n}(\widehat{\theta}_n - \theta_0) = J_n(\widetilde{\theta}_n)^{-1} n^{-\frac{1}{2}} S_n(\theta_0)$.
Regularity gives $J_n(\widetilde{\theta}_n) \xrightarrow{P} I(\theta_0)$ and $n^{-\frac{1}{2}} S_n(\theta_0) \Rightarrow N(0, I(\theta_0))$, so Slutsky gives the result.
Structural Commentary: Local likelihood curvature converts score noise into estimator error.
Key Insight: The MLE is asymptotically score divided by information. □

Remark 2.2.6. *Efficiency and Approximation*
Finite-sample efficiency means equality in the Cramer–Rao bound.
Asymptotic efficiency means $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$; finite-sample bias or boundary effects still matter.

2.3 Method of Moments and Parametric Likelihood Examples

Method 2.3.1. *Method of Moments*

(i) Let $X_1, \dots, X_n$ be independent from P_θ, and set $\mu_k(\theta) := \mathbb{E}_\theta X^k$ for $k = 1, \dots, r$.

(ii) If $\mu(\theta) = (\mu_1(\theta), \dots, \mu_r(\theta))$ identifies $\theta \in \Theta \subseteq \mathbb{R}^r$, define $\widehat{\mu}_k := n^{-1} \sum_{i=1}^n X_i^k$.

(iii) The *method of moments* estimator solves $\mu_k(\widehat{\theta}) = \widehat{\mu}_k$ for $k = 1, \dots, r$.

Key Insight: Match population moments to sample moments.

Proposition 2.3.2. *Consistency of Continuous Moment Estimates*
Let $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu_0$ *and let* g *be continuous at* μ_0 *with* $g(\mu_0) = \theta_0$. *Then* $\widehat{\theta}_n := g(\widehat{\mu}_n) \xrightarrow{P_{\theta_0}} \theta_0$. *Thus method-of-moments estimators are consistent whenever* $\widehat{\mu}_n \xrightarrow{P} \mu(\theta_0)$, *inversion map* $g = \mu^{-1}$ *is continuous at* $\mu(\theta_0)$.

Proof. Since $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu_0$ and g is continuous at μ_0, by continuous mapping theorem, $g(\widehat{\mu}_n) \xrightarrow{P_{\theta_0}} g(\mu_0) = \theta_0$.
For MoM, if the required moments are finite, the weak law gives $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu(\theta_0)$.
Structural Commentary: Moment consistency is convergence plus continuous inversion.
Key Insight: Estimate moments first, then pass through g. □

Example 2.3.3. *Moment Estimation for Small Open Probabilities*
Let $N \sim \mathrm{Binomial}(m, p)$ be the number of open channels and let $S = cN$ be total current.
Then $\mathbb{E}S = cmp$, $\mathrm{Var}(S) = c^2 mp(1-p)$, and $\frac{\mathrm{Var}(S)}{\mathbb{E}S} = c(1-p) \approx c$ for small p.
Thus independent measurements estimate c by the empirical ratio $\frac{\widehat{\mathrm{Var}}(S)}{\widehat{\mathbb{E}}S}$, without knowing m.

Example 2.3.4. *Poisson Mean*
Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ be independent. Since $\mathbb{E}_\lambda X = \lambda$, $\widehat{\lambda}_{\text{mom}} = \overline{X}$.
Also $\ell(\lambda) = \sum_{i=1}^{n} \{X_i \log \lambda - \lambda - \log(X_i!)\}$ and $\ell'(\lambda) = \frac{S}{\lambda} - n$, where $S = \sum_i X_i$, so $\widehat{\lambda}_{\text{mle}} = \frac{S}{n} = \overline{X}$.
If true value is λ_0, then $S \sim \text{Poisson}(n\lambda_0)$. Hence $\mathbb{E}_{\lambda_0}\widehat{\lambda} = \lambda_0$, $\text{Var}_{\lambda_0}(\widehat{\lambda}) = \frac{\lambda_0}{n}$, $\text{SE}_{\lambda_0}(\widehat{\lambda}) = \sqrt{\frac{\lambda_0}{n}}$, $\widehat{\text{SE}}(\widehat{\lambda}) = \sqrt{\frac{\widehat{\lambda}}{n}}$.

Example 2.3.5. *Negative Binomial Overdispersion*
For overdispersed counts, negative-binomial mass $f(x \mid m, k) = (1 + \frac{m}{k})^{-k}\Gamma(k+x)\{x!\Gamma(k)\}^{-1}(\frac{m}{m+k})^x$, $x \in \mathbb{N}_0$,
has $\mathbb{E}X = m$ and $\text{Var}(X) = m + \frac{m^2}{k}$. By MoM $\widehat{m} = \overline{X}$, $\widehat{k} = \frac{\overline{X}^2}{\widehat{\sigma}^2 - \overline{X}}$ when $\widehat{\sigma}^2 > \overline{X}$.

Example 2.3.6. *Normal Mean and Variance*
For iid $X_i \sim N(\mu, \sigma^2)$, $\mu_1 = \mu$, $\mu_2 = \mu^2 + \sigma^2$. Thus by MoM $\widehat{\mu} = \overline{X}$; $\widehat{\sigma}^2 = n^{-1}\sum_{i=1}^{n} X_i^2 - \overline{X}^2 = n^{-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$.
The normal MLEs are the same; $\widehat{\sigma}^2$ uses denominator n, not the unbiased denominator $n-1$.

Example 2.3.7. *Gamma Shape and Rate*
Let X_i be iid gamma with shape α and rate λ, so $\mathbb{E}X = \frac{\alpha}{\lambda}$ and $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.
With $\widehat{\sigma}^2 := n^{-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$, MoM gives $\widehat{\alpha}_{\text{mom}} = \frac{\overline{X}^2}{\widehat{\sigma}^2}$ and $\widehat{\lambda}_{\text{mom}} = \frac{\overline{X}}{\widehat{\sigma}^2}$.
The log-likelihood is $\ell(\alpha, \lambda) = n\alpha \log \lambda + (\alpha - 1)\sum_i \log X_i - \lambda \sum_i X_i - n \log \Gamma(\alpha)$.
The MLE satisfies $\widehat{\lambda} = \frac{\widehat{\alpha}}{\overline{X}}$ and $\log \widehat{\alpha} - \log \overline{X} + n^{-1}\sum_i \log X_i - \psi(\widehat{\alpha}) = 0$, where $\psi = \frac{\Gamma'}{\Gamma}$.
The MoM estimate is a natural starting value.

Example 2.3.8. *Angular Density*
Let $f(x \mid \alpha) = \frac{1+\alpha x}{2}$ on $[-1, 1]$, with $\alpha \in [-1, 1]$. Since $\mathbb{E}_\alpha X = \int_{-1}^{1} \frac{x(1+\alpha x)}{2}\,dx = \frac{\alpha}{3}$, MoM gives $\widehat{\alpha} = 3\overline{X}$.
Also $\text{Var}_\alpha(X) = \frac{1}{3} - \frac{\alpha^2}{9}$, so $\text{Var}_\alpha(\widehat{\alpha}) = 9\,\text{Var}_\alpha(\overline{X}) = \frac{3-\alpha^2}{n}$.

3 Hypothesis Testing and Confidence

3.1 Tests, Likelihood Ratios, and Error Control

Testing formalises decisions between hypotheses. The central quantities are the probability of false rejection under the null and the probability of rejection under alternatives.

Definition 3.1.1. *Hypotheses, Tests, Size, and Power*
(i) A *null hypothesis* is a subset $\Theta_0 \subseteq \Theta$ of the parameter space.
(ii) An *alternative* is a subset $\Theta_1 \subseteq \Theta$ disjoint from Θ_0, so $\Theta_0 \cap \Theta_1 = \emptyset$.
(iii) Randomised *test* is a measurable map $\varphi : \mathcal{X} \to [0, 1]$, where $\varphi(x)$ is rejection probability at observation x.
(iv) The *size* of φ is $\alpha(\varphi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi(X)]$.
(v) The *power function* of φ is $\beta_\varphi(\theta) = \mathbb{E}_\theta[\varphi(X)]$.

Definition 3.1.2. *Simple and Composite Hypotheses; Type Errors*
(i) A *simple* hypothesis specifies one law; a *composite* hypothesis specifies several.
(ii) A *type I error* rejects a true H_0; its probability is bounded by $\alpha(\varphi)$.
(iii) For a simple alternative P_1, a *type II error* is failure to reject H_0 under P_1, with probability $\mathbb{E}_1[1 - \varphi(X)]$.
(iv) The *power* against P_1 is $1 - \mathbb{E}_1[1 - \varphi(X)]$.

Theorem 3.1.3. *Neyman–Pearson Lemma*
Let $P_1 \ll P_0$, $\Lambda := \frac{dP_1}{dP_0}$, $\alpha \in [0, 1]$. If $c \geq 0$ and $\gamma \in [0, 1]$ make $\varphi^* := \mathbf{1}_{\{\Lambda > c\}} + \gamma \mathbf{1}_{\{\Lambda = c\}}$ satisfy $\mathbb{E}_0\varphi^* = \alpha$, then φ^* is most powerful among level-α tests: for every $\psi : \mathcal{X} \to [0, 1]$ with $\mathbb{E}_0\psi \leq \alpha$, $\mathbb{E}_1\psi \leq \mathbb{E}_1\varphi^*$.

Proof. For any level-α competitor ψ, $(\varphi^* - \psi)(\Lambda - c) \geq 0$ pointwise.
Since $dP_1 = \Lambda\, dP_0$, $\mathbb{E}_1(\varphi^* - \psi) = \mathbb{E}_0[\Lambda(\varphi^* - \psi)] \geq c\mathbb{E}_0(\varphi^* - \psi) = c(\alpha - \mathbb{E}_0\psi) \geq 0$.
Structural Commentary: Null controls size constraint, while alternative supplies power objective.
Key Insight: Use the rejection budget where $\Lambda = \frac{dP_1}{dP_0}$ is largest. □

Remark 3.1.4. *Choosing the Null*
At level a, for simple P_0 versus P_1, the *Neyman–Pearson problem* is $\sup\{\mathbb{E}_1\varphi : 0 \leq \varphi \leq 1,\ \mathbb{E}_0\varphi \leq a\}$.
Thus H_0 is the protected claim: choose it so the false-rejection probability $\mathbb{E}_0\varphi$ is the error to be bounded by a.
The choice of H_0 is substantive, not determined by $\{P_0, P_1\}$.

Definition 3.1.5. *Uniformly Most Powerful Test*
For a composite alternative Θ_1, a level-α test φ is *uniformly most powerful* if, for every $\theta \in \Theta_1$, it has power at least as large as every other level-α test at that same θ.

Example 3.1.6. *Normal Mean, Known Variance*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ^2. For $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1 > \mu_0$, the likelihood ratio is increasing in $\overline{X}$, so the level-α test is $\varphi^* = \mathbf{1}_{\{\overline{X} > \mu_0 + \frac{z_{1-\alpha}\sigma}{\sqrt{n}}\}}$, where $z_{1-\alpha}$ is the $N(0,1)$ quantile. Since the cutoff is independent of μ_1, φ^* is uniformly most powerful for $H_1 : \mu > \mu_0$; no single likelihood-ratio direction is uniformly most powerful for $H_1 : \mu \neq \mu_0$.

Definition 3.1.7. *Generalized Likelihood-Ratio Test*
For $H_0 : \theta \in \Theta_0 \subseteq \Theta$, the *generalized likelihood ratio* is $\lambda(x) := \frac{\sup_{\theta \in \Theta_0} L_x(\theta)}{\sup_{\theta \in \Theta} L_x(\theta)} \in [0, 1]$.
A GLR test rejects H_0 for small $\lambda(x)$, typically $\lambda(x) \leq c_\alpha$ with c_α chosen to give level α.

Theorem 3.1.8. *Large-Sample Likelihood-Ratio Calibration*

(i) Let $H_0 : \theta \in \Theta_0 \subseteq \Theta$, $\dim \Theta = d$, $\dim \Theta_0 = d_0$, $r := d - d_0$, *and* $X^{(n)} \sim P_{\theta_0}$ *with* $\theta_0 \in \Theta_0$.

(ii) Let $L_n(\theta) := e^{\ell_n(\theta)}$ *and* $\lambda_n := \dfrac{\sup_{\theta \in \Theta_0} L_n(\theta)}{\sup_{\theta \in \Theta} L_n(\theta)}$.

(iii) Assume local coordinates with $\theta_0 = 0$, $\Theta_0 = \{(u, 0) : u \in \mathbb{R}^{d_0}\}$, $h = (u, v) \in \mathbb{R}^{d_0} \times \mathbb{R}^r$, *and* $\ell_n(\frac{h}{\sqrt{n}}) - \ell_n(0) = h^\top Z_n - \frac{\|h\|^2}{2} + o_{P_{\theta_0}}(1)$, *where* $Z_n = (U_n, V_n) \Rightarrow N_d(0, I_d)$, $U_n \in \mathbb{R}^{d_0}$, $V_n \in \mathbb{R}^r$.

(iv) Then under H_0, $-2 \log \lambda_n \Rightarrow \chi^2_r$.

Proof.

(i) Since $L_n = e^{\ell_n}$, $\log \lambda_n = \sup_{\theta \in \Theta_0} \ell_n(\theta) - \sup_{\theta \in \Theta} \ell_n(\theta)$, hence $-2 \log \lambda_n = 2\{\sup_{\theta \in \Theta} \ell_n(\theta) - \sup_{\theta \in \Theta_0} \ell_n(\theta)\}$.

(ii) In the local coordinates, write $\theta = \frac{h}{\sqrt{n}}$ for the full model and $\theta = \frac{(u,0)}{\sqrt{n}}$ for the null model.

With $Q_n(h) := h^\top Z_n - \frac{\|h\|^2}{2}$ and $Q_{0,n}(u) := Q_n(u, 0) = u^\top U_n - \frac{\|u\|^2}{2}$, (iii) and cancellation of $\ell_n(0)$ give $-2 \log \lambda_n = 2\{\sup_{h \in \mathbb{R}^d} Q_n(h) - \sup_{u \in \mathbb{R}^{d_0}} Q_{0,n}(u)\} + o_{P_{\theta_0}}(1)$.

(iii) For $a \in \mathbb{R}^m$, $x^\top a - \frac{\|x\|^2}{2} = \frac{\|a\|^2}{2} - \frac{\|x-a\|^2}{2}$, so supremum over $x \in \mathbb{R}^m$ is $\frac{\|a\|^2}{2}$ at $x = a$.

Applying this with $a = Z_n$ and $a = U_n$ gives $\sup_{h \in \mathbb{R}^d} Q_n(h) = \frac{\|Z_n\|^2}{2}$ and $\sup_{u \in \mathbb{R}^{d_0}} Q_{0,n}(u) = \frac{\|U_n\|^2}{2}$,

hence $-2 \log \lambda_n = \|Z_n\|^2 - \|U_n\|^2 + o_{P_{\theta_0}}(1) = \|V_n\|^2 + o_{P_{\theta_0}}(1)$.

(iv) Since $Z_n = (U_n, V_n) \Rightarrow N_d(0, I_d)$, marginal convergence gives $V_n \Rightarrow Y$ with $Y \sim N_r(0, I_r)$.
Therefore $\|V_n\|^2 \Rightarrow \|Y\|^2 \sim \chi^2_r$ by continuous mapping, and Slutsky gives $-2 \log \lambda_n \Rightarrow \chi^2_r$.

Structural Commentary: The likelihood ratio compares unconstrained fit with null-constrained fit.
Key Insight: Wilks' statistic is the squared Gaussian fluctuation in the $r = d - d_0$ constrained directions. □

Example 3.1.9. *Two-Sided Normal Mean GLRT*
Let $X_1, \ldots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ^2, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.
Since $\widehat{\mu} = \overline{X}$, $\lambda(X) = \frac{L_n(\mu_0)}{L_n(\overline{X})} = \exp\{-\frac{n(\overline{X}-\mu_0)^2}{2\sigma^2}\}$, so $-2 \log \lambda(X) = \frac{n(\overline{X}-\mu_0)^2}{\sigma^2}$.
Under H_0, $\frac{\sqrt{n}(\overline{X}-\mu_0)}{\sigma} \sim N(0, 1)$; hence $-2 \log \lambda(X) \sim \chi^2_1$,
and the level-α GLR rejects iff $\frac{n(\overline{X}-\mu_0)^2}{\sigma^2} > \chi^2_{1,1-\alpha}$, equivalently $|\overline{X} - \mu_0| > \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}$.

Example 3.1.10. *Sign Test as a Binomial GLRT*
Let $X \sim \text{Binomial}(n, p)$ and test $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$.
For observed x, $\widehat{p} = \frac{x}{n}$ and $\lambda(x) = \frac{L_x(\frac{1}{2})}{L_x(\widehat{p})} = \frac{(\frac{1}{2})^n}{[(\frac{x}{n})^x(1-\frac{x}{n})^{n-x}]}$, $0^0 = 1$ at $x = 0, n$.
Since $\lambda(x)$ decreases as $|x - \frac{n}{2}|$ increases, a level-α GLR rejects on the largest values of $|X - \frac{n}{2}|$, with cutoff chosen under $X \sim \text{Binomial}(n, \frac{1}{2})$; this is the two-sided sign test.

Definition 3.1.11. *p-Value*
A *p-value* for $H_0 : \theta \in \Theta_0$ is a statistic $p : \mathcal{X} \to [0, 1]$ such that $\sup_{\theta \in \Theta_0} \mathbb{P}_\theta\{p(X) \leq u\} \leq u$ for every $u \in [0, 1]$.
Then $\varphi_\alpha(X) := \mathbf{1}_{\{p(X) \leq \alpha\}}$ has size $\alpha(\varphi_\alpha) \leq \alpha$.

Proposition 3.1.12. *Continuous Null p-Values Are Uniform*
Let $H_0 : \theta \in \Theta_0$ *and let* $T : \mathcal{X} \to \mathbb{R}$ *satisfy* $\mathbb{P}_\theta(T \leq t) = F_0(t)$ *for all* $\theta \in \Theta_0$, *where* F_0 *is continuous.*
For the right-tail convention, set $p(X) := 1 - F_0(T(X))$.
Then $p(X) \sim \text{Uniform}(0, 1)$ *under every* $\theta \in \Theta_0$, *and* $\varphi_\alpha(X) := \mathbf{1}_{\{p(X) \leq \alpha\}}$ *has size* α *for* $\alpha \in [0, 1]$.

Proof. Fix $\theta \in \Theta_0$ and set $U := F_0(T(X))$.
By probability integral transform, Proposition 2.3.6, $U \sim \text{Uniform}(0, 1)$ under $\mathbb{P}_\theta$,
hence $p(X) = 1 - U \sim \text{Uniform}(0, 1)$. Thus $\mathbb{E}_\theta \varphi_\alpha = \mathbb{P}_\theta(p(X) \leq \alpha) = \alpha \ \forall \theta \in \Theta_0$, so $\alpha(\varphi_\alpha) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta \varphi_\alpha = \alpha$.

Structural Commentary: A continuous null CDF converts the statistic to a canonical uniform scale.
Key Insight: Exact calibration is the probability integral transform applied under H_0. □

Remark 3.1.13. *p-Values Are Tail Probabilities*
A p-value is a null tail probability, e.g. $p(x) = \mathbb{P}_{H_0}\{T(X) \geq T(x)\}$ for a right-tail statistic T.
It is not $\mathbb{P}(H_0 \mid x)$; posterior probabilities require priors and an alternative likelihood.

Definition 3.1.14. *Multiple-Testing Error Counts and FDR*
For a multiple-testing rule over M hypotheses.
Let M_0 be the number of true nulls and $M_1 = M - M_0$ the number of false nulls.
Let V be false rejections, S true rejections, U true nonrejections, and T false nonrejections.
The total number of discoveries is $R = V + S$.

(i) *Family-wise error rate*: $\text{FWER} := \mathbb{P}(V \geq 1)$.
(ii) *False discovery proportion*: $\text{FDP} := \frac{V}{\max(R,1)}$.
(iii) *False discovery rate*: $\text{FDR} := \mathbb{E}[\text{FDP}]$.
(iv) *Power*: One aggregate power measure is $\mathbb{E}[S]/M_1$ when $M_1 > 0$.

Thus FWER controls the chance of any false discovery.
FDR controls the expected false fraction among the reported discoveries.

Remark 3.1.15. *Error Rates*
For one test φ, size is $\alpha(\varphi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta \varphi$. For a family of tests, let R be the number of rejections and V the number of false rejections; common targets are $\text{FWER} = \mathbb{P}\{V \geq 1\}$ and $\text{FDR} = \mathbb{E}[\frac{V}{R \vee 1}]$, while directional or selective rates replace V by the corresponding error count.

Method 3.1.16. *Fishing Expeditions and Bonferroni Control*

(i) For N null hypotheses, let R_i be the event that null i is rejected.
(ii) By the union bound, $\mathbb{P}(\bigcup_{i=1}^{N} R_i) \le \sum_{i=1}^{N} \mathbb{P}(R_i)$.
(iii) To control family-wise error at level α, test each null at level $\frac{\alpha}{N}$; then $\mathbb{P}(\text{at least one false rejection}) \le \alpha$.
(iv) Without such correction, many tests can produce small p-values even when all nulls are true.

Definition 3.1.17. *Positive FDR, Local FDR, and q-Values*

Let $Z_j = 0$ indicate a true null, $Z_j = 1$ a false null, t_j a test statistic, and Γ a rejection region.

(i) *pFDR*: For false discoveries V and discoveries R, $\mathrm{pFDR} := \mathbb{E}[V/R \mid R > 0]$.
(ii) *Mixture model*: If $\mathbb{P}(Z_j = 0) = \pi_0$, $\mathcal{L}(t_j \mid Z_j = 0) = F_0$, and $\mathcal{L}(t_j \mid Z_j = 1) = F_1$, then $\mathcal{L}(t_j) = \pi_0 F_0 + (1 - \pi_0) F_1$.
(iii) *Bayes form*: For a rejection region Γ, $\mathrm{pFDR}(\Gamma) = \mathbb{P}(Z_j = 0 \mid t_j \in \Gamma)$.
(iv) *Local FDR*: At value t_0, $\mathrm{lfdr}(t_0) := \mathbb{P}(Z_j = 0 \mid t_j = t_0)$.
(v) *q-value*: For feature j, $q_j := \inf\{\alpha : j \in \mathcal{R}_\alpha\}$, where $\mathcal{R}_\alpha$ is the discovery set at FDR level α.

Remark 3.1.18. *FDR Versus Family-Wise Error*

With false discoveries V and total discoveries R, $\mathrm{FWER} := \mathbb{P}(V \ge 1)$ controls any false rejection, while $\mathrm{FDR} := \mathbb{E}[V/\max(R, 1)]$ controls the expected false fraction among discoveries. Thus FDR is less conservative for large screening lists and measures list reliability directly.

Method 3.1.19. *Benjamini–Hochberg Step-Up Rule*

For M hypotheses with p-values $p_1, \dots, p_M$ and target level $\alpha \in (0, 1)$:

(i) *Order*: Let $p_{(1)} \le \cdots \le p_{(M)}$ be the order statistics.
(ii) *Step-up index*: Set $L := \max\{j : p_{(j)} \le \alpha j / M\}$, with $L := 0$ if the set is empty.
(iii) *Rejections*: Reject $\mathcal{R} := \{m : p_m \le p_{(L)}\}$ if $L > 0$, and reject none if $L = 0$.
(iv) *Guarantee*: If true-null p-values are independent and uniform, then $\mathrm{FDR} \le (M_0/M)\alpha \le \alpha$, where M_0 is the number of true nulls.

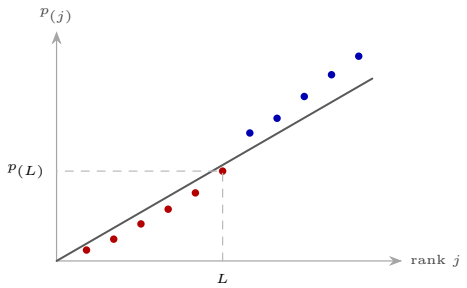

Figure 238: Benjamini–Hochberg threshold: red points reject, blue points do not; line is $\alpha j/M$.

3.2 Confidence Sets, Pivots, and Asymptotic Intervals

Confidence procedures attach uncertainty to estimators by controlling coverage probability under repeated sampling.

Definition 3.2.1. *Confidence Sets and Coverage*

A random set $C(X) \subseteq \Psi$ is a $1 - \alpha$ *confidence set* for $\psi : \Theta \to \Psi$ if $\inf_{\theta \in \Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1 - \alpha$.

The left side is the *coverage guarantee*.

Definition 3.2.2. *Pivots and Confidence Construction*

(i) A *pivot* is a statistic $Q(X, \theta)$ whose distribution under $\mathbb{P}_\theta$ is free of θ.
(ii) If constants a_α, b_α satisfy $\mathbb{P}_\theta\{a_\alpha \le Q(X, \theta) \le b_\alpha\} = 1 - \alpha$ for all $\theta \in \Theta$, then $C(X) := \{\theta \in \Theta : a_\alpha \le Q(X, \theta) \le b_\alpha\}$ is an exact $1 - \alpha$ confidence set for θ.

Corollary 3.2.3. *Hoeffding Confidence Interval for a Bernoulli Parameter*

Let $X_1, \dots, X_n$ be independent Bernoulli(p) *variables, $\widehat{p}_n := \overline{X}_n$, and $\alpha \in (0, 1)$.*

Set $\varepsilon_n := \sqrt{(2n)^{-1} \log(\frac{2}{\alpha})}$ and $C_n := [\widehat{p}_n - \varepsilon_n, \widehat{p}_n + \varepsilon_n]$.

(i) Coverage: $\mathbb{P}(p \in C_n) \ge 1 - \alpha$.
(ii) Length: $|C_n| = 2\varepsilon_n \le 0.05$ whenever $n \ge 800 \log(\frac{2}{\alpha})$.

Proof.

(i) Since $\{p \notin C_n\} = \{|\widehat{p}_n - p| > \varepsilon_n\}$, Corollary 3.3.5 gives $\mathbb{P}(p \notin C_n) \le 2e^{-2n\varepsilon_n^2} = \alpha$.
(ii) If $n \ge 800 \log(\frac{2}{\alpha})$, then $\varepsilon_n^2 \le \frac{1}{1600}$, hence $|C_n| = 2\varepsilon_n \le 0.05$.

Structural Commentary: Confidence intervals invert a concentration inequality.
Key Insight: Choose ε_n so the Hoeffding failure bound equals α. □

Proposition 3.2.4. *Test–Confidence Duality*

For each $\psi_0 \in \Psi$, let $\varphi_{\psi_0} : \mathcal{X} \to \{0, 1\}$ satisfy $\sup_{\theta : \psi(\theta) = \psi_0} \mathbb{E}_\theta \varphi_{\psi_0} \le \alpha$.

Then $C(X) := \{\psi_0 \in \Psi : \varphi_{\psi_0}(X) = 0\}$ is a $1 - \alpha$ confidence set for $\psi(\theta)$.

Proof. Fix $\theta \in \Theta$ and set $\psi_\theta := \psi(\theta)$.
By definition, $\{\psi(\theta) \notin C(X)\} = \{\varphi_{\psi_\theta}(X) = 1\}$, so $\mathbb{P}_\theta\{\psi(\theta) \notin C(X)\} = \mathbb{E}_\theta \varphi_{\psi_\theta} \le \sup_{\eta:\psi(\eta)=\psi_\theta} \mathbb{E}_\eta \varphi_{\psi_\theta} \le \alpha$.
Hence $\mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1-\alpha$ for every θ, and $\inf_{\theta\in\Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1-\alpha$.
Structural Commentary: Inverting all accepted point-null tests turns non-rejection regions into a confidence set.
Key Insight: Coverage failure equals rejection of the true null value. □

Proposition 3.2.5. *Confidence–Test Converse*
Let $C(X) \subseteq \Psi$ satisfy $\inf_{\theta\in\Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \ge 1-\alpha$.
For $\psi_0 \in \Psi$, let $\varphi_{\psi_0}(X) := \mathbf{1}_{\{\psi_0 \notin C(X)\}}$ for testing $H_0 : \psi(\theta) = \psi_0$. Then φ_{ψ_0} is level α: $\sup_{\theta:\psi(\theta)=\psi_0} \mathbb{E}_\theta \varphi_{\psi_0} \le \alpha$.

Proof. Fix θ with $\psi(\theta) = \psi_0$. Then $\mathbb{E}_\theta \varphi_{\psi_0} = \mathbb{P}_\theta\{\psi_0 \notin C(X)\} = \mathbb{P}_\theta\{\psi(\theta) \notin C(X)\} = 1 - \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \le \alpha$.
Taking $\sup_{\theta:\psi(\theta)=\psi_0}$ gives the level bound.
Structural Commentary: A confidence set tests a proposed value by checking membership.
Key Insight: Rejection of ψ_0 is exactly non-coverage when ψ_0 is true. □

Remark 3.2.6. *Acceptance Regions and Confidence Regions*
For level-α point-null tests with acceptance regions $A_{\psi_0} \subseteq \mathcal{X}$, the inversion $C(X) := \{\psi_0 \in \Psi : X \in A_{\psi_0}\}$ is the set of null values not rejected at X. Conversely, any confidence set $C(X)$ induces point-null tests $\varphi_{\psi_0}(X) = \mathbf{1}_{\{\psi_0 \notin C(X)\}}$.

Definition 3.2.7. *Wald, Score, and Likelihood Intervals*
For scalar regular θ with estimator $\widehat{\theta}$, standard error $\widehat{\mathrm{se}}(\widehat{\theta})$, score U_n, and likelihood ratio $\lambda_n(\theta_0)$:

(i) *Wald*: use $\widehat{\theta} \pm z_{1-\frac{\alpha}{2}} \widehat{\mathrm{se}}(\widehat{\theta})$.
(ii) *Score*: invert $\frac{|U_n(\theta_0)|}{\sqrt{I_n(\theta_0)}} \le z_{1-\frac{\alpha}{2}}$ over θ_0.
(iii) *Likelihood*: invert $-2\log\lambda_n(\theta_0) \le \chi^2_{1,1-\alpha}$ over θ_0.

Key Insight: Wald, score, and likelihood intervals are first-order equivalent in regular models.

Example 3.2.8. *Poisson Mean Wald Interval*
Let $X_1, \dots, X_n$ be independent Poisson(λ) variables.
Then $\widehat{\lambda} = \overline{X}$ and $\frac{\sqrt{n}(\overline{X}-\lambda)}{\sqrt{\lambda}} \Rightarrow N(0,1)$, so the plug-in standard error is $\widehat{\mathrm{se}}(\widehat{\lambda}) = \sqrt{\frac{\overline{X}}{n}}$.
The Wald interval is $C_\alpha(X) = [\overline{X} - z_{1-\frac{\alpha}{2}}\sqrt{\frac{\overline{X}}{n}}, \overline{X} + z_{1-\frac{\alpha}{2}}\sqrt{\frac{\overline{X}}{n}}]$, with asymptotic coverage $1-\alpha$.

3.3 Goodness-of-Fit and Diagnostic Plots

Goodness-of-fit methods compare an observed empirical pattern with the pattern predicted by a probability model. Formal tests give calibrated evidence against a specified model, while diagnostic plots show where the model fits or fails.

Definition 3.3.1. *Multinomial Goodness-of-Fit Model*

(i) *Observed counts*: $O = (O_1, \dots, O_m)$ records cell frequencies from n trials, with $O_i \in \mathbb{N}_0$ and $\sum_{i=1}^m O_i = n$.
(ii) *Null model*: $\mathcal{M}_0 = \{p(\theta) = (p_1(\theta), \dots, p_m(\theta)) : \theta \in \Theta_0\} \subseteq \Delta_{m-1}$ is parametric.
(iii) *Saturated model*: $\Delta_{m-1} = \{p \in [0,1]^m : \sum_i p_i = 1\}$ allows every multinomial probability vector.
(iv) *Fitted counts*: if $\widehat{\theta}$ is the null MLE, then $E_i := np_i(\widehat{\theta})$ for $i = 1, \dots, m$.
(v) *Degrees of freedom*: if $\dim\Theta_0 = k$, then $\mathrm{df} = (m-1) - k = m - k - 1$.

Definition 3.3.2. *Pearson and Likelihood-Ratio Goodness-of-Fit Statistics*

(i) *Inputs*: $O = (O_1, \dots, O_m)$ are observed counts, $E = (E_1, \dots, E_m)$ are fitted null counts with $E_i > 0$, and the null model has k free parameters.
(ii) *Pearson statistic*: $X^2 := \sum_{i=1}^m \frac{(O_i - E_i)^2}{E_i}$.
(iii) *Likelihood-ratio statistic*: $G^2 := 2\sum_{i=1}^m O_i \log(\frac{O_i}{E_i})$, with $0\log(\frac{0}{E_i}) := 0$.
(iv) *Null calibration*: under the regular null model, $X^2 \Rightarrow \chi^2_{m-k-1}$ and $G^2 = -2\log\lambda(X) \Rightarrow \chi^2_{m-k-1}$.
Moreover $G^2 - X^2 = o_P(1)$ under H_0.

Example 3.3.3. *Hardy–Weinberg Fit*
For genotype cells (M, MN, N), the Hardy–Weinberg model is $p(\theta) = ((1-\theta)^2, 2\theta(1-\theta), \theta^2)$.
With $n = 1029$ and $\widehat{\theta} = 0.4247$, the observed and fitted counts are

	M	MN	N
O_i	342	500	187
E_i	340.6	502.8	185.6

Since $m = 3$ and $k = 1$, $\mathrm{df} = m - k - 1 = 1$. Pearson statistic is $X^2 = 0.0319$, so $p = \mathbb{P}\{\chi^2_1 \ge 0.0319\}$ is large.
The data give no evidence against Hardy–Weinberg equilibrium.

Method 3.3.4. *Reading Cell Contributions*

(i) Compute signed residuals $r_i := O_i - E_i$ and Pearson contributions $c_i := \frac{(O_i - E_i)^2}{E_i}$.
(ii) Large c_i identify cells driving the rejection; the sign of r_i indicates overfit or underfit in that cell.
(iii) If some E_i are small, group adjacent or tail cells before testing so the chi-square approximation is reliable.

Remark 3.3.5. *Grouped Counts and Expected Frequencies*

Let $Y_1, \dots, Y_N$ be observations, let $A_1, \dots, A_m$ partition sample space, with observed counts $O_j := \sum_{i=1}^{N} \mathbf{1}_{\{Y_i \in A_j\}}$.

Given a fitted parametric law $P_{\widehat{\theta}}$, set $p_j(\widehat{\theta}) := P_{\widehat{\theta}}(Y \in A_j)$; the fitted expected count is $E_j := N p_j(\widehat{\theta})$.
For grouped Poisson tails, $p_j(\widehat{\theta})$ is a sum of Poisson masses, $(O_1, \dots, O_m)$ is modelled as multinomial under $P_{\widehat{\theta}}$.

Example 3.3.6. *Poisson Clumps*
For grouped bacterial clump frequencies from $n = 400$ grid squares, fit $H_0 : X \sim \text{Poisson}(\lambda)$ with $\widehat{\lambda} = 2.44$ and grouped tails so E_i are adequate. The Pearson statistic is $X^2 = 75.4$ with df $= 6$, so $p = \mathbb{P}\{\chi^2_6 \geq 75.4\} < 0.005$; the largest $c_i = \frac{(O_i - E_i)^2}{E_i}$ are in the first and last cells, indicating excess small and large counts.

Definition 3.3.7. *Overdispersed Counts*
Count data are *overdispersed* relative to a Poisson model if $\text{Var}(X) > \mathbb{E}X$; empirically, this is $\widehat{\sigma}^2 > \overline{X}$.

Definition 3.3.8. *Poisson Dispersion Test*

The data is $x_1, \dots, x_n$ are independent counts and $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$.

(i) *Models*: $H_0 : x_i \sim \text{Poisson}(\lambda)$ with common λ; the saturated alternative uses separate rates λ_i.

(ii) *Likelihood-ratio statistic*: $G^2_D := 2 \sum_{i=1}^{n} x_i \log(\frac{x_i}{\overline{x}})$, with $0 \log(\frac{0}{\overline{x}}) := 0$.

(iii) *Pearson approximation*: $T := \overline{x}^{-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$.

(iv) *Calibration*: under H_0, G^2_D and T are calibrated by χ^2_{n-1}
Large values indicate overdispersion relative to $\text{Var}(X) = \mathbb{E}X$.

Example 3.3.9. *Dispersion in Bacterial Clumps*
For bacterial clump counts with $n = 400$, $\overline{x} = 2.44$ and $s^2 = 4.59$, so $\frac{s^2}{\overline{x}} = 1.88 > 1$.
Dispersion statistic is $T \approx 752.7$ with df $= n - 1 = 399$, so $p = \mathbb{P}\{\chi^2_{399} \geq 752.7\} \approx 0$.
The data show strong overdispersion relative to a Poisson model.

Definition 3.3.10. *Hanging Rootograms and Chi-Grams*
Let n_j be observed bin counts and $\widehat{n}_j$ fitted counts in ordered bins.

(i) A *hanging histogram* plots raw residuals $r_j := n_j - \widehat{n}_j$, whose variance depends on $\widehat{n}_j$.

(ii) A *hanging rootogram* plots $q_j := \sqrt{n_j} - \sqrt{\widehat{n}_j}$, using $\text{Var}(\sqrt{n_j}) \approx \frac{1}{4}$ for count data.

(iii) A *hanging chi-gram* plots standardized residuals $z_j := \frac{n_j - \widehat{n}_j}{\sqrt{\widehat{n}_j}}$.

Runs of same-sign residuals, tail spikes, or systematic curvature indicate model departure.

Method 3.3.11. *Probability Plots*
Let $X_{(1)} \leq \cdots \leq X_{(n)}$ be order statistics and let F be the candidate continuous strictly increasing CDF.

(i) A *P-P plot* compares $F(X_{(k)})$ with $u_k := \frac{k}{n+1}$ for $k = 1, \dots, n$.

(ii) A *Q-Q plot* compares $X_{(k)}$ with $F^{-1}(u_k)$ for $k = 1, \dots, n$.

The model is plausible when the plotted points are approximately linear.
For $F(x) = G(\frac{x-\mu}{\sigma})$, Q-Q slope and intercept estimate σ and μ.

Method 3.3.12. *Probability Plots for Grouped Data*

Let bins boundaries $x_1 < \cdots < x_{m+1}$, counts $n_1, \dots, n_m$, total $n := \sum_{j=1}^{m} n_j$, cumulative counts $N_j := \sum_{i=1}^{j} n_i$.

Let G be the standardised CDF in the proposed location-scale family $F_{\mu,\sigma}(x) = G(\frac{x-\mu}{\sigma})$.

(i) Plot empirical boundary quantiles x_{j+1} against theoretical quantiles $G^{-1}(\frac{N_j}{n+1})$ for $j = 1, \dots, m$.

(ii) Use this grouped Q-Q plot when individual order statistics are unavailable.

Remark 3.3.13. *Grouped Normality Tests*
For binned normal-fit testing, use $X^2 = \sum_j \frac{(O_j - E_j)^2}{E_j}$ with $E_j = n\{F_{\widehat{\mu},\widehat{\sigma}}(b_{j+1}) - F_{\widehat{\mu},\widehat{\sigma}}(b_j)\}$.

The usual χ^2_{m-k-1} calibration requires $(\widehat{\mu}, \widehat{\sigma})$ from the grouped counts; ungrouped estimates change the limit.

Definition 3.3.14. *Normality Diagnostics*
Let $X_{(1)} \leq \cdots \leq X_{(n)}$, $\overline{X} := n^{-1} \sum_i X_i$, and $s^2 := n^{-1} \sum_i (X_i - \overline{X})^2$.

(i) *Normal Q-Q plot*: plot $X_{(k)}$ against $\Phi^{-1}(\frac{k}{n+1})$; normality predicts approximate linearity.

(ii) *Skewness*: $b_1 := n^{-1} \sum_{i=1}^{n} \frac{(X_i - \overline{X})^3}{s^3}$ measures asymmetry; large $|b_1|$ indicates nonnormality.

(iii) *Kurtosis*: $b_2 := n^{-1} \sum_{i=1}^{n} \frac{(X_i - \overline{X})^4}{s^4}$ measures tail weight and peakedness; normality gives $b_2 \approx 3$.

(iv) *Plot correlation*: Q-Q point correlation; small values indicate curvature.

3.4 Normal Sampling Pivots

For normal samples, the sample mean and sample variance separate exactly. This is the finite-sample source of the classical z, χ^2, and Student t pivots used in normal-theory confidence intervals and tests.

Corollary 3.4.1. *Student Pivot for a Normal Mean*
Let $X_1, \dots, X_n$ be independent; $X_i \sim N(\mu, \sigma^2)$, $\overline{X} := n^{-1} \sum_i X_i$, $S^2 := (n-1)^{-1} \sum_i (X_i - \overline{X})^2$; $\frac{\overline{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$.

Proof. Let $Z := \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}$ and $V := \frac{(n-1)S^2}{\sigma^2}$. By normal mean law, $Z \sim N(0,1)$. By sample-variance chi-square law, $V \sim \chi^2_{n-1}$. By mean-residual independence, $Z \perp\!\!\!\perp V$. Therefore $\frac{\overline{X}-\mu}{\frac{S}{\sqrt{n}}} = \frac{Z}{\sqrt{\frac{V}{n-1}}} \sim t_{n-1}$.

Structural Commentary: Replacing σ by S yields a Student pivot.
Key Insight: Student's t is a normal variable divided by an independent chi-square scale estimate. □

Method 3.4.2. *Exact Normal Mean Intervals*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables, with $\overline{X} := n^{-1}\sum_i X_i$ and $S^2 := (n-1)^{-1}\sum_i (X_i - \overline{X})^2$.

Below cases have exact coverage $1-\alpha$.

(i) *Known variance*: if σ is known, $C_\mu(X) = [\overline{X} - \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}, \overline{X} + \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}]$

(ii) *Unknown variance*: if σ is unknown, $C_\mu(X) = [\overline{X} - \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}, \overline{X} + \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}]$

(iii) *Variance*: $C_{\sigma^2}(X) = [\frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}]$

Example 3.4.3. *Normal Mean with Known Variance*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ.
The pivot $Z_\mu := \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1)$ gives $\mathbb{P}_\mu\{|Z_\mu| \le z_{1-\frac{\alpha}{2}}\} = 1-\alpha$.
Solving for μ gives the exact interval $C_\alpha(X) = [\overline{X} - \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}, \overline{X} + \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}]$.
Test Connection: The two-sided level-α test of $H_0 : \mu = \mu_0$ rejects iff $\mu_0 \notin C_\alpha(X)$.

Example 3.4.4. *Normal Mean and Variance with Unknown Variance*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables, and set $S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$.
The exact pivots are $T_\mu := \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{n-1}$ and $W_{\sigma^2} := \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$. Writing $t_{\nu,p}$ and $\chi^2_{\nu,p}$ for p-quantiles, pivot inversion gives $C_\mu(X) = [\overline{X} - \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}, \overline{X} + \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}]$ and $C_{\sigma^2}(X) = [\frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}]$.
Both intervals have exact marginal coverage $1-\alpha$; $C_{\sigma^2}(X)$ is asymmetric because the χ^2 pivot is not symmetric.

3.5 Two-Sample Normal Inference

Two-sample inference compares the location of two response distributions. In a controlled experiment, one sample may receive a treatment and the other a control; in an observational comparison, the same formulas may be computed but their causal interpretation depends on the design.

Definition 3.5.1. *Two Independent Samples and the Location Contrast*
$X_1, \dots, X_n \sim F$ are independent, $Y_1, \dots, Y_m \sim G$ are independent, and the two samples are independent.

(i) *Means*: $\mu_X := \mathbb{E}X_i$ and $\mu_Y := \mathbb{E}Y_j$ summarize the two average levels, assuming both exist.
(ii) *Location contrast*: $\Delta := \mu_X - \mu_Y$ measures the shift from group Y to group X.
(iii) *Estimator*: $\widehat{\Delta} := \overline{X} - \overline{Y}$ estimates Δ, where $\overline{X} := n^{-1}\sum_i X_i$ and $\overline{Y} := m^{-1}\sum_j Y_j$.

Proposition 3.5.2. *Known Common Variance Pivot*
Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be independent samples, $X_i \sim N(\mu_X, \sigma^2)$, $Y_j \sim N(\mu_Y, \sigma^2)$, known $\sigma > 0$.
Set $\Delta := \mu_X - \mu_Y$, $\widehat{\Delta} := \overline{X} - \overline{Y}$, and $\mathrm{se}_0 := \sigma\sqrt{n^{-1} + m^{-1}}$.

(i) $\widehat{\Delta} \sim N(\Delta, \mathrm{se}_0^2)$ and $Z := \frac{\widehat{\Delta}-\Delta}{\mathrm{se}_0} \sim N(0,1)$.
(ii) $C_\alpha(X,Y) := [\widehat{\Delta} - z_{1-\frac{\alpha}{2}}\,\mathrm{se}_0, \widehat{\Delta} + z_{1-\frac{\alpha}{2}}\,\mathrm{se}_0]$ satisfies $\mathbb{P}_\Delta\{\Delta \in C_\alpha(X,Y)\} = 1-\alpha$.

Proof.

(i) Since $\overline{X} \sim N(\mu_X, \frac{\sigma^2}{n})$, $\overline{Y} \sim N(\mu_Y, \frac{\sigma^2}{m})$, and $\overline{X} \perp\!\!\!\perp \overline{Y}$, $\widehat{\Delta} = \overline{X} - \overline{Y} \sim N(\Delta, \frac{\sigma^2}{n} + \frac{\sigma^2}{m}) = N(\Delta, \mathrm{se}_0^2)$. Standardization gives $Z \sim N(0,1)$.
(ii) By (i), $\mathbb{P}_\Delta\{|Z| \le z_{1-\frac{\alpha}{2}}\} = 1-\alpha$, equivalently $\mathbb{P}_\Delta\{\Delta \in C_\alpha(X,Y)\} = 1-\alpha$.

Structural Commentary: The contrast is a linear combination of independent sample means.
Key Insight: Known common variance gives an exact normal pivot for Δ. □

Definition 3.5.3. *Pooled Sample Variance*
Let $X_1, \dots, X_n \sim N(\mu_X, \sigma^2)$ and $Y_1, \dots, Y_m \sim N(\mu_Y, \sigma^2)$ be independent samples with common unknown σ^2.

(i) *Sample variances*: $S_X^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$ and $S_Y^2 := (m-1)^{-1}\sum_{j=1}^m (Y_j - \overline{Y})^2$.
(ii) *Pooled variance*: $S_p^2 := \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$.
(iii) *Degrees-of-freedom weighting*: S_p^2 weights S_X^2 and S_Y^2 by $n-1$ and $m-1$, respectively.

Theorem 3.5.4. *Pooled Two-Sample Student Pivot*
Let $X_i \sim N(\mu_X, \sigma^2)$, $i = 1, \dots, n$, and $Y_j \sim N(\mu_Y, \sigma^2)$, $j = 1, \dots, m$, be mutually independent, with unknown common $\sigma^2 > 0$. Set $\Delta := \mu_X - \mu_Y$ and $\nu := n + m - 2$. Then $T := \frac{(\overline{X}-\overline{Y})-\Delta}{S_p\sqrt{n^{-1}+m^{-1}}} \sim t_\nu$.

Proof. Let $Z := \frac{(\overline{X}-\overline{Y})-\Delta}{\sigma\sqrt{n^{-1}+m^{-1}}}$ and $W := \frac{\nu S_p^2}{\sigma^2} = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{\sigma^2}$.
Since $\overline{X} - \overline{Y} \sim N(\Delta, \sigma^2(n^{-1}+m^{-1}))$, $Z \sim N(0,1)$. Let $U_X := \frac{(n-1)S_X^2}{\sigma^2}$ and $U_Y := \frac{(m-1)S_Y^2}{\sigma^2}$.
By normal mean-residual independence and sample independence, Z, U_X, U_Y mutually independent,

with $U_X \sim \chi^2_{n-1}$, $U_Y \sim \chi^2_{m-1}$. Hence $W = U_X + U_Y \sim \chi^2_\nu$ and $W \perp\!\!\!\perp Z$. Since $S_p^2 = \frac{\sigma^2 W}{\nu}$, $T = \frac{Z}{\sqrt{\frac{W}{\nu}}} \sim t_\nu$.

Structural Commentary: The pooled variance combines the two residual sums of squares.
Key Insight: A common variance permits replacing σ by an independent pooled estimate. □

Method 3.5.5. *Pooled Two-Sample t Procedures*
Under the equal-variance normal model, set $\Delta := \mu_X - \mu_Y$, $\nu := n + m - 2$, and $\mathrm{se}_p := S_p\sqrt{n^{-1} + m^{-1}}$.

(i) *Interval*: $C_\alpha := [(\overline{X} - \overline{Y}) - t_{\nu,1-\frac{\alpha}{2}}\,\mathrm{se}_p, (\overline{X} - \overline{Y}) + t_{\nu,1-\frac{\alpha}{2}}\,\mathrm{se}_p]$ is exact $1-\alpha$ for Δ.
(ii) *Statistic*: for $H_0 : \Delta = 0$, use $t_0 := \frac{\overline{X}-\overline{Y}}{\mathrm{se}_p}$, with $t_0 \sim t_\nu$ under H_0.
(iii) *Rejection*: reject H_0 for $|t_0| > t_{\nu,1-\frac{\alpha}{2}}$ under $H_1 : \Delta \neq 0$
reject H_0 for $t_0 > t_{\nu,1-\alpha}$ under $H_1 : \Delta > 0$
reject H_0 for $t_0 < -t_{\nu,1-\alpha}$ under $H_1 : \Delta < 0$.
(iv) *Duality*: the two-sided test rejects iff $0 \notin C_\alpha$.

Example 3.5.6. *Latent Heat of Fusion: Two Methods*
For methods A and B, let $n_A = 13$, $n_B = 8$, $\overline{x}_A = 80.02$, $\overline{x}_B = 79.98$, $s_A = 0.024$, $s_B = 0.031$, and $S_p = 0.027$.
Then $\widehat{\Delta} = \overline{x}_A - \overline{x}_B = 0.04$, $\widehat{\mathrm{se}} = S_p\sqrt{13^{-1} + 8^{-1}} \approx 0.012$, pooled 95% interval for $\Delta = \mu_A - \mu_B$ is
$\approx (0.015, 0.065)$. For $H_0 : \Delta = 0$, $t_0 = \frac{\widehat{\Delta}}{\widehat{\mathrm{se}}} \approx 3.33$ with $\nu = 19$, strong evidence that method means differ.

Proposition 3.5.7. *Likelihood-Ratio View of the Pooled Test*
Let $X_i \sim N(\mu_X, \sigma^2)$, $i = 1, \ldots, n$, and $Y_j \sim N(\mu_Y, \sigma^2)$, $j = 1, \ldots, m$, be mutually independent, with unknown common $\sigma^2 > 0$. Set $\Delta := \mu_X - \mu_Y$, $\nu := n + m - 2$, and $T := \frac{\overline{X}-\overline{Y}}{S_p\sqrt{n^{-1}+m^{-1}}}$.
For $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$, $\lambda(X, Y) = (1 + \frac{T^2}{\nu})^{-\frac{n+m}{2}}$.
Hence the GLR rejects for large $|T|$; at level α, reject iff $|T| > t_{\nu,1-\frac{\alpha}{2}}$.

Proof. Set $N := n+m$, $\nu := N-2$, $d := \overline{X}-\overline{Y}$, $R_1 := \sum_i (X_i - \overline{X})^2 + \sum_j (Y_j - \overline{Y})^2$, $\widehat{\mu}_0 := \frac{n\overline{X}+m\overline{Y}}{N}$, and $R_0 := \sum_i (X_i - \widehat{\mu}_0)^2 + \sum_j (Y_j - \widehat{\mu}_0)^2$.
Under H_1, $(\widehat{\mu}_X, \widehat{\mu}_Y, \widehat{\sigma}_1^2) = (\overline{X}, \overline{Y}, \frac{R_1}{N})$; under H_0, $(\widehat{\mu}, \widehat{\sigma}_0^2) = (\widehat{\mu}_0, \frac{R_0}{N})$.
The maximised Gaussian likelihood is proportional to $\widehat{\sigma}^{-N}$, hence $\lambda = (\frac{R_1}{R_0})^{\frac{N}{2}}$.
ANOVA decomposition gives $R_0 = R_1 + \{\frac{nm}{N}\}d^2$, while $T^2 = \frac{d^2}{(\frac{R_1}{\nu})(n^{-1}+m^{-1})} = \frac{\nu\{\frac{nm}{N}\}d^2}{R_1}$.
Therefore $\frac{R_0}{R_1} = 1 + \frac{T^2}{\nu}$ and $\lambda = (1 + \frac{T^2}{\nu})^{-\frac{N}{2}}$, decreasing in T^2; small λ iff large $|T|$.
Structural Commentary: The null fit adds only the between-sample mean discrepancy to the unrestricted fit.
Key Insight: The GLR depends on the data through the pooled t statistic. □

Method 3.5.8. *Unequal-Variance Two-Sample Procedure*
For independent samples without the equal-variance assumption, set $\widehat{v} := \frac{S_X^2}{n} + \frac{S_Y^2}{m}$.

(i) *Standard error*: estimate $\mathrm{Var}(\overline{X} - \overline{Y})$ by $\widehat{v}$.
(ii) *Welch statistic*: for $H_0 : \Delta = \Delta_0$, use $T_W := \frac{(\overline{X}-\overline{Y})-\Delta_0}{\sqrt{\widehat{v}}}$.
(iii) *Degrees of freedom*: calibrate T_W by t_ν, where $\nu := \frac{\widehat{v}^2}{\frac{(S_X^2/n)^2}{n-1} + \frac{(S_Y^2/m)^2}{m-1}}$.
(iv) *Use*: the procedure avoids pooling variances, at the cost of approximate calibration.

Corollary 3.5.9. *Variance Ratio Test Under Normality*
Let $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$ be independent samples, with $n, m \geq 2$.
Set $\rho := \frac{\sigma_X^2}{\sigma_Y^2}$ and $R := \frac{S_X^2}{S_Y^2}$. Then $\frac{R}{\rho} \sim F_{n-1,m-1}$.

Proof. Let $U := \frac{(n-1)S_X^2}{\sigma_X^2}$ and $V := \frac{(m-1)S_Y^2}{\sigma_Y^2}$.
Normal residual chi-square laws give $U \sim \chi^2_{n-1}$ and $V \sim \chi^2_{m-1}$, and sample independence gives $U \perp\!\!\!\perp V$.
Thus $\frac{\frac{U}{n-1}}{\frac{V}{m-1}} = \frac{R}{\rho} \sim F_{n-1,m-1}$. If $H_0 : \rho = 1$, then $R \sim F_{n-1,m-1}$, yielding exact normal-theory tests and confidence intervals for ρ; the calibration is normality-sensitive.
Structural Commentary: The comparison standardises two independent residual variance estimates.
Key Insight: Normality turns each sample variance into an independent chi-square pivot. □

Method 3.5.10. *Approximate Power for Balanced Two-Sample z Tests*
For planning, assume balanced samples of size n, known common σ, and target effect $\Delta := \mu_X - \mu_Y > 0$.

(i) *Statistic*: use $Z := \frac{\overline{X}-\overline{Y}}{\sigma\sqrt{\frac{2}{n}}}$ and reject two-sided at level α when $|Z| > z_{1-\frac{\alpha}{2}}$.
(ii) *Shift*: under $\Delta > 0$, $Z \sim N(\delta, 1)$ with $\delta := (\frac{\Delta}{\sigma})\sqrt{\frac{n}{2}}$.
(iii) *Power*: $\pi(\Delta) := 1 - \Phi(z_{1-\frac{\alpha}{2}} - \delta) + \Phi(-z_{1-\frac{\alpha}{2}} - \delta)$.
(iv) *Sample size*: if the lower-tail term is negligible, target power $1-\beta$ gives $n \approx \frac{2\sigma^2(z_{1-\frac{\alpha}{2}} + z_{1-\beta})^2}{\Delta^2}$.

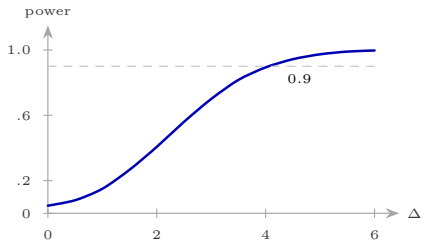

Figure 239: Power rises as the detectable mean difference Δ increases.

Remark 3.5.11. *Transformations and Normal Diagnostics*
Normal-theory two-sample procedures are often CLT-robust for large samples, but small skewed or outlier-prone samples require diagnostics. Transformations may stabilize spread and improve symmetry, but they change the estimand, e.g. a log-scale mean difference corresponds to a ratio-scale comparison.

3.6 Paired-Sample Inference

Paired comparisons arise when observations are naturally matched: before and after measurements on the same unit, littermates, matched subjects, or two measurement methods applied to the same specimen. Pairing changes the stochastic structure because the two samples are no longer independent.

Definition 3.6.1. *Paired Differences*

(i) *Pairs*: (X_i, Y_i), $i = 1, \dots, n$, are independent across i; dependence within each pair is allowed.

(ii) *Differences*: $D_i := X_i - Y_i$ and $\overline{D} := n^{-1} \sum_{i=1}^{n} D_i$.

(iii) *Mean*: if $\mathbb{E}X_i = \mu_X$ and $\mathbb{E}Y_i = \mu_Y$, then $\mathbb{E}D_i = \mu_X - \mu_Y$.

(iv) *Variance*: if $\mathrm{Var}(X_i) = \sigma_X^2$, $\mathrm{Var}(Y_i) = \sigma_Y^2$, and $\mathrm{Cov}(X_i, Y_i) = \sigma_{XY}$, then $\mathrm{Var}(D_i) = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}$.

Proposition 3.6.2. *Variance Effect of Pairing*
Let (X_i, Y_i), $i = 1, \dots, n$, *be independent pairs with* $\mathrm{Var}(X_i) = \sigma_X^2$, $\mathrm{Var}(Y_i) = \sigma_Y^2$, *and* $\mathrm{Corr}(X_i, Y_i) = \rho$.
Set $D_i := X_i - Y_i$ *and* $\overline{D} := n^{-1} \sum_i D_i$. *Then* $\mathrm{Var}(\overline{D}) = n^{-1}(\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y)$.

Proof. Since $\mathrm{Cov}(X_i, Y_i) = \rho\sigma_X\sigma_Y$, $\mathrm{Var}(D_i) = \mathrm{Var}(X_i - Y_i) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$.
Independence across pairs gives $\mathrm{Var}(\overline{D}) = n^{-2} \sum_i \mathrm{Var}(D_i) = \frac{\mathrm{Var}(D_i)}{n}$.

If $\sigma_X = \sigma_Y = \sigma$, substitution gives $\mathrm{Var}(\overline{D}) = \frac{2\sigma^2(1-\rho)}{n}$; independent unpaired samples of size n per group have $\mathrm{Var}(\overline{X} - \overline{Y}) = \frac{2\sigma^2}{n}$, so the paired-to-unpaired variance ratio is $1 - \rho$.
Structural Commentary: Pairing replaces two marginal measurements by one within-pair contrast.
Key Insight: Positive within-pair correlation cancels shared variation and lowers variance. □

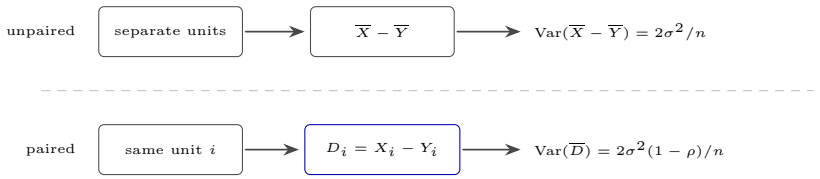

Figure 240: Pairing cancels shared variation; positive within-pair correlation lowers variance.

Method 3.6.3. *Paired t Procedure*
Let $D_1, \dots, D_n \sim N(\mu_D, \sigma_D^2)$, where $\mu_D := \mu_X - \mu_Y$.

(i) *Variance*: $S_D^2 := (n-1)^{-1} \sum_{i=1}^{n} (D_i - \overline{D})^2$.

(ii) *Pivot*: $T := \frac{\overline{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$.

(iii) *Interval*: $C_\alpha := [\overline{D} - \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}, \overline{D} + \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}]$ is exact $1 - \alpha$ for μ_D.

(iv) *Test*: for $H_0 : \mu_D = 0$, reject two-sided at level α iff $|\overline{D}| > \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}$.

For large n, use same rule by CLT; for small nonnormal differences, prefer rank or sign methods.

Example 3.6.4. *Platelet Aggregation Before and After Smoking*
For $n = 11$ before-after platelet measurements, let $D_i = \text{after}_i - \text{before}_i$ with $\overline{D} = 10.27$ and $\frac{S_D}{\sqrt{11}} = 2.40$.
Then $t_0 = \frac{\overline{D}}{\frac{S_D}{\sqrt{11}}} = 4.28$ with $\nu = 10$, so $p < 0.01$ two-sided; a 90% confidence interval is approximately $(5.9, 14.6)$, indicating a positive mean increase after smoking.

Example 3.6.5. *Mercury Measurement Methods*
For $n = 25$ paired mercury measurements, let $D_i = M_{1i} - M_{2i}$ with $\overline{D} = 0.04$ and $S_D = 0.116$.
Then $t_0 = \frac{\overline{D}}{\frac{S_D}{\sqrt{25}}} = 1.724$ with $\nu = 24$, giving $p \approx 0.094$ two-sided.
Plot of D_i against measurement level should be checked: level-dependent discrepancies not captured by $\overline{D}$ alone.

Remark 3.6.6. *Pairing Is Not Extra Independent Sampling*
If unit i contributes repeated measurements $Z_{i1}, \dots, Z_{ik}$, independence is across units, not within unit.
Treating all nk observations as iid ignores within-unit dependence and overstates the effective sample size.

4 Decision Theory

4.1 Loss, Risk, Bayes, and Minimax Rules

Decision theory compares statistical procedures by their consequences, not only by their formulas. The same data model can lead to different optimal rules under different loss functions.

Definition 4.1.1. *Statistical Decision Problem and Risk*

(i) *Problem*: a statistical decision problem is $(\{P_\theta : \theta \in \Theta\}, \mathcal{A}, L)$ with $L : \Theta \times \mathcal{A} \to [0, \infty]$.
(ii) *Rule*: a decision rule is a measurable $\delta : \mathcal{X} \to \mathcal{A}$.
(iii) *Risk*: $R(\theta, \delta) := \mathbb{E}_\theta[L(\theta, \delta(X))]$.

Definition 4.1.2. *Bayes Risk, Minimaxity, and Admissibility*

(i) *Bayes risk*: for prior Π on Θ, $r(\Pi, \delta) := \int R(\theta, \delta)\, d\Pi(\theta)$.
(ii) *Bayes rule*: δ_Π is Bayes if $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.
(iii) *Maximum risk*: $M(\delta) := \sup_{\theta \in \Theta} R(\theta, \delta)$; $\delta^\star$ is minimax if $M(\delta^\star) = \inf_\delta M(\delta)$.
(iv) *Admissibility*: δ inadmissible if $\exists \delta'$ with $R(\theta, \delta') \le R(\theta, \delta)\ \forall \theta$ and $<$ for some θ; otherwise δ admissible.

Proposition 4.1.3. *Posterior Risk Characterization*
Let Π be a prior, let $\Pi(\cdot \mid x)$ be a posterior, and set $\rho_x(a) := \int L(\theta, a)\, d\Pi(\theta \mid x)$.
If a measurable selector $\delta_\Pi(x) \in \arg\min_{a \in \mathcal{A}} \rho_x(a)$ exists, then δ_Π is Bayes: $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.

Proof. By iterated expectation, $r(\Pi, \delta) = \mathbb{E}_\Pi[\rho_X(\delta(X))]$, where X has the prior predictive law.
Since $\rho_x(\delta_\Pi(x)) \le \rho_x(\delta(x))\ \forall$ rule δ and almost every x, $r(\Pi, \delta_\Pi) \le r(\Pi, \delta)\ \forall \delta$. Thus $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.
Structural Commentary: Bayes risk is the prior-predictive average of posterior risks.
Key Insight: Global Bayes optimization reduces to pointwise posterior-risk minimization. □

Example 4.1.4. *Bayes Rules Under Common Losses*
For a real parameter and posterior distribution $\Pi(\cdot \mid x)$:

(i) Squared-error loss is minimized by the posterior mean.
(ii) Absolute-error loss is minimized by a posterior median.
(iii) Zero-one loss on a finite action set is minimized by a posterior mode.

Remark 4.1.5. *Least Favorable Priors*
If a prior $\Pi^\star$ has Bayes risk equal to the minimax value, then any Bayes rule for $\Pi^\star$ with constant risk on the support of $\Pi^\star$ is minimax. Such priors are called least favorable.

4.2 Shrinkage, Empirical Bayes, and Regularization

Risk can often be reduced by intentionally pulling noisy estimates toward a structured target. The price is bias; the benefit is variance reduction.

Definition 4.2.1. *Shrinkage Estimator*

(i) *Input*: let $\widehat{\theta} = \widehat{\theta}(X)$ be a data-based estimator.
(ii) *Target*: choose θ_{tar} as a reference value or lower-complexity model.
(iii) *Weight*: let $a(X) \in [0, 1]$ control the amount of shrinkage.
(iv) *Estimator*: $\delta(X) := a(X)\widehat{\theta} + \{1 - a(X)\}\theta_{\text{tar}}$.

Example 4.2.2. *Ridge Regression as Shrinkage*
For design matrix X, response y, and $\lambda \ge 0$, ridge regression sets $\widehat{\beta}_\lambda \in \arg\min_\beta \{\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2\}$.

If $X^\top X$ is invertible, $\widehat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$, shrinking $\widehat{\beta}_{\text{LS}}$ toward 0 most in weak eigendirections.

Theorem 4.2.3. *James–Stein Phenomenon*
Let $X \sim \mathcal{N}_d(\theta, \sigma^2 I_d)$ with $d \ge 3$, known σ^2, and squared-error risk $R(\theta, \delta) := \mathbb{E}_\theta \|\delta(X) - \theta\|^2$.
Define $\delta_{\text{JS}}(X) := (1 - \frac{(d-2)\sigma^2}{\|X\|^2})X$ for $X \ne 0$, with arbitrary value at $X = 0$.
Then $R(\theta, \delta_{\text{JS}}) < R(\theta, X) = d\sigma^2$ for every $\theta \in \mathbb{R}^d$.

Proof. Let $a := (d-2)\sigma^2$ and $g(x) := -\frac{ax}{\|x\|^2}$, so $\delta_{\text{JS}}(X) = X + g(X)$.
Stein's identity gives $R(\theta, X + g) = d\sigma^2 + \mathbb{E}_\theta[\|g(X)\|^2 + 2\sigma^2 \nabla \cdot g(X)]$.
For $x \ne 0$, $\|g(x)\|^2 = \frac{a^2}{\|x\|^2}$ and $\nabla \cdot g(x) = -\frac{a(d-2)}{\|x\|^2}$.
Thus $R(\theta, \delta_{\text{JS}}) = d\sigma^2 - (d-2)^2\sigma^4 \mathbb{E}_\theta[\|X\|^{-2}] < d\sigma^2$, with $\mathbb{E}_\theta[\|X\|^{-2}] < \infty$ since $d \ge 3$.
Structural Commentary: Shrinkage trades bias toward 0 for enough variance reduction to lower total risk.
Key Insight: In dimension $d \ge 3$, the usual estimator X is inadmissible under squared-error loss. □

Definition 4.2.4. *Empirical Bayes*

(i) *Hierarchy*: $\theta_i \mid \eta \sim \Pi_\eta$ and $X_i \mid \theta_i \sim P_{\theta_i}$ for $i = 1, \dots, m$.
(ii) *Hyperparameter*: estimate η by $\widehat{\eta} = \widehat{\eta}(X_1, \dots, X_m)$ from the marginal law of the data.
(iii) *Plug-in Bayes rule*: use the Bayes rule computed under $\Pi_{\widehat{\eta}}$.

Remark 4.2.5. *Regularization as Decision-Theoretic Structure*
Penalization, shrinkage, and empirical Bayes impose structure to reduce risk, typically by trading increased bias for reduced variance. The criterion is $R(\theta, \delta)$, not unbiasedness.

5 Bayesian Inference

5.1 Priors, Posteriors, and Conjugacy

Bayesian inference treats unknown parameters as uncertain quantities and updates a prior distribution by the likelihood supplied by the data.

Definition 5.1.1. *Prior, Posterior, and Predictive Distribution*
Let Π be a prior on Θ and $p_\theta(x)$ a dominated likelihood.

(i) *Prior*: Π encodes uncertainty about $\theta \in \Theta$ before observing x.
(ii) *Evidence*: $m(x) := \int_\Theta p_\vartheta(x)\, d\Pi(\vartheta)$, assumed finite and positive.
(iii) *Posterior*: $d\Pi(\theta \mid x) := \frac{p_\theta(x)\, d\Pi(\theta)}{m(x)}$.
(iv) *Predictive*: for future X_{new}, $\mathbb{P}(X_{\text{new}} \in A \mid x) = \int_\Theta P_\theta(A)\, d\Pi(\theta \mid x)$.

Definition 5.1.2. *Conjugate Prior Family*

(i) *Conjugacy*: $\{\Pi_\eta : \eta \in H\}$ is conjugate if $\Pi_\eta(\cdot \mid x) = \Pi_{\eta(x)}$ for some $\eta(x) \in H$.

Proposition 5.1.3. *Posterior Odds and Likelihood Ratios*
Let H_0, H_1 be simple hypotheses with $\mathbb{P}(H_i) = \pi_i > 0$ and likelihood $p_i(x) = p(x \mid H_i)$ for $i = 0, 1$.
For $m(x) := \pi_0 p_0(x) + \pi_1 p_1(x) > 0$, $\mathbb{P}(H_i \mid x) = \frac{\pi_i p_i(x)}{m(x)}$ and $\frac{\mathbb{P}(H_0 \mid x)}{\mathbb{P}(H_1 \mid x)} = (\frac{\pi_0}{\pi_1})(\frac{p_0(x)}{p_1(x)})$.

Proof. Bayes' rule gives $\mathbb{P}(H_i \mid x) = \frac{\pi_i p_i(x)}{m(x)}$ for $i = 0, 1$. Therefore $\frac{\mathbb{P}(H_0 \mid x)}{\mathbb{P}(H_1 \mid x)} = \frac{\frac{\pi_0 p_0(x)}{m(x)}}{\frac{\pi_1 p_1(x)}{m(x)}} = (\frac{\pi_0}{\pi_1})(\frac{p_0(x)}{p_1(x)})$.
Structural Commentary: Posterior odds factor as prior odds times the data likelihood ratio.
Key Insight: Bayes updating multiplies odds by $\frac{p_0(x)}{p_1(x)}$. □

Example 5.1.4. *Beta–Binomial Updating*
If $X \mid p \sim \text{Binomial}(n, p)$ and $p \sim \text{Beta}(a, b)$, then $p \mid X = x \sim \text{Beta}(a + x, b + n - x)$.
Thus $\mathbb{E}[p \mid x] = \frac{a+x}{a+b+n}$, a weighted average of $\frac{a}{a+b}$ and $\frac{x}{n}$.

Example 5.1.5. *Gamma–Poisson Updating*
Let $X_1, \dots, X_n \mid \lambda$ be iid Poisson(λ) and $\lambda \sim \text{Gamma}(\alpha, \nu)$ in rate form.
Writing $s := \sum_{i=1}^{n} x_i$, the posterior kernel is $\pi(\lambda \mid x) \propto \lambda^{\alpha+s-1} e^{-(\nu+n)\lambda}$, hence $\lambda \mid x \sim \text{Gamma}(\alpha + s, \nu + n)$.
Thus $\mathbb{E}[\lambda \mid x] = \frac{\alpha+s}{\nu+n}$, $\text{Var}(\lambda \mid x) = \frac{\alpha+s}{(\nu+n)^2}$, $\text{mode}(\lambda \mid x) = \frac{\alpha+s-1}{\nu+n}$ if $\alpha + s > 1$.

Example 5.1.6. *Normal Mean with Normal Prior*
Let $X_1, \dots, X_n \mid \mu$ be iid $N(\mu, \sigma^2)$ with known σ^2, and let $\mu \sim N(m_0, \tau_0^2)$.
Then $\mu \mid X \sim N(m_n, \tau_n^2)$, where $\tau_n^{-2} = \tau_0^{-2} + n\sigma^{-2}$ and $m_n = \tau_n^2(\tau_0^{-2} m_0 + n\sigma^{-2}\overline{X})$.

Example 5.1.7. *Known Mean and Unknown Precision*
Let $X_1, \dots, X_n \mid \xi$ be iid $N(\theta_0, \xi^{-1})$ with known θ_0, and let $\xi \sim \text{Gamma}(\alpha, \lambda)$ in rate form.
Writing $q := \sum_{i=1}^{n} (x_i - \theta_0)^2$, the posterior is $\xi \mid x \sim \text{Gamma}(\alpha + \frac{n}{2}, \lambda + \frac{q}{2})$.
With a flat gamma prior, $\sigma^2 = \xi^{-1}$ centers near the MLE $\frac{q}{n}$.

Definition 5.1.8. *Improper Prior*
An *improper prior* is a σ-finite prior measure Π on Θ with $\Pi(\Theta) = \infty$. For likelihood $p_\theta(x)$, it is usable only when $0 < m_\Pi(x) := \int_\Theta p_\theta(x)\, d\Pi(\theta) < \infty$, in which case $d\Pi(\theta \mid x) := \frac{p_\theta(x)\, d\Pi(\theta)}{m_\Pi(x)}$ is a proper posterior.

Remark 5.1.9. *Conditioning Viewpoint*
Bayesian updating is conditioning under the joint law $dP(\theta, x) = p_\theta(x)\, d\Pi(\theta)\, dx$; the posterior $\Pi(\cdot \mid x)$ is a regular conditional law of θ given $X = x$.

5.2 Computation, Prediction, and Asymptotic Bayes

Except in conjugate models, posterior summaries are integrals that usually require approximation. Monte Carlo and optimization methods from numerical computation become statistical tools here.

Remark 5.2.1. *Bayesian Integrals as a Motivation*
For prior density $\pi(\theta)$ and likelihood $\mathcal{L}(\theta)$, write $Z = \int \mathcal{L}(u)\pi(u)\, du$.
Then $\pi(\theta \mid X^n) = Z^{-1}\mathcal{L}(\theta)\pi(\theta)$ and $\mathbb{E}[\theta \mid X^n] = Z^{-1} \int \theta\, \mathcal{L}(\theta)\pi(\theta)\, d\theta$.
For $\theta = (\theta_1, \dots, \theta_k)$, high-dimensional marginalisation is often analytically impractical.

Definition 5.2.2. *Bayesian Credible Set*
Let $\Pi(\cdot \mid x)$ be the posterior on Θ.

(i) *Credible set*: $C(x) \subseteq \Theta$ has credibility $1 - \alpha$ if $\Pi(C(x) \mid x) \geq 1 - \alpha$.
(ii) *Conditioning*: credibility is a posterior probability conditional on the observed data x.

Definition 5.2.3. *Posterior Monte Carlo Integration*
Let $\Pi(\cdot \mid x)$ be the posterior on Θ.

(i) *Monte Carlo sample*: let $\theta^{(1)}, \dots, \theta^{(M)}$ be draws, or approximate draws, from $\Pi(\cdot \mid x)$.
(ii) *Posterior mean*: for integrable g, $\int g(\theta)\, d\Pi(\theta \mid x) \approx M^{-1} \sum_{m=1}^{M} g(\theta^{(m)})$.
(iii) *Dependent draws*: MCMC uses dependent samples but targets the same posterior integrals.

Method 5.2.4. *Simulating a Posterior Function*

(i) Let $\delta = g(p_1, \dots, p_k)$ be the posterior target.

(ii) Draw posterior samples $(P_1^{(i)}, \dots, P_k^{(i)})$ for $i = 1, \dots, N$.

(iii) Set $\delta^{(i)} = g(P_1^{(i)}, \dots, P_k^{(i)})$, use empirical distribution of $\delta^{(i)}$ to estimate posterior distribution of δ.

Example 5.2.5. *Two Binomial Proportions*

Suppose $X \sim \text{Binomial}(n, p_1)$ and $Y \sim \text{Binomial}(m, p_2)$ are independent, with independent flat priors on p_1, p_2.
Then $p_1 \mid X \sim \text{Beta}(X+1, n-X+1)$ and $p_2 \mid Y \sim \text{Beta}(Y+1, m-Y+1)$.
For $\delta = p_2 - p_1$, draw $P_1^{(i)} \sim \text{Beta}(X+1, n-X+1)$ and $P_2^{(i)} \sim \text{Beta}(Y+1, m-Y+1)$, set $\delta^{(i)} = P_2^{(i)} - P_1^{(i)}$, and summarize the empirical distribution of $\delta^{(i)}$.

Method 5.2.6. *Gibbs Sampling from Full Conditionals*

(i) *Kernel*: unknown mean θ, precision ξ, flat-prior posterior satisfies $\pi(\theta, \xi \mid x) \propto \xi^{\frac{n}{2}-1} \exp\{-\frac{\xi}{2} \sum_i (x_i - \theta)^2\}$.

(ii) *Decomposition*: $\sum_i (x_i - \theta)^2 = \sum_i (x_i - \overline{x})^2 + n(\theta - \overline{x})^2$.

(iii) *Mean conditional*: $\theta \mid \xi, x \sim N(\overline{x}, (n\xi)^{-1})$.

(iv) *Precision conditional*: $\xi \mid \theta, x \sim \text{Gamma}(\frac{n}{2}, \frac{1}{2} \sum_i (x_i - \theta)^2)$ in rate form.

(v) *Gibbs sampler*: alternate conditionals; post-burn-in averages estimate posterior summaries.

Method 5.2.7. *Laplace Approximation*

(i) *Setup*: let $\pi_n(\theta \mid x) \propto \exp\{-n q_n(\theta)\}$, with interior minimiser $\widehat{\theta}$.

(ii) *Curvature*: set $H_n(\widehat{\theta}) := \nabla^2 q_n(\widehat{\theta})$, assumed positive definite.

(iii) *Quadratic approximation*: $q_n(\theta) \approx q_n(\widehat{\theta}) + \frac{1}{2}(\theta - \widehat{\theta})^\top H_n(\widehat{\theta})(\theta - \widehat{\theta})$.

(iv) *Normal approximation*: $\Pi(\cdot \mid x) \approx N(\widehat{\theta}, \{n H_n(\widehat{\theta})\}^{-1})$.

Theorem 5.2.8. *Bernstein–von Mises Principle*

Let $X_1, \dots, X_n \sim P_{\theta_0}$ in a regular d-dimensional model with nonsingular information $I(\theta_0)$.
Assume the prior density π is positive and continuous near θ_0, and let $\widehat{\theta}_n$ be an efficient center.
Then $\Pi(\sqrt{n}(\theta - \widehat{\theta}_n) \in \cdot \mid X_1, \dots, X_n) \Rightarrow N_d(0, I(\theta_0)^{-1})$ in P_{θ_0}-probability.

Proof. Set $h := \sqrt{n}(\theta - \widehat{\theta}_n)$. Local asymptotic normality gives $\ell_n(\widehat{\theta}_n + \frac{h}{\sqrt{n}}) - \ell_n(\widehat{\theta}_n) = -\frac{1}{2} h^\top I(\theta_0) h + o_{P_{\theta_0}}(1)$ uniformly on bounded h. Prior continuity gives $\frac{\pi(\widehat{\theta}_n + \frac{h}{\sqrt{n}})}{\pi(\theta_0)} \to 1$ in P_{θ_0}-probability on bounded h.
After normalization, the posterior density of h converges to the $N_d(0, I(\theta_0)^{-1})$ density, with tail mass outside bounded sets negligible. Thus the displayed posterior weak convergence holds.
Structural Commentary: Locally, the likelihood is quadratic and the prior is flat to first order.
Key Insight: In regular finite-dimensional models, posterior uncertainty is asymptotically frequentist normal. □

Remark 5.2.9. *Alignment and Limits*

In regular fixed-dimensional models, Bayesian credible sets based on the Bernstein–von Mises normal approximation have asymptotic frequentist coverage. This alignment can fail in high-dimensional, nonparametric, weakly identified, or boundary problems, where the prior may remain first-order relevant.

5.3 Bayesian Normal Comparisons

The normal two-sample model also has a Bayesian interpretation. With diffuse priors, the resulting posterior formulas resemble frequentist t intervals, but the conditioning and interpretation are different.

Definition 5.3.1. *Common-Precision Two-Sample Normal Model*

(i) *Common-Precision Model*: $\theta = (\mu_X, \mu_Y, \xi) \in \mathbb{R}^2 \times (0, \infty)$ with shared precision $\xi = \sigma^{-2}$ and variance ξ^{-1}.

(ii) *Conditional model*: Given (μ_X, μ_Y, ξ), the variables $X_1, \dots, X_n, Y_1, \dots, Y_m$ are independent, with $X_i \mid (\mu_X, \xi) \sim N(\mu_X, \xi^{-1})$ and $Y_j \mid (\mu_Y, \xi) \sim N(\mu_Y, \xi^{-1})$.

(iii) *Improper prior*: $\pi(\mu_X, \mu_Y, \xi) \propto \xi^{-1}$ on $\mu_X, \mu_Y \in \mathbb{R}$ and $\xi > 0$.

(iv) *Observed data*: Write $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$.

(v) *Sample means*: $\overline{x} := n^{-1} \sum_{i=1}^n x_i$ and $\overline{y} := m^{-1} \sum_{j=1}^m y_j$.

(vi) *Sample variances*: $s_X^2 := (n-1)^{-1} \sum_{i=1}^n (x_i - \overline{x})^2$ and $s_Y^2 := (m-1)^{-1} \sum_{j=1}^m (y_j - \overline{y})^2$.

(vii) *Pooled variance*: $s_p^2 := \frac{(n-1)s_X^2 + (m-1)s_Y^2}{m+n-2}$.

(viii) *Posterior kernel*: $\pi(\mu_X, \mu_Y, \xi \mid x, y) \propto \xi^{\frac{m+n}{2}-1} \exp\left\{-\frac{\xi}{2}\left[\sum_{i=1}^n (x_i - \mu_X)^2 + \sum_{j=1}^m (y_j - \mu_Y)^2\right]\right\}$.

Proposition 5.3.2. *Conditional Posterior Factors*

For $n, m \geq 2$, put $\nu := m + n - 2$ and take Gamma *shape-rate.*
In the common-precision model, $\pi(\mu_X, \mu_Y, \xi \mid x, y) = \pi(\mu_X \mid \xi, x, y)\pi(\mu_Y \mid \xi, x, y)\pi(\xi \mid x, y)$,
where $(\mu_X, \mu_Y) \mid \xi, x, y \sim N(\overline{x}, (n\xi)^{-1}) \otimes N(\overline{y}, (m\xi)^{-1})$ and $\xi \mid x, y \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu s_p^2}{2})$. $\mu_X \perp\!\!\!\perp \mu_Y \mid (\xi, x, y)$.

Proof. Let $S_X(\mu_X) := \sum_{i=1}^n (x_i - \mu_X)^2$, $S_Y(\mu_Y) := \sum_{j=1}^m (y_j - \mu_Y)^2$, and $A := (n-1)s_X^2 + (m-1)s_Y^2 = \nu s_p^2$.
Completing squares gives $S_X(\mu_X) = (n-1)s_X^2 + n(\mu_X - \overline{x})^2$ and $S_Y(\mu_Y) = (m-1)s_Y^2 + m(\mu_Y - \overline{y})^2$.
Thus $\pi(\mu_X, \mu_Y, \xi \mid x, y) \propto \xi^{\frac{m+n}{2}-1} \exp\{-\frac{\xi A}{2}\} \exp\{-\frac{n\xi(\mu_X - \overline{x})^2}{2}\} \exp\{-\frac{m\xi(\mu_Y - \overline{y})^2}{2}\}$.

For fixed ξ, last two factors are independent normal kernels for μ_X, μ_Y with variances $(n\xi)^{-1}$, $(m\xi)^{-1}$.
Also $\int \exp\{-\frac{n\xi(\mu_X-\overline{x})^2}{2}\}\,d\mu_X \propto (n\xi)^{-\frac{1}{2}}$ and $\int \exp\{-\frac{m\xi(\mu_Y-\overline{y})^2}{2}\}\,d\mu_Y \propto (m\xi)^{-\frac{1}{2}}$.
Hence marginalising μ_X, μ_Y leaves $\pi(\xi \mid x, y) \propto \xi^{\frac{\nu}{2}-1}\exp\{-\frac{\xi\nu s_p^2}{2}\}$.
Structural Commentary: Completing squares isolates location kernels and pooled residual $A = \nu s_p^2$.
Key Insight: Marginalising μ_X, μ_Y leaves the pooled gamma law for ξ. □

Proposition 5.3.3. *Posterior t Law for a Mean Difference*
In the common-precision model with $n, m \geq 2$, set $\Delta := \mu_X - \mu_Y$, $d := \overline{x} - \overline{y}$, $c := n^{-1} + m^{-1}$, $\nu := m + n - 2$, and $T := \frac{\Delta - d}{s_p\sqrt{c}}$. Then $T \mid x, y \sim t_\nu$.

Proof. By Proposition 5.3.2, $\Delta \mid \xi, x, y \sim N(d, \frac{c}{\xi})$ and $\xi \mid x, y \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu s_p^2}{2})$.
Equivalently, take independent $E \sim N(0,1)$, $V \sim \chi^2_\nu$, $\xi = \frac{V}{\nu s_p^2}$, $\Delta = d + \sqrt{\frac{c}{\xi}}\,E$. Thus $T = \frac{\Delta - d}{s_p\sqrt{c}} = \frac{E}{\sqrt{\frac{V}{\nu}}}$, QED.
Structural Commentary: Normal given ξ; Student after mixing over ξ.
Key Insight: The t law is exactly a standard normal divided by an independent estimated-scale factor $\sqrt{\frac{V}{\nu}}$. □

Remark 5.3.4. *Same Algebra, Different Conditioning*
The statistic has the same algebra as the pooled two-sample t pivot in Theorem 3.5.4: frequentist inference treats Δ as fixed and X, Y as random, whereas the posterior treats x, y as fixed and $\Delta \mid x, y$ as random.

Example 5.3.5. *Latent Heat Posterior Comparison*
For method contrast $\Delta = \mu_X - \mu_Y$, the common-precision posterior gives $\mathbb{P}(\Delta > 0 \mid X, Y) \approx 0.998$ and posterior interval $\mathbb{P}(0.015 \leq \Delta \leq 0.065 \mid X, Y) \approx 0.95$, numerically matching the pooled normal-theory interval.

6 Nonparametric and High-Dimensional Statistics

6.1 Empirical Processes, Density, and Regression Smoothing

Nonparametric statistics replaces finite-dimensional parameter assumptions by smoothness, shape, sparsity, or complexity restrictions.

Definition 6.1.1. *Empirical Measure and Empirical Process*
Let $X_1, \ldots, X_n$ be observations in $(\mathcal{X}, \mathcal{A})$ with reference law P.

(i) *Empirical measure*: $\widehat{P}_n := n^{-1}\sum_{i=1}^n \delta_{X_i}$, so $\widehat{P}_n f = n^{-1}\sum_{i=1}^n f(X_i)$.

(ii) *Empirical process*: For measurable $\mathcal{F}$, $\mathbb{G}_n(f) := \sqrt{n}(\widehat{P}_n f - Pf)$ for $f \in \mathcal{F}$.

Proposition 6.1.2. *Fixed-Point Behavior of the Empirical CDF*
Let $X_1, \ldots, X_n$ be independent with common CDF F.
For $x \in \mathbb{R}$, set $I_i(x) := \mathbf{1}_{(-\infty, x]}(X_i)$ and $F_n(x) := n^{-1}\sum_{i=1}^n I_i(x)$. Then, for each fixed $x \in \mathbb{R}$:

(i) Law: $nF_n(x) \sim \text{Binomial}(n, F(x))$.
(ii) Mean: $\mathbb{E}F_n(x) = F(x)$.
(iii) Variance: $\text{Var}(F_n(x)) = n^{-1}F(x)(1 - F(x))$.

Proof. Fix $x \in \mathbb{R}$ and put $p := F(x)$ and $I_i := I_i(x)$.

(i) Since $I_i \sim \text{Bernoulli}(p)$ independently, $nF_n(x) = \sum_{i=1}^n I_i \sim \text{Binomial}(n, p)$.

(ii) Linearity gives $\mathbb{E}F_n(x) = n^{-1}\sum_{i=1}^n \mathbb{E}I_i = n^{-1}np = p = F(x)$.

(iii) Independence gives $\text{Var}(F_n(x)) = n^{-2}\sum_{i=1}^n \text{Var}(I_i) = n^{-2}np(1-p) = n^{-1}F(x)(1 - F(x))$.

Structural Commentary: Pointwise, $F_n(x) = n^{-1}\sum_{i=1}^n I_i(x)$ with $I_i(x) \sim \text{Bernoulli}(F(x))$ independently.
Key Insight: The empirical CDF at one point is a normalised binomial count. □

Remark 6.1.3. *Pointwise and Simultaneous CDF Uncertainty*
Proposition 6.1.2 is pointwise: x is fixed and $F_n(x)$ is a normalised binomial count.
Simultaneous uncertainty controls $\|F_n - F\|_\infty := \sup_x |F_n(x) - F(x)|$, yielding bands and diagnostics.

Proposition 6.1.4. *Two-Point Covariance of the Empirical CDF*
Let $X_1, \ldots, X_n$ be independent with common CDF F.
For $w \in \mathbb{R}$, set $I_i(w) := \mathbf{1}_{(-\infty, w]}(X_i)$, $F_n(w) := n^{-1}\sum_{i=1}^n I_i(w)$. For $u, v \in \mathbb{R}$, put $m := \min\{u, v\}$. Then:

(i) Covariance: $\text{Cov}\big(F_n(u), F_n(v)\big) = n^{-1}\big(F(m) - F(u)F(v)\big)$.
(ii) Sign: $\text{Cov}\big(F_n(u), F_n(v)\big) \geq 0$.

Proof. Fix $u, v \in \mathbb{R}$ and set $m := \min\{u, v\}$.

(i) *Covariance*: $\text{Cov}(F_n(u), F_n(v)) = n^{-2}\sum_{i,j}\text{Cov}(I_i(u), I_j(v))$.

For $i \neq j$ the term is 0 by independence, while for $i = j$,
$\text{Cov}(I_i(u), I_i(v)) = \mathbb{E}[I_i(u)I_i(v)] - \mathbb{E}I_i(u)\mathbb{E}I_i(v) = F(m) - F(u)F(v)$.
Hence $\text{Cov}(F_n(u), F_n(v)) = n^{-1}(F(m) - F(u)F(v))$.

(ii) *Sign*: If $u \le v$, then $F(m) - F(u)F(v) = F(u)(1 - F(v)) \ge 0$.
If $v \le u$, then $F(m) - F(u)F(v) = F(v)(1 - F(u)) \ge 0$.

Structural Commentary: $F_n(u)$ and $F_n(v)$ average iid Bernoulli pairs, so covariance scales by $1/n$.
Key Insight: Cross-index independence leaves only the same-observation overlap $\{X_i \le m\}$. □

Theorem 6.1.5. *Glivenko–Cantelli Theorem*
Let $X_1, X_2, \dots$ be i.i.d. real variables with CDF F, and set $\widehat{F}_n(t) := n^{-1} \sum_{i=1}^{n} \mathbf{1}_{\{X_i \le t\}}$.

Then $\sup_{t \in \mathbb{R}} \left| \widehat{F}_n(t) - F(t) \right| \to 0$ almost surely.

Proof. Add independent $R_i \sim \mathrm{Unif}[0,1]$ and set $U_i := F(X_i-) + R_i\{F(X_i) - F(X_i-)\}$.
Then U_i are i.i.d. Unif$[0,1]$, and with $H_n(u) := n^{-1} \sum_i \mathbf{1}_{\{U_i \le u\}}$ one has $\widehat{F}_n(t) = H_n(F(t))$ a.s.

Hence $\sup_t \left| \widehat{F}_n(t) - F(t) \right| \le \sup_{0 \le u \le 1} |H_n(u) - u|$.
Fix $k \ge 1$, $u_j := \frac{j}{k}$, let $M_{n,k} := \max_{0 \le j \le k} |H_n(u_j) - u_j|$.
By the SLLN, $M_{n,k} \to 0$ a.s. If $u_{j-1} \le u \le u_j$, monotonicity gives $H_n(u_{j-1}) - u_j \le H_n(u) - u \le H_n(u_j) - u_{j-1}$,
Hence $\sup_{0 \le u \le 1} |H_n(u) - u| \le M_{n,k} + k^{-1}$, and $\limsup_{n \to \infty} \sup_{0 \le u \le 1} |H_n(u) - u| \le k^{-1}$ a.s. $\forall k$. Let $k \to \infty$.

Structural Commentary: Finite-grid SLLN plus monotonicity gives uniform CDF control.
Key Insight: Monotonicity bounds between-grid drift by the mesh. □

Proposition 6.1.6. *Approximate Variance of a Sample Quantile*
Let $Y_1, \dots, Y_n$ be iid with continuous CDF F and density f. Fix $p \in (0,1)$, set $x_p := F^{-1}(p)$ with $f(x_p) > 0$, choose k with $\frac{k}{n+1} \approx p$, and put $\widehat{x}_p := Y_{(k)}$. Then:

(i) Probability Scale: $U_{(k)} := F(Y_{(k)})$ has kth uniform order-statistic law. $\mathrm{Var}(U_{(k)}) = \frac{k(n+1-k)}{(n+1)^2(n+2)} \approx \frac{p(1-p)}{n}$.

(ii) Quantile Scale: $\mathrm{Var}(\widehat{x}_p) \approx \frac{p(1-p)}{n f(x_p)^2}$.

Proof.
(i) By probability integral transform, $U_i := F(Y_i)$ are iid Uniform$(0,1)$, $U_{(k)} = F(Y_{(k)})$.
Theorem 4.1.4 gives the stated exact variance, hence $\mathrm{Var}(U_{(k)}) \approx \frac{p(1-p)}{n}$ when $\frac{k}{n+1} \approx p$.
(ii) Let $g := F^{-1}$. Since $g'(p) = \frac{1}{f(x_p)}$ and $\widehat{x}_p = g(U_{(k)})$, delta approximation $g(U_{(k)}) \approx g(p) + g'(p)(U_{(k)} - p)$
gives $\mathrm{Var}(\widehat{x}_p) \approx g'(p)^2 \mathrm{Var}(U_{(k)}) \approx \frac{p(1-p)}{n f(x_p)^2}$.

Structural Commentary: Quantile variance is probability-scale variance multiplied by $(F^{-1})'(p)^2$.
Key Insight: Low density near x_p inflates sample-quantile variability by $1/f(x_p)^2$. □

Proposition 6.1.7. *Distribution-Free Median Interval*
Let $X_1, \dots, X_n$ be independent with continuous CDF F and median η, so $F(\eta) = \frac{1}{2}$.
Let $X_{(1)} \le \cdots \le X_{(n)}$ and $1 \le k \le \lfloor \frac{n}{2} \rfloor$. Then:

(i) Interval: $C_k := [X_{(k)}, X_{(n-k+1)}]$.

(ii) Coverage: $\mathbb{P}(\eta \in C_k) = 1 - \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$.

Proof.
(i) Let $B_- := \sum_{i=1}^{n} \mathbf{1}_{\{X_i < \eta\}}$, $B_+ := \sum_{i=1}^{n} \mathbf{1}_{\{X_i > \eta\}}$. By continuity, $B_- + B_+ = n$ a.s. and $B_- \sim \mathrm{Binomial}(n, \frac{1}{2})$.
(ii) $\eta \notin C_k$ iff $B_- < k$ or $B_+ < k$. Since $k \le \lfloor \frac{n}{2} \rfloor$, these two events are disjoint.
(iii) By symmetry, $\mathbb{P}(\eta \notin C_k) = 2\mathbb{P}(B_- < k) = 2 \sum_{j=0}^{k-1} \binom{n}{j} 2^{-n} = \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$, so $\mathbb{P}(\eta \in C_k) = 1 - \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$.

Structural Commentary: The coverage depends only on the binomial count of observations below η, not on F.
Key Insight: Continuity turns median coverage into a symmetric binomial tail. □

Definition 6.1.8. *Kernel Density Estimator*
(i) *Kernel and bandwidth*: Let $\int K(u)\,du = 1$ and $h > 0$.
(ii) *Estimator*: For data $X_1, \dots, X_n$, $\widehat{f}_h(x) := (nh)^{-1} \sum_{i=1}^{n} K(\frac{x - X_i}{h})$.

Larger h lowers variance and raises bias; smaller h does the reverse.

Definition 6.1.9. *Nonparametric Regression*
(i) *Model*: $Y = m(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X] = 0$, so $m(x) = \mathbb{E}[Y \mid X = x]$.
(ii) *Weights*: For kernel K and bandwidth $h > 0$, set $w_i(x) := K(\frac{x - X_i}{h})$.
(iii) *Estimator*: If $\sum_i w_i(x) > 0$, define $\widehat{m}_h(x) := \frac{\sum_{i=1}^{n} w_i(x) Y_i}{\sum_{i=1}^{n} w_i(x)}$.

Definition 6.1.10. *Kernel-Weighted Local Average*
Let $(X_i, Y_i)_{i=1}^n$ be regression data and fix a query point x_0.
(i) *Local kernel*: For a nonnegative profile D and width $h_\lambda(x_0) > 0$, set $K_\lambda(x_0, x) := D(\|x - x_0\| / h_\lambda(x_0))$.

(ii) *Estimator*: If $\sum_{i=1}^{n} K_\lambda(x_0, X_i) > 0$, local-constant (Nadaraya–Watson) estimate is $\widehat{m}(x_0) := \frac{\sum_{i=1}^{n} K_\lambda(x_0, X_i) Y_i}{\sum_{i=1}^{n} K_\lambda(x_0, X_i)}$.

Metric windows set $h_\lambda(x_0) = \lambda$; nearest-neighbor windows set $h_\lambda(x_0) = \operatorname{dist}(x_0, X_{(k)}(x_0))$.

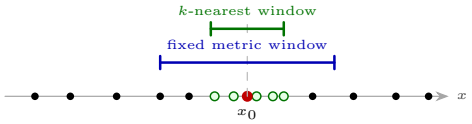

Figure 241: Fixed metric and k-nearest windows around x_0.

Example 6.1.11. *Common Local-Smoothing Kernels*
In one dimension, common compact profiles are $D_E(t) := \frac{3}{4}(1-t^2)\mathbf{1}_{\{|t|\le 1\}}$ for Epanechnikov smoothing and $D_T(t) := (1-|t|^3)^3\mathbf{1}_{\{|t|\le 1\}}$ for tri-cube smoothing. Both have support $[-1,1]$; Gaussian profiles have full support, and normalization constants cancel in regression-weight averages but matter for density estimation.

Method 6.1.12. *Local Linear Regression Smoothing*
Fix x_0 and bandwidth $h > 0$.

(i) *Weighted fit*: Set $(\widehat{\beta}_0(x_0), \widehat{\beta}_1(x_0)) \in \operatorname*{argmin}_{\beta_0,\beta_1} \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1(X_i - x_0))^2 K(\frac{X_i - x_0}{h})$.

(ii) *Estimate*: Define $\widehat{m}_h(x_0) := \widehat{\beta}_0(x_0)$.

Smaller h gives a more variable fit; larger h gives more bias.

Method 6.1.13. *Local Polynomial Regression*
Fix a query point $x_0 \in \mathbb{R}^p$ and degree $d \in \mathbb{N}_0$.

(i) *Basis*: Let $\mathcal{A}_d := \{\alpha \in \mathbb{N}_0^p : |\alpha| \le d\}$ and $b_d(u) := (u^\alpha)_{\alpha\in\mathcal{A}_d}$, where $u^\alpha := \prod_{j=1}^{p} u_j^{\alpha_j}$.

(ii) *Weights*: Set $K_i(x_0) := K_\lambda(x_0, X_i)$.

(iii) *Weighted fit*: For $\beta = (\beta_\alpha)_{\alpha\in\mathcal{A}_d}$, choose $\widehat{\beta}(x_0) \in \operatorname*{argmin}_{\beta\in\mathbb{R}^{|\mathcal{A}_d|}} \sum_{i=1}^{n} K_i(x_0)(Y_i - b_d(X_i - x_0)^\top\beta)^2$.

(iv) *Estimator*: With $\mathbf{0} \in \mathcal{A}_d$ the constant multi-index, define $\widehat{m}_d(x_0) := \widehat{\beta}_{\mathbf{0}}(x_0)$.

$d = 0$ is local constant; $d = 1$ is local linear with boundary correction.
Larger d trades lower curvature bias for higher variance.

Proposition 6.1.14. *Equivalent Kernel Moments*
Fix $x_0 \in \mathbb{R}$ and set $u_i := X_i - x_0$, $K_i := K_\lambda(x_0, X_i)$, $z_i := (1, u_i)^\top$, $Z := (z_1^\top, \dots, z_n^\top)^\top$, $W := \operatorname{diag}(K_1, \dots, K_n)$, $Y := (Y_1, \dots, Y_n)^\top$, and $G := Z^\top W Z$.
Assume G is nonsingular, let $e_1 := (1,0)^\top$, $\widehat{\beta}(x_0) := G^{-1}Z^\top WY$, and $\widehat{m}(x_0) := e_1^\top\widehat{\beta}(x_0)$.

(i) Equivalent weights: $\widehat{m}(x_0) = \sum_{i=1}^{n} \ell_i(x_0) Y_i$ with $\ell_i(x_0) := K_i z_i^\top G^{-1} e_1$.

(ii) Moment constraints: $\sum_{i=1}^{n} \ell_i(x_0) = 1$ and $\sum_{i=1}^{n} u_i \ell_i(x_0) = 0$.

(iii) Affine reproduction: If $Y_i = a + bX_i$, then $\widehat{m}(x_0) = a + bx_0$.

Proof.

(i) *Equivalent weights*: The weighted criterion $(Y - Z\beta)^\top W(Y - Z\beta)$ has normal equations $G\widehat{\beta} = Z^\top WY$, so $\widehat{\beta} = G^{-1}Z^\top WY$. Since G^{-1} is symmetric, $\widehat{m}(x_0) = e_1^\top G^{-1}Z^\top WY = \sum_{i=1}^{n} K_i e_1^\top G^{-1} z_i Y_i = \sum_{i=1}^{n} \ell_i(x_0) Y_i$.

(ii) *Moment constraints*: With $\ell(x_0) := (\ell_1(x_0), \dots, \ell_n(x_0))^\top = WZG^{-1}e_1$, $Z^\top\ell(x_0) = Z^\top WZG^{-1}e_1 = e_1$, hence $\sum_i \ell_i(x_0) = 1$ and $\sum_i u_i \ell_i(x_0) = 0$.

(iii) *Affine reproduction*: If $Y_i = a + bX_i = (a + bx_0) + bu_i$, then $\widehat{m}(x_0) = (a + bx_0)\sum_i \ell_i(x_0) + b\sum_i u_i\ell_i(x_0) = a + bx_0$.

Structural Commentary: Weighted projection onto $\operatorname{span}\{1, x - x_0\}$ imposes two weight moments.
Key Insight: $Z^\top\ell = e_1$ encodes constant reproduction and zero first moment. □

Definition 6.1.15. *Structured Kernel Metric*
Fix $x_0, x \in \mathbb{R}^p$, bandwidth $\lambda > 0$, profile $D \ge 0$, and a positive semidefinite matrix $A \succeq 0$.

(i) *Quadratic scale*: Define $\rho_A(x, x_0) := (x - x_0)^\top A(x - x_0)$.

(ii) *Kernel*: Set $K_{\lambda,A}(x_0, x) := D(\rho_A(x, x_0)/\lambda)$.

A sets coordinate weights, correlations, and active directions.
Standardise predictors first since ρ_A is unit-dependent.

Method 6.1.16. *Local Likelihood and Varying Coefficients*
Let $\ell_i(\theta)$ be the log-likelihood contribution of observation i under parameter θ.

(i) *Local likelihood*: At target x_0, set $K_i(x_0) := K_\lambda(x_0, X_i)$ and choose $\widehat{\theta}(x_0) \in \operatorname*{argmax}_\theta \sum_{i=1}^{n} K_i(x_0)\ell_i(\theta)$.

(ii) *Local parameterisation*: In generalised models, replace a constant natural parameter by $\eta_i(\gamma; x_0) := b_d(X_i - x_0)^\top\gamma$ and maximize $\sum_i K_i(x_0)\ell_i(\eta_i(\gamma; x_0))$ over γ.

(iii) *Varying coefficients*: For predictors (U, Z) with $U \in \mathbb{R}^q$, model $m(U, Z) := \alpha(Z) + \sum_{r=1}^{q} \beta_r(Z) U_r$.

(iv) *Local fit*: At z_0, choose $(\widehat{\alpha}, \widehat{\beta}) \in \operatorname{argmin}_{a,b} \sum_i K_\lambda(z_0, Z_i)(Y_i - a - b^\top U_i)^2$ and set $(\widehat{\alpha}(z_0), \widehat{\beta}(z_0)) := (\widehat{\alpha}, \widehat{\beta})$.

The coefficients are ordinary regression coefficients locally in U, but functions of the index variable Z.

Remark 6.1.17. *Bandwidth, Boundaries, and Dimension*
Let $N_\lambda(x_0) := \{i : K_\lambda(x_0, X_i) > 0\}$ and $n_\lambda(x_0) := |N_\lambda(x_0)|$.

(i) *Bias–variance*: Larger λ usually increases $n_\lambda(x_0)$, lowering variance, raising bias; smaller λ does reverse.
(ii) *Windows*: Metric windows fix radius; k-nearest-neighbor windows choose $\lambda(x_0)$ so that $n_{\lambda(x_0)}(x_0) \approx k$.
(iii) *Dimension*: In $\mathbb{R}^p$, neighborhoods must expand with p to keep $n_\lambda(x_0)$ stable, reducing local resolution.
(iv) *Cost*: Without indexing, one query evaluates $K_\lambda(x_0, X_i)$ for all i, costing $O(n)$.

Method 6.1.18. *Bandwidth Choice by Leave-One-Out Cross-Validation*

(i) *Candidates*: Fix bandwidths $h_1, \dots, h_M > 0$.

(ii) *Leave-one-out score*: Let $\widehat{m}_{h_k,-i}$ be local linear fit excluding (X_i, Y_i). Set $\mathrm{CV}(h_k) := \sum_{i=1}^{n} (Y_i - \widehat{m}_{h_k,-i}(X_i))^2$.

(iii) *Selection*: Choose $\widehat{h} \in \operatorname{argmin}_{1 \le k \le M} \mathrm{CV}(h_k)$, with a visual smoothness check.

Remark 6.1.19. *Complexity Control*
For a function class $\mathcal{F}$, estimation is stable when $\sup_{f \in \mathcal{F}} \left| \widehat{P}_n f - P f \right|$ is small with high probability or a.s.
Smoothness and entropy control this supremum in place of finite dimension.

6.2 Sparse High-Dimensional Inference

High-dimensional statistics studies regimes where the number of candidate parameters is comparable to, or much larger than, the sample size. Estimation is possible only after adding structure.

Definition 6.2.1. *High-Dimensional Regime and Sparsity*

(i) *High dimension*: A sequence has dimension $p = p_n$, often with $p_n \gg n$.
(ii) *Sparsity*: For $\theta \in \mathbb{R}^p$, $\|\theta\|_0 := |\{j : \theta_j \neq 0\}|$; θ is s-sparse if $\|\theta\|_0 \le s$.

Definition 6.2.2. *Lasso Estimator*

(i) *Lasso*: For $y = X\beta_0 + \varepsilon$ and $\lambda \ge 0$, $\widehat{\beta}_\lambda \in \arg\min_{\beta \in \mathbb{R}^p} \{(2n)^{-1} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1\}$.

The convex ℓ^1 penalty promotes sparsity.

Definition 6.2.3. *Restricted Eigenvalue Condition*

(i) *Restricted eigenvalue*: For $S \subseteq [p] := \{1, \dots, p\}$, $C(S) := \{v : \|v_{S^c}\|_1 \le 3 \|v_S\|_1\}$, and $\kappa > 0$,
X satisfies $\mathrm{RE}(S, \kappa)$ if $n^{-\frac{1}{2}} \|Xv\|_2 \ge \kappa \|v_S\|_2$ for all $v \in C(S)$.

Theorem 6.2.4. *Basic Lasso Rate Form*
Let $y = X\beta_0 + \varepsilon$, $S := \mathrm{supp}(\beta_0)$, $|S| \le s$, and let $\widehat{\beta}_\lambda$ be any lasso solution.
If X satisfies $\mathrm{RE}(S, \kappa)$ and $\lambda > 0$ with $\lambda \ge 2 \left\| \frac{X^\top \varepsilon}{n} \right\|_\infty$, then $n^{-1} \left\| X(\widehat{\beta}_\lambda - \beta_0) \right\|_2^2 \le \frac{9 s \lambda^2}{\kappa^2}$ and $\left\| \widehat{\beta}_\lambda - \beta_0 \right\|_1 \le \frac{12 s \lambda}{\kappa^2}$.

Proof. Set $\delta := \widehat{\beta}_\lambda - \beta_0$. The lasso optimality inequality gives $(2n)^{-1} \|X\delta\|_2^2 + \lambda \|\beta_0 + \delta\|_1 \le n^{-1} \varepsilon^\top X\delta + \lambda \|\beta_0\|_1$.
Since $\lambda \ge 2 \left\| \frac{X^\top \varepsilon}{n} \right\|_\infty$, $n^{-1} \varepsilon^\top X\delta \le \frac{\lambda}{2} \|\delta\|_1$, and since $\beta_{0,S^c} = 0$, $\|\beta_0\|_1 - \|\beta_0 + \delta\|_1 \le \|\delta_S\|_1 - \|\delta_{S^c}\|_1$.
Thus $(2n)^{-1} \|X\delta\|_2^2 + \frac{\lambda}{2} \|\delta_{S^c}\|_1 \le \frac{3\lambda}{2} \|\delta_S\|_1$, so $\delta \in C(S)$. By $\mathrm{RE}(S, \kappa)$, $\|\delta_S\|_1 \le \sqrt{s} \|\delta_S\|_2 \le \frac{\sqrt{s} \|X\delta\|_2}{\kappa \sqrt{n}}$.
Hence $n^{-1} \|X\delta\|_2^2 \le \frac{3\lambda \sqrt{s} \|X\delta\|_2}{\kappa \sqrt{n}}$, giving $n^{-1} \|X\delta\|_2^2 \le \frac{9 s \lambda^2}{\kappa^2}$; also $\|\delta\|_1 \le 4 \|\delta_S\|_1 \le \frac{12 s \lambda}{\kappa^2}$.
Structural Commentary: Basic inequality plus RE converts score control into prediction and ℓ^1 error bounds.
Key Insight: The ℓ^1 penalty forces the error into the RE cone. □

Method 6.2.5. *Featurewise Two-Sample Screening*
Let x_{ij} be feature $j \in [M]$ for observation i, with groups C_1, C_2 and $n_\ell := |C_\ell|$.

(i) *Group means*: For $\ell \in \{1, 2\}$, set $\bar{x}_{\ell j} := n_\ell^{-1} \sum_{i \in C_\ell} x_{ij}$.

(ii) *Pooled scale*: Set $\widehat{\sigma}_j^2 := (n_1 + n_2 - 2)^{-1} \{ \sum_{i \in C_1} (x_{ij} - \bar{x}_{1j})^2 + \sum_{i \in C_2} (x_{ij} - \bar{x}_{2j})^2 \}$ and $\mathrm{se}_j := \widehat{\sigma}_j \sqrt{n_1^{-1} + n_2^{-1}}$.

(iii) *Statistic*: Define $t_j := (\bar{x}_{2j} - \bar{x}_{1j}) / \mathrm{se}_j$; large $|t_j|$ flags a featurewise mean shift.
(iv) *Family*: Screening acts on $\{t_j : 1 \le j \le M\}$, so error control targets the family, not a single statistic.

Remark 6.2.6. *Statistical and Computational Structure*
High-dimensional inference couples concentration bounds, convex optimization, and selection error control; computation and statistics are part of the same guarantee.

6.3 Rank-Based Comparative Tests

Rank methods replace numerical values by their relative order. They are invariant under strictly increasing transformations and reduce the influence of extreme observations, which makes them natural companions to the two-sample and paired t procedures.

Definition 6.3.1. *Mann–Whitney and Rank-Sum Statistics*
Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$ be independent continuous samples.
Rank the pooled observations increasingly.

(i) *Pooled size*: $N := m + n$.
(ii) *Y-ranks*: $R_j^Y := \operatorname{rank}(Y_j)$ in the pooled sample.
(iii) *Rank sum*: $T_Y := \sum_{j=1}^{m} R_j^Y$.
(iv) *Mann–Whitney statistic*: $U_Y := T_Y - \frac{m(m+1)}{2}$.

Under continuity, $U_Y = \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{1}_{\{X_i < Y_j\}}$ a.s., so U_Y counts cross-sample pairs with $Y_j > X_i$.

Theorem 6.3.2. *Null Mean and Variance of the Rank Sum*
Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$ be independent continuous samples.
Let rank-sum T_Y and Mann–Whitney statistic U_Y. Assume $F = G$ and set $N := m + n$.

(i) Rank law: $T_Y = \sum_{r \in A} r$, where A is uniform over all m-subsets of $\{1, \dots, N\}$.
(ii) Rank-sum moments: $\mathbb{E}T_Y = \frac{m(N+1)}{2}$ and $\operatorname{Var}(T_Y) = \frac{mn(N+1)}{12}$.
(iii) Mann–Whitney moments: $\mathbb{E}U_Y = \frac{mn}{2}$ and $\operatorname{Var}(U_Y) = \frac{mn(N+1)}{12}$.

Proof.

(i) Continuity gives no ties and $F = G$ makes labels exchangeable, so the Y-rank set $A \subseteq \{1, \dots, N\}$ is uniform over all m-subsets and $T_Y = \sum_{r \in A} r$.
(ii) For ranks $1, \dots, N$, $\bar{r} = \frac{N+1}{2}$ and $S_r^2 = \frac{N^2-1}{12}$; sampling m ranks without replacement gives $\mathbb{E}T_Y = m\bar{r} = \frac{m(N+1)}{2}$ and $\operatorname{Var}(T_Y) = \frac{mS_r^2(N-m)}{N-1} = \frac{mn(N+1)}{12}$.
(iii) Since $U_Y = T_Y - \frac{m(m+1)}{2}$, shifting by a constant gives $\mathbb{E}U_Y = \frac{m(N+1)}{2} - \frac{m(m+1)}{2} = \frac{mn}{2}$ and $\operatorname{Var}(U_Y) = \operatorname{Var}(T_Y) = \frac{mn(N+1)}{12}$.

Structural Commentary: Under $F = G$, ranks reduce to simple random sampling without replacement.
Key Insight: U_Y is just a centered rank sum, so only the mean shifts. □

Method 6.3.3. *Mann–Whitney Test*
Given independent samples $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$, test $H_0 : F = G$:

(i) *Compute*: Pool and rank $m + n$ observations, use midranks for ties. Compute T_Y, $U_Y = T_Y - \frac{m(m+1)}{2}$.
(ii) *Calibrate*: For small m, n, enumerate m-subsets of $\{1, \dots, m+n\}$ to get the exact null law.
For large m, n, use $Z := \frac{U_Y - \frac{mn}{2}}{\sqrt{\frac{mn(m+n+1)}{12}}} \approx N(0,1)$, with tie correction when needed.
(iii) *Decide*: Compute the relevant one- or two-sided tail probability under the chosen null law.
Reject H_0 at level α when $p \le \alpha$.

Example 6.3.4. *Latent Heat Data by Ranks*
For the latent-heat data, take $m = 8$, $n = 13$, $N = 21$, and midranks for ties.
The smaller-sample rank sum is $R = 51$, its complement is $R' = m(N+1) - R = 125$, and $R^* = \min(R, R') = 51$.
Under equality of the two method distributions, $\mu_T = \frac{m(N+1)}{2} = 88$ and $\sigma_T = \sqrt{\frac{mn(N+1)}{12}} \approx 13.8$,
hence $z = \frac{R - \mu_T}{\sigma_T} \approx -2.68$ and the two-sided normal p-value is ≈ 0.007.
The exact rank-sum test therefore rejects equality at level 0.01.

Definition 6.3.5. *Mann–Whitney Effect Probability*
Let $X \sim F$ and $Y \sim G$ be independent, with samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$.

(i) *Effect probability*: $\pi := \mathbb{P}(X < Y)$.
(ii) *Estimator*: $\widehat{\pi} := (mn)^{-1} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{1}_{\{X_i < Y_j\}} = \frac{U_Y}{mn}$.

Thus π is the probability that a random Y exceeds a random X.

Method 6.3.6. *Shift Intervals from Pairwise Differences*
Assume the shift model $G(x) = F(x - \Delta)$.

(i) *Compute*: Set $D_{ij} := Y_j - X_i$ and order $D_{(1)} \le \cdots \le D_{(mn)}$.
(ii) *Calibrate*: Choose k so the two-sided rank test accepts a candidate shift when $k \le U_Y(\Delta) \le mn - k$.
(iii) *Invert*: Report $[D_{(k)}, D_{(mn-k+1)}]$ as the inverted shift interval.

The Hodges–Lehmann point estimate is $\widehat{\Delta}_{HL} := \operatorname{median}\{D_{ij} : 1 \le i \le n, 1 \le j \le m\}$.

Definition 6.3.7. *Signed-Rank Statistic*
Let $D_1, \dots, D_n$ be paired differences.

(i) *Nonzero index set*: $I := \{i : D_i \neq 0\}$.
(ii) *Absolute ranks*: $R_i := \operatorname{rank}(|D_i|)$ among $\{|D_\ell| : \ell \in I\}$, using midranks for ties.
(iii) *Signed-rank statistic*: $W_+ := \sum_{i \in I} R_i \mathbf{1}_{\{D_i > 0\}}$.

Theorem 6.3.8. *Signed-Rank Mean and Variance*
Under the signed-rank null, assume no zero differences and no ties in $|D_i|$.

(i) Mean: $\mathbb{E}W_+ = \frac{n(n+1)}{4}$.

(ii) Variance: $\mathrm{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}$.

Proof. Let $\mathcal{R}$ be the absolute-rank information and let I_k indicate positivity at absolute-rank k.
Then $W_+ = \sum_{k=1}^{n} kI_k$ and, under symmetry, $I_k \mid \mathcal{R}$ are independent Bernoulli($\frac{1}{2}$).

(i) *Mean*: $\mathbb{E}(W_+ \mid \mathcal{R}) = \frac{1}{2}\sum_{k=1}^{n} k = \frac{n(n+1)}{4}$; the right side is deterministic, hence $\mathbb{E}W_+ = \frac{n(n+1)}{4}$.

(ii) *Variance*: $\mathrm{Var}(W_+ \mid \mathcal{R}) = \frac{1}{4}\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}$, while $\mathbb{E}(W_+ \mid \mathcal{R})$ is deterministic.

Total variance gives $\mathrm{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}$.

Structural Commentary: Condition on absolute ranks; only independent fair signs remain.
Key Insight: W_+ is a weighted sum of Bernoulli signs with weights $1, \ldots, n$. □

Method 6.3.9. *Signed-Rank Test*
For paired differences D_i, test $H_0 : D_i \overset{d}{=} -D_i$.

(i) *Compute*: Discard zero differences, rank nonzero $|D_i|$ using midranks for ties, and compute W_+.
(ii) *Calibrate*: Conditional on the nonzero absolute ranks, enumerate the 2^r sign assignments, where $r := |I|$, to get the null law of W_+.
(iii) *Decide*: Compute the relevant tail probability and reject H_0 at level α when $p \le \alpha$.

The sign test is the coarser paired alternative using only signs, not ranks.

Example 6.3.10. *Paired Rank Analyses*
Two paired-rank calculations are:

(i) *Platelet data*: With one negative signed rank, use W_-; exact signed-rank test rejects symmetry at 0.01.
(ii) *Mercury data*: Dropping one zero difference and using midranks gives $n = 24$ and $W_+ = 194.5$. Under H_0, $\mathbb{E}_0 W_+ = \frac{24\cdot 25}{4} = 150$ and $\mathrm{Var}_0(W_+) = \frac{24\cdot 25\cdot 49}{24} = 1225$, so $Z = \frac{194.5-150}{35} \approx 1.27$ and two-sided $p \approx 0.20$; there is no strong evidence of a systematic shift.

Remark 6.3.11. *Rank Tests and Normal Theory*
Rank tests use order information, require no normal model, and are robust to outliers; under normal models their efficiency relative to t tests is often close to 1.

7 Time Series Models

7.1 Stationarity, Autocovariance, and Linear Processes

Time-series statistics studies data ordered by time, where dependence is part of the signal rather than a nuisance to be removed. The process-level notions of strict and weak stationarity are fixed in Definition 4.5.1.

Definition 7.1.1. *Autocorrelation, Partial Autocorrelation, and Spectral Density*
Let $(X_t)_{t\in\mathbb{Z}}$ be weakly stationary with autocovariance $\gamma(h) := \mathrm{Cov}(X_t, X_{t-h})$ and $\gamma(0) > 0$.
For $h \ge 2$ and $t \in \mathbb{Z}$, set $\mathcal{H}_{t,h} := \mathrm{span}\{1, X_{t-1}, \ldots, X_{t-h+1}\}$ let $P_{t,h}$ be L^2-orthogonal projection onto $\mathcal{H}_{t,h}$.
Assume $\mathrm{Var}(X_t - P_{t,h}X_t) > 0$ and $\mathrm{Var}(X_{t-h} - P_{t,h}X_{t-h}) > 0$ $\forall h \ge 2,\ t \in \mathbb{Z}$.

(i) *Autocorrelation function*: $\rho : \mathbb{Z} \to [-1, 1]$, $\rho(h) := \frac{\gamma(h)}{\gamma(0)}$.

(ii) *Partial autocorrelation function*: $\alpha : \mathbb{N}_0 \to [-1, 1]$, $\alpha(0) := 1$, $\alpha(1) := \frac{\gamma(1)}{\gamma(0)}$, and $\alpha(h) := \mathrm{Corr}(X_t - P_{t,h}X_t, X_{t-h} - P_{t,h}X_{t-h})$ $\forall h \ge 2,\ t \in \mathbb{Z}$.

(iii) *Spectral density*: If $\sum_{h\in\mathbb{Z}} |\gamma(h)| < \infty$, $f_X : [-\pi, \pi] \to \mathbb{C}$, $f_X(\lambda) := (2\pi)^{-1} \sum_{h\in\mathbb{Z}} \gamma(h)e^{-ih\lambda}$ $\forall \lambda \in [-\pi, \pi]$.

Definition 7.1.2. *White Noise and Linear Process*
Let $\sigma^2 \ge 0$, $\mu \in \mathbb{R}$, and $(\psi_j)_{j\ge 0} \subset \mathbb{R}$ satisfy $\sum_{j=0}^{\infty} |\psi_j| < \infty$; all processes below are real-valued, indexed by $\mathbb{Z}$.

(i) *White noise*: $(\varepsilon_t)_{t\in\mathbb{Z}} \subset L^2$, $\mathbb{E}\varepsilon_t = 0$ and $\mathrm{Var}(\varepsilon_t) = \sigma^2$ $\forall t \in \mathbb{Z}$, and $\mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0$ $\forall s, t \in \mathbb{Z}$ with $s \ne t$.

(ii) *Linear process*: $(X_t)_{t\in\mathbb{Z}}$ satisfies $X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ $\forall t \in \mathbb{Z}$ for some $(\varepsilon_t)_{t\in\mathbb{Z}}$ satisfying (i).

Structural Commentary: $\sum_j |\psi_j| < \infty$ gives an L^2-stable one-sided innovation filter.

Example 7.1.3. *Autoregression of Order One*
Let $(\varepsilon_t)_{t\in\mathbb{Z}}$ be white noise, variance σ^2, $|\phi| < 1$. For $X_t = \phi X_{t-1} + \varepsilon_t$, stationary solution is $X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$.
Thus $\gamma(h) = \mathrm{Cov}(X_t, X_{t-h}) = \frac{\sigma^2\phi^{|h|}}{1-\phi^2}$, $\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^{|h|}$. Autocorrelation decays geometrically at rate $|\phi|$.

Remark 7.1.4. *Process-Level Prerequisites*
Time-series inference uses filtrations $(\mathcal{F}_t)$, adapted X_t, predictors $\mathbb{E}(X_{t+h} \mid \mathcal{F}_t)$, martingale differences ε_t, and stationarity or ergodicity assumptions.

7.2 ARIMA, State-Space, and Volatility Models

Classical time-series models combine deterministic filters, stochastic innovations, and likelihood-based estimation.

Definition 7.2.1. *ARMA Model*
Let $p, q \in \mathbb{N}_0$, $\phi_i \in \mathbb{R}$ $(1 \le i \le p)$, $\theta_j \in \mathbb{R}$ $(1 \le j \le q)$, and let B denote the backshift, $(BX)_t = X_{t-1}$.
Set $\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$ and $\theta(z) := 1 + \sum_{j=1}^{q} \theta_j z^j$; empty sums are zero.
$(X_t)_{t\in\mathbb{Z}}$ is ARMA(p, q) with innovations $(\varepsilon_t)_{t\in\mathbb{Z}}$ iff (ε_t) is white noise and $\phi(B)X_t = \theta(B)\varepsilon_t$ $\forall t \in \mathbb{Z}$.
Structural Commentary: Causal iff $\phi(z) \neq 0$ for $|z| \le 1$; invertible iff $\theta(z) \neq 0$ for $|z| \le 1$.
Key Insight: AR roots control recursion; MA roots control innovation recovery.

Definition 7.2.2. *ARIMA Model*
Let $p, q, d \in \mathbb{N}_0$, $\phi_i \in \mathbb{R}$ $(1 \le i \le p)$, $\theta_j \in \mathbb{R}$ $(1 \le j \le q)$, and let B denote the backshift.
Set $\Delta^d := (1 - B)^d$, $\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$, and $\theta(z) := 1 + \sum_{j=1}^{q} \theta_j z^j$.
$(X_t)_{t\in\mathbb{Z}}$ is ARIMA(p, d, q) with innovations $(\varepsilon_t)_{t\in\mathbb{Z}}$ iff (ε_t) is white noise and $\phi(B)\Delta^d X_t = \theta(B)\varepsilon_t$ $\forall t \in \mathbb{Z}$.
Structural Commentary: $Y_t := \Delta^d X_t$ satisfies ARMA(p, q) with the same innovations.
Key Insight: $d > 0$ shifts stationarity from X_t to $\Delta^d X_t$.

Definition 7.2.3. *Filtering Problem*
Let $(Z_t, Y_t)_{t\ge1}$ be a hidden-state model with state space $(E, \mathcal{E})$, and set $\mathcal{F}_t^Y := \sigma(Y_1, \dots, Y_t)$.
For $A \in \mathcal{E}$, $\pi_t(A) := \mathbb{P}(Z_t \in A \mid \mathcal{F}_t^Y)$; the task is recursive propagation or approximation of $(\pi_t)_{t\ge1}$.
Structural Commentary: (π_t) is an $\mathcal{F}_t^Y$-adapted probability-kernel-valued process.
Key Insight: Conditioning converts hidden dynamics into an observable belief-state recursion.

Definition 7.2.4. *Linear Gaussian State-Space Model*
Let $m, r, n \in \mathbb{N}$, $F \in \mathbb{R}^{m\times m}$, $G \in \mathbb{R}^{m\times r}$, $H \in \mathbb{R}^{n\times m}$, and $Q \in \mathbb{R}^{r\times r}$, $R \in \mathbb{R}^{n\times n}$ with $Q, R \succeq 0$.
Let (η_t) and (ε_t) be mutually independent Gaussian noises with $\eta_t \sim N(0, Q)$ and $\varepsilon_t \sim N(0, R)$.
Let α_0 be independent Gaussian. $(\alpha_t, Y_t)_{t\in\mathbb{N}_0}$ is *linear Gaussian state-space* iff
$\alpha_t \in \mathbb{R}^m$, $Y_t \in \mathbb{R}^n$, $\alpha_{t+1} = F\alpha_t + G\eta_t$, $Y_t = H\alpha_t + \varepsilon_t$ $\forall t \in \mathbb{N}_0$.
Key Insight: With $\mathcal{F}_t^Y := \sigma(Y_0, \dots, Y_t)$, Kalman filtering recurses $(a_t, P_t) := (\mathbb{E}(\alpha_t \mid \mathcal{F}_t^Y), \mathrm{Var}(\alpha_t \mid \mathcal{F}_t^Y))$.

Definition 7.2.5. *ARCH and GARCH Volatility*
Let $p, q \in \mathbb{N}_0$, $\omega > 0$, $\alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p \ge 0$, and let $(z_t)_{t\in\mathbb{Z}}$ be standardized white noise.
All processes below are real-valued and indexed by $\mathbb{Z}$; set $A := \sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$.

(i) *ARCH(q)*: $(X_t, \sigma_t)_{t\in\mathbb{Z}}$, $\sigma_t \ge 0$, satisfies $X_t = \sigma_t z_t$ and $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i X_{t-i}^2$ $\forall t \in \mathbb{Z}$.

(ii) *GARCH(p, q)*: $(X_t, \sigma_t)_{t\in\mathbb{Z}}$, $\sigma_t \ge 0$, satisfies $X_t = \sigma_t z_t$ and $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i X_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$ $\forall t \in \mathbb{Z}$.

Structural Commentary: $\omega > 0$, $\alpha_i, \beta_j \ge 0$ force $\sigma_t^2 \ge 0$; by stationary GARCH, $A < 1$ has $\mathbb{E}X_t^2 = \frac{\omega}{1-A}$.

Remark 7.2.6. *Likelihood by Filtering*
If one-step densities exist, likelihood filtering has $S_t = \mathcal{T}_\vartheta(S_{t-1}, W_t)$ and $\ell_T(\vartheta) = \sum_{t=1}^{T} \log p_\vartheta(W_t \mid S_{t-1})$.
Inference depends on stability of S_t.

Method 7.2.7. *Estimating a Finite Transition Matrix*

(i) Observe finite homogeneous chain $X_0, \dots, X_n$, let $n_{ij} := \sum_{t=0}^{n-1} \mathbf{1}\{X_t = i, X_{t+1} = j\}$, $n_i := \sum_j n_{ij}$.

(ii) The row likelihood is proportional to $\prod_j p_{ij}^{n_{ij}}$ subject to $\sum_j p_{ij} = 1$.

(iii) If $n_i > 0$, the rowwise MLE is $\widehat{p}_{ij} = \frac{n_{ij}}{n_i}$.

(iv) If $n_i = 0$, row i is not identifiable without a convention or prior.

8 Sequential Inference and Resampling

8.1 Resampling, Bootstrap, and Permutation Methods

Resampling methods approximate sampling distributions by recomputing statistics on data-generated pseudo-samples.

Definition 8.1.1. *Nonparametric Bootstrap*
For $X_{1:n} \sim P^{\otimes n}$, set $\widehat{P}_n := n^{-1} \sum_{i=1}^{n} \delta_{X_i}$. Conditional on $X_{1:n}$, let $X_{1:n}^* \sim \widehat{P}_n^{\otimes n}$ and $\widehat{P}_n^* := n^{-1} \sum_{i=1}^{n} \delta_{X_i^*}$.
For any law functional T, $\mathcal{L}(T(\widehat{P}_n^*) - T(\widehat{P}_n) \mid X_{1:n})$ approximates $\mathcal{L}(T(\widehat{P}_n) - T(P))$.
Key Insight: P is replaced by $\widehat{P}_n$ before pseudo-sampling.

Method 8.1.2. *Bootstrap Standard Error*
Let $\widehat{\psi} = s(X_{1:n})$ be observed statistic, $\mathcal{R}_*(X_{1:n})$ be the chosen bootstrap law on pseudo-samples, $B \ge 2$.

(i) *Resample*: $X_{1:n}^{*b} \sim \mathcal{R}_*(X_{1:n})$ for $b = 1, \dots, B$.

(ii) *Recompute*: $\widehat{\psi}^{*b} := s(X_{1:n}^{*b})$ for $b = 1, \dots, B$.

(iii) *Estimate*: $\widehat{\mathrm{se}}_*(\widehat{\psi}) := \{(B-1)^{-1} \sum_{b=1}^{B} (\widehat{\psi}^{*b} - \overline{\psi}^*)^2\}^{\frac{1}{2}}$, where $\overline{\psi}^* := B^{-1} \sum_{b=1}^{B} \widehat{\psi}^{*b}$.

Method 8.1.3. *Bootstrap Variability for Location Summaries*
Let $T = T(x_1, \dots, x_n)$ be a location summary and let $B \geq 1$.

(i) *Resample*: For $1 \leq b \leq B$, draw $x_1^{*(b)}, \dots, x_n^{*(b)}$ iid from $\widehat{F}_n := n^{-1} \sum_{i=1}^{n} \delta_{x_i}$.

(ii) *Recompute*: Set $T_b^* := T(x_1^{*(b)}, \dots, x_n^{*(b)})$ and $\overline{T}^* := B^{-1} \sum_{b=1}^{B} T_b^*$.

(iii) *Spread*: Estimate variability by $\widehat{\mathrm{se}}_*(T) := \left((B-1)^{-1} \sum_{b=1}^{B} (T_b^* - \overline{T}^*)^2 \right)^{1/2}$.

Key Insight: This is useful for medians and trimmed means, where analytic standard errors can be tail-sensitive.

Definition 8.1.4. *Parametric Bootstrap*
Let $\{P_\theta : \theta \in \Theta\}$ be parametric model for $X_{1:n} := (X_1, \dots, X_n)$, let $\widehat{\theta} := \widehat{\theta}(X_{1:n})$, and let S be a statistic.
$X_1^*, \dots, X_n^* \mid X_{1:n} \sim P_{\widehat{\theta}}^{\otimes n}$, and $\mathcal{L}(S(X_{1:n}^*) \mid X_{1:n})$ is the fitted-model resampling law for $S(X_{1:n})$.

Key Insight: The model class $\{P_\theta\}$ is retained; only θ is replaced by $\widehat{\theta}$.

Method 8.1.5. *Parametric Bootstrap Standard Errors and Intervals*
Let $\{P_\theta : \theta \in \Theta\}$ be parametric model, $\widehat{\psi} := s(X_{1:n})$ estimate $\psi(\theta)$, fix $B \geq 2$, fix confidence level $1 - \alpha$.

(i) *Fit*: $\widehat{\theta} := \widehat{\theta}(X_{1:n})$.

(ii) *Simulate*: $X_{1:n}^{*b} \sim P_{\widehat{\theta}}^{\otimes n}$ and $\widehat{\psi}^{*b} := s(X_{1:n}^{*b})$ for $b = 1, \dots, B$.

(iii) *Standard error*: $\widehat{\mathrm{se}}_*(\widehat{\psi}) := \{(B-1)^{-1} \sum_{b=1}^{B} (\widehat{\psi}^{*b} - \overline{\psi}^*)^2\}^{\frac{1}{2}}$, where $\overline{\psi}^* := B^{-1} \sum_{b=1}^{B} \widehat{\psi}^{*b}$.

(iv) *Pivotal interval*: for empirical quantiles $q_{\frac{\alpha}{2}}^*$ and $q_{1-\frac{\alpha}{2}}^*$ of $\{\widehat{\psi}^{*b} - \widehat{\psi}\}_{b=1}^B$, use $[\widehat{\psi} - q_{1-\frac{\alpha}{2}}^*, \widehat{\psi} - q_{\frac{\alpha}{2}}^*]$.

The percentile interval uses empirical quantiles of $\{\widehat{\psi}^{*b}\}_{b=1}^B$ directly.
The pivotal and percentile intervals coincide when the bootstrap error law is symmetric about 0.

Remark 8.1.6. *Plug-In Monte Carlo*
The parametric bootstrap estimates model-dependent quantities by replacing the unknown parameter with its estimate before simulation. If $h(X, \theta)$ is a loss, error indicator, or other accuracy summary, then Monte Carlo from $P_{\widehat{\theta}}$ approximates $\mathbb{E}_\theta h(X, \theta)$ by the empirical average of $h(X^{*b}, \widehat{\theta})$.

Definition 8.1.7. *Permutation Test*
Let a finite group G act on the sample space $\mathcal{X}$, let P_0 satisfy $gX \overset{\text{dist}}{=} X$ under P_0 $\forall g \in G$, and let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic. With U_G uniform on G and independent of X, $\mathcal{L}(T(U_G X) \mid X)$, the empirical law on $\{T(gX) : g \in G\}$, is the reference law for tail probabilities.

Key Insight: Calibration follows from the invariance $gX \overset{\text{dist}}{=} X$ under P_0.

Method 8.1.8. *Permutation Plug-In FDR Estimate*
Let t_j be observed statistics for $j \in [M]$, and let $t_j^{(k)}$ be the statistic after label permutation $k \in [K]$.

(i) *Observed discoveries*: For cutpoint $C > 0$, set $R_{\text{obs}}(C) := \sum_{j=1}^{M} \mathbf{1}_{\{|t_j| > C\}}$.

(ii) *Null discoveries*: Estimate the expected false discoveries by $\widehat{\mathbb{E}}(V; C) := K^{-1} \sum_{k=1}^{K} \sum_{j=1}^{M} \mathbf{1}_{\{|t_j^{(k)}| > C\}}$.

(iii) *Plug-in FDR*: If $R_{\text{obs}}(C) > 0$, set $\widehat{\mathrm{FDR}}(C) := \widehat{\mathbb{E}}(V; C) / R_{\text{obs}}(C)$.

(iv) *Permutation p-values*: Use $\{t_j^{(k)}\}$ to compare $|t_j|$ with featurewise or pooled permutation null statistics.

Remark 8.1.9. *Asymmetric Cutpoints and SAM*
Let $t_{(1)} \leq \cdots \leq t_{(M)}$ and $\bar{t}_{(j)}^0 := K^{-1} \sum_{k=1}^{K} t_{(j)}^{(k)}$, where $t_{(j)}^{(k)}$ is the jth ordered permutation statistic.
For $d_j := t_{(j)} - \bar{t}_{(j)}^0$, $a > 0$, set $c_+(a) := \min\{t_{(j)} : d_j > a\}$, $c_-(a) := \max\{t_{(j)} : d_j < -a\}$ when nonempty.
Reject $t_j \in A(c_-, c_+) := (-\infty, c_-] \cup [c_+, \infty)$, estimate FDR by Method 8.1.8, $A(c_-, c_+)$ replace $\{|t| > C\}$.

Remark 8.1.10. *Validity Depends on the Resampling Law*
Bootstrap validity is asymptotic; permutation validity is finite-sample exact under invariance; both require the resampling law to match the target sampling variation.

8.2 Sequential Experiments, Filtering, and Online Inference

Sequential methods update decisions as observations arrive. They must account for optional stopping, adaptive sampling, and computational constraints.

Definition 8.2.1. *Sequential Test and Stopping Time*
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $I \subseteq \mathbb{N}_0$ be a time index, let $(\mathcal{F}_t)_{t \in I}$ be a filtration, and let $(D, \mathcal{D})$ be a decision space. For any $\tau : \Omega \to I \cup \{\infty\}$, set $\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \ \forall t \in I\}$.

(i) *Stopping time*: $\tau : \Omega \to I \cup \{\infty\}$ satisfies $\{\tau \leq t\} \in \mathcal{F}_t$ $\forall t \in I$.

(ii) *Sequential test*: (τ, d) with τ a stopping time and $d : (\Omega, \mathcal{F}_\tau) \to (D, \mathcal{D})$ measurable.

Key Insight: Stop and decide using only pre-stopping information.

Definition 8.2.2. *Sequential Probability Ratio Test*
Let P_0 and P_1 be simple iid models with one-observation densities p_0 and p_1 for $(X_t)_{t \geq 1}$,
let $\Lambda_0 := 1$, $\Lambda_t := \prod_{i=1}^{t} \frac{p_1(X_i)}{p_0(X_i)}$, and choose $0 < A < B < \infty$.
$\tau := \inf\{t \geq 1 : \Lambda_t \in (0, A] \cup [B, \infty)\}$, with terminal decision $d_\tau = P_0$ on $\{\Lambda_\tau \leq A\}$ and $d_\tau = P_1$ on $\{\Lambda_\tau \geq B\}$.
Key Insight: First exit of $\log \Lambda_t$ from $(\log A, \log B)$ converts cumulative likelihood evidence into a stopping rule.

Remark 8.2.3. *Sequential Cross-References*
Sequential inference uses filtrations and martingales from stochastic processes, stability of recursions from dynamical systems, and simulation methods from Monte Carlo computation.

8.3 Two-Sample Resampling Methods

Two-sample resampling procedures must preserve the source of randomness relevant to the question. Bootstrap procedures usually resample within groups to estimate sampling variability under the fitted empirical distributions; permutation procedures pool labels when the null asserts exchangeability.

Method 8.3.1. *Two-Sample Nonparametric Bootstrap*

Let $x_{1:n}$ and $y_{1:m}$ be independent samples from F and G, let $\widehat{F}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i}$ and $\widehat{G}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j}$.

Let $T(F,G)$ be a two-sample functional, and fix $B \geq 2$.

(i) *Resample*: $x^{*b}_{1:n} \sim \widehat{F}_n^{\otimes n}$ and $y^{*b}_{1:m} \sim \widehat{G}_m^{\otimes m}$ independently for $b = 1,\dots,B$.

(ii) *Recompute*: with $\widehat{F}^{*b}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i^{*b}}$ and $\widehat{G}^{*b}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j^{*b}}$, set $\widehat{T}^{*b} := T(\widehat{F}^{*b}_n, \widehat{G}^{*b}_m)$.

(iii) *Summarize*: $s^*_T := \mathrm{sd}\{\widehat{T}^{*b} : 1 \leq b \leq B\}$; $I^*_{1-\alpha} := [q^*_{\frac{\alpha}{2}}, q^*_{1-\frac{\alpha}{2}}]$, where q^*_γ is the empirical γ-quantile.

This is the two-sample version of Method 8.1.2.

Example 8.3.2. *Bootstrap for $\pi = \mathbb{P}(X < Y)$*
Let $x_{1:n} \sim F^{\otimes n}$ and $y_{1:m} \sim G^{\otimes m}$ independently.
Set $\widehat{F}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i}$, $\widehat{G}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j}$, and $\widehat{\pi} := (mn)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}_{\{x_i<y_j\}}$.
For $b = 1,\dots,B$, draw $x^{*b}_{1:n} \sim \widehat{F}_n^{\otimes n}$ and $y^{*b}_{1:m} \sim \widehat{G}_m^{\otimes m}$ independently.
Set $\widehat{\pi}^{*b} := (mn)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}_{\{x_i^{*b}<y_j^{*b}\}}$, and use $s^*_\pi := \mathrm{sd}\{\widehat{\pi}^{*b} : 1 \leq b \leq B\}$.
Key Insight: Separate resampling targets $F \otimes G$ variation; rank-sum calibration imposes $F = G$.

Method 8.3.3. *Bootstrap for a Hodges–Lehmann Shift*
For independent samples $x_{1:n}$ and $y_{1:m}$ and $B \geq 2$, set $\widehat{\Delta}_{\mathrm{HL}} := \mathrm{median}\{y_j - x_i : 1 \leq i \leq n,\ 1 \leq j \leq m\}$, resample the two groups separately for $b = 1,\dots,B$, and use the empirical spread or quantiles of the recomputed medians $\widehat{\Delta}^{*b}_{\mathrm{HL}}$ for a standard error or interval.

Method 8.3.4. *Permutation Test for a Difference of Means*

Let $x_{1:n}$ be control responses, let $y_{1:m}$ be treatment responses, set $\overline{x} := n^{-1}\sum_{i=1}^{n}x_i$ and $\overline{y} := m^{-1}\sum_{j=1}^{m}y_j$.

Let $z_{1:N}$ with $N = n + m$ be the pooled responses, and set $T_{\mathrm{obs}} := \overline{y} - \overline{x}$.
Let H_0 denote no treatment effect, set $\mathcal{S}_{m,N} := \{S \subseteq \{1,\dots,N\} : |S| = m\}$, and use $S \mid H_0 \sim \mathrm{Unif}(\mathcal{S}_{m,N})$.

(i) *Relabel*: $S \in \mathcal{S}_{m,N}$, by enumeration or Monte Carlo draws from $\mathrm{Unif}(\mathcal{S}_{m,N})$.

(ii) *Recompute*: $T^*_S := m^{-1}\sum_{i\in S}z_i - n^{-1}\sum_{i\notin S}z_i$.

(iii) *Calibrate*: $p_{\mathrm{exact}} := \binom{N}{m}^{-1}\#\{S \in \mathcal{S}_{m,N} : |T^*_S| \geq |T_{\mathrm{obs}}|\}$, Monte Carlo approximation if $\binom{N}{m}$ large.

Remark 8.3.5. *Permutation and Rank Tests*
The permutation test for means and the Mann–Whitney test both use the randomization principle that, under the null, labels are exchangeable. They differ in the statistic: the permutation mean test uses the numerical values and is sensitive to outliers, while the Mann–Whitney test first replaces values by ranks.

Part XX
Statistical Data Analysis and Applied Models

Applied statistical analysis organizes data summaries, sampling designs, contingency tables, designed experiments, and regression models into reusable workflows. The inferential tools of Part 15 supply the testing, estimation, and uncertainty language used here.

1 Survey Sampling

1.1 Finite Population Parameters and Simple Random Sampling

Survey sampling treats a finite population as fixed and the sample-selection mechanism as random. The inferential target is often a population mean, total, or proportion, while the random variables come from the sampling design.

Definition 1.1.1. *Finite Population Parameters*
Let $U := \{1, \dots, N\}$ and fix $x_i \in \mathbb{R}$, $i \in U$.

(i) *Population mean*: $\mu := \frac{1}{N} \sum_{i=1}^{N} x_i$.

(ii) *Population total*: $\tau := \sum_{i=1}^{N} x_i = N\mu$.

(iii) *Population variance*: $\sigma^2 := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^{N} x_i^2 - \mu^2$.

(iv) *Population proportion*: $\{x_i : i \in U\} \subseteq \{0, 1\} \implies p := \mu = \frac{1}{N} \sum_{i=1}^{N} x_i$ and $\sigma^2 = p(1-p)$.

Definition 1.1.2. *Simple Random Sampling*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, and fix $x_i \in \mathbb{R}$, $i \in U$.
Let S be a random n-element subset of U; for a random ordering $J_1, \dots, J_n$ of S, set $X_k := x_{J_k}$.

(i) *Simple random sample*: $\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}}$, $\forall s \subseteq U$ with $|s| = n$.

(ii) *Sample mean*: $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$.

(iii) *Total estimator*: $T := N\overline{X}$.

(iv) *Sample proportion*: $\{x_i : i \in U\} \subseteq \{0, 1\} \implies \widehat{p} := \overline{X}$.

Method 1.1.3. *Sequential Simple Random Sampling*
Fix target size $n \ge 1$, assume terminal stream length is $N \ge n$, N unknown while streaming.
Population units arrive sequentially as $1, \dots, N$. Let $S_t \subseteq \{1, \dots, t\}$ denote current reservoir after t arrivals.

(i) *Initialize*: for $1 \le t \le n$, set $S_t = \{1, \dots, t\}$.
(ii) *Admit*: for $t > n$, draw $B_t \sim \text{Bernoulli}(\frac{n}{t})$.
(iii) *Update*: if $B_t = 0$, set $S_t = S_{t-1}$; if $B_t = 1$, draw $J_t \mid S_{t-1} \sim \text{Unif}(S_{t-1})$ and set $S_t = (S_{t-1} \setminus \{J_t\}) \cup \{t\}$.
(iv) *Output*: after the stream ends, return S_N.

Then for every $t \ge n$, $S_t \sim \text{Unif}\{s \subseteq \{1, \dots, t\} : |s| = n\}$.
Hence S_N is a simple random sample without replacement.

Example 1.1.4. *Capture–Recapture Likelihood*
Suppose t population units are tagged and a later sample of size m contains $r > 0$ tagged units.
For population size n, likelihood is $L_n = \frac{\binom{t}{r}\binom{n-t}{m-r}}{\binom{n}{m}}$. Since $\frac{L_n}{L_{n-1}} = \frac{(n-t)(n-m)}{n(n-t-m+r)}$, L_n increases for $n < \frac{mt}{r}$ and decreases for $n > \frac{mt}{r}$.
Hence $\widehat{n} = \lfloor \frac{mt}{r} \rfloor$; for $(t, m, r) = (10, 20, 4)$, $\widehat{n} = 50$.

Lemma 1.1.5. *Law of One Sample Position*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Let $\mu := \frac{1}{N} \sum_{a=1}^{N} x_a$, $\sigma^2 := \frac{1}{N} \sum_{a=1}^{N} (x_a - \mu)^2$, $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.
If $(J_1, \dots, J_n)$ is a random ordering of S, set $X_k := x_{J_k}$.
Let $\zeta_1, \dots, \zeta_m$ be distinct values in $\{x_a : a \in U\}$, let $n_j := \#\{a \in U : x_a = \zeta_j\}$. For each $k \in \{1, \dots, n\}$:

(i) Distribution: the selected value X_k has the finite-population law $\mathbb{P}(X_k = \zeta_j) = \frac{n_j}{N}$ for $j = 1, \dots, m$.
(ii) Moments: the finite-population law has mean $\mathbb{E}X_k = \mu$ and variance $\text{Var}(X_k) = \sigma^2$.

Proof.

(i) For every $a \in U$, $\mathbb{P}(J_k = a) = \frac{1}{N}$, so $\mathbb{P}(X_k = \zeta_j) = \sum_{a : x_a = \zeta_j} \mathbb{P}(J_k = a) = \frac{n_j}{N}$.

(ii) From (i), $\mathbb{E}X_k = \frac{1}{N} \sum_{a=1}^{N} x_a = \mu$. By finite-population law, $\text{Var}(X_k) = \frac{1}{N} \sum_{a=1}^{N} (x_a - \mu)^2 = \sigma^2$.

Structural Commentary: Each sample position is marginally uniform on U.
Key Insight: Without replacement changes joint dependence, not the one-position marginal distribution. □

Theorem 1.1.6. *Unbiased SRS Estimators*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N} \sum_{a=1}^{N} x_a$, $\tau := \sum_{a=1}^{N} x_a = N\mu$. Let $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.
Given random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n} \sum_{k=1}^{n} X_k$, and $T := N\overline{X}$. Then:

(i) Mean estimator: the sample mean is design-unbiased for the finite-population mean, $\mathbb{E}\overline{X} = \mu$.

(ii) Total estimator: the expansion estimator is design-unbiased for the finite-population total, $\mathbb{E}T = \tau$.

Proof.

(i) For each k, $\mathbb{P}(J_k = a) = \frac{1}{N}\ \forall a \in U$, so $\mathbb{E}X_k = \frac{1}{N}\sum_{a=1}^{N} x_a = \mu$. By linearity $\mathbb{E}\overline{X} = \frac{1}{n}\sum_{k=1}^{n}\mathbb{E}X_k = \mu$.

(ii) Since $T = N\overline{X}$, $\mathbb{E}T = N\mathbb{E}\overline{X} = N\mu = \tau$.

Structural Commentary: $T = N\overline{X}$ rescales the mean estimate.
Key Insight: Equal position marginals give unbiasedness; independence is unnecessary. □

Lemma 1.1.7. *Negative Dependence Without Replacement*
Let $U := \{1, \dots, N\}$, $2 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\tau := \sum_{a=1}^{N} x_a$, $\sigma^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu)^2$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
For random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$. If $i, j \in \{1, \dots, n\}$ and $i \ne j$, then $\mathrm{Cov}(X_i, X_j) = -\frac{\sigma^2}{N-1}$.

Proof. Let $(Y_1, \dots, Y_N)$ be a uniformly random ordering of $(x_1, \dots, x_N)$.
Each Y_a has variance σ^2, and symmetry gives $\mathrm{Cov}(Y_a, Y_b) = \gamma$ for $a \ne b$.
Since $\sum_{a=1}^{N} Y_a = \tau$ is fixed, $0 = \mathrm{Var}(\sum_{a=1}^{N} Y_a) = N\sigma^2 + N(N-1)\gamma$.
Thus $\gamma = -\frac{\sigma^2}{N-1}$, and (X_i, X_j) has same law as any two distinct positions in full random ordering.
Structural Commentary: A fixed total forces negative pairwise covariance.
Key Insight: Without replacement, one large draw leaves less for the rest. □

Theorem 1.1.8. *Variance and Finite Population Correction*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N}\sum_{a=1}^{N} x_a$ and $\sigma^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu)^2$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, and $T := N\overline{X}$. Then:

(i) Mean variance: the sample mean has design variance $\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right)$.
(ii) Total variance: the total estimator has design variance $\mathrm{Var}(T) = N^2\,\mathrm{Var}(\overline{X})$.

Proof.

(i) Each X_k has variance σ^2. Let $(Y_1, \dots, Y_N)$ be a uniformly random ordering of $(x_1, \dots, x_N)$, set $\tau := \sum_{a=1}^{N} x_a$, and let γ be the common covariance of Y_a, Y_b for $a \ne b$.
Since $\sum_{a=1}^{N} Y_a = \tau$, $0 = \mathrm{Var}(\sum_{a=1}^{N} Y_a) = N\sigma^2 + N(N-1)\gamma$, so $\gamma = -\frac{\sigma^2}{N-1}$.
Therefore $\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\left(n\sigma^2 + n(n-1)\left[-\frac{\sigma^2}{N-1}\right]\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right)$.

(ii) The total estimator satisfies $T = N\overline{X}$, so $\mathrm{Var}(T) = \mathrm{Var}(N\overline{X}) = N^2\,\mathrm{Var}(\overline{X})$.

Structural Commentary: The factor $\frac{N-n}{N-1}$ is the *finite population correction*.
Key Insight: Averaging gives $\frac{1}{n}$ scaling; without replacement subtracts negative covariance. □

Corollary 1.1.9. *Estimated Standard Errors*
Let $U := \{1, \dots, N\}$, $1 < n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$, and let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$. For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $T := N\overline{X}$, and $s^2 := \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \overline{X})^2$.
Then practical plug-in estimated variances are:

(i) Mean: the plug-in estimated variance of the sample mean is $s^2_{\overline{X}} = \frac{s^2}{n}\left(1 - \frac{n}{N}\right)$.
(ii) Total: the plug-in estimated variance of the total estimator is $s^2_T = N^2 s^2_{\overline{X}}$.
(iii) Proportion: if $x_a \in \{0, 1\}\ \forall a \in U$ and $\widehat{p} := \overline{X}$, then plug-in estimated variance of $\widehat{p}$ is $s^2_{\widehat{p}} = \frac{\widehat{p}(1-\widehat{p})}{n-1}\left(1 - \frac{n}{N}\right)$.

Proof. Set $c_N := 1 - \frac{n}{N}$.

(i) Replace unknown finite-population variance by s^2, correction by c_N gives $s^2_{\overline{X}} = (\frac{s^2}{n})c_N = \frac{s^2}{n}(1 - \frac{n}{N})$.

(ii) Since $T = N\overline{X}$, variance scales quadratically, so $s^2_T = N^2 s^2_{\overline{X}}$.

(iii) If $X_k \in \{0, 1\}$ and $\widehat{p} = \overline{X}$, then $s^2 = \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \widehat{p})^2 = \frac{1}{n-1}(n\widehat{p} - n\widehat{p}^2) = \frac{n\widehat{p}(1-\widehat{p})}{n-1}$.
Substituting this into (i) gives $s^2_{\widehat{p}} = \frac{\widehat{p}(1-\widehat{p})c_N}{n-1} = \frac{\widehat{p}(1-\widehat{p})}{n-1}(1 - \frac{n}{N})$.

Structural Commentary: Plug-in SEs use s^2 and the $1 - \frac{n}{N}$ correction.
Key Insight: Binary-data variance depends only on $\widehat{p}$. □

Example 1.1.10. *Finite Population Correction in a Hospital Survey*
For a finite population of $N = 393$ hospitals, let x_i denote discharges and suppose the finite-population standard deviation is $\sigma \approx 589.7$. Under simple random sampling without replacement with $n = 32$, the sample-mean standard deviation is $\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}\sqrt{1 - \frac{n-1}{N-1}} \approx \frac{589.7}{\sqrt{32}}\sqrt{1 - \frac{31}{392}} \approx 100.0$. Since $\frac{n}{N} = \frac{32}{393}$ is small, the finite-population correction is near 1, and the standard error is mainly governed by the $\frac{1}{\sqrt{n}}$ averaging scale.

1.2 Normal Approximation and Confidence Intervals

The exact design formulas in Theorem 1.1.8 give the center and scale of the sampling distribution. For large samples, a normal approximation turns those formulas into error probabilities and confidence intervals.

Method 1.2.1. *Normal Approximation for SRS*

(i) Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, $x_i \in \mathbb{R}$ for $i \in U$, and $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.

(ii) For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := n^{-1} \sum_{k=1}^{n} X_k$, $T := N\overline{X}$, $\mu := N^{-1} \sum_{i=1}^{N} x_i$, $\sigma^2 := N^{-1} \sum_{i=1}^{N} (x_i - \mu)^2$, $\tau := N\mu$, and $\sigma^2_{\overline{X}} := \frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)$.

(iii) Let Φ be the standard normal cdf. For large n, use $\overline{X} \approx N(\mu, \sigma^2_{\overline{X}})$.

(iv) If $\sigma_{\overline{X}} > 0$, then $\mathbb{P}(|\overline{X} - \mu| \le \delta) \approx 2\Phi(\frac{\delta}{\sigma_{\overline{X}}}) - 1$ for $\delta > 0$, and $T - \tau = N(\overline{X} - \mu) \approx N(0, N^2\sigma^2_{\overline{X}})$.

Method 1.2.2. *Approximate SRS Confidence Intervals*

(i) Let S be an SRS without replacement, with targets μ, τ and estimators $\overline{X}, T$.

(ii) Fix $0 < \alpha < 1$, let Φ be the standard normal cdf, set $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$, and use plug-in SEs $s_{\overline{X}}, s_T$, and, when binary, $s_{\widehat{p}}$.

(iii) $\text{CI}_{1-\alpha}(\mu) = \overline{X} \pm z(\frac{\alpha}{2}) s_{\overline{X}}$ and $\text{CI}_{1-\alpha}(\tau) = T \pm z(\frac{\alpha}{2}) s_T$.

(iv) In the dichotomous case, with population proportion p and sample proportion $\widehat{p}$, use $\text{CI}_{1-\alpha}(p) = \widehat{p} \pm z(\frac{\alpha}{2}) s_{\widehat{p}}$.

(v) Intervals are random before sampling; after observation, parameters are fixed and coverage is procedural.

Example 1.2.3. *Generic SRS Intervals*

Suppose finite population has $N = 8000$ units. Simple random sample of $n = 100$ gives $\overline{X} = 1.6$ for variable with sample standard deviation $s = 0.8$. Use $z(0.025) \approx 1.96$, suppose sample gives $\widehat{p} = 0.12$ for binary attribute.

(i) *Mean*: $s_{\overline{X}} = \frac{s}{\sqrt{n}} \sqrt{1 - \frac{n}{N}} = \frac{0.8}{10} \sqrt{1 - \frac{100}{8000}} \approx 0.0795$, then 95% CI for μ is $1.6 \pm 1.96(0.0795) = (1.44, 1.76)$.

(ii) *Total*: $T = N\overline{X} = 8000(1.6) = 12800$ and $s_T = N s_{\overline{X}} \approx 636$, then 95% CI for τ is $(11553, 14047)$.

(iii) *Proportion*: $s_{\widehat{p}} = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n-1} \left(1 - \frac{n}{N}\right)} = \sqrt{\frac{0.12(0.88)}{99} \left(1 - \frac{100}{8000}\right)} \approx 0.0325$, then 95% CI for p is $(0.056, 0.184)$.
The estimated attribute total is $N\widehat{p} = 960$ with $N s_{\widehat{p}} \approx 260$ and interval $(451, 1469)$.

These intervals quantify SRS uncertainty for fixed finite-population quantities.

Method 1.2.4. *Sample Size from Desired Half-Width*

(i) Fix $0 < \alpha < 1$, half-width $h > 0$, planning SD $\sigma > 0$, and planned size n.
Let Φ be the standard normal cdf and set $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$.

(ii) Ignoring the finite population correction, $h \approx z(\frac{\alpha}{2}) \frac{\sigma}{\sqrt{n}}$ gives $n \approx \left(\frac{z(\frac{\alpha}{2})\sigma}{h}\right)^2$.

(iii) For total based on population proportion over N units, use planning value $p_0 \in [0, 1]$, SE $N\sqrt{\frac{p_0(1-p_0)}{n}}$.

(iv) With negligible finite population correction, halving interval width requires roughly $4n$ observations.

1.3 Ratio Estimation

Ratio estimation uses an auxiliary variable whose population mean or total is known. It is especially useful when the auxiliary variable is strongly correlated with the target variable.

Definition 1.3.1. *Population and Sample Ratios*

Let $U := \{1, \dots, N\}$ and fixed pairs $(x_i, y_i) \in \mathbb{R}^2$, $i \in U$.

Set $\mu_x := \frac{1}{N} \sum_{i=1}^{N} x_i$, $\mu_y := \frac{1}{N} \sum_{i=1}^{N} y_i$, $\tau_x := \sum_{i=1}^{N} x_i$, and $\tau_y := \sum_{i=1}^{N} y_i$, with $\mu_x \neq 0$.

For SRS pairs $(X_i, Y_i)_{i=1}^n$, set $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$ and $\overline{Y} := \frac{1}{n} \sum_{i=1}^{n} Y_i$, with $\overline{X} \neq 0$.

(i) *Population ratio*: $r := \sum_{i=1}^{N} y_i / \sum_{i=1}^{N} x_i = \frac{\mu_y}{\mu_x}$.

(ii) *Sample ratio*: $R := \frac{\overline{Y}}{\overline{X}}$.

(iii) *Ratio mean estimator*: $\overline{Y}_R := \mu_x R$ when μ_x is known.

(iv) *Ratio total estimator*: $T_R := \tau_x R$ when τ_x is known.

Key Insight: $\{i \in U : x_i = 0\} = \emptyset$ does not imply $r = \frac{1}{N} \sum_{i=1}^{N} \frac{y_i}{x_i}$.

Definition 1.3.2. *Population Covariance and Correlation*

Let $U := \{1, \dots, N\}$ and fixed pairs $(x_i, y_i) \in \mathbb{R}^2$, $i \in U$.

Set $\mu_x := \frac{1}{N} \sum_{i=1}^{N} x_i$, $\mu_y := \frac{1}{N} \sum_{i=1}^{N} y_i$, $\sigma_x^2 := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_x)^2$, and $\sigma_y^2 := \frac{1}{N} \sum_{i=1}^{N} (y_i - \mu_y)^2$.

For sample pairs $(X_i, Y_i)_{i=1}^n$, set $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$ and $\overline{Y} := \frac{1}{n} \sum_{i=1}^{n} Y_i$.

(i) *Population covariance*: $\sigma_{xy} := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_x)(y_i - \mu_y)$.

(ii) *Population correlation*: $\rho := \frac{\sigma_{xy}}{\sigma_x \sigma_y}$ when $\sigma_x, \sigma_y > 0$.

(iii) *Sample covariance*: $s_{xy} := \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})$.

Theorem 1.3.3. *Approximate Moments of the Ratio Estimator*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, *and let* $(x_a, y_a) \in \mathbb{R}^2$ *be fixed for* $a \in U$.
Set $\lambda_n := \frac{1}{n}(1 - \frac{n-1}{N-1})$, $\mu_x := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\mu_y := \frac{1}{N}\sum_{a=1}^{N} y_a$, $r := \frac{\mu_y}{\mu_x}$
with $\mu_x \neq 0$, $\sigma_x^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu_x)^2$, $\sigma_y^2 := \frac{1}{N}\sum_{a=1}^{N}(y_a - \mu_y)^2$, *and* $\sigma_{xy} := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu_x)(y_a - \mu_y)$.
Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$, *let* $(J_1, \dots, J_n)$ *be a random ordering of* S, *set* $(X_k, Y_k) := (x_{J_k}, y_{J_k})$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $\overline{Y} := \frac{1}{n}\sum_{k=1}^{n} Y_k$,
and $R := \frac{\overline{Y}}{\overline{X}}$ *on* $\{\overline{X} \neq 0\}$. *The subscript* d *denotes design moments.*

(i) Variance: first-order Taylor gives $\mathrm{Var}_d(R) \approx \frac{\lambda_n}{\mu_x^2}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$.

(ii) Mean: second-order Taylor gives $\mathbb{E}_d R \approx r + \frac{\lambda_n}{\mu_x^2}(r\sigma_x^2 - \sigma_{xy})$.

Proof. Let $(A_q, B_q)_{q=1}^N$ be a uniformly random ordering of the population pairs; the ordered sample has the law of the first n entries. For $q \neq r$, write $\gamma_x := \mathrm{Cov}_d(A_q, A_r)$, $\gamma_y := \mathrm{Cov}_d(B_q, B_r)$, and $\gamma_{xy} := \mathrm{Cov}_d(A_q, B_r)$.
The fixed sums give $0 = \mathrm{Var}_d(\sum_{q=1}^{N} A_q) = N\sigma_x^2 + N(N-1)\gamma_x$, $0 = \mathrm{Var}_d(\sum_{q=1}^{N} B_q) = N\sigma_y^2 + N(N-1)\gamma_y$,
and $0 = \mathrm{Cov}_d(\sum_{q=1}^{N} A_q, \sum_{q=1}^{N} B_q) = N\sigma_{xy} + N(N-1)\gamma_{xy}$.
Hence $\gamma_x = -\frac{\sigma_x^2}{N-1}$, $\gamma_y = -\frac{\sigma_y^2}{N-1}$, $\gamma_{xy} = -\frac{\sigma_{xy}}{N-1}$, $V_x := \mathrm{Var}_d(\overline{X}) = \lambda_n\sigma_x^2$, $V_y := \mathrm{Var}_d(\overline{Y}) = \lambda_n\sigma_y^2$,
and $C_{xy} := \mathrm{Cov}_d(\overline{X}, \overline{Y}) = \lambda_n\sigma_{xy}$. Let $\Sigma := \begin{pmatrix} V_x & C_{xy} \\ C_{xy} & V_y \end{pmatrix}$.
For $g(x, y) = \frac{y}{x}$ and $\mu := (\mu_x, \mu_y)$, $g(\mu) = r$, $\nabla g(\mu) = (-\frac{r}{\mu_x}, \frac{1}{\mu_x})^\top$, and $H_g(\mu) = \begin{pmatrix} 2r/\mu_x^2 & -1/\mu_x^2 \\ -1/\mu_x^2 & 0 \end{pmatrix}$.

(i) *Variance*: $\mathrm{Var}_d(R) \approx \nabla g(\mu)^\top \Sigma \nabla g(\mu) = \frac{\lambda_n}{\mu_x^2}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$.

(ii) *Mean*: $\mathbb{E}_d R \approx g(\mu) + \frac{1}{2}\operatorname{tr}(H_g(\mu)\Sigma) = r + \frac{\lambda_n}{\mu_x^2}(r\sigma_x^2 - \sigma_{xy})$.

Structural Commentary: The ratio estimator is locally linear for variance and locally quadratic for bias.
Key Insight: Ratio variance is scaled residual variation in $Y - rX$. □

Corollary 1.3.4. *Estimated Standard Error of a Ratio*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 < n \le N$, *and let* $(x_a, y_a) \in \mathbb{R}^2$ *be fixed for* $a \in U$.
Set $\lambda_n := \frac{1}{n}(1 - \frac{n-1}{N-1})$, $\mu_x := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\tau_x := \sum_{a=1}^{N} x_a$, $\tau_y := \sum_{a=1}^{N} y_a$, *and* $r := \frac{\tau_y}{\tau_x}$ *with* $\tau_x \neq 0$.
Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$. *For a random ordering* $(J_1, \dots, J_n)$ *of* S,
set $(X_i, Y_i) := (x_{J_i}, y_{J_i})$, $\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$, *and* $\overline{Y} := \frac{1}{n}\sum_{i=1}^{n} Y_i$.
Assume $\overline{X} \neq 0$ *and set* $R := \frac{\overline{Y}}{\overline{X}}$, $s_x^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$, $s_y^2 := \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2$, $s_{xy} := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$, *and* $q_R := R^2 s_x^2 + s_y^2 - 2Rs_{xy}$. *If* $0 < \alpha < 1$ *and* $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$ *for the standard normal cdf* Φ, *then:*

(i) Ratio variance: $s_R^2 := \frac{\lambda_n}{\overline{X}^2} q_R$ *estimates* $\mathrm{Var}_d(R)$.

(ii) Ratio interval: $\mathrm{CI}_{1-\alpha}(r) := R \pm z(\frac{\alpha}{2}) s_R$.

(iii) Mean and total: for $\overline{Y}_R := \mu_x R$ *and* $T_R := \tau_x R$, $\mathrm{SE}(\overline{Y}_R) = |\mu_x|\, s_R$ *and* $\mathrm{SE}(T_R) = |\tau_x|\, s_R$ *when the corresponding auxiliary constant is known.*

Proof.

(i) The plug-in map $(\mu_x, r, \sigma_x^2, \sigma_y^2, \sigma_{xy}) \mapsto (\overline{X}, R, s_x^2, s_y^2, s_{xy})$ in Theorem 1.3.3 yields $\widehat{\mathrm{Var}}_d(R) = \frac{\lambda_n}{\overline{X}^2}(R^2 s_x^2 + s_y^2 - 2Rs_{xy}) = s_R^2$.

(ii) If $Z_R := (R - r)/s_R \approx \mathcal{N}(0, 1)$, then $\mathbb{P}_d(|Z_R| \le z(\frac{\alpha}{2})) \approx 1 - \alpha$, equivalently $r \in R \pm z(\frac{\alpha}{2}) s_R$.

(iii) For known c, $\widehat{\mathrm{Var}}_d(cR) = c^2 s_R^2$, so $\mathrm{SE}(cR) = |c|\, s_R$; set $c = \mu_x$ and $c = \tau_x$.

Structural Commentary: Plug in sample moments to estimate the ratio variance.
Key Insight: Ratio uncertainty is residual uncertainty in $Y - RX$. □

Method 1.3.5. *When the Ratio Estimate Helps*
Let $U = \{1, \dots, N\}$ contain fixed pairs (x_i, y_i) with $\mu_x, \mu_y > 0$, $\sigma_x, \sigma_y > 0$, σ_{xy}, $\rho := \frac{\sigma_{xy}}{\sigma_x \sigma_y}$, and $r := \frac{\mu_y}{\mu_x}$.
For an SRS $(X_k, Y_k)_{k=1}^n$, set $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $\overline{Y} := \frac{1}{n}\sum_{k=1}^{n} Y_k$, $R := \frac{\overline{Y}}{\overline{X}}$, and $\overline{Y}_R := \mu_x R$.
Ignoring finite-population corrections:

(i) *Variances*: $V_0 := \mathrm{Var}(\overline{Y}) = \frac{\sigma_y^2}{n}$ and $V_R^* := \frac{1}{n}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$ approximate $\mathrm{Var}(\overline{Y}_R)$.

(ii) *Improvement*: $V_R^* < V_0 \iff 2\sigma_{xy} > r\sigma_x^2 \iff 2\rho\sigma_y > r\sigma_x$.

(iii) *Coefficient form*: with $C_x := \frac{\sigma_x}{\mu_x}$ and $C_y := \frac{\sigma_y}{\mu_y}$, $V_R^* < V_0 \iff \rho > \frac{1}{2}\frac{C_x}{C_y}$.

(iv) *Favorable case*: large positive ρ and small C_x/C_y.

Example 1.3.6. *Mortgage-Payment Ratio*
For an SRS of $n = 100$ recent house buyers, let Y_i be monthly mortgage payment and X_i gross income.
Suppose $\overline{X} = 3100$, $\overline{Y} = 868$, $s_x = 1200$, $s_y = 250$, and $\widehat{\rho} := s_{xy}/(s_x s_y) = 0.85$.

(i) *Ratio*: $R = \frac{\overline{Y}}{\overline{X}} = \frac{868}{3100} = 0.28$.

(ii) *Variance input*: $q := R^2 s_x^2 + s_y^2 - 2R\widehat{\rho} s_x s_y \approx 32596$.

(iii) *Standard error*: ignoring the finite-population correction, $s_R \approx \frac{\sqrt{q}}{\sqrt{100}\,\overline{X}} \approx 0.00582$.
(iv) *Interval*: $\mathrm{CI}_{0.95}(r) = R \pm 1.96 s_R \approx 0.28 \pm 1.96(0.00582) = (0.269, 0.291)$.

Thus the population mortgage-payment-to-income ratio is about 28%.

Remark 1.3.7. *Bias and Resampling Corrections*
For SRS from fixed pairs (x_i, y_i), set $R := \overline{Y}/\overline{X}$, $\overline{Y}_R := \mu_x R$, and $T_R := \tau_x R$.
(i) Since $g(x,y) = y/x$ is nonlinear, generally $\mathbb{E}_d R \neq r$, hence $\mathbb{E}_d \overline{Y}_R \neq \mu_y$ and $\mathbb{E}_d T_R \neq \tau_y$.
(ii) Bias corrections target $B_d(R) := \mathbb{E}_d R - r$; use resampling when Taylor bias matters.

1.4 Stratified Random Sampling

Stratified sampling partitions the population into subpopulations and samples within each one. It can guarantee representation of important groups and can reduce variance when strata are internally homogeneous or have different means.

Definition 1.4.1. *Strata and the Stratified Mean*

Let U be a finite population, $L \geq 1$, nonempty subsets $U_1, \ldots, U_L \subseteq U$, $N_l := |U_l|$, $N := \sum_{l=1}^{L} N_l$,
and fixed values $x_{il} \in \mathbb{R}$ for $i = 1, \ldots, N_l$, $l = 1, \ldots, L$.
For stratum samples of sizes n_l, write sampled values as X_{il}, $i = 1, \ldots, n_l$.

(i) *Strata*: $U = \bigsqcup_{l=1}^{L} U_l$.
(ii) *Stratum weight*: $W_l := \frac{N_l}{N}$.
(iii) *Stratum mean*: $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$.
(iv) *Stratum variance*: $\sigma_l^2 := \frac{1}{N_l} \sum_{i=1}^{N_l} (x_{il} - \mu_l)^2$.
(v) *Stratum sample mean*: $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$.
(vi) *Stratified mean*: $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$.

Structural Commentary: With $\mu := \frac{1}{N} \sum_{l=1}^{L} \sum_{i=1}^{N_l} x_{il}$, $\mu = \sum_{l=1}^{L} W_l \mu_l$.

Theorem 1.4.2. *Mean and Variance of the Stratified Estimate*

Let $L \geq 1$, $N_l > 1$, $1 \leq n_l \leq N_l$, $N := \sum_{l=1}^{L} N_l$, $W_l := \frac{N_l}{N}$, and $\lambda_l := \frac{1}{n_l}(1 - \frac{n_l - 1}{N_l - 1})$.

For fixed $x_{il} \in \mathbb{R}$, $i = 1, \ldots, N_l$, $l = 1, \ldots, L$, set $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$, $\sigma_l^2 := \frac{1}{N_l} \sum_{i=1}^{N_l} (x_{il} - \mu_l)^2$, and $\mu := \sum_{l=1}^{L} W_l \mu_l$.

Independently sample $S_l \sim \mathrm{Unif}\{s \subseteq \{1, \ldots, N_l\} : |s| = n_l\}$; for random orderings $(J_{il})_{i=1}^{n_l}$ of S_l, set $X_{il} := x_{J_{il}, l}$, $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$,

and $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$. Let $\mathbb{E}_d, \mathrm{Var}_d$ denote design moments.

(i) Mean: $\mathbb{E}_d \overline{X}_s = \mu$.
(ii) Variance: $\mathrm{Var}_d(\overline{X}_s) = \sum_{l=1}^{L} W_l^2 \lambda_l \sigma_l^2$.
(iii) Small fractions: if $n_l/N_l \ll 1$ for all l, then $\mathrm{Var}_d(\overline{X}_s) \approx \sum_{l=1}^{L} W_l^2 \frac{\sigma_l^2}{n_l}$.

Proof.

(i) For $a \in \{1, \ldots, N_l\}$, $\mathbb{P}_d(J_{il} = a) = 1/N_l$, hence $\mathbb{E}_d X_{il} = \mu_l$, $\mathbb{E}_d \overline{X}_l = \mu_l$, and $\mathbb{E}_d \overline{X}_s = \sum_{l=1}^{L} W_l \mu_l = \mu$.

(ii) Independence across strata gives $\mathrm{Var}_d(\overline{X}_s) = \sum_{l=1}^{L} W_l^2 \mathrm{Var}_d(\overline{X}_l)$.

For stratum l, let $(Y_{1,l}, \ldots, Y_{N_l,l})$ uniformly order $\{x_{il} : 1 \leq i \leq N_l\}$, put $\gamma_l := \mathrm{Cov}_d(Y_{a,l}, Y_{b,l})$ for $a \neq b$.

As $\sum_{a=1}^{N_l} Y_{a,l}$ is fixed, $0 = N_l \sigma_l^2 + N_l(N_l - 1)\gamma_l$, so $\gamma_l = -\frac{\sigma_l^2}{N_l - 1}$, $\mathrm{Var}_d(\overline{X}_l) = \frac{1}{n_l^2}(n_l \sigma_l^2 + n_l(n_l - 1)\gamma_l) = \lambda_l \sigma_l^2$.

(iii) If $n_l/N_l \ll 1$, then $1 - \frac{n_l - 1}{N_l - 1} \approx 1$, so $\lambda_l \sigma_l^2 \approx \sigma_l^2/n_l$.

Structural Commentary: Stratified estimation is a weighted sum of independent within-stratum estimators.
Key Insight: Independent strata make variance add as $\sum_l W_l^2 \mathrm{Var}_d(\overline{X}_l)$. □

Corollary 1.4.3. *Totals and Estimated Variances*

Let $L \geq 1$, $N_l > 1$, $1 < n_l \leq N_l$, $N := \sum_{l=1}^{L} N_l$, $W_l := \frac{N_l}{N}$, and $f_l := n_l/N_l$.

Fix $x_{il} \in \mathbb{R}$ for $i = 1, \ldots, N_l$, $l = 1, \ldots, L$, set $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$, $\mu := \sum_{l=1}^{L} W_l \mu_l$, and $\tau := N\mu$.

Assume independently $S_l \sim \mathrm{Unif}\{s \subseteq \{1, \ldots, N_l\} : |s| = n_l\}$; for random orderings $(J_{il})_{i=1}^{n_l}$ of S_l.

Set $X_{il} := x_{J_{il}, l}$, $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$, $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$, $T_s := N \overline{X}_s$, and $s_l^2 := \frac{1}{n_l - 1} \sum_{i=1}^{n_l} (X_{il} - \overline{X}_l)^2$.

For $0 < \alpha < 1$, let $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$.

(i) Total moments: $\mathbb{E}_d T_s = \tau$ and $\mathrm{Var}_d(T_s) = N^2 \mathrm{Var}_d(\overline{X}_s)$.

(ii) Estimated mean variance: $s^2_{\overline{X}_s} := \sum_{l=1}^{L} W_l^2(1-f_l)\frac{s_l^2}{n_l}$.

(iii) Intervals: $\mathrm{CI}_{1-\alpha}(\mu) = \overline{X}_s \pm z(\frac{\alpha}{2})s_{\overline{X}_s}$ *and* $\mathrm{CI}_{1-\alpha}(\tau) = T_s \pm z(\frac{\alpha}{2})Ns_{\overline{X}_s}$.

Proof.

(i) Since $T_s = N\overline{X}_s$ and $\tau = N\mu$, Theorem 1.4.2 gives $\mathbb{E}_d T_s = N\mu = \tau$ and $\mathrm{Var}_d(T_s) = N^2\,\mathrm{Var}_d(\overline{X}_s)$.

(ii) Let $S_l^2 := \frac{1}{N_l-1}\sum_{i=1}^{N_l}(x_{il}-\mu_l)^2$. Then $\mathrm{Var}_d(\overline{X}_l) = \frac{(1-f_l)S_l^2}{n_l}$, $\mathbb{E}_d s_l^2 = S_l^2$, so $S_l^2 \mapsto s_l^2$; $s^2_{\overline{X}_s} = \sum_{l=1}^{L} W_l^2(1-f_l)\frac{s_l^2}{n_l}$.

(iii) With $\frac{\overline{X}_s-\mu}{s_{\overline{X}_s}} \approx \mathcal{N}(0,1)$, inversion gives $\mathrm{CI}_{1-\alpha}(\mu)$; multiplying by N gives $\mathrm{CI}_{1-\alpha}(\tau)$.

Structural Commentary: Totals and intervals scale the stratified mean.
Key Insight: Estimated variance adds weighted stratum variances. □

Theorem 1.4.4. *Neyman Allocation*

Let $L \geq 1$, $W_l > 0$, $\sum_{l=1}^{L} W_l = 1$, $\sigma_l > 0$, $S_\sigma := \sum_{k=1}^{L} W_k\sigma_k$, *and* $n > 0$.

On $\mathcal{A} := \{(n_1,\dots,n_L) \in (0,\infty)^L : \sum_{l=1}^{L} n_l = n\}$, *set* $V(n_1,\dots,n_L) := \sum_{l=1}^{L}\frac{W_l^2\sigma_l^2}{n_l}$.

Then V *has the unique minimiser* $n_l^* := n\frac{W_l\sigma_l}{S_\sigma}$, $l = 1,\dots,L$.

Proof. V is strictly convex on $(0,\infty)^L$ since $\partial_{ll}V = \frac{2W_l^2\sigma_l^2}{n_l^3} > 0$ and $\partial_{lm}V = 0$ for $l \neq m$.

For $\mathcal{L}(n,\lambda) := V(n) + \lambda(\sum_{l=1}^{L} n_l - n)$, the first-order equations are $-\frac{W_l^2\sigma_l^2}{n_l^2} + \lambda = 0$, hence $n_l = \frac{W_l\sigma_l}{\sqrt{\lambda}}$.
The constraint gives $\frac{1}{\sqrt{\lambda}} = \frac{n}{S_\sigma}$, so $n_l = n_l^*$; strict convexity gives uniqueness.
Structural Commentary: The constrained optimum balances marginal variance reduction across strata.
Key Insight: A stratum receives more sample not only when it is large, but also when it is internally variable. □

Corollary 1.4.5. *Allocation Comparisons*

Let $L \geq 1$, $W_l > 0$, $\sum_{l=1}^{L} W_l = 1$, $\mu_l \in \mathbb{R}$, $\sigma_l > 0$, $\mu := \sum_{l=1}^{L} W_l\mu_l$, $\overline{\sigma} := \sum_{l=1}^{L} W_l\sigma_l$, *and* $n > 0$.

Ignoring finite-population corrections, let $\overline{X}_{so}$ *be the optimal-allocation stratified mean,* $\overline{X}_{sp}$ *the proportional-allocation stratified mean with* $n_l = nW_l$, *and* $\overline{X}$ *the ordinary SRS mean.*

(i) Allocation variances: $\mathrm{Var}(\overline{X}_{so}) \approx \frac{\overline{\sigma}^2}{n}$ *and* $\mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l\sigma_l^2$.

(ii) Optimal-allocation gain: $\mathrm{Var}(\overline{X}_{sp}) - \mathrm{Var}(\overline{X}_{so}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l(\sigma_l-\overline{\sigma})^2$.

(iii) Proportional-allocation gain: $\mathrm{Var}(\overline{X}) - \mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l(\mu_l-\mu)^2$.

Proof.

(i) With finite-population corrections suppressed, $\mathrm{Var}(\overline{X}_s) \approx \sum_{l=1}^{L}\frac{W_l^2\sigma_l^2}{n_l}$. Neyman allocation gives $n_l = \frac{nW_l\sigma_l}{\overline{\sigma}}$, hence $\mathrm{Var}(\overline{X}_{so}) \approx \frac{\overline{\sigma}^2}{n}$; proportional allocation $n_l = nW_l$ gives $\mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l\sigma_l^2$.

(ii) Since $\sum_{l=1}^{L} W_l = 1$, $\sum_{l=1}^{L} W_l\sigma_l^2 - \overline{\sigma}^2 = \sum_{l=1}^{L} W_l(\sigma_l-\overline{\sigma})^2$.

(iii) For ordinary SRS, $\mathrm{Var}(\overline{X}) \approx \frac{1}{n}\{\sum_{l=1}^{L} W_l\sigma_l^2 + \sum_{l=1}^{L} W_l(\mu_l-\mu)^2\}$; subtract the proportional-allocation variance.

Structural Commentary: Proportional handles means; optimal also handles spreads.
Key Insight: Mean gaps help stratification; spread gaps help Neyman allocation. □

Example 1.4.6. *Four-Stratum Stratification Summary*
Let $N = 393$ and strata $l \in \{A,B,C,D\}$ have rounded summaries $(N_l) = (98,98,98,99)$, $(W_l) = (.249,.249,.249,.251)$, $(\mu_l) = (182.9, 526.5, 956.3, 1591.2)$, and $(\sigma_l) = (103.4, 204.8, 243.5, 419.2)$. Let n be the total sample size, ignore finite-population corrections, and write $\overline{X}_s$ for the stratified mean.

(i) *Equal allocation*: if $n_l = n/4$, then $\mathrm{Var}(\overline{X}_s) \approx \sum_{l=1}^{4} W_l^2\sigma_l^2/(n/4) \approx 72042.6/n$, so $\mathrm{SE}(\overline{X}_s) \approx 268.4/\sqrt{n}$.

(ii) *SRS benchmark*: with overall standard deviation $\sigma \approx 589.7$, $\mathrm{SE}(\overline{X}) \approx 589.7/\sqrt{n}$.

(iii) *Neyman allocation*: $a_l := \frac{W_l\sigma_l}{\sum_{k=1}^{4} W_k\sigma_k} \approx (.106,.210,.250,.434)$ for (A,B,C,D).

Stratification reduces variance, and Neyman allocation assigns the largest share to the stratum with largest σ_l.

Remark 1.4.7. *Model Limits and Other Designs*
For a finite-population survey:

(i) *Systematic*: fixed-interval sampling after a random start is SRS-like only when list order is noninformative; periodic order can bias selection.

(ii) *Cluster*: sampling groups lowers cost but can inflate variance when within-cluster correlation is positive.

(iii) *Scope*: design-based SEs measure random selection variation, not nonresponse, coverage error, selection bias, interviewer effects, or wording effects.

2 Exploratory Data Analysis and Data Summaries

2.1 Empirical Distribution and Survival Summaries

Exploratory summaries first treat the observed values as a batch of numbers. A stochastic model may later explain their variability, but the displays themselves are useful before a model has been chosen.

Definition 2.1.1. *Empirical CDF and Empirical Survival Function*
Let $x_1, \dots, x_n \in \mathbb{R}$ be data; for survival summaries assume $x_i \geq 0$ for $1 \leq i \leq n$.

(i) *Empirical CDF*: $F_n : \mathbb{R} \to [0,1]$, $F_n(x) := n^{-1}\#\{i \in \{1,\dots,n\} : x_i \leq x\}$.
(ii) *Empirical Survival Function*: $S_n : [0,\infty) \to [0,1]$, $S_n(t) := 1 - F_n(t) = n^{-1}\#\{i \in \{1,\dots,n\} : x_i > t\}$.

Key Insight: F_n and S_n encode complementary empirical tails, with $S_n(t) = 1 - F_n(t)$ for $t \geq 0$.

Remark 2.1.2. *Step-Function Convention*
Let $v_1 < \cdots < v_m$ be the distinct observed values, with multiplicities $r_1, \dots, r_m$. Then F_n is right-continuous, constant on each component of $\mathbb{R} \setminus \{v_1, \dots, v_m\}$, and has jump $\Delta F_n(v_j) = \frac{r_j}{n}$.

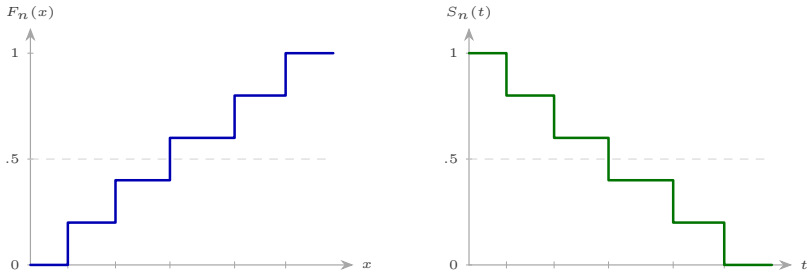

Figure 242: Empirical CDF (blue) rises as empirical survival (green) falls.

Method 2.1.3. *Empirical Log-Survival Plot*
Suppose $0 \leq T_{(1)} < \cdots < T_{(n)}$ and there are no ties. Set $p_i := \frac{i}{n+1}$ and $y_i := \log(1 - p_i)$.

(i) *Plot*: Draw $P^{(n)}_{\log S} := \{(T_{(i)}, y_i) : 1 \leq i \leq n\}$; using $n+1$ keeps y_n finite.
(ii) *Hazard Reading*: Since $(\log S)' = -h$, more negative local slopes indicate larger hazards.
(iii) *Tail Variability*: $\mathrm{Var}(\log[1 - F_n(t)]) \approx n^{-1}\frac{F(t)}{1-F(t)}$, which grows as $F(t) \uparrow 1$.

2.2 Quantile and Shape Displays

Distributional shape can be read from cumulative displays, density displays, or quantile comparisons. These views emphasize different questions: cumulative proportion, local concentration, and relative change between two distributions.

Definition 2.2.1. *Quantile–Quantile Plot*
Let F and G be continuous strictly increasing CDFs with quantile inverses $F^{-1}, G^{-1} : (0,1) \to \mathbb{R}$. For equal-sized batches $x_1, \dots, x_n$ and $y_1, \dots, y_n$, let $x_{(1)} \leq \cdots \leq x_{(n)}$ and $y_{(1)} \leq \cdots \leq y_{(n)}$.

(i) *Q–Q Plot*: $Q_{F,G} := \{(F^{-1}(p), G^{-1}(p)) : 0 < p < 1\} \subseteq \mathbb{R}^2$.
(ii) *Empirical Q–Q Plot*: $Q^{(n)}_{x,y} := \{(x_{(i)}, y_{(i)}) : 1 \leq i \leq n\} \subseteq \mathbb{R}^2$.

Key Insight: $(u, v) \in Q_{F,G}$ implies $F(u) = G(v)$, so plotted pairs match common cumulative probability levels.

Proposition 2.2.2. *Location and Scale Effects in Q–Q Plots*
Let F, G be continuous strictly increasing CDFs.
Set $x_p := F^{-1}(p)$, $y_p := G^{-1}(p)$ for $p \in (0,1)$, and $Q_{F,G} := \{(x_p, y_p) : 0 < p < 1\}$. Then:
(i) Location: If $y_p = x_p + h$ $\forall p \in (0,1)$ with $h \in \mathbb{R}$, then $G(y) = F(y - h)$ $\forall y \in \mathbb{R}$, $Q_{F,G} \subseteq \{(x,y) : y = x + h\}$.
(ii) Scale: If $y_p = cx_p$ $\forall p \in (0,1)$ with $c > 0$, then $G(y) = F(\frac{y}{c})$ $\forall y \in \mathbb{R}$, $Q_{F,G} \subseteq \{(x,y) : y = cx\}$.

Proof.

(i) *Location*: Fix $y \in \mathbb{R}$ and set $p := G(y)$. Then $p \in (0,1)$, $y = y_p = x_p + h$, and $x_p = y - h$, so $G(y) = p = F(x_p) = F(y - h)$. Also $(x_p, y_p) \in Q_{F,G}$ implies $y_p = x_p + h$.
(ii) *Scale*: Fix $y \in \mathbb{R}$ and set $p := G(y)$. Then $p \in (0,1)$, $y = y_p = cx_p$, and $x_p = \frac{y}{c}$, so $G(y) = p = F(x_p) = F(\frac{y}{c})$. Also $(x_p, y_p) \in Q_{F,G}$ implies $y_p = cx_p$.

Structural Commentary: A Q–Q plot is the graph of the quantile transport $F^{-1}(p) \mapsto G^{-1}(p)$.
Key Insight: Straight Q–Q lines encode location shifts and scale changes. □

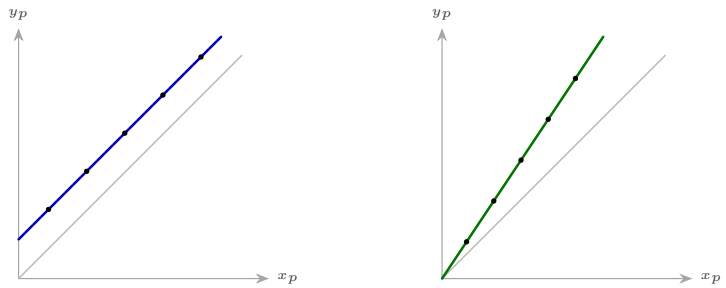

Figure 243: Q–Q lines: blue $y = x + h$ for location shift, green $y = cx$ for scale change.

Example 2.2.3. *Normal and Exponential Q–Q Lines*
For $p \in (0,1)$, set $x_p := F^{-1}(p)$ and $y_p := G^{-1}(p)$.

(i) *Normal Location*: If $F = N(0,1)$ and $G = N(1,1)$, then $y_p = x_p + 1$, so the Q–Q line is $y = x + 1$.
(ii) *Normal Location–Scale*: If $F = N(0,1)$ and $G = N(1,4)$, then $y_p = 1 + 2x_p$, so the Q–Q line is $y = 1 + 2x$.
(iii) *Exponential Rates*: If $F = \text{Exp}(\lambda_1)$ and $G = \text{Exp}(\lambda_2)$, then $x_p = -\frac{\log(1-p)}{\lambda_1}$ and $y_p = -\frac{\log(1-p)}{\lambda_2} = \frac{\lambda_1}{\lambda_2} x_p$, so the Q–Q line is $y = \frac{\lambda_1}{\lambda_2} x$.

Method 2.2.4. *Interpolated Empirical Quantiles*
Let $X_{(1)} \le \cdots \le X_{(n)}$ and set $p_k := \frac{k}{n+1}$ for $1 \le k \le n$.

(i) *Grid*: Assign $X_{(k)}$ to probability p_k.
(ii) *Interpolation*: If $p_k \le p \le p_{k+1}$ for $1 \le k < n$, set $\lambda := \frac{p - p_k}{p_{k+1} - p_k} = (n+1)p - k$, $\widehat{x}_p := (1-\lambda)X_{(k)} + \lambda X_{(k+1)}$.
(iii) *Q–Q Use*: For unequal samples, evaluate both interpolated quantile functions on a common grid $p \in (0,1)$.

Definition 2.2.5. *Histogram, Density Display, and Stem-and-Leaf Plot*
Let $x_1, \ldots, x_n \in \mathbb{R}$ be data. For disjoint intervals $B_1, \ldots, B_m$ with widths $w_j > 0$.
Set $N_j := \#\{i \in \{1, \ldots, n\} : x_i \in B_j\}$. Let K be a kernel, $h > 0$, and $b > 0$ be a chosen scale.

(i) *Histogram*: $H_j^c := N_j$, $H_j^p := \frac{N_j}{n}$, or $H_j^d := \frac{N_j}{n w_j}$ on B_j.
(ii) *Density Display*: $\widehat{f}_h : \mathbb{R} \to [0, \infty)$, $\widehat{f}_h(x) := (nh)^{-1} \sum_{i=1}^{n} K(\frac{x - x_i}{h})$.
(iii) *Stem-and-Leaf Plot*: (s_i, ℓ_i) satisfies $x_i = b s_i + \ell_i$, $s_i \in \mathbb{Z}$, $\ell_i \in [0, b)$.
Entries with common s_i are printed together in increasing s_i.

Structural Commentary: For density-scaled heights $H_j^d = \frac{N_j}{n w_j}$, $\sum_j H_j^d w_j = 1$ when the bins cover the sample. *Key Insight*: Smaller w_j or h retains more local variation; larger w_j or h increases smoothing.

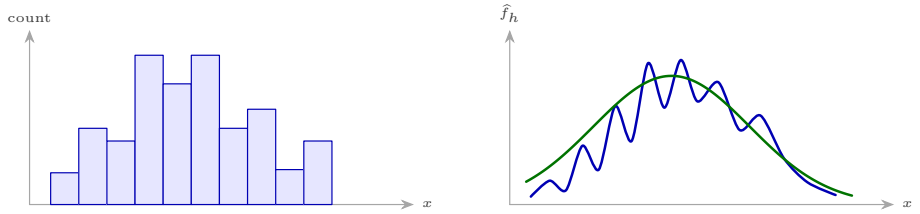

Figure 244: Shape displays: narrow bins at left; small h (blue) is rougher than large h (green).

Remark 2.2.6. *What Shape Displays Preserve*

(i) *ECDF*: Preserves ranks and supports direct quantile reading.
(ii) *Histogram*: Preserves bin counts (N_j), not exact values.
(iii) *Kernel Estimate*: Preserves smoothed local mass; bandwidth h controls resolution.
(iv) *Stem–Leaf*: Preserves coarse shape and recoverable numerical values.

2.3 Location, Scale, and Robust Summaries

Numerical summaries compress a batch more aggressively than a plot. They should therefore be chosen with attention to the feature being summarized and to the influence of extreme observations.

Definition 2.3.1. *Location Summaries for a Batch*
Let $x_1, \ldots, x_n \in \mathbb{R}$ have order statistics $x_{(1)} \le \cdots \le x_{(n)}$.
When a trim fraction is used, let $0 \le \alpha < \frac{1}{2}$ and $m_\alpha := \lfloor n\alpha \rfloor$.

(i) *Arithmetic Mean*: $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$.
(ii) *Median*: $\widetilde{x} := x_{\left(\frac{n+1}{2}\right)}$ if n is odd, and $\widetilde{x} := \frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}}{2}$ if n is even.
(iii) $100\alpha\%$ *Trimmed Mean*: $\overline{x}_\alpha := (n - 2m_\alpha)^{-1} \sum_{i=m_\alpha+1}^{n-m_\alpha} x_{(i)}$.

Structural Commentary: Under $x_i \mapsto x_i + \Delta$, $\overline{x} \mapsto \overline{x} + \frac{\Delta}{n}$; $\widetilde{x}$ and $\overline{x}_\alpha$ depend only on central order statistics.
Key Insight: Extreme magnitudes affect $\overline{x}$ directly, but affect $\widetilde{x}$ and $\overline{x}_\alpha$ mainly through rank changes.

Remark 2.3.2. *Outliers and Center*
Replacing one x_i by M sends $\overline{x}$ to $\overline{x} + \frac{M - x_i}{n}$, hence unbounded as $M \to \pm\infty$.
Once an observation is extreme, moving it farther outward leaves $\widetilde{x}$ unchanged and leaves $\overline{x}_\alpha$ unchanged if it remains trimmed. This is descriptive robustness, not an independence or model.

Remark 2.3.3. *Compare More Than One Center*
Let $\mathcal{C} := \{\overline{x}, \widetilde{x}, \overline{x}_\alpha, \widehat{\mu}_\Psi\}$. For symmetric light-tailed data, range($\mathcal{C}$) is typically small; a large range signals skewness, heavy tails, outliers, or mixture structure.

Definition 2.3.4. *Dispersion Summaries*
Let $x_1, \ldots, x_n \in \mathbb{R}$ have mean $\overline{x}$, median $\widetilde{x}$, and selected sample quartiles Q_1, Q_3.

(i) *Sample Variance*: $s^2 := (n-1)^{-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$.
(ii) *Sample Standard Deviation*: $s := \sqrt{s^2}$.
(iii) *Interquartile Range*: $\mathrm{IQR} := Q_3 - Q_1$.
(iv) *Median Absolute Deviation*: $\mathrm{MAD} := \mathrm{median}\{|x_i - \widetilde{x}| : 1 \le i \le n\}$.

Structural Commentary: s^2 is residual-based; IQR and MAD are order-based.
Key Insight: A single extreme value can dominate s, but not IQR or MAD unless it changes central ranks.

Remark 2.3.5. *Normal Scale Calibrations*
For approximately normal data, $\frac{\mathrm{IQR}}{1.35}$ and $\frac{\mathrm{MAD}}{0.675}$ are common robust scale estimates for the normal standard deviation. The constants calibrate the population normal IQR and median absolute deviation to σ.

2.4 Boxplots, Scatterplots, and Relationship Summaries

Boxplots and scatterplots are compact exploratory displays. A boxplot summarizes one batch or compares several batches; a scatterplot displays relationships between paired variables.

Method 2.4.1. *Boxplot Construction*
A standard boxplot is built from the median, lower quartile Q_1, upper quartile Q_3, and $\mathrm{IQR} = Q_3 - Q_1$:

(i) Draw a box from Q_1 to Q_3 and mark the median inside the box.
(ii) Draw whiskers to the most extreme observations lying within 1.5 IQR of the nearer quartile.
(iii) Mark observations beyond the whiskers individually as possible outliers.

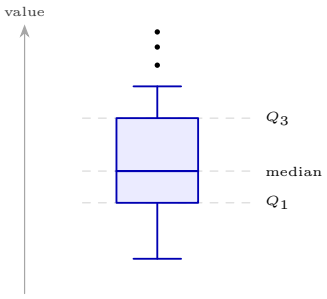

Figure 245: Boxplot: box Q_1 to Q_3, line median, dots possible outliers.

Remark 2.4.2. *Comparing Batches with Boxplots*

(i) *Location*: Differences in medians M_g indicate group shifts.
(ii) *Spread*: Differences in $\mathrm{IQR}_g = Q_{3,g} - Q_{1,g}$ indicate central-spread changes.
(iii) *Tails*: Unequal whiskers or one-sided outliers indicate skewness or tail imbalance.

Definition 2.4.3. *Scatterplot, Pearson Correlation, and Rank Correlation*
Let $(x_i, y_i) \in \mathbb{R}^2$ for $1 \le i \le n$, set $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$ and $\overline{y} := n^{-1} \sum_{i=1}^{n} y_i$.
Let R_i, S_i be chosen ranks of x_i, y_i, using average ranks for ties.
Set $\overline{R} := n^{-1} \sum_{i=1}^{n} R_i$ and $\overline{S} := n^{-1} \sum_{i=1}^{n} S_i$. Set $a_i := x_i - \overline{x}$, $b_i := y_i - \overline{y}$, $u_i := R_i - \overline{R}$, and $v_i := S_i - \overline{S}$.
Let $S_{xx} := \sum_i a_i^2$, $S_{yy} := \sum_i b_i^2$, $S_{xy} := \sum_i a_i b_i$, $S_{RR} := \sum_i u_i^2$, $S_{SS} := \sum_i v_i^2$, and $S_{RS} := \sum_i u_i v_i$.

(i) *Scatterplot*: $P := \{(x_i, y_i) : 1 \le i \le n\} \subset \mathbb{R}^2$, displayed in the xy-plane.
(ii) *Pearson Correlation Coefficient*: if $S_{xx} S_{yy} > 0$, $r := \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$.
(iii) *Rank Correlation Coefficient*: if $S_{RR} S_{SS} > 0$, $r_{\mathrm{rank}} := \frac{S_{RS}}{\sqrt{S_{RR} S_{SS}}}$.

Structural Commentary: r is affine-scale invariant; r_{rank} is monotone-transform invariant.
Key Insight: r measures linear association; r_{rank} measures rank association.

Method 2.4.4. *Scatterplot Diagnostics*
For paired data $P = \{(x_i, y_i) : 1 \le i \le n\}$:

(i) *Raw Plot*: Inspect P for outliers, curvature, clusters, and nonconstant spread.
(ii) *Transform*: If scale obscures structure, compare $(g(x_i), h(y_i))$ for domain-meaningful g, h such as log.

(iii) *Correlation*: Use r only when the displayed relation is approximately linear.
(iv) *Multiple Variables*: For $z_i \in \mathbb{R}^d$ with $d > 2$, inspect all pairwise plots before relying on pairwise summaries.

Example 2.4.5. *Changing Scale Can Reveal Structure*
If $x_i > 0$ and $\max_i x_i / \min_i x_i \gg 1$, the raw plot $P = \{(x_i, y_i)\}$ may compress variation. The transformed plot $P_{\log} := \{(\log x_i, y_i)\}$ can reveal monotone or linear structure, and generally $r(P_{\log}) \neq r(P)$ because Pearson correlation depends on the chosen coordinates.

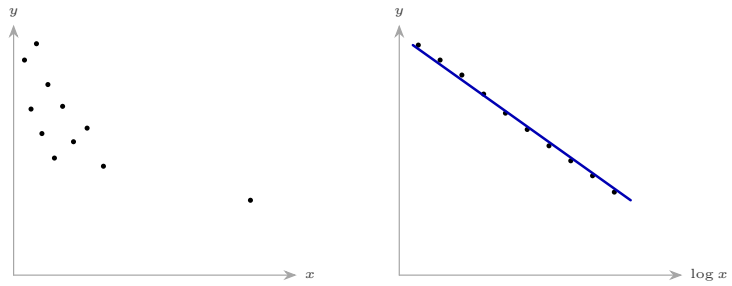

Figure 246: Scatterplots: raw x compresses points; $\log x$ reveals a linear trend.

Remark 2.4.6. *Exploration Before Formal Inference*
Exploratory displays choose the inferential target: transformations $g(X)$, influential subsets, comparable spreads IQR_g, and data-quality checks should be settled before intervals, tests, or models are trusted.

3 Categorical Data Analysis

3.1 Fisher's Exact Test and Fixed Margins

Categorical data record counts in classes rather than measurements on a continuous scale. In a two-way table, the rows and columns encode two classifications, and the cells record how many units fall in each cross-classification. The probability model is therefore built from multinomial and hypergeometric count laws rather than from normal sampling pivots.

Definition 3.1.1. *Two-Way Contingency Table*
Let $I, J \in \mathbb{N}$ and $n_{ij} \in \mathbb{N}_0$ for $1 \le i \le I$, $1 \le j \le J$. Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, and $n_{\cdot\cdot} := \sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij}$.
For corresponding random counts, write N_{ij} and use capitalized totals.
(i) *Two-way contingency table*: $(n_{ij})_{1\le i\le I,\ 1\le j\le J}$, indexed by row and column classifications.
(ii) *Margins*: $\{n_{i\cdot} : 1 \le i \le I\}$, $\{n_{\cdot j} : 1 \le j \le J\}$, and $n_{\cdot\cdot}$.
Structural Commentary: Fixed-margin exact tests condition on the margins of random table $N = (N_{ij})$.

Method 3.1.2. *Fisher's Exact Test for a 2×2 Table*
Let a 2×2 table have fixed margins

	1	2	Total
1	N_{11}	N_{12}	$n_{1\cdot}$
2	N_{21}	N_{22}	$n_{2\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	$n_{\cdot\cdot}$

Let R_1, R_2 be fixed disjoint row sets with $|R_i| = n_{i\cdot}$.
Let C be the random column-1 set with $|C| = n_{\cdot 1}$, and set $\Omega_0 := \{A \subseteq R_1 \cup R_2 : |A| = n_{\cdot 1}\}$.
(i) $H_0 : \mathbb{P}_0(C = A) = \binom{n_{\cdot\cdot}}{n_{\cdot 1}}^{-1}$ $\forall A \in \Omega_0$.
(ii) $N_{11} = |C \cap R_1|$, $N_{12} = n_{1\cdot} - N_{11}$, $N_{21} = n_{\cdot 1} - N_{11}$, and $N_{22} = n_{2\cdot} - n_{\cdot 1} + N_{11}$.
(iii) $\mathcal{S} := \{x \in \mathbb{Z} : 0 \le x \le n_{1\cdot},\ 0 \le n_{\cdot 1} - x \le n_{2\cdot}\}$ and $p_0(x) := \mathbb{P}_0(N_{11} = x) = \dfrac{\binom{n_{1\cdot}}{x}\binom{n_{2\cdot}}{n_{\cdot 1}-x}}{\binom{n_{\cdot\cdot}}{n_{\cdot 1}}}$, $x \in \mathcal{S}$.
(iv) $p_+(x_{\mathrm{obs}}) := \sum_{x\in\mathcal{S}:x\ge x_{\mathrm{obs}}} p_0(x)$, $p_-(x_{\mathrm{obs}}) := \sum_{x\in\mathcal{S}:x\le x_{\mathrm{obs}}} p_0(x)$, and $p_2(x_{\mathrm{obs}}) := \sum_{x\in\mathcal{S}:p_0(x)\le p_0(x_{\mathrm{obs}})} p_0(x)$.
Reject when the selected value is $\le \alpha$.

Example 3.1.3. *Promotion Decisions and Fixed Margins*
In randomised label experiment with 24 male-labeled,24 female-labeled files, the observed table is

	Male	Female	Total
Promote	21	14	35
Hold file	3	10	13
Total	24	24	48

Under H_0 and fixed margins $(35, 13)$, $(24, 24)$, N_{11} satisfies $\mathbb{P}_0(N_{11} = x) = \dfrac{\binom{35}{x}\binom{13}{24-x}}{\binom{48}{24}}$, $x \in \mathcal{S} := \{11, \dots, 24\}$.
At level 0.05, $R^{(2)}_{0.05} := \{11, 12, 13, 14, 21, 22, 23, 24\}$; since $x_{\mathrm{obs}} = 21 \in R^{(2)}_{0.05}$, reject H_0.

Remark 3.1.4. *What Is Exact Here*
For $N = (N_{ij})$ and $\mathcal{M} := (\{N_{i\cdot}\}_i, \{N_{\cdot j}\}_j)$, exactness means evaluating $\mathcal{L}_0(N \mid \mathcal{M})$ without asymptotics. Conditioning is design-justified when $\mathcal{M}$ is ancillary to H_0; rejection interpretation is design-dependent.

3.2 Chi-Square Tests for Contingency Tables

For an $I \times J$ table with observed counts n_{ij} and fitted null counts E_{ij}, use $X^2 := \sum_{i=1}^{I} \sum_{j=1}^{J} (n_{ij} - E_{ij})^2 / E_{ij}$; in contingency tables, E_{ij} is determined by the row-column null model.

Definition 3.2.1. *Homogeneity of Multinomial Distributions*

Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and $\Delta_{I-1} := \{\pi \in [0,1]^I : \sum_{i=1}^{I} \pi_i = 1\}$.

For independent columns $j \in [J]$, let $(N_{1j}, \dots, N_{Ij}) \sim \text{Multinomial}(n_{\cdot j}; \pi_{1j}, \dots, \pi_{Ij})$ with observed counts n_{ij}.
The column multinomial laws are *homogeneous* iff $H_0 : \exists \pi \in \Delta_{I-1}$ with $\pi_{ij} = \pi_i \ \forall (i,j) \in [I] \times [J]$; under H_0, π_i is the common category probability.

Proposition 3.2.2. *Common Probability MLE Under Homogeneity*

Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, $\Delta_{I-1} := \{\pi \in [0,1]^I : \sum_{i=1}^{I} \pi_i = 1\}$, and $N_{\cdot j} := (N_{1j}, \dots, N_{Ij})$, independently for $j \in [J]$. Under H_0, $N_{\cdot j} \sim \text{Multinomial}(n_{\cdot j}; \pi)$ for a common $\pi \in \Delta_{I-1}$.

With observed counts n_{ij}, set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$ and $n_{\cdot\cdot} := \sum_{i=1}^{I} n_{i\cdot} > 0$.

Then $\widehat{\pi}_i = \frac{n_{i\cdot}}{n_{\cdot\cdot}}$ and $E_{ij} = n_{\cdot j} \widehat{\pi}_i = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$ for $(i,j) \in [I] \times [J]$.

Proof. Up to factors independent of π, $L(\pi) \propto \prod_{j=1}^{J} \prod_{i=1}^{I} \pi_i^{n_{ij}} = \prod_{i=1}^{I} \pi_i^{n_{i\cdot}}$, hence $\ell(\pi) = \sum_{i=1}^{I} n_{i\cdot} \log \pi_i$ on Δ_{I-1}.

Stationarity of $\ell(\pi) + \lambda(1 - \sum_{i=1}^{I} \pi_i)$ gives $\frac{n_{i\cdot}}{\pi_i} - \lambda = 0$ when $n_{i\cdot} > 0$ and $\widehat{\pi}_i = 0$ when $n_{i\cdot} = 0$,
Normalisation gives $\lambda = n_{\cdot\cdot}$ and $\widehat{\pi}_i = \frac{n_{i\cdot}}{n_{\cdot\cdot}}$. Therefore $E_{ij} = \mathbb{E}_{\widehat{\pi}} N_{ij} = n_{\cdot j} \widehat{\pi}_i$.
Structural Commentary: H_0 reduces IJ cell probabilities to one shared $\pi \in \Delta_{I-1}$.
Key Insight: $\widehat{\pi}$ is the pooled row proportion, and E_{ij} redistributes each column total by $\widehat{\pi}_i$. □

Definition 3.2.3. *Homogeneity Fitted Counts*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$, and $\widehat{\pi}_i := \frac{n_{i\cdot}}{n_{\cdot\cdot}}$.
The homogeneity *fitted counts* are $E_{ij} := n_{\cdot j} \widehat{\pi}_i = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$ for $(i,j) \in [I] \times [J]$.

Method 3.2.4. *Pearson Test of Homogeneity*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$, and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$. Assume $E_{ij} > 0$ for cells used.

(i) *Statistic*: $X^2 := \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$.

(ii) *Calibration*: If $n_{\cdot\cdot}$ large, $\min_{i,j} E_{ij}$ not small, then $X^2 \overset{H_0}{\approx} \chi^2_\nu$; $\nu := J(I-1) - (I-1) = (I-1)(J-1)$.
(iii) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_{(I-1)(J-1)} \geq X^2_{\text{obs}}\} \leq \alpha$.

Example 3.2.5. *Word Counts by Text Group*
Let $W = \{a, an, this, that, with, without\}$, let $G_1, \dots, G_4$ be text groups. Observed word-count table n_{wj} is

Word	G_1	G_2	G_3	G_4
a	147	186	101	83
an	25	26	11	29
this	32	39	15	15
that	94	105	37	22
with	59	74	28	43
without	18	10	10	4
Total	375	440	202	196

For $C_A := \{1, 2, 3\}$, by homogeneity test: $X^2_A = 12.27$, $\nu_A = (6-1)(3-1) = 10$, $p_A = \mathbb{P}\{\chi^2_{10} \geq 12.27\} \approx 0.27$.
Pooling C_A and comparing it with $C_B := \{4\}$ gives $X^2_{A,B} = 32.81$, $\nu_{A,B} = 6 - 1 = 5$, and $p_{A,B} < 0.001$.
$w \in \{an, that\}$ gives the largest Pearson contributions, localising nonhomogeneity by row.

Definition 3.2.6. *Independence in a Contingency Table*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and $N = (N_{ij}) \sim \text{Multinomial}(n; (\pi_{ij})_{[I] \times [J]})$ with $n > 0$, observed counts n_{ij}.
Set $\pi_{i\cdot} := \sum_{j=1}^{J} \pi_{ij}$, $\pi_{\cdot j} := \sum_{i=1}^{I} \pi_{ij}$, $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, and $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$.
Rows and columns are *independent* iff $H_0 : \pi_{ij} = \pi_{i\cdot} \pi_{\cdot j} \ \forall (i,j) \in [I] \times [J]$; then $\widehat{\pi}_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n^2}$ and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$.

Method 3.2.7. *Pearson Test of Independence*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$, $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$. Assume $E_{ij} > 0$ for cells used.

(i) *Statistic*: $X^2 := \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$.

(ii) *Calibration*: If n is large and $\min_{i,j} E_{ij}$ is not small, then $X^2 \overset{H_0}{\approx} \chi^2_\nu$ with $\nu := (I-1)(J-1)$.
(iii) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_\nu \geq X^2_{\text{obs}}\} \leq \alpha$.

(iv) *Degrees of Freedom*: $\nu = (IJ-1) - ((I-1)+(J-1)) = (I-1)(J-1)$.

Example 3.2.8. *Binary Classifications*
Let $n = 1436$, $[n] := \{1, \dots, n\}$, and $(A_k, B_k) \in \{a_1, a_2\} \times \{b_1, b_2\}$ for $k \in [n]$.
Set $n_{ij} := \#\{k \in [n] : A_k = a_i,\ B_k = b_j\}$ and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$. The observed and independence-fitted tables are

n_{ij}

$a_i \backslash b_j$	b_1	b_2	Total
a_1	550	61	611
a_2	681	144	825
Total	1231	205	1436

E_{ij}

$a_i \backslash b_j$	b_1	b_2
a_1	523.8	87.2
a_2	707.2	117.8

Under $H_0 : A \perp B$, $X^2 = \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{(n_{ij} - E_{ij})^2}{E_{ij}} = 16.01$, $\nu = 1$, and $p < 0.001$; reject H_0.
Thus $A \not\perp B$; absent randomised A, inference is associative, not causal.

Remark 3.2.9. *Homogeneity Versus Independence*
Both tests use $E_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$, $X^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$, and $\nu = (I-1)(J-1)$.

(i) *Homogeneity*: $N_{\cdot j} \sim \text{Multinomial}(n_{\cdot j}; \pi_{\cdot j})$ independently for $j \in [J]$, with fixed $n_{\cdot j}$, $H_0 : \pi_{\cdot 1} = \cdots = \pi_{\cdot J}$.
(ii) *Independence*: $N \sim \text{Multinomial}(n_{\cdot\cdot}; (\pi_{ij})_{[I]\times[J]})$ with $H_0 : \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}\ \forall (i,j) \in [I] \times [J]$.

If $\pi_{\cdot j} > 0$, then independence iff $\pi_{i|j} := \frac{\pi_{ij}}{\pi_{\cdot j}} = \pi_{i\cdot}\ \forall (i,j) \in [I] \times [J]$.

Method 3.2.10. *Likelihood-Ratio Version*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ with null fitted counts E_{ij} for $(i,j) \in [I] \times [J]$.
Set $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$ and $\nu := (I-1)(J-1)$.
Let H_0 be the homogeneity or independence null producing E_{ij}, and assume $E_{ij} > 0$ whenever $n_{ij} > 0$.

(i) *Statistic*: $G^2 := 2 \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} \log\left(\frac{n_{ij}}{E_{ij}}\right)$.
(ii) *Zero Counts*: $n_{ij} = 0$ contributes 0 to G^2.
(iii) *Calibration*: Under the large-sample likelihood-ratio conditions of Theorem 3.1.8, $G^2 \overset{H_0}{\approx} \chi^2_\nu$.
(iv) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_\nu \geq G^2_{\text{obs}}\} \leq \alpha$.

3.3 Matched Pairs and McNemar's Test

Matched categorical data require a different table from unmatched samples. The pair, not the individual measurement, is the independent sampling unit.

Definition 3.3.1. *Matched Binary Pair Table*
Let $(Y_k^{(1)}, Y_k^{(2)}) \in \{0,1\}^2$ for $k \in [n]$, and set $R_k := Y_k^{(1)} + 1$, $S_k := Y_k^{(2)} + 1$.
For $(r,s) \in \{1,2\}^2$, set $n_{rs} := \#\{k \in [n] : (R_k, S_k) = (r,s)\}$, so $n = \sum_{r=1}^{2} \sum_{s=1}^{2} n_{rs}$.

(i) *Matched Binary Pair Table*: $N := (n_{rs})_{(r,s)\in\{1,2\}^2}$.
(ii) *Concordant Cells*: $C := \{(r,s) \in \{1,2\}^2 : r = s\} = \{(1,1),(2,2)\}$; counts $(n_{rs})_{(r,s)\in C} = (n_{11}, n_{22})$.
(iii) *Discordant Cells*: $D := \{(r,s) \in \{1,2\}^2 : r \neq s\} = \{(1,2),(2,1)\}$; counts $(n_{rs})_{(r,s)\in D} = (n_{12}, n_{21})$.
Key Insight: (n_{12}, n_{21}) determines marginal change; (n_{11}, n_{22}) records agreement.

Definition 3.3.2. *Marginal Homogeneity for Matched Pairs*
Let $\pi = (\pi_{rs})_{(r,s)\in\{1,2\}^2} \in [0,1]^4$ satisfy $\sum_{r=1}^{2} \sum_{s=1}^{2} \pi_{rs} = 1$. Set $\pi_{r\cdot} := \sum_{s=1}^{2} \pi_{rs}$ and $\pi_{\cdot s} := \sum_{r=1}^{2} \pi_{rs}$.
The matched-pair law is *marginally homogeneous* iff $H_0 : (\pi_{1\cdot}, \pi_{2\cdot}) = (\pi_{\cdot 1}, \pi_{\cdot 2})$, equivalently $\pi_{12} = \pi_{21}$.
Key Insight: H_0 constrains only the discordant probabilities π_{12}, π_{21}.

Method 3.3.3. *McNemar's Test*
Let $n_{rs} \in \mathbb{N}_0$ and cell probabilities π_{rs} for $(r,s) \in \{1,2\}^2$, with $n := \sum_{r=1}^{2} \sum_{s=1}^{2} n_{rs} > 0$. Set $d := n_{12} + n_{21}$.

(i) *Null*: $H_0 : \pi_{12} = \pi_{21}$.
(ii) *MLE Under H_0*: $\widehat{\pi}_{11} = \frac{n_{11}}{n}$, $\widehat{\pi}_{22} = \frac{n_{22}}{n}$, and $\widehat{\pi}_{12} = \widehat{\pi}_{21} = \frac{d}{2n}$.
(iii) *Statistic*: If $d > 0$, $X^2 := \frac{(n_{12} - n_{21})^2}{d}$.
(iv) *Calibration*: If d is large, then $X^2 \overset{H_0}{\approx} \chi^2_1$.
(v) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_1 \geq X^2_{\text{obs}}\} \leq \alpha$.

Example 3.3.4. *Matched Case-Control Pairs*
In a matched case-control study, 85 cases were each paired with one control, and a binary exposure was recorded for both members of each pair.

	Control	
Case	Unexposed	Exposed
Unexposed	37	7
Exposed	15	26

An unmatched chi-square test treats cases and controls as independent samples.
McNemar's test uses only discordant pairs; here $X^2 = (7-15)^2/(7+15) = 2.91$ and $p \approx 0.09$, so accounting for the matching weakens the evidence of an exposure difference.

Example 3.3.5. *Matched Case-Crossover Exposure*
In a matched case-crossover study, each subject is observed during an event window and a comparable control window, with a binary exposure recorded in both windows.

Event window	Control window		
	Exposed	Unexposed	Total
Exposed	13	157	170
Unexposed	24	505	529
Total	37	662	699

The discordant counts are 157 and 24, so $X^2 = (157-24)^2/(157+24) = 97.7$, giving strong evidence that exposure prevalence differs between the two windows. This does not by itself establish causation, since time-varying covariates may affect both exposure and event risk.

Remark 3.3.6. *Matched Does Not Mean Independent*
McNemar's null hypothesis is marginal homogeneity, not independence of paired responses. Thus the analysis must preserve the matching structure rather than treat the two margins as independent samples.

3.4 Odds Ratios and Case-Control Sampling

Odds ratios summarize association in a binary exposure–binary outcome table. They are especially useful because the same cross-product ratio can be estimated under several sampling designs, including retrospective case-control designs where risks themselves cannot be estimated.

Definition 3.4.1. *Odds and Odds Ratio*
Let A satisfy $0 < \mathbb{P}(A) < 1$.
Let X and D be binary exposure and disease events with $0 < \mathbb{P}(X) < 1$ and $0 < \mathbb{P}(D \mid X), \mathbb{P}(D \mid \overline{X}) < 1$.

(i) *Odds*: $\operatorname{odds}(A) := \frac{\mathbb{P}(A)}{1-\mathbb{P}(A)}$.
(ii) *Odds ratio*: $\Delta := \frac{\operatorname{odds}(D|X)}{\operatorname{odds}(D|\overline{X})}$.

Structural Commentary: If $o = \operatorname{odds}(A)$, then $\mathbb{P}(A) = \frac{o}{1+o}$; $p \mapsto p/(1-p)$ is a bijection $(0,1) \to (0,\infty)$.

Proposition 3.4.2. *Cross-Product Formula*
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $X, D \in \mathcal{F}$, and set $X_0 := \overline{X}$, $X_1 := X$, $D_0 := \overline{D}$, $D_1 := D$.
Let $\pi_{ij} := \mathbb{P}(X_i \cap D_j) > 0$ for $i, j \in \{0,1\}$, with margins $\pi_{i\cdot} := \pi_{i0} + \pi_{i1}$ and $\pi_{\cdot j} := \pi_{0j} + \pi_{1j}$.

	$\overline{D}$	D	
$\overline{X}$	π_{00}	π_{01}	$\pi_{0\cdot}$
X	π_{10}	π_{11}	$\pi_{1\cdot}$
	$\pi_{\cdot 0}$	$\pi_{\cdot 1}$	1

With $\Delta := \frac{\operatorname{odds}(D|X)}{\operatorname{odds}(D|\overline{X})}$, the cross-product identity is $\Delta = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$.

Proof. Since $\pi_{1\cdot} = \pi_{10} + \pi_{11}$ and $\pi_{0\cdot} = \pi_{00} + \pi_{01}$, $\mathbb{P}(D \mid X) = \frac{\pi_{11}}{\pi_{1\cdot}}$ and $1 - \mathbb{P}(D \mid X) = \frac{\pi_{10}}{\pi_{1\cdot}}$, so $\operatorname{odds}(D \mid X) = \frac{\pi_{11}}{\pi_{10}}$.
Similarly, $\operatorname{odds}(D \mid \overline{X}) = \frac{\pi_{01}}{\pi_{00}}$. Hence $\Delta = \frac{\frac{\pi_{11}}{\pi_{10}}}{\frac{\pi_{01}}{\pi_{00}}} = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$.
Structural Commentary: $\pi_{1\cdot}, \pi_{0\cdot}$ enter conditional probabilities but cancel after passage to odds.
Key Insight: Δ equals diagonal product $\pi_{11}\pi_{00}$ divided by off-diagonal product $\pi_{01}\pi_{10}$. □

Method 3.4.3. *Estimating an Odds Ratio by Design*
Let $N = (n_{ij})_{i,j\in\{0,1\}}$ be a 2×2 table for (X_i, D_j), where $X_1 := X$, $X_0 := \overline{X}$, $D_1 := D$, and $D_0 := \overline{D}$.
Assume $n_{ij} > 0$, set $n_{i\cdot} := n_{i0} + n_{i1}$, $n_{\cdot j} := n_{0j} + n_{1j}$, $n := \sum_{i=0}^{1}\sum_{j=0}^{1} n_{ij}$.

(i) *Simple Random Sampling*: $\widehat{\pi}_{ij} := \frac{n_{ij}}{n}$ estimates π_{ij}; hence $\widehat{\Delta} := \frac{n_{00}n_{11}}{n_{01}n_{10}}$ estimates Δ.
(ii) *Prospective Sampling*: $n_{0\cdot}, n_{1\cdot}$ fixed; $\frac{n_{01}}{n_{0\cdot}}$, $\frac{n_{11}}{n_{1\cdot}}$ estimate $\mathbb{P}(D \mid \overline{X})$, $\mathbb{P}(D \mid X)$; $\mathbb{P}(X)$ not identified.
(iii) *Retrospective Case-Control Sampling*: $n_{\cdot 0}, n_{\cdot 1}$ are fixed; disease risks are not identified, but $\widehat{\Delta} = \frac{n_{11}}{n_{01}} / \frac{n_{10}}{n_{00}}$ is estimable from exposure odds.

Key Insight: Design-fixed margins cancel in $\widehat{\Delta}$, so odds ratio is estimable even when risks or margins are not.

Example 3.4.4. *Unmatched Case-Control Odds Ratio*
Consider an unmatched case-control table with binary exposure:

	Exposed	Unexposed
Case	67	34
Control	43	64

The estimated odds ratio is $\widehat{\Delta} = \frac{67\cdot 64}{43\cdot 34)} = 2.93$, indicating a positive exposure–disease association.

Method 3.4.5. *Parametric Bootstrap for an Odds-Ratio Standard Error*
Let $a, b, c, d \in \mathbb{N}$ denote the observed counts in rows $D, \overline{D}$ and columns $X, \overline{X}$: $a := n_{DX}$, $b := n_{D\overline{X}}$, $c := n_{\overline{D}X}$, and $d := n_{\overline{D}\overline{X}}$. Set $m_D := a + b$, $m_{\overline{D}} := c + d$, $\widehat{p}_D := \frac{a}{m_D}$, $\widehat{p}_{\overline{D}} := \frac{c}{m_{\overline{D}}}$, and $\widehat{\Delta} := \frac{ad}{bc}$.

(i) *Bootstrap Law*: For $r \in [R]$, draw independently $A_r^* \sim \operatorname{Binomial}(m_D, \widehat{p}_D)$ and $C_r^* \sim \operatorname{Binomial}(m_{\overline{D}}, \widehat{p}_{\overline{D}})$.
(ii) *Replicate*: If $0 < A_r^* < m_D$ and $0 < C_r^* < m_{\overline{D}}$, set $\widehat{\Delta}_r^* := \frac{A_r^*(m_{\overline{D}} - C_r^*)}{(m_D - A_r^*)C_r^*}$.
(iii) *Standard Error*: With $\overline{\Delta}^* := \frac{1}{R}\sum_{r=1}^{R}\widehat{\Delta}_r^*$, set $\widehat{\operatorname{se}}_*(\widehat{\Delta}) := \left(\frac{1}{R-1}\sum_{r=1}^{R}(\widehat{\Delta}_r^* - \overline{\Delta}^*)^2\right)^{1/2}$.

Key Insight: Resample exposure counts conditional on fixed row totals; $\operatorname{sd}\{(\widehat{\Delta}_r^*)_{r\in[R]}\}$ estimates $\operatorname{se}(\widehat{\Delta})$.

Remark 3.4.6. *Odds Ratios Are Not Automatically Risks*
The odds ratio $\Delta = \text{odds}(D \mid X)/\text{odds}(D \mid \overline{X})$ differs from risk contrasts such as $\mathbb{P}(D \mid X) - \mathbb{P}(D \mid \overline{X})$.
In retrospective sampling, $n_{\cdot 0}, n_{\cdot 1}$ are design-fixed, absolute risks are not identified without population margins; Δ remains estimable because those margins cancel in the cross-product ratio.

3.5 Higher-Order Tables and Sampling Design

Two-way tables are the simplest categorical-data structure. Higher-order tables introduce several possible notions of dependence: full independence, conditional independence given a third factor, marginal homogeneity, and interactions among factors.

Definition 3.5.1. *Two-Way Log-Linear Form*
Let $I, J \in \mathbb{N}$ and let $\pi_{ij} > 0$ be two-way cell probabilities for $1 \le i \le I$, $1 \le j \le J$,
with $\pi_{i\cdot} := \sum_{j=1}^{J} \pi_{ij}$, $\pi_{\cdot j} := \sum_{i=1}^{I} \pi_{ij}$, $\alpha_i := \log \pi_{i\cdot}$, and $\beta_j := \log \pi_{\cdot j}$.

(i) *Independence form*: $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$ $\forall 1 \le i \le I$, $1 \le j \le J \iff \log \pi_{ij} = \alpha_i + \beta_j$ $\forall 1 \le i \le I$, $1 \le j \le J$.
(ii) *Interaction terms*: $\gamma_{ij} := \log \pi_{ij} - \alpha_i - \beta_j$, giving $\log \pi_{ij} = \alpha_i + \beta_j + \gamma_{ij}$.

Key Insight: $\gamma_{ij} = 0$ $\forall 1 \le i \le I$, $1 \le j \le J \iff \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$ $\forall 1 \le i \le I$, $1 \le j \le J$.

Remark 3.5.2. *Three-Way Interactions*
Let $\pi_{ijk} > 0$ for $(i, j, k) \in [I] \times [J] \times [K]$. A log-linear model with all two-factor interactions and no three-factor interaction is $\log \pi_{ijk} = \mu + \alpha_i + \beta_j + \eta_k + \delta_{ij} + \varepsilon_{ik} + \gamma_{jk}$.
The absent term λ_{ijk} encodes the constraint of no third-order interaction; nested models are compared by deleting interaction terms and using the corresponding loss of degrees of freedom.

Example 3.5.3. *Graduate Admissions and Aggregation*
In the Berkeley admissions example, the aggregate admission rate for men was higher than for women.
Within the six listed majors, however, women were not consistently admitted at lower rates.
The apparent aggregate disadvantage was driven by application patterns: women applied more often to majors with lower admission rates. Major was therefore confounded with sex in the observational comparison.

Principle 3.5.4. *Choose the Test from the Sampling Mechanism*
For a contingency table $N = (n_{ij})$, let $n_{i\cdot}$ and $n_{\cdot j}$ denote row and column margins.
Select the test from the design-fixed quantities:

(i) *Fixed Margins*: If $(n_{i\cdot})$ and $(n_{\cdot j})$ are fixed in a 2×2 table, use Fisher's exact test.
(ii) *Independent Samples*: If rows are independent multinomial samples with fixed $(n_{i\cdot})$, test homogeneity of column probabilities.
(iii) *Single Multinomial Sample*: If $N \sim \text{Multinomial}(n, (\pi_{ij}))$, test independence, i.e. $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$.
(iv) *Matched Pairs*: If observations are paired counts n_{rs}, test marginal homogeneity using discordant cells.
(v) *Case-Control Sampling*: If $(n_{\cdot 0}, n_{\cdot 1})$ are fixed by disease status, estimate odds ratios. Absolute risks require population margins.

Key Insight: The same numeric table can encode different null hypotheses under different sampling laws.

Remark 3.5.5. *Use Counts, Not Percentages*
Let $Q(O, E) := \sum_i \frac{(O_i - E_i)^2}{E_i}$. Pearson's chi-square calibration applies to counts: $Q(O/n, E/n) = \frac{1}{n} Q(O, E)$ and $Q(100O/n, 100E/n) = \frac{100}{n} Q(O, E)$. Thus relative frequencies or percentages give a rescaled statistic unless the original count scale is restored.

4 Analysis of Variance and Designed Experiments

4.1 One-Way Layouts and the F Test

Analysis of variance compares several mean levels by splitting total variation into variation within treatment groups and variation between treatment groups. The method is a multisample continuation of the equal-variance two-sample procedure, with design and randomisation assumptions kept explicit.

Definition 4.1.1. *Balanced One-Way Layout*
For $I, J \in \mathbb{N}$, one factor has levels $i = 1, \dots, I$, each with exactly J independent replicates Y_{ij}, $j = 1, \dots, J$.
The normal fixed-effects model satisfies $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $\sum_{i=1}^{I} \alpha_i = 0$, and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Structural Commentary: $\mathbb{E}Y_{ij} = \mu + \alpha_i$, and $\sum_{i=1}^{I} \alpha_i = 0$ identifies μ as the average treatment mean.

Key Insight: $H_0 : \alpha_i = 0$ $\forall i$ iff all treatment means are equal.

Definition 4.1.2. *One-Way Sums of Squares*
For a balanced one-way layout, set $\overline{Y}_{i\cdot} := \frac{1}{J} \sum_{j=1}^{J} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ} \sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij}$.

(i) *Total*: $SS_{\text{tot}} := \sum_{i=1}^{I} \sum_{j=1}^{J} (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$; grand-mean dispersion.
(ii) *Within-treatment*: $SS_W := \sum_{i=1}^{I} \sum_{j=1}^{J} (Y_{ij} - \overline{Y}_{i\cdot})^2$; residual dispersion within treatments.
(iii) *Between-treatment*: $SS_B := J \sum_{i=1}^{I} (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$; treatment-mean dispersion scaled by J.

Key Insight: $SS_{\text{tot}} = SS_W + SS_B$ separates within-treatment noise from between-treatment mean differences.

Proposition 4.1.3. *One-Way Sum-of-Squares Decomposition*
For $I, J \in \mathbb{N}$ and observations Y_{ij}, set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$,

$SS_{\text{tot}} := \sum_i\sum_j (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$, $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, *and* $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$. *Then* $SS_{\text{tot}} = SS_W + SS_B$.

Proof. Set $r_{ij} := Y_{ij} - \overline{Y}_{i\cdot}$ and $b_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$. Then $Y_{ij} - \overline{Y}_{\cdot\cdot} = r_{ij} + b_i$ and $\sum_j r_{ij} = 0\ \forall i$.

Thus $SS_{\text{tot}} = \sum_i\sum_j (r_{ij} + b_i)^2 = \sum_i\sum_j r_{ij}^2 + 2\sum_i b_i \sum_j r_{ij} + J\sum_i b_i^2 = SS_W + SS_B$.

Structural Commentary: $Y_{ij} - \overline{Y}_{\cdot\cdot} = r_{ij} + b_i$, where $r_{ij} := Y_{ij} - \overline{Y}_{i\cdot}$ and $b_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$.
Key Insight: $\sum_j r_{ij} = 0\ \forall i$ annihilates the cross term $2\sum_i b_i \sum_j r_{ij}$. □

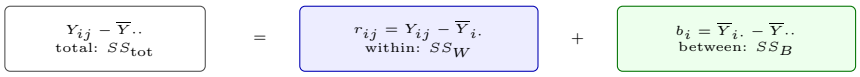

Figure 247: Squaring and summing gives $SS_{\text{tot}} = SS_W + SS_B$; large MS_B/MS_W suggests treatment effects.

Theorem 4.1.4. *Expected One-Way Sums of Squares*
For $I, J \in \mathbb{N}$, let $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = 0$, independent errors satisfying

$\mathbb{E}\varepsilon_{ij} = 0$, $\mathrm{Var}(\varepsilon_{ij}) = \sigma^2$. *Set* $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$.

Then $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$; $\mathbb{E}(SS_W) = I(J-1)\sigma^2$

and $\mathbb{E}(SS_B) = (I-1)\sigma^2 + J\sum_i \alpha_i^2$. *Hence* $S_p^2 := \frac{SS_W}{I(J-1)}$ *satisfies* $\mathbb{E}S_p^2 = \sigma^2$.

Proof. For fixed i, Lemma 3.2.8 with $\mu_{ij} = \mu + \alpha_i$ gives $\mathbb{E}\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2 = (J-1)\sigma^2$, hence $\mathbb{E}SS_W = I(J-1)\sigma^2$.

For $\overline{Y}_{i\cdot}$, $\mathbb{E}\overline{Y}_{i\cdot} = \mu + \alpha_i$, $\mathrm{Var}(\overline{Y}_{i\cdot}) = \frac{\sigma^2}{J}$, and $\frac{1}{I}\sum_i (\mu + \alpha_i) = \mu$. By Lemma 3.2.8, $(\overline{Y}_{i\cdot})_{i=1}^I$ gives

$\mathbb{E}\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2 = \sum_i \alpha_i^2 + \frac{I-1}{J}\sigma^2$, so $\mathbb{E}SS_B = J\sum_i \alpha_i^2 + (I-1)\sigma^2$. Thus $\mathbb{E}S_p^2 = \frac{\mathbb{E}SS_W}{I(J-1)} = \sigma^2$.
Structural Commentary: $\mathbb{E}SS_W$ depends only on σ^2; $\mathbb{E}SS_B$ adds $J\sum_i \alpha_i^2$.
Key Insight: Under $\alpha_i = 0\ \forall i$, $\mathbb{E}\{\frac{SS_B}{I-1}\} = \mathbb{E}\{\frac{SS_W}{I(J-1)}\} = \sigma^2$. □

Theorem 4.1.5. *Normal-Theory One-Way F Test*
For $I, J \in \mathbb{N}$, let $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = 0$ and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$, $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$.

$MS_W := \frac{SS_W}{I(J-1)}$, $MS_B := \frac{SS_B}{I-1}$; $\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}$.

If $H_0 : \alpha_i = 0\ \forall i$, then $\frac{SS_B}{\sigma^2} \sim \chi^2_{I-1}$, $SS_B \perp\!\!\!\perp SS_W$, $F := \frac{MS_B}{MS_W} \sim F_{I-1, I(J-1)}$.

Proof. Let $C_m := \mathrm{Id}_m - m^{-1}\mathbf{1}_m\mathbf{1}_m^\top$, so $C_m^2 = C_m = C_m^\top$ and $\mathrm{rank}\, C_m = m - 1$.
For each i, $Z_i := \frac{1}{\sigma}(Y_{i1} - \mu - \alpha_i, \ldots, Y_{iJ} - \mu - \alpha_i)^\top \sim N_J(0, \mathrm{Id}_J)$ independently,
$\frac{1}{\sigma^2}\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2 = Z_i^\top C_J Z_i \sim \chi^2_{J-1}$. Summing over independent treatments gives $\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}$.
Under H_0, $\widetilde{Z} := \sqrt{J}\sigma^{-1}(\overline{Y}_{1\cdot} - \mu, \ldots, \overline{Y}_{I\cdot} - \mu)^\top \sim N_I(0, \mathrm{Id}_I)$, and $\frac{SS_B}{\sigma^2} = \widetilde{Z}^\top C_I \widetilde{Z} \sim \chi^2_{I-1}$.
For each i, $C_J Z_i \perp\!\!\!\perp \mathbf{1}_J^\top Z_i$ by orthogonal normal components.
Hence $SS_W \perp\!\!\!\perp (\overline{Y}_{1\cdot}, \ldots, \overline{Y}_{I\cdot})$ and $SS_W \perp\!\!\!\perp SS_B$. Thus $F = \frac{SS_B}{I-1} / \frac{SS_W}{I(J-1)} = \frac{MS_B}{MS_W} \sim F_{I-1, I(J-1)}$.
Structural Commentary: $IJ = 1 + I(J-1) + (I-1)$ partitions df into mean, residual, and contrast subspaces.
Key Insight: Under H_0, MS_B and MS_W are independent σ^2 estimators, so $\frac{MS_B}{MS_W}$ has an F law. □

Example 4.1.6. *Generic Balanced One-Way ANOVA*
Let $I = 7$, $J = 10$, and $Y_{ij} = \mu_i + \varepsilon_{ij}$ with independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$. Suppose the ANOVA table is

Component	*df*	*SS*	*MS*	*F*
Treatment	6	.125	.021	5.66
Error	63	.231	.0037	
Total	69	.356		

Then $F = 5.66$ has null law $F_{6,63}$, $p = \mathbb{P}(F_{6,63} \ge 5.66) < .01$, and $H_0 : \mu_1 = \cdots = \mu_7$ is rejected at level .01.
If diagnostics are adequate, between-treatment variation exceeds within-treatment noise.

Method 4.1.7. *Unequal Sample Sizes*

For $I \in \mathbb{N}$, treatment i has observations Y_{ij}, $1 \le j \le J_i$, with $J_i \in \mathbb{N}$ and $N := \sum_{i=1}^I J_i$.

(i) $\overline{Y}_{i\cdot} := \frac{1}{J_i}\sum_{j=1}^{J_i} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{N}\sum_{i=1}^I\sum_{j=1}^{J_i} Y_{ij}$.

(ii) $SS_W := \sum_{i=1}^I\sum_{j=1}^{J_i} (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := \sum_{i=1}^I J_i(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$, $SS_{\text{tot}} := \sum_{i=1}^I\sum_{j=1}^{J_i} (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$
satisfy $SS_{\text{tot}} = SS_W + SS_B$.

(iii) $\mathrm{df}_W = N - I$ and $\mathrm{df}_B = I - 1$.

(iv) If $Y_{ij} = \mu_i + \varepsilon_{ij}$ with independent $\varepsilon_{ij} \sim N(0, \sigma^2)$ and $H_0 : \mu_1 = \cdots = \mu_I$, then $F := \frac{\frac{SS_B}{I-1}}{\frac{SS_W}{N-I}} \sim F_{I-1,N-I}$.

Corollary 4.1.8. *Two Treatments Give $F = t^2$*
Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ be independent normal samples with common variance, means $\overline{X}, \overline{Y}$, pooled variance $S_p^2 := \frac{SS_W}{n+m-2}$, and $t := \frac{\overline{X}-\overline{Y}}{S_p\sqrt{n^{-1}+m^{-1}}}$. The one-way ANOVA statistic for the two groups satisfies $F = t^2$.

Proof. Let $N := n + m$ and $\overline{Z} := \frac{n\overline{X}+m\overline{Y}}{N}$. Then $SS_B = n(\overline{X} - \overline{Z})^2 + m(\overline{Y} - \overline{Z})^2 = \frac{nm}{N}(\overline{X} - \overline{Y})^2$.
Since $df_B = 1$ and $MS_W = \frac{SS_W}{N-2} = S_p^2$, $F = \frac{SS_B}{MS_W} = \frac{(\overline{X}-\overline{Y})^2}{S_p^2(n^{-1}+m^{-1})} = t^2$.
Structural Commentary: Two groups leave one between-group contrast.
Key Insight: ANOVA squares the pooled t contrast. □

The same one-way decomposition also supports a random-effects interpretation, where groups are sampled from a population of possible group levels rather than fixed in advance.

Definition 4.1.9. *Balanced One-Way Random Effects Model*
A *balanced one-way random-effects model* has I groups and common replication J, with $Y_{ij} = \mu + A_i + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, independent A_i satisfying $\mathbb{E}A_i = 0$ and $\mathrm{Var}(A_i) = \sigma_A^2$, and independent errors ε_{ij} independent of (A_i) with $\mathbb{E}\varepsilon_{ij} = 0$ and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$.
Structural Commentary: $\mathrm{Var}(Y_{ij}) = \sigma_A^2 + \sigma_\varepsilon^2$, and $\mathrm{Cov}(Y_{ij}, Y_{ij'}) = \sigma_A^2$ for $j \ne j'$ within the same group.

Proposition 4.1.10. *Random-Effects Expected Mean Squares*
Assume $Y_{ij} = \mu + A_i + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, with independent A_i and ε_{ij}, $\mathbb{E}A_i = \mathbb{E}\varepsilon_{ij} = 0$, $\mathrm{Var}(A_i) = \sigma_A^2$, and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$. Then $\mathbb{E}(MS_W) = \sigma_\varepsilon^2$ and $\mathbb{E}(MS_B) = \sigma_\varepsilon^2 + J\sigma_A^2$.
The method-of-moments estimators are $\widehat{\sigma}_\varepsilon^2 = MS_W$ and $\widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{J}$, truncated at 0 if required.

Proof. Condition on (A_i). Since $Y_{ij} - \overline{Y}_{i\cdot} = \varepsilon_{ij} - \overline{\varepsilon}_{i\cdot}$, $\mathbb{E}(SS_W \mid A) = I(J-1)\sigma_\varepsilon^2$, hence $\mathbb{E}(MS_W) = \sigma_\varepsilon^2$.
Also $\overline{Y}_{i\cdot} = \mu + A_i + \overline{\varepsilon}_{i\cdot}$ has variance $\sigma_A^2 + \frac{\sigma_\varepsilon^2}{J}$.
Hence the one-way between mean square satisfies $\mathbb{E}(MS_B) = J\,\mathrm{Var}(\overline{Y}_{i\cdot}) = \sigma_\varepsilon^2 + J\sigma_A^2$.
Solving these two moment equations gives the estimators.
Structural Commentary: Within estimates error; between adds group heterogeneity.
Key Insight: The excess $MS_B - MS_W$ estimates $J\sigma_A^2$. □

Example 4.1.11. *Dye Strength Batch Variation*
Suppose $I = 6$ batches each have $J = 6$ subsamples, with $\overline{y}_{\cdot\cdot} = 93.61$
and batch means $(92.00, 90.50, 91.25, 98.25, 91.17, 98.50)$.
Given $SS_B = 415.47$ and $SS_W = 134.08$, the mean squares are $MS_B = 83.09$ and $MS_W = 4.47$.
The moment estimates are $\widehat{\sigma}_\varepsilon^2 = MS_W = 4.47$ and $\widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{6} = 13.10$.
Batch variation dominates measurement-error variation.

Remark 4.1.12. *Assumptions and Robustness*
For $Y_{ij} = \mu_i + \varepsilon_{ij}$, the one-way F calibration assumes independent $\varepsilon_{ij} \sim N(0, \sigma^2)$. Balance softens variance heterogeneity; clustered, paired, blocked, or repeated data require dependence modeling.

4.2 Multiple Comparisons After ANOVA

The one-way F test assesses $H_0 : \mu_1 = \cdots = \mu_I$; after rejection, inference targets contrasts $\mu_a - \mu_b$ with error control over the chosen comparison family.

Definition 4.2.1. *Studentized Range Quantile*
Let $Q_{I,\nu}$ denote a random variable with the studentized range distribution for I treatment means and ν error degrees of freedom. The *studentized range quantile* is $q_{I,\nu}(\alpha) := \inf\{q \in \mathbb{R} : \mathbb{P}(Q_{I,\nu} > q) \le \alpha\}$.
Structural Commentary: Balanced Tukey comparisons use pooled scale S_p.

Method 4.2.2. *Tukey Pairwise Comparisons*
In a balanced normal one-way layout, let $\mathcal{P} := \{(a,b) : 1 \le a < b \le I\}$, $\nu := I(J-1)$, and $h_\alpha := q_{I,\nu}(\alpha)S_p/\sqrt{J}$.

(i) For $(a,b) \in \mathcal{P}$, the simultaneous $100(1-\alpha)\%$ confidence interval for $\mu_a - \mu_b$ is $(\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}) \pm h_\alpha$.
(ii) Reject $H_{0,ab} : \mu_a = \mu_b$ iff $\left|\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}\right| > h_\alpha$.

Simultaneity over $\mathcal{P}$ uses the studentized range cutoff.

Example 4.2.3. *Generic Tukey Pairwise Calculation*
Consider a balanced one-way layout with $I = 7$, $J = 10$, $S_p = .06$, and ordered sample means
$(\overline{Y}_{1\cdot}, \overline{Y}_{3\cdot}, \overline{Y}_{7\cdot}, \overline{Y}_{2\cdot}, \overline{Y}_{5\cdot}, \overline{Y}_{6\cdot}, \overline{Y}_{4\cdot}) = (4.062, 4.003, 3.998, 3.997, 3.957, 3.955, 3.920)$.
With $q_{7,60}(.05) = 4.31$, the Tukey half-width is $h_{.05} = 4.31(.06)/\sqrt{10} \approx .082$.
Rejected pairs at simultaneous level .05 are $(1,4), (1,5), (1,6), (3,4)$; remaining pairwise nulls not rejected.

Method 4.2.4. *Bonferroni Pairwise Comparisons*
Let $\mathcal{C} \subseteq \{(a,b) : 1 \le a < b \le I\}$ be the planned pairwise family with $|\mathcal{C}| = k$.
Set $\nu := N - I$ and $h_{ab} := t_{\nu,1-\alpha/(2k)}S_p\sqrt{1/J_a + 1/J_b}$.

(i) For $(a,b) \in \mathcal{C}$, the Bonferroni confidence interval for $\mu_a - \mu_b$ is $(\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}) \pm h_{ab}$.
(ii) Reject $H_{0,ab} : \mu_a = \mu_b$ iff $\left|\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}\right| > h_{ab}$.

The family satisfies FWER $\le \alpha$ and allows unequal treatment sizes through J_a, J_b.

Example 4.2.5. *Generic Bonferroni Pairwise Calculation*
For $I = 7$ treatments and $k = \binom{7}{2} = 21$ planned pairwise comparisons at $\alpha = .05$, take $t_{60,1-.05/(2\cdot 21)} \approx 3.16$.
The resulting simultaneous 95% Bonferroni half-width is $h \approx .085$.
Rejected pairwise nulls are $(1,4), (1,5), (1,6)$; all other planned pairwise nulls are not rejected.

Remark 4.2.6. *The Global Test Does Not Name the Difference*
Rejecting $H_0 : \mu_1 = \cdots = \mu_I$ establishes only $\exists\, a < b$ with $\mu_a \neq \mu_b$.
It does not identify such pairs or provide simultaneous sign guarantees.
Select Tukey, Bonferroni, or planned-contrast procedures according to the preassigned comparison family.

4.3 Rank-Based One-Way ANOVA

When normality or outlier sensitivity is a concern, a one-way comparison can be made after replacing observations by pooled ranks.

Definition 4.3.1. *Kruskal–Wallis Ranks*
Let $\mathcal{Y} := \{Y_{ij} : 1 \le i \le I, 1 \le j \le J_i\}$ be an unequal-size one-way sample with $N := \sum_{i=1}^{I} J_i$.
Assign increasing pooled ranks to $\mathcal{Y}$, averaging ranks over ties.

(i) *Pooled rank*: $R_{ij} := \mathrm{rank}_{\mathrm{pool}}(Y_{ij})$.

(ii) *Treatment mean rank*: $\overline{R}_{i\cdot} := \frac{1}{J_i} \sum_{j=1}^{J_i} R_{ij}$.

(iii) *Grand mean rank*: $\overline{R}_{\cdot\cdot} := \frac{N+1}{2}$.

Method 4.3.2. *Kruskal–Wallis Test*
Let $SS_B^{(R)} := \sum_{i=1}^{I} J_i(\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2$ and $K := \frac{12}{N(N+1)} SS_B^{(R)} = \frac{12}{N(N+1)} \sum_{i=1}^{I} J_i \overline{R}_{i\cdot}^{\,2} - 3(N+1)$.

(i) The exact null law enumerates all allocations of pooled ranks to group sizes $J_1, \ldots, J_I$.

(ii) For moderate samples, $K \overset{H_0}{\approx} \chi^2_{I-1}$.

(iii) Reject $H_0 : F_1 = \cdots = F_I$ at level α when $\mathbb{P}\{\chi^2_{I-1} \ge K_{\mathrm{obs}}\} \le \alpha$.

Example 4.3.3. *Generic Kruskal–Wallis Rank Calculation*
For $I = 7$ groups, suppose the pooled ranks give $K_{\mathrm{obs}} = 29.51$.
Using $K \overset{H_0}{\approx} \chi^2_6$, the tail probability satisfies $\mathbb{P}\{\chi^2_6 \ge 29.51\} < .005$.
Reject $H_0 : F_1 = \cdots = F_7$; the group response distributions are not all equal.

Corollary 4.3.4. *Two Groups Recover Mann–Whitney*
For $I = 2$, let $T_i := \sum_{j=1}^{J_i} R_{ij}$ and $N = J_1 + J_2$. Then $K = \frac{12}{J_1 J_2 (N+1)} \left(T_1 - \frac{J_1(N+1)}{2}\right)^2$, so the Kruskal–Wallis rejection order is the two-sided Mann–Whitney rank-sum order.

Proof. Let $G := \frac{N+1}{2}$. Since $T_1 + T_2 = NG$, $\overline{R}_{1\cdot} - G = \frac{T_1 - J_1 G}{J_1}$ and $\overline{R}_{2\cdot} - G = -\frac{T_1 - J_1 G}{J_2}$.
Hence $SS_B^{(R)} = (T_1 - J_1 G)^2 (\frac{1}{J_1} + \frac{1}{J_2}) = \frac{N(T_1 - J_1 G)^2}{J_1 J_2}$, and the formula for K follows.
Since $U_1 = T_1 - \frac{J_1(J_1+1)}{2}$ is affine in T_1, both tests order evidence by $|T_1 - J_1 G|$.
Structural Commentary: Two groups leave one rank contrast.
Key Insight: K is a squared centered rank sum. □

Method 4.3.5. *Rank-Based Follow-Up Comparisons*
After rejecting global rank null, let $\mathcal{C} \subseteq \{(a,b) : 1 \le a < b \le I\}$ be planned follow-up family with $|\mathcal{C}| = k$.

(i) For $(a,b) \in \mathcal{C}$, test $H_{0,ab} : F_a = F_b$ by a two-sample rank-sum statistic and let p_{ab} be its two-sided p-value.

(ii) Reject $H_{0,ab}$ iff $p_{ab} \le \frac{\alpha}{k}$, equivalently $k p_{ab} \le \alpha$.

(iii) The selected family satisfies FWER $\le \alpha$ and is most efficient when $\mathcal{C}$ is fixed before inspection.

Remark 4.3.6. *What Rank ANOVA Tests*
The null is $H_0 : F_1 = \cdots = F_I$, not merely $\mu_1 = \cdots = \mu_I$. Under comparable distributional shapes, rejection is read as a location difference; the rank form avoids normality and reduces sensitivity to extremes.

4.4 Two-Way Layouts and Interaction

A two-way layout studies two factors at once. It estimates main effects for each factor and checks whether the effect of one factor changes across levels of the other.

Definition 4.4.1. *Two-Way Additive Parametrization*
For one observation Y_{ij} at factor levels $i = 1, \ldots, I$ and $j = 1, \ldots, J$,
set $\overline{Y}_{i\cdot} := J^{-1} \sum_{j=1}^{J} Y_{ij}$, $\overline{Y}_{\cdot j} := I^{-1} \sum_{i=1}^{I} Y_{ij}$, and $\overline{Y}_{\cdot\cdot} := (IJ)^{-1} \sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij}$.

(i) *Intercept*: $\widehat{\mu} := \overline{Y}_{\cdot\cdot}$.

(ii) *Factor A, B coefficient*: $\widehat{\alpha}_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$, $1 \le i \le I$ and $\widehat{\beta}_j := \overline{Y}_{\cdot j} - \overline{Y}_{\cdot\cdot}$, $1 \le j \le J$.

(iii) *Additive fitted value*: $\widehat{Y}_{ij} := \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j$.

(iv) *Interaction residual*: $\widehat{\delta}_{ij} := Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}_{\cdot\cdot}$, so $Y_{ij} = \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j + \widehat{\delta}_{ij}$.

Structural Commentary: Exact fits satisfy zero-sum effects and margins: $\sum_i \widehat{\alpha}_i = \sum_j \widehat{\beta}_j = \sum_i \widehat{\delta}_{ij} = \sum_j \widehat{\delta}_{ij} = 0$.

Remark 4.4.2. *Interaction and Parallel Profiles*
Under additivity, $\widehat{Y}_{ij} - \widehat{Y}_{i'j} = \widehat{\alpha}_i - \widehat{\alpha}_{i'}$ is independent of j, so factor-A profiles are parallel across B. Nonparallel profiles indicate $\delta_{ij} \neq 0$ structure: the contrast between factor levels depends on the conditioning level.

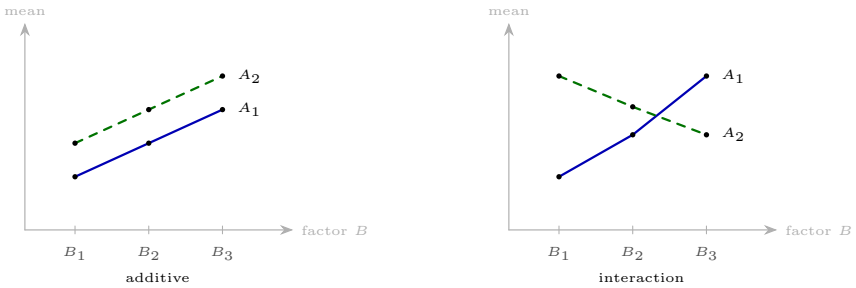

Figure 248: Parallel profiles indicate additivity; changing gaps indicate interaction.

Definition 4.4.3. *Balanced Two-Way Normal Model*
A *balanced two-way normal fixed-effects model* observes Y_{ijk} for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$, with common cell size $K > 1$, and assumes $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ for mutually independent $\varepsilon_{ijk} \sim N(0, \sigma^2)$,
under $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$, and $\sum_j \delta_{ij} = 0\ \forall i$.
Key Insight: $\delta_{ij} = 0\ \forall (i,j)$ iff the cell means are additive in the two factors.

Definition 4.4.4. *Two-Way Sums of Squares*
Let Y_{ijk} be balanced observations for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$.

(i) *Means*: $\overline{Y}_{ij\cdot} := \frac{1}{K}\sum_k Y_{ijk}$, $\overline{Y}_{i\cdot\cdot} := \frac{1}{JK}\sum_j\sum_k Y_{ijk}$, $\overline{Y}_{\cdot j\cdot} := \frac{1}{IK}\sum_i\sum_k Y_{ijk}$, $\overline{Y}_{\cdots} := \frac{1}{IJK}\sum_i\sum_j\sum_k Y_{ijk}$.

(ii) *Between-cell components*: $SS_A := JK\sum_i(\overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdots})^2$, $SS_B := IK\sum_j(\overline{Y}_{\cdot j\cdot} - \overline{Y}_{\cdots})^2$,
and $SS_{AB} := K\sum_i\sum_j(\overline{Y}_{ij\cdot} - \overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdot j\cdot} + \overline{Y}_{\cdots})^2$.

(iii) *Within-cell and total*: $SS_E := \sum_i\sum_j\sum_k(Y_{ijk} - \overline{Y}_{ij\cdot})^2$, $SS_{\text{tot}} := \sum_i\sum_j\sum_k(Y_{ijk} - \overline{Y}_{\cdots})^2$,
and $SS_{\text{tot}} = SS_A + SS_B + SS_{AB} + SS_E$.

Structural Commentary: Orthogonal projections split total variation.

Theorem 4.4.5. *Expected Two-Way Sums of Squares*
Assume $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ with independent $\varepsilon_{ijk} \sim N(0,\sigma^2)$,
and $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$; $\sum_j \delta_{ij} = 0\ \forall i$.

(i) $\mathbb{E}(SS_A) = (I-1)\sigma^2 + JK\sum_i \alpha_i^2$.

(ii) $\mathbb{E}(SS_B) = (J-1)\sigma^2 + IK\sum_j \beta_j^2$.

(iii) $\mathbb{E}(SS_{AB}) = (I-1)(J-1)\sigma^2 + K\sum_i\sum_j \delta_{ij}^2$.

(iv) $\mathbb{E}(SS_E) = IJ(K-1)\sigma^2$.

Proof. Let $Y = m + \varepsilon \in \mathbb{R}^{IJK}$ with $m_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij}$ and $\varepsilon \sim N(0, \sigma^2 I_{IJK})$.
For an orthogonal projection P, $\mathbb{E}\,\|PY\|^2 = \|Pm\|^2 + \sigma^2 \operatorname{rank} P$.

(i) $P_A m$ has value α_i repeated over j,k, so $\|P_A m\|^2 = JK\sum_i \alpha_i^2$ and $\operatorname{rank} P_A = I - 1$.

(ii) $P_B m$ has value β_j repeated over i,k, so $\|P_B m\|^2 = IK\sum_j \beta_j^2$ and $\operatorname{rank} P_B = J - 1$.

(iii) $P_{AB} m$ has value δ_{ij} repeated over k, so $\|P_{AB} m\|^2 = K\sum_i\sum_j \delta_{ij}^2$ and $\operatorname{rank} P_{AB} = (I-1)(J-1)$.

(iv) P_E centers within cells; since m_{ijk} is constant in k for fixed (i,j), $P_E m = 0$ and $\operatorname{rank} P_E = IJ(K-1)$.

Applying the projection identity to P_A, P_B, P_{AB}, P_E gives the four expectations.
Structural Commentary: Balanced orthogonality separates noise degrees of freedom from effect energy.
Key Insight: Each expectation is σ^2 times rank plus squared signal norm. □

Theorem 4.4.6. *Two-Way Normal-Theory F Tests*
Assume observations Y_{ijk} for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$ satisfy $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$
with independent $\varepsilon_{ijk} \sim N(0,\sigma^2)$, $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$, and $\sum_j \delta_{ij} = 0\ \forall i$.
Let $d_E := IJ(K-1)$ and $MS_E := \frac{SS_E}{d_E}$.

(i) $SS_A, SS_B, SS_{AB}, SS_E$ are independent, and $\frac{SS_E}{\sigma^2} \sim \chi^2_{d_E}$.

(ii) Under $H_A : \alpha_i = 0\ \forall i$, $\frac{SS_A}{\sigma^2} \sim \chi^2_{I-1}$; under $H_B : \beta_j = 0\ \forall j$, $\frac{SS_B}{\sigma^2} \sim \chi^2_{J-1}$;
under $H_{AB} : \delta_{ij} = 0\ \forall (i,j)$, $\frac{SS_{AB}}{\sigma^2} \sim \chi^2_{(I-1)(J-1)}$.

(iii) $F_A := \frac{\frac{SS_A}{I-1}}{MS_E} \sim F_{I-1,d_E}$ under H_A, $F_B := \frac{\frac{SS_B}{J-1}}{MS_E} \sim F_{J-1,d_E}$ under H_B,
and $F_{AB} := \frac{\frac{SS_{AB}}{(I-1)(J-1)}}{MS_E} \sim F_{(I-1)(J-1),d_E}$ under H_{AB}.

Proof.

(i) Vectorise $Y = m + \varepsilon \in \mathbb{R}^{IJK}$ with $\varepsilon \sim N(0, \sigma^2 I_{IJK})$.
Let P_A, P_B, P_{AB}, P_E be the mutually orthogonal projections onto the factor-A, factor-B, interaction, and within-cell subspaces, so $SS_R = \|P_R Y\|^2$ for $R \in \{A, B, AB, E\}$.

Orthogonal projections of a spherical normal vector are independent, and $\frac{\|P_R Y\|^2}{\sigma^2} \sim \chi^2_{d_R}(\lambda_R)$ with $d_R = \operatorname{rank} P_R$ and $\lambda_R = \frac{\|P_R m\|^2}{\sigma^2}$. Since $d_E = IJ(K-1)$ and $\lambda_E = 0$, claim (i) follows.

(ii) Here $d_A = I-1$, $d_B = J-1$, and $d_{AB} = (I-1)(J-1)$. Under H_A, H_B, H_{AB} respectively, $\lambda_A, \lambda_B, \lambda_{AB}$ vanish, so the central chi-square laws in claim (ii) follow.

(iii) In each null case the numerator chi-square in claim (ii) is independent of SS_E by claim (i); dividing by its degrees of freedom and by $\frac{SS_E}{d_E} = MS_E$ gives the stated F ratios in claim (iii).

Structural Commentary: Balanced normal ANOVA is an orthogonal Gaussian projection decomposition.
Key Insight: Null effects remove noncentrality, leaving independent chi-square ratios. □

Example 4.4.7. *Generic Transformed Two-Way ANOVA*
Consider balanced two-way study with factor A having 2 levels, factor B having 3 levels, $K = 18$ observations per cell. After log transformation stabilises cell spread, ANOVA table on transformed scale is

Source	*df*	*SS*	*MS*	*F*
Factor A	1	2.074	2.074	5.99
Factor B	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

For factor A, $F_A = 5.99$ gives $p < .025$, and an estimated log contrast .28 corresponds to the original-scale multiplier $e^{.28} \approx 1.32$ with approximate interval $(1.06, 1.65)$. For interaction, $F_{AB} = 1.17$ gives $p = .31$, so there is no strong evidence that the A contrast depends on the level of B.

Remark 4.4.8. *Transformations and Diagnostics*
Two-way ANOVA inherits the normal-theory assumptions of the one-way model. Residual plots, normal probability plots, profile plots, and plots of cell spread against cell mean help decide whether a transformation is needed. A transformation can stabilize variance and improve symmetry, but it also changes additive effects on the transformed scale into multiplicative or otherwise nonlinear effects on the original scale.

Remark 4.4.9. *Factorial Economy and Incomplete Designs*
Factorial designs use each observation to estimate multiple factor contrasts and interactions.
As the factor set grows, full factorials may be replaced by fractional factorials that omit cells while preserving chosen estimable effects; when blocks cannot contain all treatments, incomplete block designs retain partial blocking without all block-treatment pairs.

4.5 Randomised Blocks and Friedman's Test

Randomised block designs compare treatments within relatively homogeneous blocks. Blocking removes major background variation from the treatment comparison, much as pairing removes shared variation.

Definition 4.5.1. *Randomised Block Design*
A randomised block design has J blocks, each containing one experimental unit for each of the I treatments; within every block, treatments are assigned by an independent random permutation.
The additive block model is $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = \sum_j \beta_j = 0$.

Key Insight: Blocks absorb heterogeneity; $\alpha_i - \alpha_{i'}$ is a within-block treatment contrast.

Method 4.5.2. *Randomized Block F Test*
Treat $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ as a two-way layout with one observation per treatment-block cell and no interaction.

(i) Define $MS_A := \frac{SS_A}{I-1}$, $MS_B := \frac{SS_B}{J-1}$, and $MS_{AB} := \frac{SS_{AB}}{(I-1)(J-1)}$.

(ii) Under additivity, $\mathbb{E}(MS_A) = \sigma^2 + \frac{J}{I-1}\sum_i \alpha_i^2$, $\mathbb{E}(MS_B) = \sigma^2 + \frac{I}{J-1}\sum_j \beta_j^2$, and $\mathbb{E}(MS_{AB}) = \sigma^2$.

(iii) Test $H_0 : \alpha_1 = \cdots = \alpha_I = 0$ by $F := \frac{MS_A}{MS_{AB}}$.

(iv) Under the normal additive null, $F \sim F_{I-1,(I-1)(J-1)}$.

Remark 4.5.3. *Interaction in a Block Design*
If $\delta_{ij} \neq 0$, then MS_{AB} estimates σ^2 plus treatment-block interaction energy.
The denominator of $F = \frac{MS_A}{MS_{AB}}$ is inflated, so the additive-model treatment test is conservative.

Example 4.5.4. *Generic Randomized Block ANOVA*
Suppose $I = 7$ treatments are compared across $J = 10$ blocks, with each block receiving all treatments in randomized order. The randomized-block ANOVA table is

Source	*df*	*SS*	*MS*	*F*
Treatments	6	53013	8835	2.85
Blocks	9	103280	11476	3.71
Interaction	54	167130	3095	
Total	69	323422		

$F = 2.85$ on $(6, 54)$ degrees of freedom gives $p < .025$, rejecting equal treatment effects.
Blocking isolates nuisance block heterogeneity.

Method 4.5.5. *Friedman's Rank Test for Blocks*
For observations Y_{ij} from treatment i in block j, rank $Y_{1j}, \ldots, Y_{Ij}$ within each block.

(i) Let R_{ij} be within-block rank, use average ranks for ties, set $\overline{R}_{i\cdot} := \frac{1}{J}\sum_j R_{ij}$, $\overline{R}_{\cdot\cdot} := \frac{I+1}{2}$.

(ii) Define $Q := \frac{12J}{I(I+1)}\sum_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2$.

(iii) Under $H_0 : F_1 = \cdots = F_I$, use $Q \overset{H_0}{\approx} \chi^2_{I-1}$ for moderate J.

(iv) Reject at level α when $\mathbb{P}\{\chi^2_{I-1} \ge Q_{\text{obs}}\} \le \alpha$.

Example 4.5.6. *Generic Friedman Rank Calculation*
For $I = 7$ treatments, suppose the average within-block ranks are

Treatment	T_1	T_2	T_3	T_4	T_5	T_6	T_7
Average rank	5.10	4.90	2.30	3.50	3.05	4.90	4.25

With $\overline{R}_{\cdot\cdot} = 4$, $\sum_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2 = 6.935$ and $Q = 14.86$.
Using $Q \overset{H_0}{\approx} \chi^2_6$, the tail probability is below .025, so the treatment response distributions are not all equal.

Corollary 4.5.7. *Two Treatments Recover the Sign Test*
For $I = 2$, set $D_j := Y_{1j} - Y_{2j}$ and $B := \#\{j : D_j > 0\}$. Friedman's statistic is a function of B alone, so the two-treatment Friedman test is equivalent to the paired sign test.

Proof. Ignoring ties, $R_{1j} = 2\mathbf{1}_{\{D_j>0\}} + \mathbf{1}_{\{D_j<0\}} = 1 + \mathbf{1}_{\{D_j>0\}}$.
Hence $\sum_j R_{1j} = J + B$, so $\overline{R}_{1\cdot}$, $\overline{R}_{2\cdot}$, and Q are determined by B.
Under H_0, signs are exchangeable and $B \sim \mathrm{Bin}(J, 1/2)$, which is the paired sign-test calibration.
Structural Commentary: Two within-block ranks encode only the sign of D_j.
Key Insight: The rank-sum statistic collapses to a binomial sign count. □

5 Linear Models and Regression

5.1 Least Squares Curve Fitting and Linearization

Least squares begins as a geometric fitting rule and becomes a statistical model only after assumptions are made about the errors. The optimization step is asymmetric in the two plotted variables: ordinary regression measures vertical prediction errors.

Definition 5.1.1. *Least-Squares Fit to a Parametric Family*
Let $[n] := \{1, \dots, n\}$, let $(x_i, y_i)_{i\in[n]} \subseteq \mathcal{X} \times \mathbb{R}$, let $\Theta \subseteq \mathbb{R}^p$, and let $m : \mathcal{X} \times \Theta \to \mathbb{R}$.

(i) *Criterion*: $S : \Theta \to [0, \infty)$, $S(\theta) := \sum_{i=1}^{n} \big(y_i - m(x_i; \theta)\big)^2$.

(ii) *Estimate*: $\widehat{\theta} \in \arg\min_{\theta\in\Theta} S(\theta)$.

(iii) *Residual*: $\widehat{e}_i := y_i - m(x_i; \widehat{\theta})$, $i \in [n]$.

Key Insight: $y_i = m(x_i; \widehat{\theta}) + \widehat{e}_i$, so $\widehat{e}_i > 0$ iff $y_i > m(x_i; \widehat{\theta})$.

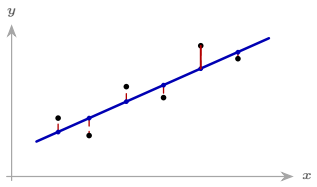

Figure 249: Least squares minimizes squared vertical residuals.

Definition 5.1.2. *Linear in the Parameters*
Let $\mathcal{X}$ be an input set, let $\Theta \subseteq \mathbb{R}^p$, and let $m : \mathcal{X} \times \Theta \to \mathbb{R}$. m is linear in β if and only if
$\exists$ fixed maps $\phi_0, \dots, \phi_{p-1} : \mathcal{X} \to \mathbb{R}$ with $m(x; \beta) = \sum_{j=0}^{p-1} \beta_j \phi_j(x)$ $\forall x \in \mathcal{X}$ and $\beta \in \Theta$.

Structural Commentary: Polynomial regression uses $\phi_j(x) = x^j$ and estimates only the coefficients β_j.

Proposition 5.1.3. *Least-Squares Line*
Let $[n] := \{1, \dots, n\}$, $(x_i, y_i)_{i\in[n]} \subseteq \mathbb{R}^2$, $\overline{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$, $\overline{y} := \frac{1}{n}\sum_{i=1}^{n} y_i$, assume $D := \sum_{i=1}^{n} (x_i - \overline{x})^2 > 0$.

For $b = (b_0, b_1) \in \mathbb{R}^2$, set $S(b) := \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$.

Then $\exists! \widehat{\beta} = (\widehat{\beta}_0, \widehat{\beta}_1) \in \arg\min_{b\in\mathbb{R}^2} S(b)$; $\widehat{y}(x) := \widehat{\beta}_0 + \widehat{\beta}_1 x$ satisfies $\widehat{\beta}_1 = \frac{1}{D}\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})$, $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$.

Proof. Set $u_i := x_i - \overline{x}$, $v_i := y_i - \overline{y}$, and $c(b) := b_0 + b_1\overline{x} - \overline{y}$.
Since $\sum_{i=1}^{n} u_i = \sum_{i=1}^{n} v_i = 0$, $S(b) = \sum_{i=1}^{n} (v_i - b_1 u_i)^2 + n c(b)^2$.
Thus $c(\widehat{\beta}) = 0$, so $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1\overline{x}$, and $\widehat{\beta}_1$ uniquely minimizes $Q(t) := \sum_{i=1}^{n} (v_i - t u_i)^2$.
Since $Q'(t) = -2\sum_{i=1}^{n} u_i(v_i - t u_i)$ and $\sum_{i=1}^{n} u_i^2 = D > 0$, the unique critical point is $t = \frac{1}{D}\sum_{i=1}^{n} u_i v_i$.
Structural Commentary: The fitted line passes through $(\overline{x}, \overline{y})$.
Key Insight: Centering turns the slope into covariance divided by variance. □

Example 5.1.4. *Linear and Nonlinear Parameters*
For fixed $\omega = (\omega_1, \omega_2)$, $f_{\theta,\omega}(t) := A\cos(\omega_1 t) + B\sin(\omega_1 t) + C\cos(\omega_2 t) + D\sin(\omega_2 t) = \theta^t \phi_\omega(t)$
with $\theta = (A, B, C, D)^t$ and $\phi_\omega(t) = (\cos(\omega_1 t), \sin(\omega_1 t), \cos(\omega_2 t), \sin(\omega_2 t))^t$.
Estimating ω makes $f_{\theta,\omega}$ nonlinear in (θ, ω).
For fixed $\lambda = (\alpha, \beta)$, $g_{\eta,\lambda}(t) := Ae^{-\alpha t} + Be^{-\beta t} = \eta^t \psi_\lambda(t)$ with $\eta = (A, B)^t$ and $\psi_\lambda(t) = (e^{-\alpha t}, e^{-\beta t})^t$.
Estimating λ makes $g_{\eta,\lambda}$ nonlinear in (η, λ).

Method 5.1.5. *Linearizing Transformations*
Let $[n] := \{1, \dots, n\}$, $K > 0$, $C > 0$, $e_A \in \mathbb{R}$, $T > 0$, and $\alpha(T) := C \exp\left(-\frac{e_A}{KT}\right)$.
For observations $(T_i, \alpha_i)_{i \in [n]} \subset (0, \infty)^2$, set $x := \frac{1}{T}$, $z := \log \alpha(T)$, $x_i := \frac{1}{T_i}$, and $z_i := \log \alpha_i$.

(i) $z = a + bx$ with $a := \log C$ and $b := -\frac{e_A}{K}$.

(ii) $(\widehat{a}, \widehat{b}) \in \underset{(a,b) \in \mathbb{R}^2}{\arg\min} \sum_{i=1}^{n} (z_i - a - bx_i)^2$, $\widehat{C} := e^{\widehat{a}}$, and $\widehat{e}_A := -K\widehat{b}$.

(iii) $\widehat{\varepsilon}_i := z_i - \widehat{a} - \widehat{b}x_i$ is diagnosed against x_i and $\widehat{z}_i := \widehat{a} + \widehat{b}x_i$.
Systematic structure indicates misspecification on the transformed scale.

5.2 Simple Linear Regression Inference and Diagnostics

Simple linear regression adds a noise model to the least-squares line. The x_i values are treated as fixed design points, and randomness enters through the observed responses.

Definition 5.2.1. *Standard Simple Linear Regression Model*
Let $[n] := \{1, \dots, n\}$, let $x_i \in \mathbb{R}$ be fixed for $i \in [n]$, let $\beta_0, \beta_1 \in \mathbb{R}$, and let $\sigma^2 > 0$.

(i) *Standard model*: $Y_i = \beta_0 + \beta_1 x_i + e_i$, $\mathbb{E}e_i = 0$, $\mathrm{Var}(e_i) = \sigma^2$, and $\mathrm{Cov}(e_i, e_j) = 0$ for $i \neq j$, $i, j \in [n]$.

(ii) *Normal model*: $Y_i = \beta_0 + \beta_1 x_i + e_i$ with $e_1, \dots, e_n$ independent and $e_i \sim N(0, \sigma^2)$, $i \in [n]$.

Structural Commentary: $(x_i)_{i \in [n]}$ is fixed; randomness enters through $(e_i)_{i \in [n]}$.

Theorem 5.2.2. *Mean and Variance of the Simple Regression Coefficients*
Let $[n] := \{1, \dots, n\}$, let $x_i \in \mathbb{R}$, and set $\overline{x} := \frac{1}{n} \sum_{i=1}^{n} x_i$ and $D := \sum_{i=1}^{n} (x_i - \overline{x})^2 > 0$.
Let $\beta := (\beta_0, \beta_1)$ and assume $Y_i = \beta_0 + \beta_1 x_i + e_i$, $\mathbb{E}e_i = 0$, $\mathrm{Var}(e_i) = \sigma^2$, and $\mathrm{Cov}(e_i, e_j) = 0$ for $i \neq j$.
If $\widehat{\beta} = (\widehat{\beta}_0, \widehat{\beta}_1)$ minimizes $\sum_{i=1}^{n} (Y_i - b_0 - b_1 x_i)^2$ over $b = (b_0, b_1) \in \mathbb{R}^2$, then $\mathbb{E}\widehat{\beta} = \beta$; $\mathrm{Var}(\widehat{\beta}) = \sigma^2 \begin{pmatrix} \frac{1}{n} + \frac{\overline{x}^2}{D} & -\frac{\overline{x}}{D} \\ -\frac{\overline{x}}{D} & \frac{1}{D} \end{pmatrix}$.

Proof. Set $u_i := x_i - \overline{x}$ and $\overline{e} := \frac{1}{n} \sum_{i=1}^{n} e_i$.

Proposition 5.1.3 gives $\widehat{\beta}_1 - \beta_1 = D^{-1} \sum_{i=1}^{n} u_i e_i$ and $\widehat{\beta}_0 - \beta_0 = \overline{e} - \overline{x}(\widehat{\beta}_1 - \beta_1)$.

Because $\sum_{i=1}^{n} u_i = 0$, $\mathbb{E}(\widehat{\beta} - \beta) = 0$, $\mathrm{Var}(\widehat{\beta}_1) = D^{-2} \sum_{i=1}^{n} u_i^2 \sigma^2 = \frac{\sigma^2}{D}$, $\mathrm{Cov}(\overline{e}, \widehat{\beta}_1) = \frac{1}{n} D^{-1} \sum_{i=1}^{n} u_i \sigma^2 = 0$, and $\mathrm{Var}(\overline{e}) = \frac{\sigma^2}{n}$. Therefore $\mathrm{Var}(\widehat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\overline{x}^2 \sigma^2}{D}$ and $\mathrm{Cov}(\widehat{\beta}_0, \widehat{\beta}_1) = -\frac{\overline{x}\sigma^2}{D}$.
Structural Commentary: The slope is a fixed linear combination of the responses.
Key Insight: Centering the design isolates the uncertainty in the slope. □

Definition 5.2.3. *Residual Sum of Squares and Error Scale*
Let $[n] := \{1, \dots, n\}$ with $n > 2$, and let $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ for $i \in [n]$.

(i) *Residual Sum of Squares*: $\mathrm{RSS} := \sum_{i=1}^{n} (Y_i - \widehat{y}_i)^2$.

(ii) *Error Scale*: $s^2 := \frac{\mathrm{RSS}}{n-2}$ and $s := \sqrt{s^2}$.

Key Insight: $n - 2$ subtracts the fitted coefficients $\widehat{\beta}_0, \widehat{\beta}_1$.

Corollary 5.2.4. *Simple Regression t Intervals and Tests*
Let $n > 2$, $[n] := \{1, \dots, n\}$, $\alpha \in (0, 1)$, $\beta \in \mathbb{R}^2$, $\sigma^2 > 0$, $x = (x_1, \dots, x_n)^t \in \mathbb{R}^n$, $X := (\mathbf{1}, x) \in \mathbb{R}^{n \times 2}$ with $\mathrm{rank}(X) = 2$, $Y = X\beta + e$, and $e \sim N(0, \sigma^2 I_n)$.
Set $\widehat{\beta} := (X^t X)^{-1} X^t Y$, $s^2 := \frac{\|Y - X\widehat{\beta}\|_2^2}{n-2}$, $\overline{x} := \frac{1}{n} \sum_{i=1}^{n} x_i$, $D := \sum_{i=1}^{n} (x_i - \overline{x})^2$, $c_0 := \frac{1}{n} + \frac{\overline{x}^2}{D}$, and $c_1 := D^{-1}$.
For $j \in \{0, 1\}$ and $\beta_{j0} \in \mathbb{R}$, set $s_{\widehat{\beta}_j} := s\sqrt{c_j}$.
Then $T_j := \frac{\widehat{\beta}_j - \beta_j}{s_{\widehat{\beta}_j}} \sim t_{n-2}$, $C_j(Y) := [\widehat{\beta}_j - t_{n-2,1-\frac{\alpha}{2}} s_{\widehat{\beta}_j}, \widehat{\beta}_j + t_{n-2,1-\frac{\alpha}{2}} s_{\widehat{\beta}_j}]$ satisfies $\mathbb{P}_\beta(\beta_j \in C_j(Y)) = 1 - \alpha$,
and the two-sided level-α test of $H_0 : \beta_j = \beta_{j0}$ rejects iff $\left| \frac{\widehat{\beta}_j - \beta_{j0}}{s_{\widehat{\beta}_j}} \right| > t_{n-2,1-\frac{\alpha}{2}}$.

Proof. Let $P := X(X^t X)^{-1} X^t$, $M := I - P$, and $G := (X^t X)^{-1} = (g_{jk})_{j,k=0}^{1}$.
Then $\widehat{\beta} - \beta = GX^t e$, $g_{00} = c_0$, $g_{11} = c_1$, $\mathrm{Var}(\widehat{\beta}_j) = \sigma^2 c_j$, and $Z_j := \frac{\widehat{\beta}_j - \beta_j}{\sigma\sqrt{c_j}} \sim N(0, 1)$.
Also $\frac{(n-2)s^2}{\sigma^2} = \frac{e^t M e}{\sigma^2}$, $M = M^t = M^2$, $\mathrm{rank}(M) = n - 2$, and $MX = 0$, hence $U := \frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2}$ and $\mathrm{Cov}(GX^t e, Me) = GX^t \sigma^2 M = 0$. Joint normality gives $Z_j \perp U$, so $T_j = \frac{Z_j}{\sqrt{\frac{U}{n-2}}} \sim t_{n-2}$.
The interval and rejection rule follow from $\mathbb{P}(|T_j| \leq t_{n-2,1-\frac{\alpha}{2}}) = 1 - \alpha$.
Structural Commentary: $T_j = \frac{Z_j}{\sqrt{\frac{U}{n-2}}}$ with $Z_j \sim N(0, 1)$, $U \sim \chi^2_{n-2}$, and $Z_j \perp U$.
Key Insight: Normality and orthogonal residual projection make replacing σ by s exact. □

Example 5.2.5. *Calibration by Peak Area*
Let $[n] := \{1, \dots, n\}$, with known standard concentrations x_i and measured peak areas A_i, $i \in [n]$.
Fit $A_i = \beta_0 + \beta_1 x_i + e_i$ by least squares, giving $\widehat{A}(x) = \widehat{\beta}_0 + \widehat{\beta}_1 x$.
For a new area A_*, the inverse calibration estimate is $\widehat{x}_* := \frac{A_* - \widehat{\beta}_0}{\widehat{\beta}_1}$ when $\widehat{\beta}_1 \neq 0$.
In the reported fit, $\widehat{\beta}_0 = .0729$, $\widehat{\beta}_1 = 10.77$, $s = .068$, $\mathrm{se}(\widehat{\beta}_0) = .0297$, and $\mathrm{se}(\widehat{\beta}_1) = .27$, hence $t_0 = 2.45$, $t_1 = 40.20$, and the reported

95% intervals are $C_0 = (.011, .135)$ and $C_1 = (10.21, 11.33)$.
Thus $\widehat{\beta}_1 > 0$ makes peak area a monotone calibration signal.

Definition 5.2.6. *Homoscedastic and Heteroscedastic Errors*
Let $[n] := \{1, \dots, n\}$, let $e = (e_1, \dots, e_n)^t$ be errors at design points x_i, and set $v_i := \mathrm{Var}(e_i)$.

(i) *Homoscedastic*: $\exists \sigma^2 \geq 0$ with $v_i = \sigma^2$, $i \in [n]$.
(ii) *Heteroscedastic*: $\exists i, j \in [n]$ with $v_i \neq v_j$.

Key Insight: $\mathrm{Var}(e) \neq \sigma^2 I_n$ invalidates formulas requiring $\mathrm{Var}(e) = \sigma^2 I_n$.

Method 5.2.7. *Residual Diagnostics*
After fitting a regression, examine residuals against the fitted values and against important predictors.

(i) A random horizontal band supports the chosen mean form.
(ii) Curvature suggests a transformation or added nonlinear term.
(iii) Changing spread suggests heteroscedasticity.
(iv) A normal probability plot checks large departures from normality and possible outliers.
(v) Points with extreme x values may have high leverage and can strongly affect the fitted line.

These checks refine the modeling assumptions; they do not replace them.

Example 5.2.8. *Diagnostic Patterns*
Let $[n] := \{1, \dots, n\}$, $r_i := Y_i - \widehat{Y}_i$, $f_i := \widehat{Y}_i$, $\ell_i \in \mathcal{G}$,
$m_r(x) := \mathbb{E}(r_i \mid x_i = x)$, $v_r(f) := \mathrm{Var}(r_i \mid f_i = f)$, and $m_a := \mathbb{E}(r_i \mid \ell_i = a)$.

(i) $m_r \not\equiv 0$ or m_r non-affine; transform variables or add nonlinear terms.
(ii) v_r nonconstant; transform Y_i or use heteroscedastic errors.
(iii) $\exists a, b \in \mathcal{G}$ with $m_a \neq m_b$ or $\exists i \neq j$ with $\mathrm{Cov}(r_i, r_j \mid \ell_i = \ell_j) \neq 0$; add group effects or correlated errors.

Proposition 5.2.9. *Correlation and the Regression Line*
Let $[n] := \{1, \dots, n\}$, *let* $(x_i, y_i) \in \mathbb{R}^2$ *for* $i \in [n]$.
Set $\overline{x} := \frac{1}{n}\sum_{i=1}^n x_i$, $\overline{y} := \frac{1}{n}\sum_{i=1}^n y_i$, $s_{xx} := \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})^2$, $s_{yy} := \frac{1}{n}\sum_{i=1}^n (y_i - \overline{y})^2$, $s_{xy} := \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})$.
If $s_{xx}, s_{yy} > 0$, $r := \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$, $(\widehat{\beta}_0, \widehat{\beta}_1)$ *minimizes* $\sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$ *over* $(b_0, b_1) \in \mathbb{R}^2$,
and $\widehat{y}(x) := \widehat{\beta}_0 + \widehat{\beta}_1 x$, *then* $\widehat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$, $r = \widehat{\beta}_1 \sqrt{\frac{s_{xx}}{s_{yy}}}$, *and* $\frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}} = r \frac{x - \overline{x}}{\sqrt{s_{xx}}}$.

Proof. Set $D := \sum_{i=1}^n (x_i - \overline{x})^2 = n s_{xx}$ and $N := \sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y}) = n s_{xy}$.
Proposition 5.1.3 gives $\widehat{\beta}_1 = \frac{N}{D} = \frac{s_{xy}}{s_{xx}}$ and $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$.
Hence $r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}} = \widehat{\beta}_1 \sqrt{\frac{s_{xx}}{s_{yy}}}$ and $\widehat{y}(x) - \overline{y} = \widehat{\beta}_1 (x - \overline{x})$, so $\frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}} = r \frac{x - \overline{x}}{\sqrt{s_{xx}}}$.
Structural Commentary: $z_x := \frac{x - \overline{x}}{\sqrt{s_{xx}}}$ and $z_{\widehat{y}} := \frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}}$ give $z_{\widehat{y}} = r z_x$.
Key Insight: $\widehat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$ becomes $r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$ after standardising both coordinates. □

Remark 5.2.10. *Regression to the Mean*
With $s_{xx}, s_{yy} > 0$ and $0 < r < 1$, set $z_x := \frac{x - \overline{x}}{\sqrt{s_{xx}}}$ and $z_{\widehat{y}} := \frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}}$.
Proposition 5.2.9 gives $z_{\widehat{y}} = r z_x$, hence $|z_{\widehat{y}}| < |z_x|$ for $z_x \neq 0$.

5.3 Matrix Linear Models and Normal Equations

Matrix notation separates the geometry of least squares from the statistical assumptions on the errors. It also makes multiple regression a direct extension of simple regression.

Definition 5.3.1. *Fixed-Design Linear Model*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta, b \in \mathbb{R}^p$, and $e \in \mathbb{R}^n$.

(i) *Model*: $Y = X\beta + e$.
(ii) *Least-Squares Estimator*: $\widehat{\beta} \in \arg\min_{b \in \mathbb{R}^p} \|Y - Xb\|_2^2$.

Structural Commentary: An intercept is encoded by $X_{\cdot 1} = \mathbf{1}$.

Theorem 5.3.2. *Normal Equations for the Linear Model*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, *and* $Q(b) := \|Y - Xb\|_2^2$ *for* $b \in \mathbb{R}^p$.

(i) $\widehat{\beta} \in \arg\min_{b \in \mathbb{R}^p} Q(b)$ *iff* $X^t X \widehat{\beta} = X^t Y$.
(ii) If $\mathrm{rank}(X) = p$, *then* $X^t X$ *is invertible and* $\arg\min_{b \in \mathbb{R}^p} Q(b) = \{(X^t X)^{-1} X^t Y\}$.

Proof.
(i) ($\Rightarrow$) If $\widehat{\beta}$ minimizes Q, then $\forall (t, h) \in \mathbb{R} \times \mathbb{R}^p$, $0 \leq Q(\widehat{\beta} + th) - Q(\widehat{\beta}) = 2th^t X^t (X\widehat{\beta} - Y) + t^2 \|Xh\|_2^2$.
Hence $\forall h \in \mathbb{R}^p$, $h^t X^t (X\widehat{\beta} - Y) = 0$, so $X^t X \widehat{\beta} = X^t Y$.
(i) ($\Leftarrow$) If $X^t X \widehat{\beta} = X^t Y$, then $\forall h \in \mathbb{R}^p$, $Q(\widehat{\beta} + h) - Q(\widehat{\beta}) = \|Xh\|_2^2 \geq 0$, so $\widehat{\beta}$ minimizes Q.
(ii) If $\mathrm{rank}(X) = p$, then $\forall u \in \mathbb{R}^p \setminus \{0\}$, $u^t X^t X u = \|Xu\|_2^2 > 0$, so $X^t X$ is invertible.
Applying (i) gives $\arg\min_{b \in \mathbb{R}^p} Q(b) = \{(X^t X)^{-1} X^t Y\}$.

Structural Commentary: $\widehat{Y} = X\widehat{\beta}$ and $\widehat{e} = Y - \widehat{Y}$ satisfy $X^t \widehat{e} = 0$, hence $\widehat{e} \perp \mathrm{col}(X)$.
Key Insight: The projection $\widehat{Y} \in \mathrm{col}(X)$ is unique; $\mathrm{rank}(X) = p$ makes $b \mapsto Xb$ injective, so $\widehat{\beta}$ is unique. □

Method 5.3.3. *Projection View of Least Squares*

(i) *Full rank*: If $\operatorname{rank}(X) = d$, define $H := X(X^\top X)^{-1}X^\top$ and $\widehat{y} := Hy$.
(ii) *Projection*: H is the orthogonal projection onto $\operatorname{col}(X)$; hence $y - \widehat{y} \perp \operatorname{col}(X)$.
(iii) *Rank deficiency*: If $\operatorname{rank}(X) < d$, $\widehat{\beta}$ may be nonunique, but $\widehat{y} = \operatorname{proj}_{\operatorname{col}(X)} y$ is unique.

Rank deficiency affects coefficient representation, not least-squares fitted values.

Definition 5.3.4. *Hat Matrix, Fitted Values, and Residuals*
Assume $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\operatorname{rank}(X) = p$, $\widehat{\beta} = (X^tX)^{-1}X^tY$, and $I = I_n$.

(i) *Hat Matrix*: $P := X(X^tX)^{-1}X^t$.
(ii) *Fitted Values*: $\widehat{Y} := X\widehat{\beta} = PY$.
(iii) *Residuals*: $\widehat{e} := Y - \widehat{Y} = (I - P)Y$.

Key Insight: $\operatorname{range}(P) = \operatorname{col}(X)$ and $PY = \widehat{Y}$.

Proposition 5.3.5. *Projection Properties of the Hat Matrix*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $I = I_n$, $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, and $\widehat{e} := (I - P)Y$.
Then $P = P^t = P^2$, $I - P = (I - P)^t = (I - P)^2$, $\operatorname{range}(P) = \operatorname{col}(X)$, $\widehat{Y} = P_{\operatorname{col}(X)}Y$, and $X^t\widehat{e} = 0$.

Proof. Since $(X^tX)^t = X^tX$, $P^t = X((X^tX)^{-1})^tX^t = P$, and $P^2 = X(X^tX)^{-1}X^tX(X^tX)^{-1}X^t = P$.
Hence $(I - P)^t = I - P$ and $(I - P)^2 = I - 2P + P^2 = I - P$.
If $z = Xb \in \operatorname{col}(X)$, then $Pz = X(X^tX)^{-1}X^tXb = z$; also $\operatorname{range}(P) \subseteq \operatorname{col}(X)$, so $\operatorname{range}(P) = \operatorname{col}(X)$.
Finally, $X^t\widehat{e} = X^t(I - P)Y = X^tY - X^tX(X^tX)^{-1}X^tY = 0$, so $\widehat{e} \perp \operatorname{col}(X)$ and $\widehat{Y} = P_{\operatorname{col}(X)}Y$.
Structural Commentary: $P = P_{\operatorname{col}(X)}$ and $I - P = P_{\operatorname{col}(X)^\perp}$.
Key Insight: Fitting decomposes $Y = PY + (I - P)Y = \widehat{Y} + \widehat{e}$ with $\widehat{Y} \perp \widehat{e}$. □

Theorem 5.3.6. *Unbiased Estimate of the Error Variance*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, $\mathbb{E}e = 0$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, $\widehat{e} := Y - \widehat{Y}$, and $s^2 := \frac{\|\widehat{e}\|_2^2}{n-p}$. Then $\mathbb{E}\|\widehat{e}\|_2^2 = (n - p)\sigma^2$ and $\mathbb{E}s^2 = \sigma^2$.

Proof. Let $M := I_n - P$. Since $PX = X$, $\widehat{e} = MY = M(X\beta + e) = Me$.
Since $M = M^t = M^2$, $\|\widehat{e}\|_2^2 = e^tM^tMe = e^tMe$. For fixed $A \in \mathbb{R}^{n\times n}$ and centered e with $\operatorname{Var}(e) = \sigma^2 I_n$,
$\mathbb{E}(e^tAe) = \operatorname{tr}(A\sigma^2 I_n) = \sigma^2 \operatorname{tr}(A)$. Thus $\mathbb{E}\|\widehat{e}\|_2^2 = \sigma^2 \operatorname{tr}(M)$,
and $\operatorname{tr}(M) = n - \operatorname{tr}(P) = n - \operatorname{tr}((X^tX)^{-1}X^tX) = n - p$. Therefore $\mathbb{E}s^2 = \frac{1}{n-p}\mathbb{E}\|\widehat{e}\|_2^2 = \sigma^2$.
Structural Commentary: $M = I_n - P = P_{\operatorname{col}(X)^\perp}$ and $\operatorname{tr}(M) = n - p$.
Key Insight: $\widehat{e} = Me$, so residual variation is noise projected onto the $n - p$ directions orthogonal to $\operatorname{col}(X)$. □

Definition 5.3.7. *Standardized Residual*
Let $[n] := \{1, \dots, n\}$, $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $P = (p_{ij}) := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, $\widehat{e} := Y - \widehat{Y}$, $s^2 := \frac{\|\widehat{e}\|_2^2}{n-p}$, and $s > 0$. The *standardized residual* is $r_i := \frac{\widehat{e}_i}{s\sqrt{1-p_{ii}}}$ for $i \in [n]$.
Key Insight: Under $\operatorname{Var}(e) = \sigma^2 I_n$, $\operatorname{Var}(\widehat{e}_i) = \sigma^2(1 - p_{ii})$.

Proposition 5.3.8. *Residuals Are Uncorrelated with Fitted Values*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, and $\widehat{e} := (I_n - P)Y$. Then $\operatorname{Cov}(\widehat{e}, \widehat{Y}) = 0$.

Proof. Since $\operatorname{Cov}(Y) = \operatorname{Cov}(e) = \sigma^2 I_n$ and $P = P^t = P^2$,
$\operatorname{Cov}(\widehat{e}, \widehat{Y}) = (I_n - P)\operatorname{Cov}(Y)P^t = \sigma^2(I_n - P)P = \sigma^2(P - P^2) = 0$.
Structural Commentary: $\widehat{Y} = PY$ and $\widehat{e} = (I_n - P)Y$ are images of Y under complementary projections.
Key Insight: Spherical error covariance makes orthogonal projection components uncorrelated. □

5.4 Inference, Prediction, and Multiple Regression

Once the linear model is written in matrix form, standard errors for coefficients and predictions are consequences of linear transformations of random vectors.

Theorem 5.4.1. *Mean and Covariance of Least-Squares Estimates*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, $\mathbb{E}e = 0$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$. Then $\mathbb{E}\widehat{\beta} = \beta$ and $\operatorname{Var}(\widehat{\beta}) = \sigma^2(X^tX)^{-1}$.

Proof. Let $A := (X^tX)^{-1}X^t$. Since $AX = I_p$, $\widehat{\beta} = A(X\beta + e) = \beta + Ae$.
Hence $\mathbb{E}\widehat{\beta} = \beta + A\mathbb{E}e = \beta$, and $\operatorname{Var}(\widehat{\beta}) = A\operatorname{Var}(e)A^t = \sigma^2(X^tX)^{-1}X^tX(X^tX)^{-1} = \sigma^2(X^tX)^{-1}$.
Structural Commentary: $\widehat{\beta} - \beta = (X^tX)^{-1}X^te$ is a fixed linear transformation of e.
Key Insight: X^tX controls precision: $\operatorname{Var}(\widehat{\beta}) = \sigma^2(X^tX)^{-1}$. □

Method 5.4.2. *Coefficient Inference in a Normal Linear Model*
Let $Y = X\beta + e$, $e \sim N_n(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $\widehat{\beta} := (X^tX)^{-1}X^tY$,
$s^2 := \frac{1}{n-p}\left\|Y - X\widehat{\beta}\right\|_2^2$, $C = (c_{jk}) := (X^tX)^{-1}$, $j \in \{1, \dots, p\}$, and $\alpha \in (0, 1)$.

(i) *Standard Error*: $s_{\widehat{\beta}_j} := s\sqrt{c_{jj}}$.
(ii) *Pivot*: $T_j := \frac{\widehat{\beta}_j - \beta_j}{s_{\widehat{\beta}_j}} \sim t_{n-p}$.
(iii) *Confidence Interval*: $C_{j,1-\alpha} := \left[\widehat{\beta}_j - t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}, \widehat{\beta}_j + t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}\right]$.
(iv) *Test Statistic*: For $H_0 : \beta_j = \beta_{j0}$, $t_j := \frac{\widehat{\beta}_j - \beta_{j0}}{s_{\widehat{\beta}_j}}$.

If a coefficient CLT holds, $T_j \approx N(0,1)$.

Remark 5.4.3. *Linear-Model Inference After Selection*

Under fixed-design $y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$, and $\operatorname{rank}(X) = d$, $\widehat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1})$ and $\widehat{\sigma}^2 := \frac{\|y - X\widehat{\beta}\|_2^2}{n-d}$.
These give standard errors, t or asymptotic z statistics, and nested-model F tests when X is fixed in advance.
After data-driven selection of X or the model, these standard errors ignore selection.

Proposition 5.4.4. *Conditional Coverage Implies Unconditional Coverage*
Let $C(X,Y)$ be a confidence set for β_j, $\alpha \in (0,1)$, and $F := \{\operatorname{rank}(X) = p\}$.
If $\mathbb{P}(F) = 1$ and $\mathbb{P}(\beta_j \in C(X,Y) \mid X) = 1 - \alpha$ on F, then $\mathbb{P}(\beta_j \in C(X,Y)) = 1 - \alpha$.

Proof. Set $A := \{\beta_j \in C(X,Y)\}$. Since $\mathbb{P}(F) = 1$, $\mathbb{P}(A) = \mathbb{E}[\mathbb{P}(A \mid X)] = \mathbb{E}[(1-\alpha)\mathbf{1}_F + \mathbb{P}(A \mid X)\mathbf{1}_{F^c}] = 1 - \alpha$. *Structural Commentary*: $\mathbb{P}(F) = 1$ and $\mathbb{P}(A \mid X) = 1 - \alpha$ on F.
Key Insight: $\mathbb{P}(A \mid X) = 1 - \alpha$ a.s. implies $\mathbb{P}(A) = 1 - \alpha$. □

Proposition 5.4.5. *Mean Response at a New Design Point*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $e \sim N_n(0, \sigma^2 I_n)$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $s^2 := \frac{1}{n-p}\left\|Y - X\widehat{\beta}\right\|_2^2$, $x_0 \in \mathbb{R}^p$, $c_0 := x_0^t(X^tX)^{-1}x_0 > 0$,
$\mu_0 := x_0^t\beta$, $\widehat{\mu}_0 := x_0^t\widehat{\beta}$, $s_{\widehat{\mu}_0} := s\sqrt{c_0}$, and $\alpha \in (0,1)$. Then $\mathbb{E}\widehat{\mu}_0 = \mu_0$, $\operatorname{Var}(\widehat{\mu}_0) = \sigma^2 c_0$, $T_0 := \frac{\widehat{\mu}_0 - \mu_0}{s_{\widehat{\mu}_0}} \sim t_{n-p}$,
and $C_{0,1-\alpha} := \left[\widehat{\mu}_0 - t_{n-p,1-\frac{\alpha}{2}} s_{\widehat{\mu}_0}, \widehat{\mu}_0 + t_{n-p,1-\frac{\alpha}{2}} s_{\widehat{\mu}_0}\right]$ has coverage $1 - \alpha$ for μ_0.

Proof. Let $A := (X^tX)^{-1}X^t$, $P := XA$, and $M := I_n - P$.
Since $AX = I_p$, $\widehat{\mu}_0 - \mu_0 = x_0^tA(X\beta + e) - x_0^t\beta = x_0^tAe$, so $\mathbb{E}\widehat{\mu}_0 = \mu_0$ and $\operatorname{Var}(\widehat{\mu}_0) = x_0^tA(\sigma^2 I_n)A^tx_0 = \sigma^2 c_0$.
Normality gives $Z := \frac{\widehat{\mu}_0 - \mu_0}{\sigma\sqrt{c_0}} \sim N(0,1)$.

Since $M = M^t = M^2$, $\operatorname{rank}(M) = n - p$, and $MA^tx_0 = 0$, $Z \perp U := \frac{e^tMe}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
Thus $T_0 = \frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}$, and $\mathbb{P}(\mu_0 \in C_{0,1-\alpha}) = \mathbb{P}(|T_0| \le t_{n-p,1-\frac{\alpha}{2}}) = 1 - \alpha$.

Structural Commentary: $\widehat{\mu}_0 - \mu_0 = x_0^t(X^tX)^{-1}X^te$ and $s^2 = \frac{e^t(I_n - P)e}{n-p}$ use orthogonal components of e.
Key Insight: Estimating a mean response adds uncertainty only through $c_0 = x_0^t(X^tX)^{-1}x_0$. □

Corollary 5.4.6. *Prediction Interval for a New Observation*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $e \sim N_n(0, \sigma^2 I_n)$.
Let $x_0 \in \mathbb{R}^p$, $e_0 \sim N(0, \sigma^2)$ with $e_0 \perp e$, and $Y_0 := x_0^t\beta + e_0$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $\widehat{\mu}_0 := x_0^t\widehat{\beta}$, $s^2 := \frac{\|Y - X\widehat{\beta}\|_2^2}{n-p}$, $c_0 := x_0^t(X^tX)^{-1}x_0$, $s_{\text{pred}} := s\sqrt{1 + c_0}$, and $\alpha \in (0,1)$.
Then $\operatorname{Var}(Y_0 - \widehat{\mu}_0) = \sigma^2(1 + c_0)$, $T_{\text{pred}} := \frac{Y_0 - \widehat{\mu}_0}{s_{\text{pred}}} \sim t_{n-p}$,
and $I_{0,1-\alpha} := \left[\widehat{\mu}_0 - t_{n-p,1-\frac{\alpha}{2}} s_{\text{pred}}, \widehat{\mu}_0 + t_{n-p,1-\frac{\alpha}{2}} s_{\text{pred}}\right]$ has coverage $1 - \alpha$ for Y_0.

Proof. Let $A := (X^tX)^{-1}X^t$, $P := XA$, and $M := I_n - P$.
Since $AX = I_p$, $Y_0 - \widehat{\mu}_0 = e_0 - x_0^tAe$, so $\operatorname{Var}(Y_0 - \widehat{\mu}_0) = \sigma^2 + \sigma^2 x_0^tAA^tx_0 = \sigma^2(1 + c_0)$.
Normality gives $Z := \frac{Y_0 - \widehat{\mu}_0}{\sigma\sqrt{1+c_0}} \sim N(0,1)$. Since $AM = 0$, $Z \perp U := \frac{e^tMe}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
Thus $T_{\text{pred}} = \frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}$, and $\mathbb{P}(Y_0 \in I_{0,1-\alpha}) = \mathbb{P}(|T_{\text{pred}}| \le t_{n-p,1-\frac{\alpha}{2}}) = 1 - \alpha$.
Structural Commentary: $Y_0 - \widehat{\mu}_0 = e_0 - x_0^t(X^tX)^{-1}X^te$ combines future noise and estimation error.
Key Insight: Prediction variance is mean-response variance plus irreducible noise, $\sigma^2 c_0 + \sigma^2$. □

Definition 5.4.7. *Multiple Regression Coefficients*

Let $[n] := \{1, \dots, n\}$, $z_i = (x_{i1}, \dots, x_{i,p-1}) \in \mathbb{R}^{p-1}$ for $i \in [n]$, $u_1, \dots, u_{p-1}$ be the standard basis of $\mathbb{R}^{p-1}$, and $m(z) := \beta_0 + \sum_{k=1}^{p-1} \beta_k z_k$
In $Y_i = m(z_i) + e_i$, the *multiple regression coefficient* β_k satisfies $\forall z \in \mathbb{R}^{p-1}$ and $\forall \delta \in \mathbb{R}$, $m(z + \delta u_k) - m(z) = \delta\beta_k$ for $k \in \{1, \dots, p-1\}$.
Structural Commentary: Polynomial regression replaces z_i by transformed design columns such as z_{ik}^2.

Definition 5.4.8. *Coefficient of Determination*

Let $[n] := \{1, \dots, n\}$, $Y, \widehat{Y} \in \mathbb{R}^n$, $\overline{Y} := n^{-1}\sum_{i=1}^n Y_i$, $\mathbf{1} \in \operatorname{col}(X)$, and $\sum_{i=1}^n (Y_i - \overline{Y})^2 > 0$.

(i) *Residual Sum of Squares*: $\text{RSS} := \sum_{i=1}^n (Y_i - \widehat{Y}_i)^2$.

(ii) *Total Sum of Squares*: $\text{TSS} := \sum_{i=1}^n (Y_i - \overline{Y})^2$.

(iii) *Coefficient of Determination*: $R^2 := 1 - \frac{\text{RSS}}{\text{TSS}}$.

Structural Commentary: $R^2 = \operatorname{corr}(Y, \widehat{Y})^2$ for an intercept model.
Key Insight: $R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}$ is the observed response variation removed by the fitted model.

Remark 5.4.9. *Adjusted Coefficients*
Let X_{-j} be X without column x_j, and set $z_j := (I - \operatorname{proj}_{\operatorname{col}(X_{-j})})x_j$.
Then $\widehat{\beta}_j = \frac{z_j^\top y}{z_j^\top z_j}$, the simple-regression coefficient of y on the adjusted predictor z_j.
Thus $\widehat{\beta}_j$ uses only the part of x_j left after adjusting for other predictors.
If $x_j \approx \operatorname{col}(X_{-j})$, then $\|z_j\|_2$ is small and $\widehat{\beta}_j$ is unstable; this is collinearity.

Remark 5.4.10. *Collinearity and Interpretation*
Let X_{-j} be X without column $X_{\cdot j}$, set $d_j := \operatorname{dist}(X_{\cdot j}, \operatorname{col}(X_{-j}))$, and let $C = (c_{kl}) := (X^tX)^{-1}$.
If $0 < d_j \ll \|X_{\cdot j}\|_2$, then X^tX is nearly singular and $s_{\widehat{\beta}_j} = s\sqrt{c_{jj}}$ can be large even when $\widehat{Y} = X\widehat{\beta}$ is stable.
If $d_j = 0$, then $\operatorname{rank}(X) < p$ and $\widehat{\beta}$ is not unique.

Example 5.4.11. *Quadratic Terms as a Lack-of-Fit Check*
Compare $M_1 : Y_i = \beta_0 + \beta_1 x_i + e_i$ with $M_2 : Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + e_i$.
If $t_2 = \widehat{\beta}_2 / s_{\widehat{\beta}_2}$ is small and residual curvature is absent, retain M_1.
If $|t_2|$ is large and residual curvature decreases under M_2, use the quadratic term.
Add x^2 only when residual structure or scale supports $\beta_2 \neq 0$.

5.5 Random Design, Bootstrap, and Extensions

The fixed-design model conditions on the observed predictor values. Many observational studies are more naturally viewed as sampling random predictor-response pairs, with the fixed-design analysis recovered after conditioning on the realised design.

Definition 5.5.1. *Random-Design Linear Regression*
Let $[n] := \{1, \dots, n\}$, $(Y_i, \xi_i)_{i \in [n]}$ be independent copies of (Y, ξ), $\xi \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\sigma^2 > 0$.
A *Random-Design Linear Regression* is an iid sample $(Y_i, \xi_i)_{i \in [n]}$ satisfying $\mathbb{E}(Y \mid \xi = x) = x^t\beta$
and $\operatorname{Var}(Y \mid \xi = x) = \sigma^2$ for $x \in \operatorname{supp}(\xi)$.
Structural Commentary: Conditioning on $X_{i\cdot} = \xi_i^t$ gives $Y_{1:n} = X\beta + e$ with $e_i = Y_i - \xi_i^t\beta$.

Remark 5.5.2. *Unconditional Variability Can Differ*
Under random design, if $\mathbb{E}(\widehat{\beta} \mid X) = \beta$, then $\mathbb{E}\widehat{\beta} = \beta$.
Moreover, $\operatorname{Var}(\widehat{\beta}) = \mathbb{E}[\operatorname{Var}(\widehat{\beta} \mid X)] + \operatorname{Var}(\mathbb{E}[\widehat{\beta} \mid X]) = \sigma^2\mathbb{E}[(X^tX)^{-1}]$, not generally $\sigma^2(X^tX)^{-1}$ for frozen design.

Method 5.5.3. *Pairs Bootstrap for Random-Design Regression*
Let $D := \{(Y_i, x_i) : i \in [n]\}$, $x_i \in \mathbb{R}^p$, and let $T(D)$ be a fitted coefficient, prediction, or smooth statistic.

(i) Draw $D_b^* := \{(Y_{i_m}, x_{i_m}) : m \in [n]\}$ with $i_1, \dots, i_n$ sampled iid from $[n]$.
(ii) Fit least squares on D_b^* and compute $T_b^* := T(D_b^*)$.
(iii) Repeat for $b \in [B]$ and estimate $\operatorname{se}(T(D))$ by $\widehat{\operatorname{se}}_*(T) := \left(\frac{1}{B-1} \sum_{b=1}^{B} (T_b^* - \overline{T}^*)^2 \right)^{1/2}$, where $\overline{T}^* := B^{-1} \sum_{b=1}^{B} T_b^*$.

Perspective 5.5.4. *Extensions Beyond Ordinary Least Squares*

(i) *Robust regression*: Replace $\sum_i \widehat{e}_i^2$ by $\sum_i \Psi(\widehat{e}_i)$; $\Psi(u) = u^2$ gives least squares.
(ii) *Variable selection*: For selected predictors $S \subseteq \{1, \dots, p\}$, $\widehat{\beta}_S$ and $\operatorname{se}(\widehat{\beta}_S)$ are conditional on S.
(iii) *Influence*: Large leverage $h_{ii} = x_i^t(X^tX)^{-1}x_i$ can make observation i dominate $\widehat{\beta}$.
(iv) *Inverse calibration*: Given a fitted response curve $\widehat{m}$ and measured response y_0,
estimating x_0 from $\widehat{m}(x_0) = y_0$ is an inverse problem with uncertainty beyond $\operatorname{se}(\widehat{\beta})$.

Part XXI
Statistical Learning and Deep Learning

Statistical learning studies prediction rules, representation maps, and trained systems as mathematical objects. The statistical foundations of Parts 15 and 16 remain the language for experiments, inference, and model assessment; here the emphasis is on risk, generalization, optimization, architectures, and evaluation of learned predictors.

1 Learning Problems and Risk

1.1 Prediction, Loss Functions, and Risk

A learning problem turns observations into an action rule. Unlike inferential problems, the primary object is often the future loss of a predictor rather than a confidence statement about a parameter.

Definition 1.1.1. *Prediction Problem and Risk*
Inputs lie in $(\mathcal{X}, \mathcal{A})$, outputs in $(\mathcal{Y}, \mathcal{B})$, and $(X, Y) \sim P$ on $\mathcal{X} \times \mathcal{Y}$. Let $(\mathcal{A}_0, \mathcal{C})$ be an action space.

(i) *Predictor*: A predictor is a measurable map $f : \mathcal{X} \to \mathcal{A}_0$.
(ii) *Loss and risk*: For loss $\ell : \mathcal{A}_0 \times \mathcal{Y} \to [0, \infty]$, define $R(f) := \mathbb{E}_P \ell(f(X), Y)$.
(iii) *Bayes risk*: $R^* := \inf_f R(f)$, over measurable $f : \mathcal{X} \to \mathcal{A}_0$.

Definition 1.1.2. *Supervised Learning Terminology*
Let $S := \{(x_i, y_i) : 1 \le i \le n\}$ with $x_i = (x_{i1}, \dots, x_{ip})^\top \in \mathcal{X}$ and $y_i \in \mathcal{Y}$.

(i) *Features*: Coordinates x_{ij} are variables, inputs, predictors, independent variables, or features.
(ii) *Response*: Values y_i are outputs, responses, targets, or dependent variables.
(iii) *Regression*: If $\mathcal{Y} \subseteq \mathbb{R}^q$, the task is quantitative prediction.
(iv) *Classification*: If $\mathcal{Y} = \mathcal{G}$ is a finite unordered label set, the task is classification.
(v) *Ordered labels*: If $\mathcal{Y}$ is ordered categorical, order matters but arithmetic on labels usually does not.
(vi) *One-hot coding*: For $|\mathcal{G}| = K$, encode class g by $t_g \in \{0, 1\}^K$ with $(t_g)_g = 1$.
Classify vector scores $s(x) \in \mathbb{R}^K$ by $\operatorname{argmax}_k s_k(x)$.

If y_i are unobserved, the problem is unsupervised and targets structure in $\{x_i\}_{i=1}^n$.

Example 1.1.3. *Representative Learning Problems*

(i) *Email filtering*: x contains word and punctuation frequencies, $y \in \{\text{spam}, \text{ham}\}$, and asymmetric loss can penalize false deletion more than missed spam.
(ii) *Clinical regression*: $y = \log(\text{PSA})$ and x contains log cancer volume, prostate weight, age, and related measurements; correlated predictors are common.
(iii) *Digit recognition*: $x \in [0, 1]^{16 \times 16}$ or $[0, 1]^{256}$ and $y \in \{0, \dots, 9\}$; action set may add rejection option $\perp$.
(iv) *Microarrays*: An expression matrix $X \in \mathbb{R}^{G \times n}$ has genes as rows and samples as columns.
Typical first tasks are unsupervised clustering of genes or samples.

Example 1.1.4. *Common Prediction Losses*

(i) Regression commonly uses squared loss $\ell(a, y) = \|a - y\|_2^2$ or absolute loss $\ell(a, y) = |a - y|$.
(ii) Binary classification with labels in $\{-1, 1\}$ uses zero-one loss $\mathbf{1}_{\{ay \le 0\}}$ for a signed score a.
(iii) Probabilistic classification uses log loss $-\log q_y$ when q is a predicted distribution on labels.

Definition 1.1.5. *Bayes Predictors*

(i) *Conditional risk*: For action a, set $r_x(a) := \mathbb{E}[\ell(a, Y) \mid X = x]$; any $f^*(x) \in \operatorname{argmin}_a r_x(a)$ is Bayes.
(ii) *Squared loss*: If $\ell(a, y) = (a - y)^2$ and $\mathbb{E}[|Y| \mid X = x] < \infty$, then $f^*(x) = \mathbb{E}[Y \mid X = x]$.
(iii) *Absolute loss*: If $\ell(a, y) = |a - y|$, any conditional median $m(x)$
with $\mathbb{P}(Y \le m(x) \mid X = x) \ge \frac{1}{2}$ and $\mathbb{P}(Y \ge m(x) \mid X = x) \ge \frac{1}{2}$ is Bayes.
(iv) *Zero-one loss*: For labels $\mathcal{G}$ and $\ell(g, y) = \mathbf{1}_{\{g \ne y\}}$, a Bayes classifier is $g^*(x) \in \operatorname{argmax}_{g \in \mathcal{G}} \mathbb{P}(Y = g \mid X = x)$.

Remark 1.1.6. *Risk as a Prediction Target*
Here risk means predictive loss $R(f) = \mathbb{E}_P \ell(f(X), Y)$ under a data law P.
Decision-theoretic risk instead studies $R(\theta, \delta) = \mathbb{E}_\theta L(\theta, \delta(X))$ over a model family.

1.2 Empirical Risk Minimization

The population risk is the target, but only a sample is observed. Empirical risk minimisation replaces the unknown expectation by an average over the training data.

Definition 1.2.1. *Empirical Risk Minimization*
Let $S := ((x_1, y_1), \dots, (x_n, y_n))$.

(i) *Empirical risk*: For loss ℓ, define $\widehat{R}_S(f) := n^{-1} \sum_{i=1}^n \ell(f(x_i), y_i)$.
(ii) *Hypothesis class*: Let $\mathcal{F}$ be the admissible predictor class.
(iii) *ERM*: Any $\widehat{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \widehat{R}_S(f)$ is an empirical risk minimiser.

Definition 1.2.2. *Regularised Empirical Risk*
Let $\Omega : \mathcal{F} \to [0, \infty]$ be a complexity penalty and $\lambda \ge 0$.
Define $Q_\lambda(f) := \widehat{R}_S(f) + \lambda \Omega(f)$ and choose $\widehat{f}_\lambda \in \operatorname{argmin}_{f \in \mathcal{F}} Q_\lambda(f)$.
The penalty Ω may encode smoothness, sparsity, norm control, margin size, or architecture.

Theorem 1.2.3. *Finite-Class ERM Bound*
Let $S = ((X_1, Y_1), \dots, (X_n, Y_n))$ be i.i.d. from P, let $\emptyset \neq \mathcal{F}$ be finite.
Assume $0 \le \ell(f(X), Y) \le 1$ for all $f \in \mathcal{F}$, and fix $\delta \in (0,1)$. Set $\varepsilon_{n,\delta} := \sqrt{\log(2|\mathcal{F}|/\delta)/(2n)}$.

(i) Uniform deviation: With probability at least $1-\delta$, $\sup_{f\in\mathcal{F}} \left|R(f) - \widehat{R}_S(f)\right| \le \varepsilon_{n,\delta}$.

(ii) ERM excess risk: If $\widehat{f} \in \operatorname{argmin}_{f\in\mathcal{F}} \widehat{R}_S(f)$, then $R(\widehat{f}) \le \inf_{f\in\mathcal{F}} R(f) + 2\varepsilon_{n,\delta}$.

Proof.

(i) For fixed f, set $Z_i(f) := \ell(f(X_i), Y_i) \in [0,1]$, so $R(f) = \mathbb{E}Z_i(f)$ and $\widehat{R}_S(f) = n^{-1}\sum_{i=1}^n Z_i(f)$.
Hoeffding and a union bound give $\mathbb{P}(\sup_{f\in\mathcal{F}} \left|\widehat{R}_S(f) - R(f)\right| > \varepsilon) \le 2|\mathcal{F}| e^{-2n\varepsilon^2}$; set $\varepsilon = \varepsilon_{n,\delta}$.

(ii) On the uniform-deviation event, choose $f^* \in \operatorname{argmin}_{f\in\mathcal{F}} R(f)$.
Since $\widehat{R}_S(\widehat{f}) \le \widehat{R}_S(f^*)$, $R(\widehat{f}) \le \widehat{R}_S(\widehat{f}) + \varepsilon_{n,\delta} \le \widehat{R}_S(f^*) + \varepsilon_{n,\delta} \le R(f^*) + 2\varepsilon_{n,\delta}$.

Structural Commentary: Union-bounded concentration gives uniform control; ERM turns it into risk control.
Key Insight: Control every gap $\left|R - \widehat{R}_S\right|$ by $\varepsilon_{n,\delta}$; the ERM comparison pays this gap twice. □

Remark 1.2.4. *Statistical and Computational Error*
In practice the selected predictor may be only an approximate minimizer of the empirical criterion.
Its excess risk is then decomposed into approximation error, estimation error, and optimisation error.

1.3 Surrogate Losses and Calibration

Many prediction losses are discontinuous or combinatorial. Learning algorithms often optimise a smoother surrogate and rely on calibration to connect that surrogate to the target loss.

Definition 1.3.1. *Surrogate Loss*
For binary labels $y \in \{-1, 1\}$, let $g : \mathcal{X} \to \mathbb{R}$ be score function. Define $m_g(x, y) := yg(x)$ and $\widehat{y}_g(x) := \operatorname{sign}(g(x))$.
(i) *Zero-one loss*: $\ell_{01}(g(x), y) := \mathbf{1}_{\{m_g(x,y)\le 0\}}$.
(ii) *Surrogate*: $\ell_\phi(g(x), y) := \phi(m_g(x, y))$ for tractable $\phi : \mathbb{R} \to [0, \infty)$.

Example 1.3.2. *Margin Surrogates*
(i) Hinge loss: $\phi(t) = \max\{0, 1 - t\}$.
(ii) Logistic loss: $\phi(t) = \log(1 + \exp(-t))$.
(iii) Exponential loss: $\phi(t) = \exp(-t)$.
(iv) Squared margin loss: $\phi(t) = (1 - t)^2$.

Definition 1.3.3. *Classification Calibration*
Let $R_\phi(g) := \mathbb{E}\phi(Yg(X))$, $R_{01}(g) := \mathbb{P}(Yg(X) \le 0)$, $R^*_\phi := \inf_g R_\phi(g)$, and $R^*_{01} := \inf_g R_{01}(g)$.
Margin surrogate ϕ is classification calibrated if $R_\phi(g_m) \to R^*_\phi$ implies $R_{01}(g_m) \to R^*_{01}$ $\forall$ score sequence (g_m).
Equivalently, $R_\phi(g) - R^*_\phi \to 0$ forces $R_{01}(g) - R^*_{01} \to 0$.

Remark 1.3.4. *Proper Scoring Rules*
A scoring rule $S(q, y)$ is proper if $\operatorname{argmin}_q \mathbb{E}[S(q, Y) \mid X = x] = \mathcal{L}(Y \mid X = x)$.
Thus log and Brier losses learn calibrated conditional probabilities.

1.4 Bias-Variance and Approximation-Estimation Tradeoffs

Learning quality depends both on the richness of the model class and on the variability introduced by fitting that class from a finite sample.

Proposition 1.4.1. *Bias-Variance Decomposition*
Let $Y = f_0(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X = x] = 0$ and $\operatorname{Var}(\varepsilon \mid X = x) = \sigma^2(x)$.
Let random S produce $\widehat{f}_S$, set $\bar{f}(x) := \mathbb{E}_S\widehat{f}_S(x)$, and let $Y_x \perp S$ be a fresh response at x.
Then $\mathbb{E}_{S,Y_x}[(Y_x - \widehat{f}_S(x))^2] = \sigma^2(x) + (f_0(x) - \bar{f}(x))^2 + \operatorname{Var}_S(\widehat{f}_S(x))$.

Proof. Write $Y_x - \widehat{f}_S(x) = \varepsilon_x + f_0(x) - \widehat{f}_S(x)$, where $\varepsilon_x := Y_x - f_0(x)$.
Since $\mathbb{E}[\varepsilon_x] = 0$ and Y_x is conditionally independent of S, $\mathbb{E}_{S,Y_x}[\varepsilon_x\{f_0(x) - \widehat{f}_S(x)\}] = 0$.
Thus $\mathbb{E}_{S,Y_x}[(Y_x - \widehat{f}_S(x))^2] = \sigma^2(x) + \mathbb{E}_S[(f_0(x) - \widehat{f}_S(x))^2]$.
Finally, $\mathbb{E}_S[(f_0(x) - \widehat{f}_S(x))^2] = (f_0(x) - \bar{f}(x))^2 + \mathbb{E}_S[(\widehat{f}_S(x) - \bar{f}(x))^2]$.
Structural Commentary: Squared prediction error equals noise plus fitted-function error.
Key Insight: Center $\widehat{f}_S(x)$ at $\bar{f}(x)$; the remaining cross term has mean zero. □

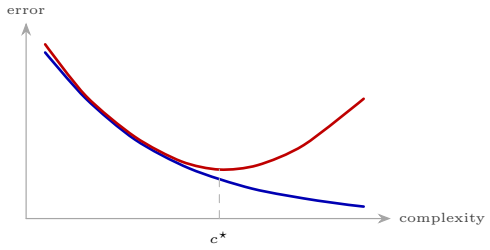

Figure 250: Bias–variance tradeoff: blue: training error, red: test error; dashed line marks test minimum.

Method 1.4.2. *Nearest-Neighbor Local Averaging*
Let $\mathcal{T} := \{(x_i, y_i) : 1 \le i \le n\}$ be training data in a metric space $(\mathcal{X}, d)$.
Let $N_k(x) \subseteq \{1, \dots, n\}$ be the indices of the k smallest distances $d(x_i, x)$.
For numerical responses, define the local average $\widehat{f}_k(x) := \frac{1}{k} \sum_{i \in N_k(x)} y_i$.
For class labels $g_i \in \mathcal{G}$, define $\widehat{p}_g(x) := \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}_{\{g_i = g\}}$ and classify by $\widehat{g}_k(x) \in \operatorname{argmax}_{g \in \mathcal{G}} \widehat{p}_g(x)$.
Thus k-nearest neighbours replace conditioning at $X = x$ by averaging over $N_k(x)$.

Remark 1.4.3. *Locality in High Dimension*
The parameter k trades local bias against variance: small k reacts strongly to S, while large k averages more but may wash out local structure. In $[0,1]^p$, a subcube with volume fraction r has side length $e_p(r) = r^{1/p}$, so $e_{10}(0.01) \approx 0.63$ and $e_{10}(0.10) \approx 0.80$. For N uniform points in the unit ball, the median nearest-neighbour radius is $d(p, N) = (1 - 2^{-1/N})^{1/p}$, about 0.52 for $(p, N) = (10, 500)$. Thus high-dimensional neighbourhoods large enough to contain data may not be local; useful local methods need structure or an adapted metric.

Definition 1.4.4. *Approximation and Estimation Errors*
Let $R^* := \inf_f R(f)$ and $R^*_{\mathcal{F}} := \inf_{f \in \mathcal{F}} R(f)$.

(i) *Approximation error*: $\mathcal{A}(\mathcal{F}) := R^*_{\mathcal{F}} - R^*$.
(ii) *Estimation error*: For a data-dependent $\widehat{f} \in \mathcal{F}$, $\mathcal{E}(\widehat{f}; \mathcal{F}) := R(\widehat{f}) - R^*_{\mathcal{F}}$.
(iii) *Excess risk*: $R(\widehat{f}) - R^* = \mathcal{A}(\mathcal{F}) + \mathcal{E}(\widehat{f}; \mathcal{F})$.

Larger $\mathcal{F}$ usually lowers $\mathcal{A}(\mathcal{F})$ and raises estimation difficulty.

Remark 1.4.5. *Model Complexity*
As $c(\mathcal{F})$ grows, $\widehat{R}_S(\widehat{f}_{\mathcal{F}})$ usually decreases, while $R(\widehat{f}_{\mathcal{F}}) - R^* = \mathcal{A}(\mathcal{F}) + \mathcal{E}(\widehat{f}_{\mathcal{F}}; \mathcal{F})$ trades smaller approximation error for larger estimation error. For zero-one loss, the relevant failure is crossing the wrong side of the decision boundary, not an additive squared-error bias–variance split.

2 Generalization Theory

2.1 Uniform Convergence and VC Theory

Generalisation asks when small training loss implies small population loss. Uniform convergence controls the gap for all predictors in a class at once.

Definition 2.1.1. *Uniform Convergence*
For samples S_n of size n and bounded loss, $\mathcal{F}$ has uniform convergence
if $\sup_{f \in \mathcal{F}} \left| R(f) - \widehat{R}_{S_n}(f) \right| \to 0$ in probability, or almost surely, as $n \to \infty$.

Definition 2.1.2. *Shattering and VC Dimension*
For $\mathcal{C} \subseteq 2^{\mathcal{X}}$, a finite $A \subseteq \mathcal{X}$ is shattered if $\{A \cap C : C \in \mathcal{C}\} = 2^A$,
and $\operatorname{VCdim}(\mathcal{C})$ is the supremum of $|A|$ over finite shattered $A \subseteq \mathcal{X}$.

Example 2.1.3. *Shattering Examples*
For $\mathcal{H}_p := \{\{x \in \mathbb{R}^p : a^\top x + b \ge 0\} : a \in \mathbb{R}^p, b \in \mathbb{R}\}$, $\operatorname{VCdim}(\mathcal{H}_p) = p + 1$; in particular, $\mathcal{H}_2$ shatters any three noncollinear points and shatters no four-point set.
Parameter count not decisive: one-parameter class $\mathcal{C} := \{\{x \in \mathbb{R} : \sin(\alpha x) > 0\} : \alpha \in \mathbb{R}\}$ has infinite VCdim.

Theorem 2.1.4. *VC Generalization Bound Form*
Let $\mathcal{F} \subseteq \{0,1\}^{\mathcal{X}}$ have $1 \le d := \operatorname{VCdim}(\mathcal{F}) < \infty$, let $S_n = ((X_i, Y_i))_{i=1}^n$ be i.i.d., and use $\ell_f(x, y) := \mathbf{1}_{\{f(x) \ne y\}}$, $R(f) := \mathbb{E}\ell_f(X, Y)$, and $\widehat{R}_{S_n}(f) := \frac{1}{n} \sum_{i=1}^n \ell_f(X_i, Y_i)$.
Define $\Pi_{\mathcal{F}}(m) := \max |\{(f(x_1), \dots, f(x_m)) : f \in \mathcal{F}\}|$, where the maximum is over $x_1, \dots, x_m \in \mathcal{X}$.

(i) Growth bound: If $m \ge d \ge 1$, then $\Pi_{\mathcal{F}}(m) \le \sum_{j=0}^{d} \binom{m}{j} \le \left(\frac{em}{d}\right)^d$.

(ii) Uniform deviation: For every $\delta \in (0,1)$ and $n \ge d$, with probability at least $1 - \delta$,

$$\sup_{f \in \mathcal{F}} \left| R(f) - \widehat{R}_{S_n}(f) \right| \le C \sqrt{\frac{d \log\left(\frac{2en}{d}\right) + \log\left(\frac{1}{\delta}\right)}{n}}$$

for a universal constant C.

Proof.

(i) *Growth bound*: Since no set of size $d+1$ is shattered, Sauer's lemma gives $\Pi_{\mathcal{F}}(m) \le \sum_{j=0}^{d} \binom{m}{j}$,
and the binomial-sum bound gives $\sum_{j=0}^{d} \binom{m}{j} \le \left(\frac{em}{d}\right)^d$ for $m \ge d$.

(ii) *Uniform deviation*: Symmetrization with an independent ghost sample and Hoeffding's inequality on the at most $\Pi_{\mathcal{F}}(2n)$ realized labelings give $\mathbb{P}(\sup_{f\in\mathcal{F}}\left|R(f)-\widehat{R}_{S_n}(f)\right|>\varepsilon)\le 8\Pi_{\mathcal{F}}(2n)\exp(-cn\varepsilon^2)$ for a universal $c>0$; applying (i) with $m=2n$ and solving the right side at level δ yields (ii).

Structural Commentary: VC dimension polynomially bounds sample labelings.
Key Insight: In finite-class concentration, use $|\mathcal{F}|\rightsquigarrow\Pi_{\mathcal{F}}(2n)\le\left(\frac{2en}{d}\right)^d$. □

Remark 2.1.5. *Growth Functions*
For $d=\mathrm{VCdim}(\mathcal{F})$, $\Pi_{\mathcal{F}}(m)$ is the effective number of labelings on m sample points. Sauer's lemma gives $\log\Pi_{\mathcal{F}}(m)\le d\log\left(\frac{em}{d}\right)$, so VC bounds replace $\log|\mathcal{F}|$ by the sample-level complexity $\log\Pi_{\mathcal{F}}(m)$.

Method 2.1.6. *Structural Risk Minimization*

(i) *Model ladder*: Choose $\mathcal{F}_1\subseteq\mathcal{F}_2\subseteq\cdots$ with complexities c_m, such as $c_m=\mathrm{VCdim}(\mathcal{F}_m)$.
(ii) *Uniform penalties*: Choose $\delta_m>0$ with $\sum_m\delta_m\le\delta$ and penalties $\psi_m(n,\delta_m)$ such that $\mathbb{P}(\sup_{f\in\mathcal{F}_m}\left|R(f)-\widehat{R}_{S_n}(f)\right|\le\psi_m(n,\delta_m))\ge 1-\delta_m$.
(iii) *SRM rule*: Choose $(\widehat{m},\widehat{f})\in\operatorname{argmin}_{m\in\mathbb{N},f\in\mathcal{F}_m}\{\widehat{R}_{S_n}(f)+\psi_m(n,\delta_m)\}$.

Larger m improves fit but raises ψ_m; SRM balances flexibility against search cost.

2.2 Rademacher Complexity and Covering Numbers

For real-valued predictors, complexity is often measured by how well the class can fit random noise.

Definition 2.2.1. *Rademacher Complexity*
Let $\mathcal{F}\subseteq\mathbb{R}^{\mathcal{X}}$, fix $S_n=(x_1,\dots,x_n)\in\mathcal{X}^n$, and let $\sigma_1,\dots,\sigma_n$ be i.i.d. Rademacher signs, $\mathbb{P}(\sigma_i=1)=\mathbb{P}(\sigma_i=-1)=\frac{1}{2}$. The empirical Rademacher complexity is $\widehat{\mathfrak{R}}_{S_n}(\mathcal{F}):=\mathbb{E}_\sigma\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\sigma_i f(x_i)$.

If $X_1,\dots,X_n\sim P_X$ i.i.d., the population complexity is $\mathfrak{R}_n(\mathcal{F}):=\mathbb{E}_{S_n}\widehat{\mathfrak{R}}_{S_n}(\mathcal{F})$.

Definition 2.2.2. *Covering Number*
For a metric space (T,d) and $\varepsilon>0$, $N(\varepsilon,T,d):=\inf\{m:\exists t_1,\dots,t_m\in T,\ T\subseteq\bigcup_{j=1}^m\overline{B}_d(t_j,\varepsilon)\}$.

The entropy $\log N(\varepsilon,T,d)$ measures metric size at scale ε, not parameter count.

Remark 2.2.3. *Entropy Methods*
Bounds on $\log N(\varepsilon,\mathcal{F},d)$ can be integrated over scales to control empirical processes.
This is the route from functional-analytic compactness to statistical-learning rates.

2.3 Concentration and Stability

Generalization can also follow from concentration of the training criterion or from stability of the learning algorithm itself.

Definition 2.3.1. *Subgaussian Random Variable*
Centered random variable Z is *subgaussian* with variance proxy σ^2 if $\mathbb{E}\exp(\lambda Z)\le\exp(\lambda^2\sigma^2/2)\ \forall\lambda\in\mathbb{R}$.
Subgaussian variables have Gaussian-type tail bounds.

Theorem 2.3.2. *Bounded Differences Inequality*
Let $X_1,\dots,X_n$ be independent, $X:=(X_1,\dots,X_n)$, and $F:\prod_{i=1}^n\mathcal{X}_i\to\mathbb{R}$.
For $x=(x_1,\dots,x_n)$, write $x^{(i\leftarrow x_i')}:=(x_1,\dots,x_{i-1},x_i',x_{i+1},\dots,x_n)$.
Assume $\left|F(x)-F(x^{(i\leftarrow x_i')})\right|\le c_i$ for all i,x,x_i', and set $v:=\sum_{i=1}^n c_i^2$.
Then, for every $t>0$, $\mathbb{P}\{F(X)-\mathbb{E}F(X)\ge t\}\le\exp\left(-\frac{2t^2}{v}\right)$ and $\mathbb{P}\{|F(X)-\mathbb{E}F(X)|\ge t\}\le 2\exp\left(-\frac{2t^2}{v}\right)$.

Proof. Let $\mathcal{G}_i:=\sigma(X_1,\dots,X_i)$, $M_i:=\mathbb{E}(F(X)\mid\mathcal{G}_i)$, and $\Delta_i:=M_i-M_{i-1}$.
Then $M_0=\mathbb{E}F(X)$, $M_n=F(X)$, and the bounded-differences hypothesis implies that, conditionally on $\mathcal{G}_{i-1}$, Δ_i has mean 0 and lies in an interval of length at most c_i. By Hoeffding's lemma, $\mathbb{E}(\exp(\lambda\Delta_i)\mid\mathcal{G}_{i-1})\le\exp\left(\frac{\lambda^2c_i^2}{8}\right)$ for $\lambda>0$. Iterating, $\mathbb{E}\exp(\lambda(F(X)-\mathbb{E}F(X)))\le\exp\left(\frac{\lambda^2v}{8}\right)$.
By Chernoff's bound, $\mathbb{P}\{F(X)-\mathbb{E}F(X)\ge t\}\le\exp\left(-\lambda t+\frac{\lambda^2v}{8}\right)$, optimised at $\lambda=\frac{4t}{v}$.
Applying the same bound to $-F$ and taking a union bound gives the two-sided inequality.
Structural Commentary: Coordinate sensitivity plus independence makes $F(X)$ subgaussian.
Key Insight: Doob increments reduce global concentration to bounds on one-coordinate changes c_i. □

Theorem 2.3.3. *Rademacher Generalization Bound*
Let $Z_i=(X_i,Y_i)\in\mathcal{Z}:=\mathcal{X}\times\mathcal{Y}$ be i.i.d. from P.
Let $S_n=(Z_1,\dots,Z_n)$, $\mathcal{G}:=\{g_f:\mathcal{Z}\to[0,1]:g_f(x,y)=\ell(f(x),y),f\in\mathcal{F}\}$.
For $g\in\mathcal{G}$, write $Pg:=\mathbb{E}g(Z)$ and $P_ng:=\frac{1}{n}\sum_{i=1}^n g(Z_i)$.

(i) Uniform deviation: $\forall\delta\in(0,1)$, with probability at least $1-\delta$, $\sup_{g\in\mathcal{G}}(Pg-P_ng)\le 2\widehat{\mathfrak{R}}_{S_n}(\mathcal{G})+3\sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$.

(ii) Risk bound: Hence, simultaneously for all $f\in\mathcal{F}$, $R(f)\le\widehat{R}_{S_n}(f)+2\widehat{\mathfrak{R}}_{S_n}(\ell\circ\mathcal{F})+3\sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$.

Proof.

(i) Let $\Phi(S_n) := \sup_{g\in\mathcal{G}}(Pg - P_n g)$. Changing one Z_i changes Φ and $\widehat{\mathfrak{R}}_{S_n}(\mathcal{G})$ by at most $1/n$, so McDiarmid gives $\Phi \le \mathbb{E}\Phi + \varepsilon$ and $\mathfrak{R}_n(\mathcal{G}) \le \widehat{\mathfrak{R}}_{S_n}(\mathcal{G}) + \varepsilon$ with probability at least $1-\delta$, for $\varepsilon := \sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2n}}$.
Symmetrisation gives $\mathbb{E}\Phi \le 2\mathfrak{R}_n(\mathcal{G})$, hence $\Phi \le 2\widehat{\mathfrak{R}}_{S_n}(\mathcal{G}) + 3\varepsilon$.

(ii) For $z = (x, y)$ and $g_f(z) = \ell(f(x), y)$, $Pg_f = R(f)$ and $P_n g_f = \widehat{R}_{S_n}(f)$.
Applying (i) to every $g_f \in \mathcal{G}$ gives the stated simultaneous bound.

Structural Commentary: Rademacher complexity measures sample-wise noise fitting.

Key Insight: Symmetrization turns $P - P_n$ into $\mathbb{E}_\sigma \sup_{g\in\mathcal{G}} \frac{1}{n}\sum_{i=1}^{n} \sigma_i g(Z_i)$. □

Definition 2.3.4. *Uniform Stability*
A learning algorithm A has *uniform stability* β_n for loss ℓ if, for any samples $S, S' \in \mathcal{Z}^n$ differing in one coordinate, $\sup_{z\in\mathcal{Z}} |\ell(A(S), z) - \ell(A(S'), z)| \le \beta_n$.

Remark 2.3.5. *Algorithm-Dependent Generalisation*
Stability controls generalisation through the map $S \mapsto A(S)$, not only through $\mathcal{F}$.
Thus regularisation or optimisation can generalise well even when the ambient class is large.

2.4 PAC-Bayes and Information-Theoretic Bounds

PAC-Bayes bounds analyze randomized predictors by comparing a data-dependent posterior distribution on hypotheses to a fixed prior.

Definition 2.4.1. *Randomised Predictor*
Let $\mathcal{H}$ be a hypothesis space and $Q \in \mathcal{P}(\mathcal{H})$. The *randomised predictor* Q samples $h \sim Q$ before prediction, with risk $R(Q) := \mathbb{E}_{h\sim Q} R(h)$ and empirical risk $\widehat{R}_S(Q) := \mathbb{E}_{h\sim Q}\widehat{R}_S(h)$.

Theorem 2.4.2. *PAC-Bayes Bound Form*
Let $S_n = (Z_1, \dots, Z_n)$ be i.i.d., let $\ell(h, Z) \in [0, 1]$, and let $\Pi \in \mathcal{P}(\mathcal{H})$ be independent of S_n.
For $Q \ll \Pi$, set $\mathrm{KL}(Q\|\Pi) := \mathbb{E}_Q \log \frac{dQ}{d\Pi}$ and $\mathrm{kl}(a, b) := a \log \frac{a}{b} + (1-a)\log\frac{1-a}{1-b}$.

(i) With probability at least $1-\delta$, every posterior $Q \ll \Pi$ satisfies $\mathrm{kl}(\widehat{R}_{S_n}(Q), R(Q)) \le \frac{\mathrm{KL}(Q\|\Pi) + \log\left(\frac{n+1}{\delta}\right)}{n}$.

(ii) On the same event, $R(Q) \le \widehat{R}_{S_n}(Q) + \sqrt{\frac{\mathrm{KL}(Q\|\Pi) + \log\left(\frac{n+1}{\delta}\right)}{2n}}$ for every $Q \ll \Pi$.

Proof.

(i) For fixed h, the bounded-loss kl moment bound gives $\mathbb{E}_{S_n} \exp(n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))) \le n+1$.
Thus $\mathbb{E}_{S_n}\mathbb{E}_{h\sim\Pi} \exp(n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))) \le n+1$, so Markov gives,
with probability at least $1-\delta$, $\mathbb{E}_\Pi \exp(n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))) \le \frac{n+1}{\delta}$.
On this event, the Donsker-Varadhan inequality with $\varphi(h) := n\,\mathrm{kl}(\widehat{R}_{S_n}(h), R(h))$ gives
$\mathbb{E}_Q\varphi \le \mathrm{KL}(Q\|\Pi) + \log \mathbb{E}_\Pi \exp\varphi \;\forall Q \ll \Pi$. By convexity of kl, $n\,\mathrm{kl}(\widehat{R}_{S_n}(Q), R(Q)) \le \mathbb{E}_Q\varphi$, proving (i).

(ii) If $R(Q) \ge \widehat{R}_{S_n}(Q)$, Pinsker gives $\mathrm{kl}(\widehat{R}_{S_n}(Q), R(Q)) \ge 2(R(Q) - \widehat{R}_{S_n}(Q))^2$.
if $R(Q) < \widehat{R}_{S_n}(Q)$, the claim is immediate. Combining with (i) gives (ii).

Structural Commentary: PAC-Bayes replaces class size by posterior-prior divergence.
Key Insight: A prior-average exponential tail plus $Q \ll \Pi$ yields all posteriors at cost $\mathrm{KL}(Q\|\Pi)$. □

Remark 2.4.3. *Prior and Posterior*
In PAC-Bayes, Π need not be subjective; it anchors the complexity term $\mathrm{KL}(Q\|\Pi)$.
Smaller $\mathrm{KL}(Q\|\Pi)$ means the data-dependent posterior Q stays closer to the fixed prior.

Definition 2.4.4. *Information-Theoretic Generalization*
Let $S_n = (Z_1, \dots, Z_n)$, $W = A(S_n)$ be learned output. *Information-theoretic generalization* bounds control $\mathrm{gen}(A) := \mathbb{E}[R(W) - \widehat{R}_{S_n}(W)]$ by dependence measures such as $I(W; S_n)$.

3 Unsupervised and Latent-Variable Learning

3.1 Clustering and Mixture Models

Unsupervised learning searches for structure in inputs without direct labels or a direct prediction loss. Clustering summarizes the input distribution by grouping observations, while mixture models replace hard groups by latent component probabilities.

Definition 3.1.1. *k-Means Clustering*
Let $x_1, \dots, x_n \in \mathbb{R}^d$ and $k \in \{1, \dots, n\}$. Put $\mathcal{C} = \{C : \{1, \dots, n\} \to \{1, \dots, k\}\}$, $\mathcal{M} = (\mathbb{R}^d)^k$,
$I_r(C) = \{i : C(i) = r\}$, and, for $m = (m_1, \dots, m_k)$, $\Phi(C, m) = \sum_{r=1}^{k} \sum_{i\in I_r(C)} \|x_i - m_r\|_2^2$.

(i) A *k-means solution* is any $(C^\star, m^\star) \in \operatorname{arg\,min}_{C\in\mathcal{C},\, m\in\mathcal{M}} \Phi(C, m)$.

(ii) Its *clusters* are $I_r^\star = I_r(C^\star)$, $r = 1, \dots, k$.

(iii) For fixed C, a *minimizing center* is $m_r = |I_r(C)|^{-1} \sum_{i\in I_r(C)} x_i$ if $I_r(C) \ne \varnothing$, and arbitrary if $I_r(C) = \varnothing$.

(iv) For fixed m, a *minimizing assignment* satisfies $C(i) \in \operatorname{arg\,min}_{1\le r\le k} \|x_i - m_r\|_2^2$ for each i.

Definition 3.1.2. *Object Dissimilarities*
Let $x_1, \dots, x_n$ be objects and let $d_{ii'} = d(x_i, x_{i'})$; when attributes are used, write $x_i = (x_{i1}, \dots, x_{ip})$.

(i) A *dissimilarity matrix* is $D = (d_{ii'}) \in [0,\infty)^{n\times n}$ with $d_{ii} = 0$ and $d_{ii'} = d_{i'i}$; directional inputs $a_{ii'}$ are symmetrized by $d_{ii'} = s(a_{ii'}, a_{i'i})$, where $s : [0,\infty)^2 \to [0,\infty)$ and $s(u,v) = s(v,u)$.

(ii) Given d_j and $w_j \geq 0$ with $\sum_{j=1}^{p} w_j = 1$, the *additive dissimilarity* is $d(x_i, x_{i'}) = \sum_{j=1}^{p} w_j d_j(x_{ij}, x_{i'j})$.

(iii) *Coordinate rules*: quantitative $d_j(a,b) = |a-b|$ or $(a-b)^2$; ordinal $r \mapsto (r-1/2)/M$ for $r = 1, \dots, M$; categorical $d_j(a,b) = L_j(a,b)$ with $L_j(a,a) = 0$ and $L_j(a,b) \geq 0$.

Remark 3.1.3. *Weights and Missing Values*
For $d(x_i, x_{i'}) = \sum_{j=1}^{p} w_j d_j(x_{ij}, x_{i'j})$, the empirical scale of coordinate j is $\bar{d}_j = \binom{n}{2}^{-1} \sum_{i<i'} d_j(x_{ij}, x_{i'j})$, so its effective contribution is $w_j \bar{d}_j$. If $O_{ii'} \subseteq \{1, \dots, p\}$ is the nonempty set of coordinates observed for both x_i and $x_{i'}$, pairwise deletion uses $d(x_i, x_{i'}) = \sum_{j \in O_{ii'}} \tilde{w}_{j,ii'} d_j(x_{ij}, x_{i'j})$ with $\tilde{w}_{j,ii'} = w_j / \sum_{\ell \in O_{ii'}} w_\ell$.
Alternatives are imputed values $\tilde{x}_{ij}$ before computing d or an added categorical level `missing`.

Definition 3.1.4. *Within- and Between-Cluster Scatter*
Fix $k \in \{1, \dots, n\}$ and a symmetric dissimilarity matrix $D = (d_{ii'})$.
Let $\mathcal{C}_k = \{C : \{1, \dots, n\} \to \{1, \dots, k\}\}$ and $I_r(C) = \{i : C(i) = r\}$.

(i) The *within-cluster scatter* is $W(C) = \frac{1}{2} \sum_{r=1}^{k} \sum_{i \in I_r(C)} \sum_{i' \in I_r(C)} d_{ii'}$ for $C \in \mathcal{C}_k$.

(ii) The *total scatter* is $T = \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} d_{ii'}$.

(iii) The *between-cluster scatter* is $B(C) = T - W(C)$ for $C \in \mathcal{C}_k$.

As T is fixed, $\arg\min_{C \in \mathcal{C}_k} W(C) = \arg\max_{C \in \mathcal{C}_k} B(C)$.

Algorithm 2: k-Means Alternating Minimization

Require: Data $x_1, \dots, x_n \in \mathbb{R}^d$, number of clusters k, initial centers or assignments
Ensure: Cluster assignments $C(i)$ and centers $m_1, \dots, m_k$
repeat
Given assignments, set each m_r to the mean of the observations with $C(i) = r$
Given centers, assign each observation by $C(i) \leftarrow \arg\min_{1 \leq r \leq k} \|x_i - m_r\|_2^2$
until the assignments no longer change

Proposition 3.1.5. *k-Means Descent and Local Minima*
Let $\mathcal{C} = \{C : \{1, \dots, n\} \to \{1, \dots, k\}\}$, $\mathcal{M} = (\mathbb{R}^d)^k$, *and* $\Phi(C, m) = \sum_{r=1}^{k} \sum_{i : C(i) = r} \|x_i - m_r\|_2^2$.
Suppose $(C^t, m^t) \in \mathcal{C} \times \mathcal{M}$ *satisfies* $m^{t+1} \in \arg\min_{m \in \mathcal{M}} \Phi(C^t, m)$ *and* $C^{t+1} \in \arg\min_{C \in \mathcal{C}} \Phi(C, m^{t+1})$ *for each* t.

(i) The objective is nonincreasing: $\Phi(C^{t+1}, m^{t+1}) \leq \Phi(C^t, m^t)$.
(ii) If $C^{t+1} = C^t = C^\star$ *and* $m^{t+1} = m^t = m^\star$, *then* $(C^\star, m^\star)$ *is coordinatewise minimal:* $m^\star \in \arg\min_{m \in \mathcal{M}} \Phi(C^\star, m)$ *and* $C^\star \in \arg\min_{C \in \mathcal{C}} \Phi(C, m^\star)$.
Coordinatewise minimal need not be global; compare initialisations by terminal Φ.

Proof.
(i) By the center update, $\Phi(C^t, m^{t+1}) \leq \Phi(C^t, m^t)$. By the assignment update, $\Phi(C^{t+1}, m^{t+1}) \leq \Phi(C^t, m^{t+1})$. Combining gives the claim.
(ii) At a fixed point, $C^{t+1} = C^t = C^\star$ and $m^{t+1} = m^t = m^\star$ in the two update inclusions, giving the two coordinatewise optimality conditions.

Structural Commentary: Exact block minimization forces descent.
Key Insight: A stopped run certifies blockwise, not global, optimality. □

Algorithm 3: k-Medoids Alternating Representatives

Require: Pairwise dissimilarities $D(x_i, x_{i'})$, number of clusters k, initial assignments
Ensure: Cluster assignments $C(i)$ and representative indices $i_1^\star, \dots, i_k^\star$
repeat
for $r = 1, \dots, k$ **do**
Set $i_r^\star \leftarrow \arg\min_{i : C(i) = r} \sum_{C(i') = r} D(x_i, x_{i'})$
end for
Assign each observation by $C(i) \leftarrow \arg\min_{1 \leq r \leq k} D(x_i, x_{i_r^\star})$
until the assignments no longer change

Remark 3.1.6. *k-Medoids Versus k-Means*
Fix an assignment C and write $I_r = \{i : C(i) = r\}$.

(i) *k-means*: $m_r \in \arg\min_{m \in \mathbb{R}^d} \sum_{i \in I_r} \|x_i - m\|_2^2$, hence $m_r = |I_r|^{-1} \sum_{i \in I_r} x_i$ when $I_r \neq \varnothing$.

(ii) *k-medoids*: $i_r^\star \in \arg\min_{i \in I_r} \sum_{i' \in I_r} D(x_i, x_{i'})$, with representative $x_{i_r^\star}$.

k-medoids only requires pairwise dissimilarities and keeps representatives inside the data set.
Its representative update is usually costlier than the k-means empirical mean.

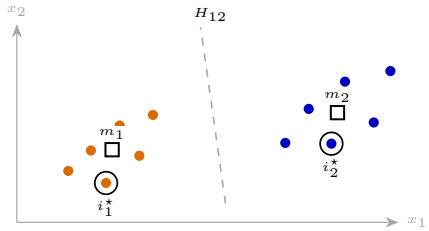

Figure 251: Open squares are centroids, circled points are medoids, the dashed line is nearest-center boundary.

Remark 3.1.7. *Choosing the Number of Clusters*
For k-means or k-medoids, choose k and an initialisation. Let W_k be terminal within-cluster scatter.
The elbow diagnostic inspects $k \mapsto W_k$; the gap statistic uses $G(k) = \mathbb{E}_0[\log W_k] - \log W_k$, where $\mathbb{E}_0$ is under a reference distribution on the data range.
With simulation standard error s'_{k+1}, one rule is $\widehat{k} = \min\{k : G(k) \geq G(k+1) - s'_{k+1}\}$.
This formalises the bend comparison but remains heuristic.

Method 3.1.8. *Hierarchical Clustering and Linkage*
Given dissimilarities $d_{ii'}$ and a linkage $L(G, H)$ between nonempty groups, agglomerative hierarchical clustering constructs nested partitions.

(i) Initialize $\mathcal{P}_0 = \{\{1\}, \ldots, \{n\}\}$.
(ii) At step t, choose distinct $G, H \in \mathcal{P}_t$ minimizing $L(G, H)$.
(iii) Set $\mathcal{P}_{t+1} = (\mathcal{P}_t \setminus \{G, H\}) \cup \{G \cup H\}$ and record merge height $L(G, H)$.
(iv) Stop at $\mathcal{P}_{n-1} = \{\{1, \ldots, n\}\}$.

For groups G, H with sizes N_G, N_H, common linkages are

(i) *Single*: $L_{\mathrm{SL}}(G, H) = \min_{i \in G,\, i' \in H} d_{ii'}$.
(ii) *Complete*: $L_{\mathrm{CL}}(G, H) = \max_{i \in G,\, i' \in H} d_{ii'}$.
(iii) *Group average*: $L_{\mathrm{GA}}(G, H) = (N_G N_H)^{-1} \sum_{i \in G} \sum_{i' \in H} d_{ii'}$.

Recorded merges form a dendrogram; divisive clustering reverses the process by recursive splits.

Remark 3.1.9. *Dendrograms and Cophenetic Distances*
For dendrogram merge height $h_T(i, i')$, the cut at height a partitions observations by $i \sim_a i'$ iff $h_T(i, i') \leq a$.
The cophenetic dissimilarity is $C_{ii'} = h_T(i, i')$, the first height at which i and i' share a cluster.
It is ultrametric: $C_{ii'} \leq \max\{C_{ih}, C_{i'h}\}$. Thus a dendrogram is an algorithmic hierarchical summary, not an exact representation of arbitrary input dissimilarities.

Method 3.1.10. *Self-Organizing Map*
A self-organizing map uses prototypes $m_1^t, \ldots, m_q^t \in \mathbb{R}^p$ indexed by grid locations $\ell_1, \ldots, \ell_q \in \mathcal{Q}_1 \times \mathcal{Q}_2$.
Let $\alpha_t > 0$ and $h_t : [0, \infty) \to [0, 1]$ be the learning rate and neighborhood kernel.

(i) Select an observation x_i.
(ii) Find a winner $j_t \in \arg\min_{1 \leq j \leq q} \left\| x_i - m_j^t \right\|_2^2$.
(iii) Update every prototype by $m_k^{t+1} = m_k^t + \alpha_t h_t(\|\ell_{j_t} - \ell_k\|_2)(x_i - m_k^t)$, $k = 1, \ldots, q$.
(iv) Increment t and repeat until the stopping rule or schedule terminates.

A common kernel is $h_t(s) = \mathbf{1}_{s \leq r_t}$ with decreasing radius r_t.
Early large neighborhoods enforce grid smoothness; if $h_t(s) = \mathbf{1}_{s=0}$, the method reduces to online k-means.

Remark 3.1.11. *Checking the Topological Constraint*
For q prototypes, compare $E_{\mathrm{SOM}} = \sum_i \left\| x_i - m_{C(i)} \right\|_2^2$ with the q-means optimum $E_q^\star = \min_{C, m} \sum_i \left\| x_i - m_{C(i)} \right\|_2^2$.
Since $E_q^\star \leq E_{\mathrm{SOM}}$, a large gap $E_{\mathrm{SOM}} - E_q^\star$ indicates an overly restrictive lattice constraint.

Method 3.1.12. *Spectral Clustering*
Given weights $w_{ii'} \geq 0$, set $W = (w_{ii'})$, $g_i = \sum_{i'=1}^{n} w_{ii'}$, $G = \mathrm{diag}(g_i)$, and $L = G - W$.
Often $w_{ii'} = s_{ii'} = \exp(-\frac{\|x_i - x_{i'}\|_2^2}{c})$ (restricted to mutual nearest-neighbor edges). For target k clusters:

(i) Construct W, G, and the Laplacian L.
(ii) Compute eigenvectors $u_1, \ldots, u_k$ for the k smallest nontrivial eigenvalues of L.
(iii) Form $U = (u_1, \ldots, u_k) \in \mathbb{R}^{n \times k}$ and represent observation i by row $U_{i\cdot}$.
(iv) Cluster the rows $U_{i\cdot}$ into k groups, typically by k-means.

$f^T L f = \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} w_{ii'}(f_i - f_{i'})^2$, so small-eigenvalue directions are smooth across high-weight edges.

Remark 3.1.13. *Connected Components and Random Walks*
Let $L = G - W$ and, when $g_i > 0$, $P = G^{-1} W$.

(i) If the graph has components $A_1, \ldots, A_m$, then $\ker L = \mathrm{span}\{\mathbf{1}_{A_1}, \ldots, \mathbf{1}_{A_m}\}$.
(ii) The matrix P is a random-walk transition matrix.
A good spectral cluster A has large within-set transition mass $\sum_{i \in A} \sum_{i' \in A} P_{ii'}$ and small cross-set mass.

Nearly separated clusters replace exact zero eigenvalues by small eigenvalues.

Definition 3.1.14. *Finite Mixture Model*

Let $Z \in \{1, \dots, k\}$ be latent with $\mathbb{P}(Z = z) = \pi_z$, $\pi_z \geq 0$, and $\sum_{z=1}^{k} \pi_z = 1$.

Given component densities $p_z(\cdot; \theta_z)$, the marginal density is $p(x) = \sum_{z=1}^{k} \pi_z p_z(x; \theta_z)$.

Definition 3.1.15. *Mixture Responsibilities*

For fitted $(\widehat{\pi}_m, \widehat{\theta}_m)_{m=1}^{k}$, define $\widehat{r}_{im} = \widehat{\mathbb{P}}(Z_i = m \mid x_i) = \widehat{\pi}_m p_m(x_i; \widehat{\theta}_m) \left(\sum_{\ell=1}^{k} \widehat{\pi}_\ell p_\ell(x_i; \widehat{\theta}_\ell) \right)^{-1}$.

Then $\widehat{r}_i$ is soft membership and $\widehat{C}(i) \in \arg\max_{1 \leq m \leq k} \widehat{r}_{im}$ is a hard assignment.

Remark 3.1.16. *Mixtures as Density Estimates*

Gaussian mixtures give flexible density estimates $p(x) = \sum_{m=1}^{k} \pi_m \phi(x; \mu_m, \Sigma_m)$. If class-conditional mixtures $\widehat{p}_g(x)$ and class priors $\widehat{\pi}_g$ are fit separately, the Bayes classifier is $\widehat{g}(x) \in \arg\max_g \widehat{\pi}_g \widehat{p}_g(x)$.

Without labels, component-label agreement is discovered structure, not supervised training.

Remark 3.1.17. *Gaussian Mixtures as Soft k-Means*

For $p_m(x) = \phi(x; \mu_m, \sigma^2 I)$, the responsibility is $r_{im} = \pi_m \exp(-\frac{\|x_i - \mu_m\|_2^2}{2\sigma^2}) \left(\sum_{\ell=1}^{k} \pi_\ell \exp(-\frac{\|x_i - \mu_\ell\|_2^2}{2\sigma^2}) \right)^{-1}$.

Thus r_{im} is distance-weighted; as $\sigma^2 \downarrow 0$, it concentrates on nearest centers, yielding hard k-means assignments.

Method 3.1.18. *Expectation-Maximization*

Given $x_{1:n}$ and $p(x) = \sum_{m=1}^{k} \pi_m p_m(x; \theta_m)$, set $\ell(\pi, \theta) = \sum_{i=1}^{n} \log(\sum_{m=1}^{k} \pi_m p_m(x_i; \theta_m))$. EM alternates:

(i) *E-step*: compute $r_{im} = \frac{\pi_m p_m(x_i; \theta_m)}{\sum_{\ell=1}^{k} \pi_\ell p_\ell(x_i; \theta_\ell)}$.

(ii) *M-step*: update $\pi_m^{\text{new}} = n^{-1} \sum_{i=1}^{n} r_{im}$ and $\theta_m^{\text{new}} \in \arg\max_{\theta_m} \sum_{i=1}^{n} r_{im} \log p_m(x_i; \theta_m)$.

For Gaussian components, the M-step gives weighted means and variances or covariances.
With an exact M-step, ℓ is nondecreasing, but convergence may be to a local maximum.

Remark 3.1.19. *Identifiability*

For any permutation τ of $\{1, \dots, k\}$, $\sum_{m=1}^{k} \pi_m p_m(x; \theta_m) = \sum_{m=1}^{k} \pi_{\tau(m)} p_{\tau(m)}(x; \theta_{\tau(m)})$.
Thus labels are identifiable only up to permutation: density is invariant, component interpretation is not.

3.2 Dimension Reduction and Manifold Learning

Dimension reduction replaces high-dimensional observations, similarities, or dissimilarities by lower-dimensional coordinates that preserve selected structure. Linear methods preserve variance or inner products; manifold methods preserve neighborhoods, graph distances, or local affine geometry.

Definition 3.2.1. *Principal Component Analysis*

Let centered observations $x_1, \dots, x_n \in \mathbb{R}^p$ be the rows of $X \in \mathbb{R}^{n \times p}$, and let $1 \leq r \leq p$.

(i) Rank-r principal component analysis chooses $V = [v_1 \ \cdots \ v_r] \in \mathbb{R}^{p \times r}$, with $V^T V = I_r$ solving $\min_{V^T V = I_r} \|X - XVV^T\|_F^2$.

(ii) The columns $v_1, \dots, v_r$ are the principal directions, $\text{im}(V)$ is the rank-r principal subspace, and $x_i \mapsto VV^T x_i$ is the fitted projection.

(iii) Equivalently, V solves $\max_{V^T V = I_r} \text{tr}(V^T X^T X V)$,

with sequential directions $v_k \in \text{argmax}\{\sum_{i=1}^{n} (v^T x_i)^2 : \|v\|_2 = 1, \ v \perp v_1, \dots, v_{k-1}\}$.

Method 3.2.2. *PCA by SVD*

Let the centered observations be the rows of $X \in \mathbb{R}^{n \times p}$, and fix $1 \leq r \leq \min\{n, p\}$.

(i) Compute an SVD $X = UDV^T$ with singular values $d_1 \geq \cdots \geq 0$.
(ii) Set $V_r = [v_1 \ \cdots \ v_r]$, the first r right singular vectors.
(iii) Set the score matrix $Z_r = XV_r = U_r D_r$.
(iv) Set the fitted matrix $\widehat{X}_r = Z_r V_r^T = XV_r V_r^T = U_r D_r V_r^T$.

The columns of V_r are the rank-r PCA directions, Z_r contains the principal-component scores, and $\widehat{X}_r$ is the orthogonal rank-r reconstruction minimizing $\left\|X - \widehat{X}\right\|_F^2$.

Method 3.2.3. *Kernel PCA*

Let k be a positive definite kernel on observations $x_1, \dots, x_n$, fix r, and set $M = n^{-1}\mathbf{1}\mathbf{1}^T$.

(i) Form the Gram matrix $K = (k(x_i, x_{i'}))_{i,i'}$.
(ii) Center it by $\widetilde{K} = (I_n - M)K(I_n - M)$.
(iii) Compute $\widetilde{K} u_j = \lambda_j u_j$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0$, and set $U_r = [u_1 \ \cdots \ u_r]$, $\Lambda_r = \text{diag}(\lambda_1, \dots, \lambda_r)$.
(iv) Return coordinates $Z = U_r \Lambda_r^{1/2}$.

Kernel PCA is spectral PCA on $\widetilde{K}$; radial kernels encode local similarity.

Method 3.2.4. *Sparse Principal Components*
Let centered data have rows x_i^T in $X \in \mathbb{R}^{n\times p}$, and let $v \in \mathbb{R}^p$ be a loading vector.

(i) *Variance form*: set $\mathcal{V}_t = \{a \in \mathbb{R}^p : a^T a = 1,\ \sum_{j=1}^{p} |a_j| \le t\}$ and choose $v \in \operatorname{argmax}_{a\in\mathcal{V}_t} a^T X^T X a$.

(ii) *Reconstruction form*: for one component, choose $(\theta, v) \in \operatorname{argmin}_{\|\theta\|_2=1} \sum_{i=1}^{n} \left\|x_i - \theta v^T x_i\right\|_2^2 + \lambda \|v\|_2^2 + \lambda_1 \|v\|_1$.

L_1 regularization sparsifies loadings; multiple components alternate sparse-loading and orthogonal-score updates.

Definition 3.2.5. *Multidimensional Scaling*
Let $D = (d_{ii'})$ be a symmetric dissimilarity matrix with $d_{ii} = 0$, fix $r < n$, and write $Z = (z_1^T, \dots, z_n^T)^T \in \mathbb{R}^{n\times r}$ with fitted distances $\delta_{ii'}(Z) = \|z_i - z_{i'}\|_2$. Multidimensional scaling chooses Z so that $\delta_{ii'}(Z)$ approximates $d_{ii'}$.

(i) *Least-squares MDS*, or Kruskal–Shepard scaling, minimizes $S_{\mathrm{MDS}}(Z) = \sum_{i\neq i'} (d_{ii'} - \delta_{ii'}(Z))^2$ over $Z \in \mathbb{R}^{n\times r}$.

(ii) *Sammon MDS*, defined when $d_{ii'} > 0$ for $i \neq i'$, minimizes $S_{\mathrm{Sammon}}(Z) = \sum_{i\neq i'} \frac{(d_{ii'} - \delta_{ii'}(Z))^2}{d_{ii'}}$ over $Z \in \mathbb{R}^{n\times r}$.

Sammon weighting penalizes relative errors more strongly for small dissimilarities.

Method 3.2.6. *Classical Scaling*
Let $H = I_n - n^{-1}\mathbf{1}\mathbf{1}^T$ and fix $r < n$.

(i) If centered similarities $S = (s_{ii'})$ are supplied, set $B = S$.
(ii) If Euclidean distances $\Delta = (\Delta_{ii'})$ are supplied, set $\Delta^{(2)} = (\Delta_{ii'}^2)$ and $B = -\frac{1}{2} H \Delta^{(2)} H$.
(iii) Compute $Be_j = \lambda_j e_j$ with $\lambda_1 \ge \cdots \ge \lambda_r > 0$, and set $E_r = [e_1\ \cdots\ e_r]$, $\Lambda_r = \operatorname{diag}(\lambda_1, \dots, \lambda_r)$.
(iv) Return $Z_r = E_r \Lambda_r^{1/2} \in \mathbb{R}^{n\times r}$, with row z_i^T as the coordinate of observation i.

The construction gives centered coordinates and fitted Gram matrix $Z_r Z_r^T = E_r \Lambda_r E_r^T$, the rank-$r$ positive-semidefinite approximation to B from its leading positive spectrum.

Method 3.2.7. *Nonmetric Scaling*
Given dissimilarities $D = (d_{ii'})$, coordinates $Z = (z_1^T, \dots, z_n^T)^T \in \mathbb{R}^{n\times r}$, and an increasing map θ, nonmetric scaling minimizes $S_{\mathrm{NM}}(Z, \theta) = \sum_{i\neq i'} (\|z_i - z_{i'}\|_2 - \theta(d_{ii'}))^2 / \sum_{i\neq i'} \|z_i - z_{i'}\|_2^2$.

(i) With θ fixed, update Z by minimizing $S_{\mathrm{NM}}(Z, \theta)$ over $Z \in \mathbb{R}^{n\times r}$.
(ii) With Z fixed, update θ by isotonic regression of the fitted distances $\|z_i - z_{i'}\|_2$ on the dissimilarities $d_{ii'}$.

Only the ordering of $d_{ii'}$ constrains $\theta(d_{ii'})$; their original scale is discarded.

Definition 3.2.8. *Encoder and Decoder*
Let $\mathcal{X}$ be a data space and $\mathcal{H}$ a representation space.

(i) An *encoder* is a map $e : \mathcal{X} \to \mathcal{H}$, with representation $h = e(x)$.
(ii) A *decoder* is a map $d : \mathcal{H} \to \mathcal{X}$, with reconstruction $\widehat{x} = d(h) = d(e(x))$.

Linear PCA is the case e, d linear with $\operatorname{im}(d)$ an orthogonal representation subspace.

Method 3.2.9. *Procrustes Shape Alignment*
Let $X_1, X_2 \in \mathbb{R}^{n\times p}$ have corresponding landmark rows.
Orthogonal Procrustes alignment solves $\min_{R^T R = I, \mu\in\mathbb{R}^p} \left\|X_2 - (X_1 R + \mathbf{1}\mu^T)\right\|_F$.

(i) Compute column means $\bar{x}_1, \bar{x}_2$ and centered matrices $\widetilde{X}_a = X_a - \mathbf{1}\bar{x}_a^T$, $a = 1, 2$.
(ii) Compute an SVD $\widetilde{X}_1^T \widetilde{X}_2 = U D V^T$.
(iii) Set $\widehat{R} = UV^T$ and $\widehat{\mu} = \bar{x}_2 - \widehat{R}^T \bar{x}_1$.
(iv) For scaled Procrustes, set $\widehat{\beta} = \operatorname{tr}(D) / \left\|\widetilde{X}_1\right\|_F^2$, $\widehat{\mu}_\beta = \bar{x}_2 - \widehat{\beta}\widehat{R}^T \bar{x}_1$, and use fitted configuration $\widehat{\beta} X_1 \widehat{R} + \mathbf{1}\widehat{\mu}_\beta^T$.

The unscaled solution uses $\widehat{\beta} = 1$ and preserves size; the scaled solution also matches overall scale.

Method 3.2.10. *Procrustes Shape Average*

For centered matched shapes $X_\ell \in \mathbb{R}^{n\times p}$, set $\Phi(M, \{R_\ell\}) = \sum_{\ell=1}^{L} \|X_\ell R_\ell - M\|_F^2$, $R_\ell^T R_\ell = I_p$.

A Procrustes shape average solves $\min_{M, \{R_\ell\}} \Phi(M, \{R_\ell\})$.

(i) Initialize $M^{(0)}$ as a centered observed shape or an initially aligned average.
(ii) Given $M^{(t)}$, compute $R_\ell^{(t)} \in \operatorname{argmin}_{R^T R = I_p} \left\|X_\ell R - M^{(t)}\right\|_F^2$ for each ℓ.
(iii) Update $M^{(t+1)} = L^{-1} \sum_{\ell=1}^{L} X_\ell R_\ell^{(t)}$.
(iv) Recenter $M^{(t+1)}$ and, when size is fixed, rescale it to the chosen norm.
(v) Repeat (ii)–(iv) until Φ changes by less than the chosen tolerance.

The template is determined only up to a common right rotation unless a final orientation convention is imposed.

Remark 3.2.11. *Manifold Hypothesis*
The manifold hypothesis asserts that observations $x_1, \dots, x_n \in \mathbb{R}^p$ satisfy $\operatorname{dist}(x_i, \mathcal{M}) \le \varepsilon$ for some r-dimensional set $\mathcal{M} \subset \mathbb{R}^p$, with $r \ll p$ and small $\varepsilon \ge 0$. For curved $\mathcal{M}$, ambient distances $\|x_i - x_j\|_2$ need not approximate intrinsic distances $d_{\mathcal{M}}(x_i, x_j)$. Manifold learning therefore estimates low-dimensional coordinates from neighborhoods, graph geodesics, or tangent spaces, rather than from global Euclidean variance alone.

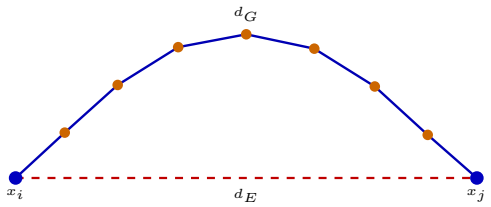

Figure 252: d_G follows nearby samples along the curve; d_E is the ambient shortcut.

Example 3.2.12. *Manifold Methods*
Isomap, locally linear embedding, Laplacian eigenmaps, and diffusion maps construct low-dimensional coordinates from a neighborhood graph or kernel on the sample.

Method 3.2.13. *Isomap*
Let $x_i \in \mathbb{R}^p$, fix r, set $H = I_n - n^{-1}\mathbf{1}\mathbf{1}^T$, and choose a connected neighborhood graph G on $[n]$.

(i) Assign edge weights $w_{ij} = \|x_i - x_j\|_2$ to neighboring pairs.
(ii) Compute shortest-path distances $d_G(i,j)$ and set $D_G^{(2)} = (d_G(i,j)^2)$.
(iii) Form the centered Gram estimate $B = -\frac{1}{2} H D_G^{(2)} H$.
(iv) Compute $Be_k = \lambda_k e_k$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0$, set $E_r = [e_1 \ \cdots \ e_r]$, $\Lambda_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$.
(v) Return $Z = E_r \Lambda_r^{1/2}$.

Isomap replaces ambient distances by graph geodesic distances; the coordinate extraction is spectral.

Method 3.2.14. *Locally Linear Embedding*
Let $x_i \in \mathbb{R}^p$, fix K, r, and let $\mathcal{N}(i)$ be the K nearest neighbors of x_i.

(i) Set $\mathcal{W}_i = \{w : \sum_k w_k = 1,\ w_k = 0\ (k \notin \mathcal{N}(i))\}$ and $w_i \in \operatorname*{argmin}_{w \in \mathcal{W}_i} \left\| x_i - \sum_k w_k x_k \right\|_2^2$.
(ii) Form $W = (w_{ik})$ and $M = (I_n - W)^T (I_n - W)$.
(iii) Compute eigenvectors $u_1, \ldots, u_{r+1}$ of M for the smallest eigenvalues, with $u_1 \propto \mathbf{1}$.
(iv) Return $Y = [u_2 \ \cdots \ u_{r+1}] \in \mathbb{R}^{n \times r}$, with row y_i^T as the coordinate of x_i.

The affine constraint makes local weights translation-invariant; discarding u_1 gives mean-centered coordinates.

Method 3.2.15. *Local MDS*
Let $\mathcal{N} \subset \{(i,i') : i < i'\}$ be symmetric nearby pairs, let $z_i \in \mathbb{R}^r$, and fix $\tau > 0$.
Local MDS minimizes $S_{\text{local}}(Z) = \sum_{(i,i') \in \mathcal{N}} (d_{ii'} - \|z_i - z_{i'}\|_2)^2 - \tau \sum_{(i,i') \notin \mathcal{N}} \|z_i - z_{i'}\|_2$.

(i) The local stress term preserves distances on nearby pairs $(i,i') \in \mathcal{N}$.
(ii) The separation term penalizes collapse of non-neighboring pairs $(i,i') \notin \mathcal{N}$.

Thus the objective preserves local metric structure while repelling unrelated manifold regions.

Definition 3.2.16. *Principal Curves and Surfaces*
Let $X \in \mathbb{R}^p$ be a random vector and let $f : \Lambda \to \mathbb{R}^p$ be a smooth parametrized set.

(i) The *projection index* is $\lambda_f(x) \in \operatorname*{argmin}_{\lambda \in \Lambda} \|x - f(\lambda)\|_2$.
(ii) A *principal curve* has $\Lambda \subset \mathbb{R}$ and satisfies $f(\lambda) = \mathbb{E}\{X \mid \lambda_f(X) = \lambda\}$.
(iii) A *principal surface* is the same definition with $\Lambda \subset \mathbb{R}^q$, $q \geq 2$, and $f = (f_1, \ldots, f_p)$.

For finite data, alternate coordinate-wise smoothing against $\lambda_f(x_i)$ and reprojection onto $f(\Lambda)$.

Remark 3.2.17. *SOMs Versus Principal Surfaces*
Both self-organizing maps and principal surfaces approximate data by low-dimensional representatives.

(i) A principal surface is a smooth map $f : \Lambda \subset \mathbb{R}^q \to \mathbb{R}^p$ with fitted coordinates $\lambda_f(x_i)$.
(ii) A self-organizing map is a finite prototype family $\{m_a\}_{a \in A} \subset \mathbb{R}^p$ indexed by a lattice or graph A.

With many prototypes and kernel neighborhood weights, batch SOM updates resemble coordinate-wise surface smoothing, but one fit is smooth-parametric and the other is finite-prototype.

Remark 3.2.18. *Local versus Global Scaling*
Let $d_E(i,j) = \|x_i - x_j\|_2$ and let $\mathcal{N}$ be a neighborhood relation.

(i) *Global scaling* uses all pairs through $d_E(i,j)$ or centered inner products, and is appropriate when ambient distances encode the target geometry.
(ii) *Local scaling* uses $\mathcal{N}$, graph distances, or local reconstruction weights to preserve neighborhood order or local shape on a curved set.

Neighborhood scale is critical: too small disconnects, too large shortcuts.

3.3 Matrix Factorization and Topic Models

Many unsupervised models express data as combinations of latent factors.

Definition 3.3.1. *Low-Rank Matrix Factorization*
Let $X \in \mathbb{R}^{m \times n}$ and $1 \leq r \leq \min\{m,n\}$.
A *rank-r matrix factorization* is pair $(U,V) \in \mathbb{R}^{m \times r} \times \mathbb{R}^{n \times r}$ with fitted matrix $\widehat{X} = UV^\top$ and $\operatorname{rank}(\widehat{X}) \leq r$.
For a loss $\mathcal{L}$, choose (U,V) to minimize $\mathcal{L}(X, UV^\top)$; the columns encode latent row and column factors.

Definition 3.3.2. *Nonnegative Matrix Factorization*
Let $X \in \mathbb{R}_+^{n\times p}$ and $1 \le r \le \min\{n,p\}$.
A *nonnegative matrix factorization* is a pair $(W,H) \in \mathbb{R}_+^{n\times r} \times \mathbb{R}_+^{r\times p}$ with fitted matrix $\widehat{X} = WH$.
For loss $\mathcal{L}$, minimize $\mathcal{L}(X, WH)$ over $W, H \ge 0$, yielding additive parts-based factors.

Method 3.3.3. *Poisson NMF Multiplicative Updates*
For count data $X = (x_{ij})$, set $\mu = WH$ and $L(W,H) = \sum_{i,j}\{x_{ij}\log\mu_{ij} - \mu_{ij}\}$, up to constants.

(i) Initialize $W, H > 0$.
(ii) Set $\rho_{ik}^W = (\sum_j \frac{h_{kj}x_{ij}}{\mu_{ij}})/(\sum_j h_{kj})$ and update $w_{ik} \leftarrow w_{ik}\rho_{ik}^W$.
(iii) Recompute $\mu = WH$, set $\rho_{kj}^H = (\sum_i \frac{w_{ik}x_{ij}}{\mu_{ij}})/(\sum_i w_{ik})$, and update $h_{kj} \leftarrow h_{kj}\rho_{kj}^H$.
(iv) Repeat (ii)–(iii) until $L(W,H)$ stabilises.

The updates are local; different initialisations may give different NMF decompositions.

Remark 3.3.4. *NMF Non-Uniqueness*
An NMF $X = WH$ is not identifiable from nonnegativity alone: if A is invertible and $WA \ge 0$, $A^{-1}H \ge 0$, then $X = (WA)(A^{-1}H)$.
Interpretability therefore depends on additional geometry, constraints, or initialization.

Definition 3.3.5. *Archetypal Analysis*
Let $X \in \mathbb{R}^{n\times p}$ have observed rows x_i^T, and fix r archetypes.

(i) Observation weights $W = (w_{ik}) \in \mathbb{R}_+^{n\times r}$ satisfy $\sum_{k=1}^{r} w_{ik} = 1$.
(ii) Archetype weights $B = (b_{ki}) \in \mathbb{R}_+^{r\times n}$ satisfy $\sum_{i=1}^{n} b_{ki} = 1$.
(iii) Archetypes are the rows of $H = BX$, hence lie in $\mathrm{conv}\{x_1,\dots,x_n\}$.
(iv) Archetypal analysis chooses W, B minimising $J(W,B) = \|X - WBX\|_F^2$.

Archetypes are convex-hull extremes rather than central prototypes.

Remark 3.3.6. *Archetypes, k-Means, and NMF*
For prototype matrix H and assignment/mixture weights W:

(i) k-means uses one-hot rows of W.
(ii) NMF uses nonnegative W, H without requiring prototype rows of H to be observed-data mixtures.
(iii) Archetypal analysis uses simplex rows of W and constrains $H = BX$ with simplex rows of B.

Archetypes are convex-hull prototypes; k-means is hard-assigned and NMF is hull-free.

Definition 3.3.7. *Topic Model*
Fix K topics and a vocabulary of size p.

(i) Topic k is a word distribution $\beta_k \in \Delta_{p-1}$.
(ii) Document d has topic mixture $\theta_d \in \Delta_{K-1}$.
(iii) Token m in document d is generated by $z_{dm} \sim \mathrm{Cat}(\theta_d)$ and $w_{dm} \sim \mathrm{Cat}(\beta_{z_{dm}})$.
(iv) Latent Dirichlet allocation adds priors $\theta_d \sim \mathrm{Dir}(\alpha)$ and $\beta_k \sim \mathrm{Dir}(\eta)$.

Remark 3.3.8. *Regularized Factors*
For a factorization $X \approx UV^\top$, constraints select an identifiable and interpretable solution class.

(i) Low rank bounds latent dimension.
(ii) Sparsity restricts active variables or factors.
(iii) Nonnegativity enforces additive parts.
(iv) Probabilistic priors encode uncertainty and shrinkage.

These restrictions convert an underdetermined algebraic fit into a statistical learning problem.

3.4 Representation Learning Objectives

Representation learning trains maps that make downstream prediction, reconstruction, comparison, or generation easier.

Definition 3.4.1. *Autoencoder*
Let $x_1,\dots,x_n \in \mathcal{X}$ and let $\mathcal{H}$ be a representation space.

(i) An *autoencoder* is an encoder-decoder pair $e : \mathcal{X} \to \mathcal{H}$ and $d : \mathcal{H} \to \mathcal{X}$.
(ii) It is trained by minimizing a reconstruction loss such as $n^{-1}\sum_{i=1}^{n} \|x_i - d(e(x_i))\|_2^2$.

Bottlenecks, noise, sparsity, or latent-variable constraints rule out the unrestricted identity solution.

Definition 3.4.2. *Sparse Representation Objective*
Let $x_i \in \mathbb{R}^p$, dictionary $W \in \mathbb{R}^{p\times r}$, codes $h_i \in \mathbb{R}^r$, and $\lambda > 0$.
A *sparse representation objective* chooses $W, h_1,\dots,h_n$ to minimize $\sum_{i=1}^{n}\{\|x_i - Wh_i\|_2^2 + \lambda\|h_i\|_1\}$.
The reconstruction term fits x_i, while the L_1 term sparsifies h_i.

Definition 3.4.3. *Contrastive Objective*
Let $g : \mathcal{X} \to \mathbb{R}^r$ be a representation map, $s(u,v)$ a similarity score, and $\tau > 0$ a temperature.

(i) For an anchor x, sample a positive x^+ and negatives $x_1^-,\dots,x_K^-$.
(ii) With $a^+ = s(g(x), g(x^+))/\tau$ and $a_k^- = s(g(x), g(x_k^-))/\tau$, min loss i.e. $\log\left\{\exp(a^+) + \sum_{k=1}^{K}\exp(a_k^-)\right\} - a^+$.

The objective raises similarity to positives and lowers relative similarity to negatives.

3.5 Association Rules and Density-Ratio Views

Association-rule analysis searches for simple descriptions of regions where the input distribution has relatively large probability. The classical market-basket setting is the binary special case, but the same idea extends to conjunctions of coordinate conditions.

Definition 3.5.1. *Conjunctive Region*
Let $X = (X_1, \dots, X_p)$ take values in $\mathcal{S} = \prod_{j=1}^{p} \mathcal{S}_j$. For $\mathcal{J} \subseteq \{1, \dots, p\}$ and $s = (s_j)_{j \in \mathcal{J}}$ with $s_j \subseteq \mathcal{S}_j$, the *conjunctive region* is $R(\mathcal{J}, s) = \{x \in \mathcal{S} : x_j \in s_j \ \forall j \in \mathcal{J}\}$, with event $\{X \in R(\mathcal{J}, s)\} = \bigcap_{j \in \mathcal{J}} \{X_j \in s_j\}$.
Coordinates in $\{1, \dots, p\} \setminus \mathcal{J}$ are unrestricted; under the extension $s_j = \mathcal{S}_j$, they add no constraint.
For $\mathcal{S}_j \subseteq \mathbb{R}$, s_j is typically an interval; for categorical $\mathcal{S}_j$, s_j is typically a subset of levels.

Definition 3.5.2. *Market-Basket Item Sets*
Let $\mathcal{I} = \{1, \dots, K\}$ and $z_i = (z_{ik})_{k \in \mathcal{I}} \in \{0,1\}^{\mathcal{I}}$ for $i = 1, \dots, n$, where $z_{ik} = 1$ iff observation i contains item k.

(i) An *item set* is any $\mathcal{K} \subseteq \mathcal{I}$.
(ii) The containment indicator is $Y_i(\mathcal{K}) = \prod_{k \in \mathcal{K}} z_{ik}$; observation i contains $\mathcal{K}$ iff $Y_i(\mathcal{K}) = 1$.
(iii) The empirical support is $T(\mathcal{K}) = n^{-1} \sum_{i=1}^{n} Y_i(\mathcal{K})$.
(iv) For finite categorical supports, take $\mathcal{I} = \{(j, a) : 1 \le j \le p,\ a \in \mathcal{S}_j\}$, hence $K = \sum_{j=1}^{p} |\mathcal{S}_j|$.

Definition 3.5.3. *Association Rule, Confidence, and Lift*
Let $A, B \subseteq \mathcal{I}$ satisfy $A \cap B = \emptyset$, $A \neq \emptyset$, $B \neq \emptyset$, $T(A) > 0$, and $T(B) > 0$.
The *association rule* with antecedent A and consequent B is $A \Rightarrow B$.

(i) The rule support is $S(A \Rightarrow B) = T(A \cup B)$.
(ii) The confidence is $C(A \Rightarrow B) = \frac{T(A \cup B)}{T(A)}$.
(iii) The lift is $L(A \Rightarrow B) = \frac{C(A \Rightarrow B)}{T(B)} = \frac{T(A \cup B)}{T(A)T(B)}$.

Interpretation: $C(A \Rightarrow B)$ estimates prevalence of B given A; $L(A \Rightarrow B)$ compares it with $T(B)$.

Example 3.5.4. *A Market-Basket Rule*
Let $A = \{\text{peanut butter}, \text{jelly}\}$ and $B = \{\text{bread}\}$.

(i) Support $T(A \cup B) = 0.03$ means all three items appear in about 3% of baskets.
(ii) Confidence $C(A \Rightarrow B) = 0.82$ means about 82% of baskets containing A also contain bread.
(iii) If $T(B) = 0.43$, then $L(A \Rightarrow B) = 0.82/0.43 \approx 1.91$.

The rule predicts bread more often than its marginal prevalence.

Algorithm 4: Apriori Association-Rule Search

Require: Binary item data $z_1, \dots, z_n$, support threshold t, confidence threshold c
Ensure: Rules with support above t and confidence above c
Compute supports for all one-item sets and retain $\mathcal{F}_1 = \{\mathcal{K} : |\mathcal{K}| = 1,\ T(\mathcal{K}) > t\}$
Set $m \leftarrow 2$
while $\mathcal{F}_{m-1} \neq \emptyset$ **do**
  Form candidate item sets of size m whose every size-$(m-1)$ subset lies in $\mathcal{F}_{m-1}$
  Scan the data to compute $T(\mathcal{K})$ for each candidate
  Retain $\mathcal{F}_m = \{\mathcal{K} : |\mathcal{K}| = m,\ T(\mathcal{K}) > t\}$
  Set $m \leftarrow m + 1$
end while
For each retained item set $\mathcal{K}$, test nonempty proper antecedents $A \subset \mathcal{K}$
Report $A \Rightarrow \mathcal{K} \setminus A$ when $C(A \Rightarrow \mathcal{K} \setminus A) > c$

Remark 3.5.5. *Apriori Monotonicity and Limits*
Support is antitone: $\mathcal{L} \subseteq \mathcal{K} \Longrightarrow T(\mathcal{L}) \ge T(\mathcal{K})$.

(i) If $T(\mathcal{L}) < t$, then every $\mathcal{K} \supseteq \mathcal{L}$ also has $T(\mathcal{K}) < t$ and may be pruned.
(ii) A support threshold t can discard rules $A \Rightarrow B$ with large $C(A \Rightarrow B)$ or $L(A \Rightarrow B)$ but small $T(A \cup B)$.

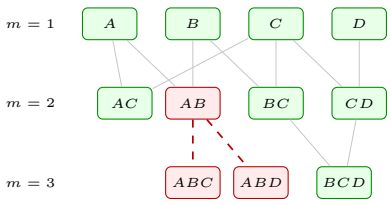

Figure 253: Apriori keeps frequent item sets; if $T(AB) \le t$, all supersets of AB are pruned.

Method 3.5.6. *Generalized Association Rules*
For $\mathcal{J} \subseteq \{1, \dots, p\}$ and $s_j \subseteq \mathcal{S}_j$, set $R(\mathcal{J}, s) = \bigcap_{j \in \mathcal{J}} \{X_j \in s_j\}$ and $\widehat{P}\{R(\mathcal{J}, s)\} = n^{-1} \sum_{i=1}^{n} \mathbf{1}\{x_{ij} \in s_j \ \forall j \in \mathcal{J}\}$.

(i) Allow s_j to be intervals for quantitative variables and subsets of levels for categorical variables.
(ii) Search over candidate regions with a heuristic rule learner.
(iii) Retain rules whose fitted score and observed support meet the chosen thresholds.

Method 3.5.7. *Unsupervised Learning as Supervised Density-Ratio Estimation*
Let g be the data density, g_0 a reference density, and $Y = 1$ label data while $Y = 0$ labels reference observations.

(i) Draw a reference sample from g_0 and pool it with the data, weighting the two classes equally.
(ii) Fit a probabilistic classifier $\widehat{\mu}(x) \approx \Pr(Y = 1 \mid X = x)$.
(iii) Use $\mu(x) = g(x)/(g(x) + g_0(x))$ to estimate the density ratio by $\widehat{g}(x)/g_0(x) = \widehat{\mu}(x)/(1 - \widehat{\mu}(x))$.

Uniform, Gaussian, and product-marginal g_0 target high density, nonnormality, and dependence, respectively.

Remark 3.5.8. *Rule Learners for Density Ratios*
After estimating $\widehat{g}/g_0$, any learner that returns regions $R = \bigcap_{j \in \mathcal{J}} \{X_j \in s_j\}$ can search for generalized item sets. Tree leaves and box-peeling regions have this conjunctive form.
The unsupervised target is large $\widehat{g}(x)/g_0(x)$ on R together with adequate empirical support, not predictive risk.

3.6 Graph Ranking and PageRank

Graph ranking turns link structure into a numerical importance score. PageRank is a spectral example: a page receives weight from incoming links, and a link from an already important page contributes more than a link from an unimportant one.

Definition 3.6.1. *PageRank Equations*

For pages indexed by $\{1, \dots, N\}$, define $L \in \{0,1\}^{N \times N}$ by $L_{ij} = 1$ iff page j links to page i, and set $c_j = \sum_{i=1}^{N} L_{ij}$.

Assume $c_j > 0$ for all j and fix $d \in (0,1)$.

A *PageRank score vector* is $p = (p_1, \dots, p_N) \in \mathbb{R}^N$ satisfying $p_i = (1-d) + d \sum_{j=1}^{N} \frac{L_{ij}}{c_j} p_j$ for $i = 1, \dots, N$.

Proposition 3.6.2. *PageRank Normalization*

Let $L \in \{0,1\}^{N \times N}$, $c_j = \sum_{i=1}^{N} L_{ij} > 0$, *and* $d \in (0,1)$.

If $p \in \mathbb{R}^N$ *satisfies* $p_i = (1-d) + d \sum_{j=1}^{N} \frac{L_{ij}}{c_j} p_j$ *for* $i = 1, \dots, N$, *then* $\sum_{i=1}^{N} p_i = N$.

Proof. Let $P = \sum_{i=1}^{N} p_i$. Summing over i gives $P = N(1-d) + dP$, since $\sum_{i=1}^{N} L_{ij} = c_j$.
Hence $(1-d)P = N(1-d)$, so $P = N$.
Structural Commentary: Column normalisation preserves total score up to the teleportation offset.
Key Insight: Summing cancels the link structure because each column has mass c_j. □

Method 3.6.3. *Eigenvector Form*
Let $\mathbf{e} = (1, \dots, 1)^T$, $D_c = \operatorname{diag}(c_1, \dots, c_N)$, and $A = (1-d)\frac{\mathbf{e}\mathbf{e}^T}{N} + dLD_c^{-1}$. The PageRank equations are $\mathbf{p} = A\mathbf{p}$ with $\mathbf{e}^T\mathbf{p} = N$. Since $\mathbf{e}^T A = \mathbf{e}^T$ and $A_{ij} \geq (1-d)/N > 0$, A is positive column-stochastic. Thus $\mathbf{p}$ is the positive eigenvector for eigenvalue 1 normalized by $\mathbf{e}^T\mathbf{p} = N$, and $\mathbf{p}/N$ is the stationary distribution in column-vector convention.

Algorithm 5: PageRank by Power Iteration

Require: Link matrix L, positive outlink counts c_j, damping parameter $0 < d < 1$, initial vector $\mathbf{p}_0 > 0$
Ensure: PageRank vector normalized by $\mathbf{e}^T\mathbf{p} = N$
  Form $A = (1-d)\mathbf{e}\mathbf{e}^T/N + dLD_c^{-1}$
  repeat
    $\mathbf{p}_k \leftarrow A\mathbf{p}_{k-1}$
    $\mathbf{p}_k \leftarrow N\mathbf{p}_k/(\mathbf{e}^T\mathbf{p}_k)$
  until the iterates stabilize

Remark 3.6.4. *Random-Surfer Normalization*

The probability-scale normalization uses $\pi_i = (1-d)/N + d \sum_{j=1}^{N} (L_{ij}/c_j)\pi_j$ and $\mathbf{e}^T\pi = 1$.

Equivalently, $p = N\pi$ satisfies the average-score form $\mathbf{e}^T\mathbf{p} = N$.
Random-surfer form: follow an outgoing link with probability d; jump uniformly with probability $1-d$.

Example 3.6.5. *Four-Page Network*
Let $d = 0.85$, $L = \begin{pmatrix} 0&0&1&0 \\ 1&0&0&0 \\ 1&1&0&1 \\ 0&0&0&0 \end{pmatrix}$, $\mathbf{c} = L^T\mathbf{e} = (2,1,1,1)^T$, and $D_c = \operatorname{diag}(\mathbf{c})$.
Solving $\mathbf{p} = (1-d)\mathbf{e} + dLD_c^{-1}\mathbf{p}$ gives $\widehat{\mathbf{p}} = (1.49, 0.78, 1.58, 0.15)^T$. Since $L_{4j} = 0$ for all j, $\widehat{p}_4 = 1 - d = 0.15$.

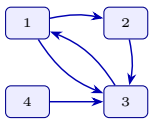

Figure 254: Four-page PageRank graph: $\widehat{\mathbf{p}} \approx (1.49, 0.78, 1.58, 0.15)^T$.

3.7 Latent Factor and Component Analysis

Latent factor methods model observed coordinates as mixtures of unobserved sources. The main question is which structure makes the sources interpretable: covariance structure, statistical independence, or non-Gaussian projections.

Definition 3.7.1. *Factor Analysis*
Let $X \in \mathbb{R}^p$ and $q < p$. A *factor-analysis model* is a representation $X = AS + \varepsilon$ with $A \in \mathbb{R}^{p\times q}$, $S \in \mathbb{R}^q$, $\varepsilon \in \mathbb{R}^p$, where: $\mathrm{Cov}(S) = I_q$, $\mathrm{Cov}(\varepsilon) = D_\varepsilon$ for diagonal $D_\varepsilon \in \mathbb{R}^{p\times p}$, and $\mathrm{Cov}(S, \varepsilon) = 0$.
The columns $a_1, \dots, a_q$ of A are the *factor loadings*, and $\Sigma := \mathrm{Cov}(X) = AA^T + D_\varepsilon$.
For any orthogonal $R \in \mathbb{R}^{q\times q}$, (A, S) and (AR^T, RS) yield the same Σ.
Hence (A, S) is identifiable only up to orthogonal rotation.

Definition 3.7.2. *Independent Component Analysis*
Let $X \in \mathbb{R}^p$. An *independent component model* is a representation $X = AS$ such that:

(i) $A \in \mathbb{R}^{p\times p}$ is nonsingular.
(ii) $S = (S_1, \dots, S_p)^T \in \mathbb{R}^p$ has mutually independent components.

If X is whitened, then $\mathrm{Cov}(X) = I_p$ and A may be taken orthogonal.
ICA estimates orthogonal A with $Y = A^T X$ approximately componentwise independent, typically via non-Gaussianity contrasts. Gaussian components are identifiable only up to orthogonal rotations.

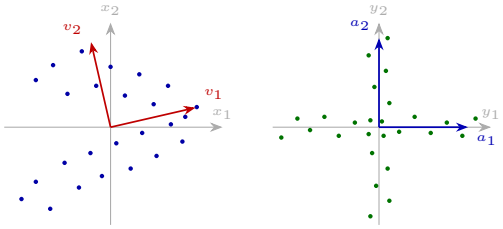

Figure 255: v_1, v_2: PCA variance axes; a_1, a_2: ICA rotation exposing independent non-Gaussian coordinates.

Method 3.7.3. *Entropy View of ICA*
For a density g, define $H(g) = -\int g(y) \log g(y)\, dy$.
Let $Y = (Y_1, \dots, Y_p)$ have joint density g_Y and marginal densities g_j.

(i) Set $H(Y) := H(g_Y)$, $H(Y_j) := H(g_j)$, and $I(Y) := \sum_{j=1}^p H(Y_j) - H(Y)$.
Then $I(Y) \geq 0$, with $I(Y) = 0$ iff $Y_1, \dots, Y_p$ are independent.
(ii) If X is whitened, A is orthogonal, and $Y = A^T X$, then $H(Y) = H(X)$.
Hence minimising $I(Y)$ over orthogonal A is equivalent to minimising $\sum_{j=1}^p H(Y_j)$.
(iii) Replace marginal-entropy minimisation by marginal non-Gaussianity maximisation,
e.g. through negentropy contrasts, since Gaussian marginals maximise entropy at fixed variance.

Remark 3.7.4. *Projection Pursuit and ICA*
Projection pursuit optimizes a non-Gaussianity index over $Y = a^T X$ or $Y = (a_1^T X, a_2^T X)$.
ICA uses the same type of index on $Y_j = a_j^T X$, but imposes an orthogonal system $(a_1, \dots, a_p)$ after whitening.

Method 3.7.5. *Product-Density ICA*
For whitened $X \in \mathbb{R}^p$ and orthogonal $A \in \mathbb{R}^{p\times p}$, set $Y = A^T X$.

(i) Assume the product density $f_Y(y) = \prod_{j=1}^p f_j(y_j)$.
(ii) Parameterize each marginal density by $f_j(s) = \phi(s)\exp(g_j(s))$ with $\int \phi(s)\exp(g_j(s))\, ds = 1$.
(iii) Estimate $A, g_1, \dots, g_p$ by penalized likelihood subject to $A^T A = I$.
For fixed A, the objective decomposes into p one-dimensional smooth density fits.

The roughness penalty shrinks g_j toward 0, the Gaussian baseline ϕ.

Algorithm 6: Product-Density ICA Alternation

Require: Whitened observations $x_1, \dots, x_n$, initial orthogonal matrix A
Ensure: Orthogonal unmixing matrix A and components $s_{ij} = a_j^T x_i$
repeat
  for each component j **do**
    Fit a smooth tilted-Gaussian density $f_j(s) = \phi(s)\exp(g_j(s))$ to the current scores $a_j^T x_i$
    Update $a_j \leftarrow \mathbb{E}\{X g_j'(a_j^T X)\} - \mathbb{E}\{g_j''(a_j^T X)\} a_j$
  end for
  Orthogonalize A: if $A = UDV^T$ is its SVD, set $A \leftarrow UV^T$
until the directions stabilize

4 Probabilistic Graphical and Causal Learning

4.1 Directed and Undirected Graphical Models

Graphical models use graph structure to encode conditional independence and to organize high-dimensional probability laws.

Definition 4.1.1. *Directed Graphical Model*
Let $G = (V, E)$ be a directed acyclic graph with $V = \{1, \dots, d\}$ and $\mathrm{pa}(j) = \{i \in V : i \to j \in E\}$.
For $x = (x_1, \dots, x_d)$, write $x_{\mathrm{pa}(j)} = (x_i)_{i\in \mathrm{pa}(j)}$. A law for $X = (X_1, \dots, X_d)$ with joint density or mass function p *factorizes according to* G if $p(x) = \prod_{j=1}^d p(x_j \mid x_{\mathrm{pa}(j)})$, with $p(x_j \mid x_\varnothing) := p(x_j)$.

Definition 4.1.2. *Markov Graph Vocabulary*
Let $G = (V, E)$ be an undirected graph, and let $A, B, C \subseteq V$ be disjoint.

(i) Vertices $j, k \in V$ are *adjacent* iff $\{j, k\} \in E$.
(ii) A *path* from u to v is $(v_0, \dots, v_m)$ with $v_0 = u$, $v_m = v$, and $\{v_{\ell-1}, v_\ell\} \in E$ for $\ell = 1, \dots, m$.
(iii) C *separates* A and B iff every path from $a \in A$ to $b \in B$ contains a vertex in C.
(iv) A *clique* is $K \subseteq V$ such that $\{j, k\} \in E$ for all distinct $j, k \in K$.
It is *maximal* iff no strictly larger clique contains K.

Definition 4.1.3. *Undirected Graphical Model*
Let $G = (V, E)$ be an undirected graph, let $\mathcal{C}$ be its maximal cliques, and let p be a positive joint density or mass function for $X = (X_j)_{j \in V}$. For $C \subseteq V$, write $x_C = (x_j)_{j \in C}$. The following are equivalent:

(i) p is *Markov with respect to* G: if $j \neq k$ and $\{j, k\} \notin E$, then $X_j \perp\!\!\!\perp X_k \mid X_{V \setminus \{j,k\}}$.
(ii) p factorizes as $p(x) = Z^{-1} \prod_{C \in \mathcal{C}} \psi_C(x_C)$ for clique potentials $\psi_C \geq 0$, with $Z = \sum_x \prod_{C \in \mathcal{C}} \psi_C(x_C)$ in the discrete case and $Z = \int \prod_{C \in \mathcal{C}} \psi_C(x_C)\, dx$ in the continuous case.

Proposition 4.1.4. *Gaussian Graphical Zeros*
Let $X \sim N_d(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Theta = \Sigma^{-1} = (\theta_{jk})$. For j, write $\theta_{-j,j} = (\theta_{\ell j})_{\ell \neq j}$ and $\mu_{-j} = (\mu_\ell)_{\ell \neq j}$. Then:

(i) For $z \in \mathbb{R}^{d-1}$, $\mathbb{E}[X_j \mid X_{-j} = z] = \mu_j - \frac{\theta_{-j,j}^T (z - \mu_{-j})}{\theta_{jj}}$.
(ii) For $j \neq k$, $X_j \perp\!\!\!\perp X_k \mid X_{-(j,k)}$ iff $\theta_{jk} = 0$.

Proof.

(i) Let $Y = X_j$ and $Z = X_{-j}$. Order coordinates as (Z, Y), with covariance block Σ_{ZZ}, cross-covariance σ_{Zj}.
Gaussian conditioning gives $\mathbb{E}[Y \mid Z = z] = \mu_j + (z - \mu_{-j})^T \Sigma_{ZZ}^{-1} \sigma_{Zj}$.
From $\Sigma\Theta = I$, the (Z, j) block gives $\Sigma_{ZZ}\theta_{-j,j} + \sigma_{Zj}\theta_{jj} = 0$, hence $\Sigma_{ZZ}^{-1}\sigma_{Zj} = -\frac{\theta_{-j,j}}{\theta_{jj}}$.
(ii) Let $A = \{j, k\}$ and $B = A^c$. The conditional Gaussian law of $X_A \mid X_B$ has precision $\Theta_{AA} = \begin{pmatrix} \theta_{jj} & \theta_{jk} \\ \theta_{jk} & \theta_{kk} \end{pmatrix}$. A bivariate Gaussian has independent coordinates iff its precision is diagonal, so $X_j \perp\!\!\!\perp X_k \mid X_B$ iff $\theta_{jk} = 0$.

Structural Commentary: Conditioning turns precision entries into regression interactions.
Key Insight: Gaussian conditional independence is zero conditional precision. □

Definition 4.1.5. *Ising Model*
Let $E \subseteq \{\{j, k\} : 1 \leq j < k \leq p\}$, set $x_0 = 1$, $\widetilde{E} = E \cup \{\{0, j\} : 1 \leq j \leq p\}$, and $\mathcal{N}(j) = \{k : \{j, k\} \in E\}$.
A *pairwise Ising model* for $X \in \{0, 1\}^p$ is specified by:

(i) Joint law $p_\Theta(x) = \exp\left\{ \sum_{\{j,k\} \in \widetilde{E}} \theta_{jk} x_j x_k - \Phi(\Theta) \right\}$, where $\Phi(\Theta) = \log \sum_{x \in \{0,1\}^p} \exp\left\{ \sum_{\{j,k\} \in \widetilde{E}} \theta_{jk} x_j x_k \right\}$.
(ii) Node conditionals $\mathbb{P}_\Theta(X_j = 1 \mid X_{-j} = x_{-j}) = \left(1 + \exp\left\{ -\theta_{0j} - \sum_{k \in \mathcal{N}(j)} \theta_{jk} x_k \right\}\right)^{-1}$ for $j = 1, \dots, p$.

For $\{j, k\} \in E$, θ_{jk} is the conditional log-odds interaction between nodes j and k.

Definition 4.1.6. *Restricted Boltzmann Machine*
Let $v \in \{0, 1\}^p$ be visible units, let $h \in \{0, 1\}^q$ be hidden units, and let $\sigma(t) = (1 + \exp(-t))^{-1}$.
A *restricted Boltzmann machine* is specified by:

(i) Graph restriction: no visible-visible or hidden-hidden edges.
(ii) Joint law $p(v, h) = Z^{-1} \exp(a^T v + b^T h + v^T W h)$ with $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$, and $W \in \mathbb{R}^{p \times q}$.
(iii) Conditional factorization: $p(v \mid h) = \prod_{j=1}^{p} \mathrm{Bern}(v_j; \sigma(a_j + W_{j\cdot} h))$ and $p(h \mid v) = \prod_{k=1}^{q} \mathrm{Bern}(h_k; \sigma(b_k + W_{\cdot k}^T v))$.

In supervised variants, v may concatenate feature and label units.

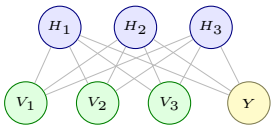

Figure 256: RBM: H_j connect only to visible features V_i and label Y; no within-layer edges.

Remark 4.1.7. *Separation*
For disjoint vertex sets A, B, C, separation is a graphical certificate for $X_A \perp\!\!\!\perp X_B \mid X_C$: DAGs use d-separation, while undirected graphs use path separation by C.
In a Markov graph, C separates A and B implies $X_A \perp\!\!\!\perp X_B \mid X_C$.
Missing edges give pairwise certificates; separation gives setwise certificates.

4.2 Inference and Learning in Graphical Models

Inference in a graphical model computes marginal, conditional, or most likely latent quantities from the joint law encoded by the graph.

Definition 4.2.1. *Inference Tasks*
Let $X = (X_j)_{j \in V}$ have joint law p, and let $A, B \subseteq V$ be disjoint. Standard *inference tasks* are:

(i) *Marginal inference*: compute $p(x_A)$.

(ii) *Conditional inference*: compute $p(x_A \mid x_B)$.
(iii) *MAP inference*: compute $x_A^\star \in \arg\max_{x_A} p(x_A \mid x_B)$, with $B = \varnothing$ for unconditional MAP.
(iv) *Latent posterior inference*: for latent Z and observed $X = x$, compute $p(z \mid x)$.

Method 4.2.2. *Message Passing*
Let $G = (V, E)$ be a graphical model with local potentials.

(i) *Tree belief propagation*: on trees, two-way edge messages yield exact node and edge marginals.
(ii) *Loopy belief propagation*: on cyclic graphs, iterated message updates yield approximate marginals.

Definition 4.2.3. *Variational Inference*
For latent-variable model $p(x, z)$, intractable $p(z \mid x)$, tractable family $\mathcal{Q}$, *variational inference* choose $q^\star \in \mathcal{Q}$:

(i) *KL projection*: $q^\star \in \arg\min_{q \in \mathcal{Q}} \mathrm{KL}(q(z) \| p(z \mid x))$.
(ii) *ELBO maximization*: equivalently, maximize $\mathcal{L}(q) := \mathbb{E}_q[\log p(x, z)] - \mathbb{E}_q[\log q(z)]$, since $\log p(x) = \mathcal{L}(q) + \mathrm{KL}(q(z) \| p(z \mid x))$.

Remark 4.2.4. *Learning Parameters and Structure*
Let G denote the graph and θ its model parameters.

(i) *Parameter learning*: for fixed G, estimate θ by likelihood or regularized likelihood.
(ii) *Structure learning*: estimate unknown G by optimizing a statistical score over graph candidates, often with structural penalties or combinatorial search.

Method 4.2.5. *Gaussian Graphical Model Learning*
For centered Gaussian data with empirical covariance S, let $\Theta = \Sigma^{-1} \succ 0$ and $\ell(\Theta) := \log\det\Theta - \mathrm{trace}(S\Theta)$.

(i) *Known graph*: for edge set E, estimate $\widehat{\Theta} \in \arg\max_{\Theta \succ 0} \ell(\Theta)$ subject to $\theta_{jk} = 0$ whenever $\{j, k\} \notin E$.
(ii) *Unknown graph*: graphical lasso solves $\widehat{\Theta} \in \arg\max_{\Theta \succ 0} \left\{ \ell(\Theta) - \lambda \sum_{j \neq k} |\theta_{jk}| \right\}$, penalise off-diagonal entries.

Graph selection is therefore shrinkage of inverse-covariance entries; zeros of $\widehat{\Theta}$ remove edges.

Figure 257: $\mathcal{D}$ is the data and S its empirical covariance. Y gives $X \perp \{Z, W\} \mid Y$; graphical lasso solves $\widehat{\Theta}_\lambda \in \arg\max_{\Theta \succ 0} \left\{ \log\det\Theta - \mathrm{tr}(S\Theta) - \lambda \sum_{j \neq k} |\theta_{jk}| \right\}$, and zeros of $\widehat{\Theta}_\lambda$ define missing edges in $\widehat{G}$.

Algorithm 7: Graphical Lasso by Block Coordinate Descent

Require: Empirical covariance $S \in \mathbb{R}^{p \times p}$, penalty $\lambda \geq 0$
Ensure: Covariance estimate W and sparse precision estimate $\widehat{\Theta} = W^{-1}$
Initialize $W \leftarrow S + \lambda I$ and keep the diagonal fixed
repeat
  for $j = 1, \ldots, p$ **do**
    Partition W into W_{11}, w_{12}, and w_{22} by isolating row and column j; partition S similarly to get s_{12}
    Define $\mathcal{S}_\lambda(z) = \mathrm{sign}(z)(|z| - \lambda)_+$
    repeat
      for $k = 1, \ldots, p-1$ **do**

$$\beta_k \leftarrow \frac{\mathcal{S}_\lambda\left(s_{12,k} - \sum_{\ell \neq k} (W_{11})_{k\ell}\beta_\ell\right)}{(W_{11})_{kk}}$$

      end for
    until β has converged for this block
    Update $w_{12} \leftarrow W_{11}\beta$ and symmetrically update row and column j of W
  end for
until W has converged
In the final cycle recover $\widehat{\theta}_{12} = -\widehat{\beta}\widehat{\theta}_{22}$ with $1/\widehat{\theta}_{22} = w_{22} - w_{12}^T\widehat{\beta}$

Method 4.2.6. *Discrete Markov Network Learning*
For a binary Markov network with edge statistic $T_{jk}(X) = X_j X_k$, likelihood equations are $\widehat{\mathbb{E}}T_{jk} = \mathbb{E}_\Theta T_{jk}$ for $\{j, k\} \in E$; exact $\mathbb{E}_\Theta$ sums over $x \in \{0, 1\}^p$.

(i) *Small graphs*: solve $\widehat{\mathbb{E}}T_{jk} = \mathbb{E}_\Theta T_{jk}$ by log-linear Poisson MLE, gradient/Newton updates, or iterative proportional fitting.
(ii) *Large graphs*: approximate $\mathbb{E}_\Theta T_{jk}$ by mean-field or Gibbs-sampling estimates.
(iii) *Hidden-node models*: use observed-data conditionals for $\widehat{\mathbb{E}}T_{jk}$; for RBMs, CD-$r$ starts at data and runs r alternating Gibbs steps $h \sim p(h \mid v)$, $v \sim p(v \mid h)$.

4.3 Causal Graphs, Interventions, and Identifiability

Causal learning asks which predictions remain valid under interventions, not merely under passive observation.

Definition 4.3.1. *Structural Causal Model*
Let J index endogenous variables $X = (X_j)_{j \in J}$ and exogenous disturbances $U = (U_j)_{j \in J}$.
A structural causal model is $M = (G, \{f_j\}_{j \in J}, P_U)$ such that:

(i) $G = (J, E)$ is a directed graph with $\mathrm{pa}(j) = \{k \in J : k \to j \in E\}$.

(ii) P_U is the joint law of U.
(iii) $X_j = f_j(X_{\mathrm{pa}(j)}, U_j)$ for every $j \in J$.
The edges $k \to j$ are the direct causal inputs to X_j.

Definition 4.3.2. *Intervention*
For $M = (G, \{f_k\}_{k\in J}, P_U)$ and $x \in \mathcal{X}_j$, set $\tilde{f}_j \equiv x$ and $\tilde{f}_k = f_k$ for $k \neq j$.
The *intervention* $\mathrm{do}(X_j = x)$ is the modified model $M_{j:=x} = (G_{j:=x}, \{\tilde{f}_k\}_{k\in J}, P_U)$, where $G_{j:=x}$ deletes all incoming edges to j; the induced law $P_{M_{j:=x}}$ is the interventional distribution.

Definition 4.3.3. *Identifiability*
Let $\mathcal{M}$ be the assumed model class and let P_M be the observational law induced by $M \in \mathcal{M}$.
A causal functional $\psi : \mathcal{M} \to \Theta$ is *identifiable* on $\mathcal{M}$ if $P_M = P_{M'}$ implies $\psi(M) = \psi(M')$ for all $M, M' \in \mathcal{M}$.

Remark 4.3.4. *Causal and Predictive Targets*
Let $\hat{y} = g(z)$ be trained under the observational law P_M. Small risk $R_{P_M}(g) = \mathbb{E}_{P_M}[\ell(Y, g(Z))]$ need not imply small interventional risk $R_{P_{M_{j:=x}}}(g)$ after $\mathrm{do}(X_j = x)$. Transport across regimes requires causal stability of the conditional mechanism used by g, not only predictive fit under P_M.

4.4 Counterfactuals and Policy Evaluation

Policy learning and evaluation combine prediction with causal questions about actions not actually taken.

Definition 4.4.1. *Potential Outcomes*
Let $A \in \mathcal{A}$ be a treatment.
(i) *Potential outcome*: $Y(a)$ is the outcome under $\mathrm{do}(A = a)$.
(ii) *Consistency*: $Y = Y(A)$.
(iii) *Individual effect*: for binary $A \in \{0, 1\}$, $\tau = Y(1) - Y(0)$.
(iv) *Average treatment effect*: $\mathrm{ATE} = \mathbb{E}[\tau]$.

Definition 4.4.2. *Ignorability and Propensity*
For binary treatment A and covariates X:
(i) *Ignorability*: $(Y(0), Y(1)) \perp\!\!\!\perp A \mid X$.
(ii) *Propensity score*: $e(x) = P(A = 1 \mid X = x)$.
(iii) *Overlap*: $0 < e(X) < 1$ almost surely.

Definition 4.4.3. *Off-Policy Evaluation*
Let $(X_i, A_i, Y_i)_{i=1}^n$ be logged under behavior policy $\mu(a \mid x)$, and let $\pi(a \mid x)$ be a target policy.
(i) *Target value*: $V(\pi) = \mathbb{E}_\pi[Y]$.
(ii) *Off-policy evaluation*: estimate $V(\pi)$ from data sampled under μ.
(iii) *Inverse-propensity weighting*: $\hat{V}_{\mathrm{IPW}}(\pi) = n^{-1} \sum_{i=1}^n \{\pi(A_i \mid X_i)/\mu(A_i \mid X_i)\} Y_i$.
(iv) *Overlap*: $\mu(a \mid x) > 0$ whenever $\pi(a \mid x) > 0$.

Remark 4.4.4. *Feedback*
If a predictor g induces actions through a policy $\pi_g(a \mid x)$, then future data are drawn from the post-deployment law P^{π_g} rather than the pre-deployment law P^0. Feedback is the dependence $g \mapsto \pi_g \mapsto P^{\pi_g}$; causal learning, reinforcement learning, and deployment monitoring all study or control this induced law.

5 Neural Network Foundations

5.1 Network Architectures and Expressivity

The migrated deep-feedforward-network notes describe neural networks as learned feature maps followed by prediction heads.

Definition 5.1.1. *Deep Feedforward Network*
Let $x \in \mathcal{X}$, set $h^{(0)} = x$, and fix depth L.
(i) *Layer recursion*: $h^{(k)} = g^{(k)}(W^{(k)} h^{(k-1)} + b^{(k)})$ for $k = 1, \ldots, L$.
(ii) *Network map*: $f_\theta : \mathcal{X} \to \mathcal{Y}$ is $f_\theta(x) = h^{(L)} = (f_L \circ \cdots \circ f_1)(x)$.
(iii) *Parameters*: $\theta = \{(W^{(k)}, b^{(k)})\}_{k=1}^L$.
(iv) *Training objective*: choose θ to minimize empirical risk $\hat{R}_n(f_\theta)$ or population risk $R(f_\theta)$.

Remark 5.1.2. *Learned Feature Maps*
Let ℓ be the task loss and let $R(f) = \mathbb{E}[\ell(Y, f(X))]$. Two representation parametrizations are:
(i) *fixed features*: $f_w(x) = a_w(\phi(x))$, with ϕ fixed and only the head parameter w optimized.
(ii) *learned features*: $f_{\theta,w}(x) = a_w(\phi_\theta(x))$, with θ and w optimized jointly.
Learning ϕ_θ adds adaptivity but usually destroys convexity.

Definition 5.1.3. *Projection Pursuit and Ridge Functions*
Let $x \in \mathbb{R}^p$.
(i) *Projection-pursuit form*: $f(x) = \sum_{m=1}^M g_m(\omega_m^T x)$ with $\|\omega_m\|_2 = 1$.
(ii) *Ridge coordinate*: $u_m = \omega_m^T x$.
(iii) *Ridge function*: $g_m(u_m)$ depends on x only through the scalar coordinate u_m.
(iv) *Learned components*: the directions ω_m and univariate profiles g_m are estimated from data.

Definition 5.1.4. *Output and Hidden Units*
Let $T \in \mathbb{R}^K$ denote output scores. Common output parametrizations are:
(i) *linear*: $f_k(x) = T_k$, for conditional means or real-valued responses.

(ii) *sigmoid*: $p(x) = \sigma(T) = \frac{1}{1+\exp(-T)}$, for Bernoulli probabilities.

(iii) *softmax*: $p_k(x) = \exp(T_k)/\sum_{\ell=1}^{K} \exp(T_\ell)$, for K-class probabilities.

Hidden units define activations $Z_m = \phi_m(a_m)$ from preactivations a_m, commonly using ReLU, leaky ReLU, maxout, tanh, or another nonlinear map.

Definition 5.1.5. *Single Hidden Layer Network*
Let $x \in \mathbb{R}^p$, $Z = (Z_1, \dots, Z_M)^T$, $T = (T_1, \dots, T_K)^T$. A single-hidden-layer feedforward network is defined by:

(i) $Z_m = \sigma(\alpha_{0m} + \alpha_m^T x)$ for $m = 1, \dots, M$.
(ii) $T_k = \beta_{0k} + \beta_k^T Z$ for $k = 1, \dots, K$.
(iii) $f_k(x) = g_k(T)$, with $g_k(T) = T_k$ for regression and $g_k(T) = \exp(T_k)/\sum_{\ell=1}^{K} \exp(T_\ell)$ for K-class classification.

In the classification case, $f(x) \in \Delta^{K-1}$.

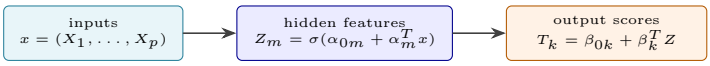

Figure 258: Single-hidden-layer feedforward network.

Remark 5.1.6. *Neural Networks as Ridge Expansions*
For scalar output, a one-hidden-layer network has $f(x) = \beta_0 + \sum_{m=1}^{M} \beta_m \sigma(\alpha_{0m} + \alpha_m^T x)$.
If $\alpha_m \neq 0$ and $\omega_m = \frac{\alpha_m}{\|\alpha_m\|_2}$, the mth term is $\beta_m \sigma(\alpha_{0m} + \|\alpha_m\|_2 \omega_m^T x)$, a ridge term in coordinate $\omega_m^T x$ with learned *direction* ω_m, *scale* $\|\alpha_m\|_2$, *threshold* α_{0m}, and *output coefficient* β_m.
Thus a shallow scalar network is a projection-pursuit expansion with parametrized ridge profiles.

Theorem 5.1.7. *Universal Approximation Theorem Form*
For compact $K \subset \mathbb{R}^d$, $h \in C(K)$, non-polynomial activation $\sigma : \mathbb{R} \to \mathbb{R}$, $\forall \varepsilon > 0$, $\exists\, M \in \mathbb{N}$, $a_m, b_m \in \mathbb{R}$, $w_m \in \mathbb{R}^d$ s.t. one-hidden-layer network $f_M(x) = \sum_{m=1}^{M} a_m \sigma(w_m^\top x + b_m)$ satisfies $\|f_M - h\|_\infty < \varepsilon$ on K.

Proof. Let $\mathcal{A}_\sigma := \operatorname{span}\{\sigma(w^\top x + b) : w \in \mathbb{R}^d,\ b \in \mathbb{R}\} \subset C(K)$.
The classical universal approximation theorem states that $\overline{\mathcal{A}_\sigma}^{\|\cdot\|_\infty} = C(K)$ for suitable non-polynomial σ.
Since $h \in C(K)$ and $\varepsilon > 0$, there is $f_M \in \mathcal{A}_\sigma$ with $\|f_M - h\|_\infty < \varepsilon$.
Thus $f_M(x) = \sum_{m=1}^{M} a_m \sigma(w_m^\top x + b_m)$ has one hidden layer and linear output.
Structural Commentary: Non-polynomial activations make ridge spans dense in $C(K)$.
Key Insight: Width supplies finite linear combinations of nonlinear ridge functions. □

Remark 5.1.8. *Depth and Expressivity*
Let $N_L(\varepsilon, h)$ be the minimum width of a depth-L network with $\|f - h\|_\infty \leq \varepsilon$ on a compact domain. Universal approximation gives $N_L(\varepsilon, h) < \infty$, not trainability; depth expressivity is $N_L(\varepsilon, h) \ll N_{L'}(\varepsilon, h)$ for some $L > L'$.

5.2 Backpropagation and Automatic Differentiation

Backpropagation is reverse-mode differentiation applied to the computational graph of the network and loss.

Definition 5.2.1. *Computational Graph*
A computational graph is a finite directed acyclic graph $G = (V, E)$ together with variables $(x_v)_{v \in V}$ and local maps $(f_v)_{v \in V \setminus S}$, where $\operatorname{pa}(v) = \{u \in V : (u, v) \in E\}$ and $S = \{v \in V : \operatorname{pa}(v) = \varnothing\}$.

(i) For $v \in S$, x_v is an input, parameter, or constant.
(ii) For $v \in V \setminus S$, $x_v = f_v((x_u)_{u \in \operatorname{pa}(v)})$.
(iii) An edge $u \to v$ records the direct dependence of x_v on x_u in f_v.

For scalar output $L = x_\ell$ and parameter sources $P \subseteq S$, reverse mode computes $(\partial L/\partial x_v)_{v \in V}$, hence $\nabla_{(x_p)_{p \in P}} L$.

Method 5.2.2. *Backpropagation*
Let $h^{(0)} = x$, $z^{(k)} = W^{(k)} h^{(k-1)} + b^{(k)}$, $h^{(k)} = \phi_k(z^{(k)})$ for $1 \leq k \leq K$, and $L = \ell(h^{(K)}, y)$.
Write $\bar{u} = \nabla_u L$ for the adjoint of u.

(i) Compute $z^{(k)}$ and $h^{(k)}$ for $k = 1, \dots, K$, then compute L.
(ii) Initialize $\bar{h}^{(K)} = \nabla_{h^{(K)}} \ell(h^{(K)}, y)$.
(iii) For $k = K, \dots, 1$, set $\bar{z}^{(k)} = J_{\phi_k}(z^{(k)})^T \bar{h}^{(k)}$ and $(\nabla_{W^{(k)}} L, \nabla_{b^{(k)}} L, \bar{h}^{(k-1)}) = (\bar{z}^{(k)} (h^{(k-1)})^T, \bar{z}^{(k)}, (W^{(k)})^T \bar{z}^{(k)})$.

Thus backpropagation is reverse-mode differentiation of L through the layerwise computational graph.

Method 5.2.3. *Single-Hidden-Layer Backpropagation Equations*
For one training case (x_i, y_i) in a single-hidden-layer network, let $m = 1, \dots, M$ index hidden units, $k = 1, \dots, K$ index output components, and $z_i = (z_{1i}, \dots, z_{Mi})^T$.

(i) Compute $a_{mi} = \alpha_{0m} + \alpha_m^T x_i$, $z_{mi} = \sigma(a_{mi})$, $t_{ki} = \beta_{0k} + \beta_k^T z_i$, and $f_k(x_i) = g_k(t_{ki})$.
(ii) Compute $R_i = \sum_{k=1}^{K} \{y_{ik} - f_k(x_i)\}^2$ and $\delta_{ki} = \frac{\partial R_i}{\partial t_{ki}} = -2\{y_{ik} - f_k(x_i)\} g_k'(t_{ki})$.
(iii) Compute $\nabla_{\beta_k} R_i = \delta_{ki} z_i$ and $\frac{\partial R_i}{\partial \beta_{0k}} = \delta_{ki}$.
(iv) Compute $s_{mi} = \frac{\partial R_i}{\partial a_{mi}} = \sigma'(a_{mi}) \sum_{k=1}^{K} \beta_{km} \delta_{ki}$.

(v) Compute $\nabla_{\alpha_m} R_i = s_{mi} x_i$ and $\frac{\partial R_i}{\partial \alpha_{0m}} = s_{mi}$.

Batch gradients are obtained by summing these per-case derivatives over i.

Remark 5.2.4. *Automatic Differentiation*
For an implemented graph map $F : \mathbb{R}^p \to \mathbb{R}^q$, automatic differentiation returns $DF(x)$ by chain-rule evaluation on the executed graph. It differentiates the floating-point computation up to roundoff, without finite differences or symbolic simplification.

5.3 Optimization for Deep Learning

The old optimization notes are migrated here only where the algorithms are used as training methods for neural networks. General numerical optimization belongs to Part 18.

Definition 5.3.1. *Neural Empirical Objective*

For data $(x_i, y_i)_{i=1}^n$, network $f_\theta : x \mapsto f(x;\theta)$, loss L, neural empirical objective is $J_n(\theta) = n^{-1} \sum_{i=1}^n L(f_\theta(x_i), y_i)$.

The population objective is $J(\theta) = \mathbb{E}[L(f_\theta(X), Y)]$ under the data-generating law.

Algorithm 8: Minibatch Backpropagation Step

Require: Minibatch B, parameters θ, learning rate η
Compute predictions $f(x;\theta)$ for all $(x, y) \in B$
Compute minibatch loss $\widehat{L}_B(\theta)$
Run reverse-mode automatic differentiation to compute $g = \nabla_\theta \widehat{L}_B(\theta)$
Update parameters $\theta \leftarrow \theta - \eta g$

Remark 5.3.2. *Deep Optimization Challenges*
For empirical loss $J_n(\theta)$, deep optimization is shaped by:

(i) *geometry*: large $\kappa(\nabla^2 J_n)$, saddles with $\nabla J_n \approx 0$ and indefinite $\nabla^2 J_n$, and plateaus with $\|\nabla J_n\| \approx 0$.
(ii) *gradient propagation*: large or small $\left\| \frac{\partial h^{(L)}}{\partial h^{(k)}} \right\|$ causes exploding or vanishing gradients.
(iii) *stochasticity*: $g_k = \nabla J_n(\theta_k) + \xi_k$, with $\mathbb{E}[\xi_k \mid \theta_k] = 0$ and $\mathrm{Var}(\xi_k \mid \theta_k) \neq 0$.
(iv) *stabilization*: initialization, normalization, gradient clipping, skip connections, and learning-rate schedules control scale, conditioning, and update variance.

These choices are part of the training specification, not post hoc implementation details.

Remark 5.3.3. *Second-Order Methods*
Let $J : \mathbb{R}^d \to \mathbb{R}$, $g_k = \nabla J(\theta_k)$, and $H_k = \nabla^2 J(\theta_k)$.

(i) *Newton*: choose p_k from $H_k p_k = -g_k$ and set $\theta_{k+1} = \theta_k + \eta_k p_k$.
(ii) *Conjugate gradient*: solve $H_k p = -g_k$ approximately using only products $H_k v$.
(iii) *BFGS/L-BFGS*: use $p_k = -B_k g_k$, where $B_k \approx H_k^{-1}$; L-BFGS stores few curvature pairs.

Large networks make explicit H_k storage or factorization prohibitive, so these methods usually act as preconditioners, curvature approximations, or analysis tools.

5.4 Regularization, Normalization, and Generalization

Deep-learning regularization combines penalties, stochastic training effects, architecture, and validation-based stopping.

Definition 5.4.1. *Parameter-Norm Regularization*
Minimize $\widetilde{J}_\lambda(\theta) = J(\theta) + \lambda \Omega(\theta)$, with $\lambda \geq 0$, $\Omega \geq 0$, and penalized non-intercept set $\mathcal{P}$.

(i) *ℓ^2 weight decay*: $\Omega_2(\theta) = \sum_{j \in \mathcal{P}} \theta_j^2$.
(ii) *ℓ^1 penalty*: $\Omega_1(\theta) = \sum_{j \in \mathcal{P}} |\theta_j|$.
(iii) *Single-hidden-layer weight decay*: for weights $\alpha_{m\ell}$ and β_{km}, $\Omega_2(\theta) = \sum_{m,\ell} \alpha_{m\ell}^2 + \sum_{k,m} \beta_{km}^2$.

Intercepts such as α_{0m} and β_{0k} are excluded unless explicitly included in $\mathcal{P}$.

Definition 5.4.2. *Dropout*
Let $h \in \mathbb{R}^d$, $q \in (0, 1]$, and $m_j \overset{\text{ind}}{\sim} \text{Bernoulli}(q)$.

(i) *Training mask*: replace h by $\widetilde{h} = \frac{m \odot h}{q}$.
(ii) *Subnetwork*: each mask $m \in \{0, 1\}^d$ selects the active coordinates $\{j : m_j = 1\}$ with shared parameters.
(iii) *Evaluation*: use the unmasked activation h when inverted dropout scaling is used in training.

Dropout optimizes an expected masked loss over tied-parameter subnetworks.

Method 5.4.3. *Early Stopping*
Let θ_t be the parameter after training step t and let $V(\theta)$ be the validation loss.

(i) Compute $V_t = V(\theta_t)$ after selected training steps t.
(ii) Select $t^\star \in \arg\min_{0 \leq t \leq T} V_t$.
(iii) Return $\widehat{\theta}_{\text{ES}} = \theta_{t^\star}$.

The stopping time $t^\star$ restricts the reachable parameter set and acts as an implicit regularization parameter.

Remark 5.4.4. *Starting Values, Scaling, and Multiple Fits*
For a nonconvex neural objective $J(\theta)$, the fitted parameter can depend on initialization and scaling.

(i) *Initialization*: avoid $\theta_0 = 0$ when symmetry identifies hidden units, and avoid large $\|\theta_0\|$ when $\sigma'(a) \approx 0$.
(ii) *Scaling*: standardize inputs, e.g. $x_j \leftarrow \frac{x_j - \hat{\mu}_j}{\hat{\sigma}_j}$, so coordinates have comparable optimization scale.
(iii) *Multiple fits*: run starts $r = 1, \ldots, R$, obtain $\widehat{\theta}^{(r)}$, and choose $\widehat{\theta}^{(r^\star)}$ by validation or penalized criterion.

Thus initialization law, preprocessing, and restart selection are part of the training specification.

Remark 5.4.5. *Hidden Units and Weight Decay*
Let M be hidden width, λ the weight-decay strength, and $t^\star$ the early-stopping time.

(i) *Small M*: the model class may have excessive approximation error.
(ii) *Large M*: overfitting can occur unless $\lambda > 0$, finite $t^\star$, or another constraint limits effective complexity.
(iii) *Selection*: choose $(M, \lambda, t^\star)$ by validation or another held-out criterion.

Weight decay and early stopping bias the fit toward smaller effective weights and simpler functions.

Definition 5.4.6. *Batch Normalization*
Let $H = (h_{ij}) \in \mathbb{R}^{b\times d}$ be minibatch activations, indexed by example i and coordinate j.

(i) *Batch statistics*: $\mu_j = \frac{1}{b}\sum_i h_{ij}$ and $s_j^2 = \frac{1}{b}\sum_i (h_{ij} - \mu_j)^2$.
(ii) *Normalization*: $\widehat{h}_{ij} = \frac{h_{ij}-\mu_j}{\sqrt{s_j^2+\varepsilon}}$.
(iii) *Affine output*: $y_{ij} = \gamma_j \widehat{h}_{ij} + \beta_j$, with learned γ_j, β_j.
(iv) *Evaluation*: replace μ_j and s_j^2 by running training estimates.

Remark 5.4.7. *Augmentation and Noise*
Let $\ell(y, f_\theta(x))$ be the task loss.

(i) *Data augmentation*: replace (X, Y) by $(T(X), Y)$ for random label-preserving transforms T.
(ii) *Label smoothing*: replace one-hot y by $y^{(\varepsilon)} = (1-\varepsilon)y + \varepsilon u$, where u is a reference class distribution.
(iii) *Noise injection*: train on $x + \xi$ or hidden activations $h + \xi$, with $\mathbb{E}\xi = 0$.
(iv) *Multitask learning*: minimize $\sum_{r=1}^{R} \omega_r \ell_r(y_r, f_{\theta,r}(x))$ with shared representation parameters.

These methods regularize through altered data, targets, perturbations, or shared representations.

6 Modern Deep Models

6.1 Convolutional and Geometric Architectures

The convolutional-network source is migrated here because convolution is an architectural prior for data with grid, group, or geometric structure.

Definition 6.1.1. *Convolution*
For $f, g : \mathbb{R} \to \mathbb{R}$ and $a, b : \mathbb{Z} \to \mathbb{R}$:

(i) *Continuous*: $f * g : D_c \ni t \mapsto \int_{-\infty}^{\infty} f(\tau) g(t-\tau)\, d\tau \in \mathbb{R}$; $D_c := \{t \in \mathbb{R} : \int_{\mathbb{R}} |f(\tau)g(t-\tau)|\, d\tau < \infty\}$.
(ii) *Discrete*: $a * b : D_d \ni n \mapsto \sum_{m\in\mathbb{Z}} a[m]b[n-m] \in \mathbb{R}$; $D_d := \{n \in \mathbb{Z} : \sum_{m\in\mathbb{Z}} |a[m]b[n-m]| < \infty\}$.

Definition 6.1.2. *Convolutional Neural Network*
Let $\Phi = L_k \circ \cdots \circ L_1$ be a neural network on array-valued activations.

(i) *Convolutional layer*: on $z : \mathbb{Z}^d \to \mathbb{R}^{C_{\text{in}}}$, $L_j z = \sigma(K * z + b)$ or $\sigma(K \star z + b)$, with finite-support K, $b \in \mathbb{R}^{C_{\text{out}}}$, pointwise σ, and cross-correlation $\star$.
(ii) *Convolutional neural network*: Φ is a CNN if at least one layer L_j is convolutional.
(iii) *Library convention*: many neural-network libraries implement $\star$ while naming the operation convolution.

Definition 6.1.3. *Sparse Interactions, Sharing, and Equivariance*
For $x : \mathbb{Z}^d \to \mathbb{R}^{C_{\text{in}}}$, finite $W \subset \mathbb{Z}^d$, kernels $K_w \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$, $b \in \mathbb{R}^{C_{\text{out}}}$, and pointwise σ,
set $Lx(u) := \sigma(\sum_{w\in W} K_w x(u-w) + b)$ and $(\tau_a x)(u) := x(u-a)$.

(i) *Sparse interaction*: $Lx(u)$ depends only on $\{x(u-w) : w \in W\}$.
(ii) *Parameter sharing*: the same $(K_w)_{w\in W}$ and b define $Lx(u)$ for every u.
(iii) *Translation equivariance*: $L(\tau_a x) = \tau_a(Lx)$ for every $a \in \mathbb{Z}^d$ where defined.

Pooling with finite U applies ρ to $\{Lx(u+w) : w \in U\}$; shift-stable ρ gives local approximate invariance.

Definition 6.1.4. *Pooling, Padding, and Stride*
Let $x : \Omega \to \mathbb{R}^C$ on finite $\Omega \subset \mathbb{Z}^d$, with $W \subset \mathbb{Z}^d$, $p \in \mathbb{N}_0^d$, $s \in \mathbb{N}^d$, offset r, statistic $\rho : (\mathbb{R}^C)^W \to \mathbb{R}^{C'}$,
and $\Omega^{(p)} := \Omega + \prod_i \{-p_i, \dots, p_i\}$.

(i) *Padding*: an extension $\tilde{x} : \Omega^{(p)} \to \mathbb{R}^C$ with $\tilde{x}|_\Omega = x$ and a boundary rule on $\Omega^{(p)} \setminus \Omega$.
(ii) *Stride*: evaluation sites $\Omega_{s,r} := \{u \in \Omega^{(p)} : \forall i,\ u_i \equiv r_i \pmod{s_i},\ u + W \subset \Omega^{(p)}\}$.
(iii) *Pooling*: $P_{\rho,W,s,r} x : \Omega_{s,r} \ni u \mapsto \rho((\tilde{x}(u+w))_{w\in W}) \in \mathbb{R}^{C'}$.

Remark 6.1.5. *Geometric Extensions*
Let X be a domain with local neighborhoods $\mathcal{N}(x) \subset X$ and symmetry group G.

(i) *Graph neural networks*: on $\mathcal{G} = (V, E)$, $h_v^{(\ell+1)} = \phi(h_v^{(\ell)}, \text{Agg}_{u\in\mathcal{N}(v)}\, m_{uv}^{(\ell)})$ with invariant Agg.
(ii) *Equivariant networks*: Φ is G-equivariant if $\Phi(\rho_{\text{in}}(g)x) = \rho_{\text{out}}(g)\Phi(x)$ for every $g \in G$.
(iii) *Manifold architectures*: for a manifold M, layers use chart, tangent, or geodesic neighborhoods $\mathcal{N}(p) \subset M$.

Each variant builds locality or symmetry into the model class.

6.2 Sequence Models and Attention

Sequence models process ordered data where context and position matter.

Definition 6.2.1. *Recurrent Neural Network*
For dim d_x, d_h, d_y, parameters $\Theta = (W_x, W_h, b, \phi, \psi, h_0)$, $T \geq 1$, an RNN is a map $\Phi_T : (\mathbb{R}^{d_x})^T \to (\mathbb{R}^{d_y})^T$:

(i) *State update*: $h_t = \phi(W_x x_t + W_h h_{t-1} + b)$ for $t = 1, \ldots, T$, with shared (W_x, W_h, b).
(ii) *Readout*: $y_t = \psi(h_t)$ and $\Phi_T(x_{1:T}) = (y_1, \ldots, y_T)$.

Remark 6.2.2. *Long-Term Dependencies*
For $h_t = F_t(h_{t-1}, x_t)$, set $J_t := \partial h_t / \partial h_{t-1}$ and $P_{s:T} := \prod_{t=s+1}^{T} J_t$.
Then $\nabla_{h_s} \mathcal{L} = P_{s:T}^{\top} \nabla_{h_T} \mathcal{L}$; $\|P_{s:T}\| \ll 1$ vanishes gradients and $\|P_{s:T}\| \gg 1$ explodes them.

(i) *Gating*: gates $g_t \in [0,1]^{d_h}$ interpolate between copy and update.
(ii) *Gradient clipping*: replace a gradient g by $\min\{1, c/\|g\|\}g$, so the clipped norm is at most c.
(iii) *Skip paths*: use additive routes such as $h_t = h_{t-1} + F_t(h_{t-1}, x_t)$, giving transition Jacobian $I + \partial F_t / \partial h_{t-1}$.

These mechanisms keep the long-horizon Jacobian product numerically controlled.

Definition 6.2.3. *Attention*
For $Q \in \mathbb{R}^{n_q \times d}$, $K \in \mathbb{R}^{n_k \times d}$, $V \in \mathbb{R}^{n_k \times d_v}$, define $\mathrm{Attn}(Q, K, V) \in \mathbb{R}^{n_q \times d_v}$ row-wise:

(i) *Scores*: $S := d^{-1/2} Q K^{\top} \in \mathbb{R}^{n_q \times n_k}$.
(ii) *Weights*: $A := \mathrm{softmax}_{\mathrm{row}}(S)$, $A_{ij} := \exp(S_{ij}) / \sum_{\ell=1}^{n_k} \exp(S_{i\ell})$.
(iii) *Output*: $\mathrm{Attn}(Q, K, V) := AV$, equivalently $(\mathrm{Attn}(Q, K, V))_{i\cdot} = \sum_{j=1}^{n_k} A_{ij} V_{j\cdot}$.

Remark 6.2.4. *Attention as Routing*
For token matrix X, set $Q = XW_q$, $K = XW_k$, $V = XW_v$, $S = d^{-1/2} Q K^{\top}$, and $A = \mathrm{softmax}_{\mathrm{row}}(S)$.

(i) *Content routing*: $y_i = \sum_j A_{ij} v_j$ routes value vectors by query-key compatibility.
(ii) *Masking*: a mask $M_{ij} \in \{0, -\infty\}$ replaces S by $S + M$, forcing $A_{ij} = 0$ for forbidden interactions.
(iii) *Positional encoding*: absolute codes use $x_i + e_i$; relative codes add a position term to S_{ij}.

Thus attention replaces a fixed recurrence neighborhood by data-dependent communication.

6.3 Transformers and Large-Scale Training

Transformers combine attention, feedforward blocks, normalization, and residual connections into architectures that scale well on sequence and multimodal data.

Definition 6.3.1. *Self-Attention*
For $X \in \mathbb{R}^{T \times d}$, head dimension d_h, and projection matrices W_Q, W_K, W_V, W_O:

(i) *Self-attention*: $Q = XW_Q$, $K = XW_K$, $V = XW_V$, and $\mathrm{SA}(X) := \mathrm{Attn}(Q, K, V)$.
(ii) *Multi-head attention*: with heads $H_r := \mathrm{Attn}(XW_Q^{(r)}, XW_K^{(r)}, XW_V^{(r)})$, set $\mathrm{MHA}(X) := [H_1 | \cdots | H_R] W_O$.

Definition 6.3.2. *Transformer Block*
For $X \in \mathbb{R}^{T \times d}$, let MHA be multi-head attention, FFN positionwise feedforward, and $\mathrm{LN}_1, \mathrm{LN}_2$ layer normalizations. A pre-norm block $B : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times d}$ is:

(i) *Attention sublayer*: $Z := X + \mathrm{MHA}(\mathrm{LN}_1 X)$.
(ii) *Feedforward sublayer*: $B(X) := Z + \mathrm{FFN}(\mathrm{LN}_2 Z)$.

Stacking blocks gives $B_L \circ \cdots \circ B_1(X)$.

Remark 6.3.3. *Large-Scale Training*
For data $\mathcal{D} = \{z_i\}_{i=1}^N$, loss $\ell_i(\theta)$, mini-batch B_t, and gradient estimate $\hat{g}_t := |B_t|^{-1} \sum_{i \in B_t} \nabla_\theta \ell_i(\theta_t)$:

(i) *Optimization*: choose update $\theta_{t+1} = U_t(\theta_t, \hat{g}_t)$, batch size $|B_t|$, and schedule η_t.
(ii) *Data*: sampling, filtering, weighting, and augmentation define the empirical training law $\widehat{P}_{\mathcal{D}}$.
(iii) *Numerics*: mixed precision uses low-precision arithmetic with scaling or high-precision accumulators.
(iv) *Parallelism*: workers compute local gradients $g_t^{(p)}$ and synchronize $\bar{g}_t = P^{-1} \sum_{p=1}^{P} g_t^{(p)}$.
(v) *Memory*: checkpointing stores selected activations and recomputes the rest during backpropagation.
(vi) *Evaluation*: validation loss and task metrics estimate out-of-sample performance.

Remark 6.3.4. *Inductive Bias*
Let $I = \{1, \ldots, T\}$ index tokens, $W_i \subset I$ be a local window, and $A_{ij}(x)$ be attention weights.

(i) *Locality*: convolution uses $y_i = F((x_j)_{j \in W_i})$.
(ii) *Global interaction*: attention uses $y_i = \sum_{j \in I} A_{ij}(x) V x_j$.
(iii) *Domain-specific bias*: impose masks, recurrence, sparsity, or $\Phi(\rho_{\mathrm{in}}(g)x) = \rho_{\mathrm{out}}(g)\Phi(x)$.

Useful bias matches the task's locality, symmetry, and dependency scale.

6.4 Generative Models

Generative modeling learns distributions, simulators, or conditional samplers rather than only point predictions.

Definition 6.4.1. *Latent-Variable Generative Model*
Let $Z \in \mathcal{Z}$, $X \in \mathcal{X}$, prior $p(z)$, decoder $p_\theta(x \mid z)$, and data law P_{data}.

(i) *Joint law*: $p_\theta(x, z) := p_\theta(x \mid z) p(z)$.
(ii) *Marginal law*: $p_\theta(x) := \int_{\mathcal{Z}} p_\theta(x \mid z) p(z)\, dz$.
(iii) *Learning*: choose θ so p_θ approximates P_{data}, e.g. by maximizing $\sum_{i=1}^n \log p_\theta(x_i)$.

Definition 6.4.2. *Variational Autoencoder*
For prior $p(z)$, decoder $p_\theta(x \mid z)$, encoder $q_\phi(z \mid x)$, and data $\{x_i\}_{i=1}^n$:

(i) *Generative model*: $p_\theta(x, z) := p_\theta(x \mid z)p(z)$.
(ii) *Variational posterior*: $q_\phi(z \mid x)$ approximates $p_\theta(z \mid x)$.
(iii) *ELBO*: $\mathcal{L}(\theta, \phi; x) := \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x) \| p(z))$.
(iv) *Training*: maximize $\sum_{i=1}^n \mathcal{L}(\theta, \phi; x_i)$ over (θ, ϕ).

Definition 6.4.3. *Generative Adversarial Network*
Let $Z \sim p_Z$, data law P_{data}, generator $G_\theta : \mathcal{Z} \to \mathcal{X}$, and discriminator $D_\psi : \mathcal{X} \to (0, 1)$.

(i) *Generator*: G_θ induces $P_\theta := (G_\theta)_{\#} p_Z$ on $\mathcal{X}$.
(ii) *Discriminator*: $D_\psi(x)$ estimates the probability that $x \sim P_{\text{data}}$ rather than $x \sim P_\theta$.
(iii) *Adversarial game*: $\min_\theta \max_\psi \mathbb{E}_{x \sim P_{\text{data}}} \log D_\psi(x) + \mathbb{E}_{z \sim p_Z} \log(1 - D_\psi(G_\theta(z)))$.

Definition 6.4.4. *Diffusion Model*
Let $X_0 \sim P_{\text{data}}$ and $X_0, \ldots, X_T$ be a Markov chain on $\mathcal{X}$.

(i) *Forward noising*: fixed kernels $q_t(x_t \mid x_{t-1})$ define $q(x_{1:T} \mid x_0) = \prod_{t=1}^T q_t(x_t \mid x_{t-1})$.
(ii) *Reverse model*: learned kernels $p_\theta(x_{t-1} \mid x_t)$ define $p_\theta(x_{0:T}) = p_T(x_T) \prod_{t=1}^T p_\theta(x_{t-1} \mid x_t)$.
(iii) *Training*: fit θ so $p_\theta(x_{t-1} \mid x_t)$ approximates the reverse conditionals of q.
(iv) *Sampling*: draw $X_T \sim p_T$ and iterate $X_{t-1} \sim p_\theta(\cdot \mid X_t)$ for $t = T, \ldots, 1$.

6.5 Reinforcement Learning and Decision Systems

Reinforcement learning studies agents whose actions affect future observations and rewards.

Definition 6.5.1. *Markov Decision Process*
An MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with:

(i) *State-action spaces*: states $s \in \mathcal{S}$ and actions $a \in \mathcal{A}$.
(ii) *Transition law*: $P(\cdot \mid s, a) \in \mathcal{P}(\mathcal{S})$ and $S_{t+1} \sim P(\cdot \mid S_t, A_t)$.
(iii) *Reward and discount*: $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ and $\gamma \in [0, 1)$.
(iv) *Policy*: $\pi(\cdot \mid s) \in \mathcal{P}(\mathcal{A})$ and $A_t \sim \pi(\cdot \mid S_t)$.

Definition 6.5.2. *Value Functions*
For policy π, define the discounted return $G_t := \sum_{k=0}^{\infty} \gamma^k r(S_{t+k}, A_{t+k})$.

(i) *State value*: $V^\pi(s) := \mathbb{E}_\pi[G_0 \mid S_0 = s]$.
(ii) *Action value*: $Q^\pi(s, a) := \mathbb{E}_\pi[G_0 \mid S_0 = s, A_0 = a]$.

Definition 6.5.3. *Bellman Equation*
For an MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ and value function $V : \mathcal{S} \to \mathbb{R}$:

(i) *Policy Bellman equation*: $V^\pi(s) = \sum_a \pi(a \mid s)(r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s'))$.
(ii) *Optimal Bellman equation*: $V^\star(s) = \max_a (r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\star(s'))$.

Remark 6.5.4. *RL as a Learning System*
Let $\tau = (S_0, A_0, R_0, \ldots)$ denote a trajectory and P^π its law under policy π.

(i) *Prediction*: estimate V^π, Q^π, or transition/reward models from trajectories.
(ii) *Feedback*: policy changes alter the data law, $\pi \mapsto P^\pi$.
(iii) *Optimization*: choose π to maximize $J(\pi) := \mathbb{E}_{P^\pi} \sum_{t \geq 0} \gamma^t R_t$.
(iv) *Exploration*: sample actions to reduce uncertainty about rewards, transitions, or values.
(v) *Evaluation*: validate under sequential dependence and policy-induced distribution shift.

7 Learning Systems and Evaluation

7.1 Data Splitting, Validation, and Benchmarking

Evaluation estimates how the trained system behaves on data not used to fit it.

Definition 7.1.1. *Training, Validation, and Test Splits*
Let $\mathcal{D} = \mathcal{D}_{\text{tr}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{te}}$ be a disjoint partition of $\{(x_i, y_i)\}_{i=1}^n$, and fix a loss ℓ.

(i) Training: for each $\lambda \in \Lambda$, fit $\widehat{f}_\lambda$ using only $\mathcal{D}_{\text{tr}}$.
(ii) Validation: set $\widehat{\lambda} \in \arg\min_{\lambda \in \Lambda} \widehat{R}_{\text{val}}(\lambda)$, where $\widehat{R}_{\text{val}}(\lambda) = |\mathcal{D}_{\text{val}}|^{-1} \sum_{(x,y) \in \mathcal{D}_{\text{val}}} \ell(\widehat{f}_\lambda(x), y)$.
(iii) Test: after $\widehat{\lambda}$ is fixed, report $\widehat{R}_{\text{te}} = |\mathcal{D}_{\text{te}}|^{-1} \sum_{(x,y) \in \mathcal{D}_{\text{te}}} \ell(\widehat{f}_{\widehat{\lambda}}(x), y)$ as the final performance estimate.

If $\mathcal{D}_{\text{val}}$ is unavailable, cross-validation or bootstrap resampling approximates validation.
Any pre-reporting use of $\mathcal{D}_{\text{te}}$ makes $\widehat{R}_{\text{te}}$ adaptive and typically optimistic.

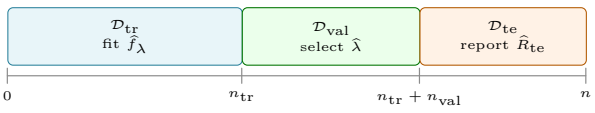

Figure 259: Train/validation/test split: $\mathcal{D} = \mathcal{D}_{\text{tr}} \sqcup \mathcal{D}_{\text{val}} \sqcup \mathcal{D}_{\text{te}}$, with $\mathcal{D}_{\text{tr}}$ fitting $\widehat{f}_\lambda$, $\mathcal{D}_{\text{val}}$ selecting $\widehat{\lambda}$, and $\mathcal{D}_{\text{te}}$ estimating final risk $\widehat{R}_{\text{te}}$.

Definition 7.1.2. *Model Selection and Model Assessment*
Let $\{\widehat{f}_\lambda : \lambda \in \Lambda\}$ be candidates, let $\widehat{R}_{\mathrm{val}}(\lambda)$ be validation error, and let $R(f) = \mathbb{E}\ell(f(X), Y)$.

(i) *Model selection*: choose $\widehat{\lambda} \in \Lambda$, typically $\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} \widehat{R}_{\mathrm{val}}(\lambda)$, and output $\widehat{f}_{\widehat{\lambda}}$.
(ii) *Model assessment*: after $\widehat{\lambda}$ is fixed, estimate $R(\widehat{f}_{\widehat{\lambda}})$ using data independent of training and selection.

Reusing selection data for assessment yields adaptive, typically optimistic error.

Definition 7.1.3. *Conditional and Expected Test Error*
Let $\widehat{f}_{\mathcal{T}}$ be trained on random $\mathcal{T}$, and let (X^0, Y^0) be an independent test point.

(i) *Conditional test error*: $\mathrm{Err}_{\mathcal{T}} = \mathbb{E}_{X^0,Y^0}[\ell(\widehat{f}_{\mathcal{T}}(X^0), Y^0) \mid \mathcal{T}]$.
(ii) *Expected test error*: $\mathrm{Err} = \mathbb{E}_{\mathcal{T}} \mathrm{Err}_{\mathcal{T}}$.

Reusable error estimators usually target Err, while $\mathrm{Err}_{\mathcal{T}}$ is the error conditional on the realized training set.

Method 7.1.4. *K-Fold Cross-Validation*
Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, loss ℓ, candidates Λ, and fold map $\kappa : \{1, \dots, n\} \to \{1, \dots, K\}$ be fixed.

(i) For each $\lambda \in \Lambda$ and $k = 1, \dots, K$, fit $\widehat{f}_\lambda^{-k}$ using only observations with $\kappa(i) \neq k$.
(ii) For each $\lambda \in \Lambda$, compute $\mathrm{CV}(\lambda) = n^{-1} \sum_{i=1}^n \ell(\widehat{f}_\lambda^{-\kappa(i)}(x_i), y_i)$.
(iii) Choose $\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} \mathrm{CV}(\lambda)$, or the one-standard-error candidate when that rule is specified.
(iv) Refit $\widehat{f}_{\widehat{\lambda}}$ on all of $\mathcal{D}$.

Leave-one-out has $K = n$; $K = 5$ or 10 often lowers variance and computation but fits on smaller training folds.

Remark 7.1.5. *What Cross-Validation Estimates*
For learning rule $\mathcal{A}$, let $\mathrm{Err}_{\mathcal{A}}(m) = \mathbb{E}_{\mathcal{T}_m,X^0,Y^0}\ell(\mathcal{A}(\mathcal{T}_m)(X^0), Y^0)$. K-fold cross-validation targets roughly $\mathrm{Err}_{\mathcal{A}}(n(1 - 1/K))$, not the conditional error of $\mathcal{A}(\mathcal{D})$. Leave-one-out has $m = n-1$ but dependent losses; fixed $K = 5$ or 10 lowers computation and often variance, with possible upward bias.

Definition 7.1.6. *Benchmark*
A benchmark is a fixed evaluation protocol $\mathcal{B} = (\mathcal{D}, \Pi, \mathcal{S}, m, \mathcal{R})$.

(i) *Data*: $\mathcal{D}$ specifies the population, sample, or data source.
(ii) *Preprocessing*: Π specifies transformations, filtering, feature construction, and allowed fitting scope.
(iii) *Splitting*: $\mathcal{S}$ specifies train-validation-test splits or resampling schemes.
(iv) *Metric*: m specifies the reported score.
(v) *Resources*: $\mathcal{R}$ specifies allowed external data, models, compute, and manual intervention.

Two reported scores are directly comparable only when computed under the same $\mathcal{B}$.

Remark 7.1.7. *Metric Choice*
Choose $m(f; \mathcal{D}_{\mathrm{te}})$ so its ordering of candidate rules matches the deployment objective $U(f)$.
Accuracy, log loss, calibration, ranking quality, cost, and latency need not agree.

Remark 7.1.8. *Data Leakage*
For protected evaluation data E, leakage is dependence of any pre-evaluation choice Z_{pre} on labels or covariates in E. Then error measured on E is adaptive, not honest out-of-sample error.

Method 7.1.9. *Cross-Validation with Screening or Preprocessing*
For a fold map $\kappa : \{1, \dots, n\} \to \{1, \dots, K\}$, set $\mathcal{D}_k = \{(x_i, y_i) : \kappa(i) = k\}$ and $\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k$.
Data-dependent pipeline choices must be fold-local:

(i) For each k, learn preprocessing, screening, feature selection, and tuning from $\mathcal{D}_{-k}$ only.
(ii) Fit $\widehat{f}^{-k}$ on $\mathcal{D}_{-k}$ after applying exactly those fold-local choices.
(iii) Predict and score $\mathcal{D}_k$ using $\widehat{f}^{-k}$ and the same fold-local choices.
(iv) Average the held-out losses over $k = 1, \dots, K$.

Full-data supervised choices leak held-out labels into $\widehat{f}^{-k}$ and bias cross-validation downward.
The protocol should state whether unsupervised preprocessing is fold-local or full-covariate.

Method 7.1.10. *Bootstrap Estimates of Prediction Error*
Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$; draw bootstrap samples S^{*b} and fit $\widehat{f}^{*b}$; set $C^{-i} = \{b : (x_i, y_i) \notin S^{*b}\}$, $I_{\mathrm{oob}} = \{i : C^{-i} \neq \varnothing\}$, and $\overline{\mathrm{err}} = n^{-1} \sum_i \ell(\widehat{f}(x_i), y_i)$.

(i) *Naive bootstrap*: $\widehat{\mathrm{Err}}_{\mathrm{boot}} = B^{-1} \sum_{b=1}^B n^{-1} \sum_{i=1}^n \ell(\widehat{f}^{*b}(x_i), y_i)$.
(ii) *Out-of-bag bootstrap*: $\widehat{\mathrm{Err}}^{(1)} = |I_{\mathrm{oob}}|^{-1} \sum_{i\in I_{\mathrm{oob}}} |C^{-i}|^{-1} \sum_{b\in C^{-i}} \ell(\widehat{f}^{*b}(x_i), y_i)$.
(iii) *.632 bootstrap*: $\widehat{\mathrm{Err}}^{(.632)} = .368\,\overline{\mathrm{err}} + .632\,\widehat{\mathrm{Err}}^{(1)}$.
(iv) *.632+ bootstrap*: $\widehat{\mathrm{Err}}^{(.632+)} = (1 - w_+)\overline{\mathrm{err}} + w_+\widehat{\mathrm{Err}}^{(1)}$, with $w_+ \in [.632, 1]$ increase with overfitting.

Naive reuse is most optimistic; out-of-bag and .632 variants reduce reuse bias.

Remark 7.1.11. *Limits of Reuse-Based Assessment*
Write $\widehat{R}_{\mathrm{reuse}}(\lambda) = R(\widehat{f}_\lambda) + b_\lambda + \varepsilon_\lambda$. Nearly constant b_λ can preserve candidate comparisons despite biased levels.
After choosing $\widehat{\lambda}$, estimating $R(\widehat{f}_{\widehat{\lambda}})$ requires data independent of all fitting, preprocessing, tuning, and iteration.

7.2 Robustness, Shift, and Distributional Uncertainty

Learning systems are deployed under distributions that may differ from the training law.

Definition 7.2.1. *Distribution Shift*
Let P be training law, Q deployment law on (X, Y). Distribution shift means $Q \neq P$; common cases are:

(i) *Covariate shift*: $Q_X \neq P_X$ and $Q_{Y|X} = P_{Y|X}$.
(ii) *Label shift*: $Q_Y \neq P_Y$ and $Q_{X|Y} = P_{X|Y}$.
(iii) *Concept shift*: $Q_{Y|X} \neq P_{Y|X}$.

Definition 7.2.2. *Adversarial Example*
Let $\mathcal{F}$ be a predictor class, ℓ a loss, $d_f : \mathcal{X} \to \mathcal{D}$ the decision rule induced by $f \in \mathcal{F}$, $\Delta(x)$ the admissible perturbation set at x, and $\mathcal{H}_y \subseteq \mathcal{D} \times \mathcal{D}$ the harmful decision-change relation for label y.

(i) *Adversarial example*: for fixed $f \in \mathcal{F}$ and (x, y), any $x' = x + \delta$ with $\delta \in \Delta(x)$ and $\big(\ell(f(x'), y) > \ell(f(x), y)\big) \vee \big((d_f(x), d_f(x')) \in \mathcal{H}_y\big)$.

(ii) *Adversarial training*: for $S = \{(x_i, y_i)\}_{i=1}^n$, choose $\widehat{f}_{\mathrm{adv}} \in \arg\min_{f \in \mathcal{F}} n^{-1} \sum_{i=1}^n \sup_{\delta \in \Delta(x_i)} \ell(f(x_i + \delta), y_i)$.

(iii) *Neural perturbation form*: minimize $\ell(y, f_\theta(x + \delta^\star))$, where $\delta^\star \in \arg\max_{\|\delta\| \le \rho} \ell(y, f_\theta(x + \delta))$.

Definition 7.2.3. *Distributionally Robust Optimization*
Let $\mathcal{F}$ be a hypothesis class, ℓ a loss, and $\mathcal{U} \subseteq \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ an uncertainty set of deployment laws.

(i) *Robust risk*: for $Q \in \mathcal{U}$ and $f \in \mathcal{F}$, $R_Q(f) = \mathbb{E}_{(X,Y)\sim Q} \ell(f(X), Y)$ and $R_{\mathcal{U}}(f) = \sup_{Q \in \mathcal{U}} R_Q(f)$.

(ii) *DRO solution*: any $\widehat{f}_{\mathrm{DRO}} \in \arg\min_{f \in \mathcal{F}} R_{\mathcal{U}}(f)$.

Remark 7.2.4. *Uncertainty*
Let $\mathcal{T}$ be the training data, P_{tr} the training law, and P_{dep} the deployment law.

(i) *Aleatoric uncertainty*: irreducible variability in $Y \mid X = x$ under the data-generating law.
(ii) *Epistemic uncertainty*: uncertainty in f from finite $\mathcal{T}$, misspecification, or nonidentifiability.
(iii) *Out-of-distribution uncertainty*: uncertainty from $P_{\mathrm{dep}} \neq P_{\mathrm{tr}}$ or inputs outside the training support.

Robust systems calibrate probabilities and abstain when uncertainty exceeds a policy threshold.

7.3 Interpretability, Fairness, and Safety

Evaluation also asks whether a model can be understood, audited, and constrained in the setting where it is used.

Definition 7.3.1. *Interpretability*
Let $\mathcal{F}, \mathcal{C}, \mathcal{E}, \mathcal{A}, \mathcal{B}$ denote predictor, query-context, explanation, audience, and audit-task spaces.

(i) *Interpretability specification*: $\mathcal{I} = (\mathcal{E}, a, b, \Phi)$ with $(a, b) \in \mathcal{A} \times \mathcal{B}$ and $\Phi : \mathcal{F} \times \mathcal{C} \to \mathcal{E}$.
(ii) *Explanation*: for $(f, c) \in \mathcal{F} \times \mathcal{C}$, $e = \Phi(f, c)$.

Typical codomains $\mathcal{E}$ encode attributions, counterfactuals, surrogate rules, or representation summaries.

Definition 7.3.2. *Fairness Criteria*
Let A be a protected attribute, $\widehat{Y}$ a decision, S a score, and Y the outcome.
A group fairness criterion imposes equalities or tolerances across values of A; common examples are:

(i) *Demographic parity*: $\mathbb{P}(\widehat{Y} = 1 \mid A = a)$ is constant in a.
(ii) *Equalized odds*: $\mathbb{P}(\widehat{Y} = 1 \mid Y = y, A = a)$ is constant in a for each y.
(iii) *Group calibration*: $\mathbb{P}(Y = 1 \mid S = s, A = a) = s$ for each relevant s and a.

These criteria need not be mutually compatible; the chosen condition should match the policy goal.

Definition 7.3.3. *Safety Constraint*
An additional requirement $f \in \mathcal{C}$ or $R_{\mathrm{safe}}(f) \le \tau$ beyond average-case loss is a *safety constraint*.
$\mathcal{C}$ encodes abstention, monotonicity, action bounds, review triggers, or worst-case risk limits.

Remark 7.3.4. *Beyond Accuracy*
For a deployed predictor f, evaluation beyond accuracy is a specification $\mathcal{E} = (\mathcal{I}, \mathcal{F}, \mathcal{C})$.

(i) *Interpretability*: $\mathcal{I}$ specifies the explanation object, audience, and audit protocol.
(ii) *Fairness*: $\mathcal{F}$ specifies protected attributes, group metrics, and required parity or error constraints.
(iii) *Safety*: $\mathcal{C}$ specifies forbidden actions, abstention rules, human review triggers, or worst-case risk bounds.

A scalar score is incomplete unless the relevant components of $\mathcal{E}$ are stated.

7.4 Deployment, Monitoring, and Feedback Loops

Deployment turns a fitted model into part of a larger stochastic system.

Definition 7.4.1. *Monitoring*
Given deployed f and stream $Z_t = (X_t, \widehat{Y}_t, Y_t^*)$, monitoring outputs (M_t, a_t) with $a_t \in \{0, 1\}$.

(i) *Distribution*: compare $P_t(X)$ and $P_t(\widehat{Y})$ with validation baselines.
(ii) *Performance*: estimate loss, calibration, and subgroup error when labels Y_t^* arrive.
(iii) *Operations*: track latency, missingness, failures, and data-pipeline changes.

An alarm $a_t = 1$ indicates that validation assumptions or service constraints may no longer hold.

Definition 7.4.2. *Feedback Loop*
A deployed policy $A_t = \pi(f(X_t), H_t)$ is a *feedback loop* if $P(X_{t+1}, Y_{t+1} \mid A_t) \neq P(X_{t+1}, Y_{t+1})$.
Future data are then endogenous to f.

Method 7.4.3. *Model Update Policy*
Given deployed f_t, data $\mathcal{D}_{\le t}$, and diagnostics M_t, a model update policy maps f_t to f_{t+1}:

(i) *Trigger*: retrain only when a scheduled time, alarm M_t, or performance threshold calls for it.
(ii) *Eligibility*: choose the data subset $\mathcal{D}_{\mathrm{upd}} \subseteq \mathcal{D}_{\le t}$ allowed for training.
(iii) *Validation*: accept candidate $\widetilde{f}$ only if it passes the benchmark and safety criteria against f_t.
(iv) *Deployment*: replace f_t by $\widetilde{f}$ under a rollout plan and rollback rule.

The policy determines future training data and evaluation conditions.

Remark 7.4.4. *Human-System Boundary*
For human-supported decisions, the deployed policy is $\pi = (f, I, A, O, M, G)$, not just predictor f.

(i) *Interface*: I determines what predictions, explanations, and uncertainty are shown.

(ii) *Action rule*: A maps model output and human input to decisions.
(iii) *Override*: O specifies when humans may ignore or modify f.
(iv) *Monitoring*: M tracks errors, drift, latency, and misuse.
(v) *Governance*: G assigns review, audit, and accountability.

System evaluation should target π rather than f alone.

Part XXII
Classical Supervised Learning

1 Linear Prediction and Generalized Linear Models

1.1 Linear Predictors and Least Squares

Linear prediction reduces inputs to scores of the form "intercept plus weighted features." Regression reads scores as predictions; classification compares or transforms them. Its appeal is geometric: projections, hyperplanes, and feature maps.

Definition 1.1.1. *Linear Predictor*
For $x \in \mathbb{R}^p$, $w \in \mathbb{R}^p$, and $b \in \mathbb{R}$, the *linear predictor* is $f_{w,b}(x) := w^\top x + b$.
Regression uses $f_{w,b}$ directly; classification transforms it by g. Score level sets are parallel hyperplanes.

Definition 1.1.2. *Least-Squares Linear Regression*
For $X \in \mathbb{R}^{n\times d}$ and $y \in \mathbb{R}^n$, *least-squares linear regression* chooses $\widehat{\beta} \in \operatorname*{argmin}_{\beta\in\mathbb{R}^d} \|y - X\beta\|_2^2$ and $\widehat{y} := X\widehat{\beta}$.

If $X^\top X$ is nonsingular, then $\widehat{\beta} = (X^\top X)^{-1}X^\top y$. Columns of X may be raw variables, transformations, indicators, interactions, or basis functions; linearity is in β.

Geometrically, least squares moves y to the closest point in $\operatorname{col}(X)$.

Method 1.1.3. *Multiple-Output Least Squares*
For $Y \in \mathbb{R}^{n\times K}$ and $B \in \mathbb{R}^{d\times K}$, set $Y = XB + E$.

(i) *Least squares*: If $\operatorname{rank}(X) = d$, then $\widehat{B} = (X^\top X)^{-1}X^\top Y$.
(ii) *Decoupling*: Writing $Y = (y_1, \dots, y_K)$ gives $\widehat{B} = (\widehat{\beta}_1, \dots, \widehat{\beta}_K)$, where each $\widehat{\beta}_k$ is the OLS fit of y_k on X.

Shared responses can be pooled by low-rank B or common-direction shrinkage.

Method 1.1.4. *Canonical Correlation and Reduced-Rank Regression*

(i) *CCA*: Choose (v_j, u_j) with large $\operatorname{corr}(Xv_j, Yu_j)$, orthogonal across j; at most $\min(d, K)$ pairs.
(ii) *Rank constraint*: Fit $Y \approx XB$ with $\operatorname{rank}(B) \le m$, restricting fitted responses to m dimensions.
(iii) *Objective*: $\widehat{B}_m^{\mathrm{rr}} \in \operatorname*{argmin}_{\operatorname{rank}(B)\le m} \sum_i (y_i - B^\top x_i)^\top \widehat{\Sigma}^{-1}(y_i - B^\top x_i)$.
(iv) *Role of m*: $m = K$ gives separate OLS; smaller m pools shared response structure.

Reduced rank assumes the predictable response variation lies in a low-dimensional subspace.

1.2 Linear Classification and Generalized Linear Models

Definition 1.2.1. *Discriminant Functions and Linear Boundaries*
For $\mathcal{G} = \{1, \dots, K\}$ and scores $\delta_k : \mathcal{X} \to \mathbb{R}$, set $\widehat{G}(x) \in \operatorname{argmax}_{k\in\mathcal{G}} \delta_k(x)$. Only score differences matter.

(i) *Pairwise boundary*: The k versus ℓ boundary lies in $\{x : \delta_k(x) = \delta_\ell(x)\}$.
(ii) *Linear boundary*: If $\delta_k - \delta_\ell$ is affine in x for all k, ℓ, then all pairwise decision boundaries are hyperplanes.
(iii) *Binary log-odds*: If $\log \frac{\mathbb{P}(G=1|X=x)}{\mathbb{P}(G=2|X=x)} = \beta_0 + \beta^\top x$, then the boundary is $\{x : \beta_0 + \beta^\top x = 0\}$.

Method 1.2.2. *Indicator-Matrix Regression Classifier*
For classes $g_i \in \{1, \dots, K\}$, set $t_k = e_k \in \mathbb{R}^K$, $Y_i = t_{g_i}$, and $\widetilde{x}_i^\top = (1, x_i^\top)$, coding labels as Euclidean targets.

(i) *Least-squares fit*: Choose $\widehat{B} \in \operatorname{argmin}_B \sum_{i=1}^n \|Y_i - B^\top \widetilde{x}_i\|_2^2$; if $\widetilde{X}^\top\widetilde{X}$ is nonsingular, $\widehat{B} = (\widetilde{X}^\top\widetilde{X})^{-1}\widetilde{X}^\top Y$.
(ii) *Scores*: For a new x, set $\widetilde{x} = (1, x^\top)^\top$ and $\widehat{f}(x) := \widehat{B}^\top\widetilde{x}$.
(iii) *Classification*: Predict $\widehat{G}(x) \in \operatorname{argmax}_k \widehat{f}_k(x)$, equivalently nearest t_k.

With an intercept, $\sum_k \widehat{f}_k(x) = 1$, but $\widehat{f}_k(x)$ may leave $[0, 1]$ and, for $K \ge 3$, *mask* middle classes.

Definition 1.2.3. *Generalized Linear Model*
A *generalized linear model* gives $Y \mid X = x$ an exponential-family law and makes a linked mean linear: $g(\mu(x)) = w^\top x + b$. Logistic regression uses the Bernoulli law and logit link $g(\mu) = \log \frac{\mu}{1-\mu}$.

Thus a GLM separates score, mean scale, and likelihood.

Definition 1.2.4. *Multiclass Logistic Regression*
With class K as baseline, set $\eta_k(x) := \beta_{k0} + \beta_k^\top x = \log \frac{\mathbb{P}(G=k|X=x)}{\mathbb{P}(G=K|X=x)}$ for $k < K$.

(i) *Softmax probabilities*: For $k < K$, $p_k(x;\theta) := \frac{\exp(\eta_k(x))}{1+\sum_{\ell=1}^{K-1}\exp(\eta_\ell(x))}$, and $p_K(x;\theta) := \frac{1}{1+\sum_{\ell=1}^{K-1}\exp(\eta_\ell(x))}$.
(ii) *Likelihood*: Fit θ by maximising $\ell(\theta) := \sum_{i=1}^n \log p_{g_i}(x_i;\theta)$.

The softmax gives $p(x;\theta) \in \Delta^{K-1}$; the reference class changes only the parametrisation.

Method 1.2.5. *Binary Logistic Regression by IRLS*
For $y_i \in \{0, 1\}$ and intercept-augmented rows $x_i^\top$, set $p_i(\beta) := (1 + \exp(-x_i^\top\beta))^{-1}$.

(i) *Derivatives*: With $p = (p_i)$ and $W := \operatorname{diag}(p_i(1-p_i))$, $\nabla\ell(\beta) = X^\top(y - p)$ and $\nabla^2\ell(\beta) = -X^\top WX$.
(ii) *Adjusted response*: At β^{old}, set $z := X\beta^{\mathrm{old}} + W^{-1}(y - p)$.
(iii) *IRLS update*: Set $\beta^{\mathrm{new}} := (X^\top WX)^{-1}X^\top Wz$.

IRLS is Newton's method as weighted least squares.
The log-likelihood is concave when the MLE is finite; step-halving controls Newton overshoot.

Method 1.2.6. *L_1-Regularized Logistic Regression*

With unpenalized intercept, maximize $\ell_\lambda(\beta_0,\beta) := \sum_{i=1}^{n}\{y_i\eta_i - \log(1+\exp(\eta_i))\} - \lambda\sum_{j=1}^{p}|\beta_j|$, where $\eta_i := \beta_0 + x_i^\top\beta$.

(i) *Probabilities*: Set $p_i := (1+\exp(-\eta_i))^{-1}$ and residual $r := y - p$.

(ii) *KKT conditions*: If $\widehat{\beta}_j \neq 0$, then $x_j^\top r = \lambda\,\mathrm{sign}(\widehat{\beta}_j)$; if $\widehat{\beta}_j = 0$, then $\left|x_j^\top r\right| \leq \lambda$.

The KKT conditions are thresholded feature–residual correlations.

1.3 Discriminant Analysis and Naive Bayes

Method 1.3.1. *Gaussian Discriminant Analysis*

If $X \mid G = k$ has density f_k and prior π_k, then $\mathbb{P}(G=k \mid X=x) = \frac{f_k(x)\pi_k}{\sum_{\ell=1}^{K} f_\ell(x)\pi_\ell}$.

(i) *Score*: If $X \mid G = k \sim N(\mu_k,\Sigma_k)$, compare $\delta_k(x) := -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^\top\Sigma_k^{-1}(x-\mu_k) + \log\pi_k$.

(ii) *LDA*: If $\Sigma_k = \Sigma$ for all k, equivalent linear scores are $\delta_k^{\mathrm{LDA}}(x) := x^\top\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top\Sigma^{-1}\mu_k + \log\pi_k$.

(iii) *QDA*: If Σ_k varies with k, the quadratic term remains and the rule is quadratic discriminant analysis.

Estimate π_k, μ_k, Σ_k from training data; pooled covariance gives LDA and classwise covariances give QDA.
Shared covariance makes the boundaries linear.

Method 1.3.2. *Regularized Discriminant Analysis*

For $0 \leq \alpha \leq 1$, set $\widehat{\Sigma}_k(\alpha) := \alpha\widehat{\Sigma}_k + (1-\alpha)\widehat{\Sigma}$.

(i) *Endpoints*: $\alpha = 0$ gives LDA and $\alpha = 1$ gives QDA.

(ii) *Scalar shrinkage*: For $0 \leq \gamma \leq 1$, replace $\widehat{\Sigma}$ by $\widehat{\Sigma}(\gamma) := \gamma\widehat{\Sigma} + (1-\gamma)\widehat{\sigma}^2 I$.

Shrinkage reduces variance when covariance estimation is unstable.

Definition 1.3.3. *Naive Bayes Classifier*

For $x = (x_1,\ldots,x_p)$, assume $f_j(x) = \prod_{k=1}^{p} f_{jk}(x_k)$ for each class j.

(i) *Posterior*: With priors π_j, $p_j(x) := \mathbb{P}(G=j \mid X=x) \propto \pi_j \prod_{k=1}^{p} f_{jk}(x_k)$.

(ii) *Additive log odds*: Relative to class J, $\log\frac{p_\ell(x)}{p_J(x)} = \log\frac{\pi_\ell}{\pi_J} + \sum_{k=1}^{p}\log\frac{f_{\ell k}(x_k)}{f_{Jk}(x_k)}$.

On the log scale, features contribute additively.
Even misspecified independence can reduce variance enough to classify well near boundaries.

Remark 1.3.4. *Least Squares Versus LDA*

For $K = 2$, least-squares regression on any distinct label coding gives $\widehat{\beta} \propto \widehat{\Sigma}^{-1}(\widehat{\mu}_2 - \widehat{\mu}_1)$, the LDA direction.
The cutpoint may differ unless classes are balanced or tuned separately.
For $K > 2$, LDA uses pooled-Mahalanobis centroid distances and avoids masking.

Remark 1.3.5. *Logistic Regression Versus LDA*

Both can yield linear log-odds. Logistic regression models class probabilities directly and leaves P_X free; LDA fits a joint Gaussian mixture for (X,G) and uses Bayes' rule.

Remark 1.3.6. *Whitening and Reduced-Rank LDA*

With common covariance W, LDA compares $W^{-1/2}x$ to whitened centroids, adjusted by $\log\pi_k$.
The whitened centroids span dimension at most $K-1$, so LDA reduces to this centroid subspace.
Fisher directions maximize $\frac{a^\top Ba}{a^\top Wa}$, equivalently solve $Ba = \lambda Wa$, where B is between-class scatter.

Remark 1.3.7. *Feature Maps*

A feature map $\phi : \mathcal{X} \to \mathcal{H}$ gives linear prediction in feature space, $f(x) = w^\top\phi(x) + b$, which is generally nonlinear in x. Kernel methods choose ϕ implicitly; neural networks learn ϕ from data.

2 Regularisation and Model Selection

2.1 Penalized Linear Estimation

Regularisation controls model complexity by penalising unstable parameter directions or by constraining the feasible set.

Definition 2.1.1. *Penalised Linear Estimation*

For empirical loss $L_n(\beta)$, a penalised estimator solves $\widehat{\beta}_\lambda \in \arg\min_\beta\{L_n(\beta) + \lambda\Omega(\beta)\}$.
The choices $\Omega(\beta) = \|\beta\|_2^2$ and $\Omega(\beta) = \|\beta\|_1$ give ridge and lasso penalties.

Definition 2.1.2. *Subset Selection, Ridge, and Lasso*

Let $y \in \mathbb{R}^n$ be centered and let $X \in \mathbb{R}^{n\times p}$ have centered, standardised columns.

(i) *Best subset*: for a size m, solve $\widehat{\beta}_m^{\mathrm{sub}} \in \underset{|\,\mathrm{supp}(\beta)|\leq m}{\arg\min} \|y - X\beta\|_2^2$

(ii) *Ridge*: for $\lambda \geq 0$, solve $\widehat{\beta}_\lambda^{\mathrm{ridge}} \in \arg\min_\beta\{\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\}$

(iii) *Lasso*: for $\lambda \geq 0$, solve $\widehat{\beta}_\lambda^{\mathrm{lasso}} \in \arg\min_\beta\left\{\frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1\right\}$

Centering separates the unpenalised intercept, estimated by the mean on the raw scale.

Remark 2.1.3. *Penalty Geometry*

Assume $X^TX = I$ and set $z_j = x_j^T y$. For subset size m, best subset selection keeps the m largest $|z_j|$.
Ridge and lasso have coordinates $\widehat{\beta}_{\lambda,j}^{\mathrm{ridge}} = \frac{z_j}{1+\lambda}$ and $\widehat{\beta}_{\lambda,j}^{\mathrm{lasso}} = \mathrm{sign}(z_j)(|z_j| - \lambda)_+$.
Thus ℓ^2 shrinkage is smooth, while ℓ^1 shrinkage can set coefficients exactly to zero.

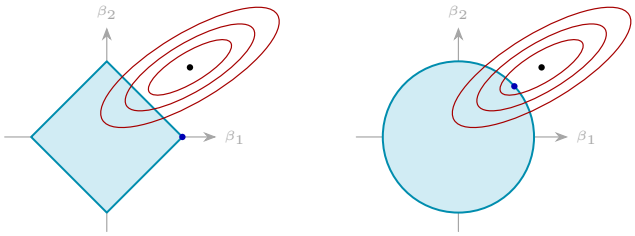

Figure 260: Left: lasso ℓ^1 diamond; right: ridge ℓ^2 disk.

Proposition 2.1.4. *Ridge Shrinkage and Effective Degrees*
Let $X \in \mathbb{R}^{n\times p}$ be centered with rank r and thin SVD $X = UDV^T$, where $D = \operatorname{diag}(d_1, \dots, d_r)$ and $d_j > 0$.
For $\lambda > 0$, set $S_\lambda := X(X^TX + \lambda I_p)^{-1}X^T$ and $s_j(\lambda) := \frac{d_j^2}{d_j^2+\lambda}$. Then $S_\lambda = U \operatorname{diag}(s_1(\lambda), \dots, s_r(\lambda))U^T$.
So $\widehat{y}_\lambda^{\text{ridge}} := X\widehat{\beta}_\lambda^{\text{ridge}} = S_\lambda y = \sum_{j=1}^{r} s_j(\lambda)u_j u_j^T y$ and $\operatorname{df}(\lambda) := \operatorname{tr}(S_\lambda) = \sum_{j=1}^{r} s_j(\lambda)$.

Proof. Since $X^TX = VD^2V^T$ and $X^T = VDU^T$, we have $V^T(X^TX + \lambda I_p)^{-1}V = (D^2 + \lambda I_r)^{-1}$.
Hence $S_\lambda = UDV^T(X^TX + \lambda I_p)^{-1}VDU^T = UD(D^2 + \lambda I_r)^{-1}DU^T = U \operatorname{diag}(s_1(\lambda), \dots, s_r(\lambda))U^T$.
Apply to y and take traces, using $U^TU = I_r$.
Structural Commentary: Ridge is diagonal in the left singular-vector basis of X.
Key Insight: Ridge keeps direction u_j by $s_j(\lambda)$; weak directions count less in the fit and $\operatorname{df}(\lambda)$. □

2.2 Validation and Selection Bias

Definition 2.2.1. *Model Selection by Validation*
Let $\mathcal{M}$ be a candidate set of models or hyperparameters.

(i) A *validation criterion* is $m \mapsto \widehat{R}_{\text{val}}(m)$; choose $\widehat{m} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_{\text{val}}(m)$.

(ii) In *K-fold cross-validation*, take folds $I_1, \dots, I_K$ and fits $\widehat{f}_m^{(-k)}$.
Define $\ell_{ik}(m) := \ell(y_i, \widehat{f}_m^{(-k)}(x_i))$, $\widehat{R}_{\text{cv}}(m) := \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in I_k} \ell_{ik}(m)$, and choose $\widehat{m}_{\text{cv}} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_{\text{cv}}(m)$.

Method 2.2.2. *One-Standard-Error Rule*
Let $(\mathcal{M}, \preceq)$ order candidates by complexity, with $\widehat{R}_{\text{cv}}, \widehat{\text{se}} : \mathcal{M} \to \mathbb{R}$.

(i) Choose $\widehat{m}_{\min} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_{\text{cv}}(m)$ and set $\widehat{R}_{\min} := \widehat{R}_{\text{cv}}(\widehat{m}_{\min})$.
(ii) Define $\mathcal{M}_{1\text{se}} := \{m \in \mathcal{M} : \widehat{R}_{\text{cv}}(m) \le \widehat{R}_{\min} + \widehat{\text{se}}(\widehat{m}_{\min})\}$.
(iii) Select $\widehat{m}_{1\text{se}}$ as the $\preceq$-minimal element of $\mathcal{M}_{1\text{se}}$; use any final test set only after this choice.

Example 2.2.3. *A Selection and Shrinkage Workflow*
Let $D = \{(x_i, y_i)\}_{i=1}^n$, where $y_i = \log(\text{PSA}_i)$ and $x_i = (\text{lcavol}_i, \text{lweight}_i, \text{age}_i, \text{lbph}_i, \text{svi}_i, \text{lcp}_i, \text{gleason}_i, \text{pgg45}_i)$.
Split indices into training and test sets T, E, standardize X_T, and choose each complexity parameter by cross-validation on T: subset size m, ridge/lasso penalty λ, or PCR/PLS dimension M.
The selected fit $\widehat{f}$ is then evaluated once on E by $\widehat{R}_{\text{test}} := |E|^{-1}\sum_{i\in E} \ell(y_i, \widehat{f}(x_i))$.

Remark 2.2.4. *Selection Bias*
If $\widehat{m} \in \arg\min_{m\in\mathcal{M}} \widehat{R}_V(m)$ is chosen on validation set V, then $\widehat{R}_V(\widehat{m})$ is optimistically biased for $R(\widehat{m})$.
Estimate the selected model with an independent set E, using $\widehat{R}_E(\widehat{m})$, or use nested validation.

2.3 Derived Input Directions and Regularization Paths

Method 2.3.1. *Derived Input Directions*
Let X, y be centered. A derived-direction method chooses $Z_M = (z_1, \dots, z_M)$ with $z_m = Xv_m$ or data-adaptive analogues, then fits $\widehat{y}_M = P_{Z_M} y$, where $P_{Z_M} := Z_M(Z_M^TZ_M)^{-1}Z_M^T$; M is selected by validation.

(i) *Principal components regression*: for $X = UDV^T$, take $v_1, \dots, v_M$ as the leading right singular vectors,
set $z_m = Xv_m = d_m u_m$, and fit $\widehat{y}_M^{\text{pcr}} = \sum_{m=1}^{M} z_m \frac{z_m^T y}{z_m^T z_m} = \sum_{m=1}^{M} u_m u_m^T y$.

(ii) *Partial least squares*: set $X^{(0)} = X$ and, for $m = 1, \dots, M$, define $z_m = X^{(m-1)}X^{(m-1)T}y$ and
$X^{(m)} = (I - P_{z_m})X^{(m-1)}$, where $P_{z_m} := \frac{z_m z_m^T}{z_m^T z_m}$. With $Z_M = (z_1, \dots, z_M)$, fit $\widehat{y}_M^{\text{pls}} = P_{Z_M} y$.

Algorithm 9: Partial Least Squares Directions

Require: Centered response y, standardized columns $x_1, \dots, x_p$, number of directions $M \le p$
Ensure: Fitted vectors $\widehat{y}^{(1)}, \dots, \widehat{y}^{(M)}$
Set $\widehat{y}^{(0)} \leftarrow 0$ and $x_j^{(0)} \leftarrow x_j$ for $j = 1, \dots, p$
for $m = 1, \dots, M$ **do**
$z_m \leftarrow \sum_{j=1}^{p} (x_j^{(m-1)})^T y\, x_j^{(m-1)}$
$\widehat{\theta}_m \leftarrow \frac{z_m^T y}{z_m^T z_m}$
$\widehat{y}^{(m)} \leftarrow \widehat{y}^{(m-1)} + \widehat{\theta}_m z_m$

for $j = 1, \dots, p$ **do**
$x_j^{(m)} \leftarrow x_j^{(m-1)} - \frac{z_m^T x_j^{(m-1)}}{z_m^T z_m} z_m$
end for
end for

Remark 2.3.2. *PLS Versus PCR*
PCR chooses v_m from $X^T X v_m = d_m^2 v_m$, so directions are ordered only by predictor variance d_m^2.
PLS chooses $z_m = X^{(m-1)} X^{(m-1)T} y$ and deflates $X^{(m)} = (I - P_{z_m}) X^{(m-1)}$, so each direction depends on both variance in $X^{(m-1)}$ and covariance with y.

Method 2.3.3. *Piecewise-Linear Path Algorithms*
Let $\widehat{\beta}(\lambda) \in \arg\min_\beta \{R(\beta) + \lambda J(\beta)\}$.

(i) If R is quadratic or piecewise quadratic and J is piecewise linear, fixed active-set and sign regions give affine KKT equations in λ, so $\widehat{\beta}(\lambda)$ is piecewise linear.
(ii) *Least angle regression*: with residual $r = y - X\beta$ and active set $\mathcal{A}$, move $\beta_\mathcal{A}$ in a direction keeping $|x_j^T r|$ tied for $j \in \mathcal{A}$ until a new variable ties or the least-squares fit is reached.
(iii) *Lasso path*: follow the same tied-correlation path and remove $j \in \mathcal{A}$ when β_j reaches 0.
(iv) *Coordinate descent*: for fixed λ, update $\beta_j \leftarrow \arg\min_t \{\frac{1}{2} \|y - X_{-j}\beta_{-j} - x_j t\|_2^2 + \lambda(|t| + \|\beta_{-j}\|_1)\}$ and warm-start along $\lambda_1 > \cdots > \lambda_L$.

Method 2.3.4. *LAR, Lasso, and Correlation Ties*
Let $\widehat{\beta}_\lambda \in \arg\min_\beta \{\frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1\}$, $r_\lambda := y - X\widehat{\beta}_\lambda$, and $\mathcal{A}_\lambda := \{j : \widehat{\beta}_{\lambda,j} \neq 0\}$.

(i) *KKT*: $x_j^T r_\lambda = \lambda \operatorname{sign}(\widehat{\beta}_{\lambda,j})$ for $j \in \mathcal{A}_\lambda$, and $|x_j^T r_\lambda| \le \lambda$ for $j \notin \mathcal{A}_\lambda$.
(ii) *Correlation ties*: active predictors satisfy $|x_j^T r_\lambda| = \lambda$; inactive predictors satisfy $|x_j^T r_\lambda| \le \lambda$.
(iii) *LAR*: with $C = \|X^T r\|_\infty$, move in the equiangular active direction preserving $|x_j^T r| = C$ for $j \in \mathcal{A}$ until some $k \notin \mathcal{A}$ reaches $|x_k^T r| = C$.
(iv) *Lasso path*: on a fixed active set and sign pattern, the KKT equations are affine in λ. When some $\widehat{\beta}_{\lambda,j}$ reaches 0, remove j and recompute the active direction.

Algorithm 10: Least Angle Regression Path

Require: Centered response y and standardized columns $x_1, \dots, x_p$ with $\|x_j\|_2 = 1$
Ensure: Piecewise-linear coefficient path $\{\beta(t)\}$
Initialize $\beta \leftarrow 0$, $r \leftarrow y$, $c \leftarrow X^T r$, $C \leftarrow \|c\|_\infty$, and $\mathcal{A} \leftarrow \{j : |c_j| = C\}$
repeat
$s_\mathcal{A} \leftarrow \operatorname{sign}(c_\mathcal{A})$, $\widetilde{X}_\mathcal{A} \leftarrow X_\mathcal{A} \operatorname{diag}(s_\mathcal{A})$, and $G_\mathcal{A} \leftarrow \widetilde{X}_\mathcal{A}^T \widetilde{X}_\mathcal{A}$
$\alpha_\mathcal{A} \leftarrow (\mathbf{1}^T G_\mathcal{A}^{-1} \mathbf{1})^{-1/2}$, $w_\mathcal{A} \leftarrow \alpha_\mathcal{A} G_\mathcal{A}^{-1} \mathbf{1}$, $u_\mathcal{A} \leftarrow \widetilde{X}_\mathcal{A} w_\mathcal{A}$, and $a \leftarrow X^T u_\mathcal{A}$
$\gamma \leftarrow \min\{C/\alpha_\mathcal{A}, \min_{j \notin \mathcal{A}}^+ \{(C - c_j)/(\alpha_\mathcal{A} - a_j), (C + c_j)/(\alpha_\mathcal{A} + a_j)\}\}$
Record $\beta_\mathcal{A}(t) = \beta_\mathcal{A} + t \operatorname{diag}(s_\mathcal{A}) w_\mathcal{A}$ for $0 \le t \le \gamma$
$\beta_\mathcal{A} \leftarrow \beta_\mathcal{A} + \gamma \operatorname{diag}(s_\mathcal{A}) w_\mathcal{A}$ and $r \leftarrow r - \gamma u_\mathcal{A}$
$c \leftarrow X^T r$, $C \leftarrow \|c\|_\infty$, and $\mathcal{A} \leftarrow \mathcal{A} \cup \{j \notin \mathcal{A} : |c_j| = C\}$
until $C = 0$ or $\mathcal{A} = \{1, \dots, p\}$

Algorithm 11: Incremental Forward Stagewise Regression

Require: Centered response y, standardized columns $x_1, \dots, x_p$, small step size $\varepsilon > 0$
Ensure: A stagewise coefficient path
Initialize $\beta \leftarrow 0$ and $r \leftarrow y$
repeat
Choose j maximizing $|x_j^T r|$
$\delta \leftarrow \varepsilon \operatorname{sign}(x_j^T r)$
$\beta_j \leftarrow \beta_j + \delta$
$r \leftarrow r - \delta x_j$
until the residual has negligible correlation with every predictor

Remark 2.3.5. *Stagewise and Lasso Paths*
Let $c = X^T r$ and $j \in \arg\max_k |c_k|$. Forward stagewise uses $\beta_j \leftarrow \beta_j + \varepsilon \operatorname{sign}(c_j)$ and $r \leftarrow r - \varepsilon \operatorname{sign}(c_j) x_j$; as $\varepsilon \downarrow 0$, it follows maximal-correlation directions, tracking LAR/lasso until a sign reversal is required.
Boosting applies the same greedy infinitesimal update to additive fits.

Algorithm 12: Pathwise Coordinate Descent for the Lasso

Require: Centered response y, standardized columns x_j with $\|x_j\|_2 = 1$, decreasing penalties $\lambda_1 > \cdots > \lambda_L \ge 0$
Ensure: Lasso coefficient vectors $\widehat{\beta}(\lambda_\ell)$ on the penalty grid
Define $S(z, \lambda) = \operatorname{sign}(z)(|z| - \lambda)_+$
Initialize $\beta \leftarrow 0$ and $r \leftarrow y - X\beta$
for $\ell = 1, \dots, L$ **do**
repeat
for $j = 1, \dots, p$ **do**
$r \leftarrow r + x_j \beta_j$
$z_j \leftarrow x_j^T r$
$\beta_j \leftarrow S(z_j, \lambda_\ell)$
$r \leftarrow r - x_j \beta_j$
end for
until the coefficient vector has converged
$\widehat{\beta}(\lambda_\ell) \leftarrow \beta$
end for

2.4 Effective Degrees and Information Criteria

Definition 2.4.1. *Effective Degrees of Freedom*
Let $\widehat{y} = S_\lambda y$ be a linear smoother with $S_\lambda = S_\lambda(X)$ independent of y.
Its *effective degrees of freedom* are $\mathrm{df}(\lambda) := \mathrm{trace}(S_\lambda)$. If $S_\lambda = P_V$ with $\dim V = m$, then $\mathrm{df}(\lambda) = m$.
For shrinkage smoothers, $\mathrm{df}(\lambda)$ need not be integral.

Remark 2.4.2. *Optimism of Training Error*
Assume $y = \mu + \varepsilon$, $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2 I$, and $\widehat{y} = S_\lambda y$ with $S_\lambda = S_\lambda(X)$.
Let $y^0 = \mu + \varepsilon^0$ be independent, $\mathrm{err}_{\mathrm{tr}}(\lambda) := n^{-1}\|y - \widehat{y}\|_2^2$, and $\mathrm{Err}_{\mathrm{in}}(\lambda) := n^{-1}\mathbb{E}_{y^0}\|y^0 - \widehat{y}\|_2^2$.
Then $\mathrm{Opt}(\lambda) := \mathbb{E}\{\mathrm{Err}_{\mathrm{in}}(\lambda) - \mathrm{err}_{\mathrm{tr}}(\lambda)\} = \frac{2}{n}\sum_{i=1}^n \mathrm{Cov}(\widehat{y}_i, y_i) = \frac{2\sigma^2}{n}\,\mathrm{trace}(S_\lambda)$.
Thus $\mathrm{df}(\lambda) = \mathrm{trace}(S_\lambda)$ is the covariance penalty.

Method 2.4.3. *AIC, C_p, and BIC Selection*
Let Λ be candidate tuning values. For a criterion $C : \Lambda \to \mathbb{R}$, select $\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} C(\lambda)$.

(i) For squared error, set $\overline{\mathrm{err}}(\lambda) := \frac{1}{n}\sum_{i=1}^n (y_i - \widehat{y}_{\lambda,i})^2$ and $d_\lambda := \mathrm{df}(S_\lambda)$; use $C_p(\lambda) := \overline{\mathrm{err}}(\lambda) + \frac{2\widehat{\sigma}^2}{n} d_\lambda$.
(ii) For likelihood fits with maximized log-likelihood ℓ_λ, use $\mathrm{AIC}(\lambda) := -\frac{2}{n}\ell_\lambda + \frac{2}{n} d_\lambda$.
(iii) Use $\mathrm{BIC}(\lambda) := -\frac{2}{n}\ell_\lambda + \frac{\log n}{n} d_\lambda$; the larger penalty favors smaller d_λ as n grows.

Remark 2.4.4. *Bayesian and MDL Readings of BIC*
For models $\mathcal{M}_m$ with dimension d_m, maximised likelihood L_m, and nearly uniform prior π_m,
set $\mathrm{BIC}_m := -2\log L_m + d_m \log n$. By laplace approximation, posterior weight is $\widehat{\mathbb{P}}(\mathcal{M}_m \mid Z) = \frac{\pi_m \exp\{-\mathrm{BIC}_m/2\}}{\sum_{\ell=1}^M \pi_\ell \exp\{-\mathrm{BIC}_\ell/2\}}$, so $\arg\min_m \mathrm{BIC}_m$ is an approximate MAP model.
In MDL form, $\frac{1}{2}\mathrm{BIC}_m$ is model code length plus fitted-discrepancy code length up to constants.

Method 2.4.5. *Generalized Cross-Validation*
Let Λ be candidate tuning values and $\widehat{y}_\lambda = S_\lambda y$.

(i) Define $e_{\lambda i} := y_i - \widehat{y}_{\lambda i}$, $h_{\lambda i} := S_{\lambda,ii}$, and $d_\lambda := \mathrm{trace}(S_\lambda)$.
(ii) The leave-one-out residual is $e_{\lambda i}^{(-i)} = y_i - \widehat{f}_\lambda^{(-i)}(x_i) = \frac{e_{\lambda i}}{1 - h_{\lambda i}}$.
(iii) *Generalized cross-validation* replaces $h_{\lambda i}$ by $\bar{h}_\lambda := d_\lambda/n$ and sets $\mathrm{GCV}(\lambda) := \frac{1}{n}\sum_{i=1}^n \left(\frac{e_{\lambda i}}{1 - d_\lambda/n}\right)^2$.
(iv) Select $\widehat{\lambda}_{\mathrm{gcv}} \in \arg\min_{\lambda\in\Lambda} \mathrm{GCV}(\lambda)$; this uses d_λ instead of the full leverage diagonal.

3 Basis Expansions, Splines, and Wavelets

3.1 Basis Expansions, Splines, and Wavelet Features

Basis expansions keep the fitting step linear while moving nonlinearity into the representation of the inputs.

Definition 3.1.1. *Linear Basis Expansion*

A *linear basis expansion* chooses $h = (h_1, \ldots, h_M) : \mathcal{X} \to \mathbb{R}^M$ and fits $f_\beta(x) := \beta^T h(x) = \sum_{m=1}^M \beta_m h_m(x)$.

Typical h_m include coordinates, powers, interactions, one-dimensional transforms, and region indicators.
The model is linear in $h(x)$, but generally nonlinear in x.

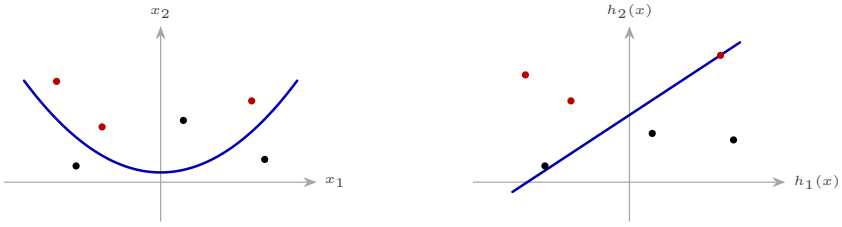

Figure 261: Basis expansion: boundary curved in x, linear in $h(x)$.

Remark 3.1.2. *Dictionaries Need Complexity Control*

For a dictionary $\mathcal{H} = \{h_m\}_{m=1}^M$, fitting $f_\beta(x) = \sum_{m=1}^M \beta_m h_m(x)$ is flexible but unstable when M is large or the h_m are nearly collinear. Complexity control restricts admissible fits: additive models impose $f(x) = \sum_j f_j(x_j)$, subset selection imposes $\|\beta\|_0 \le s$, and regularization minimizes empirical loss plus $\lambda\Omega(\beta)$.

Definition 3.1.3. *Regression Splines*
Let $\xi_1 < \cdots < \xi_K$ be knots.

(i) *Order-M regression spline* is piecewise polynomial $s : \mathbb{R} \to \mathbb{R}$ of degree $\le M-1$, $s \in C^{M-2}$ at knots.
(ii) *Cubic truncated-power basis* is basis $1, x, x^2, x^3, (x-\xi_1)_+^3, \ldots, (x-\xi_K)_+^3$ for cubic spline space.
(iii) *Natural cubic spline* is a cubic regression spline satisfying $s''(x) = 0$ for $x \notin [\xi_1, \xi_K]$.
(iv) *B-spline basis* is a locally supported basis for the same spline space.

Method 3.1.4. *Smoothing Splines as Penalized Fits*
Let $(x_i, y_i)_{i=1}^n \subset \mathbb{R}^2$, $y = (y_1, \ldots, y_n)^T$, $\lambda \ge 0$, and $\mathcal{F} := \{f : f'' \in L^2(\mathbb{R})\}$.

(i) The smoothing spline is $\widehat{f}_\lambda \in \arg\min_{f\in\mathcal{F}} \left\{\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_{\mathbb{R}} (f''(t))^2\, dt\right\}$.

(ii) $\widehat{f}_\lambda$ is a natural cubic spline with knots at the distinct observed inputs.
(iii) With $\widehat{y}_\lambda = (\widehat{f}_\lambda(x_1), \dots, \widehat{f}_\lambda(x_n))^T$, there is a smoother matrix S_λ such that $\widehat{y}_\lambda = S_\lambda y$ and $d_\lambda := \operatorname{trace}(S_\lambda)$.
(iv) For $e_{\lambda i} := y_i - \widehat{f}_\lambda(x_i)$ and $h_{\lambda i} := S_{\lambda,ii}$, leave-one-out CV is $\mathrm{CV}(\lambda) := \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_{\lambda i}}{1-h_{\lambda i}}\right)^2$.

Definition 3.1.5. *Tensor-Product and Thin-Plate Splines*
Let $\{h_{rj}\}_{j=1}^{J_r}$ be a univariate basis for coordinate x_r.
(i) A *two-way tensor-product basis* is $\{g_{jk}\}$ with $g_{jk}(x_1,x_2) := h_{1j}(x_1)h_{2k}(x_2)$ and dimension J_1J_2.
(ii) A *d-way tensor-product basis* is $\{g_{j_1,\dots,j_d}\}$ with $g_{j_1,\dots,j_d}(x) := \prod_{r=1}^{d} h_{rj_r}(x_r)$ and dimension $\prod_{r=1}^{d} J_r$.
(iii) A *thin-plate spline fit* is any $\widehat{f}_\lambda \in \arg\min_f Q_\lambda(f)$, where $Q_\lambda(f) := \sum_{i=1}^{n}(y_i - f(x_i))^2 + \lambda J(f)$, $D_{ab}f := \partial_a\partial_b f$, and $J(f) := \int_{\mathbb{R}^2}\{(D_{11}f)^2 + 2(D_{12}f)^2 + (D_{22}f)^2\}\,dx$.
(iv) A *thin-plate representer* is $\widehat{f}(x) = a_0 + a^\top x + \sum_{\ell=1}^{L} c_\ell\phi(\|x-\kappa_\ell\|)$; $\phi(r) := r^2\log r$, $\phi(0) := 0$, $\sum_{\ell=1}^{L} c_\ell(1,\kappa_\ell^\top) = 0$.

Method 3.1.6. *Spline-Filtered Logistic Features*
Let $(x_i, y_i)_{i=1}^n$ with $x_i \in \mathbb{R}^p$, $y_i \in \{0,1\}$, and let $H \in \mathbb{R}^{p\times M}$ be a smooth basis matrix with $M \ll p$.
(i) Restrict $\beta \in \mathbb{R}^p$ to $\mathrm{col}(H)$ by writing $\beta = H\theta$, $\theta \in \mathbb{R}^M$.
(ii) Define filtered features $z_i := H^T x_i$; then $x_i^T\beta = z_i^T\theta$.
(iii) Fit $\widehat{\theta}, \widehat{b} \in \arg\min_{\theta,b} \sum_{i=1}^{n}\{\log(1+\exp(b+z_i^T\theta)) - y_i(b+z_i^T\theta)\}$.
(iv) Set $\widehat{\beta} = H\widehat{\theta}$; smoothness is imposed by $\widehat{\beta} \in \mathrm{col}(H)$ and dimension drops from p to M.

Method 3.1.7. *Wavelet Shrinkage*
Let $y \in \mathbb{R}^n$ be observed on a regular lattice and let $W \in \mathbb{R}^{n\times n}$ be an orthonormal wavelet matrix.
(i) Transform y to wavelet coefficients $y^* := W^\top y$.
(ii) Estimate sparse coefficients by $\widehat{\theta}_\lambda \in \arg\min_{\theta\in\mathbb{R}^n} Q_\lambda(\theta)$, where $Q_\lambda(\theta) := \|y - W\theta\|_2^2 + 2\lambda\|\theta\|_1$.
(iii) Since $W^\top W = I$, $\widehat{\theta}_{\lambda,j} = \operatorname{sign}(y_j^*)(|y_j^*| - \lambda)_+$ for $j = 1,\dots,n$.
(iv) Reconstruct $\widehat{f}_\lambda := W\widehat{\theta}_\lambda = \sum_{j:|y_j^*|>\lambda}\operatorname{sign}(y_j^*)(|y_j^*|-\lambda)w_j$, where w_j is the jth wavelet atom.

4 Kernel Methods and RKHS Theory

4.1 Kernel Methods and RKHS Representations

Kernel methods keep the algorithm linear in an implicit feature space while computing only inner products.

Definition 4.1.1. *Positive Definite Kernel*
A map $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ is a *positive definite kernel* if:
(i) $k(x,y) = k(y,x)$ for all $x, y \in \mathcal{X}$;
(ii) for every $n \in \mathbb{N}$ and $x_1,\dots,x_n \in \mathcal{X}$, $K = (k(x_i,x_j))_{i,j=1}^n$ satisfies $a^\top Ka = \sum_{i,j=1}^{n} a_ia_jk(x_i,x_j) \ge 0\ \forall a \in \mathbb{R}^n$.

Example 4.1.2. *Common Kernels*
For $x, y \in \mathbb{R}^p$, common positive definite kernels include:
(i) *Polynomial*: for $d \in \mathbb{N}_{>0}$, $k_d(x,y) := (\langle x,y\rangle + 1)^d = \langle h_d(x), h_d(y)\rangle$, encoding monomials of degree $\le d$. If $p = 2$ and $d = 2$, one feature map is $h_2(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2) \in \mathbb{R}^6$.
(ii) *Gaussian RBF*: for $\nu > 0$, $k_\nu(x,y) := \exp(-\nu\|x-y\|_2^2)$; it is radial and translation invariant, with $k_\nu(x,y)$ decreasing as $\|x-y\|_2$ increases.

Definition 4.1.3. *Reproducing Kernel Hilbert Space*
A Hilbert space $\mathcal{H} \subseteq \mathbb{R}^{\mathcal{X}}$ is a *reproducing kernel Hilbert space* if point evaluation is continuous; explicitly:
(i) for every $x \in \mathcal{X}$, the evaluation map $E_x : \mathcal{H} \to \mathbb{R}$, $E_x(f) := f(x)$, is continuous;
(ii) by Riesz, $k_x \in \mathcal{H}$ denotes the unique representer of E_x, so $f(x) = \langle f, k_x\rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$;
(iii) the *reproducing kernel* is $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$, $k(x,y) := k_x(y) = \langle k_x, k_y\rangle_{\mathcal{H}}$, with $k_x = k(x,\cdot)$.

Theorem 4.1.4. *Representer Theorem Form*
Let $\mathcal{H}$ be an RKHS on $\mathcal{X}$ with kernel k.
Let $(x_i,y_i)_{i=1}^n \subseteq \mathcal{X}\times\mathcal{Y}$, $\lambda > 0$, $\ell : \mathbb{R}\times\mathcal{Y} \to (-\infty,+\infty]$, and $\Phi : [0,\infty) \to \mathbb{R}$ nondecreasing.
For $J(f) := \sum_{i=1}^{n}\ell(f(x_i),y_i) + \lambda\Phi(\|f\|_{\mathcal{H}})$ and $S := \operatorname{span}\{k(x_i,\cdot) : 1 \le i \le n\}$, if $f^\star \in \operatorname{argmin}_{f\in\mathcal{H}} J(f)$, then $Pf^\star \in S$ is also a minimiser, where $P : \mathcal{H} \to S$ is orthogonal projection.
Hence some minimiser has $f^\star(\cdot) = \sum_{i=1}^{n}\alpha_ik(x_i,\cdot)$; if Φ is strictly increasing, every minimiser has this form.

Proof. Since S is finite-dimensional, it is closed, so P exists.
Write $f = f_S + f_\perp$ with $f_S := Pf \in S$ and $f_\perp \in S^\perp$.
For each i, $k(x_i,\cdot) \in S$, hence $f_\perp(x_i) = \langle f_\perp, k(x_i,\cdot)\rangle_{\mathcal{H}} = 0$ and $f_S(x_i) = f(x_i)$.
Thus the empirical loss in J is unchanged by $f \mapsto f_S$.
By orthogonality, $\|f\|_{\mathcal{H}}^2 = \|f_S\|_{\mathcal{H}}^2 + \|f_\perp\|_{\mathcal{H}}^2$, so $\|f_S\|_{\mathcal{H}} \le \|f\|_{\mathcal{H}}$, hence $J(f_S) \le J(f)$ by monotonicity of Φ.
Applying this to a minimiser $f^\star$ gives $J(Pf^\star) = J(f^\star)$, so $Pf^\star$ is a minimiser in S.
Since $S = \{\sum_{i=1}^{n}\alpha_ik(x_i,\cdot) : \alpha \in \mathbb{R}^n\}$, finite expansion follows. If Φ strictly increasing, $f_\perp \ne 0$, then $\|f_S\|_{\mathcal{H}} < \|f\|_{\mathcal{H}}$ while loss is

unchanged, so $J(f_S) < J(f)$, contradicting minimality.
Structural Commentary: $E : \mathcal{H} \to \mathbb{R}^n$, $Ef = (f(x_i))_{i=1}^n$, has $E|_{S^\perp} = 0$ and $Ef = E(Pf)$; only S is data-visible.
Key Insight: Regularisation kills $S^\perp$, so optimise over $\alpha \in \mathbb{R}^n$ via $f_\alpha = \sum_{i=1}^n \alpha_i k(x_i, \cdot)$ and $Ef_\alpha = K\alpha$. □

Method 4.1.5. *Kernel Ridge Regression*
Let $(x_i, y_i)_{i=1}^n \subseteq \mathcal{X} \times \mathbb{R}$, $y := (y_1, \dots, y_n)^\top$, $\lambda > 0$, and $K_{ij} := k(x_i, x_j)$.

(i) *RKHS problem*. Solve $\widehat{f} \in \operatorname*{argmin}_{f \in \mathcal{H}} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}}^2$.

(ii) *Representer reduction*. Write $\widehat{f}(\cdot) = \sum_{i=1}^n \widehat{\alpha}_i k(x_i, \cdot)$; then $\widehat{f}(x_j) = (K\widehat{\alpha})_j$ and $\left\|\widehat{f}\right\|_{\mathcal{H}}^2 = \widehat{\alpha}^\top K \widehat{\alpha}$.

(iii) *Finite problem*. Minimise $\widehat{\alpha} \in \operatorname*{argmin}_{\alpha \in \mathbb{R}^n} \|y - K\alpha\|_2^2 + \lambda \alpha^\top K \alpha$.

(iv) *Solution*. Since $K \succeq 0$ and $\lambda > 0$, $K + \lambda I \succ 0$ and $\widehat{\alpha} = (K + \lambda I)^{-1} y$.
Hence $\widehat{y} = K\widehat{\alpha}$ and $\widehat{f}(x) = k_x^\top \widehat{\alpha}$, where $k_x := (k(x, x_1), \dots, k(x, x_n))^\top$.

Remark 4.1.6. *Localisation Kernels and RKHS Kernels*
Localisation kernel has weights $w_i(x_0) \propto K_\lambda(x_0, x_i)$. $w_i(x_0)$ is large when x_i close to x_0; small otherwise.
RKHS kernel is PSD: $K_{ij} = k(x_i, x_j)$ is Gram matrix, i.e., $k(x, y) = \langle h(x), h(y) \rangle_{\mathcal{F}}$ for some feature map h.
Thus: localisation kernels → local averaging; RKHS kernels → inner products for global regularisation.

Example 4.1.7. *Radial Basis Expansions*
Let $\xi_1, \dots, \xi_M \in \mathbb{R}^p$, $\lambda_j > 0$, $\beta \in \mathbb{R}^M$, and let $D : [0, \infty) \to [0, \infty)$ be a radial profile.

(i) *Unnormalized basis*: $\varphi_j(x) := D(\frac{\|x - \xi_j\|_2}{\lambda_j})$ and $f(x) := \sum_{j=1}^M \beta_j \varphi_j(x)$.

(ii) *Normalized basis*: for $\lambda_j = \lambda$, set $h_j(x) := D(\frac{\|x - \xi_j\|_2}{\lambda})/s(x)$, where $s(x) := \sum_{k=1}^M D(\frac{\|x - \xi_k\|_2}{\lambda}) > 0$.

Then $h_j(x) \geq 0$ and $\sum_{j=1}^M h_j(x) = 1$.

(iii) *Training-point centers*: $M = n$, $\xi_i = x_i$, $\beta_i = Y_i$ give $\widehat{f}(x) := \sum_{i=1}^n Y_i h_i(x)$ (Nadaraya–Watson local avg).

Remark 4.1.8. *Spline and Kernel Viewpoints*
Common form: $\min_f \sum_{i=1}^n \ell(f(x_i), y_i) + \lambda \|Pf\|_{\mathcal{H}}^2$, where P removes an unpenalised $\mathcal{U}$, gives $\widehat{f} = u + \sum_{i=1}^n \alpha_i k(x_i, \cdot)$ with $u \in \mathcal{U}$. Splines choose P, kernel methods choose k, and SVMs choose a margin loss ℓ.

5 Support Vector Machines and Margin Methods

5.1 Margins and SVM Primal Problems

Margin methods prefer classifiers that separate classes with a buffer, not merely with zero training error. The support-vector construction makes this preference explicit by trading margin width against margin violations.

Definition 5.1.1. *Affine Hyperplane and Signed Distance*
Let $w \in \mathbb{R}^p \setminus \{0\}$, $b \in \mathbb{R}$, and $q_{w,b}(x) := w^\top x + b$.

(i) *Affine hyperplane*: $H_{w,b} := \{x \in \mathbb{R}^p : q_{w,b}(x) = 0\}$, with normal direction w.

(ii) *Signed distance*: $d_{w,b}(x) := \frac{q_{w,b}(x)}{\|w\|_2}$.

(iii) *Scale invariance*: for $c > 0$, $H_{cw,cb} = H_{w,b}$ and $q_{cw,cb} = c q_{w,b}$.

Definition 5.1.2. *Margin*
Let $y_i \in \{-1, 1\}$ and $q(x) := w^\top x + b$.

(i) *Functional margin*: $\gamma_i(q) := y_i q(x_i)$.

(ii) *Geometric margin*: for $w \neq 0$, $\rho_i(w, b) := \frac{y_i(w^\top x_i + b)}{\|w\|_2}$.

(iii) *Correct classification*: $\gamma_i(q) > 0$; larger $\gamma_i(q)$ means larger signed confidence at fixed score scale.

Definition 5.1.3. *Hard-Margin Support Vector Machine*
For separable $(x_i, y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1, 1\}$:

(i) *Unit-margin scale*: $y_i(w^\top x_i + b) \geq 1$ for all i.

(ii) *Hard-margin primal*: $(\widehat{w}, \widehat{b}) \in \operatorname*{argmin}_{w,b} \frac{1}{2}\|w\|_2^2$ subject to $y_i(w^\top x_i + b) \geq 1$, $i = 1, \dots, n$.

(iii) *Margin width*: the slab $-1 \leq w^\top x + b \leq 1$ has width $\frac{2}{\|w\|_2}$, so the primal maximises geometric margin.

Definition 5.1.4. *Soft-Margin Support Vector Machine*
For possibly nonseparable $(x_i, y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1, 1\}$ and $C > 0$:

(i) *Slack variables*: $\xi_i \geq 0$ and $y_i(w^\top x_i + b) \geq 1 - \xi_i$, $i = 1, \dots, n$.

(ii) *Soft-margin primal*: $(\widehat{w}, \widehat{b}, \widehat{\xi}) \in \operatorname{argmin}_{w,b,\xi}\{\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n \xi_i\}$ subject to the slack constraints.

(iii) *Penalty parameter*: C prices $\sum_{i=1}^n \xi_i$; larger C penalises violations more strongly.

Remark 5.1.5. *Why the Standard Slack Scaling*
Before score normalisation, additive slack uses $y_i(w^\top x_i + b) \geq M - \xi_i$, relative slack uses $y_i(w^\top x_i + b) \geq M(1 - \xi_i)$.
Setting $M = 1$ gives convex constraint $y_i(w^\top x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$.
Thus ξ_i measures margin failure: 0 on/outside the margin, $(0, 1]$ inside but correct, and > 1 misclassified.

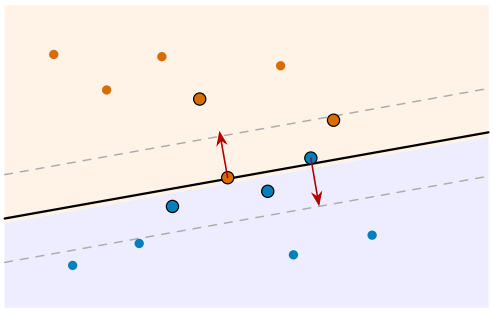

Figure 262: Soft-margin SVM: orange $y=+1$, blue $y=-1$; solid $f(x)=0$, dashed $f(x)=\pm 1$; circled support vectors, red slack arrows ξ_i.

Proposition 5.1.6. *Hinge-Loss Penalty Form*
Let $(x_i,y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1,1\}$, $C>0$, $f_{w,b}(x) := w^\top x + b$, and $\lambda := 1/C$. For soft-margin primal $\min_{w,b,\xi}\{\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n \xi_i\}$ *subject to $\xi_i \geq 0$ and $y_i f_{w,b}(x_i) \geq 1-\xi_i$, eliminating ξ and scaling by $\frac{1}{C}$ gives* $\min_{w,b}\sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+ + \frac{\lambda}{2}\|w\|_2^2$. *For fixed (w,b), minimal feasible slack is $\xi_i^\star = [1-y_i f_{w,b}(x_i)]_+$.*

Proof. Fix (w,b) and put $r_i := 1-y_i f_{w,b}(x_i)$. Feasibility gives $\xi_i \geq 0$ and $\xi_i \geq r_i$, hence $\xi_i \geq \max\{0,r_i\} = [r_i]_+$.
Since $C>0$, the objective is strictly increasing in each ξ_i, so $\xi_i^\star = [r_i]_+$.
Substitution yields $\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+$.
Multiply by $1/C>0$ preserves minimisers, $\sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+ + \frac{1}{2C}\|w\|_2^2 = \sum_{i=1}^n [1-y_i f_{w,b}(x_i)]_+ + \frac{\lambda}{2}\|w\|_2^2$.
Structural Commentary: The ξ_i decouple; eliminate them pointwise.
Key Insight: Soft-margin SVM is hinge empirical risk plus an ℓ_2 penalty on w; the intercept b is unpenalized. □

5.2 Dual Problems and Kernelization

Proposition 5.2.1. *Hard-Margin Dual and Support Vectors*
Let $(x_i,y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1,1\}$ be linearly separable. The primal $P : \min_{w,b} \frac{1}{2}\|w\|_2^2$ subject to $y_i(w^\top x_i + b) \geq 1$ has dual $D : \max_{\alpha \in \mathbb{R}_+^n} \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j x_i^\top x_j$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$.
At any optimum, $\widehat{w} = \sum_{i=1}^n \widehat{\alpha}_i y_i x_i$ and $\widehat{\alpha}_i > 0 \Rightarrow y_i(\widehat{w}^\top x_i + \widehat{b}) = 1$; these active points are the support vectors.

Proof. Let $g_i(w,b) := y_i(w^\top x_i + b) - 1$ and $\alpha_i \geq 0$. The Lagrangian is $L(w,b,\alpha) = \frac{1}{2}\|w\|_2^2 - \sum_{i=1}^n \alpha_i g_i(w,b)$.
Stationarity gives $0 = \nabla_w L = w - \sum_{i=1}^n \alpha_i y_i x_i$ and $0 = \partial_b L = -\sum_{i=1}^n \alpha_i y_i$, hence $w = \sum_{i=1}^n \alpha_i y_i x_i$ and $\sum_{i=1}^n \alpha_i y_i = 0$.
Substituting into L gives $D(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j x_i^\top x_j$, with $\alpha \geq 0$ and $\sum_i \alpha_i y_i = 0$.
Strong duality holds by convexity and strict feasibility.
Complementary slackness gives $\widehat{\alpha}_i[y_i(\widehat{w}^\top x_i + \widehat{b}) - 1] = 0$, so $\widehat{\alpha}_i > 0$ forces $y_i(\widehat{w}^\top x_i + \widehat{b}) = 1$.
Structural Commentary: Stationarity leaves α; data enter via $y_i y_j x_i^\top x_j$ and $\sum_i \alpha_i y_i = 0$.
Key Insight: $\alpha_i > 0$ marks margin support; these points form $\widehat{w}$ and fix the separator. □

Proposition 5.2.2. *Soft-Margin Dual and Support Types*
Let $(x_i,y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \{-1,1\}$ and $C>0$. The primal $P : \min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^n \xi_i$ subject to $y_i(w^\top x_i + b) \geq 1-\xi_i$ and $\xi_i \geq 0$ has dual $D : \max_{\alpha \in [0,C]^n} \sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j x_i^\top x_j$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$.
At any optimum, $\widehat{w} = \sum_{i=1}^n \widehat{\alpha}_i y_i x_i$. With $\widehat{f}(x) := \widehat{w}^\top x + \widehat{b}$, $\widehat{\alpha}_i = 0 \Rightarrow y_i\widehat{f}(x_i) \geq 1$, $0 < \widehat{\alpha}_i < C \Rightarrow y_i\widehat{f}(x_i) = 1$, and $\widehat{\alpha}_i = C \Rightarrow y_i\widehat{f}(x_i) = 1 - \widehat{\xi}_i \leq 1$. The support vectors are exactly $\{i : \widehat{\alpha}_i > 0\}$.

Proof. Let $\alpha_i \geq 0$ multiply $y_i(w^\top x_i + b) - 1 + \xi_i \geq 0$ and $\mu_i \geq 0$ multiply $\xi_i \geq 0$.
The Lagrangian is $L = \frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i - \sum_i \alpha_i[y_i(w^\top x_i + b) - 1 + \xi_i] - \sum_i \mu_i\xi_i$.
Stationarity gives $w = \sum_i \alpha_i y_i x_i$, $\sum_i \alpha_i y_i = 0$, and $C - \alpha_i - \mu_i = 0$, hence $0 \leq \alpha_i \leq C$.
Substitution gives dual; Slater gives strong duality.
Complementary slackness gives $\alpha_i[y_i f(x_i) - 1 + \xi_i] = 0$ and $(C-\alpha_i)\xi_i = 0$.
Thus $\alpha_i = 0$ implies $\xi_i = 0$ and $y_i f(x_i) \geq 1$; $0 < \alpha_i < C$ implies $\xi_i = 0$ and $y_i f(x_i) = 1$.
Hence $\alpha_i = C$ implies $y_i f(x_i) = 1 - \xi_i \leq 1$.
Structural Commentary: Slacks become box constraints $0 \leq \alpha_i \leq C$.
Key Insight: α_i labels type: 0 outside, $(0,C)$ on-margin, C inside/wrong-side. □

Method 5.2.3. *Kernelized Support Vector Classifier*
Let $(x_i, y_i)_{i=1}^n \subseteq \mathcal{X} \times \{-1, 1\}$, $C > 0$, $h : \mathcal{X} \to \mathcal{F}$, and $k(x, x') := \langle h(x), h(x') \rangle_{\mathcal{F}}$.
The kernelized dual is $\widehat{\alpha} \in \underset{\alpha \in [0,C]^n}{\operatorname{argmax}} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(x_i, x_j)$ subject to $\sum_{i=1}^n \alpha_i y_i = 0$.
The fitted score and classifier are $\widehat{f}(x) = \sum_{i=1}^n \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$ and $\widehat{G}(x) = \operatorname{sign}(\widehat{f}(x))$.
If $0 < \widehat{\alpha}_s < C$, then $\widehat{b} = y_s - \sum_{i=1}^n \widehat{\alpha}_i y_i k(x_s, x_i)$; average this over all such s for stability.

Proof. In feature space, the soft-margin primal uses $v \in \mathcal{F}$ and score $f_{v,b}(x) = \langle v, h(x) \rangle_{\mathcal{F}} + b$.
Its Lagrangian stationarity gives $v = \sum_{i=1}^n \alpha_i y_i h(x_i)$ and $\sum_{i=1}^n \alpha_i y_i = 0$, while slack stationarity gives $0 \le \alpha_i \le C$.
Substitution yields $\sum_i \alpha_i - \frac{1}{2} \left\| \sum_i \alpha_i y_i h(x_i) \right\|_{\mathcal{F}}^2 = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j k(x_i, x_j)$.
Hence $\widehat{f}(x) = \langle \widehat{v}, h(x) \rangle_{\mathcal{F}} + \widehat{b} = \sum_i \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$.
If $0 < \widehat{\alpha}_s < C$, KKT gives zero slack and $y_s \widehat{f}(x_s) = 1$, so $\widehat{f}(x_s) = y_s$ and the stated $\widehat{b}$ follows.
Structural Commentary: Coordinates vanish; optimisation uses only $K_{ij} = k(x_i, x_j)$.
Key Insight: The classifier is finite and support-vector based: only indices with $\widehat{\alpha}_i > 0$ contribute to $\widehat{f}$. □

Remark 5.2.4. *Hinge Loss and Probability-Oriented Losses*
Let $\eta(x) := \mathbb{P}(Y = 1 \mid X = x)$ and $m := yf(x)$.
(i) *Hinge*: $[1 - m]_+$ targets $\operatorname{sign}(2\eta(x) - 1)$ and is zero for $m \ge 1$.
(ii) *Logistic*: $\log(1 + e^{-m})$ targets $\log(\eta(x)/(1 - \eta(x)))$ and keeps a smooth positive tail.
(iii) *Squared*: squared loss targets $\mathbb{E}[Y \mid X = x] = 2\eta(x) - 1$ and keeps residual-magnitude leverage.

Method 5.2.5. *SVM Regularization Path View*
set $\lambda := 1/C$; as $\lambda \downarrow 0$ ($C \uparrow \infty$), hinge loss dominates norm regularisation.
(i) *Fit*: for each λ, solve for $f_\lambda(x) := w_\lambda^\top x + b_\lambda$ and multipliers $\alpha_i(\lambda)$.
(ii) *Classify states*: with $m_i(\lambda) := y_i f_\lambda(x_i)$, $m_i > 1 \Rightarrow \alpha_i = 0$, $m_i = 1 \Rightarrow 0 \le \alpha_i \le C$, and $m_i < 1 \Rightarrow \alpha_i = C$.
(iii) *Track*: fixed active set $\Rightarrow \alpha_i(\lambda)$ and f_λ are linear.
(iv) *Kernelize*: replace $w_\lambda^\top x$ by $\sum_{i=1}^n \alpha_i(\lambda) y_i k(x, x_i)$, so $f_\lambda(x) = \sum_{i=1}^n \alpha_i(\lambda) y_i k(x, x_i) + b_\lambda$.

5.3 Support Vector Regression and Perceptrons

Method 5.3.1. *Epsilon-Insensitive Support Vector Regression*
Let $(x_i, y_i)_{i=1}^n \subseteq \mathbb{R}^p \times \mathbb{R}$, $\epsilon > 0$, $\lambda > 0$, and $f_{w,b}(x) := w^\top x + b$.
(i) *Define loss*: for residual r, set $V_\epsilon(r) := [|r| - \epsilon]_+$.
(ii) *Fit*: solve $\min_{w,b} \sum_{i=1}^n V_\epsilon(y_i - f_{w,b}(x_i)) + \frac{\lambda}{2} \|w\|_2^2$.
(iii) *Read tube*: $|r_i| \le \epsilon \Rightarrow V_\epsilon(r_i) = 0$ and $|r_i| > \epsilon \Rightarrow V_\epsilon(r_i) = |r_i| - \epsilon$.
(iv) *Kernelize*: $\widehat{f}(x) = \sum_{i=1}^n (\widehat{\alpha}_i^* - \widehat{\alpha}_i) k(x, x_i) + \widehat{b}$; tube sides have opposite signs.

Method 5.3.2. *Perceptron Update as a Contrast*
Choose $(w_0, b_0) \in \mathbb{R}^p \times \mathbb{R}$ and learning rate $\rho > 0$.
(i) *Scan*: at step t, choose (x_i, y_i) and compute $m_i(t) := y_i(w_t^\top x_i + b_t)$.
(ii) *Update*: if $m_i(t) \le 0$, set $(w_{t+1}, b_{t+1}) := (w_t, b_t) + \rho y_i (x_i, 1)$.
(iii) *Keep*: if $m_i(t) > 0$, set $(w_{t+1}, b_{t+1}) := (w_t, b_t)$.
(iv) *Contrast*: perceptron may separate but is order-dependent; SVM solves a convex max-margin problem.

Remark 5.3.3. *Support Vectors*
Let $\widehat{f}(x) = \sum_{i=1}^n \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$ and $S := \{i : \widehat{\alpha}_i > 0\}$.
Then $\widehat{f}(x) = \sum_{i \in S} \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}$, and KKT gives $i \in S \Rightarrow y_i \widehat{f}(x_i) \le 1$. Thus only support vectors enter $\widehat{f}$.

Remark 5.3.4. *Margins and Generalization*
Let $m_i(f) := y_i f(x_i)$ and $\widehat{R}_\gamma(f) := n^{-1} \sum_{i=1}^n \mathbf{1}\{m_i(f) \le \gamma\}$.
Bounds control $R(f) := \mathbb{P}(Yf(X) \le 0)$ by $\widehat{R}_\gamma(f)$ plus a complexity proxy $\mathcal{C}_\gamma(f) \asymp \|f\|^2 / \gamma^2$.
For linear $f_{w,b}$ with $\min_i m_i(f) \ge 1$, the geometric margin is $\rho = 1/\|w\|_2$.

6 Trees and Ensemble Averaging

6.1 Tree Models and Bagging

Tree-based methods partition the input space into simple regions and fit a local rule in each region. Ensemble averaging methods keep this local modelling idea but stabilise it by resampling, voting, and decorrelating the individual trees.

Definition 6.1.1. *Decision Tree*
Let $\mathcal{X} \subseteq \mathbb{R}^p$ and let $\mathcal{A}$ be the prediction space.
A decision tree T is a finite binary recursive partition of $\mathcal{X}$ with a leaf value in $\mathcal{A}$ on each terminal region.
(i) *Split*: a split at node $R \subseteq \mathcal{X}$ is a feature-threshold pair (j, t) with $j \in \{1, \dots, p\}$ and $t \in \mathbb{R}$.
It produces children $R_L := R \cap \{x : x_j \le t\}$ and $R_R := R \cap \{x : x_j > t\}$.

(ii) *Terminal partition*: after splitting stops, the leaves of T are nonempty regions $R_1(T), \dots, R_M(T)$ with $\mathcal{X} = \bigcup_{m=1}^{M} R_m(T)$ and $R_m(T) \cap R_{m'}(T) = \varnothing$ for $m \neq m'$.

(iii) *Leaf rule*: a leaf assignment $a_m \in \mathcal{A}$ for $m = 1, \dots, M$ defines $\widehat{f}_T : \mathcal{X} \to \mathcal{A}$ by $\widehat{f}_T(x) := a_m$ for the unique m such that $x \in R_m(T)$.

6.2 CART Splits and Pruning

Definition 6.2.1. *CART Regression-Tree Notation and Scores*
Let $D_n = \{(x_i, y_i)\}_{i=1}^n \subseteq \mathcal{X} \times \mathbb{R}$. For $R \subseteq \mathcal{X}$, set $I_R := \{i : x_i \in R\}$ and $n_R := |I_R|$.

(i) *Cell loss*: if $n_R > 0$, define $\bar{y}_R := n_R^{-1} \sum_{i \in I_R} y_i$ and $\mathrm{SSE}(R) := \sum_{i \in I_R} (y_i - \bar{y}_R)^2$.

(ii) *Tree predictor*: if T has leaves $R_1(T), \dots, R_{|T|}(T)$, let $m_T(x)$ be the unique $m \in \{1, \dots, |T|\}$ with $x \in R_m(T)$ and set $\widehat{f}_T(x) := \bar{y}_{R_{m_T(x)}(T)}$.

(iii) *Split children*: for $a = (j, s) \in \{1, \dots, p\} \times \mathbb{R}$, set $R_L(a) := R \cap \{x : x_j \le s\}$ and $R_R(a) := R \cap \{x : x_j > s\}$.

(iv) *Admissible splits*: set $\mathcal{A}_R := \{a \in \{1, \dots, p\} \times \mathbb{R} : n_{R_L(a)} > 0, n_{R_R(a)} > 0\}$.

(v) *Split score*: for $a \in \mathcal{A}_R$, set $S_R(a) := \mathrm{SSE}(R_L(a)) + \mathrm{SSE}(R_R(a))$.

(vi) *Pruning score*: for $\alpha \ge 0$, set $C_\alpha(T) := \sum_{m=1}^{|T|} \mathrm{SSE}(R_m(T)) + \alpha |T|$.

Method 6.2.2. *CART-Style Regression Trees*

(i) *Grow*: split eligible leaves by $a_R \in \operatorname{argmin}_{a \in \mathcal{A}_R} S_R(a)$ until stopping, yielding T_0.

(ii) *Prune*: for each $\alpha \ge 0$, choose $T_\alpha \in \operatorname{argmin}_{T \preceq T_0} C_\alpha(T)$.

(iii) *Select*: choose $\widehat{\alpha} \ge 0$ by validation or cross-validation and return $T_{\widehat{\alpha}}$.

Deep growth lowers bias; the cost-complexity penalty $\alpha|T|$ controls variance by subtree selection.

Definition 6.2.3. *Classification Tree Impurity*
For node m, let $N_{mk} := |\{i : x_i \in R_m, y_i = k\}|$, $N_m := \sum_{k=1}^K N_{mk}$, $\widehat{p}_{mk} := \frac{N_{mk}}{N_m}$. Common impurity scores are:

(i) *Misclassification*: $I_{\mathrm{mis}}(m) := 1 - \max_{1 \le k \le K} \widehat{p}_{mk}$.

(ii) *Gini*: $I_{\mathrm{Gini}}(m) := \sum_{k=1}^K \widehat{p}_{mk}(1 - \widehat{p}_{mk}) = 1 - \sum_{k=1}^K \widehat{p}_{mk}^2$.

(iii) *Entropy*: $I_{\mathrm{ent}}(m) := -\sum_{k : \widehat{p}_{mk} > 0} \widehat{p}_{mk} \log \widehat{p}_{mk}$.

The usual leaf prediction is the majority class, $\widehat{k}(m) \in \operatorname{argmax}_{1 \le k \le K} \widehat{p}_{mk}$.

With unequal losses $L_{kk} = 0$, choose the class minimizing expected loss, $\widehat{\ell}(m) \in \operatorname{argmin}_{1 \le \ell \le K} \sum_{k=1}^K L_{k\ell} \widehat{p}_{mk}$.

Remark 6.2.4. *Tree Practicalities*
For categorical levels $\mathcal{A}_j$, split $B \subset \mathcal{A}_j$ versus B^c.
For binary responses, set $I_a := \{i : X_{ij} = a\}$ and order levels by $\widehat{p}_a := |I_a|^{-1} \sum_{i \in I_a} Y_i$.
Missing values can be an NA level or use a surrogate split g' maximising $\sum_i \mathbf{1}\{g'(x_i) = g(x_i)\}$.
Many candidate splits inflate gain; control growth with $N_m \ge n_{\min}$, depth limits, validation, or pruning.

6.3 Ensembles, Random Forests, and Stacking

Definition 6.3.1. *Bagging and Random Forests*
Let $D_n = \{(x_i, y_i)\}_{i=1}^n$, $B \ge 1$, and $m \le p$.

(i) *Bagging*: draw $D_1^*, \dots, D_B^*$ from D_n, fit $T_1, \dots, T_B$, and aggregate $\widehat{f}_{\mathrm{bag}}(x) := B^{-1} \sum_{b=1}^B T_b(x)$ for regression, with voting or averaged probabilities for classification.

(ii) *Random forest*: in addition to bagging, at each node R of T_b choose $M_{b,R} \subseteq \{1, \dots, p\}$ with $|M_{b,R}| = m$ and optimise the split over features in $M_{b,R}$ only.

Bagging averages bootstrap perturbations; random split-feature subsets reduce $\mathrm{corr}(T_b, T_{b'})$ before averaging.

Method 6.3.2. *Bagged Prediction and Votes*
Let $D_n := \{(x_i, y_i)\}_{i=1}^n$ and $B \ge 1$; base outputs are $\widehat{f}$ for regression and $(\widehat{g}, \widehat{p}_1, \dots, \widehat{p}_K)$ for classification.

(i) For $b = 1, \dots, B$, draw a bootstrap sample D_b^* from D_n and fit the base learner, giving $\widehat{f}^{*b}$ or $(\widehat{g}^{*b}, \widehat{p}_{b1}, \dots, \widehat{p}_{bK})$.

(ii) Return the task-specific aggregate: for regression, $\widehat{f}_{\mathrm{bag}}(x) := B^{-1} \sum_{b=1}^B \widehat{f}^{*b}(x)$; for classification, either $\widehat{g}_{\mathrm{vote}}(x) \in \operatorname{argmax}_{1 \le k \le K} \sum_{b=1}^B \mathbf{1}\{\widehat{g}^{*b}(x) = k\}$ or $\bar{p}_k(x) := B^{-1} \sum_{b=1}^B \widehat{p}_{bk}(x)$ with $\widehat{g}_{\mathrm{prob}}(x) \in \operatorname{argmax}_{1 \le k \le K} \bar{p}_k(x)$.

Averaging reduces perturbation variance for unstable base learners.
Random split-feature subsets additionally reduce $\mathrm{corr}(T_b, T_{b'})$.

Theorem 6.3.3. *Squared-Error Benefit of Averaging*
Fix x, let $Z := \widehat{f}^*(x)$ be perturbation-random and independent of $Y \mid X = x$, and set $f_{\mathrm{ag}}(x) := \mathbb{E}_* Z$.
If $\mathbb{E}_* Z^2 < \infty$ and $\mathbb{E}_Y Y^2 < \infty$, then $\mathbb{E}_{*,Y}[(Y - Z)^2] = \mathbb{E}_Y[(Y - f_{\mathrm{ag}}(x))^2] + \mathbb{E}_*[(Z - f_{\mathrm{ag}}(x))^2]$.

Proof. Let $\mu := f_{\mathrm{ag}}(x) = \mathbb{E}_* Z$.
Since $Y - \mu$ and $Z - \mu$ are independent and $\mathbb{E}_*(Z - \mu) = 0$, $\mathbb{E}_{*,Y}[(Y - \mu)(Z - \mu)] = \mathbb{E}_Y(Y - \mu)\mathbb{E}_*(Z - \mu) = 0$.
Expanding $Y - Z = (Y - \mu) - (Z - \mu)$ gives $\mathbb{E}_{*,Y}[(Y - Z)^2] = \mathbb{E}_Y[(Y - \mu)^2] + \mathbb{E}_*[(Z - \mu)^2]$.

For independent fits Z_b, the average $\widehat{f}_B(x) := B^{-1}\sum_{b=1}^{B} Z_b$ has mean μ and perturbation variance $\mathrm{Var}_*(Z)/B$.
Structural Commentary: squared loss decomposes into target risk plus perturbation variance.
Key Insight: bagging estimates $\mathbb{E}_*\widehat{f}^*(x)$; unstable base learners gain most because $\mathrm{Var}_*(Z)$ is large. □

Theorem 6.3.4. *Correlated Ensemble Variance*
Let $Z_1,\dots,Z_B$ *have* $\mathrm{Var}(Z_b)=\sigma^2>0$ *and* $\mathrm{Corr}(Z_b,Z_c)=\rho$ *for* $b\neq c$.
For $\bar{Z}_B := \frac{1}{B}\sum_{b=1}^{B} Z_b$, $\mathrm{Var}(\bar{Z}_B)=\rho\sigma^2+\frac{1-\rho}{B}\sigma^2$; *hence, if* $\rho\geq 0$, $\mathrm{Var}(\bar{Z}_B)\to\rho\sigma^2$ *as* $B\to\infty$.

Proof. Since $\mathrm{Cov}(Z_b,Z_c)=\rho\sigma^2$ for $b\neq c$, $\mathrm{Var}(\bar{Z}_B)=B^{-2}\left\{\sum_{b=1}^{B}\mathrm{Var}(Z_b)+2\sum_{1\leq b<c\leq B}\mathrm{Cov}(Z_b,Z_c)\right\}$.
Substitution gives $\mathrm{Var}(\bar{Z}_B)=B^{-2}\left\{B\sigma^2+B(B-1)\rho\sigma^2\right\}=\rho\sigma^2+\frac{1-\rho}{B}\sigma^2$.
Structural Commentary: $\mathrm{Var}(\bar{Z}_B)=\rho\sigma^2+(1-\rho)\sigma^2/B$ separates shared and idiosyncratic variance.
Key Insight: increasing B shrinks only $(1-\rho)\sigma^2/B$; Random Forest lowers ρ by split randomisation. □

Algorithm 13: Random Forest

Require: Training data $(x_i,y_i)_{i=1}^n$, number of trees B, candidate split count $m\leq p$, tree stopping rule
Ensure: Trees $T_1,\dots,T_B$ and an aggregated predictor
for $b=1,\dots,B$ **do**
Draw a bootstrap resample S^{*b} of size n
Grow a tree T_b on S^{*b}
for each nonterminal node while growing T_b **do**
Select m of the p predictors at random
Choose the best split using only those m predictors
Split the node and continue until the stopping rule is met
end for
end for
For regression, return $\widehat{f}_{\mathrm{rf}}(x)=B^{-1}\sum_{b=1}^{B}T_b(x)$
For classification, return a majority vote or averaged class-probability rule from $T_1(x),\dots,T_B(x)$

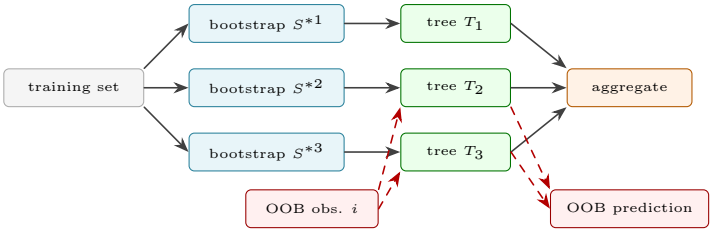

Figure 263: Blue bootstraps train green trees with m split candidates; black arrows aggregate, red dashed arrows give OOB prediction for i.

Method 6.3.5. *Out-of-Bag Error and Forest Importance*
Let I_b^* index bootstrap sample for T_b, $O_i:=\{b: i\notin I_b^*\}$, $\mathcal{I}_{\mathrm{oob}}:=\{i:|O_i|>0\}$, and L be prediction loss.

(i) For each $i\in\mathcal{I}_{\mathrm{oob}}$, construct $\widehat{h}_{\mathrm{oob}}(x_i)$: for regression, set $\widehat{h}_{\mathrm{oob}}(x_i):=|O_i|^{-1}\sum_{b\in O_i}T_b(x_i)$; for classification, use an OOB vote or averaged class-probability vector over $b\in O_i$.
(ii) Set $\widehat{R}_{\mathrm{oob}}:=|\mathcal{I}_{\mathrm{oob}}|^{-1}\sum_{i\in\mathcal{I}_{\mathrm{oob}}}L(y_i,\widehat{h}_{\mathrm{oob}}(x_i))$ and update the estimate as B changes.

For feature j, let $\mathcal{S}_{bj}$ be j-splits of T_b, with gains ΔQ_{bs} and OOB losses $R_b^{\mathrm{oob}}, R_{bj}^{\pi}$ before/after permuting X_j.
Define $I_j^{\mathrm{split}}:=\sum_{b=1}^{B}\sum_{s\in\mathcal{S}_{bj}}\Delta Q_{bs}$ and $I_j^{\pi}:=B^{-1}\sum_{b=1}^{B}(R_{bj}^{\pi}-R_b^{\mathrm{oob}})$.
Both are fitted-forest summaries; I_j^{π} is not a refit-without-X_j loss contrast.

Remark 6.3.6. *Proximities, Tuning, and Overfitting*
Let $L_b(i)$ denote the terminal leaf of x_i in T_b, and set $O_{ij}:=\{b: i,j\notin I_b^*\}$.
Define $P_{ij}:=B^{-1}\sum_{b=1}^{B}\mathbf{1}\{L_b(i)=L_b(j)\}$ and, when $|O_{ij}|>0$, $P_{ij}^{\mathrm{oob}}:=|O_{ij}|^{-1}\sum_{b\in O_{ij}}\mathbf{1}\{L_b(i)=L_b(j)\}$.
Thus P is a fitted-forest similarity; it is neither data-intrinsic nor generally a metric.
Tuning effects are parameter-specific: $m\downarrow$ lowers $\mathrm{corr}(T_b,T_{b'})$ and can raise bias; $n_{\min}\uparrow$ smooths terminal leaves; $B\uparrow$ lowers finite-ensemble Monte Carlo error but does not regularise the limiting forest.
Defaults $m_{\mathrm{cls}}\approx\sqrt{p}$ and $m_{\mathrm{reg}}\approx p/3$ are starting values only.

Method 6.3.7. *Error-Correcting Output Codes*
For training data $(x_i,y_i)_{i=1}^n$ with $y_i\in\{1,\dots,K\}$, let d_H denote Hamming distance.
A code matrix $C\in\{0,1\}^{K\times L}$ assigns class k the row $C_{k\cdot}$, and column ℓ induces binary labels $z_{i\ell}:=C_{y_i\ell}$.

(i) Choose C with distinct rows, equivalently $d_H(C_{k\cdot},C_{k'\cdot})>0$ for $k\neq k'$.
(ii) For each $\ell=1,\dots,L$, fit $\widehat{p}_\ell(x)\approx\mathbb{P}(Z_\ell=1\mid X=x)$ to $(x_i,z_{i\ell})_{i=1}^n$.
(iii) For a test point x, compute $\delta_k(x):=\sum_{\ell=1}^{L}|C_{k\ell}-\widehat{p}_\ell(x)|$ for $k=1,\dots,K$ and return $\widehat{k}(x)\in\operatorname{argmin}_{1\leq k\leq K}\delta_k(x)$.

Row separation gives error correction; redundant columns reduce variance and absorb column errors.

Method 6.3.8. *Stacking and Bumping*
Let $\widehat{f}_1, \dots, \widehat{f}_M$ be fitted candidates, $\widehat{f}_m^{-i}(x_i)$ the out-of-sample prediction for observation i from candidate m, $\mathcal{W} \subseteq \mathbb{R}^M$ the admissible weight set, and L a loss. Commonly $\mathcal{W} := \{w \in \mathbb{R}^M : w_m \geq 0, \sum_{m=1}^{M} w_m = 1\}$.

(i) *Stacking*: set $R_{\mathrm{st}}(w) := \sum_{i=1}^{n} (y_i - \sum_{m=1}^{M} w_m \widehat{f}_m^{-i}(x_i))^2$.

Choose $\widehat{w}^{\mathrm{st}} \in \operatorname{argmin}_{w \in \mathcal{W}} R_{\mathrm{st}}(w)$, and return $\widehat{f}_{\mathrm{st}}(x) := \sum_{m=1}^{M} \widehat{w}_m^{\mathrm{st}} \widehat{f}_m(x)$.

(ii) *Bumping*: fit bootstrap candidates $\widehat{f}^{*1}, \dots, \widehat{f}^{*B}$, set $R_b^* := \sum_{i=1}^{n} L(y_i, \widehat{f}^{*b}(x_i))$.

Choose $\widehat{b} \in \operatorname{argmin}_{1 \leq b \leq B} R_b^*$, and return $\widehat{f}^{*\widehat{b}}$.

Coordinate weights $w = e_m$ give cross-validated model selection; general $w \in \mathcal{W}$ gives model averaging.
Stacking averages fixed candidates; bumping selects among resampled fits.

7 Additive Models and Adaptive Basis Methods

Additive and adaptive basis methods sit between rigid linear models and fully local learners. They keep prediction interpretable by building the fitted function from one-dimensional effects, boxes, hinges, or soft routing weights.

7.1 Generalized Additive Models

Definition 7.1.1. *Generalized Additive Model*
Let $X = (X_1, \dots, X_p)$, $x = (x_1, \dots, x_p)$, $\mu(x) := \mathbb{E}[Y \mid X = x]$, $\alpha \in \mathbb{R}$, and $f_j : \mathcal{X}_j \to \mathbb{R}$.

(i) *Additive mean*: $\mu(x) = \alpha + \sum_{j=1}^{p} f_j(x_j)$.

(ii) *Generalized additive model*: for a link g on the range of μ, $g(\mu(x)) = \alpha + \sum_{j=1}^{p} f_j(x_j)$.

(iii) *Identifiability*: with data $x_i = (x_{i1}, \dots, x_{ip})$, require $\sum_{i=1}^{n} f_j(x_{ij}) = 0$ for all j; this separates α from the f_j.

(iv) *Standard links*: Gaussian $g(t) = t$; binomial $g(t) = \log\left(\frac{t}{1-t}\right)$ or $\Phi^{-1}(t)$; counts $g(t) = \log t$.

Algorithm 14: Backfitting for an Additive Regression Model

Require: Training data $(x_i, y_i)_{i=1}^n$, univariate smoothers S_j for predictors $j = 1, \dots, p$
Ensure: Centered component fits $\widehat{f}_1, \dots, \widehat{f}_p$ and intercept $\widehat{\alpha}$
  Set $\widehat{\alpha} \leftarrow n^{-1} \sum_{i=1}^{n} y_i$ and initialize $\widehat{f}_j(x_{ij}) \leftarrow 0$
  repeat
    for $j = 1, \dots, p$ **do**
      $r_{ij} \leftarrow y_i - \widehat{\alpha} - \sum_{k \neq j} \widehat{f}_k(x_{ik})$ for $i = 1, \dots, n$
      $\widehat{f}_j \leftarrow S_j(r_{1j}, \dots, r_{nj})$
      Center $\widehat{f}_j$ by subtracting $n^{-1} \sum_i \widehat{f}_j(x_{ij})$
    end for
  until the component functions change by less than the chosen tolerance

Method 7.1.2. *Local Scoring for Additive Logistic Regression*
For binary data $(x_i, y_i)_{i=1}^n$, $y_i \in \{0, 1\}$, use the additive predictor $\eta_i = \alpha + \sum_{j=1}^{p} f_j(x_{ij})$.

(i) *Mean*: set $p_i = \frac{1}{1+\exp(-\eta_i)}$ for $i = 1, \dots, n$.

(ii) *Working pair*: set $z_i = \eta_i + \frac{y_i - p_i}{p_i(1-p_i)}$ and $w_i = p_i(1 - p_i)$.

(iii) *Weighted fit*: update $(\alpha, f_1, \dots, f_p)$ by weighted backfitting, minimizing $\sum_{i=1}^{n} w_i \{z_i - \alpha - \sum_{j=1}^{p} f_j(x_{ij})\}^2$.

(iv) *Iteration*: recompute η_i and repeat (i)–(iii) until the additive predictor converges.

7.2 Adaptive Basis and Rule Methods

Method 7.2.1. *MARS: Adaptive Piecewise-Linear Basis Search*
For data $(x_i, y_i)_{i=1}^n$ and $x \in \mathbb{R}^p$, use reflected hinges $(x_j - t)_+$ and $(t - x_j)_+$ with knots $t \in \{x_{ij} : i = 1, \dots, n\}$.
A MARS fit is $\widehat{f}(x) = \widehat{\beta}_0 + \sum_{m=1}^{M} \widehat{\beta}_m h_m(x)$, with each h_m a selected hinge product of order $\leq d_{\max}$.

(i) *Forward pass*: add the admissible hinge pair or product giving the largest decrease in $\sum_{i=1}^{n} (y_i - \widehat{f}(x_i))^2$.

(ii) *Backward pass*: prune the forward sequence.

Score model λ by $\mathrm{GCV}(\lambda) = \frac{\sum_{i=1}^{n} (y_i - \widehat{f}_\lambda(x_i))^2}{\{1 - M(\lambda)/n\}^2}$, with effective size $M(\lambda)$.

(iii) *Selection*: choose the pruned model with smallest GCV.

The cap $d_{\max}$ controls interaction order; $d_{\max} = 1$ gives an additive piecewise-linear model.

Method 7.2.2. *PRIM Box Peeling*
For data $(x_i, y_i)_{i=1}^n$, a box $B = \prod_{j=1}^p [a_j, b_j]$, set $I(B) = \{i : x_i \in B\}$ and $\bar{y}(B) = |I(B)|^{-1} \sum_{i \in I(B)} y_i$.

Fix peeling fraction $\rho \in (0,1)$ and minimum size $m_{\min}$.

(i) *Initialise*: take B to be the full training box.
(ii) *Peel*: while $|I(B)| > m_{\min}$, replace B by the one-face deletion $B' \subset B$ removing a ρ fraction of $I(B)$ with maximal $\bar{y}(B')$.
(iii) *Paste*: expand a face when the resulting box B' increases $\bar{y}(B')$.
(iv) *Select*: choose B^* from the peeling-pasting path by validation or constraints; remove $I(B^*)$ and repeat for additional boxes.

PRIM returns high-mean rectangular regions rather than a recursive partition.

Algorithm 15: Patient Rule Induction Method

Require: Data $(x_i, y_i)_{i=1}^n$, peel fraction ρ, minimum size $m_{\min}$, box count q
Ensure: Selected boxes $\mathcal{S} = (B_1^*, \dots, B_q^*)$
  $D \leftarrow \{1, \dots, n\}$, $\mathcal{S} \leftarrow ()$
  for $r = 1, \dots, q$ **do**
    $I_D(B) := \{i \in D : x_i \in B\}$, $\bar{y}_D(B) := |I_D(B)|^{-1} \sum_{i \in I_D(B)} y_i$
    $B \leftarrow \prod_{j=1}^p [\min_{i \in D} x_{ij}, \max_{i \in D} x_{ij}]$, $\mathcal{P} \leftarrow \{B\}$
    repeat
      $\mathcal{C}_\rho(B) \leftarrow \{B' \subset B : |I_D(B')| \geq (1-\rho)|I_D(B)|\}$ among one-face deletions
      $B \leftarrow \operatorname{argmax}_{B' \in \mathcal{C}_\rho(B)} \bar{y}_D(B')$, $\mathcal{P} \leftarrow \mathcal{P} \cup \{B\}$
    until $|I_D(B)| \leq m_{\min}$
    $\mathcal{E}(B) \leftarrow \{B' \supset B : \bar{y}_D(B') > \bar{y}_D(B)\}$ among one-face expansions
    if $\mathcal{E}(B) \neq \varnothing$ **then**
      $B \leftarrow \operatorname{argmax}_{B' \in \mathcal{E}(B)} \bar{y}_D(B')$, $\mathcal{P} \leftarrow \mathcal{P} \cup \{B\}$
    end if
    $B_r^* \leftarrow \text{Select}(\mathcal{P})$, $\mathcal{S} \leftarrow (\mathcal{S}, B_r^*)$, $D \leftarrow D \setminus I_D(B_r^*)$
  end for

Remark 7.2.3. *Adaptive Structure Versus Tree Structure*

(i) *CART*: represents $\mathcal{X}$ by a binary-tree partition; leaves carry constant local fits.
(ii) *PRIM*: represents high-response regions by boxes $B \subset \mathcal{X}$ without requiring a tree partition.
(iii) *MARS*: represents $\widehat{f}$ by hinge products h_m and coefficients $\widehat{\beta}_m$, linear in $\widehat{\beta}$.

CART emphasizes tree interpretability; PRIM keeps box rules; MARS trades boxes for smoother hinge bases.

Remark 7.2.4. *Soft Trees and Mixtures of Experts*
Let node v have children $1, \dots, K_v$, gates γ_{vk}, leaves $\mathcal{L}$, leaf path $P(\ell)$, and expert $m_\ell(x; \theta_\ell)$.

(i) *Gate*: $g_{vk}(x) = \frac{\exp(\gamma_{vk}^\top x)}{\sum_{r=1}^{K_v} \exp(\gamma_{vr}^\top x)}$, so $\sum_{k=1}^{K_v} g_{vk}(x) = 1$.
(ii) *Leaf weight*: $\pi_\ell(x) = \prod_{(v,k) \in P(\ell)} g_{vk}(x)$.
(iii) *Prediction*: $\widehat{m}(x) = \sum_{\ell \in \mathcal{L}} \pi_\ell(x) m_\ell(x; \theta_\ell)$.
(iv) *Fitting*: estimate (γ, θ) by maximizing the mixture likelihood, often via EM-type alternating updates.

Soft trees replace hard branch indicators by input-dependent probabilities while retaining tree-local experts.

Remark 7.2.5. *Missing Predictors*
For observation i, let $O_i = \{j : x_{ij} \text{ observed}\}$ and $M_i = \{1, \dots, p\} \setminus O_i$; specify prediction from x_{i,O_i}.

(i) *Trees*: use surrogate splits or a categorical missing level.
(ii) *Single imputation*: replace x_{ij}, $j \in M_i$, by $\widehat{x}_{ij}$ before fitting additive or basis models.
(iii) *Model-based imputation*: estimate $\widehat{x}_{ij}$ from a feature model $\widehat{m}_j(x_{i,O_i})$.
(iv) *Multiple imputation*: fit on completed data sets $X^{(1)}, \dots, X^{(L)}$ to propagate imputation uncertainty.

Single imputation treats $\widehat{x}_{ij}$ as observed; multiple imputation propagates missing-data uncertainty.

8 Boosting and Additive Trees

8.1 Stagewise Boosting

Boosting is a stagewise way to build a strong predictor from simple base learners. The key object is an additive score: later terms are trained against the current fit rather than fit independently.

Definition 8.1.1. *Boosting*
Let $\mathcal{H}$ be a weak-learner class and set $F_0 \equiv 0$.

(i) *Additive score*: $F_M(x) = \sum_{m=1}^M \alpha_m h_m(x)$, with $h_m \in \mathcal{H}$ and $\alpha_m \in \mathbb{R}$.
(ii) *Stagewise update*: choose (α_m, h_m) using residuals, gradients, or weights from F_{m-1}; $F_m = F_{m-1} + \alpha_m h_m$.
(iii) *Prediction*: regression uses $F_M(x)$; binary classification uses $\operatorname{sign} F_M(x)$; multiclass scores use $\operatorname{argmax}_k F_{Mk}(x)$.

Boosting is the stagewise construction of F_M from weak learners.

Method 8.1.2. *Forward Stagewise Additive Modeling*
Let $b(x; \gamma)$ range over a basis class and let $L(y, f)$ be a pointwise loss.

(i) *Initialize*: choose f_0.

(ii) *Fit step*: at stage m, take $(\beta_m, \gamma_m) \in \operatorname{argmin}_{\beta,\gamma} \sum_{i=1}^{n} L\{y_i, f_{m-1}(x_i) + \beta b(x_i;\gamma)\}$.
(iii) *Update*: set $f_m(x) = f_{m-1}(x) + \beta_m b(x;\gamma_m)$.
(iv) *Stop*: after M stages, return f_M.

Previously added terms are fixed; for squared-error loss, step (ii) fits the current residuals.

Remark 8.1.3. *Stagewise Paths and Lasso Geometry*

Let $\mathcal{T} = \{T_1, \dots, T_K\}$ be a tree dictionary and $f_\alpha(x) = \sum_{k=1}^{K} \alpha_k T_k(x)$.

For large K, direct least squares is ill posed; a penalized target is $\min_\alpha \{\sum_{i=1}^{n} (y_i - f_\alpha(x_i))^2 + \lambda J(\alpha)\}$.

With $J(\alpha) = \sum_{k=1}^{K} |\alpha_k|$, active trees are sparse.

Small-step boosting traces a forward-stagewise path through this dictionary; shrinkage shortens each move.

8.2 AdaBoost and Exponential Loss

Algorithm 16: AdaBoost.M1

Require: Binary training data (x_i, y_i) with $y_i \in \{-1, 1\}$, weighted classification learner, iterations M
Ensure: Classifier $G(x)$

```
Initialize w_i ← 1/n for i = 1, ..., n
for m = 1, ..., M do
    Fit a classifier G_m using observation weights w_i
    err_m ← (Σ_{i=1}^n w_i 1_{y_i ≠ G_m(x_i)}) / (Σ_{i=1}^n w_i)
    α_m ← log((1 − err_m)/err_m)
    for i = 1, ..., n do
        w_i ← w_i exp{α_m 1_{y_i ≠ G_m(x_i)}}
    end for
end for
return G(x) = sign(Σ_{m=1}^M α_m G_m(x))
```

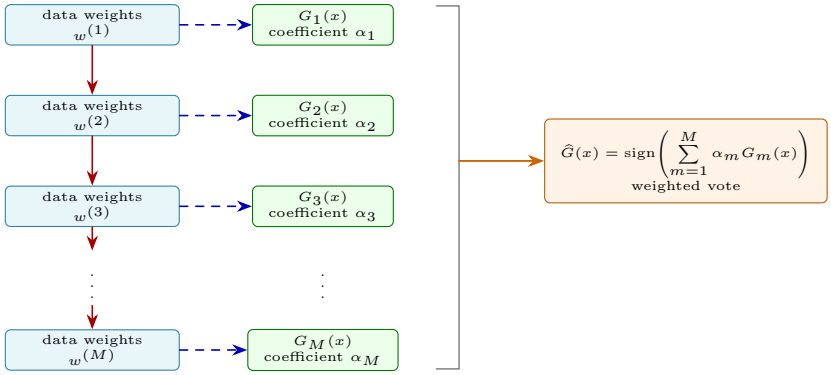

Figure 264: AdaBoost alternates fitting weak learners with updating observation weights, then aggregates them by a weighted vote.

Proposition 8.2.1. *AdaBoost as Exponential-Loss Stagewise Fitting*

For $y_i \in \{-1, 1\}$, use $L(y, f) = \exp(-yf)$. Fix stage m, current score f_{m-1}, weights $w_i^{(m)} := \exp\{-y_i f_{m-1}(x_i)\}$, and classifier $G : \mathcal{X} \to \{-1, 1\}$. Set $Q_m(\beta, G) := \sum_{i=1}^{n} L\{y_i, f_{m-1}(x_i) + \beta G(x_i)\}$ and $E_m(G) := \sum_{i=1}^{n} w_i^{(m)} \mathbf{1}_{\{y_i \neq G(x_i)\}}$.

(i) For fixed $\beta > 0$, $\operatorname{argmin}_G Q_m(\beta, G) = \operatorname{argmin}_G E_m(G)$.

(ii) If $G_m \in \operatorname{argmin}_G E_m(G)$ and $\mathrm{err}_m := E_m(G_m) / \sum_{i=1}^{n} w_i^{(m)}$, then $\beta_m = \frac{1}{2} \log\left(\frac{1-\mathrm{err}_m}{\mathrm{err}_m}\right)$ minimizes $Q_m(\beta, G_m)$.

(iii) The induced weights satisfy $w_i^{(m+1)} = w_i^{(m)} \exp\{-\beta_m y_i G_m(x_i)\}$. With $\alpha_m = 2\beta_m$, a misclassified point has relative factor $\exp(\alpha_m)$.

Remark 8.2.2. *Exponential Loss and Class Probabilities*

For $Y \in \{-1, 1\}$ and $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$, the population minimizer of $\mathbb{E}\{\exp(-Yf(X)) \mid X = x\}$ is $f^*(x) = \frac{1}{2} \log \frac{\eta(x)}{1-\eta(x)}$. Hence $\operatorname{sign} f^*(x)$ is the Bayes rule. The binomial deviance $\log\{1 + \exp(-2Yf(X))\}$ has the same minimizer but penalizes large negative margins less aggressively on finite samples.

8.3 Gradient Boosted Trees and Rule Ensembles

Definition 8.3.1. *Additive Tree Expansion*

Let $R_1, \dots, R_J$ be terminal regions, $\gamma_j \in \mathbb{R}$, and $\Theta = \{(R_j, \gamma_j) : 1 \le j \le J\}$.

(i) *Tree*: $T(x;\Theta) = \sum_{j=1}^{J} \gamma_j \mathbf{1}_{\{x \in R_j\}}$.

(ii) *Additive expansion*: $f_M(x) = \sum_{m=1}^{M} T(x;\Theta_m)$.

(iii) *Terminal fit*: with regions R_{jm} fixed, $\widehat{\gamma}_{jm} \in \operatorname{argmin}_{\gamma} \sum_{x_i \in R_{jm}} L\{y_i, f_{m-1}(x_i) + \gamma\}$.

Algorithm 17: Gradient Tree Boosting

Require: Training data (x_i, y_i), differentiable loss $L(y, f)$, tree sizes J_m, learning rate $0 < \nu \leq 1$, iterations M
Ensure: Additive predictor f_M

$f_0(x) \leftarrow \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma)$
for $m = 1, \ldots, M$ **do**
  for $i = 1, \ldots, n$ **do**
    $r_{im} \leftarrow -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f=f_{m-1}}$
  end for
  Fit a regression tree to targets r_{im}, giving terminal regions R_{jm}, $j = 1, \ldots, J_m$
  for $j = 1, \ldots, J_m$ **do**
    $\gamma_{jm} \leftarrow \arg\min_{\gamma} \sum_{x_i \in R_{jm}} L\{y_i, f_{m-1}(x_i) + \gamma\}$
  end for
  $f_m(x) \leftarrow f_{m-1}(x) + \nu \sum_{j=1}^{J_m} \gamma_{jm} \mathbf{1}_{\{x \in R_{jm}\}}$
end for
return f_M

Remark 8.3.2. *Pseudo-Residuals*
At stage m, pseudo-residuals are negative pointwise gradients $r_{im} := -\partial_f L(y_i, f)|_{f=f_{m-1}(x_i)}$.
Gradient boosting fits the next tree to (x_i, r_{im}).

(i) *Squared error*: $r_{im} = y_i - f_{m-1}(x_i)$.
(ii) *Absolute error*: $r_{im} = \operatorname{sign}\{y_i - f_{m-1}(x_i)\}$, away from zero residuals.
(iii) *Multinomial deviance*: $r_{ikm} = \mathbf{1}_{\{y_i = \mathcal{G}_k\}} - p_k(x_i)$, where p_k is the current softmax probability.

Algorithm 18: Gradient Boosting for K-Class Classification

Require: Encoded labels $y_{ik} \in \{0, 1\}$, classes $k = 1, \ldots, K$, tree sizes J_m, iterations M
Ensure: Class-score functions $\widehat{f}_1, \ldots, \widehat{f}_K$
Initialize $f_{k0}(x) \leftarrow 0$ for $k = 1, \ldots, K$
for $m = 1, \ldots, M$ **do**
  $p_k(x) \leftarrow \frac{\exp(f_{k,m-1}(x))}{\sum_{\ell=1}^{K} \exp(f_{\ell,m-1}(x))}$ for $k = 1, \ldots, K$
  for $k = 1, \ldots, K$ **do**
    Compute $r_{ikm} \leftarrow y_{ik} - p_k(x_i)$ for $i = 1, \ldots, n$
    Fit a regression tree to targets r_{ikm}, giving terminal regions R_{jkm}, $j = 1, \ldots, J_m$
    for $j = 1, \ldots, J_m$ **do**
      $\gamma_{jkm} \leftarrow \frac{K-1}{K} \frac{\sum_{x_i \in R_{jkm}} r_{ikm}}{\sum_{x_i \in R_{jkm}} |r_{ikm}|(1 - |r_{ikm}|)}$
    end for
    $f_{km}(x) \leftarrow f_{k,m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jkm} \mathbf{1}_{\{x \in R_{jkm}\}}$
  end for
end for
return $\widehat{f}_k(x) = f_{kM}(x)$ for $k = 1, \ldots, K$

Remark 8.3.3. *Tree Size, Shrinkage, and Subsampling*
Boosted-tree complexity is controlled by iterations M, terminal counts J_m, shrinkage ν, and sampling.

(i) *Iterations*: larger M adds more trees and increases model capacity.
(ii) *Tree size*: a common J caps interactions; stumps ($J = 2$) give main effects, and larger J allows order up to $J - 1$.
(iii) *Shrinkage*: use $f_m = f_{m-1} + \nu T_m$ with $0 < \nu \leq 1$; smaller ν usually requires larger M.
(iv) *Subsampling*: fit T_m on random observations to reduce cost and add averaging-like randomness.

These tuning choices trade optimization speed, interaction depth, and test risk.

Method 8.3.4. *Relative Importance and Partial Dependence*
Let f be a fitted tree ensemble. For $S \subset \{1, \ldots, p\}$, write $C = S^c$ and $x_i = (x_{iS}, x_{iC})$.

(i) *Relative importance*: for predictor j, sum the split-improvement scores over tree nodes splitting on j.
For an ensemble, average over trees and rescale if desired.

(ii) *Partial dependence*: define $f_S(x_S) = \mathbb{E}_{X_C}\{f(x_S, X_C)\}$ and estimate $\widehat{f}_S(x_S) = n^{-1} \sum_{i=1}^{n} f(x_S, x_{iC})$.

Partial dependence averages over the marginal distribution of X_C.
It equals $\mathbb{E}\{f(X) \mid X_S = x_S\}$ only under compatible dependence structure.

Principle 8.3.5. *Bet on Sparsity*

Let $\mathcal{T} = \{T_1, \ldots, T_K\}$ and $f_\alpha = \sum_{k=1}^{K} \alpha_k T_k$, with support $S(\alpha) = \{k : \alpha_k \neq 0\}$.

An ℓ^1 penalty is useful when a low-risk predictor has $|S(\alpha)| \ll K$ and $|S(\alpha)|$ is small relative to n and noise.
Dense signals with many small α_k remain hard to estimate.
Sparsity is meaningful only relative to n, noise level, and $\mathcal{T}$.

Method 8.3.6. *Lasso Post-Processing of Ensembles*
Let a fitted ensemble generate the fixed dictionary $\mathcal{T}_L = \{T_1, \dots, T_M\}$ and let L be the chosen prediction loss.

(i) *Fit path*: compute $\widehat{\alpha}(\lambda) \in \operatorname{argmin}_{\alpha}\{\sum_{i=1}^{n} L(y_i, \alpha_0 + \sum_{m=1}^{M} \alpha_m T_m(x_i)) + \lambda \sum_{m=1}^{M} |\alpha_m|\}$.
(ii) *Select*: choose λ by validation or an information criterion.
(iii) *Return*: use $\widehat{f}_\lambda(x) = \widehat{\alpha}_0 + \sum_{m:\widehat{\alpha}_m(\lambda)\neq 0} \widehat{\alpha}_m(\lambda) T_m(x)$.

The post-processed model can use fewer trees.
A useful generated dictionary is relevant and not nearly duplicated, with subsampling one way to diversify it.

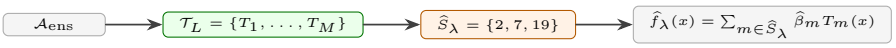

Figure 265: An ensemble generator $\mathcal{A}_{\text{ens}}$ builds the dictionary $\mathcal{T}_L$; lasso selects the active set $\widehat{S}_\lambda$ for the sparse predictor $\widehat{f}_\lambda$.

Method 8.3.7. *Rule Ensembles*
Let $T_1, \dots, T_M$ be fitted trees. A node path p defines the rule $r_p(x) = \mathbf{1}_{\{x \text{ satisfies all splits on } p\}}$.

(i) *Extract*: form $\mathcal{T}^m_{\text{rule}} = \{r_p : p \text{ is a node path in } T_m\}$.
(ii) *Aggregate*: set $\mathcal{T}_{\text{rule}} = \bigcup_{m=1}^{M} \mathcal{T}^m_{\text{rule}}$.
(iii) *Fit*: regress y on $\mathcal{T}_{\text{rule}}$ by lasso or another regularized linear method.
(iv) *Augment*: optionally add original variables x_j as linear terms.

Rules encode split interactions; added linear terms encode main linear effects.

9 High-Dimensional Supervised Learning

High-dimensional supervised learning studies prediction problems in which the number of candidate features is comparable to, or much larger than, the number of observations. The central difficulty is not merely computational size: many rules can fit the training data too well, while the scientific question often asks which features or feature groups are driving the prediction.

9.1 High-Dimensional Linear Rules

Principle 9.1.1. *Less Fitting in the $p \gg n$ Regime*
Let $X \in \mathbb{R}^{n\times p}$, $r = \operatorname{rank}(X)$, and $p \gg n$.

(i) If $r < p$, least-squares coefficients are nonunique since $\ker(X) \neq \{\mathbf{0}\}$.
(ii) Full-covariance discriminant rules are singular when covariance rank is $< p$.
Pooled rank is $\leq n - K$, and class-k rank is $\leq n_k - 1$.
(iii) Impose effective dimension $< p$ via shrinkage, diagonal covariance, row-space fitting, sparsity, or screening.

Measure complexity by $\operatorname{df}(\lambda)$, as in Proposition 2.1.4: ridge keeps coordinates; sparse rules select features.

Method 9.1.2. *Diagonal LDA and Nearest Shrunken Centroids*
For classes $k = 1, \dots, K$ and features $j = 1, \dots, p$, let $\bar{x}_{kj}$ be the class mean, $\bar{x}_{\cdot j}$ the overall mean, s_j the pooled within-class standard deviation, π_k the class prior, n_k the class size, $m_k := (\frac{1}{n_k} - \frac{1}{n})^{1/2}$, and $s_0 > 0$.

(i) *Diagonal LDA*: score x by $\delta_k(x) := -\sum_{j=1}^{p} \frac{(x_j - \bar{x}_{kj})^2}{s_j^2} + 2\log \pi_k$ and set $\widehat{G}(x) := \arg\max_k \delta_k(x)$.
(ii) *Nearest shrunken centroids*: for $\Delta \geq 0$, set $d_{kj} := \frac{\bar{x}_{kj} - \bar{x}_{\cdot j}}{m_k(s_j + s_0)}$, $d'_{kj} := \operatorname{sign}(d_{kj})(|d_{kj}| - \Delta)_+$,
and $\bar{x}'_{kj} := \bar{x}_{\cdot j} + m_k(s_j + s_0)d'_{kj}$; evaluate the diagonal LDA score with $\bar{x}'_{kj}$ in place of $\bar{x}_{kj}$.

Diagonal LDA is nearest-centroid classification after coordinatewise standardization. In nearest shrunken centroids, feature j is removed iff $d'_{kj} = 0$ for all k; increasing Δ increases shrinkage and sparsity.

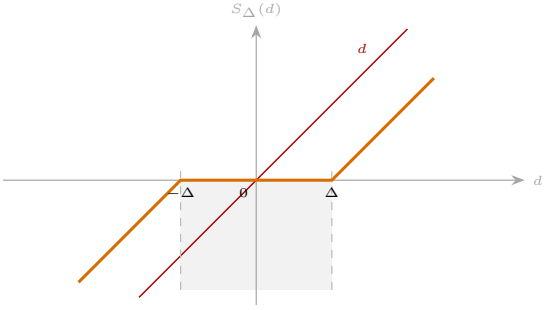

Figure 266: Soft thresholding $S_\Delta(d) = \operatorname{sign}(d)(|d| - \Delta)_+$. The shaded band $|d| \leq \Delta$ maps to zero; outside it, contrasts shrink toward zero by Δ.

Method 9.1.3. *Quadratic Regularization for Linear Rules*
Let $\eta_k(x) := \beta_{k0} + x^T\beta_k$, $\lambda > 0$, and $\gamma \in [0,1]$.

(i) *L_2 multinomial logits*: set $p_{\beta,k}(x) := \frac{\exp(\eta_k(x))}{\sum_{\ell=1}^{K}\exp(\eta_\ell(x))}$ and choose $\widehat{\beta} \in \arg\max_{\beta}\{\sum_{i=1}^{n}\log p_{\beta,g_i}(x_i) - \frac{\lambda}{2}\sum_{k=1}^{K}\|\beta_k\|_2^2\}$; leave β_{k0} unpenalized.
(ii) *Covariance shrinkage*: set $\widehat{\Sigma}_\gamma := \gamma\widehat{\Sigma} + (1-\gamma)\operatorname{diag}(\widehat{\Sigma})$ and use $\widehat{\Sigma}_\gamma$ in the discriminant score.

Both variants impose quadratic stabilization without sparsity. The endpoint $\gamma = 0$ gives diagonal LDA; increasing γ reintroduces off-diagonal covariance information.

Proposition 9.1.4. *Row-Space Reduction for Quadratic Penalties*
Let $X \in \mathbb{R}^{n\times p}$ have rank r and thin SVD $X = UDV^T = RV^T$, where $R := UD \in \mathbb{R}^{n\times r}$ and $V^TV = I_r$.
Fix $y \in \mathbb{R}^n$ and $\lambda > 0$.

(i) Ridge regression satisfies $\widehat{\beta}_\lambda^{\text{ridge}} = V(R^TR + \lambda I_r)^{-1}R^Ty$.
(ii) If $F(\beta_0,\beta) := \sum_{i=1}^{n} L(y_i, \beta_0 + x_i^T\beta) + \lambda\|\beta\|_2^2$ has a minimiser, then one satisfies $\widehat{\beta} \in \operatorname{row}(X) = \operatorname{col}(V)$.

Thus quadratic-penalty fitting may be performed in the r row-space coordinates r_i^T and mapped back by $\widehat{\beta} = V\widehat{\theta}$.

Proof. Every $\beta \in \mathbb{R}^p$ decomposes uniquely as $\beta = V\theta + z$, where $\theta = V^T\beta$ and $z \in \operatorname{col}(V)^\perp = \ker(X)$.
Then $X\beta = R\theta$ and $\|\beta\|_2^2 = \|\theta\|_2^2 + \|z\|_2^2$.

(i) For ridge, dropping z preserves fitted values and weakly decreases the penalty, so a minimiser has $z = 0$.
Thus $\widehat{\theta} \in \arg\min_{\theta}\{\|y - R\theta\|_2^2 + \lambda\|\theta\|_2^2\}$, giving $\widehat{\theta} = (R^TR + \lambda I_r)^{-1}R^Ty$ and $\widehat{\beta} = V\widehat{\theta}$.
(ii) If $(\widehat{\beta}_0, \widehat{\beta})$ minimises F and $\widehat{\beta} = V\widehat{\theta} + \widehat{z}$, then $F(\widehat{\beta}_0, V\widehat{\theta}) \le F(\widehat{\beta}_0, \widehat{\beta})$, with the same linear predictors and no larger penalty. Hence $(\widehat{\beta}_0, V\widehat{\theta})$ is also a minimiser and $V\widehat{\theta} \in \operatorname{col}(V)$.

Structural Commentary: Kernel components affect only the quadratic penalty, not the fitted values.
Key Insight: Quadratic penalties reduce fitting to $\operatorname{row}(X)$. □

9.2 Sparse and Structured Penalties

Method 9.2.1. *Sparse and Structured Penalties*
Let $\lambda, \lambda_1, \lambda_2 \ge 0$ and $\alpha \in [0,1]$. Sparse penalties replace quadratic shrinkage by nonsmooth coordinate penalties.

(i) *Lasso multinomial logits*: maximize $\sum_{i=1}^{n}\log\mathbb{P}(g_i \mid x_i) - \lambda\sum_{k=1}^{K}\sum_{j=1}^{p}|\beta_{kj}|$.
(ii) *Elastic net*: replace $|\beta_{kj}|$ by $\alpha|\beta_{kj}| + (1-\alpha)\beta_{kj}^2$.
(iii) *Fused lasso*: for ordered features, use $\lambda_1\sum_{j=1}^{p}|\beta_j| + \lambda_2\sum_{j=1}^{p-1}|\beta_{j+1} - \beta_j|$.

The L_1 term selects coordinates; the quadratic term shrinks correlated features.
Adjacent differences favor piecewise-constant coefficients.

Method 9.2.2. *Dantzig Selector*
Assume centered y and standardised columns x_j of X.
For residual $r(\beta) := y - X\beta$, the Dantzig selector has two equivalent constrained forms.

(i) *Correlation budget*: choose $\widehat{\beta}_s^{\text{DS}} \in \arg\min_{\beta}\|\beta\|_1$ subject to $\|X^Tr(\beta)\|_\infty \le s$.
(ii) *Coefficient budget*: choose $\widehat{\beta}_t^{\text{DS}} \in \arg\min_{\|\beta\|_1\le t}\|X^Tr(\beta)\|_\infty$.

Both Dantzig and lasso control $\|\beta\|_1$ and monitor residual correlations $x_j^Tr(\beta)$.
Lasso minimises residual sum of squares, while Dantzig directly bounds $\|X^Tr(\beta)\|_\infty$.

Method 9.2.3. *Grouped Lasso*
Let $\{1,\dots,p\} = G_1 \sqcup \cdots \sqcup G_L$, with $|G_\ell| = p_\ell$, design block $X_\ell \in \mathbb{R}^{n\times p_\ell}$, coefficient block $\beta_\ell \in \mathbb{R}^{p_\ell}$, and $\lambda \ge 0$.
The grouped lasso solves $(\widehat{\beta}_0, \widehat{\beta}_1, \dots, \widehat{\beta}_L) \in \arg\min_{\beta_0,\beta_1,\dots,\beta_L}\left\{\left\|y - \beta_0\mathbf{1} - \sum_{\ell=1}^{L}X_\ell\beta_\ell\right\|_2^2 + \lambda\sum_{\ell=1}^{L}\sqrt{p_\ell}\,\|\beta_\ell\|_2\right\}$.
Since $\|\beta_\ell\|_2 = 0$ iff $\beta_\ell = \mathbf{0}$, the penalty selects or removes entire groups; $\sqrt{p_\ell}$ normalizes group size.

Method 9.2.4. *Relaxed, Adaptive, and Nonconvex Lasso Variants*
Let $\widehat{\beta}_\lambda^{\text{lasso}}$ be the lasso estimate, $A_\lambda := \{j : \widehat{\beta}_{\lambda,j}^{\text{lasso}} \ne 0\}$, $\widetilde{\beta}$ an initial estimate, $\nu > 0$, and $a \ge 2$.

(i) *Relaxed lasso*: fit the lasso to get A_λ, then refit on X_{A_λ} by least squares or with penalty $\phi\lambda$, $0 \le \phi \le 1$.
(ii) *Adaptive lasso*: set $w_j := \frac{1}{|\widetilde{\beta}_j|^\nu}$ and minimize $\frac{1}{2}\|y - X\beta\|_2^2 + \lambda\sum_{j=1}^{p}w_j|\beta_j|$.
(iii) *SCAD*: use the nonconvex penalty $J_{a,\lambda}$ with $J'_{a,\lambda}(t) := \lambda\operatorname{sign}(t)\left[\mathbf{1}_{\{|t|\le\lambda\}} + \frac{(a\lambda - |t|)_+}{(a-1)\lambda}\mathbf{1}_{\{|t|>\lambda\}}\right]$.

All three reduce lasso shrinkage bias for large signals.
Relaxed and adaptive lasso keep convex refits or weighted convex fits, while SCAD is nonconvex.

Remark 9.2.5. *Three Roles for Feature Selection*
In high-dimensional learning, feature selection has three distinct roles.

(i) *Preprediction screening*: reduce variance before fitting, as in nearest shrunken centroids.
(ii) *Sparse prediction*: estimate a sparse predictive rule, as in lasso or elastic-net logistic regression.
(iii) *Scientific discovery*: report selected features for separate assessment.
Multiplicity control is then inferential, not predictive.

9.3 Latent Factors and Distance-Based Classifiers

Algorithm 19: Supervised Principal Components

Require: Standardized feature matrix X, outcome y, association scores s_j, thresholds $0 \le \theta_1 < \cdots < \theta_K$, candidate component counts $\mathcal{M}$
Ensure: A selected principal-component predictor
  for $k = 1, \dots, K$ **do**
    $A_k \leftarrow \{j : |s_j| > \theta_k\}$
    for $m \in \mathcal{M}$ **do**
      Compute the first m principal components of the submatrix X_{A_k}
      Fit the outcome model using those components as predictors
      Estimate held-out prediction error by cross-validation
    end for
  end for
  Choose (θ_k, m) by the cross-validation criterion and refit using the selected columns

Remark 9.3.1. *Screening Before Components*
Let s_j measure the marginal association between feature x_j and outcome y, and set $A_\theta := \{j : |s_j| > \theta\}$.
Supervised principal components compute PCs of X_{A_θ} rather than all of X.

(i) *Regression score*: take s_j as a standardized univariate regression coefficient for y on x_j.
(ii) *Survival score*: take s_j as the one-feature Cox score for x_j.

Ordinary PCs follow variance in X alone; screening makes the component outcome-guided.
The resulting component is a denoised latent factor from weak but aligned features.

Method 9.3.2. *Preconditioned Lasso*
Given X, y, and $\lambda > 0$, preconditioning uses a denoised response before sparse refitting.

(i) Fit supervised principal components and compute fitted values $\widehat{y} = (\widehat{y}_1, \dots, \widehat{y}_n)^T$.
(ii) Fit $\widehat{\beta}^{\mathrm{pre}} \in \arg\min_\beta \{\frac{1}{2} \|\widehat{y} - X\beta\|_2^2 + \lambda \|\beta\|_1\}$.

Stage (i) denoises the target; stage (ii) selects original features that reproduce $\widehat{y}$.

Method 9.3.3. *Inner-Product and Distance-Only Classifiers*
Let $K_{ii'} := \langle x_i, x_{i'} \rangle$ be a positive semidefinite Gram matrix, g_i class labels, $n_k = \#\{i : g_i = k\}$, and $M := \frac{1}{n}\mathbf{1}\mathbf{1}^T$.

(i) *Nearest neighbor*: compute $\|x_i - x_{i'}\|^2 = K_{ii} + K_{i'i'} - 2K_{ii'}$.
(ii) *Nearest centroid*: for test point x_0, compute $\|x_0 - \bar{x}_k\|^2 = \langle x_0, x_0 \rangle - \frac{2}{n_k} \sum_{g_i = k} \langle x_0, x_i \rangle + \frac{1}{n_k^2} \sum_{g_i = k} \sum_{g_{i'} = k} \langle x_i, x_{i'} \rangle$.
(iii) *Distance-only input*: for squared distances $\Delta_{ii'}^2$, set $B_{ii'} := -\frac{1}{2}\Delta_{ii'}^2$ and $\widetilde{K} := (I - M)B(I - M)$.

Thus nearest-neighbor, nearest-centroid, kernel PCA, and related classifiers require only inner products or centered distances, not explicit feature vectors.

Example 9.3.4. *String Kernels*
For strings over a finite alphabet, let $\mathcal{A}_m$ be the set of length-m substrings, and let $\phi_a(x)$ count occurrences of a in string x. Define $\Phi_m(x) := (\phi_a(x))_{a \in \mathcal{A}_m}$ and $K_m(x, x') := \langle \Phi_m(x), \Phi_m(x') \rangle$.
Although $|\mathcal{A}_m|$ grows exponentially in m, $K_m(x, x')$ can often be computed without forming $\Phi_m(x)$.
Thus support-vector or centroid classifiers can use sequence similarities directly.

10 Prototype and Nearest-Neighbor Methods

Prototype and nearest-neighbor classifiers are model-free rules built from distances in feature space. They can fit irregular decision boundaries and are often strong predictors, but their behavior is controlled by geometry: scaling, metric choice, neighborhood size, and storage of representative points. The local-averaging view appears in Method 1.4.2; this section records the classification and metric-design side.

10.1 Prototype Classifiers

Definition 10.1.1. *Labeled Prototype Rule*
Let $\mathcal{D}_n = \{(x_i, g_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^p$ and $g_i \in \{1, \dots, K\}$. Fix prototype counts $R_1, \dots, R_K$, distance d on $\mathbb{R}^p$.

(i) A labeled prototype set is $\mathcal{M} = \{(m_j(k), k) : m_j(k) \in \mathbb{R}^p,\ 1 \le j \le R_k,\ 1 \le k \le K\}$.
(ii) For $x \in \mathbb{R}^p$, choose $(\widehat{j}, \widehat{k}) \in \arg\min_{1 \le k \le K,\ 1 \le j \le R_k} d(x, m_j(k))$ and define $\widehat{G}(x) = \widehat{k}$.
(iii) If $\mathcal{M} = \{(x_i, g_i) : 1 \le i \le n\}$, then $\widehat{G}$ is the one-nearest-neighbor classifier.

For quantitative features, Euclidean distance requires comparable coordinate scales.

Method 10.1.2. *k-Means Prototypes*
Let $I_k = \{i : g_i = k\}$, fix R prototypes per class, and use $d(x, z) = \|x - z\|_2$.

(i) For each $k = 1, \dots, K$, choose $m_1(k), \dots, m_R(k) \in \mathbb{R}^p$ to minimize $\sum_{i \in I_k} \min_{1 \le j \le R} \|x_i - m_j(k)\|_2^2$.
(ii) For $x \in \mathbb{R}^p$, choose $(\widehat{j}, \widehat{k}) \in \arg\min_{1 \le k \le K,\ 1 \le j \le R} \|x - m_j(k)\|_2$ and set $\widehat{G}(x) = \widehat{k}$.

The stored labeled prototype set is $\mathcal{M} = \{(m_j(k), k) : 1 \le j \le R,\ 1 \le k \le K\}$.
The rule has piecewise-linear Euclidean decision boundaries; since classes are compressed independently, prototypes need not move away from competing classes near a boundary.

Algorithm 20: Learning Vector Quantization

Require: Labeled data $(x_i, g_i)_{i=1}^n$, prototypes per class R, learning rates $\epsilon_t \downarrow 0$

Ensure: Labeled prototypes $m_j(k)$
  Initialize $m_1(k), \dots, m_R(k)$ from training observations in class k, for each k
  for $t = 1, 2, \dots$ **do**
    Sample a training index i
    Choose the closest prototype (j, k) to x_i over all classes and prototype indices
    if $g_i = k$ **then**
      $m_j(k) \leftarrow m_j(k) + \epsilon_t\big(x_i - m_j(k)\big)$
    else
      $m_j(k) \leftarrow m_j(k) - \epsilon_t\big(x_i - m_j(k)\big)$
    end if
  end for

Remark 10.1.3. *Boundary-Aware Prototype Motion*
For a sampled pair (x_i, g_i), let $(j^\star, k^\star) \in \underset{1\le k\le K,\ 1\le j\le R_k}{\arg\min} d(x_i, m_j(k))$ and let $\epsilon_t > 0$.

(i) If $g_i = k^\star$, set $m_{j^\star}(k^\star) \leftarrow m_{j^\star}(k^\star) + \epsilon_t(x_i - m_{j^\star}(k^\star))$.
(ii) If $g_i \neq k^\star$, set $m_{j^\star}(k^\star) \leftarrow m_{j^\star}(k^\star) - \epsilon_t(x_i - m_{j^\star}(k^\star))$.

Thus every sampled class can move the nearest prototype of any class.
Unlike classwise k-means, LVQ is an online algorithmic rule rather than exact minimization of a fixed criterion.

Method 10.1.4. *Gaussian Mixtures as Soft Prototypes*
Let $\varphi_p(\cdot; \mu, \Sigma)$ be the p-variate Gaussian density. For class k, fix a prior ρ_k, component weights α_{kj}, centers $\mu_{kj} \in \mathbb{R}^p$, covariances $\Sigma_{kj} \succ 0$, and $j = 1, \dots, R_k$, with $\rho_k, \alpha_{kj} \ge 0$, $\sum_{k=1}^{K} \rho_k = 1$, and $\sum_{j=1}^{R_k} \alpha_{kj} = 1$.

(i) Define the class density $f_k(x) = \sum_{j=1}^{R_k} \alpha_{kj} \varphi_p(x; \mu_{kj}, \Sigma_{kj})$.

(ii) For $x \in \mathbb{R}^p$, set $Z(x) = \sum_{\ell=1}^{K} \rho_\ell f_\ell(x) = \sum_{\ell=1}^{K} \sum_{r=1}^{R_\ell} \rho_\ell \alpha_{\ell r} \varphi_p(x; \mu_{\ell r}, \Sigma_{\ell r})$.

(iii) Compute $\tau_k(x) = \frac{\rho_k f_k(x)}{Z(x)}$ and $\omega_{kj}(x) = \frac{\rho_k \alpha_{kj} \varphi_p(x; \mu_{kj}, \Sigma_{kj})}{Z(x)}$. Set $\widehat{G}(x) \in \underset{1\le k\le K}{\arg\max}\, \tau_k(x)$.

The pairs (μ_{kj}, Σ_{kj}) are soft prototypes and $\omega_{kj}(x)$ gives fractional membership.
If $\Sigma_{kj} = \sigma^2 I$ and all $\rho_k \alpha_{kj} > 0$, then as $\sigma^2 \downarrow 0$ the dominant component is nearest center μ_{kj} away from ties.

10.2 Nearest Neighbors

Method 10.2.1. *k-Nearest-Neighbor Classification*
Fix a distance d on the feature space and an integer $k \in \{1, \dots, n\}$.

(i) For a query x_0, compute $d_i = d(x_i, x_0)$ for $i = 1, \dots, n$.
(ii) Let $N_k(x_0) \subset \{1, \dots, n\}$ be indices of k smallest values among $d_1, \dots, d_n$. Set $\widehat{G}(x_0) \in \underset{g\in\{1,\dots,K\}}{\arg\max} \sum_{i\in N_k(x_0)} \mathbf{1}_{\{g_i = g\}}$.

Small k gives lower bias and higher variance; larger k smooths the decision boundary by averaging over more labels. The tuning parameter k is usually chosen by validation or cross-validation.

Proposition 10.2.2. *Asymptotic One-Nearest-Neighbor Error*
Assume $K \ge 2$ and fix x. Let $p_k = \mathbb{P}(G = k \mid X = x)$, choose $k^\star \in \underset{1\le k\le K}{\arg\max} p_k$, and set $L^\star(x) = 1 - p_{k^\star}$.

In the asymptotic one-nearest-neighbor idealisation, the query label G and nearest-neighbor label G' are independent draws from $(p_1, \dots, p_K)$, and the classifier predicts G'.

(i) $L_{1\mathrm{NN}}(x) = \mathbb{P}(G' \neq G) = \sum_{k=1}^{K} p_k(1 - p_k) = 1 - \sum_{k=1}^{K} p_k^2$.

(ii) $L^\star(x) \le L_{1\mathrm{NN}}(x) \le 2L^\star(x) - \frac{K}{K-1}\big(L^\star(x)\big)^2 \le 2L^\star(x)$.

Proof.

(i) Independence gives $\mathbb{P}(G' = G) = \sum_{k=1}^{K} p_k^2$, hence $L_{1\mathrm{NN}}(x) = 1 - \sum_{k=1}^{K} p_k^2$.

(ii) Let $p_\star = p_{k^\star}$ and $q = 1 - p_\star = L^\star(x)$.

Since $p_k \le p_\star$ for every k, $\sum_{k=1}^{K} p_k^2 \le p_\star \sum_{k=1}^{K} p_k = p_\star$, so $L_{1\mathrm{NN}}(x) \ge 1 - p_\star = L^\star(x)$.

Also $\sum_{k\neq k^\star} p_k = q$, so Cauchy-Schwarz gives $\sum_{k\neq k^\star} p_k^2 \ge \frac{q^2}{K-1}$.

Therefore $\sum_{k=1}^{K} p_k^2 \ge (1-q)^2 + \frac{q^2}{K-1}$ and $L_{1\mathrm{NN}}(x) \le 2q - \frac{Kq^2}{K-1} \le 2q = 2L^\star(x)$.

Structural Commentary: Dominant mass gives the lower bound; uniform residual mass gives the upper bound. *Key Insight*: One-nearest-neighbor error is label-disagreement probability at fixed x. □

Example 10.2.3. *Tuning Depends on the Problem*
Let $X = (X_1, \dots, X_{10})$ have independent $\mathrm{Unif}(0, 1)$ coordinates and let $Y \in \{0, 1\}$.

(i) *Halfspace*: $Y = \mathbf{1}_{\{X_1 > 1/2\}}$; the decision boundary is the single hyperplane $X_1 = 1/2$.
(ii) *Checkerboard*: $Y = \mathbf{1}_{\{\mathrm{sign}(\prod_{j=1}^{3}(X_j - 1/2)) > 0\}}$.

The planes $X_j = 1/2$ for $j = 1, 2, 3$ form eight cells with alternating labels.

Across repeated simulations, the preferred neighborhood size k or prototype count R depended on the case. For memory-based and prototype rules, tuning is geometry-dependent, not a universal complexity constant.

Example 10.2.4. *Image Scene Classification by Local Pixel Features*
For each pixel s, let $b_s \in \mathbb{R}^4$ be its four spectral-band vector and let $g_s \in \{1, \dots, 7\}$ be its land-use label.
Replace b_s by the local feature vector $z_s = (b_t : t \in \mathcal{N}_3(s)) \in \mathbb{R}^{36}$, where $\mathcal{N}_3(s)$ is the 3×3 pixel patch centered at s. A five-nearest-neighbor rule on the features z_s had reported test error about 9.5%, showing that local spatial features can make nearest-neighbor classification competitive for irregular class boundaries.

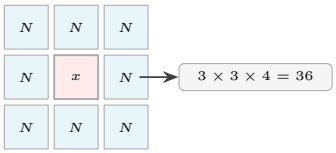

Figure 267: A 3×3 patch over four bands gives 36 features per pixel.

10.3 Invariant Distances and Adaptive Neighborhoods

Method 10.3.1. *Invariant Metrics and Tangent Distance*
Let $x, x' \in \mathbb{R}^p$, let $\mathcal{A}$ be a transformation family, and write $T_\alpha x$ for the transformed input.
For local transformations, let $V_x \in \mathbb{R}^{p \times r_x}$ span the tangent directions at x.

(i) *Orbit metric*: $d_{\text{inv}}(x, x') = \inf_{\alpha, \beta \in \mathcal{A}} \left\| T_\alpha x - T_\beta x' \right\|_2$ compares the transformation orbits of x and x'.

(ii) *Tangent metric*: $d_{\tan}(x, x') = \inf_{a,b} \|x + V_x a - x' - V_{x'} b\|_2$ compares tangent affine spaces.

The tangent metric discounts small local transformations without making large transformations cost-free.
Data augmentation encodes similar invariance by storing transformed copies, which becomes costly when many transformation directions are needed.

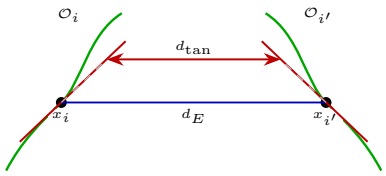

Figure 268: Tangent distance uses local tangents instead of orbits; d_E is ordinary distance.

Method 10.3.2. *Discriminant Adaptive Nearest Neighbors*
DANN replaces the fixed Euclidean metric by a query-dependent quadratic metric.
For a query x_0, choose a pilot index set $P(x_0)$, final neighbor count h, and $\epsilon > 0$. From $\{(x_i, g_i) : i \in P(x_0)\}$ estimate local class proportions π_k, class means $\bar{x}_k$, local mean $\bar{x}$, and pooled within-class covariance $\mathbf{W} \succ 0$.

(i) Compute the local between-class covariance $\mathbf{B} = \sum_{k=1}^{K} \pi_k (\bar{x}_k - \bar{x})(\bar{x}_k - \bar{x})^T$.

(ii) Form $\mathbf{B}^\star = \mathbf{W}^{-1/2} \mathbf{B} \mathbf{W}^{-1/2}$ and $\mathbf{\Sigma}_{x_0} = \mathbf{W}^{-1/2} (\mathbf{B}^\star + \epsilon I) \mathbf{W}^{-1/2}$.

(iii) With $D_{x_0}(x_i, x_0) = (x_i - x_0)^T \mathbf{\Sigma}_{x_0} (x_i - x_0)$, let $N_h(x_0)$ index h smallest distances.

(iv) Set $\widehat{G}(x_0) \in \arg\max_{g \in \{1, \dots, K\}} \sum_{i \in N_h(x_0)} \mathbf{1}_{\{g_i = g\}}$.

Whitening by $\mathbf{W}$ removes local within-class scale.
$\mathbf{B}^\star + \epsilon I$ stretches neighborhoods along directions with weak local class separation, and ϵ prevents degeneracy.

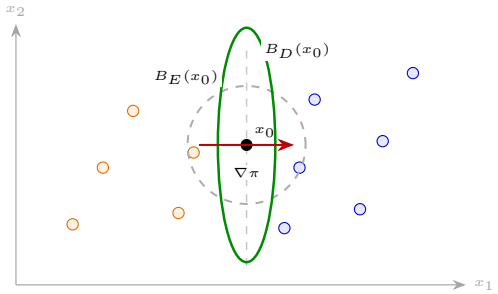

Figure 269: DANN replaces the Euclidean ball by a local ellipsoid, narrow along the class-change direction and elongated where class probabilities vary slowly.

Method 10.3.3. *Global Dimension Reduction for Nearest Neighbors*
Fix a target dimension $L \le p$. Let $\mathbf{B}_i$ be the local between-class matrix estimated at training point x_i.

(i) Compute the global class-separation matrix $\overline{\mathbf{B}} = N^{-1} \sum_{i=1}^{N} \mathbf{B}_i$.

(ii) Let $(\theta_\ell, e_\ell)_{\ell=1}^{p}$ be eigenpairs of $\overline{\mathbf{B}}$ with $\theta_1 \ge \cdots \ge \theta_p$.

(iii) Project by $P_L x = (e_1^T x, \dots, e_L^T x) \in \mathbb{R}^L$ and run nearest-neighbor classification on $P_L x_i$.

The rank-L summary $\overline{\mathbf{B}}_{[L]} = \sum_{\ell=1}^{L} \theta_\ell e_\ell e_\ell^T$ solves $\min_{\mathrm{rank}(M)=L} \sum_{i=1}^{N} \mathrm{trace}[(\mathbf{B}_i - M)^2]$.
The reduced space is selected from local class-separation information, not unsupervised variance alone.

Remark 10.3.4. *Computation and Storage*
With N stored observations in $\mathbb{R}^p$, direct nearest-neighbor search costs $O(Np)$ per query and stores $O(Np)$ coordinates. Storage can be reduced by replacing the retained set $S \subset \{1, \dots, N\}$.

(i) *Prototype compression*: store learned representatives rather than all training observations.
(ii) *Editing*: delete observations whose local neighborhoods have inconsistent labels.
(iii) *Condensing*: grow a small retained set by adding i only if it misclassifies (x_i, g_i).

Fast search reduces query cost in favorable geometries; compression, editing, and condensing reduce storage by keeping representatives or boundary-relevant observations.

11 Flexible and Mixture Discriminant Analysis

11.1 Flexible and Mixture Discriminant Rules

Linear discriminant analysis is useful partly because it is simple: each class is represented by a centroid, distances are measured in a common covariance metric, and the decision boundaries are linear. Its limitations are equally structural. A single centroid may not represent an inhomogeneous class, linear boundaries may be too rigid, and high-dimensional predictors may require smoothness or shrinkage beyond ordinary covariance pooling. Flexible, penalized, and mixture discriminant methods keep the LDA comparison geometry while changing the regression fit, the metric, or the number of class prototypes.

Method 11.1.1. *Flexible and Penalized Discriminant Analysis*
Optimal scoring represents LDA by regressions from x_i to transformed labels.
Let $g_i \in \{1, \dots, K\}$, $1 \le L \le K-1$, label scores $\theta_\ell : \{1, \dots, K\} \to \mathbb{R}$, and fitted score maps $\eta_\ell : \mathcal{X} \to \mathbb{R}$, with centering and orthogonality constraints on the θ_ℓ removing constant score directions.

(i) *Flexible discriminant analysis*: minimize $\mathcal{L}_{\mathrm{FDA}} = n^{-1} \sum_{\ell=1}^{L} \sum_{i=1}^{n} \{\theta_\ell(g_i) - \eta_\ell(x_i)\}^2 + \lambda \sum_{\ell=1}^{L} J(\eta_\ell)$ over θ_ℓ and η_ℓ from a chosen smoother class: linear, spline, additive, MARS, kernel, or similar.
(ii) *Penalized discriminant analysis*: impose $\eta_\ell(x) = h(x)^\top \beta_\ell$ and roughness penalty $\beta_\ell^\top \Omega \beta_\ell$.
Equivalently classify by $D_\lambda(x,k) = \big(h(x) - \bar{h}_k\big)^\top (\widehat{\Sigma}_{W,h} + \lambda\Omega)^{-1} \big(h(x) - \bar{h}_k\big)$, with $\bar{h}_k$ and $\widehat{\Sigma}_{W,h}$ the class-k centroid and pooled within-class covariance in h-coordinates.

For $\eta_\ell(x) = x^\top \beta_\ell$ and no roughness penalty, the fitted directions recover LDA up to scale; FDA changes the regression class to obtain nonlinear boundaries, while PDA shrinks rough or unstable h-directions.

Algorithm 21: FDA Coordinates by Optimal Scoring

Require: Class labels $g_i \in \{1, \dots, K\}$, indicator response matrix $Y \in \mathbb{R}^{n \times K}$, linear smoother S_λ for the chosen regression method
Ensure: Nonconstant discriminant coordinate functions $\eta_1, \dots, \eta_{K-1}$
Fit the multivariate regression of Y on the features: $\widehat{Y} \leftarrow S_\lambda Y$
Let $\eta^*(x)$ denote the K fitted regression functions evaluated at a new input x
Set $D_\pi \leftarrow Y^T Y / n$
Solve $Y^T \widehat{Y} \theta = \gamma D_\pi \theta$ and normalize eigenvectors so that $\Theta^T D_\pi \Theta = I$
Discard the constant scoring direction
for each remaining eigenvector θ_ℓ **do**
  Set $\eta_\ell(x) \leftarrow \theta_\ell^T \eta^*(x)$
end for

Remark 11.1.2. *FDA as LDA After a Regression Map*
Let $\eta = (\eta_1, \dots, \eta_L) : \mathcal{X} \to \mathbb{R}^L$ and $z = \eta(x)$. FDA applies LDA in z-space, so linear discriminants in z pull back to decision boundaries in x determined by the regression class for η.
Polynomial η gives algebraic boundaries, quadratic η gives quadratic boundaries, and additive, spline, or kernel η gives the corresponding structured nonlinear boundaries.

Method 11.1.3. *Mixture Discriminant Analysis*
Mixture discriminant analysis replaces the class-k LDA prototype by R_k Gaussian subclasses while sharing one covariance Σ. Let $\pi_{kr} \ge 0$, $\sum_{r=1}^{R_k} \pi_{kr} = 1$, centers μ_{kr}, class prior ρ_k, $f_k(x) = \sum_{r=1}^{R_k} \pi_{kr} \phi(x; \mu_{kr}, \Sigma)$, and $p_k(x) = \Pr(G = k \mid X = x) = \rho_k f_k(x) / \sum_{j=1}^{K} \rho_j f_j(x)$.
If $R_k = 1$ for all k, this is ordinary common-covariance Gaussian discrimination.

(i) *E-step*: for $g_i = k$, set $r_{ikr} = \pi_{kr} \phi(x_i; \mu_{kr}, \Sigma) / \sum_{s=1}^{R_k} \pi_{ks} \phi(x_i; \mu_{ks}, \Sigma)$ for $r = 1, \dots, R_k$.
(ii) *M-step*: update π_{kr}, μ_{kr}, and Σ by weighted Gaussian maximum likelihood with weights r_{ikr}.
(iii) *Blurred-response variant*: use subclass columns c_{kr} and row i equal to $(r_{ik1}, \dots, r_{ikR_k})$ on class-k columns, zero elsewhere; then fit weighted DA with optional rank restrictions, FDA, or PDA.

Remark 11.1.4. *MDA and Prototype Rules*
MDA is a probabilistic prototype classifier with prototypes μ_{kr}, weights π_{kr}, and shared metric Σ^{-1}. It assigns x by posterior scores $p_k(x) \propto \rho_k \sum_{r=1}^{R_k} \pi_{kr} \phi(x; \mu_{kr}, \Sigma)$, not by a hard nearest-center rule.
Relative to classwise k-means, it estimates π_{kr} and Σ; relative to an unconstrained Gaussian mixture, each component (k, r) is nested inside the observed class k.

Definition 11.1.5. *Kernel Density Classification*
Let $G \in \{1, \dots, J\}$ be the class label.
Estimate $\pi_j := \mathbb{P}(G = j)$ by $\widehat{\pi}_j$ and $f_j(x) := f_{X \mid G=j}(x)$ by a kernel density estimate $\widehat{f}_j(x)$.

(i) *Posterior*: Define $\widehat{p}_j(x) := \widehat{\mathbb{P}}(G = j \mid X = x) := \dfrac{\widehat{\pi}_j \widehat{f}_j(x)}{\sum_{k=1}^{J} \widehat{\pi}_k \widehat{f}_k(x)}$ whenever the denominator is positive.
(ii) *Classifier*: The plug-in Bayes rule is $\widehat{g}(x) \in \underset{1 \le j \le J}{\mathrm{argmax}}\, \widehat{p}_j(x)$.

Full density estimation is unnecessary if only the boundary is needed.

Remark 11.1.6. *Computational Scale*
Let N be the sample size, p the predictor dimension, and T_{fit} the cost of the chosen smoother.

(i) *LDA/PDA*: covariance-scale linear algebra costs about $Np^2 + p^3$ before low-rank reduction.
(ii) *FDA*: cost is T_{fit}; additive or adaptive basis fits can be nearly linear in N for fixed model size, while dense kernel or spline smoothers can cost at least an $N \times N$ solve.
(iii) *MDA*: cost multiplies the common-covariance discriminant fit by EM iterations and depends on initialization because mixture likelihoods are nonconvex.

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